OpenSourceRail Documentation Book

Complete reader edition: project, technical, software, hardware, engineering, manufacturing, country-planning, and city-model documentation.

Generated from 454 source documents and 265 city models.

Book Contents

Each source chapter prints its repository path. The city section is generated directly from each canonical design.toml model.

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Reader-edition boundary

This book includes canonical explanatory and review documentation. The generated Markdown inventory, duplicate city READMEs and catalogue tables, detailed component definitions, and signable shop travelers remain in the repository and are linked by their higher-level chapters; reproducing them here would duplicate source records.

Front Matter

OpenSourceRail

Repository source: `README.md`

nobody should be priced out of dignity

Give your city a railway it can own

OpenSourceRail is an open urban-rail reference platform for countries that want to retain design authority, software, fabrication, integration, operations and maintenance capability in-country. It connects city planning, GIS, CAD/IFC, simulation, project controls, local manufacturing, ERP, equipment supervision and assurance around the same city, asset and engineering-revision identities.

> Important
> Repository outputs are planning and engineering-screening evidence—not bids, construction releases, safety certificates, approvals or endorsements. Open topography and water data do not replace survey, geotechnical or hydraulic evidence. Simulation and formal checks do not authorize live railway command.

OpenSourceRail light-metro reference trainset

Start here: two-page brochure · complete PDF book · one-page overview · architecture · current roadmap and open work

Functional comparison

OpenSourceRail connects corridor choices to engineering revisions, local production, delivery costs and maintenance evidence in one owner-controlled workflow. Its public planning portfolio covers 265 cities in 44 developing countries, plus one European comparison model. The table compares published functions, not performance, price or railway acceptance; specialist tools can complement OSR. Reviewed 12 October 2026: names link to primary product/project sources; capabilities depend on edition and configuration. “Integrated” identifies an existing OSR component; “workflow” identifies an engineering tool used with OSR; “benchmark” implies no installed connector.

Tool / suite	Availability / relationship	Main functions	OpenSourceRail's corresponding value and limits
OpenSourceRail	Open source; connected reference platform	City GIS, service/energy simulation, CAD/IFC, quantities, local manufacture, finite-resource CPM, procurement/cashflow, ERP, supervision and maintenance evidence	Common city/asset/revision identities connect planning through repair; reproducible demonstrations exist, while physical railway qualification and live control acceptance remain open.
Siemens TPS.plan, TPS.live, TPS.trackworks	Commercial; benchmark	Infrastructure-aware timetables, traffic forecasts/conflicts, dispatching and track-work windows	OSR schedules declared physical routes around possessions and resource conflicts, with fleet/energy and staff recovery; search is bounded and live dispatch acceptance remains open.
Alstom Urbalis Vision / ICONIS supervision	Commercial; benchmark	Integrated urban traffic supervision, regulation, SCADA, incident and depot management	OSR joins OCC, simulator and observation-only FUXA views to asset/work records; its management stack is not an accepted live ATS/CBTC product.
Alstom HealthHub / FlexCare digital services	Commercial; benchmark	Fleet, infrastructure and signalling condition monitoring, predictive maintenance and service support	OSR joins condition-to-repair and serial/parts history to exposure/fault trends and held-out degradation/RUL validation; measured calibration and predictive deployment remain open.
Hitachi Rail HMAX	Commercial; benchmark	Digital asset management, harmonised health data, condition/predictive maintenance and digital twins	OSR exposes engineering provenance and condition-to-repair evidence in an adaptable owner-held stack; deployed predictive depth and support equivalence are unproven.
Hitachi Energy EAM / FSM / APM	Commercial; benchmark	Rail asset registers, track defects, work orders, field service and asset performance	OSR links assets, work and independent handback to revisions, with offline typed/photo work packs, durable synchronisation and native-state revalidation; operator commissioning remains open.
Dassault DELMIA Quintiq	Commercial; benchmark	Coupled service, fleet and crew planning, constraint optimisation and disruption replanning	OSR proposes possession-aware service/fleet/energy recovery and bounded staff rosters, retaining deadheads, charging and fixed maintenance; unrestricted optimisation depth remains unproven.
Tracsis ATTUne / TRACS / TRACS Enterprise	Commercial; benchmark	Timetables, vehicle/crew diagrams, rostering, personnel and on-day operations	OSR proposes rosters across operating and maintenance roles, checking competence, travel, rest, breaks and workload; native assignment and deployed rostering acceptance remain separate.
Wabtec RailConnect	Commercial; freight benchmark	Railcar movement, switching, mobile work, freight revenue/billing and industry exchanges	OSR's current reference is urban passenger rail with engineering-to-maintenance continuity; freight settlement and car-hire workflows would require a separate profile.
Petrosoft RAILSoft / mRails	Commercial; benchmark	Fleet planning, maintenance intervals, parts/history and mobile field execution, including offline work	OSR joins ERP asset/parts/people records to offline field evidence, operational/GTFS exchange and configured operator identity adapters; national transport integrations remain deployment-specific.
IBM Maximo Application Suite	Commercial; benchmark	Asset/work management, inspection, reliability, monitoring and predictive maintenance	OSR connects comparable asset/work record categories to public railway definitions and engineering changes; equivalent operating scale, analytics and service support require proof.
Bentley OpenRail Designer	Commercial; benchmark	Specialist alignment, cant, turnouts/yards, corridor modelling and design documentation	OSR carries screening geometry into quantities, IFC, procurement and cashflow; it does not claim equivalent detailed alignment design or issued-for-construction capability.
OSRD	Open source; benchmark	Railway infrastructure, train simulation, timetables, capacity and short-term path requests	OSR extends the workflow into vehicle/civil references, manufacture, delivery and maintenance; breadth does not establish equivalent timetable/capacity depth.
Eclipse SUMO	Open source; workflow	Microscopic multimodal simulation: road, rail, public transport and pedestrians	OSR uses SUMO for independent screening and adds rail engineering, asset and delivery context; numerical agreement requires matched inputs and declared tolerances.
QGIS	Open source; workflow	Raster/vector GIS, spatial analysis, cartography and geospatial exchange	OSR supplies city/route revisions, demand and terrain/water screening connected to service, civil quantities and delivery outputs.
FreeCAD	Open source; workflow	Parametric solid modelling, assemblies, drawings and extensible engineering tools	OSR supplies controlled train/station/civil families, product interfaces, BOMs and manufacturing/evidence gates around native CAD.
Bonsai / IfcOpenShell	Open source; workflow	Native IFC authoring, model review, coordination and programmable BIM	OSR supplies railway asset identities, IFC4.3 packages, quantities, 4D states and executable information requirements linked to revisions.

Tool / suite	Availability / relationship	Main functions	OpenSourceRail's corresponding value and limits
ERPNext / Frappe HR	Open source; integrated	Procurement, stock, manufacturing, quality, projects, assets, finance, employees and payroll	These hold OSR's native business records; OSR adds railway context, engineering-change disposition, competence/resource checks and independent inspection/handback.
FUXA	Open source; integrated	Web SCADA/HMI, process displays and industrial protocol connectivity	OSR supplies versioned observations, durable history/alarms and ERP maintenance linkage; the installed OSR gateway carries no railway command authority.
OpenProject Community	Open source core; benchmark	Project/work-package tracking, collaboration, Gantt schedules and dependencies	OSR adds finite-resource CPM, railway quantities/BOM demand, supplier order-by dates and local/import cashflow; no OpenProject connector is claimed.
CalculiX / OpenSees	Open source; workflow	Finite-element analysis; structural/geotechnical nonlinear and dynamic research	OSR drives source-bound beam–pier–foundation comparisons and retains failed cases, tradeoffs and model limits; solver convergence does not qualify a railway structure.

Why use OSR? A corridor or component change can be reviewed alongside its asset quantities, manufacturing work, procurement, cash needs and maintenance consequences, with inspectable source and retained evidence. Start with the owner/operator platform; use the competitive position and proof roadmap to judge maturity. The six-item operations implementation adds bounded resource/fleet/staff planning, offline evidence, reliability validation and operational exchange; measured/operator commissioning remains separate.

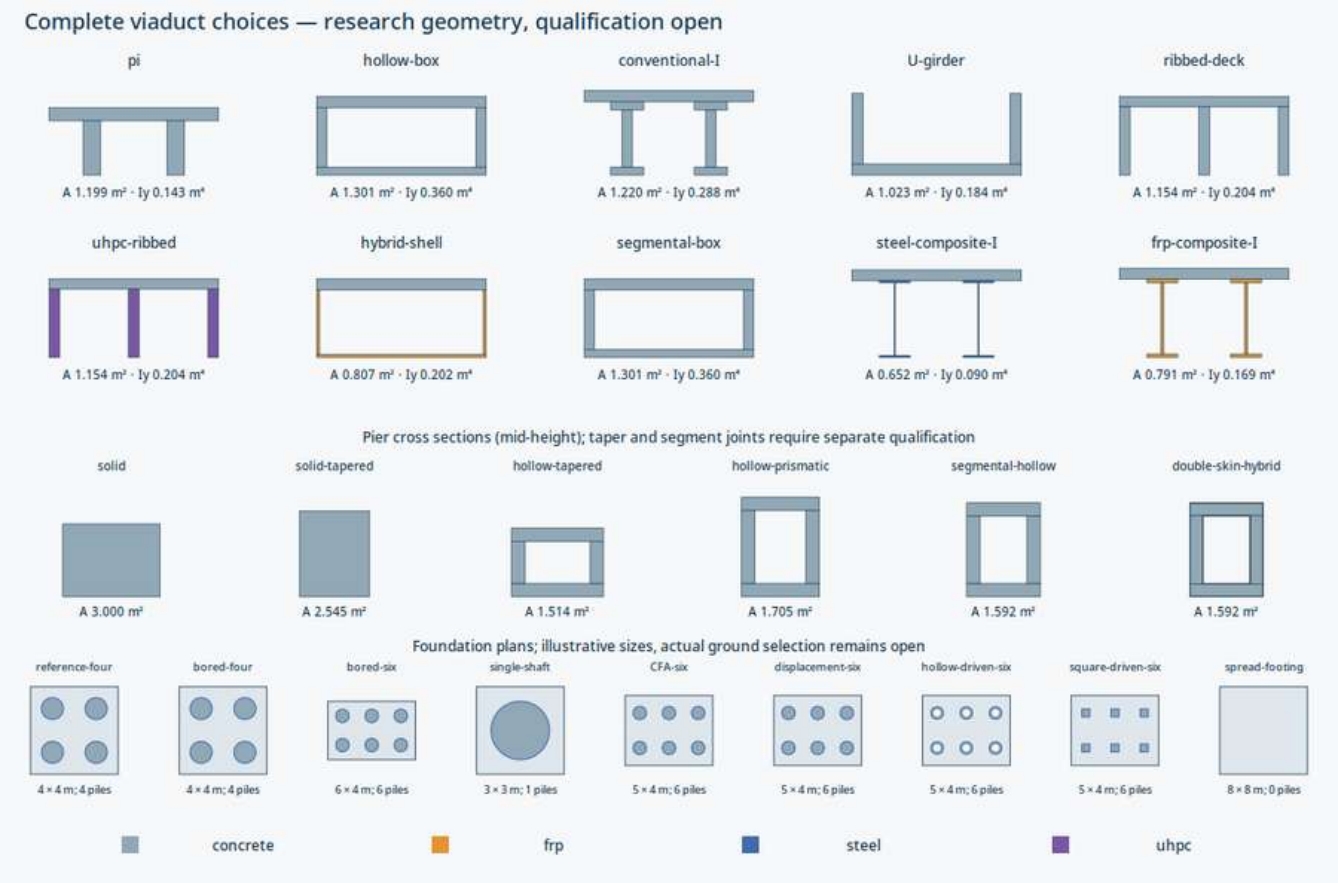
The physical reference is a 49.5 m, three-car driverless light metro with 360 AW2 / 480 AW3 capacity and 675 kWh gross / 540 kWh usable LFP storage, integrated battery/cabin thermal control, seven station archetypes and reusable civil families. A welded steel safety frame, non-structural GFRP cladding and supplier-neutral COTS subsystem interfaces support local fabrication, assembly and repair. Its 120 LM3 product rows, 26 assembly nodes, 30 tooling/mould families and 146 native FreeCAD models with matching split IFC4.3 files connect that make/buy strategy to BOMs, tooling and qualification gates.

The city catalogue uses elevated land sections in its controlled central study areas, radial tangents and curved ring connections where terrain and water permit; retained shore detours and priced bridge candidates replace long lake crossings. Outer approaches retain their actual at-grade, elevated and special-crossing mix, with no tunnels. Each city reports its current civil quantities, one full-fleet planning depot per line, graded staffing and an 18-month factory-readiness plan. These are planning concepts: property, survey, supplier, structural and operational releases remain open. Catalogue design review · Water and interchange checks · Baghdad's dedicated proposal and funding.

Complete viaduct design exploration

The civil workbench compares complete beam–pier–foundation packages, with Baghdad as the qualification target. It searches for separate low-cost, low-mass and short-construction-time choices and retains the Pareto tradeoffs, failures, changed parameters and evidence gaps. The retained LM3 verification studies and Baghdad research results have distinct load bases; Baghdad supplier axle positions are still missing.

Part of the system	Choices investigated
Beams and decks	Pi20/Pi25 controls; hollow box; prestressed I; U; ribbed; targeted UHPC ribs; FRP shell/concrete; segmental box; steel–concrete I; FRP–concrete I. Research spans include 20, 25 and 30 m.
Materials and composites	Normal, high-performance and structural lightweight concrete; targeted UHPC; steel; GFRP and CFRP orthotropic matrices. Properties remain declared research hypotheses until measured.
Piers	Solid; solid tapered; hollow tapered; hollow prismatic; segmental hollow; experimental double-skin FRP/concrete. Reinforcement, confinement, joint and cyclic acceptance remain open.
Foundations	Reference pile group; four/six bored piles; single shaft; CFA; displacement piles; hollow and square driven precast piles; spread footings. Diameter, length and cap sizes are investigated with each package.
Assembly and delivery	Simple spans, link slabs and continuity; full precast, segmental, shell/infill and in-situ construction; finite factory/rig/transport/erection resources, curing and inspection.



Parametric beam, pier and foundation research choices; colours identify materials and blue markers identify pile positions

Executed comparison: 731 distinct complete packages, 779 evaluations, 24 detailed packages, using 300 m of double track.

Conditional objective	Beam	Pier	Foundation	Synthetic installed cost	Installed study mass	Working time
Cheapest in `concrete-first`	U-girder 20 m, lightweight	hollow-tapered	CFA-six	$1,774,796	6,504.4 t	120 working days
Lightest in `concrete-first`	hybrid-shell 20 m, high-performance	hollow-prismatic	square-driven-six	$7,374,804	5,499.2 t	154.625 working days
Fastest in `concrete-first`	uhpc-ribbed 20 m, high-performance	hollow-prismatic	CFA-six	$2,640,915	7,609.1 t	63 working days

These are the best confirmed research choices found within the declared search budget and synthetic price/productivity scenarios. Supplier-priced cheapest design, railway qualification and global optimality remain unresolved. All physical and operating release flags remain false.

Reduced native models screen the packages. A diverse shortlist receives three actual-section CalculiX solid meshes and three 3D OpenSees meshes, including separate tracks, asymmetric uniform loading, braking sensitivity, transverse cap bending, torsion and coupled foundation translation/rotation. Composite-slip diagnostics expose dependence on connectors. Empty shipping/lifting units remain separate from complete service mass; driven-pile production and delivery are included in construction planning. The detailed shortlist contains 16 designs at 20 m, six at 25 m and two at 30 m. All 24 converged; 22 passed every shared provisional constraint. The two 30 m cases remain rejected diagnostics, with their failed limits retained. Every refined load case repeats the shared drift, settlement, foundation-demand, deflection and lifting checks. Force and central solid-stress convergence are recorded as well as displacement convergence. Evolution considers all three commercial scenarios, whole-life cost and erection mass; per-seed benchmarks retain unique evaluations, timing, coverage and observed Pareto quality.

Native FreeCAD 1.1.4 verification passed all 25 registered beam, pier and foundation families and all 24 complete shortlisted assemblies, including valid kernel solids, independent net volumes and agreement with the retained IFC quantities. The source-bound receipt is separate from ordinary Python CAD fallback tests. Reproduce with `FreeCADCmd tools/automation/verify-civil-native-cad.py` in a FreeCAD environment. Structural construction-stage load paths, V/Y supports, haunched sections and calibrated ground improvement remain explicit model extensions; measured inputs, physical validation and independent acceptance remain open.

Reproduce with `tools/automation/osr-python tools/automation/civil-study.py system-study` and `system-verify`; see the retained comparison below for the complete commands. Open the generated `report.html` to filter choices and compare objectives, or read the retained comparison and implementation/acceptance details. The scope includes materials, reinforcement/tendon allowances, foundations, connections, production, transport, plant, temporary

works, labour, QA, site carrying cost, maintenance and bearing/joint replacement scenarios. The shared engineering demonstrators reconcile native CAD mass/CG/inertia, hollow sections, IFC analysis identities, joint screens and evidence-gated mixed searches. Actual Baghdad supplier prices and construction commitments remain open. Static distributed planning loads support research comparisons; actual axle loading, calibrated soil/material/connection behaviour, physical tests and independent acceptance are required for railway qualification. No numerical winner automatically changes the catalogue, city finances or release status.

Feature Highlights

```
flowchart LR
  P[1. PLAN<br/>City Studio / GIS] --> E[2. ENGINEER<br/>CAD / IFC4.3 / simulation] --> D[3. DELIVER<br/>CPM / ERP / QA
/ manufacturing] --> O[4. OPERATE<br/>assets / SCADA views / evidence] --> R[maintain / renew] --> P
```

Digital standards adherence from the first design change

The Digital Standards Thread connects publisher records and deployment law to controls, requirements, hazards, design items, evidence and G1–G4 decisions. CI rejects stale reviews, broken mappings and changed hashes, while its impact index identifies reassessment after change. A green result means evidence coherence—not clause conformity, physical qualification, independent assessment or legal authorization.

1. Turn city data into a costed railway candidate. Rust planning and simulation work with deterministic Python GIS/design generators to connect source-locked city data route and station candidates reference vehicles and civil quantities service, fleet and energy checks procurement, construction schedule and cashflow. In City Studio, edit lines, stations and service over 20 layers including roads, buildings, population/demand, open-DEM elevation/slope and mapped water. Compare catchment and service implications, identify likely crossings and station exclusions, seal a candidate revision and regenerate its delivery twin. This makes alternatives and their costs reviewable from the same inputs. Surveyed walking access, turnout/siding layouts, calibrated running times and engineering release require further evidence; route coverage, battery suitability and safety headways are checked against declared inputs rather than guaranteed by automation. The population, transfer and viaduct-clearance review separates native population catchments from routing demand, includes transfers through intermediate lines, and records beam/building, support/foundation and terrain conflicts. Unknown heights and incomplete mapping remain explicit release gates.

2. Engineer the railway and its interfaces. Export OSR-ALN, GIS and IFC4.3 data for QGIS and Bonsai/IfcOpenShell; generate quantities, classifications, asset identities, 4D states, IDS requirements and BCF-oriented evidence; and use native FreeCAD train, station and civil geometry. System simulation connects trains, stations, energy, wayside, points/crossings, fares, regenerative braking and depots. Route changes flow into named compatibility checks for gradient, curvature, cant, platforms, clearances, braking, energy, crosswind, recovery and evacuation. The design-detail register controls terrain/water impacts, BIM groups, vehicle datums, mechanical interfaces, load cases and route gates. All 146 split LM3 IFC files carry provenance and interface properties; the buildingSMART IDS requirements execute 4 specifications and 945 checks. This closes information delivery—not supplier selection, calculations, tolerance stacks, structural proof, physical tests or authority acceptance.

3. Deliver through local production and accountable project controls. Turn city design into asset registers, BOM demand, finite-resource schedules, supplier candidates, order-by dates, cash requirements, QA gates and construction states. ERPNext holds native projects, procurement, inventory, manufacturing, quality, finance, people and maintenance records. The first-article baseline controls 120 product rows and 26 assembly nodes; 30 tooling families, 16 factory packages and 29 drawing-definition seeds feed nested BOMs, methods, travelers and hold points. Across 62 locally made rows, 472 fail-closed slots—including exterior finish and mass closure—prevent first-article release without required production and approval evidence.

4. Operate, maintain and retain evidence. One Workbench joins City Studio, simulation, OCC, railway works, ERPNext/Frappe HR, FUXA and lifecycle views. Nine native Rust evaluator families publish exact units, identities and provenance through a versioned observation-only contract into durable history, alarms and read-only supervision. Reviewed conditions become ERP maintenance cases and Asset Repairs with parts, technicians, downtime and evidence. End-to-end CI exercises complete-city load, queued delivery, restart persistence, backup and clean-volume recovery. ERP, FUXA and management automation have no path to movement authority, point/barrier command, safety release or engineering acceptance.

Step-by-step use cases

Scenario 1 — Compare two city corridors before funding survey work

1. Select a catalogue city in City Studio, then inspect roads, buildings, demand, existing rail and places. 2. Draw candidate lines and stations; compare OD demand, fleet and timetable implications. 3. Add source-locked open DEM and water evidence. The router classifies likely at-grade, viaduct and bridge segments and rejects stations in mapped exclusions. 4. Model constrained single-track sections with passing at stations and recovery sidings where appropriate. 5. Compile

each candidate into an immutable revision and generate its asset, quantity, cost, cashflow, critical-path and survey-gap outputs. 6. Use the comparison to commission field survey, geotechnical and hydraulic work—not to release a route or structure.

Scenario 2 — Propagate an alignment change into engineering and delivery

1 Seal the revised GIS/OSR-ALN candidate so every downstream result refers to the same bytes. 2. Regenerate civil/station IFC4.3, quantities, classifications, asset IDs and 4D construction states. 3. Run route–vehicle gates for gradient, curvature, cant, platform, clearance, braking, energy, crosswind, evacuation and recovery interfaces. 4. Validate the 146-file LM3 federation against the IDS requirements, then run deterministic FMEA/standards and baseline-drift checks. 5. Recalculate BOM demand, finite-resource CPM, suppliers, order-by dates, QA gates and local/import cashflow; expose affected open ERP work for disposition. 6. Stop at unresolved calculations, supplier data, physical tests or approvals. A green digital report is design evidence, not construction release.

Scenario 3 — Manufacture and commission a locally built subsystem

1 Start from one of 120 controlled product rows and follow its CAD/IFC, mechanical interfaces, material/process fields and nested BOM. 2. Issue a bounded supplier package separating NRE, tooling, first article, repeat units and support; treat catalogue vendors as leads until qualified. 3. Create the ERP project, purchase and receipt records; preserve batch/serial identity and inspection references. 4. Execute the factory traveler and hold points, record NCRs and accepted evidence, then install the serialized item against its planned asset identity. 5. Bind design, execution, inspection and simulation commissioning evidence. A later replacement keeps the removed serial's history and requires fresh evidence. 6. Provision a lookup-only QR only after an operator has configured a secure resolver and independently verified the physical binding; see the governance templates.

Scenario 4 — Turn a train or station fault into controlled maintenance

1 The Rust evaluator emits unit- and source-qualified observations through the command-free `osr-supervision-contract` boundary. 2. The gateway retains history and alarm state; FUXA shows condition but cannot grant movement authority or engineering release. 3. A reviewed actionable condition creates one durable ERP Issue, including response guidance and immutable incident provenance. 4. A maintainer previews the linked Asset, parts availability, technician and downtime, then creates one unsubmitted Asset Repair. 5. Native ERP records work, stock and actual downtime; separate inspection and handback evidence decide whether railway use may resume. 6. Restart, ERP outage and duplicate-delivery tests prove queued events and serial history recover without turning case closure into safety clearance.

Scenario 5 — Rehearse degraded operation and recovery

1 Run normal, peak, continuously degraded and recovery profiles against the current controlled Samawah scenario using the multi-day soak harness. 2. Exercise shared battery/cabin thermal control: pack cooling, traction derate and protection take priority over comfort. 3. Remove the network path. Stored plan plus sensor localisation may retain route intent only when two fresh, trusted inputs match exactly; disagreement holds. 4. Rehearse corridor recovery from every-third-station sidings with the fail-held shunt-robot model, while retaining separation, route locking, detected points, speed supervision and emergency braking requirements. 5. Review bounded queues, historian tiers, events, CBM state, work-order evidence and the resilience report; proceed to HIL and physical trials before any safety claim.

Scenario 6 — Use multi-model AI for administration without delegating authority

1 Freeze the ERP/SCADA context and ask the experimental executive council for an allowlisted budget, maintenance, material-request or work-order proposal. 2. Collect blind ballots from strategy, finance, operations and risk identities spanning at least two providers and three model families. 3. Require 75% endorsement; any rejection, missing perspective, stale context or diversity failure routes the proposal to human review. 4. Deterministically verify the policy, context, proposal and ballot hashes, then require an operator-keyed attestation. 5. Let ERPNext independently verify and map the result to an unsubmitted draft. Models cannot pay, contract, hire/fire, certify, release engineering, operate SCADA or command trains.

For a single runnable story joining planning, ERPNext, manufacturing, FUXA, native Rust evaluation, maintenance, restart and clean-volume recovery, follow the Samawah example-city acceptance.

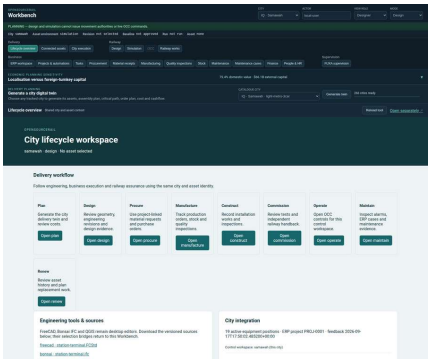
What is ready for what?

Product	Current maturity	Credible use now
Design and delivery platform	Serious demonstration	City GIS, alignment, service, cost, IFC, procurement, schedule, project twin and engineering coordination.
Train and infrastructure reference system	Engineering development	Local-manufacture planning, supplier RFQs, prototype definition, BIM coordination and civil option studies.
Open GoA 4 control system	R&D / pre-certification	Deterministic simulation, shadow mode, formal review and HIL preparation—not live railway command.

The first adoptable product does not require a new autonomous railway. Start with an existing depot, workshop or pilot corridor and deploy the non-safety owner/operator stack: asset register, project controls, simulator, Ops Core, QA, maintenance and evidence portal. See the first-adoptable-product boundary.

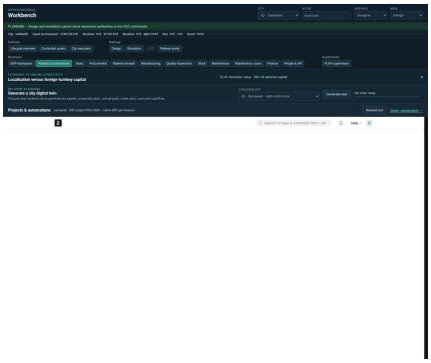
See the current system

Unified lifecycle workspace



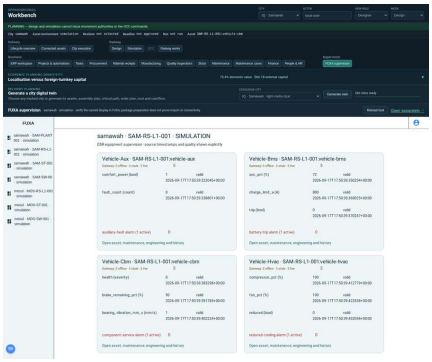
Installed Workbench lifecycle overview and city deployment status

ERPNext city execution



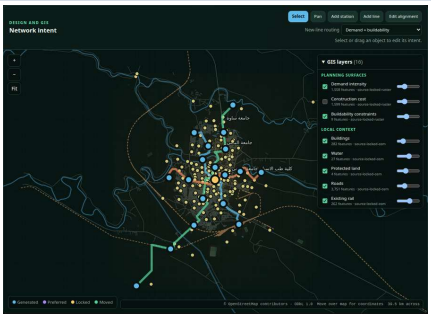
Native ERPNext project and OpenSourceRail operating actions inside Workbench

Embedded supervision



FUXA vehicle condition monitoring from Rust evaluators

City Studio

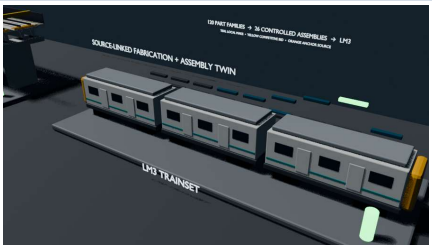


City Studio deterministic GIS workspace

Civil IFC coordination

Bonsai IFC4.3 construction sequence

Trainset assembly



LM3 complete 146-node assembly

These are captures of the installed simulation platform and generated engineering models, not UI mockups. Reproduce the Workbench captures with `capture-platform.mjs`, or watch the 88-second product assembly and 48-second civil IFC sequence.

The economic case

The platform is designed to let a public owner competitively procure ordinary civil works, vehicle structures, GFRP panels, interiors, wiring, installation and maintenance locally, importing specialist components where local suppliers are not yet qualified. The reference model uses about $0.9M per 3-car light-metro trainset (current build record: $885k), a factory-gate planning target whose BOM, labour and contingency basis remains subject to supplier/process qualification. Delivered procurement must separately price engineering, homologation, commissioning, warranty and spares. The $60k per supported vehicle/car module country factory allowance scales with fleet size; national capital uses the larger of that allowance and the physical city-order factory envelope. Workshop budgets must account for the actual buildings, equipment, tooling, test paths and production-readiness work. The Iraq strategy publishes conditional city studies; a consolidated national budget, funded deployment and realised national saving remain unestablished. Transparent code and COTS choices expose assumptions for review; independently calibrated demand, risk and comparable bids are still needed to control optimism bias.

For an illustrative $100M OpenSourceRail scope, the editable default comparison applies a 2.0× foreign-turnkey price with 90% requiring foreign currency or international capital:

Same modelled railway scope	Localisation-first OpenSourceRail	Foreign-turnkey sensitivity
Programme price	$100.0M	$200.0M
Value not requiring external capital	$79.1M	$20.0M
External-capital requirement	$20.9M	$180.0M

In that scenario, the external-capital requirement is $159.1M (88.4%) lower before interest. Across the 265-city model, about $488B—roughly 79% of programme value—is assigned to domestic activity. Scope snapshot: 7 October 2026, retained catalogue finance; excludes adoption of the revised Baghdad scenario. These are reproducible planning sensitivities, not bids, audited origin claims or financing offers. Review the assumptions, low/default/high comparisons and financing cases in the portfolio calculation.

Run it

On Debian, Ubuntu, Mint, Fedora, RHEL, Rocky, AlmaLinux, CentOS, openSUSE or Arch Linux, install, build and launch the Workbench; or run the deterministic simulator and test harness:

```
./install.sh
./osr build
./osr book
./osr
./osr sim --duration 3600 --status-every 300
./osr test
```

Open http://127.0.0.1:8090/. The local server is not an authenticated public deployment. The installer reports existing tools before changing anything and offers larger engineering applications separately. Disposable output goes under `build/` and `target/release/`; public review artifacts remain tracked outside those trees.

Find Your Way

If you are…	Start with…
A city or transport planner	City Studio, city catalogue, cost model and portfolio
A civil/BIM engineer	Civil system, Bonsai IFC workflow, BIM reference package and survey gates
A rolling-stock, supplier or manufacturing engineer	LM3 reference, buildable trainset, supplier technical-support package, CAD models and factory readiness
Researching component and system best practice	Source-backed handbook, installed design twin, design expansion review, component mapping and OSR-BP-001
An operator or maintainer	Operations, Workbench, connected lifecycle and example-city deployment
An ERP/SCADA integrator	Operating platform, ERPNext setup, supervision gateway and embedded contract
A software or assurance reviewer	Rust workspace, software architecture, formal results, safety case, assurance/authorization framework, manufactured viaduct review, controls and financial review and historical codebase review
A public owner, funder or delivery partner	October 5 cost and demand reconciliation, latest Baghdad staffing, depot, local-production and funding recalculation, complete Baghdad proposal, Baghdad-only original reference funding programme, competitive position and proof roadmap, owner–builder–operator plan, mobilisation status, JV/development framework, partnership status and supplier-support status
A contributor	Contributing guide, governance, change log and release checklist

Source Of Truth

Source-locked inputs validated candidate and deterministic generators content-addressed Git-reviewable revision GIS/OSR-ALN/CAD/IFC/cost/simulation/project-twin outputs independent evidence, named approval and operational baseline.

This root README is the only human-facing front door. A hash proves which bytes were reviewed; it does not approve them. Generated city packages still require survey, calibrated demand, utility and ground data, supplier qualification, first-article manufacture and testing, competent engineering review and national authorization. The principal editable sources are:

Concern	Source of truth	Generated/review output
Architecture and decisions	`docs/ARCHITECTURE.md`, accepted `docs/rfcs/`	Diagrams, guides and summaries
Rust behavior and simulation	`crates/` and tests	Applications, traces and coverage evidence
City synthesis and shared assumptions	`design/city-generation/`, `lib/templates/`, `lib/recipes/`, `lib/city-batches/world-sample.toml`	City catalogue, service, finance and engineering packages
Iraqi funding and current publication	`lib/templates/baghdad-programme-recalculation.toml`, `lib/templates/iraq-funding.toml`, controlled procurement schedules and `lib/templates/city-publication.toml`	Current Baghdad summary, latest scope and funding cases; original catalogue appraisals are labelled separately
Interactive city revisions	`cities/workspaces/`	Content-addressed candidates and exports
Mechanical/station/civil geometry	`design/component-catalogue/src/osr_mech/`	FreeCAD, IFC, BOM, traveler and image artifacts
Component and system practice research	`engineering/best_practice/data/`, linked to existing standards controls	Reproducible handbook, catalogue coverage and open verification/backlog plans

Concern	Source of truth	Generated/review output
ERP and supervision integration	`deployment/erpnext/`, `deployment/supervision/`, `services/integration/`	ERP workflows, FUXA views, history and maintenance evidence
Survey and released alignment	Accepted deployment GIS and OSR-ALN sources	QGIS, GeoPackage, corridor and IFC alignment evidence
Assurance and approval	`docs/certification/`, `docs/safety-case/`, `engineering/assurance/formal/`	Proof records, verification status and acceptance reports

The artifact policy explains what remains in Git. CI keeps useful CAD, IFC, PDF, image and animation evidence while enforcing a 50 MiB per-file ceiling.

License

Software is Apache 2.0. Control electronics and open physical designs use CERN-OHL-S v2. Documentation is CC-BY-SA 4.0. See LICENSE.md and LICENSES/ for the full texts.

Contributing To OpenSourceRail

Repository source: `CONTRIBUTING.md`

OpenSourceRail is pre-1.0 infrastructure software and documentation. It welcomes review, tests, corrections, reproducible data, and narrowly scoped implementation work, but it is not yet a certified railway product and should not be described as one.

Good First Contributions

- Fix stale links, unclear wording, units, assumptions, or generated documentation drift.
- Add simulator scenarios, fault-injection cases, or city data that can be regenerated.
- Improve tests for existing Rust crates and Python sidecars.
- Review operations, maintenance, manufacturing, QA, hardware, and certification evidence from real practitioner experience.
- Validate COTS hardware, BOM rows, wiring maps, CAD assumptions, or rolling-stock manufacturing steps against real supplier data.

Safety And Certification Boundaries

Safety-related changes need extra care:

- Use "SIL target" or "target assurance tier" unless an independent assessor has accepted a deployment-specific case.
- Do not remove fail-restrictive behavior without replacing the hazard argument and tests.
- Link safety claims to evidence: code, tests, formal model, hazard log, RFC, generated matrix, or field evidence.
- Keep production-service claims out of comments, docs, and release notes unless they are backed by deployment evidence.

OpenSourceRail produces reference artifacts. Operators, owners, prime integrators, assessors, insurers, and regulators carry the statutory safety case for a real railway.

Development Workflow

Use the root one-command setup. Installation behaviour is documented there only so it cannot drift between guides.

1. Open or reference an issue for behavioral changes, safety claims, architecture decisions, generated data changes, or new public docs.
2. Keep pull requests focused. Separate wording cleanup, generated artifacts, code changes, and large data refreshes when practical.
3. Follow the local style. Prefer existing helpers, templates, and RFC patterns over new parallel structures.
4. Regenerate derived artifacts when sources change.
5. Include the checks you ran in the PR description.

Useful checks:

```
python3 tools/automation/repo-health.py --quiet
python3 tools/automation/check-markdown-links.py
python3 tools/automation/generate-doc-index.py
cargo test --workspace
pytest design/city-generation/tests -q
PYTHONPATH=design/component-catalogue/src pytest design/component-catalogue/tests -q
pytest tools/automation/tests -q
python3 tools/automation/audit-project-twins.py
npm run test:frontend:unit
```

For Samawah/RFC consistency:

```
PYTHONPATH=design/city-generation/src pytest -q design/city-generation/tests/test_rfc_drift.py
```

For the reader edition:

```
python3 tools/automation/build-doc-book.py
```

Generated Artifacts

Generated city, operations, CAD, and evidence files should remain reproducible from tracked sources. If a generated file changes, include the command that produced it and avoid hand-editing generated rows unless the generator is also updated.

AI-Assisted Contributions

AI-assisted contributions are acceptable when the contributor takes responsibility for the result. Do not submit code, text, data, images, or technical claims copied from proprietary or incompatible sources. Check citations, units, and generated artifacts before opening a PR.

Licensing

The project split is:

- Software: Apache 2.0.
- Hardware designs: CERN-OHL-S v2.
- Documentation: CC-BY-SA 4.0.

By contributing, you agree that your contribution may be distributed under the applicable license for the files you change. The complete texts and path mapping are in `LICENSE.md` and `LICENSES/`.

Communication

Use GitHub issues and pull requests for now. Prefer concrete reports: file path, command, observed behavior, expected behavior, and any source data needed to reproduce the result.

OpenSourceRail Governance

Repository source: `GOVERNANCE.md`

OpenSourceRail is currently a founder-led pre-1.0 project. This governance file makes the operating model explicit while the contributor base, maintainers, and deployment partners are still forming.

Project Principles

- Safety claims are evidence-led and conservative.
- Deployment responsibility is explicit: the repository is not the operator, assessor, insurer, or regulator.
- City-specific work stays parameterized through the unified deployment model rather than becoming one-off forks.
- Vendor neutrality matters. Open standards, reproducible artifacts, and local manufacture are preferred where they are technically credible.
- The first adoptable product is the non-safety owner-operator stack: simulator, Ops Core, asset register, QA, maintenance, and evidence portal.

Roles

Role	Responsibility
Maintainer	Reviews and merges changes, keeps releases coherent, protects scope.
Domain reviewer	Reviews a specific area such as operations, civil, rolling stock, hardware, formal methods, or safety evidence.
Contributor	Proposes issues, docs, tests, data, or implementation changes.
Release manager	Freezes a milestone, runs checks, assembles release notes and assets, and publishes tags/releases.
Deployment partner	Supplies local evidence, operator review, site data, field tests, and authority engagement for a real deployment.

Until named maintainers are listed in a separate `MAINTAINERS.md`, the repository owner acts as maintainer and release manager.

Decision Process

Small fixes can be merged by maintainer review:

- typos, stale links, formatting, and screenshots;
- generated documentation drift with reproducible commands;
- tests that do not change public behavior;
- small implementation fixes within an existing RFC.

Larger changes need an issue or RFC before merge:

- public architecture or safety claims;
- new crates or subsystem boundaries;
- changes to generated city cost, finance, fleet, energy, or capacity assumptions;
- hardware host-class changes;
- rolling-stock structure, battery, braking, door, or train-control changes;
- governance, license, release, and contribution policy changes.

The default rule is rough consensus plus maintainer acceptance. Safety and certification changes require explicit evidence review, not only code review.

Releases

Milestones use tags such as `v0.2.0`. A release should include:

- release notes and known limitations;
- the reader PDF;

- Samawah case-study links;
- simulator and operations-portal instructions;
- evidence matrix and release-gap links;
- verification commands and results.

The current release checklist is `docs/releases.md`.

Conduct

This project does not yet have a separate code-of-conduct file. Until it does, contributors are expected to be precise, civil, safety-conscious, and willing to separate technical disagreement from personal criticism. Maintainers may close or moderate issues that are abusive, off-topic, spam, or unsafe.

Conflicts Of Interest

Contributors connected to vendors, operators, contractors, funders, or public authorities should disclose that context when it materially affects a recommendation, benchmark, cost claim, or deployment proposal.

Changing Governance

Governance changes should be proposed as pull requests against this file with a short rationale. Once there are multiple active maintainers, this file should be updated to list maintainer nomination, removal, and tie-break rules.

OpenSourceRail changelog

Repository source: `CHANGELOG.md`

Unreleased

- Added possession-aware physical-resource planning, energy/maintenance fleet diagrams, combined service/fleet/staff recovery and bounded competence/rest/travel/break-aware staff rostering. Workbench now includes offline typed/photo field work with persistent device/gateway queues, native ERP revalidation and immutable inspection-pending receipts. Added fault/exposure/MDBF/MTTR trends, held-out degradation/RUL validation, operational JSON and GTFS timetable/update exchange, and configured operator identity mappings. The isolated pilot and browser acceptance cover software behaviour; real topology, personnel/method data, measured predictive calibration and native-site commissioning remain open. See implementation and scope.
- Default development train protection to separate A/B processes and journals, two-out-of-two permission and either-channel trip. Added missing-channel, disagreement, replay and output-feedback scenarios; late feeds cannot hide a missed output deadline. TACS software checks now reject stale failure-analysis applicability, with a rebased unreviewed authoring baseline. Other local pairs, autonomous physical outputs and qualification remain open; no installed BOM, post-fault availability or operational release is claimed.
- Added the train-centred distributed-control process reference around existing consensus, interlocking, ATP, ATO and brake evaluators: authenticated/fenced resource ownership, frozen startup configurations, charging/departure interlocks and an output-guard contract. The two-train junction campaign uses actual process hosts. Bounded resource TLC and Rust/property tests cover retained locks, partitions, reconnection, restarts and unsafe departure prevention. Shared FMEA/change impact binds software evidence and planned HIL/track/independent release. Physical and operational deployment remain blocked; pilot defaults are unchanged.
- Hardened assurance impact and qualification acceptance against the reviewed prototype findings. Relationship-only and source-only changes reopen affected evidence. Fault tests now require normal preconditions, ordered rising responses and reviewed sampling resolution. RAMS allocations compare supported uncertainty envelopes against explicit metrics, units and conditions. Scoped graph safety, obligation and gate findings block qualification unless a current, accountable exclusion receives authenticated approvals. Calibration is checked at acquisition time, with separate retrospective instrument events. Workbench and exported dossiers expose allocation outcomes and graph findings.
- Added quantitative subsystem assurance and a controlled cooling-rig programme. Exact Boolean fault-tree/cut-set analysis retains shared dependencies and unknown systematic events; thermal, repairable availability, latent exposure, degraded-braking, queue and spares screens expose their assumptions. A qualification compiler checks measured CSV provenance, configuration, calibration, held-out model correlation, production equivalence and signed independent decisions, and exports original evidence. Workbench now shows six separate decision-readiness states. Selected hardware, physical measurements and deployment acceptance remain explicitly unresolved.
- Added a configuration-bound battery-cooling assurance example that imports the existing product hierarchy and connects specific failure propagation, requirements, interfaces, controller cases, physical-test plans, synthetic production/FRACAS records, batches and installed occurrences. Dependency impact traverses both old and new relationships; passports expose separately controlled, blocked deployment decisions. Four actual Rust controller cases have raw, source-bound evidence without advancing physical acceptance. Severity-5 FMEA modes now require safety review regardless of RPN; unknown occurrence remains an explicit evidence gap.
- Promoted digital standards adherence into a core product capability. The new Digital Standards Thread maintains 23 reviewed publisher/legal records, five assurance profiles and 16 lifecycle control objectives; maps them to 15 hashed repository checks, the complete engineering inventory and G0–G4 passports; expires stale metadata; and emits a path-level change-impact index. Its machine result is explicitly evidence coherence, while clause applicability, conformity, physical qualification, independent assessment and authorization remain open decisions. Updated ISO 9001 from the obsolete 2015 guidance reference to the current 2026 edition while preserving the explicit ISO 22163:2023 transition issue. Catalogue generation now detects stale Baghdad/all-city metrics rather than silently rewriting them in CI.
- Added a complete Baghdad concept/FEED offer built from a freshly regenerated nine-line city design, open topography/water screening, QGIS/GDAL civil package, native degraded-case simulation, SUMO timetable validation and the Baghdad project/operations twin. A deterministic nine-page PDF and three city-specific application screenshots are hash-bound to their inputs. The proposed 831-trainset fleet now identifies CRRC traction, electrical-control, bogie, suspension, coupling and brake product families as competitively procured candidate components without claiming a partnership, endorsement, quotation or selected configuration. The offer keeps surveyed depot/stabling design and every construction, supplier, safety and legal release gate open.

- Replaced the signalling-centric “certification pack” front door with one assurance, conformity and authorization model for every controlled item. Five fail-closed gates now separate definition, design assurance, implementation qualification, integration validation and independent/legal acceptance. A deterministic register emits configuration-bound passports for 279 train/station/civil/Rust items plus seven Workbench, ERP/HR, supervision, FUXA, Ops Core, AI-council and lifecycle-identity services. All 286 are baselined at G0, all G1–G4 decisions remain explicitly open, and CI rejects missing routes, sources, hashes, gates or unsupported release claims.
- Added an integrated battery/cabin thermal supervisor, modelled after compact combined rail cooling architectures while retaining isolated air and coolant circuits. Battery cooling and protection pre-empt passenger comfort, faulted coolant/sensor states request derate or isolation, Rust unit/property tests cover the controller, and the service simulator now executes and reports it.
- Added fail-closed two-of-three onboard route selection from a signed stored plan, sensor-derived topology/localisation and authenticated network command. Plan-plus-sensor agreement supports network-loss route intent, while stale, untrusted, single or disagreeing inputs hold. The selector can never grant movement authority; the existing separation, route-locking, ATP and brake chain remains authoritative and the distributed profile remains a research target pending HIL, field and independent-assessment evidence.
- Added constrained single-track generation with passing loops at stops/portals and a protected recovery siding every third station, including a bounded remote shunt-robot concept. A deterministic standards/FMEA compiler now validates evidence and component/subsystem/system coverage across six engineering and operational domains, deterministically screens all 279 controlled train/station/civil/software inventory entries on every build/test, emits hashed Markdown/JSON reports and keeps every item-specific, physical and regulatory release gate explicit.
- Added a supplier-facing technical-support layer that reuses the LM3 product tree, controlled interfaces, COTS/RFQ candidates, city duty models, first-article gates and ERP/lifecycle identities. A fail-closed template now separates three public catalogue leads from qualification; defines eight review/industrialisation/localisation/support work packages, eight cross-supplier interfaces, eight cost buckets and seven decision gates; and prevents discussions, public webpages or status labels from being reported as manufacturer authority, configuration compatibility, first-article approval, transferred local capability, all-in cost or railway acceptance. CI checks both the source references and generated public readiness output.
- Converted the external competitive and joint-venture review into an evidence-led product/commercialisation layer. The repository now distinguishes its demonstrable planning and owner/operator products from the still-open physical railway and GoA4 programme; prioritises measured operator deployments and independently comparable corridors through an unlimited, evidence-separated candidate portfolio; and provides a fail-closed partnership template for legal counterparties, accepted contributions, IP/manufacturing rights, finance separation, related-party supply, finite performance-based exclusivity and four-stage localisation. CI rejects stale commercial-readiness outputs, while the public reference records zero orders, commitments, accepted contributions and partnership authority.
- Added one fail-closed lifecycle-control system across mechanical products and assemblies, station variants, reusable civil types and every Rust crate. A validated ERP/HR/management template now defines separated roles, competence, configuration, inspection, calibration, NCR/CAPA, management review, shift handover, software-release and asset-identity evidence. A shared Python/Rust QR contract is lookup-only, rejects hidden authority or commands and emits no printable label until a real operator provides an HTTPS resolver and verifies the physical binding. All-city readiness now compiles these controls and CI checks the generated 231-record subsystem register.
- Completed a workspace-wide Rust scrutiny and integration pass. Routing now rejects malformed grids, penalties and non-finite inputs; raster ingestion honours source manifests while validating paths, geography and layer semantics; and City Studio enforces source locks on every GIS access path. A new strict, observation-only Rust supervision contract replaces the ad-hoc simulator export and is checked against the same fixture by Rust and Python. Release overflow checks, all-feature tests, doctests, RustSec auditing, property tests, scheduled long-horizon simulation and coverage reporting now broaden the verification harness. The Rust 1.92/egui 0.34 refresh removes vulnerable GUI transitive dependencies, and the unmaintained `bincode` serializer has been replaced by `postcard`. The accompanying review records the remaining hardware, certification and production-infrastructure gates. Follow-through tests execute all topology archetypes and the greedy planner, exercise malformed raster/DEM metadata, pin a representative `postcard` message to exact bytes and cover every fail-closed supervision header branch. Final local coverage is 75.45% overall, including 76.05% for routing topology; 39/39 currently tractable declared Kani properties pass, while nine resource-bound obligations remain explicitly open. Push and pull-request Kani gates now cover the 39 completing properties; manual dispatch retains the nine open failures as non-gating diagnostics, while release export still requires all 48 properties and fails closed on missing or unsuccessful evidence.
- Added an experimental multi-model AI executive council to the ERP/supervisory gateway. City-scoped proposals freeze their operational evidence, authenticated model identities vote from independent perspectives, a deterministic quorum and diversity policy exposes dissent, and terminal decisions receive an operator-keyed HMAC attestation. Four reversible ERP

draft intents are now independently reverified by ERPNext and delivered through its existing native adapters; SCADA, railway/safety, HR, payment, submission and legal authority remain explicitly non-delegable. A provider-neutral, one-identity model runner keeps other ballots and credentials isolated, rejects non-versioned responses, and records exact prompt/response hashes for Workbench review.

- Added source-locked open SRTM elevation, derived terrain slope and independent OSM water coverage to City Studio planning bundles. The designer now exposes 20 deterministic GIS layers, highlights likely bridge and terrain/surface viaduct runs, reports elevation/slope at every station, rejects new stations over predominantly mapped water, and preserves the survey/vertical-alignment release boundary.
- Added reusable authority snapshots, immutable ATP braking profiles and borrowed topology walks. ATP and interlocking determinism passed on candidate `2bf8b7717`; odometry determinism and two-train non-overlap remain open. Non-overlap now declares paired checks for Clear, missing, Unknown and Present wayside verdicts, correcting the earlier expectation of extension without a Clear verdict. All eight checks are required for closure.
- Added explicit proof time/memory budgets, whole-process-group cleanup and failure-log retention. See the execution register for tested source, outcomes and unresolved verification limits.
- Separated interlocking, ATP and odometry from topology storage through a shared read-only interface and validated immutable tables. All 41 declared Kani properties were rerun locally: 37 passed, with four remaining 300-second timeouts. Three odometry proofs now finish with unchanged inputs and assertions. Default city maps and evaluator algorithms remain in use; adapter comparisons, the full Rust workspace and evidence-packaging regressions pass. Samawah and Mosul design options each pass a fresh nominal/eight-case qualification with identical detailed outcomes; Workbench now displays the new source-bound bundles.
- Added reusable city design options for charging dwell, fleet capacity, modular storage and solar generation, with source-bound simulation qualification and a city-scoped Workbench comparison view. Service thresholds stay unchanged.
- Made retained native CAD launchers portable and regenerated both city bundles through their native ERP production checks after CI exposed host-specific paths.
- Added a reproducible native FreeCAD CalculiX nested ERP BOM production disposition manufacturing screening-evidence workflow, with a city-scoped Workbench view and retained Samawah/Mosul results. Twelve solver runs and twenty-two native ERP checks pass; engineering and physical acceptance remain open.
- Added per-line full-window mileage gates so surplus service on one line cannot hide another line's shortfall. Reports expose configured passenger capacity by timetable window while retaining observed headways, demand and operator review as separate acceptance requirements.
- Added coordinated private checkpoints and isolated fresh-volume recovery of ERP, gateway queues, FUXA, Ops Core, files, keys and immutable container images. The native rehearsal verifies ten volume inventories, signed records and duplicate-safe ERP reconciliation; production cutover and recovery objectives remain open.
- Added reusable per-city FUXA site polling, preserving equipment tags and reviewed imports; integration image `osr/integration:0.1.3`. The load runner now separates client/server processes and checks effective exported tag values. Mosul retains every sample but does not consistently meet the two-second target.
- Recorded `e9e4ff5cf287` CI: general, integrated-stack and example-city passed; Kani passed 34 of 41 properties, with seven remaining timeouts.
- Added atomic, authorized telemetry batches and native simulator publishing; gateway image `osr/integration:0.1.2` queues contending reads/writes and meets the two-second target in a five-cycle local Samawah rehearsal without lost samples. Mosul retained every sample but missed the same target; larger-city capacity remains open. CI now retains full-city load reports; production capacity acceptance remains open.
- Replaced ATP integer square-root division with 32 bounded radix-four steps. All ATP tests pass, including boundary and full-width generated comparisons; severe-overspeed and uncertainty-conservatism Kani harnesses passed locally. Seven earlier proof timeouts still require resolution and exact-commit execution.
- Removed unused topology fixtures from two unregistered-train interlocking proofs; both previously timed-out harnesses passed targeted local verification with unchanged assumptions and assertions. This reduced the earlier eleven timeouts to nine before the ATP follow-through.
- Fixed FUXA device polls loading entire city histories and cross-city traffic hiding pending ERP deliveries. Added scoped counts, cursor pagination and lifecycle controls for evidence and deliveries; gateway image `osr/integration:0.1.1`.

- Hardened Ops Core backup verification and added fresh-directory offline restore with evidence-path rebasing. Release packaging now retains and validates all declared Kani manifests, results and logs. Workbench has a generated verification and acceptance status view; a repeatable full-city HTTP load rehearsal records functional outcomes separately from polling capacity.
- Recorded exact-commit integration results and local fresh-volume ERP recovery rehearsals, including ledger records, nonempty public/private files and encrypted settings. The broader proof run has 30 passes and 11 timeouts; release packaging, coordinated whole-platform recovery and independent acceptance remain open.
- Fixed Workbench revision changes retaining an earlier baseline and simulation run. Embedded City Studio, simulator, OCC and Railway works now remove superseded context; a baseline change invalidates the preceding run. Browser handoff tests bind exact native revision/approval responses to the saved work.
- Made complete-city acceptance a commit-bound software release gate; added nine unmocked disposition browser checks and 24 native business outcome/rollback checks. Native replenishment now retains city/project identity and rejects ambiguous rules; the reproducible ERP application image is revision `osr13`.
- Added cross-domain change-review freshness and artifact checks in Workbench, controlled execution of all 41 declared Kani harnesses, independently signed acceptance verification, and a resumable city simulation runner. These do not grant physical, engineering or independent safety acceptance.
- Fixed persistent point-machine drive faults losing their stop latch at cooldown expiry; corrected signed-absolute-value guards and insufficient intrusion proof unwind bounds found by the broader proof run. Production obstacle/intrusion LIDAR and radar comparisons now handle minimum signed offsets without overflow. The supervisory simulator rechecks local enable/disconnect at command execution so changes during telemetry and polling cannot reuse an older enabled state. Removed unused topology construction from same-section/zero-wheel proof paths and corrected further unwind bounds without disabling verification checks. Added partitioned full-catalogue simulation CI with complete, current artifact validation; candidate simulation results do not grant physical acceptance. See the review follow-through.
- Added an evidence-backed inventory of 515 OSR integration functions, inputs, shared settings and workflows, with explicit partial/uncovered entries and guards against unreviewed contract changes. The example-city expansion adds 389 live/native/restoration assertions and 230 separately labelled schema checks, including concurrent configuration/telemetry, command state interactions, all native maintenance intervals, inspection outcomes and supplier payment settlement/cancellation. See the coverage register.
- Added an isolated, reproducible Samawah lifecycle acceptance deployment with a Mosul control city, persistent native business records, configuration sensitivity checks, browser screenshots and outage/restart verification. Run it with `./osr example-city setup` and `./osr example-city run`; see the example-city guide.
- Made Workbench data paths, integration origin and ERP parent navigation configurable for independent installations. FUXA asset navigation now accepts a Workbench origin bound into the reviewed deployment project. The native simulator now observes configured sampling intervals; display-only alarm clearance does not enqueue an unwanted ERP maintenance event. Alarm priority and response now reach native ERP triage while preserving subsequent operator priority changes.
- Added registry-driven independent evidence verification for every disposition action: retention, cancellation, direct amendment, production stop, corrective work with accepted inspection, performed stock inspection and material trace. Native record links, measured readings, ledger quantities and serial/batch identities are checked against the full endorsed revision exposure. Immutable evidence, repeat-safe recording and stale-state feedback appear in ERP and Workbench; these checks never perform the action or authorise railway release. Older proposals without a full exposure snapshot require a new reviewed proposal.
- Added assigned revision-disposition proposals and independent plan decisions in native ERPNext, with exact exposure checks, immutable history, repeat-safe submission, self-review rejection and Workbench feedback. Plans never execute native purchase, production or stock actions or grant railway release.
- Added reusable, revision-bound ERP exposure snapshots to the connected lifecycle view. Explicit nested BOM dependencies identify draft and unfinished production, potential purchase exposure and linked submitted stock movements. Reviews retain project/company scope, permission gaps, source timestamps and checksums; downloadable observations do not change or release business work.
- Removed the change-impact fallback that could label all project transactions as affected when an equipment Item mapping was absent.

v0.4.0 — Integrated operating software

- Corrected factory Work Order correlation to require a reviewed exact Item/BOM pair at the selected engineering revision. LM3 method views now require the city baseline's `light-metro-3car` family; other families remain unsupported.

- Made the preferred FUXA city/site explicit and preserved selected-equipment navigation. Retired history no longer interrupts active simulator telemetry.
- Separated safety-case traceability from successful result records and independent acceptance, with source/report hashes, named harnesses, tools and bounds. Generated GSN counts and corrected charging, ERP ownership and assurance claims.
- Made the readiness inventory independent of untracked local operations bundles.
- Released the v0.4.0 software baseline for the integrated ERPNext, FUXA and OSR lifecycle platform. Supervisory package updates now retain removed equipment as retired history while rejecting new telemetry and commands, fail pending commands on retirement, and support reviewed reactivation. Alarm acknowledgments are immutable per occurrence and reset on reactivation; command and alarm contracts are validated before a package is accepted.
- Added a deterministic prepared-versus-live change review across design and embedded contracts, asset topology, ERP identities, installed serials, evidence, open cases and pending commands. Workbench explains the impact and matching ERP actuals; changed simulation packages require the fresh review hash, while installed retirement, pending-command contract changes and physical remapping stay blocked.
- Extended the versioned native-controller bridge and generated FUXA packages to identified points, declared level crossings and station-aggregate fare gates. All 9,097 catalogue switches retain their asset identity; the read-only wayside views expose no movement authority or safety command, and crossings remain absent where the asset registers contain none.
- Added nine generated factory-method supervision views to applicable LM3 production plants, reusing the existing 120-product/30-tooling method source. Explicit unavailable and quality-hold fixtures create scoped ERP Issues; Workbench now correlates native Work Orders only through reviewed execution mappings and keeps inspection, disposition and manufacturing release independent.
- Added scenario-bound, all-line SUMO dwell modelling and a deterministic OSR/SUMO running-time cross-check, with explicit external junction-occupancy and authority gates.
- Added a shared city field-evidence specification and generated Samawah brief covering survey control, utilities, land, flood, ground, workshop/fleet and reality capture. City Studio can issue the revision-locked brief and empty receipt manifest through a fifth allowlisted engineering adapter.
- Added the next survey-control gate: controlled relative-path, metadata and SHA-256 receipt validation, deterministic `rnx2rtkp` execution and quality parsing, and a separate explicit survey-authority acceptance record. Pilot reports remain `awaiting-field-data`; no observations or approvals are invented.
- Added the checksum-pinned official RTKLIB 2.4.3-b34 source build to the one-command optional engineering setup, installed user-locally on supported Linux architectures without requiring a distribution-specific package.
- Added the surveyed-ground-model gate for accepted control, GeoPackage structure, terrain/orthophoto signatures, registered E57/LAS/LAZ clouds, independent-check RMSE, ODM/CloudCompare reports, voids and authority release. The pilot reports remain explicitly pending because no field dataset exists.
- Added the surveyed-alignment acceptance gate: a deterministic per-line receipt checks explicit OSR-ALN horizontal/vertical/cant content, all hard gates and stations, LandXML re-import hashes and tolerances, platform/yard/ turnout/clearance evidence, and a separate signed multi-discipline release.
- Added the following route/station-fit gate, binding utility, land, flood, accessibility, yard/intercity, road, construction-logistics and possession evidence to every live line and station while rejecting unresolved high or critical issues before coordinated authority acceptance.
- Added a drainage/ground-design gate that independently reruns received SWMM inputs, checks report/source provenance and continuity, validates per-line/ station catalogue foundation or ground-treatment schedules, requires actual deep-element lengths, and makes OpenGeoSys evidence trigger-dependent.
- Added a per-asset structural-release gate for chainaged span/pier/abutment/ foundation schedules, immutable OpenSees/CalculiX evidence, load-case and result reconciliation, independent checking and signed authority release.
- Added a deterministic all-variant station systems package: OpenSees canopy gravity/uplift, JuPedSim normal/degraded/egress and repeatable-bay SWMM drainage, plus tracked EnergyPlus and FDS runs. The adverse enclosed depot baselines are retained alongside a passing separated/open energy-compound and N+1 cooled-controls screen, with six explicit deployment work packages.
- Extended the one-command optional engineering setup to install checksum- verified EnergyPlus 26.1 and FDS 6.11.1 user-locally without editing shell startup files; the engineering smoke check now exercises both native solvers.

- Added seven complete station-variant FreeCAD and geometric IFC4.3 assemblies, covering all 45 controlled station products with deterministic validation; the FreeCAD set now carries 4,324 installed/exploded native shapes, controlled interface/datums, compact variant definitions and a bidirectional orphan-ID gate.
- Expanded the LM3 reference library to 523 primitives, detailed door/window, bogie, traction, battery and HV interfaces, and added an installed-coordinate sidecar with nominal, curve and twist review states.
- Added STEP, DXF inspection and three-view SVG references for all 46 locally manufactured LM3 rows, plus a generated CAD/IFC fidelity register.
- Expanded turnout, depot and reusable civil maintenance-interface geometry.

v0.3.1 — 2026-08-30 — CI correctness patch

Corrects the clean-checkout release gates discovered after v0.3.0: both Python jobs now share one pinned IFC runtime, generated city acceptance reports link only to tracked compact artifacts, and Kani CI runs explicit bounded properties that complete inside its controlled timeout. See the v0.3.1 patch notes.

v0.3.0 — 2026-08-30

v0.3 focuses on a credible product boundary and a reproducible integrated delivery platform. It adds the shared Python runtime/CI path, three-product maturity model, current standards baseline, safety-controller selection gate, release-selected Kani workflow, consensus refinement map, corrected halt energy scope, safer battery-enclosure definition and appropriately qualified economic claims.

See the concise release notes. Physical first articles, qualified safety hardware, formal refinement closure, independent assessment and deployment approval remain open gates rather than release claims.

v0.2 development baseline

Post-v0.1 development baseline. The repository has moved beyond the initial publication baseline: it now has a complete, buildable software + documentation surface, generated design catalogues, mechanical/CAD evidence, COTS/DIY hardware integration paths, operations material, and a regulator-facing pre-submission structure for a GoA 4 urban-rail deployment. What's not yet done is everything that requires external hands — physical pilot build, civil survey, independent assessor review, regulator engagement, operator validation, and deployment approval.

This file summarises how the repo arrived here and what each major subsystem is ready for.

Headline numbers

- 55 Rust crates with Kani harnesses, proptests, and integration tests across the safety-critical software surface.
- Two Python sidecars: `design/city-generation` (GIS + network synthesis), `design/component-catalogue` (parametric mechanical / civil / station source geometry), plus two Python tools. The current audit collects 429 Python/tool tests across their four suites (428 passing, 1 environment- dependent test skipped).
- Two egui operator GUIs: `osr-sim-gui` (designer), `osr-occ-gui` (dispatcher). Both native + WebAssembly.
- Thirty RFCs covering software architecture, rail civil engineering, operations, driverless operation, wayside intrusion detection, cybersecurity, the operator GUIs, construction QA, maintenance scheduling, and manufacturing scheduling.
- Workspace builds are expected to stay warning-free; warnings are treated as drift from the v0.2 development baseline.

What's ready

Software

- SIL-4-target onboard chain — position fusion ATP brake + derailment + fire + door-control + obstacle-detect (RFC 0015). Every SIL-4 evaluator carries Kani bounded-model proofs and proptest coverage; GSN safety-case compiler gates CI.
- SIL-4-target wayside chain — MA computer on top of Raft-derived consensus + point-machine controller + intrusion-detect (RFC 0016). Same Kani + proptest + GSN discipline.
- Two-layer GoA 4 safety envelope — wayside intrusion gate
  - onboard obstacle detect, both wired into the same brake chain, both demonstrable in the sim via scenario fault injection.

- Distributed signalling — `osr-consensus` designed against the TLA+ `SMRaft` spec and differential-tested against an independent Python twin; formal refinement remains an open certification gate.
- Message authentication — `osr-secbus` (ed25519 on `osr-crypto`) plus authenticated consensus proposal/commit paths and simulator fault-injection coverage (RFC 0017 v2/v3). Deployment key provisioning and production-transport/hardware evidence remain open.
- Automatic network generation — OSM 20 m cost grid Dijkstra population-tiered multi-line network, scales to 500-city batches.
- Two operator GUIs — see the README screenshots.

Rail engineering

- Five trainset families (urban-shuttle-1car metro-6car), cabless per RFC 0015.
- Four track-geometry presets with a machine-readable OSR-ALN interchange format, LandXML OSR-ALN converter, and a validator that enforces 8 hard gates + 3 soft gates per the format spec.
- Six station archetypes; prefab steel-portal + solar- canopy, no station building, bolt-together on-site.
- Three civil classes (at-grade / elevated / bridge) — tunnels explicitly excluded per RFC 0011.
- Three turnout tangents covering tram / urban / mainline.
- Three depot archetypes with a fleet-sizing formula.
- Worked OSR-ALN alignment instance for an earlier Samawah generated network (29 km, 22 stations, validator-clean), proving the shared civil export/validate workflow rather than a Samawah-specific civil standard.

Operations

- Operations rulebook across four shipping role families (dispatcher S1–S7, station-staff T1–T5, maintenance M1–M7, control-centre C1–C3). Cabbed operation and train-driver procedures are outside repository scope.
- First adoptable product path — Ops Core + simulator + asset register + QA/maintenance/evidence portal for an existing depot, workshop, or pilot corridor, explicitly before safety-critical train control is deployed.

Governance and release readiness

- Top-level contribution and governance docs now define project status, contribution expectations, safety-claim boundaries, decision process, release responsibilities, and the current founder-led pre-1.0 operating model.
- Deployment roles page assigns owner/operator, prime integrator, independent safety assessor, insurer, EPC/civil contractor, local workshop, hardware integrator, financing entity, and regulator responsibilities.
- v0.2 release pack identifies release notes, PDF/brochure assets, Samawah case-study links, simulator instructions, evidence matrix, GitHub metadata, and publication commands.

Hardware

- Five host classes reference-spec'd and assigned controlled service manifests: T-ECU/S (safety kernel), T-ECU/A (application), T-OBS (obstacle detect, RFC 0015), W-SBC (wayside), S-SBC (station / depot). Palette restricted to Raspberry Pi + Radxa for domestic procurement.
- Board-level specs per class: block diagram, power budget, pinouts, connector tables, safety-nets (SIL-4 target boards only; none is qualification evidence by itself.
- Pilot COTS/DIY path per RFC 0019: commodity SBCs, Pi Pico 2 boards, sensor modules, relay/HAT modules, terminal blocks, DIN-rail enclosures, and prepared image flow. KiCad is not a first-pilot prerequisite when no custom PCB is used.

Certification

- EN 62267 type-certification pre-submission pack at `docs/certification/` covering system description, 24 safety requirements, 17 hazards across 7 classes, a clause-by-clause compliance matrix, and an evidence register that links every claim to a concrete artefact.

What needs external engagement

- Pilot hardware integration evidence for the RFC 0019 COTS/SBC path: exact SKU BOMs, wiring/harness maps, connector maps, enclosure/mounting notes, power/thermal margins, SD-card image checksums, `osr-selftest` output, safety-net bench tests, and commissioning records.
- Custom-board KiCad + gerbers only where a deployment chooses OSR-specific carrier, power, safety-I/O, or sensor-interface boards. All net lists + pinouts + safety-net rules are frozen; a KiCad workflow turns those specs into volume-production files.
- Civil survey replacing the planning-grade Samawah UTM coordinates with real GNSS data.
- Independent safety-assessor review of the certification pack. Deployment partner's scope.
- Type-approval submission to a national safety authority. Deployment partner's scope.
- Practising-operator review of RFC 0013 — a working dispatcher + maintenance foreman red-lining the rule text against reality.
- RFC 0017 deployment closure — freeze the key registry and secure- element provisioning/rotation procedure, then capture production- transport and hardware-in-the-loop verification evidence.
- Revenue operation — the last item. Requires all of the above.

What this repo is not

- Not a standards body. Where good open standards exist (GTFS, NeTEx, IEEE 802.1 TSN, EN 50126/8/9, IEC 62443-4-2) we adopt them.
- Not a safety certifier. The project produces artefacts suitable for independent assessment; certification is done by national authorities.
- Not a museum. We do not aim for plug-in compatibility with every legacy vendor protocol. Migration paths are scoped; permanent legacy support is not.
- Not a vendor. There is no OSR commercial entity. Support and deployment are the responsibility of the deployment partner.

How to engage

The project is looking for, in order:

1. Practising rail operators to review the RFC 0013 rulebook and flag the places where real-world procedure differs from what we wrote.
2. Civil engineering firms in the target deployment footprint (MENA, sub-Saharan Africa, South Asia, Latin America) to pilot the OSR-ALN converter against a real survey.
3. Pilot integrators and hardware reviewers to test the RFC 0019 DIY path against a real bill of materials, wiring maps, enclosures, power/thermal margins, and self-test logs. PCB designers remain valuable for custom-board deployments and later volume production.
4. Independent safety assessors familiar with EN 62267 GoA 4 and IEC 62443-4-2 to read the `docs/certification/` pack and identify gaps we haven't seen.
5. Contributors willing to pick an open crate from RFC 0005 (cybersecurity consensus integration, full TSN transport, CBM backend) or an open RFC follow-up.

File issues with specific disagreements; send pull requests with tests.

License

- Software: Apache 2.0.
- Hardware designs: CERN-OHL-S v2.
- Documentation: CC-BY-SA 4.0.

Contribution and governance process is now in `CONTRIBUTING.md` and `GOVERNANCE.md`. Exact license texts and path-level applicability are in `LICENSE.md` and `LICENSES/`.

Milestone version

This repository is now at a v0.2 development baseline. There is no breakage policy yet — the repo is pre-1.0 and interfaces can change.

Remaining v0.2 hardening work is tracked in `docs/ROADMAP.md`. Several originally planned items have already landed after the v0.1 snapshot, including the DIY electronics cookbook, `osr-selftest`, commercial-tool gap-closing in `osr-alignment`, turnout/depot/clearance/accessibility CAD, the crashworthiness scaffold, and the two-track hardware release model. The remaining high-value v0.2 targets are:

- RFC 0017 deployment evidence — secure-element provisioning, production-transport capture, and hardware-in-the-loop verification.
- Pilot hardware integration evidence for the RFC 0019 COTS/SBC path, plus KiCad, gerber, board BOM, and assembly outputs where custom boards are chosen.
- Rolling-stock production-detail package: supplier-exact envelopes, weld maps, tolerance stacks, FEA-ready brackets, and release drawings.
- Residual-risk narrative and first external safety/operator review feedback incorporated into the certification pack.
- Reconcile distributed-control development target in RFC 0033; migrate resource lifecycle into existing interlocking and reuse ATP/ATO/brake through durable, authenticated process hosts. Physical qualification and profile release remain pending.

OpenSourceRail Licensing

Repository source: `LICENSE.md`

OpenSourceRail uses a path-based multi-license model. The complete license texts are tracked in `LICENSES/`.

Material	Default license
Software source, tests, scripts, and formal-model source	Apache License 2.0
Hardware designs, mechanical CAD source/output, fabrication definitions, and assembly drawings	CERN Open Hardware Licence Version 2 — Strongly Reciprocal
Documentation and original diagrams/media	Creative Commons Attribution-ShareAlike 4.0 International

More-specific notices in a file or directory override this default. Third-party inputs and generated artifacts retain their applicable upstream terms. In particular, OpenStreetMap-derived data must preserve its required attribution and database-license notices.

See `LICENSES/README.md` for the path mapping and contribution rule.

License Texts And Path Mapping

Repository source: `LICENSES/README.md`

This directory contains the complete license texts used by OpenSourceRail:

- `Apache-2.0.txt`
- `CERN-OHL-S-2.0.txt`
- `CC-BY-SA-4.0.txt`

Default Path Mapping

Paths/material	SPDX identifier
`crates/`, `design/city-generation/src/`, `design/component-catalogue/src/`, `tools/automation/`, `tools/`, tests, and executable portal code	`Apache-2.0`
`control-electronics/`, `design/component-catalogue/catalog/`, generated CAD, fabrication definitions, travelers, and control-electronics/mechanical drawings	`CERN-OHL-S-2.0`
`docs/`, repository Markdown, original diagrams, and original documentation media	`CC-BY-SA-4.0`
`engineering/assurance/formal/` model source	`Apache-2.0`

Generated files follow the license of the source material unless their manifest or embedded notice states otherwise. Third-party data, maps, supplier documents, fonts, and cited material are not relicensed by this mapping. OpenStreetMap-derived artifacts require OpenStreetMap attribution and remain subject to the Open Database License where it applies.

Contributors license changes under the license applicable to the path they modify. New mixed-purpose directories must include a local license notice before accepting substantive content.

Architecture, Planning and Operations

OpenSourceRail — Scope & Architecture

Repository source: `docs/ARCHITECTURE.md`

Status: Current development baseline (2026-09-27) Audience: Engineers, operators, policymakers, and funders evaluating the project. Purpose: Define what OpenSourceRail is, what it is not, and the architectural stance that distinguishes it from existing rail vendor offerings.

1. Mission

OpenSourceRail is an open reference platform for designing, testing, building, and operating urban rail systems—light metro and urban metro. Its primary beneficiaries are developing-world nations seeking transparent, locally adaptable alternatives to opaque turnkey planning and integration. Repository outputs are screening and engineering evidence, not certified or construction-ready systems.

Scope is urban transit only. Intercity passenger, long-distance, and freight/cargo rail are explicitly out of scope. The rationale is focus: urban transit is where the unit of need is highest in target markets, where the duty cycle is most favorable to the catenary-free battery-electric architecture proposed in §4 D7, and where a single reference design generalizes cleanly across many deployments. Freight and intercity have meaningfully different energy, signaling, and rolling-stock profiles that would dilute focus; they may be candidates for sibling projects or a future v2.

The project succeeds when a national railway authority, working with domestic engineering firms and qualified suppliers, can retain design authority, localise ordinary fabrication and integration, and limit imported content to specialist components that cannot yet be sourced or qualified locally.

Target deployment regions — sub-Saharan Africa, MENA, South and Southeast Asia, most of Latin America — share two properties that shape the architecture: they are capital-constrained, and they are high-insolation. Design choices throughout this document treat both as load-bearing assumptions rather than edge cases.

1.1 Non-goals

- Not a standards body. Where good open standards exist (e.g., GTFS, NeTEx, IEEE 802.1 TSN), adopt them. Where they don't, define minimal project-local interfaces — don't chase ISO ratification.
- Not a safety certifier or operator of record. The project will produce artifacts suitable for independent safety assessment, but certification is performed by national authorities and independent assessors. Deployment owners, operators, prime integrators, insurers, and entities in charge of maintenance carry the statutory safety case, liability, and operating approvals.
- No compatibility layers in the reference core. Deployment-specific adapters stay outside the shared architecture and are owned by the deployment that needs them.

2. Design Principles

1. Commodity hardware, custom software. Industrial PLCs, proprietary trainbuses, and vendor-specific radios are replaced with commodity SBCs, deterministic Ethernet, and standard radio layers. Value is captured in the software and system integration, which can be produced anywhere.
2. Appropriate tools with controlled interfaces. Rust carries the runtime, simulation and control core; Python carries GIS, engineering and CAD automation; browser applications use pinned web toolchains. Schemas and deterministic generators control the boundaries.
3. Replace and remove. Superseded reference designs leave the active repository instead of accumulating alternate paths. Every major choice records the concrete advantage of its replacement.
4. Simulation-first. No subsystem is considered designed until it runs end-to-end in a shared digital twin. Hardware pilots come after simulation sign-off.
5. Machine-checkable safety. Safety arguments are structured (GSN-style), version-controlled, and linked to formal proofs, test evidence, and code. Safety cases regenerate on every commit.

6 Small, replaceable pieces. Each subsystem is deployable independently. A country can adopt the dispatching platform without committing to the rolling-stock ECUs, and vice versa.

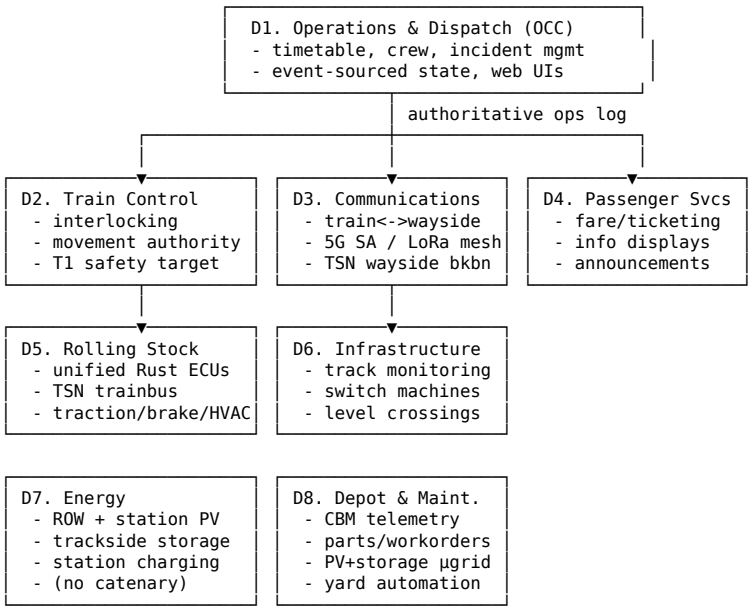
7 Defense in depth. Every wayside node is assumed to be reachable by an adversary. Signed firmware, mutually authenticated links, minimal attack surface, privilege separation at the OS level.

The Rust implementation, its present assurance level, completed hardening work, and the remaining path from reference software to deployable railway product are tracked in the `Rust codebase review and integration plan`. In particular, runtime observations cross into supervision through a versioned, observation-only contract; ERP and AI governance cannot issue movement authority or bypass independently assessed train-protection functions.

3. System-of-Systems Map

A rail system is a federation of largely independent subsystems that share a handful of critical interfaces. OpenSourceRail is organized around eight top-level domains:

For detailed backend, onboard train, station/depot, wayside waypoint, energy, manufacturing, QA, and maintenance software diagrams, see `software-architecture-diagrams.md`.



Each domain has a dedicated section below. Interfaces between domains are deliberately narrow and enumerated in §5.

3.1 Business operating platform

ERPNext and Frappe HR own projects and task execution, purchasing, inventory, manufacturing administration, routine asset maintenance, finance and HR. OSR owns railway-specific engineering, control and assurance. Shared defaults and 266 city profiles compile versioned operating packages into native ERPNext Projects, Tasks, departments, warehouses and cost centres. The same compile now binds mechanical, station, civil and Rust items to common configuration, quality, competence, release-evidence and lookup-only identity templates; QR payloads remain disabled until an operator provisions a resolver and verifies each physical binding. Permission-filtered execution snapshots return to Workbench and the Project Twin view. ERP status cannot approve railway handback or command the railway. Historic SQLite business records remain readable, with new business writes disabled.

3.2 Design And Delivery Control Plane

The OSR Workbench integrates City Studio, simulation, OCC training, ERPNext/Frappe HR, FUXA and OSR railway assurance under one origin. It carries city, actor, immutable revision, approved baseline, simulation run and selected-asset context without merging authority boundaries:

- City Studio owns planning intent and Git-reviewable revisions;
- the Workbench's allowlisted any-city generator turns a selected catalogue design into assets, family-scoped assembly tasks, finite-resource CPM, schedule of values, order-by plan and local/imported cashflow, then opens it in Operations without accepting shell text or arbitrary paths;
- GIS, OSR-ALN, IFC4.3, IDS/BCF, CAD, cost and simulation files are generated evidence tied to source and tool hashes;

- planning and training views cannot issue live OCC actions;
- live mode does not expose design or simulation controls;
- revision hashes prove identity, not approval; approval history is append-only and evidence-referenced.

The Bonsai civil workflow is the federation, inspection, drawing, quantity and 4D review path. OSR project and engineering rules remain authoritative.

4. Reference subsystem designs

For each domain, we state (a) the dominant current practice, (b) its problems for our target users, and (c) the OpenSourceRail approach.

D1. Operations & Dispatch

	Legacy	OpenSourceRail
Platform	Proprietary SCADA (Thales ARAMIS, Siemens Vicos, GE Transportation) on Windows	Event-sourced Rust services on Linux SBCs/servers, web-based UIs
State model	Periodic polling, opaque internal state, vendor-specific DBs	Append-only operations log (single source of truth), CQRS read models, standard Postgres
Observability	Proprietary HMI screens, vendor-locked historians	OpenTelemetry traces + Prometheus + Grafana; historian is just retention on the log
Integration	Per-vendor OPC/IEC 60870-5-104 gateways	Typed gRPC/NATS interfaces with OPC UA at the hardware boundary
Disaster recovery	Active/passive OCCs with manual cutover	Multi-region event log replication, any dispatcher terminal can fail over to a replica

Deprecated: Windows-based SCADA, proprietary HMIs, per-vendor historians. Why this is safe: Dispatch is non-safety-critical (safety is enforced at D2). The ops platform is free to use modern stacks.

D2. Train Control (Interlocking, ATP, ATO)

This is the heart of the safety argument and receives the most design attention.

	Legacy	OpenSourceRail
Block model	Fixed block (track circuits, axle counters) or vendor CBTC moving block	Pilot: independently detected fixed/sectional blocks. Research target: software-defined moving block via distributed consensus after independent assessment and field closure.
Interlocking	Relay panels or proprietary PLC-based (Simis, Smartlock) at $55k–$550k/site	Open Rust logic hosted on commodity wayside computers, with fail-safe outputs controlled by a separately selected and qualified safety channel. Hardware cost remains open until that channel and its integration are frozen.
Position	Track circuits + balises (Eurobalise)	Sensor fusion: GNSS + IMU + wheel odometry + low-cost UWB/beacon fixes at switches and platforms
Movement authority	Centralized zone controller issues MA to trains	Pilot: conservative wayside sectional authority bounded by independent occupancy. Consensus and onboard computation run in shadow first; distributed authority is a later gated target.
Certification path	Per-vendor SIL-4 case, years to re-certify	Open formal models + continuously regenerated safety-case evidence, assessed per deployment

Pilot boundary: Independent train detection, conservative sectional movement authority and a separately qualified fail-safe safety channel are required by the reference pilot profile. Track circuits or axle counters are therefore not excluded from a pilot. The target architecture may reduce trackside equipment only after the staged evidence gates in the pilot signalling profile close.

Any residual train-control rate in the parametric city cost model is a planning allowance, not a supplier quote or a claim that a safety system can be delivered for that amount. A pilot estimate must explicitly cover independent occupancy detection, the qualified safety channel, communications, installation, integration, assurance and contingency.

Why this is novel: "Rail as a distributed system." The repository explores a restricted consensus protocol with fixed membership and no dynamic reconfiguration in the hot path. The Rust implementation has property tests, selected bounded Kani checks and a documented TLA+ abstraction map. A complete machine-checked refinement proof does not yet exist; the implementation is therefore property-tested research software, not a proved signalling product. Key risk: regulators, insurers, and procurement authorities need more than a technical artifact. A Git repository cannot carry the safety certificate or product liability for a railway. Mitigation: publish the formal model early, solicit review from independent safety assessors, and start with deployable non-safety subsystems, yard/test-track trials, and segregated-ROW pilots before any full metro deployment.

D3. Communications

	Legacy	OpenSourceRail
Mainline radio	GSM-R (sunsetting ~2030), FRMCS over dedicated 5G	Public 5G SA with slicing where available; private 5G on n77/n78 or CBRS where not
Backup/telemetry	Dedicated leaky-feeder or TETRA	LoRa mesh for low-bandwidth safety telemetry (MA updates, position reports) — cheap, long-range, carrier-independent
Wayside backbone	Vendor-specific fiber rings, proprietary timing	Ethernet + TSN (IEEE 802.1) for determinism; PTP (1588) for time sync
Station/passenger	Per-vendor WiFi controllers	WiFi HaLow (802.11ah) for IoT/telemetry, standard WiFi 6 for passenger access
Security	Often weak (GSM-R has known weaknesses)	mTLS everywhere, Noise Protocol Framework on LoRa link, signed firmware on every radio

Deprecated: GSM-R, TETRA for rail, proprietary fiber timing protocols. Why LoRa for safety telemetry? Safety messages are small (<100 bytes) and infrequent (~1 Hz per train). LoRa gives 5–15 km line-of-sight range at sub-$10 BOM, so a mesh of wayside LoRa gateways provides coverage redundancy at negligible cost compared to a primary 5G link.

D4. Passenger Services

	Legacy	OpenSourceRail
Fare media	Smart cards (MIFARE, Calypso) with per-operator infrastructure	Account-based ticketing: mobile money (M-Pesa, UPI, etc.) + QR + optional NFC, validated via signed short-TTL tokens offline-capable
Ticket vending	Dedicated TVMs at $22k+/unit	SBC-based TVMs with commodity barcode scanner + contactless reader; <$1k/unit; most passengers use phones and don't need a TVM at all
Info displays	Proprietary LCD controllers with custom protocols	SBC + standard HDMI + minimal Rust display agent subscribing to the ops event stream
Announcements	Per-station PA controllers, manual voicing	Text-to-speech from the same event stream, multilingual by default; human override for incidents

Deprecated: Closed-loop smart card systems, dedicated TVMs as the primary sales channel. Why account-based first: Most target markets have leapfrogged cards via mobile money. Investing in a smart card ecosystem now is a decade-long dead end.

D5. Rolling Stock

	Legacy	OpenSourceRail
Trainbus	MVB + WTB (IEC 61375), proprietary silicon, decades-old	TSN Ethernet (IEEE 802.1 Qbv/Qcc) over standard copper/fiber — off-the-shelf switches, deterministic traffic
ECU diversity	20–50 ECUs from different vendors, different RTOSes, different toolchains	Single reference ECU hardware (RISC-V or ARM SoC + CAN-FD + TSN Ethernet PHY). Different "apps" (traction control, door control, HVAC, PIS) as Rust binaries on identical hardware
OS	Mix of VxWorks, QNX, proprietary	Hubris (or seL4) on safety-critical ECUs; Linux on infotainment
Energy source	Pantograph/shoe drawing from catenary/3rd rail; diesel genset on non-electrified lines	800 V-class onboard LFP. Three 225 kWh gross car packs on the promoted three-car train; automated conductive station charging; no continuous catenary.
Traction drive	Proprietary VVVF inverters from rail Tier-1s	Six supplier-neutral heavy-commercial-vehicle-class PMSM/controller sets on the promoted train, qualified against an OSR interface control document
Diagnostics	Per-vendor tools, per-train service laptops	Unified OpenTelemetry export to depot; any maintainer with a browser can inspect any train

Deprecated: MVB/WTB, per-vendor ECUs, VxWorks/QNX dependence, pantograph-as-primary-energy-source, diesel-genset rolling stock. Why a single reference ECU: Two trains of the same class today contain dozens of subtly different PCBs from a supply chain deeply entangled with specific Tier-1s. One reference board (manufactured anywhere with basic SMT capability) for all non-traction-power functions collapses the supply chain dramatically. Why onboard battery instead of continuous electrification: See D7 for the full argument. Each reference car carries one 225 kWh gross LFP pack in fire-resistant side enclosures beneath the longitudinal-seat zone, for 675 kWh gross across the three-car train. The enclosures have no saloon service opening; access and thermal-runaway venting are outward through the lower bodyside.

D6. Infrastructure Monitoring

	Legacy	OpenSourceRail
Track geometry	Dedicated measurement trains, quarterly runs	Continuous CBM: every service train carries a low-cost IMU + acoustic package; data fused server-side
Switch machines	Proprietary motor controllers with point-detection contacts	Commodity BLDC + SBC controller, dual redundant position sensors, reports to the train-control consensus log
Level crossings	Standalone controllers, minimal integration	Same SBC platform as interlocking; crossings are just wayside nodes in the consensus network
Bridge/tunnel	Periodic manual inspection	Distributed vibration/strain sensors over LoRa, anomaly detection in the ops platform

Deprecated: Dedicated measurement trains as the primary data source, isolated crossing controllers.

D7. Energy — Solar Generation, Trackside Storage, Battery Traction

This domain departs most sharply from legacy rail. OpenSourceRail systems are catenary-free by default, powered by solar PV deployed on railway-owned land with battery storage buffering the mismatch between generation and train demand.

	Legacy	OpenSourceRail
Traction energy delivery	Overhead catenary (25 kV AC / 1.5–3 kV DC) or third rail, fed from grid substations every 2–5 km	No catenary. Onboard battery traction; recharged at stations/depots from trackside storage (see D5).
Primary generation	Grid purchase at fixed tariff; dedicated traction substations	Solar PV on (a) platform/station canopies, (b) depot roofs, (c) right-of-way along track (vertical bifacial + between-rail panels)
Energy buffering	Typically none — instantaneous draw from grid	Repeatable 500 kWh stationary-LFP modules at stations, quantity proved by the timetable/energy model
Grid interface	High-power traction substations, often dedicated feeders	Smaller grid-tie inverters that import when PV + storage are insufficient and export surplus where regulations allow
Chemistry preference	N/A (no storage)	LFP reference chemistry onboard and wayside; no NMC in the baseline
Charging interface	Continuous pantograph/shoe contact along entire route	500 kW automated conductive side contact at station dwells; terminals use the same cabinet for longer
Regeneration	Often wasted in resistor banks	Fully recaptured to onboard battery and exported to trackside storage during platform charging
Control	Centralized SCADA	Per-site Rust controller + PV-storage-charge optimizer driven by the ops timetable (D1); autonomous on sub-second timescales

Deprecated by default in the reference architecture: overhead catenary, third rail, dedicated traction substations, resistor-brake dissipation, centralized thyristor rectifiers, diesel gensets on non-electrified lines.

7.1 Why catenary-free

- Capex. Catenary + support structures + paralleling posts cost $1–3M/km. Eliminating that is the single largest capex reduction available on a greenfield rail project.
- Opex and workforce. OCS maintenance requires specialized crews (live-line work, night possessions). Battery + station-charging infrastructure is maintainable by general electricians and renewable-energy technicians — skills that are more transferable and more widely available in target economies.
- Copper theft. Overhead copper is a persistent operational hazard in many developing markets. Trackside infrastructure in hardened cabinets is a much harder target.
- Phased deployment. Non-electrified lines become usable the day rolling stock arrives, with no "electrification project" preceding revenue service.
- Operational resilience. A single catenary fault halts every train on the section; a distributed storage+generation network degrades gracefully — trains can reach the next charging station on reserve capacity.

7.2 Why solar-along-ROW

- Railways own long, linear, already-cleared strips of land with predictable sun exposure. The real estate is effectively free.
- Generation is co-located with load. Losses and transmission infrastructure are minimal.
- Three complementary PV deployment patterns (pilots exist for each):

1. Platform and station canopies — straightforward, aesthetically acceptable, dual-use as weather protection.
2. Vertical bifacial PV along ROW boundaries — doubles as property fence; captures morning/evening sun at different angles than horizontal panels (smoother daily generation curve).

3. Between-rail / on-sleeper PV (Sun-Ways-style) — highest area utilization but must tolerate train passage and ballast tamping; treat as Phase 4+ after basic patterns prove out.

- First-order math at 5 peak sun-hours, 3 m usable PV width per track-km, 200 W/m² module density: ≈3 MWh/day per track-km. A light metro trainset consumes on the order of 3–5 kWh/car-km. A modestly loaded 3-car service at 20 round-trips/day consumes ≈180–300 kWh/day/km. That makes self-powered operation a plausible screening target across many OSR regions, but not a guarantee. The generated energy model must survive local solar, dust, temperature, grid, export-tariff, charger, storage, and timetable studies before a deployment claims net-positive operation. Grid import covers monsoon/cloudy stretches and overnight draw on trackside storage; grid export monetizes midday surplus where regulations allow.

7.3 LFP and the 800 V-class DC system

RFC 0021 is the controlling energy architecture. The promoted three-car train retains 675 kWh gross / 540 kWh routinely usable capacity, split into three independently serviceable 225 kWh LFP car packs. The pack is 650–700 V nominal, with its exact cell count and upper SOC voltage constrained by controller transient margin.

Stations use the same chemistry in commodity stationary modules: one 500 kWh gross module is the starting normal-station design, and the simulation may add integer modules where timetable, solar, grid, or degraded-service evidence requires them. In the three-car reference, one 500 kW DC/DC cabinet is shared between platform contacts. Four-car high-throughput deployments repeat that complete storage/cabinet/contact module three times and six-car deployments four times; the converter design itself is unchanged and the generator carries every module in CAPEX. This choice prioritizes mature, high-volume heavy-vehicle and stationary- storage supply chains. LFP is the sole repository battery chemistry; NMC and other chemistries are outside the reference design.

7.4 Scope fit

This architecture is tuned for urban transit duty cycles: short runs, frequent stops, predictable daily service patterns, and fleet sizes in the low tens of trainsets. Within that envelope the catenary-free + solar model is a credible default to test, not a universal replacement for electrification. Heavy freight and long-distance intercity service have meaningfully different energy profiles and are explicitly out of scope (§1); this section does not attempt to cover them.

7.5 Interfaces

New or modified cross-domain interfaces:

- I9 (Charging Dispatch): D1 ops informs D7 which trains are due at which charging sites and when, so trackside storage can pre-charge buffers.
- I10 (Energy Telemetry): D5 trains report state-of-charge and predicted arrival SoC to D1/D7; D1 can re-plan around a low-SoC unit the way it would re-plan around a delayed one.
- I11 (Grid Tie): D7 sites speak standard grid-interface protocols (IEEE 2030.5 / Sunspec Modbus) to the local utility.

These are additions to §5 and will be formalized in `crates/osr-core/proto/energy.proto`.

D8. Depot & Maintenance

	Legacy	OpenSourceRail
Work orders	SAP PM or proprietary EAM	Rust service on the same event-sourced platform; integrated with train telemetry
CBM	Optional, bolt-on, per-vendor	First-class; every ECU streams condition data continuously
Yard automation	Manual	Same train-control stack, restricted to yard speeds; useful proving ground
Energy	Grid-fed charging (where electric at all), diesel refueling	Depot PV+LFP-storage microgrid: rooftop solar, repeated stationary modules, overnight low-C balancing, and grid import for shortfalls.

5. Cross-Domain Interfaces

OpenSourceRail defines a minimal set of typed interfaces between domains. Everything else is internal to the owning domain.

#	Interface	Producers	Consumers	Transport	Format
I1	Track State Log	D2 interlockings	D1 ops, D5 trains	Raft-replicated, gRPC streaming	Protobuf schema `track_state.proto`
I2	Movement Authority	D2	D5 trains	mTLS gRPC over D3 radio	`movement_authority.proto`
I3	Train Position Report	D5 trains	D2	mTLS gRPC or LoRa (primary/backup)	`position_report.proto`

#	Interface	Producers	Consumers	Transport	Format
I4	Ops Event Stream	D1	D4, D6, D8	NATS JetStream or Kafka-compatible	CloudEvents JSON
I5	Telemetry	all ECUs/wayside	D1, D8	OpenTelemetry OTLP	OTLP standard
I6	Passenger-facing schedule	D1	public	GTFS-RT + GTFS static	GTFS standard
I7	Fare Token	D4 issuing service	D4 validators	Offline-verifiable JWT variant	RFC 8392 CWT
I8	Traction Setpoints	D7 controller	charging / inverter hardware	OPC UA	standard
I9	Charging Dispatch	D1 ops	D7 station charging sites	NATS JetStream	`charging_dispatch.proto`
I10	Energy Telemetry	D5 trains, D7 sites	D1 ops, D8 depot	OpenTelemetry OTLP	OTLP standard
I11	Grid Tie	D7 sites	local utility	IEEE 2030.5 / Sunspec Modbus	standard

Only I1–I3 are safety-critical. Everything else can fail without loss of safe operation (degraded service only). I9/I10 failure degrades energy planning but does not affect train control — trains carry enough reserve to reach the next charging site without active dispatch.

6. Platform Stack

6.1 Software tiers

Tier	Examples	OS	Rust flavor	Constraints
T1 — Safety kernel	Interlocking, ATP, door safety	Selected safety runtime / RTOS	`no_std`, RTIC	SIL-4 system target; selected bounded proofs only; full refinement, hardware qualification and assessment remain open; < 50k LoC per binary
T2 — Safety-adjacent	Train ECU apps, substation control	Hubris or PREEMPT_RT Linux	`no_std` or minimal std	SIL-2 target; strict coding standard; full test coverage
T3 — Supervisory	Dispatching, ops services, telemetry collectors	Linux (Debian or Yocto)	`tokio` std	High availability; not safety-critical
T4 — UX/Back-office	Dispatcher web UI, passenger apps, EAM	Linux / browser	`axum`/`leptos` Rust, TypeScript at edges	Standard web security practices

6.2 Reference hardware

- Wayside application host (W-SBC): RISC-V or ARM64 commodity compute, dual Ethernet, CAN-FD and hardware root of trust. It runs supervision, communications, logging and pilot/shadow applications; it cannot directly command a safety output.
- Train application host (T-ECU): A ruggedized host with TSN Ethernet, CAN-FD and isolated I/O for safety-adjacent and supervisory applications. Any safety function uses a separately selected channel and justified interface boundary.
- Safety channel: Not yet selected. Candidate families, evidence required for selection and the prohibition on using an SBC as the safety rationale are recorded in `control-electronics/safety-controller-selection.md`.
- Ops server (O-SRV): Commodity x86 or ARM64 server, Debian, standard datacenter gear. T3/T4.

Reference hardware has two implementation tracks under `control-electronics/`. The first-pilot track is the RFC 0019 COTS/SBC assembly path: commodity compute modules, sensor boards, relay modules, power supplies, terminal blocks, DIN-rail enclosures, prepared firmware images, wiring/harness maps, and bench evidence. It does not require KiCad or gerbers when no custom PCB is used. The volume-production track is the RFC 0007 custom-board path: OSR-specific carrier, power, safety-I/O, or sensor-interface boards captured as schematics, BOMs, layouts, and Gerbers. Custom boards must remain manufacturable by a 4-layer PCB fab with 0.15 mm trace/space and 0.3 mm vias — routine at tier-2 fabs across the target deployment footprint.

6.3 Repository layout

The canonical runtime-component map is in RFC 0005; the complete implemented inventory is checked by simulation software coverage.

```
OpenSourceRail/
├── crates/                     # Runtime, embedded, simulation and browser software
├── control-electronics/        # ECUs, SBCs, power/I/O, wiring and enclosures
├── design/
│   ├── city-generation/        # GIS, network and scenario generation
│   └── component-catalogue/    # Parametric train, track, station and civil products
├── cities/
│   ├── workspaces/             # Editable City Studio sources and revisions
│   └── catalogue/              # Published city and national evidence
```

```
├── engineering/
│   ├── analysis/                # Calculations and external-tool benchmarks
│   ├── interchange/             # IFC and station exchange generators
│   ├── models/                  # Public BIM and digital-twin review sets
│   └── assurance/               # Formal models and coverage evidence
├── tools/                       # Automation and independent utilities
├── lib/                         # Shared templates, recipes and schemas
├── deployment/                  # Deployable host compositions
└── docs/                        # Shared decisions, requirements and guidance
```

7. Safety & Certification Strategy

OpenSourceRail uses the maintained Digital Standards Thread and generated standards baseline, which distinguish publisher metadata, deployment applicability, evidence coherence and competent acceptance. Until a deployment-specific assessor and authority accept the evidence, SIL wording in this repository means target assurance class and hazard allocation, not certification. The approach is:

1. Formal models first. Signaling logic is expressed in a formal model (Kani, Creusot, or TLA+ where appropriate) before implementation. Refinement of the Rust implementation against the formal model is the intended assurance approach; the consensus refinement proof remains open.
2. Small safety kernel. T1 binaries are aggressively minimized. Everything that can be pushed out of the safety kernel is.
3. Diverse evidence where it exists; no invented redundancy. Property tests, bounded proofs, simulation, fault injection and selected cross-language differential checks challenge the same safety intent from different directions. Independent implementations, teams, toolchains or hardware channels are credited only when the exact released configuration provides them; they remain deployment design and assessment decisions.
4. Machine-checkable safety-case structure. Safety arguments are written in GSN (Goal Structuring Notation) serialized as TOML, with claim evidence pointers resolving to controlled sources and result records. CI rejects structurally missing or stale links. It does not turn a pointer, written harness or unaccepted result into proof that the safety case is true.
5. Independent assessment. The project produces artifacts. Certification is performed per-deployment by the national authority's chosen ISA, with the owner/operator and prime integrator carrying the accepted safety case, insurance, and liability. Reference safety cases from pilot deployments are published to compound assessor familiarity across countries.

8. Security Posture

- Zero trust on the wayside network. Every node authenticates every peer (mTLS, mutually verified certificates rooted in a project-operated PKI per deployment).
- Signed firmware, measured boot. TPM-backed attestation; update rollback via A/B partitions.
- Privilege separation at OS level. Hubris's "task" model gives hardware-enforced isolation; on Linux, systemd + seccomp + namespaces.
- No default credentials, ever. Commissioning generates unique per-device keys.
- Auditable by construction. Every safety-relevant event is in the append-only log; tampering is detectable.

9. Roadmap

Implemented software, design and evidence capabilities are summarised in the root feature table. Remaining work is maintained once in ROADMAP.md and the domain release checklists. Calendar predictions are deliberately excluded until an owner, funded work package and external evidence route exist.

10. Open Questions

The controlling open questions are deployment evidence rather than missing marketing claims:

1. Which eligible operator, university, integrator and independent assessor will govern the first bounded pilot?
2. Which supplier modules, host hardware and secure deployment transports pass the required qualification and lifecycle evidence?

3. Which surveyed alignment, ground, utility, climate, demand and fire data replace the planning assumptions for that pilot?
4. Which national rules determine homologation, cyber approval, accessibility, grid connection, battery end-of-life and operating responsibility?
5. Which public, research or development-finance route funds each evidence stage without treating screening outputs as an investment approval?

Detailed gaps are tracked in the certification, civil, rolling-stock, hardware and operations release checklists linked from ROADMAP.md.

11. Architecture Reading Order

Start with the root Find Your Way table. For a deeper architecture review, continue in this order:

1. This document.
2. `docs/GLOSSARY.md` — rail-domain terms for software engineers and vice versa.
3. `docs/rfcs/0001-track-state-consensus.md` — the distributed track-state log design.
4. `docs/rfcs/0002-energy-sizing.md` — quantitative sizing for the catenary-free, solar-first energy architecture.
5. `docs/rfcs/0003-samawah-reference-deployment.md` - Samawah context, pilot rationale, and risks; generated city numbers live under `cities/catalogue/west-asia/Iraq/Samawah/`.
6. `docs/rfcs/0004-osr-interlocking-plan.md` — implementation plan for the Rust SIL-4 MA computer.
7. `docs/rfcs/0005-sbc-software-architecture.md` — canonical SBC software architecture + crate map.
8. `crates/osr-core/proto/track_state.proto` — the interface definitions.
9. `engineering/assurance/formal/tla/SMRaft.tla` — TLA+ spec of the consensus protocol.

This document is a living architecture brief. Material changes go through RFCs in `docs/rfcs/`; editorial changes can be PRs directly.

Glossary

Repository source: `docs/GLOSSARY.md`

Common rail, software, hardware, and certification terms used across OpenSourceRail.

Term	Meaning
ATO	Automatic Train Operation: controls traction/braking within the movement authority.
ATP	Automatic Train Protection: safety layer that trips or limits the train when authority, speed, or position is unsafe.
AW0 / AW2 / AW3	Passenger loading cases: empty, normal loaded, and crush loaded.
BOM	Bill of materials. The repo separates planning BOMs, procurement BOMs, and board BOMs.
COTS	Common off-the-shelf component. In OSR docs this means sourceable modules with supplier evidence, not hand-designed internals.
EN 14752	European standard for bodyside doors on railway vehicles.
EN 15085	Welding standard for railway vehicles and components.
EN 15227	Crashworthiness standard for railway vehicle bodies.
EN 50126 / 50128 / 50129	RAMS, safety software, and safety-related electronic-system standards for rail.
EN 62267	Automated urban guided transport safety standard; primary GoA 4 reference for OSR certification docs.
FEA	Finite element analysis, used for structural, fatigue, and crashworthiness evidence.
GoA 4	Grade of Automation 4: unattended train operation with no driver in the train.
GSN	Goal Structuring Notation, used for the machine-checkable safety-case structure.
HVIL	High-voltage interlock loop, proving HV connectors/covers are closed before energisation.
Kani	Rust model-checking tool used for bounded safety-property proofs.
MA	Movement authority: the safe track extent a train is allowed to occupy.
MBOM	Manufacturing bill of materials, more detailed than the repository's planning/procurement BOMs.
NDT	Non-destructive testing such as visual, magnetic-particle, ultrasonic, or radiographic inspection.
OCC	Operations control centre.
PRM	Persons with reduced mobility.
PSD	Platform screen door.
SIL	Safety Integrity Level. OSR uses SIL-4 for train protection and safety-critical supervision.
TCN	Train communication network.
T-ECU/S	Train electronic control unit, safety kernel.
T-ECU/A	Train electronic control unit, application layer.
T-OBS	Train obstacle-detection host.
W-SBC / S-SBC	Wayside and station single-board-computer host classes.

Roadmap

Repository source: `docs/ROADMAP.md`

This page tracks remaining validation and hardening work beyond the v0.4.0 software baseline. It is a planning map, not a release contract. The open-source CAD, survey, analysis, and simulation selections and the evidence-producing work packages for the remaining engineering items are in the engineering design and simulation plan.

Reviewed Open Work

The six-item operations review implementation adds bounded possession/resource, fleet/energy and staff planning, offline field execution evidence, reliability/RUL validation and operational/GTFS exchange. Remaining work is real topology/energy and labour-policy calibration, provisioned native-site acceptance, measured predictive validation and country/operator transport adapters; the existing railML geometry subset is unchanged.

The 2 October platform review follow-up implements logical A/B train protection, paired outputs, the late-feed fix and FMEA applicability checks. The development policy retains two-out-of-two permission and either-channel trip; all physical channel qualification and the five other local pairs remain open. Priorities are autonomous watchdog/output execution, typed sensor/emergency integration, repeated station missions, the calibrated shared plant and sealed city export, sustained resource/storage duty, specific fault-response evidence and hardware freeze quantities. Existing city duty/energy/stabling and `osr-sim` remain the planning sources. Service arrival in the short fixture does not close these integration tasks or permit operational release.

The city deployment evidence workflow now covers the complete catalogue with soilDB civil screening, preserved field receipts, current native/SUMO timing comparisons and per-city deployment gap registers. The registers distinguish completed software/data work from outstanding site evidence. The city workforce and design packages add funded role/skills/task breakdowns, construction concurrency, climate-bound movement sensitivities and asset-level joint/finish/cleaning work. Named rosters, crew sizes and material/site qualification remain explicit closure work.

The 2026-09-09 substance review now records completed roof-drainage input/physical-balance repairs and per-site electrical import/export gates. All seven drainage variants pass the corrected planning screen. The ethos audit corrects the electrical interpretation: the reported connection exceedances occur with PV and storage disabled, while every existing coordinated daylight case meets connection limits. The repaired gate reports grid-only findings as diagnostics; all 266 solar/storage snapshots pass, with full operating energy validation still open. Stabling now accepts storage-backed stations without grid import or a 150 kW minimum, and a grid outage alone does not disable a battery-backed destination. The 2026-09-10 shared-source and charging-headroom corrections are checked with fresh passing two-day hybrid replays in all nine retained cities. Historical station-only experiments remain outside selected-plan package acceptance. The 2026-09-11 broader validation adds Uige, Quelimane, Edea, Bukavu, Soroti, Sheikhupura, Sumbawanga and Tartus to Samawah and Mosul: all ten pass two nominal runs and eight degraded cases with current source provenance. The remaining 256 cities retain their full-service refresh gate; the deployment summary tracks coverage. These findings do not prescribe grid upgrades. The depot energy quantity contract is now shared by operating tiers, layout requirements and the BOM, with all 266 cities reporting PV area, equipment-cost sensitivity and initial dispatch capacity requirements. Located equipment, installed-cost reconciliation, physical stabling allocation, procurement/calendar cashflow, battery renewal and shared factory scheduling remain open. Distributed station holding, low-C charging and morning restart now have an executable candidate and passing Samawah holding/restart behavior; declared reserve roles and every planned departure direction are now checked. Reference platform comparisons expose unresolved stabling space; verified track capacity, reserve activation and full-day timetable/energy acceptance remain open. See the station-stabling workflow. The charging-reachability gate repairs the six/ten-train overnight stranding found in Samawah's first continuous replay. Both service cycles pass placement and morning restart, with all trains at selected powered stations, but the combined replay now fails the explicit two-train station-capacity gate. Queues still reach 18 then 19 trains; recurring physical capacity, daytime headways and outage/recovery duties remain unverified. The overnight plan now assigns two revenue trains per selected station and the remaining fleet to storage on its own line. Samawah allocates 40 to stations and 37/16/15 to Line 1/2/3 storage, separately from workshop bays. The station-only replay and seven-train redistribution study remain diagnostic benchmarks. Explicit native home allocations now support same-line depot returns and morning dispatch, validated across two cycles on a connected eight-train fixture. Two-day city hybrid screens pass for Samawah, Uíge, Quelimane, Edéa, Bukavu, Soroti, Sheikhupura, Sumbawanga and Tartus, checking actual homes, storage capacity and morning launch stock. Lines operate independently, with no interline depot connections; physical yard routing and city evening placement remain subject to verification. The corrected RFC records the emitter's family-specific fleet assumptions instead of the stale uniform 10% recovery factor. Gulu, Rahim Yar Khan and Tanta already declare two charging cabinets in their design cost basis. The scenario generator now applies that count consistently to charging validation, station power and site equipment, preserving their retained charging/grid settings. This closes the energy-configuration drift; their broader scenario and operating-evidence refresh remains open.

The 2026-09-08 repository review found no failing committed software, city, project-twin, link, or browser acceptance baseline. The documentation audit also reconciled the current rolling-stock product, assembly, tooling, factory, mass and

evidence-package counts; added mechanical and civil workface controls plus the owner–builder–operator mobilisation plan; regenerated the global index and reader book; and added checks that make those front-door claims fail when their machine-readable contracts change.

The following work remains genuinely open and must not be represented as complete merely by adding documentation:

The operating-readiness audit removes one formerly implicit software gap: all 266 city ERP profiles, component packages and real-asset supervision packages now compile against tracked city evidence. Only two full ERP task payloads are stored in Git; the other 264 are deterministic on-demand outputs. Legal companies, approved calendars, ERP masters, physical bindings, supplier interfaces and commissioning remain external inputs rather than fabricated defaults.

The next engineering-change slice is now executable for supervisory packages. A read-only, checksum-bound review identifies exact design/embedded contract changes and affected component, topology, IFC, source-crate, ERP, installed- serial, evidence, open-case and pending-command records. Workbench makes the result usable per asset and correlates existing ERP execution feedback; apply rechecks it transactionally and leaves unrelated city baselines unchanged. The ERP exposure reader now follows reviewed Item/BOM mappings through explicit nested production BOMs, project purchase lines, draft/unfinished Work Orders and linked stock movements. Workbench shows permission gaps and snapshot provenance; purchase allocations and current stock are not inferred. Complete CAD assemblies, solver reruns, engineering acceptance of rework/stock dispositions and formal evidence supersession remain open. Native ERP now records assigned disposition proposals and independent endorsement/rejection with stale-exposure checks; those plans do not execute business actions. Independent verification now checks all catalogue outcomes against full reviewed exposure: retention, cancellation, direct amendment, stop, native corrective Job Cards with accepted inspections, performed stock inspections and movement tracing. Later changes make verification stale. These checks do not accept manufacturing configuration or grant railway release. See revision exposure.

Native operating supervision now also covers the catalogue's identified switches through the existing fail-restrictive points evaluator, plus one clearly labelled AFC aggregate per station. The same adapter and template contract is ready for level crossings, but zero city asset registers currently identify one; field identities and commissioned sensor bindings remain open rather than being invented. These FUXA/ERP views are maintenance-only and expose no safety command, reset, route state or movement authority.

Factory supervision now closes the duplicate-recipe and uncorrelated-dashboard gap: nine views are generated from the existing LM3 methods on production-plant identities in cities whose baseline selects `light-metro-3car`. Other families require applicable methods or reviewed module reuse. Workbench joins native Work Orders through the exact Item/BOM pair at the reviewed engineering revision. It can rehearse cell unavailability and a quality hold through the durable Issue route. Real machine adapters, capacity dispatch, physical traveler/inspection values and manufacturing acceptance remain open. The native evidence workflow now checks performed inspection readings and corrective Job Cards with accepted follow-up inspections; no simulated value closes those gates.

Boundary	Remaining evidence	Authoritative register
City and civil design	The 19 reusable IFC types now have exact accountability across six release packages and nine drawing-definition briefs, separating nine civil-owned types from ten controlled track/station/vehicle interfaces. Shared deployment gates cover field receipts, control, surveyed ground/alignment, route fit, SWMM replay, borehole-zoned foundation selection and an independent OSR/SUMO running-time comparison across every pilot line. Real observations, supplier freezes, first articles, junction-conflict evidence, signed studies, per-span calculations and local approvals remain external.	Civil type/release register, drawing briefs, Samawah field brief, alignment, route fit, operations cross-check, drainage/ground and civil checklist
Stations	The 45-product/seven-variant FreeCAD and IFC4.3 coordination geometry, BOM/traveler reconciliation and structural, passenger, drainage, thermal and fire screens are present. The failed enclosed depot baselines now have a screened separated/open compound and N+1 cooled-controls response. Supplier/site freeze, surveyed placement, project calculations, drawings, fire strategy, commissioning and approvals remain.	Station gap register, systems evidence and mitigation work packages
Rolling stock	The 120-product-row/26-assembly IFC and FreeCAD design-reference library, nine solver-backed structural screens, 41 manufacturer/research candidates covering all 56 bought-in rows, 16 factory packages, 29 individual drawing-definition seeds and one executable 13-gate first-article route are present. Drawing ownership and the production-data register now cover all 62 locally made rows, including bogie frames, door/window carriers, battery/HV hardware, configurable ends and trainline harnesses; 472 drawing, material, cut/form/NC, mould, harness, finish and inspection artifact slots remain fail-closed. The mass ledger maps 120/120 rows but correctly leaves 0/117 active rows mass-closed; all 16 drawing packages open and all production-data releases remain open. Supplier configuration freeze, production FEA/drawings, weld/NDT, calibrated mass/CG/axle-load evidence, thermal/HV/fire tests and first articles remain.	Production-data register, drawing seeds, factory readiness, mass closure, finish system, COTS/RFQ candidates, public work packages, execution pack and evidence status

Boundary	Remaining evidence	Authoritative register
Control electronics	Exact pilot SKUs, real harness/enclosure/power packs, deployable signed images, bench evidence, and KiCad fabrication data where a custom board is selected	Control-electronics release checklist
Software and operations	Ops Core now has local authenticated RBAC, managed evidence/documents, server attestations and verified backup archives. Production TLS/SSO/MFA, hosted secret/backup operations, signed live train actions, production transports, HIL and operator validation remain.	Operations Portal, certification and operations validation
Approval	Supplier qualification, physical tests, independent safety assessment and national/operator authorization	Safety case and certification

The rolling-stock packages now link to ten manufacturing and assembly controls, and the civil packages link to ten fabrication and construction controls. These fill ordinary work-sequence, inspection, handling and handback gaps. The linked inspection plans now expose 416 LM3 and 114 civil package characteristics as explicit hold, witness or review points. They do not close the supplier, drawing, calculation, site or authority evidence in the table. Blank mechanical and civil execution records prevent planned checks from being mistaken for performed work. The owner–builder–operator plan and its fail-closed mobilisation status assign an institutional route for closing those gaps.

The 2026-09-10 continuation repairs the simulator GUI's depot-return event handling, the CI soil-coordinate dependency, a stabling-test lint failure, stale Soroti/Sheikhupura engineering plot manifests and Workbench startup. Workbench readiness now checks the lightweight page endpoint instead of repeatedly compiling project data under a one-second connection timeout. The LandXML companion-file workflow now imports revision-bound station IDs, civil spans and cant with source, line, CRS and coverage checks. Vendor-specific survey/cant parsing and coordinate transforms remain outside that implementation. The operations follow-up also makes overlapping trains share one grid-import and battery-discharge allowance per simulation step, including charger losses; the previous code shared cabinet output but reapplied source limits to each request. Its replay exposed a numerical charging-gap hold in Samawah: T45 arrived at an unpowered stop about 1 Wh short of the next departure threshold. Powered departures now retain an additional 0.001 percentage point of SoC headroom; the 20% reserve gate is unchanged. The focused regression and fresh Samawah two-day replay pass. Electrical sizing and daytime timetable acceptance remain open.

Repository work that can advance without external evidence remains visible in the analysis register and workstream table: charger duty, station analyses, capacity-constrained passenger boarding and physical pedestrian capacity. City Studio now assigns OD flows to connected scheduled routes and combines directional section and station loads. Civil jobs tessellate native IFC solids into source-bound chunks for the existing object/4D viewer. The review closed the previously planned CalculiX thermal-block solver benchmark with a deterministic input, analytical acceptance check, and machine-readable result. These tasks improve screening evidence but cannot close the external release gates above.

The competitive position and proof roadmap now separates three propositions that must not be collapsed into one claim: the planning/design/delivery platform, the owner/operator product, and the physical railway/GoA4 development programme. Near-term product evidence therefore prioritises measured owner/operator deployments, independently comparable corridor candidates, supplier-neutral owner-data interfaces and closure of physical first-article gaps. Samawah and Mosul are initial examples, not a limit: any catalogue or customer corridor can enter the portfolio, but each keeps its own funding, inputs, outcomes and approvals. Adding generated cities alone is not a substitute for customer, supplier, field or acceptance evidence.

The commercial path is also fail-closed. The joint-development framework starts with a funded development agreement and gates any later equity JV on verified legal counterparties, accepted contributions, IP/manufacturing rights, separated development/project finance, related-party controls and measured localisation. The public partnership status remains open until real evidence is supplied in a controlled opportunity copy.

Potential supplier support now has a separate technical evidence path. The supplier technical-support package maps the existing LM3 product tree, interface register, COTS/RFQ candidates, first-article gates, city duty models and ERP lifecycle into eight review and delivery work packages. Its generated readiness record separately controls authority, exact configurations, cross-supplier responsibility, first-article evidence, localisation, lifecycle support and all-in cost. Public catalogues or conversations cannot close those gates.

v0.4 Workstream

Workstream	Target outcome
Documentation accessibility	One root front door and source registry are enforced; local READMEs retain only discipline/city evidence, with generated inventory and link/drift checks
Owner–builder–operator institution	Establish the public owner, ring-fenced delivery/operator company, independent assurance, accountable executives, competence pipeline, management systems and gated move from development to revenue service
First adoptable owner/operator product	Deploy one or more existing workshops/depots or bounded pilots, preserving the no-command boundary while measuring record completeness, defect-to-handback time, recovery, support effort, user errors and owner-controlled export/exit separately for each deployment
Independent corridor benchmarks	Maintain an open candidate portfolio; freeze a separate input package for every selected corridor and publish matched native/SUMO geometry, service, demand, energy, degraded-operation and civil/cost results with predeclared tolerances and adjudicated differences
Commercial partnership evidence	Qualify actual legal counterparties and customer scope; accept contributions and IP/manufacturing rights before equity; separate development from railway finance; permit only finite, performance-based exclusivity
Supplier-backed LM3 definition	Freeze one route/duty/configuration; obtain controlled bogie/brake and traction/battery/charging packages; close eight cross-supplier interfaces; assign engineering responsibility; separate NRE, tooling, first-article, repeat, transfer, spares/support, logistics and acceptance costs; accept evidence through TS0–TS6
Rolling-stock detail package	Close the 16 factory packages and 117 active mass rows with supplier-exact envelopes, production solids/masses, weld maps, tolerance stacks, harness clamp locations, FEA-ready brackets, 2D drawings, NC/flat-pattern outputs and calibrated weight/balance evidence
Mechanical CAD	The candidate layer and generated FreeCAD/IFC review geometry are implemented; next closure is supplier-returned exact configurations, mass/envelopes, released drawings and production CAD/NC data
Control-electronics integration evidence	Pilot-ready COTS/DIY integration packs for T-ECU/S, T-ECU/A, T-OBS, W-SBC, and S-SBC: exact SKUs, wiring/harness maps, connector maps, enclosure/mounting notes, power/thermal margins, SD-card images, self-test logs, and bench records
Custom-board release artifacts	KiCad capture, gerbers, board BOMs, DFM review, and assembly drawings only for deployments that choose OSR-specific carrier, power, safety-I/O, or sensor-interface boards
DIY deployment path	Prebuilt SD-card images, checksums, role-specific self-test evidence, and first external build feedback
Software integration	Workbench context plus onboard, station, intrusion and T2G-to-depot CBM/historian/analytics software-in-loop are implemented; supervisory revisions now expose source-crate, telemetry, alarm and command-contract impact and reject clock regressions/stale reviews; signed live actions, production transports, asset-specific points/crossing/fare-gate harnesses, HIL, and authenticated live GUI paths remain
ERP and supervision integration	All 266 tracked ERP/component profiles and real-asset supervision packages compile through one drift-checked readiness gate; generated factory views reuse all nine LM3 methods and reviewed mappings correlate native Work Orders; assigned disposition plans and independent review are implemented; independent native evidence verification covers all disposition catalogue actions; next add commissioned machine/quality inputs, company/calendar/ERP masters, physical mappings, permissions, recovery evidence and operator validation
City Studio	Git-backed city projects, source locks, layered offline GIS editing, complete manual line/station/alignment authoring, semantic revision comparison including BCF, OD demand, per-line IFC survey control and revision-locked field-to-structural-release gates, plus controlled engineering jobs, verified viewers and 144-check Playwright browser/restart persistence acceptance (implemented); native tessellated IFC viewing and routed OD assignment with shared section/station loads are now implemented; received field data, capacity-constrained boarding and platform/interchange pedestrian capacity remain
Certification evidence	Tool-backed/assessor-accepted consensus refinement, qualified safety-controller freeze, residual-risk narrative, independent-assessor review notes, first-article field-evidence plan, and traceability updates
Civil/station package	Close the six civil release packages with survey-grade alignments, ground models, supplier data, per-span checks, reinforcement/prestress, first articles and signed deployment releases; close the station packages against surveyed placement and project evidence

Open release gates for certification, hardware, rolling stock, civil/station, and operations are tracked in the relevant section checklists rather than left as implicit TODOs.

v0.4 Definition Of Done

- All top-level domains have local READMEs and current status notes.
- `python3 tools/automation/repo-health.py --quiet` passes on the release tree.
- Rolling-stock documentation links to the current FreeCAD/PNG/COTS package and clearly separates envelope CAD from production drawings.
- Hardware host classes have either pilot-ready COTS/DIY integration packs or explicit custom-board KiCad/gerber/BOM gaps where custom boards are actually required.

- Certification and safety-case pages identify remaining external evidence rather than implying it already exists.

Repository Media Assets

Repository source: `docs/assets/README.md`

Asset	Status	Purpose
`logo.png`	Retained original media	Project wordmark for release and outreach use
`solar-metro-trainset.png`	Current front-page image	Reference vehicle overview
`solar-metro-production-assembly.png`	Current documentation image	Production/assembly overview
`digital-twin-animation.gif`	Generated systems animation	160-frame FreeCAD/Coin3D ground/elevated operating-scene review; generated by `tools/automation/freecad-generate.sh --digital-twin-animation`

New media must record its source or generator, license, and intended consumer. The primary guided assembly MP4 and its Blender source are retained with the `engineering/models/digital-twins/` review package. The operating-scene GIF generator renders 160 source-driven frames and rejects an encoded file at or above 20 MB.

Baghdad review follow-up: controls, CI and deliverable-service sensitivity

Repository source: `docs/baghdad-ci-controls-review-2026-10-04.md`

> Historical revision review. Its figures describe the revision reviewed below. The reworked central alignment, current staffing/depot scope and current cashflows are in the Baghdad city summary and programme recalculation.

The attached review examined `aff4fbc9`; the subsequent continuation already integrated depot, energy, payroll and replacement-reserve cash. This follow-up strengthens that implementation and preserves the original financing case as a comparator. None of the engineering alternatives has field acceptance or committed finance.

Financing and service

The integrated monthly and six-month ledgers now include interim development leadership from NTP and separately purchased mentoring before opening. Existing contracted teaching/assessment is scheduled once. Added development/mentoring cash is IQD 2.582bn before inflation. The full-fleet sensitivity has USD 8.060bn capital, IQD 11.360tn peak gap debt, IQD 7.532tn terminal gap debt, and USD −4.692bn company cash NPV before finance. Approximately USD 515.831m equivalent of eligible core principal is prepaid after reserves, buffers and assumed call premiums; this does not clear gap debt.

Government remains 25% of scenario capital. Imports retain 50% government USD / 50% proposed Chinese USD credit. Remaining government cash, ordinary/green bonds, bank and gap credit, fares and commercial receipts remain IQD. There are no new commitments, subscriptions or guaranteed property proceeds. Existing variable fares and 5% annual fare/OPEX/income growth remain explicit assumptions. The IQD 1,750 sensitivity still clears debt in month 378, with negative company cash NPV and unverified demand/affordability.

The new installed-energy supply bound retains existing electrical capacity, removes purchases/connection charges for the unbuilt expansion, and reduces fares, station commercial receipts and additional commercial income. Synthetic installed charger/energy throughput is about 87.27% of allocated traction demand. Its terminal gap debt is approximately IQD 12.967tn, with IQD 2.787tn uncovered support. This is an energy-throughput upper bound; actual charging events, train circulation and a feasible timetable remain unaccepted. The firm-service case instead purchases shortage energy and requires unpriced physical upgrades. Neither may be described as a complete bankable budget or achieved service.

All integrated cases model zero dividends. Cash left after the debt/reserve/buffer waterfall stays in the company; audited after-tax distributable profit, covenant permission and shareholder approval remain unresolved. Earlier holding-company returns describe their original scope and must not be transferred to these higher-cost sensitivities.

Engineering and mobilisation

The 28 former pooled shortage hours remain available as a comparator. Site-hour diagnostics trace grid/charger limits, opening/closing stored energy and generation paths. On-site PV receives no remote wheeling loss or charge. The reference has 2,194 distinct hours with at least one bus shortfall; the larger site-hour count also includes charger overload. Those counts measure different scopes.

Retained workshop sites explicitly require line-local covered lifting, bogie exchange, wheel servicing and body overhaul. Eight zero-bay sites still need additional facilities and quotes; mobile tools do not replace covered bays, and a disconnected central workshop provides no train-transfer route. Road transport covers removed components only. The integrated case uses workload bays at all nine sites. The five missing morning-direction requirements remain an open gate: frozen-start alternatives require either an accepted smaller start-point policy or five revenue trains plus one spare and consequential engineering/funding.

The 450-train factory replay removes deferred orders and rebuilds factory resource queues, including the extra line-9 spare. Manufacturing selection, procurement cash, energy, operating payroll, maintenance and battery reserves follow the lower supply. Original expansion-capable factory/depot capital and civil opening dates remain funded; no earlier receipts or cheaper plant are invented. The case still has approximately IQD 6.525tn uncovered support, and deferred expansion remains unfunded.

Development and training schedules distinguish interim authorities, operating successors and signed handovers. Training leadership starts at month 18 and rigs must be ready by month 24, before the first operating cohorts join at month 34. Minimum purchased peak capacity is 19 teaching, 35 assessment and 90 mentoring posts, at different times. Operating recruits supply zero qualified training capacity. All 21 operating roles have lesson/coverage requirements; actual equipment, assessors, daily/rest rosters and named appointments remain absent. The anonymous OCC roster is only a pilot.

The editable metro-six-car price now feeds fleet procurement and the parent BOM allocation. Workforce conversion uses its configured FX reference; publication rejects inconsistent funding/workforce/corridor FX assumptions.

Workforce controls and CI corrections

The evaluator now checks task/location/asset/isolation scope, named authorised supervisors, qualified independent verifiers, correct calibrated tools and released part/revision/quantity. Duplicate stock records cannot count twice. Regression tests reject the review's combined wrong-line permit, missing supervisor and unqualified verifier example.

The native workforce preview validates a private attached snapshot. `snapshot_eligible` describes that snapshot; `eligible` and `live_start_check` remain false because current authoritative revocation is unresolved. Rechecking a timestamp detects expiry, not later revocation. Human task start still requires current authoritative evidence and work-release authority. No worker or assignment was created.

The six observed CI failures have corresponding changes:

- The Python test environment installs PyYAML. The hash-verified bootstrap restores declared energy/GIS/SUMO inputs and preserves differing workspace data.
- Portfolio browser expectations use the authoritative API and the UI's formatting rather than stale capital literals.
- Disabling unused postcard default features removes the heapless-CAS dependency path containing unmaintained `atomic-polyfill` and yanked `spin`; the wire fixture remains unchanged.
- TLC 1.8.0 is pinned to SHA256 `c2fe4e56e43bde19f213b4a7e441d037297fda733e503579623e859b79348239`, independently matched to upstream asset metadata and the published SHA1. A changed artifact must fail verification rather than run unverified.
- Workbench uses a viewport-sized flex layout instead of an oversized minimum frame. The native test dismisses an asynchronous ERP update notice through its visible close button when present. Proposal, review and verification retain real clicks without forced clicks.

5 October follow-up: upstream replaced the `v1.8.0/tla2tools.jar` asset with a different build, and CI correctly rejected its changed checksum. The retained TLC executable preserves the previously verified SHA256 and published SHA1 above. CI copies and verifies those exact bytes before executing the model; accepting a replacement requires a separate tool review and fresh assurance execution.

Local verification includes 124 focused tests, all 43 frontend acceptance tests, native disposition clicks with a deliberately obstructing Frappe notice, the source-bound TACS twin and bounded formal execution, and the Rust workspace/audit checks. The broad Python run passed 1,779 tests and twelve subtests, with one cache-dependent skip; its sole stale-offer failure was fixed and the four offer tests then passed. A further 124 publication/finance tests passed after regeneration. Source/output, archive, repository-health and link checks pass. Passing software checks provides no construction, railway-operation or investment release.

ERPNext now runs app 0.5.1 in the reproducible `osr14` image; deployed workforce/rules/probe hashes match repository files. Refreshing six packages updated 34 unique existing evidence Tasks, preserving native identity and operator-managed fields. A fresh SQL backup matched six inventory/schema/permission fingerprints in an isolated disposable database, which was removed. Private backups and receipts remain outside the publication.

For a source change, restore the exact ignored inputs first, then regenerate in this dependency order: Iraqi funding programme; delivery stress; viaduct rentals; qualification; financing redesign; equity; delivery baseline; delivery closure; reference offer; complete proposal; reader book. Source-bound checks reject stale intermediate packages. The proposal bootstrap also preserves differing local solver/GIS data rather than overwriting it.

Baghdad continuation: physical packages and revised financial conclusion

Repository source: `docs/baghdad-continuation-review-2026-10-04.md`

The delivery continuation completes the repository calculations and reviewable work packages following the first delivery reconciliation. Surveyed designs, quotations, actual demand, appointments, subscriptions and independent field acceptance remain open. The original 831-train, nine-line programme and factory opening dates are unchanged; new cases are explicit alternatives.

Financial finding

The full-fleet cost sensitivity replaces the USD 8m depot allowance with the USD 175.932m workload package and adds an explicit 7% incremental EPC allowance. Programme capital becomes USD 8.060bn, before still-unpriced scope. It replaces the original annual labour and solar allowances once, adds paid pre-opening mobilisation, and includes only replacement-reserve inflation shortfalls above contributions already in maintenance. Firm energy assumes physical upgrades whose capital is still unpriced.

Government covers 25%, USD 2.015bn equivalent. Its assumed imported-component downpayment is USD 913.123m; proposed Chinese credit is another USD 913.123m. Government's remaining cash, ordinary/green bonds, bank credit, supplemental credit, fares and commercial receipts are IQD. Total USD-denominated capital is USD 1.826bn, 22.7% of this scenario. At the historical model FX of IQD 1,300/USD this is about 89.9% below the requested USD 18bn all-foreign-currency comparator; distinct scope, price dates and unverified financing prohibit a like-for-like bid saving claim. Imported shares and Chinese eligibility require actual supplier/invoice evidence.

With existing variable pricing and annual 5% fare/OPEX/income growth, peak supplemental debt is IQD 11.360tn, and IQD 7.532tn remains outstanding at the 30-year full-network operating horizon. Before-finance company cash NPV is USD −4.692bn at the model's nominal discount rate. Surpluses repay USD 515.831m equivalent of eligible core principal early, after reserves/buffers and applicable call premiums, but do not eliminate the gap debt. Restricted maintenance cash cannot repay bonds or loans.

The monthly and six-month ledgers retain native currencies, bond-face/rounded-unit placement requirements, loan draws and principal/cash residuals. Conditional climate grants, rights proceeds, additional local income and cheap gap credit are uncommitted. A separate case fixes the original absolute government cash ceiling; another removes additional income and green pricing benefits. Opening-fleet and contracted-solar cases expose reduced receipts and transferred provider costs rather than treating them as free savings.

An initial IQD 1,750 base-fare sensitivity clears debt at month 378 after financial close, conditional on funding placement and the assumed demand elasticity. It still has negative company cash NPV, and 44 monthly trips consume 15.6% of the historical monthly income proxy before variable-price weighting. Higher-fare trials show repayment and affordability together; no tariff or commercial first-corridor selection is adopted. Actual OD, paid transfers, disposable-income distribution and concession funding remain required.

Prepared physical and operating packages

- Nine depot planning drawings label 533 storage positions, individual tracks/slots and 119 workload workshop bays. The yard-only slab production calculation meets cumulative installation deadlines using assumed mould cycles; its plant cost, curing qualification, reinforcement/foundations, turnouts, drainage, fire access and actual land remain open. Rectangular workshop packing is not an accepted accessible bay layout.
- Twenty-three six-car child-part/RFQ rows expand eight parent allocations within USD 1.68m/train. Unknown supplier quantities, prices, masses and drawings remain null. First article, LM3 applicability and qualification/design-training overlap remain unaccepted.
- Per-site chronological energy retains each installed storage/grid/charger limit. Reference imports are approximately 938.5GWh/year, with 6.6GWh unserved charging; firm energy services plus owned-solar maintenance cost USD 139.835m/year in the sensitivity. Synthetic allocation does not prove charger queues, charging events, outages or physical upgrade delivery.
- Task-based rolling maintenance and all 26 controlled intervals cover vehicles, stations, civil/track/structures, energy, wayside, tooling and finishes. Separate full/opening-fleet reserve ledgers show replacement dates, inflation top-ups and restricted cash without double-counting the replacement bill.
- Nine standalone opening-corridor cases compare operating break-even on a consistent basis; OD, actual catchments, shared dependencies and allocated debt remain unknown. Startup alternatives retain the frozen original dispatch points:

line 9 needs five extra revenue trains plus one spare, USD 10.08m vehicle reference before consequential costs, or an accepted smaller dispatch-point policy. Neither alternative closes the failed startup/access gate.

- Seven practical lesson cards and rest-limited anonymous pilot slots retain independent assessment and worker/task/location authorisation. No worker, assessor, permit or operational release is invented. Twelve scope-inclusion forms and supplier RFQs identify the remaining external inputs.

Native ERP and recovery evidence

The local ERP image was rebuilt with the workforce observer and bench-only administrative probes; the activated module hashes match repository files. Six existing DL90 Task descriptions were refreshed while preserving native Task identity, owner and operator-managed status. There are 34 Baghdad evidence Tasks in total; preparing or refreshing a Task accepts no evidence.

Five System Manager read probes each returned 54 permission-filtered rows from bounded native samples. The first took 0.455 seconds and subsequent samples approximately 0.008 seconds. This is a local read probe, not proof of complete-city throughput or the permissions of every production role. The workforce observer's permission and identity gates are separately tested.

Native Guest calls to the read probe, recovery fingerprint and workforce administration readiness were denied. These checks changed no worker, role, assignment or work-release record.

A native SQL backup was restored into a new MariaDB container with no network or published ports. Six data/schema/role/permission/File-inventory fingerprints matched; the temporary database was removed and the source database was never a restore destination. This verifies SQL recovery of the sampled inventories, not file decryption or production RTO/RPO. Private backups, import/deployment receipts, fingerprints and recovery receipts remain in ignored `var/erpnext/`; no native personal records or credentials enter the proposal archive.

Reproduction and acceptance

The checks below describe the preceding continuation. The CI and controls follow-up adds source-verified TLC, dependency, native-click and workforce fixes, earlier leadership/training cash and installed-throughput sensitivities.

Validation passed 1,747 Python tests and 12 subtests across city generation, components, engineering analysis, alignment/reference converters and automation. One Samawah planner test was skipped because its OSM cache is absent. Repository health, local Markdown links, source/output checks, proposal/archive checks and exact-input restoration in an empty temporary checkout passed. The full Rust, browser and target-hardware suites were not rerun for this continuation.

Run the exact-input bootstrap, then the delivery baseline/continuation generators and proposal builder in the documented dependency order. Both delivery generators and the publication support `--check`. The continuation binds source and output hashes; the proposal binds every packaged source and archive member. The complete proposal includes these packages and this review.

The six work packages remain field-open. Actual owners, surveyed parcels, supplier offers, OD/affordability evidence, Iraqi approvals, investor subscriptions and accepted engineering/operating evidence must replace the null fields. The delivered repository package is not a complete investment budget or railway release.

Baghdad cost and demand review — 5 October 2026

Repository source: `docs/baghdad-cost-and-demand-review-2026-10-05.md`

Controlled planning basis; construction prices, surveyed demand and committed financing remain unresolved.

The monetary model now excludes route-search curve penalties. Local curvature changes determine construction-product boundaries, while tangent length cannot spread a remote bend's penalty. This correction preserves Baghdad's controlled 479.0 km alignment, 186 station records, 772 six-car trainsets and nine full-fleet depots.

Same-alignment reference	Reviewed baseline	Corrected base case
Programme planning allowance, USD bn	15.233	8.312
Residual unfunded support, IQD tn	27.330	0.0326
Terminal debt, IQD tn	13.000	11.162
Company NPV before finance, USD bn	−10.591	−4.848
Permanent operating FTE	3,716	3,716

The lower allowance removes a pricing artifact. It is not a proven saving or a complete installed budget. Special structures, actual foundations, installed grid and charging upgrades, land, utilities and supplier qualification still require quantities and quotations. Maintenance is partly a capital-percentage proxy; its reduction does not demonstrate fewer physical maintenance tasks. Staffing, payroll and traction duty are retained. The source-bound reconciliation retains the original reviewed files for offline reproduction.

The reference allocates USD 1.983bn to modelled imported invoices. This includes origin shares inherited from benchmark allowances, rather than verified import quantities or supplier commitments. Government USD cash and Chinese USD credit each cover half, approximately USD 0.991bn. Other credit, bonds and cash are denominated in Iraqi dinars. Government's total capital contribution remains 25%. Imported invoices represent 23.85% of the corrected base allowance: the share rises when an inflated local-cost denominator is removed, even though modelled absolute imported exposure falls. These are uncommitted model allocations. Neither exchange-rate risk nor an outstanding IQD liability disappears through currency labelling.

Month 101, annualised from one nominal model month after the revenue ramp, gives USD 652.6m of fares and existing commercial receipts, USD 25.0m of additional assumed receipts, and USD 519.1m OPEX. The reference operating balance before debt service is USD 158.6m. This is not a calendar-year forecast or evidence of debt repayment: terminal debt remains, no debt-clearance date is recorded, and the financing case is not closed.

Integrated fares must charge a passenger journey once. The service-capacity reference assumes one boarding per paid journey and is a zero-transfer upper bound. Keeping trains, operating costs and non-fare receipts unchanged gives:

Mean boardings per paid journey	Month-101 annualised operating balance before debt, USD m
1.00 reference	158.6
1.25 sensitivity	34.2
1.50 sensitivity	−48.8
2.00 sensitivity	−152.5

These conversion factors are uncalibrated sensitivities. Population catchments do not establish fares or train loading. The demand handoff validates unique origin–destination cohorts, declared transfers and explicit tariffs, but adopts no surveyed forecast without evidence. Jobs, station entrances, pedestrian routes, route choice and peak section loads remain open. The retained road export is insufficient to claim verified entrance-based walking catchments.

The monthly financing ledger includes escalated fares and OPEX, commercial receipts, construction phasing, debt service and surplus sweeps. Mezzanine funding changes the liability profile; it does not establish sustainable repayment. Unknown construction increments must be priced before selecting a funded case.

Clearance results distinguish provisional support-to-support terrain gradients from a designed rail profile. DEM noise attribution, surveyed vertical alignment, building heights, foundation access and property rights remain unresolved. Event counts are not unique buildings, confirmed defects or procurement quantities. Depot launch and charging capacity remain separate acceptance gates. See the clearance screen and civil allowance boundary.

Across the catalogue, disconnected line networks receive candidate transfer links for survey, without invented walking paths or new accepted connections. Wau is filed under South Sudan; its retained `waw` slug is a compatibility alias. South Sudan's finance seed is explicitly generic and uncalibrated. Current city simulations and package checks bind their actual inputs and executable evidence; planning completeness does not release a physical design.

Baghdad delivery review — regenerated planning scope

Repository source: `docs/baghdad-delivery-review-2026-10-04.md`

Originally reviewed baseline: `135e249e20f0630f92f426764ea72d21e46c2a68`. The figures below are regenerated from current inputs, rather than attributed to that historical revision. This delivery-baseline accounting retains legacy depot and payroll allowances; the latest cost and demand review and expanded programme recalculation carry the current selected planning case. It is a targeted delivery review, not a line-by-line audit or investment/construction release.

The controlled city remains 14 lines, 513.3355 km, 159 stations and 809 six-car trains. The delivery-baseline city allowance is USD 7.626537bn, and one Baghdad factory including EPC brings that accounting basis to USD 7.951287bn. These amounts exclude the subsequent expanded-scope adjustments and are not the selected programme total. Its monthly capital table has 82 periods and conditional full-line openings are 40, 41, 43, 42, 44, 44, 50, 55, 59, 64, 69, 73, 78, 81 months; the facility's 18 months starts at proposed NTP. The older 347-month rollout and earlier funding totals are historical and superseded. Proposed government funding remains 25% of capital, imports 50% government USD / 50% Chinese USD loan, with other funds in IQD. No wider national revenue is added.

The delivery package reconciles baseline WBS, exclusions, depot quantities, six-car product, chronological energy, demand/fleet phasing and workload/competence. None of its unquoted alternatives is adopted into debt, fares or official capital. Depot layouts, land/utility/title, local pay/workload, OD, equipment quotations and six-car first-article release remain open. Named owners cannot be manufactured by a role generator.

The legacy depot allowance in this accounting is USD 8m. The expanded programme replaces it with nine quantity-based full-fleet depots; it must not be read as the current depot provision. Unresolved workshop sites, morning-direction checks, overlap and unpriced scope remain acceptance gates. This delivery study's storage/workshop alternatives are quantity-based, with explicit baseline-overlap and unpriced-cost fields. The legacy six-car allocation sums to USD 1.68m/train but does not imply qualified LM3 applicability or supplier quotations.

Hourly synthetic owned-solar reference purchases 942.7 GWh, costing USD 94.274m/year before separately identified wheeling/balancing/connection/O&M. These are sensitivity outputs from synthetic weather and aggregate pooling, not a Baghdad forecast. Unserved energy and per-site physical/contract constraints remain visible. Staffing uses explicit workload/shift cover rather than dividing a legacy labour allowance. The expanded programme's establishment and payroll supersede that allowance; pay, productive hours and roster authorisations remain unaccepted.

Fresh checkout: run `.venv/bin/python tools/automation/bootstrap_baghdad_tests.py`, then `.venv/bin/python -m pytest tools/automation/tests design/component-catalogue/tests -q`. The bootstrap validates the tracked proposal archive and selectively restores the exact ignored operations payload. It preserves differing workspace data and does not resynthesise designs. Install the existing test/render dependencies, including `mistune` and `reportlab`, before publication tests. Validate the delivery package and proposal with their generators' `--check` modes.

Six local ERP evidence drafts carry estimator work budgets, accountable functions, source revisions and independent exit evidence. No appointment, quote, lease, lender commitment, physical permission or railway authority is granted. Existing observation-only integration remains intact. The full Rust/browser/hardware suites are outside this targeted calculation change; verification results are reported with its commit.

Baghdad manufactured viaduct and cost review

Repository source: `docs/baghdad-manufactured-viaduct-review-2026-10-04.md`

The two supplied reviews cover a manufactured-viaduct proposal and the older `135e249e` cost baseline. The latter predates the integrated delivery and financing follow-up. Its USD 7.88bn is retained as a historical planning comparator; it is not a complete budget.

The new comparison package provides Pi20 and Pi25 with both structural-continuity and simple-span/link-slab schemes, plus an OSR-US constrained-access alternative. All use the same complete double-track comparison scope. The longest current Pi25 run is an investigation candidate; the separate 1 km study is not a surveyed alignment or selected project. No 30 m product or literature-based saving is introduced.

The civil load seed now distinguishes the 12-axle LM3 from Baghdad's six-car, 24-axle consist. Its 111 m length and tare/passenger masses reconcile with the controlled rolling-stock profile. Supplier axle positions and loaded axle distribution remain absent; repeating LM3 spacing would invent geometry. The 16 t infrastructure allowance must be applied with the complete loading pattern and adopted dynamic/combination rules.

The LM3 seed's passenger masses now use the same explicit 75 kg/person planning assumption: its previous AW2/AW3 masses silently used 60 kg. Reference axle geometry and both passenger-mass assumptions remain subject to deployment design acceptance; no physical load case is released by this correction.

Catalogue code now distinguishes a structural continuity connection from a link slab. A link slab retains eight bearings at internal twin-track piers; four shared bearings require the structural connection's staged load path. The finite support register includes both ends and reconciles with the shared catalogue function. Connections, CWR, thermal/braking/seismic/foundation forces, fatigue, replacement access and prices remain design/check releases. The existing lower civil rate is explicitly conditional on the unaccepted structural arrangement.

The comparison separates the standard allowance and routing penalty for every elevated segment, with an individual form for each exceptional alignment. Alternatives retain survey, wider curves, station movements, rights-of-way, segmental, utility/land, traffic and whole-life costs as unresolved fields. Removing a routing penalty is not an achieved saving. Existing capital and financing scenarios are preserved. A separate bearing-index cashflow sensitivity prices only existing standard Pi25 length, adds incremental EPC once and recalculates monthly/six-month funding, reserves and repayments. Government remains 25%; assumed imports retain 50% government USD / 50% Chinese USD credit, and other funding remains IQD. Invoice dates and origin shares are inherited assumptions; segmental/special designs receive no Pi25 bearing quantity or price credit. Connection and consequential costs remain unpriced.

Installed-price forms include quantities, invoice currency, price dates, origin, exclusions, supplier references and credit eligibility. Foundations need actual support ground zones and deep-element lengths. Unknown costs cannot produce a total. Ground cranes/portals and overhead launchers require configured proposals, complete member/rigging masses, route and support checks, temporary/deck loads and recovery plans. Pi25 and Pi20 have almost no bare-mass margin within their respective product targets.

The production model uses all eight fronts through survey/correction and final acceptance to calculate the slowest complete double-track bay rate. Missing cycles or resources keep that rate unknown. The existing casting cycle and 10–15-bay buffer remain planning assumptions pending measured trials. First article beam, pier/cap/connection, transport, erection and finished bay hold points provide the required measurement scope; no trial has been performed. UHPC connections and special composite crossings remain bounded alternatives, without carbon percentages being converted to cost savings.

The older review's depot/storage, hourly/site energy, six-car allocations, workforce, demand/fare sensitivities, rental alternatives and bootstrap work already have controlled outputs. Supplier-backed prices, site design, demand counts and financing commitments remain open. The artifact policy now names Baghdad's proposal/archive and distinguishes current, historical and diagnostic evidence. The front-page README includes nobody should be priced out of dignity.

Regenerate the comparison with `.venv/bin/python tools/automation/baghdad_viaduct_comparison.py`; use `--check` to verify source/output hashes. The complete proposal includes this study and its machine-readable inputs. Automated checks verify arithmetic and data boundaries; physical trials and licensed independent checks remain necessary.

Baghdad scope and industrial review — 4 October 2026

Repository source: `docs/baghdad-scope-and-industrial-review-2026-10-04.md`

> Historical review: numeric quantities and funding figures below describe the reviewed revision. Use the current city README, proposal and source-bound catalogue review for the regenerated planning basis.

> Historical revision review. Its figures describe the revision reviewed below. The reworked central alignment, current staffing/depot scope and current cashflows are in the Baghdad city summary and programme recalculation.

The programme recalculation is the latest Baghdad staffing, depot, procurement-origin and financing study. The complete proposal publishes it alongside its source files, twelve monthly cashflows, contract registers and six-month placement schedules. It supersedes the earlier staffing/depot assumptions for this study; the original catalogue design, generic city pages and financing cases remain separately identified comparators. No national borrowing or future factory sales are credited to Baghdad.

People and operating coverage

There are two concurrent station posts during two normal eight-hour shifts. That gives 728 daily shift assignments for 182 stations, rather than 728 employees covering every day. The retained 05:30–02:00 service requires an additional 4.5 hours of late cover. After weekly, leave, training, sickness and handover allowances, station cover requires 1,800 FTE within 3,762 permanent railway FTE. Role, line and recruitment-cohort registers reconcile payroll and pre-opening training. A shorter station opening day would require a separate service/revenue scenario.

The pay basis uses the employee median of IQD 614,000 in the 2021 Labour Force Survey, reported by the IMF. An assumed 5% annual index gives the editable 2026 proxy; the general grade is 1.5 times that proxy, and technical, supervisor, senior and director grades are 2.25, 3, 4 and 5 times it. Employer costs and overtime are explicit. This is not a measured current median, and the old household-income proxy is not substituted for employee wages or altered to improve fare affordability.

Construction, train manufacture, upstream component manufacture and permanent operations are separate establishments. The construction register is a configured work-package crew screen; it is not a complete contractor payroll or a schedule for the new depot/factory sites. Contract rates already contain labour. A separate labour-content stress tests an assumed 20% content with 50% wage uplift and incremental EPC once, without inventing a measured overlap credit. Factory payroll funds the full establishment through the actual scheduled order; productive labour in unit costs is deducted before adding capacity payroll.

Depots and manufacture

Each of the nine lines receives one depot sized for its full six-car fleet, including spares and cold reserve. Storage totals 831 slots at 121 m per train position, using the actual 111 m train. Workshop bays follow annual bay-hour demand independently of storage tracks. The item register prices storage/workshop/access tracks, turnouts, drainage/access, workshop shells/equipment/services, lighting/fire services, offices/stores, wash plants, wheel lathes and isolation/rescue. Existing depot PV/BESS is counted once. Land, utilities, actual ground conditions, connected access alignments and installed charging/grid upgrades remain open; depot launch/throat conflicts still require a full-service replay.

The component study follows the actual six-car BOM quantities: 9,972 bogies and motor/inverter sets each, 4,986 battery packs, 19,944 door cassettes and 29,916 window cassettes. Process cells match the train factory's required rate and include Chinese process machinery, Iraqi facilities/materials, labour, support, maintenance, qualification and residual imported inputs. Bogie fabrication, motor assembly, battery-pack assembly and door manufacture show positive whole-order margins under the unquoted inputs. Window manufacture has a negative Baghdad-only margin; future national orders are not assumed to rescue it. Imported cells/BMS and other critical parts remain. Battery pack assembly does not imply local cell manufacture.

Serial qualification within the 18-month facility target remains an assumption. A six-month supplier-delay case shifts rolling-stock invoices, line openings and operating receipts, prices idle payroll and preserves the full operating horizon. A 25% raw-input price stress tests the same selected facilities. License, yield/rework, tooling overlap and actual vendor quotations require evidence before investment.

Alignment and finance

The study permits a 40% elevated ceiling and screens 29.148 km of candidate additional viaduct around exceptional segments, reaching 20.32% elevated. Intervals include approach allowances, merge overlap and exclude existing elevated/bridge lengths. Elevation alone removes no horizontal bend. Wider-radius geometry, station moves, land, gradients, egress, geotechnical and utility constraints need design work. The 0/25/50% routing-penalty-removal cases are counterfactuals; no penalty reduction is adopted as an achieved saving.

Government funds exactly 25% of each case's capital. Imported invoices are split 50% government USD cash and 50% proposed Chinese USD credit. The remaining government appropriation and all domestic bonds, bank/gap debt and mezzanine are IQD. Machinery-heavy months follow their invoice downpayments; the total government contribution, rather than each month's share, is constrained to 25%. Foreign procurement intensity, foreign debt and reporting equivalents are separate measures. The old 148 km / USD 18bn third-party example retains its distinct scope and assumed foreign-currency funding; it is not a matched tender comparison.

Mezzanine substitutes for 10% of domestic residual capital borrowing. The tested terms are project assumptions: 6% cash coupon, 4% PIK, 2% fee, 15-year balloon and senior DSCR 1.20 before junior cash payments. Deferred coupons and PIK increase debt until maturity; unpaid balloons become overdue balances, with simple-interest sensitivity and no automatic refinancing or conversion. Gap draws or unsourced support cannot fund junior distributions. Source/use, senior principal, junior principal/PIK and overdue-interest identities are checked independently. Semiannual schedules report flows separately from closing stocks and distinguish face-denomination rounding from actual cash raised.

The revised senior case still has unsourced support and unpaid debt at the horizon. Mezzanine reduces immediate funding pressure by deferring payments but increases terminal debt and leaves 85 junior draw vintages in default. It is not recommended as the solution to the operating deficit. Indexed fares, OPEX inflation, advertising/kiosks/rentals and inherited additional receipts are already included. Green terms, grants, rights income and 2% concessional gap credit remain uncommitted; the 8% gap-rate stress shows that financing dependence. There are no dividends. Verified capital, affordable revenue or accepted scope savings are still needed.

Reproduction and release limits

The city landing page is published from the current case rather than the earlier catalogue budget. The city regeneration, batch generation and controls-refresh commands use publish-city-summary.py. Its registry identifies the current study and seventeen earlier reference documents. Original finance, depot, staffing, national and portfolio figures remain traceable with an explicit original-scope notice; document receipt updates do not change financial values or release flags. A stale study stops publication. CI checks the committed page before regeneration, then verifies the publisher after national-brief regeneration. The reader book includes the current city summary and current study before the original funding appendices.

- Assumptions: baghdad-programme-recalculation.toml.
- Generate: `.venv/bin/python tools/automation/baghdad_programme_recalculation.py`; verify with `--check`.
- Cash engine: baghdad_recalculation_finance.py.
- Tests: test_baghdad_programme_recalculation.py.
- Earlier comparators: delivery continuation and original funding programme.

The ERP payload contains planning drafts for staffing, depot projects/assets and factory workstations. It creates no actual employees, appointments, financing commitments, operational transactions or permissions. Survey, quotations, demand/timetable acceptance, wage data, manufacturing qualification, complete installed budgets and executed funding terms remain release gates.

Research incorporated into the design twin

Repository source: `docs/best-practice/digital-twin-design-expansion.md`

The interactive design review and native FreeCAD assembly contain original OSR study designs installed in the shared six-car model. The STEP detail export provides portable geometry for one car's additions and the infrastructure study. The complete twin JSON contains the actual shared model, interfaces, infrastructure and analytical cases. The machine review is the authority for counts, source hashes, mass changes and open gates.

These are concrete parameterised designs and executable model additions. Supplier selection, construction release and physical acceptance remain open. The existing catalogue and its 24 kW HVAC default have not been silently promoted to this alternative configuration.

Research traversal

The research graph records the executed search queries, reviewed primary-source nodes, followed publisher links and remaining branches. Research followed product pages into technical documents and verification context, for example:

- Liebherr modular HVAC publisher downloads MACS technical brochure and equipment-test resources;
- Roxtec rolling-stock sealing transit frames, plastic-conduit fire protection, clamping and installation;
- Pandrol fastening overview baseplate and two-pad technical sheets;
- Texas Instruments precharge reference its design guide; and
- NIST digital-twin implementation scenarios the linked implementation report.

The broader search also reviewed bearing replacement, gangway route-motion measurement, wheel-sensor installation, connector guidance, bonding processes, precast tolerance responsibilities, tool setting records, grounding and running-gear diagnostics. The expanded source register and handbook retain publisher, date, findings and limitations. Search recursion is bounded and reviewed; the tool does not ingest arbitrary web pages into a design or silently refresh an accepted configuration.

Design decisions and their implementation

Change	Concrete geometry and twin behaviour	Remaining qualification
Modular HVAC with a one-module-out study margin	Four candidate module envelopes per car, original drained carrier, mounting feet and hole pattern; thermal cases and explicit mass addition	Actual supply voltage/conversion, ambient derating, airflow, ventilation, refrigerant, roof loads and installed comfort
Replaceable door installation	Paired cassette envelopes, folded steel channel carriers, surveyed header holes and threshold datums; isolated LV power and separate control ports	Supplier cassette, body portal, seals, sensitive edges, emergency release, retention loads and physical interlocks
HV service panel and isolated LV conversion	Removable cabinet, segregation wall, contactor/fuse/resistor reservations and separate converter envelope; original circuit topology and RC study	Creepage/clearance, switching/protection ratings, bleeder discharge, converter isolation, EMC and actual controller
Cable support and barrier penetrations	Separate HV/LV trays, replaceable open transit frames and sensor reservations; configuration-bound interfaces and inspection IDs	Actual cable/connector routes, strain relief, sealed insert build-up, fire tests and sensor implementation
Maintainable rail seats	Original perforated bolted baseplates, two pads, toe insulators, bushings, washers and anchors; per-seat stiffness converted into the distributed support model	Anchor embedment, pad curves, toe/longitudinal restraint, fatigue, electrical insulation and rail-structure interaction
Bearing replacement interfaces	Fixed/guided/free reservations, accessible cap seats and jacking pads, thermal-movement demand	Actual bearing capacity/travel, reinforcement, staged lifting loads, synchronized jacking and authority release

The component schedule records physical study quantities rather than treating a car kit as a fully decomposed component. Catalogue allocation groups remain explicit. New mapped groups are marked partial original study decomposition: the unmodelled contents and supplier internals remain open.

The input file distinguishes public candidate information from OSR assumptions. The HVAC brochure provides a 1,000 × 1,600 × 220 mm single-module envelope and nominal 8 kW cooling; the product page supplies the approximate 125 kg mass. These are external candidate data, with actual purchase revision and environmental/electrical conditions open. Door, cabinet, carrier, rail-seat and jacking geometry is original OSR design. No supplier clip, patented sensor clamp, proprietary circuit board or normative standard text is copied.

A failed case led to a design change

The initial three-module arrangement has 24 kW nominal capacity. Under the OSR study assumptions of 45°C ambient, 9.5 kW combined internal/solar gain, 500 W/K conductance and a 26°C setpoint, maintaining the setpoint requires 19 kW cooling. Losing one module leaves 16 kW; the lumped model reaches about 29.17°C after 30 minutes and fails the declared 27°C study limit.

The proposed four-module arrangement retains 24 kW nominal capacity after one failure and holds 26°C in the same model. The original failed case is retained beside the proposal. This is conditional sizing evidence: fresh-air and latent loads, actual climate, supplier derating, air distribution and physical comfort tests are still required. The 27°C limit is an OSR study policy, not an ISO acceptance value.

The proposal adds one reference 125 kg module per car, or 750 kg across six cars. That increment is explicit. Other newly decomposed equipment is extracted once from the original synthetic residual-body allowance. Mass, CG, inertia, static reactions and the spatial vehicle matrices are recalculated. The three-module allowance inside that residual is a declared OSR assumption; physical mass closure still requires actual supplier and measured values.

Rail pads similarly reach the numerical model: the two assumed 60 MN/m pads in series produce 30 MN/m per rail seat. Two rails at 0.6 m spacing give a 100 MN/m² distributed foundation modulus. The historical adapter key ends in `_n_m`, but its tributary-length operation uses N/m². The conversion and dimensional meaning are recorded explicitly in the review.

Executed checks and boundaries

The native audit records installed Assembly solving, saved/reopened geometry, joint datums, actual occupied-volume checks, independent OCC primitive inertia, drawing edges and STEP round-trip volume. It does not qualify the synthetic context body's door openings, structural capacity or a fire barrier. An open frame is an installation interface, not a certified seal.

The original circuit and netlist include positive/return contactors, a separate resistive precharge path, bus capacitance, an unresolved bleeder and load. The declared 150 Ω / 6 mF study reaches 95% at about 2.70 s; its tolerance corner is about 3.11 s. Required pulse energy with the stated margin is 4,009.5 J. No resistor or contactor rating is accepted. Offline cases reject open HVIL, insulation fault, wrong feedback, movement and timeout; the model controls no physical switch. The closure decision helper is stateless. Persistent fault latching, switching sequence, observation freshness, authorised recovery and target-controller implementation remain required work.

The coupled smoke check retains four integration steps over 0.02 s with the new train masses, CG/inertia and distributed rail-pad input. It is an integration check, with its initial live-load transient and limitations retained. It does not establish time/mesh convergence, ride comfort, structural serviceability or a complete passage. The existing six-car contact-domain issue remains open. Neither that issue nor physical acceptance is closed by these design additions.

Reproduce and extend

```
tools/automation/osr-python tools/automation/design-best-practice-twin.py \
  generate --output build/engineering/my-design-study
tools/automation/osr-python tools/automation/design-best-practice-twin.py \
  verify --output build/engineering/my-design-study
tools/automation/osr-python tools/automation/design-twin-freecad.py \
  --package build/engineering/my-design-study \
  --output build/engineering/my-design-study/reference-twin.FCStd
tools/automation/osr-python tools/automation/design-twin-smoke.py \
  --package build/engineering/my-design-study \
  --output build/engineering/my-design-study/spatial-smoke.json
tools/automation/osr-python -m pytest -q engineering/analysis/tests/test_design_twin.py
```

Use fresh output paths to retain earlier evidence. Only canonical input changes are published with `--publish`; generated JSON, HTML, CSV and SVG are checked against recompilation. CI verifies the published design, native artifact hashes and meaningful software tests. Native execution is recorded separately; CI does not claim to rerun FreeCAD or perform physical tests.

Next research branches are actual contact profiles, tunnel fire/ventilation, stationary storage/PV protection, approved connector and refrigerant selections, and route-specific gangway/crash endurance. Extend the existing standards thread and OSR-BP-001 with project applicability, adopted clauses, acceptance limits, procedures and independent decisions before release.

Integration review continuation — 11 October 2026

The current twin has 414 train instances, 413 installed joints and 545 typed interfaces, including 222 new study instances and 38 infrastructure study objects. The additional 96 service instances cover conductor/insulation sections, keyed connector reservations, empty coolant pipes, replaceable pipe clamps and perimeter weather seals. They extend existing kit allocations; the coolant kit adds six partial allocations. 210 of 1,367 catalogue allocations now have explicit reference decomposition. This does not qualify their complete contents.

The service pieces have actual material sections and density-derived masses. Those masses are extracted from the residual body allowance, preserving the 201,240.108273 kg train total while recomputing CG and inertia. Cable ampacity, termination/pinout, circuit continuity, coolant/fittings/fluid mass, seal compression, fire build-up and mounting capacities remain unqualified. Their placement and occupied volumes have been checked in the saved/reopened native CAD.

The expanded twin completed a full six-car passage at 15 m/s following the wheel-yaw rolling-velocity correction. This is configuration-specific numerical coverage. The retained four-step smoke remains labelled as a smoke test. Separate refinement found a substantial coarse-mesh ride-acceleration change; full operating and convergence acceptance remains open.

The 23-item integration programme tracks the attached review separately from implementation and physical acceptance. OSR-INT-001 defines the open project evidence rules. Executable additions include native Rust controller/plant experiments, tabulated SOC/temperature battery maps, simultaneous battery mounting loads from retained passages, free-free modal transfer verification, a hollow carbody shell with four door portals, interacting station outage queues, installed lifetime cost scope, and serial/calibration-bound condition records. Research fixtures remain labelled; missing physical records, nonlinear resistance, full spatial modal/actuator insertion and independent acceptance remain open.

Engineering integration review — 11 October 2026

Repository source: `docs/best-practice/engineering-integration-review.md`

The 23-item integration programme records the remaining review work, implemented capabilities and external evidence requirements. Implementation, numerical verification, physical correlation and independent acceptance are separate states. This work does not close production, site deployment, hardware qualification or railway conformity gates.

Reproducibility and native assembly

The publication check preserves exact configuration, source and retained-artifact hashes. Recomputed solver observables have narrowly named numerical reproduction tolerances; engineering limits and acceptance flags remain exact. Baghdad's connected-build package was regenerated. The dependency scanner reports stale current packages separately from retained historical records.

The expanded train contains 414 instances, 413 joints and 545 interfaces. Conductor/insulation sections, connectors, coolant pipes/clamps and door seals are original study designs. Their masses are extracted once from residual allocations. The assembled mass is 201,240.10827341524 kg. These additions do not complete the inventory: 1,157 of 1,367 allocation slots remain unresolved, and the 210 mapped slots include partial kits. The native twin retains occupied-solid, assembly-constraint, inertia, drawing and STEP checks.

Open examples and represented physics

OSRD's physical model informs force balances and effort/resistance handling. Its retained fictional commuter example is pinned to an exact upstream commit, Git blob and byte hash. The schema's per-kilogram Davis coefficients are converted once. Conditional curves, supply mode and train-tail speed-limit clearance remain explicit. The 390,000 kg, 110 m example and its 1.5/25 kV supply modes are not adopted as OSR ratings. Its LGPL-3.0-only data notices and licence texts are retained.

CONTACT contributes nominal Manchester benchmark files: 399 wheel and 495 rail points, retained with Apache-2.0 notices, source hashes and transformation metadata. Raw ingestion does not establish the axis/side/datum transform, approved project profiles, flange or multiple-contact support. The external CONTACT solver has not been executed here. The research register also distinguishes researched RCM-DX interoperability from an implemented adapter.

Yaw rolling-reference kinematics were corrected without weakening contact-domain guards. Complete six-car reference and expanded-twin passages were executed. Separate reference time refinement and spatial refinement reach the registered 5% numerical screen at the finest retained levels. This is one configuration's numerical comparison, not an accepted operating envelope or an ISO ride rating. All 22 baseline cases were attempted: 17 completed, two exceeded the old duration budget, two left the contact domain, and the two-train run exceeded its wall budget. Fresh 414-instance rescue and maintenance passages both completed 39.18 seconds; its fresh two-train run also completed the passage. These configuration-specific records do not merge into an accepted common envelope. The contact investigation retains the older failures and explains the corrected scope. The published evidence manifest binds compact review results to actual execution receipts; large histories remain in the ignored local build evidence. Historical source scopes are labelled.

The spatial solver can now separate battery equipment mass from the carbody, solve finite six-direction retention joints from the actual mounting-hole coordinates, and insert work-conjugate carbody beam modes into the same sparse mechanical solve. Native Rust BMS, traction, brake and thermal controllers exchange states with six separate battery plants. Converged wheel speeds determine drive work, friction heat and electrical/thermal evolution; each auxiliary load is counted once. One-second experiments include a pump/fire protection fault and separate timestep comparisons. Coarse failed comparisons are retained. The final 3,200-to-6,400-step comparison passes the 5% research screen (contact force 3.683%, wheel unloading 0.7845%); spatial, operating-envelope and physical qualification remain open. These experiments use prescribed transport with bounded perturbations, assumed maps, equivalent cells and reduced beam modes. They do not transfer measured shell/bogie modes or qualify a complete service passage with native control.

The native hollow carbody shell, its four door portals and modal output remain an independent proposed structural study. Complete bogie structure, local production joints, nonlinear civil resistance and physical modal/force correlation remain open. The civil audit accounts for all 38 displayed instances and explicitly reports unreconciled sections/masses/datums and undisplayed pier/foundation domains. An association with a beam/spring is not geometric or material equivalence.

The search compiles battery topology through the shared variant/design compiler and screens complete passages. Each method/seed has an equal time quota, cache use is reported, and finalist time is reserved. Temporal and spatial refinement are separate; refined results are checked against adopted limits. Unknown limits, resistance or complete installed costs exclude qualified feasible Pareto claims.

Native software lifecycle

Coincident City Studio markers have separate ordinary click targets and leaders; canonical surveyed coordinates and intentional dragging are preserved. Native example lifecycle, browser selection, restart and disposition checks were rerun. The ERP business suite executes 35 checks covering reviewed resource/crew release, assignment strategies, attendance/payroll, procurement commitment budgets, asset opening/depreciation journals and SLA holiday calendars. Fixture transactions are rolled back and private fixture files removed. Payroll examples are not adopted jurisdictional policy or employee payments.

Submitted fleet lifecycle changes now use the existing role guard, with only the lifecycle custom field editable after submission. Cost fields remain protected. All isolated ERP application services were rebuilt/recreated with the new code. The actual production ERP deployment was not changed.

A fresh coordinated checkpoint preserves runtime temporary mounts and immutable image IDs. Its isolated rehearsal verified ten volume inventories, signing identity, ERP/encryption state, queue lost-reply reconciliation and HTTP services, then removed its own restore resources. Source writers resumed. This is a same-host fresh-volume rehearsal; separately hosted cutover, resumed destination controllers/jobs and adopted RPO/RTO remain open. Public reports contain results; private checkpoint credentials, state and archives are not published.

Remaining acceptance and release boundaries

All nine previously open Kani obligations were attempted with complete checks, 900-second timeouts and 6 GiB memory limits. Eight memory failures and one timeout establish no proof. Alternative backend errors are retained. The TLA+ to Rust consensus refinement argument remains unresolved. Passing a bounded CI subset does not close these obligations.

No supplier test records, calibrated prototype observations, selected controller hardware records or approved wheel/rail profiles were available. Accordingly, physical serial/measurement ingress, calibration validity, independent condition revisions and analysis invalidation are implemented, while physical correlation and release remain open. The 15 Baghdad deployment gates, real process windows, material grades, first articles, full civil resistance, actual targets, quotations, manufacturing rights and support commitments still require their actual records.

OSR-INT-001 is the original open internal integration standard. ISO and industry references inform methods; configuration-specific clause adoption and authority acceptance are not inferred. Station vehicle-fire HL1 claims were removed, and IEC 60439/61439 assembly applicability was corrected.

The retained local lifecycle began on a working tree under development. It is representative software evidence, not clean-commit release acceptance. The existing `example-city` CI workflow reruns the lifecycle on the pushed clean candidate and separately records release evidence. Its result must be checked; local passing reports are not relabelled as that release run.

OpenSourceRail best-practice handbook

Repository source: `docs/best-practice/handbook.md`

Research baseline 2026-10-11 · generated from the reviewed open register.

54 practice cards, 73 primary references and 356 explicitly mapped definitions. Coverage means guidance exists for the scoped inventory. Project adoption, production release and physical qualification remain open.

The design, materials, assembly, inspection and maintenance guidance below is original OSR synthesis. References support the method or application context; public standard scopes and supplier examples do not establish clause conformity or product performance.

Use OSR-BP-001 and the research plan. The component mapping, coverage audit and open verification plan are generated together.

Scope and coverage

Definition kind	Mapped
civil-type	19
software-package	60
station-assembly	10
station-product	50
station-system	7
subsystem-template	64
train-assembly	26
train-product	120

Station variants: depot-terminal, halt, interchange, interchange-elevated, major, standard, terminal.

Optional zero-quantity train products are retained as configuration alternatives; they are not counted as installed. Civil coordination envelopes retain their interface status. Station applicability records preserve titles and quantities across variants. A kit mapping does not decompose its individual bolts, terminals, seals or consumables. Software source titles may contain SIL/GoA targets; they are not achieved integrity claims. Common lifecycle cards require applicability review; physical material/tool checks do not automatically apply to pure software.

Practice index

Practice	Domain	Responsible role
Configuration, requirements and open interfaces	lifecycle	systems engineer
Reliability, availability, maintainability and safety	lifecycle	safety assurance lead
Datums, measurement and acceptance decisions	lifecycle	quality/metrology lead
Material qualification and supplier equivalence	lifecycle	materials engineer
Qualified manufacture, travelers and quality learning	lifecycle	production/quality lead
Lifecycle energy, carbon, repair and circularity	lifecycle	environment/lifecycle lead
Secure software, networks and maintenance access	lifecycle	cybersecurity/software lead
Carbody load paths, lifting and equipment mounts	train	structural/mechanical engineer
Composite shells, liners and bonded inserts	train	composites/process engineer
Coatings, graphics, seals and weathering	train	surface/materials engineer
Fasteners, captive service rails and retained fixtures	train	joint/assembly engineer
Bogie frames, guards and torque reaction	train	bogie structural engineer
Wheelsets, rail profiles and running contact	train	vehicle/track dynamics engineer
Axleboxes, bearings, lubrication and sensing	train	bearing/running-gear engineer
Primary/secondary suspension and body connection	train	suspension/dynamics engineer
Service, emergency and parking braking with adhesion	train	brake/safety engineer
Motors, gearboxes, couplings and thermal duty	train	traction engineer
Traction batteries, enclosures and safe removal	train	battery/electrical safety engineer
HV/LV power distribution, bonding and harnesses	train	electrical integration engineer
Cooling, fluids, pneumatic auxiliaries and drains	train	thermal/fluid engineer
HVAC, air distribution and passenger comfort	train	HVAC engineer
Passenger doors, carriers and boarding interlocks	train	door/control engineer
Glazing, retention, seals and replacement	train	glazing/joint engineer
Floors, seats, handrails and passenger fixtures	train	interior/accessibility engineer
Couplers, articulation, gangways and flexible services	train	coupling/articulation engineer
Train control, navigation and automated lifecycle	train	train-control/safety engineer
Train/wayside communications, timing and event records	train	communications/software engineer
Information, lighting, intercom and security equipment	train	passenger systems engineer
Fire detection, containment, evacuation and rescue	system	fire/life-safety engineer
Survey, alignment, gauge and route configuration	infrastructure	survey/track engineer
Ground investigation, foundations and settlement	infrastructure	geotechnical/foundation engineer
Precast/prestressed beams, decks, piers and platform units	infrastructure	civil structural/production engineer
Bridge bearings, joints, jacking and replaceability	infrastructure	bridge/interface engineer
Logistics, lifting, temporary works and site assembly	infrastructure	construction/temporary works engineer
Drainage, water management and climate resilience	infrastructure	drainage/environment engineer
Rails, fastenings, slabs and wheel/rail interfaces	infrastructure	track/vehicle integration engineer
Turnouts, point machines, detection and track ends	infrastructure	turnout/signalling engineer
Wayside signalling, detection and movement authority	infrastructure	signalling safety engineer
Platforms, boarding gaps and platform doors	station	station/vehicle interface engineer
Station steel, solar roofs and safe access	station	station structural/roof engineer
Station LV services, UPS, cabinets and lighting	station	station electrical/MEP engineer
Station information, fares, CCTV and help points	station	passenger systems/operations engineer
Approaches, lifts, escalators, stairs and overbridges	station	access/vertical transport engineer
Train/wayside charging and traction substations	infrastructure	traction power/charging engineer
PV, stationary storage and microgrid integration	infrastructure	energy/electrical engineer
Depot lifts, workshops, wash, lathe and isolation	operations	depot/workshop engineer
Service planning, degraded operation and incident response	operations	operator/operations lead
Maintenance, spares, condition monitoring and return to service	operations	maintenance authority
Level crossings, public interfaces and intrusion	infrastructure	crossing/public safety engineer
Integrated vehicle, track, bridge and station dynamics	system	integrated vehicle/civil engineer
Integrated train, charger, route and energy supply	system	system energy engineer
Integrated accessible passenger journey and evacuation	system	accessibility/fire/operations lead
Integrated safety control, power, doors and operations	system	system safety/control engineer
Software requirements, deterministic behaviour and target validation	software	software assurance lead

Configuration, requirements and open interfaces

Responsible role: systems engineer. Applied to 356 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-001, STD-CTRL-002, STD-CTRL-010, STD-CTRL-016. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Allocate measurable requirements and hazards to named owners. Publish mechanical, electrical, fluid and control interface revisions.

Materials: Bind candidate and released grades to distinct states. Preserve third-party licensing and source provenance.

Assembly: Use revision-bound drawings and portable JSON/CSV work records. Identify each physical part by serial or lot.

Inspection: Reconcile BOM, CAD, installation and retained evidence identities. Review changes across all affected interfaces.

Maintenance: Keep as-built and as-maintained configurations. Record approved substitutions and obsolescence.

Verification: Independent requirement-to-verification matrix review. Round-trip interface and unit reconciliation.

Open evidence and decisions: Accepted project configuration and authority. Complete physical allocation of kits and area quantities.

Basis: NASA: Systems engineering handbook: interfaces and verification, buildingSMART: IFC railway domain and open exchange, ISO: ISO 22163:2023 railway quality-management scope, NIST: ISO 23247 implementation scenarios, NIST: AMS 400-2 digital twin implementation report.

Reliability, availability, maintainability and safety

Responsible role: safety assurance lead. Applied to 356 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-001, STD-CTRL-003, STD-CTRL-004, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Define service concept and normal, degraded, rescue and maintenance modes. Allocate hazards and common-cause failures across system boundaries.

Materials: Use environment and life assumptions in failure analysis. Identify safety-significant materials and consumables.

Assembly: Identify safety holds and competence requirements. Preserve independent checks for critical installation.

Inspection: Trace mitigations to evidence and residual risk disposition. Retain failures and nonconformances.

Maintenance: Feed service failures into controlled risk review. Qualify interval changes before adoption.

Verification: Independent hazard and failure-analysis review. Fault-injection and emergency exercises under approved protocols.

Open evidence and decisions: Project risk criteria and assurance independence. Safety integrity allocation and applicable complete standards.

Basis: IEC: IEC 62278-1:2025, FTA: Public Transportation Agency Safety Plans.

Datums, measurement and acceptance decisions

Responsible role: quality/metrology lead. Applied to 356 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-012, STD-CTRL-013. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Specify functional datums and signed tolerance chains. Define acceptance limits and uncertainty decision rules before measurement.

Materials: Include thermal expansion and surface condition. Separate supplier nominal data from measured properties.

Assembly: Use qualified fixtures and calibrated tools. Record temperature, setup and measurement revision.

Inspection: Retain uncertainty and tool-validity records. Treat overlapping limits as unresolved review.

Maintenance: Recalibrate and inspect fixtures after damage or relocation. Preserve repeatability studies and drift history.

Verification: Datum repeatability and measurement-system study. Independent uncertainty-aware conformity review.

Open evidence and decisions: Released limits and uncertainty coverage. Fixture clamp capacity and load verification.

Basis: BIPM: Uncertainty-aware conformity decisions; explicit decision-rule selection, Siemens: Measurement context, temperature and calibration traceability, NIST: Diagnostic spread/capability statistics and their stability/distribution/sample assumptions, PCI: Precast tolerance responsibilities and PCI 135 transition.

Material qualification and supplier equivalence

Responsible role: materials engineer. Applied to 356 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-007, STD-CTRL-008, STD-CTRL-011, STD-CTRL-012. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Select materials against strength, fatigue, fire, exposure and repair needs. Evaluate substitutions at the installed assembly level.

Materials: Maintain grade, heat/lot, shelf-life, certificate and supplier revision. Require measured properties and representative test coupons.

Assembly: Control storage, mixing and contamination. Quarantine expired or unapproved substitutes.

Inspection: Check certificate-to-lot identity and incoming condition. Preserve rejected batches and deviations.

Maintenance: Publish repair compatibility and replacement routes. Control cleaning chemicals and ageing inspections.

Verification: Supplier equivalence matrix and representative material tests. Installed build-up fire and ageing test plan.

Open evidence and decisions: Released grades and supplier certificates. Hazard categories and acceptance values.

Basis: ISO: ISO 22163:2023 railway quality-management scope, BSI: BS EN 45545-2:2020+A1:2023, Sika: Rail bonding and coating applications.

Qualified manufacture, travelers and quality learning

Responsible role: production/quality lead. Applied to 356 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-011, STD-CTRL-012, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Design for repeatable fixtures and accessible inspections. Separate planning cycle times from observed qualified production.

Materials: Control input lot and special-process consumables. Bind process windows to material system and revision.

Assembly: Use issued procedures, competent staff and sequenced holds. Capture tool traces and approved rework in portable travelers.

Inspection: Prevent open NCRs or missing holds from passing review. Use capability statistics only with suitable stability and measurement evidence.

Maintenance: Maintain tooling and process equipment. Feed recurring defects into corrective action and instruction revisions.

Verification: First-article traveler audit and process qualification. Traceable repeated-build inspection and capability study.

Open evidence and decisions: Approved process windows and product coverage. Released drawings, competence and real production records.

Basis: Alstom: Repeatable special processes, equipment monitoring, competence and lean production, Alstom: Digital work instructions and per-joint smart torque records, ISO: ISO 3834-2:2021 fusion-welding quality scope, NIST: Diagnostic spread/capability statistics and their stability/distribution/sample assumptions, Liebherr: Development and qualification tests, Roxtec: Installation instructions and guidance, Böllhoff: RIVNUT setting-process controls, ECARV: EN 17460 interpretation and implementation, NIST: ISO 23247 implementation scenarios, ISO: ISO 21368:2022.

Lifecycle energy, carbon, repair and circularity

Responsible role: environment/lifecycle lead. Applied to 356 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-001, STD-CTRL-015, STD-CTRL-016. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Compare alternatives against the same duty, lifespan and system boundary. Prioritise replaceable modules and disassembly access.

Materials: Request supplier EPD and recycled-content evidence. Account for adhesives, batteries and mixed-material separation.

Assembly: Measure actual waste and process energy. Record material recovery and hazardous waste routes.

Inspection: Review functional unit, geography and uncertainty. Do not transfer vendor savings to OSR forecasts.

Maintenance: Track replacements and energy against duty. Assess repair before replacement and retain safe disposal records.

Verification: Independently reviewed lifecycle inventory. Baseline-versus-alternative duty-cycle comparison.

Open evidence and decisions: Supplier environmental data and adopted LCA methods. Local recycling and disposal capability.

Basis: ISO: ISO 14040:2006, Alstom: Energy, recyclability and lifecycle-aware material choices, ISO: ISO 55001:2024.

Secure software, networks and maintenance access

Responsible role: cybersecurity/software lead. Applied to 356 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-005, STD-CTRL-006, STD-CTRL-016. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Separate safety control from enterprise and passenger networks. Define trust boundaries, least privilege and recoverable update paths.

Materials: Track firmware, software bill of materials and component support life. Secure key provisioning and credential handling.

Assembly: Verify software identity and approved configuration. Restrict remote access and retain signed update/audit records.

Inspection: Test segmentation, loss of communications and rollback. Review vulnerabilities with safety and availability impacts.

Maintenance: Maintain offline recovery and revocable access. Approve patches against the safety configuration.

Verification: Threat model and controlled penetration/robustness tests. Restore-from-backup and update rollback demonstration.

Open evidence and decisions: Applicable railway cyber standards and acceptance authority. Validated security/safety interaction and actual control implementation.

Basis: NIST: SP 800-82 Rev. 3: operational technology security, IEC: IEC 62278-1:2025.

Carbody load paths, lifting and equipment mounts

Responsible role: structural/mechanical engineer. Applied to 22 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-008, STD-CTRL-011, STD-CTRL-012, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Resolve fatigue, crash, lifting, towing and equipment retention load paths. Include installed mass, CG and interface loads.

Materials: Specify traceable structural grades and corrosion isolation. Qualify weld and fastener systems for the actual material.

Assembly: Build from common datums with controlled distortion. Inspect inaccessible welds before closure.

Inspection: Survey mounting interfaces and weld disposition. Weigh completed assemblies and reconcile the mass ledger.

Maintenance: Provide inspection and replaceable bracket access. Publish lifting and rescue instructions with load limits.

Verification: Independent structural/fatigue and recovery load assessment. First-article dimensional, NDT and weighed mass/CG records.

Open evidence and decisions: Released load cases and resistance factors. Qualified crashworthiness, fatigue and recovery evidence.

Basis: ERA: LOC&PAS TSI and current application guidance, ISO: ISO 3834-2:2021 fusion-welding quality scope, IEC: IEC 61373:2010.

Composite shells, liners and bonded inserts

Responsible role: composites/process engineer. Applied to 22 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-008, STD-CTRL-011, STD-CTRL-012, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Separate cosmetic skins from structural load assumptions. Design removable modules and retained edges.

Materials: Freeze resin, fibre, core, adhesive and full coating build-up. Qualify fire, moisture, UV and cleaning compatibility.

Assembly: Control ply orientation, cure, vacuum, insert position and bond preparation. Retain batch/coupon genealogy and cure traces.

Inspection: Inspect thickness, voids, cure and insert retention using qualified methods. Test representative coated and bonded coupons.

Maintenance: Define approved patch, removal and water-ingress repairs. Inspect retention after impact or repair.

Verification: Representative laminate/insert mechanical and ageing tests. Installed fire build-up and repeated removal tests.

Open evidence and decisions: Released laminate schedules and cure windows. Repair qualification and allowable defects.

Basis: Sika: Rail bonding and coating applications, BSI: BS EN 45545-2:2020+A1:2023, Alstom: Repeatable special processes, equipment monitoring, competence and lean production, Sika: Bonding and sealing processing guideline, ECARV: EN 17460 interpretation and implementation, ISO: ISO 21368:2022.

Coatings, graphics, seals and weathering

Responsible role: surface/materials engineer. Applied to 20 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-008, STD-CTRL-011, STD-CTRL-012, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Design drainage and isolate dissimilar metals. Keep experimental radiative coatings behind qualification gates.

Materials: Freeze surface preparation, primer, coating, film and seal compatibility. Include fire and repair build-up evidence.

Assembly: Record ambient/dew-point conditions and cure dependencies. Maintain clean preparation and protected handling.

Inspection: Measure preparation, coverage, film thickness and adhesion by issued methods. Inspect edges, penetrations and trapped water.

Maintenance: Define cleaning and touch-up systems. Inspect water ingress, seal ageing and coating breakdown.

Verification: Representative corrosion/weathering and adhesion assessment. Water-ingress and repair compatibility tests.

Open evidence and decisions: Exposure category and coating system. Radiative-coating durability and actual energy benefit.

Basis: ISO: ISO 12944-2:2017, Sika: Rail bonding and coating applications, Siemens: Resource scheduling with coating/drying and quality dependencies, Sika: Bonding and sealing processing guideline.

Fasteners, captive service rails and retained fixtures

Responsible role: joint/assembly engineer. Applied to 19 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-008, STD-CTRL-011, STD-CTRL-012, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Reduce part variety using qualified interface families. Check preload, fatigue, slip and secondary retention.

Materials: Specify grade, coating and galvanic isolation. Control locking elements and single-use parts.

Assembly: Use per-joint tool traces and qualified tightening method. Approve retorque and replacement before rework.

Inspection: Inspect joint access, engagement and witness identity. Torque alone shall not be treated as measured preload.

Maintenance: Publish reusable versus replace-on-removal rules. Inspect service rails and anti-lift retainers after replacement.

Verification: Joint preload/retention qualification and repeated-service trial. Configuration-bound calibrated tightening record audit.

Open evidence and decisions: Released torque/angle or tension limits. Joint-specific friction, fatigue and service reuse evidence.

Basis: Alstom: Digital work instructions and per-joint smart torque records, NASA: Systems engineering handbook: interfaces and verification, Böllhoff: RIVNUT setting-process controls.

Bogie frames, guards and torque reaction

Responsible role: bogie structural engineer. Applied to 9 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-008, STD-CTRL-011, STD-CTRL-012, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Include wheelset, brake, motor and suspension interface loads. Check guard retention, clearance and common fatigue details.

Materials: Specify structural grades and weld-quality classes by consequence. Avoid unqualified mass substitutions.

Assembly: Use released fixture and weld sequence. Inspect alignment before installing running equipment.

Inspection: Perform qualified weld NDT and datum surveys. Check accessible guard and reaction-link retainers.

Maintenance: Define frame inspection and repair exclusions. Maintain fatigue-critical weld and mount histories.

Verification: Independent fatigue/load assessment and frame test plan. First-article NDT and running-equipment alignment records.

Open evidence and decisions: Actual suspension/motor/brake loads. Frame fatigue qualification and approved repair limits.

Basis: ERA: LOC&PAS TSI and current application guidance, ISO: ISO 3834-2:2021 fusion-welding quality scope, IEC: IEC 61373:2010.

Wheelsets, rail profiles and running contact

Responsible role: vehicle/track dynamics engineer. Applied to 7 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-008, STD-CTRL-012, STD-CTRL-013, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Select wheel and rail profiles together with gauge and turnouts. Use models within their validated contact and loading domain.

Materials: Freeze axle, wheel, disc and profile specifications. Retain heat and wheelset serial genealogy.

Assembly: Control wheel press fits, back-to-back and disc attachment. Avoid bearing and axle contamination.

Inspection: Measure profiles, dimensions and qualified axle/wheel NDT. Reject unsupported contact footprints in numerical evidence.

Maintenance: Set measured wear, reprofile and defect limits. Preserve mileage, thermal events and wheelset change records.

Verification: Actual-profile running-dynamics and braking assessment. Qualified wheelset fit, NDT and dimensional tests.

Open evidence and decisions: Released profiles and running-safety criteria. Full six-car passage remains outside the current model contact domain.

Basis: ERA: LOC&PAS TSI and current application guidance, FTA: Report 0215: track inspection and maintenance.

Axleboxes, bearings, lubrication and sensing

Responsible role: bearing/running-gear engineer. Applied to 8 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-008, STD-CTRL-012, STD-CTRL-013, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Design bearing fits, sealing and sensor placement for actual duty. Separate diagnostic monitoring from safety decisions.

Materials: Freeze bearing, lubricant, seal and sensor revisions. Control cleanliness and storage.

Assembly: Use qualified press/heating tools and fit procedures. Record serials and avoid transmitting assembly loads through rollers.

Inspection: Measure fits and inspect sealing/temperature-sensor function. Retain installation and lubricant genealogy.

Maintenance: Set intervals from supplier duty and observed evidence. Investigate temperature/vibration alarms and lubricant contamination.

Verification: Bearing application and thermal/life assessment. Installation fit and sensor fault/health tests.

Open evidence and decisions: Released fits, loads and lubrication data. Validated alarm thresholds and response policy.

Basis: SKF: Evolution of railway axlebox technology, Hitachi Rail: HMAX for signalling, Siemens Mobility: Bogie diagnostic architecture.

Primary/secondary suspension and body connection

Responsible role: suspension/dynamics engineer. Applied to 8 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-008, STD-CTRL-012, STD-CTRL-013, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Model full loaded/empty travel and emergency support. Check ride, roll, clearance and retention together.

Materials: Freeze spring, air-spring, damper and elastomer specifications. Include temperature, ageing and leak behaviour.

Assembly: Use surveyed pads and released shim/height procedures. Verify link orientation and retained fasteners.

Inspection: Measure installed height, travel and air leakage. Record component identity and characteristic curves.

Maintenance: Inspect leaks, ageing, wear and damper condition. Keep approved replacement pairing and height resets.

Verification: Measured force-displacement/damping curves. Loaded/empty dynamics and deflated-suspension tests.

Open evidence and decisions: Supplier curves and approved ride criteria. Bump-stop, emergency spring and degraded-mode qualification.

Basis: Continental: Railway secondary suspension, ERA: LOC&PAS TSI and current application guidance.

Service, emergency and parking braking with adhesion

Responsible role: brake/safety engineer. Applied to 11 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-003, STD-CTRL-004, STD-CTRL-007, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Prove friction braking when regeneration or charging is unavailable. Include low adhesion, load, gradient and degraded channels.

Materials: Freeze pads, discs, actuators and WSP sensors. Consider thermal capacity and contamination.

Assembly: Verify brake clearances, hose/harness routing and polarity. Preserve independently checked safety joints.

Inspection: Test parking hold, brake demand, release and fault handling. Retain synchronized speed/pressure/deceleration traces.

Maintenance: Maintain wear, thermal-damage and actuator inspections. Trend failures without extending intervals automatically.

Verification: Approved loaded/empty stopping and low-adhesion test campaign. Emergency, parking, WSP fault and regen-loss tests.

Open evidence and decisions: Accepted stopping envelopes and failure allocations. Qualified physical brake tests and signalling interface.

Basis: Knorr-Bremse: Adhesion and wheel-slide protection, IEC: IEC 62290-1:2025.

Motors, gearboxes, couplings and thermal duty

Responsible role: traction engineer. Applied to 12 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-003, STD-CTRL-004, STD-CTRL-007, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Size propulsion for route, gradient, load and degraded operation. Compare measured efficiency maps rather than nominal vendor gains.

Materials: Freeze winding/insulation, gear, lubricant and coupling specifications. Include contamination and thermal ageing.

Assembly: Align motor/gearbox interfaces and reaction links. Record lubrication and calibrated critical joints.

Inspection: Measure alignment, insulation and rotation direction. Test loaded thermal behaviour and protective trips.

Maintenance: Provide replaceable drive modules and safe isolation. Track wear, temperature and lubricant condition.

Verification: Duty-cycle traction/thermal and efficiency-map assessment. Alignment, overspeed/protection and degraded-drive tests.

Open evidence and decisions: Actual supplier maps and continuous/peak envelopes. Validated energy and cooling performance.

Basis: CRRC: Primary account of lifecycle automation, redundancy, running-gear monitoring and efficient traction, ERA: LOC&PAS TSI and current application guidance.

Traction batteries, enclosures and safe removal

Responsible role: battery/electrical safety engineer. Applied to 13 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-003, STD-CTRL-004, STD-CTRL-007, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Treat chemistry changes as controlled redesign. Analyse pack containment, propagation, ventilation and emergency isolation.

Materials: Freeze cell/module/BMS and enclosure build-up. Track cell lots, transport and storage conditions.

Assembly: Use protected handling, interlocked trays and keyed connections. Bind pack serials and test history to configuration.

Inspection: Check insulation, bonding, BMS sensing and shutdown. Inspect pack retention and cooling/vent routes.

Maintenance: Define health limits, isolation, safe removal and quarantine. Retain abuse, thermal and repair history.

Verification: Qualified pack electrical/mechanical/thermal and fault test plan. Removal, isolation and emergency-response demonstration.

Open evidence and decisions: Released chemistry and cell-to-pack qualification. LTO reference geometry does not release an LFP substitution.

Basis: IEC: IEC 62928:2017, IEC: IEC 61373:2010, BSI: BS EN 45545-2:2020+A1:2023, Schaltbau: Rail and industrial DC contactors.

HV/LV power distribution, bonding and harnesses

Responsible role: electrical integration engineer. Applied to 19 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-003, STD-CTRL-004, STD-CTRL-007, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Coordinate protection, grounding, creepage and stored-energy discharge. Segregate HV, safety, data and passenger circuits.

Materials: Specify cable, connector, fuse, contactor and insulation systems. Include flame, abrasion and environment evidence.

Assembly: Control crimps, terminal forces, bend radii and bonding. Use keyed circuits and verified routing.

Inspection: Perform continuity, polarity, insulation and protection tests. Record each circuit, tool and connector revision.

Maintenance: Provide safe lockout and replaceable harness sections. Inspect chafing, loosening, ingress and insulation condition.

Verification: Protection coordination and insulation assessment. Installed EMC, electrical tests and stuck-contactor fault tests.

Open evidence and decisions: Adopted voltage/altitude/pollution design basis. Qualified protection and installed-system EMC evidence.

Basis: IEC: IEC 62497-1:2010 with AMD1:2013 review required, IEC: IEC 62236-3-2:2018, IEC: IEC 61373:2010, Roxtec: Rolling-stock sealing and clamping, Roxtec: Compact replaceable transit frames, Roxtec: Rolling-stock cable clamping, Roxtec: Installation instructions and guidance, HARTING: Transportation users guide, Schaltbau: Rail and industrial DC contactors, nVent ERICO: Rail connections, bonding and protection, Texas Instruments: Precharge circuit application brief, Texas Instruments: TIDA-050063 active precharge reference, Texas Instruments: TIDA-050063 design guide.

Cooling, fluids, pneumatic auxiliaries and drains

Responsible role: thermal/fluid engineer. Applied to 16 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-003, STD-CTRL-004, STD-CTRL-007, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Separate hazardous fluids and electrical zones. Size cooling and pneumatic support for climate and failed equipment.

Materials: Freeze hose, coolant, lubricant, seal and pressure ratings. Check chemical compatibility and cleanliness.

Assembly: Control routing, clamping, bleeding and contamination. Keep leak checks before inaccessible closure.

Inspection: Measure flow, pressure, leakage and protective response. Verify drain paths and isolated maintenance points.

Maintenance: Maintain filters, dryers, drains and fluid sampling. Publish depressurisation and spill response.

Verification: Thermal/pressure duty and failure assessment. Leak, blocked-drain, pump-loss and compressor-loss tests.

Open evidence and decisions: Released flow/pressure/temperature windows. Supplier compatibility and degraded support capacity.

Basis: ISO: ISO 19659-4:2026, Continental: Railway secondary suspension.

HVAC, air distribution and passenger comfort

Responsible role: HVAC engineer. Applied to 10 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-003, STD-CTRL-004, STD-CTRL-007, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Specify route climate, solar, passenger, door-cycle and ventilation loads. Assess unit performance together with installed ducts.

Materials: Freeze refrigerant, insulation, filters and fire build-up. Check corrosion and condensate compatibility.

Assembly: Seal and support ducts without obstructing service access. Control refrigeration joints and condensate slope.

Inspection: Measure airflow, thermal comfort and noise at occupied positions. Check leakage, control response and drainage.

Maintenance: Provide replaceable filters and maintainable condensate systems. Track refrigerant leakage and actual energy demand.

Verification: Climate-chamber and installed saloon comfort tests. Ventilation, failed-unit, condensate and noise tests.

Open evidence and decisions: Project comfort values and fresh-air basis. Actual climate/load performance and cab applicability.

Basis: ISO: ISO 19659-2:2020, ISO: ISO 19659-4:2026, ISO: ISO 3381:2021, Liebherr: Modular railway HVAC, Liebherr: Transportation downloads and certificates, Liebherr: MACS 8.0 product brochure, Liebherr: Acoustic test capability, Liebherr: Development and qualification tests.

Passenger doors, carriers and boarding interlocks

Responsible role: door/control engineer. Applied to 6 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-003, STD-CTRL-004, STD-CTRL-007, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Treat cassette, carrier, sensitive edge and control as one interface. Coordinate platform alignment, entrapment and emergency release.

Materials: Freeze seals, glazing, actuators and detection devices. Include fire, fatigue and weathering.

Assembly: Survey carrier datums, shims and connector keying. Preserve door clearance through interior trim installation.

Inspection: Check obstruction detection, locking and traction inhibit. Measure alignment, water ingress and emergency opening.

Maintenance: Retain cycle counts, seal and mechanism inspections. Prohibit bypasses outside approved degraded procedures.

Verification: Approved obstruction, emergency and repeated-cycle tests. Wrong-side, sensor failure and platform-door integration tests.

Open evidence and decisions: Door forces, clearances and safety allocation. Qualified physical cassette and interlock evidence.

Basis: Knorr-Bremse / IFE: Train entrance systems, IEC: IEC 62290-1:2025.

Glazing, retention, seals and replacement

Responsible role: glazing/joint engineer. Applied to 11 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-003, STD-CTRL-004, STD-CTRL-007, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Design load, impact, retention and safe replacement together. Keep drains and secondary retention accessible.

Materials: Freeze glass, heaters, seals, setting blocks and adhesive build-up. Evaluate installed fire and ageing compatibility.

Assembly: Control bond preparation, thickness and cure or dry-seal compression. Inspect setting blocks before closure.

Inspection: Measure opening/glass fit and retained edges. Verify water ingress and heating/isolation function.

Maintenance: Publish removal and cure/return-to-service instructions. Inspect chips, delamination and seal breakdown.

Verification: Retention/impact and environmental qualification. First replacement and water-ingress demonstration.

Open evidence and decisions: Approved glazing performance and opening geometry. Bonding qualification and replacement release times.

Basis: Sika: Rail bonding and coating applications, ERA: LOC&PAS TSI and current application guidance, Sika: Bonding and sealing processing guideline.

Floors, seats, handrails and passenger fixtures

Responsible role: interior/accessibility engineer. Applied to 14 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-003, STD-CTRL-004, STD-CTRL-007, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Design slip, retention and unobstructed accessible circulation. Coordinate removable floors with underfloor hazards.

Materials: Qualify actual floor/adhesive/liner/seat build-ups. Require fire, cleaning and wear compatibility.

Assembly: Use common retained mounts and accessible captive fixings. Protect door and PRM clearance through fit-out.

Inspection: Inspect edges, ramps, floor bonds and fixture joints. Survey installed accessible spaces.

Maintenance: Publish cleaning, replacement and damage limits. Inspect retained hatches and handrails after service.

Verification: Installed fixture-load, slip/wear and fire tests. Accessible circulation and repeated hatch-removal trials.

Open evidence and decisions: Vehicle-specific access and load criteria. Installed fire build-up and fixture retention evidence.

Basis: BSI: BS EN 45545-2:2020+A1:2023, ISO: ISO 21542:2021, IEC: IEC 61373:2010.

Couplers, articulation, gangways and flexible services

Responsible role: coupling/articulation engineer. Applied to 20 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-003, STD-CTRL-004, STD-CTRL-007, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Check curves, twist, collision, rescue and passenger bridge envelopes. Select one end-interface configuration per position.

Materials: Freeze bearings, pins, bellows, wear liners and service loops. Include ageing and fire compatibility.

Assembly: Survey common end datums and retained pins. Keep flexible HV/LV/fluid paths segregated through full movement.

Inspection: Measure movement clearance, retention and leak paths. Test coupling lock and service continuity.

Maintenance: Inspect fatigue, pin wear, bellows and service-loop abrasion. Bind replacements to selected end configuration.

Verification: Full-envelope articulation and rescue load tests. Repeated movement, gangway gap and service continuity tests.

Open evidence and decisions: Released motion/load envelope and coupling compatibility. Optional train-to-train configuration transfer qualification.

Basis: ERA: LOC&PAS TSI and current application guidance, NASA: Systems engineering handbook: interfaces and verification, HÜBNER: Gangway maintenance and motion recording.

Train control, navigation and automated lifecycle

Responsible role: train-control/safety engineer. Applied to 18 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-003, STD-CTRL-004, STD-CTRL-007, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Allocate movement authority and interlocks independently of navigation. Define standby, depot, mainline, platform, rescue and wash modes.

Materials: Freeze control hardware, sensor and software revisions. Include shock, thermal, EMC and security evidence.

Assembly: Verify safety I/O identities, sensor datums and protected harnesses. Keep commissioning configuration under change control.

Inspection: Test brake/door/HV/comms interlocks and stale/lost inputs. Preserve critical lockout and bounded auxiliary recovery.

Maintenance: Use controlled diagnostics and approved manual/degraded controls. Review software changes and sensor calibration.

Verification: Independent safety architecture and integrated fault campaign. Approved physical lifecycle and movement-authority tests.

Open evidence and decisions: Real signalling architecture and safety integrity allocation. Offline Python contracts do not establish GoA4 qualification.

Basis: IEC: IEC 62290-1:2025, IEC: IEC 62278-1:2025, CRRC: Primary account of lifecycle automation, redundancy, running-gear monitoring and efficient traction.

Train/wayside communications, timing and event records

Responsible role: communications/software engineer. Applied to 21 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-006, STD-CTRL-014, STD-CTRL-015. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Specify latency, time alignment, availability and data ownership. Distinguish diagnostics from safety communications.

Materials: Freeze antenna, network device and recorder revisions. Check EMC, storage endurance and environment.

Assembly: Verify cable segregation, connector identity and antenna placement. Provision least-privilege identities and synchronized clocks.

Inspection: Test coverage, handover, time skew, link loss and record retrieval. Retain traceable event identifiers and checksums.

Maintenance: Publish open diagnostic exports and offline access. Control retention, privacy, backup and unsupported firmware.

Verification: Route communications/EMC and failover trials. Timestamped event reconstruction and recovery test.

Open evidence and decisions: Qualified safety communication channels and time budgets. Crashworthy recorder and actual coverage evidence.

Basis: NIST: SP 800-82 Rev. 3: operational technology security, IEC: IEC 62290-1:2025, Hitachi Rail: HMAX for signalling, HARTING: Transportation users guide.

Information, lighting, intercom and security equipment

Responsible role: passenger systems engineer. Applied to 14 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-006, STD-CTRL-014, STD-CTRL-015. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Coordinate visual/audio information, lighting and emergency calls. Keep emergency functions available under defined power failures.

Materials: Freeze luminaires, displays, speakers, cameras and mounts. Check fire and cleaning/ingress suitability.

Assembly: Use keyed feeders and common retained service mounts. Verify camera/speaker coverage without obstructing access.

Inspection: Test intelligibility, display accuracy, illuminance and emergency call routing. Inspect backup supply and independent lighting feeds.

Maintenance: Maintain lamps, optics, content and privacy controls. Check help points and emergency functions periodically.

Verification: Installed passenger-information/accessibility demonstration. Power-loss, audio, lighting and intercom failover tests.

Open evidence and decisions: Project lighting/intelligibility and privacy rules. Qualified emergency duration and accessibility evidence.

Basis: ISO: ISO 21542:2021, IEC: IEC 62236-3-2:2018, Knorr-Bremse / IFE: Train entrance systems.

Fire detection, containment, evacuation and rescue

Responsible role: fire/life-safety engineer. Applied to 22 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-001, STD-CTRL-003, STD-CTRL-004, STD-CTRL-014, STD-CTRL-015. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Analyse battery, electrical, interior and station fire scenarios together. Define detection, isolation, evacuation and responder access.

Materials: Qualify installed build-ups and penetration seals. Choose suppression only after hazard-specific assessment.

Assembly: Protect detector/vent routes and fire barriers. Inspect concealed seals before closure.

Inspection: Test alarms, isolation, emergency lighting and communications. Record cause/effect and barrier installation evidence.

Maintenance: Maintain detectors, barriers, extinguishers and escape routes. Train operators/responders using current approved plans.

Verification: Project fire/evacuation strategy review. Controlled emergency exercises and hazard-specific detection/suppression tests.

Open evidence and decisions: Applicable fire codes and evacuation criteria. Physical fire propagation, containment and rescue evidence.

Basis: BSI: BS EN 45545-2:2020+A1:2023, FTA: Emergency preparedness planning guide (2003), IEC: IEC 62928:2017, Roxtec: Rolling-stock sealing and clamping, Roxtec: Fire protection of plastic conduits.

Survey, alignment, gauge and route configuration

Responsible role: survey/track engineer. Applied to 18 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-009, STD-CTRL-010, STD-CTRL-012, STD-CTRL-013. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Use accepted coordinate/vertical datums and survey uncertainty. Reconcile alignment with vehicle swept and platform envelopes.

Materials: Control monument, instrument and rail-profile revisions. Preserve survey provenance and ground reference conditions.

Assembly: Set out from checked control networks. Verify coordinates and units before lifting or track installation.

Inspection: Independently check closure, gauge, level and clearance. Keep raw observations and adjusted results.

Maintenance: Monitor movement and renew damaged monuments. Update as-maintained alignment after repairs.

Verification: Independent survey adjustment and clearance audit. Loaded-envelope and track-geometry reconciliation.

Open evidence and decisions: Accepted survey control and route geometry. Project gauge/clearance limits and uncertainty.

Basis: buildingSMART: IFC railway domain and open exchange, FTA: Report 0215: track inspection and maintenance.

Ground investigation, foundations and settlement

Responsible role: geotechnical/foundation engineer. Applied to 10 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-009, STD-CTRL-010, STD-CTRL-012, STD-CTRL-013. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Base foundations on site investigation and groundwater. Include settlement, scour, seismic and adjacent utilities.

Materials: Freeze concrete, reinforcement and ground-treatment systems. Control water/soil aggressivity assumptions.

Assembly: Issue excavation, support and pile/foundation procedures. Record ground deviations before concealment.

Inspection: Inspect founding level, reinforcement and concrete evidence. Retain pile/load and settlement records as applicable.

Maintenance: Provide settlement and drainage monitoring. Inspect scour and deterioration after extreme events.

Verification: Independent geotechnical/design review and site tests. Foundation capacity and settlement verification.

Open evidence and decisions: Site ground model and design criteria. Foundation, groundwater and temporary support release.

Basis: FHWA: Bridge concrete technical resources, FHWA: Bridge preservation resources.

Precast/prestressed beams, decks, piers and platform units

Responsible role: civil structural/production engineer. Applied to 26 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-009, STD-CTRL-011, STD-CTRL-012, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Design lifting, transport, erection and final load stages. Include railway fatigue, dynamics, durability and replaceability.

Materials: Freeze concrete mix, reinforcement, tendons, inserts and grout. Keep batch/heat/strand traceability.

Assembly: Inspect forms, reinforcement and ducts before casting. Record concrete, curing, strength release and stressing/grout procedures.

Inspection: Survey dimensions, inserts, cracks and tendon/grout evidence. Quarantine damaged units pending engineer disposition.

Maintenance: Provide inspection access and repair limits. Track cracks, leakage, corrosion and bearing-zone condition.

Verification: Independent staged structural and fatigue assessment. Qualified concrete/strength, stressing/grouting and lifting records.

Open evidence and decisions: Project resistance/durability criteria and reinforcement. Released mix, strength-at-handling and prestress procedures.

Basis: FHWA: Bridge concrete technical resources, ISO: ISO 3834-2:2021 fusion-welding quality scope, Alstom: Repeatable special processes, equipment monitoring, competence and lean production, PCI: Precast tolerance responsibilities and PCI 135 transition.

Bridge bearings, joints, jacking and replaceability

Responsible role: bridge/interface engineer. Applied to 4 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-009, STD-CTRL-011, STD-CTRL-012, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Check translation, rotation, restraint and rail-structure interaction. Design safe bearing replacement and inspection access.

Materials: Freeze bearing, elastomer/sliding surface and grout systems. Check corrosion and seal compatibility.

Assembly: Survey seats and orientation with temperature context. Use released jacking, seating and grout sequence.

Inspection: Measure seat levels, movement and keeper clearances. Retain bearing serial and orientation evidence.

Maintenance: Inspect movement, debris, leaks and restraint condition. Use approved staged jacking for replacement.

Verification: Bearing/load/movement assessment. Seat survey and first replacement/jacking demonstration.

Open evidence and decisions: Released bridge movement/load envelope. Supplier bearing qualification and replacement procedure.

Basis: FHWA: Bridge preservation resources, NASA: Systems engineering handbook: interfaces and verification, Freyssinet: Bearing movement and load interfaces, Freyssinet: Bearing replacement and jacking.

Logistics, lifting, temporary works and site assembly

Responsible role: construction/temporary works engineer. Applied to 12 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-009, STD-CTRL-011, STD-CTRL-012, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Design transport routes, lifts and temporary stability stages. Schedule holds, strength release and weather constraints explicitly.

Materials: Freeze lifting accessories, anchor and grout systems. Inspect reusable lifting equipment condition.

Assembly: Use issued lift plans, competent crews and exclusion zones. Release temporary supports only after checked handover.

Inspection: Verify lifting capacity/configuration and incoming unit condition. Record surveys and critical hold completion.

Maintenance: Maintain cranes, slings, rigs and temporary works. Retain reusable fixture inspection histories.

Verification: Independent temporary-works and lift-plan review. Controlled first lift, erection survey and handover audit.

Open evidence and decisions: Site-specific lifting capacity and wind limits. Released temporary works and actual strength evidence.

Basis: FHWA: Bridge concrete technical resources, Siemens: Resource scheduling with coating/drying and quality dependencies, ISO: ISO 22163:2023 railway quality-management scope, Freyssinet: Bearing replacement and jacking.

Drainage, water management and climate resilience

Responsible role: drainage/environment engineer. Applied to 11 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-009, STD-CTRL-011, STD-CTRL-012, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Coordinate route, deck, platform, battery and canopy drainage. Design maintainable outfalls for site flood and debris scenarios.

Materials: Freeze pipe, seal, protection and ground-treatment compatibility. Include soil/water exposure.

Assembly: Survey falls and clean connections before closure. Protect drains during erection and finishing.

Inspection: Test flow, blockage access and leakage. Reconcile installed outlets with accepted outfall design.

Maintenance: Inspect before seasonal extremes and after floods. Remove debris and record erosion, scour and ingress.

Verification: Site hydrology/drainage and resilience assessment. Flow, blocked-drain and inspection-access demonstrations.

Open evidence and decisions: Site rainfall/flood climate and outfall permission. Extreme-weather operating and recovery criteria.

Basis: FHWA: Bridge preservation resources, ISO: ISO 12944-2:2017.

Rails, fastenings, slabs and wheel/rail interfaces

Responsible role: track/vehicle integration engineer. Applied to 14 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-009, STD-CTRL-011, STD-CTRL-012, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Select rail, pad, clip, insulator and support as one qualified system. Check gauge, rail stress, dynamic stiffness and vehicle profiles.

Materials: Freeze rail grade/profile, fasteners, pads and slab interfaces. Retain rail heat and fastening batch evidence.

Assembly: Use approved rail joining, stressing and fastening installation. Survey support geometry and electrical isolation.

Inspection: Measure gauge, alignment, joints and fastener condition. Check actual installed support stiffness where required.

Maintenance: Set inspected wear, geometry and defect limits. Control rail grinding, replacement and thermal interventions.

Verification: Vehicle-track dynamics and rail-structure interaction assessment. Track geometry, joining and fastening qualification.

Open evidence and decisions: Actual rail/wheel profiles and track stiffness. Approved wear/geometry limits and inspection intervals.

Basis: Pandrol: Fastclip rail fastening system, FTA: Report 0215: track inspection and maintenance, ERA: LOC&PAS TSI and current application guidance, nVent ERICO: Rail connections, bonding and protection, Pandrol: Fastclip baseplate technical sheet, Pandrol: VIPA SP technical sheet.

Turnouts, point machines, detection and track ends

Responsible role: turnout/signalling engineer. Applied to 13 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-009, STD-CTRL-011, STD-CTRL-012, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Qualify geometry, locking, detection and interlocking together. Include obstruction, run-through, manual operation and terminal overrun.

Materials: Freeze rail/frog, sleepers, actuator and detector systems. Check heater/protection environment and wear.

Assembly: Survey turnout datums and mechanically verify lock/detection. Control hand-wind isolation, cabling and heater installation.

Inspection: Test obstruction, wrong detection and loss of power. Measure switch/frog geometry and end-stop installation.

Maintenance: Inspect wear, lock, force and detector drift. Retain movement counts and approved manual intervention.

Verification: Integrated point machine/locking/interlocking fault tests. Turnout geometry, wheel passage and end-stop energy assessment.

Open evidence and decisions: Released point forces, detector logic and stopping envelope. Qualified turnout and terminal-overrun protection.

Basis: voestalpine Railway Systems: Turnout systems and lifecycle integration, FTA: Report 0215: track inspection and maintenance, IEC: IEC 62290-1:2025, Frauscher: RSR180 wheel sensor installation.

Wayside signalling, detection and movement authority

Responsible role: signalling safety engineer. Applied to 14 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-003, STD-CTRL-004, STD-CTRL-006, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Define safe authority, route locking and train detection independently of advisory telemetry. Include degraded and maintenance modes.

Materials: Freeze safety hardware/software and detection assumptions. Check vehicle EMC and axle/detection compatibility.

Assembly: Verify route-specific wiring and configured topology. Independently check safety-critical mappings.

Inspection: Test wrong route, loss of detection/comms and stale authority. Retain versioned route and event records.

Maintenance: Use controlled fault isolation and configuration restoration. Condition monitoring shall not override safe detection.

Verification: Independent signalling safety case review. Integrated route locking, train detection and authority fault tests.

Open evidence and decisions: Applicable complete signalling/safety standards. Actual safety architecture and physical train/wayside integration.

Basis: IEC: IEC 62290-1:2025, IEC: IEC 62278-1:2025, ERA: LOC&PAS TSI and current application guidance, Hitachi Rail: HMAX for signalling, Frauscher: RSR180 wheel sensor installation.

Platforms, boarding gaps and platform doors

Responsible role: station/vehicle interface engineer. Applied to 22 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-003, STD-CTRL-009, STD-CTRL-014, STD-CTRL-015. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Assess boarding against loaded/empty vehicle and curve envelopes. Coordinate evacuation, crowding and platform-door synchronization.

Materials: Freeze coping, tactile, slip-resistant and door systems. Include outdoor exposure and fire compatibility.

Assembly: Survey edge datums against actual track and vehicle. Protect accessible gaps through closure and rework.

Inspection: Measure platform height/gap with uncertainty and all relevant vehicle states. Test synchronized and degraded door opening.

Maintenance: Inspect edge movement, tactile condition and door alignment. Publish approved closure and boarding-assistance procedures.

Verification: Integrated vehicle/platform envelope and access demonstration. Platform-door obstruction, wrong-side and power-loss tests.

Open evidence and decisions: Adopted access/boarding limits and crowd assumptions. Qualified platform-door and evacuation interface.

Basis: ISO: ISO 21542:2021, Knorr-Bremse / IFE: Train entrance systems, IEC: IEC 62290-1:2025.

Station steel, solar roofs and safe access

Responsible role: station structural/roof engineer. Applied to 10 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-009, STD-CTRL-011, STD-CTRL-012, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Check wind, seismic, uplift, rain and maintenance loads. Design replaceable panels and safe roof access.

Materials: Freeze steel, fasteners, sandwich panels and bond build-up. Include corrosion, fire and UV compatibility.

Assembly: Use temporary bracing and surveyed base plates. Control roof-panel attachment and drainage sequence.

Inspection: Inspect structural joints, anchors, bonds and water ingress. Verify roof-access and edge-protection interfaces.

Maintenance: Inspect corrosion, wind damage and seal/drain condition. Plan safe module replacement with electrical isolation.

Verification: Independent structure/uplift and panel retention assessment. First erection, water and access-system tests.

Open evidence and decisions: Site loads and structural resistance basis. Qualified roof panels, bonds and safe maintenance access.

Basis: ISO: ISO 12944-2:2017, FHWA: Bridge concrete technical resources, Sika: Rail bonding and coating applications.

Station LV services, UPS, cabinets and lighting

Responsible role: station electrical/MEP engineer. Applied to 9 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-007, STD-CTRL-009, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Coordinate emergency supplies, earthing and cabinet thermal loads. Define loss-of-power and fire cause/effect.

Materials: Freeze switchgear, cable, enclosure and UPS specifications. Include protection, fire and weathering evidence.

Assembly: Verify protection settings, cable identity and bonds. Keep service access and fire penetration seals.

Inspection: Test insulation, grounding, backup duration and emergency lighting. Check cabinet thermal/ingress behaviour.

Maintenance: Inspect batteries, protection devices, ingress and terminal condition. Preserve isolation and UPS change records.

Verification: Protection/thermal and backup supply assessment. Installed electrical and emergency power-loss tests.

Open evidence and decisions: Applicable installation codes and emergency duration. Approved protection settings and actual load evidence.

Basis: IEC: IEC 62497-1:2010 with AMD1:2013 review required, ISO: ISO 21542:2021, FTA: Emergency preparedness planning guide (2003), nVent ERICO: Rail connections, bonding and protection.

Station information, fares, CCTV and help points

Responsible role: passenger systems/operations engineer. Applied to 18 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-006, STD-CTRL-014, STD-CTRL-015. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Design accessible fares, intelligible information and emergency calls. Keep passenger networks separate from safety control.

Materials: Freeze equipment, mounts and weatherproofing. Plan supplier support and privacy constraints.

Assembly: Survey accessible lanes and protected plinth cable paths. Verify speaker/camera coverage and help-point routing.

Inspection: Test transactions, power-loss egress and information consistency. Check audio/visual and accessible user journeys.

Maintenance: Maintain content, cleaning, backups and access controls. Periodically test emergency calls and gate fail states.

Verification: Accessible end-to-end passenger and fare tests. Power/comms-loss and OCC help-point failover tests.

Open evidence and decisions: Local fare/privacy rules and accessibility values. Actual speech, security and egress performance.

Basis: ISO: ISO 21542:2021, NIST: SP 800-82 Rev. 3: operational technology security, FTA: Emergency preparedness planning guide (2003).

Approaches, lifts, escalators, stairs and overbridges

Responsible role: access/vertical transport engineer. Applied to 9 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-003, STD-CTRL-009, STD-CTRL-014, STD-CTRL-015. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Provide accessible routes under lift outage and evacuation scenarios. Coordinate clear widths, fall protection and structural interfaces.

Materials: Freeze lift/escalator, surfacing, rail and structural systems. Include slip, corrosion and enclosure exposure.

Assembly: Verify shaft/anchor interfaces and competent supplier installation. Protect accessible approach gradients and drainage.

Inspection: Survey clearances and accessible routes. Test emergency stop, rescue, outage and backup arrangements.

Maintenance: Retain competent statutory service/inspection records. Keep rescue access and outage passenger assistance current.

Verification: Applicable lift/escalator and structural acceptance review. Accessible journey, rescue and evacuation demonstration.

Open evidence and decisions: National lift/escalator codes and rescue authority. Actual capacity, route geometry and outage accessibility.

Basis: ISO: ISO 21542:2021, FTA: Emergency preparedness planning guide (2003).

Train/wayside charging and traction substations

Responsible role: traction power/charging engineer. Applied to 7 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-007, STD-CTRL-009, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Coordinate charger, battery, protection and mechanical alignment. Require isolation before exposed contact and movement.

Materials: Freeze connector, cable, protection and power conversion specifications. Include arc, temperature and ingress ratings.

Assembly: Survey wayside alignment and bonding. Verify polarity, interlock I/O and protection settings.

Inspection: Test misalignment, stuck contactor, link loss and thermal limits. Confirm charger disconnect/traction-inhibit logic.

Maintenance: Inspect contacts, cable damage and cooling. Keep isolation procedures and approved firmware compatibility.

Verification: Charger/battery compatibility and protection assessment. Integrated energized/isolated, fault and movement-interlock tests.

Open evidence and decisions: Applicable fixed-installation/charging standards. Released current/duty and physical interface qualification.

Basis: IEC: IEC 62497-1:2010 with AMD1:2013 review required, IEC: IEC 62928:2017, IEC: IEC 62290-1:2025.

PV, stationary storage and microgrid integration

Responsible role: energy/electrical engineer. Applied to 14 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-007, STD-CTRL-009, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Model generation, export, backup and battery operation with actual climate. Separate stationary storage from onboard battery qualification.

Materials: Freeze PV, inverter, connector and stationary ESS systems. Require electrical, fire and weathering evidence.

Assembly: Verify connector compatibility, string polarity and bonding. Coordinate roof retention, lightning and safe isolation.

Inspection: Test protection, islanding/export policy and insulation. Record string configuration and real energy metering.

Maintenance: Inspect connectors, roof ingress, storage health and protection. Define fire response and safe module disposal.

Verification: Applicable PV/ESS/grid-code and fire assessment. Installed polarity/protection, backup and metered-energy tests.

Open evidence and decisions: Project grid rules and stationary-storage standards. Qualified ESS fire response and actual yield/lifecycle data.

Basis: ISO: ISO 14040:2006, IEC: IEC 62497-1:2010 with AMD1:2013 review required, NIST: SP 800-82 Rev. 3: operational technology security.

Depot lifts, workshops, wash, lathe and isolation

Responsible role: depot/workshop engineer. Applied to 9 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-001, STD-CTRL-003, STD-CTRL-014, STD-CTRL-015, STD-CTRL-016. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Design safe stabling, bogie change and heavy maintenance access. Coordinate recovery, wash, battery quarantine and fire separation.

Materials: Freeze lifting, tooling, service-fluid and equipment specifications. Provide calibrated inspection tools and qualified replacement parts.

Assembly: Release lifting foundations, synchronized lifts and crane interfaces. Verify workshop isolation and segregated vehicle movements.

Inspection: Test lift/crane protection, clearance and isolation. Inspect wash effluent, traction power and service interfaces.

Maintenance: Maintain competent lifting-equipment and tooling inspections. Retain work orders and controlled return-to-service records.

Verification: Site/workshop hazard and lift capacity assessment. Controlled first bogie change, isolation and recovery demonstration.

Open evidence and decisions: Applicable machinery/lifting/environmental rules. Qualified equipment and actual workshop capacity.

Basis: ISO: ISO 55001:2024, ERA: Entities in charge of maintenance, FTA: Emergency preparedness planning guide (2003).

Service planning, degraded operation and incident response

Responsible role: operator/operations lead. Applied to 30 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-001, STD-CTRL-003, STD-CTRL-014, STD-CTRL-015, STD-CTRL-016. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Plan capacity, dwell, headway and recovery against real route constraints. Define dispatch, degraded, evacuation and rescue authority.

Materials: Provide controlled consumables, spares and emergency supplies. Track replacement availability and shelf life.

Assembly: Train competent staff and rehearse handovers. Restrict unapproved bypasses and temporary configurations.

Inspection: Monitor service failures, energy and passenger outcomes. Inspect emergency readiness and accessible support.

Maintenance: Feed incidents into hazard and instruction review. Validate changes to operating envelopes and staffing.

Verification: Timetable/headway and degraded-service validation. Controlled emergency drills and operator competence assessment.

Open evidence and decisions: Approved operating rules and regulator acceptance. Actual availability, demand and emergency-response evidence.

Basis: FTA: Public Transportation Agency Safety Plans, IEC: IEC 62290-1:2025, FTA: Emergency preparedness planning guide (2003).

Maintenance, spares, condition monitoring and return to service

Responsible role: maintenance authority. Applied to 28 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-001, STD-CTRL-003, STD-CTRL-014, STD-CTRL-015, STD-CTRL-016. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Allocate maintainable units and failure detection requirements. Keep time/mileage/cycle intervals distinct from diagnostic advice.

Materials: Control spares equivalence, expiry and obsolete suppliers. Preserve serial/lot and material genealogy.

Assembly: Use issued repair tasks and independent safety checks. Bind work orders, tools, NCRs and configuration changes.

Inspection: Verify repaired function and close mandatory holds. Retain failures including no-fault-found events.

Maintenance: Extend intervals only through approved duty/condition evidence. Freeze as-maintained configuration before return to service.

Verification: Maintenance task/interval and spares review. Physical repair, inspection and return-to-service demonstration.

Open evidence and decisions: Approved supplier/project intervals and acceptance authority. Validated diagnostic thresholds and interval changes.

Basis: ERA: Entities in charge of maintenance, ISO: ISO 55001:2024, Hitachi Rail: HMAX for signalling, Liebherr: Modular railway HVAC, HÜBNER: Gangway maintenance and motion recording, Siemens Mobility: Bogie diagnostic architecture.

Level crossings, public interfaces and intrusion

Responsible role: crossing/public safety engineer. Applied to 2 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-003, STD-CTRL-004, STD-CTRL-006, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Assess separation, sighting, accessible passage and road/rail risks. Coordinate detection, protection, intrusion and emergency access.

Materials: Freeze crossing surface, barrier, lighting and detection systems. Include outdoor, wear and vandalism exposure.

Assembly: Verify approach geometry and fail-state wiring. Keep pedestrian paths and drainage clear.

Inspection: Test obstruction, protection failure and road/rail timing. Inspect visibility and accessible route condition.

Maintenance: Maintain detectors, barriers, surfaces and public information. Review near misses and changed traffic patterns.

Verification: Independent crossing/public risk assessment. Controlled fail-state, intrusion and road/rail integration tests.

Open evidence and decisions: Local crossing authority and legal protection requirements. Qualified traffic/detection and accessible crossing evidence.

Basis: IEC: IEC 62278-1:2025, IEC: IEC 62290-1:2025, FTA: Public Transportation Agency Safety Plans.

Integrated vehicle, track, bridge and station dynamics

Responsible role: integrated vehicle/civil engineer. Applied to 40 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-001, STD-CTRL-002, STD-CTRL-003, STD-CTRL-004, STD-CTRL-013, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Use one mass/load/geometry configuration across disciplines. Cover loaded/empty, braking, rescue, both tracks and degradation.

Materials: Use measured stiffness, damping, masses and profiles. Retain supplier/physical provenance and uncertainty.

Assembly: Reconcile actual datums and installed support conditions. Record deviations before dynamic trials.

Inspection: Keep model-domain failures and incomplete passages visible. Check force, energy and numerical refinement independently.

Maintenance: Review track/vehicle changes together. Monitor settlements, wear and changed service loads.

Verification: Independent coupled dynamics and complete passage campaign. Physical profile, load, modal and running-response correlation.

Open evidence and decisions: Validated contact model and complete six-car passage. Accepted safety/ride criteria and measured correlation data.

Basis: ERA: LOC&PAS TSI and current application guidance, NASA: Systems engineering handbook: interfaces and verification, FTA: Report 0215: track inspection and maintenance.

Integrated train, charger, route and energy supply

Responsible role: system energy engineer. Applied to 21 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-001, STD-CTRL-002, STD-CTRL-003, STD-CTRL-004, STD-CTRL-013, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Balance duty-cycle traction, auxiliaries, regeneration and charging. Include low state of charge, failed chargers and reserve policy.

Materials: Use released chemistry, efficiency maps and climate ageing data. Control supply/storage lifecycle boundaries.

Assembly: Reconcile actual train and wayside configurations. Verify charger compatibility before energisation.

Inspection: Use calibrated synchronized meters and energy residuals. Retain unmet-demand and curtailed-regeneration scenarios.

Maintenance: Track battery health and actual energy per duty. Reassess timetables and supply when assets degrade.

Verification: Independent energy/thermal and contingency assessment. Metered full-duty and failed-charger validation.

Open evidence and decisions: Accepted reserve, range and energy criteria. Measured efficiency/ageing maps and actual grid/charging capacity.

Basis: IEC: IEC 62928:2017, ISO: ISO 14040:2006, CRRC: Primary account of lifecycle automation, redundancy, running-gear monitoring and efficient traction.

Integrated accessible passenger journey and evacuation

Responsible role: accessibility/fire/operations lead. Applied to 29 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-001, STD-CTRL-002, STD-CTRL-003, STD-CTRL-004, STD-CTRL-013, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Validate street-to-seat and safe-return journeys together. Include crowded, lift-outage, wrong-side and emergency conditions.

Materials: Choose tactile, slip, signage and fire systems for actual use. Retain compatibility of installed passenger fixtures.

Assembly: Reconcile platform, train, doors and access routes. Protect clearance after fit-out and maintenance.

Inspection: Test complete journeys with relevant user needs. Verify information, assistance and emergency access.

Maintenance: Provide controlled outage assistance and inspection tasks. Review incident feedback and changed demographics.

Verification: Independent accessible journey and evacuation review. Physical boarding, outage assistance and emergency exercises.

Open evidence and decisions: Adopted access and evacuation acceptance rules. Representative user validation and integrated physical evidence.

Basis: ISO: ISO 21542:2021, FTA: Emergency preparedness planning guide (2003), Knorr-Bremse / IFE: Train entrance systems.

Integrated safety control, power, doors and operations

Responsible role: system safety/control engineer. Applied to 44 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-001, STD-CTRL-002, STD-CTRL-003, STD-CTRL-004, STD-CTRL-013, STD-CTRL-014. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Define safe states for stale inputs, common causes and partial power loss. Keep diagnostics, navigation and maintenance requests subordinate to authority.

Materials: Freeze combined hardware/software and security baselines. Review failure/obsolescence of shared services.

Assembly: Verify complete interface topology and fail-state mapping. Independently check configured safety I/O.

Inspection: Test combined brake, door, charger, detection and communication faults. Retain failed and interrupted trials.

Maintenance: Control overrides and updates with independent review. Prove recovery without silent safety bypass.

Verification: Independent integrated safety/cyber/interface review. Approved physical multi-fault and degraded-mode campaign.

Open evidence and decisions: Actual safety architecture, communication contracts and integrity levels. Independent qualification and operating authority.

Basis: IEC: IEC 62278-1:2025, IEC: IEC 62290-1:2025, NIST: SP 800-82 Rev. 3: operational technology security.

Software requirements, deterministic behaviour and target validation

Responsible role: software assurance lead. Applied to 60 scoped definitions. Status: recommended; project adoption open.

Existing assurance objectives: STD-CTRL-002, STD-CTRL-005, STD-CTRL-006, STD-CTRL-013, STD-CTRL-014, STD-CTRL-016. Use the Digital Standards Thread for applicability, clause adoption and evidence decisions.

Design: Derive bounded, deterministic behaviours from allocated hazards and interface contracts. Separate simulation/proof assumptions from actual target timing and I/O.

Materials: Pin source, dependencies, toolchains and reproducible artifact identities. Track firmware support, vulnerabilities and third-party licence obligations.

Assembly: Build reviewed artifacts with controlled deployment settings. Verify startup, watchdog, resource and hardware-interface configuration.

Inspection: Use independent review and requirement-derived unit/integration/formal checks. Retain failed, incomplete and unsupported-target checks.

Maintenance: Control updates, rollback and repair configuration. Reassess proof assumptions and target behaviour after dependency or interface changes.

Verification: Independent software requirement, tool and configuration review. Target-hardware timing, resource, HIL and fault-response validation.

Open evidence and decisions: Project safety/software integrity allocation and adopted clauses. Actual target I/O, timing, tool qualification and independent acceptance.

Basis: IEC: IEC 62279:2015, NIST: SP 800-82 Rev. 3: operational technology security, NASA: Systems engineering handbook: interfaces and verification.

Source limitations and refresh

All references were reviewed on the date recorded below. ISO/IEC/EN entries are public-scope discovery, with complete adopted clause review open. Regulatory scope depends on project jurisdiction and system type. Manufacturer examples contribute methods; no proprietary design, guaranteed improvement or accepted supplier specification is redistributed.

Refresh policies are OSR research rules: regulatory sources 90 days, manufacturer examples 180 days, standard scopes 365 days, method references 730 days. Checks use an explicit assessment date; a source overdue for review stays visible in the audit.

- CRRC: Primary account of lifecycle automation, redundancy, running-gear monitoring and efficient traction (manufacturer-practice; reviewed 2026-10-11). Primary account of lifecycle automation, redundancy, running-gear monitoring and efficient traction Manufacturer claims; performance objectives require OSR duty-cycle baselines and independent evidence
- Alstom: Repeatable special processes, equipment monitoring, competence and lean production (manufacturer-practice; reviewed 2026-10-11). Repeatable special processes, equipment monitoring, competence and lean production Methods adapted as original OSR controls; no robot count or productivity claim transferred
- Alstom: Digital work instructions and per-joint smart torque records (manufacturer-practice; reviewed 2026-10-11). Digital work instructions and per-joint smart torque records OSR stores open tool-neutral records; no vendor cloud implementation copied
- Siemens: Resource scheduling with coating/drying and quality dependencies (manufacturer-practice; reviewed 2026-10-11). Resource scheduling with coating/drying and quality dependencies Actual durations and process windows require released material/process data
- Siemens: Measurement context, temperature and calibration traceability (manufacturer-practice; reviewed 2026-10-11). Measurement context, temperature and calibration traceability No proprietary inspection software dependency

- Alstom: Energy, recyclability and lifecycle-aware material choices (manufacturer-practice; reviewed 2026-10-11). Energy, recyclability and lifecycle-aware material choices No environmental percentages adopted; supplier EPD and repair evidence remain required
- NIST: Diagnostic spread/capability statistics and their stability/distribution/sample assumptions (method; reviewed 2026-10-11). Diagnostic spread/capability statistics and their stability/distribution/sample assumptions No universal Cpk acceptance threshold or process qualification inferred
- BIPM: Uncertainty-aware conformity decisions; explicit decision-rule selection (method; reviewed 2026-10-11). Uncertainty-aware conformity decisions; explicit decision-rule selection Conservative interval containment is an OSR internal rule, not a universal ISO acceptance rule
- ISO: ISO 22163:2023 railway quality-management scope (standard-scope; reviewed 2026-10-11). ISO 22163:2023 railway quality-management scope Public scope reviewed; controlled clauses, amendments and conformity assessment remain open
- ISO: ISO 3834-2:2021 fusion-welding quality scope (standard-scope; reviewed 2026-10-11). ISO 3834-2:2021 fusion-welding quality scope Public scope reviewed; material/process-specific qualification remains open
- NASA: Systems engineering handbook: interfaces and verification (method; reviewed 2026-10-11). Integration and verification planning connect requirements, interfaces and evidence. A systems-engineering method reference; NASA requirements are not rail operating rules.
- IEC: IEC 62278-1:2025 (standard-scope; reviewed 2026-10-11). Railway RAMS lifecycle management scope. Public scope only; project safety allocation and the applicable complete standard series require controlled review.
- IEC: IEC 62290-1:2025 (standard-scope; reviewed 2026-10-11). Urban guided transport management and command/control concepts. No GoA4 certification, protocol specification or safety integrity level inferred from the public scope.
- IEC: IEC 62928:2017 (standard-scope; reviewed 2026-10-11). Onboard lithium-ion traction battery design and test scope. Stationary storage and charging installations need their own applicable standards; no cell-to-pack approval inferred.
- IEC: IEC 61373:2010 (standard-scope; reviewed 2026-10-11). Railway equipment shock and vibration test scope. Mounting category, test levels and clauses require project review; a CAD attachment is not qualification.
- IEC: IEC 62497-1:2010 with AMD1:2013 review required (standard-scope; reviewed 2026-10-11). Rail insulation coordination includes rolling stock, signalling and fixed installations. Adopted edition/amendment, altitude, pollution, voltage and clearances require controlled clause review.
- IEC: IEC 62236-3-2:2018 (standard-scope; reviewed 2026-10-11). Rolling-stock apparatus electromagnetic compatibility scope. Apparatus evidence does not qualify the complete vehicle or wayside system; select other series parts as applicable.
- ISO: ISO 55001:2024 (standard-scope; reviewed 2026-10-11). Asset-management requirements address lifecycle decisions. A management-system reference, not a component maintenance interval or asset certification.
- ISO: ISO 21542:2021 (standard-scope; reviewed 2026-10-11). Accessibility and usability of buildings and their approaches. Vehicle and platform boarding limits and national accessibility rules require separate adoption.
- ISO: ISO 19659-2:2020 (standard-scope; reviewed 2026-10-11). Passenger-area thermal comfort and measurement scope. Does not cover driver-cab thermal comfort; climate, loads and acceptance values need project selection.
- ISO: ISO 19659-4:2026 (standard-scope; reviewed 2026-10-11). HVAC customer design inputs and unit test/inspection scope. Public scope reviewed; unit test evidence must be integrated with saloon/cab and installed duct performance.
- ISO: ISO 3381:2021 (standard-scope; reviewed 2026-10-11). Reproducible vehicle interior noise type-test methods. No universal noise limit or whole-journey exposure result inferred.
- ISO: ISO 12944-2:2017 (standard-scope; reviewed 2026-10-11). Steel protective-coating exposure classification. Classification alone does not select a coating; preparation, application, durability and inspection require additional applicable methods.
- ISO: ISO 14040:2006 (standard-scope; reviewed 2026-10-11). Lifecycle assessment goal, scope and interpretation framework. Review amendment and other applicable LCA/EPD methods; no carbon saving transferred from a vendor statement.
- BSI: BS EN 45545-2:2020+A1:2023 (standard-scope; reviewed 2026-10-11). Rail vehicle material/component fire-behaviour scope. Hazard/category, requirement set and actual installed build-up require controlled review; no blanket fire rating.

- NIST: SP 800-82 Rev. 3: operational technology security (method; reviewed 2026-10-11). OT security must account for physical processes, availability and safety. Reference guidance; railway threat model and applicable IEC/rail-specific cyber requirements remain project decisions.
- ERA: LOC&PAS TSI and current application guidance (regulatory-scope; reviewed 2026-10-11). Vehicle/subsystem interfaces and links to current guidance and amendments. EU interoperability applicability must be decided for each project; metro/tram scope is not assumed. No global legal adoption.
- ERA: Entities in charge of maintenance (regulatory-scope; reviewed 2026-10-11). Maintenance functions require responsibility and traceability in the ECM system. EU certification applicability is project-dependent; guidance version 10.0 (2025) listed at review.
- FTA: Public Transportation Agency Safety Plans (regulatory-scope; reviewed 2026-10-11). Agency safety plans incorporate safety-management processes. US funding/oversight applicability is limited; OSR adapts the management approach, not a global legal requirement.
- FTA: Report 0215: track inspection and maintenance (method; reviewed 2026-10-11). Review of US rail-transit inspection and maintenance practice. 2022 research portal reviewed; local defect limits, intervals and current rules require independent selection.
- FTA: Emergency preparedness planning guide (2003) (method; reviewed 2026-10-11). Emergency response planning is a system lifecycle activity. Historical planning reference; current fire, evacuation and emergency-response requirements must govern each project.
- FHWA: Bridge concrete technical resources (method; reviewed 2026-10-11). Public precast/prestressed bridge construction and assessment resources. Road bridge methods need railway load, fatigue, vibration and jurisdiction transfer review; no curing duration adopted from the portal.
- FHWA: Bridge preservation resources (method; reviewed 2026-10-11). Preservation and service-life planning include assessment and maintenance. Road bridge reference; actual railway inspection criteria and access plans remain project-specific.
- SKF: Evolution of railway axlebox technology (manufacturer-practice; reviewed 2026-10-11). Axlebox bearing architecture includes sealing, lubrication and integration. Supplier architecture reference; fits, loads, lubricants, life and intervals require released supplier data.
- Continental: Railway secondary suspension (manufacturer-practice; reviewed 2026-10-11). Air-spring solutions couple vehicle support and ride behaviour. No universal stiffness, emergency capacity or maintenance interval inferred.
- Knorr-Bremse: Adhesion and wheel-slide protection (manufacturer-practice; reviewed 2026-10-11). Adhesion management and wheel-slide protection belong to integrated braking systems. No stopping-distance improvement or proprietary control algorithm adopted; qualify actual duty and degraded modes.
- Knorr-Bremse / IFE: Train entrance systems (manufacturer-practice; reviewed 2026-10-11). Entrance equipment is supplied as integrated door systems. No closing force, speed, geometry or interoperability acceptance imported from marketing.
- Sika: Rail bonding and coating applications (manufacturer-practice; reviewed 2026-10-11). Rail bonding requires application-specific design, production and material data; EN 17460 is identified. Supplier method context, not independent proof of product superiority or bonding certification; review EN 17460 separately.
- Pandrol: Fastclip rail fastening system (manufacturer-practice; reviewed 2026-10-11). Elastic fastening is a rail/pad/insulator/clip installation system. Specific vendor system is a candidate only; no generic clip force, spacing or approved interchangeability.
- voestalpine Railway Systems: Turnout systems and lifecycle integration (manufacturer-practice; reviewed 2026-10-11). Turnout geometry, actuation, signalling and lifecycle care are treated as one system. Vendor lifecycle approach; no actuator load or inspection interval adopted.
- Hitachi Rail: HMAX for signalling (manufacturer-practice; reviewed 2026-10-11). Asset monitoring supports maintenance decisions for signalling. Advisory monitoring is not safety authority; no predictive accuracy, interval extension or vendor platform dependency adopted.
- buildingSMART: IFC railway domain and open exchange (method; reviewed 2026-10-11). IFC 4.3 provides railway spatial structures for open information exchange. Export must be checked for units, identity and semantics; an IFC file is not a construction release.
- IEC: IEC 62279:2015 (standard-scope; reviewed 2026-10-11). Railway control/protection software development process scope. Full clauses, tool qualification and applicable edition transitions require project review. Source SIL/GoA target labels do not establish achieved integrity.

- Liebherr: Modular railway HVAC (manufacturer-practice; reviewed 2026-10-11). Replaceable cooling cycles support off-car repair; public module mass is 125 kg and height 220 mm. Public method/installation reference only; actual configuration, clauses, performance, material and independent qualification remain open. Vendor gains and proprietary internals are not adopted.
- Liebherr: Transportation downloads and certificates (manufacturer-practice; reviewed 2026-10-11). Primary route from product overview to technical brochures and qualification context. Public method/installation reference only; actual configuration, clauses, performance, material and independent qualification remain open. Vendor gains and proprietary internals are not adopted.
- Liebherr: MACS 8.0 product brochure (manufacturer-practice; reviewed 2026-10-11). Brochure page 2 gives one-module envelope 1000 × 1600 × 220 mm and nominal 8 kW cooling. Public method/installation reference only; actual configuration, clauses, performance, material and independent qualification remain open. Vendor gains and proprietary internals are not adopted.
- Liebherr: Acoustic test capability (manufacturer-practice; reviewed 2026-10-11). Controlled noise testing distinguishes test-cell context from installed passenger noise. Public method/installation reference only; actual configuration, clauses, performance, material and independent qualification remain open. Vendor gains and proprietary internals are not adopted.
- Liebherr: Development and qualification tests (manufacturer-practice; reviewed 2026-10-11). Combined mechanical, climatic, pneumatic and electrical conditions inform equipment test planning. Public method/installation reference only; actual configuration, clauses, performance, material and independent qualification remain open. Vendor gains and proprietary internals are not adopted.
- Roxtec: Rolling-stock sealing and clamping (manufacturer-practice; reviewed 2026-10-11). Cable/pipe penetrations connect fire, ingress, vibration and retention responsibilities. Public method/installation reference only; actual configuration, clauses, performance, material and independent qualification remain open. Vendor gains and proprietary internals are not adopted.
- Roxtec: Compact replaceable transit frames (manufacturer-practice; reviewed 2026-10-11). Replaceable frame and seal elements support configuration-specific penetrations and retrofit. Public method/installation reference only; actual configuration, clauses, performance, material and independent qualification remain open. Vendor gains and proprietary internals are not adopted.
- Roxtec: Fire protection of plastic conduits (manufacturer-practice; reviewed 2026-10-11). Plastic conduit penetrations require qualified fire sealing; an empty frame has no fire integrity. Public method/installation reference only; actual configuration, clauses, performance, material and independent qualification remain open. Vendor gains and proprietary internals are not adopted.
- Roxtec: Rolling-stock cable clamping (manufacturer-practice; reviewed 2026-10-11). Cable support and strain relief are distinct from barrier sealing. Public method/installation reference only; actual configuration, clauses, performance, material and independent qualification remain open. Vendor gains and proprietary internals are not adopted.
- Roxtec: Installation instructions and guidance (manufacturer-practice; reviewed 2026-10-11). Product-specific installation and aperture details matter to sealing performance. Public method/installation reference only; actual configuration, clauses, performance, material and independent qualification remain open. Vendor gains and proprietary internals are not adopted.
- HARTING: Transportation users guide (manufacturer-practice; reviewed 2026-10-11). Connector and cable interfaces need railway application-specific installation guidance. Public method/installation reference only; actual configuration, clauses, performance, material and independent qualification remain open. Vendor gains and proprietary internals are not adopted.
- Schaltbau: Rail and industrial DC contactors (manufacturer-practice; reviewed 2026-10-11). Switching equipment must be selected for the actual DC duty and disconnect role. Public method/installation reference only; actual configuration, clauses, performance, material and independent qualification remain open. Vendor gains and proprietary internals are not adopted.
- Freyssinet: Bearing movement and load interfaces (manufacturer-practice; reviewed 2026-10-11). Fixed, guided and free bearing functions differ; replacement access belongs in the design. Public method/installation reference only; actual configuration, clauses, performance, material and independent qualification remain open. Vendor gains and proprietary internals are not adopted.
- Freyssinet: Bearing replacement and jacking (manufacturer-practice; reviewed 2026-10-11). Bearing assessment and replacement require site-specific synchronized jacking and load control. Public method/installation reference only; actual configuration, clauses, performance, material and independent qualification remain open. Vendor gains and proprietary internals are not adopted.

- Frauscher: RSR180 wheel sensor installation (manufacturer-practice; reviewed 2026-10-11). Rail-mounted sensing separates installation, electromagnetic behaviour and safe detection assessment. Public method/installation reference only; actual configuration, clauses, performance, material and independent qualification remain open. Vendor gains and proprietary internals are not adopted.
- HÜBNER: Gangway maintenance and motion recording (manufacturer-practice; reviewed 2026-10-11). Actual route-relative movements inform gangway design and endurance testing. Public method/installation reference only; actual configuration, clauses, performance, material and independent qualification remain open. Vendor gains and proprietary internals are not adopted.
- PCI: Precast tolerance responsibilities and PCI 135 transition (standard-scope; reviewed 2026-10-11). Production, erection and interfacing tolerances need separate ownership; PCI 135-25 changes earlier MNL references. Public method/installation reference only; actual configuration, clauses, performance, material and independent qualification remain open. Vendor gains and proprietary internals are not adopted.
- nVent ERICO: Rail connections, bonding and protection (manufacturer-practice; reviewed 2026-10-11). Rail bonding, cable connections and surge protection need coordinated installation evidence. Public method/installation reference only; actual configuration, clauses, performance, material and independent qualification remain open. Vendor gains and proprietary internals are not adopted.
- Siemens Mobility: Bogie diagnostic architecture (manufacturer-practice; reviewed 2026-10-11). Running-gear monitoring supports condition-based maintenance with configured data interfaces. Public method/installation reference only; actual configuration, clauses, performance, material and independent qualification remain open. Vendor gains and proprietary internals are not adopted.
- Böllhoff: RIVNUT setting-process controls (manufacturer-practice; reviewed 2026-10-11). Tool setting, stroke and alarm records inform installation traceability. Public method/installation reference only; actual configuration, clauses, performance, material and independent qualification remain open. Vendor gains and proprietary internals are not adopted.
- Sika: Bonding and sealing processing guideline (manufacturer-practice; reviewed 2026-10-11). Joint design, surface preparation and professional processing jointly control bond quality. Public method/installation reference only; actual configuration, clauses, performance, material and independent qualification remain open. Vendor gains and proprietary internals are not adopted.
- ECARV: EN 17460 interpretation and implementation (method; reviewed 2026-10-11). Rail adhesive processes require consistent quality implementation across production and repair. Public method/installation reference only; actual configuration, clauses, performance, material and independent qualification remain open. Vendor gains and proprietary internals are not adopted.
- Texas Instruments: Precharge circuit application brief (method; reviewed 2026-10-11). A resistor and separate switching path can limit DC-link capacitor inrush before main contactor closure. Public method/installation reference only; actual configuration, clauses, performance, material and independent qualification remain open. Vendor gains and proprietary internals are not adopted.
- Texas Instruments: TIDA-050063 active precharge reference (method; reviewed 2026-10-11). Active precharge offers an alternative architecture; OSR does not copy its circuit or claim railway qualification. Public method/installation reference only; actual configuration, clauses, performance, material and independent qualification remain open. Vendor gains and proprietary internals are not adopted.
- NIST: ISO 23247 implementation scenarios (method; reviewed 2026-10-11). Digital-twin use cases connect observable elements, models, information exchange and manufacturing decisions. Public method/installation reference only; actual configuration, clauses, performance, material and independent qualification remain open. Vendor gains and proprietary internals are not adopted.
- ISO: ISO 21368:2022 (standard-scope; reviewed 2026-10-11). Adhesive structure fabrication/reporting scope complements application-specific rail bonding review. Public method/installation reference only; actual configuration, clauses, performance, material and independent qualification remain open. Vendor gains and proprietary internals are not adopted.
- Pandrol: Fastclip baseplate technical sheet (manufacturer-practice; reviewed 2026-10-11). Replaceable pads, insulation, anchor inserts and adjustable interfaces form a complete fastening system. Public technical reference; no supplier design, guaranteed result, product selection or qualification adopted. Values used by OSR remain explicitly scoped.
- Pandrol: VIPA SP technical sheet (manufacturer-practice; reviewed 2026-10-11). Two resilient interfaces provide configurable support; assembly properties depend on actual pads, restraint and installation. Public technical reference; no supplier design, guaranteed result, product selection or qualification adopted. Values used by OSR remain explicitly scoped.

- NIST: AMS 400-2 digital twin implementation report (method; reviewed 2026-10-11). Reference use cases clarify information exchange and observable manufacturing elements. Public technical reference; no supplier design, guaranteed result, product selection or qualification adopted. Values used by OSR remain explicitly scoped.
- Texas Instruments: TIDA-050063 design guide (method; reviewed 2026-10-11). Active precharge reference demonstrates component and verification responsibilities; no automotive-to-rail approval inferred. Public technical reference; no supplier design, guaranteed result, product selection or qualification adopted. Values used by OSR remain explicitly scoped.

Expansion boundaries

- tunnels (research-and-design-open): No detailed tunnel product catalogue in this inventory; fire/smoke control, lining, ventilation, waterproofing and evacuation require separate expansion.
- overhead-third-rail (research-and-design-open): The current charging architecture does not define a complete overhead/third-rail collection system; add power collection, return current and stray-current assurance if adopted.
- kits-and-consumables (decomposition-open): Every catalogued kit is mapped, but its fasteners, crimps, seals and fluids need a released piece-level BOM and scoped verification.
- other-fleets (variant-transfer-open): LM3 catalogue practices are candidates for other consist families; lengths, loads, interfaces and evidence require configuration-specific transfer review.
- full-standard-clauses (applicability-and-clause-review-open): Public scopes support discovery only; adopt controlled complete ISO/IEC/EN and local requirements with clause-level traceability before acceptance.

Best-practice research and implementation plan

Repository source: `docs/best-practice/research-plan.md`

The handbook brings component, subsystem and railway-system guidance together. Its source inputs are the practice cards, primary references, scope assignments and assurance crosswalk. The coverage audit reports the actual inventory counts rather than a separately maintained headline total.

The follow-up digital-twin design expansion incorporates recursive primary-source research into original installed CAD, mass/interface models, fault studies and infrastructure details. It retains the initial failed HVAC case and the proposed four-module alternative.

The approach follows OSR's open-hardware philosophy: use original designs, supplier-neutral interface requirements, repairable modules, local production where qualified, portable work records and retained failed results. A supplier example is a candidate method to assess against OSR duty and hazards. It does not establish a proprietary specification or guaranteed improvement.

Research decisions

CRRC's Hangzhou account supports studying lifecycle automation, redundancy and running-gear monitoring. The existing factory expansion already translates these themes into offline commissioning contracts, modular fixtures, travelers and open quality evidence. This research expands those methods across infrastructure, stations, software and operation.

Braking guidance treats low adhesion and wheel-slide protection as an integrated interface, informed by Knorr-Bremse's adhesion systems. Turnout guidance similarly joins geometry, actuation, locking, detection and lifecycle care, informed by voestalpine's integrated turnout approach. OSR adoption requires actual route-specific tests and released limits.

The ISO preference is applied by scope: thermal comfort and HVAC unit inputs use ISO 19659-2:2020 and ISO 19659-4:2026; building/access routes use ISO 21542:2021; lifecycle decisions use ISO 55001:2024 and ISO 14040:2006. Railway battery, automation and software scopes use relevant IEC references alongside the existing standards thread. Public scope review leaves complete clause adoption open.

Work packages and acceptance

Package	Deliverable and completion condition	Current state
BP-01 Inventory	Explicit guidance for every current catalogue definition, Rust package and engineering template; reasoned exclusions and optional/variant context retained	Implemented; automated drift checks
BP-02 Research	Primary-source register with dated findings, limits, source kinds and specialist lifecycle cards	Implemented; refresh and applicability remain continuing work
BP-03 Interfaces	Vehicle/track/bridge, energy/charging, accessible boarding/evacuation and safety-control guidance with existing assurance objectives	Implemented as recommendations and unperformed verification plans
BP-04 Open tooling	Deterministic handbook, component CSV, coverage audit and verification/backlog JSON; mutation tests and CI	Implemented; software validation only
BP-05 Project adoption	Complete adopted clauses, approved design basis, numerical limits, verification procedures, sampling and decision authority per configuration	Open; project inputs and competent review required
BP-06 Qualification	Supplier freeze, physical first articles, special-process qualification, integrated tests and independent acceptance	Open; research and generated records cannot substitute for tests

Every practice card names a responsible role and unresolved questions. The verification plan allocates each task to all relevant definitions without duplicating prose. Task acceptance criteria, adopted clauses, procedures, results and evidence are deliberately unfilled. Resolve them in the existing project assurance workflow, not by changing an open research task into a passing result.

Next technical priorities

1. Wheel/rail/brake/signalling integration. Freeze actual wheel/rail profiles, adhesion and braking envelopes, load cases and movement-authority interfaces. The six-car contact-domain diagnosis remains an unresolved numerical qualification limit. Complete the validated contact model and whole-passage campaign before service claims.
2. Battery/charger/fire/energy integration. Freeze chemistry, pack design, cooling and charging protection. The reference LTO geometry does not release the catalogue LFP pack. Obtain qualified pack data and physical isolation, propagation, charger and route-energy evidence.

3 Structure/material/process release. Use existing factory passports, drawings, inspection characteristics and travelers. Fill grade, process windows, calibration and installed build-up fire evidence through competent review. Validate fixtures for actual clamp and handling loads.

4 Station/vehicle/access/evacuation integration. Survey actual track and boarding envelopes for each station variant, confirm lift-outage routes and emergency supplies, then perform representative passenger and responder demonstrations.

5 Software/wayside/operations integration. Review target timing, hardware I/O, safety communications, security boundaries and rollback. Source-level tests, formal checks and declared SIL/GoA targets require actual target and independent integration evidence.

6 Piece-level and new-architecture research. Decompose kits into released fasteners, terminals, seals and fluids. Transfer guidance explicitly to other fleets. Add detailed tunnel and overhead/third-rail scopes if those architectures are adopted.

The scope is the current controlled catalogue and workspace. It does not claim to enumerate every possible rail component, every city asset instance or every piece inside a kit. Finance, appraisal and publication-only templates have explicit exclusions; they do not become physical qualification evidence.

Reproduction and contribution

```
python3 tools/automation/best-practice.py generate --as-of 2026-10-11
python3 tools/automation/best-practice.py check --as-of 2026-10-11
python3 tools/automation/best-practice.py review-sources --as-of 2026-10-11
tools/automation/osr-python -m pytest -q engineering/analysis/tests/test_best_practice.py
```

The explicit date makes historical reproduction possible. For a current review, use the actual assessment date and review any overdue sources first. CI checks the published assessment date for reproducibility and the current UTC date for source expiry. It does not browse or silently refresh research. `review-sources` checks current source age without rewriting a historical publication.

Edit authoring inputs first. Add source limitations and original OSR wording, map each new definition explicitly, connect it to existing STD-CTRL objectives, and regenerate all four outputs. A new catalogue item, crate, station variant or template without a scope decision fails the checker. Changes to input bytes invalidate the published audit even when item IDs are unchanged.

Current design across the city catalogue

Repository source: `docs/catalogue-current-design-review.md`

265 other city examples regenerated, including Lyon as a technical comparison; 264 are developing-world examples. Baghdad retains its dedicated funding and scope publication. All regenerated examples have complete source-bound planning packages. Construction and operational releases remain open.

The adopted core concepts straighten radial routes where retained water evidence permits and use elevated land sections; short water crossings remain bridge candidates. Shoreline detours keep their actual lengths and geometry review gates. Rings remain in the controlled inventory and use analytical fillets where suitable. Retained unsuitable fragments keep their geometry and special-product review gates. Cell-centre conversion is inverted explicitly, including the last raster row and column. Immutable seeds retain the original controlled geometry and capture revision.

The independent catalogue geography audit covers all 266 cities, including Baghdad: zero wet platforms, zero missed checked dry line junctions and zero unapproved water runs above the 1 km planning limit. Full-resolution centreline exports preserve shoreline detours, and map markers use actual platform coordinates. Historical WorldCover samples and retained OSM geometry remain planning evidence; footprints, bank access, transfer levels and crossing structures require separate releases. Geography audit.

Each line has one full-fleet planning depot sized to its train count and consist length. Storage slots and maintenance bays are separate. Storage roads hold up to three sets; the last road can be shorter, while the land screen retains a rectangular envelope. Depot PV/storage equipment is included once in depot capital; land, utility and installation quotations remain open. Native station dispatch does not validate a full-depot launch.

Every listed station/platform record has two posts and two normal eight-hour shifts, with additional cover for the actual service window. Interchange platform nodes are counted separately. FTE cover deducts leave, training, sickness and handover from paid hours. Wages start at 150% of the retained country income proxy; technical and management roles have higher premiums, plus employer/overtime allowances. Finance and role totals reconcile.

Factories are sized to each city order and family. Facility readiness is 18 months; qualification and serial manufacture follow it. Integrated openings wait when an 18-month facility, qualification, production or the physical test-path limit is the critical path. Infrastructure deadlines remain distinct from integrated targets. The national brief counts shared factory capital once, using its module allowance or the larger physical city-order envelope. National sequencing and concurrent capacity remain uncommitted.

Every city retains actual source-bound native outputs for the two-hour trace, full nominal service day and eight full-day degraded cases. Aggregate software planning screens pass the unchanged thresholds. Per-line service, depot launch, physical acceptance and operating qualification remain separate release gates.

Connected span, station, construction, battery and delivery-evidence studies are also generated for Baghdad, Samawah and Amarah. These city-specific pilots use the declared hypothetical fleet and Iraqi supply assumptions. They establish no supplier commitment, engineering acceptance or adopted accelerated saving in country finance.

Country finance assumptions stay local. Baghdad's government share, Chinese credit, IQD bonds and indexed monthly programme are not copied into other cities. Generic finance remains a fixed-price steady-state screen, with fares and commercial income at its recorded country assumptions. Earlier operating studies remain historical diagnostics.

Full city quantities and costs · Source-bound audit · Catalogue index

Region/country code	Other programme cities	Largest city-order factory reference USD million
AF	5	696.017
AO	9	573.285
BD	10	927.370
BF	1	735.746
BO	1	827.233
CD	7	703.195
CM	10	1018.121
EC	1	687.987
EG	19	769.053
GN	1	542.348
ID	2	347.020
IN	17	865.454
IQ	17	939.223
JO	4	723.231
KE	15	749.987
KH	1	351.691
LA	1	770.563

Region/country code	Other programme cities	Largest city-order factory reference USD million
LB	3	811.056
LK	4	759.694
MA	12	912.342
MG	1	354.399
ML	1	280.817
MM	2	579.763
MZ	10	917.104
NE	1	875.628
NG	10	877.649
NP	3	836.662
PH	1	369.888
PK	13	1018.121
PS	3	759.694
RW	3	794.580
SA	14	863.087
SD	6	507.247
SN	1	336.131
SO	1	386.365
SS	1	190.977
SV	1	484.389
SY	9	667.470
TN	1	457.239
TZ	15	910.801
UG	12	749.987
YE	9	733.269
ZA	5	863.087
ZM	1	516.904

- Core rectangles are design-centroid planning screens, not surveyed city-centre boundaries or property rights.
- Unaccepted land, foundations, suppliers, installation, commissioning and independent-check gates remain open.
- Country income proxies are retained estimates; graded wages are planning premiums, not observed payroll quotations.
- Factory capital is a shared national allowance, not city capital; independent city-order envelopes do not demonstrate simultaneous national capacity.
- Generic finance retains country terms and fixed-price steady-state sensitivities; Baghdad indexed monthly cashflows remain separate.
- Native service screens retain station dispatch/staging; full-depot yard access, launch capacity and the transition to depot dispatch require separate validation.
- Historical station-stabling studies retain their original geometry and source fingerprints; they cannot release these new layouts.

Assurance, conformity and authorization

Repository source: `docs/certification/README.md`

OpenSourceRail does not treat “certification” as one certificate or one final test. A railway is accepted through several different decisions: the design is verified, real products are qualified, installed subsystems are validated, the system safety argument is independently assessed, and legally competent organizations authorize the stated use.

> Important
> The repository has no power to certify, approve or authorize a railway. It can make evidence complete, reproducible, traceable and difficult to misstate. Suppliers, competent engineers, independent assessors, operators, infrastructure managers and national/local authorities retain their own decisions and liability.

The short answer

OpenSourceRail now uses one simplified model for every controlled item:

one passport per item five gates route-specific evidence subsystem roll-up one deployment safety/authorization decision

The companion Digital Standards Thread makes that lifecycle executable: it reviews publisher metadata, freezes open applicability inputs, maps standards to cross-domain controls and evidence, hashes the exact sources and produces a deterministic change-impact index. It is a primary product capability and a submission accelerator—not an automated certificate.

The generated all-component assurance register publishes the current item count directly from controlled sources and covers:

- train products and assemblies;
- station products and assemblies;
- reusable civil types;
- every Rust workspace crate; and
- configured owner/operator platform services covering Workbench, ERP/HR, supervision, FUXA, Ops Core, the AI council and lifecycle/QR identity.

Every listed item has a controlled identity and source baseline at G0. Item-specific G1–G4 evidence is deliberately open, and zero items are represented as certified or released.

The connected engineering example now links the physical hierarchy to battery-cooling functions, interfaces, specific failure modes, configuration-bound scenarios, production and installed occurrences. It exposes missing physical evidence and child/dependency/integration blockers while preserving separately controlled deployment decisions.

The subsystem qualification workflow adds quantitative RAMS screens, a controlled cooling-rig programme, measurement/model correlation, production-equivalence findings and Workbench decision readiness. Physical qualification remains open until real measurements and accountable decisions support the frozen configuration.

The five gates

```
flowchart LR
  G0[G0 Define<br/>identity · use · boundary] --> G1[G1 Assure design<br/>requirements · hazards · plan]
  G1 --> G2[G2 Qualify implementation<br/>real software · product · workmanship]
  G2 --> G3[G3 Validate integration<br/>HIL · FAT/SAT · site · commissioning]
  G3 --> G4[G4 Accept and authorize<br/>independent review · duty holder · authority]
  G4 --> O[Operate within conditions]
  O --> C[change · incident · expiry · field evidence]
  C --> G0
```

Gate	The decision	Typical evidence	Who decides
G0 — Define	Is the reference item, intended function, boundary, owner and configuration known?	Stable ID, parent/interface, source revision/hash, reference-use statement	Configuration manager
G1 — Assure design	Are deployment use, applicable rules, classification, requirements, hazards, interfaces and V&V plan complete?	Applicability record, requirements, ICDs, item FMEA/FMEDA, calculations/test plan	Design authority plus independent reviewer
G2 — Qualify implementation	Does the actual software, manufactured item or construction product meet the frozen design?	Released artifact/drawings, supplier/material records, raw analysis/tests, calibration, NCR closure	Quality and technical authorities with required independence
G3 — Validate integration	Does the installed item work with the real vehicle, route, environment, people and procedures?	HIL/FAT/SAT/site tests, as-built/deployed configuration, degraded modes, commissioning, maintenance information	Integrator, operator and project safety authority

Gate	The decision	Typical evidence	Who decides
G4 — Accept and authorize	Have independent and legally competent parties accepted the exact evidence and conditions of use?	Independent assessment, closed actions, accepted residual risk, operator acceptance, permits/authorization	Assessor, duty holder and competent authority

A later gate cannot hide an earlier gap. A pass belongs only to the exact configuration, use, environment, evidence revision and conditions recorded in the passport. A significant change, incident, expiry or adverse field result reopens the affected gate and every dependent gate.

Five assurance routes—not five separate systems

All routes use the same gates and passport fields. Their evidence differs because a bridge, brake controller and finance application are not certified in the same way.

Route	Items	Main qualification question	Final acceptance route
Rolling stock and onboard equipment	150	Are supplier configuration, materials/processes, structure, fire, electrical, EMC, environmental, endurance and first-article evidence adequate?	Applicable product/type and vehicle route
Stations, charging and wayside	55	Do the site interfaces, FAT, installation, SAT, accessibility, emergency modes and handover satisfy project and local rules?	Project, utility, building/fire and railway acceptance as applicable
Civil infrastructure	19	Do surveyed ground/water conditions, calculations, temporary works, materials, inspection and as-built geometry support this site?	Competent-engineer, project and construction/rail authority acceptance
Software and control	59	Is safety classification explicit, and are lifecycle, traceability, proof/test, target timing/resources, HIL and cyber evidence complete?	Exact target/system safety acceptance within a frozen authority boundary
Business, supervision and administration	7	Are non-safety boundaries, access control, integrity, recovery, support and operator acceptance proven?	Organizational production acceptance—never movement authority or safety release

Candidate standards in a passport are prompts for applicability review, not conformity claims. G1 must freeze the jurisdiction, intended use, safety classification, standard editions/national adoptions, contractual rules and assessment bodies for the deployment.

“Certified” is too vague—name the actual decision

Decision	What it means here	What it does not mean
Digital design gate	Tracked inputs and deterministic evidence are internally coherent	Physical qualification or approval
Product conformity/type evidence	A defined manufactured configuration met its applicable assessment route	Every installation or route is accepted
Construction/installation acceptance	The site-specific work and as-built state meet project/local requirements	The complete railway may operate
Independent safety assessment	An independent body reviewed the safety process, evidence and findings	The assessor becomes the operator or regulator
Placing-in-service/operating authorization	The competent party permits a stated use subject to conditions	Unlimited approval after later change
Organization/SMS or management-system acceptance	The duty holder can manage risk, competence, change and operation	A product is technically qualified merely because the organization is certified

This distinction follows the current RAMS lifecycle principle that assurance applies from complete systems down to subsystems and components, while the RAMS standard itself does not define product certification or stakeholder approval. See the official IEC 62278-1:2025 record. EU terminology also separates constituent/subsystem conformity, vehicle or fixed-installation authorization and safety authorization; the EU declaration/certificate templates are an example, not a default legal route for non-EU deployments.

How to close a component passport

1. Name the claim. State the exact item, revision, intended use, environment, parent system and requested decision.
2. Freeze applicability. Record jurisdiction, standards/rules, safety significance, assessment route, duty holders and required independence.
3. Plan evidence before building. Link requirements, hazards/FMEA, interfaces, analyses, tests, acceptance criteria, tools, specimens and responsible people.
4. Capture raw results. Retain supplier submissions, calculations, model/input versions, calibration, test data, failures, NCRs, concessions and hashes—not only a summary PDF.

5 Check configuration and independence. Evidence must match the item; authors cannot silently act as their own independent verifier or approver.

6 Roll up without averaging. A subsystem cannot pass while a safety-significant child, interface, common-cause risk or operating dependency is open or stale.

7 Record the decision. Name the decision maker, scope, basis, date, expiry, limitations, residual risk and conditions of use.

8 Control change and service evidence. Preserve superseded records and reopen affected gates after modifications, incidents, expiry or adverse trends.

Worked examples

Battery/HVAC module

- G1: freeze thermal duty, fire hazards, coolant/refrigerant separation, electrical interfaces and applicable battery/rolling-stock rules.
- G2: qualify the selected compressor, heat exchangers, pumps, hoses, pack interfaces and controller using supplier data, thermal chamber/calorimeter, leakage, EMC, shock/vibration and fault tests.
- G3: validate the installed vehicle at maximum climate, aged-battery and degraded-cooling duties, including protection/derate behavior.
- G4: independent review and vehicle/type acceptance retain conditions and limits. Passing the Rust thermal model alone never closes G2–G4.

Safety-related Rust evaluator

- G1: classify its safety role, allocate requirements/hazards and freeze its interface and target assumptions.
- G2: retain reviewed source/toolchain, unit/property/formal results, coverage, vulnerabilities, reproducible binary, WCET/stack/heap and target-hardware evidence.
- G3: run HIL, communications/clock/power faults, vehicle/wayside integration and representative commissioning cases.
- G4: obtain independent safety acceptance for the exact binary, hardware, configuration and authority boundary. A crate test cannot authorize a train.

Bridge or viaduct type

- G1: replace open-map screening with surveyed alignment, levels, hydraulics, ground, loads, durability and site-code applicability.
- G2: release checked calculations/drawings, materials, fabrication/placement procedures, temporary works and inspection/test plans.
- G3: validate foundations, concealed work, geometry, bearings/joints, drainage and as-built/load evidence on the actual site.
- G4: competent engineering and construction/railway authorities accept the structure for its stated use.

ERP, FUXA or AI administration

- G1: freeze its non-safety authority boundary, data ownership, permissions, retention, privacy, cyber and business-continuity requirements.
- G2: verify configuration, authentication/authorization, audit, backup, recovery, dependencies and fail-closed behavior.
- G3: validate real organizational roles, operating procedures, outage/recovery and human approval paths.
- G4: the organization may accept it for production administration. It still cannot gain movement authority, certify engineering, pay, contract, hire/fire or approve itself.

One evidence chain

```
component passport
  → subsystem assurance case
    → train / station / civil / software validation
      → integrated railway safety case
        → independent assessment
          → operator, project and competent-authority decisions
```

Hashes show which bytes were reviewed; they do not show that the content is correct. Simulation reduces redesign and physical-test risk; it does not replace mandatory type, routine, environmental, site, commissioning or operational evidence.

Current position

Layer	Repository position	Release meaning
Component identities	Every generated passport has a G0 baseline	Traceable, not approved
Cross-domain digital checks	Current counts and hashes are generated in the digital report	Evidence coherence only; conformity not assessed
Item design assurance	G1 item plans/reviews remain open	No item-specific design release claimed
Implementation qualification	G2 gates remain open	No supplier/physical/target qualification claimed
Integration validation	G3 gates remain open	No HIL/site/commissioning release claimed
Independent acceptance/authorization	G4 gates remain open	No certification or permission to operate claimed

The detailed closure work is in the release-gap register. The concise status view is evidence-status.md.

The assurance pack

Read only what your decision needs:

1. All items: component assurance register and its complete JSON passports.
2. System and boundary: system description and standards baseline.
3. Safety intent: safety requirements, hazard log and compiled GSN safety case.
4. Evidence: evidence register, generated digital assurance, software resilience and multi-day soak.
5. Open work: evidence status, release gaps and component RFC readiness.
6. Control profiles: pilot signalling and the research-only distributed onboard profile.
7. Clause mapping: EN/IEC 62267 compliance matrix, after the deployment confirms the applicable adoption and edition.

Deterministic checks

```
python3 tools/automation/component_assurance.py --check
python3 tools/automation/digital-assurance.py --check
python3 engineering/subsystem_control_register.py --check
./osr test
```

These commands detect missing identities, routes, sources, hashes and stale generated evidence. They cannot authenticate a signatory, judge engineering adequacy, witness a physical test or grant approval.

Deployment dossier handover

A deployment should export a controlled dossier containing the selected passports, frozen applicability/standards record, configuration index, requirements and hazard trace, supplier and construction files, raw verification/validation evidence, NCR/deviation log, competence and calibration records, safety case, independent assessment report, decisions/conditions and change history. The national or contractual submission format wraps this evidence; it must not create a second uncontrolled technical baseline.

The worked Samawah package is an example deployment evidence structure, not a special certification route and not an Iraqi approval claim.

EN 62267 compliance matrix

Repository source: `docs/certification/compliance-matrix.md`

Clause-by-clause traceability from EN 62267 (Railway applications — Automated urban guided transport — Safety requirements) to OSR implementation. Citations are to the 2009 edition.

Format per clause:

- Clause — reference.
- EN 62267 requirement summary — single-sentence paraphrase.
- OSR implementation — which crate(s) / RFC(s) / rulebook section fulfils it.
- Evidence — specific artefact to cite in the submission.

§5 — Safety principles

§5.1 — Inherent and functional safety

Requirement: system shall implement both inherent safety (design choices that make hazards impossible by construction) and functional safety (safety functions that detect + mitigate hazards that cannot be designed out).

OSR implementation:

- Inherent safety: no tunnels (RFC 0011), prefab-only civil (RFC 0011), single consist family per line (RFC 0008), fixed-geometry preset per line (RFC 0009). Each removes an entire hazard class by eliminating its precondition.
- Functional safety: SIL-4 chain per safety-requirements.md SR-01..SR-24.

Evidence: RFCs 0008–0011; safety-requirements.md.

§5.2 — Responsibility assignment

Requirement: a responsibility matrix shall document which entity owns each safety-relevant function.

OSR implementation: RFC 0013 §4 (one shared rulebook across 4 role families); `osr-safety-case` GSN tree links every claim to an owning evaluator.

Evidence: RFC 0013; GSN tree under `docs/safety-case/gsn/`.

§6 — Functional requirements (SIL-4 core)

§6.1 — Ensuring safe movement

Requirement: automated system shall ensure the train moves only when it is safe to do so.

OSR implementation: `osr-interlocking::section_available_to` with four gates (occupancy, route, override, intrusion); `osr-atp::atp_evaluate` overlays the envelope check.

Evidence: SR-01, SR-02, SR-03, SR-07; Kani harness modules on `osr-interlocking`, `osr-atp`.

§6.2 — Preventing collision

Requirement: system shall prevent collisions between trains, with infrastructure, and with obstacles.

OSR implementation: train-train via SR-01; track-worker via SR-01 + `MaintenanceOverride`; obstacle via SR-04 (onboard) + SR-07 (wayside) — two-layer defence per RFC 0015 + RFC 0016.

Evidence: H-CO-01..04 hazard log; RFC 0015 §6 safety case.

§6.3 — Preventing derailment

Requirement: system shall prevent derailment from overspeed, mis-set switch, or mechanical cause.

OSR implementation: overspeed via ATP (SR-03); switch via `osr-wayside-points` 2oo2 sensor fusion; mechanical via SR-11 onboard derailment monitor; track maintenance cadence RFC 0013 M3 + M4.

Evidence: SR-03, SR-11; hazard H-DE-01, H-DE-02.

§6.4 — Preventing falls

Requirement: system shall prevent falls from platforms.

OSR implementation: SR-24 PSDs at every boarding platform; SR-10 platform-alignment gate; station-staff T2 supervision at non-PSD depot-terminals.

Evidence: RFC 0010 §6; RFC 0015 §5.4.

§6.5 — Protecting people on track

Requirement: system shall protect people on track — workers and intruders.

OSR implementation: workers via `MaintenanceOverride` + RFC 0013 S5 + M2 protection signals; intruders via SR-07 wayside intrusion gate + physical fence (civil scope).

Evidence: SR-07; H-CO-04, H-SA-01.

§6.6 — Closing doors

Requirement: system shall ensure doors close safely.

OSR implementation: SR-08, SR-09 on `osr-door-control`; RFC 0013 T2.5 platform-clear confirmation at non-PSD stations.

Evidence: SR-08, SR-09; hazard H-PB-01, H-PB-03.

§6.7 — Starting the train

Requirement: the system shall start the train only when safe (all doors closed, MA valid, all monitors nominal).

OSR implementation: `osr-atp`'s start-inhibit composes: doors_closed_all && ma_valid && !any_monitor_trip; RFC 0013 documents the automated ATO and OCC-visible behaviour.

Evidence: SR-15, SR-19; `osr-ato` start-gate unit tests.

§6.8 — Stopping the train

Requirement: the system shall stop the train safely on any EB source.

OSR implementation: `osr-brake` emergency-source union (ATP + fire + derailment + authenticated remote assistance + obstacle); SR-15 — EB dominates.

Evidence: `osr-brake` Kani B2 + B3 harness; proptest `b2_emergency_union`.

§6.9 — Preventing hostile environments

Requirement: system shall handle weather, dust, heat per deployment climate.

OSR implementation: RFC 0013 S4 weather rulebook; `climate-adapters.toml` per-deployment tuning; `osr-energy-site` handles PV dust derating.

Evidence: RFC 0013 §6 weather rules; climate-adapter file.

§6.10 — Detecting smoke + fire

Requirement: onboard fire detection + suppression.

OSR implementation: `osr-fire-safety` (SIL-4); RFC 0008 bays each carry BaySensors; suppression by Novec 1230 or equiv.

Evidence: SR-12; H-FI-01, H-FI-02.

§6.11 — Operating in degraded mode

Requirement: system shall safely continue at reduced performance under named degraded modes.

OSR implementation: RFC 0013 §5 M1 (manual on MA) / M2 (restricted service) / M3 (evacuation) as the only three degraded modes.

Evidence: RFC 0013 §5; dispatcher rules S7 + S5.

§6.12 — Supervision and rescue of passengers

Requirement: system shall support passenger supervision + rescue without a driver.

OSR implementation: SR-14 passenger emergency intercom; SR-18 remote-assist channel; RFC 0015 §5.3 + §5.5 cabin CCTV live to OCC.

Evidence: SR-14, SR-18; RFC 0015 §5.

§7 — System performance

§7.1 — Tolerable hazard rate (THR)

Requirement: the safety integrity of each safety function shall correspond to a tolerable hazard rate.

OSR implementation: SIL-4 functions target THR ≤ 10^{-9} /h, consistent with EN 50129 (SIL-4 allocated to the safety functions per the hazard log).

Evidence: hazard-log.md severity × frequency table.

§7.2 — Systematic failures

Requirement: systematic failures (specification, design, implementation bugs) shall be controlled via the EN 50716 process.

OSR implementation: Rust `#![forbid(unsafe_code)]`; integer-only safety path; Kani bounded-model proofs; proptest randomised verification; differential twin (Rust vs Python); RFC-driven change control.

Evidence: Evidence register §1–§5; CI gates on `cargo test` + `starter_case_closes`.

§7.3 — Random hardware failures

Requirement: random hardware failures shall be detected and handled fail-restrictively.

OSR implementation: 2oo2 at T-ECU/S + T-OBS (SR-20); watchdog + supervisor (SR-21); ATECC608B trust anchor; per-rail power supervision; redundant 24 V input.

Evidence: `control-electronics/t-ecu-s/schematics/v2-spec/safety-nets.md`; `control-electronics/t-obs/schematics/v2-spec/safety-nets.md`.

§8 — Safety management

§8.1 — Safety lifecycle

Requirement: EN 50126 lifecycle from concept to decommissioning.

OSR implementation: RFC-per-major-decision; GSN claim tree under `docs/safety-case/`; operations rulebook under `docs/operations/`; `osr-safety-case` CI gate. Deployment partner handles site-specific lifecycle stages (commissioning, operations, decommissioning).

Evidence: RFC 0001–0017; this certification pack as the "safety case" deliverable.

§8.2 — Independent safety assessment

Requirement: an independent assessor reviews the safety case.

OSR implementation: this pack is prepared for external assessment. The deployment partner engages the assessor (RFC 0003 §5).

Evidence: N/A (upstream prerequisite for the assessor, not produced by this pack).

§8.3 — Competency of staff

Requirement: operating + maintenance staff competent for their role.

OSR implementation: RFC 0013 C1 role qualification (§C1.5); RFC 0013 §8 training + recurrent qualification program.

Evidence: RFC 0013 §8; deployment-specific training record.

§9 — Cybersecurity (complementary)

Although EN 62267 itself does not prescribe cybersecurity requirements (it pre-dates the CENELEC EN 50701 split), a modern GoA 4 submission is expected to carry IEC 62443-4-2 + EN 50701 evidence as a complementary cybersecurity dossier.

§9.1 — Message authenticity

OSR implementation: SR-22 `osr-secbus` (RFC 0017 v1 library; v2 wiring open).

Evidence: GSN G25–G27; `osr-secbus` Kani + proptest.

§9.2 — Component-level protection

OSR implementation: ATECC608B trust anchors on every SIL-4 host class; zero-on-drop key wrappers (`osr-crypto`); per-deployment key registry (`osr-secbus`).

Evidence: `control-electronics/t-ecu-s/schematics/v2-spec/`; `osr-crypto` + `osr-secbus` source.

Summary

- §5.1–§9.2: 19 clauses mapped.
- 16 clauses have direct implementation + evidence in tree.
- 3 clauses (§7.1 residual risk, §8.2 independent assessment, §8.3 staff competency) depend on deployment- partner input — deliberately scoped that way per README.md.
- 1 clause (§9.1 message authenticity) has library in tree but v2 wire integration open.

This matrix maps the clauses to implementation intent and current evidence. It does not close the full type-approval case by itself: message-auth wire integration, hardware release evidence, rolling-stock structural evidence, door-system certification, field sensor evidence, station-energy studies, assessor review, and deployment-specific staff competence remain tracked in `release-gap-register.md`.

All-component assurance register

Repository source: `docs/certification/component-assurance-register.md`

> Generated by `python3 tools/automation/component_assurance.py`. Do not edit this file or its JSON partner by hand.

This model records what evidence exists, what remains open, who must decide, and which use is permitted. It does not certify a component, appoint a legal duty holder, approve a design, release manufacture or construction, authorize operation, or replace an independent assessor or competent authority.

The register contains 296 assurance passports: 285 engineering inventory items, 4 connected cooling design definitions and 7 owner/operator platform services. Every item has a locked identity/source baseline at G0. All item-specific design, qualification, integration and independent-acceptance gates remain open; zero items are represented as certified or released.

One lifecycle

Gate	Decision question	Current repository state	Decision owner
G0 — Define	Is the item, reference function, boundary, owner and exact source configuration known?	`pass-baseline-recorded`	configuration manager
G1 — Assure the design	Are applicable requirements, hazards, standards, interfaces and verification plans complete for this item?	`open-item-assurance-plan`	design authority with independent reviewer
G2 — Qualify the implementation	Does the actual software, manufactured item or construction product meet the frozen design?	`open-implementation-qualification`	quality and technical authorities with required independence
G3 — Validate the integration	Does the installed item work safely with its real interfaces, environment, route and operating procedures?	`open-integration-validation`	system integrator, operator and project safety authority
G4 — Accept and authorize	Have independent reviewers and the legally competent organizations accepted the evidence for the stated use?	`open-independent-acceptance`	named independent assessor, duty holder and competent authority

A pass applies only to the named item, configuration, intended use and evidence revision. A changed identity, use, requirement, implementation, interface, environment or acceptance condition reopens the affected gate and every dependent later gate.

Assurance routes

Route	Items	Scope	Qualification emphasis	Final decision
`rolling-stock`	150	Rolling stock and onboard equipment	supplier configuration, material/process and dimensional evidence, structural, fire, electrical, EMC, environmental and endurance tests, first article and vehicle integration	product/type and vehicle acceptance route selected by the deployment jurisdiction
`station-wayside`	60	Stations, charging, wayside and passenger systems	site and utility interfaces, supplier FAT, installation inspection, SAT and integrated emergency/degraded-mode tests, accessibility and handover	project, utility, building/fire and railway acceptance as applicable
`civil-infrastructure`	19	Civil infrastructure and reusable structural types	survey and ground model, site-specific calculations, materials and temporary works, inspection/test plan, as-built survey and structural handover	competent engineer, project authority and national/local construction or railway authority
`software-control`	60	Software, embedded evaluation and control	explicit safety classification, requirements/hazard traceability, locked source/toolchain, review, test and proof results, target timing/resources, HIL, cybersecurity and deployment evidence	software/system safety acceptance for the exact target and intended authority boundary
`business-supervision`	7	Business, supervision and administrative platforms	non-safety authority boundary, identity and access control, data integrity and retention, backup/recovery, supplier/software support, operator acceptance	organizational production acceptance; never movement authority or railway safety release

Standards in each passport are candidate applicability prompts, not declarations of conformity. The deployment must freeze jurisdiction, intended use, safety classification, editions, national adoptions, contractual rules and assessment bodies at G1.

Current gate position

Gate	Baselined/passed	Open	Meaning
G0 — Define	296	0	Identity and source are recorded.
G1 — Assure design	0	296	Item applicability, classification and assurance plan remain deployment work.
G2 — Qualify implementation	0	296	Supplier, build, target, test and physical evidence remain open.
G3 — Validate integration	0	296	HIL/FAT/SAT/site/commissioning evidence remains open.
G4 — Accept and authorize	0	296	Independent acceptance and legal/contractual authorization remain open.

What each passport contains

Each JSON passport carries a stable assurance ID, controlled item ID, scope/domain, route, source/hash reference, intended-use boundary, open safety classification, existing controls, five gate states and a fail-closed release state. Route definitions hold the candidate standards and evidence focus; gate definitions hold the decision owners. The JSON is the complete item-level register; this page intentionally stays short.

How evidence rolls up

1. Close a component gate only against its exact configuration and raw evidence.
2. Check every child, interface and common-cause dependency before closing the parent subsystem gate.
3. Validate the integrated train, station, corridor and operating organization in their real environment.
4. Keep product conformity, project acceptance, safety assessment and legal authorization as separate decisions.
5. Reopen affected gates after change, incident, expiry or adverse field evidence; retain the superseded decision and conditions of use.

The full deterministic register is `component-assurance-register.json`. The certification front door explains how to use it with the safety case, release gaps and deployment dossier.

Connected engineering assurance

Repository source: `docs/certification/connected-engineering-report.md`

> Generated by `tools/automation/connected_assurance.py`.

Battery-cooling vertical example. Asset, batch, production and incident identities prefixed DEMO are synthetic; no installed railway or physical test is claimed.

Relationship checks passed; reviewed physical evidence and controlled deployment decisions remain open.

Design fingerprint: `30a01593b4bc492341219fe29d27a6487032e4a9bead231147e8c48c544b913a`. Physical release: BLOCKED.

Failure propagation

Failure	Subject/revision	Local higher system effect	Priority
FM-COOL-CONNECTOR	OSR-COOL-CONNECTOR / A	coolant inventory lost pump starvation and loop heat-transfer loss battery overheating/fire contribution	mandatory-safety-review
FM-COOL-LOOP	OSR-COOL-LOOP / A	heat removal reduced battery temperature rises battery overheating/fire or traction loss	mandatory-safety-review
FM-COOL-PACK	LM3-TRC-P040 / sha256:165dde130eb423ac20dba527e827a81684782b9875c7810c464ac50db5567b02	cell temperature rises BMS requests isolation fire contribution or loss of traction	mandatory-safety-review
FM-COOL-PUMP	OSR-COOL-PUMP / A	coolant flow falls cooling loop loses circulation battery overheating and traction isolation	mandatory-safety-review
FM-COOL-RAILWAY	OSR-RAILWAY-BOUNDARY / A	railway capacity and passenger conditions degrade emergency response required unsafe passenger conditions or loss of separation	mandatory-safety-review
FM-COOL-SENSOR	OSR-COOL-FLOW-SENSOR / A	loss of circulation remains undetected latent exposure to cooling loss late isolation permits overheating/fire	mandatory-safety-review
FM-COOL-TORQUE	OSR-COOL-CONNECTOR / A	latent leakage defect introduced cooling loop loses coolant in service battery overheating/fire contribution	mandatory-safety-review
FM-COOL-VEHICLE	LM3-CAR-A900 / sha256:957a44510ffbd3ec61fd2ac6ca08dc7f805008edf35e453b647896c08cf1334f	vehicle traction derates or is lost train stops or cannot continue train stranded between stations	mandatory-safety-review

Verification scenarios

Scenario	Scope	Controller execution
SCN-COOL-CHARGING	physical-plan-open	not-executed
SCN-COOL-HOT	controller-executable	stale
SCN-COOL-LEAK	controller-executable	stale
SCN-COOL-MASKED	physical-plan-open	not-executed
SCN-COOL-PUMP	controller-executable	stale
SCN-COOL-PUMP-HOT	controller-executable	stale
SCN-COOL-RECOVERY	deployment-plan-open	not-executed

Unresolved evidence and decisions

Record	Blockers
CFG-DEMO-COOL-A	stale design revision: LM3-CAR-A900; stale design revision: LM3-TRAINSET-A000
DEC-LM3-CAR-A900-G1	EVD-COOL-BENCH: evidence planned; EVD-COOL-CONTROLLER: evidence planned; EVD-COOL-INTEGRATION: evidence planned; EVD-COOL-TORQUE: evidence planned; FM-COOL-VEHICLE: failure analysis and residual risk unreviewed; FM-COOL-VEHICLE: failure analysis binds an earlier design revision; FM-COOL-VEHICLE: mandatory safety review requires a controlled decision; FM-COOL-VEHICLE: occurrence evidence unknown; FM-COOL-VEHICLE: operating context verification open: charging, degraded-operation, evacuation, maintenance, normal-running, shutdown, startup; REQ-COOL-RECOVERY: acceptance criteria require independent engineering review; child/dependency decision open: LM3-AUX-P010/G1; child/dependency decision open: LM3-BOG-SA610/G1; child/dependency decision open: LM3-BOG-SA620/G1; child/dependency decision open: LM3-DOOR-SA310/G1; child/dependency decision open: LM3-FIX-SA340/G1; child/dependency decision open: LM3-HV-SA510/G1; child/dependency decision open: LM3-INT-SA330/G1; child/dependency decision open: LM3-LGT-SA350/G1; child/dependency decision open: LM3-ROOF-SA410/G1; child/dependency decision open: LM3-SHELL-A200/G1; controlled deployment acceptance is not authenticated; decision open; stale design revision: LM3-CAR-A900; stale design revision: LM3-TRAINSET-A000
DEC-LM3-CAR-A900-G2	EVD-COOL-BENCH: evidence planned; EVD-COOL-CONTROLLER: evidence planned; EVD-COOL-INTEGRATION: evidence planned; EVD-COOL-TORQUE: evidence planned; FM-COOL-VEHICLE: failure analysis and residual risk unreviewed; FM-COOL-VEHICLE: failure analysis binds an earlier design revision; FM-COOL-VEHICLE: mandatory safety review requires a controlled decision; FM-COOL-VEHICLE: occurrence evidence unknown; FM-COOL-VEHICLE: operating context verification open: charging, degraded-operation, evacuation, maintenance, normal-running, shutdown, startup; REQ-COOL-RECOVERY: acceptance criteria require independent engineering review; child/dependency decision open: LM3-AUX-P010/G2; child/dependency decision open: LM3-BOG-SA610/G2; child/dependency decision open: LM3-BOG-SA620/G2; child/dependency decision open: LM3-DOOR-SA310/G2; child/dependency decision open: LM3-FIX-SA340/G2; child/dependency decision open: LM3-HV-SA510/G2; child/dependency decision open: LM3-INT-SA330/G2; child/dependency decision open: LM3-LGT-SA350/G2; child/dependency decision open: LM3-ROOF-SA410/G2; child/dependency decision open: LM3-SHELL-A200/G2; controlled deployment acceptance is not authenticated; decision open; prior gate decision open: G1; stale design revision: LM3-CAR-A900; stale design revision: LM3-TRAINSET-A000
DEC-LM3-CAR-A900-G3	EVD-COOL-BENCH: evidence planned; EVD-COOL-CONTROLLER: evidence planned; EVD-COOL-INTEGRATION: evidence planned; EVD-COOL-TORQUE: evidence planned; FM-COOL-VEHICLE: failure analysis and residual risk unreviewed; FM-COOL-VEHICLE: failure analysis binds an earlier design revision; FM-COOL-VEHICLE: mandatory safety review requires a controlled decision; FM-COOL-VEHICLE: occurrence evidence unknown; FM-COOL-VEHICLE: operating context verification open: charging, degraded-operation, evacuation, maintenance, normal-running, shutdown, startup; REQ-COOL-RECOVERY: acceptance criteria require independent engineering review; child/dependency decision open: LM3-AUX-P010/G3; child/dependency decision open: LM3-BOG-SA610/G3; child/dependency decision open: LM3-BOG-SA620/G3; child/dependency decision open: LM3-DOOR-SA310/G3; child/dependency decision open: LM3-FIX-SA340/G3; child/dependency decision open: LM3-HV-SA510/G3; child/dependency decision open: LM3-INT-SA330/G3; child/dependency decision open: LM3-LGT-SA350/G3; child/dependency decision open: LM3-ROOF-SA410/G3; child/dependency decision open: LM3-SHELL-A200/G3; controlled deployment acceptance is not authenticated; decision open; prior gate decision open: G2; reviewed physical integration evidence missing; stale design revision: LM3-CAR-A900; stale design revision: LM3-TRAINSET-A000
DEC-LM3-CAR-A900-G4	EVD-COOL-BENCH: evidence planned; EVD-COOL-CONTROLLER: evidence planned; EVD-COOL-INTEGRATION: evidence planned; EVD-COOL-TORQUE: evidence planned; FM-COOL-VEHICLE: failure analysis and residual risk unreviewed; FM-COOL-VEHICLE: failure analysis binds an earlier design revision; FM-COOL-VEHICLE: mandatory safety review requires a controlled decision; FM-COOL-VEHICLE: occurrence evidence unknown; FM-COOL-VEHICLE: operating context verification open: charging, degraded-operation, evacuation, maintenance, normal-running, shutdown, startup; REQ-COOL-RECOVERY: acceptance criteria require independent engineering review; child/dependency decision open: LM3-AUX-P010/G4; child/dependency decision open: LM3-BOG-SA610/G4; child/dependency decision open: LM3-BOG-SA620/G4; child/dependency decision open: LM3-DOOR-SA310/G4; child/dependency decision open: LM3-FIX-SA340/G4; child/dependency decision open: LM3-HV-SA510/G4; child/dependency decision open: LM3-INT-SA330/G4; child/dependency decision open: LM3-LGT-SA350/G4; child/dependency decision open: LM3-ROOF-SA410/G4; child/dependency decision open: LM3-SHELL-A200/G4; controlled deployment acceptance is not authenticated; decision open; prior gate decision open: G3; reviewed physical integration evidence missing; stale design revision: LM3-CAR-A900; stale design revision: LM3-TRAINSET-A000
DEC-LM3-HV-SA510-G1	EVD-COOL-BENCH: evidence planned; EVD-COOL-CONTROLLER: evidence planned; EVD-COOL-INTEGRATION: evidence planned; EVD-COOL-TORQUE: evidence planned; child/dependency decision open: LM3-BDY-P050/G1; child/dependency decision open: LM3-HV-P010/G1; child/dependency decision open: LM3-HV-P020/G1; child/dependency decision open: LM3-HV-P030/G1; child/dependency decision open: LM3-SAF-P010/G1; child/dependency decision open: LM3-TRC-P030/G1; child/dependency decision open: LM3-TRC-P040/G1; child/dependency decision open: LM3-TRC-P060/G1; child/dependency decision open: LM3-TRC-P070/G1; controlled deployment acceptance is not authenticated; decision open; stale design revision: LM3-CAR-A900; stale design revision: LM3-TRAINSET-A000
Record	**Blockers**
DEC-LM3-HV-SA510-G2	EVD-COOL-BENCH: evidence planned; EVD-COOL-CONTROLLER: evidence planned; EVD-COOL-INTEGRATION: evidence planned; EVD-COOL-TORQUE: evidence planned; child/dependency decision open: LM3-BDY-P050/G2; child/dependency decision open: LM3-HV-P010/G2; child/dependency decision open: LM3-HV-P020/G2; child/dependency decision open: LM3-HV-P030/G2; child/dependency decision open: LM3-SAF-P010/G2; child/dependency decision open: LM3-TRC-P030/G2; child/dependency decision open: LM3-TRC-P040/G2; child/dependency decision open:

Record	Blockers
	LM3-TRC-P060/G2; child/dependency decision open: LM3-TRC-P070/G2; controlleddeployment acceptance is not authenticated; decision open; prior gate decision open:G1; stale design revision: LM3-CAR-A900; stale design revision: LM3-TRAINSET-A000
DEC-LM3-HV-SA510-G3	EVD-COOL-BENCH: evidence planned; EVD-COOL-CONTROLLER: evidence planned; EVD-COOL-INTEGRATION: evidence planned; EVD-COOL-TORQUE: evidence planned; child/dependency decision open: LM3-BDY-P050/G3; child/dependency decision open: LM3-HV-P010/G3; child/dependency decision open: LM3-HV-P020/G3; child/dependency decision open: LM3-HV-P030/G3; child/dependency decision open: LM3-SAF-P010/G3; child/dependency decision open: LM3-TRC-P030/G3; child/dependency decision open: LM3-TRC-P040/G3; child/dependency decision open: LM3-TRC-P060/G3; child/dependency decision open: LM3-TRC-P070/G3; controlled deployment acceptance is not authenticated; decision open; prior gate decision open: G2; reviewed physical integration evidence missing; stale design revision: LM3-CAR-A900; stale design revision: LM3-TRAINSET-A000
DEC-LM3-HV-SA510-G4	EVD-COOL-BENCH: evidence planned; EVD-COOL-CONTROLLER: evidence planned; EVD-COOL-INTEGRATION: evidence planned; EVD-COOL-TORQUE: evidence planned; child/dependency decision open: LM3-BDY-P050/G4; child/dependency decision open: LM3-HV-P010/G4; child/dependency decision open: LM3-HV-P020/G4; child/dependency decision open: LM3-HV-P030/G4; child/dependency decision open: LM3-SAF-P010/G4; child/dependency decision open: LM3-TRC-P030/G4; child/dependency decision open: LM3-TRC-P040/G4; child/dependency decision open: LM3-TRC-P060/G4; child/dependency decision open: LM3-TRC-P070/G4; controlled deployment acceptance is not authenticated; decision open; prior gate decision open: G3; reviewed physical integration evidence missing; stale design revision: LM3-CAR-A900; stale design revision: LM3-TRAINSET-A000
DEC-LM3-TRAINSET-A000-G1	EVD-COOL-BENCH: evidence planned; EVD-COOL-CONTROLLER: evidence planned; EVD-COOL-INTEGRATION: evidence planned; EVD-COOL-TORQUE: evidence planned; REQ-COOL-RECOVERY: acceptance criteria require independent engineering review; child/dependency decision open: LM3-ART-SA800/G1; child/dependency decision open: LM3-CAR-A900/G1; child/dependency decision open: LM3-EIF-SA650/G1; child/dependency decision open: LM3-END-SA700/G1; child/dependency decision open: LM3-SYS-SA900/G1; controlled deployment acceptance is not authenticated; decision open; stale design revision: LM3-CAR-A900; stale design revision: LM3-TRAINSET-A000
DEC-LM3-TRAINSET-A000-G2	EVD-COOL-BENCH: evidence planned; EVD-COOL-CONTROLLER: evidence planned; EVD-COOL-INTEGRATION: evidence planned; EVD-COOL-TORQUE: evidence planned; REQ-COOL-RECOVERY: acceptance criteria require independent engineering review; child/dependency decision open: LM3-ART-SA800/G2; child/dependency decision open: LM3-CAR-A900/G2; child/dependency decision open: LM3-EIF-SA650/G2; child/dependency decision open: LM3-END-SA700/G2; child/dependency decision open: LM3-SYS-SA900/G2; controlled deployment acceptance is not authenticated; decision open; prior gate decision open: G1; stale design revision: LM3-CAR-A900; stale design revision: LM3-TRAINSET-A000
DEC-LM3-TRAINSET-A000-G3	EVD-COOL-BENCH: evidence planned; EVD-COOL-CONTROLLER: evidence planned; EVD-COOL-INTEGRATION: evidence planned; EVD-COOL-TORQUE: evidence planned; REQ-COOL-RECOVERY: acceptance criteria require independent engineering review; child/dependency decision open: LM3-ART-SA800/G3; child/dependency decision open: LM3-CAR-A900/G3; child/dependency decision open: LM3-EIF-SA650/G3; child/dependency decision open: LM3-END-SA700/G3; child/dependency decision open: LM3-SYS-SA900/G3; controlled deployment acceptance is not authenticated; decision open; prior gate decision open: G2; reviewed physical integration evidence missing; stale design revision: LM3-CAR-A900; stale design revision: LM3-TRAINSET-A000
DEC-LM3-TRAINSET-A000-G4	EVD-COOL-BENCH: evidence planned; EVD-COOL-CONTROLLER: evidence planned; EVD-COOL-INTEGRATION: evidence planned; EVD-COOL-TORQUE: evidence planned; REQ-COOL-RECOVERY: acceptance criteria require independent engineering review; child/dependency decision open: LM3-ART-SA800/G4; child/dependency decision open: LM3-CAR-A900/G4; child/dependency decision open: LM3-EIF-SA650/G4; child/dependency decision open: LM3-END-SA700/G4; child/dependency decision open: LM3-SYS-SA900/G4; controlled deployment acceptance is not authenticated; decision open; prior gate decision open: G3; reviewed physical integration evidence missing; stale design revision: LM3-CAR-A900; stale design revision: LM3-TRAINSET-A000
DEC-OSR-COOL-LOOP-G1	EVD-COOL-BENCH: evidence planned; EVD-COOL-CONTROLLER: evidence planned; EVD-COOL-INTEGRATION: evidence planned; EVD-COOL-TORQUE: evidence planned; FM-COOL-LOOP: failure analysis and residual risk unreviewed; FM-COOL-LOOP: mandatory safety review requires a controlled decision; FM-COOL-LOOP: occurrence evidence unknown; FM-COOL-LOOP: operating context verification open: degraded-operation, evacuation, maintenance, recovery, shutdown, startup; REQ-COOL-PHYSICAL: acceptance criteria require independent engineering review; child/dependency decision open: LM3-AUX-P010/G1; child/dependency decision open: OSR-COOL-CONNECTOR/G1; child/dependency decision open: OSR-COOL-FLOW-SENSOR/G1; child/dependency decision open: OSR-COOL-PUMP/G1; controlled deployment acceptance is not authenticated; decision open; stale design revision: LM3-CAR-A900; stale design revision: LM3-TRAINSET-A000
DEC-OSR-COOL-LOOP-G2	EVD-COOL-BENCH: evidence planned; EVD-COOL-CONTROLLER: evidence planned; EVD-COOL-INTEGRATION: evidence planned; EVD-COOL-TORQUE: evidence planned; FM-COOL-LOOP: failure analysis and residual risk unreviewed; FM-COOL-LOOP: mandatory safety review requires a controlled decision; FM-COOL-LOOP: occurrence evidence unknown; FM-COOL-LOOP: operating context verification open: degraded-operation, evacuation, maintenance, recovery, shutdown, startup; REQ-COOL-PHYSICAL: acceptance criteria require independent engineering review; child/dependency decision open: LM3-AUX-P010/G2; child/dependency decision open: OSR-COOL-CONNECTOR/G2; child/dependency decision open: OSR-COOL-FLOW-SENSOR/G2; child/dependency decision open: OSR-COOL-PUMP/G2; controlled deployment acceptance is not authenticated; decision open; prior gate decision open: G1; stale design revision: LM3-CAR-A900; stale design revision: LM3-TRAINSET-A000

Record	Blockers
DEC-OSR-COOL-LOOP-G3	EVD-COOL-BENCH: evidence planned; EVD-COOL-CONTROLLER: evidence planned; EVD-COOL-INTEGRATION: evidence planned; EVD-COOL-TORQUE: evidence planned; FM-COOL-LOOP: failure analysis and residual risk unreviewed; FM-COOL-LOOP: mandatory safety review requires a controlled decision; FM-COOL-LOOP: occurrence

Record	Blockers
	evidence unknown; FM-COOL-LOOP: operating context verification open:degraded-operation, evacuation, maintenance, recovery, shutdown, startup;REQ-COOL-PHYSICAL: acceptance criteria require independent engineering review;child/dependency decision open: LM3-AUX-P010/G3; child/dependency decision open:OSR-COOL-CONNECTOR/G3; child/dependency decision open:OSR-COOL-FLOW-SENSOR/G3; child/dependency decision open: OSR-COOL-PUMP/G3;controlled deployment acceptance is not authenticated; decision open; prior gatedecision open: G2; reviewed physical integration evidence missing; stale design revision:LM3-CAR-A900; stale design revision: LM3-TRAINSET-A000
DEC-OSR-COOL-LOOP-G4	EVD-COOL-BENCH: evidence planned; EVD-COOL-CONTROLLER: evidence planned; EVD-COOL-INTEGRATION: evidence planned; EVD-COOL-TORQUE: evidence planned; FM-COOL-LOOP: failure analysis and residual risk unreviewed; FM-COOL-LOOP: mandatory safety review requires a controlled decision; FM-COOL-LOOP: occurrence evidence unknown; FM-COOL-LOOP: operating context verification open: degraded-operation, evacuation, maintenance, recovery, shutdown, startup; REQ-COOL-PHYSICAL: acceptance criteria require independent engineering review; child/dependency decision open: LM3-AUX-P010/G4; child/dependency decision open: OSR-COOL-CONNECTOR/G4; child/dependency decision open: OSR-COOL-FLOW-SENSOR/G4; child/dependency decision open: OSR-COOL-PUMP/G4; controlled deployment acceptance is not authenticated; decision open; prior gate decision open: G3; reviewed physical integration evidence missing; stale design revision: LM3-CAR-A900; stale design revision: LM3-TRAINSET-A000
DEC-OSR-COOL-PUMP-G1	EVD-COOL-BENCH: evidence planned; EVD-COOL-CONTROLLER: evidence planned; EVD-COOL-INTEGRATION: evidence planned; EVD-COOL-TORQUE: evidence planned; FM-COOL-PUMP: failure analysis and residual risk unreviewed; FM-COOL-PUMP: mandatory safety review requires a controlled decision; FM-COOL-PUMP: occurrence evidence unknown; FM-COOL-PUMP: operating context verification open: charging, degraded-operation, evacuation, maintenance, recovery, shutdown, startup; REQ-COOL-PUMP: acceptance criteria require independent engineering review; child/dependency decision open: LM3-AUX-P010/G1; controlled deployment acceptance is not authenticated; decision open; stale design revision: LM3-CAR-A900; stale design revision: LM3-TRAINSET-A000
DEC-OSR-COOL-PUMP-G2	EVD-COOL-BENCH: evidence planned; EVD-COOL-CONTROLLER: evidence planned; EVD-COOL-INTEGRATION: evidence planned; EVD-COOL-TORQUE: evidence planned; FM-COOL-PUMP: failure analysis and residual risk unreviewed; FM-COOL-PUMP: mandatory safety review requires a controlled decision; FM-COOL-PUMP: occurrence evidence unknown; FM-COOL-PUMP: operating context verification open: charging, degraded-operation, evacuation, maintenance, recovery, shutdown, startup; REQ-COOL-PUMP: acceptance criteria require independent engineering review; child/dependency decision open: LM3-AUX-P010/G2; controlled deployment acceptance is not authenticated; decision open; prior gate decision open: G1; stale design revision: LM3-CAR-A900; stale design revision: LM3-TRAINSET-A000
DEC-OSR-COOL-PUMP-G3	EVD-COOL-BENCH: evidence planned; EVD-COOL-CONTROLLER: evidence planned; EVD-COOL-INTEGRATION: evidence planned; EVD-COOL-TORQUE: evidence planned; FM-COOL-PUMP: failure analysis and residual risk unreviewed; FM-COOL-PUMP: mandatory safety review requires a controlled decision; FM-COOL-PUMP: occurrence evidence unknown; FM-COOL-PUMP: operating context verification open: charging, degraded-operation, evacuation, maintenance, recovery, shutdown, startup; REQ-COOL-PUMP: acceptance criteria require independent engineering review; child/dependency decision open: LM3-AUX-P010/G3; controlled deployment acceptance is not authenticated; decision open; prior gate decision open: G2; reviewed physical integration evidence missing; stale design revision: LM3-CAR-A900; stale design revision: LM3-TRAINSET-A000
DEC-OSR-COOL-PUMP-G4	EVD-COOL-BENCH: evidence planned; EVD-COOL-CONTROLLER: evidence planned; EVD-COOL-INTEGRATION: evidence planned; EVD-COOL-TORQUE: evidence planned; FM-COOL-PUMP: failure analysis and residual risk unreviewed; FM-COOL-PUMP: mandatory safety review requires a controlled decision; FM-COOL-PUMP: occurrence evidence unknown; FM-COOL-PUMP: operating context verification open: charging, degraded-operation, evacuation, maintenance, recovery, shutdown, startup; REQ-COOL-PUMP: acceptance criteria require independent engineering review; child/dependency decision open: LM3-AUX-P010/G4; controlled deployment acceptance is not authenticated; decision open; prior gate decision open: G3; reviewed physical integration evidence missing; stale design revision: LM3-CAR-A900; stale design revision: LM3-TRAINSET-A000
DEMO-CAR-01	design changed; installed applicability requires reassessment
DEMO-CAR-02	design changed; installed applicability requires reassessment
DEMO-FRACAS-01	engineering handback and effectiveness review open
EVD-COOL-BENCH	evidence planned
EVD-COOL-CONTROLLER	evidence planned
EVD-COOL-INTEGRATION	evidence planned
EVD-COOL-TORQUE	evidence planned
FM-COOL-CONNECTOR	occurrence evidence unknown; failure analysis and residual risk unreviewed; mandatory safety review requires a controlled decision; mitigation lacks a verification scenario: REQ-COOL-TORQUE; operating context verification open: charging, degraded-operation, evacuation, maintenance, recovery, shutdown, startup
FM-COOL-LOOP	occurrence evidence unknown; failure analysis and residual risk unreviewed; mandatory safety review requires a controlled decision; operating context verification open: degraded-operation, evacuation, maintenance, recovery, shutdown, startup
FM-COOL-PACK	occurrence evidence unknown; failure analysis and residual risk unreviewed; mandatory safety review requires a controlled decision; operating context verification open: charging, degraded-operation, evacuation, maintenance, recovery, shutdown, startup
FM-COOL-PUMP	occurrence evidence unknown; failure analysis and residual risk unreviewed; mandatory safety review requires a controlled decision; operating context verification open: charging, degraded-operation, evacuation, maintenance, recovery, shutdown, startup

Record	Blockers
FM-COOL-RAILWAY	occurrence evidence unknown; failure analysis and residual risk unreviewed; mandatory safety review requires a controlled decision; operating context verification open: charging, degraded-operation, evacuation, maintenance, normal-running, shutdown, startup
FM-COOL-SENSOR	occurrence evidence unknown; failure analysis and residual risk unreviewed; mandatory safety review requires a controlled decision; operating context verification open: charging, degraded-operation, evacuation, maintenance, recovery, shutdown, startup
FM-COOL-TORQUE	occurrence evidence unknown; failure analysis and residual risk unreviewed; mandatory safety review requires a controlled decision; mitigation lacks a verification scenario: REQ-COOL-TORQUE; operating context verification open: charging, degraded-operation, evacuation, maintenance, recovery, shutdown, startup
FM-COOL-VEHICLE	failure analysis binds an earlier design revision; occurrence evidence unknown; failure analysis and residual risk unreviewed; mandatory safety review requires a controlled decision; operating context verification open: charging, degraded-operation, evacuation, maintenance, normal-running, shutdown, startup
OBL-COOL-APPLICABILITY	deployment applicability and clause selection open
REQ-COOL-HOT	acceptance criteria require independent engineering review
REQ-COOL-LEAK	acceptance criteria require independent engineering review
REQ-COOL-PHYSICAL	acceptance criteria require independent engineering review
REQ-COOL-PUMP	acceptance criteria require independent engineering review
REQ-COOL-RECOVERY	acceptance criteria require independent engineering review
REQ-COOL-TORQUE	acceptance criteria require independent engineering review

Connected engineering assurance

Repository source: `docs/certification/connected-engineering.md`

The connected engineering source extends the Digital Standards Thread with configuration-bound engineering relationships. The compiler imports the trainset product manifest, adds a battery-cooling decomposition, and emits a review report and machine-readable graph. Digital standards reports embed the same graph; component passports link their design revisions, failure modes, occurrences and deployment decisions.

This is the first vertical example. `DEMO` asset serials, batches, production records and incidents are synthetic. Supplier choices, physical results, licensed clause selection and accountable reviewers remain open.

Physical structure and configuration

The existing product manifest owns the assembly hierarchy. Imported revisions are SHA-256 identities of individual manifest definitions, the CAD baseline and referenced CAD source bytes; the full manifest hash is also retained. The overlay decomposes the cooling supplier kit into pump, connection, flow diagnostic and loop definitions, with their own IDs and revisions. It preserves the manifest's layer names and adds the railway operating boundary explicitly.

```
flowchart LR
  P[Pump / connector / flow diagnostic] --> L[Cooling loop]
  L --> K[Existing supplier cooling kit]
  K --> H[Existing HV assembly]
  H --> C[Car]
  C --> T[Trainset]
  T --> R[Railway operating boundary]
  A[Shared auxiliary supply] --> P
  A --> L
  P --> I[Flow / power / mounting interfaces]
  I --> B[Battery pack and adjacent equipment]
```

Design definitions and occurrences are separate records. Each occurrence has its installed design revision, configuration, position, serial, supplier batch, parent occurrence and maintenance history. Changing the design retains those installed revisions and reports reassessment gaps. It does not silently replace the installed equipment. Configuration snapshots enumerate exact applicable design revisions; stale analysis and mismatched occurrences remain visible.

Executable record schema

`osr-connected-engineering/1` uses globally unique IDs across these collections. The compiler validates reference types, required fields and cycle constraints.

Collection	Controlled information
`design_items`, `functions`, `interfaces`	Design revision, decomposition rationale, containment, function allocation, shared dependencies, interface endpoints and parameters
`configurations`, `occurrences`	Deployment, exact item revisions, initial state, installed serial/batch/position and maintenance history
`failure_modes`, `hazards`	Item/revision/configuration/function, operating context, mechanisms, local/higher/system effects, downstream modes, diagnostic coverage/delay/exposure, response/recovery, severity, likelihood basis, uncertainty, hazard, owner/action/deadline/reviewer and residual risk
`requirements`, `obligations`	Allocation, authored acceptance criteria, review, deployment applicability, clause identifier, verification class and decision links
`models`, `scenarios`	Hashed model sources/version, calibration/validation/envelope/limitations, exact configuration, initial inputs, fault location/onset/duration/magnitude, requirement-derived criteria and physical-validation relationship
`evidence`, `decisions`	Verification class, configuration, requirement/scenario scope, provenance, independent-review metadata, validity, child/dependency/prior gates and physical integration evidence
`production_records`, `incidents`	Affected designs and serials, torque operation, calibration and inspection hold point, failure diagnosis, corrective action, effectiveness check and engineering handback

Relationships include `part_of`, `implements_function`, `depends_on`, interface endpoints, `installed_as`, failure propagation, scenario/requirement/evidence links and dependent gate decisions. Containment and failure propagation cycles are rejected. Shared dependencies and interface traversal can contain cycles; impact traversal visits each record once.

Severity 5 always retains mandatory safety review. Unknown occurrence is stored as `unknown`; the legacy FMEA compiler emits a null RPN for it and an occurrence evidence gap for unsupported ratings. Multiplication cannot waive a safety review.

Run and inspect the example

```
# Execute four declared cases through the existing Rust thermal controller.
python3 tools/automation/connected_assurance.py --run-scenarios

# Generate reviewable records, then check drift.
python3 tools/automation/connected_assurance.py
python3 tools/automation/digital-assurance.py
python3 tools/automation/component_assurance.py
python3 tools/automation/connected_assurance.py --check

# Follow every recorded dependant of a safety-significant design.
python3 tools/automation/connected_assurance.py --changed-item OSR-COOL-PUMP

# Limit a defective-batch investigation to applicable occurrences and records.
python3 tools/automation/connected_assurance.py --batch DEMO-BATCH-A

# Compare a saved graph before and after changing the pump revision/interfaces.
python3 tools/automation/connected_assurance.py --impact-from /tmp/before.json
```

The pump query reaches requirements, failure propagation through the railway boundary, controller scenarios, planned physical tests, production instructions, both demo cars and their serials, FRACAS and dependent G1–G4 decisions. The batch query reaches one demo car, its affected serials, production record and incident. Impact outputs include a traversal path for each affected record. Comparisons compare node hashes, complete relationship records and controlled source hashes. Added, deleted or changed relationships seed both endpoints, even when every node is unchanged. Traversal uses both old and new edges so a deleted link cannot hide a former dependant. Source changes seed their bound records; a source with no recorded binding conservatively reopens all configuration, model, scenario, evidence and decision records. `changed_relationships`, `changed_sources` and `seed_reasons` retain why the review reopened. Interface traversal is intentionally conservative; an engineer must review applicability rather than interpret every path as a proven failure.

What the controller evidence establishes

The Rust runner receives declared initial conditions and injected diagnostic inputs. It returns actual controller outputs; the Python runner compares them against separately authored requirement criteria. It records the raw output, tool version, configuration identity and source hashes in the execution bundle. Source changes invalidate this evidence.

Executed case	Observed requirement
Reported pump unavailable	Isolation and derating requested; pump command zero
Reported pressure loss	Isolation requested; pump command zero
Reported hottest cell at 55 °C reference threshold	Isolation and derating requested
Reported pump unavailable and hottest cell at 60 °C	Isolation and derating requested; pump command zero

These are single controller evaluations. Physical pump seizure must first be detected and reported; detection coverage and delay remain unknown. The model does not calculate coolant flow, heat transfer, hotspots, thermal inertia, physical actuator response or target timing. High ambient is an input without calibrated physical thermal response. The masked pump/flow-sensor fault, charging/fouling and unavailable-recovery cases remain explicit plans. No bench, HIL or deployment result is invented. Acceptance criteria still need independent engineering review before they can support a deployment decision.

Evidence and gate behavior

Malformed/orphan records, missing effects, cycles, unsupported accepted states and contradictory scenario criteria fail the compiler. Missing verification, unknown likelihood, stale configurations, expired evidence, changed input/result hashes and open handback appear as blockers in an otherwise structurally valid report.

Reviewed evidence requires separate author/reviewer identities, a review reference, validity date, configuration fingerprint and hashed inputs/results. Local metadata cannot authenticate a competent person's acceptance. Decision records therefore permit `open`, `proposed`, `rejected` and `superseded`; the compiler never manufactures `accepted`. Earlier gates, significant children, shared dependencies and reviewed physical integration evidence remain explicit parent blockers. A controller run remains `generated-unreviewed` and does not advance any passport beyond G0.

The torque-process failure links to a controlled requirement, production operation, calibration gap, inspection hold point and affected serials. The synthetic incident keeps administrative closure separate from engineering handback and corrective-action effectiveness review.

Expansion

The subsystem qualification workflow now provides quantitative screens, rig/measurement definitions, production-equivalence checks and six separate Workbench readiness states. Physical evidence remains open.

Next, obtain the supplier's pump, connection and diagnostic definitions; review measurable flow, temperature and timing criteria; execute calibrated bench and HIL cases; and bind genuine manufacturing and installed records. Apply the same schema to brakes, doors, localisation and train protection, then charging, structures, civil occurrences and emergency operations. Add reviewed top-down fault trees and common-cause analyses to complement the recorded bottom-up propagation. Software defect rates are not inferred from hardware statistics.

ERPNext, IFC/FreeCAD, FUXA and Workbench can consume these stable identities and JSON records. The qualification workflow adds a Workbench readiness interface; these compilers do not write to external systems. The obligation example deliberately leaves licensed clause selection and deployment applicability unresolved.

Digital standards-adherence report

Repository source: `docs/certification/digital-assurance-report.md`

> Deterministic evidence-coherence result—not certification, conformity, construction release or permission to operate.

- Design fingerprint: `7c49232f0faf92e855f38c93861e6359487f7e01e084b734396f62af50fdccae`
- Digital evidence-coherence gate: PASS
- Standards conformity: NOT-ASSESSED
- Physical/revenue release: BLOCKED
- Scope: 23 publisher records, 5 assurance profiles, 16 control objectives and 15 repository checks
- Inventory: 285 engineering items screened; 285 item reviews remain open
- Registry reviewed: 2026-10-02; next mandatory review: 2026-10-31

What the result means

Passing means the named repository evidence is present, hashed, traceable and internally coherent. It is not a clause-by-clause conformity finding, certification, physical validation, permission to manufacture, construction release or permission to operate.

The compiler validates identities, mappings, coverage, evidence paths, hashes, review dates and fail-closed states. It deliberately leaves deployment applicability, licensed clause assessment, physical evidence and every competent decision open.

Digital thread

```
publisher record + deployment law + intended use
  → applicability decision and licensed clauses
    → OSR control objective
      → requirement / hazard / design item
        → method + acceptance criterion + configuration
          → raw evidence + hash + reviewer + validity
            → G1–G4 decision and conditions of use
              → change impact / expiry / field feedback
```

Open applicability inputs

ID	Deployment input	State
`APP-JURISDICTION`	Jurisdiction and competent authorities	`deployment-input-open`
`APP-INTENDED-USE`	System boundary, intended use and operating concept	`deployment-input-open`
`APP-LEGAL`	Railway, building, fire, accessibility, labour, privacy and environmental law	`deployment-input-open`
`APP-CONTRACT`	Employer requirements, contracts, funding conditions and adopted technical rules	`deployment-input-open`
`APP-CLASSIFICATION`	Safety, security, quality and asset criticality classifications	`deployment-input-open`
`APP-ASSESSMENT`	Conformity modules, assessment bodies, independence and authorization route	`deployment-input-open`

Assurance profiles

Profile	Standards	Decision boundary
`PROFILE-SYSTEM-SAFETY` — System RAMS, automation and authorization	5	Deployment safety authority and competent bodies freeze applicability and acceptance route.
`PROFILE-CONTROL-SOFTWARE` — Safety-related control, software and communications	6	Safety classification and national/international software route are deployment decisions.
`PROFILE-VEHICLE-EQUIPMENT` — Vehicle equipment, battery and environment	4	Product and vehicle qualification require exact selected components and physical evidence.
`PROFILE-CIVIL-BIM` — Civil structures and lifecycle information	4	The project freezes national structural rules, annexes, survey basis and information requirements.
`PROFILE-QUALITY-ASSET` — Quality, laboratories, assets and system lifecycle	5	Organization, laboratory and certification scopes remain separately assessed.

Control objectives

Control	Gate	Domains	Physical evidence	State
STD-CTRL-001 — System boundary, intended use and applicability	G1	system, operations, information, quality	depends on child evidence	mapped-not-assessed
STD-CTRL-002 — Requirements, interfaces and bidirectional traceability	G3	system, information, quality	required	mapped-not-assessed
STD-CTRL-003 — Hazard identification, FMEA and accepted risk	G4	system, mechanical, electrical, thermal, embedded-software, civil, operations	required	mapped-not-assessed
STD-CTRL-004 — RAMS allocation, automated-operation protection and safety case	G4	system, embedded-software, operations, quality	required	mapped-not-assessed
STD-CTRL-005 — Safety-related software lifecycle and target evidence	G4	embedded-software, system, quality	required	mapped-not-assessed
STD-CTRL-006 — Safety communication, cybersecurity and recovery	G4	embedded-software, electrical, cybersecurity, information, operations	required	mapped-not-assessed
STD-CTRL-007 — Battery, thermal and rolling-stock electronic equipment	G4	electrical, thermal, mechanical, embedded-software	required	mapped-not-assessed
STD-CTRL-008 — Equipment mounting, shock and vibration	G3	mechanical, electrical	required	mapped-not-assessed
STD-CTRL-009 — Civil, geotechnical, hydraulic and structural design basis	G4	civil, mechanical, system	required	mapped-not-assessed
STD-CTRL-010 — BIM and lifecycle information requirements	G3	information, civil, mechanical, operations	required	mapped-not-assessed
STD-CTRL-011 — Quality, supplier, manufacturing and special-process control	G4	quality, mechanical, electrical, civil, embedded-software, operations	required	mapped-not-assessed
STD-CTRL-012 — Test method, laboratory competence and measurement traceability	G4	quality, information, mechanical, electrical, thermal, civil	required	mapped-not-assessed
STD-CTRL-013 — Model, simulation and tool validity	G3	system, mechanical, electrical, thermal, embedded-software, civil, information	required	mapped-not-assessed
STD-CTRL-014 — Verification, validation, commissioning and operational demonstration	G4	system, mechanical, electrical, thermal, embedded-software, civil, operations, quality	required	mapped-not-assessed
STD-CTRL-015 — Asset, maintenance, competence and field-performance management	G4	operations, quality, information, system	required	mapped-not-assessed
STD-CTRL-016 — Configuration, change impact, evidence provenance and decision ledger	G4	system, information, quality, operations, embedded-software	depends on child evidence	mapped-not-assessed

Repository evidence checks

ID	Check	Controls	Machine result	Conformity
DA-001	Safety requirements and hazards remain traceable	STD-CTRL-002, STD-CTRL-003, STD-CTRL-004, STD-CTRL-016	evidence-linked-and-hashed	not-assessed
DA-002	Train separation and speed supervision remain independent of route selection	STD-CTRL-004, STD-CTRL-005	evidence-linked-and-hashed	not-assessed
DA-003	Network-loss routing fails restrictively on disagreement	STD-CTRL-004, STD-CTRL-005, STD-CTRL-006	evidence-linked-and-hashed	not-assessed
DA-004	Cabin and battery cooling allocation prioritises battery protection	STD-CTRL-003, STD-CTRL-007	evidence-linked-and-hashed	not-assessed
DA-005	Cross-domain FMEA has component, subsystem and system coverage	STD-CTRL-002, STD-CTRL-003	evidence-linked-and-hashed	not-assessed
DA-006	Civil structural screens retain survey and physical release boundaries	STD-CTRL-009, STD-CTRL-012, STD-CTRL-013, STD-CTRL-014	evidence-linked-and-hashed	not-assessed
DA-007	Recovery sidings, robots and single-track loops retain protected movement functions	STD-CTRL-003, STD-CTRL-004, STD-CTRL-014	evidence-linked-and-hashed	not-assessed
DA-008	Physical and independent-assessment gaps remain release-blocking	STD-CTRL-007, STD-CTRL-008, STD-CTRL-009, STD-CTRL-012, STD-CTRL-014, STD-CTRL-016	evidence-linked-and-hashed	not-assessed
DA-009	Consensus restart and infrastructure faults fail closed	STD-CTRL-005, STD-CTRL-006, STD-CTRL-016	evidence-linked-and-hashed	not-assessed
DA-010	Engineering benchmarks and interchange preserve units, CRS and identity	STD-CTRL-002, STD-CTRL-007, STD-CTRL-009, STD-CTRL-010, STD-CTRL-013, STD-CTRL-016	evidence-linked-and-hashed	not-assessed
DA-011	Promoted component RFCs retain traceable implementation and release gaps	STD-CTRL-002, STD-CTRL-003, STD-CTRL-007, STD-CTRL-009, STD-CTRL-011, STD-CTRL-016	evidence-linked-and-hashed	not-assessed
DA-012	Multi-day normal, peak, degraded and recovery profiles retain bounded logical state	STD-CTRL-004, STD-CTRL-005, STD-CTRL-013, STD-CTRL-014	evidence-linked-and-hashed	not-assessed
DA-013	Every controlled engineering and operational-platform item has a fail-closed assurance passport	STD-CTRL-001, STD-CTRL-003, STD-CTRL-011, STD-CTRL-015, STD-CTRL-016	evidence-linked-and-hashed	not-assessed
DA-014	BIM federation and IDS checks use current IFC4.3 lifecycle identities	STD-CTRL-010, STD-CTRL-016	evidence-linked-and-hashed	not-assessed
DA-015	Quality, lifecycle and laboratory evidence retain competence and approval boundaries	STD-CTRL-011, STD-CTRL-012, STD-CTRL-015, STD-CTRL-016	evidence-linked-and-hashed	not-assessed

Standards registry

Record	Publisher / edition	Status at review	Jurisdiction	Next review
IEC-62278-1-2025	IEC / 2025	current-publisher-record	international-reference	2027-01-02
IEC-62278-2-2025	IEC / 2025	current-publisher-record	international-reference	2027-01-02
IEC-62267-2009	IEC / 2009	current-publisher-record-stability-2028	international-reference	2027-01-02
IEC-62290-1-2025	IEC / 2025	current-publisher-record	international-reference	2027-01-02
IEC-62425-2025	IEC / 2025	current-publisher-record	international-reference	2027-01-02
IEC-62279-2015	IEC / 2015	current-publisher-record-stability-2030	international-reference	2027-01-02
EN-50716-2023	CEN-CENELEC national adoption record / 2023	current-European-baseline-transition-from-EN-50128-and-EN-50657	European-or-contractual-candidate	2026-10-31
IEC-62280-2014	IEC / 2014	current-publisher-record-stability-2028	international-reference	2027-01-02
IEC-60812-2018	IEC / 2018	current-publisher-record	international-reference	2027-01-02
IEC-62928-2017	IEC / 2017	current-publisher-record	international-reference	2027-01-02
IEC-60571-2012	IEC / 2012	current-publisher-record	international-reference	2027-01-02
IEC-61373-2026	IEC / 2026	current-publisher-record	international-reference	2027-01-02
IEC-62443-4-1-2018	IEC / 2018	current-publisher-record-edition-2-under-development	international-reference	2027-01-02
IEC-62443-4-2-2019	IEC / 2019 with COR1:2022	current-publisher-record	international-reference	2027-01-02
ISO-IEC-IEEE-15288-2023	ISO/IEC/IEEE / 2023	current-publisher-record	international-reference	2027-01-02
ISO-9001-2026	ISO / 2026	current-publisher-record-replaces-2015	management-system-candidate	2026-12-16
ISO-22163-2023	ISO / 2023 with Amd 1:2024	current-publisher-record-references-ISO-9001-2015-transition-review-required	railway-quality-candidate	2026-12-16
ISO-55001-2024	ISO / 2024	current-publisher-record	management-system-candidate	2027-01-02
ISO-19650-1-2018	ISO / 2018	current-confirmed-2024-revision-under-development	international-reference	2026-12-31
ISO-16739-1-2024	ISO / 2024	current-publisher-record-replaces-2018	international-reference	2027-01-02
ISO-IEC-17025-2017	ISO/IEC / 2017	current-confirmed-2023	conformity-assessment-candidate	2027-01-02
EN-1990-DEPLOYMENT	CEN national adoption selected by deployment / deployment-selected edition and national annex	applicability-and-national-adoption-open	European-or-contractual-candidate	2027-01-02
EU-CSM-402-2013	European Union / Regulation 402/2013 as amended	current-EU-legal-source-jurisdiction-limited	EU-or-contractually-adopted-only	2027-01-02

FMEA coverage

Domain	Component	Subsystem	System
civil	1	1	1
electrical	1	1	1
embedded-software	1	1	1
mechanical	1	1	1
operations	1	1	1
thermal	1	1	1

Evidence and change control

Every evidence object must carry 16 metadata fields. Current repository links are machine state `generated-unreviewed`; no link is silently promoted to reviewed or accepted.

The JSON report contains a path-level `change_impact_index` for 158 hashed inputs. Comparing reports identifies changed paths and reopens mapped controls rather than averaging them into a green parent score.

Interpretation

The deterministic evidence-coherence gate passed. Standards applicability, clause-level assessment, physical tests, site evidence, independent assessment and regulatory acceptance remain open.

Connected engineering

The connected engineering example and generated report bind battery-cooling failure propagation, requirement criteria, controller scenarios, planned physical tests, synthetic production records and installed occurrences to exact design revisions. The JSON includes dependency traversal and explicit blocked deployment decisions.

The subsystem qualification workflow adds quantitative RAMS screens, controlled rig measurements, model correlation, manufacturing equivalence and six separate decision-readiness states. Physical evidence and deployment decisions remain open.

The civil reference demonstration adds 20/25 m double-track bays, connection and erection controls, measured-result release interfaces and a connected construction/service FMEA. Its graph and controlled source hashes are included in this report and change-impact traversal; site inputs, physical qualification and independent release remain pending.

The train-centred prototype adds the actual process reference around existing interlocking/ATP/ATO/brake, bounded formal protocol and shared sensor-to-separation/interaction FMEA. Model and firmware changes invalidate execution evidence and traverse procedures and blocked release decisions; physical and operational qualification remain pending.

Digital Standards Thread

Repository source: `docs/certification/digital-standards-adherence.md`

OpenSourceRail turns standards work from a late document exercise into a live, version-controlled engineering thread. Every applicable rule can be connected to a control objective, requirement, hazard, design item, verification method, raw evidence object and named decision. When a hashed input changes, the affected controls can be found and reopened before fabrication or site work repeats the mistake.

This is a core product capability: design once, show the evidence chain, and know exactly what must be reassessed after change. It is digital assurance and approval preparation—not self-certification.

> Important
> A green OpenSourceRail report means the repository evidence is present, current, mapped and internally coherent. Only the applicable duty holder, competent engineer, accredited laboratory, independent assessor, certification body or public authority can make the corresponding real-world decision.

What is implemented

The machine-readable source is `digital-assurance.toml`. The compiler produces the current digital standards report, JSON evidence graph and standards baseline. CI fails when an output, source hash, evidence mapping or standards-review deadline is stale.

```
flowchart LR
  P[Publisher record] --> A[Deployment applicability]
  L[Law · contract · intended use] --> A
  A --> C[OSR control objective]
  C --> R[Requirement · hazard · interface]
  R --> D[Design item · exact revision]
  D --> V[Method · acceptance criterion]
  V --> E[Raw evidence · provenance · hash]
  E --> G[G1–G4 decision]
  G --> O[Conditions of use · operation]
  O --> X[Change · incident · expiry]
  X --> A
```

The implementation provides:

- a reviewed registry of international, European and management-system publisher records with edition status, jurisdiction boundary and a mandatory next-review date;
- five reusable profiles for system safety, control/software, vehicle equipment, civil/BIM and quality/assets;
- cross-domain control objectives covering concept through retirement;
- mandatory deployment inputs for jurisdiction, intended use, law, contract, classification and assessment route;
- deterministic links from controls to repository checks and evidence hashes;
- complete preliminary FMEA screening of the controlled engineering inventory;
- one G0–G4 assurance passport per controlled item;
- a connected battery-cooling example with exact design/installed revisions, failure propagation, requirement-derived controller cases, planned physical evidence, production/FRACAS links and blocked decisions;
- a path-level impact index that identifies the controls, checks, standards and failure modes affected by changed bytes; and
- fail-closed states: generated evidence cannot silently become reviewed, accepted, certified or released.

The three layers of truth

Layer	Question answered	Current machine authority
Publisher and applicability	Which edition and legal/contractual route might apply here?	Metadata and open applicability prompts only
Engineering evidence	Is the exact claim connected to current, reproducible evidence?	Deterministic coherence result
Acceptance and authorization	Is that evidence adequate for this product, site and use?	Named human/organizational decision only

This separation prevents two common errors: treating a list of standards as a compliance matrix, and treating a passing simulation as approval of a physical railway.

Deployment workflow

1. Freeze the claim and system boundary

Identify the city, railway, subsystem or item; intended use; operating concept; interfaces; environment; configuration; owner; and requested decision. A claim without an exact subject and use cannot enter G1.

2. Freeze applicability

Record national and local law, adopted standards and editions, National Annexes, contractual rules, funder requirements, safety/security classification, conformity route, competent authorities and independence requirements. Obtain licensed normative text where required. A publisher catalogue page is evidence of metadata, not evidence that every clause applies.

3. Build the clause-to-control overlay

The public registry stores no copyrighted normative text. A deployment overlay may store permitted clause identifiers, a locally authored obligation summary, applicability rationale, owner, verification method, evidence type, acceptance criterion and decision gate. Each clause is classified as applicable, not-applicable-with-rationale, informative or superseded; silence is an open gap.

4. Plan evidence before design release

Connect each control to requirements, hazards/FMEA, interfaces, design objects, calculations, simulations, reviews and physical tests. Define specimens, sampling, environments, instruments, calibration, competence, independence and pass/fail criteria before executing the work.

5. Capture an evidence object

Every evidence object needs, at minimum:

Field group	Required content
Identity	Stable evidence ID, claim, controlled item and configuration
Method	Procedure/model, acceptance criterion, tool and version, environment
Provenance	Source revision, raw result, SHA-256, author and creation time
Review	Reviewer, independence, disposition and limitations
Validity	Stated use, validity/expiry trigger and supersession relationship

Repository evidence begins as `generated-unreviewed`. Only an accountable workflow can move it to `reviewed` or `accepted-for-stated-use`. Rejection, expiry and supersession are retained rather than overwritten.

6. Roll up through G0–G4

The component assurance model keeps definition, design assurance, implementation qualification, integration validation and acceptance/authorization separate. A parent cannot pass while a significant child, interface, common-cause dependency or required independent decision is open. Scores and percentages never average away a release blocker.

7. Reopen on change

Compare a previous JSON report with the current tree:

```
python3 tools/automation/digital-assurance.py \
  --impact-from path/to/previous-digital-assurance-report.json
```

The result lists changed paths and affected controls, checks, standards and failure modes, together with dependency traversal through the connected engineering graph. Both previous and current relationships participate so removed links retain their impact. The engineering change record then decides which requirements, analyses, tests and G1–G4 decisions must repeat. The algorithm finds impact; it does not decide technical adequacy.

Worked use cases

Alignment or station change

A revised route invalidates its source hash. The impact review reaches terrain, water, clearance, civil structure, station, vehicle compatibility, evacuation, BIM, quantity, cost and delivery evidence. Open DEM and mapped water may rerun immediately; survey, geotechnical, hydraulic and competent structural decisions stay open.

Battery/HVAC component substitution

A changed compressor, chiller, pump or battery module reopens the exact product passport, thermal model assumptions, interface requirements, FMEA, supplier evidence and vehicle integration controls. Digital duty-cycle and fault tests reduce redesign risk; calorimeter, chamber, leakage, EMC, environmental and vehicle tests remain physical evidence.

Rust control release

A source, dependency, compiler or target change invalidates the relevant software and provenance hashes. Unit, property, differential, formal, resilience and soak checks can rerun deterministically. Target timing/resources, HIL, production communications, cybersecurity, commissioning and independent safety acceptance remain separate gates.

Supplier and first article

The supplier receives the applicable requirements, interface baseline, evidence schema and acceptance plan rather than an ambiguous standards list. Returned drawings, certificates, raw measurements, NCRs and test records bind to the exact item and revision. A catalogue page, declaration or summary PDF cannot silently close first-article qualification.

Operational modification

An altered maintenance interval, procedure, competence scope or software patch is treated as a controlled system change. The thread links asset performance, work records, hazard controls, training/authorization and any assessor or authority condition that must be reconsidered.

Standards maintenance and licensing

The tracked baseline uses authoritative publisher or legal records and records when each was checked. CI deliberately expires the metadata review. It does not download or reproduce copyrighted standards, monitor every national adoption, or infer local law.

Known transition points are explicit. ISO 9001:2026 replaced ISO 9001:2015 in September 2026. ISO 22163:2023 remains the current railway QMS record but still references ISO 9001:2015, so a deployment must agree and document its transition basis. EN 50716:2023 is the current OSR European software baseline while legacy EN 50128/EN 50657 references require a recorded national or contractual reason.

Claims suitable for proposals

OpenSourceRail may accurately say:

- “standards-aware evidence is generated and checked on every controlled change”;
- “the system exposes missing, stale, unreviewed and physical evidence instead of hiding it”;
- “change impact is traced before manufacture and site work”; and
- “the dossier is structured for supplier, assessor and authority review.”

It must not say “certified,” “compliant,” “approved,” “type-approved” or “physical tests are unnecessary” unless the exact scope, edition, configuration, use, evidence and competent decision genuinely support that statement.

Handover dossier

A real submission exports the frozen applicability overlay, licensed-clause working record, configuration index, requirements and hazard trace, passports, supplier and construction files, model/tool validation, raw V&V evidence, calibration and competence records, deviations/NCRs, safety case, independent assessment, decisions, conditions of use and change history. National forms wrap this evidence; they do not create a second uncontrolled baseline.

Run the gate

```
python3 tools/automation/digital-assurance.py --check
python3 tools/automation/connected_assurance.py --check
python3 tools/automation/component_assurance.py --check
python3 engineering/subsystem_control_register.py --check
./osr test
```

The JSON output is intended for Workbench, ERP, supplier portals and future authority-specific exporters. The Markdown output is the review and marketing view of the same deterministic data.

Distributed Onboard Control Profile

Repository source: `docs/certification/distributed-onboard-control-profile.md`

Status: development target; not the deployable pilot baseline.

This profile reduces dependence on centralised wayside computation and permits continued route selection during a communications outage. It does not remove the safety functions commonly implemented by interlocking, ATS or ATP. Those functions may move between onboard and distributed equipment only after a hazard analysis, independent assessment and deployment approval demonstrate equivalent or better safety.

The deployable reference remains the conservative sectional-authority profile in pilot-signalling-profile.md.

Executable train-centred milestone

RFC 0033 and the TACS assurance package use the existing committed interlocking MA and ATP/ATO/brake components through thin, authenticated process hosts. The reference includes A/B agents for each of two trains, three static voters, point I/O and two station/charging interfaces. The development default requires both protection channels to permit and either to trip; loss stops the affected function. Train logical pairs are executed; point/crossing/door/charger/obstacle pairs and physical independence remain open. Raft replication cannot substitute for safety outputs. Generated software evidence remains unreviewed. HIL, measured braking/integrity, field recovery, independent assessment and a named approved installed bundle are required to change the deployable profile. The conservative pilot remains the default.

Three independent route inputs

The onboard selector compares:

1. a signed, versioned plan stored onboard;
2. route and position inferred from onboard sensors against immutable topology;
3. an authenticated network command from operations control.

An input is eligible only while trusted, fresh and internally valid. Exactly matching route, next-section and plan-epoch values from two eligible sources are required. If all three disagree, only one source is eligible, or the two available sources disagree, the result is `Hold`. A network outage can produce `OnboardAutonomous` only when the onboard plan and sensor-derived input agree.

Route selection is not permission to move. Every accepted selection carries an explicit `movement_authority_required` condition and is passed through the independent authority, localisation, speed-envelope and emergency-brake path.

Functions that must remain

Any implementation must retain, even if equipment placement changes:

- movement-authority generation and expiry;
- train separation and opposing-move exclusion;
- route locking and detected/locked point position;
- occupancy or safety-integrity localisation;
- overspeed supervision and emergency braking;
- intrusion/obstacle gates required by the operating envelope;
- degraded-mode supervision, event recording and accountable override.

“Eliminating interlocking/ATS/ATP” is therefore interpreted as reducing duplicated central equipment, not deleting their safety outcomes. A model, sensor fusion algorithm or radio link cannot independently claim equivalence.

Narrow single-track operation

Short single-track sections are permitted only on exclusive guideway and with passing loops at passenger stops or protected portals. Entry requires locked route exclusivity, detected points, clear occupancy, a valid authority and a proved timetable/degraded-mode strategy. Loss of required state prevents entry; it does not rely on two trains negotiating informally over the radio.

Recovery sites and shunt robots

The reference planner reserves a recovery siding at or immediately beyond every third passenger station. Each site has a protected mainline interface, independent stop detection and remote isolation. Its small shunt robot is remote-controlled, has local emergency stop and fail-held braking, and cannot autonomously enter the main line. Radio loss inhibits traction. Mainline entry, towing and return to service remain controlled recovery operations.

Evidence gates

Before this profile can replace any pilot equipment, the project must freeze the safety allocation and demonstrate source independence/common-cause analysis, target-hardware timing, localisation accuracy, radio-loss behaviour, point and occupancy interfaces, cybersecurity, HIL fault injection, shadow running, representative field trials, recovery drills and independent safety assessment. The deterministic report records design coherence; it does not close these physical or regulatory gates.

Evidence register

Repository source: `docs/certification/evidence-register.md`

Complete inventory of verification evidence produced by the OpenSourceRail project, indexed to the safety requirements from safety-requirements.md and the hazards from hazard-log.md.

1. Formal-methods evidence (Kani bounded-model checking)

Each SIL-4 evaluator carries Kani harnesses under `crates/<name>/src/kani_proofs.rs`, gated on `cfg(kani)` so `cargo test` skips them and `cargo kani` runs them. These are written proof harnesses. A successful Kani run can establish the stated property within its recorded bounds; the presence of a harness or a passing `cargo test` run does not establish that result.

Crate	Harness module	Properties	SR coverage
`osr-atp`	`kani_proofs.rs`	A1–A7	SR-01, SR-02, SR-03
`osr-brake`	`kani_proofs.rs`	B1–B5	SR-15
`osr-odometry`	`kani_proofs.rs`	O1–O5	SR-10
`osr-wayside-points`	`kani_proofs.rs`	W1–W6	SR-01, H-CO-02
`osr-interlocking`	`kani_proofs.rs`	P1–P5	SR-01, SR-02, H-CO-01, H-CO-04
`osr-obstacle-detect`	`kani_proofs.rs`	O1–O5	SR-04, SR-05, SR-06
`osr-intrusion-detect`	`kani_proofs.rs`	I1–I5	SR-07, H-CO-03, H-SA-01
`osr-secbus`	`kani_proofs.rs`	S1, S3	SR-22, H-DI-01

Inventory: named-property harnesses across the eight evaluators above. Harnesses written, runs passed, results current and results independently accepted are separate states. The two selected CI properties are documented in the safety-case README; wider proofs remain open. Rerun with `cargo kani --package <name>` on a host with Kani installed (Kani is not part of the `cargo test` default path).

2. Randomised property testing (proptest)

Every Kani property has a matching proptest that exercises the same invariant across a random input space (typically 256–1000 cases per property per run). Proptest runs on `cargo test` by default.

Crate	Test file	Props	Typical cases per run
`osr-atp`	`tests/proptest_atp.rs`	A1–A7 + bonus	256
`osr-brake`	`tests/proptest_brake.rs`	B1–B5	256
`osr-odometry`	`tests/proptest_odom.rs`	O1–O5	256
`osr-wayside-points`	`tests/proptest_wayside.rs`	W1–W6	256
`osr-interlocking`	`tests/proptest_ma.rs`	P1–P5	1000
`osr-interlocking`	`tests/proptest_determinism.rs`	derive-state determinism	1000
`osr-interlocking`	`tests/differential.rs`	Rust Python twin agreement	256
`osr-consensus`	(inline + suite)	5 Raft properties; refinement open	2000
`osr-obstacle-detect`	`tests/proptest_obstacle.rs`	O1–O5 + severity + classifier downgrade	256
`osr-intrusion-detect`	`tests/proptest_intrusion.rs`	I1–I5	256
`osr-secbus`	`tests/proptest_secbus.rs`	S1–S3 + roundtrip	256
`osr-crypto`	(inline)	C1–C4 + ed25519 determinism	256

Total proptest properties: 60+ across the SIL-4 + SIL-2 crates. Rerun with `cargo test --workspace`; keep the resulting workspace test count in the release log rather than baking a drifting number into this register.

3. Differential testing (Rust Python)

The `tools/reference-ma/` directory carries a stdlib-only Python re-implementation of `osr-interlocking`. `crates/osr-interlocking/tests/differential.rs` generates random log prefixes, computes the MA in both Rust and Python, and asserts byte-identical JSON output. Catches spec- level bugs in either implementation.

- Coverage: ring lines, switch observations, route grants, speed restrictions, partial sections.
- Run: `cargo test -p osr-interlocking --test differential`.

4. Integration-level evidence (sim shadow stack)

`crates/osr-sim` runs the full Samawah network under the ATP + brake + obstacle-detect + fire + derailment shadow stack per-tick per-train. Fault injection exercises each SIL-4 evaluator's restrictive path end-to-end.

Scenario	Fault kinds	What it demonstrates
`cities/catalogue/west-asia/Iraq/Samawah/samawah.toml`	None	Nominal operation: zero spurious emergencies over multi-hour revenue sim
Built-in dust-storm fixture (sim CLI)	PV dust, grid outage, charging-pad outage	Energy-system fault-tolerance
Built-in obstacle-fault fixture (sim CLI)	LIDAR/radar/ultrasonic/peer-disagreement (RFC 0015)	O1–O5 all fire through `BrakeInputs::obstacle_emergency`
Built-in wayside-intrusion fixture (sim CLI)	Present/Unknown on specific sections (RFC 0016)	Interlocking gate (d) withholds MA without a single train violating

Rerun with `cargo run --release --bin osr-sim -- --config scenarios/<name>.toml`.

5. GSN safety-case tree

Goal-Structuring-Notation argument tree under `docs/safety-case/gsn/`, compiled by the `osr-safety-case` CI gate. Every safety goal links to evidence (Kani harness, proptest file, sim scenario); the CI job fails if any goal is added without an evidence pointer. It does not run the pointed-to proof or grant acceptance.

File	Goals	Strategy
`00-top.toml`	G0	Root claim
`10-non-overlap.toml`	G1	Non-overlap of MAs
`20-fail-safety.toml`	G2	Fail-restrictive default
`30-consistency.toml`	G3	Log consistency
`40-onboard-sil4.toml`	G4	Onboard SIL-4 layer
`50-wayside-sil4.toml`	G5–G14	Wayside SIL-4 + infra
`60-obstacle-detect.toml`	G15–G19	Obstacle detection (RFC 0015)
`70-intrusion-detect.toml`	G20–G24	Intrusion detection (RFC 0016)
`80-message-authentication.toml`	G25–G27	Message auth (RFC 0017)

Generated case inventory: 32 goals, 6 strategies, 78 solutions. These counts describe traceability, not successful or accepted proofs.

CI gate: the `starter_case_closes` test in `crates/osr-safety-case/tests/` fails the build if a goal lacks structural traceability. The separate result gate checks execution records, source/report hashes and independent acceptance records.

6. Operational-controls evidence (RFC 0013 rulebook)

The operations rulebook under `docs/operations/` defines the unattended- operation role families. Each rule is one sentence plus a `Why:` paragraph, cross-referenced to the specific crate it relies on.

Role	File count	Rule count
Dispatcher	S1–S7	35
Station staff	T1–T5	29
Maintenance	M1–M7	42
Control centre	C1–C3	16

Total: 122 operational rules backing the safety-case claims.

7. Hardware-level evidence

Per-board spec documents under `control-electronics/<class>/schematics/v2-spec/`:

Board	Complete spec
T-ECU/S	(block diagram, power budget, pinout RP2350, pinout CM5, safety-nets, connector tables)
T-OBS	(block diagram, power budget, pinout RP2350, safety-nets, connector tables)
T-ECU/A	pending (see option 3 of the current sprint)
W-SBC	pending
S-SBC	pending

For pilot deployments, the RFC 0019 COTS/SBC path is a valid hardware implementation route and does not require KiCad or gerbers when no custom PCB is used. Its missing evidence is the integration pack: exact SKUs, wiring/harness maps, connector maps, enclosure and mounting notes, power/thermal margins, SD-card image checksums, `osr-selftest` output, and bench/commissioning records.

KiCad schematic capture + gerber generation remain custom-board milestones for any deployment that chooses OSR-specific carrier, power, safety-I/O, or sensor-interface boards.

8. Evidence status and release gaps

Source inventories and successful test runs are useful development evidence. They do not imply all formal proofs have run or all safety goals are accepted. Record the exact source scope, named harness, tool version, bounds, result and report hash for each run, then obtain independent acceptance of that record. The result gate reports missing or stale records as open, even when all GSN pointers resolve. It does not authenticate reviewers or replace an assessor.

Current release limitations remain in the release-gap register and safety-case proof status. No zero-gap safety or physical-release claim follows from the traceability gate.

The generated all-component assurance register adds one configuration-bound passport to every controlled engineering item and owner/operator platform service. It routes this evidence through G0–G4 but does not convert the evidence pointers above into item qualification, integration validation, independent acceptance or authorization.

Evidence Status Matrix

Repository source: `docs/certification/evidence-status.md`

This matrix separates what is already evidenced in the repository from what remains deployment or assessor work. It is not a safety approval; it is a coherence map for the pre-submission pack.

The all-component assurance register is the authoritative item-level view: its generated passports use the same G0–G4 lifecycle while retaining route-specific evidence. This page summarises system-level evidence without duplicating a count that can drift.

For closure criteria on each open item, see `release-gap-register.md`.

Area	Current repository evidence	Status	Next action
All-component assurance	The generated register covers every current engineering inventory item and configured business/supervision platform; every identity/source baseline is recorded	G0 baselined; G1–G4 open	Freeze deployment use, jurisdiction, classification, standards applicability, assessor and evidence plan item by item
Movement authority non-overlap	`osr-interlocking` unit/proptest/differential tests; RFC 0004	Implemented + tested	Add assessor-reviewed trace from hazards to tests
Three-source onboard routing	`osr-onboard-routing` two-of-three selector, unit/proptests and simulator execution; selection cannot create authority	Implemented + simulated	Independent-source/common-cause analysis, HIL and network-loss field trials
Consensus log safety	TLA+ model, `osr-consensus` simulation/proptests	Implemented + modeled	Refinement argument from TLA+ spec to Rust harness
Consensus persistence and recovery	Versioned/checksummed stable-state envelope, atomic disk-full/partial-write behavior, fail-closed restore and deterministic crash/partition/clock/telemetry trace	Implemented + tested in process	Repeat on selected storage, processor, network and clock hardware; add power-cut/endurance/WCET evidence
Onboard obstacle detection	RFC 0015, `osr-obstacle-detect`, sim fault injection	Implemented + simulated	First-article sensor dataset and calibration report
Wayside intrusion detection	RFC 0016, interlocking gate, sim integration	Implemented + simulated	Pilot installation evidence on representative sections
Message authentication	`osr-crypto`, `osr-secbus`, authenticated `osr-consensus` ingress/commit consumer, simulator integration, and adversarial integration tests	Implemented + simulated	Freeze deployment key registry/provisioning and verify on the selected production transport/hardware
Hardware safety nets	RFC 0007 v2 specs, RFC 0019 DIY path, and hardware docs	Specified	Pilot integration pack, bench test records, and custom-board KiCad/Gerber/BOM only where custom boards are used
Rolling-stock mechanical concept	RFC 0008/0021/0022, `design/component-catalogue` source plus FreeCAD review artifacts	Parametric reference	FEA, crashworthiness simulation, supplier drawings
Integrated cabin/battery thermal control	`osr-hvac::integrated_thermal`, property tests and rolling-stock thermal template prioritise battery protection while isolating air/coolant circuits	Implemented + tested model	Supplier sizing, calorimeter/thermal-chamber, leakage, EMC and vehicle integration evidence
Cross-domain FMEA	`system-fmea.toml` supplies component/subsystem/system scenarios across mechanical, electrical, thermal, embedded software, civil and operations; the generator screens every current controlled train, station, civil and Rust inventory entry	Baseline + complete inventory screen	Close each deliberately open item-specific analysis with accountable owners/actions, supplier data, tests and field feedback
Digital Standards Thread	`digital-assurance.py` validates publisher-review deadlines, applicability inputs, control coverage, evidence paths/hashes and change impacts and emits Markdown/JSON on every build/test	Evidence-coherence gate	Freeze deployment clauses and close technical, physical, assessor and authority evidence without converting coherence into conformity
Engineering interchange gate	Seven analytical references plus LandXML/OSR-ALN, IFC identity/envelope and duty-cycle solver projections run deterministically	Implemented regression gate	Repeat live solver suite on a second controlled machine and retain signed environment manifest
Single track and recovery	Planner emits constrained segment track counts, passing loops and every-third-station recovery sites; corridor template constrains protected shunt operation	Planning implementation	Site alignment, operations proof, turnout/occupancy HIL, tow tests and recovery drill
Station charging energy	RFC 0002, generated city energy feasibility tables	Planning-grade	Site-specific solar yield, grid-tie, and charger thermal study
Operations rulebook	RFC 0013 and `docs/operations/`	Drafted	Operator review and local authority adaptation
Certification pack	`docs/certification/` and GSN claims	Pre-submission scaffold	Independent assessor review and evidence freeze

The repo health gate (`tools/automation/repo-health.py`) guards generated design coherence. It is deliberately not a substitute for IEC 62278-1, EN 50716/50129 assessment evidence, hardware bench records, structural FEA, or deployment-specific field evidence.

Hazard log

Repository source: `docs/certification/hazard-log.md`

GoA 4 Unattended Train Operation hazards identified against the system boundary in system-description.md. Organised by hazard class: collision, runaway, derailment, passenger-boarding, fire, sabotage, data-integrity.

Each row carries:

- Severity — using the EN 50126 scale: Catastrophic (multiple fatalities) / Critical (single fatality or multiple serious injuries) / Marginal (single serious injury or multiple minor) / Insignificant.
- Frequency without controls — order-of-magnitude estimate for how often the hazard would occur absent any mitigation.
- Residual frequency — the frequency that remains once the listed controls are in place; the per-hazard "Residual acceptance" rows below argue why each residual is tolerable.
- Mitigating controls — cross-referenced to requirements and artefacts.

Frequencies are expressed on a qualitative EN 50126 scale: Frequent (≥ 1/yr) / Probable (1/10 yr) / Occasional (1/100 yr) / Remote (1/1 000 yr) / Improbable (1/10 000 yr) / Incredible (< 1/10 000 yr).

H-CO — Collision hazards

H-CO-01 — Rear-end collision between two trains

Field	Value
Description	Two trains occupy the same block; the following train collides with the leading.
Severity	Catastrophic
Frequency without controls	Frequent
Controls	SR-01 section-occupancy gate; SR-02 MA validity window; SR-03 MA conservatism; 2oo2 AND-gate at brake actuator (SR-20).
Residual frequency	Incredible

H-CO-02 — Head-on collision through mis-set switch

Field	Value
Description	Switch commanded Normal but observed Reverse (or Unknown); train enters a conflicting route.
Severity	Catastrophic
Frequency without controls	Probable
Controls	`osr-wayside-points` 2oo2 sensor fusion; `osr-interlocking::section_available_to` checks switch state before MA grant; fail-restrictive `Unknown` classification.
Residual frequency	Incredible

H-CO-03 — Collision with obstacle on track (debris, vehicle, person)

Field	Value
Description	A person, animal, car, or large debris enters the track envelope between trains; next train hits it.
Severity	Critical to Catastrophic
Frequency without controls	Occasional (target climate: dust storms deposit debris; urban ROW crossings possible)
Controls	SR-07 wayside intrusion gate (proactive — RFC 0016); SR-04 onboard obstacle detection (reactive — RFC 0015); SR-24 PSDs at every boarding platform; physical fence (deployment civil scope).
Residual frequency	Remote

H-CO-04 — Collision with a track worker under MaintenanceOverride

Field	Value
Description	Track worker on foot under an active `MaintenanceOverride`; train enters the blocked section.
Severity	Critical
Frequency without controls	Probable
Controls	SR-01 section-occupancy gate refuses MA on any section carrying an active override; protection signals (RFC 0013 S5.3); physical trap point (site-specific); 30-minute block-renewal cadence (S5.4) catches silent over-runs.
Residual frequency	Incredible

H-RA — Runaway hazards

H-RA-01 — Runaway on grade after brake release

Field	Value
Description	Train parked on a grade; parking brake fails or is released in error.
Severity	Catastrophic
Frequency without controls	Occasional
Controls	Parking-brake engagement below `park_brake_max_speed_mmps` (SR-20 `osr-brake` B5 property); EB latches on any monitor trip; both SIL-4 channels required to release.
Residual frequency	Improbable

H-RA-02 — Overspeed through brake failure

Field	Value
Description	Train exceeds the civil-speed envelope due to a regen/friction-blend failure.
Severity	Critical
Frequency without controls	Occasional
Controls	ATP overspeed trip (SR-03); regen-shortfall falls back to friction; WSP subtractive-only (B4).
Residual frequency	Remote

H-DE — Derailment hazards

H-DE-01 — Derailment through undetected switch Unknown

Field	Value
Description	Switch observation fails or reads `Unknown`; train enters the switch without MA.
Severity	Catastrophic
Frequency without controls	Probable
Controls	Switch observation fail-restrictive (`Unknown` rejects MA); `osr-wayside-points` 2oo2 sensor fusion; RFC 0013 M4 30-day switch inspection.
Residual frequency	Incredible

H-DE-02 — Derailment through lateral forces

Field	Value
Description	Excessive lateral acceleration (track-twist, excessive speed into curve) causes derailment.
Severity	Catastrophic
Frequency without controls	Occasional
Controls	SR-11 onboard derailment monitor (`osr-derailment`); RFC 0013 M3 track-geometry inspection cadence (90/60/45 days per preset); RFC 0009 preset-bounded curve radius + cant.
Residual frequency	Improbable

H-PB — Passenger-boarding hazards

H-PB-01 — Passenger struck by closing doors

Field	Value
Description	Door closes on a passenger limb / bag.
Severity	Marginal (rare Critical)
Frequency without controls	Frequent
Controls	SR-09 door-obstruction motor-current + sensor; reopen on detection; SR-08 closing-interlock above 5 km/h.
Residual frequency	Remote

H-PB-02 — Passenger falls between platform and train

Field	Value
Description	Passenger falls into the platform gap during boarding.
Severity	Critical
Frequency without controls	Frequent (legacy no-PSD metros)
Controls	SR-24 PSDs at every boarding platform; RFC 0010 platform-gap design; station-staff T2 supervision.
Residual frequency	Improbable

H-PB-03 — Train departs with passenger caught in door

Field	Value
Description	Passenger's limb / bag remains trapped in closed door; train begins to move.
Severity	Critical
Frequency without controls	Probable
Controls	SR-09 door-sensor + SR-08 interlock; station-staff T2.5 platform-clear confirmation at non-PSD stations; PSD interlock at PSD stations (SR-24).
Residual frequency	Improbable

H-FI — Fire hazards

H-FI-01 — Battery thermal runaway

Field	Value
Description	Cell thermal runaway propagates to adjacent cells; fire on-board.
Severity	Critical
Frequency without controls	Occasional
Controls	RFC 0021 LFP module compartmentation; outward vent; temperature and off-gas detection; charge inhibit; proved string/car-pack isolation; per-car localized water-mist cooling with reservoir/pump/pressure/flow diagnostics; controlled movement to the nearest safe platform unless containment, crash state, traction/braking availability, or fire growth requires immediate emergency braking; IEC 62928-based design and test plan.
Residual frequency	Remote

H-FI-01A — Battery fire propagates after visible flame reduction

Field	Value
Description	Suppression reduces visible flame but cell heating continues and propagates into an adjacent module.
Severity	Critical
Frequency without controls	Occasional
Controls	Water mist is credited for cooling/propagation delay, not guaranteed cell extinguishment; thermal barriers, outward venting, adjacent-module nozzle coverage, propagation test, continuous post-discharge temperature/off-gas monitoring, evacuation and fire-service handover.
Residual frequency	Remote — pending propagation test evidence

H-FI-01B — Battery mist unavailable or ineffective

Field	Value
Description	Empty reservoir, failed pump, blocked nozzle, pipe rupture, lost pressure, or absent flow prevents localized cooling.
Severity	Critical
Frequency without controls	Probable
Controls	Independent system per car; reservoir-level, pump-current, pressure and flow monitoring; startup self-test; maintenance proof-flow; fault reported to TCMS/OCC; fire response does not rely on mist to preserve electrical containment or evacuation.
Residual frequency	Remote — pending component reliability and proof-flow evidence

H-FI-01C — Unsafe stop strategy during onboard fire

Field	Value
Description	Immediate braking strands passengers between stations, or continued movement worsens a fire/crash hazard.
Severity	Critical
Frequency without controls	Occasional
Controls	Staged `osr-fire-safety` outputs distinguish controlled safe-platform stop from emergency braking; independent immediate-danger and safe-movement inputs; OCC alert; route/platform availability check; emergency override for loss of containment, essential control, braking, or safe route.
Residual frequency	Remote — pending integrated scenario and evacuation evidence

H-FI-02 — Electrical fire in traction inverter

Field	Value
Description	Short or arc in the SiC inverter; fire.
Severity	Critical
Frequency without controls	Occasional
Controls	SR-12; `osr-traction` isolation on fault; RFC 0008 §3 EN 45545 HL2 R1 material compliance.
Residual frequency	Remote

H-SA — Sabotage + intrusion hazards

H-SA-01 — Deliberate obstacle placement on track

Field	Value
Description	Attacker places a large obstacle on the ROW between trains.
Severity	Critical to Catastrophic
Frequency without controls	Occasional
Controls	SR-07 wayside intrusion gate (proactive, detects placement); SR-04 onboard obstacle detection (reactive); fence-line contact detection (RFC 0016 fence sensor); CCTV coverage.
Residual frequency	Remote

H-SA-02 — Station-platform intrusion

Field	Value
Description	Attacker gains access to the trackside from a platform.
Severity	Critical
Frequency without controls	Probable (stations are public)
Controls	SR-24 PSDs block direct platform trackside access; station-staff T2.1 platform walk; CCTV.
Residual frequency	Remote

H-DI — Data-integrity hazards

H-DI-01 — Forged consensus entry

Field	Value
Description	Attacker with network access injects a forged `SectionIntrusion::Clear` or `RouteGrant`.
Severity	Critical to Catastrophic (depends on forged content)
Frequency without controls	Remote (requires network access + cryptographic sophistication)
Controls	SR-22 consensus entry authenticity (RFC 0017 — library in tree, v2 wiring open); physical network isolation (deployment scope); IEC 62443-4-2 component-level controls.
Residual frequency	Improbable (after SR-22 v2) / Remote (today, pending SR-22 wire integration)

H-DI-02 — Replay of a valid entry

Field	Value
Description	Attacker replays an old `RouteRelease` or `MaintenanceOverride` at an inconvenient time.
Severity	Critical
Frequency without controls	Remote
Controls	Derived-state fold is idempotent in `entry_id` (RFC 0017 §3.4); 60 s freshness window on timestamp_ns; Raft already rejects duplicate entry_ids.
Residual frequency	Improbable

Hazard summary

Class	Count	Worst-case severity	After controls
Collision	4	Catastrophic	Incredible–Remote
Runaway	2	Catastrophic	Improbable–Remote
Derailment	2	Catastrophic	Incredible–Improbable
Passenger-boarding	3	Critical	Improbable–Remote
Fire	2	Critical	Remote
Sabotage / intrusion	2	Catastrophic	Remote
Data integrity	2	Catastrophic	Improbable
Total	17		

Every hazard carries ≥ 2 independent controls. No single control failure produces a single-fault hazard of residual frequency worse than Remote.

Pilot Signalling Profile

Repository source: `docs/certification/pilot-signalling-profile.md`

`OSR-PILOT-1` is the conservative deployment baseline for a segregated pilot. It separates what can be exercised now from the research target so that a city does not need to accept novel moving-block authority on day one.

The machine-readable source is `lib/templates/signalling.toml` under `profiles.osr-pilot-1`.

Live Pilot Boundary

Function	Pilot treatment
Occupancy	Independently detected by assessor-accepted axle counters or track circuits. A loss, disagreement or unknown state is occupied.
Movement authority	Conservative wayside sectional/fixed-block authority. No authority extends beyond independently proved clear and locked sections.
Points	Independently detected, locked and fail-safe before authority is issued.
OSR ATP	Active only inside the conventional sectional authority and selected safety-controller boundary.
OSR ATO	Supervised; ATP and the conventional authority remain the hard limit.
Consensus and onboard MA	Recorded, compared and alarmed in shadow mode; unable to drive a safety output.
Right of way	Segregated pilot only, with local operating rules and independently assessed degraded modes.

The application host is not the safety controller. The deployment must freeze a qualified safety channel using the controller selection gate, including watchdog, power, I/O, common-cause, environmental and HIL evidence.

Staged Transition

1. Shadow: compare OSR position, consensus state and calculated authority against the conventional pilot authority without controlling movement.
2. Supervised sectional: permit OSR ATO under active ATP, while independent occupancy and sectional movement authority remain authoritative.
3. Sectional GoA 4: remove the onboard attendant only after operations, obstacle detection, evacuation, communications and safety assessment close.
4. Moving-block trial: exercise distributed authority on a closed test track with independent fallback and an assessor-approved test plan.
5. Revenue moving block: only after machine-checked refinement or an assessor-accepted equivalent argument, full bounded-proof evidence, selected hardware qualification, HIL/field results and national authority approval.

No stage inherits approval from the previous stage automatically. Each city must record its own hazards, authority, configuration, evidence and rollback criteria. The target profile in the template is a research configuration, not a construction or revenue-service release.

Certification Release Gap Register

Repository source: `docs/certification/release-gap-register.md`

This register is the release-facing list of evidence still needed before the current repository package can support a revenue-service GoA 4 type-approval submission. It complements `evidence-status.md`: that page summarises status by area; this page names the closure evidence expected for each gap.

The simplified component assurance model groups these details into G0 definition, G1 design assurance, G2 implementation qualification, G3 integration validation and G4 independent acceptance. A gap may affect more than one gate; the strictest affected gate controls release.

Gate	Current position	Principal gap class
G0 — Define	Every current generated passport has a reference identity and source baseline	Deployment use and applicability move into G1
G1 — Assure design	All item plans remain open	Jurisdiction, classification, requirements, interfaces, item FMEA/FMEDA and V&V plans
G2 — Qualify implementation	All qualifications remain open	Suppliers, detailed design, production/target artifacts, calibrated analysis and physical tests
G3 — Validate integration	All validations remain open	HIL, FAT/SAT, site/as-built, commissioning, degraded modes and operating interfaces
G4 — Accept and authorize	All acceptances remain open	Independent assessment, residual-risk acceptance, duty-holder decisions and authority approval

Gap	Current in-tree state	Closure evidence
All-component applicability and classification	The deterministic register gives every current item a stable G0 passport, route, source hash and fail-closed state; standards remain candidate prompts	For every deployed item, close G1 with intended use, jurisdiction, safety classification, applicable standards/rules and editions, interfaces, hazards, V&V plan, owners and independence requirements
Independent safety assessment	Pre-submission pack is structured and traceable, but not assessor-reviewed	Named assessor review report, action log, and accepted residual-risk statement
Residual-risk acceptance	Hazards and mitigations are listed, but no national-authority threshold is applied	Deployment-specific ALARP / tolerability criterion signed by authority or assessor
Consensus refinement	TLA+ spec, Rust proptests and a documented abstraction/action map exist, but no machine-checked refinement proof connects them	Assessor-accepted refinement argument or tool-backed proof from `SMRaft.tla` to `osr-consensus` behavior
Consensus storage and recovery qualification	Stable state is versioned, checksummed, atomically replaced and exercised against restart, disk-full, partial-write, corruption, network and clock/telemetry faults in a deterministic in-process harness	Selected-storage power-cut and endurance campaigns, processor/RTOS timing and resource evidence, production network/clock fault injection, signed binary/update evidence and independent review
ATP bounded-proof completion	CI proves the fast expired-authority property; topology-heavy ATP harnesses exist but their full-run unwind bounds are not frozen release evidence	Review and freeze topology bounds and Kani/CBMC versions, complete every ATP harness without unwinding failures, and archive the reports
Interlocking bounded-proof completion	CI proves the movement-authority validity-window arithmetic; topology-heavy P1–P4 and whole-function P5 harnesses do not complete inside the hosted-runner limit and are not represented as a green CI gate	Refactor the proof model or freeze reviewed topology bounds, complete the full interlocking set without unwinding failures, and archive versioned reports
Safety-controller freeze	RP2350/Radxa pilot rigs are specified, while qualified safety-channel candidates and selection requirements are documented but not selected	Assessor-reviewed MCU/channel selection, safety manual and FMEDA inputs, common-cause analysis, frozen schematics and bench/HIL evidence
Signed safety-log deployment evidence	Authenticated consensus ingress/commit verification and simulator fault-injection tests reject forged, altered, replayed, stale/future, unknown-issuer, metadata-mismatched, and category-downgraded proposals	Frozen deployment key registry, secure-element provisioning/rotation procedure, production-transport capture, and hardware-in-the-loop evidence showing only verified entries reach state derivation
Pilot hardware integration evidence	RFC 0019 COTS/SBC path and board-level specs exist; controlled pilot integration evidence is not frozen	Exact SKU BOMs, wiring/harness maps, connector maps, enclosure/mounting notes, power/thermal margins, SD-card image checksums, bring-up logs, safety-net bench tests, and commissioning records
Custom-board release evidence	Custom boards are optional for a first pilot and required only when a deployment chooses OSR-specific carrier, power, safety-I/O, or sensor-interface boards; KiCad, gerbers, and board BOMs are not released	For any custom-board deployment: KiCad projects, gerber zips, board BOMs, bring-up logs, safety-net bench tests, and DFM/DFT review
Rolling-stock structural evidence	Nine CalculiX beam/space-frame screens and raw solver outputs cover chassis, bogie, body and articulation gross load paths. They are screening models, not production shell/solid FEA or physical correlation.	Released EN 15227 / EN 12663 / EN 13749 model and reports, convergence and weld/joint treatment, independent review, plus first-article proof/fatigue correlation and weld/NDT records
Integrated thermal-system evidence	A deterministic controller and property tests allocate common refrigerant capacity while keeping battery coolant and cabin air physically isolated; battery derate/protect states dominate comfort demand	Supplier heat-rejection and pressure-drop model, selected components, thermal chamber/calorimeter tests, leakage/contamination analysis, EMC/environmental tests, battery fault trials and vehicle-level validation
Cross-domain FMEA/FMEdA completion	The generated report covers the controlled cross-domain risk scenarios and deterministically screens the complete current train, station, civil and Rust inventory. Every item review remains explicitly release-blocking; the generic screen is not a supplier FMEDA.	Complete item-specific modes, local/end effects, causes, rankings and interfaces against the frozen product/asset breakdown; assign owners/dates/actions; incorporate supplier reliability/FMEdA data; demonstrate mitigations by analysis/fault injection/physical test; and obtain independent review
Distributed onboard-control release	Two-of-three route selection is implemented and tested; route selection remains subordinate to authority and ATP, and the profile is not deployable	Safety allocation and common-cause analysis, qualified hardware, timing/WCET, localisation and radio-loss HIL, cybersecurity, shadow running, representative field trials and assessor acceptance

Gap	Current in-tree state	Closure evidence
Single-track and recovery-site release	Planner/template outputs define loops, protected interfaces and every-third-station shunt sites	Site-specific alignment/evacuation/access approval, timetable and degraded-mode proof, point/occupancy interface tests, tow/brake/coupler/radio-loss tests and full recovery drill
Door-system evidence	RFC 0023, build definitions and an IFE supplier-family anchor define the boundary; the exact cassette/order code and EN 14752 route remain open	Controlled supplier drawings/data, actuator/lock/obstruction/cycle/emergency/water tests, configuration record, notified-body route and vehicle integration tests
Obstacle/intrusion field evidence	Logic and simulations exist; no representative first-article sensor dataset	Calibration report, hot-weather/soiling/night/rain datasets, false-positive/false-negative analysis
Station charging and site energy	The tracked Samawah summary freezes eight passing full-day cases: separate and combined end-of-life battery/maximum-climate duty, reduced contacts, two consecutive unavailable charging stops, a late-running three-times-dwell shared-charger overlap, a ten-hour cabinet/pad outage and all-site grid outage. Overlapping/constrained requests and conservation errors are explicit evidence fields.	Re-run the matrix for each deployment timetable, then complete site solar yield, grid interconnect, charger thermal study, protection settings, train/charger interface tests and utility approval.
Measured traction duty	A strict solver-neutral schema and planning fixture preserve sample identity, time, units and power semantics across battery, grid and traffic projections; planning/simulated sources cannot become acceptance-eligible through conversion	Instrumented representative train/route measurements, calibrated sensors, frozen configuration, raw records/checksums, uncertainty review and accepted normal/degraded/aged duty set
Operations validation	Rulebook is drafted in English	Operator workshop minutes, translated deployment rulebook where needed, competence/training records
LM3 first-article evidence control	All 81 unresolved product rows have controlled closure state and one or more routes through the 13-package evidence register. The candidate register covers all 56 bought-in rows with 41 official manufacturer/research options; the mass ledger maps all 120 rows while leaving all 117 active row masses open; the factory readiness register keeps all 16 drawing/tooling/verification packages open and assigns every locally made row to at least one drawing; and the execution pack defines the ordered RFQ/rig/test/weigh route. Zero supplier/physical/mass packages are marked accepted.	Supplier responses and accountable submissions with exact configuration/procedure, calibrated equipment, raw artifacts and checksums, plus independent review and accepted work-package closure

These are not code-style TODOs. They are release gates for the first deployment package.

Safety requirements specification

Repository source: `docs/certification/safety-requirements.md`

Functional safety requirements (SR-01 … SR-24) derived from EN 62267 §6 (functional requirements for GoA 4 operation) and cross-referenced to OSR artefacts (crates, RFCs, rulebook sections). Requirements are grouped by the EN 62267 function.

Each requirement states what the system must do and names the evidence that it does so. Detailed evidence lives in evidence-register.md.

1. Train separation (EN 62267 §6.1)

SR-01 — Safe separation between trains

The system shall at all times prevent two trains from occupying the same track section.

- Implementation: `osr-interlocking::section_available_to` gate (a) refuses MA into any section whose `section_occupancy` record names a different train.
- Evidence: GSN G1 (via `docs/safety-case/gsn/10-non-overlap.toml`); property P2 proptest at `crates/osr-interlocking/tests/proptest_ma.rs`; Kani harness `kani_a5_head_past_ma_end_trips_same_section`.

SR-02 — MA validity window

Every movement authority issued to a train shall carry an absolute expiry time and shall not be extended without a fresh position report.

- Implementation: `MA_VALIDITY_WINDOW_NS` bound on every MA; ATP property A2 trips on expiry.
- Evidence: Kani `kani_a2_expired_ma_trips`; property P5 proptest.

SR-03 — MA conservatism under uncertainty

Widening any sensor-uncertainty or shortening any MA input shall never move the ATP verdict toward a less-restrictive brake command.

- Implementation: ATP property A7.
- Evidence: Kani `kani_a7_uncertainty_widening_is_conservative`; property A7 proptest.

2. Train protection (EN 62267 §6.2)

SR-04 — Stopping before obstacle (GoA 4 substitution for driver)

The system shall detect obstacles inside the current stopping- distance envelope and command Emergency Brake within the ATP tick.

- Implementation: `osr-obstacle-detect::evaluate` property O1; any LIDAR / radar / ultrasonic detection inside the envelope forces `ObstacleVerdict::EmergencyBrake`, which flows into `BrakeInputs::obstacle_emergency`.
- Evidence: Kani `o1_ultrasonic_return_forces_eb`; proptest `o1_ultrasonic_return_forces_eb`; sim built-in obstacle-fault fault-injection fixture.

SR-05 — Fail-restrictive on sensor failure

If any safety-primary obstacle-detect sensor is stale beyond 100 ms, the verdict shall be at least `RestrictedSpeed` (O4b) and never `Clear`.

- Implementation: O2 + O4b in `osr-obstacle-detect`.
- Evidence: Kani `o2_stale_ultrasonic_forces_eb`, `o4b_lidar_offline_with_radar_healthy_restricts_speed`.

SR-06 — 2oo2 cross-check on obstacle detection

The two redundant obstacle-detect channels (RP2350 A + B on the T-OBS board) shall both agree on `Clear` before the trainset may proceed at full MA-permitted speed.

- Implementation: O3 property; any peer disagreement EB.

- Evidence: Kani `o3_peer_disagreement_forces_eb`.

SR-07 — Wayside intrusion gate

The system shall withhold MA on any track section whose latest wayside intrusion verdict is not `Clear`.

- Implementation: `osr-interlocking::section_available_to` gate (d) introduced by RFC 0016 v2.
- Evidence: `intrusion_present_blocks_section`, `intrusion_unknown_is_fail_restrictive`, `latest_intrusion_verdict_wins` unit tests in `crates/osr-interlocking/src/ma.rs`.

3. Door operation (EN 62267 §6.3)

SR-08 — Door closed above threshold speed

The system shall interlock door opening so that a door cannot be commanded open while `speed ≥ 5 km/h`.

- Implementation: `osr-door-control` closing interlock (SIL-4 per RFC 0005 §4.2).
- Evidence: `osr-door-control` proptests.

SR-09 — Door obstruction detection

The system shall detect door obstructions via motor current + dedicated sensor and reopen the affected door on detection.

- Implementation: `osr-door-control` obstruction handler.
- Evidence: `osr-door-control` integration tests.

SR-10 — Platform-side alignment at standstill

Door opening shall be enabled only when the train is at standstill (`speed < threshold`) and reporting position within ± 0.5 m of the platform reference mark.

- Implementation: Door-enable gate uses `osr-odometry` position + `osr-ato` stop-detection; PSD interlock adds a second layer at PSD-equipped stations.
- Evidence: `osr-ato` stop-tolerance tests; RFC 0010 §7 platform geometry.

4. Obstacle and derailment detection (EN 62267 §6.4)

SR-11 — Derailment detection

Lateral g-forces + axle-rotation asymmetry monitored in 2oo2 shall trip Emergency Brake on any indication of derailment.

- Implementation: `osr-derailment::derailment_evaluate`; feeds `BrakeInputs::derailment_emergency`.
- Evidence: `osr-derailment` proptests.

SR-12 — Onboard fire detection

Smoke + temperature in battery / traction / HVAC bays shall trigger suppression + EB.

- Implementation: `osr-fire-safety::fire_evaluate`; feeds `BrakeInputs::fire_emergency`.
- Evidence: `osr-fire-safety` proptests.

SR-13 — Hot-axle box detection

Axle bearing temperature above threshold shall produce an advisory event; OCC triages per RFC 0013 S3.

- Implementation: `osr-hot-axle` (onboard, SIL-2) + `osr-hot-axle-wayside` (wayside, SIL-4).
- Evidence: `osr-hot-axle-wayside` proptests.

5. Emergency detection and management (EN 62267 §6.5)

SR-14 — Passenger emergency intercom

Passengers shall be able to trigger a controlled brake + open an audio + video channel to OCC by pressing an emergency intercom; the brake shall bring the train to a safe stop at the next station (not immediate EB, to avoid stranding).

- Implementation: `osr-passenger-assist` + RFC 0015 §5.3. The normal intercom action contributes a 500 ppt service-brake request; only an authenticated OCC escalation feeds the emergency input.
- Evidence: state-machine tests cover immediate latch, link loss, authenticated release/escalation, and the `osr-sim` integration test carries the request into `osr-brake` within one simulation tick. Release closure still requires real media, event-recorder, and incident-workflow evidence in release-gap-register.md.

SR-15 — EB dominates all other brake commands

Emergency Brake shall not be reducible by any other input; service-brake or release commands during an EB shall be ignored until EB is explicitly released.

- Implementation: `osr-brake::brake_evaluate` B2 property.
- Evidence: Kani `kani_b2_emergency_union`, `kani_b3_emergency_completeness`.

SR-16 — Passenger fire-evacuation on EB + fire

If EB is driven by a fire trip, the train shall continue to the next station platform where safe, then open doors + announce evacuation.

- Implementation: `osr-ato` fire-response mode; `osr-fire-safety` coordinates with `osr-brake`; RFC 0013 S3.6 rulebook procedure.
- Evidence: Rulebook; deployment-specific simulator and first-article evacuation evidence are release gaps.

6. Operations supervision (EN 62267 §6.6)

SR-17 — Dispatcher accountability

Every MA grant, MaintenanceOverride, and degraded-mode transition shall be attributable to a named dispatcher via their OCC login.

- Implementation: RFC 0013 S1.1 rule (personal credential); `osr-occ` console logs.
- Evidence: Rulebook; audit-log structure in `osr-historian`.

SR-18 — Remote-assist for passenger incidents

OCC shall maintain an audio + video channel to any trainset under remote-assist; dispatcher may hold the train, request medical, or release to continue.

- Implementation: `osr-passenger-assist` owns the call lifecycle, media-channel request, latched hold, and authenticated OCC accept/release/emergency commands; RFC 0013 C2 owns the procedure.
- Evidence: state-machine and simulator brake-path tests. The actual SIP/audio/video transport, device hardware, and staffed workflow remain release gaps.

7. Fault management (EN 62267 §6.7)

SR-19 — Fail-restrictive default

Every SIL-4 evaluator shall fail in the restrictive direction — on any uncertainty, ambiguity, or missing input the verdict is at least the second-most-severe (usually a brake, sometimes a speed cap).

- Implementation: All O-series (obstacle-detect), I-series (intrusion-detect), A-series (ATP), P-series (interlocking), O-odom, B-brake properties carry fail- restrictive clauses.
- Evidence: 7 Kani harness modules across the SIL-4 crates; GSN G1–G24 closure.

SR-20 — 2oo2 hardware voting at output stage

Every brake command to the actuator shall pass through a 2oo2 AND-gate relay driven by both RP2350 channels of the T-ECU/S.

- Implementation: Hardware AND-gate relay stage per `control-electronics/t-ecu-s/schematics/v2-spec/safety-nets.md`.
- Evidence: Safety-nets spec; DRC pass at board production (deployment-specific).

SR-21 — Watchdog-driven reset

Loss of heartbeat from either RP2350 shall drive the TPS3701 supervisor to reset that channel; concurrent loss of both heartbeats shall drop the EB relay directly (bypassing the AND gate).

- Implementation: `control-electronics/t-ecu-s/schematics/v2-spec/safety-nets.md` watchdog rules.
- Evidence: Safety-nets spec.

8. Cybersecurity (EN 62267 §6.8, complementary IEC 62443-4-2)

SR-22 — Consensus entry authenticity

Every entry committed to the track-state consensus log shall be signed by its originating entity and verified by every consumer before it affects derived state.

- Implementation: `osr-consensus::authenticated::ProposalVerifier` verifies signed proposal metadata, category, freshness, replay state, and issuer before proposal and again before committed-entry decode/apply.
- Evidence: `osr-secbus` Kani/proptests plus consensus ingress rejection and simulator integration tests. Production provisioning and transport/HIL evidence remain open.

SR-23 — Fare-token authenticity

Every fare transaction token shall carry an HMAC-SHA256 MAC keyed to a per-deployment secret that is rotatable without re-issuing existing tokens.

- Implementation: `osr-crypto::hmac_sha256_verify`; used by `osr-afc`.
- Evidence: `osr-crypto` proptests; `osr-afc` integration.

9. Passenger interaction (EN 62267 §6.9)

SR-24 — Platform screen door mandatory at GoA 4

Every passenger boarding platform in a GoA 4 deployment shall be equipped with platform screen doors (PSDs) per RFC 0010 §6 + RFC 0015 §5.4.

- Implementation: RFC 0010 station-design standard with PSD in every non-`halt` archetype; RFC 0015 §5.4 tightens the default to PSD-mandatory for every boarding platform (except depot-terminal, where access is staff-only).
- Evidence: RFC 0010, RFC 0015; deployment-specific design report.

10. Distributed onboard route selection

SR-25 — Route selection cannot authorize movement

The route selector shall never create or extend movement authority. Every selected route shall remain conditional on a separately valid authority, localisation verdict, speed envelope and train-protection output.

- Implementation: `osr-onboard-routing::select_route` always returns `movement_authority_required = true`; the simulator evaluates selection before the independent interlocking/ATP path.
- Evidence: unit, property and onboard integration tests; deterministic assurance check DA-002.

SR-26 — Offline route selection requires two-source agreement

With the network command absent, continued route selection shall require an eligible signed onboard plan and eligible sensor-derived route to match on route, next section and epoch. Stale, untrusted, single-source or disagreeing inputs shall produce `Hold`.

- Implementation: `osr-onboard-routing` deterministic two-of-three selector.
- Evidence: exhaustive unit cases, property tests and DA-003. Hardware, localisation and field-loss evidence remain open.

Requirement summary table

ID	Summary	SIL	Evidence kind
SR-01	Section-occupancy gate	4	Kani + proptest + GSN G1
SR-02	MA validity window	4	Kani + proptest
SR-03	MA conservatism	4	Kani + proptest
SR-04	Obstacle detection in envelope	4	Kani + proptest + sim
SR-05	Fail-restrictive on sensor failure	4	Kani + proptest
SR-06	2oo2 obstacle-detect cross-check	4	Kani
SR-07	Wayside intrusion gate	4	Unit test + GSN G20–G24
SR-08	Door-interlock above threshold	4	Proptest
SR-09	Door-obstruction reopen	4	Integration test
SR-10	Platform-alignment gate	4	ATO tests + RFC 0010
SR-11	Derailment detection	4	Proptest
SR-12	Onboard fire detection	4	Proptest
SR-13	Hot-axle-box detection	4 (wayside) / 2 (onboard)	Proptest
SR-14	Passenger emergency intercom	2	State machine + sim brake-path test; control-electronics/media evidence open
SR-15	EB dominates	4	Kani
SR-16	Evacuation on fire-EB	2	Rulebook + sim pending
SR-17	Dispatcher accountability	—	Rulebook S1.1
SR-18	Remote-assist channel	—	Call/control state machine; media/HW evidence open
SR-19	Fail-restrictive default	4	8 Kani modules + GSN G1–G24
SR-20	2oo2 hardware voting	4	Safety-nets spec
SR-21	Watchdog + reset	4	Safety-nets spec
SR-22	Consensus entry auth	2	Kani + proptest + GSN G25–G27
SR-23	Fare-token auth	2	Proptest
SR-24	PSDs at every boarding platform	—	RFC 0010 + RFC 0015 §5.4
SR-25	Route selection never grants movement authority	4 boundary	Unit + proptest + sim
SR-26	Offline routing requires plan/sensor agreement	4 boundary	Unit + proptest + HIL open

Deterministic Software Resilience Report

Repository source: `docs/certification/software-resilience-report.md`

> In-process design evidence only. Hardware, filesystem, network, clock, HIL and independent-assessment evidence remain required.

- Result: PASS
- Deterministic seed: `24301`
- Fault/recovery actions: 34
- Fail-restrictive safety proposal rejections: 3
- Runtime invariant failures: 0

Checked Boundaries

- node crash, stable-state restart and post-heal convergence;
- asymmetric message loss, bounded delay, partition and healing;
- atomic checkpoint behavior under disk-full and partial writes;
- fail-closed rejection of corrupted, truncated, wrong-node and inconsistent stable state;
- clock-offset, time-source and telemetry health gating for safety proposals; and
- election, log-matching, leader-completeness, state-machine and fail-restrictive consensus invariants after every action.

Trace

Step	Action	Outcome	Leader	Commit	Safe inputs
0	`Tick { duration_ns: 100000000 }`	`ticked`	N0	0	yes
1	`Tick { duration_ns: 100000000 }`	`ticked`	N0	0	yes
2	`Propose { value: [98, 97, 115, 101, 108, 105, 110, 101], category: Advisory }`	`proposed`	N0	0	yes
3	`Checkpoint { node: NodeId(2) }`	`checkpoint-committed`	N0	0	yes
4	`Delay { node: NodeId(2), ticks: 3 }`	`delay-scheduled`	N0	0	yes
5	`DropFrom { node: NodeId(1), enabled: true }`	`asymmetric-egress-updated`	N0	0	yes
6	`Tick { duration_ns: 100000000 }`	`ticked`	N0	0	yes
7	`Heal { node: NodeId(1) }`	`healed`	N0	0	yes
8	`Tick { duration_ns: 100000000 }`	`ticked`	N0	1	yes
9	`Tick { duration_ns: 100000000 }`	`ticked`	N0	1	yes
10	`DiskFull { node: NodeId(0) }`	`checkpoint-rejected-disk-full`	N0	1	yes
11	`PartialWrite { node: NodeId(1), bytes: 7 }`	`checkpoint-rejected-partial-write`	N0	1	yes
12	`Checkpoint { node: NodeId(2) }`	`checkpoint-committed`	N0	1	yes
13	`Crash { node: NodeId(2) }`	`crashed`	N0	1	yes
14	`Tick { duration_ns: 100000000 }`	`ticked`	N0	1	yes
15	`Restart { node: NodeId(2) }`	`restarted`	N0	1	yes
16	`Telemetry { available: false }`	`telemetry-updated`	N0	1	no
17	`Propose { value: [116, 101, 108, 101, 109, 101, 116, 114, 121, 45, 114, 101, 106, 101, 99, 116, 101, 100], category: Safety }`	`rejected-fail-restrictive`	N0	1	no
18	`Telemetry { available: true }`	`telemetry-updated`	N0	1	yes
19	`ClockOffset { offset_ns: 8000000000 }`	`clock-offset-updated`	N0	1	no
20	`Propose { value: [99, 108, 111, 99, 107, 45, 114, 101, 106, 101, 99, 116, 101, 100], category: Safety }`	`rejected-fail-restrictive`	N0	1	no
21	`ClockOffset { offset_ns: 0 }`	`clock-offset-updated`	N0	1	yes
22	`TimeSource { available: false }`	`time-source-updated`	N0	1	no
23	`Propose { value: [116, 105, 109, 101, 45, 115, 111, 117, 114, 99, 101, 45, 114, 101, 106, 101, 99, 116, 101, 100], category: Safety }`	`rejected-fail-restrictive`	N0	1	no
24	`TimeSource { available: true }`	`time-source-updated`	N0	1	yes
25	`Partition { node: NodeId(0) }`	`partitioned`	N0	1	yes
26	`Tick { duration_ns: 100000000 }`	`ticked`	N0	1	yes
27	`Tick { duration_ns: 100000000 }`	`ticked`	N0	1	yes

Step	Action	Outcome	Leader	Commit	Safe inputs
28	`Heal { node: NodeId(0) }`	`healed`	N0	1	yes
29	`Tick { duration_ns: 100000000 }`	`ticked`	none	1	yes
30	`Tick { duration_ns: 100000000 }`	`ticked`	none	1	yes
31	`Checkpoint { node: NodeId(0) }`	`checkpoint-committed`	none	1	yes
32	`Checkpoint { node: NodeId(1) }`	`checkpoint-committed`	none	1	yes
33	`Checkpoint { node: NodeId(2) }`	`checkpoint-committed`	none	1	yes

Remaining Release Evidence

Repeat these cases on the selected storage, processor, RTOS/OS, production transport and clock sources. Add power-cut injection, storage endurance, target-hardware wall-clock soak, WCET/stack/heap measurements, signed binary/update evidence and independent review before release. The separate deterministic multi-day simulator soak is design evidence only.

Deterministic Multi-Day Software Soak Report

Repository source: `docs/certification/software-soak-report.md`

> Software design evidence only. This accelerated simulation is not target-hardware endurance, WCET evidence, certification, or permission to operate.

- Result: PASS
- Duration: 2 simulated days per profile
- Fixed simulation step: 5 seconds
- Profiles: normal, peak, degraded and recovery
- Growth measure: retained logical state at the halfway and final checkpoints

Profile Results

Profile	Seed	Result	Min SoC	Train-km	Faults	Invariant failures
normal	24301	PASS	0.200	36328.3	0	0
peak	24302	PASS	0.200	49460.4	0	0
degraded	24303	PASS	0.200	23443.8	2	0
recovery	24304	PASS	0.200	27206.6	3	0

Retained-State Growth

Every final value is checked against the declared bound. A smaller checkpoint value may grow only until that cap is reached.

Profile	Resource	Halfway	Final	Bound
normal	detailed events	0	0	0
normal	event-count keys	8	8	9
normal	event records	540672	540672	540672
normal	T2G payloads	0	0	540672
normal	historian metrics	660	660	660
normal	historian samples	2708640	3278880	8078400
normal	CBM components	4356	4356	4356
normal	work orders	0	0	4096
normal	compact result bytes	171643	171987	2000000
peak	detailed events	0	0	0
peak	event-count keys	6	6	9
peak	event records	540672	540672	540672
peak	T2G payloads	0	0	540672
peak	historian metrics	660	660	660
peak	historian samples	2708640	3278880	8078400
peak	CBM components	4356	4356	4356
peak	work orders	0	0	4096
peak	compact result bytes	173108	173601	2000000
degraded	detailed events	0	0	0
degraded	event-count keys	8	8	9
degraded	event records	540672	540672	540672
degraded	T2G payloads	540672	540672	540672
degraded	historian metrics	0	0	660
degraded	historian samples	0	0	8078400
degraded	CBM components	0	0	4356
degraded	work orders	0	0	4096
degraded	compact result bytes	173997	174744	2000000
recovery	detailed events	0	0	0
recovery	event-count keys	8	8	9
recovery	event records	540672	540672	540672
recovery	T2G payloads	540672	0	540672
recovery	historian metrics	660	660	660
recovery	historian samples	13860	2866380	8078400
recovery	CBM components	4356	4356	4356
recovery	work orders	0	4096	4096

Profile	Resource	Halfway	Final	Bound
recovery	compact result bytes	175369	610340	2000000

Assertions

normal

- PASS — detailed event retention disabled while aggregate counts remain available
- PASS — event recorder and T2G queues stay within fixed per-train capacities
- PASS — historian raw and decimated tiers stay within fixed per-metric capacities
- PASS — CBM component state and detailed work-order evidence stay within explicit capacities
- PASS — every generated work order is represented by retained or dropped-record accounting
- PASS — compact result payload stays within the deterministic evidence envelope
- PASS — operating reserve and simulator invariants remain satisfied
- PASS — normal service completes repeated service/stabling cycles without injected faults
- PASS — checkpoint-to-final retained-state growth remains inside every declared cap

peak

- PASS — detailed event retention disabled while aggregate counts remain available
- PASS — event recorder and T2G queues stay within fixed per-train capacities
- PASS — historian raw and decimated tiers stay within fixed per-metric capacities
- PASS — CBM component state and detailed work-order evidence stay within explicit capacities
- PASS — every generated work order is represented by retained or dropped-record accounting
- PASS — compact result payload stays within the deterministic evidence envelope
- PASS — operating reserve and simulator invariants remain satisfied
- PASS — peak profile uses three-minute-or-better published headways and continues accumulating fleet distance after the checkpoint
- PASS — checkpoint-to-final retained-state growth remains inside every declared cap

degraded

- PASS — detailed event retention disabled while aggregate counts remain available
- PASS — event recorder and T2G queues stay within fixed per-train capacities
- PASS — historian raw and decimated tiers stay within fixed per-metric capacities
- PASS — CBM component state and detailed work-order evidence stay within explicit capacities
- PASS — every generated work order is represented by retained or dropped-record accounting
- PASS — compact result payload stays within the deterministic evidence envelope
- PASS — operating reserve and simulator invariants remain satisfied
- PASS — continuous radio loss saturates, but never exceeds, the bounded store-and-forward queue
- PASS — checkpoint-to-final retained-state growth remains inside every declared cap

recovery

- PASS — detailed event retention disabled while aggregate counts remain available
- PASS — event recorder and T2G queues stay within fixed per-train capacities
- PASS — historian raw and decimated tiers stay within fixed per-metric capacities
- PASS — CBM component state and detailed work-order evidence stay within explicit capacities
- PASS — every generated work order is represented by retained or dropped-record accounting

- PASS — compact result payload stays within the deterministic evidence envelope
- PASS — operating reserve and simulator invariants remain satisfied
- PASS — radio/grid/CBM degradation clears and queued telemetry drains before the final checkpoint
- PASS — checkpoint-to-final retained-state growth remains inside every declared cap

Remaining Release Evidence

- selected-target WCET, stack and heap measurement
- wall-clock endurance and storage wear on production hardware
- power-cut, network and clock HIL with production transports
- independent safety assessment and authority acceptance

Current standards applicability baseline

Repository source: `docs/certification/standards-baseline.md`

> Generated from `lib/templates/digital-assurance.toml`; do not edit by hand.

This registry records current publisher metadata and how OpenSourceRail routes evidence. It does not reproduce normative clauses or claim applicability or conformity. A deployment must obtain licensed text where required and freeze law, jurisdiction, national adoptions, editions, contracts, intended use and assessment bodies.

Metadata was reviewed on 2026-10-02. The next fail-closed review is due 2026-10-31.

Publisher records

Record	Scope used by OSR	Evidence that remains outside the digital gate
IEC-62278-1-2025	Lifecycle, system definition, RAMS requirements, change impact and safety-case traceability	site validation, commissioning, operational demonstration, independent assessment
IEC-62278-2-2025	Safety process, risk assessment, requirement allocation, integrity allocation and independence	independent assessment, validation evidence, accepted residual risk
IEC-62267-2009	GoA4 functional safety and degraded-operation traceability	obstacle and intrusion trials, evacuation, rescue, operational validation
IEC-62290-1-2025	Continuous supervision, onboard localisation and command/control functional boundary	communications coverage, localisation calibration, end-to-end train protection test
IEC-62425-2025	Safety-case structure, integrity allocation, reuse and safety-related tool boundary	qualified controller, HIL, environmental tests, independent safety assessment
IEC-62279-2015	Safety-related software lifecycle, competence, configuration, verification and tool evidence	target-hardware integration, timing and resource evidence, HIL, commissioning
EN-50716-2023	European software lifecycle, verification, configuration and tool evidence	target-hardware integration, HIL, commissioning
IEC-62280-2014	Threat and error model for safety-related communication over transmission systems	production-network fault injection, coverage and latency tests, key and configuration evidence
IEC-60812-2018	Failure-mode identification, effects, causes, controls, ranking and maintenance	supplier FMEDA, inspection, fault injection, field feedback
IEC-62928-2017	Battery design, operating limits, data exchange and test-plan traceability	type tests, routine tests, thermal propagation, pack and vehicle integration
IEC-60571-2012	Electronic-equipment requirements and deterministic test mapping	environmental tests, power transients, EMC, target hardware
IEC-61373-2026	Mounting category, load case and permitted simulation/test-exemption traceability	shock test, random vibration test, post-test functional inspection
IEC-62443-4-1-2018	Secure development, verification, vulnerability, patch and end-of-life process	product security assessment, deployment hardening, penetration testing, patch exercise
IEC-62443-4-2-2019	Component security capability and foundational-requirement evidence mapping	configured component test, network segmentation test, credential and update validation
ISO-IEC-IEEE-15288-2023	Common system life-cycle process and acquirer/supplier information exchange framework	project plans, reviews, verification and validation records, transition and disposal evidence
ISO-9001-2026	Organization-wide quality, documented information, operational control, performance evaluation and improvement	implemented QMS, competence records, supplier and production records, internal and external audit
ISO-22163-2023	Railway-sector quality management, supplier, project and production evidence structure	implemented railway QMS, process audits, supplier controls, production and special-process qualification
ISO-55001-2024	Asset-management objectives, lifecycle decisions, information, performance and improvement	implemented asset management system, asset data and decisions, audits, field performance
ISO-19650-1-2018	Whole-life information management, exchange, recording, versioning and organization	project information requirements, accepted information deliveries, as-built and asset information
ISO-16739-1-2024	Open IFC exchange for buildings and infrastructure including railways and bridges	accepted model-view and information requirements, as-built model validation, handover verification
ISO-IEC-17025-2017	Competence, impartiality, method, calibration, traceability and result-record requirements for test evidence	laboratory scope and accreditation where required, method validation, calibration traceability, raw test records
EN-1990-DEPLOYMENT	Ultimate/serviceability limit-state, reliability, durability and design-working-life evidence structure	material certificates, geotechnical investigation, proof/load tests where specified, as-built survey
EU-CSM-402-2013	Significance, hazard record, risk acceptance principles, interfaces and independent assessment for changes when applicable	proposer risk-management file, assessment-body report, hazard closure and accepted change

Transition notes

- ISO 9001:2026 is the current ISO quality-management edition as of the review date and replaces ISO 9001:2015.
- ISO 22163:2023 with Amendment 1:2024 remains the current railway QMS publisher record and still names ISO 9001:2015; deployments must agree and record the transition basis rather than silently rewriting its normative reference.
- EN 50716:2023 is the current OSR European software baseline; legacy EN 50128/EN 50657 mappings require a recorded national or contractual reason during the transition.

- EN 1990 is deliberately edition-neutral here because the deployment must select its national adoption and National Annex.
- EU Regulation 402/2013 is included only as an EU/contractually adopted example; it is not automatically applicable outside that jurisdiction.

Machine-readable controls

The generated digital standards report maps 23 records through 16 control objectives to 15 hashed repository checks. Its result is evidence coherence only. The all-component assurance register carries the resulting G0–G4 decision boundary to each controlled item.

Subsystem qualification and quantitative RAMS

Repository source: `docs/certification/subsystem-qualification.md`

The battery-cooling workflow connects the assurance graph to quantitative screening, controlled rig measurements, manufacturing equivalence and signed review decisions. Workbench's Decision readiness view shows requirements, design verification, physical qualification, integration, independent review and operating conditions separately.

The current package is a reference programme with no selected physical rig, supplier parameters or measured qualification results. Numerical examples are labelled assumptions. They cannot establish an operating restriction, inspection interval, staffing commitment or accepted railway use.

Controlled sources

Source	Responsibility
Qualification definition	RAMS allocation, hazards/interfaces, assessment agreement, rig, instruments, test criteria, production controls and reviews
Quantitative inputs	Parameter ranges, units, evidence basis, common causes, repair/inspection/logistics assumptions and alternatives
RAMS models	Numerical models with bounded problem sizes and finite input validation
Qualification compiler	Measurement checks, model correlation, manufacturing findings, signed-review verification and export
Generated readiness	Results and unresolved decisions; JSON also feeds standards reports and passports

The definition allocates reliability, availability, maintainability and safety targets from the reference railway boundary to the cooling loop and its flow and power interfaces. Targets remain null until responsible engineers allocate and review them. Operator, independent assessor and agreed assessment reference are explicit prerequisites.

Each allocation declares an allowlisted `metric`, exact `result_path`, `unit`, `comparison` (<= or >=), target, review state and applicable `operating_conditions`. Conditions must match the linked result; mission probability also binds `mission_h`. The input study declares its condition envelope in `operating_conditions`. The current metric links cover pump failure rate, stationary function availability, pump mean repair time and physical thermal time to limit. These metrics retain their particular scope and do not constitute a quantified battery-fire risk or railway service-availability claim.

`allocation_results` reports `met`, `not-met` or `undetermined`, with result bounds, numerical outcome and reasons. The required `all-bounds` treatment returns `met` only if the entire supported range satisfies the target, `not-met` if the entire range fails, and `undetermined` if it overlaps. Unsupported input evidence, unknown dependencies, unvalidated physical predictions, missing targets or mismatched units/conditions cannot establish `met`. Every result other than `met` remains an acceptance blocker, even after target review.

Quantitative methods

Question	Method	Practical limits
Time after cooling loss	Analytic lumped heat balance, `C dT/dt = Q - UA(T-Ta)`, with corner sensitivity and temperature trajectories	Constant heat/conductance/ambient; no hydraulic, hotspot, spatial, phase-change or runaway model. Non-crossing cases stay explicit.
Failure combinations	Exact Boolean state enumeration and minimal cut sets	Shared/repeated events count once. Unknown software/process events retain uncertainty, with no invented hardware rates. Maximum 16 primitive events.
Repairable availability	Same Boolean structure with stationary down fraction `lambda*MTTR/(1+lambda*MTTR)`	Independent exponential failure/repair processes and explicit common causes. Queue competition, wear-out and logistics are outside this availability screen.
Latent exposure	Exact time-average undetected probability under periodic proof tests; invert against an exposure budget	Constant hardware rate, perfect detection/restoration and uniform demand timing. Actual coverage, repair exposure and demand need evidence.
One braking channel unavailable	Constant effective deceleration and reaction delay against available stopping distance	Grade, adhesion, brake development, target timing and operational authority must support actual restrictions.
Spares and repair capacity	M/M/c mean queue wait and Poisson replenishment pipeline	Concurrent crews are not roster headcount. Shift cover, skills, correlated failures and stockout-driven downtime need additional modelling. This does not prove service availability.
Design versus cost	Declared failure-rate and residual-cooling alternatives	Cost/performance inputs are illustrative study units. Overlapping uncertainty does not select a supplier or release a design.

The exponential/HPP assumption follows the NIST reliability handbook. Boolean cut-set reasoning follows the NASA fault-tree handbook. These methodological references are not railway conformity decisions. The thermal, braking and queue screens are tested against analytic cases.

Every interval has `low`, `high`, `unit`, `basis` and `evidence_ids`. Basis is `unknown`, `illustrative-assumption`, `supplier-data` or `measured`. Unknown inputs have null bounds. Supplier/measured parameters require controlled, hashed

evidence with review references and validity dates in `input_evidence`. Corner ranges are sensitivity envelopes, not confidence intervals or validated physical bounds. Without a declared primitive-independence assumption, probability and availability quantification remain unresolved.

Controlled rig programme

Freeze selected pump, connection, circuit, sensors, controller and representative thermal load with supplier identities and serials. Bind the as-built rig to the engineering configuration fingerprint. Record the expected serials in `rig.specimen_serials`; every run must observe those same specimens. Define duty, ambient, flow, pressure and instrument calibration before execution. Fault injection, containment, independent shutdown and protection arrangements require competent review and a recorded hold point.

Test	Observations
Normal duty	Temperature and flow at allocated operating points
Pump seizure/circulation loss	Flow loss, detection delay, physical isolation and thermal response
Leak/pressure loss	Pressure, leakage response, detection and protection
Pump fault plus flow reading stuck high	Remaining diagnostics/protection versus the actual fault
Charging, high ambient and fouling	Representative duty and isolation within the approved envelope
Shared auxiliary loss	Circulation, primary diagnostics and independent protection/integration response

Each test has a controlled procedure revision, requirement/scenario links, acceptance limits, duration, repetitions and model-error tolerance. Values are intentionally unresolved in the reference programme. Review must establish that the rig, duty and sampling represent the claim.

Raw measurements use UTF-8 CSV with fixed units/channels:

```
time_s,temperature_c,ambient_c,flow_l_min,pressure_kpa,fault_active,detected,isolation,derate
```

Times strictly increase; values are finite; digital channels are 0/1. `fault_onset_s` records commanded onset. Fault scenarios require pre-onset normal operation: inactive fault, diagnostics and protection, reviewed minimum flow and baseline duration. `preconditions.digital_state`, `min_flow_l_min` and `minimum_duration_s` declare that starting state. `response_sequence` declares the required ordered rising transitions after observed fault onset, normally diagnosis followed by isolation. A signal already asserted before the fault, missing baseline, absent transition, reversed order or transitions recorded in the same sample cannot demonstrate that sequence. Such records retain failed validation and null delays; an already-isolated rig does not prove fault response.

Every test includes a reviewed `maximum_sample_gap_s` limit. Actual sample gaps must meet it, and the declared resolution cannot exceed the tightest response timing criterion. Normal-operation tests cannot include an injected fault or asserted protection. `response_validation` preserves transition times and findings; reported delays are observed upper bounds from commanded onset, without sub-sample interpolation. Sampling and baseline criteria remain null until competent review determines the resolution and operating state needed. `isolation` must mean a physical energy-isolation observation, rather than a controller request.

Each `measurement_runs` entry includes run/test ID, CSV path/hash, configuration fingerprint, rig and all observed component serials, physical/synthetic origin, operator, timezone-aware timestamp, procedure revision, fault onset, channel-to-instrument provenance and signal semantics. Calibration records include instrument ID, certificate path/hash, `valid_from`, `valid_until` and temperature uncertainty. `performed_at` is acquisition start; CSV times are elapsed seconds from it. Calibration must cover the entire acquisition interval, including the last sample. Date-only certificate bounds include the full UTC day; explicit timestamps require a timezone. Later expiry does not invalidate a measurement that was valid at acquisition. Missing starts, future calibration, expiry during acquisition and changed certificate bytes remain blockers.

`calibration_events` separately records retrospective instrument findings with ID, `instrument_id`, `discovered_at`, `affected_from`, `affected_until` and reference. Events match the certificate's `instrument_id` (or its record ID) and invalidate overlapping runs. `calibration_impact` lists the affected runs; non-overlapping runs retain their original calibration result. New valid measurements are needed to replace affected evidence; deleting a finding is not a revalidation process. Mismatched configuration, missing physical identity and unresolved/failed criteria stay blocked. Synthetic fixtures exercise the software and cannot qualify a physical assembly.

Close the prediction and measurement loop

1. Agree requirement-derived criteria and the assessment approach.
2. Freeze rig configuration, procedure, model parameters and valid envelope.
3. Predict normal/fault temperatures and retain source/input hashes.
4. Capture raw physical temperatures, flows and protection signals.

5 Record maximum model discrepancy, including measurement uncertainty.

6 Investigate differences and revise the design/model under change control.

7 Repeat affected tests against the revised frozen configuration.

8 Validate on separate held-out runs and obtain independent review.

Per-test `thermal_inputs` can define different duty/fault envelopes; otherwise the compiler uses `physical_thermal`. Those physical parameters are initially unknown. The compiler compares supplied parameters and does not fit them to validation data. Calibration and validation run IDs must be disjoint, with held-out coverage of every declared normal/fault case.

Production and operational assurance

Critical characteristics allocate torque, flow, pressure and diagnostic latency to operations and inspection hold points. Production batches bind supplier batches, serials, process revision, configuration fingerprint, per-serial results and original inspection evidence. The same checks apply to assemblies 2, 20 and 200. Unmeasured values, different processes/configurations and synthetic records remain findings.

NCRs retain affected serials, disposition, independent review and requalification status. Replacement rules bind compatible revisions to supplier evidence and maintenance instructions; design changes reopen compatibility. Operational records retain asset serial, failure mode, corrective action, effectiveness and engineering handback. Administrative closure does not establish handback.

The records use existing manifest and ERP identities. This milestone imports and checks dossier data and exports a review package; it does not create ERPNext production orders or approve supplier dispositions.

Scoped readiness and signed decisions

Workbench recompiles current inputs through a read-only endpoint. It shows configuration, accepted use, changes, six decision areas, uncertainty, measurement gaps and manufacturing findings. Reference evidence is explicitly separate from a city's acceptance. A real package for another city/environment is withheld from that view.

Existing Ed25519-signed reviews can be verified against an operator-controlled policy passed through `--review-policy`, or the server's `OSR_QUALIFICATION_REVIEW_POLICY`. The envelope schema is `osr-subsystem-review/1`, with reviewer, authorized role, disposition, reference, validity date, exact evidence/configuration fingerprints and accepted use where appropriate. The policy enables reviewers, authorizes roles and pins public-key hashes. Signers must differ from the package owner, executing identities and exclusion authors.

The qualification subject's assurance scope is derived from its contained components and dependencies, then followed through the graph to propagated failure analyses, requirements, evidence, applicability obligations and dependent gates. `assurance_scope.additional_record_ids` can widen this scope. Every relevant graph blocker feeds the corresponding readiness area and prevents subsystem acceptance. The report retains the complete record list, graph fingerprint and individually identified findings.

An exclusion is a review proposal, not a way to silently remove a finding. `assurance_scope.exclusions` must identify the exact finding ID, record ID and current graph fingerprint, with rationale, accountable owner, author, distinct reviewer and review reference. Both authenticated assessor and design-authority envelopes must explicitly list the finding in `approved_exclusion_ids`; the assessor identity and reference must match the exclusion's accountable review. Missing approvals or changed baselines leave the blocker open. A scoped exclusion does not accept the graph's railway gate or establish a wider operating release.

Integration records use `kind` values `hil`, `vehicle`, `infrastructure` and `operations`. Each area requires its own configuration-bound evidence; a passing HIL record alone leaves the other areas unresolved.

Source, measurement, criteria, configuration and process changes invalidate the signed scope. Assessor and design-authority decisions are both required. Review pointers are excluded from the signed evidence fingerprint so adding the signature does not change its own subject. Every readiness area must also have current evidence before supplied subsystem acceptance can be displayed. Vehicle/railway authorization remains separate and `release_ready` stays false. No trusted reviewer policy or acceptance signatures are supplied by default.

Reproduce and export

```
python3 tools/automation/subsystem_qualification.py
python3 tools/automation/subsystem_qualification.py --check
python3 tools/automation/subsystem_qualification.py --stdout
python3 tools/automation/subsystem_qualification.py --plan path/to/definition.json --stdout
python3 tools/automation/subsystem_qualification.py --export build/qualification/cooling-dossier.zip
python3 tools/automation/digital-assurance.py
python3 tools/automation/component_assurance.py
```

The export contains the effective definition, readable results and original checksummed input/evidence bytes. It checks for changes during export.

To close the physical milestone, supply the selected supplier/rig definition, agreed targets/criteria, calibration certificates, normal/fault measurements, production inspections, held-out validation and independently reviewed results. The current software makes those missing inputs explicit and provides the executable assessment process.

System description

Repository source: `docs/certification/system-description.md`

1. What OSR is

OpenSourceRail is an open-source urban-rail engineering and research stack — software, control-electronics reference designs and operational rules — with a target GoA 4 architecture. It is not a certified train-control product. The first physical pilot uses conservative sectional protection, independent occupancy detection and a separately qualified safety channel while OSR runs in shadow and then supervised mode. Each deployment follows the same unified deployment model: shared OSR stack, city-specific inputs, generated design outputs, and deployment-specific evidence. The Samawah instance (RFC 0003) applies that model to a brownfield rail-yard/workshop context (RFC 0027).

The stack is the entirety of a modern metro:

- Rolling stock — the frozen LM3 three-car first article is the current engineering focus; the other consist families remain planning variants.
- Track — four geometry presets per RFC 0009 covering gauge 1435 mm, radius, grade, cant, and rail profile.
- Civil — at-grade + elevated + bridge only per RFC 0011; no tunnels.
- Stations — six archetypes per RFC 0010, prefab steel- portal + solar-canopy with no station building.
- Signalling — conventional sectional pilot protection with independent detection; distributed consensus-based control remains a staged research target under RFC 0001.
- Energy — catenary-free; onboard LFP batteries + station/depot PV per RFC 0002.
- Operations — one shared ≤ 60-page rulebook across four role families per RFC 0013.

For the intended assurance case, the relevant research subset is partitioned across the following onboard and wayside host classes. SIL labels express the development target, not achieved integrity or certification:

- T-ECU/S — onboard safety kernel (2× RP2350 + RPi CM5 in 2oo2), runs `osr-atp`, `osr-brake`, `osr-derailment`, `osr-door-control`, `osr-fire-safety`, `osr-odometry`.
- T-ECU/A — onboard application tier (RPi CM5), runs `osr-ato`, `osr-tcms`, `osr-event-recorder`, `osr-tcn`.
- T-OBS — onboard obstacle detector (RFC 0015), runs `osr-obstacle-detect`.
- W-SBC — wayside application-compute candidate; a Radxa CM5 is one development host, not the selected safety controller. It runs `osr-interlocking`, `osr-consensus`, `osr-wayside-points`, `osr-intrusion-detect` (RFC 0016).

2. System boundary

2.1 Intended assurance scope

The repository structures evidence for the following future claims; it does not itself certify them:

- Onboard safety chain — from sensor-fusion position through ATP envelope computation to brake command actuation.
- Wayside safety chain — from track-occupancy + switch- observation + intrusion detection through MA computation to MA delivery to the train.
- Distributed signalling consensus — the Raft-derived target protocol, initially evaluated only in simulation and pilot shadow mode.
- Obstacle detection + intrusion detection — the RFC 0015 onboard sensor fusion and RFC 0016 wayside sensor fusion that together substitute for a driver's-eye detection.
- Operational rules — the RFC 0013 rulebook that binds the safety-case solutions to human procedures (dispatcher, station staff, maintenance, OCC).

2.2 Outside the repository assurance boundary

- Crashworthiness — EN 15227 certification is the rolling- stock builder's scope (RFC 0008 §3).
- Fire-protection material performance — EN 45545 is material-certification scope.

- Civil structure — structural engineering of elevated viaducts and at-grade formation is the deployment partner's civil scope (RFC 0011).
- Electrical supply — MV/LV distribution from the grid connection to the rectifier substation is the deployment partner's electrical scope.
- Cybersecurity above SIL-2 — IEC 62443-4-2 message authentication is in tree at SIL-2 (RFC 0017) and integrates as complementary evidence; component-level cybersecurity certification is tracked separately from EN 62267.

3. Operational envelope

- Maximum civil speed: 80 km/h (22 m/s) at `standard-urban` preset; 100 km/h on `mainline-mixed`.
- Minimum curve radius: preset-dependent per RFC 0009 §1.
- Maximum gradient: preset-dependent per RFC 0009 §1.
- Climate envelope: −10 °C to +55 °C ambient per RFC 0003 §4.5 (deployment-adaptable).
- Crashworthiness envelope: EN 15227 Cat C-II up to 25 km/h metro-to-obstacle collision.
- Communication envelope: 5G primary + LoRa backup for TRG; Cat 6a + TSN for wayside back-office (RFC 0005 §5).

Operation outside any of these envelopes triggers a degraded- mode response per RFC 0013 §5 — the deployment continues safely at reduced performance; no passenger is carried under a violated envelope.

4. Interfaces to the external world

- Passengers — at station platforms (PSDs per RFC 0010) and inside the trainset (seats, doors, emergency intercoms per RFC 0015 §5.3).
- Track workers — via `MaintenanceOverride` grants issued from OCC per RFC 0013 S5; physical protection signals per RFC 0013 S5.3.
- Emergency services — via OCC radio liaison per RFC 0013 C2.4; passenger emergency intercom direct to OCC per RFC 0015 §5.3.
- National safety authority — via the OCC incident- notification path per RFC 0013 S3.5 (I5 within 2 h, I6 immediately).
- Power grid — at each rectifier substation, bidirectional per RFC 0002 §6.
- OSM + Civil 3D / OpenRail / Trimble — design-time only, via the OSR-ALN alignment interchange format (RFC 0009 v2).

5. Non-interfaces

OSR does not interface with:

- Third-party CBTC zone controllers — RFC 0001 replaces them with distributed consensus.
- Magnetic smartcard fare systems — RFC 0013 and `osr-afc` use account-based mobile money + QR.
- GSM-R radio — RFC 0007 uses 5G + LoRa.
- Catenary power distribution — RFC 0002 is onboard- battery-only.

Rolling-stock energy and fire boundary

RFC 0021 defines the promoted train as three electrically self-contained cars. Each has a 225 kWh gross LFP pack on a 650–700 V nominal DC link, two independent motor controllers, isolated LV DC/DC domains, direct-HV DC HVAC, roof MPPT, local pack contactors/insulation monitoring, and an independent battery water-mist system. The train exposes module/string hazard identity but only credits isolation at the level proved by the commissioned pack topology.

The fire controller produces distinct charge-inhibit, car-HV-isolate, traction-restrict, OCC-alert, controlled-safe-platform-stop, and emergency- brake outputs. Traction and HVAC electrical enclosures have detection and isolation but no automatic suppression. The passenger saloon has neither automatic sprinklers nor CO2 flooding.

Stations provide a 500 kW DC/DC cabinet shared by up to two interlocked platform contacts and stationary LFP in repeated 500 kWh gross modules. A non-isolated candidate converter does not remove the deployment integrator's responsibility for isolation/earthing analysis, insulation monitoring, contactors, touch-safe sequencing, enclosure cooling, and emergency isolation.

A deployment that must interface with any of the above forks the repository — no upstream compatibility is offered.

6. Quantitative footprint

Metric	Value
Rust crates in the workspace	55
Total Rust tests, current workspace	753
SIL-4 evaluators with Kani harnesses	7 (`osr-atp`, `osr-brake`, `osr-odometry`, `osr-wayside-points`, `osr-interlocking`, `osr-obstacle-detect`, `osr-intrusion-detect`)
GSN goals currently closed against evidence	G1–G27 across 8 TOML files under `docs/safety-case/gsn/`
Operations rulebook page count (target)	≤ 60
Operations rulebook role families	4 (dispatcher, station staff, maintenance, control centre)

City soil inputs and deployment evidence

Repository source: `docs/city-deployment-evidence.md`

The catalogue now materialises the survey/civil readiness chain and native/SUMO journey-time comparison for every city. Each city's `engineering/deployment/` register identifies the remaining work, responsible function, closure evidence and current result. The catalogue index aggregates actual open gates. Generated documents do not count as received surveys, completed construction or operating approval.

City workforce and design development adds per-city organisation, skills/tasks, construction work profiles, joint and finish maintenance, and climate-informed movement calculations.

Soil inputs used for civil planning

The historical extraction before the current core-layout regeneration covered 67,033 scope locations: all 10,215 station features and 10,597 civil segments, using 54,594 distinct coordinates. That extraction had complete profiles at 64,805 locations; 2,228 had missing values, spread across 126 cities. Current geometry and soil samples are bound together in each city's `engineering/soil/summary.json`; historical coverage totals do not describe the regenerated layouts. Repeated coordinates at shared scope boundaries are sampled once and retained against each applicable station or segment.

OpenLandMap soilDB supplies shallow pedological predictions, including uncertainty. The importer uses its published 2020–2022 mean, p16 and p84 layers at a common 120 m resolution, for clay, sand, silt, fine-earth bulk density, organic carbon and pH. Depth intervals are 0–30, 30–60 and 60–100 cm. The nominal p16–p84 interval describes prediction uncertainty; it is not an engineering characteristic value or a range of proven site conditions.

The source lock pins the upstream Git revision and 54 layer URLs. The original layer catalogue is retained with its hash. HTTP range requests read only relevant raster blocks. Each city retains raw sampled values in engineering units, source ETags, raster scales, nodata values, coordinate system and input/output hashes. The dataset is attributed to Hengl et al., DOI 10.5194/essd-2025-336, under CC BY 4.0.

Sample locations include every station, each civil segment's ends and midpoint, and additional locations with no more than 1 km of mapped route between them. The route and station GIS layers determine locations; one city-centre value is not extrapolated across the railway. Raster pixel size and route sample spacing are separate quantities. Native raster pixels are sampled without smoothing or nearest-valid-pixel substitution.

The upstream bulk-density filename encodes g/cm³ and its TIFF scale is 0.01, while the catalogue unit column says kg/m³. The importer applies the TIFF scale once, then multiplies by 1,000 to report kg/m³: raw 150 becomes 1,500 kg/m³. Soil pH is scaled once to ordinary pH units. Nodata becomes an empty CSV value and an explicit coverage flag. The source excludes deserts and permanent ice; no-data can also occur locally within mapped cities.

Each city receives:

File under `engineering/soil/`	Use
`samples.csv`	Location, chainage, depth, six properties and three statistics
`source-receipt.json`	Input and sample hashes, source URLs and raster metadata
`sample-locations.geojson`	GIS map with location-specific investigation triggers
`civil-investigation-plan.json`	Every station and civil segment mapped to samples and field work
`summary.json` and `README.md`	Coverage, units, triggers and modelling boundaries

The civil plan prioritises plasticity/shrink-swell tests when the clay upper prediction reaches 35%, moisture/erosion review when silt reaches 50%, granular density/groundwater testing when sand reaches 70%, organic/compressibility tests when organic carbon reaches 50 g/kg, and durability testing when the pH lower prediction is below 5.5. These are transparent OSR investigation triggers, not regulatory thresholds, soil hazard diagnoses or foundation selection rules. High sand content alone does not establish liquefaction; clay content alone does not establish swelling. Texture fractions remain independent predictions.

Bearing capacity, CBR, friction, cohesion, groundwater, contamination, sulfate/ chloride attack and deep stratigraphy are not supplied by these maps. Those fields remain unknown. SoilDB values inform investigation locations and lab scope; they do not manufacture boreholes or automatically resize foundations, change excavation quantities or increase civil CAPEX.

The Samawah soil report and deployment register show the complete handoff. The drainage/ground readiness report references the soil summary and investigation-plan hashes while retaining its field-evidence gate.

Other gaps repaired

- Survey/civil readiness and operations cross-checks were present for only two cities. They are now generated for the complete catalogue.

- Survey and alignment regeneration could overwrite received manifests. Existing receipt files are now preserved; their contents are still validated.
- Most retained simulation summaries lacked native reference timing fields. A separately generated, source-bound native kinematic report now supplies them. Its one-second execution extracts the reference calculation; it is explicitly not a full service or endurance replay.
- SUMO evidence is refreshed with current station dwell times before comparison. Missing or stale timing inputs cannot silently pass, and the comparison remains separate from conflict-aware capacity and operating acceptance.
- The provenance review exposed 264 full-service reports bound to older scenarios or validator inputs, with all 266 predating the current simulator binary. Current timing comparisons are separate from the full-service freshness gate; stale service acceptance is not carried forward. Following the 2026-09-10 shared-source and charging-headroom corrections, Samawah and Mosul pass fresh two-run nominal validation and all eight degraded cases each. The 2026-09-11 expansion repeats that complete suite for Uige, Quelimane, Edea, Bukavu, Soroti, Sheikhupura, Sumbawanga and Tartus: ten cities now have current passing full-service evidence. Six expanded scenarios now explicitly declare the shared trainset systems contract that they previously inherited from simulator defaults; operating values are unchanged. Their dependent hybrid, engineering and deployment reports were regenerated. The other 256 cities retain an explicit full-service replay-refresh gate; earlier passes remain bound to their original binaries. Continuous two-day station/depot replay now passes in nine cities: Samawah, Uige, Quelimane, Edea, Bukavu, Soroti, Sheikhupura, Sumbawanga and Tartus.
- Soil, timing and deployment evidence is included in city package inventories and source-hash checks. Regeneration keeps these outputs connected to the inputs.

Reproduce

```
# Fetch the pinned soil layers for the complete current catalogue.
.venv/bin/python engineering/analysis/city_soils.py --all --fetch --jobs 6

# Regenerate soil reports offline from retained, hash-checked samples.
.venv/bin/python engineering/analysis/city_soils.py --all

# Build native references, refresh civil gates and compare current SUMO evidence.
.venv/bin/python tools/automation/generate-deployment-evidence.py --all --reuse-sumo

# Rebuild the catalogue deployment summary.
.venv/bin/python engineering/analysis/city_deployment.py --all
```

For one changed city use `--design <city>/design.toml`; add `--fetch-soils` to the deployment command when soil sample locations or source inputs have changed. The normal city and full-package regeneration workflows include this step. Existing receipt manifests survive regeneration. Newly required deliveries or changed alignment scope appear as unresolved receipt checks until supplied.

Remaining site work includes survey/land/utilities, geotechnical investigation, local hydrology, support and erection design, physical stabling and depot layouts, installed-cost reconciliation and operator acceptance. Remaining operating-model work includes the solar/storage replenishment duty and continuous stabling evidence beyond the nine retained two-day replays. Per-city fleet shortfalls remain explicit; neither interline transfers nor a default grid upgrade is introduced to hide them.

OSR City Studio

Repository source: `docs/city-studio.md`

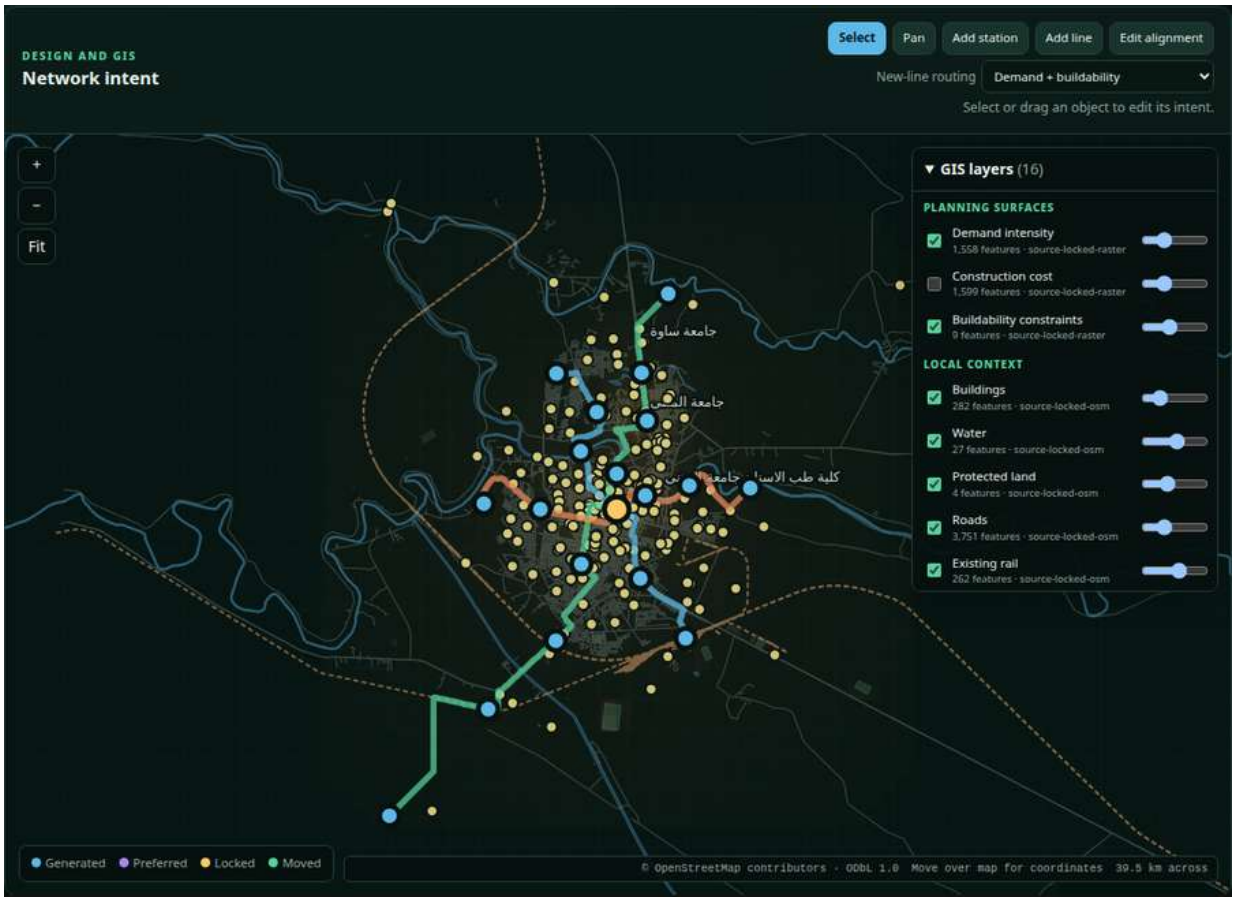
OSR City Studio is the Git-backed design and service-planning interface for OpenSourceRail. The initial vertical slice loads Samawah, displays its geographic corridors and stations, records station locks or movements, configures service by line/day/time, calculates fleet and capacity screens, authors period-specific origin–destination demand and screens it against scheduled line capacity, persists civil construction-system choices and their derived bearing/joint interfaces, compiles a deterministic candidate, and materializes a revision for GitHub review.

The civil panel controls the 20/25 m standard span, four/five-span expansion unit, reinforced-soil approach height, long-open slipform and constrained ST6 methods, maturity-controlled mould-cycle target, and comparison of road grade-separation alternatives. Values are validated, stored in `project.osr.toml`, included in the candidate revision hash, and restored after restart. The Studio displays the shared design-derived civil cost contract. Authoritative CAD quantity changes can regenerate that contract and every downstream city price; changing project method intent alone does not claim a new rate unless the calibrated driver model maps that input to a quantity.

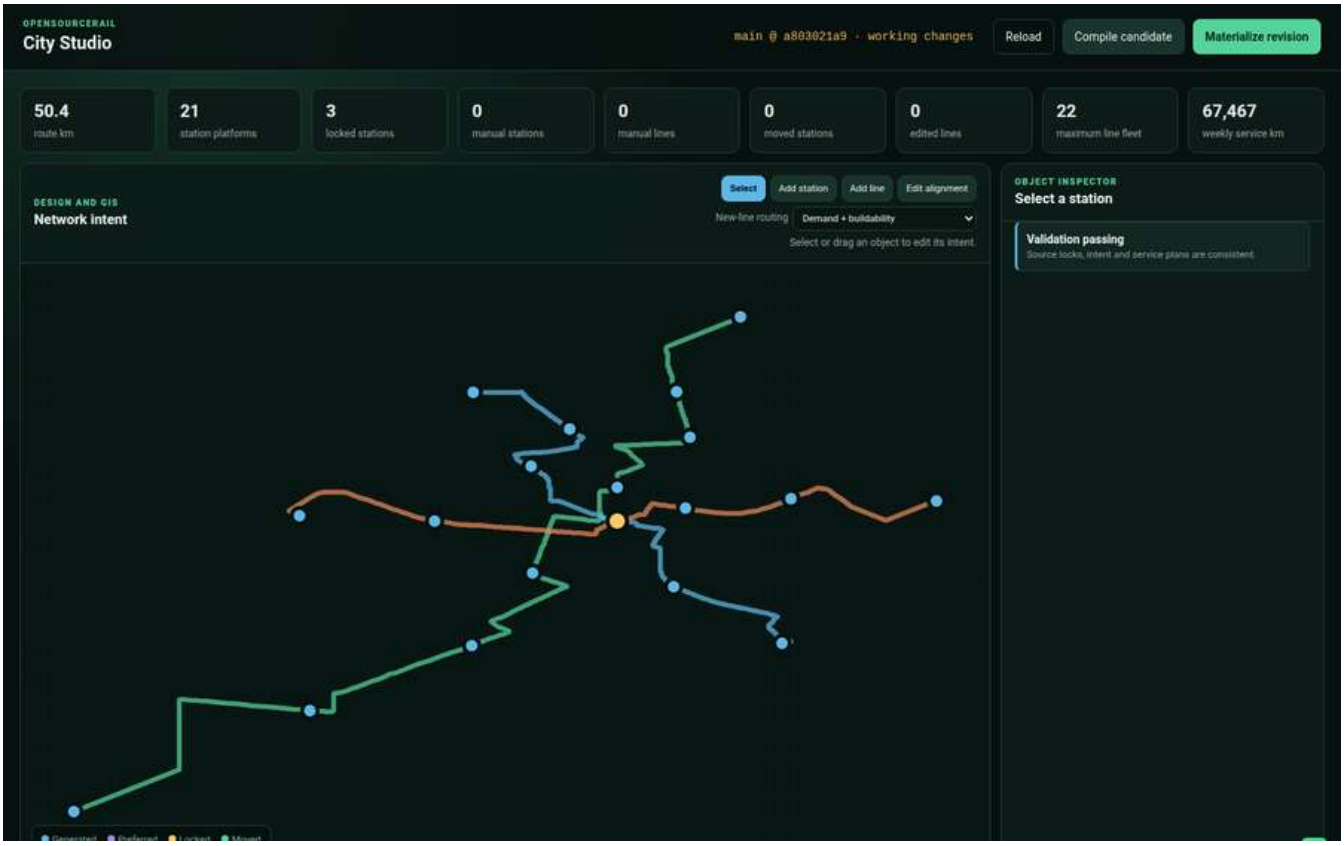
The controlling design decision is RFC 0031.

Declared cross-line interchange complexes appear as one junction on the map. Select a junction to expose its individual platforms for editing, or choose a platform directly from the station picker. Smaller dots retain the actual platform locations; connecting strokes identify complex membership. Platform coordinates and line chainages remain controlled separately.

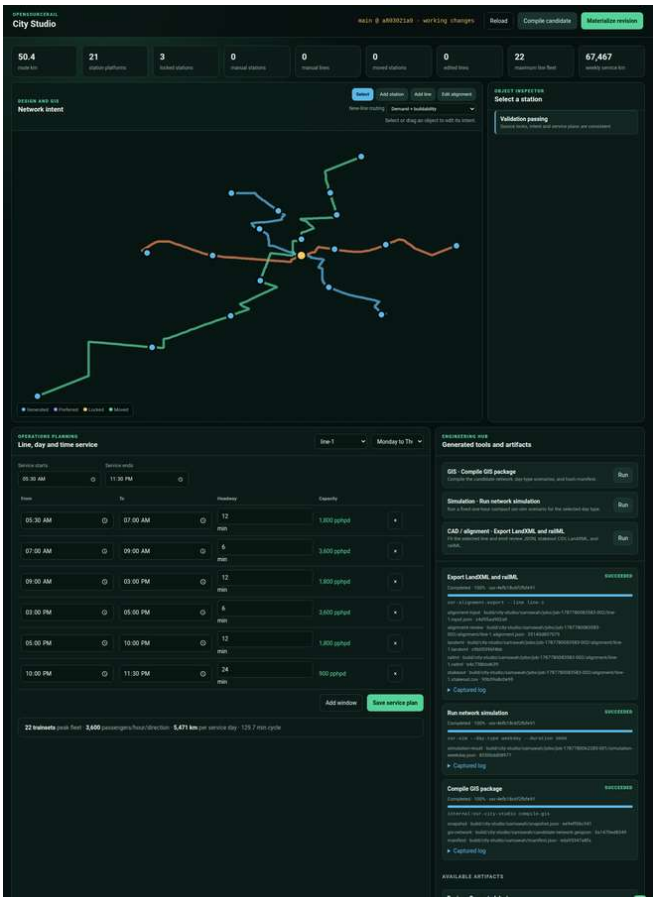
Interface



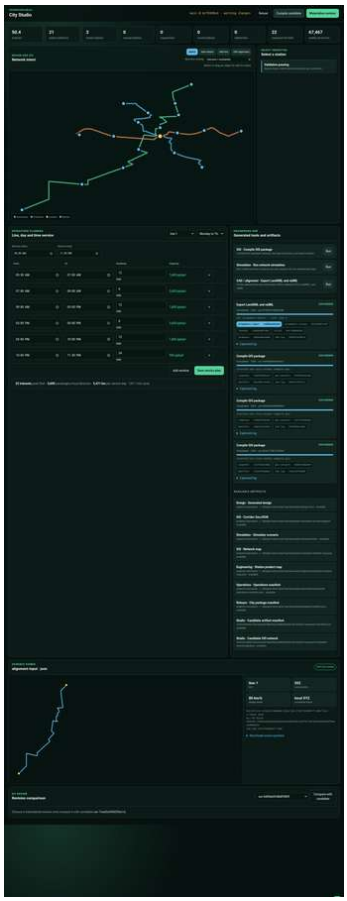
City Studio deterministic layered GIS workspace



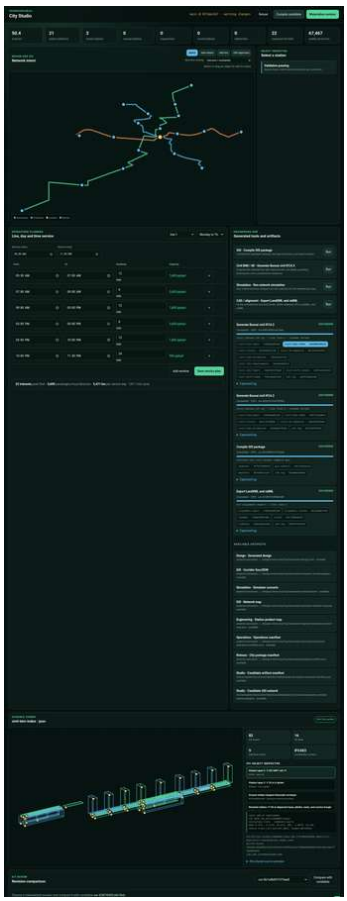
City Studio network authoring, routing strategy, validation, and revision controls



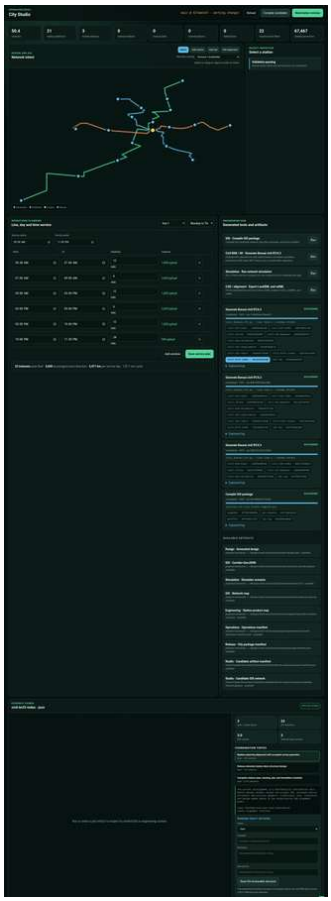
City Studio line/day/time service planning and controlled engineering jobs



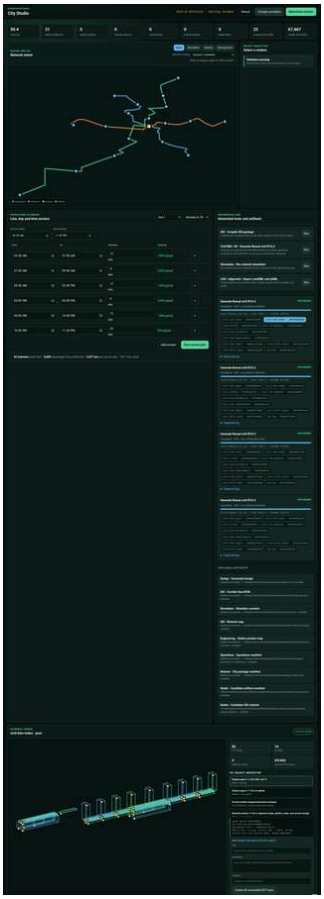
City Studio SHA-256-verified alignment and GIS evidence viewer



City Studio SHA-256-verified IFC object inspector with IDS and BCF evidence



City Studio BCF coordination decision form backed by Git-reviewable project intent



City Studio creating a deterministic BCF topic from a selected IFC asset

Browser-tested complete City Studio with civil construction controls, OD capacity planning, bulk service controls, interactive civil 4D review, approval history, and multi-asset IFC selection

Run

From the repository root:

```
cargo run -p osr-city-studio -- serve
```

Open http://127.0.0.1:8090/.

Use another project or port with:

```
cargo run -p osr-city-studio -- \
  --project cities/workspaces/samawah serve --port 8091
```

The server binds only to localhost by default. The initial interface has no authentication and must not be exposed as a shared or public service.

To carry revisions into simulation, OCC training and Ops Core, use the same-origin Workbench instead of starting Studio alone.

Command line

Validate source locks, station intent, calendars, all line/day plans, and OD demand references:

```
cargo run -p osr-city-studio -- validate
```

Write a deterministic working snapshot under build/city-studio/samawah/:

```
cargo run -p osr-city-studio -- compile
```

Write cities/workspaces/samawah/revisions/osr-<hash>.json:

```
cargo run -p osr-city-studio -- revision
```

Inspect the branch, parent commit, and uncommitted paths:

```
cargo run -p osr-city-studio -- git-status
```

List immutable project revisions or compare one with the working candidate:

```
cargo run -p osr-city-studio -- revisions
cargo run -p osr-city-studio -- compare osr-1f41358e43a86600
```

Validate every committed city project:

```
python3 tools/automation/validate-city-projects.py
```

Install through the root one-command setup, then run every browser front end with `npm run test:frontend`.

For the 122-check Studio-only workflow, use `node tools/automation/test-city-studio-gui.mjs`. Playwright uses one worker, a fixed viewport, locale and timezone, and a disposable project. Its JSON report and screenshot are written to `build/gui-acceptance/`; committed Samawah inputs are not edited.

Revision workflow

1. Create a branch.
2. Use the Studio to edit station intent, service plans, and OD demand.
3. Compile and resolve every validation error.
4. Materialize a revision.
5. Review the Git diff, including semantic changes in the revision JSON.
6. Commit the project inputs and revision together.
7. Push and open a GitHub pull request.
8. Append the reviewer decision with its PR/evidence reference.
9. After approval and merge, create the suggested protected or signed tag.

Materializing a revision does not run git add, commit, push, or contact GitHub. Remote repository changes always remain an explicit user action.

Current editing scope

Implemented:

- source hash locks;
- an offline layered GIS workspace with pan, zoom, fit, coordinates, scale, feature inspection, visibility and opacity controls;
- content-hashed local roads, buildings, water, protected land, existing rail, demand, construction-cost, buildability, destination, civil, energy, depot, interchange, issue, published-network and live-candidate layers exposed through a deterministic GIS API;
- stable station ids;
- generated/preferred/locked/retired station intent;
- drag-to-move stations;
- click-to-create manual stations with stable content-derived ids;
- manual station naming, archetypes, movement, and retirement;
- two-click manual line creation with stable endpoint-derived ids;
- automatic terminal platforms and day-type service plans for new lines;
- coordinated manual-line naming, alignment editing, and retirement;
- selectable source-locked demand/buildability routing or explicit direct planning chords for new manual lines;
- line-level routing provenance, demand weight, and raster source ids in the candidate, GIS export, and immutable revision;
- locked-anchor-aware local corridor regeneration after station movement;
- click-to-create and drag-to-edit alignment control points;
- weekly day-type calendars;
- per-line contiguous time windows and headways;
- one-operation headway adjustment across every time window for the current line or every active route, using one validated atomic write, plus complete service-plan copying between day types;
- indicative cycle, fleet, capacity, daily and weekly service metrics;

- source-controlled planning periods and OD flows with deterministic stable IDs, active-station/day-type validation, create/edit/delete controls, and a conservative capacity/utilization screen over every overlapping service window;
- validation findings;
- deterministic candidate and revision hashes;
- in-GUI semantic comparison of station, line, service, demand, coordination, per-line IFC survey control, and summary changes;
- append-only approval/changes-requested decisions tied to existing immutable revisions, including reviewer role, decision date, rationale, and review or pull-request reference;
- day-type-specific simulator scenarios and a hash-addressed artifact manifest;
- allowlisted field-evidence brief, GIS compilation, one-hour simulator, and LandXML/railML alignment jobs plus IFC4.3 civil federation with persistent progress, command display, captured logs, exit state, and SHA-256 artifact records;
- an integrated evidence viewer for GeoJSON, alignment JSON, LandXML, railML, stakeout CSV, civil IFC object indices/raw STEP, simulator JSON, manifests, snapshots, and captured logs;
- searchable IFC asset inspection and deterministic coordination topics that can select several civil assets in one issue;
- reusable IFC type counts and stable type identity in the civil artifact metrics, search, and object detail;
- native IFC material-family counts, source authority, and visibly unresolved grade/design status in the civil artifact inspector;
- native section-profile counts, profile-derived rail extrusion coverage, and review-versus-mill geometry status in the artifact inspector;
- native IFC presentation-layer and coordination-group catalogues, membership, semantic boundaries, and visibility filters;
- native qualitative interface-constraint intent, current evaluation, source, scope, and explicit no-fabricated-metric boundary;
- native OSR property/quantity dictionaries, applicability, typed fields, definition linkage, and custom-versus-standard naming boundary;
- native IFC4.3 horizontal and vertical planning segments, gradient-curve representation, stationing, and explicit unresolved transition/cant status;
- a native USD SCHEDULE0FRATES containing the generated at-grade, elevated, and bridge planning rates, source hash, cost drivers, and no-estimate boundary;
- hash-verified 4D sequence loading alongside the IFC object index, with federation rotation, native layer visibility, stage scrubbing/playback, and task/QA-hold context;
- visibility of existing GIS, engineering, simulation, operations, and release artifacts.

Next:

- IDS editing and a general-purpose BIM editing environment;
- capacity-constrained passenger boarding, observed OD calibration, and platform/interchange pedestrian capacity and evacuation analysis;
- object-aware Git merge assistance.

Approval records

The Git review panel can append an Approved or Changes requested record for an existing `osr-…` revision. Records are stored in `cities/workspaces/<slug>/approvals/reviews.toml` and receive deterministic `approval-…` IDs. Submitting identical content is idempotent. Decisions cannot refer to a missing revision and are not edited or deleted through City Studio; a later decision is appended as additional history.

Approval metadata is deliberately excluded from the design content hash. The record refers to an already immutable revision, so adding the decision does not manufacture a different design merely because that design was reviewed. The approval file, review/PR reference, and revision remain ordinary GitHub review material.

Map authoring

The map has five explicit tools. Select inspects and drags existing objects or reads GIS properties. Pan moves the viewport; wheel or +/− controls zoom and Fit restores the project extent. Add station inserts a manual station where a line is clicked, assigns a stable id, and opens its name/archetype inspector. Add line takes two endpoints and uses the selected routing strategy. Demand + buildability snaps each endpoint to the nearest feasible cell and runs deterministic least-cost search over the project's locked planning bundle. Direct planning chord remains an explicit fallback. Both strategies create two terminal platforms and copy the existing day-type policy into editable service plans. Edit alignment adds a local corridor control point to either generated or manual lines. Manual station/line additions and retirements rebuild both the candidate GeoJSON and the station, line, fleet, dispatch, and ordered topology records in every generated simulator scenario.

The layer catalogue comes from `/api/gis/manifest`; each layer is retrieved by stable ID from `/api/gis/layers/<id>` with a SHA-256 digest, feature count, provenance class and default presentation. Planning rasters are aggregated into display cells at least 500 m wide, with at most 10,000 tiles per surface, without changing the locked 20 m routing inputs. Layer properties state both resolutions; aggregate water coverage cannot permit a station site. The map does not call a public tile service, so a revision or test cannot drift with a remote basemap. The Samawah bundle includes source-locked NASA/NGA SRTM elevation from the AWS Open Data Terrain Tiles service, derived terrain slope, and a retained water mask combining OSM geometry with independently sampled ESA WorldCover 2021 permanent-water classes. The source receipt retains the actual source tiles, grid samples and hashes. These drive visible likely bridge/viaduct segments, warn on steep station sites, and prohibit a station in any cell with detected water or unknown coverage. Deployment teams may add survey, parcel, utility or licensed local basemap extracts to the same locked project workflow; the included DEM, place anchors and planning surfaces are not survey evidence or a final vertical alignment.

The tracked Samawah context is regenerated from the pipeline's processed OSM snapshot with `python3 tools/automation/export-gis-context.py .cache/osr-pipeline/osm/samawah.json cities/workspaces/samawah/gis`. The exporter writes stable GeoJSON, ODbL attribution and fetch provenance, then refreshes the five matching entries in `sources.lock.json`.

The Samawah routing bundle is a committed 20 m planning surface retaining the pipeline's 20 m cost, demand, buildability, water, elevation and slope rasters. Every component, upstream terrain tile and derivation record is SHA-256 locked. This is appropriate for comparing planning alternatives, not survey or detailed civil design. Designers can shape either route strategy further with alignment control points.

The revision comparison panel compares any committed revision JSON with the current working candidate. It reports object additions, removals, movements, line length/station-count changes, and service/fleet/capacity deltas before a new revision is materialized. It also reports OD flow additions, removals, and passenger-rate changes.

Demand planning

`cities/workspaces/<slug>/demand/od-matrix.toml` defines named periods by service day type and clock interval, followed by optional station-to-station passenger flows. The GUI creates a stable `od-…` ID from the period, origin, and destination, while passengers/hour remains editable without changing object identity. Changes are written atomically, included in the candidate hash, and survive process restart.

The compiler assigns each flow to a deterministic shortest scheduled-time route. Adjacent stations form directed ride sections; rings include their closing section. Travel time uses the declared planning speed and station dwell, with half the conservative scheduled headway at boarding. Transfers require both stations to be tagged as interchanges. Co-located platforms within one metre add five minutes; platforms in the same explicitly declared junction group within 700 m add five minutes plus walking at the planning assumption of 80 m/min. Both add the next line's expected wait. Group access paths and measured walking times still require project verification. Nearby unconnected stations do not create a route. A line must have service for the whole period, including intervals crossing midnight.

Loads from every OD flow accumulate on each directed section. Opposing flows remain separate, and each flow's utilization is the highest shared-section utilization along its assigned route. The GUI shows the station sequence, journey time, section loads, boardings, alightings, transfer boardings and arrivals over one maximum scheduled headway. These outputs and their method participate in the revision hash; old snapshots deserialize with empty assignment fields.

This is an all-or-nothing planning assignment, not a ridership forecast or a capacity-constrained boarding simulation. Overloads remain visible; passengers are not silently dropped or assumed to board. Arrivals per headway exclude accumulated denied-boarding queues. Platform usable areas, directional circulation, transfer walking measurements, arrival variability and egress still require project-specific pedestrian analysis.

Controlled engineering jobs

The engineering hub exposes five fixed adapters:

- Issue field-evidence brief locks the current candidate revision and generates survey, utility, land, flood, geotechnical, workshop/fleet and reality-capture requirements, an explicitly unaccepted receipt manifest, and the corresponding survey-control, ground-model and per-line surveyed- alignment, route-fit, drainage/ground and structural readiness reports and empty controlled receipts;
- Compile GIS package publishes the candidate snapshot, GeoJSON, day-type scenarios, and manifest;
- Run network simulation executes a fixed one-hour compact `osr-sim` run for the selected day type and rejects invariant violations;
- Export LandXML and railML converts the selected candidate line to local engineering coordinates and runs `osr-alignment-export`, producing review JSON, stakeout CSV, LandXML, and railML;
- Generate Bonsai civil IFC4.3 combines the selected immutable line reference with the checked civil kit and produces native IFC4.3 objects, quantities, provenance, an object index, two typed `IfcVehicle/ROLLINGSTOCK` trainsets, 18 linked 4D construction tasks with 134 physical outputs and 45 separately identified virtual review interfaces, a twenty-specification IDS 1.0 audit, 36 native typed bearings and explicit virtual foundation/jacking interfaces, 27 native support connections with 60 bearing realizations, an internal 15-reference OSR asset classification, five native source-zone coordination groups, 15 hash-locked native source-document records including a native link to nine objectives and six numeric metrics, a nine-item external engineering decision register, six native functional systems (three specialized as `IfcBuiltSystem`), seven system-to-railway-part references, four native presentation layers, nine native interface constraints with six numeric `IfcMetric` benchmarks, three qualitative-only gates, and 107 scoped project/asset/group/system evidence links, 16 native property/quantity dictionaries, native planning alignment layouts and stationing, one native schedule containing three generated planning rates, and a BCF 3.0 package of open release issues with IFC GUID selections. If the selected line has accepted survey control in `[[civil.ifc_georeferencing]]`, the job also emits its native `IfcProjectedCRS` and `IfcMapConversion`; unmatched lines remain explicitly on the unresolved local grid. OSR remains the alignment/geometry authority; Bonsai is the downstream federation and detail-review environment.

Jobs serialize through one engineering slot. Their records, immutable evidence copies, and full logs live under `build/city-studio/<slug>/jobs/<job-id>/`; the browser displays status, progress, the effective command, a bounded log tail, and every output hash. Project edits serialize against the running job, and a queued job fails if its recorded revision has become stale. Interrupted records are marked failed when the server restarts. Adapter ids, arguments, durations, and binaries are allowlisted in Rust—the API never accepts an executable name or arbitrary shell text.

Portfolio delivery generation is exposed one level above this project editor: Workbench Generate a city digital twin can select any catalogue city and open its assembly, CPM, order, costing and cashflow result in Operations. City Studio remains the geometry/service editor for the currently opened workspace.

Artifact buttons open the evidence viewer. Before returning content, the server canonicalizes the path beneath the City Studio build root, enforces a 4 MB preview limit, rejects unknown formats, and recalculates SHA-256. GeoJSON and alignment/stakeout geometry and tessellated IFC solids are plotted directly. Individual civil objects expose their stable ID, IFC GUID, class, discipline, bounds, detail mode, source, internal asset class, and applicable hash-locked source documents. The searchable inspector also exposes the native classification, property-dictionary, alignment, planning-rate, interface-constraint and its source-document linkage, presentation-layer, coordination-group, functional-system, and document registers, type/direct assignment mode, semantic boundaries, repository locations, revisions, scope, and association counts. IDS specifications and BCF topics have selectable audit/issue cards; IFC STEP, IDS XML, BCF containers, and JSON also receive format-specific metrics and a bounded structured-source preview.

When the civil object index is selected, City Studio also loads the companion `civil-4d-sequence` through the same SHA-256-verifying artifact endpoint. The reviewer can rotate the projected federation, isolate track, substructure, above-track, or lineside native layers, toggle any native review group or functional system, scrub the construction task sequence, or play it automatically. The current task title and QA hold are shown beside the visible-asset count. This is an interactive coordination view of the source IFC solids. Civil jobs use IfcOpenShell geometry processing to apply object placements and tessellate native representations in metre-based engineering coordinates. A source-bound manifest and JSON chunks below the 4 MB preview limit are loaded through the verified artifact endpoint; every mesh retains its IFC GUID and asset ID for picking, filtering and BCF selection. The viewer rejects missing chunks, source/index hash mismatches and duplicate or invalid object geometry. Virtual objects without representations and older jobs remain explicitly identified envelope views. Map conversion is not applied again to these engineering coordinates. Bonsai remains the IFC editing and detailed engineering environment.

BCF review decisions are written to `cities/workspaces/<slug>/coordination/issues.toml`, not into the selected job artifact. One or more filtered IFC assets in the object inspector can seed a new topic with a title, description, assignee, and stable asset selection. City Studio derives a content-addressed `custom-…` issue ID; the civil exporter derives stable BCF topic and viewpoint UUIDs and rejects selections that do not resolve in the new IFC federation. Open and in-progress issues may be assigned without closure evidence. Resolved or closed issues require a substantive resolution and reviewer. The project compiler includes those decisions in its content hash and revision comparison; rerunning Generate Bonsai civil IFC4.3 creates

the new immutable BCF state.

See the Bonsai/IFC4.3 civil workflow for the engineering authority boundary, standalone generator, Blender review scene, and construction animation.

City workforce, design refinements and remaining development

Repository source: `docs/city-workforce-design-development.md`

The city index links each city's organisation, workforce spreadsheet, construction work profile and mechanical/civil refinement package. These are generated software outputs, tied by hashes to finance, operations work orders, soil inputs and shared design templates. The local team can combine functions where competence and shift cover permit; the template does not require a separate organisation department for each function.

Organisation and workers

The workforce template replaces obsolete per-train driver hiring with 21 functions for GoA 4 operations: control and remote assistance, platform accessibility, passenger service and cleaning, mechanical and battery maintenance, civil/track, solar/storage, wayside systems, management, assurance, training, procurement and people/finance administration. Each has a reporting line, skills, tasks and practical competence evidence.

```
flowchart TD
    Board[Local owner board] --> Director[City accountable lead]
    Board --> Assurance[Quality and safety: independent escalation]
    Director --> OCC[Control, dispatch and remote assistance]
    Director --> Engineer[Engineering authority]
    Director --> Support[Training, procurement, finance and people]
    OCC --> Stations[Station teams and passenger assistance]
    Engineer --> Fleet[Line-local fleet maintenance and service crews]
    Engineer --> Infrastructure[Civil, track, solar, storage and wayside teams]
    Shared[Shared domestic design, software, factories and laboratories] -.-> Engineer
    Shared -.-> Support
```

The generator splits the existing driverless financial model's six labour groups using explicit weights, function minima and largest-remainder integer allocation with stable role-ID tie breaking. Total FTE and annual labour cost remain equal to the existing model. The single city-director function receives one FTE; quality/safety has independent escalation to the owner board. Shares are editable planning choices, not locally measured workload. Existing country-income labour allowances remain visible; no new imported salary schedule is introduced.

The deployment register now tracks a separate workforce-duty/workload gate. It requires a duty/leave roster, local employment/rest inputs and measured workloads, especially simultaneous train service arrivals. The planned 12-minute turnaround does not demonstrate that a particular cleaning crew can complete its work. The new cleaning calculation accepts measured area, productivity, cycles and setup time and returns person-hours. Canopy area is not substituted for floor area.

Construction work-centre profiles come from the current dated manufacturing and civil work orders. Inclusive start/finish events calculate concurrency and resource-days. Missing crew counts stay unknown; a scheduled task slot is not a worker. Skills and package IDs identify exactly which crew definitions must be completed. National rolling-stock factories, design/software and specialist laboratories are shared domestic capabilities; their teams are not repeatedly added to every city's permanent operating headcount. Temporary works and operating staff are reported separately because their programmes overlap differently.

Joints and civil interfaces

Every elevated segment receives its own semi-continuous unit and interface count. Separate elevated fragments are never joined across an at-grade gap. The current 25 m / four-span catalogue planning arrangement supplies bearing/link-slab counts; short fragments use their actual rounded planned unit length for movement checks. These quantities do not locate surveyed pier foundations.

The city report computes free thermal movement using its climate preset and explicit design overrides, including a cold-side case where a lower extreme is available. The 8/10/12 µstrain/K expansion coefficients are declared sensitivity cases, not measured properties. Annual mean ambient is a comparison reference, not erection temperature. Solar-heated material temperatures, gradients, shrinkage/creep, restraint, seismic movement and rail interaction remain required. FHWA specifically recommends measuring concrete CTE instead of accepting generic literature defaults. FHWA concrete CTE guidance.

The reusable interface calculations also calculate worst-case seal compression from thickness/land tolerances and fastener preload bounds from torque tolerance and measured nut-factor bounds. For example, a purely illustrative 10 ±1 mm seal in an 8 ±1 mm land gives 0–36.36% compression. This exposes loss of compression margin; it does not qualify that seal. A numeric torque requires actual joint loads, proof strength, thread and bearing checks, lubrication/finish testing and preload-loss evidence. NASA's public fastener manual provides background on these dependencies; it is a public engineering reference, not a rail-specific acceptance standard or software. NASA Fastener Design Manual.

Mapped soil inputs remain location-specific investigation priorities. Fine-soil, organic, granular, moisture and acidic-soil flags identify tests and inspection needs. They do not establish foundation capacity or chloride/sulfate corrosion exposure. See the soil methodology.

Paints, finishes and cleaning

The shared finish system and joint schedule remain the controlled reference. Steel preparation/primer/stripe/topcoat, GFRP gelcoat or qualified paint, and optional livery film require substrate-specific qualification. Mask bond lands, seals, drains, inspection marks and active PV. Dry-film thickness, cure windows and wash pressures follow the qualified product and interface; no universal figure is assigned to all products or climates. Radiative roof paint remains a trial with no assumed operating energy benefit.

Four new maintenance work packages expand onto actual trainsets, civil/station assets and energy sites: finish/seal/joint defects, controlled vehicle washing, civil joint/drain/finish checks and PV-soiling cleaning. They specify condition or event triggers, owners and work-order evidence. Joint openings are recorded with component temperature. Cleaning trials cover EPDM, gelcoat, paint, film, glazing, connectors and PV; they record person-minutes, water, recovery and effluent handling. PV cleaning is driven by measured soiling and safe access while preserving station storage/top-up availability. Cleaning can itself abrade PV surfaces, which is why material trials and before/after normalised yield matter. NREL soiling and cleaning research.

Additional operating evidence

Two-day station/depot replay now includes Soroti, Sheikhupura, Sumbawanga and Tartus, alongside the existing five cities. Soroti and Sheikhupura also received nominal full-service and eight disruption cases each. Initial validation exposed missing explicit component configuration in their old scenario files. The shared rolling-stock configuration was restored and both complete validations rerun; the service/SoC acceptance floors were unchanged.

All nine retained two-day candidates pass. The two additional full validations pass, including all sixteen added disruption cases. These runs test software under specified loads and weather; they do not supply a final duty roster, physical stabling geometry or seasonal solar/storage endurance. The deployment index remains the current source of city-by-city open gates and stale evidence.

New two-day replay	Trainsets	Minimum train SoC	Result
Soroti	23	90.011%	Pass
Sheikhupura	37	56.114%	Pass
Sumbawanga	40	86.781%	Pass
Tartus	60	83.757%	Pass

The full disruption runs have different initial placement and fault conditions from the two-day candidates. Soroti's lowest disruption SoC was 57.406%; Sheikhupura's was 20.0048%, very close to the protected 20% floor. Sheikhupura completed 95.09% of scheduled train-km during its ten-hour all-site grid-outage case. This demonstrates the configured protective behaviour, with little energy margin in that case; it does not establish weather or seasonal robustness or justify an assumed grid upgrade. Site PV/storage replenishment and real charging duty remain the next energy evidence to resolve.

Remaining gaps and resources that can help

Each next artifact below addresses a real missing input or analysis. Existing integrations are extended before adding another dependency. None requires interline depot links, per-train drivers or a grid-led traction architecture.

Gap	Available open resource	Existing OSR starting point	Next useful artifact and inputs
Named shift/leave cover and simultaneous cleaning/inspection workload	OR-Tools scheduling	City role spreadsheet and dated work orders	Feasible local roster with skills, shift limits, leave and measured task person-minutes; solver used offline, core dispatch remains deterministic
Unsized construction crews and shared factory bottlenecks	OR-Tools scheduling	Manufacturing work-centre concurrency and production plans	Named crew mixes, observed cycle times and cross-city factory order schedule; staffing and cashflow reconciliation
Joint contact, fatigue and assembly tolerance release	FreeCAD FEM / CalculiX	Buildable joint schedule, CAD and CalculiX thermal benchmark	Load-bound joint models with friction/preload sensitivity, mesh convergence and coupon correlation; supplier material/contact data required
Soil/foundation flexibility, seismic restraint and rail interaction	OpenSees and soilDB	Structural screening and route-specific soil investigation plans	Route support models using received stratigraphy, groundwater, stiffness, loads and hazard inputs; maps guide investigation, not strength selection
Storm drainage, flood level and wash-water routing	EPA SWMM	Retained SWMM runoff benchmark and civil readiness manifests	Surveyed catchment/outfall model with local rainfall, blockage and receiving-water scenarios; water quality/disposal evidence separate

Gap	Available open resource	Existing OSR starting point	Next useful artifact and inputs
Adverse-weather station PV/storage replenishment	pvlib	Existing pvlib/microgrid screening and site controllers	Multi-day/year site PV plus train top-up duty with weather, shading, losses, soiling and storage ageing; quantify explicitly declared residual backup duty
Finish areas, replaceable-part quantities and asset handover	IfcOpenShell	Buildable station/trainset catalogues, IFC work and operations asset IDs	Measured surface/cleaning quantities linked to module IDs, substrates, finish batches and work orders; verify exclusions and openings
Conflict-aware passenger and service recovery acceptance	Existing SUMO and JuPedSim	City journey-time comparisons, native replay and station-corridor benchmark	Route/station-specific conflicts, arrivals, accessible evacuation and incident cases with explicit demand and geometry; refresh stale full-service reports

Actual corrosion exposure, material qualification, cleaning compatibility and worker competence still require local observations and trials. Open software can process these records; it cannot supply them. The repository now identifies the responsible function and concrete missing evidence instead of treating a generated document as completion.

Reproduction

Run the normal city regeneration pipeline, or refresh finance and operations, then run `engineering/analysis/city_delivery.py --all` and `engineering/analysis/city_deployment.py --all`. The package manifest checks the new report's source hashes. Existing operator receipts are preserved.

Civil And Alignment

Repository source: `docs/civil/README.md`

This folder holds civil-engineering interfaces and planning-grade alignment material. It is the bridge between generated city designs, survey/CAD tools, and the mechanical civil catalogue.

Civil design artifacts remain planning-grade until local survey, geotechnical, utility, permit, and structural checks are complete. The release gates are tracked in `deployment-release-checklist.md`.

Key Files

File	Scope
design-exploration.md	Reproducible complete civil-system research, native moving-load/flexible-support models and the staged acceptance backlog
line-based-foundation-and-assembly-planning.md	Per-support ground/load/install packets, directed line/front bay assemblies and inspection/handover dependencies
network-junction-and-residential-integration.md	Bounded interchanges, crossings/shared corridors, source-bound residential infill and branch/feeder planning
integrated Baghdad plans	Nine line plans, offline support/junction viewer, individual foundation/bay registers and 266-city geometry/coverage audit
civil-works-master-plan.md	Connected detailed civil programme with 25 m average-span planning, individual span orders, stations, all other civil scope and complete handover
viaduct-construction-method.md	Workfront release, foundations, columns/caps, production, launcher stages, paired beam erection, completion and relocation
station-construction-method.md	Elevated/at-grade station structures, spreading approaches, shafts/access, roofs, systems and integrated handover
other-civil-works-methods.md	Utilities, earthworks/treatment, retaining/track, drainage, specials, depots, energy sites and reinstatement
construction-logistics-plan.md	Existing-facility audits, yards, production positions, heavy haulage, buffers, material flow and launcher transfers
civil-programme-and-controls.md	Fourteen-package WBS, finite resources, calendars, workforce, change, cash and complete-line opening logic
civil-inspection-and-handover.md	Site-stage hold/witness/review plan, first repetition, following-trade release and final acceptance
Baghdad/engineering/civil-works/	Source-locked line quantities, 549 run packages, 186 station packages, logistics sensitivities and open review findings
marketplace-cost-anchors.md	Marketplace audit trail and retained benchmarks for at-grade track, elevated guideway, bridges, stations, depots, and charging interfaces
civil-cost-calibration.toml	Reviewed benchmark quantities and cost shares; the human-edited input to design-derived civil rates
civil-cost-model.toml	Generated rate contract consumed by city synthesis, finance reports, IFC metadata, and city READMEs
rapid-implementation-materials-review.md	Internet review of rapid ballastless track, modular stations, and recycled or lower-carbon materials
slab-trackforms.md	Reference ballastless slab designs for at-grade and elevated guideway sections
depot-bogie-change-interface.md	Shared LM3 jack-point datum, synchronized depot lift, pit, and bogie-extraction assembly contract
wayside-rerailing-access-interface.md	Optional selected-node hardstanding, offload, bridge-bearing, handling-route, and exclusion-zone contract for portable LM3 rerailing
../../design/component-catalogue/catalog/buildable-civil/	Generated accountability for all 19 reusable IFC types, six civil release packages, nine drawing-definition briefs, and the empty project release record
../../design/component-catalogue/catalog/buildable-civil/fabrication-and-construction-controls.md	Ten reference workface controls for survey, precast production, transport, erection, at-grade work, interfaces and NCR handback
../../design/component-catalogue/catalog/buildable-civil/evidence/construction-control-record-template.json	Blank per-workfront sequence, hold/NCR, inspection and next-trade handback record
../../design/component-catalogue/catalog/buildable-civil/inspection-and-test-plan.md	114 package-specific hold, witness and review characteristics covering all reusable IFC types
viaduct-substructure-kit.md	Controlled pier/abutment EBOMs, interfaces, assembly sequences, and deployment release gates
viaduct-design-basis.md	Rapid Viaduct Kit catalogue boundary, structural actions, geometry rules, and release evidence
viaduct-load-model.toml	LM3 12-axle and Baghdad 24-axle load/action seeds; supplier spacing and distribution remain required
viaduct-kinematic-egress-envelope.md	Train, walkway, parapet, and straight-span curve compatibility gates
viaduct-bearing-and-movement-schedule.md	Interior/end bearing counts and project movement-schedule requirements
viaduct-transport-and-erection-envelope.md	Permit-load transport, lifting, launcher, and temporary-stage release gates
viaduct-first-article-test-plan.md	First mould, girder, cap, lift, erected bay, and track/egress hold points
viaduct-quantity-cost-model.toml	Per-kilometre quantity seed and cost-estimate classification
osr-aln-format.md	OSR-ALN alignment interchange format and validation gates
bonsai-ifc-workflow.md	IFC4.3/Bonsai civil federation, IDS delivery audit, BCF coordination, quantities, 4D sequencing, and engineering-authority boundary
Samawah/engineering/alignment/	Current three-line Samawah planning OSR-ALN package, provenance, and survey replacement gates
deployment-release-checklist.md	Survey, geotechnical, structure, station, energy-site, and permit release gates
../owner-builder-operator-setup.md	Default public owner, ring-fenced delivery/operator company, independent assurance and staged mobilisation

Related Tools

Tool	Purpose
Civil reference demonstration	20/25 m double-track bays, connection briefs, complete mass gate, foundation/trackform comparisons, real solver exercises and connected civil FMEA
Numerical evidence workflow	Reviewed asset register, actual numerical outputs, per-support comparisons and hydraulic scenario acceptance exchanges
Civil reference compiler	Compile or check the controlled demonstration; `--run-solvers` reproduces OpenSees/CalculiX/SWMM examples
`crates/osr-alignment`	Alignment geometry, quantities, exports, and stake-out data
`tools/osr-aln-convert`	Companion converter for external alignment formats
`design/component-catalogue/src/osr_mech/civil/`	Parametric girders, piers, abutments, slab/guideway edges, elevated platform units, and civil CAD source
`tools/automation/bonsai-civil.sh`	Deterministic IFC/IDS/BCF generation, Bonsai import, saved review scene, 48-second MP4/GIF, and milestone screenshots
`tools/automation/buildable-civil.sh`	Reconcile the reusable IFC type set with civil-owned and interdisciplinary release packages; fail on unclassified geometry changes
`tools/automation/generate-civil-cost-model.py`	Recalculate planning rates from current CAD quantities and reject stale output with `--check`
`tools/automation/audit-project-twins.py`	Reconcile every city civil/finance bucket, CPM and cashflow source hash with the validated reference IFC and mechanical package

The IFC is a generated coordination output, not a competing geometry source. Change the parametric CAD/BIM source or reviewed calibration, then regenerate:

```
CAD geometry → per-route-km quantity model → civil cost contract
            ↘ IFC4.3 quantities/provenance   ↘ city CAPEX and READMEs
```

Project-specific Bonsai detail edits that are intended to become authoritative must first be reviewed and promoted into the parametric source; otherwise the next deterministic regeneration intentionally replaces them.

The city delivery twin schedules civil work by actual line length, civil class, predecessors and finite crews. Its `civil` budget comes from the same generated rate contract as city CAPEX; charging work is assigned to energy assets, not an overhead placeholder. Survey, geotechnical, utilities, local temporary works, permits and engineer release remain recorded gates rather than invented data.

Population-led additional lines and country regeneration describes immutable baselines, actual station coverage, native junction checks and finite construction plant.

Automated reference parts and full-train variants

Repository source: `docs/civil/automated-reference-parts.md`

Current generated review · original generated review · native and operating evidence · complete native assembly · native CAD receipt · saved-file audit · industry source register · default choices · compiler · OSR-ENG-002 generation requirements.

The compiler generates a complete planning train and regenerates its reference part geometry, installed datums, mass/CG/inertia, spatial dynamics inputs, mechanical/electrical/thermal/control interfaces, battery quantities, inspection requirements and coverage register from one set of design choices. It uses the existing catalogue part IDs and authoritative operational family definition.

The default six-car variant contains 192 instances, 191 joints, 12 bogies, 24 wheelsets and 48 wheel contacts. Its computed planning mass is 200,490.108 kg. This is a reference design study with residual synthetic mass allocations, rather than a released supplier train.

Industry examples and their actual use

Primary source	Adopted input or practice	Remaining evidence
Toshiba railway application	Type3-23 nominal 27.6 V, 45 Ah, approximately 15 kg, 190 × 361 × 125 mm module envelope	Purchased-part revision, tolerances, impedance/OCV maps, current limits, installation, rail/fire/vibration release
SKF axlebox architecture	Separate axlebox/bearing and wheelset responsibilities; provision for condition sensing	Bearing dimensions/fits/load ratings, supplier mass, speed/temperature sensor installation
Continental secondary suspension	Air-spring and auxiliary-support installation architecture	Force/displacement/damping curves, emergency operation, actual envelopes and life
Pandrol Fastclip	Captive fastenings and configurable replaceable pad stiffness inform infrastructure coverage requirements	Clip/pad/insulator geometry, tested properties, supplier loads and inspection data
NASA systems engineering handbook	Units, coordinate systems, endpoint responsibilities and requirements-to-verification traceability	Project-owned interface approval and measured verification

Only Toshiba's listed nominal module data supplies numerical supplier-reference inputs. The other manufacturer sources guide architecture; their proprietary part geometry or numerical load ratings have not been reconstructed. Reference axleboxes, springs, motors and gearboxes use explicitly assumed mass and simplified OSR envelopes. The existing wheelset, frame and body CAD remain reference models.

Consistent battery generation

The default battery is a 28-series × 4-parallel arrangement of 112 modules per car: 772.8 V nominal, 180 Ah, 139.104 kWh nominal and 111.2832 kWh over the declared 10–90% SOC window. It is an LTO study alternative to the existing catalogue's LFP battery. Its part ID is an allocation link, not approval to substitute chemistry or reuse the LFP supplier release. Toshiba's Type3 system-components description identifies the lithium-titanate cell architecture.

Each installation includes a generated base with eight clearance holes, lid, four enclosure walls and eight reference fastener stacks. Every module and enclosure/hardware feature has a cutlist identity. The compiler sums independently declared module mass and assumed-density OSR enclosure/hardware mass, with analytic primitive inertia and the parallel-axis theorem. It does not turn the enclosing module volume into active-material mass.

Changing the parallel-string count regenerates width, installed mass/inertia, Ah/kWh, pack resistance, current limits, heat capacity, cooling conductance, SOC histories and conditional range/recharge estimates. The pack's DC model solves `P = I (V - I R)` and retains electrical and thermal energy balances. Discharge consumes chemical energy; regeneration increases it while generating positive resistive heat. A lumped temperature uses exact constant-power Newton cooling within each duty segment.

Resistance, current limits, cooling conductance, thermal capacity, SOC window and temperature limit are OSR research assumptions, listed separately from Toshiba's nominal data. This is a constant-OCV, equal-current-sharing screening model. It does not resolve cell hot spots, protection logic, coolant hydraulics, age-dependent impedance or SOC-dependent voltage. Conditional range uses an explicit assumed 20 kWh/train-km; charging assumes 600 kW/train. These estimates do not establish route autonomy, traction capability or station dwell feasibility.

Expanded running gear and mass reconciliation

Each wheelset allocation expands into the wheelset and two separately identified axleboxes. Each bogie expands primary and secondary suspension kits into four primary spring/guide envelopes and two air/auxiliary spring envelopes. Powered bogies additionally instantiate two motors and two gearboxes. Their masses are deducted from the original wheelset/frame aggregates before dynamics assembly. Axlebox mass stays unsprung; frame-mounted component mass stays in its bogie group.

The six-car total reconciles exactly to the frozen planning tare plus the change from the old 2,500 kg/car battery allocation. Body/frame residual masses still represent components that have not been individually resolved. Their inertia remains a uniform research approximation, not an as-built tensor. The suspension parts map to the existing reduced six-direction joint laws; generating envelopes does not derive spring rates from geometry.

Coverage and interfaces

The six-car catalogue produces 1,367 required allocation slots. The compiler maps 114 to explicit reference components and reports 1,253 unresolved slots. The earlier published 737-slot allocation belongs to the three-car catalogue; these totals use different families. Coverage records every required slot, representation, evidence state, joint/interface mappings, acceptance requirements and model limitations. Unresolved kit/area/length allocations remain visible. Planning mass closure is distinct from complete production component coverage.

All 191 physical joints have mechanical interface records that identify local datums, units, signs and endpoint responsibilities. Battery interfaces add electrical, thermal/fluid and control contracts. Unknown connector/protection, hydraulic and latency capabilities remain open. Validation rejects stale model identity, missing/unknown endpoints, mismatched joint mappings, wrong units and invalid declared limit intervals. It cannot approve compatibility where supplier port capabilities are absent.

Geometry validation rejects malformed primitives and invalid/overlapping holes. Battery bay screening rejects overlap with the reference inner-wheel/bogie envelopes and excess width/height. This is a specific installation check, not a complete train clearance, cable-routing or removal-envelope certification.

Generate, verify and run

Use fresh output folders to retain previous evidence:

```
tools/automation/osr-python tools/automation/compile-reference-parts.py generate \
  --output build/engineering/my-reference-parts

tools/automation/osr-python tools/automation/compile-reference-parts.py verify \
  --output build/engineering/my-reference-parts

tools/automation/shared-engineering-freecad.sh \
  --model build/engineering/my-reference-parts/model.json \
  --output build/engineering/my-reference-train.FCStd

tools/automation/osr-python tools/automation/verify-reference-parts-freecad.py \
  --model build/engineering/my-reference-parts/model.json \
  --cad build/engineering/my-reference-train.FCStd \
  --output build/engineering/my-reference-train.audit.json

tools/automation/osr-python tools/automation/shared-service-envelope.py run \
  --model build/engineering/my-reference-parts/model.json \
  --config build/engineering/my-reference-parts/spatial_configuration.json \
  --output build/engineering/my-reference-service --cases empty-15mps
```

Pass `--choices path/to/choices.json` to generate another architecture. For the shorter three-car planning modules, reduce series modules to fit the bay; 20 series modules is a tested example across all five authoritative families. Invalid variants fail before outputs or analysis can be accepted.

The package includes `model.json`, `spatial_configuration.json`, `battery.json`, `interfaces.json`, `coverage.json`, `manufacturing.json`, `mass_properties.json`, `joint_register.json`, `verification.json`, source inputs and a hash-bound receipt. Verification checks retained output/source hashes and recompiles every result. CAD uses the same embedded primitive definitions and actual installed datums; the native solver and projected drawing edges are checked independently. The elevation/plan scale adjusts to the complete family length. The native audit reopens the saved file, solves the restored assembly, checks all installed instance placements and tests generated reference solids against independent OpenCASCADE volume/inertia integrals. Legacy body/frame/wheelset geometry remains under its existing reference-model limitations.

Research packaging and mass checks, numerical completion, engineering constraint approval and physical validation remain separate states. Inspection methods and nominals are generated with acceptance limits open. The automated battery retention pattern is ready for simultaneous six-resultant bolt-group screening; preload, plate prying, fatigue and assembly-specific native submodels still require actual materials, mounting geometry and loading evidence.

The full operating matrix, independent full-family time/mesh convergence, geometry-specific local strength/fatigue, traction/brake maps, coolant/controller coupling and physical asset measurements remain required. Historical two-car proofs retain their original producing commit and are not relabelled as tests of this new full-train variant.

Retained verification outcome

The focused suite passed 108 tests, including 24 new reference-part tests. Native FreeCAD 1.1.4 generated and reopened the 192-instance assembly, solved all 191 joints and produced four nonempty drawing projections. The saved-file audit checked 164 distinct reference primitives against independent analytic volume and inertia calculations. The drawing remains unissued.

FreeCAD emits subshape-reference restoration diagnostics for the detached whole-feature joints. The audit checks the restored solver and actual installed placements independently; these pass. This does not establish supplier fits or physical attachment capacity.

The fresh complete six-car passage attempt used `dt=0.005 s`, two deck elements per span and 2 m rail spacing. It failed nonlinear convergence at 9.775 s, with a reported trial residual of approximately 2.0493e7 N. The campaign retains all converged history through 9.770 s and its failed-case receipt. No full-train passage or time/mesh convergence is accepted from that attempt. The other 21 planned cases remain unexecuted. Resolving this coarse-run failure and independently refining the full-train operating cases is the next numerical qualification task, separate from the completed part-generation and CAD checks.

The subsequent contact-domain investigation finds an unsupported Hertz footprint at 9.130 s, before the reported non-convergence. The current solver rejects unsupported converged states before committing their contact state/history. The original review retains its producing commit; a separate current compiler review records the updated implementation.

Baghdad civil qualification inputs

Repository source: `docs/civil/baghdad-qualification.md`

The user-selected qualification target is Baghdad. The retained civil workbench uses a 100 m double-track comparison segment, separate from the city's 54-line planning inventory. This selection does not convert reference-software results into qualification evidence.

The controlled sources are `viaduct-load-model.toml`, `rolling-stock.toml`, and the Baghdad design, soil and survey readiness records. The selected profile has six cars, 24 axles and a 111 m planning length. Its retained mass envelopes are 204 t tare, 258 t AW2, 276 t AW3 and a separate 384 t infrastructure allowance. None specifies actual loaded axle forces. LM3's three-car axle spacing must not be repeated.

The readiness packet records open inputs and accountable disciplines. Its JSON counterpart hashes each controlled source. Supplier quotes, actual materials, support-zone ground profiles, adopted project limits, surveyed alignment, physical testing and independent authority acceptance remain open. The retained desktop soil summary reports 30 missing profiles; stiffness hypotheses cannot supply those measurements.

Run `civil-study.py qualification` to generate a fresh readiness packet. Default `run` and `programme` write the packet and return status 2 without invoking native analysis when the supplier train pattern is missing. To replay the existing software controls, select `--deployment reference` explicitly.

The first execution dependency is a supplier six-car drawing and case-specific loaded axle schedule. A strict supplier record binds the original document by repository path and SHA-256. All 24 positions must be unique, ordered and inside the stated train length; axle forces must sum to the declared loaded mass under 9.81 m/s² gravity. A source-bound pattern enables moving-force research, while every other qualification gate stays open. It does not establish engineering acceptance of the supplier data or the structure.

The full coupled Baghdad programme additionally needs supplier suspension, sprung/unsprung masses, bogie/articulation and contact data, and an adapter for those relationships. The existing uniform LM3 coupled diagnostic is retained as reference verification. Costs remain unknown until quotations and their scope are supplied. Physical and operating release remain false.

Commands, resumption behaviour and model limits are documented in the workbench README.

Bonsai / IFC4.3 Civil Workflow

Repository source: `docs/civil/bonsai-ifc-workflow.md`

Status: implemented design-reference workflow
Baseline: Bonsai 0.8.5, Blender 5.2, IfcOpenShell/IfcTester/BCF 0.8.5, IFC4.3

Decision

Bonsai is better suited than FreeCAD alone for the federated civil BIM part of OpenSourceRail: native IFC authoring, rail-domain object classification, property and quantity inspection, drawing/coordination work, and construction sequence review. It is not currently a better source for deterministic railway alignment engineering. The OSR alignment engine and parametric civil catalogue therefore remain authoritative; Bonsai receives a generated IFC4.3 model and must not silently override its route or engineering rules.

Information	Authority	Bonsai role
Horizontal/vertical alignment, transitions, cant and rule checks	OSR-ALN plus accepted survey/GIS inputs	Inspect the exported `IfcAlignment` reference and coordinate issues
Pier, girder, trackform, platform and canopy parameters	`design/component-catalogue/src/osr_mech/`	Federate, classify, inspect, annotate and develop deployment detail against stable IDs
Structural capacity, reinforcement, bearings, foundations and temporary works	Reviewed analysis decks and engineer-released drawings	Coordinate released results; never infer approval from visible geometry
IFC object hierarchy, property sets, quantities and 4D task links	Generated IFC checked by IfcOpenShell schema/EXPRESS validation and IDS	Native GUI authoring/review environment
Civil planning rates	Generated `lib/templates/civil-cost-model.toml` from CAD quantities and reviewed benchmark shares	Inspect the model hash, rate properties and quantity basis; do not treat them as a quote
Revision identity	City Studio content hash and Git revision	Carry revision and source hashes in IFC properties and the saved Blender scene

This boundary reflects current upstream limitations: Bonsai's road/rail guide describes its PI CSV alignment import as a temporary stop-gap, while the IfcOpenShell alignment API is still under development and explicitly does not select engineering design parameters.

Generate and review

Check the installed stack:

```
tools/automation/bonsai-civil.sh --check
```

Generate deterministic IFC and JSON evidence without opening Blender:

```
tools/automation/bonsai-civil.sh --generate
```

Generate, import through Bonsai, save a native review scene, and render it:

```
tools/automation/bonsai-civil.sh --render
```

Render the normalized 48-second construction sequence, native scene, MP4, GIF and milestone screenshots as well:

```
tools/automation/bonsai-civil.sh --animate
```

To bind the federation to a City Studio line, use the Generate Bonsai civil IFC4.3 job in the engineering hub. The adapter converts the selected line to the local engineering frame and supplies that immutable reference axis to the same exporter. The civil settings panel can store an accepted map conversion per stable line ID; it is Git-visible, revision-hashed, shown in semantic revision comparison, and passed only when that line is regenerated.

Ordinary jobs are written beneath `build/engineering/bonsai-civil/`. The same generator publishes the named GitHub-review set beneath `engineering/models/bim/reference/`:

- `civil-coordination.ifc` — native IFC4.3 model;
- `civil-coordination.index.json` — object-aware browser index and authority boundary;
- `civil-construction-sequence.json` — tasks, dates, QA holds and product assignments;
- `civil-coordination.blend` — native scene with selectable IFC assets and construction keyframes;
- `civil-construction-sequence.mp4` / `.gif` — full sequence and GitHub preview;
- `civil-information-requirements.ids` — buildingSMART IDS 1.0 delivery requirements;
- `civil-information-requirements.report.json` — complete deterministic IDS audit evidence;
- `civil-coordination-issues.bcf` — BCF 3.0 package with object-linked release issues;
- `civil-coordination-issues.index.json` — browser-safe issue and IFC selection index;

- `civil-coordination.validation.json` — reopen, IDS, BCF-link, and exact artifact-hash checks.

The named scene, review video and GIF are tracked with the reference federation; the three milestone screenshots are under `docs/screenshots/civil/`. Intermediate PNG frames and repeated project renders remain under `build/` and are removed after encoding.

Model content

The current reference federation contains 185 stable assets and 19 reusable component types organised below an `IfcRailway` and four `IfcRailwayPart` containers: track, substructure, above-track systems, and lineside/clearance. Civil objects use IFC4.3 types including `IfcRail`, `IfcBeam`, `IfcBearing`, `IfcColumn`, `IfcSlab`, `IfcRoof`, and an `IfcAlignment`. The selected line's control-point polyline is represented by native `IfcAlignmentHorizontal` `LINE` segments, `IfcAlignmentVertical` `CONSTANTGRADIENT` segments, an `IfcGradientCurve`, and stationing referents. This improves civil-tool interoperability without inventing curve radii, transitions, or cant that the planning input does not contain. Each asset carries:

- stable OSR asset ID and class;
- canonical source and source hash;
- revision ID and lifecycle status;
- inspectable coordination dimensions and source net volume;
- explicit detail mode when a complex assembly is represented by its review envelope.

One hundred and thirty-eight occurrences are linked to an `IfcTypeProduct` derived from their exact asset class and source-geometry recipe. Types carry stable `OSR-TYPE-…` identity and `OSR_Type`; occurrence placement and geometry remain authoritative. The 47 virtual clearance, foundation, and jacking interfaces remain untyped because IFC4.3 has no `IfcVirtualElementType`.

Each of the nine source pier compounds is split without overlapping geometry into one column, one `IfcBeam/PIERCAP`, four `IfcBearing/ELASTOMERIC` products, four bearing-replacement jacking interfaces, and one foundation interface. The bearings retain their measured 0.6 × 0.5 × 0.1 m source envelopes and explicitly unresolved supplier/load/movement schedules. Foundation and jacking zones use `IfcVirtualElement`: the source intentionally withholds the actual foundation type and depth, so exporting `IfcFooting` or `IfcDeepFoundation` would assert engineering that does not exist.

Face contact in the checked envelopes also generates 27 `IfcRelConnectsWithRealizingElements` relationships: 24 beam-end connections realized by two bearings each and three station-deck connections realized by four bearings each. All 36 bearings participate in the resulting 60 realizations. The relationships record physical topology only; connection geometry, stiffness, loads, movement capacity, supplier selection, and release remain deliberately unset.

Three native `IfcMaterial` family declarations cover 46 occurrences through five type associations: running-rail steel, prestressed beam concrete, and precast platform-unit concrete. `OSR_MaterialStatus` identifies the source authority and explicitly records that grade, design, supplier certification, and release remain unresolved. Mixed or interface-only assemblies receive no material association until their constituent specification is authoritative.

The 32 straight running-rail occurrences use one native `IfcMaterialProfileSet` and cardinal-point-5 `IfcMaterialProfileSetUsage`. Their bodies are `IfcExtrudedAreaSolid` geometry driven by the same simplified 60E1 polygon as the CAD source, replacing coordination box meshes. The indexed section area and extrusion volume are checked against the source solid. This remains a straight-line review polygon; procurement still requires the full mill profile, fillets, tolerances, released steel grade, and certificates.

Native OSR asset classification

One internal `IfcClassification` contains 15 lightweight references using the existing OSR asset codes. One hundred and thirty-eight occurrences inherit their reference from 19 reusable types; the 47 untyped virtual interfaces receive their class directly. All 185 assets therefore remain natively queryable without duplicating type-level relationships. This classification supports OSR automation only. Its metadata and City Studio inspector explicitly state that country/client mappings remain unnominated and require an approved deployment crosswalk.

The five source layouts are also native `IfcGroup` coordination groups, with all 185 assets assigned exactly once. Their properties state that they are separated review layouts—not surveyed spatial zones or functional engineering systems—so Bonsai and City Studio can filter them without asserting false `IfcSpatialZone` semantics or confusing review geography with system function.

Six native `IfcSystem`-family records provide the separate functional view: track and running way, guideway structure, station interfaces, rolling-stock reference, clearance assurance, and unreleased civil interfaces. IFC4.3 specializations are used where

the source supports them: track is IfcBuiltSystem/RAILWAYTRACK, guideway is IfcBuiltSystem/LOADBEARING, and the station interface is IfcBuiltSystem/USERDEFINED; clearance and rolling-stock references remain generic IfcSystem. Foundation and jacking interfaces use a generic system because virtual elements cannot be members of an IfcBuiltSystem. The 15 authoritative OSR asset classes map without overlap, so all 185 occurrences have exactly one membership.

Four IfcRelReferencedInSpatialStructure relationships provide seven explicit links from the systems to the track, substructure, above-track, or lineside IfcRailwayPart they cover. The station system legitimately spans two parts. These are non-hierarchical spatial-service references, not containment. The systems remain design-reference groupings—not surveyed zones, commissioned operational systems, or safety releases—and track number, usage, electrification, and other unavailable operational properties stay unset.

The two representative LM3 trainsets use native IfcVehicle/ROLLINGSTOCK occurrences and one reusable IfcVehicleType/ROLLINGSTOCK, replacing the former generic building proxies. Each vehicle carries the standard Qto_VehicleBaseQuantities Length, Width, and Height measured directly from the deterministic geometry. Capacity, mass, availability, serial identity, manufacturer, and operational release are intentionally not inferred.

Four IfcPresentationLayerAssignment records assign each asset's IfcShapeRepresentation to track, substructure, above-track, or lineside. They provide native visibility filters for Bonsai and layer-oriented CAD/BIM tools. They deliberately carry no engineering meaning beyond simple geometry grouping; object identity and semantics remain in IFC objects, types, classification, and coordination groups.

The nine deterministic integration gates are native project-level IfcObjective constraints with hard DESIGNINTENT semantics, source path, fixed evaluation time, current observation, and PASS/FAIL state. Six checks also carry native nested IfcMetric records with EQUALTO length benchmarks; their observed and target values come from structured integration data, not parsed prose. The other three gates remain explicitly qualitative because their sources provide no defensible numeric benchmark.

All custom properties now use the standards-compliant OSR_ prefix; the IFC Pset_ prefix is reserved for property sets defined by buildingSMART. Sixteen native IfcPropertySetTemplate dictionaries declare 99 property and quantity field types across occurrence, type, material, profile, and quantity data. Applicable templates link 483 IfcPropertySet/IfcElementQuantity definitions via IfcRelDefinesByTemplate; material and profile resources cannot use that relationship, so their four definitions match the declared material/profile templates by name and template type. All 16 templates are declared in the IFC project context and indexed for City Studio inspection.

Fifteen native IfcDocumentInformation/IfcDocumentReference records expose the actual repository sources used by the federation. Complete SHA-256 revisions and repository-relative locations cover the component generators, integration source, exporter, alignment contract, cost contract, and this release-boundary workflow. IfcRelAssociatesDocument links all 185 occurrences and their reusable types to the applicable source modules. These are source records, not issued drawings or a substitute for a deployment CDE.

Dimensions, source volume, and representation count are native IfcElementQuantity values in OSR_CoordinationEnvelopeQuantities, with the calculation method declared on the quantity set. The four discipline containers also declare the IFC4.3-required VERTICAL railway-part usage. The exporter runs IfcOpenShell schema and EXPRESS rules after writing and rejects any issue; the result is retained in civil-coordination.validation.json.

The default model deliberately remains on the named local engineering grid. When accepted survey/GIS control exists, use City Studio's per-line IFC survey control form or add this optional block to the civil alignment input:

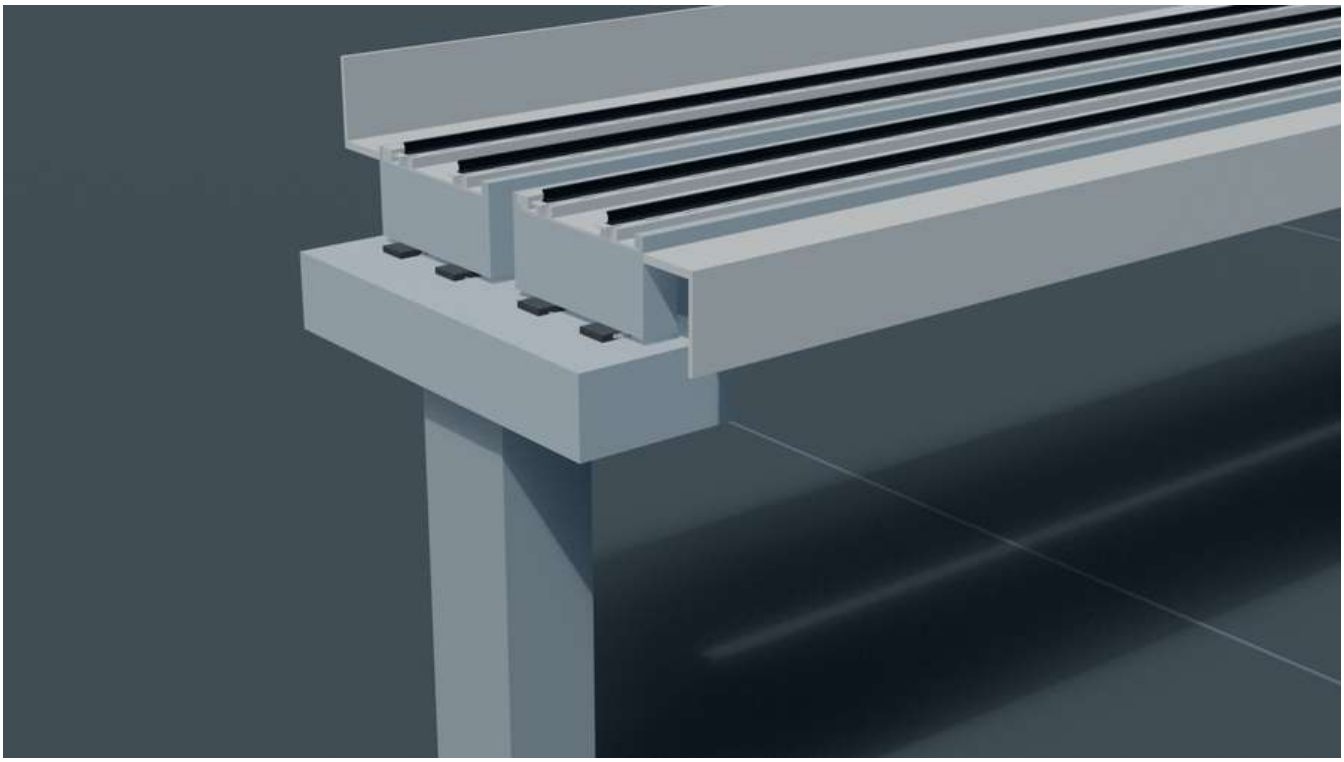
```
{
  "georeferencing": {
    "crs_name": "EPSG:9306",
    "eastings": 198765.4,
    "northings": 431234.5,
    "orthogonal_height": 18.25,
    "x_axis_abscissa": 0.999847695,
    "x_axis_ordinate": -0.017452406,
    "scale": 0.99995,
    "source": "Accepted survey control revision S-04"
  }
}
```

Use the actual project CRS and survey transform; the example values are only a shape example. With the block present, the IFC contains IfcProjectedCRS and IfcMapConversion. Without it, the index and project properties explicitly record project-crs-unresolved rather than inventing map coordinates. For a 3D model, use a compound CRS that identifies the vertical datum as required by IFC4.3.

The IFC project also carries OSR_CostModel: the generated contract hash, maturity and current class rates. Its JSON index includes the full per-route-km quantity basis. Editing a reviewed parametric dimension and regenerating first updates the quantity model and cost contract, then emits both into IFC and city CAPEX. Direct Bonsai edits remain coordination changes until promoted back into the authoritative parametric source.

The same generated contract is also represented by one native USD `IfcCostSchedule` with `SCHEDULEOFRATES` semantics and three `IfcCostItem` alternatives: at-grade, elevated, and bridge. Each `IfcCostValue` uses a 1,000-project-metre unit basis. The items intentionally have no cost quantities or product assignments, so consumers cannot mistake mutually exclusive planning rates for a selected scope, multiplied project total, bill, tender, or quotation. The cost-contract document is hash-locked to both the project and this schedule.

Eighteen track, station, and viaduct tasks are embedded in an `IfcWorkSchedule`. Five tasks now carry stage-specific outputs: track completion, station structural assembly, foundation/pier/cap completion, bearing/beam erection, and viaduct trackform/egress finishing. Native `IfcRelAssignsToProduct` relationships distinguish 134 physical construction outputs from 45 virtual foundation/jacking review interfaces by relationship name and description. Virtual interfaces therefore participate in their QA gate without appearing as constructed products in the 4D animation. Predecessors, planning durations, hold points, and evidence remain in the companion sequence index.



Bonsai support-end review of twin OSR-Pi25 decks and the reduced common cap

The support-end render is generated by the same command. It shows the coordination relationship between the two narrow track decks, bearing lines, 7 m common cap and independent outer walkway/containment cassettes; it does not represent released reinforcement, prestress or connection detailing.

IDS delivery gate and BCF review loop

The exporter writes an IDS 1.0 contract and immediately reopens it with IfcTester against the written IFC. Twenty specifications cover all 185 assets, the 19-type catalogue, the source-backed material subset, alignment authority, the native rail-profile assignments, OSR asset classification, native document register, five coordination groups, six functional systems, four presentation layers, 36 native bearings and their connectivity records, nine virtual foundation interfaces, two native vehicles and their standard base quantities, nine interface constraints, six native metrics, typed property dictionaries, the planning schedule of rates, and project provenance. The current reference exchange passes all 3,340 entity-level checks. The audit is deterministic: entity evidence is sorted before hashing, so a repeat build produces byte-identical IFC, IDS, audit, BCF, indexes, sequence, and validation files.

Passing IDS means that the information delivery is complete for this declared coordination purpose. It does not mean the design is structurally or legally released. The generated BCF 3.0 package therefore carries three open issues with viewpoints and selected IFC GUIDs:

1. replace the planning axis with accepted survey and alignment geometry;
2. release the elevated station deck structural design;
3. complete the viaduct span, bearing, pier, foundation, and reinforcement work.

City Studio exposes the requirement report and issue index alongside the IFC. Bonsai can open the native BCF package for discipline coordination and issue closure against the same stable objects.

City Studio also pairs the object index with the hash-verified construction sequence. Its interactive review controls rotate the projected federation, toggle any of the four native presentation layers, five coordination groups, or six functional systems, and scrub or play the 18-task 4D sequence while showing the current QA hold, physical-output count, virtual-review count, and visible-asset count. These projected envelopes support rapid coordination and BCF selection; native tessellated geometry and authoritative IFC editing remain in Bonsai.

City Studio stores a review decision in the project's committed `coordination/issues.toml`. It never rewrites a prior BCF artifact. Closing or resolving a topic requires both resolution evidence and a reviewer; the next civil job reads the new content-addressed revision and emits a fresh BCF with the same deterministic topic/viewpoint identities and updated status, assignee, resolution, and reviewer.

The IFC object inspector can also create a new coordination topic against a selected stable asset. Custom issues use a deterministic content-derived ID; their asset IDs must resolve in the regenerated federation. The exporter then creates deterministic BCF topic/viewpoint UUIDs and a selected-component viewpoint. Unknown assets fail the civil job instead of producing a dangling coordination reference.

Engineering release boundary

The generated model is deliberately labelled design-reference / not for construction. Its nine current gates check interface geometry such as rail/platform datums, clearance gaps, pier bearing top to girder soffit, and station/viaduct track-support continuity. It does not release geotechnical parameters, reinforcement, prestress, bearing schedules, seismic restraints, drainage capacity, construction loads, or local-code compliance. Those remain deployment analyses and competent-engineer decisions linked back into IFC only after review.

Each gate remains governed by the `IfcProject`, while its single `IfcRelAssociatesConstraint` also identifies the evidence it evaluates. Across the nine objectives this produces 91 asset, six review-group, one functional- system, and nine project links. City Studio exposes those IDs directly. This scope is traceability, not a claim that the qualitative checks replace structural analysis or approval.

The registered, SHA-256-locked civil-integration `IfcDocumentReference` is also attached to all nine objectives and their six metrics by one `IfcExternalReferenceRelationship`. This is the IFC4.3 resource-level link intended for constraints, which do not inherit from `IfcRoot`; it is provenance only and does not represent approval, issue, or engineering release.

Capability closure and external decisions

All IFC/Bonsai work supported by the current repository inputs is implemented. The generated index reports `source-supported-ifc-work-complete` with zero implementable open tasks. Nine external-decision records identify the named authority, evidence, blocked capability, and safe current state for work that cannot be completed truthfully from repository data alone.

IFC capability	Decision
Native quantity take-off	Implemented now; it replaces quantity-shaped generic properties and enables later parametric IFC costing.
CRS/map conversion	Implemented now as validated opt-in input; unresolved projects remain visibly local rather than receiving a guessed EPSG code.
Alignment layouts and linear placement	Native horizontal `LINE` and vertical `CONSTANTGRADIENT` planning segments, gradient-curve geometry, and stationing are implemented from the selected line polyline. Design radii, transition spirals, vertical curves, cant, and product linear placement remain deferred until an accepted OSR-ALN design supplies those parameters; LandXML, railML, and stakeout CSV remain the detailed handoff.
Native `IfcCostSchedule`	Implemented as a USD `SCHEDULEOFRATES` with three generated route-kilometre alternatives and a 1,000 m unit basis. It has no product assignments, cost quantities, or project total; element-level estimating remains deferred until approved rates and selected scope exist.
Reusable object types	Implemented for 138 safely typable occurrences using exact source recipes. Type geometry maps are deliberately omitted so occurrence geometry remains authoritative.
Material families	Implemented for the 46 occurrences whose source explicitly declares a safe single-material family. Every declaration is visibly grade/design unresolved.
Native rail profile	Implemented for all 32 straight 60E1 rail occurrences using a shared material profile set, occurrence usages, and matching extruded solids.
Native rolling-stock vehicles	Implemented for both representative trainsets with one reusable `IfcVehicleType/ROLLINGSTOCK` and standard measured Length, Width, and Height quantities. Operational, manufacturer, capacity, mass, and availability fields remain unset until authoritative inputs exist.
Bearings and foundation interfaces	The 36 source bearings are native typed `IfcBearing/ELASTOMERIC` assets with measured envelopes and explicit supplier/load/movement release gates. Nine foundation and 36 jacking envelopes remain `IfcVirtualElement`; promote them only after geotechnical and supplier design.
Bearing connectivity	Implemented with 27 native `IfcRelConnectsWithRealizingElements` relationships and 60 bearing realizations derived from exact source-envelope face contact. No analytical condition or supplier performance is inferred.

IFC capability	Decision
Mixed materials, other profiles, and reinforcement	Do not flatten mixed assemblies or populate generic placeholders. Add constituent/profile and reinforcement data when released specifications and deployment engineering are available.
Native OSR asset classification	Implemented with one internal system, 15 lightweight references, type inheritance for 138 occurrences, and direct references for 47 untyped virtual interfaces.
Country/client classification mapping	Defer Uniclass, OmniClass, national, or client codes until the deployment nominates an edition and approves a crosswalk; no global mapping is guessed.
Native coordination groups	Implemented with five deterministic `IfcGroup` records and exactly one group membership per asset. They preserve source review layouts without claiming surveyed space or system function.
Native functional systems	Implemented with six deterministic `IfcSystem`-family records and exactly one membership per asset, derived from the complete 15-class OSR asset classification. Three use supported `IfcBuiltSystem` subtypes, and seven non-hierarchical references connect all systems to the railway parts they cover. Virtual civil interfaces remain in a generic system.
Surveyed spatial zones	Add `IfcSpatialZone` only when accepted boundaries and survey control exist; the current separated review layout is insufficient evidence.
Native presentation layers	Implemented with four stable layers and exactly one `IfcShapeRepresentation` assignment per asset for simple visibility control. Object semantics remain elsewhere.
Native interface constraints	Implemented with nine hard `IfcObjective` records and 107 scoped project/asset/group/system evidence links. Six checks have native nested `IfcMetric`/`EQUALTO` length benchmarks; three remain explicitly qualitative.
Stage-specific 4D product semantics	Implemented across five output tasks: 134 physical assets animate at their actual completion stage, while 45 virtual foundation/jacking interfaces are separately related as review gates and never treated as constructed output.
Native OSR property dictionaries	Implemented with 16 project-declared templates, 99 typed fields, 483 direct definition links, and four name/type-matched material/profile definitions. Custom sets use `OSR_`, never the reserved `Pset_` prefix.
Native source-document register	Implemented with 15 hash-locked repository sources, direct project/cost/type/occurrence associations, and one resource-level relationship linking the civil source to all nine objectives and six metrics.
External engineering decision register	Implemented with nine indexed decisions. Survey, geotechnical, structural, bearing, material, classification, commercial, rolling-stock, and CDE promotion remains blocked until the named authority supplies the listed evidence.
Issued drawings and CDE document control	Defer sheets, issue/transmittal states, and CDE URLs until a deployment selects its naming, approval, and common-data-environment convention.

Upstream references

- Bonsai introduction and native IFC workflow
- Bonsai road and rail alignment guide
- IfcOpenShell alignment API and current design limitations
- IFC4.3 `IfcAlignmentHorizontal`
- IFC4.3 `IfcAlignmentVertical`
- IFC4.3 `IfcReferent`
- IfcOpenShell georeferencing API
- IFC4.3 `IfcCostSchedule`
- IFC4.3 `IfcCostValue`
- IfcOpenShell cost API
- IFC4.3 `IfcProjectedCRS`
- IFC4.3 `IfcMapConversion`
- IFC4.3 railway-part organisation and usage
- IFC4.3 native quantity sets
- IFC4.3 object typing
- IfcOpenShell type assignment
- IFC4.3 material sets
- IfcOpenShell material assignment
- IFC4.3 material profile usage
- IfcOpenShell arbitrary profiles

- IFC4.3 classification systems
- IfcOpenShell classification API
- IFC4.3 `IfcGroup`
- IFC4.3 `IfcSpatialZone`
- IFC4.3 `IfcSystem`
- IFC4.3 `IfcVehicle`
- IFC4.3 vehicle types
- IFC4.3 vehicle base quantities
- IFC4.3 `IfcBearing`
- IFC4.3 bearing types
- IFC4.3 element connectivity
- IFC4.3 connections with realizing elements
- IFC4.3 `IfcFooting` and shallow/deep-foundation boundary
- IFC4.3 `IfcBuiltSystem`
- IFC4.3 built-system types
- IFC4.3 spatial service connectivity
- IfcOpenShell group API
- IfcOpenShell system API
- IFC4.3 `IfcPresentationLayerAssignment`
- IfcOpenShell layer API
- IFC4.3 constraint resource
- IFC4.3 `IfcObjective`
- IFC4.3 `IfcMetric`
- IFC4.3 `IfcRelAssociatesConstraint`
- IFC4.3 `IfcExternalReferenceRelationship`
- IFC4.3 `IfcDocumentReference`
- IfcOpenShell constraint API
- IFC4.3 custom property-set naming
- IFC4.3 property-set templates
- IfcOpenShell property-template API
- IFC4.3 document information
- IFC4.3 document associations
- IfcOpenShell document API
- IfcOpenShell schema validation
- IfcOpenShell parametric cost-quantity links
- IfcTester IDS API
- IfcOpenShell BCF 3 topic API
- buildingSMART IFC4.3 rail domain

Civil inspection, testing and handover plan

Repository source: `docs/civil/civil-inspection-and-handover.md`

Apply to the master plan and method-specific asset register. The existing catalogue ITP retains reusable-product characteristics; this plan adds site/workface and full-line interfaces. Project designers define test methods, sampling, acceptance values and independent-review scope. Empty criteria or unsigned results cannot close a hold point.

H = stop until authorized release; W = notified witness opportunity under the agreed ITP; R = controlled record review. Each row must name responsible inspector, independent/owner role where required, current asset/design/method revision, actual test/result, acceptance basis, date and signed disposition. Proposed role titles are not approvals.

Stage	Type	Required evidence before following work
Survey/utility/land	H	Verified control, support/station/site coordinates, trial-hole and isolation records, rights and construction permissions
Ground and foundation method	H	Site ground/water model, released foundation/ground-treatment design, trial method and specified verification tests
Working platform/temporary works	H	Design/check, actual ground/setup inspection, limits and permit to use/load; subsequent change checks
Foundation installation	W/R	Strata/installation geometry, actual length/count, driving/drilling/treatment logs, materials and adverse events
Foundation acceptance	H	Required load/integrity/settlement results, dimensions and connection datum, resolved NCRs
Reinforcement/inserts before casting	H	Released drawing/bar schedule, cover/position, tendons/ducts/inserts, earthing and lifting interfaces
Concrete and grout	W/R	Mix/batch traceability, fresh properties, weather/temperature/curing, specimens/maturity, pour continuity and tests
Prestress transfer/demould/handling	H at each distinct release	Stress/calibration records, stage-specific strength, cracks/camber/dimensions, approved lifting/support design
Factory acceptance	H	Serialized complete member, materials/geometry, mass/CG and handling envelope, traveller and NCR closure
Transport dispatch	H	Accepted item, actual vehicle/axle/load restraint, surveyed route/permits, appointments and receiving readiness
Receiving	W/H for damage or discrepancy	Correct identity/condition, transport inspection, supported storage and traceability; quarantine discrepancies
Column/cap/bearing support ready	H	Foundation, connections/strength, survey, bearing type/orientation and next launcher/delivery-load release
Launcher assembly/commissioning	H	Supplier configuration, stage checks, assembly inspection/tests, maintained equipment and authorised crew
Each beam placement and bay release	H	Correct span/track/member, lift plan/load/gear, seats/contact/survey, temporary restraint and completed stage path
Continuity/joints and restraint removal	H	Selected structural scheme, reinforcement/grout/concrete /bolts, movement datum, strength and designer removal permission
Waterproofing/ducts/drainage concealment	H	Inspected continuity, protection, levels, joint and clean/pull/water tests, surveyed as-built routes
At-grade formation/track support	H	Treatment/compaction/settlement, drainage, line/level and bedding/slab/panel records
Station structure/access	H	Structure/fixing/shaft/roof records, platform/train/PSD interfaces, continuous accessible circulation, equipment commissioning
Special crossing	H	Site-specific independent design/stage acceptance, road/water permits, closure fit and final structural inspection
Depot/energy civil handover	H	Actual yard/equipment layout, foundations/pits/ducts/earthing/drainage, fire/replacement/maintenance access, tests and load restrictions
Reinstatement/public handback	H	Utility/road owner acceptance, accessible routes, drainage, waste/property records and defects
Structural/civil handover	H	As-builts, quantities, installed serials, complete inspections, monitoring, manuals, warranties and controlled remaining restrictions
Complete railway opening	H	Civil plus station/track/energy/depot/fleet/testing/approv al acceptance for the same installed configuration

Test planning and first repetition

Select representative foundation/ground zones and temporary-loading configurations, Pi20/Pi25/cap/column production and transport/lifts, complete first bay with connection/track/walkways, first station primary structure and access equipment, first at-grade slab/panel zone, special crossing and depot/pit/energy interfaces. Test both ordinary installation and credible degraded/recovery conditions. The representative unit does not qualify unlike ground zones, structural products or equipment configurations without a reviewed extension.

Record baseline calibration, test setup/instrumentation, operator/inspector competence, measured values, uncertainty, raw files, criteria and configuration. Review results before shortening cure cycles, increasing erection rates, reducing restraints/buffers or repeating a changed method. Failed results stay in the package and trigger investigation/rework/retest under a controlled NCR.

Following-trade and final handover

Use a signed boundary packet for each run/station/site: allowed loads/work/access, residual temporary works, edge protection, drainage/earthing, penetrations, installed component IDs, inspection state and outstanding restrictions. Systems contractors accept the civil interface before mounting, pulling cables or energising. No shared folder or completed-task flag substitutes for that packet.

At final handover, reconcile every purchased/installed item to the asset and drawing revision; retain survey/model/IFC references, supplier certificates, material/installation/test records and NCR/BCF closure. Transfer inspection/maintenance intervals, bearing/joint replacement and recovery access, drainage cleaning, warranties/spares and monitoring baselines. Record owner/operator/independent and local-authority acceptance separately.

Store native ERP references for Project, Task, Asset, construction release and competence/reservations, with access-controlled personal records. Public planning artifacts contain proposed roles and unaccepted evidence templates; they do not publish invented inspector identities or customer approvals.

Civil programme and workface controls

Repository source: `docs/civil/civil-programme-and-controls.md`

This is the scheduling and execution interface for the civil master plan. The generated package graph includes all fourteen civil packages per line and links the complete-line opening requirements for every current line. It supplies logic and quantities; dates stay unknown where surveyed scope, measured rates, contract calendars or release dates are missing.

Work-breakdown and asset records

Use programme city line/zone continuous run, station, special, depot or energy site work package approved Task serialized installed asset. Keep running span/support IDs, station/platform/access IDs and special/utility/drainage/yard references stable. Model station and transition scope explicitly rather than duplicating it in running bays. Bill quantities and costs by this hierarchy; link drawing/method/design revisions and suppliers.

Each Task carries location and IDs, method revision, design/temporary-works reference, predecessor release, measured quantity, required role quantities, reserved equipment/materials/crew, allowed shift/calendar, estimated/measured duration, hold points, completion evidence and independent acceptance requirements. Method authoring status cannot issue a construction permit. Use current native ERP eligibility and future reservations before assignment; revalidate at start and after a relevant change.

Programme stages

1. Mobilisation/design: appoint roles; land/survey/utility/ground investigations; approve worksite/temporary-works strategy and procurement scopes; qualify representative methods and trial products.
2. Enabling: create access, drainage and working platforms; divert/protect services; obtain road/water/land permissions and construction power/water.
3. Parallel production: qualify factories/first articles; construct verified foundations and substructure ahead; build station and special packages; book delivery and erection paths.
4. Running erection: draw from accepted stock and support buffers; complete two-beam bays with checked restraints and stage handover; track delivery/rejection and resource hours.
5. Interfaces and finishing: station approaches/special closures, continuity/joints, walkways/drainage/waterproofing, track support/rail, station enclosure/access, depot/energy civil and public realm.
6. Handover/testing: inspect complete assets, clear NCRs/restrictions, issue as-builts and systems access, complete integrated railway commissioning and independent/local approvals.

Foundations, columns/caps, beam supply and erection operate as linked crews and queues. Use a location/time chart to preserve the tested safe lead/lag and buffer, then a finite-resource CPM for stations, specials, utilities and handovers. Do not take the maximum of optimistic independent rates as the accepted programme.

Calendars and staffing

The proposed erection calendar is six days/week with an eight-hour shift before approved leave, holiday, heat/weather, maintenance and traffic-window deductions. No night work is credited without approval, supervision, lighting and relief. Factory curing and delivery operate on their own elapsed-time and permitted-shift calendars; a weekend erection stop can increase accepted stock and overflow a yard if production continues.

For the explicit single-shift capacity screen, 80% productive time less 0.5 h handover and 0.5 h maintenance leaves 5.4 productive hours/day. The existing connected reference has two 2 h placements, two 1 h securing operations and 2 h advance: an 8 h serial bay cycle, with two shifts in its original scenario. At one bay per day, a single-shift cycle would need to fit 5.4 h; at 1.5 bays, 3.6 h. Those targets therefore require measured method improvements or additional authorised shift hours. The generated review records this mismatch; it does not transfer the original two-shift rate into the one-shift calendar.

Calculate role-hours from actual task cycles and simultaneous fronts. Reserve qualified survey, foundation, pier/cap, launcher operator, two riggers where the method requires them, lift supervisor, independent inspection and handover resources. Add production/logistics, station trades and specialist installation crews by their own workloads. Protect supervisor/inspector independence, rest, breaks, leave, training, sickness and handover. A nominal role present does not satisfy required quantities or concurrency.

Recruit backwards from first-article/readiness dates through lead time, induction, training batches, supervised practice and practical equipment/task assessments. Supplier commissioning/technical support and equipment-authorised operators are separate dependencies. Training completion cannot substitute for work authorisation. The workload/recruitment model

leaves missing measurements and funded appointments open.

Running-front control and reassignment

Before the weekly plan, verify design/support/utility release horizons, accepted beam SKU stock, production slots, delivery appointments, equipment condition, crew competence and permissible work hours. Plan only tasks that can satisfy those constraints; show withheld packages and limiting resource. Review daily queue/buffer levels, cycle components, weather/traffic losses, NCRs and plant maintenance.

Treat each of the 548 disconnected Baghdad running intervals as a separate access/launcher-boundary review. Survey support spacing and actual relocation routes. Reuse a launcher sequentially only after dismantling or checked passage, transport, new-front readiness, reassembly/inspection and commissioning. Assess radial-to-ring redistribution with all resources and full-line milestones, then evaluate transfer to the next city. Record equipment exclusivity during every movement.

Cost and cash programme

Time-phase accepted quotation scopes against fabrication, factory acceptance, dispatch/arrival, installed milestones, tests and handover. Separate advance payments/security, retention, provisional items, recurring plant costs and final account. Finance ties to the same WBS and quantities; cash need may precede installation by months. Link delay/rejection/route changes to storage, demurrage, standby, interest and replacement, preserving unknown values rather than setting them to zero.

Use existing finance as a comparator until complete quantities/prices, scope credit and accepted dates are supplied. A higher launcher rate does not establish an installed-rate saving. Systems/fleet/testing/approval milestones remain separate from civil handover, and paid passenger revenue starts only through the accepted opening/service scenario.

Inspection, change and reporting

No next-stage release passes an open hold point or unresolved NCR affecting safety, load path or concealed work. Retain drawing/method/configuration hashes, observations, measured results, equipment/worker authority, reviewer identity/date and sign-off scope. Reinspect after transport damage, abnormal load, weather event, utility change or failed test. Quarantine changed/rejected items and reopen affected tasks/handovers.

Publish weekly planned/completed quantities by package, accepted stock, unique released supports, active fronts, installed span/beam IDs, station milestone state, labour/plant hours, rejects/rework, closures/disruption, forecast dependencies, costs/cash and decisions. Separate physical progress, documentary readiness, conditional forecast and accepted opening. Keep a reviewable change record showing input quantities, resource assumptions and conclusions that changed.

Complete opening

CW-14 civil handover requires CW-03/04/06/07/08/09/10/11/12/13 completion as applicable and controlled enabling/production records. Link it to running structures, stations, specials/transitions, track, energy and depot/stabling opening packages. Fleet, integrated testing and approvals are additional gates; the railway opens only after all are accepted for the same line/configuration. Missing package dates or costs keep complete opening date and cash unknown.

Civil works master plan

Repository source: `docs/civil/civil-works-master-plan.md`

Revision basis: 8 October 2026. Detailed construction planning and method-development package; project drawings, temporary works, permits, prices and site acceptance remain required before execution.

This plan connects viaducts, stations, ground-supported railway, special crossings, depots, energy sites, utilities, drainage, access and reinstatement. Use 25 m average span for strategic production and logistics sizing. Purchase and erect against the identified span schedule: shorter catalogue bays, station transitions and separately designed closures retain their own geometry. Do not replace that schedule with rounded kilometres multiplied by a standard beam count.

The generated Baghdad civil works package reconciles the current alignment, running-span register, station topology and depot scope. It provides line quantities, workfront and station registers, production/logistics sensitivities, package dependencies, inspection gates and source locks. Run the civil planning compiler after a controlled alignment or method revision; `--check` rejects stale outputs. Other cities can apply the methods after substituting their own identified assets and local evidence.

Package structure and responsibilities

Package	Accountable role	Controlled outputs and handover
CW-01 surveys, land and utilities	survey/consents lead	Survey control, corridor boundaries, utility trial-hole register, property/access permissions, verified site model
CW-02 enabling and temporary works	construction manager and temporary-works coordinator	Worksite/traffic layouts, haul roads, platforms, drainage, hoardings, design/check register, permits to load/use/dismantle
CW-03 foundations and ground treatment	geotechnical authority	Zone design, support schedule, tested method, actual installed lengths, verification and settlement records
CW-04 viaduct substructure	bridge construction lead	Columns, caps, abutments, bearing seats, dimensional/strength records and released support packets
CW-05 precast production and delivery	precast/logistics manager	Existing-facility audits, shop drawings, accepted serialized components, load/route plans and receiving records
CW-06 viaduct erection and completion	appointed lift planner and bridge lead	Configured plant, erection stages, temporary restraints, connections, waterproofing, barriers, drainage and structural handover
CW-07 station civil and access	station construction lead	Structure, platforms, shafts, stairs, concourse, roofs, street routes, equipment interfaces and public access acceptance
CW-08 at-grade track and approaches	track/earthworks lead	Released formation, treatment, retained approaches, slab/panel zones, crossings, track geometry and drainage
CW-09 special crossings and closures	special-structures authority	Site-specific bridge/closure design, access and erection study, water/road interfaces, independent check
CW-10 drainage and flood protection	drainage authority	Catchments, receiving outfalls, attenuation, cross drainage, scour/flood controls, clean/test records
CW-11 depot and stabling civil	depot design/construction lead	Surveyed yard, buildings, pits, pavements, tracks, charging foundations, workshop and fire/access interfaces
CW-12 energy and railway systems civil	systems interface manager	Ducts, foundations, equipment rooms, earthing interfaces, fire separation, pull/test and systems handover
CW-13 roads, public realm and reinstatement	urban works lead	Permanent traffic arrangements, accessible paths, road/utility restoration and local handback
CW-14 integrated acceptance	owner handover manager	Asset/as-built records, outstanding defect controls, independent acceptance and line-opening packet

Roles are proposed responsibilities, not appointments. The owner appoints designers, principal delivery leadership, independent checking and local approving bodies before corresponding releases. A contractor programme cannot grant its own independent acceptance.

Scope and 25 m average-span basis

The expanded Baghdad planning inventory has 54 lines and 863.9238 route-km: 273.8781 at grade, 545.9111 elevated and 44.1346 bridge/water-crossing. Running elevated intervals cover 407.4884 km; the remaining 138.4227 km of elevated station/transition scope requires station and interface design. These are distinct chainage scopes. Special bridge class is additional to unresolved short closures in running elevated intervals.

The identified running layout contains 15,888 catalogue bays (14,248 Pi25 and 1,640 Pi20), 31,776 catalogue single-track beams and 747 separately designed spans. Its 16,635 running spans average approximately 24.50 m. There are 17,782 proposed running support chainage records across 1,147 disconnected runs. The integrated line-based plan resolves closed-ring start/end aliases into physical support packets. Its candidate foundation count requires field design; support sharing, boundaries and specials prevent deriving foundations as bays plus one. The generated civil quantities and line

assembly summary control later revisions.

For a reference uninterrupted kilometre, 25 m means 40 double-track bays and 80 single-track full-span beams. Count 41 support positions only for that isolated kilometre; adjoining sections share boundaries. Four-span structural-continuity units and simple-span link slabs require different bearing counts and stage designs. Neither a 25 m average nor a reference-kilometre count establishes a foundation, pile length or released beam order.

The station scope includes 600 stations, 747 physical platforms and 1,303 boarding faces. There are 452 elevated and 148 at-grade station layouts, with 1,014 each of lifts, shafts, stairs and escalators. Layout counts do not size foundations, fire stairs, concourses or accessible street routes. Each station gets an individual work package and interface review.

Programme logic and working calendar

Use the programme and workface controls with viaduct, station, other civil works, logistics, and inspection/handover methods.

Start with two candidate erection fronts per line and an eighteen-machine study pool, subject to independently feasible released fronts. Expanded inventories queue later fronts behind the previous assignment of each physical machine, including transport/recommissioning gates. Foundation and substructure crews work ahead; precast stock is accepted before dispatch. Station structures and specials are parallel critical-path packages with explicit crossing/relocation interfaces. When radial fronts finish, evaluate additional ring fronts before moving machines to another city. Extra launchers help only when access, foundations, accepted stock, deliveries, crews and temporary works support them.

The resource envelope uses a proposed six-day erection week, an eight-hour shift and explicit holiday/maintenance/weather deductions. Production curing uses elapsed time, not an erection working-day counter. Illustrative rates of 0.5, 1.0 and 1.5 accepted bays per launcher-working-day test capacity; none is a measured production commitment. The minimum quantity/rate duration is a lower bound. Full dates require site-specific quantities, calendars, finite resources, release dates and the complete network of predecessors. Use the regenerated finite-pool programme for conditional running-bay comparisons; full line openings retain all station, special, depot, systems and acceptance gates.

Existing industry, procurement and commercial plan

Iraq has many existing precast facilities and production expertise. Begin with factory capability and spare-slot audits, adaptable beds/moulds, prestressing/handling equipment, concrete and reinforcement supply, and regional delivery routes. Use the potential vendor register for screening; a listing gives no compatible product, spare capacity or agreement. A dedicated new civil factory is an option only after comparing existing contracted capacity, upgrades, site and transport costs.

Issue configuration-specific packages for Pi20/Pi25, caps/columns, panels/walkways, bearings/expansion joints, lifting equipment, heavy haulage, station modules, drainage, track support and systems foundations. Separate design/IP support, tooling, qualification/first article, repeat supply, freight/permits, installation, commissioning, spares, maintenance and taxes. Require supplier drawings and declared complete mass, CG, lifting/support positions, release strengths, tolerances and traceability. Record rejected items, NCR rework and replacement capacity in delivery planning.

Cost, payment and opening interfaces

Build the estimate from named packages and measured quantities. Record foundations by actual type/count/length, columns by height, caps by selected type, beams by span ID, station structures by engineered take-off, and earthworks/ground treatment by surveyed volume/area. Include temporary works, access/traffic, factory tooling, spoil disposal, hauling/escorts, storage/handling, crane or launcher mobilisation, testing, utility diversions, special crossings, flood works, systems civil interfaces, overheads, escalation, risk and contingency.

The retained route-km rates and USD 9m launcher purchase allowance are comparator scopes; they are not a complete installed quotation. Credit embedded erection or plant allowances only after reconciling invoice scope. Milestone payments follow accepted releases/deliveries/installation and contractual certification, with retention and final handback separate. Unknown prices remain unknown; no sum of known rows becomes the full adopted CAPEX.

For each line, opening requires accepted running structures, stations, specials/transitions, track, energy, depot/stabling, fleet, testing and approvals. Construction handover includes safe access for following trades; opening additionally needs integrated railway and operating acceptance. The coupled programme study retains these gates and conditional finance months. No date, revenue or saving is adopted from a running-span finish alone.

Review and release plan

Review the first representative complete bay, station access/structure package, at-grade treatment/track zone, and depot/energy foundation package before repetition. Check design interfaces, sequencing, working platforms, ground/temporary loads, manufacturing and logistics capacity, tolerance stack-up, traffic/access, rescue arrangements and full installed costs. Feed measured first-cycle durations, rejected quantities and disruption back into the finite-resource programme.

The generated review register distinguishes repository quantity consistency from site approval. It currently flags unresolved survey/structural releases, no qualified supplier or delivery-route capacity, missing station/special take-offs, inconsistent illustrative buffer sizes, and incomplete depot/stabling provision. Close each through a dated configuration-bound record; do not turn an authoring review into independent construction acceptance.

Reference practice

International guidance supports method development; Iraqi authorities and the project design basis determine adoption. FHWA prefabricated bridge elements describes off-site fabrication, transport and site assembly. FHWA accelerated bridge construction treats planning, geotechnical and structural placement choices together. HSE temporary works guidance describes coordination, designs and checking. HSE lifting-operation planning supports competent planning and supervision. These references do not supply Baghdad geometry, capacities, acceptance values or permits.

Complete viaduct system exploration

Repository source: `docs/civil/complete-system-exploration.md`

Review corrections and remaining sequence

The review of `4a20d183` found four baseline weaknesses. Construction scheduling now hashes its imported `tools/automation/project_twin.py` implementation, and a regression test verifies that changing it rejects campaign resume. The dedicated civil workflow runs its bounded numerical tests on every main push and includes the imported model, parser, geometry, scheduler and reference dependencies in its pull-request filters. General CI and station selection are checked separately.

Screening and detailed confirmation use the same provisional constraint evaluator: relative deflection, settlement, pier drift, foundation axial demand, lifting mass and gross deck/pier stresses where a limit is actually declared. Unknown strength limits stay open. Detailed 3D checks retain individual support reactions, offset reaction couples and pier motion under two-track, one-track and braking cases in both ground scenarios. Convergence covers these demands, cap bending, deck torsion and gross stresses. Solid stress convergence uses a volume-weighted integration-point RMS in the central half-span, avoiding support singularities; the unrestricted peak stresses are still retained.

The shortlist reserves two cases per registered span before filling commercial leaders and beam families. A span with no screening passes contributes its least violating designs for detailed diagnostics. Their failed constraints stay visible and exclude them from confirmed winners. No limits are relaxed to obtain coverage.

Evolution now uses installed cost, whole-life cost and working time in every declared commercial scenario, plus installed mass and maximum erection mass. Per-seed benchmarks report distinct packages, new evaluations, measured evaluation time, feasible span/family coverage and normalized additive epsilon against the pooled observed Pareto front. This comparison does not establish a global optimum.

The next model extensions remain purpose-specific: V/Y supports, variable haunch and tendon geometry, ground treatment/load transfer, assembled nonlinear cyclic behaviour, and structural construction stages. Fabrication/curing/erection schedules do not validate prestress transfer, temporary support stability or joint locking. Their adoption requires explicit model adapters and independently benchmarked load paths. Measured Baghdad axle, soil/pile, material/connector and equipment records, supplier quotations, physical tests and independent acceptance remain open. The ordinary Python CAD facade remains a fallback; native kernel verification must carry a separate FreeCAD receipt.

The workbench now investigates complete packages rather than varying one hollow beam/pier with a fixed foundation. It extends the original Word action plan and preserves the C01–C14 acceptance audit. The root README exposes the choices, executed comparison and model limits.

Scope and choices

The common comparison uses 300 m of double track, allowing whole 20, 25 and 30 m bays with consistent end supports. This is a research segment, separate from the initial 100 m reference milestone and the complete Baghdad inventory. Existing Pi20/Pi25 shapes remain controls. Ten beam families, six pier families, nine foundation layouts and three connection schemes are registered before search. Each carries allowed parameters/materials, ownership and uncovered failure modes. More families are not silently treated as qualified catalogue products.

Normal/HPC/lightweight concrete selection changes the deck's density and stiffness. It does not automatically lighten piers or foundations. Separate support and foundation material records govern those quantities. Targeted UHPC, structural steel and GFRP/CFRP occupy disjoint actual material regions. FRP is orthotropic; neither its matrix nor its shell is credited as a replacement for required seismic reinforcement. All initial property sets are explicitly synthetic hypotheses, with no supplier mix, laminate or measured strength adopted.

Foundation geometry includes solid circular shafts/piles, square precast piles, annular precast piles and zero-pile spread footings. Net voids survive CAD and IFC export. Pile diameter/length and cap dimensions are sampled with the beam and pier. Diameter changes update cap plan size proportionally to the declared layout; this preserves its unqualified spacing ratio rather than inventing a site spacing acceptance. Exact capacity and cap reinforcement remain open.

The native pile model calculates three cap unit-load responses with elastic Timoshenko piles, axial/lateral soil springs, toe springs and a rigid cap. Reciprocity, positive energy and at least three discretisations are checked before condensation. Horizontal–rotation coupling is retained in the system model. Refinement continues from 16 to 32/64 elements per pile when the declared stiffness convergence check needs more resolution; unresolved cases retain a numerical-failure status. Both longitudinal and transverse axes are checked. Separate longitudinal/transverse condensed responses drive the 3D model. Shallow support springs use the declared rectangular footing and soil bed. Installation methods receive no automatic soil improvement credit. These are declared linear research soil laws, not Baghdad stratigraphy or a substitute for calibrated nonlinear group, cyclic or liquefaction response.

Loading and numerical confirmation

The six-car infrastructure planning allowance is divided by the retained train length to create a uniform static stiffness sensitivity. Axle concentration, train dynamics and combinations are not inferred from this average. The static profile contains no LM3 or invented Baghdad axle array. Actual six-car axle positions/forces remain blocked in the qualification workflow.

Reduced native screening includes gravity, uniform service and the declared braking sensitivity, finite bearings, material-region stiffness, piers and foundation compliance. Provisional response/lifting/nominal soil-resistance limits are tagged research constraints. They do not establish concrete capacity, prestress, fatigue, site foundation resistance or adopted railway limits. Failures remain distinguishable from missing evidence and from solver failures.

The diverse shortlist receives three actual C3D20R solid meshes with common gravity/service pressure, source-bound native displacement/stress fields and an explicit displacement convergence criterion. The 3D shear-flexible frame uses separate tracks, flexible hollow transverse caps, column/cap offsets, finite bearing translations/roll and coupled foundation stiffness. Both-track, single-track and braking sensitivities run at three meshes in both soil cases. Vertical/horizontal force and moment equilibrium are checked independently. Saint-Venant rectangular torsion and closed-cell Bredt torsion are explicitly bounded formulations; open-section warping/distortion remain uncovered.

A two-layer composite diagnostic adds distributed shear-slip energy to axial and bending behaviour. Independent no-bond and strong-connector analytical limits verify it. Unmeasured connector stiffness sweeps show possible loss of the perfect-bond advantage; they cannot establish connector resistance or debonding/fatigue/fire behaviour. Existing nonlinear fibre, pile, P-delta, friction, orthotropic and coupled-vehicle component benchmarks are rerun. Shortlisted piers also receive a native P-delta diagnostic using the minimum effective section and worst reduced axial/shear actions. Its independently checked prismatic cantilever domain does not establish full flexible-base stability, taper/joint/cyclic capacity or accepted candidate resistance.

Construction and complete cost

Each method has actual shipping/lifting units derived from canonical regions: full members, beam segments, empty structural shells, pier segments/shells, caps and precast pile segments. Rebar/tendon/inserts remain declared quantity allowances, not fabricated bar/tendon geometry or accepted rigging weights. Completed service mass is distinct from empty transport mass. Driven piles include production, curing, delivery, handling/splice time and installation. Site concrete delivery is separate from material supply.

The existing `project_twin.apply_resource_cpm` engine schedules whole-hour slots with finite rigs, beds, curing yard, transport, erection, site crews and QA. Procurement, foundation tests, production, support readiness, infill, joints, curing and final inspection have explicit dependencies. External gates remain unresolved, with no assumed zero-duration acceptance. Reported working time starts conditionally after those gates and uses conservative whole-hour rounding; it is not an adopted Baghdad calendar or delivery commitment.

Every package has complete material-separated quantities and rows for material, reinforcement/tendons/inserts, bearings/joints, piling plant, delivered concrete, precast transport, crane plant/mobilisation, temporary works, labour, bed/curing/QA/tooling, track/walkway/drainage interfaces, reinstatement and site overhead. Synthetic annual maintenance and bearing/joint replacement scenarios give discounted whole-life sensitivities. Actual rates/charts/availability and maintenance intervals remain open.

Three synthetic price/productivity cases expose ranking reversals. They are not Baghdad market estimates. Hypothetical crane tiers expose step changes but are not accepted crane charts. Actual supplier-price totals remain null when any applicable cost row is missing. The cheapest source-priced qualified design remains unresolved. Carbon totals remain unknown without controlled EPD factors.

Search, records and review

Random and Pareto-evolution searches use equal attempt budgets and multiple seeds. Every registered beam family has engineering seeds and early coverage. Evolution changes bounded dimensions, materials, pier/foundation choice and construction method within compatible families. Cost, installed mass and working time remain separate objectives; installed/whole-life/lifting tradeoffs remain reviewable. The best found choices are bounded by the declared search, not a proof of global optimality. Reduced leaders and converged-shortlist leaders are reported separately.

Candidate IDs bind geometry/material/foundation data. Package IDs also bind assembly/construction choices; input snapshots bind loads, scenarios, code and native binaries. Modification events retain parent packages and reasons. Checkpoints verify all saved results, raw schedules, costs and source bytes. Failed native/refinement attempts retain their logs. The full shortlist is packed in immutable sub-50-MiB archive parts and restored with member hash verification. A sealed change proposal names the existing change/release workflows while keeping acceptance false; it never promotes itself into the catalogue.

```
tools/automation/osr-python tools/automation/civil-study.py system-study \
  --output build/engineering/civil-studies/baghdad-complete-systems-v2
tools/automation/osr-python tools/automation/civil-study.py system-verify \
  build/engineering/civil-studies/baghdad-complete-systems-v2
tools/automation/osr-python tools/automation/export-civil-system-review.py \
  build/engineering/civil-studies/baghdad-complete-systems-v2 --update-readme
```

`--resume` requires matching configuration, dependencies, native environment and all checkpoint hashes. `--skip-detail` explicitly creates reduced evidence only and cannot be exported as the retained confirmed review. Use `--seeds`, `--evaluations-per-method` and `--shortlist` for bounded pilots. Configurations and quote sources must be retained repository files.

The compact retained comparison and JSON have a named regression consumer. The scratch `report.html` is a self-contained filter/sort viewer with quantities, failures and evidence gaps; `comparison.csv`, material-section SVG and Pareto SVG support external review.

What cannot be completed from software alone

The C01–C14 software/evidence audit is generated with every comparison. Project/supplier load records, adopted requirements, measured mix/laminate/bond properties, reinforcement/tendon/anchorage designs, support-zone ground/survey, installation/equipment trials, quoted cost/productivity, physical instrumented tests and independent authority acceptance remain actual dependencies. V/Y support systems, site ground-improvement load transfer, haunched members and novel recycled primary structures remain purpose-specific model/adoption work, with explicit owners/data needs. The current search does not count them as implemented or qualified families. Engineering acceptance of all document items therefore remains incomplete.

Research sources and limits

FHWA's lightweight concrete primer supports investigating density and stiffness together; its highway bridge guidance does not qualify a local metro mix. FHWA's UHPC waffle-deck work includes connection and field studies; the targeted-rib option here is a distinct research geometry, not that tested bridge product.

MxV Rail's hybrid-beam service observations show why connection deterioration and inspection access need explicit evidence. The straight shell/I options here do not reproduce the tested concrete-arch HCB. FHWA's CFA/displacement-pile guidance identifies installation-dependent behaviour and limitations; no automatic capacity benefit or constructability acceptance is adopted from it. The OpenSees 3D Timoshenko interface defines the native shear-flexible formulation used; equation checks and solver agreement provide numerical verification, not physical validation.

Civil construction logistics plan

Repository source: `docs/civil/construction-logistics-plan.md`

The master plan uses 25 m average spans for logistics sizing and identified spans for orders. The generated logistics envelope distinguishes assumptions, required capacity, available evidence and adopted capacity. No surveyed delivery route or contracted supplier capacity is currently entered.

Factory and regional-yard network

Audit existing Iraqi precast facilities and production expertise before selecting expansion or a new factory. For each plant record bed usable length/width, mould positions, prestressing capacity, mix/curing control, cage production, handling/stacking capacity, available shifts, current commitments, inspection staff, power/water, raw-material suppliers, environmental permits and delivery access. Assess Pi20/Pi25, column/cap shells, walkways and ST6 separately; a suitable panel factory is not automatically a prestressed-beam supplier.

Select regional yards by permitted land, road/bridge access, geotechnical bearing/drainage, safe handling, flood risk, distance and delivery restrictions. Separate raw materials, cages/moulds, curing, accepted stock, quarantine, dispatch lanes and returnable gear. Layout access for the longest vehicle, fire/rescue and crane replacement. Design dunnage, restraint and storage stacking; the actual layout determines gross area and maximum stock. A count of beams cannot establish a yard area without supports, spacing and handling aisles.

Capacity equations and scenario boundaries

Requirement	Equation and qualification
Erection supply	`active launchers × accepted bays/launcher-working-day × 2 beams/bay`
Production positions	`ceil(accepted daily beam demand × mould cycle hours / production hours per day / first-pass yield)`; a position casts one complete beam per cycle
Weekly balance	`erection working days/week × daily demand`; compare with actual factory/haulage working days, cure time and outages
Fresh concrete	Beam-specific accepted count × released cast volume, adjusted for production rejects; foundation/station/other concrete is additional
Beam movements	One complete beam per proposed load; verify transporter configuration, axle/bridge capacity, load/support restraint and route permissions
Round-trip vehicle cycle	Loading + outward journey + unloading + return + delays/driver breaks; reserve loading/unloading equipment concurrently
Trailer requirement	`ceil(required trips/day / executable trips/vehicle/day)`; derive trips from the route's permitted window and complete cycle, then add independently justified resilience
Beam buffer	`2 × 10–15 released bays × active fronts`, limited by remaining continuous run, storage capacity and accepted handling/access
Ready support buffer	All unique supports for the next 10–15 bays, each released for the planned machine/beam load; endpoint sharing is explicit

The generated six-day scenarios use illustrative 48-hour production cycles, 24 production-calendar hours/day, 90% first-pass yield, a 12-hour haulage window and an eight-hour complete route cycle (including a declared allowance for return and delays). These are sizing assumptions, not supplier or route evidence. A 12-hour window is not permission for a driver to work without statutory rest. Verified driver/crew calendars further constrain executable trips.

Daily rates also need a compatible erection cycle. The programme controls deduct productivity, handover and maintenance from the proposed single shift; the existing eight-hour serial reference cycle does not support the higher one-shift targets. Resource rows show the cycle that would be required and retain achieved rate as unknown.

At eighteen active launchers and one accepted bay per launcher-working-day, the requirement is 36 accepted beams and 450 equivalent route-metres per erection day. Using the Pi25 complete-member study, beam cargo alone is about 3,078 t/day; trailer tare, rigging and all other materials are additional. Production at 90% yield needs 40 gross beams/day and 80 one-beam mould positions under the illustrative 48-hour/24-hour basis. These are capacity requirements to compare with available audited beds, not a proposal for eighty new moulds or a commitment to supply.

The 10–15-bay buffer requires 360–540 accepted beams across eighteen sufficiently long fronts. Existing conditional route examples use twelve beams per front, equivalent to only six double-track bays. The generated review flags that mismatch. Reconcile buffer policy, site/yard storage and front lengths before assigning a build rate; do not silently claim that six bays meets a ten-bay requirement.

Route survey and transport release

Create one route record per supplier–yard–front movement, with GIS path and direction, surveyed dimensions, gradients/crossfall, bends/swept path, weak bridge/culvert/pavement limits, overhead utilities, temporary widening, load permits, closures/escorts, delivery windows and alternates. Record the actual vehicle, axle spacing/load distribution, gross mass, loaded height/width/length, ground clearance, cargo CG, support/restraint design and braking/grade constraints. Gross tonnage alone cannot assess a bridge or pavement.

Load using inspected equipment and released beam/support points; verify member ID, mass, release strength, dimensions, restraints and travel documentation. Inspect after loading and at receipt. Use wheel/ground restraints and segregated receiving zones; place on designed storage supports or supply directly to the released erection position. Missing route, equipment or access evidence blocks dispatch even when a beam is accepted.

Release lift, transport and storage separately. The current Pi25 manufactured study is 85.51 t, versus 89.51 t suspended with illustrative gear. Transport gross mass also includes trailer/tractor and carried accessories. Neither the bare 75 t product target nor nominal equipment tonnage clears those loads. The handling envelope and supplier actual configuration govern release.

Booking, stock and material flow

Allocate each beam to span and track, factory slot, inspection gate, route, receiving yard/front, delivery appointment and installed record. Dispatch in erection sequence, including Pi20 and special interfaces; maintain location/condition and reservations in ERP. Reserve trailers, loading/unloading cranes, competent workers and route/closure slots before commitment. Keep rejected/awaiting-test stock separate from accepted available stock.

Coordinate concrete trucks, reinforcement/cage deliveries, spoil, station modules, lifts/escalators, rail/panels, fuel and systems equipment in the same access calendar. Check opposing flows and shared loading gates. Keep deliveries off pedestrian/emergency routes and avoid unplanned roadside queuing. Provide supplier/driver check-in, receiving inspection, safe standby, washout/spill and emergency procedures.

Record daily opening stock, accepted production, rejects/rework, loaded/dispatched, in transit, received, installed and closing stock by SKU/location. Balance mass and count; no item can be simultaneously available at the factory and workfront. Front saturation or a stopped crane/launcher pauses dispatch. Test loss of a supplier, road closure, trailer breakdown, rejected load, weather stop, buffer overflow and extended curing, with identified safe storage and controlled resequencing.

Material quantities, spoil and environment

Develop take-offs for concrete/rebar/prestress, foundations, station modules, track/panels/rails, bearings/joints, drainage/ducts, fill/ground treatment and road reinstatement. Convert them to delivery lots by approved vehicle/material constraints. The generated beam-only concrete quantity excludes other civil scope; do not present it as total project concrete or infer rebar from a generic kg/m³ study as a purchased schedule.

Plan permitted quarries/batching, water quality, standby concrete supply, curing/shade/temperature measures, pumping reach and rejected concrete disposal. Reuse suitable excavated material only after tests; measure outgoing spoil and transport conditions. Record disposal receipts, dust/noise/vibration controls, wheel cleaning, runoff/spill containment and packaging/returnable-rack recovery.

Launcher and station-module movements

Provide discrete transport/assembly/commissioning packages for each launcher transfer. Check dismantled component weights, unloading platforms, escorts/access, assembly crane and tested foundations/deck supports. Reserve transfer crew and equipment; assess additional ring fronts against remaining critical-path demand. The launcher is unavailable to both old and new fronts during transfer and recommissioning.

Station modules have their own load/route/installation packages and often share access with running beams. Reserve roof/platform/shaft lifts, maintain following-trade and emergency routes, and deliver access equipment before closing necessary apertures. Under-deck storage or commercial use requires fire, flooding, impact and inspection-access review; it is not automatically free logistics space.

Civil Construction-System Selection

Repository source: `docs/civil/construction-system-selection.md`

Status: deterministic planning rules; every selected product remains subject to project structural, geotechnical, rail-system and construction release.

The machine-readable source is `lib/templates/civil-construction-systems.toml`. It changes the method used to construct the existing civil product family. The retained 3/12/18 M USD per route-km benchmarks remain visible for audit; the active planning rates are regenerated from the corresponding CAD quantity drivers in `civil-cost-model.toml`.

Zone condition	Default selected method	Deterministic exception
Long open at-grade run	Continuous slipformed twin-track slab	ST6 single-track panels in tagged utility, street, flood, transition or short-possession zones
Normal 20/25 m viaduct	Four-span semi-continuous Pi-beam unit	Five spans may be assessed inside the 80–125 m movement range; special products remain separate
Viaduct approach up to 4.5 m	Reinforced-soil retained embankment	Reject for flood/scour, severe settlement, insufficient ROW or failed cyclic deformation check
Soft ground beneath formation/embankment	Zone-selected rigid inclusions, deep mixing, stone columns, lightweight fill or stabilisation	Full project-specific solution where catalogue applicability fails
Ordinary pier with clear piling access	Driven pile bent without a buried pile cap	Integral bored shaft in vibration-restricted ground; pile group for high lateral/seismic/scour cases
Major road conflict	Compare road over/under/relocation and short rail bridge with long rail elevation	Discard infeasible options; score direct cost plus construction penalties

Semi-continuous reference kilometre

`semi_continuous_unit_plan(1000)` resolves 40 Pi25 spans into ten four-span units. Thirty internal joints require defined connections; ten deck gaps remain. One bearing line at an internal unit support gives four bearings; a unit boundary keeps two independent lines and eight bearings. The complete reference kilometre therefore has 200 bearings. No kilometre-long continuity is implied. The 200-bearing planning quantity requires structural continuity and a shared load path. Simple spans connected by link slabs retain 320 bearings/km. The active USD 9.748m/km target is conditional on that unaccepted structural alternative; restoring 320 bearings gives USD 10.198m/km under the same cost index, before connection and other consequential costs. No quotation or accepted saving is established by either index.

Release requires continuous-welded-rail interaction, braking/traction, temperature/shrinkage, seismic displacement, foundation-flexibility and connection fatigue/waterproofing analyses. Fully integral internal piers are a future option, not the default.

Geotechnical-zone record

Every zone chooses either a foundation product for a support or a ground- improvement product for ground-supported construction. Quantities are stored as actual pile/shaft length and count, treated area/volume and column geometry, installation time, verification result and installed cost. The model never converts route length into a generic volume of foundation concrete.

Manufacturing controls

The production target is a 24-hour reusable-mould cycle using self- consolidating concrete, maturity-based release, standard welded cages, controlled inserts and serialised component travellers. The schedule retains the validated 48-hour planning cycle until a mix, sensor, mould and first- article trial releases the target. Small closure pours and grouted sockets are preferred; rare 3D-printed formwork and exotic materials are not standard primary members.

Cost validation

The current CAD-indexed contract gives 2.584 M USD/km at-grade and 9.748 M USD/km elevated against the retained 3 M and 12 M benchmarks; bridge remains 18 M USD/km pending a bridge-specific redesign. These are planning targets, not quotations. Rail-specific prototypes, the project foundation schedule, installation trials and supplier prices are still required before a project budget is released. Each accepted revision must include its changed assumptions, source evidence, quantity diff and test outputs.

Civil And Station Deployment Release Checklist

Repository source: `docs/civil/deployment-release-checklist.md`

Generated alignments and station packages are planning-grade until site evidence replaces catalogue assumptions. This checklist defines what a deployment partner must add before civil/station design freeze.

Gate	Required evidence	Closure criterion
Survey-grade alignment	Topographic survey, utility survey, property boundary check, flood levels, and control points	Generated route geometry is replaced or confirmed by survey-grade alignment
Ground model	Geotechnical boreholes, groundwater, bearing capacity, corrosivity, and settlement risk	Foundations and at-grade sections are sized by local engineer
Structures check	Project issue of the `viaduct-design-basis.md`, complete deployment train (24 axles in Baghdad), kinematic/egress study, per-span 3-D analysis, seismic/wind/fatigue checks, prestress/reinforcement, foundations, drainage, and independent check	Licensed structural/geotechnical engineers sign the selected catalogue variants or issue OSR-US/OSR-SP designs
Bearings and movement	Eight-bearing interior/four-bearing end schedule, fixed/guided/free axes, deck gaps, jacking, thermal/creep/shrinkage, and CWR bridge interaction	Responsible engineer and track engineer jointly release the movement schedule
Transport and erection	Permit-load route, transporter, lifting points, temporary bracing, crane/launcher reactions, wind limits, contingency landing, and first-article trial	Temporary-works designer, lifting appointed person, and structural engineer close hold point
Station fit	Platform length, access route, drainage, lighting, PSD/edge protection, emergency egress, accessibility audit	Station archetype matches local code and passenger flow
Energy site	Solar yield and storage replenishment under actual charging duty, charger thermal study, battery/fire separation; interconnect and utility approval where residual grid backup is selected	Station charging assumptions are replaced by site-specific design
Permits and stakeholders	Road closures, land access, environmental clearance, emergency-services access, operator acceptance	Permit register has owner, due date, and approval status

For every city, this checklist defines the evidence needed to close survey-grade alignment replacement and deployment-specific assumptions. Generating the checklist does not close those gates.

The city soil and deployment workflow supplies soilDB predictions and uncertainty at planned station and civil-segment sample locations. Its investigation plan feeds the drainage/ground readiness report; it supplies regional context, not accepted bearing capacity or deep stratigraphy.

Issue the project information request from the canonical `field-evidence.toml` template. The Samawah field-evidence brief shows the required packages and provisional accuracy schedule. Its receipt manifest is intentionally empty: only the appointed authorities can approve the CRS, vertical datum, delivery and mobilisation. The linked control-processing report then verifies receipt metadata and hashes, runs the frozen RTKLIB profile when real inputs exist, and keeps technical screening separate from survey-authority acceptance. The subsequent ground-model gate checks the master GeoPackage, DTM, orthophoto, registered point cloud, independent residuals, processing reports and signed release without storing raw imagery or scans in Git. The following surveyed-alignment gate then requires one checked OSR-ALN, LandXML export and deterministic round-trip report per line, plus exact platform reconciliation, yard/turnout/clearance dispositions and a signed alignment-designer/survey/track/information-management acceptance record. Planning GIS traces cannot satisfy it. The subsequent route/station-fit gate requires hash-locked utility, property/access, flood/drainage, station access, intercity/yard, road/traffic, compound/lifting and possession/staging evidence. It reconciles every current line and station and rejects unresolved high or critical issues before coordinated authority acceptance. The drainage/ground gate then replays the project SWMM input, checks accepted rainfall/level provenance, requires borehole-zoned foundation or ground-treatment schedules for every line and station, and invokes OpenGeoSys evidence only when a reviewed groundwater/ coupling decision warrants it. The structural-release gate then binds every chainaged span, pier, abutment, foundation and special structure to converged OpenSees/CalculiX evidence, load cases, fatigue/seismic/ wind/bearing results, closed independent-check comments and signed release.

LM3 Depot Bogie-Change Interface

Repository source: `docs/civil/depot-bogie-change-interface.md`

This design-reference contract ties the LM3 underframe lifting hardware to the main-heavy depot civil and equipment assembly. Both models consume `lm3_bogie_change_datum()`; changing a jack centre on one side without changing the other therefore fails the cross-domain geometry test.

It is a coordination baseline, not a certified lifting design. Vehicle mass cases, asymmetric support cases, foundation reactions, reinforcement, column capacity, controls integrity, proof loads, and the equipment supplier's exact installation remain release inputs.

Controlled geometry

Interface	Design-reference value
Rail gauge	1,435 mm
LM3 car envelope	16,500 × 2,850 mm
Bogie centre spacing	12,300 mm
Jack-point longitudinal spacing	11,200 mm
Jack-point transverse spacing	2,360 mm
Lift-column transverse spacing	4,400 mm
Inspection-pit clear envelope	16,000 × 1,400 × 1,500 mm
Bogie extraction clear width	5,000 mm
Nominal lift stroke reserved	1,500 mm

Coordinates use car centre as X/Y origin and top of rail as the local vertical datum. The four lift heads are at X = ±5,600 mm and Y = ±1,180 mm. Lift-column bodies are outside the car at Y = ±2,200 mm; retractable arms reach the vehicle pads while preserving the pit and bogie-drop path.

Assembly boundary

The rolling-stock side contains four underframe load spreaders, four replaceable pad faces, surveyed J1–J4 targets, retained lifting-eye bosses, two towing and rerailing lugs, and external recovery instruction plates. `LM3-BDY-P120` is the packed car kit; `underframe-jacking-recovery-interface` is its installed review state within the full mechanical-interface assembly.

The main-heavy depot side contains the reinforced pit base and edge beams, running rails, four lift-column foundation/anchor pockets, supplier column envelopes, retractable mechanically locked arms, four controlled lift heads, two transverse extraction paths, transfer-table guide rails, a restrained bogie parking zone, local controls, emergency stop, and trapped-key isolation. The secondary and layup archetypes do not inherit overhaul lifting capability.

Release sequence and hold points

1. Freeze the weighed vehicle load cases, centre-of-gravity envelope, allowable jack reactions, and permitted support combinations.
2. Release underframe load paths, pad drawings, weld maps, NDT, proof-load values, and the physical four-point interface gauge.
3. Complete site geotechnical work and the pit, foundation, drainage, reinforcement, settlement, and accidental-action calculations.
4. Freeze the synchronized lift and transfer-table suppliers, including stroke, mechanical locks, loss-of-level limits, controls integrity, emergency lowering, and rescue provisions.
5. Survey the installed rail and J1–J4 head coordinates with the same gauge used on the vehicle; record the as-built results against the asset IDs.
6. Prove no-load motion, mechanical locks, emergency stop and isolation, then conduct the approved staged proof and asymmetric-load tests.
7. Perform a complete bogie disconnect, lift, drop, transverse extraction, restraint, replacement, lowering, reconnection, brake test, and vehicle release rehearsal before operational acceptance.

The generated station product row `STN-DEP-P060` carries these acceptance gates. Deployment structural, lifting-equipment, electrical-control, occupational safety, and local approval records remain mandatory external evidence.

Civil design exploration implementation and acceptance

Repository source: `docs/civil/design-exploration.md`

The action plan starts from commit `64549d8`; the implementation starts from `42c8bf8f0f`, after catalogue regeneration. The first executable workbench is `engineering/civil_exploration`. The original Word plan is an external planning input, not a model source.

Executed software workflow

One command freezes candidate and scenario contracts, replays the existing native reference, verifies analytical benchmarks and compares complete Pi20/Pi25, hollow-deck, hollow-pier and combined packages. Native outputs include reaction balance, dead/axle/braking cases, flexible support response, handling overhangs, modes, moving-force histories and mesh/time refinements. Its quantities preserve fabricated, transported, suspended and installed mass distinctions.

The first campaign uses 100 m of double track and the retained LM3 three-car planning axle pattern. Baghdad six-car supplier geometry is not extrapolated. Foundation dimensions/stiffness, damping and material/steel allowances are explicit scenario inputs. Supplier costs and accepted railway limits are absent. These gaps prevent engineering-feasible rankings regardless of numerical passes.

The expanded `civil-study.py programme --deployment reference` command now connects eight geometric families, material-region matrices, whole-assembly CAD/IFC, actual C3D20R solids, nonlinear sections/piles/P-delta, cyclic friction, coupled suspension/contact, thermal/continuity, complete costing interfaces, multi-seed Pareto search, range sensitivity, safe artifact retrieval, physical-test planning and the existing structural release gate. The generated C01–C14 audit records executed software separately from outstanding engineering acceptance.

The complete-system extension adds ten beam, six pier and nine foundation choices, lightweight/HPC/UHPC/steel/GFRP/CFRP matrices, variable pile/cap sizes, complete construction-unit/CPM/cost scopes, geometry-specific coupled foundation stiffness, 3D caps/tracks/bearings, composite slip and multi-family cost/mass/time search. Its native shortlist and root README are generated from the sealed research campaign. Accepted criteria, actual supplier/site inputs and physical/independent validation remain open.

Baghdad is now the active deployment qualification target. Its source-bound input readiness record retains the six-car planning profile and open supplier/site inputs. Default `run/programme` commands stop before numerical execution when these inputs are missing. The existing LM3 results remain explicit reference verification evidence.

Existing baseline verification found stale Baghdad proposal evidence and missing catalogue manifests. The workbench isolates its source-bound civil reference; it does not reinterpret those unrelated outputs as accepted engineering evidence.

Work packages and ownership

Roles below are accountable disciplines, not appointed people or acceptance signatures. Assign named implementers and independent reviewers before accepting engineering criteria. Software regression results are not engineering acceptance.

ID	Owner / acceptance reviewer	Current delivery	Remaining acceptance evidence
C01	Structural lead / independent structural checker	Frozen study, native reference replay, explicit train/ground/maturity gaps	Adopted code editions, approved load combinations and limits, measured inputs
C02	Analysis/CI engineer / separate software reviewer	Strict material/geometry contracts, source/native identities, child reasons and complete ancestor records	Independent contract review and actual material/connection specifications
C03	Structural/CAD lead / specialist checker	Pi, box, conventional I, U, ribbed, UHPC, hybrid and segmental geometry; existing foundation catalogue linked	Supplier-defined reinforcement, joints, fabrication limits and qualified domains
C04	CAD engineer / quantity checker	Disjoint whole-corridor CAD solids, stable IFC4.3 products/net quantities, native element/material regions	Actual bar/tendon geometry and accepted fabrication/installation drawings
C05	Analysis engineer / structural checker	Analytical native shear/support/modal/vehicle checks, fibre tangent, nonlinear reaction balance, orthotropic coupon and friction limit	Measured material/connection/soil datasets and physical applicability validation
C06	Analysis/CI engineer / separate reviewer	Native nodal exchange uses existing parsers; versioned integration-point stress parser; raw mode/time fields and stale-source checks	Accepted project criteria and reviewed output specifications
C07	Structural/geotechnical leads / independent checker	Native Concrete02/steel sections, nonlinear p-y/t-z/q-z piles and second-order columns	Site-calibrated group interaction, cyclic properties, scour/liquefaction and seismic selection
C08	Vehicle/structural/construction leads / checker	Moving forces, coupled vertical suspension/contact, handling supports, thermal eigenstrain, simple/link-slab/continuous models	Supplier train properties, asymmetric/3D interaction and site-specific erection/thermal cases

ID	Owner / acceptance reviewer	Current delivery	Remaining acceptance evidence
C09	Manufacturing/cost lead / cost checker	Complete scope bill, material-separated quantities, source-bound rates/charts, transport and discounted maintenance/replacement	Actual quotes, equipment/support proposals, routes, productivity and maintenance intervals
C10a	Analysis/CI engineer / software reviewer	Bounded native workers, source snapshots, recorded failures, attempt budgets, resumable search and CI	Independent clean-environment replay and agreed operational compute budgets
C10b	Cost/construction leads / checker	Immutable sub-50-MiB multipart archive and safe hash-verified restore; exercised retrieval	Project artifact custody, retention and hosted backend decision if required
C11	Optimisation engineer / structural lead	Multi-seed constrained Pareto evolution, equal-budget random baseline, native shortlist and range/ranking sensitivity	Accepted constraints/costs and candidate-specific independent qualification
C12	Structural/material specialists / independent checker	Actual-section three-mesh solids, orthotropic material matrices, nonlinear shortlist sections, prestress-loss/camber and friction/fatigue diagnostics	Calibrated hybrid bond/contact, anchorage, deterioration and cyclic failure models
C13a	Materials/geotechnical leads / test authority	Detailed protocol/measurement/stop-condition templates and source-bound calibration/holdout processing	Named laboratories, source rights, specimens, procurement and reviewed test loads/limits
C13b	Test leads / independent engineering reviewer	Physical results remain absent and explicit	Instrumented coupons, joints, piles and representative assemblies; blind validation
C14	Configuration owner / engineering authority	Sealed impact proposal invokes existing structural gate; research qualification cannot be bypassed	Actual independent check, candidate-specific physical evidence and authority acceptance

Corrected sequencing

C10a begins alongside C02–C06 rather than waiting for every commercial adapter. The early demonstration provides geometry, quantities and bounded response verification. C07–C09 and C10b progressively add calibrated systems, load coverage and complete pricing. C11 may explore provisional responses, but confirmed shortlists require applicable C12 models. C13 procurement starts during C01; validation follows the relevant model and specimen definition.

The 24-week programme remains an indicative software schedule. Per-package effort, named staffing, benchmark availability, storage and compute costs must be estimated before committing to the complete option space. Physical validation and supplier procurement retain separate evidence-dependent schedules.

Model acceptance policy

Register the observable, units, analytical/test reference, uncertainty, convergence tolerance, applicable domain and acceptance owner before treating a capability as qualified. Independent solver agreement is verification; measured tests provide validation. Native failure, timeout, uncovered physics and a violated requirement remain distinguishable. Null prices cannot produce a cheapest-design claim.

The current report records research outcomes with engineering feasibility `unresolved` and physical/operating release `false`. Existing civil evidence, structural-release and sealed engineering-change gates remain binding for any future catalogue adoption. Source snapshots and raw solver outputs stay in `build/engineering/`; intentional compact review artifacts follow the normal 50 MiB-per-file artifact policy.

Foundation And Civil Production System

Repository source: `docs/civil/foundation-and-production-system.md`

Status: deterministic planning and release-control model; not a geotechnical or structural design.

Foundation selection

`lib/templates/foundation-catalog.toml` standardises the pier-to-foundation interface and selection logic, not one universal footing. The deterministic selector uses ground/access classes:

Condition	Catalogue interface
Dense gravel, rock, strong cemented soil	Shallow spread footing with column socket
Urban alluvium or vibration restriction	Integral bored shaft
Uniform soft ground with clear access	Cap-free driven prestressed pile bent
Weak/liquefiable ground or high lateral load	Project-specific pile group and thin cap shell
Viaduct end or low embankment	Reinforced-soil abutment where validated

Ground-supported at-grade and embankment zones use a separate product set: rigid-inclusion load-transfer platforms, deep soil mixing, stone columns where settlement permits, lightweight fill, or lime/cement formation stabilisation. Each zone records treated geometry and verification rather than assuming "foundation concrete per kilometre".

Deep foundations have no default pile or shaft length. Quantity and cost calls fail until an actual project length is supplied. Each support record carries the type, element count and length, concrete/rebar quantity, installation time, test result and installed cost. A representative shaft or pile is load-tested in every geotechnical zone before production piling. `foundation_installed_record()` validates and emits this auditable per-support record; deep-foundation records cannot be created through the quantity path without their actual installed length.

Resource-driven production

`osr_mech.civil.construction` calculates line programmes from route geometry:

- elevated bays = `ceil(elevated metres / selected 20 m or 25 m span)`;
- beams = `2 × elevated bays`;
- foundations = `bays + 1`;
- slipformed metres = open at-grade metres not tagged as constrained;
- ST6 panels = `2 × ceil(constrained at-grade metres / 6 m)`;
- beam days = mould cycles required × cure cycle;
- foundation, erection and panel days = quantity ÷ active resource rate;
- the elevated critical path includes a 10–15 bay foundation/beam buffer.

The controls live in `lib/templates/manufacturing-schedule.toml`. Operators can change moulds, curing cycle, piling rigs, foundation rate, launcher count, bay rate, panel gantries, panel rate and working calendar. Generated operations data publishes the calculated plan separately for every line and records the equation behind every track-section duration.

Production and erection flow

```
flowchart TD
    A[Survey, utilities, CPTu and boreholes] --> B[Freeze span and foundation zones]
    B --> C[Foundations, pier shells and hollow caps]
    B --> D[Pi20/Pi25, ST6 and walkway production]
    C --> E[Two Pi-beam lifts per bay]
    D --> E
    E --> F[Walkways, trackform, rail and systems]
```

The primary plant is one long-line Pi20/Pi25 prestressing bed with movable stop ends, two initial beam moulds, adjustable column/cap-shell forms, small ST6 and walkway moulds, cage jigs, dimensional survey and concrete maturity control. The reference launcher target is one complete double-track bay per gantry per shift after learning. These are planning assumptions until supplier and first- article evidence release them. A 24-hour mould cycle is the maturity-controlled target; the planning schedule stays at the validated 48-hour cycle until the mix, sensor and production trial releases the shorter cycle.

Cost boundary

The generated civil contract currently carries a 9.748 million USD per route-km elevated planning target, indexed from the retained 12 million USD benchmark by the checked beam, bearing and deck-gap quantities. This is an unquoted design target, not an installed-rate promise. A project estimate still requires the supplier-designed section, actual foundation schedule, local installed rates, erection study, utilities/access, risk, independent checking and contingency. Change the calibration or CAD quantity model and regenerate `civil-cost-model.toml`; do not edit a downstream city price independently.

Full-train contact-domain verification

Repository source: `docs/civil/full-train-contact-domain.md`

Original six-car evidence · independent profile replay · original parts review provenance · current parts compiler review · timestep and guarded-run comparison · current native/service evidence.

The first six-car run reported nonlinear non-convergence at 9.775 s. Independent replay shows that the supplied wheel profile was already insufficient for the computed contact footprint at 9.130 s. A smaller timestep cannot establish profile validity by itself, even if Newton iterations converge.

The affected contact is `train-A / car-6/bogie-2/axle-1 / side -1`. At the first footprint exceedance:

Quantity	Retained value
Normal force	43,776.549 N
Wheel contact coordinate	56.064200 mm
Distance to supplied wheel-profile boundary	3.935800 mm
Hertz lateral semi-axis	3.944283 mm
Footprint beyond the declared boundary	0.008483 mm

The contact centre was inside the supplied table while part of its ellipse lay outside it. Checking only the minimum-gap point therefore missed the first domain violation. The original history contains 119 positive-load samples with unsupported footprints. Subsequent reference-location substitution introduces a discontinuous contact response; no forces beyond the supported footprint are qualified by that history.

Changed solver contract

The spatial adapter compares the full lateral semi-axis with both wheel and rail profile margins. Using the full semi-axis in horizontal profile coordinates is conservative for an inclined local tangent plane. This is a numerical domain check, not an ISO resistance, derailment or wheel-profile acceptance criterion.

A Newton trial may temporarily use the reference tangent after an unsupported profile lookup. The converged state must pass the actual profile, footprint, joint-motion, patch-size and small-angle checks before any contact state is committed or history sample emitted. An invalid converged state raises `SpatialDomainError`; it is retained as `outside-model-domain` by the service runner. Non-convergence remains a separate failure with residual, iteration count and any final-trial domain violations.

The change prevents a substituted reference patch from entering a supposedly valid history. It does not extend measured or supplied geometry, introduce an unqualified flange model, or make the original train a qualified operating design.

Reproduce the independent replay

```
tools/automation/osr-python tools/automation/audit-contact-profile-domain.py \
  --campaign build/engineering/reference-full-train-passage \
  --revision 349cf69d3cc0607924ac6671a59b7f851eff9395 \
  --output build/engineering/my-profile-domain-audit.json
```

The reader validates every retained campaign output hash and the recorded solver sources against the producing commit, checks wheel identity/completeness and ordered timestamps, then recomputes profile margins from the saved contact coordinates and ellipse. It reports the first/worst unsupported footprint. It does not infer a complete passage from a partial history or requalify the old numerical environment. Reference-location fallback cannot be identified unambiguously from the old coordinate fields alone.

Use the current service runner to generate fresh domain-checked evidence:

```
tools/automation/osr-python tools/automation/shared-service-envelope.py run \
  --model build/engineering/reference-parts-profile-guard/model.json \
  --config build/engineering/reference-parts-profile-guard/spatial_configuration.json \
  --output build/engineering/my-domain-checked-passage --cases empty-15mps
```

The original parts review and failed campaign remain bound to commit `349cf69d3cc0607924ac6671a59b7f851eff9395`. The current compiler review is a separate artifact; its source hashes do not rewrite the historical evidence.

Required next physics evidence

Complete operating cases require supported new/worn wheel and rail profiles, qualified treatment when flange or multiple contact occurs, and defensible running-gear suspension/creep parameters. The current synthetic conicity, bristle friction and equivalent suspension laws cannot supply that qualification. Profile-domain rejection must remain an explicit constraint in optimisation and calibration. Full-passage temporal/spatial convergence, railway acceptance and physical holdouts remain open until the operating states stay inside a supported contact domain or are assessed by an appropriately verified extended adapter.

Fresh verification outcome

The preserved implementation was rerun with `dt=0.0025 s`, retaining the same two deck elements per span and 2 m rail spacing. Its first unsupported footprint also occurs at 9.130 s; its later Newton failure occurs at 9.6775 s. The normal force at the first boundary differs from the coarse replay by approximately 0.00376%. This diagnoses a profile-domain limit; it is not a complete-passage time/mesh convergence result.

The retained pre-correction guarded run stops at 9.130 s with `outside-model-domain` and retains the last supported sample at 9.125 s. Independent replay checks 87,600 contacts across 1,825 supported samples, with no unsupported footprint in the committed history. The failed state is retained separately in the case diagnostic. All three campaign receipts and the producing sources for the archived timestep probe were checked before publication.

The focused suite passes 124 tests, including contact-centre/ellipse boundary cases, rejection before contact-state commitment, corrupt-history rejection, service failure classification and retained-proof hash checks. No complete six-car service passage or physical qualification is claimed.

Steering correction and retained passage comparisons — 11 October 2026

The lateral rolling-reference velocity previously omitted steady wheel yaw. The current small-rotation model projects body steering into the rolling direction, while excluding the wheel spin coordinate. Independent rotation and Jacobian checks exercise that term. The finite profile, patch-footprint, contact opening and multiple-patch guards remain active.

With that correction, the 192-instance reference six-car train completed its 13.06 s passage at 15 m/s. The expanded 414-instance design also completed the same passage. A cylindrical-profile verification benchmark completed separately; it is not a supplier profile. Experimental convected-bristle runs still leave the finite profile domain and are retained as rejected research cases.

Halving the reference time step from 0.005 to 0.0025 s changed maximum wheel force from 44,123.906 to 44,128.648 N. Refining deck mesh 2 to 4 and rail step 2 to 1 m at the smaller time step changed peak carbody acceleration from 0.351569 to 0.247309 m/s². This substantial spatial change prevents a convergence claim. Finer meshes and the broader load/speed/degradation matrix are separate retained runs. Approved profiles, creep-law comparisons, suspension data and physical contact/ride correlation are still required.

Line-based foundation and viaduct assembly planning

Repository source: `docs/civil/line-based-foundation-and-assembly-planning.md`

The integrated Baghdad package is the line-specific implementation of the civil master plan. Its construction plans for every current line, support register, bay register, directed launcher fronts, junction packages and stage library share actual line/span/component identities. The offline viewer lets reviewers select a line, zoom to individual supports and inspect the coordinate, adjacent spans, desktop investigation reference and unresolved design values.

One physical support, one foundation packet

Every unique running support has one `FND:<support-id>` record, a proposed route-interpolated position, line chainage and the adjoining bay IDs. Adjacent bays reference that packet; they do not create two footings at the same support. Geographic positions remain planning datums requiring survey control, utilities and property/access release. Coincident positions on different lines enter a common junction/site review; a shared foundation is never assumed merely from XY coincidence.

Station/concourse/shaft, approach, water bridge, depot/pit and energy-site foundations are separate design scopes in the other-structure register. Their counts and geometry remain unknown where no structural/site design exists. Reconcile station/running support ownership before adding quantities; do not price both as independent foundations for the same load path.

Ground investigation and design brief

The support packet links its nearest same-line desktop-soil sample by chainage, including the offset and investigation flags. Those mapped topsoil properties do not establish deep stratigraphy, bearing capacity, groundwater, settlement or pile length. Use them to organise field investigation, not to select a footing.

Design task	Required support/zone evidence
Survey and existing assets	Surveyed support/track coordinates and elevations, utilities/trial holes, property, adjacent buildings and traffic/access
Ground and groundwater	Borehole/CPT/test-pit programme, verified stratigraphy, groundwater and laboratory classification/strength/compressibility/durability; zone limits and uncertainty
Load transfer	Selected beam/continuity scheme, columns/caps/bearings, full train and lateral/dynamic actions, launcher/delivery/asymmetric stages, wind/seismic/collision/flood/scour
Foundation options	Shallow footing where verified competent ground permits; bored shaft for suitable constrained/vibration conditions; driven bent where installation is acceptable; pile group for selected lateral/scour/ground cases
Geometry and verification	Actual element count/size/depth, socket/cap/column details, settlement/rotation limits, installation and proof/integrity/load testing, independent checking
Construction interface	Working platform/excavation/dewatering/slurry/spoil method, reinforcement/connection, concrete/curing, tolerance/inspection and support-ready release

The register reports an unqualified simple-span beam-deadload screen from complete-member study masses: two single-track beams per bay, half of each at each end. It excludes cap/column, train, continuity redistribution, construction combinations and lateral/environmental actions. It is not an adopted foundation axial load or allowable bearing pressure; complete design axial/lateral/moment values remain unknown. Special-member mass also remains unknown.

Prepare a foundation design and installation method for each verified ground/load zone. Trial and test representative foundations before repetition, with a reviewed extension to unlike zones. Select concrete, reinforcement, corrosion/durability and test criteria through the project design basis. Actual lengths, drilling/driving logs, concrete volumes, specimens/maturity, integrity/load tests, dimensions and NCRs become installed records. No universal six-metre pile or generic foundation-concrete allowance is used.

Directed workfront plan

Two candidate launcher fronts per line are inherited from the running-bay sensitivity, with finite launcher-pool predecessors for queued reuse. Positive fronts install toward increasing chainage; negative fronts begin at the high-chainage end and install backward. The existing scheduler interprets direction internally; the new exports make the execution and dispatch order explicit. All identified catalogue spans are assigned to one front and one sequence position. Special spans retain a separately engineered package and create no ordinary Pi25 order.

For each front, release actual access/assembly space, working platforms, configured machinery, delivery route, competent rested crew and the first consecutive ready supports. Maintain 10–15 accepted bays of support and beam readiness only where continuous run and storage capacity permit. Factory slots, inspection, accepted stock, trailers, receiving plant and installation appointments follow the directed SKU/component order. Shared equipment/crews cannot be reserved to two fronts simultaneously.

Disconnected intervals, stations and specials require a checked passage or dismantle/transport/reassembly/recommission method. The predecessor is the previous executed bay, not simply the preceding ascending row in a database. Foundations used by adjoining bays share the same stage packet. Released radial launchers can transfer to additional ring fronts only after access, foundations, supply, crews and recommissioning are evaluated together.

Assembly-stage dependencies

Stage	Entry and exit
F0–F3 foundation	Survey/utility/ground/load design trial and temporary-platform release actual selected installation accepted tests/dimensions/NCR closure
P0–B0 support	Column/cap and connection/temporary restraint surveyed bearing seats and actual next-stage loading release
M0–M3 manufacturing	Frozen shop/handling configuration mould/cage/prestress/cast/curing separate strength gates and inspection accepted serialized member
H0–H1 delivery	Actual vehicle/route/load restraint and appointment receiving/damage survey, designed storage and reservation
E0 preparation	Both support B0 packets, both beam H1 records, predecessor/relocation and relevant junction J2 release
E1 first beam	Configured controlled lift, seating, balance and required lateral/torsional restraint
E2 paired bay	Second controlled lift, securing, surveyed geometry and bearing contact
E3 connections	Selected connection/joint/strength, bearing movement and NCR acceptance; no shared bearing reduction without structural continuity
E4 advance	Accepted current-stage path, checked advance/delivery loads, next supports, weather and protected access
E5 following trades	Walkway/barrier, waterproofing/drainage, track interface, permitted loads/access and signed handover

The stage library stores these reusable chains; the bay register binds them to actual component and parent IDs. Junction-affected member orders remain on design hold until level/clearance/support/stage releases close. Installation is never inferred from an accepted factory beam or a completed documentary task.

Inspection and complete opening

Apply the site ITP and handover plan. Record source configuration, personnel/equipment authority, measured results, test criteria and signatures before closing each hold. After transport damage, abnormal loading, changed temporary works, utilities or adverse weather, inspect and re-release affected stages.

Complete line opening requires stations, specials/transitions, track, energy, depot/stabling, commissioned fleet, integrated testing and approvals in addition to running-bay completion. The plan does not adopt an opening date, field foundation quantity, installed quote or paid-passenger uplift from illustrative capacity or coverage.

Civil Marketplace Cost Anchors

Repository source: `docs/civil/marketplace-cost-anchors.md`

Planning-grade OSR city costs use USD as the source currency. Generated `design.toml` files keep `*_eur` mirrors at 0.92 USD->EUR for schema compatibility. The reviewed civil benchmarks and cost-share assumptions live in `civil-cost-calibration.toml`; the active CAD-indexed rates are generated in `civil-cost-model.toml`. General CAPEX units remain in `capex-costs.toml`. This note is the civil marketplace audit trail.

These are not turnkey EPC bid prices. The benchmarks and generated targets assume direct procurement, local fabrication, self-performed assembly, repeatable QA, and the no-tunnel/no-catenary OSR civil discipline.

At-Grade Double Track

Active design target: 2,584,000 USD per route-km. Retained marketplace benchmark: 3,000,000 USD per route-km.

Component	Marketplace anchor	Planning basis
UIC60/60E1 rail	Alibaba UIC60 rail listings show about 600-700 USD/t for 60E1/UIC60 rail: https://www.alibaba.com/showroom/railway-rail-uic60.html	Double track uses ~241 t/km of rail, so rail steel lands near 145-170 k USD/km before freight
Slab/plinth concrete and reinforcement	Local concrete, rebar, mesh, formwork, and precast suppliers rather than a rail-specific commodity	Ballastless urban track shifts spend from sleepers/ballast/tamping into slab, plinth, embedment, and drainage works
Direct-fixation fasteners, baseplates, pads	Alibaba rail-fastening listings show clips around 0.5-2.5 USD and assemblies/baseplates around 4-6 USD: https://www.alibaba.com/showroom/rail-fastening-system.html	Double-track fastening hardware, pads, baseplates, weld kits, and small parts budgeted as a bundled allowance
Ballastless slab/embedded trackform, drainage, cable trough, installation	Local concrete/rebar/precast and civil supplier item rather than rail-specific Alibaba commodity	Residual allowance plus local labour/equipment, freight, QA, welding, direct-fixation installation, drainage, and urban possession logistics formed the retained 3.0 M USD/km benchmark; current slipform/CAD quantities index it to 2.584 M USD/km

Elevated Guideway And Bridges

Civil class	Design target	Retained benchmark	Marketplace anchors and included scope
Elevated viaduct	9.748 M USD/km	12.0 M USD/km	Conditional structural-continuity index: 40 Pi25 bays, 80 beams, 30 structural connections, ten periodic deck gaps and 200 bearings/km. Simple spans with link slabs retain 320 bearings/km and give a 10.198 M USD/km index before consequential costs. `Quantity/cost scope` includes trackform, walkways, zone foundations, yard, transport, erection, utilities, traffic management, checking, testing and contingency; all remain unquoted
Bridge / water crossing	18.0 M USD/km	18.0 M USD/km	Bridge/trestle anchors plus pile/foundation equipment examples: https://www.alibaba.com/showroom/reinforced-concrete-pile.html; bridge-specific redesign and foundation schedules remain outstanding
Elevated interchange premium	4.5 M USD/site	—	Uses the elevated guideway stack plus stacked-platform and approach complexity where an interchange must grade-separate

Stations

Station costs assume prefab portal-frame canopies, ground-level platform slabs with lowered guideway channels, direct pedestrian access, simple MEP, lighting, signs, fare gates, CCTV, and local assembly. Commodity lifts/escalators and overbridges appear only where an elevated/stacked interchange or local road-barrier override requires them. No underground concourse, no bespoke architectural cladding, no continuous traction-power plant.

Archetype	Unit
`halt`	600 k USD
`standard`	2.5 M USD
`major`	4.5 M USD
`terminal`	4.5 M USD
`depot-terminal`	5.0 M USD
`interchange`	8.0 M USD
`interchange-elevated`	12.0 M USD

Marketplace anchors:

- Prefab steel canopy / portal-frame buildings: Alibaba steel-canopy and prefab steel-structure listings show common ranges around 15-65 USD/m2 for simple steel structures: https://www.alibaba.com/showroom/steel-canopy-structures.html
- Escalators: Alibaba escalator listings show common public/commercial units in the 9,000-30,000 USD band, with some lower small-unit entries: https://www.alibaba.com/showroom/escalator.html
- Lifts/elevators: Alibaba outdoor/passenger lift listings show small lift entries from roughly 1,300-4,000 USD and broader commercial elevator bands above that: https://www.alibaba.com/showroom/elevator-outdoor.html
- Footbridges: station foot-overbridge examples range from sub-1 M USD low-cost-market steel packages to multi-million USD rail overbridges once access cores, possessions, utilities, and crowd-flow width are included. Examples include a 12 m wide Guindy station FOB reported at about Rs 6.5 crore (https://www.newindianexpress.com/cities/chennai/2023/mar/14/new-foot-overbridge-at-guindy-railway-station-to-ease-passenger-flow-2555815.html) and UK station-footbridge commentary noting costs above GBP 4 M each (https://www.ingenia.org.uk/articles/the-flat-pack-footbridge-for-train-stations/). OSR uses a modular low-cost floor but no longer treats the access bridge as free.

Depots And Charging Interfaces

Depot costs were also brought onto the same direct-procurement basis because the station/track recalculation otherwise left the fixed-asset stack mixed. The current units include the distributed overnight-stabling policy: healthy trainsets can sleep at powered stations, so layups are station stabling kits and the main depot no longer needs parking capacity for every set.

Depot archetype	Unit
`main-heavy`	8.0 M USD
`secondary-medium`	3.0 M USD
`layup-minimal`	400 k USD

Station/depot charging microgrid interface:

Stop archetype	Unit
`halt`	0 USD unless energy-model promoted
`standard`	100 k USD
`major`	100 k USD
`terminal`	100 k USD
`interchange`	100 k USD
`interchange-elevated`	125 k USD
`depot-terminal`	250 k USD

This bucket includes the standard 500 kWh stationary-LFP module, shared 500 kW DC/DC cabinet, conductive contacts, switchgear, short cable runs, protection/control, and a local installation allowance. PV panels, train batteries, depot-scale storage, building/civil work, and onboard hardware are counted elsewhere. The equipment-only RFQ target is $55k–75k per normal site; the planning unit carries integration and uncertainty.

Network junction and residential integration

Repository source: `docs/civil/network-junction-and-residential-integration.md`

The integrated network model coordinates line crossings, shared corridors, station complexes and underserved residential areas with line-specific foundation/assembly packages. A visually overlapping route is not automatically a rail junction, walkable transfer or completed civil connection.

Crossing and overlap packages

Situation	Required planning and design treatment
Two independent rail lines cross	Grade-separated structure with actual rail levels, gradients, clearances, support/cap and construction load paths; a passenger interchange is a separate access decision
Routes occupy a common XY corridor	Compare a coordinated four-track footprint with laterally separate guides and shared station/access structure, or reviewed separate vertical levels; do not duplicate coincident decks or footings
A line revisits the same location	Preserve distinct chainage legs, then choose a real grade-separated passage or reroute; do not collapse both visits into one node or imply a switch
A short A–B–A ring spur	Remove the spurious one-cell reversal without introducing a new land/water shortcut; regenerate route chainages, stations, civil quantities, scenario and dependent evidence
An apparent near-miss interchange	Preserve platform positions; identify the actual concourse/street/vertical connection, barriers, rights, walking distance and timing before acceptance

The Baghdad ring repair removes its 20 m outward-and-return spur, shortening the planning route by 40 m. It does not erase a district-serving branch or insert an unreviewed chord. The source repair leaves longer excursions and supplied protected cells for explicit review.

Every junction design package identifies participating lines and chainage legs, affected spans and support packets, design alternatives and the J0–J3 arrangement/profile/site/stage/handover sequence. Upper/lower rail levels, vertical clearance and graded approaches stay unselected until a real profile and independent structural review exist. Ordinary member orders that touch the interface are held until J2; current drawings and existing route-km budgets cannot establish an accepted junction cost or final geometry.

Bounded station complexes

Grouping uses the maximum pairwise platform separation, with a 600 m planning screen. A chain of nearby stations does not make its distant ends one complex, and an additional same-line amalgamation cannot silently enlarge the diameter. Original oversized groups remain in the review register with the bounded replacement concepts and their actual platform IDs. No platform is shifted to an averaged centre point.

The diameter is a geometry screen, not a walking path. Rivers, roads, walls, level changes, stairs/lifts and entry permissions need actual accessible routes and transfer times. A close pair can still fail. Concourse/platform capacity, outage operation, assisted rescue, fire, PSD/train interfaces and structural loading use the installed arrangement. The proposed grouping does not alter the native timetable or invent a physical track switch.

Population-led coverage

Use retained native 2020 WorldPop count pixels and count each person once in a union of distance circles. Keep the population year, bbox and nodata explicit; never multiply a routing-demand score by a current city-population headline. The coverage model separates existing station proximity, corridor proximity, infill sensitivity and branch/feeder investigation.

Infill candidates sit on current lines and are ranked by incremental previously unserved retained population, subject to minimum same-line spacing and ring wraparound. Candidate stations require water/footprint, property, access, platform/approach, timetable/fleet/energy and complete installed-cost checks. A population-rich point is not a released station site.

Large residential gaps outside current corridors cannot be solved by infill alone. The priority areas and expansion corridors identify source-linked district/nearby place names, existing station connection, route/access gaps, and an arterial investigation path. Compare an engineered rail branch/extension with a frequent feeder and accessible transfer. Road vertices define an undirected preliminary graph; traffic direction, stopping sites, grade connections, bridge limits and permissions are unverified. Do not issue driver navigation or a rail alignment from this screen.

The combined 500 m feeder-stop and 1 km rail-station circle cases are conditional sensitivities, not measured walking coverage or funded service. Shared candidate road segments are counted once in the corridor-edge register. Priority-area circles can overlap and are never summed as unique residents. Actual travel-time/OD, affordability, service/headway, capacity, operating cost and finance remain required before selecting and adopting expansion.

ITDP People Near Transit provides a population/proximity assessment reference; ITDP physical integration addresses interchange/access development. WorldPop's model-method description explains the constrained/unconstrained dataset assumptions. These references do not turn retained circles or road lines into accepted Baghdad walksheds.

Catalogue review and adoption

The 266-city audit reports declared transfer components, self-intersecting lines, cross-line overlap pairs, oversized interchange groups and retained 1 km population fractions. Missing geometry/population stays unavailable. Audit findings require local geometry, site and service investigation; they are not automatically deployed changes to every city.

Use one controlled adoption revision for selected corridors, junctions and stations. Regenerate design/scenario, civil and mechanical quantities, foundation/assembly, geography/clearance, energy/service, finance, staffing, deployment and publication in dependency order. Execute actual native nominal/degraded cases for changed inputs, preserve source-bound receipts, and reopen affected independent/site gates. The original supporting archive and new companion planning archive keep the earlier comparator and new model accessible without exceeding the repository's per-file limit.

OSR-ALN — alignment interchange format (v1.0)

Repository source: `docs/civil/osr-aln-format.md`

Status: v2 deliverable of RFC 0009. Purpose: a tool-agnostic text schema that civil-engineering software (Civil 3D, Bentley OpenRail, Trimble, open-source tools) can export to and the OSR simulator / emitter can import from.

Why OSR-ALN

Every civil-engineering firm uses a different authoring tool. Every tool has a different native file format. LandXML covers the generic case but is verbose and has many optional fields that don't apply to urban rail.

OSR-ALN is a minimal text format that round-trips a rail-ready alignment to / from any tool. It is a 100%-plain- text TOML schema (no XML, no binary) with one row per `HorizontalPoint` + one row per `VerticalPoint`. A civil firm produces OSR-ALN from their native format as part of the deployment deliverable; the OSR pipeline validates it against RFC 0009's compatibility matrix and feeds `osr-sim` + `osr-routing` without further adaptation.

File layout

One alignment per line per file. Filename convention: `<city>-<line>.aln.toml`. Example: `sample-city-line-a.aln.toml`.

```
# OSR-ALN v1.0 — alignment interchange format per RFC 0009 v2.
[meta]
schema_version = "1.0"
line_id        = "sample-city-line-a"
design_date    = "2026-04-22"
surveyor       = "<firm name>"
preset         = "standard-urban"       # RFC 0009 §1 preset key
consist        = "light-metro-3car"     # RFC 0008 §1 family key
crs            = "EPSG:32638"           # Coordinate Reference System (UTM Zone 38N for Iraq)
units          = "metric"               # SI only — OSR is SI
is_ring        = false

# ---------------------------------------------------------------------
# Horizontal alignment points — tangent + circular + transition elements
# expressed as a sequence of PI (Point of Intersection) with before /
# after bearings. The standard alignment reconstructs the clothoid
# transition curves deterministically from adjacent PIs + the declared
# radii + transition lengths.
# ---------------------------------------------------------------------

[[horizontal]]
station_m          = 0.0
easting_m          = 477_500.00       # UTM
northing_m         = 3_467_100.00
bearing_in_deg     = 128.7
bearing_out_deg    = 128.7
curve_radius_m     = 0.0              # 0 = tangent
transition_length_m = 0.0

[[horizontal]]
station_m          = 1_000.0
easting_m          = 478_135.50
northing_m         = 3_466_522.00
bearing_in_deg     = 128.7
bearing_out_deg    = 123.7            # Curve begins here
curve_radius_m     = 26_300.0
transition_length_m = 2.0              # Clothoid transition length

# ... more rows one per PI or tangent endpoint ...

# ---------------------------------------------------------------------
# Vertical alignment points — elevation along station_m. Interpolation
# is linear between adjacent points with a circular vertical curve of
# radius `vc_radius_m` at each grade-change point.
# ---------------------------------------------------------------------

[[vertical]]
station_m     = 0.0
elevation_m   = 6.0                    # at top of rail
vc_radius_m   = 0.0                    # 0 = tangent (no vertical curve here)

[[vertical]]
station_m     = 1_000.0
elevation_m   = 6.0                    # still level
vc_radius_m   = 0.0

[[vertical]]
station_m     = 1_150.0
elevation_m   = 11.25                   # top of ramp into viaduct
vc_radius_m   = 300.0                  # sag curve

# ...
```

```
# ---------------------------------------------------------------------
# Civil-class spans — per-class segment between two station_m values.
# Must cover the full [0, total_length_m] range with no gaps and no
# overlaps.
# ---------------------------------------------------------------------

[[civil]]
from_station_m = 0.0
to_station_m   = 1_000.0
class          = "at-grade"

[[civil]]
from_station_m = 1_000.0
to_station_m   = 2_300.0
class          = "elevated"

[[civil]]
from_station_m = 4_100.0
to_station_m   = 5_300.0
class          = "bridge"

# ...

# ---------------------------------------------------------------------
# Stations on this alignment — references the station ids in the
# deployment's design.toml. station_m pinpoints the platform
# centreline stop mark on this alignment.
# ---------------------------------------------------------------------

[[station]]
id          = "samawah-rws"
station_m   = 0.0
platform_length_m = 59.5              # 49.5 m consist + 10 m RFC 0010 clearance

[[station]]
id          = "north-gate"
station_m   = 1_000.0
platform_length_m = 59.5

# ...

# ---------------------------------------------------------------------
# Superelevation (cant) per curved span. One row per cant-change PI.
# ---------------------------------------------------------------------

[[cant]]
from_station_m  = 1_000.0
to_station_m    = 2_300.0
max_cant_mm     = 30
transition_in_m = 50.0                 # cant development length in
transition_out_m = 50.0                # cant runoff length out
```

Required fields per section

[meta]

Field	Type	Required	Notes
`schema_version`	string	yes	Always `"1.0"` for this format
`line_id`	string	yes	Matches a `[[lines]] name` in design.toml
`design_date`	date	yes	ISO 8601
`surveyor`	string	yes	Civil firm of record
`preset`	string	yes	Must be one of the RFC 0009 §1 presets
`consist`	string	yes	Must be one of the RFC 0008 §1 families
`crs`	string	yes	EPSG code or WKT; UTM zone typical
`units`	string	yes	Always `"metric"`
`is_ring`	bool	yes	Whether the line loops back

[[horizontal]]

Sequence of horizontal points along the alignment. `station_m` values must be monotonic-ascending. Bearings are in compass degrees (0° = north, 90° = east). Tangent segments have `curve_radius_m = 0.0`. Transition curves (clothoids) are declared on the entry/exit of each circular curve via `transition_length_m`.

[[vertical]]

Sequence of vertical points. `station_m` monotonic-ascending. `elevation_m` is top-of-rail elevation above the local datum (typically mean sea level, per `crs`). `vc_radius_m` declares a circular vertical curve at grade-change points; 0 = sharp grade break (discouraged above 2.5 ‰).

[[civil]]

Civil-class spans covering the full length. No gaps, no overlaps. `class` must be one of `at-grade`, `elevated`, `bridge` per RFC 0011 — no `tunnel` (v2 validator rejects).

[[station]]

Platform centrelines along the alignment. Every station id must match a `[[stations]]` block in the deployment's `design.toml`.

[[cant]]

Optional superelevation ramp + full-cant section + runoff. Absent cant rows mean `max_cant_mm = 0` for that span (tangent track).

Validator semantics

A validator ships at `tools/osr-aln-convert/src/osr_aln/validate.py` (exposed as the `osr-aln-validate` CLI after `pip install -e .`). It checks:

Hard gates (reject on fail)

1. `meta.preset` is one of the four RFC 0009 presets.
2. `meta.consist` is in the preset's `compatible_consists`.
3. `[[civil]]` spans are contiguous and cover the full length.
4. No `[[civil]]` has `class = "tunnel"`.
5. Every station id has a matching entry in the deployment's design.toml.
6. Every curve's `curve_radius_m` ≥ preset's `min_curve_radius_m`.
7. Every vertical grade ≤ preset's `max_gradient_per_mille`.
8. Every `max_cant_mm` ≤ preset's `max_cant_mm`.

Soft gates (warn, don't reject)

1. Elevated share > 30 % RFC 0011 §8 soft gate.
2. Any curve radius ≤ 2× preset minimum flag for speed restriction.
3. Any grade > 80 % of preset maximum flag.

Round-trip consistency

Round-tripping OSR-ALN simulator OSR-ALN must produce a byte-identical file. The simulator may compute derived quantities (stopping distance, power draw, etc.) but must not modify the alignment.

Tool support

Converter at `tools/osr-aln-convert/` — stdlib-only Python; installs with `pip install -e .` and exposes one CLI per source format.

Firms using:

- Autodesk Civil 3D — export via LandXML, then transform with `landxml-to-osr-aln`. v1 (shipping).
- Bentley OpenRail — export via LandXML (OpenRail emits a compatible subset), transform with `landxml-to-osr-aln`. Explicit cant can be supplied through the revision-bound TOML companion; the vendor-specific <Cant> extension is not parsed directly.
- Trimble Business Center — export via LandXML, transform with `landxml-to-osr-aln`. A TCL CSV reader is v1.1. v1 via LandXML.
- Open-source (QGIS + rail-path plugins) — LandXML export supported. v1.

Generated Planning Exports

LandXML imports also accept `--sidecar mapping.toml --design design.toml`. The companion-file contract maps every current line station, contiguous civil spans and explicit cant against hashes of the XML and design inputs. The importer preserves the declared vertical datum and all input hashes under `[meta]` and `[provenance]`, and runs the hard gates before writing. These imports remain subject to survey and engineering acceptance.

`current-network-to-osr-aln` can join a generated city `design.toml` and corridor GeoJSON into a deterministic per-line package. Such output is a traceability and interface artifact, not a surveyed rail-ready alignment. It must identify itself as planning-only and explicitly mark unfitted horizontal curves, placeholder vertical levels, and undesigned cant. The current example is the three-line Samawah package.

Passing the validator proves schema, catalogue compatibility, civil coverage, station/line references, and declared envelope limits. It does not replace survey, curve fitting, constructability, geotechnical, drainage, structural, or independent design checks.

Validation Example

A deployment exports one `.aln.toml` file per surveyed line and validates it against the current generated `design.toml` for that city:

```
$ osr-aln-validate exports/survey/samawah-line-1.aln.toml \
                   --design-toml cities/catalogue/west-asia/Iraq/Samawah/design.toml
2 soft-gate warning(s):
  ⚠ S3: grade between [[vertical]] #1 and #2 is 35.00 ‰ — within 80 %
       of preset maximum 40.0 ‰, flag for review
  ⚠ S3: grade between [[vertical]] #5 and #6 is 35.00 ‰ — within 80 %
       of preset maximum 40.0 ‰, flag for review
```

Both warnings are expected (the two viaduct ramps at 35 ‰); no hard gates fire.

Version history

Version	Date	Changes
1.0	2026-04-22	Initial spec
1.1	2026-08-11	Current-network planning export, input provenance hashes, and line-id validation
1.2	2026-09-10	Revision-bound LandXML station/civil/cant companion imports; OSR-ALN wire schema remains 1.0

What The Current Toolchain Does NOT Include

- A "delta file" format for alignment updates during construction (v4, if ever needed).
- Drainage profiles — handled separately at civil-design time, not part of the alignment.

Other civil works construction methods

Repository source: `docs/civil/other-civil-works-methods.md`

These methods complete the civil master plan. Define project quantities and release criteria from survey, design and local approvals; route length is not an earthworks, pipe, pile or building take-off. Use the inspection and handover plan for concealed work and following-trade release.

Surveys, enabling, demolition and utilities

Establish verified horizontal/vertical control and a maintained survey network. Survey ground, existing roads/drains, water levels, structures, property, trees and utility routes; combine records with detection and trial holes. Resolve clashes in the common model and issue utility-owner approved protection/diversion/isolation plans. Locate working platforms, stockpiles and access outside protected services.

Before demolition or site clearance, establish ownership, hazardous-material/waste findings, structural sequence, adjacent support, service isolation, segregated removal route and disposal/recovery destinations. Maintain traffic, pedestrians and emergency access; record vibration, dust/noise and settlement where relevant. Protect retained assets and survey adjoining property before intrusive work. Restore survey control disturbed by construction.

Utility diversions precede conflicting excavation/piling. Construct replacement ducts/pipes and chambers, test, switch service under owner control, verify the new route, then remove/protect the obsolete asset. Record actual depths/offsets and access in as-built data before concealment. A drawing clash marked resolved is not field clearance.

Earthworks, formation and ground improvement

Strip and segregate reusable topsoil; test excavated material and decide reuse, treatment or disposal. Install temporary drainage and designed cut/excavation support before bulk works. Excavate/fill to surveyed profiles with inspected lifts, moisture/compaction control, test locations and proof/formation acceptance appropriate to the design. Track quantities using survey surfaces and truck/weighbridge records; account for swell, shrinkage and rejected material in haul planning.

Select rigid inclusions/load-transfer platforms, soil mixing, stone columns, stabilisation or lightweight fill by zone-specific settlement, drainage, cyclic loads, liquefaction and utilities/access constraints. Trial the method and verify equipment/material supply before production. Retain installation depths/geometry, mixing or compaction parameters, tests and settlement monitoring. The foundation/production system supplies selection interfaces, not site treatment depths.

Release formation only after drainage, bearing/settlement and survey acceptance. Protect it from delivery traffic and weather; repair and reverify damage before trackwork. Sequence treatment and trenches so later excavation does not remove or undermine accepted treatment.

Retaining structures, ramps and transitions

Review reinforced-soil walls, embankments, cast/precast retaining walls and abutments using land width, flood/scour, bearing/settlement, seismic and construction access. Install foundations and drainage, then facing/reinforcement and controlled fill in the designer's sequence. Keep heavy compaction/plant away from faces according to the approved method. Inspect hidden reinforcement, filters and outlets before each closure.

At grade-to-viaduct transitions, coordinate stiffness/settlement, rail geometry, movement, approach slabs, drainage and maintenance access. Survey as-built levels and monitor settlement through the selected stabilization/acceptance period. The 4.5 m retained-approach selection envelope is a catalogue planning rule, not permission to retain an unsuitable flood or soft-ground site.

At-grade and elevated track civil interfaces

For long accessible at-grade runs, compare the released continuous slipform method with constrained ST6 panel zones. Install tested formation, subbase and service/drainage crossings first. Survey line/level and reinforcement/inserts; cast and cure the selected slab or prepare bedding and erect panels on designed supports. Inspect bedding/grout, panel joints, anchors, drainage and stray-current interfaces. Use project-defined trial panels/slipform lengths before repetition.

Keep closure pours at station ends, turnouts, transitions and surveyed corrections separately designed. Record panel IDs and final coordinates. The slab-trackform catalogue has a 6 m single-track module; double track requires two rows. Panel mass is a bare study envelope; complete lifting/transport includes embeds, gear and configuration. Do not buy a rounded panel length that extends outside an identified method zone.

For elevated track, release deck geometry, movement, waterproofing and drainage before track supports. Fit local plinths/alignment layers and adjustable fasteners to the reviewed rail/structure design. Protect prestressing and

waterproofing during fixing. Rail welding/stressing, alignment/cant and fastening acceptance follow the railway specification; the civil plan supplies foundations and interfaces, not invented rail neutral temperatures or tolerances.

Drainage, water management and flood protection

Prepare catchments, storm basis, surveyed outfalls, receiving-system capacity, water-quality requirements and flood/scour scenarios. Design temporary drainage through all earthwork and foundation stages; avoid redirecting water into occupied streets or neighboring foundations. Sequence permanent culverts, cross-drains, attenuation and outfall protection ahead of dependent formation.

Construct foundations/bedding and pipes/channels, inspect joint levels, watertightness and access, then backfill in tested layers. Keep service separations and chamber access. Clean/CCTV or otherwise inspect the specified system; verify gravity falls and pumps/alarms/backups where required. Test before concealment and again before handover. Elevated deck drains, station roofs/shafts, abutment drainage and depot contaminated/washwater systems need separate catchments and controlled discharge.

Protect water crossings against construction runoff, debris, washout and accidental spills. Investigate erosion/scour, flood levels and safe construction windows. Temporary causeways/cofferdams, bank loading and navigational restrictions require hydrological/geotechnical design and permissions; no blanket dry-season assumption is adopted.

Special bridges, closure spans and road conflicts

The bridge/water civil class and unresolved running closures are separate inventories. Survey each conflict and compare rail bridge, road relocation/grade separation, longer-span steel/composite or segmental solutions, short verified products and alignment alternatives. Include road clearance, navigation/flood, utilities/property, foundations, track geometry, temporary works, traffic disruption and whole installed cost.

Issue a site-specific staged design and supplier package before buying special members. Define lifting/segmental or launching sequence, supports, stability, post-tensioning/joints where selected, closure geometry and rail/structure interaction. Coordinate permissions and shutdowns with adjacent running/station packages. A residual 23.3 m interval does not become a Pi25 catalogue beam, and a short segment does not automatically satisfy shipping width.

Depot, workshop and stabling civil

Freeze actual yard tracks, usable berth lengths, switches/leads, clearances, departure queues, charging positions and maintenance/recovery access. Reconcile declared storage against the depot scope. Baghdad's one declared heavy depot and 116 stalls do not establish physical overnight provision for all 772 trainsets; workshop bays and passenger platforms are not verified parking.

Sequence land/utility release, ground/drainage treatment, underground services, building/pit and equipment foundations, superstructure/enclosure, slab/pavements, yard track/turnouts, equipment installation, MEP/fire and commissioning. Inspect workshop pits for support, groundwater/waterproofing, ventilation/access/rescue and fall protection; coordinate jacking/bogie extraction and crane reactions with equipment suppliers. Do not load slabs or pits with plant before release.

Separate dirty washwater/oil drainage from stormwater; provide selected treatment and permitted discharge. Design stores, delivery/road turning, fire routes, staff access, waste, workshop extraction and equipment replacement. Fit charging/PV/battery bases, ducts and protected separation only after actual energy/fire/thermal requirements and usable layout are reviewed. Conduct yard-route and repeated-day commissioning with the final rail/charging graph.

Energy, equipment-room and systems civil

Survey every station/depot/wayside energy site. Freeze supplier foundation reactions, equipment dimensions, access/replacement envelopes, electrical isolation, fire separation, earthing and flood levels. Construct foundations, plinths, retaining and drainage; install ducts/chambers/cable bridges, sealed penetrations and equipment-room structure/enclosure. Inspect concealed earthing/ducts, reinforcement/inserts, water/weather protection and pull routes before backfill/closure.

Provide civil handover by equipment site: dry clean access, foundations and anchors surveyed, ducts cleaned/proved, load capacities and penetration rules, drainage, barriers, lighting and isolation boundaries. Grid energisation, BMS commissioning, signalling proving and fire acceptance are separate system gates. Protect battery equipment against construction dust, water, impact and unauthorized storage; supplier-specific fire/thermal data determines the permanent layout.

Roads, access, public realm and reinstatement

Create staged road/pedestrian plans for piling, deliveries and lifting, with emergency routes and local business/property access. Use booked off-peak windows only where approved; stop work if the approved segregation or access cannot be maintained. Plan diversions, crossing marshals and barrier/lighting inspections for every shift. Retain route access through future maintenance and recovery, not only construction.

After underground and structural completion, restore utility covers, pavements, drainage, kerbs, crossings, accessible paths, lighting, street furniture and landscape to local-owner standards. Survey and inspect reinstatement, remove temporary obstructions, repair monitored damage and close waste/land records. Hand back streets and property independently of railway opening; keep defect/maintenance responsibilities and warranties explicit.

Population-led additional lines and country regeneration

Repository source: `docs/civil/population-led-network-regeneration.md`

The controlled route inventory now includes additional lines selected from residential gaps, rather than leaving all additions in a separate sensitivity map. Each added route connects to a retained corridor at an actual common grid cell. Near-ring diversions preserve selected residential sites and avoid new self-intersections, protected cells, unknown water and unapproved long crossings. Native crossing, platform, terminal and bounded-interchange checks still apply.

The shared policy uses retained native 2020 population counts, a working 80% station-circle target and a 1 km radius. Native station placement follows the local demand bands: 1.6 km in the core, 3 km in the urban area and 7 km on suburban/outer approaches. Sampled planning stops do not override these bands. Ordinary neighbours within an interchange's local spacing envelope are removed after the actual platforms and operational endpoints are established, provided removal preserves the outer station-gap limit. Distinct physical crossings remain explicit site-review exceptions.

New corridors must be at least 4 km long. Additions are limited to half the retained trunk count (rounded up), at most eight lines, and 25% of the retained route length. At least 45% of each added corridor's sampled routing cells must be 1.2 km away from the committed network, allowing shared junction approaches while rejecting parallel feeders with duplicate catchments. Population gains are recomputed after every addition. Retained DEM slope penalizes hills and excludes missing terrain from new routing; core straightening cannot introduce a steeper terrain conflict or cross protected/unknown cells. Water crossing bounds, native geometry and connectivity checks still apply. These are ground-terrain planning screens, not surveyed track grades. Country and city finance inputs remain local; Baghdad retains its selected-population and sector policy.

The network map shows one main junction symbol on an actual platform for each bounded interchange complex, with smaller dots at other platform locations. Connecting strokes identify complex membership schematically; they do not depict surveyed walking paths or rail switches. Platforms and their line chainages remain separate engineering records.

After final junction and bank consolidation, a separate spacing pass restores only intervals above the outer 7 km limit, selecting dry cells on the existing route. A hard final check includes the closing interval of rings. Local stops and mandatory crossing platforms keep their positions; unresolved excess gaps stop generation instead of being hidden in a map.

Closed out-and-back excursions on retained radial routes are removed before stations and junctions are generated. This preserves the endpoints and uses existing route edges, avoiding additional shortcuts through terrain or water. The alignment report records each removed retrace; final route quantities, stations, coverage and costs follow the emitted inventory.

Controlled revision and actual coverage

Each city retains `engineering/alignment/network-expansion-baseline.json.gz`, preserving the original design bytes and their hash. Its `residential-line-expansion.json` identifies added line IDs, priority cells, actual corridor connections, route quantities, controlled original-line junction changes and unresolved site/geometry reviews. Mandatory native geometry checks can reject a candidate; a failed diversion never creates a fictional transfer.

Selection credits retained baseline stops and explicit residential priority sites. Ordinary sampled stations are not credited as delivered coverage because native spacing and junction consolidation can remove them. The separate `residential-expansion-evaluation.json` audits the actual emitted station coordinates after generation, using the union of native population-count pixels. Remaining gaps and a missed target are reported. A target is never declared reached from a renderer, a routing-demand score or a sum of overlapping district circles.

Where population evidence is missing or contradicts the controlled source/jurisdiction finding, population-led additions remain unavailable. Those cities still receive the current route, junction, civil, fleet, depot and economic logic. No residents or coverage fraction are invented. Existing unsuitable geometry remains an explicit structural/profile/access review where a controlled replacement is infeasible.

Complete regeneration

1. Rework controlled core geometry, connect reviewed junctions and add population-led routes; retain immutable baselines and constraints.
2. Emit actual native stations, bounded complexes, civil classifications, fleet/depot quantities, operating overrides and scenarios. New native spacing owns placement; the older infill overlay becomes a comparator and records unresolved gap reviews.
3. Refresh actual local civil costing before freezing the design and scenario inputs. Execute the full nominal day and degraded native software cases, retaining their actual source-bound outputs and receipts.

4 Regenerate SUMO/GIS/energy, desktop ground and site-evidence packages, timing references, operations, finance, depots, stabling, factory, delivery and deployment. Reconcile operations and finance in dependency order; a funding source mismatch stops publication.

5 Audit emitted-station population and connectivity, geography and clearance. Update city packages, country briefs, portfolio, construction/assembly, readiness, catalogue audits and publications from the final inventory.

6 Verify source/output hashes, complete archives, links, file limits, repository health and generated drift before merging.

Use `tools/automation/regenerate-all.sh --current-design-logic --prepare-only --jobs N` for the layout phase. The batch receipt explicitly says engineering refresh is required; it does not mark a planning package complete. The complete run uses actual engineering and native validation, or verified native CI artifacts from the exact frozen source revision.

Finite construction plant

Two candidate erection fronts per line do not create two additional purchased machines. The connected construction study assigns fronts to a finite launcher pool, with the preceding assignment as a required dependency. Support-release teams also follow finite pool queues and working calendars. Optional ring transfers use only the final donor assignment; a machine cannot be taken from a queued line.

Per-line foundation, bay, beam and launcher records retain actual identities and directed installation order. Soil investigation, foundation selection, loading combinations, vertical profiles, access/property rights, temporary works, suppliers, prices and independent construction/operating acceptance remain required. Station-circle proximity is not current census, surveyed walking access, observed ridership or funded passenger service.

Rapid Implementation And Recycled Materials Review

Repository source: `docs/civil/rapid-implementation-materials-review.md`

This note reviews current public rail-infrastructure practice for faster delivery of urban rail track and stations, with attention to recycled or lower-carbon materials. It is intended to guide OSR civil and station standards, not to replace local structural approval, geotechnical design, or railway product acceptance.

> Implementation status: The core trackform and elevated-guideway recommendations in this review are incorporated into the parametric civil catalogue; station-material and circular-procurement extensions remain identified below. Its original 3/12/18 M USD per route-km assumptions are retained as calibration benchmarks; active CAD-indexed planning targets are generated in `civil-cost-model.toml`.

Executive Takeaways

1. Use ballastless urban track as the default, but keep it boring. The strongest near-term pattern is continuous slipformed or precast ballastless slab track with adjustable direct-fixation rail fasteners. This fits urban corridors better than ballast, avoids tamping cycles and loose aggregate, and is cleaner around crossings and platforms. The speed gain comes from repeatable geometry, mechanised placement and fast rail adjustment, not from inventing a new trackform.
2. Split the track kit into two approved production methods. Use continuous slipform on long open, machine-accessible sections. Use transportable ST6 single-track panels in constrained, replaceable, utility, transition and short-possession zones. Local closure pours remain available where survey tolerance or interfaces make either repeated method unsuitable.
3. Make elevated guideway a factory product, not a bridge project. Low-cost elevated rail depends on a short span catalogue, single-column pier lines, transportable decked Pi20/Pi25 or special segmental spans, standard bearings, and repeatable erection. It should be used where it avoids demolition, severe junction conflicts, or tunnelling, but the model should not assume elevated is cheap where foundations, utilities, stations, or special spans dominate.
4. Treat recycled plastic sleepers as a specialist tool, not the main running-track answer. Recycled composite sleepers are now real railway products, with Network Rail approval and metro applications. They are most attractive for bridges, tunnels, turnouts, temporary works, depots, and brownfield repairs where timber would otherwise be used. For new OSR urban lines, the main at-grade track should remain ballastless slab/direct fixation.
5. Stations can benefit immediately from modular construction. Precast platform units, modular GRP/composite deck systems, recycled rubber platform blocks, and standardized station "kit of parts" design all have real market examples. OSR should keep the current station strategy: small number of repeatable platform/canopy/service modules, factory QA, minimal bespoke architecture.
6. Recycled materials should be specified by performance class. Good candidates now: recycled steel, recovered/recycled aggregate in non-critical concrete, low-carbon cement replacements, recycled rubber vibration mats, recycled rubber platform blocks, recycled plastic sleepers in non-slab applications, and recycled plastic access modules. Avoid mandating a material recipe where local standards or supply are immature.
7. The procurement system matters as much as the material. Circular material depots, material passports, batch traceability, pre-approved supplier lists, and standard test packs are what make recycled materials fast rather than risky.

Track Technologies

Technology	Readiness	Speed effect	Recycled / low-carbon potential	OSR recommendation
Precast ballastless slab panels	High	High where geometry is repetitive and survey control is strong	Use low-carbon concrete, recycled aggregate where approved, EAF/recycled steel reinforcement	Adopt as standard option for tangent and broad-curve urban sections
Cast-in-place direct-fixation slab/plinths	High	Medium; slower concrete work but flexible around utilities and tight geometry	Low-carbon concrete mixes, recycled aggregate in base layers	Adopt as fallback and station/turnout/default constrained-section option
Adjustable direct-fixation fasteners	High	High for tolerance recovery and maintenance	Rubber pads may include recycled content if certified	Adopt; require vertical/lateral adjustment and replaceable pads
Slab mats / embedded resilient blocks	High	Medium; adds material but reduces later vibration mitigation work	Recycled rubber can be used where tested	Adopt in vibration-sensitive streets, hospitals, schools, and bridges
Recycled composite sleepers	Medium-high	High for spot replacements and timber-substitute locations	High recycled plastic content	Use for bridges, depots, temporary panels, and turnouts where approved; not core OSR slab track
Recycled rubber underlay	Medium	Medium; can reduce maintenance on ballasted track, less central to slab track	High recycled tyre content	Watchlist for depots and transition zones
Geopolymer / rubberized concrete sleepers	Research / pilot	Low near-term for OSR because OSR is slab-first	High cement reduction and tyre-rubber reuse potential	Watchlist; pilot only after local code acceptance

Ballastless Slab Track

The market already offers several forms of ballastless track that align with OSR's urban environment: prefabricated slabs, cast-in-place slabs, and direct-fixation plinth systems. WEGH's Arianna system is a prefabricated prestressed slab system for ballastless lines, and its EPD describes the slabs being laid on a bedding slab to even out foundation irregularities. SYSTRA and STRADAL describe a low-carbon ballastless slab track that can use existing ballastless-track installation equipment, including gantry cranes, to protect schedule.

OSR should not copy a proprietary system outright. It should define an interface envelope:

- Standard slab panel widths, lift points, recesses, duct routes, and drainage interfaces.
- Rail seat spacing and baseplate anchor pattern.
- Tolerance stack: subbase, panel, grout/bedding, baseplate adjustment, rail weld geometry.
- Precast QA: compressive strength, dimensional survey, anchor pull-out, embedded conduit continuity, and batch traceability.

Direct-Fixation Fasteners

Fast, accurate installation depends on adjustability. Pandrol's Fastclip Baseplate is explicitly for ballastless applications on precast concrete, direct-pour concrete, or steel structures, with quick vertical and lateral adjustment. Pandrol's DRS baseplate is positioned for metro, light rail, bridge, viaduct, tunnel, and slab-track use where vibration reduction matters.

OSR should require:

- At least +/-10 mm lateral adjustment at rail-seat level where possible.
- Vertical shim range sufficient to recover slab tolerances without breaking out concrete.
- Replaceable pads and clips.
- Electrical insulation suitable for track circuits only if the local signalling design needs it; OSR's onboard-first design reduces this dependency but does not remove bonding/earthing requirements.

Recycled Composite Sleepers

Network Rail's approval of recycled plastic composite sleepers, and Sicut's product acceptance, show that recycled-plastic sleepers are no longer purely experimental. Network Rail's public material also frames the move as part of circular economy and hardwood replacement. Victoria's Duratrack trials show the approval route can be narrower: low-speed siding and yard applications first, then broader use as standards mature.

For OSR:

- Use composite sleepers for depot yards, temporary construction track, bridges where timber would otherwise be used, and brownfield pilots.
- Do not use recycled-plastic sleepers as the primary support for new urban running track, because OSR has moved to ballastless slab.

- Require product acceptance evidence for screw pull-out, thermal movement, creep, fire/smoke, UV, dynamic stiffness, and fastening compatibility.

Elevated Guideway Technologies

Technology	Readiness	Speed effect	Recycled / low-carbon potential	OSR recommendation
Rail-designed decked Pi20/Pi25 beams	Prototype design required; established bridge family	High on 20–25 m repetitive spans	Low-carbon concrete after early-strength, creep and bond trials; reusable long-line mould	Adopt as the OSR reference after structural prototype and first article; one ≤3 m, ≤75 t beam per track
Full-span precast U-girders	High	High on 20–30 m repetitive spans with wide/heavy transport access	Low-carbon concrete, EAF/recycled rebar, reusable moulds	Retain as an acoustic or project-specific special product
Precast segmental U/box guideway with launching gantry	High	Very high on long viaduct packages and constrained streets	Low-carbon concrete and recycled rebar; lower traffic disruption	Adopt for long continuous elevated packages or where crane access is poor; avoid for short isolated stubs
Single-column median piers / classified foundations	High	High; reduces excavation width and utility diversion	Low-carbon concrete and recycled rebar in substructure	Select spread, bored, driven or pile-group interface by geotechnical/access zone
Precast pier caps or pier-cap shells	Medium-high	High; removes complex formwork above traffic	Low-carbon concrete, reusable forms	Adopt as a standard option for viaduct and elevated-station piers
Steel modular special spans	High	High for road/rail crossings and emergency works	Recycled steel content can be high	Use only for special crossings; concrete remains the default for repetitive spans
UHPC / fibre-reinforced concrete viaduct elements	Medium	Medium; may extend spans or reduce section depth	Lower material volume but higher material and QA burden	Watchlist; use only where local supply and structural review support it

Preferred Elevated Kit

The lowest-cost elevated portions should use the same discipline as the rest of OSR: one repeatable kit, many repetitions. SYSTRA's U-shaped metro viaduct work is a useful benchmark: Dubai Metro used 58.7 km of elevated guideway mainly formed from U-shape precast segmental superstructure, and SYSTRA frames the U-shaped viaduct as rapid, economical, and material- efficient. FHWA's accelerated bridge construction guidance points in the same direction for bridge-like work: prefabricate structural elements offsite or near-site, transport them in, and install rapidly to reduce onsite time and disruption.

For OSR, the elevated kit should be:

- Pi20/Pi25 standard spans, with a controlled special-span variant for road, railway, river, and utility crossings.
- One 2.9 m decked Pi-beam per track, stems below the rails, with separate outer walkway/barrier cassettes and a small inner service closure.
- Single-column piers in a road median or other clear line, with no bespoke pier architecture.
- Standard elastomeric bearings, expansion joints, drainage scuppers, inspection walkway brackets, earthing/bonding points, and cable trays.
- Ballastless direct-fixation track on the deck, with adjustable baseplates to absorb casting and erection tolerances.
- No overhead catenary envelope where battery operation is retained; this reduces visual clutter and vertical clearance pressure, but does not remove evacuation, walkway, drainage, bonding, or maintenance access.

Production Strategy

Elevated guideway becomes rapid only when foundations, precast production, and erection are overlapped as separate work fronts:

1. Early utility and geotechnical release: fix the pier line before girder production. Foundation surprises erase the factory advantage.
2. Near-site precast yard: cast Pi-beams, pier caps, ST6 panels, platform edge units, parapets, cable troughs, and walkway modules close enough that trucking does not become the bottleneck. Surrey Langley SkyTrain is a current example of guideway segments being made in a dedicated precast facility and trucked to the route.
3. Standard formwork battery: one mould family with reusable steel forms, surveyed casting beds, embedded inserts, and barcode/batch traceability.
4. Parallel erection: foundations stay several spans ahead, pier construction follows, and the launcher or crane crew erects spans while track and systems follow behind.
5. Use the right erection method: mobile crawler cranes are simplest for short elevated packages with street access. Launching gantries make sense when the elevated package is long enough to amortize setup or when traffic/utility

constraints prevent repeated crane closures. Freyssinet reports span-by-span launching gantry methods as a fast method for multi-span viaducts, while precast girder erection is simpler and can place a span per shift.

Cost Rules

The retained $12.0M/route-km elevated benchmark, and the current $9.748M/route-km design-derived target, describe a repetitive package rather than a promise for every elevated metre. The target is most plausible when the alignment has:

- Long runs of repeated 20-30 m spans.
- Median or public-land pier locations.
- Shallow-to-moderate foundations with no major utility diversion.
- At-grade or simple elevated stations, not stacked architectural structures.
- Few special spans, few skewed crossings, and no custom aesthetic deck.
- A local precast yard with enough repetition to amortize moulds, stressing beds, transport fixtures, and launching equipment.

The model should add risk or avoid elevated where:

- Soft ground, seismic demand, flood scour, or deep utilities dominate.
- The line needs many elevated stations close together.
- Special spans exceed the standard catalogue repeatedly.
- Curves are tight enough to force many unique segment geometries.
- The project asks for signature architecture, enclosed viaducts, or double-deck structures.

Elevated Stations

Elevated stations can destroy the cost advantage if each one becomes a custom building. The station should be designed as a widened guideway bay, not as a separate bridge:

- Keep side platforms on a regular 30 m column grid where possible.
- Use repeated pier caps/portal frames rather than deep transfer structures.
- Put stairs, lifts, and services in bolt-on modules outside the guideway load path.
- Keep paid concourse and plant rooms compact; avoid full-width enclosed concourses unless climate, security, or fare control requires them.
- Use the same canopy bay, platform edge, screen-door interface, cable trench, and drainage details as at-grade stations.

Recycled And Low-Carbon Content In Elevated Works

Use recycled and lower-carbon content where it does not compromise the prestressed primary span:

- Use EAF/recycled-content steel for rebar, brackets, walkways, cable trays, and canopies where certified.
- Use GGBS, fly ash, limestone calcined clay, or other approved low-carbon cement blends in piers, pile caps, pier caps, and station modules.
- Keep prestressed Pi-beams and special U-girders conservative until the precast supplier can prove early strength, durability, shrinkage, creep, and tendon bond.
- Use recycled aggregate in non-prestressed substructure or backfill only where local standards allow.
- Use GRP/composite walkway grating, cable covers, and access panels where fire/smoke and UV ageing evidence exists.
- Crush rejected precast elements and demolition concrete into subbase or non-critical aggregate rather than sending them to landfill.

Station Technologies

Technology	Readiness	Speed effect	Recycled / low-carbon potential	OSR recommendation
Precast platform L-walls / slabs / TT units	High	High	Low-carbon concrete and recycled aggregate can be specified	Keep as primary station platform structure
Modular GRP/FRP deck systems	Medium-high	High; supplier claims major installation time savings	Recyclability varies; low weight reduces transport and plant	Use for light platforms, extensions, temporary stops, and low-load station elements
Recycled rubber modular platforms	Medium	High; manual-handled blocks reduce plant and possessions	Very high recycled tyre content	Pilot for halt platforms, temporary stations, and extensions; require fire/slip/drainage tests
Modular station kit-of-parts	High	High across programme; less useful for one-off showpiece stations	Compatible with timber, recycled steel, low-carbon concrete	Adopt as OSR default
Recycled plastic bus/platform access modules	Medium	High for temporary accessibility works	High recycled plastic content	Use for temporary or feeder stops, not primary heavy rail platforms without approval
Mass timber station buildings	Medium-high	Medium; fast above-platform assembly	Renewable, low embodied carbon	Watchlist for dry climates and non-critical buildings; fire/termite/moisture details matter

Modular Platform Systems

There is a mature market for precast railway platform elements. Moore Concrete supplies front wall units, L-walls, edging units, slabs, and copings for new platforms and platform extensions. HERING's modular platform systems are designed around prefabricated foundations, slabs, and substructure components, including variants suited to tight closure times and lighter lifting equipment. Older modular platform research such as StepSafe also supports the basic direction: a small number of factory made components, simple foundations, and repeatable installation sequence.

OSR should keep the station platform as a product family:

- Halt: short precast edge beam + modular deck + minimal canopy.
- Standard: L-wall or TT slab modules + portal canopy + cable/service trench.
- Major/interchange: same modules repeated with larger foundations, more circulation width, and more MEP.

Recycled Rubber And Composite Platforms

Arcadis and partners developed the Footprint Modular Platform using 100% recycled rubber blocks for platform walls. Arcadis reports a typical 36 m platform extension can save about 25 tonnes of carbon and that the blocks reduce heavy lifting needs. This is highly relevant to fast extensions, temporary stations, and low-platform pilots.

The important caution is product acceptance. Rubber-block platforms need:

- Fire/smoke classification.
- Slip resistance when dusty, wet, oily, and hot.
- UV/heat ageing data for hot climates.
- Drainage and trapped-water details.
- Load tests for crowding, maintenance carts, and edge impact.
- A repair method for damaged blocks.

GRP / FRP Platform Decks

Dura Composites reports GRP platform systems already installed across rail stations and claims installation time can be cut by up to 65%. These systems are useful where speed, corrosion resistance, and low weight matter. They are less obviously circular than recycled rubber or plastic systems unless the supplier can provide end-of-life takeback and recycled-content evidence.

OSR should allow GRP decks as an option for:

- Temporary platforms.
- Footbridges and access ramps.
- Low-load platform extensions.
- Coastal/corrosive environments.

Modular Station Design

Network Rail's HUB Station and station design guidance both point toward kit-of-parts station design. The US FTA's Los Angeles modular station case is older, but still useful: standardize boxes, ancillary rooms, MEP, structural elements, circulation, and platform layouts while leaving site-specific entrances and public realm flexible. Arup's modular station handbook for Indian station redevelopment reaches the same conclusion for live rail environments: standardize typical components, prebuild offsite, and reduce redevelopment time.

For OSR, that means:

- One platform edge detail across all station archetypes.
- One canopy bay module with repeatable columns, purlins, PV rails, and drainage.
- One electrical/telecom cabinet family.
- One ramp/stair kit.
- One ticketing/gate plinth, even when fare policy changes.
- Common MEP service routes and quick-disconnect interfaces.

Recycled And Lower-Carbon Materials

Near-Term Approved Material Families

Material	Where to use now	Notes
Recycled steel / EAF steel	Rails only with certified new rail supply; station steelwork, canopy frames, rebar	Reused rail should not become running rail without full traceability and ultrasonic/profile acceptance
Recovered ballast / aggregate	Subbase, drainage aggregate, non-critical concrete where standards allow	Less central to ballastless OSR, still useful in subbase and civil works
Recycled concrete aggregate	Platform foundations, non-prestressed slabs, drainage works	Use performance specs; keep prestressed slab track conservative until approved
GGBS / fly ash / limestone calcined clay / low-carbon cement blends	Precast station elements, platform foundations, track slabs where supplier can certify strength and durability	Local supply varies strongly
Recycled rubber	Vibration mats, platform blocks, temporary access surfaces	Check fire, heat, smoke, creep, and slip
Recycled plastic composites	Sleepers/bearers in approved applications, access modules, cable trough covers, temporary works	Check thermal expansion and fire/smoke
Reused station components	Canopy steel, cabinets, benches, cable trays, lighting poles	Needs material passport and condition grading

Circular Procurement

Network Rail, Deutsche Bahn, SNCF, UIC, and Victoria's Big Build all point in the same direction: reuse and recycled content need a system, not isolated experiments. OSR should build procurement around:

- A material passport for every major batch: steel, concrete, rubber, plastic, rail, fasteners, and PV/canopy components.
- A recovered-material hierarchy: reuse as-is, remanufacture, recycle into equal-grade infrastructure, then downcycle.
- Regional material depots near construction packages.
- Contractor tender scoring for recycled content, takeback, and embodied carbon, but only where standards allow the material.
- A local product-acceptance register, not one-off waivers.

Implementation Recommendations For OSR

Adopt Now

1. At-grade urban trackform: ballastless direct-fixation slab; the current CAD-indexed target is $2.584M/route-km against the retained $3.0M benchmark.
2. Elevated guideway kit: repeatable Pi20/Pi25 or special segmental guideway, single-column piers, standard bearings, standard drainage, and one erection playbook.
3. Two at-grade construction methods: continuous slipform for long open runs and ST6 single-track precast panels for constrained or replaceable zones.
4. Adjustable baseplates: required for all ballastless track to speed survey recovery and maintenance.

5. Station kit-of-parts: precast platform units, steel portal canopy, modular ramp/stair/cabinet/service trench.
6. Circular procurement clauses: require recycled-content disclosure, takeback options, and batch traceability.
7. Recycled-material pilots: rubber platform extension, composite sleepers in depot/temporary track, low-carbon concrete in non-critical station elements.

Pilot Before Standard Use

- Recycled rubber modular platform walls for permanent passenger platforms.
- GRP platform decks in high heat and high footfall.
- Recycled aggregate in prestressed track slabs.
- Geopolymer or graphene-enhanced concrete sleepers/slabs.
- Recycled rubber slab mats under main running track.

Avoid As Core Standard For Now

- Reused rail as new running rail without full certification.
- Fully bespoke station architecture per city.
- Uncertified plastic structural modules in fire-critical station areas.
- One proprietary slab-track system that prevents local fabrication.
- Heavy recycled-content requirements that local suppliers cannot certify.

Design Recommendations And Status

1. Implemented: the civil catalogue includes the ballastless urban slab system with:
   - continuous-slipform and constrained ST6 variants,
   - direct-fixation fastener envelope,
   - slab mat option,
   - cable trough and drainage interfaces.
2. Recommended extension: add a modular platform kit with:
   - precast concrete base option,
   - recycled rubber extension option,
   - GRP deck option,
   - standard service trench and cabinet plinth.
3. Implemented: the rapid elevated guideway catalogue includes:
   - full-span Pi20/Pi25 and segmental-launch variants,
   - 20 m / 25 m standard spans,
   - controlled special-span interface,
   - single-column pier and pier-cap shell catalogue,
   - elevated-station widened-bay interface.
4. Recommended extension: add a procurement-release checklist covering:
   - recycled content certificate,
   - product acceptance certificate,
   - fire/smoke/slip/UV/thermal test pack,
   - repair method,
   - end-of-life takeback or recycling route.

5 Retain $3.0M/route-km as the at-grade calibration benchmark and treat the current $2.584M CAD-indexed value as an unquoted planning target. Recycled content can reduce embodied carbon and site time, but should not be assumed to reduce first cost until supplier pricing is proven.

6 Retain $12.0M/route-km as the elevated calibration benchmark and treat the current $9.748M CAD-indexed value as a repetitive-package planning target only. Short elevated stubs, soft-ground foundations, many elevated stations, or repeated special spans should carry a local risk premium.

Source Links

- Network Rail: HUB Station modular kit-of-parts
- Network Rail Station Design Guidance
- FTA: Modular Station Design
- Arup: Design Handbook for Modular Railway Stations
- HERING: modular railway platform construction systems
- HERING: low-carbon platform concrete and recycled aggregate
- Moore Concrete: precast railway platform components
- Dura Composites: GRP railway station platforms
- Arcadis: Footprint recycled-rubber modular platform
- Arcadis project page: low-carbon rail platform
- Sicut: Network Rail approval for recycled plastic sleepers
- Sicut: Network Rail hardwood replacement with composite sleepers
- Sustainability Victoria: Duratrack recycled plastic sleeper trials
- Pandrol Fastclip Baseplate for ballastless track
- Pandrol DRS Baseplate for non-ballasted metro/light rail
- WEGH Arianna ballastless slab track system
- EPD International: Arianna slab track EPD
- SYSTRA and STRADAL low-carbon ballastless slab track
- FHWA: prefabricated bridge elements and accelerated bridge construction
- Freyssinet: precast segmental erection methods
- SYSTRA: bridges and U-shaped viaducts
- SYSTRA: Dubai Metro U-shaped precast elevated guideway
- TRID: Dubai Metro light rail viaducts superstructure
- Surrey Langley SkyTrain: precast concrete facility
- Surrey Langley SkyTrain: launching gantries
- MRTA Orange Line East: elevated structures and elevated stations
- ANM Riyadh Metro Line 3: 3,000 precast segments
- edilon)(sedra slab track mats
- Network Rail: reducing waste and material reuse
- Network Rail: Whitemoor recycling centre
- Deutsche Bahn: ballast recycling
- SNCF: ballast recycling
- UIC circular economy programme
- Victoria Big Build recycled materials resources
- Victoria Big Build environment and Recycled First

- Research: sustainable rubberized geopolymer concrete sleepers
- Research: reducing carbon footprint of railway sleepers using recycled plastics

Civil reference and numerical evidence workflow

Repository source: `docs/civil/reference-demonstration.md`

Civil demonstration A contains two adjoining double-track bays (20 m and 25 m), an at-grade transition, connection briefs, bearing and foundation options, construction hold points, a conventional I-girder/deck comparison, and a connected civil FMEA. Its independent review, site design and construction release are pending. The recorded OpenSees, CalculiX and SWMM exercises demonstrate calculations and rejection cases; they do not qualify the structures.

Mass, lifting and option selection

`osr_mech.civil.reference.lifting_budget` requires every net member detail and rigging mass plus the reviewed equipment configuration, chart and radius. It compares dynamic hook demand with capacity at that radius and permitted utilisation. A 75 t nominal equipment label is insufficient. Member manufacturing mass and lifted mass are separate values. `member_target_met`, `equipment_capacity_met` and `overall_accepted` are separate outputs; `lifting_check_passed` requires both product-envelope acceptance and equipment capacity. A heavier-product `product_deviation` requires `decision=accepted`, different `engineer/checker`, `signed_at`, `controlled_reference`, `rationale` and the exact `mass_budget_sha256` emitted by the complete budget. Mass changes invalidate it. The generated transport summary shows unresolved details and equipment selection. Additional diaphragm mass is net of the bare section; the CAD design zones overlap and cannot be summed as fabricated concrete. Density-based reinforcement adjustment must have a documented basis.

`foundation_candidates` returns alternative methods with no selected design. `compare_foundations` requires capacity, total/differential settlement, groundwater, chemistry, liquefaction/scour applicability, utilities and access, plus an independently reviewed selection and comparison rationale. The legacy `select_foundation().id` is retained as a planning preference for existing callers; `selected_id` stays empty. It is unsuitable for project release. The ground-design gate now reconciles the reviewed per-support comparison with the schedule's capacity and settlement values. Calcium-based soil treatment needs a chemistry compatibility review. Every ground-treatment selection remains subject to site design.

Both the Pi and conventional-girder candidates need supplier section/prestress/reinforcement details, complete installed quantities, foundations, transport, erection, temporary stability, access and maintenance evidence. Ballast, slipformed slab and panels remain at-grade alternatives. No installed-cost or whole-life-cost winner has been established.

Civil asset register

Both civil release manifests now require a `civil_asset_register` JSON receipt. Its schema is `osr-civil-assets/1` with:

- `design_sha256`: the received design file digest.
- `review`: `decision=accepted`, different named `producer` and `checker`, `signed_at`, and a `controlled_reference` to the independently reviewed layout.
- `assets`: unique `asset_id`, `line_id`, `asset_type`, `from_station_m`, `to_station_m`, `foundation_ref` matching the submitted structural schedule, and explicit `support_ids`. Spans/at-grade structures include supports at both endpoints. Point piers/abutments/foundations can have equal start/end chainage.
- `supports`: unique `support_id`, `scope_type` (`line` or `station`), `scope_id`, `chainage_m` and `zone_id`, matching the foundation schedule exactly.
- `coverage_intervals`: `line_id`, `coverage_group`, start/end chainage. Associated asset intervals carry the same group, cover its whole extent and have no gap or overlap. Groups use `track-1` through `track-N`. The received design independently declares each line's `length_m`, optional `from_station_m`, and positive integer `civil_track_count`. These authoritative extents must match the register exactly. Missing design extents block the screen. The demonstration uses a separate route basis; shortening the register cannot hide omitted route.

A line-scoped support must belong to the asset's line; station-scoped supports must belong to a station affiliated with that line in the design (`line`, `line_id`, or `lines`). Cross-scope sharing requires one `shared_support_relationships` row for each explicit `asset_id`/`support_id` pair and its own independent accepted `review`. Matching chainage alone is insufficient.

The concept register shows the structure with review deliberately pending. Expected IDs must come from the reviewed layout, rather than deriving expectations from the submitted schedule. One row labelled with a line ID cannot satisfy omitted spans or supports.

Structural numerical exchange

Solver reports retain status, convergence, version, input digest and model revision. They additionally require:

- `output_hashes`: map of actual relative file paths to SHA-256 digests, resolved under the controlled evidence root. Missing, empty, changed or escaping files fail.
- `native_output_paths`: nonempty list of hashed native solver files.
- `numerical_results_path`: a hashed CSV with `asset_id`, `load_case_id`, `metric`, `value`, `unit`.

The independently reviewed `load_case_register` must bind `civil_asset_register_sha256`, include the same `review` fields as above, and define `required_load_cases` and `criteria`, each keyed by `opensees` and `calculix`. Every criterion has `asset_id`, `load_case_id`, `metric`, `unit`, `stage` (`service` or `construction`), finite `limit` and `operator` (`max`, `min` or `abs-max`). Criteria cover every asset/load-case combination; actual CSV rows match exactly, including metrics and units. Required cases, both lifecycle stages and numerical limits are checked. A negative displacement cannot evade an `abs-max` criterion.

The load-case register also supplies `native_output_spec[solver]`: `parser_id`, `model_sha256`, `model_units`, `files` and `selectors`. The report supplies `native_parser` from `solver_results.parser_provenance`, including parser ID/version/source hash, specification hash and model hash. `extract_results` generates the CSV, and the gate repeats extraction and reconciles every asset/case/metric/unit/value (relative tolerance 1e-9, absolute 1e-12). A CSV showing 1 mm cannot pass when native displacement is 100 mm, even if all file hashes match.

The initial adapters support OpenSees ASCII Node recorder displacement/reaction output with time, explicitly ordered nodes/DOFs, and CalculiX nodal displacement/force `.dat` tables with step, set, time, node and DOF selectors (CalculiX manual). Native outputs do not declare physical units: the independently reviewed specification pins displacement (`m/mm`) and force (`N/kN`) model units and converts to criterion units. Each selector names `asset_id`, `load_case_id`, `metric`, `path`, `node_id`, `dof`, `time`, `quantity` and `operation` (`signed/abs`); CalculiX also requires `step` and `set_name`. OpenSees file definitions declare `nodes`, `dofs`, `time_column=true` and `response` (`disp/reaction`). The recorded beam exchanges provide reproducible examples. Unsupported response formats, ambiguous samples and missing selectors fail closed.

The automated checker establishes native/exchange agreement, provenance, coverage and compliance with supplied limits. The engineer and independent checker remain responsible for selecting sufficient metrics/cases and validating loads, mesh, soil springs, prestress, reinforcement, railway criteria and construction-stage assumptions. Signed acceptance continues to bind the evidence digests.

Foundation comparison exchange

`ground_design_verification_report.support_comparisons` is keyed by every scheduled support ID, exactly once. Each entry supplies ground class and actual `vibration_restricted`, `clear_access`, `high_lateral_load` screening conditions; `site` inputs; and a `comparisons` row for every candidate. Site actions are `axial_demand_kN`, `lateral_demand_kN`, `settlement_limit_mm` and `differential_settlement_limit_mm`. Site groundwater, chemistry, liquefaction, scour, utilities and construction access need documented references or reviewed applicability decisions.

Each comparison includes `id`, `axial_capacity_kN`, `lateral_capacity_kN`, `settlement_mm`, `differential_settlement_mm`, `constructable`, `chemistry_compatible`, `liquefaction_checked`, `scour_checked` and `calculation_reference`. `site.selection_review` supplies accepted decision, selected ID, different engineer/checker, signed date, controlled reference and comparison rationale. Schedule `design_capacity` is in kN and settlement in mm. Catalogue foundation concrete is in m³, reconciled against actual finite lengths and integer counts; zero cannot replace a deep-member count. Ground improvement comparisons cover the catalogue alternatives and use the reviewed zone actions/deformations, quantities and compatibility evidence.

Drainage numerical exchange

The accepted hydrology basis needs independent `review`, `scenario_model_hashes` for `normal`, `blocked-drain` and `backwater`, `scenario_assumptions` describing the adverse cases, and `performance_limits` for each case. The SWMM report's `scenarios` maps those names to `model_path` (relative to the report) and `sha256`; the normal model matches the received SWMM input. Scenario models must be distinct and match the reviewed basis digests.

The initial adverse-model adapter requires dynamic-wave routing, an actual reduction of a conduit opening for blockage, and a fixed tailwater above the normal outfall head for backwater. Comment-only changes and mislabeled cases fail. Other blockage mechanisms or time-varying boundaries need a checkable adapter before using this gate; scenario engineering assumptions still need independent review.

Each set of limits supplies `flow_units` and `system_units` matching replay; `nodes` and `links` keyed by all simulated IDs. Node limits are maximum `peak_head`, `flooding_volume`, `surcharge_duration`, plus `outfall_peak_flow` for outfalls. Conduit limits are `peak_depth` and `surcharge_duration`. Head/depth use m (SI) or ft (US), flooding volume m³ or ft³, duration hours, and flow the SWMM flow units. Results use cumulative engine statistics over every routing step rather than report-interval samples. Continuity is checked for all three cases alongside hydraulic acceptance. All values must be finite.

The hydraulic gate checks numerical performance against the reviewed limits, not the engineering suitability of the storms, freeboard, blockage assumptions or outfall boundary. Site catchments, culverts, discharge consents, erosion/scour, cleaning access and transition settlement still need design. Use additional 2D flood assessment where overflow pathways materially affect the case.

Shared assembly assurance and change propagation

The civil assembly configuration uses `osr-connected-engineering/1` through the existing component validator, including required ownership, effects, detection/mitigation, risk, frozen item revisions, evidence classes, requirement-derived scenario criteria, hierarchy/cycle checks and closure decisions. A layout or mass change updates quantities and item revisions, flags stale configuration/FMEA records and reopens dependent evidence and gates. Unknown installed costs remain empty; benchmark rates cannot produce a validated option price.

The blockage chain links drainage, formation, foundation, bearing, track and operational failures, each to an identified planned asset, owner, calculation/observation requirement and response. The graph includes physical hierarchy, interfaces, planned production controls and maintenance feedback. No synthetic occurrence or planned record proves an installed asset, concrete strength or completed work. Prestress transfer, storage, transport, actual lift positions, stability, fatigue, torsion and rail interaction remain engineering work requiring new analysis and physical evidence.

Reproduction and migration

```
.venv/bin/python tools/automation/civil_reference.py --run-solvers
.venv/bin/python tools/automation/civil_reference.py --check
.venv/bin/python tools/automation/civil_reference.py --verify-native-replay
.venv/bin/python -m pytest -q engineering/analysis/tests design/component-catalogue/tests tools/automation/tests
```

Replay requires OpenSeesPy, PySWMM and native CalculiX (`ccx`). Check mode validates recorded native/exchange agreement and source digests without requiring those solvers. `--verify-native-replay` runs in a temporary directory and compares reproduced beam values and hydraulic outcomes without rewriting the recorded evidence. Existing pending city manifests gain the new register role during normal civil-gate regeneration. Previously declaration-only submitted reports need the new exchanges and reviews before they can pass. No existing authority decision is manufactured or carried forward as acceptance of a changed evidence package.

Shared trainset and viaduct engineering model

Repository source: `docs/civil/shared-train-viaduct.md`

Retained numerical review · comparison figure · spatial numerical review · spatial comparison figure · OpenSourceRail standard OSR-ENG-001 · native car-pair assembly · native joint/drawing receipt.

The automated reference-parts compiler now adds a separate complete-family variant with 192 reference instances, 191 joints and all 24 six-car wheelsets. Its parameterised battery architecture, coverage register, typed interfaces and manufacturing/QA outputs extend this foundation. The retained two-car demonstrations below continue to describe their original configurations. The full-train contact-domain investigation adds explicit supplied-profile footprint checks and distinguishes domain failures from numerical non-convergence.

The first demonstrator connects a representative articulated car pair to a three-span viaduct, flexible rail, finite bearings, axial piers and foundation springs. Component mass, CG and full inertia tensors feed carbody/bogie heave and pitch, individual wheelsets, suspension and articulation. Wheel forces deform the rail and bridge; their displacement changes the vehicle contact forces. The same instance definition supplies native FreeCAD Assembly joints, an unissued TechDraw view, cost quantities and the verification dependency register.

This is a synthetic numerical verification model. Its mass allocations, inertias, stiffnesses and track properties are declared research inputs. It is not a supplier train, a production drawing package or Baghdad qualification. The two represented cars use the six-car operational family's planning module dimensions; they do not represent all 24 axles of the complete Baghdad train.

One family definition

`lib/templates/rolling-stock.toml` owns family counts and overall lengths. Both catalogue CAD and the FreeCAD review generator consume the `family definition`. Their planning car modules are 21 m, 19.5 m, 16.5 m, 18.75 m and 18.5 m for the one-, two-, three-, four- and six-car families. The six-car length is 111 m, with 12 bogies and 24 axle positions. The detailed bogie and family placement share their 2.1 m wheelbase datum.

Equal car modules, two bogies per car and the reference inset are planning assumptions. Supplier-defined lengths, articulation arrangements and installation geometry remain required before these datums can become railway load evidence.

Instance and evidence contract

The published schema and instance implementation extend the existing product and assembly IDs. The normal buildable-trainset generator emits `engineering-instance-definition.json` alongside the existing manifest. The unfilled slots retain quantities and parent assemblies; kits and area allocations still need splitting into production parts. Their mass fields remain null until data is supplied.

The retained instance allocation contains 737 open slots for existing active product rows, including explicit non-counted kit/area allocations. It adds no invented closed mass records.

Each instance identifies its part/revision, parent, serial or material batch, geometry classification, material regions, thickness/process slots, named datums, rigid transform, attachment and physical mass scope. Design, supplier and measured properties occupy separate slots with an explicit selected source. Supplier declarations and measurements require retained files and SHA-256 bindings. Released/supplier installation solids require a retained geometry source; the initial native adapter reads BREP solids. A supplier envelope's enclosing volume never creates a material mass estimate.

The model embeds the family definition used for that configuration. A new as-designed model must match the current authoritative profile. As-built and as-maintained records retain their frozen family geometry when the master profile changes. Their own serial/batch and revision identities still apply; changing the master profile does not rewrite an already manufactured vehicle.

The mass calculation rotates component tensors and applies the parallel-axis theorem about the combined CG. It calculates asymmetric static wheel reactions while conserving weight and longitudinal/lateral moments. Independent mass and position bounds produce conservative wheel-load, CG and inertia intervals. Position uncertainty encloses the combined CG/datum location error; transforms are otherwise exact. Inertia uncertainty is an absolute spectral bound. Correlation losses are stated. Missing quantities or tensors remain open and cannot start the coupled solver.

Joint instances reference actual instance datums and retain revision, permitted motion, mechanical properties, friction/stops, definition fields and inspection IDs. Native fixed, slider and revolute joints establish placement/motion. Their mechanical stiffness is a separate record, not a consequence of CAD solving. Detailed bolt stacks, weld procedures, laminate/bond data, fits, preload and nonlinear suspension curves still require actual design or supplier evidence.

Run and verify

From the repository root, using the engineering environment:

```
PYTHONPATH=design/component-catalogue/src tools/automation/osr-python \
  tools/automation/shared-train-viaduct.py \
  --output build/engineering/shared-train-viaduct \
  --speeds 15,20,25 --dt .0025 --mesh 8

PYTHONPATH=design/component-catalogue/src tools/automation/osr-python \
  tools/automation/shared-train-viaduct.py \
  --verify build/engineering/shared-train-viaduct
```

The component catalogue's `analysis` optional dependencies provide NumPy, SciPy, JSON Schema checks and OpenSeesPy. Use a new output directory for each campaign. The supplied example changes one battery mass by 10%, secondary stiffness by −20%, articulation stiffness by a factor of two, and deck modulus by −10%. `--model` accepts a filled definition; it requires `--baseline-only` because the named variants are specific to the synthetic example.

Outputs retain every axle contact-force history, axle unloading, body acceleration, joint force/displacement, rail displacement, deck acceleration, bearing and foundation reaction. Independent temporal and spatial refinements use a declared 5% numerical screening tolerance. That tolerance is an implementation check, not a railway acceptance limit. Native OpenSees independently replays the deck and finite-support equations. Force, moment and instantaneous work mapping are checked separately. Source, numerical library and output hashes bind the receipt. Contact tension or exceeding a joint stop fails the linear adapter's domain.

The first adapter resolves straight-track vertical heave/pitch and linear compressive contact. Its rail mesh resolves the rail-on-pad bending length independently of the deck mesh, and elastic approach track removes moving-support force jumps. The spatial adapter described below adds lateral/roll/yaw motion, curves/cant, separate wheel contacts and both tracks. The retained vertical comparison remains a distinct benchmark with its own scope and refinements. Project acceptance limits remain external required inputs.

Generate the native assembly through the installed FreeCADCmd/Flatpak runtime:

```
tools/automation/shared-engineering-freecad.sh \
  --model build/engineering/shared-train-viaduct/baseline/definition.json \
  --output build/engineering/shared-car-pair.FCStd
```

The retained native receipt checks solver return, named-datum position/orientation residuals and geometry source hashes. Drawing generation leaves `DrawingIssued=false`. Inspection evidence must be performed and reviewed separately.

Export the compact review and standalone figure after a verified campaign:

```
PYTHONPATH=design/component-catalogue/src tools/automation/osr-python \
  tools/automation/export-shared-engineering-review.py \
  build/engineering/shared-train-viaduct \
  --native design/component-catalogue/models/cad/shared-car-pair.native.json \
  --output engineering/civil_exploration/examples/shared-train-viaduct-review.json \
  --plot engineering/civil_exploration/examples/shared-train-viaduct-comparison.png
```

The figure exporter additionally requires Matplotlib. It verifies retained source/output hashes and native datum/geometry bindings before publishing.

Changes and manufacturing identity

The verification register binds instance/joint inputs to mass properties, vehicle assembly, coupled vehicle response, bridge demand, cost, drawings and inspection characteristics. A deck change invalidates vehicle response through feedback. Unaffected instance drawings retain their dependency fingerprints. Old results are preserved; a new fingerprint never reaccepts an old inspection.

The ERP characteristic adapter and engineering proposal compiler bind the exact instance-definition artifact, design/part revision, asset, serial/batch, joint IDs and inspection units. Native work-order, quality inspection, material batch, calibration and nonconformance references accompany a measured result. The existing execution preview validates and preserves that map; feedback exposes it under native read permissions. Comparison with defined limits remains separate from engineering acceptance. No live ERP records are written by the demonstrator.

Civil quantities use the existing equal-scope double-track bill, explicitly labelled separately from the single-track dynamic benchmark. Vehicle component counts and mass update with the instance definition. Installed cost remains null without rates, quotations and full scope: a mass change does not invent a saving.

Spatial, material, joint and correlation workflows

The second adapter uses six translations/rotations per rigid body, bogie and wheelset, full component inertia tensors, two individually flexible rails per track, shear-flexible 3D decks, transverse caps, piers and six-direction foundation springs. It solves the full sparse coupled equations; its diagnostic modes do not truncate the transient calculation. The route supplies circular horizontal and vertical alignment, grade and a cant ramp. Passenger counts come from the frozen family profile; asymmetric payloads update CG and inertia.

The spatial solver uses supplied wheel/rail profiles to locate one nonconformal contact patch, elliptical Hertz normal contact and passive stick/slip traction. Contact can open and reclose. It retains wheel spin, gyroscopic terms, reference alignment acceleration, braking/traction demand, articulation loads, wheel forces, unloading, lateral/vertical ratios, carbody acceleration, rail/deck response, foundation reactions and all six structural member resultants. Primary wheel spin is a free joint coordinate. Nonlinear supplied force/displacement and damping curves and unilateral bump stops have separate tangents in the implicit solve.

Straight, curved/canted/braking/uneven, two-track crush/defect, and degraded suspension/adhesion cases run from explicitly synthetic inputs. Simple spans, continuous idealised members and finite six-direction link-slab connectors are available. The mixed search includes native planar pile-group condensation in both directions, retaining horizontal/rotation coupling. Its torsional spring is separately declared, rather than claimed as a planar solver result. Site soil, cyclic degradation and actual connection resistance remain required inputs.

```
tools/automation/osr-python tools/automation/shared-engineering-campaign.py \
  spatial --output build/engineering/spatial-campaign

tools/automation/osr-python tools/automation/shared-engineering-campaign.py \
  search --output build/engineering/shared-mixed-search --seeds 11,23

tools/automation/osr-python tools/automation/shared-engineering-campaign.py \
  joint --output build/engineering/shared-joint \
  --model build/engineering/spatial-campaign/definition.json \
  --dynamic-result build/engineering/spatial-campaign/straight-empty/dynamics.json

tools/automation/osr-python tools/automation/shared-engineering-campaign.py \
  verify --output build/engineering/spatial-campaign
```

Each campaign freezes source, solver environment, inputs, outputs and protocols. Verification rejects changed outputs, sources, configuration evidence or calibration records. Failed runs keep their partial evidence. A supplied hardware model requires its own spatial configuration; it receives no synthetic material strengths. Optional `--material-laws` supplies the actual region records.

The material adapter uses the actual candidate regions and simultaneous member demands. Supplied age/temperature curves, effective creep modulus, shrinkage/thermal eigenstrain, compressive softening and energy-regularised tensile cracking enter fiber section equilibrium. Steel has a plastic return law; composite layups retain ABD coupling and local ply-face stress checks. The separate cohesive law uses mixed-mode fracture energy and irreversible damage, retaining compression after debonding. Supplied tendon losses alter section axial force and moment. Signed member histories feed rainflow spectra and supplied S-N curves. Missing strengths, interfaces or S-N coverage stay open. Section equilibrium does not certify shear, torsion, local buckling, anchorage or history-dependent crack redistribution.

The physical joint register keeps explicit fields for bolt patterns/stacks/preload, weld procedures, bonds, bearing fits, suspension curves and articulation features. Unspecified production fields remain null. Native TechDraw now includes elevation, plan, a powered-bogie view, a section, named datums and inspection/joint references; generation checks that every projection has visible geometry. Drawing issue remains separate.

The solid submodel meshes two holed plates and a bolt/head/nut fixture with native Gmsh and solves quadratic tetrahedra, face contact, friction and pretension with native CalculiX. It first applies force pretension, then locks the retained pretension displacement for the external-load step. Loads come from one simultaneous dynamic joint vector; nodal allocation preserves force and moment. Reactions and shank stress are retained for equilibrium/refinement review. The 80×60 mm coupon, 22 mm holes, 20 mm shank and cylindrical 34 mm head/nut are synthetic verification geometry, not a released bogie connection. Actual installation solids, grip stacks, fasteners, preload, locking, fits, procedures and limits require engineering data.

The mixed search reuses the existing civil option library for geometry, materials, foundations, connections and construction. Random and Pareto evolution consume equal trial budgets and identical deterministic uncertainty envelopes. Battery mass and suspension settings also vary. It retains rejected designs, native foundation calculations, logistics, scope quantities and finer finalist dynamics. The initial dynamics are short startup screens, not complete operating or resonance envelopes. Objective values are provisional; installed costs and the qualified feasible Pareto set remain open without full rates, lifecycle scope, accepted capacities and measured uncertainty distributions.

The correlation adapter checks exact model/revision/serial, retained measurement and calibration hashes, timestamps, SI units and positive measurement uncertainty. Bounded inverse fitting uses calibration specimens; different specimens form the holdout set. Results retain sensitivity rank, local parameter covariance, holdout bias/RMSE and measurement bands.

Synthetic recovery tests cannot become physical validation. The command-line forward evaluator calls the spatial solver for supplied contact channels, with explicit stiffness/contact parameters:

```
tools/automation/osr-python tools/automation/shared-engineering-campaign.py \
  correlate --output build/engineering/prototype-correlation \
  --model path/to/definition.json --config path/to/spatial-input.json \
  --measurements path/to/measurements.json --parameters path/to/parameters.json \
  --holdout-specimens prototype-B
```

The retained synthetic correlation inputs exercise the same spatial forward solver. The planted primary stiffness factor is 0.9; the noisy calibration recovered about 0.910, with a separate specimen held out. This remains an algorithm recovery test.

The generated four-level protocols cover parts, subassemblies, the train and train/infrastructure passages. They require calibrated instruments, approved test loads/routes/speeds, stop conditions, adopted project limits, specimen identities and independent holdout/review records. No physical tests have been performed.

The OpenSourceRail internal process standard now governs numerical verification and the prepared protocols, with applicable ISO method references. Its numerical thresholds are OpenSourceRail rules. The retained assessment keeps ISO conformity and physical release open while evaluating the numerical evidence.

Complete-passage planning and motion histories

Retained complete-passage review. The empty represented car pair completes an 8.13-second passage at 15 m/s. Every recorded contact enters and clears the modelled track; separate time-step and mesh checks pass the 5% numerical screen. The other planned cases are explicitly unexecuted, and the full six-car operating envelope remains open.

The service-envelope workflow now places every represented wheel before the modelled approach track and computes the time needed for the last wheel to clear it. The case matrix includes empty/nominal/crush/uneven loads, operating speeds, acceleration, service/emergency braking, rescue, maintenance, curves/cant/vertical curves, irregularity/defects, synthetic wear, degraded suspension/ground/bearings, and simultaneous traffic with different speeds and arrival offsets. A supplied fine speed band adds resonance screening cases; generating a plan does not execute or qualify those cases. Braking that stops before clearance is a stopping test.

```
tools/automation/osr-python tools/automation/shared-service-envelope.py \
  plan --output build/engineering/service-plan --resonance-band 14:16:.25

tools/automation/osr-python tools/automation/shared-service-envelope.py \
  run --output build/engineering/service-passages --cases empty-15mps \
  --dt .005 --mesh 2 --rail-step 2

tools/automation/osr-python tools/automation/shared-service-envelope.py \
  verify --output build/engineering/service-passages
```

Coverage uses each recorded wheel contact on both sides at every timestep, its entry into the structure and clearance of the approach. Duration metadata alone cannot establish a passage. These are the linearised adapter's nominal contact interpolation stations; finite contact motion and actual operational train clearance need the appropriate geometry and validation. The demonstration still represents two cars of the six-car planning family; it does not qualify the full 111 m train. Unexecuted cases remain explicit in the campaign report.

Motion histories retain each body's global and nominal-body-axis acceleration, angular acceleration, velocity, rotation perturbation and reference frame at the rigid CG. Unweighted RMS, crest factor and fourth-power VDV are computed from the actual time series. They are not ISO frequency-weighted comfort results or seat/body-interface measurements. Body linear/angular acceleration and vertical wheel-force channels can now be selected by the inverse-calibration adapter. Invalid units, invalid solver domains and relabelled specimen serials are rejected.

Every converged sample is streamed to a deterministic compressed JSONL file. Nonconvergence and wall-budget termination preserve the converged prefix. The normal gap now includes lateral profile movement once, and sliding traction has tangents for displacement direction and normal-force dependence. Lateral/vertical ratios use the vertical force component; normal force is retained separately. Powered axles receive the equivalent positive-traction demand; braking uses all axles. Wheel-spin reference acceleration is included. Actual motor/brake control laws, torque limits and finite moving-patch geometry remain required for release.

Finalist confirmation now uses the same uneven payload and adverse conditions as its search scenarios. Time and mesh refinements are separate; a failed refinement or domain check prevents a numerical confirmation claim. Assembly/capacity and physical acceptance remain separate.

Published evidence versions

The earlier vertical/spatial reviews and standards assessment are retained unchanged as research evidence produced by commit `bcb21593fb`. Their source hashes are not rewritten to match new solver code. Verify the producing source version:

```
tools/automation/osr-python tools/automation/verify-shared-review-history.py \
  engineering/civil_exploration/examples/shared-spatial-engineering-review.json \
```

```
--revision bcb21593fb
```

This checks the publication and recorded code against the immutable Git commit. It reports whether the current implementation matches, and does not requalify the current solver or claim to verify unavailable bulk campaign outputs. New service campaign receipts bind their own current source/environment and retained traces.

Numerical scope and remaining release gates

Independent tests cover spherical Hertz contact/pressure, stick/slip passivity, crack-band/cohesive energy, concrete small-strain stiffness and creep consistency, steel yield return, laminate coupling/ply stress, bolt-group equilibrium, section resultants and independent OpenSees 3D shear/bending/axial/torsional response. Campaigns separately retain time and mesh refinements; a failed observable remains failed. The 5% screening tolerance is numerical research policy, not a project acceptance criterion.

The contact adapter rejects flange, multiple-patch and conformal contact; finite body rotations and detailed railway rolling-contact reference comparison still need appropriate adapters and benchmarks. Patch geometry checks do not establish elastic material validity or accepted wheel pressure. Structural dynamics are linear about the declared alignment, with Euler–Bernoulli consistent bending mass and Timoshenko stiffness; component flexibility, plastic redistribution, soil degradation and thermal/settlement histories require verified models and project inputs. Startup transients are explicitly identified in the outputs.

FreeCAD, Gmsh, CalculiX and OpenSees are the executed core. Chrono, CONTACT, Code_Aster/SALOME, OpenGeoSys and FMI/preCICE remain conditional choices; no unavailable tool is represented as executed or railway-validated. The shared identity and retained force/work histories define the boundary for those adapters.

Production release still requires released supplier geometry and properties, actual manufacturing/QA records, adopted project standards and limits, full operating/erection load cases, instrumented prototypes, independent holdout correlation and the existing authority acceptance process. Software cannot manufacture that evidence or release the drawings by filling synthetic slots.

Native joint semantics follow the FreeCAD Assembly documentation. Solid pretension follows the CalculiX user manual. preCICE coupling configuration describes convergence mechanisms for a future partitioned adapter; the current prototype solves the coupled equations monolithically.

Ballastless Slab Trackform Designs

Repository source: `docs/civil/slab-trackforms.md`

This note records the OSR planning-grade slab trackforms for urban at-grade and elevated sections. The parametric CAD source lives in `design/component-catalogue/src/osr_mech/civil/slab.py`.

The designs are reference envelopes for cost, clash, fabrication, and constructability review. Deployment partners still need local structural checks for concrete grade, reinforcement, geotechnical support, drainage, stray-current control, thermal movement, seismic load, flood/scour, and product acceptance of the direct-fixation fasteners.

At-Grade Urban Slab

Long open, machine-accessible corridors use a continuously slipformed slab by default. The table below defines the six-metre single-track ST6 product kept for constrained streets, utility crossings, short possessions, flood-prone and replaceable zones. Both methods retain the same controlled track support section and concrete quantity basis; a project method-zone schedule determines panel counts.

Item	Reference value
Trackform	OSR-ST6 single-track transportable panel; two rows form double track
Panel length	6.0 m
Overall panel width	2.9 m
Track-centre spacing	3.5 m
Base slab thickness	250 mm
Rail plinth width	380 mm
Rail plinth height above base slab	160 mm
Direct-fixation pitch	Maximum 650 mm; centred 600 mm on the 6 m reference module
Rail seats per 6 m panel	20
Nominal concrete per 6 m panel	5.08 m3
Planning bare panel mass	12.7 t

Use ST6 where the corridor is in a constrained street, utility or transition zone, short possession, flood-prone location, station interface, depot throat, or other place where ballast dust, tamping possessions, and loose aggregate are unwanted.

The slab uses two edge troughs for drainage, low-voltage/data routing, and maintainable cable access. The direct-fixation plinths are continuous so baseplates can recover small alignment tolerances without breaking out concrete.

Elevated Deck Slab

Item	Reference value
Trackform	Local direct-fixation plinths over thin non-structural alignment layer
Panel length	6.0 m
Overall alignment-layer width	2.7 m
Deck compatibility	Fits the 2.9 m OSR-Pi20/Pi25 flange
Alignment layer thickness	40 mm; may be omitted by project design
Rail plinth width	380 mm
Rail plinth height above base slab	160 mm
Direct-fixation pitch	Maximum 650 mm; centred 600 mm on the 6 m reference module
Rail seats per 6 m panel	20
Nominal concrete per 6 m panel	1.38 m3

Use one elevated trackform run per single-track Pi-beam. A double-track elevated section therefore uses two parallel Pi-beam/trackform assemblies.

The Pi-beam upper flange is the structural deck. The alignment layer provides local tolerance recovery only and may be omitted where surveyed casting plus adjustable baseplates can recover rail geometry. A full-width reinforced or floating slab is reserved for locations where a vibration/noise and structural study justifies its dead weight.

The cable/drainage trough is installed on the inner/non-egress side. The separate outer 1.0 m walkway/barrier cassette remains clear of service covers, cabinets, screen posts and other permanent obstructions.

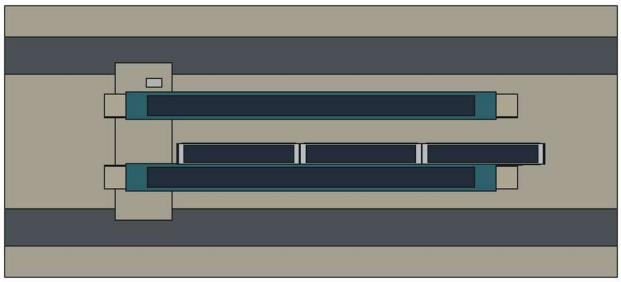
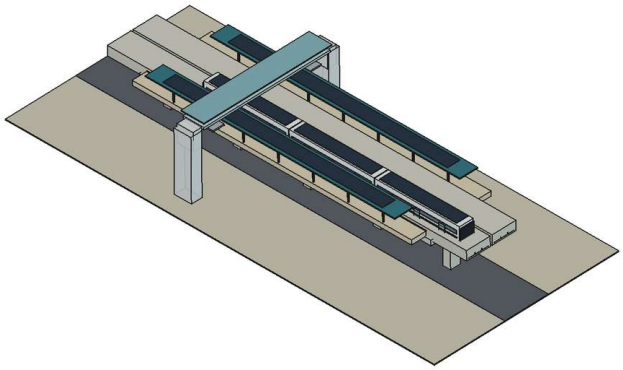
Iraqi manufacture and complete seat kits

The Baghdad detailed engineering plan adds the local slab production route, prototype inspection, lifting/bedding requirements and a complete fastener child schedule. The 6 m module has ten seats per rail centred at 300, 900, …, 5700 mm, avoiding the former pad projection beyond the panel ends. Supplier anchor edge distances still require release. Continuous-run quantities retain the existing 650 mm planning basis; precast method zones use their own panel/seat installation schedule.

Concrete volume and bare mass cover base/plinths only. The CAD service troughs are allocation envelopes; covers, drainage sections, steel, inserts, bedding and rigging need separate schedules. Make panels, cages, moulds, ducts and suitable metalwork in Iraq; source qualified fastener/anchor systems initially and qualify any localisation through system fatigue, stiffness, insulation and process tests. The catalogue does not issue a rebar schedule, mix grade, lifting approval or supplier drill pattern.

FreeCAD Scene Renders

The slab trackforms are shown below in station context with driverless rolling stock and the station access elements that affect civil cost.

At-grade ballastless station track	Elevated deck station track
	
At-grade ballastless station track with driverless train	Elevated deck station track with driverless train

The source scene document is `design/component-catalogue/models/cad/station-scenes.FCStd` and is regenerated with `design/component-catalogue/scripts/freecad_station_scenes.sh`.

Fabrication And Installation Rules

- Slipform long accessible runs over the released machine-controlled formation. Keep the 6 m ST6 module for constrained and replaceable zones; use closure pours only at turnouts, station ends, curves that require local survey correction, and transitions to bridge/special spans.
- Cast embedded inserts, earthing/bonding points, drainage scuppers, and lifting anchors in the precast yard.
- Require surveyed casting beds and a dimensional QA record for every batch.
- Use adjustable baseplates on all running rails.
- Keep prestressed elevated girders conservative; recycled aggregate and aggressive low-carbon mixes can be used first in non-prestressed slabs, piers, pile caps, and station modules.
- Treat direct-fixation pads, anchors, and baseplates as replaceable wear/interface parts, not as buried civil structure.

Cost Model Link

These slab designs feed the generated civil planning contract:

Civil class	Design-derived target	Retained benchmark
At-grade ballastless slab	2.584 M USD/route-km	3.0 M USD/route-km
Elevated guideway with local-plinth trackform	9.748 M USD/route-km	12.0 M USD/route-km
Bridge/water-crossing guideway	18.0 M USD/route-km	18.0 M USD/route-km

The active values come from `civil-cost-model.toml` and are planning targets, not tender or turnkey EPC estimates. Build the project estimate from `viaduct-quantity-cost-model.toml` plus site rates, utilities, access, risk, programme, independent checking, and contingency. Short elevated stubs, stations, soft ground, deep utilities, tight curves, or repeated

special spans carry project-specific premiums.

Station civil and access construction method

Repository source: `docs/civil/station-construction-method.md`

Apply with the civil master plan, other civil works, logistics and inspection plan. Each station has a generated individual package. Physical platforms, boarding faces and access assemblies are separate quantity bases. The current layouts identify equipment and interfaces; they do not release station structures, fire design or street entrances.

Station package and interface freeze

Freeze station ID/chainage, elevation and topology, physical platform lengths/widths, track spread/return, concourse and roof geometry, boarding faces, train/PSD clearance, shafts/stairs/escalators, accessible routes, utilities and drainage. Review actual boarding/alighting/transfer pulses, queues and equipment-outage capacity against the chosen layout. Include fire separation, evacuation and assisted rescue, cleaning/maintenance and emergency access. Confirm land rights for every entrance and pedestrian connection, rather than only the station centre point.

Coordinate structural spans/supports with the running viaduct and turnouts. Station widenings, lift shafts, concourse loads, roof/wind actions and launcher passage require a station-specific structural and temporary-stage design. Do not assume the running Pi25 deck or standard cap carries all station loads. Freeze load-transfer boundaries and dimensional datums before commissioning shop drawings.

The station lead prepares a delivery package with civil, architectural, MEP, fire/access, track, power/earthing, signalling/PSD and operations interface owners. Record sequencing constraints, tolerances and hold points. Select modular/precast platforms, stairs, shaft shells or steelwork only after installation access, lifting, connections, fire/durability and complete cost are reviewed.

Elevated station sequence

Stage	Construction method and coordination	Exit evidence
Enabling	Maintain roads, shop/pedestrian and emergency access; locate/divert utilities; construct isolated working platforms and approved crane/haul areas	Land/utility releases, traffic plan, ground/temporary-works checks
Foundations/substructure	Install station-specific foundations and verify; cast/erect columns, caps and shaft/concourse supports with temporary restraint	Actual foundation/test records, pre-pour inspections, strength and dimensional release
Primary structure	Erect beams/steelwork/platform support and concourse in checked stages; release bearings, connections and bracing	Stage-load checks, lifting records, welding/bolting/grout and survey acceptance
Platforms and edges	Install slabs/units, drainage, edge/PSD support and walkways; survey platform height/gap against track/train dynamic interface	Slab/connection records, tolerances and complete clear circulation envelope
Shafts and stairs	Build pits/shafts/landings and stair structures; fit temporary covers/barriers; keep access for equipment installation	Lift-maker interface, pit waterproofing, openings/loads, fire/access and fall-protection review
Roof and enclosure	Assemble checked roof/canopy/containment with restraints; complete flashings, drainage and weatherproofing	Structural/fixing and water tests, clear evacuation and maintenance routes
MEP and railway systems	Install ductwork, cables, earthing, plant bases, lighting, drainage, fire and passenger/PSD equipment against approved penetrations	Civil/system handover, isolation and inspection/test records
Access equipment	Supplier installs/commissions lifts and escalators after shaft/power/clearance acceptance; preserve stair access and protection during work	Certificates, functional/rescue/outage tests and independent access acceptance
Finishes and street routes	Install nonslip surfaces, wayfinding, lighting, barriers/crossings and accessible approaches; reinstate construction areas	Walk-through, drainage/lighting/finish checks and local street handback
Integrated handover	Test evacuation, assisted rescue, queues, PSD/train alignment, maintainability and outage operation with the final installed configuration	Accepted as-built station packet, resolved defects/restrictions and opening approval

Coordinate roof, platform-edge, shaft and track work so no crew works beneath another uncontrolled load or open edge. Temporary public routing cannot use an unfinished platform or uncommissioned lift. Define safe supplier installation paths for replacement motors/cabs/PSD modules; close permanent apertures only after large equipment has passed.

At-grade station sequence

Release land, crossings, utility isolation and traffic/pedestrian diversion. Construct drainage and reviewed earthworks/ground treatment; verify settlement and finished formation. Install services and foundations before slabs/platforms; provide protected draw pits and crossings. Cast or erect platforms and access structures with surveyed track/edge datum and bedding/support control.

Lay the selected slab/trackform and rail; complete platform faces, PSD support, canopy and systems. Maintain a continuous step-free street-to-platform path with drainage and crossings; isolate construction traffic. Complete lighting, signage, fencing, barriers and road/public-realm restoration. Test train/platform and access interfaces using the actual installed configuration and outage/rescue plan.

At-grade topology does not imply zero station civil cost. Soil treatment, retained levels, canopy/wind, service trenches, drainage/outfalls, safe street crossings and accessibility may govern the package. Elevated and ground-level interchange platforms need a reviewed transfer route and shared-resource capacity assessment.

Track spreading and running-viaduct interfaces

The generated running-span schedule excludes station/transition intervals. Resolve each excluded chainage with the station support plan, track-spread geometry, curves/cant, platform loads, movement joints and switch/closure interfaces. Confirm the station-end pier/cap boundary is counted once. Reconcile the approach lengths and support/beam take-off before ordering; do not fill the interval with rounded Pi25 bays.

Maintain a station-interface register covering datum, loads, tolerances, bearing/joint roles, waterproofing/drainage, egress, earthing/stray current, cable routes and erection access. The bridge and station authorities sign the structural/temporary-stage boundaries; track and systems leads sign their following-trade interfaces. Record whether the launcher crosses, parks short, or relocates around the station.

Logistics and workforce

Deliver bulky station structures and access equipment in installation order. Reserve off-road receiving space, checked crane/platform positions, shaft access and damage-free storage; keep lift/electrical components dry and secure to supplier requirements. Check dimensions and CG for stair/shaft/canopy modules independently of viaduct beam haulage. Plan returnable racks, packaging and rejection/replacement movements.

Reserve separate survey, concrete/precast, steelwork, roofing, access-equipment, MEP, track/PSD and independent inspection crews. Resource-share only when actual shift and task reservations allow it. Include work at height, rigging quantities, confined/pit access where applicable, lifting equipment authority, rescue competence, relief, heat management and handovers in the workload plan.

Hold points and handover

Stop before foundation production, primary lifting, temporary-support removal, covering reinforcement/ducts, waterproofing concealment, rail/PSD fixing and public access. Inspect platform equipment containment and continuous circulation paths after installation, including cabinets, covers and barriers. Test one-lift-out routing and throughput, assisted rescue, emergency power where selected, evacuation and smoke/fire assumptions; graph reachability alone is insufficient.

As-built deliverables include support/structure and service drawings, surveyed rail/platform geometry, installed component serials, material/connection/test records, operations/rescue and maintenance manuals, replacement access, defects, BCF/NCR closure and independently accepted station release. A civil handover for systems installation does not authorize passenger opening.

Viaduct Bearing And Movement Schedule

Repository source: `docs/civil/viaduct-bearing-and-movement-schedule.md`

Compare two twin-track schemes over the same four-span movement unit. Bearings remain replaceable in both. The reduced arrangement is an unaccepted structural alternative, not a saving established by a link slab.

Support	Tracks	Bearing lines	Bearings
Abutment/end support	2	1	4
Internal pier, structural continuity connection	2	1	4
Internal pier, simple spans with link slab	2	2	8
Expansion-unit boundary	2	2	8

Internal bearings are in one longitudinal row; adjacent unit ends retain two independent rows at an expansion boundary. The supplier schedule must assign fixed, guided, and free directions for every span end and show design reactions, translations, rotations, uplift restraint if required, replacement loads, installation temperature, and identification marks. Permanent jacking shelves and a checked load path are mandatory.

The 25 m reference places a deck gap every four spans, nominally 100 m; the permitted four/five-span family stays within 80–125 m. A structural connection must establish the shared support load path, including construction stages, prestress, reinforcement and fatigue. A link slab may remove a deck joint while each girder retains its independent end bearings and permitted rotation. For the periodic 25 m reference kilometre these alternatives require 200 and 320 bearings respectively. A finite 40-span structural layout also has 200 bearings: 30 internal supports, nine boundaries and two end supports. It has 41 supports and nine expansion gaps; the periodic index has 40 supports and ten gaps. Do not mix periodic and finite corridor quantities. The project schedule must close thermal, creep, shrinkage, seismic, braking, bearing-friction, connection fatigue and differential-settlement actions; provide covers over walkways and cable routes; and complete continuous-welded-rail/bridge interaction. Rail expansion devices are installed only where that analysis requires them.

The FHWA UHPC connection guidance describes a link slab over simple-span girders with rotating/moving bearings. It is a highway precedent, not a released rail design. Connection cost, foundation forces, maintenance access and bearing replacement must accompany both bids before whole-life selection.

Viaduct construction method

Repository source: `docs/civil/viaduct-construction-method.md`

Planning method for the civil master plan. Normal strategic span average is 25 m; use the identified Baghdad spans for orders and erection. Pi20, station approaches, closures, tight curves and special crossings stay individually resolved. Refer to the design basis, substructure kit, bearing schedule and transport envelope.

1. Release a complete workfront

The construction manager assembles a package for each continuous run: line/chainage, span and support IDs, survey datums, accepted design revision, land and access, trial holes and utility isolation/diversion, ground model, temporary works, traffic/pedestrian arrangements, environmental limits, approved lifting route and emergency access. Record station/junction/water/road boundaries and whether the launcher can pass, cross with a checked support arrangement, or dismantle/relocate. A visually continuous alignment is not proof of machine passage.

Stake support centres and offsets from redundant control; check against track alignment, utilities and adjoining structures. Establish settlement/vibration monitoring where excavations, piling or temporary loads affect property. Release piling/crane platforms with verified bearing, drainage, edges and underground-service exclusion. Install hoarding, vehicle/pedestrian segregation, lighting, welfare, spill response, washout and spoil management before production.

2. Select and construct foundations by ground zone

Select shallow foundations, bored shafts, driven bents, pile groups or ground treatment using the foundation system. The selection must consider actual axial/lateral actions, settlement, seismic/collision, groundwater, scour, nearby buildings and installation access. Record support-specific geometry and installed quantities. Do not assign one pile depth to the route or assume a generic foundation volume per kilometre.

Method	Proposed sequence	Release and retained records
Shallow footing/socket	Verify excavation/support and groundwater method; excavate to design level; inspect founding material; install blinding, reinforcement/inserts and formwork; cast, cure, survey and backfill in tested layers	Formation acceptance, level/bearing basis, reinforcement and concrete records, drainage/settlement protection
Bored shaft	Trial method and test foundation; prepare working platform; drill using design-selected casing/slurry; record strata and geometry; clean/inspect base; lower checked cage; place concrete by the released dry-hole or submerged method; inspect/test and trim for connection	Drilling and slurry records, cage position, concrete continuity/volume, integrity/load tests, accepted actual length
Driven bent/piles	Verify vibration/property constraints and platform; trial installation; lift and pitch with approved restraints; drive/splice to the accepted criteria; test and survey; trim and connect	Driving/hammer/penetration records, damage/splice checks, verification tests and final pile level/length
Pile group/cap	Install and verify group; excavate/shoring/dewater as designed; prepare pile heads; fit reinforcement/cap shell and column interface; cast/cure and restore drainage	Group effects and lateral design, pile/cap records, connection survey, excavation and backfill release

For submerged bored-shaft concreting, use the designer's tremie procedure, verified mix workability/retention and continuous placement records; no generic embedment, strength or acceptance number is issued here. FHWA's drilled-shaft manual is a method-development reference for drilling, concrete placement and inspection. Site trial, project specification and independent review determine the applicable procedure and test criteria.

Treat groundwater disposal and drilling spoil as planned waste streams. Maintain excavation support and dewatering until the designer permits removal; monitor drawdown and adjacent settlement. Failed integrity/strength/dimension results enter NCR control before a support is released.

3. Columns, caps, abutments and bearings

Compare in-situ columns with precast columns or shells using complete connections, temporary bracing, access and installed costs. Survey foundation/interface levels; prepare sockets or starter reinforcement; install the column/form with checked temporary restraint; inspect reinforcement, inserts, access and earthing; cast/grout and cure to the connection-specific strength. Maintain bracing until the designed permanent path is established and signed off.

For a precast or hollow-shell cap, verify complete mass, CG and lifting anchors against the configured crane/launcher and actual ground reactions. Survey seating and reinforcement/socket interfaces; erect, restrain, connect and complete infill/grouting under the approved stage design. Do not treat the 25–35 t catalogue cap target as a measured lifting weight. In-situ caps require independently checked formwork/falsework, pre-pour inspection and staged striking criteria.

Construct abutments, retained approaches and wing walls with released earth pressure, settlement, drainage and flood/scour design. Survey bearing seats and orientation. Install fixed/guided/free bearings, anchorages and grout to supplier/project

instructions, including temperature datum, permitted movements and replacement-jacking interfaces. A support-ready packet includes foundation verification, column/cap strength, geometry, bearing inspection, connection status and the actual next launcher/delivery reactions.

4. Produce and accept full-span beams

Use suitable existing Iraqi factories after auditing available production slots and the complete process. Freeze drawings, tendons/anchors, reinforcement, embeds, lifting/support locations and shop traveller against each beam/track/span ID. Survey mould and bed; inspect cages, ducts/inserts and calibrated stressing plant; cast the released mix with batch traceability and compaction method; retain curing temperature/maturity and test specimens.

Prestress transfer, demoulding, lifting, storage and installation are distinct strength gates. Release time comes from the qualified mix and test/maturity relationship, not elapsed hours alone. The existing 48-hour planning cycle stays a capacity assumption until trials validate it. A 24-hour target requires a new qualified cycle and equipment/thermal assessment. Sequence transfer and lifting to the supplier design; inspect cracks, camber, bearing/lifting zones and dimensions after transfer and handling.

The current Baghdad complete-member studies are approximately 69.68 t for Pi20 and 85.51 t for Pi25; with the illustrative 4 t gear, suspended studies are 73.68 and 89.51 t. These are unqualified budgeted masses from the connected beam register. Bare-section targets are a different scope. Require actual complete-member mass/CG and lifting design before selecting machinery; a nominal 100 t machine does not establish configured capacity.

Accepted members enter segregated serialized storage on designed supports with verified ground/stacking/restraint. Quarantine NCR items. Preserve strength, dimensions, lifting-anchor certificates, support positions and dispatch readiness in the traveller. Transport handling adds inspections at loading and receipt; acceptance at the factory does not clear transport damage.

5. Erect and commission the launcher

Deliver components against the logistics plan. Verify assembly space, ground bearing, traffic closure, lifting equipment and the launcher foundation/deck support arrangement. Assemble to supplier instructions; inspect structural/pin/bolted connections, electrical/hydraulic systems, brakes, limit devices and emergency arrangements. Complete required tests and documented commissioning at the actual configuration before use.

The stage design covers assembly, loaded/unloaded operation, launch/advance, turning or relocation, one-beam eccentricity, gust exposure, parked/storm state, delivery vehicle positions, breakdown and dismantling. Check every loaded pier, cap, bearing and previously erected span. Over-deck deliveries require construction load paths and protection against unauthorized additional plant. Ground-delivered erection requires a separately released access path and unloading position.

6. Repeat the bay cycle

1. Confirm daily/shift release: next span/support IDs, current beam traveller, work permit, weather/visibility within supplier limits, exclusion/traffic arrangements, healthy equipment, competent rested crew and communications. Recheck changes since the previous shift.
2. Verify seats, bearing orientation/levels, lifting gear and beam identity. Place transport in the permitted position; secure against movement. Review load, CG, lift path, rigging and contingency landing location with the team.
3. Attach only the released lifting points/gear. Apply the controlled trial-lift/check from the lift plan. Confirm balance, connections and clearances before travelling; stop for any unexpected movement or communication loss.
4. Place the first single-track beam on surveyed seats. Maintain the designed lateral/torsional restraints and required crane/launcher support. Do not release the load merely because it touches the bearings.
5. Place the second beam using the checked asymmetric sequence. Survey alignment, bearing contact and joint gaps; complete permanent/temporary restraints and necessary connection/grout strength gates before releasing the next stage.
6. Complete the bay inspection/accepted record. Advance only when the construction-load path is released, next supports are ready, delivery and next-machine positions are checked, and the worksite remains segregated.
7. Hand over a stable inspected bay to following trades; record outstanding restrictions, access/edge protection and permitted construction loads. Update stock, installed component IDs, cycle time, NCRs, weather and equipment hours.

Measure preparation, loading/arrival, each beam lift and securing, inspection, advance, handover and interruptions separately. Two lifts plus launcher advance are one double-track bay cycle. The plan does not prescribe wind limits, lifting capacity, rigging angles or concrete strengths: the approved equipment/stage design supplies them. Complex or tandem lifting needs

its own competent written plan; HSE lifting-operation guidance is a practice reference.

7. Connections and railway finish

Construct the selected continuity connection or link slab only to the released reinforcement, pour, cure, bearing and stage sequence. A link slab preserves independent girder-end support; it does not justify reducing bearing lines. Retain temporary restraints and movement controls until their removal is authorized. Inspect grout, concrete, joints, waterproofing continuity and drainage paths.

Install independent walkway/barrier cassettes, fall containment, cable/drainage troughs and earthing interfaces with their own handling and fixing designs. Keep the escape walkway clear. Survey track datum; prepare waterproofing/alignment interfaces; install plinths or the selected trackform, adjustable fasteners and rail; complete rail/structure interaction, movement and geometry checks. Install systems only under an agreed load/penetration/earthing interface; unreviewed drilling into prestressed members is prohibited by the proposed method.

8. Run boundaries, specials, recovery and demobilisation

Before each station, closure or disconnected run, decide whether to pass using a designed temporary support, bridge with an engineered launcher configuration, or dismantle and reassemble. Model closure permissions, access, transport, setup/commissioning time and resource reservations. A shorter span is not automatically launcher-compatible. Special spans require their own product and order; no Pi25 order is generated for a residual unsupported length.

Prepare recovery for hydraulic/electrical failure, suspended-load stall, damaged bearing, unseated beam, loss of support readiness and weather stop. Define a safe supported/parked condition, exclusion and communications, engineered rescue/repair, inspection and re-release. Stop deliveries when stock/access is saturated. No improvisational re-rigging, unreviewed props or moving a suspended load over the public is authorized by this plan.

After a front finishes, check the remaining critical path and candidate ring fronts. Inspect/dismantle, transport, verify new supports/access, recommission and reassess the crew before reassignment. Release a machine to another city only after the complete Baghdad programme and resource review. Close final bearing/structure surveys, as-built records, drainage and access, defects and monitoring before structural handover.

OSR Rapid Viaduct Kit — Design Basis

Repository source: `docs/civil/viaduct-design-basis.md`

Status: controlled planning basis; not a construction design
Primary products: OSR-Pi25 and OSR-Pi20 prestressed decked single-track beams Special products: OSR-U25-SPECIAL acoustic/project-specific trough, OSR-US segmental U/box, and OSR-SP special crossing

Catalogue boundary

Two identical 2.9 m-wide decked pi-beams share a single-column pier at the same 3.5 m tangent track centres used at grade. Each beam has two stems beneath the rail lines and a full-depth upper flange forming the structural track deck. Separate 1 m outer walkway/barrier cassettes give a controlled 8.5–9.0 m twin-track envelope. Pi20 and Pi25 come from one long-line mould with movable stop ends. A 30 m beam is not in the catalogue.

Product	Controlled use
`OSR-Pi25`	Tangent and broad-curve normal span; target bare mass 65–75 t
`OSR-Pi20`	Closure bays, junctions and utility avoidance; target bare mass 50–60 t
`OSR-U25-SPECIAL`	Project-specific acoustic/containment trough after transport, lift and structural release
`OSR-US`	2.5–3.0 m match-cast segments where access or curvature prevents full-span erection
`OSR-SP`	Separately engineered I-girder, steel-composite, or segmental crossing over 30 m; turnouts and exceptional crossings

The OSR-US review model places each 2.5–3.0 m chord on a surveyed circular arc, with rotated match-cast joint planes and chord-following tendon corridors; it is no longer represented by a straight full-span extrusion with markers.

The default Python geometry is `osr_mech.civil.decked_pi`; legacy troughs stay in `osr_mech.civil.ugirder` as special products. Flange/stem proportions, diaphragms, anchorages, tendons, lifting zones, reinforcement, fatigue, derailment, vibration and rail–structure interaction remain supplier design and independent-check releases.

Actions and design situations

Use the deployment country's adopted standards and combinations. The design must include the complete deployment train: the LM3 reference has 12 axles; Baghdad's six-car baseline has 24 axles. Its supplier axle positions and loaded axle distribution must be frozen before moving-load analysis. A 16 t infrastructure axle allowance does not replace the full consist loading pattern. Also include dynamic amplification, braking, traction, nosing, centrifugal force, wind on empty and loaded trains, derailment/containment, collision, maintenance and rescue vehicles, temperature, creep, shrinkage, fatigue, seismic/foundation interaction, and every lifting, transport, temporary-bracing, and erection stage. The machine-readable seed is `viaduct-load-model.toml`.

Decked pi sections require three-dimensional analysis for torsion, transverse distribution, asymmetric loading, local flange/stem actions, end-zone bursting, camber, creep, shrinkage, fatigue and staged prestress. Final reinforcement, tendons, anchorages, bearings, foundations, rail/bridge interaction, and temporary works require independent checking by licensed bridge and geotechnical engineers.

Geometry and constructability gates

- A straight 25 m full-span unit is preferred at radii of about 300 m or more.
- Below about 300 m, use Pi20, OSR-US, a shorter verified product, an I-girder deck, or realign. A 90 m system curve is exceptional elevated geometry.
- Four normal spans may form a semi-continuous unit. Four bearings at internal piers require a checked structural continuity connection and its staged load path. A link slab joining simply supported decks retains eight bearings at internal piers; eliminating a deck joint does not eliminate bearing lines. Both configurations have four bearings at end supports. Price both using the bearing schedule, including connection, CWR, temperature, braking, seismic and foundation effects.
- Parapet/containment height is measured above the finished escape walkway, not above the structural soffit or pre-track floor.
- The trackform is local direct-fixation plinths over the waterproofed structural floor. A full structural topping is a project-specific acoustic or vibration option, not the default.
- The shared hollow/precast-shell cap is 6.5–7.5 m wide and targets 25–35 t.

- Foundation selection uses `foundation-catalog.toml`. Deep-element geometry and cost require actual site length; no six-metre placeholder is permitted.
- Foundations and accepted beam stock are held 10–15 bays ahead of erection.

Automated catalogue checks live in `osr_mech.civil.viaduct`. Passing them is necessary but not sufficient for design release.

Minimum release evidence

The release package must close survey and utilities, ground model, design basis and combinations, global and local analysis, prestress/reinforcement, fatigue and durability, bearings and movement, rail/bridge interaction, drainage and egress, transport/erection, first-article testing, independent check, and deployment-authority approvals.

Viaduct First-Article Test Plan

Repository source: `docs/civil/viaduct-first-article-test-plan.md`

The first production mould, first Pi20/Pi25 unit, first selected foundation in each geotechnical zone, first cap shell, first erected bay, and first completed track/egress bay are hold points. Production may not free-run until the responsible engineer and independent checker accept the applicable evidence.

Hold point	Minimum evidence
Foundation zone	Utilities/CPTu/boreholes, selected catalogue type, actual pile/shaft count and length, installation time and cost, concrete/rebar quantity, representative load test and as-built survey
Mould and casting bed	Survey, stop-end settings, tendon/rebar/inserts, release agent, concrete process, calibration
First girder	Materials, cube/cylinder strength, maturity, prestress force/elongation, transfer sequence, dimensions, camber, cover, NDT, drainage test
Trial lift and transport	Certified portal/launcher or actual crane load chart, rigging, radius, wind and ground-pressure release, load test where required, strain/deflection observations, transporter trial and route clearance
Cap/column connection	Reinforcement and connection record, geometry, bearing-seat survey, grout/concrete acceptance, jacking interface
First erected bay	Bearing identity/orientation, role-selected one/two-row survey, reactions/level, girder geometry, gaps, stability, diaphragm/end-zone inspection
First selected connection unit	Four-span staged analysis correlation; distinguish structural continuity/shared bearings from simple spans/link slab; cage and closure record, maturity/grout release, waterproofing, fatigue detail, CWR/braking/temperature/seismic/foundation-flexibility closure
Track and egress	Waterproofing, local plinth geometry, fastener proof/torque, drainage flow, 1.0 m clear walkway, 1.4 m containment datum, gap covers, earthing
Structural acceptance	Correlated analysis; static/dynamic proof test if required by the design basis; residual deflection/cracking criteria; independent-check closure

Nonconformities require disposition, repaired-item acceptance, root-cause review, and demonstrated recurrence controls before the production hold point is released.

Record the complete double-track bay cycle: accepted foundations and caps, casting/strength release, transport, both track beams or all segments, closures, survey and tolerance corrections, waterproofing, track/egress and acceptance. Include crew/equipment hours, curing delays, rejects/rework, stock buffers and traffic possessions. Beam lifts per day alone cannot validate the programme.

Viaduct Kinematic And Egress Envelope

Repository source: `docs/civil/viaduct-kinematic-egress-envelope.md`

The controlled planning width is derived from the train and evacuation need:

Component	Width
Dynamic train envelope	2.97 m
Clear escape walkway	1.00 m
Opposite-side clearance	0.25 m
Sway, handrail, tolerance, and study reserve	0.22 m
Minimum tangent planning clear width	4.44 m
25 m straight-span chord allowance at 300 m radius	0.261 m
OSR reference clear width	4.72 m

The 4.72 m value is a planning seed, not a substitute for a deployment-specific kinematic simulation, cant/throw calculation, evacuation flow study, handrail envelope, or accessibility review. The escape ledge must remain 1.0 m clear after cable covers, drainage, screen posts, and construction tolerances.

For a straight span on a circular curve, mid-chord offset is:

```
offset = radius - sqrt(radius^2 - (span/2)^2)
```

Span	Offset at 90 m radius
15 m	0.31 m
20 m	0.56 m
25 m	0.87 m
30 m	1.26 m

The reference 25 m straight unit retains about 0.28 m after the train, walkway, opposite-side clearance, and protected 0.22 m reserve. It therefore passes the catalogue screen at 300 m radius without spending that reserve but fails at 90 m. OSR-US or another curved/special deck is then required unless the alignment is relaxed.

Measure the 1.4 m containment/parapet datum above the finished escape walkway. Cover every structural gap with a maintainable evacuation-path plate and keep cable routes continuous without blocking drainage or bearing inspection.

Viaduct building, support and terrain clearance

Repository source: `docs/civil/viaduct-obstacle-clearance.md`

Status: controlled planning screen; physical release blocked. Straight horizontal routes can overfly low buildings only with a verified vertical alignment and property/air rights. Elevation does not remove a tall obstacle or provide a place to build a pier.

`tools/automation/audit-viaduct-clearance.py --all --retain-inputs` retains source footprints near the actual full-resolution corridor and any available height tags and terrain. `--all --check` repeats the calculation offline and rejects altered sources or stale corridor evidence. Every city's `engineering/clearance/README.md` reports beam/building, support/foundation and terrain findings separately; the compressed register records actual chainages. Unknown heights, omitted footprints, invalid geometry, missing terrain, utilities and property approvals remain unresolved even if a reference screen finds no collision. These reports do not alter the accepted scenario or prices.

The reference swept envelope includes both tracks, deck/walkways and a lateral allowance. A building below a provisional soffit is a conditional overflight, not a released design. Each pier's entire foundation envelope is tested separately. A terrain crest is tested against the straight beam between its supports; provisional support elevations are checked for excessive railway grade. SRTM-derived tiles can contain surface/roof/canopy returns and must not be represented as a precise bare-earth survey.

The current screen uses these explicit planning assumptions:

Item	Assumption
Twin-track structure/walkway envelope	9.0 m
Extra lateral building allowance	2.0 m each side
Foundation reference radius / setback	3.0 m / 2.0 m
Rail height above support ground	12.0 m
Conservative rail-to-soffit depth allowance	1.6 m
Minimum provisional roof-to-soffit allowance	2.0 m
Pi25 / Pi20 reference span	25 m / 20 m
Reference maximum gradient	3.5%

The Pi20/Pi25 bare structural section is 1.155 m deep in `design/component-catalogue/src/osr_mech/civil/decked_pi.py`. The 1.6 m rail-to-soffit screening allowance leaves 0.445 m for the unverified track/rail stack. Special U/box/crossing products need their own supplier geometry; this Pi reference cannot establish their clearance.

These values require project approval and supplier/site design. They are not deployment-country statutory clearances. Building storeys times 3.5 m provide an explicitly uncertain estimate only; they do not replace a height survey.

To resolve a conflict, reroute or secure an approved property solution; raise the railway with grade-compliant approaches and revised structure costs; move support lines using verified Pi20/Pi25 spans; change independently checked foundation/cap arrangements; or release a project-specific crossing. The catalogue contains no ordinary 30 m full-span beam. Boundary and special span layouts remain unresolved until independently designed.

Release requires a surveyed building/roof/plant and ground model, longitudinal and cross-section profiles, utilities and protected zones, property/air rights, fire and rescue access, flood levels, vertical curves, railway/bridge movement, pile-group and cap layout, temporary works and each transport/erection stage. Any adopted geometry change requires regeneration of stations, quantities, costs, schedule, financial cash flows and native service evidence.

OSM height tags and Overture building/part height fields can supplement footprints, but neither source establishes mapping completeness. See the OSM height definition, Overture building documentation and Terrain Tiles source.

Viaduct pier and abutment kit

Repository source: `docs/civil/viaduct-substructure-kit.md`

The controlled planning geometry is generated by `osr_mech.civil.substructure`. A standard elevated cross-section uses two single-track decked Pi-beams at 3.5 m centres on one shared hollow/precast-shell cap. These are interface and quantity envelopes, not stamped foundations.

Pier EBOM

ID	Product	Quantity basis	Release gate
`CIV-PIER-P010`	Soil/access-selected foundation catalogue interface	1 set	Actual length/cost, site geotechnical and foundation test release
`CIV-PIER-P020`	1.5 × 2.0 m RC pier column	Selected height, 5–12 vertical m	Reinforcement, seismic, and collision calculation
`CIV-PIER-P030`	7 m hollow/precast-shell shared cap	1	25–35 t target, lifting, reinforcement, connection and cap calculation
`CIV-PIER-P040`	Fixed/guided/free bearing	4 at internal support; 8 at expansion-unit boundary	Supplier and bearing/movement schedule freeze
`CIV-PIER-P045`	Bearing-replacement jacking interface	4	Jacking and maintenance-load calculation
`CIV-PIER-P050`	Drainage, earthing, access, and identification	1 kit	Site services and inspection release

Assembly sequence: release survey/geotechnics and temporary works; construct the foundation; cast or erect the column and cap; install and survey the role-selected bearing line(s); lift the paired Pi-beams; then complete the selected connection after the erection hold point closes. Structural continuity must establish the shared-bearing load path; a simple-span link slab retains two girder-end bearing lines. Catalogue pier geometry and BOMs support both schemes.

Abutment EBOM

ID	Product	Quantity	Release gate
`CIV-ABT-P010`	Foundation interface	1 set	Site geotechnical and foundation release
`CIV-ABT-P020`	Bearing shelf and backwall	1	Structural and drainage calculation
`CIV-ABT-P030`	RC wing wall	2	Earth-pressure/retaining-wall release
`CIV-ABT-P040`	Elastomeric/PTFE bearing	4	Supplier and bearing schedule freeze
`CIV-ABT-P050`	Replaceable expansion joint	1	Thermal movement and supplier freeze
`CIV-ABT-P060`	Approach slab, drainage, earthing, and access	1 kit	Settlement, drainage, and inspection release

Assembly sequence: release earthworks, geotechnics, drainage and thermal movement; construct foundation, shelf, backwall, wing walls, and approach; install bearings and expansion joint; then survey seats, clearances, settlement datum, drainage, and access before girder erection.

Deployment-specific pile type/count/actual length and cost, scour protection, utility diversions, reinforcement, seismic/collision design, soil pressure, and erection temporary works remain outside the upstream catalogue.

Viaduct Transport And Erection Envelope

Repository source: `docs/civil/viaduct-transport-and-erection-envelope.md`

The bare OSR-Pi25 envelope is 74,937.5 kg, leaving only 62.5 kg below the 75 t product target before net details. Pi20 is 59,950 kg. Complete-member and hook masses remain unresolved. Use the generated mass and equipment summary and its controlled budget before selecting transport or erection plant. The 3.0 m shipping-width and 75 t member targets require product compliance or an independently reviewed deviation bound to the actual mass budget.

Compare the Baghdad complete-package alternatives before selecting plant. Pi20's bare mass is also only 50 kg below its 60 t product target; smaller is not automatically compliant. Price the extra supports, foundations and complete-bay cycles alongside lighter handling. Overhead delivery over completed spans adds construction loads and launch reactions requiring their own deck, pier and temporary-works analysis.

A portal, launching frame or certified strand-jack arrangement is an erection option awaiting selection and qualification. Its required capacity comes from complete member, rigging and dynamic demand at the reviewed configuration; an approximate 80 t label cannot establish suitability. The number and weight of lifts come from the actual asset layout.

Before release, record the casting-yard route, swept path, bridge and culvert capacities, overhead/side clearances, pavement and temporary works, transporter axle configuration, escort/closure plan, launcher reactions, strand-jack synchronization, wind limits, lifting points, temporary bracing, bearing survey, contingency landing position and rescue/recovery method.

For isolated packages a mobile or crawler crane may be used only against its actual configured load chart. The release records load, rigging, lift radius, wind limit, outrigger/crawler reactions and verified ground-bearing pressure; nominal crane tonnage is not accepted as proof.

Where the route or worksite cannot accept Pi20/Pi25, use OSR-US match-cast 2.5–3.0 m segments with an overhead launcher only after verifying every transport constraint. The current OSR-US coordination section exceeds the primary 3 m shipping-width target: shorter segments do not resolve restricted width. A supplier redesign or reviewed oversize route is required. Its release adds segment geometry control, match-cast records, epoxy joints, shear keys, post-tensioning, grouting, launcher stages and specialist QA. The legacy OSR-U25-SPECIAL trough is retained only where project acoustic/containment needs justify its wider and heavier transport envelope.

LM3 Wayside Rerailing Access Interface

Repository source: `docs/civil/wayside-rerailing-access-interface.md`

Status: optional design-reference envelope for selected recovery nodes. It is not a universal requirement to pave the whole alignment and is not a released foundation or lift design.

Purpose

Vehicle portability does not by itself make a derailment easy to recover. The hydraulic cylinders and transverse bridge need level bearing, road-accessible offload space, an unobstructed route to the vehicle, and an exclusion zone. `civil-wayside-rerailing-access-interface` reserves those interfaces at strategic locations such as depot approaches, terminal throats, junctions and long or operationally difficult line sections.

Controlled Envelope

Interface	Design-reference value
Rerailing hardstanding	18,000 x 4,000 mm
Equipment offload/staging pad	6,000 x 4,000 mm
Temporary exclusion zone	22,000 x 12,000 mm
Transverse bridge length	4,000 mm
Bridge-bearing zones	X = +/-5,600 mm at the LM3 transverse jack pairs
Rail gauge	1,435 mm

The CAD source shows one-sided road access. Deployment designs may mirror it, but must preserve the cross-track equipment path and verify that the response vehicle can stop and unload without fouling another open railway or public traffic lane.

Site Selection

Recovery nodes should be selected from an operational response-time study, not at arbitrary uniform spacing. Prefer locations with all-weather road access, low-gradient tangent track, drainage, lighting/power opportunities, communications coverage and enough space to establish railway and public exclusion zones. Avoid locating the only planned node where a bridge parapet, tunnel wall, platform, live conductor, utility corridor, steep cutting or soft embankment prevents lateral translation or equipment deployment.

Civil Release Requirements

1. Survey track, cess, access-road, drainage, buried services, overhead/live electrical equipment and adjacent property constraints.
2. Establish allowable ground bearing and settlement for the released jack, bridge, cribbing and response-vehicle reactions, including saturated and post-incident ground conditions.
3. Design the slab, reinforced formation or deployable spreader solution from those reactions; the catalogue envelope contains no assumed soil capacity.
4. Preserve two 1.6 m wide cross-track handling routes at the LM3 transverse jack pairs and keep covers, cabinets and drainage hardware removable or out of those routes.
5. Define traffic management, railway protection, traction-power isolation, lighting, communications and emergency access arrangements.
6. Trial the production equipment and representative vehicle at one first article node, then survey and record the as-built bearing/staging datums.

The paired vehicle/equipment load cases and prohibited configurations are in the LM3 portable field-rerailing concept.

Codebase review and regenerated Iraqi examples — 3 October 2026

Repository source: `docs/codebase-and-iraq-review-2026-10-03.md`

> Historical review: numeric quantities and funding figures below describe the reviewed revision. Use the current city README, proposal and source-bound catalogue review for the regenerated planning basis.

Historical review — current-state Iraqi city figures superseded. The Samawah and Mosul rows also precede the catalogue alignment, fleet and depot regeneration. Use their current city summaries for the active quantities. The 347-month rollout, earlier factory gate, capital/funding figures and phased openings below describe the reviewed historical revision. Use the current delivery review, current Baghdad programme and proposal for today's source-bound figures. This document remains an audit record; its historical values are not the current estimate.

Original reviewed baseline: `4951645a287b0de0d9071b473ef68c417ed9f4b5`. This reconciliation follows committed baseline `1603d0511`, retaining the earlier review findings and refreshing affected controls and financial evidence for the default `main` branch. This review combines repository-wide automated checks with inspection of the architecture, assurance boundaries, regeneration pipeline, cost models, project controls and documentation. It does not claim manual inspection of every source line or physical acceptance of a railway.

The three requested cities are complete planning examples against the current controlled layouts and software. They contain current engineering, operations, financial and simulation evidence. Construction and operational release remain blocked by their recorded physical gates. The Iraqi funding structure is proposed and uncommitted.

Findings and corrections

Finding	Correction and evidence
A routine rebuild could replace an existing layout using unrelated local corridor caches. In the review trial this removed Mosul's ring and relocated Samawah's interchange.	Retain controlled layouts by default. `--resynthesise-design`, `--resynthesise-corridors` or `--from-scratch` explicitly request replacement. Regenerated examples retain the original station identities and controlled workspace inputs. A regression verifies preservation of a ring despite an unrelated cache.
Resynthesis omitted Samawah's explicit hot-axle detection policy.	Retain its controlled HABD setting through `design-overrides.toml`; validate override scope and values before emitting scenarios. Positive HABD passage tests remain in place.
Cost recalculation did not refresh station, depot and fleet unit prices, and could rewrite historical execution hashes and workspace locks.	Refresh every affected cost bucket and EPC using the current catalogue. Record a cost migration receipt; never reattest old runs, approvals or source locks. A regression changes prices while preserving an execution record. The portfolio's existing fleet prices were already current: the audit changed zero designs.
Project controls called 30-working-day buckets calendar months.	Convert working-day milestones using the explicit 260-working-day annual assumption and 30 working days before NTP. Regenerate all 266 project twins, construction cash requirements and operations manifests. The approved Iraqi holiday/calendar baseline remains open.
Generic five-year, uniform debt financing obscured actual Iraqi draw dates, currencies and public support.	Add schedule-linked Chinese component credit, government capital, sovereign IQD bonds and IQD bank credit. Separate interest, principal, fees, reserves, subsidies and cash gaps. Retain generic calculations only as labelled comparators.
Finance and twin records could appear current despite stale additional financial inputs.	Validate all declared input hashes, including the Iraq terms, financing implementation and retained schedule projection. The portfolio twin audit checks current budgets, bucket reconciliation and source bindings.
City EPC overhead was assigned wholly to the initial five-day baseline-freeze task.	Spread the unchanged allowance across direct-work budgets and milestones. Refresh all 266 city controls, delivery/deployment records, operations manifests and documentation.
Gross cash injections and future retained cash could be mistaken for the net funding deficit; extra borrowing had no priced repayment model.	Reconstruct original principal, interest and fees independently; disclose gross, retained and net cash separately. Price supplemental IQD credit, cap outstanding balances, sweep surplus to repayment and expose remaining debt. Publish six-month native-currency bond/loan envelopes and clearly uncommitted green/grant/rights sensitivities.
Large ignored procurement CSVs left funding schedules unavailable in a fresh checkout.	Track a compact `funding-input.csv` projection of actual budget, timing and procurement-origin fields. Finance validates it against current CAPEX before using it.
Package completion conflated a complete planning example with accepted infrastructure.	Add explicit `planning_example_complete` and `operational_release=false`. Depot and stabling placement/cost/physical failures remain visible; selected simulation failures cannot be excused. Default package status remains `incomplete` while physical gates are open.
Native resilience reruns discarded completed executions and processed independent cases in rigid batches.	Reuse actual runs only when simulator binary, scenario, duration, options and output hashes match; reject damaged cache entries. Execute independent cases through a shared worker pool. No simulation physics or safety acceptance criterion was weakened.
The subsystem control register omitted the current dual-channel source.	Recompile its generated register against the actual source set; retain qualification boundaries.
RFC 0001's current amendment disagreed with its older body about voter sizing, snapshots, witnesses and service availability. Its witness intersection also ambiguously took a maximum speed restriction.	Align the body with the three-voter reference and unimplemented logical snapshots/witnesses; separate original timing/cost targets from measured behavior. Preserve occupied resources after permission expiry, identify required-channel loss as a stop, and specify the minimum permitted maximum speed on compatible routes. Remove unsupported blanket vendor-cost and availability comparisons.
Station rebuilds could omit installed EnergyPlus/FDS tools because their directory was absent from the inherited search path.	Export the configured engineering toolchain path and execute both tools again. Baseline failing thermal/fire cases remain recorded alongside the proposed mitigations.
City documents and the Baghdad offer contained stale or fixed quantities.	Generate current quantities, schedule, costs, funding and screenshots from controlled records; rebuild the evidence-linked Baghdad PDF. Update Iraq/national briefs, portfolio summaries, catalogue index and documentation inventory. Repository health now checks the structured Iraqi disclosure rather than requiring obsolete generic financing rows.
Browser acceptance omitted the new civil register in its drainage/ground and structural role counts. Its temporary workspaces also appeared as missing public documentation during concurrent checks.	Require ten drainage/ground and eleven structural roles, explicitly verifying the missing civil register remains a gate. Ignore disposable hidden browser workspaces and align book-manifest tests with the reader's existing hidden-directory exclusion.
The reader-book rebuild failed on qualification table rows taller than one page. Its city briefs also truncated networks to eight lines and presented design-base cost without the separately sized solar provision.	Enable row pagination, respect PDF frame padding, show every line and use current design-bound reconciled city finance totals. Add a regression that builds a PDF with a multi-page table row and check all 265 brief totals/line counts. Include the three Iraqi funding appraisals and consolidated programme in the book.

Superseded station-only experiments remain diagnostic records, with their original source bindings and failed results. Their hashes were not rewritten to imply acceptance of the selected station/depot plan. Other cities' historical solver evidence is retained and explicitly marked unverified where stale; this task executes the full city refresh for Baghdad, Samawah and Mosul.

Current city examples

City	Lines	Stations	Trainsets	City CAPEX, USD m	Capital-cash horizon, calendar months	Minimum operating DSCR before public support
Baghdad	9	182	831	7,555.74	347	0.05
Samawah	3	21	108	415.34	48	1.20
Mosul	6	69	256	1,926.74	104	0.18

These totals include timetable-sized solar provision and exclude the separately budgeted Baghdad plant. Geometry is a planning layout, not a surveyed route. All three nominal simulations and eight mandatory resilience scenarios pass.

Samawah's selected 48-hour hybrid stabling replay passes both overnight allocation and morning-launch checks: 40 trainsets at stations and 68 at depots. This does not resolve surveyed depot access, land, detailed yard geometry, costs or physical acceptance.

The recalculated schedule uses earliest-feasible ready work rather than whole-fleet stage batching, within unchanged finite resource capacities. The proposed Baghdad plant gates train production for 520 working days; civil works proceed in parallel. Full-network capital cash now spans 347 months. The separate phased sensitivity starts line revenue in month 66 and reaches all lines in month 346. These heuristic planning dates require an approved resource calendar, commissioning and actual acceptance; they are not a short construction promise. The funding programme is now Baghdad-only; Samawah and Mosul remain separate engineering examples.

Baghdad-only sources, uses and cashflows

The Baghdad-only funding programme includes Baghdad and one manufacturing plant, counted once at USD 299.16 million plus USD 20.94 million EPC. Total capital uses are USD 7,875.84 million. Samawah, Mosul and every other city are excluded.

Proposed capital source	USD million	Share of total capital
Chinese USD export buyer credit	896.07	11.38%
Government capital	1,968.96	25.00%
Domestic IQD bonds	3,758.11	47.72%
IQD bank term credit	1,252.70	15.91%
Total	7,875.84	100%

The editable Iraq assumptions apply a Baghdad override: government covers 25% of total capital, not 25% of capital after Chinese proceeds. The imported-parts budget of USD 1,792.15 million is now split 50% government USD cash / 50% proposed Chinese USD credit. This includes the original solar, bogie, battery, window, door and tooling categories plus other imports pending supplier-origin and lender qualification. Full-basket eligibility is an unconfirmed scenario assumption. The USD 896.07 million government import cash is inside its 25% contribution; the remaining USD 1,072.89 million equivalent government capital is IQD 1.395 trillion. The residual after government and Chinese proceeds is split 75% bonds and 25% bank credit. Other Iraqi city standalone appraisals retain their prior assumptions.

Proposed terms remain USD Chinese credit at 5% with 48 months' grace and 180 months' repayment; IQD bonds at 8% with 24 months' grace and 180 months' repayment; and IQD bank credit at 9% with immediate amortisation over 84 months. Each draw starts its own grace/maturity clock. Rates, tenors, allocations and guarantees are unconfirmed appraisal assumptions. Construction cohorts require separate approved facilities; the long programme is not a promised lender draw window.

If 25% of capital is also the cap on all government cash, the recalculated full-network-only case leaves USD 9,759.64 million of additional nominal liquidity requirements. The conditional phased-opening case reduces this to USD 8,255.74 million. Capital remains USD 7,875.84 million; these are separate cash requirements, including debt principal repayments during construction, interest, fees, reserves and operating deficits after modelled receipts. They are not extra construction CAPEX or approved contributions.

The previous USD 9,890.27 million result imposed a whole-fleet stage barrier, withheld all fares until the last capital milestone and funded reserve deposits with public cash even when project cash was available. Ready-work dispatch now pipelines production, includes the new plant availability constraint, and uses available project cash and eligible reserve releases before requesting external support. The full-network-only case is retained as a conservative comparator.

Phased revenue and variable OPEX follow each line's share of controlled trainsets, with a separate 50% / 75% / 100% revenue ramp. From first opening, 25% of full-network OPEX is fixed and 75% scales with opened fleet. These editable, uncalibrated proxies need surveyed demand and an accepted staffing/operating plan. Depot and shared system work gate every line; planned completion does not establish safety acceptance.

Peak annual capped government capital is USD 302.88 million; peak annual additional requirement in the full-network-only case is USD 605.69 million. Conditional lifetime public cash is USD 11,728.60 million in that case, or USD 10,224.70 million with phased opening. These amounts exceed the agreed direct capital contribution and are not approved spending. The phased case's annual additional-liquidity peak is USD 777.35 million in year 6, higher than the full-network-only peak because reserve funding and fixed OPEX begin earlier. A lower lifetime injection total does not imply a lower annual financing requirement.

The phased city ledger retains USD 8,188.24 million at the end of its horizon. Gross required cash injections are therefore not net lifetime loss. That later nominal surplus is neither returned to the sponsor nor available to meet earlier obligations without an approved bridge; the separate priced liquidity sensitivities below sweep later cash to repay their facility and expose terminal unpaid debt. Factory manufacturing income and OPEX remain outside the capital-financing appraisal. City project NPV is USD -4,573.02 million at an 8% discount rate under the phased low-demand assumptions; retained distant surplus does not establish present-value viability.

Phased monthly programme cashflow and phased annual cashflow retain USD government/loan flows and native IQD local capital, debt service, revenue, OPEX and additional requirements. The programme JSON also retains 40% lower demand (USD 9,428.40 million additional funding) and two-year delayed openings (USD 8,538.11 million).

The monthly programme cashflow and annual cashflow distinguish the capped contribution and unfunded additional requirements. The Baghdad city appraisal contains component allocations, native-currency debt, charts and downside cases. Samawah and Mosul appraisals remain available as standalone examples outside this programme.

Demand reduction, CAPEX increases, IQD depreciation, interest increases, delayed commissioning, declined Chinese credit, delayed public payments, bullet-bond redemption and combined downside remain modelled. Capital reconciliation and conditional debt repayment establish calculation consistency, not committed funding or affordability. The 347-month capital horizon and conditional phased results make commissioning, production capacity and phase-specific demand material appraisal decisions.

Independent reconciliation, additional finance and six-month tranches

The independent reconstruction rebuilds each original principal payment, interest charge and fee from native draws. It fixes an additional timing error: city EPC overhead was all paid against the initial five-day baseline-freeze task. EPC now follows the budgets and milestones of direct works, preserving total capital uses. Every city's project controls are refreshed against that correction.

For pooled Baghdad/plant cash, lifetime operating receipts of USD 18,252.764 million face OPEX of USD 8,044.868 million, original capital principal repayments of USD 5,906.884 million, interest of USD 4,325.820 million and fees of USD 42.220 million. Their difference is USD 67.027 million net nominal deficit before gap-finance costs. Capital sources already pay for construction; repayment is not extra construction CAPEX.

USD 8,255.266 million gross early cash needs minus USD 8,188.239 million retained later cash equals that net deficit. The component-based phased total above retains city/plant ledgers separately, while this independent result pools cash; the USD 0.473 million difference is factory cash reused by the city. Such pooling requires creditor consent and permitted transfers. Retained future cash cannot finance earlier debt service without a priced facility.

Tickets, kiosk/shop leases and advertising are already included. At full-network steady operation they contribute respectively IQD 547.335, 12.984 and 20.175 billion annually. Retail assumes 88% occupancy, advertising 85%; rates and paid trips are uncalibrated planning proxies. Phased receipts follow each line's opening and revenue ramp. New commercial proceeds must be additional and net of costs, rather than counting these rents a second time.

The reference six-month placement schedule and blended schedule state native government cash, Chinese USD draw requirements, IQD bond face/unit counts, IQD bank credit, repayment windows, reserves and required supplemental funding. Each half-year is an envelope settling monthly against expenditure at par; an upfront six-month sale requires different carry/interest modelling. No demand, mandate or auction success is asserted.

Green bonds replace qualified conventional debt; the green label alone changes neither sources nor pricing. The separate candidate tests IQD green debt at 4% plus fees, a USD 25m-equivalent climate capital grant reducing domestic debt, USD 300m-equivalent net development-rights proceeds and USD 25m-equivalent yearly new net local receipts. All are uncommitted targets. GCF/accredited-entity, CBI renewable-finance, guarantees, sukuk, rights, sponsorship and telecom routes are documented with primary sources; no bank product or donor allocation is presumed available for this railway. Foreign concessional money requires confirmed IQD on-lending or a priced hedge to preserve the currency strategy.

The supplemental facility pays its own interest and origination costs and sweeps surplus to repay itself. Its illustrative maximum outstanding balance is IQD 13 trillion, not market capacity. Commercial 8% gap credit still needs IQD 39.260 trillion of uncovered external cash and leaves IQD 13.000 trillion unpaid. The candidate 2% blended case stays inside the cap but leaves IQD 9.400 trillion unpaid. With constant nominal fares and OPEX and its other assumptions, eliminating both uncovered cash and terminal debt requires approximately USD 134.06 million/year genuinely new NET receipts, scaled with opened fleet. That is a funding requirement, not a verified income forecast or proposed fare increase. It does not establish discounted project viability.

Variable ticketing and operating-cost inflation

The requested additional sensitivity raises both fares and OPEX 5% annually, in separate explicit indices from financial close. First service still starts in month 66; no fare receipts are credited before opening. With income growing 5% too, real fare relative to income stays unchanged, so the illustrative -0.30 price elasticity does not reduce trips. Existing rents remain nominally flat unless a separate lease-indexation case is selected; the additional grant/rights/new-net-income targets retain their prior uncommitted amounts.

The paired 5% blended case peaks at IQD 10.774 trillion supplemental debt, incurs IQD 10.962 trillion of gross facility draws, and ends with zero uncovered cash and zero unpaid supplemental debt. It still needs an available early IQD facility, the

conditional climate grant/development receipts and assumed nominal financing terms. Annual price growth changes repayment capacity, not whether a lender has committed. The constant-price USD 134.06m/year net-income threshold above belongs to a different sensitivity and is not also required in this indexed case.

Average nominal tickets are IQD 1,681 at first opening and IQD 5,164 at full opening in month 346. Forty-four trips stay 11.7% of the indexed income proxy with 5% income growth. If income grows only 2%, that full-opening burden rises to 26.4%, reducing paid demand in the model. These are conditional affordability calculations, not wage or inflation forecasts.

Variable peak/off-peak pricing assumes 40% of baseline trips at a 25% premium and the rest at a 10% discount; each tier receives its own price response before revenue is combined. With the same 5% indices, peak supplemental debt becomes IQD 10.613 trillion and terminal debt clears. Household concessions, transfer integration, commuter caps, demand switching and collection costs need a separately funded tariff design. Discount-induced trips cannot exceed the unchanged practical capacity.

If OPEX instead rises 7% while fares and income rise 5%, the case again leaves IQD 11.768 trillion uncovered cash and IQD 13.000 trillion terminal debt. Inflation cannot simply be applied to revenues alone. At 5% general inflation, the 8% real discount assumption becomes 13.4% nominal; paired-case unlevered NPV is USD -4.121 billion equivalent, excluding new grant/rights/net-income targets. Repaying the conditional loan does not make discounted project value positive. Capital escalation, lifecycle cost escalation, floating rates and future FX still need full appraisal.

Paired 5% six-month financing, monthly prices and affordability, and variable-ticket financing preserve the native cashflow and price assumptions.

Surplus-funded early loan and bond repayment

This follows baseline `f79ab4c` without changing Baghdad CAPEX, the 25% government contribution, or the imported 50:50 USD cash/Chinese-credit split. Six new comparisons use identical paired 5% fare/OPEX/income assumptions and the existing conditional blended finance. They first pay OPEX and scheduled debt service, fund debt-service reserves and retain three months of current OPEX; then surplus can retire principal early. No supplemental draw or uncovered cash funds discretionary principal in the same month.

Repayment strategy	All debt cleared, month after close	Nominal financing saving versus buffered gap-only, USD equivalent m	Assumed early-payment premiums, USD equivalent m
Buffered gap-only	634	0.000	0.000
Gap first, then core facilities	416	73.287	2.882
Loans before bonds	415	189.758	6.791
Interest-rate priority	414	293.309	11.311
Interest-rate priority, zero premiums	414	308.763	0.000
Interest-rate priority, bonds noncallable	634	129.259	4.520

Interest-rate priority pays 9% bank credit, 8% ordinary bonds, 5% Chinese credit, green bonds at 4% plus a 0.5% annual guarantee charge, and finally 2% gap credit. It saves USD 293.309m equivalent after premiums and clears debt 220 months earlier than the comparable buffered gap-only case. These are nominal financing costs, not discounted gains or principal savings. The original no-buffer case remains separate; its metrics are unchanged. All six new cases retain IQD 0.994tn operating buffer at the horizon, separately from unrestricted cash. Cost-priority peak gap debt remains IQD 10.868tn; early retirement does not eliminate construction-period financing.

With retained contractual instalments, the cost-priority bank/Chinese/ordinary/green facilities make their final principal payments in months 352 / 358 / 370 / 370. The gap facility makes its final payment in month 414. Buffered gap-only retains original core final payment months 430 / 574 / 550 / 634. Current-month reserve targets reflect mandatory debt service before voluntary principal; subsequent months release reserves as that service falls. Factory reserve service uses its actual tagged debt vintages rather than the unprepaid original schedule.

The contracts are uncommitted. Assumed premiums are 1% bank/Chinese/green and 2% ordinary bonds, with minimum draw ages of 6, 12 and 24 months respectively. Eligible vintages are repaid oldest first. Notice, compensation, investor consent, call rights and market buyback prices require actual terms. A noncallable comparison forbids voluntary bond repayment; cost ordering is a heuristic rather than a globally optimal treasury solution. The detailed report links primary examples of prepayment and bond buyback mechanisms without asserting Iraqi issuer powers or Chinese contractual rights.

The monthly repayment ledger, six-month tranches and all six complete calculations expose scheduled principal, voluntary principal, premiums, interest and restricted/unrestricted cash separately. Six-month outstanding balances now use the final month's balance rather than adding six balances. Tranche contractual maturity windows are explicitly distinguished from actual early principal dates. Native debt balances independently reconcile draws minus scheduled and voluntary repayments; interest is checked against prior closing debt plus half-month new draws. The terminal cash improvement equals net financing savings when buffers and all other flows are held identical.

Repayment allocation does not change unlevered project NPV. The paired-case negative NPV, uncalibrated demand/income, capital escalation, future FX, lifecycle provision and required uncommitted facilities remain appraisal limits.

Currency, historical proposal and local benefits

Only the proposed Chinese loan is USD debt, but half of imports also require direct government USD cash. Combined USD capital funding is USD 1.792 billion, or 22.75% of uses. The other 77.25% uses IQD government cash, IQD bonds and IQD bank credit. The 25% direct budget contribution is not a cap on sovereign bond liabilities, possible guarantees or lifetime public cash needs. Programme monthly and annual files retain both USD loan/government flows and native IQD flows. USD equivalent reporting is distinct from funding currency.

The updated programme compares this 516.5 km / 182-station planning layout against the July 2024 reported proposal: 148 km, seven lines, 64 stations, USD 18 billion. The NIC notice requires a bidder funding plan; entirely USD foreign loans and government cash are the requested comparison scenario, not established final terms. Under that scenario, combined USD capital funding is about 90.0% lower. Different scopes, price dates, qualification maturity and schedules prevent this planning comparison from establishing a like-for-like bid saving.

Modelled average fare is IQD 1,317 per paid trip. At fixed low-case demand, OPEX-only neutrality requires IQD 485; the first complete operating year including debt and fees requires about IQD 1,482 in the full-network-only case, excluding factory debt and reserve deposits. These are cash thresholds, not recommended tariffs. Thirty paid trips use 8% of the retained income proxy and 44 commuter trips 11.7%. The 46.4% anchor score implies a 4.54 million-resident proxy, not measured walk-access coverage or a census result. The older reported 80% city-coverage ambition has no comparable denominator.

Potential local procurement is USD 6.084 billion equivalent, or 77.25% of capital, under unverified procurement-origin allocations. Iraqi body fabrication, fit-out, wiring, civil/precast production, erection, installation and maintenance can retain skills, wages, supplier income and repair capability. These budgets are not payroll, GDP added, or verified local content. The model carries 2,350 indicative operating FTE and IQD 19.50 billion annual labour cost; construction and manufacturing jobs need validated hours, wages and productivity. The third-party plan's Iraqi labour share is unknown and cannot be assumed zero.

In the full-network-only comparator, additional cash divides into USD 9,281.11 million before full-network revenue, USD 59.92 million during operations/debt tail, and USD 418.61 million for plant financing. Phasing brings receipts forward but does not close the cash gap. Currency matching alone does not establish financial sustainability.

Primary sources and remaining decisions

The IMF 2025 Article IV release supplies the historical 2024 average of 1,300 IQD/USD and fiscal context. The model uses that historical anchor, not a verified October 2026 dealing rate. Existing income/demand inputs are explicitly labelled planning proxies rather than current surveyed Iraqi statistics.

The CBI fiscal-agent description and Injaz issuance example support a proposed Ministry of Finance sovereign issuance route. This review does not establish municipal borrowing powers or market acceptance of the proposed long amortising bonds. China Exim's export buyer-credit product supports the instrument concept; it does not verify this model's advance, interest rate or tenor.

Financial close needs a controlled sponsor and borrowing authority, signed term sheets and guarantees/insurance where required, appropriation and subsidy agreements, bond placement/redemption arrangements, supplier-origin evidence, quotations, FX access, tax/duty/land/utility pricing, surveyed demand and an accepted phased resource calendar. Base costs and revenues are nominal without escalation; imported maintenance exposure and factory operating budgets need separate appraisal. The factory ledger isolates capital financing and does not represent a complete factory business case.

Existing platform boundaries remain: process-level paired control is a software reference; independent physical outputs/watchdogs/power, calibrated train and infrastructure interfaces, formal refinement, HIL and independent assessment remain open. The city simulator is not an accepted export of a complete city into the bounded two-train TACS reference. Passing software tests does not close those gaps.

Validation record

Check	Result
Rust workspace, all targets/features, and documentation tests	974 passed, zero failures, one ignored; Clippy passes with warnings denied.
Original review: Python suites across city generation, component catalogue, engineering and automation	Previously recorded: 1,499 passed, one skipped, 12 subtests passed; the additional book pagination regression was checked separately after its fix. The single warning is the deliberately duplicated ZIP entry in an invalid-backup regression.
Frontend unit and browser acceptance	Ten unit tests passed; 42 browser cases passed in the full run, followed by a passing City Studio rerun after correcting its stale required-role assertions. All 43 cases are verified, including all 145 City Studio checks.
Three-city native simulation	Nominal screens and all eight mandatory resilience cases pass for each city.
Samawah hybrid stabling	Both overnight cycles and morning launches pass in the selected 48-hour replay.
Station thermal/fire and analytical screens	Actual EnergyPlus and FDS runs completed; mitigation comparisons retained without physical release.
Financial/project controls audit	All 266 cities reconcile; zero findings.
Prior Baghdad funding follow-up	Current Python suites (city generation, component catalogue, engineering analysis and automation): 1,441 passed, one skipped, 12 subtests passed. This includes 15 finance/programme regressions; government total capital remains 25%, imports reconcile 50% USD cash / 50% USD loan, native-currency sources and monthly/annual cap gaps reconcile. Other cities are excluded from the programme.
This reconciliation, pricing and OPEX-inflation revision	1,457 passed, one skipped and 12 subtests passed across city generation, component catalogue, engineering analysis and automation. Independent debt/fee reconstruction, supplemental facility costs/caps, grants replacing debt, all 15 financing/pricing ledgers and six-month envelopes reconcile. Three new pricing regressions cover OPEX inflation, real-price demand response and tier-specific capacity.
Surplus repayment follow-up from `f79ab4c`	1,463 passed, one skipped and 12 subtests passed across the same four Python suites. Six added tests cover premium cash budgets, shorter maturities, native principal/interest conservation, reserve and OPEX-buffer priority, minimum ages, noncallable bonds, borrowing exclusions and actual factory reserve service. All six monthly/six-month repayment cases reconcile, and the previous 15 scenarios retain their numeric results. Repository health and 1,990 maintained README/brief checks pass. Programme and offer source hashes match; the rebuilt 15-page offer's early-repayment page was visually checked.
Current generated evidence and documentation	All 266 project twins audit cleanly; repository health, 1,990 README/brief checks, national/portfolio/overview drift, programme and offer source bindings, package planning status and whitespace checks pass. Baghdad offer rebuilt to 14 pages with native funding, receipts, inflation and affordability sensitivity charts.
City packages	All three have zero missing artifacts, no stale analysis sources, current operations bundles and `planning_example_complete=true`; depot/stabling physical gates remain open.
Documentation and hygiene	Maintained/generated README checks, national/portfolio/overview/catalogue drift checks, Markdown inventory, repository health, artifact-size and whitespace checks pass. CSV CRLF is recognised as its existing format.

The reader book includes this review, the Baghdad-only programme, Baghdad funding appraisal and independent reconciliation with six-month funding requirements. Samawah and Mosul appraisals are in a separate standalone chapter outside the programme. The original codebase and prior follow-up validation above predate this ready-task/reserve/phasing recalculation; follow-up finance tests and controls checks are recorded with the updated scenario. The Baghdad concept/FEED offer is also rebuilt with current source-bound figures and visually checked screenshots/charts. Tests establish the recorded software and planning behavior; they do not close the financial, physical or approval decisions above.

Ready-task and cashflow recalculation validation

The new regressions verify production pipelining without lane overlaps, deterministic schedules and cycle rejection; reserve funding from project cash; per-line revenue ramps and unchanged debt grace; identical programme capital shares across phased cases; complete line projections; cash conservation, repayment and programme source freshness. Validation: the full Python suites across city generation, component catalogue, engineering analysis and automation passed 1,446 tests and 12 subtests, with one skipped test. The only warning is the deliberate duplicate ZIP entry used to test invalid-backup rejection. The source/budget/cashflow audit passed for all 266 cities; repository health, all 1,990 maintained README/brief checks, generated national/portfolio/overview/catalogue checks and whitespace checks passed. All three selected planning packages have zero missing artifacts and zero stale analysis sources, with operational release false. The 12-page Baghdad offer has current input/PDF hashes and its updated financing page was visually checked. The reader book is rebuilt from these final documents and checked for current figures and size.

Joint Development And Delivery Framework

Repository source: `docs/commercial/joint-development-framework.md`

This framework turns a possible industrial partnership into evidence-bearing work. The recommended first instrument is a funded development agreement with a defined route to an equity joint venture, not an immediate broad partnership announcement or permanent territorial exclusivity.

> This is a technical and commercial readiness framework, not legal, tax, procurement, financing or investment advice. Local counsel and the relevant customer/authority must select and approve the actual structure.

Proposed Purpose

Develop and deliver affordable urban-rail products and owner/operator systems with open engineering information, qualified specialist equipment, progressively capable local production and continuing support.

The business can earn revenue before a complete railway is available:

Offer	Customer receives
Engineering development	Bounded studies, requirements, reference adaptation, design packages and procurement preparation
Owner/operator platform	Configured asset, work, procurement, competence, maintenance and assurance workflows
Prototype/pilot delivery	Equipment, integration, local assembly and an agreed test programme
Vehicle/module supply	Configuration-controlled accepted products, spares and support
Factory and workforce preparation	Tooling, methods, qualification, training and competence transfer
Long-term support	Maintenance assistance, software support, engineering changes and fleet expansion

Non-recurring development, factory/tooling, first-of-class acceptance and repeat-production equipment remain separate prices.

Three-Layer Structure

Layer	Default form	Retained purpose
OpenSourceRail project	Public development under the existing software, hardware/design and documentation licences	Maintain the reusable public platform, evidence and governance
Commercial development/delivery venture	Contractual collaboration first; an equity entity only after accepted commitments justify it	Employ engineers, fund work, contract, integrate, support and carry allocated commercial responsibility
Country delivery/manufacturing	Customer contracts and local workshops first; local subsidiary/JV when funded volume justifies it	Perform site-specific delivery, develop local capability and maintain the railway

The local railway owner is normally the customer and asset owner, not merely a sales channel. It retains its statutory duties, requirements, project records and acceptance decisions.

Counterparty Qualification

A brand, website relationship or product catalogue is not manufacturer backing. Before negotiation passes gate P0, record:

- the exact legal entity becoming a party or shareholder;
- registration, beneficial ownership and authority to commit;
- named engineers, facilities and availability;
- each manufacturer it is contractually authorised to represent;
- who owns and can license each drawing, software item and method;
- who warrants the delivered configuration and carries product liability;
- conflicts, commissions, related parties and sanctions/export-control review;
- the proposed procurement role: owner adviser, prospective supplier or integrated contractor.

RailMac describes itself as a SinoMac sub-brand and states that SinoMac seeks country partners through branches or joint ventures. That makes a proposal plausible, but is not evidence that RailMac, SinoMac, CRRC or any named manufacturer has committed resources to OpenSourceRail. See the current RailMac site and require direct contractual evidence.

Contribution Before Equity

Negotiate ownership from accepted value and continuing obligations rather than from an assumed 50:50 or 51:49 split.

Every contribution records:

1. contributing legal party;
2. cash, people, equipment, facility, technology/right or customer/site input;
3. bounded quantity, timing and valuation method;
4. acceptance criteria and non-conflicted acceptance owner;
5. evidence of delivery and treatment if it is late, defective or withdrawn;
6. whether the JV pays separately for the same item.

Future support promises do not count as delivered equity. Equipment invoiced at full commercial price is not also an in-kind equity contribution. The public code is already non-exclusive; commercial value comes from maintained configurations, engineering capacity, qualified production, implementation, support and accepted references.

Use the machine-readable `commercial-partnership-readiness.toml` as the contribution and decision register. The tracked reference remains blank; confidential copies live in the controlled opportunity workspace.

Governance And Reserved Matters

Routine work follows an approved budget and management delegation. Joint or shareholder approval should be confined to material matters such as:

- purpose, localisation commitments and approved business plan;
- disposal or exclusive licensing of material technology, data, tooling or brand;
- major borrowing, guarantees, new shares and dilution;
- large or unusual related-party purchases;
- entry into a materially different business, sale or winding-up;
- changes to owner continuity, open-platform or alternative-supplier rights.

The shareholders' agreement needs a practical deadlock, capital-shortfall, default, withdrawal, valuation, transfer and termination route. Majority ownership alone does not solve these issues.

IP, Data And Manufacturing Rights

WIPO describes equity and contractual JVs and recommends early agreement on background and new IP. The project therefore separates six records:

Material	Required treatment
Existing public OpenSourceRail work	Existing licences remain effective; it is not converted into an exclusive asset
Supplier background technology	Used only under rights the supplier is authorised to grant
New reusable development	Ownership, publication, licence, maintenance and patent treatment agreed before work starts
Customer-specific information	Delivery, access, confidentiality, retention, security and exit rights defined
Brands/trademarks	Ownership and permitted use separate from copyright licences
Manufacturing information	Editable drawings, tooling data, processes, inspection/diagnostics and training usable for the agreed local work and continuity

Apache-2.0 allows commercial services and separates trademark rights. CERN-OHL-S v2 is a strongly reciprocal hardware licence. Component boundaries and available component treatment must be reviewed before public hardware definitions and supplier proprietary material are combined. See the Apache licence and CERN OHL source.

Technology transfer is complete only when named local people can perform the agreed task with the delivered information, tooling, rights, inspection method and authorization—not when a folder of PDFs changes hands.

Shareholder-Supplier Controls

An industrial shareholder may also supply equipment. The venture therefore requires:

- agreed related-party pricing and benchmarkable major quotations;

- disclosure of rebates, commissions and conflicts;
- non-conflicted approval and audit rights;
- exact configuration, acceptance, warranty and liability allocation;
- open-book treatment of development versus repeat-unit charges;
- the right to qualify alternatives against objective performance;
- step-in, data, tooling, spares and termination assistance.

Default to non-exclusive cooperation. Any exclusivity must name territories or opportunities, have a finite term, measurable investment/delivery/support conditions and termination rights. There is no permanent blanket exclusivity for the “developing world” in the reference template.

Separate The Two Financing Problems

Capital need	Pays for	Evidence
Venture/development capital	Engineers, prototypes, tests, customer development and working capital	Shareholder/customer commitments, budget, cash controls and contribution acceptance
Railway project finance	Land, civil works, stations, fleet, commissioning and operating mobilisation	Customer/project approval, procurement strategy, affordability, finance and risk allocation

Formation of a JV does not finance a railway. The first venture contract should not combine unproven product-development, construction, demand, currency and long-term fare risk unless the parties explicitly price, finance and allocate them.

Where development-bank procurement may apply, decide the venture's role before paid preparation begins. Owner-side consultancy, prospective supply and an integrated design-build/turnkey role have different conflict implications. The World Bank September 2025 procurement regulations are one relevant reference; the actual financing institution and procurement documents govern the project.

Localisation As Accepted Capability

Stage	Capability	Acceptance evidence
LOC-1 — Assembly and maintenance	Assemble selected equipment, inspect it, diagnose faults and replace modules	Authorized people, completed assemblies, inspection and maintenance demonstrations
LOC-2 — Selected fabrication	Produce suitable structures, panels, interiors, brackets and harnesses	Released drawings/fixtures, qualified processes and accepted first articles
LOC-3 — Broader production	Repeat additional assemblies when capability and volume justify it	Conformity, cost, delivery and qualified-supplier records
LOC-4 — Continuing engineering	Evaluate change, control configuration and qualify replacements	Competent roles, controlled records and accepted decisions

Measure local payroll and invoices, origin evidence, tasks performed without overseas intervention, supplier qualifications, first-pass yield, rework, lead-time and cost trends. Contract labels or local final assembly alone are not proof of design capability.

Alstom reports that its Gibela programme combines a 600-train customer order, local manufacturing, long support/spares obligations and more than 65% local content. The transferable lesson is that localisation was attached to a defined customer programme and measurable industrial obligations—not that its volume or percentage applies to OSR. See Alstom's South Africa programme.

Reference Candidate Portfolio

Maintain an open-ended portfolio rather than limiting the venture to one example. Candidates can include integrated corridors, existing workshop/depot deployments, manufacturing first articles, supplier-neutral brownfield integrations and other customer-defined pilots. Iraq is a logical source of initial examples because repository evidence exists for Samawah and Mosul, but the repository contains no customer order for either city.

Screen any number of potential candidates in parallel. Select a candidate for a funded project only when it has:

1. an identifiable customer and decision-maker;
2. a funded bounded scope;
3. a usable site, workshop or corridor;
4. a defined engineering, procurement and approval route;
5. evidence reusable by later projects.

Possible packages include a non-safety owner/workshop deployment, an independently compared corridor, supplier review of an LM3 first article, or a combination. Several may proceed where each has its own customer, funding, scope, evidence and approval route. Evidence is never pooled in a way that makes one candidate appear ready because another passed.

The machine-readable template includes four illustrative screening rows and accepts additional `[[reference_candidate]]` entries without a fixed maximum. Once selected, each project receives a controlled project-specific copy so confidential records and decisions remain separated.

First 90 Days

Period	Work	Required decision evidence
Days 1–30	Qualify legal parties, authority, named resources, candidate customers and proposed contributions	P0 counterparty/opportunity record, candidate portfolio and draft heads of terms
Days 31–60	Execute the bounded supplier technical-support package: review LM3 scope, supplier authority, component information, cross-supplier interfaces, local facility and production tasks	Machine-checked technical-support readiness, design-review findings, supplier-backed BOM/interface gaps and prototype/localisation plan
Days 61–90	Price development, allocate responsibility, separate funding, agree IP/data/manufacturing rights and candidate propositions	Costed development agreement and board basis to form, defer or reject an equity JV and select zero or more funded candidates

The 90 days define and fund work. They do not promise a production-ready or approved train.

Gated Decision Sequence

Gate	Decision
P0	Are the real counterparties, customer problem and procurement role verified?
P1	Is a costed development agreement funded with contributions and IP treatment accepted?
P2	Do delivered commitments justify forming/capitalising an equity JV?
P3	Is at least one bounded customer-funded candidate project ready for authorization in its own record?
P4	Do audited outcomes justify replication or wider territorial rights?

Current public status is generated at commercial partnership readiness. It deliberately shows zero accepted contributions and zero accepted gates until real evidence is entered in a controlled deployment copy.

The commercial record does not substitute for engineering acceptance. The linked supplier-support record independently holds component configuration, interface, first-article, local capability, lifecycle and separated-cost gates.

Commercial Partnership Readiness

Repository source: `docs/commercial/partnership-readiness.md`

> Status: unfilled-no-commercial-authority — no partnership, supplier backing, customer order, financing, certification or operating authority is implied.

This template records evidence for a possible development agreement or joint venture. It is not legal advice, a partnership announcement, an offer, a customer order, supplier authorization, financing, procurement approval, certification or authority to construct or operate a railway.

Opportunity ready: no · Parties: 0/3 · Contributions: 0/8 · IP items: 0/6 · Controls: 0/4 · Localisation stages: 0/4 · Gates: 0/5

Reference candidates active: 4/4 · Shortlisted: 0 · Selected and evidenced: 0

Funding separated and agreed: no · Exclusivity controlled: yes · Public claims evidenced: no

Development agreement ready: no · Equity JV ready: no · Any candidate project ready: no · Replication ready: no

Reference Candidate Portfolio

The candidate array is open-ended. Screening status records a possible example, not customer interest, funding or authorization. Give every selected project its own controlled copy before adding confidential evidence.

Candidate	Type	Country / site	Selection evidence	Status	Selected
`CAND-SAMAWAH-CORRIDOR`	integrated-corridor	Iraq / Samawah	0/5	`screening`	no
`CAND-MOSUL-CORRIDOR`	integrated-corridor	Iraq / Mosul	0/5	`screening`	no
`CAND-EXISTING-WORKSHOP`	owner-operator-workshop	to be nominated / existing depot or workshop	0/5	`screening`	no
`CAND-LM3-FIRST-ARTICLE`	manufacturing-first-article	to be nominated / qualified local production facility	0/5	`screening`	no

Parties

Party	Role	Legal identity	Authority	Ready
`PARTY-OSR`	platform-and-system-development	open	open	no
`PARTY-INDUSTRIAL`	industrial-engineering-and-supply	open	open	no
`PARTY-LOCAL`	customer-or-local-delivery	open	open	no

Contributions

Contribution	Party	Milestone	Evidence	Status	Accepted
CONTRIB-OSR-PLATFORM — Maintained OpenSourceRail baseline, systems integration and configuration leadership	PARTY-OSR	funded-development-start	0/3	proposed	no
CONTRIB-OSR-DEVELOPMENT — Named systems, software, civil, vehicle and delivery-integration work	PARTY-OSR	reference-development-review	0/3	proposed	no
CONTRIB-IND-ENGINEERS — Named vehicle, mechanical, electrical, manufacturing and commissioning engineers	PARTY-INDUSTRIAL	reference-development-review	0/3	proposed	no
CONTRIB-IND-MANUFACTURING — Manufacturing review, production facilities, tooling access and supplier coordination	PARTY-INDUSTRIAL	first-article-plan	0/3	proposed	no
CONTRIB-IND-PROTOTYPE — Prototype equipment, test access or development cash contributed at risk	PARTY-INDUSTRIAL	prototype-authorization	0/3	proposed	no
CONTRIB-LOCAL-FACILITY — Local workshop, metrology, stores, training and first-article production capacity	PARTY-LOCAL	localisation-baseline	0/3	proposed	no
CONTRIB-LOCAL-WORKFORCE — Local production, maintenance and engineering counterparts with a competence plan	PARTY-LOCAL	localisation-baseline	0/3	proposed	no
CONTRIB-CUSTOMER-SCOPE — Funded bounded requirement, usable site and defined engineering and approval route	PARTY-LOCAL	reference-project-selection	0/3	proposed	no

IP And Manufacturing Rights

Item	Material	Rights / publication evidence	Status	Ready
IP-OSR-BACKGROUND	Existing public OpenSourceRail software, designs and documentation	open	unagreed	no
IP-SUPPLIER-BACKGROUND	Existing supplier products, drawings, software and know-how	open	unagreed	no
IP-FOREGROUND	Reusable work funded by the development programme	open	unagreed	no
IP-CUSTOMER	Customer, site, operational and commercial information	open	unagreed	no
IP-BRAND	OpenSourceRail and commercial service names and marks	open	unagreed	no
IP-MANUFACTURING	Editable drawings, tooling, process, inspection, diagnostic and training information	open	unagreed	no

Commercial Controls

Control	Evidence	Status	Ready
CTRL-RELATED-PARTY-PRICE — Shareholder and related-party pricing	0/3	open	no
CTRL-REBATES — Rebates, commissions and conflicts disclosure	0/3	open	no
CTRL-WARRANTY — Configuration, warranty and liability allocation	0/3	open	no
CTRL-ALTERNATIVES — Alternative supplier and continuity rights	0/3	open	no

Localisation Capability

Stage	Outputs	Status	Complete
`LOC-1` — Assembly and maintenance	0/3	`not-started`	no
`LOC-2` — Selected fabrication	0/3	`not-started`	no
`LOC-3` — Broader repeat production	0/3	`not-started`	no
`LOC-4` — Continuing engineering	0/3	`not-started`	no

Decision Gates

Gate	Evidence	Decision	Accepted
`P0` — Counterparty and opportunity qualification	0/3	`open`	no
`P1` — Funded development agreement	0/4	`open`	no
`P2` — Equity joint-venture formation	0/4	`open`	no
`P3` — Selected-candidate project authorization	0/4	`open`	no
`P4` — Replication and territorial expansion	0/4	`open`	no

Use the joint-development framework before tailoring this record. Use the separate supplier technical-support readiness record for component configuration, interface, first-article, localisation and engineering-acceptance evidence. Copy `commercial-partnership-readiness.toml` into a controlled opportunity workspace; do not fill the public reference template with confidential negotiation material. An accepted status without the required evidence, named acceptance or date remains open.

Supplier Technical-Support Package

Repository source: `docs/commercial/supplier-technical-support-package.md`

Purpose

This package converts a potentially useful supplier discussion into controlled engineering work for the LM3. It is designed for RailMac/SinoMac, a directly authorised manufacturer, another qualified supplier, an independent engineering house, or a combination. It does not assume that any named organisation has committed support.

The desired outcome is not a general endorsement. It is a configuration-specific set of reviewed requirements, component selections, interface decisions, production information, first-article evidence, quotations and local-capability demonstrations that can update OpenSourceRail's existing source-of-truth records.

Use the machine-checked supplier technical-support readiness record to control the engagement. The public template is deliberately unfilled; copy it to a controlled opportunity workspace for legal names, confidential drawings, prices, signatures and test records.

The coordinated industrialisation programme links vehicle interfaces, staged bogie localisation and the viaduct production chain. Its possible vendor register adds official-source component and equipment leads, including HÜBNER and CRRC as proposed partners. These candidates use this same configuration-specific engagement process; a public product listing does not select a part or grant manufacturing rights.

What The Public Information Establishes

RailMac's current site describes a SinoMac railway offering spanning pre-design, component selection and supply, installation, commissioning and after-sales support. It also presents train, traction, battery, bogie, brake, component and maintenance-equipment categories. The tourism-transport site presents related rail-mounted vehicles associated with CRRC Special Purpose Equipment.

These are useful catalogue and capability leads. They do not establish:

- the exact legal entity able to contract or make an engineering commitment;
- whether a representative can bind a manufacturer or provide manufacturer engineers, drawings, test evidence or warranty;
- which published component or vehicle configuration is current and offered;
- compatibility with LM3 speed, loading, duty, thermal environment, geometry, interfaces, local manufacture or unattended operation;
- a price, delivery schedule, customer order or railway acceptance.

The public 1435 sightseeing-train material is therefore a discussion reference, not an LM3 design input. Any inconsistency between introductory copy and a public technical table must be resolved by a controlled supplier datasheet, drawing and signed deviation record.

Existing OpenSourceRail Inputs

The engagement begins from existing controlled work rather than an empty brief:

Existing capability	Use in the supplier package
LM3 requirements and interfaces	Freeze the train envelope, formation, loading, performance intent and system boundaries
120-row product definition	Map offered products, assemblies, quantities, masses and supplier identities without creating a parallel BOM
Supplier anchors and COTS candidates	Issue comparable configuration-specific enquiries and retain alternatives
Mechanical/BIM detail register	Connect supplier datums, loads, keep-outs and evidence to the twelve controlled vehicle interfaces and route gates
First-article execution pack	Reuse the 13 evidence gates, travellers, serialized configuration, calibrated test and independent acceptance route
City route and duty models	Reconcile gradients, curves, ambient conditions, timetable, charging, energy reserve and degraded operation against the offered configuration
Lifecycle and ERP model	Load accepted products, documents, inspections, maintenance, spares, competence and changes into owner-held identities
Commercial framework	Control authority, funding, IP, manufacturing rights, related-party procurement, localisation and exit

A supplier return is useful when it closes or changes one of these records through the normal configuration process. A detached presentation, generic brochure or unmapped spreadsheet does not become a second engineering baseline.

Requested Work Packages

TS-WP-01 — Frozen LM3 System Review

Provide one review basis containing the exact repository commit, LM3 configuration, train envelope, passenger loading, 90 km/h target, duty cycle, energy architecture, environment and localisation objective. For every reviewed requirement, return one of four dispositions:

1. suitable as proposed;
2. suitable with a named change;
3. a different component or engineering approach is required;
4. a test or missing input is required before conclusion.

Each disposition needs an owner, rationale, affected interface, evidence and due decision. Silence is not acceptance.

TS-WP-02 — Running Gear And Braking

Return exact, mutually compatible bogie, wheelset, bearing, suspension, brake and air-supply configurations. Include order codes, controlled drawings, masses, loads, wheel/rail compatibility, curve and speed envelope, brake allocation, adhesion and degraded cases, thermal duty, maintenance, RAMS, spares and relevant test evidence.

The output must name who accepts the assembled system. Individually acceptable parts do not prove combined braking, axle load, ride, clearance or fatigue performance.

TS-WP-03 — Traction, Battery, Charging And Thermal System

Treat motor, gearbox, converter, battery, protection, BMS, cooling, charging and grid interface as one configuration. Require:

- voltage, current, torque, speed, power and energy envelopes;
- efficiency, loss, derating and thermal maps rather than headline ratings;
- mounting, shaft, cooling, electrical, network and software interfaces;
- short-circuit, selectivity, earthing, isolation and HVIL coordination;
- charge, regeneration, reserve, parked-soak and degraded-cooling behavior;
- fire/propagation, vibration, EMC, ingress, RAMS and maintenance evidence.

Run the returned maps through the existing city duty and energy models before freezing battery capacity or charging power. A related battery train at 40 km/h does not validate the 90 km/h LM3 duty.

TS-WP-04 — Manufacturing And Design For Production

Review chassis, carbody, GFRP modules, equipment attachments, joints, welding and bonding access, harness/fluid routing, assembly sequence, maintenance access, lifting, tooling, fixtures, gauges and inspection. Return a production-data gap schedule that distinguishes:

- editable released CAD and drawings;
- interface drawings and tolerance stacks;
- weld, bond, fastener and material/process specifications;
- NC, flat-pattern, mould, jig and fixture data;
- work instructions, travellers and inspection plans;
- supplier-proprietary information that cannot be published or locally modified.

TS-WP-05 — First Article And Test Evidence

Define what the first article is: complete train, car, body, bogie, subsystem rig or staged combination. Price and schedule it separately. Use serialized parts, accepted travellers, calibrated instruments, raw results, photographs, deviations, NCR/rework records and named hold-point decisions. The existing first-article gates remain open until independent reviewers accept their required evidence.

TS-WP-06 — Local Workshop And Technology Transfer

Audit the intended facility and define which tasks local staff will assemble, fabricate, inspect, test, diagnose, maintain and eventually engineer. For each task identify delivered rights, editable information, tooling, metrology, facility changes, training, supervised practice and independent demonstration.

Measure capability through accepted work, first-pass yield, rework, time, cost and ability to perform without overseas intervention—not a local-content label alone.

TS-WP-07 — Maintenance, Spares And Support

Return the serialized parts structure, maintenance and overhaul tasks, limits, special tools, diagnostic access, consumables, initial spares, repair route, obsolescence process, warranty response, field-support levels and training. Owner records must remain usable after an adapter or supplier service is removed.

TS-WP-08 — Cross-Supplier Integration

Create one signed responsibility matrix and ICD set across wheel/rail, bogie/body, brake/control, motor/gearbox/wheelset, battery/converter/motor/charger/grid, thermal/fire/environment, vehicle/platform/evacuation and diagnostics/ERP. Every open interface receives an owner, due evidence and operating limitation.

Data Intake Into The Core Platform

```
flowchart LR
  S[controlled supplier return] --> Q[authority / configuration / evidence validation]
  Q --> I[interface and responsibility review]
  I --> C[OSR configuration change]
  C --> P[product tree / CAD / IFC / mass]
  C --> D[city duty / energy / route compatibility]
  C --> E[ERP products / inspection / spares / maintenance]
  C --> A[first-article and assurance gates]
  A --> H[human acceptance / hold / reject]
```

The intake process must preserve the supplier document identifier, revision, configuration, issue purpose, licence/confidentiality, hash, receipt date and review decision. Public repository changes contain only material that may lawfully be published; controlled storage retains proprietary originals and signed records.

No supplier data crosses the existing safety or command boundary. ERP and SCADA may display configuration, health, work and evidence; they do not convert a supplier statement into train-control authority.

Cost And Commercial Return

Require separate prices for:

Cost bucket	Minimum scope statement
Non-recurring engineering	Design review, calculations, drawings, configuration, responsibility and project management
Tooling and workshop preparation	Jigs, fixtures, moulds, gauges, metrology, installation and facility changes
First article	Material, construction, inspection, tests, rework allowance and acceptance support
Repeat train	Exact configuration, quantity, Incoterm, production rate, validity and escalation basis
Training and transfer	Courses, paired work, documents, editable data, licences and task authorizations
Spares and support	Initial stock, special tools, warranty, field response, software and long-term support
Logistics and local charges	Packing, freight, insurance, duties, taxes, currency and local handling
Acceptance support	Independent review, type evidence, homologation interfaces and commissioning

Every price lists inclusions and exclusions. Factory-gate repeat price must not be compared with an all-in railway cost, and equipment sold at full commercial price must not also be counted as an in-kind equity contribution.

Decision Sequence

Gate	Stop/go question
TS0	Is the exact contracting entity authorised, with a verified path to the necessary manufacturer and engineers?
TS1	Is one funded LM3 baseline frozen with duty, environment, scope and schedule?
TS2	Are exact configurations supported by controlled drawings, masses, ratings, limits and evidence?
TS3	Are interfaces and whole-system responsibilities accepted, with deviations controlled?
TS4	Are production data, facility, people, processes, tests, hold points and funding sufficient to authorize a first article?
TS5	Can named local people perform accepted tasks with delivered rights, information, tools and support?
TS6	Do first-article, quality, cost, schedule and customer evidence justify repeat production?

Passing TS6 still does not grant railway acceptance. Project engineering, safety, independent assessment, customer acceptance and the relevant national authorities remain separate.

First Contact Deliverables

For an initial bounded engagement, issue only the frozen review pack and request:

1. the exact legal and technical delivery chain;
2. named engineering disciplines and available time;
3. a priced TS-WP-01 review plus options for TS-WP-02 to TS-WP-08;
4. the proposed controlled-data exchange and confidentiality route;
5. a sample component/interface return demonstrating the promised document depth;
6. separate NRE, tooling, first-article, repeat, training, spares/support and acceptance-support prices;
7. a clear statement of public, customer-controlled and supplier-proprietary outputs and local manufacturing/maintenance rights.

This package can be issued to multiple potential partners. Comparison should be against common requirements and evidence, not the breadth of a marketing catalogue or the identity of a proposed shareholder.

Supplier Technical-Support Readiness

Repository source: `docs/commercial/supplier-technical-support-readiness.md`

> Generated from `lib/templates/supplier-technical-support-readiness.toml`; do not edit this report directly.

> Authority boundary: This register prepares a supplier-backed LM3 design-review and first-article package. Public webpages and discussions are screening inputs only. They do not prove an engineering commitment, manufacturer authorization, a compatible product configuration, accepted design responsibility, price, warranty, railway approval or permission to manufacture or operate.

Current Decision

Technical-support ready: NO. The baseline is not frozen, 0/3 catalogue leads are qualified, 0/8 work packages and 0/8 interfaces are accepted, 0/8 cost buckets are accepted, and 0/7 gates have passed.

Public supplier pages are useful discovery inputs. They do not count as a commitment, controlled configuration, manufacturer authorization or LM3 evidence. A controlled opportunity copy must carry confidential offers and acceptance records.

Baseline

Field	Value
`configuration_id`	LM3-FA-001
`osr_commit`	—
`freeze_record_ref`	—
`duty_cycle_ref`	—
`route_environment_ref`	—
`status`	not-frozen

Catalogue Leads

Lead	Published scope	Status	Qualified
`LEAD-RAILMAC-INTEGRATION` — RailMac / SinoMac railway integration offering	Pre-design, product selection, component supply, installation, commissioning and after-sales support across trains, traction, bogies, brakes, components and maintenance equipment	screening-source-only	no
`LEAD-CRRC-SPECIAL-EQUIPMENT` — CRRC Special Purpose Equipment tourism-transport offering	Supplier-published rail-mounted sightseeing vehicles and tourism transport systems	screening-source-only	no
`LEAD-OPEN-ALTERNATIVES` — Additional qualified suppliers and engineering partners	Open supplier-neutral candidate set for exact components, systems, laboratories and production support	screening-source-only	no

Technical Work Packages

Package	Discipline	Existing OSR inputs	Evidence / required	Accepted
`TS-WP-01` — Frozen LM3 system review	systems-engineering	3	0 / 5	no
`TS-WP-02` — Bogie, wheelset, suspension and brake configuration	running-gear-and-braking	4	0 / 6	no
`TS-WP-03` — Traction, battery, charging and thermal configuration	traction-energy	4	0 / 6	no
`TS-WP-04` — Manufacturing and design-for-production review	industrialisation	3	0 / 6	no
`TS-WP-05` — First-article construction and evidence	prototype-and-test	3	0 / 6	no
`TS-WP-06` — Local workshop capability and technology transfer	localisation	3	0 / 6	no
`TS-WP-07` — Maintenance, spares and lifecycle support	supportability	3	0 / 6	no
`TS-WP-08` — Cross-supplier system integration and configuration control	integration	4	0 / 6	no

Cross-Supplier Interfaces

Interface	Work packages	Evidence / fields	Accepted
`TS-IF-01` — Wheel rail axle load and route geometry	`TS-WP-01`, `TS-WP-02`, `TS-WP-08`	0 / 5	no
`TS-IF-02` — Bogie carbody articulation and ride	`TS-WP-02`, `TS-WP-04`, `TS-WP-08`	0 / 5	no
`TS-IF-03` — Brake train control and adhesion	`TS-WP-02`, `TS-WP-08`	0 / 5	no
`TS-IF-04` — Motor gearbox wheelset and mechanical drive	`TS-WP-02`, `TS-WP-03`, `TS-WP-08`	0 / 5	no
`TS-IF-05` — Battery converter motor charger and grid	`TS-WP-03`, `TS-WP-08`	0 / 5	no
`TS-IF-06` — Thermal fire ventilation and environmental exposure	`TS-WP-03`, `TS-WP-04`, `TS-WP-05`	0 / 5	no
`TS-IF-07` — Vehicle platform structure and evacuation	`TS-WP-01`, `TS-WP-08`	0 / 5	no
`TS-IF-08` — Diagnostics maintenance ERP and owner data	`TS-WP-06`, `TS-WP-07`, `TS-WP-08`	0 / 5	no

Separated Cost Statement

Bucket	Status	Amount	Scoped	Accepted
`COST-NRE` — Non-recurring engineering and design responsibility	unquoted	—	no	no
`COST-TOOLING` — Tooling, fixtures, metrology and workshop preparation	unquoted	—	no	no
`COST-FIRST-ARTICLE` — First-article material, construction, inspection and testing	unquoted	—	no	no
`COST-REPEAT-UNIT` — Repeat-production train at stated quantity and configuration	unquoted	—	no	no
`COST-TRAINING-TRANSFER` — Training, documentation and technology-transfer rights	unquoted	—	no	no
`COST-SPARES-SUPPORT` — Initial spares, special tools, warranty and continuing support	unquoted	—	no	no
`COST-LOGISTICS-TAX` — Packing, logistics, duties, taxes, currency and local delivery	unquoted	—	no	no
`COST-ACCEPTANCE` — Independent review, type evidence, homologation and commissioning support	unquoted	—	no	no

Decision Gates

Gate	Decision	Evidence / required	Passed
`TS0` — Counterparty and technical-access qualification	open	0 / 4	no
`TS1` — Frozen review baseline and funded scope	open	0 / 4	no
`TS2` — Controlled component configurations	open	0 / 4	no
`TS3` — Interface and design-responsibility acceptance	open	0 / 4	no
`TS4` — First-article authorization	open	0 / 4	no
`TS5` — Local capability and lifecycle acceptance	open	0 / 4	no
`TS6` — Repeat-production decision	open	0 / 4	no

Controlled Claims

Claim	Recorded status	Evidence	Claim evidenced
supplier_support_committed	unverified	—	no
manufacturer_access_authorized	unverified	—	no
lm3_configuration_compatible	unproven	—	no
first_article_authorized	not-authorized	—	no
local_manufacture_transferred	unproven	—	no
all_in_repeat_cost	unproven	—	no
railway_acceptance	not-granted	—	no

How This Uses The Existing Platform

The eight work packages consume the existing LM3 requirements, product tree, supplier anchors, RFQ candidates, mechanical interface register, first-article gates, manufacturing controls, city duty-cycle evidence and ERP/lifecycle identities. Accepted supplier data should update those sources through configuration-controlled changes; it must not be copied into a detached supplier spreadsheet and treated as a second design baseline. The supplier intake contract is `design/component-catalogue/catalog/buildable-trainset/evidence/rfq-response-template.json` schema 2.0; it carries authority, frozen-baseline, separated-cost, interface-responsibility, local-capability and lifecycle-import fields.

Use `python3 tools/automation/validate-supplier-technical-support.py --check` to reject stale output or invalid references.

Competitive Position And Proof Roadmap

Repository source: `docs/competitive-position-and-proof-roadmap.md`

Baseline reviewed: repository `main` after the 2026-09-28 lifecycle release

Decision use: product positioning and evidence planning—not tender evaluation, supplier prequalification or a claim of equivalent railway acceptance

Position In One Sentence

OpenSourceRail is presently strongest as an open, owner-controlled railway planning, engineering, delivery and asset-information platform. Its reference vehicles, infrastructure and GoA4 control system remain development programmes and must not be sold as equivalent to commissioned commercial railways.

That distinction produces three separate products with separate claims:

Product	Evidence now	Honest use now	Evidence still needed
Planning, design and delivery platform	City Studio, source-locked GIS, native/SUMO simulation, FreeCAD/IFC, quantities, asset identities, schedules, procurement and cashflow	Corridor screening, reference design, option comparison and delivery coordination	Surveyed corridor validation, independent numerical benchmarks, issued design deliverables and customer deployment evidence
Owner/operator and maintenance platform	ERPNext/Frappe HR, Ops Core, governed asset/work/inspection/maintenance records, observation-only FUXA integration and recovery exercises	Existing workshop, depot or pilot-corridor implementation with no railway command	Operator master data, production identity/TLS/MFA, commissioned equipment adapters, measured adoption outcomes and service support
Physical railway and GoA4 control	LM3 and reusable civil reference designs, deterministic evaluators, formal checks and staged pilot boundaries	Supplier engagement, prototype definition, shadow/HIL research and evidence planning	Supplier-frozen configurations, released production drawings, first articles, physical/HIL/field tests, independent assessment and national/operator authorization

The 265-city developing-country portfolio demonstrates repeatable model generation. It is not 265 deployments, orders or approved railway projects.

What Competes With Each Product

No single reviewed alternative maps to the entire repository. The useful comparison is by job to be done.

Benchmark	Strong benchmark for	OpenSourceRail distinction to prove	Do not claim
OSRD	Open railway infrastructure, timetable, capacity and train simulation	Engineering revision, product, procurement, construction and maintenance consequences carried beyond planning	That breadth implies equivalent railway-planning depth or institutional maturity; OSRD publishes an active roadmap
Eclipse SUMO	Microscopic multimodal traffic, pedestrian/public-transport and network interaction	A common city revision connected to rail engineering and delivery packages	That OSR native results are independently validated until matched inputs and tolerances are published
Netzgrafik-Editor	Regular-interval timetable editing and connections	Downstream asset, financial and construction effects	That a broad platform replaces every specialist timetable tool
EULYNX / railML / RCM-DX	Signalling interfaces and railway data exchange	Owner-held integration and lifecycle records	That interface standards or partial simulators are turnkey signalling products
Bentley OpenRail Designer and connected commercial tooling	Specialist alignment, cant, turnout, corridor, documentation and multidisciplinary CDE workflows	Inspectable source, adaptable workflow and owner-held configuration	That commercial products are uniformly closed, obsolete or non-interoperable
IBM Maximo and specialist EAM/APM systems	Deployed asset, work, inspection, inventory and reliability management	A railway-specific, locally adaptable implementation connected to public engineering definitions	Equivalent supportability, scale or operating history before a real deployment measures it
Commercial train/CBTC/turnkey suppliers	Accepted product configurations, system integration, commissioning and lifecycle support	A different ownership, localisation and supplier-replacement proposition	Equivalent safety, availability, delivery price or passenger service before physical evidence and like-for-like offers exist

The practical open-source competitor is also a competent integrator assembling QGIS, FreeCAD/Bonsai, ERPNext and FUXA directly. OpenSourceRail earns its place only when its common identities, revisions, quantities, interfaces, evidence and repeatable deployment workflow measurably reduce reconciliation work and errors.

Claims Register

Claim	Current status	Evidence required before strengthening it
The repository joins planning through maintenance	Demonstrated in reproducible software and generated evidence	Retain exact-commit integration and recovery results
The owner can retain inspectable source and records	Supported by repository licences and architecture	Deployment contracts must preserve data export, configuration, tooling and step-in rights
The owner/operator stack is adoptable before train control	Credible product proposition	One real workshop/depot deployment with baseline, user, integrity, recovery, support and outcome measurements
The railway is cheaper than commercial alternatives	Unproven; mechanisms and editable sensitivities exist	Same corridor, scope, capacity, civil mix, risk, acceptance, warranty, spares, renewals and financing; independent or tendered prices
The LM3 can be locally manufactured	Development objective with controlled work packages	Supplier-backed BOM, production drawings, closed mass/axle loads, qualified processes, first articles and repeated conformity
GoA4 is ready for passenger operation	False	Frozen safety architecture, production hardware/transports, HIL and field evidence, independent assessment and authorization
An industrial partner or manufacturer supports OSR	Unverified until contracted	Exact legal entity, authority, named resources, equipment responsibility, warranty and contribution evidence

The cost model must keep factory-gate repeat-unit targets separate from non-recurring engineering, factory/tooling, supplier qualification, homologation, commissioning, warranty, spares, renewals and finance. Import content measures foreign-exchange exposure; it does not by itself determine external borrowing.

Four Proof Programmes

1. Prove The Owner/Operator Product

Deploy the first adoptable product in an existing workshop, depot or bounded pilot environment. Measure:

- asset and work-record completeness before and after migration;
- elapsed time and reconciliation errors from defect to inspected handback;
- evidence retrieval, backup and clean-volume recovery;
- inventory, calibration, competence and authorization exceptions;
- implementation/support hours and operator-owned exit/export results;
- user review time, error rates and overrides where AI-assisted drafts are used.

The comparison baseline is the operator's actual prior process or a stated configured alternative—not an invented manual-work estimate.

2. Publish Rigorous Corridor Comparisons

Maintain multiple potential examples rather than treating one city as the permanent reference. Samawah and Mosul are useful repository-backed starting points; any of the catalogue cities—or a customer-supplied corridor—may enter the candidate portfolio. For each selected study, obtain the relevant owner's agreement, freeze its own input package and compare OSR native simulation with SUMO or another independent railway tool for:

Evidence family	Minimum publication
Geometry and service	Source/CRS/revision, line lengths, gradients, curves, station/turnback layout, dwell and timetable
Demand and capacity	OD source/calibration, boarding rules, sectional loads, platform/turnback constraints and crowding
Motion and energy	Train mass/configuration, resistance, traction/braking, auxiliaries, charging losses and full-day state of charge
Reliability	Delay distributions, charger/train/station outages, degraded headways, recovery rules and stranded-service outcomes
Civil and cost	Survey/topography basis, at-grade/elevated/bridge quantities, station scope, supplier quotes and excluded risk
Comparison	Versioned tools, identical inputs where possible, explained model differences, tolerance and adjudication owner

A candidate pass means results agree within predeclared tolerances or each difference is explained and accepted for its decision use. It is not a railway approval. Comparisons remain separate evidence records so a strong result in one city cannot conceal a weak or inapplicable result in another.

3. Make The Owner Platform Supplier-Neutral

Prove that a deployment can buy accepted trains, signalling or maintenance equipment from another supplier while retaining OSR asset, project, configuration and evidence workflows. Required demonstrations are:

- stable owner asset/product identities mapped to supplier identities;
- import/export and revision reconciliation in documented open formats;
- a supplier adapter that cannot cross the observation-only authority boundary;
- owner-held work, defect, inspection and handback history after adapter removal;
- documented exit, archive, backup and alternative-supplier qualification paths.

4. Close Physical Evidence, Not Catalogue Volume

Prioritise supplier-frozen configurations, released drawings, mass/CG/axle-load closure, manufacturing qualifications, first articles, calibrated test results, production hardware and independent field evidence. Adding another generated city does not close any of those gates.

Use the supplier technical-support package to turn an industrial discussion into configuration-specific work. It reuses the existing LM3 product tree, interface register, city duty models, first-article evidence gates and ERP identities while requiring named responsibility across running gear/brakes, traction/energy/charging, thermal/fire, manufacturing, local capability and lifecycle support. Its readiness output remains open until controlled evidence replaces catalogue claims.

Evidence-Led Release Sequence

Stage	Product decision	Exit evidence
C0 — Position	Decide which of the three products is being offered	Named customer problem, bounded scope and accurate maturity statement
C1 — Benchmark	Establish a fair alternative and baseline	Common requirements, input package, comparison method and accepted tolerances
C2 — Demonstrate	Run the product in a controlled customer environment	Exact configuration, observed outcomes, failures, recovery and user acceptance
C3 — Contract	Allocate delivery, data, support and liability	Legal counterparties, acceptance, warranty, IP/data, change, exit and support terms
C4 — Replicate	Reuse only what evidence supports	Audited cost, schedule, reliability, capability and lessons from the applicable reference projects

Commercial or partnership work follows the separate joint-development framework and its fail-closed readiness record. Physical supplier work follows the independent technical-support readiness record, so an equity or development agreement cannot silently count as component compatibility or first-article acceptance.

Primary Sources Used For The Framework

- OSRD product and governance and published roadmap
- Eclipse SUMO product and railway simulation documentation
- Bentley OpenRail Designer
- IBM Maximo for travel and transportation
- WIPO technology-transfer and joint-venture guidance
- RailMac published railway equipment and technical-support scope

Product pages describe their publishers' capabilities and are not independent comparative evaluations. A procurement must verify the applicable version, configuration, licence, support, references and offered terms directly.

Component RFC Implementation Readiness

Repository source: `docs/component-rfc-readiness.md`

> Machine-checked implementation/procurement completeness only. It is not supplier freeze, manufacturing release, certification, or permission to operate.

- Digital completeness: PASS
- RFC packages checked: 5
- Physically release-ready: 0

Package	Scope	Owner	Digital package	Physical release	Open blockers
RFC-PKG-0023	Door cassette, operator and train interface	rolling-stock mechanical lead	PASS	BLOCKED	3
RFC-PKG-0024	Battery and cabin integrated thermal system	rolling-stock thermal and battery lead	PASS	BLOCKED	4
RFC-PKG-0025	Turnout, locking, detection and point machine	trackwork and signalling interface lead	PASS	BLOCKED	4
RFC-PKG-0026	Automated side conductive charging interface	traction power and charging lead	PASS	BLOCKED	4
RFC-PKG-0027	Brownfield asset recovery and workshop integration	brownfield recovery and depot lead	PASS	BLOCKED	4

Every package names requirements, ICDs, hazards, controlled product/BOM IDs, drawings, assembly steps, analysis inputs, required tests, assumptions, accountable role and explicit blockers. Evidence hashes are retained in the JSON report.

These are configuration-controlled implementation and procurement packages. Open supplier, physical, site and authority evidence remains release-blocking.

OpenSourceRail Cost Model

Repository source: `docs/cost-model.md`

This file is generated by `tools/automation/generate-cost-model.py`. Do not hand-edit the numbers here; change the source data and regenerate.

Sources Of Truth

Number family	Source
Non-civil CAPEX unit rates, retained civil benchmarks, USD/EUR reporting views, EPC, solar, charging	`lib/templates/capex-costs.toml`
Reviewed civil benchmark quantities and cost shares	`lib/templates/civil-cost-calibration.toml`
Generated CAD-indexed civil rate contract	`lib/templates/civil-cost-model.toml`
Light-metro 3-car BOM line items	`docs/rolling-stock/light-metro-3car/bom-skeleton.md`
Generated rolling-stock BOM CSV	`build/bom/rolling_stock_bom.csv` via `tools/automation/export-light-metro-bom.py`
Recalculated LM3 build cost	`design/component-catalogue/catalog/buildable-trainset/trainset-build-cost.json`
Country finance and fare assumptions	`lib/templates/country-finance.toml`
Schedule-linked Iraq funding assumptions, native-currency debt and downside cashflow	`lib/templates/iraq-funding.toml` and `osr_scenario/iraq_finance.py`
Broad-benefit assumptions	`lib/templates/economic-benefits.toml`

Generated `*_eur` fields are converted reporting views at 0.92 USD->EUR. The detailed civil marketplace anchors live in `docs/civil/marketplace-cost-anchors.md`.

Rolling Stock

Rolling stock is budgeted by local-owner trainset-family planning unit, not by multiplying an inflated per-car price. A 3-car `light-metro-3car` trainset is $900 k per trainset with `*_eur` reporting views generated at 0.92 USD->EUR.

The current `light-metro-3car` build-cost estimate uses the promoted design candidate cost of 682,431 USD, adds 5,524 h of direct labour at $10/h, then applies a 20% unexpected-cost premium. That gives a local factory-gate planning target of 885,205 USD per 3-car consist. City CAPEX keeps the rounded 900,000 USD trainset unit so there is still a small nominal QA/acceptance and local handover margin, while fixtures/tooling sit in the railway production plant and warranty, spares, homologation, supplier qualification, first-of-class engineering, and routine commissioning support remain outside that factory-gate comparison.

The direct material/supplier-module bucket already includes the requested passenger fit-out, HVAC, windows, and doors: $18 k for seats/floor systems/grab rails/interior lighting, $17 k for three roof HVAC units, and $112 k for side windows, powered side doors, door sill/emergency kits, and panoramic end glass. The included requested-scope subtotal is $147 k before labour and premium, so no extra $20k interior allowance is added unless a deployment chooses to carry a separate contingency.

3-car trainset cost bucket	Basis	Cost
Direct material and supplier modules	Promoted design candidate cost metric: frame, panels, glazing, doors, articulation/gangways, end couplers, bogies, suspension air supply, traction, batteries, HVAC, electronics, interiors	$682 k
Direct labour	5,524 h explicit first-article/final-assembly plan at $10/h	$55 k
Unexpected-cost premium	20% for rework, logistics, consumables, local fabrication variation, and shop learning	$148 k
Recalculated build estimate	Direct modules + labour + unexpected-cost premium	$885 k
Nominal QA + handover rounding margin	Acceptance evidence, test dossier, local movement, manuals/training handover; warranty/spares stay in OPEX	$15 k
Total per 3-car trainset	Local-owner production planning unit	$900 k

Family	USD / trainset	EUR mirror
`urban-shuttle-1car`	$280 k	EUR 0.258 M
`tram-2car`	$560 k	EUR 0.515 M
`light-metro-3car`	$900 k	EUR 0.828 M
`metro-4car`	$1.1 M	EUR 1.030 M
`metro-6car`	$1.7 M	EUR 1.546 M

Only `light-metro-3car` currently has the detailed 120-product, 26-assembly, tooling and shop-traveller release reference. Other family values—including tram-2car, metro-4car and metro-6car—are planning units: their city twins keep the correct fleet cost but mark detailed rolling-stock work and procurement as `family-specific-detail-required` instead of reusing LM3 part numbers.

The base value assumes direct procurement, local final assembly, common bogie modules, one-metre clip-on fiberglass non-structural cladding, COTS doors/windows/HVAC/interior modules, open control electronics, and no proprietary CBTC onboard bundle. It includes labour, shop overhead, nominal per-unit QA/acceptance evidence, and local handover logistics. It does not repeat fixtures, tooling, production-readiness, warranty, initial spares, or routine commissioning support inside every trainset. Fixtures and tooling are carried in the production-plant line; spares, warranty response, and routine commissioning support are OPEX.

The rolling-stock BOM carries line-level low/base/high bands in generated path `build/bom/rolling_stock_bom.csv`. For the `light-metro-3car`, the direct-material band is 516,027-1,031,521 USD before labour; adding the BOM's 28% assembly allowance gives a 660,515-1,320,347 USD older marketplace-floor consist band, with the base case landing at 848,115 USD. This remains an audit lower bound; the current build estimate above supersedes it for trainset planning.

Railway Production Plant

Each country carries one shared railway production-plant setup allowance; cities do not duplicate the factory in city CAPEX. The national plant is sized to the largest single-city fleet programme and reused through a phased rollout. The base case is $60 k USD per vehicle/car module, not per trainset; the earlier $120 k USD per vehicle/car module value is kept as a high sensitivity check rather than the default.

This line covers lean local production/assembly setup: reusable one-metre panel moulds, clip/drill gauges, basic steel fixtures, plant services, commissioning bay setup, material handling, homologation/production-readiness work, and first-article support. It is deliberately separate from the trainset unit above, so procurement costs and national plant setup remain auditable instead of being hidden in one large rolling-stock number.

The generated LM3 pilot factory plan sizes the minimum enclosed building at about 3,515 m2 (37,835 ft2), plus 2,200 m2 of outside yard/test apron and a separate short depot/test track. Its rough machinery and setup list totals $1.0 M, including 20% equipment setup contingency. This one-time national factory setup remains separate from the per-trainset build estimate.

Example	Base plant allowance	High sensitivity
1-car vehicle module	$60 k	$120 k
3-car `light-metro-3car` trainset	$180 k	$360 k

Procurement Origin and Capital Boundary

Each generated city and national brief separates imported value from local value. Imported value is the minimum foreign-currency / international capital requirement; local value can be funded with domestic-currency bonds, public equity, or other local sources. Until a country supplier audit is available, the controlled planning shares are:

CAPEX bucket	Imported share	Local share
`civil`	15%	85%
`stations`	20%	80%
`depots`	25%	75%
`rolling_stock`	35%	65%
`production_plant`	20%	80%
`solar_plant`	45%	55%
`signalling`	50%	50%
`charging_microgrid`	40%	60%
`epc_overhead`	15%	85%

These are localization-first targets: standard structures, fabrication, installation, wiring/cabinets, software integration, and project services are assigned locally, while specialist cells, power electronics, control hardware, PV equipment, and initial machinery remain imported. They are planning assumptions until replaced by a country rules-of-origin and supplier-capability audit.

Foreign-Turnkey Comparator

The city finance summaries and national briefs include a controlled, editable foreign-company turnkey sensitivity. It is not a vendor quotation or a claim about any named supplier. The comparator applies a cost multiplier to the same OSR network, fleet, service, and energy scope, then estimates the share of that price requiring foreign currency or international capital.

Variable	Controlled value
`low` cost multiplier	1.50× OSR CAPEX
`default` cost multiplier	2.00× OSR CAPEX
`high` cost multiplier	3.00× OSR CAPEX
Foreign-turnkey external-capital share	90%

Illustrative variable benchmark for an equivalent foreign-company turnkey delivery. It excludes tunnels, land, tax/duty, utility relocation, financing fees, and escalation on both sides; it does not represent a received bid or named vendor price. The reported savings are calculated as foreign-turnkey external capital minus OSR imported content; annual savings use the same country construction period. Replace the multiplier and foreign-capital share with normalized bids before procurement or investment approval. Lifetime external-interest savings hold financing terms constant: both OSR and the foreign-turnkey case use the same country external rate, construction interest period, and repayment tenor. The comparator treats its external capital requirement as debt-financed. Interest comprises interest-only payments during construction plus total level debt service after grace less principal. The headline lifetime saving is avoided external capital plus avoided external interest; it does not include local-bond interest or OPEX.

Civil Works

Civil work is costed as a design-derived planning target for standard-gauge, double-track OSR alignments:

Civil class	Unit cost	Included scope
At-grade	$2.6 M / route-km	UIC60 rail, slipformed ballastless trackform, direct-fixation fasteners, drainage, cable troughs, local installation
Elevated	$9.7 M / route-km	Decked pi-beams in semi-continuous units, piers, zone-selected foundations, fewer bearings/joints, trackform and erection
Bridge	$18.0 M / route-km	Longer-span/water-crossing system; benchmark retained pending bridge-specific redesign and foundation schedule
Elevated-interchange premium	$4.5 M / site	Added stacked-platform and approach complexity where an interchange must grade-separate

These are design-derived planning targets, not supplier quotations. The generated civil cost contract indexes the retained $3.0 M / $12.0 M / $18.0 M benchmark against CAD quantities; bridge remains unchanged. Values are also below turnkey metro-bid benchmarks because OSR excludes tunnels, overhead catenary, proprietary signalling civil plant, bespoke station architecture, and contractor-led EPC margin.

Stations

Station costs are prefab portal-frame canopy + ground-level platform slab/guideway channel + direct pedestrian access + simple MEP/signs/CCTV/fare gates. Overbridges, lifts, stairs, and concourses are not the default at-grade station assumption; they appear only where an elevated/stacked interchange or local road-barrier override requires them.

Station archetype	Unit cost
`halt`	$600 k
`standard`	$2.5 M
`major`	$4.5 M
`terminal`	$4.5 M
`depot-terminal`	$5.0 M
`interchange`	$8.0 M
`interchange-elevated`	$12.0 M

Depots

Depot archetype	Unit cost
`main-heavy`	$8.0 M
`secondary-medium`	$3.0 M
`layup-minimal`	$400 k

Depot scope is maintenance rather than fleet-wide parking: main-heavy workshop/inspection roads, pits, lifting and wheel tooling, wash/defect functions, and local PV/storage tie-in. Healthy sets stable and recharge at powered passenger stations overnight; secondary and layup sites are site-specific exceptions.

Charging Microgrids

There is no route traction-power system in the OSR baseline: no OCS, third rail, feeder substations, or continuous traction distribution along the railway. The energy infrastructure cost in city designs is therefore station/depot charging microgrid interface CAPEX.

Station archetype	Unit cost	Included scope
`halt`	$0	No charger or trackside energy site; shelter electrical service is in station CAPEX

Station archetype	Unit cost	Included scope
`standard`	$100 k	500 kW class conductive charger, switchgear, inverter interface
`major`	$100 k	Larger queueing/anchor-stop charger and buffer tie
`terminal`	$100 k	End-of-line charger with higher turnback utilization
`depot-terminal`	$250 k	Passenger-stop charger plus depot/yard charging interface
`interchange`	$100 k	Multi-platform charger/switchgear allowance
`interchange-elevated`	$125 k	Elevated multi-platform charger/switchgear allowance

Station PV canopies, stationary LFP packs, depot buildings, and train batteries are not re-billed here. They appear in station, energy-site/depot, and rolling-stock scopes respectively.

Dedicated Solar Plant

Generated city READMEs add a separate utility-scale solar plant or contracted offsite solar PPA asset when the timetable traction-energy model exceeds station/depot PV generation. The plant is sized from the annual shortfall after on-site PV, with a 115% planning coverage margin, and uses:

Item	Planning rate
Utility PV field	$700/kW
Grid interconnection / PPA tie-in	$100/kW
Annual plant O&M	1.5% of plant CAPEX

This plant is carried as infrastructure CAPEX. Its O&M is carried in annual traction-energy OPEX; grid/PPA energy purchases are charged only for any residual import after on-site PV plus the dedicated plant.

Train-Control Wayside

Residual train-control wayside is budgeted at $50 k per route-km. The expensive ATP/ATO function lives onboard in the trainset cost. The wayside scope is sparse W-Nodes at switches/stations, passive balises, validation beacons, LoRa gateways, and OCC interfaces.

Revenue Neutrality

City READMEs include a post-opening operating-neutral revenue case. The model uses an 8% median-income monthly pass for the stronger service/revenue case, derives annual paid trips from practical system capacity and the configured low/high `capacity_utilization_*` bracket, and adds station shop leases plus advertising boards. The operating-neutral column solves the capacity utilisation needed so:

```
farebox + station-shop leases + advertising
= annual OPEX
```

The default farebox-recovery reference in `country-finance.toml` is 50%. OPEX uses the generated fleet schedule for train-km. On-site PV generation offsets traction demand first, the dedicated solar plant covers the remaining planned shortfall, and only residual import is charged as grid/PPA energy using `grid_energy_usd_per_kwh` from `lib/templates/country-finance.toml`. Driverless labour is no longer a flat route-km scalar: the README roster scales with service hours, lines, revenue fleet, station archetypes, high-case paid trips, annual train-km, depots, and the RFC 0015 shift of safety staff from train cabs to OCC and platform posts.

For Baghdad only, the primary structured funding programme replaces the generic uniform-construction funding screen. Imported purchases are split 50% government USD cash / 50% proposed Chinese USD credit, assuming full-basket supplier-origin and lender eligibility can qualify. Government contributes 25% of total capital including that USD cash; its remaining contribution is IQD; the residual after that contribution and Chinese proceeds is split 75% IQD bonds and 25% IQD bank term credit. Other cities are excluded. Additional cash requirements are unfunded if public cash is capped at the capital contribution. Monthly native-currency balances, staged draws, fees, cash reserves, public support and uncovered downside gaps are explicit. Rates and tenors are assumptions, not offers. The retained Iraq income figure is a planning proxy, not a verified current household median.

The maintenance percentages are planning-cost envelopes. The actual work content and inspection intervals are controlled by RFC 0029 and `lib/templates/maintenance-schedule.toml`, covering rolling stock, stations, track/civil, structures, energy, signalling/comms, depot equipment, and railway production-plant tools.

Construction-period local equity and interest-only grace payments on the repayable tranches remain public capital commitments. The base finance boundary assumes no climate/development grant: imported value is the minimum external climate/MDB or foreign-currency requirement, while 80% of local value is assigned to domestic-currency bonds and the balance to local public equity or another domestic source. The external tranche is a placeholder for a lender term sheet, not

evidence of an available loan. Plausible channels include MDB lending blended with climate funds such as GCF or CIF, or an equivalent national development bank / IsDB route where eligible. The operating-neutral case applies only to steady-state operations after opening. Where the capacity-use scenario produces revenue above OPEX, that operating surplus is netted against repayable-debt support in the government commitment summary; the gross post-grace external and local-bond debt-service figures remain visible.

Broad Economic Benefits

Generated city READMEs include a `Broad economic benefits` screening section sourced from `lib/templates/economic-benefits.toml`. It is not a formal benefit-cost analysis; it is a transparent first-pass calculation for channels that matter to cities and development lenders but do not appear as railway revenue.

The annual benefit/activity proxy quantifies:

Channel	Model basis
Travel time + reliability	Annual paid trips from capacity use x minutes saved x median-income value-of-time proxy
Congestion relief	Paid trips x average trip length x road mode-shift share / vehicle occupancy
Environmental effect	Avoided road CO2e minus rail residual-grid CO2e, valued at $80/t social-carbon proxy
Local road externalities	Avoided road vehicle-km x air/noise/safety proxy
Commerce and entertainment	Relevant trip shares x a median-income local-spend proxy

The access table reports education, healthcare, commerce, and entertainment/community access-events per year. It uses station anchors (`anchor_kind` / `anchor_name`) where available, with conservative base shares so sparse-OSM cities do not report zero service-access benefit.

The CAPEX recirculation table estimates how much of the initial capital programme is retained locally through civil works, station fabrication, depot works, shared national railway production-plant setup, rolling-stock assembly, charging microgrids, EPC labour, and solar-plant delivery. The retained CAPEX is then multiplied by the 1.6 construction local-supplier / wage multiplier and converted to approximate construction job-years using the country median-income table and the 4.0x job-output multiple. These rows are economic-activity indicators, not fiscal income.

EPC

EPC integration and project management is 7% of subtotal:

```
city: civil + stations + depots + rolling_stock
+ residual_train_control_wayside + charging_microgrids
national: one shared railway_production_plant
```

Dedicated solar plant CAPEX is then added as a separate infrastructure bucket when the generated energy plan requires it.

Country labour/material multipliers are applied downstream through `lib/templates/country-costs.toml` when a local tender view is needed.

Automatic mapping: population, streets and traffic

Repository source: `docs/data/automatic-mapping-sources.md`

Reviewed 11 October 2026. Baghdad now has a source-selected mapping evidence package alongside its retained 2020 access audit. The compact corridor planner uses native WorldPop Global2 2026 counts and reserves two additional investigation corridors for the northeast planning sector and two for the west, within the eight-line addition budget. It retains the original nine trunks and the original native 1.6/3/7 km density-dependent station settings. Sector reservations guide selection; they do not establish construction, walking access or operating acceptance.

Source	Use in OpenSourceRail	Availability and limits
WorldPop Global2 Iraq, 2026, 100 m	Imported native count pixels; selected for compact route prioritisation and sector audits	CC BY 4.0; R2025A v1 alpha modelled projection, not a census. Approximate 100 m means 3 arc-seconds; physical east–west width varies with latitude.
JRC GHS-POP R2023A	Imported 2025 native 3 arc-second tile R6 C23; separate coverage comparison	Reuse with attribution. Modelled projection; different allocations and epochs. Do not average grids, add populations together or interpret their difference as observed growth.
Iraq 2024 census announcement, UNFPA	Preferred future administrative calibration when compatible small-area records and boundaries are obtained	National announcement verifies a real census, but does not supply georeferenced Baghdad enumeration blocks for route placement. Never scale a city bbox to a governorate or national total.
Geofabrik Iraq OSM extract	Imported dated 2026-10-09 PBF; retained city pedestrian topology, street geometry and building inventory	ODbL 1.0, attribution to OSM contributors and Geofabrik. Shared OSM node IDs define connections; overlapping coordinates do not connect bridges or tunnels. Mapping completeness varies.
Overture Maps	Next supplementary source for building footprints, named destinations and transportation cross-checks	Bbox downloads and pinned releases are available. Licences differ by theme and source; transportation is ODbL. Not imported into the current selector, and not a traffic-measurement service.
Microsoft Global ML Building Footprints	Alternative settlement-completeness cross-check	CDLA Permissive 2.0. Imagery dates vary; an update date is not the observation date of every building. Footprints do not establish dwellings, occupancy or current population. Not imported into current evidence.
TomTom classic Traffic API coverage, Orbis coverage	Screened and excluded as a Baghdad measured-traffic input	Both reviewed tables omit Iraq and state that omitted markets are unsupported. Map display coverage does not establish traffic API coverage. No API account was created.
TomTom flow-segment semantics	Reference for future speed/travel-time import semantics in supported markets	Speed, travel time, direction and confidence are separate from vehicle counts and passenger OD demand. Product-specific rights must permit retention and publication.
HERE Traffic API	Possible commercial investigation	Baghdad product coverage and usable retention/redistribution terms have not been confirmed; no credentials, traffic records or coverage claim are incorporated.

WorldPop's country census input metadata identifies Iraq population estimates/projections for 2003 and 2018, with the second timepoint used for modelling. Consequently, the newer grid must not be described as incorporating the 2024 Iraqi census. Updated settlement covariates and projections make it a useful planning scenario, with source disagreement retained visibly. Its release statement explicitly describes the alpha status and limitations.

Executable integration

`refresh-city-mapping.py --city baghdad --fetch` downloads the explicitly selected sources, retains native bbox pixels, extracts the pedestrian graph and regenerates the comparison. `--check` reproduces the report offline and fails on altered pixels, graph, bbox, source selection or producer code. Fetching requires `rasterio` and `osmium`; the retained-data check uses NumPy/SciPy and does not require network access or OSM parsing. Country files are cached outside Git; only bounded city evidence is published.

The original 2020 evidence remains a separate dated source. It is never relabelled, its receipt is not rewritten to claim a newer epoch, and it still governs the existing catalogue access measure. Mapping reports expose all three source/epoch estimates. Native counts are summed once per catchment union and remain separate from smoothed routing scores, financial trips and fares.

The compact planner performs northeast, west and citywide phases, accounting for already-selected stop circles before subsequent gains. Sector-specific counts guide reserved phases, while the reported incremental gains are recomputed on the full selected population union. Candidate paths retain existing water, buildability, backtracking, ring-junction and route-length guards. The final emitted station inventory is checked independently; a priority cell is not proof that a native stop survived consolidation. Native synthesis retains the existing validated cost/demand/water/buildability routing grid. The new population source changes bounded corridor selection; it has not replaced every native station-density input or the complete road-cost raster. Fresh settlement/destination and routing-grid adoption requires its own reproducible regeneration.

The pedestrian screen uses shortest paths on actual OSM topology with a virtual station source preserving each station's own snap cost. Both station and population snaps are bounded at 150 m. These connections, entrances, sidewalks, access permissions and accessibility remain unverified. The screen never calls an arbitrary river-crossing snap an accepted connection. Border clipping can underestimate access, while missing restrictions and unsurveyed short snaps can overestimate it. A graph-access result is therefore neither a surveyed walkshed nor a lower-bound guarantee.

Evidence needed for traffic-led routing

No verified open Baghdad road-level traffic counts, speeds or OD table were obtained in this review. The pipeline records these as unavailable. Road classes, building counts and synthetic SUMO flows cannot fill that evidence gap.

An eventual import must retain the observation time and interval, coordinate reference system, road/segment ID and direction, vehicle/person units, count versus speed classification, sample/confidence information, provenance and publication rights. Snap observations to a dated street network with explicit unmatched records. Keep peak/off-peak and weekday/weekend evidence separate; never convert road speeds directly into rail passengers. Household travel surveys, jobs, schools and actual transit demand are needed to justify mode share and fares.

Surveyed entrances and legal pedestrian crossings, administrative census boundaries, compatible 2024 small-area census records and traffic observations remain open inputs. Their absence does not prevent numerical planning, but it prevents claiming validated local access or demand.

Selected mapping policy · Retained Baghdad evidence.

Unified Deployment Model

Repository source: `docs/deployment-model.md`

OpenSourceRail has one deployment model. A city such as Samawah, Baghdad, Karachi, or Kinshasa is an instance of the same pipeline, not a fork of the architecture.

Public portfolio evidence uses the developing-world scope. Any European comparison design remains available for technical inspection but is not included in those totals or examples.

Rule

No deployment gets a special OSR stack. City-specific information is limited to inputs and generated artifacts:

- city record in `lib/city-batches/world-sample.toml`;
- country, cost, finance, climate, regulatory, and workforce templates under `lib/templates/`;
- optional recipe adjustments in `lib/recipes/`;
- generated design outputs under `cities/catalogue/<region>/<country>/<city>/`;
- deployment-specific survey, asset, operator, and assessor evidence.

Everything else stays shared: train families, track geometry presets, station archetypes, hardware host classes, operations rules, safety requirements, certification evidence structure, and regeneration tools.

Pipeline

1. Add or update the city record and country assumptions.
2. Select the city in Workbench Generate a city digital twin (the same deterministic generator remains available to automation).
3. Review the generated network, scenario, cost, energy, fleet, station, depot, and quality outputs.
4. Export civil alignment from the generated design through OSR-ALN.
5. Build the pilot hardware through the RFC 0019 COTS/SBC integration track, or choose the RFC 0007 custom-board track when volume or deployment constraints justify it.
6. Fill the deployment evidence gaps: survey, supplier quotes, workshop audit, bench logs, first-article tests, operator review, independent assessor review, and regulator submission.

Samawah's Role

Samawah is the worked Iraq instance because it combines a generated city network with a plausible brownfield rail-yard/workshop asset base. It is not a special architecture, special train, special hardware path, or special certification path.

When a city-specific document disagrees with the generated city design, the generated design wins. Superseded worked examples should be removed rather than kept as parallel plans.

Artifact	Role
`cities/catalogue/west-asia/Iraq/Samawah/`	Authoritative generated city model and public planning numbers
`docs/rfcs/0003-samawah-reference-deployment.md`	Context, pilot rationale, risks, and next steps
`docs/stations/standard-archetype/`	Shared station-archetype worked example; city-specific station lists come from generated designs

Outputs To Keep Unified

Area	Shared source	City-specific output
City design	`osr-design`, `design/city-generation`, `lib/templates/`	`cities/catalogue/<region>/<country>/<city>/design.toml` and generated README
Civil alignment	RFC 0009 + OSR-ALN tools	per-line `.aln.toml`, survey replacement, civil-class summary
Rolling stock	RFC 0008 families + rolling-stock packages	selected family and fleet count
Stations	RFC 0010 archetypes	station archetype assignments and site adaptations
Energy	RFC 0002/0021/0024/0026	station/depot/solar plant sizing and utility study
Hardware	RFC 0019 integration or RFC 0007 custom boards	SKU freeze, wiring, enclosures, bench records
Certification	`docs/certification/`	assessor report, field evidence, national submission
Delivery controls	manufacturing/resource templates, BOM and finance model	family-scoped assembly plan, CPM, order-by plan, schedule of values, local/imported cashflow and recorded actuals

The detailed 120-product/26-assembly manufacturing release currently applies to `light-metro-3car`. Other-family cities receive their correct family cost, fleet and programme scope, but rolling-stock work packages are explicitly held at `family-specific-detail-required`; they never borrow LM3 part numbers.

Anti-Patterns

- Creating a city-specific train family when an existing family fits.
- Hard-coding Samawah, Baghdad, or any other city into shared tools.
- Treating a worked civil package as the source of truth after the generated design changes.
- Publishing cost, fleet, or energy figures outside generated outputs without a regeneration path.
- Forking safety requirements or operations rules by city instead of parameterising deployment evidence.

Deployment Planning Reference

Repository source: `docs/deployment-planning-reference.md`

This page defines the common interpretation of every generated OpenSourceRail city README and national brief. Location folders contain only local inputs, calculated results, exceptions and evidence links. Shared assumptions belong here or in the linked machine-readable templates—not in hundreds of generated copies.

Generation And Review Boundary

The city catalogue is deterministic:

```
city catalogue + OSM/raster evidence + shared templates
  → routed network + stations + service + fleet
  → energy, civil, CAPEX and finance calculations
  → engineering and simulation evidence
  → concise city README and national aggregation
```

Generated READMEs are indexes and decision summaries. The authoritative local inputs and results remain `design.toml`, the expanded scenario TOML, and the JSON/GeoJSON evidence inside each city folder. Git records changes to inputs, generators and outputs together. Do not hand-edit a generated city or country README; change its source or a shared template and regenerate it.

Network And Station Planning

`osr_geo` rasterises source-locked OpenStreetMap roads, buildings, water, protected areas and demand anchors onto the planning grid. `osr-design` uses a demand-rewarded least-cost route, places stations against the demand surface, and classifies each segment as at-grade, elevated or bridge. Tunnels are not in the upstream catalogue.

The standard spacing policy is 1.6 km in central areas, 3 km in ordinary urban areas, and up to 7 km on low-demand outer approaches. Cross-line platforms inside the 600 m station-complex envelope are consolidated as one interchange. The retained `*.design-quality.yaml` coverage score is the fraction of high-demand cells within 20 routing cells of a track, irrespective of station access. It is not anchor-weighted resident coverage and must not be multiplied by the catalogue population. Each city's `engineering/access/README.md` instead separates direct transfers, reachability through intermediate lines, and native population counts within the union of station circles at 500/800/1,000/1,500/2,000 m. Population year, source-pixel resolution, bbox denominator and missing evidence remain explicit. Wider radii are access/feeder sensitivities; radial distance does not prove a walkable route across rivers or other barriers or create fare demand. No current population total is inferred by rebasing a 2020 raster.

Straight viaducts also require separate beam, support/foundation and vertical terrain checks. Every city retains a source-bound `engineering/clearance` screen; missing heights or mapping prevent an obstacle-clearance claim. See the viaduct obstacle-clearance policy.

Service, Fleet And Capacity

The default service is GoA 4 battery-electric operation with line-specific time windows and headways. Fleet sizing covers peak revenue service, planned maintenance spares and cold reserve. Off-peak surplus performs terminal or depot inspections without creating a second service-rotation fleet unless a local schedule proves one is required.

Healthy trainsets may stable overnight at protected powered stations near their first departure. Red defects, overdue heavy maintenance, isolation failures or security failures return a set to the main depot. Station stabling requires charging, CCTV, remote traction isolation, emergency access and an OCC-assigned track slot.

Capacity uses the selected trainset family's AW2 planning capacity, the actual generated timetable and a controlled practical load factor. The annual paid-trip range is a capacity-use scenario, not a ridership forecast. Replace it with a surveyed demand model before an investment decision.

Energy Method

Each city calculation combines:

- trainset energy per car-km and the generated timetable;
- climate and non-revenue movement factors;
- opportunity charging at powered stations;
- station/depot PV and stationary LFP storage;

- a dedicated solar plant or equivalent PPA asset for the remaining annual traction-energy requirement; and
- residual grid import after those assets.

The generator checks every line's powered-stop gap, traversal energy, delivered charge, protected onboard reserve and operating margin. These are feasibility screens; grid connection, protection, harmonics, thermal, fire and local solar yield studies remain deployment work.

Civil And Cost Method

Civil geometry is generated from the parametric source and exported to IFC4.3 for Bonsai coordination. Checked quantities feed the generated `civil-cost-model.toml`, which is then consumed by city CAPEX, finance, IFC properties and documentation. Current design targets are 2.584 M USD/route-km at-grade, 9.748 M USD/route-km elevated and 18 M USD/route-km for bridges. The retained 3/12/18 M benchmarks remain visible for audit.

These figures are unquoted planning targets. Land, utilities, tax and duty, site geotechnics, project risk, programme, independent checking and local supplier rates must be added or validated before procurement. The complete shared cost basis is in the generated cost model.

Capital And Finance Method

City CAPEX includes civil works, stations, depots, rolling stock, residual train control, charging interfaces, timetable-sized dedicated solar and EPC/project services. The national trainset factory is excluded from cities and counted once in each national brief, sized to the largest single-city fleet programme.

The catalogue's controlled core concepts adopt elevated land sections and straight radial tangents where analytical geometry permits. Water crossings remain bridges; retained unsuitable fragments keep their geometry review gates. Core boundaries are design-centroid planning screens requiring survey, property and obstacle confirmation. Immutable seeds preserve the original line inventory, including rings.

The current line-depot scope stores the full fleet, including spares, in one planning depot per line. Train length sets storage track quantities; measured maintenance workload remains a release requirement for workshop bays. Depot PV/storage reference equipment is included once in depot capital. Land, utility upgrades, installation and site acceptance remain open.

Station staffing uses two posts, two normal eight-hour shifts and additional cover for the actual service window. Paid hours are reduced for leave, training, sickness and handover. Country income proxies are retained estimates: basic wages start at 150% of that proxy, technical grades at 225% and management at 300%, plus 25% employer and 10% overtime allowances. Baghdad's dedicated revised scope retains its separate indexed Iraqi wage evidence.

City-order factory requirements use the actual train family and consist. Facility readiness is 18 months; qualification and serial manufacture follow it. Short civil programmes explicitly wait for fleet readiness. The national factory allowance is the larger of the module-based allowance and independent city-order physical envelopes, counted once. Concurrent national loading and sequencing require a separate accepted programme.

The procurement-origin model separates imported value—the minimum foreign currency or international-capital requirement—from locally supplied labour, materials, fabrication and services that can use domestic funding. Country finance parameters define construction period, debt rates, tenor and the local bond/equity split.

For Iraq, the primary proposal is the schedule-linked funding programme. It assigns proposed Chinese buyer credit only to assumed eligible imported component invoices, then splits the remainder among government appropriations, IQD bonds and IQD bank credit. The controlled assumptions remain uncommitted. Monthly native-currency debt schedules, interest on staged draws, fees, reserves, public support and downside cash gaps are explicit. Construction uses the actual resource-constrained procurement milestones under an assumed 260-working-day year; the generic five-year financing comparator does not define the Iraq construction schedule. Physical and operational release gates remain open even when all planning example artifacts are regenerated.

The foreign-turnkey comparison is an editable like-for-like sensitivity, not a vendor quotation. Both cases use the same network, fleet, service, energy scope and financing schedule. Replace the multiplier, external share and all funding terms with scope-normalised bids and signed lender terms before approval.

OPEX, revenue, NPV, IRR and DSCR outputs are planning screens. Capacity-use scenarios are not demand forecasts, and no grant, tax treatment, land-value capture or lender commitment exists unless explicit controlled evidence says otherwise.

QA, Maintenance And Assurance

Shared construction QA gates, manufacturing travellers and maintenance intervals are generated from the controlled templates. City-specific asset counts and schedules are materialised in the local operations bundle. The common system covers civil works, trackform, stations, rolling stock, passenger systems, energy, signalling/comms, depots and integrated trial running.

Generated evidence proves internal consistency and reproducibility—not railway certification. Survey, geotechnical release, structural design, fire/egress, product acceptance, calibrated operations modelling, independent safety assessment and authority approval remain deployment responsibilities.

Local Evidence Contract

Each city README links the current status of these local artifacts:

Artifact	Purpose
`design.toml`	Authoritative generated network, fleet and CAPEX design
`<slug>.toml`	Expanded simulator scenario
`<slug>.design-quality.yaml`	Coverage, routing and civil-quality gates
`engineering/finance/summary.json`	Reconciled CAPEX, OPEX and finance cases
`engineering/simulation/validation-summary.json`	Nominal and degraded simulation acceptance
`engineering/gis/summary.json`	GIS package provenance and layer counts
`engineering/sumo/summary.json`	Executed timetable screening
`engineering/energy/summary.json`	Grid, charging and solar screening
`operations/`	Asset, QA, manufacturing and maintenance evidence

The README generator verifies source hashes and passing status before publishing these evidence links. Repository health also regenerates city and country pages and rejects drift.

Regeneration

```
# One city
tools/automation/regenerate-city.sh samawah

# Complete catalogue, using current caches
tools/automation/regenerate-all.sh

# Documentation, links and generated drift
python3 tools/automation/check-readmes.py
python3 tools/automation/check-markdown-links.py
python3 tools/automation/repo-health.py --quiet
```

Changes to common assumptions should normally touch one template or this page, then regenerate affected outputs. Location folders should change only when their local input, calculated result, evidence or explicit exception changes.

Deployment Roles

Repository source: `docs/deployment-roles.md`

OpenSourceRail separates open reference artifacts from the organizations that carry legal, financial, and safety responsibility for a real railway. A deployment should name each role before any safety-critical trial begins.

Role Map

Role	Carries	Typical evidence or decision
Owner / public sponsor	Budget, public objectives, asset ownership, residual public risk	Business case, funding approvals, land access, public-service obligations
Operator	Operating rules, competence, service delivery, degraded-mode response	Rulebook acceptance, staffing plan, training records, trial-running reports
Entity in charge of maintenance	Maintenance system, asset condition, spares, inspection records	Maintenance plan, work-order closeout, defect/NCR history, handback signatures
Prime integrator	System integration, interface control, technical delivery, safety-case assembly	Interface control documents, configuration baseline, integration tests, evidence index
Independent safety assessor	Independent review of safety argument and evidence	Assessment report, action log, residual-risk findings
Regulator / national safety authority	Legal approval for trial running and revenue operation	Authorization, conditions, operating certificate, accepted residual-risk decision
Insurer / liability underwriter	Insurability, claims framework, liability exclusions	Insurance terms, risk survey, required mitigations
EPC / civil contractor	Survey, civil works, structures, stations, depots, trackform, construction QA	Survey package, method statements, ITPs, as-built records, handover certificates
Rolling-stock production plant	Local manufacture, tooling, fixtures, first article, production QA	BOM, routings, weld/NDT records, first-article inspection, configuration records
Hardware integrator	COTS/SBC or custom-board build, wiring, enclosures, bench tests, commissioning	SKU freeze, harness maps, power/thermal margins, self-test logs
Energy and utility partner	PV, BESS, chargers, grid/PPA, protection settings, energy operations	Grid study, interconnect approval, charger thermal study, tariff/PPA
Financing entity	Debt, grant, guarantee, disbursement conditions	Term sheet, covenants, eligibility evidence, procurement conditions

OSR Project Boundary

OpenSourceRail can provide:

- reference architecture and RFCs;
- simulator and generated city designs;
- operations portal, asset registers, QA, maintenance, manufacturing, and evidence templates;
- safety-case scaffolds, tests, formal models, and release-gap registers;
- hardware and rolling-stock reference packages.

OpenSourceRail does not provide:

- statutory operating approval;
- product liability cover;
- sovereign guarantees or project finance;
- independent safety assessment;
- construction warranties;
- operator competence certification.

Those duties belong to the deployment roles above.

Minimum Role Closure For A Pilot

A non-safety digital pilot can start with only an owner/operator sponsor and an implementation team. A closed-track or depot movement pilot needs more:

Gate	Roles that must be named
Digital-only shadow mode	Owner/operator, implementation lead, data owner
Depot or closed-track movement	Owner/operator, prime integrator, entity in charge of maintenance, safety reviewer, insurer or risk owner
Public trial running	Owner/operator, prime integrator, independent safety assessor, regulator, insurer, civil/EPC, maintainer

Gate	Roles that must be named
Revenue operation	All roles in the role map, with accepted safety case and operating authorization

Relationship To Other Docs

- Organisational mobilisation, governance, staffing and stage gates: `owner-builder-operator-setup.md`
- Fail-closed role, independence and G0–G7 evidence state: `owner-builder-operator-mobilisation-status.md`
- Unified city/deployment pipeline: `deployment-model.md`
- First adoptable product: `first-adoptable-product.md`
- Certification gaps: `certification/release-gap-register.md`
- Acceptance evidence basis: `certification/evidence-status.md`

Engineering Design And Simulation Plan

Repository source: `docs/engineering-design-simulation-plan.md`

Status: active implementation and evidence plan Reviewed: 2026-09-04 Scope: repository-owned engineering work and external deployment-evidence gates across COMP-014 and COMP-016 through COMP-019

This plan selects an open-source engineering toolchain and turns the remaining mechanical, station, civil, energy, and system-validation gaps into evidence-producing work packages. It does not treat a solver result as design approval: survey data, supplier data, applicable-code checks, and competent local engineering review remain release inputs.

Decision

No single application should own the whole design. The repository keeps the following authority boundaries:

Design information	Authority	Review or analysis handoff
Parametric rolling-stock, component, station, and civil-kit geometry	`design/component-catalogue/src/osr_mech/`	Native FreeCAD `.FCStd`, STEP, drawings, solver meshes
Surveyed route, levels, control, parcels, utilities, and terrain	Deployment GeoPackage in QGIS; approved OSR-ALN for railway alignment	LandXML, GeoJSON, GeoTIFF, LAS/LAZ/E57
Product structure, quantities, and assembly sequence	Generated repository BOMs, travelers, and assembly documents	IFC property sets and drawing schedules reference stable OSR part IDs
Operational behavior and measured duty cycle	OSR Rust simulator and logged field evidence	CSV/Parquet or versioned JSON input decks for specialist tools
Federated station/civil coordination model	IFC4.3 export checked with IfcOpenShell and reviewed in Bonsai	BCF/IDS findings, quantities, drawings, and 4D task-product links
Analysis evidence	Versioned input deck, assumptions, solver/version, convergence log, and reviewed summary	Results are evidence, never a replacement for canonical geometry or requirements

FreeCAD is the baseline component-CAD, assembly-review, drawing, and first-pass FEA environment for mechanical, station, and reusable civil-kit parts. It is not the authority for surveyed routes or the federated civil model. QGIS remains the primary geospatial and alignment-review environment. The FreeCAD Road/Trails family may be evaluated for visual coordination, but it must not become the surveyed alignment authority until an OSR-ALN round-trip benchmark passes without station, curve, level, cant, or CRS loss. Bonsai is the civil federation, detail-review, quantity, and 4D construction environment, not the route-design authority. The implemented Bonsai/IFC4.3 workflow carries stable OSR IDs, checked review geometry, an `IfcAlignment` reference, and construction tasks without duplicating alignment or structural design rules inside Blender.

Selected Open-Source Toolchain

The “baseline” tools should receive reproducible runners and benchmark cases. “Conditional” tools are introduced only when their listed question cannot be answered adequately by the baseline. “Evaluation” tools must pass a small interchange trial before project data is committed to them.

Work area	Selection	Class	Intended use and boundary
Parametric component CAD, assemblies, drawings	FreeCAD	Baseline	Mechanical, station and reusable civil-kit geometry; Assembly/Part Design/TechDraw and IFC/STEP handoff. The generated evidence records the actual tool version used.
First-pass structural and thermal FEA	FreeCAD FEM, CalculiX, and Gmsh	Baseline	Repeatable component, frame, bracket, platform, and catalogue-structure screening; FreeCAD calls external meshers/solvers, so each evidence pack records all three versions
Survey/GIS/alignment review	QGIS and GDAL	Baseline	Control, terrain, parcels, utilities, flood layers, station siting, CRS control, and LandXML/OSR-ALN review
GNSS processing	RTKLIB	Baseline when field data arrives	Static/kinematic GNSS processing and reproducible control reports; a licensed surveyor owns control acceptance
Photogrammetry and point clouds	OpenDroneMap and CloudCompare	Baseline when field data arrives	Orthophoto/DEM generation, registration, cleaning, cross-sections, and clearance comparison; raw captures stay outside Git
IFC federation and checking	IfcOpenShell/Bonsai	Baseline	IFC4.3 rail hierarchy, deterministic civil geometry, stable OSR IDs, quantities, BCF/IDS coordination and 4D sequencing; IFC is a coordination artifact rather than a second geometry or alignment authority
Drainage and station runoff	EPA SWMM	Baseline	Design-storm runoff, inlet/channel/pipe/storage routing, surcharge, and station/depot drainage scenarios
Global structures and soil-structure response	OpenSees	Baseline for COMP-017 structures	Per-span frame response, bearings, pier families, nonlinear/seismic cases, and soil springs using local design inputs

Work area	Selection	Class	Intended use and boundary
Detailed nonlinear structural cross-check	Code_Aster	Conditional	Concrete, contact, fatigue, or thermomechanical cases that exceed the verified CalculiX template; use on a benchmark before project models
Groundwater and coupled ground response	OpenGeoSys	Conditional	Settlement/groundwater/thermal-hydraulic questions supported by boreholes and laboratory parameters; it does not replace a ground investigation
Passenger movement and egress	JuPedSim	Baseline	Normal, degraded, interchange, and evacuation pedestrian scenarios using demand ranges and measured/calibrated flow assumptions
Fire and smoke	NIST FDS and Smokeview	Baseline for enclosed/high-risk spaces	Smoke, heat, tenability, battery-room and station fire scenarios; use JuPedSim separately for egress because FDS+Evac is no longer the selected workflow
Building energy and HVAC	EnergyPlus	Baseline	Station/depot loads, envelope, ventilation and HVAC energy; detailed CFD is added only when a lumped model cannot answer the design question
Battery electrochemistry, heat, and degradation	PyBaMM	Baseline after supplier data	Reproduce the selected LFP cell against supplier/lab curves, then apply measured OSR duty and high-ambient cases; a generic lithium-ion parameter set must not stand in for the selected cell
Solar yield	pvlib-python	Baseline	Site-specific plane-of-array yield, temperature derating, inverter clipping, and uncertainty cases
Site/depot electrical network	pandapower	Baseline	Load flow, voltage drop, transformer/cable loading, fault-level screening, charger coincidence, PV and storage scenarios
Rail operations and multimodal interaction	Eclipse SUMO	Baseline independent check	Timetable, junction, road interaction, feeder and pedestrian-demand sensitivity; OSR simulator remains authoritative for OSR control behavior
Railway timetable/capacity design	OSRD	Evaluation	Trial import of one Samawah line for running-time, conflict, capacity, and timetable comparison; adopt only after data round-trip and maintainability review
Commercial railway timetable/capacity cross-check	OpenTrack	Optional licensed evaluation	Do not replace SUMO or `osr-sim`; after lawful procurement, trial Samawah Line 1 for running time, block occupation, minimum headway and seeded robustness per the OpenTrack evaluation
Vehicle multibody dynamics	Project Chrono	Evaluation/conditional	Suspension, articulation and component motion studies; rail wheel/contact behavior requires an OSR benchmark and is not available merely by importing the CAD model

The supported toolchain uses isolated Flatpaks, a pinned Python environment, and versioned user-local native releases. Exact adopted versions, licenses, source URLs and checksums are in `engineering/toolchain/tool-manifest.toml`. OpenDroneMap remains gated on survey mobilisation; the conditional/evaluation solvers remain uninstalled until their adoption gate identifies a concrete design question.

Common Evidence Contract

Every analysis package must contain, either directly or through a generated manifest:

1. the requirement, hazard, drawing, assembly, BOM, and OSR part IDs being checked;
2. canonical source revision and input checksums;
3. tool, solver, mesher, plug-in, and operating-environment versions;
4. units, coordinate reference system, material/supplier data, loads, boundary conditions, uncertainties, and exclusions;
5. mesh/time-step/sensitivity study appropriate to the model;
6. convergence and warning logs, plus a hand calculation or published benchmark before a new solver family is trusted;
7. acceptance criterion, result, margin, reviewer, date, and disposition;
8. a compact tracked summary and input deck. Large raw captures, caches, and disposable result fields follow `repository-artifact-policy.md` and stay out of Git unless explicitly accepted as evidence.

Neutral interchange is limited deliberately:

Exchange	Required content
OSR-ALN/LandXML	horizontal and vertical geometry, cant, station IDs, CRS/vertical datum, source hash

Exchange	Required content
GeoPackage	control, alignment, stations, utilities, property, flood, boreholes, issues, provenance
IFC 4.x	stable OSR part/assembly IDs, placement, classification, status, material, drawing/BOM references
STEP	geometry handoff only; never trusted for requirements, material, fastener, or revision metadata
Analysis input deck	human-reviewable parameters and stable references back to canonical source
Results summary	machine-readable values plus a concise reviewed Markdown report

Execution Plan

Wave 0 — Reproducible tool and interchange baseline

- [x] ENG-TOOL-001 — Create the environment manifest. Pin FreeCAD, CalculiX, Gmsh, QGIS/GDAL, Python libraries, and each adopted solver. Add license, download source, checksum/container or package reference, and a command that prints the installed version.
- [x] ENG-TOOL-002 — Add repository runners. Provide scripts that place generated work under `build/engineering/`, refuse missing inputs, capture versions/logs, and write atomic summaries. Do not require GUI state for a reproducible analysis.
- [x] ENG-TOOL-003 — Add solver benchmarks. Start with a cantilever, thermal block, simple drainage network, one-zone energy model, four-bus electrical network, corridor evacuation, and one-line timetable. Compare analytical or published answers and set tolerances.
- [x] ENG-TOOL-004 — Add interchange drift tests. Check OSR-ALN LandXML/QGIS, OSR part IDs IFC, CAD geometry analysis mesh, and measured-duty export PyBaMM/pandapower/SUMO inputs.
- [x] ENG-TOOL-005 — Add an analysis register. Record model status as planned, screening, calibrated, independently checked, or accepted. Reject unlabelled screenshots and unconverged output as closure evidence.

Implementation status on 2026-09-29: ENG-TOOL-001 through ENG-TOOL-005 are closed at repository design-evidence level. The repository runner captures exact versions/hashes and exercises IFC creation, an analytical OpenSees case, pandapower, pvlib, PyBaMM, SWMM, EnergyPlus and FDS. It also runs a deterministic JuPedSim station corridor and a 12-service SUMO timetable generated from all three canonical Samawah lines. The generalized batch runner has now generated decks and station/product mappings for all 266 catalogue cities: 1,022 lines and 14,636 station occurrences, backed by seven shared geometric station-archetype IFC assemblies. The full live SUMO batch completed all 4,052 scheduled screening services with zero simulation failures. Its execution gate passes; its input-quality gate remains open for the endpoint findings below. ENG-TOOL-002/003 are closed for the repository-owned deterministic baseline; live-solver reruns and second-machine reproduction remain environment evidence, not source-level release claims. The CalculiX thermal block passes its analytical temperature and flux checks. ENG-TOOL-004 now passes station manifest ID, positive-volume representation, semantic class, property-set and assembly-hierarchy round trips for all seven archetypes. The fast tracked baseline now adds independent analytical forms for a cantilever, thermal block, drainage balance, one-zone energy balance, four-bus electrical network, corridor evacuation and one-line timetable. It also fails on LandXML/OSR-ALN unit or CRS drift, IFC identity/hash/envelope drift, and battery/grid/traffic duty-projection drift. Atomic report replacement prevents a failed run from being mistaken for fresh evidence. Live solver and second-machine results remain environment evidence rather than source claims; survey-coordinate/alignment drift and deployment-specific IDS/BCF checks remain open.

The civil federation now generates a byte-deterministic IFC4.3 model with 185 stable assets, 19 source-recipe component types, three source-backed material families covering 46 occurrences, one native 60E1 profile driving 32 rail extrusions, two typed `IfcVehicle/ROLLINGSTOCK` trainsets with standard base quantities, 36 native typed bearings, nine pier caps, nine columns, and 45 explicitly virtual foundation/jacking interfaces, 27 native support connections with 60 bearing realizations, 15 hash-locked native source documents associated to all assets and the civil source natively linked to all nine objectives and six numeric metrics, one internal classification with 15 references covering all 185 assets, four `IfcRailwayPart` disciplines, five native coordination groups, six native functional systems (three specialized as `IfcBuiltSystem`), seven system-to-railway-part references, and four native presentation layers covering all assets exactly once in each scheme, an `IfcAlignment` with native horizontal/vertical planning segments, gradient-curve representation and stationing, 18 construction tasks with five stage-specific output tasks, 134 physical product links and 45 separately identified virtual review-interface links, native quantities, provenance, and nine passing checks exposed as native interface constraints with six numeric `IfcMetric` benchmarks, three qualitative-only gates, and 107 scoped project/asset/group/system evidence links. Sixteen project-declared property/quantity dictionaries expose 99 typed fields without misusing the reserved `Pset_` prefix. One native USD `SCHEDULEOFRATES` exposes the three generated planning alternatives without product assignments, multiplied quantities, or a project total. Its twenty-specification IDS 1.0 contract passes 3,340/3,340 checks; deterministic BCF 3.0 topics retain object links and Git-reviewed decisions. Bonsai 0.8.5 imports the model headlessly and saves the review/animation scene. All source-supported IFC implementation work is closed; nine indexed decisions identify the survey,

supplier, engineering, client, commercial, and CDE evidence still required.

The city package now also converts canonical catalogue inputs into QGIS/GDAL GeoPackages, shapes SUMO edges from the corridor GeoJSON, and runs per-city pandapower/pvlib electrical and PV screens. Samawah and Mosul are the primary package acceptance cases and Songea is a portability check. Current nominal and degraded full-service simulation evidence now covers ten cities, as tracked in the deployment evidence workflow. Samawah's Line 1 corridor geometry is 2.3% shorter than declared chainage and its maximum station-to- corridor offset is 75.3 m; these remain explicit coordination findings pending survey-grade alignment. Its peak grid-only case overloads 14 planning transformers, while the coordinated daylight case converges without loading or voltage-band findings. Neither the mapping nor the energy result is survey, utility, or measured-weather acceptance evidence. The Samawah design now uses the `hot-desert` climate preset; project weather files and authority design temperatures remain external inputs.

The compact catalogue still contains inherited layouts emitted by older generator revisions. Catalogue-wide ring/interchange and station-cluster reports keep that migration backlog explicit. The current generator itself now fails closed on sub-600 m same-line spacing, ungrouped cross-line near misses, incomplete interchange complexes, and ring/radial approaches without a shared transfer before emitting a design. Batch regeneration therefore cannot create a new package with those layout defects; the retained compact snapshots must be migrated progressively and the reports refreshed after each batch. The catalogue-wide result cannot be marked input-quality-passed until their canonical alignments or station extents are corrected.

Exit: one deterministic benchmark per adopted baseline tool, a clean re-run on a second machine, and no loss of units, CRS, stable IDs, or source revision through the interchange tests.

Wave 1 — COMP-016 station variants and assembly reconciliation

- [x] COMP-016-01 — Freeze the station variation matrix. Compare standard, halt, major, interchange, interchange-elevated, terminal, and depot interfaces. Mark every envelope, service, access, compliance, structure, equipment, and assembly field as shared, parameterized, or unique.
- [x] COMP-016-02 — Generate all FreeCAD envelopes. All seven native files contain complete installed and hidden exploded geometry states. Their hash-locked sidecars carry coordinate-bearing track/train/optional-PSD/edge interfaces, maintainability and lifting/installation bounds, plus controlled track-centreline, top-of-rail, platform-face and boarding datums. Product geometry also exposes equipment maintenance envelopes; site swept-path and lift-plan approval remain release evidence rather than assumed results.
- [x] COMP-016-03 — Complete variant documentation. The generated compact pages under `catalog/buildable-stations/variants/` link every variant to the shared envelope/services/accessibility/compliance pages and 43-drawing register, record the complete parameter delta, and enumerate all unique products and assembly relationships.
- [ ] COMP-016-03A — Release deployment drawings. Produce the missing envelope, services, accessibility, compliance, drawing-register, and assembly drawings for each selected site; parameter-only variants use the governing standard artifact plus their generated delta.
- [x] COMP-016-04 — Reconcile product structure. The generated station reconciliation register and CI tests now fail on an orphan in either direction across manifest, BOM, definition/drawing register, traveler, FreeCAD state map, IFC product and assembly IDs. Product-derived `DRW` and applicable `CONN` identifiers preserve identity while remaining explicit deployment deliverables rather than falsely released drawings.
- [x] COMP-016-05 — Federate and check IFC. All seven archetypes round-trip every manifest part/assembly ID, positive-volume design-reference geometry, semantic IFC class, OSR property set and assembly relationship through IfcOpenShell. Project survey placement, grade-specific materials and any resulting deployment IDS/BCF findings remain COMP-017 release work.
- [x] COMP-016-06 — Run station analyses. Use FreeCAD/CalculiX or OpenSees for selected structures, JuPedSim for normal/degraded/egress flows, EnergyPlus for heat/ventilation/load, SWMM for drainage, and FDS for enclosed or battery/charger fire scenarios. The shared 22 m canopy OpenSees load path, all-variant JuPedSim route cases and per-bay SWMM roof drainage are solver-backed and tracked. EnergyPlus and FDS also execute the depot comparison decks. The ventilation-only/enclosed-room baselines retain their adverse 52.6 C and 170.9 C / 1.0 m findings. The proposed separated, cooled controls room and open-sided energy compound screen at 35.0 C and 21.0 C / 30 m respectively; these are design-direction results, not project thermal or fire approval.
- [x] COMP-016-07 — Release the design-reference package. Product/BOM, traveler, FreeCAD and IFC identity is reconciled without orphan IDs; shared structural, accessibility/egress, drainage, thermal and fire screens are tracked. The depot catalogue now carries the screened separation, N+1 cooling, remote-isolation and evidence response. Six controlled deployment work packages keep supplier, site, local-code, detailed-design, commissioning and approval items open rather than misrepresenting them as construction release.

Exit: every station variant is either a controlled parameter delta or has a complete unique package, with zero orphan CAD/BOM/drawing/assembly IDs.

Civil exploration alongside the surveyed pilot

The civil exploration implementation adds a source-bound complete-system campaign to the existing civil geometry and native reference checks. The programme now also executes nonlinear fibre/pile/P-delta, orthotropic and friction benchmarks, actual-section solids, coupled vertical suspension/contact, thermal/continuity, commercial scenarios, multi-seed search, range sensitivity and safe evidence retrieval. Supplier/site data, calibration, full-system applicability and physical validation remain separate acceptance work. The research runner cannot promote a product or change structural release status.

Wave 2 — COMP-017 surveyed Samawah pilot and civil checks

- [x] COMP-017-01 — Issue the field evidence brief. The shared controlled template defines control, provisional accuracy, CRS/vertical-datum approval, utilities, property, flood, ground investigation, workshop/fleet audit, photography/scan coverage, ownership and acceptance. The generated Samawah brief and empty receipt manifest are also available as a revision-locked City Studio job. Mobilisation, field capture and acceptance remain external.
- [ ] COMP-017-02 — Process and accept control. Process GNSS observations with RTKLIB, retain raw and processing reports in controlled project storage, and have the deployment survey authority accept the network. The deterministic receipt/hash validator, RTKLIB execution profile, solution- quality screen and separate authority-record gate are implemented; the tracked pilot reports correctly remain `awaiting-field-data` until real observations and signed acceptance arrive.
- [ ] COMP-017-03 — Build the surveyed ground model. Use QGIS, ODM and CloudCompare to register terrain/point clouds and create the master GeoPackage. Record residuals, voids, epochs, vertical datum, and uncertainty. The shared receipt, file-signature/GeoPackage inspection, independent- checkpoint RMSE, processing-report and authority-record gates are now implemented. Samawah and Mosul remain `awaiting-ground-model-data`; source capture, processing and acceptance require the deployment survey campaign.
- [ ] COMP-017-04 — Confirm or replace all three alignments. Fit survey- grade horizontal/vertical geometry and cant, verify platform/yard/turnout interfaces, export OSR-ALN, and pass hard gates plus a LandXML round trip. The shared receipt and deterministic gate now validate every design line, exact station reconciliation, explicit horizontal/vertical/cant content, the OSR-ALN hard gates, converter-derived LandXML re-import hashes, round-trip/interface tolerances and a separate signed acceptance record. The pilots remain `awaiting-surveyed-alignments` until accepted ground data, fitted geometry, interface evidence and project-authority approvals arrive.
- [ ] COMP-017-05 — Confirm route and station fit. Resolve utilities, property, flood level, access, intercity-station/yard integration, road interaction, construction compounds, lifting paths, and possession/staging. The shared gate now hash-locks each discipline package, verifies exact coverage of every current line and station, checks that all eight domain statuses are resolved, rejects open high/critical issues, and keeps the coordinated authority record separate. The pilots remain `awaiting-route-fit-evidence` until the real studies and approvals arrive.
- [ ] COMP-017-06 — Complete drainage and ground design. Run SWMM with accepted storms and levels; size ground/foundation variants from borehole evidence. Use OpenGeoSys only where groundwater/coupled behavior warrants it. The shared gate now deterministically reruns the received SWMM input, checks continuity and provenance, validates catalogue foundation/ground-treatment selections and actual deep-element lengths, and requires exact line/station coverage. A reviewed trigger record makes OpenGeoSys evidence conditional; the pilots remain `awaiting-drainage-ground-evidence` until project inputs, calculations and approvals exist.
- [ ] COMP-017-07 — Complete per-span structural checks. Instantiate the actual span/pier/abutment/foundation schedule. Use OpenSees for global, seismic and soil-spring cases and verified FreeCAD/CalculiX templates for component checks; use Code_Aster only for identified nonlinear/detail cases. The shared release gate now requires a chainaged asset schedule, immutable OpenSees/CalculiX inputs and converged reports, exact per-asset foundation, wind, seismic, fatigue and bearing/movement results, closed independent-check comments and signed release. The pilots remain `awaiting-structural-evidence`.
- [ ] COMP-017-08 — Cross-check operations. Compare OSR running times and junction occupancy with SUMO. Trial one-line OSRD import separately. An OpenTrack comparison is optional and licensed, following `opentrack-evaluation.md`; neither evaluated tool can become a release dependency until interchange and repeatability pass. The shared SUMO generator now covers every line, binds both `design.toml` and the expanded scenario hash, uses the actual energy-derived station dwells, and completes opposed services. OSR independently emits section-speed, consist-performance and dwell-derived arrival times. The deterministic cross-check passes all three Samawah lines (3.0–3.5% difference) and all six Mosul lines (4.0–4.8% difference). Surveyed geometry, a conflict-capable and independently reviewed junction-occupancy model, road interactions and the optional OSRD evaluation remain open; neither screen is authority release.

- [ ] COMP-017-09 — Obtain deployment sign-off. Close the civil checklist for survey alignment, ground, structures, station fit, energy site, permits, stakeholders, and constructability. Keep every unreceived external item open.

Exit: signed survey and ground inputs, approved three-line OSR-ALN package, site-fitted station/yard schedule, drainage model, per-span calculation register, constructability plan, and named external approvals.

Wave 3 — COMP-018 battery, charging, and site energy

RFC 0021 fixes the planning architecture for this wave: 675/540 kWh gross/ usable LFP split into three 225 kWh car packs, a 650–700 V nominal DC link, six PMSM/controller sets, one shared 500 kW station cabinet, and stationary storage in repeated 500 kWh LFP modules. “Freeze” below means qualify the supplier implementation and close the provisional limits; it does not reopen the selected architecture or introduce a second voltage or charger tier.

- [ ] COMP-018-01 — Freeze chemistry and supplier evidence. Select actual cells/modules/BMS/charger and obtain dimensions, mass, electrical limits, temperature limits, cycle/calendar data, propagation evidence, warranties, communication details, and change control.
- [ ] COMP-018-02 — Export measured duty. Define a versioned duty-cycle schema and collect representative traction, auxiliary, dwell/charge, ambient, passenger-load, degraded-operation, and ageing cases. Planning defaults remain sensitivity inputs, not the acceptance case.
- [ ] COMP-018-03 — Calibrate the battery model. Reproduce supplier/lab curves in PyBaMM, document whether the chosen chemistry is supported, fit parameters without hiding uncertainty, and validate against independent test data before applying line duty.
- [ ] COMP-018-04 — Close pack and charger thermal design. Use the FreeCAD geometry plus verified thermal models for normal, hot-soak, fast-charge, cooling fault and degraded cases. Confirm temperatures, derating, ventilation, enclosure clearances, sensing and maintainability.
- [ ] COMP-018-05 — Complete fire/venting evidence. Define cell-to-pack test evidence and use FDS for site enclosure smoke/heat/ventilation and separation questions. Obtain fire-authority review; simulation alone does not establish propagation safety.
- [ ] COMP-018-06 — Size solar and grid connection. Use pvlib with accepted weather/soiling/temperature data and pandapower for charger coincidence, transformer/cable loading, voltage, faults, storage and degraded-grid cases. Use EnergyPlus for station/depot building loads and ventilation energy.
- [ ] COMP-018-07 — Release the connector package. Complete the mating geometry, tolerance stack, keying, touch protection, earthing, interlock, communications, thermal/current tests, insertion-cycle test, inspection, replacement and assembly instructions; reconcile all BOM and ICD IDs.
- [ ] COMP-018-08 — Obtain utility and authority evidence. Record the accepted grid study, metering/protection settings, solar permissions, emergency isolation, fire/ventilation approval, and operating constraints.
- [ ] COMP-018-09 — Close traction and DC-link compatibility. Select the cell/string count, normal top voltage and transient margin; prove contactor, precharge, HVIL, IMD, creepage/clearance and fault interruption; then derive gearbox ratio and validate AW0–AW3 acceleration, grade, wet adhesion, overspeed, regen and repeated-stop duty at 50 °C. Retain the 1.8 MW control cap until this closes.
- [ ] COMP-018-10 — Close shared-cabinet service scenarios. Verify two- contact arbitration, 825 A and 500 kW limits, conversion/contact losses, high-ambient derating, 60-second stops, longer terminal dwell, low-solar, weak-grid, charger outage and minimum route SOC. Promote station storage only in integer 500 kWh modules when the model demonstrates need.
- [ ] COMP-018-11 — Close staged fire response. Qualify temperature/off- gas detection, string/pack isolation topology, outward vents, propagation barriers and per-car mist flow/pressure. Exercise failed reservoir/pump/ nozzle/sensor cases and prove the controlled-safe-platform versus emergency- brake decision with evacuation and fire-service review.

Exit: supplier-frozen pack and charger, calibrated duty/thermal/degradation evidence, site-specific yield and network studies, released connector/assembly package, and external utility/fire decisions.

Wave 4 — COMP-019 component RFC implementation packages

- [ ] COMP-019-01 — RFC 0023 doors. Create shall-requirements, body/door ICD, hazard links, tolerance and kinematic model, FreeCAD assembly, actuator/sensor/edge BOM, manual release, obstruction/degraded tests, endurance plan, assembly traveler, and acceptance matrix. Use Project Chrono only if the simple kinematic/contact model cannot answer a named risk.

- [ ] COMP-019-02 — RFC 0024 battery thermal. Reference the frozen COMP-018 configuration; add thermal-control requirements, sensor/duct/ enclosure ICDs, limits, fault response, test fixtures and pass/fail matrix.
- [ ] COMP-019-03 — RFC 0025 points. Complete rail/blade/stock-rail, stretcher, actuator, locking/detection, drainage and maintainability ICDs; reconcile the FreeCAD kit, BOM and assembly traveler; add force, wear, obstruction, water/dust, manual-operation and endurance tests.
- [ ] COMP-019-04 — RFC 0026 charging connector. Promote the released COMP-018 connector configuration into normative requirements, electrical/ mechanical/control ICDs, manufacturing package and traceable acceptance.
- [ ] COMP-019-05 — RFC 0027 brownfield recovery. Define the survey and condition-grading schema, QGIS/point-cloud workflow, material/test coupons, reuse/rework/reject decision rules, chain of custody, resulting BOM status, and verification after recovery.
- [x] COMP-019-06 — Enforce RFC completeness. Add a repository check that each promoted RFC names its requirements, ICDs, hazards, part/BOM IDs, drawings, assembly steps, analysis/test evidence, unresolved assumptions, owner and acceptance status.

Exit: RFCs 0023–0027 are traceable implementation or procurement packages, not architecture/cost sketches.

Repository implementation status on 2026-09-29: the generated `component-rfc-readiness.md` now closes COMP-019-06. All five packages must name requirements, ICDs, hazards, controlled part/BOM and drawing IDs, assembly steps, analysis/test evidence, assumptions, owner role and acceptance state; missing paths, invented parts, unknown hazards or premature acceptance claims fail CI. COMP-019-01 through 05 remain open until their listed supplier, detailed-design, physical-test and authority evidence closes; the digital packages deliberately remain blocked.

Parallel software-assurance track — COMP-014

- [x] COMP-014-01 — Build a deterministic fault harness. Reuse the Rust integration stack and add controllable process kill/restart, node loss, network delay/loss/partition/heal, clock offset, disk-full, corrupt config, and telemetry-loss adapters. Prefer Linux `tc netem`, constrained filesystems and in-process fault points over a new orchestration platform initially.
- [x] COMP-014-02 — Specify recovery invariants. Define safe state, consensus non-overlap, durable state, event continuity, bounded recovery, alarm visibility, configuration rejection, and rollback criteria before running long tests.
- [x] COMP-014-03 — Add restart and corruption tests. Cover clean and unclean restart at each persistence boundary, partial writes, stale state, incompatible/corrupted configuration and upgrade/rollback.
- [x] COMP-014-04 — Add partition and clock tests. Exercise minority/ majority isolation, asymmetric loss, healing, jitter, clock step/slew and loss of time source while checking the invariants.
- [x] COMP-014-05 — Add multi-day soak profiles. Run normal, peak, degraded and recovery cycles with bounded resource-growth assertions and deterministic seeds. Keep a short CI profile and schedule the full profile.
- [x] COMP-014-06 — Publish safety evidence. Store scenario manifests, exact revisions/seeds, summarized timelines, invariant results and failure triage; link accepted evidence into the safety-case register.

Exit: every requested fault has a repeatable scenario, all recovery invariants are machine-checked, the full-duration run meets resource/recovery limits, and the safety case points to reviewable evidence.

Repository implementation status on 2026-09-29: the deterministic in-process harness and tracked report close COMP-014-01 through 04 and 06 for software design evidence. The stable-state envelope rejects corruption, truncation, wrong-node restore and commit/log inconsistency; disk-full and partial writes preserve the last good image; restart discards volatile leadership/quorum claims. Safety proposals are rejected while clock, time-source or telemetry health is invalid. The tracked two-day normal, peak, degraded and recovery profile now compares halfway/final retained state against explicit event, queue, historian, CBM, work-order and compact-result bounds; the scheduled seven-day profile publishes its own workflow artifact. This closes COMP-014-05 for deterministic repository design evidence. All cases still require selected-hardware HIL, wall-clock power-cut/endurance, WCET/stack/heap measurement and independent assessment before release.

Priority And Dependencies

Priority	Work that can start now	Dependency that prevents closure
1	LM3 factory drawings/interfaces, product mass-evidence intake and COMP-014 fault/recovery harness	Supplier configuration, production solids, physical build and independent review for closure
2	Maintain field receipts, QGIS/control/ground/alignment gates and reusable SWMM/OpenSees checks	Samawah survey partner, permissions, control and geotechnical campaign
3	Maintain station solver decks and advance battery/charging duty, thermal and protection templates	Accepted passenger demand, supplier loss/cell data, local code, fire scenarios, utility and weather data
4	Complete supplier-neutral connector/door/points ICD and test templates	Selected cells, charger, actuators, connectors and supplier evidence
5	Site-calibrated civil and energy analysis, final production assembly/mass release	Survey, measured duty, supplier freeze, local authority and independent review

The reusable tool baseline, station family and COMP-017 field brief are now in place. The repository-owned critical path is the LM3 factory/production-detail package, mass-properties intake and COMP-014 fault harness, maintained in parallel with the external Samawah survey and supplier campaigns. Final geometry, BOM, mass, assembly instructions and acceptance evidence are reconciled only after the relevant surveyed or supplier configuration is frozen and real evidence is independently accepted.

Software Adoption Gates

A tool moves from evaluation to baseline only when all of the following pass:

- open-source license and redistributable dependencies are recorded;
- a maintained release/source and a reproducible installation path exist;
- command-line or scripted execution can capture the full configuration;
- the OSR benchmark meets a documented tolerance;
- input/output can be reviewed without a proprietary application;
- stable OSR part, station, alignment, requirement, and evidence IDs survive;
- failure, warning, convergence, and unit/CRS errors cause a non-zero check;
- a named maintainer owns updates and replacement/migration.

These gates deliberately leave FreeCAD Road/Trails, OSRD, Project Chrono, Code_Aster, and OpenGeoSys conditional until they answer a concrete design question and pass an OSR benchmark. This keeps the core toolchain small while preserving open routes for advanced analysis.

Audit of today's corrections against the OpenSourceRail design

Repository source: `docs/ethos-audit-2026-09-09.md`

Reviewed revision: `b5e3b301a`. Scope: all 13 commits dated 9 September 2026, from `3515839c1` through `b5e3b301a`, compared with `5d5f25b43`. This audit examines changed source logic, templates, tests and report consumers; generated city results were checked by aggregation and against their inputs. It is not a new physical design or a complete verification of every subsystem.

The controlling rationale is in Architecture §2 and D7, RFC 0002, RFC 0021, and the user's operating instructions: solar generation on railway land, local stationary storage, brief train top-ups, grid backup for residual demand, repeatable components, domestic design and manufacturing capability, two revenue sets per selected station, overflow storage on each train's own line, and no interline depot connections.

Repairs completed

The findings below describe the audited revision. The subsequent repair:

- accepts station storage/source capability with any positive installed charging rating; grid import and a 150 kW minimum are no longer prerequisites;
- caps low-C top-ups by installed station and shared cabinet/source limits;
- checks current battery/source availability during charger reachability, while retaining onboard reserve checks and metering actual shared charging;
- separates grid-only contingency findings from solar/storage snapshot results, includes charger DC/DC efficiency, and prevents battery discharge into a PV surplus; full operating-energy validation remains explicitly open;
- retains historical stabling experiments and their hashes as diagnostics, including source drift, without using them to accept or reject the selected line-local hybrid configuration;
- permits a deployment-specific mobilisation scope, role IDs, management systems, work packages and evidence requirements, validating their references and dependencies rather than fixed job titles or arbitrary document counts;
- checks package-status honesty and selected-evidence provenance in repository health, while leaving incomplete physical deployment packages visible.

All 266 energy snapshots were regenerated. The same 3,976 grid-only connection exceedances remain visible; all 266 coordinated solar/storage snapshots pass. Samawah, Uíge, Quelimane and Edéa pass new continuous two-day hybrid replays. The Rust simulator/controller suites pass 127 tests with one ignored test. Focused regressions cover off-grid lower-power stabling, battery-backed grid outages versus empty storage/failed pads, diagnostic package scope and a small workshop's tailored mobilisation record. City equipment and fleet inputs are unchanged. Native converter/transformer energy-boundary reconciliation and complete solar/storage endurance remain modelling limitations, not mandates for grid expansion.

Confirmed drift at the audited revision

Finding	What changed or was misinterpreted	Required correction
Grid-only contingency became a general design failure	Today's electrical connection gates feed both snapshots into `passed = solver_passed and not findings`. The grid-only case explicitly sets PV and battery discharge to zero. The review then recommended connection upgrades from those results.	Retain connection measurements, distinguish contingency requirements from normal solar/storage operation, and evaluate the actual buffered duty. The blanket upgrade recommendation is withdrawn.
Station stabling requires a grid connection	Today's Python candidate selector and Rust loader require `grid_import_kw > 0`; the new Python regression even treats removal of grid import as removal of all power. The template's pre-existing `requires_grid_connection = true` was propagated into executable logic.	Station eligibility must use the declared charging/storage capability. A grid connection cannot be a universal prerequisite for a solar/storage station.
Grid outage is treated as charger outage	Today's `can_reach_stabling_charger` rejects a destination when `grid_disabled_at(next)` is true, regardless of the station battery. The energy subsystem itself can still deliver battery energy during a grid outage.	Evaluate usable charging energy and equipment availability separately from grid status, preserving onboard reserve and deterministic allocation. Do not promise future battery energy without accounting for competing train demand.
Superseded station-only experiments still block the adopted plan's package	The package generator adds every present stabling screen to `required`, then treats any `passed: false` as a package failure. Samawah's passing hybrid replay coexists with three failed station-only/redistribution experiments in `failed_summaries`.	Separate historical diagnostics from acceptance of the selected operating configuration. A rejected experiment must not become a requirement to enlarge stations or alter the adopted station/depot split.

Sources: electrical generator, candidate selector, native loader, reachability logic, candidate tests, package generator, Samawah package.

Defaults that must not become universal requirements

- 150 kW stabling eligibility: the new selectors turn the existing low-C top-up rate into a hard minimum installed rating. A target charging rate is not by itself evidence that a lower-rated station cannot meet its overnight duty. This needs an energy/time check, without inventing a replacement rate.
- Depot size: today's quantity reconciliation carries the already-existing 5,000 kWp / 40,000 kWh main-depot scenario into the physical scope register. This removed conflicting quantities; it did not prove that every deployment needs that equipment. The area and cost differences are conditional on those retained inputs. They must not drive a larger central depot by default.
- Fixed organisational template: the new mobilisation validator locks thirteen role IDs, eleven management-system IDs and eighteen work packages, with minimum counts of documents. Accountable responsibilities and actual inspection evidence support domestic capability. Fixed job titles, committee structures and document counts need deployment tailoring; they are not universal requirements for independently adopting a small subsystem. This validator currently checks the reference template, not train dispatch. It does not establish that thirteen separate executives must be hired.
- Reference drainage levels: correcting SWMM columns and rainfall totals was valid. The added 0.4 m pipe fall over 40 m is explicitly a planning assumption, not a measured site correction or mandatory civil layout.

Sources: depot quantity model, depot scope, mobilisation validator, drainage model.

Corrections that remain consistent with the project

Change group	Assessment
Local Ops Core persistence, revision conflicts, latest-inspection/approval checks, private-file protection and visible failed saves	Keep. They protect accountable local records using the existing SQLite/browser stack. The loopback restriction covers unauthenticated Workbench mutation routes; it does not introduce a cloud dependency.
SWMM rainfall, imperviousness and mass-balance repairs	Keep the arithmetic/input fixes and independent checks. Preserve the disclosed planning boundary for levels, rainfall and outfall.
Converter/transformer loss measurements and separate import/export limits	Keep the measurements. Correct their operating interpretation and reconcile the separate DC/DC loss.
Depot quantity source consolidation	Keep one quantity contract with explicit site overrides. It must remain subordinate to verified distributed operating needs.
Two station homes, reserve inventory, continuous battery/position state, same-line home returns and station launch priority	Keep. They implement the user's operating model with explicit deterministic rules. Yard geometry and reserve activation remain incomplete.
Final removal of interline depot assignments	Keep. The current allocator uses an existing powered service location on the train's own line and assigns no additional heavy workshop to the overflow storage locations.
Source hashes, stale-result labels and repeatability tests	Keep provenance checks. Correct which evidence is applicable before aggregating pass/fail; a hash cannot validate a wrong requirement.
Local production data, inspection plans and actual measurement fields	Keep. These make domestic fabrication inspectable and reproducible. Blank records must remain blank until work is done; supplier-exact bought-in interfaces do not require foreign turnkey delivery.
Cashflow/calendar, battery renewal, passenger demand and shared-factory scheduling findings	Retain as model-integration gaps. They do not prescribe new rolling stock, enterprise software, interline tracks or imported management systems.

The active architecture and conservative pilot signalling profile were not changed by these 13 commits. This audit does not attribute those earlier decisions to today's corrections or remove physical safety evidence merely because it is also needed by other railway designs.

Rechecked power figures

All 266 energy reports match their current scenario and electrical-generator hashes. They contain 9,328 sites, with 16 distinct combinations of the electrical inputs used by this snapshot model.

Existing model result	Verified result
Sites exceeding import limits with PV and storage disabled	3,976 across 262 cities
Those flagged sites with stationary storage installed	3,976
Connection exceedances with the existing coordinated daylight dispatch	0 across 266 cities
Samawah exceedances in the grid-only contingency	7
Mosul exceedances in the grid-only contingency	26

The coordinated case stipulates 60% PV nameplate and 50% battery discharge rating. These are existing model inputs, not a newly proposed operating policy or a measured state of charge. Its passing snapshot does not prove endurance.

Each flagged Samawah station has 500 kWh gross stationary storage, 500 kW maximum battery discharge, 300 kWp PV and a 500 kW grid-import limit. The existing snapshot calculates 519.020 kW import with solar/storage disabled, versus 73.213 kW

under its coordinated dispatch. Calling these installations "grid-only sites" was incorrect.

The snapshot omits the configured 98% train-charger DC/DC loss. Holding the existing transformer ratings and dispatch fractions fixed, applying that loss changes those two figures to 529.647 kW and 83.748 kW respectively. Replaying all sixteen electrical configurations with that correction leaves the counts unchanged: 3,976 grid-only exceedances, zero coordinated-daylight exceedances. These are conditional calculations, not a revised equipment specification.

For the station battery alone, 500 kW of source discharge for 60 seconds uses 8.333 kWh and delivers about 8.167 kWh at 98% charger efficiency. This is consistent with the RFC's approximately 8 kWh top-up. Two contacts share the cabinet/source budget; neither receives an independent 500 kW allowance.

The native simulator includes charger efficiency but does not separately apply the electrical screen's rectifier and transformer losses. Its controller uses PV/storage before residual grid import; grid-to-storage refill is not implemented. That absence is not a reason to add routine grid charging by default. Reconcile the energy boundaries with the solar/storage architecture.

The retained Samawah 48-hour-30-minute hybrid run reports 128.400 MWh station PV generation, 254.418 MWh grid import and 345.252 MWh delivered from sites to trains. Initial station storage is 29.000 MWh; at the end all nineteen ordinary station batteries are effectively empty while main-depot storage is at 84.24%. These are outputs of the existing model, not new grid requirements. They show that the current replay cannot establish the intended solar/storage energy balance. Its passing stabling result only establishes its declared home and morning-start checks. The raw JSON hash matches the tracked hybrid report. Its duration field is 174,601 seconds, including the final sampled second.

The architecture also includes generation along the railway right-of-way. The city energy screen consumes declared site PV values; it does not calculate additional generation from route length. Additional ROW energy must be traced to declared site inputs and physical quantities before it can be credited. This audit adds no assumed PV acreage, grid duty, station battery or depot.

Audit coverage and change status

Today's commits	Subject reviewed
`3515839c1`	Mobilisation records, exact role/system/work-package validation, template versus deployment scope
`aae98e42c`, `5205c40d2`	Civil/trainset inspection plans, manufacturing records and per-part production data
`a1248699b`	Portal integrity/access fixes, drainage, electrical checks, provenance and CI
`2b5038b72`	Depot energy quantity contract, conditional area/cost scope and regenerated reports
`7d999e075`, `565f7d978`	Station stabling candidates, native holding, charging eligibility and allocations
`255b99148`, `596e9d92a`, `614bfe96c`	Continuous cycles, reserve roles, morning coverage and charger reachability
`482adf174`, `53ebc6136`, `b5e3b301a`	Two-train capacity, depot storage, actual home returns, line independence and repeatability

No city design/scenario TOML files changed across this commit range. Today's review therefore did not install larger grids, add solar arrays or increase the retained city fleet inputs. The drift is in requirements, software gates and interpretation; those can still steer later design decisions incorrectly.

The written review, runtime gates, candidate selector, package aggregation and mobilisation validator have now been repaired as described above. Historical experiments retain their original fail results and source hashes; current selected evidence was regenerated using the repaired software.

Checks during this audit: catalogue energy/source aggregation, pandapower replay of the sixteen independent electrical configurations, raw hybrid-output hash verification, and 19 filtered simulator energy/stabling tests passed. The audit exposed a candidate test that expected solar/storage stations without grid import to be rejected. That expectation has been replaced with regressions accepting stored-energy operation and rejecting sites without usable capability.

Civil allowances and routing scores

Repository source: `docs/finance/civil-allowance-boundary.md`

The civil estimate prices route length at controlled at-grade, elevated and bridge planning rates. A curvature deterrent is a route-search score, never a supplier price or a monetary multiplier. Local three-point radii use a 100 m route window on each side; elevated construction-method runs split at local product changes. The radius remains inferred raster geometry, rather than a surveyed design.

`refresh-local-civil-costs.py` calls the native classifier against the published full corridor geometry and preserves existing civil class boundaries, routes, station positions and fleets. It records an actual cost migration; it does not rewrite previous solver receipts. Changes to Rust or controlled designs require genuine new full-day and degraded planning runs.

The displayed capital is a base planning allowance, not a complete installed estimate. Tangent Pi25/Pi20 quantities can use their respective catalogue envelopes. OSR-US and exceptional curves require their own supplier geometry, connection, support, erection and foundation quantities. Missing rates or quantities remain unknown; they are never zero-cost scope or credited savings. Each special crossing is separately engineered. Actual foundations need surveyed ground zones and deep-element lengths. Transport and erection require complete mass budgets and configured equipment proposals.

Removing the old segment-wide search penalty corrects accounting. It establishes no saving against a buildable, priced alternative. Land, rights, utilities, special-structure increments, installed grid/charging upgrades, taxes/duties, escalation, risk and financing must close before the base allowance becomes a complete delivery budget. Maintenance charged on a smaller allowance is a conditional calculation, not evidence of reduced maintenance workload. Staffing, energy, train-km and replacement costs must retain their independent quantities.

Demand coverage, capacity and receipts are separate: population pixels describe residents near stations; network walksheds need verified entrances, accessible paths, crossings and vertical access; surveyed origins/destinations and jobs determine route choice and paid journeys. A transfer creates another boarding, not another paid journey under an integrated fare.

First Adoptable Product

Repository source: `docs/first-adoptable-product.md`

The first adoptable OpenSourceRail product should be the non-safety owner-operator stack:

```
ERPNext + Frappe HR + railway Ops Core + simulator + asset/evidence portal
for an existing depot, workshop, or pilot corridor
```

This can be useful before any train-control software is certified. It helps an operator organize assets, work orders, manufacturing packages, QA hold points, maintenance schedules, defects/NCR, and acceptance evidence while the safety-critical stack remains in simulation or shadow mode.

Target User

- A public railway owner evaluating a depot, workshop, or pilot corridor.
- A local workshop preparing rolling-stock or infrastructure packages.
- An operator trying to structure maintenance and QA records.
- A university, assessor, or funder reviewing whether OSR evidence is coherent enough for a physical pilot.

Product Scope

Included	Not included in the first product
SQLite-backed Ops Core work orders	Live movement authority or interlocking control
Generated asset register	Certified train-control functions
ERPNext business applications and Frappe HR, with OSR planning references	Automatic migration of an operator's ledgers or staff records
QA gate register and evidence links	Legal certification or safety approval
Maintenance schedule per train, station, track section, waypoint, depot, energy site, and tool	Insurance, finance, or procurement guarantee
Authenticated RBAC, managed photos/files, independent approvals, controlled document revisions and sealed audit trail	Qualified signatures, enterprise SSO/MFA or full offline mobile field app
Simulator and scenario outputs	Full telemetry-driven predictive maintenance
Acceptance evidence matrix	Regulator submission by itself

Deployment Shape

1. Select an existing depot, workshop, yard, or short pilot corridor.
2. Load or generate the asset register.
3. Import the baseline manufacturing, QA, and maintenance schedules.
4. Set up the ERPNext operating platform for business records and run the SQLite-backed railway assurance portal.
5. Create work orders from QA, manufacturing, and maintenance rows.
6. Upload managed evidence and close work through authenticated inspection and independent handback.
7. Use defects/NCR and audit records to build an acceptance evidence pack.
8. Keep train-control in simulator or shadow mode until the deployment safety case is independently assessed.

Current Implementation

Capability	Current artifact
Browser portal	`operations-portal/README.md`
Ops Core model	`operations-portal/ops-core.md`
SQLite server	`../tools/automation/ops-core-server.py`
Generated operations data	`build/generated-operations/samawah/` after running the generator
Acceptance status	`certification/evidence-status.md`
Gap analysis	`operations-portal/gap-analysis.md`
Construction QA RFC	`rfcs/0028-construction-quality-assurance.md`
Maintenance schedule RFC	`rfcs/0029-maintenance-schedule-system.md`
Manufacturing schedule RFC	`rfcs/0030-manufacturing-schedule-system.md`

Run the integrated product through the root one-command setup. Standalone portal development and data-generation commands remain with the operations portal, where they are maintained.

Acceptance Criteria

A credible first deployment should be able to show:

- every asset has a stable id and owner;
- every manufacturing package has material/BOM refs and QA verification;
- QA and maintenance rows can become work orders;
- failed inspections create defects/NCR and audit records;
- authenticated roles and city scopes are enforced by the server;
- inspection, approval, controlled-document and audit records are sealed against silent editing;
- managed evidence survives a verified SQLite/evidence backup;
- predecessor work blocks successor release until pass evidence exists;
- evidence can be exported as CSV/JSON and reviewed outside the portal;
- the system runs without commanding trains.

Reference Deployment Measurement

Agree the baseline, observation window and decision owner before configuring the product. A reference deployment should publish or independently retain:

Measure	Evidence method
Asset/work-record completeness	Sample the same agreed population before migration and after acceptance
Defect-to-handback time	Timestamp issue, triage, work, inspection, independent handback and rejected/rework paths
Record quality	Count missing identities, uncontrolled attachments, stale approvals, duplicate work and reconciliation errors
Recovery	Restore a signed backup into fresh volumes and reconcile record, file and configuration inventories
User effort	Record training, data preparation, support, review and exception-handling hours by role
AI assistance	Compare draft preparation/review time, dissent, overrides and errors; retain human accountability
Owner continuity	Export open records, remove a test adapter and prove history remains usable without supplier access

The comparison baseline is the operator's real prior process or a named configured alternative. Do not infer savings from a generic manual-work assumption. Personal, employee, supplier and security-sensitive results stay in the controlled deployment evidence store; public summaries must be aggregated and approved.

Why This Is The Wedge

Rail procurement is not won by software novelty alone. This product gives operators, funders, and assessors something useful and low-risk to adopt: better asset, QA, maintenance, and acceptance discipline. It also creates the evidence trail that any later certified OSR train-control or rolling-stock deployment will need.

This reference deployment is the first proof programme in the competitive position and proof roadmap.

Connected OpenSourceRail lifecycle

Repository source: `docs/lifecycle/README.md`

The integrated view connects plan design procure manufacture construct commission operate maintain renew through stable equipment identities, versioned engineering evidence and native ERP transactions.

Open the local lifecycle view. Deployment and recovery.

Read performance and history

FUXA polls only its active device's current measurements and alarms. It does not load whole-city evidence or command history. Lifecycle snapshots retrieve latest measurements in one indexed query and include at most 20 recent evidence records per asset. Load older evidence retrieves earlier records with a stable cursor.

The delivery panel requests the selected asset's city/environment queue, displays its total and pending counts, and provides Pending only and Load older deliveries controls. Other-city traffic cannot evict a pending event from this view. Authenticated `/outbox` and `/evidence` GET endpoints accept `limit` (1–100) and `before` cursors; the gateway still enforces the caller's city/environment scope.

See the load and recovery review for measured results and the remaining two-second capacity finding.

Implemented connections

Owning system	Integration
OSR city generator	Uses existing station/depot IDs and engineering revision; shared templates with 266 city profiles
Embedded Rust	`osr-energy-site` example runs the existing evaluator; gateway converts its watts and state of charge into the common telemetry contract
Station software	`StationScadaOutput` / `LightingZoneStatus` adapter preserves native lighting semantics
FreeCAD	Versioned FCStd object metadata; selected charger object opens the city asset view
Bonsai / IFC	Existing IFC GlobalIds retained; checksum-bound sidecar maps the reference model to city positions
QGIS	Existing GeoPackage/layer evidence and original station feature IDs map to the same position
Other engineering tools	Versioned evidence manifests for SUMO, OSR analysis, OpenSees, EnergyPlus, FDS, SWMM, JuPedSim, CloudCompare, Blender and verification outputs
ERPNext	Reviewed Item/BOM mapping, partial delivery/outstanding quantities, currency-separated commitments/invoices, native manufacturing actuals, condition-driven Issues and reviewed native Asset Repairs
FUXA	Generated station screens for charger, battery, PV and facilities, including quality, timestamp and native gateway status
Factory methods	Nine read-only views generated from the 120-product LM3 method coverage and attached to each real `depots-production` identity; reviewed ERP mappings correlate native Work Orders
OSR assurance	Existing works/inspection/handback/defect records shown alongside the asset; independent authority retained

Samawah and Mosul have operating simulation pilots. Their reference station geometry is not a site-approved cabinet manufacturing design. Supplier drawings, manufacturing recipes, commissioned device maps and operational acceptance remain real engineering inputs, not automatically generated approvals.

Engineering packages and desktop tools

```
./osr supervision engineering cities/catalogue/west-asia/Iraq/Samawah/operations/engineering-integration.json --output
build/supervision/samawah/engineering.json
python3 tools/automation/desktop-links.py samawah
```

An engineering manifest records city, asset, input revision, assumptions, owning tools/versions and actual file checksums. IFC objects, FCStd object names, QGIS layers and GIS feature IDs are extracted from existing artifacts. A tool's absent version remains explicitly unknown; it must be established during review.

`build/supervision/<city>/desktop-links.json` connects the existing reference charger `STN-CHG-P010` to the pilot station position. Shared library models are qualified by city and artifact; they are not mistaken for unique installed assets. For a multi-station model, provide reviewed object-to-position bindings; ambiguous selection is rejected.

In FreeCAD, select a charger part and run `opensource_rail.FCMacro`. Choose the generated city link file. The macro opens Workbench without editing CAD.

For Bonsai's Python console:

```
import sys
```

```
sys.path.insert(0, '/path/to/OpenSourceRail/tools/integration')
import desktop_bridge
# Select the charger IFC object before running:
desktop_bridge.bonsai('/path/to/OpenSourceRail/build/supervision/samawah/desktop-links.json')
```

For QGIS's Python console, after selecting the existing station feature:

```
import sys
sys.path.insert(0, '/path/to/OpenSourceRail/tools/integration')
import desktop_bridge
desktop_bridge.qgis('/path/to/OpenSourceRail/build/supervision/samawah/desktop-links.json', iface)
```

These bridges carry stable references. CAD changes return to FreeCAD; coordination changes return through Bonsai/BCF to the design owner. Live telemetry and ERP transactions stay outside IFC. Desktop selection resolution is tested with the actual reference object identities; graphical interaction inside each desktop application still requires a user session.

Engineering-to-execution boundary

An execution proposal must explicitly map component type/revision to ERP Item, stock unit, production BOM, drawing, inspection requirement and reviewed conversion rule. The conversion rule describes production allowances and process inputs; a CAD parts list alone does not satisfy it.

```
./osr supervision execution-proposal build/supervision/samawah/engineering.json reviewed-item-mapping.json --output
build/supervision/samawah/execution.json
```

`osr_erpnext.integration.preview_execution` verifies the native Item and submitted active BOM and returns a fingerprint. `apply_execution` records an immutable `OSR Execution Mapping` after review. Neither submits orders. Use the existing manufacturing component to create draft Work Orders from the reviewed BOM. Native ERP procurement, quality, stock and accounting workflows remain in force.

Manufacturing-method supervision

A production-plant asset in a `light-metro-3car` city receives one view for each of the nine validated LM3 methods. This reuses the generated 120-product coverage, 30 tooling families, work centres, crew/cycle planning, steps, hold points and release gates; it does not copy recipes into a second configuration. FUXA and Workbench show simulated cycle progress, cell unavailability and quality hold with timestamp/quality. The Workbench filters native Work Orders through immutable `OSR Execution Mapping` rows for the selected method or its product IDs and current engineering revision, requiring the exact Item and BOM pair from the same mapping. A matching Item with a different BOM revision is excluded. When no reviewed mapping exists, it explicitly reports zero matching work orders instead of treating every project production record as affected.

The quality-hold fixture rehearses a method excursion and creates a deduplicated ERP Issue after persistence. It deliberately does not invent a process temperature, create/complete a Quality Inspection, change Work Order state, accept or reject product, or satisfy the method release gate. Actual travelers, observations, NCR/rework disposition, inspection evidence and accountable manufacturing release remain native ERP/QMS and OSR evidence work.

Before a prepared supervisory revision replaces a live simulation baseline, `review-package` now traces exact changed values through existing component, parent/source-asset, IFC, Rust-crate and ERP identities. It includes installed serials, append-only lifecycle evidence, open alarm/maintenance cases and pending commands. The Workbench presents the same review beside matching ERP purchase, receipt and production feedback. Evidence is marked for reassessment rather than edited or invalidated, and no order is automatically cancelled or changed.

Apply is bound to the accepted baseline hash and the complete review hash; new evidence or other in-scope lifecycle state makes an older review stale. Unrelated city/environment baselines remain outside the transaction. Automatic apply is blocked for component substitution, an installed-position retirement, a pending command-contract change and physical remapping. Native BOM exposure is now available alongside this supervisory slice; full CAD/solver dependency graphs remain open. The bounded native checks below now cover every disposition action in the ERP catalogue; they do not constitute manufacturing or railway release.

Engineering revision exposure

Engineering & delivery Engineering revision exposure follows the selected equipment's component or factory product/tooling IDs into its reviewed execution mappings. It includes the current engineering revision and, when a prepared change exists, the old and proposed revisions. City, company and project must match. Missing mappings do not fall back to every transaction in the project.

The reusable ERP reader follows explicit nested `BOM Item.bom_no` references, never the Item's current default BOM. It reports:

- Draft and submitted project purchase lines for the mapped Item and its BOM materials. These are potential shared-item exposures, not proven revision allocations; closed orders remain labelled historical.

- Draft, unfinished and completed Work Orders matching an exact Item/BOM pair. Planned, produced, remaining and native material quantities stay separate.
- Submitted Stock Entries linked to those Work Orders, retaining warehouses, stock units, serial/batch bundle and quality-inspection references. Movements do not establish current available inventory, accepted WIP or railway release.

The export preserves native read permissions and shows unreadable/missing BOMs, cyclic references and traversal limits as coverage warnings. All records are limited to the selected project and company. Other projects and unlinked stock need separate review. The snapshot observation time is visible, and Download revision exposure exports the same checksummed observations for review.

Refresh with `./osr erp snapshot`, or the existing five-minute feedback timer. These snapshots cannot put work on hold, cancel procurement or supersede engineering evidence. The native workflow below records a proposed disposition and its independent review. The supervisory apply checksum does not cover these separately timed ERP exports.

Reviewed disposition plans

Inside Workbench's native ERP city project, choose OpenSourceRail Revision dispositions. Pick a reviewed engineering mapping and a purchase line, exact Item/BOM Work Order or linked Stock Entry from its visible exposure. A proposal requires a responsible ERP user with project access, due date, rationale and versioned evidence references. Its preview shows the affected native record; recording creates one immutable OSR Revision Disposition and a native ToDo. Repeating the same request returns the existing record; changed content with the same identity is rejected.

The record-specific choices are requests to retain/review, amend or cancel a purchase, stop or rework production, or inspect/trace material movements. These are plans, not native transaction commands. A second user, different from both proposer and responsible person, can preview and record Endorse plan or Reject plan with review evidence. Endorsement requires the same current ERP exposure checksum and no coverage warnings. A stale proposal can be rejected; it needs a new proposal identity for a revised plan. Each proposal has at most one immutable decision, preserved in OSR Disposition Decision.

Manufacturing Manager, Projects Manager and System Manager roles can create/read these records. Existing project and exposure permissions still govern the workflow. Server endpoints derive actor identities; generic document insertion, editing and deletion cannot bypass the reviewed workflow. Workbench displays the decision separately from whether its exposure is still current. Native ToDo completion does not verify execution of the requested business action. Reviewers also need the native source-document read roles: for example, Manufacturing User grants Work Order access, while Manufacturing Manager alone does not. Partial exposure remains explicit and cannot receive an endorsement.

Upgrade after a backup: `./osr erp build`, `./osr erp up`, `./osr erp bench migrate`, then `./osr erp snapshot`. Migration installs all three disposition record types. The same workflow uses each city's existing project, company, mappings and permissions without city-specific code or invented organisation/approval identities.

The workflow rechecks visible ERP data during preview and recording; it does not freeze native orders or stock against concurrent or later changes. Such changes make the recorded plan stale in refreshed feedback. An endorsed plan alone does not verify its execution.

Independent native outcome verification

For an endorsed plan, perform business work through the native ERPNext workflow, then choose Verify native outcome in Revision dispositions. The action registry supplies the appropriate native document fields. Enter rationale and versioned evidence references, inspect the displayed record details and readings, then record the independent verification. Open verification evidence exposes the immutable observation and its underlying native document identities.

Requested action	Native evidence and verification boundary
Retain for review	Unchanged full exposure and unchanged native target; records a retention review without imposing a hold or release
Cancellation	Cancelled Purchase Order, with original reviewed quantities and received quantities preserved
Amendment	Submitted direct `amended_from` successor of the cancelled order, all lines in the same project, same Items/units, no receiving yet; the reviewer sees replacement quantities, prices, dates and totals
Production stop	Submitted Work Order with native status `Stopped`, unchanged quantities, BOM and materials
Rework	Completed native corrective Job Card linked to the original submitted Job Card, exact project/Work Order/BOM/Item, performed time logs and quantity, zero process loss, and its linked submitted accepted Quality Inspection
Inspection	Submitted Quality Inspection for every reviewed Stock Entry line, exact Item/row references, sample sizes, inspectors and performed readings; accepted and rejected results remain distinct
Material trace	Submitted Stock Entry reconciled to permission-visible Stock Ledger Entries and referenced serial/batch bundles; identifies this movement, not current inventory or installed configuration

The verifier must differ from the proposer, responsible person, native record's last updater and recorded evidence updaters/inspectors. Existing project and source-document permissions apply. In the pinned ERP version, Work Order access requires Manufacturing User, Quality Inspection access requires Quality Manager, and Stock Ledger Entry access requires Stock User (or another native role granting that access). Disposition roles alone do not grant access to source evidence.

Checks compare the complete saved, checksummed exposure with current visible exposure. Only the selected outcome's allowed differences are excluded: cancelled or directly amended purchase orders, a stop status, inspection links, and the Work Order timestamp updated by corrective costing. Other BOM, order, quantity or material changes require a fresh disposition review. Amended orders, corrective cards, inspections and traced stock entries must be submitted. Cancellation, stop, amendment, inspection and corrective evidence must be updated after endorsement. Retention and trace intentionally read existing records. Item substitution needs a new engineering review and mapping.

Recording adds an immutable OSR Disposition Execution, bound to the observed native data, evidence selection and verifier. Identical retries return the existing record; changed observations invalidate a preview. Refreshed feedback reports the action-specific outcome, Verification stale, or Verification unavailable, retaining the verifier and history. Resuming work, cancelling an inspection or changing evidence invalidates the earlier observation. A verified rejected inspection means the requested inspection was performed: it never means accepted material. The original exposure may correctly show changed while the expected native outcome is verified. Refresh `./osr erp snapshot` after business changes.

Proposals created before full exposure capture need a new reviewed proposal; they cannot be retroactively verified from a checksum alone. Preview and record recheck observations but do not freeze concurrent native transactions. Native inspection and corrective evidence cover only the displayed quantities and samples, not all WIP, all defects or engineering conformity. The workflow does not execute business actions, complete ToDos or authorise manufacturing/railway release.

The native acceptance test purchases ten units, receives four, reports six outstanding, manufactures one assembly from its reviewed BOM and checks actual production. It also tests draft and partially completed production, mixed-project purchase lines, linked stock movements, a restricted ERP reader, independent all catalogue outcomes, native corrective work, rejected inspections, batch movement traces and stale feedback after source changes. Its temporary transactions are rolled back. No test production or purchase is presented as real city progress.

Maintenance and configuration feedback

The simulation demonstrates fault FUXA alarm one ERP Issue reviewed native Asset Repair part consumption, actual downtime and repair feedback. The Issue links back to evidence/trends, and its native project is retained. From the Issue, Prepare repair previews the commissioned Asset, technician, expected downtime, part warehouse availability and optional serial/batch bundle. Applying the review creates a draft Asset Repair plus native assignment, without submitting it or closing the case. The repair preserves a checksum-bound request, condition-event history and references to the existing OSR configuration/serial evidence.

The native acceptance test closes the Issue while its Asset remains out of order, then completes the repair, consumes a stocked part through ERPNext and confirms feedback still reports `required-in-osr`. The supervision rehearsal replaces `SIM-CHARGER-001` with `SIM-CHARGER-002`, retains the removed serial in affected- asset search, and requires a fresh independent commissioning test/release before maintenance evidence. Inspections, restrictions and handback continue in OSR: an Issue closure, alarm clearance or completed Asset Repair grants no operational release.

Installation/serial/batch records and replacement history support affected-assets queries. Simulation evidence can rehearse as-designed, released-for-execution, as-built, as-commissioned and as-maintained transitions. A physical installation must instead obtain its release from the existing OSR assurance workflow.

Delivery, invoicing, production, installation and engineering acceptance remain separate measures. The feedback does not invent accepted quantities, domestic value added or manufacturing origin when source evidence is absent.

Extension points and remaining deployment work

The generated operating-readiness audit compiles real-asset simulation packages for all 266 city profiles rather than validating them against a placeholder station. It also compiles every ERP/component profile and records which full task payloads are locally materialised. This closes the repository configuration/asset-coverage ambiguity; it does not turn simulation templates into commissioned physical mappings.

Those packages now reuse all declared catalogue switch identities for read-only points condition views and derive one explicitly aggregate AFC view from each station. Native level-crossing evaluation and templates are present, but no city currently declares a crossing asset, so the audit creates none. FUXA/ERP expose no point or barrier command, route state, protection reset or movement authority. Samawah also reuses its production-plant identity for nine LM3 method views; Mosul's `metro-4car` family receives none until applicable methods or a reviewed module-reuse mapping exist. These views are provided without claiming that a planned method, simulated hold or ERP Work Order is a performed traveler, accepted

product or released first article.

The contract and evidence adapters are reusable; the currently running equipment binding is the Rust simulation gateway. Physical Modbus/OPC UA/MQTT/NATS adapters need the actual supplier interfaces and deployed broker configuration. No new broker or Node-RED installation is needed for this HTTP pilot.

FUXA full-project replacement is now guarded by a deterministic deployment manifest and a live-versus-desired review. The review exposes every device/view addition, replacement and removal and becomes stale if either side changes. The previous project and applied review are backed up before import. Reviewed custom display inputs are not yet a separate configuration type, so the manifest records an empty set and treats live-only edits as explicit removals rather than silently claiming they are preserved.

The later commercial, fare settlement, land agreements, supplier localisation, multi-location repair-pool and contractor valuation proposals require their own reviewed rules and real master data. This implementation establishes their stable identities and execution evidence; it does not claim those domain workflows are complete.

Evidence and traceability in the Workbench

Connected assets Record lifecycle evidence supports design review, execution release, installation/replacement, commissioning tests, simulation commissioning release, maintenance, renewal review, analysis and change proposals. The shared evidence catalogue builds the form for every city. Review shows the exact asset, environment, engineering revision, immutable record ID and versioned references before submission. A scoped evidence credential is required; the gateway enforces roles and prerequisites. A failed reply can be retried with the same reviewed ID. Changing the target discards that review and clears the evidence credential. Successful recording also clears the credential.

Engineering & delivery Prepared change impact is a read-only usability surface. It shows the proposal-wide summary and then narrows to the selected equipment position, including exact changed paths, affected records, embedded source crates and required human review activities. “Preview only” and any apply blockers remain visible; the view cannot approve or apply a package.

Evidence is not a file-upload or electronic-signature service. References identify existing versioned sources. Inspector and reviewer must be different identities for simulation commissioning release; physical release still belongs to OSR assurance. ERP case closure remains separate.

Trace serial or batch searches current and removed installations within the selected city and environment. Results show revision and replacement history, and can open a matching connected asset. A search on both fields requires both to match; it does not search other cities or bypass credential scopes.

Cross-domain change context

Prepared change impact also checks the proposed revision against ERP observations scoped to city, company and project, with a five-minute freshness limit, and verifies linked engineering artifact hashes. Missing or changed inputs are listed explicitly. The context checksum binds these observations; it does not extend the supervisory apply transaction to ERP. Native business actions still revalidate their current exposure, and CAD/solver, rework acceptance and formal supersession remain open. See the review follow-through.

Native embedded software, ERPNext and FUXA

Repository source: `docs/lifecycle/embedded-integration.md`

The operating platform consumes selected Rust evaluator outputs through a versioned, observation-only simulation contract. It preserves the distinction between controller state, supervisory observations, condition alarms, business maintenance and railway release.

```
flowchart LR
  I[Explicit simulation inputs] --> R[Native OSR Rust evaluators]
  R --> C[Strict observation-only contract]
  C --> G[Scoped integration gateway]
  G --> H[Timestamped equipment history]
  G --> F[FUXA station, vehicle and wayside displays]
  G --> Q[Durable condition-event queue]
  Q --> E[ERPNext maintenance Issue]
  E --> W[Workbench asset lifecycle]
  H --> W
  W --> A[OSR inspection and handback evidence]
```

Implemented mappings

Native crate	Equipment and output	Operating use
`osr-energy-site`	Station PV/pad power and input battery SoC	Energy supervision; temperature and lifetime counters remain explicit sensor fixtures
`osr-station-scada`	Effective lighting enabled state and station fault count	Bounded simulated lighting requests and station fault maintenance cases
`osr-bms`	Vehicle battery SoC, current limit and trip state	Battery supervision and a maintenance case on a sustained trip; no remote contactor/reset command
`osr-aux-power`	Comfort-branch availability and fault count	Auxiliary maintenance case; its output gates the HVAC evaluator
`osr-hvac`	Compressor/fan demand and reduced mode	Comfort supervision and investigation of unavailable supply
`osr-onboard-routing`	Route-intent vote, source agreement and hold state	T-ECU/S input to independent movement-authority and ATP evaluation; never a movement permission
`osr-cbm-onboard`	Native health classification, brake remaining fraction and bearing vibration	Sustained Service classification creates one linked ERP Issue per incident
`osr-wayside-points`	Fused detected position and motor activity for each catalogue switch identity	A sustained fail-restrictive `Unknown` detection creates a maintenance case; no point command is exposed
`osr-level-crossing`	Barrier state, warning and fault state	Ready for identified `level-crossing` assets; no catalogue city currently declares one, so none is fabricated
`osr-afc`	Station-level aggregate gate state and grant/denial counters	Passenger-flow visibility from the fare evaluator; it is an aggregate, not an invented count of physical gates

The Rust adapter uses JSON lines over stdin/stdout and keeps BMS, auxiliary-power, HVAC, point, crossing and fare-gate state per city/asset key. It emits the exact `osr-supervision-contract` schema: bounded unique observations with units, source-crate provenance and the literal authority `observation-only`. The crate deliberately contains no command, reset, movement-authority, protection, ERP action or executive-decision type. Battery protection therefore propagates to auxiliary/HVAC availability, and native controller latches survive successive samples. Input time must increase. Each process has a bounded key registry; restarting it resets these simulation states. This harness uses explicit small-pack, comfort, sensor, barrier and token fixtures, not commissioned calibration, key material or actual sensor readings.

A serialized frame fixture records the normal output contract. Rust asserts its native output equals that fixture; Python independently rejects field, identity, range, enum, unit, provenance and authority drift.

The Python projection checks `osr-supervisory-observations/1`, requires the simulation environment and `observation-only` authority, and accepts exactly the reviewed identities, ranges, discrete values, units and source-crate sets. Rust performs native-unit conversion at the contract boundary. Missing or additional observations are invalid, never invented healthy zeroes. The controller process has a reply timeout. Existing source ownership, monotonic sequence, timestamp, staleness and alarm-persistence checks remain at the gateway.

Each equipment package names its `source_crates`; Workbench shows them with the asset. Station/depot, rolling-stock and wayside templates use existing catalogue asset IDs, so `SAM-RS-L1-001:vehicle-cbm` retains its vehicle parent and `SAM-SW-001:points` retains its real switch identity. Selecting a station follows the asset register's parent links to its switch children. The fare-gate view is explicitly a station aggregate. Engineering artifacts are shown only when their reviewed asset binding matches the equipment or its parent. No ERP or FUXA record becomes train-control authority: point and crossing commands, resets, `crossing_safe_for_train`, route state and movement authority remain exclusively in the safety controllers/interlocking and consensus boundary.

The gateway also contains an experimental AI executive council. Its independently authenticated model perspectives can collectively endorse a recommendation and prepare a small allowlist of unsubmitted ERP drafts. Each proposal is bound to the current supervisory asset, telemetry, alarm, case and pending-command digest. No council outcome enters the controller

command queue or changes the engineering, commissioning, employment, payment or legal authority boundaries above.

Reproduce a station, vehicle and wayside pilot

First follow service setup, including ERP project binding and scoped principals. Then:

```
./osr supervision prepare samawah --first-site --first-vehicle
./osr supervision prepare mosul --first-site --first-vehicle
# Initial installation: apply each generated package.
./osr supervision apply build/supervision/samawah/simulation/package.json
./osr supervision apply build/supervision/mosul/simulation/package.json
./osr supervision preview-fuxa \
  build/supervision/samawah/simulation/package.json \
  build/supervision/mosul/simulation/package.json \
  --output build/supervision/fuxa-import-review.json
./osr supervision import-fuxa \
  build/supervision/samawah/simulation/package.json \
  build/supervision/mosul/simulation/package.json \
  --review build/supervision/fuxa-import-review.json
./osr supervision simulate
```

For an existing installation, take `./osr supervision backup`, generate the machine review and apply only that exact proposal:

```
./osr supervision review-package build/supervision/samawah/simulation/package.json \
  --output build/supervision/samawah/simulation/change-review.json
./osr supervision apply build/supervision/samawah/simulation/package.json \
  --expected PREVIOUS_SHA256 \
  --review build/supervision/samawah/simulation/change-review.json
```

The review names changes to source crates, measurement units/ranges/scaling, alarm rules, command bounds/permissives, device bindings and engineering revision. It also carries the affected installed serials, evidence, open cases and pending commands. Apply regenerates it under the database write lock, so a package or lifecycle-record change after review is rejected. Clock regression, future-clock jumps, replay, wrong units, disconnects, stale data and invalid values all have adapter/gateway regression coverage. Physical mapping changes remain blocked. The FUXA preview binds the live project checksum to the desired project and records included package/revision hashes plus device/view additions, changes and removals, along with other project-setting changes. Import requires that unchanged review, then backs up both it and the previous project before replacement. Include every package whose displays must remain. Untracked live display edits appear explicitly as replacements or removals; move wanted changes into a reviewed generator input before import. The generated asset links return to the Workbench in the same browser window.

Without selection flags, preparation expands the templates across applicable station, depot, rolling-stock, switch and declared level-crossing assets. `--first-site --first-vehicle` deliberately limits the Samawah and Mosul local demonstrations to ten equipment positions each: five station views, four vehicle views and the first station's real switch. The all-city readiness compile reuses the current station and switch identities already in the asset registers. No generated package includes a crossing until its city declares a stable `level-crossing` asset. Generic rules live in generic.json; city `operations/supervision.json` overrides select sites, project/company and supplier bindings. No per-city controller fork is needed.

Condition-to-maintenance acceptance

verify-embedded.py injects explicitly simulated Samawah brake-wear and dual-sensor disagreement conditions through the Rust CBM and points crates. It checks native Service/Unknown results, separate deduplicated ERP Issues, isolation from Mosul and condition clearance while both Issues stay open. It restores the previous private simulator fixture file in a `finally` block. The simulation Issues remain audit evidence; the test does not close maintenance records automatically.

The private simulator control file supports `cooling_fault`, `station_fault`, `battery_trip`, `aux_fault`, `cbm_service`, `points_detection_fault`, `crossing_motor_fault`, `faregate_denial`, `disconnected` and `local_remote_disabled`, with optional `cities` overrides. These are simulation fixtures, not public control APIs. BMS, point and crossing states retain their native behaviour; removing a fixture is not a protection-reset command.

The installed native UI acceptance checks ERP permissions and project filters, FUXA city displays, scoped evidence recording and a lighting request through controller output. Controller-result retries are idempotent for the same owner, state and result.

Revised simulation packages retain omitted equipment as retired history while rejecting its new telemetry and commands; reactivation requires another reviewed package. Alarm acknowledgments bind to the displayed occurrence and reset when a cleared condition activates again.

Physical MQTT/OPC UA/Modbus or device-bus adapters, persistent commissioned controller state, supplier calibration and independent railway acceptance remain separate deployment work. Existing ATP/interlocking/traction/brake authority is not routed through ERPNext or FUXA.

OpenSourceRail — one-page overview

Repository source: `docs/open-source-rail-overview.md`

> Build rail locally · retain skills · reduce foreign-capital dependence

OpenSourceRail light-metro reference trainset

An open urban-rail platform designed to keep ordinary engineering, fabrication, integration, software and long-term maintenance capability in the adopting country.

The same deterministic workspace connects city design, GIS, CAD/IFC, simulation, ERPNext business execution, FUXA supervision, costs and railway assurance. Its 265 developing-world planning models estimate 79% domestic value; all values remain planning sensitivities rather than bids or funding commitments.

about $488B	265 cities	$900k	120 rows
Roughly 79% modeled domestic value across 44 country programmes	Developing-world public evidence models; one European model is comparison-only	Local factory-gate LM3 planning target; generated build record $885k	Traceable LM3 parts and assemblies with visible supplier and release gaps

Why local delivery changes finance

Open design lets a country competitively procure ordinary civil work, vehicle structures and interiors, wiring, software, integration and maintenance locally, importing specialist components only where domestic suppliers are not yet qualified.

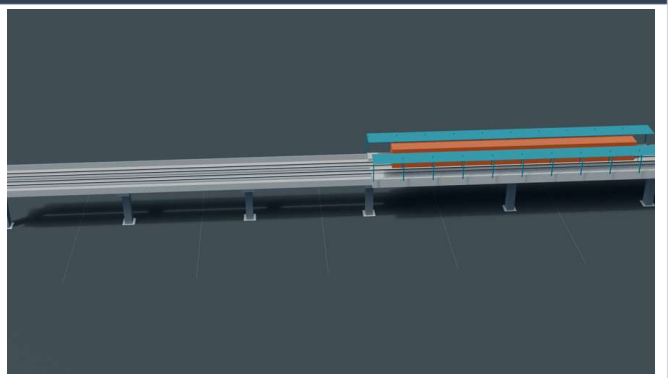
For the same modelled railway scope, suppose the OpenSourceRail case is $100M. The foreign-turnkey column uses the editable default sensitivity: a 2.0× delivered price with 90% requiring foreign currency or international capital.

Where the money goes	Localisation-first OpenSourceRail	Foreign-vendor turnkey sensitivity
Total programme price	$100.0M	$200.0M
Value not requiring external capital	$79.1M	$20.0M
External-capital requirement	$20.9M ■■■■■■■■■■	$180.0M ■■■■■■■■■■

That is $159.1M less external capital (88.4%) before interest. When debt-financed, the external requirement becomes loan principal; interest depends on country terms. This is a controlled sensitivity—not a vendor quotation or financing offer.

Design, regenerate and operate

- Edit lines, stations, alignments, demand and line/day/hour service over local GIS.
- Generate IFC4.3, CAD, quantities, costs and Git-reviewable city packages.
- Run deterministic train, station, energy, wayside, point/crossing and depot software together.
- Use one Workbench for engineering, simulation, OCC, ERPNext/Frappe HR, FUXA and railway assurance.
- Trace 23 publisher records through 16 cross-domain controls and 15 hashed repository checks; change impact reopens affected evidence.

City Studio	Civil IFC coordination
City Studio deterministic browser acceptance	 Bonsai IFC4.3 civil coordination model

Buildable and standards-aware

Reference packages cover all 120 LM3 product rows, nine timed manufacturing methods, 30 mould/tooling families, modular rolling stock, stations, civil works, battery traction, renewable charging, operations and assurance. The Digital Standards Thread spans 23 publisher records, 16 cross-domain controls and 15 hashed repository checks. Supplier freeze, licensed clause assessment, detailed drawings, proof testing, certification and authority approval remain explicit release gates.

Review or collaborate

Review the assumptions, reproduce the generators, open a technical issue or propose an evidence-backed contribution through the public repository.

For offline printing, download the landscape HTML edition.

OpenTrack Evaluation

Repository source: `docs/opentrack-evaluation.md`

Status: accepted architecture decision
Reviewed: 2026-08-14

Decision

OpenTrack does not replace SUMO or the native OpenSourceRail simulator in the default toolchain.

OpenTrack may be used as an optional, independently procured railway-capacity cross-check after a Samawah interchange trial. OpenSourceRail will incorporate the useful analysis classes natively, using open schemas and deterministic tests, rather than depending on OpenTrack at build, CI, city-generation, or deployment time.

This preserves three deliberately different roles:

Tool	Authority in OSR	Role
`osr-sim`	Authoritative for OSR control and energy behaviour	800 V battery/charging, BMS reserve, roof PV, depot service, adaptive headways, movement-authority and onboard safety monitors
Eclipse SUMO	Open, reproducible independent screen	Geometry-shaped timetable execution, opposed services, junction/road interaction, feeder and multimodal sensitivity
OpenTrack	Optional licensed independent review	Rail-specific running time, headway, block occupation, timetable robustness, delay propagation, signalling and capacity comparison

Why It Is Useful

The current OpenTrack product describes support for metro, light rail and tram systems; continuous speed/distance movement constrained by signalling and timetables; line/station capacity and minimum-headway calculation; block, moving-block, CBTC, ATP and ATO studies; infrastructure and train failures; Monte-Carlo delay runs; and power/energy outputs. It also represents gradients, speed limits and train tractive-effort/speed characteristics. Those capabilities are better aligned with rail capacity work than SUMO's generic microscopic traffic model.

Official product information also lists railML 1.0 and 2.2 import/export for infrastructure, timetables and rolling stock, plus OpenTrack XML/ASCII and a SOAP-over-HTTP API. This makes a controlled comparison technically feasible.

Sources:

- OpenTrack Railway Simulation product description
- ETH Zürich archived OpenTrack project description
- OpenTrack API description

Why It Is Not The Default

The official site presents a separately distributed full application and a two-train-limited Light version, with access and API details handled through the vendor. No open-source licence or public source distribution is identified there. OSR therefore treats OpenTrack as proprietary external software unless a procurement review proves otherwise. That inference must be rechecked before any trial.

Making it the default would prevent an unlicensed contributor or CI runner from reproducing a city package. Its advertised railML support also stops at 2.2, whereas `osr-alignment` currently exports a limited railML 3.2 infrastructure document. A direct round trip is not available today.

SUMO must remain because it is open and automatable and answers road, pedestrian, feeder and multimodal questions that are outside OpenTrack's main railway-capacity role. `osr-sim` must remain because neither external tool is the authority for OSR's battery charging, BMS reserve, distributed movement authority, onboard obstacle detection, depot rotation or safety monitors.

Current Capability Gap

Analysis	`osr-sim` now	SUMO city deck now	OpenTrack trial value	OSR disposition
Timetable/headway execution	Yes, scheduled and energy-adaptive	Yes, small deterministic service sample	Detailed timetable construction and robustness	Keep both open implementations; compare results
Continuous train motion	Rest-to-rest trapezoidal/triangular profile	Generic rail vehicle kinematics	Differential speed/distance with train characteristics	Add gradients, resistance and tractive-effort curves natively
Safe separation	OSR section occupancy and movement authority	Rail signals on per-line screening edges	Fixed/short/moving blocks and critical-headway analysis	Add occupation/headway evidence export natively
Interchange conflicts	Logical transfer groups; no full junction timetable model	Current lines are largely independent	Route occupation and station capacity	Add explicit route/conflict graph before acceptance
Disturbances	Named equipment/safety faults	Not a robustness study	Initial/station delays, failures and Monte Carlo	Add seeded stochastic timetable robustness runner
Energy	OSR-specific battery, charging, PV and sites	Not authoritative	General train power/energy	Keep OSR authoritative; compare mechanical duty only
Multimodal/road interaction	Out of scope	Appropriate independent tool	Not the primary purpose	Retain SUMO

Interchange Trial

The first trial is limited to Samawah Line 1 and must not modify the canonical design.

1. Define a simulator-neutral OSR operations exchange containing topology, routes, station chainage, gradients, speed limits, dwell, timetable, consist mass/length, acceleration/braking limits and tractive-effort data.
2. Add a railML 2.2 export adapter specifically for the OpenTrack trial. Do not downgrade or replace the existing railML 3.2 alignment export.
3. Obtain a lawful full-version licence and written automation/API terms. The two-train Light limit is not sufficient for capacity or robustness work.
4. Import the same canonical exchange into `osr-sim`, SUMO and OpenTrack; use identical stop patterns and operating assumptions.
5. Compare running time, minimum headway, block/route occupation, energy at wheel, delay recovery and timetable conflicts. Record model differences, not merely percentage agreement.
6. Export results to a vendor-neutral JSON summary with tool version, input hashes, random seeds and acceptance tolerances. Never require an OpenTrack binary to read retained evidence.

Adoption fails if the trial loses station/route identifiers, needs unreproducible GUI edits, cannot run with controlled seeds, cannot export reviewable results, or requires OpenTrack-specific data to become canonical.

Native OSR Work Packages

The following functions are approved for implementation independently of an OpenTrack purchase:

- `OPS-SIM-001`: simulator-neutral network, vehicle and timetable exchange;
- `OPS-SIM-002`: gradient, curve resistance, adhesion and tractive-effort/speed integration in `osr-sim`;
- `OPS-SIM-003`: route locking plus block/route occupation and minimum-headway reports;
- `OPS-SIM-004`: seeded initial-delay, station-delay and equipment-failure ensembles with percentile punctuality and recovery metrics;
- `OPS-SIM-005`: timetable conflict, knock-on delay and turnback recovery analysis;
- `OPS-SIM-006`: Samawah cross-tool benchmark and documented tolerances.

Until OPS-SIM-001 through OPS-SIM-005 pass their benchmarks, city engineering summaries must describe timetable and capacity outputs as screening evidence, not calibrated operational acceptance.

OpenSourceRail operating platform

Repository source: `docs/operating/README.md`

ERPNext is the business system of record. Frappe HR supplies personnel and payroll on the same Frappe site. OSR retains railway engineering, control and assurance. The operating page launches native ERPNext modules; business forms, permissions and ledgers are maintained by upstream apps.

For the generic configuration, all 266 city override profiles, native operating tasks and digital-twin feedback, use the city operating platform. The basic import below remains available for older planning packages.

The generated operating-readiness audit validates each tracked asset register, operations manifest and project twin, then compiles all 266 ERP profiles, reusable-component packages and real-asset supervision packages. It also compiles the lifecycle governance, HR/admin and QR templates against every asset identity while deliberately producing no printable label. The audit separates reproducible large payloads from legal-company, calendar, ERP master and physical-binding data that the repository cannot safely invent. Supervision reuses station, vehicle and all declared switch identities; fare gates are labelled station aggregates, and zero crossing positions are generated until real level-crossing assets exist. Wayside views remain read-only maintenance surfaces and cannot issue safety commands or movement authority.

What replaces what

Existing OSR component or requirement	Operating owner	OSR responsibility retained
Project Twin execution, progress and task administration	ERPNext Project, Task, Timesheet	Design generator, finite-resource engineering baseline, source revision and scenario evidence
Ops Core purchase orders and procurement actuals	Material Request, Supplier Quotation, Purchase Order	Generated procurement requirements and source BOM references
Deliveries and material records	Purchase Receipt, Item, Warehouse, Stock Entry, Serial No / Batch	Railway asset identity and engineering configuration
Invoices, payments, budgets and cashflow actuals	Purchase Invoice, Payment Entry, Budget, Cost Center, accounts	Parametric estimates and finance sensitivities remain planning evidence
Manufacturing administration	BOM, Workstation, Work Order, Job Card, Quality Inspection	Railway manufacturing definitions, travellers and assurance hold points
Routine maintenance administration	Asset, Asset Maintenance, Asset Maintenance Log, Asset Repair	CBM telemetry, defect severity, railway work evidence and independent handback
HR and ordinary workforce administration	Frappe HR Employee, Recruitment, Shift Assignment, Attendance, Leave, Payroll, Expense Claim	Railway competence/fitness acceptance and safety-critical roster constraints
Ordinary business attachments and approvals	Frappe File, roles, permissions, workflows and document history	Controlled engineering revisions, sealed inspections, NCRs and railway release evidence
OCC, signalling, ATP/ATO, train electronics, depot movements, AFC, historian	OpenSourceRail railway components	Entire railway-specific function

ERPNext's Work Order means manufacturing production. Routine maintenance uses Asset Maintenance / Asset Repair, not a manufacturing Work Order. Frappe HR is a separate app; installing ERPNext alone does not supply payroll.

Condition maintenance now joins those native records without replacing them. The supervisory gateway deduplicates a condition into an Issue. A maintainer with native permissions can preview and create an Asset Repair from that Issue, including part requirements, a native technician assignment, expected downtime and links to the existing condition/configuration evidence. The repair is deliberately a draft; ERPNext owns completion, stock consumption, serial/batch selection, costs and actual downtime. OSR continues to own inspection, restrictions and railway handback.

Local installation

Requires Docker Engine with Compose and Buildx. The local composition binds only `127.0.0.1:8080`; MariaDB and Redis have no published ports.

```
./osr erp init
./osr erp build
./osr erp up
./osr erp setup
./osr erp status
```

Open http://127.0.0.1:8080. Log in as `Administrator` using the generated `ADMIN_PASSWORD` in `var/erpnext/local.env` (mode 0600, ignored by Git and not served by OSR). Complete ERPNext's setup wizard for the actual organisation. Do not replace a real legal entity with the evaluation fixture below.

The image pins ERPNext `v15.121.2` and Frappe HR `v15.64.0` on the maintained v15 line. It extends the official ERPNext image with the local `osr_erpnext` app. Frappe comes from that image; `./osr erp bench list-apps` reports the exact installed versions. The app adds reference fields and a workspace using installation/migration hooks, with no edits to upstream source.

For rootless Docker installed in `~/bin`, the launcher finds Docker there. The user service starts at login; persistence without a login requires the host administrator to configure user lingering. No rootful daemon is needed.

Run `./osr` and select Operating · ERPNext. `OSR_ERP_URL` can point the OSR server at an existing ERP site. Navigation opens a separate ERP login; OSR actor/role URL parameters grant no ERP permissions. No API keys enter browsers.

Import a city baseline

```
python3 tools/automation/erpnext-plan.py \
  cities/catalogue/west-asia/Iraq/Samawah/operations/samawah-operations.json.gz \
  --output build/erpnext/samawah-plan.json

# After creating your company in ERPNext:
./osr erp import build/erpnext/samawah-plan.json --company 'Your Company'
```

For a local evaluation only, create the explicitly named sample company:

```
./osr erp bench execute osr_erpnext.setup.create_evaluation_company
./osr erp import build/erpnext/samawah-plan.json --company 'OpenSourceRail Evaluation'
```

The importer creates one native Project per company/city/revision and native Tasks for manufacturing and maintenance planning rows. Original row JSON, city, railway asset ID, revision and SHA-256 references remain attached. Unique source keys make retries skip existing tasks, preserving operator edits. Changed content under an existing revision fails; a new revision creates a separate planning project for explicit review. The import is transactional.

`--kind manufacturing` or `--kind maintenance` limits a prepared plan. The importer carries relative working-day offsets, dependencies, CAD/BOM refs and cadences in source evidence. It does not convert those into approved dates, recurrence, task dependencies or released production/maintenance orders. Map the approved calendar, dependencies and resources in ERPNext before execution. Generated assets are proposed engineering objects, so the importer does not automatically capitalize them or fabricate Item, Supplier or accounting masters.

Authenticated integrations can POST `{plan, company}` to `/api/method/osr_erpnext.api.import_plan` using Frappe's token authentication. The endpoint checks Project/Task creation permissions and Company access. Use dedicated users with appropriate Company User Permissions and project access; cities are provenance fields, not automatic tenancy boundaries.

Cutover and existing records

1. Back up SQLite/evidence with `ops-core-backup.py` and preserve its signing key separately. Keep historic records for audit; do not delete them.
2. Configure actual companies, users, company permissions, currency, chart of accounts, fiscal years, warehouses, suppliers, items and asset categories.
3. Import a reviewed engineering baseline. Map actual assets and material/BOM definitions; commission maintenance schedules and manufacturing orders.
4. Reconcile historic Ops Core purchases, deliveries, invoices, payments and progress with ERPNext using its Data Import and accountant-reviewed opening records. These collections remain readable in OSR; their mutation is now rejected server-side. No automatic ledger migration is supplied.
5. Use ERPNext for all new business records. OSR railway work records continue to carry inspections, NCRs, controlled documents and independent handback.

Permission-filtered business snapshots return to the Workbench through the city feedback bridge. No reverse writes to railway state are installed. ERP completion, quality acceptance, employee attendance or stock receipt never grants railway release authority. An ERP outage must not stop interlocking, dispatch or railway assurance reads. This change supplies business applications and a planning handoff, not a completed migration of an operator's financial or personnel records.

Operations and recovery

```
./osr erp logs
./osr erp backup
./osr erp stop
```

Backups are written to the site's `private/backups` directory in the `sites` volume. Copy them to private off-host storage along with site configuration and its encryption key. Verify restore in a separate Compose project before upgrading. `stop` preserves all volumes. Do not use `down -v` on real records. For upgrades, back up first, review compatible upstream app versions, rebuild, then migrate the site. Restore the pre-upgrade database/files with the previous image for rollback; changing the image alone cannot undo schema migrations.

Before shared hosting, use a dedicated TLS domain, named ERP users, tested role and company permissions, scheduled backups and restore drills. Configure local payroll/tax rules for each legal entity. The checked-in localhost stack is an evaluation/development installation, not a commissioned shared service.

Further integration and automation

Reusable operating components provide dynamic forms and city configuration packages for stock, production, quality, maintenance, training, budgets, service issues and assignment rules.

Lifecycle governance and QR templates define the role separation, competence/authorization, inspection, calibration, configuration, management-review and lookup-only asset-identity records that connect those native forms to mechanical, civil and software evidence.

See the function review and implementation map for procurement draft creation, automatic work-readiness checks, purchasing lifecycle feedback and the next stock, production, maintenance and workforce integrations.

Upstream basis and licensing

- ERPNext source and supported business modules
- Frappe HR source and separate-app installation
- Official container custom-image guide
- Frappe REST API and authentication
- Docker rootless installation

ERPNext and Frappe HR retain GPL-3.0; Frappe Framework retains MIT. OSR's local integration source follows the repository Apache-2.0 software license. Upstream source/license obligations apply when distributing the combined container; the OSR license does not relicense upstream applications.

Connected lifecycle and equipment supervision

Lifecycle integration connects the existing ERP platform to FUXA, embedded OSR controllers, FreeCAD, IFC/Bonsai and QGIS. Run the supervisory pilot.

AI executive council for ERP and supervisory decisions

Repository source: `docs/operating/ai-executive-council.md`

OpenSourceRail now has an experimental, city-scoped governance contract for an “AI CEO” and supporting AI finance, operations, maintenance, procurement and workforce managers. These are software offices, not legal persons, company directors, railway duty holders or substitutes for licensed and accountable people. Their useful role is to analyse the same evidence, expose reasoning and prepare routine ERP work at low marginal cost.

Authority does not come from a persuasive model answer. It comes from an operator-approved delegation policy, independently authenticated model identities, a deterministic verifier and an operator-held attestation key.

```
flowchart LR
  S[Immutable SCADA / ERP context digest] --> P[Executive proposal]
  P --> A[Strategy model]
  P --> B[Finance model]
  P --> C[Operations model]
  P --> D[Risk model]
  A --> V[Deterministic policy verifier]
  B --> V
  C --> V
  D --> V
  V -->|quorum + independence + no dissent| H[HMAC-attested decision]
  V -->|missing evidence or dissent| R[Human review]
  H --> E[Bounded ERP draft packet]
  H -. never .-> X[SCADA actuation / railway authority]
```

Why this is more authoritative than “one AI decides”

Each ballot identity is taken from the authenticated service principal, not from the submitted JSON. One principal and one model identity can cast only one immutable ballot. The default policy requires four perspectives—strategy, finance, operations and risk—at least two providers, at least three model families and a 75% endorsement ratio. Any rejection records visible dissent and forces human review. Four wrappers around the same model therefore do not satisfy the independence test.

Every proposal is bound to a transactionally captured operational context: accepted asset/package state, latest telemetry, alarms and ERP case status, pending controller commands and pending ERP condition events. The context, proposal, each ballot, policy and final decision have SHA-256 identities. A successful terminal decision is signed with the gateway's operator-owned HMAC key. This proves what the configured council and policy decided; it does not prove that the recommendation is correct.

The gateway deliberately contains no provider client. A deployment must connect separately secured model runners and give each one an `executive-model` principal with fixed `model_id`, `model_family`, `provider` and `perspective` metadata. This prevents a caller from claiming a different model in its ballot and keeps provider credentials outside the gateway.

The repository now includes a provider-neutral `executive-model-runner.py`. Run one process and one secret set per model identity; do not place all model tokens in a central “council” process. A runner polls only its configured scopes, skips proposals it has already voted on, fetches the immutable context, calls one HTTPS (or loopback HTTP) adapter and posts the result using its own gateway token. It never receives other ballots, so an earlier answer cannot anchor its analysis. The gateway checks the runner's expected model/perspective against the authenticated principal and records hashes of the exact adapter request and raw response.

A runner configuration contains environment-variable names, never secrets:

```
{
  "schema": "osr-executive-model-runner/1",
  "gateway_url": "http://127.0.0.1:8092",
  "adapter_url": "https://independent-risk-model.example/evaluate",
  "model_id": "deployment-risk-model-v1",
  "perspective": "risk",
  "gateway_token_env": "OSR_RISK_GATEWAY_TOKEN",
  "adapter_token_env": "OSR_RISK_ADAPTER_TOKEN",
  "cities": ["samawah"],
  "environments": ["simulation"],
  "poll_seconds": 30
}
```

The adapter receives `osr-executive-model-request/1`: the proposal, frozen operational context, its digest and separately labelled decision rules. All proposal/context strings are explicitly untrusted data. It must return only:

```
{
  "schema": "osr-executive-model-response/1",
  "verdict": "endorse",
  "rationale": "Bounded explanation tied to the supplied evidence.",
  "claims": ["One independently checkable claim."]
}
```

Start a single identity with `python tools/automation/executive-model-runner.py PRIVATE_CONFIG.json`; add `--once` for commissioning checks. The adapter is the deployment-specific boundary to a local open-source model or hosted

provider. OSR does not assume a provider API, place provider credentials in tracked files, or silently fall back to another model.

Delegation constitution

Result	What the system may do
`advisory-endorsed`	Publish a traceable recommendation only.
`delegated-erp-draft-authorized`	Place an attested packet in the local ERP-draft handoff queue. It remains unsubmitted.
`human-approval-required`	Preserve the recommendation for a named approver; create no action packet.
`disputed-human-review-required`	Preserve every ballot and dissent; create no action packet.
`insufficient-verification`	Keep collecting ballots; do not seal or act.
`expired-unverified`	Seal the expired record without an action packet.

The code-level allowlist contains only reversible draft preparation:

- draft budget scenarios, maintenance plans, material requests and work orders.

Everything else is outside delegation. In particular, an AI office cannot issue or relay SCADA commands, movement authority, signalling or safety release; hire, dismiss, discipline or set compensation; make payments; sign contracts; submit ERP documents; certify competence; accept engineering; or hand back an asset. Adding one of these strings to a runtime configuration is rejected because the configuration allowlist must itself be a subset of the compiled safe list.

This boundary makes the HR-cost proposition testable rather than rhetorical: measure time removed from evidence collation, draft preparation and routine coordination while retaining the legally and operationally accountable roles. Do not book a head-count saving until a pilot has measured review time, error rates, provider cost, exceptions and required oversight.

API and records

The authenticated gateway provides:

Endpoint	Role	Effect
`POST /executive/proposals`	`executive-secretary`	Validate a proposal and freeze its current city/environment context.
`POST /executive/ballots`	`executive-model`	Add one immutable ballot using the model metadata on the principal.
`POST /executive/finalize`	`executive-auditor`	Apply the deterministic policy and, for a terminal result, attest and seal it.
`GET /executive/decisions`	any scoped principal	List decisions or fetch one by `?id=` with context, ballots, dissent and action packet.

Workbench Connected lifecycle includes a read-only AI executive council section for the selected city/environment. It shows quorum and diversity counts, every attributed ballot and dissent, evidence/decision hashes, attestation key and signature, the explicit authority boundary, and ERP delivery or rejection state. The browser receives only the server-side scoped viewer capability and exposes no secretariat, auditor, model or attestation credential.

`./osr supervision init` creates private secretary/auditor identities, a fresh attestation key and the default council policy. It does not invent model identities. Configure real model runners explicitly, for example:

```
{
  "subject": "council-risk-runner",
  "role": "executive-model",
  "model_id": "deployment-model-id",
  "model_family": "deployment-family",
  "provider": "deployment-provider",
  "perspective": "risk",
  "token": "PRIVATE_RANDOM_TOKEN",
  "cities": ["samawah"],
  "environments": ["simulation"]
}
```

Create and finalize reviewed files without copying credentials into shell history:

```
./osr supervision council-propose proposal.json --output received-proposal.json
# Allow the separately deployed model identities to post their ballots.
./osr supervision council-finalize PROPOSAL_ID --output decision.json
```

The ERP handoff is a durable `executive_actions` queue with bounded retry. The ERPNext receiver independently recalculates the proposal and decision hashes, verifies the HMAC against a privately provisioned key ID, confirms the action is on its own compiled allowlist and checks city/project scope. It then reuses the existing permission-checked preview/apply adapter to create or return exactly one native draft. `OSR Executive Decision` retains the packet-to-document mapping. A lost response is safe to retry. The receiver rejects any post-decision change, non-draft result or non-allowlisted action and never submits the document. Native ERP permissions remain a second gate: the integration provisioner does not grant finance, stock, manufacturing or maintenance manager roles. Operators must commission least-privilege access for only the enabled draft types and referenced masters; permission or validation failures are recorded as rejected.

Operating controls

- Keep the attestation key in the private deployment secret and rotate it with a new key ID. A decision retains the key ID used to sign it.
- Keep model providers in distinct security and billing accounts where possible. Provider diversity is weak if all providers share one upstream model or prompt.
- Pin model/version and prompt artifacts in each runner's own evidence. The gateway authenticates the configured identity but does not remotely attest a provider binary.
- Treat SCADA text, alarms, work descriptions and attachments as untrusted input. Model runners should isolate retrieved content from system instructions.
- Review abstentions and dissent. Do not retry models until a preferred answer appears; proposals and ballots are immutable, so a changed case needs a new ID.
- Monitor correlated recommendations, drift, missing perspectives, stale proposals, provider outages and the ratio of drafts rejected by humans.
- Back up the gateway database with the existing supervision backup. Council proposals, contexts, ballots, decisions, actions and audit events are included.

The approach aims for transparent institutional memory and lower administrative cost, not the fiction that management accountability can be outsourced to a language model.

ERPNext integration and automation review

Repository source: `docs/operating/automation-review.md`

This document preserves the initial adapter review and commissioning sequence. Its “next” recommendations describe that increment, not the current completion status. Use integration status and the review follow-through for current capabilities, executed acceptance checks and remaining gaps. The connected lifecycle now includes native purchasing quantities, production/serial records, repairs and revision dispositions beyond the initial document-count feedback described below.

Reusable adapters for the eight recommended areas are now implemented. See Operating components for their dynamic forms, city profiles, preview/apply API and activation behaviour. The master-data requirements below still apply when commissioning each component.

Reviewed against the installed ERPNext 15.121.2, Frappe 15.120.1 and Frappe HR 15.64.0. All 62 native document types checked across the ten areas below exist in this installation. Current upstream documentation can also describe newer versions; this implementation uses the installed v15 schemas and native validation.

Implemented in this increment

1. Procurement requirement draft Material Request. Open an imported procurement Task and choose OpenSourceRail Material request. Search by requirement ID, description or railway asset; select a real ERP Item, enter the verified quantity in its stock unit, a required date and a city warehouse. The native request retains the source requirement, project and cost centre. Repeating the same mapping returns the existing request and preserves edits. Different inputs for an already converted requirement require review of that request. Requests remain unsubmitted; no supplier orders or emails are sent.
2. Purchasing lifecycle feedback. The twin counts native Material Requests, Purchase Orders, Purchase Receipts and Purchase Invoices linked through either document headers or item lines. A document appears once per project even if several lines match. Parent-document permissions and company scope apply. Work Orders and project Budgets also appear. These are document counts, not ledger amounts or quantities combined across incompatible units.
3. Automatic work-readiness checks. Every snapshot counts overdue, undated and unassigned unfinished Tasks. Completed/cancelled work is excluded; missing dates are not classified as late. Counts overlap. The existing five-minute feedback timer refreshes these checks without assigning staff or sending notifications. ERP uses its configured site date for lateness.
4. Condition Issue reviewed Asset Repair. An Issue linked to a commissioned Asset now previews the native repair, parts and city-store availability, technician assignment and expected downtime. Apply is repeat-safe and creates only a draft; native submission owns part consumption and actual repair data. Feedback returns repair/part/assignment/evidence state while always marking railway handback as external and still required.

These additions work for every applied city baseline without changing its engineering/configuration checksum. Native operator transactions sit alongside that immutable baseline. The generic/company/city warehouse configuration is reused; no separate per-city implementation is needed.

An experimental AI executive council now adds a multi-model recommendation and attestation contract above these integrations. It can prepare only an allowlisted, unsubmitted ERP draft packet after quorum, provider/model-family diversity, all required perspectives and no dissent. It does not create employees, approve budgets, submit documents, pay suppliers, sign contracts or exercise railway/SCADA authority. ERPNext independently checks the attestation and content binding, then uses the existing native component or procurement adapter to create exactly one unsubmitted draft.

Original integration recommendations

These are implementation recommendations based on the available native functions, rather than claims that all integrations are already enabled.

Area / native components	OpenSourceRail input and useful automation	City/operator data needed	Recommended order
Purchasing: Material Request, RFQ, Supplier Quotation, Purchase Order, Receipt, Invoice, Supplier Scorecard	Convert verified BOM requirements into requests; follow the native quotation/order/receipt process; compare promised and actual delivery against required-by dates	Approved Item/UOM mapping, verified quantities, delivery dates, supplier qualification and purchasing authorities	Now: draft conversion and lifecycle counts installed; bulk mappings and delivery-risk checks next
Inventory: Item Reorder, Warehouse, Stock Entry, Serial No, Batch	Maintain depot spare levels; use native automatic reordering; associate serialised parts and batches with receipt/installation evidence	City stores, stock opening balances, replenishment levels and quantities, traceability policy	Next, after the Item catalogue

Area / native components	OpenSourceRail input and useful automation	City/operator data needed	Recommended order
Manufacturing: BOM, Operation, Workstation, Production Plan, Work Order, Job Card, Subcontracting Order	Convert approved production recipes into Work Orders and Job Cards; feed material consumption and actual production time back to the twin	Verified multi-level BOMs, output Items, operations, workstations, capacities and calendars	Next, one representative assembly first
Quality: Quality Inspection Template/Inspection, Quality Procedure, Goal, Review, Action	Require incoming/in-process inspections on native stock/production documents; carry batch/serial and rejected-material references into OSR evidence	Test methods, tolerances, sampling, qualified inspectors and real receipt/job-card references	Next, with procurement and production
Assets: Asset, Maintenance Team, Maintenance, Maintenance Log, Repair, Movement	Map commissioned assets; generate native scheduled maintenance logs; convert condition Issues into reviewed Repairs with part/assignment feedback; track repair costs/movements	Real asset identities, commissioning dates, approved intervals, service teams and accounting treatment	Now: calendar servicing and condition-repair transition installed; movement/repair-pool commissioning next
Finance: Budget, Cost Center, Timesheet, Expense Claim, Payment Entry, Bank Transaction	Apply native project/account budget controls; collect labour costs and expenses; reconcile commitments and actuals	Fiscal year, account mappings, approved budgets, currency, costing rates and access controls	Now: project Budget visibility; budget commissioning next
People: Staffing Plan, Job Requisition, Job Opening, Onboarding, Training Program/Event/Result, Employee Skill Map, Shift Type/Assignment	Derive role-demand and training proposals from task roles; use native recruitment, onboarding and training workflows; compare available skills to work demand	Actual establishment, named employees, competent assessors, training outcomes and agreed calendars	Next: training/role templates; staff assignment after validation
Service: Issue, SLA, Warranty Claim, Maintenance Schedule/Visit	Track facility defects, supplier warranty cases and contracted service response; link tickets to city/project/asset references	Service providers, contract terms, priority rules, response hours and authorised communication channels	After assets and supplier contracts
Contracts/revenue: Contract, Quotation, Sales Order, Sales Invoice	Track EPC/service milestones, customer claims and contractual billing	Signed contract structure, customers, tax/accounting setup and milestone acceptance	Later, per operating business model
Workflow: Workflow, Assignment Rule, Notification, Auto Repeat, Webhook	Route business approvals, distribute work by city/department, repeat eligible business documents, publish authenticated integration events	Named approvers/users, segregation rules, escalation policy, approved recipients and event consumers	Configure after the responsible teams are known

Boundaries that matter

- A manufacturing planning task is not automatically a valid production BOM or Work Order. Scope descriptions and quantity-basis strings need verified units and executable recipes before they become stock transactions.
- OSR maintenance triggers include calendar, mileage and condition logic. Native calendar maintenance and the condition-Issue-to-Repair transition are installed. Mileage/cycle thresholds and deployment-specific approved maintenance rules remain to be commissioned with trustworthy meter/reset data.
- ERP inspection acceptance, training completion and asset status do not grant railway competence, signalling authority, safety-case acceptance or handback.
- The city CAPEX estimate is a planning reference, not an approved ERP Budget. Budget controls need account/fiscal-year/currency mapping and operator approval.
- Native Issues can cover business/facility service. Train dispatch and railway incident authority remain in OSR. Frappe Helpdesk, CRM and Learning are separate apps, not assumed to be installed by this integration.
- Payroll, bank payments, email delivery and external webhooks need real operator configuration. This review does not activate them.

Reproducible commissioning sequence

For each city, apply its operating baseline, then maintain the following reviewed mapping set alongside its deployment records: Items/UOMs and quantities; stores and reorder policy; production BOMs/workstations; commissioned assets and service intervals; roles/training requirements; fiscal/account/budget mapping; business approval rules. Reuse shared templates and override only the city-specific data. Keep credentials and employee personal data out of tracked city profiles.

The first practical sequence is items and procurement stock and incoming quality production commissioned asset maintenance workforce and finance controls. Each stage should use native ERP documents and return its execution state through the existing city/revision-linked twin.

API and operational behaviour

- POST `osr_erpnext.executive.ingest_draft`: accepts only a complete attested council packet. It independently verifies the HMAC, decision/proposal hashes, compiled action allowlist and provisioned city/project scope, then calls the existing native preview/apply adapter. The result is repeat-safe and must have `docstatus = 0`; an `OSR Executive Decision` records the immutable mapping.

- GET `osr_erpnext.procurement.candidates`: parameters `task`, optional `search` and `offset`; returns at most 50 matching source requirements.
- POST `osr_erpnext.procurement.create_material_request`: parameters `task`, `requirement_id`, `item_code`, `quantity`, `schedule_date`, `warehouse`. Requires Task read, Project write, Material Request create and master-data read access. Warehouse must belong to that city's configured company/stores. Quantity is explicit, finite and positive; required date cannot be in the past.
- GET `osr_erpnext.city_runtime.city_status`: adds `readiness` and extends `business_documents`. Existing snapshot fields remain compatible.
- POST `osr_erpnext.integration.preview_repair`: parameters `issue`, `proposal`; validates the linked city Asset, technician, Items/warehouses, optional serial/ batch bundles and availability without writing.
- POST `osr_erpnext.integration.apply_repair`: adds the preview fingerprint and creates or returns the repeat-safe draft Asset Repair and native assignment. It does not submit, close the Issue or authorise handback.

A request identity is unique per project, procurement task and source requirement. The database unique constraint prevents concurrent duplicates; a competing call can receive a duplicate error and safely retry. Cancellation/amendment remains an explicit native workflow. Repeating a conversion returns the existing record and its document status; it does not silently recreate a cancelled request.

Verification

Run the pure rule tests:

```
tools/automation/osr-python -m pytest -q tools/automation/tests/test_erpnext_automation.py
```

On the local evaluation site with Samawah and Mosul applied, run the native transaction check:

```
docker compose --env-file var/erpnext/local.env -f deployment/erpnext/compose.yaml \
  exec -T backend env/bin/python < deployment/erpnext/tests/verify-procurement.py
```

It creates a temporary Item and draft request inside a rolled-back transaction, checks line-linked twin counts, preserved edits, conflicting inputs, wrong-city warehouses and denied guest access. It does not submit or commit its fixtures.

Upstream references

- Material Requests and automatic stock replenishment.
- Work Orders, Quality Inspection and Asset Maintenance.
- Budgets, Training Programs and Job Requisitions.
- Issues and SLAs and Assignment Rules.

Reproducible city operating twins

Repository source: `docs/operating/city-platform.md`

Each city has a reproducible business operating baseline alongside its railway engineering baseline. ERPNext provides the native operating records. The OSR integration compiles planning inputs and returns permission-filtered execution snapshots; it never issues railway release authority.

Configuration layers

Layer	Location	Purpose
Shared configuration	generic.toml	Departments, warehouses, task categories, programme tasks and planning calendar defaults
City overrides	`cities/catalogue/<region>/<country>/<city>/operations/erpnext.toml`	Identity, local calendar, organisation, operating release and additional city tasks
Compiled package	`build/erpnext/cities/<slug>/`	Effective configuration, complete task plan, dependency graph and checksummed summary
ERP execution	Native Project, Task, Department, Warehouse, Cost Center	Actual work, assignments, dates, timesheets, costing and business document workflows
Twin feedback	Private `var/erpnext/operating-twins.json`	Observed task/category/asset status, recorded hours, task costs and project-linked business document counts

All 266 catalogue cities have specific override files. The compiler derives their actual task and asset scope from their own generated operations bundle; it does not reuse Samawah's tasks for another city. See the Samawah profile and Mosul profile for local timezones, illustrative six-day construction calendars and distinct city work packages. Other profiles inherit the shared planning assumptions until a city operator supplies overrides.

The catalogue readiness gate validates all 266 configurations against their own tracked asset, manifest and project-twin evidence. Supervision can compile directly from those reviewable assets for every city. Full ERP task plans additionally need the deterministic compressed operations payload: Samawah and Mosul keep it locally in Git, while the other 264 materialise it with `./osr city <slug>` before import. A manifest is not treated as a substitute for the task records it hashes.

Tables merge recursively. Lists explicitly replace the shared list. Unknown keys fail validation. `city_programme` adds city tasks to the shared programme; each task needs a unique ID. Removing a department requires updating every task category/programme reference that used it.

The generic company, calendar start date and timezone are unset. The shared Monday–Saturday calendar and the Iraq examples are planning assumptions, not claims about an operator's actual roster or approved public holidays. No employee, supplier, bank, payroll, tax or commissioned asset data is invented.

Repeatable deployment

```
# Creates missing profiles; existing overrides are preserved.
./osr erp city init-configs
./osr erp city validate
./osr readiness --check

# The target company must already exist in ERPNext.
./osr erp city prepare samawah --company 'OpenSourceRail Evaluation'
./osr erp import build/erpnext/cities/samawah/plan.json \
  --company 'OpenSourceRail Evaluation'

./osr erp city prepare mosul --company 'OpenSourceRail Evaluation'
./osr erp import build/erpnext/cities/mosul/plan.json \
  --company 'OpenSourceRail Evaluation'
./osr erp snapshot
```

Replace either slug with any catalogue city. If its generated operations bundle is missing, run `./osr city <slug>` first, or supply a generated bundle with `prepare --bundle <path>`. Generation and preparation do not contact ERP. Review `effective-config.json`, `summary.json` and `plan.json` before import.

To make company binding permanent, add `[organisation] company = "…"` to the city profile. CLI `--company`, `--start-date` and `--release` overrides are captured in the effective configuration and its checksum. No credentials belong in tracked city profiles; use separate private site configuration.

When updating the integration code on an existing site:

```
./osr erp backup
./osr erp build
./osr erp up
./osr erp bench migrate
```

Functions supplied by the city package

- Manufacturing delivery: native Tasks with OSR manufacturing identity, task types, baseline day offsets and native predecessor relationships.
- Maintenance planning: fleet work maps to Fleet Maintenance; other assets map to Infrastructure Maintenance. Source cadences, triggers and evidence remain attached. Operators turn approved schedules and actual commissioned assets into native Asset Maintenance/Repair workflows.
- Assurance coordination: native Tasks for QA actions retain the railway hold point, asset, evidence and release-authority references. Completing them in ERPNext does not close OSR NCRs or approve railway handback.
- Procurement packages: requirements are grouped by sourcing route, with their earliest order-by and latest required-by day. All original BOM and sourcing requirements remain in the source record. Use native Material Request/Quotation/Purchase Order forms after resolving real items, quantities, suppliers, currency and approvals. The procurement automation also converts a selected requirement into a native draft Material Request once the operator provides its Item, verified stock-unit quantity, date and city warehouse.
- Programme and people: tasks for governance, business masters, workforce planning, maintenance readiness, assurance responsibilities and recovery, plus the city's own additional tasks.
- Organisation: stable native departments, warehouse locations and a city cost centre, scoped by company and city. Task Types are shared. No existing roles or user permissions are expanded by the configuration process.
- Digital-twin feedback: native Project forms gain an OpenSourceRail Operating twin action. Workbench displays matching ERP execution in the Project Twin view, and the operating page compares city baselines and asset workload. Hours/costs come from native Task costing; these are not the full general ledger or the city's generated CAPEX estimate.

Project and Task forms keep their source evidence, checksums and configuration under an OpenSourceRail reference tab, alongside the native business tabs.

Relative working days map to dates only when `calendar.start_date` is set. Weekends and explicit holidays are skipped, including negative mobilisation offsets. A non-working anchor rolls forward to the next working day. The compiler rejects missing dependencies, cycles, duplicate IDs, invalid calendars, mixed-city records and checksum mismatches. Native ERP dependencies retain ERPNext's business semantics, including its treatment of cancelled tasks.

Baselines, upgrades and operator edits

The first operating release (`release = "1"`) upgrades the earlier Project/Task import in place when source identity and hashes match. It fills empty type, department and date fields, adds dependencies only when none exist, and preserves existing task IDs, status, progress, assignments, priorities and operator-entered planning fields. It also creates the new work categories. Subsequent identical imports skip configured tasks and reuse city masters.

A changed configuration or source package cannot silently replace an applied baseline. Increment the city's `release` and prepare again to create a separate reviewable project baseline. Existing records stay intact. There is no automatic transfer of actual costs or approval of a new release. Workbench's engineering view shows execution only when exactly one exported ERP project matches the city and engineering revision; multiple candidates require explicit review on the operating page. No release is silently selected as authoritative.

Imports use native document validation, one database transaction and a final project-total recalculation. A failure rolls back new masters/tasks and metadata. The importer preloads permitted task identities and uses ERPNext's native project bulk-update flag to avoid recalculating thousands of records repeatedly. Use the bench-backed CLI for large city packages; a synchronous HTTP import can exceed a deployment's request timeout. Native task-tree updates can hold database locks during large imports, so schedule them outside active task editing.

Refreshing the integrated twin

```
# One private, atomic export. Previous valid feedback survives export failures.
./osr erp snapshot

# Linux user timer: refresh every five minutes, using the local Docker/bench site.
./osr erp feedback start
./osr erp feedback status
./osr erp feedback stop
```

The timer and service are installed under `~/.config/systemd/user/` as `osr-erp-feedback`. The user session and container runtime must be running. The snapshot file is mode 0600 in a private directory; OSR's static server does not expose it. Only the loopback Workbench exposes the aggregate snapshot endpoint. Shared/network deployments need an authenticated integration service and their own permissions; this local bridge is not a shared-host auth scheme. The bundled single-site proxy validates browser WebSocket origins and routes ERP notification authentication through the internal Compose network. Review that proxy configuration when adding a public domain or TLS termination.

ERP callers can GET `/api/method/osr_erpnext.city_runtime.city_status?project=…`. The method checks Project access and uses native permission-filtered lists for Tasks and business documents. Unavailable document categories are marked unavailable, not reported as zero. No employee personal details are exported. The local bench export runs with the site's administrative visibility.

Snapshots show their observation time; the UI labels snapshots older than one hour. Refresh view reads the latest exported file; it does not claim to query ERP directly. ERP unavailability leaves the previous snapshot available with its original timestamp and does not affect railway control.

Verification

```
tools/automation/osr-python -m pytest -q \
  tools/automation/tests/test_erpnext_city.py \
  tools/automation/tests/test_erpnext_planning.py
./osr erp city validate
```

The tests cover configuration inheritance, every city's identity, working-day calendars, dependency rejection, deterministic packages, import fingerprints, legacy keys and company/release separation. Live validation should additionally exercise native imports, repeated imports, permission-filtered feedback and the Project/Workbench views against a private ERP site.

Upstream behaviour: ERPNext Tasks, Asset Maintenance.

Reusable operating components

Repository source: `docs/operating/components.md`

Eight typed adapters connect city operating baselines to native ERPNext and Frappe HR records. Their shared registry drives the API, validation, interactive forms and repeatable configuration packages. Adding another city requires data, not another implementation of the adapters.

These execution adapters sit underneath the lifecycle governance, HR/admin and QR templates. The templates specify evidence and separation of duties; they do not fabricate employees, permissions, approvals or native ERP master data.

Use from ERPNext

Open a city Project and choose OpenSourceRail Operating components. Select a component, supply a stable instance key and its inputs, then Preview. The review shows the native record and its effects. Apply component validates again and opens the resulting native record. Warehouse choices are restricted to the city's configured stores. Quality inspection references provide an item line picker. Account and user lists use editable table controls.

Component	Native result	Behaviour after apply
Stock replenishment	An Item Reorder row on an existing stock Item	Live configuration; native stock settings/scheduler govern automatic requests
Production order	Work Order populated from an active submitted BOM	Draft; native submission and Job Card/stock workflows follow
Incoming inspection	Quality Inspection using an approved parameter template and a project-linked receipt/invoice line	Draft; inspectors enter measurements and submit; native reference-link hooks still run
Asset servicing	Asset Maintenance, native log and assignment	Schedule and team-member assignment start on save; requires a submitted, commissioned city Asset
Training programme	Training Program	Scheduled programme; events, attendance, results and competence decisions are separate
Project budget	Budget with native account allocations and purchasing/expense controls	Draft; controls take effect after native submission
Facility service	Issue, optionally with a native SLA	Open issue; native business response rules apply
City task assignment	Assignment Rule, scoped to project and work category	Disabled initially; review users and enable in ERPNext

An Item can have replenishment rules for several cities. Existing rules for a warehouse are preserved; the adapter refuses to replace an unmanaged rule. Native maintenance provides one schedule document per Asset; edit its native child task table when extending an existing schedule. Native approvals, stock transactions, invoice/payment processing and railway release remain in their respective systems.

Generic configuration and city overrides

- Shared defaults/instances: components.json.
- City overrides: `cities/catalogue/<region>/<country>/<city>/operations/erp-components.json`.
- Compiled packages: `build/erpnext/cities/<slug>/components.json`.
- Schemas/adapters: component_catalogue.py and components.py.

Defaults merge by component and input name. Lists such as budget account rows replace the inherited list. Instances merge by `(component, key)`; a city can override an inherited instance or set `enabled: false` to omit it. `${city}` and `${project}` are literal substitutions; profiles contain no executable code. Unknown inputs, duplicate instance keys, invalid numbers/dates and unsupported components fail validation.

The shared profile creates a city operating induction programme, with distinct Samawah and Mosul title overrides. Other components require verified native master records and operator inputs. None of those identities, stock quantities, approved BOMs, asset service intervals, budgets or personnel are invented. All 266 merged profiles and their induction packages are compiled by the operating-readiness audit; that compile proves schema and package integrity, not the existence of the native ERP masters needed to enable the other seven adapter types.

```
{
  "schema": "osr-components/1",
  "city": "samawah",
  "defaults": {"assignment": {"strategy": "Load Balancing"}},
  "instances": [
    {
      "component": "replenishment",
      "key": "depot-bearing-stock",
      "enabled": false,
      "inputs": {"level": 12, "quantity": 24}
    }
  ]
}
```

This example is deliberately incomplete and disabled. Supply a verified ERP `item` and city `warehouse`, review quantities in its stock unit, then enable it when ready to configure native replenishment. Do not put credentials or employee personal records in tracked city files.

Repeatable commands

```
./osr erp components catalogue
./osr erp components init-configs   # preserves existing overrides
./osr erp components validate
./osr readiness --check
./osr erp components prepare samawah --project PROJ-0001
./osr erp components preview build/erpnext/cities/samawah/components.json
./osr erp components apply build/erpnext/cities/samawah/components.json
./osr erp snapshot
```

A package is bound to an explicit ERP Project and city. Apply first previews all its enabled instances, then creates their records in one database transaction. A validation failure rolls back that transaction. Native external hooks or integrations must be commissioned with their own delivery guarantees. The package's effective defaults are saved on the Project so interactive forms use the same city defaults. This configuration is separate from the immutable engineering/task baseline and does not force its re-import.

Identity, permissions and repeatability

Each native record carries its project, component ID, instance key and input checksum. The database enforces a unique component identity. Identical calls reuse records, preserving operator edits; changed inputs with the same key are rejected. Review the native record or deliberately choose a new key. A new key cannot bypass native uniqueness constraints, such as one maintenance schedule per Asset or one reorder rule per Item/warehouse. Concurrent attempts can return a duplicate error and safely retry; Item updates also acquire a row lock.

The preview fingerprint covers the component version, inputs, derived native values and project baseline. Apply rebuilds and compares it. Preview does not insert records or execute native document hooks. Apply uses native document permissions and validation; it never uses `ignore_permissions` or submits a financial/production document. Linked companies, city warehouses, receipt item project, asset baseline membership, maintenance team and user access are checked. Assignment conditions are constructed by code, not accepted as user expressions.

Feedback and API

The existing five-minute twin refresh includes counts and states for all eight components. It uses native parent permissions, including Item permissions for reorder child rows. It does not expose staff lists or imply railway authority. Only records managed by these components are counted in this section; the existing purchasing/project feedback also covers other native business records.

- GET `osr_erpnext.components.catalogue?project=…`
- GET `osr_erpnext.components.inspection_lines?project=…&reference_type=…&reference=…`
- POST `osr_erpnext.components.preview`: `project`, `component`, `key`, `inputs`.
- POST `osr_erpnext.components.apply`: the same inputs plus preview `fingerprint`.

Validation

Run `tools/automation/osr-python -m pytest -q tools/automation/tests/test_erpnext_components.py` and `./osr erp components validate`. Native integration checks run against the local evaluation site using temporary fixtures and transaction rollback. New adapters need schema tests, native validation coverage and city-scope checks.

```
docker compose --env-file var/erpnext/local.env -f deployment/erpnext/compose.yaml \
  exec -T backend env/bin/python < deployment/erpnext/tests/verify-components.py
```

This local check expects the Samawah and Mosul evaluation baselines. It mutes test emails, creates its own Items/BOM/receipt/Asset/team, exercises all eight adapters and rolls back the transaction, including temporary user permissions.

Integrated operating software: completion and deployment inputs

Repository source: `docs/operating/integration-status.md`

This register covers the requested ERPNext/FUXA/OpenSourceRail software integration and its reusable city workflows. The published v0.4.0 baseline is immutable; the disposition enhancements below are recorded in Unreleased. Software acceptance uses simulation and temporary ERP transactions. Physical deployment and independent acceptance have their own evidence gates.

Implemented software

Task	Implementation and acceptance
ERP business backbone	Pinned ERPNext/Frappe HR deployment, native projects/tasks, procurement, stock, manufacturing, quality, assets, maintenance, finance and workforce components; deployment and native component/procurement tests
Generic and city-specific configuration	Shared component/equipment templates, per-city project/company mappings, 266 validated profiles and reproducible task payload generation; city platform and readiness
Operations planning and field work	Six-item management implementation: declared resource/possession planning, fleet/staff recovery, offline work evidence, reliability validation and operational/GTFS exchange; isolated gateway/browser acceptance, with native-site and measured calibration gates open
One connected UI	Workbench embeds authenticated ERP and FUXA and carries city/equipment context to OSR controls, engineering and lifecycle screens; guide and screenshots
Engineering tools and digital twin	Reviewed execution mappings, FreeCAD/IFC/Bonsai/GIS evidence references, stable equipment identities, installed serial history and revision impact; lifecycle guide
Rust and embedded supervision	Shared controller contracts, native simulation bridge, quality/timestamps, command request/result states and maintained historian; embedded contract
Alarm to maintenance and handback	Durable condition-event delivery, deduplicated Issues, reviewed Asset Repairs, native parts/actuals, and independent OSR inspection/handback; native lifecycle and controller acceptance
Engineering revision disposition	Permission-filtered BOM/order/material exposure, assigned immutable proposals, independent endorsement/rejection, stale review rejection and evidence download
All disposition outcome types	Registry-driven retention, cancellation, direct amendment, production stop, corrective work plus accepted inspection, performed inspection and reconciled movement trace; exact checks and limits
Reproducible acceptance and recovery	Unit/browser tests plus disposable native stack workflow, including service outage/recovery; CI definition
Documentation and publication	Root README, current deployment/workflow guides, changelog and generated reader book; published v0.4.0 remains the historical release baseline

The outcome registry is in `outcome_contract.py`. It drives native document selectors in the ERP dialog. Native readers enforce permissions; pure checks validate document relationships and measurements. Verification records bind the evidence selection, observed data, reviewer and exposure checksum. The same implementation uses each city's existing project, company and engineering mapping; no city-specific verifier code is required.

Complete example city

The isolated Samawah scenario adds a retained end-to-end city lifecycle and settings matrix, with Mosul as a control. It includes actual native ERP records and unmocked browser checks. Its report separates full planning coverage, selected live equipment and physical acceptance limits. The push/PR/manual CI workflow publishes public reports and screenshots while keeping credentials private. The recorded 18 September 2026 local run passed 98 lifecycle/settings/recovery checks, including ten real-service browser checks and a full stop/start. Its separate whole-city contract sweep covered 573 equipment positions and 1,610 measurements. This records local verification; candidate CI evidence must be checked separately by exact commit.

The additional `expand-check` passed 389 live/native/restoration assertions and a separate 230-check schema matrix. Its coverage register tracks 531 integration entries: 36 scenario, 64 varied, 135 partial and 296 without qualifying evidence. Payment posting/cancellation is exercised transactionally; other remaining workflows are listed explicitly. Passing examples do not establish coverage of every function or variable combination.

`business-check` adds 24 native outcome/rollback checks for actual budget enforcement, assignment, replenishment and training. `disposition-check` adds nine unmocked browser checks using three native identities, including stale verification after a subsequent native action. The release gate now requires this city workflow and validates its commit-bound report hashes. See the review follow-through for cross-domain freshness, controller fixes, broader proof execution and open gates.

The implementation candidate `92b20ff32` passed general CI, integrated-stack and complete-city CI. Its broader formal run passed 30 of 41 declared Kani harnesses; 11 timed out, so release packaging remains blocked. Local fresh-volume ERP restore rehearsals checked database records, ledger entries, public/private files and encrypted settings. A later coordinated

recovery rehearsal verifies cold ERP/FUXA/gateway/OSR data recovery; operational cutover and independent acceptance remain open. CI at `e9e4ff5cf287` passed general, integrated-stack and complete-city checks and confirmed 34 Kani properties, with seven remaining timeouts. The review record links the exact runs and explains the recovery scope.

Acceptance commands

The CI workflow creates a disposable deployment and runs:

- `deployment/erpnext/tests/verify-components.py`, `verify-procurement.py` and `verify-lifecycle.py` inside the ERP backend. Transactional business fixtures roll back, including native cancellation, amendment, corrective Job Cards, inspections and manufactured batch trace.
- `deployment/workbench/tests/verify-dispositions.mjs` against real ERP Desk widgets with mocked business responses, exercising dynamic evidence fields and explicit preview/record. The native Python tests separately verify actual ERP records and permissions.
- FUXA, pilot, embedded, native Workbench and outage checks in the sequence specified by the integrated-stack workflow.
- `tools/automation/tests/test_outcome_contract.py` and `test_outcome_evidence.py` for invalid identities, scope, quantities, readings, missing evidence and unrelated source changes; lifecycle browser tests for current/stale status and city scoping.

Run these against a disposable or local simulation installation. Pilot/controller tests record explicitly labelled simulation evidence; they are not physical commissioning procedures. Site credentials remain in private local configuration.

Inputs that remain open for deployment

Required input or acceptance	Why software cannot close it
Legal companies, operating calendars, approved ERP masters and role assignments	Must be supplied and approved by each owner/operator
Surveyed alignment, released geometry, supplier configurations and engineering evidence	Generated reference geometry and analysis links do not replace received site/supplier data or design approval
Physical controller addresses, protocols, calibration and machine/quality inputs	Simulation bindings are not commissioned equipment bindings
Production identity/TLS/MFA, hosted backup operation and recovery acceptance	The localhost evaluation stack is not an operated shared production environment
Performed manufacturing inspections, accepted configuration and first articles	Native test readings demonstrate software behaviour; real production evidence must be received and independently accepted
Signed physical commands, HIL, operator validation and railway release	ERP completion and native outcome verification do not authorise railway service

The broader roadmap retains operating-capacity/energy validation, full CAD/solver change propagation and the physical release workstreams. Completion of this software integration does not mark those separate workstreams complete.

Lifecycle governance, ERP/HR administration and QR identity

Repository source: `docs/operating/lifecycle-governance-and-qr.md`

OpenSourceRail uses one evidence-routing template across mechanical production, stations, civil construction, Rust releases, asset maintenance and management operations. The tracked source is `lifecycle-governance.json`; the generated subsystem control register shows which controls apply to every defined subsystem item.

The template does not appoint directors, employ people, release designs, accept construction, certify competence or authorize railway operation. It prepares a consistent record structure so the accountable operator can do those things in its legal, contractual and regulatory context.

Control chain

```
flowchart LR
  B[Controlled baseline] --> W[Mechanical / civil / software work]
  W --> I[Inspection, test and calibration]
  I --> N{Conforming?}
  N -->|no| C[NCR, containment, correction and CAPA]
  C --> W
  N -->|yes| E[Named release evidence]
  E --> H[Accountable human review / handback]
  E --> A[Asset identity and as-built history]
  A --> Q[Lookup-only QR resolver]
  Q -. no authority .-> H
```

Configuration, measurement, competence, nonconformance and release evidence are separate records. A status in one cannot silently stand in for another. In particular, training attendance is not competence authorization, a passed inspection is not engineering release, ERP submission is not railway handback, and a QR scan is not permission to work.

ERP and HR administration templates

Template	Native ERPNext/Frappe HR records	Required additional control
Employee and role administration	Employee, user/role permissions, department and employment records	Verify real identity, contract, organization scope and local employment/privacy requirements; personal data stays out of Git.
Training and competence	Training Program, Training Event, Training Result	Record the role/scope assessment, fitness prerequisite, accountable authorizer, validity, expiry and suspension separately from attendance.
Shift and handover	Shift Assignment, Employee Checkin, Issue and Task	Record incoming/outgoing leads, open permits, isolations, alarms, degraded modes and explicit acceptance; live control remains elsewhere.
Quality inspection	Quality Inspection, Quality Inspection Template and Job Card	Bind requirement revision, method, instruments, results, inspector, verifier and disposition to a named release gate.
Calibration	Asset, Asset Maintenance and certificate attachment	Record range, accuracy, traceability basis, status and due date; an overdue or failed instrument invalidates dependent evidence until reviewed.
Nonconformance and CAPA	Quality Inspection, Issue and Task	Preserve affected IDs, containment, root cause, correction, corrective action, effectiveness review and concession authority.
Configuration baseline	Project, Document and Task	Freeze scope, revision, source hashes, distribution, approval evidence and supersession without claiming technical adequacy.
Management review	Project, Meeting, Issue and Task	Freeze objectives, performance, audits, risks/opportunities, resources, decisions, owners and due dates.

The mappings follow native product behavior rather than creating parallel shadow ledgers. Frappe HR Training Events record attendance and results, while OSR deliberately keeps the railway authorization decision separate. ERPNext Quality Inspection provides incoming, in-process and outgoing inspection records, with engineering release retained as an independent gate.

Roles and management operation

The generic template defines accountable executive, design authority, quality, configuration, asset, operations, maintenance, competence, safety assurance, software security and data-steward roles. A deployment maps these functions to named people and may combine roles only after checking independence, conflicts and applicable rules. The tracked template never invents a person or approval.

Four maximum review intervals establish an operating rhythm:

Cadence	Maximum interval	Minimum focus
Shift	1 day	Handover, alarms, isolations, service state and staffing.
Weekly	7 days	Safety, service, maintenance, quality, materials and workforce actions.

Cadence	Maximum interval	Minimum focus
Monthly	31 days	Objectives, asset condition, cost, schedule, risk, competence and suppliers.
Quarterly	92 days	Audit, performance, risk/opportunity, data quality, security and improvement.

Each review records inputs, decisions, owners, resources and effectiveness. The structure reflects the lifecycle and management-review emphasis of ISO 55001:2024, the quality-management and documented-information requirements in ISO 9001:2026, and the project/quality controls in the FTA Quality Management System Guidelines. Using those sources as a template basis is not ISO certification, contractual conformity or regulatory acceptance.

Mechanical, station and civil workflows

Mechanical work moves from released baseline through receipt, calibrated-tool checks, manufacture, in-process inspection, NCR review, final inspection and serialized as-built evidence. Material traceability, calibration status, datum/interface acceptance and independent release are explicit hold points.

Civil work moves through released design, survey set-out, ground/material acceptance, temporary works, construction, inspection/test, as-built survey and asset handover. Current permits/design, accepted survey control, witnessed concealed work and closed as-built/punch-list evidence are hold points. Open DEM, OSM water and planning IFC are never substituted for project survey, geotechnical, hydraulic, temporary-works or construction evidence.

The register currently covers 146 LM3 products/assemblies, seven station variants and 19 reusable civil types. All remain `evidence-open-not-released`; the register routes the required closure records without pretending that a physical item exists or is accepted.

Rust release workflow

Every workspace crate receives a software-release control record. The minimum route is requirements and threat/hazard review, implementation, peer review, automated verification, dependency review, reproducible build, signed release and a separate deployment approval. The evidence list includes locked source and toolchain, all-feature tests and Clippy, vulnerability disposition, artifact provenance and the crate's authority boundary.

This adapts the NIST Secure Software Development Framework to repository evidence. Passing software checks demonstrates the reviewed build; it cannot certify or commission railway use. `osr-supervision-contract` remains observation-only, while `osr-lifecycle-identity` is lookup-only.

QR and physical-asset identity

The canonical identity contains only:

```
{
  "schema": "osr-asset-identity/1",
  "city": "samawah",
  "asset_id": "SAM-ST-001",
  "engineering_revision": "rev:abc123",
  "resolver_path": "/id/osr/samawah/asset/SAM-ST-001",
  "authority": "lookup-only"
}
```

No personal data, credential, token, password, command, movement authority or approval may appear in the record or QR payload. The payload is generated only after deployment supplies a bounded HTTPS origin. The tracked repository has no resolver and therefore produces zero printable labels.

Deployment sequence:

1. The operator establishes the asset register, retention policy and stable ID.
2. It provisions an HTTPS resolver with access control, monitoring, backup and an offline/stale-state presentation rule.
3. An authorized person binds the identifier to one physical asset, records its class/revision/location and verifies the applied label.
4. A second check confirms the scan resolves to that asset and makes no current release, isolation or safety claim.
5. Replacement retains the supersession history and binds a replacement identity; labels are never casually re-used.

The URI pattern follows the resolver principle in GS1 Digital Link. An operator may map a properly licensed and governed GS1 GIAI, but OSR internal IDs must not be presented as GS1 identifiers. For IFC, the stable operator identity is kept distinct from an IFC instance identifier, consistent with buildingSMART's AssetIdentifier definition.

AI management relationship

The AI executive council may collate this evidence and prepare an allowlisted, attested ERP draft after independent multi-model review. It cannot fill human names, attest competence, close a hold point, approve a baseline, accept an NCR concession, release a product, print a QR label, hire or dismiss, submit a transaction, pay, contract, operate SCADA or command a train. Management accountability stays with the deployment's named officers.

Compile and verify

```
python3 engineering/subsystem_control_register.py
./osr readiness
./osr test
```

`./osr readiness` compiles a deterministic governance package for every city and every tracked asset identity. `./osr test` checks the generated register and all-city source digest, exercises Python/Rust parity and tests authority, resolver, tampering, duplicate and fail-closed cases.

Coordinated platform recovery

Repository source: `docs/operating/platform-recovery.md`

The reusable recovery runner captures ERPNext, Redis, the gateway outbox, FUXA, Ops Core, application files, encryption/signing keys and the example city's Workbench context at one cold checkpoint. It restores them into new isolated volumes and checks their consistency before removing the rehearsal containers.

Reproduce

Use a running, populated example city. The host needs Linux `/proc`, Docker Compose, the repository Python environment, and enough free space for the images, volume archives and restored volumes. New checkpoints archive every container image by immutable ID before stopping writers. Recovery loads those archived images and verifies their IDs, without replacing mutable repository tags. The destination must support their platform and architecture. Legacy checkpoints without an image archive still require all captured image IDs to exist locally.

```
tools/automation/osr-python tools/automation/platform-recovery.py checkpoint \
  --profile deployment/recovery/example-city.json \
  --output var/recovery/example-checkpoint \
  --report build/recovery-checkpoint.json

tools/automation/osr-python tools/automation/platform-recovery.py verify \
  var/recovery/example-checkpoint

tools/automation/osr-python tools/automation/platform-recovery.py rehearse \
  var/recovery/example-checkpoint --report build/recovery-result.json
```

The checkpoint destination must be new and under `var/`. It contains credentials, private files and keys; retain it as a protected backup, never as a public CI artifact. Use only trusted checkpoints. Checksums detect changes but do not provide an external signature or establish who created a backup.

Capture stops the matching host simulator and Workbench, then source containers, and resumes them in a `finally` block. This introduces downtime. A failed capture must not be used; check both `passed` and `source_resumed` in its report and inspect private logs if either fails. The host process registry is refreshed on restart.

For another installation, copy the JSON profile and supply its Compose project names, private root, Ops Core database, ERP site, cities, auxiliary context and health URLs. Service names must be unique across projects. The current runner supports named volumes and regular-file bind mounts; the host process discovery expects the repository Workbench and simulator. Other writers need explicit quiescing support before their data can share this checkpoint.

What the rehearsal checks

- Exact checkpoint membership and checksums; import and verification of archived immutable container images. Rehearsals never pull a replacement image.
- Fresh destination volumes, with file bytes, paths, owners, modes and symlink targets matching the captured inventories before services start.
- Restored Ops Core database/evidence and verification with the original signing key.
- Native ERP record hashes, selected ledger fields, files and encrypted settings.
- Gateway alarm-to-Issue links and replay of a delivered event after simulating a lost reply: the original Issue is reused and checked ERP records remain unchanged.
- Restored ERP, gateway, FUXA, Workbench lifecycle and Ops Core HTTP responses, including the correct city scope for both cities.

The clone uses an internal Docker network with no published container ports. Fixed-destination tunnels through `docker exec` expose only local verification ports (default 28880–28884). All source volume identities are replaced. The runner removes its temporary containers and volumes in `finally`; private logs, configuration and restored host files remain under `var/recovery/` for inspection.

Recorded result and limits

The 18 September 2026 rehearsal verified ten volume inventories, ten signed Ops Core records and 25 linked ERP cases. The checked ERP snapshot included 11,571 Tasks, 26 Issues, six GL entries, ten stock-ledger entries and two files. The lost-reply replay returned the same Issue without changing those checked records. Restored services and both city contexts passed. Archived image import through native verification took 40.519 seconds; initial checkpoint integrity verification, capture and cleanup are excluded. The complete capture-and-resume cycle, including image archiving before writer shutdown, took 121.868 seconds. All source services resumed and the temporary recovery containers/volumes were removed. The source-bound report records the checkpoint hash, source hashes, base commit and dirty working-tree state.

This is coordinated data recovery, not an accepted operational cutover. Controllers, queue workers and schedulers stay stopped in the clone. FUXA's saved links retain the source origin; clone navigation and resuming jobs/controllers require a reviewed origin/identity migration and a further native browser run. Production recovery-time and recovery-point objectives, a separate-host rehearsal, off-host backup retention, interrupted capture/process failure injection and independent acceptance remain open. No recovery-time guarantee follows from one local rehearsal.

Operating platform readiness

Repository source: `docs/operating/readiness.md`

> Generated by `./osr readiness`. Do not edit this file or `readiness.json` by hand.

This gate compiles the tracked ERP, reusable-component and supervision inputs against each city's own engineering evidence. It separates repository-complete work from information that only a real operator, supplier or commissioning team can provide.

Catalogue result

Gate	Ready
Design, scenario, asset register, manifest and project twin	266/266
ERP city profile	266/266
Reusable ERP component package	266/266
Real-asset simulation supervision package	266/266
Lifecycle governance, HR/admin and QR identity template	266/266
Full ERP task plan from a local operations payload	2/266

The supervision result covers 363,032 equipment records derived from the 266 tracked city asset registers. A compressed operations payload is no longer required to prepare FUXA/gateway packages.

The lifecycle-governance compile covers 152,650 asset identity templates across the catalogue. It produces 0 printable labels: labels stay blocked until a real operator provisions an HTTPS resolver, binds the physical asset and verifies the label. Scanning is lookup-only and never grants work, isolation, release or movement authority.

Full ERP task-plan compilation is exercised for `mosul`, `samawah`. For the other 264 cities, the tracked manifest, asset register and compact project twin are validated here, while the large deterministic task payload is materialised on demand with `./osr city <slug>` before ERP import. This is a storage/materialisation boundary, not permission to substitute another city's tasks.

Deployment-owned inputs

Input recorded in tracked profiles	Configured
Legal ERP company	0/266
Approved calendar start	0/266
Local timezone	2/266
ERP project link for supervision	2/266
Physical equipment bindings	0

Blank deployment inputs are deliberate fail-closed states. This audit does not invent companies, dates, holidays, employees, suppliers, ERP masters, serial numbers, controller addresses, field results or approval evidence.

Reproduce

```
./osr readiness
./osr readiness --check
./osr readiness --deep-check
./osr supervision validate
./osr erp city validate
./osr erp components validate
```

The machine-readable per-city paths, counts, revisions, package hashes and deployment-input states are in readiness.json.

Integration review follow-through — 18 September 2026

Repository source: `docs/operating/review-follow-through.md`

This follows the review of `ec7a2023`. It distinguishes implemented controls, executed software tests and work that still requires evidence. The complete railway programme is not accepted or complete. Published v0.4.0 remains immutable.

The subsequent scaling and recovery review records telemetry and polling improvements, measured capacity limits, coordinated cold recovery and complete proof-artifact packaging. Its retained source-bound reports distinguish implemented data recovery from production cutover acceptance.

Recommendations and disposition

Review recommendation	Implemented response	Remaining closure
Make complete-city testing a release gate	`example-city` runs on push, PR and manual dispatch. Release packaging requires four successful workflows on the exact commit and validates the downloaded city's report hashes and clean scenario-start commit. Missing, failed, empty and altered reports are rejected.	A candidate needs its own successful CI runs; older local reports cannot qualify.
Unmocked disposition UI	Nine browser assertions use real ERP responses and three native users, through proposal, endorsement, native stop, independent verification and stale detection after resume. Self-review and premature verification are rejected.	Other action types retain native transactional tests; every browser/action/concurrency permutation is not covered.
Controlled proof evidence and independent acceptance	The runner executes every declared Kani harness, records actual exit codes/logs and conservative dependency hashes, and rejects source changes during execution. CI covers eight packages and all 41 declared harnesses. Ed25519 verification binds an independent authorized reviewer to exact result bytes, scope and reports.	Timeouts remain unproved. Runner provenance and reviewer key ownership require external trust review. No independent acceptance has been issued. Citation, property-test and other safety-case results remain separate.
Cross-domain engineering changes	Workbench binds the proposed package to city/company/project ERP observations and linked artifact hashes. Missing, ambiguous, older-than-five-minute, future-dated or revision-mismatched observations and changed artifacts are flagged; the context checksum changes with business observations.	This is a point-in-time review, not a cross-service transaction. The controlled member example now executes native CAD and solver reruns through ERP production and supersedes screening evidence. Complete dependency discovery, accepted rework and formal-evidence impact assessment remain open. Supervisory apply does not close these gates.
Broader operating validation	A resource-bounded, resumable runner supports selected cities or the full catalogue, nominal or degraded full-window runs, and generated candidates. Catalogue CI partitions the workload and rejects missing, duplicate, failed, stale or altered city artifacts.	Canonical full-service, continuous stabling, morning fleet, site-energy and physical acceptance need their own qualifying evidence; generated candidates do not promote canonical packages.
Manufacturing and production deployment	Existing supplier, factory, inspection, HIL and operating-release gates remain fail-closed. No nominal calculation or simulated ERP transaction is relabelled as performed evidence.	Approved supplier configurations, site surveys, inspected first articles, measured mass, HIL, production identity/TLS and independently accepted restore/recovery evidence are required.

Defects found by the expanded execution

- Complete-city browser execution exposed a document-navigation race while reading the native Task filter. Polling now retries only destroyed/detached document contexts, as the simulator/OCC readiness check already did; the authenticated user and exact city Project assertions still have to pass.
- A revision change could retain the preceding baseline and simulation run in Workbench context. Changing the revision now clears both; changing the baseline clears the run. City Studio, the simulator, OCC and Railway works also remove fields omitted from the shell's current snapshot, and a missing baseline removes live mode. Browser regressions exercise this propagation. These UI references do not establish independent engineering or safety acceptance.
- The connected browser journey now waits for actual revision materialization and approval responses, then compares their exact revision and baseline hash with the downstream railway work record. Matching the format of an already-visible revision was insufficient synchronization on slower clean deployments.
- Complete-city CI exposed a simulator command race: a local disable arriving during telemetry/polling could be missed by command execution. The simulator now rereads local enable/disconnect inputs immediately before each command; deterministic regression cases reproduce the old acceptance and verify rejection. The city test checks this independently of its shortened command-expiry probe.
- ERPNext's native reorder scheduler omitted the city Project from generated Material Request lines. The OSR hook now derives it from the exact Item/warehouse rule, preserves explicit values, and rejects ambiguous or stale scope. The reproducible ERP image is `osr/erpnext:15.121.2-hr15.64.0-osr14`.
- The point-machine controller could clear a persistent drive fault when its cooldown expired. Active over-temperature/drive faults now refresh the latch and inhibit motor movement, including clock saturation. Regression tests and all seven declared points harnesses passed after the correction.

- Signed `abs()` in brake, odometry and intrusion proof assumptions could itself overflow before the input constraint applied. Guards now use `unsigned_abs()`. Two intrusion harnesses needed unwind 10 to traverse the eight-slot LIDAR array; unwinding assertions remain enabled. All five intrusion and five brake harnesses passed in targeted execution. Full dependency-bound results are generated by the final broader sweep; these targeted checks alone do not close the safety case.
- Production obstacle/intrusion LIDAR and radar comparisons also used signed `abs()`. A regression test reproduced a panic at `i32::MIN`; unsigned magnitude comparisons now handle both signed extremes consistently with the off-profile rule. The updated five-crate unit/property suites pass. Superseded source-bound proof runs cannot qualify as evidence for these revised evaluators.
- ATP's same-section overrun harness and odometry's zero-wheel balise harness constructed topology their evaluated paths never read. Removing that fixture work, and bounding unreachable ATP loops with unwinding assertions enabled, allowed both proofs to finish with their original input ranges and assertions. The overspeed/conservatism harnesses also exposed insufficient square-root unwind bounds; these are increased to 32, with unresolved executions still treated as unproved. Graph-dependent harnesses keep their topology fixtures.
- Kisumu and Najaf's checked-in scenarios lacked the current train-system fields. Regeneration also exposes their explicit point-machine identities. Their original nominal runs failed the configuration contract even though operational checks ran. The batch runner can test regenerated candidates without overwriting their canonical city package or falsely promoting dependent evidence.

Executed results and limits

- Catalogue CI 35344461975 passed all 266 generated candidates, each over 90,000 simulated seconds, at `7b744e124`. The downloaded reports, source hashes and aggregate were verified locally. These are nominal runs; canonical packages were not promoted.
- Two further generated candidates were selected from that result by their lowest service-completion and battery-charge margins. Hofuf passed all eight degraded cases; Nampula passed seven of eight. Nampula's combined 80% aged battery and maximum HVAC case completed 89.68% of scheduled service, failing the 90% requirement even with the existing 0.2 percentage-point numerical tolerance. Its controller invariant count was zero. This is an open capacity/timetable validation finding, not a reason to lower the acceptance threshold. Full local records are generated under `build/city-validation/review-margin-resilience/`. Two isolated Nampula alternatives subsequently passed nominal and all eight degraded cases: adding 30 seconds to each line's charging dwell raises the combined-case result to 90.32%, with hardware settings unchanged; the two-cabinet alternative reaches 90.85%, with larger generator-derived charger, grid and storage assumptions. Neither promotes the canonical package or grants operating acceptance. Exact settings and results are recorded under `build/city-validation/nampula-alternatives/`.
- Broader Kani CI 35353021756 executed all 41 declared harnesses at `92b20ff32`: 30 passed and 11 timed out at 300 seconds each. Their manifest, report and source hashes were verified locally. Timeouts do not establish the properties, and no independent acceptance has been issued. The required release workflow therefore remains unsuccessful.

Integration candidate and ERP recovery

The implementation at `92b20ff32` passed general CI, integrated-stack and complete-city acceptance. The downloaded city's seven report hashes and clean scenario origin were checked against that exact commit. The Kani failure above remains a release blocker. Development integration is separate from release packaging and independent railway acceptance; the published v0.4.0 tag and assets remain unchanged.

Fresh-volume ERP restore rehearsals also passed locally. Each used a disposable Docker project with an internal network, restored an actual database backup and public/private file archives, migrated the site, checked encrypted settings and native HTTP availability, then removed the temporary containers and volumes. Selected native field hashes and record counts matched the source, including six GL entries, ten stock-ledger entries, four Work Orders, four Stock Entries and one submitted Purchase Invoice in the retained example-city fixture. A separate two-stage rehearsal backed up and restored nonempty public/private file fixtures and verified every archive entry's content hash.

These checks cover ERP database, files and encrypted settings. They do not cover coordinated restoration of FUXA, the gateway and OSR operations data, production recovery time objectives or independent recovery acceptance. Private backups and credentials are excluded from the public evidence bundle. Full local reports are `build/review-fresh-restore.json`, `build/review-fresh-restore-files.json` and `build/review-fresh-restore-example.json`; they are execution records, not CI results.

These are results for the recorded inputs and commits, not approval of a later candidate. Complete-city release artifacts must still pass their exact-commit gate.

Repeatable commands

```
./osr example-city business-check
./osr example-city disposition-check
./osr example-city coverage
# Only after a full scenario started on the same clean candidate commit:
./osr example-city release-evidence

PATH="$HOME/.cargo/bin:$PATH" tools/automation/osr-python \
  tools/automation/assurance-evidence.py run \
  --output build/assurance/review --timeout 300

# All cities, including degraded full-window scenarios; bounded parallelism.
tools/automation/osr-python tools/automation/validate-city-batch.py \
  --all --jobs 2 --timeout 1800 --resume
# Examine current-generator candidates separately from canonical packages.
tools/automation/osr-python tools/automation/validate-city-batch.py \
  --city kisumu --city najaf --nominal-only --regenerate \
  --output build/city-validation/candidates

# Optional catalogue CI: 16 disjoint partitions, at most eight jobs at once.
gh workflow run city-catalogue.yml --ref YOUR_CANDIDATE_REF \
  -f basis=generator-candidate -f resilience=false
```

The catalogue workflow also runs on explicitly pushed `validation/cities/**` branches. Each partition uses two workers; its collector verifies every current catalogue city, commit, dependency scope and report/scenario/design hashes. It publishes failed diagnostics as well as successes. This optional modelling check is separate from the four required integrated software-release workflows.

Timeouts and failed runs retain diagnostics and exit unsuccessfully. `--resume` never reuses failed results or results for changed inputs. `--nominal-only` excludes resilience and is recorded in the evidence; it cannot satisfy a resilience run.

Integration coverage and evidence still needed

The coverage register records 519 integration entries: 36 scenario, 64 varied, 135 partial and 284 gaps. The additional business scenario has 24 native outcome/rollback checks; the browser scenario has nine unmocked disposition checks. Neither count represents all ERPNext functions or all possible variable combinations.

Payroll posting, jurisdiction-specific tax, capitalization/depreciation, SLA calendars/escalation, independent competence decisions, desktop regeneration, production load and recovery cutover retain explicit gaps. The coordinated fresh-volume rehearsal now covers restored data and isolated recovery checks. The native engineering-change example retains a real CAD edit, six solver runs and eleven native production checks; these separate reports do not increase the mapped coverage counts. Physical and manufacturing closure still needs approved evidence and independent reviewers. The tracked deployment summary retains its city-level gates until qualifying canonical evidence is generated and reviewed.

The stale RFC release label, “unbounded” property-testing wording and operations stabling description have been corrected. Samawah's adopted planning allocation is 40 station-held plus 68 depot/terminal-held trains, with physical fit separate.

Recommended development sequence: current position

Step	Implemented or observed	Still required
FUXA scaling, queues and backup verification	Atomic telemetry, city-scoped queues, site polling and checksummed recovery rehearsals are retained.	Mosul does not consistently meet the two-second polling target.
Proof timeouts and release evidence	Read-only topology adapters separate evaluator logic from map storage. A complete local source-bound run passes 37/41 declared properties and retains every outcome/log; three prior odometry timeouts are closed with unchanged bounds.	Four timeouts remain, including 900-second local follow-up attempts. Local evidence is unattested; exact-candidate CI, consensus refinement and independent acceptance remain open.
Full-city load and coordinated restore	Separate-process full-network load and isolated restoration of ten volumes and five archived images are exercised.	Simultaneous full-city native FUXA/ERP/operator capacity, production cutover and recovery objectives remain open.
CAD analysis BOM production assurance	Samawah and Mosul execute increasing/decreasing native geometry, twelve solver runs and twenty-two native ERP checks through one Workbench module.	Complete dependency discovery, formal impact assessment, fabrication geometry and independent engineering acceptance remain open.
Canonical city promotion	Catalogue qualification now requires per-line mileage; reports expose configured passenger capacity by service window. Samawah and Mosul nominal service passes; stricter degraded checks reveal four Samawah failures and one Mosul failure. A retained Samawah 35% trigger candidate resolves one shortfall but leaves three and reduces emergency delivery. The design-options workflow now provides versioned dwell, fleet, storage and solar alternatives; its 300-second / 20-train Samawah option and Mosul modular-energy option each pass nominal and all eight degraded cases.	Review candidate resource/cost and multi-day effects, remaining city shortfalls, observed peak headways, calibrated passenger demand/crowding and operator acceptance before promotion.
Vehicle, station and civil engineering packages	Existing controlled part, drawing, analysis and release registers remain available; the new member workflow binds actual native artifacts and production observations.	Complete production geometry, supplier/material choices, connections and load combinations, construction stages, inspected articles and independent acceptance for the representative packages.

The analysis register now contains 29 analyses: 17 screening and 12 planned. None is independently accepted. These implementation and screening results do not close the engineering-package or operating-release gates.

Gateway scaling, backup integrity and evidence follow-through

Repository source: `docs/operating/scaling-and-recovery-review.md`

This implements the first software fixes from the broader review of `a99af132`. It does not close the entire programme or the remaining formal, engineering, capacity and deployment gates. Verification and acceptance status keeps recorded commits, test scopes, maturity and canonical gates together and is available through Workbench's Verification & gates module.

Findings and changes

Finding	Implemented correction	Verification and limit
One FUXA device read loaded the entire city and history	Dedicated active-device read with three SELECTs; one indexed statement retrieves latest measurements. Whole-city snapshots use nine SELECTs regardless of equipment count.	Regression enforces query budgets and preserves values, quality, timestamps, active configuration and city/environment boundaries. This is query scaling, not production capacity acceptance.
Other-city traffic hid pending ERP deliveries	Persistent city/environment/asset scope, forward migration from existing event payloads, filtering before LIMIT, pending/total counts and cursor pagination.	More than 100 newer other-city events no longer hide the original city's queue. Tests cover migration, pagination, pending state and forbidden HTTP scopes. Delivery retry ordering remains unchanged.
Ops Core accepted an unchecked invalid database	Strict manifest schema and membership, mandatory checksummed database, unique safe paths, content checksums, SQLite header/integrity/foreign-key checks and application evidence-file checks.	Original exploit fails. Negative tests cover missing, duplicate, extra, corrupt and unsafe entries. Fresh-directory restore retains application records and rebases evidence paths; it cannot overwrite a deployment.
Unbounded lifecycle evidence response	Snapshot contains the latest 20 records per asset, with totals and an older-record cursor. Authenticated pages and UI controls retrieve earlier evidence.	Tests traverse all 45 fixture records without duplication or cross-city access. Existing records remain stored.
Broader Kani artifacts absent from release pack	Export and validate every declared harness's manifest, TOML result and log, bound to the exact commit, workflow run and current dependency hashes.	Missing, stale, duplicated, failed and altered results are rejected. The archive preserves original report paths and bytes. No independent acceptance is inferred.
Fragmented status presentation	Generated status page in Workbench combines historical CI, generated catalogue results, integration coverage, canonical gates and analysis maturity.	Each CI result retains its commit and source link. Generated candidates never become accepted canonical deployments by aggregation.

Executed load rehearsal

The reusable command creates an isolated gateway database and real local HTTP servers using the production handlers. It interleaves telemetry writes, FUXA tag adapter polls and multiple operator snapshot/queue reads:

```
tools/automation/osr-python tools/automation/supervision-load.py \
  --city samawah --rounds 3 --workers 16 --operators 4 \
  --output build/supervision-load/samawah
```

The output directory must be new. The package, source hashes, timings, request errors, received sample count and latest effective values are retained in its report. The command exits unsuccessfully on functional failure. The separate `polling_target_met` field records whether complete cycles met two seconds; `functional_passed` alone does not establish that capacity.

The initial review run covered 573 equipment positions and 1,610 measurements, with 4,830 telemetry writes, 1,719 FUXA-adapter polls and 24 operator reads. All requests completed, all expected samples were retained, changed final values matched, and SQLite integrity passed. Device reads had approximately 8 ms p95 response time, while complete cycles took 3.94–4.38 seconds. The two-second cycle target was not met. These local results include host contention and are not a controlled production sizing benchmark.

This runs the FUXA polling API, not hundreds of FUXA browser sessions or concurrent ERP posting. Sustained load, longer historian retention, real controllers, ERP transactions and coordinated production sizing remain required. Do not lower the acceptance target or relabel this result as production-ready.

Bounded telemetry batches and measured follow-through

Gateway image `osr/integration:0.1.2` adds authenticated `POST /telemetry/batch` with body `{"readings": [<telemetry message>, ...]}`. A batch contains 1–128 readings and uses the existing 4 MB request bound. Every member is authorized against the controller's city/environment and source binding. Readings run in order in one transaction, with the same sequence, timestamp, units, quality, commissioning and alarm checks as `/telemetry`. A rejected member rolls back all new readings, alarm changes, audit and delivery events. A retry must preserve each original message, including its timestamp and sequence; identical messages return `duplicate: true` without repeating effects. Response `results` entries correspond to the input order. Clients divide larger sets into batches; atomicity applies to each batch, not a whole sampling cycle.

The native Rust supervisory simulator now publishes these bounded batches. Telemetry writers queue within the process before taking SQLite's write lock. Device/snapshot transactions also run one at a time to avoid short concurrent cursors competing for the Python interpreter; writers use a separate lock and SQLite retains cross-process isolation. Each view reads current database state. There is no response cache, added freshness interval or changed quality rule. The deployed HTTP server and load runner now share the same 128-connection listener backlog.

The follow-through rehearsal exercised the same 573 positions, 1,610 measurements, 16 clients and four operators over five cycles. All 8,050 readings, 2,865 device polls and 40 operator reads completed correctly. Cycles took 1.43–1.76 seconds; the two-second target passed on this host. Device-read p95 was 49 ms, snapshot p95 458 ms, and batch-write p95 66 ms. These are bounded local observations, not production sizing or concurrent native ERP/FUXA browser acceptance.

A second city run covered Mosul: 1,422 positions and 4,009 measurements. All 12,027 readings across three cycles were retained and final values matched, but cycles took 4.49–4.63 seconds, missing the same two-second target. It ran on the shared development host during other verification work. This establishes functional reuse of the runner, not sufficient capacity for the larger city; controlled sizing, partitioning and native multi-operator load remain open.

The command defaults to batches of 128. Use `--batch-size 1` for the original single-reading comparison, and `--require-polling-target` to fail if any cycle exceeds two seconds. CI retains source-bound load reports and fails on functional errors; hosted-runner timings remain observations rather than a capacity approval. Both local pilots were upgraded with private consistent gateway backups and preserved asset configurations.

ATP envelope square root now uses 32 division-free radix-four steps while retaining floor rounding. Boundary checks, 10,000 generated full-width `u64` comparisons against Rust's standard implementation, and all ATP tests pass. Targeted controlled Kani runs passed severe overspeed (`E4.1f`, 140.1 seconds) and conservative speed uncertainty (`E4.1g`, 96.5 seconds). Their symbolic input ranges and assertions are unchanged; unwind increases to 33 to cover all 32 steps and the loop exit. Local proof reports remain unattested and do not replace exact-commit CI or independent review of the safety-path arithmetic change.

Optional site polling and current capacity evidence

Gateway `osr/integration:0.1.3` supports `fuxa_polling_scope: "site"` in city configuration, while the shared default stays `"asset"`. Site polling combines transport requests within one city/environment/site. Equipment measurement and alarm tag identities, timestamps, quality and asset links remain separate. Package changes and whole-project FUXA imports still require hash-bound reviews. See the configuration guide.

```
tools/automation/osr-python tools/automation/supervision-load.py \
  --city mosul --rounds 5 --workers 16 --operators 4 \
  --polling-scope site --output build/supervision-load/mosul-site
```

The runner now starts the gateway in a separate process, matching the deployed client/server separation; `--in-process` retains the earlier benchmark layout. It polls actual generated FUXA URLs and checks final tag values and quality against persisted readings. Reports retain the exported project hash and source hashes. Comparing these results directly with the earlier shared-interpreter numbers would combine a benchmark change with the transport improvement.

With the same separate-process layout, the Mosul asset-polling control took 2.17–2.64 seconds per cycle. Site polling reduces 1,422 requests to 375 without removing any of the 4,009 measurements. An earlier five-cycle run took 1.24–1.77 seconds, but the final source-bound run took 1.50–2.24 seconds: all 20,045 samples were retained and final FUXA values matched, while the two-second target was not met consistently. These shared-host observations support the transport change, not production capacity acceptance. Retained reports: asset control and site polling.

The example-city scenario uses Mosul site polling and Samawah asset polling. The native browser check compares Mosul's rendered power with current gateway telemetry, as well as checking city switching and the existing lifecycle handoff. All ten checks in the local native browser journey passed, and the automation suite passed 340 tests plus 12 subtests. These ran on the recorded dirty working tree; they do not replace clean-candidate CI. Full-city native FUXA/ERP/operator concurrency remains a separate acceptance task.

Scope still open

- CI at `e9e4ff5cf287` completed all 41 declared Kani harnesses: 34 passed, seven timed out. A later complete local run using validated static topology tables passes 37/41: forward non-regression, uncertainty monotonicity and GNSS conservatism now complete with unchanged input ranges, assertions and unwind bounds. Four 300-second timeouts remain: interlocking non-overlap and determinism, ATP determinism and odometry determinism. Each also timed out in a 900-second shared-host follow-up. Full logs and source hashes are retained. Local runs are unattested; exact-candidate CI, consensus refinement and independent acceptance remain open.
- Coordinated cold data recovery now passes the checks described in the recovery guide. Production cutover, resuming jobs and controllers, recovery objectives and production identity migration remain open.

- The controlled cross-bearer workflow now executes native CAD, quantities, solver reruns, nested BOM revisions, production stop and revised manufacturing, retaining superseded screening evidence. Full dependency discovery, formal impact assessment and independently accepted engineering changes remain open.
- Canonical promotion still requires regenerate validate review publish, per-line service, peak headways, passenger/capacity measures and accepted operating evidence. The nominal generated-city catalogue does not close these gates. `validate-city-service.py` gates full-window and degraded mileage separately for every line, and the catalogue collector rejects missing or failed per-line evidence. The retained Samawah generated candidate passes nominal line screens at 90.86%, 103.87% and 103.87% of scheduled mileage. Its configured peak ceiling is 7,200 passengers/hour/direction; observed peak departures, calibrated demand and crowding remain open. These results do not promote the candidate or validate every service window. The existing `validate-city-simulation.py` reports retain their original aggregate software scope. The batch runner now uses the stricter qualification adapter, which reuses that suite and its native per-line observations; an older aggregate pass cannot satisfy the new catalogue collector. The canonical Samawah qualification fails four degraded cases: line 1 delivers 87.26% under maximum climate load, 87.54% with combined ageing/heat, 88.77% during the single-pad outage and 88.34% with consecutive missed charging stops. Each is below the 89.8% threshold including numerical tolerance, despite the aggregate software screen passing. These remain baseline failures; tested design alternatives are recorded separately in City design options. No acceptance threshold or canonical design was changed. A 35% adaptive-service trigger candidate changes only `normal_service_soc` from 0.40 to 0.35 in the retained design and scenario. It preserves the 20% reserve floor and improves line 1's single-pad case from 88.77% to 90.01%, but the heat, combined ageing/heat and missed-stop cases still fail. All-site outage delivery falls from 78.27% to 74.55% on that line (above its separate 60% emergency floor). The candidate is not promoted. This comparison demonstrates effective settings changes and their trade-offs. Mosul's stricter qualification passes all six nominal lines and seven degraded cases. The all-site grid outage fails on line 2 at 51.81%, below its 59.8% emergency threshold including tolerance. The five canonical degraded-case failures are visible in Workbench Verification & gates. The canonical configurations remain unchanged. Separate controlled design options now pass both complete suites: Samawah adds 20 line-1 trainsets and increases charging dwell to 300 seconds; Mosul adds 13.5 MWh storage and 2.7 MWp solar across nine line-2 sites. Costs, site feasibility, repeatable multi-day charging, peak headways and passenger demand still require review before promotion.

Reproduce the stricter qualification without overwriting legacy aggregate evidence:

```
tools/automation/osr-python tools/automation/validate-city-service.py \
  --scenario cities/catalogue/west-asia/Iraq/Samawah/samawah.toml \
  --resilience --full-only --output build/samawah-service-qualification.json
# A failing line returns nonzero and retains the complete diagnostic report.
```

Remaining package and programme gates:

- Supplier freezes, inspected articles, measured mass, calibrated analyses, physical transports/HIL, construction-stage civil/station checks and independently accepted vehicle/site packages require their own evidence and responsible reviewers.
- Commercial reconciliation of exclusions, supplier quotes, commitments and actuals, plus named owner/builder/operator appointments and competence, remains open.

The coverage register now has 519 entries: 36 scenario, 64 varied, 135 partial and 284 gaps. The two new GET routes and batch POST route have regression/HTTP tests but retain gaps in the complete-city scenario register until qualifying scenario evidence is mapped. Existing programme gaps are not marked complete by these software fixes.

OpenSourceRail — Operations Rulebook

Repository source: `docs/operations/README.md`

This directory holds the rule text referenced by RFC 0013. Every rule sits inside the one-sentence-one-decision style the RFC commits to.

Target length, rendered PDF: ≤ 60 pages. If the full compiled book exceeds that, something gets cut before anything is added.

Layout

Path	Audience	RFC 0013 section
`dispatcher/`	OCC dispatcher	§4.1 (S1–S7)
`station-staff/`	Station agent / inspector	§4.2 (T1–T5)
`maintenance/`	Depot + MOW worker	§4.3 (M1–M7)
`control-centre/`	OCC supervisor / engineer	§4.4 (C1–C3)

Each subdirectory has one file per rule block, numbered per the RFC. The rulebook now contains drafted one-decision rules with `Why:` rationales across dispatcher, station-staff, maintenance, and control-centre role families. Practising-operator review is tracked in `validation-checklist.md` and remains a deployment release gate.

Maintenance intervals across rolling stock, stations, track/civil, structures, energy, signalling/comms, depots, and production-plant tools are consolidated in `../rfcs/0029-maintenance-schedule-system.md` and the machine-readable `../../lib/templates/maintenance-schedule.toml` template.

Manufacturing scheduling for trains, waypoints/W-SBCs, track, switches, stations, depots, energy sites, and production plant tasks is documented in `../rfcs/0030-manufacturing-schedule-system.md` and the machine-readable `../../lib/templates/manufacturing-schedule.toml` template.

The browser front door for those generated city operations records is `../operations-portal/`. It combines the asset register, manufacturing schedule, QA actions, maintenance schedule, and launch panels for the existing OCC/simulator/back-office crates.

The integrated Workbench can generate this complete delivery twin for any catalogue city without a shell. Its Project Twin tab shows the resource-loaded critical path, budget work packages, order-by dates, supplier-anchor/local-equivalent basis, controlled manufacturer candidate IDs, selection states and monthly local/imported cash needs. New purchase orders, deliveries, invoices, payments and business progress are recorded in ERPNext. Historic Ops Core business records remain read-only; railway works, inspection and handback remain OSR records. A generated plan never becomes an issued commercial record by itself.

The institutional route for turning those records into an accountable railway is the owner–builder–operator setup plan. It defines the retained public owner, ring-fenced delivery/operator company, independent assurance, responsible executives, competence build and staged transition from development through revenue service.

Healthy trainsets use a hybrid of station and depot/terminal overnight stabling. The adopted Samawah planning allocation places 40 trainsets at stations and 68 at depot/terminal storage. Station charging may use grid, solar or storage-backed supply; it has no universal minimum grid connection. The selected plan must verify energy, morning dispatch and physical berth/routing capacity separately. Depot facilities retain the inspection and maintenance functions: station stabling is not distributed maintenance.

How to add a rule

1. Find the right block (e.g. S3 — Incident handling).
2. Append a numbered rule at the bottom of that file in the template:

```
### D3.4 — <short title>

<one sentence, one decision>

**Why:** <one short paragraph>
```

3. The numbering is `<block>.<n>`; next n = the last +1.
4. Every rule has exactly one author — credit in the commit message, not the rule body.

Non-mission-critical reminders for contributors

- One sentence per rule. Compound rules break testability and become accretion. Split.

- Every rule has a `Why:`. An unexplained rule rots. The `Why:` block is a contract with the future driver who must follow the rule.
- Cite the RFC / safety-case solution the rule supports. This is how the rulebook feeds the GSN safety case (`docs/safety-case/gsn/`).

Language policy (RFC 0013 §13)

- English is the canonical rulebook.
- Per-country translations land under `docs/operations/<iso-lang>/` as a parallel tree once the first deployment needs one. Until then, only English is authoritative.

C1 — OCC watch roles

Repository source: `docs/operations/control-centre/c1-occ-watch-roles.md`

Scope: the three on-shift roles: dispatcher, supervisor, engineer-on-call.

Cross-refs: RFC 0013 §4.5 C1, `osr-occ`.

Rules

C1.1 — Dispatcher

The dispatcher owns all dispatching decisions for the shift: route grants, timetable adherence, headway regulation, driver radio authorisations, work-block coordination. The dispatcher's authority ends at override approval and incident escalation.

Why: the dispatcher is the operational executive — fast decisions, clear authority. Splitting dispatching across multiple shift personnel creates decision collisions (two trains authorised onto the same block).

C1.2 — Supervisor

The supervisor approves `MaintenanceOverride` extensions beyond S5.4 defaults, handles NRSA and media liaison, countersigns incident reports, and makes the call on degraded-mode transitions. The supervisor does not dispatch except as the dispatcher's relief during short breaks.

Why: the supervisor is the quasi-legal authority on shift — the person accountable to the regulator, to whom the dispatcher escalates. Keeping the supervisor clear of dispatching prevents decision-role collisions at the moment an incident demands both.

C1.3 — Engineer-on-call

The engineer-on-call handles software and hardware alerts that exceed the dispatcher's scope: consensus-cluster degradations, `osr-routing` / `osr-interlocking` anomalies, wayside-points drift alarms. Engineer-on-call is a 24×7 phone rotation, escalation target for the dispatcher.

Why: the dispatcher is not a systems engineer. Routing a technical alert to a named on-call engineer moves the diagnosis to the person trained for it without blocking the dispatcher's operational duties.

C1.4 — Role rotation

Shifts rotate on a 12-hour cadence with a 15-minute overlap for the handover. The handover overlap is paid time and is not compressible — a late-arriving relief does not shorten the overlap.

Why: 12 hours matches the circadian pattern for alertness (with the 8-hour extension used in some deployments); the 15-minute overlap is the floor for the S1/S6 and C3 handover protocols to complete without handover shortcuts.

C1.5 — Minimum qualification

Each role has a minimum qualification: dispatcher (certified by the NRSA-equivalent after 6 months floor training + exam), supervisor (3 years dispatcher experience

- supervisor exam), engineer-on-call (software-engineer role
- deployment-specific rotation). No role may be filled by someone below the minimum.

Why: qualification is the hard floor of competence. An under-qualified supervisor making the NRSA call is how incidents become scandals; an under-qualified dispatcher is how incidents happen.

C1.6 — Fatigue policy

Dispatcher slots are capped at 10 hours; a Category I5 or I6 incident triggers a mandatory 30-minute break after the immediate response phase, with supervisor relief. Double shifts are prohibited.

Why: fatigued dispatchers miss radio calls and make timetable errors. The 10-hour cap and post-incident break is the compensating control — recovery time when it matters most.

C2 — Communications

Repository source: `docs/operations/control-centre/c2-communications.md`

Scope: radio discipline, PA authority, external-party notification (NRSA, emergency services, press liaison).

Cross-refs: `osr-t2g`, `osr-pis-onboard`, `osr-pis-station`, RFC 0013 §4.5 C2.

Rules

C2.1 — Radio discipline

Radio transmissions open with call sign (dispatcher + unit being called), cover one topic per transmission, and use the standard phraseology (Proceed / Stop / Hold / Recall / Rescue). Ambiguous phrases ("kinda go" / "maybe stop") are rephrased on the spot.

Why: radio is the fallback when everything else is broken. Standard phraseology removes ambiguity when stress is high and accents vary across a multi-nationality operator workforce.

C2.2 — PA authority

PA authority is partitioned: the dispatcher owns network- wide PA (all stations or all trains simultaneously), station staff own station-local PA, drivers own in-train PA. A role may not make a PA outside its partition without handover.

Why: overlapping PAs produce cacophony at the station where a passenger trying to understand one announcement gets cut off by another. The partition matches authority to audience.

C2.3 — NRSA notification

NRSA notification windows: Category I5 (derailment, collision) within 2 hours; Category I6 (mass casualty) immediately; Categories I3 and I4 within 24 hours; I1 and I2 are consolidated into the monthly safety summary. The supervisor makes the call; the dispatcher does not.

Why: NRSA windows are regulatory — missing them is an operator-level finding. Centralising on the supervisor ensures each window is tracked once, not N times by N dispatchers each assuming the other called.

C2.4 — Emergency services coordination

Emergency-services coordination (fire, ambulance, police) is through the OCC supervisor as the single point of contact. The dispatcher may make the initial 112 / 999 call (per deployment dialling) but hands the ongoing liaison to the supervisor within the first 5 minutes.

Why: fire chiefs and ambulance incident commanders need one rail-side point of contact; a dispatcher juggling services calls and train dispatching does neither well. The handover within 5 minutes keeps the initial speed without compromising continuity.

C2.5 — Press and social media

Press and social-media statements are made only by an authorised operator-level spokesperson (CEO, press officer, or nominated deputy). OCC staff do not make public statements, including on personal social media, about incidents in progress or under investigation.

Why: in-progress incident statements by line staff contradict themselves (because information is incomplete) and are quoted against the operator for years. The centralised spokesperson is the standard mitigation.

C2.6 — International coordination

Incidents involving foreign nationals or cross-border operations (rare in Samawah but possible in other deployments) are coordinated through the supervisor with a qualified translator on the bridge, including for consular notifications.

Why: consular notifications and cross-border regulator contacts have specific diplomatic protocol that is outside the dispatcher's remit. The supervisor + translator pairing is the compensating control.

C3 — Shift handover

Repository source: `docs/operations/control-centre/c3-shift-handover.md`

Scope: written log, outstanding items, on-call list for the incoming watch.

Cross-refs: S6 (dispatcher handover), RFC 0013 §4.5 C3.

Rules

C3.1 — Written handover log

The OCC handover log covers, for every shift: open incidents (with category and owner), active `MaintenanceOverride` entries, active speed restrictions, weather overlays, fleet status (revenue / maintenance / reserve), and any CBM red alerts. The log is created before the handover and reviewed by both outgoing and incoming watch.

Why: writing the log forces the outgoing watch to synthesise the shift; reading it forces the incoming watch to absorb. Verbal-only handovers lose details; the written log is the persistent record.

C3.2 — On-call engineer list

The on-call engineer list is updated for the incoming shift: name, phone, escalation backup. An engineer going off-call confirms the next engineer has accepted handover before going off-duty.

Why: "I thought X was on call" is the classic escalation failure. The explicit acceptance step closes the gap.

C3.3 — Overnight reduced staffing

Overnight watches run on reduced staffing: one supervisor

- one dispatcher minimum; the engineer-on-call is phone- available but not on-site. Overnight dispatcher scope is limited to monitoring, alarm response, and the M2 work- block authorisation for overnight possessions.

Why: overnight activity is low (no revenue service, limited movements); full-staff daytime structure wastes the labour. The minimum crew keeps the watch safe-walked without over-staffing.

C3.4 — I5/I6 full brief

Any Category I5 or I6 incident ongoing across a shift boundary gets a full brief at handover — not summarised, not presumed-known. The outgoing supervisor briefs the incoming supervisor verbally for at least 15 minutes.

Why: the biggest incidents span many shifts and develop over hours. A silent transfer loses the investigator's working hypotheses; the full verbal brief restates them for the person now responsible for acting on them.

C3.5 — System health dashboard

At handover, the OCC captures a screenshot of the system health dashboard (fleet status, consensus-cluster health, station SCADA roll-up, backhaul link state) and files it to the shift folder. The incoming watch confirms the screenshot reflects reality before acknowledging.

Why: the screenshot is a timestamped snapshot of "what was true at handover" — useful for any later investigation that needs to establish what the incoming watch knew at the start of their shift.

S1 — Shift start

Repository source: `docs/operations/dispatcher/s1-shift-start.md`

Scope: OCC dispatcher handover, login, fleet inventory, schedule ready-check.

Cross-refs: `osr-occ`, RFC 0013 §4.2 S1.

Rules

S1.1 — OCC login

The incoming dispatcher logs into the OCC console with a personal credential (not a shared shift account). The console records the login timestamp in the shift-audit log.

Why: every radio authorisation, MA release, and degraded- mode transition is attributed to a person. Shared accounts break the chain of custody that post-incident review depends on.

S1.2 — Read handover log

Before accepting control, the incoming dispatcher reads the outgoing shift's handover log in full: outstanding faults, trainsets held out of service, planned possessions, weather advisories, radio-coverage gaps.

Why: the first 30 minutes of a shift is when handover context is dense and perishable. Skimming now costs more later when an ongoing situation escalates and the new dispatcher doesn't have the thread.

S1.3 — Fleet inventory

The dispatcher confirms the fleet roster against the CBM- backend overnight report: trainsets in revenue service, in maintenance, in cold reserve. Discrepancies (e.g. a trainset marked revenue that is still on a depot road) are resolved with the depot foreman before first service.

Why: the schedule builder allocates trainsets to services; if the roster is wrong, the first attempt to dispatch that service hits a no-trainset error in the middle of peak boarding.

S1.4 — Schedule ready-check

The dispatcher verifies today's roster matches the `service-hours.toml` window (first revenue service, last revenue service, headway steps). Mismatches (e.g. a public holiday schedule loaded on a weekday) are corrected before first service.

Why: schedule mismatches in small systems are detected by the travelling public, which is the worst failure mode. The ready-check is the last place to catch them.

S1.5 — Network status

The dispatcher reviews any active degraded-mode bulletins: speed restrictions, single-track working, substations on reduced output, stations in bypass. Each active item is countersigned on the shift sheet.

Why: a degraded mode left over from the previous shift that the new dispatcher isn't aware of is the shortest path to a rules violation. The countersignature makes "I didn't know" impossible.

S1.6 — Acknowledge handover

The dispatcher countersigns the handover log, which closes the outgoing shift and transfers accountability. The OCC console prompts for the countersignature; until it is entered, the outgoing dispatcher remains the accountable operator.

Why: there must be exactly one accountable dispatcher at all times. A gap between "I'm logging out" and "I'm logged in" creates a window where nobody owns the radio.

S2 — Routine dispatching

Repository source: `docs/operations/dispatcher/s2-routine-dispatching.md`

Scope: timetable adherence, route grants, headway maintenance during nominal operation.

Cross-refs: `osr-occ`, `osr-interlocking`, RFC 0013 §4.2 S2.

Rules

S2.1 — Route grant on schedule

The dispatcher issues each route grant via `osr-interlocking` at the scheduled call time minus the interlocking's route-setup lead. The OCC console pre-fills the grant from the timetable; the dispatcher confirms.

Why: manual route picking defeats the timetable and desynchronises the whole line. The confirmation step keeps a human in the loop for the unusual case (terminal swap, emergency diversion) without burdening the nominal case.

S2.2 — Headway drift

When headway drift exceeds 60 s for any pair of adjacent trainsets, the dispatcher engages the recovery pattern: shorten dwells on the leading trainset, extend dwells on the trailing trainset, and hold the next-following at an upstream station until the gap closes.

Why: bunching is self-reinforcing — a late train picks up more passengers, dwells longer, falls further behind, and empties the following train. Catching it at the 60 s threshold prevents a 20-minute unravel later.

S2.3 — Dwell extensions

Dwell extensions beyond the driver's local 15 s authority (D4.4) require dispatcher authorisation. The dispatcher may grant up to 60 s cumulative per station per service; beyond that, the service is regulated (short-turned or held at a terminal) rather than extended further.

Why: chronic over-dwelling on one service drags the whole line. The 60 s ceiling forces a regulation decision instead of letting the problem propagate silently.

S2.4 — Terminal turnback dispatch

At each terminal, the dispatcher confirms the cab-transfer report from the driver (ATO handover, brake test, doors closed) before releasing the reverse-direction route grant.

Why: a turnback with an unconfirmed cab transfer risks dispatching a trainset whose active cab is still the inbound end — a wrong-end movement that the interlocking will block but that burns a dispatch slot.

S2.5 — Crowding response

When a station reports a boarding rate exceeding its archetype capacity (per RFC 0010 §10), the dispatcher may hold the next-following trainset at the previous station to create an empty-train gap that absorbs the surge.

Why: a single overcrowded train generates a long dwell, which generates the next overcrowded train. Holding the follower briefly costs a headway but breaks the feedback loop.

S3 — Incident handling

Repository source: `docs/operations/dispatcher/s3-incident-handling.md`

Scope: dispatcher response to in-service incidents. Covers all six categories from RFC 0013 §7 (I1 self-recovery I6 mass casualty).

Cross-refs: RFC 0013 §7, `osr-event-recorder`, `osr-occ`.

Rules

S3.1 — I1 single-train fault

For a single-train fault that clears within one dwell (lost ATO link recovers, door-close recycle succeeds), the dispatcher logs the event to `osr-event-recorder`, notifies the depot for trend analysis, and issues no passenger PA unless service-minute impact exceeds 2 minutes.

Why: a PA on every blip trains passengers to ignore PAs, which is bad when a real PA matters. The 2-minute threshold keeps announcements credible.

S3.2 — I2 service disruption

For a service disruption lasting more than 5 minutes (trainset unable to move under own power, signalling fault requiring staff talk-through), the dispatcher initiates the rescue- coupling protocol (D7), issues a passenger PA every 3 minutes, and logs a timetable recovery plan (short-turns, skip-stops) to the shift log.

Why: disruptions longer than a few minutes need both a technical response (rescue) and a communications response (PA cadence). Running them as a single protocol reduces the chance that one is forgotten while the dispatcher focuses on the other.

S3.3 — I3 short safety event

For a short safety event (EB triggered, SPAD, overspeed trip), the dispatcher orders the driver not to move until the event recorder is preserved, initiates a 72-hour investigation with a minute-by-minute timeline, and holds the trainset out of service pending the investigation outcome.

Why: moving a trainset after an EB contaminates the event record with recovery data and obscures the initiating sequence. The 72-hour window matches RFC 0013's safety- case evidence retention requirement.

S3.4 — I4 passenger injury

For a passenger injury (reported by driver, station staff, or passenger emergency intercom), the dispatcher places a medical call, coordinates with station staff for a station handover of the casualty, and files the passenger-injury report form within 24 hours.

Why: the injured passenger needs a named owner on the OCC side so the ambulance arrives at the right platform. The 24-hour form is the insurance record, which is expensive to reconstruct from memory.

S3.5 — I5 collision or derailment

For a collision or derailment, the dispatcher orders a full line shutdown, notifies the National Rail Safety Authority within 2 hours (or the equivalent regulator per the deployment), and issues an incident-site preservation order (no movement, no cleanup) until the NRSA investigator releases the site.

Why: the first instinct after a major incident is to clear the site to restore service — which destroys the evidence the regulator needs to determine cause. The preservation order makes the cost of not clearing explicit.

S3.6 — I6 mass casualty

For a mass-casualty event (fire, explosion, structural failure), the dispatcher activates the evacuation protocol for the affected stations, escalates to emergency services (fire + ambulance + police), notifies NRSA and regional media liaisons, and coordinates shutdown boundaries to contain the incident.

Why: mass-casualty response is a multi-agency operation where the rail dispatcher is one voice among many. The role is to own the rail side of the response — tracks de-energised, trains stopped, stations evacuated — and hand off everything else to the authorities trained for it.

S3.7 — Onboard battery fire or off-gas alarm

On a confirmed battery-compartment event, the dispatcher verifies the reported car/module/string identity, charge inhibit, HV isolation request, mist-system status, train movement-safety assessment, nearest usable platform, platform evacuation route, and emergency-service rendezvous point.

When `osr-fire-safety` requests a controlled stop and the train remains safe to move, OCC protects the route to the nearest suitable platform, clears that platform, prevents another train entering it, and prepares immediate door release/evacuation on the side approved by the incident assessment. OCC does not command the train to continue beyond that platform.

When immediate danger, loss of containment, crash damage, essential-control loss, unsafe route, or emergency-brake output is present, OCC accepts the immediate stop, protects both directions, isolates nearby charging equipment, and directs evacuation only after trackside hazards and the safer egress side are established. The water-mist indication is treated as propagation-control status, not proof that the cell event is extinguished.

S4 — Weather and environmental

Repository source: `docs/operations/dispatcher/s4-weather-and-environmental.md`

Scope: network-wide environmental overlays per RFC 0013 §6.

Cross-refs: RFC 0013 §6 + `lib/templates/climate-adapters.toml`.

Rules

S4.1 — Dust storm (haboob)

On a declared dust-storm event (visibility below 200 m or PM10 above the climate-adapter threshold), the dispatcher imposes a network-wide 50 % speed cap, issues a passenger PA on every line, and alerts the depot to schedule post-event axle-box and switch-machine inspections.

Why: dust defeats optical sensors and accelerates switch- machine wear. The 50 % cap preserves stopping-distance margin against a reduced-adhesion wheel-rail contact; the post-event inspection catches the wear before it becomes a failure.

S4.2 — Heatwave

On a sustained ambient above 45 °C for 2 hours or more, the dispatcher imposes a 15 % speed cap on continuous-welded- rail spans (per RFC 0009 presets marked CWR) and schedules a pre-dawn recalibration of switch-machine detection sensors.

Why: CWR buckling risk rises sharply above 45 °C rail temperature; the speed cap reduces dynamic load. Switch- sensor drift at high temperature is recoverable by recalibration but misreads in the meantime can jam a route.

S4.3 — Flooding

On measured rainfall exceeding 50 mm per hour, the dispatcher places an inspection hold on at-grade civil spans (no movement until walked by MOW); elevated and bridge spans continue per RFC 0011 §5.

Why: at-grade track is vulnerable to ballast washout and drainage-ditch overtopping, which neither `osr-wayside- points` nor the driver can see from the cab. Elevated and bridge spans drain by design.

S4.4 — Lightning

On lightning strikes within 2 km of the alignment, the dispatcher monitors radio backhaul closely. If both 5G and LoRa backhaul are lost for more than 10 s, the dispatcher accepts degraded-mode operations (manual block working per D6) until backhaul recovers.

Why: lightning induces surges that can knock out multiple backhaul paths simultaneously. The 10 s window distinguishes a momentary flap from a sustained outage that warrants the degraded-mode switch.

S4.5 — Sandstorm recovery

After a sandstorm clears, the dispatcher holds full-speed operations until MOW has walked each affected section and confirmed switches are clear of drift. Restricted-speed (50 %) operation may resume as soon as visibility lifts.

Why: drift in switch throats jams the points mechanism and can be missed by detection sensors that have been abraded by the same event. The walk catches it; the speed cap keeps passengers moving while the walk is in progress.

S5 — Maintenance interlock

Repository source: `docs/operations/dispatcher/s5-maintenance-interlock.md`

Scope: granting work-on-track authorities, blocking sections for maintenance, and restoring to revenue after work completes.

Cross-refs: `osr-interlocking` `MaintenanceOverride` entries, RFC 0013 §4.2 S5.

Rules

S5.1 — Formal work-block authorisation

The dispatcher issues a work block by committing a `MaintenanceOverride` entry to `osr-interlocking` with a specific work-crew id, section ids, and an expiry timestamp (default 60 min, extendable per S5.4). The OCC console prompts for each field; blank fields reject the commit.

Why: open-ended blocks with vague scope are how work crews get dispatched onto live track. The structured form makes every block uniquely identifiable, time-bounded, and traceable to a named crew.

S5.2 — Train clear verified

Before the `MaintenanceOverride` commit is accepted, `osr-interlocking` verifies no trainset footprint overlaps the requested section. The dispatcher cannot override this check.

Why: granting a work block while a trainset is in the section puts the crew in front of a moving train. The interlocking's section-occupancy logic is the same one that enforces Movement Authority — so if it can't prove the section is clear, the block waits.

S5.3 — Protection signals

Before work begins, the dispatcher requests protection signals (red aspect + trap point if fitted) at both ends of the blocked section. The wayside crew radio confirms each protection signal is visible and effective.

Why: the interlocking prevents new MA into the section, but a trainset already holding MA through the section must still see a stop signal. Protection is the last defence against a runaway movement.

S5.4 — Block extension

Block extensions are granted in 30-minute increments, not open-ended. The dispatcher renews the `MaintenanceOverride` expiry at each increment with a fresh radio check to the crew.

Why: a 4-hour block that overruns silently leaves the track blocked after the crew has gone home. The 30-minute cadence forces regular radio contact, which catches silent overruns.

S5.5 — Block clearance

The dispatcher clears the block only after: crew withdrawal confirmed on radio, protection signals cleared, and track walk-back confirmed clear of tools and material. The `MaintenanceOverride` expiry is then committed — which returns the section to revenue availability.

Why: tools left on the track are how the first trainset back into the section derails. The walk-back is the non- negotiable final step; the dispatcher is the gatekeeper.

S6 — Shift end

Repository source: `docs/operations/dispatcher/s6-shift-end.md`

Scope: end-of-shift handover, outstanding-event log, system state summary for the incoming dispatcher.

Cross-refs: RFC 0013 §4.2 S6.

Rules

S6.1 — Write the handover log

The outgoing dispatcher writes the handover log 30 minutes before shift end. The log covers: outstanding incidents, active `MaintenanceOverride` entries, active speed restrictions, degraded-mode flags, depot slot availability, and any passenger-affecting events from the shift.

Why: the 30-minute margin gives time for the outgoing dispatcher to answer questions while still on shift, rather than leaving the incoming dispatcher to decode a hastily- written log alone.

S6.2 — Confirm fleet overnight status

Before close, the outgoing dispatcher assigns every healthy trainset to a named powered passenger-station track near its first morning duty or to the main depot, and attaches a named fault to every set held out of service.

Why: a trainset "somewhere" overnight is a trainset that gets missed at first service next morning. The explicit stabling assignment makes the morning roster pre-solved.

S6.3 — Accept a passenger-station stabling slot

The dispatcher accepts a station slot only when the set is telemetry-healthy and the track meets the accepted charging-duty and morning-readiness criteria for that set, with CCTV, remote traction isolation, protected emergency access, and a confirmed passenger-area closure plan.

Why: overnight station parking is cheaper and starts morning service across the route, but only if security, electrical isolation, fire response, and public access are controlled as deliberately as a depot road; charging power is derived from the accepted duty and shared-charger schedule, without a universal minimum (see distributed stabling).

S6.4 — Verify simultaneous morning release

Before handover, the dispatcher records each station-stabled set's target SoC, scheduled self-test, first duty, and fallback set; any failed charge, security, or self-test condition routes that set to the main depot or reserve plan.

Why: distributed stabling saves no time if morning dispatch discovers an unready set only when the first trip is due.

S6.5 — Carry over open incidents

Any I2/I3/I4/I5/I6 incident still open at shift end is listed explicitly in the handover log, with the investigation owner, deadline, and next action. Incidents are never closed by "shift change" — only by resolution.

Why: incidents closed implicitly by handover disappear from the investigation record. The 72-hour and 2-hour regulatory clocks run on wall-clock time, not shift time.

S6.6 — Log off

The outgoing dispatcher logs off the OCC console only after the incoming dispatcher countersigns the handover log (per S1.6). Until countersignature, the outgoing dispatcher remains accountable.

Why: leaving early while the incoming dispatcher is still catching up creates a window where no qualified person is at the console. The countersignature closes that window explicitly.

S7 — Intrusion response

Repository source: `docs/operations/dispatcher/s7-intrusion-response.md`

Scope: OCC dispatcher response to wayside track-intrusion detections raised by `osr-intrusion-detect`. Per-section verdicts arrive on the OCC dashboard as `Present` / `Unknown` / `Clear` (RFC 0016 §5.2).

Cross-refs: `osr-intrusion-detect`, `osr-interlocking` `SectionIntrusion` entries, RFC 0016 §7.1.

Rules

S7.1 — First-response track patrol

On the first `Present` verdict on any section, the dispatcher dispatches track-patrol within 10 minutes and notifies the engineer-on-call. While the verdict stands, the interlocking withholds MA on that section automatically — no manual hold is required.

Why: the interlocking already enforces the safety restriction — there is no "move faster before the gate drops" concern. The dispatcher's job is to resolve the underlying cause (animal, debris, trespasser) and return the section to `Clear`, which is a ground-level task.

S7.2 — Sustained Unknown treated as equipment failure

A `Unknown` verdict persisting for more than 5 minutes is treated as a wayside-equipment fault. The dispatcher opens a work ticket against the affected junction box, holds or diverts any trains whose route requires the section, and notifies maintenance per M7.

Why: a brief `Unknown` is a sensor flap; five minutes of `Unknown` is a hardware problem that will not resolve itself. The ticket routes the fix into the CBM stream rather than the dispatcher improvising a repair.

S7.3 — Return to service

The dispatcher returns the section to revenue only after: (a) track-patrol confirms the section clear of obstruction, (b) wayside sensors have produced at least one fresh `Clear` verdict, and (c) any affected PSDs and stations are ready. The "return to service" action is an OCC-console command that logs the dispatcher's confirmation to the shift log.

Why: clearing the patrol check but not the sensor check is the shortest path to immediately re-blocking the same section (e.g., debris missed on walk-through). The three-gate sequence keeps the system honest.

S7.4 — Multi-section outage escalation

When three or more sections on the same line carry non-Clear verdicts concurrently, the dispatcher treats it as a line- wide event — escalates to the supervisor, considers cutting revenue service on the affected line, and activates the network-disruption passenger communications protocol.

Why: three concurrent intrusions are almost always either a coordinated event (storm, security incident) or a wayside comms failure that's affecting multiple junction boxes. Both cases call for line-level intervention, not per-section management.

S7.5 — No override

The dispatcher has no authority to force a section to `Clear` by dispatcher action. The interlocking's `section_available_to` gate (d) is a hard interlock.

Why: if the dispatcher could clear the verdict from the console, the safety case for wayside detection is broken — the interlocking would be advisory rather than authoritative. Real clearance comes from track-patrol + sensor agreement per S7.3, not from a dispatcher button.

Station and depot overnight stabling

Repository source: `docs/operations/distributed-stabling.md`

Two revenue trainsets stay at each selected powered station for coordinated morning starts. The remaining revenue trains and reserves stay at declared storage on their own line. Depot parking tracks are sized separately from maintenance and inspection bays.

For Samawah, the plan is:

Location	Revenue	Spare	Cold reserve	Total
20 stations, two trains each	40	0	0	40
Line 1 main depot	32	4	1	37
Line 2 service-terminal storage	13	2	1	16
Line 3 service-terminal storage	12	2	1	15
Fleet	97	8	3	108

The station allocation covers all 34 planned morning departure directions. The three storage locations require 4,046 m of usable slots in total at 49.5 m per train plus 10 m clearance: 2,201.5 m on Line 1, 952 m on Line 2, and 892.5 m on Line 3. The main depot's 17 workshop bays remain separate. Lines 2 and 3 use their existing powered service terminals, with the reusable layup block defining storage requirements. No additional heavy workshop is inferred. Storage layout and energy sizing still need engineering verification.

Each line operates independently. Trains stay on their own line for station launch, depot storage and return movements. The allocator adds no interline connections, new route geometry or cross-line train transfers. This follows the simplicity principle: use the existing line and service point, with explicit, repeatable rules.

Generated plans

Every catalogue city's `engineering/stabling/summary.json` includes a `hybrid_allocation`: station and depot assignments by line and fleet role, depot storage positions, usable slot lengths and unresolved access requirements. The allocator first assigns revenue stock to morning departure directions, then fills station provision up to two trains, and assigns the remainder and reserves to storage on the same line. It prefers an existing depot, then the line's designated powered service terminal. Stations shared between lines count once. Where a line lacks enough revenue stock for all morning directions, the report identifies the uncovered directions instead of creating trains.

`allocation_passed` checks fleet accounting, the station limit, planned depot positions and morning direction coverage. Depot positions are requirements, not surveyed capacity. Physical release remains open until track lengths, turnouts, charging access, security and inspection arrangements are verified.

```
.venv/bin/python tools/automation/generate-stabling-plan.py --all
```

Native hybrid operation

The simulator accepts explicit `overnight_allocations` for every fleet, with station, departure heading, location type, role and train count. Depot stations also declare `depot_stabling_positions`; `is_depot` and workshop bays alone provide no storage allowance. The loader checks complete fleet/role inventory, at most two station homes per shared station, depot capacity, powered dispatch locations and access on the train's own line. Station batteries can supply charging without grid import; a grid outage does not itself disable a charger. Positive installed charging/source capability selects a candidate. The continuous replay, rather than a universal 150 kW minimum, tests its duty.

At closing, active trains finish their journeys along real sections until they reach their assigned homes. Return moves retain energy and movement-authority checks and use the line's existing peak-service headway for spacing, independently of the closed passenger timetable. The event log records these empty movements as `ReturnToStabling`. Passenger service closes at 02:00; empty returns can continue until trains are stabled before the morning start. At home, trains berth with the declared morning heading and top up under the shared site limits. CSV `stabling_location` distinguishes station, depot and in-transit stock. The cabinet output, grid import and battery discharge each have one budget per simulation step, shared by every train at that site. Source budgets include charger losses and reset only at the next site tick. The 2026-09-10 correction removes the former per-request reuse of grid and discharge allowances; a larger cabinet cannot make a smaller source connection available separately to each waiting train. Station launch trains get priority, followed by waiting depot revenue trains, then other trains; train ID resolves ties. Schedule, faults and energy still gate morning departures. Reserves remain in depot storage.

The connected eight-train fixture passes two continuous service cycles: two trains at each of three stations, one revenue train and one spare at the depot, real depot departures/returns, charging and no movement-authority invariant violations. Train positions and batteries are retained between cycles. A repeat-run test also compares the full per-train CSV byte for byte, event

sequence, distance and energy totals. With the same simulator build, input ordering, scheduled faults and time step, both runs match exactly. This establishes repeatability for the tested model; it is not a proof of real-world timing or bitwise equivalence across platforms. Depot berthing and heading changes are abstract node operations; yard roads, throats and turnback geometry remain outside the physical graph.

The generator emits a separate `*-hybrid.toml` where the station launch allocation is complete. The current catalogue has 182 native candidates; 84 plans still lack revenue trains for every selected morning direction. Samawah now has a runnable candidate with all 108 train homes on their own lines. Candidate generation is separate from continuous operating acceptance.

`screen-hybrid-stabling.py` runs full service days and checks each train against its explicit home, separating station and depot capacity. It checks the passenger-service closure, battery reserve, parked reserves and station launch directions. Empty returns during 02:30–05:30 are reported explicitly. Depot departures cannot substitute for missing station launch stock. The first 30 minutes of depot revenue departures are reported without requiring every stored train to join service immediately.

```
.venv/bin/python tools/automation/screen-hybrid-stabling.py --design cities/catalogue/west-asia/Iraq/Samawah/design.
toml --days 2
cargo test -p osr-sim --test hybrid_stabling
cargo run --release --bin osr-sim -- --config lib/examples/hybrid-stabling.toml --duration 173701 --time-step 5 --ma-
check-every 0 --csv-out /tmp/hybrid-stabling.csv
```

City operating evidence

The 2026-09-10 replays use the corrected shared grid and battery source limits. All nine line-local hybrid screens pass two continuous service days:

City	Station / depot trains each night	Station directions restarting within 60 s, each morning
Samawah	40 / 68	34 / 34
Uíge	12 / 15	10 / 10
Quelimane	10 / 13	8 / 8
Edéa	10 / 10	8 / 8
Bukavu	44 / 86	38 / 38
Soroti	12 / 11	8 / 8
Sheikhupura	18 / 19	14 / 14
Sumbawanga	20 / 20	14 / 14
Tartus	26 / 34	20 / 20

Samawah passes both nights with 40 station and 68 depot trains and all 34 station launch directions on both mornings. The shared-source correction initially exposed a numerical hold: T45 reached unpowered `line-1-0814-0268-s019260` about 1 Wh short of the next section's reserve-protected departure threshold. Powered departures now require an extra 0.001 percentage point of stored SoC (6.75 Wh for the 675 kWh LM3), protecting against accumulation in per-tick floating-point updates. This is additional charge at the source; the 20% section-entry reserve is unchanged and unpowered stops do not require another top-up margin. A focused charging-gap regression and the fresh full two-day replay pass.

All nine runs preserve the reserve within the existing 0.001-percentage-point numerical tolerance, keep reserves parked, conserve site energy and record no passenger departures after closure or movement-authority invariant violations. Empty return movements after 02:30 remain visible in each report. Physical layouts, electrical acceptance and daytime headway delivery remain open.

Retained station-only operating evidence

The existing native candidate is a station-only benchmark. It implements station holding, shared charging up to 150 kW per train toward 95% SoC, schedule/headway/movement-authority gated starts, and sufficient departure energy to reach the next selected charger with the 20% reserve. Spare and cold-reserve roles remain parked and do not enter routine service. Automatic reserve activation and physical yard routing remain to be implemented.

Samawah's short overnight replay and continuous two-day replay pass their holding and restart checks: all 34 planned directions restart within 60 seconds, with no routine reserve departures or invariant violations. The continuous replay retains positions, train batteries and site storage between days. Its largest overnight station queues reach 18 and 19 trains, so its combined acceptance result fails the two-train station limit.

Those failures describe historical station-only benchmarks. Their source hashes remain attached to the original experiments; the package manifest records them as diagnostics, including any source drift, outside selected-plan acceptance. Those failures describe the station-only benchmark. They do not invalidate the 40-station/68-depot planning split or demonstrate its execution. The earlier seven-train redistribution study is also a station-only counterfactual, rather than the adopted overnight plan.

```
.venv/bin/python tools/automation/screen-stabling-plan.py --design cities/catalogue/west-asia/Iraq/Samawah/design.toml
.venv/bin/python tools/automation/screen-stabling-cycles.py --design cities/catalogue/west-asia/Iraq/Samawah/design.
toml --days 2
```

Runnable benchmarks and raw traces live under `build/engineering/stabling/`. Compact reports record source, candidate, binary and raw-output hashes.

Remaining implementation

Detail storage tracks and access within each line's service location, then validate the same-line home-return policy against city evening and morning timetables while retaining two station launch trains. The shared source-rate budgets are enforced in software; verify the resulting charging duty, daytime headways, inspection release, reserve activation and defective-train recovery against those duties. Reassess depot and station energy sizing from the resulting operation.

M1 — Depot safety

Repository source: `docs/operations/maintenance/m1-depot-safety.md`

Scope: hazard awareness, PPE, lone-worker protocol, pit work, crane operations.

Cross-refs: RFC 0014 §5 (main-heavy layout), RFC 0013 §4.4 M1.

Rules

M1.1 — PPE baseline

All depot workers on the shop floor wear hi-vis vest, steel- toe footwear, hearing protection in designated noise zones, and eye protection when grinding, welding, or handling compressed gas. Entry to the shop without baseline PPE is refused by the gate.

Why: the baseline is enforced at the gate because mid- shop "just popping in" is when PPE is forgotten and how eye injuries happen. The gate is the single chokepoint that catches it.

M1.2 — Pit work

Pit work (under-car inspection or maintenance) requires: pit lighting on, pit edge rails in place, no-unaccompanied rule (minimum 2 workers, one at ground level), and the trainset power-isolated and chocked.

Why: a worker falling into a pit or a trainset rolling over a pit worker is catastrophic. The paired-work rule means help is on site; the chock rule means the trainset can't move.

M1.3 — Crane operations

Overhead crane (40 t in main-heavy, 10 t in secondary-medium) operations require a qualified operator (licensed per deployment regulator), a clear area below the hook path (no workers in the shadow), and a spotter in radio contact with the operator.

Why: a lifted load swinging into a worker kills; the spotter's job is specifically to see what the operator can't from the cab. The qualification requirement is the legal minimum and the insurance precondition.

M1.4 — Lock-out tag-out (LOTO)

Before working on energised systems, the responsible technician applies a personal LOTO tag to each energy source: each car traction-battery/string isolator, station-contact isolation, isolated LV DC/DC inputs, roof-PV isolator, compressed-air reservoir bleed valve. Each tag carries the technician's name; only they can remove it.

Why: "someone else will check" is how people get electrocuted. The named-tag-per-source rule is defence in depth; tag removal by anyone other than the owner is a disciplinary offence.

M1.5 — Lone-worker

After-hours lone work (single technician on depot floor) is permitted only with an active radio check-in every 30 minutes to the OCC night watch. A missed check-in triggers a welfare call and, if unanswered, dispatch of a second worker to investigate.

Why: a lone worker injured and unable to call for help is invisible for the rest of the shift otherwise. The 30-minute cadence bounds the worst-case discovery delay.

M1.6 — Chemical safety

Cleaning agents, lubricants, and gels are stored in a designated chemical cabinet with labelled containers and Safety Data Sheets (SDS) accessible at the cabinet. Mixing chemicals or transferring to unlabelled containers is prohibited.

Why: decanted-into-water-bottle solvent is the classic depot poisoning; mixed bleach + ammonia is the classic depot gassing. The cabinet and labels defeat both.

M2 — Work-on-track authorisation

Repository source: `docs/operations/maintenance/m2-work-on-track-authorisation.md`

Scope: formal work blocks, protection signals, access to active ROW. Paired 1:1 with the dispatcher's S5 rules.

Cross-refs: S5 (dispatcher) + `osr-interlocking` `MaintenanceOverride`, RFC 0013 §4.4 M2.

Rules

M2.1 — Request work-block

Work-block requests are submitted to the OCC at least 48 hours in advance for planned work; 2 hours minimum for urgent (safety-affecting) work. The request names the crew lead, section ids, scope of work, and planned duration.

Why: 48 hours allows the dispatcher to slot the block without service disruption; the 2-hour floor for urgent work balances speed against coordination with other movements. Unplanned drop-in requests strand crews.

M2.2 — Protection signals set

On arrival at the work site, the crew lead confirms protection signals (per S5.3) are set at both ends of the block and radios the OCC to commit the `MaintenanceOverride`. Work does not start until the commit is acknowledged.

Why: arriving and working before the commit means the crew is on track with no interlocking protection. The radio handshake is the moment the crew is formally protected.

M2.3 — Stay inside the block

During the block, crew members remain inside the protected section except for a single "roamer" permit per crew (one person authorised to step outside for inspection or tool retrieval). The roamer carries a high-visibility armband and a personal radio.

Why: crew members outside the block are not protected by the block. Limiting this to a single named roamer keeps the accounting tractable and ensures the roamer knows they are exposed.

M2.4 — Half-hourly check-in

The crew lead radios the OCC every 30 minutes with a crew headcount and progress update. Missed check-ins trigger the OCC welfare protocol (S5.4 extension check).

Why: a crew in trouble (medical, fall, entrapment) needs help fast; the 30-minute cadence bounds the OCC's discovery delay and matches the S5.4 block-extension cadence on the other side.

M2.5 — Block clearance

On work completion, the crew lead confirms: all tools withdrawn, all personnel accounted for by name, track walked clear of debris, protection signals cleared. The OCC is then radioed to release the `MaintenanceOverride` per S5.5.

Why: a hammer left on a rail derails a trainset. The named-personnel accounting prevents leaving a worker behind; the walk-clear is the final sweep.

M2.6 — No trains through a block

No rolling-stock movement is authorised through a section under active `MaintenanceOverride`, regardless of written orders, radio instruction, or apparent urgency. The block is a hard interlock.

Why: historical incidents almost all involve "just one train through" at a supposedly secure block. Making M2.6 a no-exception rule removes the ambiguity — if a train must go through, the block must first be cleared, not overridden.

M3 — Routine track inspection

Repository source: `docs/operations/maintenance/m3-routine-track-inspection.md`

Scope: walking the line, visual checks, geometry-recording-car cadence per RFC 0009 §9.

Cross-refs: RFC 0009 §9, RFC 0013 §4.4 M3.

Rules

M3.1 — Visual walk

Every section is walked visually every 7 days in the direction of traffic. Walkers check for: missing fasteners, cracked sleepers, rail-head spalling, ballast shoulder loss, and vegetation encroachment.

Why: visual inspection catches the failures that measurement misses (missing bolts, cracked concrete sleepers) and measurement catches what visual misses (geometry drift). Both cadences are non-redundant.

M3.2 — Geometry recording car

The geometry recording car runs per RFC 0009 §9 preset cadence: `standard-urban` every 90 days, `standard-metro` every 60 days, `mainline-mixed` every 45 days. Output is reconciled against the previous run to detect trend deterioration.

Why: geometry drift (gauge, cant, alignment, twist) is linear-predictable from trend data; a single reading is less informative than the delta. The preset-specific cadence reflects the loading severity of each preset.

M3.3 — Post-weather inspection

Flooding, dust, or heatwave events (per S4.1–S4.3) trigger an immediate post-event walk of affected sections before full-speed service resumes. The walk is additional to, not a substitute for, the 7-day routine walk.

Why: extreme weather produces failure modes (ballast washout, drift-jammed switches, rail buckles) that the 7-day walk won't catch in time. Post-event walks insert an extra data point at the moment risk is elevated.

M3.4 — Defect classification

Defects are classified at the point of discovery as QN1 (quality note, alert) or QN2 (immediate action). QN1 is logged and scheduled for the next maintenance window; QN2 raises a work block under M2 and halts service through the affected section until fixed.

Why: the classification is the inspector's judgement call but it's binary — alert or act. Grey-zone intermediates (semi-urgent, action-next-week) are how a QN2 becomes a derailment.

M3.5 — Turnout inspection

All turnouts are inspected monthly for bolt tightness, weld cracks, switch-point closure, and detection-sensor alignment. Findings are logged per switch id to `osr-wayside-points`.

Why: turnouts concentrate wear and are the single largest derailment-risk feature on the railway. Monthly cycling keeps the inspection window shorter than the failure window for the known modes.

M3.6 — Fence + intrusion walk

Every 30 days, the fence line is walked for breaks, damaged gates, and evidence of animal intrusion. Breaks are recorded, GPS-tagged, and scheduled for repair within 7 days.

Why: a fence break lets in both humans (trespass risk) and animals (collision risk). The 30-day cycle catches seasonal changes (grown-in gaps closing or new gaps opened by flooding) before an incident.

M4 — Switch maintenance

Repository source: `docs/operations/maintenance/m4-switch-maintenance.md`

Scope: blade lubrication, detection-sensor cleaning, mechanical inspection per RFC 0012 §4.

Cross-refs: RFC 0012 §4, `osr-wayside-points`, RFC 0013 §4.4 M4.

Rules

M4.1 — 30-day switch inspection

Every switch is inspected every 30 days per RFC 0012 §4.1: blade closure at toe (≤ 1.5 mm gap), stretcher-bar tightness, switch-rail wear limit, stock-rail wear limit, detection-sensor alignment, and cable-gland integrity.

Why: the 30-day cycle matches the switch-machine CBM curve — beyond that, probability of silent drift into QN2 territory rises above the acceptable envelope.

M4.2 — Self-lubricating slide chair

The RFC 0012 self-lubricating slide-chair spec requires no routine re-greasing, but the 30-day inspection confirms slide chairs are free of sand contamination, dust clumping, and debris. Contaminated chairs are cleaned but not re-greased.

Why: greasing over contaminated chairs traps the contamination in the slide path, which is worse than the contamination alone. Cleaning without re-grease is the correct maintenance action per the chair datasheet.

M4.3 — Stretcher-bar tightness

Stretcher-bar bolts are torque-checked to the RFC 0012 spec (200 N·m, per RFC 0012 §3 kit) every 30 days. Loose bolts are tightened and flagged for follow-up on the next cycle.

Why: a stretcher-bar failure is a classic derailment mode — the blade opens under a passing wheel. Torque-checking is the only way to catch creeping loosening before it becomes mechanical play.

M4.4 — Detection-sensor cleaning

A/B inductive detection-sensor faces are cleaned every 30 days with a dry cloth. Sensors showing drift in their response curve (logged to `osr-wayside-points`) are replaced at the next 30-day cycle rather than cleaned again.

Why: detection sensor reliability is the `osr-wayside- points` safety case's key assumption. Cleaning maintains the reliability; replacement at drift onset is cheaper than the MA-blocking fault that a failed sensor triggers.

M4.5 — Switch-machine motor

Switch-machine electric motors are inspected every 90 days for brush wear, commutator condition, and current draw at throw (compared against the baseline). Motors outside the current-draw envelope are replaced.

Why: motor current rise precedes mechanical failure by several weeks; catching it at the 90-day cycle prevents mid-operation failure. The baseline comparison makes the drift detectable earlier than a single snapshot.

M4.6 — Points heating

For deployments in climate zones where frost is a concern (not Samawah, but the RFC 0012 spec covers cold-climate adaptations), thermostats for points heating are checked at the start of cold season: calibration check, heating- element resistance check, contactor operation.

Why: frozen points are a winter-morning incident in every cold-climate system; a heating fault discovered on the first frost is a reliability failure. The pre-season check moves discovery into a controlled window.

M4.7 — Post-dust-storm

After any S4.1 dust-storm event, all switches in the affected area are inspected and blade-cleaned within 48 hours, independent of the 30-day cycle.

Why: dust-jammed switches are the single largest post- haboob failure mode in the Middle East climate adapter. The 48-hour window ensures the cleaning happens while the drift is fresh, before wheel traffic packs it into the slide chair.

M5 — Fleet maintenance

Repository source: `docs/operations/maintenance/m5-fleet-maintenance.md`

Scope: inspection intervals at the depot — daily, 7-day, 30-day, overhaul. Per RFC 0014 §5.1.

Cross-refs: RFC 0014 §§5.1, 5.2, `osr-cbm-onboard`, `osr-cbm-backend`, RFC 0013 §4.4 M5.

Rules

M5.1 — Daily inspection

Every trainset returning to depot is inspected daily at a `main-heavy` or `secondary-medium` depot for: wheel and tread visual (no flats, no cracks), door sealing (no visible gaps), HVAC nominal, external body damage, and coupler face integrity.

Why: daily inspection is the fastest feedback loop on service wear. Catching a tread flat within one day of formation prevents the flat from becoming a wheel spall that requires reprofiling.

M5.2 — 7-day inspection

Every trainset is inspected on a 7-day cycle for: wheel-wear measurement (flange thickness, tread depth), friction-brake pad thickness, battery-pack cell voltage spread, and under- car wire harness chafe.

Why: 7 days is the cadence at which wheel wear becomes measurably distinguishable from noise, and where a pad approaching its wear limit can still be service-scheduled without an emergency swap.

M5.3 — 30-day inspection (EN 50126 A-class)

Every trainset is taken out of revenue service monthly for a 30-day inspection: BMS deep-scan (every cell, every temperature sensor, every contactor), motor-bearing vibration spectrum, HVAC filter replacement, and aux-inverter output quality check.

Why: the deep-scan items are diagnostic gold — they catch failure precursors (cell imbalance, bearing harmonics) that the daily and 7-day scans are not instrumented to see. EN 50126 A-class is the safety-case-anchored interval.

M5.4 — Wheel reprofiling

Wheels are reprofiled on the depot lathe at 150 000 km cumulative service, or earlier if the 7-day inspection records wheel wear exceeding the reprofile limit.

Why: 150 000 km is the RFC 0014 §5.2 CBM curve intersection of wheel wear and reprofile cost. Earlier reprofiling wastes wheel; later reprofiling risks flange failure.

M5.5 — Bogie overhaul

Bogies are overhauled at 600 000 km cumulative service at `main-heavy` depots only. Secondary-medium depots do not carry the jigs required for bogie overhaul.

Why: bogie overhaul requires frame-straightening jigs and a wheelset press that would double the footprint of a secondary-medium depot for a task that occurs once every 10+ years per trainset. Centralising at main-heavy is the RFC 0014 archetype rationale.

M5.6 — Body overhaul

The carbody undergoes overhaul at 10 years in service and every 10 years thereafter: body shell corrosion inspection, interior refurbishment, cable replacement, HVAC system swap.

Why: body overhaul is a 20-40 year fleet life extension activity. Skipping it shortens the fleet life from 40 years to around 20 — a capital-planning catastrophe for the deployment.

M5.7 — CBM-triggered intervention

When `osr-cbm-backend` raises an amber condition on a trainset component, the technician schedules intervention within 7 days; a red condition triggers intervention within 24 hours, with the trainset held out of service in the interim.

Why: CBM exists to move interventions from reactive to scheduled. The 7-day and 24-hour windows are the RFC 0013 §4.4 thresholds calibrated against failure-mode progression curves for the component classes in scope.

M6 — Incident site access

Repository source: `docs/operations/maintenance/m6-incident-site-access.md`

Scope: response-crew access to derailment / fire / crash sites. Coordination with emergency services and site preservation.

Cross-refs: RFC 0013 §§4.4 M6 + §7 incident categorisation.

Rules

M6.1 — Site access authorisation

Access to an incident site is authorised only under a formal M2 work block, or under an explicit emergency authorisation issued by the OCC (name + timestamp + scope). Self- dispatched recovery crews are turned away at the protection signal.

Why: well-intentioned crews arriving unannounced are a hazard to emergency services and contaminate the incident scene. The authorisation channels all access through the OCC, which keeps the site count controllable.

M6.2 — Crash-site preservation

At a crash or derailment site, rolling stock is left as- found for the NRSA investigator. Movement is authorised only after the investigator releases the site, or in the specific case where movement is necessary to free trapped passengers (in which case a named OCC authorisation with timestamp is recorded).

Why: moving evidence is how cause determination gets closed as "undetermined". The trapped-passengers exception is the only override; it is rare and must be documented.

M6.3 — Fire site

At an onboard fire, civil fire services lead the response on arrival. The maintenance crew's role is limited to: confirm `osr-fire-safety` onboard systems have actuated (suppression discharged, traction isolated), provide traction-battery isolation information to the fire officer, and brief the LFP chemistry, affected module/string, outward vent direction, mist reservoir/pump/pressure/flow state, and the last recorded module temperatures/off-gas readings. No person opens a battery compartment, restores HV, or assumes the event is extinguished because visible flame has stopped without fire-service authority and a documented thermal monitoring period.

Why: fire services are trained for fire; rail crews are trained for rail. The crew's job is to give the fire officer the information they need to fight the fire safely without pretending to lead a task the crew is not trained for.

M6.4 — Crane operations at site

On-site crane operations (recovering a derailed trainset) require a qualified operator (per M1.3), an additional lift plan reviewed by the M&E engineer, and spotters at both ends of the lift. Multi-lift (two cranes on one load) requires a specific multi-lift plan.

Why: on-site crane work is higher-risk than depot crane work — uneven ground, time pressure, media attention. The additional plan requirement is the compensating control.

M6.5 — Evidence chain

Before any recovery move, the crew photographs the site comprehensively (GPS-tagged, timestamped), extracts the event recorder per D8, and makes a sealed copy for the investigator. The chain-of-custody form is countersigned by the crew lead and OCC duty officer.

Why: recovery destroys the spatial evidence; photographs and the event recorder preserve the information content of the site. The chain-of-custody form makes the evidence admissible.

M6.6 — Recovery move

A recovery move (derailed or crashed trainset being moved off the line) operates at 15 km/h maximum, with physical escort fore and aft, and carries no passengers or personnel inside the trainset being recovered.

Why: recovery moves use compromised rolling stock on possibly compromised track — the speed cap keeps kinetic energy low. The no-passengers rule prevents a bad move becoming a mass-casualty event.

M7 — Wayside sensor maintenance

Repository source: `docs/operations/maintenance/m7-wayside-sensor-maintenance.md`

Scope: per-section wayside intrusion-detection packs — fence-line contacts, ROW-mounted LIDAR, ROW-mounted radar, CCTV with AI classifier. See RFC 0016 §4 for the pack composition.

Cross-refs: `osr-intrusion-detect`, RFC 0016 §7.2 M7, RFC 0013 §4.4 (maintenance rulebook).

Rules

M7.1 — 30-day sensor-pack walk-through

Every wayside sensor pack is walked at 30-day intervals: LIDAR and radar housings visually inspected for physical damage and lens fouling, cable runs checked for rodent damage, fence-line contact sensors spot-tested by induced vibration, junction box door sealed.

Why: 30 days matches the M3 track-inspection cadence so the walkthroughs can be combined and the crew makes one trip instead of two. Most sensor faults develop over weeks, not days — monthly is a reasonable detection floor.

M7.2 — Post-weather inspection

After any RFC 0013 S4 weather event (haboob, heatwave, flooding, lightning), maintenance inspects the wayside packs in the affected sections within 48 hours. LIDAR + radar lens cleaning is routine after haboob; fence-line contact re-set is routine after lightning.

Why: weather is the single largest source of wayside- sensor false positives and silent failures. Getting in within 48 hours catches the obvious (dust-fouled lens) before it becomes a sustained `Unknown` that an S7.2 ticket is already addressing.

M7.3 — Sustained-Unknown ticket response

A work ticket opened by dispatcher S7.2 (sustained `Unknown` verdict) triggers an on-site maintainer visit within 4 hours of ticket creation. The maintainer performs a calibration + self-test on each sensor, replaces any failing unit from the depot spare pool, and issues a "returned to service" confirmation to the dispatcher.

Why: the interlocking holds the section against revenue traffic for the entire duration of the `Unknown`. Four hours is the network-service cost budget: anything longer starts to degrade timetable adherence to the point where the dispatcher is re-routing or short-turning.

M7.4 — Calibration after replacement

A replaced wayside sensor is re-calibrated against the section's known rail-profile envelope before being committed back into the `osr-intrusion-detect` evaluator. Calibration uses the depot's reference retro-reflector at the published lateral offsets from rail centreline; the first `Clear` verdict after calibration is logged to the shift record.

Why: a replacement unit out of the box reads lateral offsets off its own factory axis, not the section's rail centreline. Skipping calibration is how sensor replacement turns into a false-`Present` storm that reproduces the failure on the next train.

M7.5 — Fence-line continuity check

Every 90 days, the fence-line contact run is end-to-end continuity-tested under sensor simulator stimulus. Any continuity loss drives a segmented walk to isolate the fault within 500 m.

Why: a quiet fence-line is ambiguous — it could mean "no breach" or "sensor run broken for weeks." The 90-day stimulus test distinguishes the two, which is what makes the fence contact trustworthy as an I3 safety input.

T1 — Station opening

Repository source: `docs/operations/station-staff/t1-station-opening.md`

Scope: daily station opening — power up, PIS check, fare-gate test, PSD test, CCTV + PA handover.

Cross-refs: `osr-pis-station`, `osr-afc`, `osr-tvm`, `osr-psd`, `osr-station-scada`, RFC 0013 §4.3 T1.

Rules

T1.1 — Power-up sequence

Station staff power up subsystems in order: station SCADA PIS fare (AFC + TVM) PSD. Each subsystem must reach "ready" on the station console before the next is started.

Why: PIS depends on SCADA time-sync; fare depends on PIS for passenger display; PSD depends on fare for emergency-release interlock. Out-of-order power-up races the dependencies and produces spurious amber faults that waste the opening window.

T1.2 — PIS display self-test

Staff trigger the PIS self-test from the station console. The test cycles each platform display through a colour pattern + a sample "next train" message and reports OK per display to `osr-pis-station`. Any display that fails self-test is logged as a T1 fault and replaced or bypassed.

Why: a display that lights up but shows stale data is worse than a dark display — passengers trust the wrong information. The self-test exercises the end-to-end rendering path, not just the backlight.

T1.3 — Fare-gate test

Staff test each fare gate through a full open / close / emergency- release cycle using a staff test card. Gates that fail to close within 3 s or fail emergency release are taken out of service and flagged for depot repair.

Why: a stuck-closed gate traps passengers in peak; a stuck- open gate loses revenue and undermines fare enforcement. The daily test catches both before passengers arrive.

T1.4 — TVM test

Staff run a test purchase on each TVM: QR-code display, mobile-money confirm, receipt print. Failed TVMs are taken out of service with a paper "out of order" sign placed physically on the machine.

Why: a TVM whose receipt printer has run out of paper sells tickets passengers can't prove they bought, which breaks fare enforcement downstream. The daily test catches the mundane failures (paper, ribbon, mobile-money API key expired).

T1.5 — PSD test

At PSD-equipped stations, staff trigger a platform-side PSD open / close / emergency-release cycle via the station console. PSDs that fail open or close are locked open (not closed) pending repair.

Why: a failed-closed PSD can trap a passenger on the platform side of the door after train departure. Locking failed PSDs open is the fail-safe choice — passengers can still reach the platform but the PSD is visibly out of service.

T1.6 — CCTV handover

Staff assume CCTV monitoring from the OCC night watch via a radio handover: confirmed cameras-OK count, outstanding events logged overnight, any area under intrusion alert.

Why: OCC night watch covers many stations at low attention; morning station staff cover one station at high attention. The handover makes sure any overnight intrusion alert is followed up on the ground, not filed and forgotten.

T1.7 — Ready-to-OCC

After T1.1–T1.6, staff post a "station ready" status via `osr-occ`, which clears the station's morning-open flag in the OCC roster. The dispatcher will not dispatch a first revenue service to a station that has not posted ready.

Why: an unready station takes a first-service trainset full of passengers with no working PIS, no open fare gates, and possibly no PSDs. The explicit ready-post is the handshake that prevents that.

T2 — Passenger boarding

Repository source: `docs/operations/station-staff/t2-passenger-boarding.md`

Scope: platform supervision during revenue hours. PSD monitoring, wheelchair assistance, queue regulation.

Cross-refs: RFC 0010 §§5, 7 (access/circulation, accessibility), RFC 0013 §4.3 T2.

Rules

T2.1 — Platform walk

During peak hours, staff walk the length of the platform at 15-minute intervals, checking for unattended items, fallen passengers, blocked emergency exits, and fare-gate queue overflow. Walks are logged on the station console.

Why: stationary platform staff see the area around their post; walking staff see the whole platform. Peak hours are when things go wrong fastest, so the walk cadence tightens then.

T2.2 — Wheelchair boarding

On a wheelchair boarding request (pre-notified or spot), staff position the boarding ramp at door 2 of car 1 (aligned with the reference stop mark per D4.1), assist the passenger across the platform gap, and radio the driver to extend dwell per D4.4.

Why: the gap between platform and trainset is wider than most wheelchairs can clear unassisted, and the dwell is shorter than unassisted boarding needs. The ramp + extension is the accessibility guarantee that RFC 0010 §7 commits to.

T2.3 — Overcrowding

When platform density exceeds the archetype capacity threshold (per RFC 0010 §10), staff start queue management (rope dividers, PA announcements regulating flow to the platform) and notify the dispatcher, who may hold the next-following trainset per S2.5.

Why: platform crowding is a crush risk. Staff have the local signal the dispatcher cannot see; the dispatcher has the fleet lever the staff cannot pull. Splitting the response matches the information to the authority.

T2.4 — Unattended item

On an unattended item report, staff clear a 5 m radius around the item, notify the OCC, and call police / security per the deployment's protocol. The trainset approaching the platform is held at the previous station until the item is cleared.

Why: a suspicious item is rarely a bomb but always a disruption. The 5 m clear preserves the scene; the hold prevents a new trainset-load of passengers arriving into it.

T2.5 — Platform clear confirmation

At non-PSD stations with no train-camera live feed, staff confirm the platform is clear of passengers in the door zone before the driver commands door close. Confirmation is by radio; absence of confirmation defers door close by up to 15 s (one dwell extension) before the driver must interrogate.

Why: the driver's side mirror doesn't cover the full platform length. Staff are the second pair of eyes that catches a passenger running for the train at door close. At PSD-equipped stations the PSD handles this mechanically, so the rule scopes to non-PSD only.

T3 — Passenger incidents

Repository source: `docs/operations/station-staff/t3-passenger-incidents.md`

Scope: station-level response to passenger-side incidents — medical, behavioural, fare dispute, lost property.

Cross-refs: RFC 0013 §§4.3 T3 + §7 incident categorisation.

Rules

T3.1 — Medical

On a medical incident, staff render first aid to the level of their training, request emergency services via the OCC (who calls I4 per S3.4), and preserve a 3 m area around the casualty clear of other passengers.

Why: first responder staff don't diagnose — they stabilise and wait. The clear area is both dignity for the casualty and room for ambulance crew to work when they arrive.

T3.2 — Behavioural / altercation

On an altercation or threatening behaviour, staff de-escalate verbally, call transit security (where provided by the deployment) or local police, and log the incident to the station SCADA event log. Staff do not physically intervene except to protect an identified victim.

Why: station staff are not trained for physical intervention and are usually outnumbered. De-escalation plus summoning trained responders is the evidence-based protocol.

T3.3 — Fare dispute

On a fare dispute at the gate, staff do not debate the fare at the gate — they issue a supplement ticket or a dispute slip and direct the passenger to the documented appeal route (station office or online). The gate is cleared within 30 s.

Why: a fare dispute at the gate blocks every passenger behind. The dispute process happens off the critical path; the passenger is neither presumed guilty nor let through free.

T3.4 — Lost property

Lost-property items are logged to the station log with description, finder, and time, secured in the station office locker for 30 days, and entered into the network lost-property database accessible to other stations. After 30 days, items are donated or disposed of per deployment policy.

Why: passengers tracking lost items often call or visit more than one station. The network database makes the find discoverable from any station; the 30-day ceiling keeps the locker from filling with old umbrellas.

T3.5 — Child separated

On a child-separated-from-caregiver report, staff issue a platform PA with description, initiate CCTV review from the last known time and place, and notify the OCC within 5 minutes if the child is not found. Station exit gates are watched manually until the child is located.

Why: separated children are found most often within the first 10 minutes. The staged response (PA CCTV OCC escalation) scales effort to duration without burning the network on a 2-minute reunion.

T4 — Fare enforcement

Repository source: `docs/operations/station-staff/t4-fare-enforcement.md`

Scope: on-board inspector work, non-payment handling, appeals.

Cross-refs: `osr-afc`, RFC 0013 §4.3 T4.

Rules

T4.1 — Inspector identification

Fare inspectors wear a visible identity badge with photograph and inspector number, and rotate routes daily so that the same inspector does not work the same line consecutively.

Why: visible ID legitimises the stop for the passenger. Daily rotation limits the opportunity for harassment (by or of) any one inspector becoming personal.

T4.2 — Fare check

On stopping a passenger, the inspector scans the passenger's ticket (paper QR, mobile QR, or tap card) via the handheld which verifies the HMAC-signed token against `osr-afc`'s active-tokens list. Verification is online or cached (cached ≤ 15 min old is accepted).

Why: the HMAC check catches forged tickets deterministic- ally. The 15-minute cache keeps inspectors working through short backhaul flaps without falsely accusing paying passengers.

T4.3 — Fare-evasion record

On a failed check, the inspector issues a supplement-fare notice (printed on the handheld) stating the infringement, date, and appeal route. The passenger's identifier (name + document number per deployment law) is logged to the infringement database. No physical detention occurs — the passenger may leave the train at the next station.

Why: detention escalates the situation and puts both parties at risk. The paper trail is sufficient legal basis for the penalty; physical enforcement is a police function.

T4.4 — Appeals

Supplement-fare appeals may be filed within 14 days of issue, in writing or via the deployment's online portal. The OCC reviews the appeal against the AFC record and inspector log; outcome issued within 30 days.

Why: the 14-day window is long enough for a passenger returning from travel and short enough to keep the AFC record fresh in the database. The OCC-not-inspector review keeps the decision independent of the person who issued.

T4.5 — Inspector safety

Inspectors work in pairs during late-evening slots (after 21:00) and on routes flagged as elevated-risk by the OCC. Lone working is permitted only during peak daytime hours on mainline routes.

Why: lone inspectors on quiet late services are an assault risk. Pair working is not about enforcement productivity (which drops) but about the inspector going home safely.

T5 — Station closure

Repository source: `docs/operations/station-staff/t5-station-closure.md`

Scope: end-of-day shutdown. Evacuation (normal last-train passenger clear), equipment powerdown, CCTV handover to OCC night watch.

Cross-refs: RFC 0013 §4.3 T5.

Rules

T5.1 — Last-train clear

After the last revenue service departs, staff confirm the platform is clear of alighting passengers and that no passenger is attempting to board a non-revenue trainset. The last-train clear is timestamped to the station log.

Why: the last train is where passengers are most likely to miss a stop or fall asleep. The explicit clear establishes that the platform was empty at a known time, which is important for any later incident report.

T5.2 — Platform + concourse walk

Staff walk the platform, stairs, concourse, and fare-paid area checking for unattended items, sleeping passengers, and equipment left out. Sleeping passengers are woken and escorted out; unattended items follow T2.4.

Why: station closure without a walk leaves people inside a locked station overnight, which has happened at many real systems. The walk is the non-negotiable final sweep.

T5.3 — Fare-gate + TVM lockout

Fare gates are locked in the closed-no-entry position; TVMs are switched to "out of service" display and physically secured. Night-mode lighting (50 % level) is switched on.

Why: open fare gates overnight invite entry to a station with no staff presence; lit TVMs invite tampering. The lockout and the out-of-service display remove the temptation.

T5.4 — PIS off-hours banner

The PIS is switched to the off-hours banner (first-train timetable for tomorrow, emergency contact, no moving content). Displays that would otherwise show "no service" are put into standby instead.

Why: "no service" on a PIS overnight conditions passengers to ignore PIS in the morning when "no service" would be real. The dedicated off-hours banner is unambiguous.

T5.5 — PSD overnight position

At PSD-equipped stations, PSDs are driven to the fully-closed position for overnight. The PSD fault log is flushed to `osr-station-scada` so morning staff can review overnight anomalies.

Why: closed PSDs overnight prevent unauthorised platform access from the concourse and reduce track-side dust accumulation on PSD guide rails.

T5.6 — CCTV handover

Staff hand over CCTV monitoring to the OCC night watch via radio, acknowledged by the OCC night operator with a countersigned entry in the station log.

Why: CCTV without a watcher is CCTV for post-hoc review only. The handover ensures someone is always watching, and the countersignature makes the moment of handover auditable.

T5.7 — SCADA low-power mode

The station SCADA is left in low-power mode (displays off, core services running, heartbeat to the OCC active). Services that can be off overnight (PA, TVM, non-critical lighting) are switched off.

Why: the station still needs to respond to an overnight event (fire, intrusion, flood) so the core SCADA stays on; everything else reduces overnight load on the station-solar battery bank.

Operations Validation Checklist

Repository source: `docs/operations/validation-checklist.md`

The rulebook is drafted in English. It becomes deployment-ready only after operator and authority review. This checklist defines the review evidence required for a deployment-controlled operations release.

Gate	Required evidence	Closure criterion
Operator workshop	Dispatcher, maintenance, station, control-centre, and emergency-service walkthrough notes	Every rule block has accepted, changed, or rejected status
Local authority adaptation	Regulator comments, required terminology, incident reporting law, police/medical/fire contact protocol	Local deviations are documented without weakening safety requirements
Training package	Role syllabus, practical drills, assessment form, recurrent interval, competence matrix	Staff competence evidence can be linked from certification §8.3
Simulation drills	Degraded mode, fire, passenger incident, intrusion, point failure, comms loss, station evacuation	Drill outcomes map to rule IDs and simulator scenarios
Translation if needed	Country-language rulebook and controlled terminology table	English remains canonical; translation is reviewed by operator and safety assessor
Handover to OCC tooling	Console action labels, alarm names, incident categories, and audit-log fields align with rule text	GUI/live-system terminology no longer diverges from the rulebook

Close this checklist against dispatcher, station, maintenance, and control-centre roles. The repository supports unattended operation only.

OSR Operations Portal

Repository source: `docs/operations-portal/README.md`

> Operating architecture update: ERPNext + Frappe HR now own business execution, purchasing, stock, finance and HR. OSR retains railway planning and assurance. Historic SQLite business records are read-only.

Static browser portal for city-level operations data:

- Authenticated OSR Ops Core work orders, managed inspection photos/files, independent handback, controlled document revisions, defects/NCR and sealed audit records persisted in SQLite.
- Manufacturing schedule for trains, waypoints/W-SBCs, track sections, switches, stations, depots, production plant fixtures, energy sites, and station systems.
- Construction QA gates and asset-level QA actions.
- Maintenance schedule expanded to trainsets, stations, track sections, waypoints, switches, structures, energy sites, signalling/comms nodes, depots, and production-plant tools.
- Asset register with stable ids.
- Revisioned project digital twin with finite-resource CPM, critical path, supplier/order-by planning, schedule of values, monthly cash requirements, and an IFC/animation-ready construction-state timeline.
- Launch/status panels for existing OCC, simulator, CBM, AFC, historian, and Ops Core tooling.

The simplified operating model is documented in `ops-core.md`.

Configured deployments enforce city-scoped roles for planners, maintainers, inspectors, approvers, document controllers and auditors. Passing work requires a different authenticated inspector and handback approver; rejection returns work for rework and open NCRs block closeout. The server seals immutable records with HMAC-SHA256 content attestations. These provide tamper evidence inside the deployment but are not qualified electronic signatures or a substitute for an organisation's legal signature policy.

For the integrated City Studio → simulation → OCC replay → work-order flow, run the OSR Workbench. Workbench-created records retain their revision, approved baseline, run and selected-asset references.

Generate Data

```
python3 tools/automation/generate-qa-maintenance-data.py
```

The default input is the generated Samawah design and scenario:

- `cities/catalogue/west-asia/Iraq/Samawah/design.toml`
- `cities/catalogue/west-asia/Iraq/Samawah/samawah.toml`

Outputs land in `build/generated-operations/samawah/`:

- `samawah-operations.json.gz` plus its small integrity manifest
- `samawah-assets.csv`
- `samawah-manufacturing-schedule.csv`
- `samawah-manufacturing-materials.csv`
- `samawah-manufacturing-verification.csv`
- `samawah-procurement-plan.csv`
- `samawah-budget-work-packages.csv`
- `samawah-cashflow-requirements.csv`
- `samawah-construction-timeline.json`
- `samawah-acceptance-evidence-matrix.csv`
- `samawah-maintenance-schedule.csv`
- `samawah-qa-register.csv`

The Manufacturing tab is generated from `../../lib/templates/manufacturing-schedule.toml` and documented in `../rfcs/0030-manufacturing-schedule-system.md`. Track-section durations are calculated from route metres or single-track ST6 panel quantities, production rate and resource count. The JSON bundle also contains a line-level `civil_production` plan derived from civil-segment metres, Pi20/Pi25 bays, foundations, mould/cure cycles, piling rigs, launcher rate, panel gantries and the working calendar. Each schedule row includes the asset id, project-day schedule window, work center, crew roles, staff tasks, controlled BOM/material refs, dependencies, deliverables, evidence required, release authority, QA action link, verification row, and priority.

The Project Twin tab uses the same records. Work-centre capacities produce a finite-resource baseline; a CPM backward pass adds late dates, float and the critical flag. BOM requirements are deduplicated per asset, connected to their required-on-site task, and given a source-class planning lead time. Authoritative city CAPEX is allocated into schedule-of-values work packages, so changing a task or resource capacity moves its order-by and monthly cash requirement on regeneration. The cashflow reconciles to the finance summary; it is not an invoice or funding commitment. Month `0` contains mobilisation requirements before notice to proceed; negative order-by days identify long-lead actions that must be resolved before baseline approval.

For a repository city, the compact Git-reviewable output is `engineering/project-twin/summary.json`. The complete records live in the reproducible compressed operations bundle. Planned purchase-order rows are explicitly `planned-not-issued`; review and issue actual purchasing records in ERPNext. The portal retains the generated requirement and historic records for reference.

The generated material table links rolling-stock packages to `build/bom/rolling_stock_bom.csv` and `build/bom/rolling_stock_cots_fitout_bom.csv`. Infrastructure packages use controlled `project_kit:*` refs until detailed civil/station/energy BOMs exist. The verification table links every manufacturing row to a QA action and marks the QA hold point that blocks successor work.

The acceptance/accreditation evidence basis is generated as `build/generated-operations/samawah/acceptance-evidence-report.md`. It summarizes the controlled material/BOM basis, QA gate coverage, release blocking logic, and links to the evidence matrix CSV.

Run With SQLite

Serve the repository root and persist Ops Core work orders, inspections, defects, audit events and railway project revisions to SQLite. Historic business collections remain readable; new purchasing and financial writes are rejected:

```
python3 tools/automation/ops-core-server.py --port 8008
```

The server creates the database and schema automatically if they do not exist. The default database path is:

```
var/ops-core.sqlite3
```

Then open:

```
http://127.0.0.1:8008/docs/operations-portal/
```

Localhost starts in clearly labelled trusted-development mode. Before shared or network deployment, create private user accounts and start with the user store:

```
python3 tools/automation/ops-user-admin.py var/ops-users.json inspector1 --display-name "Inspector One" --roles
inspector --cities samawah
python3 tools/automation/ops-core-server.py --host 0.0.0.0 --port 8008 --users var/ops-users.json
```

Create separate approver and document-controller identities the same way. The server refuses a non-localhost bind without a configured user store. Put TLS in front of it for any network deployment; keep `var/ops-users.json` and the generated signing key private.

Evidence, Document Control and Backup

Inspection files and photos are stored content-addressed under `var/ops-evidence/`, with their SHA-256, uploader and immutable record retained in SQLite. The Controlled Documents panel requires a document id and revision; a later revision must name the record it supersedes.

Create and verify a consistent SQLite/evidence backup with:

```
python3 tools/automation/ops-core-backup.py create ../osr-private-backups/ops-core.zip
python3 tools/automation/ops-core-backup.py verify ../osr-private-backups/ops-core.zip
# Destination must not exist; restores offline, never over the live database.
python3 tools/automation/ops-core-backup.py restore ../osr-private-backups/ops-core.zip ../osr-restored
```

The archive deliberately excludes the password store and server signing key. Keep backup archives outside the repository's public asset directories. The server serves only explicit public roots and does not expose directory listings. Back those up separately in the deployment's secret vault. Recovery remains an operator-controlled procedure so the tool cannot overwrite a live database. Verification requires exact archive/manifest membership, a checksummed database, unique safe file paths,

valid sizes/hashes, SQLite integrity and foreign-key checks, and every registered evidence file. Duplicate, unlisted, missing or symlink entries are rejected. Restore writes a new directory and rebases evidence paths to it; application records and evidence URLs remain unchanged. Start an isolated server against `data/ops-core.sqlite3` and `data/evidence` there after provisioning secrets. This does not establish coordinated ERP/FUXA/gateway recovery or recovery time objectives.

Samawah is the default dataset. To open another generated city, pass its repository-relative operations bundle in the `data` query parameter.

Save Conflicts And Recovery

Use the localhost Ops Core server for demonstrations and the authenticated server for shared consoles. A server rejection or connection failure keeps the portal attached to SQLite and displays the error. Edits remain an unsaved draft; they do not become accepted records in browser storage. Retry a recoverable failure, or use Download draft and reload server records before reviewing and reapplying changes. Reauthenticate after a session expires.

Every save carries the city revision read by the client. SQLite checks it in the same transaction as the write and returns HTTP 409 for stale snapshots. Inspection and approval order is assigned by the server; an older pass cannot release work after a newer failed inspection.

Manufacturing schedule rows can generate OSR assurance work orders with `source_type = manufacturing`, sharing the evidence, defect/NCR, audit and reconciliation path used by QA and railway maintenance. ERPNext remains the production-transaction authority; OSR closeout controls these assurance tasks. The portal blocks successor manufacturing work until predecessor rows are closed with pass evidence, and it blocks manufacturing closeout until pass evidence exists for the selected work order.

Reconcile Local Records

If a console was used in static/local mode before SQLite was available, open the portal through `ops-core-server.py` and use Storage Reconciliation in the Ops Core tab. It compares SQLite with the browser-local fallback records and can merge local-only records into SQLite while keeping the newest record when ids conflict.

Open:

```
http://127.0.0.1:8008/docs/operations-portal/
```

Historical OSR Operations Portal Gap Analysis — August 2026

Repository source: `docs/operations-portal/gap-analysis.md`

> Operating architecture update: ERPNext + Frappe HR now own business execution, purchasing, stock, finance and HR. OSR retains railway planning and assurance. Historic SQLite business records are read-only.

Reviewed: 2026-08-31. Historical comparison; the capability statements below do not describe the current platform. Use the current integration status, native ERP/FUXA ownership, asset/BIM lifecycle and deployment readiness for implemented capabilities and remaining acceptance evidence. Legacy SQLite execution has been superseded by native ERP records; demonstrated telemetry and asset bindings do not establish a commissioned production deployment.

This retained note compares the August 2026 OSR Operations Portal against adjacent commercial and open-source systems:

- Rail/EAM/APM: IBM Maximo, HxGN EAM, Siemens Railigent X, Bentley AssetWise Rail Condition Analytics
- Construction QA / common data environment: Procore Quality & Safety, Autodesk Construction Cloud quality management, Oracle Aconex
- Open-source / lighter CMMS comparators: openMAINT, Odoo Maintenance, ERPNext Maintenance Schedule
- Standards posture: ISO 55001 asset-management system, ISO 19650 BIM/information management

August 2026 OSR Position

The portal is a strong planning and generated-register tool:

- Deterministically expands a city design into asset ids.
- Generates spreadsheet-friendly asset, manufacturing, QA, and maintenance CSVs.
- Gives each trainset, station, track section, switch, energy site, waypoint/W-SBC, signalling node, depot, and production-plant tool a stable row.
- Renders manufacturing packages, QA gates, and maintenance intervals in one browser UI.
- Keeps the design-to-operations link explicit and reproducible.

It is now a lightweight owner-operator execution tool, but it is not a full CMMS/EAM, construction-management platform, or manufacturing MES.

Historical Gap List

Priority	Gap	Why it matters	Comparator signal
P0	Work-order lifecycle depth	OSR has authenticated city-scoped RBAC, assignable work orders, status transitions, holds, rejection/rework, independent handback, SQLite persistence and server-enforced closeout. Routed queues and deployment-configurable authority matrices remain open.	HxGN EAM work requests become approved work orders, can be dispatched, scheduled, rescheduled, and status-managed.
P0	Inspection forms/checklists	The portal has pass/watch/fail evidence, readings, authenticated inspector identity, managed photo/file upload, automatic NCR creation and server content attestations. Template-specific typed checksheets and offline capture remain open.	Procore/Autodesk emphasize mobile inspection checklists, field observations, photos, signatures, and issues for nonconforming items.
P0	NCR/CAPA workflow	Failed inspection evidence creates an NCR and blocks work; open NCRs block closeout. Root-cause taxonomy, corrective/preventive actions, retest, waiver/deviation authority and effectiveness review remain missing.	Construction QA tools center observations, corrective actions, and issue resolution.
P0	Document control / CDE	Managed content-addressed evidence, append-only document revisions, supersession links and sealed audit records are implemented. RFI/submittal transmittals, organisation ownership and full ISO 19650 CDE workflows remain open.	Oracle Aconex sells complete project records, process workflows, document ownership, and audit trail as core features.
P0	Roles and permissions	Password-authenticated users, city scopes, RBAC, CSRF protection and identity-based segregation of inspection/approval duties are enforced server-side. HMAC server attestations are not qualified signatures; SSO/MFA, organisation federation and jurisdiction-specific legal-signature policy remain deployment work.	Aconex/Procore/Maximo all rely on accountable users, permissions, and auditable actions.
P1	Calendar and crew scheduling	Manufacturing now has project-day windows and staff-task roles, while maintenance still uses textual next-due basis. There is no actual opening date, crew assignment roster, depot capacity, possession window, or workload balancing.	HxGN EAM includes daily scheduling, labor/parts availability, labor utilization, routing, and workload balancing.
P1	Meter/telemetry ingestion	The schedule has km/condition triggers, but no live service-km, BMS, wheel, vibration, charger, W-SBC, or SCADA feed.	Maximo and Railigent emphasize condition data, sensor data, predictive maintenance, and health states.
P1	Defect and failure history	There are no defect codes, failure modes, root causes, MTBF/MTTR, bad-actor assets, or reliability trends.	Odoo computes maintenance statistics such as MTBF/MTTR; Maximo and AssetWise emphasize reliability/performance analytics.
P1	Spares, inventory, tools, warranty	Tasks list owners but do not reserve parts/tools, track stock, manage warranty claims, or roll costs against assets.	HxGN EAM work orders include labor, materials, tools, permits, safety docs, and warranty/claims tracking.
P1	Linear asset analytics	Track sections are simple inter-station rows, not true linear-referencing assets with chainage, measurements, geometry defects, renewal prioritization, or state-of-good-repair scoring.	Bentley AssetWise Rail Condition Analytics focuses on rail condition data, linear analytics, predictive renewal decisions, and SGR.
P1	GIS/BIM/digital twin linkage	Assets have lat/lon or chainage text, but no map view, BIM object ids, as-built model references, or ISO 19650-style information requirements.	ISO 19650 expects secure lifecycle information management for built assets; openMAINT mentions GIS and BIM.
P2	Mobile/offline field app	The browser supports work-order and inspection entry with local fallback, but is not an offline-first technician app with sync conflict policy, QR scans, managed photos or device identity.	Procore, Odoo, ERPNext, and CMMS tools all lean on field/mobile execution.
P2	Regulator/opening evidence package	QA evidence exists as planned rows, not a packaged safety-case/submittal bundle with signoff status and dependencies.	Aconex/Autodesk tie documents, RFIs, issues, assets, commissioning, and handover.
P2	Cost and lifecycle management	There is no asset lifecycle-cost rollup, replacement forecast, depreciation, renewal budget, or capex/opex optimization.	ISO 55001 focuses on balancing performance, risk, and expenditure across the asset lifecycle.

Suggested Build Order

1. Production identity integration. Add OIDC/SSO, MFA, organisation boundaries, routed approval queues and deployment-defined authority policy above the implemented local authenticated RBAC baseline.
2. Inspection/checksheet templates. Expand manufacturing, maintenance, and QA rows into fillable forms with typed fields, pass/fail, numeric readings, photo/evidence slots, signature, and generated NCR on failure.

3. Opening-date scheduler. Convert `next_due_basis` into actual due dates from a city opening date, plus km-based placeholder counters.
4. Defect/NCR/CAPA depth. Add defect severity, root cause, corrective action, responsible party, due date, retest, and waiver.
5. Telemetry bridge. Feed `osr-cbm-backend`, energy-site telemetry, and simulation outputs into the same asset ids.
6. Inventory/tools module. Add parts, tools, calibration records, stock min/max, reservations, and warranty/claim fields.
7. Linear/GIS view. Add map/chainage view for track sections, switches, structures, stations, and energy sites.
8. CDE workflow depth. Add RFI, submittal and transmittal workflows around the implemented managed-file, document-revision and immutable-audit core.

Product Decision

The portal should stay intentionally lightweight and open. Its current small transactional core is:

```
asset register + work orders + inspection evidence + independent handback + NCR + audit log
```

This is a deployable workshop/pilot reference with a materially stronger trust boundary, not a complete enterprise EAM or hosted security service. A real deployment must add TLS, secret management, monitored backups, SSO/MFA where required, data-retention policy and an independent security review. The simplified implementation is captured in `ops-core.md` and exposed through the portal's Ops Core tab.

OSR Ops Core

Repository source: `docs/operations-portal/ops-core.md`

> Operating architecture update: ERPNext + Frappe HR now own business execution, purchasing, stock, finance and HR. OSR retains railway planning and assurance. Historic SQLite business records are read-only.

OSR Ops Core is the railway assurance and evidence layer for the portal. It keeps the generated asset register, manufacturing schedule, QA gates, and maintenance schedule as the source of truth, then adds only the records needed to run a real railway day without copying a full enterprise EAM system.

Operating Loop

1. Generate due work from the city maintenance schedule.
2. Open extra work orders from any manufacturing row, maintenance row, QA gate, or manual finding.
3. Move work through `open`, `assigned`, `in_progress`, `ready_to_close`, `hold`, and `closed`.
4. Record inspection evidence against the selected work order.
5. Put failed work on hold and raise a defect / NCR.
6. Have a different authenticated verifier approve or reject handback.
7. Resolve defects, close approved work, and export the evidence trail.

Core Records

Record	Required fields	Purpose
Asset	asset id, type, name, line/location	Stable reference for every train, station, track section, switch, energy site, system node, depot, and tool group.
Work order	id, source row, asset id, owner, due date, priority, status	Turns a planned manufacturing row, maintenance row, or QA gate into accountable work.
Inspection	work order id, result, reading/reference, managed photos/files, authenticated inspector, timestamp and server attestation	Captures proof that work was done and whether it passed.
Handback approval	work order, latest passing inspection, decision, authenticated approver, declaration, timestamp and server attestation	Requires an identity-distinct decision before closeout.
Defect / NCR	defect id, work order id, asset id, severity, finding, owner, due date, status	Tracks failures until resolved or formally waived.
Controlled document	document id, revision, status, superseded record, file SHA-256 and controller attestation	Keeps evidence and drawing revisions append-only and traceable.
Audit event	timestamp, action, reference, detail and server attestation	Keeps the operating history exportable, reviewable and tamper-evident.

Included

- Asset-specific work orders for trains, waypoints/W-SBCs, stations, track, switches, energy, signalling/comms, depots, and production-plant tooling.
- One-click conversion from generated manufacturing tasks, maintenance tasks, and QA gates.
- Manufacturing work orders carry BOM/material counts, QA action ids, verification ids, predecessor ids, and release evidence requirements.
- Manufacturing successor work is blocked until predecessor work orders are closed with pass evidence.
- Manufacturing closeout is blocked until the work order has pass evidence.
- Daily, weekly, or full-schedule work generation.
- Pass/watch/fail inspection evidence.
- Automatic hold plus defect/NCR creation on failed evidence.
- Password-authenticated, city-scoped RBAC for planner, maintainer, inspector, approver, document-controller, auditor and administrator roles.
- Identity-based handback segregation; the authenticated inspector cannot approve their own inspection, and open NCRs block closeout server-side.

- Managed content-addressed uploads, append-only document revisions and HMAC-SHA256 server attestations for inspections, approvals, documents and audit events.
- Online SQLite snapshots plus checksummed evidence backup archives.
- CSV export for work orders, defects, and audit events.
- SQLite storage through `tools/automation/ops-core-server.py` for a simple owner-operator deployment.
- Atomic city revisions reject stale saves; rejected edits remain an explicit unsaved draft with retry and download/reload controls.

Not Included

- Heavy procurement, finance, HR, or ERP workflows.
- Large CDE workflows with RFIs, submittals and formal transmittals.
- Multi-party contract administration.
- Predictive analytics beyond schedule and condition-trigger placeholders.
- Enterprise organisation hierarchies, OIDC/SSO or MFA.
- Qualified electronic signatures. Server HMAC attestations prove integrity within the deployment; they are not a legal trust service.

ERPNext and Frappe HR provide the ordinary business workflows outside this railway assurance layer. CDE and qualified-signature requirements need separate deployment decisions; they are not supplied by the planning integration.

Storage

SQLite is the store for railway assurance records because it gives the portal a durable file that can be backed up, copied to another console, or inspected with normal SQLite tools. The default database is:

```
var/ops-core.sqlite3
```

The browser still receives the same simple JSON state:

```
workOrders + inspections + approvals + defects + documents + audit + counters
```

That keeps the UI easy to understand while allowing the server to keep indexed SQLite tables for work orders, defects, inspection evidence, controlled documents, evidence-file metadata and audit events.

The server creates `var/ops-core.sqlite3` and its tables automatically if they do not exist. If a browser has local fallback records from earlier static use, the Ops Core tab's Storage Reconciliation panel compares local records with SQLite records and can merge local-only records into SQLite.

Owner–Builder–Operator Mobilisation Status

Repository source: `docs/owner-builder-operator-mobilisation-status.md`

> Status: unfilled-all-gates-open — not legal, spending, construction, safety or operating authority.

Project: `unassigned template`

Scope: Reference full railway delivery and operation; tailor responsibilities and evidence to the selected deployment

The reference organisation and programme are editable. Validate the selected responsibilities, evidence and dependencies; job titles, committee structures and document counts are not universal requirements.

Entity model complete: no · Roles ready: 0/13 · Independent parties ready: 0/3 · Management systems ready: 0/11 · Work packages complete: 0/18 · Gates accepted: 0/8

This template organises mobilisation evidence. It does not create a legal entity, delegate a statutory duty, appoint a competent person, approve expenditure, release construction, certify safety or authorise railway operation.

Accountable Roles

ID	Role	Appointment	Competence	Ready
ROLE-OWNER	owner accountable officer	open	open	no
ROLE-CEO	chief executive	open	open	no
ROLE-PROGRAMME	programme director	open	open	no
ROLE-ENGINEERING	chief engineer and design authority	open	open	no
ROLE-SAFETY	safety and assurance director	open	open	no
ROLE-OPERATIONS	operations director	open	open	no
ROLE-ASSET	asset and maintenance director	open	open	no
ROLE-MANUFACTURING	manufacturing director	open	open	no
ROLE-INFRASTRUCTURE	infrastructure director	open	open	no
ROLE-COMMERCIAL	commercial and local-content director	open	open	no
ROLE-FINANCE	finance director	open	open	no
ROLE-PEOPLE	people and competence director	open	open	no
ROLE-DIGITAL	digital and configuration director	open	open	no

Independent Parties

ID	Party	Organisation / appointment	Ready
IND-ASSESSOR	independent safety assessor	open	no
IND-CHECKER	independent design checker	open	no
IND-AUDITOR	external auditor	open	no

Management-System Readiness

ID	System	Accountable	Required by	Evidence	Status	Ready
MS-ASSET	asset maintenance spares warranty and return to service	ROLE-ASSET	G5	0/4	not-established	no
MS-COM	commercial supplier and local-content management	ROLE-COMMERCIAL	G3	0/4	not-established	no
MS-COMP	competence authorisation fitness and fatigue	ROLE-PEOPLE	G3	0/4	not-established	no
MS-DIG	digital information cybersecurity backup and OT boundary	ROLE-DIGITAL	G2	0/4	not-established	no
MS-ENG	requirements engineering interfaces configuration and change	ROLE-ENGINEERING	G2	0/4	not-established	no
MS-ENV	environment social land utilities and construction impacts	ROLE-INFRASTRUCTURE	G2	0/4	not-established	no
MS-FIN	finance treasury affordability and renewals reserve	ROLE-FINANCE	G1	0/4	not-established	no
MS-GOV	governance delegations risk and audit	ROLE-OWNER	G1	0/4	not-established	no
MS-PC	programme scope schedule cost risk and benefits control	ROLE-PROGRAMME	G2	0/4	not-established	no
MS-QUA	factory site quality nonconformance and calibration	ROLE-MANUFACTURING	G3	0/4	not-established	no
MS-SAF	safety assurance human factors and emergency management	ROLE-SAFETY	G2	0/4	not-established	no

Mobilisation Gates

Gate	Window	Accountable roles	Evidence	Decision	Accepted
G0 — public mandate	month 0-3	ROLE-OWNER	0/5	open	no
G1 — entity and controls	month 2-6	ROLE-OWNER, ROLE-CEO, ROLE-FINANCE	0/5	open	no
G2 — development baseline	month 4-12	ROLE-PROGRAMME, ROLE-ENGINEERING, ROLE-OPERATIONS, ROLE-SAFETY	0/5	open	no
G3 — procure and prototype	month 9-18	ROLE-PROGRAMME, ROLE-COMMERCIAL, ROLE-MANUFACTURING, ROLE-INFRASTRUCTURE	0/5	open	no
G4 — first articles and enabling works	month 15-30	ROLE-MANUFACTURING, ROLE-INFRASTRUCTURE, ROLE-ENGINEERING, ROLE-SAFETY	0/5	open	no
G5 — construction and operational readiness	programme-derived	ROLE-PROGRAMME, ROLE-OPERATIONS, ROLE-ASSET, ROLE-SAFETY	0/5	open	no
G6 — trial and revenue service	authority-derived	ROLE-OWNER, ROLE-OPERATIONS, ROLE-SAFETY	0/5	open	no
G7 — stable operation and replication	first 12-24 service months	ROLE-CEO, ROLE-OPERATIONS, ROLE-ASSET, ROLE-FINANCE	0/5	open	no

Default Mobilisation Work Programme

Planning horizon: month 0–60. Overlap is intentional; local approvals, procurement and construction determine the actual baseline.

Work package	Months	Planning FTE	Accountable	Depends on	Gate	Evidence	Status
MOB-010 — secure public mandate and accountable sponsorship	0–3	2–4	ROLE-OWNER	start	G0	0/5	not-started
MOB-020 — establish legal entity and owner-company agreement	1–6	2–5	ROLE-OWNER	MOB-010	G1	0/5	not-started

Work package	Months	Planning FTE	Accountable	Depends on	Gate	Evidence	Status
MOB-030 — stand up governance finance audit and procurement controls	1–6	3–6	ROLE-FINANCE	MOB-010	G1	0/5	not-started
MOB-040 — recruit and authorise core accountable leadership	1–9	2–5	ROLE-CEO	MOB-010	G2	0/5	not-started
MOB-050 — deploy CDE configuration cybersecurity and asset-ID controls	1–8	3–7	ROLE-DIGITAL	MOB-010	G2	0/5	not-started
MOB-080 — establish safety assurance security and authorisation strategy	2–18	4–9	ROLE-SAFETY	MOB-010, MOB-040	G2	0/5	not-started
MOB-090 — develop affordability funding cashflow and benefits baseline	2–15	4–8	ROLE-FINANCE	MOB-010, MOB-030	G2	0/5	not-started
MOB-060 — freeze concept of operations and development requirements	3–12	8–15	ROLE-ENGINEERING	MOB-010, MOB-040	G2	0/5	not-started
MOB-070 — complete survey ground utility land and consent evidence plan	3–15	6–12	ROLE-INFRASTRUCTURE	MOB-010	G2	0/5	not-started
MOB-120 — build workforce apprenticeship and competence pipeline	4–30	4–10	ROLE-PEOPLE	MOB-040, MOB-060	G3	0/5	not-started
MOB-100 — package procure and qualify delivery supply chain	8–18	6–12	ROLE-COMMERCIAL	MOB-060, MOB-070, MOB-080, MOB-090	G3	0/5	not-started
MOB-110 — mobilise factory depot OCC and civil production facilities	8–20	6–14	ROLE-MANUFACTURING	MOB-060, MOB-070	G3	0/5	not-started
MOB-130 — execute and accept vehicle and reusable-product first articles	15–30	12–25	ROLE-MANUFACTURING	MOB-100, MOB-110, MOB-120	G4	0/5	not-started
MOB-140 — deliver enabling works and site-specific civil construction	15–36	10–20	ROLE-INFRASTRUCTURE	MOB-070, MOB-100	G4	0/5	not-started
MOB-150 — build operator and complete operational readiness trials	18–36	10–22	ROLE-OPERATIONS	MOB-080, MOB-110, MOB-120	G5	0/5	not-started
MOB-160 — establish maintenance asset information spares and handover	18–36	8–16	ROLE-ASSET	MOB-130, MOB-140	G5	0/5	not-started
MOB-170 — obtain trial and revenue-service authorisation	30–42	6–12	ROLE-OWNER	MOB-130, MOB-140, MOB-150, MOB-160	G6	0/5	not-started
MOB-180 — stabilise service and release replication baseline	42–60	5–10	ROLE-CEO	MOB-170	G7	0/5	not-started

Use the setup plan to mobilise the organisation. Copy `owner-builder-operator-mobilisation.toml` into a controlled deployment workspace, fill it with evidence references, and retain this repository file as the blank default. A gate is accepted only when its decision, decision-maker/date and at least one reference for every required evidence item are recorded.

Owner–Builder–Operator Setup Plan

Repository source: `docs/owner-builder-operator-setup.md`

OpenSourceRail needs an institution that can own the design, procure openly, build locally, accept evidence and operate the railway for decades. The default is a public asset owner with a ring-fenced delivery and operating company, not a temporary project office and not a turnkey vendor holding the design data.

This is a jurisdiction-neutral mobilisation plan. Local government must obtain legal, labour, tax, procurement, railway-safety, environmental and insurance advice before creating entities or delegating statutory duties.

Recommended Default

```
elected government / public sponsor
└── public asset owner and funding authority
    ├── audit and risk committee
    ├── safety, environment and assurance committee
    ├── design/data authority (retained client capability)
    └── ring-fenced rail delivery and operating company
        ├── programme, commercial, finance and land
        ├── system integration and configuration
        ├── infrastructure delivery
        ├── rolling-stock/local-manufacture delivery
        ├── operations-readiness and railway operations
        ├── engineering, maintenance and asset management
        └── customer, stations and community relations

independent of delivery reporting:
independent safety assessor • independent design checker • external auditor
regulator/authorising body • emergency services • insurer
```

The public owner keeps the corridor, assets, open technical baseline, data, configuration rights, safety duties that law does not permit it to delegate, and the ability to retender individual packages. The company employs the people who integrate, build, commission, operate and maintain the system. Contractors, factories and specialist suppliers deliver bounded work packages to that company.

For a small local authority, the company can begin as a ring-fenced municipal business unit and incorporate after the development gate. For several cities, a national or regional manufacturing and design company can supply vehicles and standard civil products while each city retains its own asset owner/operator.

Entity Choices

Model	Use when	Advantage	Main risk / control
Ring-fenced authority directorate	Earliest feasibility or a single small pilot	Fastest start; existing public controls	Skills can disperse; give it a dedicated budget, accountable officer and configuration/data ownership
Wholly public corporation	Default for construction and operation	Stable employer, clearer accounts, can serve multiple corridors	Political or commercial drift; protect public-service objectives and reserved owner decisions in its constitution
Public–public joint venture	Several authorities share a factory, depot or operator	Pools scarce engineering and procurement capability	Ambiguous accountability; one named owner for each asset and safety duty
Mixed delivery joint venture	A local industrial partner brings plant/capability	Accelerates technology and workforce transfer	Vendor lock-in; owner retains design/data, audit, step-in and competitive procurement rights
Long concession	Only where a funded authority can regulate outputs for decades	Transfers defined service obligations	Financing and information asymmetry; not the default for a first open system

Do not create one legal entity for every work package. Use one accountable public owner and one delivery/operator company unless local law requires separation. Keep manufacturing as an internal business unit, subsidiary or competitively procured joint venture only when its workload and liabilities justify that step.

Reserved Decisions And Independence

The owner board reserves route and service outcomes, total budget, borrowing, land acquisition, major contract awards, safety-policy acceptance, entry into public trial/revenue service, disposal of assets and changes to open-design/data rights. The company board controls delivery within those limits.

Four lines must remain distinct even in a lean organisation:

1. Delivery does the work. Production and construction managers own output.
2. Engineering accepts technical conformity. The design authority controls requirements, interfaces, calculations, drawings and deviations.
3. Quality verifies the record. Quality can stop work and does not report to the production manager whose work it inspects.

4 Independent assurance challenges the case. The assessor/checker has direct access to the owner board and is not paid on successful approval alone.

The regulator authorises where law requires; it is never treated as the project's designer. The project must not self-certify merely because a role exists on an organisation chart.

Accountable Executive Team

Accountable role	Owns	Cannot delegate away
Owner accountable officer	Public value, funding, land, statutory sponsor duties	Business-case truth, board reporting and reserved decisions
Chief executive	Company performance and organisational competence	Clear accountability and adequate resources
Programme director	Scope, schedule, interfaces and benefits	Integrated baseline and recovery decisions
Chief engineer / design authority	Requirements, technical baseline, interfaces, concessions	Fitness of engineering decisions and competent approval routes
Safety and assurance director	Safety-management system, hazard record, assurance plan	Escalation of unacceptable risk and assessor access
Operations director	Rulebook, OCC, stations, service delivery and emergency response	Operational readiness and competent operating staff
Asset and maintenance director	Maintenance system, depot, spares and whole-life performance	Safe asset condition and maintenance handback
Manufacturing director	Vehicle/product factory output, tooling and industrialisation	Production conformity and first-article discipline
Infrastructure director	Survey, utilities, civil/track/station delivery	Workfront release, temporary states and construction handover
Commercial/local-content director	Packaging, competition, suppliers and claims	Fair procurement, conflicts and auditable local-value evidence
Finance director	Controls, cash, treasury, audit and affordability	Solvency, segregated funds and truthful forecasts
People/competence director	Workforce plan, labour relations and competence records	Role authorisation, fatigue and training-system integrity
Digital/configuration director	CDE, asset IDs, cybersecurity, software and data custody	Reproducible baselines, access control, backup and audit trail

One person may cover several roles during feasibility, but each decision still records which hat was worn. Chief engineer, quality lead and safety/assurance lead must not be subordinated to production output when accepting their own gates.

Decision And Handshake Matrix

Decision / deliverable	Accountable	Does the work	Required concurrence
Sponsor requirements and affordability envelope	Owner accountable officer	Strategy/finance	Operator, community/land and treasury/funder
System requirements and configuration baseline	Chief engineer	Systems integration	Operations, maintenance, safety and configuration
Reusable vehicle/civil product release	Chief engineer	Discipline design authority	Manufacturing/infrastructure quality and independent check as classified
Site workfront release	Infrastructure director	Contractor/site team	Designer, survey/geotechnics, quality and temporary-works authority
First vehicle/product acceptance	Manufacturing director	Factory and test teams	Chief engineer, quality, safety and independent assessor for credited evidence
Trial-operation readiness	Operations director	ORAT team	Chief engineer, maintainer, safety director, emergency services, insurer and regulator
Revenue-service entry	Owner board	Operator/company	Regulator/authorising body and independent assurance report
Maintenance return to service	Asset/maintenance director	Authorised maintainer	Independent inspection where the maintenance classification requires it
Design change after opening	Chief engineer	Configuration/change team	Operations, maintenance, safety and affected asset owner

Capability Build: People And Facilities

These ranges are mobilisation allowances for one pilot corridor and one LM3 first-article stream, not universal staffing standards. The service plan, local law, automation level, shifts, contracting strategy and emergency arrangements must produce the approved establishment.

Phase / permanent capability	Core client/company people	Delivery workforce outside the core	Purpose
Feasibility and entity setup	8–15	5–15 advisers/survey specialists	Sponsor, finance, route, engagement, engineering and legal/procurement control
Development and procurement	35–60	20–60 designers/investigators	Requirements, surveys, reference adaptation, safety plan, packages and approvals
LM3 pilot factory	20–30 engineering/quality/planning	80–140 production, stores and test over one shift	Tooling, local parts, vehicle assembly, first article and production learning

Phase / permanent capability	Core client/company people	Delivery workforce outside the core	Purpose
Infrastructure delivery client	20–35	Contractor workforce from quantities/programme	Workfront release, interface control, site assurance and handover
Pre-opening operator/maintainer	35–60 leadership, engineering, trainers and planners	Recruits/trainees sized from timetable and rosters	Rulebook, OCC/depot/stations, trials, emergency exercises and asset takeover
Steady operation	25–45 central functions	Frontline establishment from service, station, depot and shift model	Safe service, maintenance, customer support, renewals and continuous improvement

Use local apprenticeships and paired roles from the first design package: every imported specialist has a named local counterpart, deliverable training record and exit criterion. Count local value from payroll, invoices and origin evidence, not contract labels.

Minimum initial facilities are a controlled project office/CDE, survey and field stores, a metrology/quality room, training and simulator room, secure software and configuration environment, and—when authorised—the pilot factory, depot/OCC and civil production/laydown sites defined by their engineering packages.

Management Systems To Establish

Create these as working processes with owners and records before buying large packages:

- corporate delegations, board calendar, conflicts, whistleblowing and internal audit;
- requirements, interface, configuration, document and design-change control;
- safety, hazard, human-factors, security and emergency-management systems;
- competence, role authorisation, medical/fatigue controls and just-culture reporting;
- procurement, supplier qualification, local-content evidence and counterfeit-part control;
- project controls: scope, estimate, schedule, risk, change, contingency and earned progress;
- factory/site quality: ITPs, travellers, material/serial traceability, calibration and NCRs;
- asset information, maintenance, spares, warranty, obsolescence and return-to-service;
- environmental/social consent, land, utilities, traffic, community complaints and accessibility;
- treasury, invoice/commitment control, external-currency exposure and whole-life renewals reserve;
- cybersecurity, identity/access, backup/restore, incident response and operational-technology boundary.

Start with the repository's configuration, QA, maintenance and evidence records, then qualify the deployed CDE/ERP/EAM/identity/signature systems. A Git commit or server attestation is useful provenance but is not a legal signature or approval. The mobilisation status now carries these eleven systems as accountable, gate-linked records. Each stays `not-established` until all required controlled documents, evidence references and named approval are present.

Commercial And Industrial Structure

Package procurement so local firms can qualify without giving one bidder the entire system:

Package family	Default commercial route	Owner-retained control
Survey, ground, utilities and enabling works	Local professional/services and works lots	Control network, data licence, accepted ground model
Repetitive precast/steel/civil products	Framework with first-article then call-offs	Product definition, mould/tool ownership, inspection rights
Vehicle structures, GFRP and interiors	Local factory or JV with transparent make/buy plan	CAD/source, tooling, BOM, configuration and right to qualify alternatives
Specialist doors, brakes, bearings, traction, batteries and control hardware	Competitive performance/interface packages	Interface drawings, diagnostics, spares/data, change notification and escrow where needed
System integration and assurance	Retained internal authority plus bounded specialist support	Requirements, hazard/configuration records and final acceptance
Operations and maintenance	Public company by default; competitively source support/outcomes	Rulebook, asset history, competence and step-in continuity

Every contract should identify deliverables, acceptance evidence, open formats, background/foreground IP, tooling ownership, cybersecurity access, audit rights, training, local counterpart, spares/obsolescence, warranties, change control, termination assistance and safe handback. Avoid debt or supply terms that require an unnamed future monopoly supplier.

Where a commercial development or manufacturing JV is proposed, use the joint-development framework and its machine-checked partnership readiness record. That route qualifies the exact legal counterparty, accepted cash/in-kind work, background and foreground IP, manufacturing information, shareholder-supplier pricing, procurement conflicts, finance separation and local capability before any partnership, cost, manufacturer-support or territorial-right claim is made. The

preferred sequence is a funded development agreement followed by an equity decision after contributions are delivered—not equity awarded for unmeasured future promises.

Stage-Gated Mobilisation

Gate	Indicative timing	What must be true before passing
G0 — Public mandate	Month 0–3	Named sponsor, problem/service outcome, initial funding, corridor protection and accountable officer
G1 — Entity and controls	Month 2–6	Legal form, board/committees, delegations, bank/audit/procurement controls, design/data rights and initial executives
G2 — Development baseline	Month 4–12	Survey plan, stakeholder/land strategy, concept of operations, system requirements, cost/risk baseline and safety/assurance plan
G3 — Procure and prototype	Month 9–18	Competitively packaged work, supplier qualification route, factory/site plans, training partnerships and accepted first-article plans
G4 — First articles and enabling works	Month 15–30	Tooling accepted, first vehicle/civil products tested, site-specific designs checked, NCR learning incorporated and operating organisation recruiting
G5 — Construction and operational readiness	Programme-derived	Released workfronts, progressive handover, maintainability/spares data, trained rosters, trial timetable and emergency exercises
G6 — Trial and revenue service	Authority-derived	Accepted safety case, independent assessment, insurance, regulator authorisation, asset acceptance and owner-board decision
G7 — Stable operation and replication	First 12–24 service months	Reliability/maintenance evidence, defects closed, audited benefits/local value, renewals reserve and controlled lessons fed into the reusable baseline

Timing overlaps intentionally; passing a later gate never waives an earlier open condition. Build schedule and headcount from the selected corridor's resource-loaded project twin after survey and procurement strategy replace the planning assumptions. The reference organisation is editable: responsibilities may be combined where compatible, while required independence remains explicit. A workshop adopting the non-safety tools does not have to create this full organisation.

The machine-readable mobilisation status turns this sequence into 18 dependency-checked work packages over the default month 0–60 planning horizon and eleven management-system records; every role, system, package and gate is open in the template.

First 90 Days

1. Name the owner accountable officer and an interim chief engineer; record all statutory duties and decisions that cannot be delegated.
2. Choose the entity model, approve reserved matters, form audit/risk and safety/assurance committees, and publish conflicts/procurement rules.
3. Secure the OpenSourceRail technical baseline, CDE, asset-ID/configuration rules, repository mirror, backups and data/IP licence in the owner's control.
4. Recruit the programme, commercial, finance, operations, maintenance, safety, quality, survey/land and digital leads; appoint independent assurance through a conflict-checked route.
5. Freeze a 12-month development budget and cash-control process, including external-currency exposure and explicit contingency authority.
6. Commission corridor survey/ground/utility work, stakeholder engagement, concept of operations and the local legal/regulatory route.
7. Turn the generated mechanical, civil, station and city packages into a package responsibility matrix; assign an owner and closure date to every open gate.
8. Agree apprenticeship/university/technical-college partners and pair local staff with every external specialist from their first deliverable.

Twelve-Month Definition Of Done

By month 12 the sponsor should be able to show, without relying on a vendor slide deck:

- the legal owner/operator path and named accountable executives;
- an approved concept of operations, requirements baseline and route evidence plan;

- a cost, funding, cashflow, risk and external-capital envelope with sensitivities;
- a procurement/package plan that preserves competition and owner-held design data;
- a safety/assurance strategy agreed in principle with the authorising route;
- factory, depot, OCC, civil production and construction mobilisation plans;
- a competence and workforce pipeline tied to actual roles and shifts;
- an immutable open-gate register linking every claim to evidence and an owner;
- a board decision either to proceed to first articles, re-scope, pause or stop.

Repository Handoffs

- Machine-readable mobilisation status — thirteen accountable roles, three independent parties, eleven management systems and gates G0–G7, deliberately open in the repository template.
- Mobilisation source template — copy to the controlled deployment workspace and bind decisions to evidence.
- Deployment role boundary — legal and evidence-carrying roles.
- Deployment model — city selection through operations.
- LM3 manufacturing controls — factory work instructions and stop rules.
- Civil construction controls — precast/site sequence and handback.
- Certification and authorisation — evidence gaps and release route.
- Operations rulebook and maintenance system — operating organisation inputs.
- Project twin — resource, procurement, cash and evidence baseline.

This plan establishes capability; it does not grant permission to construct or operate. Each deployment remains responsible for competent design, independent checking, supplier qualification, physical testing, insurance and statutory approval.

Validate a tailored deployment

Copy the mobilisation TOML, declare `deployment_scope`, and select the roles, management systems, evidence, work packages and dependencies needed for that scope. `entity_model.required_fields` declares applicable entity evidence. The validator checks references and evidence against that selection; it does not enforce the reference job titles, committee structure or document counts.

```
.venv/bin/python tools/automation/validate-owner-builder-operator-mobilisation.py --source /path/to/deployment.toml --
output-dir /path/to/deployment-status
```

Portfolio capital summary

Repository source: `docs/portfolio-summary.md`

> Original catalogue or earlier scope reference. The figures and policies below retain their original assumptions; they are not the latest Baghdad staffing, depot, procurement or funding basis. See the current Baghdad recalculation and current city summary. National/portfolio totals have not been repriced with that conditional Baghdad option.

This is the current aggregation of generated city models in the developing-world evidence scope, including one shared trainset factory per country. European comparison designs remain in the engineering catalogue but are excluded here. This is a planning screen, not a financing commitment, audited origin declaration, supplier quotation, or vendor bid.

Machine-readable values: `portfolio-summary.json`.

265-city / 44-country catalogue	Planning value	Annual construction draw across country programmes
External capital for imported components and machinery	$128.82B (20.9%)	$20.17B/year
Local capital for domestic value	$487.98B (79.1%)	$76.49B/year
of which planned local-currency bond issuance	$390.38B	—
local public equity / other domestic funding	$97.60B	—
Total national programmes	$616.80B	$96.65B/year

Foreign-turnkey sensitivity

The comparison holds the modelled railway scope and each country's financing terms constant. It changes only the foreign-turnkey price multiplier; all cases assume 90% of that price needs foreign currency or international capital:

```
turnkey price            = OpenSourceRail CAPEX × price multiplier
turnkey external capital = turnkey price × 90%
external capital avoided = turnkey external capital − OpenSourceRail imports
```

Case	Price multiplier	Turnkey total	Turnkey external capital	External capital avoided	Capital + external interest avoided
Low	1.5×	$925.19B	$832.68B	$703.85B (84.5%)	$1.59T
Default	2.0×	$1.23T	$1.11T	$981.41B (88.4%)	$2.21T
High	3.0×	$1.85T	$1.67T	$1.54T (92.3%)	$3.46T

The default row is the front-page illustration at portfolio scale: $981.41B (88.4%) less external capital and $1.23T less external interest. The comparator treats all foreign-turnkey external capital as debt; the OpenSourceRail case retains its generated grant/debt split. Debt on both sides uses the same country construction periods, rates and repayment tenors.

Individual city imported shares range from 17.8% to 22.7%. Replace the imported shares, cost multiplier and financing terms with audited supplier capability, normalized bids and signed lender terms before an investment decision.

Regenerate with `python3 tools/automation/generate-portfolio-summary.py` after city costs or country financing inputs change.

Historical Samawah planning reference

Repository source: `docs/reference/history/samawah-pre-core-planning.md`

Retained from revision `23878c15c5b20e42da587764865ae4101f9c56b3`. Numerical forecasts and unsurveyed brownfield observations below are historical assumptions. They do not describe or release the regenerated network. See the current reference RFC and current city example.

RFC 0003 — Samawah Context And Pilot Rationale

Historical design snapshot. Fleet, route, station and distributed-stabling figures below describe the earlier controlled revision. The current Samawah example adopts the regenerated central alignment and full-fleet line depots. These earlier figures and operating experiments cannot release the new layout.

Status: Draft Date: 2026-04-20 Depends on: docs/ARCHITECTURE.md, RFC 0001, RFC 0002, RFC 0028, RFC 0029

1. Summary

This RFC instantiates the unified deployment model for Samawah (As-Samawah), capital of Al-Muthanna Governorate, Iraq. Samawah is a worked Iraq instance, not a special OSR fork. The current source of truth is `cities/catalogue/west-asia/Iraq/Samawah/design.toml`, emitted by `osr-design` from the same OpenStreetMap, WorldPop, country, cost, finance, and template pipeline used by every generated city. This RFC is context and pilot rationale; it is not a second Samawah network plan. Every inter-station polyline is generated by the routing pipeline rather than sketched "as the crow flies". The current generated network is:

- Line 1 — 25.6 km / 14-station southwest-north radial.
- Line 2 — 12.8 km / 8-station southeast-northwest radial with the generated main depot-terminal.
- Line 3 — 12.0 km / 8-station west/east connector.

Total generated network: ~50 km of double-track light metro, 21 unique stations, and 108 trainsets: 97 peak, no dedicated depot-service rotation, 8 planned spares, and 3 cold reserves.

The previous draft of this RFC proposed a 2-line design (14 km radial + 16 km ring, 30 km total). That geometry placed Al-Muthanna University ~8 km south-east of its real location, placed "Samawah Railway Station" in the north-west when the real station is south-east, and routed the Halqa ring through palm groves + agricultural land that wasn't built-up. That design was retired in favour of the OSM-grounded network below.

This is a planning-grade concept design, not a surveyed alignment. It exists to make the unified architecture concrete for one city while keeping the deployment path parameterised for any city in the catalogue.

2. Why Samawah

Factor	Samawah	What it means for OSR
Population	~373,770 (As-Samawah Subdistrict, 2024 Iraq census; canonical figure in `lib/city-batches/world-sample.toml`)	Right-sized for light metro — too small for a major vendor to bother with, ideal for an open-source first deployment
Existing rail	Station on Baghdad–Basra mainline + rail yard with 300–800 stored wagons + adjacent rolling-stock workshop (see §2.1)	Rail culture, potential intercity interface, and a brownfield-recovery anchor that turns this from a reference scenario into a deployable pilot
Climate	Hot desert; generated model uses a conservative 5.0 peak-sun-hour planning proxy	Catenary-free + solar remains practical while leaving room for dust, heat, and seasonal derates
Grid reliability	Limited; frequent outages	Storage autonomy has real operational value, not just a planning formality
Existing electrification	None on Iraqi mainlines	No legacy catenary, no vendor-locked OCS — greenfield on D7
Development context	Al-Muthanna is among Iraq's lower-income governorates	Capital sensitivity is high; the ~66% capex reduction in RFC 0002 is decisive, not marginal
Institutional anchors	Al-Muthanna University, healthcare, government, market, and neighbourhood centres	Major daily demand generators that the generated radial lines use as anchor rewards
Workforce	Engineering graduates from Al-Muthanna University	Potential domestic design/build partnership — the core promise of OSR is "workers who design and build the systems in the country"

The combination is close to optimal: genuine need, favorable physics, no legacy to migrate, and domestic technical capacity to engage. If OSR cannot make sense here, it likely cannot make sense anywhere.

2.1 Physical-asset anchor — Samawah is brownfield, not greenfield

Satellite imagery review (2026-04-26) of the rail yards immediately adjacent to Samawah Train Station ( )
identifies two stockpiles of dormant rolling stock:

- Northern yard (along Samawah Train Station): ~4–6 parallel sidings packed end-to-end. Estimated ~120–300 wagons.
- Southern yard (diagonal NE–SW sidings): ~5–8 parallel sidings with similar density, plus a large workshop / shed building at the southern end. Estimated ~150–480 wagons.
- Total estimate: ~300–800 wagons. Predominantly standard 4-axle freight stock (covered wagons, hoppers, flatbeds) — passenger coaches a small fraction.

The southern yard's workshop building was the target of a 2011 Iranian Waxon Park rehabilitation deal (per Iraqi Business News historical reporting). Outcome of that deal is unclear and is part of the RFC 0027 Phase 1 assessment.

National context (Iraqi Republic Railways / IRR /):

- National fleet: 10,326 freight wagons, 255 passenger coaches
- Gauge: 1 435 mm standard nationwide fully compatible with RFC 0009
- Couplers: SA-3 (Soviet automatic) + screw/buffer combo incompatible with the Scharfenberg Type 10 spec'd by RFC 0008 §3.1, but the recovered drawgear is resaleable to legacy operators (RFC 0027 §5.3)

What this means for a Samawah pilot: the pilot is no longer a hypothetical greenfield deployment. It is the systematic conversion of an existing rail-yard + workshop complex into the first OSR production site, anchored on:

1. A dormant fleet with recoverable wheelsets, axleboxes, brake gear, and structural steel worth $8–15 M in component value against $1–3 M in recovery operations cost (per RFC 0027 §3).
2. An existing workshop building with foundations, pit tracks, and crane gantries that would otherwise require €5–8 M to build greenfield (RFC 0014).
3. An active rail corridor and yard/workshop complex through the city. The current generated city model does not yet stitch this into a shared metro/intercity transfer node; that is now an explicit alignment-refinement task rather than an assumed feature.
4. Standard gauge matching OSR spec — no gauge conversion required.
5. A national rail authority (IRR) under reconstruction pressure — the political context favours engagement.

Conversion strategy is in RFC 0027 — Brownfield Pilot: Existing-Asset Recovery & Workshop Integration, which carries the per-phase doctrine, recovery yield estimates, workshop conversion matrix, and risk register. RFC 0027 is intended to apply to every brownfield deployment, not just Samawah; the candidate-city list is in §9 there.

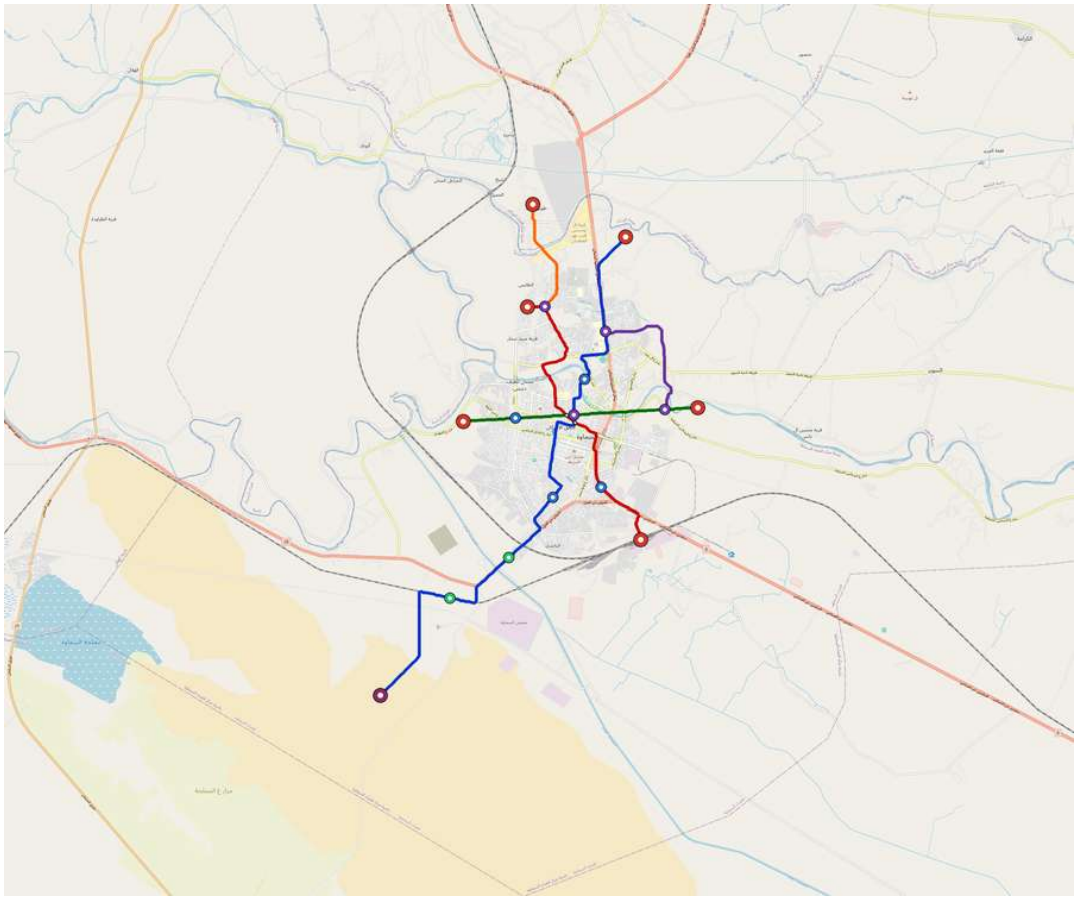
Status of the brownfield assessment at time of writing:

- Phase 1 site visit + fleet census — not yet started; gating item
- Workshop tooling audit — pending Phase 1
- NDT campaign on a sample wheelset population — pending Phase 1
- MoU draft with IRR / Iraqi Ministry of Transport — pending diplomatic / technical-community introduction

The single highest-value next step on the OSR programme is getting a real Phase 1 census on the ground. Without it, the Samawah deployment costs in §1 of RFC 0021 and the depot CAPEX in RFC 0014 are greenfield estimates; with it, the per-consist mechanical CAPEX drops by ~$30–60 k and the depot CAPEX drops by a factor of 5–10 per RFC 0027.

3. Geography and Indicative Alignment

Samawah sits on the Euphrates in southern Iraq. The city stretches roughly east–west along the river with residential suburbs fanning north and south. The mainline railway enters from the northwest, the city centre sits on the river bend, and the major newer developments (hospital, university campus, expanding residential districts) are south and east of the historic core.



Samawah network — 3 auto-planned lines on OpenStreetMap

Three generated lines on OSM tiles. Station markers are coloured by archetype (red terminal, purple depot-terminal, blue major, orange interchange, white standard). Every polyline comes from the `osr-design` routing pipeline. Regenerate with `tools/automation/regenerate-city.sh samawah`; see §3.5 Realism Notes for what surveying would still change.

> Important caveat. The alignment below is indicative. Precise routing, station siting, and ROW acquisition are out of scope for this RFC — those depend on surveying, stakeholder consultation, and local planning processes that only Samawah's own planners and Al-Muthanna Governorate can lead. What this RFC provides is a credible skeleton that demonstrates how an OpenSourceRail network would be organized for a city of Samawah's size and form.

3.1 Current Generated Lines

The current source-of-truth line table is the generated city README and `design.toml`. RFC 0003 no longer carries hand-maintained station tables because those went stale as the routing pipeline improved.

Line	Length	Stations	Trainsets	Termini band	Planning role
line-1	25.6 km	9	53	N Mid SW Outer	Long southwest-north radial through the strongest generated anchor chain
line-2	12.8 km	6	28	SE Mid N Mid	Southeast radial, including the generated depot-terminal / main maintenance anchor
line-3	12.0 km	6	27	E Mid W Inner	Shorter connector filling the west/east catchment
Total	50.4 km	21 unique	108		

3.2 Transfers And Interchange Caveat

The generated README reports 1 interchange-class station and 100% multi-line transfer reachability, meaning every line-pair shares at least one generated station ID in the current graph. In plain terms: the current output now has a single station-complex transfer pattern, while the exact surveyed platform arrangement still needs local alignment review.

That is a planning gap, not a Samawah-specific design rule. The next alignment iteration should decide whether the three radials should meet at shared platforms, adjacent paid-area transfers, or remain separate services linked by local bus/walk transfers.

3.3 Revenue Model Contribution

The generated capacity-led revenue bracket is system-wide: 73.3–117.3 M annual paid trips, based on 50–80% use of practical service capacity under the 3-minute peak / 6-minute off-peak timetable. The RFC should use that system value rather than maintaining separate line-level ridership sketches.

3.4 System totals

Metric	Value
Route-km (double track)	~50.4 km
Stations (unique)	21
Lines	3 (auto-planned by `osr-design`)
Interchange-class stations	1 shared multi-line interchange complex
Multi-line transfer reachability	100% of line-pairs sharing a station ID in the current generated graph
Fleet (revenue)	97 × 3-car trainsets
Fleet (dedicated depot-service rotation)	0 — off-peak uses surplus peak fleet
Fleet (spare + cold-reserve)	11 × 3-car trainsets
Average inter-station spacing	Generated from `SpacingConfig`: 1.6 km central, 3 km urban, and up to 7 km on suburban approaches and the lowest-demand outer fringe
Demand surface	Anchor (POI) + WorldPop residential population blend per `build_demand_surface` — lines reach population centres without mapped POIs
Service hours	05:30 – 02:00, ~20.5 hours
Target annual paid trips (steady-state)	73.3 M – 117.3 M passenger-trips, equal to 50–80% use of practical service capacity

> The table above is derived from `design.toml`; regenerate with `python -m osr_scenario.stats --format markdown`. A drift test (`design/city-generation/tests/test_rfc_drift.py`) fails CI if this table ever contradicts the design file. The full regeneration pipeline is one command:
>
> ```
> tools/automation/regenerate-city.sh samawah
> ```
>
> which emits the sim scenario (`cities/catalogue/west-asia/Iraq/Samawah/samawah.toml`), the network map PNG, runs the drift + round-trip tests, and prints the summary stats above.

3.5 Realism Notes

Honest assessment of the current generated 3-line design against real Samawah OSM data:

What's real. Station coordinates in `cities/catalogue/west-asia/Iraq/Samawah/design.toml` are generated from the same OSM/WorldPop/country catalogue pipeline as every other city model. Named stations correspond to mapped anchors where the source has one; generated `line-*` IDs mark algorithmic stations placed against the cost/demand surface. The 373,770 population figure, bounding box, Euphrates position, and Baghdad–Basra mainline context are all catalogue/OSM-consistent.

What's not real yet. Things a surveyor would still catch:

- Inter-station polylines follow the cheapest-road shortest path through the OSM network (arterials weighted 0.6–0.8, residentials 1.8). That's planning-grade, not engineering-grade — curve radii, grades, and the ROW width needed for double track aren't checked. A real alignment survey will adjust station coordinates by tens of metres to snap to stakeable centrelines + may replace some segments with elevated viaduct where the road is too narrow.
- Transfer stitching still needs local design. The current output has one interchange-class station and 100% line-pair station sharing. A local planning pass must decide whether that becomes a true shared-platform transfer, adjacent paid-area transfer, or a separately phased stop.
- Some stations are algorithmic placements rather than named POIs. These are legitimate planning placeholders, but a survey/local workshop should replace weak `line-*` IDs with buildable sites.
- Railway-station / yard integration is not yet in the generated topology. The brownfield yard remains a strong pilot anchor, but it should be explicitly added or connected in the next design iteration rather than assumed from older RFC text.
- 50 km / 21 stations / 108 trainsets is a full multi-phase build-out target for a 373,770-population city. A first physical validation step should prove the unified OSR hardware, rolling stock, safety case, and operations stack on a smaller segment before committing the full network.

How to take this to the next fidelity level:

1. Re-run the `design/city-generation` routing + rendering pipeline with a one-command refresh:

   ```
   tools/automation/regenerate-city.sh samawah
   ```

which fetches Samawah OSM (cached), routes every line on the road graph, emits `corridor.geojson` + `cities/catalogue/west-asia/Iraq/Samawah/samawah.toml`

- the map PNGs, and runs the scenario + RFC drift tests.

2 Feed the stationed alignment + `corridor.geojson` into `osr-alignment-export` to produce LandXML + railML for import into Bentley OpenRail or Civil 3D.

3 Cross-reference against Iraqi Ministry of Transport + Al- Muthanna Governorate road + cadastre datasets.

4 Review with Al-Muthanna University transportation faculty and the Governorate planning office.

4. Service Plan

4.1 Headways

Period	line-1	line-2	line-3
AM peak (07:00–09:00)	3 min	3 min	3 min
Midday (09:00–15:00)	6 min	6 min	6 min
PM peak (15:00–17:00)	3 min	3 min	3 min
Evening (17:00–23:30)	6 min	6 min	6 min
Late evening (23:30–02:00)	12 min	12 min	12 min

The generated scenario uses the same service window and headway profile on all three lines. Line-specific service tuning should come from a proper origin-destination and timetable study, not from the older hand-authored "central interchange / railway branch" narrative.

4.2 Anchor-driven demand

Unusually for light metro, ridership here is dominated by a few large institutional anchors rather than by distributed demand. This has design consequences:

- Education anchors: sharp peaks 07:30 and 15:00 during term; low on Fridays and summer. Platforms near university/college anchors need surge-capacity checks.
- Hospitals and clinics: steady demand through the day, with a visiting-hours peak in mid-afternoon. Longer service span may be justified for hospital access.
- Railway yard / intercity station: bunched demand and brownfield industrial value remain important, but the current generated topology has not yet encoded a metro/intercity transfer station.
- Souq and centre anchors: evenings and weekends, counter-cyclical to the education peak, which helps fleet utilization.

4.3 Rolling stock

Reference platform per ARCHITECTURE §4 D5:

- 3-car, 49.5 m consist with low-floor centre door zones (light metro class)
- LFP onboard battery, 540 kWh usable / 675 kWh gross per trainset (180 / 225 kWh per car), sized for roughly one route length with station opportunity charging; the current 25.6 km longest line uses ~307 kWh one-way at 12 kWh/train-km, leaving ~233 kWh of usable margin above the protected reserve
- Six heavy-vehicle PMSM/controller channels with Rust supervisory control
- TSN Ethernet trainbus
- HVAC sized for 50 °C ambient, which is load-bearing in Samawah summers (see §5.4)

4.4 Construction QA and maintenance basis

Samawah uses the system-wide construction QA and maintenance schedules, not a one-off pilot checklist:

- RFC 0028 gates locally built trainsets through carbody, bogie, traction/brake/battery, passenger systems, onboard control, first-article, and trial-running acceptance. It also gates infrastructure through survey/geotech, earthworks, trackform, structures, stations/depots/plant, energy, wayside/comms, and integrated opening evidence.
- RFC 0029 creates the baseline asset schedule for 108 trainsets, 21 stations, 50.4 route-km, energy sites, W-SBCs, switches, sensors, depots, and production-plant tools. Condition-based escalation overrides calendar intervals when telemetry or inspections raise amber/red defects.

5. Energy: Generated City Instance

Applying RFC 0002 to this generated city instance, with city and climate parameters supplied through the shared templates and generated design outputs.

5.1 Demand

Component	kWh/day
Scheduled traction service (23,436 train-km/day × 3 cars × 4 kWh/car-km)	≈ 281,000
Depot/deadhead allowance (+8%)	≈ 22,000
Generated traction-energy model	≈ 304,000 kWh/day = 304 MWh/day

This replaces the earlier 85-110 MWh/day sketch. The 3-minute peak / 6-minute off-peak plan changes the duty cycle materially: Samawah now needs roughly 110.9 GWh/year of traction energy before station/building auxiliaries. The current generated timetable schedules 1,395 one-way train journeys per day across the three lines, equal to 23,436 train-km/day before the depot/deadhead factor.

5.2 PV generation

The generated city model uses a 6.0 peak-sun-hour planning proxy for the committed README figures. It no longer relies on the older hand sketch of 30 km of ROW-mounted panels.

Onboard roof PV is now also present in the simulator scenario: each 3-car trainset carries 15.12 kW nameplate at 65% usable, with an optional 0.9 kW low-pressure air-cleaner load that recovers 75% of dust-driven roof-PV loss in the simulation model.

The generated infrastructure model now closes that gap with a dedicated utility-scale solar plant or contracted offsite solar PPA asset:

Component	Generated Samawah planning value
On-site station/depot PV	13.7 MW / 30.0 GWh/yr
Traction demand	110.9 GWh/yr
Shortfall before supplemental plant	81.0 GWh/yr
Dedicated solar plant, incl. 115% coverage margin	42.5 MW / 93.0 GWh/yr
Residual grid/PPA import in normal-year proxy	0.0 GWh/yr

Samawah therefore remains catenary-free and net-positive in the planning case, with the grid/PPA contract retained as backup import and export settlement rather than as the primary energy supply. The headroom case still matters for:

1. Dust-storm resilience. Peak soiling events in Muthanna can cut yield 40–60% for several days.
2. Grid export opportunity. Iraq's grid is energy-constrained; feed-in under a state-utility PPA could be a material revenue line once generation exceeds service demand.

5.3 Storage

Given grid unreliability, Samawah's generated station/depot storage is substantial but should still be stress-tested against dusty, overcast, and grid-outage sequences.

- Station/depot stationary storage: 54.5 MWh LFP distributed across 30 energy sites.
- Main depot-class site: 40 MWh.
- Standard/major/interchange/terminal sites: 500 kWh each by template tier.

Onboard: 108 trainsets × 675 kWh = 72.9 MWh gross fleet battery capacity (58.3 MWh usable), of which 97 trainsets are peak-revenue fleet and 11 are planned spares/cold reserve. This keeps the current longest-line run above dispatch reserve, but it does not make the line independent of station charging for repeated service; terminal and intermediate charging remain part of the operating assumption.

Powered stations and the single main-heavy depot are represented as charging locations. During normal turnback the train uses the shared 500 kW passenger-platform cabinet; during a service hold it can use a 150 kW low-C service charger, topping toward 95% SoC where the timetable permits. Healthy trains stable overnight at powered passenger stations; red defects, failed isolation, failed security, or heavy-maintenance-due sets return to the main-heavy depot.

One powered service point on each line is the designated turnaround-service point. During the 3-minute morning and afternoon peaks, trains make the normal quick turnaround and are allowed greater battery depletion while the 20% reserve gate stays active. In the 6- and 12-minute off-peak windows they receive a 12-minute concurrent clean, safety-release inspection, fault-log download, and recharge. The peak fleet's off-peak surplus covers this work, so it requires zero additional trainsets and zero incremental fleet CAPEX. The generated all-in planning CAPEX is $461 M with all 108 sets included.

5.4 Thermal considerations

Samawah summers routinely exceed 45 °C; record highs approach 55 °C. This has three design consequences:

1. HVAC energy dominates. Per-car traction + HVAC at 50 °C ambient can reach 6 kWh/car-km. The 4 kWh/car-km reference figure is already rolled up with an average HVAC; the RFC's +25% summer uplift reflects realistic worst-case.
2. Battery conditioning. The selected LFP cells prefer operation below 45 °C. Trackside battery enclosures need insulation + passive ventilation minimum; active cooling for high-C discharge periods is justified for the main depot. Onboard packs sit under the seats in thermally managed, fire-resistant compartments with outward venting and localized water mist.
3. PV temperature derating. Panels in Samawah can reach 70+ °C module temp on a summer afternoon. Bifacial + vertical mounting runs cooler than horizontal ballasted arrays and is preferred on the ROW for this reason as much as the area one.

5.5 Capex summary (generated city instance)

Applying the current cost templates to the generated 50.4 km city instance:

Item	Cost (USD)
Dedicated 42.5 MW solar plant + interconnection	$34 M
Station/depot charging microgrids	$3.3 M
Station/depot PV + stationary storage	included in station/depot archetype CAPEX
Onboard batteries	included in rolling-stock CAPEX
Visible energy adders in generated CAPEX	≈ $37 M

The dedicated plant adds about $34 M to infrastructure CAPEX and about $0.51 M/year of plant O&M, but removes the earlier ~$10 M/year recurring grid/PPA energy purchase in the Samawah README. For comparison, the older catenary sketch used roughly $6 M/km for 25 kV AC OCS plus traction substations; scaling that to 50.4 km would be on the order of $300 M before local feeder works. The OSR energy approach remains materially below the catenary case while also avoiding a large recurring import bill.

6. Mapping to OpenSourceRail Architecture

This is where the city instance connects back to the shared domain designs:

D1. Operations & Dispatch

- One Operations Control Centre at the generated main depot-terminal, with a backup dispatch workstation at a hardened station or municipal facility selected during local design.
- Dispatcher interface: web-based; full observability via the OpenTelemetry + Prometheus stack; no proprietary HMI.
- Integration with intercity rail remains a planning requirement, but the current generated network has not yet encoded a shared metro/intercity transfer station.

D2. Train Control

- Three consensus regions: one per generated line, with station W-Nodes, turnback control, and depot/yard throat control derived from `design.toml`.
- Region boundaries and transfer handoffs are a future alignment task because the current generated graph has no shared station IDs between lines.
- Interlockings cover line turnbacks plus the main depot and layup throats emitted by the generator.
- No track circuits: position is sensor-fused per RFC 0001 §5.1, with beacon-fix sites at every switch and every platform.

D3. Communications

- Public 5G SA for primary trainwayside link — Zain Iraq, Asiacell, or Korek presence in Samawah; one or more operators will likely carry the project with a network-slicing arrangement.
- LoRa mesh gateways at every W-Node site as the safety-telemetry fallback, giving citywide coverage independent of carrier uptime.
- TSN Ethernet backbone along the ROW (single fiber pair in a ruggedized conduit, ring topology per line).

D4. Passenger Services

- Account-based fare per ARCHITECTURE §4 D4:
  - Iraqi mobile money rails (zainCash, AsiaHawala) as primary.
  - QR tickets via a local web-app (no native app required — important for feature-phone access).
  - Optional reloadable NFC card for those who prefer it, but not the default channel.
- Trilingual passenger info (Arabic, Kurdish, English) on displays and announcements; Arabic is primary for Samawah but the project infrastructure should generalize.
- Station screens: SBC + standard displays per the D4 design, with prayer-time integration for culturally appropriate PA silence windows.

D5. Rolling Stock

- 97 peak-revenue + 11 spare/cold-reserve 3-car trainsets, 324 cars total; depot service uses off-peak peak-fleet surplus.
- Reference T-ECU hardware across all non-traction-power functions.
- Onboard battery at 540 kWh usable / 675 kWh gross per trainset, LFP.
- Roof PV at 15.12 kW/trainset with low-pressure air-cleaner support in dusty service.
- HVAC designed for 50 °C ambient, verified in summer commissioning.

D6. Infrastructure

- Switch machines: turnbacks and depot/yard throats generated from the current design file; final quantities wait for surveyed track layout.
- Level crossings and street-running conflict points are not final until civil alignment export; every retained conflict point must be instrumented into the consensus log the same way stations are.
- Track geometry CBM: every revenue service train carries the sensor package from D6.

D7. Energy

- See §5. Samawah is an archetypal D7 deployment: sunny, grid-constrained, catenary-free.

D8. Depot & Maintenance

- Generated depot set: 1 main-heavy depot-terminal. The main depot carries the heavy workshop, CBM, training wing, and 40 MWh storage role; healthy trainsets stable overnight at powered passenger stations rather than at a depot parking yard.

7. What This Means for `osr-sim`

Samawah is the primary reference scenario for `osr-sim`. The current simulator already imports the generated 3-line scenario, runs the 108-trainset timetable, applies station/depot charging and roof PV, enforces the clean/inspection service slot, and can emit CSV/JSON traces and README screenshots. Development priorities now move from "stand up the first topology" to fidelity:

1. Transfer and station stitching. Represent the selected local transfer design once planners decide how the three radials should connect.
2. Demand model. Origin-destination matrix calibrated to the generated anchor classes with time-of-day, Friday, school-term, and hospital/market curves.
3. Annual climate model. Seasonal PSH, ambient temperature, dust, and HVAC load, not just a single representative day.
4. Fleet and charging validation. Stress the 97+11 trainset fleet, 540 kWh usable / 675 kWh nameplate packs, station chargers, roof PV, and air-cleaner parasitic load under peak, degraded, and overnight service.
5. Fault injection. W-Node crashes, radio outages, grid outages, dust events, and sensor faults. Tests should show graceful degraded operation rather than only nominal service.

The existence of a concrete reference case is important for discipline: every feature added to `osr-sim` should demonstrably improve fidelity of the Samawah simulation, not just be abstractly useful.

8. Stakeholders and Partnerships

This is early and speculative, but the architecture only makes sense if it's eventually deployable. Plausible stakeholders to engage, ordered by likely path-opening value:

1. Al-Muthanna University, College of Engineering — natural partner for design-phase engineering work, student internships, and a credible domestic technical voice.
2. Al-Muthanna Governorate — transit planning authority and likely project sponsor.
3. Iraqi Republic Railways (IRR) — operator of the existing mainline; intercity interchange is via them.
4. Iraqi Ministry of Transport — for regulatory approval of novel signaling architecture.
5. GIZ / German development cooperation — given the new German Hospital anchor, there may be a natural coordination path with German-Iraqi development relations.
6. Multilateral development banks (World Bank, IsDB, AIIB) — likely financing routes for a novel urban-rail pilot.
7. Domestic EPC and electronics industry — the path to workforce domestication runs through companies that would actually fabricate PCBs, build switchgear, and integrate stations.

No engagement is being proposed here. This list exists so that when code is running and the simulator is working, there is a clear trajectory toward a conversation, not a blank map.

9. Open Questions

1. Urban planning compatibility. The indicative alignment in §3 must be compared against Samawah's urban plan before it can be taken seriously as a routing proposal. This is a desk exercise that a motivated student at Al-Muthanna University could do productively.
2. Demand reality check. The generated finance model now uses the system's practical capacity as the revenue basis: 73.3–117.3 M annual paid trips at 50–80% capacity use. A proper local demand study would refine that capacity-use bracket and its line-by-line distribution. Fleet capacity now has margin; the generated energy plan closes the on-site PV gap with a dedicated solar plant, but its land/PPA counterpart, interconnection, outage reserve, and export-settlement terms need local feasibility work.
3. Transfer and intercity stitching. Should Samawah get shared metro platforms, adjacent paid-area transfers, or separated radial stops? How should the IRR station and brownfield yard connect to the first OSR segment?
4. Civil constraints. Which generated segments need elevated guideway, road reallocation, bridge work, or local rerouting once survey and cadastre data replace OSM planning geometry?
5. Construction phasing. Full network in one go is unlikely. Plausible phasing should start with a unified OSR pilot segment that proves rolling stock, station charging, roof PV/cleaner behavior, onboard train control, safety case, and maintenance workflow before expanding to the full 58.4 km network.
6. Land acquisition. ROW is the single biggest planning-phase risk and is entirely local-authority territory; OSR technology cannot help here, only not make it worse (the catenary-free design needs no overhead easements, which is a small help).
7. Women-only car policy. Common in the region; implementable as a carriage-designation display change, but the social design of the service should be discussed with local stakeholders rather than assumed.
8. Prayer-time service patterns. Friday midday service should be adjusted for mosque access; Ramadan schedules differ from the standard day. These are timetable questions, not infrastructure.

10. Next Steps

With this RFC in place:

1. Replace planning OSR-ALN with surveyed controls. The current three-line package is reproducibly exported and validated, but its GIS horizontal trace, zero-datum vertical placeholder, and absent cant must be replaced by signed survey/CAD deliverables before construction release.
2. Resolve transfer/intercity topology. Decide whether the generated interchange-class stations become true transfers and how the IRR yard / station becomes part of the pilot story.
3. Run annual simulation cases. Extend the current simulator beyond short traces into annual PSH/HVAC/dust/grid-outage scenarios and use those runs to tune storage and charger sizing.

4 Publish an accessible summary (Arabic + English) aimed at a non-technical audience at Al-Muthanna University and in governorate outreach, so that the project is visible to potential domestic contributors from the beginning rather than being announced after the fact.

This RFC treats Samawah as a design target, not a prediction. Whether OpenSourceRail actually deploys here depends on decisions made by the people and institutions of Samawah and Iraq. What we commit to is: the reference design will be kept honest, the simulator will match the real geography, and the work will be done in public so that local engineers can contribute from day one.

OpenSourceRail v0.3.1

Repository source: `docs/release-v0.3.1.md`

v0.3.1 is the corrective patch for clean-checkout failures discovered after v0.3.0 was published. It does not change the product or certification boundary.

Corrections

- Python and browser jobs install one pinned IFC runtime, including the schedule and EXPRESS-validation dependencies used at runtime.
- All 266 acceptance reports link only to tracked compact asset registers and manifests; reproducible high-volume local outputs are named without broken GitHub links.
- CI checks that the report links and IFC dependency source cannot drift.
- Kani CI proves explicit fast ATP and interlocking properties. Larger topology-backed proofs remain controlled open evidence instead of repeatedly timing out while appearing to be a release gate.

The complete Rust, Python/generated, browser and bounded Kani workflows passed before tagging. Physical qualification, supplier freeze, independent safety assessment and deployment approval remain external gates.

OpenSourceRail v0.3.0

Repository source: `docs/release-v0.3.md`

v0.3 makes the project boundary more credible while strengthening the most adoptable product: deterministic city design, delivery planning and owner/ operator evidence management.

Release highlights

- City Studio, civil IFC generation and CI use one portable Python-runtime resolver; browser acceptance provisions the pinned IFC stack it exercises.
- The root front door presents three products with distinct maturity and keeps the first adoption path non-safety.
- Headline economics use appropriately rounded domestic-value figures; the exact foreign-turnkey comparison remains a reproducible sensitivity model.
- LM3 battery terminology now defines externally accessed, saloon-isolated side enclosures beneath the seat zone with outward-only drainage and venting.
- The cost and station sources agree that a `halt` has no charger or energy site; powered station archetypes retain their charging allowance.
- EN 50716:2023, IEC 62278-1:2025, IEC 62267:2009 and IEC 62290-1:2025 are organised in one maintained standards baseline.
- Commodity Raspberry Pi/Radxa equipment remains the pilot/application path, while revenue-service safety channels now have a separate assessor-controlled industrial safety-MCU selection gate.
- Fast ATP expired-authority and interlocking validity-window arithmetic proofs run in CI. Topology-heavy ATP and interlocking proofs remain bounded formal evidence, not ordinary-CI claims, until their unwind limits are frozen and reviewed.
- A documented SMRaft-to-Rust abstraction map improves reviewability without misrepresenting it as a machine-checked refinement proof.

Deliberately not claimed

v0.3 is not a safety certificate, construction release, production drawing set, supplier freeze or revenue-service GoA 4 system. Physical train/civil proof, qualified safety hardware, combined deployment charging cases, structural and geotechnical validation, independent assessment and regulator acceptance remain explicit gates in the certification and discipline checklists.

Assessment disposition

Recommendation	v0.3 decision
Restore integration health and share one Python runtime	Implemented and covered by Rust, Python and browser acceptance.
Separate commodity application hosts from safety channels	Implemented as an architecture and controller-selection gate; hardware is not frozen.
Refresh standards and formal evidence	Implemented baseline, abstraction map and release-selected ATP proof; full refinement and bounded ATP completion remain open.
Clarify battery, halt-charging and economic claims	Implemented in source generators, generated evidence and the public front door.
Test combined degraded charging operation	Existing deterministic cases retained; deployment timetable combinations remain an explicit gap.
Add more viaduct families	Rejected for v0.3: retain Pi25, Pi20 and the documented segmental fallback; resolve haul/lift choice and foundations from city logistics and geotechnical evidence.
Claim production-ready train/civil packages	Rejected until supplier freeze, structural/bench proof, first articles and deployment QA records exist.
Trace CAD through physical evidence and cost actuals	Accepted as the next vertical delivery method, not represented as completed evidence.

Compatibility

The project is pre-1.0. Generated packages should be regenerated from the v0.3 sources. Git revisions remain the review/history mechanism; deployment actuals remain controlled separately in Ops Core.

Use the setup and verification commands from the root README. A tag is created only after the release checklist is fully satisfied on a clean commit.

OpenSourceRail v0.4.0

Repository source: `docs/release-v0.4.md`

v0.4 recognizes ERPNext business execution and FUXA/OSR equipment supervision as part of the connected OpenSourceRail software platform. The Workbench now carries city and asset context across planning, engineering, procurement, manufacturing, construction, commissioning, operations, maintenance and renewal while each service retains its own permissions and authority.

Included baseline

- Reproducible ERPNext city projects, component imports and native project, procurement, stock, manufacturing, quality, finance, HR and maintenance views.
- Generated FUXA devices and displays backed by the OSR historian, with explicit quality and timestamp tags and no FUXA write-through to railway controls.
- Native Rust simulation bindings for station energy/SCADA, vehicle BMS, auxiliary power, HVAC and condition monitoring, points, level crossings and station-aggregate fare gates. Existing switch identities are reused; crossing positions remain absent until a city declares real assets.
- Nine production-plant views for cities selecting the LM3 family, generated directly from its validated method source, covering the 120-product union and 30 tooling families. Explicit cell and quality-hold fixtures route accountable ERP Issues; reviewed Item/BOM mappings require the exact Item/BOM pair at the engineering revision. Other families receive no implied LM3 production coverage, and QA release stays separate.
- Durable, ordered and idempotent condition-event delivery to ERP maintenance Issues, plus case-status reconciliation without automatic railway release.
- A reviewed, repeat-safe Issue-to-Asset-Repair transition with native part/stock consumption, technician assignment, expected/actual downtime, linked condition and serial evidence, permission-filtered feedback and no automatic handback.
- City-scoped Workbench navigation, trends, alarms, serial/batch history, lifecycle evidence and bounded simulation-only lighting requests.
- Revision-safe supervisory updates: removed equipment is retained as retired history, pending commands fail, telemetry and new commands are rejected, and a later reviewed package can reactivate the stable asset identity.
- A review-bound package-change workflow and Workbench surface that classify design, embedded, telemetry, alarm, command, ERP and display changes; trace installed serials and append-only evidence; reject stale reviews; isolate city baselines; and block unsafe installed-position or pending-command changes.
- Review-bound FUXA replacement with a package/device/view manifest, explicit add/change/remove sets, stale-review rejection and backups of both the previous project and the applied review.
- Occurrence-bound alarm acknowledgment, including stale-view rejection and a fresh acknowledgment requirement when a cleared condition activates again.
- A generated catalogue-readiness gate that compiles all 266 ERP/component profiles and real-asset supervision packages, preserves per-city hashes and distinguishes on-demand task payloads from operator/commissioning inputs.
- A disposable native-stack CI gate covering ERP transactions and feedback, FUXA/gateway/controller behaviour, Workbench navigation and service outage/recovery, condition-to-repair execution, retained removed-serial history and independent replacement re-release, alongside version-pinned Kani proofs.

Review corrections and release verification

- Default FUXA displays follow the configured city/site, while explicit equipment selection takes precedence. Adding a factory cannot change a city's default.
- Retired equipment retains history without stopping telemetry for active assets.
- Safety-case traceability, current successful result records and independent acceptance are separate. Result checks bind named harnesses, tools, bounds, input/report hashes and acceptance fingerprints; they do not execute proofs or authenticate reviewers. Missing controlled results remain open.

- GSN counts are generated from TOML. Operations rules use accepted charging duty and morning readiness, new business transactions belong to ERPNext, and formal model refinement remains an assurance objective rather than a completed proof.
- Catalogue readiness is independent of untracked local generated bundles.

The software release gates are the complete Rust/Python/browser CI workflow, the two pinned Kani properties, and the disposable ERPNext/FUXA/controller/ Workbench workflow, including service loss and recovery. The attached release evidence manifest records the exact commit and successful workflow runs; its checksums bind the published reader book, overview and evidence assets.

Only ATP rejection of expired authority and interlocking validity-window arithmetic are selected Kani release gates. Passing these does not close the remaining topology proofs, all 71 declared safety-case solution records, or independent safety acceptance. See result validation.

Upgrade and compatibility

Back up ERPNext, the integration database and FUXA project before upgrading. Rebuild the ERP application image and run site migration, rebuild the gateway, regenerate each city package, review the exact old/new package hashes, then apply and reimport the reviewed FUXA project. Restart the native simulator. Use deployment instructions for the commands.

Factory metadata now records its rolling-stock family. Earlier packages must be regenerated; incompatible factory positions become retired history through the normal reviewed package-change process. Installed-position removals remain blocked for independent disposition. Samawah's bounded simulation package has 19 positions; Mosul's has 10, with no LM3 factory methods. Historical records, serials, Issues and evidence must not be erased to make an upgrade pass.

Boundary

The local stack is an integration and simulation baseline. Supervisory points and crossing views are maintenance/status surfaces only: they expose no movement authority, point/barrier command, protection reset or interlocking bypass. Production TLS, organisation identity/MFA, commissioned physical MQTT/OPC UA/Modbus or device-bus adapters, supplier calibration, signed live actions, HIL, operator validation and independent railway acceptance remain deployment work. ERP case closure, alarm clearance and local lifecycle evidence do not grant a railway operational release. Likewise, simulated factory progress and quality holds are workflow demonstrations: performed travelers, process observations, NCR/rework disposition, Quality Inspections and manufacturing acceptance require accountable real records.

The repository remains pre-1.0. A v0.4.0 software version is not a safety certificate, construction release, production drawing set or deployment approval.

Release Policy And Checklist

Repository source: `docs/releases.md`

OpenSourceRail is pre-1.0 and uses semantic `vMAJOR.MINOR.PATCH` tags.

- `Cargo.toml` carries the full software version.
- `VERSION` carries the release line (`MAJOR.MINOR`) used by generators.
- A tag is immutable. A published release pack becomes a historical record.
- Generated assets must be produced from the tag, checksummed and attached to that release.
- Safety approval, hardware release and deployment acceptance are separate controlled baselines; a software tag does not imply them.

v0.4.0 software baseline

The v0.4.0 notes define the ERPNext, Frappe HR, FUXA, OSR integration gateway and shared Workbench release scope, review corrections, upgrade procedure and remaining limits. The GitHub release is the authoritative publication record and asset catalogue.

The release evidence manifest identifies the exact commit, source tree, successful CI runs and asset hashes. The two selected Kani properties are bounded software checks; they are not independent safety acceptance. Wider modelling, supplier, manufacturing and physical commissioning work remains in the roadmap and controlled release registers.

Release procedure

1. Review the intended changes, public API/schema compatibility, upgrade path, open audit work and documentation. Keep external acceptance gates explicit.
2. Align `Cargo.toml`, `Cargo.lock`, `VERSION`, package metadata, changelog and release notes. Regenerate the reader book, overview, catalogues and evidence summaries with `./osr build`, `./osr readiness` and the safety-case summary.
3. Run `./osr test`, the declared pinned Kani harnesses and native integrated acceptance, including the complete example-city lifecycle, restart, expansion, business and unmocked disposition checks. Review generated changes, commit the complete baseline and push it for CI.
4. Require successful `ci`, `kani`, `integrated-stack` and `example-city` workflows on that exact commit. The integrated workflow must reach its final outage/recovery test. The city run must start on a clean candidate commit and publish its hashed `city-evidence.json`; release packaging downloads and validates that run's reports.
5. With a clean tracked working tree, create the immutable annotated tag at the verified commit. Never move an existing release tag to repair a failure.
6. Build the publication assets from that tag:

   ```
   python3 tools/automation/release-evidence.py --tag v0.4.0 \
     --output build/releases/v0.4.0/publish
   ```

   The builder checks all four workflows, rebuilds the reader PDF with links pinned to the tag, embeds the overview images, exports the readiness report and release notes, and writes the evidence manifest and `SHA256SUMS`. It also downloads all eight `safety-execution-*` artifacts from the successful Kani run. Every declared harness must match the commit, runner, dependency hashes, result and log; missing, duplicate, failed and changed evidence is rejected. `OpenSourceRail-Kani-Evidence-<tag>.zip` retains all original manifests, TOML results and logs under their `build/assurance/<package>/` paths. The release manifest enumerates all declared harnesses; independent acceptance remains separate.
7. Publish the tag and versioned assets. Verify the uploaded assets against the local SHA-256 checksums and preserve the exact commit and workflow evidence.

Safety, hardware, rolling-stock, civil and deployment release remain independent of this software-publication procedure. A failed or missing software check blocks publication; an explicitly open physical gate must never be marked complete by publishing software.

Published v0.3.1 record

v0.3.1 was tagged and published on 2026-08-30. Its scope and claim boundaries are recorded in the patch notes and v0.3 notes. The tag records that completed software baseline.

Repository Artifact Policy

Repository source: `docs/repository-artifact-policy.md`

Git contains the useful source and review artifacts needed to understand, inspect, build, test and change OpenSourceRail without specialist regeneration. That includes CAD, IFC, PDFs, images and animations when they help a GitHub user inspect or build the system. Reproducible scratch output still belongs in `build/`; CI enforces a straightforward 50 MiB limit per tracked file.

Tracked

- Rust, Python, shell, HTML, CAD-generator, and configuration source.
- RFCs, engineering requirements, safety evidence authored by people, and stable schemas.
- Small test fixtures with a named consumer.
- The checksum-pinned TLC assurance executable, retained as a named CI input because its upstream download URL is mutable. Its embedded licences and reviewed byte identity are preserved.
- The compact machine-readable city core for every entry in `lib/city-batches/world-sample.toml`: `design.toml`, simulator scenario, quality result, corridor GeoJSON, and station JSON. Routed designs are retained because regeneration can depend on changing external OSM and population data.
- Catalogue-wide ring/interchange and station-cluster validation reports. These reports are allowed to record inherited planning failures, but their source and validator hashes must remain current; reviewed references must pass both validators.
- Compact per-city soil/deployment and workforce/design review packages, including workforce CSVs and their input hashes. Large local operations bundles are bound through their tracked compressed-hash manifests; checking a compact report does not require downloading a complete operations bundle.
- One generated `NATIONAL-BRIEF.md` per country, aggregating city CAPEX, imported/external capital, local funding, and the single shared trainset factory.
- Two complete end-to-end acceptance references, Samawah (`light-metro-3car`) and Mosul (`metro-4car`), including map, README, alignment, simulation, energy, finance, QA, maintenance, manufacturing, acceptance evidence, and a fail-closed package manifest. Their simulation evidence includes nominal operation plus end-of-life battery, maximum planning-climate load, reduced charging-contact availability, all-site grid outage, and single-pad outage cases.
- One compact high-throughput architecture reference, Basra (`metro-6car`), adding a map and README without duplicating the complete acceptance bundles.
- Baghdad's complete planning proposal, supporting-data archive, source/output manifests and compact engineering/finance studies, including the manufactured viaduct comparison. Each tracked file remains below 50 MiB. The supporting archive preserves hash-bound ignored inputs for the clean-checkout bootstrap; it is not field acceptance or an independent source of revised assumptions. Current generated studies, explicitly superseded reviews and diagnostic sensitivities retain distinct labels; historical prices and returns do not describe a later scope.
- The Samawah OSR-ALN fixture used by converter tests.
- Compact FreeCAD and Blender review assemblies, including the LM3 selectable manufacturing-tool set; the named civil and LM3-manufacturing IFC4.3 review federations, native civil construction scene and complete sub-50-MiB review animations under `engineering/models/bim/reference/`; stable documentation images, and the root `OpenSourceRail-Book.pdf` reader edition. These are published review aids generated from authoritative sources, not parallel inputs.

Not tracked

- Rust/Python build products, logs, caches, temporary data, and screenshots that are not part of the stable documentation review set.
- High-volume generated city engineering and operations inputs except the named Mosul/Samawah acceptance payloads and Baghdad's hash-bound supporting archive. Baghdad's extracted ignored inputs remain local after bootstrap.
- Generated BOM exports, simulation traces, release-only brochures, unnamed or repeated videos, CAD/BIM job outputs, and other release binaries.
- Files that only preserve an obsolete schema, chemistry, voltage, or API name.

Output locations

Output	Location
Full city package	`cities/catalogue/`
Repeated simulation and engineering job output	`build/engineering/`
Public CAD review set	`design/component-catalogue/models/cad/`
Public BIM coordination review set	`engineering/models/bim/reference/`
BOM exports	`build/bom/`
Published reader book	`OpenSourceRail-Book.pdf`
Release-only media	`build/releases/` before tagged release storage
Tagged deliverables	Release storage with version and checksum

Use `tools/automation/regenerate-city.sh <slug>` for a full city package in the canonical `cities/catalogue/` tree. Catalogue changes must include the corresponding current-generator scenario and pass the catalogue-completeness health check; generated engineering and operations directories remain tied to the city folder that produced them.

Rules

1. Change canonical source before regenerating output.
2. Do not hand-edit generated output.
3. Every retained fixture must name its generator and test consumer.
4. CI compares compact fixtures and regenerates larger evidence when required.
5. Never commit credentials, caches, absolute local paths, or live operational databases.
6. A complete generated city package is not release-ready merely because its manifest passes: survey, calibrated demand, local ground/fire/utility data, supplier freeze, and independent safety/construction release remain external project gates.
7. Tracked reports link only to tracked review artifacts; local high-volume outputs are named as code, not presented as GitHub links.
8. The root reader PDF is generated, never hand-edited. Its Markdown and city model sources remain authoritative and are printed in the book.
9. Keep useful CAD/BIM/media artifacts in Git when each file is at most 50 MiB. `tools/automation/check-tracked-file-sizes.py` enforces the ceiling. A future history optimisation must preserve access and requires an explicit maintainer decision; it is not part of routine cleanup.

Documentation ownership

- The root `README.md` is the only human-facing navigation page and owns the common setup instructions and source-of-truth map.
- A discipline or folder README contains only material local to that subject; it links to common guidance instead of copying it.
- City and country pages contain only local values and evidence. Shared planning interpretation lives in `docs/deployment-planning-reference.md`.
- `docs/INDEX.md` is a generated inventory for search and CI, not a competing guide or technical authority.
- When a claim can be derived, generate it from the authoritative source and test for drift rather than maintaining a second prose value.

RFC 0001 — Track State Consensus

Repository source: `docs/rfcs/0001-track-state-consensus.md`

Status: Draft Date: 2026-04-20 Depends on: docs/ARCHITECTURE.md §4 D2

Current development amendment

The current reference uses three static voters (five is supported by the model), selected for failure tolerance and independent power/network placement. Additional station I/O is non-voting. Historical 5–20-node regional sizing is not the implemented reference. Disk-journal checkpoint compaction retains the complete Raft log; logical snapshot/log truncation remains unimplemented. Resource decisions and positions are Safety entries, committed before outputs. The prototype verifies a static member's committed-prefix assertion under crash-fault assumptions; the historical two-node independent witness is not implemented or claimed by this reference and remains a separate assessment obligation. See RFC 0033 for the proposed responsibility map, evidence boundaries and release gates.

1. Summary

OpenSourceRail explores a distributed, replicated log of track state maintained by a static consensus group. Each train computes its own Movement Authority (MA) from validated committed state. The independent wayside witnesses described below are a proposed extension. This draft records the original schemas, algorithms and verification plan; its proposed timing, handover and witness behavior must not be read as implemented or qualified runtime behavior. RFC 0033 and the executable evidence define the current reference.

Safety integrity must be allocated by the assessed hazards and deployment profile. The original sub-€5k site cost and sub-200 ms p99 latency were unverified research targets, not installed-system estimates or measurements. No SIL qualification is established by this RFC.

2. Motivation

Conventional CBTC (Siemens Trainguard, Alstom Urbalis, Thales SelTrac) concentrates authority in a Zone Controller (ZC) — a redundant pair of proprietary computers per line or per large segment. The ZC:

- Centralises authority calculation within a defined operating area.
- Uses deployment-specific redundant equipment, integration interfaces and recovery arrangements.
- Requires a controlled safety case and supplier/operator integration evidence.

A distributed log lets N reference voter nodes collectively hold authoritative state. A surviving quorum can continue replication under the documented crash-fault assumptions. Service also requires valid train inputs, physical proving and both safety-output channels; loss of one required channel stops the affected function. A quorum is not a service-availability guarantee. State is a function of committed entries, and the state machine is public.

Local ownership and auditability motivate the approach. Cost, availability and safety claims require matched-service comparisons and independently accepted evidence; a complete runtime refinement proof remains open.

3. Non-goals

- Not a general-purpose database. The log carries a bounded, well-typed set of rail-domain facts. We do not aim for SQL, secondary indexes, or arbitrary queries.
- Not Byzantine-fault-tolerant in v1. Wayside nodes are operated by a single authority, run signed firmware, and attest at boot (see ARCHITECTURE.md §8). We trust them to follow protocol or crash. Byzantine resilience may be added in v2 but it is not required to close the SIL-4 safety case.
- Not a replacement for ATP onboard. Onboard Automatic Train Protection (emergency brake on MA violation) remains the last line of defense. The log defines what is allowed; onboard ATP enforces what actually happens.
- Not globally consistent across regions. Consensus is per region (§6.1). Region boundaries are handled via a handoff protocol (§7.4), not global consensus.

4. Concepts and Vocabulary

Term	Meaning
Region	An operating area covered by one static consensus group; the reference uses three voters, with five supported by the model. Physical extent and non-voting I/O are deployment-specific.
Wayside Node (W-Node)	SBC running the consensus + interlocking stack at a physical site (station, junction, substation).
Train Agent	Software on a train's T-ECU that publishes position reports, computes MA, and enforces ATP.
Log	The append-only, totally-ordered, quorum-committed sequence of facts about the region.
Entry	A single typed record in the log (§5).
Movement Authority (MA)	A permission for a specific train to occupy a specific set of track sections up to a specific position, valid until a specific time.
Safety Envelope	The union of track sections a train may occupy without colliding with anything, given current log state. A train's MA is always a subset of its safety envelope.
Commit	An entry has been accepted by a quorum majority and is durable.
Fail-safe direction	In rail, the fail-safe state is "stopped." Any ambiguity must resolve toward more restrictive MA, not less.

5. The Log

The log is a sequence of entries. Each entry has:

- `entry_id`: monotonic u64, unique within region
- `term`: consensus term (Raft-style)
- `timestamp`: TAI nanoseconds, PTP-synced wall clock at the leader
- `kind`: one of the entry types below
- `payload`: typed body
- `signature`: ed25519, signed by the leader's hardware key (for audit; not load-bearing for consensus)

5.1 Entry types

```
TrainPositionReport {
  train_id: TrainId,
  position: Position,           // track-ID + offset, sub-meter resolution
  uncertainty_m: f32,           // 95% CI half-width
  speed_mps: f32,
  speed_uncertainty_mps: f32,
  heading: Heading,
  source_mask: u8,              // which sensors contributed (GNSS/IMU/odo/beacon)
  onboard_time_ns: u64,
}

SwitchCommand {
  switch_id: SwitchId,
  requested_position: SwitchPosition,
  requested_by: EntityId,       // dispatcher or automatic route setter
  lock_until_entry: Option<EntryId>,
}

SwitchObservation {
  switch_id: SwitchId,
  observed_position: SwitchPosition,
  confidence: Confidence,       // Locked | Transitioning | Unknown
  observed_at_ns: u64,
}

RouteRequest {
  route_id: RouteId,
  requested_by: EntityId,
  entry_point: TrackRef,
  exit_point: TrackRef,
  train_id: Option<TrainId>,    // bound when granted
}

RouteGrant {
  route_id: RouteId,
  train_id: TrainId,
  locked_switches: Vec<SwitchId>,
  locked_sections: Vec<SectionId>,
  expires_at_ns: u64,
}

SpeedRestriction {
  section: TrackRef,
  max_speed_mps: f32,
  reason: RestrictionReason,    // permanent | temporary | emergency
  effective_from_ns: u64,
  effective_until_ns: Option<u64>,
}

TrainRegistration {
  train_id: TrainId,
  consist: ConsistDescriptor,   // length, mass, braking curve, classification
  cert_chain: CertChain,
}

Heartbeat {
  from_entity: EntityId,
  health: HealthStatus,
}

MaintenanceOverride {
  section: TrackRef,
  granted_to: EntityId,
  granted_until_ns: u64,
  signed_by: DispatcherSignature,
}
```

All payload schemas live in `crates/osr-core/proto/track_state.proto`. Types are enums with explicit discriminants; forward-compat is handled by `reserved` fields and a strict versioning rule (see §12).

5.2 Derived state

From the log, any W-Node (or train agent) deterministically computes:

- Section occupancy map: for each section, which train (if any) currently occupies it, with confidence interval.
- Switch state map: for each switch, last observed position and lock status.
- Protected routes/resources: reservations and occupied resources retained until their clearance and release conditions are proved. Permission expiry inhibits further entry; it is not proof that a resource is empty.
- Speed restriction map: keyed by section.
- Known trains: registered consists and last-known positions.

Derived state is a pure function of the committed prefix of the log. Two nodes with the same committed prefix must compute identical derived state (this is a verified property).

6. Consensus Protocol

6.1 Overview

We use Static-Membership Raft with Fail-Restrictive Timeout (SMRaft) — a narrow variant of Raft with three modifications tuned for safety-critical use:

1. Static membership. Membership changes require a maintenance window; there is no online reconfiguration. This eliminates the trickiest part of Raft (joint consensus) and the hardest part of its correctness proof.
2. Fail-restrictive timeout. When a node cannot confirm liveness of a quorum within `T_safe` (default 2 s), it stops emitting new RouteGrants and notifies trains via a "shrinking authority" signal. Existing MAs remain valid until their own expiry. This is the fail-safe direction.
3. Log capacity boundary. Logical Raft snapshots and log truncation are unimplemented. Disk-journal checkpoint compaction retains the complete logical log; reaching the bounded reference capacity stops publication. Snapshot, replay-fence and safe truncation work needs separate evidence.

SMRaft is Raft with parts removed, not added. The underlying correctness argument rides on Raft's well-studied safety proof (Ongaro's TLA+ spec, verified refinements in Ironfleet).

6.2 Region sizing

- Three reference voters; five supported by the model. Select membership from assessed failure tolerance and independent placement. Three voters tolerate one unavailable voter; five tolerate two under the quorum assumptions. Historical 5–20-node sizing is not a current default.
- Quorum: `floor(N/2) + 1`. Standard Raft quorum.
- Node placement: independent voter power/network domains require deployment assessment. Station or junction I/O does not automatically add a voter; collocation must not be counted as independent failure tolerance.

6.3 Timing parameters

Parameter	Default	Rationale
`heartbeat_interval`	50 ms	Allows election timeout much shorter than rail dynamic timescales.
`election_timeout`	300–600 ms (randomized)	Median failover ~400 ms.
`T_safe` (fail-restrictive)	2 s	Trains can coast safely for 2 s; anything longer risks MA expiry stacking up.
`commit_latency_p99_target`	50 ms	LAN RTT ×2 plus processing. Achievable on TSN backbone.
`MA_refresh_period`	500 ms	How often a train reevaluates its MA against log tail.
`MA_validity_window`	3 s	How long an MA is valid after issue. Short enough to be self-expiring, long enough to tolerate transient network hiccups.

These are original planning assumptions, not qualified defaults or current measured runtime deadlines. Use the exact controlled runtime configuration and its output-lease/deadline evidence for an executable reference.

6.4 Leader responsibilities

- Serialize writes and replicate to followers.
- Detect stale followers and emit catch-up stream.
- On term change: followers invalidate any in-flight MA computations and recompute from newly committed prefix.
- The leader does not compute MAs. MA computation is decentralized (§7); the leader only maintains the log.

6.5 What the protocol guarantees

Given quorum liveness:

- Agreement: all nodes agree on the committed prefix.
- Total order: entries are totally ordered by `(term, entry_id)`.
- Durability: committed entries survive minority failures.

- Monotonic progress: committed prefix only grows.

Given loss of quorum:

- No new commits. The log stops growing.
- Existing MAs remain valid until expiry. Trains with valid MAs continue until the MA window elapses, then stop.
- No silent split-brain. A minority partition cannot produce new grants because it cannot reach quorum. This is Raft's defining safety property.

7. Train Agent and MA Computation

7.1 Train agent architecture

Each train runs a `TrainAgent` on its T-ECU. It has four coordinating tasks:

1. Publisher: builds `TrainPositionReport` every 200 ms from sensor fusion; submits to the local W-Node via a regional proxy.
2. Subscriber: streams the committed log tail from any two independent W-Nodes (`primary` and `witness`).
3. MA Computer: recomputes self-MA on every new committed entry in relevant sections (debounced to `MA_refresh_period`).
4. ATP Enforcer: supervises actual train motion against self-MA + wayside-MA; triggers emergency brake on violation.

These four tasks run in isolated Hubris tasks with hardware-enforced memory isolation. Only the ATP Enforcer can command the brake.

7.2 The MA computation algorithm

The self-MA is computed deterministically from the log. The algorithm is small enough to fit on a page:

```
fn compute_self_ma(
    train_id: TrainId,
    log_prefix: &[Entry],      // committed entries visible to this train
    now_ns: u64,
) -> MovementAuthority {
    let state = derive_state(log_prefix);

    // 1. Where am I?
    let my_pos = state.position_of(train_id)
        .unwrap_or_else(|| fail_safe_no_position());

    // 2. What sections does my consist currently occupy?
    let my_footprint = footprint_from(my_pos, state.consist(train_id));

    // 3. Which adjacent sections are free to me?
    let candidate_extension = state.forward_chain(my_pos.direction(), MAX_MA_DISTANCE);

    // 4. Clip at first blocker.
    let mut ma_end = my_pos.head();
    for section in candidate_extension {
        if !section_available_to(train_id, section, &state) {
            break;
        }
        ma_end = section.far_end(my_pos.direction());
    }

    // 5. Apply speed restrictions within the authority.
    let restrictions = state.speed_restrictions_between(my_pos.head(), ma_end);

    // 6. Apply time bound.
    let valid_until = now_ns + MA_VALIDITY_WINDOW_NS;

    MovementAuthority {
        train_id,
        end: ma_end,
        restrictions,
        valid_until,
        derived_from_entry_id: log_prefix.last().entry_id,
    }
}

fn section_available_to(train_id: TrainId, s: Section, state: &State) -> bool {
    // A section is available if and only if:
    //   (a) no other train occupies it (with uncertainty margin), AND
    //   (b) it is part of a RouteGrant currently assigned to this train,
    //       OR it is explicitly unreserved and free, AND
    //   (c) every switch in it is observed-locked to the correct position, AND
    //   (d) no MaintenanceOverride blocks it, AND
    //   (e) all adjacent opposing-direction routes are clear.
    // Any uncertainty (unknown switch, unobserved train, stale heartbeat) => NOT available.
```

```
    ...
}
```

Five properties must be proven about this function:

- P1 (determinism): same input same output, across nodes and train agents.
- P2 (non-overlap): two trains A, B at the same instant: `ma_end(A)` and `ma_end(B)` do not share any section unless `A == B`.
- P3 (consist-fit): MA extension accounts for full consist length including safety margin.
- P4 (conservatism under missing data): any missing, stale, or uncertain input produces a more-restrictive MA, never less-restrictive.
- P5 (time bounded): `valid_until - now` is bounded above by `MA_VALIDITY_WINDOW_NS`; no path through the function produces an unbounded authority.

These are the core safety properties we formally verify (§11).

7.3 Proposed independent witness cross-validation

This witness protocol is not implemented by the current reference. Its committed-prefix checks and paired output guards do not establish two independent wayside witnesses.

The train agent subscribes to two independent W-Nodes. Each independently:

- Streams the log to the train.
- Runs the same MA computation.
- Emits a signed `MovementAuthorityWitness` to the train.

The train agent accepts the intersection of:

- Its own self-computed MA,
- Primary W-Node's witness,
- Secondary W-Node's witness.

"Intersection" means the most restrictive compatible extent, the minimum permitted maximum speed and the earliest validity limit. The position comparison must use a common route and direction; a minimum raw coordinate is insufficient on opposing routes.

Disagreement logs an incident and degrades to the most restrictive value. Persistent disagreement (>3 cycles) triggers a service brake and notifies dispatch — this indicates either a compromised node or a bug, both of which warrant intervention.

7.4 Region boundaries

A train traversing a region boundary runs two train agents in parallel for a transition window:

1. `T_in` (approaching boundary): subscribes to region A's log only.
2. `T_cross` (entering approach zone ~500 m before boundary): subscribes to both A and B. Both regions must see a consistent `TrainRegistration` for the train. Both regions compute MAs; train accepts the intersection.
3. `T_out` (fully in region B): unsubscribes from A after A has logged a `TrainDeparture` entry.

This handoff is not consensus across regions. It's coordinated publish/subscribe with cross-region authentication. The safety argument is local to each region plus the handoff protocol itself, which is much simpler to verify than global consensus would be.

8. Fault Model and Behavior

Fault	Frequency	System response
Single W-Node crash	Occasional	Quorum holds; no operational impact. Failed node restarted, rejoins as follower.
Leader crash	Occasional	Election in ~300–600 ms; brief commit pause; no MA impact if pause < `MA_VALIDITY_WINDOW`.
Quorum loss (partition)	Rare	No new commits. Existing MAs expire over ~3 s; trains coast to stop. Service resumes when quorum reforms.
Radio link degradation	Common	Missing/stale evidence cannot increase permission. Required-channel loss inhibits release; any degraded operation requires a separately approved profile.
Position sensor failure (one modality)	Uncommon	Sensor fusion degrades gracefully; position uncertainty grows; MA shrinks proportionally.
Position sensor failure (all modalities)	Rare	Train reports "position lost" to log; all other trains treat section containing unknown train as blocked; the train itself brakes to stop.
Byzantine W-Node	Out of scope for v1	See §3.
Train agent crash	Rare	Watchdog brake. Train stops. Onboard spare can be hot-swapped in depot.
Rogue dispatcher command	Mitigated	Dispatcher commands go through `MaintenanceOverride` which requires a signed second authorization for safety-critical operations.

Invariant under all enumerated faults: no train ever receives an MA that includes a section occupied by or reserved for another train. This is the central property the safety case argues.

9. Latency Budget

End-to-end MA update (position changes train has updated MA):

Stage	Budget (p99)	Implementation note
Onboard sensor fusion update	20 ms	Runs at 200 ms cadence; jitter <20 ms
Position report W-Node (5G uplink)	40 ms	Commercial 5G SA; LoRa fallback is slower and not in this path
W-Node ingress leader	5 ms	TSN backbone
Leader commit (quorum replication)	30 ms	LAN RTT ×2 + processing; budget includes worst-case node
Commit train subscribers	40 ms	5G downlink to two witnesses in parallel
MA recomputation (onboard)	5 ms	Deterministic fixed-cost algorithm
Witness cross-check	10 ms	In-memory
Total end-to-end	150 ms	
Target p99	200 ms	50 ms headroom

If 5G uplink is unavailable, position reports fall back to LoRa (carrier-independent, but ~500 ms latency). In that mode, MA refresh rate drops and MAs shrink defensively — not a safety concern, but throughput drops until 5G restores.

10. Formal Verification Plan

Four layers of verification, each proving different properties:

10.1 Protocol layer (TLA+)

A TLA+ specification of SMRaft, refining Ongaro's Raft spec, proves:

- Election safety: at most one leader per term.
- Log matching: identical committed prefixes across nodes.
- Leader completeness: committed entries persist across terms.
- Fail-restrictive liveness: under quorum loss, no node emits new RouteGrants; existing MAs are allowed to expire.

10.2 State-machine layer (Kani)

Bounded model checking of `derive_state` and `compute_self_ma` with Kani:

- Determinism (P1): two invocations on the same prefix produce byte-identical outputs.
- Non-overlap (P2): for any pair of registered trains with committed positions, the computed MAs do not share sections.

- Conservatism (P4): mutating an input entry to be "more uncertain" produces an MA whose end is no further than the original.
- Time boundedness (P5): all MA outputs have `valid_until - now ≤ MA_VALIDITY_WINDOW_NS`.

Bounds: up to 8 trains, up to 50 log entries per harness — enough to exercise all code paths. Larger systems rely on the TLA+ protocol argument plus the property that state derivation composes over log suffixes.

10.3 Rust-code layer (Creusot where feasible, tests elsewhere)

- Creusot contracts on the pure functions (`section_available_to`, `forward_chain`, `footprint_from`) discharging obligations from the Kani layer.
- Property-based tests (`proptest`) for algebraic invariants (e.g., "MA end is always ≥ current head position").
- Differential testing against a reference interpreter written in a simpler style (likely Python) for the same log inputs.

10.4 System layer (simulator)

- `osr-sim` runs full regional simulations with injected faults (node crashes, partitions, sensor glitches, Byzantine-ish data corruption short of full Byzantine fault).
- Invariant monitors in the sim flag any state where two trains could collide within their MAs. Any run that flags an invariant is a test failure.
- Soak tests for 30-day simulated operation with realistic traffic patterns before any hardware pilot.

10.5 Evidence linkage

Each Kani harness, TLA+ module, Creusot proof, and sim run emits a structured evidence artifact consumed by the safety-case compiler. The safety case (GSN) claims "P2 holds" link to the specific evidence artifacts that discharge the claim. If a commit invalidates any artifact, CI breaks the build.

11. Schema Evolution

Log entries are protobuf. Evolution rules:

- Never reuse field numbers. Deprecated fields are `reserved` and held forever.
- Never change a field's type. Introduce a new field with a new number.
- Log format version is part of the log. The first entry of every region's log is a `FormatVersion`. Entries past a version bump are rejected by nodes running the older version — they must upgrade or be replaced. (Upgrade is a maintenance-window activity.)
- Schema changes require an RFC. No in-repo schema changes without review.

The strict discipline is worth the friction. A rail signaling log will be read by tools we haven't written yet, in decades.

12. Open Questions

1. Raft vs. alternatives. Is there a protocol that's even simpler to formally verify while meeting our needs? Candidates: single-decree Paxos per entry (more messages but simpler proofs), a static-primary protocol with only failover consensus. Needs a focused comparison RFC.
2. How conservative is "conservative enough" for uncertainty handling? The exact error model for sensor fusion inputs determines how much MA margin trains leave; this will be tuned empirically in simulation and pilot.
3. Ed25519 vs. post-quantum signatures. The log is append-only and will be read for decades. Should signatures be post-quantum from day one? Tentative yes — Dilithium2 costs us ~2 KB per entry, affordable.
4. Dispatcher override safety. Maintenance overrides require dispatcher signatures. What's the exact protocol for "I, a human dispatcher, authorize this train into a blocked section for rescue"? Needs its own RFC.
5. Time source integrity. PTP sync is assumed. What if the GPS clock source is spoofed or jammed? Trains use their own clocks for MA expiry; disagreements between train-local and log-local time produce restrictive behavior automatically, but this needs to be verified.
6. Scale testing. Largest region we've thought through is ~50 km / 20 nodes. What's the upper bound before we need regional subdivision for latency?

13. Prior Art

- Raft (Ongaro, 2014) — consensus protocol we specialize from.
- Ironfleet (Hawblitzel et al., 2015) — formally verified replicated state machines in Dafny; inspiration for our verification layering.
- PALS / Hurdle — formal work on distributed real-time control systems in avionics; informs timing argument.
- Thales SelTrac, Siemens Trainguard, Alstom Urbalis — the centralized ZC designs we replace.
- Alstom Axonis, Siemens Trainguard MT with DCS-M — recent vendor moves toward more decentralized architectures; validate that the industry is trending this way, but still not open.
- Sun-Ways (track-PV), Stadler FLIRT Akku (BEMU), Hitachi Masaccio — related innovations on the energy side (see ARCHITECTURE.md §4 D5, D7).

This RFC is a draft. Substantive changes go through PR review with at least one reviewer from each of: distributed-systems, formal-methods, and rail-signaling background.

RFC 0002 — Energy Sizing Method

Repository source: `docs/rfcs/0002-energy-sizing.md`

Status: Current Depends on: RFC 0021, RFC 0024, RFC 0026

1. Purpose

This RFC defines the calculation method for onboard energy, station storage, charging, solar generation, and grid backup. Equipment selection is governed by RFC 0021.

2. Canonical equipment units

Unit	Reference
Onboard gross energy	225 kWh LFP per car
Onboard routinely usable energy	180 kWh per car
Normal stationary storage module	500 kWh gross LFP
Station charger cabinet	500 kW bidirectional DC/DC
Cabinet current limit	825 A
Platform contacts	2 sharing one cabinet budget
Nominal train DC range	650–700 V
Charger efficiency used in planning	98% unless qualified data replaces it

Additional capacity is built by repeating these units. The generator does not create a separate high-power charger product.

3. Sizing sequence

For every line and service window:

1. Calculate car-km from route length, headway, fleet turns, deadhead, and depot movements.
2. Apply the selected train's kWh/car-km and climate/HVAC factor.
3. Simulate every station arrival, dwell, contact conflict, conversion loss, current ceiling, battery acceptance limit, and charger outage.
4. Protect 20% onboard SOC unless a stricter deployment rule applies.
5. Add terminal/depot balancing dwell before adding stationary modules or cabinets.
6. Size PV from local hourly yield, temperature, soiling, shading, and maintenance availability.
7. Size grid import/export for the residual energy and resilience duty.

The design passes only when the full service window meets SOC reserve, service-completion, thermal, and failed-equipment criteria. Daily energy balance alone is insufficient.

4. Reference train

The promoted three-car set contains 675 kWh gross / 540 kWh usable energy. The other catalogue families use the same 225/180 kWh per-car module basis. Energy intensity starts at 2.4 kWh/car-km before deployment climate uplift; the simulator and measured duty replace that planning seed as evidence matures.

5. Station and grid rules

- Passenger stations start with one 500 kWh module and one 500 kW cabinet.
- Terminals gain energy through dwell, not different hardware.
- Four-car high-throughput families use three complete modules/cabinets and six-car families use four. These counts keep energy-derived dwell near the peak-service envelope in the canonical large-city simulations.
- Contacts never each receive the full shared-cabinet rating simultaneously.
- Grid AC is rectified once into the station DC bus; the train never consumes station AC.
- Solar is an energy source, not a guarantee of instantaneous charging power.
- Grid outage, charger outage, low-solar, and simultaneous-arrival cases are mandatory.

6. Evidence

Every generated deployment records input hashes, equipment-unit counts, minimum SOC, service completion, curtailed/exported energy, grid import, charger contention, adaptive-service events, and invariant violations.

RFC 0003 — Samawah Current Reference Deployment

Repository source: `docs/rfcs/0003-samawah-reference-deployment.md`

Status: Controlled planning example; construction and operating releases remain open.

Regenerate with `python3 tools/automation/generate-reference-city-rfc.py`. The city design and scenario are the controlled inputs; this RFC reports their current quantities. Earlier planning figures and brownfield observations remain explicitly historical.

1. Purpose

Samawah is an Iraqi reference instance of the shared deployment model. Its retained catalogue population is 373,770; the population and income proxies are planning inputs. The example uses the same geometry, energy, fleet, staffing and factory rules as other catalogue cities. Baghdad retains its own funding programme; its government share, Chinese USD credit and IQD bond structure are not applied to Samawah.

Current city summary · Controlled design · Scenario

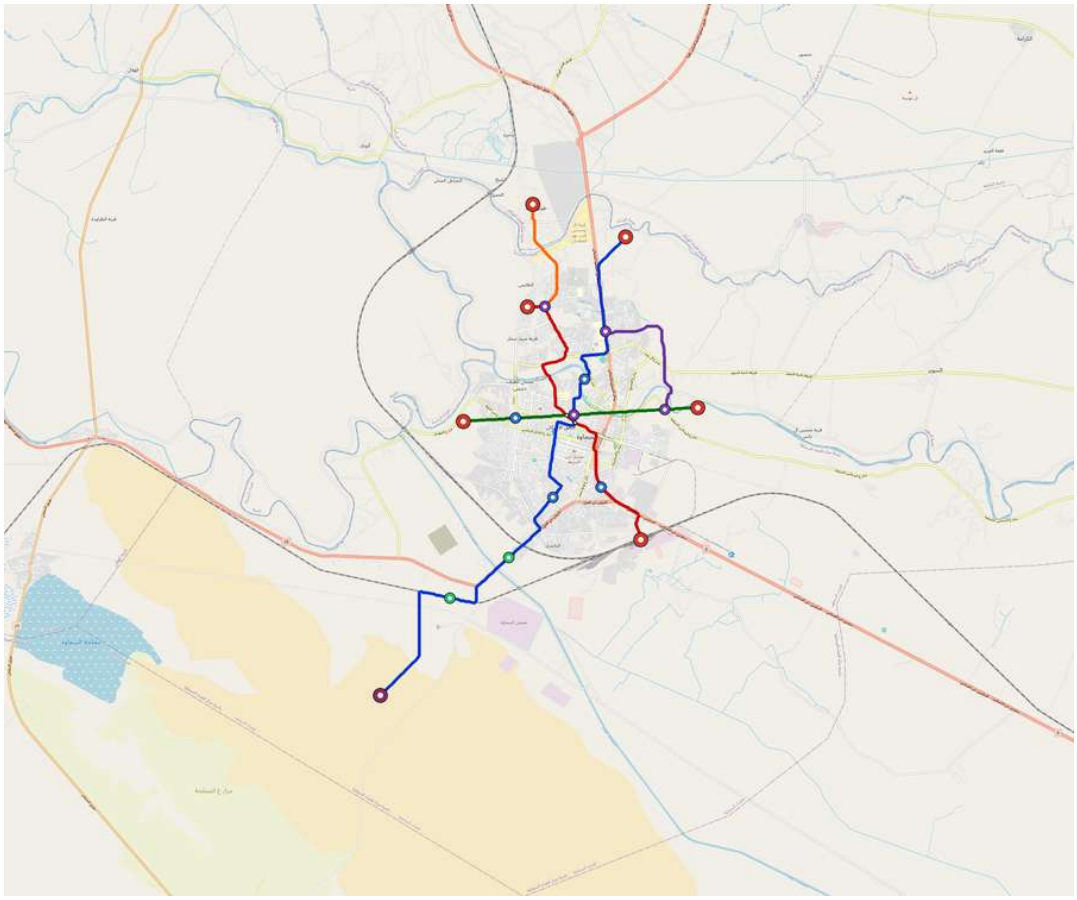
2. Existing assets and local investigation

The earlier railway-yard and workshop observations are candidates for a site census, tooling audit, ownership/access review and material testing. They do not establish usable capacity, recovered-component acceptance or a capital saving. Any brownfield proposal must close the releases in RFC 0027. The current full-fleet depot requirement is retained until a verified site alternative replaces it.

3. Current generated network

3.1 Lines

Line	Route km	Platform records	Total trainsets
line-1	26.603	10	81
line-2	12.756	6	40
line-3	8.799	6	28
line-4	5.446	2	18
line-5	4.283	2	15



Current Samawah network

The central planning concept straightens and elevates suitable core corridors. Analytical curve controls are recorded where verified by the generator; unsuitable fragments retain explicit geometry-review gates. Outer corridors retain their controlled routing seeds. Property rights, utilities, foundations, clearances and station footprints remain project investigations.

3.2 Transfers and platform siting

The current design records 4 multi-line interchange groups. Platform IDs remain distinct on their own lines. Declared transfers and bounded dry-bank moves preserve the checked walking legs; they do not demonstrate a surveyed paid-area connection, accessible footprint or stable bank. Water screen · Alignment review.

3.3 Revenue and finance

Use the current finance ledger for fare assumptions, commercial income, graded payroll, energy and financing sensitivities. These are planning estimates, rather than observed ridership or committed funding. Generic country finance is a fixed-price steady-state screen; Baghdad’s indexed monthly cashflow studies remain separate.

3.4 System totals

Metric	Value
Route-km (double track)	57.887 km
Stations (unique)	26
Lines	5
Multi-line interchanges	4
Fleet (revenue)	163 × trainsets
Fleet (spare + cold-reserve)	19 × trainsets
Fleet (total)	182 × trainsets
Depots	5
Best peak headway	3 min

Station totals count the unique listed platform records, including individual interchange platforms. Revenue fleet is the controlled peak requirement; reserve roles and any service-rotation stock remain separate in the design. A full-fleet planning depot is provided for each line and sized to the actual train count and consist length.

3.5 Civil quantities

Civil class	Route km
at-grade	8.662
elevated	44.883
bridge	4.341

These are route kilometres for a double-track system. Geometry and unit-cost targets do not release pi-beam prestress, complete-member lifting masses, foundations, bridge interaction or temporary works. Civil standard.

4. Operations, trainsets and labour

The controlled consist uses 3 cars and 675 kWh gross battery capacity. Its 20% protected reserve leaves 540 kWh above that reserve before operational derates. Climate/HVAC, ageing, charging availability and service recovery are checked separately; a static battery calculation cannot establish full-day performance.

Timetables, dwell and fleet roles come from the scenario. Operating evidence records the actual model inputs, executable and outputs. Native station dispatch remains a software planning screen; full-depot access, launch throughput, local operating qualification and physical acceptance remain releases.

Each listed station/platform has two posts and two normal eight-hour shifts, plus cover for the actual service window, weekends, leave, training and sickness. Role-based wages start at 150% of the configured country income proxy, with higher technical/management grades and employer allowances. Operating-scope controls and the finance ledger retain the reconciled headcounts and payroll.

5. Factories, energy and acceptance

The factory plan is sized to this city's actual order and vehicle family. Facility readiness is 18 months; qualification, serial manufacture and integrated acceptance follow it. A civil completion date is not automatically an opening date. National factory sharing requires a separately committed production sequence.

Use the current energy study, line-depot quantities, and operations package. Depot PV/storage is priced once per site. Buffered snapshots, grid-only diagnostics and native duty-cycle evidence have distinct scopes. Supplier, installation, utility, maintenance, independent-check and authority approvals remain open.

Architecture · Construction QA · Maintenance system

RFC 0004 — osr-interlocking Implementation Plan

Repository source: `docs/rfcs/0004-osr-interlocking-plan.md`

Status: Draft — planning only, no code yet Date: 2026-04-20 Depends on: RFC 0001 Track State Consensus

Current development amendment

The interlocking is implemented. Its existing compute_self_ma path remains the sole MA calculator. ResourceControl extends the committed fold with retained ownership, qualified clearance, epoch recovery, and original-evidence freshness bounds. The historical future/no-code wording below describes the original plan, not current implementation or acceptance. See RFC 0033 for the proposed responsibility map, evidence boundaries and release gates.

1. Summary

This RFC lays out the plan for the `osr-interlocking` crate — the Rust implementation of the Movement Authority (MA) computer and the SIL-4 safety kernel described in RFC 0001 §7. It is a planning RFC, not an implementation RFC: it enumerates the modules, the safety properties to verify, the tools to use, and the order of work. No code ships as part of this RFC. The intent is to set a concrete starting line before the next dedicated implementation session.

The reason for separating the plan from the implementation is scope. A credible first pass at `osr-interlocking` includes five loosely coupled pieces — data structures, pure MA computation, property-based tests, Kani-verified bounded models, and integration with `osr-sim` — that together are a multi-session effort. This document makes the structure explicit so the work can begin concretely next time.

2. Non-goals

- Not a full SIL-4 safety case. That requires independent assessment and deployment-specific evidence; the crate produces artifacts consumable by a safety case, but it is not one.
- Not a complete consensus layer. The MA computer is deterministic given a log prefix; the log itself is maintained by the future `osr-consensus` crate, which is a separate piece of work.
- Not a Byzantine-fault-tolerant design. Explicitly out of scope for v1 per RFC 0001 §3.
- Not a replacement for onboard ATP. The MA computer decides what is allowed; onboard ATP enforces what actually happens. They are separate crates.

3. Crate structure

```
crates/
└── osr-interlocking/
    ├── Cargo.toml
    ├── src/
    │   ├── lib.rs              # facade
    │   ├── state.rs            # derived-state types (track, trains, switches)
    │   ├── log.rs              # in-memory log prefix representation
    │   ├── ma.rs               # compute_self_ma + helpers (pure functions)
    │   ├── witness.rs          # signed MovementAuthorityWitness
    │   └── invariants.rs       # runtime-checkable invariants (debug builds)
    ├── harnesses/
    │   ├── kani_determinism.rs
    │   ├── kani_non_overlap.rs
    │   ├── kani_conservatism.rs
    │   ├── kani_time_bounds.rs
    │   └── kani_fit.rs
    └── tests/
        ├── proptest_invariants.rs  # algebraic properties
        └── differential.rs         # against a reference Python interpreter
```

Conventions: no `unsafe`, no allocator calls in the hot path (pre-allocated `Vec`s and `SmallVec`s), all pure functions annotated with `#[must_use]`, all public types `Debug + Clone + PartialEq` for test ergonomics.

4. Properties to verify

From RFC 0001 §7.2, the five safety properties:

- P1 (determinism): same log prefix byte-identical MA output across every node and train agent. Verified by Kani enumeration over small bounded states; asserted via proptests at larger scale by comparing outputs of random but identical inputs.

- P2 (non-overlap): for any two registered trains with committed positions, the computed MAs do not share any section (unless both are the same train). Verified by Kani on up to 8 trains × 50 entries; asserted by proptests with independent interpreter for larger N.
- P3 (consist-fit): MA extension accounts for full consist length plus a safety margin. Verified by Kani by construction of the `footprint_from` function; asserted by proptests that vary consist length randomly.
- P4 (conservatism): mutating any input entry to be "more uncertain" (higher position uncertainty, `SwitchPosition::Unknown`, missing heartbeat) produces an MA whose end is no further than the original. This is the hardest property — encoded as a monotone refinement lemma.
- P5 (time bounded): all MA outputs have `valid_until - now ≤ MA_VALIDITY_WINDOW_NS`. Verified trivially by Kani; also by ensuring the code path never constructs an MA with `valid_until` greater than `now + MA_VALIDITY_WINDOW_NS`.

Each property gets its own Kani harness file under `harnesses/`. The CI pipeline runs all harnesses on every PR.

5. Tool chain

Layer	Tool	Why
State-machine bounded model check	Kani	Rust-native, strong Rust semantic support, SMT-backed. Already used in the consensus RFC's verification plan.
Property-based testing	`proptest`	Scales to larger inputs than Kani; catches regressions cheaply.
Differential testing	A companion Python reference interpreter	Independent implementation of the same state machine; differential fuzzing catches bugs in either direction.
Formal model refinement	TLA+ (already stood up in `engineering/assurance/formal/tla/SMRaft.tla`)	The rail state machine is a refinement of the SMRaft-maintained log. Refinement proofs connect the two.
Contracts on pure functions	Creusot (optional, v2)	If Kani coverage is insufficient for some helpers, Creusot can discharge function contracts directly.

6. Milestones

The work breaks cleanly into five milestones. Each is a separate session or PR; none should be attempted as a single contiguous effort.

M1 — Types and log (done 2026-04-20)

- Port the protobuf entry schema (`crates/osr-core/proto/track_state.proto`) to Rust types in `osr-interlocking::state` and `osr-interlocking::log`.
- Implement `derive_state(log_prefix: &[Entry]) -> State` as a pure function.
- Proptest: `derive_state` is a pure function (two invocations on the same input produce identical outputs).

Delivered:

- `crates/osr-interlocking` scaffolded with Cargo workspace integration.
- `Entry` + `EntryPayload` (12 variants) and all payload types in `log.rs`, mirroring `track_state.proto` with integer-only safety-path units (mm, mm/s, ns, SoC in parts-per-thousand).
- `DerivedState` with `BTreeMap`-based deterministic ordering and `derive_state(&[Entry]) -> DerivedState` + `DerivedState::apply(&Entry)` in `state.rs`.
- Four added IDs in osr-core: `RouteId`, `EntityId`, `EntryId`, `RegionId`.
- Ten passing tests (6 unit + 4 proptests over 1024 random cases): P1 determinism, batch-vs-incremental composition, prefix-extension monotonicity, totality of `derive_state` under malformed input.

M2 — MA computation (§7.2 of RFC 0001) (done 2026-04-20)

- Implement `compute_self_ma(train_id, log_prefix, now)` per the pseudocode in RFC 0001 §7.2.
- Implement helper functions `section_available_to`, `footprint_from`, `forward_chain`, `apply_speed_restrictions`.
- Add debug-build invariants via `invariants.rs` that check P1, P3, P5 at runtime.
- Proptest: random log prefixes never produce MAs that violate P3 or P5.

Delivered:

- topology.rs: network traversal helpers — locate_section, forward_chain (budgeted, honors linear-vs-ring semantics, stops at section boundaries to stay fail-restrictive), footprint_from (backward walk by consist length), far_end_of. Includes 6 unit tests.
- ma.rs: MovementAuthority data type + compute_self_ma(train_id, log_prefix, network, now_ns) implementing the RFC 0001 §7.2 pseudocode. Fail-restrictive by default when the train has no registration or no known head position. Debug-build debug_assert!s verify P3 (end not behind head) and P5 (validity window bounded) on every call.
- Four MA-specific proptests: P1 determinism (256 cases), P5 validity bound (256 cases), totality (256 cases), MA-end structural validity (256 cases).
- Fixed two P3 edge cases during development:
  1. forward_chain originally extended past MAX_MA_DISTANCE_MM on the last section; now it stops at the previous section boundary. MAs always end at station boundaries — more conservative and simpler to reason about.
  2. Flawed P4 sketch proptest removed after discovering it was false in general (other trains' position reports can release occupancy and extend an MA). The correct P4 (mutation-based monotonicity) is Kani territory, explicitly deferred to M3.
- Five unit tests in ma.rs covering: no-registration fail-restrictive, single-train full extension, other-train blocking, P2 non-overlap on a specific two-train case, P5 validity window.

Workspace now has 43 passing tests (osr-sim 18, osr-interlocking 25).

M3 — Kani harnesses (written 2026-04-21; scaling bounds pending)

- Write one harness per safety property (§4). Bounds: 8 trains, 50 entries, 100 sections. Sufficient to exercise all code paths.
- Add a CI workflow that runs cargo kani on every PR.
- Document harness outputs as the first set of safety-case evidence files (linked from docs/safety-case/).

Delivered:

- crates/osr-interlocking/src/kani_proofs.rs contains six #[kani::proof] harnesses covering all five properties:
  - kani_p1_determinism (P1)
  - kani_p2_non_overlap_two_trains (P2, tiny-network bound)
  - kani_p3_consist_fit_single_train (P3)
  - kani_p4_fail_restrictive_is_not_less_restrictive_than_known (P4, fail-restrictive path only — full conservatism refinement proof is deferred to a mutation-based harness once Kani's state space allows)
  - kani_p5_time_bounded, kani_p5_time_bounded_with_known_position (P5)
- .github/workflows/kani.yml runs cargo kani -p osr-interlocking on every push.
- docs/safety-case/README.md documents the verifier, how to run it, and what's pending.

Pending for full closure:

- Scaling the P2 / P4 bounds toward the RFC's 8-train / 50-entry / 100-section target. Current bounds are 2 sections and 1–2 trains — enough to exercise the control flow but small enough to discharge in minutes.
- Full mutation-based P4 (conservatism): compare compute_self_ma on an input and on a "more uncertain" mutation of that input, assert the MA end never advances. Harder for Kani due to the doubled state space; deferred to a dedicated harness.
- Machine-readable evidence export (Kani already produces JSON; wire into the GSN safety case when the compiler lands).

M4 — Differential interpreter + proptest (done 2026-04-22)

- Reference interpreter in Python (~500 lines) implementing the same state machine in a more readable style. Placed under tools/reference-ma/.
- Differential test: random log prefixes compare Rust output byte-for-byte to Python output. Any divergence fails the test suite.
- Fuzz corpus under tests/fuzz-corpus/ for regression catches.

Delivered:

- `tools/reference-ma/` — stdlib-only Python package (~820 LoC across `types`, `log`, `topology`, `state`, `ma`, and `__main__` CLI). Mirrors the five-module shape of `crates/osr-interlocking/src/` for reviewability.
- CLI accepts the exact serde JSON shape Rust produces (`u64` IDs via `#[serde(transparent)]`, externally-tagged `EntryPayload`, nested `Network` with stringified-int keys) and writes `MovementAuthority` JSON that round-trips back through `serde_json::from_str::<MovementAuthority>()`.
- `crates/osr-interlocking/tests/differential.rs` — Rust harness with 3 smoke cases (no-registration / single-train / other-train-blocks) + a proptest run of 16 random log prefixes over a 4-station linear network. Each case serialises to JSON, shells out to `python3 -m reference_ma`, parses the output back as a Rust `MovementAuthority`, and asserts equality against the native Rust result. Skipped cleanly when Python3 is unavailable or `OSR_SKIP_PY_DIFF=1` is set.
- `tools/reference-ma/tests/test_basic.py` — 9 Python-side unit tests mirroring the key cases from `ma.rs` and `topology.rs` so the Python impl can be exercised standalone.
- 100 % agreement on every case tested so far: Rust and Python produce byte-identical `MovementAuthority` JSON.

Scope notes:

- The proptest generator currently stays inside a fixed 4-station linear network with 1..=3 trains. Extending to ring lines, switch observations, route grants, and speed restrictions is a straightforward follow-up — every primitive is already implemented in both twins.
- No standalone `tests/fuzz-corpus/` directory yet; the proptest's `proptest-regressions` file serves that role. A separate corpus is only warranted once external fuzzing (afl/libfuzzer) surfaces cases proptest alone would miss.

M5 — Integration with `osr-sim` (done 2026-04-22)

- Replace `osr-sim`'s `OccupancyMap` with a thin shim that queries the `osr-interlocking` MA computer against a synthesized log.
- Sim now emits `Entry` objects as trains move; the MA computer derives authorities; conflicts become real invariant violations rather than in-memory occupancy check failures.
- Proves the MA computer works end-to-end on a generated city instance scenario.

Delivered:

- `OccupancyMap` deleted from `osr-core`; the `BTreeMap<SectionId, TrainId>` in `DerivedState::section_occupancy` is the single source of truth.
- The sim's `MaLogBackend` (in `crates/osr-sim/src/sim.rs`) grew a `section_available_to(train_id, section)` predicate that delegates to `osr_interlocking::section_available_to` — the same primitive `compute_self_ma` uses when clipping the forward chain.
- `enter_next_section` (`sim.rs`) gates every section entry through that predicate. On authorisation, `register_and_enter` emits a `TrainPositionReport` entry into the log whose head offset equals the consist length, so `DerivedState.apply_position` marks exactly the new section as occupied (and clears the old one).
- `ConsensusBackend` caches a `DerivedState` from the committed prefix; gate checks stay O(1) in log length.
- `ConsensusBackend::propose` now calls `cluster.run_until_committed` synchronously (30 ms × 30 ticks = 900 ms budget, well under the 3 s MA validity window) so the gate sees fresh state even when the Raft cluster is driving commits. The consensus integration test (`crates/osr-sim/tests/consensus_integration.rs`) proves the consensus-backed simulator produces clean MA summaries.
- `ma_check::run_check_state` (formerly `run_check_entries`) is now a pure fleet-wide health sweep — no cross-check, because occupancy is owned by the MA computer itself, so any cross-check would be tautological. The periodic sweep populates `checks_run`, `total_mas_computed`, and `fail_restrictive_mas` in the run report.
- Generated Samawah reference scenario runs with the consensus backend and produces zero invariant violations.

After M5, the project has a property-tested MA computer running continuously in simulation. CI carries selected bounded Kani evidence; the topology-heavy proofs, TLA+Rust refinement and assessor acceptance remain open in the release gap register.

7. Safety case anchors

The five properties above anchor the safety case. Each property maps to a GSN goal:

```
G1: "No train occupies a section simultaneously with another train"
  └ G1.1: "MA computation enforces section exclusivity"  ← P2 (Kani)
  └ G1.2: "Train obeys its MA via ATP"                   ← future crate
  └ G1.3: "Consensus provides consistent log prefix"     ← RFC 0001 §11.1

G2: "Under faults, system degrades safely (fail-restrictive)"
  └ G2.1: "Loss of quorum prevents new Safety entries"   ← SMRaft TLA+
  └ G2.2: "MA expires after MA_VALIDITY_WINDOW"          ← P5

G3: "MA is consistent across witnesses"
  └ G3.1: "Same log prefix produces identical MA"        ← P1
```

The structured safety case (GSN serialised as TOML) lives under `docs/safety-case/` once M3 lands and CI begins producing evidence.

8. Pitfalls and decisions

These were flagged during planning; leaving them explicit so the implementation session doesn't rediscover them:

- Floating point. Distances are in mm (i64), speeds in mm/s (i64); no floats in the safety path. Kani cooperates much better with integers.
- Undefined-state handling. The RFC says "any uncertainty produces more-restrictive MA". Concretely: `SwitchPosition::Unknown`, `PositionSource` with zero mass, stale `Heartbeat` all short-circuit `section_available_to` to `false`. Property P4 verifies this.
- Test bounds. Kani's depth is the limit, not coverage. Small bounded models cover all code paths; larger N relies on the log-composition property (state derivation is monotonic in log prefix).
- `no_std` compatibility. Target `no_std` from day one to keep the crate usable on the safety-kernel OS (Hubris / seL4). `std`-dependent tests are gated behind a `std` feature flag.
- Proof fragility. Verification tooling evolves. Pin Kani version in `Cargo.toml` and document the known-working toolchain in `docs/safety-case/`.

9. Open questions (deferred to implementation sessions)

1. Exact numeric type for `entry_id` and `term` — `u64` is obvious but saturation arithmetic semantics need to be spelled out.
2. Representation of the "safety envelope" intermediate term — do we materialise it or is it implicit in the computation?
3. Error handling: should `compute_self_ma` return a `Result` (with fail-restrictive defaults) or always a valid MA? Trade-off between explicitness and brittleness.
4. How does `osr-sim`'s stepping interact with MA validity windows when the sim clock advances by `dt_s = 1` but MA window is 3 s? Need a clear contract for "when is the MA re-evaluated".
5. Where do `TrainAgent` and `ATP Enforcer` live — same crate or separate? Leaning toward separate: `osr-interlocking` is pure compute; `osr-train-agent` handles sensor fusion and brake enforcement.

10. Done criteria for this planning RFC

- [x] Crate layout documented
- [x] Properties enumerated and mapped to tools
- [x] Milestones ordered and scoped
- [x] Pitfalls and decisions explicit
- [x] Safety-case anchors identified

The next session picks up at M1 — Types and log.

RFC 0005 — SBC Software Architecture

Repository source: `docs/rfcs/0005-sbc-software-architecture.md`

Status: Implemented software baseline; production transport/hardware evidence remains Date: 2026-04-21 Depends on: RFC 0001 Track State Consensus, RFC 0004 `osr-interlocking` Plan Supersedes for repo layout: ARCHITECTURE.md §6.3

Current development amendment

Processor selections follow control-electronics/README.md and its safety-controller-selection gate: commodity CM5 application hosts and separately qualified safety channels; RP2350 rigs are pilot hardware. Earlier processor examples below are historical design alternatives. Functional safety allocations derive from hazards and physical interfaces, not blanket crate labels; unintended door opening and failure to open require distinct analyses. Thin train/wayside hosts do not qualify safety-output hardware. See RFC 0033 for the proposed responsibility map, evidence boundaries and release gates.

1. Summary

This RFC enumerates the Rust crates that run on OpenSourceRail hardware — onboard train ECUs, wayside SBCs, station and depot controllers, and the OCC back-office — and fixes their boundaries, buses, SIL tiers, and rollout order before the first of them is written.

The workspace now implements the crate map below. `osr-sim` connects the onboard safety, vehicle, TCMS, recorder, CBM, hot-axle, and T2G evaluators, plus station PSD/PIS/SCADA and wayside intrusion detection, then carries CBM through T2G into the depot backend, historian, and analytics. A TCMS trip inhibits dispatch or section progress on the following one-second control cycle. Explicit physical HABDs run the trackside evaluator, cap warning-band trains until the next station, and latch stop orders until a named post-inspection reset is accepted. This is all timestamped by a real `osr-ptp` state machine that acquires the simulator's deterministic shared clock. This is integration evidence, not certification or production hardware-I/O evidence.

Without a map, those crates would grow ad hoc, buses would be redesigned repeatedly, SIL boundaries would be fuzzy, and the safety case would be rebuilt from scratch several times. This document is the map.

The crate map remains the architectural contract. Implemented links are tracked by `lib/simulation-component-coverage.toml`; hardware and production transport evidence remains governed by the later RFCs and release checklists.

2. Non-goals

- Not a hardware reference design. Schematics, BOMs, and ratings are the domain of RFC 0007 and the `control-electronics/` directory. This RFC specifies which software targets which hardware class, not what that hardware looks like.
- Not a formal safety case. SIL targets here are allocations, not claims. The evidence that a crate meets its allocation is produced per crate (cf. RFC 0004 §7 for the interlocking pattern).
- Not a commitment to build every crate listed. Items beyond Phase 2 of §11 are placeholders; inclusion here is architectural scoping, not a roadmap promise.
- Not a redefinition of domains D1–D8 in ARCHITECTURE.md §3. Domains are kept; this RFC decomposes each domain into crates.

3. Design context

3.1 Hardware classes

Five hardware classes host OpenSourceRail software. Every crate is tagged with the class(es) it targets. The reference designs for each class — SoC, peripherals, baseboard, manufacturability — are in RFC 0007.

Class	Acronym	Environment	Typical SoC	Reference OS
Train ECU — safety kernel	T-ECU/S	EN 50155, −25…+70 °C, vibration, dual-redundant	RISC-V (MilkV Jupiter-class) or ARM Cortex-R52	Hubris or seL4, `no_std`
Train ECU — app	T-ECU/A	EN 50155, single-redundant	ARM64 (CM5-class)	PREEMPT_RT Linux
Wayside SBC	W-SBC	IP67 cabinet, −40…+70 °C, dual-PSU	RISC-V or ARM64	Hubris (safety) / Debian (non-safety)
Station / depot SBC	S-SBC	indoor/sheltered, commodity	ARM64	Debian or Yocto
Ops server	O-SRV	datacenter / colo	x86-64 or ARM64	Debian

T-ECU/S and W-SBC (safety role) are the hosts for the SIL-4 tier; everything else is SIL-2 or lower. The safety kernel binaries stay small (cf. ARCHITECTURE.md §6.1, "< 50k LoC per binary") by design.

3.2 Buses

Bus	Standard	Role	Determinism
TCN-E	IEC 61375-2-3 (Ethernet TRDP) on IEEE 802.1 Qbv TSN	On-train backbone	Hard real-time
CAN-FD	ISO 11898-1	Short-haul ECU sub-segments (door, brake panel, lighting)	Soft real-time
LIN	ISO 17987	Low-cost switch/sensor strings (cab panels, exterior lighting)	Best effort
TRG-1	Public 5G SA with slicing, or private 5G on n77/n78/CBRS	Train-to-ground primary	Statistical
TRG-2	LoRa mesh per ARCHITECTURE.md §D3	Train-to-ground backup for safety telemetry	Statistical, short messages
WAY-E	Ethernet + TSN + PTP (IEEE 1588)	Wayside backbone	Hard real-time
NATS-JS	NATS JetStream over TLS	Ops event stream, non-safety	Best effort
gRPC/mTLS	HTTP/2 + TLS 1.3	Control-plane RPC between wayside nodes and the OCC	Best effort

All onboard publishers and subscribers use TCN-E topics.

3.3 SIL allocation rules

Three rules govern where each crate falls:

1. Anything that can command or suppress a brake application is SIL-4. This is the traditional rail safety partition and we adopt it.
2. Anything whose failure can cause a train to exceed its Movement Authority is SIL-4. (Consequence of rule 1 via ATP.)
3. Anything else is SIL-2 or SIL-0. SIL-2 for systems that can cause operational disruption but not injury (e.g. door control opening, traction control limiting); SIL-0 for passenger-experience systems (PIS, lighting, HVAC comfort) where failure degrades comfort only.

Door closing interlock and wheel-slide protection sit in SIL-4 under rule 1. Door opening sits in SIL-2 because failing to open is not a safety hazard — it is a service disruption.

4. Crate inventory

Thirty-five crates across the eight domains. Each row shows the owning domain, the crate name, the hardware class it targets, the SIL allocation, and a one-line scope. Within each domain crates are ordered roughly by build dependency.

4.1 D2 — Train control (onboard)

These crates live on the train and are the direct consumers of the `osr-interlocking` MA output. All four SIL-4 crates here must share the same toolchain pin and formal-methods harness discipline as the interlocking crate.

Crate	Class	SIL	Scope
osr-atp	T-ECU/S	4	Onboard ATP. Receives MA from `osr-interlocking`, computes a time-based speed envelope from the braking curve, commands emergency brake on violation. Pure function of `(MA, consist, position, speed, now)`; no I/O in the core.
osr-ato	T-ECU/A	2	Automatic Train Operation. Generates traction and service-brake setpoints inside the envelope `osr-atp` permits. Station-stopping, energy-optimal driving, dwell management. GoA 4 unattended operation is the only supported profile per RFC 0015.
osr-odometry	T-ECU/S	4	Sensor fusion: wheel tachometers + IMU + GNSS + balise fixes train head/tail position and confidence. Feeds `osr-atp` and emits `TrainPositionReport` entries.
osr-obstacle-detect	T-OBS	4	NEW (RFC 0015 v1, 2026-04-22): onboard obstacle detection for GoA 4. Fuses ultrasonic + LIDAR + mmWave radar + stereo camera into an `ObstacleVerdict` per tick (Clear / CrawlOnly / EmergencyBrake). Five SIL-4 properties O1–O5 with Kani harnesses + proptest coverage. Substitutes for the driver's eyes in the unattended-operation safety case.

Crate	Class	SIL	Scope
osr-tcms	T-ECU/A	2	Train Management System. Aggregates ECU state onto TCN-E; records the event recorder stream; publishes passenger PIS status. Non-safety subscribers; does not mediate between `osr-atp` and the brake.
osr-event-recorder	T-ECU/A	2	Onboard "black box". Crash-survivable circular storage of TCN-E traffic at a deterministic sample rate. Feeds incident investigation.

4.2 D5 — Rolling stock ECUs

Actuators and comfort systems. Traction, brake, BMS, and door-closing interlock are SIL-4; everything else is SIL-2 or SIL-0.

Crate	Class	SIL	Scope
osr-traction	T-ECU/S	4	Field-oriented control for the RFC 0021 commercial-vehicle PMSM/controller channel; torque command from `osr-ato`; anti-slip; anti-slide.
osr-bms	T-ECU/S	4	LFP battery management. Cell balancing, SoC/SoH estimation, thermal limits, contactor control, off-gas/fire isolation, and cell-out derating. Exposes pack state to `osr-ato` for energy-optimal driving.
osr-brake	T-ECU/S	4	EP brake controller. Blends regen (commanded via `osr-traction`) with friction brake; integrates WSP (wheel-slide protection); parking brake; emergency brake. Primary consumer of `osr-atp`'s brake-apply command.
osr-regen	T-ECU/A	2	Regen arbitration. Decides whether excess regen current goes to the pack (via BMS), to trackside storage (via pantograph at charging pads, when contacted), or to a dump resistor. Not SIL-4 because `osr-brake` can always apply friction brake if regen is refused.
osr-aux-power	T-ECU/A	2	Isolated 24 V and 110 V DC conversion plus direct-HV comfort-branch control. Load-shed logic under low SoC.
osr-door-control	T-ECU/S	4	Door controller. Closing-interlock above 5 km/h is SIL-4 (rule 1 in §3.3); opening is SIL-2 but lives in the same crate to keep the interlock logic colocated. Obstacle detection via motor current + dedicated sensor.
osr-hvac	T-ECU/A	0	HVAC control loops. Climate setpoint (20–24 °C) per RFC 0003 hot-climate uplift. Publishes thermal state to `osr-tcms`. Failure degrades comfort.
osr-lighting	T-ECU/A	0	Interior and exterior lighting, emergency egress lighting, headlight/taillight PWM. Emergency-lighting battery backup is handled at hardware level; this crate only drives the normal-mode controller.
osr-pis-onboard	T-ECU/A	0	Passenger Information System — onboard displays, audio announcements, next-station logic. Consumes the ops event stream through `osr-t2g`.

4.3 D5 — Onboard safety monitors

Detect conditions that require an emergency brake application independently of ATP. All SIL-4.

Crate	Class	SIL	Scope
osr-fire-safety	T-ECU/S	4	Aspirating smoke detection + suppression activation in battery bay, traction bay, HVAC plenum. Emergency brake + passenger alert on detection.
osr-derailment	T-ECU/S	4	Lateral accelerometer + tilt sensor. Threshold-based trigger for emergency brake per EN 50159 envelopes.
osr-hot-axle	T-ECU/A	2	Onboard axle-bearing temperature monitoring. Not SIL-4 because wayside HABD provides the primary trip; onboard detection is backup and maintenance-signalling.

4.4 D5 — Condition-based maintenance (onboard)

Not safety-critical; streams telemetry to the depot for offline analysis.

Crate	Class	SIL	Scope
osr-cbm-onboard	T-ECU/A	0	Continuous monitoring of bearing vibration, brake pad wear, motor temperature, wheel profile (accelerometer-based), pantograph state if present. Streams to `osr-cbm-backend` over TRG.

4.5 D3 — Communications

Crate	Class	SIL	Scope
osr-tcn	T-ECU/*	2	TCN-E (IEC 61375-2-3) stack. Topic-addressed pub/sub over TSN Ethernet. Shared by every onboard crate that speaks on the bus.
osr-t2g	T-ECU/A	2	Train-to-ground link abstraction. Multipath over 5G (TRG-1) and LoRa (TRG-2); mTLS on both; Noise Protocol for LoRa. Exposes a simple `send(topic, bytes)` / subscribe interface to the rest of the train.
osr-ptp	T-ECU/* & W-SBC	2	PTP (IEEE 1588) time sync for TSN determinism. Wrapped into the TCN stack but broken out as a separate crate because the wayside uses the same code.

4.6 D6 — Wayside infrastructure

Crate	Class	SIL	Scope
osr-wayside-points	W-SBC	4	Power switch (point) machine controller. Commodity BLDC motor driver + dual redundant position sensors. Publishes `SwitchObservation` entries to the consensus log; accepts `SwitchCommand` entries.
osr-balise	W-SBC	2	Balise/transponder reader. Provides position fixes at fixed points (platform edges, switches) for `osr-odometry` to consume. Passive balises preferred; active balises used only where a data payload is needed.
osr-level-crossing	W-SBC	4	Level-crossing controller. Same SBC family as the interlocking; crossings are consensus-log participants.
osr-hot-axle-wayside	W-SBC	4	Wayside Hot Axle Box Detector (HABD). IR sensor array reads bearing temperatures on passing trains; publishes a speed-restriction entry or a stop-order if thresholds exceeded.
osr-consensus	W-SBC & O-SRV	4	Raft implementation of the SMRaft spec (see `engineering/assurance/formal/tla/SMRaft.tla`). Maintains the track-state log consumed by `osr-interlocking`. Separate crate because every safety-critical wayside node hosts a replica.

4.7 D4 — Passenger services

Crate	Class	SIL	Scope
osr-psd	S-SBC	2	Platform Screen Door controller. Synchronises with train-side `osr-door-control`; reports state to the OCC. Failure-to-open is a service failure; failure-to-close in presence of a passenger is a safety issue and that interlock lives in hardware.
osr-afc	S-SBC	0	Automatic Fare Collection gate firmware. QR + NFC + account-based tokens per I7 in ARCHITECTURE.md §5.
osr-tvm	S-SBC	0	Ticket Vending Machine firmware. Commodity barcode scanner + contactless reader; accepts mobile-money QR and cash where applicable.
osr-pis-station	S-SBC	0	Station PIS — next-arrival displays, PA, emergency announcements. Consumes the ops event stream directly.
osr-station-scada	S-SBC	2	Station SCADA — escalators, lifts, HVAC, lighting, CCTV NVR integration. Monitoring-only for SIL-rated station subsystems; vendor safety systems remain self-contained, this crate only observes them.

4.8 D1 & D8 — Ops, depot, backend

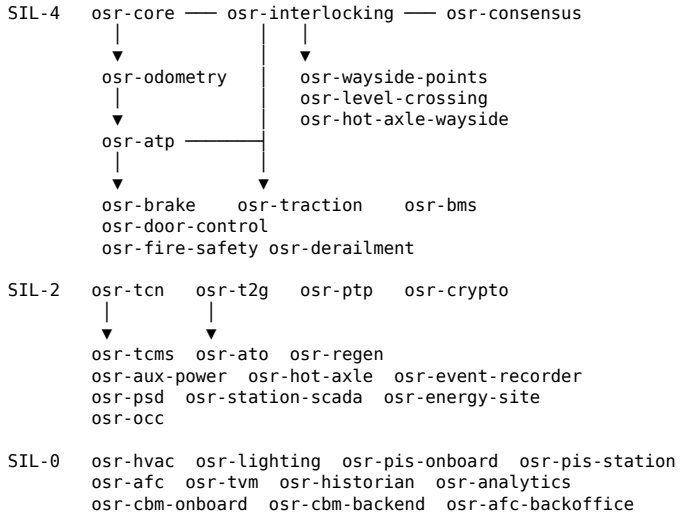
Crate	Class	SIL	Scope
osr-occ	O-SRV	2	Operations Control Centre. Fleet roster, incident management, dispatch holds, timetable editor and ATS (Automatic Train Supervision). Its deterministic core runs in `osr-sim`; the production event stream remains the deployment boundary.
osr-historian	O-SRV	0	Time-series ingest + retention. Prometheus remote-write + a typed Parquet archive for audit.
osr-analytics	O-SRV	0	KPIs (MDBF, availability, kWh/km, regen %, occupancy heatmaps). Consumes the historian; produces Grafana-compatible datasets and monthly PDF reports.
osr-cbm-backend	O-SRV	0	Depot-side analysis of `osr-cbm-onboard` telemetry. Predictive-maintenance triggers, work-order generation.
osr-afc-backoffice	O-SRV	0	Fare settlement, revenue reconciliation, MaaS API, fraud detection.
osr-energy-site	W-SBC	2	Per-site PV, stationary LFP, 500 kW DC/DC and grid-tie controller per RFC 0002.

4.9 Cross-cutting support crates

Crate	Class	SIL	Scope
osr-proto	all	n/a	Generated types for the protobuf schemas in `crates/osr-core/proto/`. Pulled out of `osr-core` once `osr-consensus` demands the wire format.
osr-crypto	all	2	Thin wrapper around RustCrypto primitives and `rustls` configured for IEC 62443-4-2 conformance. Every safety-relevant network node links this crate rather than rolling its own TLS/KEM usage.
osr-safety-case	tooling	n/a	GSN (Goal Structuring Notation) compiler. TOML claim files in, rendered safety case out. CI fails if the case no longer closes. Starter case under `docs/safety-case/gsn/` — 11 goals / 3 strategies / 14 solutions, all linked to real evidence in-tree.

5. Dependency graph

Build dependencies flow bottom-up; SIL-4 crates sit at the bottom so they have no dependencies on higher-SIL crates (a SIL-4 crate must not link SIL-0 code).



The binding rule: a crate at tier N must not `[dependencies]` any crate at tier >N. This is enforced by a `cargo-deny` policy added in the first implementation PR (RFC 0006).

6. Interface contracts

Five on-train interfaces and four train-to-ground interfaces. Each is defined by a message schema in `crates/osr-proto/` (new location — moved out of `osr-core` in Phase 2a, see §11).

6.1 On-train

#	Topic	Producer	Consumers	Rate	Size
O1	`osr.train.ma`	`osr-atp` (mirror of `osr-interlocking` output)	`osr-ato`, `osr-event-recorder`	On change, ≤ 1 Hz	~200 B
O2	`osr.train.position`	`osr-odometry`	`osr-tcms` TRG; `osr-atp` (internal)	10 Hz	~80 B
O3	`osr.train.setpoint`	`osr-ato`	`osr-traction`, `osr-brake`	50 Hz	~30 B
O4	`osr.train.brake_apply`	`osr-atp`, `osr-fire-safety`, `osr-derailment`, `osr-passenger-assist`, `osr-obstacle-detect`	`osr-brake`	On event	~20 B
O5	`osr.train.telemetry.*`	every crate	`osr-tcms` TRG	1–10 Hz	variable

O4 is safety-critical. It is a union of commands from every SIL-4 monitor; the brake crate applies the emergency brake if any subscriber asserts it. This is the single point where safety partitioning is enforced on the train; all other interfaces are for coordination, not safety.

6.2 Train-to-ground

#	Topic	Producer	Consumers	Transport	Cadence
G1	`TrainPositionReport` (log entry)	`osr-odometry` via `osr-t2g`	`osr-consensus`	TRG-1 primary, TRG-2 fallback	1 Hz
G2	Movement Authority subscription	`osr-interlocking` at wayside	`osr-atp` onboard	TRG-1 + TRG-2 replicated	On change
G3	Ops event stream	`osr-occ`	`osr-pis-onboard`, `osr-tcms`	TRG-1	As produced
G4	Telemetry + CBM stream	`osr-tcms`, `osr-cbm-onboard`	`osr-historian`, `osr-cbm-backend`	TRG-1	1–10 Hz

G1 and G2 are safety-critical. G2 is replicated on both radios; onboard `osr-atp` uses the most recent valid MA from either path. Loss of both paths for more than `MA_VALIDITY_WINDOW_NS` (3 s) forces fail-restrictive behaviour per RFC 0001.

7. SIL-4 coding standard

Every SIL-4 crate in §4 follows the conventions established for `osr-interlocking` (RFC 0004 §3):

- `#![forbid(unsafe_code)]` at crate root.
- Integer-only units in the hot path: millimetres, millimetres-per-second, nanoseconds, parts-per-thousand for SoC. No floats on the safety path.
- No allocator calls in the control loop. Pre-allocated buffers, bounded collections, `heapless`/`SmallVec` where dynamic sizing is genuinely needed.
- All public API types `Debug + Clone + PartialEq` for test ergonomics.
- `no_std` compatible by default. Tests that need `std` are gated behind a `std` feature.
- Kani harnesses live under `crates/<name>/harnesses/`, one file per named property, bounded by the same small-model sizes used in `osr-interlocking`.
- Proptests live under `crates/<name>/tests/` and run in CI unbounded.
- Every SIL-4 crate pins the same Rust toolchain version (`rust-toolchain.toml` at workspace root).

SIL-2 crates follow the same standard except for Kani (encouraged but not mandatory) and `no_std` (optional).

SIL-0 crates follow ordinary Rust idiomatic style; the goal there is velocity.

8. Deployment topology

This is the target electronics/software partition, not an approved hardware installation. SBC names in this RFC are development candidates. The pilot signalling profile keeps OSR wayside control in shadow/supervised operation behind independent occupancy detection and a separately qualified safety channel until its staged assessment gates close.

One trainset hosts: two redundant T-ECU/S units (primary + hot spare, running `osr-atp`, `osr-odometry`, `osr-traction`, `osr-brake`, `osr-bms`, `osr-door-control`, `osr-fire-safety`, `osr-derailment`), and one or more T-ECU/A units (running `osr-ato`, `osr-tcms`, `osr-pis-onboard`, `osr-hvac`, `osr-lighting`, `osr-event-recorder`, `osr-aux-power`, `osr-regen`, `osr-cbm-onboard`, `osr-hot-axle`, `osr-t2g`).

One wayside site (a station, a switch, or a level crossing) hosts: one to three W-SBCs running the relevant subset of `osr-wayside-points`, `osr-level-crossing`, `osr-balise`, `osr-hot-axle-wayside`, plus `osr-consensus` and `osr-interlocking` on sites that are safety participants. Non-safety stations also host `osr-energy-site` if the station has PV and storage.

One station hosts: one S-SBC for `osr-psd`, one to four for `osr-pis-station`, `osr-station-scada`, and as many as needed for `osr-afc` and `osr-tvm`.

The OCC hosts: `osr-occ`, `osr-historian`, `osr-analytics`, `osr-cbm-backend`, `osr-afc-backoffice`, and a replica of `osr-consensus`.

9. Cybersecurity posture

Per ARCHITECTURE.md §8 and IEC 62443-4-2 / TS 50701:

- Every TCN-E and WAY-E node holds a unique key pair provisioned at commissioning; no shared credentials.
- All inter-node traffic is mTLS (gRPC) or Noise-protected (LoRa). `osr-crypto` is the only TLS-configuring crate.
- Network segmentation on the wayside: safety plane (consensus, interlocking, points) is physically separated from operational plane (SCADA, telemetry) and the passenger plane (Wi-Fi, PIS). The three planes meet only at `osr-occ`, across audited gateways.
- Signed firmware everywhere; measured boot on T-ECU/S and W-SBC.
- The safety plane is considered air-gapped from the public internet in normal operation; updates are staged and pushed manually via a cleanroom.

10. Reference boundary

The reference stack uses TCN-E over TSN, the two ECU hardware classes in §4, onboard odometry plus the shared track-state log, OTLP condition monitoring, and account-based fare collection. Deployment-specific adapters are outside this crate map.

11. Rollout order

Five phases, chosen to front-load the safety-critical onboard stack while allowing non-safety passenger-facing work to proceed in parallel once the safety partition is firm.

Phase 2a — Onboard safety partition (starts immediately)

Prerequisite: RFC 0004 M1+M2 are done; M3 (Kani) can run in parallel.

1. `osr-atp` — RFC 0006, this week. First SBC crate. Consumes `MovementAuthority` from `osr-interlocking`, produces brake-apply commands. Pure function + integration stub for simulator-in-the-loop.
2. `osr-odometry` — RFC 0007. Sensor-fusion kernel; emits `TrainPositionReport` entries.
3. `osr-brake` — RFC 0008. EP brake blending + WSP.

Phase 2b — Onboard traction partition

1. `osr-bms` — LFP pack management and fire-isolation interface.
2. `osr-traction` — VVVF firmware against a SPICE-simulated inverter first, hardware second.
3. `osr-ato` — consumes ATP envelope, drives traction + service brake within it.

Phase 2c — On-train bus and non-safety app layer

1. `osr-tcn`, `osr-ptp` — bus stack.
2. `osr-tcms`, `osr-event-recorder`.
3. `osr-door-control`, `osr-hvac`, `osr-lighting`, `osr-aux-power`, `osr-pis-onboard`.
4. `osr-t2g` — radio abstraction (5G SA SIM emulated first).
5. `osr-fire-safety`, `osr-derailment`, `osr-hot-axle`, `osr-regen`, `osr-cbm-onboard`.

Phase 2d — Wayside and consensus

1. `osr-consensus` — Raft against the TLA+ spec.

2. `osr-wayside-points`, `osr-level-crossing`, `osr-balise`, `osr-hot-axle-wayside`. The simulator derives one stable passive-balise record per directed section, audits each crossing through `osr-balise`, and passes only valid fixes into the real `osr-odometry` evaluator.
3. `osr-energy-site` (already scoped in RFC 0002; crate created here).

Phase 2e — Stations and back-office

1. `osr-psd`, `osr-pis-station`, `osr-station-scada`.
2. `osr-afc`, `osr-tvm`, `osr-afc-backoffice`. The simulator executes the complete signed-token and settlement path as a representative station workload, including signature-tamper denial and fraud-rate evidence; it does not treat that workload as demand forecast.
3. `osr-occ`, `osr-historian`, `osr-analytics`, `osr-cbm-backend`.

Phase 3 — Hardware integration (overlapping)

Hardware-in-the-loop bench for each SIL-4 crate as it lands. Full-system HIL with simulator-driven wayside + real T-ECU by end of Phase 2c.

12. Relationship to existing work

- `osr-interlocking` is the producer of MAs; `osr-atp` (Phase 2a, crate 1) is the first real consumer. The two crates share no mutable state; `osr-atp` treats MAs as immutable inputs.
- `osr-sim` remains the digital twin. In Phase 2a, `osr-atp` runs inside the simulator against `osr-sim`'s kinematics, consuming MAs derived from consensus log entries via `osr-interlocking`. Station and intrusion controllers also run against deterministic physical/sensor shadows; hardware remains a separate gate.
- `osr-core` stays as it is; the protobuf types migrate into a new `osr-proto` crate when `osr-consensus` lands (Phase 2d) so that cross-language consumers (the Python reference interpreter, future tooling) can link the schema without linking domain logic.
- `engineering/assurance/formal/tla/SMRaft.tla` is the spec that `osr-consensus` implements. The TLA+ work is already done; only the Rust refinement is pending.

13. Pitfalls and decisions

- Two T-ECU/S units, not three. Standard SIL-4 practice is triple-modular redundancy. We adopt dual-redundant ("composite fail-safe") because the consensus layer provides a third opinion at the system level — any disagreement between the two onboard ECUs falls back to the wayside-authoritative MA. This is a design bet; validated in the safety case per deployment.
- Regen is not SIL-4. Rule 1 in §3.3 might be read as forcing it, but the brake crate can always substitute friction brake; regen refusing to accept current is an energy-efficiency issue, not a safety one. Listed here so the allocation is auditable.
- Door opening is SIL-2, closing is SIL-4. Same crate. This bifurcation is the one place where a single crate spans two SIL tiers. The alternative — splitting the crate — leaves the interlock logic divided between two binaries, which is worse. The crate's internal partitioning (separate tasks under Hubris) is the mitigation.
- `osr-tcn` is SIL-2, not SIL-4. Justification: the bus is a shared mechanism; the safety argument lives in the payload crates (atp, brake, bms, etc.) which validate every message. A corrupted frame fails validation and is discarded; a missing frame triggers fail-restrictive timeouts in the consumer. TCN-E itself does not need SIL-4 treatment.
- SIL-4 crates must not depend on SIL-2 code at runtime. `osr-atp` cannot `use osr_tcn`. Instead, the TCN plumbing delivers MA bytes to a memory-mapped region that `osr-atp` reads; any error in that delivery manifests as MA expiry, which `osr-atp` handles safely. This pattern is consistent across every SIL-4 crate.
- OS choice per tier is deferred. Hubris vs. seL4 for T1 remains open (ARCHITECTURE.md §10 open question 3). SIL-4 crates are written to be OS-agnostic `no_std` libraries; a thin OS-specific task wrapper lives outside the crate.
- Cybersecurity is not a crate. It is a set of invariants enforced by `osr-crypto` plus configuration discipline plus the measured-boot hardware stack. A single "osr-cyber" crate would be an anti-pattern.

14. Open questions

1. Crate naming: hyphens or underscores in the wire-topic namespace? The topics in §6 use dotted names (`osr.train.ma`); crate names use hyphens. The mapping is unambiguous but warrants a one-line style rule in the contribution guide.
2. Does `osr-atp` own the envelope math or delegate to a companion `osr-brake-model` crate? Leaning toward owning it; the braking curve is already in `osr-core::consist::BrakingCurve`. Revisit if the curve model needs to support non-trivial consists (articulated, heterogeneous).
3. Pantograph monitoring. If a deployment instance adds opportunity-charging pads, a `osr-pantograph` crate is needed. Not in §4 because pads are Phase 4+ (post-Samawah baseline).
4. Wayside relay interlocking retrofit shim. Some brownfield pilots may need to bridge to an existing relay plant. Shim crate or per-deployment tooling? Deferred to first retrofit engagement.
5. Hot-axle threshold tuning. Value varies by bearing vendor and ambient. Parameterised in TOML config; default per RFC 0003 climate envelope.
6. Repository layout change. The ARCHITECTURE.md §6.3 tree is superseded by §4 here (more crates, cleaner domain split). A follow-up editorial PR updates ARCHITECTURE.md to point at this RFC for the crate list.

15. Done criteria

- [x] Every hardware class tagged (§3.1)
- [x] Every bus named and standardised (§3.2)
- [x] SIL allocation rules explicit (§3.3)
- [x] All thirty-five crates enumerated with owner-domain, class, SIL, scope (§4)
- [x] Dependency graph closed under the "no higher-SIL-on-lower" rule (§5)
- [x] On-train and train-to-ground interface contracts defined (§6)
- [x] SIL-4 coding standard carried forward from RFC 0004 (§7)
- [x] Deployment topology stated (§8)
- [x] Cybersecurity posture linked to ARCHITECTURE.md §8 (§9)
- [x] Reference boundary stated (§10)
- [x] Rollout order fixed (§11)
- [x] Pitfalls and open questions captured (§13, §14)

The next session picks up at Phase 2a, crate 1 — `osr-atp` (RFC 0006).

RFC 0006 — osr-tcn design

Repository source: `docs/rfcs/0006-osr-tcn-design.md`

Status: Draft 1.1 (2026-04-22) — v1 crate landed, v1.5 UDP transport landed Depends on: RFC 0005 SBC Software Architecture
Supersedes part of: ARCHITECTURE.md §D5 trainbus description

1. Summary

`osr-tcn` is the on-train communications stack. It implements a narrow, deterministic publish-subscribe layer over IEC 61375-2-3 Ethernet TRDP, providing the bus on which every onboard crate exchanges state. This RFC specifies the topic namespace, payload encoding, traffic classes, time synchronisation, and API shape — before any code is written.

This is Phase 2c's biggest remaining design decision and the only RFC-level question left for the onboard stack. Every existing onboard crate produces or consumes data that will flow over TCN-E; getting the bus shape right once is worth an explicit design pass.

2. Goals and non-goals

2.1 Goals

1. Deterministic delivery of safety-relevant messages within a bounded latency, using IEEE 802.1 TSN Qbv time-aware scheduling.
2. Topic-addressed publish/subscribe — producers don't know consumers by name; the topic is the contract.
3. Zero-allocation publish path on the Rust side. Safety-tier crates (T1/T2 per RFC 0005 §6.1) publish into pre-allocated frames without heap traffic.
4. Compact binary payloads — integer-only wire format matching the safety-path discipline of the producer crates.
5. Testable in software — a mock transport drives the same crate API so unit tests exercise real publish/subscribe without hardware.
6. PTP time sync fan-out so every onboard clock reads the same `now_ns` to within TSN's accuracy budget (sub-microsecond for the safety partition, sub-millisecond for applications).

2.2 Non-goals

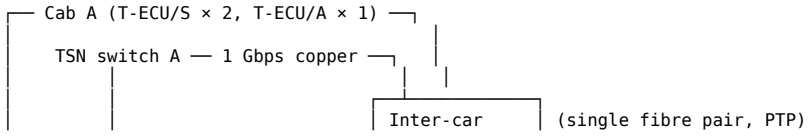
- Security beyond network segmentation. mTLS / signing sits in `osr-crypto` (future) on top of TCN-E. TCN itself is physical-layer secured by the partition isolation described in RFC 0005 §9.
- Dynamic membership. The consist's ECU inventory is fixed at commissioning. No hotplug; no membership negotiation at runtime.
- Protocol bridging. Deployment-specific adapters are outside TCN-E.
- Train-to-train coupling. TCN-E is intra-consist. Multi-unit coupling is a separate RFC.

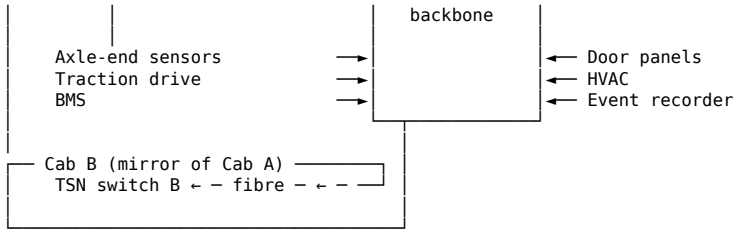
3. Standards alignment

- IEC 61375-2-3 — Train Communication Network, Ethernet TRDP profile. Defines the message layout and service classes.
- IEEE 802.1Qbv — Time-Aware Shaper. Gates output ports on a scheduled calendar so safety traffic is never blocked behind elephant flows.
- IEEE 802.1Qbu / 802.3br — Frame Preemption. Allows a safety frame to preempt an in-flight best-effort frame.
- IEEE 1588-2019 / 802.1AS — PTP time sync. Sub-microsecond accuracy across the bus.
- IEC 61375-3-4 — Ethernet consist network (bounds the physical topology: backbone switches + stub links to ECUs).

4. Physical topology (reference)

A 3-car light-metro consist per RFC 0003:





Every ECU has exactly one link (1 Gbps copper) to its cab's TSN switch; cab switches link to each other via a fibre pair running the length of the consist. The backbone tolerates a single fibre cut (either A or B still reaches everyone via the intact link); both cut puts the consist into a safe-state stop via ATP time-out.

5. Topic namespace

Topics are hierarchical dotted strings. The first segment is the producer domain, the second is the message class, subsequent segments are qualifiers:

```
osr.train.atp.envelope
osr.train.atp.command
osr.train.brake.apply
osr.train.odom.position
osr.train.odom.speed
osr.train.traction.torque_setpoint
osr.train.traction.current_estimate
osr.train.bms.soc
osr.train.bms.limits
osr.train.bms.fault
osr.train.door.status          (per-door idx: osr.train.door.status.3)
osr.train.door.interlock
osr.train.hvac.status
osr.train.lighting.status
osr.train.pis.display
osr.train.pis.announcement
osr.train.tcms.consist_status
osr.train.monitors.fire
osr.train.monitors.derailment
osr.train.monitors.hot_axle
osr.train.monitors.aux_power
osr.train.event.record         (event-recorder entries)
```

Design choice: dotted hierarchical topics, not numeric IDs. Hierarchy lets subscribers use prefix filters (`osr.train.monitors.*`) without a lookup table. The cost — slightly larger frames — is not the bottleneck on a 1 Gbps link with ~100 topics.

On the wire the topic is interned to a 16-bit ID by the publisher's TCN stack (`osr-tcn` maintains a deterministic string → `u16` map populated at boot from a static config file); consumers see both the interned ID and, on-demand, the full string. The intern table is part of the bring-up handshake and must match across every ECU in the consist.

6. Payload encoding

6.1 Rule

Every payload is a fixed-size, endian-normalised (little-endian) byte layout defined as a Rust `#[repr(C)]` struct in a shared payload crate (`osr-tcn-payloads`, or a module under `osr-tcn`). Producers serialise by `core::mem::transmute`-less byte-copying (`bytemuck::bytes_of(&payload)`); consumers the reverse. This gives the safety crates a zero-allocation, zero-parsing publish path.

6.2 Why not protobuf / CBOR / JSON

- Variable-size encodings complicate bounded-latency proofs. TSN Qbv reserves a fixed number of bytes per safety slot; a fixed-size payload is a natural fit.
- Schema evolution happens at the topic level (new topic = new schema); we don't need per-field tagging.
- Test ergonomics: Rust struct == wire layout makes round-trip testing trivial.
- Certification: a safety case that says "this byte at this offset is the MA validity-window" is far easier to audit than one that threads through a protobuf decoder.

6.3 Alignment and endianness

- All integer fields little-endian.
- All struct fields `#[repr(C)]`, explicit padding where needed.

- Maximum payload size per TRDP PD (Process Data) message: 1432 bytes. Most payloads are 16–64 bytes.
- Larger payloads (e.g., CCTV keyframes) go over TRDP MD (Message Data) in a separate channel, not addressed in this RFC.

6.4 Example payload (MA envelope)

```
#[repr(C)]
#[derive(Copy, Clone, Debug, PartialEq, Eq, bytemuck::Pod, bytemuck::Zeroable)]
pub struct AtpEnvelopePayload {
    pub now_ns: u64,
    pub envelope_mmps: i32,
    pub distance_to_end_mm: i64,
    pub has_known_position: u8,
    pub _pad: [u8; 3],
    pub trigger_reason: u8,
    pub _pad2: [u8; 7],
}
// sizeof == 32
```

`bytemuck::Pod` + `Zeroable` give safe transmute semantics without `unsafe`. The workspace-level `forbid(unsafe_code)` is preserved.

7. Traffic classes

Three classes, mapped to 802.1Q priority codes per Qbv gate schedule:

Class	PCP	Slot type	Example topics
Safety (S)	7	Guaranteed slot every 1 ms	`atp.envelope`, `brake.apply`, `odom.position`, `monitors.*`
Control (C)	5	Guaranteed slot every 10 ms	`traction.torque_setpoint`, `bms.limits`, `door.status`
App (A)	3	Best-effort, preemptible	`hvac.status`, `pis.display`, `event.record`

A safety-class publish uses a pre-allocated buffer and transmits in its next Qbv slot — bounded latency < 1 ms. The publisher's Rust API must not spin-wait; if the TCN stack's internal queue is full (which should be unreachable under a correctly sized bus), the publish returns an error and the caller records a `TcnError::QueueFull` event.

8. Time synchronisation

- Grandmaster: the primary T-ECU/S in cab A. Runs a PPS-locked oscillator if available; otherwise free-runs its TCXO.
- Secondary: the primary T-ECU/S in cab B. Takes over on grandmaster timeout (standard IEEE 1588 BMC algorithm).
- Slaves: every other ECU. Maintain ≤ 1 µs offset to grandmaster for safety-tier timestamps.

The `now_ns` parameter that every evaluator (ATP, ATO, BMS, etc.) consumes is read from the local PTP-disciplined clock. Callers must not mix PTP time with monotonic `std::time::Instant` on the publish path.

9. Crate API

9.1 Publisher side

```
pub struct Tcn {
    // implementation detail
}

impl Tcn {
    pub fn publish<P: TcnPayload>(
        &self,
        topic: TopicId,
        payload: &P,
    ) -> Result<(), TcnError>;
}

pub trait TcnPayload: bytemuck::Pod {
    const CLASS: TrafficClass;
    const SIZE: usize;
}
```

9.2 Subscriber side

Two modes:

- Blocking read (unit tests, non-hot-path): `recv(topic) -> Option<&P>`.

- Callback registration (hot path, safety crates): at init time, subscribe with a `fn(&P)` that the stack invokes on receipt. The callback runs in the ECU's deterministic tick, not in an interrupt handler.

9.3 Mock transport for tests

```
pub struct MockTcn { /* in-memory bag of (topic, payload) tuples */ }
impl MockTcn {
    pub fn publish<P>(&mut self, topic: TopicId, payload: &P);
    pub fn drain<P>(&mut self, topic: TopicId) -> Vec<P>;
}
```

The mock drives the real crate API — every existing onboard crate can be fed into it without change. This is how `osr-sim`'s shadow stack will exercise the full publish/subscribe chain in the simulator.

10. Scope of v1 implementation

The first cut ships:

1. Topic ID interning (static config file `TopicId` enum).
2. Payload trait + derive macro (or explicit `bytemuck::Pod` impls for now — macro is follow-up).
3. In-memory mock transport with traffic-class awareness.
4. Proptest harness verifying:
   - Round-trip encode/decode produces byte-identical output.
   - Safety-class messages are delivered in publish order.
   - Class-A messages may be dropped under backpressure; class-S never drops.

Out of scope for v1:

- Real TSN hardware driver — needs a kernel driver story (DPDK vs AF_XDP vs stock socket). Separate RFC.
- PTP client — use a userspace `ptp4l` + Linux PHC until we write a bare-metal one for Hubris.
- `osr-tcn-payloads` split out of `osr-tcn` — reasonable at scale, premature now.

11. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	`osr-tcn` crate with mock transport + topic registry + proptests	—
v1.5	Real network transport (`UdpTcn`) on commodity UDP — drop-in API replacement for `MockTcn`, loopback round-trip tested; the simplest thing that works for a multi-host bench (done 2026-04-22)	v1
v2	`osr-sim` publishes the odometry snapshot and consumes it at the ATP boundary through `MockTcn`; transport failure becomes unknown position and trips fail-restrictively	v1
v3	Real TSN driver (Linux AF_XDP first, Hubris bare-metal later)	reference hardware RFC
v4	PTP client + grandmaster fail-over	v3

v1.5 is deliberately a small step past v1. It preserves the MockTcn contract so downstream crates never see the transport; what changes is only whether bytes travel through a BTreeMap or through the kernel UDP stack. This gives operators a working multi-host bench without pulling in DPDK / AF_XDP complexity, which stays in v3 where the hardware story is fixed.

12. Relationship to `osr-crypto` and `osr-t2g`

- `osr-crypto` provides HMAC-signing of safety-class payloads for defense-in-depth against a compromised ECU. Sign/verify wraps the Pod layout; wire format stays fixed-size.
- `osr-t2g` sits outside TCN and takes its inputs from specific TCN topics. TCN frames do not leave the consist; ground-bound telemetry is re-serialised (likely in a bandwidth-efficient encoding like CBOR) before transmission.

13. Open questions

1. Topic ID collisions across firmware versions. The intern map must be version-stable within a consist or the bus silently routes to the wrong subscriber. Solution sketch: the intern map is derived from a canonical SHA-256 of the sorted topic list; consists bootstrap-verify that all ECUs agree.
2. Payload schema evolution. Today a new topic requires a new schema and all consist ECUs must agree on the topic-map hash before entering service.
3. Multicast vs unicast. Most TRDP PD is multicast; control flows (e.g., `door.command` from a central door controller to specific doors) could be unicast. Current plan: everything multicast with subscriber filtering, simplifies the stack.
4. No-std on Hubris. `bytemuck::Pod` is `no_std`-friendly; the current `std` usage in `osr-tcn` would need the same treatment the other SIL-4 crates receive in the no_std migration (RFC 0005 §7 ongoing).

14. Done criteria

- [ ] RFC ratified by reviewer(s).
- [x] Expand the landed v1 crate from 4 to at least 6 proptest-verified transport/registry properties.
- [ ] Existing SIL-4 crates compile against the `TcnPayload` trait for at least one topic each (e.g., `osr-atp` exports `AtpEnvelopePayload`).

The v1, v1.5, and v2 implementations have landed. The cross-crate payload criterion remains open: the adapter currently owns the `TcnPayload` wrapper so SIL-4 evaluator crates do not acquire a runtime dependency on the SIL-2 bus, as required by RFC 0005 §13.

RFC 0007 — Control Electronics Reference Designs

Repository source: `docs/rfcs/0007-control-electronics-reference-designs.md`

Status: Draft v1.2 — planning baseline; production safety controller not frozen Date: 2026-08-30 (v1.2 separates commodity application hosts from safety channels) Depends on: RFC 0005 SBC Software Architecture, RFC 0006 `osr-tcn` design

Hardware interface review — 3 October 2026

The revised host specifications and reference integration correct the earlier draft's unverified wiring and complete-host assumptions. CM5 uses dual 100-pin connectors; Pico 2 needs external CAN/Ethernet controllers; TPS3701 is a voltage detector, not a heartbeat watchdog. T-OBS requires the recalculated carrier/sensor/startup envelope and a complete harness/protection BOM. Exact SKUs, prices, pin assignments, relay failure behaviour, cooling and qualification remain open. Historical custom-board cost envelopes below are planning allowances, not complete qualified procurement totals. Refer to RFC 0019 for the corrected bench scope.

1. Summary

This RFC fixes the custom-board reference designs for the physical host classes on which OpenSourceRail software runs — T-ECU/S, T-ECU/A, T-OBS, W-SBC, S-SBC — before a schematic is drawn. It picks the SoC, the module form factor, the baseboard constraints, and the per-board BOM envelope. It does not commit the schematics; those land as a separate editorial PR once the decisions below are ratified. For first-article and pilot deployments, the commodity-module path in RFC 0019 is the preferred implementation route and does not require KiCad or gerbers when no custom PCB is used.

The O-SRV class is commodity server hardware — any x86-64 or ARM64 Linux box that meets the operator's procurement rules. It is out of scope for this RFC.

Why a planning RFC first: hardware decisions are sticky. A T-ECU/S reference design outlives three software refreshes and four deployment generations, because the PCB, the environmental compliance testing, and the depot spares-and-repairs pipeline all calcify around the choice. Getting the class selection right costs a session of writing; getting it wrong costs an order of magnitude more in re-compliance later.

Three constraints shape every decision below:

1. Local manufacturability. Every board must be producible on a 4-layer PCB with 0.15 mm trace/space, 0.3 mm vias, standard SMT (0402 minimum passive), and through-hole where rail vibration/shock demands it. No micro-vias, no blind/buried vias, no 0201 passives, no exotic stackups. These are routine tier-2 board shops across MENA, sub-Saharan Africa, South and Southeast Asia, and Latin America — the target deployment footprint.
2. Commodity System-on-Module (SoM) first. Wherever a ready-made compute module with the right environmental rating exists, we adopt it and design only the baseboard. This reduces the domestic engineering burden to a carrier board plus enclosure — a fraction of the work of a from-scratch SoC design.
3. Simple application palette; qualified safety-channel exception. Raspberry Pi and Radxa are the application, diagnostic and pilot hosts. Revenue-service safety channels may use an industrial safety MCU selected through the controlled safety-controller gate. This exception avoids making commodity computers the certification premise while keeping local procurement and spares simple everywhere else.

2. Non-goals

- Not a full hardware BOM. This RFC specifies reference choices; the per-deployment BOM is owned by the operator and tuned to local availability.
- Not a certification strategy. EN 50155 / IEC 61373 / EN 50121-3-2 compliance testing is in scope for the safety case (`osr-safety-case`), not this document.
- Not a redundancy topology. The logical redundancy (dual T-ECU/S, 3-node W-SBC consensus cluster) is already fixed by RFC 0001 and RFC 0005 §8. This RFC is about one board of each class; stitching them into a consist or a wayside cabinet is deployment-level.
- Not a hard-real-time bus design. TSN / PTP details are RFC 0006. This RFC references those choices but does not re-litigate them.

3. Common platform choices

These apply to every class below unless overridden.

Aspect	Choice	Rationale
PCB	4-layer FR-4, 0.15 mm trace/space, 0.3 mm min via	Routine at tier-2 fabs worldwide.
Passives	0402 minimum	SMT-line friendly, hand-rework possible.
Connectors	M12 for field I/O, board-to-board for modules	M12 is the rail industry standard for trackside and consist wiring (EN 50467).
Isolation	1.5 kV galvanic on every external I/O	EN 50155 demands it for rolling stock; W-SBC gains the same treatment for consistency.
Power input	24 V DC nominal, 16.8–30 V range	EN 50155 wide-input class. On trains this comes from the battery bus via `osr-aux-power`. At wayside sites it comes from the site supply.
Power sequencing	TI LM5164 buck 5 V per-rail LDO	Single widely-stocked part, no exotic controllers.
Firmware storage	eMMC (on-module) + external SPI-NOR for boot	eMMC is the reference OS filesystem; SPI-NOR holds the measured-boot root.
Trust anchor	Microchip ATECC608 or NXP A1006 SE	Either is commodity, EN 50155 rated, well-documented. ATECC608 is our baseline because ed25519 support is native.

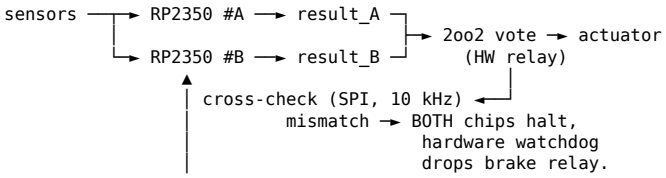
4. Class T-ECU/S — Train Safety Kernel

Role: Hosts the SIL-4 crates — `osr-atp`, `osr-odometry`, `osr-traction` (MCU bring-up), `osr-bms`, `osr-brake`, `osr-door-control`, `osr-fire-safety`, `osr-derailment`. One primary + one hot-standby per consist.

Environment: EN 50155 OT4 (−40…+85 °C), IEC 61373 Cat 1 Class B vibration + shock, EN 50121-3-2 EMC.

4.1 Safety architecture — prototype 2oo2 evidence rig

The pilot board carries two Raspberry Pi RP2350 MCUs wired in a 2-out-of-2 arrangement. It exercises independent inputs, comparison, watchdogs and fail-safe outputs at low cost:



- Both chips receive the same raw sensor inputs on electrically separate pins.
- Both chips execute identical Rust `no_std` code (Hubris or a bespoke bounded scheduler).
- Each tick they exchange their computed safety decisions over a dedicated SPI link and compare byte-for-byte.
- The final actuator-drive output is produced by a hardware AND of both chips' drive lines through an external 2oo2 relay network. A single chip can only permit motion; it cannot command it alone.
- On disagreement, both chips set themselves to fail-safe (drive lines open) and a downstream hardware watchdog asserts the emergency brake relay directly.

This rig is not the SIL-4 argument. Identical chips and identical software have systematic and common-cause obligations, while bounded proofs do not replace a qualified hardware architecture. Production channels must pass the shared safety-controller selection gate, including assessor review, safety manual/FMEDA inputs and a system-level redundancy/diversity argument.

4.2 SoC picks

- Prototype MCU (×2 per board): Raspberry Pi RP2350. It contains two processor sockets; each socket selects Cortex-M33 or Hazard3 at boot and the unused implementation is held in reset. It has 520 KB SRAM and runs at 150 MHz. Arm TrustZone-M for secure boot, hardware random number generator, and a single signed ROM loader. Published datasheet, globally available through the Raspberry Pi channel and every major distributor. Per-chip price ≈ €1 in 1k volume.

  Why RP2350 remains in the pilot:

  - low cost and broad availability make it useful for repeatable HIL rigs;
  - its selectable ISA lets developers exercise two toolchain targets, but selection within the same silicon does not establish hardware diversity;
  - Available in QFN-56 (0.4 mm pitch) — hand-solderable under a loupe, trivial on any SMT line.
- Application processor (×1 per board): Raspberry Pi CM5. Handles TCN-E, logging, OTA, diagnostics, and hosts the event recorder. Runs mainline Linux. The app processor is never on the safety path — it subscribes to a one-way SPI feed from the safety MCUs and cannot preempt their decisions.

4.3 Peripherals (baseboard)

Peripheral	Qty	Purpose
CAN-FD	4	Door, brake, traction, BMS buses (two CAN-FDs per RP2350, identical)
TSN Ethernet (on CM5 + external PHY KSZ9031)	2	TCN-E (A + B, redundant paths per RFC 0006)
Isolated DI	8	Emergency plunger, cab door switches, depot enable, deadman handle (each routed to both RP2350s)
Isolated DO	4	Emergency brake relay, traction cut relay, park brake solenoid, fire-suppression trigger (via 2oo2 relay network)
Tachometer inputs	2	Wheel encoders (quadrature, 5–24 V, each read by both RP2350s)
IMU	1	Bosch BMI088 6-axis (derailment + odometry) — read by both RP2350s via separate SPI chip-selects
GNSS	1	u-blox NEO-F10N over UART to CM5
Balise reader	1	PN5180 NFC front-end over SPI (passive HF balise per RFC 0003)
Thermistors	4	Battery bay, traction bay, HVAC plenum, enclosure ambient
ATECC608B SE	2	One per RP2350 — trust anchor for signed firmware + cross-check keys

Every input reaches both RP2350s on physically separate pins; every output is AND-gated through the external 2oo2 relay stage.

4.4 Target BOM (volume 100+)

Item	Unit cost (€)	Per board
RP2350 × 2	1.10	2.20
Raspberry Pi CM5 (4 GB)	85	85
Baseboard (4-layer, 160 × 100 mm)	40	40
CAN-FD transceivers × 4	3	12
Ethernet PHYs + mags	12	12
Isolated DI / DO (ADUM + discrete drivers)	35	35
BMI088 IMU + NEO-F10N GNSS + PN5180	45	45
ATECC608B × 2, passives, connectors, power	45	45
Total	—	~€280

Two pilot boards per consist = €560 per trainset for the evidence rig. The production safety-controller allowance is frozen only after selection and supplier quotation; this pilot number must not be used as that allowance.

4.5 Form factor

160 × 100 mm Eurocard, DIN-rail mountable in the cab cabinet. Conduction-cooled to the DIN rail; no fans. Conformal coated (MG Chemicals 419) before integration. Sealed to IP54.

5. Class T-ECU/A — Train Application

Role: Hosts SIL-2 / SIL-0 app-tier crates — `osr-ato`, `osr-tcms`, `osr-pis-onboard`, `osr-hvac`, `osr-lighting`, `osr-aux-power`, `osr-event-recorder`, `osr-regen`, `osr-hot-axle`, `osr-cbm-onboard`, `osr-t2g`.

Environment: EN 50155 OT4, single-redundant (two units per consist is standard but a single unit is tolerable for non-safety functions).

SoC baseline: Raspberry Pi Compute Module 5 (BCM2712, Cortex-A76 4-core, 4–16 GB LPDDR4X, eMMC on-module). RPi CM-class is the obvious commodity module — wide availability, mature Linux support, global stock, SO-DIMM-style connector.

Alternative drop-in: Radxa CM5 (RK3588S — Cortex-A76 + A55 big.LITTLE on the same SO-DIMM footprint). Radxa's module is pin-compatible with our baseboard; the only code delta is the device-tree blob. Operators can source whichever is locally available at procurement time.

Peripherals (baseboard):

Peripheral	Qty	Purpose
TSN Ethernet	2	TCN-E A/B (redundant)
CAN-FD	2	HVAC + lighting buses
USB-C (host + device)	2	Depot console + diag loader
HDMI	1	Commissioning display output
USB 2.0	2	Diagnostics and PIS service
M.2 2280 NVMe	1	Event recorder ring buffer + OTA staging

Peripheral	Qty	Purpose
M.2 2230 Cat.22 5G	1	TRG-1 (primary ground radio)
LoRa SX1276 on SPI	1	TRG-2 (backup ground radio)
CCA-B2B to T-ECU/S	1	Mailbox bridge to the safety side
ATECC608 SE	1	Trust anchor

Target BOM: €220 per board (CM5 module ≈ €85, baseboard ≈ €75, radios ≈ €60 volume).

Form factor: 160 × 100 mm Eurocard, same DIN mount as T-ECU/S so a cabinet can accept either without re-designing the slot.

5.5 Class T-OBS — Train Obstacle-Detection ECU

Role: Hosts the SIL-4 `osr-obstacle-detect` evaluator per RFC 0015 §5.2. Fuses the nose- cone sensor suite (ultrasonic × 4, LIDAR, mmWave radar, stereo camera) into an `ObstacleVerdict` bus message consumed by the T-ECU/S 2oo2 AND-gate relay stage.

Two modules per trainset — one at each nose. Only the leading module is active for a given direction of travel; the trailing module self-tests continuously and publishes an `Inactive` status.

Environment: EN 50155 OT4, dual-redundant (both nose-end modules duplicate the safety evaluator; either can drive the brake chain).

Pilot architecture: mirrors the T-ECU/S rig: two Raspberry Pi RP2350 MCUs running the `osr-obstacle-detect` evaluator in a 2oo2 cross-check plus a Raspberry Pi CM5 application processor for sensor fusion, classifier inference, and the non-safety data path. The safety-critical verdict is produced inside the evaluator pair; the CM5 supplies pre-processed detection lists but cannot emit a `Clear` on its own. Production evaluator channels pass the same safety-controller selection gate.

Peripherals (baseboard):

Peripheral	Qty	Purpose
Ultrasonic transceivers (40 kHz, 200 µs drive, 10-bit ADC)	4	Close-range safety belt per RFC 0015 §5.1
CAN-FD to mmWave radar (TI AWR1843 or eq.)	1	All-weather long-range
Vendor-specific Ethernet to LIDAR (HAP TX: 100BASE-TX; HAP T1: 100BASE-T1)	1	Candidate-specific interface; other models separately verified
MIPI-CSI to stereo camera pair (IMX477 / IMX219)	2	Classification
TSN Ethernet	2	TCN-E A/B back to T-ECU/S + T-ECU/A
CCA-B2B to T-ECU/S	1	Direct brake-demand line (2oo2 AND-gate stage)
ATECC608 SE	1	Trust anchor (shared with T-ECU/S)

Power budget: ~18 W under full load (LIDAR dominates at ~12 W); 24 V DC in with the same EN 50155 isolation as the rest of the cab.

Target BOM: $850 per board module, dominated by the LIDAR unit (~$550 for a Livox-class sensor in volume) + mmWave radar (~$550 / $275 volume) + stereo camera pair (~$165) + baseboard plus 2× RP2350 + CM5 (~$240 per the T-ECU/A BOM). Trainset total (2 modules): ~$1 700 per consist, well below the ~$150 k cab capex removed.

Form factor: 180 × 120 mm Eurocard — slightly larger than T-ECU/S to accommodate the four ultrasonic analog front-ends. Mounted behind the RF-transparent nose panel; cooling is ambient-air natural convection.

Detailed v2 spec: scaffold at `control-electronics/t-obs/schematics/v2-spec/` follows the T-ECU/S v2 template (block diagram, power budget, pinouts, connector table, safety-net routing rules). KiCad capture is v3 per §11 rollout.

6. Class W-SBC — Wayside

Role: Hosts `osr-consensus`, `osr-interlocking`, `osr-wayside-points`, `osr-level-crossing`, `osr-hot-axle-wayside`, `osr-balise` (reader side), `osr-energy-site`. Deployed in pole-mounted IP67 cabinets along the corridor and at each junction.

Environment: IEC 60529 IP67, EN 50121-4 EMC, −40…+70 °C, pole-mount vibration profile.

SoC baseline: Radxa CM5 (RK3588S — Cortex-A76 4-core + Cortex-A55 4-core big.LITTLE, 4/8/16 GB LPDDR4X, eMMC on-module). Radxa offers the industrial-temperature variant of the CM5 (−20…+85 °C grade) which matches the IP67-cabinet thermal envelope; the consumer-grade module is the drop-in for mild-climate deployments.

Rationale:

- Same SO-DIMM connector as the Raspberry Pi CM5, so a baseboard built around Radxa CM5 also accepts RPi CM5 (with the device tree recompiled). One carrier design, two module options.

- RK3588S is widely second-sourced (Rockchip is one of the few merchant SoC vendors with strong domestic semiconductor availability across the target regions).
- RK3588S ships with mainline Linux kernel support; Debian and Yocto both build without vendor forks.
- For safety-related wayside functions, Radxa hosts the application, diagnostics and communications. A separately qualified controller owns field proving, watchdogs and fail-safe outputs. Core pinning or a separation kernel remains useful defence in depth but is not the safety boundary.
- For non-safety sites (balise reader, energy site) the same module runs Debian straight-through.

Why not the RPi CM5 for wayside too: Raspberry Pi publishes a narrower industrial-temp range than Radxa (commercial 0…+50 °C vs Radxa's industrial +85 °C upper). Pole-mount cabinets in MENA / sub-Saharan routes routinely hit cabinet internals above +60 °C at midday; Radxa's variant has the margin.

Peripherals (baseboard, one SKU fits all wayside roles):

Peripheral	Qty	Purpose
TSN Ethernet	2	WAY-E A/B (redundant)
RS-485	4	Switch-machine motor drive, level-crossing barrier, HABD sensor bus, balise reader I/F
Isolated DI	8	Switch end-of-travel sensors (A+B), crossing strike detectors, field-fault inputs
Isolated DO	4	Motor directional contactors, crossing barrier lift/drop
4-wire PT100	4	Hot-axle IR temperature measurement
802.11ax AP	1	Depot maintenance console only; disabled in revenue service by default
Dual 24 V DC-in	2	Primary + backup site supplies
ATECC608 SE	1	Trust anchor

The same baseboard populates selectively: a switch site leaves the PT100s unpopulated; a HABD site leaves the switch-driver DOs unpopulated. One SKU, one spares stock, one support burden — the simplicity bet pays off in the depot.

Target BOM: €280 per board, €550 with IP67 enclosure and mounting hardware.

Form factor: 120 × 80 mm, inside a DIN-rail-mountable aluminium extrusion with IP67 M12 penetrations. Conduction-cooled to the enclosure.

7. Class S-SBC — Station / Depot

Role: Hosts `osr-psd`, `osr-pis-station`, `osr-station-scada`, `osr-afc`, `osr-tvm`. Deployed indoors; no rail-industry environmental rating required.

SoC baseline: Raspberry Pi CM5 on a commodity carrier (e.g. Waveshare CM5-IO). No custom baseboard needed — the station enclosure carries the I/O. This explicitly deviates from the other classes; indoor environments don't require the custom engineering.

Peripherals (commodity carrier):

- 2 × Gigabit Ethernet (one to station LAN, one to PSD/gate network)
- 4 × USB 3.0 for QR scanner, NFC reader, receipt printer, TVM bill acceptor
- HDMI for the passenger-facing display
- USB-C console
- ATECC608 on a small add-on board (I²C)

Target BOM: €130 per complete station-role SBC, including the CM5 and the add-on SE board.

8. Manufacturing envelope

Every reference PCB respects these constraints so that a tier-2 fab in any target region can produce it:

Constraint	Value	Source
Layer count	4	Lowest yield-loss tier.
Trace / space	0.15 mm / 0.15 mm	Routine at tier-2 fabs (6 mil).
Minimum via	0.3 mm drilled, 0.55 mm annular ring	No micro-vias.
Aspect ratio	≤ 8:1	Standard.
Min hole-to-pad	0.2 mm	Standard.
Passive minimum	0402	Hand-reworkable, SMT-line friendly.

Constraint	Value	Source
BGA	≤ 0.8 mm pitch, escape on 4 layers	CM5 module uses 0.65 mm BGA internally but exposes a SO-DIMM connector — no direct BGA escape needed on our baseboards.
SMT line min width	10 mm component edge-to-edge	Fits on any SMT line with ≥ 0.5 m conveyor.
Conformal coating	Acrylic (MG Chemicals 419)	Single-can hand coating acceptable for low volume; dip-coat line at high volume.

Nothing on the BOM requires any restricted-export component. Every part is a catalog item at Mouser/Digi-Key/LCSC. Local second-source is identified for every component in the Phase 3 procurement RFC (out of scope for this doc).

9. Reliability envelope

Class	MTBF target	MDBF target	Notes
T-ECU/S	≥ 200 000 h	≥ 500 000 km	Safety role — bigger than the rest. Figure taken from EN 50126 for on-vehicle SIL-4 nodes.
T-ECU/A	≥ 100 000 h	≥ 200 000 km	Single-redundant; failure is a comfort / diagnostic issue, not a safety issue.
W-SBC	≥ 150 000 h	n/a	Ten years of unattended pole-mount operation is the minimum acceptable.
S-SBC	≥ 50 000 h	n/a	Indoor environment; consumer-grade components acceptable.

Derating rules per Telcordia SR-332 Issue 4: thermal derate all active devices to ≤ 80 % of rated junction temperature at worst-case ambient plus 10 °C margin.

10. Pitfalls and decisions

These were flagged during planning — they're recorded here so the implementation session doesn't rediscover them.

- Two RP2350s on T-ECU/S, not one safety MCU with internal lockstep. Rule 3.3 of RFC 0005 requires SIL-4 code not to depend on SIL-2 code at runtime. Internal lockstep on a single die is a single-vendor claim; two physically-separate RP2350s cross-checking over an external SPI link are independently verifiable. The €2.20 BOM cost for two chips is trivial; the cross-domain auditability is worth orders of magnitude more in the safety case. Future v2 can move one of the two RP2350s to its Hazard3 RISC-V cores for genuine ISA-diverse redundancy without a BOM change.
- RPi CM5 for consumer-grade hosts, Radxa CM5 for industrial-grade hosts. T-ECU/A and S-SBC use RPi CM5; W-SBC uses Radxa CM5 for its wider industrial temp range. Both are SO-DIMM compatible, so a baseboard designed for either accepts the other with a device-tree swap. Operators in mild climates can single-source RPi CM5; operators in hot climates spec Radxa CM5 everywhere if they'd rather single-source the other way. The palette is two vendors only, by policy.
- No custom ASICs. An OpenSourceRail-branded silicon die is explicitly out of scope for v1 and v2. The cost is borne by accepting some commodity-module markup; the benefit is zero ASIC design NRE and a three-year faster path to first deployment.
- No NXP / ST / MilkV / StarFive on the BOM. Earlier drafts of this RFC used NXP S32K344 + i.MX 93 for the safety board and MilkV Jupiter for wayside. The two-vendor-only restriction (§3 constraint 3) replaced those with the RPi + Radxa palette above. The previous choices were technically sound; the new palette trades a small amount of safety-MCU certifiability for dramatic procurement simplicity and a tighter alignment with the two vendors' multi-year support commitments.
- Conformal coating is mandatory. Rail environments are humid, dusty, and occasionally sprayed. Every production board ships coated; lab samples for development use are coated before integration testing, not after, to catch solder-mask defects early.
- No fans. Fan bearings are a top-quartile failure mode in trackside equipment. Every reference design is conduction-cooled to its enclosure. Thermal headroom is the binding constraint, not CPU performance.
- Isolation on every external I/O. Digital I/O that leaves the board crosses a galvanic barrier. This is an EMC + safety bet: one lightning surge at the wayside must not take out a pole-mount W-SBC via an unpowered sensor line.

11. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	Bring-up board for each class on vendor dev kits (RPi CM5 dev kit, Radxa CM5 dev kit, Raspberry Pi Pico 2 — which carries the RP2350) — validate software boots and all peripherals enumerate. Runbooks are colocated with each host under `control-electronics/` (done 2026-04-22)	—
v2 (spec)	Board-level specifications for every host class: T-ECU/S at `control-electronics/t-ecu-s/schematics/v2-spec/` (done 2026-04-22), T-OBS at `control-electronics/t-obs/schematics/v2-spec/` (done 2026-04-22), T-ECU/A at `control-electronics/t-ecu-a/schematics/v2-spec/`, W-SBC at `control-electronics/w-sbc/schematics/v2-spec/`, S-SBC at `control-electronics/s-sbc/schematics/v2-spec/` (all done 2026-04-23). Each covers block diagram, power budget, connector tables, and (for SIL-4 boards) safety-nets. KiCad capture + gerbers + BOM are custom-board deliverables under CERN-OHL-S v2, not prerequisites for an RFC 0019 pilot build.	v1
v3	Compliance-test campaign for T-ECU/S (EN 50155 OT4, IEC 61373 Cat 1B, EN 50121-3-2) on one v2 prototype	v2
v4	Quantity-100 procurement package published — canonical BOM with Mouser / Digi-Key / LCSC lines, pick-and-place + stencil files, assembly drawings	v2
v5	First-article approval by a deployment operator	v4, RFC 0003

This RFC's v0 deliverable is only the plan. Every later phase is a dedicated piece of work owned by the hardware working group. The session that implements v1 starts by ordering three dev kits — Raspberry Pi Pico 2 (RP2350), Raspberry Pi CM5 IO Board, and Radxa CM5 IO Board — and boots the relevant Rust crates on each: the onboard safety stack on a pair of Pico 2 boards, the onboard app stack on RPi CM5, and `osr-consensus` + `osr-interlocking` + `osr-wayside-points` on Radxa CM5. The sim already runs every crate without hardware, so the bring-up is a migration, not a redesign.

12. Relationship to existing work

- Software stays unchanged. Every crate in `crates/` is designed to be hardware-agnostic Rust. The reference designs here are the first physical host; a second or third SoC family is additive, not breaking, so long as it exposes the same TCN-E / CAN-FD / GPIO surface specified in §§4–7.
- `control-electronics/` directory. The top-level directory now carries one subdirectory per class, board-level v2 specifications where available, RFC 0019 DIY assembly docs, and an explicit `control-electronics/release-checklist.md` with separate pilot-integration and custom-board fabrication gates. A class is only pilot-ready after the integration gate is closed, and only fabrication-ready after the custom-board gate is closed.
- Licensing. All hardware designs under CERN-OHL-S v2 per ARCHITECTURE.md §9. Documentation (this RFC) under CC-BY-SA 4.0. Firmware ports of existing crates keep their Apache-2.0 origin.

13. Open questions

1. Does the 2oo2 RP2350 pattern need an independent type-approval? Unlike internal lockstep silicon, the 2oo2 claim is a system-level argument that lives in the safety case. Our line is that an operator runs per-deployment type approval on the full T-ECU/S board anyway; the silicon-level certificate becomes a building block of the board certificate. Open: does this match EN 50129 appendix D's expectations for diverse redundancy? Resolve during v3 compliance-test planning.
2. ISA-diverse 2oo2 — when does it land? RP2350 can boot either Arm M33 or Hazard3 RISC-V. Running the two chips on different ISAs (identical Rust source, two different codegen back-ends) strengthens the diversity argument against common-mode compiler bugs. Not a v1 goal; track for v2 after the baseline 2oo2 is in service.
3. Which balise protocol? RFC 0003 proposed passive 27 MHz transponders; the safety board's footprint reserves a PN5180 NFC front-end (13.56 MHz) for now. Committing the passive-balise frequency band is a deployment-level choice — different operators have different RFID-spectrum allocations.
4. How does the energy-site W-SBC physically interface to the PV inverter and BESS? Modbus-TCP over an isolated Ethernet segment is the RFC 0002 v1 assumption; the same 4-port wayside baseboard already has it.
5. Repairability requirement. Should the target markets demand a specific MTTR (mean time to repair) at the field level? A 5-minute module swap is the baseline the DIN-rail + M12 strategy enables; whether that's sufficient for a country's availability SLA is a per-deployment conversation.

14. Done criteria

- [x] Every class's SoC and SoM pick is justified (§§4–7)
- [x] One baseboard per class, with a peripheral count (§§4–7)
- [x] Manufacturing envelope fits a tier-2 fab (§8)
- [x] Reliability targets allocated per class (§9)
- [x] Pitfalls and alternatives captured (§10)
- [x] Rollout ordered and scoped (§11)
- [x] Relationship to existing software + licensing named (§12)

The next session picks up at v1 — bring-up boards on vendor carriers. The expected sequence is (a) order the three dev kits, (b) bring up the Rust stack on each, (c) validate peripherals against this RFC's §§4–7, (d) file v1 deliverable as a `control-electronics/` folder update.

RFC 0008 — Rolling-Stock Reference Families

Repository source: `docs/rfcs/0008-rolling-stock-reference-design.md`

Status: Current Depends on: RFC 0020, RFC 0021, RFC 0022, RFC 0023

1. Common platform

Every OSR train family is assembled from self-contained, driverless car modules. A car has one powered bogie, one trailer bogie, one 225 kWh gross / 180 kWh usable LFP pack, two motor/controller channels, direct-HV DC HVAC, isolated low-voltage domains, and roof-PV MPPT.

The welded S355 underframe/spaceframe is the safety load path. Side and roof weather skins use CNC-trimmed fiberglass modules on a common 1,000 mm longitudinal pitch. Keyed hooks, captive over-centre clips, independent anti-lift retainers, and dry EPDM seals attach them to the frame without a production adhesive cure. The front/rear fiberglass shell remains sacrificial over the steel crash structure.

2. Families

Canonical values live in `lib/templates/rolling-stock.toml`.

Family	Cars	Gross / usable energy	Primary use
`urban-shuttle-1car`	1	225 / 180 kWh	Low-demand shuttle
`tram-2car`	2	450 / 360 kWh	Street/urban tram
`light-metro-3car`	3	675 / 540 kWh	Promoted reference train
`metro-4car`	4	900 / 720 kWh	Higher-capacity metro
`metro-6car`	6	1,350 / 1,080 kWh	High-throughput metro

The population-to-family mapping is a planning default. Platform, depot, alignment, demand, evacuation, and operating studies may select another family without changing its internal interfaces.

3. Promoted three-car train

Parameter	Reference
Cars	3 repeated 16.5 m modules
Overall length	49.5 m over couplers
Bogies	3 powered + 3 trailer
Powered axles	6
Installed motor capability	2.1 MW short duration
Operational traction cap	1.8 MW
Passenger capacity	360 AW2 / 480 AW3
Seats	60 longitudinal
Door arrangement	2 pairs per side per car
Low-floor centre	approximately 10 m per car
Maximum axle load	14 t at AW3 target
Maximum speed	90 km/h

The end saloons have panoramic glazing and no driving cab. T-OBS sensing, lighting, washer/heater, and service access are mounted in identical end-cowl modules.

4. Modular body rules

- 3-2-1 datum location with an asymmetric anti-reversal feature;
- 1,000 mm standard longitudinal mould/clip pitch, with shorter steel end-ring transitions outside the module grid;
- captive mechanical clips, anti-lift retention, and dry replaceable seals for side/roof modules;
- mechanical fail-safe retention;
- replaceable side, portal, skirt, ceiling, roof, and nose cassettes;
- molded empty ducts, drains, raceways, and mounting bosses; and
- no encapsulated cable, pipe, sensor, fan, filter, light, or connector.

For the promoted three-car family, 144 prefinished side/roof modules may be installed on three painted and dimensionally released frames in one eight-hour shift by six parallel two-person crews. This timing excludes doors, glazing, systems, bogies,

commissioning, homologation, and first-article testing.

5. Compatibility

The generator selects a track preset that lists the family in `compatible_consists`, derives platform length from the consist, and rejects incompatible station/track/family combinations. The source templates, rather than prose copies, are authoritative for numerical compatibility values.

6. Release gates

Release requires supplier envelopes, mass closure, axle loads, kinematic gauge, crash analysis, structural/fatigue FEA, weld and composite process qualification, fire/smoke/toxicity evidence, door/HVAC/traction integration, accessibility, EMC, maintainability, and first-article running evidence.

RFC 0009 — Track Design Standard

Repository source: `docs/rfcs/0009-track-design-standard.md`

Status: Draft — planning only, no civil drawings ship with this RFC Date: 2026-04-22 Depends on: RFC 0008 Rolling-Stock Reference Design

1. Summary

OpenSourceRail commits to three track-geometry presets, corresponding 1:1 to the rolling-stock families in RFC 0008. Every line in every deployment picks exactly one preset. Deviating from a preset means stepping outside the project's tested envelope — the simulator, the safety case, and the auto-gen pipeline all assume one of these three.

Preset	Gauge	Min radius	Max grade	Cant max	Rail profile	Compatible consists
`standard-urban`	1 435 mm	90 m	50 ‰	150 mm	UIC60	`tram-2car`, `light-metro-3car`
`standard-metro`	1 435 mm	200 m	35 ‰	160 mm	UIC60	`light-metro-3car`, `metro-4car`, `metro-6car`

> No-tunnel invariant. Per RFC 0011, every alignment is at-grade, elevated, or (at water crossings) bridge. The three presets above assume above-ground operation only; underground running is explicitly out of the project's upstream catalogue. Where a corridor cannot fit at-grade, the solver routes it elevated, never under. | `mainline-mixed` | 1 435 mm | 400 m | 25 ‰ | 180 mm | UIC60E1 | `metro-4car`, `metro-6car` |

The presets already exist as a schema in `lib/templates/track-geometry.toml`. This RFC promotes them into a committed engineering envelope with the civil rationale for every number.

2. Non-goals

- Not a civil drawing standard. This alignment/track-geometry RFC does not itself select slab versus ballast. RFC 0011 and the default OSR civil recipe select ballastless direct fixation for urban at-grade and elevated sections; a deployment may override that choice with lifecycle and local-capability evidence while remaining inside this RFC's geometry envelope.
- Not an earthworks rulebook. Cut/fill, drainage, embankment stability are owned by the deploying operator's civil team; this RFC specifies the alignment envelope they must respect.
- Not a signalling-plan document. Where signals / markers / balises physically go along the track is a deployment artefact — covered at a deployment level, not here.
- Not a tunnel/bridge standard. Those have their own envelopes (gauge clearance, ventilation, emergency egress) handled by `lib/templates/structures.toml` and a future structures RFC.
- Not a standards body. We reference EN 13848 (track geometry quality), UIC 505-1 (structure gauge), UIC 510 (wheel profile), UIC 720 (transition curves). We do not publish new ones.

3. Why three presets, not a continuum

A continuum of allowable geometry (pick your own radius, pick your own grade) doesn't fit the project's "simple for countries to implement" constraint. The sim's validation, the sensor-tuning (odometry, hot-axle), the brake-curve coefficients, and the station-archetype platform lengths are all calibrated against a specific envelope. Going off-preset means redoing every one of those calibrations at the deployment level.

Three presets cover the actual range of urban-rail deployments in the target regions:

- `standard-urban` is street-integrated light metro — the Samawah reference (RFC 0003).
- `standard-metro` is dedicated-ROW medium and large metro, the dominant deployment category worldwide.
- `mainline-mixed` is the edge case where the OSR-branded light metro shares track with existing heavy rail. Intentionally conservative — if you need this preset, you are outside the primary mission scope, and the envelope is deliberately close to a normal mainline spec to minimise surprises.

4. Gauge

1 435 mm standard gauge is the gauge for all upstream presets. It is the only gauge for which off-the-shelf wheelsets, rail profiles, switches, and fasteners are widely available from multiple global suppliers and from domestic producers in MENA, sub- Saharan Africa, South and Southeast Asia, and Latin America.

No Iberian gauge (1 668 mm), no Russian gauge (1 520 mm), no other non-standard upstream gauges. Operators in those regions are free to fork the project and add them; OpenSourceRail's upstream envelope stays at 1 435 mm.

5. Horizontal alignment

5.1 Minimum radius

Preset	Min radius	Derivation
`standard-urban`	90 m	UIC60 rail + `light-metro-3car` standard-bogie articulated consist + 22 m/s max speed — radius comes from balanced-cant condition at 15 m/s through the curve (2/3 of max), 150 mm max cant.
`standard-metro`	200 m	`metro-4car` + `metro-6car` consists are less forgiving; 200 m allows 25 m/s through a well-canted curve with unbalanced cant ≤ 60 mm (EN 13803-1 normal).
`mainline-mixed`	400 m	Mainline conservative. Allows 33 m/s (120 km/h) at 180 mm cant.

Transition curves (clothoid) are required at every curve-tangent interface per UIC 720. Minimum transition length is `L ≥ 0.5 × cant_mm × speed_m_s / 1000` (normal) — the auto-gen solver already enforces this via the civil-class inference.

5.2 Maximum gradient

Preset	Max grade	Derivation
`standard-urban`	50 ‰ (5 %)	`light-metro-3car` at 50 % powered gives enough tractive effort; 5 % is the practical limit above which wheelslip becomes frequent in wet conditions.
`standard-metro`	35 ‰ (3.5 %)	`metro-4car` / `metro-6car` fully-loaded (AW3) at 75–100 % powered axles; 3.5 % is the inflection where regenerative-braking downhill doesn't saturate the battery.
`mainline-mixed`	25 ‰ (2.5 %)	Mainline conservative; allows a light-metro consist to operate on a corridor sometimes used by heavier stock.

Gradient transitions (vertical curves) are not less than `R_v ≥ 0.4 × v²` where v is design speed in m/s — standard UIC metro practice, embedded in the auto-gen pipeline's civil-class inference.

5.3 Cant (superelevation)

Preset	Max cant	Rationale
`standard-urban`	150 mm	Street-integrated with some dedicated ROW.
`standard-metro`	160 mm	Dedicated ROW, standard metro number.
`mainline-mixed`	180 mm	Mainline standard.

Cant deficiency (unbalanced cant at design speed):

- Normal: ≤ 100 mm for `standard-urban`, ≤ 110 mm for `standard-metro`, ≤ 130 mm for `mainline-mixed`.
- Exceptional (with extra driver notice, slow orders): up to 150 mm.

5.4 Ruling speed

Per-preset `nominal_speed_mps` in the TOML is the design speed for the alignment; actual line speed is constrained by the smallest curve + grade + consist limit. This matches EN 13803-1 intent.

6. Vertical alignment

- Minimum crest/sag radius = `0.4 × v²` m with v in m/s.
- Vertical grades interpreted at station throats are limited to ≤ 2.5 ‰ regardless of preset — stopping on a grade is a long-tail fault that every operator tolerates badly.
- A grade is applied only in simple one-slope segments; compound grades are a civil-complexity red flag surfaced by the auto-gen's `quality.yaml`.

7. Track structure

7.1 Rail

Preset	Profile	Mass	Notes
`standard-urban`	UIC60 (60E2)	60 kg/m	Standard vignole; 260 Brinell grade.
`standard-metro`	UIC60 (60E2)	60 kg/m	Same as urban; heavier profile unnecessary for 14 t axles.
`mainline-mixed`	UIC60E1	60 kg/m	Equivalent mass, different head geometry — better wear profile when shared with freight.

7.2 Sleepers / slab

The geometry preset does not itself fix slab-track versus ballast. The upstream civil baseline in RFC 0011 is ballastless direct fixation for urban at-grade and elevated sections; deployments record any approved override. Constraints the presets do fix:

- Sleeper spacing: 600 mm typical, 550 mm in curves below 2× min radius for all ballasted presets.
- Fastener: Pandrol fastclip family or equivalent elastic-rail fastener; no rigid K-type fasteners (heavy maintenance).
- Direct-fixation fasteners must remain adjustable whether the support is an OSR-ST6 panel, a decked Pi-beam or a project-specific cast-in-place plinth.

7.3 Turnouts

Preset	Typical turnout tangent	Divergent speed
`standard-urban`	1:9	40 km/h
`standard-metro`	1:9 (station ends) / 1:14 (mainline crossovers)	40 / 60 km/h
`mainline-mixed`	1:14 / 1:18.5	60 / 80 km/h

The `osr-wayside-points` software already handles any tangent; this RFC pins the civil choice.

8. Structure gauge

All three presets respect UIC 505-1 structure gauge (static envelope at 3.15 m wide × 4.32 m tall above top-of-rail), with a dynamic outline adding ± 80 mm lateral and ± 50 mm vertical for the design-maximum cant + sway. Per-consist structure gauges come from RFC 0008 §3.1; no family violates the static envelope.

9. Track geometry quality

Track geometry quality is measured and maintained per EN 13848:

Indicator	Class QN1 (alert)	Class QN2 (immediate action)
Longitudinal level D1	6 mm	10 mm
Alignment D1	5 mm	7 mm
Cant	15 mm	20 mm
Twist (2 m base)	4 mm/m	7 mm/m

Measurement cadence per preset:

Preset	Recording-car cadence	Hand-measurement after each weather event
`standard-urban`	90 days	Yes
`standard-metro`	60 days	Yes
`mainline-mixed`	45 days	Yes

A recording car is explicitly not required to be OpenSourceRail hardware; any EN 13848-compliant recording trolley from the commercial market is acceptable. Track-geometry ingestion belongs in `osr-alignment`; standard GPM/RECD import remains a follow-on parser.

10. Self-consistency constraints

The auto-gen pipeline enforces the compatibility matrix from RFC 0008 §4. Every preset declares `compatible_consists`; the emitter picks a rolling-stock family first (by population band) and then selects the tightest geometry preset that still lists that family as compatible:

pop band	chosen family	chosen geometry preset
≤ 300 k	tram-2car	standard-urban

```
300 k … 1 M      light-metro-3car        standard-urban
1 M … 3 M        metro-4car              standard-metro
≥ 3 M            metro-6car              standard-metro OR mainline-mixed
```

The emitter default for each band is the first entry in the choice list; the per-deployment override is a single line in the city recipe.

11. Pitfalls and decisions

- We pick UIC60 over IRJ-equivalents. Rail profile is where we take the standard path rather than the simple path — UIC60's global supply chain is well-stocked and second-source. A simpler lighter-profile would save mass but lose that supplier depth.
- Geometry stays neutral; the civil baseline is explicit. RFC 0011 selects ballastless OSR-ST6/direct fixation upstream. A deployment changes it only through a recorded lifecycle, supply-chain and maintainability decision.
- No mixed-gauge track. Dual-gauge trackwork is explicitly out of scope. Operators with that legacy have to choose one gauge to extend outside the upstream OSR envelope.
- No tilting rolling stock. The rolling-stock families in RFC 0008 do not tilt; cant-deficiency limits are therefore conventional. If a future family adds tilting, the presets gain a `tilting_allowed` attribute and more permissive cant deficiency.
- Maximum axle load is the consist, not the track. Track is designed for ≥ 16 t axle load across every preset; the rolling stock stays under 14 t by design. This gives a future maintenance / utility vehicle a headroom to share the track.

12. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	Track preset compatibility matrix ratified for generated city designs; superseded hand-authored alignment examples were removed so deployments export fresh OSR-ALN from the current city model and survey package.	RFC 0003
v2	Emitter enforces geometry/consist compatibility + emits `geometry` per line in auto-gen output (done 2026-04-22). OSR-ALN alignment interchange format at `docs/civil/osr-aln-format.md` — tool-agnostic TOML schema civil firms export to.	v0, RFC 0008 v2
v3	Parametric track components at `design/component-catalogue/src/osr_mech/track/` (done 2026-04-22), including the controlled 1:9 turnout kit. Civil-tool bridge: `tools/osr-aln-convert/` — stdlib-only Python toolchain: `landxml-to-osr-aln` reads LandXML 1.2 exports from Civil 3D / Bentley OpenRail / Trimble BC / QGIS; `current-network-to-osr-aln` deterministically exports the generated three-line Samawah planning package with source hashes; `osr-aln-validate` enforces the 8 hard gates + 3 soft gates and resolves both line and station IDs against a deployment's design.toml. 26 passing tool tests. Remaining for v3.1: Trimble TCL CSV reader, Bentley cant-extension parser, and CERN-OHL-S v2 relicensing.	v1
v4	`osr-track-geometry` crate that ingests EN 13848 recording data and feeds the CBM pipeline	v3
v5	First-article track constructed and recorded at a pilot deployment	v1, v3

13. Relationship to existing work

- `lib/templates/track-geometry.toml` — the Lego-block schema that this RFC ratifies.
- `crates/osr-routing/src/civil.rs` — already classifies every 20 m cell as at-grade / elevated / bridge / bored tunnel; the civil inference respects minimum curve and max grade from the preset but uses hard-coded numbers today. Emitter v2 reads them from this RFC.
- `crates/osr-design` — emits `geometry = "standard-urban"` as a hardcoded line attribute today; v2 picks per §10 above.
- `osr-sim` — its `Section` type has `length_mm` and `max_speed_mps`; gradient and cant are unexposed today and land as v4 when the simulator gains a kinematic model beyond the current trapezoidal profile.

14. Open questions

1. Should `standard-urban` allow steeper grades in short segments (say, 70 ‰ over < 100 m) for ramping to elevated structures? Opens a per-segment override vs clean preset. Resolve when a surveyed alignment is produced for the first pilot.
2. Metric vs imperial-historical gauge. Does the mission scope ever need to support Cape gauge (1 067 mm) for southern Africa legacy? Revisit after the first sub-Saharan pilot.
3. Slab-track cost model — the per-site BOM comparison between ballast + slab + fastener + sleeper over 30 years. A planning-grade spreadsheet that the operator can localise would help — candidate output of v3.
4. Expansion joints / CWR. Continuously welded rail is the default; the RFC should commit on stress-free-temperature profile per climate band. Defer to a separate climate-adapter addendum.

15. Done criteria

- [x] Four presets committed with engineering rationale (§§3–7)
- [x] Gauge policy fixed (§4)
- [x] Horizontal + vertical alignment limits stated (§§5–6)
- [x] Rail / sleeper / turnout envelope named (§7)
- [x] Structure gauge named (§8)
- [x] Geometry-quality thresholds fixed (§9)
- [x] Rolling-stock geometry compatibility matrix (§10)
- [x] Pitfalls and alternatives explicit (§11)
- [x] Rollout ordered (§12)
- [x] Relationship to software + templates named (§13)

The next session picks up at v2 — emitter compatibility, a single change in `crates/osr-design/src/emit.rs`.

RFC 0010 — Station Design Standard

Repository source: `docs/rfcs/0010-station-design-standard.md`

Status: Draft — planning only, no architectural drawings ship with this RFC Date: 2026-04-22 Depends on: RFC 0008 Rolling-Stock Reference Design, RFC 0009 Track Design Standard

1. Summary

OpenSourceRail commits to six base station archetypes plus the controlled `interchange-elevated` variant covering every stop in every deployment. The catalogue entries already exist as a schema in `lib/templates/stations.toml`; this RFC promotes them into a committed architectural envelope with platform geometry, passenger-flow throughput, accessibility, fire egress, and canopy-PV integration spelled out.

Station placement follows a light-metro spacing rule: ordinary same-line stations target 1.6 km spacing in the highest-demand centre, with ordinary spacing widening to 3 km in the wider urban area and up to 7 km on suburban approaches / the lowest-demand outer fringe. This is selected from the demand raster rather than applied as a blanket suburban spacing, unless a deployment explicitly selects a tram-like stop pattern. In city centres, nearby cross-line platforms should be consolidated into one interchange complex rather than emitted as separate station assets. The validation gate uses a 600 m walkable transfer envelope for ordinary cross-line approaches and a 600 m station-complex consolidation rule.

Archetype	Platforms	Layout	Canopy	Charging	Dwell	Role
`halt`	1	side	solar-canopy	none	60 s	rural / low-demand stop; no trackside energy site
`standard`	2	side	solar-canopy	500 kW	60 s	default urban / suburban
`major`	2	island	solar-canopy	500 kW	60 s	anchor stop (hospital / mall / centre)
`interchange`	4	same-grade transfer	solar-canopy	500 kW	60 s	two-line junction
`interchange-elevated`	4	stacked	solar-canopy	500 kW	60 s	elevated variant of `interchange`
`terminal`	2	side	solar-canopy	1 000 kW	60 s	end-of-line with turnback
`depot-terminal`	2	side	solar-canopy	1 000 kW	240 s	end-of-line + fleet depot

Platform length is not fixed by the archetype — it is derived at design-emission time from the line's rolling-stock choice:

```
platform_length_m = consist.length_m + station.platform_clearance_m
```

with `platform_clearance_m` defaulting to 6 m for `halt` and 10 m elsewhere. This keeps archetypes consist-agnostic and prevents drift between the rolling stock, the station, and the line plan.

2. Non-goals

- Not a full architectural spec. Column spacing, façade materials, HVAC layout, and wayfinding graphics are architect-of-record decisions per deployment. This RFC fixes the functional envelope the archetype must meet.
- Not a fare-system spec. `osr-afc` + `osr-tvm` handle the ticketing logic. This RFC specifies the number of gate lanes and TVMs per archetype, nothing about payment UX.
- Not a passenger-experience spec. Seat count per platform, vending machine placement, retail integration — operator choices.
- Not an emergency-management plan. Emergency egress calculation is specified; the incident response procedures that use those egress paths live in the operations rulebook (separate, per deployment).
- Not a land-use spec. Parking, bus-bay integration, TOD (transit-oriented development) are municipal planning work outside OSR's scope.

3. Why six base archetypes, not a continuum

A continuum of allowable station designs — pick your platform length, pick your canopy, pick your fare gate count — fractures the per-spare-part economics and the simulator validation. Six base archetypes and one controlled elevation variant cover the real distribution of stops:

- `halt` for rural / low-demand points where a full station would over-build.
- `standard` is the bread-and-butter urban stop.
- `major` for generator stops (hospital, mall, transit square).

- `interchange` for where two OSR lines meet. The default is a same-grade transfer with direct pedestrian access; the controlled `interchange-elevated` variant lifts one line when geometry or road conflicts make that unavoidable (see RFC 0011 §7).
- `terminal` for end-of-line with turnback tracks.
- `depot-terminal` for end-of-line + the fleet parking + the maintenance shop.

On the current generated Samawah model (RFC 0003), the station mix is 3 `halt`, 6 `standard`, 15 `major`, 5 `terminal`, 1 `depot-terminal`, and 3 `interchange-elevated` stops. The archetype catalogue stays shared; each generated city only selects counts and site adaptations.

4. Platform

4.1 Length

Derived, not fixed. For every consist/archetype pair:

Consist (RFC 0008)	consist length	clearance	platform length
`urban-shuttle-1car`	21 m	6 m (halt) / 10 m (others)	27 / 31 m
`tram-2car`	39 m	6 / 10 m	45 / 49 m
`light-metro-3car`	49.5 m	6 / 10 m	55.5 / 59.5 m
`metro-4car`	75 m	10 m	85 m
`metro-6car`	111 m	10 m	121 m

4.2 Height

Defined by the rolling-stock family the line uses:

- Every OSR rolling-stock family uses the same low-floor centre door zone at 350 mm above top-of-rail (ToR), with raised floor over the standard bogies. For at-grade stations, the platform walking surface is set at the local pedestrian pavement level; the rail/top-of-rail datum sits 350 mm below that surface inside the drained guideway channel. In other words, the platform is flat with the street/sidewalk, while level boarding is kept by lowering the rail datum through the station bay rather than raising passengers onto a separate platform structure.
- Elevated or stacked stations keep the same 350 mm platform-to-ToR boarding datum, but those cases are the exception and are driven by alignment conflicts rather than the station archetype default. Gap-fillers are only needed on tight curved platforms where the horizontal gap exceeds the accessibility limit.

Mixed-consist lines are not supported (RFC 0008 §5 fixes one family per line). A line that carries mixed heights is a safety red flag rejected at design-emission time.

4.3 Width

Archetype	Minimum usable width	Notes
`halt`	2.5 m	Single-platform, assumes ≤ 5 boardings/min at peak.
`standard` (side)	3.5 m per side platform	Per-platform; total station width with two side platforms + track = 3.5 + 1.435 + 3.5 = 8.4 m minimum.
`major` (island)	6.0 m island platform	One platform serves both directions; drops footprint versus side-side.
`interchange` (same-grade transfer)	4.5 m per platform	Transfer paths stay at pedestrian grade where the alignment permits.
`terminal`	4.0 m per side platform	Accommodates peak alighting surge from inbound terminal runs.
`depot-terminal`	4.0 m per side platform	Same as terminal.

Minimum widths are derived from peak boarding flow at the archetype's design pphpd plus egress (§6) and wheelchair circulation (§7). Operators may go wider; below the minimum is rejected.

4.4 Edge treatment

- Platform edge coping: tactile paving 600 mm deep along the full platform edge per ISO 23599 (compliant with regional disability-access regulations).
- Horizontal gap: ≤ 75 mm between platform edge and vehicle sill at the consist's door — UIC 741 normal.
- Vertical gap: ≤ 50 mm between platform and sill — same.

- Platform screen doors (PSD): optional on every archetype, required on `metro-4car` and `metro-6car` from v1. Full-height PSDs preferred over half-height for environmental reasons (HVAC doesn't bleed to open-air). `osr-psd` (crates/osr-psd) handles the controller.

5. Access and circulation

At-grade stations default to direct pedestrian-level access: sidewalk / forecourt fare line or validator plinth platform, with no lift, stair, ramp, or overbridge. A local ramp is only required where the surrounding street itself has a level mismatch; it is not part of the OSR standard station kit.

Grade-separated circulation is required only for elevated stations, stacked transfer sites, or road/ROW constraints that a local authority will not permit as a protected same-grade crossing:

Mode	Capacity	When required
Direct paved path	Site-limited	Default for `halt`, `standard`, `major`, `terminal`, and `depot-terminal` when at grade.
Protected pedestrian crossing	≤ 60 pax/min per 1.5 m width	Used when passengers must cross a track/road at pedestrian level; interlocked with the signalling system.
Ramp (max 1:20 preferred, 1:12 absolute max)	≤ 60 pax/min per 1.5 m width	Local street/forecourt level mismatches only.
Stair	≤ 80 pax/min per 1.5 m width	Elevated or stacked stations only.
Escalator (one-way, 0.5 m/s)	100 pax/min per unit	Elevated/stacked `interchange`, `terminal`, or high-demand local overrides.
Elevator (≥ 1 100 × 1 400 mm car)	10 pax/min per car	Elevated or stacked stations; not required for flat at-grade platforms.

Total upward capacity at peak must exceed expected alighting rate + 25 % reserve. The auto-gen emitter computes this at design time.

6. Egress

Emergency egress capacity is fixed at 4 minutes from the furthest passenger on the platform to a safe point of discharge. This matches NFPA 130 and the industry consensus for tunnel/station design.

Minimum effective egress width per direction:

- `halt`: 1.0 m (one direction of travel only — the passenger exits to street level directly).
- `standard`: 2.0 m aggregate.
- `major`: 3.0 m aggregate.
- `interchange`: 4.0 m aggregate, redundant on each level.
- `terminal`: 3.0 m aggregate, with distinct primary + emergency routes.
- `depot-terminal`: same as terminal.

A single exit is never the only path. Even a `halt` must have two evacuation directions — either two opposite-side street accesses or a platform exit + a walk-path along the rail reserve.

7. Accessibility

All archetypes meet these minima (per the accessibility template `lib/templates/accessibility.toml`):

- Step-free path from street to platform on every approach.
- Tactile paving at every platform edge, validator/fare-line approach, and stair/escalator landings where such vertical circulation exists.
- Audio + visual train-arrival announcements on every platform.
- Seating on every platform — `halt` excepted.
- Wheelchair space at 1 per 30 m of platform length; co-located with door-2 of every consist for quickest dwell.
- Ticket gate accessibility: at least one gate per direction is a wide wheelchair gate (≥ 900 mm).

8. Fare gates + TVMs

Archetype	Fare-gate lanes per direction	TVMs per direction
`halt`	0 (open)	0 — buy via app / TVM at neighbouring standard station
`standard`	2	1
`major`	3	2
`interchange`	4 (each level)	2 per level
`terminal`	3	2
`depot-terminal`	3	2

Gates are driven by `osr-afc`; TVMs by `osr-tvm`. At `halt` stations the line is still revenue-positive because fare inspection happens on-board by a roving inspector or via QR-scan-on-boarding (GoA 4).

9. Canopy and solar

Every archetype carries a solar canopy over the platform — the primary PV source the `osr-energy-site` controller dispatches from. Site canopy/PV area targets from the template:

Archetype	Canopy area
`halt`	400 m²
`standard`	1 800 m²
`major`	2 000 m²
`interchange`	2 800 m²
`interchange-elevated`	3 200 m²
`terminal`	2 400 m²
`depot-terminal`	1 500 m² (smaller canopy — yard PV dominates)

The area target is the total station-site shade/PV allowance. The parametric passenger-platform canopy covers the circulation zone; auxiliary forecourt or concourse modules close any difference to the site target and must be listed separately in the station BOM.

The controlled auxiliary product is an 8.5 m × 22 m (187 m²) solar-roof bay derived from the reviewed station-spanning truss envelope. Adjacent bays share 22 m transverse frames: `N` roof modules use `N + 1` frames and `2 × (N + 1)` column foundations. The BOM rounds module count up, so installed shade/PV area never falls below the template target. Parametric review geometry and quantity functions live in `design/component-catalogue/src/osr_mech/station/auxiliary_canopy.py`.

Canopy design envelope:

- Prefabricated bolt-together steel portal frames. No on-site welding, no wet concrete beyond pad footings. One standard 6 m bay module (HEA 200 column + HEA 180 cantilever rafter + SHS 100 bracing) replicates along the platform. Hot-dip galvanised in the fabrication shop. Detailed parametric geometry at `design/component-catalogue/src/osr_mech/station/portal.py`; FreeCAD/CAD-review artifacts under `design/component-catalogue/catalog/`.
- Solar-integrated roof sandwich panel. One factory-bonded panel per bay: CIGS or lightweight c-Si PV on standing-seam galvanised steel, polyurethane foam core, white-coated underside. Delivered pre-terminated with MC4 connectors; bays plug into each other to form a single DC string. ~200 W/m², ~20 kg/m². See `design/component-catalogue/src/osr_mech/station/solar_roof.py`.
- Cantilever-from-rear geometry. Columns sit at the platform rear edge only (no columns at the platform edge); the rafter cantilevers 3.5 m + 0.7 m eave over the platform. Middle third of the platform is column-free for passenger circulation by construction, not by careful placement.
- Roof: 1:15 mono-pitch (3.8°) toward the track shoulder. Rainwater runs off directly onto the track drainage; integrated gutter only at the rear column face. In the dry target regions we do not bank on rain; the mono-pitch handles the occasional storm without integrated gutters at all.
- Shading coefficient target: ≥ 90 % midday-summer coverage of the platform.
- No station building. Fare gates on rolled-steel plinths at the platform entry; PIS / CCTV / lighting / radio mount to the canopy columns. The whole station is the canopy + the ground-level platform slab/guideway channel + the plinth — no masonry, no curtain wall, no on-site architecture.

Erection: the current `standard × light-metro-3car` parametric kit has 10 bays per platform. Each platform frame contains approximately 3.3 t of steel before roof panels and services and fits within a normal articulated-lorry payload. It is erected in 3–5 days with a small crew and a 30 t crawler crane. No structural engineer is needed on site — the deployment

partner picks the kit size from the catalogue; load envelopes are published in the source docstrings.

Canopy-PV sizing math lives in RFC 0002; `design/component-catalogue/src/osr_mech/station/canopy.py` exposes `canopy_kwp(archetype, consist)` which `osr-energy-site` can consume directly. One `standard × light-metro-3car` platform canopy (10 bays) produces ~43 kWp; a two-side-platform station therefore has about 86 kWp over its passenger platforms before auxiliary site PV.

10. Passenger-flow model

The auto-gen emitter computes expected peak passenger flow per station using:

```
boarding_per_hour = demand_at_station × population_capture_radius
                    × modal_share × peak_factor
```

with planning-grade defaults:

- `demand_at_station`: from the `osr-routing` raster (0..1 normalised).
- `population_capture_radius`: 500 m walking, 1 500 m bike+feeder bus.
- `modal_share`: 8 % of catchment population per day.
- `peak_factor`: 0.12 (peak hour share of all-day boardings).

Outputs per station:

- Expected peak boardings/minute.
- Required platform width (from §4.3) — must fit.
- Required vertical-circulation capacity (from §5) — must fit.
- Required egress capacity (from §6) — must fit.

Mismatch surfaces in `design-quality.yaml` as a hard gate, just like station-spacing does today.

11. Thermal + environmental

- Outdoor stations (the common case in the target regions): passive cooling via canopy shade + cross-ventilation. No HVAC on the platform; just on enclosed spaces such as a TVM kiosk, staff room, or control cabinet where the operator chooses to add one.
- Underground stations (where the civil segment classifies as tunnel): mechanical ventilation required; emergency smoke extraction per NFPA 130; cooling by chiller or absorption unit fed from the depot-scale PV.
- Interior noise (outdoor platform): ≤ 80 dB(A) during train approach at 80 km/h per ISO 3095-1.
- Lighting: 100 lx on platform at night per EN 12464-2.

12. Self-consistency with RFC 0008 + 0009

The auto-gen pipeline enforces:

- `station.platform_length_m` = `consist.length_m + station.platform_clearance_m`
- `station.platform_to_tor_height_mm` = `consist.floor_height_mm`
- At-grade station walking surfaces remain at the local pedestrian datum; only the rail/guideway channel drops to satisfy the platform-to-ToR height.
- Every `interchange` archetype on line A also exists in line B's station list (otherwise the operator has two adjacent but disconnected stations, a common error).
- Every `terminal` archetype is at a line endpoint (not mid-line).
- Every `depot-terminal` sits at a line endpoint and is tagged `is_depot = true`.

Mismatch fails the `design-quality.yaml` hard gate.

13. Pitfalls and decisions

- Level boarding is a hard requirement. Some operators push back on the platform-height cost (either low-floor needs floor adaption on existing infrastructure, or high-floor needs taller platforms). We absorb that cost because level boarding is the single biggest accessibility win — it's not optional.

- Solar canopy is mandatory on non-halt archetypes. This is a project-wide bet, not a station-level choice. Operators who want to skip the canopy must fork the project; upstream keeps the bet.
- Platform screen doors are optional below `metro-4car`. PSDs are a top-quartile CAPEX line; for `tram-2car` and most `light-metro-3car` cases the simpler yellow-line + tactile edge is sufficient. Above that capacity, PSDs become mandatory — the risk/benefit crosses over.
- No island platform at `standard`. Island platforms halve platform width for equal footprint but require passengers to choose direction before descending, which slows wayfinding. We reserve them for `major` and above where the throughput win dominates.
- Depot-terminal is a single archetype. Some operators have depot-only facilities (freight, maintenance) separate from passenger terminals; those are out of scope — OSR deployments co-locate the two. This is a simplicity bet; a future depot-only archetype is opt-in.

14. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	Architectural envelope + canopy structural first-pass + accessibility + services + compliance matrix for the shared `standard` archetype at `docs/stations/standard-archetype/` (done 2026-04-22). Deployment station lists come from the generated city model before procurement.	RFC 0003
v2	Emitter: terminal / interchange auto-detection + platform-length derivation from consist + depot-terminal promotion at the farthest radial endpoint (done 2026-04-22). Architectural drawing register for the `standard` archetype at `docs/stations/standard-archetype/drawing-register.md` — 43 drawings across A/S/M/E/F/T disciplines with scale + size + v1-envelope cross-refs. Quality-gate failure on compatibility mismatch is deferred to v3.	v0, RFC 0008 v2, RFC 0009 v2
v3	Parametric prefab catalogue landed at `design/component-catalogue/`: platform and auxiliary canopy modules, solar-roof panels, ground-level slab/guideway-channel edge modules, elevated L-units, and separate fare/TVM plinths all resolve into generated BOMs and travelers. `STANDARD × light-metro-3car` emits a 10-bay / ~43 kWp canopy per platform plus seven 187 m² auxiliary bays. Tracked CAD review artifacts use compact FreeCAD/PNG outputs; local neutral exports can be generated when a partner toolchain needs them.	v1
v4	Platform-flow simulator extension in `osr-sim` — peak-hour passenger flow against archetype capacity	v2
v5	First-article station for a deployment instance	v1, RFC 0003 §5

15. Relationship to existing work

- `lib/templates/stations.toml` — the Lego-block schema this RFC ratifies.
- `lib/templates/platform-doors.toml` — PSD calibration defaults.
- `lib/templates/accessibility.toml` — the accessibility envelope §7 references.
- `crates/osr-design/src/emit.rs` — already picks a station archetype (halt/standard/major) from demand thresholds. v2 adds terminal/interchange detection, per-archetype platform length, and the compatibility gate.
- `crates/osr-psd`, `crates/osr-afc`, `crates/osr-tvm`, `crates/osr-pis-station`, `crates/osr-station-scada` — the station-side software already targets this envelope.

16. Open questions

1. `interchange` at 3-line junctions. Stacked 4-platform works for 2 lines crossing. A 3-line junction needs a different archetype; extend to `mega-interchange` when an actual deployment needs it.
2. Outdoor air-quality filtering for stations near heavy road traffic (common in target corridors). HEPA-grade canopies are a real market niche; candidate v4 addendum.
3. Depot-terminal scaling. For deployments with more than 20 trainsets, the 1 500 m² canopy + yard footprint may under-serve. Introduce a `mega-depot` archetype at that threshold.
4. Retail footprint. Some operators depend on retail concessions for revenue. The `standard` archetype should accommodate ≈ 50 m² optional retail without redesign; confirm at v3.

5 Overnight stabling at `terminal` vs `depot-terminal`. The line's fleet parks overnight at the depot-terminal, not at a normal terminal. How does the auto-gen pipeline enforce this when the population distribution picks two terminals and no obvious depot site?

17. Done criteria

- [x] Six base archetypes plus the controlled elevated-interchange variant committed (§1)
- [x] Platform geometry envelope (§4) + access/circulation (§5) + egress (§6)
- [x] Accessibility minima (§7) + fare-gate sizing (§8) + canopy-PV sizing (§9)
- [x] Passenger-flow model (§10)
- [x] Self-consistency rules with RFC 0008/0009 (§12)
- [x] Pitfalls + alternatives explicit (§13)
- [x] Rollout ordered (§14)
- [x] Relationship to existing software + templates (§15)

The next session picks up at v2 — emitter upgrade, a shared deliverable with RFC 0008 v2 and RFC 0009 v2 (same `emit.rs` file).

RFC 0011 — Civil Infrastructure Design Standard

Repository source: `docs/rfcs/0011-civil-infrastructure-design-standard.md`

Status: Draft — planning only, no structural drawings ship with this RFC Date: 2026-04-22 Updated: 2026-08-17 — OSR Rapid Viaduct Kit correction Depends on: RFC 0008 Rolling-Stock Reference Design, RFC 0009 Track Design Standard, RFC 0010 Station Design Standard

1. Summary

OpenSourceRail commits to two civil classes only — at-grade and elevated — with bridges for water crossings as a narrow third class. No tunnels. Every corridor the auto-gen pipeline produces, and every hand-authored corridor the operator commits, will sit on one of:

Class	Share (target)	Description
at-grade	≥ 70 %	Track sits on prepared subgrade at natural ground level (possibly with minor cut/fill ≤ 3 m). The cheapest and fastest class by 1–2 orders of magnitude.
elevated	≤ 25 %	Track sits on a viaduct structure 5–12 m above ground (precast OSR-Pi20/Pi25 decked beams, standardised span).
bridge	≤ 5 %	Over a water crossing or deep gap. Separately engineered special span with site-specific substructure.

The fourth class the first-pass auto-gen inference used — tunnel (cut-and-cover and bored) — is removed from the catalogue. Where an alignment previously would have gone underground, it now goes elevated over the same footprint, or deviates around the constrained zone on at-grade. Operators who deliberately want a tunnel are forking the project; OpenSourceRail upstream does not carry tunnel designs, tunnel cost models, or tunnel-specific signalling / ventilation / evacuation engineering.

2. Non-goals

- Not a tunnel standard. By construction. The project's cost model, the simulator's energy budget, the sensor-tuning for GNSS + balise, and the operations rulebook (RFC 0013) are all calibrated against above-ground operation. Going underground reopens every one of those calibrations.
- Not a geotechnical rulebook. Soil survey, pile type, foundation depth, seismic detailing — owned by the deploying operator's civil team. This RFC fixes the functional envelope (span range, clearance, load class) not the method.
- Not a bridge design manual. Spans above 30 m, water crossings, turnouts, and exceptional crossings use a separately engineered segmental, I-girder, or steel-composite product. They are not a stretched version of the standard full-span trough.
- Not a land-acquisition policy. How a deployment acquires right-of-way (ROW) — expropriation, purchase, existing-rail corridor reuse, new-build — is municipal work.
- Not a standards body. We reference EN 1990 (Eurocode basis), EN 1991-2 (traffic loads on bridges / viaducts), EN 1998 (seismic), UIC 777-2 (infrastructure gauge), local building codes where deployed. We do not publish new ones.

3. Why no tunnels

A planning-grade cost comparison for a 1 km segment in typical dense-urban conditions across the target regions:

Class	OSR target / retained benchmark (USD/km)	Build time	Post-build maintenance
At-grade dedicated ROW	2.584 M / 3.0 M	6–12 months	Low — ballastless slab/embedded trackform
Elevated viaduct	9.748 M / 12.0 M	12–18 months	Medium — bearing + expansion-joint inspection
Bridge over water	18.0 M / 18.0 M	18–24 months	Medium — same as viaduct + scour inspection
Cut-and-cover tunnel	60–120 M conventional	30–48 months	High — ventilation + pumping + egress drills
Bored tunnel	90–200 M conventional	48–72 months	High — same as above, plus tunnel-boring-machine commissioning

Numbers are planning-grade, calibrated against generated deployment instances including Samawah context (RFC 0003 §2.3).

A tunnel segment is 10–40× the CAPEX and 4–8× the build-time of an equivalent at-grade segment. It also brings a permanent operational overhead: mechanical ventilation, flood pumping, fire suppression, emergency evacuation plans, and radio-coverage retrofit. Every one of those is an additional subsystem the deploying operator must staff and maintain.

The mission is "urban rail affordable enough that a developing nation can finance it domestically." A corridor with any tunnelling in it is outside that envelope. The design decision is: if the alignment can't go at-grade or elevated, the alignment should deviate around the constraint. Even at a 20 % longer corridor, the detour is cheaper than the tunnel it avoids.

4. At-grade — the 70 % case

4.1 Envelope

- Subgrade: compacted fill or in-situ soil graded to +/-25 mm of design. Capping layer of 150 mm crushed stone, then ballastless reinforced slab / embedded direct-fixation plinths for the running rails. The default urban trackform avoids ballast migration, tamping cycles, and loose aggregate at crossings.
- Earthworks: cut / fill up to 3 m depth only. Beyond 3 m the segment is either:
  - Re-classified to elevated (bridge over the gap / viaduct over the hill), or
  - Deviated around the constraint.
- Drainage: transverse slope >= 2 % either side; longitudinal channels every 50 m; culverts at every flow-path crossing. Drainage is not optional: a flooded slab undermines fastener integrity, corrodes embedded hardware, and damages the subgrade.
- Fencing: dedicated ROW is fenced the full length on both sides. No uncontrolled pedestrian crossings. Level crossings only where a city street intersects and an underpass / overpass is prohibitively expensive. See RFC 0012 for the level-crossing equipment standard.
- ROW width: minimum 10.5 m for double-track at-grade (two tracks × 3.5 m wide plus 1 m each side clearance plus fencing allowance); 7 m for single-track.

4.1.1 Narrow old-town single-track option

Some historic cores and market streets cannot fit a 10.5 m dedicated double-track reservation without demolition, but also do not justify viaduct intrusion. OpenSourceRail therefore permits a constrained single-track at-grade segment as an at-grade variant, not a fourth civil class.

Rules:

- Use only for short constrained sections where double track would require demolition or permanent removal of the street's essential pedestrian/service function.
- Fit the running way inside the 7 m single-track envelope, with protected pedestrian edges and no mixed running with road traffic.
- Provide double-track passing loops only at stations, portals, or immediately outside the constrained street.
- Interlocking treats the whole single-track section as one opposing-move exclusion zone unless a deployment-specific signalling design divides it into shorter protected blocks.
- Timetables must prove the desired headway through the single-track section before the option is accepted. At 3-minute peak headway, a single-track section longer than roughly 800-1,000 m will usually force either lower frequency, directional tidal operation, or a bypass.
- Emergency walkways, evacuation gates, drainage, charging isolation, and intrusion detection remain mandatory; the narrow option saves width, not safety systems.

The auto-generation pipeline should prefer this option before elevated only when a tagged old-town / constrained-street corridor is short, station-loop spacing is feasible, and the resulting timetable still meets the line's published service class. Otherwise it should stay double-track or go elevated/deviate.

4.2 When at-grade is inappropriate

Even with the default bias to at-grade, certain conditions push toward elevated:

- The alignment crosses a river wider than 30 m bridge.
- The alignment crosses a highway or mainline rail at grade that cannot be broken by a level crossing (too many crossings per hour would make the crossing barrier close more than open) elevated.

- The alignment runs through a dense built-up core where ROW acquisition would require demolition of more than 3 buildings per 100 m of corridor elevated or deviate.
- The alignment runs along a street where at-grade light-rail running would conflict with road vehicles at every junction elevated or route to a parallel street.

The auto-gen pipeline's civil inference (§6) now maps these conditions to `Elevated` where it previously would have emitted `BoredTunnel`.

5. Elevated — the 25 % case

5.1 Envelope

- Structure: paired single-track prestressed decked pi-beams. OSR-Pi25 is the primary product and OSR-Pi20 is the closure/utility- avoidance product. Each beam is 2.9 m wide with stems beneath the rail lines; independent outer walkway/barrier cassettes produce an 8.5–9.0 m controlled twin-track envelope. The catalogue contains no 30 m Pi beam.
- Piers: single-column reinforced concrete, 1.5 m × 2.0 m rectangular, heights 5–12 m, with a 6.5–7.5 m hollow/precast-shell cap. Foundation type comes from the soil/access catalogue and actual deep-element lengths.
- Deck clearance: 5.0 m vertical under the girder soffit over any road; 5.5 m over a truck route; 7.5 m over a rail corridor (UIC 777-2 minimum).
- Pier spacing: 25 m nominal, using 20 m closure bays where constrained. Spans above 30 m use OSR-SP or another separately engineered structure.
- Bearings: scheduled fixed/guided/free bearings. A semi-continuous internal pier has four bearings in one longitudinal line; an expansion-unit boundary has eight in two independent lines; an end support has four. Permanent checked jacking interfaces permit replacement.
- Movement: four normal spans form a short semi-continuous unit using a released reinforced diaphragm or link slab. Expansion gaps occur every 80–125 m, never as one kilometre-long continuous structure. The project issues a bearing/movement schedule and continuous-welded-rail interaction, braking, temperature, seismic and foundation-flexibility analyses.
- Parapets: 1.4 m solid concrete panel, continuous, with a handrail on the inspection walkway.
- Walkway: separate outer inspection/escape cassette, 1.0 m clear, escape ramps every 250 m down to at-grade.
- Acoustic screens: fit only where the project noise model demonstrates a requirement; use replaceable secondary panels rather than widening the primary member by default.
- PV integration: concentrate PV at stations, depots and canopies. Normal viaduct spans do not carry a default PV attachment.

5.2 Rationale for one reference girder

- One long-line mould family across the whole project. The precast yard casts one Pi-beam section with movable Pi20/Pi25 stop ends; every normal span of an OpenSourceRail deployment uses that shape. This is the biggest single CAPEX / schedule lever in elevated construction and matches the "one spares pool, one CAD reuse" principle from RFC 0008 / 0009 / 0010.
- Double-track substructure, repeated single-track beams. One shared hollow/precast-shell cap carries two identical Pi-beams at 3.5 m centres. A single-track initial line still builds the shared substructure and may defer the second girder until capacity demands it.
- Short semi-continuity, not kilometre-long continuity. Transport and erect ordinary individual beams, then connect four spans with a released link slab or diaphragm. This removes most deck gaps and one bearing line at ordinary internal supports while retaining manageable thermal units.

6. Bridge — the 5 % case

- Reference class: separately engineered OSR-SP segmental, I-girder, or steel-composite span. The standard full-span U catalogue ends at 30 m.
- Pier: when a pier has to sit in the water, foundation is a reinforced-concrete caisson or a pile group with a protection cutwater. Scour assessment per EN 1991-2 §13 with a 100-year design flood.

- Navigation clearance: per the navigable waterway's published class. The operator commissions the bridge design and the standard OSR-U catalogue ends at the approach where the site demands a non-catalogue structure.
- Thermal: project-specific bearing, joint, and rail/bridge interaction design for the complete temperature, creep, shrinkage, and seismic range.

7. Stations-as-structures

Station archetypes from RFC 0010 map to civil class as follows:

Archetype	Typical civil class	Notes
`halt`	at-grade	No canopy PV. Single side platform on direct-ground footing.
`standard`	at-grade	Solar canopy + side platforms.
`major`	at-grade	Island platform, larger canopy.
`interchange`	elevated + at-grade	One line is on the viaduct, the other at-grade. The two platforms stack vertically at the junction. Never two elevated lines stacked — the second level's wind and seismic loads rule that out of the upstream catalogue.
`terminal`	at-grade	End-of-line; turnback tracks sit at grade.
`depot-terminal`	at-grade	Yard footprint is large; elevating it is prohibitively expensive.

Underground stations are not in the catalogue. The previous draft of RFC 0010 used the word "stacked" to describe `interchange` layouts; this RFC clarifies that stacked means vertical separation between at-grade and elevated levels at the junction, not between two underground levels.

8. Auto-gen pipeline — the civil inference, revised

The civil-class inference in `crates/osr-routing/src/civil.rs` currently classifies a cell based on the underlying cost surface. Under the no-tunnel invariant:

- Cells with cost ≤ 40 at-grade (unchanged).
- Cells with cost 40..100 at-grade (was already; unchanged).
- Cells with cost 100..400 (water) bridge (unchanged).
- Cells with cost ≥ 400 (building / dense built-up) elevated (was `BoredTunnel` now elevated).

This aligns with the rule: "if the corridor can't be at-grade, it goes above, not below."

Two downstream consequences, surfaced in `design-quality.yaml`:

1. Elevated fraction as a soft gate. If the share of elevated structure for a line exceeds 30 %, the quality file raises a warning. Designs with > 50 % elevated either need deviation or are inherently expensive and should prompt review.
2. No tunnel class. Quality output reports only at-grade, elevated, and bridge fractions.

9. Cost-anchor lookup

Total CAPEX drives the national-treasury conversation. The auto-gen pipeline emits a full `[costs]` block per city in `design.toml`, broken down by bucket below.

The source currency is USD because the marketplace and country-finance templates quote in USD. Generated `*_eur` fields are converted reporting views at 0.92 USD->EUR. Non-civil units and retained civil benchmarks live in `capex-costs.toml`. The active civil rates are generated from `civil-cost-calibration.toml` and CAD quantities into `civil-cost-model.toml`; this RFC keeps the narrative summary.

9.1 Civil works (USD/km × civil mix)

Class	Active USD planning target	Retained benchmark
at-grade	2 584 000 / route-km	3 000 000 / route-km
elevated	9 748 000 / route-km	12 000 000 / route-km
bridge	18 000 000 / route-km	18 000 000 / route-km
elevated-interchange premium	4 500 000 / site	—

9.2 Stations (RFC 0010 archetype catalogue)

Archetype	USD planning-grade
`halt`	600 000
`standard`	2 500 000
`major`	4 500 000
`terminal`	4 500 000
`depot-terminal`	5 000 000
`interchange`	8 000 000
`interchange-elevated`	12 000 000

The elevated-junction premium in §9.1 covers the viaduct + upper deck at interchanges; the table above covers the at-grade lower platform, ground-level pedestrian access, and protected crossing equipment where needed. Overbridges, lifts, and stairs are not the standard station assumption; they are carried only by elevated/stacked sites or local road-barrier overrides.

9.3 Depots (RFC 0014 archetype catalogue)

Archetype	USD planning-grade
`main-heavy`	8 000 000
`secondary-medium`	3 000 000
`layup-minimal`	400 000

These rates assume RFC 0014 distributed overnight stabling: healthy trainsets may sleep at powered passenger stations and layups, so the main-heavy no longer carries parking capacity for every train every night and `layup-minimal` is a powered-station stabling/isolation kit rather than a small standalone yard.

9.4 Rolling stock (RFC 0008 local-owner production × fleet count)

Rolling-stock city CAPEX uses a local-owner trainset-family planning unit, not an inflated per-car multiplier and not the raw marketplace BOM floor. The unit includes direct material, local assembly/labour, nominal per-train QA/acceptance evidence, and modest local handover logistics. Fixtures/tooling live in the railway production plant line; warranty, spares, and routine commissioning support are OPEX.

Family	USD / trainset local-owner unit
`urban-shuttle-1car`	280 000
`tram-2car`	560 000
`light-metro-3car`	900 000
`metro-4car`	1 120 000
`metro-6car`	1 680 000

Fleet count = peak-revenue + spare + cold-reserve per RFC 0014 §4.

9.5 Railway production plant

Each country carries one lean shared railway production-plant setup allowance for all city fleets it will build or assemble locally. Cities do not duplicate this asset in city CAPEX. The national factory is sized to the largest single-city fleet programme, costed per supported vehicle/car module, reused through a phased rollout, and kept separate from rolling-stock procurement.

Item	Planning-grade basis
Base local plant setup	60 000 USD / vehicle-car module
High sensitivity check	120 000 USD / vehicle-car module

The allowance covers reusable 1 m fiberglass panel moulds, clip/drill gauges, basic steel fixtures, plant services, commissioning bay setup, material handling, and production-readiness work. It excludes a full-length body mould and production adhesive cure hall. Six parallel two-person crews can install and release a three-car exterior body in one eight-hour shift after painted-frame dimensional release; doors, glazing, equipment, bogies, commissioning, homologation, and first-article testing remain separate work. Distributed overnight stabling reduces the number of depot-centred commissioning and parking bays needed in the production/depot complex. A three-module increment of supported factory capacity therefore carries 180 000 USD of base plant allowance, while the high sensitivity is 360 000 USD.

9.6 Systems

Item	Planning-grade basis
Residual train-control wayside (RFC 0015 GoA 4)	50 000 USD / route-km
Station/depot charging microgrid interfaces	120 000–1 000 000 USD / stop, by station archetype

RFC 0015 mandates battery-electric operation with station charging: there is no overhead catenary, third rail, feeder substation, or route traction-power budget. The energy infrastructure bucket covers the conductive chargers, local switchgear, inverter interface, and station/depot microgrid tie-in. PV/storage quantities are sized in the energy-site catalogue rather than as a continuous per-km rail-power system.

9.7 EPC overhead

Integration + project management is 7 % of the subtotal across §9.1 – §9.6.

Rates are deployment-localised via the `country-costs.toml` template (labour, materials, finance cost adjustments). The base rates above are what the upstream catalogue quotes; actual numbers in any given deployment can be ± 40 %.

`total_usd` is the procurement-basis headline per city. `total_eur` is also emitted for compatibility with earlier generated schemas.

10. Constructability constraints

- Concrete supply. Precast concrete yards exist in every target region; the Pi section needs a supplier-designed railway prototype and a reusable long-line mould fabricated from established formwork systems (PERI, Meva, Doka). Operators without a nearby precast yard bootstrap one — a 5 km² yard with 20 formworks produces enough girders for 50 km of viaduct in 18 months.
- Steel reinforcement. All target regions have domestic rebar mills in the 12–40 mm range we use.
- Post-tensioning tendons. Commodity; local supply exists in every target region.
- Lifting and transport. Pi25 is 2.9 m wide and targets 65–75 t. The project must release the route, transporter, lifting points, portal/launcher or actual crane load chart, synchronized lift controls, ground pressure, wind limits and contingency landing method.
- Launching. Where full-span access is constrained, OSR-US uses match-cast 2.5–3.0 m segments, post-tensioning, and an overhead launcher with specialist geometry, epoxy-joint, tendon-grouting, and staged-load QA.

No restricted-export equipment. No bespoke machinery. The construction envelope matches what a mid-size regional civil contractor can mobilise.

11. Pitfalls and decisions

- No tunnels is the non-negotiable decision. An operator who believes their corridor absolutely requires a 3 km tunnel under a city centre has exactly two options: (a) deviate the alignment; (b) fork the project. Upstream stays on the at-grade / elevated / bridge envelope.
- Choose the civil mix from the actual corridor. Use at grade where land, access and geometry permit. Controlled central viaduct concepts trade higher civil costs for more direct routes and fewer street conflicts. Report the resulting quantities, property needs and whole-scope costs; the earlier 70% at-grade aspiration is retained as a comparison assumption.
- Use the controlled viaduct families. OSR-Pi20/Pi25 decked beams are the primary products; OSR-US match-cast segments and OSR-SP special crossings address project constraints. Product geometry, complete lift masses, load combinations and independent checking remain release gates.
- Special crossings are separate products. OSR-Pi25 is not stretched to 40 m. OSR hands off to OSR-SP or a local engineering tender at the approach.
- Noise mitigation is structural, not operational. Acoustic screens on viaducts > operational speed restrictions. A 2 m concrete panel costs almost nothing and avoids a permanent speed-limit loss. Operators sometimes skip this; upstream builds it in.
- Seismic detailing is per deployment. The reference viaduct envelope handles PGA ≤ 0.3 g; above that, the deployment's structural team ups the rebar + bearing spec per EN 1998. The shape stays the same.

12. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	`osr-routing::civil` updated to drop tunnel classes (in this RFC's commit)	v0
v2	Emit `[costs]` block (at_grade / elevated / bridge / total_eur) per city in the auto-gen output, using the §9 rate table (done 2026-04-22)	v1
v2.1	Extend `[costs]` to the full CAPEX stack — stations (RFC 0010 archetypes), depots (RFC 0014 archetypes), rolling stock (RFC 0008 families), GoA 4 onboard autonomy/residual wayside train control, station/depot charging microgrids, and EPC overhead. USD is the procurement basis and EUR values are explicit converted views.	v2
v2.2	Move the source cost basis to USD direct-procurement planning rates, add `*_usd` fields, keep `*_eur` mirrors, and drop EPC overhead to 7 % (done 2026-06-04).	v2.1
v2.3	Add the railway production-plant setup bucket, now 60 000 USD per vehicle/car module with 120 000 USD retained as a high sensitivity check after 1 m clip-on fiberglass body and distributed-stabling scope reductions (updated 2026-08-15).	v2.2
v2.4	Replace duplicated city production plants with one shared national factory sized to the largest city fleet programme; city CAPEX is zero for this bucket and national briefs carry the asset once. Add imported-value and local-capital financing boundaries.	v2.3
v3	Parametric OSR-Pi20/Pi25 decked beams, transportable OSR-ST6 panels, 3.5 m elevated track centres, 6.5–7.5 m hollow caps, soil/access foundation catalogue, geometry-derived quantities and resource-driven production equations are controlled under `design/component-catalogue/src/osr_mech/civil/`. Legacy U-troughs remain special products; OSR-US and OSR-SP cover exceptional geometry. The controlled design-basis, load, egress, movement, erection, first-article, foundation, production and quantity packages live under `docs/civil/`.	v0
v4	Surveyed civil design package for the first pilot line, generated from the current city model	v3, RFC 0003
v5	First-article viaduct erected for a deployment instance	v3

13. Relationship to existing work

- `lib/templates/structures.toml` — the Lego-block schema. v1 of this RFC removes the `cut-and-cover-tunnel` and `bored-tunnel` entries.
- `crates/osr-routing/src/civil.rs` — the inference function. v1 of this RFC collapses the top threshold to `Elevated` (was `BoredTunnel`) and removes the `CutAndCoverTunnel` and `BoredTunnel` variants from `CivilClass`.
- `crates/osr-design/src/emit.rs` — the quality YAML emitter. v1 stops counting tunnel length (always zero now) and adds the elevated-fraction soft gate.
- RFC 0009 §5 and RFC 0010 §13 reference tunnel alignment / stacked-underground interchange respectively — both amended in this commit to match the no-tunnel invariant.
- `lib/templates/platform-doors.toml` references "tunnel stations"; that line is updated to describe full-height PSDs as an optional upgrade for climate-sealed stations regardless of civil class.

14. Open questions

1. Grade-separated level crossings — a road underpass below an at-grade rail corridor counts as a structure; does it get its own civil class? v1 folds it into the road authority's scope, not rail's.
2. Cut-and-fill extent limits. 3 m is arbitrary. Real deployments in hilly terrain may need 5 m cuts. Revisit with a hilly-pilot case study.
3. Acoustic screen effectiveness in hot climates. Concrete screens expand/contract; joints need seasonal inspection. How does that feed the CBM schedule? Candidate `osr-cbm-*` extension.
4. Wooden / bamboo viaduct for ultra-low-cost deployments? Real regional-rail projects in parts of South and Southeast Asia have used timber structures historically. Out of scope upstream; a local fork can layer that on.
5. Green-roof viaducts — planting integrated into the deck for heat-island mitigation. Candidate v4 addendum.

15. Done criteria

- [x] Two classes only — at-grade + elevated — with bridges narrow (§1)
- [x] No-tunnel rationale explicit with cost comparison (§3)
- [x] At-grade envelope specified (§4)
- [x] Elevated envelope specified, one reference girder (§5)
- [x] Bridge class as a viaduct variant (§6)
- [x] Stations-as-structures mapping (§7)
- [x] Civil inference updated (§8)
- [x] Cost anchors named (§9)
- [x] Constructability constraints stated (§10)
- [x] Pitfalls + alternatives (§11)
- [x] Rollout ordered (§12)
- [x] Relationship to existing code + templates (§13)

The next session picks up at v2 — cost estimates in auto-gen output (a short follow-on to the v1 tunnel-removal already landed in this commit).

RFC 0012 — Switches & Crossings (Physical)

Repository source: `docs/rfcs/0012-switches-and-crossings.md`

Status: Draft — planning only, no mechanical drawings ship with this RFC Date: 2026-04-22 Depends on: RFC 0009 Track Design Standard, RFC 0011 Civil Infrastructure Design Standard

1. Summary

OpenSourceRail commits to three turnout tangents (1:9, 1:14, 1:18.5) across every deployment. This RFC fixes the physical hardware standard — switch machine, stretcher bars, detectors, frog, heating — that the `osr-wayside-points` controller operates.

Tangent	Divergent speed	Preset	Use case
1:9	40 km/h	`standard-urban`, `standard-metro` (station ends)	The bread-and-butter metro turnout
1:14	60 km/h	`standard-metro` (mainline crossovers), `mainline-mixed`	High-speed crossover between parallel tracks
1:18.5	80 km/h	`mainline-mixed` only	Reserved for shared-corridor operation

Minimise switch count. Prefer trailing-point moves over facing-point moves wherever possible. Every switch is a maintenance liability; every facing-point switch on a revenue corridor is a SIL-4-rated safety asset.

2. Non-goals

- Not a rail-industry standards body. References EN 13232 (turnout geometry), EN 13145 (sleepers), EN 13146 (fastening tests). No new ones published.
- Not a level-crossing-gate policy manual. Legal requirements for warning times, signage, barrier type vary per country; this RFC specifies the equipment envelope the deployment chooses from.
- Not a points-heating policy for arctic climates. Target regions hit ≤ 0 °C briefly in some mountain areas; a simple resistive points-heating circuit is specified. Sub-arctic deployments customise locally.
- Not a ballast-cleaner or tamper spec. Maintenance-of-way equipment is separately sourced; the turnout standard doesn't constrain its choice.

3. Why three tangents, not a continuum

Picking three standard turnout geometries covers the full operational envelope while keeping tooling, inspection, and spares repeatable:

- 1:9 handles every station-throat and depot turnout. 40 km/h divergent speed is plenty for a metro that dwells 30 s at every stop.
- 1:14 handles mid-line crossovers on dedicated metro ROW where the trailing-point move is routine.
- 1:18.5 handles the shared-corridor edge case (`mainline-mixed` preset). Operators never specify intermediate tangents (1:7, 1:10, 1:12). Simplicity pays: one turnout mould, one stretcher-bar part number, one switch-machine part number per tangent; three total across the whole project.

4. Switch architecture

4.1 Mechanical

Aspect	Choice
Rail type	UIC60E2 (60 kg/m) for 1:9 / 1:14 / 1:18.5. Matches the parent rail profile from RFC 0009.
Switch blade	Flexible switch blade, continuous machined profile; no bolt-on tip. The controlled planning envelopes are 7.8 m for 1:9, 11.8 m for 1:14, and 16.2 m for 1:18.5; the supplier drawing freezes the final machined length.
Stock rail	Standard-profile rail for length of the switch panel, bolted to the next panel. No welded joints inside the switch area.
Slide chair	Self-lubricating (graphite-filled bronze) — no daily greasing. Inspection every 30 days.
Stretcher bars	Two stretcher bars per switch (front + heel). Front bar integrates the position detector per §5.
Frog	Cast manganese steel, 13 % Mn, no moveable-point frog upstream — adds complexity and a failure mode without enough speed gain at our tangents. Swing-nose frog is an opt-in v2 extension for 1:14 / 1:18.5.
Check rails	Bolted to the running rail opposite the frog. Fixed-gap (no cold-weather compensation).
Sleepers	Pre-stressed concrete switch sleepers per EN 13145; spacing compressed from the 600 mm mainline standard to 540 mm through the switch panel.

4.2 Switch machine

Single vendor / single machine family across every deployment:

Aspect	Choice
Type	Electro-mechanical, BLDC motor + planetary gearbox + crank. No hydraulic switch machines — hydraulic packs are top-quartile maintenance items.
Throw force	6 kN nominal, 12 kN peak (matches standard metro load at ≤ 25 °C ambient).
Throw time	≤ 3 s from rest to locked, including detection settling.
Detection	Dual-redundant position sensors (A + B, 2oo2 agreement) at the front stretcher bar. Integrated with `osr-wayside-points`'s `SwitchObservation` reporting.
Locking	Mechanical lock at the tip via the front stretcher bar; blade cannot move when locked. No electrical-only locking.
Heating	3 kW resistive heating strip along the slide chair + blade foot, thermostatically switched at +5 °C ambient. Powered from the trackside 230 V AC supply; dropped to LVDC aux in depot areas.
Cabinet	IP67 aluminium enclosure matching the W-SBC form factor (RFC 0007 §6). Motor + controller share the cabinet.

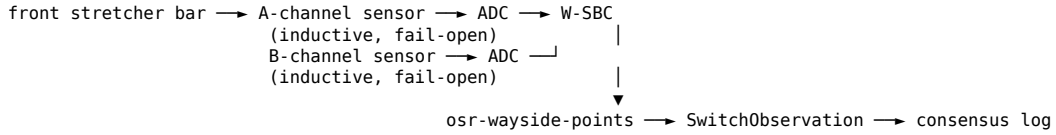
4.3 Facing vs trailing

- Facing-point switch (FPS): a switch the approaching train has to trust to guide it onto the right track. SIL-4 asset. Every FPS is a permanent W-SBC deployment (`osr-wayside-points`) with consensus-log-backed position observation.
- Trailing-point switch (TPS): a switch a train merges into from one of two converging tracks. The train flips the blade on its way through; no pre-positioning required. SIL-2 asset — the controller still reports position for dispatch awareness, but the safety case doesn't rely on it being at any specific position when a train approaches.

Every deployment counts and minimises FPSes. A typical linear deployment instance (RFC 0003) has 0 FPSes on a linear line (only crossovers at depot entry, which are TPSes on the way in and FPSes only if an operator reverses a train on the mainline — normally handled by the turnback at the terminal station). Any deployment with > 8 FPSes per line is a design red flag.

5. Position detection + integration

Per RFC 0007 §6, every wayside site including switch sites hosts a W-SBC. The switch-detection chain is:



Two channels are redundant by construction. Any disagreement, or either channel reading "indeterminate," yields `DetectedPosition::Unknown` — the fail-restrictive behaviour already encoded in `osr-wayside-points`'s W2 proptest + Kani property.

No track circuits in the switch zone. Detection is by dedicated stretcher-bar sensors. Axle-counters sit outside the switch panel as a redundant train-detection sensor — not as the switch-position detector.

6. Level crossings

Level crossings remain a specific category handled by `osr-level-crossing`. The physical equipment envelope:

Element	Spec
Barrier type	Half-barrier, counterweighted, 4 m arm length (8 m arm at 2-lane crossings). Pneumatic or electro-mechanical drive; no hydraulic.
Barrier descent time	8–12 s (meets UIC 762 and most national rules).
Warning lights	Alternating red flashers + audible warning bell, 30 s pre-barrier activation.
Road approach warning	150 m advance warning sign + surface markings.
Train detection	Redundant axle-counter pairs at entry + exit; feeds `osr-level-crossing`'s five-state FSM.
Emergency stop strip	600 mm rubber strip on the road at the rail approach — survives being struck, triggers an emergency barrier lift only via a dispatcher OK.
Canopy / shelter	Optional on the road side for the approach warning cabinet; not on the rail side.

Approach warning time is 30 s minimum, per `osr-level-crossing`'s invariant, irrespective of design speed. Operators who want a longer approach configure per-crossing; upstream defaults to 30 s.

7. Crossings (diamonds)

Rail-on-rail diamonds (one track crossing another at grade) are not in the standard catalogue. They are:

- Expensive to maintain (high impact load at the flangeway gaps).
- A top-quartile derailment site.
- A needless complication on a new-build corridor where junctions are trivially re-routed around parallel alignments.

If a deployment's geometry seems to demand a diamond, the alignment should be revisited. Grade-separated crossings (elevated over at-grade) are the catalogue answer. Per RFC 0011 §5 those are standard viaduct spans.

8. Turnout geometry table (per tangent)

Planning-grade per EN 13232. Exact radii / lead lengths per manufacturer's standard drawing.

Tangent	Angle	Switch lead length	Total envelope	Blade envelope	Through-radius	Turnout sleepers	Divergent speed
1:9	6.3°	23 m	27 m	7.8 m	190 m	42	40 km/h
1:14	4.1°	37 m	43 m	11.8 m	500 m	68	60 km/h
1:18.5	3.1°	49 m	60 m	16.2 m	900 m	94	80 km/h

Any deployment using a non-listed tangent is outside the upstream envelope.

9. Auto-gen pipeline

At deployment-generation time, the emitter needs to know where switches go. Today's pipeline:

- Terminal stations: the emitter already tags first/last stations as `terminal` (RFC 0010 §12). A future v2 of the emitter can emit a `turnback_switch` entry at every `terminal` using tangent 1:9.
- Depot entries: the `depot-terminal` archetype gets a depot fan of 1:9 turnouts. The fan geometry is parametric in the depot stall count (see RFC 0014).
- Interchange stations: no switches. Lines cross grade- separated; see RFC 0011 §7.

Emitter v2 work is the short follow-on that pairs with RFC 0010 v2 / RFC 0011 v2.

10. Pitfalls and decisions

- Only three tangents in the catalogue. A deployment team pushing for a 1:10 or 1:12 to "match their legacy network" is asking for bespoke mould work. The cost math almost always favours standardising on 1:9 + slightly-reduced divergent speed where needed.
- Electro-mechanical switch machines, not hydraulic. Hydraulic is smoother and quieter; it is also top-quartile for maintenance and has a top-quartile environmental liability (pack-oil leaks into track ballast). Electro-mechanical wins the mission alignment.
- Fixed frog, not moveable-point. Moveable-point frogs are smoother at 80 km/h+; at our tangent speeds they don't pay back the complexity. Opt-in at 1:14 / 1:18.5 in v2 if the acoustic argument becomes dominant.

- No diamonds. Grade-separate, or re-route. Operators used to network-style legacy rail sometimes push for diamonds on psychological grounds ("my country's rail has them"); we do not.
- Level crossings are a necessary evil. Every road-rail level crossing added is a maintenance, safety, and service-punctuality liability. The auto-gen emitter counts them per km of line; deployments with > 3 level crossings per km are flagged as a soft gate for review.

11. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	`lib/templates/switches.toml` updated to match this RFC	v0
v2	Emitter emits turnback at every terminal archetype + one `no-9-mainline` per stall at every `depot-terminal` (yard fan) (done 2026-04-22)	v0, RFC 0010 v2
v3	Reference CAD drawings for the 1:9 turnout under CERN-OHL-S v2	v1
v4	Generated switch-bill for the current pilot city model, including terminal turnbacks, depot fans, and any required mainline crossovers	v1, RFC 0003

12. Relationship to existing work

- `lib/templates/switches.toml` — the Lego-block schema. v1 amends to drop any non-catalogue tangents.
- `crates/osr-wayside-points` — the W-SBC software controller. Already W1–W6 verified (RFC 0004 §M4 work). No software change for v0 of this RFC.
- `crates/osr-level-crossing` — the level-crossing 5-state FSM. Already deployed; this RFC pins the physical equipment envelope that feeds it.
- `crates/osr-interlocking` — consumes `SwitchObservation` log entries. Semantics unchanged.

13. Open questions

1. Swing-nose frog opt-in rule. At what line speed / ridership does the acoustic / rail-wear benefit cross over the added mechanical complexity? Revisit at v3.
2. Shared-corridor freight impact. On `mainline-mixed`, a freight train's 22.5 t axle load over a 1:9 switch daily accelerates wear substantially. The catalogue's assumption is that shared corridors use 1:14 minimum. Verify with a real mixed-corridor pilot.
3. Seasonal-heated switches in hot climates. +50 °C summer ambient can rail-expand beyond the standard switch tolerance. Temperature-compensated switch geometry is a future addendum for deployment-scale turnout counts.
4. Single-vendor switch-machine risk. The project's simplicity bet locks the fleet to one machine family. What happens if that vendor exits the market? Mitigation: the reference drawings under CERN-OHL-S v2 let any replacement machine vendor fit into the footprint.

14. Done criteria

- [x] Three (plus tram) tangents committed (§1)
- [x] Switch architecture (§4) + detection chain (§5)
- [x] Level-crossing equipment envelope (§6)
- [x] Diamonds out of scope with rationale (§7)
- [x] Geometry table per tangent (§8)
- [x] Auto-gen implications (§9)
- [x] Pitfalls + alternatives (§10)
- [x] Rollout ordered (§11)
- [x] Relationship to software + templates (§12)

The next session picks up at v1 — `lib/templates/switches.toml` update, a short editorial PR.

RFC 0013 — Operations Rulebook

Repository source: `docs/rfcs/0013-operations-rulebook.md`

Status: Draft — planning only Date: 2026-04-22 Depends on: RFC 0008 Rolling-Stock Reference Design, RFC 0009 Track Design Standard, RFC 0010 Station Design Standard, RFC 0011 Civil Infrastructure Design Standard, RFC 0012 Switches & Crossings

1. Summary

Every OpenSourceRail deployment inherits one baseline operations rulebook — the standing rules for drivers, dispatchers, station staff, and maintenance workers. The rulebook is deliberately short (target ≤ 60 pages in the rendered PDF) and shared across deployments. Operators adapt it for national regulatory overlays but do not re-invent it.

This RFC fixes the rulebook's table of contents and the governing rule-granularity principle. The rule text itself lives in markdown under `docs/operations/` and is owned by a standing committee once the project has one; until then, the project maintainer ratifies additions.

2. Non-goals

- Not a national regulatory document. Every country that adopts OSR has a national rail safety authority (NRSA or equivalent) that must accept the rulebook. This RFC produces a baseline that a deployment's local legal counsel adapts.
- Not a driver training manual. Training content comes from the rulebook, but the pedagogy (lesson plans, sim scenarios, exam questions) is a per-deployment deliverable handled by workforce materials (`lib/templates/workforce.toml`).
- Not a safety-case evidence file. The rulebook is an operational artefact; the safety case (`docs/safety-case/`) references rules as supporting evidence, not the other way around.
- Not an incident-response playbook. Playbooks are scenario- specific and owned per-deployment (fires, passenger illness, derailment, signal failure); this RFC specifies the categories each playbook must exist for, not the specific response.

3. The simplicity principle

Every rule in the rulebook must pass this test:

> A new driver who has never seen this system before can read the whole rulebook in one morning and have an accurate enough mental model to operate safely under supervision by the end of the day.

This means:

- Every rule is one sentence, one decision. No compound rules.
- Every rule has exactly one author (the project maintainer until the standing committee exists).
- Every rule is numbered; operators cite rules by number in incident reports.
- Every rule has a why: a short justification paragraph. No rule is an unexplained ritual.

The target is a ≤ 60-page PDF. Rail industry rulebooks that hit 200 + pages are artefacts of 150 years of accretion — we're not accruing that.

4. Table of contents — one section per role

The rulebook is organised by role, not by scenario. A driver reads one section; a dispatcher reads another; the sections overlap only where the roles interact.

4.1 Dispatcher rulebook (target ≤ 15 pages)

Section	Scope
S1	Shift start — OCC login, fleet inventory check, schedule ready
S2	Routine dispatching — route grants, timetable adherence, headway maintenance
S3	Incident handling — single-train fault, section unavailable, mass-casualty
S4	Weather and environmental — dust storms, heatwave, flood warning
S5	Maintenance interlock — work-on-track grants, blocking a section from revenue
S6	Shift end — handover, outstanding-event log, system state summary
S7	Intrusion response (RFC 0016) — track-patrol dispatch, return-to-service gating, sustained-Unknown escalation

4.2 Station staff rulebook (target ≤ 10 pages)

Section	Scope
T1	Station opening — power up, PIS check, fare-gate test
T2	Passenger boarding — platform supervision, PSD status, wheelchair assist
T3	Passenger incidents — medical, fare dispute, unattended item
T4	Fare enforcement — inspector on-board, non-payment, appeal
T5	Station closure — evacuation, power down

4.3 Maintenance worker rulebook (target ≤ 10 pages)

Section	Scope
M1	Depot safety — hazard awareness, PPE, lone-worker protocol
M2	Work-on-track authorisation — formal blocks, protection signals, access to active ROW
M3	Routine track inspection — walking the line, visual checks, GPM/RECD recording-car cadence (per RFC 0009 §9)
M4	Switch maintenance — blade lubrication, detection-sensor cleaning, 30-day check
M5	Fleet maintenance at depot — inspection intervals (daily / 7-day / 30-day / overhaul)
M6	Incident site access — derailment response, fire site, crane operations
M7	Wayside sensor maintenance (RFC 0016) — 30-day walkthrough, post-weather inspection, sensor replacement + calibration

4.4 Control-centre rulebook (target ≤ 5 pages)

Section	Scope
C1	OCC watch roles — dispatcher, supervisor, engineer-on-call
C2	Communications — radio discipline, PA authority, external-party notifications
C3	Shift handover — written log, outstanding items, emergency-on-call list

Total target: 40 pages across four sections.

5. Degraded-mode operations — the important part

Modern CBTC metros routinely run in "normal mode" (full ATO under a fresh MA) and have short, mission-critical manual-restoration procedures when the signalling degrades. OpenSourceRail keeps exactly three degraded modes:

Mode	Trigger	Operator action
M1 — Automated restricted MA	ATO is healthy and ATP can enforce a shortened MA	OCC authorises reduced-speed automatic movement; ATP remains fully active and speed is capped at 80 % of normal.
M2 — Controlled stop	Consensus cannot issue a trustworthy MA	Every affected train remains stopped while OCC establishes a work block and restores the signalling quorum. No written order may bypass ATP.
M3 — Protected depot shunt	Movement is required inside an isolated depot worksite	An authenticated local maintenance pendant requests automatic movement at ≤ 15 km/h under ATP, direct CCTV observation, and a locked work zone.

No intermediate modes. No "sort-of-ATP" states. Either the safety envelope holds (modes M1 / M3) or the train remains stopped (mode M2). This is a forcing decision: ambiguous degraded modes are the dominant cause of metro-world incidents.

6. Weather-dependent rules

Target regions routinely experience weather conditions that challenge legacy rolling stock:

- Dust storm (haboob): visibility < 100 m automatic speed cap at 50 % (`osr-ato` enforces). Dispatcher issues a network-wide speed restriction at the OCC. Passenger announcements via `osr-pis-onboard`. See also `lib/templates/climate-adapters.toml`.
- Heatwave (> 45 °C sustained): rail-expansion risk triggers a pre-emptive 15 % speed cap on `mainline-mixed` and `standard-metro` presets with continuous-welded-rail. Switch detection sensors recalibrate at dawn.
- Flooding: any rainfall > 50 mm / h triggers immediate inspection of at-grade segments; the dispatcher can preventively block sections via `osr-interlocking`'s maintenance-override. Elevated segments per RFC 0011 §5 are unaffected by surface flooding — deliberate design bet.
- Lightning: nearest-strike distance < 2 km Cat.22 5G backhaul and LoRa backhaul are both watched for outage; if both fail for > 10 s the operator notifies the OCC and cooperates with `osr-atp`'s default fail-restrictive MA.

7. Incident categorisation

Every incident fits one of six categories. This drives the notification chain and the regulatory report:

Category	Example	Dispatcher action
I1 — fleet self-recovery	Door interlock fault on a single train, resolved at next terminal	Log in event recorder; notify depot; no regulatory report.
I2 — service disruption	Single-train immobile; next train coupled to rescue it	Passenger PA, timetable recovery plan; 24h incident report.
I3 — short safety event	Emergency brake triggered by spurious trip; no injury	Log, investigate root cause, 72h report.
I4 — passenger injury	Slip, fall, medical emergency on board	Medical call, depot handover; formal incident report to NRSA within 7 days.
I5 — collision / derailment	Train-to-train, train-to-obstacle, train off-rail	Full network shutdown of affected line; NRSA notification within 2 hours; incident-site preservation.
I6 — mass-casualty / fire	Onboard fire not extinguished, large-scale passenger injury	Immediate evacuation protocol; emergency services; NRSA + national media escalation.

The categories map directly to `osr-event-recorder` tags, so the post-incident analysis starts from the same data the categories describe.

8. Training and recurrent qualification

- Initial driver qualification: 4 weeks classroom + 2 weeks simulator + 4 weeks supervised revenue service. Final qualification exam in written + simulator + on-road segments.
- Recurrent: 2 days refresher every 12 months + 1 simulator session per quarter.
- Degraded-mode drills: every driver runs modes M1 / M2 / M3 in the simulator every quarter. The `osr-sim` sim is the primary training platform.
- Dispatcher qualification: 8 weeks OCC-side classroom + simulator + supervised desk time.
- Maintenance: role-specific certification per EN 50126 / per national NRSA rules.

Training content is a deployment deliverable; this RFC fixes the minimum hours + structure.

9. Auto-gen implications

The rulebook does not directly feed the corridor-generation pipeline — but it does feed two outputs:

1. Depot capacity: `osr-sim`'s fleet-sizing needs to include enough spare trains for the rulebook's degraded-mode operations. A deployment with 10 trainsets on rostered revenue needs ≥ 1 spare in depot for maintenance (M5) and ≥ 1 cold- reserve for I-category incidents. See RFC 0014 §4 for the formula.
2. Workforce sizing: the driverless financial model supplies group FTE; `lib/templates/workforce.toml` allocates those totals into city roles, skills and tasks. The city delivery generator writes the organisation and spreadsheet. Named duty/leave cover must still be checked against the rulebook's shift structure (§4.5).

Changing this prose does not change software parameters automatically. Update the controlled model/template inputs, regenerate and check fleet and workforce evidence when operating rules change.

10. Pitfalls and decisions

- One rulebook across deployments is a big bet. Legacy metros each have their own 200-page rulebook; consolidating to one is the simplicity gain. National NRSAs will initially resist. The counter: the process of NRSA approval works on a per-document basis, not per-system basis — a shorter document approves faster. The operator carries the regulatory overlay in a thin appendix.
- Six incident categories, not more. Operators unused to this regime sometimes want a I7 or I8 for niche scenarios. We absorb niche scenarios into the existing six rather than proliferate.
- Paper written orders are allowed in mode M2. This is not anti-digital — paper is explicitly the fallback when the digital system degrades. Written orders remove ambiguity about authority when the digital system can't issue it. Rare enough that the paperwork tax is tiny.
- No "caution mode" between M1 and M2. Legacy metros have a smorgasbord of half-ATP-half-manual modes; every one is a source of operator confusion. We commit to the hard boundary.
- ≤ 60 pages is a hard target. The rulebook is reviewed on every minor-version bump; if it exceeds 60 pages, something gets cut. Adding text is easier than subtracting.

11. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	`docs/operations/` scaffold with per-role stub files for dispatcher, station staff, maintenance, and control-centre roles, each with scope + cross-refs + proposed rule outline (done 2026-04-22)	v0
v2	Full unattended-operations rulebook drafted across all four role families. Every rule is one sentence + a `Why :` paragraph, cross-referenced to the relevant crate and safety-case solution. Practising-operator review is v2.1 per-deployment work.	v1
v3	NRSA submission pack for Samawah (rulebook + safety case cross-reference)	RFC 0003 §5, v2
v4	`osr-sim` scenario library covering every degraded-mode scenario for operator refresher training	v2

12. Relationship to existing work

- `lib/templates/workforce.toml` — shift-structure + headcount inputs. v3 of this RFC aligns the two documents.
- `lib/templates/service-hours.toml` — service hours drive the rosterable workforce; the rulebook sets the rostering envelope.
- `docs/safety-case/` — references specific rulebook rules as "solution" evidence for goals around fail- restrictive operation, evacuation readiness, and training competence. GSN goals for "operational fail-restrictive behaviour" land in v4.
- `crates/osr-occ` — the dispatcher's OCC UI embeds the dispatcher rulebook's decision-support prompts. Software change in v3.

13. Open questions

1. Bilingual rulebooks. Every target-region deployment operates in at least one local language plus English. Do we maintain both upstream? v2 commits on this: likely yes, with English as the canonical and translations as separate branches in `docs/operations/<lang>/`.
2. Printable-PDF vs living-document. Rail NRSAs typically want a static PDF they can archive. The project wants a live document. Compromise: every minor-version bump produces a sealed PDF, and the commit hash is the canonical reference.
3. Simulator-based competence testing. `osr-sim` can evaluate driver actions against MA envelope in software. Can NRSAs accept simulator certification in lieu of on-road supervised hours? Per-country, probably not at v1.

14. Done criteria

- [x] Table of contents per role (§4)
- [x] Degraded-mode operations (§5)
- [x] Weather rules (§6)
- [x] Incident categorisation (§7)
- [x] Training envelope (§8)
- [x] Auto-gen implications (§9)
- [x] Pitfalls (§10)
- [x] Rollout ordered (§11)
- [x] Relationship to existing work (§12)

The next session picks up at v1 — `docs/operations/` scaffold, short editorial PR that seeds the per-role files. Actual rule text lands in v2.

RFC 0014 — Depot Design Standard

Repository source: `docs/rfcs/0014-depot-design-standard.md`

Status: Draft — planning only, no site drawings ship with this RFC Date: 2026-04-22 Depends on: RFC 0008 Rolling-Stock Reference Design, RFC 0010 Station Design Standard, RFC 0011 Civil Infrastructure Design Standard, RFC 0013 Operations Rulebook

1. Summary

The current city catalogue adopts one full-fleet planning depot per line, sized to that line's complete stock and actual train length. Storage tracks and workload-sized maintenance bays are separate quantities. Depot PV/storage is priced once; property, access alignment, yard topology, installed quotations and morning launch capacity remain releases. The controlled generic quantities are in `city-operating-scope.toml` and each city's `engineering/line-depots/summary.json`. Baghdad retains its dedicated revised scope and financing ledger.

The distributed-stabling pattern below is retained as the earlier reference and simulation comparison. Its passenger-platform parking assumptions do not reduce the current full-fleet depot capital requirement.

1.1 Retained distributed-stabling reference

OpenSourceRail commits to three depot archetypes covering the full range of deployment sizes. Every deployment has exactly one main depot plus optional secondaries and layups. The main depot co-locates with a `depot-terminal` station (RFC 0010 §1).

The default operating pattern is distributed overnight stabling: clean, telemetry-healthy trainsets may finish service at powered passenger stations or layups near their first morning trip instead of all running empty to a terminal depot. This reduces depot stall requirements, avoids wasteful dead mileage, and lets first departures begin across the line at the same time. The main-heavy remains the maintenance authority. Defective trains are held and recovered to a service point on their own line; maintenance planning must not assume an interline rail connection. Two revenue trains per selected station form the morning launch stock; remaining revenue trains and reserves stable at a powered service point on their own line. Storage tracks are sized separately from workshop bays, using reusable depot/layup blocks.

Archetype	Catalogue max stalls	Heavy maintenance	Overhaul throughput	Notes
`main-heavy`	20	Yes — overhaul bay, wheelset lathe	≤ 40 trainsets	Larger workshops require a controlled site-specific extension
`secondary-medium`	12	Light only — brake disc, door actuator, HVAC modules	—	Second depot on a long line
`layup-minimal`	6	None — overnight stabling only	—	Remote-terminal layup

The catalogue max is the reference layout envelope. The per-deployment workshop requirement comes from §4 (`max(4, ceil(fleet × 0.15))`). It is not an overnight parking allowance. Requirements beyond that envelope need an explicit site design; a template ceiling must not silently truncate them.

Plus one optional type for edge cases:

Archetype	Purpose
`training-wing`	Co-located operations-training facility — OCC dispatcher-console simulators, maintenance-technician workshop mock-ups, and recovery-mode-crew briefing rooms. OSR is GoA 4 driverless (RFC 0015), so there are no revenue-service drivers to train — the wing exists for dispatchers, maintainers, station staff, and the recovery-mode crew who operate the manual-fallback joystick (RFC 0015 §8.2). Opt-in on `main-heavy`; not a standalone depot.

All depots are at-grade. Per RFC 0011 §1 depots are never elevated — the footprint is too large to economically put on viaduct. A deployment that can't site its depot at-grade has an alignment problem, not a depot problem.

2. Non-goals

- Not a workshop management system. How the depot schedules its work — CMMS, parts management, labour planning — is owned by the operator and overlays with the `osr-cbm-backend` output.
- Not an off-line overhaul facility. Major overhauls (wheel reprofiling at 150 000 km, bogie overhaul at 600 000 km, body overhaul at 10 years) happen in the `main-heavy` depot's overhaul bay. Deployments without a main-heavy either contract overhaul out or depot-trip their fleet to a sister network.
- Not a yard-design manual. Track yard geometry within the depot boundary uses RFC 0009's `standard-urban` or (for large shunting yards) a depot-specific geometry variant. Detailed yard layout per facility.

- Not an HR / workforce document. Shift patterns and headcount from the ops rulebook (RFC 0013) and the workforce template.
- Not a brownfield-rehabilitation scope. Where a deployment inherits an existing rail workshop (Samawah, Khartoum-Atbara, Karachi Cantt, Maputo Machava, etc.), RFC 0027 governs the conversion of the existing site into an OSR depot. This RFC's archetypes are the greenfield envelope; the brownfield path swaps greenfield depot CAPEX (use the current cost template and reconcile energy inclusions) for workshop rehabilitation (~$0.5–2 M building + $1–3 M new OSR-specific tooling) — an order-of-magnitude saving that the per-deployment plan should commit to whenever assets are real.

3. Why three archetypes

Depots are where the simplicity bet pays off most. The three- archetype set covers:

- `main-heavy` — the one non-negotiable facility per deployment, where heavy maintenance happens.
- `secondary-medium` — for deployments whose geography makes defect and light-maintenance access to the main depot impractical.
- `layup-minimal` — storage at a line's powered service location for stock beyond the two-train station provision; no separate heavy workshop or interline connection is implied.

The catalogue provides reference envelopes. A workshop exceeding the reference bay count needs a controlled extension at the deployment main depot, including access, equipment and staffing capacity.

4. Fleet-sizing formula

Historical distributed-stabling comparison. The workshop and overnight allocation formulas below belong to the earlier reference. Current full-fleet line depots use the workload, storage and price controls linked in §1; the current Samawah quantities are generated in RFC 0003.

Workshop bays are sized for concurrent maintenance and inspection. Two revenue sets stable at each selected powered passenger station where fleet permits; remaining revenue trains and reserves use separate depot stabling tracks:

```
main_depot_workshop_bays = max(4, ceil(total_fleet × 0.15))
secondary_depot_bays = site-specific exception
service_rotation = 0  # depot service is prohibited in peak windows
total_fleet = peak_revenue_trainsets + service_rotation + spare + cold_reserve
station_overnight_positions = 2 × unique_selected_station_count
depot_overnight_trainsets = total_fleet - station_overnight_trainsets
```

The two-train station provision is an overnight allocation constraint. Allocate remaining revenue trains and all reserves to same-line storage, preserving total fleet inventory. Depot storage positions and usable track lengths are sized separately from workshop bays. The earlier Samawah comparison allocated 40 trains to stations and 68 to storage on their own lines: 37 on Line 1, 16 on Line 2 and 15 on Line 3. These comprise 57 revenue, eight spare and three cold reserve trains. The storage requirement totals 4,046 m of usable slots at 59.5 m per train, separately from the existing 17 main-depot workshop bays. Lines 2 and 3 reuse their powered service terminals. Local storage layouts need detailed design; interline depot connections are not part of the operating requirement.

- `peak_revenue_trainsets` = what the schedule requires at peak. Sized from the physical round-trip cycle vs. the peak headway:

  ```
  traversal_energy_kwh = line_length_km × cars × 2.4 × 1.25 × 1.25
  charging_dwell_s = ceil_30s(3600 × traversal_energy_kwh / sum(charger_kw))
  charging_dwell_s = clamp(charging_dwell_s, 120, 600)
  added_dwell_min  = charging_stops × max(charging_dwell_s / 60 - 1, 0)
  stop_count       = station_count - 2  # radial; station_count for a ring
  one_way_min      = (line_length_km / cruise_speed_kmh) × 60
                     + stop_count × stop_penalty_min
  radial_cycle_min = 2 × one_way_min + 2 × 3 + 2 × added_dwell_min
  ring_cycle_min   = one_way_min + added_dwell_min
  peak_revenue_trainsets = max(2, ceil(cycle_min × recovery_factor / headway_min))
  ```

  Current native emitter parameters:

Family	Cruise km/h	Stop penalty min	Recovery factor	500 kW cabinets per powered station
tram-2car	48	1.45	1.10	1
light-metro-3car	62	1.35	1.62	1
metro-4car	72	1.30	1.30	3
metro-6car	72	1.30	1.45	4

  Radials use a three-minute peak headway and a three-minute turnback at each end; rings use a six-minute peak headway. The stop penalty includes the one-minute reference dwell, and only charging dwell above that reference is added. The energy balance uses a 25% climate uplift and a separate 25% charging margin. The larger recovery factors include the emitter's retained onboard-energy resilience allowance; they are planning assumptions requiring city-specific service and energy evidence, not a uniform 10% timetable margin.

These parameters replace the former written 35 km/h proxy and uniform 1.10 recovery factor. Existing city fleet inventories remain declared inputs; reconcile regeneration changes with timetable, energy, cost and physical stabling evidence before adopting a different count. This formula does not enforce two trainsets per station. Station capacity must be checked separately.

- `spare` = max(1, floor(peak_revenue_trainsets / 10)), for planned maintenance rotation.
- `cold_reserve` = 1 per line, for unplanned incidents per RFC 0013 §7.
- `service_rotation` = zero dedicated sets. Depot service is prohibited in the 3-minute peak windows and uses surplus peak-fleet capacity when the timetable relaxes to 6 or 12 minutes.

The 12-minute service happens at one designated service depot per line only in lower-frequency windows. During 07:00–09:00 and 15:00–17:00, trains make the normal quick terminal turnback, can use the platform connector, and may deplete the battery further while the 20% section-entry reserve remains mandatory. Interior cleaning, exterior/running-gear walk-around, door, coupler and emergency-equipment release checks, fault-log download, and a 150 kW low-C recharge run concurrently. A red defect removes the set from rotation and uses a planned spare; a clear inspection releases it straight back to revenue service. The opposite terminal may still top up or stable a train, but does not repeat the full inspection on the same round trip.

The 15% workshop concurrency allowance covers overhaul, defect repair, incoming inspection, and planned maintenance. Growth beyond the 20-bay catalogue envelope requires a controlled workshop extension; it does not restore fleet-wide depot parking.

Distributed overnight stabling does not increase the service-rotation fleet. End-of-service dispatch assigns healthy trainsets to powered stations with CCTV monitoring, remote isolation, and at least low-C charging access capped by installed equipment. The 150 kW rate is a top-up target, not a minimum station rating. Morning dispatch releases those sets from their stabled stations so outer stops do not wait for trains to run out from the terminal depot. The main-heavy stall formula remains sized for maintenance, inspection, overhaul, spares, and growth rather than for parking every healthy set every night.

Historical Samawah comparison (September 2026, 3-min peak headway):

- The earlier figures are retained in the historical Samawah reference. The current city summary adopts the full-fleet line-depot requirement.
- The September 2026 design has fleets of 53, 28 and 27 trainsets: 108 total. The depot reconciliation derives initial placements and slot-length requirements from the stored scenario. Physical overnight allocation remains unverified.
- `design.toml` emits the depot archetype, terminal assignment, and `fleet_stalls` as workshop/inspection bays from the same run. Older 1.25× fleet-parking examples and hand-calculated 6/4/8 trainset examples were removed from the docs and must not be copied into current deployment documents.

The emitter (RFC 0014 v2) computes this from the design.toml's `fleet_sizing` block.

5. `main-heavy` envelope

5.1 Functions

- Workshop holding for inspection, scheduled work and defective trainsets (bay count per §4); overnight fleet allocation requires a separate plan.
- Daily inspection of every incoming trainset.
- Weekly inspection (7-day service).
- 30-day inspection (A-class service per EN 50126).
- Light repair (door actuator, HVAC module, brake-disc replacement, battery module swap).
- Heavy overhaul: bogie swap, wheel reprofiling on depot lathe, body-work repaint.
- Wash: automated two-pass wash track.
- Battery-module inspection, coolant service, and sealed-module exchange.

5.2 Physical layout

Area	Target footprint (per consist bay)
Stabling track	1× consist length + 5 m clearance at each end
Daily-inspection bay	Roofed, pit-equipped, overhead power, compressed air, data sync drop (TCN-E monitoring)
Weekly bay	Pit + roof walkway for upper-body access
Overhaul bay	Pit + overhead crane (40 t) + wheel-lathe trench
Wash track	Two-pass automatic with water recycling
PV canopy	4 000 m² reference allowance; full operating PV inventory needs additional located area (§5.3)
Classroom + office	≥ 200 m² for operators + trainers
Stores	≥ 600 m² with rack storage for spare modules

Total `main-heavy` footprint: ~5 000 m² gross for a 10-stall facility, ~8 000 m² for a 20-stall. These historical workshop envelopes do not include the complete operating PV/storage inventory and cannot be used as total deployment land requirements.

5.3 Energy

- `depots.toml` selects the canonical `energy-sites.toml` tier. `main-heavy` uses `depot-main`: 5,000 kWp PV and 40,000 kWh storage, comprising 80 × 500 kWh stationary modules. Scenario, BOM and layout requirements resolve the same inventory; explicit site overrides must remain traceable. These are retained operating assumptions, not validated depot sizing. Reassess energy duties with distributed overnight station charging before freezing procurement or depot land requirements.
- The old 4,000 m² / 600 kWp / 2,000 kWh scope is a reference allowance. At the catalogue density of 0.15 kWp/m², 5,000 kWp requires 33,333 m² of modules, before access, setbacks and packing. Additional land/canopy and the battery compound are not yet located by the reference CAD.
- Annual PV generation must come from the city energy study with declared weather and loss assumptions. A nameplate rating alone is not annual yield.
- Stationary storage feeds the RFC 0021 DC charging system. Supplier dimensions, conversion/protection equipment, fire separation and installed cost require explicit reconciliation with the fixed depot allowance.
- Grid import/export must satisfy the declared connection limits and utility study. Per RFC 0002, every depot is one physical `osr-energy-site`, including when co-located with a passenger station.

5.4 Signalling

- A single W-SBC plus an interlocking zone boundary at the depot entry.
- Inside the depot, speed limit 15 km/h, manual operation under mode M3 (RFC 0013 §5).
- Shunting signals at each stall entry — mechanical dwarf signals driven by `osr-wayside-points` extensions.
- No ATP envelope inside the depot; `osr-atp` disables below 15 km/h in depot-tagged geofence, per a safety-bounded configuration. The operational safety envelope inside the depot is line-of-sight + hand-signal coordination.

5.5 Access

- Personnel access: turnstile-gated with ID-card entry, separate from the fleet access.
- Fleet access: one track connection to the line, TPS-style merge per RFC 0012 §4.3 (trains go in — it's a trailing move).
- Road access: for parts delivery, emergency services, depot staff parking. Minimum one 4 m-wide road connection.
- Rail access for deliveries of new trainsets is via the same line connection (new rolling stock rolls in on its own wheels).

6. `secondary-medium` envelope

- Functions: overnight stabling for ≤ 16 trainsets, daily inspection, weekly inspection, light module swap.
- No overhaul bay. No wheel lathe. No bogie-lift. Anything heavier is trips back to the main-heavy.
- Footprint: ~2 500 m² for 8 stalls.
- Reference PV canopy: 1,500 m² / 225 kWp, with 500 kWh baseline storage.

- Operating `depot-secondary` tier: 1,500 kWp PV and 5,000 kWh storage (10 × 500 kWh modules); additional placement and installed cost remain open.
- Signalling: same as main-heavy but smaller zone.
- Classroom: none. Reuse the nearest `major` station's community room if needed.

7. `layup-minimal` envelope

- Functions: overnight stabling, low-C top-up, cleaning, and a ground-level safety-release walk-around. Trainsets drive in at end of service, can pass through the same 12-minute turnaround service if the site is designated for its line, sit overnight, and drive out at morning service start. There is no pit inspection, repair work, or heavy maintenance.
- Footprint: ~1 000 m² for 4 stalls.
- PV canopy: 600 m² 90 kWp.
- Operating `depot-layup` tier: one 150 kWh stationary package for opportunity charging during dwell; the main-heavy is primary.
- Train top-up: 150 kW low-C per occupied stabling road, used while a set is held between duties and overnight; passenger-platform fast charging remains a separate terminal function.
- Signalling: single W-SBC at the layup entry; rest is manual line-of-sight.
- No personnel on-site overnight; the OCC monitors via CCTV.

8. Integration with the auto-gen pipeline

Today's emitter picks one `depot-terminal` archetype (`crates/osr-design/src/emit.rs`, `compute_archetypes`, priority rule). That choice is where the main-heavy sits. v2 of the emitter extends this:

1. Main depot: co-located with the longest radial line's far terminal. The station archetype is `depot-terminal`; the depot archetype is `main-heavy`.
2. Powered station stabling: every other terminal and selected intermediate powered stations receive healthy sets overnight; they do not emit depots.
3. Secondary/layup exception: emit only after a site-specific security, utility, access, or defect-recovery study shows station stabling is not viable.

Every depot writes a record into `design.toml`:

```
[[depots]]
station          = "line-a-east"
archetype        = "main-heavy"
fleet_stalls     = 4  # illustrative workshop bays; calculate per §4
energy_site_tier = "depot-main"  # optional: selected by the archetype
```

The scenario generator resolves depot energy tiers into `[[sites]]` and operating service settings into the simulator input. The simulator (`osr-sim`) models service countdowns and energy flows. A future physical distributed-stabling model must satisfy these requirements:

- Charges each trainset at depot pad power while held and overnight.
- Treat powered passenger stations and layups as valid overnight stabling points when the distributed-stabling policy is enabled in `lib/templates/depots.toml`.
- Holds a returning train for the configured clean/inspect/recharge slot at one designated depot per line and emits service-start/service-complete evidence.
- Preserves the zero dedicated service-rotation policy and distinguishes planned spares and cold reserve from revenue fleet requirements.
- Feeds the depot battery bank into the network-wide energy balance.

8.1 Executable station-stabling candidate

The simulator supports `station_stabling = true` in scenario `[[fleets]]`, with selected powered-station `dispatch_points`. Stations may use solar/storage without grid import. Waiting trains use up to 150 kW, capped by installed station and shared cabinet/source limits, for low-C top-up to 95% SoC; after closing, returning trains hold at selected stations and resume under schedule, energy and movement-authority gates. This does not require returning healthy trains to the main depot. Before departure, a distributed train must retain the 20% reserve over the route to the next selected charging station, including unpowered stops and terminal reversals. A failed pad or unavailable energy source excludes a destination. A grid outage alone does not disable a battery-backed charger. The gate checks current source availability; it does not reserve future energy against other arrivals. Actual charging debits shared storage, and each next departure rechecks the onboard reserve. Continuous replay verifies the duty. Optional `spare_count` and `cold_reserve_count` are included in each fleet's total inventory and default to zero. Candidates use the city's declared counts; these trains remain parked and can charge without routine dispatch.

The distributed-stabling workflow generates separate candidates for all catalogue cities and records the Samawah overnight comparison, checking each planned line/station/direction at opening. Plans compare allocated train lengths with the reference platform envelope. The planning/replay acceptance gate also limits overnight allocation to two trainsets per selected station, including reserves. Shared station IDs count once; a four-berth reference platform does not override this provision. The simulator exposes over-capacity queues for diagnosis, and the combined replay fails even when holding, charging and morning departures otherwise pass. Verified physical slots, security, inspection release, reserve activation, defective-train routing, full-day energy duty and deployment acceptance remain open.

9. Pitfalls and decisions

- At-grade only, per RFC 0011. Depots are large; elevating the footprint is prohibitively expensive. The auto-gen emitter rejects depot siting on an elevated section.
- One reference `main-heavy` everywhere. Operators with unique geography (coastal, mountainside) will push for bespoke depots. The upstream catalogue stays at one shape; the deployment's civil team handles the site-specific adaptation.
- Wheel lathe at the main-heavy only. Decentralising lathes is tempting (saves depot-trip time) but the operator depth for wheel-reprofiling is thin in target regions; centralising at one site lets that thin team do all the work.
- No pit at `layup-minimal`. A pit adds cost, safety maintenance, and lighting — just for overnight parking. Skip.
- Depot simulation in `osr-sim`. The sim models depot top-up while held plus an explicit service countdown at the designated depot. Cleaning, release inspection, diagnostics, and recharge are concurrent; deeper maintenance remains a separate availability/CMMS model.
- Distributed stabling is not distributed maintenance. Overnight passenger-station stabling is allowed only for healthy sets with remote isolation and CCTV. Any trainset needing inspection beyond the daily release walk-around, fault rectification, wheel work, or battery-module exchange goes to the main-heavy.

10. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	Shared energy tier, BOM quantities and layout requirements reconciled (2026-09-09); full equipment placement and installed cost remain open	v0
v2	Retained Rust seed and distributed-stabling studies; catalogue post-processing now prices one full-fleet depot per line	v0, RFC 0010 v2
v3	`osr-sim` turnaround-service state (clean / inspect / diagnostics / recharge), service countdown and event evidence (done 2026-08-12)	v2
v4	Accepted site/access/topology plans for all current Samawah line depots, with separately verified launch and maintenance capacity	Current full-fleet scope, v3
v5	Reference depot CAD under CERN-OHL-S v2	v4

11. Relationship to existing work

- `lib/templates/depots.toml` — schema this RFC ratifies.
- `crates/osr-design/src/emit.rs` — auto-gen emitter. `depot-terminal` detection today; the `[[depots]]` block emission is v2.
- `crates/osr-energy-site` — every depot is one `osr-energy-site` instance; existing integration holds.

- `crates/osr-sim` — modest extension for v3; charging semantics already work.
- RFC 0010 `depot-terminal` archetype pairs 1:1 with this RFC's `main-heavy` or a mix of main-heavy + layup.

12. Open questions

1. Co-located vs detached training wing. Some operators prefer the training simulator in a quiet city building (closer to drivers' homes). The catalogue default is co- located with the main depot; per-deployment override.
2. Decentralised battery swapping. A future opt-in where trainsets exchange their battery pack at depot instead of recharging — halves the charging-pad occupancy. Out of scope upstream until battery-swap mechanical designs mature.
3. Night-shift head count. Ops rulebook §4.5 (RFC 0013) fixes the shift structure; what's the minimum overnight staff headcount at a `layup-minimal`? Currently zero (CCTV-monitored); confirm when the first `layup-minimal` deploys.
4. Depot expansion path. A `secondary-medium` that outgrows its 8 stalls — does it upgrade to `main-heavy` in place, or is a second-depot-nearby preferred? Likely the latter given construction-on-live-facility complexity.

13. Done criteria

- [x] Three archetypes committed (§1)
- [x] Fleet-sizing formula (§4)
- [x] Main-heavy envelope (§5) + secondary-medium (§6) + layup-minimal (§7)
- [x] Auto-gen pipeline integration (§8)
- [x] Pitfalls (§9)
- [x] Rollout ordered (§10)
- [x] Relationship to existing software + templates (§11)

Next: locate the full PV/storage inventory, reconcile installed cost, and provide track-by-track overnight allocation and conflict-aware run-in/run-out evidence. The quantity contract alone does not satisfy those release gates.

RFC 0015 — Driverless operation (GoA 4) as the sole train profile

Repository source: `docs/rfcs/0015-driverless-operation.md`

Status: accepted · 2026-04-22. Authors: OSR project. Superseded: nothing (new). Amends: RFC 0008 (cab), RFC 0007 (hardware), RFC 0005 (crate map), RFC 0013 (operations).

Current development amendment

Train-centred control retains onboard ATP, obstacle, emergency brake and physical departure functions. The new reference exercises existing ATP/ATO/brake evaluators; it does not demonstrate the complete GoA 4 sensor/door/PSD stack or approve driverless operation. OCC intentions and local privileged stopped recovery cannot bypass physical protection. See RFC 0033 for the proposed responsibility map, evidence boundaries and release gates.

1. Purpose

Every OpenSourceRail deployment ships as a GoA 4 (Unattended Train Operation, UTO) system from day one. The rolling stock has no driver cab and the operations model has no driver role.

This RFC sets the safety, hardware, and operations envelope for that commitment — specifically, the onboard obstacle-detection sensor suite + ECU that lets a trainset run without a driver's eyes in the loop.

2. Why driverless by default

Three non-novelty reasons:

1. Lower total cost. A cab costs ~$270k per trainset when wiring, cab displays, HVAC, controls, and emergency exits are summed. Over a 40-trainset fleet that is ~$11 M of capex. Drivers cost ~$110 k/year fully loaded; a 50-driver roster for round-the- clock operation is ~$5.5 M/year in wages alone.
2. Simpler trainset. Removing the cab takes out roughly 40 kg of wiring, four cab HMIs (status, mirror monitors, rear-view, comms), the master controller + dead-man handle, cab HVAC, cab door + exit, and all the diagnostic plumbing that terminates in the cab. The nose becomes a sensor cowl plus a coupler; the floor plan becomes full passenger capacity end-to-end. ~5 % more seats per trainset.
3. Simpler operations. Safety responsibilities are assigned directly to OCC, station staff, maintenance, and the onboard automation stack; the repository carries no train-driver procedure family.

The tradeoff is concentrated in the obstacle-detection problem: what a driver's eyes used to do, a sensor suite must now do. The rest of this RFC is mostly about that.

3. Scope

In scope of this RFC:

- Obstacle-detection sensor suite (ultrasonic + radar + camera, LIDAR optional) and the onboard ECU that fuses them.
- Removal of cab-only software and interfaces.
- New operations primitives: remote assist, platform staff role expansion, passenger emergency intercom.
- Amendments to RFC 0008, 0007, 0005, 0013.

Out of scope:

- Cab-retained variants. They are outside the project scope.
- Passenger-facing accessibility (covered by RFC 0010 §7 and not changed by driverless).
- Cybersecurity of the obstacle-detect data path (covered by the pending `osr-secbus` RFC).

4. What goes away

Removed	Reason	Replacement (if any)
Driver cab + windscreen	No driver	Nose-cone sensor cowl
Cab touchscreen	No driver	Remote-assist web console (OCC)
Master controller + dead-man	No driver	ATO + emergency plunger (intercom-triggered)
Cab HVAC + cab door	No driver	—
Cab-to-ground radio handset	Voice comms are now OCC-mediated	Cabin PA + passenger emergency intercom
Train-driver procedures	No driver	Dispatcher S1–S7 + Station-staff T1–T5 absorb
Dead-man controller	No driver to monitor	—
Driver display software	No driver to display to	Passenger PIS already exists

5. What becomes required

5.1 Obstacle-detection sensor suite

Two identical sensor packs per trainset — one behind each nose panel. The trainset has no permanent "leading end" — either end can lead on the current run. The active pack is always the one facing the direction of travel; the trailing pack self-tests continuously and stands ready for the return trip.

Each pack contains:

Sensor	Role	Range	Update rate	Unit price (2026)	Why
4 × ultrasonic (Murata MA40H1S-R or equivalent, 40 kHz)	Close-range safety belt	0.2 – 20 m	20 Hz	~$25 each	Platform gap, depot yard, emergency crawl — cheap, solid-state, unaffected by darkness, failure mode self-detectable from echo pattern
1 × solid-state LIDAR (Livox HAP / Tele-series / RoboSense M1 / Leishen CH-series — all Chinese OEMs now in the $550–2,200 bracket)	Mid-range 3D primary	5 – 200 m	10 Hz	$550 – $2,200	3D point cloud, day/night, 150–300 m published range, IP67 industrial variants available
1 × mmWave radar (TI AWR1843 or eq., 77 GHz)	All-weather validation	5 – 200 m	20 Hz	~$550	Penetrates dust, fog, and the sandstorms the LIDAR point cloud degrades under — independent physics for a true disagreement vote
1 × stereo camera pair	Object classification	0 – 100 m	30 Hz	~$325	Distinguishes human / animal / debris; informs verdict severity, not primary safety

The suite is multi-sensor by design — obstacle detection is too consequential to rest on a single physics. A sensor-disagreement verdict is explicit in the evaluator (§10.1) and fail-restrictive.

5.1.1 Degraded-sensor speed policy

The evaluator exposes a dedicated `RestrictedSpeed` verdict so that partial sensor failures produce a smooth speed reduction instead of an EB:

Configuration	Verdict (no obstacles detected)	Effect
Ultrasonic + LIDAR + radar all healthy	`Clear`	Full MA-permitted speed
LIDAR offline / stale · radar healthy	`RestrictedSpeed`	ATO caps at 40 km/h (the ultrasonic-safe envelope)
LIDAR healthy · radar offline	`Clear`	Full speed — radar is a validation channel, not primary
Both long-range offline · train ≤ 40 km/h	`RestrictedSpeed`	ATO holds ≤ 40 km/h
Both long-range offline · train > 40 km/h	`EmergencyBrake`	No coverage beyond ultrasonic band at this speed — full EB
Any ultrasonic channel stale / faulted	`EmergencyBrake`	Ultrasonic is always required (close-range safety belt)
Peer 2oo2 channel disagrees	`EmergencyBrake`	Fail-restrictive

The policy is asymmetric on purpose: LIDAR is the primary long-range channel; radar is its all-weather validator. Losing LIDAR restricts speed; losing radar (while LIDAR still sees 3D at 5–200 m) does not. Both gone EB if speed would outrun the ultrasonic envelope.

Why LIDAR failing caps at 40 km/h rather than EB. At 40 km/h the stopping distance is ≈ 18 m, inside the ultrasonic reliable band. The trainset keeps moving (avoiding a stranded-passengers incident) but at a speed that ultrasonic can safely cover if radar also fails in the next tick. Braking smoothly with the service brake to 40 km/h from mainline speed is a passenger-comfort decision that costs no safety — the remaining sensors still cover the revised stopping distance.

Why ultrasonic is primary for the close-range envelope. Ultrasonic transducers are the cheapest, simplest, and most failure-predictable sensor in the suite. A quadrant (four transducers with overlapping cones) costs ~€100, draws < 1 W, and has a failure mode that is detectable from its own echo pattern. For the platform-approach envelope (< 40 km/h, < 20 m stopping distance) it is sufficient on its own — which is why RFC 0015 makes it the primary safety belt for that speed band. Depot and workshop movements run permanently inside the ultrasonic band.

Why LIDAR is primary mid-range, not optional. The 2025– 2026 market shift in Chinese solid-state LIDAR put the Livox-class unit (150–300 m range, IP67, automotive-grade) into the $550–2 200 bracket — an order of magnitude below where it was three years ago. At that price, LIDAR becomes the cheapest way to cover the 20 m – 200 m obstacle envelope with full 3D range data; it displaces radar as the primary mid-range sensor and pushes radar into the all-weather-validation role. Any target-region dust-storm event (see RFC 0013 S4.1) degrades LIDAR point density, which is exactly when the radar's mmWave penetration matters.

Why radar stays in the suite even with LIDAR promoted. The point of the multi-sensor architecture is that each sensor fails under a different physics. LIDAR fails in heavy dust or direct sun; radar fails on small stationary objects; ultrasonic fails at speed; cameras fail at night. A 2oo2-style safety vote across independent physics is stronger than any one sensor's specified SIL rating.

Why cameras are classifier-only. Cameras inform the severity of a detection (human vs. animal vs. debris) which feeds the verdict severity (`CrawlOnly` for a paper bag, `EmergencyBrake` for a person). They are not the safety-primary channel for detection — detection comes from ultrasonic / LIDAR / radar. Cameras are included at cheap cost ($325/stereo pair) because classifier data improves operational availability (fewer spurious EBs on windblown debris).

5.2 The T-OBS ECU

A new onboard host class: T-OBS (Train Obstacle-Detection ECU). Architecture mirrors T-ECU/S:

- 2 × Raspberry Pi RP2350 in a 2oo2 composite fail-safe voting arrangement, each running identical `no_std` Rust obstacle-detection code.
- 1 × Raspberry Pi CM5 app processor for sensor fusion and classifier inference (non-safety path).
- Inputs: 4 × ultrasonic (analog-in per channel), radar (CAN-FD), stereo camera (MIPI-CSI to CM5), LIDAR (Ethernet, optional).
- Outputs: a single `ObstacleVerdict` bus message published every 50 ms, evaluated into the main T-ECU/S 2oo2 chain as a brake-demand input. Verdict `Emergency` EB via the same 2oo2 AND-gate relay stage as the ATP.

Two T-OBS modules per trainset, one at each end. Only the leading end's verdict is active; the trailing module self-tests and publishes an `Inactive` status.

Detailed hardware spec lives at `control-electronics/t-obs/schematics/v2-spec/` (scaffold only; v3 KiCad capture per RFC 0007 v3 rollout).

5.3 Passenger emergency intercom

Every car carries at least 4 passenger emergency intercoms (one per car end, both doors). Pressing an intercom:

1. Triggers an immediate brake application to 50 % braking effort (not EB) — the trainset slows to a controlled stop at the next station.
2. Opens an audio + video channel to OCC remote-assist.
3. Logs the event to `osr-event-recorder` with timestamp, car id, and intercom id.

OCC can release the hold and resume normal operation remotely, or escalate to full EB if the situation warrants.

Why braking to the next station, not EB: unplanned EB in a driverless trainset with no driver to manage the aftermath strands the passengers between stations — a self-inflicted mass-incident risk. Braking controlled to the next station puts the passengers where they can be helped.

5.4 Platform screen doors upgraded to mandatory

Earlier station concepts made PSDs optional for `halt` archetypes. Under GoA 4, PSDs are mandatory at every station with passenger boarding — no driver means no human last-line- of-defence between a passenger and the track. RFC 0010 v2.1 will reflect this.

Exception: `depot-terminal` platforms accessible only by staff do not require PSDs; staff training (M1) substitutes.

5.5 Enhanced CCTV coverage

Every car carries at least 4 cabin cameras, live to OCC. Platform cameras already exist per RFC 0010; coverage is now contiguous end-to-end so OCC can see any point in the system on demand.

6. Safety case

6.1 Equivalence claim

The driverless trainset makes its safety case through:

- SIL-4 obstacle detection replacing driver's-eye detection, with a lower false-negative rate on rail-relevant obstacle classes (validated against a 10 000-object test set during type certification).
- Lower reaction latency (radar-to-brake ~50 ms vs typical driver reaction of 1.5 s).
- No fatigue, no distraction, no panic. Driverless trains don't miss red signals because they were texting. The statistical argument is well-established in global metro operations (including DLR, Copenhagen, and Singapore NSL).

6.2 New SIL-4 claims for the safety case

RFC 0015 adds five safety goals to the `osr-safety-case` GSN tree:

Goal	Claim	Evidence
G15	`osr-obstacle-detect` produces a restrictive verdict when any sensor reports an obstacle inside the current stopping-distance envelope; confident light-debris classifications may downgrade to `CrawlOnly`, all other in-envelope detections produce `EmergencyBrake`	O1 Kani harness + proptest
G16	Sensor disagreement between the 2oo2 channels produces `Emergency` (fail-restrictive)	O3 Kani harness
G17	Stale sensor data (> 100 ms since last update) produces `Emergency`	O2 Kani harness
G18a	With every long-range sensor offline above 40 km/h, the verdict is `EmergencyBrake`	O4a Kani harness
G18b	With LIDAR offline (radar healthy), the verdict is at least `RestrictedSpeed` and never `EmergencyBrake` from the O4 branch alone	O4b Kani harness
G19	Passenger-intercom-triggered stop is delivered within 3 s of press	`osr-passenger-assist` state-machine tests + `osr-sim` intercom-to-brake integration test (one simulation tick)

6.3 Residual risks explicit

Driverless operation carries residual risks that must be handled without an onboard operator. Each one is explicitly acknowledged:

- Obscured obstacle above sensor plane (e.g., a person standing upright but only the torso is visible above the platform edge): mitigated by PSD mandatory (§5.4) + ultrasonic under-body sweep at platform approach.
- Suicide on track: the most common rail-fatality mode in the industry, driverless or not. Not uniquely worsened by driverless; PSDs eliminate the station-platform variant entirely.
- Slipping rail, wheelspin: detected by `osr-odometry` slip-flag, triggers EB. No driver judgement involved.
- Terrorism / sabotage on track: track-intrusion detection at wayside (out of RFC 0015 scope; RFC 0016 candidate).
- Passenger medical emergency mid-journey: intercom to OCC; OCC dispatches medical assistance to the next station.

6.4 Certification pathway

Type certification to EN 62267 (UGTMS — Urban Guided Transport Management + Command/Control Systems) at GoA 4 is the target. Each deployment additionally certifies to the local regulator (NRSA equivalent).

The certification package draws from the existing safety-case tree plus the five new goals in §6.2.

7. Operations implications

7.1 Unattended roles only

RFC 0013 assigns all operational decisions to the onboard automation, OCC dispatcher, station staff, maintenance, or control-centre roles. It contains no onboard train-driver role or mixed-fleet procedure set.

7.2 OCC role expands

The OCC gains two new staff roles (amendment to RFC 0013 §4.5 C1):

- Remote-assist operator. Handles passenger intercom calls, takes manual override of a stopped trainset if needed, and coordinates with emergency services for on-board medical events. One operator per 5–10 trainsets in service.
- Fleet-health supervisor. Monitors the live CCTV + sensor feeds for trainset anomalies (smoke, unusual passenger behaviour, unattended items) that a driver would previously have noticed.

7.3 Station staff role expands

Station-staff T2 (Passenger boarding) gains two additional responsibilities:

- Platform dispatch ready-check. Staff confirm PSD closed + platform clear before the dispatch signal is released. This substitutes for the driver's final platform check (D4.5).
- Boarding assistance. Staff are now the sole human presence during boarding; wheelchair assist, passenger inquiries, and boarding-rate management all sit at platform level.

7.4 No loss of safety personnel — they move where risk is

The driverless pivot does not reduce safety staffing. The heads-per-hour shifted from driver-in-cab to OCC + platform: same people, different posts, generally safer posts.

8. Rolling-stock implications (amends RFC 0008)

8.1 Cab eliminated

- Front wall: just a nose cone with the §5.1 sensor cowl + a coupler interface.
- No windscreen, no cab display, no master controller, no cab door, no cab HVAC, no rear-view mirror.
- Both ends of the trainset are symmetric — no leading / trailing distinction at the rolling-stock level.

8.2 Recovery-mode lockable cabinet

Every trainset retains a recovery-mode keyswitch cabinet — a steel-locked enclosure behind the nose, accessible only with a physical key held by depot recovery crew. Inside:

- A wired pendant with: forward / reverse select, 0–15 km/h movement request, emergency stop.
- The keyswitch enables a slow-speed (≤ 15 km/h) protected shunt under ATP supervision — used only to position a stopped trainset for rescue coupling inside an isolated work zone.

This is the only manual-control path. It is physical, locked, slow-speed, and single-purpose. No full cab required.

8.3 Floor plan extends to both ends

Passenger seating extends to the full car length. The additional ~3 m × 2.4 m of floor at each end of the lead + rear cars = about 14 extra seats per trainset on the `light-metro-3car` family.

8.4 Mass + capex savings (per trainset)

Rough figures for the `light-metro-3car` reference:

Item	Saved mass (kg)	Saved capex (USD)
Cab structure (front wall, door, windscreen)	~800	~$33 000
Cab controls + displays + wiring	~150	~$87 000
Cab HVAC	~120	~$16 000
Cab seat + interior trim	~80	~$9 000
Mirror / wiper / horn package	~40	~$4 000
Cab lighting + accessories	~20	~$3 000
Total	~1 210 kg / cab × 2 cabs	~$150 k × 2
Added (T-OBS + sensors + intercoms)	~60 kg × 2 ends	~$27 k × 2
Net	~2 300 kg saved	~$250 k saved

9. Hardware implications (amends RFC 0007)

A new host class T-OBS joins the palette:

Class	SoC	Role
T-ECU/S	2× RP2350 + RPi CM5	Train safety kernel (ATP, brake, etc.)
T-ECU/A	RPi CM5	Train application (TCMS, PIS, TCN)
T-OBS	2× RP2350 + RPi CM5	Obstacle detection + sensor fusion
W-SBC	Radxa CM5	Wayside
S-SBC	RPi CM5	Station

T-OBS is two modules per trainset (one at each end). Architecture and PCB family are re-used from T-ECU/S; only the carrier board (sensor breakouts, CAN-FD transceiver, analog-in front-end, MIPI-CSI camera headers) differs.

10. Software implications (amends RFC 0005)

10.1 New crate

`osr-obstacle-detect` — pure-function SIL-4 evaluator. Runs on T-OBS. Takes a `SensorFrame` (ultrasonic ranges × 4, LIDAR return list, radar return list, camera classifier output), current train speed, current Movement Authority end, and the peer-channel clear bit, and returns an `ObstacleVerdict`:

```
enum ObstacleVerdict {
    Clear,
    RestrictedSpeed,   // ATO caps at 40 km/h (LIDAR degraded)
    CrawlOnly,         // ATO caps at 15 km/h (soft obstacle)
    EmergencyBrake,
}
```

Verdicts are strictly ordered by severity: `Clear < RestrictedSpeed < CrawlOnly < EmergencyBrake`. A detection inside the envelope can only escalate severity; a clearer sensor frame can only de-escalate.

Five SIL-4 safety properties (O1, O2, O3, O4a+O4b, O5) mapped to Kani harnesses; proptest coverage across the same properties.

10.2 Exclusive unattended train image

`osr-trainset-image` is the single onboard integration entry point. It re-exports the GoA 4 stack (ATP, ATO, brake, derailment, door control, fire safety, obstacle detection, odometry, passenger assistance, and TCMS). There is no cab feature flag, driver-display crate, or dead-man controller in the workspace.

11. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	`osr-obstacle-detect` crate at `crates/osr-obstacle-detect/` — SIL-4 pure-function evaluator with O1/O2/O3/O4a/O4b/O5 Kani harnesses + 8 proptests (26 tests total); GSN safety-case goals G15–G19 at `docs/safety-case/gsn/60-obstacle-detect.toml` close against real Kani + proptest evidence (done 2026-04-22).	v0
v2	`osr-sim` shadow onboard stack now calls `osr_obstacle_detect::evaluate` at every tick: each `OnboardShadow` carries an `obstacle_out` field, feeds an all-clear synthetic `SensorFrame` into the evaluator, threads `ObstacleVerdict::EmergencyBrake` through `BrakeInputs::obstacle_emergency`, and feeds `RestrictedSpeed` / `CrawlOnly` into the ATO cruise target as 40 km/h / 15 km/h caps. Per-verdict tick counters roll up in `OnboardSummary::total_obstacle_{restricted,crawl,emergency}_ticks`; the simulator summary prints them under "Onboard shadow stack". `BrakeInputs`, `EmergencySources` (in both `osr-brake` and `osr-tcms`), and the brake evaluator carry `obstacle_emergency` end-to-end. v2.1 (2026-04-23): scenario-driven sensor-fault injection — four new `FaultKind` variants (`LidarOffline`, `RadarOffline`, `UltrasonicChannelStale`, `ObstaclePeerDisagreement`) with per-train + fleet-wide scope; parser in `scenario_file.rs`; a demonstrator fault-injection scenario exercises every O-series path through the shadow stack. Run confirms RestrictedSpeed / EmergencyBrake verdicts fire without invariant violations. (done 2026-04-23)	v1
v3	T-OBS v2 schematic specification at `control-electronics/t-obs/schematics/v2-spec/` — block diagram + per-rail power budget + safety-nets with 2oo2 AND-gate design + RP2350 A/B pinouts + 12-entry M12 connector table (done 2026-04-22). KiCad capture is v3.1, deferred alongside the RFC 0007 hardware KiCad rollout.	v0
v4 (v1)	Type-certification pre-submission pack at `docs/certification/` — README + system-description + safety-requirements (SR-01..SR-24) + hazard-log (17 hazards, 7 classes) + evidence-register + EN 62267 clause-by-clause compliance-matrix. Every SR cross-referenced to a Kani harness / proptest / GSN goal / rulebook rule. Remaining for v4.1 (open): residual-risk narrative, per-clause compliance prose for §5–§9, and independent-assessor review (deployment-partner scope). (done 2026-04-23)	v2, v3
v5	First driverless revenue service on an OSR deployment instance	v4

12. Relationship to existing RFCs

- RFC 0003 (Samawah context and pilot rationale) — shows how one city can instantiate the GoA 4 operating model.
- RFC 0005 (SBC software architecture) — crate map amended in §10.
- RFC 0007 (control-electronics reference designs) — new T-OBS class in §9.
- RFC 0008 (rolling stock) — cab elimination in §8.
- RFC 0010 (station design standard) — PSDs become mandatory (§5.4); v2.1 amendment.
- RFC 0013 (operations rulebook) — OCC and station staff own the operational responsibilities (§7).

13. What this RFC does NOT include

- Automated track-intrusion detection at wayside — covered by RFC 0016.
- Autonomous platform-side boarding (still a human station-staff function; automation would require RFC 0015.1).
- Autonomous depot shunting under GoA 4 (treated as a separate RFC scope because the obstacle-detection envelope at depot speeds is different).

RFC 0016 — Wayside track-intrusion detection

Repository source: `docs/rfcs/0016-wayside-track-intrusion.md`

Status: accepted · 2026-04-23. Authors: OSR project. Complements: RFC 0015 (onboard obstacle detection). Amends: RFC 0005 (crate map), RFC 0007 (hardware palette adds one SKU variant), RFC 0001 (interlocking consumes intrusion state).

Current development amendment

Resource ownership never substitutes for an intrusion verdict. Asset-scoped infrastructure observations remain independent inputs to the interlocking; unknown or missing required proving restricts movement. The process reference supplies synthetic independent sensor ports. Hardware common causes, observation age and qualified physical detection remain explicit acceptance gates. See RFC 0033 for the proposed responsibility map, evidence boundaries and release gates.

1. Purpose

Driverless trains (RFC 0015) detect obstacles in their own path with the onboard `osr-obstacle-detect` evaluator. This RFC complements that with wayside track-intrusion detection — fence-line + ROW-mounted sensors that detect people, animals, debris, or vehicles entering the track envelope before a train arrives.

The goal is not redundancy with the onboard detector. The two systems serve different safety arguments:

- Onboard detector sees what's in front of the moving train, reactively — once something enters the current envelope, brake hard.
- Wayside detector sees what's in the ROW at any time, proactively — if the section is not clear, the interlocking withholds Movement Authority and the train never arrives at the obstacle.

Wayside detection is what lets the interlocking say "this section is safe" rather than "this section was safe the last time a train passed through." In GoA 4 operation, with no driver to see irregularities between trains, this is a necessary safety property — not a nice-to-have.

2. Why now, not in RFC 0015

RFC 0015 §13 explicitly listed "automated track-intrusion detection at wayside" as a future RFC candidate. The two RFCs partition cleanly:

- RFC 0015 owns the onboard sensor suite, the T-OBS ECU, and the driverless rolling-stock profile.
- RFC 0016 owns the wayside sensor suite, a new `osr-intrusion-detect` evaluator on the W-SBC host class, and the feed into `osr-interlocking` that withholds MA on intrusion-flagged sections.

Splitting keeps each RFC focused on one SIL-4 evaluator + one host class, matching the RFC-per-crate-family pattern of the project.

3. Scope

In scope:

- Wayside sensor-suite spec (fence contact, ROW LIDAR, ROW radar, CCTV AI).
- `osr-intrusion-detect` evaluator: per-section `IntrusionVerdict` (Clear / Unknown / Present), with fail-restrictive on Unknown.
- Integration with `osr-interlocking`: intrusion state is a new gating input alongside switch-position and block-occupancy.
- Operations amendments: how the dispatcher handles a section under active intrusion.

Out of scope:

- Fencing construction (deployment-partner civil scope).
- Video-analytics AI model training (training data is deployment- specific; the classifier runs as a black box per §5.3).
- Long-range perimeter surveillance outside the ROW (deployment partner's security concern, not rail safety).

4. What the wayside sensor suite looks like

Per-section sensor pack (one pack per `[SEC]` id in the interlocking). Installed at ROW junction boxes; each junction box feeds a W-SBC running `osr-intrusion-detect`.

Sensor	Role	Typical unit	Unit price (2026)	Why
Fence-line contact	Perimeter breach	Vibration + cut-wire sensor (e.g., Senstar FlexZone)	$325–550 / 100 m	Instant alarm on fence cut or climb; mature tech, low false-positive rate
ROW LIDAR (pole-mount, every ~200 m)	3D silhouette in ROW	Livox Mid-360 (360° FoV, 70 m range) or equivalent	$875–1 300	3D presence detection, day/night, IP67 industrial — same supply-chain story as RFC 0015 §5.1
ROW radar (pole-mount, every ~500 m)	All-weather presence	TI AWR1843 or eq. 77 GHz	~$550	Penetrates fog/dust that degrades LIDAR; redundant physics
CCTV with AI classifier	Classification + human oversight	Commodity 4 K IP camera + edge TPU (Google Coral or eq.)	$435–875 per camera	Per-station already required by RFC 0010; AI classification reuses existing cameras

Sensor spacing is driven by section length (RFC 0001):

- `at-grade` sections: LIDAR every 200 m, radar every 500 m, fence-line continuous.
- `elevated` sections: LIDAR every 200 m; fence-line not applicable (viaduct edge is the barrier); radar optional.
- `bridge` sections: LIDAR at each end + mid-span; no fence-line applicable.

A typical `standard-urban` 1 km `at-grade` section carries: 5 LIDAR + 2 radar + 1 CCTV + 1 km fence-line. Total hardware cost ≈ $7 600 per section — small fraction of the $3.0 M per km civil cost.

5. The `osr-intrusion-detect` evaluator

5.1 Pure-function API

```
pub fn evaluate(
    frame: &WaysideSensorFrame,
    now_ns: u64,
    params: &IntrusionParams,
) -> IntrusionOutcome;
```

Inputs:

- `WaysideSensorFrame` — one tick of sensor data per section: fence-line state, LIDAR detection list (range + lateral), radar detection list, camera classification + confidence.
- `now_ns` — sim / system time for staleness checks.
- `IntrusionParams` — per-deployment thresholds (lateral rail- profile width, freshness timeout, classifier confidence floor).

Output:

- `IntrusionVerdict` — `Clear` / `Unknown` / `Present`.
- `TriggerReason` — which sensor + which class, for logging.

5.2 The three verdicts

- `Clear` — the section has no detected intrusion. MA may cross this section.
- `Unknown` — at least one safety-primary sensor is stale or offline, and no `Present` detection is confirmed. MA is withheld (fail-restrictive). This is the crucial property: a broken sensor does not imply safe track.
- `Present` — an intrusion is confirmed by at least one sensor of sufficient confidence. MA is withheld; dispatcher is alerted.

5.3 Safety properties (I1–I5)

Paralleling the O-series from RFC 0015:

Property	Claim
I1	Any LIDAR detection inside the ±1500 mm rail profile `Present`.
I2	Any sensor frame stale beyond `MAX_SENSOR_STALE_MS` `Unknown` (not `Clear`).
I3	Fence-line breach `Present` regardless of other sensor state.

Property	Claim
I4	Camera classifier alone (no LIDAR / radar corroboration) cannot emit `Clear` — the safety-primary physics must be alive.
I5	Refreshing a sensor never moves the verdict in a less-restrictive direction (monotone under freshness).

Each property anchors a Kani harness + a proptest — same pattern as the O-series.

5.4 Integration with `osr-interlocking`

A new `SectionIntrusion` entry on the consensus log carries the verdict per section. `osr-interlocking::section_available_to` reads this alongside the existing gates (block occupancy, switch position) and refuses to grant MA across a section whose latest `SectionIntrusion` is not `Clear`.

This is a v2 deliverable of this RFC; v1 scaffolds the evaluator and a first Kani suite.

6. Hardware implications

`osr-intrusion-detect` runs on the W-SBC (Radxa CM5 RK3588S) per RFC 0007. No new host class — the W-SBC already hosts `osr-interlocking` and `osr-wayside-points`, so adding intrusion-detect is a same-crate-family deployment.

Baseboard additions per junction box:

- 8× fence-line contact inputs (optically isolated).
- 2× CAN-FD (radar + spare).
- 1× 1000BASE-T to LIDAR.
- 1× MIPI-CSI to CCTV edge-TPU (or Ethernet if the TPU is separate).

No change to the T-ECU/S or T-OBS boards — this is purely a wayside addition.

7. Operations implications

7.1 Dispatcher (S rulebook amendment)

A new rule group S7 — Intrusion response:

- S7.1 On first `Present` verdict, the dispatcher holds the section and dispatches track-patrol within 10 min.
- S7.2 On `Unknown` verdict persisting > 5 min, the dispatcher treats it as a functional failure of wayside equipment and opens a work ticket; trains are held or diverted.
- S7.3 A section cleared by track-patrol and returned to `Clear` may re-enter revenue service.

7.2 Maintenance (M rulebook amendment)

- M7 Wayside sensor maintenance: 30-day walk-through inspecting each LIDAR/radar housing, cable runs, fence contacts. Same cadence as M3 track inspection.

Rulebook text added when RFC 0016 v2 lands.

8. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	`osr-intrusion-detect` crate at `crates/osr-intrusion-detect/` — pure-function evaluator with 5 Kani harnesses + proptest properties I1–I5. GSN safety-case file at `docs/safety-case/gsn/70-intrusion-detect.toml` closes against real evidence (done 2026-04-23).	v0
v2	Integration with `osr-interlocking`: `EntryPayload::SectionIntrusion` and `IntrusionState` (`Clear`/`Unknown`/`Present`) feed gate (d); only `Clear` permits movement, while `Present`, `Unknown`, or a missing verdict withhold MA.	v1, RFC 0001
v3	Sim integration: `FaultKind::WaysideIntrusion { section, state }` emits committed consensus entries. A demonstrator scenario stages Present / Unknown / Present events across three sections without any train entering a flagged section.	v2
v4	Rulebook amendments — dispatcher S7 at `docs/operations/dispatcher/s7-intrusion-response.md` (5 rules: S7.1 track-patrol, S7.2 sustained-Unknown, S7.3 return-to-service, S7.4 multi-section escalation, S7.5 no-override) and maintenance M7 at `docs/operations/maintenance/m7-wayside-sensor-maintenance.md` (5 rules: M7.1 30-day walk, M7.2 post-weather, M7.3 sustained-Unknown ticket, M7.4 replacement calibration, M7.5 fence continuity test). RFC 0013 §4.2 and §4.4 indices updated. (done 2026-04-23)	v2
v5	Deployment at Samawah: first-article sensor packs at 3 sections on Line 1	RFC 0003, v2

9. Relationship to existing RFCs

- RFC 0001 (consensus) — adds `SectionIntrusion` log entry.
- RFC 0005 (software architecture) — adds `osr-intrusion-detect` to the W-SBC crate map.
- RFC 0007 (hardware) — junction-box baseboard adds sensor headers; no new host class.
- RFC 0013 (operations) — S7 + M7 rule groups (v2+).
- RFC 0015 (driverless) — complements onboard obstacle- detect; the two-layer defence (wayside proactive, onboard reactive) is the full GoA 4 safety-envelope story.

10. What this RFC does NOT include

- Passive fence line without contact-sensor instrumentation — standard civil scope.
- Personnel-identification against a whitelist (crew vs. intruder) — handled by CCTV + human review at the OCC, not automated in this RFC.
- Anti-climb / anti-cut physical deterrents on the fence — civil scope per deployment-partner security spec.
- Section-level live-radar tracking of authorised track workers (under a `MaintenanceOverride`, RFC 0013 S5 rules apply; the intrusion-detect evaluator flags `Present` and the dispatcher cross-references against the active work-block manifest).

RFC 0017 — Cybersecurity: message authentication on the safety log

Repository source: `docs/rfcs/0017-cybersecurity-message-authentication.md`

Status: accepted · 2026-04-23. Authors: OSR project. Complements: RFC 0001 (consensus), RFC 0005 §4.9 (crate map). Amends: `osr-crypto` (adds ed25519), `osr-interlocking` (envelope on `Entry`), consensus wire format.

Current development amendment

The process reference reuses Ed25519/secbus and signed consensus proposals, with role/asset permissions, exact configuration hash, registered session, durable sequence/replay fences, freshness and size bounds. All position/protection events are Safety. Public deterministic test keys are synthetic and unsuitable for operation. A signed voter view assumes crash-fault behavior, not Byzantine quorum certification; authenticated erroneous observations remain in the FMEA. See RFC 0033 for the proposed responsibility map, evidence boundaries and release gates.

1. Purpose

Every entry on the track-state consensus log carries safety-critical authority — a forged `MaintenanceOverride` or `SectionIntrusion::Clear` could unlock a blocked section and put a train in front of a work crew. Today the protobuf schema reserves a `leader_signature` field but the code path that would verify it is not implemented: entries are trusted because they came from a running process in tree.

This RFC closes that gap. Every entry on the committed log is signed by the originating entity (train, wayside controller, dispatcher console) and every consumer verifies the signature before applying the entry to derived state.

2. Threat model

In scope:

- Forged entries. An attacker with network access to the consensus cluster proposes a crafted `SectionIntrusion::Clear` or a fake `SwitchObservation` hoping the interlocking accepts it.
- Replay. An attacker re-submits an old legitimate entry to confuse the derived state (e.g., replay a `RouteRelease`).
- Key compromise. A single wayside box is physically captured; its key is now under the attacker's control.

Out of scope for v1:

- Side-channel attacks on the signing key (handled at hardware level by the ATECC608B trust anchors on T-ECU/S, W-SBC, S-SBC).
- Quantum-era attacks. Ed25519 is 128-bit-classical-secure; a post-quantum migration is an RFC 0017.1 candidate once NIST finalises a PQ signature scheme in the rail-relevant performance bracket.
- TLS session setup on TRG and backhaul links — that's RFC 0018 (not yet drafted).

3. Design

3.1 Signature scheme

Ed25519 — curve25519 in the twisted-Edwards form, per RFC 8032.

- 64-byte signatures, 32-byte keys. Small enough to attach to every consensus entry without bloating the wire format.
- Batch verification available — a consensus cluster can verify N entries in sub-N time when under load.
- Deterministic signatures — no per-signature nonce required (eliminates the ECDSA Sony-PS3 class of bugs).
- Open-source reference implementations — `ed25519-dalek` in Rust is widely audited and actively maintained.

3.2 Entity keys

Every entity (train, wayside box, station, OCC console) holds one long-lived ed25519 keypair in its ATECC608B secure element. Public keys are published in a per-deployment key registry at bootstrap; private keys never leave the secure element.

Key ids mirror the existing `EntityId` scheme so the `Entry.leader_signature` maps one-to-one to `Entry` authorship.

3.3 Envelope

A signed entry is encoded as:

```
SignedEntry {
    entry_bytes: Vec<u8>,   // serde-json serialisation of Entry
    issuer: EntityId,       // which entity signed
    sig: [u8; 64],          // ed25519 signature over entry_bytes
}
```

Verification:

1. Look up `issuer` in the `KeyRegistry`. If absent → reject (`VerifyError::UnknownIssuer`).
2. Verify `sig` over `entry_bytes` with the registry's public key. If invalid → reject (`VerifyError::BadSignature`).
3. Otherwise return OK, and only then deserialise and apply the entry.

Verification is before deserialisation so a hostile payload can't exploit a parser bug before the signature check.

3.4 Replay protection

Each `Entry` already carries `entry_id` (monotonic) and `timestamp_ns` (PTP-synced). The derived-state fold is idempotent in `entry_id`: an entry already applied is a no-op when re-applied. Replay does not trick the derived state into a bad configuration.

The `osr-secbus` v1 crate does not perform this freshness check: it authenticates opaque bytes against a deployment key registry before deserialisation. The consensus wire integration in rollout v2 is responsible for deserialising verified entries, checking monotonic `entry_id`, and rejecting entries whose `timestamp_ns` is more than 60 seconds older than the cluster's current leader clock.

3.5 Key rotation (v2)

v1 treats keys as long-lived. v2 introduces a `KeyRotation` entry type that transfers authority from an old key to a new one; the registry follows the latest rotation. Deferred because it adds operational complexity that is not needed for first-article deployment.

4. What this RFC delivers (v1)

- `osr-crypto` extended with ed25519 sign + verify primitives.
- `osr-secbus` — new SIL-2 crate with `KeyRegistry`, `SignedBytes` envelope, and the verification API.
- Three SIL-2 properties (S1–S3) with Kani harnesses + proptest.
- GSN goals G25–G27 on the safety case, anchored to the properties.

5. Safety properties (S1–S3)

- S1 (deterministic verify): `verify(pk, msg, sig)` is a pure function — identical inputs → identical output.
- S2 (reject bad signature): any single-bit mutation of `sig` causes `verify` to return `false`.
- S3 (reject missing issuer): a `SignedBytes` whose `issuer` is not in the `KeyRegistry` rejects with `VerifyError::UnknownIssuer`.

6. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	`osr-crypto` ed25519 primitives + `osr-secbus` crate with key registry, envelope, verify; S1–S3 Kani + proptest; GSN G25–G27 close. (done 2026-04-23)	v0
v2	Authenticated consensus application ingress signs entry id, timestamp, category, and track-entry bytes as one body; the leader rejects unknown, forged, malformed, stale, future, replayed, and metadata-mismatched proposals before append; committed consumers independently verify the stored envelope and signed category before deserialising the track entry. (done 2026-08-11)	v1, RFC 0001
v3	`osr-sim::ConsensusBackend` uses the authenticated proposal and committed-consumer paths. End-to-end tests cover forged signature, altered payload, replay, stale/future timestamp, unknown issuer, metadata mismatch, and category downgrade. (done 2026-08-11)	v2
v4	Key rotation (`KeyRotation` entry type) + deployment tooling to mint and distribute keys	v2
v5	Hardware integration: ATECC608B provisioning pipeline; per-entity key generation + transfer at factory	v2, RFC 0007

7. Relationship to existing RFCs

- RFC 0001 (consensus) — adds the verify step between receive and apply; no changes to Raft itself.
- RFC 0005 (software architecture) — `osr-secbus` joins the Phase 2f infrastructure crate family alongside `osr-crypto` and `osr-historian`.
- RFC 0007 (hardware) — the ATECC608B role is already in the T-ECU/S + T-OBS + W-SBC specs; this RFC defines what that chip actually signs.
- RFC 0015 + RFC 0016 — every `SectionIntrusion` and `MaintenanceOverride` is now signed by its issuing W-SBC (RFC 0016) or OCC console (RFC 0013 S5). Prevents the forged- `Clear`-verdict attack that would otherwise defeat the safety gate.

8. What this RFC does NOT include

- TLS on TRG radios or OCC backhaul — RFC 0018 candidate.
- Encryption of the log at rest — the entries are integrity- protected; confidentiality is not claimed.
- Perfect-forward-secrecy session keys — long-lived ed25519 keys are sufficient for entry authentication; PFS is a session-layer concern (RFC 0018).
- Post-quantum migration — deferred pending NIST standardisation in a performance bracket suitable for embedded rail controllers.

RFC 0018 — Operator GUIs: simulator + operations control

Repository source: `docs/rfcs/0018-operator-guis.md`

Status: accepted · 2026-04-23. Authors: OSR project. Complements: RFC 0003 (sim scenarios), RFC 0013 (OCC procedures), RFC 0017 (entry authentication).

1. Purpose

Two human-facing GUIs that have been implicit in the project but not built:

- Simulator GUI (`osr-sim-gui`) — runs `osr-sim` against a scenario TOML, renders the network + trains + MAs + faults live, and lets a designer inspect any entity by click-through. The audience is the deployment engineer validating their design.toml before commissioning, and the OSR developer verifying new scenarios.
- Operations-control GUI (`osr-occ-gui`) — the dispatcher console that RFC 0013 §4.2 has always assumed exists. Renders the live network state (from the consensus log), shows every RFC 0013 alert (S3 incident categories, S4 weather overlays, S7 intrusion verdicts), and exposes the specific actions the rulebook allows a dispatcher to take (route grant, MaintenanceOverride, degraded-mode transition).

2. Non-goals

- Passenger-facing displays. PIS rendering is `osr-pis-onboard`
  - `osr-pis-station` — separate codepath, separate build.
- Maintenance workbench GUI. CBM dashboards are a v2 scope of `osr-cbm-backend`, not this RFC.
- Mobile apps. Neither GUI needs to run on a phone.
- 3D rendering. 2D network schematic is sufficient — legacy metros have proved that.
- Legacy CBTC operator-console emulation. OSR operates its own consensus log; there's no back-compat requirement.

3. Technology stack

egui (+ eframe) in pure Rust.

Why:

- Type-consistent with the rest of the workspace. One language from SIL-4 kernel to dispatcher console; fits the project's ethos of no-JavaScript-stack, no-proprietary-SCADA.
- Immediate-mode rendering matches the per-tick update cadence of the sim + consensus log — redraw cheaply, don't reason about widget state machines.
- Works natively + WebAssembly. A deployment partner can host the OCC GUI as a web app for remote supervision without rewriting it.
- Conduction-cooled SBC friendly. egui renders at 10 Hz on a 400 MHz single-core ARM — not that we need that, but it guarantees the OCC GUI runs on any reasonable workstation.
- No Electron. An Electron OCC app would pull in 200 MB of Chromium; egui's eframe binary is ~30 MB static-linked.

Rejected alternatives:

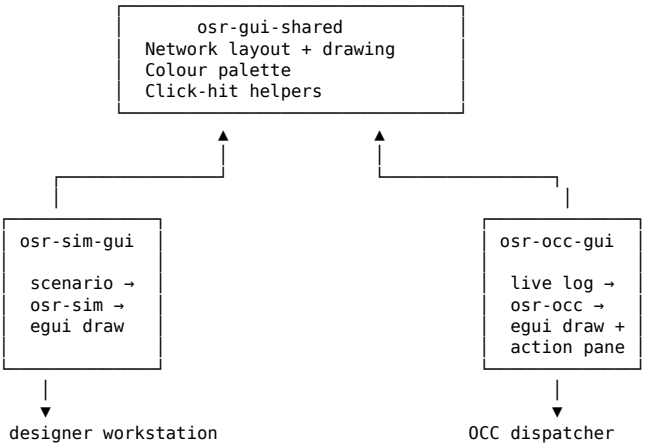
- React/Vue + Tauri — excellent for complex UIs, but adds a full JavaScript toolchain to the build + another set of security primitives to audit. Tauri remains a fallback if egui runs into a specific limitation.
- Iced / Slint — also pure Rust, but egui's immediate-mode model is a better fit for the "redraw the whole network each tick" workload than retained-mode widget trees.
- GTK / Qt — C/C++ deps outside the project's sandbox.

4. Architecture

Three new crates:

- `osr-gui-shared` — library. Network-layout + drawing primitives + colour palette. Consumed by both GUI binaries so the two have a consistent look. Not intended for direct use outside the GUIs.

- `osr-sim-gui` — binary. Wraps `osr-sim` + `osr-gui-shared`. Loads a scenario TOML, runs the sim per-tick in the same thread as the UI, renders the state each frame.
- `osr-occ-gui` — binary. Wraps `osr-occ` + `osr-gui-shared`. Reads from a live `ConsensusBackend`, renders the latest derived state, exposes dispatcher actions via modal dialogs that emit signed consensus entries (RFC 0017).



5. Shared rendering — `osr-gui-shared`

Scope:

- `NetworkLayout` — per-line horizontal strip, stations placed along the strip by cumulative distance. Same layout strategy as the existing HTML `osr-vis`, reused so dispatchers
  - designers see an identical schematic.
- `draw_network` / `draw_train` / `draw_section_state` — egui paint helpers that take a `Painter`, a `NetworkLayout`, and the current state to render.
- `Palette` — colours for every train phase, section state, intrusion verdict, obstacle verdict, fault state.

6. `osr-sim-gui`

Minimum viable feature set:

- Scenario loader. Filebrowser pick a `scenarios/*.toml` load.
- Playback controls. Play / pause / step / speed (1×, 10×, 100×).
- Network render. Generated Samawah schematic with every train drawn as a dot at its current `station_m`, colour-coded by phase.
- Inspector pane. Click a train sidebar shows ATP verdict, brake cmd, SoC, obstacle-detect verdict.
- Event log. Scrolling list of `EventKind` emissions — Dispatched, ArriveStation, DepartStation, etc.
- Fault indicators. Active faults from the scenario's `[[faults]]` section render as coloured badges on the affected station / section / train.

Post-v1 (deferred):

- Timeline scrubber for replay
- Record / export runs
- Differential compare of two scenarios side-by-side
- Remote-monitor mode (read the sim from another host)

7. `osr-occ-gui`

Minimum viable feature set for v1:

- Log-source selector. Pick a file replay (for exercises) or connect to a live `ConsensusBackend`.
- Network render. Same layout as the sim GUI.

- Section-state panel. `SectionIntrusion` verdicts from RFC 0016, colour-coded; click a section to see the latest `MaintenanceOverride`, `RouteGrant`, or `SectionIntrusion` on that section.
- Train panel. List of every train under the dispatcher's control, with the same colour coding as the sim GUI.
- Alert feed. Intercom presses, EB events, weather overlays, CBM red alerts.
- Action modal stubs. The four dispatcher actions from RFC 0013 §4.2 (issue route grant S2.1, commit `MaintenanceOverride` S5.1, extend dwell S2.3, declare degraded mode) exist as modal dialogs that validate the input but emit no-ops for v1.

Post-v1 (deferred):

- Live command emission to the consensus cluster with RFC 0017 signed envelopes
- Multi-operator role partitioning (dispatcher vs supervisor vs engineer-on-call) with per-role action gates
- Full audit-log display
- Video feed from station CCTV + cabin CCTV (RFC 0015 §5.5)

8. Security model

The OCC GUI is a privileged actor. Its action emissions must be signed by the logged-in dispatcher's personal key (RFC 0013 S1.1, RFC 0017). v2 of this RFC wires the signing key OCC UI login flow; v1 is read-only so the security model is trivially "read-only view of committed log."

9. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	`osr-gui-shared` network renderer + `osr-sim-gui` + `osr-occ-gui` scaffolded; both compile cleanly in the workspace, open an egui window, render the Samawah network, and expose the v1 minimum feature set listed above.	v0
v2	`osr-sim-gui` full sim integration (live tick stream, click-to-inspect)	v1, RFC 0003
v3	`osr-occ-gui` live consensus integration (read + write via RFC 0017 signed envelopes)	v1, RFC 0017 v2
v4	WebAssembly builds for remote access to both GUIs	v2, v3
v5	First OSR deployment instance uses `osr-occ-gui` in revenue service	v3

10. What this RFC does NOT include

- PIS (passenger-facing) displays. Separate `osr-pis-*` crates.
- CBM / maintenance dashboards. Separate `osr-cbm-backend` UI.
- Electron / web-framework-native versions. egui's WASM target covers the "browser access" use case.
- 3D rendering. Schematic 2D is explicitly the goal.
- Mobile apps. Not a target deployment form factor for either audience.

RFC 0019 — DIY plug-and-play electronics

Repository source: `docs/rfcs/0019-diy-electronics.md`

Status: accepted prototype integration path; hardware and operational qualification open. Review: 3 October 2026 — corrects the April candidate wiring and price assumptions. Amends: RFC 0007.

1. Purpose

Commodity modules provide a useful bench and first-article integration path for T-ECU/S, T-ECU/A, T-OBS, W-SBC and S-SBC. A custom PCB is optional for a bench prototype. The accepted path does not release a railway control assembly, a safety integrity level, a complete BOM or a bootable target image.

The detailed host requirements, reference integration source and release checklist control the current design stage. Exact supplier, pin, power and environment data must be frozen and tested before an operator can use the hardware.

2. Host classes

Host	Bench reference	Integration work still required
T-ECU/S	Dual evaluator channels and optional application carrier	External bus controllers, separately protected power/watchdogs, output feedback and real field-input mapping
T-ECU/A	CM5-compatible application carrier	Peripheral/modem compatibility, protected power, network/radio and thermal design
T-OBS	Dual evaluators, application carrier and sensor candidates	Sensor/harness choice, corrected startup power, timed ingress and measured fault/coverage tests
W-SBC	Commodity wayside controller	Cabinet mains/DC/UPS, earthing, field I/O, output and resource-control qualification
S-SBC	Commodity station controller	Station services, UPS, network, AFC/PIS/SCADA and degraded-operation design

3. Candidate components and corrected interfaces

Exact order codes, lifecycle, alternates and prices remain unverified in the historical candidate catalogue. Use vendor documentation and a complete board/harness BOM before ordering.

Earlier assumption	Required correction
CM5/CM5IO codes SC1124/SC1125 released as exact selections	Freeze actual memory/storage/carrier variant and vendor order code
USB isolator for SPI or two Pico USB-device endpoints	Directionally correct isolated SPI with controller/peripheral roles and separate supplies
Native Pico CAN-FD or Ethernet/RMII	External controllers/transceivers and measured bus/deadline budget
TPS3701 heartbeat watchdog	Voltage detector only; separate external heartbeat watchdog functions
HDR-60-24 powers 5/12 V from a 24 V onboard bus	AC/high-voltage DC input to 24 V output; protected onboard DC/DC branches required
ADS1115 captures HC-SR04 digital echo	Timer/PIO capture with voltage conversion; raw piezo requires excitation/receiver AFE
Murata raw piezo labelled rail-grade plug-in replacement	Candidate-specific driver, lifecycle and environmental qualification; no rail-grade claim
HAP USB/Gigabit/12 V PoE interface	TX uses 100BASE-TX; T1 uses 100BASE-T1; vendor 9–18 V supply/cable separately
Relay board proves 2oo2 safety	Bench simulated load only; driver/contact feedback, welded faults and common-cause analysis required

The CM5 carrier alone has a 25 W input envelope. The T-OBS bench candidate calculation allocates 71.5 W simultaneous output and 99.31 W input capacity with efficiency/margin, selecting a 100 W reference envelope. This is sizing rather than measured sustained consumption; no fuse or converter part is thereby qualified. Exact power and thermal behaviour must be tested on the selected revision.

4. Procurement and cost

The former DIY/custom price comparison and host subtotals are withdrawn as complete assembly comparisons. They omitted DC/DC conversion, watchdogs, output feedback, external controllers, harness/connectors, enclosure thermal provisions and qualification. A low-price module does not price the installed system. Keep unknown costs open and reconcile a quoted complete BOM against the city allowance before changing CAPEX or finance.

The Baghdad detailed register allocates reference quantities and Iraqi integration routes for the six-car fleet. These are nested engineering requirements, not additional purchases on top of complete priced assemblies or live ERP Items.

5. Assembly and evidence

1. Freeze host/board revision, vendor interfaces, exposed/package pins and supported peripheral resources.

2. Complete load, inrush, protection, isolation, thermal and EMC calculations.
3. Issue a cavity-by-cavity harness, terminal/wire list, enclosure and simulated-load output truth table.
4. Build and inspect a bench unit; measure voltage domains, boot/default outputs, power loss, heartbeat failures, peer faults and contact feedback.
5. Bind actual HAL/drivers, protocol/configuration and signed image to the revision; test stale data, timing, environmental exposure and recovery.
6. Close the appropriate release checklist and independent assessment before any field or train actuation.

Production SD images remain absent. Software/library tests, the same MCU silicon or a relay schematic do not prove qualified hardware independence. Do not connect a bench assembly to real brake/traction actuators as a result of this RFC.

RFC 0020 — Crashworthiness (EN 15227)

Repository source: `docs/rfcs/0020-crashworthiness.md`

Status: Draft — parametric spec only, no FEA ships with this RFC Date: 2026-04-24 Depends on: RFC 0008 Rolling-Stock Reference Design, RFC 0015 Driverless Operation

1. Summary

Every OSR trainset must survive the EN 15227 crash-reference scenarios C-I (train-to-train) and C-III (train-to-obstacle) without passenger-compartment intrusion. Because OSR is cabless (RFC 0015 §5.1), the impact-absorption zone that commercial cabbed designs put in the driver's cab has to live somewhere else. This RFC publishes the energy-absorption-zone allocation that replaces it — the values the design/component-catalogue parametric car body tracks and that vendors must meet when supplying an OSR-compatible shell.

The current published baseline still does not ship a full finite-element crash simulation. What it ships is:

- A parametric energy budget per consist family.
- An allocation of that budget to three absorption zones on each end of the trainset.
- A published set of geometric constraints (crumple-zone length, anti-climber interface height, under-frame load path) the car body must honour.

A deployment must commission an EN 15227-certified FEA vendor to verify the final design; this RFC gives that vendor a well-defined target rather than a moving spec.

2. Non-goals

- Not an FEA solver. We specify targets; we do not solve the dynamics. Simcenter / Abaqus / LS-DYNA remains the right tool.
- Not a material choice. The body material is aluminium per RFC 0008 §3.2; the RFC applies to any ductile alloy that meets EN 15085 weld class CP-C.
- Not an obstacle-deflection spec. Track-clearing plough (EN 15227 §6.5) lives in `osr-obstacle-detect`'s physical hardware spec, not here.

3. Reference scenarios

Scenario	Description	Closing speed	Target
C-I	Collision with an identical trainset on tangent track	36 km/h	No passenger-compartment intrusion; decel ≤ 7.5 g peak
C-III	Collision with a 15-tonne rigid obstacle (road vehicle at a level crossing)	36 km/h	No passenger-compartment intrusion; anti-climber engages
C-IV (informational)	Collision at a level crossing with a light road vehicle	Not binding on urban metro per RFC 0013 §9	—

Scenarios C-II (obstacle on track) and C-V (tram-style) are covered by deployment-specific assessments. Every OSR trainset must certify C-I + C-III.

4. Energy budget per consist

Under the C-I scenario, two identical trainsets close at 36 km/h relative closing energy is $0.5 \cdot 2 \cdot m \cdot v^2$ where m is the consist mass and v is 5 m/s per trainset. For the reference families (RFC 0008 §3.3 seated + standing + interior fit-out):

Consist	Mass tare + 6 p/m²	C-I closing energy
tram-2car	~66 t	1.65 MJ
light-metro-3car	~120 t	3.00 MJ
metro-4car	~160 t	4.00 MJ
metro-6car	~240 t	6.00 MJ

These totals govern the absorption-zone sizing in §5.

5. Three-zone absorption layout

Each end of the trainset splits its contribution to the closing energy across three sequential zones, each contributing a known fraction of the total. This is the standard EN 15227 arrangement; what's specific to OSR cabless is that zones 1 + 2 previously lived inside a driver's cab and now live in the sensor- cowl + forward under-frame section.

Zone	Location	Share	Mechanism
Zone 1 — Sacrificial cowl	Sensor cowl front + anti-climber	30 %	Controlled-crush foam + metal honeycomb within the cowl; sensors are lost but absorb ~0.45 MJ per end for a 3-car consist
Zone 2 — Under-frame crumple	Forward 2.5 m of under-frame between cowl and first bogie	40 %	Tubular steel crumple cans; cuts into depot access but stays clear of passenger envelope
Zone 3 — Anti-override interface	End wall between under-frame crumple and first door	30 %	Reinforced bulkhead; redistributes remaining energy into the body torsion ring

Each end of every consist must provide the summed per-end budget (half the closing-energy total for C-I, full for C-III).

6. Parametric geometry constraints

The design/component-catalogue car body + sensor cowl enforce these constraints via the `Crashworthiness` helper in osr_mech.rolling_stock.car_body (added in v0.1). The CAD geometry reserves the zone envelopes — vendor-supplied shells that fit in must meet the mechanical targets below.

Constraint	Value	Rationale
Zone 1 crumple length (sensor cowl)	≥ 900 mm	Honeycomb deceleration band at 150 kN/m² crush stress
Zone 2 crumple length (under-frame)	≥ 2 000 mm	Tubular can progressive collapse
Anti-climber pin height above rail	760 mm (± 25)	EN 15227 §6.4 pin-on-pin interface
End bulkhead thickness	≥ 40 mm aluminium or equivalent	Load-path continuity
Passenger compartment survival space (post-crash)	Body length ≥ 95 % of original	Egress path integrity
Peak longitudinal decel (passenger floor)	≤ 7.5 g	EN 15227 occupant criterion

7. Verification path

1. Parametric check (in-repo, pytest): the car body geometry reserves the Zone 1 + Zone 2 volumes. `test_tier1_additions.py` covers the envelope reservation.
2. Static FEA (external): linear analysis of the crumple-zone load paths by a licensed FEA vendor. Outputs: peak stress, first-yield energy.
3. Dynamic FEA (external): LS-DYNA / Abaqus explicit for each scenario. Outputs: decel curves, crush displacement, passenger- envelope intrusion check.
4. Full-scale test (per EN 15227): one test specimen per consist family, at the closing speed. Carried out by the type-approval authority; pass/fail gate for revenue service.

The current baseline closes step 1 and documents the plan for steps 2–4. A deployment budget must include the external FEA + physical-test cost — typically EUR 1.5–3.0 M per consist family, amortisable across every trainset in the family.

8. Non-OSR obstacle-plough interface

The obstacle-deflection plough (EN 15227 §6.5) fits onto the sensor-cowl front within the Zone 1 envelope. The plough's 6° deflection angle + 200 kN rating clears track-level debris without triggering the cowl's sacrificial crush zone. Interface:

- 4× M24 bolts on a 400 × 200 pattern, pre-tapped in the cowl front.
- Plough mass ≤ 120 kg (included in Zone 1 budget).
- Catch-pan and drain geometry to be frozen in the controlled vehicle drawing release; it is not yet a released manufacturing artifact.

9. Deployment checklist

- [ ] Commission an EN 15227 FEA analysis from a certified vendor (Politecnico di Milano, Newrail, Bombardier Legacy, or equivalent) against the §5 zone allocation.
- [ ] Confirm the vendor's crumple-can dimensions fit inside the parametric Zone 1 + Zone 2 envelopes.
- [ ] Budget for one full-scale test per consist family.

- [ ] File the type-approval evidence pack with the national rail regulator before first revenue day.

10. Revision history

Date	Version	Change
2026-04-24	v1	Initial draft. Three-zone layout + energy budget + parametric constraints.

RFC 0021 — 800 V-Class LFP Battery and Traction Architecture

Repository source: `docs/rfcs/0021-battery-traction.md`

Status: Current reference architecture Date: 2026-08-14 Depends on: RFC 0008, RFC 0020, RFC 0024, RFC 0026, RFC 0028

1. Decision

OpenSourceRail uses one catenary-free, 800 V-class LFP architecture. The promoted three-car train operates at approximately 650–700 V nominal and has:

- 675 kWh gross / 540 kWh routinely usable LFP;
- three independently serviceable 225 kWh gross car packs;
- two contactor-isolated strings per car, subject to the final pack study;
- six heavy-commercial-vehicle PMSMs and six matched controllers;
- 2.1 MW installed short-duration capability with a 1.8 MW control cap;
- direct-HV DC HVAC and isolated 110/48/24 V DC/DC domains;
- one protected roof-PV MPPT per car; and
- no traction transformer, train-wide auxiliary AC bus, or continuous wayside electrification.

Named products are RFQ reference candidates, not sole-source selections or proof of railway suitability. Supplier qualification, railway environmental tests, integration evidence, warranty, spares, and delivered scope remain mandatory.

2. Per-car topology

```
roof PV -> MPPT -------------------------+
station contact -> protected DC interface +--> 225 kWh LFP car pack
regenerative braking ---------------------+          |
                                                      +-> controller -> PMSM axle 1
                                                      +-> controller -> PMSM axle 2
                                                      +-> isolated LV DC/DC
                                                      +-> direct-HV DC HVAC
```

Each car protects and reports its own HV source. Train-level software may coordinate power limits but cannot replace local BMS protection or assume that either neighbouring car pack is available.

Required interface values include minimum/nominal/maximum voltage, continuous and pulse current, regeneration overvoltage margin, contactor interrupt rating, precharge energy, insulation thresholds, creepage/clearance, cooling duty, EMC, and 50 °C derating. The provisional normal upper voltage is 740 V until the selected controller and switching transients are qualified together.

3. Reference traction candidates

- EVE-class liquid-cooled high-power/railway LFP cells and modules;
- Inovance HM47-class PMSM and LD32-class controller, one controller per motor; and
- Longertek-class direct-HV DC rail HVAC.

Equivalent equipment is acceptable when it satisfies the OSR interface control document and qualification plan.

The gearbox ratio is derived from the selected motor torque-speed map; no historic ratio is inherited. Acceptance covers AW0–AW3 acceleration, grade, wet-rail adhesion, wheel wear, motor/controller overspeed, regenerative braking, one-channel unavailable operation, and repeated-stop thermal duty at 50 °C ambient.

4. Station DC architecture

```
PV -> MPPT DC/DC -> station DC bus <-> 500 kWh LFP module(s)
grid AC -> rectifier --------------------^
                                         |
                              500 kW DC/DC --+--> interlocked platform contacts
```

The normal three-car station starts with one 500 kWh gross stationary-LFP module, one 500 kW bidirectional DC/DC cabinet, and two mechanically separate contacts sharing that single cabinet budget. Terminals use the same hardware with longer dwell. At 700 V, 500 kW is about 714 A; at 650 V it is about 769 A. Power, current, efficiency, cable/contact loss, temperature

derating, battery charge acceptance, and simultaneous arrivals are enforced together.

A 60-second charge adds approximately 8 kWh after loss. It is a top-up, not a full recharge. The onboard pack absorbs route-to-route imbalance and terminal or depot dwell restores the daily energy balance.

The reference station power stage is a SINEXCEL PDS1-750K-H-class bidirectional DC/DC converter configured to 500 kW. The station separately provides isolation/earthing, insulation monitoring, cooling, contactors, emergency isolation, and safe contact sequencing.

Four-car high-throughput deployments use three complete 500 kWh storage modules and 500 kW cabinet/contact units; six-car deployments use four. The counts are the lowest standard-module quantities that keep the canonical large-city EOL-battery and maximum-climate simulations within their service floor. Storage power and the grid interface scale with the repeated modules; generated CAPEX carries every module. High-throughput fleet sizing also retains the aggregate onboard reserve demonstrated by the degraded-energy suite rather than shrinking solely because charging dwell becomes shorter. This is replication of one standard module, not a separate 1 MW product tier.

5. Battery fire protection

IEC 62928 is a design and test input. Each under-seat module has a fire-resistant enclosure, outward gas vent, temperature monitoring, and off-gas detection. The pack must limit propagation for the required evacuation interval.

Each car has an independent localized cooling system:

```
water reservoir -> monitored DC pump -> stainless distribution pipe
                -> battery-compartment mist nozzles
```

The mist cools adjacent modules and restricts propagation; it does not claim to stop cell-level thermal runaway. Passenger saloons have no automatic sprinklers or CO2 flooding. Controllers, DC/DC equipment, and HVAC electronics use fire-resistant metal enclosures, detection, and electrical isolation.

The staged response is:

1. Validate the event and inhibit charging.
2. Isolate the affected string or car pack according to the proved topology.
3. Restrict traction, alert TCMS/OCC, and request the nearest safe platform.
4. Discharge localized mist around the affected compartment.
5. Apply emergency braking if containment or essential control is lost.
6. Release doors only where the train/platform hazard assessment permits.

Reservoir level, pump current, line pressure, flow, nozzle continuity, sensor disagreement, and isolation state are diagnosable.

6. Body and service interfaces

The welded S355 underframe and spaceframe carry bogie, coupler, crash, anti-climber, and longitudinal loads. Composite side, portal, skirt, nose, ceiling, and roof cassettes use:

- 3-2-1 datum location with an asymmetric anti-reversal key;
- standardized captive fasteners;
- a controlled adhesive/sealant groove plus mechanical retention;
- defined bonding points; and
- keyed service connectors.

Composite cassettes may contain empty HVAC ducts, drains, cable raceways, clips, and equipment bosses. Cables, pipes, sensors, fans, filters, evaporators, dampers, and lights remain removable through access covers and are never encapsulated in laminate.

7. Procurement targets

Planning targets, pending comparable delivered-scope RFQs:

Equipment	Target
Complete 675 kWh traction-battery system	65–80 kUSD/train
Six motor/controller sets	48–72 kUSD/train
HV distribution/contactors	8–12 kUSD/train

Equipment	Target
LV DC/DC equipment	5–10 kUSD/train
Roof-PV MPPT/DC equipment	2–4 kUSD/train
500 kWh stationary-LFP module, BMS/cooling/enclosure/protection	35–40 kUSD/site
500 kW DC/DC power stage	10–20 kUSD/site
Contact/protection/control package	10–15 kUSD/site

These are affordability gates, not quotations. RFQs separate hardware, qualification, software/interface support, warranty, spares, delivery, and local integration.

8. Release gates

The architecture is not released for manufacture until these close:

- cell/string count and voltage/transient margin;
- isolation, earthing, HVIL, precharge, and fault interruption;
- gearing, acceleration, adhesion, grade, overspeed, and thermal duty;
- dwell charging, simultaneous arrivals, charger outage, low-solar days, terminal balancing, and minimum route SOC;
- propagation, vent routing, mist flow/pressure, failed-suppression behavior, detection latency, and evacuation time;
- composite fire/smoke/toxicity, insert pull-out, bond durability, water ingress, module-removal time, and repair method;
- mass, centre of gravity, axle load, crash load path, and FEA; and
- supplier RFQs and qualified-equivalent interface evidence.

RFC 0022 — Bogie and Traction Drive

Repository source: `docs/rfcs/0022-bogie-traction-drive.md`

Status: Current Depends on: RFC 0008, RFC 0021

1. Decision

Each car has one two-axle powered bogie and one two-axle trailer bogie. The powered bogie carries two independent heavy-commercial-vehicle-class PMSM and controller channels. The promoted three-car train therefore has three powered bogies, six powered axles, and six controllers.

2. Reference envelope

Parameter	Reference
Gauge	1,435 mm
Wheelbase	2,100 mm
Wheel diameter	760 mm new / 680 mm worn
Candidate motor	Inovance HM47-class or qualified equivalent
Candidate controller	Inovance LD32-class or qualified equivalent
Candidate short peak	350 kW per motor
Train installed peak	2.1 MW
Initial control cap	1.8 MW
DC operating range	650–700 V nominal, 740 V provisional upper limit

The bogie frame, suspension, brake mounts, axlebox, wheelset, and carbody interfaces are common wherever axle load and kinematic-envelope evidence permit.

3. Gearbox

No fixed ratio is released. The integrator selects the ratio from the actual motor torque-speed map, new/worn wheel diameters, 90 km/h service speed, overspeed margin, grade, acceleration, adhesion, and thermal duty. Packaging CAD values are explicitly non-authoritative seeds.

4. Acceptance

Evidence covers frame static/fatigue load, wheel/axle/bearing qualification, AW0–AW3 acceleration, wet/dry adhesion, wheel-slip protection, braking blend, regen at high SOC, motor/controller thermal soak, one-channel unavailable operation, ride dynamics, noise, ingress, EMC, lubrication, and wheel-lathe access.

RFC 0023 — Door System Reference Design

Repository source: `docs/rfcs/0023-door-system-reference-design.md`

Status: Current — v2A architecture, certification evidence open Date: 2026-04-26 Depends on: RFC 0008 Rolling-Stock Reference Design, RFC 0015 Driverless Operation, RFC 0007 Control Electronics Reference Designs

1. Summary

RFC 0008 §3.3 commits each 16.5 m car to two large double-leaf plug-door cassettes per side: twelve exterior cassettes, twenty-four leaves, and a nominal 1 500 mm clear opening per cassette on the `light-metro-3car` consist. It did not previously specify the door operator — the electromechanical actuator that opens, closes, locks, and senses obstruction. This was a meaningful omission. Door operators are a top-three reliability and operating-cost item on every metro, and the rail-certified vendor market (IFE / Knorr-Bremse, Faiveley / Wabtec, Nabtesco) prices a single operator at $20 k–$40 k per cassette at OECD list — $120 k–$240 k per `light-metro-3car` consist, of which 60–70 % is certification margin re-billed to each operator.

This RFC commits an electric linear-actuator door operator built from commodity industrial silicon, certified once at the project level under EN 14752, and integrated with `osr-door-control`.

2. Non-goals

- Not a pneumatic door system. Pneumatic plug doors (the legacy default) require a compressor + air-treatment loop; the rolling-stock design (RFC 0008 §3.2) has already removed pneumatics from the brake system, and adding them back for doors would re-introduce the maintenance line item we eliminated. Electric only.
- Not a sliding-pocket-door design. OSR commits to plug doors per RFC 0008 — this RFC covers the operator, not the door geometry.
- Not a platform-screen-door (PSD) interface spec. PSD coupling protocol stays in `osr-psd`.

3. Architecture

Aspect	Choice	Rationale
Actuator	24 V DC brushless linear actuator, 900-1 200 mm stroke, 3-5 kN continuous	Commodity industrial silicon (Thomson PC-series, Festo EPCO, SKF CASM, regional copies). Single-vendor risk avoided by spec'ing to interchangeable form factor.
Drive	Cog-belt + slide-rail, no pneumatic	Same architecture as modern electric metro plug doors at the mechanical level; the complexity is in EN 14752 certification, not in the parts.
Controller	BLDC servo controller, CAN-FD interface to `osr-door-control`, hardwired closed-and-locked loop to T-ECU/S	Roboteq / RoboClaw class for pilot; in-house RFC 0007-class board for production. Safety output does not depend on CAN alone.
Obstruction detection	Light-curtain across opening (Banner / SICK rail-grade) + pinch-current sensing in actuator	Two independent detection paths; either alone halts the close.
Lock	Mechanical hook lock at full-close, monitored by independent micro-switch	EN 14752 mandates positive-locked status sense.
Emergency egress	Manual unlock handle inside + outside, mechanical override of lock	EN 14752 + UIC 560.
Position sense	Linear encoder on actuator + redundant Hall on the lock cam	Dual-channel agreement to `osr-door-control` over CAN-FD.

4. Interface Contract

Interface	Requirement
Structural aperture	Door cassette installs into the side-frame portal defined by `LM3-DOOR-200`; supplier changes may not move primary steel datums
Power	24 V DC control power plus protected motor feed from the car auxiliary cabinet
Data	CAN-FD command/status to `osr-door-control`; diagnostic Ethernet optional
Safety loop	Independent hardwired closed-and-locked contact per cassette, wired into the train safety chain
Drainage	Threshold tray drains outside the saloon; drain path must remain inspectable after interior trim installation
Emergency release	Interior release at the doorway and exterior release behind a tamper-evident cover
Gap filler	Door-sill flap/skirt permissive is interlocked with door-open command and platform-side selection

5. Certification

EN 14752 (Bodyside doors for rail vehicles) is the governing standard. The certification surface is real but is a one-time engineering item: the project absorbs it once and amortises across every deployment, in contrast to the current commercial-vendor model where every operator re-pays the certification load embedded in the per-door price.

Certification artefacts land under `docs/certification/door-system/`. The release gap is tracked in `docs/certification/release-gap-register.md`.

6. Cost model (indicative, 2026, USD OECD-base)

Component	DIY-path BOM	Vendor reference
Linear actuator	$400	(bundled)
BLDC servo controller	$250	(bundled)
Slide rail + cog belt + carrier	$300	(bundled)
Light curtain	$400	(bundled)
Lock + position-sense + harness	$400	(bundled)
Integration + harness + commissioning	$250	(bundled)
Per-door total	~$2 000	$20 k–$40 k

`light-metro-3car` consist (12 exterior cassettes): ~$24 k DIY operator BOM vs $240 k–$480 k vendor before certification and qualification overhead. Country-cost multipliers from `lib/templates/country-costs.toml` apply to the labour share downstream.

7. Release Checklist

Gate	Closure artifact
Door envelope freeze	Supplier drawing proving the cassette fits `LM3-DOOR-200` without primary steel changes
EN 14752 plan	Notified-body or assessor route, test plan, acceptance criteria, and sample count
Lock-loop validation	Tests proving a false closed-and-locked status cannot be produced by one sensor fault
Obstruction validation	Light-curtain, current-sense, reopen timing, force limit, and degraded-mode tests
Emergency release	Interior/exterior release force, tamper evidence, reset procedure, and signage
PSD/platform interface	Door-side select, platform-screen-door interlock, gap-filler permissive, and failure states
Environmental tests	Heat, dust, rain ingress, washer/cleaning chemical compatibility, and low-temperature grease variant

8. Open questions

1. EN 14752 certification path — UK / EU notified-body engagement vs self-certification by the project consortium.
2. PSD interfacing — does the actuator need to expose a hardline interlock to the platform PSD, or is the CAN-FD bridge through `osr-psd` sufficient?
3. Cold-climate variant — at the ≤ −25 °C end of the rolling-stock envelope (RFC 0008 §3.5), actuator grease and light-curtain fogging are open issues.
4. Recovery path on actuator failure — whether the door reverts to manual-only or whether a redundant secondary actuator is mandated.

9. Revision history

Date	Version	Change
2026-04-26	v0	Stub. Architecture sketch, BOM model, certification frame. Detailed elaboration pending.
2026-05-20	v1	Added interface contract and release checklist.
2026-08-14	v2	Aligned the RFC with the promoted 16.5 m car and twelve-cassette trainset layout.

RFC 0024 — LFP Thermal Design for High Ambient Temperature

Repository source: `docs/rfcs/0024-battery-thermal-high-ambient.md`

Status: Current Depends on: RFC 0021

1. Envelope

The reference train and station equipment must operate in 50 °C ambient and survive the deployment's declared parked/solar-soak condition. Supplier cell, module, controller, converter, HVAC, coolant, and enclosure limits replace generic curves before release.

2. Thermal architecture

- Every 225 kWh car pack uses monitored liquid cold plates.
- Each car can isolate its pack and cooling branch.
- Stationary 500 kWh modules use their qualified enclosure cooling system.
- Pumps, fans, valves, pressure, flow, coolant temperature, and leak detection are diagnosable.
- Battery cooling is preserved ahead of comfort HVAC during load shedding.
- Fire-mist water and battery coolant are separate circuits.

3. Control rules

Charge and discharge current are derated from measured cell temperature, coolant state, SOC, and supplier limits. Charging is inhibited before any qualified cell-temperature ceiling is crossed. A failed cooling branch cannot be hidden by train-level averaging.

The simulator applies ambient/HVAC energy uplift and converter derating, but does not substitute for cell electrothermal modelling or a climatic-chamber test.

4. Evidence

- selected-cell electrothermal model correlated to supplier/lab data;
- repeated acceleration, regen, and 500 kW dwell-charge duty at 50 °C;
- parked solar soak and restart;
- loss of pump, fan, flow, coolant, sensor, and station cooling;
- thermal propagation and outward-vent interaction;
- passenger-surface temperature and saloon heat ingress; and
- cooling energy, mass, service access, and leak containment.

RFC 0025 — DIY Switch and Point-Machine Production

Repository source: `docs/rfcs/0025-diy-switch-and-point-machine.md`

Status: Draft — extends RFC 0012 §4 Date: 2026-04-26 Depends on: RFC 0012 Switches & Crossings, RFC 0007 Control Electronics Reference Designs

1. Summary

RFC 0012 §4 already commits a non-proprietary switch architecture — flexible machined switch blade, cast-manganese frog, electro-mechanical BLDC + planetary-gearbox + crank point machine, 2oo2 detection. What it does not commit is a production path for that architecture in target regions, nor specific commodity SKUs for the actuator. The result in practice: deployments still buy $120 k turnouts from VAE / Vossloh / Schwihag because RFC 0012 is silent on the make-vs-buy decision.

This RFC commits a regional switch-shop bootstrap as the primary production path, with vendor procurement as the alternate for deployments that don't have the volume to justify the shop.

For a 100 km deployment with ~80 switches, the procurement-vs-DIY delta is ~$8–10 M USD, the largest single affordability lever in any OSR civil build.

2. Non-goals

- Not a re-spec of RFC 0012 geometry. The three turnout tangents (1:9 / 1:14 / 1:18.5) and the switch architecture remain canonical.
- Not a re-spec of `osr-wayside-points`. The detection, observation, and consensus path stay in `osr-wayside-points`.
- Not a re-spec of the safety case. A DIY switch and a vendor switch carry the same SIL-4 safety case — the SIL-4 logic lives in software (`osr-wayside-points` + `osr-interlocking`), not in the actuator. The mechanical part is below SIL.

3. Switch-shop bootstrap (primary path)

3.1 Tooling and capex

Equipment	Spec	New	Used
CNC milling machine	4 m × 1.5 m × 1 m bed, 3-axis minimum	$200 k	$80 k
Plate-cutting (plasma or fibre laser)	4 m × 8 m bed	$80 k	$30 k
Forge press	200 t hydraulic	$80 k	$35 k
Heat-treatment furnace	2 m × 1 m, 1 200 °C	$60 k	$25 k
Welding bay	MAG/MIG to EN 15085 CP-C2	$30 k	$15 k
Inspection table + CMM	4 m × 2 m granite + portable CMM	$50 k	$25 k
Misc tooling, jigs, fixtures	—	$80 k	$30 k
Total switch-shop CAPEX		~$580 k	~$240 k

3.2 Output and unit economics

- Production rate: ~2 switches/week steady-state, ~100/year.
- Per-switch BOM (1:9 turnout):
  - Switch blades (machined from rail blank): $1 800
  - Stock rails: $1 200
  - Stretcher bars (forged): $400
  - Frog (cast manganese, sub-contracted to regional foundry): $1 500
  - Check rails: $300
  - Switch sleepers (per RFC 0009): $2 000
  - Point machine assembly (per §4): $2 200
  - Heating + cabinet + harness: $800
  - Total BOM: ~$10 200
- Vendor 1:9 reference: $120 k (matches `lib/templates/switches.toml` `kits.no-9-mainline` `cost_usd`).

- DIY savings per 1:9: ~$110 k.
- 80 switches: ~$8.8 M saved, vs ~$580 k shop CAPEX ~15 × payback.

3.3 Reference designs

Switch-blade and frog geometry are public at AREMA standard or UIC 60E1 equivalent. Drawings circulate in the public domain and cover every tangent in RFC 0012.

4. Point machine — commodity actuator

The RFC 0012 §4.2 architecture (BLDC + planetary + crank, 6 kN nominal / 12 kN peak, ≤ 3 s throw) is satisfied by commodity industrial linear actuators with no rail-specific silicon:

SKU class	Vendor	Stroke	Force	List	Notes
PC-Series	Thomson Linear	200 mm	5–10 kN	$1 800	IP67, BLDC+planetary, regional distributor everywhere
EPCO	Festo	150–300 mm	5–8 kN	$2 000	Same
CASM-100	SKF	200 mm	8 kN	$1 600	Same
LinMot P10	LinMot AG	240 mm	5 kN	$2 200	Direct-drive linear (no gearbox) — alternate
Regional copies	Various Chinese / Indian	200 mm	5–10 kN	$700	Need qualification testing — see §6

Vendor-branded rail point machines (Siemens M3, Alstom M58 EBI Switch, Schwihag): $25 k–$45 k each.

Per OSR switch: ~$1 800 vs $25 k $23 k saved per switch on the actuator alone.

5. Detection and integration

Unchanged from RFC 0012 §5 and `osr-wayside-points`. The DIY switch reports position over the same 2oo2 channel as a vendor switch; the W-SBC (RFC 0007 §6) sees no difference.

6. Qualification and safety

The DIY-switch SIL-4 case is the same as the vendor-switch case because:

- Position sense, locking confirmation, and FPS observation are all in `osr-wayside-points` (already SIL-4 qualified).
- The mechanical interlock is via the front stretcher bar, identical to the vendor architecture.
- The actuator below the lock has no SIL claim — it is "best effort" with the SIL-4 verdict coming from sensed position, not commanded position.

What does need a per-design qualification:

1. Throw force margin — proof test to 12 kN peak with a 1.5 × safety factor (18 kN ultimate).
2. Throw time — measured throw time ≤ 3 s from rest to locked per RFC 0012 §4.2.
3. Endurance — 10^6 cycles at full force without lock failure (industry standard for points: ~3-month bench test).
4. Environmental — IP67, −25 °C to +55 °C operation.

These are mechanical qualifications, not safety-case qualifications. A regional university mechanical engineering department can run them in 6 months.

7. Procurement alternate

For deployments with < 20 switches/year volume, vendor procurement remains the right answer. Alternate suppliers in priority order:

Supplier	Region	Notes
Voestalpine VAE	Austria / Brazil / India	reference, premium pricing, MENA distribution
Vossloh Cogifer	France / Germany	reference
AnaSteel / Texmaco Rail	India	mid-tier, good for South Asia + East Africa
CRRC Yangtze	China	volume, ~50 % of European pricing
Voestalpine VAE Brasil	Brazil	regional Latin America

8. Open questions

1. Manganese-steel frog foundry — does the project source a regional foundry per deployment, or maintain a small list of qualified foundries?
2. Switch-blade-stock supply — head-hardened 60E1 rail blanks need to come from one of the rail mills in RFC 0009.
3. Whether the actuator qualification (per §6) is a per-deployment activity or a one-time project-level activity.

9. Revision history

Date	Version	Change
2026-04-26	v0	Stub. Switch-shop CAPEX model, commodity actuator SKUs, DIY-vs-vendor unit economics, qualification frame.

RFC 0026 — Automated Conductive Charging Interface

Repository source: `docs/rfcs/0026-charging-connector-reconciliation.md`

Status: Current Depends on: RFC 0010, RFC 0021

1. Decision

The standard passenger interface is an automated side conductive contact fed by the 500 kW station DC/DC cabinet. Up to two platform contacts share one cabinet budget. Terminals use identical hardware with longer dwell.

Depot maintenance stalls use a separate low-C plug-in service connector for balancing, diagnostics, and long holds. It is not a passenger-platform power tier and does not define the train DC architecture.

2. Electrical interface

Parameter	Reference
Train voltage	650–700 V nominal
Normal cabinet power	500 kW total
Cabinet current ceiling	825 A
Contacts per cabinet	2, interlocked
Planning efficiency	98%
Normal dwell transfer	approximately 8 kWh in 60 seconds

Four-car high-throughput deployments use three complete storage/cabinet/contact modules and six-car deployments use four. Aggregate power is never represented as a different converter product.

3. Sequence

1. Train proves stopped, correctly aligned, braked, and movement-inhibited.
2. Wayside and train exchange identity, voltage, SOC, temperature, and charge limits over the safety-related control link.
3. Protective earth/bond, insulation, HVIL, and contact-clear checks pass.
4. The mechanism closes, weld detection and precharge complete, then the BMS permits current.
5. Either side can abort; current reaches zero before mechanical separation.
6. The mechanism proves stowed before movement authority can be released.

4. Mechanical and safety requirements

The interface is touch-safe when open, tolerant of the qualified alignment envelope, protected against water/dust/debris, guarded from passengers, and replaceable without civil reconstruction. Breakaway, welded-contact, unexpected-voltage, comms-loss, train-movement, emergency-stop, and two-train contention cases fail safe.

5. Acceptance

FAT and SAT cover voltage/current, insulation, earthing, temperature rise, contact resistance, cycle endurance, alignment, contamination, ingress, emergency isolation, aborted sequences, vehicle movement interlock, simultaneous arrivals, cabinet outage, and rescue/recovery access.

RFC 0027 — Brownfield Pilot: Existing-Asset Recovery & Workshop Integration

Repository source: `docs/rfcs/0027-brownfield-pilot-asset-recovery.md`

Status: Draft — proposed Date: 2026-04-26 Depends on: RFC 0003 Samawah Context And Pilot Rationale, RFC 0008 Rolling-Stock Reference Design, RFC 0014 Depot Design Standard, RFC 0022 Bogie + Traction Drive, RFC 0025 DIY Switch and Point-Machine

1. Summary

Most OSR target countries — Iraq, Sudan, Pakistan, Mozambique, Angola, Venezuela, several CIS states — already own large quantities of dormant Soviet-era / British-era / American-aid-era rolling stock and rail workshops, mostly stored or partially functional. RFC 0003's Samawah context is the worked example: satellite imagery identifies 300–800 wagons stored in the rail yards adjacent to Samawah Train Station, plus a rolling-stock workshop building (historically the target of a 2011 Iranian Waxon Park rehabilitation deal whose outcome is unclear).

Stored fleet + dormant workshop + active mainline through the city could make Samawah a brownfield pilot rather than a greenfield one. This RFC commits a systematic doctrine for converting brownfield assets into OSR-compatible production capacity, and is intended to apply to every deployment that begins with an existing asset stockpile.

The doctrine has three phases:

Phase	Scope	Duration
P1 — Asset assessment	Site visit, fleet inventory, workshop tooling audit, bogie/wheelset NDT sample, IRR/national-rail-authority disposition negotiation	3–6 months
P2 — Component recovery	Strip 50–100 wagons for wheelsets, axleboxes, selected brake hardware, fasteners, and couplers; sell scrap steel; train workforce on the strip operation	6–12 months
P3 — First-article OSR trainset	Build fresh OSR powered/trailer bogie frames (RFC 0022 §2.2) + accepted wheelset components recovered in P2, OSR car body (RFC 0008 Track A or B), commission at the rehabilitated workshop. Workshop becomes the regional rolling-stock factory.	12–24 months

This RFC does NOT replace any greenfield-spec RFC. It runs in parallel to RFC 0008 / RFC 0014 / RFC 0022. Where a deployment has no brownfield assets, the greenfield path remains the default. Where it has them, this RFC is the contract for converting them.

2. Non-goals

- Not a wagon-acquisition policy. The legal framework for buying, leasing, or jointly operating recovered wagons is per-deployment and lives in RFC 0029 (operator-financial model, separate).
- Not a replacement for RFC 0008. OSR rolling stock is still the light-metro / metro families per RFC 0008 §1. Recovered wagons are donor stock — the OSR trainset is built fresh, using recovered components.
- Not a heritage-preservation effort. Wagons that are historically significant should be retained for museum/preservation use; this RFC's scope is the operationally serviceable bulk remainder.

3. Why brownfield pilots beat greenfield where assets exist

Asset class	Greenfield CAPEX	Brownfield CAPEX	Saved
Wheelsets (760 mm forged, refurbished)	$2 800–4 200 / pair (RFC 0022 §10 tier-2 sourcing)	$300–800 / pair (NDT + reprofile)	~$2 500 / pair × 4 pairs / consist = ~$10 k / consist
Bogie machine tools (CNC mill 4 m, jig table, MAG welder bay, plasma cutter)	$400 k–700 k for greenfield switch shop (RFC 0025 §3.1)	$50–200 k for tool refurbishment + commissioning at existing workshop	~$250–500 k per shop
Workshop building (10 000 m² portal-frame industrial shed, pit tracks, traverser foundations, crane gantries)	$5–8 M for greenfield depot (RFC 0014)	$0.5–2 M for rehabilitation of dormant workshop	~$3–6 M per workshop
Rail-grade structural steel (recoverable from non-restorable wagon bodies)	$1 100–1 400 / t (RFC 0009 sourcing)	$150–250 / t scrap-recovery cost	~$900 / t × ~5 t / wagon × 200 wagons = ~$900 k
Coupler / brake hardware (SA-3, vacuum / air-brake fittings)	$5 k–12 k / unit	$200–500 / unit (refurbish or scrap)	Highly site-specific
Trained workforce (rail mechanical fitters, welders, machine operators)	18–24 months training pipeline (RFC 0027 — workforce plan, planned)	Existing workforce continues + cross-trains on OSR-specific kit	Salary-savings + early-deployment time-savings of 1–2 years

For a brownfield starter pilot (for example 16 trainsets, a single light-metro line, and one main depot), asset recovery saves on the order of $8–15 M in mechanical/civil capital plus 18–24 months of schedule — a meaningful fraction of the RFC 0021 §6

€435 M total CAPEX, and an even larger fraction of the time budget which is usually the binding constraint.

4. Phase 1 — Asset assessment (3–6 months)

4.1 Required deliverables

Artefact	Owner	Scope
Fleet inventory	OSR design team + national rail authority	Wagon-by-wagon list with VIN, year, type (covered / hopper / flatbed / passenger coach), recorded last service date, structural condition (1–5 scale), owning entity, legal disposition status
Workshop tooling audit	Mechanical engineering team (local + visiting)	Per-tool register: machine type, capacity envelope, last calibration / operation date, replacement/repair cost, missing components
Bogie / wheelset NDT campaign	Independent NDT contractor	Magnetic-particle inspection on a representative ~5 % sample of stored wheelsets; ultrasonic testing on a ~2 % sample of axles. Report flaw severity per EN 13262 class.
Civil survey of workshop site	Local civil engineering firm	Foundation condition, water/sewer connections, electrical-supply capacity, road/rail access, contamination assessment
Disposition framework	Government-relations lead	Memorandum-of-understanding draft with national rail authority covering ownership, lease terms, joint-venture options, indemnity for legacy environmental liabilities

4.2 Sample budget

Phase 1 typically runs $200 k–500 k depending on stockpile size. Major line items:

Line item	Sample cost
Fleet-inventory team (4 engineers × 4 months on-site)	$80–120 k
NDT contractor (sample wheelsets / axles)	$40–80 k
Workshop tooling audit + civil survey	$30–60 k
MoU legal / negotiation support	$20–60 k
Per-diem, travel, equipment hire	$30–80 k
Contingency 20 %	$40–80 k

This is a small fraction of the total project CAPEX and the findings feed directly into Phase 2 / Phase 3 sizing.

5. Phase 2 — Component recovery (6–12 months)

5.1 Recovery operation

The recovery operation runs at the existing workshop, using the existing workforce. Wagons are pulled into the workshop one at a time, stripped to component level, and the components are sorted into:

- OSR-reusable components (wheelsets, axleboxes, bearings, brake cylinders, coupler bodies, structural-steel sections, hand brakes, electrical contactors, lighting fixtures).
- Resale components (couplers + drawgear in working order go to freight operators in the region; HVAC and water systems if any).
- Scrap stream (steel body sheets, broken castings, weather- damaged wood).

5.2 Wheelset recovery — the highest-value stream

Per RFC 0022 §3, OSR uses 760 mm new / 680 mm worn wheelsets. Iraqi Republic Railways (IRR) and most former-Soviet-aligned networks use 920 mm / 1050 mm wheelsets — these are NOT directly OSR-compatible. However, the axleboxes, bearings, springs, and brake gear are reusable, and the axles themselves can be re-machined to OSR diameter if their rough-bar diameter is above the 130 mm minimum.

So the recovery yield is:

- Wheelsets directly usable: ~5–10 % (where rolling stock was already 760 mm, e.g. some passenger-coach trucks)
- Axles re-machinable: ~50–60 % of stored stock
- Axleboxes / bearings / spring sets: ~70–80 % (subject to NDT)
- Brake hardware (cylinders, slack adjusters): ~60–70 %
- Couplers (SA-3 + drawgear): ~70 % directly resaleable to legacy freight operators in the region

Strip rate at a properly tooled workshop is ~1 wagon per crew per shift. With two crews and a single shift, ~10 wagons / week 250 / year. This is sufficient to support a 16-trainset OSR fleet at 4 wheelsets / car × 3 cars / consist × 16 = 192 wheelsets, with the wheelset balance going to the spares pipeline and to subsequent deployments.

5.3 Coupler retrofit — the catch

OSR commits to Scharfenberg Type 10 automatic couplers per RFC 0008 §3.1. Recovered wagons use SA-3 (former-Soviet automatic) plus screw/buffer. These are incompatible, so the brownfield path does NOT involve re-coupling recovered wagons into the OSR consist directly. Every OSR car ships with new Scharfenberg couplers; the recovered SA-3 couplers go to the resale stream.

This is correct: a brownfield recovery should NOT compromise the single-coupler-standard discipline that lets any OSR consist rescue any other OSR consist on the network.

6. Phase 3 — First-article OSR trainset (12–24 months)

The recovered components feed forward into the standard OSR build process:

OSR component	Source from recovery
Bogie frame (RFC 0022 §2.2)	New, fabricated in the rehabilitated workshop using its CNC + welding capability
Wheelset (RFC 0022 §3)	Recovered axle + new (or refurbished 760 mm) wheel forging press-fit at the workshop
Axlebox + bearing	Recovered housing only if NDT/metrology pass; bearing replaced or re-certified by supplier, with new gaskets / seals
Primary suspension (chevron rubber)	New (RFC 0022 §5.1) — recovered rubber elements are usually weathered
Secondary suspension (air-spring)	New (RFC 0022 §5.2) — recovered air-spring assemblies are usually leak-prone after >5 y storage
Traction motor (PMSM)	New (RFC 0022 §4.1, sourced from RFC 0022 §10 tier-2 supplier)
Inverter (SiC)	New (RFC 0008 §3.2 DIY path or RFC 0022 §4 sourced)
Brake disc + caliper	New (per RFC 0022 §6) — recovered cast-iron discs are usually unsuitable for the regenerative-friction blend RFC 0021 expects
Body shell (alu-extrusion or steel space-frame)	New (RFC 0008 §3.1 Track A or RFC 0023 §1.1 Track B) — built in the workshop
LFP battery and per-car fire-control package	New per RFC 0021
Electronics (DIY safety stack)	New per RFC 0019
Coupler	New Scharfenberg (RFC 0008 §3.1)
Lighting / interior	Mix — recovered wiring harness + COTS interior per RFC 0008 §3.3

In other words: the brownfield component recovery yields ~30–40 % of the per-consist mechanical mass and ~$30–60 k per consist in saved component cost, but the OSR architecture is preserved verbatim. No SA-3 couplers, no 920 mm wheelsets, no Soviet-era electrical systems on the OSR consist itself.

7. Workshop conversion — what stays, what goes

The Samawah workshop is presumed to have:

Existing capability	OSR fit	Action
Wheelset turning / re-profiling	Direct fit	Keep + commission to RFC 0022 NDT spec
Bogie pit tracks	Direct fit	Keep + extend to OSR bogie wheelbase (2 100 mm — RFC 0022 §3)
Overhead bridge crane	Useful but RFC 0014 prefers stinger track	Keep where existing; RFC 0014 stinger-track is a greenfield-only choice
Steam + diesel locomotive shed equipment	Not applicable to OSR (battery-electric, RFC 0021)	Decommission or convert to general-purpose mechanical workshop
SA-3 coupler tooling	Not applicable	Decommission or repurpose for resale-stream operations
Air-brake testing rig	Compatible if convertible to electric brake	Repurpose for OSR EP-brake commissioning (RFC 0008 §3.2)
Wagon-frame welding line	Direct fit	Keep + commission to EN 15085 CP-C2 (the lowest rail welding class — fits the existing workforce)
Painting + livery shop	Direct fit	Keep + retrain on RFC 0008 §3.5 livery scheme

The conversion isn't free — typically $0.5–2 M for the rehabilitation itself plus $1–3 M for net-new tooling required for OSR-specific operations (PMSM/controller commissioning, 800 V-class DC test rigs, LFP battery pack assembly). But that's an order of magnitude less than greenfield `main-heavy` depot CAPEX of $12.0 M (RFC 0014).

8. Risk register

Risk	Likelihood	Mitigation
Phase 1 inventory reveals stored fleet is too degraded to recover	Medium	Phase 1 cost is bounded ($200–500 k); decision-gate at end of Phase 1 — abandon brownfield if NDT pass-rate < 30 %
National rail authority refuses to negotiate disposition	Low–medium	Pre-screen via diplomatic + technical-community channels before committing to Phase 1
Workshop legacy environmental liability (asbestos, lead paint, fuel-tank contamination)	Medium	Civil survey in Phase 1 explicitly includes environmental scoping; remediation cost goes into the disposition negotiation, not the project budget
Workforce knowledge gap on OSR-specific equipment (SiC inverters, PMSM motors, DIY safety electronics)	High	Cross-training in Phase 2, with project-funded secondment of OSR-experienced engineers from earlier deployments. RFC 0027 (workforce plan, planned) will own this.
SA-3 coupler resale market dries up before recovery completes	Low	Resale stream is a bonus, not a critical path. Scrap value remains.
Recovered axles fail post-machining NDT	Medium	Build Phase 3 demand schedule against the post-NDT survival rate, not raw recovery numbers. Carry a 30 % buffer.
Mid-project change of national-rail-authority leadership delays disposition	Medium–high	MoU should commit ownership at Phase 1 close, before Phase 2 capital is exposed

9. Applicability beyond Samawah

This RFC's Samawah example is illustrative. The same doctrine applies wherever a country has:

- An active mainline running through (or near) a candidate OSR deployment city.
- A dormant rolling-stock stockpile of standard-gauge (1 435 mm) freight / passenger stock — even if the stock is otherwise incompatible with OSR.
- A rail workshop / depot building, even partially functional.
- A national rail authority capable of negotiating disposition.

Cities with documented brownfield potential (preliminary, not verified):

City	Country	Stockpile rumour / status
Samawah	Iraq	Confirmed by 2026-04-26 satellite review — 300–800 wagons, workshop building
Khartoum	Sudan	Sudan Railways stored fleet at Atbara workshop (Sudan Railways / Sudan Railway Workshops)
Karachi	Pakistan	Pakistan Railways Karachi Cantonment yard — large freight stockpile
Maputo	Mozambique	CFM stored fleet at Machava
Luanda	Angola	CFL stockpile (Caminho de Ferro de Luanda)
Tashkent	Uzbekistan	UTY workshops at Tashkent + significant Soviet-era stockpile

Each of these warrants a city-specific Phase 1 assessment before greenfield-vs-brownfield decision. The decision should default to brownfield where assets are real and recoverable.

10. Open questions

1. IRR (Iraqi Republic Railways) disposition framework — ownership of stored stock, lease terms, joint-venture options for the Samawah workshop. Requires direct contact with the Iraqi Ministry of Transport ( ) and IRR / .
2. Waxon Park 2011 deal outcome — did the Iranian rehabilitation agreement complete? What was delivered? Is the equipment intact?
3. Wagon-by-wagon census — currently we have a satellite-image estimate of 300–800 wagons. A real census on the ground is the single highest-value Phase 1 deliverable.
4. Environmental liability assessment at the workshop — historical diesel + steam-loco operations may have left contamination that needs remediation before OSR commissioning.
5. Legal pathway for Iraqi (or other national) rolling-stock transfer to a project-controlled vehicle — sovereign-immunity, competition law, public-procurement compliance.

11. Revision history

Date	Version	Change
2026-04-26	v0	Stub. Three-phase doctrine (assess / recover / build), Samawah worked example, applicability matrix. Triggered by 2026-04-26 operator review identifying the Samawah rail-yard stockpile + workshop from satellite imagery.

RFC 0028 — Construction Quality Assurance System

Repository source: `docs/rfcs/0028-construction-quality-assurance.md`

Status: Draft — proposed Date: 2026-06-12 Depends on: RFC 0008 Rolling-Stock Reference Design, RFC 0009 Track Design Standard, RFC 0010 Station Design Standard, RFC 0011 Civil Infrastructure Design Standard, RFC 0013 Operations Rulebook, RFC 0014 Depot Design Standard, RFC 0015 Driverless Operation, RFC 0021 Battery Traction, RFC 0022 Bogie + Traction Drive, RFC 0023 Door System Reference Design, RFC 0029 Maintenance Schedule System

1. Summary

OpenSourceRail commits to an owner-controlled construction QA system for both locally built rolling stock and fixed infrastructure. The system is gate-based: a package cannot move to the next stage until the relevant hold point is released by the owner, owner engineer, or named acceptance board.

The machine-readable gate register is `lib/templates/construction-qa.toml`. Generated city READMEs render the same register so every deployment shows the required evidence before passenger opening.

This is deliberately not a vendor warranty model. OSR assumes local production, local ownership, and local maintenance. The quality system therefore makes the owner the evidence holder from day one.

2. Non-goals

- Not a replacement for local law or regulator approval. Local building, railway, labour, fire, and electrical approvals still apply.
- Not a standards rewrite. Welding, civil, electrical, fire, accessibility, and software standards are referenced by the component RFCs and local authority requirements.
- Not a paperwork-only system. A hold point is only released when physical inspection, test evidence, and open-defect disposition agree.
- Not a reason to inflate train CAPEX. Per-train QA/acceptance is a small nominal production cost; fixtures, tooling, calibration systems, and commissioning bays sit in the railway production plant or OPEX.

3. Quality Record Structure

Every QA record uses the same minimum fields:

Field	Requirement
Asset id	Trainset, car, bogie, station, span, switch, charger, W-SBC, or package id
Drawing/config version	Released drawing, firmware hash, test procedure, template id
Inspector	Named person and organisation
Result	Pass, pass with minor defects, hold, reject
Evidence link	Test sheet, photo, measurement file, certificate, log export
Defect id	Required when result is not clean pass
Release authority	Named role that accepted the hold point
Handover target	Operations, maintenance, safety case, or asset register

The evidence register in `docs/certification/evidence-register.md` is the safety-case view. The construction QA register is the production view. The two must cross-reference each other but do not duplicate every line item.

4. Gate Register

The mandatory gate ids are:

Gate	Scope	Release condition
`qa-00-design-freeze`	Whole railway	Drawings, interfaces, hazards, inspection/test plans, and supplier/material registers baselined
`qa-10-carbody-structure`	Carbody, underframe, crash structure, coupler pockets	Weld/material traceability, dimensional survey, NDT, coating, and load evidence accepted
`qa-11-bogie-wheelset`	Bogies, wheelsets, suspension, brake rigging	Bogie NDT, wheel/axle inspection, bearing certificate, brake static test, and torque logs accepted
`qa-12-traction-brake-battery`	Motors, inverters, brake blending, onboard battery, thermal system	Bench and trainset functional tests accepted before dynamic movement
`qa-13-passenger-systems`	Doors, HVAC, lighting, interiors, accessibility, passenger information	Door cycles, HVAC performance, PRM check, emergency equipment, and saloon inspection accepted
`qa-14-onboard-control`	TCN-E, ATP/ATO, odometry, radios, sensors	Hardware identity, firmware hashes, authentication, simulator replay, and trainline tests accepted

Gate	Scope	Release condition
`qa-15-first-article-trainset`	Complete trainset	Static, dynamic, endurance, braking, charging, evacuation, rescue, and maintainability acceptance complete
`qa-20-survey-geotech`	Alignment, ROW, utilities, geotechnical	Survey, utility, borehole/test-pit, drainage/flood, and ROW constraints accepted
`qa-21-earthworks-drainage`	Earthworks, subgrade, drainage, fencing	Compaction, material, drainage, culvert, and fence/gate evidence accepted
`qa-22-trackform-rail`	Slab track, rail, welds, fasteners, turnouts, crossings	Geometry, weld NDT, fastener torque, turnout detection, and crossing tests accepted
`qa-23-structures`	Viaducts, bridges, bearings, expansion joints, parapets, walkways	Concrete/rebar, bearings, joints, drainage, access, and structural inspection accepted
`qa-24-stations-depots-plant`	Stations, depots, production plant, public realm	Platform gauge, canopy, fire/life safety, accessibility, depot bay, and tool calibration accepted
`qa-25-power-energy`	PV, stationary storage, chargers, grid/PPA tie, earthing	PV string, BESS, charger, relay, earthing, isolation, and islanding tests accepted
`qa-26-wayside-comms-safety`	W-SBCs, switches, intrusion sensors, comms, fare/PIS/CCTV	Identity register, coverage, switch proof, sensor calibration, passenger-system, and cyber tests accepted
`qa-30-integrated-trial-running`	Whole railway	Trial running, degraded modes, emergency exercise, maintenance handback, and safety release accepted

5. Rolling-Stock QA

Rolling-stock QA is organised by trainset, car, and replaceable module. The owner must be able to trace a defect from service all the way back to the production record that released the affected component.

5.1 Carbody and Crash Structure

Required records:

- Material certificates for underframe, side frame, roof, end cowl support, crash-energy components, coupler pocket, and door posts.
- Weld map and welder qualification record.
- Dimensional survey after welding and after bogie marriage.
- NDT sampling plan and results.
- Corrosion protection, paint, and sealant record.
- Static proof or calculation pack for AW3 loading, jacking, lifting, and recovery tow cases.

5.2 Bogie, Wheelset, Suspension, and Brake

Required records:

- Bogie frame weld and NDT pack.
- Wheel, axle, bearing, and axlebox certificates.
- Wheel profile and back-to-back measurement.
- Torque records for suspension, brake, axlebox, and motor mounts.
- Brake static and dynamic test records.
- Post-run inspection after the first dynamic movement.

5.3 Traction, Battery, and Thermal System

Required records:

- Motor and inverter factory acceptance or bench-test result.
- Insulation resistance and protective bonding result.
- BMS cell map, serial numbers, initial capacity sample, and contactor test.
- Thermal soak result for high-ambient operation.
- Charging interface test with station/depot chargers.
- Brake blending and regenerative-friction transition curves.

5.4 Passenger and Accessibility Systems

Required records:

- Door-cycle counter test and obstacle detection.
- HVAC cooling/heating result for the deployment climate preset.
- Emergency lighting, PA, passenger information, CCTV, and call-point test.
- PRM bay, tactile, grab-pole, wheelchair, and boarding-gap checklist.
- Fire-load and interior-material compliance evidence.

5.5 Onboard Control and Software

Required records:

- Hardware serial register.
- Firmware hash and signing record.
- Message-authentication configuration.
- TCN-E continuity and latency test.
- Odometry, sensor, and degraded-mode simulation replay.
- Cybersecurity checklist before line testing.

5.6 First Article and Batch Release

The first trainset of each local production run must complete:

- Full static acceptance.
- Full dynamic acceptance.
- Braking curves in dry and reduced-adhesion proxy conditions.
- Charging at representative station and depot chargers.
- Rescue/coupling drill.
- Passenger evacuation drill.
- Maintainability demonstration for battery module, door actuator, brake pad, HVAC filter, and W-SBC/onboard computer replacement.
- At least 1,000 km fault-free trial running before series release.

Series trainsets may use a reduced evidence pack only after the first article is accepted and the production process is unchanged.

6. Infrastructure QA

Infrastructure QA is organised by physical work package and asset id. Every station, line section, switch, structure, energy site, and systems cabinet must enter the asset register before handover.

6.1 Survey, Geotechnical, and ROW

Required records:

- Survey control and alignment setout.
- Utility scan and clash register.
- Borehole or test-pit log by geotechnical zone.
- Flood path, drainage, and culvert basis.
- Cadastral and ROW constraint register.

6.2 Earthworks, Drainage, and Fencing

Required records:

- Subgrade and capping-layer compaction tests.
- Material gradation and source record.
- Drainage as-built and flow direction.
- Culvert inspection and cover level.
- Fence, gate, and intrusion-risk punch list.

6.3 Trackform, Rail, Turnouts, and Crossings

Required records:

- Slab/plinth pour records.
- Fastener torque logs.
- Weld NDT and rail-stress record.
- Track geometry run before trial service.
- Turnout detection and point-machine proof.
- Level-crossing barrier, warning, and obstacle-detection test where fitted.

6.4 Structures

Required records:

- Concrete, reinforcement, precast, and post-tensioning records.
- Bearing installation and survey.
- Expansion-joint record.
- Drainage, parapet, walkway, and access inspection.
- Bridge scour check for water crossings.

6.5 Stations, Depots, and Production Plant

Required records:

- Platform gauge and boarding-gap survey.
- Canopy, roof PV, drainage, and fixings inspection.
- Fire/life-safety signoff.
- Accessibility audit.
- Depot pit, stinger, wash, lathe, lifting, and isolation checks.
- Production plant fixture and tool calibration.

6.6 Energy, Signalling, Comms, and Passenger Systems

Required records:

- PV string test, BESS commissioning, charger load test, relay settings, earthing, and emergency isolation drill.
- W-SBC identity register and switch proof test.
- Radio coverage and backup-link test.
- Intrusion sensor calibration.
- OCC, passenger information, fare, CCTV, and public-address tests.
- Cybersecurity closeout and firmware baseline.

7. Nonconformance Control

Defects are classified:

Class	Meaning	Rule
NCR-A	Minor quality note	May be accepted with owner engineer approval and logged for future maintenance
NCR-B	Correct-before-next-stage	The package cannot pass its current gate until reworked and re-inspected
NCR-C	Safety or configuration blocker	The asset is rejected or isolated until root cause, corrective action, and re-test are complete

Repeated NCR-B or any NCR-C triggers a production pause for the affected process. The pause is local to the process, not necessarily the whole programme, unless the defect indicates a shared design or supplier failure.

8. Trial Running and Opening

Passenger service is not allowed until `qa-30-integrated-trial-running` is released. Minimum evidence:

- Timetable trial running over representative service hours.
- Emergency exercise with local responders.
- Degraded-mode drill for OCC, station staff, and maintenance.
- Possession and handback demonstration.
- Maintenance defect response demonstration.
- Open-defect register showing no safety blockers.
- Safety-case release note referencing the accepted evidence.

9. Samawah Application

For Samawah, the QA system applies to:

- 96 locally built 3-car trainsets.
- 58.4 km of generated route.
- 31 unique stations.
- Station/depot charging microgrids and the dedicated solar plant/PPA interface.
- The brownfield yard/workshop conversion if RFC 0027 Phase 1 confirms usable assets.

The first-article trainset and first completed station should be treated as programme learning gates: they set the acceptance standard for the remaining batch.

RFC 0029 — Maintenance Schedule System

Repository source: `docs/rfcs/0029-maintenance-schedule-system.md`

Status: Draft — proposed Date: 2026-06-12 Depends on: RFC 0008 Rolling-Stock Reference Design, RFC 0009 Track Design Standard, RFC 0010 Station Design Standard, RFC 0011 Civil Infrastructure Design Standard, RFC 0013 Operations Rulebook, RFC 0014 Depot Design Standard, RFC 0016 Wayside Track Intrusion, RFC 0021 Battery Traction, RFC 0028 Construction Quality Assurance System

1. Summary

OpenSourceRail commits to a single maintenance schedule system for rolling stock, stations, track/civil works, structures, energy assets, signalling/comms, depots, and the railway production plant.

The system has three layers:

1. Baseline intervals by asset class, stored in `lib/templates/maintenance-schedule.toml`.
2. Condition-based escalation from telemetry, inspections, and defect reports.
3. Work-order closeout that proves the asset is safe to return to service.

Generated city READMEs render the baseline schedule with that city's fleet, station, route-km, and train-km context.

2. Non-goals

- Not a CMMS product choice. A deployment may use open-source CMMS, a national operator system, or a simple database during pilot service.
- Not a parts catalogue. Spare parts lists live with the asset and BOM documents.
- Not a replacement for inspections mandated by local law. Local statutory inspections are additive.
- Not a vendor maintenance contract. OSR assumes the owner maintains the railway; outside specialists can support but do not own the maintenance evidence.

3. Asset Register

Every maintainable asset must have an id before revenue service:

Asset class	Minimum id structure
Trainset	trainset id, car ids, bogie ids, battery pack ids, major module serials
Station	station id, platform ids, canopy/PV id, public systems, emergency equipment
Track section	line id, chainage range, civil class, drainage and fence assets
Switch/crossing	switch id, point machine id, detection channel ids
Structure	span id, pier id, bearing id, expansion joint id
Energy site	PV strings, BESS rack ids, charger ids, protection devices
Signalling/comms	W-SBC id, radio/cabinet ids, sensor ids, firmware baseline
Depot/plant	bay id, pit id, tool/fixture id, lathe/lift/calibration device id

The construction QA system in RFC 0028 creates the starting asset register. This RFC keeps it alive through operations.

4. Work-Order Rule

Every maintenance action produces a work order with:

- Asset id.
- Trigger: calendar, km, cycle count, telemetry, defect, event, or regulatory inspection.
- Task procedure/version.
- Findings and severity.
- Parts used and removed.
- Test/inspection evidence.
- Return-to-service authority.
- Next due date or next due km/cycle.

No asset returns to service after a red defect without a signed return-to-service record.

5. Baseline Schedule

The baseline schedule is:

Group	Cadence	Scope
Rolling stock	Daily / each depot return	Walk-around, wheel/tread visual, doors, HVAC, coupler face, saloon damage, emergency kit, fault-log download
Rolling stock	7 days	Wheel wear, brake pads, BMS cell spread, harness chafe, roof/PV cleaner, door obstruction sensors
Rolling stock	30 days	BMS deep scan, motor-bearing vibration, inverter thermal log, HVAC filters, PIS/CCTV, firmware inventory
Rolling stock	150,000 km or wear limit	Wheel reprofiling
Rolling stock	600,000 km	Bogie overhaul
Rolling stock	10 years	Body/interior/HVAC/cable overhaul
Stations	Daily	Cleaning, lighting, platform edge, PA/PIS/fare equipment, CCTV, emergency equipment
Stations	7 days	Canopy/fixings, drainage, accessibility path, charger cabinet visual, extinguishers, signage
Stations	30 days	Emergency lighting, fire alarm, UPS, lift/escalator where fitted, access control, comms cabinets
Stations	12 months	Structural survey, canopy PV fixings, drainage capacity, accessibility audit, passenger-flow review
Track/civil	7 days	Track walk, fasteners, slab cracking, drainage, fencing, vegetation
Track/civil	60-90 days	Geometry run and trend comparison
Switches/crossings	30 days	Point closure, detection, lubrication, bolts, weld cracks, drive inspection
Structures	12 months	Viaduct/bridge visual inspection, bearings, joints, parapets, walkways, drainage, scour
Energy	Daily remote	SCADA alarms, charger availability, SOC/temperature, PV yield anomaly, grid import/export
Energy	30 days	Charger sample load test, BESS thermal inspection, PV soiling, cabinet seals, protection status
Energy	12 months	Relay test, earthing resistance, emergency isolation drill, BESS capacity sample, PV insulation
Signalling/comms	Daily remote	OCC/W-SBC health, radio alarms, time sync, authentication failures, CCTV/fare/PIS fault queue
Signalling/comms	30 days	Switch proof test, sensor calibration sample, comms cabinet inspection, backup link, UPS
Signalling/comms	90 days	Firmware inventory, vulnerability review, backup restore, degraded-mode drill, simulator replay
Depot/production plant	30-180 days	Tool calibration, lifting equipment, pits/stingers, wheel lathe, welding fixture survey

The calendar intervals are the floor. Hot desert, monsoon, flood, coastal corrosion, dust, high ridership, or poor grid quality can only shorten intervals, never lengthen them without owner-engineer approval.

6. Condition-Based Escalation

Telemetry and inspection findings override the calendar:

Condition	Response
Green	Continue normal service; next baseline interval applies
Amber	Schedule work within 7 days; asset may stay in service if risk assessment permits
Red	Hold asset out of service or isolate the affected section until rectified
Unknown on safety asset	Treat as red when the asset protects movement authority, intrusion, braking, switching, charging isolation, or emergency response

Examples:

- Battery cell spread over amber limit schedules a 7-day intervention; red limit holds the trainset.
- Sustained intrusion-sensor `Unknown` holds or restricts the affected section per RFC 0016 and the operations rulebook.
- Track geometry trend that crosses amber schedules possession work; red geometry holds the track section.
- Charger insulation or earthing failure isolates that charger until re-tested.

7. Rolling-Stock Maintenance

Rolling-stock maintenance is depot-led:

- `main-heavy` performs daily, weekly, monthly, wheel reprofiling, bogie overhaul, body overhaul, and major component rebuild.
- `secondary-medium` performs daily, weekly, monthly light work and module swaps.
- `layup-minimal` performs visual checks and fault reporting only.

The maintenance reserve in `lib/templates/fleet-sizing.toml` exists so scheduled work can happen without cancelling service.

8. Stations and Passenger Assets

Station maintenance covers public safety and revenue equipment:

- Platforms, tactile paving, ramps, edge markings, lighting, signage.
- Canopy, roof PV, drainage, gutters, fixings.
- PA, passenger information, CCTV, fare validators/gates, Wi-Fi where fitted.
- Fire extinguishers, emergency lighting, call points, first-aid kits.
- Lifts/escalators only where the station archetype requires them.

Station staff perform daily observations. Maintenance staff perform the weekly/monthly technical checks. The owner engineer performs or accepts the annual structural/accessibility condition report.

9. Track, Civil, and Structures

Maintenance of way covers:

- Track walks and defect classification.
- Geometry measurement and trend comparison.
- Switches and point machines.
- Drainage, culverts, vegetation, and fencing.
- Viaducts, bridges, bearings, expansion joints, parapets, walkways, and bridge scour.

Post-weather inspections are additional to the baseline schedule. Flood, dust storm, heatwave, lightning, collision, derailment, trespass, or reported structure strike triggers event-based inspection before normal speed resumes.

10. Energy and Charging

Energy maintenance covers:

- PV strings and cleaning/soiling checks.
- Stationary storage, thermal management, racks, contactors, and BMS.
- Train chargers and charger rails/connectors.
- Protection relays, earthing, isolation, and grid/PPA interface.
- Daily SCADA review for yield, SOC, temperature, charger uptime, and grid import/export anomalies.

Energy defects can affect both service and safety. Charger faults may reduce timetable capacity; earthing, isolation, or BESS thermal faults hold the affected energy site.

11. Signalling, Comms, and Software

Systems maintenance covers:

- OCC health, dashboards, logs, backups, and time sync.
- W-SBCs, switches, sensors, cabinets, UPS, and comms links.
- Radio coverage and backup communications.
- CCTV, passenger information, fare systems, and platform systems.
- Firmware inventory, vulnerability review, restore drills, and simulator replay.

Software maintenance is not casual patching. Any firmware or safety software change requires a versioned work order, rollback plan, replay test, and updated configuration baseline.

12. Depot and Production Plant

Depot and plant maintenance covers:

- Pit tracks, stinger supply, isolation equipment, wash plant, stores, compressed air, cranes/lifts where fitted.
- Wheel lathe and measuring equipment.
- Welding fixtures, jigs, calibration tools, torque tools, test benches.
- Lifting equipment statutory inspection.
- Production plant fixtures and commissioning bay equipment.

Tool calibration defects can block new train acceptance and heavy maintenance release. The production plant is therefore part of the railway asset base, not a one-off construction convenience.

13. Samawah Application

For the current Samawah generated instance, the maintenance system starts with:

- 96 trainsets: 86 peak-revenue, 7 planned spares, and 3 cold reserves; depot turnaround service uses peak-fleet surplus in off-peak windows.
- 31 stations.
- 58.4 route-km.
- 29,515 scheduled train-km/day before depot/deadhead factor.
- 29 station/depot energy sites plus the dedicated solar plant/PPA interface.

The daily service intensity makes train-km and telemetry-based triggers as important as calendar triggers. The battery adequacy warning in the Samawah README should also be tracked as a maintenance-risk item: operating near reserve limits increases charger, battery, and timetable discipline sensitivity.

RFC 0030 - Manufacturing Schedule System

Repository source: `docs/rfcs/0030-manufacturing-schedule-system.md`

Status: v2 accepted project-twin baseline

1. Decision

OpenSourceRail uses one generated manufacturing schedule system for locally built trains and infrastructure. The same city asset register that feeds QA and maintenance also feeds production work packages for:

- trainsets
- waypoints / W-SBC nodes
- track sections
- switches and crossings
- stations
- depots and the railway production plant
- energy sites
- station systems and signalling/comms nodes
- whole-system baseline and trial-running release

The authoritative template is `lib/templates/manufacturing-schedule.toml`. The generated, spreadsheet-friendly city outputs are created by `tools/automation/generate-qa-maintenance-data.py`. They include the schedule, material/BOM control rows, stage QA verification, finite resources, CPM, procurement requirements, schedule of values, cashflow and visualization events.

2. Why

The project is an owner-builder system, so manufacturing cannot be treated as a black-box supplier milestone. A city needs a practical schedule that answers:

- what is being built
- which asset id it belongs to
- where the work happens
- which crew roles are accountable
- what staff tasks must be completed
- what materials, inputs, and predecessors are required
- what evidence is needed for QA and commissioning
- which rows should become work orders

The system is deliberately lighter than ERP/MES software. It is designed to be useful on day one in a local railway production plant, depot, station site office, or owner engineer console.

3. Data Model

Each generated manufacturing row contains:

Field	Purpose
`manufacturing_uid`	Stable city + asset + package id
`asset_id`, `asset_name`, `asset_type`, `line`	Links work to the generated asset register
`package_id`, `phase`, `sequence`	Work package and ordering
`work_center`	Plant, depot, site, bench, or crew location
`duration_days`	Calculated or fixed planning duration
`duration_model`	`fixed-days`, `route-metres`, or `single-track-panels`
`quantity_basis`	Auditable quantity/resource equation used for the duration
`resource_count`	Crews, gantries, rigs or other parallel-resource basis where rate-derived
`planned_start_day`, `planned_finish_day`	Project-day schedule window

Field	Purpose
predecessors, predecessor_uids	Package dependencies and resolved predecessor rows
work_order_title, work_order_detail	Text used when opening an Ops Core work order
staff_roles, staff_tasks	Crew accountability and task detail
bom_refs, material_count, material_status	Controlled material/BOM references and generated kit status
materials_or_inputs, deliverables	Human-readable material/input summary and produced output
evidence_required	QA, commissioning, or handover proof
release_authority	Person/role allowed to release the work
qa_gate_hint, qa_uid, verification_uid	Link to the relevant construction QA action and verification row
priority, status	Work-order priority and schedule state

The generator also emits:

Table	Purpose
manufacturing_materials	One controlled material/BOM row per manufacturing package reference, including BOM source, line id/category, description, quantity basis, cost basis, traceability requirement, and receiving evidence.
manufacturing_verifications	One QA hold-point row per manufacturing package, linked to the generated QA action by qa_uid, with required result, release authority, evidence, and successor-blocking rule.

Rolling-stock packages reference the generated BOM files:

- build/bom/rolling_stock_bom.csv
- build/bom/rolling_stock_cots_fitout_bom.csv

Infrastructure packages use controlled project_kit:* references until detailed civil, station, depot, wayside, switch, and energy BOM files are added. They still produce material-control rows and traceability evidence.

4. Scheduling Rule

The baseline schedule uses project-day offsets from notice-to-proceed. Those offsets are placeholders until a city baseline is approved; they can be converted to real dates in a spreadsheet or future calendar view.

The accepted dependency graph is scheduled from project day zero into finite lanes for each work centre. Capacity is controlled in the same TOML template. Resource-lane predecessors are made explicit, then a backward pass calculates late dates, total float and critical tasks. Unresolved external decisions are reported as baseline release gates instead of silently receiving zero duration.

The schedule remains a deterministic planning baseline, not a contractor's approved programme. A city must replace planning capacities and calendars with its accepted delivery resources before construction release.

5. Work Orders

The Operations Portal has a Manufacturing tab. Any row can be opened as an Ops Core work order with:

- source type manufacturing
- source uid equal to manufacturing_uid
- owner from release_authority
- priority from the package
- asset id and asset type from the asset register
- title from package_id plus work_order_title

Once opened, the work order follows the same lifecycle as maintenance and QA work:

```
open -> assigned -> in_progress -> ready_to_close -> closed
```

It can also be placed on hold, receive evidence records, create defects/NCRs, and appear in the audit trail.

The portal enforces two release rules:

- A manufacturing work order cannot be opened while resolved predecessor manufacturing rows have not been closed with pass evidence.
- A manufacturing work order cannot move to ready_to_close or closed until it has pass evidence and material/BOM references.

6. SQLite Storage

Manufacturing work orders are stored in the existing Ops Core SQLite tables. No separate manufacturing database is required.

The server creates the database automatically at:

```
var/ops-core.sqlite3
```

If a browser is used offline or through a static server, local records can later be reconciled into SQLite through the Storage Reconciliation panel.

7. Waypoints

The generator now creates `waypoint` assets for route sections. These are the practical W-Node / W-SBC production units used by wayside safety, radio, beacon, intrusion, and section-state functions. They receive:

- manufacturing packages
- QA coverage through `qa-26-wayside-comms-safety`
- maintenance coverage through systems maintenance intervals

Waypoint install packages resolve their track-section predecessor where the generated asset relationship identifies the parent track section.

8. Commercial and Actual Records

The generated twin creates planned order candidates, not issued orders. It calculates required-on-site and order-by days, exposes supplier anchors or the local-equivalent rule, carries controlled COTS/RFQ candidate IDs and selection states, and marks rows requiring quotations. The source register and first-article execution pack are hashed twin inputs. City CAPEX becomes schedule-of-values contracts with task-linked milestone cash requirements.

Ops Core stores actual-side purchase orders, deliveries, invoices, payments, progress updates and project revisions separately. This preserves deterministic regeneration while allowing planned-versus-actual comparison. Payroll, stock decrementing, tax/customs calculation, bank integration and approved supplier terms still require the city's real commercial data.

RFC 0031 — OSR City Studio and Git-Based Design Revisions

Repository source: `docs/rfcs/0031-city-studio-git-revisions.md`

Status: accepted, initial vertical slice implemented · 2026-08-26
Authors: OSR project
Complements: RFC 0018 operator GUIs, RFC 0028 construction QA, RFC 0029 maintenance schedules, RFC 0030 manufacturing schedules

1. Decision

OpenSourceRail will provide OSR City Studio, a browser-accessible design and service-planning environment over the existing deterministic generator, GIS sidecars, simulator, engineering toolchain, CAD generators, and operations packages.

Git is the authoritative revision and approval history. The Studio edits small, reviewable project-intent files. It never treats its working database, browser state, or generated build directory as the source of truth.

The generated city design.toml remains a compiler output. Interactive edits are recorded as intent overrides, constraints, geometry locks, and service plans. This prevents the next catalogue regeneration from silently replacing an accepted station movement.

2. Goals

- Reproduce a city revision from versioned inputs without hidden GUI state.
- Edit station intent and service levels by line, day type, and time window.
- Preserve stable object ids through manual review and regeneration.
- Preview changes before accepting a candidate.
- Pin source data, tool versions, parameters, and generated artifact hashes.
- Review revisions through ordinary GitHub branches and pull requests.
- Publish approved revisions as signed or protected tags/releases.
- Keep planning changes outside the live safety-critical OCC command path.

3. Non-goals of the initial slice

- Reimplement FreeCAD, QGIS, or detailed B-rep editing in the browser.
- Send route grants or other safety-critical commands to a live railway.
- Automatically push branches, open pull requests, or create remote tags with a user's GitHub credentials.
- Commit every pointer movement or autosave.
- Store raw imagery, solver scratch directories, or large simulation traces in ordinary Git history.

4. Project package

Each city has a committed package under cities/workspaces/<slug>/:

```
cities/workspaces/<slug>/
├── project.osr.toml
├── sources.lock.json
├── network/
│   └── overrides.toml
├── services/
│   └── service-plan.toml
├── coordination/
│   └── issues.toml
├── demand/
│   └── od-matrix.toml
├── approvals/
│   └── reviews.toml
└── revisions/
    ├── README.md
    └── osr-<content-hash>.json
```

project.osr.toml declares identity, paths, planning assumptions, the default branch, revision policy, and approved tag prefix. Paths are resolved relative to the project package so a checkout remains portable.

sources.lock.json records a SHA-256 for every authoritative generated or external input. Compilation fails validation if the bytes differ. Updating a source is therefore an explicit project change that must update its lock.

network/overrides.toml records stable station ids and one of:

- generated: the generator may replace it;
- preferred: regeneration should penalize movement but may propose it;
- locked: regeneration must preserve the accepted location;
- manual: created by a designer rather than the generator;
- retired: retained in history but excluded from the candidate.

The compiler supports generated, preferred, locked, and retired states for source stations, plus manual and retired states for designer-created stations. Moving a station deterministically warps its affected corridor inside the project regeneration radius, holds accepted locked stations as zero-displacement anchors, recalculates the affected route length and simulator inter-station distances, and preserves unaffected line geometry. The map also supports stable, click-to-create alignment control points. A designer can drag a point, choose its regeneration influence radius, and mark it preferred or locked; the same deterministic corridor and simulator regeneration then applies. Manual station insertion or retirement rebuilds the effective simulator station catalogue and ordered per-line topology as well as the GIS network. A two-endpoint manual-line workflow creates a stable line id, deterministically sampled initial corridor, two terminal platforms, a service plan for every day type, and simulator line/fleet/dispatch records. Local alignment controls can then shape the corridor. Manual lines can use either an explicit direct chord or a deterministic least-cost route over SHA-256-locked demand/buildability planning surfaces; the revision records the method, source IDs, and demand weight.

services/service-plan.toml contains a seven-day calendar mapped to named day types and a plan for every line/day-type pair. Each plan has a service span and contiguous headway windows. The compiler calculates cycle time, indicative peak fleet, capacity per hour per direction, daily service kilometres, and weekly service kilometres.

demand/od-matrix.toml contains source-controlled planning periods and optional origin–destination flows in passengers/hour. Periods refer to service day types and time intervals; flows refer only to active stable station IDs. A content-derived ID preserves flow identity while the passenger rate is edited. The compiler screens each record against the lower scheduled passengers/hour/direction capacity across all service windows overlapping the period and records direct/cross-line status and indicative utilization. This early screen is deliberately not a passenger assignment, ridership forecast, platform-flow model, or interchange/egress analysis.

coordination/issues.toml stores review decisions for deterministic civil BCF topics. Status, assignee, resolution, and reviewer are project intent rather than edits to a generated job artifact. Resolving or closing an issue requires a substantive resolution and reviewer. The compiler validates the issue set, includes it in the project hash and immutable revision snapshot, and reports semantic issue-state changes in revision comparison. A new civil BIM job then emits a new BCF reflecting that revision; earlier BCF files remain immutable evidence.

Designers may seed a new issue from a selected IFC asset. City Studio validates the title, description, and one-to-fifty stable OSR asset IDs and derives a deterministic `custom-<content-hash>` identifier. The civil adapter rejects unknown assets, then derives stable BCF topic and viewpoint UUIDs from that identifier. Re-entering identical content is rejected as a duplicate rather than silently creating parallel issues.

5. Revision lifecycle

```
working draft
    | save intent/service files
    ▼
Git-visible working tree
    | compile and validate
    ▼
candidate snapshot in build/
    | materialize revision
    ▼
cities/workspaces/<slug>/revisions/osr-<hash>.json
    | branch + commit + pull request
    ▼
cities/workspaces/<slug>/approvals/reviews.toml
    | append reviewer decision + review reference
    ▼
reviewed merge
    | protected/signed tag
    ▼
approved design baseline
```

Autosaves do not create commits. Materialize revision writes an immutable, content-addressed JSON record only when validation has no errors. It includes:

- project identity and schema version;
- source locks and their observed hashes;
- effective lines and stations;
- complete weekly service plans and fleet/capacity metrics;
- planning periods, OD flows, and conservative scheduled-capacity metrics;
- civil coordination issue status, assignment, resolution, and reviewer;
- semantic station movements;
- validation findings;
- exact project-input SHA-256, compiler version, and embedded compiler/UI source SHA-256;
- content SHA-256;
- parent Git commit.

The content hash excludes wall-clock time and Git branch state. Recompiling the same project bytes produces the same content hash and revision id. The revision record separately pins the Git commit from which the candidate was created.

The Studio suggests a branch and tag; users or CI perform remote GitHub operations. This avoids embedding personal tokens or silently rewriting repository history.

Approval decisions are append-only records referring to an existing immutable revision. Each records approved or changes-requested status, reviewer and role, decision date, rationale, and a pull-request or evidence reference. A stable content-derived `approval-…` id makes identical submissions idempotent. Approval metadata is intentionally excluded from the design content hash: otherwise approving a revision would create a different revision requiring a new approval. The TOML approval history remains Git-reviewed alongside the revision and can survive migration away from GitHub.

6. Determinism contract

OSR uses three determinism classes:

1. Byte deterministic: project files, JSON/TOML/GeoJSON/CSV outputs, revision manifests, and schedules.
2. Geometry deterministic: CAD/GIS exports may contain tool metadata or archive timestamps but must be geometrically equivalent within a declared tolerance.
3. Analysis deterministic: numerical solvers must reproduce accepted results within a declared tolerance using pinned inputs and tool versions.

Remote OSM, population, building, terrain, imagery, and survey inputs must be captured or checksum-locked. Running against an unpinned live endpoint is an exploration, not a releasable design revision.

7. Architecture

The initial implementation is crates/osr-city-studio:

- a Rust project compiler and validator;
- deterministic source hashing and revision ids;
- semantic station movement detection;
- manual station authoring and simulator-topology regeneration;
- manual line authoring with terminal, service, GIS, and simulator synthesis;
- semantic comparison between materialized revisions and the working candidate;
- weekly service metrics;
- deterministic OD demand intent, validation, scheduled-capacity screening, and semantic flow comparison;
- a local HTTP API;
- an embedded browser UI with an editable, layered local SVG/GIS view whose source catalogue, feature counts and SHA-256 digests are served with the project;
- artifact visibility for current design, GIS, simulation, engineering, operations, and release outputs.

The server binds to localhost by default. It reloads committed project files for each request so external Git changes are visible. Writes are serialized and published atomically.

The longer-term tool adapter boundary is:

```
declared inputs + parameters + tool version
                  |
                  v
            allowlisted adapter
                  |
                  v
artifact + logs + findings + content hashes
```

The first implemented adapters compile the candidate GIS package, run a fixed one-hour `osr-sim` day-type scenario, and export a selected line through `osr-alignment-export` to LandXML, railML, review JSON, and stakeout CSV. They serialize through one job slot, persist records and full logs, expose progress and bounded log tails, and hash every output. Adapter names, binaries, and arguments are selected by Rust code; the browser cannot supply an executable or unrestricted shell text. Future adapters may wrap osr-design, osr-scenario, GDAL/QGIS, FreeCAD/IFC, SUMO, pandapower/pvlib, finance, and operations generation under the same boundary.

Successful jobs copy their review evidence into the job-specific directory so later candidate compilations cannot overwrite the recorded files. The browser artifact endpoint canonicalizes the selected record path beneath the City Studio build root, limits preview size, accepts only known JSON/XML/CSV/text formats, and verifies the recorded SHA-256 before returning content. The GUI plots GeoJSON, local alignment points, LandXML geometry, and stakeout data and summarizes railML and simulator evidence. For civil review it joins the verified object index and 4D sequence to provide deterministic envelope rotation, native IFC presentation-layer visibility, construction-stage playback, task/QA context, and stable BCF object picking. These previews are not an authoritative CAD or safety-assurance tool; native IFC detail remains in Bonsai.

8. GitHub review policy

An approved city revision should use:

- protected default branches;
- pull requests rather than direct pushes;
- required City Studio validation;
- engineering and operator review appropriate to the changed objects;
- immutable release tags;
- signed tags where the deployment governance supports them;
- an in-repository approval record for decisions that must survive migration away from GitHub.

Text, compact geometry, summaries, manifests, and approval records belong in Git. Large FreeCAD/Blender/IFC packages, imagery, rasters, videos, and raw solver output belong in Git LFS, GitHub Releases, or deployment-controlled artifact storage. Their hashes and generating inputs remain in the revision.

Geo and service files use stable ids, sorted records, and canonical serialization. The Studio will add object-aware merge conflict reporting before concurrent multi-user editing is promoted.

9. Safety boundary

City Studio produces planning candidates and approved configuration packages. It cannot issue a movement authority, route grant, maintenance override, or degraded-mode command.

The OCC may consume an approved, signed baseline. Live actions remain governed by RFC 0013, RFC 0017, RFC 0018, deployment roles, and the deployment safety case. Sharing a map renderer or object id does not merge those authorities.

10. Implemented acceptance fixture

The Samawah project is the first committed acceptance fixture. It:

- pins the current design, corridor, and simulator scenario;
- locks all three platforms in the accepted CBD interchange;
- defines weekday, Friday, and weekend service for all three lines;
- produces nine line/day-type service metric records;
- exposes 21 station platforms in the Studio map;

- regenerates affected GIS corridor geometry and simulator distances after a station movement while preserving locked station anchors;
- supports direct line edits through stable alignment control points with editable influence radii;
- creates, moves, renames, classifies, and retires manual stations while regenerating GIS and simulator topology;
- creates and retires complete manual lines, including terminal platforms, weekly services, fleet sizing, dispatch points, and candidate geometry;
- creates manual lines through either source-locked demand/buildability least-cost search or an explicitly identified direct planning chord;
- records routing method, demand weight, and every raster/provenance source id on the line and in candidate GIS output;
- compares old revision records with the working candidate using engineering tolerances for serialized floating-point values;
- persists civil coordination decisions as Git-reviewable intent, rejects unsupported or unevidenced closures, and carries accepted state into a newly generated BCF without mutating prior job evidence;
- creates deterministic custom BCF topics from selected IFC assets and rejects duplicate content, invalid asset IDs, and dangling selections;
- loads the hash-verified civil 4D sequence with discipline filters, rotation, task-stage playback, and stable object selection;
- appends deterministic approval decisions only for existing immutable revisions and proves that approval does not alter the design hash;
- emits separate weekday, Friday, and weekend simulator scenarios plus a SHA-256 artifact manifest;
- runs allowlisted field-evidence, GIS, simulator, and LandXML/railML alignment jobs with persistent status, captured logs, and SHA-256 output records;
- retains immutable per-job artifact copies and rejects altered evidence before browser preview;
- compiles without validation errors;
- produces an identical content hash across repeated compilations.

The next acceptance increment is native IFC geometry streaming plus demand/OD and interchange-capacity inputs.

RFC 0032 — Train-centred control prototype (superseded)

Repository source: `docs/rfcs/0032-train-centred-control.md`

Status: superseded by RFC 0033.

The original prototype introduced a separate `osr-tacs` movement-authority and protection implementation. It is retired in favour of resource contracts in `osr-core`, committed lifecycle handling in `osr-interlocking`, and thin `osr-runtime` process hosts around the existing consensus/MA/ATP/ATO/brake path. The previous synchronous twin results are replaced by actual process-reference execution. Bounded resource-model evidence remains explicitly limited to its physical-proving assumptions. No hardware or operating release is claimed.

RFC 0033 — TACS runtime and committed resource control

Repository source: `docs/rfcs/0033-tacs-runtime-and-resource-control.md`

Status: proposed development target; software evidence generated/unreviewed. Supersedes: RFC 0032's separate authority/protection implementation. Deployment: synthetic reference only; sectional-authority pilot retained.

Responsibility and interface map

Function	Decision owner	Physical or deployment boundary
Shared railway identities/configuration	`osr-core::deployment`	Exact model-byte hash; frozen startup parameters; designed revision distinct from installed and approved records
Authoritative resource lifecycle	`osr-interlocking::resources`, `resource_log`, `DerivedState`	Only verified committed `ResourceControl` entries mutate ownership
Ordered authoritative decisions	Existing `osr-consensus::step`	Three static voters; independent power/network placement must be designed and assessed
Physical proving	Identified point/station I/O hosts	Asset-scoped occupancy, integrity, points, restrictions and qualified no-reentry proof; synthetic ports in the reference
Onboard movement authority	Existing `compute_self_ma_from_state`	Validated committed view, exact frozen model, bounded full train footprint and evidence age
Speed and emergency protection	Existing `osr-atp::atp_evaluate`	Grade-adjusted illustrative braking profile; measured braking/adhesion/latency remains open
Mission and station requests	Existing `osr-ato::ato_evaluate`, `station_step`	Doors, charger isolation, connector clearance and energy independently gate departure
Final brake request	Existing `osr-brake::brake_evaluate`	Application request to separately qualified safety-output channel; software host is not qualified output hardware
Authentication and message contracts	`osr-proto`, `osr-crypto`, `osr-secbus`, consensus proposal verifier	Registered issuer, asset permissions, configuration/session/sequence/time checks before state effects
Process execution and persistence	Thin `osr-runtime` hosts	Framed local ports, event scheduling, disk writes before outbound effects, conservative cold start
OCC	Service intentions and observations	No authority, ownership release or latch-clear bypass
ERPNext/FUXA	Existing observation/maintenance integrations	Observation only; no new railway command interface

There is one MA implementation and one ATP implementation, executed in each logical protection channel. Resource contracts from the earlier prototype move into existing crates; `osr-tacs` and its parallel MA/protection algorithms are retired. `osr-runtime` contains orchestration and adapters. Route selection remains intent, per `osr-onboard-routing`; agreement cannot supersede missing permission or physical proving. Logical peer observations may traverse common infrastructure; no direct radio or `osr-t2t` crate is added.

Configuration and communications

`railway-model.json` schema `osr-tacs-railway/2` supplies resource IDs, existing section IDs, conflicts, extents, asset identities, train/session/formation parameters, station locations, static voters and role-scoped public keys. `deployment.json` is an exact-byte-hash-bound generated startup export. `FrozenRailway` rejects unsupported versions, invalid topology/identities, changed startup parameters and operational activation. This source is a designed synthetic configuration; `installed_state=not-installed` and `approved_operating_profile=prototype-unapproved` are separate explicit facts. Changes require a new bundle and restart, never live digital-twin mutation.

The reference transport uses signed `osr-runtime/1` packets, postcard encoding, existing signed consensus proposals and explicit internal interlocking entries. Legacy protobuf/postcard `osr-proto::Entry` remains a separate versioned format; it must not be deserialised directly into internal entries. Float-based legacy position inputs require checked unit conversion at an explicitly selected adapter. The runtime path publishes integer measured speed and bounded, non-zero uncertainty. Unsupported/malformed packets fail closed.

The envelope binds configuration hash, registered session, monotonic sequence, original observation time and message kind. Limits: 512 KiB application payload, 768 KiB signed payload, 4 MiB framed command, 16 publications per batch, 12,000 log entries and a 256-message reference queue. One-second input freshness and 100 ms maximum configured clock error are prototype assumptions; PTP and hardware timing must establish their operational applicability. Future-dated packets are rejected. Send/receive counters persist across restart. Asset permissions are separate from signature validity. All load-bearing events, including positions, use `Safety`. The legacy simulator consensus publisher also classifies occupancy-changing position reports as `Safety` and retains its last committed view when quorum confirmation is unavailable. No advisory route/position can unlock track.

The published deterministic reference keys are test fixtures. Operational provisioning, protected key storage, rotation, revocation distribution, offline replacement and recommissioning require a controlled maintenance pack. Changing keys or sessions changes the configuration hash; interoperability stops until the complete bundle is reconciled. Revoked participants

are rejected. Crash-fault Raft is assumed: a signed member's committed-prefix assertion is not a Byzantine quorum certificate. Authenticated erroneous sensors and common software, configuration, power and clock causes remain FMEA concerns.

Ownership and evidence freshness

Unknown qualified clear available reserve/prove/lock occupy explicit withdrawal qualified no-reentry/clearance available. Identified occupied startup admission can bind the proved train/session when no conflicting owner exists; it cannot transfer another owner's lock. Leases time out to `ReleasePending`, retaining their owner. Restart increments the committed controller epoch, retains owners as unknown and quarantines empty clearance for the maximum prior permission plus clock error. Identified retained owners can be reconciled without transferring protection. Blocked possessions cannot be cleared by these recovery APIs.

An empty detector does not prove cancellation safe. Qualified release requires bounded full rear clearance with integrity, or committed withdrawal and a proved stop/protected approach that prevents old-permission re-entry. These are physical interface assumptions to validate. Platform resources remain owned at arrival. Depot/work resource is explicitly protected; rescue, coupled formations, turnbacks, arbitrary topology and large fleets remain separate packages.

MA expiry is the minimum of the new MA window and the original position, local proof and held permission deadlines. Recalculation cannot refresh evidence. Every traversed section must have the correct active owner/configuration/epoch, current point proof and integrity. Local full-footprint uncertainty checking adds a restrictive gate before ATP. Recovery requires stopped, valid local facts and a privileged local procedure, never a dashboard acknowledgement. Civil conditions enter through asset-scoped signed restrictions/closures; urgent protection must have an engineered path independent of alarm viewing. Removal requires qualified inspection/handback, not an ERP status change.

Persistence and process reference

`osr-consensus::disk::DiskJournal` has one OS-locked writer. Frames carry node, configuration, term, vote, complete log deltas, commit index, application state and replay fences, with length and CRC bounds. Every host persists and fsyncs file and directory before exposing messages or actuator requests. Torn/corrupt journals inhibit startup; repair never silently discards an accepted lock. Storage compaction atomically replaces a full-log checkpoint after fsync. Logical Raft snapshotting/log truncation is not implemented. Capacity exhaustion inhibits further publication; offline servicing requires retained-protection and configuration reconciliation. A power interruption during rename/fsync requires storage/hardware qualification; SIGKILL tests are software evidence.

The default Raft batch remains one entry. The reference host uses bounded contiguous batches of twelve; Rust handles each entry under the same checked prefix semantics. This is outside the one-entry formal refinement and needs independent refinement review. Quorum confirmation records fresh per-peer responses in the current term, including empty-log heartbeats; stale historical replication indices cannot refresh a quorum. Durable reboot restores no leader role, quorum confirmation or volatile acknowledgements.

Run `python3 tools/automation/tacs_reference.py --output /tmp/reference.json`. The coordinator spawns four `osr-train-agent` processes (A/B for each train) and six `osr-wayside-agent` processes: three voters, point I/O and west/east station I/O. Charger requests and local proving pass through the station process ports; their asset-scoped interlocks can inhibit power and cannot grant movement. The depot/work resource is committed blocked and remains blocked across expiry and controller restart. Two further `osr-safety-output` processes each host the paired `osr-brake::dual::DualGuard` comparator. Both channels must agree to permit; either emergency, missing/stale/replayed request, disagreement or unhealthy feedback latches an emergency output with zero torque. The reference feeds at 50 ms, before the 100 ms output deadline. A late feed cannot replace an overdue request to hide a missed interval; only stopped, privileged, fresh recovery can clear the latch. The deterministic twin explicitly selects `--clock virtual`. The default output host now has an independent monotonic-clock sampling thread, bounded input parsing and timestamped output snapshots. Idle/partial input and blocked consumption cannot suspend that sampler. Source issue timestamps and process-clock epochs remain distinct from receipt time. Native process and expiry tests are software evidence; independent physical timers, scheduling bounds, source authentication and de-energised hardware outputs remain unqualified. See the runtime contract. Processes exchange signatures through asynchronous bounded queues; publication does not wait for commitment. Simulation truth enters only own-train sensors and local infrastructure proving ports, never the MA state. Real journals and process kill/restart are used. The same frozen bundle feeds every process. The physics fixture checks its supported extents, conflict identities, formation/braking and berth/energy parameters before startup; unsupported bundles require a new calibrated fixture. `osr-sim`'s legacy synchronous cluster publisher remains isolated from this reference; it is not claimed as the deployed asynchronous path. Full integration of its scenarios/physics, shared clocks, radio/T2G/TCN failover and direct-peer performance is future work.

Station sequence is stop/secure exchange/charge isolate physical departure proof ready. The reference measures synthetic energy and closed-loop speed; per-car BMS/thermal/door/PSD/traction integrations remain pending HIL. Failed

charging can hold service without erasing track ownership. Emergency departure cannot bypass a connected charger or unlocked door. Missing application output must trip a separately qualified deadline/output channel; the isolated reference output ports model deadlines and latches but do not qualify that hardware function.

Default redundant protection and platform review follow-up

The executable policy sets two logical channels, two-out-of-two permission and either-channel trip as the development default for train protection, point proving/locking, crossings, doors/platforms, charging isolation and obstacle protection. Loss of either channel stops the affected function. Automatic single-channel operation or takeover is excluded. Availability after a channel failure requires a separately assessed architecture. Three Raft voters replicate authoritative state; they are not protection outputs. The conservative deployable pilot profile and its release gates remain in force.

Each train A/B host uses a separate OS process, journal and restored committed view. The binary requires `--safety-channel A|B`; a persisted journal cannot change channel on restart. A publishes the train's network proposals; B evaluates the same committed domain inputs without publishing duplicate proposals. Loss of A does not authorise B takeover. Pair comparison uses exact synthetic cycle sequence/time, brake and torque agreement. It has no end-to-end source-issued output identity yet. Shared software, sensor fixture, host, test keys, coordinator clock and comparator are explicit common causes. Physical power/output separation, channel-local clocks and independent feedback must be engineered and qualified. Only train protection's logical pair is executed; the policy explicitly keeps all five other local pairs open. Two processes are not installed board quantities.

The 2 October 2026 review of `17b31f055c9e16b5439c18b4a4d83f004890bcad` identified a stale FMEA applicability check and a concealed deadline gap. This change rejects stale failure subjects as well as stale configuration baselines, retains the late-feed regression and adds missing-A, missing-B, disagreement, replay and feedback-failure process cases. Each new case must first permit motion, actually inject its fault, trip within one 50 ms reference step, stop with retained protection and remain latched. These are virtual-time software criteria, not measured hardware response times. The existing TLC abstraction covers resource ownership only; it does not prove the new paired output function.

Next work closes named integration gaps in this order. Owners and reviewers are roles to appoint; acceptance is still open.

Work	Owner / independent reviewer	Required exit evidence
Autonomous watchdog and output paths	Embedded/electronics / safety-I/O reviewer	Coordinator/application/comparator freeze and kill, delayed scheduling, clock rollback/skew, power loss and stuck feedback; measured traction inhibition and effective braking without a new application command
Sensors and emergency union	Train integration / protection reviewer	Typed adapters to existing localisation, fire, derailment, obstacle and BMS/thermal evaluators; wheel slip/bias and simultaneous resource faults independently reach the final output; shared bad sensors do not establish independence
Repeated station missions	ATO/station integration / operations reviewer	Explicit mission/stop identity and arrival/departure transitions; at least three stops, short charge, delayed doors, turnback/resume, per-car charging and no moving door release
Remaining local protection pairs	Wayside/station / safety-I/O reviewer	Point/crossing proving, door/PSD, charger isolation and obstacle channels; independent release and inspected handback
Calibrated plant and city conversion	Simulation/configuration / braking and site reviewers	Sealed city revision converted to supported resources and shared `osr-sim` plant with mass/load, adhesion, gradients, build-up/delay, thermal/energy limits; full LM3 stopping envelope under uncertainty sweeps
Duty, recovery and storage	Simulation/runtime / operations and storage reviewers	Representative city's station capacity, overnight storage, morning dispatch, charging outage/recovery; separate rescue/tow/split/reverse/handover missions and sustained duty without capacity exhaustion or unbounded restart; safe snapshots/replay fences retain locks
Formal and scenario evidence	Assurance / independent formal reviewer	Property-to-code/test mapping, overlapping multi-resource routes, sequences/epochs/crash points, counterexample replay and specific hazard activation/deadline/excursion/queue/CPU/storage/recovery metrics
Integration and lifecycle quantities	Platform/hardware / configuration and cost reviewers	Observation-only A/B health/discrepancy/feedback/age/degraded/revision data through supervision gateway to FUXA/ERP; no acknowledgement resets; actual serial/power/firmware/calibration/replacement records and matched-service redundancy/spares/training/assessment costs
Physical release	Hardware/operations / independent assessor	HIL, measured braking/integrity/radio, power interruption, closed-track recovery and maintenance drills before profile approval

`station_step(Ready, fresh berthed/secured arrival)` currently remains Ready; the host targets the final stop. Nominal arrival therefore does not demonstrate another passenger exchange or a multi-station mission. The bounded two-train fixture retains geometry checks; expanding it cannot substitute for calibrated city conversion. Civil closures must bind precise operational resources and inspected handback separately from ERP repair completion. No duplicated ERP/GUI or

comfort software is implied by protection redundancy. Commodity hosts, the existing small Rust evaluators, reusable local controllers and open replacement/ calibration procedures remain the implementation approach. Hardware quantities and hot-standby/two-MCU mappings must be reconciled at the hardware freeze.

Evidence and staged delivery

The shared connected-engineering compiler carries part subassembly controller function train/infrastructure hazard control evidence operating response. Source changes stale execution hashes and propagate through configuration-bound claims and blocked release decisions. TLC checks a bounded resource abstraction with qualified physical proving assumptions; it is neither a Rust refinement proof nor an unbounded liveness result. G0–G4 decisions remain blocked pending review and measurements.

Owners/reviewers below are accountable roles to appoint, not fabricated named approvals. No package is accepted merely because a generated checkbox is true.

Package	Technical owner	Independent reviewer	Exit evidence / dependency
1 Architecture	Control architect	Independent railway safety engineer	Reviewed responsibility map; current baseline
2 Shared configuration	Configuration/lifecycle engineer	Independent configuration auditor	Identity/schema/mixed-version tests; 1
3 Resource/durability	Interlocking/consensus engineer	Independent formal/storage reviewer	Lifecycle, crash and restart evidence; 2
4 Authority/protection	ATP/localisation engineer	Independent braking engineer	Freshness/uncertainty/integrity properties; 2–3
5 Communications	Communications/security engineer	Independent security/timing reviewer	Replay/loss/restart/vendor-neutral transport tests; 2
6 Runtime	Embedded runtime engineer	Independent safety-I/O reviewer	Multi-process reference and physical power-loss tests; 3–5
7 Station	Station/energy integration engineer	Independent doors/charging reviewer	Complete station-cycle and per-car tests; 4–6
8 Twin/campaign	Simulation engineer	Independent test engineer	Reproducible raw results; starts after 2
9 FMEA/GSN	Assurance engineer	Independent hazard reviewer	Connected change impact and applicability; continuous from 1
10 HIL/hardware	Electronics integration engineer	Independent hardware assessor	Measured output/timing/fault records; 6–9
11 Closed track	Test/operations engineer	Independent field safety reviewer	Braking, integrity, recovery drills; 10
12 Profile release	Railway release authority	Independent assessor	Named gates, rollback and approved installed bundle; 9–11

Hardware preparation proceeds with software and assurance using existing `t-ecu-s`, `t-ecu-a`, `t-obs`, `w-sbc`, `s-sbc` families. Each pack needs exact module/SKU/substitutes, wiring/connectors/enclosure, power/thermal/environmental budgets, fail-safe outputs/watchdogs, calibration, controlled firmware, fixtures, bench records, and complete-unit replacement/recommissioning instructions. Existing COTS and custom-board release tracks both remain valid; safety-channel selection follows `control-electronics/safety-controller-selection.md`. This change supplies no invented SKUs, installed serials or physical test records.

Cost comparison uses equivalent service and availability, separately counting train controllers, static voters, non-voting sites, radios/antennas/power, retained detection, installation, spares/calibration/maintenance, integration, assessment, recovery/disruption and local/import content. No equipment removal or city-model savings is authorised without replacement function and evidence. The reference's ten processes are software roles, not a bill of materials. Cost quantities/prices remain deployment-specific inputs; an unknown input must keep the comparison incomplete. Track duplicate safety implementations, configuration variants, core dependencies and physical interfaces alongside money.

RFC 0034 — Connected station, production and battery model

Repository source: `docs/rfcs/0034-connected-station-production-and-battery-model.md`

Status: proposed planning design; 2026-10-06. The draft mentioned against `5f5d9fd` was not present in this checkout; this RFC reconciles the requested position with the current source rather than assuming that an unpublished text is authoritative. No vendor quotation, structural release, railway approval, construction contract or funding commitment is created.

The shared model joins station topology, civil quantities, supplier and logistics contracts, shift scheduling, scope reconciliation, battery profiles and construction fleet/crew contracts. Baghdad is a generated application of these shared contracts, not a separate set of design rules.

Related decisions: RFC 0010 stations, 0011 civil, 0021 battery traction, 0024 thermal, 0026 charging, 0028 construction QA, 0029 maintenance, 0030 manufacturing, and 0002 energy sizing.

Shared decisions

Subject	Proposed position	Release boundary
Ordinary elevated station	One central island, two boarding faces and two tracks	Site exceptions require a recorded reason; accessibility, fire, passenger-flow and structure qualification
Running spans	Existing single-track OSR-Pi20/Pi25 family	Prestressing and all permanent/construction stages require structural release
Erection	Whole-beam launcher on constrained continuous fronts; mobile cranes where access is advantageous	Separate equipment, lifting and closures approval
Segmental	Separate geometry/transport option	Match-casting, temporary suspension, joints, stressing and tendon protection; separate gantry and productivity
Baghdad	18 launchers, two separate crews/shifts per day	User-selected sensitivity; independent access and ready supports required
Precast supply	Existing qualified Iraqi factories plus necessary tooling/bed upgrades	Current contracted capacity, delivery and acceptance evidence; candidate capability grants no capacity
Batteries	LFP reference plus separate stationary/onboard sodium-ion study profiles	Cell/pack, electrical, thermal, life, software calibration and charging qualification
Equipment reuse	Individually identified transferable construction fleet	Compatibility, availability, shipping and recommissioning checks in each city

Every new input carries units, source, date, confidence and qualification. User-selected scenarios, calculated quantities, documented historical capability and supplier quotations are different evidence kinds. Stable input-content revisions and file hashes identify generated packages, including modified working-tree inputs, without depending on their containing commit. Unknown installed prices, reaction designs or spare capacities remain unknown; zero is reserved for a measured/declared absence.

Station and civil contract

Elevation is independent of archetype. `major` is a physical island, not two side platforms. Each platform, face, track direction, boarding datum and door/PSD side comes from one layout. Stacked interchanges have individual platform levels and explicit concourse transfer connections. CAD, IFC, assembly datums and the EBOM consume this contract. A physical platform count never substitutes for boarding faces or train berths.

The initial elevated island study has two lifts, two escalators, two lift shafts and two separated stair routes. It is not an approved universal minimum. Equipment has stable identities, served levels, envelopes, entrances and concourse connections. The layout rejects overlapping equipment and insufficient clear widths. Bidirectional flow, one-lift-unavailable operation, independent evacuation, assisted rescue, road barriers and additional street equipment remain separate assessments. Additional equipment/entrances can be supplied by site configuration.

Access CAPEX uses installed item quantities and associated works; energy, annual maintenance and replacement have separate outputs. Net island savings deduct wider deck, changed supports and spreading approaches. Unknown structural costs prevent a net saving claim.

Station decks, widened pier caps and foundations never inherit running-viaduct width. Approach offsets use a quintic transition and a curvature screen, followed by alignment/cant/gradient qualification. Construction requires wider launcher support and clearance checks, delivery access, released supports and an erection sequence. Changed entrances have geographic coordinates and trigger a coverage review; existing centroid coverage cannot be certified for newly proposed entrances.

Manufacturing, suppliers and erection

Pi-beam specifications record concrete, steel, prestressing and inserts; manufactured study mass and centre of gravity; lifting/storage/transport positions; strength gates; bearing/restraint interfaces; tolerances; and acceptance records. The 60/75-tonne figures are bare sections. Study suspended load includes local diaphragms, reinforcement, prestressing, embedded items and lifting gear. Equipment approval fails until the arrangement and strength/tolerance records are released. Launcher support reactions, asymmetric placement torsion and delivery-vehicle deck loads have separate construction-stage screening outputs.

The capacity register records products, prestressing capability, beds, moulds, handling dimensions/mass, demonstrated cycles, total versus contracted output, commitments, storage/dispatch, prices, terms, qualification and upgrades. Iraq already has many precast facilities and production expertise; this is the user-provided sourcing basis. No named facility or unverified spare capacity is adopted. Each factory must enter the shared register with product-specific evidence and contracted capacity before it can supply the programme. Slabs, walkways, ducts and suitable pier parts can be allocated separately; aggregate allocations cannot exceed a product's contracted capacity.

The shared chronological supply chain now feeds contracted completions, acceptance holds, manufacturing rejects, dispatch and transport directly into erection. Factory buffers count both awaiting-acceptance and accepted stock. Full buffers pause casting without banking unused whole production capacity; rejected or damaged beams create replacement demand while installed quantities remain fixed. Shared transport fleets cannot be multiplied across routes. In-transit loads reserve front storage and receive credit only after arrival; an available qualified alternative route can serve a blocked primary route.

Delivery routes constrain journey/window/loading/unloading times, trailers, payload/bridge limits, long-load/turning access, accepted/rejected components, order and buffer space, plus alternative routes/suppliers. Unknown/rejected route approval supplies zero output. Front start/end chainage, direction, launcher, delivery access, support readiness, interruptions, start/finish/relocation dates and work intervals are explicit. “Two per line” initializes the scenario; assignments may change with lengths/readiness. Shared obstructed routes and sequential erection paths cannot receive simultaneous output credit.

The simulator counts integer beams and complete paired-track bays. Daily production is productive shift time divided by a complete bay cycle, with handovers, maintenance, weather, windows, ramp-up, station interruptions and relocations. Extra launchers on one sequential front are spares, not extra independent paths. Casting, acceptance, transport and released supports cap progress. Each front reports the model days limited by each resource, and the package separately records production, acceptance, shipment, arrival and rejection totals. The conditional pool describes a capacity requirement and requires explicit study opt-in; it is not inserted into the real supplier register. The shared `precast-logistics.json` register contains actual route evidence when supplied. Fractional 1.5/1.8/2.0 bays/launcher/day are averages, implying 54/64.8/72 beams/day across 18 machines. Two shifts do not automatically double output. A conditional forecast assumes the necessary supply; the evidence-backed forecast remains unreleased when no suppliers/contracts/supports are qualified.

Costs, equipment and people

Make/buy separates delivered purchases, project tooling, upgrades/beds, storage/dispatch and new factories. Supplier amortisation and project tooling cannot both be charged. Rolling-stock assembly and electronics factory allowances remain until their scope is reconciled; replacing civil supply is not permission to delete them.

The civil calibration is edited and the cost file regenerated. Supplied products, erection equipment, transport, temporary works, labour/supervision, night shifts, inspection, maintenance/relocation, traffic management and supports have explicit study allocations. Station exceptions and special structures stay separate. The $9m central purchase allowance for 18 whole-beam launchers remains unquoted, with transporter, lifting gear, freight/duties, assembly, commissioning, supports, spares, crews, relocation and station-modification exclusions. It is not a segmental-gantry quote. The installed civil rate's embedded erection allowance remains unverified: purchase expenditure creates no additional claimed saving until an explicit scope-matched removal is justified.

ERP drafts identify each launcher, transporter and lifting frame as an Asset, with purchase/commissioning records, operating configurations, inspection/maintenance, spares, competence, location, availability, allocation and dismantle/ship/recommission tasks. Cash purchase, project cost allocation and residual value stay separate; initial Baghdad cash is not eliminated by reuse. Transfers cannot overlap allocations and require subsequent-city compatibility.

Department unit crew qualified worker task is a validated chain. Each front and shift requires supervision/lifting, operators, riggers/signalling, survey/QA, mechanical/electrical maintenance, delivery coordination and factory acceptance. Workers cannot fill both shifts concurrently. Handovers, relief/leave coverage, competence expiry, supervised commissioning and equipment-specific assessment are explicit; unfilled drafts do not grant permission to work.

Battery and chronological energy contract

Profiles specify supplier/cell identity and evidence revision, gross/usable kWh, complete mass/volume, electrical configuration/range, temperature/SOC/age-dependent charge and discharge limits, efficiency/cooling, life, installed/replacement price and qualification. Catalogue sodium values are illustrative pack sensitivities, not asserted vendor performance. Vehicle tare changes by the difference from the included LFP pack; that mass propagates to traction energy and costs. Cases hold usable capacity and service constant: LFP throughout, sodium stationary/LFP onboard, sodium both. The small onboard pack remains a separate duty-cycle option.

The network duty uses declared fleets, dispatch termini, headways, station distances, dwell and installed site capacities, with an explicit average-speed timing assumption and stock-readiness checks. Minute chronology allocates concurrent arrivals across shared cabinet power, subtracts setup time, applies charge taper/temperature/SoH, accounts for regen acceptance, seasonal solar/cleaning, hot-weather auxiliaries, missed visits, outages, reserves, degradation and replacement. Report minimum SOC, unserved duties and missed charges, not annual totals alone. The study needs a calibrated timetable and full seasonal chronological validation before service acceptance.

`osr-bms` requires an explicit commissioned profile with matching pack, chemistry, evidence revision, qualification and valid calibration. A chemistry-string change cannot select or derive sodium protection from LFP defaults. Existing trip, latch, cooldown, off-gas/fire isolation and derating behaviour are preserved and shared by the validated evaluator. Planning profiles are not BMS commissioning profiles.

Regeneration and acceptance

The outstanding review is traced in `remaining-work.json`, covering all 20 software, engineering, commercial, workforce, deployment and governance areas. Source-hashed records may be entered through `connected-delivery-evidence.json`; a receipt is not authority authentication or acceptance. No supplier, appointment or qualification is invented. The user-provided Iraqi precast capability basis remains qualitative, without selecting a named facility.

`beam-stage-assurance.json` adds gross-section properties, uniform-selfweight moments, lift/store/transport support reactions, sling forces and launcher-position envelopes. Geometry comes from the canonical Pi family. These are statics screens with explicit mass/support/dynamic assumptions. Prestress, reinforcement, connections, material/release strength, fatigue/vibration, anchors, bearings, stability and manufacturer load charts remain required for an independently checked design.

`station-capacity.json` calculates the union of shared-layout equipment envelopes on each platform and its remaining area and contiguous lane. Lift-out capacity and circulation/density comparisons use source-entered criteria; an empty register yields unknown throughput. Multilevel demand allocation, street barriers, surveyed entrance walking routes, evacuation/rescue and station/launcher structures require site assessment.

`procurement-evidence.json` separates dated preliminary, firm and contracted quotations from factory capacity and purchase commitment. Its catalogue beam requirements reference identified span components; special spans remain outside ordinary orders. Tooling already amortised in a unit price cannot be added as project tooling. The USD 9m allowance remains purchase cash only. `cashflow-sensitivities.json` links those identified payments to each conditional schedule at assumed 4/8/12% simple ACT/365 interest with full borrowing of the identified cash. Unknown costs, scope-removal credit, complete financing and accepted revenue remain null, so these sensitivities cannot be published as complete capital or net savings.

The generic front splitter can reassign the available radial donors in another city without requiring six donors or fabricating 18 active fronts. Configured spare launchers add no independent output. Other-city runs remain independent, unqualified planning scenarios; transfer from Baghdad is not authorised before its critical path and compatibility/recommissioning are accepted.

ERP execution records now bind lifecycle release to the actual project/front. Required transfers need transfer, compatibility and recommissioning evidence. Actual work start/close timestamps are controlled; active work cannot be reopened to hide an allocation, and cancellation requires reviewed handback. Completed/cancelled work history continues to participate in worker-rest checks. These business controls grant no railway engineering acceptance.

The identified span register partitions running intervals at millimetre precision into proposed 20 m/25 m catalogue spans and explicitly unresolved special spans. Ordinary beam requirements have stable component and span IDs; special spans receive no ordinary-beam order or invented design. Proposed supports require survey and structural acceptance. Disconnected sections incur a stated relocation allowance instead of receiving continuous launcher progress. Fixed allocation and a conditional six-machine transfer to additional ring fronts use the same 18 launchers, contracted-throughput requirement and 72-transporter fleet. Added ring access and support readiness are assumptions, not feasibility evidence.

Ring stock serves both headings at the declared dispatch stations, including controlled wrap distances in reverse. The energy study continues through three service days without resetting train or storage state. A declared study strategy permits grid replenishment within each site's remaining import capacity. Native section timing comes from `osr-sim`'s scenario loader and rest-to-rest movement model. The fixed-assignment diagnostic and coupled dispatch controller are separate: the controller keeps an energy-held train at its actual station, continues charging and removes it from subsequent dispatch availability until recovery. It reports distinct held journeys, hold minutes and missed opportunities separately. Minute discretisation, berth/depot access, transient traction, conflict authorities and field calibration remain open; no service or chemistry acceptance follows from these screens.

Explicit working calendars now gate erection, casting, acceptance, deliveries and foundation release on closed days. Holidays, dated maintenance, available windows and permission for the erection night shift are configurable. A six-day sensitivity retains the same fleet and installed quantities. Daily supplier-specific shifts, permitted night work and named relief rosters need actual evidence; no statutory holiday list or permission is invented.

Native simulator commissioning uses `[[battery_commissioning]]` records, each naming train, profile ID, pack identity, chemistry, evidence revision, nominal pack voltage, model capacity and a complete `CommissioningProfile`. Every train must be covered if any binding is supplied; duplicate packs, mismatched chemistry/capacity and unqualified profiles fail. The onboard evaluator uses the selected immutable calibration, cell count and voltage and exposes identity in its report. Legacy absent bindings remain labelled by zero commissioned trains and are synthetic planning shadows. Catalogue chemistry profiles contain no accepted commissioning records.

ERP native Task/ToDo transitions now require a submitted `OSR Construction Release`, reviewed by a user with HR and project authority roles. Evidence is private, task-bound and hashed. Controlled Employee authorisations and Asset lifecycle records require appropriate HR/project roles to change and are re-read under resource locks. Work checks active company/department identity, unrevoked competence, asset configuration/commissioning, inspection/maintenance coverage, unresolved repairs, native shifts, approved leave and holiday lists, reviewed night permission, relief/handover, overlap and rest across completed shifts. Completing work additionally requires a reviewed completion record. Revoked reviews and changed attachments cannot grant work. Installation and local tests establish code paths, not a deployed workforce, actual competence or physical railway release.

Station-specific assignment accepts sourced OD cohorts with declared station legs, headings, transfers and a peak window. It counts each paid journey once, additional transfer boardings separately, section loads and station waiting accumulation against that window's declared service. Empty inputs produce unknown demand. The common 3,000 pax/h input is a separately labelled stress case; measured lift-out throughput, circulation, rescue and entrance-based catchments remain evidence gates.

The Iraq national brief and funding document use one current country basis. The USD 8.312bn `local_positive` comparator is distinct from the unpriced accelerated case and the historical USD 7.581bn funding calculation. No debt-clearance month, national budget or realised saving is claimed for the current case. Existing Iraqi precast facilities and expertise are acknowledged without selecting a named facility or assuming spare contracted capacity.

Run `.venv/bin/python tools/automation/connected-build-study.py --city baghdad --refresh-cad` and `--check`. Regenerate station kits, native FreeCAD station libraries and IFCs with their established generators, and generate `civil-cost-model.toml` from calibration. The connected package contains quantities, fronts, resource and procurement/payment sensitivities, conditional opening dates, battery/operating/replacement costs, financing gaps, proposed entrance geography, ERP asset/crew drafts and a hashed source/output manifest. Prior adopted cost/finance/coverage packages remain identified comparators while unresolved supplier, structural and access inputs prevent promotion to an accepted budget or programme.

Regression gates: one island/two faces; access equipment matches geometry; shifts change resources not physical quantities; no erection beyond accepted/delivered beams or supports; contracted supplier limits; mutually exclusive access and asset transfers; battery mass/energy/cost propagation; and source/assumption provenance. The implementation sequence is station/civil topology, suppliers/logistics/shifts, costs/reuse, batteries and city regeneration.

RFC Index

Repository source: `docs/rfcs/README.md`

RFCs are the design-decision record for OpenSourceRail. They describe what the project chose, what it deliberately excluded, and which implementation milestones prove the choice in code, CAD, or documents.

By Theme

Theme	RFCs
Core architecture and software	0001, 0004, 0005, 0006, 0017, 0018, 0031, 0032 (superseded), 0033
Energy and train systems	0002, 0008, 0020, 0021, 0022, 0023, 0024, 0026
Civil, track, stations, depots	0009, 0010, 0011, 0012, 0014, 0025
Operations and safety	0013, 0015, 0016, 0027, 0028, 0029, 0030
Control electronics and DIY deployment	0007, 0019

The historical v0.3 release-candidate summary is in `../../CHANGELOG.md`, and remaining validation work is tracked in `../ROADMAP.md`.

The current published software baseline is v0.4.0; subsequent work is recorded in the changelog.

- RFC 0033 — TACS runtime and resource control — proposed development target; existing control components and multi-process reference.
- RFC 0034 — Connected station, production and battery model — proposed shared topology, contracted supply, shift scheduling, scope costs and battery alternatives.

Rolling Stock

Repository source: `docs/rolling-stock/README.md`

This folder contains the handoff documentation for train designs. The current mid-size-city vehicle is the `light-metro-3car`: a cabless, driverless, battery-electric trainset with roof solar, identical multi-part fiberglass end cowls with single panoramic glass faces, LED headlamps, repeated self-contained cars, one powered bogie and one trailer bogie per car, high-floor bogie zones, low-floor centre door/PRM zones, and externally accessed, saloon-isolated side battery enclosures beneath the seat zone for the 300 k to 1 M population networks. Samawah is one generated instance of this shared family, not the reason the family exists.

Packages

Package	Scope
`design-system.md`	Top-down / bottom-up design hierarchy and automatic candidate-iteration loop
`light-metro-3car/`	General arrangement, fiberglass end cowl, body, bogie, traction, interfaces, BOM, fabrication plan, compliance, and drawing register

Related Artifacts

Use the generated buildable package as the practical handoff from this documentation to drawing, RFQ, shop routing, QA, and first-article planning: manifest first, definitions for drawing/RFQ, travelers for operation routing, material/process control, and signoff templates.

Artifact	Location
Generated CAD screenshots	`../screenshots/`
Generated design iteration scorecards	`../../design/component-catalogue/catalog/design-system/`
Generated buildable trainset manifest and review	`../../design/component-catalogue/catalog/buildable-trainset/`
Product-tree definitions for drawing/RFQ/material/process specs	`../../design/component-catalogue/catalog/buildable-trainset/definitions/index.md`
Unsigned shop traveler templates with process controls	`../../design/component-catalogue/catalog/buildable-trainset/travelers/index.md`
Canonical parametric source	`../../design/component-catalogue/src/osr_mech/rolling_stock/`
Generated FreeCAD review artifacts	`../../design/component-catalogue/models/cad/`
Concept image	`../assets/solar-metro-trainset.png`
Control-electronics integration matrix	`../../control-electronics/rolling-stock-integration.md`
Interior COTS catalogue	`../../control-electronics/trainset-interiors.md`

Rolling-stock top-down / bottom-up design system

Repository source: `docs/rolling-stock/design-system.md`

This is the design authority layer above the parametric CAD. It gives the project a systemic way to move from whole-train intent down to bought-in parts, fabricated parts, subassemblies, final assemblies, and then back up through scoring and evidence.

The rule of thumb is simple:

1. Start top-down with requirements: platform fit, axle load, range, traction margin, thermal margin, cost, mass, manufacturability, and safety evidence.
2. Decompose into the owned design hierarchy: external components, fabricated parts, subassemblies, final assemblies.
3. Iterate candidate definitions automatically inside an explicit design space.
4. Promote only feasible candidates into FreeCAD/FEM regeneration.
5. Freeze only when the requirements scorecard, supplier evidence, CAD, FEM, BOM, and manufacturing plan agree.

Hierarchy

Layer	Meaning	Examples	Source of truth
Top-level requirements	What the train must achieve	platform margin, range, axle load, cost, mass, traction/HVAC margin	`osr_mech.design_definition.REQUIREMENTS`
External components	Bought-in or BID/SOURCE modules	traction motors, HVAC, batteries, doors, glazing, sensors, couplers	`ExternalComponent` records plus supplier evidence
Fabricated parts	Parts OSR owns and regional workshops make	shell, bogie frame, motor cradle, cowl rings, articulation adapters	`FabricatedPart` records plus CAD templates
Subassemblies	Integration units that can be inspected	car module, powered bogie, trailer bogie, cowl, articulation, roof services	`Subassembly` records plus FreeCAD review groups
Final assemblies	Released trainset configurations	promoted light-metro 3-car trainset	`FinalAssembly` records plus generated CAD/FEM/BOM

Automated iteration

The first implementation is deterministic and dependency-free:

```
tools/automation/design-iterate.sh
```

It exhausts the declared discrete design space for the default `light-metro-3car` family, evaluates every candidate, and writes:

- `design/component-catalogue/catalog/design-system/design-iteration.json`
- `design/component-catalogue/catalog/design-system/design-iteration-summary.md`

To turn the winning candidate into a buildable product tree and review the current CAD/BOM baseline against it:

```
tools/automation/buildable-trainset.sh
```

That writes the buildable manifest and buildability review under `design/component-catalogue/catalog/buildable-trainset/`.

To run a bounded exploratory pass:

```
tools/automation/design-iterate.sh --family metro-4car --max-iterations 250
```

The current scorecard is intentionally planning-grade. It is good enough to rank candidate architecture choices before spending FreeCAD/FEM time, but it is not a certification result. As the project matures, replace or augment the simple metrics with:

- FreeCAD-derived mass properties and package clearances;
- CalculiX margins from `tools/automation/freecad-generate.sh --fem`;
- supplier quotations and evidence status;
- lifecycle energy and maintenance models;
- risk penalties from the hazard log and compliance matrix.

Promotion loop

Use the generated best candidate as a design review input, then run:

```
tools/automation/freecad-generate.sh --models --assemblies --fem
tools/automation/freecad-generate.sh --screenshots --station-scenes
```

If CAD/FEM/supplier evidence rejects the candidate, update the component catalogue or constraints in `osr_mech.design_definition`, rerun `tools/automation/design-iterate.sh`, and promote the next best feasible candidate. That is the closed loop: requirements drive definitions, definitions drive CAD/FEM, evidence updates definitions.

Executed trainset industrialisation expansion

Repository source: `docs/rolling-stock/industrialisation-execution.md`

Implementation plan · generated review · execution evidence · industry source register · machine-readable programme · OSR-MFG-001 · native battery datum fixture · native fixture receipt.

The six repository work packages in the plan are implemented. They expand the existing shared train model and factory definitions with callable screening, planning, trace-validation and offline commissioning tools. The current six-car reference yields 117 material passports, 117 process routes and 756 assembly operations, linked to 16 existing factory packages, 10 manufacturing controls and 416 first-article characteristics. Physical qualification remains open.

Delivered implementation

Package	Implemented behaviour	Verification and physical boundary
IM-01 design	Eight source-backed improvements bind automation, redundancy, monitoring, traction, HVAC, doors, acoustics and modular materials to existing part IDs	Missing sensor/latency/common-cause evidence remains visible; CCTV is not reclassified as an obstacle safety sensor
IM-02 materials	Per-product passports preserve existing material/process candidates; coupon mass/stiffness, layered conduction and duty-energy tools compare explicit inputs	No grade/strength/fire/fatigue/EPD release is inferred from a lighter material or supplier envelope
IM-03 assembly	Six-restraint fixture, signed tolerance stacks, parent/child preparation and handback dependencies	Native locator positions/volumes and two drawing projections checked; clamp capacity, deflection, calibration and physical repeatability remain required
IM-04 production	Open travelers, CSV exports, offline work instructions and capacity/wait-aware scheduling	Actual operation times and qualified process windows remain null; synthetic scheduling demonstrations are not factory forecasts
IM-05 quality	Exact identity/revision/lot checks, calibrated range/expiry, environment, uncertainty decisions, torque/angle traces, process-window sampling and NCR blocking	Outputs only prepare independent review; they never approve physical work or change ERP status
IM-06 commissioning	Offline lifecycle and critical-fault contract, bounded auxiliary reset and movement inhibition	Computational fault cases pass; real TCMS/ATP/ATO, redundancy timing, sensor coverage and GoA4 evidence remain open

The reference fixture now places all six locators on pads, supported brackets, outriggers and base members. Its kinematic rank is six. These solids are an inspectable datum study rather than released load-bearing shop drawings.

Materials examples are explicitly synthetic. For the same coupon geometry, lower-density materials reduce mass while lower modulus increases deflection. Changing thickness updates both mass and bending stiffness; an assumed strength or carbon factor is not silently supplied. The duty comparison conserves energy and exposes effects of declared mass, traction/regeneration efficiencies, resistance and auxiliaries. It does not verify CRRC's energy-saving claim.

Run and inspect

Use a fresh output folder for each controlled generation:

```
tools/automation/osr-python tools/automation/factory-modernisation.py generate \
  --output build/engineering/my-factory-programme

tools/automation/osr-python tools/automation/factory-modernisation.py verify \
  --output build/engineering/my-factory-programme

tools/automation/osr-python tools/automation/factory-fixture-freecad.py \
  --fixture build/engineering/my-factory-programme/assembly_package.json \
  --output build/engineering/my-battery-fixture.FCStd
```

Supply `--model` and `--battery` together to bind another compiled train variant. The package retains those inputs, source hashes, JSON requirements/passports/ operations/protocols, the blank traveler, synthetic method demonstrations, `assembly-operations.csv`, `material-passports.csv`, `work-instructions.md` and an immutable receipt. Verification checks outputs and recompiles generated data.

For an actual work order, retain the generated blank template and create a separate controlled execution package/record. Bind it to the identified as-built configuration, issued family-specific drawings, complete part allocations, released acceptance criteria and qualified process windows. Fill actual physical serials/lots, operators, independent inspectors, timestamps, calibrated evidence and NCR dispositions. Reconcile its revision/hash whenever the approved package changes. Reference aggregate rows cannot substitute for resolved physical parts.

```
tools/automation/osr-python tools/automation/factory-modernisation.py assess \
  --output build/engineering/my-factory-programme --record path/to/actual-record.json
```

The default reference package remains blocked: it is as-designed, has no physical asset identity and lacks complete production allocations, released criteria and special-process data. Assessment of that template cannot become an approval.

Design and factory handoff still required

The detailed LM3 manufacturing packages must undergo family-transfer review for six-car geometry, loads, datums and supplier interfaces. The reference LTO battery is also an alternative to the catalogue's LFP candidate, requiring an approved substitution/design release. No selected adhesive, cure recipe, weld authority, torque setting or fire-rated material is issued by this expansion.

Execute physical pilots in the plan's order: battery mounting, representative bogie, carbody/door/HVAC bay, then full-train commissioning. Collect first-pass yield, scrap, rework causes, actual cycles, repeatability and repair/removal time before claiming factory improvements. The existing full-train contact-domain rejection at 9.13 s remains a service gate. Supplier, process, physical and independent railway/safety evidence cannot be closed by software demonstrations.

Retained verification

42 focused tests pass, including 30 new tests for this expansion and the existing fabrication-twin and production-schedule checks. They cover identity, stale/tampered evidence, expired tools/materials, uncertainty margins, process sampling, NCR/predecessor holds, fixture rank, resource limits, energy balance, critical fault lockout and withdrawn movement qualification.

The generated package was verified by source/output hashes and recompilation. Native FreeCAD 1.1.4 checked all 15 fixture solids and their positions/volumes, and generated two nonempty projections with 49 plan and 36 elevation edges. The independent six-restraint check passes. All 13 offline commissioning cases pass their software contracts; physical commissioning and fixture qualification remain unperformed.

OpenSourceRail trainset improvement and industrialisation plan

Repository source: `docs/rolling-stock/industrialisation-expansion-plan.md`

Revision 1 · 2026-10-11 · implementation and verification programme.

The programme expands design, materials, assembly, production and quality using the existing product tree, shared train definition and factory release packages. It will deliver executable planning, inspection and commissioning tools with retained evidence. Physical process qualification, supplier release and railway approval remain separate deliverables requiring actual facilities and measurements.

Evidence and interpretation

The requested Railway-News article was published on 12 May 2026. CRRC's primary account describes a Line 12 train completed on 17 April 2026. It identifies automated lifecycle functions, redundant critical subsystems, obstacle/running-gear monitoring, permanent-magnet traction and passenger comfort improvements. Its reported energy-saving objective is a manufacturer/project claim; OSR will require a defined duty cycle and measured baseline rather than adopt that number. The AH specification and vendor implementation are not an open design recipe.

Source	Practice to adapt	OSR interpretation
Alstom smart manufacturing	Skilled process teams, repeatable special processes, monitored equipment and lean flow	Qualified manual/semi-automatic work first; add robots only after repeatability, access and throughput justify them
Alstom digital maintenance	Digital instructions and connected torque records	Tool-neutral per-joint records and offline instructions, with revisions, calibrated tools and independent review
Siemens railcar coating schedule	Coordinate coating resources, drying and QA timing	Resource-constrained assembly planning; material-specific wait times must be explicit
Siemens measurement quality	Associate dimensional inspection with measurement-room conditions	Retain temperature, calibration and uncertainty with measurements
Alstom environmental design	Consider energy, recyclability and lifecycle effects	Compare mass, stiffness, repair, fire/corrosion evidence and lifecycle data together
NIST capability methods	Capability requires suitable data and process assumptions	Calculate diagnostic statistics; withhold capability approval without stability, measurement-system and distribution evidence
BIPM JCGM 106	Measurement uncertainty affects conformity decisions	Use an explicit project-approved decision rule; offer a conservative interval guard as an internal method
ISO 22163, ISO 3834-2	Railway quality and welding quality requirements	Retain applicability, controlled editions, competence and qualification evidence; no certification inferred from generated records

OpenSourceRail constraints

Preserve the repository's path-based licensing: Apache-2.0 software, CERN-OHL-S hardware and CC-BY-SA documentation. Manufacturer sources remain references under their own terms. Do not redistribute proprietary CAD, process recipes or AH train specifications as OSR designs.

Keep interfaces, part identity, inspection definitions and exports open in JSON, CSV, Markdown and native CAD. Each supplier substitute must pass the same functional interface and evidence requirements. Support local fixtures, common fasteners, repairable modules, documented removal paths and offline work instructions. Do not require a manufacturer's cloud or proprietary torque tool.

The detailed factory packages describe the LM3 reference. Transferring their methods to another family requires its own length/datums, loading, drawing and supplier compatibility checks; the six-car planning generator is not that release. The existing 9.13 s contact-domain rejection remains an explicit operating gate.

Execution packages

ID	Work package	Repository implementation	Completion evidence
IM-01	Source-backed design expansion	Feature/requirement register binds automation, redundancy, running-gear health, traction, HVAC, doors and acoustic interfaces to catalogue IDs	Every feature has source, limits, part/interface ownership, verification and unresolved evidence
IM-02	Material and energy trade-offs	Material qualification passports; geometric mass/stiffness/thermal screening and energy balance tools	Independent formula tests; no qualified selection without fire, fatigue, corrosion, repair and supplier evidence
IM-03	Assembly quality by design	Datum fixture constraint checks, tolerance stacks, per-instance assembly operations and hold points	Six independent fixture restraints; worst-case fit checks; cycles/unknown predecessors rejected
IM-04	Production execution	Open travelers and resource scheduling with explicit cure/dry waits, competencies and process requirements	Capacity/predecessor tests; immutable work-order/configuration/evidence bindings
IM-05	Quality at source	Calibration/uncertainty-aware measurement decisions, material genealogy, torque/process traces, NCR blocking and diagnostic capability statistics	Negative tests for stale revisions, expired tools, missing evidence, reused identities and unresolved defects
IM-06	Automated commissioning contracts	Offline lifecycle/interlock/fault-injection model and commissioning protocols	Door/obstacle/brake/depot/washing/reset invariants tested; no live control or GoA4 qualification claimed

The implementation will reuse the existing factory packages, manufacturing controls, first-article inspection plan, product material/process specifications and shared engineering instance IDs. Generated plans and blank travelers cannot serve as execution evidence. Factory acceptance requires actual controlled documents, physical records, independent quality review and the designated authority.

Pilot sequence and physical work

1. Battery installation and equipment mounts: kit by joint/lot, verify mating datums, validate tolerance/thermal clearances, record fastener stacks and tool traces, inspect routing/isolation and perform approved load/leak/electrical tests.
2. Representative bogie: material receipt, qualified weld/fixture sequence, distortion and NDT, axlebox fit and bearing trace, suspension settings, brake and sensor installation, then an independently witnessed rolling/loaded test.
3. Carbody/door/HVAC bay: survey portals/curbs, qualify composite/bond/seal processes, verify doors and drains, assess heat/noise/fire evidence, and demonstrate sequential removal and interchangeable replacement.
4. Full train: close genealogy and configuration, execute integration/fault protocols, then complete supported wheel–rail/service, braking, thermal, comfort and safety qualification with physical holdouts.

Record first-pass yield, rework by defect cause, scrap/yield, calibration failures, trace completeness, measured cycle time, WIP and replacement time. Establish baselines before claiming savings. Automation investment follows repeatability and measured production demand; avoid transferring another factory's robot count, cycle times or quality claims to OSR.

Acceptance and execution reporting

The repository implementation is complete when IM-01 through IM-06 are callable, their generated products reconcile with existing identities, their tests and demonstrations are retained, and the review states the remaining physical work. Executable protocol/software completion is distinct from train or factory qualification. Report each package as implemented, verified in software, awaiting supplier/process data, or awaiting physical validation.

light-metro-3car trainset package

Repository source: `docs/rolling-stock/light-metro-3car/README.md`

This directory holds the dimensioned specification for the light-metro-3car trainset — the default family for populations 300 k – 1 M per RFC 0008 §5. Samawah is one generated city instance that selects this family; the trainset package itself is shared.

This package started as the v1 RFC 0008 bid specification and now also links the current envelope-level FreeCAD/PNG outputs and generated buildable trainset handoff. A domestic rolling-stock fabricator can read the dimensions, masses, interfaces, sub-assembly tree, fabrication sequence, procurement BOM skeleton, and design-review CAD. The design deliberately favours a modern low-capex factory: COTS rail subsystems wherever possible, simple cut/bend/weld fabrication for the primary frame, and composite non-structural panels for sides, roof fairings, multi-part fiberglass cabless cowls, and interior liners. The side and roof skin now uses a common 1 m mould/clip pitch, allowing the three-car exterior body to be installed in one eight-hour shift after the frames pass paint and dimensional release.

Contents

File	Scope
`general-arrangement.md`	Overall envelope, gauge clearance, consist diagram, floor heights, door positions
`fabrication-plan.md`	Cut-bend-weld primary structure, composite cladding, COTS module installation sequence
`bogie.md`	2-axle standard bogie spec, wheel profile, suspension, brake mount
`body.md`	Welded steel underframe/spaceframe, composite side panels, end bulkheads, articulation interface frames
`dedicated-parts-and-moulds.md`	Discrete chassis/body, fascia/glass/lamp and interior fitout parts with dedicated moulds, trim nests, gauges and release evidence
`modular-fiberglass-body.md`	1 m clip-on fiberglass side/roof modules, keyed retainers, dry seals, one-shift trainset body route, and cost basis
`roof-fitout.md`	Roof skin/fairing mould family, HVAC curb and skirts, PV/gland/antenna closeouts, walkways, keep-outs and assembly tests
`exterior-finish-process.md`	Simplified white base plus pre-cut livery film and trial-only calcium-carbonate radiative roof coating qualification route
`end-cowl.md`	Identical A/B-end fiberglass cowl kit, cast split, laminate/tooling rules, glazing/sensor interfaces
`cabin-fiberglass.md`	Cabin FRP/phenolic ceiling liners, sidewall/window reveals, battery strake covers, vestibule/PRM trims, tooling, lay-up, trim, inspection, and repair instructions
`articulation.md`	Inter-car and train-to-train articulation/gangway modules: lower spherical pivot, upper links, bellows, turntable, trainline routing, and configurable open-end option
`cots-integration.md`	COTS/fabricated interface diagrams, part delineation, and assembly sequence
`traction.md`	800 V-class LFP packs, six PMSM/controller sets, direct-DC auxiliaries, station charging, and verification gates
`localization-plan.md`	Staged make/buy policy for bogies, doors, articulation, windows, seats, battery, traction, and body
`interfaces.md`	Coupler, station charging, articulation, platform gap, TCN-E connector, aux power
`bom-skeleton.md`	Procurement BOM lines and marketplace listed-price consist totals
`marketplace-price-anchors.md`	Alibaba/AliExpress line-by-line price anchors and qualification caveats
`compliance.md`	Standards matrix: EN 15227, EN 45545, EN 14363, EN 50155, ISO 3095, EN 12299
`drawing-register.md`	v2 drawing IDs, supplier documents, inspection evidence, release gates
`v2-release-checklist.md`	Fabrication-release gates: supplier envelopes, FEA, weld maps, drawings, NC data, harness routing, first-article evidence

The governing visual/layout reference is `solar-metro-trainset.png`: white/silver body, green waist band, dark skirts, single dark panoramic-glass ends, mixed bonded/rail-mounted roof PV, two low-floor door pairs per side per car, repeated self-contained cars with one powered bogie and one trailer bogie each, and batteries under longitudinal seats. The end glazing is an open driverless passenger view through heated RF-transparent laminated glass with LED headlamp and marker-light clusters below it. Because the cars use standard bogies, each car has ~3 m high-floor end decks over the bogies and a 350 mm, ~10 m low-floor centre door/PRM zone. The front and rear passenger ends are not walled off by cab bulkheads; the end saloon looks through the single panoramic glass pane at both ends of the driverless train. The front/back exterior module is the same multi-part fiberglass cast kit at both ends; its controlled design is in `end-cowl.md`.

The companion production graphic is `solar-metro-production-assembly.png`. It shows the intended manufacturing story: a welded 16.5 m datum frame, bolt/bond COTS side, roof, door, glazing, battery-seat, and HVAC modules, then lower the car onto standard bogies and repeat the module to make the consist.

COTS passenger-facing modules are controlled by the supplier-neutral envelope catalogue at `control-electronics/trainset-interiors.md`. Locally moulded or CNC-trimmed cabin fiberglass/phenolic liners, battery strake covers, and vestibule trims are controlled in `cabin-fiberglass.md`. The current vendor-fit shortlist is captured in `bom-skeleton.md` and names rail product families for doors, windows, HVAC, flooring, seats, lighting, PIS/audio, batteries, traction motors, brakes, couplers, and front/back sensing while keeping the CAD envelope swap-friendly.

Canonical Parametric Source

The parametric Python source files are the design basis for envelopes, interfaces, and generated review views. The styled fiberglass end-cowl A-surfaces are controlled by the surface-modelled LM3-BDY-155 release CAD described in `end-cowl.md`; those surfaces are exported back into the drawing pack as neutral CAD.

Source	Controls
`trainset.py`	Family length, car count, repeated car motorisation, cowl/body/bogie assembly
`car_body.py`	16.5 m body module as layered CAD subassemblies: primary structure, exterior/glazing/doors, interior, HVAC ducts, LV/data routing, HV/PV/thermal/fire paths
`modular_fiberglass_body.py`	Sixteen 1 m cladding bays per car, side/roof module geometry, clips, anti-lift retainers, dry gaskets, and the eight-hour body schedule
`sensor_cowl.py`	Identical A/B-end multi-part fiberglass cowl kit envelope with one dark panoramic glass pane, LED headlamps, marker lights, and T-OBS visual interface
`systems.py`	Couplers, detailed articulations/gangways, batteries, rooftop solar/MPPT package, DC charge/protection racks, doors, electronics, charging, T-OBS packs
`bogie/`	Powered and trailer bogie assemblies
`cad_templates/rolling_stock.py`	Sheet-metal/chassis manufacturing templates
`buildable_trainset.py`	Product-tree manifest, definitions, assembly integration design, and unsigned shop travelers
`exterior_finish.py`	Machine-readable finish zones, prohibited coverage, livery-film process, CaCO3 qualification gates and review geometry

Buildable handoff package

The generated buildable package lives under `design/component-catalogue/catalog/buildable-trainset/`. Use it as the bridge from concept CAD to build planning:

This is specifically the `light-metro-3car` detailed reference. The integrated city twin uses these parts, moulds and travellers only when that family is selected. Other-family cities retain their generated family cost and programme but show an explicit family-definition hold until equivalent detailed packages are engineered and released.

Artifact	Use it for
`buildable-trainset-manifest.md`	Parts subassemblies assemblies trainset tree, quantities, parentage, acceptance gates
`train-end-interface.md`	Single configurable train-end interface that can select either the panoramic glass front/end or the optional mid open train-to-train connection
`critical-path.md`	Rough first-train critical path, parallel fabrication plan, labour estimate, and minimum space model
`factory-plan.md`	Pilot factory sizing: chassis fabrication, bogie integration, moulding shop, final assembly, enclosed area, yard, machinery, and rough equipment prices
`factory-drawings/index.md`	Twenty-nine individual drafting/checking briefs binding the sixteen packages to products, reference envelopes, sources, tooling, views, controls and verification without claiming manufacturing issue
`factory-release-readiness.md`	Open issue/approval state across all 16 drawing, product-configuration, tooling and verification packages
`trainset-build-cost.md`	Recalculated 3-car trainset budget, including the explicit seats/floors/lighting/HVAC/windows/doors scope already inside the direct-module bucket
`mass-budget.md`	Reconciled 75.308 t modeled subtotal, 3.442 t engineering reserve, and 78.75 t controlled planning tare
`mass-closure-ledger.md`	All 120 product rows mapped to the nine mass categories, evidence routes for 117 active rows, the 73.376 t lightest existing design-space study, and recovery-reaction linkage
`joint-control-schedule.md`	Machine-readable joining classes, torque authority, and release state for all 108 integration joints
`manufacturing-and-assembly-controls.md`	Ten default work instructions covering datum control, steel, bolting, bonding/glazing, GFRP/fascia moulding, roof, interiors, lifting/recovery, finish and NCR handback
`first-article-inspection-plan.md`	416 package-specific hold, witness and record-review characteristics with every execution result left unperformed
`definitions/index.md`	Drawing/RFQ definitions for every fabricated part, external component, subassembly, assembly, and trainset node, including structured material and process specs
`travelers/index.md`	Unsigned shop travelers with material/process controls, operation routers, labor estimates, tooling IDs, QA gates, revision approvals, signoff blocks, and NCR/deviation logs
`supplier-anchors.md`	Real manufacturer-family anchor, procurement state, known fit gaps and controlled local-equivalent route for every bought-in product row
`manufacturing-methods.md`	All 120 product links, timed mould/film/coating/seal/floor/fixture/motor methods, joining parts, hold points and external design references
`FreeCAD part/assembly library`	120 individual native part documents plus 26 subassembly/car/final-assembly documents, hashes and reachability checks
`IFC4.3 part/assembly library`	Matching split geometric IFC files with retained product classes and hierarchy
`lm3-manufacturing-tooling.FCStd`	Selectable FreeCAD review objects for 30 mould, trim-nest, glass/lamp gauge, finish, stand and test-tool families

Artifact	Use it for
lm3-manufacturing-reference.ifc	IFC4.3 product hierarchy, typed doors/windows/floor/furniture/motor, 59 tasks, method assignments and multi-part tooling geometry
current-design-buildability-review.md	Current green/yellow/red closure status before first steel cut

Typical first-article workflow:

1. Start at the buildability review and resolve release blockers.
2. Use the manifest to choose the next build cell or RFQ package.
3. Open that node's definition to create controlled drawings, supplier envelopes, material specs, process specs, or fixture requirements.
4. Open the matching traveler to plan operation sequence, labor, tooling, QA gates, and signoff responsibility.
5. During a real build, fill approval/signature fields and attach actual inspection evidence; generated traveler templates are intentionally unsigned.

Current CAD / PNG Design Outputs

The whole-train part map and COTS interface diagrams are in cots-integration.md.

The present train design now has envelope-level definitions for the major train assemblies, sub-assemblies, repeated components, and body service layers. The car_body() assembly is deliberately nested as components -> subassemblies -> final car body, so CAD review can hide/show the primary steel shell, exterior solar-train skin, passenger interior, HVAC ducting, LV/data harnesses, high-voltage traction/PV routing, thermal-management pipes, and battery fire vent paths. The models are supplier-neutral CAD geometry in design/component-catalogue/src/osr_mech/rolling_stock and render to these design-review and manufacturing-method PNGs:

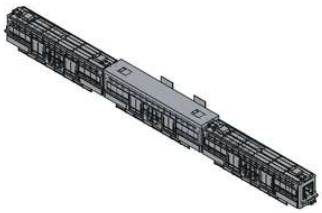
Output	Scope
trainset-light-metro-3car.png	Final 3-car trainset assembly with car bodies, bogies, single panoramic-glass cowls, couplers, inter-car articulation, and train systems
end-glass-cowl.png	Cabless trainset end close-up with multi-part fiberglass cowl casts, one heated laminated panoramic glass pane, bonded edge frame, demist busbars, and washer/service hardware
trainset-car-detail.png	16.5 m layered car body: structure, door/window openings, glazing, livery, roof PV/HVAC, interior, ducts, LV/data and HV/thermal routes
trainset-car-body-structure.png	Primary fabricated structure: shell, 10 m low-floor pan, side sills, crossmembers, roof cantrails, door portals, window posts, end rings, and bogie clearance envelopes
trainset-car-body-bogie-subassembly.png	Single-car structure with standard motor/trailer bogies under the ~3 m raised high-floor end zones
trainset-car-body-exterior.png	Solar-train exterior layer: glazing, door leaves, livery band, mixed bonded/rail-mounted roof PV/HVAC, and service skirts
trainset-roof-solar-system.png	Per-car rooftop solar package with bonded flexible laminates, raised rigid panels, rails, clamps, junction boxes, fire isolators, MPPT combiner, and downlink gland
trainset-car-body-interior.png	Passenger interior layer: under-seat battery strakes, benches, PRM bays, grab poles, handrails, and PIS
trainset-car-body-services.png	Service layers: HVAC ducts, LV/data trays, lighting, CCTV/intercoms, HV/PV routing, coolant, and battery fire vents
trainset-interior-fit-out.png	COTS passenger fit-out envelopes inside the car body
trainset-body-sheet-metal-kit.png	Sheet-metal/chassis manufacturing kit: underframe, bolsters, coupler pockets, side posts, rails, roof bows, end rings
trainset-factory-layout.png	Pilot factory layout with one 55 m final bay, chassis/body-frame fixtures, GFRP moulding/trim, bogie assembly, interior/HVAC kits, corrosion/base-finish cells, stores, QA, yard staging, and short test access
trainset-assembly-method-flow.png	Parallel first-article method plan tying critical-path tasks to chassis/body frame, GFRP modules, bogies, interior kits, doors/windows/roof, HV/electrical work, articulation/static testing, and dynamic release
trainset-gfrp-moulding-method.png	One-metre glass-fibre module method: mould/cure, demould/trim, insert/seal, master-frame dry fit, and clip-on installation to the painted carbody
trainset-bogie-marriage-method.png	Bogie-to-carbody marriage method with mobile lifting columns, accepted motor/trailer bogies, centre-pivot/air-spring datum checks, and static release hold points
trainset-car-systems.png	One self-contained car systems package: four door cassettes, platform interlocks, batteries, rooftop PV package, traction/charge power rack, charging connector, and accessibility/safety reservations
trainset-battery-pack.png	Eight LFP module envelopes plus HV contactor, fuse, BMS cabinet, outward vent, off-gas detection, and localized mist interfaces per car
trainset-door-system.png	Door cassette pair with sill gap fillers, locks, and external emergency releases
trainset-electronics-cabinet.png	Per-end T-ECU/S, T-ECU/A, and crashworthy event recorder, two sets per trainset
trainset-end-coupler.png	Scharfenberg Type 10 coupler, electric-head carrier, and EN 15227 crash absorber envelope
trainset-inter-car-articulation.png	Detailed inter-car articulation: lower spherical joint, anti-lift keeper, upper links, double-wall bellows, turntable floor, trainline routing, and kinematic envelopes; the same architecture now has a generated train-to-train open-end option in the buildable handoff

Output	Scope
`trainset-tobs-sensor-pack.png`	T-OBS LIDAR, mmWave radar, stereo camera, and ultrasonic sensor envelopes
`bogie-motor.png`	Powered bogie assembly with frame, wheelsets, motors, gearboxes, suspension, and brakes
`bogie-trailer.png`	Trailer bogie assembly using the common frame and suspension envelope

FreeCAD Assembly And FEA Screenshot Review

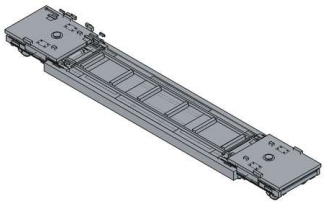
These screenshots are generated from the FreeCAD `.FCStd` review documents and the FreeCAD/CalculiX screening-output document. The assembled and exploded states are view groups in the FreeCAD files, not separate hand-positioned drawings.

Trainset FreeCAD review



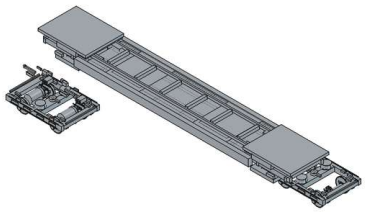
FreeCAD trainset light metro 3-car

Chassis + bogie assembled



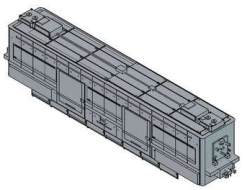
Chassis bogie assembled

Chassis + bogie exploded



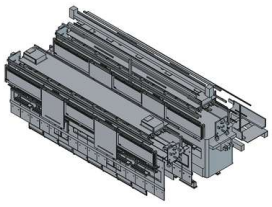
Chassis bogie exploded

Full body assembled



Full body assembled

Full body exploded



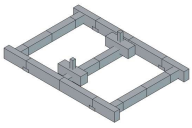
Full body exploded

FEA screening models

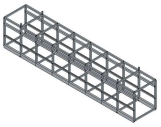
FEA screening models

Chassis FEA

Bogie FEA

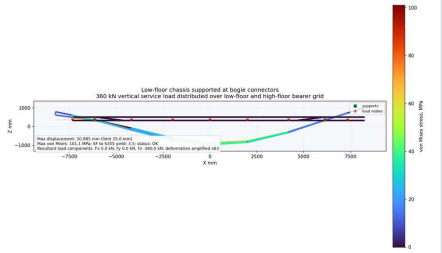
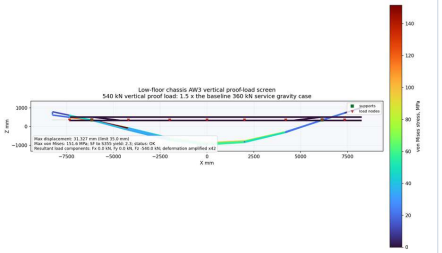
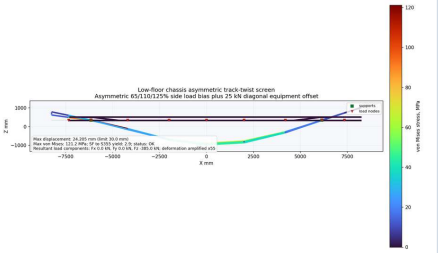


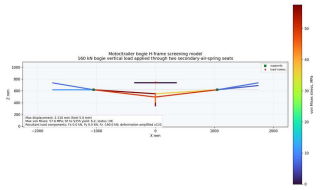
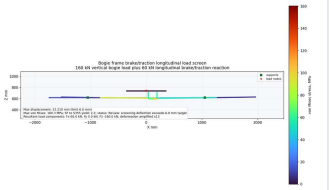
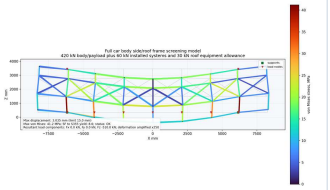
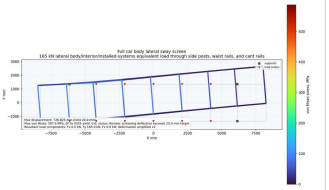
Body FEA

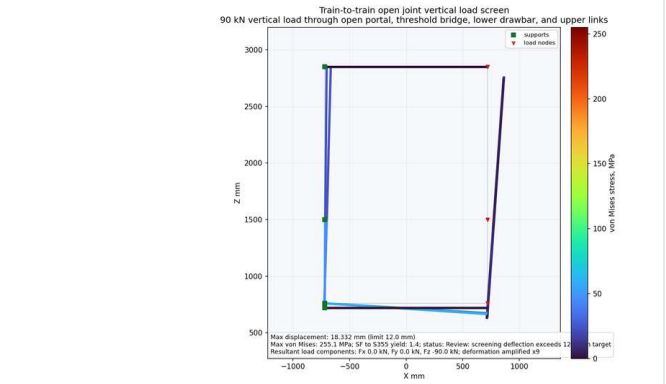
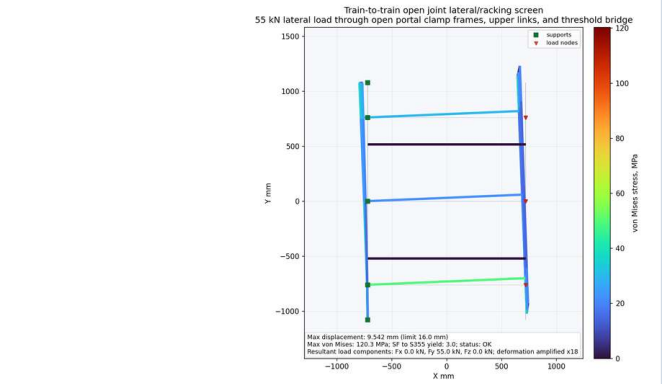


FEA screening models	Chassis FEA	Bogie FEA	Body FEA
	Chassis FEA	Bogie FEA	Body FEA

The actual solver-result PNGs below are generated from CalculiX `.dat` fields: deformed beam shape, support/load markers, and von Mises stress colour scale.

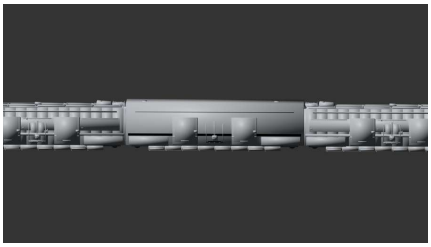
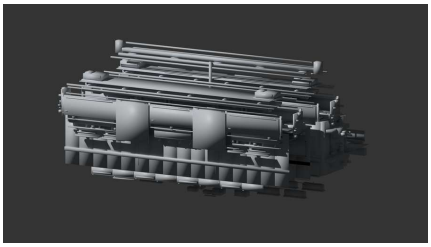
Chassis service gravity	Chassis AW3 proof	Chassis track twist
		
Chassis service FEA result	Chassis AW3 proof FEA result	Chassis track twist FEA result

Bogie vertical	Bogie brake/traction	Full body vertical	Full body lateral sway
			
Bogie vertical FEA result	Bogie brake traction FEA result	Full body vertical FEA result	Full body lateral sway FEA result

Train-to-train joint vertical	Train-to-train joint lateral sway
	
Train-to-train joint vertical FEA result	Train-to-train joint lateral sway FEA result

The latest screening summary is `design/component-catalogue/catalog/fea/screening-summary.md`. The current FEA catalog includes solver-backed train-to-train local joint cases: vertical passenger/gangway loading and lateral/racking. The source FreeCAD review documents are catalogued in `design/component-catalogue/models/cad/README.md`, and raw CalculiX output folders are catalogued in `design/component-catalogue/catalog/fea/README.md`.

The same `.FCStd` documents can also be exported to local STL render meshes and rendered with Blender/Cycles for README-grade engineering-clay views:

Trainset clay render	Full-body clay render	Chassis/bogie clay render
		
Blender Cycles trainset render	Blender Cycles full body render	Blender Cycles chassis bogie render

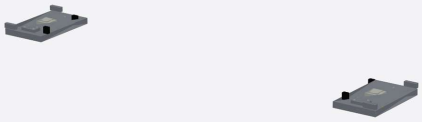
After the low-floor chassis rework, the chassis screen is inside the 25 mm deflection target: 9.8 mm maximum displacement under the 360 kN service-load screen. The broadened lateral body sway screen is now inside the 20 mm screening target at 18.1 mm maximum displacement, with stress still low at 40.2 MPa. The local train-to-train joint screens are also inside target after the deeper end-ring, portal, threshold, drawbar, and upper-link section update: 4.4 mm vertical displacement against a 12

mm target and 2.3 mm lateral displacement against a 16 mm target.

Mechanical Interface Component Gallery

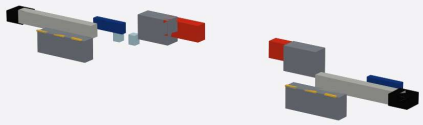
These component images are generated from `mechanical_interfaces.py` and match the tracked FreeCAD assembly-review documents under `design/component-catalogue/models/cad/`.

Bogie to chassis



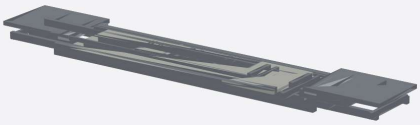
Bogie to chassis connector

Bogie to motor



Bogie to motor connector

Low-floor chassis

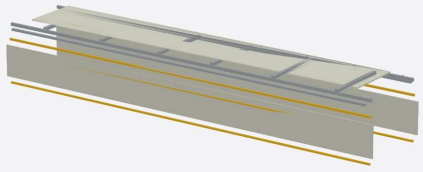


Low-floor chassis

Side body frame

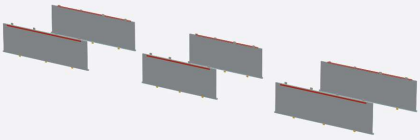
Side body frame attachments

Composite body and roof



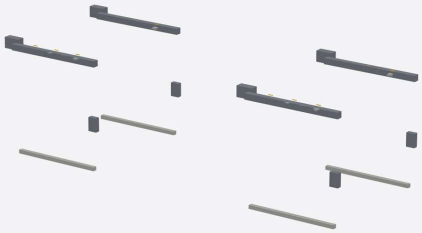
Composite body roof attachments

Window installations



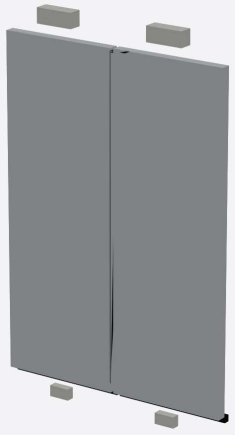
Window installations

Door mounts



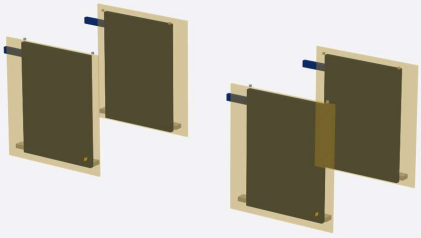
Door mounts

Door design



Door design

Door installations

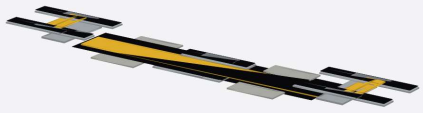


Door installations

Door to body

Door to body installations

Cabin flooring



Cabin flooring

Battery installations

Battery installations

Benches over batteries

Bench on battery installations

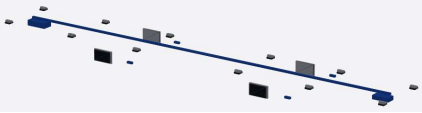
Internal lighting

Internal lighting installation

HVAC roof and ducting

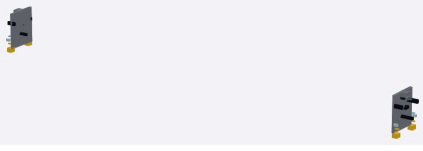
HVAC roof ducting installation

Screens and speakers



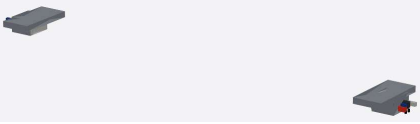
Screen speaker mountings

External lighting and LIDAR



External lighting and LIDAR

Train connector mounts



Train connector mounts

Complete mechanical interface package

Complete mechanical interface package

The tracked generated CAD review artifacts are: `trainset-light-metro-3car.FCStd`, `chassis-bogie-assembly-states.FCStd`, `full-body-assembly-states.FCStd`, and `fea-screening-models.FCStd`. The matching electronics host-class quantities are mirrored in `control-electronics/rolling-stock-integration.md`.

The cross-domain depot bogie-change interface now ties the installed LM3 jack pads to the main-heavy depot lift heads through one tested datum contract. It remains a design-reference coordination assembly until vehicle reactions, underframe calculations, foundation design, selected lifting equipment, proof tests, and local safety approval are released.

The portable field-rerailing concept now screens the controlled 78.75 t tare and lighter sensitivity cases against a rail-rated 200 kN hydraulic cylinder envelope. It also pairs the J1--J4 vehicle interface with an optional wayside access interface for hardstanding, equipment offload, transverse bridge bearing and controlled access. Automotive scissor jacks and unilateral lifts are explicitly outside the design basis.

The generated product-level mass ledger keeps the lighter body/bogie option honest: its modeled subtotal is 73.376 t, or 76.818 t with the current reserve retained, but it remains an unpromoted study until all active row masses, individual-car weights, axle loads, centres of gravity and affected structural/dynamic evidence are accepted.

The remaining gaps are not missing assemblies in the train envelope; they are controlled detail-design and release tasks: frozen supplier exact envelopes, one-car-first manufacturing drawings, weld maps, NDT acceptance sheets, FEA-ready brackets, harness clamp locations, manufacturing tolerances, weight-and-balance evidence, and release drawings listed in `drawing-register.md` and `v2-release-checklist.md`.

For supplier work, the generated `default-product-specifications.md` now gives all 58 `BID/SOURCE` rows a costable reference configuration, published manufacturer facts, the current CAD fit envelope, localisation basis, and explicit override triggers. These values start comparable RFQs; they are not nominated suppliers or released order codes.

Reference envelope (from RFC 0008 §1)

Parameter	Value
Cars	3 (articulated)
Overall length (over couplers)	49.5 m
Tare mass target	78.75 t controlled planning tare: 75.308 t modeled subtotal + 3.442 t engineering reserve; drawing-package mass closure still required
Axle load (AW3 crush)	≤ 14 t
Max speed	25 m/s (90 km/h)
Seats	60 longitudinal seats
Passenger capacity (AW2)	360 (seated + standing)
Passenger capacity (AW3 crush)	480 short-duration crush load
Onboard battery	540 kWh usable / 675 kWh gross LFP (180 / 225 kWh per car), in externally accessed, saloon-isolated side enclosures beneath the seat zone with outward venting
Peak onboard motor output	1.8 MW operational cap (2.1 MW installed short-duration capability)
Floor height (above ToR)	350 mm, ~10 m low-floor centre door/PRM zone; 760 mm, ~3 m high-floor end decks over standard bogies
Gauge	1 435 mm standard gauge

What v1 does NOT include

- Production-detail MCAD files, selected supplier internals, tolerance stacks, manufacturing drawings, and custom-board KiCad files only where a deployment chooses custom electronics (v2).
- Detailed finite-element analysis (v3 — homologation phase).
- Operator-specific livery artwork and approved colour masters; the shared base-protection, pre-cut film and trial-only cool-roof process is defined.

How to execute this package

1. A fabricator reads `general-arrangement.md`
   - `interfaces.md` to size their production line tooling.
2. `fabrication-plan.md` defines the shop route: tube/plate cutting, press-brake bends, welding fixtures, composite bonding, and COTS module installation.
3. Use the generated buildable manifest to see the current product-tree quantities and parent/child assembly structure.
4. `assembly-plan.md` defines how those parts and subassemblies join: weld, bolt, bond, gasket, harness, coolant, articulation, bogie marriage, end-cowl, and final trainset sequence.
5. Use the generated definition pack to start controlled drawings, RFQs, material specs, process specs, and fixture requirements.
6. Use the generated shop traveler pack to plan operation routers, labor, tooling IDs, QA gates, signoff responsibilities, and NCR/deviation records.
7. `bom-skeleton.md` gives the procurement team the source-identified parts (off-the-shelf commodity) and the TBD parts (where the fabricator bids on make-or-buy).
8. `drawing-register.md` turns the current CAD package and v2 detail work into controlled drawing IDs, supplier document requirements, and release gates.
9. `compliance.md` lists the test campaigns the type-approval needs; each is a separately-tendered scope with an accredited test house.
10. The current FreeCAD/PNG package supports design review. The v2 production CAD pack adds cut-lists, NC code, flat patterns, tolerance-controlled drawings, and welding-robot path artifacts for the shop floor.

Licensing

v1 specification: CC-BY-SA 4.0. v2 CAD + drawings: CERN-OHL-S v2, matching the hardware licensing from ARCHITECTURE.md §9.

Inter-car articulation - light-metro-3car

Repository source: `docs/rolling-stock/light-metro-3car/articulation.md`

The `light-metro-3car` uses two semi-permanent articulated gangway modules, one between Car A/B and one between Car B/C. This is not a Jacobs-bogie train: every 16.5 m car still has two standard two-axle bogies. The articulation module carries inter-car structural loads, keeps the passenger passage sealed, and routes trainlines while allowing the bodies to yaw, pitch, and roll independently on curves and vertical profiles.

Reference architecture

The detailed CAD envelope is generated by `inter_car_articulation()`.

Subsystem	Design intent
Lower central joint	Maintenance-light radial spherical bearing at 720 mm above top-of-rail, with anti-lift keeper and vertical anti-climb shear keys
Semi-permanent drawbar	Forged or welded drawbar between underframe anchor castings; carries traction/brake buff and draft loads within the consist
Upper articulation	Twin upper roll-yaw-pitch links at the gangway roof line to stabilise body relative motion without carrying floor load
Passenger gangway	Double-wall corrugated bellows, bolted clamp frames, flexible side walls and ceiling, segmented anti-slip turntable floor
Service routing	Separated HV jumper, redundant TCN-E Ethernet jumpers, CAN-FD/safety-loop jumper, coolant hose loops, HVAC air sleeve, drain channels
Maintenance access	Removable turntable service hatch, quick-access bellows clamp bolts, drain traps, replaceable wear plates and bearing inspection covers

Reference product families used for sizing are HÜBNER-class tram/metro articulation and gangway systems and Schaeffler-class compact central articulated pivots. HÜBNER's public material describes modular lower and upper articulation systems, bellows, turntables, energy guidance, and air-duct integration for urban rail vehicles; Schaeffler's central pivot reference describes a lower central joint using a radial spherical plain bearing with anti-lift function for turn, tilt, and pitch movement under operating loads.

Geometry envelope

Parameter	Value	Notes
Modules per 3-car consist	2	Car A/B and Car B/C
Module length envelope	1 120 mm	Straddles the carbody interface; does not add to the 49.5 m consist length
Passenger passage width	1 650 mm	Clear gangway walk-through target before trim tolerances
Gangway floor height	760 mm above ToR	Matches the raised bogie-end deck height at each car end
Lower joint height	720 mm above ToR	Aligned with the existing coupler/drawgear datum
Bellows outside width	2 300-2 400 mm	Inside the 2 850 mm body envelope with clamp-frame allowance
Bellows outside height	2 620-2 690 mm	Clear above 2 100 mm passenger headroom
Floor turntable panels	5 segments	Removable centre service hatch over lower joint

Kinematic envelope

The CAD carries explicit framed clearance envelopes so reviewers can see the motion allowance in PNG/FreeCAD views without hiding the joint.

Motion	Design clearance	Basis
Yaw	+/- 12 deg	Covers the RFC 0009 `standard-urban` 90 m minimum curve with margin
Pitch	+/- 6 deg	Covers vertical curves, lifting, and bogie height variation
Roll	+/- 5 deg	Covers cant, suspension roll, uneven loading, and maintenance lift cases

The clearance envelope is owned by OSR. The supplier may use a different internal link or bearing arrangement if it keeps the same carbody interfaces, passage envelope, and motion envelope.

Train-to-train open end option

The same end-of-car structural datum is now treated as a configurable train-end interface. Each end position must be built as exactly one of two options:

End option	Use	Installed modules
Panoramic glass front/end	Normal outer A and C ends of the reference three-car trainset	`LM3-EIF-SA650` common interface, `LM3-END-SA700` closed end cowl/coupler/crash/sensor assembly
Mid open train-to-train connection	Optional end treatment when two train modules are semi-permanently married into one longer walk-through consist	`LM3-EIF-SA650` common interface, `LM3-TTART-SA850` open-end articulation/gangway cassette

The open connection is not a second carbody design. It replaces only the selected end dress-out: no panoramic glass, fiberglass nose cowl, or nose-mounted T-OBS sensor pack is fitted at the joined end. The common carrier ring, end bolt grid, underframe load datums, drain lands, seal datums, and protected service routes remain the same. The supplier gangway cassette then provides the lower joint/drawbar, upper links, bellows, turntable floor, and trainline/service jumpers across the two train modules.

The generated design record is `train-end-interface.md`. The CAD envelope is generated by `train_to_train_articulation()`.

Load path

Load case	Primary path	Detail requirement
Traction and braking draft	Underframe anchor casting -> drawbar -> lower spherical bearing -> opposite anchor	Proof to consist draft/brake load from traction and emergency braking
Buff/compression	Anti-climb shear keys and drawbar washer stack	No override into passenger gangway trim
Vertical relative movement	Lower spherical joint + upper links	Anti-lift keeper must retain bodies during jacking/lifting cases
Yaw/curve movement	Lower spherical joint with upper roll-yaw-pitch links	No binding at +/- 12 deg yaw
Gangway passenger load	Segmented turntable floor into both adapter frames	5 kN/m2 service load target before supplier-specific proof
Crash isolation	End crash energy remains at outer couplers; articulation protects inter-car passage and trainline integrity	No articulation-mounted crush can in v1 unless supplier requires it

The primary steel at each car end must include an articulation adapter frame, underframe anchor casting, upper clevis brackets, bellows clamp frame nutplates, drain lands, and service hatch aperture. These are fabricator-owned steel details; the gangway module remains a tendered COTS/BID package.

Trainline routing

The articulation separates high-energy, data, safety, coolant, and HVAC paths so a single abrasion or water-ingress failure cannot remove all services.

Route	Location	Notes
HV traction jumper	Right-hand protected conduit	Shielded, orange, bonded at both carbody frames; HVIL loop routed adjacent
TCN-E redundant Ethernet	Upper right protected conduit	Two physically separated TSN Ethernet ports continue the consist ring
CAN-FD + emergency-brake safety loop	Mid-height right conduit	Hardwired emergency-brake loop remains electrically independent
Coolant flexible hose loops	Low left service bay	Drip tray below, quick-disconnect only in depot
HVAC air sleeve	Upper left duct	Flexible sleeve, condensate drainage, no external add-on duct
Drain channels	Both floor edges	Water traps accessible through gangway floor hatch

Maintenance and replacement

Task	Interval target	Access
Visual bellows and clamp-frame inspection	Daily walkaround / depot entry	Exterior and interior gangway faces
Drain trap clean-out	Monthly or local dust/rain interval	Floor service hatch
Bearing/keeper inspection	6-month A-check	Centre hatch and underframe access
Upper-link spherical eye inspection	6-month A-check	Ceiling hatch / roofline service cover
Bellows replacement	10-15 year overhaul planning interval	Clamp frames unbolt; module removed as a cassette
Full articulation module replacement	Heavy overhaul	Cars supported on jacks; drawbar unloaded; service jumpers disconnected

Procurement closure

The v1 BOM treats the complete gangway/articulation module as a BID line and the trainline energy-guidance kit as a SOURCE/MAKE line. At v2 release, the supplier package must provide:

- carbody adapter frame drilling pattern and shim pack limits,

- lower central joint bearing rating, anti-lift proof, and inspection method,
- upper link rating, spherical eye replacement process, and rubber/metal fire-smoke evidence,
- bellows, turntable, and interior panel EN 45545 evidence,
- energy guidance bending-radius limits and harness retention method,
- water ingress, drain, and cleaning instructions,
- articulation motion test report at the OSR yaw/pitch/roll envelope.

Assembly and joining plan — light-metro-3car

Repository source: `docs/rolling-stock/light-metro-3car/assembly-plan.md`

This plan answers the shop-floor question: do the parts and subassemblies actually connect, and how are they joined?

The current product tree is connected by parent/child IDs in the generated buildable manifest: `design/component-catalogue/catalog/buildable-trainset/buildable-trainset-manifest.md`. This document adds the missing single-page join doctrine: datum stack, joining method, sequence, and acceptance hold point for each subassembly and final-assembly interface.

The machine-readable parent/child join classes, torque authority, and release state are generated in `joint-control-schedule.md`.

It is still a planning document, not a released shop drawing. Every structural weld, bolted joint, adhesive bond, HV connection, and coolant joint still needs a released interface-control drawing, the numeric torque or supplier procedure required by the joint schedule, a weld map, surface-preparation procedure, and signed traveler before first article build.

Join-class rules

Use the least exotic joint that preserves structure, inspection, and serviceability:

Join class	Used for	Default method	Release evidence
Structural weld	Primary steel underframe, side frames, bolsters, coupler pockets, end rings, bogie frames	Fixture tack, WPS-controlled MIG/MAG weld, controlled cool/stress relief where required, post-weld machining of critical datums	Material heat trace, WPS/WPQR, welder ID, weld map, VT plus MT/UT where classed, post-weld datum survey
Bolted structural datum	Bogie marriage, coupler/crash absorber, articulation bearings, door cassette frames, HVAC curbs, battery trays	Classed fasteners into machined pads/weld nuts/captive inserts; shim only at released shim packs; torque stripe after final torque	Bolt grade certificate, torque record, locking method, shim map, witness marks, re-torque rule
Qualified adhesive/bonded interface	Glazing lands, selected non-service cowl seams, PV bonded pads, and supplier-required local seams; not LM3-BDY-160 side/roof modules	Abrade/clean/prime, controlled adhesive batch and pot life, clamped cure, witness coupon where required	Surface-prep record, adhesive batch/shelf-life, cure time/temperature, coupon/pull evidence, water/leak test
Gasketed removable panel	One-metre GFRP side/roof modules, cowl hatches, skirts, service covers, battery access lids, and roof fairings	Keyed hooks, captive clips, retained fasteners, or quarter-turn retainers with continuous gasket and drain path	Insert pull-out, fastener retention, gasket compression witness, clip/anti-lift witness marks, water test, service-removal check
Electrical / data	HV battery, inverter, charge rack, roof PV, TCN-E, CAN-FD, safety loop, door and HVAC looms	Keyed connectors, HVIL where HV, segregated clipped routing, bonding jumpers, strain relief and service loops	Continuity, insulation resistance, HVIL, bond continuity, network enumeration, bend-radius and clamp-pitch inspection
Fluid / thermal	Battery/inverter coolant, HVAC drains, washer tubes, fire suppression interfaces	Quick-disconnects or compression fittings on supported pipe/hose; no unsupported hose spans across moving joints	Pressure/leak test, drain-flow test, rub/chafe inspection, service isolation procedure

Assembly datum stack

Build from hard datums outward:

1. Track/fixture centreline.
2. Underframe centre spine and bogie-centre datums.
3. Side-frame door/window aperture datums.
4. Roof rail and HVAC/PV datum plane.
5. End-ring, coupler face, and cowl glass/sensor datums.
6. Systems routing datums: HV, LV/data, coolant, drains.
7. Passenger-facing trim and service covers.

No supplier module may force rework of an upstream structural datum. If a supplier envelope does not fit the frozen datum, the accepted actions are: released shim pack, adapter bracket revision, or supplier alternate. Grinding/drilling a released primary datum in final assembly is a nonconformance.

Subassembly connection plan

Node	Connects to parent by	Critical parent datum	Acceptance hold point
`LM3-BDY-SA110` underframe datum weldment	WPS-controlled welds between side sills, centre spine, cross bearers, bolsters, coupler pockets, and tray rails; post-weld machined bogie/coupler datums	Fixture centreline, bogie centres, coupler pocket centre plane	Material release, fixture tack survey, weld/NDT release, post-weld datum survey
`LM3-BDY-SA120` carbody spaceframe	Welded/bolted side frames and roof bows onto underframe datum pads; door/window portal reinforcement tack-welded before closure	Door aperture planes, window carrier rings, roof-rail pitch	Door/window aperture survey, roof rail survey, carbody dimensional report
`LM3-SHELL-A200` painted frame and body skin	Corrosion-protected steel frame plus moulded 1 m clip-on GFRP side/roof modules, glazing lands, and removable skirt/cowl interfaces	Painted steel clip rails, window apertures, end-ring cowl datum	Corrosion report, mould/cure/coupon records, sealed-edge record, clip/anti-lift witness map, eight-hour route record, water-ingress pre-test; cure record only for glazing/cowl seams
`LM3-WIN-SA320` side glazing cassette	Bonded or gasketed laminated-glass cassette into `LM3-BDY-P110` carrier ring; mechanical anti-drop retention where supplier requires	Window carrier ring and primer/bond land	Aperture gauge, bond/gasket procedure, water-ingress test
`LM3-DOOR-SA310` door cassette	Bolted COTS door cassette into `LM3-BDY-P100` portal frame with shim pack, threshold beam, drains, closed/locked loop, and emergency release	Door portal plane, sill height, lock-loop bracket	Door gauge fit, obstruction test, closed-and-locked test
`LM3-INT-SA330` interior fit-out	FRP/phenolic liners and battery strake covers on potted inserts, clip grids, retained fasteners, and limited non-service adhesive/sealant; lighting/PIS/CCTV plugs into LV looms	Saloon floor datum, ceiling rail, window reveal datum, PRM aisle gauge	Egress check, fire-material pack, liner/trim fit survey, lighting/PIS/CCTV static test
`LM3-ROOF-SA410` roof systems	HVAC bolted to gasketed roof curb; PV modules either bonded to prepared pads or clamped to raised rails; antennas/lights through sealed glands; bonding jumpers to roof earth studs	Roof rail pitch, HVAC curb plane, PV keep-out zones, cable-gland locations	Roof leak test, HVAC drain test, PV isolation/bonding check
`LM3-HV-SA510` traction/HV/cooling	Battery packs bolted into under-seat trays with isolation mounts; inverter/charge rack bolted to underframe rails; HV cables in covered trays; coolant quick-disconnects and supported pipes	Battery tray rail, HV segregated route, side-pin charger datum, coolant manifold support	HVIL test, insulation resistance, coolant pressure test, first-energisation release
`LM3-BOG-SA610` powered bogie	Welded H-frame plus bolted wheelsets, bearings, primary suspension, brake hardware, traction motors, gearbox, torque links, and earth straps	Bogie frame pivot, axle centres, motor-cradle datums	Frame NDT, wheelset/bearing certificate, motor/gearbox alignment, static brake test
`LM3-BOG-SA620` trailer bogie	Welded H-frame plus bolted wheelsets, bearings, suspension, brake package, wear plates, and height-setting shims	Bogie frame pivot, axle centres, air-spring pad plane	Frame NDT, wheelset/bearing certificate, ride-height setup, static brake test
`LM3-CWL-SA710` fiberglass cowl cast kit	FRP cowl shell dry-built on cowl fixture; potted/captive inserts; bolted removable hatches; adhesive/sealant only at non-service sealing surfaces	Backing-ring flange, glass carrier land, lamp/sensor hatch datum	Laminate coupon release, insert pull-out, trim/drill survey, A/B-end dry-build water test
`LM3-END-SA700` train-end assembly	Coupler/crash absorber bolted into steel pocket; end ring/anti-climber bolted/welded to carbody datum; cowl kit bolted/gasketed to backing ring; sensors/lights bolted to datum brackets	Coupler face, anti-climber datum, cowl backing ring, sensor optical/radar datum	A/B end interchange, coupler datum survey, sensor calibration, recovery interface check
`LM3-ART-SA800` articulation/gangway	Lower spherical bearing and drawbar pinned/bolted to underframe anchor castings; upper links bolted to end frames; bellows clamp frames bolted/gasketed; service bundles through drag-chain	Inter-car joint centre, yaw/pitch/roll envelope, trainline bend-radius route	Motion-envelope proof, trainline continuity, water-ingress/drain test
`LM3-SYS-SA900` train control/electronics	T-ECU/A and cabinets bolted to DIN/cabinet rails; TCN-E/CAN/safety-loop connectors plugged and labelled; firmware/configuration loaded after harness continuity	LV cabinet rail, trainline backbone, event-recorder access	Network enumeration, firmware record, self-test, event-recorder write/read test
`LM3-CAR-A900` complete car	Shell receives doors, windows, roof, interior, HV, powered bogie, and trailer bogie; bogies married through centre pivot/air springs/yaw dampers; all earth bonds closed before energisation	Carbody bogie centres, ride-height datum, door/platform datum, HV isolation boundary	Car weigh, door/HVAC/static systems test, bogie marriage report, low-speed yard movement
`LM3-TRAINSET-A000` trainset	Three complete cars joined with two semi-permanent articulations; A/B end assemblies installed at outer ends; train-control network configured across the consist	Train centreline, articulation centres, coupler faces, trainline continuity	Trainset weigh, static brake/door/HVAC/HV tests, FEM screening accepted, dynamic-test release

Final assembly route

1. Release child subassemblies

- Confirm every child assembly has a signed traveler, accepted material pack, and matching revision in the manifest.

- Quarantine any COTS module whose supplier envelope, mass, connector, or evidence pack does not match the interface-control drawing.
- Stage A-car, B-car, C-car kits separately. The cars are repeated, but the outer-end fit-out is only installed on the two trainset ends.

2. Build and release each car module

1. Put `LM3-SHELL-A200` on leveled stands using the underframe datum.
2. Install `LM3-DOOR-SA310` and `LM3-WIN-SA320`; complete water and door-loop tests before interior trim blocks access.
3. Install `LM3-ROOF-SA410`; close roof leak, drain, PV isolation, and roof-bonding checks.
4. Install `LM3-HV-SA510`; leave HV locked out until bond continuity, coolant pressure, and insulation tests pass.
5. Install `LM3-INT-SA330`; prove PRM aisle, hatch access, fire-label, rattle, and egress checks.
6. Marry `LM3-BOG-SA610` and `LM3-BOG-SA620` to the carbody. Set ride height and record shim packs.
7. Weigh the car and complete low-speed yard movement before it enters trainset assembly.

Each one-metre side/roof body module is already moulded, cured, demoulded, CNC-trimmed, edge-sealed, fitted with inserts/clips/gaskets, labelled, and dry-fitted to a master frame before the car reaches final assembly. Installation then follows the LM3-BDY-160 dry cycle: clean and inspect the datum rail, fit the EPDM seal, engage the asymmetric hook, close captive clips, engage the independent anti-lift retainer, and record the visible witness marks. Six two-person crews install the 144 modules across three released frames in one eight-hour shift. Glazing and end-cowl non-service seams retain their separate qualified adhesive cycles. No clip or fastener may pull an out-of-tolerance frame or panel into position.

3. Join the three-car consist

1. Place Car A, Car B, and Car C on level track with coupler/articulation centres on the train centreline.
2. Confirm the end-interface configuration record for each end position: the reference three-car trainset selects panoramic glass at the two outer ends and open mid connections only at internal or train-to-train joints.
3. Install `LM3-ART-SA800` between A-B and B-C:
   - pin/bolt lower drawbar and spherical bearing;
   - install anti-lift keeper;
   - install upper roll/yaw/pitch links;
   - fit bellows clamp frames and turntable floor;
   - route HV, LV/data, coolant, HVAC sleeve, and drains through the released drag-chain path.
4. Perform articulation yaw/pitch/roll sweep before connecting final trim covers.
5. Complete trainline continuity, safety-loop, Ethernet ring, coolant pressure, and water/drain tests across both joints.

4. Select and install end assemblies

1. Survey `LM3-EIF-SA650`, the common configurable end-interface set, before fitting any end dress-out.
2. For a panoramic outer end, install `LM3-END-SA700`.
3. Bolt coupler/crash absorber into the steel pocket and torque stripe.
4. Fit fiberglass cowl kit to the backing ring with gasketed retained fasteners; seal only at released non-serviceable seams.
5. Install panoramic glass, lamps, T-OBS sensors, washer/heater lines, rescue cabinet, and electrical jumper.
6. Survey coupler face and sensor datum. Do not use cowl shimming to compensate for coupler-pocket error.
7. For a train-to-train open mid connection, do not install the panoramic cowl/glass/sensor stack. Install `LM3-TTART-SA850` only after the two train modules are aligned on the final assembly track, then fit the open portal clamp frame, bellows, threshold bridge, turntable, service-jumper cassette, drain trays, and blanking/transition covers.
8. Complete the train-to-train motion sweep, threshold level check, trainline continuity, service segregation, and water/drain test before closing passenger trim.

5. Configure train control and perform static tests

- Install `LM3-SYS-SA900` cabinets and train-control equipment.
- Enumerate TCN-E devices car by car, then end to end.
- Load firmware/configuration and record hashes.
- Run static brake, door, HVAC, charger, roof-PV isolation, HVIL, event-recorder, passenger information, CCTV/intercom, and emergency release tests.

6. FEM/static proof gates before running

Before dynamic running, release the FEM/static screens for:

- underframe supported at bogie interfaces;
- coupler pocket and crash-load path;
- bogie frame and motor cradle;
- battery tray/service-lid rails;
- door portal and threshold;
- roof equipment rails/PV mounts;
- articulation anchor brackets;
- train-to-train open joint vertical and lateral/racking loads if `LM3-TTART-SA850` is selected.

Any screen above the service-stress threshold or with local joint stress concentration requires a detailed local model and drawing revision before the trainset leaves the commissioning cell.

7. Dynamic running release

Run in this order:

1. Low-speed yard movement: 0-10 km/h propulsion, brake apply/release, door inhibit, and emergency-stop checks.
2. Articulation sweep on curves and vertical transitions.
3. Brake bedding and static-to-dynamic brake correlation.
4. Charger docking at the standard side-pin interface, including two-train shared-cabinet arbitration and emergency isolation.
5. Roof PV charge acceptance and isolation under motion/dwell states.
6. Passenger-systems shake/rattle check.
7. Fault-injection run: door interlock fault, HVIL open, TCN device missing, charger abort, battery off-gas, failed mist pump/flow, and obstacle- detection degraded state.

Release to trial running only after the signed trainset traveler, FEM/static proof pack, NCR closure list, and as-built interface-control drawings agree.

Open release gaps

- Join class, torque authority, and release state are emitted for every product-tree integration joint; numerical torques still require the named supplier instruction or released joint calculation.
- Weld symbols/classes exist as process intent, but not as released dimensioned drawings.
- Adhesive bead dimensions, open time, clamp plan, and cure fixtures need supplier-specific procedures.
- Electrical harness clamp coordinates and coolant support coordinates need a released routing drawing.

Body structure — light-metro-3car

Repository source: `docs/rolling-stock/light-metro-3car/body.md`

The body architecture is deliberately simple to manufacture in a regional workshop: a welded carbon-steel primary structure with clipped one-metre fiberglass cladding and COTS passenger-facing modules. The design avoids large aluminium extrusion dies, stainless body pressings, and bespoke door/window/HVAC development.

Architecture

Each 16.5 m car is split into four production zones:

Zone	Construction	Function
Stepped underframe ladder	Cut/bend/weld S355 steel box and folded plate	Carries buff/compression, bogie pivots, battery boxes, coupler loads; drops through the centre door/PRM zone and rises over standard bogies
Side/roof spaceframe	S355 rectangular hollow section	Door posts, window rails, roof equipment rails, composite panel support
Composite exterior	1 m-wide fire-rated fiberglass sandwich modules	Clipped, replaceable weather skin, insulation, aerodynamics, livery surface
Interior cassette	COTS panels, seats, lights, grab rails, PIS, HVAC ducting	Passenger finish and replaceable equipment

Steel is the certified load path. Composite panels are non-structural: they may stiffen local skins, but crashworthiness, jacking, lifting, coupler loads, and bogie loads are closed through the welded frame.

The CAD implementation follows the same hierarchy in `car_body.py`:

CAD subassembly	Included layers
`car_body_structure()`	Welded shell, 350 mm and ~10 m low-floor centre pan, 760 mm and ~3 m high-floor bogie-end decks, transition ramps/steps, side sills, raised bogie plinths, bogie sweep/drop envelopes, crossmembers, window posts, waist rails, end rings, anti-climber beams, roof cantrails, door portal posts and headers
`car_body_exterior()`	Sixteen 1 m fiberglass side/roof bays with clips and dry seals, plus glazing, door leaves, livery, skirts, roof PV, and HVAC
`car_body_interior()`	Under-seat battery strakes, longitudinal seats, PRM bays, grab poles, handrails, passenger information displays
`car_body_services()`	HVAC supply/return/drop ducts, LV/TCN cable trays, LED lighting, CCTV, intercoms, door harness loops, HV traction/PV routing, coolant pipes, battery fire vent paths

`car_body()` returns the complete nested assembly. FreeCAD and PNG review artifacts are generated from that source; they are not the design authority.

Primary steel structure

Underframe

The underframe is a jig-welded ladder:

Member	Section	Material	Notes
Side sill	200 × 100 × 8 mm RHS	S355J2 or local equivalent	Continuous between end modules
Centre spine	250 × 100 × 8 mm RHS	S355J2	Battery-box and equipment support
Cross bearer	160 × 80 × 6 mm RHS	S355J2	750 mm nominal pitch
Bogie bolster	Folded/welded 12 mm plate box	S355J2	Machined centre-pivot insert after weld
Coupler pocket	16 mm folded plate + crush can interface	S355J2	EN 15227 energy absorber bolts on
Battery tray rail	80 × 60 × 5 mm RHS	S355J2	Under-seat module cassette support
Articulation adapter frame	12 mm folded plate + machined inserts	S355J2	Lower spherical joint, bellows clamp, and upper-link clevis datums

The floor is not flat. The two centre doors per side and wheelchair bay sit in a ~10 m low-floor zone at 350 mm above top-of-rail for platform boarding. The end saloon zones rise to 760 mm over the standard ~3 m bogies, with short interior steps and ramp panels between the zones. The body sides therefore use a lowered side sill through the centre door bay and raised plinth rails over the bogie zones.

All primary welds are made in a rotating fixture so flat/horizontal weld positions dominate. The v2 drawing pack assigns EN 15085 weld classes per joint; the v1 assumption is CL1 for bolsters, coupler pockets, and crash boxes, CL2 for side sills/cross bearers, CL3/CL4 for panel tabs and interior brackets.

Articulation interface frames

Each inner car end carries an articulation adapter frame rather than a simple drawbar bracket. The frame includes:

- underframe anchor casting for the semi-permanent articulated drawbar,
- machined land for the lower spherical bearing and anti-lift keeper,
- vertical anti-climb shear-key pockets beside the lower joint,
- upper clevis brackets for the twin roll-yaw-pitch stabilising links,
- bellows bolted clamp frame and replaceable nutplates,
- drain channel lands and a floor service-hatch aperture,
- separated HV/data/coolant/HVAC bracket datums for the energy-guidance routes through the gangway.

The articulation supplier owns the internal bearing, bellows, turntable, and link details. The fabricator owns the steel adapter frame, weld map, machined datum survey, corrosion protection, and shim pack. See `articulation.md` for the motion envelope and `interfaces.md` for the trainline interface.

Side frame

The side frame is a light spaceframe welded to the underframe:

- Door posts: 120 × 80 × 6 mm RHS.
- Window posts: 80 × 50 × 4 mm RHS.
- Waist rail: 100 × 50 × 5 mm RHS.
- Cant rail: 100 × 50 × 5 mm RHS.
- Roof bows: 80 × 40 × 4 mm RHS at 1 000 mm pitch.
- Equipment rails: bolted stainless inserts for HVAC, antennas, marker lights, CCTV, and cable trays.

Door openings are not cut from a finished shell. They are built as open bays around the COTS door cassette so the door supplier's frame, threshold, drainage tray, and emergency-release hardware install as a single bolted module.

Composite exterior panels

Exterior body modules hang from keyed rails and are mechanically retained to the steel frame. The common design is controlled in `modular-fiberglass-body.md`:

Panel	Construction	Attachment
Side skin	28 mm nominal fire-rated GFRP sandwich; 1,000 mm pitch / 994 mm finished width	Keyed hook, captive over-centre clips, independent anti-lift retainer, dry EPDM joint
Roof fairing	28 mm nominal GFRP sandwich; 1,000 mm pitch / 994 mm finished width	Same clip grid; removable above HVAC/equipment
Nose/sensor cowl	Multi-part fiberglass (GFRP) cast kit over steel crash frame	Identical at both train ends; bolted, sacrificial, and replaceable; single heated RF-transparent panoramic end glass pane with LED headlamp/marker clusters
Skirts	Composite or aluminium removable panels	Quarter-turn service fasteners
Interior liners	EN 45545 HL2 FRP or phenolic panels	Clip/bolt to secondary rails

The composite supplier owns laminate coupons, resin selection, fire test evidence, and repair manuals. The OSR body drawing owns only panel envelopes, edge radii, insert locations, grounding/bonding points, and removal clearances.

The front and rear exterior ends use the same A/B-end fiberglass kit: upper brow, left/right cheek casts, lower apron, lamp/service hatches, and backing-ring flanges around the single glass carrier over the steel crash frame. See `end-cowl.md` for the cast split, laminate schedule, mould/tooling rules, and glass/sensor service interfaces.

Cabin fiberglass/phenolic parts are controlled separately because they are passenger-facing and have different fire, edge-radius, rattle, and service-removal rules from the exterior cowls. See `cabin-fiberglass.md` for the ceiling liner, sidewall/window reveal, battery strake cover, vestibule kick-panel, PRM transition, tooling, lay-up/trim, inspection, and repair instructions.

COTS windows

Windows are bought as certified rail/bus glazing modules rather than fabricated locally:

- Laminated safety glass to EN 15152 or equivalent.
- Typical opening: 900 × 1 200 mm.
- Bonded-in or gasketed cassette depending on supplier.

- Emergency ventilation only where required by the authority having jurisdiction; default HVAC assumes fixed glazing.
- Drainage path and adhesive bead are per supplier installation drawing, not re-engineered by OSR.

COTS doors

Door modules are commercial off-the-shelf electric sliding-plug or plug-outward rail doors:

- Clear opening: 1 250 mm × 2 000 mm.
- Electric actuation; no trainwide pneumatic door supply.
- Supplier-provided threshold, guide rail, controller, obstacle detection, emergency release, seals, and manual isolation.
- OSR provides 24 V DC / 110 V DC power, Ethernet/CAN-FD command, hardwired closed-and-locked loop, drainage, and mounting datum.

The preferred procurement pattern is to qualify two door suppliers against the same envelope. The carbody must not depend on a unique proprietary door pocket.

COTS roof and passenger systems

The body reserves bolted envelopes for:

- Roof HVAC: one 24 kW nominal packaged unit per car, hot-climate variant for +50 °C ambient.
- LED lighting: 24 V DC rail-rated strip or troffer modules.
- PIS displays and speakers: off-the-shelf rail/bus units.
- CCTV: PoE cameras in serviceable composite housings.
- Intercom: COTS EN 50155 rail intercom stations.
- Seats: longitudinal rail/bus bench modules bolted to under-seat battery cover rails.
- Grab rails: stainless tube systems with off-the-shelf clamps.

HVAC, lighting, PIS, CCTV, intercom, seats, and grab rails are procurement modules. OSR fixes the envelope, fastener grid, power budget, data interface, fire rating, and maintainability clearance. Supplier-neutral envelopes and evidence requirements live in `control-electronics/trainset-interiors.md`.

The full interface map, including how the COTS modules join to the fabricated steel/composite body, is in `cots-integration.md`.

Keyed composite cassette interface

The steel underframe and spaceframe are the permanent datum skeleton. Side and roof weather modules use the dry LM3-BDY-160 interface; door, window, skirt, nose, and interior cassettes retain their component-specific interfaces:

- a 3-2-1 locating scheme, with two primary locating pins and an asymmetric third locator that prevents reversed installation;
- captive clips and a standard 1,000 mm longitudinal pitch;
- a replaceable EPDM compression seal, drained secondary rail, and visible latch witness mark for side/roof modules;
- independent mechanical anti-lift retention;
- dedicated bonding/grounding studs; and
- keyed quick connectors located outside wet drainage paths.

Fastener holes clamp the cassette; they do not establish body geometry. Location is owned by the datum features. Production adhesive is not used at the removable side/roof body-to-frame joint. Glazing and released non-service end-cowl seams retain their qualified adhesive processes. The fiberglass nose remains a sacrificial shell over the steel crash structure.

Molded service infrastructure

Composite parts may incorporate empty HVAC supply/return geometry, condensate drains, cable raceways, conduit paths, clip lands, lighting recesses, and equipment bosses. Consumable services remain removable:

- no cable, coolant pipe, sensor, lamp, fan, filter, evaporator, damper, or connector is encapsulated in laminate;

- passenger-side service covers provide removal access without cutting the panel;
- wiring is installed as keyed, labelled, pre-terminated looms; and
- main roof air-duct geometry may be molded into the sandwich, but every active HVAC component remains a separate service module.

The cassette release test includes datum repeatability, insert pull-out, bond durability, water ingress, bond continuity, fire/smoke/toxicity, and a timed service-removal demonstration.

Floor

- 5-6 mm steel or aluminium deck panels over cross bearers.
- Removable hatches above battery and HV equipment zones.
- Phenolic or aluminium honeycomb floor boards in the saloon.
- Slip-resistant vinyl or rubber floor covering, EN 45545-2 R5.
- ~10 m low-floor centre door zone; ~3 m raised end decks over bogies.

Corrosion protection

Steel primary structure receives:

- Shot blast to Sa 2.5.
- Zinc-rich epoxy primer.
- Seam sealer at composite/steel interfaces.
- Cavity wax in closed sections after weld inspection.
- Polyurethane topcoat where visible or exposed below floor.
- Drain holes in every closed bay that can collect condensation.

Composite/steel joints use isolation tape or coated inserts to avoid galvanic corrosion where aluminium brackets or stainless fasteners are present.

Fire compliance

Every passenger-facing material must meet EN 45545-2 HL2 or the deployment authority's accepted equivalent.

Item	Requirement
Steel primary frame	Non-combustible; coating fire data required
Composite exterior panels	R1 HL2 or better
Interior panels	R1 HL2 or better
Floor covering	R5 HL2 or better
Seats	R7 HL2 or better
Cable insulation	R15/R24 HL2 or better
Door/window seals	R23 HL2 or better

v2 deliverables

- Welded frame FreeCAD assembly and 2D drawings.
- Flat pattern DXF for folded plates.
- Tube cut list with mitre angles and fixture datum references.
- Weld map with EN 15085 classes, WPS references, and inspection hold points.
- Composite panel envelope drawings and insert maps.
- 3-2-1 locator, asymmetric key, fastener-grid, sealant-groove, bonding, and quick-connector interface drawings.
- Door/window/HVAC supplier interface drawings.
- Static proof-load and fatigue FEA report.
- EN 15227 crashworthiness simulation report.

Bogies — light-metro-3car

Repository source: `docs/rolling-stock/light-metro-3car/bogie.md`

Six bogies per consist: three powered bogies and three trailer bogies, arranged as one powered bogie plus one trailer bogie under each repeated 16.5 m car module. All bogies use the RFC 0022 suspension, wheelset, brake, and pivot geometry; the trailer variant omits motors and gearbox.

Bogie envelope (all six, common)

Parameter	Value
Wheelbase (axle-to-axle)	2 100 mm
Wheel diameter, new / worn	760 / 680 mm
Wheelset type	Monobloc, machined per RFC 0022
Wheel profile	S1002 on standard-gauge rail
Axle load (AW3, both axles)	≤ 14 t each
Gauge	1 435 mm standard gauge
Track-brake coil (magnetic track brake)	not fitted — per RFC 0008 §3.2 electric-only brake

Powered bogie (three per consist)

Frame

- Welded H-frame, steel S355JR per EN 10025, EN 15085 CL1 welds.
- Length (axle-to-axle) 2 100 mm; width 2 400 mm; depth 300 mm.
- 4 suspension pedestals — 1 per axle end.

Primary suspension

Chevron rubber springs (one per axle end):

- Vertical stiffness: 1 500 kN/m nominal per spring.
- Lateral stiffness: 5 000 kN/m.
- No shock absorbers at primary level (the chevron's internal damping is sufficient).
- Service interval: replace every 10 years or on inspection finding; visual inspection every 7 days per RFC 0013 M5.

Secondary suspension

Twin-bellows air spring + lateral damper (one pair per bogie):

- Air spring: Continental / ContiTech CF-series class, 55 °C continuous bellows rating.
- Levelling valves hold the centre door-zone floor within ± 5 mm from empty to AW3.
- Damper: KONI 32-series linear, 6 kNs/m at nominal velocity.
- Auxiliary rubber emergency bearer supports the car after air loss.

Motor mount

Two PMSM axle-hung motors per powered bogie, one per axle. Motor details in `traction.md`.

Brake

- Service + emergency: axle-mounted brake disc on each axle. Caliper electromagnetic actuation — two calipers per disc.
- Parking: spring-applied, electromagnetic-release on the same caliper; cut-out spring force ≥ 12 kN per caliper.
- WSP: wheel-slide protection via the per-axle tacho input + `osr-brake` modulation (proptest-verified B4 conservative).

Trailer bogie (three per consist)

The trailer bogie is a fresh OSR bogie frame built to the same datum scheme as the powered bogie. Differences from the powered bogie:

- No motor. All axles idle.
- Brake: same 600 mm disc on each axle for service / emergency / park.
- Mass: lower than the powered bogie by the omitted motors, gearboxes, and inverter cabling.

Brownfield recovered-component policy

Recovered freight bogie frames are not spliced into the OSR consist. Where a deployment has usable donor stock, only component-level recovery is allowed after quarantine, cleaning, dimensional survey, and NDT:

- Axles may be re-machined/reprofiled only if UT/MT inspection and metrology pass the supplier acceptance sheet.
- Axleboxes and bearing housings may be reused only after crack inspection, bore measurement, and traceable bearing replacement or bearing supplier re-certification.
- Chevron primary springs, air springs, dampers, brake calipers, yaw links, and motor/gearbox mounts are new qualified parts.

This keeps the mechanical release on one fresh bogie-frame drawing while still allowing brownfield cost recovery where the evidence is credible.

Interfaces

- Bogie-to-body pivot: centre pin ring bearing, 400 mm diameter, PTFE slider.
- Yaw restraint: two longitudinal links per bogie with elastomeric bushes. No hydraulic yaw dampers (tangent speeds ≤ 25 m/s don't demand them on the RFC 0009 preset radii).
- Power electrical: 650–700 V nominal traction DC plus isolated 110 V and 24 V DC services through a cableguide at the centre pin; not through the pivot itself.
- Air: local secondary-suspension levelling only, fed by one compact compressor/dryer/reservoir/isolation manifold per car module. No trainwide pneumatic brake or door supply per RFC 0008 §3.2.

Interchangeability

Every powered bogie in every OSR deployment of this family is identical. The trailer bogie shares:

- Frame geometry (drilling pattern).
- Suspension parts.
- Brake disc + caliper.
- Wheel monobloc.

So most bogie spares are shared across powered and trailer variants.

Weld classes (EN 15085)

Assembly	Class
Frame primary welds	CL1
Motor mount welds	CL1
Brake caliper mount	CL1
Cable-guide mount	CL3
Non-structural covers	CL4

Per EN 15085, every CL1 weld is radiographically inspected at the certified shop.

v2 deliverables (not in v1)

- Dimensioned assembly drawings per EN 15016 / ISO 128.
- FEA: frame stress under AW3 + dynamic accelerations.
- FRA: fatigue-endurance curves over 1M simulated load cycles.
- Weld-quality records template (EN 15085 compliance file).

Procurement BOM skeleton — light-metro-3car

Repository source: `docs/rolling-stock/light-metro-3car/bom-skeleton.md`

This is the source-identified parts list a fabricator uses to price the consist. Parts split three ways:

- SOURCE — commodity off-the-shelf. Distributor, catalogue, or marketplace quote line available; no engineering work required.
- MAKE — fabricator produces in-house. Single-supplier tender.
- BID — multi-supplier tender. The deployment's procurement office runs the bid.

Costs are BASE USD volume-100. Country cost factor from `lib/templates/country-costs.toml` scales them per deployment.

Scope note: this is a consist-level procurement BOM, not a final manufacturer MBOM. It now covers every train-level subassembly needed to build, commission, and maintain the v1 trainset concept; supplier drawings still break many lines into child parts during v2.

City CAPEX uses this BOM as an audit lower bound, not as the final rolling-stock price. The deployable trainset cost is carried in `lib/templates/capex-costs.toml` as a local-owner production planning unit, and the shared national railway production-plant setup allowance is 60 k USD per supported vehicle/car module base, with 120 k USD per vehicle/car module retained as a high sensitivity check.

Generated CSV outputs:

- `build/bom/rolling_stock_bom.csv` adds low/base/high cost bands to every BOM line.
- `build/bom/rolling_stock_cots_fitout_bom.csv` is generated directly from the source-shaped COTS fit-out catalogue in `design/component-catalogue`.

Cost confidence:

- SOURCE lines use public catalogue/distributor or commodity quote bands until procurement replaces them with current supplier quotes.
- MAKE lines use local fabrication estimate bands until a shop route, fixture plan, and labour-rate pack are issued.
- BID lines are tender-only rail components. Public product pages inform shape, interface, and supplier class; exact price requires RFQ.

Concept alignment

This BOM is keyed to the visual and layout concept in `docs/assets/solar-metro-trainset.png`:

- 3 cars, each 16.5 m, 49.5 m over couplers.
- Each car repeats the same self-contained architecture: one powered bogie, one trailer bogie, one 180 kWh usable under-seat battery pack, two independent motor controllers, isolated LV DC/DC, and protected station/PV DC interfaces.
- 3 powered bogies and 3 trailer bogies total.
- 6 traction motors total; the planning profile caps consist traction peak at 1.8 MW.
- Roof solar arrays on each car, with compact end HVAC modules.
- Wide passenger windows, identical multi-part fiberglass driverless end cowls with one large dark panoramic glass pane and LED headlamp/marker clusters, white/silver body shell, green waist band, dark skirts, and yellow door thresholds.
- Standard bogies under every car, requiring ~3 m high-floor end decks over the bogies and a ~10 m low-floor centre zone with two door openings per side.
- Batteries remain under longitudinal seats, not on the roof or deep underframe.

Supplier fit-in references

The generated COTS/RFQ candidate register is the only current selection input. It maps every bought-in LM3 product row to an official manufacturer source, selection maturity, published facts, fit work and a local-equivalent route. The broader supplier-anchor register defines supplier-neutral boundaries. Neither register is an order, a quotation or approval; exact configurations close only through the first-article execution pack.

Historical price floor

The compact marketplace-price ledger preserves the older low-cost sensitivity calculation only. It is not a component-selection source and cannot close a supplier, engineering or safety gate. Regenerated city twins use the controlled manufacturer candidate IDs while retaining the existing cost as a planning allowance until an RFQ response replaces it.

Body + interior

Line	Desc	Qty per consist	Source	Base USD	Notes
B1	S355 RHS tube, underframe + side/roof frame	3.2 t	SOURCE	3 000	Alibaba S355 RHS tube anchor; cut/bend/weld locally
B2	S355 folded plate, bolsters/coupler pockets/brackets	2.0 t	SOURCE	2 000	Alibaba S355 plate anchor; press-brake + weld
B3	CNC cutting, drilling, fixture consumables	set	MAKE	6 000	Alibaba fabrication-service proxy; per consist allocation
B4	Weld consumables + shielding gas	set	SOURCE	2 000	Alibaba welding-wire/gas anchor; EN 15085 WPS-controlled
B5	Shot blast + zinc-rich primer + cavity wax	set	SOURCE	4 000	Alibaba zinc-rich primer/blast proxy; corrosion package
B6	Clip-on fiberglass side body modules, fire-rated	96 × 1 m modules	BID	7 000	One common 1,000 mm mould pitch with CNC-trimmed solid/window /door variants; GFRP needs EN 45545 evidence
B7	Clip-on fiberglass roof modules, retainers, seals + skirts	48 × 1 m roof modules + kit	BID	4 000	Reusable short moulds, keyed captive clips, anti-lift retainers, dry EPDM joints, and removable service skirts
B8	Multi-part fiberglass A/B-end sensor cowl cast kits	2	BID	4 000	Alibaba FRP/composite proxy; CWL-FRP-01 through CWL-FRP-06 cast kit accepts one bonded panoramic glass pane, LED lamp clusters, washer/heater service hatches, and T-OBS aperture
B9	Complete inter-car articulation/gangway module	2	BID	30 000	Alibaba train-gangway anchor; lower pivot, upper links, bellows, turntable floor
B10	COTS laminated safety-glass window cassette	18	SOURCE	6 000	Alibaba laminated transport-glass anchor; bonded/gasketed rail-style cassette
B11	COTS electric plug/sliding door cassette	12	BID	90 000	Alibaba sliding-door-system proxy; door, controller, obstruction sensors, seals, emergency release
B12	Stepped floor board + hatch system, EN 45545	135 m²	SOURCE	6 000	Marketplace honeycomb/composite-board anchor; 350 mm low centre, 760 mm bogie-end decks
B13	Vinyl/rubber floor covering, EN 45545 R5	135 m²	SOURCE	1 500	Alibaba bus/train vinyl-floor anchor
B14	COTS longitudinal seat modules, EN 45545 R7	60 seats	SOURCE	6 000	Alibaba bus/metro seat anchor; rail fire-test evidence still required
B15	COTS grab rail + stanchion kit	set	SOURCE	2 500	Alibaba bus handrail fittings anchor; stainless modular system
B16	COTS interior LED lighting kit	3 cars	SOURCE	1 500	Alibaba/AliExpress 24 V bus-light anchor; emergency mode
B17	COTS exterior marker + head/taillight kit	set	SOURCE	1 200	Alibaba bus lamp anchor; sealed head/tail/marker class
B18	COTS PIS display + speaker kit	set	SOURCE	3 000	Alibaba bus/metro display anchor; interior/exterior
B19	COTS CCTV + intercom kit	set	SOURCE	2 500	Alibaba bus CCTV plus AliExpress PoE camera anchor; passenger intercom included
B20	Paint + 2K urethane topcoat	set	SOURCE	3 000	Alibaba automotive coating proxy; white/silver base coat under concept livery
B21	Interior FR panels + trim clips	set	SOURCE	5 000	Alibaba FR/composite panel anchor; supplier-certified HL2 required
B22	Scharfenberg Type 10 end coupler + electric head	2	BID	8 000	Alibaba railway-coupler proxy; rescue unlock handle, certified coupler RFQ required

Line	Desc	Qty per consist	Source	Base USD	Notes
B23	EN 15227 crash energy absorber / crush-can set	2	BID	4 000	Alibaba draft-gear/crash-part proxy; bolts to coupler pocket
B24	Inter-car energy-guidance trainline and service-loop kit	2	SOURCE	4 000	Commodity HV jumper/cable-chain anchor; TCN-E, CAN-FD, 24 V, 110 V, EB loop, coolant, HVAC sleeve
B25	Door sill gap-filler + external emergency release kit	12	BID	6 000	Commodity actuator anchor plus door-safety qualification; per exterior door cassette
B26	Jacking, lifting, towing, and recovery fittings	set	MAKE	3 000	Alibaba machined/welded steel proxy; welded pads, tow eyes, labels
B27	Single laminated panoramic end-glass assemblies	2	BID	10 000	Alibaba train curved/laminated glass anchor; RF-transparent heated/de-iced end glazing still needs RFQ
B28	Concept livery package: white/silver body, green band, yellow thresholds	set	SOURCE	2 000	Alibaba vinyl/paint-mask anchor; exterior decals and safety edging
B29	Articulation adapter-frame steel, anti-lift, and shim kit	2	MAKE	6 000	Marketplace machined-steel proxy; underframe anchors, clevis brackets, shear-key datums, machined shims
Body + interior subtotal				233 200	

Bogies (6 per consist)

Line	Desc	Qty	Source	Base USD	Notes
G1	Welded common bogie frame, powered build	3	MAKE	15 000	Alibaba train-bogie fabrication proxy; EN 15085 CL1 required
G2	Welded common bogie frame, trailer build	3	MAKE	9 000	Alibaba bogie-frame/rail-part proxy; fresh OSR frame, no recovered freight-frame splice
G3	Wheelset monobloc (RFC 0022, S1002)	12	BID	18 000	Alibaba railway wheelset anchor; inspection and profile verification required
G4	Axle bearing box (SKF / FAG)	24	SOURCE	12 000	Alibaba railway axle-bearing/box anchor with certified bearing uplift
G5	Primary chevron spring	24	SOURCE	4 800	Alibaba rail spring/rubber anchor; qualified compound required
G6	Secondary air spring + levelling valve	12	SOURCE	6 000	Alibaba railway air-spring anchor; matches RFC 0022 twin-bellows spec
G7	Secondary damper	12	SOURCE	3 000	Alibaba railway hydraulic-damper proxy
G8	Brake disc	12	SOURCE	12 000	Alibaba rail brake-disc/brake-part proxy; EN material traceability required
G9	Electromagnetic brake caliper	24	BID	12 000	Alibaba caliper proxy; rail electromagnetic caliper RFQ required
G10	Park-brake spring assembly	24	SOURCE	2 400	Alibaba rail spring anchor; caliper integration remains G9
G11	Centre-pin ring bearing + PTFE slider	6	SOURCE	3 000	Alibaba heavy bearing/PTFE slider anchor
G12	Yaw-restraint link + bushes	12	SOURCE	1 800	Alibaba rail bushing/link anchor
G13	Cable-guide + centre-pin assembly	6	MAKE	3 000	Marketplace rail-part fabrication proxy
G14	Wheel-tach (quadrature encoder)	12	SOURCE	600	Alibaba/AliExpress encoder anchor; rail harnessing in G17
G15	Axle bearing temp sensor	24	SOURCE	500	Commodity sealed temperature-sensor anchor
G16	Brake pad / friction lining kit	12 axle sets	SOURCE	1 500	Alibaba rail brake-pad anchor; commissioning set, per disc
G17	WSP + brake control harness	6	MAKE	3 000	Tacho, caliper command, temp sensor loom
G18	Motor cradle + torque-link weldment	3 powered bogies	MAKE	6 000	Alibaba machined/welded steel proxy; uses RFC 0022 motor mount datum
G19	Flexible gear coupling / torque flange set	6	BID	3 000	Alibaba industrial coupling anchor; motor-to-reduction gear interface
G20	Bogie inspection covers and guards	6	MAKE	3 000	Removable non-structural covers
G21	Suspension air-supply compressor, dryer, reservoir, and isolation manifold	3 car sets	SOURCE	6 000	Local secondary-suspension air only; no trainwide pneumatic brake, door, or coupler supply
Bogies subtotal				125 600	

Traction + power

Line	Desc	Qty	Source	Base USD	Notes
T1	HM47-class PMSM axle motor (350 kW short peak candidate)	6	BID	36 000	RFQ planning share of the $48k–72k six-set motor/controller target; rail duty evidence excluded
T2	Single-stage reduction gear, ratio to be selected	6	BID	15 000	Current 6.5:1 CAD seed is not released; select from torque-speed, wheel, adhesion, grade and thermal analysis
T3	LD32-class independent motor controller	6	BID	24 000	One per motor; 400–750 V candidate envelope requires transient-margin qualification
T4	Cold-plate + traction cooling branch	6	SOURCE	6 000	One monitored branch per axle controller/motor set
T5	LFP under-seat car pack (180 kWh usable / 225 kWh gross)	3	BID	75 000	EVE-class RFQ target; includes liquid-cooled module hardware but not unsupported commodity-pri ce claims
T6	BMS electronics (pack-level)	3	BID	4 500	Commodity pack-BMS anchor; rail battery supplier must verify HVIL, thermal, and fire-path interfaces
T7	Pack cooling plate set	3	SOURCE	2 000	Alibaba battery cold-plate anchor
T8	Under-seat aluminium module enclosure set	3	MAKE	6 000	Alibaba aluminium battery-box fabrication proxy
T9	Battery temperature/off-gas detection plus electrical-en closure smoke detection	3 car sets	BID	10 800	Qualified detector response, calibration, poisoning and false-positive evidence required
T10	Localized battery water-mist reservoir, DC pump, stainless pipe, nozzles and feedback sensors	3 car sets	BID	9 000	Battery compartments only; includes reservoir level, pump, pressure and flow diagnostics
T11	800 V-class HV distribution, contactors, IMD and bus bars	3	BID	10 000	Train-level RFQ target $8k–12k; interrupt rating and isolation architecture must close
T12	Standard station charging side-pin connector	3	BID	4 500	Touch-safe sequence, HVIL, weld detection, contact monitoring and emergency isolation
T13	Isolated 800 V to 110/48/24 V DC/DC set	3	BID	7 500	No central 400 V AC bus; safety domains remain independent
T14	Direct-HV DC packaged roof HVAC unit (24 kW each, per car)	3	BID	17 250	650–700 V input, internal motor electronics, 50 °C curve and rail evidence required
T15	Regen dump resistor (roof-mount)	3	SOURCE	1 000	Alibaba braking-resistor proxy; one per car, consist cost allowance shown
T16	Battery module service disconnect, fuses, and contactors	24 module positions	SOURCE	2 400	Alibaba/AliExpress HV contactor/fuse anchor; 8 modules per car per traction.md
T17	Battery vent duct + fire exhaust path	3 cars	MAKE	3 000	Alibaba sheet-metal duct proxy; side vent, intumescent edge seals
T18	HV cable, bonding strap, and EMC filter set	3 cars	MAKE	12 000	800 V-class pack/controller /station/PV paths with physical segregation
T19	Station-charge connector actuator + contact monitor	3	BID	3 000	Commodity actuator/senso r anchor; completes side-pin connector line T12
T20	Coolant pump, hose, manifold, and bleed kit	3 cars	SOURCE	2 500	Alibaba EV coolant-pump/h ose anchor; battery, inverter, motor, HVAC tie-in
T21	Commodity full-roof solar panel set	3 cars	SOURCE	6 000	Alibaba 400 W rigid/flexible solar-panel anchors; twelve modules per car / 36 modules per trainset
T22	Roof solar mounts, MPPT combiner, isolation, and roof harness	3 cars	SOURCE	6 000	Alibaba roof-mount/combin er/MPPT proxies; bond pads, raised rails, edge clamps, junction boxes, fire-isolation labels

Line	Desc	Qty	Source	Base USD	Notes
T23	Per-car roof MPPT and station-DC protection/controller	3	BID	3 000	Station owns the 500 kW DC/DC power conversion; car rack provides PV MPPT, protection, precharge and arbitration
Traction + power subtotal				266 450	

Electronics + safety

Line	Desc	Qty	Source	Base USD	Notes
E1	T-ECU/S board (2× RP2350 + CM5)	2	BID	1 200	Alibaba PCB-assembly proxy; custom baseboard per RFC 0007 §4
E2	T-ECU/A board (CM5)	2	BID	1 500	Alibaba PCB-assembly proxy; RFC 0007 standard, two units per consist
E3	ADIS16505 IMU (or BMI088)	2	SOURCE	200	AliExpress/PriceArchive IMU-module proxy
E4	u-blox NEO-F10N GNSS	2	SOURCE	100	AliExpress/PriceArchive GNSS module proxy
E5	PN5180 NFC balise reader	1	SOURCE	20	AliExpress/PriceArchive PN5180 proxy
E6	ATECC608B SE chip (on T-ECU/S + T-ECU/A carriers)	6	SOURCE	20	AliExpress/PriceArchive secure-element proxy; T-OBS anchors included inside E18
E7	Cat.22 5G M.2 module	1	SOURCE	200	AliExpress/PriceArchive 5G module proxy
E8	LoRa SX1276 breakout	1	SOURCE	20	PriceArchive SX1276 LoRa module anchor
E9	NVMe SSD 256 GB (event recorder)	1	SOURCE	40	AliExpress/PriceArchive NVMe proxy
E10	10.1" maintenance/service touchscreen	2	SOURCE	300	AliExpress/Alibaba industrial touchscreen anchor; hidden end-cabinet HMI, not a driver cab
E11	Depot manual-control pendant + keyed enable	2	BID	500	Alibaba industrial pendant/switch proxy; normally stowed
E12	Emergency plunger (hardwired)	2	SOURCE	40	AliExpress/Alibaba emergency switch proxy
E13	Depot enable key-switch and guarded motion consent	2	SOURCE	100	Alibaba keyed-switch/guarded-switch proxy
E14	PIS display (exterior + interior)	12	SOURCE	3 000	Alibaba bus/metro LCD/LED display anchor
E15	CCTV camera (forward + door-sill + in-car)	20	SOURCE	600	Alibaba bus CCTV plus AliExpress PoE camera anchor
E16	2oo2 relay stage (per safety output)	4	SOURCE	200	Alibaba safety-relay proxy
E17	Cable harness (pre-terminated, per car)	3	MAKE	9 000	Alibaba wiring-harness fabrication proxy
E18	T-OBS complete nose module	2	BID	8 000	Alibaba industrial-LIDAR/camera/radar proxy; rail T-OBS supplier RFQ required
E19	Sensor-window heater, washer, and service cover kit	2	SOURCE	1 000	Commodity washer/heater actuator anchor; keeps nose sensors inspectable in service
E20	Door-sill camera / obstruction sensor harness	12 doors	MAKE	3 000	Alibaba harness/camera proxy; ties doors to CCTV + door control
E21	Roof antenna kit	set	SOURCE	300	Alibaba/AliExpress multi-antenna anchor; 5G MIMO, LoRa, GNSS, public-safety blank
E22	DIN cabinets, power distribution, and terminal blocks	3 cars	MAKE	3 000	Alibaba DIN cabinet/terminal-block proxy; T-ECU, BMS, door, HVAC cabinets
E23	Crashworthy event-recorder memory module	1	SOURCE	2 000	Alibaba rugged recorder/SSD proxy; complements E9 NVMe operational recorder
Electronics + safety subtotal				34 340	

Safety, accessibility, and maintainability

Line	Desc	Qty	Source	Base USD	Notes
A1	Wheelchair bay fixtures + passenger call buttons	6 bays	SOURCE	1 000	Alibaba accessibility fixture/call-button anchor; two PRM spaces per car
A2	Tactile / visual safety signage and labels	set	SOURCE	500	Alibaba safety-label/signage anchor; doors, emergency, high-voltage, lift points
A3	Emergency lighting + exit marker kit	3 cars	SOURCE	1 000	Alibaba/AliExpress emergency LED anchor; battery-backed, EN 45545 materials
A4	Fire extinguisher, first-aid, and incident-seal kit	set	SOURCE	500	Marketplace consumables anchor; operator restocks consumables
Safety/accessibility/maintainability subtotal				3 000	

Consist total

Bucket	Subtotal (USD)
Body + interior	233 200
Bogies (6 per consist)	125 600
Traction + power	266 450
Electronics + safety	34 340
Safety, accessibility, and maintainability	3 000
Total direct-material consist	662 590

Generated cost band from the same line items:

Case	Direct material	+28% labour / assembly	Planning total
Low	516 027	144 488	660 515
Base	662 590	185 525	848 115
High	1 031 521	288 826	1 320 347

Marketplace listed-price per-consist floor (volume 100): 0.70-1.40 M USD with assembly allowance, base approximately 0.90 M USD.

This table is an audit lower bound for raw procurement. City CAPEX uses the local-owner rolling-stock planning unit in `lib/templates/capex-costs.toml`: 1.0 M USD per `light-metro-3car` trainset. That unit includes local assembly/labour, nominal per-train QA/acceptance evidence, and modest local handover logistics. Fixtures/tooling live in the railway production plant line; warranty, spares, and routine commissioning support are OPEX.

For comparison, legacy-vendor light-metro trainsets in the target regions typically land 4-6 M USD each. The marketplace floor shows how cheap the commodity hardware can be, but it is not a certified rail quote; supplier qualification, local production overhead, plant setup, QA, and acceptance testing will lift any production purchase above this floor.

Per-deployment customisation

The operator's own procurement office fills in:

- Livery paint scheme (B20).
- Seat fabric colour (B14).
- TRG-1 5G carrier SIM spec (per-country mobile network).
- Public-safety radio band (E-: not in base BOM).
- Station contact adapter geometry only where a legacy pilot cannot use the standard side contact; voltage and cabinet architecture are not customized.

v2 deliverables (not in v1)

- Supplier shortlist per BID line with qualification criteria.
- Lead-time analysis per SOURCE / MAKE / BID.
- Risk log: single-source parts + mitigation.

- Weight budget with per-line tare contribution + final target vs actual.

Cabin fiberglass and phenolic interior parts

Repository source: `docs/rolling-stock/light-metro-3car/cabin-fiberglass.md`

This page controls the non-structural fiberglass, phenolic, and glass/basalt-fibre composite parts inside the `light-metro-3car` passenger cabin. The steel body remains the certified load path. These parts are sacrificial, replaceable, fire-rated passenger liners and fairings.

The design intent is simple: the same regional composite cell that can make or repair the end cowls can also make the cabin liners, battery strake covers, and vestibule trim without expensive rail-OEM tooling.

Controlled cabin fiberglass items

Product-tree ID	Item	Qty basis	Function	Parent
`LM3-INT-P020`	FRP/phenolic ceiling liner, light trough, and HVAC plenum cover set	1 car kit/car	Covers roof bows, carries lighting trough openings, and provides removable HVAC/service access	`LM3-INT-SA330`
`LM3-INT-P030`	FRP/phenolic sidewall liner, window reveal, and cable-cover panel set	2 side kits/car	Hides secondary structure and cable trays while preserving window replacement access	`LM3-INT-SA330`
`LM3-INT-P040`	FRP battery strake covers, seat-base fairings, and service-hatch shells	1 car kit/car	Covers under-seat battery strakes, protects passengers from HV bay covers, and keeps battery access removable	`LM3-INT-SA330`
`LM3-INT-P050`	FRP vestibule kick panels, PRM ramp/step covers, and door-pocket trims	4 door-zone kits/car	Protects high-wear door/PRM transition zones and trims the low-floor/high-floor step faces	`LM3-INT-SA330`

All items are non-structural. They may carry local light fittings, labels, small service covers, or trim clips, but they do not carry seat loads, grab-rail loads, jacking loads, crash loads, battery loads, or door loads.

The buildable product tree further separates light troughs (`LM3-INT-P021`), HVAC/service ceiling pieces (`P022`), window reveals (`P031`), cable covers (`P032`), seat fairings (`P041`), PRM/step pieces (`P051`), and door-pocket trims (`P052`). The complete part/mould schedule is in `dedicated-parts-and-moulds.md`.

Material system

Use one of these material routes per part family, frozen by the v2A drawing and supplier evidence pack:

Route	Use	Nominal construction	Notes
Hand lay-up or vacuum-infused FRP	Curved ceiling, sidewall, battery strake, and door-pocket shapes	Fire-retardant E-glass or basalt cloth with vinyl-ester or phenolic-compatible resin	Good for modest tooling and local repair
Flat phenolic composite board	Flat liners, cable covers, access doors, and kick panels	EN 45545 HL2 candidate phenolic/glass board, CNC trimmed	Lowest risk where single curvature is not required
Lightweight sandwich FRP	Broad ceiling or sidewall panels where oil-canning is visible	Thin FRP skins over fire-rated foam/honeycomb core	Keep solid laminate at all inserts and edges

Passenger-facing materials need EN 45545-2 HL2 or the deployment authority's accepted equivalent. Do not install uncertified decorative films, foams, sealants, adhesives, or edge trims in the cabin.

Design rules

- Minimum visible edge radius: 3 mm; 6 mm preferred in standing areas.
- No exposed glass fibres: all cut edges are sealed after trimming.
- Use removable panels wherever the panel covers lights, HVAC drops, cable trays, battery access, fire detection, or emergency equipment.
- Use potted stainless, brass, or GFRP inserts only in solid laminate pads. Do not put inserts directly into foam/honeycomb core.
- Keep panel gaps at 3-5 mm nominal with dark shadow seals; avoid tight cosmetic gaps that become rattles after thermal cycling.
- Use captive or retained fasteners in service panels so hardware does not fall into the saloon.
- Use anti-slip surfacing and yellow/contrast nosing on PRM ramp, threshold, and step-cover pieces.
- Keep HV warning labels and service-lock labels on the removable battery strake cover, not on separate loose trim.
- Do not bond cabin trim permanently across any battery, HV, fire, HVAC, door, or window service path.

Tooling package

Tooling ID	Tool	Applies to	Acceptance
LM3-TL-INT-10	Ceiling liner buck and trim fixture	LM3-INT-P020	Light trough and HVAC openings within ±1.0 mm
LM3-TL-INT-11	Sidewall/window reveal buck and drill fixture	LM3-INT-P030	Window reveal gap 3-5 mm; fastener pitch within ±0.75 mm
LM3-TL-INT-12	Battery strake cover mould and service-hatch gauge	LM3-INT-P040	Hatch opens without seat removal; HV label visible
LM3-TL-INT-13	Door/PRM transition trim fixture	LM3-INT-P050	PRM transition and anti-slip witness accepted

The first article may use low-cost MDF/epoxy tooling or machined foam patterns sealed with epoxy tooling coat. Production tooling should be female moulds for curved visible parts and CNC trim nests for flat phenolic panels.

Making instructions

1. Release package

1. Freeze the panel envelope, insert map, visible gap, service-removal direction, and fire-rating route.
2. Check material certificates for resin, reinforcement, board stock, adhesive, sealant, decorative coating, and edge trim.
3. Issue the traveler and verify the parent carbody/interior datum.
4. Cut a witness coupon for every laminate batch and cure cycle.

Hold point: fire-material certificates and laminate/panel batch records accepted before lay-up or CNC trimming.

2. Mould or board preparation

1. Clean the mould/fixture and inspect for scratches, release buildup, or previous repair.
2. Apply release wax/PVA/semi-permanent release per supplier procedure.
3. For phenolic boards, stage the CNC spoilboard and vacuum hold-down without contaminating passenger-facing surfaces.
4. Mark every blank with product ID, car number, side, and batch.

Hold point: mould/fixture release record complete.

3. Lay-up or trim

1. Cut dry reinforcement plies with fibre directions marked.
2. Lay visible gelcoat or interior finish coat where the process uses in-mould finish.
3. Lay outer plies, core where allowed, local solid-laminate pads, and inner plies.
4. Wet out by hand lay-up or vacuum infusion; avoid resin-rich pools around inserts and tight returns.
5. Cure under controlled temperature/time. Record ambient temperature, resin batch, pot life, vacuum level if used, and cure time.
6. Demould only after the resin supplier's minimum cure condition.
7. Trim, drill, and edge-radius the panel in its trim fixture.

Hold point: coupon, cure, visual void/dry-fibre inspection, and trim-line gauge accepted.

4. Inserts, edges, and finish

1. Pot threaded inserts into solid laminate pads or bonded doubler pads.
2. Seal all drilled and trimmed edges.
3. Apply decorative paint/film only after fire-material compatibility is accepted.
4. Add anti-slip witness panels, yellow/contrast nosing, labels, and QR/part ID labels.
5. Fit gaskets, clips, retained fasteners, and access-panel tethers.

Hold point: insert pull-out sample, edge-seal inspection, and label placement accepted.

5. Trial fit and release

1. Dry-fit each panel in the carbody/interior fixture before installing seats, grab rails, and lighting.
2. Check window replacement access, battery hatch removal, lighting service removal, HVAC access, cable-tray access, and emergency equipment visibility.
3. Shake/rattle check by hand before the first static electrical test.
4. Record shim packs, clip counts, fastener torque where applicable, and any approved deviations.

Hold point: egress gauge, sharp-edge inspection, service-removal trial, rattle check, and fire-material pack complete.

Repair instructions

- Cosmetic scratches: sand, clean, fill with compatible fire-rated repair resin/filler, refinish, and record repair location.
- Cracked non-service panel: remove and replace if crack crosses an insert, edge return, access door, or passenger touch zone.
- Battery strake or PRM panel damage: replace the panel, then repeat HV-label, service-hatch, anti-slip, and sharp-edge checks.
- Fire, smoke, or heat exposure: quarantine the panel and replace unless the material supplier provides a signed repair disposition.

v2A drawing outputs

The v2A drawing pack shall release:

- panel envelope and visible-gap drawings,
- laminate/board schedule,
- insert and clip map,
- CNC trim DXF or mould trim curves,
- service-removal direction and access envelope,
- edge radius and edge-seal procedure,
- anti-slip/nosing/label map,
- first-article inspection form,
- repair and replacement limits.

Compliance matrix — light-metro-3car

Repository source: `docs/rolling-stock/light-metro-3car/compliance.md`

Standards the consist is designed against. Every row is a type- approval campaign the operator runs pre-revenue. Upstream provides the design envelope; each deployment owns its own test-house relationship.

Structural + mechanical

Standard	Scope	Evidence for v1 v2
EN 15227 Cat C-II	Crashworthiness at 25 km/h metro scenario	FEA simulation report (v2)
EN 12663 Cat P-III	Static + fatigue body loading	FEA report (v2)
EN 14363	Dynamic behaviour + safety against derailment	On-track test campaign (v3)
EN 13749	Bogie frame fatigue + structural	FEA + test-rig (v2/v3)
EN 15085 CL1–CL4	Weld quality per assembly	Per-welder EN ISO 9606 certificates (v2)
UIC 510-2	Wheel profile + wear limits	Monobloc certificate from supplier (v1 source)
UIC 505-1	Structure gauge	General-arrangement compliance shown (v1)

Fire + materials

Standard	Scope	Evidence
EN 45545-2 HL2	Fire behaviour of materials	Per-material test reports from an accredited lab (v2 campaign)
Applied to: body, seats, flooring, cabling, insulation, door rubbers		

Electrical + electronic

Standard	Scope	Evidence
EN 50155 OT4	On-board electronic equipment ambient class	Per-PCB compliance (RFC 0007 v1 bring-up)
EN 50121-3-2	Rolling-stock EMC	Pre-compliance sweep on test car (v3)
EN 50126	RAMS lifecycle	SIL-4 safety case — links into `docs/safety-case/`
EN 50716:2023	Railway software development — SIL-4 target	Coding standard per RFC 0005 §7; evidence per crate; EN 50128/50657 retained only where a national transition mapping requires them
EN 50129	Safety-related electronic systems — SIL-4	Composite fail-safe argument for T-ECU/S (RFC 0007 §4.1)

Passenger experience + accessibility

Standard	Scope	Evidence
EN 12299	Ride-comfort indices	Dynamic-ride test (v3)
ISO 3095	External + internal noise	Acoustic measurement (v3)
UIC 741	Platform-to-train gap	General arrangement + gap-filler flap spec (v1)
EN 14752	Door systems	IFE type-4 door or equiv. (v1 source-identified)
EN 16586	Accessibility PRM	Wheelchair-space layout and COTS fit-out envelopes (v1 — see COTS catalogue)
EN 15152	Windows	Laminated safety glass (v1 source-identified)

Onboard safety systems

Standard	Scope	Evidence
IEC 62278-1 / EN 50716 / EN 50129 as above for the SIL-4 target	Every onboard safety evaluator has proptest + Kani evidence	Per-crate solution nodes in the GSN safety case; physical, process and assessor evidence remain open
EN 14033	Rail infrastructure works — applies to self-propelled work consists (n/a for revenue service)	n/a

Target-region overlays (per-deployment)

Each country's NRSA (or equivalent) has its own approval process. Common overlays:

- Iraq (RFC 0003 Samawah): Iraqi Railways directorate approval.
- GCC region: GCC rail-authorities common spec.
- Sub-Saharan Africa: per-country; AU railways coordination underway.

v1 does NOT enumerate overlays — that is the operator's NRSA-liaison scope. Upstream provides the EN-standard base.

Test-campaign ordering

A typical sequence from design freeze to revenue:

1. v2 CAD release: FEA reports (EN 15227, EN 12663, EN 13749).
2. v2A drawing register release: controlled drawings, supplier installation documents, and evidence gates issued per `drawing-register.md`.
3. First-article prototype: EN 45545 material test campaigns.
4. Static test on bogie rig: EN 13749 frame fatigue.
5. Dynamic test (pilot deployment): EN 14363 derailment safety; EN 12299 ride comfort; ISO 3095 noise; EN 50121-3-2 EMC.
6. Safety-case sign-off: EN 50126 lifecycle file complete.
7. NRSA approval.
8. Revenue service.

Steps 1–5 are fabricator + accredited-test-house scope. Steps 6–7 are operator + project-safety-team scope.

What v1 commits to

- Source-identified parts where such parts exist (Scharfenberg coupler, IFE door, PZ pantograph, EN 45545 materials).
- BID scope clearly delimited (motor, inverter, battery pack).
- MAKE scope matched to local fabricator capability (cut/bend/weld body frame, bogie frames, brackets, wiring harness).
- No proprietary/closed interfaces anywhere in the consist.

What v1 does NOT commit to

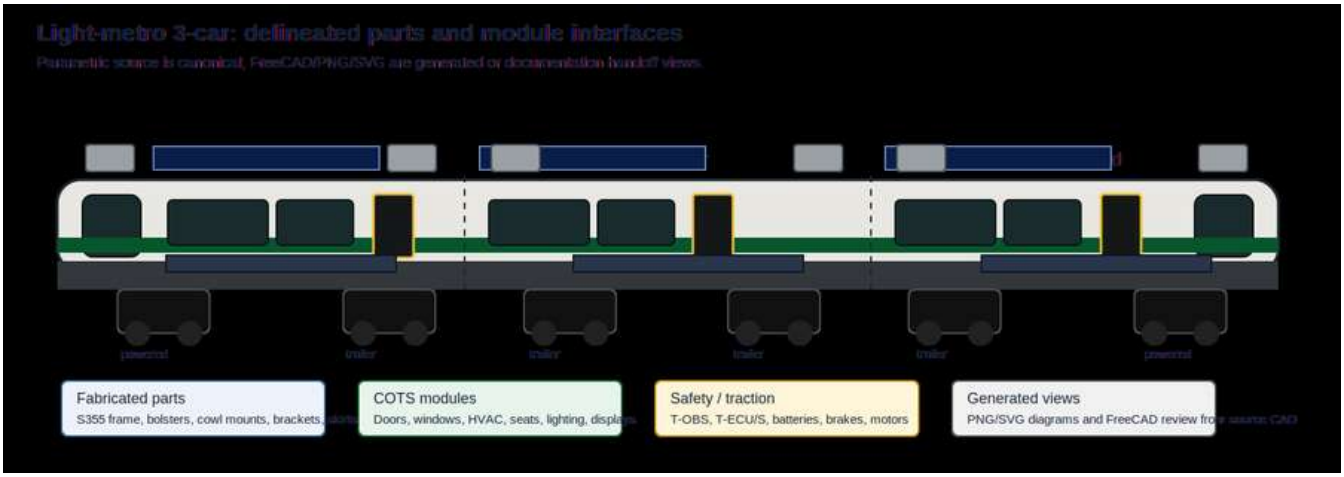
- Specific test-house accreditation (per-country choice).
- Specific national-regulatory overlay fitments (flag signage, emergency PA language, fare-gate integration — all per-deployment).
- Second-source data for BID parts (v2).

COTS Integration And Part Delineation

Repository source: `docs/rolling-stock/light-metro-3car/cots-integration.md`

This page shows how off-the-shelf rail/bus components join to the fabricated `light-metro-3car` body, bogies, cowls, and electrical systems. The parametric source remains the design basis; these diagrams are the readable integration map.

Whole-Train Part Map

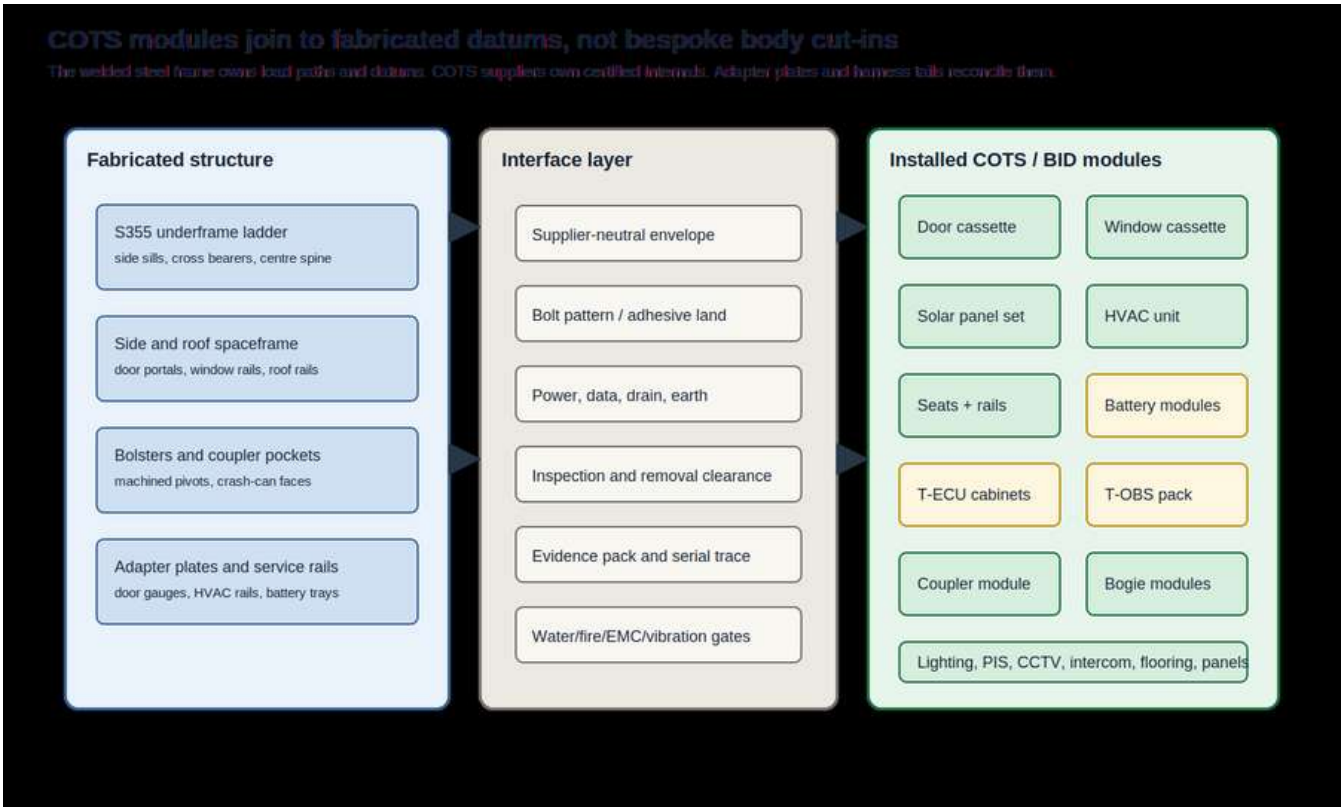


Annotated trainset part map

The train is built from three repeated 16.5 m car modules:

Zone	Fabricated parts	COTS / BID modules	Parametric source
End cowls	Steel crash frame, multi-part fiberglass A/B-end cowl casts, cowl backing ring, single panoramic glass carrier, LED lamp brackets	Dark RF-transparent laminated end glass pane, T-OBS LIDAR/radar/camera/ultrasonic pack, LED headlamps, marker/DRL bars, washer/heater	`end-cowl.md`, `sensor_cowl.py`, `systems.py`
Body side	S355 side frame, two door portals per side, window rail, waist rail, composite skin	Window cassettes, door cassettes, green livery band, yellow thresholds	`car_body.py`
Stepped floor	Dropped ~10 m low-floor centre pan, ~3 m raised bogie-end deck supports, side plinth rails, transition steps	Phenolic/aluminium floor boards, rubber covering, step nosings, PRM floor finish	`car_body.py`, `cad_templates/rolling_stock.py`
Roof	Roof bows, PV/HVAC rails, bonded-pad lands, cable tray brackets, composite fairings	Bonded flexible PV laminates, raised rigid panels on clamped rails, MPPT combiner, compact HVAC units, antennas	`car_body.py`, `systems.py`
Under-seat bay	Battery tray rails, service covers, outward vent path, mist-pipe/nozzle datums, seat support rail	225 kWh-per-car LFP modules, BMS, fuses, contactors, mist pump/sensors, longitudinal seat modules	`car_body.py`, `systems.py`
Underframe	Side sills, centre spine, cross bearers, bolsters, jacking pads	Six motor controllers, isolated LV DC/DC, protected station/PV DC interfaces, HV cabling, cooling loops, brake/WSP harnesses	`cad_templates/rolling_stock.py`, `systems.py`
Running gear	Bogie adapter, motor cradle, torque-link brackets	Three powered bogies, three trailer bogies, wheelsets, brakes, bearings, air springs	`bogie/`
Train ends and inter-car	Coupler pocket, shear plate, articulation adapter frames, underframe anchor castings, upper clevis brackets, trainline brackets	Scharfenberg Type 10 couplers, crash absorbers, lower spherical articulation joint, upper links, bellows, turntable floor, energy-guidance chains	`systems.py`, `articulation.md`

Interface Stack

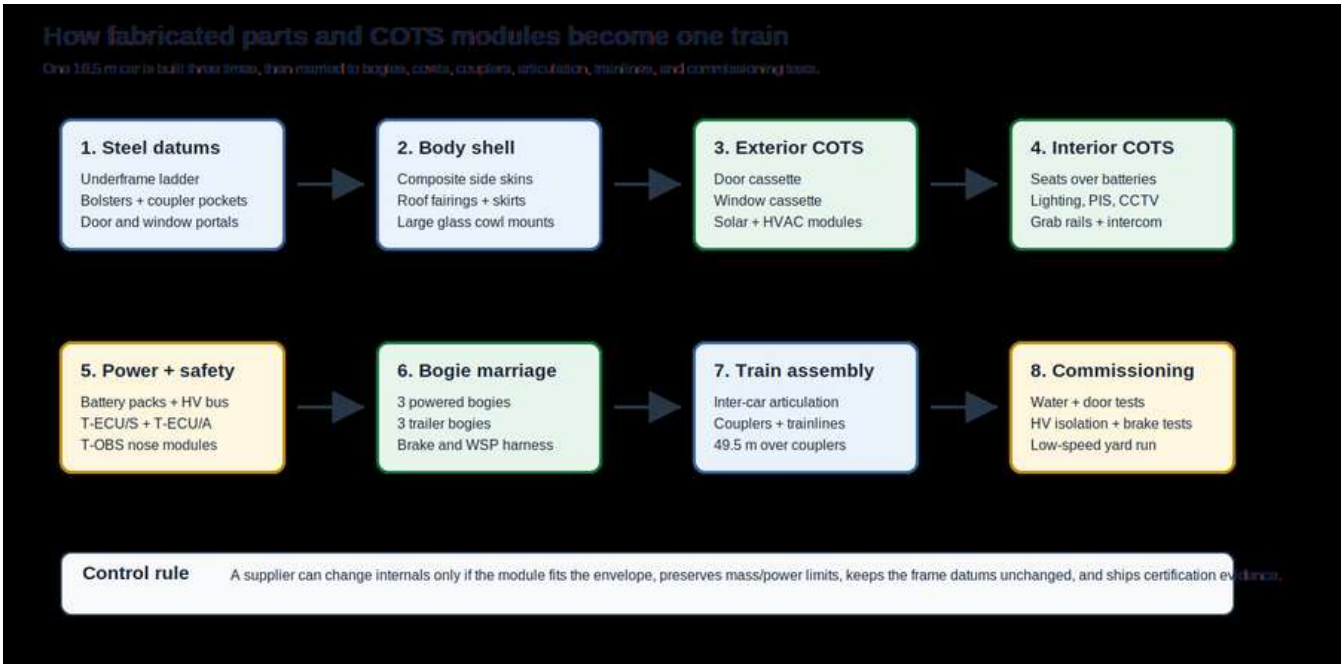


COTS modules joining fabricated structure

Every COTS module is installed through a supplier-neutral interface. OpenSourceRail owns the envelope, datum, power/data/drain/earth interfaces, maintainability clearance, and evidence requirements. The supplier owns detailed internals, service procedure, lifecycle tests, and certification data.

COTS module	Fabricated datum	Interface closure	Evidence required
Window cassette	Window rail and bonded aperture	Adhesive bead or gasket, drain path, bonded earth if heated	EN 15152 or accepted equivalent, fire/smoke data for seals
Door cassette	Door portal, threshold tray, lock-loop bracket	Bolted frame, 24/110 V DC, Ethernet/CAN, hardwired closed/locked loop, drain	EN 14752 or equivalent, obstruction detection, lifecycle test
Platform/PSD interface	Door centre datum and sill edge	ATO stopping target, platform screen-door alignment, closed/locked interlock, intrusion sensor sightline	Station integration test, door-open permissive proof, degraded-mode procedure
Rooftop PV, mounts, and air cleaner	Bonded-pad lands, roof PV rails, edge-clamp datums, cable gland, blower/nozzle hardpoints	Six bonded flexible laminates and six raised rigid modules per car, isolated combiner, MPPT feed, fire disconnect, filtered low-pressure air-knife cleaner	IEC PV module data, adhesive/bond coupon test, rail vibration mount evidence, blower IP/noise data, filter access proof, soiling-recovery test
Station/PV DC interfaces	Underframe charge-rack rails and cooling ports	Per-car roof MPPT, protected station DC input, contactors, precharge, CAN/Ethernet	Rail power-electronics datasheet, insulation/earthing analysis, EMC/vibration evidence, charge-state validation
HVAC unit	Roof equipment rails and condensate drain	Bolted rail pattern, molded/removable duct adapter, direct 650–700 V DC, CAN/Ethernet diagnostics	50 °C hot-climate curve, vibration, EMC, refrigerant record
Seat module	Battery-cover seat rail	M10 cantilever bracket, service-lid clearance below cushion	EN 45545 R7, static strength, vandal-resistance data
Battery module	Under-seat tray and vent plenum	HV connector, coolant quick-disconnect, BMS harness, side vent	Cell/pack test report, isolation, fire containment, lifting procedure
T-ECU cabinet	End electronics cabinet rail	DIN rail, 24/110 V DC, TCN-E, CAN-FD, EB loop, earth braid	Board spec, commissioning self-test, serial and firmware record
T-OBS pack	Nose cowl optical/radar datum	Heated window, service fasteners, power/data, safety verdict to T-ECU/S	Sensor calibration, 2oo2 verdict test, washer/heater test
Coupler	Coupler pocket and shear plate	M24 bolt pattern, drawgear carrier, brake/electric head clearance	EN 15227 absorber data, rescue procedure, inspection interval
Articulation/gangway	Adapter frame, lower anchor casting, upper clevis brackets	Lower spherical pivot, anti-lift keeper, upper links, bellows clamp frame, turntable floor, separated HV/data/coolant/HVAC routes	Motion-envelope test, bearing proof, EN 45545 bellows/turntable data, water ingress and drain test
Bogie	Bolster pivot and air-spring pads	Centre pin, PTFE slider, yaw links, brake/WSP/traction harness	Wheelset certs, brake test, frame weld records, suspension data

Assembly Sequence



Assembly sequence from fabricated parts to complete train

The sequence is intentionally ordinary for a regional manufacturer: fabricate datum structure first, preserve supplier-neutral interfaces, install COTS modules late, then close the train with test evidence.

Delineated Part Register

Part family	Visible in design	Procurement line(s)	Source of truth
Windows	Dark side glazing rectangles in side walls	B10	`car_body.py`, `cots_equipment.py`
Fiberglass end-cowl casts	Identical A/B-end upper brow, cheek casts, lower apron, backing-ring flanges, and lamp/service hatches	B8	`end-cowl.md`, `sensor_cowl.py`
Doors	Two double-leaf black doors with yellow thresholds per car side	B11, B25	`car_body.py`, `systems.py`
Solar panels, air cleaner, and MPPT	Mixed bonded/raised roof PV modules, filtered air pump/manifold, roof combiner, MPPT in every car systems rack	T21, T22, T23	`car_body.py`, `systems.py`
HVAC systems	Compact grey roof modules at car ends	T14	`car_body.py`, `cots_equipment.py`
Batteries	Under-seat blue module rows, eight per car in CAD	T5-T8, T16-T20	`car_body.py`, `systems.py`
Seats	Longitudinal COTS benches over battery covers	B14, A1	`cots_equipment.py`
Grab rails	Stainless saloon stanchions and rails	B15	`cots_equipment.py`
PIS / CCTV / intercom	Saloon and doorway information/safety modules	B18, B19, E14, E15, E20	`cots_equipment.py`, `systems.py`
Electronics cabinets	End-cabinet T-ECU/S, T-ECU/A, recorder modules	E1, E2, E22, E23	`systems.py`
T-OBS sensor packs	Nose LIDAR/radar/camera/ultrasonic module	E18, E19	`sensor_cowl.py`, `systems.py`
Couplers and crash absorbers	End coupler head, shear plate, absorber cartridge	B22, B23	`systems.py`
Articulation	Lower spherical joint, drawbar, anti-lift keeper, upper links, bellows, turntable, drag-chain and service loops	B9, B24, B29	`systems.py`, `articulation.md`
Powered bogies	Three powered bogies, one under each repeated car module	G1, G18, G19, T1-T3	`bogie/`, `trainset.py`
Trailer bogies	Three trailer bogies, one under each repeated car module	G2-G17, G20	`bogie/`, `trainset.py`

Generated Design Views

The same part families are visible in the generated screenshots:

View	What to inspect
`trainset-light-metro-3car.png`	Whole-train layout, repeated self-contained cars, single panoramic-glass cowls, roof PV, bogies

View	What to inspect
trainset-car-detail.png	Complete layered car body with windows, doors, solar array, HVAC, interior, electrical, and thermal routes
trainset-car-body-structure.png	Primary fabricated shell, ~10 m floor, side sills, crossmembers, roof rails, window posts, end rings, bogie envelopes, and door portals
trainset-car-body-bogie-subassembly.png	Single-car body structure mounted over standard motor/trailer bogies, showing raised ~3 m end decks and the low-floor centre zone
trainset-car-body-exterior.png	Exterior glazing, door leaves, livery band, skirts, mixed bonded/rail-mounted solar array, and compact HVAC roof units
trainset-roof-solar-system.png	Rooftop PV package: bonded flexible laminates, raised rigid panels, mounting rails, edge clamps, junction boxes, isolators, MPPT combiner, and downlink gland
trainset-car-body-interior.png	Under-seat battery strakes, seats, wheelchair bays, grab poles, handrails, and PIS
trainset-car-body-services.png	HVAC ducting, LV/data trays, lighting, CCTV/intercoms, HV/PV routing, coolant, and fire vent paths
trainset-interior-fit-out.png	COTS passenger fit-out envelopes inside the body reservation
trainset-car-systems.png	Batteries, four door cassettes, platform safety interfaces, roof PV/MPPT package, DC charge/protection rack, charging connector, wheelchair bays, systems layout
trainset-door-system.png	Door leaves, operator rail, lock/release, gap filler
trainset-battery-pack.png	Battery module set, HV contactor, fuse, BMS cabinet
trainset-electronics-cabinet.png	T-ECU/S, T-ECU/A, event recorder, power distribution
trainset-inter-car-articulation.png	Inter-car lower spherical joint, upper links, bellows, turntable floor, energy guidance, and kinematic clearance envelopes
trainset-tobs-sensor-pack.png	LIDAR, radar, stereo camera, ultrasonic sensors
bogie-motor.png	Powered bogie with motors, gearbox, suspension, brakes
bogie-trailer.png	Trailer bogie envelope

Dedicated body, chassis, fascia and interior parts

Repository source: `docs/rolling-stock/light-metro-3car/dedicated-parts-and-moulds.md`

This page is the design-reference split of the LM3 body and fitout into parts that a local factory can identify, fixture, inspect, replace and cost. It is not a released drawing pack: material grades, laminate schedules, weld classes, fastener values, structural calculations and supplier interfaces still pass through their stated release gates.

Chassis and body steel

Product ID	Dedicated part family	Manufacturing boundary	Principal fixture/evidence
`LM3-BDY-P010`	left/right side sills	Longitudinal primary members	`LM3-TOOL-STEEL-FIXTURE`; heat and straightness trace
`LM3-BDY-P020`	centre spine and longitudinal load-path kit	Kept separate from transverse members until fixture load	Spine straightness and bogie-centre datum survey
`LM3-BDY-P021`	cross-bearers, door outriggers and equipment brackets	Nested and station-marked repeated transverse parts	Cross-bearer profile gauge, station map and tack survey
`LM3-BDY-P030`	bolster boxes, spring pads and pivot inserts	Machined datum set after qualified weld	Line-bore/NDT and spring-datum survey
`LM3-BDY-P060`	low-floor centre pan and service-floor supports	Low saloon floor and removable equipment access	PRM-height, aisle and pan-weld inspection
`LM3-BDY-P061`	raised bogie-end decks, transition ramps and hatch frames	Two removable end-zone sets per car	Deck/transition gauge and hatch-removal trial
`LM3-BDY-P070`	side posts, door portals, waist and cant rails	Two side frames per car	Door/window gauges and side-frame survey
`LM3-BDY-P080`	roof bows and equipment/cable rails	Structural roof datum below removable skins	HVAC/PV pitch and bracket proof evidence

The steel frame remains the certified crash, buff, passenger, bogie, battery, jacking and recovery load path. GFRP, liners, film and fairings never substitute for these members.

One-metre exterior part family

The 144-module trainset count is retained, but the engineering tree no longer hides all four variants in one line.

Product ID	Variant	Reference quantity/trainset	Controlled distinction
`LM3-BDY-P130`	solid side	48	Unpierced side/weather panel; final mix follows the bay map
`LM3-BDY-P131`	window edge	24	Sealed reveal edge, window drain and cassette-removal clearance
`LM3-BDY-P132`	door edge	24	Door-pocket closeout, threshold drain and door-service sweep
`LM3-BDY-P133`	roof skin/fairing	48	HVAC, PV, antenna, access and drain trim variants

All variants retain the 1,000 mm mould pitch, 994 mm finished width, clip datum, independent anti-lift retainer, 6 mm dry EPDM joint and master-frame dry fit. Production quantities are configuration-controlled by the final door/window bay map; the four reference quantities sum to 144.

Interior fitout split

Product ID	Serviceable part family	Must remain accessible
`LM3-INT-P020`	main ceiling liners	Light, diffuser, detector and service apertures
`LM3-INT-P021`	light-trough bezels and diffuser carriers	Plug-in main and independent emergency lights
`LM3-INT-P022`	HVAC transitions, detector bezels and ceiling hatches	Air balance, detectors, cables and drain inspection
`LM3-INT-P030`	main sidewall liners	Side structure and window interface
`LM3-INT-P031`	four-piece window reveals	Glass edge, setting blocks and complete cassette removal
`LM3-INT-P032`	waist cable covers and LV/data access lids	Segregated harnesses and retained service loops
`LM3-INT-P040`	battery-strake covers	HV barrier labels and exterior-only pressure relief
`LM3-INT-P041`	seat-base fairings and equipment hatches	Structural seat saddles, isolation and cleaning access
`LM3-INT-P050`	vestibule kick and threshold closeouts	Door drains and threshold hardware
`LM3-INT-P051`	PRM transition/step covers, nosing and anti-slip pieces	Structural floor datum and trip-edge inspection
`LM3-INT-P052`	door-pocket liners and jamb covers	Certified door sweep, sensitive edge and emergency access

Every interior moulding is non-structural. Seats and handrails connect through calculated saddles and `OSR-RAIL-42`, never through the finish panel. All cabin materials, films, adhesives and edge treatments require the project fire/smoke evidence route.

Fascia, panoramic glass and lamps

The six cowl casts remain LM3-CWL-P011 through P016. Three dedicated fitout kits now define what sits behind them:

Product ID	Fitout	Design rule	Release check
LM3-FAS-P010	panoramic glass carrier, setting-block pockets and secondary retention	Transfers pane loads to the steel-backed ring; cowl skin is not structural	Carrier survey, edge-clearance gauge, retention proof and pane-removal trial
LM3-FAS-P020	reversible lamp cassette tray and aiming adjusters	Same physical end kit leads or trails; service hatch removal must not lose datum	Lamp gauge, aim/retention, harness clearance and removal trial
LM3-FAS-P030	EPDM seals, drain rail, washer sleeves and edge closeouts	Replaceable weather boundary with visible earths/connectors	Seal batch, compression map, drain flow and water test

The panoramic glass supplier still owns glass construction, heater/busbar and its adhesive or cassette process. The lamp supplier owns photometry, EMC/IP and thermal evidence. The OSR parts own the repeatable mechanical datums and the ability to remove either system without destroying the cowl.

Dedicated mould and fixture set

Tool ID	Purpose
LM3-TOOL-SIDE-MOULD	common 1 m side-module female surface
LM3-TOOL-SIDE-VARIANT-NEST	solid/window/door CNC trim and drill variants
LM3-TOOL-ROOF-MOULD	common 1 m roof skin surface
LM3-TOOL-ROOF-FAIRING-MOULD	HVAC curb, PV gland, antenna and hatch inserts
LM3-TOOL-COWL-MOULD	split brow, cheek, apron, hatch and backing-flange mould family
LM3-TOOL-GLASS-CARRIER-NEST	glass carrier, seal compression, sill drain and setting-block gauge
LM3-TOOL-LAMP-AIM	reversible cassette datum and photometric target carrier
LM3-TOOL-INT-CEILING-MOULD	ceiling, light and HVAC aperture family
LM3-TOOL-INT-SIDE-MOULD	sidewall, window reveal and waist-cover family
LM3-TOOL-INT-STRAKE-MOULD	battery strake, seat fairing and hatch family
LM3-TOOL-INT-DOOR-PRM-MOULD	vestibule, PRM/step and door-pocket family

First-article MDF/epoxy or sealed machined-foam patterns are acceptable only when the released resin/process permits them. Production visible surfaces use female tools, removable inserts and CNC trim nests. A mould drawing must fix draft, split lines, trim curves, flange returns, insert locations, vacuum/resin ports, cure support and a dimensional inspection method.

Remaining locally made systems and their drawing homes

The factory package now assigns every local MAKE row to at least one drawing brief. This prevents brackets, carriers and harnesses that sit between major assemblies from being left as unwritten shop-floor decisions.

Local scope	Product IDs	Drawing seeds	Controlled result
Coupler, battery, door and window primary interfaces	LM3-BDY-P040, P050, P090, P100, P110	LM3-BDY-130, LM3-BDY-140	Calculated load paths, machined datums, weld/NDT map, supplier keep-outs and replacement clearances
Replaceable door/window carriers	LM3-DOOR-P010, LM3-WIN-P010	LM3-DOOR-200, LM3-WIN-210	Adjustable carriers, setting blocks, keyed dry seals, drains, gauges and timed cassette replacement
Configurable end and articulation adapter	LM3-END-P060–P062, LM3-END-P030, LM3-ART-P010	LM3-END-650, LM3-SYS-170	One common end datum, one selected option per position, anti-lift retention, service access and full-motion proof
Powered bogie local structure/services	LM3-BOG-P010, P030, P050, P060	LM3-BOG-400	New H-frame, motor cradle/torque link, guards, sensor brackets and protected harness route
Trailer bogie local structure/services	LM3-BOG-P020, P031, P061	LM3-BOG-410	New H-frame, guards, brake/WSP brackets and protected harness route
Battery, HV and coolant local hardware	LM3-HV-P010–P030	LM3-HV-310, LM3-HV-320	Sliding trays/restraints, exterior vent/drain, segregated covered HV route, bonds, clamps, bleed/drain and test points
LV trainline and distribution	LM3-CTRL-P040	LM3-ELC-300	Harness-board definition, wire/terminal schedule, branch and clamp coordinates, service loops, labels and 100% continuity test

These are drafting and verification homes, not released dimensions. Supplier interfaces, structural calculations, cable/pipe selections, tolerances and signed first-article records remain mandatory inputs to the corresponding factory package.

Assembly handoff

1. Release and survey the underframe, floor, side and roof structural datums.

2 Dry-build the four exterior variants on the master frame; prove clips, anti-lift devices, seals, drains and window/door service sweeps.

3 Build the six cowl casts, glass carrier and lamp tray on the end fixture; prove A/B interchange before fitting glass or lamps.

4 Pre-fit all interior panels on the bare service rails; demonstrate every window, light, HVAC, cable, seat-base and door service path.

5 Apply the controlled finish system, install bought-in equipment, complete water/rattle/functional tests, and attach signed evidence to the serialized car configuration.

Drawing and evidence register — light-metro-3car

Repository source: `docs/rolling-stock/light-metro-3car/drawing-register.md`

This register is the bridge between the v1 Markdown specification and the v2 CAD/shop-drawing release. It lists the controlled drawings, supplier documents, and test evidence a fabricator must produce before the first article can be released for static and dynamic testing.

Document IDs use `LM3` for the `light-metro-3car` family. Revision `A` is the first v2 release candidate.

The generated `factory-drawings` seed index now provides an individual drafting/checking brief for each of the 29 drawing IDs controlled by the 16 factory packages. Those seeds cover every locally manufactured product row and organize scope; they remain non-dimensioned and unissued until this register's release gates close.

Controlled drawing set

ID	Title	Owner	v1 source	v2 release content
LM3-GA-000	Trainset general arrangement	OSR/mechanical	`general-arrangement.md`	2D GA, clearance envelope, platform interface, mass table
LM3-MASS-010	Controlled trainset mass budget and product closure ledger	OSR/mechanical + suppliers	generated `mass-budget.md`, generated `mass-closure-ledger.md`	Modeled category subtotal, explicit engineering reserve, all 120 product responsibility assignments, supplier/production-CAD/weigh evidence state, lightweighting comparison, individual-car/axle/CG closure
LM3-JNT-020	Joint and fastener control schedule	OSR/mechanical + integrator	generated `joint-control-schedule.md`	Parent/child joint IDs, joining classes, torque authority, locking/re-torque release requirement, numeric values, inspection record
LM3-BDY-100	Carbody primary steel assembly	Fabricator	`body.md`	FreeCAD/neutral CAD package, 2D assembly, weld map, datum scheme
LM3-BDY-110	Underframe ladder assembly	Fabricator	`body.md`	Tube cut list, plate flat patterns, fixture drawing
LM3-BDY-120	Side/roof spaceframe assembly	Fabricator	`body.md`	RHS cut list, door/window aperture datums
LM3-BDY-130	Coupler pocket and crash-can interface	Fabricator + coupler supplier	`body.md`, `interfaces.md`	Machined inserts, bolted energy absorber interface
LM3-BDY-140	Battery tray and under-seat enclosure	Fabricator + battery supplier	`body.md`, `traction.md`	Service hatches, vent path, HV isolation clearances
LM3-BDY-150	Exterior glass-fibre material, mould, and trim-envelope control	Composite supplier + fabricator	`body.md`, `modular-fiberglass-body.md`	Laminate schedule, reusable mould drawings, core/insert maps, CNC trim envelopes, edge radii, fire evidence, repair zones
LM3-BDY-155	Identical A/B-end fiberglass cowl cast kit	Composite supplier + OSR/mechanical	`end-cowl.md`, `sensor_cowl.py`	Surface-modelled exterior A-surface and derived B-surface/flange/trim/mould neutral CAD, CWL-FRP-01 through CWL-FRP-06 mould/trim drawings, laminate schedule, insert map, steel backing-ring datum, glass/lamp/sensor hatch service access
LM3-BDY-160	One-metre clip-on glass-fibre body module system	Composite supplier + fabricator	`modular-fiberglass-body.md`, `modular_fiberglass_body.py`	Common 1,000 mm side/roof moulds, mould release and cure records, solid/window/door/roof trim variants, clip rail and insert map, anti-lift retainer, EPDM joint/drain, module numbering, master-frame dry fit, timed one-shift installation route
LM3-BDY-165	Exterior module variant trim and configuration map	Composite supplier + fabricator	generated `factory-release-work-packages.md`	Solid/window/door/roof CNC trim and drill definitions, serialized bay map, clip/seal datums, replacement and repair zones
LM3-FAS-180	Panoramic glass carrier, seal and drainage interface	OSR/mechanical + glazing supplier	`end-cowl.md`, generated `factory-release-work-packages.md`	Steel-backed carrier segments, setting blocks, secondary retention, pane-edge clearances, compression map, drain/washer/heater routes and removal path
LM3-FAS-185	Reversible front-lamp cassette and aiming interface	OSR/mechanical + lamp supplier	`end-cowl.md`, generated `factory-release-work-packages.md`	Common A/B tray, retained adjusters, optical-axis datums, harness/earth/drip loops, thermal/service clearances and interchange record
LM3-END-650	Configurable train-end interface	Fabricator + integrator	`articulation.md`, generated `train-end-interface.md`	Common end carrier ring, option bolt grid, seal/drain datums, panoramic closeout option, open mid-connection portal option, configuration record
LM3-SYS-160	End coupler and crash-energy assembly	Coupler supplier + integrator	`interfaces.md`, BOM	Scharfenberg head, electric-head carrier, crash absorber envelope, recovery/tow interface
LM3-SYS-170	Inter-car articulation and trainline assembly	Integrator + articulation supplier	`articulation.md`, `body.md`, `interfaces.md`	Lower spherical pivot, anti-lift keeper, upper roll-yaw-pitch links, bellows, turntable floor, drag-chain, TCN-E/CAN-FD/HV/coolant/HVAC interfaces
LM3-SYS-175	Train-to-train open-end articulation	Integrator + articulation supplier	`articulation.md`, generated `train-end-interface.md`	Open-end gangway cassette, semi-permanent drawbar, lower/upper articulation, turntable threshold bridge, service-jumper transition, blanking covers, motion sweep
LM3-DOOR-200	Door cassette installation	Door supplier	COTS catalogue	Mounting datums, threshold, drainage, lock-loop wiring

ID	Title	Owner	v1 source	v2 release content
LM3-WIN-210	Window cassette installation	Glazing supplier	COTS catalogue	Bond/gasket land, drain path, replacement method
LM3-HVAC-220	Roof HVAC installation	HVAC supplier	COTS catalogue	Roof rails, ducting, condensate, service clearance
LM3-ROOF-225	Roof fairing, penetration, equipment and service-zone coordination	OSR/mechanical + integrator	`roof-fitout.md`, generated `factory-release-work-packages.md`	Curb/fairing/rail/pad/gland/bond drawings, penetration schedule, HVAC/PV/antenna removal paths, finish/anti-slip/heat/electrical keep-outs
LM3-INT-230	Interior fit-out installation	Integrator	COTS catalogue	Seats, rails, floor boards, hatches, panels, signage
LM3-INT-231	Interior moulded-panel, trim and service-access family	Interior fabricator + integrator	`dedicated-parts-and-moulds.md`, generated `factory-release-work-packages.md`	Ceiling/light/HVAC, sidewall/reveal/cable, battery/seat, threshold/PRM/door-pocket part drawings, moulds, trim nests and removal sequence
LM3-FIX-235	Common rail, fastener and fixture-adapter family	OSR/mechanical + integrator	generated `small-component-standard.md`, generated `factory-release-work-packages.md`	Rail/foot definition, seat/handrail/equipment adapter variants, grip/locking/torque schedule, installed coordinate map and fixture-specific proof route
LM3-FIN-240	Pre-cut livery film artwork, application and repair pack	Operator + finish supplier + integrator	`exterior-finish-process.md`, generated `factory-release-work-packages.md`	Bay-numbered cut files, substrate/application records, seam/edge/keep-out map, batch/coupon trace and local repair/removal instructions
LM3-FIN-245	Radiative roof coating qualification and one-car trial pack	Materials/test authority + integrator	`exterior-finish-process.md`, generated `factory-release-work-packages.md`	Controlled formulation/application, coupon ageing/optical plan, paired roof-zone thermal trial, maintenance/repair procedure and baseline-finish fallback
LM3-REC-270	Vehicle jacking, lifting, towing and field-rerailing interface	OSR/mechanical + recovery engineer	`field-rerailing-concept.md`, generated `factory-release-work-packages.md`	J1–J4 and adapter drawings, support combinations/reactions, stop conditions, isolation/brake/bogie-retention diagrams and proof/maintenance schedule
LM3-ELC-300	Low-voltage harness routing	Integrator	`interfaces.md`	Harness schedule, connector list, segregation, labels
LM3-HV-310	HV battery/traction routing	Integrator	`traction.md`	HVIL loop, busbars/cables, insulation clearances
LM3-HV-320	Per-car battery pack and charging assembly	Battery + traction suppliers	`traction.md`, BOM	LFP module envelope, HV contactor/BMS cabinet, direct-DC PV/station charge interface, side-pin charge connector, coolant/vent/mist paths
LM3-HV-325	Rooftop PV and charge-input assembly	Solar + traction suppliers	`traction.md`, `interfaces.md`, BOM	Bonded flexible laminates, raised rigid panels, roof rails, edge clamps, MPPT combiner, fire isolators, downlink gland, air-cleaner pump/nozzle manifold, bonding/earthing details
LM3-OBS-330	T-OBS nose sensor-pack installation	T-OBS supplier + integrator	RFC 0015, BOM	LIDAR, radar, stereo camera, ultrasonic transducers, heated sensor windows, cleaning access
LM3-BOG-400	Powered bogie assembly	Bogie fabricator	`bogie.md`	Frame drawing, motor/gearbox/brake interfaces
LM3-BOG-410	Trailer bogie assembly	Bogie fabricator	`bogie.md`	Frame drawing, brake/suspension interfaces
LM3-TRC-500	Traction package installation	Traction supplier	`traction.md`	Motor, gearbox, inverter, cooling, EMC bonding
LM3-COM-600	Train communication and antenna install	Integrator	`interfaces.md`	TCN-E, radios, GNSS, CCTV, PIS, intercom

Manufacturing evidence

ID	Evidence	Required before	Notes
LM3-EV-CFG-000	Configuration baseline sheet	v2A design freeze	Confirms 3 repeated 16.5 m cars, 3 powered bogies, 3 trailer bogies, 540 kWh usable battery, 78.75 t controlled planning tare (75.308 t modeled subtotal + 3.442 t engineering reserve), and no city-specific train variant
LM3-EV-MAT-001	Steel mill certificates and heat traceability	G0 material release	Covers RHS, plate, machined inserts
LM3-EV-REC-005	Recovered axle/axlebox acceptance pack	G0B recovered-component release	Quarantine, cleaning, dimensional survey, UT/MT/NDT, bearing replacement or supplier re-certification
LM3-EV-WLD-010	WPS/PQR register and welder qualifications	First production weld	EN 15085 / EN ISO 9606 basis
LM3-EV-WLD-020	Weld inspection and NDT report	G1 frame complete	VT + MT/UT per weld class
LM3-EV-DIM-030	Frame dimensional survey	G1 frame complete	Bogie centres, door/window apertures, coupler height
LM3-EV-COR-040	Blast, primer, topcoat, cavity-wax report	G2 corrosion complete	Includes DFT readings
LM3-EV-CMP-050	Composite material and mould-process certificate pack	G3 shell complete	EN 45545 evidence, resin/fibre/core/gelco at batch trace, mould release record, cure record, coupon trace, and repair method
LM3-EV-CMP-055	Fiberglass cowl cast first-article report	G3 shell complete	Laminate coupons, insert pull-out, split-line fit, glass-carrier land survey, and A/B-end interchange check
LM3-EV-CMP-057	One-metre body module first-article report	G3 shell complete	Mould inspection, cure and demould record, laminate coupons, CNC trim report, edge-seal record, master-frame dry fit, clip/anti-lift proof, insert pull-out, water/vibration test, and timed removal/refit
LM3-EV-DIM-058	Dedicated factory drawing/interface package review	v2A design freeze	Use the generated factory-release readiness record to show all 16 packages have approved drawings, frozen inputs, product revisions, released tooling, characteristic lists and named verification owners; generation alone does not constitute approval
LM3-EV-BND-060	Adhesive batch, surface-prep, cure records	G3 shell complete	Includes witness coupons
LM3-EV-GLZ-065	Front glass carrier, seal and service trial	G4 COTS systems fitted	Retention proof, edge/compression survey, drainage/water result, heater isolation and timed pane replacement
LM3-EV-LMP-067	Reversible front-lamp fit and aim report	G4 COTS systems fitted	A/B interchange, optical-axis/aim range, lock/vibration retention, harness access and supplier photometric evidence
LM3-EV-WTR-070	Water ingress test	G4 COTS systems fitted	Doors, windows, roof equipment
LM3-EV-FIN-075	Livery film first-car application and repair report	G4 COTS systems fitted	Actual-substrate coupon, batch/application record, edge/seam map, wash compatibility, patch and one-metre module removal trial
LM3-EV-RCF-077	Radiative roof-coating qualification disposition	Optional coating production release	Fire/material acceptance, new/aged optical results, durability/cleaning evidence, one-car trial and accepted baseline-finish fallback
LM3-EV-ELC-080	Bonding/earthing and insulation resistance	G5 electrical safe	Includes HV and LV separation checks
LM3-EV-HV-090	HVIL and battery isolation report	G5 electrical safe	Battery supplier + integrator sign-off
LM3-EV-DOOR-100	Door obstruction, release, lock-loop test	G6 static complete	Per door cassette serial number
LM3-EV-HVAC-110	HVAC cooling and drain test	G6 static complete	+50 C performance evidence may be supplier lab data
LM3-EV-INT-115	Interior moulding, fitout and service-access report	G6 static complete	Fire trace, dry-fit/gap/rattle/sharp-edge results, floor/PRM evidence and sequential service-removal demonstrations
LM3-EV-BRK-120	Static brake and park-brake test	G6 static complete	Links to bogie/brake supplier records
LM3-EV-MASS-130	Weighing and axle-load report	G7 dynamic ready	Empty and simulated AW2/AW3 cases
LM3-EV-RIDE-140	Bogie alignment and ride-height report	G7 dynamic ready	Before dynamic testing
LM3-EV-REC-150	Jacking, lifting and field-rerailing interface report	Recovery release	Individual-car mass/CG cases, structural proof, equipment freeze, four-point gauge, asymmetric/loss-of-pressure trials and trained-crew demonstration
LM3-EV-FEM-175	Train-to-train joint FEM screening pack	Optional open-end release	Train-to-train joint vertical and train-to-train joint lateral/racking beam-model screens

Supplier document index

Package	Supplier document required	Acceptance rule
Doors	Installation manual, maintenance manual, lifecycle test, EN 14752 evidence	Fits LM3-DOOR-200 without primary steel change
Windows	Glazing certificate, adhesive/gasket procedure, replacement manual	Fits LM3-WIN-210 and passes water test
HVAC	Datasheet, +50 C derating curve, wiring manual, refrigerant record	Fits LM3-HVAC-220 and aux-power budget
One-metre GFRP body modules	Mould split/datum drawings, laminate schedule, fire test, cure record, CNC trim report, repair manual, insert pull-out data, seal/gasket procedure	Modules remain non-structural, clipped and gasketed to LM3-BDY-160 without side/roof production adhesive
Fiberglass end cowls	Mould split/trim drawings, laminate coupons, fire test, insert pull-out data, gasket/seal procedure, repair manual	Same cast kit fits both A-end and B-end without steel-frame change
Rail livery film	Rail-use statement, film/ink/overlaminate/edge-system data, substrate/application/cleaning limits and fire evidence	Fits LM3-FIN-240 seams and keep-outs, adheres to the actual cured base finish, and passes repair/removal trial
Radiative roof coating	Controlled formulation, fire/chemical/material compatibility, application limits, new/aged optical and durability evidence	May be used only within LM3-FIN-245 trial zones after accepted one-car disposition; qualified light base finish is the fallback
Configurable train-end interface	End carrier ring drawing, option bolt-grid gauge, panoramic/open-mid configuration procedure, seal/drain test, selected-option record	Each end position selects exactly one of the panoramic glass end or the mid open train-to-train connection
Seats/rails	Fire certificate, pull-load certificate, cleaning/repair manual	Fits LM3-INT-230 insert grid
Lighting/PIS/CCTV/intercom	EMC/vibration evidence, firmware version, wiring manual	Enumerates on car network and passes static test
Battery	Cell/module certificate, BMS manual, vent/fire containment data	Fits LM3-BDY-140 and LM3-HV-310
Traction and charging	Motor/controller/gearbox datasheets, station DC-charge interface data, onboard HV protection/IMD data, PV MPPT data, cooling and EMC instructions	Fits LM3-TRC-500, LM3-HV-320, and LM3-HV-325
Rooftop solar and air cleaner	PV module datasheets, adhesive/bond process, rail/clamp vibration evidence, fire-isolation switch data, blower/nozzle IP and vibration evidence, filter-service access, soiling-recovery test	Fits LM3-HV-325 without roof-spaceframe redesign
Bogie parts	Wheelset, bearing, spring, damper, brake supplier certificates	Fits LM3-BOG-400/410
Articulation/gangway	Bearing rating, motion-envelope proof, bellows/turntable fire evidence, maintenance manual	Fits LM3-SYS-170 and optional LM3-SYS-175 without carbody adapter redesign

Release gates

Gate	Minimum drawing/evidence state
v2A design freeze	LM3-EV-CFG-000 and LM3-EV-DIM-058 complete; all controlled drawings issued at rev A; supplier envelopes frozen
First steel cut	LM3-BDY-100/110/120, LM3-EV-MAT-001, LM3-EV-WLD-010 complete
First carbody shell complete	LM3-EV-WLD-020, LM3-EV-DIM-030, LM3-EV-COR-040 complete
First COTS fit-out complete	Door/window/HVAC/interior/film supplier docs accepted; LM3-EV-GLZ-065, LM3-EV-LMP-067, LM3-EV-WTR-070, LM3-EV-FIN-075 and LM3-EV-INT-115 complete
First energisation	LM3-EV-ELC-080 and LM3-EV-HV-090 complete
Static test release	LM3-EV-DOOR-100, LM3-EV-HVAC-110, LM3-EV-BRK-120 complete
Dynamic test release	LM3-EV-MASS-130 and LM3-EV-RIDE-140 complete

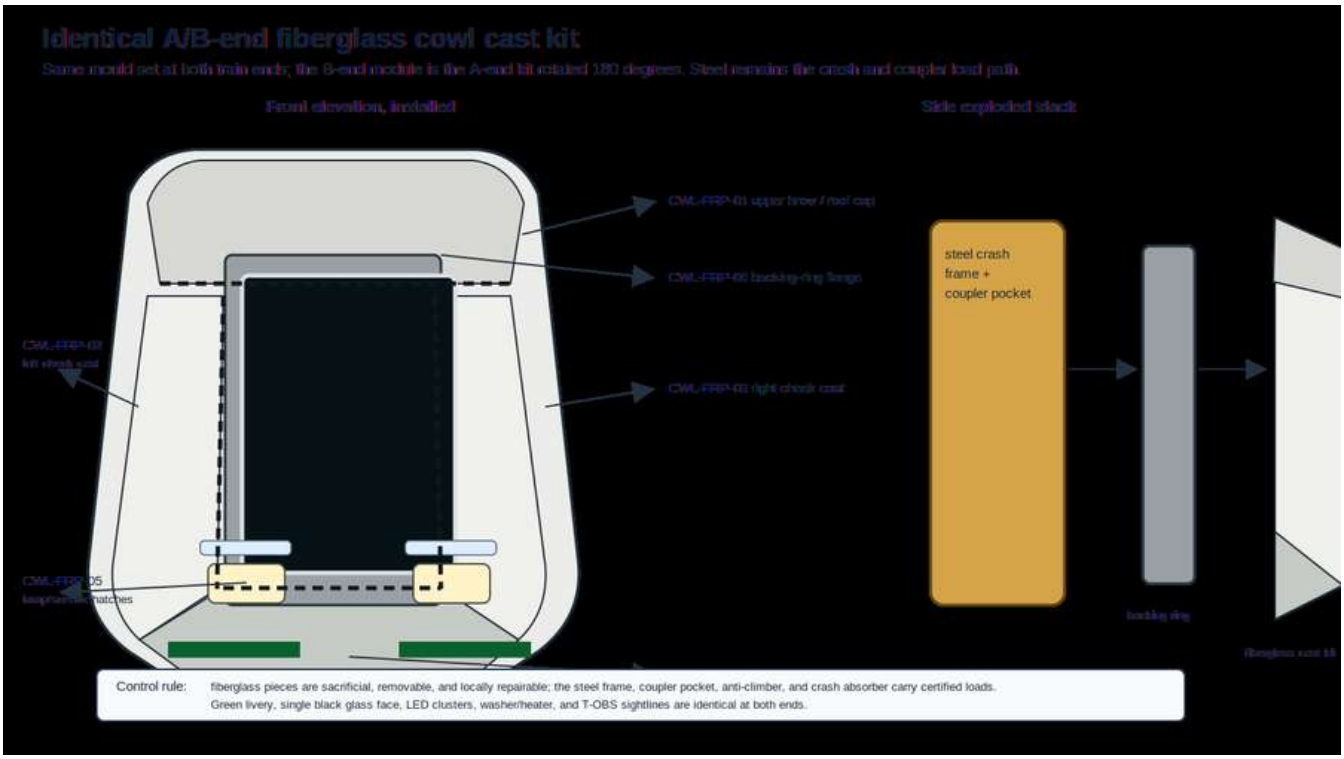
Change control

- Any supplier swap that fits the reserved envelope is a document revision, not a structural redesign.
- Any change to side sills, bolsters, coupler pockets, door posts, or crash-can interfaces reopens FEA and static proof evidence.
- Any mass increase above a module row limit requires a new axle-load report before dynamic testing.
- Any power increase above a module row budget requires an aux-power and cooling-capacity review.

Identical fiberglass end-cowl design

Repository source: `docs/rolling-stock/light-metro-3car/end-cowl.md`

This page controls the `light-metro-3car` front/back end module. The train is driverless and reversible, so the "front" and "back" are the same part kit: one A/B-end fiberglass cowl assembly installed at each outer end, with the B-end kit rotated 180 degrees about the vertical axis. There are no handed cab moulds and no rear-only cosmetic parts.



Multipart fiberglass end-cowl cast kit

The form language follows the README reference image: white/silver rounded ends, one uninterrupted dark panoramic glass face, green waist band carried onto the cowl flanks, low dark skirt, and LED head/marker lights below the glass. The cowl is a sacrificial weather/aero/sensor fairing over a steel crash frame; it is not the buff, collision, coupler, or anti-climber load path.

The production shape should be authored in a surface-modelling package suited to styled fiberglass work. The parametric `sensor_cowl.py` geometry is an envelope and interface proxy; it is not intended to be the final class-A exterior surface. The released cowl CAD should come from surface/NURBS or SubD modelling, then be exported as neutral STEP or IGES surfaces with separate mould, trim, flange, and inspection geometry.

Envelope

Parameter	Value	Notes
Module role	identical A-end/B-end kit	Same mould set and drawing set for both train ends
Along-track length	1 800 mm	Overlay inside the 17 m end-car envelope
Car interface width	2 850 mm	Matches `CarDimensions.body_width_mm`
Car interface height	3 450 mm	Matches `CarDimensions.body_height_mm`
Leading-face width	1 800 mm	Rounded/tapered visual face
Leading-face height	2 800 mm	Allows roof cap transition into body
Panoramic glass opening	1 500 mm x 1 780 mm	One heated laminated pane
Exterior corner target radius	>= 200 mm	Keep edges easy to laminate and repair
Nominal panel mass target	<= 180 kg per end kit	Excludes glass, T-OBS, coupler, and crash absorber

Fiberglass cast split

The end cowl is built as a multi-part fiberglass (GFRP) kit so it can be made in modest female moulds, repaired locally, and removed after minor collision damage without cutting the steel body.

ID	Cast part	Qty per end	Function	Main interfaces
CWL-FRP-01	Upper brow and roof cap cast	1	Blends roof into the black glass surround and sheds rain above the pane	Roof bow tabs, upper backing ring, washer nozzle covers

ID	Cast part	Qty per end	Function	Main interfaces
CWL-FRP-02	Left cheek side-return cast	1	Tapered side surface, green livery continuation, side split-line closure	Left side frame tabs, backing ring, lower apron
CWL-FRP-03	Right cheek side-return cast	1	Mirror of CWL-FRP-02	Right side frame tabs, backing ring, lower apron
CWL-FRP-04	Lower apron and anti-climber cover cast	1	Lower rounded nose, lamp recess support, removable access to coupler recovery space	Lower crash-frame tabs, lamp pods, skirt edge
CWL-FRP-05	Lamp/washer service hatch casts	2	Replaceable local covers around LED clusters, washer manifold, and service fasteners	M6 captive fasteners, gasketed edge, lamp bracket
CWL-FRP-06	Sectional backing-ring flange casts	4	Bond/bolt land behind glass carrier and cast split lines; carries seals, not crash loads	Steel crash ring, glass carrier, EPDM seals

The cheek casts may use mirrored moulds, but they keep the same datum scheme and hole pattern. The full end kit is duplicated at the other end of the train without changing sensors, lamps, glazing, wiring exits, or maintenance access.

Laminate and materials

Region	Nominal construction	Notes
Broad brow/cheek/apron skins	Fire-rated E-glass or basalt-fibre/vinyl-ester sandwich, 16-24 mm total	Foam or honeycomb core only in broad low-load panels
Split-line flanges and bolt lands	Solid GFRP, 6-10 mm	No core at fasteners, seals, jacking-prone edges, or repair trim lines
Insert pads	Local solid laminate build-up plus potted stainless or GFRP hard points	M6/M8 service fasteners; M10 only where the steel bracket owns load
Exterior finish	White/silver gelcoat or paint over primer	Must accept green livery film/paint band
Inner finish	Light-colour fire-rated coating	Allows crack and water-ingress inspection
Seals	EPDM or silicone rail-rated gasket/sealant	Continuous drain path below glass and hatch seams

All passenger-facing or exposed composite materials need EN 45545-2 HL2 or locally accepted equivalent evidence. The composite supplier owns resin selection, laminate coupons, fire/smoke/toxicity evidence, insert pull-out data, repair instructions, and batch traceability.

Tooling rules

- Build the visible A-surface first in the surface modeller, using the README image and package envelope as references.
- Derive B-surfaces, nominal laminate thickness, return flanges, split flanges, hatch offsets, and trim curves from that controlled A-surface.
- Export neutral surface CAD for design review and mould manufacture; keep the parametric FreeCAD/Python model as the envelope, integration, and documentation proxy.
- Use female moulds with removable split flanges; no one-piece bathtub mould that traps the part.
- Minimum mould draft: 3 degrees on return flanges and hatch pockets.
- Minimum exterior radius: 75 mm on cosmetic edges, 125 mm preferred at roof and cheek transitions, 200 mm at the large outer corners.
- Keep split lines on visual shadow lines: roof-to-glass brow, cheek-to-front face, apron-to-skirt, and lamp hatch surrounds.
- Add 35-50 mm internal lap flanges at cast-to-cast seams.
- Provide witness holes or bond-line tell-tales where adhesive closure is hidden after assembly.
- First article may be hand lay-up or vacuum infusion; production should prefer vacuum infusion where the supplier can control resin fraction and repeatability.

Structural interfaces

The steel end frame remains the certified load path. The fiberglass kit attaches to secondary tabs and a bolted backing ring:

- M8 stainless/captive fastener grid at 180-250 mm pitch around the steel cowl ring.
- Slotted holes only in fiberglass flanges, never in the steel crash datum.
- Nylon/GFRP isolation washers where stainless fasteners meet coated steel or mixed-metal inserts.
- Structural adhesive or elastic sealant between fiberglass and secondary rails for weather sealing and vibration control.

- Separate earth straps for heated glass, lamps, washer heater, and T-OBS hardware; do not rely on composite continuity.
- Drain holes at the lower apron and glass carrier, routed away from coupler electronics and the T-OBS backplate.

Glazing, lamps, and sensors

The cowl fitout is now separated into `LM3-FAS-P010` panoramic glass carrier, `LM3-FAS-P020` reversible lamp cassette/aiming tray, and `LM3-FAS-P030` seal/drain/washer closeout kit. Their checking nest and lamp aiming jig are defined in `dedicated-parts-and-moulds.md`.

The cowl carries one large heated RF-transparent laminated glass pane in a dark bonded carrier. The carrier bolts to the steel-backed ring through the fiberglass flange, so a damaged cast can be removed without discarding the glass if the pane is intact. Any necessary structural support, heater traces, sensor brackets, or busbars must sit behind the black edge band or read visually as part of the dark glass, not as visible vertical dividers.

The LED headlamp/marker-light cassette is symmetrical: either train end can lead or trail. Software decides head/tail aspect; the physical lamp and harness are identical. The T-OBS LIDAR, radar, stereo camera, and ultrasonic pack sits behind the same glass/sensor aperture at both ends, with washer nozzles and heater busbars accessible through CWL-FRP-05 and the upper brow service covers.

Assembly sequence

1. Survey the steel end ring and secondary cowl tabs after body corrosion coating.
2. Trial-fit CWL-FRP-06 backing-ring flange sections and shim to the steel datum.
3. Fit CWL-FRP-01, CWL-FRP-02, CWL-FRP-03, and CWL-FRP-04 dry with temporary fasteners; check split gaps and door/platform clearance.
4. Surface-prepare steel and fiberglass bond/seal lands.
5. Install backing-ring flanges, cheek casts, upper brow, lower apron, and hatch casts with adhesive/sealant and retained fasteners.
6. Install the glass carrier, laminated panes, heater/washer harness, LED lamp cassettes, and T-OBS module.
7. Complete water-ingress, heater, washer, lamp aspect, T-OBS calibration, and removable-hatch access tests.

Making instructions

1. Surface and mould release

1. Freeze the LM3-BDY-155 A-surface, B-surface, split lines, trim curves, flange returns, glass-carrier land, hatch openings, and insert map.
2. Manufacture female moulds for CWL-FRP-01 through CWL-FRP-06. Use removable mould flanges at cheek/apron/brow split lines.
3. Polish, seal, and release each mould per the tooling-resin supplier procedure. Record mould ID, release system, operator, date, and number of pulls since last reconditioning.

Hold point: mould surface and release record accepted before lay-up.

2. Lay-up / infusion

1. Cut dry glass/basalt reinforcement plies and core sheets from the released ply book; mark fibre direction and part ID.
2. Lay gelcoat or primer-compatible surface coat where the process uses in-mould finish.
3. Lay outer plies, core only in broad low-load skins, solid-laminate flange/insert zones, and inner plies.
4. Fit potted inserts and doubler pads only into solid laminate build-up zones; do not place inserts into core.
5. Hand lay-up or vacuum infuse. Record resin batch, catalyst/hardener batch, pot life, ambient temperature, vacuum level if used, and cure time.
6. Cure and post-cure per resin supplier procedure. Keep one laminate coupon with every resin batch and cure cycle.

Hold point: coupon, cure, resin batch, and visual void/dry-fibre checks accepted before demould.

3. Demould, trim, and drill

1. Demould without prying on visible A-surfaces or glass-carrier lands.
2. Trim parts in the cowl trim fixture; never freehand trim a split line or glass land.
3. Drill fastener, insert, drain, washer, lamp, and service-hatch holes from controlled drill bushes.
4. Seal all cut edges and drilled holes.
5. Dry-build CWL-FRP-01 through CWL-FRP-06 on LM3-TL-COWL-09 and measure split gaps, flange alignment, lamp pocket position, and glass-carrier land.

Hold point: trim-line survey, insert pull-out sample, edge-seal inspection, A/B interchange check, and dry-build water path accepted.

4. Finish and repair sample

1. Release the qualified white/silver gelcoat or base coating. Apply the green waist-band continuation as pre-cut rail-use film where its substrate, fire, edge, cleaning and removal evidence is accepted; otherwise use the released paint route.
2. Fit gaskets, retained fasteners, drain sleeves, washer-nozzle covers, and hatch tethers.
3. Make a sacrificial cheek/apron repair coupon from the same laminate and demonstrate sand/fill/recoat before accepting the first article.

Hold point: finish DFT/visual record, gasket compression, repair coupon, and removable-hatch trial accepted.

Inspection and release evidence

Evidence	Acceptance intent
Laminate coupon pack	Fire rating, resin/fibre fraction, cure record, material batch trace
Insert pull-out test	Confirms service fasteners and flange hard points survive repeated removal
End-ring dimensional survey	Confirms the cowl kit fits the steel datum without forcing the glass carrier
Split-line and water test	No water path into T-OBS, glass heater busbar, coupler head, or saloon floor
Lamp and sensor calibration	Both A-end and B-end kits meet identical sightline and aspect checks
Repair demonstration	Sand/fill/recoat of a damaged cheek/apron sample without replacing steel

v2 drawing outputs

The v2 drawing pack shall issue these as controlled drawings under LM3-BDY-155:

- mould split and trim drawings for CWL-FRP-01 through CWL-FRP-06,
- controlled exterior A-surface, B-surface, flange, trim, and mould surface exports from the surface modeller,
- laminate schedule and ply drop/overlap maps,
- insert and fastener map,
- steel backing-ring and shim interface drawing,
- glass carrier and seal land drawing,
- lamp/sensor hatch service drawing,
- repair-zone and allowable damage map,
- first-article dimensional inspection form.

Simplified exterior finish: base protection, livery film and cool roof

Repository source: `docs/rolling-stock/light-metro-3car/exterior-finish-process.md`

The simplified process removes most masked multi-colour painting, not the protective systems beneath it. The production sequence is:

1. qualified corrosion protection on steel;
2. white/light-neutral in-mould gelcoat or compatible base coating on GFRP;
3. pre-cut, replaceable graphic film for the green waist band, logos and local colour blocks;
4. a separately qualified high-reflectance roof coating on exposed roof fairings only.

What film can and cannot replace

Film can replace decorative colour-band masking and repainting on smooth, fully cured, accepted substrates. It cannot replace steel primer/topcoat, cavity wax, edge sealing, composite gelcoat/base protection, fire barriers, anti-slip surfaces, structural adhesive, weather seals, safety markings or electrical insulation/bonding.

The reference procurement row is `LM3-FIN-P010`: a bay-numbered, pre-cut film kit with controlled overlaps, edge treatment, datum marks and repair patches. Use a current film system whose manufacturer explicitly supports rail-vehicle graphics. 3M's current rail bulletin requires pre-installation inspection, clean substrate preparation, application controls, quality inspection, maintenance and removal procedures; its rail catalogue includes IJ180-class film. That makes it a useful procurement anchor, not an automatic LM3 approval.

Primary references:

- 3M Application: Rail instruction bulletin
- 3M Controltac Graphic Film 180mC product page

The selected film, ink, overlaminate and edge system still require project fire, adhesion, temperature, UV/weather, cleaning-chemical and removal evidence on the actual cured GFRP/steel base finish.

Film manufacturing and application route

Seq.	Operation	Evidence
10	Freeze colour artwork, module/bay seams, door/window/cowl breaks, logos and prohibited zones	Approved livery map and cut files
20	Complete and sign substrate inspection; verify full paint/gelcoat cure, repair defects and clean by selected supplier process	Per-car pre-installation record
30	Plot/cut and label left/right/bay/cowl pieces on `LM3-TOOL-FILM-TEMPLATE`; make an adhesion/removal coupon	Cut-kit map and accepted coupon
40	Dry-apply from controlled datum; squeegee air to open edges; keep knives off the vehicle finish	Installer and environment record
50	Finish seams/edges and post-heat only where the selected system requires it	Edge/overlap inspection map
60	Water/wash inspect and demonstrate local patch plus complete 1 m module removal	Appearance, ingress and repair record

Film terminates before EPDM joints, drains, captive fasteners, glass edges, lamp apertures, bonding points, safety labels and access lands. Prefer each graphic piece to stay on one replaceable 1 m module so collision repair does not require stripping an adjacent good panel.

Calcium-carbonate radiative roof candidate

Purdue's 2020 open paper demonstrated a single-layer CaCO3-acrylic research paint with 95.5% solar reflectance and 0.94 normal emissivity in the atmospheric window. Those values are useful initial screening targets, not guaranteed vehicle values. The paper does not establish rail fire compliance, adhesion to the LM3 laminate, flexibility under vibration, erosion, depot-wash durability, soiling behaviour, repairability or aged optical performance.

Primary research:

- Li et al., Full Daytime Sub-Ambient Radiative Cooling in Commercial-like Paints
- Purdue-hosted open paper PDF

`LM3-FIN-P020` is therefore `concept`/trial-only. It cannot be promoted to a trainset production finish until this gate is passed:

Gate	Minimum evidence
Formulation	Controlled binder, CaCO3 particle-size distribution/concentration, additives, shelf life, mixing and application window
Rail material	Fire/smoke/toxicity and chemical compatibility accepted for the deployment case
Substrate	Adhesion, flexibility/thermal cycling, edge compatibility and repair on the actual GFRP base system
Exposure	UV, humidity, heat, rain, dust/sand, abrasion and depot-wash ageing
Optical	Measured solar reflectance and thermal/sky-window emittance when new and after ageing/soiling/cleaning
Operations	Glare/visibility, runoff, cleaning, damage inspection and local repair procedure
Vehicle trial	One-car A/B roof-zone temperature, HVAC energy, soiling and maintenance comparison through representative hot service

The trial mask excludes PV surfaces/clamps, HVAC air and heat-exchanger paths, antennas/sensors, walkways, lifting covers, earth bonds, glands, drains, seals, labels and service lands. No HVAC-energy or passenger-temperature saving is claimed until the one-car trial measures it.

Factory simplification result

The body cell retains one small controlled coating capability for steel repair, GFRP edge repair and the roof candidate. It no longer needs to mask and spray the full green/silver livery scheme. White GFRP parts can leave the mould with their qualified base surface, while numbered film kits move decorative work to a clean, low-dust application bay and make one-metre collision repairs local.

Fabrication plan — light-metro-3car

Repository source: `docs/rolling-stock/light-metro-3car/fabrication-plan.md`

This plan turns the v1 rolling-stock package into a buildable factory route for the first `light-metro-3car` trainset. It favours ordinary regional manufacturing capability: saw/laser/plasma cutting, press-brake bending, MIG/MAG welding, dry clip-on fiberglass body modules, limited qualified adhesive bonding, bolted module installation, and COTS rail/bus subsystems.

It is not a homologated shop drawing pack. The v2 pack must add dimensioned drawings, weld symbols, tolerances, FEA, supplier installation manuals, and authority-approved inspection records.

Solar metro production assembly concept

The production concept is deliberately modular: make the steel datum structure locally, install certified supplier modules late, and repeat the same 16.5 m car body to form the trainset.

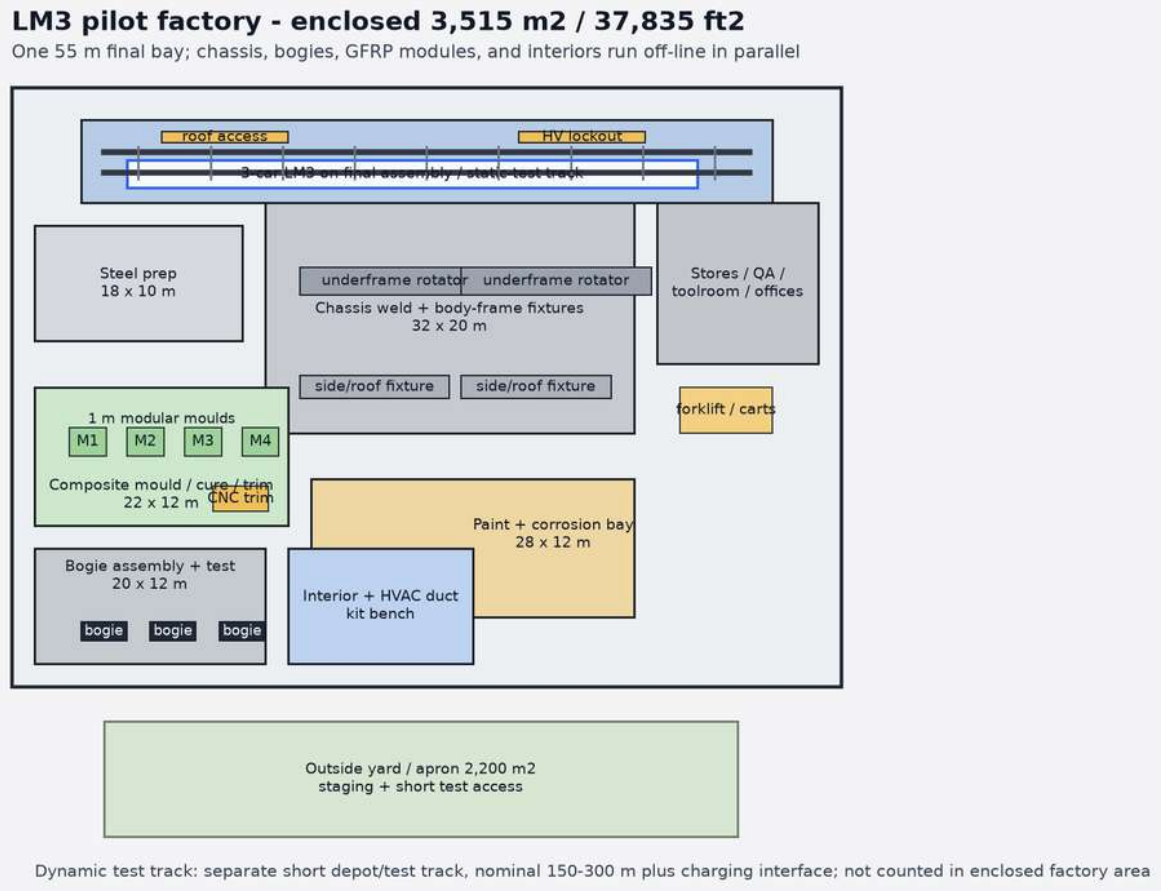
Manufacturing doctrine

Scope	Default decision	Reason
Primary structure	MAKE: cut/bend/weld S355 steel	Available in rail, truck, ship, oilfield, and bridge workshops
Side/roof body modules	MAKE/BID: 1 m-wide clipped fiberglass modules from reusable short moulds	No full-car mould or production adhesive cure; corrosion resistant and locally replaceable
Nose panels	BID/SOURCE: multi-part fiberglass end-cowl casts	Styled sacrificial shell retains its separate gasket/adhesive process over the steel crash frame
Doors	BID: COTS electric rail door cassette	Certified obstacle detection, locking, seals, emergency release
Windows	SOURCE/BID: COTS bonded/gasketed glazing	Avoids bespoke glass certification
HVAC	SOURCE/BID: packaged roof unit	Mature hot-climate bus/rail supply chain
Lighting	SOURCE: rail/bus LED modules	Commodity, low-voltage, easy replacement
Passenger interior	SOURCE/BID: bus/rail seats, rails, PIS, CCTV	Keeps OSR focused on interfaces and safety evidence
Cabin fiberglass liners/trims	MAKE/BID: fire-rated FRP or phenolic ceiling, sidewall, battery-strake, and vestibule panels	Same low-cost composite cell as the cowls; replaceable passenger-facing trim
Bogie frame	MAKE fresh OSR powered/trailer frames under controlled weld procedure	Structural rail part; can be local but must be qualified; recovered freight frames are not used
Wheelsets, bearings, brakes	BID/SOURCE	Safety-critical rotating/brake parts stay supplier-certified
Traction, batteries, BMS	BID/SOURCE	Supplier qualification and test evidence required

Factory cells

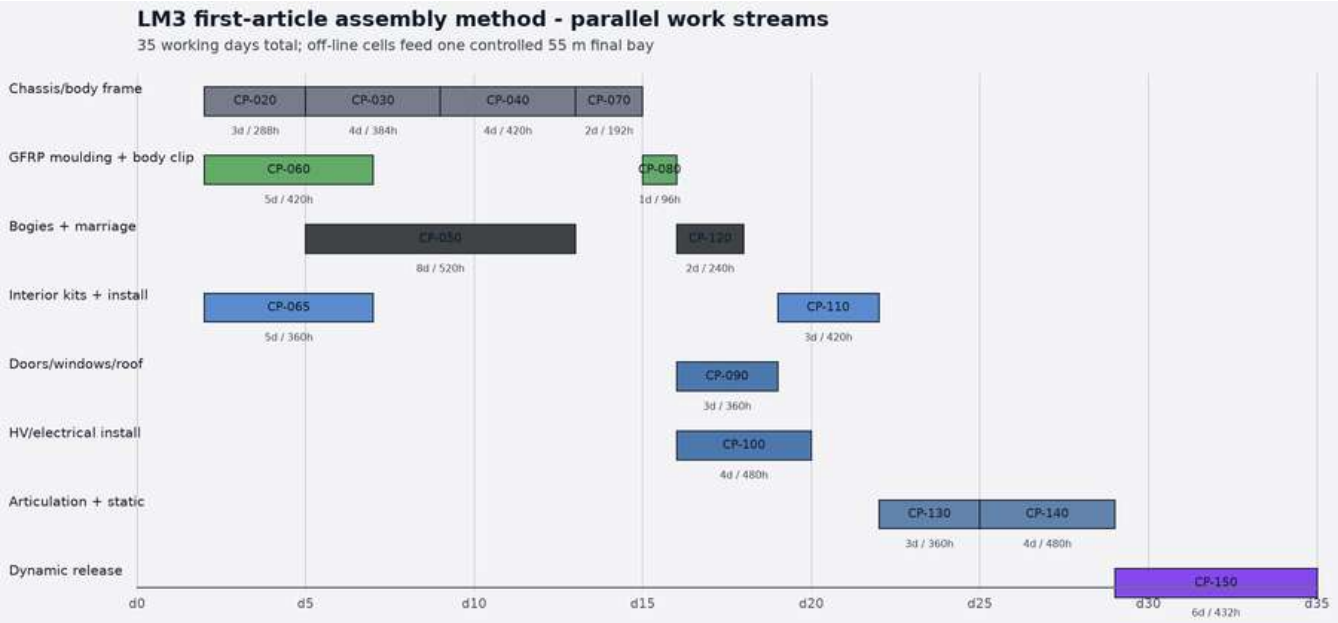
The first factory needs five cells, not a full legacy rolling-stock plant:

The generated numeric sizing is controlled in `factory-plan.md`: 3,515 m2 enclosed pilot factory, 2,200 m2 outside yard/test apron, 13 working days for chassis/painted frame fabrication, 13 working days for bogie build and bogie-to-carbody integration, 14 working days for GFRP moulding plus clip-on body installation, and 13 working days for final assembly plus static commissioning inside the first-article 35-working-day network. It also lists the rough machinery package at about $1.02M including setup contingency, excluding building/land and homologation laboratory equipment.



LM3 pilot factory layout

The generated layout keeps the long final track for accepted kits only: steel cutting, chassis/body-frame fixtures, moulding, bogie assembly, interior/HVAC kit work, corrosion protection/base finish, stores, QA, and yard staging run beside it instead of blocking it.



The first-article method plan splits the 35-working-day build into parallel streams so the constrained final bay receives released chassis, GFRP, bogie, interior, door/window, roof, HV, articulation, and static test work at the latest practical point.

1. Steel prep cell
   - CNC saw or tube laser for RHS.
   - Plasma/laser/waterjet table for plate.
   - Press brake for folded bolsters, coupler pockets, trays, and brackets.
   - Drill/mill for datum holes and bolted inserts.
2. Weld and fixture cell
   - Underframe rotating fixture.
   - Side-frame fixture.
   - Bogie-frame fixture.
   - Certified MIG/MAG equipment.
   - Weld fume extraction and preheat capability where WPS requires.
3. Paint and corrosion cell
   - Shot blast or outsourced blast booth.
   - Zinc-rich epoxy primer.
   - Seam sealing and cavity wax station.
   - Topcoat booth for exposed steel and brackets.
4. Composite and glazing cell
   - Reusable one-metre side-module and roof-module moulds with released A-surfaces, local-core maps, solid clip lands, and insert boss features.
   - One-metre module trim/drill gauge and master-frame trial-fit stand.
   - Resin, fibre, core, gelcoat/paint-primer, release-system, cure-record, and witness-coupon control for each moulded batch.
   - End-cowl cast trim/drill jigs or incoming-inspection fixtures for supplier-made fiberglass casts.
   - Clip/anti-lift proof fixture and dry-seal inspection tools.
   - Small controlled cure area for glazing and end-cowl non-service seams only.
   - Window bonding/gasket installation.
5. Final assembly and commissioning cell
   - Level track or assembly stands.
   - Bogie drop/lift equipment.
   - Roof access for HVAC and antennas.
   - HV isolation tools and test equipment.
   - Door calibration, lighting/PIS/CCTV checks, brake tests.

Sheet-metal tooling package

The v2 drawing pack shall release the following tooling before first steel cut. The tooling is intentionally ordinary: press-brake tooling, laser-cut fixture plates, bolted datum towers, and removable clamps rather than legacy carbody megajigs.

Tooling ID	Tool	Purpose	Required accuracy
LM3-TL-UNDERFRAME-01	Underframe rotating weld fixture	Holds side sills, cross-bearers, bolsters, coupler pockets, and battery trays	±1.5 mm on bogie centres; ±2.0 mm on door datum plane
LM3-TL-BOLSTER-02	Bolster box subfixture	Pre-welds folded bolster boxes, air-spring pads, and pivot boss backing plates	±0.5 mm before line-bore
LM3-TL-COUPLER-03	Coupler pocket fixture	Keeps crash-can plate, shear plate, and pocket inserts square to train centreline	±0.75 mm on coupler face datum

Tooling ID	Tool	Purpose	Required accuracy
LM3-TL-SIDE-04	Side-wall post/rail fixture	Locates door portals, window posts, waist rail, and cant rail	±1.0 mm at door cassette opening
LM3-TL-ROOF-05	Roof bow fixture	Holds roll-formed roof bows and HVAC rail inserts	±1.5 mm on HVAC rail pitch
LM3-TL-PANEL-06	One-metre side/roof module mould and trim/drill fixture	Moulds GFRP side/roof modules, locates solid clip lands and potted inserts, then CNC trims solid/window/door/roof variants from the common datum	±0.5 mm on clip grid; ±1.0 mm on trim edge
LM3-TL-DOOR-07	Door cassette fit-up gauge	Confirms COTS door cassette envelope, threshold, and lock-loop bracket clearances	Supplier installation tolerance
LM3-TL-HV-08	Battery tray/service-lid gauge	Checks tray drainage, lid gasket land, HV cable gland reach, and seat-base clearance	±1.0 mm on service-lid gasket land
LM3-TL-COWL-09	Fiberglass end-cowl cast fit-up fixture	Holds CWL-FRP-01 through CWL-FRP-06 to the steel backing-ring datum and checks split-line gaps, insert pitch, glass carrier land, and lamp/sensor hatch access	±1.0 mm on glass-carrier land; ±2.0 mm on cast split gaps
LM3-TL-INT-10	Cabin ceiling liner buck and trim fixture	Moulds/trims ceiling liners, light troughs, HVAC plenum covers, and service lids	±1.0 mm on light/HVAC apertures
LM3-TL-INT-11	Cabin sidewall/window reveal drill fixture	Locates sidewall liners, window reveals, cable covers, clips, and retained fasteners	±0.75 mm on clip/fastener pitch
LM3-TL-INT-12	Battery strake cover and hatch gauge	Checks under-seat FRP covers, service hatch removal, HV labels, and seat-base fairing clearances	Hatch removable without seat removal
LM3-TL-INT-13	Door/PRM transition trim fixture	Checks vestibule kick panels, step covers, threshold trims, anti-slip strips, and contrast nosing	PRM transition accepted by gauge

The current CAD manufacturing templates generated from `design/component-catalogue` are tracked as parametric source and FreeCAD review artifacts:

Template/source	Use
`main_frame()`	Underframe ladder, formed sills, cross-bearers, bolsters, battery trays, coupler pockets
`body_sheet_metal_kit()`	Whole body/chassis sheet-metal kit: underframe, side posts, door portals, waist/cant rails, roof bows, end rings
`fiberglass_cladding_system()` / `modular_fiberglass_body.py`	One-metre GFRP side/roof module grid, clip hardware, EPDM joints, module numbering, and one-shift installation route
`sandwich_panel()`	Legacy sandwich-panel aperture and retainer template retained only as a detail reference for non-structural local panels
Surface-modelled LM3-BDY-155 cowl CAD	Final fiberglass cowl A-surface, B-surface, flanges, trim curves, and mould surfaces
`sensor_cowl()`	Identical A/B-end fiberglass cowl envelope, single panoramic glass, lamp/service hatches, and T-OBS visual interface
`door_leaf()`	COTS-style door leaf shell, bonded glazing, EPDM seals, hanger rollers
`chassis_interface_assembly()`	Bolster, bogie adapter, motor cradle, guide blocks, fasteners, service strut, connector interfaces

For a visual map of which COTS modules bolt, bond, plug, or slide into these fabricated datums, see `cots-integration.md`.

Build sequence per car

1. Kitting

- Issue tube cut list, plate flat patterns, brackets, inserts, and supplier interface drawings.
- Export NC data directly from the v2 flat-pattern drawings; each part receives a QR/etched part ID before forming.
- Mark every primary steel member with heat/batch traceability.
- Stage COTS modules by car number: doors, windows, HVAC, lighting, seats, intercom, CCTV, PIS, battery pack, inverter, auxiliaries.
- If a brownfield deployment proposes donor axles or axlebox hardware, quarantine them in a separate incoming-inspection lot until cleaning, dimensional survey, UT/MT inspection, and supplier acceptance are complete.

Hold point: material certificates and supplier certificates accepted before cutting.

2. Underframe

- Cut side sills, centre spine, cross bearers, battery rails, and folded plate boxes.
- Press-brake side sills, bolster webs, coupler-pocket plates, battery-tray lips, and service-lid gutters using matched V-dies.
- Tack in underframe fixture from centre datum outward.

- Weld side sills, cross bearers, bolster boxes, coupler pocket, and battery/equipment supports.
- Fit temporary anti-pull bars across door openings and side sills before long weld runs.
- Stress-relieve or controlled-cool where required by WPS.
- Machine or line-bore centre-pivot and critical mounting inserts after welding.

Hold point: dimensional survey, visual weld inspection, MT/UT on CL1/CL2 joints, bogie-centre datum report.

3. Side frame and roof frame

- Weld door posts, window posts, waist rail, cant rail, and roof bows on the side-frame fixture.
- Trial-fit a COTS door cassette gauge before welding the portal reinforcement closed.
- Lift side frames onto underframe datum pads.
- Weld/bolt side frames to underframe as specified by v2 drawings.
- Install roof equipment rails and cable tray brackets.

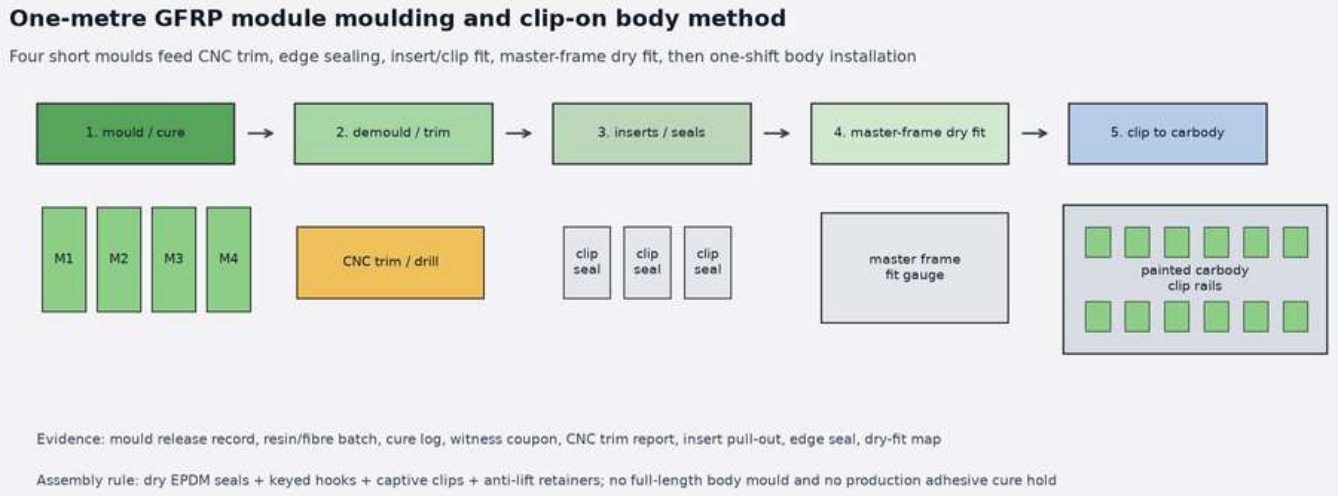
Hold point: door aperture survey against chosen COTS door cassette; window aperture survey against chosen glazing cassette.

4. Corrosion protection

- Blast exposed steel to Sa 2.5.
- Apply zinc-rich primer and topcoat system.
- Seal lap joints and composite bond interfaces.
- Apply cavity wax after all hot work and inspection is complete.

Hold point: dry-film thickness report and sealed-drain checklist.

5. One-metre composite exterior



GFRP module moulding and clip-on body method

- Before final assembly, fabricate each 1 m side and roof module in the released mould: clean and inspect the mould, apply release system and UV-stable gelcoat/paint-primer, cut glass-fibre plies and local core, lay up solid clip lands and insert bosses, infuse or wet-lay the laminate, cure, demould, retain witness coupons, CNC trim/drill the selected solid/window/door/roof variant, and seal all cut edges.
- Fit potted/captive inserts, keyed hooks, anti-lift features, drain details, and EPDM seals to each module, then dry-fit against the master frame before kitting by car side, roof bay, and serial number.
- Stage 96 numbered side modules and 48 roof modules for the three-car trainset. Verify all CNC-trim variants against the common 1,000 mm gauge.
- Put all three painted, dimensionally released frames in the body cell and assign two two-person crews per car.
- Fit dry EPDM seals, hang each module from its asymmetric keyed hook, close captive over-centre clips, and engage the independent anti-lift retainer.

- Record visible witness marks against the digital module map; do not use a clip to pull an out-of-tolerance frame or panel into position.
- Complete the three-car side/roof installation, water test, rattle check, and snag release inside one eight-hour shift per LM3-BDY-160.
- Separately trial-fit skirts and the identical A/B-end fiberglass cowl kit.
- Dry-build CWL-FRP-01 upper brow, CWL-FRP-02/03 cheek casts, CWL-FRP-04 lower apron, CWL-FRP-05 lamp/service hatches, and CWL-FRP-06 backing-ring flanges on the end-cowl fit-up fixture.
- Use the module drill jig to install captive clip receptacles and anti-lift hardware. Side/roof body-to-frame joints receive no production adhesive.
- Fit removable skirt panels with quarter-turn or captive fasteners. Fit cowl hatches with retained fasteners and continuous gasketed drain paths.

Hold point: signed module/clip map, laminate coupons, insert and clip proof, edge-seal record, mould release and cure records, eight-hour route record, water/rattle test, split-line cowl water test, applicable cowl/glazing adhesive cure records, and removable-panel access check.

6. COTS doors and windows

- Install door cassette into prepared structural bay.
- Install threshold, drainage, seals, and emergency release hardware.
- Install fixed or vented glazing cassettes.
- Connect door power, hardwired closed-and-locked loop, Ethernet/CAN, and local service connector.

Hold point: water ingress test, door obstruction test, emergency release test, closed-and-locked circuit test.

7. Roof systems

- Install packaged HVAC units onto roof rails.
- Install antennas, marker lights, CCTV housings, and regen resistor.
- Connect ducting to interior cassette.
- Fit removable composite roof fairings around service envelopes.

Hold point: roof leak test, HVAC condensate drain test, lifting and fall-arrest access check.

Roof and side modules locate on precision keyed datums. The asymmetric hook prevents reversal; captive clips provide clamping and the separate anti-lift feature provides fail-safe retention. Record the locator survey, latch witness map, earth-bond resistance, water test, and timed module removal. Active services are installed in moulded empty routes and are never laminated into the composite.

8. Interior cassette

- Install floor boards and removable hatches.
- Install fire-rated FRP/phenolic ceiling liners, light troughs, sidewall/window reveals, cable covers, under-seat battery strake covers, vestibule kick panels, PRM ramp/step covers, and door-pocket trims from `cabin-fiberglass.md`.
- Install under-seat battery enclosure covers and service locks.
- Install seats, grab rails, lighting, PIS, speakers, CCTV, intercom, emergency signage, and fire extinguishers if locally required.
- Install cable harnesses as pre-terminated looms, not loose field wiring.

Hold point: egress width check, sharp-edge check, service-panel removal trial, rattle check, lighting lux test, fire-material certificate pack complete.

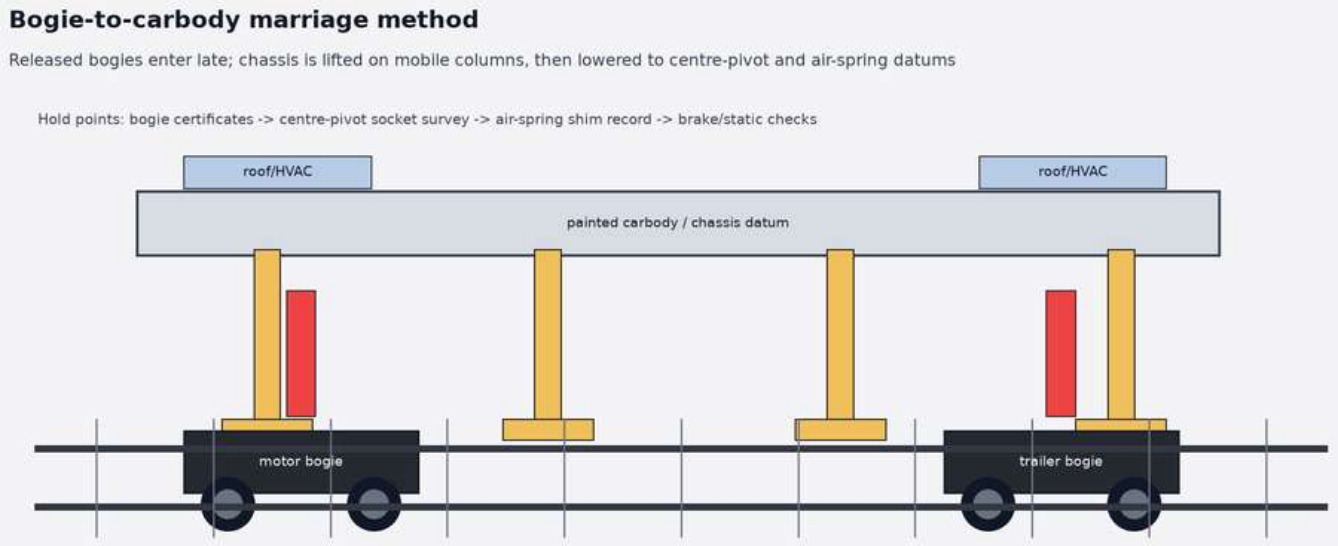
9. Power and traction equipment

- Install battery packs in under-seat cassettes.
- Install six motor controllers, isolated LV DC/DC converters, HV contactors, cooling loops, temperature/off-gas detection, outward module vents, and the per-car battery water-mist reservoir/pump/pipe/nozzles.

- Pressure-test coolant loops before energisation.
- Perform insulation resistance and HV interlock loop tests.

Hold point: HV safety sign-off before first energisation.

10. Bogie marriage and commissioning



Bogie-to-carbody marriage method

- Assemble powered and trailer bogies separately.
- Build one powered bogie and one trailer bogie through full weld, dimensional, NDT, wheelset, brake, suspension, and harness inspection before committing the remaining four bogies.
- Reject recovered freight bogie frames. Only accepted donor axles or axlebox hardware may enter the OSR bogie build, and only with the inspection records linked to the vehicle serial file.
- Drop carbody onto bogies using surveyed bolster datums.
- Connect traction, brake, WSP, tachometer, bearing-temperature, and suspension interfaces.
- Run static brake test, door test, HVAC test, lighting/PIS/CCTV test, battery charge/discharge test, and low-speed yard movement test.

Hold point: first-article inspection release before dynamic testing.

COTS interface rules

The carbody must reserve supplier-neutral interfaces:

Module	OSR fixes	Supplier owns
Door cassette	Opening envelope, structural datum, power, data, hardwired lock loop, drainage path	Door mechanics, controller, seals, obstacle detection, emergency release
Window cassette	Opening envelope, bond/gasket land, drainage, replacement access	Glass laminate, frame, adhesive/gasket, certification
HVAC	Roof rail pattern, power budget, duct opening, condensate route, service clearance	Refrigerant circuit, controls, compressor, fans, fire certificate
Battery mist	Reservoir/pump bracket, stainless-pipe route, nozzle datums, drain and service access	Qualified pump/nozzle hardware, flow/pressure sensors, component certificates
Lighting	24 V DC rail, mounting pitch, lux target, emergency mode input	Luminaire, diffuser, driver electronics
Seats/grab rails	Floor/sidewall insert grid, strength target, envelope	Seat shell, upholstery, rail clamps, fire certificates
PIS/CCTV/intercom	Power/data connector, field of view or visibility target	Device hardware, firmware, certificates

No primary steel member should be redesigned when a COTS supplier is changed. Supplier changes should require only adapter plates, harness tails, software configuration, and certification paperwork updates. The detailed envelopes and incoming-inspection evidence are maintained in the `control-electronics/trainset-interiors.md` catalogue.

Inspection gates

Gate	Evidence
G0 material release	Mill certificates, supplier certificates, incoming inspection
G0B recovered-component release	Quarantine record, cleaning record, dimensional report, UT/MT/NDT evidence, supplier sign-off for any donor axle/axlebox item
G1 frame complete	Dimensional survey, weld map, NDT report, traceability register
G2 corrosion complete	Blast record, coating DFT report, cavity wax checklist
G3 shell complete	Composite cure records, water test, removable-panel access report
G4 COTS systems fitted	Door/window/HVAC/lighting supplier acceptance records
G5 electrical safe	Insulation resistance, bonding/earthing, HVIL, functional safety I/O
G6 static complete	Brake, door, HVAC, lighting, battery, communications, fire detection
G7 dynamic ready	Weighing, ride-height, bogie alignment, low-speed movement release

First-article constraints

- Build one carbody first, not a full three-car batch.
- Build one powered bogie and one trailer bogie first, not all six bogies as an uncontrolled batch.
- Do not commit to production tooling until G1 and G3 are passed on the first carbody.
- Freeze COTS supplier envelopes only after fit-up review with actual door, window, HVAC, and seat samples.
- Treat every locally made structural weld procedure as a certifiable manufacturing process, not a prototype craft operation.
- Keep composite panels sacrificial and replaceable. They should fail cheaply without damaging the steel load path.

LM3 Portable Field-Rerailing Concept

Repository source: `docs/rolling-stock/light-metro-3car/field-rerailing-concept.md`

Status: design-reference coordination basis. This is not a certified recovery method, equipment selection, ground-bearing calculation, or authority to lift a damaged vehicle.

Design Decision

LM3 shall be designed for recovery by a small, specifically trained response crew using portable rail-rated hydraulic equipment. Ordinary automotive scissor jacks are outside the concept. The normal kit uses synchronized telescopic cylinders, wide baseplates, keyed/tilting vehicle adapters, mechanical cribbing, and a transverse rerailing bridge.

The controlled source assemblies are:

- `portable-field-rerailing-kit` for the recovery equipment envelope;
- `underframe-jacking-recovery-interface` for the installed J1–J4 pads, lifting-eye bosses, tow/rerailing lugs, and labels; and
- `civil-wayside-rerailing-access-interface` for optional hardstanding and equipment access at selected road-accessible response nodes.

All three use the same jack centres: X = +/-5,600 mm and Y = +/-1,180 mm from the car centreline/top-of-rail coordinate system.

Mass Sensitivity

The controlled planning tare remains 78,750 kg until supplier-frozen, detailed CAD, and weighed evidence closes the mass budget. The lower cases below are sensitivity studies, not released tare values.

Case	Train tare	Mean car tare	Ideal four-point static reaction
Controlled planning tare	78,750 kg	26,250 kg	64.36 kN per point
10% lower sensitivity	70,875 kg	23,625 kg	57.92 kN per point
20% lower sensitivity	63,000 kg	21,000 kg	51.48 kN per point

The generated mass-closure ledger confirms that the current design-space search does not establish a large mass saving. Its light-body/light-bogie candidate is 73.376 t before reserve and 76.818 t with the reserve retained, only about 1.932 t below the selected case. A materially lighter train therefore requires a new structural/material design and full crash, fatigue, fire, running-dynamics, braking, and manufacturability closure.

Preliminary Load Screen

For coordination only, each permitted load case applies a 1.35 unequal-load factor and a 1.50 action factor. The one-end case assigns 60% of the car tare to the raised end to screen longitudinal centre-of-gravity movement.

Load case	Active points	Supported tare	Required capacity per point	200 kN equipment-envelope margin
Complete car with running gear retained	4	100%	130.32 kN	69.68 kN
One end with its bogie retained	2	60%	156.39 kN	43.61 kN

Passing this screen only shows that a portable 200 kN cylinder class is worth developing. It does not release the car structure, the jack, or the lift. The final reactions shall use individual-car masses and centre-of-gravity envelopes for every passenger/load condition, suspension state, bogie-retained condition, tilt, damaged-point availability, and supplier load path.

Recovery Modes

Condition	Preferred response
Healthy wheels, train immobilized	Isolate, release/pipe the brake as designed, fit the recovery coupling, and tow or propel under the operating rulebook. Do not lift merely because traction power is unavailable.
One locked or locally damaged wheelset	Use the released wheel dolly/skate and rescue coupling if gauging, braking, axlebox condition, and infrastructure permit.
Upright, minor derailment with intact lift points	Stabilize and crib; lift one end or the complete car on a synchronized transverse pair/four-point set; translate on the rerailing bridge; lower to rail; inspect before any movement.
Pad inaccessible, unstable/rolled vehicle, structural damage, poor ground, bridge/tunnel constraint, or hazardous battery condition	Escalate to the specialist heavy-recovery plan and crane or alternative engineered method. The portable method is prohibited.

Equipment Envelope

- Four rail-rated telescopic cylinders, at least 200 kN rated capacity each and no more than 30 kg per cylinder envelope for two-person/manual team handling.
- Four wide-area baseplates/ground spreaders and four keyed, angular-tolerant adapters that positively locate on J1--J4.
- Two modular transverse rerailing bridges with locking traverse sleds.
- Mechanical cribbing and secondary retention. Hydraulic pressure is never the sole means of supporting a vehicle while personnel enter a hazard zone.
- A separately metered four-circuit pump/control unit allowing synchronized or deliberate individual correction from outside the exclusion zone.
- Hoses, rupture protection, gauges/load monitoring, lighting, bonding and isolation equipment, chocks, wheel dollies/skates, rescue coupler, and the controlled recovery manual.

Commercial technology demonstrates that portable equipment in this class is feasible: LUKAS publishes a 24 kg internal rerailing jack with 204 kN lifting force and transverse bridge systems; Holmatro publishes compact rerailing cylinders with hose-rupture-controlled lowering. These are capability references, not selected suppliers.

Vehicle Detail Requirements

1. Prove complete-car and one-end lifting with running gear mechanically retained, including articulation anti-lift and service-loop conditions.
2. Keep J1--J4 visible and accessible after suspension collapse; provide removable skirt panels without personnel entering beneath an unsupported vehicle.
3. Key every portable adapter so it cannot be mistaken for a depot column head or applied to non-structural cladding.
4. Provide externally readable mass, centre-of-gravity range, permitted support combinations, isolation, brake-release, bogie-retention, and lifting diagrams at both car sides and in the digital recovery manual.
5. Keep the rescue coupler, tow lugs, brake connection, dead-battery controls, and HV isolation accessible after loss of normal auxiliary power.
6. Treat a missing/damaged required pad, lateral single-side lift, or work under hydraulic support without mechanical retention as a stop condition.

Release Hold Points

1. Replace planning mass with weighed individual-car values and signed centre-of-gravity/load-condition envelopes.
2. Release underframe, articulation, bogie-retention and coupler calculations; weld/NDT schedules; adapter drawings; and pad proof loads.
3. Freeze a cylinder, bridge, pump, hose, control, cribbing and dolly supplier set with inspection and maintenance requirements.
4. Complete geotechnical/ground-bearing checks for designated access nodes and define the field method for non-designated locations.
5. Conduct no-load, representative ballast, asymmetric, loss-of-pressure, emergency-lowering, full-car, one-end and translate/lower trials.
6. Train and assess the recovery crew using the production vehicle, timed deployment, communications, exclusion-zone, battery and evidence-preservation procedures.
7. Issue vehicle-specific jacking/lifting diagrams and instructions before entry into service.

Reference Requirements And Capability Examples

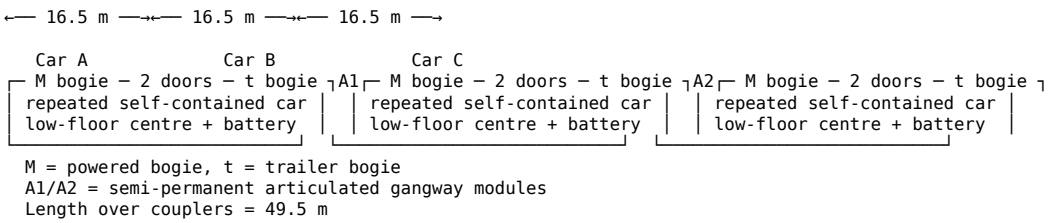
- UK LOC&PAS NTSN, Issue 2 (May 2025)
- RSSB RIS-2780-RST Issue 1.1 — Rail Vehicle Structures
- LUKAS HP 21 / 300 R rerailing jack
- LUKAS rerailing systems

- Holmatro rerailing cylinders

General arrangement — light-metro-3car

Repository source: `docs/rolling-stock/light-metro-3car/general-arrangement.md`

Consist diagram



- Three self-contained cars, joined by two semi-permanent articulated gangway modules.
- No driving cabs — GoA 4 identical A/B-end multi-part fiberglass sensor cowls at both ends per RFC 0015, with heated RF-transparent glass, T-OBS sensors, and LED headlamp / marker-light clusters.
- No cab bulkhead — the front and rear passenger ends are not walled off; the saloon remains open to the single panoramic glass end pane so passengers see through both driverless ends.
- Six bogies total — two standard 2-axle bogies per car.
- Two articulations total — lower spherical pivot/drawbar, upper roll-yaw-pitch links, double-wall bellows, segmented turntable floor, and separated trainline routing at each carbody interface.
- Powered wheelsets: one powered bogie per car, giving three powered bogies and six powered axles across the consist.
- Battery: 180 kWh usable / 225 kWh gross LFP per car under the longitudinal seats. The 10 m centre door zone stays low-floor and clear.
- Roof: PV strip per car, split around compact end HVAC modules.

Key dimensions

Parameter	Value	Source
Length over couplers	49 500 mm	promoted v2A envelope: 3 × 16.5 m car modules
Car A length	16 500 mm	modular car envelope
Car B length	16 500 mm	modular car envelope
Car C length	16 500 mm	modular car envelope
Body width (outer)	2 850 mm	concept envelope, fits UIC 505-1 gauge
Height over rail (roof)	3 450 mm	concept envelope, before roof equipment
Overall running height	3 900 mm controlled envelope; current CAD reaches 3 868 mm	includes PV, HVAC, mounts, and production tolerance
Floor height (above top-of-rail)	350 mm, 10 m low-floor centre door/PRM zone; 760 mm, ~3 m high-floor end decks over bogies	low-floor access with standard bogies
Bogie wheelbase	2 100 mm	RFC 0022 single-SKU bogie
Wheel diameter (new / worn)	760 / 680 mm	RFC 0022
Inter-bogie distance (within a car)	12 300 mm	16.5 m module with 2.1 m bogie inset from each end
Articulation module length envelope	1 120 mm	straddles each carbody interface; does not add to consist length
Articulation passage width	1 650 mm	walk-through gangway target before trim tolerances
Articulation yaw/pitch/roll clearance	+/- 12 deg / +/- 6 deg / +/- 5 deg	detailed in `articulation.md`
Coupler face height (above ToR)	720 mm	Scharfenberg Type 10 dim.
Headroom at doors	2 000 mm	
Headroom at seats	2 100 mm	
Passenger-compartment interior width	2 700 mm	

The 16.5 m promoted car module is intentional rather than a placeholder. With ~3 m bogie zones at each end it leaves about 10 m of low-floor centre span for two door pairs, wheelchair turning circles, and standing space, while keeping the 3-car train to 49.5 m for the derived 59.5 m OSR standard platform. Longer 19-20 m cars remain possible as a later capacity variant, but the v1 base favours workshop-friendly modules and curve overhang margin.

Door positions

Two large 1 400 mm-wide double-leaf plug doors per car side in the 10 m low-floor centre zone (6 door pairs per side for the consist). The two openings fit between the ~3 m high-floor bogie decks at each end, leaving the middle span for PRM circulation and standing room. Positioned:

- Each car: door centres at 5.50 m and 11.00 m from the car end.

All doors are plug-outward doors per RFC 0008 §3.3 with a 1 400 mm nominal structural opening and at least 1 250 mm clear passage after seals/thresholds at 2 000 mm height.

Gauge compliance

UIC 505-1 static envelope:

- Width at floor level: 2 850 mm ≤ 3 150 mm limit
- Overall height over ToR: 3 900 mm ≤ 4 320 mm limit
- Dynamic outline (controlled lateral sway and +30 mm vertical excursion): 2 970 × 3 930 mm — still inside the 3 150 × 4 320 mm boundary

No interference with UIC 505-1 at any of the RFC 0009 track presets (`standard-urban` / `standard-metro` / `mainline-mixed`).

Platform clearance

- Low-floor centre door and PRM zone at 350 mm ToR platform height. Each car has about 10 m of low-floor length between the standard bogie zones. The ~3 m raised end decks sit directly over the bogies; those raised zones are inside the saloon, away from the boarding thresholds.
- Horizontal gap at door sill: ≤ 75 mm per UIC 741. Achieved by a 40 mm sliding skirt at each door that extends on dwell (retracted in motion).
- On curved platforms at RFC 0009 `standard-urban` minimum radius (90 m): the door-to-platform horizontal gap grows to ~110 mm — requires an 80 mm gap-filler flap at the door sill.

Mass distribution (AW2, nominal loaded)

Location	Mass (t)
Car A incl. powered + trailer bogie	33.45
Car B incl. powered + trailer bogie	33.45
Car C incl. powered + trailer bogie	33.45
Total consist (tare + AW2 passengers)	100.35

Controlled planning tare = 78.75 t: 75.308 t optimizer subtotal plus a 3.442 t engineering reserve pending drawing-package mass closure. AW2 load = 21.6 t (360 passengers × 60 kg average).

Distribution per axle under AW3 (crush load, 480 passengers × 60 kg = 28.8 t):

- Powered and trailer bogies: ~10–12 t per axle depending on seated battery-side loading.

All ≤ 14 t per RFC 0008 §3.1 axle-load constraint .

Structural loading cases (for v2 FEA)

The design envelope the v2 CAD model is checked against:

- EN 15227 Cat C-II collision: 25 km/h train-to-obstacle. Cab crush zone absorbs. Anti-climbing features at both ends.
- EN 12663 Cat P-III car-body static: compressive end load 640 kN; diagonal 600 kN. Welded S355 underframe/spaceframe carries the load path; composite side and roof panels are non-structural weather/aero skins.
- Vertical accel: ± 0.5 g normal service; ± 2 g emergency brake / collision.
- Lateral accel: ± 0.3 g through max-cant curve.

The current screening FEA is linked from the package README. The full-body lateral-sway screen is still a v2 closure item: the next body release must add diaphragm/knee-brace detail at the side-frame and waist-rail load path, then rerun the shell/beam model before first steel cut.

Cross-refs

- Bogie detail `bogie.md`
- Body structure `body.md`
- Identical fiberglass end cowl `end-cowl.md`
- Traction `traction.md`
- Interfaces `interfaces.md`

Interfaces — light-metro-3car

Repository source: `docs/rolling-stock/light-metro-3car/interfaces.md`

The interfaces this trainset exposes to the outside world. Every interface is pinned to a commodity standard; no bespoke connectors.

Coupler

Parameter	Value
Type	Scharfenberg Type 10 automatic
Vertical coupling compatibility	Any Scharfenberg-family consist at 720 mm coupler face height
Mechanical strength	1 000 kN static tensile
Electric coupler	128 pins: 24 V DC aux, CAN-FD (2 pairs), Ethernet (TSN 1 Gbps), emergency-brake loop, video (MIPI over fibre)
Pneumatic coupler	None. No pneumatic brake / air-horn compressed-air coupling.
Activation	Electrical lock/unlock from recovery cabinet or OCC command
Cut-out / rescue mode	Mechanical unlock handle accessible at ground level

Every OpenSourceRail consist in every deployment carries the same coupler — any trainset can rescue any other.

Inter-car articulation and gangway

The 3-car reference consist has two semi-permanent articulated gangway modules. They are not end couplers and they are not Jacobs bogies; they are fixed intra-consist structural/passenger/service interfaces between self-contained standard-bogie cars.

Parameter	Value
Modules per 3-car consist	2
Lower joint	Spherical central articulation bearing with anti-lift keeper
Draft/buff path	Underframe anchor casting -> articulated drawbar -> lower joint -> opposite anchor
Upper control	Two roll-yaw-pitch links with spherical bearing eyes
Passenger passage	1 650 mm target clear width, segmented anti-slip turntable floor
Bellows	Double-wall corrugated gangway with bolted clamp frames
Kinematic envelope	+/- 12 deg yaw, +/- 6 deg pitch, +/- 5 deg roll
Trainline routing	Separated HV, TCN-E, CAN-FD/safety-loop, coolant, HVAC air sleeve, and drains
Replacement	Module cassette at heavy overhaul; bellows and floor hatch serviceable in depot

Detailed structure, service routing, supplier evidence, and maintenance intervals are in `articulation.md`.

Station charging connector

Parameter	Value
Primary type	Side-pin conductive connector per RFC 0026
Alternate type	Pantograph-down dock where platform geometry requires it
Operating voltage	Train pack voltage, approximately 650–700 V nominal and 740 V normal maximum
Contact / insulation rating	At least 1 000 V DC
Contact current (charging)	500–1 000 A station class
Actuation	Electro-mechanical (no pneumatic)
Location	One train-side connector per car, platform side
ATO interface	Extend/retract via the `osr-ato` + `osr-aux-power` pairing on the T-ECU/A
Onboard power electronics	Per-car HV contactor/protection/IMD charge rack; the station-side DC/DC regulates dock current and the roof PV uses its own MPPT stage

The station battery buffer supplies the charge pulse. Inter-car coordination and per-car current limits are handled by `osr-aux-power`; pack current and temperature limits come from `osr-bms`.

Platform gap

Per RFC 0010 §4.4:

Parameter	Value
Horizontal gap (straight platform, door-to-sill)	≤ 75 mm (target 40 mm with retractable skirt)
Horizontal gap (curved platform at 90 m radius)	≤ 110 mm with 80 mm gap-filler flap
Vertical gap (floor-to-platform)	≤ 50 mm
Door sill retractable skirt	40 mm, electrically actuated at dwell-open

Parameter	Value
Door sill gap-filler flap	80 mm, drops on PSD-open signal at curved platforms (identified per-deployment)

Gap-filler flap is a simple solenoid-driven plate — no pneumatic, no gear train. Failure mode: flap stuck up `osr-door-control` blocks door-open; failure flap stuck down visual inspection at depot next cycle.

Platform automation and PSD interface

The train has two door openings per side per car. On the 3-car reference consist that is six door centres per platform side, all in the 350 mm low-floor zones. Station platform screen doors or platform edge doors align to the train-side door centres.

Parameter	Value
Door centres per car	5.67 m and 11.33 m from the car end
Door centres per 3-car platform side	6
ATO stopping target	± 300 mm normal, ± 500 mm degraded before door-open inhibit
Door-open permissive	Train stopped + correct side selected + PSD/edge-door ready + closed/locked loop healthy
Intrusion sensing	Door-sill camera/obstruction harness plus station edge sensor input
Short-platform operation	Selective door opening by car and side via `osr-door-control`

TCN-E connector (intra-consist, via articulation)

Standard Ethernet + power bundle through each articulation joint:

Signal	Cable	Notes
TCN-E port A	Cat 6 RJ45, TSN-capable	Per RFC 0006 §4; single-ring backbone
TCN-E port B	Cat 6 RJ45, TSN-capable	Redundant ring
CAN-FD (doors + HVAC)	Twisted pair per ISO 11898	
LIN (local panels)	Twisted pair per ISO 17987	
24 V DC aux + control	2.5 mm² × 2 wires	Aux + return
110 V DC backup	4 mm² × 2 wires	Emergency battery backup
Safety loop (emergency brake)	Hardwired pair; normally-closed	Electrically independent of TCN-E

Connection through the articulation bellows via drag-chain; no disconnects — this is an intra-consist fixed bus.

Station charging dock

The charging interface to the station:

Parameter	Value
Connector	Side-pin primary; pantograph-down alternate
Voltage	Follows the train pack: approximately 650–700 V nominal, 740 V normal maximum; hardware rated at least 1 000 V DC
Max current	825 A equipment limit; approximately 714–769 A at the 500 kW operating point
Ramp rate	Controlled by the station DC/DC under train BMS limits relayed by `osr-aux-power`
Connector extend / retract signals	From `osr-aux-power` via TCN-E
Isolation	Dock-side mechanical isolator; electrical monitoring via `osr-energy-site`
Onboard sink	Traction battery through per-car HV contactor, fuse, IMD/HVIL, and BMS-controlled charge rack; no onboard AC conversion

Full RFC 0002 energy-sizing compatibility.

Roof PV charge input

The roof solar input is an onboard source, not a wayside interface, but it shares the same charge rack and isolation logic as the station dock.

Parameter	Value
Modules per car	12
Mount styles	6 bonded flexible laminates; 6 raised rigid panels on rails and edge clamps
String hardware	Module junction boxes, two string raceways, MPPT combiner, fire-isolation switches
Battery interface	PV downlink cable gland into the per-car MPPT/DC charge rack
Control	`osr-aux-power` selects PV harvest, station charge, regen absorb, or shed-load mode against `osr-bms` limits

Emergency door release (external)

- Each exterior door has an external unlock per EN 14752.
- Mechanical; no electronics on the external side.
- Engages only when trainset speed ≤ 5 km/h OR on emergency- brake trip (latched state).

Ground-radio (external antennas)

Antenna	Location	Band	Notes
TRG-1 5G main	Car A roof	n77 / n78 / CBRS	
TRG-1 5G diversity	Car C roof	Same band	MIMO pair
TRG-2 LoRa	Car B roof centre	868 MHz (EU) / 915 MHz (US/MEA)	
GNSS	Car A roof + Car C roof	L1/L2	
Public-safety (optional)	Car A roof	700 MHz	Per-deployment

All antennas use a single-type N-male to RP-SMA termination, no bespoke sockets.

Mass & structure compatibility matrix

Rolling-stock × track-geometry × station-archetype compatibility per RFCs 0008 §4 / 0009 §10 / 0010 §12:

Compatible with	Reason
`standard-urban` geometry	Yes — 49.5 m consist, 90 m min curve OK, 50 ‰ max grade OK
`standard-metro` geometry	Yes — plenty of margin
`mainline-mixed` geometry	Yes — meets 25 ‰ max grade (lower than the consist's 5 % capability)
`halt` station	Yes — short-platform door select required
`standard` station	Yes — 59.5 m derived platform for the 49.5 m consist
`major` station	Yes
`interchange` station	Yes
`terminal` station	Yes — side-pin dock fits
`depot-terminal` station	Yes

v2 deliverables (not in v1)

- Coupler electrical connector pinout drawing.
- Station charging dock mechanical envelope (for station civil design).
- Gap-filler flap mechanism drawings.
- Antenna radiation-pattern simulations at the consist roof.

Make/Buy and Localization Plan

Repository source: `docs/rolling-stock/light-metro-3car/localization-plan.md`

This plan implements RFC 0021's principle: own the interfaces and evidence, localize ordinary fabrication early, and localize safety-critical catalogue parts only after process capability and qualification are demonstrated.

System	First production	Early local work	Later localization gate
Bogies	Buy complete qualified bogies for the prototype and approximately first 20–30 trainsets	Inspection, wheel reprofiling, brake service, harnesses, final vehicle marriage	Local frame welding/final assembly after stable demand, EN 15085-quality welding, dimensional capability, NDT, fatigue evidence, and supplier audit; continue buying qualified wheels, axles, bearings, suspension, brakes, motors and gearboxes
Doors	Buy drive, lock, sensing, emergency release and controller kits	Fabricate leaves, frames, rails, housings, thresholds, adapters and harness tails to the OSR cassette ICD	Localize drive parts only after lifecycle, obstruction, lock integrity, environmental and software-interface qualification
Articulation	Buy spherical bearing, yaw/pitch joint, anti-lift elements, elastomers and bellows	Fabricate steel adapter frames, drawbar support, floor bridge, covers, trays and service guides	Localize joints/bearings only after proof, fatigue, wear and fire evidence
Windows	Buy complete laminated solar-control rail/bus safety-glass cassettes	Local aperture frames, drains, adapter plates and installation	Local glazing only after optical, impact, fragmentation, thermal and adhesive-system qualification; acrylic/polycarbonate remains limited to noncritical covers/interiors
Seats	Buy or locally fabricate simple qualified longitudinal bench shells	Local steel/aluminium frames, mounts and replaceable trim	Release only with material fire/smoke/toxicity certificates and mount/static proof-load evidence
Battery	Buy qualified cells/modules, BMS sensing, contactors and fire-detection components	Local enclosure frames, vent ducts, cooling manifolds, mist pipework, harnesses and pack integration under supplier controls	Cell manufacture is out of scope until a separate industrial program exists; pack integration requires IEC 62928-based test evidence
Traction	Buy matched motor/controller sets	Local brackets, coolant manifolds, cables and connectorized harnesses	Motor/controller localization only after dynamometer, insulation, EMC, vibration and functional-safety qualification
Body	Local from first article where capability exists	Welded S355 datum skeleton, modular GFRP cassettes, sacrificial nose, interior panels and final assembly	Production release after weld procedure/NDT, composite fire/bond/insert, water ingress, tolerance and FEA gates

The semi-permanent drawbar carries longitudinal train loads separately from the passenger gangway. Door, glazing, HVAC, traction, battery and articulation suppliers shall fit published supplier-neutral envelopes rather than drive changes into the primary steel structure.

All localization decisions are serial-number traceable. A lower purchase price never overrides missing material certificates, process qualification, inspection records, software/configuration control, spares support, or the deployment safety case.

Marketplace Price Proxies (cost studies only)

Repository source: `docs/rolling-stock/light-metro-3car/marketplace-price-anchors.md`

> Do not use this page to select build parts. The controlled supplier-anchor register assigns a real manufacturer family, known fit gaps and local-equivalent route to every bought-in LM3 product. This older ledger remains only to reproduce the low-cost sensitivity in the BOM; marketplace listings are not engineering or procurement approval.

Checked 2026-06-04 against public Alibaba marketplace pages and AliExpress item history through PriceArchive where direct AliExpress search pages are unstable. These are listed-price anchors, not certified rail procurement quotes. They exclude freight, duty, supplier audits, rail fire/smoke/toxicity reports, EN/TSI evidence packs, warranty terms, and acceptance testing unless the BOM line says otherwise.

The updated BOM therefore represents a marketplace-only procurement floor for a three-car standard-gauge consist. BID lines still need supplier RFQs before a build can be treated as qualified rolling stock.

RFC 0021 procurement baseline

The T-series rows below preserve the marketplace evidence ledger, but RFC RFC 0021 defines the procurement baseline. Current RFQs shall target:

- three 225 kWh gross liquid-cooled LFP car packs: $65k–80k per train;
- six heavy-vehicle-class PMSM/controller sets: $48k–72k per train;
- HV distribution/contactors/insulation monitoring: $8k–12k per train;
- isolated LV DC/DC equipment: $5k–10k per train; and
- roof MPPT/DC electronics: $2k–4k per train.

These are target bands, not observed quotations. Candidate reference evidence includes EVE's railway LFP application, Inovance HM47/LD32 product data, Longertek direct-DC rail HVAC, and the SINEXCEL PDS1-750K-H datasheet. None of those pages demonstrates compliance with the complete OSR railway duty; the RFQ must price qualification, software/interface support, spares, warranty, freight, and acceptance testing separately.

Anchor Index

Line(s)	Marketplace anchor	Listed price observed	BOM base used	Qualification note
B1	Alibaba S235/S355 rectangular steel tube	380-720 USD/t	3,000	Includes 3.2 t steel, waste, local handling, and cut premium.
B2	Alibaba S355 steel plate and S355 price table	435-790 USD/t	2,000	Includes 2.0 t plate plus material loss and freight allowance.
B3	Alibaba steel fabrication services	quoted per piece/order	6,000	Fixture consumables and cut/drill setup allowance; real quote needs DXF pack.
B4	Alibaba welding wire	0.86-1.50 USD/kg	2,000	Consumables only; certified welding labour remains outside direct material.
B5	Alibaba zinc-rich coating	2.11-4.21 USD/kg	4,000	Primer, blast media, cavity wax, masking, and waste allowance.
B6, B7, B21	Alibaba fire-rated composite panel	2-50 USD/sq m	9,000; 4,000; 5,000	Commodity panels priced low; EN 45545 certificate must be supplied or tested.
B8	Alibaba FRP/composite panel proxy	5-40 USD/sq m	4,000	Multi-part fiberglass end-cowl cast kit only; glazing and sensor hardware are separate lines.
B9	Alibaba train gangway suppliers	3,500-15,999 USD/set	30,000	Covers two articulation/gangway sets at listed rail-part levels.
B10, B27	Alibaba train tempered laminated glass	5-90 USD/sq m; rail curved glass 38.50-41.50 USD/sq m	6,000; 10,000	Side window cassettes are commodity anchored; heated curved end panes need RFQ.
B11	Alibaba automatic sliding door systems	145-200 USD/operator; 890-2,988 USD/hermetic door proxy	90,000	Marketplace door hardware is a proxy; passenger door certification is still BID.
B12	Alibaba floor sandwich / composite board proxy	5-50 USD/sq m	6,000	Structural floor board, hatches, inserts, and fire rating require supplier pack.
B13, B28	Alibaba bus vinyl flooring and vehicle decal/vinyl proxies	5-45 USD/sq m	1,500; 2,000	Flooring and exterior livery film are commodity-priced.
B14	Alibaba bus seats	15-160 USD/seat	6,000	Seat shell/listed transit seating anchor; rail upholstery/fire evidence needed.
B15	Alibaba bus handrail fittings	1.15-10.40 USD/fitting	2,500	Stainless tube, joints, brackets, and installation hardware.
B16, A3	Alibaba 12/24 V bus interior lamps	3-15 USD/lamp	1,500; 1,000	Emergency lighting battery packs are included in A3 allowance.
B17	Alibaba bus lighting and marker lamps	1-150 USD/lamp	1,200	Exterior head/tail/marker kit with sealed lamp allowance.
B18, E14	Alibaba bus LED sign/display and destination sign suppliers	55-350 USD/display	3,000; 3,000	Includes interior/exterior display mix and controller allowance.
B19, E15	Alibaba bus CCTV systems and AliExpress PoE camera history	37-580 USD/kit; cameras from a few USD	2,500; 600	CCTV/intercom kit uses commodity camera anchor; safety outputs remain separate.
B20	Alibaba automotive urethane coating proxy	2-15 USD/kg	3,000	Paint material only; booth labour and rework outside direct material.
B22	Alibaba railway coupler proxy	RFQ/listed parts around low thousands	8,000	Rescue-capable coupler still requires rail supplier RFQ before build.
B23	Alibaba railway draft gear proxy	RFQ/listed parts around low thousands	4,000	EN 15227 crush-can qualification remains a tender item.
B24, T18, E17, E20	Alibaba wiring harness services and HV cable proxies	quoted per loom/order	4,000; 12,000; 9,000; 3,000	Harness direct material anchor only; drawings and tests must be quoted.
B25, T19	Alibaba linear actuator proxy	20-300 USD/actuator	6,000; 3,000	Door and charging actuators need safety interlock validation.
B26, B29	Alibaba machined steel fabrication proxy	quoted per part/order	3,000; 6,000	Jacking/lifting fittings and articulation adapter frames need final drawings.
G1, G2, G13, G17, G18, G20	Alibaba train railway bogie parts	900-15,000 USD/frame/part	10,000; 12,000; 3,000; 3,000; 4,000; 3,000	Fabricated bogie items are anchored to marketplace rail-part fabrication.
G3	Alibaba railway wheel price and locomotive wheelset	500-2,000 USD/wheelset	18,000	Twelve wheelsets at listed wheelset range with inspection allowance.
G4	Alibaba railway bearing proxy	40-150 USD/bearing; higher for certified units	12,000	Uses marketplace bearing housings plus certified bearing uplift.
G5, G10, G12	Alibaba rail spring/bushing proxies	20-125 USD/spring or link proxy	4,800; 2,400; 1,800	Elastomer and spring compounds require certificate traceability.
G6	Alibaba railway air spring	300-1,500 USD/unit	6,000	Twelve air springs at low listed rail range.
G7	Alibaba railway hydraulic damper proxy	50-300 USD/unit proxy	3,000	Damping curve and temperature envelope need supplier confirmation.

Line(s)	Marketplace anchor	Listed price observed	BOM base used	Qualification note
G8, G16	Alibaba railway brake pads and rail brake-disc proxies	pads 13-100 USD; discs via RFQ/proxy	12,000; 1,500	Braking friction material must be qualified with the selected caliper.
G9	Alibaba brake caliper proxy	industrial calipers 40-215 USD; rail RFQ higher	12,000	Rail electromagnetic caliper remains BID despite commodity proxy.
G11	Alibaba heavy bearing / PTFE slider proxy	50-500 USD/unit proxy	3,000	Centre-pin ring and slider stack need load rating.
G14, G15	AliExpress/Alibaba encoder and temperature sensor proxies	single to tens of USD	600; 500	Rail harnessing and environmental protection covered elsewhere.
G19	Alibaba flexible gear coupling proxy	100-800 USD/set	2,000	Torque rating needs confirmation against final motor/gearbox.
T1	Alibaba PMSM motor proxies	3,200-11,150 USD/motor; supplier table 4,500-7,000 USD	36,000 motor share	Six HM47-class 350 kW short-peak candidates; controller share is T3 and rail duty remains unqualified.
T2	Alibaba industrial gearbox proxy	500-4,000 USD/unit proxy	10,000	Rail duty cycle and noise target need gearbox RFQ.
T3	Alibaba EV inverter / PCS proxy	4,999-5,450 USD for PCS proxy	24,000	Six independent LD32-class controllers; generic PCS listings are lower-bound evidence only.
T4, T7, T20	Alibaba EV cooling and pump proxies	tens to hundreds USD per pump/plate	6,000; 2,000; 2,500	Cooling loops need thermal validation.
T5	Alibaba 200 kWh battery packs	marketplace BESS and EV-pack listings vary widely by certification scope	75,000	Three 225 kWh gross / 180 kWh usable LFP car packs; liquid cooling, segmentation, IEC 62928 test evidence and propagation controls require RFQ.
T6	Alibaba BMS proxy	hundreds to low thousands USD/system	4,500	Pack-level BMS electronics only.
T8, T17	Alibaba aluminium enclosure / duct fabrication proxies	quoted per part/order	6,000; 3,000	Under-seat enclosure and vent ducts need final sheet-metal drawings.
T9	Alibaba aspirating smoke detector	1,095-1,850 USD/unit proxy	10,800	Six detector channels at commodity detector pricing.
T10	Alibaba water-mist component proxy	component listings vary by pump/nozzle scope	9,000	Three localized battery-only systems with reservoir, DC pump, stainless pipe, nozzles and feedback; propagation testing excluded.
T11, T16	Alibaba 1000 V DC contactors and AliExpress contactor history	7-100 USD/contactors; larger vacuum units 290-621 USD	10,000; 2,400	Main 800 V-class HV distribution requires interrupt, weld, IMD, vibration and supplier evidence.
T12, T23	Alibaba 30 kW DC charge modules	480-510 USD/module; PCS 4,999-5,450 USD	4,500; 3,000	Train has a protected station-DC interface and roof MPPT; the 500 kW conversion stage is wayside.
T13	Alibaba auxiliary DC/DC proxy	100-1,500 USD/unit proxy	7,500	Isolated 800 V to 110/48/24 V DC domains; no central AC inverter.
T14	Alibaba bus rooftop air conditioning	2,237-6,500 USD/bus roof unit	12,000	Three roof HVAC units at bus commodity pricing.
T15	Alibaba braking resistor proxy	50-1,000 USD/unit proxy	1,000	Regen dump resistor needs thermal enclosure design.
T21	Alibaba 400 W solar panels and flexible 400 W panels	rigid 41-79 USD/panel; flexible 50-139 USD/panel	6,000	Full-roof panel material for 36 mixed rigid/flexible modules.
T22	Alibaba solar mounting/combiner proxies	low dollars per clamp/rail; combiner/MPPT low hundreds	6,000	Bond pads, raised rails, clamps, junction boxes, MPPT, and isolation.
E1, E2	Alibaba PCB assembly proxy	quoted per board/order	1,200; 1,500	Custom boards include assembly allowance; SIL evidence excluded.
E3	AliExpress IMU proxy	tens to hundreds USD/module	200	ADIS-grade part can exceed commodity proxy; base uses low-cost alternate.
E4	AliExpress GNSS module proxy	tens of USD/module	100	Uses commodity GNSS board anchor.
E5	AliExpress NFC reader proxy	single to tens of USD/module	20	Reader board only.
E6	AliExpress secure-element proxy	single USD/module	20	Chip/module allowance only.
E7	AliExpress 5G M.2 modem proxy	tens to hundreds USD/module	200	Carrier approval and antenna certification excluded.
E8	PriceArchive SX1276 LoRa modules	5.62-19.04 USD/module; 55.66 USD/10-pack	20	One radio plus spares.
E9	AliExpress NVMe proxy	tens of USD	40	Operational recorder storage only; crash memory is E23.
E10	PriceArchive 10.1 inch touchscreens	100-255 USD/display class	300	Two maintenance screens.
E11, E12, E13	Alibaba industrial pendant / switch proxies	low to hundreds USD each	500; 40; 100	Depot-only manual controls, plungers, and guarded keyed switches.
E16	Alibaba safety relay proxy	tens of USD/module	200	Relay stage material only; safety case excluded.

Line(s)	Marketplace anchor	Listed price observed	BOM base used	Qualification note
E18	Alibaba industrial LIDAR proxy	598-3,289 USD/sensor proxy	8,000	Two nose modules with LIDAR/cameras/radar allowance; rail T-OBS supplier RFQ needed.
E19	Alibaba washer/heater actuator proxies	tens to hundreds USD/system	1,000	Keeps glass sensor windows serviceable.
E21	Alibaba 5G/GNSS antenna proxy	tens to hundreds USD/antenna	300	Roof antenna set material only.
E22	Alibaba DIN rail cabinet proxy	hundreds to low thousands USD/cabinet	3,000	Three electronics cabinets and distribution hardware.
E23	Alibaba rugged SSD/event-recorder proxy	hundreds to low thousands USD	2,000	Crashworthy event recorder still requires qualification.
A1	Alibaba bus call button/accessibility proxy	single to tens USD/device	1,000	Wheelchair bay fixtures, buttons, trims, and guards.
A2	Alibaba safety label/signage proxy	cents to low dollars per label	500	Full train labels and tactile/visual signage.
A4	Alibaba fire extinguisher / first aid proxy	tens of USD per kit	500	Operator consumables and seals.

Cost Interpretation

The BOM numbers were reduced where the marketplace anchor directly covered the quantity. Safety-critical, custom, and certified rail parts remain marked BID or MAKE where the listed price is only a proxy. The direct-material base is therefore useful for open-source design trade studies and commodity purchasing discussions, but it should not be used as a production-ready certified trainset quote.

One-metre clip-on fiberglass body — LM3-BDY-160

Repository source: `docs/rolling-stock/light-metro-3car/modular-fiberglass-body.md`

The train exterior is a replaceable, non-structural weather skin made from simple fiberglass modules on a 1,000 mm longitudinal pitch. The welded S355 underframe, side posts, door portals, window carriers, roof bows, end rings, anti-climbers, and crash structure remain the designated primary load path; structural qualification and acceptance remain separate release gates.

Each 16.5 m car has sixteen full 1 m cladding bays between two 250 mm steel end-ring transitions. A three-car train therefore uses 144 main modules: 96 side modules and 48 roof modules. The product tree now identifies solid-side `LM3-BDY-P130`, window-edge `P131`, door-edge `P132`, and roof/fairing `P133` parts separately. They share the same mould width, clip datums, trim datum, seals, and repair method. CNC trimming creates the aperture variants after moulding.

Controlled geometry

Item	Design value
Longitudinal pitch	1,000 mm
Finished module width	994 mm
Dry joint	6 mm replaceable EPDM compression seal
Nominal sandwich depth	28 mm
Bays per 16.5 m car	16
Side modules per car	32
Roof modules per car	16
Solid/window/door/roof reference split	48 / 24 / 24 / 48 per trainset
Main modules per trainset	144
End transition outside module grid	250 mm at each car end

The parametric source is `modular_fiberglass_body.py`. It generates side modules with the released door/window cuts, roof modules, clip hardware, anti-lift retention, and joint gaskets. The machine-readable module and assembly manifest is generated under `design/component-catalogue/catalog/modular-fiberglass-body`.

Mould fabrication route

Each side, window-edge, door-edge, and roof variant is made from the same short glass-fibre body tooling philosophy: reusable 1,000 mm mould pitch, variant-specific CNC trim, and a master-frame dry-fit before kitting.

Seq	Operation	Required evidence
10	Release mould, trim fixture, laminate schedule, core map, insert map, and module serial range	Traveler revision, material certificates, mould release
20	Clean mould, inspect A-surface, apply release system, and apply UV-stable gelcoat or paint-primer layer	Mould surface record, release-system lot
30	Cut glass reinforcement and local core; lay up solid clip lands, pot insert bosses, and bag or close mould	Ply/core/insert checklist
40	Infuse or wet-lay laminate, control cure, demould, post-cure where specified, and retain witness coupons	Cure time/temperature, coupon ID, demould inspection
50	CNC trim door/window/roof variant, drill clip grid from datum, seal all cut edges, and mark serial/revision	Trim gauge, hole-position gauge, edge-seal record
60	Fit captive inserts, keyed hooks, clips, anti-lift features, drain details, and EPDM seals; dry-fit to master frame	Insert pull-out lot, gasket compression witness, master-frame fit

Supplier BOM rows fund qualified fibre, resin, core, gelcoat/paint-primer, release systems, seal stock, and coupon material. They do not introduce a bonded full-side or full-roof panel. The body module becomes a released part only after mould/cure records, CNC trim evidence, sealed-edge inspection, insert proof, and dry-fit evidence are closed.

Decorative colour is not part of the laminate load path. The preferred production route is a qualified white/light in-mould base finish followed by bay-numbered rail-use livery film. See `exterior-finish-process.md`; film does not replace corrosion, fire, cut-edge or composite surface protection.

Retention and sealing

Every module hangs on an asymmetric keyed upper hook and closes onto captive over-centre clips. An independent anti-lift retainer prevents a panel leaving the vehicle if a clip is not fully latched. Clip witness marks are visible from the service side. The fasteners remain captive when open.

The weather joint is a dry, replaceable EPDM compression gasket over a drained secondary rail. Production adhesive is not used between the body module and frame, so there is no cure delay and a damaged 1 m section can be exchanged. Adhesive

remains permitted inside a supplier-controlled fiberglass sandwich, for glazing, and at released non-service seams in the separate end cowl.

The clip, insert, rail, and backing-plate loads still require calculation and first-article proof. The release pack must include fire/smoke/toxicity evidence, laminate coupons, insert pull-out tests, clip proof load, fatigue/vibration, water ingress, aerodynamic pressure, and timed removal/refit evidence.

One-day exterior-body route

Six two-person crews work on three released frames in parallel, two crews per car. All parts are moulded, trimmed, painted, labelled, gasketed, and kitted before the frames enter the body station.

Seq	Activity	Elapsed	Crews	Release evidence
10	Verify frame datums and kit revisions	0.5 h	3	Frame/module map signed
20	Fit dry seals and inspect clip receptacles	1.0 h	6	Seals continuous; drains open
30	Hang and latch both side skins	3.0 h	6	All clip witness marks visible
40	Hang and latch roof modules	1.5 h	6	Anti-lift retainers closed
50	Fit closures, skirts, labels, and earth bonds	1.0 h	6	Continuity and access checks
60	Water/rattle test and close snags	1.0 h	3	Signed eight-hour body release
Total elapsed		8.0 h		

This is an exterior-body installation claim, not a claim that an uncertified train can go from bare steel to passenger service in one day. Doors, glazing, roof equipment, interiors, traction equipment, batteries, bogies, static and dynamic commissioning, homologation, and first-article testing remain their own controlled operations.

Cost and plant effect

The repeated short mould removes the full-car mould, large cure fixture, long bonding operation, and bonded-panel rework allowance from the baseline. The CAPEX source therefore uses a 28% local assembly allowance, a $900 k three-car trainset planning unit, and $60 k per supported car module for the shared national production-plant setup allowance. The high plant sensitivity is $120 k per car. These remain planning values until supplier quotes and timed first-article builds replace them.

Sandwich-panel development

The hybrid panel development package adds explicit fibre directions, skin/core/bond/finish layers, reinforced edges, potted inserts, curved roof CAD and an aperture-based 144-module mass takeoff. Its interactive comparison varies skin, core, module/support spacing, crown and ribs under stated pressure and ageing assumptions. The steel/aluminium comparisons require equivalent reinforcement and performance evidence before claiming savings. Material coupons, clipped-module/roof-bay tests and complete-cost quotations remain open. These development results do not reduce the frame, adopt a new train mass, qualify a fire stack or replace the B6/B7 planning allowance.

Roof moulds, HVAC, solar and service fitout

Repository source: `docs/rolling-stock/light-metro-3car/roof-fitout.md`

The LM3 roof is a layered, serviceable assembly. Structural roof bows and equipment rails (`LM3-BDY-P080`) carry loads. Clip-on roof skins (`LM3-BDY-P133`) and local fairings (`LM3-ROOF-P030/P040`) provide weather, aerodynamic and maintenance closeout only.

Part and zone definition

Item	Function	Must not carry
`LM3-BDY-P133`	1 m roof skin and trim variants	HVAC, PV, lifting or walkway structural loads
`LM3-ROOF-P010`	bolted/gasketed HVAC curb, duct collar, condensate tray and drains	Loads outside the released HVAC interface
`LM3-ROOF-P020`	PV rails/pads, clamps, bonds and isolation labels	Unverified adhesive-only module retention
`LM3-ROOF-P030`	removable HVAC intake/exhaust skirt, curb fairing and access hatches	Curb, lifting or fall-protection loads
`LM3-ROOF-P040`	PV/gland plinths, antenna closeouts and walkway edges	Antenna/PV proof loads or worker load
`LM3-EXT-P040`	supplier roof HVAC unit	Carbody loads beyond frozen feet/curb
`LM3-EXT-P050`	supplier PV module and clamp kit	Roof-skin load; rails resolve into structure
`LM3-EXT-P070`	antennas, walkway pads, lifting covers and labels	Unreleased roof penetrations
`LM3-FIN-P020`	trial radiative coating on exposed fairings	Any structural, sealing, bonding or anti-slip role

Mould family

The common roof mould fixes crown, 1,000 mm pitch, side returns, clip lands and dry-joint geometry. Replaceable inserts in `LM3-TOOL-ROOF-FAIRING-MOULD` produce:

- plain roof skin;
- HVAC curb/fairing and removable skirt;
- PV rail/pad and junction/gland plinth;
- antenna/radio closeout;
- retained inspection or lifting cover;
- walkway edge and drain closeout.

Openings are CNC trimmed after cure from the common datum. No roof penetration is hand-cut on the vehicle. Solid laminate surrounds fasteners and cut edges; all trimmed edges are sealed, and every removable part has retained hardware.

Arrangement and keep-outs

The released roof coordination drawing must freeze HVAC feet, supply/return drops, condensate drains, PV rails/clamps, cable glands, resistor heat zone, antennas, lightning/bonding network, lifting covers, walkways and fall-safe access. It must also reserve hand/tool access and a removal path for each unit.

Radiative coating is allowed only on the remaining exposed roof/fairing area. It is prohibited on PV active surfaces and clamps; HVAC intake, exhaust and heat exchangers; antennas/sensors; walkways; lifting covers; earth bonds; glands; drains; seals; safety labels; and all service or adhesive lands.

Controlled assembly route

1. Survey roof bows and rails before corrosion-coating closeout.
2. Install and leak-test the HVAC curb, duct collar, tray and drains.
3. Fit PV rails/pads, bonds and cable-gland plinths; perform pull, pitch and continuity checks before modules obscure them.
4. Dry-fit plain/fairing roof mouldings, hatches, skirts and closeouts; check airflow, thermal and service envelopes.
5. Apply qualified exposed-roof finish with the controlled mask map.
6. Install HVAC, PV, antennas, resistor shields, walkways and covers.
7. Complete roof water, condensate-flow, bonding, PV isolation, antenna, airflow and service-removal tests.

The HVAC and PV product selections remain supplier freezes. These moulds and interfaces let the local factory progress without pretending that a generic box is the final certified unit.

Curved sandwich development and heat qualification

The development CAD and volume verification separates ply skins, adhesive, foam and solid edge lands on a parametric curved roof. Its 50 mm crown is an assumption pending the coordination drawing, and curved insert lands still require local mounting detail. The test and quotation register requires an equipment-adjacent heat-soak bay beneath PV, aged bonds/inserts, water ingress, pressure, fatigue, clip-loss retention and electrical bonding. Core thermal stability and resin glass-transition temperature do not establish an allowable temperature for the assembled panel. No equipment or personnel load is transferred to the weather skin by these development calculations.

Traction and DC Power — 800 V-Class Reference

Repository source: `docs/rolling-stock/light-metro-3car/traction.md`

Controlling decision: RFC 0021

The three-car train is three electrically self-contained cars coupled into one coordinated consist. Each car carries one 225 kWh gross / 180 kWh routinely usable liquid-cooled LFP pack, two independent traction controllers, two PMSMs, isolated low-voltage DC/DC equipment, direct-HV DC HVAC, and one roof-PV MPPT.

The pack modules sit in fire-separated side enclosures beneath the longitudinal-seat zone. They have no service opening into the saloon: removal, drainage, pressure relief and thermal-runaway venting are all outward through the lower bodyside. Interior fairings are non-structural passenger barriers, not battery access hatches.

```
Car A: roof PV -> MPPT -----+
       station contact -----+-> 225 kWh LFP -> controller -> PMSM axle 1
       regenerative braking +             -> controller -> PMSM axle 2
                                             -> isolated LV DC/DC
                                             -> direct-HV DC HVAC

Cars B and C repeat the same package.
```

There is no traction transformer, continuous catenary, train-wide AC distribution, or central auxiliary inverter. Internal electronic motor drives inside qualified HVAC equipment remain permitted.

Battery and DC link

Parameter	Reference value
Chemistry	LFP
Consist gross / routinely usable	675 / 540 kWh
Per-car gross / routinely usable	225 / 180 kWh
Architecture class	800 V-class
Nominal operating voltage	650–700 V DC
Provisional normal upper voltage	740 V DC
Side-enclosure modules beneath seat zone	8 per car, 4 per side
Electrical segmentation	2 contactor-isolated strings per car, subject to topology study
Cooling	Liquid cold plates, with independent pack limits at 50 °C ambient
Monitoring	Cell/module voltage and temperature plus compartment off-gas detection

The 740 V provisional limit preserves operating margin below a 750 V-class candidate controller ceiling. Cell count and SOC window are not released until controller transient withstand, regeneration overvoltage, contactor interrupt duty, precharge, isolation monitoring, creepage/clearance, and charger compatibility are jointly verified.

An identified module hazard causes string- or pack-level isolation according to the proved topology. The design does not claim individual series-module bypass.

Traction equipment

Parameter	Reference envelope
Motor count	6, two per powered bogie
Controller count	6, one per motor
Type	Heavy-commercial-vehicle-class PMSM/controller set
Candidate motor	Inovance HM47-class or qualified equivalent
Candidate controller	Inovance LD32-class or qualified equivalent
Candidate motor short peak	350 kW for supplier-qualified duration
Installed hardware peak	2.1 MW
Initial train control cap	1.8 MW
Cooling	Per-axle liquid loop branch with monitored flow and temperature

Supplier names define RFQ and packaging references, not sole-source requirements. The interface control document shall state voltage/current, CAN message mapping, torque accuracy, safe-torque-off behavior, thermal limits, ingress protection, shock/vibration, EMC, fault reporting, mounting, coolant, and connector requirements.

Gearbox and adhesion closure

The previous 6.5:1 gearbox ratio is withdrawn. The selected ratio must be derived from the qualified motor torque-speed map, 760/680 mm new/worn wheel diameters, 90 km/h service speed, overspeed margin, grade requirement, and thermal duty.

Analysis shall cover:

- AW0 through AW3 acceleration and jerk;
- dry and wet rail, including coefficient-of-adhesion sweeps at 0.05, 0.10, and 0.15;
- maximum wheel torque at new and worn diameter;
- field-weakening and motor/controller overspeed;
- one-controller and one-car-pack unavailable operation;
- regenerative-braking acceptance at high SOC;
- continuous grade and repeated 1 km stop duty at 50 °C ambient; and
- gearbox bearing, lubrication, sealing, and maintainability.

The 1.8 MW control cap remains the release value until these analyses show a larger command improves timetable performance without exceeding adhesion, battery, converter, or cooling limits.

Station charging interface

- Primary mechanical interface: automated conductive side contact.
- Station cabinet: one 500 kW bidirectional DC/DC stage.
- Cabinet current ceiling: 825 A reference.
- Contacts: up to two platform contacts sharing one interlocked cabinet budget; they are never each allocated 500 kW simultaneously.
- Normal dwell: approximately 60 seconds, giving about 8 kWh after losses.
- Terminal: same 500 kW hardware with a longer balancing dwell.
- Train protection: HVIL, insulation monitoring, precharge, weld detection, touch-safe sequencing, emergency isolation, and BMS charge limit.

At 700 V the ideal current at 500 kW is about 714 A; at 650 V it is about 769 A. The simulation and controller enforce power, current, battery charge acceptance, conversion loss, and ambient derating at the same time.

Rooftop PV

The existing approximately 15.12 kWp train roof array is retained as an auxiliary-energy offset and SOC-maintenance source, not the primary traction supply. Each car has its own MPPT, isolation switch, combiner, protected DC interface, and removable wiring route. Solar cables are not embedded in the roof laminate.

Both bonded flexible and rail-mounted rigid module packaging remain prototyping candidates until wind, vibration, fire, bonding, water-ingress, and replacement evidence selects one production arrangement.

Auxiliary power

Domain	Architecture
HVAC	Direct 650–700 V DC input rail unit with internal motor electronics
Safety controls	Redundant isolated 24/48 V DC/DC domains
Doors and lighting	Isolated LV DC distribution with local protection
PIS, CCTV and compute	Separate isolated service domain with load shedding
Pumps and cooling	Monitored DC supplies; fire-mist pump independent per car

Loss of a nonessential service domain must not remove braking, door release, fire detection, emergency lighting, communications, or pack isolation.

Fire protection interface

Each battery module is enclosed, monitored for temperature and off-gas, and vented outward. Each car carries a small water reservoir, DC pump, monitored stainless distribution line, and mist nozzles serving only battery compartments. The system cools adjacent modules and slows propagation; it does not claim to terminate cell-level thermal runaway.

Traction controllers, DC/DC equipment, and HVAC electronics use fire-resistant metal enclosures, detection, and electrical isolation. They do not have automatic gaseous or water-mist suppression. The saloon has no automatic sprinkler or CO2 flooding system.

Release evidence

- Pack cell/string topology and voltage/transient analysis.
- Motor/controller/gearbox duty-cycle simulation and dynamometer evidence.
- AW3 adhesion, grade, acceleration, regen, and stopping analysis.
- 50 °C train and station converter thermal/derating tests.
- Charger shared-contact arbitration and failed-cabinet scenarios.
- Battery-module propagation, vent and mist flow/pressure tests.
- EMC, shock/vibration, ingress, insulation and high-potential tests.
- Supplier-neutral interface control drawings and acceptance procedures.

Light-Metro 3-Car V2 Release Checklist

Repository source: `docs/rolling-stock/light-metro-3car/v2-release-checklist.md`

The v1 package is a complete concept envelope and procurement BOM. The v2 package is the fabrication release. This checklist defines the missing artifacts that must exist before the trainset can be treated as production-ready.

Gate	Required artifacts	Closure criterion
Design baseline freeze	Signed LM3 configuration sheet covering 3 repeated 16.5 m cars, 3 powered bogies, 3 trailer bogies, 540 kWh usable onboard battery, 78.75 t controlled planning tare (75.308 t modeled subtotal + 3.442 t engineering reserve), two door pairs per side per car, and no city-specific train variant	Drawing register, mass budget, product mass-closure ledger, BOM, RFC 0008, RFC 0021, RFC 0022, and `trainset.py` all agree
Supplier envelope freeze	Door, window, HVAC, articulation, coupler, battery, inverter, motor, brake, bearing, sensor, and charging connector interface drawings	Every selected supplier part fits without changing primary steel datums
Structural analysis	EN 15227 crashworthiness, EN 12663 carbody static/fatigue, EN 13749 bogie frame FEA, plus closure of the current lateral-sway screen review	Reports reviewed and linked from `compliance.md`; revised body model is <= 20 mm lateral displacement or has an approved target change
Weld package	Weld maps, WPS/PQR register, weld classes, NDT plan, fixture-control plan	EN 15085 CL1/CL2 joints have inspection method and acceptance criteria
Bogie frame and recovered-component acceptance	Fresh powered/trailer bogie-frame drawings, supplier certificates for new safety-critical parts, and any donor axle/axlebox quarantine + NDT + metrology sheets	No recovered freight bogie frame enters the OSR consist; recovered components are accepted only line-by-line
Manufacturing drawings	2D controlled drawings for body, bogie, brackets, panels, battery trays, door portals, coupler pockets, HVAC rails, and identical fiberglass end-cowl casts	Drawing register reaches rev A with owner, material, tolerance, and inspection method
Factory drawing/interface packages	Sixteen generated work packages covering every locally made row across chassis, structural interfaces, exterior modules, panoramic glass, lamps, configurable ends, bogies, battery/HV hardware, trainlines, roof equipment, interiors, fixture rails, livery film, trial roof coating and recovery	Every package has frozen inputs, approved controlled outputs, characteristic list, verification owner and evidence route; the generated package itself is not an approval
Flat-pattern and NC output	DXF/neutral CAD flat patterns, bend tables, nesting sheets, drilling/cutting programs	Shop can cut and bend one carbody kit without interpreting 3D envelopes
Harness and pipe routing	Clamp locations, bend radii, connector access, HV segregation, coolant bleed/drain points, EMC bonding plan	Routing pack passes maintainability and electrical safety review
Weight and balance	Close all 117 active rows in generated `mass-closure-ledger.md`; per-line mass rollup, individual-car and complete-train calibrated weighs, axle loads by AW0/AW2/AW3, roof equipment CG, battery mass distribution	Nine categories reconcile, the signed tare/CG report supports the controlled mass, and <= 14 t AW3 axle load is verified; the 73.376 t modeled lightweight candidate cannot change tare before its separate substantiation gates close
First-car build hold point	One 16.5 m carbody shell, one powered bogie, one trailer bogie, one battery enclosure set, one door bay, and one roof/HVAC/PV bay built and inspected before committing all three cars	Dimensional/NDT findings are closed or fed back into rev B tooling before repeat production
First-article inspection	Dimensional survey, water ingress, door fit, HVAC drain, battery isolation, brake/static tests	Results accepted into certification evidence register

COTS CAD Rule

Parametric CAD and FreeCAD review models may stay supplier-neutral until procurement freeze. After freeze, each selected supplier module gets a named envelope model with revision, datasheet source, keep-out zones, and service-removal path. Generic envelope models remain useful for early design review but do not close v2.

Rust codebase review and OpenSourceRail integration plan

Repository source: `docs/rust-codebase-review-and-integration-plan.md`

Review date: 2026-09-27 Scope: the complete Cargo workspace, its Python/GIS and supervision boundaries, CI, formal-evidence hooks, and the ERP/SCADA/AI architecture.

Executive assessment

The Rust workspace stands up well as a deterministic engineering, simulation and controller-evaluation codebase. It does not yet stand up as a commissioned or certified railway control product. Those statements are compatible: the repository has meaningful fail-restrictive logic, property tests, integration tests, differential tests and selected bounded proofs, while still lacking the physical I/O, persistent service deployment, calibration, independent safety assessment and operating evidence required for live authority.

At the start of this review the workspace contained 56 crates and about 72,000 lines of Rust under `src/` and `tests/`. The workspace test suite and Clippy with warnings denied passed. The review therefore concentrated on incorrect trust assumptions and integration seams rather than rewriting stable evaluator logic.

This review added a 57th crate, `osr-supervision-contract`, to make the Rust-to-operating-platform boundary explicit and observation-only. The follow-on lifecycle-control pass adds the 58th, `osr-lifecycle-identity`, so Rust and the ERP compiler enforce the same lookup-only physical-asset identity contract.

What is credible now

Area	Evidence in the repository	Current credible use
Pure control evaluators	Integer-oriented ATP, brake, BMS, points, door, fire, obstacle and related evaluators with unit and property tests	Simulation, design review, shadow execution and HIL preparation
Interlocking and consensus	Deterministic replay, partitions/healing tests, authenticated ingress, Python differential checks and selected Kani harnesses	Research and bounded pilot/shadow evidence; not certified movement authority
Integrated simulation	Cross-crate train, station, wayside, energy, fare, condition and OCC scenarios	Software-in-the-loop regression and city design validation
Design and topography	Source-locked routing rasters, open DEM elevation/slope, water masks, civil classification, station exclusions and City Studio layers	Reproducible planning screening; not survey or construction release
Security primitives	HMAC/Ed25519 primitives, authenticated envelopes, replay/freshness checks and unsafe-code prohibition	Building blocks and simulation evidence; not a deployed PKI or key ceremony
Assurance tooling	Structured safety cases, source/result hashes, Kani result capture and explicit evidence acceptance	Evidence management; not independent certification
Operator interfaces	OCC and simulation GUIs plus a local City Studio server	Demonstration and training, not a resilient production OCC
Business integration	A scoped gateway connects observations to history, FUXA and ERP maintenance; ERP cannot release the railway	Demonstration and bounded business workflow

Findings from scrutiny

Strengths

1. The workspace centrally forbids unsafe Rust and pins a minimum toolchain.
2. Safety-oriented evaluators are generally pure and deterministic, which makes replay, property testing and formal analysis practical.
3. The simulation exercises real crate APIs rather than parallel mock control laws.
4. Failure behavior is frequently conservative: unknown positions, stale authority, sensor disagreement and lost quorum restrict operation.
5. The documentation usually distinguishes research, planning and simulation evidence from certification and field acceptance.
6. City inputs and generated engineering evidence have extensive content-lock and provenance mechanisms.

Defects and trust gaps found in this review

Priority	Finding	Consequence	Resolution in this plan
P0	Routing bounds were inferred from a flattened index	An out-of-range column could alias into a later row	Explicit row and column checks plus regression/property tests
P0	Penalty length was only a debug assertion	A malformed caller could panic in an optimized build	Runtime shape/value validation and typed errors
P0	Release arithmetic used Rust's default unchecked-overflow profile	Optimized builds could wrap accidental integer overflow	Workspace release overflow checks enabled
P1	The raster loader ignored the sidecar's declared file path	Metadata and consumed bytes were not one enforced contract	Safe declared-path loading with traversal rejection
P1	Raster/grid/topography/anchor semantics were weakly validated	NaN, invalid coverage, malformed dimensions or bad anchors could leak into planning	Bounds, finiteness, domain, overflow and identity validation
P1	City Studio GIS reads could call the bundle loader without enforcing project source locks	A layer could be called source-locked while loading tampered bytes	One verified routing-bundle method for routing, compilation and GIS
P1	The Rust bridge exposed an ad hoc serialization of internal controller types	Python depended on implementation layout and authority was implicit	A strict normalized observation contract with exact units and source crates
P1	CI did not test all features, doc tests or dependency advisories	Feature-specific drift and known vulnerable dependencies could escape the normal gate	All-feature Clippy/tests, doc tests and pinned RustSec audit
P1	The first RustSec run found vulnerable GUI transitive dependencies and unmaintained `bincode`/font-parser dependencies	Known denial-of-service, argument-injection, unsoundness and maintenance risks were present in the lockfile	Rust 1.92 and egui 0.34 migration, patched transitive versions, and `postcard` wire serialization; the strict audit now passes without exceptions
P2	The long-horizon simulator check was ignored in normal CI and there was no recurring coverage artifact	Slow regressions and unexercised areas were less visible	Tracked two-day/four-mode CI evidence, weekly seven-day soak and workspace LLVM coverage artifact

Target integration architecture

```
flowchart LR
  GIS[Locked OSM, water and open DEM evidence] --> Route[Rust routing and City Studio]
  Route --> Revision[Content-addressed design revision]
  Revision --> Sim[Rust system simulation]
  Sensors[Future commissioned physical adapters] -. reviewed contract .-> Eval[Rust evaluators]
  Sim --> Eval
  Eval --> Obs[osr-supervision-contract\nobservation-only]
  Obs --> Gate[Scoped integration gateway]
  Gate --> Hist[History and alarms]
  Gate --> FUXA[Read-only supervision views]
  Gate --> ERP[ERP maintenance and draft business records]
  Hist --> Council[Governed multi-model executive evidence]
  ERP --> Council
  Council --> Draft[Attested allowlisted ERP drafts]
  Draft --> ERP
  Human[Licensed operators, engineers and accountable officers] --> Control[Railway command and release]
  Eval --> Control
  Council -. no command path .-> Control
  ERP -. no release path .-> Control
```

The critical rule is structural, not promotional: the observation crate has no command, reset, movement-authority, protection, ERP-action or executive-decision type. AI models see an immutable gateway snapshot derived from observations. They cannot call a Rust controller. Even a quorum decision can only prepare an allowlisted, unsubmitted ERP draft. Human and independently assessed railway paths retain command, protection, engineering release and legal accountability.

Executed implementation plan

Phase 1 — establish the baseline and maturity boundary

Actions:

- inventory every workspace member and its tests;
- run all-target workspace tests before modification;
- run all-feature Clippy with warnings denied;
- inspect architecture, SBC allocation, City Studio, topography, simulation, supervision, ERP and executive-council documentation;
- distinguish defects from acknowledged physical/certification work.

Exit criterion: a review based on executable behavior and documented claims, not crate names. Completed.

Phase 2 — harden topography and route synthesis

Actions:

- reject empty, overflowing or internally inconsistent grids;
- validate demand weights and penalty rasters before Dijkstra execution;
- check row and column bounds independently;
- consume the safe relative paths declared by raster sidecars;
- validate grid geography, raster byte counts, cost/demand/buildability, 0–100 water coverage, finite elevation, non-negative slope and anchors;
- use one source-lock-verifying City Studio bundle loader for routing and GIS;
- retain the planning boundary: open DEM and OSM water identify likely viaduct/bridge/station-exclusion areas but do not become surveyed alignment.

Exit criterion: malformed or tampered routing inputs fail with an error rather than aliasing, panicking, silently drifting or being shown as locked. Completed.

Phase 3 — create a stable Rust-to-supervision contract

Actions:

- introduce `osr-supervision-contract` with strict Serde fields;
- require schema, simulation environment, observation-only authority, bounded identity/time and a bounded unique observation set;
- attach units and originating Rust crates to every measurement;
- normalize nine evaluator families into the contract in the native bridge;
- make the Python gateway require the exact identities, ranges, integer enums, units and source-crate mappings;
- remove Python's dependency on serialized internal Rust state layouts;
- retain a shared JSON fixture asserted by both Rust and Python tests.

Exit criterion: an extra command field, altered authority, missing or duplicate reading, wrong unit/source, non-finite value, fractional enum or physical frame fails closed. Completed.

Phase 4 — integrate with ERP, SCADA and governed AI

Actions:

- feed only validated observations into the existing scoped gateway;
- keep FUXA as a read-only view of gateway state;
- keep condition-to-ERP delivery in the durable, idempotent maintenance path;
- let executive proposals reference the gateway's immutable context digest;
- prohibit council outcomes from entering the controller command queue;
- document that source-crate names are provenance, not proof of commissioning;
- preserve native ERP permissions and independent railway handback.

Exit criterion: Rust data improves operating and management evidence without granting ERP, FUXA or AI a safety or release capability. Completed in the software architecture; physical adapters remain a commissioning activity.

Phase 5 — broaden verification and release gates

Actions:

- add routing property tests for path bounds/connectivity and water semantics;
- execute every population archetype plus greedy topology synthesis against a connected, bounded route contract;
- add malformed-grid, path-traversal, topography-domain and source-lock tests;

- reject mismatched raster dtype/shape/byte order/length, unpaired DEM layers, invalid geographic references and duplicate anchors in regression fixtures;
- add observation-contract unit tests and adversarial Python projection tests;
- bind the Rust output byte structure to the cross-language fixture;
- pin a representative `postcard` entry to exact bytes so a serializer or enum layout change cannot silently alter the replicated-log wire format;
- run Clippy and tests with all features and add doc tests;
- add a pinned `cargo-audit` RustSec gate;
- pin Rust 1.92 in the workspace, installer, CI and `rust-toolchain.toml`;
- upgrade the GUI dependency chain and replace unmaintained `bincode` with `postcard`, then require the advisory audit to pass without an allowlist;
- compile both operator GUI libraries for their no-default-feature WebAssembly target as a distinct CI gate;
- enable checked arithmetic in optimized releases;
- run a tracked two-day normal/peak/degraded/recovery resource-bound profile in CI and a seven-day release profile on the weekly schedule;
- generate a recurring LLVM source-coverage artifact.

Exit criterion: fast PR checks cover structure and contracts; expensive soak and coverage work runs recurringly and can be invoked manually. Completed.

Phase 6 — documentation and reproducible handoff

Actions:

- update the Rust workspace map and embedded-integration contract;
- update architecture diagrams to show the observation boundary and AI non-authority path;
- record exact remaining deployment gates below;
- run repository-wide Rust and Python regression suites and hygiene checks.

Exit criterion: documentation describes the implementation that tests execute, and does not imply certification or a physical deployment. Completed, subject to final regression results recorded in the change handoff.

Test strategy after this review

Layer	Harness	Purpose
Function	Rust unit tests	Exact evaluator and validation behavior
Invariant	Proptest suites	Broad input-space determinism, bounds and fail-restrictive properties
Cross-crate	`osr-sim` integration tests	Real subsystem composition and fault propagation
Differential	Interlocking Rust/Python comparison	Independent reference agreement
Formal	Kani workflow	Bounded selected safety properties with preserved results
Cross-language	Rust bridge fixture plus Python rejection tests	Schema, unit, provenance and authority compatibility
Long horizon	Tracked two-day four-mode report plus scheduled release-mode seven-day simulation	Energy reserve, invariant health, degraded/recovery behavior and bounded retained state over time
Supply chain	RustSec audit of `Cargo.lock`	Known dependency advisories
Static	rustfmt and all-feature Clippy with warnings denied	Consistency and lint debt prevention
Coverage	Scheduled LLVM LCOV artifact	Identify unexercised Rust paths without inventing an arbitrary safety threshold

Coverage percentage is not a safety claim. A high line-coverage number cannot replace requirements traceability, boundary-value analysis, fault injection, formal arguments, HIL or independent assessment.

Verification snapshot after follow-through

The final local tree was exercised again after the review changes:

- `cargo clippy --workspace --all-features --all-targets -- -D warnings` passed;

- `cargo test --workspace --all-features --all-targets` passed; the explicitly deterministic two-day report is checked in CI and the seven-day profile is covered by the scheduled release workflow and was also run successfully during the main review;
- `cargo llvm-cov 0.9.1 --workspace --all-features --all-targets` passed and reported 28,075/37,209 instrumented lines, or 75.45%. This increased from the pre-follow-through 72.12% snapshot. The deliberately targeted changes moved routing topology from 1.78% to 76.05%, raster validation to 95.68%, and both `osr-proto` and `osr-supervision-contract` to 100%;
- the source-bound Kani 0.67.0 runner executed 39 tractable declared properties and all 39 passed, with a 600-second timeout and 4096 MiB per-process address-space cap. The manifests cover ATP, brake, points, obstacle detection, intrusion detection, secure bus, four bounded odometry properties and three bounded interlocking properties;
- the remaining nine of 48 declared properties are not closed: odometry determinism retains its recorded 1,800-second timeout and all eight interlocking non-overlap partitions retain their recorded memory exhaustion. These are open proof obligations, not observed counterexamples and not implicit passes.

The generated LCOV file and eight execution manifests are under `build/assurance/rust-review-final/`. They are ignored local working artifacts, identify the runner as `local-unattested`, set independent acceptance to false, and must not be presented as certified or independently accepted evidence.

Remaining work that cannot be honestly completed by repository code alone

These are deployment and assurance gates, not deferred software TODOs to hide:

1. Commission physical MQTT/OPC UA/Modbus/device-bus adapters against named suppliers, exact firmware, units, scaling, freshness and local permissives.
2. Replace process-lifetime demonstration state with qualified persistent controller/service state where the deployment safety concept requires it.
3. Establish device identity, PKI, key injection/rotation/revocation, secure boot, measured update and incident-response ceremonies.
4. Run processor, bus, sensor, actuator, timing, EMC, environmental and degraded communications HIL campaigns on the selected hardware.
5. Complete requirements-to-test traceability and the deployment-specific hazard log with an independent safety assessor.
6. Prove or qualify the selected compiler, target, RTOS/OS, build pipeline and reproducible binary/signing process for the claimed safety integrity.
7. Receive licensed survey, geotechnical, hydrology, utilities, land, navigation and structural evidence before bridge/viaduct/station release.
8. Commission resilient production event storage, disaster recovery, observability, capacity and cyber monitoring for the OCC and historian.
9. Appoint human statutory directors, accountable railway managers, safety authorities and employment/finance/legal signatories. The AI council remains advisory and policy-bound; it is not a legal person or safety authority.

Until these gates close, the correct description remains: serious open engineering and simulation software with explicit pilot/shadow paths—not a certified autonomous railway or an autonomous corporate officer.

Safety-case artifacts

Repository source: `docs/safety-case/README.md`

This directory collects the machine-checkable evidence that supports OpenSourceRail's safety claims. Per ARCHITECTURE.md §7 the safety case is structured in GSN (Goal Structuring Notation), serialised as TOML, and regenerated on every commit.

The TOML lives under `gsn/` and is compiled by the `osr-safety-case` crate (RFC 0005 §4.9). CI gate: `tests/starter_case_closes.rs` fails the build if any goal no longer traces to evidence — that is, adding a safety-relevant claim or deleting an evidence file breaks CI until traceability is restored. This checks graph links and source-path presence, not proof execution, current results or independent acceptance.

Compile the rendered case locally with:

```
cargo run -q --bin osr-safety-case -- docs/safety-case/gsn
```

What's verified today

RFC 0004 §4 names five safety properties of the Movement Authority computer:

Property	Harness in `crates/osr-interlocking/src/kani_proofs.rs`	Status
P1 — determinism	`kani_p1_determinism`	bounded local pass on `2bf8b7717`; see records
P2 — non-overlap	`kani_p2_non_overlap_two_trains` and seven paired companions	open; all eight partitions required
P3 — consist-fit	`kani_p3_consist_fit_single_train`	written
P4 — conservatism	`kani_p4_fail_restrictive_is_not_less_restrictive_than_known`	written (fail-restrictive path only)
P5 — time-bounded	`kani_p5_time_bounded`, `kani_p5_time_bounded_with_known_position`	written

Also proptest-checked (sampling larger input domains with finite generated tests) in `tests/proptest_ma.rs` and `tests/proptest_determinism.rs`.

Continuously exercised in `osr-sim` on every scenario run: per RFC 0004 M5 the simulator emits `Entry` objects as trains move, and every section entry is gated by `osr_interlocking::section_available_to`. The generated Samawah reference run produces 2400 MAs / 2 h with zero invariant violations under both the in-memory backend and a 3-node Raft cluster.

Cross-checked against an independent Python reference interpreter (`tools/reference-ma/`) per RFC 0004 M4: `crates/osr-interlocking/tests/differential.rs` serialises random log prefixes, shells out to `python3 -m reference_ma`, and asserts byte-identical `MovementAuthority` JSON. Any divergence between the two twins is a bug in at least one of them.

Execution status: the declared harnesses run through a source-bound evidence runner. ATP determinism and interlocking determinism passed on candidate `2bf8b7717`; odometry determinism and the complete two-train non-overlap obligation remain open. See the retained execution records for commits, budgets, input hashes and failures. These are bounded fixture proofs; scaling toward RFC 0004's targets and independent safety acceptance remain open.

What the other SIL-4 crates check today

Crate	Properties	Verification
osr-atp	A1–A7 (determinism, expired MA trips, unknown position trips, train mismatch trips, head past MA end trips, overspeed trips, conservatism)	proptest for A1–A7; Kani harnesses for A1–A7 in `src/kani_proofs.rs`
osr-brake	B1–B5 (determinism, emergency union, emergency completeness, WSP conservative, park safe)	proptest + Kani for B1–B5 in `src/kani_proofs.rs`
osr-odometry	O1–O5 (determinism, forward non-regression, uncertainty monotone, balise reset, GNSS conservative)	proptest + Kani for O1–O5 in `src/kani_proofs.rs`
osr-wayside-points	W1–W6	proptest + Kani for W1–W6 in `src/kani_proofs.rs`
osr-consensus	TLA+ invariants (ElectionSafety, LogMatching, LeaderCompleteness, StateMachineSafety, FailRestrictive)	proptest over cluster harness; TLA+ TLC on the spec, not yet a refinement proof

Running Kani

Install

```
cargo install --locked kani-verifier --version 0.67.0
cargo kani setup
```

`cargo kani setup` downloads ~200 MB of CBMC tooling on first run.

Run a single harness

```
cargo kani -p osr-interlocking --harness kani_p5_time_bounded_arithmetic
```

Run all harnesses in a crate

```
cargo kani -p osr-interlocking
```

Run all harnesses across the workspace

```
cargo kani --workspace
```

The arithmetic harness completes in seconds. Whole-function and topology-backed harnesses can exceed ordinary CI limits because standard-library map operations expand the bounded state space. Treat their bounds and reports as controlled formal evidence; never disable unwinding checks merely to obtain a green result.

CI

The Kani workflow runs on pushes to `main`, pull requests and manual dispatch. It checks ATP rejection of an expired movement authority and interlocking validity-window arithmetic, then gates the 39 declared harnesses that currently complete within the controlled budget across eight packages with Kani 0.67.0. Manual dispatch additionally re-runs the nine open resource-bound obligations as explicitly non-gating diagnostics and preserves their failures, timeouts or any future closure. The default budget is 600 seconds and 4096 MiB of virtual address space per process. Manual dispatch can select 1800 seconds and/or 8192 MiB. All eight paired non-overlap harnesses remain required for proof closure; each retains both trains and symbolic head positions for its wayside condition. See the partition rationale.

Actual successes, failures and timeouts are archived with source hashes. Timeouts and memory failures do not establish a property. The runner terminates the entire verifier process group and limits memory to preserve subsequent results. A runner shutdown can still prevent artifact upload; an incomplete run cannot qualify a release.

The results register separates historical results from the latest candidate. A green push/PR workflow is a merge-regression signal, not full formal closure. Release packaging still requires successful exact-commit artifacts for all 48 declarations and rejects missing, failed or duplicated results; therefore the nine open obligations still prevent formal release qualification. General CI, integrated-stack and example-city acceptance are also required. Merging development work does not grant release or safety acceptance, and an earlier pass does not automatically qualify a later commit.

What's planned

From RFC 0004 §M3 and the cross-crate safety plan:

1. Scale the P2 / P4 bounds toward the RFC's 8 trains × 50 entries × 100 sections target. Requires either smarter abstractions (reducing state-space explosion) or a fleet of dedicated CI workers.
2. Kani harnesses for the other SIL-4 crates — done. The four onboard/actuator evaluators (ATP, brake, odometry, wayside-points) now ship Kani harnesses alongside their proptest suites. The remaining work here is verification throughput — scaling bounds toward larger state spaces on dedicated CI.
3. A refinement proof from the TLA+ `SMRaft` spec to the Rust `osr-consensus` implementation. TLC already checks the spec's safety properties; refinement would formally connect the spec to the code. TLAPS or a separate tool is a candidate.
4. GSN safety case compiler — done. The `osr-safety-case` crate compiles the TOML claim files in `gsn/` and the `starter_case_closes` test gates traceability in CI on every commit. Counts are generated below; evidence pointers alone do not establish that a claim has been verified.

Generated case inventory: 32 goals, 6 strategies, 78 solutions. These counts describe traceability, not successful or accepted proofs.

Directory contents

- `gsn/` — GSN TOML claim files. Each file is a coherent slice of the case (top-level goals, non-overlap decomposition, fail-safety decomposition, consistency decomposition). Compiled by `osr-safety-case`.
- `evidence/` — machine-readable Kani and proptest results (controlled runner imports remain deployment work; see result validation).
- `hazard-log/` — identified hazards and mitigations (planned).

Safety evidence: traceability, results and acceptance

Repository source: `docs/safety-case/result-validation.md`

The default `osr-safety-case` command checks the GSN graph and evidence paths. It reports traceability complete, not safety-case acceptance. A source file containing a proof harness is not a proof result.

Two additional, conservative gates assess every declared solution:

```
cargo run -q --bin osr-safety-case -- docs/safety-case/gsn \
  --results build/assurance/results.toml --require-verified
cargo run -q --bin osr-safety-case -- docs/safety-case/gsn \
  --results build/assurance/results.toml --require-accepted
```

Missing results fail these gates. The repository does not manufacture passing records from its existing source pointers. Each supplied record must identify its solution, kind, source and exact anchor; for Kani the named function must exist in that source. It must also contain a successful execution status and zero exit code, tool/version, reviewed bounds, runner identity, source/input hashes and a separate report whose bytes match the recorded SHA-256. Changed source inputs or report bytes invalidate the result. Bounds must describe the actual run, including unwind limits, assumptions and tool configuration.

A trusted runner exports the following TOML contract (values below are explanatory placeholders and intentionally do not constitute usable evidence):

```
schema = "osr-evidence-results/1"

[[result]]
solution = "E1.1a"
kind = "kani"
path = "crates/osr-interlocking/src/kani_proofs.rs"
anchor = "kani_p2_non_overlap_two_trains"
tool = "kani"
tool_version = "record the actual version"
bounds = "record exact unwind, assumptions and command"
executed_by = "controlled-runner-identity"
status = "passed"
exit_code = 0
report = "build/assurance/p2-report.txt"
report_sha256 = "sha256 of actual report bytes"
[result.inputs]
"crates/osr-interlocking/src/kani_proofs.rs" = "sha256 of actual source"
# Include implementation, manifests, lockfiles and all reviewed dependencies.

[result.acceptance]
status = "accepted"
reviewer = "independent-reviewer-identity"
reference = "controlled acceptance record"
result_sha256 = "ResultRecord::fingerprint() from the reviewed result"
```

Acceptance binds the entire result record through a deterministic fingerprint, including input hashes, tool version and bounds; it cannot survive a changed execution record. The reviewer must differ from the executing identity. A passing result without this acceptance is reported separately.

This validates recorded evidence, not proof mathematics or reviewer identity. It does not parse arbitrary vendor reports, invoke proof tools, establish that an input list is complete, authenticate a runner, or grant railway release. Controlled runner exports, dependency-scope review and authenticated independent acceptance remain required. Citation solutions also need a reviewed result and local input/report artifacts. The strict gate requires all solutions rather than allowing one passing alternative to conceal missing declared evidence.

Inventory counts are generated from GSN TOML by `python3 tools/automation/safety-case-summary.py`; CI runs it with `--check`. The release-gap register remains the place to track physical and deployment acceptance work.

Controlled execution and signed acceptance

`tools/automation/assurance-evidence.py run` discovers the 41 declared Kani solutions from GSN, invokes each exact harness with Kani 0.67.0, and exports `results.toml`, per-harness logs and `execution.json`. A nonzero exit, timeout or missing successful verification summary remains failed. Inputs include workspace Rust (including untracked local source), manifests, lockfiles, GSN, runner and CI workflow. Changed inputs during execution invalidate the run. CI runs all eight packages and uploads diagnostics even on failure.

```
PATH="$HOME/.cargo/bin:$PATH" tools/automation/osr-python \
  tools/automation/assurance-evidence.py run \
  --output build/assurance/review --timeout 300
```

Local runs identify themselves as `local-unattested`. A GitHub run reference is recorded as a declared executor, not authenticated by environment variables. `runner_authenticated` remains false: verifying the actual CI run, toolchain, dependency provenance and operating separation remains an independent task.

The separate `verify-acceptance` command authenticates an authorized reviewer's Ed25519 signature over exact JSON envelope bytes, checks the result/report/input hashes and current conservative dependency scope, rejects failed results, duplicate solutions and mismatched declared harnesses, and rejects an executor/reviewer identity match. It does not create a signature or decide that a review occurred.

A separately controlled policy supplies `reviewers`, keyed by reviewer identity, with `enabled`, `public_key` (relative to the policy file) and `public_key_sha256`. Keep that trust policy outside the contributor-controlled checkout. The signed envelope contains:

```
{
  "schema": "osr-independent-acceptance/1",
  "status": "accepted",
  "reviewer": "authorized-independent-reviewer",
  "reference": "controlled-review-record",
  "results_sha256": "hash-of-exact-reviewed-results.toml",
  "solutions": ["exact-reviewed-solution-identifiers"],
  "dependency_scope_reviewed": true
}
```

These are explanatory placeholders, not an acceptance record. The reviewer must review actual runner provenance and independence; different text labels alone do not establish different people. After an authorized reviewer supplies the envelope and detached signature:

```
tools/automation/osr-python tools/automation/assurance-evidence.py verify-acceptance \
  --results build/assurance/review/results.toml \
  --envelope /controlled/review/acceptance.json \
  --signature /controlled/review/acceptance.sig \
  --policy /controlled/trust/reviewers.json
```

Successful signature verification covers only the listed execution records. It does not satisfy the complete GSN case, authenticate physical evidence, or grant railway release. Other evidence kinds and the existing complete-case validator retain their own requirements. No independent acceptance was produced by the demonstration.

Simulation software coverage

Repository source: `docs/simulation-software-coverage.md`

`osr-sim` executes connected software layers against the same run state:

- `onboard`: odometry, ATP/ATO, BMS, traction, brake, obstacle, fire, derailment, passenger assistance, and regenerative-current arbitration. The arbiter's refusal output is converted into friction effort;
- `vehicle_systems`: doors, auxiliary power, HVAC, lighting, and onboard PIS;
- `embedded`: TCMS, event recording, hot-axle monitoring, CBM sampling, and bounded T2G failover/store-and-forward. A TCMS trip inhibits dispatch or section progress on the next deterministic control cycle;
- `infrastructure_systems`: PSD, station PIS and SCADA per station, plus intrusion detection per wayside section and consensus verdict transitions;
- `habd_systems`: real trackside hot-axle evaluation at explicit bidirectional route-book locations, with next-station warning speed limits, latched stops, and inspected authority resets;
- `balise_systems`: a stable topology-derived passive-balise registry, real sighting audit, and accepted absolute fixes delivered to onboard odometry;
- `fare_systems`: signed single-ride issuance through the real TVM, gate validation, and back-office ledger/fraud reconciliation at every station;
- `occ_systems`: real TCMS-report ingestion into the OCC fleet roster; a critical train alarm opens a line incident and holds new dispatches until the line has recovered, without stopping a train already in a section;
- `proto_systems`: each live position report crosses the deterministic track-state codec; the decoded section, timestamp, speed and SoC feed OCC, and acceptance rejects decode loss or semantic drift;
- `energy_sites`: the standalone site controller governs every integer-watt PV/storage/grid/pad split; per-call conservation is checked in result data;
- `wayside_asset_systems`: every declared point machine executes against dual normal-position sensors and gates departures at its station; each declared level crossing runs the warning/barrier state machine and blocks its sections until closed. Zero crossings is an explicit not-applicable result for grade-separated networks;
- `backend_systems`: radio-delivered CBM ingestion, work-order generation, bounded history, and analytics over retained metrics;
- `time_sync`: the IEEE 1588 slave state machine acquires and retains the shared deterministic clock used by timestamped controller traffic.
- `selftest_systems`: all five deployment roles run their real software known-answer suites before service; a failure holds every dispatch.

Each layer emits deterministic result evidence. Auxiliary loads remain in the calibrated energy intensity and are not charged twice.

Coverage is deliberately narrower than “start every binary.” The simulator runs `osr-selftest` software checks, but physical wiring, sensor, relay and trust-anchor commissioning remains external evidence. The machine-readable contract at `lib/simulation-component-coverage.toml` classifies every entry in `deployment/components.toml` exactly once and fails if the inventory drifts. `scenario_model` entries would identify aggregate substitutes (none remain in the current inventory); `external_boundary` entries would remain explicit gaps (none remain in the current inventory).

Nominal OCC acceptance requires positive controller, telemetry and roster evidence with no incident or dispatch hold left active. A degraded integration test drives a TCMS emergency through the same core and verifies hold/release.

The HABD scenario contract follows the ERA infrastructure-register model: detector existence, location, and direction are explicit route assets. A trip remains latched until an identified inspection reset because the RSSB operating rule requires the train to stop and be examined before further movement. See the ERA RINF application guide and RSSB Rule Book module TW5.

Balise identifiers and positions are regenerated from directed section IDs. Nominal acceptance requires every crossing opportunity to produce a registry-validated odometry fix; missed or mismatched reports remain visible fault-injection outcomes and are never accepted as fixes.

The fare workload uses one representative transaction per station-minute to exercise software deterministically. Sales, grants, and ledger entries must reconcile, but the workload is not used as passenger-demand or revenue forecast evidence. `fare_token_tamper` exercises cryptographic denial and repeated-probe fraud signalling.

Run the inventory check with:

```
python3 tools/automation/validate-simulation-components.py
```

To verify tick evidence as well:

```
cargo run -p osr-sim --bin osr-sim -- \
  --config cities/catalogue/west-asia/Iraq/Samawah/samawah.toml \
  --duration 3600 --status-every 0 --json-out /tmp/osr-result.json
python3 tools/automation/validate-simulation-components.py --result /tmp/osr-result.json
```

City acceptance additionally hashes the rolling-stock template, buildable trainset manifest, and small-component standard. A stale or inconsistent `[consist.systems]` block therefore fails regeneration evidence.

Software Architecture Diagrams

Repository source: `docs/software-architecture-diagrams.md`

> Operating architecture update: ERPNext + Frappe HR replace custom business administration. The diagrams below describe railway and existing assurance components; ERP cannot issue railway release authority.

These diagrams expand the system map in `ARCHITECTURE.md` and the crate allocations in `RFC 0005`. They are intended as editable architecture drawings for implementers, operators, and reviewers.

1. Deployed Demonstration Platform

Solid arrows below describe the localhost demonstration. The Rust evaluators run against simulation fixtures; this is not a commissioned railway deployment. Native ERPNext and FUXA accounts retain their own permissions inside Workbench.

```
flowchart LR
  UI["Workbench: city and asset context"]
  Design["City Studio / FreeCAD / Bonsai IFC / QGIS"]
  Twin["Versioned engineering and city packages"]
  ERP["ERPNext + Frappe HR: native business transactions"]
  Gateway["OSR integration gateway: scoped API, SQLite history and durable outbox"]
  Contract["Rust supervisory contract: observation-only"]
  FUXA["FUXA: generated supervision views"]
  Native["Rust station / vehicle / wayside evaluators"]
  Council["Governed AI executive council: multi-model recommendations"]
  Fixture["Explicit simulation fixtures; factory-method rehearsal"]
  Railway["OSR simulation / OCC / railway works and handback"]
  Legacy["Historic Ops Core business records: read-only"]
  UI --> Design
  Design --> Twin
  Twin --> Gateway
  UI --> ERP
  UI --> FUXA
  UI --> Railway
  UI --> Gateway
  Fixture --> Native
  Native --> Contract
  Contract --> Gateway
  Fixture --> Gateway
  Gateway -->|deduplicated Issue / reviewed Asset Repair| ERP
  ERP -->|permission-filtered execution feedback| UI
  Gateway -->|REST tags / source quality| FUXA
  Gateway -->|immutable evidence context| Council
  ERP -->|permission-filtered business state| Council
  Council -->|attested allowlisted unsubmitted drafts only| ERP
  Legacy --> UI
```

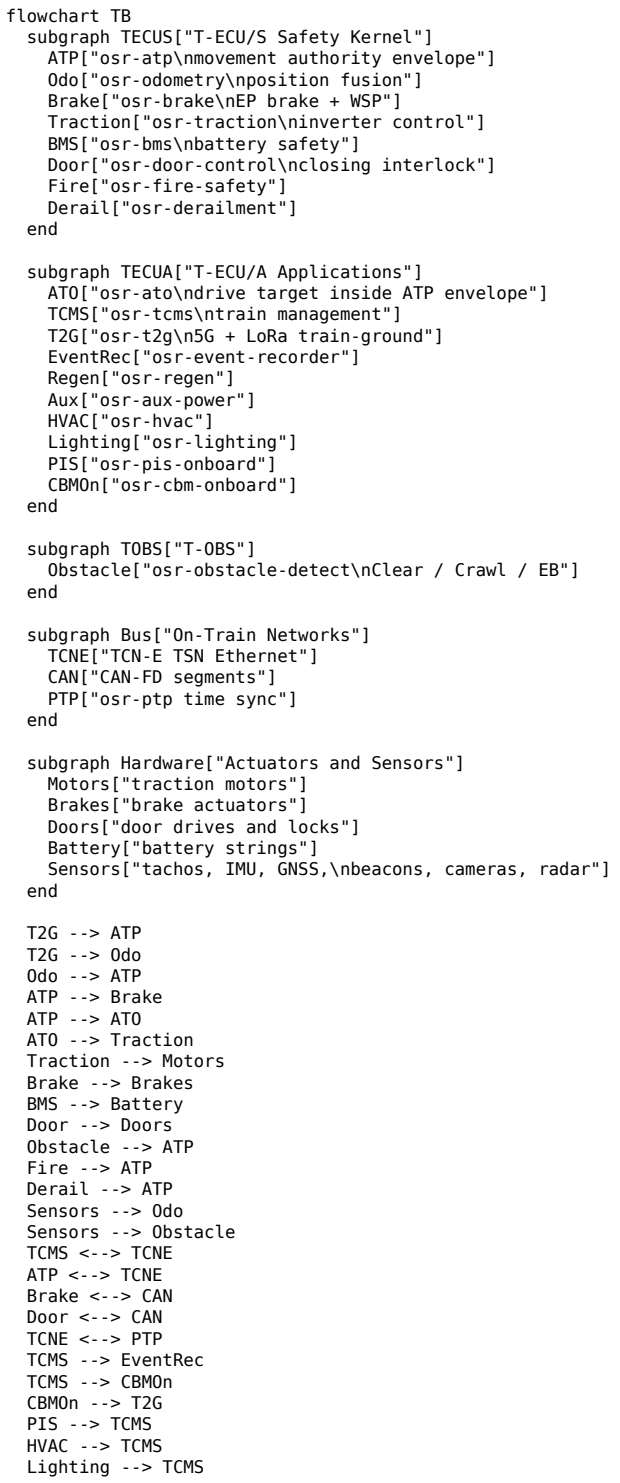
ERP completion and alarm clearance cannot grant railway release or movement authority. LM3 factory views apply only to the matching city family; cross-family module reuse needs a separate reviewed mapping before it can be represented.

2. Integration Boundaries And Future Interfaces

```
flowchart TB
  Controllers["Native simulation controllers"]
  Contract["Versioned observation-only Rust contract"]
  API["Implemented scoped HTTP integration API"]
  History["Gateway-owned SQLite historian and alarm/command audit"]
  Queue["Durable maintenance outbox"]
  ERP["ERPNext Issue / Asset Repair / stock / assignment"]
  FUXA["FUXA REST polling; DAQ disabled"]
  Hardware["Future commissioned supplier sensors and machines"]
  Transport["Future reviewed MQTT / OPC UA / Modbus adapters"]
  Bus["Future production event transport, e.g. NATS"]
  Assurance["Independent railway inspection and handback"]
  Controllers --> Contract
  Contract --> API
  API --> History
  History --> FUXA
  API --> Queue
  Queue --> ERP
  Hardware -.-> Transport
  Transport -.-> API
  API -.-> Bus
  ERP -.->|records for review; no release authority| Assurance
```

Dashed interfaces are proposed or require physical commissioning. NATS is not the transport of the deployed ERPNext/FUXA demonstration. Shared identity, TLS, backup operations, supplier bindings and HIL evidence remain deployment work. The remaining diagrams describe intended railway allocations and interfaces; they are not evidence that the depicted hardware paths have been commissioned.

3. Onboard Train Software



```
flowchart TB
  subgraph TECUS["T-ECU/S Safety Kernel"]
    ATP["osr-atp\nmovement authority envelope"]
    Odo["osr-odometry\nposition fusion"]
    Brake["osr-brake\nEP brake + WSP"]
    Traction["osr-traction\ninverter control"]
    BMS["osr-bms\nbattery safety"]
    Door["osr-door-control\nclosing interlock"]
    Fire["osr-fire-safety"]
    Derail["osr-derailment"]
  end

  subgraph TECUA["T-ECU/A Applications"]
    ATO["osr-ato\ndrive target inside ATP envelope"]
    TCMS["osr-tcms\ntrain management"]
    T2G["osr-t2g\n5G + LoRa train-ground"]
    EventRec["osr-event-recorder"]
    Regen["osr-regen"]
    Aux["osr-aux-power"]
    HVAC["osr-hvac"]
    Lighting["osr-lighting"]
    PIS["osr-pis-onboard"]
    CBMOn["osr-cbm-onboard"]
  end

  subgraph TOBS["T-OBS"]
    Obstacle["osr-obstacle-detect\nClear / Crawl / EB"]
  end

  subgraph Bus["On-Train Networks"]
    TCNE["TCN-E TSN Ethernet"]
    CAN["CAN-FD segments"]
    PTP["osr-ptp time sync"]
  end

  subgraph Hardware["Actuators and Sensors"]
    Motors["traction motors"]
    Brakes["brake actuators"]
    Doors["door drives and locks"]
    Battery["battery strings"]
    Sensors["tachos, IMU, GNSS,\nbeacons, cameras, radar"]
  end

  T2G --> ATP
  T2G --> Odo
  Odo --> ATP
  ATP --> Brake
  ATP --> ATO
  ATO --> Traction
  Traction --> Motors
  Brake --> Brakes
  BMS --> Battery
  Door --> Doors
  Obstacle --> ATP
  Fire --> ATP
  Derail --> ATP
  Sensors --> Odo
  Sensors --> Obstacle
  TCMS <--> TCNE
  ATP <--> TCNE
  Brake <--> CAN
  Door <--> CAN
  TCNE <--> PTP
  TCMS --> EventRec
  TCMS --> CBMOn
  CBMOn --> T2G
  PIS --> TCMS
  HVAC --> TCMS
  Lighting --> TCMS
```

4. Station and Depot Software

```
flowchart TB
  subgraph SSBC["S-SBC Station / Depot Host"]
    StationAgent["station agent\nhealth and config"]
    PISStation["osr-pis-station\nPIS and PA"]
    AFC["osr-afc\nfare gates"]
    TVM["osr-tvm\nticket vending"]
    PSD["osr-psd\nplatform doors"]
    Scada["osr-station-scada\nlifts, HVAC, CCTV"]
    EnergySite["osr-energy-site\nPV, BESS, chargers"]
    SelfTest["osr-selftest\nrole checks"]
  end

  subgraph LocalDevices["Local Devices"]
    Displays["platform displays"]
    Speakers["PA speakers"]
    Gates["fare gates"]
    Chargers["station chargers"]
    BESS["station battery"]
    Lifts["lifts and escalators"]
    CCTV["CCTV / NVR"]
    FirePanel["fire panel"]
    DepotTools["depot tools and fixtures"]
  end

  subgraph Backhaul["Backhaul"]
    OCC["OCC event stream"]
    Gateway["OSR integration gateway"]
    ERP["ERPNext maintenance / tooling records"]
    Railway["OSR inspection and handback"]
    AFCBack["AFC back office"]
    Historian["historian"]
  end

  PISStation --> Displays
  PISStation --> Speakers
  AFC --> Gates
  TVM --> Gates
  PSD --> Displays
  Scada --> Lifts
  Scada --> CCTV
  Scada --> FirePanel
  EnergySite --> Chargers
  EnergySite --> BESS
  StationAgent --> SelfTest
  StationAgent --> OCC
  EnergySite --> Historian
  Scada --> Historian
  AFC --> AFCBack
  SelfTest -.-> Gateway
  DepotTools -.-> Gateway
  Gateway --> ERP
  Gateway --> Railway
```

5. Wayside / Waypoint Node Software

In OSR language, a "waypoint" is any fixed railway location that helps localize, command, observe, or protect the railway: balise/beacon, switch, crossing, platform edge, intrusion sensor, hot-axle detector, or energy/charger site.

```
flowchart TB
  subgraph WSBC["W-SBC Cabinet"]
    Consensus["osr-consensus\ntrack-state replica"]
    Interlocking["osr-interlocking\nMA and route safety"]
    Points["osr-wayside-points\npoint command and detection"]
    Balise["osr-balise\nfixed position reference"]
    Crossing["osr-level-crossing"]
    HABD["osr-hot-axle-wayside"]
    Intrusion["osr-intrusion-detect"]
    PTP["osr-ptp"]
    Crypto["osr-crypto\nmTLS and signed firmware"]
    SelfTest["osr-selftest"]
  end

  subgraph FieldIO["Field I/O"]
    PointMotor["BLDC point motor"]
    PointSensors["dual point sensors"]
    Beacon["balise / UWB beacon"]
    Barrier["crossing barriers"]
    Thermal["hot axle IR array"]
    Fence["intrusion sensors"]
    UPS["cabinet PSU / UPS"]
  end

  subgraph Network["Networks"]
    WayE["WAY-E TSN Ethernet"]
    TrainRadio["train-ground radio\n5G / LoRa"]
    OCC["OCC / ops event stream"]
  end
```

```
  WayE <--> Consensus
  Consensus <--> Interlocking
  Interlocking --> Points
  Points --> PointMotor
  PointSensors --> Points
  Balise --> Beacon
  Crossing --> Barrier
  Thermal --> HABD
  Fence --> Intrusion
  PTP --> WayE
  Crypto --> WayE
  SelfTest --> OCC
  Interlocking <--> TrainRadio
  HABD --> Consensus
  Intrusion --> Consensus
  Crossing --> Consensus
  UPS --> SelfTest
```

6. Control and Data Flow

```
sequenceDiagram
  participant Dispatcher as Dispatcher Console
  participant OCC as osr-occ
  participant Log as Ops Event Log
  participant W as OSR shadow/supervised train-control candidate
  participant Train as Train T-ECU/S + T-ECU/A
  participant Hist as OSR historian / condition gateway
  participant Core as ERPNext maintenance
  participant Assurance as OSR independent handback

  Dispatcher->>OCC: request route / timetable action
  OCC->>Log: append dispatch intent
  Log->>W: route proposal
  W->>W: verify track state and point locks
  W-->>Train: candidate authority (shadow) / supervised request
  Train->>Train: ATP computes permitted envelope
  Train->>Train: ATO commands traction/brake within envelope
  Train->>W: position report and health heartbeat
  Train->>Log: telemetry and service events
  W->>Log: switch state, intrusion, HABD, route state
  Log->>Hist: telemetry archive
  Hist->>Core: condition trigger / work-order suggestion
  Core->>Dispatcher: maintenance case and repair status
  Core-->>Assurance: reviewed repair evidence for inspection
  Note over Core,Assurance: ERP completion grants no railway release
```

7. Energy and Charging Software

```
flowchart LR
  subgraph Planning["Planning and Supervision"]
    OCC["osr-occ\ntimetable and train location"]
    EnergyPlanner["charging dispatch\nI9"]
    Historian["historian\nenergy telemetry I10"]
  end

  subgraph TrainEnergy["Train"]
    BMS["osr-bms\nSoC, SoH, thermal limits"]
    ATO["osr-ato\nenergy-aware driving"]
    Regen["osr-regen\nregen routing"]
    ChargerIF["charge interface"]
  end

  subgraph Site["Station / Depot Energy Site"]
    EnergySite["osr-energy-site\nsite controller"]
    PV["PV inverter"]
    BESS["station BESS"]
    Charger["platform / depot charger"]
    Grid["grid tie\nIEEE 2030.5 / SunSpec"]
  end

  OCC --> EnergyPlanner
  BMS --> Historian
  EnergySite --> Historian
  Historian --> EnergyPlanner
  EnergyPlanner --> EnergySite
  EnergyPlanner --> ATO
  ATO --> BMS
  BMS --> ChargerIF
  Regen --> ChargerIF
  PV --> EnergySite
  BESS --> EnergySite
  EnergySite --> Charger
  Charger --> ChargerIF
  EnergySite <--> Grid
```

8. Manufacturing, QA, Maintenance, and Evidence Flow

```
flowchart TB
  Design["Versioned city design, assets and project twin"]
  Methods["LM3 methods, products, tooling and hold points"]
  Family["Selected city family applicability gate"]
  Mapping["Reviewed engineering revision + exact Item/BOM pair"]
  ERP["ERPNext: projects, procurement, production, stock, quality and repairs"]
  Workbench["Workbench: execution feedback and equipment context"]
  Gateway["OSR gateway: history, alarm queue and lifecycle evidence"]
  FUXA["FUXA: simulation method and equipment supervision"]
  Railway["OSR railway works, inspection and independent handback"]
  Legacy["Historic business records in Ops Core: read-only"]
  Physical["Open: performed travelers, machine adapters, physical measurements and rework disposition"]
  Design --> Mapping
  Mapping --> ERP
  Design --> Family
  Methods --> Family
  Family --> Gateway
  Gateway --> FUXA
  Gateway -->|condition-driven Issue| ERP
  ERP -->|permission-filtered actuals| Workbench
  Gateway --> Workbench
  Railway --> Workbench
  Legacy --> Workbench
  Physical -.-> ERP
  Physical -.-> Railway
```

Work Order correlation requires the Item and exact BOM from a single reviewed mapping at the selected engineering revision. Factory views show method context and simulation fixtures; they do not perform inspection, disposition or release.

9. Safety and Security Boundaries

```
flowchart TB
  subgraph SIL4["T1 / SIL-4 Target Safety Kernel"]
    Interlocking["osr-interlocking"]
    Consensus["osr-consensus"]
    ATP["osr-atp"]
    Odo["osr-odometry"]
    Brake["osr-brake"]
    DoorClose["door closing interlock"]
    Obstacle["osr-obstacle-detect"]
    Points["osr-wayside-points"]
  end

  subgraph SIL2["T2 / SIL-2 Target Safety Adjacent"]
    ATO["osr-ato"]
    TCMS["osr-tcms"]
    TCN["osr-tcn"]
    T2G["osr-t2g"]
    PSD["osr-psd"]
    Energy["osr-energy-site"]
    StationSCADA["osr-station-scada"]
  end

  subgraph SIL0["T3/T4 Supervisory and UX"]
    OCC["osr-occ UI"]
    Portal["operations portal"]
    Historian["osr-historian"]
    Analytics["osr-analytics"]
    PIS["passenger information"]
    AFC["fare collection"]
  end

  subgraph Security["Security Controls"]
    Crypto["osr-crypto"]
    PTP["osr-ptp"]
    SelfTest["osr-selftest"]
    Firmware["signed firmware /\nmeasured boot"]
  end

  OCC -->|requests only| ATO
  Portal -->|work orders only| StationSCADA
  ATO -->|bounded commands| ATP
  TCMS -->|bounded train commands| Brake
  Energy -->|charging limits| BMS
  Interlocking -->|movement authority| ATP
  ATP -->|safe state| TCMS
  TCMS -->|events| OCC
  StationSCADA -->|events| Portal
  Crypto --> SIL4
  Crypto --> SIL2
  PTP --> SIL4
  SelfTest --> SIL2
  Firmware --> SIL4
  Firmware --> SIL2
```

OSR-BP-001: Source-backed component and system practice

Repository source: `docs/standards/OSR-BP-001.md`

Version 1.0.0 · adopted 2026-10-11 · internal OpenSourceRail research and engineering-process standard.

This standard defines how OSR turns public industry examples and standards discovery into original, open guidance. It supplements OSR-ENG-001, OSR-ENG-002 and OSR-MFG-001. It governs research completeness and provenance; it does not supply universal product acceptance values.

The practice register and source register are the authoring inputs. The existing Digital Standards Thread remains the authority for deployment applicability, adopted editions/clauses, evidence and G0–G4 decisions. Each practice links to its existing STD-CTRL objectives; those links do not close a gate.

Rule	Mandatory internal requirement	Automated check
OSR-BP-01	Every scoped product, assembly, reusable civil type, station variant, Rust workspace package and engineering template shall have explicit specialist guidance.	Exact inventory/mapping equality; no keyword fallback
OSR-BP-02	Every shared template shall be classified as in scope or excluded with a reason.	New/deleted/ambiguous template detection
OSR-BP-03	Guidance shall cover design, materials/dependencies, assembly/build, inspection, maintenance, verification and unresolved decisions, with a responsible role.	Required nonempty card fields
OSR-BP-04	Technical research shall record a primary publisher, public URL, reviewed date, source kind, finding and limitation.	Source identity/type/date and reference validation
OSR-BP-05	Standard scopes, regulatory scopes, manufacturer examples and method references shall retain distinct interpretations.	Source-kind checks and generated limitations
OSR-BP-06	Public metadata shall not be represented as complete clause review, product qualification or achieved safety integrity.	Unsupported acceptance/physical/clause states rejected
OSR-BP-07	Cross-system interfaces shall have guidance and verification tasks in addition to local part checks.	Vehicle/track, energy, boarding and safety-control practice allocations
OSR-BP-08	Guidance shall use existing assurance control identities; project adoption shall resolve jurisdiction, edition, clauses, limits and authority.	Complete practice-to-STD-CTRL crosswalk with known IDs
OSR-BP-09	Unperformed verification plans shall retain open acceptance criteria, procedures, results and evidence.	Generated task states remain open and unperformed
OSR-BP-10	Kit contents, optional configurations, station variants and future scopes shall remain visible.	Preserved applicability records and explicit expansion backlog
OSR-BP-11	Published guidance shall be reproducible from exact source bytes and an explicit assessment date.	SHA-256 inputs and regenerated JSON/CSV/Markdown comparison
OSR-BP-12	Research shall be refreshed after source expiry or an applicability-changing revision.	Date-based refresh findings; stale outputs fail checking

OSR research refresh intervals are 90 days for regulatory sources, 180 days for manufacturer examples, 365 days for public standard scopes and 730 days for method references. These are internal review policies, not ISO or IEC requirements. A safety-relevant change triggers earlier review. Refreshing a date requires reviewing the source and recording any changed finding or limit; editing a date to make a check pass does not satisfy this rule.

Use ISO references where their scope applies, then applicable railway-specific IEC/EN methods and local authority requirements. For example, ISO 19659-4:2026 provides an HVAC-unit design-input and test scope, while IEC 62928:2017 addresses onboard traction batteries. Complete adopted clauses and installed-system evidence are required before project acceptance. Supplier web pages support discovery and method selection; they do not freeze a purchase specification.

Practice wording is original OSR synthesis and follows the repository's path-based licensing. References retain their publishers' terms. Proprietary drawings, normative text and control algorithms are not redistributed by this register. Portable data and supplier-neutral interfaces support local manufacture, repair and independent assessment.

OpenSourceRail engineering standard OSR-ENG-001

Repository source: `docs/standards/OSR-ENG-001.md`

Version 1.0.0, effective 2026-10-10. This is an active OpenSourceRail internal process standard for shared engineering definitions, numerical verification, manufacturing evidence and physical correlation. The machine-readable policy owns its version, reference editions and numerical thresholds.

This standard applies to OpenSourceRail research and engineering workflows. It does not declare ISO certification, railway approval or production release. Applicable law and authority requirements take precedence, followed by the accepted project design basis, applicable adopted ISO methods, and these internal rules. An approved project may impose stricter limits.

Applicable ISO references

The references below were checked against ISO's official catalogue on the effective date. Select the applicable material, process, product and test scope. Before claiming conformity, retain the controlled edition/amendments, applicable clauses, test procedure and completed conformity review. Catalogue descriptions identify scope; they do not substitute for normative method requirements.

Domain	Reference and intended use
Railway quality	ISO 22163:2023: railway quality management; bind customer, statutory and regulatory requirements to evidence.
Measurements	ISO/IEC 17025:2017: competent testing/calibration and retained measurement evidence.
Structural design basis	ISO 2394:2015: reliability and risk principles; project limit states and resistance factors remain required.
Ride evaluation	ISO 2631-4:2001: fixed-guideway relative comfort guidance; select the applicable ISO 2631-1 weighting and evaluation method.
Datums and tolerances	ISO 1101:2017: geometrical specification; ISO 286-1:2010: applicable fits and size-tolerance system.
Fasteners	ISO 898-1:2013: applicable steel fastener mechanical properties; ISO 16047:2005: applicable torque/clamp-force tests.
Welds	ISO 5817:2023: applicable fusion-weld workmanship; ISO 15614-1:2017: applicable welding-procedure qualification.
NDT personnel	ISO 9712:2021: relevant industrial NDT qualification/certification.
Concrete coupons	ISO 1920-4:2020: hardened-concrete strength test procedures.
Composite coupons	ISO 14125:1998: composite flexural properties; additional tests must establish the actual layup and interface laws.
Metallic fatigue	ISO 12107:2012: fatigue experiment planning/statistical analysis.
Pile tests	ISO 22477-1:2018: maintained axial static compression tests; lateral and cyclic behaviour require separate procedures.

ISO 2631-4 addresses relative comfort rather than supplying a universal absolute comfort threshold. Raw peak carbody acceleration from a short startup simulation shall not be labelled an ISO ride result. A workmanship quality level, fastener property class or coupon result shall not substitute for assembled strength, fatigue resistance, actual preload or joint capacity.

Mandatory OpenSourceRail rules

1. Configuration: retain existing product/joint identities, exact revisions, datums, material batches, serials, geometry sources, selected property sources and file hashes. Freeze as-designed, as-built and as-maintained configurations. Changed inputs shall invalidate dependent calculations and inspection evidence.
2. Model scope: declare idealisations, unresolved properties, validity ranges, initial conditions and intended load cases. Missing data shall remain open. Envelope volume shall not establish material mass. Synthetic data shall never establish supplier release, measured properties or physical validation.
3. Numerical verification: use independent analytical/native benchmarks, force/moment/work checks and separate temporal/spatial refinement. Retain unsuccessful cases. A converged nonlinear solve alone shall not establish discretisation accuracy or physical validity.
4. Material and connection resistance: identify all material regions, reinforcement, tendons, interfaces, fasteners, welds, bearings and soil laws in the load path. Evaluate applicable ultimate, service, fatigue, long-term, temperature and construction conditions. Unknown resistance or an uncovered failure mode shall block an engineering acceptance claim.
5. Normal service: normal wheel contact shall remain strictly positive for the accepted service envelope. Use geometry/friction-specific railway force limits, actual ride evaluation methods and approved project deformation limits. Distress/defect tests shall retain contact loss and stop conditions separately.

6 Coverage: include complete passages, operating/resonance speeds, empty, nominal, crush and uneven loading, braking, rescue/maintenance, degraded conditions, both tracks and arrival offsets. Short startup comparisons shall remain screening evidence.

7 Calibration: retain timestamps, SI units, instrument calibration, specimen and configuration identity, positive measurement uncertainty and bounded parameters. Holdout specimens shall be separate from calibration specimens. Report sensitivity rank, parameter covariance, residual bias/RMSE, uncertainty, correlation and extrapolation limits. Physical acceptance requires actual tests and independent review.

8 Manufacturing: drawings shall specify functional datums, tolerances, fits, processes, connection definitions and inspection characteristics before issue. Bind work orders, calibrations, batches, serials and nonconformances to the exact revision. Successful CAD generation shall not issue a drawing or clear a defect.

9 Search and cost: compare equal service/route scope and trial budgets; retain failures and identical uncertainty cases across methods. Confirm selected designs with finer coupled/local models. Include foundations, track, transport, temporary works, inspection, maintenance and replacement; missing prices shall remain null.

10 Release: numerical verification, ISO method conformity, physical validation, drawing issue and authority acceptance shall remain separate decisions. Only the existing controlled release process may authorise production or operation.

Internal numerical and calibration thresholds

These are OpenSourceRail rules, chosen as reproducible screening criteria. They are not ISO-prescribed tolerances or manufacturing specifications.

Check	Internal criterion	Reason and scope
Independent refinement	Each declared observable changes by at most 5% between final levels	Makes resolution effects visible before comparing candidates; refine time and mesh separately.
Native load equilibrium	Relative force and moment error at most 0.1%	Detects allocation/postprocessing errors; moment uses a force-times-length scale for near-zero applied moments.
Contact work mapping	Absolute residual at most 10^{-6} W in the retained benchmark	Checks the opposing work maps in this benchmark; choose documented scaling for materially different solver units/scales.
Analytical/native benchmark	Relative error at most 10^{-6}	Checks implementations against independent equations, without claiming physical accuracy.
Section equilibrium	Relative error at most 10^{-5}	Checks fiber resultants; it does not establish uncovered limit states.
Native datum solve	Position residual at most 10^{-5} mm; orientation residual at most 10^{-8} rad	Numerical assembly closure only; these values are not production tolerances.
Holdout residual	Uncertainty-normalised RMSE at most 2; absolute normalised bias at most 1	Candidate calibration screen; actual model discrepancy, correlated samples and operational prediction intervals require independent assessment.

Acceptance limits for wheel force ratios, weighted ride, fatigue, clearance, deflection, settlement, bearing pressure, material resistance and proof loads shall be linked to their applicable method, geometry and project requirement. This internal process standard does not invent universal values for those limits.

Execute and review

```
tools/automation/osr-python tools/automation/check-shared-engineering-standard.py \
  --review engineering/civil_exploration/examples/shared-spatial-engineering-review.json \
  --protocols build/engineering/spatial-campaign-converged/protocols.json \
  --output build/engineering/osr-standard-assessment
```

The assessment binds the standard version, review, protocol and implementation hashes. It produces governed protocols and separate numerical pass/fail and open physical/ISO gates. The published synthetic review remains unqualified even when its numerical rules pass. Revisions to this standard shall identify changed rules and their affected evidence; a new standard version shall not silently reaccept old inspections or physical tests.

OSR-ENG-002: Automated reference part generation

Repository source: `docs/standards/OSR-ENG-002.md`

Version 1.0.0 · internal process standard · adopted 2026-10-10.

This OpenSourceRail standard governs automated reference design studies. It supplements OSR-ENG-001. It grants no supplier release, ISO conformity, production authority or railway operating approval. Project and legal requirements retain the precedence established by OSR-ENG-001.

The industry register records primary sources, review date, numerical data actually adopted and separate OSR assumptions. Public summaries guide method selection; full ISO clause review and project applicability remain open. In particular, ISO 12405-4:2018 covers road-vehicle battery performance tests. Any railway use requires explicit applicability review. ISO 1101, 898-1, 16047 and 17025 method/traceability references follow OSR-ENG-001; their public catalogue descriptions do not supply manufacturing tolerances, bolt preload or acceptance limits.

Requirement	Internal mandatory rule	Executable verification
OSR-AP-001	A generated family shall retain the authoritative count, length, bogie and axle datums.	Full-family count/position tests for all five families; complete-passage planner
OSR-AP-002	Each physical mass shall belong to one nonoverlapping instance scope; expanded kit mass shall be removed from its aggregate.	Shared scope validator, independent total weight/reaction balance and passenger loading test
OSR-AP-003	Battery architecture changes shall regenerate geometry, quantities, mass/inertia, energy, current, resistance and thermal inputs together.	Parallel-string change propagation and cutlist conservation tests
OSR-AP-004	Generated reference geometry shall be valid within its declared primitive and packaging domain.	Positive/finite dimensions, balanced nonoverlapping holes and battery-bay rejection tests; native FreeCAD solids
OSR-AP-005	Every physical joint shall bind known instance datums and a typed mechanical interface with units and signs.	Interface/model identity, endpoint, joint coverage, unit and limit tests; native assembly datum reconciliation
OSR-AP-006	Every required catalogue allocation shall have a coverage record stating representation, evidence and unresolved limitations.	Unique complete coverage and unresolved-slot tests
OSR-AP-007	Electrical and thermal calculations shall retain separate energy balances and current/SOC/temperature constraint states.	Independent $P=VI$, I^2R, exact cooling and charge/discharge energy checks
OSR-AP-008	Generated quantities and QA characteristics shall retain instance identity; absent acceptance limits shall remain open.	Cutlist sums and QA open-limit tests
OSR-AP-009	Source and generated-output hashes shall bind each package; verification shall reproduce its compiled results.	CLI generate/verify and altered-output rejection tests
OSR-AP-010	Reference, numerical, engineering and physical acceptance states shall remain distinct.	Coverage/review false-release and null engineering-approval tests

Energy checks use a 1e-7 J absolute residual tolerance in the controlled unit tests. Train mass accounting uses 1e-12 relative reconciliation tolerance. These are internal numerical implementation checks, not ISO or railway acceptance limits. Native assembly residuals and refinement thresholds remain governed by OSR-ENG-001.

Generated inspection nominal values are design targets. Acceptance intervals, measurement calibration/uncertainty, joint capacity, fatigue, fire protection, electromagnetic compatibility and operating cases require their own controlled evidence. None can be closed by a CAD placement constraint or a numerical mass sum.

For OSR-AP-004/010, a positive-load Hertz footprint shall fit inside the supplied wheel and rail profile tables. An unsupported converged contact shall be rejected before state/history commitment and reported as outside the model domain. The contact-domain investigation records this applicability clarification; it adds no ISO or railway acceptance limit.

OSR-INT-001 — configuration-bound integration evidence

Repository source: `docs/standards/OSR-INT-001.md`

Revision 1, 11 October 2026. OpenSourceRail internal engineering policy. This is an original open project standard. It is not an ISO standard, a supplier qualification or a railway conformity certificate.

1 A design change shall compile into physical component identities, occupied geometry, material assumptions, mass, CG, inertia, installed datums and typed interfaces. Equivalent envelopes and residual allocations shall be identified. The same mass shall appear once in the assembled system.

2 Retained input, configuration, implementation and output checksums shall be exact. Recomputed floating-point results may use only a named reproduction tolerance, with units and magnitude stated. Reproduction tolerance shall not alter an engineering limit, source identity or acceptance decision.

3 A passage claim shall show every represented wheel on both sides crossing the infrastructure. A stopped braking case shall retain its different scope. Time and spatial refinement shall be separate comparisons. An unstable or unsupported profile, joint, material or operating case shall remain failed.

4 An optimisation candidate shall use the shared design compiler. Battery module counts, packaging, mass, inertia and energy shall remain linked. Research shortlists shall be separate from designs that satisfy all adopted limits. Unknown capacities, costs or operating limits shall remain unknown.

5 Mounting loads shall preserve simultaneous force and moment vectors and their time, part identity and load-history provenance. Compliance, clearance, preload, contact, member stiffness and fatigue shall state their represented domains. A coupon or elastic shank model shall not qualify an installation.

6 Flexible models shall contain material sections, openings, joints and mounting load paths appropriate to their stated fidelity. Rigid modes, mass, numerical convergence and work-conjugate attachment transfer shall be verified. Reduced beam verification and physical shell modal correlation shall remain separately identified.

7 Electrical and thermal coupling shall account for each physical load once. Current sign and units shall be explicit at controller boundaries. SOC, temperature and ageing maps shall have retained provenance and bounded interpolation domains. Missing calibration shall prevent a calibrated claim.

8 Installed condition shall bind a serial, manufactured revision, configuration, quantity, units, uncertainty, measurement record and instrument calibration valid at measurement time and across the measured range. Condition revisions shall be independent of manufactured revisions. Updating condition shall invalidate dependent analyses before any release decision.

9 Installed civil studies shall address each relevant component and limit state, including construction stages. Lifecycle comparisons shall share duration, route, traffic, access and service scope. Unpriced manufacture, transport, erection, temporary works, access, inspection, maintenance, replacement, disruption or decommissioning shall prevent complete-cost ranking.

10 Station verification shall combine boarding, route capacity, accessibility, equipment and power outages, fire scenarios and responder interactions. Queue conservation alone shall not establish evacuation/fire tenability.

11 Native manufacturing and ERP evidence shall use project-scoped work orders, inspections, materials, tools, timestamps, independent review and NCRs. Simulation records shall be labelled. Payroll examples shall not invent statutory policy; depreciation schedules shall not imply capitalisation journals were posted.

12 Controller qualification shall bind an actual target, wiring, I/O electrical characteristics, timing, watchdog, power-cut and storage evidence. Software execution shall remain distinct from hardware-in-loop and physical release. Formal proofs shall retain complete checks, source bounds and resource limits; timeouts, tool errors and memory exhaustion shall not become passes.

13 Recovery acceptance shall bind a separately hosted destination, identity migration, writer cutover, resumed controllers/jobs and measured RPO/RTO. A same-host fresh-volume rehearsal shall state its narrower scope.

14 Supplier selection and deployment closure shall require actual quotes, rights, support and authority records. Numerical, administrative and physical acceptance shall be reported separately. No generated template shall count as an executed test or approval.

15 Each displayed civil instance shall identify its analysed domain, geometry, mass, material and datum equivalence or an explicit unresolved association. Structural analysis domains absent from the display shall be reported. Homogenised springs shall not qualify discrete fasteners or anchorages.

16 Search methods and seeds shall receive equal declared evaluation and time allocations, with separate finalist-confirmation time. Shared cache use, incomplete trials and budget exhaustion shall be visible. Refined results shall satisfy all adopted limits before a numerical feasibility claim; physical and authority qualification remain separate.

ISO 10007 configuration-management, ISO 23247 manufacturing digital-twin and ISO/IEC 17025 laboratory practices inform the existing project standards thread. Applicability, purchased clauses, local law and independent acceptance remain in that thread. The reviewed method sources include NASA verification/validation, GENSYS wheel-profile verification, DfT station accessibility, and native ERP asset accounting boundaries. These references inform method selection, rather than asserting project conformity.

OSR-MFG-001: Open factory execution and quality evidence

Repository source: `docs/standards/OSR-MFG-001.md`

Version 1.0.0 · adopted 2026-10-11 · internal OpenSourceRail process standard.

This standard supplements OSR-ENG-001 and OSR-ENG-002. It governs generated manufacturing routes, travelers, material traceability, measurement decisions and commissioning protocols. It does not certify an ISO quality system, approve a manufacturing process, release a product or authorise railway operation.

ISO 22163:2023 and ISO 3834-2:2021 identify railway and fusion-welding quality scopes. OSR-ENG-001 supplies the applicable measurement, fastener, datum, welding-procedure, NDT and material test references. Full controlled clause review, amendments, material/process applicability and national requirements remain project responsibilities.

Rule	Mandatory internal requirement	Executable control
OSR-MFG-01	Work shall identify configuration, part revision, physical serial/lot, work order, operator and independent inspector.	Traveler and measurement identity checks
OSR-MFG-02	Material grade, certificate, shelf life, substitution and repair compatibility shall be controlled.	Material passport and lot/expiry checks; unknown released grade stays open
OSR-MFG-03	Functional datums shall be common to fixture, part and inspection; workholding capacity is assessed separately.	Six-restraint rank and native locator checks; rank does not qualify clamps/load capacity
OSR-MFG-04	Tolerance chains shall include signed dimensions, bounded tolerances and declared thermal offsets.	Worst-case interval; RSS only for supplied independent standard uncertainties
OSR-MFG-05	Special processes shall use qualified procedures, competent personnel, released windows and complete traces.	Process-window duration/sampling checks; blank windows cannot pass
OSR-MFG-06	Critical fasteners shall have per-joint/tool/revision traces; rework requires authority.	Torque/angle sample ordering, calibrated-range and rework checks; no preload inference from torque alone
OSR-MFG-07	Measurements shall use valid calibration, recorded environment, uncertainty and an explicit decision rule.	Conservative expanded-uncertainty interval containment for declared limits
OSR-MFG-08	Open NCRs, unperformed holds, reused unscoped measurements and stale revisions shall block review readiness.	Traveler predecessor, evidence and NCR checks
OSR-MFG-09	Production forecasts shall distinguish planning times from released cure/dry windows and observed capacity.	Resource scheduling with wait times; missing actual durations remain unplanned
OSR-MFG-10	Capability statistics shall not replace stability, distribution, measurement-system or individual inspection evidence.	Diagnostic Pp/Ppk; conditional Cp/Cpk requires explicit study records; no automatic threshold
OSR-MFG-11	Automated lifecycle recovery shall preserve critical lockout and movement interlocks.	Offline fault campaign; actual safety control and GoA4 qualification remain separate
OSR-MFG-12	Interfaces and work instructions shall remain portable, supplier-neutral and revision/hash bound.	JSON/CSV/Markdown/native CAD exports and reproducible package verification

The measurement decision rule is an OSR internal conservative interval rule, informed by JCGM 106. It does not define a universal ISO consumer/producer risk. Selected projects must approve its uncertainty coverage and suitability. Overlap with a limit requires review; an unknown acceptance limit is never a pass.

NIST capability methods inform the diagnostic implementation. Fifty independent suitable samples is a screening policy, not an ISO qualification rule. Confidence, stability and distribution evidence and acceptance thresholds remain project-defined.

Generated operations, source claims, fixture studies and offline fault tests are design/protocol evidence. Only actual controlled records and the designated authority can disposition physical work. These tools perform no ERP submission, product release or live train-control action.

Stations

Repository source: `docs/stations/README.md`

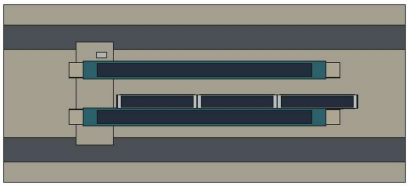
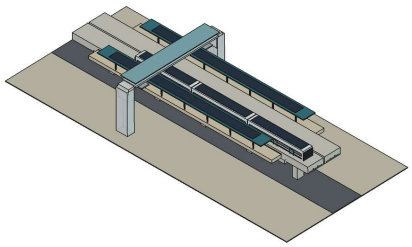
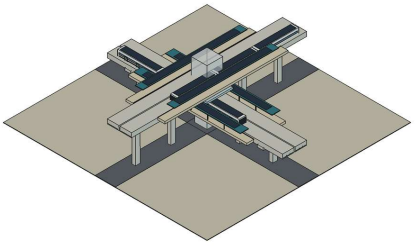
This folder is the station front door. The controlled source now generates positive-volume FreeCAD and IFC4.3 assemblies for all seven station archetypes; site coordinates, surveys, calculations and authority approval remain external release gates. The default architecture is a prefab steel portal frame with a solar canopy, platform edge systems, flat ground-level passenger slabs, and minimal station-building scope. Overbridges, lifts, and stairs are reserved for elevated/stacked sites or local road barriers.

Packages

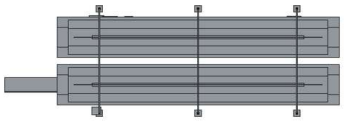
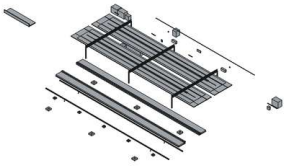
Package	Scope
`standard-archetype/`	Shared `standard` station archetype envelope, canopy, accessibility, services, compliance, and drawing register
`../../design/component-catalogue/catalog/buildable-stations/`	Generated EBOM/MBOM and matched assembly travelers for every station catalogue entry
`../../design/component-catalogue/models/cad/stations/`	Native FreeCAD assemblies for halt, standard, major, two interchange forms, terminal and depot-terminal
`../../engineering/models/bim/reference/stations/`	Matching geometric IFC4.3 assemblies plus deterministic validation summary
`../../design/component-catalogue/catalog/buildable-stations/station-product-reconciliation.md`	Bidirectional ID check across BOM, traveler, variant drawing register, installed/exploded FreeCAD states and IFC4.3
`../../design/component-catalogue/catalog/buildable-stations/factory-release-work-packages.md`	Nine bounded drawing, tooling, supplier and deployment handoff packages covering all 45 station product families
`../../design/component-catalogue/catalog/buildable-stations/factory-release-readiness.md`	Intentionally open readiness view for the 18 drawing/interface IDs and 22 tool/gauge families
`../../design/component-catalogue/catalog/buildable-stations/default-product-specifications.md`	Affordable supplier-neutral concept/RFQ defaults and override triggers for all 29 open product families
`../../design/component-catalogue/catalog/buildable-stations/factory-drawings/index.md`	Eighteen non-issued drafting/checking briefs spanning reusable fabrication, supplier interfaces and deployment drawings
`../../engineering/analysis/stations/screening-summary.md`	Solver-backed structural, passenger, drainage and baseline/proposed depot thermal/fire comparisons
`../../engineering/analysis/stations/mitigation-work-packages.md`	Owned evidence gates for the depot cooling, separation, protection, commissioning and approval response
`../../engineering/models/model-coverage.md`	Product-by-product geometry fidelity and release-evidence boundary

FreeCAD Station Scene Renders

These planning-review renders show the station access assumptions in context with ballastless track and driverless rolling stock.

At-grade side-platform station	Elevated side-platform station	Elevated interchange station
		
At-grade side-platform station with ballastless track and driverless train	Elevated side-platform station with ballastless track and driverless train	Elevated interchange station with stacked tracks and driverless trains

The standard product assembly also has two genuine selectable geometry states; the exploded view moves complete civil, canopy, MEP, passenger, access and charging work packages while retaining the installed coordinates in the same file.

Installed coordination state	Exploded work-package state
	
Standard station installed FreeCAD assembly	Standard station exploded FreeCAD assembly

Related Artifacts

Artifact	Location
Station canopy CAD source	`../../design/component-catalogue/src/osr_mech/station/`
Generated station FreeCAD scene document	`../../design/component-catalogue/models/cad/station-scenes.FCStd`
Generated station FreeCAD screenshots	`../screenshots/stations/`
Station scene generation script	`../../design/component-catalogue/scripts/freecad_station_scenes.sh`
Station BOM/traveler generator	`../../tools/automation/buildable-stations.sh`
All-variant FreeCAD generator	`../../design/component-catalogue/scripts/freecad_station_library.sh`
All-variant IFC generator	`../../engineering/interchange/station_ifc.py`
Product-structure reconciliation	`../../engineering/station_product_reconciliation.py`
Release evidence template	`../../design/component-catalogue/catalog/buildable-stations/evidence/factory-release-record-template.json`
Generated per-archetype BOMs	`build/bom/stations/` after running the generator
Station design RFC	`../rfcs/0010-station-design-standard.md`
Rapid implementation and recycled materials review	`../civil/rapid-implementation-materials-review.md`

Standard Station Archetype Worked Example

Repository source: `docs/stations/standard-archetype/README.md`

Scope: architectural envelope + canopy first-pass structural calc for the shared `standard` station archetype. The worked dimensions use a 75 m safeguarded platform and hot-climate assumptions; each deployment overrides climate, orientation, and parcel interfaces through the generated city model and local survey package.

Reference consist: `light-metro-3car` per RFC 0008. Platform clearance: 10 m per RFC 0010 §4.1. Generic platform length: 49.5 + 10 = 59.5 m. Instance build allowance: 75 m safeguarded civil length for local clearance and later fleet growth.

Contents

File	Scope
`envelope.md`	Plan + section geometry, ground-level access, egress
`canopy.md`	Solar canopy structural envelope + PV sizing + first-pass load calc
`accessibility.md`	Step-free path, tactile paving, wheelchair zones, audio+visual per EN 16586
`services.md`	Lighting, fire, drainage, small service cabinet, signage
`compliance.md`	Standards matrix (NFPA 130 egress, EN 12464-2 lighting, etc.)

Why this is repeatable

The whole point of RFC 0010's six-base-archetype catalogue and its controlled elevated-interchange variant is to avoid bespoke station design per deployment. This envelope is applied unchanged wherever the generated design selects the `standard` archetype. The operator's architect-of-record adapts it only for:

- Site-specific property-line constraints.
- Cardinal orientation (to aim the canopy PV south/north of equator; south-facing in the Iraq worked climate case).
- Pedestrian-connection geometry to the surrounding street network.
- Local material availability for façade finishes (all structural choices stay fixed).

Licensing

v1 specification: CC-BY-SA 4.0. v2 architectural drawings: CERN-OHL-S v2.

Accessibility - standard Archetype Worked Example

Repository source: `docs/stations/standard-archetype/accessibility.md`

Per RFC 0010 §7 + `lib/templates/accessibility.toml`. Designed against EN 16586 (PRM — Persons with Reduced Mobility) and the corresponding Iraqi accessibility overlay.

Step-free path — requirements

Every passenger, including wheelchair users, must have a step-free path from street to any door on any train:

1. Street / footpath platform: direct paved path at local pedestrian level. Local kerbs are ramped by the municipal footpath standard; the station kit itself adds no raised access.
2. Fare line paid zone: wide fare gate (min 900 mm clear, per RFC 0010 §8). One per direction on this archetype.
3. Platform train: the platform walking surface is flat with the adjacent pedestrian pavement, while ToR sits 350 mm below it in the station guideway channel. This matches the consist floor level for level boarding per RFC 0008 §3.3. Horizontal gap is ≤ 75 mm with the retractable skirt on the consist.

Tactile paving

Per ISO 23599 / Iraqi national code:

- Warning tactile: 600 mm deep along the full platform edge. Bright-colour contrast against the platform deck for vision-impaired users.
- Directional tactile: 300 mm wide strips guiding from footpath entries and fare gates to platform access points.
- Decision-point tactile: 1 500 × 1 500 mm at fare gates, emergency exits, and any local site ramp if one is needed.

Audio + visual information

Per RFC 0010 §7:

- Visual: next-train display (dot-matrix LED + color LCD) on each platform at 2 m height, 3 units per platform at 25 m intervals. Driven by `osr-pis-station`.
- Audio: PA speakers at 10 m intervals on each platform, weatherproof (IP65), driven by `osr-pis-station`. Volume capped at 85 dB(A) at 1 m to protect hearing.
- Audio-visual synchronisation: PA + visual announce the same next-train at the same time (tested in `osr-pis-station`'s proptest suite).

Wheelchair spaces

- On platform: 1 marked wheelchair waiting area per 30 m of platform length. For the 75 m worked-example safeguarded platform: 3 marked areas per platform × 2 platforms = 6 per station.
- On train: 2 wheelchair spaces per car × 3 cars = 6 per consist (per RFC 0008 §3.3).

Waiting area location: colocated with door 2 of the nearest car when the train stops — so a wheelchair user doesn't have to travel along the platform to reach the wheelchair-spaced door.

Vertical circulation

None for the at-grade `standard` archetype. The platform is already at pedestrian level, so lifts, stairs, and station ramps are not part of the base package. If a local site creates a level mismatch, the local approach path must meet ISO 21542 gradients and landing requirements before handover.

Seating on platform

Per RFC 0010 §7 accessibility (only `halt` archetype omits):

- 3 × 2-seat benches per platform, anchored to the platform back wall (under the canopy).
- Armrest on aisle side to aid stand-up.
- No fixed leg-rests — interferes with wheelchair approach.

Baggage + mobility aid storage

Not provided at `standard` archetype — passengers carry with them. `major` and `interchange` archetypes add storage lockers (v2 doc).

Companion-animal policy

Service dogs unrestricted; other pets per operator policy (Iraq typical: service dogs only). No rule in the archetype.

Accessible toilet

Not at `standard` archetype (per-operator choice). `major` and `terminal` archetypes have accessible toilets (v2).

Emergency communication

- Help buttons on each platform at 2 m height, 25 m intervals — directly connected to the OCC via the station SCADA. Wheelchair-user reach height max 1 200 mm.
- Audio-induction loop at ticket-gate staff position for hearing-aid users.

Compliance checklist (pre-opening)

Item	Status at v1
Step-free path from street to platform	specified
Platform-edge tactile paving 600 mm	
Directional tactile to platform	
Lift ≥ 1 100 × 1 400 mm	n/a at flat at-grade `standard` station
Audio + visual PIS	
3 wheelchair waiting areas per platform	
Wide fare gate ≥ 900 mm	
Accessible seating on platform	
Accessible help buttons	
Braille + raised-character signage	specified
Accessible toilet	n/a at `standard` archetype
Changing Places toilet	n/a at `standard` archetype (only at `major` / `terminal`)
Audio-induction loop at staffed position	specified

v2 deliverables (not in v1)

- Room data-sheet per staff position (reception, security).
- Signage typography + wayfinding ISO 7001 pictogram set.
- Accessibility-focused acceptance test with a local disability-rights advocacy group.

Solar canopy — first-pass structural envelope

Repository source: `docs/stations/standard-archetype/canopy.md`

> Status: controlled auxiliary-module structural envelope. The passenger platform baseline remains the repeatable 6 m cantilever kit. The 85 × 22 m worked roof below has been decomposed into repeatable 8.5 × 22 m auxiliary bays with shared transverse frames by `osr_mech.station.auxiliary_canopy`. It remains subject to deployment structural/foundation approval and is not a stamped site drawing.

Per RFC 0010 §9, every non-`halt` archetype carries a solar canopy over the platform. The worked `standard` archetype target is 1 800 m² (per `lib/templates/stations.toml`).

This document is the first-pass structural envelope the civil engineer bids against. v2 is the full FEA + wind-tunnel + stamped drawings.

Geometry

Parameter	Value
Canopy footprint (roof area)	~1 870 m² worked-example safeguarded roof; 1 800 m² archetype template target
Length (along track)	85 m (75 m safeguarded platform + 5 m each end eave)
Width	~22 m spanning both platforms + track between
Height above platform	2 200 mm clear (min), 3 800 mm roof peak above ToR
Pitch	7° equator-facing (south-facing in the northern-hemisphere worked case)
Column spacing along track	8.5 m (11 shared transverse frames over 10 bays / 85 m)
Column spacing across tracks	none — roof spans the whole station column-free between outer columns

Structural system

Steel truss roof, tied to steel columns at the platform-back edges. Not concrete — we need the structure to be erectable in a tight urban site by a commodity SMT-style crane (no form shoring). Welded steel plus bolted-steel truss is the simplest build.

Columns

- Size: HSS 200 × 200 × 8 mm, S355 grade.
- Count: 2 rows × 11 frames = 22 columns for the 10-bay worked row.
- Height above platform: 3.85 m (platform deck to truss bottom).
- Foundation: isolated pad, 1.8 × 1.8 × 0.8 m reinforced concrete, four #25 rebar anchors.
- Anchorage: base plate 400 × 400 × 20 mm, 4 × M30 anchor bolts per column.

Truss spans

- Span: 22 m (clear between outer-row columns).
- Truss type: Warren truss, 2.0 m depth, 2.5 m panel spacing.
- Top + bottom chords: HSS 150 × 150 × 6 mm S355.
- Diagonals: L90 × 90 × 8 mm angle, S355.
- Truss spacing along track: 8.5 m (matches column spacing).
- Total: 11 shared transverse trusses for 10 adjacent roof bays.

Roof panels

- Standing-seam sheet steel, 0.5 mm galvanised + powder-coat white.
- PV modules mounted flush on the south-facing slope.
- Sheet metal: 550 × 8 500 mm each (standing-seam stock), 2 200 sheets per station.

Photovoltaic layout

Per RFC 0010 §9 target 90 % platform shading:

Parameter	Value
Factory roof/PV bay	8.5 × 22 m, 187 m² gross
Roof/PV bays per worked canopy	10
Effective nameplate after 15 % packing allowance	~318 kWp
Cell/module count	Supplier-controlled within the factory-bonded roof envelope
Self-consumption (charging trains + station services)	~20 %
Export to grid	~80 %

The controlled calculation uses 200 W/m² gross density and a 15 % packing allowance, matching `station/solar_roof.py`. A deployment energy model applies temperature, orientation, soiling, inverter, and availability losses separately.

Load cases (first-pass)

Per EN 1991-1 basis. City-instance values for the hot-desert worked example:

Load	Characteristic value	Source
Dead load (roof + PV + services)	0.6 kN/m²	Material build-up
Snow load	0 kN/m²	Hot-desert worked case; override per city
Wind load (50-year return)	1.2 kN/m²	Iraq hot-desert baseline; override per site
Dust accumulation	0.15 kN/m²	50 mm of desert dust, post-haboob
Live load (maintenance)	0.5 kN/m²	Per EN 1991-1-1 Cat H
Seismic (PGA)	0.1 g	Iraq hot-desert baseline; override per site

Governing load combination (first-pass)

1. Wind uplift: 1.2 kN/m² up + 0.6 kN/m² self-weight down net uplift 0.6 kN/m². Governs roof tie-down + column base anchorage.
2. Gravity (hot-day dust + maintenance): 0.6 + 0.15 + 0.5 = 1.25 kN/m² down. Governs truss chord sizing.
3. Seismic: 0.1 g × 360 t total station structure mass = 360 kN horizontal. Governs the base connection — additional cross-bracing in the longitudinal direction.

Truss check (first-pass)

At 22 m span, 8.5 m tributary width, 1.25 kN/m² UDL:

- Peak moment M = (1.25 × 8.5) × 22² / 8 = 642 kN·m.
- Truss top-chord axial = M / depth = 642 / 2.0 = 321 kN.
- HSS 150 × 150 × 6 mm S355 capacity: 1 250 kN axial. Ratio 0.26 — plenty of margin.

Column check (first-pass)

At 8.5 m tributary × 22 m/2 (half-bay) × 1.25 kN/m² = 117 kN axial + lateral wind 1.2 × 8.5 × 3.85 / 2 = 20 kN lateral.

- HSS 200 × 200 × 8 mm S355 capacity: 1 800 kN axial, 90 kN·m bending. Ratios 0.07 axial, 0.42 bending — comfortable.

Foundation check (first-pass)

Isolated pad 1.8 × 1.8 × 0.8 m on 200 kPa bearing (typical alluvial soil worked case):

- Column ultimate load: 117 × 1.5 = 175 kN.
- Pad bearing pressure: 175 / (1.8 × 1.8) = 54 kPa — well below 200 kPa capacity.

Uplift check

Net uplift 0.6 kN/m² over the 1 870 m² footprint = 1 100 kN total lift. Distributed across 22 columns = 50 kN per column.

- Anchor bolt pullout (4 × M30 anchors × 380 kN capacity each) = 1 520 kN per column. Ratio 0.036.

Materials quantities (planning-grade BOM)

Line	Qty	Unit
HSS 200×200×8 mm column	84.7 m	22 × 3.85 m, S355
HSS 150×150×6 mm truss chord	484 m	two 22 m chords × 11 frames, S355
Truss web/diagonal set	11 frame sets	final cut lengths from released frame drawing
Plate 400×400×20 mm base plate	22 ea	S355
Plate 450×450×25 mm foundation plate	22 ea	S355
M30 anchor bolt, 800 mm long	88 ea	grade 8.8
C25/30 reinforced concrete (foundations)	57.2 m³	planning 2.6 m³ × 22; site calculation governs
#25 rebar	~2.6 t	planning scale; site calculation governs
Factory-bonded 8.5×22 m solar-roof bay	10 ea	supplier panel/cell schedule
PV combiner/isolation/downlink group	1 set	maximum ten roof bays per group
Gutter/downpipe/lightning/access kit	10 bay kits	coordinated to site drainage

Cost remains open until the frame calculation, factory roof supplier, site foundation design, and electrical string schedule are frozen.

Per-deployment customisation

The operator's architect-of-record adjusts:

- Cardinal orientation if site geometry forces east-west (acceptable performance loss ~15 %).
- Column finish (paint scheme / cladding — no structural implication).
- PV module brand / wattage (same racking footprint).
- Lightning protection class per local code.

What v1 doesn't include

- Full FEA / buckling analysis (v2).
- Wind-tunnel or CFD study for vortex shedding (v2 — matters only if slender columns are reduced; the envelope above is conservative).
- Foundation-specific geotech (v2 per-site).
- Detailed PV wiring + string layout (electrical-engineer scope separate from structural).

Compliance Matrix - standard Archetype Worked Example

Repository source: `docs/stations/standard-archetype/compliance.md`

Candidate standards and verification routes for the planning envelope. No configuration-specific conformity or authority approval is established here. The Digital Standards Thread governs deployment applicability, current adopted editions/clauses, evidence and independent decisions. National construction, fire, access and railway requirements remain project-specific inputs.

Structural + fire

Standard	Scope	Evidence status (v1 v2)
EN 1990 / EN 1991	Basis of structural design; loads on structures	Load cases enumerated in `canopy.md` (v1) full FEA (v2)
EN 1993	Design of steel structures	First-pass column + truss check in `canopy.md` (v1) EN 1993 stamped calc (v2)
EN 1997	Geotechnical design	Planning-grade bearing capacity (v1) site-specific geotech (v2)
EN 1998	Seismic design of structures	Iraq PGA 0.1 g baseline (v1) site-specific seismic microzonation (v2)
NFPA 130	Candidate fixed-guideway transit fire/egress route where adopted	Simple planning checks in `envelope.md` do not establish interacting crowd, fire or disruption-case compliance; adopted clauses and fire-engineer assessment open
EN 45545 (parts 2 + 3)	Railway-vehicle materials and fire barriers	Vehicle interface only where applicable. Outdoor stations have no automatic HL1 classification; building/station fire rules, installed material build-ups and test evidence require separate adoption
EN 1838	Emergency lighting	Spec'd in `services.md` (v1) product selection + test (v2)

Accessibility

Standard	Scope	Evidence status
EN 16586	Vehicle PRM boarding interface candidate; applicability review required	Planning intentions in `accessibility.md`; actual vehicle/platform compatibility and physical access tests open
ISO 23599	Tactile walking-surface indicators	Candidate tactile layout in `accessibility.md`; installed assessment open
ISO 7001	Public-information pictograms	Candidate signage basis in `services.md`; user and installed evidence open
ISO 21542	Building construction — accessibility	Flat pedestrian-grade access; local ramps only where site grading requires them
EN 81-70	Lifts accessible to persons with disabilities	n/a for flat at-grade `standard`; applies to elevated/stacked overrides

Electrical + MEP

Standard	Scope	Evidence status
IEC/EN 61439 series	Low-voltage switchgear/controlgear assemblies; select the relevant product part with Part 1	Exact assembly, adopted edition, design/routine verification and installation evidence open; IEC 60439-1 is replaced
EN 61000-6-3	Candidate generic EMC emission scope; railway/system applicability review required	LED/PIS compliance and installed compatibility unassessed
EN 12464-2	Candidate outdoor workplace lighting scope	Planning illuminance targets in `services.md`; adopted values and installed measurements open
IEC 60947-5-1	Auxiliary contacts for emergency-stop circuits	Used on emergency help buttons
ISO 14644	Cleanrooms / contamination control	n/a for transit station — not specified

Environmental

Standard	Scope	Evidence status
ISO 3095	Acoustics — railway applications, noise emitted	Pass-by noise target 80 dB(A) at platform (v1)
WHO community-noise guidelines	Night-time outdoor noise	Target 45 dB(A) at façade (RFC 0011 §5.1 acoustic screens on elevated)
EN 12101 (applicable parts)	Smoke/heat control candidate	Open-air configuration does not automatically exempt cabinets, enclosed approaches or future variants; applicability assessment open
Applicable UPS and emergency-power requirements	Station backup power, battery/fire protection and load endurance	UPS specification, critical-load allocation, product scope, installed runtime and authority decisions open; no NFPA 110 classification claimed

Cybersecurity + physical security

Standard	Scope	Evidence status
IEC 62443 (parts 2-4 + 4-2)	Industrial automation + control systems cybersecurity	`osr-station-scada` designed per `osr-crypto` discipline
EN 50159	Safety-related railway communications candidate	`osr-station-scada` design intent; channel classification, adopted clauses, configuration-specific tests and acceptance open

National overlays (Iraq — per-deployment)

Confirm the actual national/local electrical, fire, accessibility and railway rules, identifiers, editions, competent authorities and approval route. This planning document does not establish their current legal status or adoption. Arabic signage and representative user needs belong in the project access plan.

Publisher scope review: BSI EN 45545-1 concerns onboard vehicle fire protection. IEC 60439-1 is replaced; IEC 61439-1:2020 is a general assembly scope used with the applicable product part. These public records do not establish full-clause conformity.

Acceptance-test outline (pre-opening)

Unperformed candidate test outline. Adopt configuration-specific procedures, limits, calibration, representative scenarios and independent acceptance before using any entry as a pre-opening sign-off:

1. Structural: inspector visual walk-through + spot-check on column anchorage, truss welds.
2. Accessibility: actual wheelchair user rides from street to train door on both platforms — no step, no gap > spec.
3. Emergency lighting: simulate mains-fail; confirm 1-hour battery operation of exit signs + egress lighting.
4. Fire alarm: test services-cabinet detector OCC alarm ≤ 30 s latency.
5. PIS: next-train information updates correctly from the OCC's mock schedule.
6. Fare gates: open on valid QR + NFC; deny on invalid; emergency-release works.
7. Platform-screen-door coordination: (if fitted at this station — not `standard` default but per operator choice) door command / confirmation round-trip within `osr-psd`'s FSM.
8. CCTV: OCC receives all cameras at nominal resolution + frame rate.
9. Drainage: pour 1 m³ of water into the low end; confirm it drains to the municipal connection in ≤ 5 min.
10. Noise: train pass-by at design speed ≤ 80 dB(A) at platform (confirm by sound-level meter at the waiting area).

Checklist lands with actual pass/fail values in the deployment's `acceptance/` tree.

What v1 doesn't commit to

- Stamped structural calculations (v2).
- Product-specific equipment selections (operator discretion).
- Specific lift-service contracts (only for elevated/stacked overrides).
- National approval-body sign-off procedure (per-deployment).

standard station — architectural drawing register (v2)

Repository source: `docs/stations/standard-archetype/drawing-register.md`

Status: v2 deliverable of RFC 0010. Purpose: a complete list of the architectural drawings an architect-of-record (AOR) produces for a deployment instance of the shared `standard` station archetype. Every drawing is scoped, sized, and keyed to the sections of the v1 envelope documents (envelope.md, canopy.md, accessibility.md, services.md, compliance.md).

Convention

- All drawings in the register are A1 size at stated scales, except section details which may be A3.
- Revision letters: A = first issue, B = post-coordination revision, C = construction issue.
- Every drawing has a title block referencing this register + the commit hash of the v1 envelope docs.
- Drawing numbers follow a stable scheme:

  `OSR-STD-<discipline>-<sequence>`
- discipline codes: A (architectural), S (structural), M (mechanical), E (electrical), F (fire), T (tactile + signage), L (landscape — if applicable).
- sequence: zero-padded 3 digits within each discipline.

A per-deployment station uses the same numbers; only the title block's site field varies.

Drawing list

A — Architectural (15 drawings)

No.	Title	Scale	Size	Input
OSR-STD-A-001	Site plan + ROW	1:500	A1	Envelope §Plan
OSR-STD-A-002	Platform plan (both platforms)	1:100	A1	Envelope §Plan
OSR-STD-A-003	Fare/TVM plinth plan	1:50	A1	Envelope §Sub-areas
OSR-STD-A-004	Typical cross-section	1:50	A1	Envelope §Section
OSR-STD-A-005	Longitudinal section	1:100	A1	Envelope §Sub-areas
OSR-STD-A-006	West-end elevation (fare/TVM plinths)	1:50	A1	Envelope §Sub-areas
OSR-STD-A-007	East-end elevation (emergency egress)	1:50	A1	Envelope §Egress
OSR-STD-A-008	Canopy plan	1:100	A1	Canopy §Geometry
OSR-STD-A-009	Canopy section	1:50	A1	Canopy §Structural system
OSR-STD-A-010	Fare-zone plan (paid / unpaid)	1:100	A1	Envelope §Fare paid/unpaid
OSR-STD-A-011	Platform-edge detail (tactile + coping)	1:10	A3	Accessibility §Tactile paving
OSR-STD-A-012	Local approach grading details (if needed)	1:20	A3	Envelope §Access modes
OSR-STD-A-013	Equipment plinth + cabinet enclosure schedule	n/a (table)	A1	Services §Signage
OSR-STD-A-014	Entrance, validator, and fare-plinth elevations	1:20	A1	Envelope §Sub-areas
OSR-STD-A-015	Canopy, cabinet, plinth, and platform finish schedule	n/a (table)	A1	Canopy §Per-deployment customisation

S — Structural (8 drawings)

No.	Title	Scale	Size	Input
OSR-STD-S-001	Foundation plan	1:100	A1	Canopy §Foundation check
OSR-STD-S-002	Column + footing detail	1:20	A1	Canopy §Columns
OSR-STD-S-003	Canopy truss elevations (typ.)	1:50	A1	Canopy §Truss spans
OSR-STD-S-004	Canopy truss plan	1:100	A1	Canopy §Truss spans
OSR-STD-S-005	Weld details + bolt tables	1:5	A3	Canopy §Structural system
OSR-STD-S-006	Platform slab reinforcement	1:100	A1	
OSR-STD-S-007	Fare/TVM plinth anchorage	1:50	A1	
OSR-STD-S-008	Seismic-bracing plan (when PGA > 0.2 g)	1:100	A1	Canopy §Seismic load

M — Mechanical (5 drawings)

No.	Title	Scale	Size	Input
OSR-STD-M-001	Services-cabinet cooling layout	1:50	A1	Services §HVAC
OSR-STD-M-002	Platform drainage — trough + downspout	1:100	A1	Services §Drainage
OSR-STD-M-003	Canopy-drainage layout	1:100	A1	Services §Drainage
OSR-STD-M-004	Local ramp/kerb interface spec (if needed)	n/a	A3	Envelope §Access modes
OSR-STD-M-005	Accessible-toilet plumbing (if fitted per operator)	1:20	A3	Services §Utilities

E — Electrical (8 drawings)

No.	Title	Scale	Size	Input
OSR-STD-E-001	Incoming MSB + sub-board schematic	n/a	A1	Services §Electrical
OSR-STD-E-002	Platform lighting layout	1:100	A1	Services §Lighting
OSR-STD-E-003	Canopy PV layout + string diagram	1:100	A1	Canopy §PV layout
OSR-STD-E-004	Canopy PV combiner + DC cabinet	1:20	A1	Canopy §PV layout
OSR-STD-E-005	Emergency lighting + UPS	1:100	A1	Services §Fire + life safety
OSR-STD-E-006	CCTV camera positions + cable routing	1:100	A1	Services §Communications
OSR-STD-E-007	Help-button + PA loudspeaker layout	1:100	A1	Accessibility §Audio + visual
OSR-STD-E-008	Lightning-protection finial + down conductor	1:50	A1	Canopy §BOM

F — Fire + life safety (4 drawings)

No.	Title	Scale	Size	Input
OSR-STD-F-001	Fire-detection zone plan	1:100	A1	Services §Fire + life safety
OSR-STD-F-002	Fire-extinguisher placement	1:100	A1	Services §Fire + life safety
OSR-STD-F-003	Emergency-egress plan (NFPA 130 compliance)	1:100	A1	Envelope §Egress + Compliance §NFPA 130
OSR-STD-F-004	Evacuation-signage layout	1:100	A1	Services §Signage

T — Tactile + signage (3 drawings)

No.	Title	Scale	Size	Input
OSR-STD-T-001	Tactile paving layout	1:100	A1	Accessibility §Tactile paving
OSR-STD-T-002	Wayfinding signage positions + typographic spec	1:100 + swatches	A1	Services §Signage
OSR-STD-T-003	Accessible signage (Braille + raised-character) details	1:5	A3	Accessibility §Compliance checklist

Drawing count + delivery phase

Discipline	Drawing count
A — Architectural	15
S — Structural	8
M — Mechanical	5
E — Electrical	8
F — Fire	4
T — Tactile + signage	3
Total per station	43

For each generated `standard`-archetype station, the same 43-drawing register is reused. Only the site-plan (OSR-STD-A-001) varies meaningfully between sites; the other 42 drawings are near-identical, with per-site parameters (parcel geometry, north arrow, foundation specifics) as the only delta.

Interaction with per-site adaptation

The v1 envelope docs list per-deployment adaptations that the AOR implements:

- Site plan (A-001): always site-specific.

- Orientation (A-008, E-003): rotate to face equator; swap if southern hemisphere.
- Façade finish (A-015): operator choice reflected in the schedule.
- Parcel-line setbacks (A-001, A-006, A-007): per- municipality.
- Foundation depth (S-001, S-002): per-site geotech.
- Seismic (S-008): applicable only if PGA > 0.2 g.

Everything else is archetype-fixed: one drawing set is reused for every generated `standard` station in a deployment.

Delivery sequence (typical)

1. Design Development (DD) — all A + S drawings at rev A.
2. Coordination — M + E + F + T drawings at rev A + integrate back into A + S rev B.
3. Construction Documents (CD) — everything at rev C.
4. Tender — operator's procurement team issues CD to construction-firm bidders.
5. Shop drawings — fabricator's own drawings (steel fabrication, PV module mounts) land as annexes.

Acceptance criteria for v2

The v2 deliverable of this doc = the architect's drawing register for the deployment, matching the list above drawing-for-drawing. Per-drawing content is the architect's scope; the register + envelope specs are the upstream input.

For a per-deployment tender, the AOR:

1. Downloads this register.
2. Fills in the title-block fields (site, project, AOR firm name).
3. Produces the drawings.
4. Files the commit hash of the envelope docs against which they drew.

What v2 does NOT include

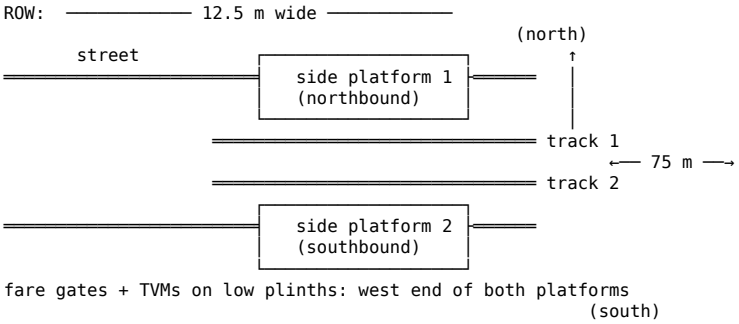
- Actual CAD files (AutoCAD / Revit / ArchiCAD — AOR's tool choice).
- Printed plot files (per-deployment plot shop).
- Renderings (optional operator deliverable).
- Cost estimates (per-deployment QS scope).

standard Archetype Envelope

Repository source: `docs/stations/standard-archetype/envelope.md`

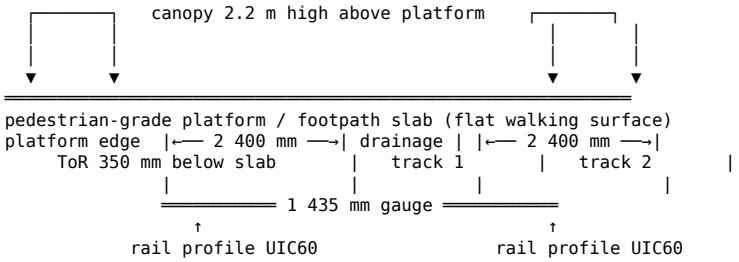
Plan + section geometry of the shared `standard` station archetype, shown with the hot-climate worked-example platform allowance. Every number here is a buildable dimension.

Plan



- Configuration: two side platforms, double-track.
- Platform length: 75 m safeguarded worked-example build allowance. The generic `light-metro-3car` standard is 59.5 m (49.5 m consist
  - 10 m clearance); this package deliberately reserves more civil length for local clearance and later fleet growth.
- Platform width: 3.5 m each (per RFC 0010 §4.3).
- Track centres: 4.0 m (standard double-track with centre drain).
- Total station ROW width: 3.5 + 4.0 + 3.5 = 11.0 m; +1.5 m buffer for drainage + ROW-line clearance = 12.5 m.
- Pedestrian datum: platform walking surfaces are flat with the adjacent footpath/forecourt. The rail datum drops through the station bay; passengers do not climb onto a raised platform.

Section through platform



- Platform walking surface: flush with adjacent pedestrian pavement. Top-of-rail (ToR) sits 350 mm below that surface inside the drained guideway channel, matching the `light-metro-3car` low-floor height for level boarding without raising the platform above street level.
- Platform-to-track horizontal gap at door: ≤ 75 mm (UIC 741).
- Canopy clearance: 2 200 mm minimum height over the platform deck. Roof is at 3 800 mm above ToR (matches the consist's max height envelope).

Sub-areas

Area	Location	Size (m²)	Function
Platform 1 (northbound)	75 m × 3.5 m	263	Passenger wait + board
Platform 2 (southbound)	75 m × 3.5 m	263	Passenger wait + board
Cross-platform canopy	85 m × 22 m	1 870	Solar canopy + rain shelter (incl. edges)
Fare/TVM plinths (west end)	2 × 10 m × 2 m	40	Fare gates + TVMs + validator line
Services cabinet room	4 m × 3 m	12	Station SCADA + sub-MSB + UPS

Total gross floor area: ~530 m² open platform + ~50 m² compact fare/services footprint. No default ticket hall, lift room, or stair well is included for an at-grade `standard` station.

Access modes (per RFC 0010 §5)

Each side platform has:

1. Direct paved access from the adjacent footpath/forecourt to the platform slab. Gradient is governed by the surrounding street, not by the station kit.
2. Fare line / validator plinth at the west end, with a wide gate and bypass gate for PRM users.
3. Secondary direct egress at the east end. If the station sits between roads, the crossing is protected and interlocked rather than grade-separated.

No lift, stair, escalator, or overbridge is part of the at-grade `standard` archetype. Those are local overrides for elevated/stacked stations or unavoidable road-barrier sites.

Egress (per RFC 0010 §6 and NFPA 130 4-minute rule)

Platform occupancy sizing:

- Planning-grade peak: 150 passengers × 2 platforms = 300.
- NFPA 130 egress: must reach a "safe point" within 4 minutes.

Egress paths (per platform):

- Primary: direct west-end paved exit to the footpath/forecourt (≤ 11 m travel from the fare line).
- Secondary: direct east-end paved exit to the footpath/forecourt.

Effective egress width:

- West exit 2.0 m + east exit 2.0 m.
- Total effective width: ≥ 4.0 m per platform.

At 300 pax per platform and NFPA 130 flow 1.3 pax/m/s:

- Clearance time = 300 / (4.0 m × 1.3 pax/m/s) ≈ 58 s. Well under the 4 min target.

Fare paid / unpaid zones

- Unpaid zone: footpath/forecourt, TVMs, outside fare gates.
- Paid zone: everything platform-side of the fare gates.

`standard` archetype gets 2 fare-gate lanes per direction per RFC 0010 §8 (so 4 total — 2 in, 2 out; 1 wide-gate for wheelchairs). TVMs: 1 per direction (2 total).

Property-line setbacks (for operator adaptation)

- ROW centreline to property line: ≥ 3 m both sides (municipal code typical).
- Ticket hall at west end can extend into unused ROW corners; operator adapts per parcel.
- If a constrained site can only accept one public entrance, keep the opposite platform end as a normally closed emergency gate. A `standard` archetype still requires at least 2 independent egress directions per platform.

Site footprint

Total approximate footprint a `standard` station occupies on the ground:

- Platform envelope: 75 × 12.5 = 940 m² main rectangle.
- Fare/TVM west plinths: 40 m² inside the main rectangle.
- Service cabinet: 12 m² west-end addition if it cannot sit under the canopy.
- Canopy shadow (extends slightly beyond platform): 85 × 22 = 1 870 m² — but this is the roof, not ground footprint.

Usable ground footprint: ~950-1 000 m² per station.

Interface with the simulator

`osr-sim` consumes these archetype numbers via the `lib/templates/stations.toml` schema. The v1 envelope does not require any sim change; all parametric sizing already lives in templates.

What this doesn't commit to

- Façade finish (brick, stucco, aluminium panel — operator choice).
- Signage typeface / bilingual arrangement (operator choice).
- Paid-zone toilets or vending (operator discretion, not an archetype feature).
- Retail concessions (excluded from `standard`; see `major` archetype at v2).

Station Services - standard Archetype Worked Example

Repository source: `docs/stations/standard-archetype/services.md`

HVAC, lighting, fire, drainage, signage — the MEP (mechanical / electrical / plumbing) envelope for the station.

Per RFC 0010 §11 an outdoor hot-climate station is passively cooled by the canopy. The at-grade `standard` archetype has no default ticket hall or lift rooms; only the services cabinet gets thermal control if the selected equipment requires it.

HVAC

- System: none for passenger areas. Optional 1-2 kW cabinet cooling for SSBC, UPS, and fare equipment at design ambient 50 °C.
- Refrigerant: R-32 (commodity).
- Fresh-air intake: n/a for open platform; filtered cabinet intake if active cooling is fitted.
- Heating: none in the hot-climate worked case; if used, reverse-cycle heat pump on the same unit.
- Filters: MERV 8 standard; MERV 14 for heavy-dust days per operator discretion.
- Drain: to condensate pipe station storm drain.
- Control: `osr-station-scada` with operator override.

Platform ventilation

None. Platform is open-air under the canopy. Natural cross-ventilation works year-round; on haboob days, platform shelter relies on the canopy, equipment plinth backs, and local wind screens if fitted.

Lighting

Per EN 12464-2:

Area	Illuminance	Colour temp	Control
Platform (night)	100 lx	4 000 K	Dusk-dawn sensor + schedule
Fare/TVM plinth	200 lx	4 000 K	Time schedule + occupancy
Platform (day)	canopy ambient	daylight	none
Emergency egress	20 lx (1 h battery)	4 000 K	EN 1838 self-test
Exterior approach	50 lx	4 000 K	Dusk-dawn
Services cabinet room	500 lx	4 000 K	Maintenance switch

All LED, EN 61000-6-3 EMC class B. Total connected load per station: ~4 kW.

Fire + life safety

Planning fire/life-safety intentions. Applicable station/building rules, adopted NFPA 130 clauses where relevant and competent fire-authority requirements must be resolved through the standards workflow:

- Smoke detection: ceiling detector in services cabinet; open platform has no enclosed smoke zone.
- Heat detection: services cabinet (rate-of-rise).
- Passive protection: qualify installed station material and barrier build-ups against the adopted building/station fire strategy. EN 45545 is a railway-vehicle reference; being outdoors does not establish a station HL1 classification or an exemption from smoke/fire assessment.
- Fire extinguishers: 1× CO_2 at services cabinet, 1× water per platform end.
- Fire alarm signal: goes directly to the OCC via `osr-station-scada`; cannot be silenced without OCC OK.
- Emergency power: 1-hour UPS (60 kWh Li-ion battery in services cabinet) keeps PIS + emergency lighting + OCC comms alive.
- Evacuation: direct paved exits at both platform ends (per `envelope.md` §egress).

Drainage

- Platform: 2 % transverse fall to centre trough drain; longitudinal 0.5 % fall to the low end.

- Track drain: centre trough between the two tracks, 300 mm wide, 400 mm deep, to perforated pipe below the ballast.
- Canopy runoff: gutter + downspout at each column base, discharges to station storm drain.
- Capacity: sized for 50 mm/h 10-year return storm. The hot-desert worked case rarely hits this, but when a sandstorm is followed by rain, the first event of the year can overload the drain — catch pit at storm drain connection.

Signage

- External: station name + route-map + OSR brand panel at each entrance.
- Internal: wayfinding per ISO 7001 pictograms; bilingual (Arabic + English) text. Signs at 2.5 m height (overhead)
  - 1.2 m height (lateral, for accessibility).
- Line / route diagram: generated route strip for the deployment network at every platform-edge support column.
- Real-time information: `osr-pis-station` drives the LED
  - LCD displays.
- Emergency exit signs: EN 1838 photo-luminescent + battery backup.

Utilities

Electrical

- Mains: 3-phase 400 V AC at 50 Hz from the municipal connection (or from the nearest `osr-energy-site` if in island mode).
- Connected load at peak: cabinet cooling 2 kW + lighting 4 kW + services + fare equipment + PIS = ~10 kW total.
- Design demand: 15 kW + 50 % future-growth headroom = 23 kW incoming supply.
- Distribution: MCB board in the services cabinet, RCD- protected sub-circuits.

Water + waste

- Water: town water for platform cleaning; no drinking fountain at `standard` archetype.
- Waste: municipal connection for storm + sanitary (if the operator adds a future toilet).

Communications

- Ethernet from the station's S-SBC to the OCC (primary + LoRa backup).
- CCTV camera feeds bonded to the station SCADA, then backhauled to OCC.

Environmental

- Noise: platform ambient target ≤ 70 dB(A) day, ≤ 55 dB(A) night during no-train condition. Train pass-by peak ≤ 80 dB(A) per ISO 3095.
- Visual impact: canopy form factor + lighting strategy follow the operator's urban-design guidelines.

v2 deliverables (not in v1)

- Detailed MEP drawings (plans + sections).
- Load schedule per sub-circuit.
- Fire-detection zone plan with alarm-panel layout.
- Lift-room machinery spec.
- PA loudspeaker aiming + acoustic coverage plots.

Engineering and delivery substance review — 2026-09-09

Repository source: `docs/substance-review-2026-09-09.md`

Historical design snapshot. Fleet, route, station and distributed-stabling figures below describe the earlier controlled revision. The current Samawah example adopts the regenerated central alignment and full-fleet line depots. These earlier figures and operating experiments cannot release the new layout.

This review follows the repository access, persistence and provenance repairs. It examines what the models actually demonstrate, rather than counting files, drawings, tests or completed solver runs. The drainage repairs and electrical connection measurements below have been regenerated. The later ethos audit corrects this review's interpretation of the electrical results: a grid-only stress failure does not establish a shortfall in the solar/storage operating design.

The repository has useful reproducible planning models and explicitly open release registers. The electrical screen now separates grid-only contingency findings from solar/storage snapshot acceptance. Full operating energy validation remains open. Delivery cashflow, depot scope and lifecycle cost also need stronger connections to the physical quantities and operating plan.

1. Roof-drainage input and validation repairs completed

Priority: high. Repository model defects repaired; site design remains open.

The original SWMM generator placed width in the imperviousness column. The standard branch represented a 10% impervious roof with a 1 m flow width and 100% slope. Its sparse five-minute rainfall records supplied only 12.5 mm, half the declared 25 mm storm. The correct column order and rainfall format are documented in the EPA SWMM manual.

The investigation also found zero pipe fall. SWMM substituted a minimum elevation drop; correcting only the roof and storm left a distorted kinematic-wave peak. The repaired reference model declares a 0.4 m fall over 40 m, uses dynamic-wave routing, and reduces the runoff/hydraulic steps to 10 s / 1 s. This gradient is an explicit planning assumption requiring survey, not an approved site level.

All seven regenerated variants pass independent checks for 25 mm rainfall, zero infiltration, rainfall-volume/runoff balance, C=1 roof peak, conduit peak, continuity, no flooding and inlet depth. Continuity is captured after SWMM finalizes the simulation. Runoff-volume error is below 0.02% across the family. The standard-station result changes from 5.62 L/s to 37.77 L/s aggregate branch flow, matching the 37.77 L/s C=1 rational calculation. The old comparison used C=0.95 despite the intended zero-loss roof.

The station report records the assumptions and observed quantities. New solver regressions reject incorrect roof columns, missing rainfall intervals and the former flat-pipe kinematic-wave setup; additional cases independently check rainfall × area and intensity × area at three roof sizes.

Still required: local rainfall/return period, surveyed levels, header and municipal-network capacity, tailwater, blockage and exceedance design. The model repair does not close those deployment requirements.

2. Electrical measurements verified; grid-only acceptance premise withdrawn

Priority: high. Correct the operating premise before prescribing equipment.

The microgrid model applies a 97% grid rectifier efficiency. A 500 kW residual DC-bus load requires 515.46 kW AC before transformer losses. This is not the grid requirement when PV or station storage supplies the load. It also excludes the separate 98% train-charger DC/DC loss. The original model enlarged the transformer as necessary and declared a pass on solver convergence without checking the connection. The initial all-city arithmetic identified 3,976 sites across 262 cities with insufficient declared import for the grid-only case.

Each site now records solved HV import/export, its separate declared limits, and the exceedance in kW. The test includes transformer and converter losses, with only 0.001 kW numerical tolerance. `solver_passed` records convergence; `passed` additionally requires no connection, transformer, voltage or climate findings. The CLI fails a design with unresolved findings, and batch resume cannot treat it as successful. README tables and engineering plots show the design result and finding count separately from solver convergence.

The audit of the earlier reports confirmed 3,976 sites across 262 cities exceed declared connections only in the case with both PV and battery discharge disabled. Every flagged site has stationary storage. The existing coordinated daylight case has zero connection exceedances across all 266 cities. Its stipulated PV/storage power fractions do not establish endurance or timetable acceptance. The earlier combined gate passed only four cities because it required the grid-only case too. That gate is repaired: all 266 solar/storage snapshots pass, and the 3,976 grid-only findings remain separate contingency diagnostics.

The Samawah result originally recorded seven connections in the grid-only case requiring 519.02 kW each at the HV connection against a declared 500 kW. These stations are not grid-only installations: each has 500 kWh storage, 500 kW battery discharge and 300 kWp PV. Their original coordinated-case import was 73.21 kW each. Including the previously omitted 98% charger DC/DC efficiency gives 529.65 kW grid-only and 83.75 kW coordinated import under the same transformer model. Mosul has 26 exceedances in the same grid-only contingency. Regression tests check import, export, converter and transformer losses, nonconvergence, and the distinction between design failure and convergence. All-city provenance and pass consistency are also checked.

Correction: the earlier instruction to resolve every finding through a connection upgrade or charging change was unsupported and is withdrawn. Evaluate solar generation, stored energy, shared charging, replenishment and the actual service duty together. Grid backup covers residual demand under the project architecture; full charger demand from grid alone is not a default requirement. The software now distinguishes snapshot acceptance from contingency findings and explicitly leaves `operating_energy_validated` false. A passing snapshot does not establish replenishment or endurance. No equipment or connection quantities were increased to obtain these results. The common ideal 33 kV bus does not demonstrate feeder routes, upstream network capacity, protection or utility approval. The stipulated storage snapshot does not establish state of charge or endurance.

3. Reconcile depot energy scope across the physical and operating models

Priority: high. Quantity contract repaired; placement and installed cost remain open.

The depot template now selects a tier from the energy-site template. The scenario generator, layout requirements and station BOM consume this common inventory: 5,000 kWp PV and 40,000 kWh storage (80 × 500 kWh modules) for `main-heavy`. The former 600 kWp / 2,000 kWh figures are explicitly retained as baseline allowances, not a competing equipment definition. Conflicting legacy per-depot quantities now fail generation. All 266 generated scenario outputs are unchanged by this repair; stored operating capacities were preserved. This reconciles what the model assumes, not whether that capacity is necessary. Reassess depot and station PV/storage against distributed overnight placement and charging duties before treating the operating inventory as an equipment requirement.

Every city now has a source-hashed depot reconciliation, linked from the catalogue. The Samawah report requires 33,333 m² of PV modules, compared with the 4,000 m² reference canopy. Gross land area and battery compound dimensions still require a controlled layout and supplier/fire-engineering inputs. Existing CAD geometry does not place this complete inventory.

The fixed depot allowance remains $8M, with a $250k station charging interface allowance. The additional 4,400 kWp and 38,000 kWh above the old baseline represent a $5.93M equipment sensitivity at existing repository rates ($700/kW PV and $75/kWh storage). This is neither a supplier quotation nor an amount added to CAPEX: baseline inclusions, conversion equipment, civil works, protection, installation and renewal remain unreconciled. The reports therefore keep physical/cost/stabling acceptance failed.

Remaining amendments: locate the full equipment inventory, obtain itemised installed scope and reconcile allowance inclusions before changing the budget; carry the resulting quantities into procurement and replacement events.

4. Prove that the distributed stabling plan fits the railway

Priority: high. Operating candidate implemented; physical allocation remains open.

The intended policy is to keep healthy trains at powered passenger stations near their first morning trips, allowing coordinated starts across the network and limiting depots to maintenance needs. The review preserves that policy; it does not recommend concentrating the fleet at depots. RFC 0014 describes simultaneous morning starts from those locations. However, the simulator fleet initializer distributes trains round-robin over configured dispatch points, without a stabling-track capacity allocation.

The retained canonical scenario still uses endpoint dispatch. Samawah has 108 trainsets, with fleets of 53, 28 and 27, each assigned to just two dispatch points. The first line consequently initializes 27 and 26 trainsets at its two endpoints. Those queues diagnose an incomplete simulator initialization; they are not the intended overnight allocation. The design includes 17 concurrent main-depot workshop bays; workshop capacity is not evidence of adequate overnight parking. The simulator's abstract awaiting-dispatch queue does not establish that these trains fit on passenger tracks or can reach their morning starting locations without conflicts.

Quantification added: all 266 city reports reproduce initial dispatch allocations and calculate train-body and clearance-inclusive slot lengths. Samawah's 27-set queue requires 1,336.5 m of train bodies, or 1,606.5 m of individual slots with the RFC's 5 m clearance at each end. These are capacity requirements, not located tracks or a proven operating arrangement. No workshop bay or passenger platform is credited as verified overnight space. The legacy depot CAD also clamps requested bays to catalogue maxima and retains a ten-bay main-depot default, while the template minimum is four; it

needs explicit oversize handling and city-specific workshop geometry. The legacy Python planner still emits depot exceptions by default and needs alignment with the current Rust emitter policy.

Operating implementation added: the native simulator now supports selected powered-station holding, 150 kW charging to 95% SoC, closing-time stops and schedule/energy/MA-gated morning departures. All 266 cities have separate runnable allocation candidates. The Samawah replay keeps all 108 trains at 20 stations overnight, reduces the largest queue from 27 to eight and restarts all 34 planned line/station/direction combinations within 60 seconds, with no invariant violations. The candidate uses the declared 97 revenue, eight spare and three cold-reserve trains; none of the 11 reserves enters routine service. Its overnight hold has zero departures; the retained endpoint scenario has four. This is a 01:30–06:00 comparison, not full-day energy acceptance. Canonical endpoint scenarios remain retained; candidates do not claim physical capacity or close package gates. See the implementation and limits.

All catalogue plans now compare allocated train lengths and clearances with the reference passenger-platform envelope. Samawah has 62 positions beyond that envelope (3,689 m of usable slots under the initial allocation). These remain unverified space requirements, pending existing-track surveys and allocation changes; they do not justify additional depots by themselves.

Charging-reachability repair: the initial two-day check left six then ten revenue trains at an unpowered Line 1 halt. The distributed departure gate now requires energy through to the next selected charger with the 20% reserve intact, accounting for direction, terminal reversal and active pad/grid outages. The refreshed two-day replay passes its behavior checks: all 108 trains finish both nights at selected powered stations and all 34 departure directions restart. Maximum queues are 18 then 19 trains, with 63 positions beyond the reference platform envelope on each night. The short replay's eight-train maximum cannot establish recurring capacity. The check uses elapsed-time snapshots and recalculates space from observed allocations. Physical capacity, daytime headway delivery, storage-only outage resilience and failed-train recovery remain open; package gates remain failed.

Station/depot allocation corrected: two revenue trains per selected station provide Samawah's morning launch stock (40 sets); the remaining 68 stay on their own lines: 37 on Line 1, 16 on Line 2 and 15 on Line 3 (57 revenue, eight spare and three cold reserve in total). The previous interpretation treated those 68 as unresolved station overflow. The new allocation conserves all 108 trains and passes the two-train station check, with 68 explicit depot storage positions requiring 4,046 m of usable slots, separately from 17 workshop bays. Lines 2 and 3 use existing powered service terminals. No interline depot connection is required.

All 266 city plans now include explicit station/depot allocations and depot storage requirements. Missing morning directions, where a line has insufficient revenue stock, remain visible in the allocation report. The native station-only replays and seven-train redistribution study remain diagnostic benchmarks; they do not simulate this hybrid allocation. Their capacity failures must not be interpreted as a need to park the entire fleet at stations or reduce it.

Native home-return implementation added: explicit per-fleet overnight homes and depot storage counts now drive initialization, same-line return movements after closing, depot holding and morning dispatch. A connected 8-train fixture passes two continuous cycles with six station and two depot trains, preserved inventory and parked reserves. The CSV separates depot stock from station queues. Capacity/role mismatches and disconnected depot homes fail loading. Line-local storage now enables 182 native city candidates, including Samawah; the remaining 84 report incomplete morning allocations. The hybrid city screen checks full-day home returns and station launch stock. Station launch trains have priority over depot stock, with train IDs resolving ties; empty returns use the existing peak-service spacing and have a distinct event kind, so overnight passenger departures cannot be hidden among return moves. Two-day screens pass for Samawah, Uíge, Quelimane and Edéa. Samawah preserves 40 station and 68 line-local depot trains and restarts all 34 directions on each morning; a raw minimum SoC of 0.19999853 is within the city validator’s 0.001-percentage-point numerical tolerance. Yard geometry, turnbacks and city-scale evening timetables remain unverified.

Fleet-sizing documentation drift: RFC 0014's former 35 km/h commercial-speed proxy, fixed 10% recovery factor and 10% charging margin did not match the current emitter. The emitter uses family-specific cruise/stop parameters, recovery factors up to 1.62 and a 25% charging margin. The RFC is corrected to describe that implementation. Retained city fleet counts require a separate regeneration/reconciliation audit before changing inventory or treating fleet size as consistent with a two-train station capacity.

Remaining additions: detailed station and depot storage layouts with usable track lengths, train lengths, clearance points, charger sharing and access; evening run-in, morning run-out and failed-train recovery schedules; reserve activation; maintenance possessions; and an occupancy-constrained simulation. First assess powered station tracks and the morning service pattern. Cost additional station sidings or secure service locations only where that allocation demonstrates a need; fleet-wide depot parking is not the default.

5. Make cashflow follow supplier terms and the working calendar

Priority: high for delivery planning. Model integration gap.

The project-twin generator publishes COTS terms of 30% order / 60% delivery / 10% acceptance and a 120-day lead time. Its cashflow calculation does not consume purchase orders. It instead allocates budget contracts using 10% mobilisation / 55% progress / 30% completion / 5% retention, with mobilisation 30 days before task start. Thus the timing and amount of supplier deposits need not appear in the published monthly requirement, even though total cash reconciles to CAPEX.

Budget allocation is weighted by task duration rather than derived from each task's priced quantity. The schedule is labelled in working days, while cashflow maps offsets to months using `floor(day / 30) + 1`. The six-day working week in the manufacturing template is not applied by that mapping. Samawah's example has 1,012 programme working days and 38 cashflow buckets; these are not yet an approved calendar-month financing schedule.

Amendments: derive procurement cash events from quantities, supplier terms and order dates; allocate labour and other residual budget separately to avoid double counting; introduce an explicit project start date and work calendars; map supplier lead times and payment dates consistently. Add a test in which an early supplier deposit changes the appropriate month's cash requirement.

6. Turn the battery renewal allowance into a lifecycle calculation

Priority: medium. Unsupported sensitivity relationship.

The finance generator describes a 12-year train-battery renewal cycle, but computes rolling-stock maintenance as 4% of fleet acquisition cost. The 12-year value is output metadata; it does not drive a replacement schedule, reserve calculation or discounted cash event. Stationary renewals are similarly described as included in a general fixed-asset maintenance percentage.

This is a transparent planning allowance, not demonstrated lifetime cost. Changing battery life, utilisation, replacement price or end-of-life capacity does not currently produce the corresponding renewal cashflow. A cheaper factory-gate vehicle also reduces the percentage-based maintenance allowance without proving a proportional reduction in physical maintenance needs.

Additions: separate onboard and stationary battery inventories; calendar and throughput life assumptions with sources; replacement cost, labour and disposal; capacity degradation in the service-energy case; and explicit renewal/reserve cashflows. Reconcile the resulting totals with ordinary maintenance to prevent double counting.

7. Distinguish catalogue completeness from pilot evidence completeness

Priority: medium. Publication status repaired; evidence coverage remains open.

Before the repair refresh, 264 city manifests retained a finance-summary hash that no longer matched their source. Those same 264 cities lacked the field-evidence brief required by the current package generator, while their old manifests still reported a passed package. The general repository health check applies the full package-manifest checks only to Samawah and Mosul.

The repair regenerated all twin/operations outputs and their enclosing manifests. That first refresh produced 2 complete packages and 264 incomplete packages. After the electrical correction, the two pilots also have failed energy summaries: the current full-package result is 0 complete and 266 incomplete. Missing evidence remains listed in each manifest; the catalogue now displays this distinction alongside its separate topology results. Completeness does not validate the modelling assumptions identified elsewhere in this review.

The follow-up refresh also found 264 GIS summaries with stale scenario hashes. Those GIS packages and all 266 engineering plot sets were regenerated. It then found 264 native-simulation reports referencing both older scenarios and an older validator. These retained reports require full resilience reruns; their old passing flags are not current operating evidence. The two pilots retain current native-simulation provenance.

The package generator now checks analysis provenance and records each stale source separately. Strict README generation still rejects stale evidence. An explicit `--allow-stale-evidence` audit mode can publish a warning and mark retained passing rows unverified while preserving failed results. That mode was used for the 264 affected catalogue pages; it does not alter solver reports or close package gates.

Amendments: define explicit catalogue-screening and pilot-evidence scopes; validate the current scope and generator for every city; extend the catalogue with resilience/cross-check coverage. The catalogue now exposes each city's electrical result, number of sites exceeding connection limits, and full-package status with links to the detailed findings. The current technical inspection count must not imply equal evidence maturity for all 266 models.

An additional generation comparison found existing energy-site drift in Gulu, Rahim Yar Khan and Tanta: their retained charging/grid settings differed from fresh generation. The 2026-09-10 follow-up traced this to the scenario generator ignoring their existing `costs.technology_basis.station_charging_cabinet_count = 2`. Charging validation, station power and site equipment now use that declared quantity, and regression checks preserve the retained station/site tables for all

three cities. Their broader scenario/evidence refresh remains open; preserving equipment quantities does not establish operating acceptance.

8. Extend station analysis from repeatable routes to actual passenger loads

Priority: medium. Acknowledged gap with a concrete next model.

The passenger screen uses 80 people for normal/degraded cases and 120 for egress, with each platform route simulated independently. The elevated/interchange cases reuse the same straight-route result where platform length matches. Stairs, lifts, gates, merging flows and passengers arriving from trains are excluded.

The rolling-stock profiles span nominal train capacities up to 720 and crush capacities up to 960. A 120-person route screen cannot establish evacuation performance for a loaded train plus waiting passengers. The current report discloses this limitation, but the next useful addition is a coupled demand/train/station case rather than another identical route run.

Additions: station-specific arrival and interchange demand; simultaneous train unloading; bottlenecks and platform occupancy; degraded routes and assisted evacuation; and deployment-specific acceptance criteria. Keep route benchmark results separate from station-capacity or evacuation acceptance.

9. Schedule the shared national factory across city programmes

Priority: medium. Portfolio delivery assumption not yet enforced.

The capital template costs one shared national plant sized to the largest single-city fleet, assuming phased reuse. Each city twin independently receives the default rolling-stock plant and fabrication resource capacities. There is no country-level resource schedule coordinating concurrent city programmes.

Additions: country programme start windows, shared plant/mould/tooling capacity, supplier bottlenecks, training ramp-up and maintenance downtime; allocate city orders against that resource pool; compare phased delivery with expanded-plant scenarios. Until then, the national factory cost assumption and individual city schedules should be presented as separate planning cases.

Work order for the next iteration

1. Validate the full solar/storage/top-up operating duty before changing equipment quantities. The grid-only gate, grid-dependent stabling checks and historical-experiment package gates identified in the ethos audit are repaired.
2. Close depot equipment placement, installed cost and physical stabling allocation against the repaired quantity contract and new per-city requirements.
3. Connect procurement, calendars, renewal events and shared factory resources to the delivery and financial model.
4. Extend passenger/station and junction-conflict cases using controlled demand and operating inputs.

Supplier configuration, production drawings, mass/CG measurement, physical testing, site survey, HIL and independent approvals remain the external gates already identified in the roadmap. This review adds specific repository-side work; it does not replace or satisfy those external gates.

Review method and limits

The review traced source templates, generators, emitted scenarios and tracked reports; checked all-city grid-limit arithmetic and manifest coverage; and replayed the drainage deck with the installed solver using temporary files. The EPA manual was consulted for the SWMM input definition. It did not include supplier quotations, field surveys, a complete structural reassessment, a physical test campaign or an independent railway safety assessment. The original diagnostic replays were followed by the controlled generator repairs, regression tests and regenerated evidence described above. Full deployment packages remain incomplete where physical or operating evidence is open. Repository health now checks truthful package status and current selected evidence; it does not require physical deployment gates to be closed merely to maintain a coherent planning repository.

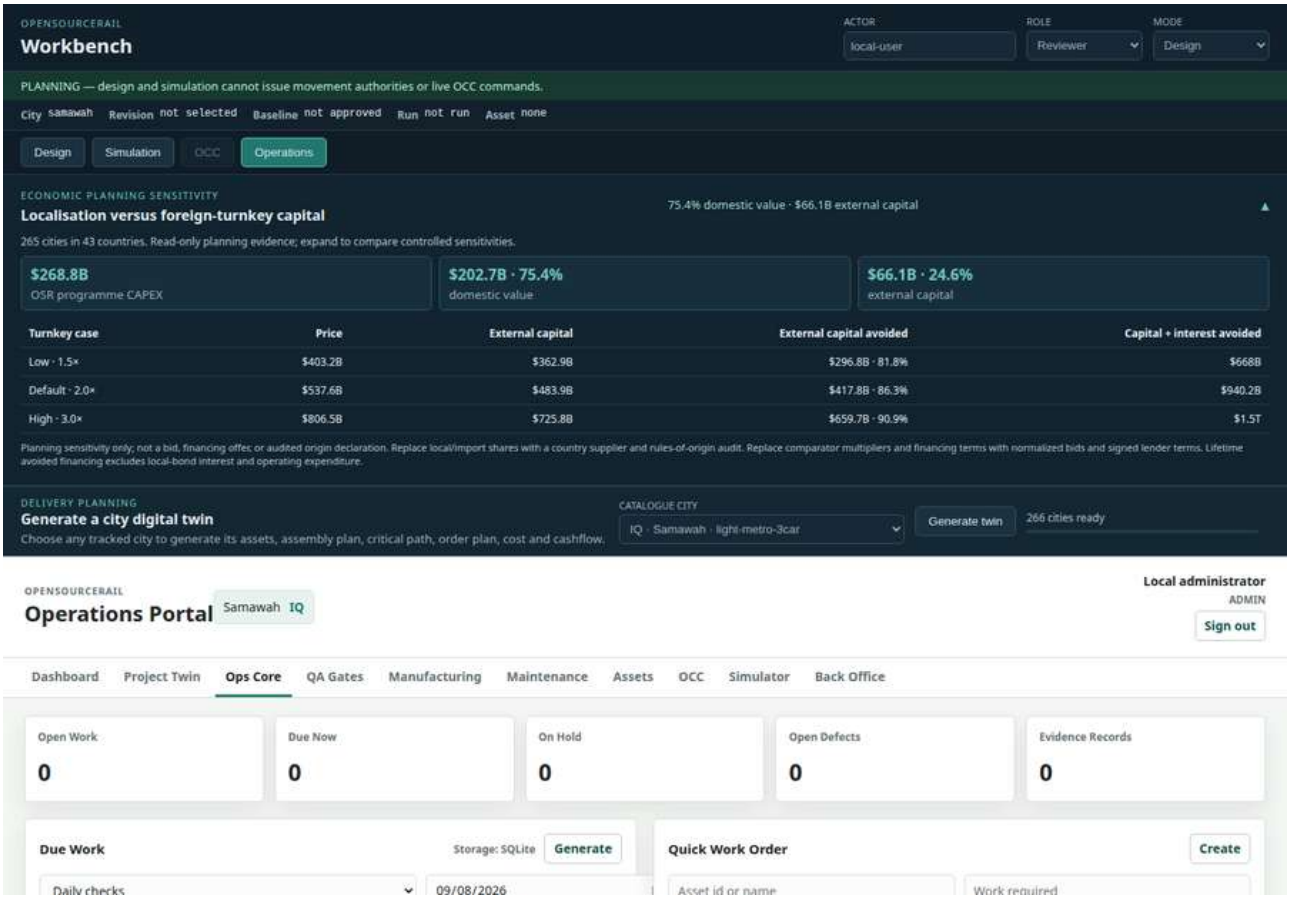
OSR Workbench

Repository source: `docs/workbench/README.md`

The Workbench presents City Studio, simulation, OCC replay, ERPNext/Frappe HR, FUXA supervision and OSR railway assurance under one origin while preserving their authority boundaries. Shared URL context links the city, immutable revision, approved baseline, simulation run and selected asset.

The collapsible Economic planning sensitivity panel reads the generated `portfolio-summary.json` through a read-only Workbench endpoint. It shows the current OSR domestic/external capital split and low/default/high foreign-turnkey comparisons without creating or approving a financing record.

The Generate a city digital twin bar lists all 266 tracked catalogue cities. One button creates that city's asset register, family-scoped assembly plan, finite-resource critical path, budget work packages, supplier/order-by plan, monthly local/imported cash requirements, QA and maintenance records, then opens the result in Operations. Generation is an allowlisted background job: the browser cannot supply a command or filesystem path. Output under `build/workbench/project-twins/` is disposable; reviewed compact baselines stay with each city under `engineering/project-twin/summary.json`.



Workbench any-city digital-twin generator

Install and launch it through the root one-command setup.

Open http://127.0.0.1:8090/. Workbench binds only to localhost; its trusted development identity is unsuitable for a shared network endpoint. Use the authenticated standalone Ops Core server for shared operations records. Planning and training modes cannot emit live OCC commands; live mode does not expose design or simulation modules. Actor and role fields provide navigation context, not authentication. The versioned contract is `context-contract.schema.json`. Simulator and OCC replays expose deterministic onboard, infrastructure, and depot-data evidence for the same run; this is software-in-loop, not hardware.

Operating platform

Decision readiness opens the subsystem qualification view. It recompiles current engineering and measurement inputs, separates six decision areas, and displays quantitative uncertainty, physical-test gaps, production-equivalence findings and changes since review. The reference cooling package is explicitly separate from city acceptance. The qualification workflow explains the controlled rig, model correlation and independent-review inputs.

City execution opens the business operating platform inside the shell. ERPNext and Frappe HR own business records and authentication. Railway works retains the OSR engineering baseline, inspections, NCRs and handback evidence. Set `OSR_ERP_URL` before starting Workbench to use another ERPNext installation.

The city operating twin joins generic and city-specific ERP configurations to each engineering baseline. The local `/api/operating/twins` endpoint reads private ERP feedback snapshots; the operating page shows city, revision, timestamp, task categories, actual task costs and asset workload. The Project Twin panel uses an exact city/revision match and requires a unique operating baseline. Run `./osr erp feedback start` for five-minute refreshes.

Integrated lifecycle workspace

The Workbench now embeds Connected assets, City execution, the native ERPNext/Frappe HR screens and FUXA supervision alongside City Studio, simulation, OCC and railway works. The default lifecycle overview links plan, design, procurement, manufacture, construction, commissioning, operation, maintenance and renewal. The generic module registry supplies navigation for every city. Existing generic ERP/supervision templates and each city's `operations/` configuration continue to own deployment-specific settings.

The City selector carries city, environment and asset identity through the shell. Changing city clears revision, baseline, run and asset context. Changing the revision clears its preceding baseline and run; changing the baseline clears the preceding run. Embedded tools remove those invalidated references when the shell sends its current context. Without a baseline, live mode returns to training. These context references do not independently authenticate an engineering approval. Railway control and design views are available only for the workspace passed to `workbench-server.py --project`; selecting another city does not rebind its controller. Any catalogue city can use delivery generation and its own operating bundle. Missing ERP, engineering or supervision packages are shown as unavailable. ERP project-linked lists use city filters where supported; City execution opens the exact linked ERP project and its OpenSourceRail Operating components actions. Native FUXA offers its own city/display selector.

Business and maintenance-case links open inside the shell. ERP Project and Issue Connected lifecycle buttons return to it. A linked condition Issue also offers Prepare repair: a native ERP review of Asset, parts/availability, technician and expected downtime followed by a draft Asset Repair. Its completion remains separate from the Workbench's railway inspection and handback evidence. Direct tool links remain available for a separate window. URL context includes an optional `environment` for simulation/physical asset inspection, independent of railway control mode.

Authentication and supervisory actions

ERPNext and FUXA keep their native accounts, sessions and permissions. This is a shared interface, not single sign-on. The ERP image permits framing only by itself and the named local Workbench origins in configure-nginx.py. Websocket origin checks remain enabled. For another deployment, configure `OSR_ERP_URL` and `OSR_FUXA_URL` and explicitly allow its Workbench origin at the service; cross-site cookie policies may require a same-site HTTPS deployment.

Connected assets exposes alarm acknowledgement and commands declared by the asset package. Enter an independently provisioned, scoped integration credential in Operator actions; it stays in page memory and is cleared when leaving. The gateway enforces identity, city, environment, role, command bounds, lifetime and controller ownership. The current pilot permits simulated station lighting only. Requested, accepted, completed, rejected and failed states remain visible in command history. A transport error requires checking that history before retrying. Workbench view roles do not grant these permissions.

FreeCAD, Bonsai and QGIS remain desktop editors. The overview exposes versioned, hash-checked engineering downloads; their selection bridge now opens the selected asset inside the Workbench. Browser execution of those desktop applications and physical control deployment are not provided here.

Verification

Browser tests cover city isolation, embedded links, stale telemetry, command credential handling and the existing design simulation `OCC works` workflow. `test_workbench_actions.py` verifies explicit credentials, same-origin mutation requests, allowlisted endpoints and preservation of gateway denials. The installed-stack check verify-native.mjs signs in through native ERP/FUXA forms, follows an ERP project back to its asset, and checks a simulation lighting request through controller completion. It requires the local pilot services and private credentials and intentionally changes simulated station lighting to 60% and restores it to 75%; it performs no physical command. The `integrated-stack` CI workflow now creates those services from a clean checkout, runs this check with the ERP transaction, embedded-condition and FUXA checks, verifies outage recovery, and destroys the disposable data volumes.

City-aware navigation and deployment inventory

The overview now includes searchable connected equipment, observation timestamps and an explicit inventory refresh. It distinguishes local profile files, prepared engineering/supervision packages, ERP feedback and actual telemetry. Prepared packages do not establish imported FUXA configuration or physical connectivity.

Native Projects & automations opens the uniquely linked city project. Tasks, procurement, receipts, manufacturing, stock, cases and invoices use native Project filters. These filters apply to the document's project field; allocations made only on child lines remain visible in City execution actuals. Quality, maintenance and HR retain their organisation-wide native scope. Missing or ambiguous feedback prevents a city-filtered workflow from silently opening an unrelated project's records. Native permissions still apply to every record.

FUXA navigation requests the generated view for the chosen city, environment and site, preferring the selected asset's site. Verify its displayed city/site name: FUXA may fall back to its default if the prepared view has not been imported. Before import, `preview-fuxa` lists the exact package checksums and all device/view additions, replacements and removals; `import-fuxa` rejects a stale review. The installed-stack check verifies actual Samawah/Mosul display switching, ERP project filtering and a simulation analysis evidence record in addition to its supervisory command check.

Independent example deployments

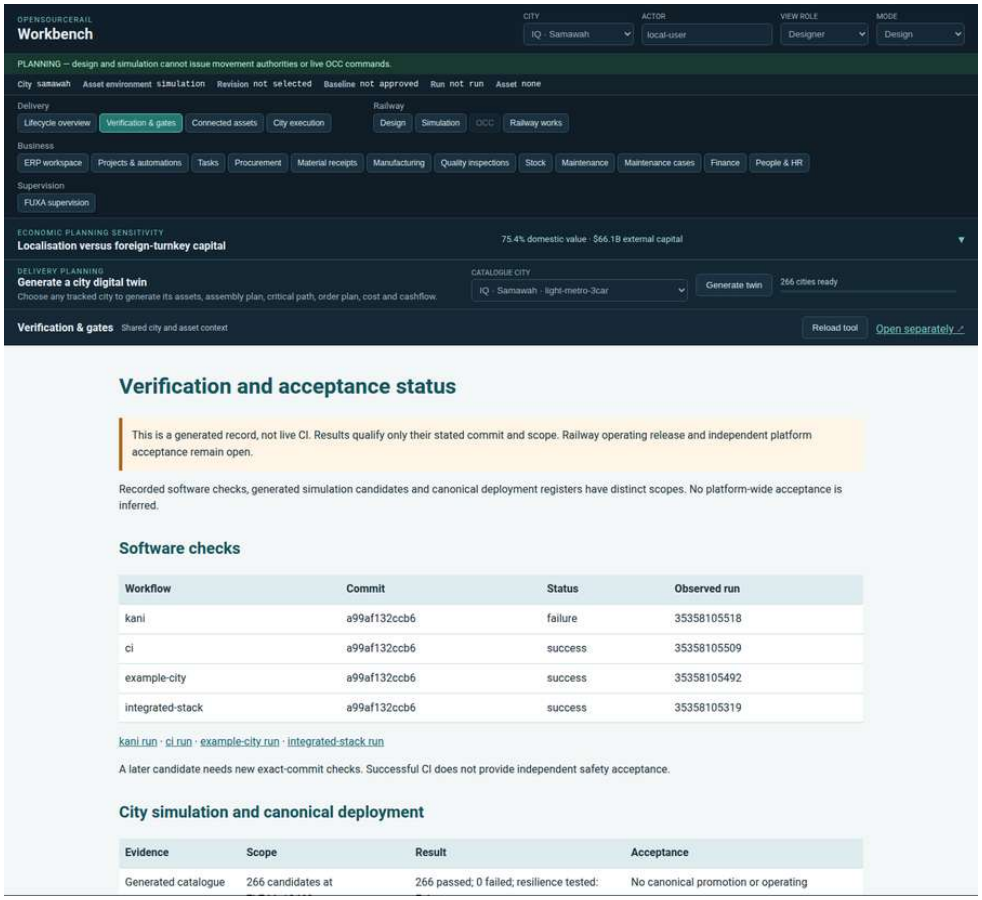
The Samawah acceptance setup runs its own ERP, FUXA, gateway, controller and Workbench without sharing operating records. Workbench accepts these operator-defined environment variables:

Variable	Purpose
`OSR_ERP_URL`, `OSR_FUXA_URL`	Native application origins
`OSR_INTEGRATION_URL`	Gateway origin for scoped reads and explicit operator requests
`OSR_SUPERVISION_CONFIG`	Private integration credential file
`OSR_SUPERVISION_ROOT`	Prepared city packages and engineering manifests
`OSR_SUPERVISION_PROFILES`	Directory of per-city JSON project/company/preferred-site profiles
`OSR_ERP_SNAPSHOT`	Private ERP operating feedback file

Defaults continue to use the regular local installation. ERP's site configuration `osr_workbench_origins` controls trusted parent navigation; the frontend's `frame-ancestors` policy must allow the same origins. The example setup configures both for port 8190. The simulator accepts matching `--config`, `--controls` and `--url` arguments, and follows the accepted city's `historian.sampling_seconds`.

Verification and gates

The Verification & gates module shows recorded CI commits and scopes, generated city candidates, canonical deployment gates, integration coverage and analysis maturity in one view. It is a generated record, not a live CI monitor. Update `docs/operating/status/evidence.json` from actual workflow observations, then run `tools/automation/osr-python tools/automation/platform-status.py`. The generator reads the current coverage, city deployment and analysis registers; every workflow keeps its original commit and link. Later commits require their own checks.



Workbench verification status with exact CI commits and separate canonical deployment gates

Service And Field Work

Open Service & field work for possession-aware planning, fleet/staff recovery, offline work packs, reliability analysis and operational/GTFS exchange. Use an explicit city-scoped integration credential; the shell view role grants no access. Device evidence survives offline reloads, while gateway/native ERP checks reject stale revisions, serials and assignments on reconnect. See the implementation, pilot and commissioning scope.

Software, Deployment and Developer Tools

Editable City Workspaces

Repository source: `cities/workspaces/README.md`

This directory contains the small, Git-reviewable source packages used by OSR City Studio. Generated catalogue outputs remain under cities/catalogue/.

Every project pins its input design/GIS/scenario bytes, records manual network intent separately from generator output, defines weekly service plans, and can materialize content-addressed revision records for pull-request review. Manual lines, stations, and alignment controls live in `network/overrides.toml`; adding or retiring one therefore remains an ordinary, reviewable Git diff. New-line day-type plans are stored beside the other services in `services/service-plan.toml`.

Samawah also commits a compact planning raster under `routing/`. Cost, demand, buildability, water coverage, open-DEM elevation, derived terrain slope, anchors, and derivation provenance are independently SHA-256 locked so demand-aware routes and civil screens are reproducible in a fresh clone.

Start with the Samawah project and the City Studio guide.

Samawah Design Revisions

Repository source: `cities/workspaces/samawah/revisions/README.md`

OSR City Studio writes deterministic candidate revision JSON files here only when a user selects Materialize revision. Autosaves and map movements do not create Git history by themselves.

Review a candidate in a branch, commit the project inputs and revision file, then submit a pull request. After approval, tag the merge using the prefix `city/samawah/design/` declared in `project.osr.toml`.

The regenerated core alignment uses new station positions and IDs. The older revision JSON and reviews remain historical records; they do not approve the current candidate. Previous platform locks are retained separately in `../network/historical-pre-core-overrides.toml` and are inactive.

The current source lock adopts the 5 October local civil-cost correction on the retained alignment. Earlier revision source hashes and approval records are unchanged; they do not approve the corrected candidate. Compile and validate the current project before reviewing any new revision.

Samawah City Studio routing bundle

Repository source: `cities/workspaces/samawah/routing/README.md`

This directory contains the source-locked planning surface used when City Studio creates a demand-aware manual line. The raster files are raw little-endian arrays despite their `.npy` suffix, matching `osr-routing`.

The bundle retains the pipeline's 20 m Samawah planning resolution (factor 1). This avoids treating a platform on a narrow dry river bank as a station in a predominantly wet 100 m aggregate cell. Aggregation preserves any buildable cell, water coverage percentage, the minimum buildable cost and maximum demand. Elevation is averaged and slope is recalculated on the retained grid to avoid amplifying one-pixel DEM noise. The derivation record pins the SHA-256 of every upstream file. Elevation comes from the public NASA/NGA SRTM-derived Skadi HGT tiles in the AWS Open Data Terrain Tiles bucket; it is planning evidence, not a surveyed ground model.

Add or refresh open terrain and independent water evidence before downsampling:

```
python3 tools/automation/augment-routing-terrain.py \
  --osm-json .cache/osr-pipeline/osm/samawah.json \
  --sidecar .cache/osr-pipeline/rasters/samawah.grid.json \
  --terrain-cache .cache/osr-pipeline/terrain
```

Regenerate or verify it from a populated local pipeline cache:

```
python3 tools/automation/downsample-routing-bundle.py \
  .cache/osr-pipeline/rasters/samawah.grid.json \
  cities/workspaces/samawah/routing --slug samawah --factor 1

python3 tools/automation/downsample-routing-bundle.py \
  .cache/osr-pipeline/rasters/samawah.grid.json \
  cities/workspaces/samawah/routing --slug samawah --factor 1 --check
```

This surface supports deterministic comparison of city-scale alternatives. It is not survey data, a final alignment, or evidence of detailed constructability.

Rust Workspace

Repository source: `crates/README.md`

The `crates/` tree is the OpenSourceRail software stack. It contains the simulator, safety evaluators, onboard and wayside services, operator GUIs, design synthesis tools, and evidence generators.

Common Commands

```
cargo test --workspace
cargo run --release --bin osr-sim -- --duration 3600 --status-every 300
cargo run --release --bin osr-design -- --help
```

Main Groups

Area	Crates
Shared foundations	`osr-core`, `osr-proto`, `osr-crypto`, `osr-secbus`, `osr-supervision-contract`, `osr-lifecycle-identity`
Signalling and wayside safety	`osr-consensus`, `osr-interlocking`, `osr-wayside-points`, `osr-level-crossing`, `osr-intrusion-detect`, `osr-hot-axle-wayside`
Reference process hosts	`osr-runtime` (development target; qualification pending)
Onboard safety	`osr-atp`, `osr-onboard-routing`, `osr-odometry`, `osr-brake`, `osr-bms`, `osr-door-control`, `osr-fire-safety`, `osr-derailment`, `osr-obstacle-detect`
Train systems	`osr-tcms`, `osr-ato`, `osr-traction`, `osr-aux-power`, `osr-hvac`, `osr-lighting`, `osr-regen`, `osr-tcn`, `osr-t2g`, `osr-event-recorder`, `osr-cbm-onboard`
Stations and passengers	`osr-afc`, `osr-afc-backoffice`, `osr-pis-onboard`, `osr-pis-station`, `osr-psd`, `osr-station-scada`
Simulation and design	`osr-sim`, `osr-design`, `osr-alignment`, `osr-routing`, `osr-analytics`, `osr-energy-site`, `osr-balise`, `osr-tvm`
Operator tooling	`osr-gui-shared`, `osr-sim-gui`, `osr-occ-gui`, `osr-occ`, `osr-historian`
Evidence and images	`osr-safety-case`, `osr-selftest`, `osr-trainset-image`, `osr-cbm-backend`

The architecture rationale is in `docs/ARCHITECTURE.md`, with the crate-level software map in `docs/rfcs/0005-sbc-software-architecture.md`. The evidence-backed maturity review and executed integration plan are in `docs/rust-codebase-review-and-integration-plan.md`.

osr-city-studio

Repository source: `crates/osr-city-studio/README.md`

Rust compiler, validator, local HTTP service, and embedded browser interface for Git-backed OSR city projects.

The crate deliberately operates above the safety-critical control stack. It creates design and service-planning candidates; it cannot issue operational movement or route-control commands.

Run the Samawah vertical slice from the repository root:

```
cargo run -p osr-city-studio -- serve
```

The interface can create/move/retire manual stations, create complete manual lines from two endpoints using locked demand/buildability surfaces or an explicit direct chord, edit local alignment controls, plan service by line/day/time over an offline GIS canvas with 20 content-hashed layers, including source-locked open-DEM elevation/slope, an independent water and station-exclusion mask, and likely bridge/viaduct screening, pan/zoom, visibility/opacity controls and feature inspection, and adjust every time-window headway for one line or every route in an atomic day-type transaction, copy a complete plan between day types, author period-specific origin–destination flows, screen them against the conservative scheduled capacity bottleneck, and edit or remove those flows, edit project-level civil construction intent, persist accepted per-line IFC survey/map conversions, and inspect derived thermal-unit, deck-gap and bearing quantities, regenerate GIS and simulator inputs, and compare the working candidate with immutable Git revision records. Its engineering hub runs only allowlisted field-evidence, GIS, simulator, alignment-exchange and civil-BIM adapters and persists their progress, logs, exit state, and immutable artifact copies. The field-evidence adapter also exposes control through per-asset structural-release readiness without claiming missing field data or approvals. The civil BIM adapter emits an IFC4.3 rail federation, object index, quantities, provenance, 19 reusable source-recipe component types, three source-backed material families, one native section profile driving 32 rail extrusions, two typed `IfcVehicle/ROLLINGSTOCK` trainsets with standard measured base quantities, a linked construction sequence that separates 134 physical outputs from 45 virtual review interfaces, 36 native typed bridge bearings and explicit virtual foundation/jacking interfaces, 27 native support connections with 60 bearing realizations, one internal classification with 15 references covering all assets, five native coordination groups, six native functional systems (three specialized as `IfcBuiltSystem`), seven system-to-railway-part references, four native presentation layers, nine native interface constraints with six numeric `IfcMetric` benchmarks, three qualitative-only gates, and 107 scoped project/asset/group/system evidence links, 16 native property/quantity dictionaries with 99 typed fields, native horizontal/vertical planning alignment segments and stationing, one native USD schedule of three generated planning rates, 15 hash-locked native source documents including a link to all nine objectives and six metrics, a machine-readable register of nine external engineering decisions, IDS audit, and BCF 3.0 release issues for Bonsai. The integrated evidence viewer re-verifies each hash before plotting GeoJSON, alignment JSON, civil IFC object tessellated solids with object picking, asset-class inheritance, review-group and functional-system membership, interface-constraint evidence, property-template applicability and definition linkage, alignment layout, stationing, and unresolved cant/transition evidence, planning-rate provenance, quantity drivers, and no-estimate boundaries, source-document records and hashes, IDS specifications, BCF topics, LandXML, railML, stakeout CSV, or simulation results. The civil job also produces source-bound native IFC triangle chunks, loaded through the verified artifact endpoint with stable object identities. The civil viewer joins the verified IFC object index with its verified 4D task sequence, supporting view rotation, presentation-layer, coordination-group, and functional-system visibility, construction-stage scrubbing/playback, task/QA-hold context, and stable object picking for BCF.

BCF decisions are saved to project intent and included in Git revision hashes; immutable job evidence is never edited in place. Resolution and closure require recorded evidence and a reviewer before a regenerated BCF can carry that state. New topics can be authored from one or more searched/selected IFC assets; content-derived issue, topic, and viewpoint IDs keep repeated builds and Git review deterministic. Materialized revisions can also receive append-only approval or changes-requested records with a reviewer, role, date, rationale, and review/PR reference. These records live in project TOML but remain outside the design hash, so approving an immutable revision cannot create a circular new revision.

Demand periods and OD flows live in `demand/od-matrix.toml`. Flow IDs are derived from period/origin/destination identity, capacity screens are rebuilt from the corresponding line/day/time service windows, and both intent and metrics participate in deterministic revision hashes and semantic comparison. The compiler assigns shortest scheduled-time routes through connected, colocated interchanges, combines OD flows on directed sections and reports station boarding, alighting and transfer loads. Periods require full service coverage. This is an all-or-nothing planning assignment; observed OD calibration, capacity-constrained boarding and physical pedestrian acceptance remain open.

Civil settings live in `[civil]` in `project.osr.toml`. The API and GUI reject values outside the controlled product family before atomically persisting the standard span, expansion-unit length, reinforced-soil height, slipform/ST6 methods, mould-cycle target and road crossing-comparison rule. Optional `[[civil.ifc_georeferencing]]` records bind a single accepted EPSG/map conversion and evidence source to a stable line ID. They participate in revision hashes and semantic comparison; the civil job passes only the record for its selected line to the IFC exporter.

Run the complete browser and cross-GUI acceptance suite with:

```
npm run test:frontend
```

Tests use disposable City Studio projects and isolated SQLite databases; the committed Samawah intent is never edited.

See the City Studio guide and RFC 0031.

osr-occ-gui — operations-control dispatcher console (RFC 0018)

Repository source: `crates/osr-occ-gui/README.md`

Pure-Rust egui front-end for the OCC. v1 ships as a read-only console that attaches to a pre-recorded sim run for demonstration; v3 (per RFC 0018 §9) wires live consensus + RFC 0017 signed action emission.

Features (v1)

- Live-style playback (speed controls 1× / 10× / 60×) against a pre-recorded sim run.
- Network schematic with per-section intrusion overlays (Present / Unknown colour-coded per RFC 0016).
- Train panel listing every in-service trainset with phase + SoC.
- Embedded telemetry roll-up from the attached deterministic recording: TCMS trip movement holds, CBM service flags, and T2G channel/transmission counts.
- IEEE 1588 acquisition and lock evidence from the same recording.
- Station/wayside roll-up for PSD states, controller ticks, and intrusion verdict transitions.
- Physical HABD passage, warning restriction, trip, active-stop, and inspected-reset evidence.
- Balise registry, accepted odometry fixes, and sighting-audit findings.
- Signed TVM sales, AFC decisions, settlement, and fraud evidence.
- Real OCC roster ingestion, incident lifecycle, and active dispatch-hold evidence from the attached recording.
- Energy-site controller conservation and regenerative-current arbiter evidence from the same deterministic run.
- Track-state wire encode/decode counts and semantic-parity failures.
- Explicit point-machine and applicability-aware level-crossing controller evidence; no crossing is invented for a grade-separated city.
- Five-role software preflight results; any failed known-answer check blocks simulated dispatch. Physical commissioning evidence remains separate.
- Depot roll-up for received CBM payloads, historian samples, analytics, and maintenance work orders.
- Alert feed (info / warn / crit) with category filter checkboxes, pre-seeded with SoC-warning alerts harvested from the recorded run and dispatcher-action entries.
- Dispatcher-action modals with input validation:
  - S2.1 Issue route grant — requires `T<number>` train id + comma-separated numeric section ids.
  - S5.1 Commit MaintenanceOverride — requires numeric section id, non-empty crew id, and expiry 15..=240 min (per RFC 0013 S5.4).
  - Release inspected HABD stop — requires a train, named qualified authority, inspection reference, and explicit examination/line-clear confirmation.
  - Declare degraded mode — selector for Normal / M1 Manual- on-MA / M2 Restricted / M3 Evacuation per RFC 0013 §5.
- Test-mode intrusion injection buttons so a dispatcher can rehearse Present / Clear transitions on SEC1001.

Run (native)

```
cargo run --release -p osr-occ-gui -- --operator "dispatcher-alpha"
```

Click Attach recording in the left sidebar to run an hour of the Samawah scenario and stream it into the UI.

Run (WebAssembly)

Same toolchain as `osr-sim-gui`:

```
rustup target add wasm32-unknown-unknown
trunk --version # must report trunk 0.21.8
cd crates/osr-occ-gui
trunk serve web/index.html --open
```

From the repository root, `npm run test:frontend` builds this release bundle and verifies it in pinned Playwright Chromium.

Security note

v1 is read-only — every dispatcher action emits a stub log line instead of a signed RFC 0017 envelope. A visible "READ-ONLY v1" watermark sits in the bottom-right of the map. v3 wires the operator's signing key from their ATECC608B-backed login and every committed action reaches the consensus cluster as a `SignedBytes` envelope that the interlocking verifies before applying.

osr-runtime

Repository source: `crates/osr-runtime/README.md`

Thin synthetic process hosts for existing consensus, committed interlocking, ATP, ATO and brake evaluators. No second authority or protection calculator. See RFC 0033 and the process-reference package.

`osr-train-agent` and `osr-wayside-agent` use newline-framed JSON local I/O ports; the application messages exchanged through them are signed versioned postcard packets and existing signed consensus proposals. Filesystem writes precede outbound effects. Reference keys are intentionally public fixtures; operational deployment is rejected. Hardware safety outputs and production radio adapters remain explicit qualification work.

Train binaries require `--safety-channel A|B` and separate journal paths. A owns network publication; B evaluates committed inputs without duplicate proposals. The paired-only safety-output port accepts `feed_pair` with optional A/B feeds; missing either channel trips. `DualGuard` requires exact requests from both and latches discrepancy, stale/replayed feeds, emergency and either feedback failure. Only stopped authorised fresh recovery clears a trip. The coordinator feeds at 50 ms against a 100 ms deadline. A late feed latches even without prior sampling.

The output port defaults to an autonomous monotonic-clock sampler, on a separate thread from bounded input parsing and output writes. It samples even when input is incomplete or its consumer stops reading. Output snapshots carry the process clock epoch and monotonic time. Source requests must retain their original issue time in that epoch; stale, replayed, future or previous-epoch requests cannot renew a deadline. EOF and malformed input fail closed. Clients must also reject stale/missing output frames; a stopped watchdog process cannot actuate hardware.

`--clock virtual` is an explicit deterministic replay mode. The TACS twin uses that mode; its virtual timing does not prove real-time scheduling. Native tests exercise idle input, blocked consumption, caller-time injection, source expiry and stopped-only recovery. Rust's Instant and bounded receive timeout provide the process clock and polling primitive; OS scheduling, suspend behaviour and hardware reaction require measured qualification.

Channel-local independent clocks, authenticated source-issued output identities, power/output isolation and physical braking response remain open integration work. See the default policy; other local protection pairs remain open. No availability claim follows from A loss.

osr-sim-gui — simulator GUI (RFC 0018)

Repository source: `crates/osr-sim-gui/README.md`

Pure-Rust egui front-end for `osr-sim`. Loads a scenario, runs the sim once to completion, builds a 1 Hz `SimTimeline`, then animates the run at arbitrary playback speed.

Features (v1)

- Scenario selection via `--scenario <path>` flag (or the Samawah built-in default).
- Play / pause / reset + 0.5× / 1× / 10× / 60× speed controls.
- Timeline scrubber (drag to any second).
- Network schematic with trains coloured by phase (traveling / dwelling / charging / SoC-warning).
- Click-to-inspect sidebar with line, phase, `station_m`, SoC, and last-event details.
- Event log with per-kind filter checkboxes (dispatched, arrive, depart, charging, turnaround, SoC warning).
- Fault-active badges for every `[[faults]]` entry that's firing at the current playback time.
- Embedded evidence for TCMS trip movement holds, event recording, CBM, hot-axle, and T2G primary/backup/offline operation.
- IEEE 1588 acquisition and lock evidence for the shared simulation clock.
- Station and wayside evidence for PSD, PIS, SCADA, and intrusion detection.
- Physical HABD passage, warning restriction, trip, latched-stop, and inspected-reset evidence.
- Balise registry, accepted odometry fixes, and sighting-audit findings.
- Signed TVM sales, AFC decisions, settlement, and fraud evidence.
- Depot evidence for radio-delivered CBM, history, analytics, and work orders.

Run (native)

```
cargo run --release -p osr-sim-gui
# or with a specific scenario
cargo run --release -p osr-sim-gui -- --scenario cities/catalogue/west-asia/Iraq/Samawah/samawah.toml
```

Run (WebAssembly)

The same code ships as a WASM app. Build + serve with trunk:

```
rustup target add wasm32-unknown-unknown
trunk --version # must report trunk 0.21.8
cd crates/osr-sim-gui
trunk serve web/index.html --open
```

The HTML file at `web/index.html` pulls the crate's library target, builds it to WASM with `wasm-bindgen`, loads it into the `<canvas id="osr_sim_canvas">`, and the same `SimApp` runs inside a browser tab.

From the repository root, `npm run test:frontend` builds this release bundle and verifies it in pinned Playwright Chromium.

Host Deployment Baseline

Repository source: `deployment/README.md`

This directory is the reproducible software-composition baseline for the five hardware classes described by RFC 0005. It answers which workspace packages belong on each host, their startup order, commissioning check, log policy, and update/rollback contract.

- `hosts.toml` is the source of truth for host images.
- `components.toml` gives every Cargo workspace package a disposition.
- `config/*.toml` contains the non-secret configuration shipped in an image.
- `tools/automation/validate-host-manifests.py` validates the model against Cargo metadata and rejects incomplete or legacy-contaminated default images.

The current packages are linked evaluator/control libraries, not independently supervised daemons. `osr-selftest` is the executable commissioning boundary. An image builder must statically link the listed packages into its OS-specific tasks and retain the manifest and Git revision beside the executable. Secrets, device identities, trust anchors, calibration values, and site addresses are provisioned after imaging and must never be committed here.

The manifests are an integration and release control. They are not evidence that an image has passed a hardware-in-the-loop, EMC, environmental, or safety assessment.

Validation

```
python3 tools/automation/validate-host-manifests.py
cargo test -p osr-trainset-image --all-targets
```

The update contract is deliberately fail-safe: stage and verify an image, activate it once, require the role self-test and external watchdog heartbeat, and automatically return to the last-known-good slot if either fails.

Business operating services

ERPNext deployment runs on an ordinary business server, separate from the five railway control host classes. It provides the operating platform without adding ERP dependencies to onboard or wayside images.

Equipment supervision

ERPNext + FUXA lifecycle deployment provides generated city equipment packages, a durable telemetry/event gateway and a simulated station pilot. Its prepared-package review traces design and embedded contract changes through installed/evidence/business identities before a baseline can be replaced. The operating-readiness audit compiles all 266 tracked city asset registers through the supervision package and all 266 ERP and component profiles, while retaining operator and physical commissioning inputs as explicit fail-closed states.

The `integrated-stack` CI workflow builds a clean, disposable Samawah/Mosul deployment from these checked-in inputs. It exercises native ERP transactions, the gateway, embedded simulator, pinned FUXA application and Workbench together, including a gateway outage and recovery, then removes the evaluation volumes. This is software integration evidence; it does not replace commissioning or hardware-in-the-loop acceptance.

ERPNext operating deployment

Repository source: `deployment/erpnext/README.md`

This composition runs ERPNext, Frappe HR, the OSR reference integration, MariaDB, Redis, workers, scheduler, websocket server and nginx frontend.

Use the root `./osr erp` launcher. Setup, ownership, import and backup procedures are in the operating platform guide. Secrets and local configuration live under `var/erpnext/`, outside this tree.

MariaDB stores its data in the persistent `db-data` volume and temporary InnoDB files in a separate bounded 512 MiB `/tmp` filesystem (mode 1777). A healthy container and a successful `/api/method/ping` response are separate checks from the city planning validators. If MariaDB reports missing temporary files, retain the data volume and restore temporary storage; do not reinitialise the database.

The Baghdad detailed engineering register lists reference parts and ERP handover requirements. These rows are not live Items, submitted BOMs or approved Work Orders; import only a released, priced family MBOM with real company, warehouse, operator and supplier inputs. The generated catalogue readiness report validates all city profiles and component packages and identifies which full task payloads still need on-demand materialisation before import.

The local Workbench embeds native ERP screens with native authentication. The proxy allows framing only by the explicitly listed Workbench origins; see the Workbench integration guide.

A condition-created Issue with a reviewed ERP Asset exposes OpenSourceRail Prepare repair. Preview validates the Issue/project/Asset identity, technician, city warehouse, stock Item, optional serial/batch bundle and current availability. Apply creates one unsubmitted native Asset Repair and ToDo assignment; it never submits stock consumption, closes the Issue or grants railway handback. Repeating the same reviewed request returns the existing record and preserves operator edits. The native repair form remains responsible for actions performed, completion, actual downtime, parts consumption and accounting. Its OSR provenance fields link back to condition history and the existing lifecycle/serial evidence.

The private city feedback export also carries engineering revision exposure: explicit nested BOMs, draft/unfinished production, potential purchases and linked stock movements. It uses native document permissions and preserves coverage warnings. Workbench displays and downloads these observations. The native Project's OpenSourceRail Revision dispositions workflow adds assigned, immutable proposals and independent endorsement/rejection records. Stale or incomplete exposure cannot receive an endorsement. Recording never executes the proposed ERP action or grants railway release. Independent native-outcome verification covers retention, cancellation, direct amendments, production stops, corrective Job Cards with accepted inspections, performed stock inspections and native movement traces. Evidence remains immutable and later changes make it stale; see the verification workflow and upgrade limits.

After a backup, rebuild/restart the app, run `./osr erp bench migrate` to install the three disposition record types, then `./osr erp snapshot` to refresh feedback. Manufacturing Manager, Projects Manager and System Manager roles can use the workflow subject to project/exposure access; reviewer, proposer and responsible person must satisfy the independent-review checks. Inspection and trace readers also need native Quality Manager and Stock User access, respectively.

The complete example-city test installs a separate site and checks actual planning, purchasing, manufacturing, serial movements, maintenance and Workbench/FUXA behavior. Native regression scripts accept `OSR_TEST_SITE` (default `osr.localhost`) so the same checks can run in that site.

Condition routing maps `low`, `medium` and `high` alarm priority to native Issue priority when creating a case. Later events preserve operator triage changes. Rules without a priority use `medium`; display-only rules do not create cases on activation or clearance unless an earlier enabled rule already routed the same incident.

The Baghdad qualification package provides ten source-bound evidence Tasks, including the independently operated first-section option. Preview with `bench --site SITE execute osr_erpnext.qualification.import_tasks --kwargs '{"package_path":"/path/to/erpnext-tasks.json"}'`; set `"apply": True` for a reviewed local import; this installed Bench CLI parses Python literals in `--kwargs`. The importer creates open native Tasks and refreshes descriptions by stable subject. It preserves actual assignment, project, dates, priority and status when repeating an import; it never completes an evidence task or grants acceptance. Assign a named accountable person through ERP; role labels are not user assignments. Physical result verification uses independently supplied reviewer keys and signed source/criteria-bound results. Generated templates contain no measurements, quotations, signatures or operating release. Local task mappings belong in `var/erpnext/`, rather than the public proposal archive.

The Baghdad workforce pilot maps workload, recruitment and practical assessment to native HR records. Run `bench --site SITE execute osr_erpnext.workforce.administration_readiness` to inspect local schemas and the caller's read permissions. This does not prove that anyone has been appointed or authorised. The bench-only `osr_erpnext.workforce.preview_assignment` accepts `project`, `task`, `employee` and `authorisation_file`; the

private File must belong to that active Employee and its controlled JSON packet must match the Project baseline and Task. HR Manager or System Manager plus native read permissions are required. This validates the attached snapshot: `snapshot_eligible` describes those assertions, while `eligible` and `live_start_check` remain false. Timestamp rechecks detect expiry; they do not resolve subsequent revocation of authoritative records. Before task start, the human work-release process must resolve current permits, competence, availability, calibration and stock release. The preview creates no assignment, permit, competence record or railway release. Keep native worker records and private attachments out of public exports. The six delivery evidence drafts use the existing Task importer and preserve actual owners/status.

Local administrative and recovery probes

`bench --site SITE execute osr_erpnext.delivery_admin.read_probe` performs bounded permission-aware native schema/record reads and reports elapsed times; it does not prove city-scale throughput or staffing. `osr_erpnext.delivery_admin.recovery_fingerprint` hashes the evidence Task, DocType, role membership, permission and File inventories for comparison with an isolated SQL restore. Both require System Manager, are bench-only and create no assignments or work-release authority. Rebuild the local image to retain these modules across container recreation.

Use the isolated restore tool with a trusted private SQL backup and matching private native fingerprint. Its temporary database has no network or published ports and is deleted after verification. SQL inventory checks complement the existing full platform recovery procedure; they do not prove encrypted file recovery, production RTO/RPO or field acceptance. Keep backups and receipts in ignored `var/erpnext/`.

Reproducible example-city acceptance

Repository source: `deployment/example-city/README.md`

This scenario builds a separate Samawah simulation deployment, with Mosul as an independent control city. It runs actual ERPNext, Frappe HR, FUXA, the integration gateway, the native Rust controller bridge and the shared Workbench. Business records are retained for inspection after a successful run. Samawah uses individual asset polling; Mosul uses site polling, with native rendered power checked against the gateway reading. Both retain individual equipment and maintenance identities.

Run

Use the repository's pinned Docker, Rust, Node/Playwright and Trunk prerequisites in the ERP deployment guide and Workbench guide. Docker must be running. The first ERP build and city imports can take tens of minutes.

```
npm ci
npm run frontend:build
cargo build -p osr-sim --example operating_bridge
PLAYWRIGHT_BROWSERS_PATH="$HOME/.local/share/opensource-rail/toolchains/playwright" npx playwright install chromium
./osr example-city setup
./osr example-city run
./osr example-city restart-check
./osr example-city expand-check
./osr example-city business-check
./osr example-city disposition-check
./osr example-city coverage
```

Open http://127.0.0.1:8190 after success. The report, native record identifiers, source hashes, full-network contract observations, before/after settings observations, browser screenshots and phase logs are under `build/city-example/`. Open `report.html` for the readable summary. The final `report.json` must contain `"passed": true`; an incomplete or failed run is never reported as a pass.

Service	Example port
Workbench	8190
City Studio	8191
ERPNext / Frappe HR	8180
Integration gateway	8192
FUXA	1981

Private credentials are generated under `var/city-example/` with restricted file permissions. ERP uses `Administrator` and the generated `ADMIN_PASSWORD` in `local.env`; FUXA uses `operator` and `operator_password` in `fuxa.json`. Integration roles are in `integration.json`. These files are not test artifacts.

```
./osr example-city stop    # stop only example services; retain records
./osr example-city start   # reopen the retained successful example
./osr example-city reset   # remove only example services, volumes and local outputs
./osr example-city setup
./osr example-city run     # repeat from a clean city deployment
```

Save reports/screenshots before reset. A run refuses to reuse an existing run marker: this avoids treating records left by a previous attempt as fresh evidence. Setup is repeat-safe and reapplies the source configuration. The regular deployment on ports 8080/8090/8092/1881 has different Compose projects, volumes and credentials.

CI also captures and rehearses a coordinated cold recovery after the native scenarios. Only public result summaries are uploaded; checkpoints, credentials and restored private files remain under `var/`. The rehearsal keeps controllers and background ERP jobs stopped in its isolated clone.

Connected lifecycle

1. Import the full generic-plus-city planning baselines for Samawah and Mosul. Sweep every configured Samawah equipment position and measurement through the native evaluator/declared fixtures and production historian; save a separate full-network contract report, lower/upper boundary checks for every measurement and a clock-controlled retention change using the production historian.
2. Reopen and hash tracked FreeCAD, IFC, GIS and analysis artifacts. Bind the exact engineering revision to a native Item, production BOM and equipment Asset.
3. Purchase ten cooling modules, receive six, invoice the receipt and observe four outstanding. Manufacture two serialized chargers against a three-unit Work Order, consuming two cooling modules per charger; record an accepted inspection.
4. Transfer a produced serial into the native installed-equipment warehouse, install it at the modeled charger position and record separate design, execution, inspection and simulation commissioning evidence.

5 Inject a controller cooling fault. Observe the alarm, scoped acknowledgement, durable event and native ERP Issue. Complete an Asset Repair consuming a spare.

6 Close the ERP Issue, clear the fault and record maintenance. Replace the serial, transfer the removed unit back to workshop stock, retain its history, reject reuse of the old commissioning test and require fresh independent simulation evidence. Reconcile remaining raw stock and both finished serials against the receipt, production and repair quantities.

7 Inspect these same records in Workbench and native ERP/FUXA without browser request mocks. Send a lighting request and observe controller completion and measured feedback. Seal a City Studio simulation baseline, run the native simulator, hand the run to OCC replay and persist a linked OSR railway work record through the same Workbench.

8 Stop ERP during another fault, recover its queued event without duplication, then restart the gateway and check retained serial history and resumed telemetry. Run `restart-check` to stop/start all example services and verify serial history, OSR work, the retained City Studio configuration and newly received telemetry.

Findings addressed

The expanded checks led to five integration changes: accepted sampling intervals now control native simulator publication; clearing a maintenance-disabled alarm does not create an ERP case; alarm priority and response reach native ERP triage without overwriting later operator decisions; and separate Workbench deployments can select their own data, gateway and trusted ERP parent origins. Generated FUXA asset links also use the configured Workbench origin, with that choice bound into the reviewed project hash.

Expanded function and variable coverage

After a successful base run, `./osr example-city expand-check` can be repeated against the retained example. It pauses the example simulator, changes reviewed Samawah configurations, checks actual HTTP effects and restores the baseline and fresh telemetry in a `finally` block. It runs native ERP variations inside a rolled-back transaction and independently compares record counts, the original invoice balance and Item inspection settings before/after. Run this in the isolated example, without another operator editing its configuration during the check.

The recorded expansion passed 389 assertions: 342 live HTTP observations, three explicit HTTP controller-result fixtures, 43 native ERP/rollback observations and one schema-suite summary. The separate generated schema suite passed 230 checks. Schema validation and controller-result fixtures are not physical or native-controller acceptance.

Added variations include:

- Exact minimum, maximum and outside-range readings for all 56 selected measurement contracts; 72 combinations of three positive scales, two offsets, three raw values and four source qualities. Zero/negative scales are rejected.
- Null measurements, malformed sequences, timestamp regression, duplicate concurrent ingestion, five unauthorized roles and competing configuration reviews.
- Pending-command configuration blockers, illegal result transitions, terminal retries, missing permissives and expiry while the source is paused.
- Invalid quality interrupting alarm persistence, inability to clear an active fault with invalid readings, repeated occurrences and stale acknowledgements.
- All eight native maintenance intervals and their expected dates; both assignment strategies across five categories; three production/reorder quantities; all three service priorities; training text changes; both submitted budget policies.
- Native receipt/invoice inspection references with accepted and rejected readings; partial supplier payment, settlement and cancellation, with balanced GL entries and restored outstanding amounts. These temporary transactions are rolled back.

The coverage register is the acceptance boundary

`coverage-register.md` is the recorded readable register. `./osr example-city coverage` regenerates current evidence under `build/city-example/coverage.json` and `coverage.md`, plus the discovered contracts in `coverage-inventory-current.json`. It inventories 526 registered functions, inputs, shared settings, route prefixes and named workflows. It explicitly excludes an exhaustive inventory of upstream ERPNext/FUXA, every city's override, desktop tool parameters and internal Rust functions.

The recorded classification is 36 scenario, 64 varied, 135 partial and 291 gap entries. These are inventory entries, not counts of broken features or an overall coverage percentage. A gap means no qualifying mapped evidence; partial coverage never counts as a finished workflow. Native budget enforcement, enabled assignment and scheduled replenishment now have varied outcome evidence. The register lists remaining payroll, independent training competence, SLA, capitalization, tax,

desktop engineering, load, backup restoration and physical acceptance work explicitly.

`coverage-plan.json` binds each entry to named evidence and a reviewed contract signature. New entries, removed entries and changed defaults/options fail the inventory check. Review the source change and its tests, then update only the reviewed IDs, signatures and evidence rules using `coverage-inventory-current.json`. Do not regenerate the review list just to silence a failure. Failed/incomplete scenario reports provide no passing coverage credit. After reviewing a new recorded run, refresh this checked-in register with:

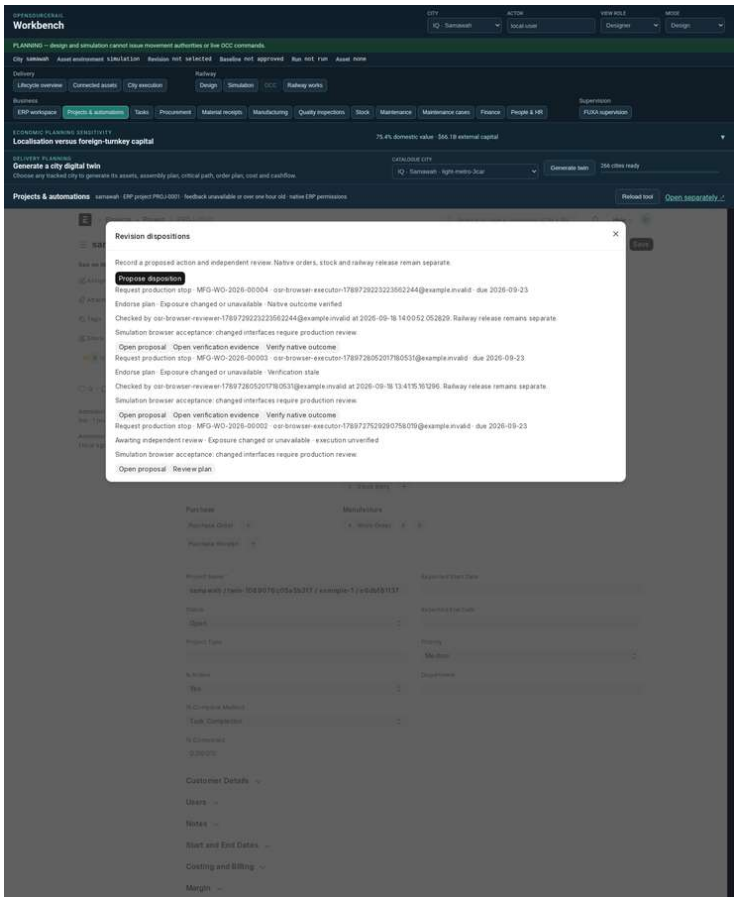
```
cp build/city-example/coverage.md deployment/example-city/coverage-register.md
```

The push/PR/manual CI workflow runs the expansion and always generates the coverage report, including on failure. Regular Python CI guards the reviewed inventory and verifies that missing evidence and newly changed contracts cannot be reported as covered. Local verification does not imply the GitHub workflow has run.

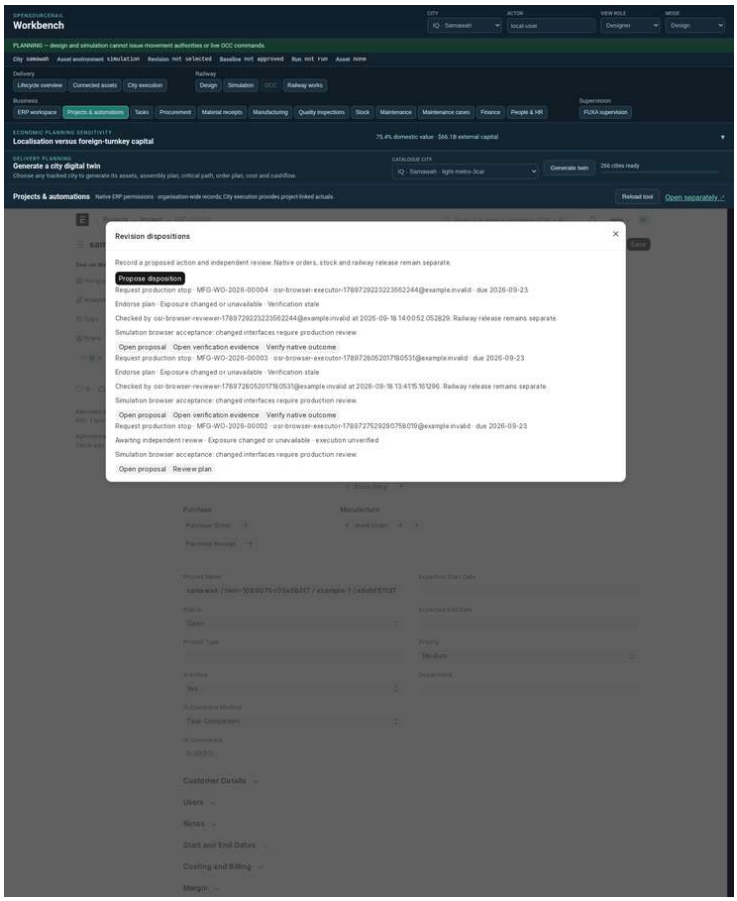
Native business outcomes and independent browser disposition

`business-check` executes 24 checks, including rollback verification: native Stop/Warn budgets at 50/100/101 against a 100 limit; enabled Round Robin and Load Balancing assignment; scheduler-generated purchase requests at stock 3/2/1 against level 2; preserved city/project identity; conflicting/stale replenishment rules; submitted training events, attendance, Pass/Fail results and invalid time periods. These are simulation fixtures. Training completion does not grant competence.

`disposition-check` executes 9 unmocked browser checks through Workbench's real ERP frame. Three independent native accounts propose, endorse, perform and verify a Work Order stop. Self-endorsement, self-verification and verification before action are rejected. Resuming the Work Order makes its prior verification visibly stale. Fixture accounts and records remain in the isolated example; passwords remain private.



Independent native disposition verification



The same verification becomes stale after a native change

Commit-bound release evidence

After all checks, `./osr example-city release-evidence` binds the required reports to the clean commit recorded at scenario start. Dirty local demonstrations and reports from an earlier commit are rejected. CI publishes `city-evidence.json`, and release packaging downloads the artifact from the exact successful candidate run. Missing, failed, changed or empty reports fail the gate. This is a software release gate; the review follow-through records the remaining engineering, formal, operating and physical work.

Configuration coverage

Area	Variations and observed effects
Planning	Five task-family switches in generated plans; changed calendar start, working weekdays, holiday, programme offsets, department and warehouse in actual ERP; repeat import preserves edited progress
Business execution	Ordered/received/outstanding quantities, invoice actuals, BOM ratio, planned/produced quantity, native serials, accepted inspection, spare consumption and downtime
Native reusable components	Transactional native component, procurement and lifecycle regression suites, including HR/training, budgets, quality, replenishment, maintenance and independent disposition verification
Telemetry	Every measurement in the selected equipment package at an in-range value and above maximum; scale/offset, unit, future clock, duplicate sequence, stale/disconnected quality and configured sampling interval
Native controllers and factory	Station, BMS, auxiliaries, HVAC, CBM, points and faregate fault inputs through Rust; all nine deployed factory methods with changed cycle progress, availability and quality holds, separate ERP cases and restoration
Alarms	Trigger/clear thresholds, persistence, repeat suppression, hysteresis, disabled maintenance routing and changed priority/response in native ERP triage
Commands	Several lighting levels with actual measured feedback; configured minimum, maximum, expiry, local-enable rejection, role and environment restrictions
Scope and revisions	Restricted city reader, independent Mosul package, stale review rejection and immutable serial/evidence history
Browser and recovery	Native login, city navigation, engineering download, ERP records, FUXA live values, UI controller command, ERP outage and gateway restart

This is a complete representative software lifecycle, with full city planning and selected live station, vehicle, wayside and factory equipment. It does not exercise every possible setting combination or every city equipment instance through the live HTTP/ERP/FUXA path. The whole-city contract sweep is sequential and does not establish full-network real-time throughput or FUXA load capacity. Artifact parsing/hashes do not execute desktop FreeCAD, Bonsai or QGIS editors. Simulator fixtures,

ERP sample inspections and simulation releases are explicitly separate from physical controller, manufacturing, site and railway acceptance. Restarting the simulator resets its volatile controller outputs to fixture defaults; ERP records, equipment history and the retained City Studio workspace persist.

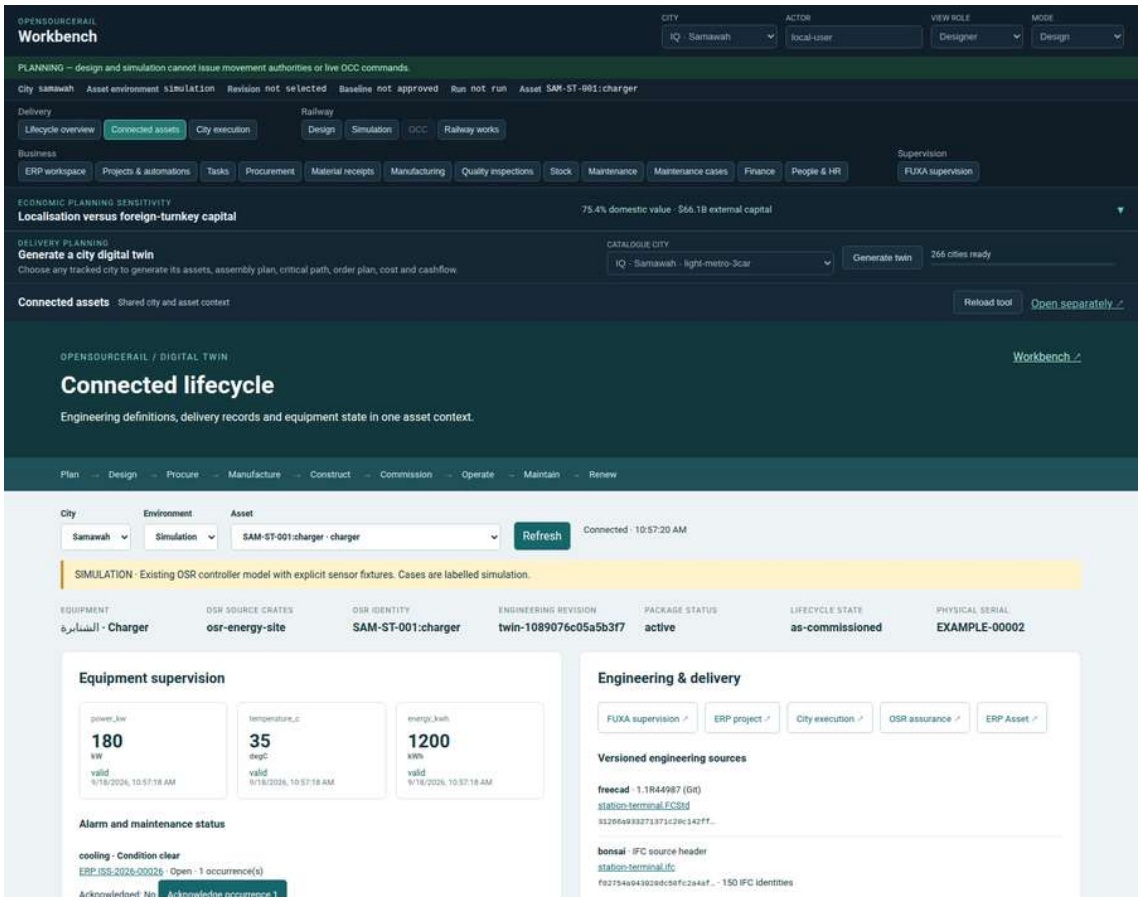
The scenario sources are `scenario.py`, `matrix.py`, `erp-scenario.py`, `network.py`, `restart.py`, `verify-ui.mjs` and `engineering.json`. The expansion and coverage sources are `expansion.py`, `erp-expansion.py`, `contracts.py`, `coverage.py` and `coverage-plan.json`. They reuse the production adapters; fixture setup and assertions are kept in this directory.

Recorded run and screenshots

The local run on 18 September 2026 passed all 98 lifecycle, configuration and recovery checks, including ten real-service browser assertions and a full stop/start. The regression suite also passed 186 Python tests, 27 frontend browser tests and ten frontend unit tests. It imported 3,254 Samawah tasks and 8,310 Mosul tasks. The whole-city contract sweep checked 573 equipment positions, 1,610 measurements and 6,440 native/boundary sample attempts. The live deployment used 19 Samawah and 10 Mosul equipment positions. Its browser checks used real services without intercepted responses, including FUXA-to-Workbench navigation. The configuration and lifecycle observations are saved with before/after values in `build/city-example/report.html` and `report.json`.

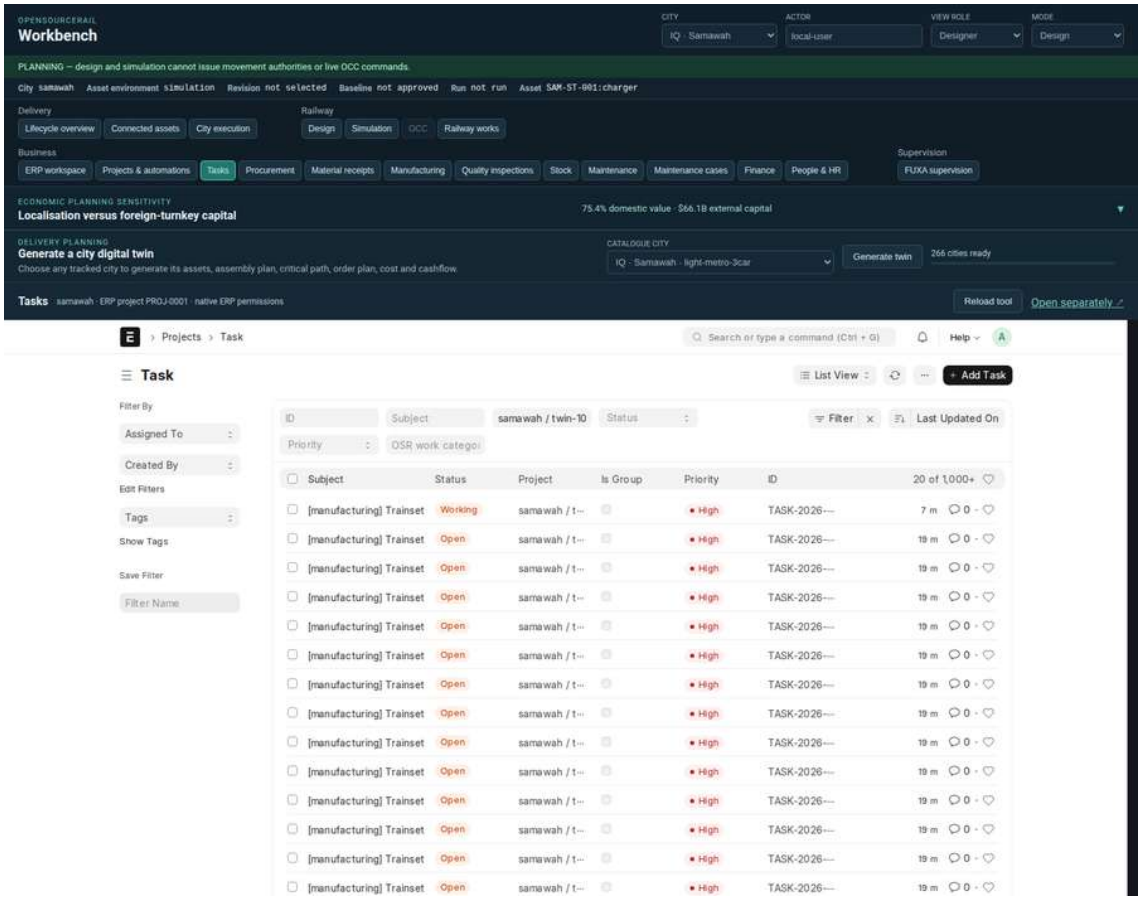
The native capital Asset is a seeded evaluation fixture; this scenario does not demonstrate accounting capitalization of the manufactured serial, tax filing or payroll execution. Supplier payment is now covered by the additional rolled-back native payment sequence above. Fault-recovery cases remain open for inspection: clearing telemetry does not automatically close ERP maintenance work.

Installed replacement serial and engineering context



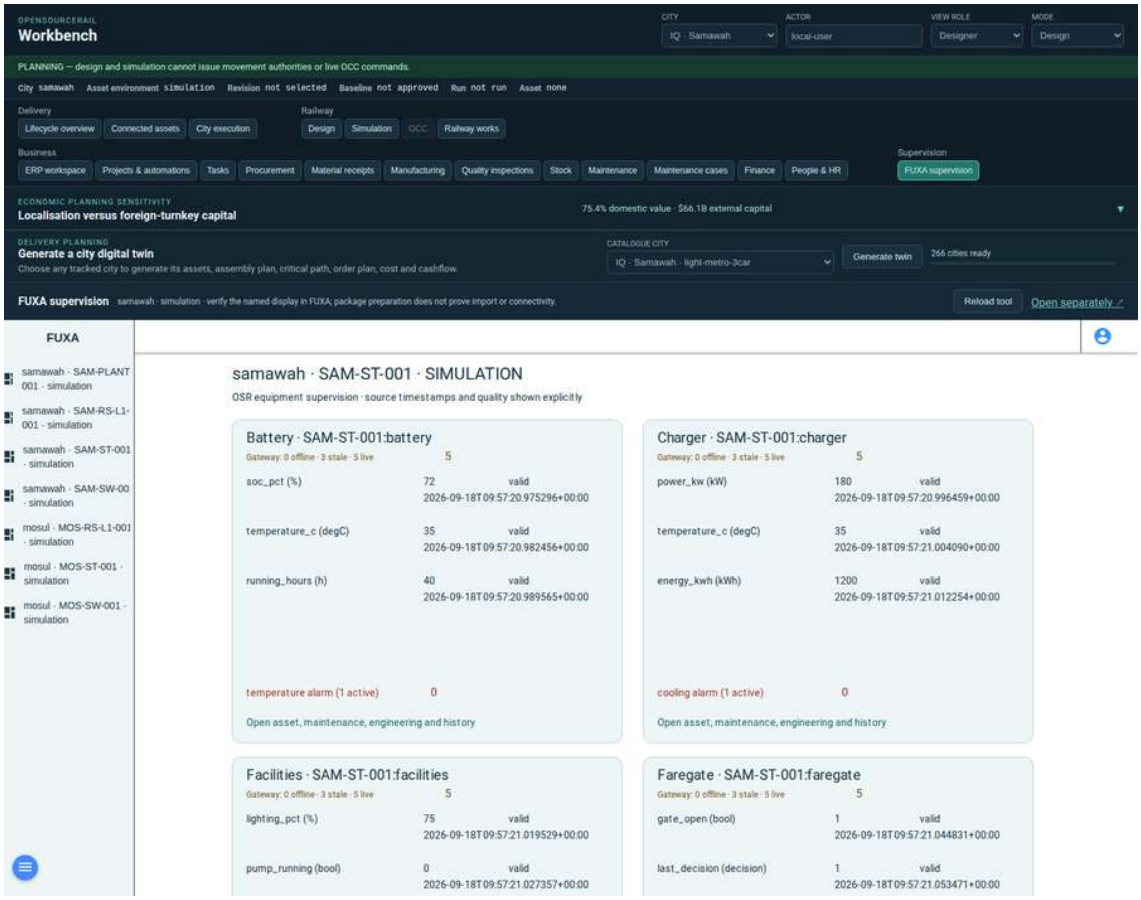
Samawah replacement serial, live telemetry and engineering evidence in Workbench

Native ERP tasks scoped to the example city



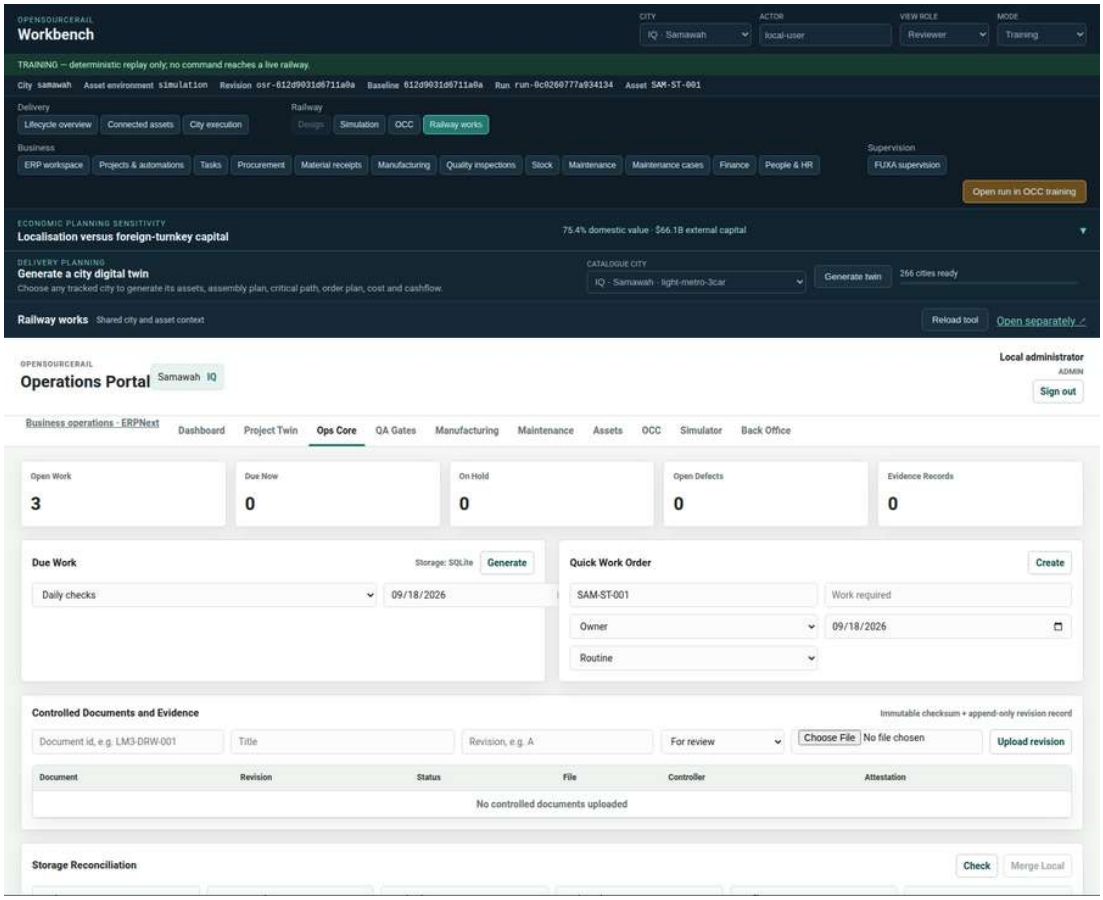
Samawah native ERP task list inside Workbench

Live FUXA supervision in the same UI



Samawah FUXA supervision inside Workbench

OSR railway work with the simulation revision and run



Samawah railway work and context in Workbench

Integration coverage register

Repository source: `deployment/example-city/coverage-register.md`

Audited scope: OSR registered ERP adapters and their input schemas, whitelisted OSR ERP entry points, declared gateway/Workbench/OSR HTTP route prefixes, shared city/supervision settings, simulator controls and named lifecycle workflows. This is not an inventory of every upstream ERPNext/FUXA API, every city override, desktop-tool parameter or internal Rust function.

This is a coverage register, not a claim of exhaustive testing. Each source-discovered entry is tracked even when it has no passing scenario. New entries fail the inventory review check until classified. Evidence requires a completed passing report; individual successes from a failed run are not credited.

Status: `scenario` means the function was exercised; `varied` means named values were changed and checked; `partial` means only part of the behavior is demonstrated; `gap` means no qualifying evidence. None means all combinations are covered.

Native ERP expansion checks run real hooks and ledger posting inside a rolled-back transaction. The live telemetry matrix pauses the example simulator and includes explicitly labelled controller-result fixtures.

Counts: gap: 296, partial: 135, scenario: 36, varied: 64. Inventory consistent: true.

Remaining workflow gaps

Workflow	Status	Next acceptance requirement
Enabled rule to actual task assignment and reassignment	varied	Actual enabled Round Robin and Load Balancing assignment to two users. Holidays, disabled users and concurrent dispatch remain.
Restore ERP database, files, gateway and FUXA into a fresh deployment	gap	A separate source-bound cold recovery rehearsal verifies fresh volumes and cross-service data (docs/operating/platform-recovery.md); this register does not yet map that report. Operational cutover, jobs/controllers and recovery objectives remain.
Submitted budget to blocked/warned overspend transaction	varied	Submitted actual expenses at/below/above Stop/Warn budgets. Purchase commitments, fiscal-period boundaries and simultaneous posting remain.
Manufactured serial to capital Asset and depreciation	gap	Connect received/manufactured serialized equipment to capitalization, depreciation and retirement entries.
Seven revision-disposition decisions and independent native outcome checks	scenario	Unmocked browser with three independent accounts covers proposal, endorsement, native stop, verification and stale detection; all target/action/concurrency permutations remain.
Edit FreeCAD/Bonsai/QGIS and propagate regenerated outputs to ERP and controllers	gap	Run real FreeCAD/Bonsai/QGIS edits, regenerate artifacts, then propagate changed quantities and mappings downstream.
Physical controller bindings, calibration, permissives and independently accepted site tests	gap	Commission real device bindings and calibration, perform HIL/site tests and independently accept results.
Simultaneous whole-city telemetry, operators and FUXA displays	gap	Run simultaneous full-city equipment and multiple browser/operator clients; measure latency, loss and resource ceilings.
Independent planning, ERP records, gateway packages and UI city context	scenario	Two-city scenario; more city families, mixed timezones and concurrent multi-city load remain.
Partial supplier payment, settlement, cancellation and ledger balance	varied	Real partial/full/cancelled payments and ledger balancing in a rollback transaction; multiple currencies, bank reconciliation and bank transport remain.
Employee attendance, payroll calculation, posting and payment	gap	Create approved employee/payroll fixtures, attendance/leave variations, salary calculation, posting and payment.
City Studio revision to simulator, OCC replay and linked OSR work	scenario	City Studio to simulator/OCC/OSR work in simulation; operating permutations and physical acceptance remain.
Stock threshold to scheduled native purchase request	varied	Native scheduler at/below/above threshold, quantities, city identity and stale/ambiguous rules. Multiple suppliers, lead times and live concurrent scheduler cycles remain.
Working calendar, response deadline and SLA escalation	gap	Vary working calendars, priorities and elapsed time; observe deadlines and escalation.
Tax configuration, calculation, statutory filing and correction	gap	Add jurisdiction-specific approved fixtures, tax calculation, returns and corrections; no statutory filing is performed by these tests.
Training events, attendance, assessment and competence decision	partial	Submitted training, attendance and Pass/Fail results; invalid periods rejected. Independent competence decisions and authorization remain.

Function and setting inventory

Entry	Status	Evidence	Remaining coverage
components.profile.defaults.assignment.strategy	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
components.profile.defaults.budget.action	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
components.profile.defaults.quality.reference_type	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
components.profile.defaults.training.description	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
components.profile.defaults.training.title	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
components.profile.instances[].component	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
components.profile.instances[].inputs	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
components.profile.instances[].key	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
components.profile.schema	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
controller.aux_fault	varied	base: Native fault controls change vehicle-aux.fault_count	Native fixture on/off observations; simultaneous fault combinations and physical hardware remain.
controller.battery_trip	varied	base: Native fault controls change vehicle-bms.trip	Native fixture on/off observations; simultaneous fault combinations and physical hardware remain.
controller.cbm_service	varied	base: Native fault controls change vehicle-cbm.health	Native fixture on/off observations; simultaneous fault combinations and physical hardware remain.
controller.cooling_fault	varied	base: Controller fault creates ERP Issue for the manufactured/installed charger	Native fixture on/off observations; simultaneous fault combinations and physical hardware remain.
controller.crossing_motor_fault	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
controller.disconnected	varied	base: Source loss becomes disconnected without reusing valid values	Native fixture on/off observations; simultaneous fault combinations and physical hardware remain.
controller.factory_cell_unavailable	varied	base: Factory method LM3-MFG-010 routes cycle, availability and quality independently; base: Factory method LM3-MFG-020 routes cycle, availability and quality independently; base: Factory method LM3-MFG-030 routes cycle, availability and quality independently; +6 observations (JSON)	All nine Samawah factory methods with selected progress/hold combinations; physical process data and full Cartesian combinations remain.
controller.factory_cycle_progress_pct	varied	base: Factory method LM3-MFG-010 routes cycle, availability and quality independently; base: Factory method LM3-MFG-020 routes cycle, availability and quality independently; base: Factory method LM3-MFG-030 routes cycle, availability and quality independently; +6 observations (JSON)	All nine Samawah factory methods with selected progress/hold combinations; physical process data and full Cartesian combinations remain.
controller.factory_method	varied	base: Factory method LM3-MFG-010 routes cycle, availability and quality independently; base: Factory method LM3-MFG-020 routes cycle, availability and quality independently; base: Factory method LM3-MFG-030 routes cycle, availability and quality independently; +6 observations (JSON)	All nine Samawah factory methods with selected progress/hold combinations; physical process data and full Cartesian combinations remain.
controller.factory_process_excursion	varied	base: Factory method LM3-MFG-010 routes cycle, availability and quality independently; base: Factory method LM3-MFG-020 routes cycle, availability and quality independently; base: Factory method LM3-MFG-030 routes cycle, availability and quality independently; +6 observations (JSON)	All nine Samawah factory methods with selected progress/hold combinations; physical process data and full Cartesian combinations remain.
controller.faregate_denial	varied	base: Native fault controls change faregate.last_decision	Native fixture on/off observations; simultaneous fault combinations and physical hardware remain.
controller.local_remote_disabled	varied	base: Controller local-enable setting rejects an otherwise valid command	Native fixture on/off observations; simultaneous fault combinations and physical hardware remain.

Entry	Status	Evidence	Remaining coverage
`controller.points_detection_fault`	varied	base: Native fault controls change points.detection_unknown	Native fixture on/off observations; simultaneous fault combinations and physical hardware remain.
`controller.station_fault`	varied	base: Native fault controls change facilities.fault_count	Native fixture on/off observations; simultaneous fault combinations and physical hardware remain.
`engineering.artifact`	partial	base: CAD, IFC and GIS artifacts reopen with exact content hashes; base: Exact engineering revision maps to native Item and production BOM; base: IFC selection resolves the same physical position and ERP Asset	Actual tracked artifacts and linked execution/IFC identities are checked. Desktop regeneration and all file/geometry transformations remain.
`engineering.execution_proposal`	partial	base: CAD, IFC and GIS artifacts reopen with exact content hashes; base: Exact engineering revision maps to native Item and production BOM; base: IFC selection resolves the same physical position and ERP Asset	Actual tracked artifacts and linked execution/IFC identities are checked. Desktop regeneration and all file/geometry transformations remain.
`engineering.ifc_overlay`	partial	base: CAD, IFC and GIS artifacts reopen with exact content hashes; base: Exact engineering revision maps to native Item and production BOM; base: IFC selection resolves the same physical position and ERP Asset	Actual tracked artifacts and linked execution/IFC identities are checked. Desktop regeneration and all file/geometry transformations remain.
`engineering.package`	partial	base: CAD, IFC and GIS artifacts reopen with exact content hashes; base: Exact engineering revision maps to native Item and production BOM; base: IFC selection resolves the same physical position and ERP Asset	Actual tracked artifacts and linked execution/IFC identities are checked. Desktop regeneration and all file/geometry transformations remain.
`erp.api.api.import_plan`	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
`erp.api.city_runtime.city_status`	partial	base: Isolated native procurement adapters and negative cases pass with rollback	Native procurement regression checks readiness and Material Request feedback; all status combinations remain.
`erp.api.components.apply`	scenario	base: Isolated native components adapters and negative cases pass with rollback	All eight adapters through native preview/apply and repeat-safe checks; every HTTP permission/prerequisite combination remains.
`erp.api.components.catalogue`	scenario	expansion: erp.catalogue	Actual city component/warehouse catalogue; other roles and city profiles remain.
`erp.api.components.inspection_lines`	varied	expansion: erp.inspection-lines.Purchase Receipt; expansion: erp.inspection-lines.Purchase Invoice	Both receipt and invoice row selection; cross-city and permission variants remain.
`erp.api.components.preview`	scenario	base: Isolated native components adapters and negative cases pass with rollback	All eight adapters through native preview/apply and repeat-safe checks; every HTTP permission/prerequisite combination remains.
`erp.api.disposition.catalogue`	partial	base: Isolated native lifecycle adapters and negative cases pass with rollback	Native regression suite exercises this adapter family; endpoint-specific HTTP permissions and all argument variants are not individually established.
`erp.api.disposition.preview`	partial	base: Isolated native lifecycle adapters and negative cases pass with rollback	Native regression suite exercises this adapter family; endpoint-specific HTTP permissions and all argument variants are not individually established.
`erp.api.disposition.preview_decision`	partial	base: Isolated native lifecycle adapters and negative cases pass with rollback	Native regression suite exercises this adapter family; endpoint-specific HTTP permissions and all argument variants are not individually established.
`erp.api.disposition.record`	partial	base: Isolated native lifecycle adapters and negative cases pass with rollback	Native regression suite exercises this adapter family; endpoint-specific HTTP permissions and all argument variants are not individually established.
`erp.api.disposition.record_decision`	partial	base: Isolated native lifecycle adapters and negative cases pass with rollback	Native regression suite exercises this adapter family; endpoint-specific HTTP permissions and all argument variants are not individually established.
`erp.api.disposition_execution.preview`	partial	base: Isolated native lifecycle adapters and negative cases pass with rollback	Native regression suite exercises this adapter family; endpoint-specific HTTP permissions and all argument variants are not individually established.
`erp.api.disposition_execution.record`	partial	base: Isolated native lifecycle adapters and negative cases pass with rollback	Native regression suite exercises this adapter family; endpoint-specific HTTP permissions and all argument variants are not individually established.
`erp.api.executive.ingest_draft`	gap	—	Pure contract tests cover checksum, HMAC, allowlist and mutation rejection; a complete example with provisioned external model runners and all four native draft types remains.

Entry	Status	Evidence	Remaining coverage
erp.api.field_operations.ingest	gap	—	Isolated operations Python/gateway/browser checks exist; native complete-city ERP delivery and deployed combinations remain unexercised.
erp.api.integration.apply_execution	scenario	base: Exact engineering revision maps to native Item and production BOM	Representative connected lifecycle; full input permutations and all permission combinations remain.
erp.api.integration.apply_repair	scenario	base: Native repair consumes the purchased spare and keeps railway release separate	Representative connected lifecycle; full input permutations and all permission combinations remain.
erp.api.integration.case_status	scenario	base: ERP recovery delivers a new fault once, without reusing the closed case	Representative connected lifecycle; full input permutations and all permission combinations remain.
erp.api.integration.condition_event	scenario	base: Controller fault creates ERP Issue for the manufactured/installed charger	Representative connected lifecycle; full input permutations and all permission combinations remain.
erp.api.integration.preview_execution	scenario	base: Exact engineering revision maps to native Item and production BOM	Representative connected lifecycle; full input permutations and all permission combinations remain.
erp.api.integration.preview_repair	scenario	base: Native repair consumes the purchased spare and keeps railway release separate	Representative connected lifecycle; full input permutations and all permission combinations remain.
erp.api.integration.repairs_for_issue	scenario	base: Native repair consumes the purchased spare and keeps railway release separate	Representative connected lifecycle; full input permutations and all permission combinations remain.
erp.api.procurement.candidates	partial	base: Isolated native procurement adapters and negative cases pass with rollback	Native regression suite exercises this adapter family; endpoint-specific HTTP permissions and all argument variants are not individually established.
erp.api.procurement.create_material_request	partial	base: Isolated native procurement adapters and negative cases pass with rollback	Native regression suite exercises this adapter family; endpoint-specific HTTP permissions and all argument variants are not individually established.
erp.component.assignment	partial	expansion: erp.assignment.RoundRobin.manufacturing; expansion: erp.assignment.RoundRobin.maintenance; expansion: erp.assignment.RoundRobin.qa; +7 observations (JSON)	Both strategies across five categories create disabled rules. Enablement, real allocation, reassignment and multi-user fairness remain.
erp.component.budget	partial	expansion: erp.budget.Stop; expansion: erp.budget.Warn	Native Stop/Warn expense enforcement also has business-report evidence; commitments and fiscal-period interactions remain.
erp.component.maintenance	partial	expansion: erp.maintenance.Daily; expansion: erp.maintenance.Weekly; expansion: erp.maintenance.Monthly; +5 observations (JSON)	All eight intervals produce expected due dates and native logs. Calendar exceptions, overdue servicing and reassignment remain.
erp.component.manufacturing	partial	expansion: erp.manufacturing.quantity.1; expansion: erp.manufacturing.quantity.2; expansion: erp.manufacturing.quantity.5	Draft quantity and BOM requirements vary; base scenario submits and produces once. Other BOMs, scrap, routing and cost combinations remain.
erp.component.quality	partial	expansion: erp.quality.PurchaseReceipt.35; expansion: erp.quality.PurchaseReceipt.99; expansion: erp.quality.PurchaseInvoice.35; +1 observations (JSON)	Both receipt/invoice references, sample sizes and accepted/rejected readings are exercised. Inspector permissions and rejection disposition combinations remain.
erp.component.replenishment	partial	expansion: erp.replenishment.0; expansion: erp.replenishment.2; expansion: erp.replenishment.10	Native scheduler behavior also has business-report evidence; supplier, lead-time and concurrency permutations remain.
erp.component.service	partial	expansion: erp.service.priority.Low; expansion: erp.service.priority.Medium; expansion: erp.service.priority.High	All three priorities create native Issues. SLA calendars, escalation and customer/service variants remain.
erp.component.training	partial	expansion: erp.training.0; expansion: erp.training.1	Programme title and objectives vary. Events, attendance, results and competence decisions remain.
erp.input.assignment.category	varied	expansion: erp.assignment.RoundRobin.manufacturing; expansion: erp.assignment.RoundRobin.maintenance; expansion: erp.assignment.RoundRobin.qa; +7 observations (JSON)	Both strategies across five categories create disabled rules. Enablement, real allocation, reassignment and multi-user fairness remain.
erp.input.assignment.strategy	varied	expansion: erp.assignment.RoundRobin.manufacturing; expansion: erp.assignment.RoundRobin.maintenance; expansion: erp.assignment.RoundRobin.qa; +7 observations (JSON)	Both strategies across five categories create disabled rules. Enablement, real allocation, reassignment and multi-user fairness remain.
erp.input.assignment.users	partial	contracts: contract.erp.input.assignment.users.missing; contracts: contract.erp.input.assignment.users.boolean; contracts: contract.erp.input.assignment.users.empty; +2 observations (JSON)	Schema boundaries are exercised; a field-specific native change/effect is not established for this entry.
Entry	**Status**	**Evidence**	**Remaining coverage**
erp.input.assignment.users[].user	partial	contracts: contract.erp.input.assignment.users[].user.missing; contracts: contract.erp.input.assignment.users[].user.	Schema boundaries are exercised; a field-specific native change/effect is not established for this entry.

Entry	Status	Evidence	Remaining coverage
		boolean; contracts: contract.erp.input.assignment.users[].user.max-length; +2observations (JSON)	
erp.input.budget.accounts	partial	contracts: contract.erp.input.budget.accounts.missing; contracts: contract.erp.input.budget.accounts.boolean; contracts: contract.erp.input.budget.accounts.empty; +2 observations (JSON)	Native Stop/Warn expense enforcement also has business-report evidence; commitments and fiscal-period interactions remain.
erp.input.budget.accounts[].account	partial	contracts: contract.erp.input.budget.accounts[].account.missing; contracts: contract.erp.input.budget.accounts[].account.boolean; contracts: contract.erp.input.budget.accounts[].account.max-length; +2 observations (JSON)	Native Stop/Warn expense enforcement also has business-report evidence; commitments and fiscal-period interactions remain.
erp.input.budget.accounts[].amount	varied	expansion: erp.budget.Stop; expansion: erp.budget.Warn	Native Stop/Warn expense enforcement also has business-report evidence; commitments and fiscal-period interactions remain.
erp.input.budget.action	varied	expansion: erp.budget.Stop; expansion: erp.budget.Warn	Native Stop/Warn expense enforcement also has business-report evidence; commitments and fiscal-period interactions remain.
erp.input.budget.fiscal_year	partial	contracts: contract.erp.input.budget.fiscal_year.missing; contracts: contract.erp.input.budget.fiscal_year.boolean; contracts: contract.erp.input.budget.fiscal_year.max-length; +2 observations (JSON)	Native Stop/Warn expense enforcement also has business-report evidence; commitments and fiscal-period interactions remain.
erp.input.maintenance.asset	partial	contracts: contract.erp.input.maintenance.asset.missing; contracts: contract.erp.input.maintenance.asset.boolean; contracts: contract.erp.input.maintenance.asset.max-length; +2 observations (JSON)	Schema boundaries are exercised; a field-specific native change/effect is not established for this entry.
erp.input.maintenance.assignee	partial	contracts: contract.erp.input.maintenance.assignee.missing; contracts: contract.erp.input.maintenance.assignee.boolean; contracts: contract.erp.input.maintenance.assignee.max-length; +2 observations (JSON)	Schema boundaries are exercised; a field-specific native change/effect is not established for this entry.
erp.input.maintenance.description	partial	contracts: contract.erp.input.maintenance.description.missing; contracts: contract.erp.input.maintenance.description.boolean; contracts: contract.erp.input.maintenance.description.max-length; +2 observations (JSON)	Schema boundaries are exercised; a field-specific native change/effect is not established for this entry.
erp.input.maintenance.periodicity	varied	expansion: erp.maintenance.Daily; expansion: erp.maintenance.Weekly; expansion: erp.maintenance.Monthly; +5 observations (JSON)	All eight intervals produce expected due dates and native logs. Calendar exceptions, overdue servicing and reassignment remain.
erp.input.maintenance.start	partial	contracts: contract.erp.input.maintenance.start.missing; contracts: contract.erp.input.maintenance.start.boolean; contracts: contract.erp.input.maintenance.start.invalid-date	Schema boundaries are exercised; a field-specific native change/effect is not established for this entry.
erp.input.maintenance.team	partial	contracts: contract.erp.input.maintenance.team.missing; contracts: contract.erp.input.maintenance.team.boolean; contracts: contract.erp.input.maintenance.team.max-length; +2 observations (JSON)	Schema boundaries are exercised; a field-specific native change/effect is not established for this entry.
erp.input.maintenance.title	partial	contracts: contract.erp.input.maintenance.title.missing; contracts: contract.erp.input.maintenance.title.boolean; contracts: contract.erp.input.maintenance.title.max-length; +2 observations (JSON)	Schema boundaries are exercised; a field-specific native change/effect is not established for this entry.
erp.input.manufacturing.bom	partial	contracts: contract.erp.input.manufacturing.bom.missing; contracts: contract.erp.input.manufacturing.bom.boolean; contracts: contract.erp.input.manufacturing.bom.max-length; +2 observations (JSON)	Schema boundaries are exercised; a field-specific native change/effect is not established for this entry.
erp.input.manufacturing.fg_warehouse	partial	contracts: contract.erp.input.manufacturing.fg_warehouse.missing; contracts: contract.erp.input.manufacturing.fg_warehouse.boolean; contracts: contract.erp.input.manufacturing.fg_warehouse.max-length; +2 observations (JSON)	Schema boundaries are exercised; a field-specific native change/effect is not established for this entry.
erp.input.manufacturing.quantity	varied	expansion: erp.manufacturing.quantity.1; expansion: erp.manufacturing.quantity.2; expansion: erp.manufacturing.quantity.5	Draft quantity and BOM requirements vary; base scenario submits and produces once. Other BOMs, scrap, routing and cost combinations remain.
Entry	**Status**	**Evidence**	**Remaining coverage**
erp.input.manufacturing.source_warehouse	partial	contracts: contract.erp.input.manufacturing.source_warehouse.missing; contracts: contract.erp.input.manufacturing.source_warehouse.boolean; contracts:	Schema boundaries are exercised; a field-specific native change/effect is not established for this entry.

Entry	Status	Evidence	Remaining coverage
		contract.erp.input.manufacturing.source_warehouse.max-length; +2 observations(JSON)	
erp.input.manufacturing.start	partial	contracts: contract.erp.input.manufacturing.start.missing; contracts: contract.erp.input.manufacturing.start.boolean; contracts: contract.erp.input.manufacturing.start.invalid-date; +1 observations (JSON)	Schema boundaries are exercised; a field-specific native change/effect is not established for this entry.
erp.input.manufacturing.wip_warehouse	partial	contracts: contract.erp.input.manufacturing.wip_warehouse.missing; contracts: contract.erp.input.manufacturing.wip_warehouse.boolean; contracts: contract.erp.input.manufacturing.wip_warehouse.max-length; +2 observations (JSON)	Schema boundaries are exercised; a field-specific native change/effect is not established for this entry.
erp.input.quality.inspector	partial	contracts: contract.erp.input.quality.inspector.missing; contracts: contract.erp.input.quality.inspector.boolean; contracts: contract.erp.input.quality.inspector.max-length; +2 observations (JSON)	Schema boundaries are exercised; a field-specific native change/effect is not established for this entry.
erp.input.quality.line	partial	contracts: contract.erp.input.quality.line.missing; contracts: contract.erp.input.quality.line.boolean; contracts: contract.erp.input.quality.line.max-length; +2 observations (JSON)	Schema boundaries are exercised; a field-specific native change/effect is not established for this entry.
erp.input.quality.reference	partial	contracts: contract.erp.input.quality.reference.missing; contracts: contract.erp.input.quality.reference.boolean; contracts: contract.erp.input.quality.reference.max-length; +2 observations (JSON)	Schema boundaries are exercised; a field-specific native change/effect is not established for this entry.
erp.input.quality.reference_type	varied	expansion: erp.quality.Purchase Receipt.35; expansion: erp.quality.Purchase Receipt.99; expansion: erp.quality.Purchase Invoice.35; +1 observations (JSON)	Both receipt/invoice references, sample sizes and accepted/rejected readings are exercised. Inspector permissions and rejection disposition combinations remain.
erp.input.quality.report_date	partial	contracts: contract.erp.input.quality.report_date.missing; contracts: contract.erp.input.quality.report_date.boolean; contracts: contract.erp.input.quality.report_date.invalid-date	Schema boundaries are exercised; a field-specific native change/effect is not established for this entry.
erp.input.quality.sample_size	varied	expansion: erp.quality.Purchase Receipt.35; expansion: erp.quality.Purchase Receipt.99; expansion: erp.quality.Purchase Invoice.35; +1 observations (JSON)	Both receipt/invoice references, sample sizes and accepted/rejected readings are exercised. Inspector permissions and rejection disposition combinations remain.
erp.input.quality.template	partial	contracts: contract.erp.input.quality.template.missing; contracts: contract.erp.input.quality.template.boolean; contracts: contract.erp.input.quality.template.max-length; +2 observations (JSON)	Schema boundaries are exercised; a field-specific native change/effect is not established for this entry.
erp.input.replenishment.item	partial	contracts: contract.erp.input.replenishment.item.missing; contracts: contract.erp.input.replenishment.item.boolean; contracts: contract.erp.input.replenishment.item.max-length; +2 observations (JSON)	Native scheduler behavior also has business-report evidence; supplier, lead-time and concurrency permutations remain.
erp.input.replenishment.level	varied	expansion: erp.replenishment.0; expansion: erp.replenishment.2; expansion: erp.replenishment.10	Native scheduler behavior also has business-report evidence; supplier, lead-time and concurrency permutations remain.
erp.input.replenishment.quantity	varied	expansion: erp.replenishment.0; expansion: erp.replenishment.2; expansion: erp.replenishment.10	Native scheduler behavior also has business-report evidence; supplier, lead-time and concurrency permutations remain.
erp.input.replenishment.warehouse	partial	contracts: contract.erp.input.replenishment.warehouse.missing; contracts: contract.erp.input.replenishment.warehouse.boolean; contracts: contract.erp.input.replenishment.warehouse.max-length; +2 observations (JSON)	Native scheduler behavior also has business-report evidence; supplier, lead-time and concurrency permutations remain.
erp.input.service.description	partial	contracts: contract.erp.input.service.description.missing; contracts: contract.erp.input.service.description.boolean; contracts: contract.erp.input.service.description.max-length; +2 observations (JSON)	Schema boundaries are exercised; a field-specific native change/effect is not established for this entry.
erp.input.service.priority	varied	expansion: erp.service.priority.Low; expansion: erp.service.priority.Medium; expansion: erp.service.priority.High	All three priorities create native Issues. SLA calendars, escalation and customer/service variants remain.
erp.input.service.sla	partial	contracts: contract.erp.input.service.sla.missing; contracts: contract.erp.input.service.sla.boolean; contracts: contract.erp.input.service.sla.max-length; +2 observations (JSON)	Schema boundaries are exercised; a field-specific native change/effect is not established for this entry.

Entry	Status	Evidence	Remaining coverage
erp.input.service.subject	partial	contracts: contract.erp.input.service.subject.missing; contracts: contract.erp.input.service.subject.boolean; contracts: contract.erp.input.service.subject.max-length; +2 observations (JSON)	Schema boundaries are exercised; a field-specific native change/effect is not established for this entry.
erp.input.training.description	varied	expansion: erp.training.0; expansion: erp.training.1	Programme title and objectives vary. Events, attendance, results and competence decisions remain.
erp.input.training.title	varied	expansion: erp.training.0; expansion: erp.training.1	Programme title and objectives vary. Events, attendance, results and competence decisions remain.
fuxa.deployment_manifest	scenario	base: Reviewed FUXA import covers both city packages	Reviewed combined city import and browser navigation. All native FUXA settings, deletion/replacement and concurrent editing combinations remain.
fuxa.deployment_review	scenario	base: Reviewed FUXA import covers both city packages	Reviewed combined city import and browser navigation. All native FUXA settings, deletion/replacement and concurrent editing combinations remain.
fuxa.project	scenario	base: Reviewed FUXA import covers both city packages	Reviewed combined city import and browser navigation. All native FUXA settings, deletion/replacement and concurrent editing combinations remain.
fuxa.validate_deployment_review	scenario	base: Reviewed FUXA import covers both city packages	Reviewed combined city import and browser navigation. All native FUXA settings, deletion/replacement and concurrent editing combinations remain.
gateway.GET./affected	scenario	base: Replacement preserves removed serial history and needs fresh independent release	Removed serial tracing is exercised; batch, missing and all cross-city query combinations remain.
gateway.GET./controller/commands	scenario	expansion: command.expiry-while-source-offline	Used by the live scenario; exhaustive filter, permission and failure permutations remain.
gateway.GET./evidence	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
gateway.GET./executive/decisions	gap	—	Unit and local HTTP tests cover authenticated model identity, scope, immutability, quorum, diversity, dissent and authority boundaries; the complete example deployment has no configured external model runners.
gateway.GET./health	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
gateway.GET./history	scenario	base: Sampling interval changes actual controller arrival times	Sampling history observed; full query, pagination/limit and access combinations remain.
gateway.GET./operations/	gap	—	Isolated operations Python/gateway/browser checks exist; native complete-city ERP delivery and deployed combinations remain unexercised.
gateway.GET./outbox	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
gateway.GET./snapshot	scenario	expansion: telemetry.interaction.1; expansion: telemetry.interaction.2; expansion: telemetry.interaction.3; +69 observations (JSON)	Used by the live scenario; exhaustive filter, permission and failure permutations remain.
gateway.POST./alarms/acknowledge	varied	expansion: alarm.stale-acknowledgement	Named live HTTP scenarios only; request shape limits and all principal/scope/state combinations remain.
gateway.POST./commands	varied	expansion: command.repeat-idempotency; expansion: command.changed-retry; expansion: command.pending-blocks-rule-change; +12 observations (JSON)	Named live HTTP scenarios only; request shape limits and all principal/scope/state combinations remain.
gateway.POST./controller/result	partial	expansion: command.transition.accepted; expansion: command.transition.completed	Named live HTTP scenarios only; request shape limits and all principal/scope/state combinations remain.
gateway.POST./evidence	scenario	base: Replacement preserves removed serial history and needs fresh independent release	Representative installation, test, release, maintenance and replacement; full role/state/revision permutations remain.
gateway.POST./executive/ballots	gap	—	Unit and local HTTP tests cover authenticated model identity, scope, immutability, quorum, diversity, dissent and authority boundaries; the complete example deployment has no configured external model runners.
gateway.POST./executive/finalize	gap	—	Unit and local HTTP tests cover authenticated model identity, scope, immutability, quorum, diversity, dissent and authority boundaries; the complete example deployment has no configured external model runners.

Entry	Status	Evidence	Remaining coverage
gateway.POST./executive/proposals	gap	—	Unit and local HTTP tests cover authenticated model identity, scope, immutability, quorum, diversity, dissent and authority boundaries; the complete example deployment has no configured external model runners.
gateway.POST./operations/	gap	—	Isolated operations Python/gateway/browser checks exist; native complete-city ERP delivery and deployed combinations remain unexercised.
gateway.POST./packages	varied	expansion: configuration.concurrent-review	Named live HTTP scenarios only; request shape limits and all principal/scope/state combinations remain.
gateway.POST./packages/preview	varied	expansion: command.pending-blocks-rule-change	Named live HTTP scenarios only; request shape limits and all principal/scope/state combinations remain.
gateway.POST./telemetry	varied	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-aux.comfort_power.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-aux.comfort_power.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-aux.comfort_power.max; +305 observations (JSON)	Named live HTTP scenarios only; request shape limits and all principal/scope/state combinations remain.
gateway.POST./telemetry/batch	gap	—	Atomic rollback, bounds, replay and per-member authorization have regression coverage; complete-city native batch failure and recovery scenarios remain.
ops.GET./api/operating	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
ops.GET./api/ops-auth/session	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
ops.OPTIONS./api/	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
ops.POST./api/ops-auth/login	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
ops.POST./api/ops-auth/logout	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
planning.calendar.holidays	varied	base: Calendar start, weekday mask and holiday change actual ERP dates	One combined calendar change reaches actual ERP dates; timezone/DST and other calendars remain.
planning.calendar.start_date	varied	base: Calendar start, weekday mask and holiday change actual ERP dates	One combined calendar change reaches actual ERP dates; timezone/DST and other calendars remain.
planning.calendar.timezone	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
planning.calendar.working_weekdays	varied	base: Calendar start, weekday mask and holiday change actual ERP dates	One combined calendar change reaches actual ERP dates; timezone/DST and other calendars remain.
planning.city_programme	varied	base: City-specific programme, department and additional warehouse reach ERP	One city-specific programme/master change; retirement, renaming and all existing-record conflicts remain.
planning.departments.infrastructure_maintenance	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
planning.departments.maintenance	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
planning.departments.manufacturing	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
planning.departments.procurement	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
planning.departments.programme	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
planning.departments.qa	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
planning.organisation.company	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
planning.organisation.departments	varied	base: City-specific programme, department and additional warehouse reach ERP	One city-specific programme/master change; retirement, renaming and all existing-record conflicts remain.
planning.organisation.warehouses	varied	base: City-specific programme, department and additional warehouse reach ERP	One city-specific programme/master change; retirement, renaming and all existing-record conflicts remain.

Entry	Status	Evidence	Remaining coverage
planning.programme[].department	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
planning.programme[].depends_on	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
planning.programme[].finish_day	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
planning.programme[].id	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
planning.programme[].start_day	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
planning.programme[].title	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
planning.release	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
planning.schema_version	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
planning.tasks.maintenance	varied	base: Planning task switch changes generated maintenance work	On/off changes are checked in generated plans; native import/removal effects for every switch combination remain.
planning.tasks.manufacturing	varied	base: Planning task switch changes generated manufacturing work	On/off changes are checked in generated plans; native import/removal effects for every switch combination remain.
planning.tasks.procurement	varied	base: Planning task switch changes generated procurement work	On/off changes are checked in generated plans; native import/removal effects for every switch combination remain.
planning.tasks.programme	varied	base: Planning task switch changes generated programme work	On/off changes are checked in generated plans; native import/removal effects for every switch combination remain.
planning.tasks.qa	varied	base: Planning task switch changes generated qa work	On/off changes are checked in generated plans; native import/removal effects for every switch combination remain.
supervision.fuxa_polling_scope	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.historian.owner	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.historian.retention_days	varied	base: Complete city equipment contracts reach the native evaluator and historian	Clock-controlled production historian sweep only; long-running live retention and disk exhaustion remain.
supervision.historian.sampling_seconds	varied	base: Sampling interval changes actual controller arrival times; expansion: configuration.concurrent-review	2-to-4-second live change and competing updates; sustained loads and all sampling/staleness interactions remain.
supervision.schema	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.template_revision	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.battery.alarms[].clear_below	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.battery.alarms[].delay_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.battery.alarms[].high	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.battery.alarms[].id	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.battery.alarms[].maintenance	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.battery.alarms[].measurement	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.battery.alarms[].priority	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.battery.alarms[].repeat_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.

Entry	Status	Evidence	Remaining coverage
supervision.templates.battery.alarms[].response	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.battery.component_type_id	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.battery.label	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.battery.measurements.running_hours.max	partial	expansion: telemetry.boundary.SAM-ST-001:battery.running_hours.below; expansion: telemetry.boundary.SAM-ST-001:battery.running_hours.min; expansion: telemetry.boundary.SAM-ST-001:battery.running_hours.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.battery.measurements.running_hours.min	partial	expansion: telemetry.boundary.SAM-ST-001:battery.running_hours.below; expansion: telemetry.boundary.SAM-ST-001:battery.running_hours.min; expansion: telemetry.boundary.SAM-ST-001:battery.running_hours.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.battery.measurements.running_hours.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.battery.measurements.running_hours.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.battery.measurements.running_hours.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.battery.measurements.running_hours.unit	partial	expansion: telemetry.boundary.SAM-ST-001:battery.running_hours.below; expansion: telemetry.boundary.SAM-ST-001:battery.running_hours.min; expansion: telemetry.boundary.SAM-ST-001:battery.running_hours.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.battery.measurements.soc_pct.max	partial	expansion: telemetry.boundary.SAM-ST-001:battery.soc_pct.below; expansion: telemetry.boundary.SAM-ST-001:battery.soc_pct.min; expansion: telemetry.boundary.SAM-ST-001:battery.soc_pct.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.battery.measurements.soc_pct.min	partial	expansion: telemetry.boundary.SAM-ST-001:battery.soc_pct.below; expansion: telemetry.boundary.SAM-ST-001:battery.soc_pct.min; expansion: telemetry.boundary.SAM-ST-001:battery.soc_pct.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.battery.measurements.soc_pct.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.battery.measurements.soc_pct.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.battery.measurements.soc_pct.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.battery.measurements.soc_pct.unit	partial	expansion: telemetry.boundary.SAM-ST-001:battery.soc_pct.below; expansion: telemetry.boundary.SAM-ST-001:battery.soc_pct.min; expansion: telemetry.boundary.SAM-ST-001:battery.soc_pct.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.battery.measurements.temperature_c.max	partial	expansion: telemetry.boundary.SAM-ST-001:battery.temperature_c.below; expansion: telemetry.boundary.SAM-ST-001:battery.temperature_c.min; expansion: telemetry.boundary.SAM-ST-001:battery.temperature_c.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.battery.measurements.temperature_c.min	partial	expansion: telemetry.boundary.SAM-ST-001:battery.temperature_c.below; expansion: telemetry.boundary.SAM-ST-001:battery.temperature_c.min; expansion: telemetry.boundary.SAM-ST-001:battery.temperature_c.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.battery.measurements.temperature_c.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.battery.measurements.temperature_c.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.battery.measurements.temperature_c.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.

Entry	Status	Evidence	Remaining coverage
supervision.templates.battery.measurements.temperature_c.unit	partial	expansion: telemetry.boundary.SAM-ST-001:battery.temperature_c.below; expansion: telemetry.boundary.SAM-ST-001:battery.temperature_c.min; expansion: telemetry.boundary.SAM-ST-001:battery.temperature_c.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.battery.source_crates	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.charger.alarms[].clear_below	varied	base: Changed clear threshold clears the alarm	Changed charger rule plus explicit invalid-quality/persistence interactions; other equipment rules and values remain.
supervision.templates.charger.alarms[].delay_seconds	varied	base: Changed alarm threshold and delay activate at the new value	Changed charger rule plus explicit invalid-quality/persistence interactions; other equipment rules and values remain.
supervision.templates.charger.alarms[].high	varied	base: Changed alarm threshold and delay activate at the new value	Changed charger rule plus explicit invalid-quality/persistence interactions; other equipment rules and values remain.
supervision.templates.charger.alarms[].id	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.charger.alarms[].maintenance	varied	base: Maintenance-disabled rule creates no ERP case	Changed charger rule plus explicit invalid-quality/persistence interactions; other equipment rules and values remain.
supervision.templates.charger.alarms[].measurement	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.charger.alarms[].priority	varied	base: Changed alarm priority and response reach native ERP triage	Changed charger rule plus explicit invalid-quality/persistence interactions; other equipment rules and values remain.
supervision.templates.charger.alarms[].repeat_seconds	varied	base: Repeat interval suppresses duplicate occurrences	Changed charger rule plus explicit invalid-quality/persistence interactions; other equipment rules and values remain.
supervision.templates.charger.alarms[].response	varied	base: Changed alarm priority and response reach native ERP triage	Changed charger rule plus explicit invalid-quality/persistence interactions; other equipment rules and values remain.
supervision.templates.charger.component_type_id	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.charger.label	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.charger.measurements.energy_kwh.max	partial	expansion: telemetry.boundary.SAM-ST-001:charger.energy_kwh.below; expansion: telemetry.boundary.SAM-ST-001:charger.energy_kwh.min; expansion: telemetry.boundary.SAM-ST-001:charger.energy_kwh.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.charger.measurements.energy_kwh.min	partial	expansion: telemetry.boundary.SAM-ST-001:charger.energy_kwh.below; expansion: telemetry.boundary.SAM-ST-001:charger.energy_kwh.min; expansion: telemetry.boundary.SAM-ST-001:charger.energy_kwh.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.charger.measurements.energy_kwh.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.charger.measurements.energy_kwh.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.charger.measurements.energy_kwh.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.charger.measurements.energy_kwh.unit	partial	expansion: telemetry.boundary.SAM-ST-001:charger.energy_kwh.below; expansion: telemetry.boundary.SAM-ST-001:charger.energy_kwh.min; expansion: telemetry.boundary.SAM-ST-001:charger.energy_kwh.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.charger.measurements.power_kw.max	partial	expansion: telemetry.boundary.SAM-ST-001:charger.power_kw.below; expansion: telemetry.boundary.SAM-ST-001:charger.power_kw.min; expansion: telemetry.boundary.SAM-ST-001:charger.power_kw.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.charger.measurements.power_kw.min	partial	expansion: telemetry.boundary.SAM-ST-001:charger.power_kw.below; expansion: telemetry.boundary.SAM-ST-001:charger.power_kw.min; expansion: telemetry.boundary.SAM-ST-001:charger.power_kw.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.charger.measurements.power_kw.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.

Entry	Status	Evidence	Remaining coverage
supervision.templates.charger.measurements.power_kw.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.charger.measurements.power_kw.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.charger.measurements.power_kw.unit	partial	expansion: telemetry.boundary.SAM-ST-001:charger.power_kw.below; expansion: telemetry.boundary.SAM-ST-001:charger.power_kw.min; expansion: telemetry.boundary.SAM-ST-001:charger.power_kw.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.charger.measurements.temperature_c.max	varied	expansion: telemetry.interaction.1; expansion: telemetry.interaction.2; expansion: telemetry.interaction.3; +69 observations (JSON)	72 declared scale/offset/raw/quality combinations; remaining numeric domain and cross-setting combinations are not exhausted.
supervision.templates.charger.measurements.temperature_c.min	varied	expansion: telemetry.interaction.1; expansion: telemetry.interaction.2; expansion: telemetry.interaction.3; +69 observations (JSON)	72 declared scale/offset/raw/quality combinations; remaining numeric domain and cross-setting combinations are not exhausted.
supervision.templates.charger.measurements.temperature_c.offset	varied	expansion: telemetry.interaction.1; expansion: telemetry.interaction.2; expansion: telemetry.interaction.3; +69 observations (JSON)	72 declared scale/offset/raw/quality combinations; remaining numeric domain and cross-setting combinations are not exhausted.
supervision.templates.charger.measurements.temperature_c.scale	varied	expansion: telemetry.interaction.1; expansion: telemetry.interaction.2; expansion: telemetry.interaction.3; +69 observations (JSON)	72 declared scale/offset/raw/quality combinations; remaining numeric domain and cross-setting combinations are not exhausted.
supervision.templates.charger.measurements.temperature_c.stale_seconds	varied	base: Changed stale threshold changes displayed quality	One live stale/disconnected threshold change; all sampling and clock interactions remain.
supervision.templates.charger.measurements.temperature_c.unit	partial	expansion: telemetry.boundary.SAM-ST-001:charger.temperature_c.below; expansion: telemetry.boundary.SAM-ST-001:charger.temperature_c.min; expansion: telemetry.boundary.SAM-ST-001:charger.temperature_c.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.charger.source_crates	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.facilities.alarms[].clear_below	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.facilities.alarms[].delay_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.facilities.alarms[].high	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.facilities.alarms[].id	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.facilities.alarms[].maintenance	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.facilities.alarms[].measurement	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.facilities.alarms[].priority	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.facilities.alarms[].repeat_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.facilities.alarms[].response	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.facilities.commands.set_lighting.max	varied	base: Changed command maximum is enforced	Representative changed limit; pending configuration, expiry and permission tests supplement it.
supervision.templates.facilities.commands.set_lighting.max_ttl_seconds	varied	base: Changed command lifetime is enforced	Representative changed limit; pending configuration, expiry and permission tests supplement it.
supervision.templates.facilities.commands.set_lighting.min	varied	base: Changed command minimum is enforced	Representative changed limit; pending configuration, expiry and permission tests supplement it.
supervision.templates.facilities.commands.set_lighting.parameter	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.facilities.commands.set_lighting.required_conditions	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.

Entry	Status	Evidence	Remaining coverage
supervision.templates.facilities.commands.set_lighting.simulation_only	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.facilities.component_type_id	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.facilities.label	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.facilities.measurements.fault_count.max	partial	expansion: telemetry.boundary.SAM-ST-001:facilities.fault_count.below; expansion: telemetry.boundary.SAM-ST-001:facilities.fault_count.min; expansion: telemetry.boundary.SAM-ST-001:facilities.fault_count.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.facilities.measurements.fault_count.min	partial	expansion: telemetry.boundary.SAM-ST-001:facilities.fault_count.below; expansion: telemetry.boundary.SAM-ST-001:facilities.fault_count.min; expansion: telemetry.boundary.SAM-ST-001:facilities.fault_count.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.facilities.measurements.fault_count.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.facilities.measurements.fault_count.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.facilities.measurements.fault_count.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.facilities.measurements.fault_count.unit	partial	expansion: telemetry.boundary.SAM-ST-001:facilities.fault_count.below; expansion: telemetry.boundary.SAM-ST-001:facilities.fault_count.min; expansion: telemetry.boundary.SAM-ST-001:facilities.fault_count.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.facilities.measurements.lighting_pct.max	partial	expansion: telemetry.boundary.SAM-ST-001:facilities.lighting_pct.below; expansion: telemetry.boundary.SAM-ST-001:facilities.lighting_pct.min; expansion: telemetry.boundary.SAM-ST-001:facilities.lighting_pct.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.facilities.measurements.lighting_pct.min	partial	expansion: telemetry.boundary.SAM-ST-001:facilities.lighting_pct.below; expansion: telemetry.boundary.SAM-ST-001:facilities.lighting_pct.min; expansion: telemetry.boundary.SAM-ST-001:facilities.lighting_pct.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.facilities.measurements.lighting_pct.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.facilities.measurements.lighting_pct.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.facilities.measurements.lighting_pct.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.facilities.measurements.lighting_pct.unit	partial	expansion: telemetry.boundary.SAM-ST-001:facilities.lighting_pct.below; expansion: telemetry.boundary.SAM-ST-001:facilities.lighting_pct.min; expansion: telemetry.boundary.SAM-ST-001:facilities.lighting_pct.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.facilities.measurements.pump_running.max	partial	expansion: telemetry.boundary.SAM-ST-001:facilities.pump_running.below; expansion: telemetry.boundary.SAM-ST-001:facilities.pump_running.min; expansion: telemetry.boundary.SAM-ST-001:facilities.pump_running.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.facilities.measurements.pump_running.min	partial	expansion: telemetry.boundary.SAM-ST-001:facilities.pump_running.below; expansion: telemetry.boundary.SAM-ST-001:facilities.pump_running.min; expansion: telemetry.boundary.SAM-ST-001:facilities.pump_running.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.facilities.measurements.pump_running.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.facilities.measurements.pump_running.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.facilities.measurements.pump_running.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.

Entry	Status	Evidence	Remaining coverage
supervision.templates.facilities.measurements.pump_running.unit	partial	expansion: telemetry.boundary.SAM-ST-001:facilities.pump_running.below; expansion: telemetry.boundary.SAM-ST-001:facilities.pump_running.min; expansion: telemetry.boundary.SAM-ST-001:facilities.pump_running.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.facilities.source_crates	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.faregate.alarms	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.faregate.asset_types	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.faregate.commands	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.faregate.component_type_id	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.faregate.label	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.faregate.measurements.denial_count.max	partial	expansion: telemetry.boundary.SAM-ST-001:faregate.denial_count.below; expansion: telemetry.boundary.SAM-ST-001:faregate.denial_count.min; expansion: telemetry.boundary.SAM-ST-001:faregate.denial_count.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.faregate.measurements.denial_count.min	partial	expansion: telemetry.boundary.SAM-ST-001:faregate.denial_count.below; expansion: telemetry.boundary.SAM-ST-001:faregate.denial_count.min; expansion: telemetry.boundary.SAM-ST-001:faregate.denial_count.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.faregate.measurements.denial_count.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.faregate.measurements.denial_count.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.faregate.measurements.denial_count.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.faregate.measurements.denial_count.unit	partial	expansion: telemetry.boundary.SAM-ST-001:faregate.denial_count.below; expansion: telemetry.boundary.SAM-ST-001:faregate.denial_count.min; expansion: telemetry.boundary.SAM-ST-001:faregate.denial_count.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.faregate.measurements.gate_open.max	partial	expansion: telemetry.boundary.SAM-ST-001:faregate.gate_open.below; expansion: telemetry.boundary.SAM-ST-001:faregate.gate_open.min; expansion: telemetry.boundary.SAM-ST-001:faregate.gate_open.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.faregate.measurements.gate_open.min	partial	expansion: telemetry.boundary.SAM-ST-001:faregate.gate_open.below; expansion: telemetry.boundary.SAM-ST-001:faregate.gate_open.min; expansion: telemetry.boundary.SAM-ST-001:faregate.gate_open.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.faregate.measurements.gate_open.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.faregate.measurements.gate_open.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.faregate.measurements.gate_open.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.faregate.measurements.gate_open.unit	partial	expansion: telemetry.boundary.SAM-ST-001:faregate.gate_open.below; expansion: telemetry.boundary.SAM-ST-001:faregate.gate_open.min; expansion: telemetry.boundary.SAM-ST-001:faregate.gate_open.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.faregate.measurements.grant_count.max	partial	expansion: telemetry.boundary.SAM-ST-001:faregate.grant_count.below; expansion: telemetry.boundary.SAM-ST-001:faregate.grant_count.min; expansion: telemetry.boundary.SAM-ST-001:faregate.grant_count.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.

Entry	Status	Evidence	Remaining coverage
supervision.templates.faregate.measurements.grant_count.min	partial	expansion: telemetry.boundary.SAM-ST-001:faregate.grant_count.below; expansion: telemetry.boundary.SAM-ST-001:faregate.grant_count.min; expansion: telemetry.boundary.SAM-ST-001:faregate.grant_count.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.faregate.measurements.grant_count.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.faregate.measurements.grant_count.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.faregate.measurements.grant_count.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.faregate.measurements.grant_count.unit	partial	expansion: telemetry.boundary.SAM-ST-001:faregate.grant_count.below; expansion: telemetry.boundary.SAM-ST-001:faregate.grant_count.min; expansion: telemetry.boundary.SAM-ST-001:faregate.grant_count.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.faregate.measurements.last_decision.max	partial	expansion: telemetry.boundary.SAM-ST-001:faregate.last_decision.below; expansion: telemetry.boundary.SAM-ST-001:faregate.last_decision.min; expansion: telemetry.boundary.SAM-ST-001:faregate.last_decision.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.faregate.measurements.last_decision.min	partial	expansion: telemetry.boundary.SAM-ST-001:faregate.last_decision.below; expansion: telemetry.boundary.SAM-ST-001:faregate.last_decision.min; expansion: telemetry.boundary.SAM-ST-001:faregate.last_decision.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.faregate.measurements.last_decision.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.faregate.measurements.last_decision.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.faregate.measurements.last_decision.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.faregate.measurements.last_decision.unit	partial	expansion: telemetry.boundary.SAM-ST-001:faregate.last_decision.below; expansion: telemetry.boundary.SAM-ST-001:faregate.last_decision.min; expansion: telemetry.boundary.SAM-ST-001:faregate.last_decision.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.faregate.source_crates	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.alarms[].clear_below	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.alarms[].delay_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.alarms[].high	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.alarms[].id	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.alarms[].maintenance	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.alarms[].measurement	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.alarms[].priority	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.alarms[].repeat_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.alarms[].response	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.asset_types	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.commands	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.

Entry	Status	Evidence	Remaining coverage
supervision.templates.level-crossing.component_type_id	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.label	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.measurements.fault.max	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.measurements.fault.min	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.measurements.fault.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.measurements.fault.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.measurements.fault.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.measurements.fault.unit	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.measurements.state.max	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.measurements.state.min	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.measurements.state.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.measurements.state.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.measurements.state.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.measurements.state.unit	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.measurements.warning_active.max	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.measurements.warning_active.min	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.measurements.warning_active.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.measurements.warning_active.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.measurements.warning_active.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.measurements.warning_active.unit	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.level-crossing.source_crates	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.points.alarms[].clear_below	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.points.alarms[].delay_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.points.alarms[].high	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.points.alarms[].id	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.points.alarms[].maintenance	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.points.alarms[].measurement	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.

Entry	Status	Evidence	Remaining coverage
supervision.templates.points.alarms[].priority	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.points.alarms[].repeat_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.points.alarms[].response	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.points.asset_types	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.points.commands	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.points.component_type_id	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.points.label	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.points.measurements.detected_position.max	partial	expansion: telemetry.boundary.SAM-SW-001:points.detected_position.below; expansion: telemetry.boundary.SAM-SW-001:points.detected_position.min; expansion: telemetry.boundary.SAM-SW-001:points.detected_position.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.points.measurements.detected_position.min	partial	expansion: telemetry.boundary.SAM-SW-001:points.detected_position.below; expansion: telemetry.boundary.SAM-SW-001:points.detected_position.min; expansion: telemetry.boundary.SAM-SW-001:points.detected_position.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.points.measurements.detected_position.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.points.measurements.detected_position.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.points.measurements.detected_position.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.points.measurements.detected_position.unit	partial	expansion: telemetry.boundary.SAM-SW-001:points.detected_position.below; expansion: telemetry.boundary.SAM-SW-001:points.detected_position.min; expansion: telemetry.boundary.SAM-SW-001:points.detected_position.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.points.measurements.detection_unknown.max	partial	expansion: telemetry.boundary.SAM-SW-001:points.detection_unknown.below; expansion: telemetry.boundary.SAM-SW-001:points.detection_unknown.min; expansion: telemetry.boundary.SAM-SW-001:points.detection_unknown.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.points.measurements.detection_unknown.min	partial	expansion: telemetry.boundary.SAM-SW-001:points.detection_unknown.below; expansion: telemetry.boundary.SAM-SW-001:points.detection_unknown.min; expansion: telemetry.boundary.SAM-SW-001:points.detection_unknown.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.points.measurements.detection_unknown.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.points.measurements.detection_unknown.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.points.measurements.detection_unknown.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.points.measurements.detection_unknown.unit	partial	expansion: telemetry.boundary.SAM-SW-001:points.detection_unknown.below; expansion: telemetry.boundary.SAM-SW-001:points.detection_unknown.min; expansion: telemetry.boundary.SAM-SW-001:points.detection_unknown.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
Entry	**Status**	**Evidence**	**Remaining coverage**
supervision.templates.points.measurements.motor_active.max	partial	expansion: telemetry.boundary.SAM-SW-001:points.motor_active.below; expansion: telemetry.boundary.SAM-SW-001:points.m	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.

Entry	Status	Evidence	Remaining coverage
		otor_active.min; expansion:telemetry.boundary.SAM-SW-001:points.motor_active.max; +1 observations (JSON)	
supervision.templates.points.measurements.motor_active.min	partial	expansion: telemetry.boundary.SAM-SW-001:points.motor_active.below; expansion: telemetry.boundary.SAM-SW-001:points.motor_active.min; expansion: telemetry.boundary.SAM-SW-001:points.motor_active.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.points.measurements.motor_active.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.points.measurements.motor_active.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.points.measurements.motor_active.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.points.measurements.motor_active.unit	partial	expansion: telemetry.boundary.SAM-SW-001:points.motor_active.below; expansion: telemetry.boundary.SAM-SW-001:points.motor_active.min; expansion: telemetry.boundary.SAM-SW-001:points.motor_active.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.points.source_crates	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.pv.alarms	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.pv.component_type_id	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.pv.label	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.pv.measurements.energy_kwh.max	partial	expansion: telemetry.boundary.SAM-ST-001:pv.energy_kwh.below; expansion: telemetry.boundary.SAM-ST-001:pv.energy_kwh.min; expansion: telemetry.boundary.SAM-ST-001:pv.energy_kwh.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.pv.measurements.energy_kwh.min	partial	expansion: telemetry.boundary.SAM-ST-001:pv.energy_kwh.below; expansion: telemetry.boundary.SAM-ST-001:pv.energy_kwh.min; expansion: telemetry.boundary.SAM-ST-001:pv.energy_kwh.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.pv.measurements.energy_kwh.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.pv.measurements.energy_kwh.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.pv.measurements.energy_kwh.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.pv.measurements.energy_kwh.unit	partial	expansion: telemetry.boundary.SAM-ST-001:pv.energy_kwh.below; expansion: telemetry.boundary.SAM-ST-001:pv.energy_kwh.min; expansion: telemetry.boundary.SAM-ST-001:pv.energy_kwh.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.pv.measurements.power_kw.max	partial	expansion: telemetry.boundary.SAM-ST-001:pv.power_kw.below; expansion: telemetry.boundary.SAM-ST-001:pv.power_kw.min; expansion: telemetry.boundary.SAM-ST-001:pv.power_kw.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.pv.measurements.power_kw.min	partial	expansion: telemetry.boundary.SAM-ST-001:pv.power_kw.below; expansion: telemetry.boundary.SAM-ST-001:pv.power_kw.min; expansion: telemetry.boundary.SAM-ST-001:pv.power_kw.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.pv.measurements.power_kw.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.pv.measurements.power_kw.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.pv.measurements.power_kw.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.

Entry	Status	Evidence	Remaining coverage
supervision.templates.pv.measurements.power_kw.unit	partial	expansion: telemetry.boundary.SAM-ST-001:pv.power_kw.below; expansion: telemetry.boundary.SAM-ST-001:pv.power_kw.min; expansion: telemetry.boundary.SAM-ST-001:pv.power_kw.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.pv.source_crates	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-aux.alarms[].clear_below	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-aux.alarms[].delay_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-aux.alarms[].high	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-aux.alarms[].id	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-aux.alarms[].maintenance	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-aux.alarms[].measurement	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-aux.alarms[].priority	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-aux.alarms[].repeat_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-aux.alarms[].response	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-aux.asset_types	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-aux.component_type_id	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-aux.label	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-aux.measurements.comfort_power.max	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-aux.comfort_power.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-aux.comfort_power.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-aux.comfort_power.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-aux.measurements.comfort_power.min	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-aux.comfort_power.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-aux.comfort_power.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-aux.comfort_power.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-aux.measurements.comfort_power.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-aux.measurements.comfort_power.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-aux.measurements.comfort_power.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-aux.measurements.comfort_power.unit	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-aux.comfort_power.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-aux.comfort_power.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-aux.comfort_power.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-aux.measurements.fault_count.max	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-aux.fault_count.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-aux.fault_count.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-aux.fault_count.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.

Entry	Status	Evidence	Remaining coverage
supervision.templates.vehicle-aux.measurements.fault_count.min	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-aux.fault_count.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-aux.fault_count.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-aux.fault_count.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-aux.measurements.fault_count.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-aux.measurements.fault_count.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-aux.measurements.fault_count.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-aux.measurements.fault_count.unit	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-aux.fault_count.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-aux.fault_count.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-aux.fault_count.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-aux.source_crates	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-bms.alarms[].clear_below	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-bms.alarms[].delay_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-bms.alarms[].high	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-bms.alarms[].id	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-bms.alarms[].maintenance	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-bms.alarms[].measurement	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-bms.alarms[].priority	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-bms.alarms[].repeat_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-bms.alarms[].response	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-bms.asset_types	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-bms.component_type_id	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-bms.label	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-bms.measurements.charge_limit_a.max	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-bms.charge_limit_a.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-bms.charge_limit_a.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-bms.charge_limit_a.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-bms.measurements.charge_limit_a.min	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-bms.charge_limit_a.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-bms.charge_limit_a.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-bms.charge_limit_a.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-bms.measurements.charge_limit_a.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-bms.measurements.charge_limit_a.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-bms.measurements.charge_limit_a.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.

Entry	Status	Evidence	Remaining coverage
supervision.templates.vehicle-bms.measurements.charge_limit_a.unit	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-bms.charge_limit_a.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-bms.charge_limit_a.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-bms.charge_limit_a.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-bms.measurements.soc_pct.max	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-bms.soc_pct.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-bms.soc_pct.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-bms.soc_pct.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-bms.measurements.soc_pct.min	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-bms.soc_pct.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-bms.soc_pct.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-bms.soc_pct.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-bms.measurements.soc_pct.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-bms.measurements.soc_pct.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-bms.measurements.soc_pct.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-bms.measurements.soc_pct.unit	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-bms.soc_pct.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-bms.soc_pct.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-bms.soc_pct.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-bms.measurements.trip.max	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-bms.trip.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-bms.trip.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-bms.trip.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-bms.measurements.trip.min	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-bms.trip.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-bms.trip.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-bms.trip.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-bms.measurements.trip.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-bms.measurements.trip.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-bms.measurements.trip.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-bms.measurements.trip.unit	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-bms.trip.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-bms.trip.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-bms.trip.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-bms.source_crates	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-cbm.alarms[].clear_below	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-cbm.alarms[].delay_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-cbm.alarms[].high	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-cbm.alarms[].id	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-cbm.alarms[].maintenance	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-cbm.alarms[].measurement	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-cbm.alarms[].priority	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.

Entry	Status	Evidence	Remaining coverage
supervision.templates.vehicle-cbm.alarms[].repeat_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-cbm.alarms[].response	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-cbm.asset_types	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-cbm.component_type_id	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-cbm.label	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-cbm.measurements.bearing_vibration_mm_s.max	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-cbm.bearing_vibration_mm_s.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-cbm.bearing_vibration_mm_s.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-cbm.bearing_vibration_mm_s.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-cbm.measurements.bearing_vibration_mm_s.min	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-cbm.bearing_vibration_mm_s.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-cbm.bearing_vibration_mm_s.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-cbm.bearing_vibration_mm_s.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-cbm.measurements.bearing_vibration_mm_s.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-cbm.measurements.bearing_vibration_mm_s.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-cbm.measurements.bearing_vibration_mm_s.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-cbm.measurements.bearing_vibration_mm_s.unit	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-cbm.bearing_vibration_mm_s.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-cbm.bearing_vibration_mm_s.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-cbm.bearing_vibration_mm_s.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-cbm.measurements.brake_remaining_pct.max	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-cbm.brake_remaining_pct.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-cbm.brake_remaining_pct.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-cbm.brake_remaining_pct.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-cbm.measurements.brake_remaining_pct.min	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-cbm.brake_remaining_pct.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-cbm.brake_remaining_pct.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-cbm.brake_remaining_pct.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-cbm.measurements.brake_remaining_pct.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-cbm.measurements.brake_remaining_pct.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-cbm.measurements.brake_remaining_pct.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-cbm.measurements.brake_remaining_pct.unit	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-cbm.brake_remaining_pct.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-cbm.brake_remaining_pct.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-cbm.brake_remaining_pct.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-cbm.measurements.health.max	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-cbm.health.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-cbm.health.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-cbm.health.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-cbm.measurements.health.min	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-cbm.health.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-cbm.health.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-cbm.health.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.

Entry	Status	Evidence	Remaining coverage
supervision.templates.vehicle-cbm.measurements.health.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-cbm.measurements.health.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-cbm.measurements.health.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-cbm.measurements.health.unit	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-cbm.health.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-cbm.health.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-cbm.health.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-cbm.source_crates	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-hvac.alarms[].clear_below	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-hvac.alarms[].delay_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-hvac.alarms[].high	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-hvac.alarms[].id	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-hvac.alarms[].maintenance	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-hvac.alarms[].measurement	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-hvac.alarms[].priority	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-hvac.alarms[].repeat_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-hvac.alarms[].response	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-hvac.asset_types	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-hvac.component_type_id	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-hvac.label	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-hvac.measurements.compressor_pct.max	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-hvac.compressor_pct.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-hvac.compressor_pct.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-hvac.compressor_pct.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-hvac.measurements.compressor_pct.min	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-hvac.compressor_pct.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-hvac.compressor_pct.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-hvac.compressor_pct.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-hvac.measurements.compressor_pct.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-hvac.measurements.compressor_pct.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-hvac.measurements.compressor_pct.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-hvac.measurements.compressor_pct.unit	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-hvac.compressor_pct.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-hvac.compressor_pct.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-hvac.compressor_pct.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.

Entry	Status	Evidence	Remaining coverage
supervision.templates.vehicle-hvac.measurements.fan_pct.max	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-hvac.fan_pct.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-hvac.fan_pct.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-hvac.fan_pct.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-hvac.measurements.fan_pct.min	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-hvac.fan_pct.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-hvac.fan_pct.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-hvac.fan_pct.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-hvac.measurements.fan_pct.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-hvac.measurements.fan_pct.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-hvac.measurements.fan_pct.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-hvac.measurements.fan_pct.unit	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-hvac.fan_pct.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-hvac.fan_pct.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-hvac.fan_pct.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-hvac.measurements.reduced.max	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-hvac.reduced.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-hvac.reduced.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-hvac.reduced.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-hvac.measurements.reduced.min	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-hvac.reduced.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-hvac.reduced.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-hvac.reduced.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-hvac.measurements.reduced.offset	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-hvac.measurements.reduced.scale	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-hvac.measurements.reduced.stale_seconds	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
supervision.templates.vehicle-hvac.measurements.reduced.unit	partial	expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-hvac.reduced.below; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-hvac.reduced.min; expansion: telemetry.boundary.SAM-RS-L1-001:vehicle-hvac.reduced.max; +1 observations (JSON)	Live min/max/beyond samples exercise this contract; arbitrary changes to the configured field remain.
supervision.templates.vehicle-hvac.source_crates	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
workbench.GET./api/assurance/readiness	gap	—	Local browser tests exercise reference scope, city switching, mismatched context and changed-evidence rejection. Map a complete example-deployment report before claiming scenario coverage; physical qualification remains open.
workbench.GET./api/lifecycle/	scenario	base: Workbench, ERP, FUXA, engineering evidence and lifecycle agree in the browser	Real browser lifecycle and railway handoff; per-route error/permission and all payload variants remain.
workbench.GET./api/operating/twins	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
workbench.GET./api/operations/	gap	—	Isolated operations Python/gateway/browser checks exist; native complete-city ERP delivery and deployed combinations remain unexercised.
workbench.GET./api/portfolio	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
workbench.GET./api/twins/catalogue	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
workbench.GET./api/twins/jobs/	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.

Entry	Status	Evidence	Remaining coverage
workbench.GET./api/workbench	scenario	base: Workbench, ERP, FUXA, engineering evidence and lifecycle agree in the browser	Real browser lifecycle and railway handoff; per-route error/permission and all payload variants remain.
workbench.GET./api/workbench/city	scenario	base: Workbench, ERP, FUXA, engineering evidence and lifecycle agree in the browser	Real browser lifecycle and railway handoff; per-route error/permission and all payload variants remain.
workbench.GET./api/workbench/services	scenario	base: Workbench, ERP, FUXA, engineering evidence and lifecycle agree in the browser	Real browser lifecycle and railway handoff; per-route error/permission and all payload variants remain.
workbench.POST./api/lifecycle/	scenario	base: Workbench, ERP, FUXA, engineering evidence and lifecycle agree in the browser	Real browser lifecycle and railway handoff; per-route error/permission and all payload variants remain.
workbench.POST./api/operations/	gap	—	Isolated operations Python/gateway/browser checks exist; native complete-city ERP delivery and deployed combinations remain unexercised.
workbench.POST./api/ops-auth/	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
workbench.POST./api/ops-core/	scenario	base: Workbench, ERP, FUXA, engineering evidence and lifecycle agree in the browser	Real browser lifecycle and railway handoff; per-route error/permission and all payload variants remain.
workbench.POST./api/twins/generate/	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
workbench.setting.--city-port	scenario	base: Full stop/start preserves the retained City Studio project configuration	Independent example paths/ports used and retained through restart; arbitrary deployment permutations and invalid values remain.
workbench.setting.--db	scenario	base: Full stop/start preserves the retained City Studio project configuration	Independent example paths/ports used and retained through restart; arbitrary deployment permutations and invalid values remain.
workbench.setting.--host	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
workbench.setting.--isolated-project	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
workbench.setting.--port	scenario	base: Full stop/start preserves the retained City Studio project configuration	Independent example paths/ports used and retained through restart; arbitrary deployment permutations and invalid values remain.
workbench.setting.--project	scenario	base: Full stop/start preserves the retained City Studio project configuration	Independent example paths/ports used and retained through restart; arbitrary deployment permutations and invalid values remain.
workbench.setting.--reset-db	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
workbench.setting.OSR_ERP_SNAPSHOT	scenario	base: Full stop/start preserves the retained City Studio project configuration	Independent example paths/ports used and retained through restart; arbitrary deployment permutations and invalid values remain.
workbench.setting.OSR_ERP_URL	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
workbench.setting.OSR_FUXA_URL	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
workbench.setting.OSR_INTEGRATION_URL	scenario	base: Full stop/start preserves the retained City Studio project configuration	Independent example paths/ports used and retained through restart; arbitrary deployment permutations and invalid values remain.
workbench.setting.OSR_QUALIFICATION_REVIEW_POLICY	gap	—	Signature verification has isolated regression coverage. Exercise a server-configured trusted reviewer policy and invalid policy paths in the complete example deployment before claiming scenario coverage.
workbench.setting.OSR_SUPERVISION_CONFIG	scenario	base: Full stop/start preserves the retained City Studio project configuration	Independent example paths/ports used and retained through restart; arbitrary deployment permutations and invalid values remain.
workbench.setting.OSR_SUPERVISION_PROFILES	gap	—	No mapped executed scenario for this inventory entry; add explicit boundary, permission and interaction checks.
workbench.setting.OSR_SUPERVISION_ROOT	scenario	base: Full stop/start preserves the retained City Studio project configuration	Independent example paths/ports used and retained through restart; arbitrary deployment permutations and invalid values remain.

Entry	Status	Evidence	Remaining coverage
workflow.assignment-execution	varied	business: business.assignment.Round Robin.0; business: business.assignment.Round Robin.1; business: business.assignment.distribution.Round Robin; +3 observations (JSON)	Actual enabled Round Robin and Load Balancing assignment to two users. Holidays, disabled users and concurrent dispatch remain.
workflow.backup-restore	gap	—	A separate source-bound cold recovery rehearsal verifies fresh volumes and cross-service data (docs/operating/platform-recovery.md); this register does not yet map that report. Operational cutover, jobs/controllers and recovery objectives remain.
workflow.budget-enforcement	varied	business: business.budget.Stop.50; business: business.budget.Stop.100; business: business.budget.Stop.101; +3 observations (JSON)	Submitted actual expenses at/below/above Stop/Warn budgets. Purchase commitments, fiscal-period boundaries and simultaneous posting remain.
workflow.capitalization	gap	—	Connect received/manufactured serialized equipment to capitalization, depreciation and retirement entries.
workflow.disposition	scenario	disposition: disposition.distinct-authenticated-users; disposition: disposition.browser-proposal-recorded; disposition: disposition.self-endorsement-rejected; +6 observations (JSON)	Unmocked browser with three independent accounts covers proposal, endorsement, native stop, verification and stale detection; all target/action/concurrency permutations remain.
workflow.engineering-desktop	gap	—	Run real FreeCAD/Bonsai/QGIS edits, regenerate artifacts, then propagate changed quantities and mappings downstream.
workflow.hardware	gap	—	Commission real device bindings and calibration, perform HIL/site tests and independently accept results.
workflow.load	gap	—	Run simultaneous full-city equipment and multiple browser/operator clients; measure latency, loss and resource ceilings.
workflow.multi-city-isolation	scenario	base: Control city native ERP tasks and transactions remain unchanged; expansion: scope.control-package-unchanged	Two-city scenario; more city families, mixed timezones and concurrent multi-city load remain.
workflow.payments	varied	expansion: erp.payment.remaining.40; expansion: erp.payment.remaining.0; expansion: erp.payment.cancel.40; +1 observations (JSON)	Real partial/full/cancelled payments and ledger balancing in a rollback transaction; multiple currencies, bank reconciliation and bank transport remain.
workflow.payroll	gap	—	Create approved employee/payroll fixtures, attendance/leave variations, salary calculation, posting and payment.
workflow.railway-handoff	scenario	base: Workbench, ERP, FUXA, engineering evidence and lifecycle agree in the browser	City Studio to simulator/OCC/OSR work in simulation; operating permutations and physical acceptance remain.
workflow.replenishment-scheduler	varied	business: business.replenishment.threshold.3; business: business.replenishment.threshold.2; business: business.replenishment.quantity.2; +5 observations (JSON)	Native scheduler at/below/above threshold, quantities, city identity and stale/ambiguous rules. Multiple suppliers, lead times and live concurrent scheduler cycles remain.
workflow.service-sla	gap	—	Vary working calendars, priorities and elapsed time; observe deadlines and escalation.
workflow.tax	gap	—	Add jurisdiction-specific approved fixtures, tax calculation, returns and corrections; no statutory filing is performed by these tests.
workflow.training-competence	partial	business: business.training.Fail; business: business.training.Pass; business: business.training.invalid-period	Submitted training, attendance and Pass/Fail results; invalid periods rejected. Independent competence decisions and authorization remain.

Equipment supervision and lifecycle integration

Repository source: `deployment/supervision/README.md`

ERPNext owns business execution; FUXA displays equipment; OSR owns engineering, controller behaviour and railway assurance. The integration joins their identities without treating a delivery, alarm clearance or ERP case closure as railway release.

Run the local pilot

Start the existing ERP stack first (`./osr erp up`). Then:

```
./osr supervision init
./osr supervision up
./osr supervision setup-fuxa
./osr supervision connect-erp samawah mosul
./osr supervision prepare samawah --first-site --first-vehicle --first-plant
./osr supervision prepare mosul --first-site --first-vehicle --first-plant
./osr supervision apply build/supervision/samawah/simulation/package.json
./osr supervision apply build/supervision/mosul/simulation/package.json
./osr supervision preview-fuxa \
  build/supervision/samawah/simulation/package.json \
  build/supervision/mosul/simulation/package.json \
  --output build/supervision/fuxa-import-review.json
./osr supervision import-fuxa \
  build/supervision/samawah/simulation/package.json \
  build/supervision/mosul/simulation/package.json \
  --review build/supervision/fuxa-import-review.json
./osr supervision simulate
```

For a separate Workbench, pass `--workbench-url http://127.0.0.1:8190` to `prepare`, `preview-fuxa` and `import-fuxa`. Use the same origin for preview and import: generated asset links and the origin are bound into the reviewed project hash. The default remains `http://127.0.0.1:8090`.

- Workbench lifecycle: http://127.0.0.1:8090/docs/lifecycle/
- FUXA: http://127.0.0.1:1881/home
- ERPNext: http://127.0.0.1:8080/app/opensourcerail
- Integration health: http://127.0.0.1:8092/health

`./osr supervision init` also installs the private policy-engine/secretariat identities and attestation key for the experimental AI executive council. It creates no pretend model accounts: each real model runner must be separately configured with a scoped `executive-model` principal and fixed provider, family and perspective. The council can emit only inspectable, unsubmitted ERP draft packets; it has no path to the controller command API. `connect-erp` privately provisions the council verification key into ERPNext. The gateway retries authorized draft packets until ERPNext has independently verified and idempotently mapped them to a native unsubmitted document. The service account still needs the least-privilege native create/read permissions for each enabled document type and its referenced masters. Missing permission or native validation rejects the packet visibly; provisioning does not silently add finance, stock, manufacturing or maintenance manager roles.

Use one separately secured `executive-model-runner.py` process per model identity. Its JSON configuration names credential environment variables; it contains no token. The runner accepts only the versioned OSR adapter response contract and records exact prompt/response hashes with its ballot. Submit and seal proposals with `./osr supervision council-propose` and `./osr supervision council-finalize`; finalization cannot manufacture missing quorum or erase dissent.

`var/supervision/fuxa.json` contains editor (`admin`) and operator (`operator`) passwords. Private integration tokens are in `var/supervision/integration.json`. No credentials are included in generated FUXA project exports. The operator cannot change the project. Workbench reads use a scoped viewer token kept on the server. All published ports bind to loopback. This is a local deployment; remote operation requires the organisation's TLS, identity, network and access configuration.

FUXA 1.3.4 and the Python runtime are pinned by image digest. The adapter generates FUXA's project export format and imports it through `/api/project`, also used by its UI. It does not modify FUXA's database. Global tag IDs are unique across cities. FUXA's native connection-status tags hide values when the gateway is stale/offline. Import is review-bound: the preview records every included city/environment package, revision, checksum, site, device and view, then lists live devices/views that will be added, replaced, retained or removed and identifies changes to remaining project settings such as charts/navigation. Because the current generator has no separate custom-display input, `reviewed_display_customisations` is explicitly empty. Move an intentional live edit into a reviewed generator input before import; otherwise the preview correctly identifies it as a replacement or removal.

ERP integration principal

`./osr supervision connect-erp <city> [<city> ...]` reads each city's configured ERP project, validates the city/project pairing, creates a dedicated Support Team/Projects User and copies its credentials privately into the gateway config. It extends the gateway's simulation city scopes and restarts it. `osr_integration_users` in ERP site config restricts the service principal to those projects and simulation. The service URL is `http://frontend:8080` on the existing ERP network. Neither account is an ERP administrator.

Reproduce for another city

1. Use its tracked operations asset register and compact project-twin revision. Regenerate the city with `./osr city <slug>` only when those sources are stale.
2. Edit `cities/catalogue/.../<city>/operations/supervision.json`. Omitted fields inherit shared templates; dictionaries merge and lists replace. Set company/project and, optionally, site IDs and equipment bindings.
3. `./osr supervision prepare <slug>` generates every applicable station, depot, vehicle, points, declared-crossing and production-method view; `--first-site` includes that station's real child switch/plant assets, while `--first-plant` can select only the first production asset for a bounded factory pilot. The command no longer needs the large compressed operations payload. `./osr supervision validate` compiles every real asset register, and the generated catalogue audit records counts and hashes.
4. Review package, devices, source bindings, units, ranges, alarm limits, response references and generated screens. Generic limits are simulation examples.
5. Extend the private principal city scopes before applying. For an existing baseline, run `review-package PACKAGE --output REVIEW`, inspect its exact field/category/dependency and affected-record sets, then run `apply PACKAGE --expected PREVIOUS_SHA256 --review REVIEW`. The gateway regenerates the review inside the apply transaction and rejects changed packages, evidence, installations, open cases or pending commands. An identical apply does nothing. An omitted simulation position is retained as retired history; an installed position cannot be retired automatically and a command contract cannot change while a request is pending. Physical mapping changes require a separately commissioned migration.
6. Preview the combined set of desired city packages with `preview-fuxa`, review its manifest and add/change/remove sets, then supply that exact file to `import-fuxa --review`. Any intervening live-project or package change is rejected. Import replaces the complete project and privately saves both the previous export and applied review first.

`--environment physical` produces a commissioning-required package. The service rejects telemetry and commands for those uncommissioned bindings. Vendor register maps, verified limits, real device adapters and OSR operational release must be supplied before physical activation. No railway or protection command is exposed.

Polling scope

Shared configuration defaults to `"fuxa_polling_scope": "asset"`. A city's `operations/supervision.json` can set `"fuxa_polling_scope": "site"`, or use `./osr supervision prepare mosul --polling-scope site` for a reviewed export. The site setting groups active equipment into one private WebAPI request per city/environment/site. Measurement and alarm tag IDs, quality, timestamps and individual equipment links are retained; one connection-status tag describes each shared transport. It does not change sampling, historian or alarm rules.

Switching scope replaces transport device IDs and changes affected display bindings. Apply the package review and then preview/import the combined FUXA project using the steps above; keep all other cities in that combined import. Existing packages without the field retain asset polling. The example-city scenario exercises both settings. See the measured capacity results.

Data and failure behaviour

- Stable key: city + environment + planned asset ID. Design revision, ERP project baseline, physical serial, IFC GlobalId and supplier device mapping are separate references. Registry history survives revision changes.
- Telemetry: asset/measurement/source IDs, sequence, source UTC timestamp, reception time, engineering unit, normalized value and quality. Replay conflicts, unknown sources/units, non-finite values and future timestamps are rejected. Stale data cannot activate or clear an alarm.
- SQLite is the pilot's single telemetry historian, sampled every two seconds, retained for seven days; latest values remain marked stale/disconnected. FUXA DAQ is disabled. ERP receives actionable events, never the measurement stream.

- Alarm delay, hysteresis and repeat suppression are persisted. One incident maps to one native Issue. Clearing the signal preserves the case. A later new fault after case closure starts a new incident. Acknowledgement is immutable for one occurrence; a fresh activation resets it, and a stale browser cannot accidentally acknowledge the newer occurrence. Acknowledgement is separate.
- A linked commissioned ERP Asset allows a permission-checked maintainer to preview parts, city-store availability, technician and expected downtime, then create one draft native Asset Repair. ERP completion/stock consumption feeds back to the project, but does not close the Issue or alter railway handback state.
- Outbox events persist across restarts, retry with bounded backoff and preserve per-incident order. ERP's unique incident key and event hashes protect against duplicate requests and lost replies. Reconciliation reads native case status.
- Commands carry request ID, operator identity, parameters, created/expiry times and required conditions. Requested/accepted/rejected/completed/failed states are recorded separately. The pilot supports simulated facility lighting only. The controller checks local enable and bounds; expired requests are never replayed.
- Engineering and commissioning rehearsals are append-only evidence records. The native-stack rehearsal replaces the pilot charger serial, proves the removed serial remains queryable through `/affected`, and requires a new independent test and release. Physical release cannot be granted through this API.

The independent existing OSR assurance records are displayed by Workbench using its existing authenticated city access. No integration event changes them.

Prepared change impact

The read-only package review compares a prepared package with the accepted live city/environment baseline. It lists exact changed paths and classifies design, asset-topology, embedded-runtime, telemetry, alarm/maintenance, command, business-execution and operator-display consequences. Existing parent/source asset IDs, component type, IFC GlobalId, source crates, ERP project/Item/Asset, installed serials, append-only evidence, open alarm/ERP cases and pending commands are reused as dependencies rather than copied into a new registry. Workbench's Prepared change impact view adds matching native ERP purchase, receipt and production feedback when available.

Preview does not mutate the baseline. Apply requires both the expected prior package hash and the freshly generated review hash, and the audit records both. Another city/environment is outside the transaction. A component-type substitution, retirement of a currently installed serial, physical remapping or pending-command contract change remains blocked for an explicit replacement or commissioned migration workflow. This is configuration impact control; complete CAD/BOM dependency graphs, open-order disposition and released-analysis reruns remain engineering/ERP work.

Backup and recovery

```
./osr supervision backup
./osr erp backup
```

The first command uses SQLite's consistent backup API, exports the FUXA project and copies private deployment credentials/settings into a private timestamped folder. Restore the SQLite backup into the `integration-data` volume with integration stopped, restore the private settings, then start and import the saved FUXA project. Recreate its users with `setup-fuxa`; they are declared in private configuration. The default pilot uses built-in graphics; separately back up the `fuxa-images` volume if operators add images. FUXA logs are diagnostic, not the command audit; the authoritative command/event audit is in the integration backup.

Database restore equivalence and pending-event survival are covered by tests. FUXA export/import and editor/operator permissions are also exercised against the running pinned version. ERP retains its separate native backup/recovery process.

Sources

- FUXA 1.3.4 source
- FUXA project import/export
- Frappe REST API

Validation commands

```
tools/automation/osr-python -m pytest -q tools/automation/tests/test_supervision.py
cargo test -p osr-energy-site --all-targets
npx playwright test tests/frontend/lifecycle.spec.mjs
node deployment/supervision/tests/verify-fuxa.mjs
python3 deployment/supervision/tests/verify-pilot.py
```

The `integrated-stack` GitHub workflow creates a clean disposable ERPNext site, imports both pilot baselines, starts the gateway and pinned FUXA release, runs the native Rust controller bridge and Workbench, then executes the ERP component, procurement, lifecycle, FUXA, embedded-condition, browser and outage/recovery checks together. It destroys its evaluation volumes after collecting failure logs.

`node deployment/supervision/tests/verify-outage.mjs` deliberately disconnects the simulated source, then temporarily stops/restarts the local integration container. It verifies FUXA's disconnected state and hides cached values on gateway loss. Use it only against this evaluation deployment. The script restores the simulator and gateway in its cleanup block.

Station, vehicle, wayside and factory integration

The simulation source now executes nine existing Rust crates for station energy, station SCADA, vehicle BMS, auxiliary power, HVAC, condition monitoring, points, level crossings and fare gates. Use `prepare CITY --first-site --first-vehicle --first-plant` for the 19-position Samawah or 10-position Mosul pilot: five station views, four vehicle views, the station's existing switch and, for the LM3 family only, nine factory-method views on the real production-plant identity. Across the catalogue, 9,097 real switch IDs are reused. Crossing support remains dormant because no tracked city yet has a `level-crossing` asset; the package builder refuses to invent one. Fare gates are labelled station aggregates rather than claims about physical gate counts. See the native embedded integration contract for mappings, reproducibility, state lifetime and the controller-to-ERP test. `simulate` rebuilds the adapter and restarts its user service so source updates actually take effect. Rebuild/reapply reviewed packages and reimport FUXA views when changing generic measurement templates; take a backup first.

The factory views are generated from the validated `manufacturing-methods.json` rather than a second manually maintained recipe list. They cover the source's 120-product union, nine methods and 30 tooling families, and retain work centre, crew, cycle, step/hold-point, source and release-boundary metadata. The simulator defaults to an idle cell. `factory_method`, `factory_cycle_progress_pct`, `factory_cell_unavailable` and `factory_process_excursion` in the private `simulator-control.json` are explicit test fixtures; no cure temperature or performed quality result is invented. Unavailability or quality-hold persistence can create an ERP Issue. The Workbench lists native Work Orders only when an immutable reviewed execution mapping joins the method/product identity to its ERP Item/BOM. A hold does not modify a Work Order, create a Quality Inspection, accept/reject output or grant manufacturing/railway release.

The Workbench default is `preferred_supervision_site` in the city profile, validated against the prepared package; otherwise a vehicle site is preferred. An explicitly selected equipment position takes precedence. An unavailable configured preference never silently opens another site. Factory applicability is read from the city project-twin family; a profile cannot override that baseline. Cross-family method reuse is blocked pending a reviewed mapping.

City Generation

Repository source: `design/city-generation/README.md`

This Python package owns the deterministic geospatial side of city generation: OSM acquisition and caching, demand/cost/buildability/water rasters, source-locked open-DEM elevation and terrain slope, batch catalogue generation, station and line planning, and simulator-scenario emission. Rust routing and design crates consume its outputs.

Package	Responsibility
`osr_osm`	OSM/Geofabrik acquisition with a local cache
`osr_geo`	Demand, cost, buildability, water and open-terrain raster preparation
`osr_batch`	Catalogue batch import and regeneration
`osr_planner`	Lines, stations, anchors and deterministic network synthesis
`osr_scenario`	City design to simulation/finance/README outputs

After using the root setup instructions, test this package with:

```
pytest design/city-generation/tests -q
```

Editable inputs live in `../../cities/workspaces/`; published results live in `../../cities/catalogue/`.

Machine-Readable Inputs

Repository source: `lib/README.md`

The `lib/` tree holds the data that drives generated city designs, simulator examples, and reusable planning assumptions. These files are intended to be edited, reviewed, and versioned alongside the code.

Contents

Folder	Purpose
`templates/`	Shared TOML assumptions for rolling stock, stations, track geometry, signalling, demand, costs, finance, climate, and regulatory context
`city-batches/`	Batch lists for generated city catalogues, including the world sample and calibration set
`recipes/`	End-to-end recipe inputs that combine templates into deployment designs
`examples/`	Small hand-authored simulator/design examples
`fault-injection-scenarios/`	Scenario files for degraded-mode and safety-path demonstrations

The generated outputs from these inputs live under `cities/catalogue/`, and the design synthesis crate is `crates/osr-design`.

The shared `templates/field-evidence.toml` drives the pre-mobilisation survey/site brief for any city. Generated briefs remain unaccepted until the named project authorities approve the requirements and field deliveries. `templates/survey-control-processing.toml` defines the corresponding GNSS receipt roles and provisional RTKLIB screen; neither automated processing nor a quality code is survey acceptance. `templates/ground-model-processing.toml` then defines the QGIS/ODM/CloudCompare derivative, checkpoint, master- GeoPackage and authority gates without duplicating raw field data in Git. `templates/surveyed-alignment-processing.toml` adds the subsequent per-line OSR-ALN, LandXML round-trip, platform/yard/turnout interface and multi-discipline acceptance contract. `templates/route-station-fit-processing.toml` then defines the utility, land, flood, access, integration, road, construction- logistics, possession, issue-closure and coordinated route-fit contract. `templates/drainage-ground-design-processing.toml` defines the subsequent SWMM replay, hydrology, geotechnical zoning, foundation/ ground-treatment schedule, conditional OpenGeoSys and authority gate. `templates/structural-release-processing.toml` defines the subsequent per-asset OpenSees/CalculiX, load-case, fatigue, movement, independent-check and authority-release contract.

`templates/digital-assurance.toml` and `templates/system-fmea.toml` define the deterministic cross-domain pre-build assurance gate. The generated Markdown and JSON reports hash their sources and evidence, require component/subsystem/system coverage, and deliberately keep physical tests and independent acceptance open. `templates/corridor-resilience.toml` controls every-third-station recovery sites, remote shunt limits, constrained single-track passing loops and the retained protection functions for the distributed onboard-control research profile.

Civil cost inputs are split deliberately: edit `templates/civil-cost-calibration.toml` and the parametric civil geometry, then run `python3 tools/automation/generate-civil-cost-model.py`. The generated `templates/civil-cost-model.toml` is the contract used by city synthesis, finance, IFC provenance and documentation.

Scenario files

Repository source: `lib/examples/README.md`

OpenSourceRail simulator scenarios are plain-text TOML files describing a city's rail network: stations, lines, fleets, schedules, and climate. This directory holds the reference scenarios. Copy one and edit it to design a network for your own city.

Running a scenario

```
cargo run --release --bin osr-sim -- --config scenarios/my-city.toml \
    --duration 3600 --status-every 300
```

Useful flags:

Flag	Purpose
`--config PATH`	Path to a scenario TOML file
`--duration SECS`	How long to simulate (sim-seconds). Multi-day runs are supported — the clock wraps at midnight, dispatch schedules re-arm at service_start, and PV cycles through subsequent days. 86400 = 1 day, 604800 = 1 week.
`--time-step SECS`	Simulation tick length (default 1 s)
`--status-every SECS`	Print a status snapshot every N sim-seconds; 0 disables
`--json-out PATH`	Write the full event trace + summary as JSON
`--csv-out PATH`	Write per-train periodic snapshots as CSV, including SoC, cumulative kWh, section position, speed, acceleration, battery-draw power, roof PV, and station-charge power
`--csv-every SECS`	CSV snapshot interval (default 60 sim-seconds)

Visualization

A companion tool `osr-vis` turns a scenario TOML into a self-contained HTML file showing the network diagram:

```
cargo run --release --bin osr-vis -- --config scenarios/my-city.toml \
    --out /tmp/my-city.html
```

No runtime dependencies — open the HTML file in any browser. Useful for reviewing station placement, line structure, and energy-site distribution at a glance.

Reference scenarios

File	What it is
`example-simple.toml`	Three-station shuttle, one train. The smallest viable config — copy this as a template for a new city.
`minimal-city.toml`	Slightly richer fixture exercising every required block.
Default Samawah scenario (omit `--config`)	Compiled canonical Samawah input. Use `--config` to select another file; the CLI has no `--scenario` flag.
`hybrid-stabling.toml`	Connected three-station fixture: six station launch trains and two depot trains, for continuous return/restart tests.
`cities/catalogue/west-asia/Iraq/Samawah/samawah.toml`	The auto-planned three-line `light-metro-3car` Samawah network emitted by `osr-design` from real OSM + WorldPop data. Pass via `--config` to `osr-sim`.

File format

A scenario has the required blocks below plus optional consist, energy, fault, HABD, and inspection-reset arrays. Here's the minimum viable shape:

```
[scenario]
name = "My City Metro"
start_time = "06:00"        # sim wall-clock start, HH:MM

[climate]
ambient_c = 30.0             # degrees Celsius
peak_sun_hours = 5.0         # used by station/depot PV and onboard roof PV

[[stations]]
id = "terminal-a"            # string ID, used by lines & fleets below
name = "Terminal A"          # human-readable
charging_power_kw = 1000     # 0 or omit for no charging pad
dwell_seconds = 60
is_terminal = true           # trains turn around here (linear lines)
# is_depot = false           # optional; marks a depot location

# ... more [[stations]] ...

[[lines]]
id = "line-1"
name = "Line 1"
is_ring = false              # true for a closed loop
```

```
# ring_wrap_length_m = 1600  # required if is_ring = true
stations = [
    { id = "terminal-a", distance_from_prev_m = 0 },
    { id = "stop-1",     distance_from_prev_m = 1500 },
    { id = "terminal-b", distance_from_prev_m = 1500 },
]

[[fleets]]
line = "line-1"              # must match a line id above
trainset_count = 4
dispatch_points = [          # where trains start AND where dispatches are
    { station = "terminal-a", heading = "forward" },
    { station = "terminal-b", heading = "reverse" },
]                            # throttled to honor the schedule
service_start = "05:30"
service_end   = "23:30"
schedule = [
    { from = "05:30", to = "07:00", headway_min = 10 },
    { from = "07:00", to = "22:00", headway_min = 5 },
    { from = "22:00", to = "23:30", headway_min = 15 },
]
```

Field reference

`[scenario]`

- `name` — any string, shown at run start.
- `start_time` — `"HH:MM"`, the sim's wall clock at tick 0. Set it earlier than the schedule's `service_start` to watch the morning startup; set it later to focus on midday or evening service.

`[climate]`

- `ambient_c` — ambient temperature in °C. Drives the HVAC uplift applied to traction energy consumption.
- `peak_sun_hours` — average daily peak sun-hours used by trackside station/depot PV and onboard roof PV.
- `hvac_uplift_frac` (optional) — override the automatic calculation of HVAC load. If omitted, a linear curve from 25 °C baseline, capped at 25 % at 50 °C, is used.

`[consist]` **(optional)**

Overrides the reference 3-car light-metro consist. All fields are optional; omitted ones fall back to the reference values.

```
[consist]
car_count = 3
length_m = 49.5
mass_kg = 78750
max_speed_kmh = 90
battery_capacity_kwh = 540
passenger_capacity = 360
seat_count = 60
crush_capacity = 480

[consist.roof_pv]             # optional trainset-level onboard solar
nameplate_kw = 15.12
usable_factor = 0.65          # heat/MPPT/cable/soiling derate
charges_while_moving = true
charges_while_dwelled = true

[consist.roof_pv.air_cleaner]  # optional low-pressure air-knife cleaner
enabled = true
compressor_power_kw = 0.9      # trainset-level parasitic draw
dust_loss_recovery_frac = 0.75 # fraction of dust loss recovered
```

`[consist.roof_pv]` applies the shared PV curve from `[climate]`, then `usable_factor`, then any fleet-wide dust derate. If `[consist.roof_pv.air_cleaner]` is enabled, the simulator recovers the configured fraction of dust-driven PV loss and subtracts the compressor load from the net PV credited to the train battery. Energy is credited every tick while moving or dwelled according to the boolean flags. The CSV trace reports this as `roof_pv_charged_kwh`, `roof_pv_kw`, and `roof_pv_cleaner_power_kw`; JSON summaries include `total_roof_pv_charged_kwh`.

`[[stations]]`

Declare each station once. Referenced by `id` from lines and fleets.

- `id` — string ID, unique within the scenario.
- `name` — human-readable; appears in logs and abbreviations.
- `dwell_seconds` — how long trains pause here. Longer at interchanges and terminals; short (20–30 s) at mid-line stops.

- `charging_power_kw` (default 0) — opportunity-charging capability. Terminals typically 1000; mid-line charging stations 300–500.
- `is_terminal` (default false) — trains flip direction here on linear lines. Rings have no terminals.
- `is_depot` (default false) — marks a maintenance/layup site. Held trains can receive low-C top-up toward 95% SoC, subject to energy-site supply and charging-pad availability.

`[[lines]]`

- `id`, `name` — identification.
- `is_ring` (default false) — true for a closed loop.
- `stations` — inline array of `{ id, distance_from_prev_m }`. The first entry must have `distance_from_prev_m = 0`. Subsequent distances are metres between consecutive stations.
- `ring_wrap_length_m` — required for rings, forbidden for linear lines. The distance of the closing segment that connects the last station back to the first.

`[[fleets]]`

- `line` — must match a line `id`.
- `trainset_count` — total train inventory assigned to this line, including reserves.
- `spare_count`, `cold_reserve_count` (default 0 each) — included in the total inventory; their sum must not exceed `trainset_count`. These trains remain parked and can charge, but do not enter routine service. Automatic reserve activation is not yet modelled.
- `station_stabling` (default false) — enables overnight holding and low-C charging at the selected powered dispatch stations. Departure also requires enough energy to reach a selected charging station along the route while retaining the 20% reserve; active pad/grid outages exclude that destination. See the station-stabling workflow for location requirements and evidence limits.
- `overnight_allocations` (optional) — explicit `{ station, heading, location_type, service_role, trainset_count }` homes. `location_type` is `"station"` or `"depot"`; roles are `"revenue"`, `"spare"`, `"cold_reserve"`. Requires `station_stabling` and a complete allocation for every fleet. Station homes total at most two per physical station. Depot homes require explicit `depot_stabling_positions` on an existing `is_depot = true` station, and every home must be a powered dispatch point on the fleet's own line. Assigned trains return home after closing and appear as `station` or `depot` in CSV `stabling_location`. Empty returns emit `ReturnToStabling` events and use the line's peak-service headway. Station launch stock precedes depot stock, with train ID resolving ties; yard tracks and turnbacks remain abstract.
- `dispatch_points` — an array of `{ station, heading }`. Trains are distributed round-robin across this list at start-up unless explicit overnight homes are supplied, and each entry acts as a throttle point: trains arriving or re-dispatching here must wait their scheduled slot. `heading` is `"forward"` or `"reverse"`, interpreted relative to the line's station order.
- `service_start`, `service_end` — `"HH:MM"`. Outside this window, passenger dispatches are blocked; explicitly assigned home-return moves may continue with energy, movement-authority and return-headway checks. If `service_end` is earlier than `service_start`, the service window crosses midnight, e.g. `05:30` to `02:00`.
- `schedule` — ordered array of `{ from, to, headway_min }` windows. Any point in time within service hours should be covered by exactly one window. A schedule window may also cross midnight, e.g. `{ from = "23:30", to = "02:00", headway_min = 12 }`.

`[[sites]]`

Trackside energy sites — each co-located with a station and providing the energy that the station's charging pads deliver to trains. A site has PV, battery storage, and an optional grid tie. Stations without a matching site cannot deliver charging energy.

```
[[sites]]
station = "east-depot"          # must match a station id with charging
pv_nameplate_kw = 5000          # solar nameplate (AC)
storage_capacity_kwh = 40000    # stationary LFP battery size
storage_max_charge_kw = 10000   # battery inbound rate limit
storage_max_discharge_kw = 10000 # battery outbound rate limit
storage_initial_soc = 0.6       # optional; default 0.5
grid_import_kw = 3000           # 0 = no grid tie
grid_export_kw = 3000           # 0 = no export capability
```

Energy flow per tick:

1. PV generation — sinusoidal curve peaking at solar noon, integrated so the daily total equals `peak_sun_hours × pv_nameplate_kw` kWh.
2. Storage charging — PV goes into the battery first, rate-limited by `storage_max_charge_kw` and capped by remaining capacity.
3. Excess PV — goes to `grid_export_kw` if configured; otherwise curtailed (and reported as waste).
4. Train charging — draws from battery first (rate-limited by `storage_max_discharge_kw`); any shortfall pulled from `grid_import_kw`.

PV sources not modeled yet: right-of-way vertical bifacial panels and between-rail ("Sun-Ways-style") PV are treated as future extensions. For now, allocate the ROW PV generation to nearby stations' `pv_nameplate_kw` or to a dedicated depot site. Onboard roof PV is modeled under `[consist.roof_pv]`.

`[[habd_detectors]]` and `[[habd_resets]]` (optional)

HABDs are explicit bidirectional route-book assets. `after_station` identifies the lower-chainage end of an interstation section; `offset_m` must lie inside it. A trip stays latched until a named inspection reset occurs after the wayside overheat clears.

```
[[habd_detectors]]
id = "line-1-terminal-approach"
line = "line-1"
after_station = "stop-1"
offset_m = 1000

[[habd_resets]]
at = "08:15"
train = "T4"
authorised_by = "rolling-stock-technician"
inspection_reference = "inspection-2026-0042"
```

`[[faults]]` (optional)

Inject scheduled energy, onboard, station, or wayside faults. Every fault has `name`, `kind`, `from`, `to`, and optional `day`; its remaining fields select the affected asset and stimulus.

```
[[faults]]
name = "morning dust storm"
kind = "dust_event"
from = "07:00"
to   = "11:00"
day  = 1                       # optional, defaults to 1
pv_output_factor = 0.3         # 30% of normal PV → 70% loss

[[faults]]
name = "substation fire"
kind = "grid_outage"
from = "13:00"
to   = "16:00"
# station = "east-depot"       # omit for system-wide outage

[[faults]]
name = "AMU pad service"
kind = "charging_pad_outage"
from = "14:00"
to   = "18:00"
station = "al-muthanna-university"

[[faults]]
name = "T4 early battery warning"
kind = "battery_off_gas"
from = "14:10"
to   = "14:15"
train = "T4"                  # omit for every train

[[faults]]
name = "T4 mist commissioning failure"
kind = "battery_mist_failure"
from = "14:10"
to   = "14:15"
train = "T4"

[[faults]]
name = "T4 containment escalation"
kind = "battery_fire_escalation"
from = "14:13"
to   = "14:15"
train = "T4"
```

Fault kinds:

- `dust_event` — scales PV output at affected sites. `pv_output_factor` is required (0.0 to 1.0). Omit `station` for a system-wide event.

- `grid_outage` — disables grid import and export at affected sites. Sites ride through on PV + storage. Omit `station` for a whole-grid failure.
- `charging_pad_outage` — disables a station's charging pad. Requires `station`. Trains dwelling there can't charge.
- `battery_off_gas` — trips the affected train's battery detection path, inhibits charging, isolates car HV, starts localized mist when healthy, alerts OCC, and requests a controlled stop.
- `battery_mist_failure` — makes the per-car reservoir/pump/pressure/flow checks fail. Combine it with `battery_off_gas` to exercise suppression failure without changing the safe-stop policy.
- `battery_fire_escalation` — declares immediate danger and continued movement unsafe, escalating the controlled stop to emergency braking.
- `t2g_primary_offline` — removes 5G and verifies backup-radio failover.
- `t2g_all_offline` — queues telemetry until either radio recovers.
- `hot_axle_overheat` — injects a dual-channel bearing-temperature trip.
- `wayside_habd_warning` — injects a warning-band bearing temperature at a physical HABD; the affected train is capped at 40 km/h until its next station.
- `wayside_habd_overheat` — injects a hot bearing only when the affected train crosses a physical HABD, exercising the authoritative trackside stop path.
- `cbm_degradation` — injects service-level bearing, motor, pad, and wheel data.
- `balise_missed` — suppresses the affected train's expected absolute fix.
- `balise_position_mismatch` — offsets the claimed balise position; the registry audit rejects it rather than passing it to odometry.
- `fare_token_tamper` — flips a TVM-issued token signature before the AFC gate. Optional `station`; omission affects every station and repeated denials exercise back-office fraud signalling.
- `platform_door_obstruction` — asserts PSD obstruction sensing. Optional `station`; omission applies it to every station.
- `station_scada_failure` — degrades reference SCADA equipment. Optional `station`; omission applies it to every station.
- `wayside_intrusion` — runs the configured section's detector with `state = "present"` or `"unknown"`, then publishes verdict transitions to consensus. Requires numeric `section_id`.
- Onboard fault kinds accept an optional `train = "T4"`; omit it to affect the fleet. Other onboard kinds are `lidar_offline`, `radar_offline`, `ultrasonic_channel_stale`, `obstacle_peer_disagreement`, and `passenger_intercom_press`. The four embedded faults above use the same optional train scope.

Fault timing:

- `from` and `to` are `"HH:MM"` on the specified `day` (1-based).
- `day = 1` means the first day of the sim; `day = 2` means the second, etc.
- Faults with windows entirely before sim start are silently skipped.

Compositional behavior:

- Multiple simultaneous dust events compose multiplicatively.
- The per-fault effects are independent: a dust event and a grid outage at the same site both apply.

At end of run, each fault that fired is listed in the ■■■■■■ `Faults` ■■■■■■ section of the summary. The per-site table shows the effects: sites with blocked grid export show zero in the `grid→` column; sites with dust events show reduced `PV kWh`.

Common patterns

A simple shuttle

See `example-simple.toml`. Two terminals, one mid stop, one train doing lazy back-and-forth.

A proper metro line

Build it like Samawah's Line 1: ~10 stations, two terminals marked `is_terminal = true`, a couple of mid-line `charging_power_kw = 500` pads, a higher-power pad at each terminal.

A ring

See line-2 in the built-in `samawah` scenario, or any auto-planned megacity design at `cities/catalogue/.../<City>/<slug>.toml` whose population band carries a ring (e.g. Baghdad's line-9). Set `is_ring = true` and `ring_wrap_length_m` to close the loop. Typical ring fleets dispatch both clockwise and counterclockwise from 1–2 major interchange stations.

Multiple lines sharing interchanges

An interchange station is defined once and referenced by `id` in both lines' `stations` arrays. In Samawah, Eastern Bridge and Al-Muthanna University appear in both `line-1` and `line-2`.

Troubleshooting

Validation errors are printed with the context of what went wrong. Common ones:

- "duplicate station id" — the same `id` appears twice in `[[stations]]`.
- "station '…' referenced by … is not defined" — a line or fleet refers to an `id` you didn't declare.
- "line '…' is a ring but missing ring_wrap_length_m" — set the wrap distance.
- "dispatch point station '…' is not on line '…'" — the `station` in `dispatch_points` must be in the line's `stations` list.
- "schedule window …–… has headway_min = 0" — headways must be positive.
- "site for station '…' has storage_initial_soc=X.Y; must be in [0.0, 1.0]" — SoC is a 0–1 fraction, not a percentage.
- "duplicate site for station '…'" — each station may have at most one `[[sites]]` entry.
- "unknown fault kind '…'" — use one of the fault kinds listed above.
- "fault '…' (dust_event) requires pv_output_factor in [0.0, 1.0]" — dust events must specify how much PV is lost (as a fraction, not a percentage).
- "fault '…' (charging_pad_outage) requires a 'station'" — pad outages must name the specific station.

Contributing scenarios

If you develop a scenario for a real city you'd like to share, PRs to this directory are welcome. Please include:

- A header comment naming the city and the design intent.
- A note on what data the alignment and station properties are based on (surveyed, desk study, or indicative).
- Expected ridership if you have it.

The goal is to build up a library of concrete urban rail designs that other cities can learn from.

Fault-injection scenarios

Repository source: `lib/fault-injection-scenarios/README.md`

Reference simulation scenarios that stage specific failure modes for exercising the safety-critical subsystems. Each file is a complete scenario (currently built on top of the Samawah network in `cities/catalogue/west-asia/Iraq/Samawah/samawah.toml`) with an additional `[[faults]]` / event-schedule section that the sim replays against the network.

File	RFC	What it exercises
`dust-storm.toml`	RFC 0002 §5.4 + RFC 0001 §8	AM-peak dust storm with PV derating + late-afternoon depot grid outage — tests degraded-mode energy operation.
`obstacle-fault.toml`	RFC 0015	Staged onboard sensor failures — LIDAR offline, ultrasonic stale, peer 2oo2 disagreement, LIDAR+radar offline at mainline speed. Validates the five O-series SIL-4 properties of `osr-obstacle-detect`.
`wayside-intrusion.toml`	RFC 0016	Three staged section-intrusion events (Present, Unknown, Present) on Samawah Line 1 — validates the interlocking's gate (d) `section_intrusion_permits` check prevents MA issuance on non-Clear sections.

Current status

These scenarios are Samawah-based full scenarios — they replicate the whole Samawah network in-file, then append the fault schedule. A future refactor should decompose each into a small overlay (just the `[[faults]]` + schedule sections) that can be merged on top of any city's base scenario, so the same fault fixtures exercise e.g. the Baghdad network. Until then, re-running the fault scenarios against a different city means copying the base scenario's content in.

Running

```
cargo run --release --bin osr-sim -- \
    --config lib/fault-injection-scenarios/dust-storm.toml \
    --duration 54000 --status-every 3600
```

End-of-run summary prints per-fault tick counts under the scenario's safety-subsystem monitors.

Tools

Repository source: `tools/README.md`

Area	Kind	Purpose
`automation/`	Internal orchestration	Build, regeneration, publishing, validation and repository health
`osr-aln-convert/`	Installable utility	OSR-ALN/LandXML conversion and validation
`reference-ma/`	Installable reference utility	Independent movement-authority reference model

User workflows start with `../osr`; domain rules stay in their owning Rust or Python packages rather than accumulating in automation scripts.

Repository Automation

Repository source: `tools/automation/README.md`

This folder contains small orchestration helpers for regeneration, documentation publishing, and repository health checks. Scripts should stay thin: domain logic belongs in Rust crates, `design/city-generation`, or `design/component-catalogue`.

Commands

Script	Purpose
release-evidence.py	Require a clean software tag and successful exact-commit CI, Kani and integrated-stack runs; publishable PDF, overview, readiness, evidence manifest and checksums
safety-case-summary.py	Generate or check GSN inventory counts without claiming successful proofs or independent acceptance
build-all.sh	One-command regeneration of shared product/cost/catalogue data, browser and native applications, BOM/IFC packages, the root reader PDF, and documentation checks; invoked as `./osr build`
../../osr	Root user command for the Workbench, build, simulator, tests and engineering workflows
verify-civil-native-cad.py	Run with FreeCADCmd to verify native solids and independent net volumes for all registered civil families; retains a source-bound kernel receipt
regenerate-city.sh	Regenerate one city design from the batch catalogue
regenerate-all.sh	Cached design synthesis plus complete city-package refresh under `cities/catalogue/`; use `--from-scratch` to force source-data rebuilding
generate-city-packages-fast.py	Resynthesise designs, then refresh scenarios, maps, engineering, resilience simulation, screenshots, operations, READMEs, and completeness manifests
generate-design-index.py	Rebuild the complete city catalogue index in `cities/catalogue/README.md`; `--check` rejects stale metrics and rows
digital-assurance.py	Compile the Digital Standards Thread: publisher review deadlines, profiles, control coverage, FMEA/inventory screening, evidence hashes, change-impact index and generated Markdown/JSON/baseline; `--impact-from` compares a prior report
component_assurance.py	Generate one fail-closed G0–G4 passport for every current engineering inventory item and configured owner/operator platform service
connected_assurance.py	Compile the configuration-bound battery-cooling graph; `--changed-item`, `--impact-from` and `--batch` traverse dependencies; `--run-scenarios` captures actual Rust controller results with physical evidence and deployment gates remaining open
subsystem_qualification.py	Compile quantitative RAMS screens and six separate readiness states; validate rig measurements, model correlation, production equivalence and scoped signed reviews; `--export` packages original evidence bytes for review
generate-cost-model.py	Rebuild `docs/cost-model.md` from the CAPEX template, generated civil rate contract, finance/benefit assumptions, and rolling-stock BOM
generate-civil-cost-model.py	Convert canonical CAD quantities and reviewed benchmark shares into the generated civil planning-rate contract; `--check` detects drift
generate-acceptance-evidence-report.py	Build the acceptance/accreditation evidence-basis report and matrix from the operations bundle
export-light-metro-bom.py	Export the rolling-stock BOM CSV from the Markdown BOM source plus the generated COTS fit-out cost/source CSV
generate-qa-maintenance-data.py	Generate operations portal assets, manufacturing schedule/materials/verification, QA register, maintenance CSVs, and a deterministic gzip JSON bundle with integrity manifest
audit-project-twins.py	Reconcile city family scope, source hashes, finance buckets, CPM/cashflow totals and the mechanical/civil reference evidence
operating-readiness.py	Compile every tracked city ERP/component/supervision/lifecycle-governance profile against its own asset and twin evidence, report exact on-demand and operator-owned inputs, and detect report drift
component_rfc_readiness.py	Validate and render requirements, ICD, hazard, BOM/drawing, assembly, test, owner, assumption and release-blocker coverage for RFCs 0023–0027
baghdad_delivery_stress.py	Audit frozen production/civil lanes, fixed process holds, priced recoveries, funding pauses and reconciled native-currency cash/debt; `--check` validates inputs and outputs
baghdad_qualification.py	Build temporary-facility RFQs, source-bound ERP evidence tasks, denied-placement gates and a separately operated first section; verify measured results with independent signatures
build-baghdad-proposal.py	Publish the complete Baghdad proposal, detailed registers and evidence archive, with a separately reconciled future Iraqi programme; `--check` validates hashes and archive integrity
build-doc-book.py	Build the complete reader-edition PDF, including current reconciled city totals, every route, and the Baghdad-only Iraqi funding appraisal; paginate long evidence rows
generate-doc-index.py	Rebuild the exhaustive Markdown inventory used for search and CI diagnostics; it is not a second documentation guide
render-sim-screenshots.py	Generate city-local simulator screenshots from any scenario
render-city-engineering.py	Render hash-linked QGIS engineering-layer and SUMO validation visuals for city READMEs
validate-ring-interchanges.py	Fail close ring/radial approaches without a shared transfer, flag radial corridors that double back on themselves, and report genuinely disconnected route layouts
validate-station-clusters.py	Fail same-line spacing below 1.2 km, unmerged cross-line stops within the 600 m station-complex envelope, and missing explicit interchange-complex records
design-iterate.sh	Iterate the rolling-stock design hierarchy across external components, fabricated parts, subassemblies, and final assemblies
buildable-trainset.sh	Generate the LM3 product tree, supplier/COTS registers, 29 factory drawing briefs covering all 62 locally made rows, package readiness, first-article work packages and evidence status
buildable-civil.sh	Reconcile all 19 reference IFC types into six accountable civil/interface release packages and nine non-issued drawing briefs; reject unclassified type-hash changes
freecad-generate.sh	Repository-level FreeCAD/Blender generator for mechanical review models, assemblies, FEM screens, screenshots, and animated digital twins

Script	Purpose
bonsai-civil.sh	Generate deterministic IFC4.3 civil federations with IDS audits and BCF 3.0 release issues, import through Bonsai, and render the linked 4D construction review scene
engineering-toolchain.sh	Install/check the engineering environment; run atomic baseline assurance, smoke tests, JuPedSim/SUMO benchmarks, analysis-register validation, and station IFC interchange checks
generate-city-engineering.py	Generate city-local QGIS packages, geometry-shaped SUMO runs, pandapower/pvlib energy screens and station-to-product mappings
generate-city-finance.py	Reconcile CAPEX; split localization-first external/local capital; compare variable foreign-turnkey cases; emit OPEX, revenue, NPV/IRR/DSCR, renewal, and risk screens
generate-iraq-funding-programme.py	Reconcile Baghdad and one plant only; publish native USD/IQD capital, priced gap finance, tariff/OPEX sensitivities and monthly/six-month early repayments
baghdad_financing_redesign.py	Compare opening-linked credit, 15-year insured tenor, station rights, energy/factory/developer ledgers and consolidated appraisal; emit six-month placements and a 90-day evidence programme
baghdad_equity.py	Model mixed holding-company ordinary shares, staged primary/secondary cash, dilution, tax/accounts, dividends, failed issues and six-month capital gates; `--check` validates sources and outputs
baghdad_viaduct_rentals.py	Screen civil-linked retained premises, generate pilot units/parts/hazards, and price leasing, deposits, vacancy/refurbishment/tax scenarios without assuming eligibility; `--check` rejects source/output drift
baghdad_delivery_baseline.py	Reconcile scope, depot/stabling quantities, six-car BOM/mass/labour, hourly energy, opening fleet, workload/competence and Baghdad governance; unquoted alternatives preserve the published funding baseline
bootstrap_baghdad_tests.py	Restore the exact ignored operations input from the tracked proposal archive for clean-checkout tests; rejects drift and preserves differing workspace data
refresh-city-controls.py	Refresh current finance, operations, project controls and documentation while preserving retained solver evidence and open release gates
apply-city-overrides.py	Apply controlled city operating policy before scenario emission; retains Samawah HABD configuration through synthesis
refresh-city-design-costs.py	Refresh controlled layout costs without letting an unreviewed corridor cache remove rings or replace station identities; resynthesis requires an explicit flag
generate-national-briefs.py	Generate concise country-specific city/factory/capital aggregates linked to the common deployment planning reference
generate-portfolio-summary.py	Aggregate the current city and national-factory models into human-readable `docs/portfolio-summary.md` and machine-readable `docs/portfolio-summary.json` evidence
generate-public-overview.py	Generate and drift-check the GitHub-native Markdown brief and landscape HTML/print edition from the design catalogue and LM3 build-cost record
recalculate-city-capex.py	Recalculate civil, station, depot, charging and family fleet CAPEX from controlled rates; write a cost migration receipt without rewriting historical evidence hashes
validate-city-simulation.py	Run nominal and mandatory degraded-energy OSR simulations on distinct physical cores, including combined aged/hot, consecutive missed-charge and late-running charger-overlap cases, and write compact reproducible evidence
check-tracked-file-sizes.py	Keep useful GitHub artifacts in-tree while enforcing the 50 MiB per-file repository ceiling
generate-lm3-first-article-work.py	Freeze `LM3-FA-001` and export its 81 gaps with controlled closure state and evidence routes
validate-lm3-first-article-evidence.py	Reject missing, unaccountable or checksum-drifted LM3 physical/supplier evidence submissions
publish-lm3-work-packages.py	Preview or idempotently reconcile LM3 work packages with public GitHub issues
ops-user-admin.py	Create/update a private PBKDF2 Ops Core user store with city-scoped roles
ops-core-backup.py	Create and verify a consistent SQLite plus managed-evidence backup archive
validate-simulation-components.py	Fail closed unless every deployed software component has one explicit simulation treatment and every tick-linked component remains an `osr-sim` dependency
generate-city-package-manifest.py	Fail closed unless a full city package contains passing design, engineering, simulation, resilience, operations, and hash-linked acceptance artifacts
operations-pilot.py	Reproduce the combined possession/train/charger/staff recovery pilot or serve an isolated gateway/Workbench for offline field acceptance
repo-health.py	Check generated artifact drift, required files, and repository hygiene
check-markdown-links.py	Check that local links in tracked Markdown resolve inside the repository
platform-recovery.py	Capture private coordinated platform checkpoints and rehearse fresh-volume recovery; see the recovery guide
check-readmes.py	Enforce titles, whitespace, provenance, concise size limits, complete Markdown indexing, and current LM3 contract counts/links across tracked READMEs and national briefs
export-gis-context.py	Convert a cached city OSM snapshot into deterministic local roads, buildings, water, protected-land and existing-rail GeoJSON layers, refreshing project source locks when present
validate-host-manifests.py	Validate all five host compositions and the complete Cargo component inventory
validate-city-projects.py	Compile and validate every Git-backed OSR City Studio project, including source locks and weekly service plans
build-web-frontends.sh	Build City Studio and the simulator/OCC production WASM bundles with pinned Trunk 0.21.8
workbench-server.py	Serve the same-origin Workbench, proxy City Studio, mount simulator/OCC WASM, and persist Ops Core records
test-city-studio-gui.mjs	Drive the real City Studio GUI with Playwright Chromium, run every adapter, restart the server, and verify project/job persistence in an isolated disposable project

For a single city, nominal runs and resilience cases are assigned to distinct physical cores. The default all-city run resynthesises `design.toml` using current raster/corridor caches where available, then regenerates every remaining artifact required by the Samawah package manifest. It also fails if the compact engineering or operations review package for any generated city is excluded by Git ignore rules. Missing source caches are created automatically; `--from-scratch` forces them to be rebuilt. Compact simulator output retains acceptance counters while omitting the detailed event trace; validation's explicit `--ma-check-every 0` uses the same movement-authority gates with bounded derived state instead of retaining three full Raft histories. `regenerate-all.sh --jobs N` automatically limits each city to one simulator when `N > 1`, leaving CPU scheduling to the operating system and preventing nested parallelism.

Typical verification:

```
python3 tools/automation/repo-health.py --quiet
python3 tools/automation/generate-civil-cost-model.py --check
python3 tools/automation/validate-host-manifests.py
tools/automation/design-iterate.sh
tools/automation/buildable-trainset.sh
tools/automation/freecad-generate.sh --check
tools/automation/bonsai-civil.sh --check
```

Build the complete documentation book with one command:

```
./osr book
```

The book includes canonical project and technical prose, software, deployment, control-electronics, formal-assurance, engineering and high-level manufacturing documentation, 44 developing-country briefs, and concise briefs generated from all 266 city models. `./osr book --list-sources` prints the exact validated manifest. Generated search indexes, duplicate city pages, component definitions and signable travelers remain linked repository records rather than repeated chapters. The illustrated build fails if a referenced local image is missing or unreadable. Chapter source bars link to GitHub and the PDF outline provides part, chapter, region, and country navigation.

The Samawah worked example is regenerated separately because it consumes the city-local alignment, station, energy, simulation, and fleet package:

```
tools/automation/freecad-generate.sh --samawah-line-twin
tools/automation/freecad-generate.sh --fabrication-twin
tools/automation/bonsai-civil.sh --render
```

Baghdad financing redesign

The twelve-case redesign preserves the existing baseline and distinguishes core unlevered appraisal, company cash, public transfers and consolidated resources. New term sheets, equity, development values and public payments are sensitivities. Native ERP imports of its six evidence tasks use the existing qualification importer and preserve actual owner/status; no model run accepts those tasks.

When shared debt logic changes, regenerate in dependency order:

```
.venv/bin/python tools/automation/generate-iraq-funding-programme.py
.venv/bin/python tools/automation/baghdad_delivery_stress.py
.venv/bin/python tools/automation/baghdad_qualification.py
.venv/bin/python tools/automation/baghdad_viaduct_rentals.py
.venv/bin/python tools/automation/baghdad_financing_redesign.py
.venv/bin/python tools/automation/baghdad_equity.py
.venv/bin/python tools/automation/baghdad_delivery_baseline.py
.venv/bin/python tools/automation/baghdad_delivery_closure.py
.venv/bin/python tools/automation/build-baghdad-proposal.py
```

The last six tools support `--check` to reject source/output drift. Delivery stress also supports `--check`. Monthly company cash and principal balances, six-month capital funding and internal transfers must reconcile before publication. Land opportunity cost and extra availability payments remain visible outside the government's 25% original-capital contribution.

The ordinary-equity study proposes a 100%-owned subsidiary holding group, retaining open licences. Government shares reclassify existing money; only primary subscriptions fund company invoices. Failed subscriptions cannot expand capital-credit envelopes. Consolidated accounts, conservative separate-business tax, investor returns and an indicative Iraqi liabilities threshold remain conditional, with no accepted incorporation/admission evidence. Its six native ERP tasks use the same source-bound qualification importer.

The retained viaduct-premises package has a civil-linked candidate register, open independent unit, 30-unit draft pilot, hazards and leasing/deposit/renewal ledgers. No area is surveyed or accepted. The large unmapped area is blocked; smaller cases feed both the ring-fenced and wholly owned financing alternatives. Equity outputs distinguish actual dividends from an unapproved coverage policy and independent terminal-cash diagnostics. Its six evidence Tasks require actual local market/engineering/quote/lease evidence; model output grants no occupancy or operating permission.

The delivery reconciliation exposes unresolved scope and overlap alongside quantity/rate/workload sensitivities. Run its generator after physical/finance inputs, then regenerate the proposal. `--check` validates its inputs, outputs and current-review document. Alternatives and uncalibrated risk quantiles are not adopted budgets. The native ERP workforce

observer reads a private Employee authorisation attachment and checks project/task/company and read permissions; it creates no assignment or railway authority.

For a fresh checkout, install the existing test/render dependencies and run `.venv/bin/python tools/automation/bootstrap_baghdad_tests.py` before `.venv/bin/python -m pytest tools/automation/tests design/component-catalogue/tests -q`. The hash-bound archive supplies the exact operations input; resynthesis is not required for tests. `--check` verifies an existing payload without writing.

The delivery continuation supplies nine depot/slot drawings, yard-slab mould capacity, child-part RFQs, per-site hourly energy, all maintenance intervals, restricted renewal cash, opening corridor break-even, practical assessment forms and six reconciled finance sensitivities. `baghdad_delivery_closure.py --check` rejects changed sources or outputs. These remain unquoted alternatives with unaccepted field evidence.

`verify_baghdad_erp_restore.py --backup PRIVATE.sql.gz --expected PRIVATE-fingerprint.json --receipt PRIVATE-receipt.json` restores a trusted native backup into a new network-isolated disposable MariaDB container, compares six native data/schema/permission/file-inventory fingerprints and removes the container. It never restores into the source database. Receipts contain counts/hashes, remain private and do not establish full file recovery or production RTO/RPO. The corresponding bench-only probe is `osr_erpnext.delivery_admin.recovery_fingerprint`; `read_probe` measures bounded permission-aware reads, not city-scale capacity.

osr-aln-convert — civil-tool bridge for OSR-ALN

Repository source: `tools/osr-aln-convert/README.md`

RFC 0009 v3 deliverable. Reads the alignment file every civil- engineering firm already produces and emits the OSR-ALN TOML that the OSR pipeline ingests.

What converters ship today

Source tool	Export format	Converter	Status
Autodesk Civil 3D	LandXML	`landxml-to-osr-aln`	v1
Bentley OpenRail	LandXML	`landxml-to-osr-aln`	v1 (same converter — OpenRail emits a compatible LandXML subset)
Trimble Business Center	LandXML	`landxml-to-osr-aln`	v1
Trimble Business Center	CSV	`tcl-to-osr-aln`	pending v1.1
QGIS + rail-path plugins	LandXML via plugin	`landxml-to-osr-aln`	v1
OSR generated city design	`design.toml` + corridor GeoJSON	`current-network-to-osr-aln`	planning export

One converter covers the three LandXML-emitting tools; they share the `<Alignments>` + `<Profiles>` schema that Civil 3D defined and Bentley + Trimble adopted.

Usage

```
# One-time install (stdlib-only converter; no wheels to build).
pip install -e .

# Convert a LandXML dump from Civil 3D / OpenRail / Trimble.
landxml-to-osr-aln \
    --input  ./exports/samawah-line1.xml \
    --output ./cities/catalogue/west-asia/Iraq/Samawah/engineering/alignment/samawah-line1.aln.toml \
    --line-id samawah-line1 \
    --preset standard-urban \
    --consist light-metro-3car \
    --crs EPSG:32638 \
    --surveyor "Samawah Civil Associates"
```

The `--line-id`, `--preset`, `--consist`, `--crs`, `--surveyor` flags populate the `[meta]` block in the output TOML; the rest of the document is derived from the LandXML geometry. Station placeholder platform lengths default from the selected `--consist` per RFC 0008/0010; use `--platform-length-default` when a civil package deliberately safeguards a longer local platform.

To regenerate a current OSR planning network directly from checked-in city artifacts:

```
current-network-to-osr-aln \
    --design ../../cities/catalogue/west-asia/Iraq/Samawah/design.toml \
    --geojson ../../cities/catalogue/west-asia/Iraq/Samawah/samawah.corridor.geojson \
    --output-dir ../../cities/catalogue/west-asia/Iraq/Samawah/engineering/alignment \
    --design-date 2026-08-12
```

This path emits one file per current line, verifies line/station IDs and civil coverage against `design.toml`, and embeds input hashes for drift review. It is explicitly planning-only: the generated GIS trace has no fitted curves, surveyed vertical profile, or designed cant. See the `Samawah package notice` for the replacement gates.

Explicit station, civil and cant companion data

Use `--sidecar line.mapping.toml --design design.toml` with the LandXML command above to import reviewed station IDs, civil spans and cant. The companion is TOML, so one file holds the three mappings and their provenance. It selects a named alignment, including from a multi-alignment XML file.

```
schema_version = 1
# Use cant = [] here only when the supplied design explicitly has no cant.

[source]
landxml_sha256 = "<SHA-256 of the exact XML bytes>"
design_sha256 = "<SHA-256 of the exact design.toml bytes>"
alignment_name = "Survey A"
line_id = "line-1"
crs = "EPSG:32638"
vertical_datum = "<datum of the supplied elevation data>"

[[station]]
id = "<station ID on line-1 in design.toml>"
source_name = "<matching LandXML Station name>"
station_m = 100.0
platform_length_m = 61.0

[[civil]]
from_station_m = 0.0
```

```
to_station_m = 1000.0
class = "at-grade"

[[cant]]
from_station_m = 400.0
to_station_m = 600.0
max_cant_mm = 30
transition_in_m = 50.0
transition_out_m = 50.0
```

This is a schema illustration, not deployment data. Replace the placeholders, include every station on the selected design line, and cover the complete alignment with contiguous civil spans. Every XML station name must map once, at its original chainage. Additional stations may supply chainages without `source_name` when the XML has no corresponding station object. Civil and cant spans cannot overlap; cant limits follow the selected preset, with nonzero ramps fitting inside each nonzero-cant span.

The importer checks the two source hashes, line/consist/preset/ring identity, CRS agreement, finite values, metric metre units and zero-origin chainage. It rejects station equations, nonzero chainage origins, incomplete mappings and unknown fields. The existing OSR-ALN hard gates also run before output is written. CRS and vertical datum are declarations of the supplied coordinates; the converter does not transform or independently verify them. Output retains all three input hashes and `deployment_release_ready = false`. Passing these checks does not replace engineering review or authority approval.

Without `--sidecar`, the existing placeholder workflow and golden fixture remain unchanged.

What the converter reads

From the input LandXML:

- `<Alignments>/<Alignment>` — one alignment per output file.
- `<Alignment>/<CoordGeom>/<Line>` + `<Curve>` + `<Spiral>` — horizontal alignment elements, emitted as `[[horizontal]]` rows.
- `<Alignment>/<Profile>/<ProfAlign>/<PVI>` — vertical profile points, emitted as `[[vertical]]` rows. Sag/crest curves (`<CircCurve>`) carry `vc_radius_m`.
- The companion-file path rejects `<Alignment>/<StaEquations>`; station-equation transforms are not implemented. The legacy path does not interpret them and must not be used for those exports.

Remaining format limits

- Station pin-pointing. LandXML has `<Station>` on an alignment but most civil tools write stations as offsets on a separate `Survey` object. In v1 the converter emits a placeholder `[[station]]` for each input `<Station>`; the deployment engineer can use the companion file above to map names to design.toml station IDs or supply externally derived station chainages. Direct Survey-object interpretation and CSV mapping are not implemented.
- Civil classes (at-grade / elevated / bridge). LandXML has no civil-class annotation. The converter emits a placeholder `[[civil]]` covering the full length as `at-grade`; the companion file replaces this placeholder with explicit RFC 0011 spans.
- Cant (superelevation). Bentley OpenRail writes cant into a non-standard `<Cant>` extension; Civil 3D writes it as a separate file. The companion file imports explicit `[[cant]]` rows; direct interpretation of vendor cant extensions remains unimplemented.

The companion file closes the explicit station/civil/cant mapping task while keeping vendor-specific parsing and survey acceptance separate.

Synthetic round-trip fixture (not deployment data)

The converter ships with a deliberately abbreviated golden fixture at `samples/samawah-line1.xml` and the corresponding golden output at `samples/samawah-line1.aln.toml`. The historical name is retained to avoid disguising fixture provenance, but the 3 km illustrative geometry is not a surveyed or current Samawah alignment; see the `samples/` fixture notice. The test suite (`tests/test_round_trip.py`) runs the converter against the sample and asserts byte-identical output. Run with `pytest`.

Coding-standard compliance

Per the `reference-ma` precedent: stdlib only, no numpy, no pydantic. Every byte of behaviour should be readable by a civil-engineering reviewer without chasing into a framework. The one imported module outside the stdlib is `pytest`, dev-only.

Converter Golden Fixtures

Repository source: `tools/osr-aln-convert/samples/README.md`

The files in this directory are synthetic, abbreviated golden fixtures for `osr-aln-convert`. They are retained because the round-trip test compares the generated OSR-ALN output byte-for-byte.

Despite the historical `samawah-line1` filename and plausible coordinates, these files are not survey data, not the current Samawah network, and not a deployment alignment. The geometry is only 3 km long, station IDs are placeholders, and the civil class is intentionally all at grade so every converter review hook remains visible.

Deployment-controlled alignments live under `docs/civil/` and require the survey and approval fields specified by `docs/civil/osr-aln-format.md`.

reference-ma — Python reference interpreter for osr-interlocking

Repository source: `tools/reference-ma/README.md`

Independent implementation of the Movement Authority state machine specified in RFC 0001 §7 and RFC 0004 §M4. The Rust crate `osr-interlocking` is the production code; this Python twin exists so that differential fuzzing against both implementations can catch bugs in either one.

The public surface mirrors `osr-interlocking`:

- `derive_state(entries) DerivedState`
- `compute_self_ma(train_id, entries, network, now_ns) MovementAuthority`
- `section_available_to(train_id, section, state) bool`
- `forward_chain`, `footprint_from`, `far_end_of`, `locate_section`

Wire format is whatever `serde_json` produces for the Rust types — IDs are plain `u64`s (`#[serde(transparent)]`), enums are externally tagged when they carry data, and every struct is field-by-field. The `__main__.py` CLI reads a case JSON on stdin and writes the same `MovementAuthority` JSON shape to stdout. The Rust differential test in `crates/osr-interlocking/tests/differential.rs` serialises random log prefixes and compares byte-for-byte.

Running

Stdlib-only — no dependencies. Quick sanity run:

```
cd tools/reference-ma
PYTHONPATH=src python3 -m unittest discover -s tests -v
```

Full differential round-trip from the workspace root:

```
cargo test -p osr-interlocking --test differential
```

The Rust test skips itself if `python3` is missing or if `OSR_SKIP_PY_DIFF=1` is set, so CI on hosts without a Python toolchain still passes.

CLI

Input (stdin or `--input case.json`):

```
{
  "network":  { "stations": {...}, "sections": {...}, "lines": [...] },
  "entries":  [ { "entry_id": 1, "term": 1, "timestamp_ns": 0,
                  "payload": { "TrainRegistration": {...} } }, ... ],
  "train_id": 1,
  "now_ns":   1000000000
}
```

Output (stdout):

```
{
  "train_id": 1,
  "end": { "section": 1001, "offset_mm": 1000000, "direction": "Forward" },
  "applicable_restrictions": [],
  "valid_until_ns": 4000000000,
  "derived_from_entry_id": 3,
  "has_known_position": true
}
```

Status

Entry variant	Supported	Notes
`TrainRegistration`	yes	full
`TrainPositionReport`	yes	full
`TrainDeparture`	yes	clears occupancy + route grants
`SwitchObservation`	yes	full
`RouteGrant` / `Release`	yes	full
`SpeedRestriction`	yes	v1 heuristic matches Rust
`MaintenanceOverride`	yes	full
`Heartbeat`	yes	monotonic-seq guard matches Rust
`FormatVersion`	yes	stores `current`
`SwitchCommand`	no-op	advisory; matches Rust
`RouteRequest`	no-op	advisory; matches Rust

The differential test currently covers registrations and position reports under a fixed linear network. Expanding to include switches, route grants, and multi-line networks is a straightforward next pass — the primitives are all in place.

Engineering, Control Electronics and Formal Assurance

OpenSourceRail Control Electronics

Repository source: `control-electronics/README.md`

The reference integration source and revised host bench requirements include the corrected power, watchdog, bus and pin interfaces. The Baghdad detail register adds six-car fleet quantities and required supplier/qualification evidence. No complete onboard target image or qualified host BOM is released.

This directory owns the physical electronic hosts for the software stack: processors, safety I/O, sensors, power interfaces, wiring, connectors and enclosures. It does not contain embedded software, rolling-stock mechanical parts, track products or civil works. Embedded implementations remain under `../crates/`; mechanical and civil component geometry remains under the `../design/component-catalogue/`.

Reference electronic designs are scoped in RFC 0007; schematics and board-release artifacts land under each subdirectory as the RFC 0007 rollout matures.

Control electronics have two valid release tracks:

1. Pilot / DIY integration track. RFC 0019 assembles commodity SBCs, Pi Pico 2 boards, sensor modules, HATs, relays, power supplies, terminal blocks, DIN-rail enclosures, and prepared SD-card images. No KiCad or gerbers are required when the build uses only COTS modules. The release evidence is the controlled integration pack: exact SKUs, wiring/harness maps, connector maps, enclosure drawings, power and thermal margins, firmware images, self-test output, and bench records.
2. Custom-board fabrication track. RFC 0007 board-level specs can be captured into KiCad when a deployment needs OSR-specific carrier boards, power conditioning, safety I/O, or sensor-interface boards. That track needs KiCad, gerbers, board BOMs, DFM/DFT review, assembly drawings, and board bring-up evidence.

Commodity application hosts; separately qualified safety channels. Raspberry Pi and Radxa remain the simple pilot, application and diagnostic palette. Safety-output hardware follows the assessor-controlled `safety-controller-selection.md` gate and may use an industrial safety-MCU family. A commodity processor never bypasses the external fail-safe output path.

Class	Role	SoC baseline	RFC 0007 §
`t-ecu-s/`	SIL-4-target train safety kernel	Qualified safety-channel pair to be frozen; 2 × RP2350 pilot rig + CM5 app processor	§4
`t-ecu-a/`	Train application (SIL-2/-0)	Raspberry Pi CM5 (Radxa CM5 drop-in)	§5
`t-obs/`	SIL-4-target obstacle verdict interface	Qualified evaluator channels to be frozen; 2 × RP2350 pilot rig + CM5 pre-processor	RFC 0015 §5
`w-sbc/`	Wayside application host	Radxa CM5; safety outputs through a separately qualified controller	§6
`s-sbc/`	Station / depot (SIL-0)	Raspberry Pi CM5 + commodity carrier	§7

0-SRV (ops server) is a commodity computer and has no reference design in tree.

Layout

Each class subdirectory has the same shape:

```
<class>/
├── README.md          class overview, SoC rationale, peripherals list
├── diy-assembly/      COTS/SBC pilot assembly notes per RFC 0019
└── schematics/        board-level specs; KiCad projects once custom boards are drawn
```

`bom/` and `gerbers/` are created only when they contain real release artifacts; empty placeholder directories are not retained.

Current status:

Class	Pilot / COTS integration	Custom-board package
`t-ecu-s`	`diy-assembly/` present; build evidence pending	Complete `schematics/v2-spec/` package; KiCad / gerbers / board BOM pending
`t-ecu-a`	`diy-assembly/` present; build evidence pending	`schematics/v2-spec/` overview and block diagram; KiCad / gerbers / board BOM pending
`t-obs`	`diy-assembly/` present; build evidence pending	Complete `schematics/v2-spec/` package; KiCad / gerbers / board BOM pending

Class	Pilot / COTS integration	Custom-board package
`w-sbc`	`diy-assembly/` present; build evidence pending	`schematics/v2-spec/` overview and block diagram; KiCad / gerbers / board BOM pending
`s-sbc`	`diy-assembly/` present; build evidence pending	`schematics/v2-spec/` overview; custom board usually unnecessary

Bring-up procedures are colocated with their host: `T-ECU/S`, `T-ECU/A`, `T-OBS`, `W-SBC` and `S-SBC`.

The release gates for both tracks are tracked in `release-checklist.md`. Until a host class meets the pilot track it is a design concept, not integration-qualified pilot hardware. Until it meets the custom-board track it is not released fabrication hardware.

For trainset quantities and the boundary between control-electronics records, mechanical CAD envelopes, and the rolling-stock procurement BOM, see `rolling-stock-integration.md`.

Licensing

Electronic designs (schematics, layouts, gerbers, board BOMs, and integration drawings) are licensed under CERN-OHL-S v2. Firmware stays under Apache-2.0 (inherited from the Rust crates). Documentation under CC-BY-SA 4.0. See ARCHITECTURE.md §9 for the project-wide licensing rationale.

Contributing

High-leverage contributions are DIY/COTS build feedback against `diy-assembly/`, integration BOM completion, wiring and enclosure review, KiCad capture where custom boards are justified, and board DFM review for the custom-board track.

Bench assembly workflow

Repository source: `control-electronics/diy-assembly/README.md`

Start with the host requirements, reference integration, candidate catalogue and release checklist. These specify integration work; no qualified plug-and-play railway assembly or downloadable target image is released.

1. Freeze exact compute/carrier, peripherals, interfaces and target firmware.
2. Calculate bus range, converter capacity, inrush, fuse selectivity and thermal/EMC design.
3. Release harness cavities, cable/terminal list, isolation topology and enclosure drawings.
4. Build a simulated-load bench unit; verify continuity, voltage domains and fault states before connecting power.
5. Run measured device/HAL, boot, heartbeat, output, sensor, thermal and EMC tests; retain hashes and instrument records.
6. Review qualification evidence and supplier substitutions before authorising any field function.

A USB isolator does not connect two USB-device Pico boards or carry SPI. Use a correctly specified SPI isolator and controller/peripheral link. An AC-input HDR-60-24 supplies 24 V output, not a 24-to-5 V conversion. Board-package pins cannot be copied into a Pico header wiring table. See image status before preparing boot media.

DIY parts catalogue

Repository source: `control-electronics/diy-assembly/parts-catalogue.md`

Every SKU used across the five host classes, consolidated for sourcing in bulk. Every item has ≥ 2 distributors so supply- chain disruption at one source does not block a deployment.

Prices are 2026 retail in USD, at quantities of 1–10 units. Bulk (50+) pricing typically 20–40 % lower per unit.

Compute modules

Part	SKU	Distributor 1	Distributor 2	Retail (USD)
Raspberry Pi Pico 2	SC1630	Official Pi Foundation	Mouser, DigiKey	$5
Raspberry Pi CM5 8 GB Lite	CM5 configuration/SKU pending freeze	Official Pi Foundation	Mouser	$85
Raspberry Pi CM5 IO Board	CM5IO vendor SKU pending freeze	Official Pi Foundation	Pimoroni	$35
Radxa CM5 industrial-temp	rock-cm5-industrial	Radxa store	ALLNET china	$110
Radxa CM5 IO Board	rock-cm5-io	Radxa store	ALLNET china	$40

Communications HATs

Part	SKU	Distributor 1	Distributor 2	Retail (USD)
Waveshare 2-CH CAN-FD HAT	2-CH-CAN-FD-HAT	Waveshare	AliExpress-commodity	$28
Waveshare SX1262 LoRa HAT (868 / 915 MHz)	SX1262-LoRa-HAT	Waveshare	Mouser	$18
Quectel RM500Q-GL 5G modem (M.2)	RM500Q-GL	DigiKey	AliExpress	$80
Quectel antennas (2 × 4 MIMO)	YQ500-B7B	DigiKey	Mouser	$35 / set

SIL-4 actuator path

Part	SKU	Distributor 1	Distributor 2	Retail (USD)
8-channel 24 V relay board, opto-isolated	SainSmart SSR-8DC24	SainSmart	Amazon / AliExpress	$12
Adafruit USB isolator	Adafruit 2107	Adafruit	Mouser	$25
SparkFun ADUM1401 isolator breakout	BOB-14712	SparkFun	DigiKey	$25
Adafruit ATECC608B breakout	Adafruit 4374	Adafruit	DigiKey	$4
ADS1115 16-bit ADC breakout	Adafruit 1085	Adafruit	DigiKey	$15
MCP23017 I/O expander breakout	Adafruit 732	Adafruit	DigiKey	$7

Obstacle-detect sensors (RFC 0015)

Part	SKU	Distributor 1	Distributor 2	Retail (USD)
HC-SR04 ultrasonic transceiver (dev grade)	HC-SR04	AliExpress	Amazon	$2 each
Murata MA40H1S-R (historical candidate; lifecycle and environmental qualification open)	MA40H1S-R	Mouser	DigiKey	$25 each
TI AWR1843BOOST 77 GHz radar eval	AWR1843BOOST	ti.com	Mouser	$500
Livox HAP solid-state LIDAR	LIVOX-HAP	Livox direct	B&H Photo	$1500
Raspberry Pi Camera Module 3	SC0872	Official Pi Foundation	Mouser	$35
ArduCam stereo camera bracket	B0203	ArduCam	Amazon	$18

Wayside-intrusion sensors (RFC 0016)

Part	SKU	Distributor 1	Distributor 2	Retail (USD)
Senstar FlexZone fence-line (per 100 m)	FlexZone-100	Senstar dealers	—	$300
Livox Mid-360 360° LIDAR	LIVOX-MID-360	Livox direct	—	$900
TI AWR1843BOOST (ROW-mounted)	AWR1843BOOST	ti.com	Mouser	$500
Reolink RLC-810A 4 K IP camera	RLC-810A	Reolink	Amazon	$130
Coral USB Accelerator (TPU)	coral-usb	Coral	Mouser	$60

Enclosures + mechanical

Part	SKU	Distributor 1	Distributor 2	Retail (USD)
UCTRONICS DIN rail Pi 5 case	U6277	UCTRONICS	Amazon	$25
Radxa CM5 industrial DIN enclosure	rock-cm5-din-industrial	Radxa store	—	$45
Phoenix Contact 12-position 24 V terminal block strip	UT 2.5 DIN	DigiKey	Mouser	$8
IP67 pole-mount cabinet 400 × 300 × 200 mm	RBM-5	DigiKey	Allied Electronics	$110
Cat 6a patch cable 1 m (TSN grade)	generic	Amazon / DigiKey	Any IT supplier	$5

Field-wiring consumables

Part	SKU	Distributor 1	Distributor 2	Retail (USD)
Insulated ferrules kit (0.5 / 0.75 / 1.5 mm²)	Knipex 97-99-910	DigiKey	Amazon	$30
Knipex ferrule crimper	97-62-145-A	DigiKey	Amazon	$60
Stranded wire 1.5 mm² (500 m spool, colour-coded)	LAPP 4520015	LAPP catalogue	Amazon	$180

Total first-article deployment cost

For one Samawah line-1 trainset + adjacent 1-km wayside section:

Bucket	Unit cost (USD)	Units	Subtotal
T-ECU/S per trainset	240	2	480
T-ECU/A per trainset	300	2	600
T-OBS per trainset (includes LIDAR)	2 410	2	4 820
W-SBC per junction box	350	2	700
Fence-line sensor (1 km)	3 000	1	3 000
ROW LIDAR × 5 + radar × 2	4 500 + 1 000	1	5 500
CCTV + TPU	190	2	380
Enclosures, cabling, consumables	—	—	1 500
Total			~$17 000

Scale per trainset (electronics + sensors only): ~$5 900. Scale per km of wayside (section + sensors): ~$10 000.

For a full Samawah deployment (10 trainsets + 30 km of track = ~30 sections): ~$360 000 in electronics. Compares to a legacy CBTC system at ~€50M for an equivalent network — two orders of magnitude difference dominated by NRE on the legacy side.

Interface correction, 3 October 2026

This is a historical candidate/retail list, not a complete qualified procurement BOM. Reference integration and revised host assembly requirements supersede earlier wiring and subtotals. Exact supplier SKUs, lifecycle and alternates require verification. A raw piezo transducer is not a plug-compatible ultrasonic ranging module; an ADS1115 is not a timed digital-echo capture device. USB isolation does not implement an SPI link. Include external heartbeat watchdogs, DC/DCs, driver feedback, connectors, fuses, harness, thermal and test hardware before quoting totals.

DIY SD-card images

Repository source: `control-electronics/diy-assembly/sd-card-images.md`

The planned application-host deployment boots from a prepared microSD card carrying a minimal Linux base, an OSR role runner, `osr-selftest`, and the per-deployment configuration. This page distinguishes that target from the artifacts actually shipped today.

Current source-build boundary

No signed OSR SD-card or Pico firmware image is published by v0.3.1. The Rust workspace provides reusable control logic, simulation integration and the `osr-selftest` software harness; it does not yet provide a board-support package, bootable image recipe, hardware drivers or deployable RP2350 binary.

The commands below validate that the host-side integration libraries compile. They do not create the `/usr/local/bin/osr-trainset-image` executable or a `.uf2` file described by older drafts:

```
# On a build host (x86-64 Linux dev machine):
cd OpenSourceRail

# Cross-check the onboard integration library for the CM5 target.
rustup target add aarch64-unknown-linux-gnu
cargo build --release --target aarch64-unknown-linux-gnu \
    -p osr-trainset-image
```

Before any image can be flashed to pilot hardware, a deployment must add and review all of the following:

- an operating-system image recipe with a pinned upstream base and packages;
- a role runner that drives the relevant libraries through reviewed hardware interfaces;
- CM5/Radxa board-support, device-tree and hardware-driver configuration;
- a genuine `no_std` RP2350 application and its watchdog/safety-I/O drivers;
- secure provisioning, update, rollback and key-rotation procedures;
- hardware bench tests, signed artifacts and reproducible-image checksums.

These are tracked as release gates in `../release-checklist.md`. A deployment may use an upstream operating-system image for bench development, but that is not an OSR release image and must not command railway actuators.

Planned signed artifacts

When the hardware gates close, each release must use filenames containing the actual host, semantic version, hardware revision and content checksum. Paired safety-channel firmware must identify its channel and refuse an incompatible peer. Do not reserve fictional filenames in advance.

Every published image must be:

- Signed with the OSR project's ed25519 release key.
- SHA-256 checksummed for supply-chain verification.
- Reproducible — given the tagged workspace commit and pinned base-image digest, anyone can rebuild identical bytes.

Only flash an artifact after verifying its release signature, checksum, target host and hardware revision. The release procedure will carry the exact command; this document deliberately does not provide a destructive dd example for an artifact that does not exist.

Configuration

The planned image boots to a default config that will not proceed past self-test until `/etc/osr/config.toml` is populated with:

- `deployment_id` — string identifier for the operator.
- `entity_id` — the unique ATECC608B-backed id for this specific unit (RFC 0017).
- `role` — host class (`t-ecu-s-A`, `t-ecu-s-B`, `t-ecu-a`, etc.).
- Deployment key registry — path or URL to the per- deployment `KeyRegistry` (RFC 0017 §3.2).
- Network endpoints — TSN neighbour list, OCC address.

Sample configs ship in each `<class>/diy-assembly/sample-config.toml`.

Boot flow + self-test

The target systemd unit at `/etc/systemd/system/osr.service` will:

1. Wait for hardware-clock sync (PTP, if available, else NTP).
2. Validate the deployment config — reject if missing required fields.
3. Load the ATECC608B trust anchor — reject if absent or disagreeing with `deployment_id`.
4. Run `osr-selftest`:
   - Ping both Pico 2 channels (for 2oo2 boards).
   - Exercise each sensor's health check.
   - Confirm every RFC 0005 crate for this role loads.
5. Emit "ready" on the LED + TSN heartbeat.
6. Enter the main role loop.

If any step fails the unit holds in a red-LED fault state and refuses to drive actuators. The custom-PCB v2 spec's watchdog supervisor has an equivalent role; the DIY path relies on systemd + the Pico 2's own watchdog.

DIY assembly — tools + materials

Repository source: `control-electronics/diy-assembly/tooling.md`

Required tools

Tool	Purpose	Typical source
PH1 + PH2 Phillips screwdriver	Enclosure + terminal-block + DIN-rail screws	Any hardware store
2.5 mm flat-head screwdriver	Small terminal block screws	Any hardware store
Wire cutter / stripper for 0.5 – 2.5 mm² stranded	Field wiring + terminal-block jumpers	Any electronics supplier
Ferrule crimper for 0.5 – 2.5 mm² ferrules	Stranded-wire termination into terminal blocks (do not tin with solder)	Knipex / generic
Micro-HDMI HDMI cable	First-boot commissioning via CM5's micro-HDMI + a monitor	Generic
Keyboard (USB)	First-boot commissioning	Generic

That is the entire tool list. No soldering iron, no hot-air station, no benchtop power supply, no logic analyser, no oscilloscope. The safety-case argument (RFC 0019 §7) does not require per-unit scope capture — the Pico 2's own self-test plus `osr-selftest` provides the verification.

Optional but recommended

Tool	Purpose
Multimeter	Spot-check 24 V power continuity at the DIN-rail terminal
ESD wrist strap	Good practice around the CM5 module; not strictly required once it's seated in the IO Board
Labelmaker	Cable labels per the wiring map; saves hours at commissioning
Torque screwdriver (0.5 Nm preset)	For enclosure screws on shock-rated field units (EN 50155 OT4)

Consumables (per host class)

Consumable	Spec	Qty per unit
Ferrules	0.5 / 0.75 / 1.5 mm² insulated colour-coded	~50
Cable ties	Nylon 2.5 mm × 100 mm	~20
Heat-shrink sleeve	Ø 3 mm + Ø 6 mm, any colour	~5 pieces
Labels	Adhesive printable	~30

Hygiene rules

1. Never power up with the CM5 not fully seated. The SO-DIMM pins will short on a partial insertion.
2. Strip to the ferrule's depth, no more. Over-strip = exposed copper outside the terminal = creepage violation.
3. Separate safety-critical wiring. The 2oo2 A-channel and B-channel wiring must route through different cable trays / different bundles. A common single-point cable-bundle damage incident defeats the 2oo2 safety argument. Use different insulation colours (A = black, B = brown, per RFC 0007 §safety-nets convention).
4. Label everything at assembly time. `A-CROSS-CHECK-IN`, `A-CROSS-CHECK-OUT`, `B-CROSS-CHECK-IN`, `B-CROSS-CHECK-OUT` — at 3 AM during a field incident nobody can guess.
5. Terminal-block torque is 0.5 Nm. Factor applies to every screw terminal in every host class.

Hardware Release Checklist

Repository source: `control-electronics/release-checklist.md`

This checklist turns the hardware gap into concrete release gates for each host class. OSR has two hardware release tracks:

- Pilot / DIY integration release: commodity SBCs, Pi Pico 2 boards, sensor modules, HATs, relay modules, power supplies, terminal blocks, and DIN-rail enclosures per RFC 0019. This track does not require KiCad or gerbers when no custom PCB is used.
- Custom-board fabrication release: OSR-specific carrier, power, safety-I/O, or sensor-interface boards per RFC 0007. This track requires KiCad, gerbers, board BOMs, DFM/DFT review, and board bring-up evidence.

Pilot / DIY Integration Track

Host class	COTS BOM	Wiring / harness map	Enclosure + mounting	Power / thermal margin	SD image + self-test	Bench / safety evidence
T-ECU/S	Pending freeze	Pending	Pending	Pending	Pending	2oo2 watchdog, relay-stage, cross-check, and power-fault bench logs
T-ECU/A	Pending freeze	Pending	Pending	Pending	Pending	Non-safety host self-test and communications fault-injection logs
T-OBS	Pending freeze	Pending	Pending	Pending	Pending	2oo2 obstacle-clear output, sensor-fault injection, heater/washer tests
W-SBC	Pending freeze	Pending	Pending	Pending	Pending	Wayside power, network isolation, intrusion/points IO fault-injection logs
S-SBC	Pending freeze	Pending	Pending	Pending	Pending	Station/depot host self-test and safe-failure procedure evidence

Pilot Integration Artifacts

Each pilot integration release creates:

- `control-electronics/<class>/diy-assembly/README.md` with assembly order and role-specific commissioning steps.
- `control-electronics/<class>/diy-assembly/bom-vN.csv` with exact SKUs, alternates, lifecycle status, supplier links, and substitution rules.
- `control-electronics/<class>/diy-assembly/wiring-map-vN.md` with terminal blocks, connector pinouts, cable labels, ferrule sizes, and harness routing.
- `control-electronics/<class>/diy-assembly/enclosure-vN.md` with DIN-rail, cabinet, sensor-mount, ingress-protection, and service-clearance notes.
- `control-electronics/<class>/diy-assembly/power-thermal-vN.md` with input ranges, fuse ratings, rail loads, heat budget, and derating limits.
- `control-electronics/<class>/bring-up/vN.md` with SD-card image checksum, boot log, `osr-selftest` output, smoke-test results, and known issues.
- `control-electronics/<class>/evidence/vN/` with photos, oscilloscope captures where relevant, watchdog/reset tests, safety-output truth tables, sensor-fault logs, and commissioning sign-off.

Custom-Board Fabrication Track

Host class	KiCad capture	Gerbers	Board BOM	Bring-up evidence	Safety evidence
T-ECU/S	Pending	Pending	Pending	Pending	2oo2 watchdog, relay-stage, cross-check, and power-fault bench logs
T-ECU/A	Pending	Pending	Pending	Pending	Non-safety host self-test and communications fault-injection logs
T-OBS	Pending	Pending	Pending	Pending	2oo2 obstacle-clear output, sensor-fault injection, heater/washer tests
W-SBC	Pending	Pending	Pending	Pending	Wayside power, network isolation, intrusion/points IO fault-injection logs
S-SBC	Usually unnecessary	Usually unnecessary	Usually unnecessary	Pending	Station/depot host self-test and safe-failure procedure evidence

Custom-Board Artifacts

Each custom-board release creates:

- `control-electronics/<class>/schematics/vN-kicad/` with the KiCad project.
- `control-electronics/<class>/gerbers/vN-rev-X/` with the fabrication ZIP and drill files.
- `control-electronics/<class>/bom/vN-rev-X.csv` with manufacturer part numbers, alternates, lifecycle status, and supplier links.
- `control-electronics/<class>/bring-up/vN-rev-X.md` with power-rail checks, programming steps, smoke-test results, and known issues.
- `control-electronics/<class>/evidence/vN-rev-X/` with oscilloscope captures, watchdog/reset tests, safety-output truth tables, and DFM/DFT review.

Release Rule

A host class may be referenced by the certification pack as `pilot integration hardware` after the Pilot / DIY Integration Track is closed for the same revision. It may be referenced as `released fabrication hardware` only after the Custom-Board Fabrication Track is closed for the same revision.

Rolling-stock hardware integration

Repository source: `control-electronics/rolling-stock-integration.md`

This file is the hardware-side bridge to the `light-metro-3car` trainset BOM and mechanical CAD. It keeps electronic host-class documentation in `control-electronics/`, while the consist-level quantities and physical envelopes remain owned by:

- `docs/rolling-stock/light-metro-3car/bom-skeleton.md`
- `docs/rolling-stock/light-metro-3car/README.md`
- `design/component-catalogue/src/osr_mech/rolling_stock/systems.py`

Canonical light-metro-3car fit

Hardware class	Consist quantity	Mechanical envelope	BOM lines	Notes
T-ECU/S safety kernel	2	Per-end electronics cabinet	E1, E3-E6, E22	One A-end unit and one B-end unit; active/hot-standby at train level
T-ECU/A application	2	Per-end electronics cabinet	E2, E7-E12, E22	One A-end unit and one B-end unit; non-safety, single-redundant
T-OBS obstacle-detection ECU	2	Nose sensor pack	E18, E19	One behind each cabless nose cowl
Trainset interior COTS modules	3 car sets	Car systems + fit-out envelopes	B12-B20, A1-A5	Quantities scale per self-contained car unless BOM says per consist
W-SBC wayside controller	route dependent	Wayside cabinet/pole, not trainset CAD	Wayside BOM	Not part of trainset material BOM
S-SBC station/depot host	station dependent	Station/depot cabinet, not trainset CAD	Station BOM	Not part of trainset material BOM

Ownership rule

- `control-electronics/` owns board architecture, DIY/COTS integration packs, schematics, custom-board gerbers, board-level BOMs, wiring maps, enclosure notes, and bring-up evidence.
- `design/component-catalogue/` owns physical CAD envelopes, FreeCAD review artifacts, and generated screenshots for train components.
- `docs/rolling-stock/light-metro-3car/bom-skeleton.md` owns procurement quantities per consist.

If a board changes size, connector side, heat dissipation, or service clearance, update the matching mechanical envelope and drawing register. If a consist quantity changes, update the rolling-stock BOM first, then mirror that quantity here and in the host-class README.

S-SBC — Station / Depot

Repository source: `control-electronics/s-sbc/README.md`

Role: Host for `osr-psd`, `osr-pis-station`, `osr-station-scada`, `osr-afc`, `osr-tvm`.

Indoor deployment; no rail-industry environmental rating required.

SoC + baseboard

Raspberry Pi CM5 on a commodity carrier (e.g. Waveshare CM5-IO). No custom baseboard. The station enclosure carries the I/O.

This is a deliberate deviation from the other classes: sheltered indoor environments don't warrant the custom engineering, and every hour saved on baseboard bring-up goes into the passenger-facing work that actually differentiates an OpenSourceRail deployment.

See RFC 0007 §7 for the commodity-carrier rationale.

Peripherals (from the commodity carrier)

- 2 × Gigabit Ethernet (station LAN + PSD/gate network)
- 4 × USB 3.0 (QR scanner, NFC reader, receipt printer, TVM bill acceptor)
- HDMI for passenger-facing display
- USB-C console
- ATECC608 add-on over I²C (the only custom piece — a small 30 × 20 mm PCB that plugs onto the CM5 I/O header)

Target BOM

~€130 total — CM5 module + carrier + small SE add-on.

Status

Pilot / DIY track: the station-side software stack already runs on stock Waveshare carriers in the development lab. Integration evidence is still pending.

Custom-board track: usually unnecessary for S-SBC. The only custom piece is the small ATECC608 add-on (two ICs + passives), tracked as a v2 deliverable alongside the other classes for consistency.

S-SBC bring-up — RPi CM5 on Waveshare CM5-IO

Repository source: `control-electronics/s-sbc/bring-up.md`

Goal: confirm the station / depot host runs `osr-psd` + `osr-afc` + `osr-tvm` + `osr-pis-station` + `osr-station-scada` on a stock commodity carrier, with the small ATECC608 add-on as the only custom-PCB piece.

Dev kits + parts

Part	Qty	Notes
Raspberry Pi CM5 (4 GB + 32 GB eMMC)	1	
Waveshare CM5-IO (commodity)	1	
ATECC608 breakout on I²C	1	The only custom PCB — a 30 × 20 mm carrier for the SE chip.
QR scanner (USB)	1	
NFC reader (USB, ACR122U or equivalent)	1	
Thermal receipt printer (USB)	1	
10-inch HDMI screen	1	PIS display.
Test bench	—	

Steps

X1 — Boot Debian

Same as T-ECU/A step A1, same image, same hostname pattern (`ssbc-devkit`). No RT configuration — station role is SIL-0.

X2 — Workspace build

Subset build: station-side crates only.

```
cargo build --release -p osr-psd -p osr-afc -p osr-tvm \
    -p osr-pis-station -p osr-station-scada
```

X3 — ATECC608 add-on enumeration

```
i2cdetect -y 1        # see ATECC608 at 0x60
```

Expected: chip responds. Provision a test key via the `osr-crypto` provisioning tool.

X4 — Fare gate (osr-afc)

Connect QR scanner + NFC reader to USB. Run the AFC gate simulator:

```
cargo run --release -p osr-afc --example gate-demo
```

Present a test QR code; present a test NFC token.

Expected: both decode + produce `FareAccepted` events within 100 ms.

X5 — Ticket vending (osr-tvm)

Connect the receipt printer. Run:

```
cargo run --release -p osr-tvm --example tvm-demo
```

Drive a scripted purchase: scan product QR, submit mock mobile- money payment, confirm printed receipt.

Expected: receipt prints within 3 s of payment confirmation.

X6 — Passenger information (osr-pis-station)

Connect the HDMI screen. Run:

```
cargo run --release -p osr-pis-station --example display-demo
```

Expected: next-arrival list displays; mock incoming MA updates refresh the arrival times.

X7 — PSD controller (osr-psd)

With no physical PSD on the bench, use the crate's software-only simulator. Run:

```
cargo run --release -p osr-psd --example psd-demo
```

Expected: the five-state PSD FSM transitions correctly on simulated train-arrival + door-open / close events.

X8 — Station SCADA

Launch:

```
cargo run --release -p osr-station-scada --example scada-demo
```

Simulate HVAC / lighting / escalator status inputs.

Expected: status dashboard populates; threshold alarms fire within one tick of exceeding setpoint.

X9 — Sustained run

Run all five station crates simultaneously for 1 hour.

Expected: CPU load ≤ 30 %, RAM ≤ 1 GB, no process restarts.

Bring-up report template

```
# S-SBC bring-up report

- Runbook commit: <hash>
- Date: <YYYY-MM-DD>
- CM5 serial: <nnn>

## Results
| Step | Status |
|---|---|
| X1 (boot)     | PASS / FAIL-\* |
| X2 (build)    | PASS / FAIL-\* |
| X3 (ATECC608) | PASS / FAIL-\* |
| X4 (AFC gate) | PASS / FAIL-\* |
| X5 (TVM)      | PASS / FAIL-\* |
| X6 (PIS)      | PASS / FAIL-\* |
| X7 (PSD)      | PASS / FAIL-\* |
| X8 (SCADA)    | PASS / FAIL-\* |
| X9 (1 h run)  | PASS / FAIL-\* |
```

S-SBC bench assembly requirements

Repository source: `control-electronics/s-sbc/diy-assembly/README.md`

Status: bench integration reference; hardware BOM, image, wiring and qualification are open. The release checklist governs the assembly. Exact SKUs and total prices are not frozen; previous retail subtotals omitted integration parts.

Block	Required detail before assembly release
Compute and carrier	Exact CM5 configuration, compatible carrier, storage and cooling; 5 V input supplied correctly
Field power	Approved input range, DC/DC, fuse/selectivity, surge/EMC, inrush and hold-up
I/O and network	External CAN controller/transceiver where required, Ethernet switching, isolation, termination and pin map
Security	Channel-local I2C secure element, provisioning and recovery process
Harness/enclosure	Mating connectors, contacts, terminals, ferrules, wire cut list, clamps, labels, grounding and thermal design
Output chain	Simulated loads for bench; qualified drivers, feedback and actuator interfaces before deployment
Firmware/image	Real target HAL/driver mapping, immutable build, configuration/signature hashes and measured startup behaviour

Mean Well HDR-60-24 converts AC/high-voltage DC input to 24 V output. It may supply a suitable isolated laboratory or mains-fed wayside setup, but cannot convert a 24 V field bus to 5 V. Onboard hosts require protected DC/DC supplies. No wiring from a mains terminal is released here.

For the safety-host bench use the validated exposed-pin allocation as a proposed starting point and freeze any host-specific changes. SPI needs a digital isolator and controller/peripheral wiring; a USB isolator does not provide SPI or a direct link between two USB devices. Pico 2 GPIO30/31 and chip-package-only pins are not available on its header. Do not assume that a Pi CAN HAT independently serves both Pico channels.

For application/wayside/station hosts, the modem needs a compatible carrier and power/USB interface; multiple HATs require connector and pin conflict review. Size UPS, enclosure and converter loads from the complete host BOM. Application, station and wayside software disposition comes from deployment/components.toml; TACS runtime remains a reference tooling host until qualified hardware composition is released.

Confirm GPIO voltage domains, loss-of-power state, wiring continuity/insulation, bus timing, EMI, thermal soak, image boot and recorded self-test on the actual revision. Procurement, construction and train actuation require the released package, not this list.

S-SBC v2 schematic specification

Repository source: `control-electronics/s-sbc/schematics/v2-spec/README.md`

Status: RFC 0007 v3 deliverable. Specification for the station / depot SBC per RFC 0007 §7.

Scope

S-SBC hosts the station-tier and depot-tier crates, all SIL-2 or SIL-0 — station operation is not safety-critical because passenger safety on the platform is delegated to PSDs (SIL-4 hardware at the door itself) + station-staff supervision per RFC 0013 T2, with the interlocking handling train-side safety autonomously.

- `osr-psd` — PSD controller (SIL-2 logic; the SIL-4 mechanical interlock is inside each PSD unit).
- `osr-station-scada` — station SCADA / HMI.
- `osr-pis-station` — passenger-information displays.
- `osr-afc` — automatic fare collection logic.
- `osr-tvm` — ticket-vending-machine controller.
- `osr-historian` — metric ring-buffer storage (when station hosts a regional historian).
- `osr-cbm-backend` — deployment-side CBM for depot-adjacent S-SBCs.

At depots, the S-SBC also hosts the depot-automation crate that orchestrates overnight trainset movements (stall allocation, charging-pad assignment, pre-service brake tests).

SoC

Raspberry Pi Compute Module 5 (BCM2712) on a commodity carrier — no custom baseboard required.

- 4-core Cortex-A76, 4–16 GB LPDDR4X.
- eMMC on-module.
- Wide availability, mature Linux, broad spares pool in target deployment markets.

S-SBC stays on the RPi CM5 (not Radxa RK3588S industrial) because station cabinets are temperature-controlled — the commercial OT3 temperature grade is sufficient indoors. For outdoor TVMs in extreme climates, deploy a Radxa CM5 drop-in replacement under the same carrier pinout (§3.2 of RFC 0007).

Architecture

Non-safety, non-redundant per station. Typical station carries one S-SBC with cold-spare kept at the depot. If a station's S-SBC fails:

- PSDs lock open (fail-safe for passenger egress) per `osr-psd`'s PSD-controller contract.
- Fare gates default to open (paid-area unlocked).
- PIS displays show the emergency-only banner.
- OCC dispatches a maintainer per RFC 0013 T5 ticket.

Recovery is swap-in of the cold spare (5–15 minute task per RFC 0013 M5 depot procedure).

Peripherals (commodity carrier)

Peripheral	Qty	Purpose
Gigabit Ethernet	2	Backhaul to W-SBC + OCC; station-local PIS LAN
USB 3.0	4	TVMs, fare gates, PSD console
HDMI	2	Station-master console + PIS display (small stations)
MIPI-CSI	2	Station-monitoring cameras
microSD	1	Boot image (eMMC is primary)
ATECC608B (optional)	1	Trust anchor for station-originated consensus entries (e.g., PSD-state)
12/24 V DC input	1	Either building DC or adapter from 230 VAC mains

Power budget

Rail	Voltage	Max power
CM5 core + RAM	5 V	~10 W
USB downstream (TVM, gates)	5 V	~8 W (at 2 × TVM idle)
Camera MIPI-CSI	3.3 V	~1 W
Fans, lights	12 V	deployment-specific
Baseboard total	—	~20 W sustained

Cooling: natural-convection in a temperature-controlled station equipment room. Fan-forced air in TVM outdoor kiosks.

Environment

- Typical: indoor station equipment room, 0–45 °C.
- Outdoor TVM: −10 to +55 °C with sun-shield; consider Radxa drop-in for the RK3588S industrial-temp variant if local climate is harsher than that.
- IP43 for indoor cabinet, IP65 for outdoor TVM.

Form factor

Commodity off-the-shelf RPi CM5 IO Board (official Raspberry Pi carrier) is sufficient for most deployments. A custom low-cost carrier is v3.1 if a particular deployment wants a single integrated SKU.

Target BOM

~€150 per board total (CM5 ≈ €85 + commodity carrier ≈ €65).

Expected v2-spec file set

- `block-diagram.md` — functional block diagram.
- `power-budget.md` — station + depot power profile.
- `connector-tables.md` — HMI + PIS + fare-gate connector pinouts.
- `deviations-log.md` — design-review notes.

No safety-nets.md because S-SBC does not drive any SIL-4 actuator — the PSD SIL-4 interlock is inside the PSD hardware, not on this board.

Safety-controller selection gate

Repository source: `control-electronics/safety-controller-selection.md`

OSR has not frozen a revenue-service safety MCU. Raspberry Pi and Radxa hosts remain the inexpensive pilot and non-safety application baseline; they are not the certification rationale for a SIL-4-target function.

Required selection envelope

A deployment safety controller must be selected with the independent safety assessor and demonstrate:

- manufacturer safety manual, diagnostic-coverage evidence and FMEDA inputs;
- lifecycle and supply assurance appropriate to the fleet life;
- independent watchdog, power, clock, I/O isolation and fail-safe output path;
- a justified redundancy, diversity and common-cause strategy at system level;
- qualified compiler/runtime, traceable configuration and hardware-in-loop tests;
- environmental, EMC, vibration and board-level evidence for the actual host.

No MCU establishes SIL-4 by part number alone. SIL capability, hardware fault tolerance and systematic capability must be argued for the complete subsystem.

Candidate families—not a supplier freeze

Family	Appropriate use	Available manufacturer evidence	v0.3 decision
Infineon AURIX TC3xx	Candidate safety channel for train or wayside controller	IEC 61508 SIL 3 / ISO 26262 ASIL D capability, safety manual and FMEDA support package	Shortlist; assessor and lifecycle review required
TI Hercules TMS570/RM	Candidate safety channel, including long-lifecycle variants	Dual lockstep cores, ECC/BIST and IEC 61508 SIL 3 collateral	Shortlist; assessor and lifecycle review required
Raspberry Pi RP2350	Software bring-up, low-cost I/O rigs and prototype 2oo2 experiments	Public datasheet; two processor sockets each boot-select Cortex-M33 or Hazard3, with the unused implementation held in reset	Prototype only; not the default certification baseline

Primary manufacturer references:

- Infineon AURIX TC3xx functional-safety documentation
- Infineon IEC 61508 certification support
- TI TMS570 safety manual
- Raspberry Pi RP2350 datasheet

For the v0.3 pilot, the existing RP2350 boards remain useful evidence rigs. For a revenue-service design, the operator records the selected channels, safety package revision and architectural rationale in its controlled hardware baseline. CM5/Radxa processors may host diagnostics, logging and applications, but cannot bypass or command a safety output.

Development redundancy default

The controlled policy requires two logical protection channels: both permit, either trips, and loss stops the affected function. Train A/B software processes share code, sensor fixtures, host, clock and a synthetic comparator; they do not establish independent hardware or availability after a channel failure. Remaining local pairs are explicitly open. Reconcile existing hot-standby boards/two-MCU quantities with real serials, power domains, firmware, calibration, independent watchdogs and output feedback at hardware freeze. No process count is an installed BOM.

T-ECU/A — Train Application

Repository source: `control-electronics/t-ecu-a/README.md`

Role: Host for the SIL-2 / SIL-0 onboard crates — `osr-ato`, `osr-tcms`, `osr-pis-onboard`, `osr-hvac`, `osr-lighting`, `osr-aux-power`, `osr-event-recorder`, `osr-regen`, `osr-hot-axle`, `osr-cbm-onboard`, `osr-t2g`, `osr-tcn`.

Canonical `light-metro-3car` fit: 2 boards per trainset, one in the A-end electronics cabinet and one in the B-end electronics cabinet. T-ECU/A is single-redundant and non-safety; the T-ECU/S + T-OBS safety chain continues without it.

Environment: EN 50155 OT4, single-redundant.

SoC

Raspberry Pi CM5 (BCM2712, Cortex-A76 4-core, 4–16 GB LPDDR4X, eMMC on-module) on a custom baseboard. Radxa CM5 (RK3588S) is the pin-compatible drop-in via the same SO-DIMM footprint — only the device-tree blob changes. Operators may source whichever module is locally stocked.

See RFC 0007 §5.

Peripherals

Peripheral	Qty	Purpose
TSN Ethernet	2	TCN-E A/B
CAN-FD	2	HVAC + lighting
USB-C host+device	2	Depot console + diag loader
HDMI	1	Commissioning display
USB 2.0	2	Diagnostics and PIS service
M.2 2280 NVMe	1	Event recorder + OTA staging
M.2 2230 Cat.22 5G	1	TRG-1 primary radio
LoRa SX1276 on SPI	1	TRG-2 backup radio
SPI bridge to T-ECU/S	1	One-way feed from the safety board
ATECC608B SE	1	Trust anchor

Form factor

160 × 100 mm Eurocard (same DIN slot as T-ECU/S). Heatsink enclosure conduction cooling, no fans.

Target BOM

~€220 per board (CM5 ≈ €85, baseboard ≈ €75, radios ≈ €60). The consist-level procurement quantity is tracked in `../rolling-stock-integration.md` and line E2 of the rolling-stock BOM.

Status

- Pilot / DIY track: `diy-assembly/` uses a stock RPi CM5 IO Board + Waveshare M.2 HAT + generic Cat.22 5G M.2 module to validate peripheral enumeration before any custom baseboard is drawn. Integration evidence is still pending.
- Custom-board track: `schematics/v2-spec/` is the board-level v2 specification overview and block diagram. `gerbers/` and `bom/` remain pending until a KiCad layout and board BOM are released.

T-ECU/A bring-up — Raspberry Pi CM5 IO Board

Repository source: `control-electronics/t-ecu-a/bring-up.md`

Goal: validate the application-tier host runs every non-safety onboard crate and the TCN-E + TRG radios enumerate on a stock CM5 carrier.

Dev kits + parts

Part	Qty	Notes
Raspberry Pi CM5 (4 GB + 32 GB eMMC)	1	
Raspberry Pi CM5 IO Board	1	
M.2 Hat+ (Waveshare or Pimoroni)	1	For NVMe + 5G modem.
NVMe SSD 256 GB	1	Event recorder ring buffer target.
Cat.22 5G M.2 module (unlocked)	1	TRG-1 primary radio.
LoRa SX1276 breakout on SPI	1	TRG-2 backup radio.
Mikroe CAN-FD Click 3 (x2)	2	Onto the IO Board's mikroBUS slots.
USB-C PD 5V/3A supply	1	

Steps

A1 — Boot mainline Debian

1. Write Debian 13 `arm64` (with RPi kernel) to the CM5 eMMC via `rpiboot`.
2. First-boot: set hostname `tecus-devkit`, expand rootfs to NVMe.
3. Install Rust toolchain: `curl ... | sh`, `rustup default stable`.

Expected: ssh login within 60 s of power-on.

A2 — Clone + build the workspace

```
git clone https://github.com/modernecotech/OpenSourceRail.git
cd OpenSourceRail
cargo build --release --workspace
```

Expected: clean build; 46 crates compile. Wall-clock ≤ 20 min on a 4 GB CM5.

A3 — Enumerate peripherals

Confirm each peripheral shows up in the kernel:

```
ip link show                # two Ethernet interfaces
ls /sys/class/net           # eth0, eth1
lspci                       # NVMe, 5G M.2
ls /dev/ttyAMA*             # UARTs
ls /dev/spidev*             # SPI buses
ls /sys/bus/i2c/devices     # I2C (TCN-E + ATECC608)
```

Tick each off in the report. Missing devices check device-tree overlay; common culprit is missing `dtoverlay=pcie2-1lane` or equivalent in `config.txt`.

A4 — TCN-E mock loopback

```
cargo test -p osr-tcn --lib -- --nocapture
cargo test -p osr-tcn --test roundtrip_single_payload -- --nocapture
```

Expected: both green. The UDP-over-loopback path (RFC 0006 v1.5) is the surrogate for the eventual TSN link.

A5 — TRG-1 (5G) bring-up

Insert the 5G module into the M.2 slot. Install `ModemManager` and `NetworkManager`.

1. Confirm `mmcli -L` lists the modem.
2. Activate a test APN: `mmcli -m 0 --simple-connect="apn=..."`.
3. Confirm IPv4 connectivity to a public endpoint.

Expected: working PDP context within 60 s of modem boot.

Carrier choice: bring-up is carrier-agnostic. For deployment use, the `osr-t2g` SIM-profile spec (RFC 0005 §3.2) handles the operator's choice.

A6 — TRG-2 (LoRa) bring-up

Connect SX1276 breakout to SPI0.

1. Build the `osr-t2g` diagnostic binary.
2. Run `osr-t2g-diag --link lora --tx-power 14dbm --channel 868.1`.
3. Confirm a second LoRa node (can be another dev board) receives the test packets with RSSI ≥ −110 dBm at 50 m line-of-sight.

Expected: ≥ 99 % packet delivery at 10 Hz test rate.

A7 — CAN-FD bus test

Install two CAN-FD Clicks on the mikroBUS slots. Connect via a CAN-FD transceiver test harness.

1. `ip link set can0 up type can bitrate 500000 dbitrate 2000000 fd on`.
2. `cansend can0 100##1.01020304`.
3. `candump can0` on the second Click receives the frame.

Expected: no frame loss at 1 kHz for 5 minutes.

A8 — Shadow onboard stack in `osr-sim`

From the workspace:

```
cargo run --release --bin osr-sim -- \
    --scenario cities/catalogue/west-asia/Iraq/Samawah/samawah.toml --duration 900 --status-every 60
```

Expected: the shadow onboard stack ran on every tick on the CM5, consuming ≤ 20 % CPU, with zero invariant violations.

A9 — Event recorder

```
osr-event-recorder-cli --tail 100 /var/log/osr/event.bin
```

Expected: the last 100 TCN-E frames from step A4 are extractable. Ring buffer is working.

Bring-up report template

```
# T-ECU/A bring-up report

- Runbook commit: <hash>
- Date: <YYYY-MM-DD>
- CM5 serial: <nnn>

## Results
| Step | Status |
|---|---|
| A1 (Debian boot) | PASS / FAIL-\* |
| A2 (workspace build) | PASS / FAIL-\* |
| A3 (peripherals) | PASS / FAIL-\* |
| A4 (TCN-E loopback) | PASS / FAIL-\* |
| A5 (5G) | PASS / FAIL-\* |
| A6 (LoRa) | PASS / FAIL-\* |
| A7 (CAN-FD) | PASS / FAIL-\* |
| A8 (sim shadow) | PASS / FAIL-\* |
| A9 (event recorder) | PASS / FAIL-\* |
```

T-ECU-A bench assembly requirements

Repository source: `control-electronics/t-ecu-a/diy-assembly/README.md`

Status: bench integration reference; hardware BOM, image, wiring and qualification are open. The release checklist governs the assembly. Exact SKUs and total prices are not frozen; previous retail subtotals omitted integration parts.

Block	Required detail before assembly release
Compute and carrier	Exact CM5 configuration, compatible carrier, storage and cooling; 5 V input supplied correctly
Field power	Approved input range, DC/DC, fuse/selectivity, surge/EMC, inrush and hold-up
I/O and network	External CAN controller/transceiver where required, Ethernet switching, isolation, termination and pin map
Security	Channel-local I2C secure element, provisioning and recovery process
Harness/enclosure	Mating connectors, contacts, terminals, ferrules, wire cut list, clamps, labels, grounding and thermal design
Output chain	Simulated loads for bench; qualified drivers, feedback and actuator interfaces before deployment
Firmware/image	Real target HAL/driver mapping, immutable build, configuration/signature hashes and measured startup behaviour

Mean Well HDR-60-24 converts AC/high-voltage DC input to 24 V output. It may supply a suitable isolated laboratory or mains-fed wayside setup, but cannot convert a 24 V field bus to 5 V. Onboard hosts require protected DC/DC supplies. No wiring from a mains terminal is released here.

For the safety-host bench use the validated exposed-pin allocation as a proposed starting point and freeze any host-specific changes. SPI needs a digital isolator and controller/peripheral wiring; a USB isolator does not provide SPI or a direct link between two USB devices. Pico 2 GPIO30/31 and chip-package-only pins are not available on its header. Do not assume that a Pi CAN HAT independently serves both Pico channels.

For application/wayside/station hosts, the modem needs a compatible carrier and power/USB interface; multiple HATs require connector and pin conflict review. Size UPS, enclosure and converter loads from the complete host BOM. Application, station and wayside software disposition comes from deployment/components.toml; TACS runtime remains a reference tooling host until qualified hardware composition is released.

Confirm GPIO voltage domains, loss-of-power state, wiring continuity/insulation, bus timing, EMI, thermal soak, image boot and recorded self-test on the actual revision. Procurement, construction and train actuation require the released package, not this list.

T-ECU/A v2 schematic specification

Repository source: `control-electronics/t-ecu-a/schematics/v2-spec/README.md`

Status: RFC 0007 v3 deliverable (partial). This folder holds the board-level specification (net list, pinouts, power budget, connector tables) for the T-ECU/A v2 application-tier ECU per RFC 0007 §5.

KiCad capture + gerbers + BOM are v3.1, deferred alongside T-ECU/S and T-OBS.

Scope

T-ECU/A hosts the SIL-2 + SIL-0 onboard application crates:

- `osr-ato` — Automatic Train Operation (GoA 4 default).
- `osr-tcms` — Train Control and Management System.
- `osr-event-recorder` — black-box circular storage.
- `osr-tcn` — TSN Ethernet trainbus (RFC 0006).
- `osr-t2g` — train-to-ground radio adapter.
- `osr-regen`, `osr-aux-power`, `osr-hvac`, `osr-lighting` — comfort + auxiliary systems.
- `osr-pis-onboard` — passenger information displays.
- `osr-hot-axle` — onboard hot-axle-box advisory (SIL-2).
- `osr-cbm-onboard` — condition-based-maintenance telemetry.

Architecture

Single-redundant at each end: the standard trainset fit carries two T-ECU/A units, one behind each nose. Unlike T-ECU/S this is not 2oo2: its crates are non-safety-critical. A failure of one T-ECU/A degrades comfort (HVAC, lighting, PIS) and costs diagnostics visibility, but the safety chain (T-ECU/S + T-OBS + brake relay stage) continues to function.

Block	Qty	Role
RPi CM5 / Radxa CM5	1	Application processor (4-core A76, 4–16 GB LPDDR4X)
TSN Ethernet switch	1	88E6321 — redundant TCN-E A/B
CAN-FD transceivers	2	HVAC bus + lighting bus (TCAN1462)
5G modem socket	1	M.2 Cat.22 — TRG primary
LoRa radio	1	SX1276 on SPI — TRG backup
NVMe slot	1	M.2 2280 for event recorder + OTA staging
USB-C	2	Depot console + diag loader
HDMI	1	Commissioning display output
ATECC608B	1	Trust anchor
Power input	1 (redundant)	24 V DC from aux bus

Power budget (target)

Rail	Voltage	Load	Max power
CM5 core	5 V	CM5 + RAM under load	~12 W
3.3 V	3.3 V	Ethernet PHY + CAN + ATECC	~2 W
1.8 V	1.8 V	MIPI + TSN	~1 W
5G radio	5 V	M.2 modem at full tx	~8 W
LoRa	3.3 V	SX1276 tx	~0.5 W
Total at 24 V in	—	—	~28 W peak, ~18 W sustained

Natural-convection cooling. Conduction path to DIN rail chassis for the CM5 heatsink. No fans.

Environment

- EN 50155 OT4 (-25 °C to +70 °C with +85 °C 10-minute peak).
- IP54 enclosed; DIN-rail mount.
- Conformal coated (MG Chemicals 419).

Form factor

160 × 100 mm Eurocard — same footprint as T-ECU/S so a cabinet slot can host either without re-mounting.

Target BOM

~€220 per board (CM5 ≈ €85, baseboard ≈ €75, radios ≈ €60 volume).

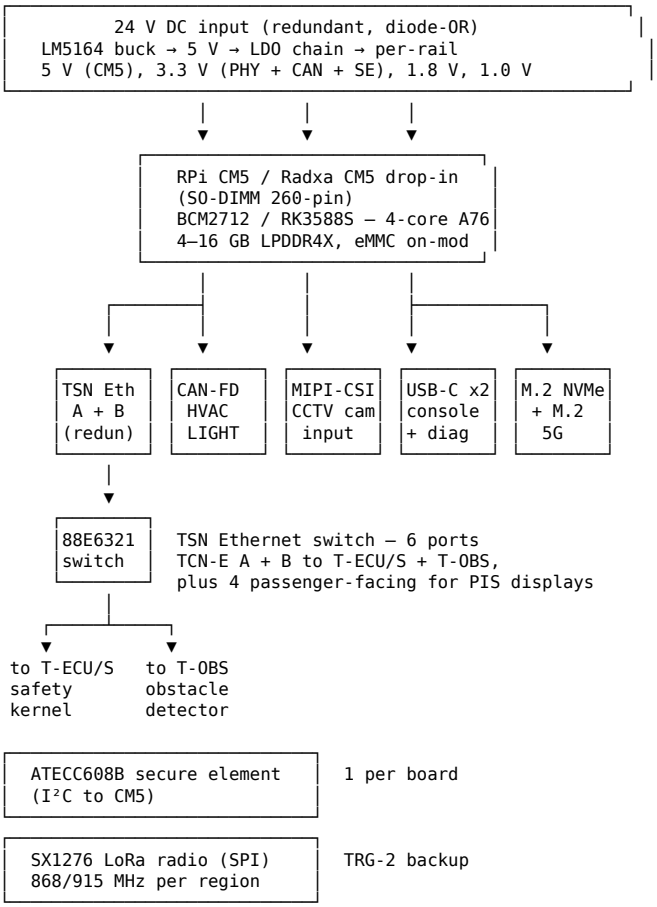
Expected v2-spec file set

- `block-diagram.md` — functional block diagram.
- `power-budget.md` — rail-by-rail load + headroom.
- `pinout-cm5.md` — CM5 SO-DIMM pin assignments.
- `connector-tables.md` — external M12 + FFC pinouts.
- `deviations-log.md` — design-review notes.

This README provides the summary; individual files fill in per-block detail as the v2-spec milestone matures.

T-ECU/A v2 — block diagram

Repository source: `control-electronics/t-ecu-a/schematics/v2-spec/block-diagram.md`



Block inventory

Block	Ref	Purpose
Power input A	J1	24 V DC primary
Power input B	J2	24 V DC redundant
Main buck	U1	LM5164 245 V, 4 A
LDO 3.3 V	U2	TPS7A52 53.3 V for PHY/CAN/SE
LDO 1.8 V	U3	For MIPI + TSN PHY
CM5 module	U4	RPi CM5 on SO-DIMM (Radxa CM5 pin-compatible)
TSN switch	U5	Marvell 88E6321 — 6-port TSN-capable
Ethernet PHY ×2	U6, U7	88E1512 on TSN A/B uplink to T-ECU/S
CAN transceiver A	U8	TCAN1462 — HVAC bus
CAN transceiver B	U9	TCAN1462 — lighting bus
USB hub	U10	USB 3.1 Gen 1 — depot console + diag loader
M.2 slot (NVMe)	M1	2280 — event recorder ring buffer
M.2 slot (5G)	M2	2230/3042 Cat.22 modem
LoRa radio	U11	SX1276 on SPI from CM5
ATECC608B	U12	Trust anchor, I²C
HDMI header	J3	Commissioning display output
TSN connector A	J4	M12 X-code — TCN-E A
TSN connector B	J5	M12 X-code — TCN-E B
USB-C	J6, J7	Depot console + diagnostics
M.2 antenna ports	J8, J9	5G (4 × 4 MIMO)
LoRa antenna	J10	868/915 MHz

Non-safety architecture

T-ECU/A is single-redundant, not 2oo2. A failure degrades comfort and diagnostics but not safety:

- TCMS stops publishing rollup state OCC supervisor sees "trainset silent" and dispatches recovery per RFC 0013 C1.3.
- HVAC fails passenger discomfort, not a safety event.
- Event recorder fails forensic gap, recoverable from the T-ECU/S black-box mirror.
- TRG silent `osr-atp` continues on the last-known MA until the validity window expires, then trips EB.

The safety chain (T-ECU/S + T-OBS + W-SBC interlocking) does not depend on T-ECU/A being healthy.

T-ECU/S — Train Safety Kernel

Repository source: `control-electronics/t-ecu-s/README.md`

Role: Host for the SIL-4 onboard crates — `osr-atp`, `osr-odometry`, `osr-brake`, `osr-bms`, `osr-fire-safety`, `osr-derailment`, `osr-door-control`, `osr-traction` (MCU bring-up only).

Canonical `light-metro-3car` fit: 2 boards per trainset, one in the A-end electronics cabinet and one in the B-end electronics cabinet. The train-level role is one primary + one hot-standby per consist; neither unit depends on a driver cab being present.

Environment: EN 50155 OT4 (−40…+85 °C), IEC 61373 Cat 1 Class B, EN 50121-3-2 EMC.

Pilot 2-out-of-2 evidence rig

The current low-cost board carries two Raspberry Pi RP2350 MCUs in a 2oo2 voting arrangement. Both chips receive the same sensor inputs on separate pins, both run identical Rust `no_std` code, both outputs are AND-gated through an external 2oo2 relay network. Each tick they cross-check their computed safety decisions over SPI; on mismatch, both chips fail-safe open and a hardware watchdog asserts the emergency brake relay directly.

This is useful for software, I/O and fail-safe-output experiments. It does not by itself establish SIL-4: identical devices/software carry systematic and common-cause obligations. Revenue-service hardware must pass the `safety-controller-selection.md` gate.

SoC picks

- Prototype MCU × 2: Raspberry Pi RP2350. Two processor sockets; each selects Cortex-M33 or Hazard3 at boot while the unused implementation is held in reset. 520 KB SRAM, 150 MHz, TrustZone-M, HW RNG. ~€1.10 in 1k volume. QFN-56 0.4 mm pitch — routine SMT.
- App processor × 1: Raspberry Pi CM5 (BCM2712). Hosts TCN-E, logging, OTA, event recorder — never on the safety path. Subscribes to a one-way SPI feed from the RP2350 pair.

Peripherals

Peripheral	Qty	Purpose
CAN-FD	4	Door, brake, traction, BMS (each on both RP2350s)
TSN Ethernet (CM5 + KSZ9031 PHY)	2	TCN-E A/B
Isolated DI	8	Plunger, cab, depot enable, deadman (routed to both RP2350s)
Isolated DO	4	EB relay, traction cut, park, fire (via 2oo2 relay stage)
Tach inputs (quadrature 5–24 V)	2	Wheel encoders (both RP2350s)
Bosch BMI088 IMU	1	Derailment + odometry, SPI (both RP2350s)
u-blox NEO-F10N GNSS	1	UART to CM5
PN5180 NFC balise reader	1	SPI to CM5
PT100 thermistors	4	Battery, traction, HVAC, ambient
Microchip ATECC608B SE	2	Trust anchor per RP2350

Form factor

160 × 100 mm Eurocard, DIN-rail mount, conduction-cooled. No fans. Conformal-coated (MG Chemicals 419) before integration. IP54.

Target BOM (volume 100+)

~€280 per prototype board. This is a pilot estimate, not the frozen revenue-service safety-controller cost. The consist-level quantity is tracked in `../rolling-stock-integration.md` and line E1 of the rolling-stock BOM.

Status

- Pilot / DIY track: `diy-assembly/` uses two Raspberry Pi Pico 2 boards (each carrying one RP2350) in a 2oo2 test jig, plus a stock RPi CM5 IO Board for the app processor. Boots `osr-atp` + `osr-brake` + `osr-odometry` + the SPI cross-check harness. Integration evidence is still pending.

- Custom-board track: `schematics/v2-spec/` is the board-level v2 specification: block diagram, power budget, connector tables, CM5/RP2350 pinouts, and safety-net rules. `gerbers/` and `bom/` remain pending until a KiCad layout and board BOM are released.

T-ECU/S bring-up — two Raspberry Pi Pico 2 boards in 2oo2

Repository source: `control-electronics/t-ecu-s/bring-up.md`

Goal: validate that the RFC 0007 §4 2oo2 safety architecture works on real silicon before we draw the custom baseboard. After this runbook, a pair of Pico 2 boards runs `osr-atp` + `osr-brake` + `osr-odometry` in composite fail-safe mode with per-tick SPI cross-check.

Dev kits + parts

Part	Qty	Notes
Raspberry Pi Pico 2	2	Each carries one RP2350 — the safety MCU.
ST-Link V3 or Pico probe	2	SWD debugger per Pico.
Breadboard + jumpers	1 set	SPI cross-link + shared inputs.
3-phase quadrature encoder sim	1	Can be an Arduino-generated square-wave pair.
BMI088 breakout	1	Feeds both Picos via separate SPI chip selects.
24 VDC bench supply	1	Via an ADuM isolator IC for input simulation.
Relay 2oo2 stage (optional)	1	Two SPDT relays in AND — validates the final actuator gating.

Steps

S1 — Build the harness

Wire each Pico to:

- Shared inputs (both Picos read the same signal, on separate pins): one pulse input (sim wheel tach), one digital input (sim emergency plunger), IMU chip-select line.
- Private SPI cross-check link: Pico-A SPI0 Pico-B SPI0 (CLK, MOSI, MISO, CS). ~1 MHz clock.
- Debug UART: each Pico's UART0 to a USB-serial adapter for log capture.

Write the topology into the bring-up report at step R0.

S2 — Flash the Pico test firmware

From the workspace:

```
cargo build --release --target thumbv8m.main-none-eabihf \
  --bin tecus-bringup
```

(The `tecus-bringup` binary is a dev-only test harness in `deployments/devkit/tecus-bringup/` — planned for landing alongside this runbook; for v1 of this document the binary is a future session's deliverable.)

Flash each Pico with the same binary via SWD. Both Picos boot into the identical code path; 2oo2 runs at identical-software redundancy.

Expected: green LED blinks at 1 Hz on both Picos. Red LED off.

FAIL-HW if either Pico fails to boot: replace Pico. FAIL-SW if the binary fails to flash: check toolchain pin.

S3 — Verify the SPI cross-check

Connect a logic analyser between the two Picos on the SPI lines.

1. Power both Picos.
2. In the analyser, expect a message exchange every 10 ms: frame format is `[seq: u32][osr_atp_output: u32][osr_brake_output: u32][crc16: u16]`.
3. Ground-truth: both messages must match byte-for-byte after the first 5 exchanges.

PASS: 100 consecutive 10 ms frames match.

FAIL-SW if frames diverge: inspect the `tecus-bringup` log output via UART; there will be a divergence report.

S4 — Simulate an input disagreement

On the harness, break one of the shared input lines — e.g. cut the wire to Pico-B's wheel-tach input.

Expected within 100 ms: both Picos:

1. Halt their computed brake output (emit `BrakeCommand::Emergency`).
2. Drop their relay-drive outputs.
3. The relay 2oo2 stage (if wired) opens, releasing the test brake pressure.
4. Both Picos' red LEDs solid.

PASS: emergency-brake actuation within 100 ms of the input disagreement.

FAIL-SW if either Pico keeps driving its output: bug in the cross-check invariant — file against `osr-atp` + `osr-brake`.

S5 — Verify the wheel-tach input

Drive the shared pulse input at 500 Hz (≈ 4.2 m/s with a 410 pulses/m calibration).

Expected: UART log from both Picos shows `odom.speed_mmps ≈ 4200`; the two speeds must match within 50 mm/s.

FAIL-SW if speeds diverge > 50 mm/s with identical input: likely a timing-quantization bug — file against `osr-odometry`.

S6 — Verify the emergency plunger

Assert the sim plunger input.

Expected: both Picos immediately command `BrakeCommand::Emergency` on their output pins. The 2oo2 AND gate in the harness sees both high drives the relay drop test brake applies.

PASS: relay drop latched within 50 ms.

S7 — CM5 handshake (optional)

If an RPi CM5 IO Board is also present, wire an SPI-slave link between Pico-A and the CM5. Boot the CM5 with the app-side bring-up test binary.

Expected: CM5 receives the same per-tick cross-check messages over SPI and logs them in JSON to `/var/log/osr/ecus.log`. This proves the one-way safety app data path (RFC 0007 §4.2).

S8 — Performance envelope

Run the 2oo2 cross-check harness for 1 hour. Measure:

- Cross-check message rate: must stay at 100 Hz ± 1 Hz.
- Missed frames: 0.
- Cross-check mismatches: 0.
- Average tick jitter: ≤ 1 ms.

A 1-hour clean run is the "boot certifiable" bar for the v2 baseboard proceed-gate.

Bring-up report template

```
# T-ECU/S bring-up report

- Runbook commit: <hash>
- Date: <YYYY-MM-DD>
- Engineer: <name>
- Pico A serial: <nnn>
- Pico B serial: <nnn>

## Results
| Step | Status |
|---|---|
| S1 (harness) | PASS / FAIL-\* |
| S2 (flash)   | PASS / FAIL-\* |
| S3 (cross-check) | PASS / FAIL-\* |
| S4 (input disagreement) | PASS / FAIL-\* |
| S5 (wheel tach) | PASS / FAIL-\* |
| S6 (plunger) | PASS / FAIL-\* |
| S7 (CM5 handshake) | PASS / FAIL-\* / N-A |
| S8 (1 h run) | PASS / FAIL-\* |

## Notes
<observations, anomalies, links to logs>
```

T-ECU-S bench assembly requirements

Repository source: `control-electronics/t-ecu-s/diy-assembly/README.md`

Status: bench integration reference; hardware BOM, image, wiring and qualification are open. The release checklist governs the assembly. Exact SKUs and total prices are not frozen; previous retail subtotals omitted integration parts.

Block	Required detail before assembly release
Compute and carrier	Exact CM5 configuration, compatible carrier, storage and cooling; 5 V input supplied correctly
Field power	Approved input range, DC/DC, fuse/selectivity, surge/EMC, inrush and hold-up
I/O and network	External CAN controller/transceiver where required, Ethernet switching, isolation, termination and pin map
Security	Channel-local I2C secure element, provisioning and recovery process
Harness/enclosure	Mating connectors, contacts, terminals, ferrules, wire cut list, clamps, labels, grounding and thermal design
Output chain	Simulated loads for bench; qualified drivers, feedback and actuator interfaces before deployment
Firmware/image	Real target HAL/driver mapping, immutable build, configuration/signature hashes and measured startup behaviour

Mean Well HDR-60-24 converts AC/high-voltage DC input to 24 V output. It may supply a suitable isolated laboratory or mains-fed wayside setup, but cannot convert a 24 V field bus to 5 V. Onboard hosts require protected DC/DC supplies. No wiring from a mains terminal is released here.

For the safety-host bench use the validated exposed-pin allocation as a proposed starting point and freeze any host-specific changes. SPI needs a digital isolator and controller/peripheral wiring; a USB isolator does not provide SPI or a direct link between two USB devices. Pico 2 GPIO30/31 and chip-package-only pins are not available on its header. Do not assume that a Pi CAN HAT independently serves both Pico channels.

For application/wayside/station hosts, the modem needs a compatible carrier and power/USB interface; multiple HATs require connector and pin conflict review. Size UPS, enclosure and converter loads from the complete host BOM. Application, station and wayside software disposition comes from deployment/components.toml; TACS runtime remains a reference tooling host until qualified hardware composition is released.

Confirm GPIO voltage domains, loss-of-power state, wiring continuity/insulation, bus timing, EMI, thermal soak, image boot and recorded self-test on the actual revision. Procurement, construction and train actuation require the released package, not this list.

T-ECU-S board design requirements

Repository source: `control-electronics/t-ecu-s/schematics/v2-spec/README.md`

Status: reference integration requirements; exact supplier BOM, schematic capture, pin/connector freeze, enclosure/thermal design and qualification evidence are open. This is not sufficient information to fabricate a railway control board.

File	Current purpose
block-diagram.md	Functional paths and remaining common-cause boundaries
power-budget.md	Capacity calculation or required complete load study; no released converter/fuse rating
pinout-rp2350.md	Exposed-pin bench allocation or custom-board allocation requirements
connector-tables.md	Interface/harness schedule; supplier cavities and exact connector variants open
safety-nets.md	Hardware watchdog, permission/feedback and end-to-end fault contract

See reference integration, host bench requirements and release checklist. T-ECU/S additionally carries CM5 carrier requirements in its own folder.

A custom-board release needs the actual KiCad schematic/layout, ERC/DRC, controlled fabrication/BOM files, supplier interfaces and instrumented bring-up/fault/thermal/EMC evidence. Create fabrication directories when those artifacts exist. Do not manufacture from historical guessed pinouts or treat passing software/library tests as hardware acceptance.

T-ECU/S reference functional diagram

Repository source: `control-electronics/t-ecu-s/schematics/v2-spec/block-diagram.md`

```
Actual auxiliary bus -> protected channel A power -> evaluator A + external watchdog A
                     -> protected channel B power -> evaluator B + external watchdog B
                     -> protected application branch -> CM5 carrier

Qualified field sensors and external bus controllers -> validated timestamped A/B ingress
Evaluator A <-> isolated peer cross-check <-> evaluator B
Local permission AND watchdog health -> independent driver A / driver B
Series monitored contacts -> qualified brake / traction actuator interface
Contact feedback / power monitoring -> fault latch and event recording
```

This is a requirements diagram. Shared inputs, grounds, buses and application-derived data remain common-cause paths requiring analysis. Native Pico CAN/RMII, fictitious CM5 SODIMM wiring, single-voltage-detector watchdogs and unqualified relay lifetime/safety claims are withdrawn. See power, MCU allocation, carrier, harness and fault contract.

T-ECU/S external interface schedule

Repository source: `control-electronics/t-ecu-s/schematics/v2-spec/connector-tables.md`

Status: connector series and cavity-by-cavity harness drawings open. The earlier invented HSD/M12 pin tables are not a compatible train harness release.

Interface	Required design record
Power A/B	Actual bus range, protected input, connector/current/derating, return/earth, mating halves and fuse coordination
Safety output A/B	Driver/series contact/feedback nets, actuator polarity and load, loss-of-power truth table
Brake/traction/door/fire and field inputs	Supplier signal/voltage, isolation, diagnostic coverage, channel topology and protected harness
CAN-FD	External controller/transceiver, termination, speed/topology, shield and supplier connector
Redundant network	Full Ethernet pair map, switch/controller topology, actual connector; Gigabit copper needs four pairs
Tachometer/IMU/temperature	Actual sensor type, excitation and data interface, protection, shielding and calibrated transfer
Recovery/debug	Separate keyed SWD/UART/USB interfaces, voltage, access controls and approved recovery role

Every connector record needs part/variant, mating half, cavity view, terminal/seal, wire ID/gauge/length, shield/chassis termination, pull/reset state and continuity/insulation test. Do not share a debug header, shield or ground path across safety channels without reviewing the resulting fault path. Vendor diagrams and the actual protection study control pin assignment and rating.

See MCU allocation, carrier interface, output contract and bench workflow.

T-ECU/S application carrier interface requirements

Repository source: `control-electronics/t-ecu-s/schematics/v2-spec/pinout-cm5.md`

CM5 uses dual 100-pin connectors, not a 260-pin DDR4 SODIMM. Use the official module/carrier documentation and exact selected module revision; none of the previous fictitious connector pin assignments release a PCB. The official CM5IO provides one Gigabit Ethernet port. A redundant train network needs an explicitly designed second interface or suitable switch/controller topology, not two assumed native Ethernet MACs.

Define carrier power/boot/recovery, storage variant, Ethernet topology, buses, GPIO voltage domains, clock/reset, debug and isolation in a controlled interface drawing. I2C secure elements and PCIe/USB peripherals need supported buses, driver/role mapping and a bandwidth/power budget. SPI peripheral/slave support and one-way isolation must be proven on the chosen host/driver; controller direction or an isolator alone does not prove that application faults cannot affect protection.

Freeze boot permissions, supply sequencing and output inhibition from actual module reset/power signals. The prior 2 ms MCU and 60 s Linux boot numbers are not acceptance limits. Timing comes from measured revision-specific evidence. Protect access to boot/recovery and keys; record image and configuration hashes.

Official CM5 hardware, power requirements, safety contract.

T-ECU/S MCU allocation requirements

Repository source: `control-electronics/t-ecu-s/schematics/v2-spec/pinout-rp2350.md`

Status: exact custom-board pin allocation open. The prior native CAN0/CAN1, SPI2, dedicated PIO pins and package-independent output map are withdrawn. RP2350 uses GPIO-multiplexed supported peripherals; PIO state machines do not add physical pins. External CAN-FD controllers and transceivers are required. ATECC608B uses its selected I2C/single-wire interface, not SPI.

The Pico 2 bench allocation is an exposed-pin example for peer SPI, I2C trust element, external CAN controller, permission and watchdog functions. It belongs to the T-OBS bench; T-ECU/S must additionally allocate IMU, tachometers, brake/traction/door/fire inputs, feedback and temperature sensing. Freeze a host-specific resource map before wiring.

The board designer must provide: package and silicon revision; pin/pad/header views; alternate function; direction and voltage domain; reset pull/state; controller/peripheral role; isolation; required peripheral/interrupt/DMA resources; channel A/B separation; harness/net identity; and test point. Validate that each required function has a physical pin and that pins, hardware peripherals and interrupt resources are not allocated twice.

Do not copy the Pico header into a custom RP2350A/B board or add cab/deadman functions to the cableless train without an approved requirement. Permission outputs default low, with external watchdog inhibition and contact feedback; see safety contract. Firm deadline and fault tests bind the released HAL to the exact board revision.

Primary source: Raspberry Pi chip documentation.

T-ECU/S host power design requirements

Repository source: `control-electronics/t-ecu-s/schematics/v2-spec/power-budget.md`

The former chip-level budget and converter/inrush claims are withdrawn as a complete host design. A CM5IO carrier alone has a 5 V / 5 A input envelope; dual MCU boards, secure elements, external bus controllers, isolators, relay drivers, watchdogs and field I/O must be added. The T-OBS sizing calculation provides a reproducible example; its sensor loads do not automatically apply to T-ECU/S.

Design record	Required input / calculation
Input contract	Actual auxiliary range, loss-of-power requirement, surge, reverse polarity and hold-up
Per-branch load sheet	Complete vendor assembly input, peak/inrush and sustained measured duty; do not add chip internal rails to an already counted module
Capacity	Simultaneous required output / actual efficiency, derating and margin; each surviving leg sized for its required failure duty
Protection	Fault current, fuse coordination, connector/contact rating, wire voltage drop and transient suppression
Independent channels	Separately controlled power and watchdog inhibition, with documented common-cause boundary
Monitoring	Rail-specific UV/OV detectors plus separate external heartbeat watchdogs; TPS3701 has no watchdog function
Thermal	Hot ambient, enclosure conduction, solar/dust exposure, full duty and abnormal modes
EMC and evidence	Actual revision burst/surge, conducted/radiated and immunity tests, instrument results and configuration hashes

No converter, capacitor, fuse or certified thermal/EMC rating is selected from this requirements table. Inrush cannot be cleared by an uncalculated bulk capacitor or fixed delay. Complete the power/harness drawings and bench fault tests before board release. CM5 carrier specification, TPS3701 specification.

T-ECU-S hardware permission and fault contract

Repository source: `control-electronics/t-ecu-s/schematics/v2-spec/safety-nets.md`

Status: requirements for a board designer; schematic, safety analysis and bench evidence remain open. Two evaluator channels are a reference architecture, not proof of a safety integrity level or independent hardware.

Function	Required behaviour	Evidence before release
Channel A/B permission outputs	Default low at boot/reset; valid fresh agreement and local health required for high	GPIO startup, stuck-high/low and timeout traces
Independent heartbeat watchdog A/B	Directly inhibit the respective hardware driver on missing or invalid heartbeat	Window/deadline design; early/late/stuck tests
Voltage monitoring A/B	Detect local rail UV/OV; inhibit local output and reset as designed	Threshold, hysteresis, delay and transient tests
Peer cross-check	Isolated SPI, age/sequence/integrity checks; disagreement restricts permission	CRC/age/babble/open/short tests
Series output contacts	Both must close for permission; opening either removes permission	Actual series wiring, driver fault and welded-contact tests
Contact feedback	Detect discrepancy before movement and in periodic proof tests	Diagnostic coverage and proof-test interval analysis
Brake/traction interface	End-to-end fail-restrictive behaviour established with supplier actuator	Polarity, load, loss-of-power and recovery truth table

TPS3701 has no heartbeat watchdog input. It is a voltage window detector and may be considered for voltage sensing only. Specify separate external watchdog functions for A and B, with channel-local reset/inhibition. A global reset, if used, cannot be the only safety output control. See TI's specification.

No relay part number, coil current, lifetime, EN 50155 qualification or failure-rate claim is released. A commodity SSR board is a bench simulated-load interface only. Select contacts and drivers from actual load and failure requirements, add suppression without unacceptable release delay, and diagnose welded contacts. The previous claim that two relays always fail safe is withdrawn: common power/ground, shared sensors, fused application data, wiring shorts, welded contacts and common firmware require explicit analysis.

Physical segregation, clearance/creepage, connectors, routing and insulation must follow the voltage/environment and fault study. A 1.5 mm layout target or successful KiCad DRC cannot establish independence, EMC or a safety case. Complete FMEA/common-cause analysis, fault injection, output proof tests and the board-specific release checklist.

The bench integration source defines a Pico 2 allocation and reference output semantics. Custom RP2350A/B pins require a package-specific allocation; no native CAN-FD or Ethernet/RMII peripheral is assumed.

T-OBS — Train Obstacle Detection

Repository source: `control-electronics/t-obs/README.md`

Role: Nose-mounted obstacle-detection ECU for cabless GoA 4 operation. Each module fuses ultrasonic transducers, solid-state LIDAR, mmWave radar, and a stereo camera pair, then sends a fail-restrictive brake-demand/verdict interface to T-ECU/S.

Canonical `light-metro-3car` fit: 2 modules per trainset, one behind the A-end nose cowl and one behind the B-end nose cowl. The matching mechanical envelope is `trainset-tobs-sensor-pack.png` and `design/component-catalogue/src/osr_mech/rolling_stock/systems.py`.

Architecture

The pilot T-OBS mirrors the T-ECU/S evidence rig: two RP2350 evaluator channels cross-check each other, while a CM5 performs non-safety sensor pre-processing and logging. The RP2350 pair owns the final obstacle-detection verdict. This is not a SIL qualification claim; production channels follow the shared safety-controller selection gate.

See:

- `schematics/v2-spec/` for the board-level v2 schematic specification.
- `diy-assembly/` for the off-the-shelf first-article build.
- `../rolling-stock-integration.md` for trainset quantities and BOM alignment.

Target BOM

The historical custom-board planning allowance is carried by rolling-stock BOM line E18. The former ~$2.4k DIY subtotal is withdrawn as an assembled-host price: power, watchdogs, output feedback, harness and environmental integration were incomplete. Freeze and quote the complete assembly before reconciling the planning allowance. See reference integration.

Status

- Pilot / DIY track: `diy-assembly/` uses commodity SBC, Pi Pico 2, and sensor-evaluation modules. Integration evidence is still pending.
- Custom-board track: `schematics/v2-spec/` is the board-level v2 specification. `gerbers/` and `bom/` will be created when a KiCad layout and v2 board BOM are released.

T-OBS bench integration design

Repository source: `control-electronics/t-obs/diy-assembly/README.md`

Status: supplier and hardware freeze open; no train operation authority. Two end assemblies represent one trainset. The component and pin schedules are in reference-integration.json; the Baghdad register expands fleet quantities and missing integration parts.

Block	Per end host	Required integration
Pico 2 evaluator	2	Separately protected supply branches, hardware watchdogs and permission outputs
CM5 and compatible carrier	1	Exact memory/storage/carrier SKU to freeze; compatible 5 V supply and camera flex cables
I2C secure element	2	Separate local channel wiring, provisioning and credentials
SPI digital isolator	1 link	Channel A controller and channel B peripheral; correct directionality and separately powered sides
External CAN-FD controller/transceiver	Per network design	Pico 2 has no native CAN-FD; a Pi HAT does not automatically connect both evaluators
Radar	1 candidate	AWR1843BOOST is an EVM; adapter harness, 5 V barrel input, micro USB debug and J3 CAN interface
Lidar	1 candidate	Select HAP TX for 100BASE-TX; separate 9–18 V power and manufacturer cable; T1 requires an automotive Ethernet adapter
Stereo cameras	2	Rigid calibration bar and CM5IO-compatible 22-pin cable family; exact cable length to freeze
Ultrasonic sensing positions	4	Weatherproof transducers/AFE to select; HC-SR04 is a laboratory candidate, not qualified outdoor equipment
Hardware permission chain	2 series contacts	Independent drivers/watchdog inhibition and contact feedback; relay part and failure analysis open
Harness/enclosure/protection	1 kit	Fuses, TVS/filter, DC/DCs, connectors, glands, terminals, labels, heat spreader and test points

Historical retail subtotals are withdrawn: they excluded essential power, isolation, harness and watchdog parts and did not identify a qualified complete assembly. Unit costs stay open until the complete supplier BOM is quoted.

Bench signal path

Use the shared GPIO allocation with its pin names, not physical header numbers. A and B have separate local I2C buses and power branches. SPI peer wiring includes clock, select and correctly crossed TX/RX through a suitable isolator; a USB isolator cannot carry SPI or connect two USB device endpoints directly.

HC-SR04-style modules need VCC, GND, TRIG and timed digital ECHO. Level-shift each echo to the Pico's 3.3 V domain. Capture pulse duration using timer/PIO, not an ADS1115 analog sample. Raw piezo transducers need an excitation driver, receiver/AFE and sampling/timing design; they are not interchangeable with a four-wire ranging module. Shared transducers or an application-generated fused list are common inputs; duplicating evaluators does not make them independent sensing channels.

Mount ultrasonic acoustic faces in qualified apertures. Do not put them behind the radar radome or claim that an 8 mm polycarbonate panel is acoustically transparent. Radar needs a tested radome; lidar and cameras need separate optical windows and contamination controls. Verify alignment, weather sealing, vibration, field overlap and stopping-distance coverage on a representative nose.

Power and outputs

Use the power envelope and output contract. No connection to actual brake or traction actuators is permitted before qualification; bench outputs drive simulated loads. Exact wire gauge, fuse, connector rating, coil suppression and contact rating follow the protection and load study.

Run bench tests for boot-low permission, stale/disagreeing sensor reports, both single-channel power failures, stuck-high/low heartbeat, welded contact, driver short, bus faults, reverse polarity, brownout, inrush and hot soak. Retain instrument traces and configuration hashes; the software self-test alone cannot establish those results.

T-OBS board design requirements

Repository source: `control-electronics/t-obs/schematics/v2-spec/README.md`

Status: reference integration requirements; exact supplier BOM, schematic capture, pin/connector freeze, enclosure/thermal design and qualification evidence are open. This is not sufficient information to fabricate a railway control board.

File	Current purpose
block-diagram.md	Functional paths and remaining common-cause boundaries
power-budget.md	Capacity calculation or required complete load study; no released converter/fuse rating
pinout-rp2350.md	Exposed-pin bench allocation or custom-board allocation requirements
connector-tables.md	Interface/harness schedule; supplier cavities and exact connector variants open
safety-nets.md	Hardware watchdog, permission/feedback and end-to-end fault contract

See reference integration, host bench requirements and release checklist. T-ECU/S has its own carrier interface requirements; its custom pinout is not inherited by T-OBS.

A custom-board release needs the actual KiCad schematic/layout, ERC/DRC, controlled fabrication/BOM files, supplier interfaces and instrumented bring-up/fault/thermal/EMC evidence. Create fabrication directories when those artifacts exist. Do not manufacture from historical guessed pinouts or treat passing software/library tests as hardware acceptance.

T-OBS reference functional diagram

Repository source: `control-electronics/t-obs/schematics/v2-spec/block-diagram.md`

Status: integration requirements, not a released PCB. See machine-readable integration, power envelope, harness schedule and output/fault contract.

```
Nominal 24 V auxiliary inputs
  -> protected DC/DC branches -> evaluator A + external watchdog A
                              -> evaluator B + external watchdog B
                              -> application carrier and sensors

Radar EVM -> external CAN interface -> timestamped, validated ingress
HAP TX -> 100BASE-TX + separate power -> application preprocessing
Stereo pair -> compatible CSI flex -> application preprocessing
Ultrasonic positions -> qualified driver/AFE or bench timed-echo inputs

Evaluator A <-> directionally isolated SPI peer link <-> evaluator B
Local valid agreement + watchdog health -> independent permission drivers
Permission contact A -- series -- permission contact B -> simulated brake interface
Contact feedback + rail monitoring -> diagnostics / output inhibition
```

The production brake-demand interface may also carry a diagnostic network verdict, but the network does not replace the qualified physical output path. Shared sensors, preprocessing, buses and supplies require common-cause analysis. Neither direct CAN-FD nor Ethernet/RMII is a native Pico 2 interface. Converter, transducer AFE, board/chip package, relay and cooling selection remain open until supplier and measured bench evidence close their gates.

T-OBS connector and harness design schedule

Repository source: `control-electronics/t-obs/schematics/v2-spec/connector-tables.md`

Status: interface schedule; connector series, contact numbers and cable assemblies are not released. All external connectors need mating-half, keying, wire/contact rating, sealing, shield termination and label definitions. The former fabricated M12/PoE pinouts are withdrawn.

Interface ID	Function	Design requirement
PWR-A / PWR-B	Nominal 24 V input branches	Power-keyed connectors, individually protected; sized from corrected load/inrush/derating, not an Ethernet D-code connector
NET-A / NET-B	Redundant train network	Approved Ethernet connector and full pair map; 1000BASE-T needs four data pairs; controller/switch topology remains open
RAD-DATA	Radar EVM CAN	AWR1843BOOST J3 CAN adapter; vendor board revision and termination; external connector chosen separately
RAD-PWR	Radar EVM supply	5 V barrel interface; dedicated protected branch; micro USB is debug/configuration
LID-DATA	HAP TX data	100BASE-TX with the vendor cable; HAP T1 needs a different physical interface
LID-PWR	HAP supply	Separate vendor 9–18 V input, 12 V reference branch; no 12 V PoE over spare Gigabit contacts
US-01..04	Ultrasonic position	Candidate-specific excitation/receive or TRIG/ECHO, level shifting and shield; raw piezo is not plug compatible
CAM-L / CAM-R	Stereo CSI	CM5IO 22-pin family with the appropriate camera adapter; internal flex length and cable orientation frozen
PERMIT / FEEDBACK	Channel permissions and contact monitoring	Segregated channel circuits, load/polarity truth table, test point and actuator supplier interface
DEBUG-A / DEBUG-B	MCU SWD	Package/board-specific keyed debug connector; production access control

Do not fabricate a harness from this table. Release a cavity-by-cavity drawing, wire cut list, ferrule/terminal/seal list, formboard and continuity/insulation test after vendor freeze. The TI EVM does not ship a standard M12 radar interface. Radar needs an RF-qualified radome; ultrasonic faces need acoustic apertures and lidar/cameras need qualified optical paths. Fixed mounting angles, bracket hole patterns and cable limits require coverage/clearance studies.

See bench integration and reference source. Primary interfaces: TI EVM guide, Livox specification, CM5 hardware.

T-OBS pin allocation and peripheral boundary

Repository source: `control-electronics/t-obs/schematics/v2-spec/pinout-rp2350.md`

The machine-readable Pico 2 bench allocation is the reference for both A and B local channels. It uses exposed GPIOs only. It replaces the former GP30–39 Pico allocation, GP14–17 ADC inputs, SPI secure-element and native CAN/RMII claims.

GPIO	Bench function
0 / 1	I2C SDA / SCL for the channel-local ATECC608B breakout
2 / 3 / 4 / 5	SPI0 peer SCK / TX / RX / CS; suitable isolator, A controller and B peripheral
10 / 11 / 12 / 13	SPI1 external CAN-FD controller SCK / TX / RX / CS
14	Permission output to simulated-load driver
15	External hardware watchdog heartbeat
16 / 17 / 18 / 19	Laboratory ultrasonic triggers
20 / 21 / 22 / 26	Corresponding level-shifted timed digital echoes

This is a proposed bench allocation. Peripheral IRQs, Ethernet controller bus, custom analog transducer AFE, debug pins and exact protocol/firmware mapping must be frozen before a board or image is released. Bench GPIO14 is not ADC0: Pico 2 exposes analog inputs on GPIO26–28. The custom chip package has a separate pinout; use its datasheet and never extrapolate the Pico header.

Pico 2 needs an external CAN controller plus transceiver. Ethernet also needs an external controller with a demonstrated data-rate/deadline budget. ATECC608B uses the selected I2C or single-wire interface, not SPI. Peer firmware and hardware tests must prove timing and fault behaviour; listing source filenames does not mean device drivers have been implemented.

Sources checked 3 October 2026: Pico 2 datasheet, Microchip ATECC608B.

T-OBS reference power and thermal envelope

Repository source: `control-electronics/t-obs/schematics/v2-spec/power-budget.md`

The source is reference-integration.json. This is a sizing envelope for the bench candidates, not measured train duty or a qualified converter/fuse selection.

Load	Output allocation	Basis
CM5IO with attached cameras	25 W	Carrier 5 V / 5 A interface envelope
AWR1843BOOST radar EVM	12.5 W	5 V / 2.5 A interface envelope
Livox HAP startup	26 W	Vendor startup specification; 12 W typical
Dual evaluators and integration	8 W	Unmeasured engineering allowance
Total simultaneous output envelope	71.5 W	Sum of the four branches
Input at 90% conversion efficiency	79.44 W	71.5 / 0.90
Capacity including 25% margin	99.31 W	79.44 × 1.25; 100 W reference supply envelope

At an assumed 18 V minimum bus, the input envelope is 4.41 A, or 5.52 A including the capacity margin. The real auxiliary bus range, inrush, hold-up, cable drop and selective branch protection require a released electrical load study. Each surviving power leg must support the complete required fault-state load if redundant operation is claimed. Do not reuse the former 1 A fuse or claim that a 40 W host allowance covers this assembly.

The 100 W value is capacity, not sustained energy consumption: no OPEX/traction-energy uplift is calculated from it. Update the train auxiliary energy model after measured duty and production sensor choice. Do not double-count cameras included within the carrier allocation. The 18 V sizing assumption is not a claimed train-bus standard.

Mean Well HDR-60-24 is an AC/high-voltage DC input supply with 24 V output. It does not convert a 24 V train bus to 5 V or 12 V. Choose protected DC/DC branches with approved voltage range, isolation, derating and transient behaviour.

Passive cooling, junction temperatures and EMC compliance remain unverified. Calculate conduction, enclosure thermal resistance, solar load and dirty-filter performance; test startup and sustained operation at the declared hot-desert ambient. Keep sensing and event capture powered during a commanded brake when the fault permits; thermal throttling or application-fusion loss must be handled as stale data rather than assumed harmless.

Primary specifications checked 3 October 2026: CM5 carrier, TI radar EVM, Livox HAP, HDR-60.

T-OBS hardware permission and fault contract

Repository source: `control-electronics/t-obs/schematics/v2-spec/safety-nets.md`

Status: requirements for a board designer; schematic, safety analysis and bench evidence remain open. Two evaluator channels are a reference architecture, not proof of a safety integrity level or independent hardware.

Function	Required behaviour	Evidence before release
Channel A/B permission outputs	Default low at boot/reset; valid fresh agreement and local health required for high	GPIO startup, stuck-high/low and timeout traces
Independent heartbeat watchdog A/B	Directly inhibit the respective hardware driver on missing or invalid heartbeat	Window/deadline design; early/late/stuck tests
Voltage monitoring A/B	Detect local rail UV/OV; inhibit local output and reset as designed	Threshold, hysteresis, delay and transient tests
Peer cross-check	Isolated SPI, age/sequence/integrity checks; disagreement restricts permission	CRC/age/babble/open/short tests
Series output contacts	Both must close for permission; opening either removes permission	Actual series wiring, driver fault and welded-contact tests
Contact feedback	Detect discrepancy before movement and in periodic proof tests	Diagnostic coverage and proof-test interval analysis
Brake/traction interface	End-to-end fail-restrictive behaviour established with supplier actuator	Polarity, load, loss-of-power and recovery truth table

TPS3701 has no heartbeat watchdog input. It is a voltage window detector and may be considered for voltage sensing only. Specify separate external watchdog functions for A and B, with channel-local reset/inhibition. A global reset, if used, cannot be the only safety output control. See TI's specification.

No relay part number, coil current, lifetime, EN 50155 qualification or failure-rate claim is released. A commodity SSR board is a bench simulated-load interface only. Select contacts and drivers from actual load and failure requirements, add suppression without unacceptable release delay, and diagnose welded contacts. The previous claim that two relays always fail safe is withdrawn: common power/ground, shared sensors, fused application data, wiring shorts, welded contacts and common firmware require explicit analysis.

Physical segregation, clearance/creepage, connectors, routing and insulation must follow the voltage/environment and fault study. A 1.5 mm layout target or successful KiCad DRC cannot establish independence, EMC or a safety case. Complete FMEA/common-cause analysis, fault injection, output proof tests and the board-specific release checklist.

The bench integration source defines a Pico 2 allocation and reference output semantics. Custom RP2350A/B pins require a package-specific allocation; no native CAN-FD or Ethernet/RMII peripheral is assumed.

Trainset interiors and closures — COTS equipment catalogue

Repository source: `control-electronics/trainset-interiors.md`

The `light-metro-3car` body is a welded steel primary frame with composite non-structural side/roof/nose panels. Passenger-facing equipment is not bespoke OSR hardware: doors, windows, HVAC, lighting, seats, grab rails, PIS, CCTV, and intercom are COTS rail or heavy-duty bus modules installed into reserved envelopes.

This file is the supplier-neutral catalogue promised by RFC 0008. The envelope is the contract; named SKU classes are examples only. Any replacement is acceptable if it fits the envelope, stays inside the mass/power budget, exposes the required interface, and arrives with usable certification evidence.

Related rolling-stock package:

- `body.md`
- `fabrication-plan.md`
- `bom-skeleton.md`

Qualification classes

Class	Meaning	Typical procurement path
COTS-R	Rail-certified COTS module	Preferred for doors, windows, HVAC, intercom
COTS-B	Heavy-duty bus module with rail-compatible evidence	Acceptable for seats, lighting, PIS, CCTV where authority allows
COTS-I	Industrial module needing enclosure and vibration evidence	Acceptable for non-safety displays/cameras only
MAKE-ADAPT	Local adapter plate, harness tail, duct, or bracket	Allowed when it does not change the primary steel frame

Safety-critical rotating, braking, traction, battery, and door-lock parts are not downgraded to generic commodity components. They remain supplier-certified modules with incoming inspection and type-test evidence.

Reserved module envelopes

Per 17 m self-contained car:

Module	Qty / car	Envelope	Mass limit	Power budget	Interface
Door cassette	2	clear opening 1 250 × 2 000 mm; cassette ≤ 1 850 W × 2 450 H × 260 D mm	180 kg each	600 W peak each	24/110 V DC, Ethernet/CAN, hardwired closed/locked loop
Side glazing	4	nominal 900 × 1 200 mm aperture	35 kg each	0	Bonded/gasketed cassette, drain path
Roof HVAC	1	≤ 2 700 L × 1 900 W × 450 H mm	420 kg	20 kW cooling electrical allowance	Direct-HV DC input, condensate drain, CAN/Ethernet diagnostics
Interior lighting	set	two 17 m ceiling runs + door-zone lights	45 kg	350 W	24 V DC, emergency-light input
Exterior marker/head/tail lights	set	nose/cowl mounted	25 kg	150 W	24 V DC, hardwired marker functions
PIS displays + speakers	set	above-door and saloon ceiling mounts	45 kg	250 W	Ethernet, 24 V DC, audio line
CCTV	set	door sill, saloon, and nose service views	20 kg	180 W	PoE or 24 V DC + Ethernet
Passenger intercom	4	recessed wall modules	12 kg	80 W	Ethernet/SIP, 24 V DC, OCC call button
Longitudinal seats	20 seats	under-window modules over battery covers	120 kg	0	M10 floor/side inserts
Grab rails/stanchions	set	centre-door and vestibule zones	45 kg	0-80 W	M8/M10 floor/ceiling inserts
Interior wall/ceiling panels	set	FRP/phenolic cassettes	220 kg	0	Clip/bolt to secondary rails
Floor boards + hatches	set	full saloon floor	280 kg	0	Removable panels over battery/HV zones

Per `light-metro-3car` consist, multiply by three cars. HVAC dominates auxiliary power; the aux converter and station-energy models should assume roughly 60 kW HVAC peak plus lighting, PIS, CCTV, intercom, control electronics, and battery thermal loads.

COTS Reference Baselines

The CAD now uses COTS-inspired integration geometry rather than plain boxes. These references are not sole-source selections; they define the type of commodity/supplier evidence the v2 drawing pack must accept or replace.

Function	Reference family	CAD/design implication
Automatic coupler	Dellner / Voith Scharfenberg Type 10 family	Coupler head, guide horns, electrical-head carrier, brake-pipe hoses, bolted shear plate, EN 15227 crash absorber pocket
Door cassette	Knorr-Bremse/IFE, Wabtec/Faiveley, Kangni electric rail sliding or plug door systems	Top operator rail, hanger rollers, controller/diagnostics box, two sliding leaves, bonded glass, obstruction light curtain, anti-pinch edge, lock/release unit, threshold drainage, gap-filler hinge
Nose sensing	Rail Vision MainLine / Ouster OS1 / LSLiDAR train-end intrusion class	Rail-grade LIDAR shroud, heated optical window, wide/narrow/thermal camera pods, radar, wash/wipe, service fasteners, 24 V feed allowance
Radar	TI AWR1843BOOST class 77 GHz radar module	Nose-centred radar envelope, radome window, CAN-FD harness clearance
Power connector	Anderson SB50 class connector	Keyed service/traction connector envelope with cable strain-relief allowance
Access fastener	Camloc quarter-turn class fastener	Quarter-turn retainers on removable skirts, panels, and service lids
Linear guide	HIWIN HG class guide block	Guide-block envelope for removable bogie/motor service adapters
Gas spring	Stabilus LIFT-O-MAT class gas spring	Rod/cylinder/end-fitting envelope for supported service flaps

The matching mechanical reference models are generated under `design/component-catalogue/catalog/fixtures/` and the rolling-stock system assemblies under `design/component-catalogue/catalog/rolling_stock/`.

The passenger fit-out source now generates a separate cost/source CSV at `build/bom/rolling_stock_cots_fitout_bom.csv`. Each row carries quantity per car, quantity per consist, low/base/high unit cost, mass, power, public source URL, and the geometry basis used for the supplier-neutral CAD envelope.

Reference datasheets/product pages used for the current supplier-neutral envelope geometry: Dellner automatic coupler Type 10, Knorr-Bremse/IFE train entrance systems, Rail Vision MainLine, Ouster OS1 LIDAR, LSLiDAR train-end railway intrusion detection, TI AWR1843BOOST, Anderson SB50 datasheet, HIWIN HGW25CC catalogue page, and Stabilus LIFT-O-MAT.

Vendor fit-in shortlist

These supplier families are now reflected in the CAD labels and BOM notes. The exact SKU remains a v2 procurement decision.

Module	Reference families	Current envelope reservation
Door cassette	Knorr-Bremse/IFE entrance systems, Wabtec/Faiveley, Kangni	1 250 × 2 000 mm clear opening; cassette controller, light curtain, pressure edge, drain scuppers
Side glazing	AGC Lamisafe/Heatlight W, Pilkington rail glazing, Saint-Gobain transport glazing	1 400 × 900 mm laminated cassette with heated busbar option and drain channel
Floor covering	Altro Transflor Tungsten, Forbo transport flooring	2 mm pre-cut anti-slip sheets bonded over removable floor and hatch panels
Seats	Compin-Fainsa SB09, Kiel, Grammer, McConnell	Cantilevered longitudinal bench over the under-seat battery strake
Lighting	Teknoware rolling-stock lighting, Luminator rail lighting	Continuous main/emergency ceiling channels plus sealed exterior head/tail/marker cassettes
PIS and audio	Luminator onboard displays/audio, Televic TRACS/GSP	Above-door VESA plates, PA speaker backboxes, amplifier/controller trays
HVAC	Liebherr rail HVAC, Knorr-Bremse Merak, Wabtec/Faiveley, Hispacold	Roof curb, compact/split unit envelope, pressure damper, filter cassette, drains, drop ducts
Battery package	EVE or qualified equivalent high-power LFP system	Under-seat sealed modules with cold plates, BMS/HVIL harness, outward vent, off-gas detection, and water-mist interface
Motor/brake	TSA/ABB/Skoda PMSM motors; Knorr-Bremse WheelAct/AxleAct or Wabtec Faiveley brakes	Axle-hung motor package and compact caliper package with mounting/diagnostic details
Coupler	Dellner Type 10, Voith Scharfenberg class	Crashworthy pocket, gas-hydraulic/deformation unit allowance, D-REX Ethernet and pneumatic heads
T-OBS nose module	Rail Vision MainLine, Ouster OS1, LSLiDAR train-end intrusion	LIDAR, thermal/wide/narrow camera pods, radar, ultrasonic, heater/wash/wipe service hardware

Generated fit-out cost band

Generated from `design/component-catalogue/src/osr_mech/rolling_stock/cots_equipment.py` for one `light-metro-3car` consist:

Module	Qty / consist	Base USD	Low-high USD	Shape basis
Side glazing panel	18	27 000	17 100-46 800	Laminated/heated rail glazing cassette
Rooftop HVAC unit	3	75 000	54 000-126 000	Compact/split passenger-saloon rail HVAC
Continuous LED ceiling strip	6	9 000	5 400-15 000	Serviceable main/emergency lighting rail

Module	Qty / consist	Base USD	Low-high USD	Shape basis
Passenger-information LCD	12	9 600	5 400-21 600	Luminator rail display datasheet dimensions
Longitudinal bench run	18	24 000	14 400-45 000	SB09-style removable-pad light-alloy bench
Vertical grab pole	24	7 992	3 840-16 800	Stainless modular stanchion with flanges
Emergency intercom / help-point	6	4 800	2 100-10 800	Recessed SIP/audio help-point module

Candidate supplier classes

Named suppliers are examples of the class of product, not a locked vendor list.

Module	Example class	Required evidence
Door cassette	Knorr-Bremse/IFE, Wabtec/Faiveley, Fuji, Nanjing Kangni rail door family	EN 14752 or equivalent, obstruction detection, lifecycle test, emergency release
Side glazing	AGC/Pilkington/Saint-Gobain rail laminated safety glass	EN 15152 or equivalent, impact, fire/smoke data for seals
HVAC	Liebherr, Knorr-Bremse Merak, Wabtec/Faiveley, Hispacold roof HVAC	EN 50155/50121 where applicable, +50 °C performance curve, refrigerant data
Lighting	Teknoware/Luminator rail LED strip or troffer modules	EN 45545 material data, EMC evidence, emergency-mode behaviour
Seats	Compin-Fainsa, Kiel, Grammer, McConnell, local bus/rail equivalent	EN 45545 R7, static strength, vandal-resistance data
Grab rails	Stainless modular rail/bus stanchion system	Pull-load data, corrosion grade, fastener spec
PIS displays	Luminator/Televic/IEI/Litemax rail LCD class	EN 50155 preferred or vibration/temperature evidence
CCTV	Axis/Hikvision/industrial PoE camera class in rail enclosure	EMC/vibration evidence, cybersecurity hardening path
Intercom	Televic TRACS, Luminator audio, Zenitel/Vingtor, Commend rail SIP intercom class	EN 50155 preferred, audio intelligibility, emergency-call behaviour
Floor/interior panels	Altro/Forbo rail flooring plus phenolic/FRP rail interior panel supplier	EN 45545 HL2 R1/R5 evidence, cleanability, repair method

Supplier-change rule

A supplier change is valid without changing the carbody if all are true:

1. The replacement fits inside the reserved envelope.
2. Mass is no more than the row limit, or the weight budget is updated and bogie axle-load margins still pass.
3. Power draw is no more than the row budget, or aux-power sizing is updated and thermal checks still pass.
4. Mounting can be handled by adapter plates or harness tails.
5. The primary steel frame, door posts, bolsters, and coupler pockets are unchanged.
6. Required fire, EMC, vibration, lifecycle, and maintainability evidence is present.

If any supplier change needs new primary steel, it is a body-design change and must go through v2 CAD/FEA release.

Incoming inspection

Module	Incoming checks
Door cassette	Dimensional check, serial/cert pack, lock loop continuity, manual release
Window cassette	Glass marking/cert pack, edge damage inspection, seal/adhesive shelf life
HVAC	Nameplate, refrigerant record, factory test sheet, drain and duct fit
Lighting/PIS/CCTV/intercom	Power-on test, firmware version, MAC/serial register, mounting hardware
Seats/grab rails	Fire certificate, pull-test certificate or batch test, finish defects
Panels/floor	Fire certificate, flatness, insert locations, repair kit present

Installation evidence

Each car receives a COTS fit-out evidence pack:

- Supplier certificate and installation manual for every COTS-R/B item.
- Envelope compliance sheet signed by production engineering.
- Mounting torque record for seats, rails, doors, HVAC, and hatches.
- Water ingress test for doors/windows/roof equipment.

- HVAC drain test.
- Door obstruction and emergency-release test.
- Lighting lux test and emergency-light mode test.
- PIS/CCTV/intercom network enumeration report.
- Material fire certificate index.

This evidence pack becomes part of the trainset technical file and feeds the compliance matrix in `docs/rolling-stock/light-metro-3car/compliance.md`.

W-SBC — Wayside

Repository source: `control-electronics/w-sbc/README.md`

Role: Host for `osr-consensus`, `osr-interlocking`, `osr-wayside-points`, `osr-level-crossing`, `osr-hot-axle-wayside`, `osr-balise` (reader side), `osr-energy-site`.

Deployed in pole-mounted IP67 cabinets along the corridor and at junctions.

Environment: IP67, EN 50121-4 EMC, −40…+70 °C.

SoC

Radxa CM5 (RK3588S — Cortex-A76 4-core + Cortex-A55 4-core big.LITTLE, 4/8/16 GB LPDDR4X, eMMC on-module) on a custom baseboard. Radxa's industrial-temperature variant (−20…+85 °C rated) is the baseline; pole-mount cabinets in hot climates routinely hit +60 °C internal, so the extra margin over the consumer-grade CM5 matters.

Raspberry Pi CM5 is the pin-compatible drop-in on the same SO-DIMM slot for mild-climate deployments.

For safety-related wayside logic, the CM5 hosts application, diagnostics and communications only. A separately qualified controller owns field proving, watchdogs and fail-safe actuator outputs; see the selection gate. Core pinning or a separation kernel on a shared SBC is defence in depth, not a SIL-4 boundary. Non-safety sites (balise-only, energy-site) run Debian straight-through.

See RFC 0007 §6.

Peripherals — one SKU fits every wayside role

Peripheral	Qty	Role
TSN Ethernet	2	WAY-E A/B
RS-485	4	Switch motor, crossing barrier, HABD bus, balise reader
Isolated DI	8	Switch end-of-travel (A+B), strike detectors, field faults
Isolated DO	4	Motor direction contactors, barrier lift/drop
4-wire PT100	4	Hot-axle IR temperature
802.11ax AP	1	Depot maintenance (disabled in service)
Dual 24 V DC-in	2	Primary + backup site supplies
ATECC608B SE	1	Trust anchor

Selective population per role — switch site leaves PT100 inputs unpopulated, HABD site leaves motor-direction DOs unpopulated. One SKU, one spares stock.

Form factor

120 × 80 mm PCB inside an IP67 aluminium extrusion with M12 penetrations. Conduction-cooled.

Target BOM

~€280 per board, ~€550 with enclosure + mounting hardware.

Status

- Pilot / DIY track: `diy-assembly/` uses the Radxa CM5 IO Board + a SparkFun RS-485 breakout to validate the `osr-consensus` + `osr-interlocking` + `osr-wayside-points` stack on a real RK3588S module. Integration evidence is still pending.
- Custom-board track: `schematics/v2-spec/` is the board-level v2 specification overview and block diagram. `gerbers/` and `bom/` remain pending until a KiCad layout and board BOM are released.

W-SBC bring-up — Radxa CM5 IO Board

Repository source: `control-electronics/w-sbc/bring-up.md`

Goal: validate the wayside host runs `osr-consensus` + `osr-interlocking` + `osr-wayside-points` on a RK3588S Radxa CM5, with the A55 little cluster isolated for the SIL-4 partition and the A76 big cluster carrying non-safety services.

Dev kits + parts

Part	Qty	Notes
Radxa CM5 (8 GB + 32 GB eMMC, industrial-temp)	1	
Radxa CM5 IO Board	1	
SparkFun RS-485 breakout	2	One for switch-motor sim, one for HABD sim.
Test bench: 5-node consensus cluster	—	4 more CM5 carriers OR a docker-compose simulator.
24 VDC bench supply via ADuM isolator	1	

Steps

W1 — Boot Debian with RK3588S BSP

1. Write Radxa Debian 13 image to eMMC via `rkdeveloptool`.
2. Expand rootfs, ssh login.
3. Update `/etc/default/cpufrq`: pin A55 cluster to 1.4 GHz, A76 to ondemand.
4. Install Rust toolchain (same as T-ECU/A).

Expected: ssh login within 90 s of power-on.

W2 — Workspace build

Same as T-ECU/A step A2. Build times on 8 GB A76 ~10 min.

W3 — RT partition setup (safety role)

Configure the kernel for PREEMPT_RT on A76 and the A55 cluster isolated for the safety partition:

```
isolcpus=0-3          # reserve A55 cores for safety
nohz_full=0-3         # tickless on safety cores
rcu_nocbs=0-3
```

Reboot. Verify `/proc/cmdline` contains the flags and `htop` shows no processes scheduled on CPU 0-3.

W4 — Consensus cluster bring-up

This step proves the W-SBC can host a consensus replica.

1. On the dev kit, run a local 3-node simulated cluster:

   ```
   cargo run --release -p osr-consensus --example local-cluster
   ```

2. Confirm leader election within 1 s.
3. Drive 100 proposals through; confirm all commit on every follower.
4. Kill one node; confirm re-election and continued progress.

Expected: clean leader election + commit progression + crash recovery on the RK3588S silicon.

W5 — Interlocking on the safety partition

Pin `osr-interlocking`'s MA computer to the isolated A55 cores:

```
taskset -c 0-3 cargo run --release --bin osr-sim -- \
  --config cities/catalogue/west-asia/Iraq/Samawah/samawah.toml --duration 600
```

Expected: the 10-minute Samawah run completes with zero invariant violations; the MA computer's tick jitter (measured via `osr-sim`'s internal timing) ≤ 5 ms.

W6 — Wayside-points driver over RS-485

1. Connect one SparkFun RS-485 to a motor-driver sim (can be an Arduino reading the bus and toggling a digital output).

2. Build the `osr-wayside-points` standalone:

   ```
   cargo run --release -p osr-wayside-points --example switch-drive
   ```

3. Send alternating `SwitchCommand::Normal` / `SwitchCommand::Reverse` every 5 seconds.

Expected: motor drive toggles faithfully, detection (simulated end-of-travel sensors) mirrors the command within the 5 s RFC 0012 §4.2 throw budget.

W7 — HABD input over RS-485

Simulate a hot-axle reading (raw temperature values) over the second RS-485 link. Build the `osr-hot-axle-wayside` driver:

```
cargo run --release -p osr-hot-axle-wayside --example demo
```

Expected: temperature thresholding fires at configured setpoint; the crate emits a `SpeedRestriction` event to the consensus log simulator.

W8 — Industrial-temp headroom

Place the Radxa CM5 in a 65 °C test oven (pre-deployment IP67 cabinet simulator) for 30 minutes with the W5 workload running.

Expected: no throttling, no CPU errors, no log anomalies. RK3588S junction temp stays < 90 °C per the on-die sensor.

FAIL-HW if the board throttles or errors: the industrial- temp variant may not be the one installed; confirm the P/N.

W9 — PoE or dual-PSU input (optional)

If the cabinet-form-factor of the IO Board is present, validate the dual-24V redundancy:

1. Run both PSU inputs; cut one; verify no service interruption.
2. Re-apply; confirm re-detection.

Expected: < 5 ms dropout at changeover.

Bring-up report template

```
# W-SBC bring-up report

- Runbook commit: <hash>
- Date: <YYYY-MM-DD>
- Radxa CM5 serial: <nnn>

## Results
| Step | Status |
|---|---|
| W1 (boot)        | PASS / FAIL-\* |
| W2 (build)       | PASS / FAIL-\* |
| W3 (RT + isol)   | PASS / FAIL-\* |
| W4 (consensus)   | PASS / FAIL-\* |
| W5 (interlocking)| PASS / FAIL-\* |
| W6 (points)      | PASS / FAIL-\* |
| W7 (HABD)        | PASS / FAIL-\* |
| W8 (thermal)     | PASS / FAIL-\* |
| W9 (dual-PSU)    | PASS / FAIL-\* / N-A |
```

W-SBC bench assembly requirements

Repository source: `control-electronics/w-sbc/diy-assembly/README.md`

Status: bench integration reference; hardware BOM, image, wiring and qualification are open. The release checklist governs the assembly. Exact SKUs and total prices are not frozen; previous retail subtotals omitted integration parts.

Block	Required detail before assembly release
Compute and carrier	Exact CM5 configuration, compatible carrier, storage and cooling; 5 V input supplied correctly
Field power	Approved input range, DC/DC, fuse/selectivity, surge/EMC, inrush and hold-up
I/O and network	External CAN controller/transceiver where required, Ethernet switching, isolation, termination and pin map
Security	Channel-local I2C secure element, provisioning and recovery process
Harness/enclosure	Mating connectors, contacts, terminals, ferrules, wire cut list, clamps, labels, grounding and thermal design
Output chain	Simulated loads for bench; qualified drivers, feedback and actuator interfaces before deployment
Firmware/image	Real target HAL/driver mapping, immutable build, configuration/signature hashes and measured startup behaviour

Mean Well HDR-60-24 converts AC/high-voltage DC input to 24 V output. It may supply a suitable isolated laboratory or mains-fed wayside setup, but cannot convert a 24 V field bus to 5 V. Onboard hosts require protected DC/DC supplies. No wiring from a mains terminal is released here.

For the safety-host bench use the validated exposed-pin allocation as a proposed starting point and freeze any host-specific changes. SPI needs a digital isolator and controller/peripheral wiring; a USB isolator does not provide SPI or a direct link between two USB devices. Pico 2 GPIO30/31 and chip-package-only pins are not available on its header. Do not assume that a Pi CAN HAT independently serves both Pico channels.

For application/wayside/station hosts, the modem needs a compatible carrier and power/USB interface; multiple HATs require connector and pin conflict review. Size UPS, enclosure and converter loads from the complete host BOM. Application, station and wayside software disposition comes from deployment/components.toml; TACS runtime remains a reference tooling host until qualified hardware composition is released.

Confirm GPIO voltage domains, loss-of-power state, wiring continuity/insulation, bus timing, EMI, thermal soak, image boot and recorded self-test on the actual revision. Procurement, construction and train actuation require the released package, not this list.

W-SBC v2 schematic specification

Repository source: `control-electronics/w-sbc/schematics/v2-spec/README.md`

Status: RFC 0007 v3 deliverable (partial). Per-board specification of the wayside SBC per RFC 0007 §6.

Scope

W-SBC hosts the wayside SIL-4 crates + non-safety energy-site management:

- `osr-interlocking` — Movement Authority computer (SIL-4).
- `osr-consensus` — Raft log, distributed across 3 W-SBC nodes minimum per line.
- `osr-wayside-points` — switch-machine controller (SIL-4).
- `osr-balise` — balise registry + sighting audit (SIL-2).
- `osr-level-crossing` — SIL-4 five-state LX controller.
- `osr-hot-axle-wayside` — SIL-4 HABD at wayside.
- `osr-intrusion-detect` — RFC 0016 wayside intrusion evaluator (SIL-4).
- `osr-energy-site` — PV + battery + grid-tie dispatch (SIL-0).

One physical W-SBC may host multiple roles depending on its location — a junction box near an LX + switch + intrusion pack runs three SIL-4 crates concurrently.

SoC

Radxa CM5 (RK3588S industrial-temp variant) — Cortex-A76 × 4 + A55 × 4, 4–16 GB LPDDR5, eMMC on-module. Industrial temp grade is the important selector — the RPi CM5 BCM2712 is OT3 commercial; the RK3588S-industrial is rated for +85 °C ambient which is needed for pole-mounted and cabinet-mounted wayside service in hot climates.

Architecture

Single-role W-SBC is not 2oo2 at the CPU level; safety redundancy lives in the 3-node Raft consensus cluster per line. Any single W-SBC losing its CPU means the cluster still has quorum.

For safety-critical roles (level-crossing, switch-machine), the actuator-output stage is still hardware-2oo2 via external relay stages on the same board: the W-SBC drives both sides of an external AND-gate exactly like T-ECU/S, even though the consensus layer provides the cross-CPU redundancy.

Peripherals (baseboard, selectively populated)

Peripheral	Qty	Purpose
TSN Ethernet	2	Backbone to adjacent W-SBCs + back-office
CAN-FD	2	Switch-machine motor drivers, LX barriers
Isolated DI	8	Fence-line contact sensors, LX inductive loops, SIL-4 switch position
Isolated DO	4	Relay drives to switch motors + LX barriers
1000BASE-T LIDAR	1	ROW-mounted LIDAR (RFC 0016)
CAN-FD radar	1	ROW-mounted mmWave radar (RFC 0016)
MIPI-CSI	1	ROW-mounted CCTV camera (optional)
GPIO (3.3 V tolerant)	16	General I/O — balise triggers, axle-counter inputs
PTP PHY	1	1588v2 time sync for consensus cluster
M.2 NVMe	1	Local log shadow + OTA staging
USB-C	1	Depot console + OTA
24 V DC input	2	Redundant, diode-OR
ATECC608B	1	Trust anchor — signs SectionIntrusion / SwitchObservation entries

Power budget

Depends on populated roles. Bounds:

Configuration	Sustained	Peak
Interlocking + consensus only	~8 W	~12 W

Configuration	Sustained	Peak
+ switch-machine control (one switch)	~12 W	~60 W (during throw)
+ level-crossing control (4 barriers)	~18 W	~50 W (during barrier transition)
+ intrusion-detect (LIDAR + radar + camera)	+14 W	+22 W
Worst-case combined	~44 W	~90 W

Input range 9 – 36 V DC to accommodate both the 24 V wayside cabinet bus and the 12 V battery-backup mode. An LTC7803 wide- input buck handles the range.

Environment

- Industrial temp: −40 °C to +85 °C ambient.
- IP67 enclosure — pole-mountable; no cabinet required for the harsh-climate configuration.
- EN 50121-4 EMC compliance for wayside equipment.
- EN 50125-3 environmental conditions for signalling.
- Vibration + shock: IEC 61373 Cat 2 (wayside).
- Conformal coat: MG Chemicals 419 before integration.

Form factor

180 × 130 mm — slightly larger than T-ECU/S to accommodate the wide 24 V + 12 V PSU and the 8+4 isolated I/O pool. DIN-rail + pole-mount bracket options. No fans.

Target BOM

~€340 per board (CM5 ≈ €110 industrial-temp premium, baseboard ≈ €95, I/O isolators ≈ €55, RF + sensor headers ≈ €30, PSU + passives ≈ €50).

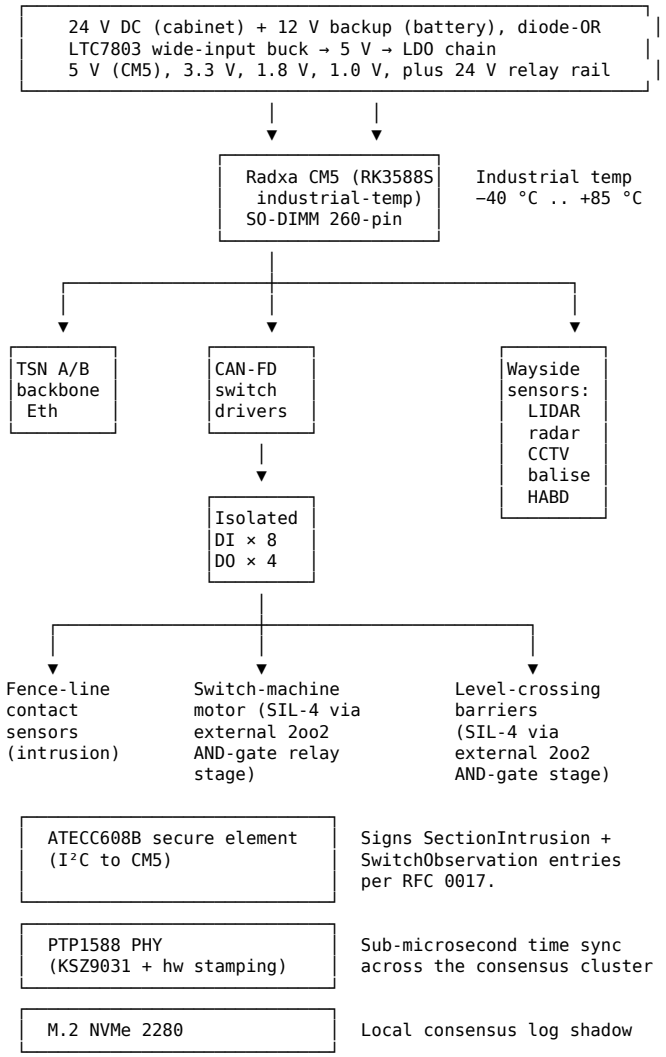
A typical wayside deployment uses three W-SBCs per line as the minimum Raft quorum; larger networks scale the cluster up as needed.

Expected v2-spec file set

- `block-diagram.md` — functional block diagram.
- `power-budget.md` — per-role power profile.
- `pinout-cm5.md` — CM5 SO-DIMM pin assignments.
- `safety-nets.md` — switch-motor + LX actuator safety-critical nets with 2oo2 AND-gate relay stages.
- `connector-tables.md` — M12 + terminal-block pinouts.
- `deviations-log.md` — design-review notes.

W-SBC v2 — block diagram

Repository source: `control-electronics/w-sbc/schematics/v2-spec/block-diagram.md`



Safety-critical actuator path

All SIL-4 actuator outputs (switch throw, LX barrier drop) flow through the same 2oo2 AND-gate relay pattern as T-ECU/S: the CM5 drives both channels via a pair of independent GPIOs through optoisolators, and the relay stage is passive-AND. Loss of either channel drops the actuator.

The CM5 is not itself 2oo2 in silicon. Safety redundancy at the CPU level is provided by the 3-node consensus cluster — if this CM5 fails, the other two nodes continue, and the cluster re-elects a leader within 1.5 s. Actuator state on this board may freeze at the last-committed value but cannot enter an unsafe state because the CM5 no longer drives the AND gate's input (fail-restrictive).

Non-safety accessory path

The non-safety crates (`osr-energy-site`, non-critical `osr-balise` logging) share the same CM5 but do not drive any actuator. Their faults are diagnostic, not safety-critical.

Interfaces to sensor suite (RFC 0016)

The intrusion-detect sensor pack lives near the W-SBC on the same pole or in the same cabinet:

- LIDAR connects via the 1000BASE-T Ethernet (using a separate PHY from the TSN backbone, so LIDAR traffic does not steal backbone bandwidth).
- Radar connects via CAN-FD.
- Stereo camera connects via MIPI-CSI-2 (optional — many junction boxes omit the camera to reduce cost).

- Fence-line contact sensors feed in via the 8 isolated digital inputs.

The CM5 runs `osr-intrusion-detect::evaluate` at 5 Hz and emits one `SectionIntrusion` consensus entry per tick (signed by the ATECC608B).

Engineering Integration and Assurance

Repository source: `engineering/README.md`

This domain combines reusable design sources with city alignment and accepted engineering inputs. It owns analysis and exchange evidence, not the canonical component geometry.

Area	Contents
`analysis/analysis-register.toml`	Machine-readable source of truth for current analysis status and remaining evidence
`analysis/`	Field-to-drainage/ground deployment gates, station evidence and solver inputs
`interchange/`	Deterministic IFC4.3 and station-interchange generators; `ifc_mesh.py` converts indexed native solids into bounded, source-bound browser geometry
`models/bim/reference/`	Public IFC, IDS, BCF, validation and coordination review set
`models/digital-twins/`	Source-linked fabrication and construction review scenes
`assurance/`	Formal specifications and simulation-component coverage evidence
`baghdad_detail.py`	Source-bound six-car Baghdad parts, software/ERP allocation and electronics sizing register; design and Iraqi slab manufacture
`toolchain/`	Pinned external engineering applications and validation commands

Structural release calculations, surveys, ground data, supplier drawings and jurisdictional approvals remain deployment-specific gates.

City `engineering/depot-scope/` reports reconcile the operating depot inventory with the station BOM, reference canopy, cost allowances and initial dispatch requirements. See the Samawah reconciliation. These reports quantify outstanding work; they do not establish equipment placement, an installed budget or overnight stabling feasibility.

City organisation and design work

Repository source: `engineering/analysis/city-delivery-index.md`

Generated from current city finance, work orders, climate presets and mapped soil. All figures remain planning quantities pending local workforce and design qualification.

City	Operating FTE	Organisation, skills, tasks and design refinements
bukavu	539	city work package
goma	497	city work package
kananga	286	city work package
kinshasa	7215	city work package
kisangani	434	city work package
lubumbashi	914	city work package
mbuji-mayi	754	city work package
benguela	518	city work package
huambo	461	city work package
lobito	390	city work package
luanda	2846	city work package
lubango	434	city work package
malanje	248	city work package
namibe	390	city work package
soyo	348	city work package
uige	116	city work package
eldoret	528	city work package
garissa	265	city work package
kakamega	393	city work package
kisii	267	city work package
kisumu	486	city work package
kitale	358	city work package
machakos	280	city work package
malindi	282	city work package
meru-ke	269	city work package
mombasa	2314	city work package
nairobi	3199	city work package
naivasha	390	city work package
nakuru	483	city work package
nyeri	327	city work package
thika	613	city work package
antananarivo	2310	city work package
beira	449	city work package
chimoio	327	city work package
lichinga	159	city work package
maputo	1367	city work package
nacala	435	city work package
nampula	455	city work package
pemba-mz	285	city work package
quelimane	128	city work package
tete	535	city work package
xai-xai	286	city work package
huye	389	city work package
kigali	1242	city work package
rubavu	380	city work package
mogadishu	862	city work package
waw	262	city work package
arusha	531	city work package
dar-es-salaam	3763	city work package
dodoma	485	city work package
iringa	296	city work package
kigoma	333	city work package
mbeya	505	city work package
morogoro	467	city work package
moshi	335	city work package
mwanza	1730	city work package
shinyanga	315	city work package

City	Operating FTE	Organisation, skills, tasks and design refinements
songea	182	city work package
sumbawanga	227	city work package
tabora	254	city work package
tanga	504	city work package
zanzibar-city	431	city work package
arua	357	city work package
entebbe	370	city work package
fort-portal	340	city work package
gulu	529	city work package
hoima	302	city work package
jinja	458	city work package
kampala	1331	city work package
lira	373	city work package
masaka	324	city work package
mbale	298	city work package
mbarara	488	city work package
soroti	140	city work package
lusaka	1973	city work package
lyon	1531	city work package
la-paz	1409	city work package
cuenca	578	city work package
san-salvador	1694	city work package
agadir	667	city work package
beni-mellal	312	city work package
fez	762	city work package
kenitra	537	city work package
khouribga	252	city work package
marrakech	1458	city work package
meknes	375	city work package
nador	317	city work package
oujda	377	city work package
safi	430	city work package
tangier	1096	city work package
tetouan	478	city work package
el-obeid	441	city work package
kassala	313	city work package
khartoum	2974	city work package
nyala	469	city work package
omdurman	1749	city work package
port-sudan	333	city work package
tunis	1343	city work package
bloemfontein	581	city work package
durban	2971	city work package
east-london-za	572	city work package
nelspruit	317	city work package
polokwane	442	city work package
herat	445	city work package
jalalabad-af	452	city work package
kabul	1659	city work package
kandahar	445	city work package
mazar-e-sharif	563	city work package
barisal	538	city work package
chittagong	2441	city work package
comilla	464	city work package
gazipur	1740	city work package
khulna	1613	city work package
mymensingh	429	city work package
narayanganj	702	city work package
rajshahi	409	city work package
rangpur	419	city work package
sylhet	453	city work package
agra	1060	city work package
bhopal	1291	city work package

City	Operating FTE	Organisation, skills, tasks and design refinements
coimbatore	2534	city work package
indore	2458	city work package
jodhpur	1171	city work package
kanpur	2687	city work package
lucknow	2317	city work package
madurai	1458	city work package
meerut	893	city work package
patna	3601	city work package
raipur	1079	city work package
rajkot	958	city work package
ranchi	1476	city work package
vadodara	1152	city work package
varanasi	1350	city work package
vijayawada	1312	city work package
visakhapatnam	1510	city work package
biratnagar	294	city work package
kathmandu	1344	city work package
pokhara	686	city work package
bahawalpur	450	city work package
faisalabad	1321	city work package
gujranwala	1141	city work package
hyderabad-pk	1214	city work package
karachi	3487	city work package
larkana	331	city work package
multan	882	city work package
peshawar	1215	city work package
quetta	701	city work package
rahim-yar-khan	536	city work package
sheikhupura	564	city work package
sialkot	529	city work package
sukkur	480	city work package
colombo	2551	city work package
galle	560	city work package
jaffna	465	city work package
kandy	574	city work package
phnom-penh	2030	city work package
bandung	2054	city work package
surabaya	2799	city work package
vientiane	584	city work package
mandalay	1523	city work package
yangon	3946	city work package
davao	1984	city work package
ouagadougou	1457	city work package
bafoussam	588	city work package
bamenda	485	city work package
bertoua	324	city work package
douala	1407	city work package
edea	116	city work package
garoua	342	city work package
kumba	420	city work package
maroua	495	city work package
ngaoundere	340	city work package
yaounde	1375	city work package
conakry	569	city work package
bamako	1790	city work package
niamey	1207	city work package
aba-ng	374	city work package
benin-city	933	city work package
ibadan	962	city work package
ilorin	442	city work package
jos	417	city work package
kano	3160	city work package
maiduguri	1123	city work package

City	Operating FTE	Organisation, skills, tasks and design refinements
onitsha	1247	city work package
port-harcourt	1219	city work package
uyo	353	city work package
dakar	2015	city work package
arish	165	city work package
asyut	454	city work package
beni-suef	376	city work package
damanhur	444	city work package
damietta	626	city work package
fayoum	590	city work package
hurghada	379	city work package
ismailia	514	city work package
kafr-el-sheikh	311	city work package
luxor	444	city work package
mahalla	380	city work package
mansoura-eg	486	city work package
minya	431	city work package
port-said	310	city work package
qena	423	city work package
sohag	386	city work package
suez	529	city work package
tanta	637	city work package
zagazig	438	city work package
amarah	482	city work package
baghdad	3460	city work package
baqubah	580	city work package
basra	2074	city work package
diwaniyah	440	city work package
duhok	528	city work package
erbil	1056	city work package
fallujah	501	city work package
hillah	572	city work package
karbala	1143	city work package
kirkuk	1021	city work package
kut	467	city work package
mosul	1417	city work package
najaf	1375	city work package
nasiriyah	494	city work package
ramadi	450	city work package
samawah	465	city work package
sulaymaniyah	804	city work package
amman	2599	city work package
aqaba	379	city work package
irbid	441	city work package
zarqa	643	city work package
beirut	1101	city work package
sidon	310	city work package
tripoli-lb	442	city work package
gaza-city	374	city work package
hebron	487	city work package
nablus	584	city work package
abha	499	city work package
al-kharj	611	city work package
buraidah	601	city work package
dammam	1915	city work package
hail	550	city work package
hofuf	666	city work package
jeddah	2579	city work package
jizan	556	city work package
khamis-mushait	637	city work package
mecca	1475	city work package
medina	1249	city work package
najran	511	city work package

City	Operating FTE	Organisation, skills, tasks and design refinements
tabuk	599	city work package
taif	560	city work package
aleppo	1238	city work package
damascus	1266	city work package
deir-ez-zor	528	city work package
hama	475	city work package
homs	399	city work package
idlib	304	city work package
latakia	345	city work package
raqqa	480	city work package
tartus	288	city work package
aden	545	city work package
dhamar	276	city work package
hodeidah	344	city work package
ibb	429	city work package
lahij	322	city work package
mukalla	821	city work package
sanaa	1776	city work package
sayun	287	city work package
taiz	441	city work package

Catalogue deployment evidence

Repository source: `engineering/analysis/deployment-summary.md`

City registers distinguish completed screening and data work from field investigation, physical design and operating release. Soil means and uncertainty are planning inputs.

City	Soil complete / sampled locations	Open deployment gates	Soil inputs	Deployment work
bukavu	452 / 463	13	soil	register
goma	177 / 177	12	soil	register
kananga	933 / 933	13	soil	register
kinshasa	5693 / 5744	14	soil	register
kisangani	343 / 343	13	soil	register
lubumbashi	613 / 613	13	soil	register
mbuji-mayi	281 / 283	14	soil	register
benguela	170 / 203	13	soil	register
huambo	110 / 110	12	soil	register
lobito	244 / 250	13	soil	register
luanda	1123 / 1123	13	soil	register
lubango	308 / 308	12	soil	register
malanje	56 / 56	12	soil	register
namibe	78 / 209	13	soil	register
soyo	92 / 92	12	soil	register
uige	21 / 21	12	soil	register
eldoret	140 / 140	12	soil	register
garissa	70 / 70	12	soil	register
kakamega	115 / 115	12	soil	register
kisii	89 / 89	12	soil	register
kisumu	226 / 228	13	soil	register
kitale	159 / 159	12	soil	register
machakos	87 / 87	12	soil	register
malindi	133 / 133	12	soil	register
meru-ke	129 / 129	12	soil	register
mombasa	2290 / 2309	14	soil	register
nairobi	1849 / 1849	13	soil	register
naivasha	84 / 84	12	soil	register
nakuru	287 / 287	12	soil	register
nyeri	241 / 241	12	soil	register
thika	212 / 212	12	soil	register
antananarivo	1887 / 1895	14	soil	register
beira	185 / 185	12	soil	register
chimoio	78 / 78	12	soil	register
lichinga	98 / 98	12	soil	register
maputo	1296 / 1296	13	soil	register
nacala	218 / 220	13	soil	register
nampula	161 / 161	12	soil	register
pemba-mz	115 / 132	13	soil	register
quelimane	48 / 48	12	soil	register
tete	524 / 526	13	soil	register
xai-xai	84 / 84	12	soil	register
huye	125 / 125	12	soil	register
kigali	815 / 815	13	soil	register
rubavu	136 / 143	13	soil	register
mogadishu	2404 / 2413	14	soil	register
waw	94 / 94	12	soil	register
arusha	408 / 408	12	soil	register
dar-es-salaam	4169 / 4170	14	soil	register
dodoma	324 / 324	12	soil	register
iringa	148 / 148	12	soil	register
kigoma	78 / 78	12	soil	register
mbeya	157 / 157	12	soil	register
morogoro	356 / 356	12	soil	register
moshi	363 / 363	12	soil	register
mwanza	1023 / 1030	14	soil	register

City	Soil complete / sampled locations	Open deployment gates	Soil inputs	Deployment work
shinyanga	90 / 90	12	soil	register
songea	25 / 25	12	soil	register
sumbawanga	48 / 48	12	soil	register
tabora	80 / 80	12	soil	register
tanga	396 / 396	12	soil	register
zanzibar-city	482 / 482	12	soil	register
arua	134 / 134	12	soil	register
entebbe	190 / 190	12	soil	register
fort-portal	105 / 105	12	soil	register
gulu	124 / 124	12	soil	register
hoima	64 / 64	12	soil	register
jinja	161 / 179	13	soil	register
kampala	1190 / 1190	13	soil	register
lira	144 / 144	12	soil	register
masaka	70 / 70	12	soil	register
mbale	170 / 170	12	soil	register
mbarara	339 / 339	12	soil	register
soroti	23 / 23	12	soil	register
lusaka	824 / 824	13	soil	register
lyon	1740 / 1765	14	soil	register
la-paz	731 / 733	14	soil	register
cuenca	240 / 240	12	soil	register
san-salvador	946 / 946	13	soil	register
agadir	273 / 276	13	soil	register
beni-mellal	62 / 62	12	soil	register
fez	473 / 473	13	soil	register
kenitra	148 / 152	13	soil	register
khouribga	52 / 52	12	soil	register
marrakech	2586 / 2659	14	soil	register
meknes	88 / 88	12	soil	register
nador	134 / 134	12	soil	register
oujda	58 / 58	12	soil	register
safi	94 / 97	13	soil	register
tangier	663 / 668	14	soil	register
tetouan	159 / 159	12	soil	register
el-obeid	98 / 98	12	soil	register
kassala	317 / 317	12	soil	register
khartoum	2336 / 2353	14	soil	register
nyala	127 / 127	12	soil	register
omdurman	2042 / 2063	14	soil	register
port-sudan	111 / 116	13	soil	register
tunis	871 / 873	14	soil	register
bloemfontein	126 / 126	12	soil	register
durban	3404 / 3409	14	soil	register
east-london-za	232 / 232	12	soil	register
nelspruit	81 / 81	12	soil	register
polokwane	112 / 112	12	soil	register
herat	161 / 164	13	soil	register
jalalabad-af	216 / 230	13	soil	register
kabul	1097 / 1101	14	soil	register
kandahar	110 / 111	13	soil	register
mazar-e-sharif	178 / 184	13	soil	register
barisal	891 / 893	13	soil	register
chittagong	3262 / 3272	14	soil	register
comilla	372 / 372	12	soil	register
gazipur	2061 / 2062	14	soil	register
khulna	2217 / 2230	14	soil	register
mymensingh	756 / 757	13	soil	register
narayanganj	1573 / 1585	13	soil	register
rajshahi	297 / 297	12	soil	register
rangpur	133 / 133	12	soil	register
sylhet	409 / 409	12	soil	register

City	Soil complete / sampled locations	Open deployment gates	Soil inputs	Deployment work
agra	741 / 741	13	soil	register
bhopal	777 / 779	14	soil	register
coimbatore	1270 / 1273	14	soil	register
indore	1562 / 1562	13	soil	register
jodhpur	1799 / 1799	13	soil	register
kanpur	1235 / 1235	13	soil	register
lucknow	1092 / 1092	13	soil	register
madurai	1238 / 1238	12	soil	register
meerut	439 / 439	13	soil	register
patna	2503 / 2505	14	soil	register
raipur	492 / 492	13	soil	register
rajkot	668 / 669	13	soil	register
ranchi	771 / 771	13	soil	register
vadodara	495 / 495	13	soil	register
varanasi	968 / 982	14	soil	register
vijayawada	1222 / 1234	14	soil	register
visakhapatnam	1211 / 1215	14	soil	register
biratnagar	902 / 902	12	soil	register
kathmandu	1411 / 1411	13	soil	register
pokhara	1045 / 1045	12	soil	register
bahawalpur	146 / 146	12	soil	register
faisalabad	641 / 641	13	soil	register
gujranwala	561 / 561	13	soil	register
hyderabad-pk	883 / 893	14	soil	register
karachi	1957 / 1973	14	soil	register
larkana	83 / 83	12	soil	register
multan	370 / 370	13	soil	register
peshawar	651 / 651	13	soil	register
quetta	359 / 359	13	soil	register
rahim-yar-khan	170 / 170	12	soil	register
sheikhupura	76 / 76	12	soil	register
sialkot	131 / 131	12	soil	register
sukkur	167 / 168	13	soil	register
colombo	3165 / 3207	14	soil	register
galle	216 / 217	13	soil	register
jaffna	163 / 169	13	soil	register
kandy	834 / 835	13	soil	register
phnom-penh	1768 / 1806	14	soil	register
bandung	2665 / 2665	13	soil	register
surabaya	3190 / 3473	14	soil	register
vientiane	196 / 197	13	soil	register
mandalay	1835 / 1917	14	soil	register
yangon	2375 / 2420	14	soil	register
davao	2033 / 2073	14	soil	register
ouagadougou	734 / 739	14	soil	register
bafoussam	380 / 380	12	soil	register
bamenda	154 / 154	12	soil	register
bertoua	63 / 63	12	soil	register
douala	1066 / 1079	14	soil	register
edea	26 / 30	13	soil	register
garoua	386 / 386	12	soil	register
kumba	153 / 153	12	soil	register
maroua	320 / 320	12	soil	register
ngaoundere	151 / 151	12	soil	register
yaounde	940 / 940	13	soil	register
conakry	1220 / 1231	14	soil	register
bamako	3770 / 3793	14	soil	register
niamey	785 / 792	14	soil	register
aba-ng	82 / 82	12	soil	register
benin-city	722 / 722	13	soil	register
ibadan	646 / 646	13	soil	register
ilorin	191 / 191	12	soil	register

City	Soil complete / sampled locations	Open deployment gates	Soil inputs	Deployment work
jos	347 / 347	12	soil	register
kano	1239 / 1239	13	soil	register
maiduguri	1105 / 1105	13	soil	register
onitsha	829 / 869	13	soil	register
port-harcourt	1001 / 1018	14	soil	register
uyo	74 / 74	12	soil	register
dakar	1317 / 1336	14	soil	register
arish	27 / 29	13	soil	register
asyut	126 / 132	13	soil	register
beni-suef	124 / 147	13	soil	register
damanhur	102 / 102	12	soil	register
damietta	259 / 260	13	soil	register
fayoum	139 / 162	13	soil	register
hurghada	131 / 132	13	soil	register
ismailia	249 / 275	13	soil	register
kafr-el-sheikh	64 / 64	12	soil	register
luxor	251 / 266	13	soil	register
mahalla	129 / 129	12	soil	register
mansoura-eg	548 / 548	12	soil	register
minya	109 / 113	13	soil	register
port-said	76 / 79	13	soil	register
qena	100 / 104	13	soil	register
sohag	93 / 100	13	soil	register
suez	166 / 289	13	soil	register
tanta	161 / 161	12	soil	register
zagazig	134 / 134	12	soil	register
amarah	126 / 145	13	soil	register
baghdad	4865 / 4881	14	soil	register
baqubah	168 / 168	12	soil	register
basra	964 / 1585	14	soil	register
diwaniyah	127 / 129	13	soil	register
duhok	229 / 229	12	soil	register
erbil	313 / 313	12	soil	register
fallujah	81 / 113	13	soil	register
hillah	272 / 272	12	soil	register
karbala	522 / 555	14	soil	register
kirkuk	456 / 477	14	soil	register
kut	130 / 131	13	soil	register
mosul	2546 / 2644	14	soil	register
najaf	556 / 652	14	soil	register
nasiriyah	142 / 163	13	soil	register
ramadi	121 / 141	13	soil	register
samawah	130 / 246	13	soil	register
sulaymaniyah	479 / 479	13	soil	register
amman	1152 / 1168	14	soil	register
aqaba	194 / 247	13	soil	register
irbid	89 / 89	12	soil	register
zarqa	182 / 217	13	soil	register
beirut	965 / 969	14	soil	register
sidon	112 / 112	12	soil	register
tripoli-lb	129 / 129	12	soil	register
gaza-city	72 / 72	12	soil	register
hebron	94 / 94	12	soil	register
nablus	130 / 130	12	soil	register
abha	196 / 196	12	soil	register
al-kharj	99 / 151	13	soil	register
buraidah	86 / 111	13	soil	register
dammam	1178 / 1295	14	soil	register
hail	116 / 188	13	soil	register
hofuf	103 / 161	13	soil	register
jeddah	1089 / 1198	14	soil	register
jizan	136 / 195	13	soil	register

City	Soil complete / sampled locations	Open deployment gates	Soil inputs	Deployment work
khamis-mushait	446 / 459	13	soil	register
mecca	426 / 564	14	soil	register
medina	670 / 739	14	soil	register
najran	109 / 118	13	soil	register
tabuk	104 / 155	13	soil	register
taif	91 / 138	13	soil	register
aleppo	1059 / 1059	13	soil	register
damascus	968 / 970	14	soil	register
deir-ez-zor	187 / 198	13	soil	register
hama	421 / 421	12	soil	register
homs	166 / 166	12	soil	register
idlib	78 / 78	12	soil	register
latakia	155 / 158	13	soil	register
raqqa	89 / 91	13	soil	register
tartus	114 / 114	12	soil	register
aden	176 / 464	13	soil	register
dhamar	61 / 61	12	soil	register
hodeidah	50 / 146	13	soil	register
ibb	122 / 122	12	soil	register
lahij	51 / 59	13	soil	register
mukalla	295 / 461	13	soil	register
sanaa	565 / 566	14	soil	register
sayun	41 / 48	13	soil	register
taiz	110 / 110	12	soil	register

Depot thermal and fire mitigation work packages

Repository source: `engineering/analysis/stations/mitigation-work-packages.md`

Generated from `mitigation-work-packages.toml`, which is the canonical owner/evidence/closure register.

Design response: Keep charger power stages and stationary storage in a physically separated outdoor/open-sided energy compound; keep controls and switchgear in a separate room with 2 x 30 kW packaged DX cooling (one duty, one standby).

> The deterministic comparison supports layout development only. No package closes without project inputs, signed supplier evidence and competent local approval.

ID	Package	Owner role	Related products	State
`STN-MIT-001`	Charger and battery supplier evidence freeze	depot electrical lead	`STN-DEP-P040`, `STN-DEP-P050`	`open-deployment`
`STN-MIT-002`	Energy-compound site and fire-strategy release	architect/engineer of record and fire engineer	`STN-DEP-P010`, `STN-DEP-P040`, `STN-DEP-P050`	`open-deployment`
`STN-MIT-003`	Controls-room cooling detailed design	mechanical services lead	`STN-DEP-P070`	`open-deployment`
`STN-MIT-004`	Energy-compound detection and protection integration	electrical and fire systems leads	`STN-DEP-P040`, `STN-DEP-P050`, `STN-DEP-P070`	`open-deployment`
`STN-MIT-005`	Factory and site acceptance evidence	depot commissioning manager	`STN-DEP-P040`, `STN-DEP-P050`, `STN-DEP-P070`	`open-deployment`
`STN-MIT-006`	Independent review and operating handover	operator design authority	`STN-DEP-SA850`	`open-deployment`

Evidence required

`STN-MIT-001` — Charger and battery supplier evidence freeze

- selected manufacturer/model and controlled datasheets
- maximum-ambient derating and charger loss/duty map
- cell-to-pack propagation and credible heat-release data
- fault response, remote isolation and maintenance clearances

`STN-MIT-002` — Energy-compound site and fire-strategy release

- surveyed compound location and code-compliant separation distances
- wind-sensitive fire/smoke cases and occupied-building exposure assessment
- outward access, fencing, emergency response and firefighting provisions
- spill/firewater drainage, containment and environmental approval

`STN-MIT-003` — Controls-room cooling detailed design

- project weather and envelope model
- selected 2 x 30 kW units with site-ambient derating
- duty/standby controls, high-temperature alarm and loss-of-cooling response
- refrigerant, condensate, maintainability and power-failure design

`STN-MIT-004` — Energy-compound detection and protection integration

- detection technology and coverage drawings
- cause-and-effect matrix including alarm, charger abort and remote isolation
- protection coordination, earthing and emergency-stop design
- interface test procedure with OCC and depot maintenance systems

`STN-MIT-005` — Factory and site acceptance evidence

- charger, battery and cooling FAT records
- installed thermal-load and duty/standby changeover test

- alarm, abort, isolation and loss-of-comms test
- fire-system integrated test witnessed by the approving parties

`STN-MIT-006` — Independent review and operating handover

- closed design comments and statutory approvals
- approved emergency and degraded-operation procedures
- as-built model, asset register, spares and maintenance plan
- training, drill and operator acceptance records

Station systems screening

Repository source: `engineering/analysis/stations/screening-summary.md`

Deterministic design-reference checks for all seven station variants. These are reproducible engineering screens, not construction release or authority approval.

- Screening execution: passed
- Deployment release: not ready
- Manifest: `design/component-catalogue/catalog/buildable-stations/station-kit-manifest.json` (`b6a0f2be36ac…`)

Results

Domain	Engine	Scope	Result	Remaining gate
Structure	OpenSeesPy 3.8.0.0	22 m shared canopy truss, gravity + uplift	PASS	Site loads, 3D stability, joints, foundations and code combinations
Passenger flow	JuPedSim 1.4.2	normal, degraded and egress route for 7 variants	PASS	Calibrated demand, conflicts, assisted evacuation and authority criteria
Roof drainage	EPA SWMM via PySWMM 2.1.0	7 canopy catchments, 75 mm/h input storm	PASS	Local rainfall, survey, tailwater, blockage and exceedance
Depot thermal	EnergyPlus	baseline room + separated/cooled controls-room comparison	input-prepared	Project climate, supplier losses, detailed HVAC, controls and commissioning
Depot fire	FDS	enclosed room + separated/open compound comparison at prescribed 250 kW	input-prepared	Supplier fire data, separation/wind cases, suppression and fire-engineer acceptance

Structural cases

Case	Load kN	Displacement / limit mm	Stress / allowable MPa	Result
`gravity`	205.7	18.65 / 91.67	62.9 / 213.0	PASS
`wind_uplift`	224.4	20.34 / 91.67	68.6 / 213.0	PASS

Passenger-flow cases

Variant	Normal s	Degraded s	Egress s	Result
`halt`	58.50	61.80	68.20	PASS
`standard`	62.75	65.15	71.35	PASS
`major`	62.75	65.15	71.35	PASS
`interchange`	62.75	65.15	71.35	PASS
`interchange-elevated`	62.75	65.15	71.35	PASS
`terminal`	62.75	65.15	71.35	PASS
`depot-terminal`	62.75	65.15	71.35	PASS

Drainage cases

The storm supplies 25 mm over 20 minutes. The roof and rational check both use C=1 (no losses). Each 300 mm branch assumes a 0.4 m fall over 40 m; these are planning levels requiring site survey. Dynamic-wave routing uses a 1 s hydraulic step and 10 s runoff step. Rainfall, zero infiltration, runoff volume/peak, conduit peak, continuity and no flooding are separate pass gates.

Variant	Catchment / branches	SWMM branch / aggregate L/s	Rational aggregate L/s	Rain mm / volume error %	Inlet depth m	Result
`halt`	439.0 m² / 11	0.83 / 9.15	9.15	25.00 / 0.019	0.020	PASS
`standard`	1813.0 m² / 27	1.40 / 37.77	37.77	25.00 / -0.009	0.025	PASS
`major`	2115.0 m² / 19	2.32 / 44.06	44.06	25.00 / -0.050	0.032	PASS
`interchange`	2921.0 m² / 31	1.96 / 60.85	60.85	25.00 / -0.038	0.030	PASS
`interchange-elevated`	3348.0 m² / 32	2.18 / 69.75	69.75	25.00 / -0.046	0.031	PASS
`terminal`	2561.0 m² / 31	1.72 / 53.35	53.35	25.00 / -0.025	0.028	PASS
`depot-terminal`	1626.0 m² / 26	1.30 / 33.88	33.88	25.00 / -0.004	0.024	PASS

Release boundary

A passing row means only that the deterministic catalogue assumption completed and met its stated screening threshold. Before procurement or construction, the deployment team must substitute surveyed/site inputs, local statutory criteria, supplier performance data, detailed connections and independent competent review. EnergyPlus/FDS solver completion and a mitigation screen pass confirm only input execution and the direction of the catalogue design response. The baseline findings, supplier evidence, project-specific design and independent approvals remain open release gates.

Population access, transfers and viaduct obstacle review

Repository source: `engineering/assurance/access-clearance/README.md`

All 266 cities retain separate population/transfer and beam/support/terrain reports. 247 have usable native WorldPop 2020 count evidence; 19 have explicit missing or zero-count evidence. No routing score is multiplied by city population.

243 line networks are fully connected through declared interchanges; 23 retain disconnected components. Direct line-pair transfers are reported separately. Geometry crossing alone creates no passenger transfer.

The historical demand score counts high-demand cells near tracks, including areas between stations; it is not population access. Native station catchment unions can be higher or lower. Radial distance ignores walking barriers; wider radii require feeder/walking evidence and do not create extra fare revenue. Source year, bounding-box denominator, native pixel resolution and nodata remain explicit.

Input finding — Wau: Wau is in South Sudan. The canonical jurisdiction and national aggregation now use SS; the historical slug waw remains an alias. Finance uses an explicitly uncalibrated generic scenario, with SSP currency; Sudan SD/SDG assumptions are excluded. Evidence.

Terrain is retained for 266 cities; gaps remain visible elsewhere. Zero cities are physically obstacle-released. Beam overflight, foundations and terrain are checked separately. Known provisional clashes and unknown heights require survey and design changes; no accepted route clearance or unpriced height increase is invented.

City	Routing-demand cells (legacy)	Population access	Direct transfers	Reachable line pairs	Clearance
Bukavu	60.1%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Goma	42.3%	available; radius sensitivities	40.0%	100.0%	physical release blocked
Kananga	80.4%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Kinshasa	62.5%	available; radius sensitivities	46.2%	100.0%	physical release blocked
Kisangani	48.2%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Lubumbashi	57.8%	available; radius sensitivities	42.9%	100.0%	physical release blocked
Mbuji-Mayi	62.0%	available; radius sensitivities	40.0%	100.0%	physical release blocked
Benguela	42.6%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Huambo	70.7%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Lobito	71.5%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Luanda	60.6%	available; radius sensitivities	47.3%	100.0%	physical release blocked
Lubango	53.6%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Malanje	58.2%	available; radius sensitivities	100.0%	100.0%	physical release blocked
Namibe	52.8%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Soyo	62.2%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Uige	47.0%	available; radius sensitivities	100.0%	100.0%	physical release blocked
Eldoret	78.8%	available; radius sensitivities	60.0%	100.0%	physical release blocked
Garissa	68.8%	available; radius sensitivities	33.3%	50.0%	physical release blocked
Kakamega	72.4%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Kisii	95.7%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Kisumu	72.5%	available; radius sensitivities	40.0%	100.0%	physical release blocked
Kitale	79.3%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Machakos	66.7%	available; radius sensitivities	33.3%	50.0%	physical release blocked
Malindi	66.4%	available; radius sensitivities	33.3%	50.0%	physical release blocked
Meru-Ke	82.7%	available; radius sensitivities	50.0%	50.0%	physical release blocked

City	Routing-demand cells (legacy)	Population access	Direct transfers	Reachable line pairs	Clearance
Mombasa	74.4%	available; radius sensitivities	44.4%	100.0%	physical release blocked
Nairobi	54.6%	available; radius sensitivities	45.1%	100.0%	physical release blocked
Naivasha	69.6%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Nakuru	54.9%	available; radius sensitivities	40.0%	100.0%	physical release blocked
Nyeri	70.6%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Thika	59.8%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Antananarivo	75.5%	available; radius sensitivities	46.2%	100.0%	physical release blocked
Beira	39.6%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Chimoio	51.9%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Lichinga	47.0%	available; radius sensitivities	0.0%	0.0%	physical release blocked
Maputo	66.8%	available; radius sensitivities	47.2%	100.0%	physical release blocked
Nacala	70.9%	available; radius sensitivities	83.3%	100.0%	physical release blocked
Nampula	69.2%	available; radius sensitivities	40.0%	100.0%	physical release blocked
Pemba-Mz	52.1%	available; radius sensitivities	33.3%	33.3%	physical release blocked
Quelimane	43.6%	available; radius sensitivities	100.0%	100.0%	physical release blocked
Tete	78.4%	available; radius sensitivities	60.0%	100.0%	physical release blocked
Xai-Xai	50.3%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Huye	64.2%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Kigali	67.1%	available; radius sensitivities	47.2%	100.0%	physical release blocked
Rubavu	44.5%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Mogadishu	60.8%	available; radius sensitivities	53.3%	100.0%	physical release blocked
Wau	79.7%	available; radius sensitivities	33.3%	33.3%	physical release blocked
Arusha	46.1%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Dar-Es-Salaam	65.1%	available; radius sensitivities	45.1%	100.0%	physical release blocked
Dodoma	69.0%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Iringa	73.2%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Kigoma	52.3%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Mbeya	48.3%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Morogoro	48.6%	available; radius sensitivities	60.0%	100.0%	physical release blocked
Moshi	69.5%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Mwanza	76.3%	available; radius sensitivities	55.6%	100.0%	physical release blocked
Shinyanga	79.5%	available; radius sensitivities	33.3%	50.0%	physical release blocked
Songea	38.3%	available; radius sensitivities	100.0%	100.0%	physical release blocked
Sumbawanga	61.6%	available; radius sensitivities	33.3%	33.3%	physical release blocked
Tabora	64.4%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Tanga	52.6%	available; radius sensitivities	60.0%	100.0%	physical release blocked
Zanzibar-City	59.1%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Arua	82.4%	available; radius sensitivities	66.7%	100.0%	physical release blocked

City	Routing-demand cells (legacy)	Population access	Direct transfers	Reachable line pairs	Clearance
Entebbe	57.3%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Fort-Portal	83.8%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Gulu	67.5%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Hoima	81.7%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Jinja	82.8%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Kampala	61.4%	available; radius sensitivities	52.8%	100.0%	physical release blocked
Lira	58.0%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Masaka	60.8%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Mbale	84.7%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Mbarara	77.0%	available; radius sensitivities	100.0%	100.0%	physical release blocked
Soroti	69.0%	available; radius sensitivities	100.0%	100.0%	physical release blocked
Lusaka	62.2%	available; radius sensitivities	42.4%	100.0%	physical release blocked
Lyon	57.7%	available; radius sensitivities	50.0%	100.0%	physical release blocked
La-Paz	52.6%	available; radius sensitivities	47.2%	100.0%	physical release blocked
Cuenca	48.0%	available; radius sensitivities	40.0%	100.0%	physical release blocked
San-Salvador	52.1%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Agadir	56.4%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Beni-Mellal	79.6%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Fez	73.2%	available; radius sensitivities	53.3%	100.0%	physical release blocked
Kenitra	59.2%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Khouribga	64.4%	available; radius sensitivities	33.3%	33.3%	physical release blocked
Marrakech	72.4%	available; radius sensitivities	55.6%	100.0%	physical release blocked
Meknes	58.0%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Nador	69.6%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Oujda	43.9%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Safi	64.6%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Tangier	61.3%	available; radius sensitivities	46.4%	100.0%	physical release blocked
Tetouan	69.4%	available; radius sensitivities	66.7%	100.0%	physical release blocked
El-Obeid	57.4%	available; radius sensitivities	40.0%	100.0%	physical release blocked
Kassala	69.0%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Khartoum	53.5%	available; radius sensitivities	39.6%	100.0%	physical release blocked
Nyala	58.0%	available; radius sensitivities	60.0%	100.0%	physical release blocked
Omdurman	58.7%	available; radius sensitivities	52.8%	100.0%	physical release blocked
Port-Sudan	41.9%	available; radius sensitivities	33.3%	50.0%	physical release blocked
Tunis	59.6%	available; radius sensitivities	60.7%	100.0%	physical release blocked
Bloemfontein	46.5%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Durban	76.4%	available; radius sensitivities	38.5%	100.0%	physical release blocked
East-London-Za	40.5%	available; radius sensitivities	50.0%	100.0%	physical release blocked

City	Routing-demand cells (legacy)	Population access	Direct transfers	Reachable line pairs	Clearance
Nelspruit	54.5%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Polokwane	57.3%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Herat	51.1%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Jalalabad-Af	60.2%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Kabul	57.0%	available; radius sensitivities	40.0%	100.0%	physical release blocked
Kandahar	64.4%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Mazar-E-Sharif	57.8%	available; radius sensitivities	60.0%	100.0%	physical release blocked
Barisal	57.5%	available; radius sensitivities	70.0%	100.0%	physical release blocked
Chittagong	74.8%	available; radius sensitivities	43.9%	100.0%	physical release blocked
Comilla	70.2%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Gazipur	46.7%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Khulna	59.6%	available; radius sensitivities	41.7%	100.0%	physical release blocked
Mymensingh	68.4%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Narayanganj	52.5%	available; radius sensitivities	40.0%	100.0%	physical release blocked
Rajshahi	61.0%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Rangpur	70.7%	available; radius sensitivities	40.0%	100.0%	physical release blocked
Sylhet	82.6%	available; radius sensitivities	83.3%	100.0%	physical release blocked
Agra	47.3%	unavailable; radius sensitivities	100.0%	100.0%	physical release blocked
Bhopal	59.3%	unavailable; radius sensitivities	93.3%	100.0%	physical release blocked
Coimbatore	70.5%	unavailable; radius sensitivities	80.6%	100.0%	physical release blocked
Indore	47.8%	unavailable; radius sensitivities	82.1%	100.0%	physical release blocked
Jodhpur	53.6%	unavailable; radius sensitivities	90.0%	100.0%	physical release blocked
Kanpur	50.7%	unavailable; radius sensitivities	78.6%	100.0%	physical release blocked
Lucknow	44.8%	unavailable; radius sensitivities	82.1%	100.0%	physical release blocked
Madurai	70.6%	unavailable; radius sensitivities	86.7%	100.0%	physical release blocked
Meerut	68.4%	unavailable; radius sensitivities	100.0%	100.0%	physical release blocked
Patna	54.3%	unavailable; radius sensitivities	100.0%	100.0%	physical release blocked
Raipur	41.1%	unavailable; radius sensitivities	66.7%	100.0%	physical release blocked
Rajkot	63.6%	unavailable; radius sensitivities	100.0%	100.0%	physical release blocked
Ranchi	49.2%	unavailable; radius sensitivities	93.3%	100.0%	physical release blocked
Vadodara	61.2%	unavailable; radius sensitivities	80.0%	100.0%	physical release blocked
Varanasi	61.5%	unavailable; radius sensitivities	80.0%	100.0%	physical release blocked
Vijayawada	77.5%	unavailable; radius sensitivities	86.7%	100.0%	physical release blocked
Visakhapatnam	49.5%	unavailable; radius sensitivities	86.7%	100.0%	physical release blocked
Biratnagar	49.4%	available; radius sensitivities	33.3%	50.0%	physical release blocked
Kathmandu	54.8%	available; radius sensitivities	52.8%	100.0%	physical release blocked
Pokhara	59.3%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Bahawalpur	53.9%	available; radius sensitivities	50.0%	100.0%	physical release blocked

City	Routing-demand cells (legacy)	Population access	Direct transfers	Reachable line pairs	Clearance
Faisalabad	75.5%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Gujranwala	69.2%	available; radius sensitivities	46.4%	100.0%	physical release blocked
Hyderabad-Pk	67.8%	available; radius sensitivities	55.6%	100.0%	physical release blocked
Karachi	70.4%	available; radius sensitivities	45.1%	100.0%	physical release blocked
Larkana	44.6%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Multan	61.1%	available; radius sensitivities	42.9%	100.0%	physical release blocked
Peshawar	60.7%	available; radius sensitivities	46.4%	100.0%	physical release blocked
Quetta	41.9%	available; radius sensitivities	40.0%	100.0%	physical release blocked
Rahim-Yar-Khan	88.5%	available; radius sensitivities	70.0%	100.0%	physical release blocked
Sheikhupura	53.8%	available; radius sensitivities	100.0%	100.0%	physical release blocked
Sialkot	50.4%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Sukkur	57.5%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Colombo	66.6%	available; radius sensitivities	39.6%	100.0%	physical release blocked
Galle	54.1%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Jaffna	73.3%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Kandy	37.0%	available; radius sensitivities	60.0%	100.0%	physical release blocked
Phnom-Penh	72.3%	available; radius sensitivities	61.1%	100.0%	physical release blocked
Bandung	60.0%	unavailable; radius sensitivities	93.3%	100.0%	physical release blocked
Surabaya	63.2%	unavailable; radius sensitivities	86.1%	100.0%	physical release blocked
Vientiane	51.0%	available; radius sensitivities	60.0%	100.0%	physical release blocked
Mandalay	54.8%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Yangon	57.9%	available; radius sensitivities	48.4%	100.0%	physical release blocked
Davao	48.2%	available; radius sensitivities	52.8%	100.0%	physical release blocked
Ouagadougou	57.0%	available; radius sensitivities	52.8%	100.0%	physical release blocked
Bafoussam	30.3%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Bamenda	40.6%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Bertoua	55.3%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Douala	57.1%	available; radius sensitivities	39.3%	100.0%	physical release blocked
Edea	49.7%	available; radius sensitivities	100.0%	100.0%	physical release blocked
Garoua	57.4%	available; radius sensitivities	33.3%	50.0%	physical release blocked
Kumba	57.5%	available; radius sensitivities	83.3%	100.0%	physical release blocked
Maroua	69.3%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Ngaoundere	83.9%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Yaounde	55.8%	available; radius sensitivities	42.9%	100.0%	physical release blocked
Conakry	85.7%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Bamako	75.7%	available; radius sensitivities	55.6%	100.0%	physical release blocked
Niamey	66.8%	available; radius sensitivities	52.8%	100.0%	physical release blocked
Aba-Ng	68.3%	available; radius sensitivities	66.7%	100.0%	physical release blocked

City	Routing-demand cells (legacy)	Population access	Direct transfers	Reachable line pairs	Clearance
Benin-City	74.7%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Ibadan	40.7%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Ilorin	55.2%	available; radius sensitivities	60.0%	100.0%	physical release blocked
Jos	42.7%	available; radius sensitivities	30.0%	60.0%	physical release blocked
Kano	55.8%	available; radius sensitivities	46.2%	100.0%	physical release blocked
Maiduguri	46.8%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Onitsha	80.5%	available; radius sensitivities	46.4%	100.0%	physical release blocked
Port-Harcourt	36.7%	available; radius sensitivities	42.9%	100.0%	physical release blocked
Uyo	65.5%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Dakar	67.1%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Arish	56.3%	available; radius sensitivities	100.0%	100.0%	physical release blocked
Asyut	61.0%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Beni-Suef	59.1%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Damanhur	81.8%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Damietta	61.6%	available; radius sensitivities	60.0%	100.0%	physical release blocked
Fayoum	69.7%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Hurghada	55.3%	available; radius sensitivities	83.3%	100.0%	physical release blocked
Ismailia	67.8%	available; radius sensitivities	83.3%	100.0%	physical release blocked
Kafr-El-Sheikh	77.5%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Luxor	55.4%	available; radius sensitivities	40.0%	60.0%	physical release blocked
Mahalla	47.3%	available; radius sensitivities	33.3%	33.3%	physical release blocked
Mansoura-Eg	44.7%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Minya	63.9%	available; radius sensitivities	83.3%	100.0%	physical release blocked
Port-Said	63.8%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Qena	73.0%	available; radius sensitivities	40.0%	100.0%	physical release blocked
Sohag	46.9%	available; radius sensitivities	16.7%	16.7%	physical release blocked
Suez	67.1%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Tanta	68.2%	available; radius sensitivities	70.0%	100.0%	physical release blocked
Zagazig	59.0%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Amarah	55.5%	available; radius sensitivities	60.0%	100.0%	physical release blocked
Baghdad	40.5%	available; radius sensitivities	39.6%	100.0%	physical release blocked
Baqubah	56.1%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Basra	71.9%	available; radius sensitivities	45.5%	100.0%	physical release blocked
Diwaniyah	46.3%	available; radius sensitivities	40.0%	60.0%	physical release blocked
Duhok	58.5%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Erbil	57.3%	available; radius sensitivities	46.4%	100.0%	physical release blocked
Fallujah	59.7%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Hillah	67.3%	available; radius sensitivities	40.0%	100.0%	physical release blocked

City	Routing-demand cells (legacy)	Population access	Direct transfers	Reachable line pairs	Clearance
Karbala	53.7%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Kirkuk	61.3%	available; radius sensitivities	39.3%	100.0%	physical release blocked
Kut	45.2%	available; radius sensitivities	40.0%	100.0%	physical release blocked
Mosul	51.3%	available; radius sensitivities	52.8%	100.0%	physical release blocked
Najaf	54.7%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Nasiriyah	46.5%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Ramadi	59.5%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Samawah	53.8%	available; radius sensitivities	60.0%	100.0%	physical release blocked
Sulaymaniyah	49.8%	available; radius sensitivities	46.7%	100.0%	physical release blocked
Amman	66.8%	available; radius sensitivities	44.0%	100.0%	physical release blocked
Aqaba	62.0%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Irbid	57.0%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Zarqa	51.3%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Beirut	57.3%	available; radius sensitivities	44.4%	100.0%	physical release blocked
Sidon	74.8%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Tripoli-Lb	65.5%	available; radius sensitivities	100.0%	100.0%	physical release blocked
Gaza-City	43.8%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Hebron	66.9%	available; radius sensitivities	83.3%	100.0%	physical release blocked
Nablus	61.0%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Abha	29.4%	available; radius sensitivities	33.3%	50.0%	physical release blocked
Al-Kharj	55.3%	available; radius sensitivities	40.0%	100.0%	physical release blocked
Buraidah	46.1%	available; radius sensitivities	40.0%	100.0%	physical release blocked
Dammam	53.7%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Hail	47.1%	available; radius sensitivities	40.0%	100.0%	physical release blocked
Hofuf	48.5%	available; radius sensitivities	40.0%	100.0%	physical release blocked
Jeddah	75.6%	available; radius sensitivities	37.4%	100.0%	physical release blocked
Jizan	61.7%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Khamis-Mushait	56.5%	available; radius sensitivities	30.0%	40.0%	physical release blocked
Mecca	41.2%	available; radius sensitivities	44.4%	100.0%	physical release blocked
Medina	46.4%	available; radius sensitivities	38.9%	100.0%	physical release blocked
Najran	54.4%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Tabuk	39.9%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Taif	56.5%	available; radius sensitivities	40.0%	100.0%	physical release blocked
Aleppo	60.1%	available; radius sensitivities	47.2%	100.0%	physical release blocked
Damascus	77.8%	available; radius sensitivities	55.6%	100.0%	physical release blocked
Deir-Ez-Zor	59.5%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Hama	59.7%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Homs	60.6%	available; radius sensitivities	83.3%	100.0%	physical release blocked

City	Routing-demand cells (legacy)	Population access	Direct transfers	Reachable line pairs	Clearance
Idlib	72.3%	available; radius sensitivities	83.3%	100.0%	physical release blocked
Latakia	51.3%	available; radius sensitivities	30.0%	40.0%	physical release blocked
Raqqa	65.4%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Tartus	77.6%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Aden	59.8%	available; radius sensitivities	40.0%	100.0%	physical release blocked
Dhamar	68.5%	available; radius sensitivities	33.3%	50.0%	physical release blocked
Hodeidah	65.1%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Ibb	23.2%	available; radius sensitivities	30.0%	60.0%	physical release blocked
Lahij	64.2%	available; radius sensitivities	66.7%	100.0%	physical release blocked
Mukalla	66.9%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Sanaa	71.2%	available; radius sensitivities	43.6%	100.0%	physical release blocked
Sayun	63.8%	available; radius sensitivities	50.0%	100.0%	physical release blocked
Taiz	65.3%	available; radius sensitivities	50.0%	100.0%	physical release blocked

Machine-readable aggregate and source hashes · Common clearance policy.

Current Baghdad test inputs

Repository source: `engineering/assurance/baghdad-test-inputs/README.md`

This snapshot contains the exact current operations, solver and GIS inputs used by repository tests. Its manifest binds every member, archive part and producer source by SHA-256. It is separate from the historical Baghdad proposal snapshot. Physical and operating release remain false.

Restore a clean checkout with:

```
tools/automation/osr-python tools/automation/bootstrap_baghdad_tests.py
```

After a deliberate input update, regenerate from current local files with:

```
tools/automation/osr-python tools/automation/pack-baghdad-test-inputs.py
```

The bootstrap preserves differing local inputs and rejects changed source, archive or member receipts. Archive parts remain below 50 MiB.

Battery-cooling qualification readiness

Repository source: `engineering/assurance/battery-cooling/qualification-report.md`

> Generated by `tools/automation/subsystem_qualification.py`.

Package `QP-COOLING-001` / configuration `CFG-DEMO-COOL-A`. QUALIFICATION-OPEN.

Quantitative screens retain their stated input evidence basis. Physical qualification and accepted use require the separately listed measurements and authenticated decisions.

Decision area	State	Open findings
requirements	blocked	ALLOC-COOL-availability: RAMS allocation/target review open; ALLOC-COOL-availability: RAMS target undetermined; Allocated operating conditions do not match the analysed conditions.; RAMS target, comparison or allocation review remains open.; Result uses assumptions, unknown events or an unvalidated model.; ALLOC-COOL-maintainability: RAMS allocation/target review open; ALLOC-COOL-maintainability: RAMS target undetermined; Allocated operating conditions do not match the analysed conditions.; RAMS target, comparison or allocation review remains open.; Result uses assumptions, unknown events or an unvalidated model.; ALLOC-COOL-reliability: RAMS allocation/target review open; ALLOC-COOL-reliability: RAMS target undetermined; Allocated operating conditions do not match the analysed conditions.; RAMS target, comparison or allocation review remains open.; Result uses assumptions, unknown events or an unvalidated model.; ALLOC-COOL-safety: RAMS allocation/target review open; ALLOC-COOL-safety: RAMS target undetermined; Allocated operating conditions do not match the analysed conditions.; RAMS target, comparison or allocation review remains open.; Quantitative result is unresolved.; Result uses assumptions, unknown events or an unvalidated model.; Assurance graph DEC-LM3-CAR-A900-G1: EVD-COOL-BENCH: evidence planned (open); Assurance graph DEC-LM3-CAR-A900-G1: EVD-COOL-CONTROLLER: evidence planned (open); Assurance graph DEC-LM3-CAR-A900-G1: EVD-COOL-INTEGRATION: evidence planned (open); Assurance graph DEC-LM3-CAR-A900-G1: EVD-COOL-TORQUE: evidence planned (open); Assurance graph DEC-LM3-CAR-A900-G1: FM-COOL-VEHICLE: failure analysis and residual risk unreviewed (open); Assurance graph DEC-LM3-CAR-A900-G1: FM-COOL-VEHICLE: failure analysis binds an earlier design revision (open); Assurance graph DEC-LM3-CAR-A900-G1: FM-COOL-VEHICLE: mandatory safety review requires a controlled decision (open); Assurance graph DEC-LM3-CAR-A900-G1: FM-COOL-VEHICLE: occurrence evidence unknown (open); Assurance graph DEC-LM3-CAR-A900-G1: FM-COOL-VEHICLE: operating context verification open: charging, degraded-operation, evacuation, maintenance, normal-running, shutdown, startup (open); Assurance graph DEC-LM3-CAR-A900-G1: REQ-COOL-RECOVERY: acceptance criteria require independent engineering review (open); Assurance graph DEC-LM3-CAR-A900-G1: child/dependency decision open: LM3-AUX-P010/G1 (open); Assurance graph DEC-LM3-CAR-A900-G1: child/dependency decision open: LM3-BOG-SA610/G1 (open); Assurance graph DEC-LM3-CAR-A900-G1: child/dependency decision open: LM3-BOG-SA620/G1 (open); Assurance graph DEC-LM3-CAR-A900-G1: child/dependency decision open: LM3-DOOR-SA310/G1 (open); Assurance graph DEC-LM3-CAR-A900-G1: child/dependency decision open: LM3-FIX-SA340/G1 (open); Assurance graph DEC-LM3-CAR-A900-G1: child/dependency decision open: LM3-HV-SA510/G1 (open); Assurance graph DEC-LM3-CAR-A900-G1: child/dependency decision open: LM3-INT-SA330/G1 (open); Assurance graph DEC-LM3-CAR-A900-G1: child/dependency decision open: LM3-LGT-SA350/G1 (open); Assurance graph DEC-LM3-CAR-A900-G1: child/dependency decision open: LM3-ROOF-SA410/G1 (open); Assurance graph DEC-LM3-CAR-A900-G1: child/dependency decision open: LM3-SHELL-A200/G1 (open); Assurance graph DEC-LM3-CAR-A900-G1: controlled deployment acceptance is not authenticated (open); Assurance graph DEC-LM3-CAR-A900-G1: decision open (open); Assurance graph DEC-LM3-CAR-A900-G1: stale design revision: LM3-CAR-A900 (open); Assurance graph DEC-LM3-CAR-A900-G1: stale design revision: LM3-TRAINSET-A000 (open); Assurance graph DEC-LM3-HV-SA510-G1: EVD-COOL-BENCH: evidence planned (open); Assurance graph DEC-LM3-HV-SA510-G1: EVD-COOL-CONTROLLER: evidence planned (open); Assurance graph DEC-LM3-HV-SA510-G1: EVD-COOL-INTEGRATION: evidence planned (open);

Decision area	State	Open findings
		Assurance graph DEC-LM3-HV-SA510-G1:EVD-COOL-TORQUE: evidence planned (open); Assurancegraph DEC-LM3-HV-SA510-G1: child/dependency decisionopen: LM3-BDY-P050/G1 (open); Assurance graphDEC-LM3-HV-SA510-G1: child/dependency decision open:LM3-HV-P010/G1 (open); Assurance graphDEC-LM3-HV-SA510-G1: child/dependency decision open:LM3-HV-P020/G1 (open); Assurance graphDEC-LM3-HV-SA510-G1: child/dependency decision open:LM3-HV-P030/G1 (open); Assurance graphDEC-LM3-HV-SA510-G1: child/dependency decision open:LM3-SAF-P010/G1 (open); Assurance graphDEC-LM3-HV-SA510-G1: child/dependency decision open:LM3-TRC-P030/G1 (open); Assurance graphDEC-LM3-HV-SA510-G1: child/dependency decision open:LM3-TRC-P040/G1 (open); Assurance graphDEC-LM3-HV-SA510-G1: child/dependency decision open:LM3-TRC-P060/G1 (open); Assurance graphDEC-LM3-HV-SA510-G1: child/dependency decision open:LM3-TRC-P070/G1 (open); Assurance graphDEC-LM3-HV-SA510-G1: controlled deploymentacceptance is not authenticated (open); Assurance graphDEC-LM3-HV-SA510-G1: decision open (open); Assurancegraph DEC-LM3-HV-SA510-G1: stale design revision:LM3-CAR-A900 (open); Assurance graphDEC-LM3-HV-SA510-G1: stale design revision:LM3-TRAINSET-A000 (open); Assurance graphDEC-LM3-TRAINSET-A000-G1: EVD-COOL-BENCH:evidence planned (open); Assurance graphDEC-LM3-TRAINSET-A000-G1: EVD-COOL-CONTROLLER:evidence planned (open); Assurance graphDEC-LM3-TRAINSET-A000-G1: EVD-COOL-INTEGRATION:evidence planned (open); Assurance graphDEC-LM3-TRAINSET-A000-G1: EVD-COOL-TORQUE:evidence planned (open); Assurance graphDEC-LM3-TRAINSET-A000-G1: REQ-COOL-RECOVERY:acceptance criteria require independent engineeringreview (open); Assurance graphDEC-LM3-TRAINSET-A000-G1: child/dependency decisionopen: LM3-ART-SA800/G1 (open); Assurance graphDEC-LM3-TRAINSET-A000-G1: child/dependency decisionopen: LM3-CAR-A900/G1 (open); Assurance graphDEC-LM3-TRAINSET-A000-G1: child/dependency decisionopen: LM3-EIF-SA650/G1 (open); Assurance graphDEC-LM3-TRAINSET-A000-G1: child/dependency decisionopen: LM3-END-SA700/G1 (open); Assurance graphDEC-LM3-TRAINSET-A000-G1: child/dependency decisionopen: LM3-SYS-SA900/G1 (open); Assurance graphDEC-LM3-TRAINSET-A000-G1: controlled deploymentacceptance is not authenticated (open); Assurance graphDEC-LM3-TRAINSET-A000-G1: decision open (open);Assurance graph DEC-LM3-TRAINSET-A000-G1: staledesign revision: LM3-CAR-A900 (open); Assurance graphDEC-LM3-TRAINSET-A000-G1: stale design revision:LM3-TRAINSET-A000 (open); Assurance graphDEC-OSR-COOL-LOOP-G1: EVD-COOL-BENCH: evidenceplanned (open); Assurance graphDEC-OSR-COOL-LOOP-G1: EVD-COOL-CONTROLLER:evidence planned (open); Assurance graphDEC-OSR-COOL-LOOP-G1: EVD-COOL-INTEGRATION:evidence planned (open); Assurance graphDEC-OSR-COOL-LOOP-G1: EVD-COOL-TORQUE: evidenceplanned (open); Assurance graphDEC-OSR-COOL-LOOP-G1: FM-COOL-LOOP: failureanalysis and residual risk unreviewed (open); Assurancegraph DEC-OSR-COOL-LOOP-G1: FM-COOL-LOOP:mandatory safety review requires a controlled decision(open); Assurance graph DEC-OSR-COOL-LOOP-G1:FM-COOL-LOOP: occurrence evidence unknown (open);Assurance graph DEC-OSR-COOL-LOOP-G1:FM-COOL-LOOP: operating context verification open:degraded-operation, evacuation, maintenance, recovery,shutdown, startup (open); Assurance graphDEC-OSR-COOL-LOOP-G1: REQ-COOL-PHYSICAL:acceptance criteria require independent engineeringreview (open); Assurance graphDEC-OSR-COOL-LOOP-G1: child/dependency decisionopen: LM3-AUX-P010/G1 (open); Assurance graphDEC-OSR-COOL-LOOP-G1: child/dependency decisionopen: OSR-COOL-CONNECTOR/G1 (open); Assurancegraph DEC-OSR-COOL-LOOP-G1: child/dependencydecision open: OSR-COOL-FLOW-SENSOR/G1 (open);Assurance graph DEC-OSR-COOL-LOOP-G1:child/dependency decision open: OSR-COOL-PUMP/G1(open); Assurance graph DEC-OSR-COOL-LOOP-G1:controlled deployment acceptance is not authenticated(open); Assurance graph DEC-OSR-COOL-LOOP-G1:decision open (open); Assurance graphDEC-OSR-COOL-LOOP-G1: stale design revision:LM3-CAR-A900 (open); Assurance graphDEC-OSR-COOL-LOOP-G1: stale design revision:LM3-TRAINSET-A000 (open); Assurance graphDEC-OSR-COOL-PUMP-G1: EVD-COOL-BENCH: evidenceplanned (open); Assurance graphDEC-OSR-COOL-PUMP-G1:

Decision area	State	Open findings
		EVD-COOL-CONTROLLER:evidence planned (open);Assurance graphDEC-OSR-COOL-PUMP-G1:EVD-COOL-INTEGRATION:evidence planned (open);Assurance graphDEC-OSR-COOL-PUMP-G1:EVD-COOL-TORQUE: evidenceplanned (open); AssurancegraphDEC-OSR-COOL-PUMP-G1: FM-COOL-PUMP:failureanalysis and residual risk unreviewed (open);Assurancegraph DEC-OSR-COOL-PUMP-G1:FM-COOL-PUMP:mandatory safety review requires acontrolled decision(open); Assurance graphDEC-OSR-COOL-PUMP-G1:FM-COOL-PUMP: occurrenceevidence unknown (open);Assurance graphDEC-OSR-COOL-PUMP-G1:FM-COOL-PUMP: operatingcontext verification open:charging, degraded-operation,evacuation, maintenance,recovery, shutdown, startup(open); Assurance graphDEC-OSR-COOL-PUMP-G1:REQ-COOL-PUMP: acceptancecriteria require independentengineering review (open);Assurance graphDEC-OSR-COOL-PUMP-G1:child/dependency decisionopen: LM3-AUX-P010/G1(open); Assurance graphDEC-OSR-COOL-PUMP-G1:controlled deploymentacceptance is not authenticated(open); Assurance graphDEC-OSR-COOL-PUMP-G1:decision open (open);Assurance graphDEC-OSR-COOL-PUMP-G1: stale designrevision:LM3-CAR-A900 (open); AssurancegraphDEC-OSR-COOL-PUMP-G1: stale designrevision:LM3-TRAINSET-A000 (open); AssurancegraphOBL-COOL-APPLICABILITY: deploymentapplicability andclause selection open (open); AssurancegraphREQ-COOL-HOT: acceptance criteria requireindependentengineering review (open); AssurancegraphREQ-COOL-LEAK: acceptance criteria requireindependentengineering review (open); AssurancegraphREQ-COOL-PHYSICAL: acceptance criteriarequireindependent engineering review (open); AssurancegraphREQ-COOL-PUMP: acceptance criteriarequireindependent engineering review (open); AssurancegraphREQ-COOL-RECOVERY: acceptance criteriarequireindependent engineering review (open); AssurancegraphREQ-COOL-TORQUE: acceptance criteriarequireindependent engineering review (open);Frozenengineering configuration is stale.;REQ-COOL-PHYSICAL:requirement review remains open;REQ-COOL-RECOVERY:requirement review remains open

Decision area	State	Open findings
design verification	blocked	Assurance graph CFG-DEMO-COOL-A: stale design revision: LM3-CAR-A900 (open); Assurance graph CFG-DEMO-COOL-A: stale design revision: LM3-TRAINSET-A000 (open); Assurance graph DEC-LM3-CAR-A900-G2: EVD-COOL-BENCH: evidence planned (open); Assurance graph DEC-LM3-CAR-A900-G2: EVD-COOL-CONTROLLER: evidence planned (open); Assurance graph DEC-LM3-CAR-A900-G2: EVD-COOL-INTEGRATION: evidence planned (open); Assurance graph DEC-LM3-CAR-A900-G2: EVD-COOL-TORQUE: evidence planned (open); Assurance graph DEC-LM3-CAR-A900-G2: FM-COOL-VEHICLE: failure analysis and residual risk unreviewed (open); Assurance graph DEC-LM3-CAR-A900-G2: FM-COOL-VEHICLE: failure analysis binds an earlier design revision (open); Assurance graph DEC-LM3-CAR-A900-G2: FM-COOL-VEHICLE: mandatory safety review requires a controlled decision (open); Assurance graph DEC-LM3-CAR-A900-G2: FM-COOL-VEHICLE: occurrence evidence unknown (open); Assurance graph DEC-LM3-CAR-A900-G2: FM-COOL-VEHICLE: operating context verification open: charging, degraded-operation, evacuation, maintenance, normal-running, shutdown, startup (open); Assurance graph DEC-LM3-CAR-A900-G2: REQ-COOL-RECOVERY: acceptance criteria require independent engineering review (open); Assurance graph DEC-LM3-CAR-A900-G2: child/dependency decision open: LM3-AUX-P010/G2 (open); Assurance graph DEC-LM3-CAR-A900-G2: child/dependency decision open: LM3-BOG-SA610/G2 (open); Assurance graph DEC-LM3-CAR-A900-G2: child/dependency decision open: LM3-BOG-SA620/G2 (open); Assurance graph DEC-LM3-CAR-A900-G2: child/dependency decision open: LM3-DOOR-SA310/G2 (open); Assurance graph DEC-LM3-CAR-A900-G2: child/dependency decision open: LM3-FIX-SA340/G2 (open); Assurance graph DEC-LM3-CAR-A900-G2: child/dependency decision open: LM3-HV-SA510/G2 (open); Assurance graph DEC-LM3-CAR-A900-G2: child/dependency decision open: LM3-INT-SA330/G2 (open); Assurance graph DEC-LM3-CAR-A900-G2: child/dependency decision open: LM3-LGT-SA350/G2 (open); Assurance graph DEC-LM3-CAR-A900-G2: child/dependency decision open: LM3-ROOF-SA410/G2 (open); Assurance graph DEC-LM3-CAR-A900-G2: child/dependency decision open: LM3-SHELL-A200/G2 (open); Assurance graph DEC-LM3-CAR-A900-G2: controlled deployment acceptance is not authenticated (open); Assurance graph DEC-LM3-CAR-A900-G2: decision open (open); Assurance graph DEC-LM3-CAR-A900-G2: prior gate decision open:

Decision area	State	Open findings
		G1 (open); Assurance graph DEC-LM3-CAR-A900-G2:stale design revision: LM3-CAR-A900 (open); Assurancegraph DEC-LM3-CAR-A900-G2: stale design revision:LM3-TRAINSET-A000 (open); Assurance graphDEC-LM3-HV-SA510-G2: EVD-COOL-BENCH: evidenceplanned (open); Assurance graph DEC-LM3-HV-SA510-G2:EVD-COOL-CONTROLLER: evidence planned (open);Assurance graph DEC-LM3-HV-SA510-G2:EVD-COOL-INTEGRATION: evidence planned (open);Assurance graph DEC-LM3-HV-SA510-G2:EVD-COOL-TORQUE: evidence planned (open); Assurancegraph DEC-LM3-HV-SA510-G2: child/dependency decisionopen: LM3-BDY-P050/G2 (open); Assurance graphDEC-LM3-HV-SA510-G2: child/dependency decision open:LM3-HV-P010/G2 (open); Assurance graphDEC-LM3-HV-SA510-G2: child/dependency decision open:LM3-HV-P020/G2 (open); Assurance graphDEC-LM3-HV-SA510-G2: child/dependency decision open:LM3-HV-P030/G2 (open); Assurance graphDEC-LM3-HV-SA510-G2: child/dependency decision open:LM3-SAF-P010/G2 (open); Assurance graphDEC-LM3-HV-SA510-G2: child/dependency decision open:LM3-TRC-P030/G2 (open); Assurance graphDEC-LM3-HV-SA510-G2: child/dependency decision open:LM3-TRC-P040/G2 (open); Assurance graphDEC-LM3-HV-SA510-G2: child/dependency decision open:LM3-TRC-P060/G2 (open); Assurance graphDEC-LM3-HV-SA510-G2: child/dependency decision open:LM3-TRC-P070/G2 (open); Assurance graphDEC-LM3-HV-SA510-G2: controlled deploymentacceptance is not authenticated (open); Assurance graphDEC-LM3-HV-SA510-G2: decision open (open); Assurancegraph DEC-LM3-HV-SA510-G2: prior gate decision open:G1 (open); Assurance graph DEC-LM3-HV-SA510-G2:stale design revision: LM3-CAR-A900 (open); Assurancegraph DEC-LM3-HV-SA510-G2: stale design revision:LM3-TRAINSET-A000 (open); Assurance graphDEC-LM3-TRAINSET-A000-G2: EVD-COOL-BENCH:evidence planned (open); Assurance graphDEC-LM3-TRAINSET-A000-G2: EVD-COOL-CONTROLLER:evidence planned (open); Assurance graphDEC-LM3-TRAINSET-A000-G2: EVD-COOL-INTEGRATION:evidence planned (open); Assurance graphDEC-LM3-TRAINSET-A000-G2: EVD-COOL-TORQUE:evidence planned (open); Assurance graphDEC-LM3-TRAINSET-A000-G2: REQ-COOL-RECOVERY:acceptance criteria require independent engineeringreview (open); Assurance graphDEC-LM3-TRAINSET-A000-G2: child/dependency decisionopen: LM3-ART-SA800/G2 (open); Assurance graphDEC-LM3-TRAINSET-A000-G2: child/dependency decisionopen: LM3-CAR-A900/G2 (open); Assurance graphDEC-LM3-TRAINSET-A000-G2: child/dependency decisionopen: LM3-EIF-SA650/G2 (open); Assurance graphDEC-LM3-TRAINSET-A000-G2: child/dependency decisionopen: LM3-END-SA700/G2 (open); Assurance graphDEC-LM3-TRAINSET-A000-G2: child/dependency decisionopen: LM3-SYS-SA900/G2 (open); Assurance graphDEC-LM3-TRAINSET-A000-G2: controlled deploymentacceptance is not authenticated (open); Assurance graphDEC-LM3-TRAINSET-A000-G2: decision open (open);Assurance graph DEC-LM3-TRAINSET-A000-G2: prior gatedecision open: G1 (open); Assurance graphDEC-LM3-TRAINSET-A000-G2: stale design revision:LM3-CAR-A900 (open); Assurance graphDEC-LM3-TRAINSET-A000-G2: stale design revision:LM3-TRAINSET-A000 (open); Assurance graphDEC-OSR-COOL-LOOP-G2: EVD-COOL-BENCH: evidenceplanned (open); Assurance graphDEC-OSR-COOL-LOOP-G2: EVD-COOL-CONTROLLER:evidence planned (open); Assurance graphDEC-OSR-COOL-LOOP-G2: EVD-COOL-INTEGRATION:evidence planned (open); Assurance graphDEC-OSR-COOL-LOOP-G2: EVD-COOL-TORQUE: evidenceplanned (open); Assurance graphDEC-OSR-COOL-LOOP-G2: FM-COOL-LOOP: failureanalysis and residual risk unreviewed (open); Assurancegraph DEC-OSR-COOL-LOOP-G2: FM-COOL-LOOP:mandatory safety review requires a controlled decision(open); Assurance graph DEC-OSR-COOL-LOOP-G2:FM-COOL-LOOP: occurrence evidence unknown (open);Assurance graph DEC-OSR-COOL-LOOP-G2:FM-COOL-LOOP: operating context verification open:degraded-operation, evacuation, maintenance, recovery,shutdown, startup (open); Assurance graphDEC-OSR-COOL-LOOP-G2: REQ-COOL-PHYSICAL:acceptance criteria require independent engineeringreview (open); Assurance graphDEC-OSR-COOL-LOOP-G2: child/dependency decisionopen: LM3-AUX-P010/G2 (open); Assurance graphDEC-OSR-COOL-LOOP-G2: child/dependency decisionopen: OSR-COOL-CONNECTOR/G2 (open); Assurancegraph DEC-OSR-COOL-LOOP-G2: child/dependencydecision open: OSR-COOL-FLOW-SENSOR/G2 (open);Assurance graph

Decision area	State	Open findings
		DEC-OSR-COOL-LOOP-G2:child/dependency decisionopen: OSR-COOL-PUMP/G2(open); Assurance graphDEC-OSR-COOL-LOOP-G2:controlled deploymentacceptance is not authenticated(open); Assurance graphDEC-OSR-COOL-LOOP-G2:decision open (open);Assurance graphDEC-OSR-COOL-LOOP-G2: prior gatedecision open: G1(open); Assurance graphDEC-OSR-COOL-LOOP-G2: staledesign revision:LM3-CAR-A900 (open); AssurancegraphDEC-OSR-COOL-LOOP-G2: stale designrevision:LM3-TRAINSET-A000 (open); AssurancegraphDEC-OSR-COOL-PUMP-G2: EVD-COOL-BENCH:evidenceplanned (open); AssurancegraphDEC-OSR-COOL-PUMP-G2:EVD-COOL-CONTROLLER:evidence planned (open);Assurance graphDEC-OSR-COOL-PUMP-G2:EVD-COOL-INTEGRATION: evidence planned (open);Assurance graphDEC-OSR-COOL-PUMP-G2:EVD-COOL-TORQUE: evidenceplanned (open); AssurancegraphDEC-OSR-COOL-PUMP-G2: FM-COOL-PUMP:failureanalysis and residual risk unreviewed (open);Assurancegraph DEC-OSR-COOL-PUMP-G2:FM-COOL-PUMP:mandatory safety review requires acontrolled decision(open); Assurance graphDEC-OSR-COOL-PUMP-G2:FM-COOL-PUMP: occurrenceevidence unknown (open);Assurance graphDEC-OSR-COOL-PUMP-G2:FM-COOL-PUMP: operatingcontext verification open:charging, degraded-operation,evacuation, maintenance,recovery, shutdown, startup(open); Assurance graphDEC-OSR-COOL-PUMP-G2:REQ-COOL-PUMP: acceptancecriteria require independentengineering review (open);Assurance graphDEC-OSR-COOL-PUMP-G2:child/dependency decisionopen: LM3-AUX-P010/G2(open); Assurance graphDEC-OSR-COOL-PUMP-G2:controlled deploymentacceptance is not authenticated(open); Assurance graphDEC-OSR-COOL-PUMP-G2:decision open (open);Assurance graphDEC-OSR-COOL-PUMP-G2: prior gatedecision open: G1(open); Assurance graphDEC-OSR-COOL-PUMP-G2: staledesign revision:LM3-CAR-A900 (open); AssurancegraphDEC-OSR-COOL-PUMP-G2: stale designrevision:LM3-TRAINSET-A000 (open); AssurancegraphEVD-COOL-CONTROLLER: evidence planned(open);Assurance graph FM-COOL-CONNECTOR: failureanalysisand residual risk unreviewed (open); AssurancegraphFM-COOL-CONNECTOR: mandatory safety reviewrequiresa controlled decision (open); AssurancegraphFM-COOL-CONNECTOR: mitigation lacks averificationscenario: REQ-COOL-TORQUE (open);Assurance graphFM-COOL-CONNECTOR: occurrenceevidence unknown(open); Assurance graphFM-COOL-CONNECTOR:operating context verificationopen: charging,degraded-operation, evacuation,maintenance, recovery,shutdown, startup (open);Assurance graphFM-COOL-LOOP: failure analysis andresidual riskunreviewed (open); Assurance graphFM-COOL-LOOP:mandatory safety review requires acontrolled decision(open); Assurance graphFM-COOL-LOOP: occurrenceevidence unknown (open);Assurance graphFM-COOL-LOOP: operating contextverification open:degraded-operation, evacuation,maintenance, recovery,shutdown, startup (open);Assurance graphFM-COOL-PACK: failure analysis andresidual riskunreviewed (open); Assurance graphFM-COOL-PACK:mandatory safety review requires acontrolled decision(open); Assurance graphFM-COOL-PACK: occurrenceevidence unknown (open);Assurance graphFM-COOL-PACK: operating contextverification open:charging, degraded-operation,evacuation, maintenance,recovery, shutdown, startup(open); Assurance graphFM-COOL-PUMP: failure analysisand residual riskunreviewed (open); Assurance graphFM-COOL-PUMP:mandatory safety review requires acontrolled decision(open); Assurance graphFM-COOL-PUMP: occurrenceevidence unknown (open);Assurance graphFM-COOL-PUMP: operating contextverification open:charging, degraded-operation,evacuation, maintenance,recovery, shutdown, startup(open); Assurance graphFM-COOL-RAILWAY: failureanalysis and residual riskunreviewed (open); Assurancegraph FM-COOL-RAILWAY:mandatory safety reviewrequires a controlled decision(open); Assurance graphFM-COOL-RAILWAY: occurrenceevidence unknown(open); Assurance graphFM-COOL-RAILWAY: operatingcontext verification open:charging, degraded-operation,evacuation, maintenance,normal-running, shutdown,startup (open); Assurancegraph FM-COOL-SENSOR:failure analysis and residualrisk unreviewed (open);Assurance graphFM-COOL-SENSOR: mandatory safetyreview requires acontrolled decision (open); AssurancegraphFM-COOL-SENSOR: occurrence evidence unknown(open);Assurance graph FM-COOL-SENSOR: operatingcontextverification open:

Decision area	State	Open findings
		charging,degraded-operation,evacuation, maintenance,recovery,shutdown, startup(open); AssurancegraphFM-COOL-TORQUE: failureanalysis and residualriskunreviewed (open); AssurancegraphFM-COOL-TORQUE:mandatory safety reviewrequires acontrolled decision(open); AssurancegraphFM-COOL-TORQUE: mitigationlacks averificationscenario: REQ-COOL-TORQUE(open);Assuranc e graphFM-COOL-TORQUE: occurrenceevidenceunknown (open);Assurance graphFM-COOL-TORQUE:operating contextverification open:charging,degraded-operation,eva cuation, maintenance,recovery,shutdown, startup(open); AssurancegraphFM-COOL-VEHICLE: failureanalysis and residualriskunreviewed (open); AssurancegraphFM-COOL-VEHICLE:failure analysis binds an earlierdesignrevision (open);Assurance graphFM-COOL-VEHICLE:mandatory safetyreview requires acontrolled decision(open); AssurancegraphFM-COOL-VEHICLE: occurrenceevidenceunknown(open); Assurance graphFM-COOL-VEHICLE:operatingcontext verification open:charging,degraded-operation,evacuation,maintenanc e,normal-running, shutdown,startup(open);CCA-COOL-AUX: common-causeassessmentopen;CCA-COOL-CONFIG: common-causeassessmentopen;Calibrated model and held-outvalidation reviewremainopen.; Frozen engineeringconfiguration is stale.;RIG-AUX:model correlationtolerance unresolved;RIG-CHARGING:model correlationtolerance unresolved;RIG-LEAK: modelcorrelationtolerance unresolved;RIG-MASKED: modelcorrelationtolerance unresolved;RIG-NORMAL: modelcorrelationtolerance unresolved;RIG-PUMP: modelcorrelationtolerance unresolved

Decision area	State	Open findings
physical qualification	blocked	Assurance graph DEC-LM3-CAR-A900-G3: EVD-COOL-BENCH: evidence planned (open); Assurance graph DEC-LM3-CAR-A900-G3: EVD-COOL-CONTROLLER: evidence planned (open); Assurance graph DEC-LM3-CAR-A900-G3: EVD-COOL-INTEGRATION: evidence planned (open); Assurance graph DEC-LM3-CAR-A900-G3: EVD-COOL-TORQUE: evidence planned (open); Assurance graph DEC-LM3-CAR-A900-G3: FM-COOL-VEHICLE: failure analysis and residual risk unreviewed (open); Assurance graph DEC-LM3-CAR-A900-G3: FM-COOL-VEHICLE: failure analysis binds an earlier design revision (open); Assurance graph DEC-LM3-CAR-A900-G3: FM-COOL-VEHICLE: mandatory safety review requires a controlled decision (open); Assurance graph DEC-LM3-CAR-A900-G3: FM-COOL-VEHICLE: occurrence evidence unknown (open); Assurance graph DEC-LM3-CAR-A900-G3: FM-COOL-VEHICLE: operating context verification open: charging, degraded-operation, evacuation, maintenance, normal-running, shutdown, startup (open); Assurance graph DEC-LM3-CAR-A900-G3: REQ-COOL-RECOVERY: acceptance criteria require independent engineering review (open); Assurance graph DEC-LM3-CAR-A900-G3: child/dependency decision open: LM3-AUX-P010/G3 (open); Assurance graph DEC-LM3-CAR-A900-G3: child/dependency decision open: LM3-BOG-SA610/G3 (open); Assurance graph DEC-LM3-CAR-A900-G3: child/dependency decision open: LM3-BOG-SA620/G3 (open); Assurance graph DEC-LM3-CAR-A900-G3: child/dependency decision open: LM3-DOOR-SA310/G3 (open); Assurance graph DEC-LM3-CAR-A900-G3: child/dependency decision open: LM3-FIX-SA340/G3 (open); Assurance graph DEC-LM3-CAR-A900-G3: child/dependency decision open: LM3-HV-SA510/G3 (open); Assurance graph DEC-LM3-CAR-A900-G3: child/dependency decision open: LM3-INT-SA330/G3 (open); Assurance graph DEC-LM3-CAR-A900-G3: child/dependency decision open: LM3-LGT-SA350/G3 (open); Assurance graph DEC-LM3-CAR-A900-G3: child/dependency decision open: LM3-ROOF-SA410/G3 (open); Assurance graph DEC-LM3-CAR-A900-G3: child/dependency decision open: LM3-SHELL-A200/G3 (open); Assurance graph DEC-LM3-CAR-A900-G3: controlled deployment acceptance is not authenticated (open); Assurance graph DEC-LM3-CAR-A900-G3: decision open (open); Assurance graph DEC-LM3-CAR-A900-G3: prior gate decision open: G2 (open); Assurance graph DEC-LM3-CAR-A900-G3: reviewed physical integration evidence missing (open); Assurance graph DEC-LM3-CAR-A900-G3: stale design revision: LM3-CAR-A900 (open); Assurance graph DEC-LM3-CAR-A900-G3: stale design revision: LM3-TRAINSET-A000 (open); Assurance graph DEC-LM3-HV-SA510-G3: EVD-COOL-BENCH: evidence planned (open); Assurance graph DEC-LM3-HV-SA510-G3: EVD-COOL-CONTROLLER: evidence planned (open); Assurance graph DEC-LM3-HV-SA510-G3: EVD-COOL-INTEGRATION: evidence planned (open); Assurance graph DEC-LM3-HV-SA510-G3: EVD-COOL-TORQUE: evidence planned (open); Assurance graph DEC-LM3-HV-SA510-G3: child/dependency decision

Decision area	State	Open findings
		open: LM3-BDY-P050/G3 (open); Assurance graphDEC-LM3-HV-SA510-G3: child/dependency decision open:LM3-HV-P010/G3 (open); Assurance graphDEC-LM3-HV-SA510-G3: child/dependency decision open:LM3-HV-P020/G3 (open); Assurance graphDEC-LM3-HV-SA510-G3: child/dependency decision open:LM3-HV-P030/G3 (open); Assurance graphDEC-LM3-HV-SA510-G3: child/dependency decision open:LM3-SAF-P010/G3 (open); Assurance graphDEC-LM3-HV-SA510-G3: child/dependency decision open:LM3-TRC-P030/G3 (open); Assurance graphDEC-LM3-HV-SA510-G3: child/dependency decision open:LM3-TRC-P040/G3 (open); Assurance graphDEC-LM3-HV-SA510-G3: child/dependency decision open:LM3-TRC-P060/G3 (open); Assurance graphDEC-LM3-HV-SA510-G3: child/dependency decision open:LM3-TRC-P070/G3 (open); Assurance graphDEC-LM3-HV-SA510-G3: controlled deploymentacceptance is not authenticated (open); Assurance graphDEC-LM3-HV-SA510-G3: decision open (open); Assurancegraph DEC-LM3-HV-SA510-G3: prior gate decision open:G2 (open); Assurance graph DEC-LM3-HV-SA510-G3:reviewed physical integration evidence missing (open);Assurance graph DEC-LM3-HV-SA510-G3: stale designrevision: LM3-CAR-A900 (open); Assurance graphDEC-LM3-HV-SA510-G3: stale design revision:LM3-TRAINSET-A000 (open); Assurance graphDEC-LM3-TRAINSET-A000-G3: EVD-COOL-BENCH:evidence planned (open); Assurance graphDEC-LM3-TRAINSET-A000-G3: EVD-COOL-CONTROLLER:evidence planned (open); Assurance graphDEC-LM3-TRAINSET-A000-G3: EVD-COOL-INTEGRATION:evidence planned (open); Assurance graphDEC-LM3-TRAINSET-A000-G3: EVD-COOL-TORQUE:evidence planned (open); Assurance graphDEC-LM3-TRAINSET-A000-G3: REQ-COOL-RECOVERY:acceptance criteria require independent engineeringreview (open); Assurance graphDEC-LM3-TRAINSET-A000-G3: child/dependency decisionopen: LM3-ART-SA800/G3 (open); Assurance graphDEC-LM3-TRAINSET-A000-G3: child/dependency decisionopen: LM3-CAR-A900/G3 (open); Assurance graphDEC-LM3-TRAINSET-A000-G3: child/dependency decisionopen: LM3-EIF-SA650/G3 (open); Assurance graphDEC-LM3-TRAINSET-A000-G3: child/dependency decisionopen: LM3-END-SA700/G3 (open); Assurance graphDEC-LM3-TRAINSET-A000-G3: child/dependency decisionopen: LM3-SYS-SA900/G3 (open); Assurance graphDEC-LM3-TRAINSET-A000-G3: controlled deploymentacceptance is not authenticated (open); Assurance graphDEC-LM3-TRAINSET-A000-G3: decision open (open);Assurance graph DEC-LM3-TRAINSET-A000-G3: prior gatedecision open: G2 (open); Assurance graphDEC-LM3-TRAINSET-A000-G3: reviewed physicalintegration evidence missing (open); Assurance graphDEC-LM3-TRAINSET-A000-G3: stale design revision:LM3-CAR-A900 (open); Assurance graphDEC-LM3-TRAINSET-A000-G3: stale design revision:LM3-TRAINSET-A000 (open); Assurance graphDEC-OSR-COOL-LOOP-G3: EVD-COOL-BENCH: evidenceplanned (open); Assurance graphDEC-OSR-COOL-LOOP-G3: EVD-COOL-CONTROLLER:evidence planned (open); Assurance graphDEC-OSR-COOL-LOOP-G3: EVD-COOL-INTEGRATION:evidence planned (open); Assurance graphDEC-OSR-COOL-LOOP-G3: EVD-COOL-TORQUE: evidenceplanned (open); Assurance graphDEC-OSR-COOL-LOOP-G3: FM-COOL-LOOP: failureanalysis and residual risk unreviewed (open); Assurancegraph DEC-OSR-COOL-LOOP-G3: FM-COOL-LOOP:mandatory safety review requires a controlled decision(open); Assurance graph DEC-OSR-COOL-LOOP-G3:FM-COOL-LOOP: occurrence evidence unknown (open);Assurance graph DEC-OSR-COOL-LOOP-G3:FM-COOL-LOOP: operating context verification open:degraded-operation, evacuation, maintenance, recovery,shutdown, startup (open); Assurance graphDEC-OSR-COOL-LOOP-G3: REQ-COOL-PHYSICAL:acceptance criteria require independent engineeringreview (open); Assurance graphDEC-OSR-COOL-LOOP-G3: child/dependency decisionopen: LM3-AUX-P010/G3 (open); Assurance graphDEC-OSR-COOL-LOOP-G3: child/dependency decisionopen: OSR-COOL-CONNECTOR/G3 (open); Assurancegraph DEC-OSR-COOL-LOOP-G3: child/dependencydecision open: OSR-COOL-FLOW-SENSOR/G3 (open);Assurance graph DEC-OSR-COOL-LOOP-G3:child/dependency decision open: OSR-COOL-PUMP/G3(open); Assurance graph DEC-OSR-COOL-LOOP-G3:controlled deployment acceptance is not authenticated(open); Assurance graph DEC-OSR-COOL-LOOP-G3:decision open (open); Assurance graphDEC-OSR-COOL-LOOP-G3: prior gate decision open: G2(open); Assurance graph DEC-OSR-COOL-LOOP-G3:reviewed physical integration

Decision area	State	Open findings
		evidence missing (open);Assurance graphDEC-OSR-COOL-LOOP-G3: stale designrevision:LM3-CAR-A900 (open); AssurancegraphDEC-OSR-COOL-LOOP-G3: stale designrevision:LM3-TRAINSET-A000 (open); AssurancegraphDEC-OSR-COOL-PUMP-G3: EVD-COOL-BENCH:evidenceplanned (open); AssurancegraphDEC-OSR-COOL-PUMP-G3:EVD-COOL-CONTROLLER:evidence planned (open);Assurance graphDEC-OSR-COOL-PUMP-G3:EVD-COOL-INTEGRATION: evidence planned (open);Assurance graphDEC-OSR-COOL-PUMP-G3:EVD-COOL-TORQUE: evidenceplanned (open); AssurancegraphDEC-OSR-COOL-PUMP-G3: FM-COOL-PUMP:failureanalysis and residual risk unreviewed (open);Assurancegraph DEC-OSR-COOL-PUMP-G3:FM-COOL-PUMP:mandatory safety review requires acontrolled decision(open); Assurance graphDEC-OSR-COOL-PUMP-G3:FM-COOL-PUMP: occurrenceevidence unknown (open);Assurance graphDEC-OSR-COOL-PUMP-G3:FM-COOL-PUMP: operatingcontext verification open:charging, degraded-operation,evacuation, maintenance,recovery, shutdown, startup(open); Assurance graphDEC-OSR-COOL-PUMP-G3:REQ-COOL-PUMP: acceptancecriteria require independentengineering review (open);Assurance graphDEC-OSR-COOL-PUMP-G3:child/dependency decisionopen: LM3-AUX-P010/G3(open); Assurance graphDEC-OSR-COOL-PUMP-G3:controlled deploymentacceptance is not authenticated(open); Assurance graphDEC-OSR-COOL-PUMP-G3:decision open (open);Assurance graphDEC-OSR-COOL-PUMP-G3: prior gatedecision open: G2(open); Assurance graphDEC-OSR-COOL-PUMP-G3:reviewed physical integrationevidence missing (open);Assurance graphDEC-OSR-COOL-PUMP-G3: stale designrevision:LM3-CAR-A900 (open); AssurancegraphDEC-OSR-COOL-PUMP-G3: stale designrevision:LM3-TRAINSET-A000 (open); AssurancegraphDEMO-CAR-01: design changed; installedapplicabilityrequires reassessment (open); AssurancegraphDEMO-CAR-02: design changed; installedapplicabilityrequires reassessment (open); AssurancegraphEVD-COOL-BENCH: evidence planned (open);Assurancegraph EVD-COOL-INTEGRATION: evidenceplanned(open); Assurance graph EVD-COOL-TORQUE:evidenceplanned (open); CC-COOL-FLOW: productionlimits andinspection release open; CC-COOL-PRESSURE:productionlimits and inspection release open;CC-COOL-SENSOR:production limits and inspectionrelease open;CC-COOL-TORQUE: production limits andinspectionrelease open; Controlled rig specimen serialsaremissing.; Fault injection, independent shutdown andrigprotection review remain open.; Frozenengineeringconfiguration is stale.; Nocontrolledproduction-equivalence records forserialisedassemblies.; OSR-COOL-PUMP: replacementcompatibilityor instructions open; RIG-AUX: acceptancelimits, durationor repeat count unresolved; RIG-AUX:representativephysical repetitions missing;RIG-CHARGING: acceptancelimits, duration or repeatcount unresolved;RIG-CHARGING: representative physicalrepetitionsmissing; RIG-LEAK: acceptance limits, durationor repeatcount unresolved; RIG-LEAK: representativephysicalrepetitions missing; RIG-MASKED: acceptancelimits,duration or repeat count unresolved;RIG-MASKED:representative physical repetitions missing;RIG-NORMAL:acceptance limits, duration or repeat countunresolved;RIG-NORMAL: representative physicalrepetitions missing;RIG-PUMP: acceptance limits,duration or repeat countunresolved; RIG-PUMP:representative physical repetitionsmissing;Representative rig duty and environmentalenvelopeunresolved.; Rig as-built configuration is notbound to thefrozen design.; Selected components andreal rig serialare missing.
Decision area	**State**	**Open findings**
integration	blocked	Assurance graph DEC-LM3-CAR-A900-G4: EVD-COOL-BENCH: evidence planned (open); Assurance graph DEC-LM3-CAR-A900-G4: EVD-COOL-CONTROLLER: evidence planned (open); Assurance graph DEC-LM3-CAR-A900-G4: EVD-COOL-INTEGRATION: evidence planned (open); Assurance graph DEC-LM3-CAR-A900-G4: EVD-COOL-TORQUE: evidence planned (open); Assurance graph DEC-LM3-CAR-A900-G4: FM-COOL-VEHICLE: failure analysis and residual risk unreviewed (open); Assurance graph DEC-LM3-CAR-A900-G4: FM-COOL-VEHICLE: failure analysis binds an earlier design revision (open); Assurance graph DEC-LM3-CAR-A900-G4: FM-COOL-VEHICLE: mandatory safety review requires a controlled decision (open); Assurance graph DEC-LM3-CAR-A900-G4: FM-COOL-VEHICLE: occurrence evidence unknown (open); Assurance graph DEC-LM3-CAR-A900-G4: FM-COOL-VEHICLE: operating

Decision area	State	Open findings
		context verification open: charging, degraded-operation,evacuation, maintenance, normal-running, shutdown,startup (open); Assurance graph DEC-LM3-CAR-A900-G4:REQ-COOL-RECOVERY: acceptance criteria requireindependent engineering review (open); Assurance graphDEC-LM3-CAR-A900-G4: child/dependency decision open:LM3-AUX-P010/G4 (open); Assurance graphDEC-LM3-CAR-A900-G4: child/dependency decision open:LM3-BOG-SA610/G4 (open); Assurance graphDEC-LM3-CAR-A900-G4: child/dependency decision open:LM3-BOG-SA620/G4 (open); Assurance graphDEC-LM3-CAR-A900-G4: child/dependency decision open:LM3-DOOR-SA310/G4 (open); Assurance graphDEC-LM3-CAR-A900-G4: child/dependency decision open:LM3-FIX-SA340/G4 (open); Assurance graphDEC-LM3-CAR-A900-G4: child/dependency decision open:LM3-HV-SA510/G4 (open); Assurance graphDEC-LM3-CAR-A900-G4: child/dependency decision open:LM3-INT-SA330/G4 (open); Assurance graphDEC-LM3-CAR-A900-G4: child/dependency decision open:LM3-LGT-SA350/G4 (open); Assurance graphDEC-LM3-CAR-A900-G4: child/dependency decision open:LM3-ROOF-SA410/G4 (open); Assurance graphDEC-LM3-CAR-A900-G4: child/dependency decision open:LM3-SHELL-A200/G4 (open); Assurance graphDEC-LM3-CAR-A900-G4: controlled deploymentacceptance is not authenticated (open); Assurance graphDEC-LM3-CAR-A900-G4: decision open (open); Assurancegraph DEC-LM3-CAR-A900-G4: prior gate decision open:G3 (open); Assurance graph DEC-LM3-CAR-A900-G4:reviewed physical integration evidence missing (open);Assurance graph DEC-LM3-CAR-A900-G4: stale designrevision: LM3-CAR-A900 (open); Assurance graphDEC-LM3-CAR-A900-G4: stale design revision:LM3-TRAINSET-A000 (open); Assurance graphDEC-LM3-HV-SA510-G4: EVD-COOL-BENCH: evidenceplanned (open); Assurance graph DEC-LM3-HV-SA510-G4:EVD-COOL-CONTROLLER: evidence planned (open);Assurance graph DEC-LM3-HV-SA510-G4:EVD-COOL-INTEGRATION: evidence planned (open);Assurance graph DEC-LM3-HV-SA510-G4:EVD-COOL-TORQUE: evidence planned (open); Assurancegraph DEC-LM3-HV-SA510-G4: child/dependency decisionopen: LM3-BDY-P050/G4 (open); Assurance graphDEC-LM3-HV-SA510-G4: child/dependency decision open:LM3-HV-P010/G4 (open); Assurance graphDEC-LM3-HV-SA510-G4: child/dependency decision open:LM3-HV-P020/G4 (open); Assurance graphDEC-LM3-HV-SA510-G4: child/dependency decision open:LM3-HV-P030/G4 (open); Assurance graphDEC-LM3-HV-SA510-G4: child/dependency decision open:LM3-SAF-P010/G4 (open); Assurance graphDEC-LM3-HV-SA510-G4: child/dependency decision open:LM3-TRC-P030/G4 (open); Assurance graphDEC-LM3-HV-SA510-G4: child/dependency decision open:LM3-TRC-P040/G4 (open); Assurance graphDEC-LM3-HV-SA510-G4: child/dependency decision open:LM3-TRC-P060/G4 (open); Assurance graphDEC-LM3-HV-SA510-G4: child/dependency decision open:LM3-TRC-P070/G4 (open); Assurance graphDEC-LM3-HV-SA510-G4: controlled deploymentacceptance is not authenticated (open); Assurance graphDEC-LM3-HV-SA510-G4: decision open (open); Assurancegraph DEC-LM3-HV-SA510-G4: prior gate decision open:G3 (open); Assurance graph DEC-LM3-HV-SA510-G4:reviewed physical integration evidence missing (open);Assurance graph DEC-LM3-HV-SA510-G4: stale designrevision: LM3-CAR-A900 (open); Assurance graphDEC-LM3-HV-SA510-G4: stale design revision:LM3-TRAINSET-A000 (open); Assurance graphDEC-LM3-TRAINSET-A000-G4: EVD-COOL-BENCH:evidence planned (open); Assurance graphDEC-LM3-TRAINSET-A000-G4: EVD-COOL-CONTROLLER:evidence planned (open); Assurance graphDEC-LM3-TRAINSET-A000-G4: EVD-COOL-INTEGRATION:evidence planned (open); Assurance graphDEC-LM3-TRAINSET-A000-G4: EVD-COOL-TORQUE:evidence planned (open); Assurance graphDEC-LM3-TRAINSET-A000-G4: REQ-COOL-RECOVERY:acceptance criteria require independent engineeringreview (open); Assurance graphDEC-LM3-TRAINSET-A000-G4: child/dependency decisionopen: LM3-ART-SA800/G4 (open); Assurance graphDEC-LM3-TRAINSET-A000-G4: child/dependency decisionopen: LM3-CAR-A900/G4 (open); Assurance graphDEC-LM3-TRAINSET-A000-G4: child/dependency decisionopen: LM3-EIF-SA650/G4 (open); Assurance graphDEC-LM3-TRAINSET-A000-G4: child/dependency decisionopen: LM3-END-SA700/G4 (open); Assurance graphDEC-LM3-TRAINSET-A000-G4: child/dependency decisionopen: LM3-SYS-SA900/G4 (open); Assurance graphDEC-LM3-TRAINSET-A000-G4: controlled deploymentacceptance is not authenticated (open); Assurance graphDEC-LM3-TRAINSET-A000-G4: decision

Decision area	State	Open findings
		open (open);Assurance graphDEC-LM3-TRAINSET-A000-G4: prior gatedecision open:G3 (open); Assurance graphDEC-LM3-TRAINSET-A000-G4:reviewed physicalintegration evidence missing (open);Assurance graphDEC-LM3-TRAINSET-A000-G4: staledesign revision:LM3-CAR-A900 (open); AssurancegraphDEC-LM3-TRAINSET-A000-G4: stale designrevision:LM3-TRAINSET-A000 (open); AssurancegraphDEC-OSR-COOL-LOOP-G4: EVD-COOL-BENCH:evidenceplanned (open); AssurancegraphDEC-OSR-COOL-LOOP-G4:EVD-COOL-CONTROLLER:evidence planned (open);Assurance graphDEC-OSR-COOL-LOOP-G4:EVD-COOL-INTEGRATION: evidence planned (open);Assurance graphDEC-OSR-COOL-LOOP-G4:EVD-COOL-TORQUE: evidenceplanned (open); AssurancegraphDEC-OSR-COOL-LOOP-G4: FM-COOL-LOOP:failureanalysis and residual risk unreviewed (open);Assurancegraph DEC-OSR-COOL-LOOP-G4:FM-COOL-LOOP:mandatory safety review requires acontrolled decision(open); Assurance graphDEC-OSR-COOL-LOOP-G4:FM-COOL-LOOP: occurrenceevidence unknown (open);Assurance graphDEC-OSR-COOL-LOOP-G4:FM-COOL-LOOP: operatingcontext verification open:degraded-operation, evacuation,maintenance, recovery,shutdown, startup (open);Assurance graphDEC-OSR-COOL-LOOP-G4:REQ-COOL-PHYSICAL:acceptance criteria requireindependent engineeringreview (open); AssurancegraphDEC-OSR-COOL-LOOP-G4: child/dependencydecisionopen: LM3-AUX-P010/G4 (open); AssurancegraphDEC-OSR-COOL-LOOP-G4: child/dependencydecisionopen: OSR-COOL-CONNECTOR/G4 (open);Assurancegraph DEC-OSR-COOL-LOOP-G4:child/dependencydecision open:OSR-COOL-FLOW-SENSOR/G4 (open);Assurance graphDEC-OSR-COOL-LOOP-G4:child/dependency decisionopen: OSR-COOL-PUMP/G4(open); Assurance graphDEC-OSR-COOL-LOOP-G4:controlled deploymentacceptance is not authenticated(open); Assurance graphDEC-OSR-COOL-LOOP-G4:decision open (open);Assurance graphDEC-OSR-COOL-LOOP-G4: prior gatedecision open: G3(open); Assurance graphDEC-OSR-COOL-LOOP-G4:reviewed physical integrationevidence missing (open);Assurance graphDEC-OSR-COOL-LOOP-G4: stale designrevision:LM3-CAR-A900 (open); AssurancegraphDEC-OSR-COOL-LOOP-G4: stale designrevision:LM3-TRAINSET-A000 (open); AssurancegraphDEC-OSR-COOL-PUMP-G4: EVD-COOL-BENCH:evidenceplanned (open); AssurancegraphDEC-OSR-COOL-PUMP-G4:EVD-COOL-CONTROLLER:evidence planned (open);Assurance graphDEC-OSR-COOL-PUMP-G4:EVD-COOL-INTEGRATION: evidence planned (open);Assurance graphDEC-OSR-COOL-PUMP-G4:EVD-COOL-TORQUE: evidenceplanned (open); AssurancegraphDEC-OSR-COOL-PUMP-G4: FM-COOL-PUMP:failureanalysis and residual risk unreviewed (open);Assurancegraph DEC-OSR-COOL-PUMP-G4:FM-COOL-PUMP:mandatory safety review requires acontrolled decision(open); Assurance graphDEC-OSR-COOL-PUMP-G4:FM-COOL-PUMP: occurrenceevidence unknown (open);Assurance graphDEC-OSR-COOL-PUMP-G4:FM-COOL-PUMP: operatingcontext verification open:charging, degraded-operation,evacuation, maintenance,recovery, shutdown, startup(open); Assurance graphDEC-OSR-COOL-PUMP-G4:REQ-COOL-PUMP: acceptancecriteria require independentengineering review (open);Assurance graphDEC-OSR-COOL-PUMP-G4:child/dependency decisionopen: LM3-AUX-P010/G4(open); Assurance graphDEC-OSR-COOL-PUMP-G4:controlled deploymentacceptance is not authenticated(open); Assurance graphDEC-OSR-COOL-PUMP-G4:decision open (open);Assurance graphDEC-OSR-COOL-PUMP-G4: prior gatedecision open: G3(open); Assurance graphDEC-OSR-COOL-PUMP-G4:reviewed physical integrationevidence missing (open);Assurance graphDEC-OSR-COOL-PUMP-G4: stale designrevision:LM3-CAR-A900 (open); AssurancegraphDEC-OSR-COOL-PUMP-G4: stale designrevision:LM3-TRAINSET-A000 (open); Frozenengineeringconfiguration is stale.; hil: integrationevidence absent.;infrastructure: integration evidenceabsent.; operations:integration evidence absent.; vehicle:integration evidenceabsent.
independent review	blocked	Authenticated assessor decision missing.; Authenticated design-authority decision missing.; Frozen engineering configuration is stale.; Independent review policy and signed decisions are not supplied.; Operator/assessor assessment approach is not agreed.

Decision area	State	Open findings
operating conditions	blocked	Assurance graph DEMO-FRACAS-01: engineering handback and effectiveness review open (open); Frozen

Decision area	State	Open findings
		engineering configuration is stale.; Operating envelope,restrictions and handback conditions are unresolved.;Reference configuration has no deployment-specificaccepted use.

RAMS allocation outcomes

Allocation / metric	Target	Result bounds	Outcome
ALLOC-COOL-reliability / pump_failure_rate	<= None 1/h	[1e-05, 5e-05]	undetermined — Allocated operating conditions do not match the analysed conditions.; RAMS target, comparison or allocation review remains open.; Result uses assumptions, unknown events or an unvalidated model.
ALLOC-COOL-availability / function_availability	>= None probability	[0.9996700950723632, 0.9999979999740007]	undetermined — Allocated operating conditions do not match the analysed conditions.; RAMS target, comparison or allocation review remains open.; Result uses assumptions, unknown events or an unvalidated model.
ALLOC-COOL-maintainability / pump_mean_repair_time	<= None h	[2.0, 6.0]	undetermined — Allocated operating conditions do not match the analysed conditions.; RAMS target, comparison or allocation review remains open.; Result uses assumptions, unknown events or an unvalidated model.
ALLOC-COOL-safety / thermal_time_to_limit	>= None s	None	undetermined — Allocated operating conditions do not match the analysed conditions.; RAMS target, comparison or allocation review remains open.; Quantitative result is unresolved.; Result uses assumptions, unknown events or an unvalidated model.

Scoped graph findings

Graph `30a01593b4bc492341219fe29d27a6487032e4a9bead231147e8c48c544b913a`; 79 scoped records. Every scoped blocker requires closure or an authenticated, baseline-bound exclusion.

Record	Finding	Disposition
CFG-DEMO-COOL-A	stale design revision: LM3-CAR-A900	open
CFG-DEMO-COOL-A	stale design revision: LM3-TRAINSET-A000	open
DEC-LM3-CAR-A900-G1	EVD-COOL-BENCH: evidence planned	open
DEC-LM3-CAR-A900-G1	EVD-COOL-CONTROLLER: evidence planned	open
DEC-LM3-CAR-A900-G1	EVD-COOL-INTEGRATION: evidence planned	open
DEC-LM3-CAR-A900-G1	EVD-COOL-TORQUE: evidence planned	open
DEC-LM3-CAR-A900-G1	FM-COOL-VEHICLE: failure analysis and residual risk unreviewed	open
DEC-LM3-CAR-A900-G1	FM-COOL-VEHICLE: failure analysis binds an earlier design revision	open
DEC-LM3-CAR-A900-G1	FM-COOL-VEHICLE: mandatory safety review requires a controlled decision	open
DEC-LM3-CAR-A900-G1	FM-COOL-VEHICLE: occurrence evidence unknown	open
DEC-LM3-CAR-A900-G1	FM-COOL-VEHICLE: operating context verification open: charging, degraded-operation, evacuation, maintenance, normal-running, shutdown, startup	open
DEC-LM3-CAR-A900-G1	REQ-COOL-RECOVERY: acceptance criteria require independent engineering review	open
DEC-LM3-CAR-A900-G1	child/dependency decision open: LM3-AUX-P010/G1	open
DEC-LM3-CAR-A900-G1	child/dependency decision open: LM3-BOG-SA610/G1	open
DEC-LM3-CAR-A900-G1	child/dependency decision open: LM3-BOG-SA620/G1	open
DEC-LM3-CAR-A900-G1	child/dependency decision open: LM3-DOOR-SA310/G1	open
DEC-LM3-CAR-A900-G1	child/dependency decision open: LM3-FIX-SA340/G1	open
DEC-LM3-CAR-A900-G1	child/dependency decision open: LM3-HV-SA510/G1	open
DEC-LM3-CAR-A900-G1	child/dependency decision open: LM3-INT-SA330/G1	open
DEC-LM3-CAR-A900-G1	child/dependency decision open: LM3-LGT-SA350/G1	open
DEC-LM3-CAR-A900-G1	child/dependency decision open: LM3-ROOF-SA410/G1	open
DEC-LM3-CAR-A900-G1	child/dependency decision open: LM3-SHELL-A200/G1	open
DEC-LM3-CAR-A900-G1	controlled deployment acceptance is not authenticated	open
DEC-LM3-CAR-A900-G1	decision open	open
DEC-LM3-CAR-A900-G1	stale design revision: LM3-CAR-A900	open
DEC-LM3-CAR-A900-G1	stale design revision: LM3-TRAINSET-A000	open
DEC-LM3-CAR-A900-G2	EVD-COOL-BENCH: evidence planned	open

Record	Finding	Disposition
DEC-LM3-CAR-A900-G2	EVD-COOL-CONTROLLER: evidence planned	open
DEC-LM3-CAR-A900-G2	EVD-COOL-INTEGRATION: evidence planned	open
DEC-LM3-CAR-A900-G2	EVD-COOL-TORQUE: evidence planned	open
DEC-LM3-CAR-A900-G2	FM-COOL-VEHICLE: failure analysis and residual risk unreviewed	open
DEC-LM3-CAR-A900-G2	FM-COOL-VEHICLE: failure analysis binds an earlier design revision	open
DEC-LM3-CAR-A900-G2	FM-COOL-VEHICLE: mandatory safety review requires a controlled decision	open
DEC-LM3-CAR-A900-G2	FM-COOL-VEHICLE: occurrence evidence unknown	open
DEC-LM3-CAR-A900-G2	FM-COOL-VEHICLE: operating context verification open: charging, degraded-operation, evacuation, maintenance, normal-running, shutdown, startup	open
DEC-LM3-CAR-A900-G2	REQ-COOL-RECOVERY: acceptance criteria require independent engineering review	open
DEC-LM3-CAR-A900-G2	child/dependency decision open: LM3-AUX-P010/G2	open
DEC-LM3-CAR-A900-G2	child/dependency decision open: LM3-BOG-SA610/G2	open
DEC-LM3-CAR-A900-G2	child/dependency decision open: LM3-BOG-SA620/G2	open
DEC-LM3-CAR-A900-G2	child/dependency decision open: LM3-DOOR-SA310/G2	open
DEC-LM3-CAR-A900-G2	child/dependency decision open: LM3-FIX-SA340/G2	open
DEC-LM3-CAR-A900-G2	child/dependency decision open: LM3-HV-SA510/G2	open
DEC-LM3-CAR-A900-G2	child/dependency decision open: LM3-INT-SA330/G2	open
DEC-LM3-CAR-A900-G2	child/dependency decision open: LM3-LGT-SA350/G2	open
DEC-LM3-CAR-A900-G2	child/dependency decision open: LM3-ROOF-SA410/G2	open
DEC-LM3-CAR-A900-G2	child/dependency decision open: LM3-SHELL-A200/G2	open
DEC-LM3-CAR-A900-G2	controlled deployment acceptance is not authenticated	open
DEC-LM3-CAR-A900-G2	decision open	open
DEC-LM3-CAR-A900-G2	prior gate decision open: G1	open
DEC-LM3-CAR-A900-G2	stale design revision: LM3-CAR-A900	open
DEC-LM3-CAR-A900-G2	stale design revision: LM3-TRAINSET-A000	open
DEC-LM3-CAR-A900-G3	EVD-COOL-BENCH: evidence planned	open
DEC-LM3-CAR-A900-G3	EVD-COOL-CONTROLLER: evidence planned	open
DEC-LM3-CAR-A900-G3	EVD-COOL-INTEGRATION: evidence planned	open
DEC-LM3-CAR-A900-G3	EVD-COOL-TORQUE: evidence planned	open
DEC-LM3-CAR-A900-G3	FM-COOL-VEHICLE: failure analysis and residual risk unreviewed	open
DEC-LM3-CAR-A900-G3	FM-COOL-VEHICLE: failure analysis binds an earlier design revision	open
DEC-LM3-CAR-A900-G3	FM-COOL-VEHICLE: mandatory safety review requires a controlled decision	open
DEC-LM3-CAR-A900-G3	FM-COOL-VEHICLE: occurrence evidence unknown	open
DEC-LM3-CAR-A900-G3	FM-COOL-VEHICLE: operating context verification open: charging, degraded-operation, evacuation, maintenance, normal-running, shutdown, startup	open
DEC-LM3-CAR-A900-G3	REQ-COOL-RECOVERY: acceptance criteria require independent engineering review	open
DEC-LM3-CAR-A900-G3	child/dependency decision open: LM3-AUX-P010/G3	open
DEC-LM3-CAR-A900-G3	child/dependency decision open: LM3-BOG-SA610/G3	open
DEC-LM3-CAR-A900-G3	child/dependency decision open: LM3-BOG-SA620/G3	open
DEC-LM3-CAR-A900-G3	child/dependency decision open: LM3-DOOR-SA310/G3	open
DEC-LM3-CAR-A900-G3	child/dependency decision open: LM3-FIX-SA340/G3	open
DEC-LM3-CAR-A900-G3	child/dependency decision open: LM3-HV-SA510/G3	open
DEC-LM3-CAR-A900-G3	child/dependency decision open: LM3-INT-SA330/G3	open
DEC-LM3-CAR-A900-G3	child/dependency decision open: LM3-LGT-SA350/G3	open
DEC-LM3-CAR-A900-G3	child/dependency decision open: LM3-ROOF-SA410/G3	open
DEC-LM3-CAR-A900-G3	child/dependency decision open: LM3-SHELL-A200/G3	open
DEC-LM3-CAR-A900-G3	controlled deployment acceptance is not authenticated	open
DEC-LM3-CAR-A900-G3	decision open	open
DEC-LM3-CAR-A900-G3	prior gate decision open: G2	open
DEC-LM3-CAR-A900-G3	reviewed physical integration evidence missing	open
DEC-LM3-CAR-A900-G3	stale design revision: LM3-CAR-A900	open
DEC-LM3-CAR-A900-G3	stale design revision: LM3-TRAINSET-A000	open
DEC-LM3-CAR-A900-G4	EVD-COOL-BENCH: evidence planned	open
DEC-LM3-CAR-A900-G4	EVD-COOL-CONTROLLER: evidence planned	open
DEC-LM3-CAR-A900-G4	EVD-COOL-INTEGRATION: evidence planned	open
DEC-LM3-CAR-A900-G4	EVD-COOL-TORQUE: evidence planned	open
DEC-LM3-CAR-A900-G4	FM-COOL-VEHICLE: failure analysis and residual risk unreviewed	open

Record	Finding	Disposition
DEC-LM3-CAR-A900-G4	FM-COOL-VEHICLE: failure analysis binds an earlier design revision	open
DEC-LM3-CAR-A900-G4	FM-COOL-VEHICLE: mandatory safety review requires a controlled decision	open
DEC-LM3-CAR-A900-G4	FM-COOL-VEHICLE: occurrence evidence unknown	open
DEC-LM3-CAR-A900-G4	FM-COOL-VEHICLE: operating context verification open: charging, degraded-operation, evacuation, maintenance, normal-running, shutdown, startup	open
DEC-LM3-CAR-A900-G4	REQ-COOL-RECOVERY: acceptance criteria require independent engineering review	open
DEC-LM3-CAR-A900-G4	child/dependency decision open: LM3-AUX-P010/G4	open
DEC-LM3-CAR-A900-G4	child/dependency decision open: LM3-BOG-SA610/G4	open
DEC-LM3-CAR-A900-G4	child/dependency decision open: LM3-BOG-SA620/G4	open
DEC-LM3-CAR-A900-G4	child/dependency decision open: LM3-DOOR-SA310/G4	open
DEC-LM3-CAR-A900-G4	child/dependency decision open: LM3-FIX-SA340/G4	open
DEC-LM3-CAR-A900-G4	child/dependency decision open: LM3-HV-SA510/G4	open
DEC-LM3-CAR-A900-G4	child/dependency decision open: LM3-INT-SA330/G4	open
DEC-LM3-CAR-A900-G4	child/dependency decision open: LM3-LGT-SA350/G4	open
DEC-LM3-CAR-A900-G4	child/dependency decision open: LM3-ROOF-SA410/G4	open
DEC-LM3-CAR-A900-G4	child/dependency decision open: LM3-SHELL-A200/G4	open
DEC-LM3-CAR-A900-G4	controlled deployment acceptance is not authenticated	open
DEC-LM3-CAR-A900-G4	decision open	open
DEC-LM3-CAR-A900-G4	prior gate decision open: G3	open
DEC-LM3-CAR-A900-G4	reviewed physical integration evidence missing	open
DEC-LM3-CAR-A900-G4	stale design revision: LM3-CAR-A900	open
DEC-LM3-CAR-A900-G4	stale design revision: LM3-TRAINSET-A000	open
DEC-LM3-HV-SA510-G1	EVD-COOL-BENCH: evidence planned	open
DEC-LM3-HV-SA510-G1	EVD-COOL-CONTROLLER: evidence planned	open
DEC-LM3-HV-SA510-G1	EVD-COOL-INTEGRATION: evidence planned	open
DEC-LM3-HV-SA510-G1	EVD-COOL-TORQUE: evidence planned	open
DEC-LM3-HV-SA510-G1	child/dependency decision open: LM3-BDY-P050/G1	open
DEC-LM3-HV-SA510-G1	child/dependency decision open: LM3-HV-P010/G1	open
DEC-LM3-HV-SA510-G1	child/dependency decision open: LM3-HV-P020/G1	open
DEC-LM3-HV-SA510-G1	child/dependency decision open: LM3-HV-P030/G1	open
DEC-LM3-HV-SA510-G1	child/dependency decision open: LM3-SAF-P010/G1	open
DEC-LM3-HV-SA510-G1	child/dependency decision open: LM3-TRC-P030/G1	open
DEC-LM3-HV-SA510-G1	child/dependency decision open: LM3-TRC-P040/G1	open
DEC-LM3-HV-SA510-G1	child/dependency decision open: LM3-TRC-P060/G1	open
DEC-LM3-HV-SA510-G1	child/dependency decision open: LM3-TRC-P070/G1	open
DEC-LM3-HV-SA510-G1	controlled deployment acceptance is not authenticated	open
DEC-LM3-HV-SA510-G1	decision open	open
DEC-LM3-HV-SA510-G1	stale design revision: LM3-CAR-A900	open
DEC-LM3-HV-SA510-G1	stale design revision: LM3-TRAINSET-A000	open
DEC-LM3-HV-SA510-G2	EVD-COOL-BENCH: evidence planned	open
DEC-LM3-HV-SA510-G2	EVD-COOL-CONTROLLER: evidence planned	open
DEC-LM3-HV-SA510-G2	EVD-COOL-INTEGRATION: evidence planned	open
DEC-LM3-HV-SA510-G2	EVD-COOL-TORQUE: evidence planned	open
DEC-LM3-HV-SA510-G2	child/dependency decision open: LM3-BDY-P050/G2	open
DEC-LM3-HV-SA510-G2	child/dependency decision open: LM3-HV-P010/G2	open
DEC-LM3-HV-SA510-G2	child/dependency decision open: LM3-HV-P020/G2	open
DEC-LM3-HV-SA510-G2	child/dependency decision open: LM3-HV-P030/G2	open
DEC-LM3-HV-SA510-G2	child/dependency decision open: LM3-SAF-P010/G2	open
DEC-LM3-HV-SA510-G2	child/dependency decision open: LM3-TRC-P030/G2	open
DEC-LM3-HV-SA510-G2	child/dependency decision open: LM3-TRC-P040/G2	open
DEC-LM3-HV-SA510-G2	child/dependency decision open: LM3-TRC-P060/G2	open
DEC-LM3-HV-SA510-G2	child/dependency decision open: LM3-TRC-P070/G2	open
DEC-LM3-HV-SA510-G2	controlled deployment acceptance is not authenticated	open
DEC-LM3-HV-SA510-G2	decision open	open
DEC-LM3-HV-SA510-G2	prior gate decision open: G1	open
DEC-LM3-HV-SA510-G2	stale design revision: LM3-CAR-A900	open
DEC-LM3-HV-SA510-G2	stale design revision: LM3-TRAINSET-A000	open
DEC-LM3-HV-SA510-G3	EVD-COOL-BENCH: evidence planned	open
DEC-LM3-HV-SA510-G3	EVD-COOL-CONTROLLER: evidence planned	open
DEC-LM3-HV-SA510-G3	EVD-COOL-INTEGRATION: evidence planned	open
DEC-LM3-HV-SA510-G3	EVD-COOL-TORQUE: evidence planned	open

Record	Finding	Disposition
DEC-LM3-HV-SA510-G3	child/dependency decision open: LM3-BDY-P050/G3	open
DEC-LM3-HV-SA510-G3	child/dependency decision open: LM3-HV-P010/G3	open
DEC-LM3-HV-SA510-G3	child/dependency decision open: LM3-HV-P020/G3	open
DEC-LM3-HV-SA510-G3	child/dependency decision open: LM3-HV-P030/G3	open
DEC-LM3-HV-SA510-G3	child/dependency decision open: LM3-SAF-P010/G3	open
DEC-LM3-HV-SA510-G3	child/dependency decision open: LM3-TRC-P030/G3	open
DEC-LM3-HV-SA510-G3	child/dependency decision open: LM3-TRC-P040/G3	open
DEC-LM3-HV-SA510-G3	child/dependency decision open: LM3-TRC-P060/G3	open
DEC-LM3-HV-SA510-G3	child/dependency decision open: LM3-TRC-P070/G3	open
DEC-LM3-HV-SA510-G3	controlled deployment acceptance is not authenticated	open
DEC-LM3-HV-SA510-G3	decision open	open
DEC-LM3-HV-SA510-G3	prior gate decision open: G2	open
DEC-LM3-HV-SA510-G3	reviewed physical integration evidence missing	open
DEC-LM3-HV-SA510-G3	stale design revision: LM3-CAR-A900	open
DEC-LM3-HV-SA510-G3	stale design revision: LM3-TRAINSET-A000	open
DEC-LM3-HV-SA510-G4	EVD-COOL-BENCH: evidence planned	open
DEC-LM3-HV-SA510-G4	EVD-COOL-CONTROLLER: evidence planned	open
DEC-LM3-HV-SA510-G4	EVD-COOL-INTEGRATION: evidence planned	open
DEC-LM3-HV-SA510-G4	EVD-COOL-TORQUE: evidence planned	open
DEC-LM3-HV-SA510-G4	child/dependency decision open: LM3-BDY-P050/G4	open
DEC-LM3-HV-SA510-G4	child/dependency decision open: LM3-HV-P010/G4	open
DEC-LM3-HV-SA510-G4	child/dependency decision open: LM3-HV-P020/G4	open
DEC-LM3-HV-SA510-G4	child/dependency decision open: LM3-HV-P030/G4	open
DEC-LM3-HV-SA510-G4	child/dependency decision open: LM3-SAF-P010/G4	open
DEC-LM3-HV-SA510-G4	child/dependency decision open: LM3-TRC-P030/G4	open
DEC-LM3-HV-SA510-G4	child/dependency decision open: LM3-TRC-P040/G4	open
DEC-LM3-HV-SA510-G4	child/dependency decision open: LM3-TRC-P060/G4	open
DEC-LM3-HV-SA510-G4	child/dependency decision open: LM3-TRC-P070/G4	open
DEC-LM3-HV-SA510-G4	controlled deployment acceptance is not authenticated	open
DEC-LM3-HV-SA510-G4	decision open	open
DEC-LM3-HV-SA510-G4	prior gate decision open: G3	open
DEC-LM3-HV-SA510-G4	reviewed physical integration evidence missing	open
DEC-LM3-HV-SA510-G4	stale design revision: LM3-CAR-A900	open
DEC-LM3-HV-SA510-G4	stale design revision: LM3-TRAINSET-A000	open
DEC-LM3-TRAINSET-A000-G1	EVD-COOL-BENCH: evidence planned	open
DEC-LM3-TRAINSET-A000-G1	EVD-COOL-CONTROLLER: evidence planned	open
DEC-LM3-TRAINSET-A000-G1	EVD-COOL-INTEGRATION: evidence planned	open
DEC-LM3-TRAINSET-A000-G1	EVD-COOL-TORQUE: evidence planned	open
DEC-LM3-TRAINSET-A000-G1	REQ-COOL-RECOVERY: acceptance criteria require independent engineering review	open
DEC-LM3-TRAINSET-A000-G1	child/dependency decision open: LM3-ART-SA800/G1	open
DEC-LM3-TRAINSET-A000-G1	child/dependency decision open: LM3-CAR-A900/G1	open
DEC-LM3-TRAINSET-A000-G1	child/dependency decision open: LM3-EIF-SA650/G1	open
DEC-LM3-TRAINSET-A000-G1	child/dependency decision open: LM3-END-SA700/G1	open
DEC-LM3-TRAINSET-A000-G1	child/dependency decision open: LM3-SYS-SA900/G1	open
DEC-LM3-TRAINSET-A000-G1	controlled deployment acceptance is not authenticated	open
DEC-LM3-TRAINSET-A000-G1	decision open	open
DEC-LM3-TRAINSET-A000-G1	stale design revision: LM3-CAR-A900	open
DEC-LM3-TRAINSET-A000-G1	stale design revision: LM3-TRAINSET-A000	open
DEC-LM3-TRAINSET-A000-G2	EVD-COOL-BENCH: evidence planned	open
DEC-LM3-TRAINSET-A000-G2	EVD-COOL-CONTROLLER: evidence planned	open
DEC-LM3-TRAINSET-A000-G2	EVD-COOL-INTEGRATION: evidence planned	open
DEC-LM3-TRAINSET-A000-G2	EVD-COOL-TORQUE: evidence planned	open
DEC-LM3-TRAINSET-A000-G2	REQ-COOL-RECOVERY: acceptance criteria require independent engineering review	open
DEC-LM3-TRAINSET-A000-G2	child/dependency decision open: LM3-ART-SA800/G2	open
DEC-LM3-TRAINSET-A000-G2	child/dependency decision open: LM3-CAR-A900/G2	open
DEC-LM3-TRAINSET-A000-G2	child/dependency decision open: LM3-EIF-SA650/G2	open
DEC-LM3-TRAINSET-A000-G2	child/dependency decision open: LM3-END-SA700/G2	open
DEC-LM3-TRAINSET-A000-G2	child/dependency decision open: LM3-SYS-SA900/G2	open
DEC-LM3-TRAINSET-A000-G2	controlled deployment acceptance is not authenticated	open
DEC-LM3-TRAINSET-A000-G2	decision open	open
DEC-LM3-TRAINSET-A000-G2	prior gate decision open: G1	open
DEC-LM3-TRAINSET-A000-G2	stale design revision: LM3-CAR-A900	open

Record	Finding	Disposition
DEC-LM3-TRAINSET-A000-G2	stale design revision: LM3-TRAINSET-A000	open
DEC-LM3-TRAINSET-A000-G3	EVD-COOL-BENCH: evidence planned	open
DEC-LM3-TRAINSET-A000-G3	EVD-COOL-CONTROLLER: evidence planned	open
DEC-LM3-TRAINSET-A000-G3	EVD-COOL-INTEGRATION: evidence planned	open
DEC-LM3-TRAINSET-A000-G3	EVD-COOL-TORQUE: evidence planned	open
DEC-LM3-TRAINSET-A000-G3	REQ-COOL-RECOVERY: acceptance criteria require independent engineering review	open
DEC-LM3-TRAINSET-A000-G3	child/dependency decision open: LM3-ART-SA800/G3	open
DEC-LM3-TRAINSET-A000-G3	child/dependency decision open: LM3-CAR-A900/G3	open
DEC-LM3-TRAINSET-A000-G3	child/dependency decision open: LM3-EIF-SA650/G3	open
DEC-LM3-TRAINSET-A000-G3	child/dependency decision open: LM3-END-SA700/G3	open
DEC-LM3-TRAINSET-A000-G3	child/dependency decision open: LM3-SYS-SA900/G3	open
DEC-LM3-TRAINSET-A000-G3	controlled deployment acceptance is not authenticated	open
DEC-LM3-TRAINSET-A000-G3	decision open	open
DEC-LM3-TRAINSET-A000-G3	prior gate decision open: G2	open
DEC-LM3-TRAINSET-A000-G3	reviewed physical integration evidence missing	open
DEC-LM3-TRAINSET-A000-G3	stale design revision: LM3-CAR-A900	open
DEC-LM3-TRAINSET-A000-G3	stale design revision: LM3-TRAINSET-A000	open
DEC-LM3-TRAINSET-A000-G4	EVD-COOL-BENCH: evidence planned	open
DEC-LM3-TRAINSET-A000-G4	EVD-COOL-CONTROLLER: evidence planned	open
DEC-LM3-TRAINSET-A000-G4	EVD-COOL-INTEGRATION: evidence planned	open
DEC-LM3-TRAINSET-A000-G4	EVD-COOL-TORQUE: evidence planned	open
DEC-LM3-TRAINSET-A000-G4	REQ-COOL-RECOVERY: acceptance criteria require independent engineering review	open
DEC-LM3-TRAINSET-A000-G4	child/dependency decision open: LM3-ART-SA800/G4	open
DEC-LM3-TRAINSET-A000-G4	child/dependency decision open: LM3-CAR-A900/G4	open
DEC-LM3-TRAINSET-A000-G4	child/dependency decision open: LM3-EIF-SA650/G4	open
DEC-LM3-TRAINSET-A000-G4	child/dependency decision open: LM3-END-SA700/G4	open
DEC-LM3-TRAINSET-A000-G4	child/dependency decision open: LM3-SYS-SA900/G4	open
DEC-LM3-TRAINSET-A000-G4	controlled deployment acceptance is not authenticated	open
DEC-LM3-TRAINSET-A000-G4	decision open	open
DEC-LM3-TRAINSET-A000-G4	prior gate decision open: G3	open
DEC-LM3-TRAINSET-A000-G4	reviewed physical integration evidence missing	open
DEC-LM3-TRAINSET-A000-G4	stale design revision: LM3-CAR-A900	open
DEC-LM3-TRAINSET-A000-G4	stale design revision: LM3-TRAINSET-A000	open
DEC-OSR-COOL-LOOP-G1	EVD-COOL-BENCH: evidence planned	open
DEC-OSR-COOL-LOOP-G1	EVD-COOL-CONTROLLER: evidence planned	open
DEC-OSR-COOL-LOOP-G1	EVD-COOL-INTEGRATION: evidence planned	open
DEC-OSR-COOL-LOOP-G1	EVD-COOL-TORQUE: evidence planned	open
DEC-OSR-COOL-LOOP-G1	FM-COOL-LOOP: failure analysis and residual risk unreviewed	open
DEC-OSR-COOL-LOOP-G1	FM-COOL-LOOP: mandatory safety review requires a controlled decision	open
DEC-OSR-COOL-LOOP-G1	FM-COOL-LOOP: occurrence evidence unknown	open
DEC-OSR-COOL-LOOP-G1	FM-COOL-LOOP: operating context verification open: degraded-operation, evacuation, maintenance, recovery, shutdown, startup	open
DEC-OSR-COOL-LOOP-G1	REQ-COOL-PHYSICAL: acceptance criteria require independent engineering review	open
DEC-OSR-COOL-LOOP-G1	child/dependency decision open: LM3-AUX-P010/G1	open
DEC-OSR-COOL-LOOP-G1	child/dependency decision open: OSR-COOL-CONNECTOR/G1	open
DEC-OSR-COOL-LOOP-G1	child/dependency decision open: OSR-COOL-FLOW-SENSOR/G1	open
DEC-OSR-COOL-LOOP-G1	child/dependency decision open: OSR-COOL-PUMP/G1	open
DEC-OSR-COOL-LOOP-G1	controlled deployment acceptance is not authenticated	open
DEC-OSR-COOL-LOOP-G1	decision open	open
DEC-OSR-COOL-LOOP-G1	stale design revision: LM3-CAR-A900	open
DEC-OSR-COOL-LOOP-G1	stale design revision: LM3-TRAINSET-A000	open
DEC-OSR-COOL-LOOP-G2	EVD-COOL-BENCH: evidence planned	open
DEC-OSR-COOL-LOOP-G2	EVD-COOL-CONTROLLER: evidence planned	open
DEC-OSR-COOL-LOOP-G2	EVD-COOL-INTEGRATION: evidence planned	open
DEC-OSR-COOL-LOOP-G2	EVD-COOL-TORQUE: evidence planned	open
DEC-OSR-COOL-LOOP-G2	FM-COOL-LOOP: failure analysis and residual risk unreviewed	open
DEC-OSR-COOL-LOOP-G2	FM-COOL-LOOP: mandatory safety review requires a controlled decision	open

Record	Finding	Disposition
DEC-OSR-COOL-LOOP-G2	FM-COOL-LOOP: occurrence evidence unknown	open
DEC-OSR-COOL-LOOP-G2	FM-COOL-LOOP: operating context verification open: degraded-operation, evacuation, maintenance, recovery, shutdown, startup	open
DEC-OSR-COOL-LOOP-G2	REQ-COOL-PHYSICAL: acceptance criteria require independent engineering review	open
DEC-OSR-COOL-LOOP-G2	child/dependency decision open: LM3-AUX-P010/G2	open
DEC-OSR-COOL-LOOP-G2	child/dependency decision open: OSR-COOL-CONNECTOR/G2	open
DEC-OSR-COOL-LOOP-G2	child/dependency decision open: OSR-COOL-FLOW-SENSOR/G2	open
DEC-OSR-COOL-LOOP-G2	child/dependency decision open: OSR-COOL-PUMP/G2	open
DEC-OSR-COOL-LOOP-G2	controlled deployment acceptance is not authenticated	open
DEC-OSR-COOL-LOOP-G2	decision open	open
DEC-OSR-COOL-LOOP-G2	prior gate decision open: G1	open
DEC-OSR-COOL-LOOP-G2	stale design revision: LM3-CAR-A900	open
DEC-OSR-COOL-LOOP-G2	stale design revision: LM3-TRAINSET-A000	open
DEC-OSR-COOL-LOOP-G3	EVD-COOL-BENCH: evidence planned	open
DEC-OSR-COOL-LOOP-G3	EVD-COOL-CONTROLLER: evidence planned	open
DEC-OSR-COOL-LOOP-G3	EVD-COOL-INTEGRATION: evidence planned	open
DEC-OSR-COOL-LOOP-G3	EVD-COOL-TORQUE: evidence planned	open
DEC-OSR-COOL-LOOP-G3	FM-COOL-LOOP: failure analysis and residual risk unreviewed	open
DEC-OSR-COOL-LOOP-G3	FM-COOL-LOOP: mandatory safety review requires a controlled decision	open
DEC-OSR-COOL-LOOP-G3	FM-COOL-LOOP: occurrence evidence unknown	open
DEC-OSR-COOL-LOOP-G3	FM-COOL-LOOP: operating context verification open: degraded-operation, evacuation, maintenance, recovery, shutdown, startup	open
DEC-OSR-COOL-LOOP-G3	REQ-COOL-PHYSICAL: acceptance criteria require independent engineering review	open
DEC-OSR-COOL-LOOP-G3	child/dependency decision open: LM3-AUX-P010/G3	open
DEC-OSR-COOL-LOOP-G3	child/dependency decision open: OSR-COOL-CONNECTOR/G3	open
DEC-OSR-COOL-LOOP-G3	child/dependency decision open: OSR-COOL-FLOW-SENSOR/G3	open
DEC-OSR-COOL-LOOP-G3	child/dependency decision open: OSR-COOL-PUMP/G3	open
DEC-OSR-COOL-LOOP-G3	controlled deployment acceptance is not authenticated	open
DEC-OSR-COOL-LOOP-G3	decision open	open
DEC-OSR-COOL-LOOP-G3	prior gate decision open: G2	open
DEC-OSR-COOL-LOOP-G3	reviewed physical integration evidence missing	open
DEC-OSR-COOL-LOOP-G3	stale design revision: LM3-CAR-A900	open
DEC-OSR-COOL-LOOP-G3	stale design revision: LM3-TRAINSET-A000	open
DEC-OSR-COOL-LOOP-G4	EVD-COOL-BENCH: evidence planned	open
DEC-OSR-COOL-LOOP-G4	EVD-COOL-CONTROLLER: evidence planned	open
DEC-OSR-COOL-LOOP-G4	EVD-COOL-INTEGRATION: evidence planned	open
DEC-OSR-COOL-LOOP-G4	EVD-COOL-TORQUE: evidence planned	open
DEC-OSR-COOL-LOOP-G4	FM-COOL-LOOP: failure analysis and residual risk unreviewed	open
DEC-OSR-COOL-LOOP-G4	FM-COOL-LOOP: mandatory safety review requires a controlled decision	open
DEC-OSR-COOL-LOOP-G4	FM-COOL-LOOP: occurrence evidence unknown	open
DEC-OSR-COOL-LOOP-G4	FM-COOL-LOOP: operating context verification open: degraded-operation, evacuation, maintenance, recovery, shutdown, startup	open
DEC-OSR-COOL-LOOP-G4	REQ-COOL-PHYSICAL: acceptance criteria require independent engineering review	open
DEC-OSR-COOL-LOOP-G4	child/dependency decision open: LM3-AUX-P010/G4	open
DEC-OSR-COOL-LOOP-G4	child/dependency decision open: OSR-COOL-CONNECTOR/G4	open
DEC-OSR-COOL-LOOP-G4	child/dependency decision open: OSR-COOL-FLOW-SENSOR/G4	open
DEC-OSR-COOL-LOOP-G4	child/dependency decision open: OSR-COOL-PUMP/G4	open
DEC-OSR-COOL-LOOP-G4	controlled deployment acceptance is not authenticated	open
DEC-OSR-COOL-LOOP-G4	decision open	open
DEC-OSR-COOL-LOOP-G4	prior gate decision open: G3	open
DEC-OSR-COOL-LOOP-G4	reviewed physical integration evidence missing	open
DEC-OSR-COOL-LOOP-G4	stale design revision: LM3-CAR-A900	open
DEC-OSR-COOL-LOOP-G4	stale design revision: LM3-TRAINSET-A000	open
DEC-OSR-COOL-PUMP-G1	EVD-COOL-BENCH: evidence planned	open

Record	Finding	Disposition
DEC-OSR-COOL-PUMP-G1	EVD-COOL-CONTROLLER: evidence planned	open
DEC-OSR-COOL-PUMP-G1	EVD-COOL-INTEGRATION: evidence planned	open
DEC-OSR-COOL-PUMP-G1	EVD-COOL-TORQUE: evidence planned	open
DEC-OSR-COOL-PUMP-G1	FM-COOL-PUMP: failure analysis and residual risk unreviewed	open
DEC-OSR-COOL-PUMP-G1	FM-COOL-PUMP: mandatory safety review requires a controlled decision	open
DEC-OSR-COOL-PUMP-G1	FM-COOL-PUMP: occurrence evidence unknown	open
DEC-OSR-COOL-PUMP-G1	FM-COOL-PUMP: operating context verification open: charging, degraded-operation, evacuation, maintenance, recovery, shutdown, startup	open
DEC-OSR-COOL-PUMP-G1	REQ-COOL-PUMP: acceptance criteria require independent engineering review	open
DEC-OSR-COOL-PUMP-G1	child/dependency decision open: LM3-AUX-P010/G1	open
DEC-OSR-COOL-PUMP-G1	controlled deployment acceptance is not authenticated	open
DEC-OSR-COOL-PUMP-G1	decision open	open
DEC-OSR-COOL-PUMP-G1	stale design revision: LM3-CAR-A900	open
DEC-OSR-COOL-PUMP-G1	stale design revision: LM3-TRAINSET-A000	open
DEC-OSR-COOL-PUMP-G2	EVD-COOL-BENCH: evidence planned	open
DEC-OSR-COOL-PUMP-G2	EVD-COOL-CONTROLLER: evidence planned	open
DEC-OSR-COOL-PUMP-G2	EVD-COOL-INTEGRATION: evidence planned	open
DEC-OSR-COOL-PUMP-G2	EVD-COOL-TORQUE: evidence planned	open
DEC-OSR-COOL-PUMP-G2	FM-COOL-PUMP: failure analysis and residual risk unreviewed	open
DEC-OSR-COOL-PUMP-G2	FM-COOL-PUMP: mandatory safety review requires a controlled decision	open
DEC-OSR-COOL-PUMP-G2	FM-COOL-PUMP: occurrence evidence unknown	open
DEC-OSR-COOL-PUMP-G2	FM-COOL-PUMP: operating context verification open: charging, degraded-operation, evacuation, maintenance, recovery, shutdown, startup	open
DEC-OSR-COOL-PUMP-G2	REQ-COOL-PUMP: acceptance criteria require independent engineering review	open
DEC-OSR-COOL-PUMP-G2	child/dependency decision open: LM3-AUX-P010/G2	open
DEC-OSR-COOL-PUMP-G2	controlled deployment acceptance is not authenticated	open
DEC-OSR-COOL-PUMP-G2	decision open	open
DEC-OSR-COOL-PUMP-G2	prior gate decision open: G1	open
DEC-OSR-COOL-PUMP-G2	stale design revision: LM3-CAR-A900	open
DEC-OSR-COOL-PUMP-G2	stale design revision: LM3-TRAINSET-A000	open
DEC-OSR-COOL-PUMP-G3	EVD-COOL-BENCH: evidence planned	open
DEC-OSR-COOL-PUMP-G3	EVD-COOL-CONTROLLER: evidence planned	open
DEC-OSR-COOL-PUMP-G3	EVD-COOL-INTEGRATION: evidence planned	open
DEC-OSR-COOL-PUMP-G3	EVD-COOL-TORQUE: evidence planned	open
DEC-OSR-COOL-PUMP-G3	FM-COOL-PUMP: failure analysis and residual risk unreviewed	open
DEC-OSR-COOL-PUMP-G3	FM-COOL-PUMP: mandatory safety review requires a controlled decision	open
DEC-OSR-COOL-PUMP-G3	FM-COOL-PUMP: occurrence evidence unknown	open
DEC-OSR-COOL-PUMP-G3	FM-COOL-PUMP: operating context verification open: charging, degraded-operation, evacuation, maintenance, recovery, shutdown, startup	open
DEC-OSR-COOL-PUMP-G3	REQ-COOL-PUMP: acceptance criteria require independent engineering review	open
DEC-OSR-COOL-PUMP-G3	child/dependency decision open: LM3-AUX-P010/G3	open
DEC-OSR-COOL-PUMP-G3	controlled deployment acceptance is not authenticated	open
DEC-OSR-COOL-PUMP-G3	decision open	open
DEC-OSR-COOL-PUMP-G3	prior gate decision open: G2	open
DEC-OSR-COOL-PUMP-G3	reviewed physical integration evidence missing	open
DEC-OSR-COOL-PUMP-G3	stale design revision: LM3-CAR-A900	open
DEC-OSR-COOL-PUMP-G3	stale design revision: LM3-TRAINSET-A000	open
DEC-OSR-COOL-PUMP-G4	EVD-COOL-BENCH: evidence planned	open
DEC-OSR-COOL-PUMP-G4	EVD-COOL-CONTROLLER: evidence planned	open
DEC-OSR-COOL-PUMP-G4	EVD-COOL-INTEGRATION: evidence planned	open
DEC-OSR-COOL-PUMP-G4	EVD-COOL-TORQUE: evidence planned	open
DEC-OSR-COOL-PUMP-G4	FM-COOL-PUMP: failure analysis and residual risk unreviewed	open
DEC-OSR-COOL-PUMP-G4	FM-COOL-PUMP: mandatory safety review requires a controlled decision	open
DEC-OSR-COOL-PUMP-G4	FM-COOL-PUMP: occurrence evidence unknown	open

Record	Finding	Disposition
DEC-OSR-COOL-PUMP-G4	FM-COOL-PUMP: operating context verification open: charging, degraded-operation, evacuation, maintenance, recovery, shutdown, startup	open
DEC-OSR-COOL-PUMP-G4	REQ-COOL-PUMP: acceptance criteria require independent engineering review	open
DEC-OSR-COOL-PUMP-G4	child/dependency decision open: LM3-AUX-P010/G4	open
DEC-OSR-COOL-PUMP-G4	controlled deployment acceptance is not authenticated	open
DEC-OSR-COOL-PUMP-G4	decision open	open
DEC-OSR-COOL-PUMP-G4	prior gate decision open: G3	open
DEC-OSR-COOL-PUMP-G4	reviewed physical integration evidence missing	open
DEC-OSR-COOL-PUMP-G4	stale design revision: LM3-CAR-A900	open
DEC-OSR-COOL-PUMP-G4	stale design revision: LM3-TRAINSET-A000	open
DEMO-CAR-01	design changed; installed applicability requires reassessment	open
DEMO-CAR-02	design changed; installed applicability requires reassessment	open
DEMO-FRACAS-01	engineering handback and effectiveness review open	open
EVD-COOL-BENCH	evidence planned	open
EVD-COOL-CONTROLLER	evidence planned	open
EVD-COOL-INTEGRATION	evidence planned	open
EVD-COOL-TORQUE	evidence planned	open
FM-COOL-CONNECTOR	failure analysis and residual risk unreviewed	open
FM-COOL-CONNECTOR	mandatory safety review requires a controlled decision	open
FM-COOL-CONNECTOR	mitigation lacks a verification scenario: REQ-COOL-TORQUE	open
FM-COOL-CONNECTOR	occurrence evidence unknown	open
FM-COOL-CONNECTOR	operating context verification open: charging, degraded-operation, evacuation, maintenance, recovery, shutdown, startup	open
FM-COOL-LOOP	failure analysis and residual risk unreviewed	open
FM-COOL-LOOP	mandatory safety review requires a controlled decision	open
FM-COOL-LOOP	occurrence evidence unknown	open
FM-COOL-LOOP	operating context verification open: degraded-operation, evacuation, maintenance, recovery, shutdown, startup	open
FM-COOL-PACK	failure analysis and residual risk unreviewed	open
FM-COOL-PACK	mandatory safety review requires a controlled decision	open
FM-COOL-PACK	occurrence evidence unknown	open
FM-COOL-PACK	operating context verification open: charging, degraded-operation, evacuation, maintenance, recovery, shutdown, startup	open
FM-COOL-PUMP	failure analysis and residual risk unreviewed	open
FM-COOL-PUMP	mandatory safety review requires a controlled decision	open
FM-COOL-PUMP	occurrence evidence unknown	open
FM-COOL-PUMP	operating context verification open: charging, degraded-operation, evacuation, maintenance, recovery, shutdown, startup	open
FM-COOL-RAILWAY	failure analysis and residual risk unreviewed	open
FM-COOL-RAILWAY	mandatory safety review requires a controlled decision	open
FM-COOL-RAILWAY	occurrence evidence unknown	open
FM-COOL-RAILWAY	operating context verification open: charging, degraded-operation, evacuation, maintenance, normal-running, shutdown, startup	open
FM-COOL-SENSOR	failure analysis and residual risk unreviewed	open
FM-COOL-SENSOR	mandatory safety review requires a controlled decision	open
FM-COOL-SENSOR	occurrence evidence unknown	open
FM-COOL-SENSOR	operating context verification open: charging, degraded-operation, evacuation, maintenance, recovery, shutdown, startup	open
FM-COOL-TORQUE	failure analysis and residual risk unreviewed	open
FM-COOL-TORQUE	mandatory safety review requires a controlled decision	open
FM-COOL-TORQUE	mitigation lacks a verification scenario: REQ-COOL-TORQUE	open
FM-COOL-TORQUE	occurrence evidence unknown	open
FM-COOL-TORQUE	operating context verification open: charging, degraded-operation, evacuation, maintenance, recovery, shutdown, startup	open
FM-COOL-VEHICLE	failure analysis and residual risk unreviewed	open
FM-COOL-VEHICLE	failure analysis binds an earlier design revision	open
FM-COOL-VEHICLE	mandatory safety review requires a controlled decision	open
FM-COOL-VEHICLE	occurrence evidence unknown	open

Record	Finding	Disposition
FM-COOL-VEHICLE	operating context verification open: charging, degraded-operation, evacuation, maintenance, normal-running, shutdown, startup	open
OBL-COOL-APPLICABILITY	deployment applicability and clause selection open	open
REQ-COOL-HOT	acceptance criteria require independent engineering review	open
REQ-COOL-LEAK	acceptance criteria require independent engineering review	open
REQ-COOL-PHYSICAL	acceptance criteria require independent engineering review	open
REQ-COOL-PUMP	acceptance criteria require independent engineering review	open
REQ-COOL-RECOVERY	acceptance criteria require independent engineering review	open
REQ-COOL-TORQUE	acceptance criteria require independent engineering review	open

Quantitative screening


```
{
  "degraded_braking": {
    "limitations": "Constant effective deceleration and fixed delay/distance; grade, adhesion, target brake timing,
uncertainty and operating authority need deployment substantiation.",
    "maximum_speed_km_h": [
      45.16761923499234,
      66.17934152448353
    ],
    "state": "illustrative-restriction-screen"
  },
  "design_options": [
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      "id": "baseline",
      "illustrative_cost": 0.0,
      "time_to_limit_s": [
        266.8945281716327,
        1188.0
      ],
      "title": "Reference pump and shared auxiliary supply",
      "top_event_probability": [
        1.6001151872690434e-05,
        0.00047990400614577087
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    {
      "decision": "comparison-only; uncertain supplier performance and cost require evidence",
      "id": "supplier-pump-improvement",
      "illustrative_cost": 1500.0,
      "time_to_limit_s": [
        266.8945281716327,
        1188.0
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      "title": "Illustrative lower pump failure rate",
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      "id": "residual-cooling-improvement",
      "illustrative_cost": 3000.0,
      "time_to_limit_s": [
        258.54467140827836,
        1188.0
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      "title": "Illustrative improved residual heat path",
      "top_event_probability": [
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  ],
  "evidence_basis": "illustrative-assumptions-no-qualified-parameters",
  "fault_tree": {
    "limitations": "Independent primitive event processes with explicit shared-cause events; exponential operating
failure times and independent repairs. No repair-resource competition, coverage, wear-out or systematic-defect rate
inference.",
    "minimal_cut_sets": [
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      ],
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        "EV-PUMP"
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        "EV-PUMP"
      ]
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    ],
    "state": "unknown-event-inputs",
    "steady_state_availability": [
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    ],
    "unknown_events": [
      "EV-CONFIG"
    ]
  },
```

```
  "inspection": {
    "average_latent_exposure": [
      1.1999904000581374e-05,
      0.0008395297975016813
    ],
    "limitations": "Perfect periodic proof-test detection and restoration, constant hardware rate, uniform demand
timing; actual diagnostic coverage and repair exposure remain open.",
    "maximum_interval_h_at_upper_rate": 200.13344454527154,
    "state": "screening-unvalidated"
  },
  "maintenance": {
    "limitations": "M/M/c mean queue wait and independent Poisson replenishment pipeline with fixed mean turnaround.
Crews are concurrent service teams, not roster headcount; correlated fleet outages, shifts, skills and logistics
require separate modelling.",
    "minimum_spares": [
      1,
      2
    ],
    "repair_crews": [
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      1
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  "physical_thermal": {
    "predictions": [],
    "state": "input-evidence-open",
    "time_to_limit_s": null
  },
  "release_authority": false,
  "thermal": {
    "limitations": "Single lumped heat capacity and constant heat/conductance/ambient; no hydraulic, spatial, phase-
change or runaway model.",
    "predictions": [
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          28.0,
          32.0
        ],
        "time_s": 0
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          28.681818181818183,
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        "time_s": 600
      }
    ],
    "some_cases_do_not_reach_limit": false,
    "state": "screening-unvalidated",
    "time_to_limit_s": [
      266.8945281716327,
      1188.0
    ],
    "uncertainty": "Parameter-corner sensitivity; no statistical confidence or validated hotspot bound."
  },
  "top_event": "Circulation and primary flow diagnostic protection unavailable; this precursor is not a quantified
battery fire hazard. Independent BMS temperature protection remains a separate assessment."
}
```

Controlled rig tests

Test	Procedure	Acceptance inputs
RIG-NORMAL — Normal operation at allocated duty	draft-A	max_temperature_c, min_flow_l_min, maximum_sample_gap_s; duration, repetitions and model tolerance require review
RIG-PUMP — Pump shaft seizure / circulation loss	draft-A	max_temperature_c, max_detection_delay_s, max_isolation_delay_s, max_fault_flow_l_min, maximum_sample_gap_s; duration, repetitions and model tolerance require review
RIG-LEAK — Leak / loss of pressure	draft-A	max_temperature_c, max_detection_delay_s, max_isolation_delay_s, maximum_sample_gap_s; duration, repetitions and model tolerance require review
RIG-MASKED — Seized pump plus flow signal stuck high	draft-A	max_temperature_c, max_detection_delay_s, max_isolation_delay_s, maximum_sample_gap_s; duration, repetitions and model tolerance require review
RIG-CHARGING — Charging with cooling loss, high ambient and fouling	draft-A	max_temperature_c, max_isolation_delay_s, maximum_sample_gap_s; duration, repetitions and model tolerance require review
RIG-AUX — Loss of shared auxiliary power	draft-A	max_temperature_c, max_detection_delay_s, max_isolation_delay_s, maximum_sample_gap_s; duration, repetitions and model tolerance require review

Catalogue water and junction placement review

Repository source: `engineering/assurance/catalogue-geography/README.md`

The current audit checks all 266 cities against retained, independently sampled water evidence and their actual exported route geometry. All checks pass. These are controlled planning examples; construction and operating approvals remain open.

Check	Previous revision	Current layouts
Cities with findings	256	0
Platform records in detected water cells	249	0
Missing grouped platforms at checked dry line junctions	1,868	0
Unapproved contiguous water runs above 1 km	197	0

The baseline audit uses revision `1fa30889464105bacfc634b87594866840575561` and the same retained water inputs. Crossing counts are checks of pairs of lines, rather than unique station complexes or passenger transfers.

Bukavu previously had four platform records in detected water cells, five missed junction checks and two long lake crossings. Its regenerated example has 21 platform records and none of those findings. Goma previously missed two checked junctions; its 19 current platforms cover both, with no wet platforms.

The shared pipeline restores complete OSM multipolygon geometry, including island holes; samples ESA WorldCover 2021 permanent water on each controlled 20 m grid; excludes every detected water cell and unknown coverage from platform sites; and checks between sparse route vertices. Short crossings are priced bridge candidates. Longer water stretches require a retained shore detour or an explicitly bounded change in endpoint scope.

Full-resolution GeoJSON preserves the actual centreline without display offsets. Platform markers use actual coordinates. Shared route sections have junctions at their entries and exits; each intermediate bend is not another interchange. Native checks and a separate Shapely audit require grouped platforms within 60 m of dry junctions at the controlled grid precision.

The changed layouts also require fresh full-day nominal and degraded software screens and current fleet, depot, workforce and finance quantities. Kut, Lubango and Khamis Mushait use costed two-cabinet station charging requirements. Their actual reruns retain the 90% floor for ordinary degraded cases and achieve minimum completion of 91.54%, 93.69% and 92.92%, respectively. Khamis Mushait completes 89.69% during the ten-hour all-site grid outage, above that exceptional case's unchanged 60% curtailment floor. The requirements and release boundaries are recorded in each city's `design-overrides.toml`; charging supplier, electrical and thermal acceptance remain open.

Kinshasa's enlarged 1,440-trainset example uses an actual local invocation of the same source-bound native runner, with four independent case workers and a larger wall-time allowance. Its execution receipt identifies the local environment. All nominal/degraded durations, service floors and native output hash checks are retained. The other 265 current executions are verified GitHub outputs. The selected-city recovery workflow also allows additional wall time for large networks. These are software planning results.

Planning capital retains the existing constructability multiplier for elevated curves below 300 m radius. Restoring full geometry can materially raise that cost screen. Realignment, survey and supplier quotations must validate it before an investment decision; earlier capital and return headlines require recalculation.

Endpoint changes in Douala, Zanzibar, Mwanza, Dar es Salaam and Tanga retain city-specific bounds and reasons in their water-route policies and alignment reports. Douala excludes the offshore end of one radial through a roughly 12 km water stretch. Recalculations use the shortened shore route. Recorded shoreline backtracking remains an open geometry review, with its measured excursion and the original 750 m threshold retained.

Each city publishes its land-cover samples, supplementary OSM water geometry, combined mask, exact grid, buildability constraints and source receipt. The receipt records actual source tile URLs and hashes, sampled windows and retained input hashes. Checks recompute the mask from these retained inputs; an empty legacy water relation is not treated as proof of dry land.

The historical 10 m classification comes from ESA WorldCover 2021 v200, class 80, with four quarter-cell samples per 20 m planning cell. Attribution: © ESA WorldCover project 2021 / Contains modified Copernicus Sentinel data (2021) processed by ESA WorldCover consortium. Data licence: CC BY 4.0.

These checks do not establish current surveyed shorelines, platform footprints, bank stability, transfer access or levels, property rights, bridge spans, continuous curve geometry or physical rail–structure acceptance. Those remain project releases even where the software planning checks pass.

Regenerate or verify with:

```
python tools/automation/refresh-city-water-evidence.py --check --jobs 4
python tools/automation/audit-city-geography.py
python tools/automation/audit-city-geography.py --check
```

Civil reference demonstration A

Repository source: `engineering/assurance/civil-reference/README.md`

Status: awaiting site design, physical evidence and independent review. Release is blocked.

The reference contains 4 beam assets with 20/25 m spans and 20 m of adjoining at-grade transition. It provides quantities, connection briefs, construction hold points, option comparisons and a connected civil FMEA. It is a review package for engineers and suppliers; it has no accepted construction drawings.

Layout and quantities

Asset	Family	Chainage (m)	Supports
PI20-T1	Pi20	0–20	SUP-00, SUP-20
PI20-T2	Pi20	0–20	SUP-00, SUP-20
PI25-T1	Pi25	20–45	SUP-20, SUP-45
PI25-T2	Pi25	20–45	SUP-20, SUP-45
TRANS-01	trackform-comparison-required	45–65	SUP-45, SUP-65
TRANS-02	trackform-comparison-required	45–65	SUP-45, SUP-65

4 bare beams contain 107.91 m³ of envelope concrete over 261.0 m² of deck. Foundation, reinforcement, prestress and installed cost quantities remain unresolved. 3 elevated support lines are identified. The candidate arrangement has 16 bearing seats; capacities and fixity are unresolved.

Mass and lifting

Family	Bare mass (kg)	Margin under 75 t (kg)	Complete lift
Pi20	59,950.0	15,050.0	blocked-incomplete-mass-or-lift
Pi25	74,937.5	62.5	blocked-incomplete-mass-or-lift

Net diaphragms, bulk-density reinforcement adjustment, prestress steel, anchorages, embedded lifting hardware, attachments, retained temporary works, slings, spreader and hook/block masses must be measured or substantiated. The overlapping CAD diaphragm zones cannot be summed as additional fabrication concrete. The actual crane/launcher chart, configuration, radius, dynamic allowance, utilisation, centre of gravity, ground support and lateral stability must be reviewed.

Foundation and trackform comparison

Candidate foundations: bored-shaft, pile-group. None is selected. A measured comparison must cover axial/lateral resistance, total and differential settlement, groundwater, chemistry, liquefaction/scour where relevant, utilities and installation constraints. Calcium-based treatment needs reviewed chemistry compatibility. Reinforced-soil approaches need railway deformation, stability and scour justification.

Ballasted track, slipformed slab and precast panels remain explicit alternatives. Compare settlement, drainage, local material and maintenance capability, repair possessions and whole-life cost for the layout-defined transition before selection.

Conventional girder comparator

Locally manufactured prestressed I-girders with a separate railway deck is retained for the 20/25 m span families. Supplier section, girder count, deck quantities and price are pending; no system is declared cheapest.

- moulds and prestress beds.
- reinforcement/end zones and transfer strength.
- girder/deck quantities and foundation reactions.
- transport and number/weight of lifts.
- temporary bracing and deck erection.
- railway deck and connection compatibility.
- bearings, jacking and replacement access.
- inspection and maintenance.
- installed time and cost.
- whole-life cost and uncertainty.

Use equivalent railway actions, geometry and service-life assumptions for both systems. Record initial capital, construction possessions, inspection, cleaning, bearing/track replacement, discount rate and uncertainty before calculating whole-life net present cost.

Controlled connection briefs

Numerical loads, tolerances, resistance and movement limits are pending for every connection. Each requires an engineered drawing, first-article qualification and independent review.

C-01 — Pier to foundation

Loads: axial, moment, shear, seismic cyclic demand.

Install: Survey starter bars/socket; trial fit; brace; grout/cure before loading.

Inspect: Survey alignment; reinforcement/coupler record; grout strength and fill evidence.

Repair/replacement: Engineer-designed local repair or replacement; retain access.

C-02 — Cap to column

Loads: vertical reactions, torsion, transverse loads and seismic.

Install: Brace cap; verify interface; complete continuity/grout; strength hold point.

Inspect: Joint dimensions, engagement and void checks; as-built survey.

Repair/replacement: Accessible repair ports and approved temporary support.

C-03 — Girder bearings and restraints

Loads: vertical uplift, braking, lateral, thermal, seismic and derailment.

Install: Survey seats; set orientation and movement datum; place beam; engage restraints.

Inspect: Seat levels, bearing rotation/contact, travel, fixity and restraint witness.

Repair/replacement: Isolate traffic; use reviewed jacking seats; replace bearing and resurvey.

C-04 — Walkway and containment attachments

Loads: egress crowd, containment impact and fatigue.

Install: Cast inserts; gauge fit; install panels and barriers with movement allowance.

Inspect: Anchor engagement/proof evidence, gaps, containment continuity and access.

Repair/replacement: Remove replaceable panels and bolts from protected work area.

C-05 — Track plinth and fastener anchors

Loads: vertical axle, lateral, braking, fatigue and rail interaction.

Install: Survey rail seats; qualify anchor/grout system; fit and gauge fasteners.

Inspect: Gauge, level, pull-out qualification, torque and electrical insulation.

Repair/replacement: Exchange fasteners or local plinth repair with track possession.

C-06 — Drainage and utility crossings

Loads: water pressure, utility weight, thermal movement and accidental loads.

Install: Set falls and movement loops; separate utilities; seal joints; commission drainage.

Inspect: Survey falls; leak/flow and blockage-access checks; utility clearance.

Repair/replacement: Accessible clean-outs and replaceable flexible connections.

C-07 — Jacking and bearing replacement

Loads: jacking reactions, temporary redistribution, lateral stability.

Install: Approve possession/supports; install synchronised monitored jacks at designed seats.

Inspect: Jack calibration, seat capacity, lift limits and stable temporary load path.

Repair/replacement: Replace bearing; controlled lowering; movement datum and track survey.

Construction sequence and hold points

Stage	Activity	Hold point
ST-01	Freeze layout, railway loads and surveyed ground/drainage basis	Independent reviewed design and safe working constraints
ST-02	Compare foundations and ground treatment; install/test supports ahead of erection	Measured load tests, settlement and chemistry compatibility; released temporary works
ST-03	Fabricate beams and connection mock-ups	Reviewed reinforcement/prestress, transfer strength, mass and lifting hardware records
ST-04	Survey bearing seats and install bearings/restraints	Seat levels, orientation, capacities and movement datum accepted
ST-05	Transport and lift first beam; brace before releasing rigging	Reviewed actual lift configuration, stability and equipment support; witness first article
ST-06	Erect paired beam and finish connections, walkway and containment	Connection strength/fill, engagement, survey and temporary support release
ST-07	Construct approach/trackform, catchment and drainage interfaces	Settlement/drainage/scour criteria and blockage access checks
ST-08	Commission, inspect and establish maintenance access	Geometry, drainage scenarios, jacking/replacement rehearsal; independent and authority acceptance

Calculations and drainage demonstration

Native model/result files preserve the reproducible elastic calculation and drainage replay. The beam comparison uses E = 30 GPa, bulk-envelope self-weight, an illustrative 20 kN/m additional service load and endpoint supports. OpenSees uses the Pi-section area/inertia; CalculiX uses an equal-area/inertia rectangular surrogate. Prestress, reinforcement, torsion, fatigue, derailment, seismic response, actual lift support positions and rail interaction are unresolved.

The elastic hand check is $5wL^4/(384EI)$. This checks software and units; its deflections are not accepted railway limits.

Span / stage	Hand deflection (mm)	OpenSees (mm)	CalculiX (mm)
20 m / construction	14.280	14.280	14.363
20 m / service	23.993	23.993	24.132
25 m / construction	34.864	34.864	34.980
25 m / service	58.576	58.576	58.772

The synthetic drainage system uses a 1 ha catchment, a 50 m conduit and a storm independent of any site. The blockage sensitivity reduces the 0.5 m conduit to 0.1 m; the backwater sensitivity fixes tailwater at 1.5 m. Reviewed project blockage/debris and flood boundaries may require different cases.

- backwater: routing continuity error 0%; 6 hydraulic limit exceedances in the illustrative screen.
- blocked-drain: routing continuity error 0%; 1 hydraulic limit exceedances in the illustrative screen.
- normal: routing continuity error 0%; 0 hydraulic limit exceedances in the illustrative screen.

Adequate continuity does not establish adequate drainage. Flooding volume, surcharge duration, peak head, conduit depth and outfall flow are checked using engine statistics across every routing step. Culvert debris, outfall consent, erosion/scour, flood pathways and cleaning access need site design. Consider ANUGA when 2D surface overflow changes the risk.

Connected civil FMEA

ID / stage	Failure chain	Controls and evidence
CF-01 / service	blocked drain saturated formation foundation movement bearing displacement track geometry deterioration operating restriction	Accessible drains; redundant route where justified; geometry monitoring and intervention. Evidence: Blockage/backwater replay, cleaning trial, settlement model and geometry intervention review. Open.
CF-02 / construction	eccentric lifting girder roll or lateral instability dropped member or collapse	Reviewed sling geometry, centre of gravity, temporary bracing and exclusion zone. Evidence: Weighed member, lift chart/radius, stability calculation and witnessed first lift. Open.
CF-03 / construction	wrong orientation or level restraint or loss of bearing contact unexpected forces or support loss	Surveyed seats and movement/fixity datum; witness installation. Evidence: Bearing schedule, seat survey and installation inspection. Open.
CF-04 / construction	voids or incomplete engagement connection resistance loss progressive cracking or collapse	Qualified grouting procedure, mock-up, fill witness and strength hold. Evidence: Batch/strength records, void/engagement inspection and connection qualification. Open.
CF-05 / construction	early prestress transfer anchorage/end-zone cracking loss of prestress or handling failure	No transfer or handling before reviewed measured strength. Evidence: Specimen maturity/strength, transfer procedure and end-zone inspection. Open.
CF-06 / construction	weak support or excessive ground pressure support settlement/overturning equipment and member collapse	Ground investigation, outrigger/launcher foundation and temporary works review. Evidence: Support loadcase, bearing/settlement calculations, survey and trial load. Open.
CF-07 / service	outfall erosion or transition stiffness change differential movement track geometry deterioration operating restriction or derailment hazard	Scour protection, graded stiffness and drainage access. Evidence: Flood/scour assessment, transition settlement model and inspection trigger. Open.

The shared schema allocates the coupled drainage chain to these planned assets and responsibilities:

Failure / asset	Owner	Calculation or observation	Required response
CF-01 / DRAIN-01	Drainage maintainer	engineering/assurance/civil-reference/calculations/drainage-sanity.json	Clean/inspect drain and invoke reviewed adverse-rain response
CF-01-BEARING / BEARING-20	Bridge maintainer/structural engineer	bearing inspection and rail-structure movement calculation required	Inspect bearing travel/fixity and review jacking/repair plan
CF-01-FOUNDATION / FOUND-20	Structural/geotechnical engineers	site support survey and geotechnical movement calculation required	Survey support position; impose reviewed movement intervention
CF-01-OPERATION / CIVIL-SYSTEM	Operator/control centre	controlled operator decision and effectiveness check required	Apply restriction or stop service pending engineering handback
CF-01-SATURATION / FORMATION-45	Geotechnical engineer	site groundwater/formation observation and coupled settlement calculation required	Inspect groundwater and formation; reassess settlement model
CF-01-TRACK / TRACK-20	Track engineer	track geometry observation and adopted railway limit calculation required	Measure geometry; compare approved limits; request restriction

The physical hierarchy connects parts to subassemblies, the reference bay and the civil system. Foundation–bearing–track interfaces and the saturation common cause are explicit. Planned member occurrences link concrete mix revision, constituent batches, curing, transfer/lifting strength, test records and nonconformances to the member. The planned drainage inspection links its work order, model reassessment, effectiveness check and engineering handback to the same failure chain. These templates contain no manufactured members, measured strengths or completed maintenance.

Machine-readable report contains the typed graph and drain-change trace. It uses the existing connected-assurance traversal, including both old and new relationships. The civil assembly is validated through the same configuration, ownership, effect, mitigation, evidence and decision schema as the component example. A drain change reaches formation, foundation, bearing and track failures, required operator response and blocked civil gates. All physical and independent evidence remains missing.

Unresolved site inputs

- Adopted railway standards and code editions.
- Survey/alignment, curvature, skew and rail movement datum.
- Railway service, derailment, fatigue, wind and seismic actions.
- Boreholes, groundwater, axial/lateral soil response and settlement.
- Sulfate/chloride exposure where present and treatment compatibility trials.
- Liquefaction, flood levels, scour and surface flood pathways where relevant.
- Supplier reinforcement/prestress, end zones, transfer/handling strengths and mould/bed limits.
- Measured net detail masses, centre of gravity, transport axle loads and route clearances.
- Lifting chart/configuration/radius, rigging, temporary stability and support-ground capacity.

- Accepted rainfall/storm ensemble, catchments, culvert sizing and outfall consent/tailwater.
- Connection tolerances, loads, movement and inspection acceptance values.
- Supplier-installed cost/time, service life, maintenance capability and discount basis.
- Independent check, physical first article evidence and signed authority release.

Public references and reuse rights

No external drawing or code was copied into this package. Record drawing-specific rights before incorporation; software licences do not grant rights to unrelated drawings.

Reference	Intended use	Rights/access status
FHWA-PBES	Prefabrication, connections, tolerances and erection practice; highway examples need railway adaptation	Reference only; individual drawing/third-party reuse rights not established; page reviewed
WSDOT-DRAWINGS	Conventional bridge detail benchmark	Reference only; drawing reuse permission pending; 403 on review; obtain controlled drawing set
PGSUPER	Conventional prestressed girder comparison; agency/highway assumptions need railway adaptation	Alternate Route Open Source License 1.1; no code incorporated; license reviewed; application not executed
RDSO	Railway bridge arrangement and detail references	Reference only; drawing access and reuse permission pending; portal unavailable on review; obtain applicable set
XC	Candidate additional structural check after validating one model	No code incorporated; check pinned license before adoption; repository reference only; not integrated
SWMM	Drainage network simulation	Existing tool integration; no external drawing incorporated; repository/API reviewed; existing solver executed
ANUGA	Optional 2D surface flood assessment when overflow pathways matter	No code incorporated; check pinned license before adoption; repository reviewed; not required or executed
FHWA-GROUND	Investigate soil chemistry before calcium-based ground treatment	Reference only; no extracts or drawings copied; primary reference supplied; site testing required
FHWA-GRS	Candidate reinforced-soil approaches; railway deformation, stability and scour review required	Reference only; no drawings copied; primary reference supplied; not a selected design

Reproduction

From the repository root:

```
.venv/bin/python tools/automation/civil_reference.py --run-solvers
.venv/bin/python tools/automation/civil_reference.py --check
.venv/bin/python tools/automation/civil_reference.py --verify-native-replay
```

OpenSeesPy, PySWMM and native `ccx` are needed for replay. The default compiler and `--check` validate recorded files and provenance without running native solvers. The release gates require a separately reviewed, design-hash-bound asset register, per-asset numerical outputs and accepted limits. This concept register is deliberately pending and cannot satisfy them.

Civil calculation records

Repository source: `engineering/assurance/civil-reference/calculations/README.md`

Small native CalculiX input/result files and OpenSees nodal results accompany `beam-sanity.json`. `drainage-sanity.json` records SWMM engine statistics for all three synthetic scenarios. Their source and output digests are validated by the civil compiler.

Reproduce from the repository root with `.venv/bin/python tools/automation/civil_reference.py --run-solvers`. The package describes the elastic formulas, surrogate section and synthetic hydraulic limits. These are software/calculation demonstrations with project acceptance explicitly false.

Synthetic drainage scenarios

Repository source: `engineering/assurance/civil-reference/drainage/README.md`

These small models adapt the repository's SWMM installation benchmark for the civil demonstration. They retain the synthetic 1 ha catchment and storm and use dynamic-wave routing at a one-second routing step.

- `normal.inp`: 0.5 m circular conduit and free outlet.
- `blocked-drain.inp`: 0.1 m effective conduit diameter as a blockage sensitivity.
- `backwater.inp`: fixed 1.5 m outlet stage.

The reduced diameter is an illustrative sensitivity, not a calibrated debris model. Rainfall, catchment geometry, flood levels, culvert/outfall design, blockage and scour assumptions need site-specific review. Recorded statistics show hydraulic failures despite acceptable continuity.

Transport and erection mass summary

Repository source: `engineering/assurance/civil-reference/transport-and-erection.md`

Equipment selection and lifting release are pending.

This summary is generated from the layout and complete mass-budget fields. The 75 t product envelope and equipment capacity are separate requirements. A larger crane cannot waive the member envelope; an independently reviewed, mass-basis-bound deviation is required.

Family	Bare member (kg)	Complete member (kg)	Missing fields	Equipment check	Overall
Pi20	59,950.0	Unresolved	net_diaphragms, reinforcement_adjustment, prestress_steel, anchorages, embedded_lifting_hardware, curbs_and_attachments, retained_temporary_works, slings, spreader, hook_and_blocks, lift.equipment_id, lift.chart_reference, lift.configuration, lift.radius_m, lift.capacity_at_radius_kg, lift.dynamic_factor, lift.permitted_utilisation, lift.review_reference, lift.support_check_reference, lift.stability_check_reference, lift.mass_basis_reference	Unresolved	blocked-incomplete-mass-or-lift
Pi25	74,937.5	Unresolved	net_diaphragms, reinforcement_adjustment, prestress_steel, anchorages, embedded_lifting_hardware, curbs_and_attachments, retained_temporary_works, slings, spreader, hook_and_blocks, lift.equipment_id, lift.chart_reference, lift.configuration, lift.radius_m, lift.capacity_at_radius_kg, lift.dynamic_factor, lift.permitted_utilisation, lift.review_reference, lift.support_check_reference, lift.stability_check_reference, lift.mass_basis_reference	Unresolved	blocked-incomplete-mass-or-lift

Confirm net detail masses, centre of gravity, transport axle loads/clearances, storage and transport support positions, lifting anchors/strength, lateral stability, temporary bracing, actual chart radius/configuration and ground support before release. Benchmark beam calculations do not qualify these stages. Earlier approximate handling guidance cannot establish feasibility of a complete member.

Complete controlled package · Machine-readable budgets and provenance

Formal Models

Repository source: `engineering/assurance/formal/README.md`

The `engineering/assurance/formal/` tree holds machine-checkable specifications that support the safety case, including signalling consensus models and bounded evaluator verification.

Contents

Folder	Purpose
`tla/`	TLA+ `SMRaft` model and TLC configurations for the restricted consensus protocol
`topology-adapters.md`	Shared evaluator interface, validated static tables and bounded proof scope
`consensus-refinement.md`	State/action abstraction map, assumptions and the remaining proof obligation

See `tla/README.md` for model-checking commands and the connection to `osr-consensus`.

SMRaft-to-Rust refinement argument

Repository source: `engineering/assurance/formal/consensus-refinement.md`

This document records the v0.3 abstraction map between `SMRaft.tla` and `osr-consensus`. It is a reviewable refinement argument and test plan, not a machine-checked refinement proof. That distinction remains a release gate for any stronger safety claim.

Abstraction relation

One TLA server is one Rust `RaftNode`. `Server` is `Config.peers`; TLA sequences are Rust `Vec<Entry>` with `LogIndex` preserving the model's one-based index; `messages` are the network's queued `Action::Send` values; and `allCommitted` is observed through committed actions/prefixes in a simulated cluster.

TLA+ state/action	Rust representation or handler	Current evidence
`currentTerm`, `votedFor`, `state`	`RaftNode.current_term`, `voted_for`, `role`	Unit and cluster property tests
`log`, `commitIndex`	`log`, `commit_index`	Invariant and randomized cluster tests
`nextIndex`, `matchIndex`	`next_index`, `match_index`	Replication tests
`votesGranted`	`votes_granted`	Election tests
`lastQuorumConfirmedTerm`	`last_quorum_confirmed_term`; Rust also has bounded wall-clock freshness	Fail-restrictive tests
`Timeout`, `RequestVote`, `BecomeLeader`	`on_tick`, `start_election`, `emit_request_votes`, `become_leader`	Rust tests plus TLC model exploration
`HandleRequestVote*`	`handle_request_vote*`	Rust tests plus TLC model exploration
`AppendEntries`, `HandleAppendEntries*`	`emit_append_entries*`, `handle_append_entries*`	Rust tests plus TLC model exploration
`AdvanceCommitIndex`	`advance_commit_index`	Safety-invariant tests
`ClientRequest`	`propose`	Category and quorum-freshness tests

Assumptions that must remain explicit

- static membership; no joint consensus or online reconfiguration;
- no log compaction in this module;
- persistent term, vote and log writes complete atomically before dependent network output in the integrating host;
- authenticated transport and issuer policy are outside the abstract TLA model;
- Rust time is monotonic and the deployment validates timer bounds;
- finite TLC bounds establish bounded model results, not an unbounded theorem.

v0.3 closure work

1. Run TLC for the checked configuration and archive its result.
2. Run fast release-selected interlocking and fail-restrictive ATP properties in CI; archive full bounded runs separately once topology unwind limits and tool versions have been frozen and reviewed.
3. Maintain this action/state map under code review.
4. Add a tool-backed trace/refinement proof or an assessor-accepted refinement argument covering initialization, every step action and stuttering.

Items 1–3 are repository evidence. Full ATP/interlocking bounded-proof completion and item 4 remain open in the certification gap register. They prevent the project from describing bounded coverage as exhaustive or consensus refinement as formally proved.

Proposed process-reference extension (RFC 0033)

The runtime reference calls the existing Rust step function with bounded contiguous append batches of twelve (default remains one). It records volatile current-term per-peer acknowledgement ages, including empty-log responses, for quorum freshness. Historical replication indices alone cannot refresh quorum. The original one-entry TLA+ action map does not establish refinement of these extensions. Independent review, response-age abstraction and batching simulation relations remain open. DiskJournal adds actual exclusive-writer filesystem persistence and full-log checkpoint compaction; SIGKILL, torn-file and fsync ordering tests are software evidence. Flash power-loss behavior and persistent output timing require hardware qualification. TACSResources is a separate bounded resource abstraction with qualified no-reentry/clearance assumptions; its checked invariants do not prove this combined implementation.

Bounded control proof results

Repository source: `engineering/assurance/formal/results/README.md`

Current follow-through — 19 September 2026

Two of the four previously open obligations have passing executions.

Original obligation	Recorded result
ATP determinism — E4.1a	Passed locally and in CI on `2bf8b7717`
Interlocking determinism — E3.1a	Passed locally on `2bf8b7717`; repeated with the memory-capped runner
Odometry determinism — E4.4a	Still open; CI timed out after 1800 seconds
Two-train non-overlap — E1.1a and its seven paired companions	Still open; all eight local partitions exhausted their 4096 MiB address-space allowance

The summary and complete retained evidence include successes, failures, input hashes, commands, manifests, source changes and regression logs. No diagnostic run with omitted checks or unverified contracts is counted as a pass.

CI run 35403962318 on commit `2bf8b7717aea09436cbcd588fa1e084ef907e1f9` retained seven package artifacts: 36 passes and one odometry timeout across 37 properties. Its interlocking runner received a shutdown signal before uploading an artifact. That interruption is not a counterexample or a passing execution.

The subsequent local interlocking run retained all 11 declared interlocking results: three passes and eight memory failures. Its Rust and GSN inputs match the CI candidate; only the source-bound runner and workflow inputs differ because they add recorded memory limits. These groups are recorded separately. They are not a successful, complete CI run on one final commit and do not satisfy the release gate or independent acceptance.

Reproduce

For the CI candidate, check out `2bf8b7717`, install Kani 0.67.0 and run:

```
python3 tools/automation/assurance-evidence.py run \
  --timeout 1800 --output build/assurance/reproduced-candidate
```

To reproduce the memory-capped local run, extract the archive at that checkout and apply `build/assurance/closure-capped/source.patch`, then run:

```
python3 tools/automation/assurance-evidence.py run \
  --package osr-interlocking --timeout 600 --memory-mib 4096 \
  --output build/assurance/reproduced-interlocking
```

Use a new output directory so historical evidence is preserved. The source patch covers the runner/workflow changes, and each manifest binds the complete Rust, Cargo and GSN inputs. The JSON lists the archive and member SHA-256 values.

The next unresolved work is a verified decomposition of variable-time odometry arithmetic and two-train state replay, or a verifier configuration that completes the full existing obligations. The isolated arithmetic and contract experiments did not establish closure and were not adopted. Increasing time alone did not resolve odometry. A separate non-overlap diagnostic with a 12,800 MiB address-space allowance also timed out after 300 seconds; its complete log is retained without counting it as a controlled pass. See the component and partition contracts.

Historical topology proof rehearsal

The summary records 37 passes and four 300-second timeouts across all 41 declared Kani properties. Each of those four properties was then rerun with a 900-second budget and timed out again; their original manifests, results and logs are included under `extended-budget/`. The archive preserves the original logs, per-package TOML results and execution manifests, plus the Rust workspace, Clippy and 27 evidence-packaging regression-test outputs. The JSON lists the archive SHA-256 and every archived member's SHA-256.

Reproduce

The execution started from checkout `0a694f8cf7af9a77fcc49dee0c342553057c1297` with the recorded Rust changes. Restore that checkout into a separate working directory, extract the archive there and apply `build/assurance/static-final/source.patch`. Every execution input is listed in the summary's `source_sha256`; the patch reproduces those source bytes. Install the reviewed Kani 0.67.0 toolchain, then run into a new output directory:

```
python3 tools/automation/assurance-evidence.py run \
  --timeout 300 --output build/assurance/reproduced
```

The command exits unsuccessfully if any proof fails or times out, while retaining all completed outcomes. The extended attempts use the same command with `--solution <ID> --timeout 900`. Runtime depends on the host and concurrent

load. These runs shared a development host with other proofs and simulations; they are not a controlled sizing benchmark. The archive is a local, unattested rehearsal and includes failures. It cannot satisfy the clean-commit CI release gate or independent safety acceptance.

See topology adapters for the fixture bounds, storage contract and limitations of the comparison tests.

Formal specs (TLA+)

Repository source: `engineering/assurance/formal/tla/README.md`

This directory holds TLA+ specifications that anchor OpenSourceRail's formal safety argument.

Files

File	Purpose
`SMRaft.tla`	Core protocol spec: Static-Membership Raft + Fail-Restrictive Timeout. Referenced by RFC 0001 §11.1.
`MCSmall.tla`	Minimum-viable TLC harness (3 nodes, 2 values).
`MCSmall.cfg`	TLC config for the small harness.
`TACSResources.tla` / `TACSResources.cfg`	Separate train-centred resource model: retained approach rights, occupancy, expiry, restart, qualified release and blocking. Two trains, two conflicting resources, two epochs.

Running TLC

Install the TLA+ toolbox (or `tla2tools.jar` from https://github.com/tlaplus/tlaplus).

```
# from this directory
java -jar tla2tools.jar -config MCSmall.cfg MCSmall.tla -workers auto -deadlock
```

Expected outcome on the small harness: all invariants hold, state-space exploration completes in minutes on a developer workstation.

What this spec does and does not prove

Proves (via TLC on small bounded models + TLAPS on the general spec, once we write the TLAPS proofs):

- `ElectionSafety`, `LogMatching`, `LeaderCompleteness`, `StateMachineSafety` — the classical Raft safety properties, inherited from Ongaro's proof and re-checked under our specialization.
- `FailRestrictive` — the novel property introduced by this variant: no Safety-category entry is committed under a leader whose quorum confirmation is stale.

Does not prove (out of scope here; covered elsewhere):

- Refinement from the Rust implementation to this spec (future `SMRaftImpl` module once `crates/osr-consensus` exists).
- The rail-domain state machine (non-overlap of MAs, etc.) — that is Kani territory, see RFC 0001 §11.2.
- Byzantine behaviour — explicitly out of scope for v1 (RFC 0001 §3).
- Log compaction / snapshotting — orthogonal to safety; separate module.

Known limitations of the current spec

1. `QuorumConfirmationExpires` is modeled as an explicit action; in the real system it is a wall-clock timeout. A timed refinement would sharpen the liveness argument but is not needed for the safety properties above.
2. The message bag is unbounded in principle; TLC terminates only because of the `MaxTerm` / `MaxLogLen` bounds. Production verification should use TLAPS for the unbounded spec.
3. No explicit modeling of crash + restart persistence. Standard Raft arguments apply; add a `Restart` action when we want to cover this explicitly.

Executed TACS invariant checks

The TACS package records the actual pinned-TLC run and source/log hashes for `TACSResources`. It checks six invariants in a finite model; qualified physical no-reentry and clearance are explicit assumptions. It proves neither Rust refinement nor unbounded liveness. The Rust core has separate bounded transition and property tests. See RFC 0032 for contracts.

Read-only topology adapters

Repository source: `engineering/assurance/formal/topology-adapters.md`

The movement-authority, ATP braking and odometry evaluators share a small `TrackTopology` interface: line count, borrowed forward/reverse section ordering, ring status and section lookup. This separates the safety calculation from how a city stores its geometry. A bounded fixture can use prevalidated constants, so the verifier spends less work constructing unrelated map and station metadata before it reaches the property being checked.

Two storage forms, one evaluator

- `Network` retains the existing city schema, maps, station metadata and normal simulator/control callers. Its adapter delegates section lookup to the original method and borrows the original line arrays.
- `StaticTopology` borrows immutable section and line tables. Construction and lookup allocate nothing. `try_new` rejects duplicate section IDs and dangling forward/reverse references. It can validate runtime slices or run at compile time for constant tables. Private fields prevent bypassing that constructor.

The constructor checks ID uniqueness and reference integrity. Surveyed alignment, section dimensions, direction pairing and operating speed limits still require their own engineering validation.

Both preserve unknown-section lookup failure. The static adapter performs a linear section search; it is useful for small fixed configurations and bounded verification, not a measured performance improvement for the full city simulator. It does not make the existing crates `no_std`, remove allocations from the evaluators, or constitute a deployed embedded image.

The public interface requires stable line views and unique section resolution. A new adapter needs its own validation. No ERPNext document, FUXA tag, city schema, railway command format or deployed control setting is changed by choosing a Rust storage implementation.

Verification scope

The Kani fixtures use constant, validated tables with the same two/three forward sections, paired reverse sections, lengths, directions and speed limits as their previous network fixtures. Station business metadata is outside the evaluator's interface. The topology migration preserves the original fixture geometry. Subsequent proof partitions and configuration compilation are described below. The normal static lookup and normal evaluator code execute under Kani; there are no stubs, unverified function contracts or disabled safety/unwinding checks.

A proof over these fixtures establishes the property within their stated bounds. It is not a proof for every topology adapter, arbitrary city geometry or the Rust standard library's map implementation. The default `Network` adapter is additionally compared with static tables through generated tests of movement authorities, braking outcomes, track advancement and sensor fusion. These cover reversed table order, forward/reverse tracks, direction mismatches, rings, position/speed uncertainty, expired authorities and GNSS/balise inputs. Constructor tests reject malformed tables. Finite comparison tests are not a universal refinement proof.

Proof outcomes are recorded by the existing source-bound runner:

```
python3 tools/automation/assurance-evidence.py run \
  --package osr-odometry --timeout 600 --output build/assurance/my-odometry
```

Timeouts and failed checks remain unproved. Independently accepted safety evidence, consensus refinement, hardware integration and operating release remain separate. Historical city simulation bundles still verify as historical artifacts; changed Rust dependencies require new qualification before claiming current-source results. Both successful Samawah and Mosul design options were requalified against commit `a5edbbe84`: nominal service and all eight degraded cases passed. Those bundles are historical evidence for that commit; they do not qualify the later state-storage and arithmetic changes.

Reusing the interface

Existing calls with `&Network` keep their behaviour and city serialization. A consumer with fixed geometry can build borrowed tables and pass the checked adapter to those same functions:

```
use osr_core::{StaticTopology, TrackLine};

// `network` is an existing, validated city Network.
let sections: Vec<_> = network.sections.values().cloned().collect();
let lines: Vec<_> = network.lines.iter().map(TrackLine::from).collect();
let topology = StaticTopology::try_new(&lines, &sections)?;
// compute_self_ma(..., &topology, ...)
// atp_evaluate(..., &topology, ...)
// odom_step(..., &topology)
```

The example allocates its input vectors; the adapter itself only borrows them. Fixed deployments can instead use constant arrays, as the Kani fixtures do. Storage selection is a Rust integration choice, not an operator setting or a change to the ERP/FUXA control boundary.

Reusable calculation and state components

`AuthoritySnapshot::from_log` folds a committed prefix once. Call `authority` for multiple trains against that owned state, then apply the next committed entry with `apply_committed_entry` or append a committed suffix with `append_committed`. The caller retains consensus ordering; this API does not validate or commit log entries. Exclusive mutable access prevents concurrent reads during an update. Generated tests compare batching and append operations with complete normal log replay and movement-authority calculation.

Derived-state tables use `OrderedMap`, a sorted vector of keys and boxed values. It preserves unique sorted keys and the existing JSON map representation. Lookup is O(log n); insertion and removal are O(n), and repeated removal during `retain` is $O(n^2)$ in the worst case. Each stored value adds an allocation. These tradeoffs are explicit: no city-size bound is introduced, but throughput and allocation behavior must be measured before deployment. Generated operation sequences compare lookup, mutation, removal, iteration and serialization with `BTreeMap`.

ATP accepts a reusable immutable `BrakeProfile`. `from_consist` applies the same conservative braking-curve conversion as the existing consist adapter; `try_new` rejects nonpositive deceleration. Compile a new profile when vehicle configuration changes. Existing calls with `ConsistDescriptor` continue to compile the profile for each evaluation. The determinism harness checks the fixture's compiled profile against the real reference consist before evaluating twice. It retains the original position, speed and time domains and compares complete outputs.

`forward_chain_view` borrows validated topology tables without allocating a section vector. ATP distance and odometry advancement use it. The original vector-returning walk remains available; generated comparisons cover linear and ring traversal, direction mismatches, offsets and distance budgets. Movement authority calculates its forward chain and footprint once per evaluation.

Odometry retains exact integer rounding and fallback behavior across full-width inputs. Fast paths use narrower arithmetic only when it is equivalent to the original expression. Boundary and generated tests compare speed and uncertainty with the previous arithmetic, including zero elapsed time and extreme timestamp, calibration and pulse values.

Non-overlap proof partitions

The three-forward-section fixture contains both registered trains and both symbolic head positions in every partition. One pair of harnesses checks the following and leading train for each of four wayside conditions: missing, Clear, Unknown and Present. Both downstream verdicts have the same condition in each pair; arbitrary mixed verdicts are outside this bounded fixture.

The following train must end on section 1000. The leading train must end on 1002 when the downstream section is Clear, and on 1001 otherwise. On the linear fixture these bounds imply disjoint authority sections. All eight partitions are required to close this declared non-overlap evidence. Passing one half is insufficient. The earlier fixture expected extension into 1002 without providing a Clear verdict; that expectation contradicted fail-restrictive operation. Ordinary tests now exercise clearance and occupancy independently.

Execution budgets

The controlled runner defaults to 600 seconds per harness. Push and pull-request jobs gate the 39 declarations that currently complete within that envelope. Manual workflow dispatch also runs the nine unresolved declarations as non-gating diagnostics and can select 1800 seconds for each of the 48 declared harnesses across eight packages. The default 4096 MiB per-process address-space limit can be raised to 8192 MiB in manual dispatch (or configured with `--memory-mib` locally); the limit is inherited by verifier children and recorded in the evidence. Non-gating means that an already-recorded resource limitation does not make every development merge red; it does not turn a failure into a pass. Memory exhaustion remains a failed proof, and core dumps are disabled. Timeouts fail the run and kill the verifier's entire process group, including solver children left behind by the driver. Every completed outcome retains its command, input hashes and log; interrupted runners may lack a complete artifact and must be rerun. Kani remains pinned to 0.67.0. Release export still demands successful evidence for all 48 declarations and fails while any of the nine open obligations is absent or unsuccessful.

Current verification follow-through

ATP determinism and interlocking determinism passed on candidate `2bf8b7717`. Odometry still timed out with 1800 seconds in CI. All eight non-overlap partitions remain unresolved because the local model exhausted its recorded memory budget. See the execution register for full records, the interrupted CI interlocking job, reproduction instructions and source boundaries.

Historical topology-migration verification

The complete local 300-second run executed all 41 declared properties with Kani 0.67.0: 37 passed and four timed out. The newly completed properties are odometry forward non-regression (`E4.4b`, 14.3 seconds), uncertainty monotonicity (`E4.4c`, 15.4 seconds), and GNSS conservatism (`E4.4e`, 52.9 seconds). The existing balise-reset proof also passes. Compared with the previous 34/41 result, input ranges, assertions and unwind limits are unchanged.

The machine-readable summary and complete log archive retain all 41 outcomes, eight execution manifests, input hashes, the source patch and regression-test logs. See the reproduction notes.

At that historical checkpoint, remaining timeouts were odometry determinism (`E4.4a`), ATP determinism (`E4.1a`), interlocking non-overlap (`E1.1a`) and interlocking determinism (`E3.1a`). All four also timed out in separate 900-second attempts on the shared development host. These are unresolved proof obligations, not demonstrated counterexamples. Alternative solver/cache experiments did not establish closure; Z3 also encountered an internal verifier error on ATP. The reviewed default solver remains unchanged.

The full Rust workspace tests, Clippy with warnings denied, and 27 evidence and release-packaging regression tests passed. These executions are local and unattested; exact-commit CI and independent acceptance remain separate.

Remaining proof work

The remaining properties exercise substantially more than fixture construction. Odometry determinism compares complete states, including variable-time integer speed division. Non-overlap also replays two trains into derived occupancy and train maps. The compiled braking profile and map refactor closed the bounded ATP and interlocking determinism checks; they did not close these other obligations.

The next design step is to isolate these calculations and state-replay contracts with explicit input/output invariants, prove the contracts, then verify their composition. Any such partition must preserve every original input case and safety check. An assumed contract or a smaller input range cannot be counted as closure of an existing obligation. The existing full harnesses remain the reference checks while that work is developed.

Subsystem lifecycle-control register

Repository source: `engineering/assurance/subsystem-control-register.md`

> Generated by `python3 engineering/subsystem_control_register.py`. Do not edit this file or its JSON partner by hand.

This register joins every controlled mechanical product and assembly, station variant, reusable civil type and Rust workspace crate to the same configuration, quality, competence, management and identity controls. It routes evidence; it does not claim fabrication, construction, software deployment or railway release.

Coverage

Domain	Controlled records
LM3 mechanical products and assemblies	146
Station variants	7
Reusable civil types	19
Rust crates	60
Total	232

The common control system contains 11 role templates, 9 record templates, 6 workflow templates and 4 management cadences. There are 172 physical identity candidates. There are deliberately zero printable QR labels until a deployment binds real assets to an operator-controlled HTTPS resolver and checks the physical labels.

Best-practice basis

Basis	Authoritative source	Applied here as
`FTA-QMS-2019`	Quality Management System Guidelines	Configuration, document, inspection, test, calibration, nonconformance and closeout record controls.
`FTA-PCMG-2025`	Project and Construction Management Guidelines	Scope, schedule, cost, safety, quality and accountable project-review templates.
`ISO-55001-2024`	Asset management system requirements	Lifecycle decision, value, risk/opportunity, information, management-review and predictive-action fields.
`ISO-9001-2026`	Quality management systems — requirements	Quality culture, documented information, competence, operational control, performance evaluation and improvement records.
`NIST-SSDF-1.1`	Secure Software Development Framework	Rust release provenance, protected source, review, test, vulnerability and change records.
`BUILDINGSMART-IFC43-ASSET-ID`	IFC 4.3 AssetIdentifier semantic definition	Separate stable operator asset identity from database and IFC instance identifiers.
`GS1-DIGITAL-LINK-1.7`	GS1 Digital Link URI Syntax	Resolver-based QR design and optional licensed GIAI mapping for individual assets.
`FRAPPE-HR-TRAINING`	Training Event	Native Training Program, Event, attendance, result and certificate workflow mapping.
`ERPNEXT-QUALITY-INSPECTION`	Quality Inspection	Incoming, in-process and outgoing inspection-template mapping.

These sources inform the templates; they do not certify OSR or replace the contract, applicable law, national standards, competent engineers, operators or independent assessors.

Control model

- Mechanical and station records require configuration, inspection/test, calibration, nonconformance and asset-identity evidence.
- Civil records add survey/ground, temporary-works, concealed-work, as-built and handover evidence.
- Rust records require locked source/toolchain, peer review, all-feature tests and linting, vulnerability disposition, reproducible artifacts and an explicit deployment-authority boundary.
- The QR contract is lookup-only. It carries no personal data, credential, command, approval, isolation or movement authority.

The complete per-record controls, source paths, maturity states and evidence requirements are in subsystem-control-register.json. The editable governance source is `lifecycle-governance.json`.

Distributed-control process reference

Repository source: `engineering/assurance/tacs/README.md`

RFC 0033 is the proposed development target. Existing `osr-consensus`, `osr-interlocking`, ATP, ATO and brake components supply the decisions; `osr-runtime` supplies thin hosts. The earlier separate `osr-tacs` engine is retired.

Four train processes (A/B for each of two trains), three static voters, one point interface and two station interfaces exchange authenticated messages through bounded asynchronous queues. Each decision/communications process has a filesystem journal. Two additional output processes each enforce two-out-of-two agreement and either-channel trip using `DualGuard`. A publishes proposals; B only evaluates. Each host has its own journal. Missing-A/B, disagreement, replay and feedback faults must activate while moving and trip within one 50 ms virtual step. The 100 ms deadline cannot be concealed by a late feed. Shared sensors/code/host/clock/keys/comparator remain common causes; autonomous real-clock outputs and hardware independence remain pending. The policy keeps point/crossing/door/charging/obstacle local pairs explicitly open. The harness provides synthetic local physics/proving inputs without supplying global truth to onboard authority. The physics fixture rejects unsupported geometry/formation/braking bundles before startup and checks moving footprints against retained protection. Positions use measured speed and non-zero bounded uncertainty. Platform locks remain owned at arrival. Process restart uses SIGKILL and the same journal.

```
cargo test -p osr-runtime -p osr-consensus -p osr-interlocking
python3 tools/automation/tacs_reference.py --output /tmp/reference.json
python3 tools/automation/redundancy_policy.py
python3 tools/automation/tacs_assurance.py --check
python3 tools/automation/tacs_assurance.py --verify-replay

# Explicit new development bundle and actual execution evidence
python3 tools/automation/tacs_assurance.py --generate-deployment --run-twin
python3 tools/automation/tacs_assurance.py --run-formal build/toolchain/tla/tla2tools-1.8.0.jar
python3 tools/automation/tacs_cost.py engineering/assurance/tacs/cost-model.json
```

report.md and raw execution records contain unreviewed software results. assurance.json uses the shared hierarchical FMEA and change-impact compiler. Source changes invalidate recorded execution and frozen configuration and failure-subject applicability; compilation does not rebase FMEA records or create passing run evidence or independent acceptance. Reference keys are publicly reproducible test keys. PIDs differ on replay; all semantic results and traces must match.

Designed, installed and approved configuration states are distinct. This bundle is synthetic and not installed; operational activation is rejected. Storage compaction retains the full Raft log. Torn storage inhibits startup; physical power loss/flash behavior remains to qualify. The formal resource model checks two trains, two resources and two epochs under qualified clearance/no-reentry assumptions; batching and filesystem/runtime refinement remain independent-review work. No complete safety proof is claimed.

HIL must establish independent safe outputs/deadlines, sensor/integrity/points, clock and radio bounds, per-car charging/doors/PSD/BMS/traction integration, real braking/adhesion and storage power-loss behavior. Closed-track recovery, independent assessment and named G0–G4/profile-release decisions follow. Depot, rescue, turnback, following trajectories, arbitrary topology and large fleets are protected/unimplemented service extensions, not passed operating cases.

cost-model.json records equipment and lifecycle inputs for equivalent service/availability. Missing values remain unknown; no equipment removal, percentage saving or city-cost reduction is approved.

TACS prototype assurance

Repository source: `engineering/assurance/tacs/report.md`

Generated by `tools/automation/tacs_assurance.py`. Software results are unreviewed; physical and operational release remain blocked.

Software milestone passed: true.

TLC checked 282 distinct bounded states. Physical clearance/no-reentry is an explicit assumption; this is not an unbounded proof or Rust refinement.

Case	Passed	Arrivals	Conflicting occupancy	Unsafe departure
None	True	2	0	0
RadioPartition	True	0	0	0
RadioReconnection	True	2	0	0
ControllerRestart	True	2	0	0
VoterRestart	True	2	0	0
FrozenOutput	True	0	0	0
IntegrityLoss	True	0	0	0
PositionUncertainty	True	0	0	0
PointsDetectionLost	True	0	0	0
ChargerStuck	True	0	0	0
IncompleteCharge	True	0	0	0
ChargerMisaligned	True	0	0	0
EmergencyDepartureConnected	True	0	0	0
CivilClosure	True	0	0	0
CivilDataStale	True	0	0	0
MissingChannelA	True	0	0	0
MissingChannelB	True	0	0	0
ChannelDisagreement	True	0	0	0
ChannelReplay	True	0	0	0
OutputFeedbackFailure	True	0	0	0

Open release stages

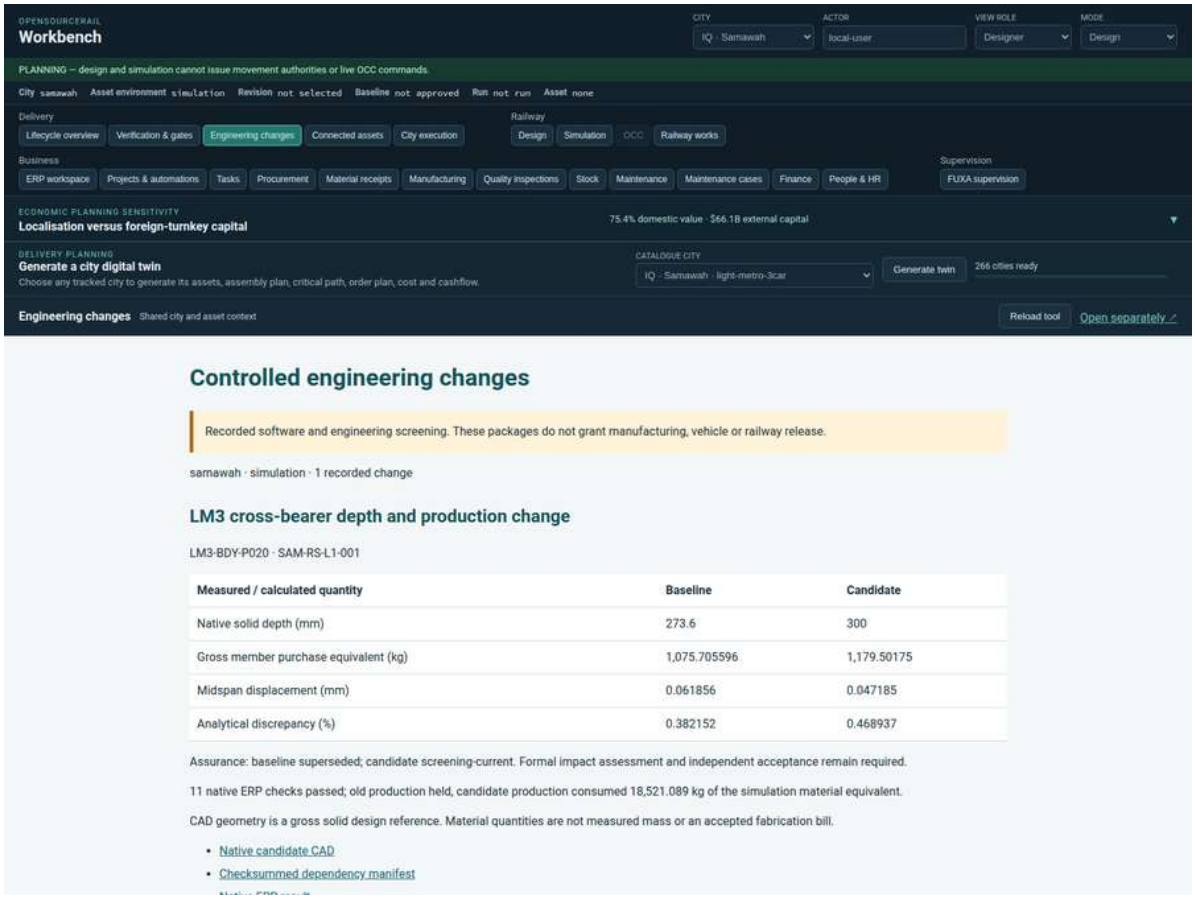
- Independent software/requirements review and Rust/formal refinement
- Hardware-in-the-loop with real output, clock, durable epoch, radio and proving interfaces
- Closed-track measured braking/integrity and recovery
- Independent assessment, approved operating/maintenance procedures and authenticated deployment release

Shared connected-engineering FMEA retains unknown occurrence evidence, unreviewed residual risks and blocked decisions. Source changes stale the execution evidence and traverse proofs, simulations, procedures and approval nodes.

Controlled engineering changes

Repository source: `engineering/changes/README.md`

Workbench Engineering changes connects retained native FreeCAD geometry, CalculiX screening, quantities, submitted ERPNext BOMs, production disposition, manufacturing transactions and superseded evidence. City and environment selection filter the recorded packages. Native document links are enabled only for the recorded ERP deployment; authentication still applies. Search links include each run's unique project/item identity so a fresh database at the same address cannot substitute reused sequential record IDs.



Workbench CAD-to-production evidence

Executed Samawah example

The retained bundle changes one cross-bearer in tracked `LM3-BDY-P020.FCStd`, reopens the candidate, checks all other leaf solids remain unchanged, and executes three CalculiX meshes for each revision. The canonical CAD remains unchanged. FreeCAD 1.1.4 and CalculiX 2.23 executed this example; complete decks, solver output, native candidate and checksums are retained.

Quantity	Baseline	Candidate
Native depth, mm	273.6	300
Gross member purchase equivalent, kg	1,075.705596	1,179.501750
Calculated midpoint displacement, mm	0.06185623	0.04718545
Analytical discrepancy	0.382%	0.469%
Final mesh displacement change	0.192%	0.237%

The solver uses B32 quadratic beams, a 10 kN central point load and simply supported constraints. The analytical check includes rectangular-section shear deformation. It is a displacement benchmark, not a strength, fatigue, buckling, connection or complete vehicle analysis. See the CalculiX manual for the element formulation and FreeCAD's topology API for native shape operations.

The CAD uses gross solid design-reference geometry. Density times volume, including a configured 5% purchase allowance, is a simulation material equivalent; it is not measured mass or an accepted fabrication BOM. The ERP material valuation is a test value, not a procurement estimate.

The native ERP result retains eleven passing checks: two submitted child/parent BOM revisions and work orders; old nested production exposure; self-endorsement and premature outcome rejection; separate software-test reviewer endorsement;

actual old work-order stop and verification; unchanged original BOM/mapping; candidate production and material consumption. The candidate consumed 18,521.089 kg equivalent, within the native 0.0005 kg rounding tolerance of its exploded BOM. New records use an isolated project, warehouse and items in the example-city service. They remain inspectable.

The Mosul profile reuses the same workflow with a 250 mm candidate depth and its own city/asset/project. Its six solver runs produce 0.08071362 mm candidate displacement (0.302% analytical discrepancy), and the separate native ERP run passes the same eleven checks with reduced material quantities. Together the two profiles exercise both increasing and decreasing geometry through the production workflow. Neither is a city-specific structural design approval.

Reproduce or configure another city

`config/generic.json` defines the shared screening assumptions. A versioned city profile overlays city, asset and any changed assumptions. This example supports an axis-aligned rectangular cross-bearer; other geometry needs a suitable analysis adapter. Reusing this method does not validate an arbitrary component.

```
# Requires FreeCADCmd and ccx, or the org.freecad.FreeCAD Flatpak with ccx.
# Use a new output folder so previous evidence is retained.
tools/automation/osr-python tools/automation/engineering-change.py prepare \
  --profile engineering/changes/config/samawah.json \
  --output build/engineering-change/samawah-new
tools/automation/osr-python tools/automation/engineering-change.py verify \
  build/engineering-change/samawah-new

# After ./osr example-city setup; use that setup's Samawah project ID.
# This creates and submits simulation records in the example ERP database.
tools/automation/osr-python tools/automation/engineering-change.py execute \
  build/engineering-change/samawah-new --reference-project PROJ-0001 \
  --report build/engineering-change/samawah-native.json
```

Use `--compose-project`, `--compose-file` and `--site` for another isolated ERP deployment. The reference project's city must match the profile. Execution refuses changed dependencies, missing/extra/altered bundle files or release claims. Reports must be new files outside the sealed bundle. Private execution diagnostics remain under `var/engineering-change/`; public reports contain record identifiers, checks and source hashes. Checkpoint evidence before removing example volumes.

To publish a new recorded example, retain its sealed bundle and native report and add a city/environment entry to `catalogue.json` with the matching ERP origin. The current receipts were rerun in a separate `osr-design-consistency` ERP site, reachable at `http://127.0.0.1:8284` when that local test site is running. Previous examples remain in Git history. Recorded results are historical observations; verification checks source currency, not the present state of mutable ERP records. CI exercises the retained bundle against a fresh example ERP and uploads its new report. Native CAD regeneration requires the tools above and is separately reproducible.

Acceptance still required

The old evidence is retained as `superseded`; the new result is `screening-current`, bound to CAD, quantities and configuration hashes. Neither status grants engineering or physical release. Software-test identities verify role separation but do not constitute independent engineering reviewers. Formal impact assessment, supplier/material selection, fabrication drawings, connection/load/fatigue checks, measured mass, first-article evidence and independent acceptance remain explicit gates. This example closes the connected software journey for one design-reference member, not a complete released subassembly.

Civil system exploration

Repository source: `engineering/civil_exploration/README.md`

This workbench investigates complete span–pier–foundation packages using the existing component catalogue and native OpenSeesPy. Its first campaign compares Pi20/Pi25 controls, a hollow deck, a hollow stepped tapered pier, and their combination over 100 m of double-track viaduct. Existing canonical designs, costs, production records and engineering release statuses remain separate.

From the repository root, using the normal engineering Python environment and native `ccx` installed for the existing reference replay:

```
tools/automation/osr-python tools/automation/civil-study.py run --deployment reference
tools/automation/osr-python tools/automation/civil-study.py verify \
  build/engineering/civil-studies/first-campaign
```

`run --deployment reference` freezes the input contract, copies governing source files, reproduces the existing native beam/drainage reference, verifies independent new benchmarks, executes isolated bounded workers and writes `comparison.md`, `comparison.csv` and `comparison.json`. The repository commit and source hashes identify the baseline even when the implementation has not yet been committed. Set `--output` to a fresh directory for each campaign.

Use `run --deployment reference --resume` to reuse only hash-verified results from the same study, code and native environment. `--retry-failed` creates another attempt while retaining the failed one. A terminated attempt's job/log files remain in its attempt directory. `verify --historical` checks retained artifact binding without claiming that old dependencies are current. A checksum is not a reviewer signature.

Full investigation programme

```
tools/automation/osr-python tools/automation/civil-study.py programme \
  --deployment reference --search-evaluations 96 --output build/engineering/civil-studies/programme
```

This executes numerical component benchmarks, eight registered deck families, CAD/IFC quantity checks, commercial/equipment adapters, continuity/thermal cases, a coupled sprung/unsprung train diagnostic, three-seed constrained optimisation with an equal-budget random baseline, declared uncertainty scenarios, three actual-section solid meshes per detailed shortlist, physical-test protocols, artifact retrieval and a sealed promotion proposal. It also repeats the full reference transient campaign. `--quick` omits that last repeat.

`programme.json` and `programme.md` audit C01–C14 separately from engineering acceptance. Actual suppliers, measured materials/soil, physical tests and named independent acceptance are still required; the software cannot manufacture them.

More realistic response models

The shared trainset–viaduct demonstrator adds one instance definition for family geometry, mass/inertia, joints, native Assembly placement, ERP inspection identity and coupled car/bogie/rail/multi-span response. It retains explicit synthetic inputs, source-bound change propagation and independent convergence/native equation checks. Supplier/physical release remains separate from the numerical demonstration.

The complete `system-study` workflow now compares ten beam, six pier and nine foundation layouts, GFRP/CFRP matrices, material/geometry variants, continuity and construction methods. It uses 300 m of double track and a distinctly labelled static Baghdad planning sensitivity, without a substituted axle array. It reruns component checks, searches cost/mass/working-time Pareto tradeoffs, retains failures/checkpoints, and confirms a diverse shortlist with three native solid meshes and three 3D frame meshes. The frame includes separate tracks, flexible transverse caps, finite bearing roll and coupled foundation stiffness; single-track and braking sensitivities are retained. Composite shear-slip diagnostics show dependence on still-unmeasured connection properties.

```
tools/automation/osr-python tools/automation/civil-study.py system-study \
  --output build/engineering/civil-studies/baghdad-complete-systems-v2
tools/automation/osr-python tools/automation/civil-study.py system-verify \
  build/engineering/civil-studies/baghdad-complete-systems-v2
```

This writes a self-contained `report.html`, CSV/Markdown comparisons, section and Pareto SVGs, candidate/input/source records, native fields, method-specific CPM schedules and complete cost rows. Actual Baghdad prices remain unknown. Three synthetic rate/productivity cases give conditional comparisons; their hypothetical crane tiers are not crane charts. Empty/segment shipping and lifting units differ from service mass. Precast pile production, delivery/splices and site concrete transport are included. Carbon totals remain unknown without controlled EPDs. Read the full workflow and remaining acceptance.

Use `--resume` only with matching frozen inputs/source/native hashes. `--skip-detail` supplies reduced evidence only and cannot create the retained confirmed review. `--evaluations-per-method`, `--seeds` and `--shortlist` bound pilots; the full default uses three seeds, equal random/evolution attempt budgets and 24 detailed packages. Numerical-library threads

are fixed at one; every CalculiX call has a 180-second limit and retains failures. Search wall/evaluation limits are in `config/system-options.json`.

Export a fully checked review, optionally updating the requested root README:

```
tools/automation/osr-python tools/automation/export-civil-system-review.py \
  build/engineering/civil-studies/baghdad-complete-systems-v2 --update-readme
```

All C01–C14 engineering acceptance gates remain open. The new workflow does not give material capacity, prestress, cyclic/fire/fatigue, surveyed ground or actual train dynamics an invented pass. The existing reference models below retain their original scope; their limitations should be read separately from the additional 3D and coupled-pile diagnostics.

- Actual disjoint section regions drive CAD, quantities, area, centroid and both bending inertias. Solid end diaphragm regions are counted once and have their own stiffness/mass. The hollow deck is a research box, not a qualified product.
- Native elastic Timoshenko elements include shear deformation, consistent mass and rotary inertia. Each span retains its own rotations and finite horizontal and vertical bearings. Shared piers, rigid cap offsets, pier self-weight and foundation-cap mass complete the 2D load path.
- Two explicitly uncalibrated ground stiffness scenarios provide finite axial, lateral and rotational compliance. Illustrative pile-group dimensions support equal-scope takeoff; they do not establish geotechnical resistance.
- The actual retained 12-axle LM3 AW2 planning pattern crosses both tracks synchronously. Static positions, Newmark moving-force histories at two speeds, modal shapes, acceleration and braking response are retained. The reference moving-force campaign does not include suspension feedback; the separate coupled-vehicle diagnostic does. Asymmetric loading and Baghdad six-car geometry need their actual supplier inputs. A mesh-independent 2 m triangular rail-distribution footprint avoids an instantaneous axle-load jump at model entry. The footprint is an explicit unverified track assumption; load and moment conservation are independently checked and its physical applicability needs validation.
- Handling models use the fabricated beam's nonuniform mass and overhanging supports at 20/80% span, rather than the service bearing arrangement. The complete suspended study mass includes a separate rigging allowance.

Three static/handling meshes and independent dynamic mesh/time refinements check response sensitivity. Failure of a declared convergence tolerance remains a failed numerical screen. Research lift limits are separate provisional constraints. Null railway limits and quotations remain unresolved. Elastic fibre stress is a gross-section response, not prestressed capacity or fatigue evidence.

Independent benchmarks cover Timoshenko UDL deflection/reactions, point loading with support settlement, a flexible-base pier, Rayleigh-limit modal frequencies, and moving-force histories against analytical modal superposition. See `benchmarks`. These verify equations and implementation; they provide no physical validation.

Contracts, history and resource limits

`config/reference.json` owns the initial input values. `schemas` publish strict study and candidate contracts. Unknown keys, wrong unit field names, booleans as measurements, nonfinite values, duplicate JSON keys, incompatible geometry and stale input hashes fail closed.

Design IDs bind configuration and geometry. Evaluation IDs additionally bind the complete study, scenario, analysis implementation and native library bytes. Results and review state are separate records. Each attempt retains its job, compressed native response fields, modal shapes, governing conditions, convergence checks, logs, failures and output hashes. The append-only ledger is also hash-bound.

Modify a design by creating a definition containing only `deck` and `pier`, then:

```
tools/automation/osr-python tools/automation/civil-study.py derive \
  --parent build/engineering/civil-studies/first-campaign/candidates/CANDIDATE_ID.json \
  --definition my-revised-definition.json --output build/engineering/civil-studies/child.json \
  --reason 'Investigate a thicker bottom flange'
```

The child retains its parent ID and change reason; the parent is untouched. Put the child definition, `parents` and `modification_reason` in a new campaign's candidate row and the complete parent records in `lineage_records`. Ancestors are validated and retained; missing parents and lineage cycles are rejected. Result histories can be inspected independently; native fields remain scratch artifacts under `build/engineering/`. Compact intentional review evidence may be retained under the repository artifact policy.

Workers run sequentially, use one numerical-library thread, have a 4 GiB address space limit and a registered timeout. Candidate count is checked against the campaign budget before outputs are created. Resume does not bypass source or native-environment checks. An execution failure is not structural infeasibility.

Remaining model and evidence work

The reference system is linear and planar. The programme additionally runs nonlinear Concrete02/steel fibre sections, native PySimple1/TzSimple1/QzSimple1 piles, P-delta columns, Coulomb friction bearings, orthotropic coupons and actual C3D20R solid sections. These have explicit, limited verification domains. It does not qualify prestress losses, cracking, shear/torsional resistance, connections, transverse cap bending, cyclic pier response, nonlinear pile/soil behaviour, uplift, scour, fire, fatigue, track irregularity or actual lift stability. Rayleigh damping and finite-domain entry/exit effects require further study. Actual train supplier records, material tests, soil data, quotes and independent review remain inputs to later phases.

Controlled evidence can be supplied through `evidence_refs`. A measured, supplier-verified or site-calibrated flag requires its role's source and checksum; those raw records are copied into the sealed study. Source currency is checked again after execution. Such a flag does not itself grant engineering acceptance.

Baghdad deployment inputs

Baghdad is the active qualification target. The commands `run` and `programme` default to this target, report missing evidence and exit with code 2 before any solver runs. Existing LM3 software verification requires explicit `--deployment reference`. Missing quotations remain open with unknown costs.

```
tools/automation/osr-python tools/automation/civil-study.py qualification \
  --output build/engineering/civil-studies/baghdad-qualification
```

This writes source-bound JSON and a readiness table with owners. The retained profile is six cars, 24 axles and 111 m long, with planning masses of 204 t tare, 258 t AW2 and 276 t AW3. The 384 t infrastructure allowance is an envelope; it does not define axle forces. All 54 city lines select this train family. The desktop ground inventory has 30 missing profiles and is not site calibration.

A supplier can provide a retained repository JSON record conforming to `schemas/supplier-train.json`. It must bind an original drawing/load document by relative path and SHA-256, list all 24 sorted axle positions from a leading-end datum increasing towards the trailing end, and provide positive case-specific axle forces balanced against loaded mass. Tests use synthetic records only to verify this contract; no synthetic record is installed as Baghdad input.

```
tools/automation/osr-python tools/automation/civil-study.py run \
  --train-record path/to/retained-supplier-record.json \
  --output build/engineering/civil-studies/baghdad-moving-force
```

With valid inputs this creates a six-car moving-force research profile and runs the selected geometry/ground cases. Use `--config` for a different study contract; the Baghdad pattern replaces that contract's train. `--resume` and `--retry-failed` preserve the existing attempt history and source checks. Verify the resulting `evaluation/` directory with the normal `verify` command.

The complete Baghdad `programme` remains blocked even after this record is supplied: its coupled diagnostic needs actual per-axle suspension, bogie and contact data and an appropriate adapter. Project load combinations, lateral/3D response, material/ground calibration, costs, physical validation and acceptance also remain open. Reference models are never relabelled as Baghdad qualification. See Baghdad qualification.

See the implementation and acceptance backlog. The compact retained campaign review keeps numerical summaries and exact provenance available without committing high-volume fields. The constrained optimiser uses verified reduced models within a declared hollow-deck/pier domain. It retains each mutation, failure, cache hit and source identity. Different random seeds and equal evaluation budgets remain visible; no global optimum or validated cost saving is asserted. No candidate belongs to a qualified feasible Pareto set and no result triggers catalogue, procurement or ERP release.

The native formulations follow the official OpenSees documentation for Timoshenko elements and transient analysis.

Individual tools

Command	Purpose
`prepare` / `generate`	Strict study/candidate contracts and source snapshots
`benchmark`	Concrete/steel, nonlinear pile, P-delta, friction and orthotropic/coupled-vehicle verification
`optimise`	Multi-seed Pareto evolution and equal-budget random comparison
`detail`	Actual-section CalculiX solids at three mesh sizes
`cost`	Complete quantities, controlled rate sources, crane/transport and lifecycle scenarios
`export-ifc`	Full span–pier–foundation solids and stable IFC product/quantity IDs
`archive` / `restore`	Immutable multipart checksummed storage and safe retrieval
`validate-data`	Separate calibration/holdout specimens and measurement uncertainty
`test-plan`	Candidate-specific laboratory/field protocols and unresolved owners/limits
`promotion-check`	Sealed impact proposal; existing structural release inspection when supplied

Use `--help` for each command. Search `--resume` verifies its checkpoint and replays the deterministic trajectory using saved results; modified inputs/code or tampered files cannot silently reuse it. Reference retries consume the registered attempt budget. Commercial totals remain null unless every scope item and currency conversion is priced; quoted sources are hash-checked.

Actual-section solids are linear elastic, with explicitly idealised supports and a bulk steel mass allowance. Nonlinear sections, pile response and friction are separate diagnostics; they do not confer full-system nonlinear or physical qualification. Unmeasured material matrices and range scenarios are labelled research inputs. Hybrid bond/slip, calibrated cyclic degradation, scour and 3D vehicle lateral interaction need project data and model selection.

The service-envelope workflow adds full represented-pair passage coverage, individual body histories and unweighted ride diagnostics. It distinguishes executed cases from plans and keeps full-family, ISO comfort and physical acceptance open. Earlier published reviews retain their producing source commit and can be checked with `tools/automation/verify-shared-review-history.py`.

Retained civil campaign review

Repository source: `engineering/civil_exploration/examples/README.md`

`first-campaign.json` is the compact intentional review output of `tools/automation/civil-study.py run --deployment reference`, exported with:

```
tools/automation/osr-python tools/automation/export-civil-study-review.py \
  build/engineering/civil-studies/first-campaign
```

Its named consumer is `engineering/analysis/tests/test_civil_exploration_review.py`. It retains the frozen inputs, exact dependency/native identities, independent benchmarks, quantity/response summaries, convergence, failures and native output checksums. Full native fields remain under `build/engineering/civil-studies/`; regenerate them with the workbench. These are research results with unresolved engineering feasibility and no physical or operating release.

Run the exporter with `--check` against the same sealed bundle to verify this compact derivative. Historical review inputs stay explicit; future material, train or model changes require a new campaign and intentional fixture refresh.

`programme-review.json` and `programme-review.md` retain the expanded C01–C14 software audit, component benchmarks, multi-seed search, range sensitivity, native solid/nonlinear shortlist checks, commercial unknowns, safe retrieval and the blocked promotion proposal. Generate them with:

```
tools/automation/osr-python tools/automation/export-civil-programme-review.py \
  build/engineering/civil-studies/completed-software-programme
```

The named consumer is `engineering/analysis/tests/test_civil_exploration_programme_review.py`. Every external acceptance item remains visible; actual physical measurements, supplier commitments and authority signatures are not supplied by these records.

`baghdad-qualification/qualification.json` and `.md` retain the user-selected deployment readiness and source hashes. They are input-gap records, with no solver results or release claim. Their generator is `civil-study.py qualification` and their named consumer is `engineering/analysis/tests/test_civil_qualification.py`. Regenerate into a fresh scratch directory, compare to the retained pair, and refresh intentionally if the controlled inputs change.

`complete-system-review.json` and `.md` retain the expanded whole-package search, source/native identities, conditional cost/mass/time leaders, failed/provisional outcomes, diverse 3D/solid confirmations and all external gates. `system-choices.svg` shows canonical sections/footprints; `pareto.svg` shows synthetic scenario tradeoffs. Actual prices remain null and no qualified/global optimum is claimed. The named consumer is `engineering/analysis/tests/test_civil_system_choices.py`; the generator is:

```
tools/automation/osr-python tools/automation/export-civil-system-review.py \
  build/engineering/civil-studies/baghdad-complete-systems-v2 --update-readme
```

Baghdad civil qualification input readiness

Repository source: `engineering/civil_exploration/examples/baghdad-qualification/qualification.md`

Target: Baghdad, six cars, 24 axles, 111 m retained planning length. Planning masses: 204 t tare, 258 t AW2 and 276 t AW3.

Supplier axle positions and loaded distribution are required before a Baghdad moving-load run. LM3 axle spacing is never repeated or relabelled.

Status: `blocked-missing-supplier-axle-inputs`. Physical release and engineering qualification remain false.

Input	Owner	Status	Required evidence
TRAIN-AXLE-POSITIONS	rolling-stock supplier	open	six-car supplier drawing with all 24 axle positions
TRAIN-LOADED-DISTRIBUTION	rolling-stock supplier/structural lead	open	case-specific loaded axle forces and balanced loaded mass
TRAIN-SUSPENSION	rolling-stock supplier	open	per-axle sprung/unsprung masses, suspension, contact and bogie/articulation relationships
TRAIN-OTHER-CASES	supplier/structural lead	open	AW3, infrastructure, braking/rescue/maintenance, lateral and dynamic combinations
GROUND-CALIBRATION	geotechnical lead	open	support-zone boreholes, calibrated axial/lateral/group/settlement parameters and pile tests
MATERIAL-AND-TENDONS	materials/structural lead	open	actual concrete/composite batches, reinforcement, tendons, anchorage and losses
CODES-AND-LIMITS	engineers of record	open	adopted Iraqi/project standards, load combinations and accepted limits
SURVEY-AND-ERECTION	survey/construction leads	open	accepted alignment/support locations, temporary stages, actual rigging/routes and equipment charts
QUOTATIONS	cost/manufacturing owner	open	Baghdad-local supplier quotations, productivity and maintenance/replacement inputs
PHYSICAL-AND-INDEPENDENT	independent checker/authority	open	instrumented validation, independent review and existing controlled authority acceptance

The 100 m double-track comparison segment is separate from the complete city inventory. Desktop soil screens are not calibrated site resistance. Existing LM3 studies remain software verification references. The coupled programme requires supplier suspension/bogie data and a suitable per-axle adapter.

Complete viaduct system exploration

Repository source: `engineering/civil_exploration/examples/complete-system-review.md`

Baghdad target. Common scope: 300 m, double track. 731 distinct packages, 779 evaluation attempts, 175 provisional screening passes.

Actual supplier costs are unknown. No engineering-qualified cheapest design or global optimum is claimed. The static load is the retained infrastructure full-train allowance spread over the retained train length. It is not an actual axle pattern or dynamic envelope.

Winner basis: converged native shortlist and all shared provisional constraints. Price and productivity scenarios are synthetic assumptions; masses follow canonical material regions and full package quantities. Material strength/prestress/fatigue capacity remains unresolved. Gross stresses are retained without an invented material-independent acceptance limit.

Scenario	Objective	Beam / pier / foundation / method	Installed cost USD	Installed mass t	Working days	Largest lift t
concrete-first	cheapest	U-girder 20 m / lightweight / hollow-tapered / CFA-six / in-situ / link-slab	1,774,796	6,504.4	120	0.0
concrete-first	lightest	hybrid-shell 20 m / high-performance + GFRP / hollow-prismatic / square-driven-six / precast-full / link-slab	7,374,804	5,499.2	154.625	43.3
concrete-first	fastest	uhpc-ribbed 20 m / high-performance / hollow-prismatic / CFA-six / precast-full / simple-span	2,640,915	7,609.1	63	64.9
lightweight-industrial	cheapest	U-girder 20 m / lightweight / hollow-tapered / CFA-six / in-situ / link-slab	1,723,577	6,504.4	120	0.0
lightweight-industrial	lightest	hybrid-shell 20 m / high-performance + GFRP / hollow-prismatic / square-driven-six / precast-full / link-slab	4,034,997	5,499.2	137.5	43.3
lightweight-industrial	fastest	uhpc-ribbed 20 m / high-performance / hollow-prismatic / CFA-six / precast-full / simple-span	2,320,060	7,609.1	63	64.9
resource-constrained	cheapest	conventional-I 25 m / lightweight / hollow-tapered / CFA-six / precast-full / continuous	2,061,077	8,033.3	92.375	64.5
resource-constrained	lightest	hybrid-shell 20 m / high-performance + GFRP / hollow-prismatic / square-driven-six / precast-full / link-slab	7,977,508	5,499.2	214.25	43.3
resource-constrained	fastest	uhpc-ribbed 20 m / high-performance / hollow-prismatic / CFA-six / precast-full / simple-span	2,853,243	7,609.1	78.125	64.9

Family coverage

Beam family	Native completed packages	Model / evidence maturity
pi	105	Research; gross elastic properties, actual supplier details and physical validation open
hollow-box	78	Research; gross elastic properties, actual supplier details and physical validation open
conventional-I	77	Research; gross elastic properties, actual supplier details and physical validation open
U-girder	114	Research; gross elastic properties, actual supplier details and physical validation open
ribbed-deck	42	Research; gross elastic properties, actual supplier details and physical validation open
uhpc-ribbed	130	Research; gross elastic properties, actual supplier details and physical validation open
hybrid-shell	41	Research; gross elastic properties, actual supplier details and physical validation open
segmental-box	41	Research; gross elastic properties, actual supplier details and physical validation open
steel-composite-I	66	Research; gross elastic properties, actual supplier details and physical validation open
frp-composite-I	37	Research; gross elastic properties, actual supplier details and physical validation open

Detailed confirmation

24 shortlisted packages; 24 completed converged three-mesh solids and 3D frame checks. 3D models include two separate tracks, flexible transverse hollow caps, finite bearing translations/roll and coupled foundation translation/rotation. Both-track and single-track uniform loading are retained at three deck meshes for both soil scenarios. Composite connector sensitivity compares noncomposite, partial interaction and strong-connector behaviour; connector values are unmeasured assumptions. These checks do not supply bond failure or cyclic/fire evidence.

Document work packages and remaining acceptance

Package	Implemented software / retained output	External acceptance still required
C01	source-bound Baghdad planning inputs and explicit qualification blockers	project inputs, applicable domain and responsible independent engineering acceptance
C02	strict candidate inputs, package/candidate IDs, immutable results and modification events	project inputs, applicable domain and responsible independent engineering acceptance
C03	ten beam, six pier, nine foundation options, material compatibility and bounded search	project inputs, applicable domain and responsible independent engineering acceptance
C04	shared disjoint CAD/IFC solids, net material quantities, shipping/lift unit centres	project inputs, applicable domain and responsible independent engineering acceptance
C05	analytical/native component checks, foundation reciprocity/energy and three-level refinement	actual instrumented tests, laboratories, independent reviewer and acceptance
C06	native CalculiX nodal/stress parsers and complete 3D force/displacement fields	project inputs, applicable domain and responsible independent engineering acceptance
C07	native nonlinear component benchmarks and coupled geometry-specific elastic pile groups	project inputs, applicable domain and responsible independent engineering acceptance
C08	uniform planning sensitivity, separate 3D tracks/caps, method/stage schedules and construction units	project inputs, applicable domain and responsible independent engineering acceptance
C09	complete material/plant/labour/overhead/maintenance/replacement bills and three conditional scenarios	project inputs, applicable domain and responsible independent engineering acceptance
C10	bounded seeded runs, checked checkpoints, sealed raw data, multipart archive and verified restore	project inputs, applicable domain and responsible independent engineering acceptance
C11	equal-budget random/evolution search, Pareto sets, scenario reversals and a diverse shortlist	project inputs, applicable domain and responsible independent engineering acceptance
C12	three-mesh actual-section solids, orthotropic materials and connector-slip/torsion diagnostics	project inputs, applicable domain and responsible independent engineering acceptance
C13	candidate-specific instrumented physical-test protocols and existing calibration/holdout workflow	actual instrumented tests, laboratories, independent reviewer and acceptance
C14	sealed change proposal and pointers to the existing controlled change/structural release process	actual instrumented tests, laboratories, independent reviewer and acceptance

Open qualification inputs

- TRAIN-AXLE-POSITIONS (rolling-stock supplier): six-car supplier drawing with all 24 axle positions — open.
- TRAIN-LOADED-DISTRIBUTION (rolling-stock supplier/structural lead): case-specific loaded axle forces and balanced loaded mass — open.

- TRAIN-SUSPENSION (rolling-stock supplier): per-axle sprung/unsprung masses, suspension, contact and bogie/articulation relationships — open.
- TRAIN-OTHER-CASES (supplier/structural lead): AW3, infrastructure, braking/rescue/maintenance, lateral and dynamic combinations — open.
- GROUND-CALIBRATION (geotechnical lead): support-zone boreholes, calibrated axial/lateral/group/settlement parameters and pile tests — open.
- MATERIAL-AND-TENDONS (materials/structural lead): actual concrete/composite batches, reinforcement, tendons, anchorage and losses — open.
- CODES-AND-LIMITS (engineers of record): adopted Iraqi/project standards, load combinations and accepted limits — open.
- SURVEY-AND-ERECTION (survey/construction leads): accepted alignment/support locations, temporary stages, actual rigging/routes and equipment charts — open.
- QUOTATIONS (cost/manufacturing owner): Baghdad-local supplier quotations, productivity and maintenance/replacement inputs — open.
- PHYSICAL-AND-INDEPENDENT (independent checker/authority): instrumented validation, independent review and existing controlled authority acceptance — open.

Further model domains

- moving/coupled Baghdad train requires supplier records.
- nonlinear 3D cyclic/seismic and scour/group calibration.
- reinforcement/prestress/joint/contact capacity and stage locking.
- physical fire/durability/fatigue and independent acceptance.
- V-piers/Y-caps: inclined/multiple-leg 3D cyclic and cap-joint qualification; requires-purpose-specific-adapter-and-data.
- ground-improvement-shallow: site treatment trial/calibration and inclusion-platform load-transfer model; requires-purpose-specific-adapter-and-data.
- recycled-primary-members: qualified material fatigue/durability datasets; secondary nonstructural use remains separate; requires-purpose-specific-adapter-and-data.
- haunched-variable-section: supplier section/tendon/diaphragm detailing and stage qualification; requires-purpose-specific-adapter-and-data.

Actual soil, supplier train records, material/connection details, quotations, physical tests and independent acceptance determine whether any shortlisted package is adoptable. Physical and operating release remain false.

C01–C14 executed software and remaining acceptance

Repository source: `engineering/civil_exploration/examples/programme-review.md`

This audit records the executed research software. It does not close physical or engineering acceptance.

ID	Software evidence	Remaining acceptance
C01 — project requirements and actual loads	implemented research workflow	adopted project codes/limits; actual train/ground/material inputs; named acceptance owners
C02 — contracts and lineage	implemented research workflow	independent contract/configuration review
C03 — family compatibility/material matrices	implemented research workflow	supplier sections, material properties and connection qualifications
C04 — CAD/IFC and quantities	implemented research workflow	accepted reinforcement/tendon/fabrication drawings
C05 — numerical benchmark domains	implemented research workflow	measured model applicability and blind physical validation
C06 — native result exchanges	implemented research workflow	accepted project output specifications and limits
C07 — nonlinear pier/soil components	implemented research workflow	site calibration, nonlinear group/scour/liquefaction/seismic selection
C08 — moving/thermal/continuity/construction cases	implemented research workflow	supplier suspension/axle records and site-specific 3D/erection cases
C09 — complete package pricing/equipment/lifecycle	implemented research workflow	actual charts, routes, productivity, quotations and maintenance/replacement costs
C10 — runner/CI/artifact retrieval	implemented research workflow	project custody/retention and independent environment replay
C11 — multi-seed constrained search and random baseline	implemented research workflow	accepted constraints, costs and candidate-specific qualification
C12 — native solids/material/joint refinements	implemented research workflow	calibrated interface/anchorage/cyclic/deterioration models and independent checks
C13 — physical datasets/tests and independent validation	implemented test/data workflow	laboratories, specimens, procurement and actual instrumented tests
C14 — sealed proposal and existing release gate	implemented research workflow	candidate-specific physical evidence, independent review and authority acceptance

Native reference attempts: 10; component checks: 10; unique native search designs: 540.

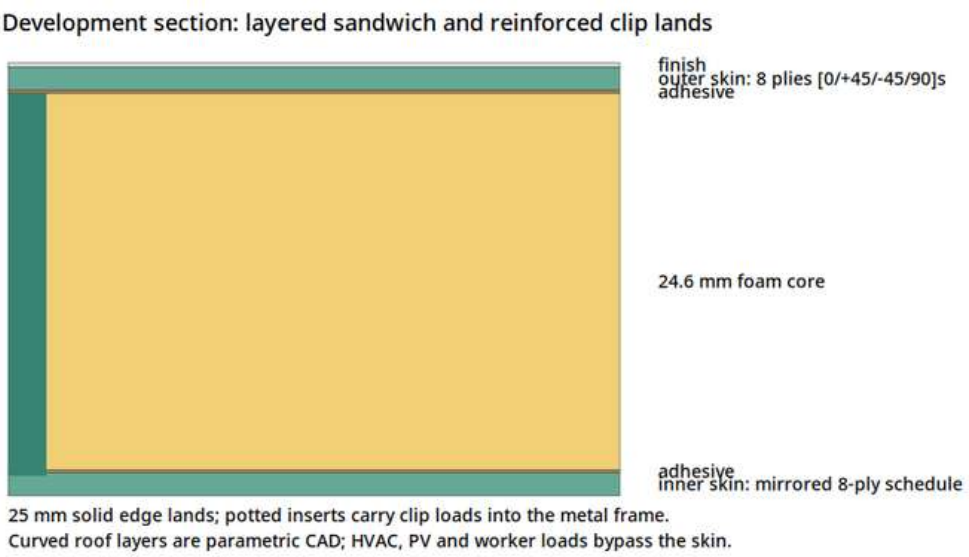
Actual physical results received: zero. The sealed promotion proposal remains blocked.

The companion `programme-review.json` retains exact source and native identities, evidence hashes, constraints, failure/domain states and conditional numerical results.

Hybrid GFRP exterior-panel development

Repository source: `engineering/composite_panels/reference/README.md`

2592 conditional panel/support combinations; no candidate is released or changes the train mass/BOM.



Layered development section

Interactive comparison · Complete-scope RFQ prepared for review.

The metal frame remains the main structural load path. HVAC, PV, lifting and personnel loads pass into its dedicated supports. The reference section has two 1.5 mm eight-ply skins, two 0.2 mm bonds, 24.6 mm core and a separate 0.3 mm finish. Material values, pressure, ageing factors and roof crown are explicit development assumptions.

Complete comparison varies skin, core depth/density, module width, clip/support span, roof crown and hat-rib depth. Rib mass is counted; rib stiffness and bond benefits await evidence. Mass includes solid edge lands, potting, inserts, clips, retention, seals and finish. Curvature earns no stiffness credit. Classical laminate ABD and one-way pressure-strip bending plus core shear are numerical screens, not plate/clip fatigue or debonding qualification. Density selects a coupled supplier-grade property record at 23 C. Every support pitch generates steel rails, brackets and fasteners with installed mass; grade-specific hot/wet, fire and attachment qualification remains open. An 85 C roof exceeds the selected grades at their 70 C continuous-temperature datum and cannot pass thermal selection.

Reference panels · Actual aperture-based 144-module inventory · Steel/aluminium comparisons · All candidates · Shared support geometry. Shared-joint seal quantities remain a gross allowance; the cutout clip map and final roof crown remain unreleased. Bare 2 mm metal sheets are unequal-performance comparators. Separate metal plate assemblies are thickness-sized to the same pressure/support/deflection surrogate and include common attachment, finish and insulation allowances. This is not an optimum metal stiffener design or demonstrated equivalent rail behaviour. No whole-train saving is claimed.

Prototype/quotation status derives from the editable evidence register. Record candidate ID, dry/hot-wet condition, metre-valued measured deflection, uncertainty and original-file SHA-256 for correlation. Tests must reproduce the candidate pressure/support/material configuration. Quotations require the complete named scope. Empty records stay unknown; the $11,000 B6/B7 allowance is a comparator, not a quote.

Coupon tests precede a clipped side module and an equipment-adjacent roof bay. The register covers aged stiffness/bond/pull-out, pressure, clip loss, vibration/fatigue, impact, heat beneath PV, ingress, UV/sand/wash, bonding and measured replacement. Fire requirement set/hazard level and the exact complete stack remain unfrozen.

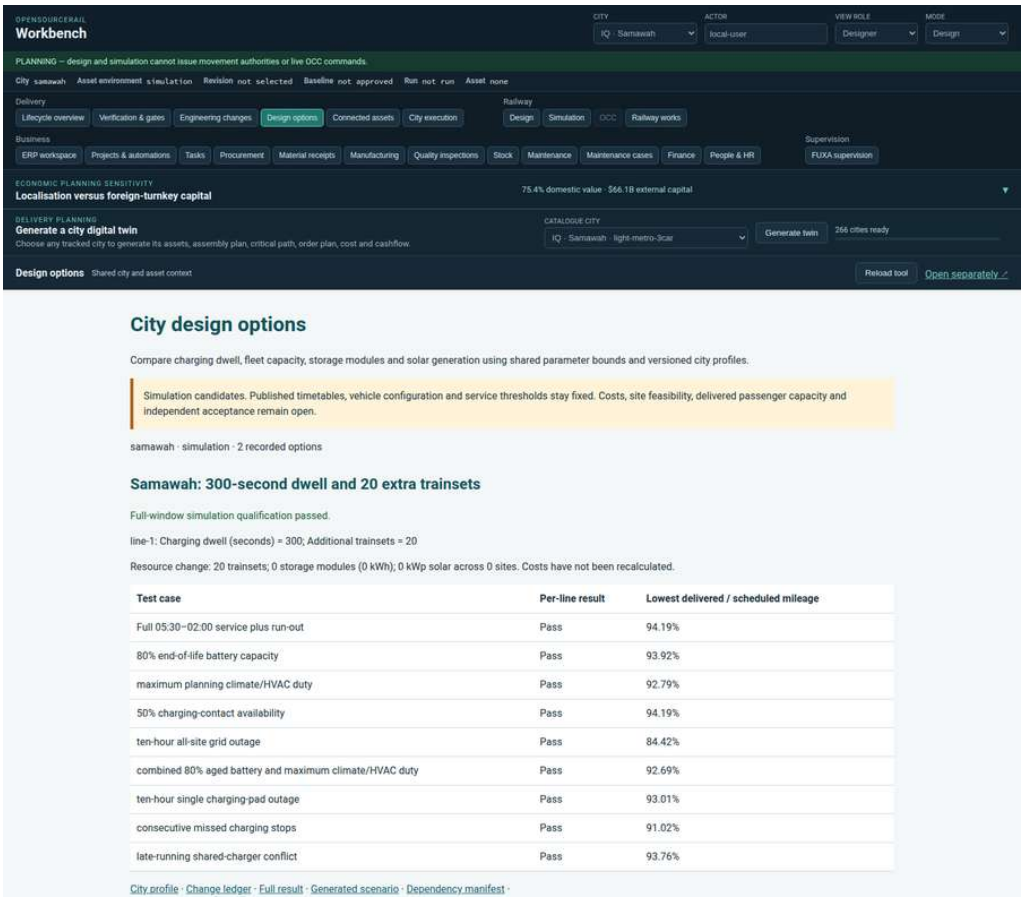
Regenerate: `.venv/bin/python tools/automation/composite-panel-study.py`; use `--check` for source/output drift. Native layered side and curved roof CAD validates nonoverlapping solids and independent volumes; curved roof insert lands remain separate development. Native CAD export/volume verification uses `tools/automation/verify-composite-panel-cad.py` in FreeCAD. Native exports do not establish physical performance.

Supplier and standards references: DIAB coupled core grades, May 2026, BSI EN 45545-2:2020+A1:2023, Sicomin SR1124, its datasheet, AIREX T90, Mat4Rail results. Supplier resin Tg and core thermal stability are separate from allowable panel service temperature.

City design options

Repository source: `engineering/design-options/README.md`

Workbench Design options compares recorded, city-scoped simulation candidates. A shared parameter policy supplies defaults and bounds; small city profiles select changes by line. The same workflow works with any catalogue city supported by the scenario generator. Canonical city packages remain controlled baselines until their separate promotion requirements are satisfied.



Workbench city design options

Adjustable design

Setting	Effect	Review consequence
`charging_dwell_seconds`	Charging dwell at powered stops on the selected line	Longer journeys, station occupation, timetable and fleet circulation
`additional_trainsets`	Adds service-rotation trainsets to that line's fleet	Purchase, stabling, maintenance, staffing and energy demand
`additional_storage_modules`	Adds whole modules at every energy site on that line, using each site's own module size	Battery footprint, fire protection, lifecycle and purchase costs
`additional_pv_kw`	Adds solar nameplate capacity at every energy site on that line	Land/roof area, grid connection and calibrated solar resource

The bounds are software exploration limits, not approved hardware ratings. Shared charger/discharge power limits remain effective: extra kWh increases endurance, not charger power. Every site is retained when explicit overrides are generated; other lines' settings remain unchanged. Duplicate depot planning quantities stay consistent with their site. Unsupported keys, unknown lines, non-finite values, fractional modules and ignored settings are rejected.

Published timetable windows/headways, vehicle configuration, reserve policy and qualification thresholds remain fixed. Additional trains cannot reduce scheduled mileage in the denominator. Nominal and seven degraded cases retain the 90% floor; the ten-hour all-site grid outage retains its existing 60% floor. The existing 0.2 percentage-point numerical tolerance is unchanged. Full-window per-line mileage is still not a measured peak-headway or passenger-capacity guarantee.

Reproduce a candidate

Requires the normal OSR Python and Rust simulation toolchains; no additional service or Python package is needed. Run from the repository root:

```
tools/automation/osr-python tools/automation/design-option.py prepare \
  --profile engineering/design-options/config/mosul-energy.json \
  --output build/design-options/my-mosul-energy
OSR_RESILIENCE_JOBS=4 tools/automation/osr-python \
  tools/automation/design-option.py qualify build/design-options/my-mosul-energy
tools/automation/osr-python tools/automation/design-option.py verify \
  build/design-options/my-mosul-energy
```

Qualification runs 90,000 simulated seconds for nominal service and each of the eight degraded cases. It returns exit 1 for a failed qualification and preserves the result. Use a new output folder for each experiment; existing evidence is not overwritten. Runtime depends on city size and available CPU resources.

To configure another city, copy a profile, set its catalogue slug and option ID, and specify only the required line overrides. `"lines": {}` is a generator-baseline control. An omitted setting uses the generic default (existing dwell or zero added resources). `prepare` resolves and checks the actual generated scenario before writing its bundle. Generated inline TOML is a complete machine-readable snapshot; the JSON profile is the human-editable source.

Each bundle contains its profile, generic policy, resolved settings, change ledger, generator baseline, candidate design/scenario and dependency manifest. Qualification binds the report to these files, Rust/generator/template dependencies and the native simulator hash. `verify` checks current source currency; `verify --historical` checks retained artifact/result binding without asserting current source currency. Checksums detect changes; they are not independent signatures or engineering approval.

To retain a result, copy its complete bundle to `examples/<option-id>` and add a city/environment entry to `catalogue.json`. Workbench verifies artifact/report hashes and scope before displaying it. It shows failure results as failures and hides other cities and physical environments. Baseline differences from the current generator are explicitly flagged. Resource deltas are recorded; canonical cost figures are not recalculated or accepted for the candidate.

Recorded experiments

See results and design tradeoffs for executed trials and resource impacts. Passing candidates are simulation evidence, not canonical promotion.

Closing the remaining programme tasks

Remaining task	Design or execution route	Closure evidence
City service shortfalls	Versioned dwell, fleet and energy options, followed by the unchanged full qualification	Per-line nominal/degraded results, then observed headways and calibrated demand
FUXA/full-city concurrency	Existing configurable city polling, batching and bounded workers; measure gateway, FUXA and ERP together	Simultaneous native full-city load meeting the stated latency/loss targets
Recovery/cutover	Existing isolated coordinated restore plus deployment-specific recovery objectives	Fresh-volume recovery, then accepted production cutover and measured recovery times
Remaining formal-proof timeouts	Simplify equivalent implementation/harness paths or partition properties with complete coverage	Every declared property proved with original scope, unwinding checks and exact source evidence
CAD-to-production acceptance	Portable native CAD/solver adapters, controlled BOM revisions and production disposition	Supplier/material choices, full load cases, inspections and independent engineering review
Vehicle, station and civil packages	Reusable package structure with city/site-specific loads, interfaces and configuration	Controlled drawings, qualified analyses, physical evidence and authorized acceptance

Configuration makes alternatives repeatable. It cannot substitute for proof results, site measurements, actual recovery execution or an independent acceptance decision.

Design option results — 18 September 2026

Repository source: `engineering/design-options/results.md`

These experiments address per-line mileage failures. They do not establish peak-headway delivery, passenger capacity, multi-day energy sustainability or engineering/operating acceptance. All service thresholds and timetable windows remain unchanged. The generator baselines match the checked-in scenarios.

The original runs qualify their recorded source hashes. After the topology-adapter change, both successful options were fully requalified against the new Rust source; see the fresh bundles below. Earlier bundles remain historical evidence.

Baseline and unsuccessful charging-only trials

The retained Samawah baseline passes nominal service but fails four degraded cases on line 1. Its 270-second charging dwell is increased in two exploratory trials:

Line 1 dwell	Hot climate	Aged battery + heat	Missed charging stops	Overall
270 seconds (baseline)	87.26%	87.54%	88.34%	Failed; pad outage also fails
300 seconds	88.82%	86.83%	87.69%	Failed
330 seconds	87.64%	86.65%	87.53%	Failed

Values are delivered/scheduled line mileage. Longer stops do not monotonically improve service: charging time competes with circulation and station occupation. The charging-only trials pass the pad-outage case but leave three failures. Their 300-second and 330-second reports retain full input snapshots and validator hashes. They predate the sealed option runner; they are exploratory evidence, not promoted candidates.

Controlled fleet and energy options

The city-scoped Workbench view loads the sealed bundles listed in catalogue.json, including unsuccessful options. Each bundle records generic/city settings, effective input changes, resource deltas, exact source dependencies and the full nominal/eight-case qualification.

Samawah: fleet capacity as well as charging time

Option on line 1	Nominal	Hot climate	Aged + heat	Missed charging	Degraded cases passed
300 seconds + 12 trainsets	93.01%	89.25%	89.58%	89.95%	6/8
300 seconds + 20 trainsets	94.19%	92.79%	92.69%	91.02%	8/8

The 12-train option fails both heat cases. The 20-train option passes the complete nominal and eight-case suite across all three lines. Its line-1 fleet increases from 53 to 73; the whole city increases from 108 to 128. Each powered line-1 stop gains 30 seconds of charging dwell. The grid-outage line-1 result is 84.42%, above its separate 60% requirement. These are demonstrated alternatives, not proof that 20 trains is the minimum addition or the least-cost solution.

Mosul: outage endurance

The controlled energy option adds three storage modules and 300 kWp solar at each of nine line-2 sites: 27 modules, 13,500 kWh storage and 2,700 kWp solar in total. Existing charge/discharge rates, charger power, fleet and timetable remain unchanged. Line 2's ten-hour all-site outage delivery rises from 51.81% to 88.78%. All six lines pass nominal service and all eight degraded cases; the lowest outage result is line 4 at 62.97%, above the existing 60% emergency floor. The separate exploratory energy run also passed before the sealed workflow rerun.

Across both cities, the original six candidate suites contain 54 runs of 90,000 simulated seconds: three exploratory and three controlled. The controlled 12-train Samawah option fails; the controlled 20-train Samawah and Mosul energy options pass. Two subsequent full requalifications add 18 runs, bringing the total to eight suites and 72 runs. This remains a targeted design comparison, not an exhaustive parameter sweep, calibrated demand study or least-cost optimization.

Requalification after the topology refactor

Both successful options pass nominal service and all eight degraded cases on the new source. Every detailed nominal/degraded outcome field matches its preceding run, including service, energy, incidents and subsystem checks. Passenger-capacity configuration, timetable windows and trainset contracts also match. The fresh bundles bind the changed Rust dependencies and simulator binary, and both pass current-source `design-option.py verify`.

Workbench now points to these fresh bundles. The original Samawah and Mosul bundles remain retained for comparison. This provides whole-city regression evidence for the storage change; physical deployment and canonical promotion still require their separate gates.

Resource counts are planning deltas, not approved purchase orders. Added trainsets also add initially charged onboard energy; a 25-hour run does not establish a repeatable multi-day charging/stabling cycle. Added solar generation requires real area and calibrated weather assumptions. Extra storage remains constrained by the existing site discharge and charger power limits. Cost, maintenance, supplier, site and independent operator reviews are still required.

Shared engineering definitions and demonstrators

Repository source: `engineering/design_consistency/README.md`

These research packages link proposed manufacturing geometry, analytical reductions and civil coordination by physical part identity, revision and geometry hash. FreeCAD is the native geometry authority; IFC carries the civil arrangement and an explicitly scoped analytical deck model. They are not issued manufacturing designs or qualified infrastructure.

Demonstrator	Implemented package	Evidence still required
Clipped side bay	Trimmed ply/bond/core/solid-land regions, bored potted inserts, drilled rails, clips, brackets, fasteners and aperture frame	Local pressure/contact/peel response, joint stiffness, trim details, retention, complete-stack fire and prototype correlation
Curved roof bay	Exact circular layered solids, oriented support rails, brackets, clips and fasteners	Curved contact lands and insert arrangement; 85°C roof case fails the selected foam grades' 70°C continuous-temperature datum
Bogie–underframe interface	Hollow manufactured sills/carriers, drilled plates, dimensional bolt shanks/heads/nuts, preload/slip/separation screen, actual assembly mass/CG/inertia and reduced modes	Threads/fits, local contact/prying/weld fatigue, supplier suspension laws, equipment/fluids and measured modes; this is not a whole-train modal basis
Viaduct bay and construction front	Deck, longitudinal reinforcement, two pier/cap/bearing sets, eight piles with explicit embedded bars, tendon reservations, full LM3 twelve-axle two-track elastic sweep, lifting/jacking cases and one-mould resource calendar	Transverse cages, anchorage, prestress history/losses, cracking/shear/torsion/cyclic resistance, calibrated ground layers, lifting/local stability, crane charts and quoted costs

Generated definitions and correspondence bind SI dimensions and material regions to each representation. Native audits independently integrate occupied solids for mass, CG and inertia, reject material overlaps, reopen each FCStd, check inspection projections and reimport STEP exports. Inspection drawings remain unissued. IFC exports exact civil extrusions, verifies tessellated volumes against material quantities and defines a deck member, pin/roller connections and a dead-load activity. Its deck idealisation excludes the uncalibrated pier/bearing/ground compliance.

Actual CalculiX/Gmsh section benchmarks compare B32R BOX hollow-beam bending with independently meshed C3D10 solids at two resolutions. The outer dimensions and walls drive both models. Thin-wall torsion is labelled separately from the polar area moment; this bending benchmark does not qualify torsion or a joint. The legacy static FEA generator now also emits explicit hollow study geometry and matching BOX sections, using proposed 8 mm research walls. Historical solid-envelope results are retained in Git history; they are not evidence for the revised sections.

Interface modes transfer each physical part's mass, CG and inertia into a kinetic matrix. A rigid-motion check independently reconciles total assembly mass/inertia, and exported modes are mass normalised. Gross-section stiffness, plate reductions, rigid equipment mounts and suspension springs are explicit assumptions. No complete vehicle basis is adopted.

Joint screen calculates bolt/member elastic stiffness, retained clamping, slip resistance, separating load and a fatigue stress range. Assumed friction and prying cannot establish bolt, weld or fatigue qualification. Civil stages retain failed deflection checks and unknown detailed resistance. Construction front counts installed deck reinforcement/tendon mass, checks transport/lift limits and rejects conflicting resource assignments. The 20 m arrangement exceeds the declared truck payload; its calendar is blocked and its quoted installed cost is unknown.

Mixed search uses real core depths, integer support counts, categorical supplier grades and binary roof geometry. Installed mass and attachment-count complexity are separate objectives. Three seeds each receive an equal-budget random benchmark and a 3.5 kPa holdout. Failed mandatory checks exclude a candidate; unknown fire, fatigue, joint, manufacturing and batch data remain unknown. The conditional frontier is not a verified frontier or a global optimum. Quoted cost, lifecycle cost, repair time and controlled carbon stay unresolved.

Foam grades select coupled density, compressive modulus, shear modulus/strength, conductivity and temperature records from the DIAB May 2026 datasheet. Public nominal/minimum values do not substitute for project batches, hot/wet ageing, fire tests or supplier quotes. Normal and shear properties are independent; an isotropic constitutive fit is not implied. Roof processing-temperature limits are not continuous service allowables.

Regenerate the Python/IFC/search/solver package:

```
.venv/bin/python tools/automation/design-consistency-study.py --ifc --search --solvers
flatpak run --filesystem=/home/hayder/Documents/OpenSourceRail --command=FreeCADCmd org.freecad.FreeCAD tools/
automation/verify-design-consistency-cad.py
.venv/bin/python tools/automation/design-consistency-study.py --check
.venv/bin/python -m pytest -q engineering/analysis/tests/test_design_consistency.py
```

Install the isolated analysis dependencies from requirements.txt. Native verification requires FreeCAD with Part and TechDraw; solver benchmarks require CalculiX. The recorded tool versions describe executed research evidence, not approved toolchains. The IFC model uses IfcOpenShell's structural API; search uses pymoo's mixed-variable support. Code_Aster, MBDyn and topology optimisation remain follow-on integrations and are not reported as executed.

The clipped-panel response refinement now adds a trimmed plate, discrete attachments, actual rail bending, all single-clip-loss locations and combined pressure/fault checks, with five native manufactured arrangements. The first delivery establishes shared definitions and bounded numerical/native checks. Remaining engineering work is deliberately

concentrated in these arrangements: mesh/contact response for the trimmed laminate, calibrated joints and equipment inventory, an assembled full-vehicle modal/contact/service envelope, detailed reinforced/prestressed civil resistance and construction stability, then measured prototype correlation. Every physical-release field remains false.

Linked research demonstrators

Repository source: `engineering/design_consistency/examples/README.md`

Manifest binds the generated physical contracts and reduced responses to their sources. Correspondence reconciles dimensions/material mass, CG and inertia; detailed local stiffness and physical qualification remain open.

- Side bay and roof bay: layered parts and support hardware; the roof thermal condition fails.
- Interface, joint screen and modes: proposed geometry and bounded mechanical response.
- Viaduct, IFC, IFC audit, civil stages and construction front: coordinated research geometry, explicit deck analysis identity and unresolved ground/resistance/construction gates.
- Mixed search: three seeds, random benchmarks and independent pressure holdouts; no verified Pareto candidates.
- Native CAD and executed solvers: independent numerical geometry/response checks, without physical release.

Reproduce using the commands in the parent README. Historical civil/service campaigns retain their original bindings.

Native manufacturing research geometry

Repository source: `engineering/design_consistency/examples/cad/README.md`

Verification records per-part mass/CG/inertia agreement, nonoverlapping occupied material, reopened native documents, unissued inspection projections and reimported STEP volumes.

Assembly	Native FreeCAD	STEP
Trimmed side bay	side.FCStd	side.step
Curved roof bay	roof.FCStd	roof.step
Underframe interface	interface.FCStd	interface.step
Viaduct bay	viaduct.FCStd	viaduct.step

Part identities and dimensions come from the shared definitions. Clearances, local contact/weld/anchor details and material acceptance require further engineering. These drawings and solids do not authorize manufacture or construction.

Executed hollow-section benchmarks

Repository source: `engineering/design_consistency/examples/solver/README.md`

Receipt binds two Gmsh/CalculiX executions and their source/input/output hashes. A clamped hollow cantilever receives the same 1 kN tip force in the B32R BOX and C3D10 solid reductions.

Coarse and refined meshes compare bending response and mesh refinement. This establishes a numerical consistency screen for that load case; torsion, local contact, fatigue and physical correlation remain open.

Hollow cantilever bending execution

Repository source: `engineering/design_consistency/examples/solver/coarse/README.md`

Benchmark records actual beam and solid responses, the hollow section and mesh size. Beam input, beam output and beam solver log retain the B32R BOX run. Solid input, solid output and solid solver log retain the C3D10 run.

The source-bound parent receipt also checks refinement. Results are numerical research evidence with no physical/manufacturing release.

Hollow cantilever bending execution

Repository source: `engineering/design_consistency/examples/solver/refined/README.md`

Benchmark records actual beam and solid responses, the hollow section and mesh size. Beam input, beam output and beam solver log retain the B32R BOX run. Solid input, solid output and solid solver log retain the C3D10 run.

The source-bound parent receipt also checks refinement. Results are numerical research evidence with no physical/manufacturing release.

Coordinated industrialisation programme

Repository source: `engineering/industrialisation/README.md`

Status: proposed programme; products, rights, commercial terms and physical acceptance remain open. Vehicle platform, bogie localisation and the viaduct system share one configuration and evidence chain. HÜBNER and CRRC are proposed discussion partners; no exact product or manufacturing-transfer agreement is selected. Controlled programme.

Vehicle configuration and civil demand

Vehicle-to-civil trace preserves the current three-car, six-bogie/twelve-axle reference. Its 78,750 kg tare is an unclosed planning control. Equal car loading, midpoint CG and equal bogie axle sharing are explicitly assumed. Bogie locations follow the existing CAD rule. The shared-bogie alternative has an explicit four-support graph with unknown redesign masses. Buying an articulation creates no bogie-count, mass, energy or cost reduction. Exact joint/bogie selection, low-floor clearances, yaw/pitch/roll, stiffness/damping, fatigue/crash and maintenance/recovery require a joint interface freeze.

The same axle pattern feeds sampled static Pi20/Pi25 live-load demand. Dynamic augmentation, permanent capacity, prestress, bearings and launcher loads need independent design. Identified Baghdad spans remain the quantity authority. No vehicle/civil loading baseline is overwritten.

Bogie localisation and first articles

First article repeat assembly with qualified bought-in subcomponents released frame cutting/forming/welding/machining/NDT/coating further component localisation where economics justify qualification. Initially buy precision articulation assemblies and make approved surrounding adapters locally. Assess existing Iraqi fabrication and machining processes. Prioritise fixtures, metrology, weld quality, inspection and test equipment. Transfer rights, fees, tooling, qualification, imported components, labour, rejects, warranty and repeat prices have separate scopes.

Civil catalogue and urban delivery

Two single-track full-span beams share a pier: 25 m primary, 20 m secondary, and separately engineered exceptional crossings. Standardise column/cap connections, bearing/jacking/lifting seats, direct-fixation datums, removable walkway/service cassettes and island-station transitions. Ground zones, depth, capacity, utility clearances and working platforms remain site-specific.

Support packets and urban controls keep foundation mobilisation and launcher support acceptance distinct. Compare precast columns/caps with reusable-form in-situ columns/precast caps by measured accepted supports/week and complete installed cost. Require inspectable connections, temporary stability, grout and verified release strength. Maintain shop/pedestrian/emergency access, appointment-based deliveries/spoil, booked lifting exclusions and prompt reinstatement. Lane-hours, access interruptions and occupation days are unioned by resource and bound to accepted completed bays; absent observations stay unknown.

Production chain and 18-launcher scenario

Production balance treats one accepted double-track bay as the planning unit. Eighteen active independent fronts at 1/1.5/2 bays per working day give 18/27/36 bays, 36/54/72 beams and 450/675/900 m under the illustrative all-25 m assumption. Actual 20/25 m mixes and special spans retain identified quantities. At 54 beams/day, 48-hour occupation needs 108 casting positions before availability/reject allowances; a long-line bed can hold several positions. A five-working-day buffer needs 270 beams. None is a production commitment.

Released foundations accepted piers accepted serialized beams transport erection completed-bay acceptance limit output together. Casting alone cannot release a bay. Mobilise only against a sufficiently long released run, qualified supply, access, authorised crew and compatible inspected machinery. Coordinate existing Iraqi precast facilities and expertise; allocate qualified products/slots and add only demonstrated necessary tooling/capacity.

Work, training, programme gates and finance

Linked work packages and ERP handoffs connect the existing supplier, civil and execution structures. The first 30 and 30–90 day windows are proposed planning targets, followed by accepted pilot trials, controlled ramp-up and up-to-18-machine deployment. Responsible roles are named; appointments, funded recruitment and dated rosters remain open. Training attendance is separate from assessed equipment-specific work authority. Future reviewed reservations must undergo the same live machinery/worker exclusivity checks.

Installed costs and delivered-service finance separates priced scopes and missing amounts. The Baghdad delivery/finance bridge joins each line's conditional running-span date, retained finance opening month/fleet deadline, controlled required fleet and energy/service diagnostics. A common financial-close anchor, supplier vehicle deliveries and accepted station/special/system handovers are still missing. Dispatch completion does not automatically scale paid journeys. Running-bay completion is not railway revenue opening. Unique paid journeys/collection and operating acceptance are required before revised receipts can be considered. Transfers do not multiply paid journeys. Further elevation comparisons expose marginal civil reference allowances for explicit scope sensitivities, with complete installed costs, disruption values, financing and net benefit unknown. No additional alignment, beam order, price or accelerated saving is adopted.

Possible component and equipment vendors

Official manufacturer product and reference pages; narrow published scope only. Historic references are not current quotations. No supplier was contacted. Related group brands are not independent offers. Product-family listings establish neither OSR fit nor regional availability, rights, prices, capacity or acceptance. Gangways are not automatically structural joints; segmental gantries are not whole-beam launchers. Machine-readable candidates · RFQ comparison CSV.

Possible vendor	Scope	Official product/reference source
HÜBNER	structural-articulation	FXpro, FXcore and vehicle-specific adapters
HEMSCHEIDT Engineering	upper-articulation	HEUAS upper articulation systems
CRRC Zhuzhou	bogie-transfer	Vehicle and localisation engineering discussion
Siemens Mobility	complete-bogie	MoComp bogies and rail components
Alstom	complete-bogie	Flexx Urban / Flexx bogie portfolio
Lucchini RS	wheelset	Rail wheelsets, wheels and axles
GHH-BONATRANS	wheelset	Complete wheelsets and wheel/axle components
SKF	bearing-axlebox	Axlebox bearings, bearing units and monitoring
Continental	suspension	Rail secondary air springs
Knorr-Bremse	braking	Metro bogie brakes, controls and air supply
Wabtec / Faiveley	braking	Transit bogie brakes
ABB	traction-power	AMXM rail traction motors
Ingeteam	traction-power	INGETRAC traction converters
Voith	gearbox-coupling	Rail gear units and drive couplings
Dellner	gangway-coupler	Gangways, automatic and semi-permanent couplers
Forsee Power	traction-battery	PULSE PLUS RAIL high-power LTO battery
Schunk Transit Systems	charging-contacts	Rail contact/grounding and charging technology
Merak	hvac	Rail heating ventilation and air conditioning
Liebherr	hvac	Passenger and cab rail HVAC
IFE	doors	Rail entrance systems
duagon	control-electronics	Embedded computing, gateways and safe control systems
Kontron	control-electronics	Railway embedded computing / KBox R-101
HARTING	connectors-harness	Rail power and data connectors and cables
Moxa	networking	ToughNet TN-4500A rail Ethernet switches
DEAL	whole-beam-erection	Launching girders for full-span erection
DLT Engineering	whole-beam-erection	Bespoke full-span erection gantry engineering
Cimolai Technology	special-segmental-erection	Segment launching gantry
Freyssinet	prestressing-bearings	Prestressing, bearing and connection systems
mageba	bearings-joints	RESTON-SPHERICAL structural bearings
MAURER	bearings-joints	Structural bearings and expansion joints
Elematic	precast-tooling	Frame line for precast columns and beams
BAUER Maschinen	foundation-equipment	BG drilling-rig family

Regeneration

Run `.venv/bin/python tools/automation/industrialisation-study.py`; add `--check` for source/output drift. Manifest records exact inputs and output bytes. Missing supplier or engineering evidence never becomes acceptance.

engineering/integration_programme/data/contact/LICENSE.md

Repository source: `engineering/integration_programme/data/contact/LICENSE.md`

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Topography redesign, BIM and mechanical detail register

Repository source: `engineering/models/bim/design-detail-register.md`

This generated register says what the new terrain/water evidence changes, what remains provisional, and what information must be closed before design or fabrication release.

> Design-reference coordination data only. Open DEM and water data are screening evidence, not survey, geotechnical, hydraulic or navigation evidence. Candidate dimensions, loads and tolerances are not fabrication or construction release values until their stated closure evidence is accepted by the competent design authority.

Redesign decision

Changed now: alignment/civil disposition; bridge/viaduct need; station siting/configuration; route compatibility evidence; BIM information requirements.

Not automatically changed: LM3 three-car architecture; car length/width concept; battery/traction topology; bogie and articulation concept.

A vehicle parameter changes only after a route-compatibility gate fails and a controlled whole-system trade study accepts that change.

Delivery plan and completed implementation

1. Classify terrain/water evidence and trace each trigger to affected design domains.
2. Define BIM objects, information, relations and acceptance checks for each asset type.
3. Establish the LM3 datum hierarchy, mechanical ICDs, load cases and route gates.
4. Publish the register into IFC property sets and the model-coverage report.
5. Enforce ID integrity, source traceability, deterministic output and IFC round-trip tests.

All five steps are implemented in the repository; engineering evidence remains open where the status below says supplier, survey, calculation, test or authority acceptance is required.

Design impacts

ID	Disposition	Trigger	Domains
`OSR-IMP-TOPO-001`	`redesign-required`	terrain elevation or slope changes the feasible vertical alignment	alignment, earthworks, drainage, structures, traction-energy
`OSR-IMP-WATER-002`	`bridge-or-reroute-required`	route intersects mapped water or its hydraulic/flood influence zone	alignment, bridge, hydrology, environment, operations
`OSR-IMP-STRUCT-003`	`asset-family-and-site-design-required`	planner selects viaduct, bridge, retained or transition civil class	substructure, superstructure, track, egress, maintenance
`OSR-IMP-STN-004`	`station-configuration-redesign-required`	terrain, water, structure or access evidence moves or elevates a station	platform, accessibility, fire-life-safety, MEP, urban-interface
`OSR-IMP-LM3-005`	`compatibility-requalification-before-vehicle-redesign`	released route profile, structures or platforms differ from the vehicle design basis	rolling-stock, wheel-rail, traction, braking, energy, evacuation
`OSR-IMP-DEPOT-006`	`site-layout-revalidation-required`	topography or alignment changes depot access, stabling or recovery routes	depot, maintenance, logistics, energy, recovery

BIM information requirements

corridor-and-alignment

- Objects: alignment, profile, terrain, water, civil-class segment, exclusion volume
- Information: CRS and vertical datum, chainage, source/license/date/resolution, confidence, civil disposition, design status
- Relations: segment-to-source-cell, segment-to-structure, station-to-exclusion, alignment-to-revision
- Acceptance: georeference round-trip; no station intersects exclusion volume; every route segment has exactly one civil disposition

bridge-and-viaduct

- Objects: foundation, pier/abutment, bearing, deck/span, trackform, walkway, parapet, drainage, movement joint
- Information: span/skew/curvature, datum and reactions, load model, movement range, exposure, inspection zone, calculation/evidence IDs
- Relations: foundation-to-ground-model, bearing-to-movement-unit, track-to-deck, asset-to-work-package
- Acceptance: reaction and movement closure; kinematic/egress clash check; constructability and access review

station

- Objects: platform edge, track, access route, lift/stair, canopy, MEP/fire/charging equipment, drainage, foundation
- Information: platform gap/step basis, accessible route, occupancy/egress, asset tags, maintainer, inspection and replacement clearances
- Relations: platform-to-track, equipment-to-system, space-to-egress route, station-to-catchment
- Acceptance: gap/step analysis; inclusive-access and flow check; MEP/structure clash review; no unresolved terrain/water exclusion

rolling-stock

- Objects: carbody, bogie, wheelset, door, articulation, coupler, battery, traction, brake, HVAC, service/removal envelope
- Information: product ID, configuration, mass/CG status, datum, interface/load/tolerance status, material/process, verification evidence
- Relations: product-to-parent assembly, product-to-interface, interface-to-datum, product-to-analysis/evidence
- Acceptance: complete controlled product graph; interface and datum coverage; route compatibility closure; configuration-specific release boundary

manufacturing-resource

- Objects: fixture, gauge, mould, tool, work centre, inspection zone
- Information: tool ID, controlled product/interface, calibration, capacity, revision, release status
- Relations: tool-to-process, gauge-to-datum/interface, process-to-product, record-to-built serial
- Acceptance: tooling coverage; calibration evidence; first-article correlation; as-built record linkage

LM3 datum hierarchy

ID	Datum	Parent	Origin and axes
`LM3-DAT-000`	trainset track datum	`ROOT`	lead-end car centreline at top-of-rail reference plane; +X forward, +Y left, +Z up
`LM3-DAT-100`	carbody primary datum	`LM3-DAT-000`	car longitudinal centre at nominal low-floor structural plane; parallel to trainset axes
`LM3-DAT-110`	body-to-bogie interface datum	`LM3-DAT-100`	secondary-suspension centre at nominal ride height; X longitudinal, Y axle direction, Z suspension reaction
`LM3-DAT-120`	door and platform datum	`LM3-DAT-100`	door opening centre at finished threshold; X along car, Y outboard, Z up
`LM3-DAT-130`	roof equipment rail datum	`LM3-DAT-100`	roof rail grid intersection at car centre; X rail pitch, Y cross-car, Z mounting normal
`LM3-DAT-140`	under-seat battery cassette datum	`LM3-DAT-100`	battery tray locating-pin centre; X withdrawal, Y cassette row, Z support reaction
`LM3-DAT-200`	wheelset and rail datum	`LM3-DAT-110`	axle centre at nominal wheel/rail contact geometry; X travel, Y axle, Z radial/up
`LM3-DAT-300`	articulation interface datum	`LM3-DAT-100`	articulation yaw-axis centre at nominal ride height; X connection chord, Y left, Z yaw axis
`LM3-DAT-400`	train-end and coupler datum	`LM3-DAT-100`	coupler centreline at nominal ride height; X coupling axis, Y left, Z up

Mechanical interface control

LM3-ICD-001 — wheel to rail and route geometry

- Datums: `LM3-DAT-000` `LM3-DAT-200`

- Products: `LM3-BOG-SA611`, `LM3-BOG-SA621`, `LM3-BOG-P040`, `LM3-BOG-P041`
- Geometry: 1435 mm standard-gauge planning basis; profile, back-to-back, wear and contact tolerances require supplier and track-system freeze
- Loads: static/dynamic axle, curving, traction, braking and derailment cases from accepted vehicle/track dynamics
- Services: none
- Verification: wheelset dimensional report; gauging and dynamics analysis; instrumented route acceptance
- Route gate: gradient, curvature, cant, vertical curve, rail profile or track tolerance changes
- Status: `design-reference-definition-complete; release-evidence-open`; tolerance `allocation-open-until-stack-and-supplier-freeze`.

LM3-ICD-002 — carbody to powered bogie

- Datums: `LM3-DAT-100 LM3-DAT-110`
- Products: `LM3-BDY-P030`, `LM3-BOG-SA610`, `LM3-BOG-P046`
- Geometry: pivot, air-spring, yaw-link and damper datums controlled as one replaceable interface set
- Loads: AW0/AW2/AW3 vertical, lateral, yaw, traction, braking, anti-lift and jacking cases
- Services: pneumatic suspension, height sensing, earth bond
- Verification: datum survey; load calculation; ride-height and full-motion test
- Route gate: curve/cant/gradient, axle-load, crosswind or structure-deflection envelope changes
- Status: `design-reference-definition-complete; release-evidence-open`; tolerance `allocation-open-until-stack-and-supplier-freeze`.

LM3-ICD-003 — carbody to trailer bogie

- Datums: `LM3-DAT-100 LM3-DAT-110`
- Products: `LM3-BDY-P030`, `LM3-BOG-SA620`, `LM3-BOG-P061`
- Geometry: common body connection with configuration-controlled trailer-bogie adapter details
- Loads: AW0/AW2/AW3 vertical, lateral, yaw, braking, anti-lift and jacking cases
- Services: pneumatic suspension, height sensing, earth bond
- Verification: A/B interchange gauge; load calculation; ride-height and full-motion test
- Route gate: curve/cant/gradient, axle-load, crosswind or structure-deflection envelope changes
- Status: `design-reference-definition-complete; release-evidence-open`; tolerance `allocation-open-until-stack-and-supplier-freeze`.

LM3-ICD-004 — door cassette, body portal and platform

- Datums: `LM3-DAT-100 LM3-DAT-120`
- Products: `LM3-DOOR-SA310`, `LM3-BDY-P100`, `LM3-DOOR-P010`, `LM3-EXT-P010`
- Geometry: 1500 x 2050 mm design-reference carrier gauge; final aperture, threshold, seal compression, leaf sweep and platform gap/step by frozen door and route ICD
- Loads: door equipment, passenger crowd, abuse, aerodynamic pressure and emergency-operation cases
- Services: LV power, safety loop, communications, drainage
- Verification: carrier proof; gap/step analysis; obstruction/closed-and-locked/water/replacement tests
- Route gate: platform offset/height, cant, curvature or suspension-deflection changes
- Status: `design-reference-definition-complete; release-evidence-open`; tolerance `allocation-open-until-stack-and-supplier-freeze`.

LM3-ICD-005 — articulation structure and passenger gangway

- Datums: `LM3-DAT-100 LM3-DAT-300`

- Products: `LM3-ART-SA800`, `LM3-ART-SA810`, `LM3-ART-SA820`, `LM3-ART-SA830`
- Geometry: yaw/pitch/roll axes, anti-lift path, bellows, bridge, turntable and pinch keep-outs controlled through full motion
- Loads: buff/draft, vertical shear, torsion, curving, emergency braking, derailment and recovery cases
- Services: HV trainline, LV/data, pneumatic, earth bond, drainage
- Verification: structural proof; full-motion sweep; gap/pinch gauge; water and passenger bridge load tests
- Route gate: minimum curve, reverse curve, vertical curve, cant transition or recovery regime changes
- Status: `design-reference-definition-complete; release-evidence-open`; tolerance `allocation-open-until-stack-and-supplier-freeze`.

LM3-ICD-006 — train end, coupler and rescue vehicle

- Datums: `LM3-DAT-100 LM3-DAT-400`
- Products: `LM3-END-SA700`, `LM3-EIF-SA650`, `LM3-END-P050`, `LM3-END-P060`
- Geometry: coupler axis, anti-climber, energy absorber, carrier ring and rescue adapter share a surveyed end datum
- Loads: buff/draft, coupling impact, crash, rescue tow, lifting and asymmetric recovery cases
- Services: pneumatic brake, LV/data jumper, earth bond
- Verification: end survey; coupler/adapter proof; rescue compatibility and recovery trial
- Route gate: maximum gradient, minimum curve, disabled-train mass or rescue fleet changes
- Status: `design-reference-definition-complete; release-evidence-open`; tolerance `allocation-open-until-stack-and-supplier-freeze`.

LM3-ICD-007 — battery cassette, tray and carbody

- Datums: `LM3-DAT-100 LM3-DAT-140`
- Products: `LM3-HV-SA510`, `LM3-BDY-P050`, `LM3-HV-P010`
- Geometry: locating pins, retention, withdrawal path, lid/gasket, vent, drain and fire barrier controlled as a replaceable cassette
- Loads: mass/CG, longitudinal/lateral acceleration, shock/vibration, crash, thermal-runaway pressure and lifting cases
- Services: traction HV, HVIL, LV/data, cooling/fire connection, vent/drain
- Verification: tray proof; retention and withdrawal tests; isolation/HVIL; thermal propagation and drainage tests
- Route gate: route energy, gradient, ambient/flood exposure or duty-cycle changes
- Status: `design-reference-definition-complete; release-evidence-open`; tolerance `allocation-open-until-stack-and-supplier-freeze`.

LM3-ICD-008 — traction motor, gearbox, axle and bogie frame

- Datums: `LM3-DAT-110 LM3-DAT-200`
- Products: `LM3-TRC-SA615`, `LM3-TRC-P010`, `LM3-TRC-P020`, `LM3-BOG-P050`
- Geometry: shaft, foot, torque-link, coupling, oil-port and removal envelopes frozen by supplier RFQ and bogie ICD
- Loads: peak/continuous torque, gear mesh, axle motion, torque reaction, shock, thermal and overspeed cases
- Services: traction HV, temperature/speed sensing, earth bond, lubricant service
- Verification: alignment and runout; mount/torque-link proof; thermal map; rotation and removal trial
- Route gate: gradient, adhesion, braking blend, speed or thermal duty changes
- Status: `design-reference-definition-complete; release-evidence-open`; tolerance `allocation-open-until-stack-and-supplier-freeze`.

LM3-ICD-009 — roof HVAC and equipment rail

- Datums: `LM3-DAT-100 LM3-DAT-130`

- Products: `LM3-ROOF-SA410`, `LM3-ROOF-P010`, `LM3-HV-P020`, `LM3-EXT-P040`
- Geometry: rail pitch, curb flatness, drop duct, condensate, keep-out and removal lift envelope controlled together
- Loads: equipment mass/CG, acceleration, crosswind, roof snow/maintenance, lifting and fatigue cases
- Services: HV/LV power, controls/data, supply/return air, condensate drain, earth bond
- Verification: rail/curb survey; attachment proof; air/leak/drain tests; removal trial
- Route gate: crosswind, ambient temperature, dust/rain or tunnel/structure clearance changes
- Status: `design-reference-definition-complete; release-evidence-open`; tolerance `allocation-open-until-stack-and-supplier-freeze`.

LM3-ICD-010 — brake, WSP and wheelset

- Datums: `LM3-DAT-110 LM3-DAT-200`
- Products: `LM3-BOG-SA611`, `LM3-BOG-SA621`, `LM3-BOG-P060`, `LM3-BOG-P061`
- Geometry: disc/caliper, sensor target, cable sweep and wear/removal clearances controlled over suspension travel
- Loads: service/emergency/parking braking, low adhesion, thermal fade, wheel-slide and failed-channel cases
- Services: pneumatic brake, LV power, WSP/speed data, earth bond
- Verification: static brake/WSP test; thermal capacity; stopping-distance and low-adhesion route tests
- Route gate: gradient, line speed, adhesion, station spacing or stopping-margin changes
- Status: `design-reference-definition-complete; release-evidence-open`; tolerance `allocation-open-until-stack-and-supplier-freeze`.

LM3-ICD-011 — vehicle envelope to structure and evacuation walkway

- Datums: `LM3-DAT-000 LM3-DAT-100`
- Products: `LM3-TRAINSET-A000`, `LM3-CAR-A900`, `LM3-SHELL-A200`
- Geometry: 2850 mm body-width concept envelope plus dynamic movement, throw, wear, construction and maintenance allowances; released swept envelope is open
- Loads: crosswind, cant deficiency, suspension failure, passenger load and structure movement combinations
- Services: emergency lighting/communications interaction
- Verification: static/dynamic gauging; BIM swept-volume clash; evacuation and detraining demonstration
- Route gate: all alignment, platform, parapet, tunnel, bridge movement or crosswind changes
- Status: `design-reference-definition-complete; release-evidence-open`; tolerance `allocation-open-until-stack-and-supplier-freeze`.

LM3-ICD-012 — lifting, jacking and field rerailing

- Datums: `LM3-DAT-000 LM3-DAT-100`
- Products: `LM3-CAR-A900`, `LM3-BDY-P010`, `LM3-END-P050`
- Geometry: four-point lifting/jacking, transverse bridge and equipment keep-outs tied to measured car mass/CG
- Loads: normal/asymmetric lift, one-point pressure loss, rerailing side load, tiedown and damaged-vehicle cases
- Services: hydraulic/mechanical equipment, bonding and communications
- Verification: point survey; structural calculation; proof load; trained-crew recovery demonstration
- Route gate: route access, viaduct parapet/walkway, depot equipment or final mass/CG changes
- Status: `design-reference-definition-complete; release-evidence-open`; tolerance `allocation-open-until-stack-and-supplier-freeze`.

Load cases

ID	Case	Closure status
LM3-LC-001	AW0/AW2/AW3 static mass, axle load and CG	measured-mass-and-loading-definition-required
LM3-LC-002	traction, service/emergency braking and low adhesion	released-route-and-brake-supplier-model-required
LM3-LC-003	curve, cant, vertical curve and full suspension motion	vehicle-dynamics-model-and-route-profile-required
LM3-LC-004	crosswind, aerodynamic pressure and exposed structures	site-wind-and-vehicle-aerodynamic-evidence-required
LM3-LC-005	buff/draft, coupling, rescue tow and crash energy	structural-and-crash-authority-definition-required
LM3-LC-006	fatigue and vibration duty spectrum	route-spectrum-and-joint-classification-required
LM3-LC-007	battery/HV thermal, fault, fire and environmental exposure	supplier-data-and-hazard-analysis-required
LM3-LC-008	lifting, jacking, rerailing and damaged-vehicle recovery	measured-mass-cg-and-recovery-method-required
LM3-LC-009	passenger, accessibility, door and evacuation operation	loading-and-operational-rules-required
LM3-LC-010	thermal, dust, rain, flood splash and water ingress	route-environment-and-ingress-classification-required

Route compatibility gates

LM3-RCG-001

Does the released static/dynamic swept envelope clear track, platforms and structures?

Decision rule: redesign geometry only if gauging cannot be closed by infrastructure tolerance or operational control.

Evidence: surveyed alignment; structure/platform as-built model; vehicle gauging model.

LM3-RCG-002

Do gradient, curvature, cant and vertical curves remain within vehicle capability?

Decision rule: change route first where reasonable; otherwise requalify bogie, articulation, traction or braking.

Evidence: released profile; vehicle dynamics; traction/braking results.

LM3-RCG-003

Does the timetable duty fit usable battery energy and thermal limits with reserve?

Decision rule: adjust timetable/charging before changing battery capacity; redesign only after whole-life comparison.

Evidence: route simulation; supplier battery/motor/inverter/HVAC maps; degraded-mode cases.

LM3-RCG-004

Are axle/load spectra accepted by track, bridge and viaduct designs?

Decision rule: iterate vehicle mass and civil structures through one controlled load model.

Evidence: measured mass/CG; dynamic augmentation; structural calculations.

LM3-RCG-005

Are crosswind and exposed-water/viaduct operations acceptable?

Decision rule: define wind monitoring/operating limits or vehicle aerodynamic changes through hazard assessment.

Evidence: site wind study; vehicle characteristic wind curve; operating rules.

LM3-RCG-006

Can passengers evacuate and responders recover a failed train everywhere?

Decision rule: coordinate walkway, detraining and access provision; change vehicle emergency interfaces only where necessary.

Evidence: evacuation model/drill; BIM access review; recovery demonstration.

LM3-RCG-007

Do station platform gap, step and door-zone flows meet the accessibility basis?

Decision rule: coordinate platform and suspension control; modify threshold/door system only after system trade study.

Evidence: platform survey; suspension cases; gap/step and passenger-flow assessment.

Verification matrix

ID	Scope	Method	Interfaces	Load cases	Evidence
LM3-VER-001	product graph and configuration	inspection	—	—	manifest/IFC ID reconciliation and configuration record
LM3-VER-002	datum and tolerance-chain closure	analysis+inspection	LM3-ICD-001, LM3-ICD-002, LM3-ICD-003, LM3-ICD-004, LM3-ICD-005, LM3-ICD-006, LM3-ICD-007, LM3-ICD-008, LM3-ICD-009, LM3-ICD-010, LM3-ICD-011, LM3-ICD-012	—	released ICDs, stack calculations, calibrated surveys and NCR disposition
LM3-VER-003	vehicle/route dynamic compatibility	analysis+test	LM3-ICD-001, LM3-ICD-002, LM3-ICD-003, LM3-ICD-005, LM3-ICD-011	LM3-LC-001, LM3-LC-002, LM3-LC-003, LM3-LC-004	independently checked simulations and instrumented route acceptance
LM3-VER-004	carbody, bogie, articulation and end structure	analysis+test	LM3-ICD-002, LM3-ICD-003, LM3-ICD-005, LM3-ICD-006, LM3-ICD-012	LM3-LC-001, LM3-LC-005, LM3-LC-006, LM3-LC-008	calculations, material/process qualification, NDT, proof and fatigue evidence
LM3-VER-005	door/platform/accessibility	analysis+demonstration	LM3-ICD-004	LM3-LC-003, LM3-LC-009	gap/step, obstruction, emergency release, water and passenger-flow evidence
LM3-VER-006	traction, braking and energy	analysis+test	LM3-ICD-001, LM3-ICD-008, LM3-ICD-010	LM3-LC-002, LM3-LC-003, LM3-LC-007, LM3-LC-010	supplier maps, HIL, dynamometer and route test results
LM3-VER-007	battery/HV fire and environmental	analysis+test	LM3-ICD-007	LM3-LC-006, LM3-LC-007, LM3-LC-010	hazard analysis, propagation/containment, ingress, isolation and emergency-response evidence
LM3-VER-008	HVAC/roof equipment	analysis+test	LM3-ICD-009	LM3-LC-004, LM3-LC-006, LM3-LC-010	attachment, airflow, thermal, EMC, ingress and removal evidence
LM3-VER-009	gauging, bridge/viaduct evacuation and recovery	analysis+demonstration	LM3-ICD-006, LM3-ICD-011, LM3-ICD-012	LM3-LC-004, LM3-LC-008, LM3-LC-009	swept-volume BIM review plus representative evacuation/recovery drills

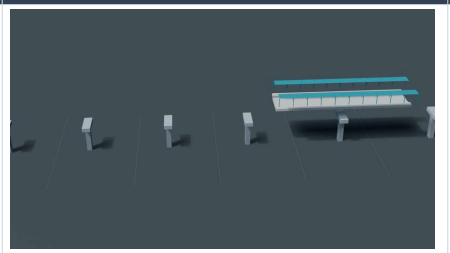
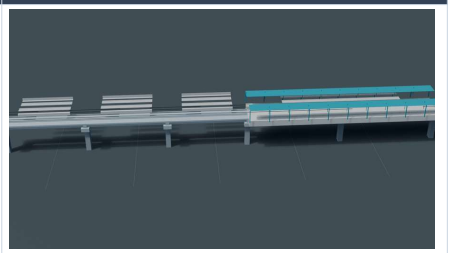
The complete machine-readable record is `design-detail-register.json`.

Civil and LM3 BIM Reference Package

Repository source: `engineering/models/bim/reference/README.md`

This is the tracked, deterministic public-review federation. It lets GitHub users inspect the current IFC4.3 model and its requirements/evidence without installing the complete engineering toolchain.

48-second civil IFC construction sequence

Substructure	Superstructure	Track and station interfaces
		
IFC substructure construction milestone	IFC superstructure erection milestone	IFC track and station milestone

File	Purpose
`civil-coordination.ifc`	IFC4.3 rail/civil/station/rolling-stock coordination model
`civil-coordination.index.json`	Stable IDs, classifications, source revisions, quantities and relationships
`civil-coordination.validation.json`	IFC schema and project validation summary
`civil-information-requirements.ids`	Machine-checkable information requirements
`civil-information-requirements.report.json`	Complete IDS result report
`civil-coordination-issues.bcf`	Open coordination issues in BCF 3.0 form
`civil-coordination-issues.index.json`	Reviewable BCF issue index
`civil-construction-sequence.json`	Deterministic construction stages and object assignments
`civil-coordination.blend`	Native Bonsai/Blender review scene with assigned IFC assets, task IDs and construction keyframes
`civil-construction-sequence.mp4`	Full 48-second H.264 4D construction review
`civil-construction-sequence.gif`	Complete GitHub-viewable preview of the same sequence
`lm3-manufacturing-reference.ifc`	IFC4.3 LM3 product hierarchy, semantic doors/windows/fixtures/motors, manufacturing methods, sequenced tasks and 30 tooling families
`lm3-manufacturing-reference.index.json`	IFC counts, source hashes, semantic classes and deterministic validation summary
`lm3-information-requirements.ids`	Machine-checkable LM3 provenance, product graph, assembly and mechanical-interface requirements
`lm3-information-requirements.report.json`	Complete passing IDS result for the federated LM3 reference model
`lm3-parts/`	120 separate geometric IFC4.3 product files, including round running-gear meshes and semantic doors, windows, lights, furniture and motor classes
`lm3-assemblies/`	26 hierarchy-preserving IFC4.3 subassembly/car/train files; the final trainset contains every active descendant product

File	Purpose
lm3-product-library.index.json	Exact split-library file hashes, representation coverage and final-assembly reachability test
stations/	Seven geometric station IFC4.3 assemblies covering every controlled product in the halt, standard, major, interchange, elevated-interchange, terminal and depot-terminal variants; IDs are cross-checked by the station reconciliation register
../../model-coverage.md	Generated LM3/station fidelity, source, analysis and release-evidence register
../design-detail-register.md	Terrain/water redesign impacts, BIM information requirements, LM3 datums, mechanical ICDs, load cases, route gates and verification matrix

Regenerate it from the authoritative component geometry and engineering interchange code:

```
tools/automation/bonsai-civil.sh --animate \
  --out-dir engineering/models/bim/reference \
  --revision-id repository-reference
python3 engineering/design_detail_register.py
python3 engineering/interchange/trainset_manufacturing_ifc.py
python3 engineering/interchange/lm3_product_ifc_library.py
python3 engineering/interchange/station_ifc.py --all-variants \
  --output-dir engineering/models/bim/reference/stations
python3 engineering/model_coverage.py
```

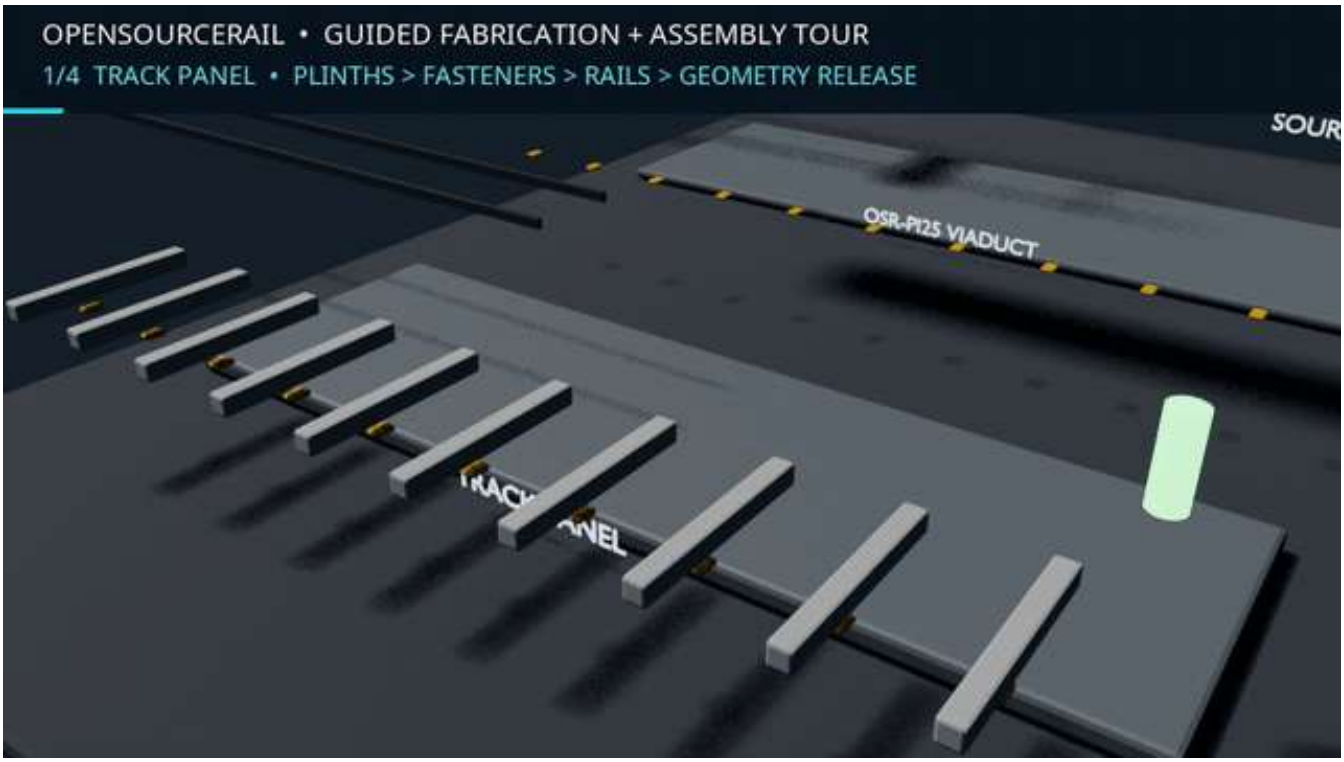
The 4D review moves the assigned IFC objects themselves: substructure rises, bearings and roofs descend, beams arrive from the erection side, and track feeds longitudinally according to the checked task assignments. It is a deterministic construction-state simulation, not crane physics, temporary-works verification, clash-free path planning or an approved method statement.

These models are coordination, product-structure and manufacturing-method evidence, not a construction release. The LM3 and station files expose their complete current product graphs and inspectable design-reference geometry. The federated LM3 IFC is checked against its IDS, and the split LM3 part/assembly IFCs carry the same design-detail source hash plus applicable datum/interface controls. These checks do not substitute for supplier-frozen interfaces, installed-coordinate/tolerance analyses, released shop drawings, qualified weld/laminate processes or NC surfaces. Repeated City Studio jobs and render intermediates remain under `build/`; this named review set is retained in Git.

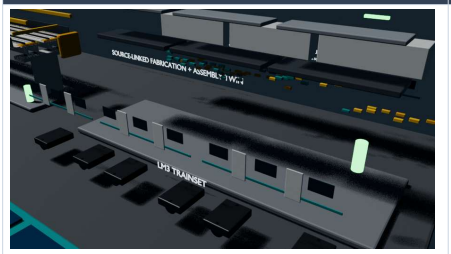
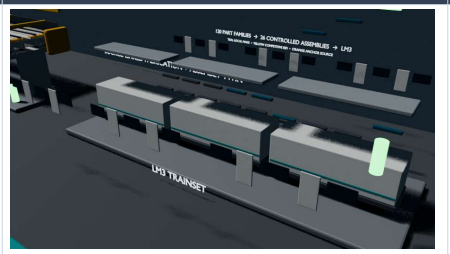
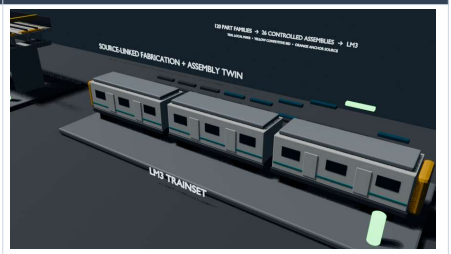
Fabrication and assembly digital twin

Repository source: `engineering/models/digital-twins/fabrication-assembly/README.md`

This source-linked planning twin follows four representative products from material release through fabrication, assembly, quality hold points, and handover: a 6 m track panel, a standard twin-platform station, an OSR-Pi25 twin-track viaduct bay, and a 49.5 m three-car LM3 trainset.



Animated fabrication and assembly digital twin

Product rows staged	Controlled subassemblies	Complete LM3 node
		
LM3 product rows in deterministic staging	LM3 subassemblies following the product dependency graph	LM3 complete assembly state

The animation is an 88-second guided assembly tour. The LM3 chapter instantiates all 120 controlled product rows and all 26 assembly nodes from subassembly to final trainset; each object is separately selectable and carries its ID, parent/children, quantity, route and dependency timing. Components move from deterministic staging positions only after their inputs are available, then the tour finishes with a completed-products overview. The chapters cover track, station, viaduct and LM3 assembly separately so small fixtures and installation order remain visible. The MP4 is the primary review artifact; the lower-frame-rate GIF is an inline preview of the same sequence.

The 146 LM3 nodes are coordination representations, not invented supplier shop geometry: repeated physical occurrences stay explicit as quantity metadata and the adjacent train geometry supplies the recognisable installed context. The tour is explanatory rather than a project schedule. The JSON retains the actual concurrent planning basis, durations, 25 stages, predecessors, work centres, inputs, outputs, hold points, evidence, cross-stream interfaces, and hashes of the controlled source documents.

The viaduct route now includes tested foundation construction, recorded pier column delivery, erection and seat survey of the compact hollow cap, two sub-75 t decked-beam lifts by portal/strand-jack launcher, an actual-load-chart fallback gate, a maturity-controlled link-slab/diaphragm connection stage, and separate walkway/containment cassettes.

Artifact	Purpose
`fabrication-assembly-digital-twin.json`	Machine-readable production route, complete 146-node LM3 DAG/timing, live-state snapshots, relationships, source hashes, and validation checks

Artifact	Purpose
fabrication-assembly-digital-twin.blend	Native Blender scene with separately selectable staged components and animation
fabrication-assembly-digital-twin.mp4	Full-resolution 88-second H.264 guided assembly tour
fabrication-assembly-digital-twin.gif	Complete 88-second inline preview below the 20 MB repository limit

Regenerate all outputs with:

```
design/component-catalogue/scripts/blender_fabrication_assembly_twin.sh
```

This is a fabrication-planning and first-article control twin, not released shop drawings or an approved construction method. Numeric weld procedures, fastener torques, lift studies, temporary works, supplier instructions, and signed travelers remain mandatory release inputs.

CAD and IFC model coverage

Repository source: `engineering/models/model-coverage.md`

This generated register is the concise fidelity view for the LM3 and station models. Product maturity, geometry fidelity and release approval are separate states; no row is a construction release merely because geometry exists.

Summary

- LM3 product models: 120
- Unique station product models: 50
- Complete station variant assemblies: 7
- LM3 objects with datum/interface control: 39 (20 products + 19 assemblies)
- Controlled mechanical interfaces: 12
- LM3 IDS checks: 945 across 4 specifications
- Geometry levels: `coordination-envelope`=92, `design-reference-detail`=23, `interface-detailed`=55

Meaning

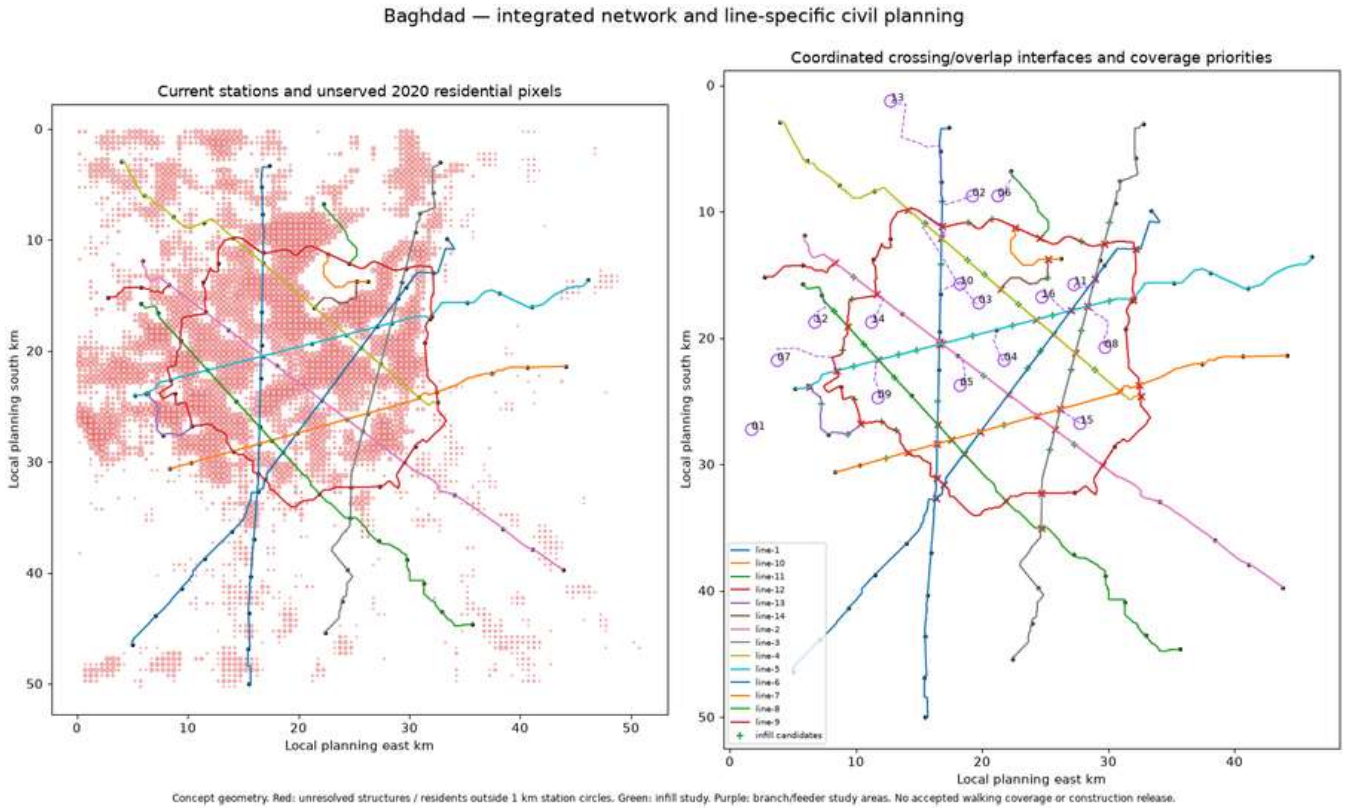
`coordination-envelope` controls space and interfaces for sourced equipment; `manufacturing-envelope` adds OSR manufacturing intent; `design-reference-detail` adds inspectable subcomponents; `interface-detailed` models repeatable datums, connections or service routes. `fabrication-detailed` and `released` require controlled drawings, tolerances and accepted evidence. Datum/interface coverage means the object is linked to the design-detail register; it does not mean that open tolerances, calculations or physical evidence are accepted.

The complete machine-readable per-product mapping, analysis IDs, evidence gates and FreeCAD/IFC/neutral-output paths are in `model-coverage.json`.

Baghdad integrated network, foundations and assembly plan

Repository source: `engineering/network-planning/baghdad/README.md`

Coordinated design-development concept; field construction and operating adoption remain open.



Network interfaces and residential gaps

Interactive line / support / junction viewer lets reviewers zoom, select a line and inspect individual support packets and interface records offline.

One source-bound asset graph connects each line, actual span, shared support, beam, launcher front, junction interface, station complex and residential intervention. The current native timetable/finance remains a comparator; this package does not claim new operating service or accepted structural profiles.

The plan assigns 10,985 support-specific foundation packets, 9,973 catalogue bay assemblies, 404 specials and 28 workfronts sharing 18 configured launchers. 10 fronts have preceding asset assignments; a front record creates no additional machine. Support packets have coordinates, chainage, adjacent spans, desktop-soil investigation references and required loads/tests. Pile type/count/depth, groundwater and bearing capacity remain unselected until field design.

Line	Foundation packets	Catalogue bay assemblies	Special packages	Candidate infill
line-1	918	770	32	3
line-2	1165	988	76	3
line-3	932	863	29	4
line-4	1040	943	44	5
line-5	952	874	35	7
line-6	974	846	60	2
line-7	867	813	21	2
line-8	915	811	39	3
line-9	2423	2280	61	8
line-10	242	238	2	0
line-11	155	153	1	0
line-12	48	46	1	0
line-13	127	123	2	2
line-14	227	225	1	1

Junction and overlap coordination

46 crossing/shared-corridor/self-crossing interfaces need explicit structural treatment. 0 original interchange groups exceed the 600 m full-diameter screen. The new bounded grouping prevents transitive long-distance complexes and preserves platform coordinates. It does not create a track switch, measured walking path or accepted vertical profile.

Integrated geometry and complex register · GIS junction/residential layer · Foundation coordinate conflicts.

Residential coverage and expansion

Retained 2020 population within 1 km station circles is 27.7%. 40 population-ranked infill candidates give a conditional radial sensitivity of 45.7%. 2,173,888 currently unserved retained residents are also over 1 km from sampled existing corridors; infill cannot close those gaps. 16 named/located priority areas therefore require branch/extension or funded feeder/access studies. Area circles can overlap and are never summed as unique residents.

These are native population-count and distance screens, not current census, surveyed walksheds, ridership or funded service. Candidate sites need water/footprint/property/access, station structure, timetable, fleet, energy and full installed cost review.

Expansion corridor plans provide 15 source-bound routes along the retained arterial graph, tied to existing station connections, with access gaps and proposed stopping sites. Compare rail branches with frequent feeders; traffic direction, bridge grades/loads, railway curvature and walk routes are unverified. The combined infill/feeder 500 m stop-circle sensitivity is 51.4%; the rail 1 km sensitivity is 58.8%. Shared corridor edges count overlapping candidate road segments once. No corridor is adopted or funded.

Foundation and assembly execution

Foundation register has one row per unique proposed support. Span assembly register has identified component pairs and both foundation parents, actual line/run, unique front assignment, sequence and predecessor. Launcher fronts preserve disconnected runs and relocation gates. Stage library defines foundation/column/cap/bearing, factory/delivery, first/second beam securing, advance and handover dependencies. Specials and junctions use their own design/erection chains.

Decode compressed CSV using `python -c "import gzip; print(gzip.open('foundation-register.csv.gz','rt').read())"`. Coordinates are interpolated from the retained planning route; desktop soil does not select a foundation, and no pile depth or allowable bearing value is invented.

Complete per-line summary · Wider catalogue audit · Catalogue findings and provenance.

Companion data archive contains this planning package and new controlled inputs. The original Baghdad proposal archive retains the baseline city/engineering input evidence. Use both for the full planning review; the companion does not imply physical release.

Run `python tools/automation/integrated-network-plan.py`; `--check` validates source/output hashes. `--retain-context` creates the immutable retained OSM road/name input from the cached extraction. Native route and financial assumptions change only through a subsequent fully regenerated, validated adoption revision.

line-1 — foundation and assembly construction plan

Repository source: `engineering/network-planning/baghdad/line-1-construction-plan.md`

Line-specific design-development package; survey, ground, supplier and staged-load releases are still required.

Route length 49.068 km. 918 unique proposed support packets; 770 identified catalogue bay assemblies and 32 special packages. Each ordinary bay has two named beam components and both foundation parents. The 25 m average is a sizing basis; actual Pi20/Pi25/closure geometry controls installation.

Front	Launcher	Previous asset assignment	Direction	Initial assembly chainage m	Bay assemblies
line-1-front-1	launcher-01	initial mobilisation	1	408.3	385
line-1-front-2	launcher-02	initial mobilisation	-1	44150.1	385

Foundation workface plan

Filter the foundation register by `line-1`. For every support, verify the proposed coordinate/chainage, rights, utilities and the referenced desktop sample; commission field ground, groundwater, durability and load investigations. Review shallow footing, bored shaft, driven bent and pile-group alternatives against actual ground, axial/lateral/settlement/scour and construction loads. Desktop topsoil does not select a pile depth, count or allowable bearing pressure.

Release trial and working-platform design (F1), install the selected foundation with actual geometry/material/installation records (F2), then verify integrity/load/settlement/dimensions and close the foundation packet (F3). Construct/erect column and cap with independently checked connections/temporary restraint (P0). Survey and inspect bearings, strength and the next launcher/beam/delivery stage before support release (B0). Keep 10–15 consecutive bays of accepted supports and beam stock ahead only where the run and storage permit.

Directed assembly and logistics sequence

Before mobilisation, release the predecessor asset assignment and its dismantle/transport/recommission package. The positive front installs toward increasing chainage; the negative front starts at its high-chainage end and installs backward. Factory/dispatch plans must follow this directed sequence, not the ascending raw span register. A foundation used by two bays is one packet; it is not built or priced twice.

For each front, identify the first complete run, permitted delivery/assembly site, temporary support/ground platform, configured plant and authorised shift/crew. Reserve both beam travellers, transport/receiving equipment, an operator, required riggers, lift supervision and inspection. Verify source strength, complete mass/CG, lifting points, rigging, bearings, weather limits and contingency landing/recovery before any lift.

Both support B0 packets and beam receiving H1 records precede E0. Place and restrain beam 1 (E1); then place/securing beam 2 (E2). Survey and accept connection/strength/bearings/NCRs (E3), release advance loads and the next supports before moving the launcher (E4), then complete walkways/barriers/waterproofing/drainage/track interface and following-trade handover (E5). The next bay depends on the preceding advance release. Following trades may overlap only with protected, agreed load/access boundaries.

Run boundaries, stations and specials require a passage or dismantle/transport/reassembly/recommission package. No discontinuity becomes an assumed launchable gap. Junction-affected orders remain on a design hold until J2 freezes the profile, clearance, support/cap geometry and staged-load/utility/access design. A physical crossing does not create a rail switch.

Initial bay packages

Front	Sequence	Span	Product	Foundation A	Foundation B	Prior span
line-1-front-1	1	line-1-run-0001-span-0001	OSR-Pi20	FND:line-1-support-000000408300	FND:line-1-support-000000428300	mobilisation
line-1-front-1	2	line-1-run-0001-span-0002	OSR-Pi20	FND:line-1-support-000000428300	FND:line-1-support-000000448300	line-1-run-0001-span-0001
line-1-front-1	3	line-1-run-0002-span-0001	OSR-Pi25	FND:line-1-support-000000588300	FND:line-1-support-000000613300	line-1-run-0001-span-0002
line-1-front-1	4	line-1-run-0002-span-0002	OSR-Pi25	FND:line-1-support-000000613300	FND:line-1-support-000000638300	line-1-run-0002-span-0001
line-1-front-1	5	line-1-run-0002-span-0003	OSR-Pi25	FND:line-1-support-000000638300	FND:line-1-support-000000663300	line-1-run-0002-span-0002
line-1-front-1	6	line-1-run-0002-span-0004	OSR-Pi25	FND:line-1-support-000000663300	FND:line-1-support-000000688300	line-1-run-0002-span-0003
line-1-front-2	1	line-1-run-0114-span-0001	OSR-Pi20	FND:line-1-support-000044130100	FND:line-1-support-000044150100	mobilisation
line-1-front-2	2	line-1-run-0113-span-0001	OSR-Pi20	FND:line-1-support-000042948700	FND:line-1-support-000042968700	line-1-run-0114-span-0001
line-1-front-2	3	line-1-run-0112-span-0082	OSR-Pi20	FND:line-1-support-000042754500	FND:line-1-support-000042774500	line-1-run-0113-span-0001
line-1-front-2	4	line-1-run-0112-span-0081	OSR-Pi25	FND:line-1-support-000042729500	FND:line-1-support-000042754500	line-1-run-0112-span-0082
line-1-front-2	5	line-1-run-0112-span-0080	OSR-Pi25	FND:line-1-support-000042704500	FND:line-1-support-000042729500	line-1-run-0112-span-0081
line-1-front-2	6	line-1-run-0112-span-0079	OSR-Pi25	FND:line-1-support-000042679500	FND:line-1-support-000042704500	line-1-run-0112-span-0080

The compressed span register provides the full sequence, individual beam IDs, predecessors and junction holds. Stage library supplies the method dependencies. Front package lists every assigned span and actual disconnected working interval.

Junctions, stations and residential interfaces

Coordinated structural interface IDs on this line: crossing-4761735b935e, crossing-596aa3d726d4, crossing-9090e87568d6, crossing-92b255c18da1, crossing-d02bcca1ecd2, crossing-e1b0b7941672, crossing-efa108fc0f05, shared-corridor-ea625bfc9e95.

Evaluate grade-separated crossings, shared four-track civil footprints or separate deck levels, and bounded platform/concourse/access complexes with the station authority. Freeze actual vertical profiles and gradients before selecting supports and ordinary spans. Preserve rail/PSD/egress/waterproofing/earthing interfaces. Proposed residential infill sites on this line require station/approach structures and revised service, fleet, power and full installed prices; they are not existing paid journeys.

Candidate infill IDs: infill-line-1-037500, infill-line-1-042500, infill-line-1-026500.

Inspection, handover and programme

Use the foundation/production/lift/connection/concealed-work hold points from the master ITP. Capture installed component IDs, geometry, tests, personnel/equipment authority, NCR disposition and signed stage release. Update actual shift cycles, losses, stocks and resource reservations; do not infer an opening date from bay throughput. Station/special/track/energy/depot/fleet/testing/approval releases close the whole line. Unknown ground quantities, installed costs and dates stay unknown.

Integrated master package · Interactive support/junction inspection.

line-10 — foundation and assembly construction plan

Repository source: `engineering/network-planning/baghdad/line-10-construction-plan.md`

Line-specific design-development package; survey, ground, supplier and staged-load releases are still required.

Route length 6.804 km. 242 unique proposed support packets; 238 identified catalogue bay assemblies and 2 special packages. Each ordinary bay has two named beam components and both foundation parents. The 25 m average is a sizing basis; actual Pi20/Pi25/closure geometry controls installation.

Front	Launcher	Previous asset assignment	Direction	Initial assembly chainage m	Bay assemblies
line-10-front-1	launcher-01	line-1-front-1	1	210.5	119
line-10-front-2	launcher-02	line-1-front-2	-1	6572.7	119

Foundation workface plan

Filter the foundation register by `line-10`. For every support, verify the proposed coordinate/chainage, rights, utilities and the referenced desktop sample; commission field ground, groundwater, durability and load investigations. Review shallow footing, bored shaft, driven bent and pile-group alternatives against actual ground, axial/lateral/settlement/scour and construction loads. Desktop topsoil does not select a pile depth, count or allowable bearing pressure.

Release trial and working-platform design (F1), install the selected foundation with actual geometry/material/installation records (F2), then verify integrity/load/settlement/dimensions and close the foundation packet (F3). Construct/erect column and cap with independently checked connections/temporary restraint (P0). Survey and inspect bearings, strength and the next launcher/beam/delivery stage before support release (B0). Keep 10–15 consecutive bays of accepted supports and beam stock ahead only where the run and storage permit.

Directed assembly and logistics sequence

Before mobilisation, release the predecessor asset assignment and its dismantle/transport/recommission package. The positive front installs toward increasing chainage; the negative front starts at its high-chainage end and installs backward. Factory/dispatch plans must follow this directed sequence, not the ascending raw span register. A foundation used by two bays is one packet; it is not built or priced twice.

For each front, identify the first complete run, permitted delivery/assembly site, temporary support/ground platform, configured plant and authorised shift/crew. Reserve both beam travellers, transport/receiving equipment, an operator, required riggers, lift supervision and inspection. Verify source strength, complete mass/CG, lifting points, rigging, bearings, weather limits and contingency landing/recovery before any lift.

Both support B0 packets and beam receiving H1 records precede E0. Place and restrain beam 1 (E1); then place/securing beam 2 (E2). Survey and accept connection/strength/bearings/NCRs (E3), release advance loads and the next supports before moving the launcher (E4), then complete walkways/barriers/waterproofing/drainage/track interface and following-trade handover (E5). The next bay depends on the preceding advance release. Following trades may overlap only with protected, agreed load/access boundaries.

Run boundaries, stations and specials require a passage or dismantle/transport/reassembly/recommission package. No discontinuity becomes an assumed launchable gap. Junction-affected orders remain on a design hold until J2 freezes the profile, clearance, support/cap geometry and staged-load/utility/access design. A physical crossing does not create a rail switch.

Initial bay packages

Front	Sequence	Span	Product	Foundation A	Foundation B	Prior span
line-10-front-1	1	line-10-run-0001-span-0001	OSR-Pi25	FND:line-10-support-0000210500	FND:line-10-support-0000235500	mobilisation
line-10-front-1	2	line-10-run-0001-span-0002	OSR-Pi25	FND:line-10-support-0000235500	FND:line-10-support-0000260500	line-10-run-0001-span-0001
line-10-front-1	3	line-10-run-0001-span-0003	OSR-Pi25	FND:line-10-support-0000260500	FND:line-10-support-0000285500	line-10-run-0001-span-0002
line-10-front-1	4	line-10-run-0001-span-0004	OSR-Pi25	FND:line-10-support-0000285500	FND:line-10-support-0000310500	line-10-run-0001-span-0003
line-10-front-1	5	line-10-run-0001-span-0005	OSR-Pi25	FND:line-10-support-0000310500	FND:line-10-support-0000335500	line-10-run-0001-span-0004
line-10-front-1	6	line-10-run-0001-span-0006	OSR-Pi25	FND:line-10-support-0000335500	FND:line-10-support-0000360500	line-10-run-0001-span-0005
line-10-front-2	1	line-10-run-0002-span-0027	OSR-Pi20	FND:line-10-support-0006552700	FND:line-10-support-0006572700	mobilisation
line-10-front-2	2	line-10-run-0002-span-0026	OSR-Pi20	FND:line-10-support-0006532700	FND:line-10-support-0006552700	line-10-run-0002-span-0027
line-10-front-2	3	line-10-run-0002-span-0025	OSR-Pi20	FND:line-10-support-0006512700	FND:line-10-support-0006532700	line-10-run-0002-span-0026
line-10-front-2	4	line-10-run-0002-span-0024	OSR-Pi25	FND:line-10-support-0006487700	FND:line-10-support-0006512700	line-10-run-0002-span-0025
line-10-front-2	5	line-10-run-0002-span-0023	OSR-Pi25	FND:line-10-support-0006462700	FND:line-10-support-0006487700	line-10-run-0002-span-0024
line-10-front-2	6	line-10-run-0002-span-0022	OSR-Pi25	FND:line-10-support-0006437700	FND:line-10-support-0006462700	line-10-run-0002-span-0023

The compressed span register provides the full sequence, individual beam IDs, predecessors and junction holds. Stage library supplies the method dependencies. Front package lists every assigned span and actual disconnected working interval.

Junctions, stations and residential interfaces

Coordinated structural interface IDs on this line: crossing-057c84aa0956, crossing-c6c859f1b2b3.

Evaluate grade-separated crossings, shared four-track civil footprints or separate deck levels, and bounded platform/concourse/access complexes with the station authority. Freeze actual vertical profiles and gradients before selecting supports and ordinary spans. Preserve rail/PSD/egress/waterproofing/earthing interfaces. Proposed residential infill sites on this line require station/approach structures and revised service, fleet, power and full installed prices; they are not existing paid journeys.

Candidate infill IDs: no qualifying candidate under this screen.

Inspection, handover and programme

Use the foundation/production/lift/connection/concealed-work hold points from the master ITP. Capture installed component IDs, geometry, tests, personnel/equipment authority, NCR disposition and signed stage release. Update actual shift cycles, losses, stocks and resource reservations; do not infer an opening date from bay throughput.
Station/special/track/energy/depot/fleet/testing/approval releases close the whole line. Unknown ground quantities, installed costs and dates stay unknown.

Integrated master package · Interactive support/junction inspection.

line-11 — foundation and assembly construction plan

Repository source: `engineering/network-planning/baghdad/line-11-construction-plan.md`

Line-specific design-development package; survey, ground, supplier and staged-load releases are still required.

Route length 6.884 km. 155 unique proposed support packets; 153 identified catalogue bay assemblies and 1 special packages. Each ordinary bay has two named beam components and both foundation parents. The 25 m average is a sizing basis; actual Pi20/Pi25/closure geometry controls installation.

Front	Launcher	Previous asset assignment	Direction	Initial assembly chainage m	Bay assemblies
line-11-front-1	launcher-03	line-2-front-1	1	210.5	76
line-11-front-2	launcher-04	line-2-front-2	-1	4035.5	77

Foundation workface plan

Filter the foundation register by `line-11`. For every support, verify the proposed coordinate/chainage, rights, utilities and the referenced desktop sample; commission field ground, groundwater, durability and load investigations. Review shallow footing, bored shaft, driven bent and pile-group alternatives against actual ground, axial/lateral/settlement/scour and construction loads. Desktop topsoil does not select a pile depth, count or allowable bearing pressure.

Release trial and working-platform design (F1), install the selected foundation with actual geometry/material/installation records (F2), then verify integrity/load/settlement/dimensions and close the foundation packet (F3). Construct/erect column and cap with independently checked connections/temporary restraint (P0). Survey and inspect bearings, strength and the next launcher/beam/delivery stage before support release (B0). Keep 10–15 consecutive bays of accepted supports and beam stock ahead only where the run and storage permit.

Directed assembly and logistics sequence

Before mobilisation, release the predecessor asset assignment and its dismantle/transport/recommission package. The positive front installs toward increasing chainage; the negative front starts at its high-chainage end and installs backward. Factory/dispatch plans must follow this directed sequence, not the ascending raw span register. A foundation used by two bays is one packet; it is not built or priced twice.

For each front, identify the first complete run, permitted delivery/assembly site, temporary support/ground platform, configured plant and authorised shift/crew. Reserve both beam travellers, transport/receiving equipment, an operator, required riggers, lift supervision and inspection. Verify source strength, complete mass/CG, lifting points, rigging, bearings, weather limits and contingency landing/recovery before any lift.

Both support B0 packets and beam receiving H1 records precede E0. Place and restrain beam 1 (E1); then place/securing beam 2 (E2). Survey and accept connection/strength/bearings/NCRs (E3), release advance loads and the next supports before moving the launcher (E4), then complete walkways/barriers/waterproofing/drainage/track interface and following-trade handover (E5). The next bay depends on the preceding advance release. Following trades may overlap only with protected, agreed load/access boundaries.

Run boundaries, stations and specials require a passage or dismantle/transport/reassembly/recommission package. No discontinuity becomes an assumed launchable gap. Junction-affected orders remain on a design hold until J2 freezes the profile, clearance, support/cap geometry and staged-load/utility/access design. A physical crossing does not create a rail switch.

Initial bay packages

Front	Sequence	Span	Product	Foundation A	Foundation B	Prior span
line-11-front-1	1	line-11-run-0001-span-0001	OSR-Pi25	FND:line-11-support-00 0000210500	FND:line-11-support-00 0000235500	mobilisation
line-11-front-1	2	line-11-run-0001-span-0002	OSR-Pi25	FND:line-11-support-00 0000235500	FND:line-11-support-00 0000260500	line-11-run-0001-span-0001
line-11-front-1	3	line-11-run-0001-span-0003	OSR-Pi25	FND:line-11-support-00 0000260500	FND:line-11-support-00 0000285500	line-11-run-0001-span-0002
line-11-front-1	4	line-11-run-0001-span-0004	OSR-Pi25	FND:line-11-support-00 0000285500	FND:line-11-support-00 0000310500	line-11-run-0001-span-0003
line-11-front-1	5	line-11-run-0001-span-0005	OSR-Pi25	FND:line-11-support-00 0000310500	FND:line-11-support-00 0000335500	line-11-run-0001-span-0004
line-11-front-1	6	line-11-run-0001-span-0006	OSR-Pi25	FND:line-11-support-00 0000335500	FND:line-11-support-00 0000360500	line-11-run-0001-span-0005
line-11-front-2	1	line-11-run-0001-span-0153	OSR-Pi25	FND:line-11-support-00 0004010500	FND:line-11-support-00 0004035500	mobilisation
line-11-front-2	2	line-11-run-0001-span-0152	OSR-Pi25	FND:line-11-support-00 0003985500	FND:line-11-support-00 0004010500	line-11-run-0001-span-0153
line-11-front-2	3	line-11-run-0001-span-0151	OSR-Pi25	FND:line-11-support-00 0003960500	FND:line-11-support-00 0003985500	line-11-run-0001-span-0152
line-11-front-2	4	line-11-run-0001-span-0150	OSR-Pi25	FND:line-11-support-00 0003935500	FND:line-11-support-00 0003960500	line-11-run-0001-span-0151
line-11-front-2	5	line-11-run-0001-span-0149	OSR-Pi25	FND:line-11-support-00 0003910500	FND:line-11-support-00 0003935500	line-11-run-0001-span-0150
line-11-front-2	6	line-11-run-0001-span-0148	OSR-Pi25	FND:line-11-support-00 0003885500	FND:line-11-support-00 0003910500	line-11-run-0001-span-0149

The compressed span register provides the full sequence, individual beam IDs, predecessors and junction holds. Stage library supplies the method dependencies. Front package lists every assigned span and actual disconnected working interval.

Junctions, stations and residential interfaces

Coordinated structural interface IDs on this line: crossing-c4ea0b93414e.

Evaluate grade-separated crossings, shared four-track civil footprints or separate deck levels, and bounded platform/concourse/access complexes with the station authority. Freeze actual vertical profiles and gradients before selecting supports and ordinary spans. Preserve rail/PSD/egress/waterproofing/earthing interfaces. Proposed residential infill sites on this line require station/approach structures and revised service, fleet, power and full installed prices; they are not existing paid journeys.

Candidate infill IDs: no qualifying candidate under this screen.

Inspection, handover and programme

Use the foundation/production/lift/connection/concealed-work hold points from the master ITP. Capture installed component IDs, geometry, tests, personnel/equipment authority, NCR disposition and signed stage release. Update actual shift cycles, losses, stocks and resource reservations; do not infer an opening date from bay throughput.
Station/special/track/energy/depot/fleet/testing/approval releases close the whole line. Unknown ground quantities, installed costs and dates stay unknown.

Integrated master package · Interactive support/junction inspection.

line-12 — foundation and assembly construction plan

Repository source: `engineering/network-planning/baghdad/line-12-construction-plan.md`

Line-specific design-development package; survey, ground, supplier and staged-load releases are still required.

Route length 6.586 km. 48 unique proposed support packets; 46 identified catalogue bay assemblies and 1 special packages. Each ordinary bay has two named beam components and both foundation parents. The 25 m average is a sizing basis; actual Pi20/Pi25/closure geometry controls installation.

Front	Launcher	Previous asset assignment	Direction	Initial assembly chainage m	Bay assemblies
line-12-front-1	launcher-05	line-3-front-1	1	210.5	23
line-12-front-2	launcher-06	line-3-front-2	-1	1355.5	23

Foundation workface plan

Filter the foundation register by `line-12`. For every support, verify the proposed coordinate/chainage, rights, utilities and the referenced desktop sample; commission field ground, groundwater, durability and load investigations. Review shallow footing, bored shaft, driven bent and pile-group alternatives against actual ground, axial/lateral/settlement/scour and construction loads. Desktop topsoil does not select a pile depth, count or allowable bearing pressure.

Release trial and working-platform design (F1), install the selected foundation with actual geometry/material/installation records (F2), then verify integrity/load/settlement/dimensions and close the foundation packet (F3). Construct/erect column and cap with independently checked connections/temporary restraint (P0). Survey and inspect bearings, strength and the next launcher/beam/delivery stage before support release (B0). Keep 10–15 consecutive bays of accepted supports and beam stock ahead only where the run and storage permit.

Directed assembly and logistics sequence

Before mobilisation, release the predecessor asset assignment and its dismantle/transport/recommission package. The positive front installs toward increasing chainage; the negative front starts at its high-chainage end and installs backward. Factory/dispatch plans must follow this directed sequence, not the ascending raw span register. A foundation used by two bays is one packet; it is not built or priced twice.

For each front, identify the first complete run, permitted delivery/assembly site, temporary support/ground platform, configured plant and authorised shift/crew. Reserve both beam travellers, transport/receiving equipment, an operator, required riggers, lift supervision and inspection. Verify source strength, complete mass/CG, lifting points, rigging, bearings, weather limits and contingency landing/recovery before any lift.

Both support B0 packets and beam receiving H1 records precede E0. Place and restrain beam 1 (E1); then place/securing beam 2 (E2). Survey and accept connection/strength/bearings/NCRs (E3), release advance loads and the next supports before moving the launcher (E4), then complete walkways/barriers/waterproofing/drainage/track interface and following-trade handover (E5). The next bay depends on the preceding advance release. Following trades may overlap only with protected, agreed load/access boundaries.

Run boundaries, stations and specials require a passage or dismantle/transport/reassembly/recommission package. No discontinuity becomes an assumed launchable gap. Junction-affected orders remain on a design hold until J2 freezes the profile, clearance, support/cap geometry and staged-load/utility/access design. A physical crossing does not create a rail switch.

Initial bay packages

Front	Sequence	Span	Product	Foundation A	Foundation B	Prior span
line-12-front-1	1	line-12-run-0001-span-0001	OSR-Pi25	FND:line-12-support-00000210500	FND:line-12-support-00000235500	mobilisation
line-12-front-1	2	line-12-run-0001-span-0002	OSR-Pi25	FND:line-12-support-00000235500	FND:line-12-support-00000260500	line-12-run-0001-span-0001
line-12-front-1	3	line-12-run-0001-span-0003	OSR-Pi25	FND:line-12-support-00000260500	FND:line-12-support-00000285500	line-12-run-0001-span-0002
line-12-front-1	4	line-12-run-0001-span-0004	OSR-Pi25	FND:line-12-support-00000285500	FND:line-12-support-00000310500	line-12-run-0001-span-0003
line-12-front-1	5	line-12-run-0001-span-0005	OSR-Pi25	FND:line-12-support-00000310500	FND:line-12-support-00000335500	line-12-run-0001-span-0004
line-12-front-1	6	line-12-run-0001-span-0006	OSR-Pi25	FND:line-12-support-00000335500	FND:line-12-support-00000360500	line-12-run-0001-span-0005
line-12-front-2	1	line-12-run-0001-span-0046	OSR-Pi20	FND:line-12-support-00001335500	FND:line-12-support-00001355500	mobilisation
line-12-front-2	2	line-12-run-0001-span-0045	OSR-Pi25	FND:line-12-support-00001310500	FND:line-12-support-00001335500	line-12-run-0001-span-0046
line-12-front-2	3	line-12-run-0001-span-0044	OSR-Pi25	FND:line-12-support-00001285500	FND:line-12-support-00001310500	line-12-run-0001-span-0045
line-12-front-2	4	line-12-run-0001-span-0043	OSR-Pi25	FND:line-12-support-00001260500	FND:line-12-support-00001285500	line-12-run-0001-span-0044
line-12-front-2	5	line-12-run-0001-span-0042	OSR-Pi25	FND:line-12-support-00001235500	FND:line-12-support-00001260500	line-12-run-0001-span-0043
line-12-front-2	6	line-12-run-0001-span-0041	OSR-Pi25	FND:line-12-support-00001210500	FND:line-12-support-00001235500	line-12-run-0001-span-0042

The compressed span register provides the full sequence, individual beam IDs, predecessors and junction holds. Stage library supplies the method dependencies. Front package lists every assigned span and actual disconnected working interval.

Junctions, stations and residential interfaces

Coordinated structural interface IDs on this line: crossing-5564106417eb.

Evaluate grade-separated crossings, shared four-track civil footprints or separate deck levels, and bounded platform/concourse/access complexes with the station authority. Freeze actual vertical profiles and gradients before selecting supports and ordinary spans. Preserve rail/PSD/egress/waterproofing/earthing interfaces. Proposed residential infill sites on this line require station/approach structures and revised service, fleet, power and full installed prices; they are not existing paid journeys.

Candidate infill IDs: no qualifying candidate under this screen.

Inspection, handover and programme

Use the foundation/production/lift/connection/concealed-work hold points from the master ITP. Capture installed component IDs, geometry, tests, personnel/equipment authority, NCR disposition and signed stage release. Update actual shift cycles, losses, stocks and resource reservations; do not infer an opening date from bay throughput.
Station/special/track/energy/depot/fleet/testing/approval releases close the whole line. Unknown ground quantities, installed costs and dates stay unknown.

Integrated master package · Interactive support/junction inspection.

line-13 — foundation and assembly construction plan

Repository source: `engineering/network-planning/baghdad/line-13-construction-plan.md`

Line-specific design-development package; survey, ground, supplier and staged-load releases are still required.

Route length 8.042 km. 127 unique proposed support packets; 123 identified catalogue bay assemblies and 2 special packages. Each ordinary bay has two named beam components and both foundation parents. The 25 m average is a sizing basis; actual Pi20/Pi25/closure geometry controls installation.

Front	Launcher	Previous asset assignment	Direction	Initial assembly chainage m	Bay assemblies
line-13-front-1	launcher-07	line-4-front-1	1	210.5	61
line-13-front-2	launcher-08	line-4-front-2	-1	3719.0	62

Foundation workface plan

Filter the foundation register by `line-13`. For every support, verify the proposed coordinate/chainage, rights, utilities and the referenced desktop sample; commission field ground, groundwater, durability and load investigations. Review shallow footing, bored shaft, driven bent and pile-group alternatives against actual ground, axial/lateral/settlement/scour and construction loads. Desktop topsoil does not select a pile depth, count or allowable bearing pressure.

Release trial and working-platform design (F1), install the selected foundation with actual geometry/material/installation records (F2), then verify integrity/load/settlement/dimensions and close the foundation packet (F3). Construct/erect column and cap with independently checked connections/temporary restraint (P0). Survey and inspect bearings, strength and the next launcher/beam/delivery stage before support release (B0). Keep 10–15 consecutive bays of accepted supports and beam stock ahead only where the run and storage permit.

Directed assembly and logistics sequence

Before mobilisation, release the predecessor asset assignment and its dismantle/transport/recommission package. The positive front installs toward increasing chainage; the negative front starts at its high-chainage end and installs backward. Factory/dispatch plans must follow this directed sequence, not the ascending raw span register. A foundation used by two bays is one packet; it is not built or priced twice.

For each front, identify the first complete run, permitted delivery/assembly site, temporary support/ground platform, configured plant and authorised shift/crew. Reserve both beam travellers, transport/receiving equipment, an operator, required riggers, lift supervision and inspection. Verify source strength, complete mass/CG, lifting points, rigging, bearings, weather limits and contingency landing/recovery before any lift.

Both support B0 packets and beam receiving H1 records precede E0. Place and restrain beam 1 (E1); then place/securing beam 2 (E2). Survey and accept connection/strength/bearings/NCRs (E3), release advance loads and the next supports before moving the launcher (E4), then complete walkways/barriers/waterproofing/drainage/track interface and following-trade handover (E5). The next bay depends on the preceding advance release. Following trades may overlap only with protected, agreed load/access boundaries.

Run boundaries, stations and specials require a passage or dismantle/transport/reassembly/recommission package. No discontinuity becomes an assumed launchable gap. Junction-affected orders remain on a design hold until J2 freezes the profile, clearance, support/cap geometry and staged-load/utility/access design. A physical crossing does not create a rail switch.

Initial bay packages

Front	Sequence	Span	Product	Foundation A	Foundation B	Prior span
line-13-front-1	1	line-13-run-0001-span-0001	OSR-Pi25	FND:line-13-support-000000210500	FND:line-13-support-000000235500	mobilisation
line-13-front-1	2	line-13-run-0001-span-0002	OSR-Pi25	FND:line-13-support-000000235500	FND:line-13-support-000000260500	line-13-run-0001-span-0001
line-13-front-1	3	line-13-run-0001-span-0003	OSR-Pi25	FND:line-13-support-000000260500	FND:line-13-support-000000285500	line-13-run-0001-span-0002
line-13-front-1	4	line-13-run-0001-span-0004	OSR-Pi25	FND:line-13-support-000000285500	FND:line-13-support-000000310500	line-13-run-0001-span-0003
line-13-front-1	5	line-13-run-0001-span-0005	OSR-Pi25	FND:line-13-support-000000310500	FND:line-13-support-000000335500	line-13-run-0001-span-0004
line-13-front-1	6	line-13-run-0001-span-0006	OSR-Pi25	FND:line-13-support-000000335500	FND:line-13-support-000000360500	line-13-run-0001-span-0005
line-13-front-2	1	line-13-run-0002-span-0011	OSR-Pi20	FND:line-13-support-000003699000	FND:line-13-support-000003719000	mobilisation
line-13-front-2	2	line-13-run-0002-span-0010	OSR-Pi25	FND:line-13-support-000003674000	FND:line-13-support-000003699000	line-13-run-0002-span-0011
line-13-front-2	3	line-13-run-0002-span-0009	OSR-Pi25	FND:line-13-support-000003649000	FND:line-13-support-000003674000	line-13-run-0002-span-0010
line-13-front-2	4	line-13-run-0002-span-0008	OSR-Pi25	FND:line-13-support-000003624000	FND:line-13-support-000003649000	line-13-run-0002-span-0009
line-13-front-2	5	line-13-run-0002-span-0007	OSR-Pi25	FND:line-13-support-000003599000	FND:line-13-support-000003624000	line-13-run-0002-span-0008
line-13-front-2	6	line-13-run-0002-span-0006	OSR-Pi25	FND:line-13-support-000003574000	FND:line-13-support-000003599000	line-13-run-0002-span-0007

The compressed span register provides the full sequence, individual beam IDs, predecessors and junction holds. Stage library supplies the method dependencies. Front package lists every assigned span and actual disconnected working interval.

Junctions, stations and residential interfaces

Coordinated structural interface IDs on this line: crossing-532e0b6bb63c, shared-corridor-96ce2230245a.

Evaluate grade-separated crossings, shared four-track civil footprints or separate deck levels, and bounded platform/concourse/access complexes with the station authority. Freeze actual vertical profiles and gradients before selecting supports and ordinary spans. Preserve rail/PSD/egress/waterproofing/earthing interfaces. Proposed residential infill sites on this line require station/approach structures and revised service, fleet, power and full installed prices; they are not existing paid journeys.

Candidate infill IDs: infill-line-13-006000, infill-line-13-001500.

Inspection, handover and programme

Use the foundation/production/lift/connection/concealed-work hold points from the master ITP. Capture installed component IDs, geometry, tests, personnel/equipment authority, NCR disposition and signed stage release. Update actual shift cycles, losses, stocks and resource reservations; do not infer an opening date from bay throughput.
Station/special/track/energy/depot/fleet/testing/approval releases close the whole line. Unknown ground quantities, installed costs and dates stay unknown.

Integrated master package · Interactive support/junction inspection.

line-14 — foundation and assembly construction plan

Repository source: `engineering/network-planning/baghdad/line-14-construction-plan.md`

Line-specific design-development package; survey, ground, supplier and staged-load releases are still required.

Route length 6.047 km. 227 unique proposed support packets; 225 identified catalogue bay assemblies and 1 special packages. Each ordinary bay has two named beam components and both foundation parents. The 25 m average is a sizing basis; actual Pi20/Pi25/closure geometry controls installation.

Front	Launcher	Previous asset assignment	Direction	Initial assembly chainage m	Bay assemblies
line-14-front-1	launcher-09	line-5-front-1	1	210.5	112
line-14-front-2	launcher-10	line-5-front-2	-1	5815.5	113

Foundation workface plan

Filter the foundation register by `line-14`. For every support, verify the proposed coordinate/chainage, rights, utilities and the referenced desktop sample; commission field ground, groundwater, durability and load investigations. Review shallow footing, bored shaft, driven bent and pile-group alternatives against actual ground, axial/lateral/settlement/scour and construction loads. Desktop topsoil does not select a pile depth, count or allowable bearing pressure.

Release trial and working-platform design (F1), install the selected foundation with actual geometry/material/installation records (F2), then verify integrity/load/settlement/dimensions and close the foundation packet (F3). Construct/erect column and cap with independently checked connections/temporary restraint (P0). Survey and inspect bearings, strength and the next launcher/beam/delivery stage before support release (B0). Keep 10–15 consecutive bays of accepted supports and beam stock ahead only where the run and storage permit.

Directed assembly and logistics sequence

Before mobilisation, release the predecessor asset assignment and its dismantle/transport/recommission package. The positive front installs toward increasing chainage; the negative front starts at its high-chainage end and installs backward. Factory/dispatch plans must follow this directed sequence, not the ascending raw span register. A foundation used by two bays is one packet; it is not built or priced twice.

For each front, identify the first complete run, permitted delivery/assembly site, temporary support/ground platform, configured plant and authorised shift/crew. Reserve both beam travellers, transport/receiving equipment, an operator, required riggers, lift supervision and inspection. Verify source strength, complete mass/CG, lifting points, rigging, bearings, weather limits and contingency landing/recovery before any lift.

Both support B0 packets and beam receiving H1 records precede E0. Place and restrain beam 1 (E1); then place/securing beam 2 (E2). Survey and accept connection/strength/bearings/NCRs (E3), release advance loads and the next supports before moving the launcher (E4), then complete walkways/barriers/waterproofing/drainage/track interface and following-trade handover (E5). The next bay depends on the preceding advance release. Following trades may overlap only with protected, agreed load/access boundaries.

Run boundaries, stations and specials require a passage or dismantle/transport/reassembly/recommission package. No discontinuity becomes an assumed launchable gap. Junction-affected orders remain on a design hold until J2 freezes the profile, clearance, support/cap geometry and staged-load/utility/access design. A physical crossing does not create a rail switch.

Initial bay packages

Front	Sequence	Span	Product	Foundation A	Foundation B	Prior span
line-14-front-1	1	line-14-run-0001-span-0001	OSR-Pi25	FND:line-14-support-00000210500	FND:line-14-support-00000235500	mobilisation
line-14-front-1	2	line-14-run-0001-span-0002	OSR-Pi25	FND:line-14-support-00000235500	FND:line-14-support-00000260500	line-14-run-0001-span-0001
line-14-front-1	3	line-14-run-0001-span-0003	OSR-Pi25	FND:line-14-support-00000260500	FND:line-14-support-00000285500	line-14-run-0001-span-0002
line-14-front-1	4	line-14-run-0001-span-0004	OSR-Pi25	FND:line-14-support-00000285500	FND:line-14-support-00000310500	line-14-run-0001-span-0003
line-14-front-1	5	line-14-run-0001-span-0005	OSR-Pi25	FND:line-14-support-00000310500	FND:line-14-support-00000335500	line-14-run-0001-span-0004
line-14-front-1	6	line-14-run-0001-span-0006	OSR-Pi25	FND:line-14-support-00000335500	FND:line-14-support-00000360500	line-14-run-0001-span-0005
line-14-front-2	1	line-14-run-0001-span-0225	OSR-Pi20	FND:line-14-support-00005795500	FND:line-14-support-00005815500	mobilisation
line-14-front-2	2	line-14-run-0001-span-0224	OSR-Pi20	FND:line-14-support-00005775500	FND:line-14-support-00005795500	line-14-run-0001-span-0225
line-14-front-2	3	line-14-run-0001-span-0223	OSR-Pi20	FND:line-14-support-00005755500	FND:line-14-support-00005775500	line-14-run-0001-span-0224
line-14-front-2	4	line-14-run-0001-span-0222	OSR-Pi20	FND:line-14-support-00005735500	FND:line-14-support-00005755500	line-14-run-0001-span-0223
line-14-front-2	5	line-14-run-0001-span-0221	OSR-Pi25	FND:line-14-support-00005710500	FND:line-14-support-00005735500	line-14-run-0001-span-0222
line-14-front-2	6	line-14-run-0001-span-0220	OSR-Pi25	FND:line-14-support-00005685500	FND:line-14-support-00005710500	line-14-run-0001-span-0221

The compressed span register provides the full sequence, individual beam IDs, predecessors and junction holds. Stage library supplies the method dependencies. Front package lists every assigned span and actual disconnected working interval.

Junctions, stations and residential interfaces

Coordinated structural interface IDs on this line: crossing-057c84aa0956, crossing-69a6a5b5c4a4.

Evaluate grade-separated crossings, shared four-track civil footprints or separate deck levels, and bounded platform/concourse/access complexes with the station authority. Freeze actual vertical profiles and gradients before selecting supports and ordinary spans. Preserve rail/PSD/egress/waterproofing/earthing interfaces. Proposed residential infill sites on this line require station/approach structures and revised service, fleet, power and full installed prices; they are not existing paid journeys.

Candidate infill IDs: infill-line-14-004500.

Inspection, handover and programme

Use the foundation/production/lift/connection/concealed-work hold points from the master ITP. Capture installed component IDs, geometry, tests, personnel/equipment authority, NCR disposition and signed stage release. Update actual shift cycles, losses, stocks and resource reservations; do not infer an opening date from bay throughput.
Station/special/track/energy/depot/fleet/testing/approval releases close the whole line. Unknown ground quantities, installed costs and dates stay unknown.

Integrated master package · Interactive support/junction inspection.

line-2 — foundation and assembly construction plan

Repository source: `engineering/network-planning/baghdad/line-2-construction-plan.md`

Line-specific design-development package; survey, ground, supplier and staged-load releases are still required.

Route length 52.851 km. 1,165 unique proposed support packets; 988 identified catalogue bay assemblies and 76 special packages. Each ordinary bay has two named beam components and both foundation parents. The 25 m average is a sizing basis; actual Pi20/Pi25/closure geometry controls installation.

Front	Launcher	Previous asset assignment	Direction	Initial assembly chainage m	Bay assemblies
line-2-front-1	launcher-03	initial mobilisation	1	2228.8	494
line-2-front-2	launcher-04	initial mobilisation	-1	51689.8	494

Foundation workface plan

Filter the foundation register by `line-2`. For every support, verify the proposed coordinate/chainage, rights, utilities and the referenced desktop sample; commission field ground, groundwater, durability and load investigations. Review shallow footing, bored shaft, driven bent and pile-group alternatives against actual ground, axial/lateral/settlement/scour and construction loads. Desktop topsoil does not select a pile depth, count or allowable bearing pressure.

Release trial and working-platform design (F1), install the selected foundation with actual geometry/material/installation records (F2), then verify integrity/load/settlement/dimensions and close the foundation packet (F3). Construct/erect column and cap with independently checked connections/temporary restraint (P0). Survey and inspect bearings, strength and the next launcher/beam/delivery stage before support release (B0). Keep 10–15 consecutive bays of accepted supports and beam stock ahead only where the run and storage permit.

Directed assembly and logistics sequence

Before mobilisation, release the predecessor asset assignment and its dismantle/transport/recommission package. The positive front installs toward increasing chainage; the negative front starts at its high-chainage end and installs backward. Factory/dispatch plans must follow this directed sequence, not the ascending raw span register. A foundation used by two bays is one packet; it is not built or priced twice.

For each front, identify the first complete run, permitted delivery/assembly site, temporary support/ground platform, configured plant and authorised shift/crew. Reserve both beam travellers, transport/receiving equipment, an operator, required riggers, lift supervision and inspection. Verify source strength, complete mass/CG, lifting points, rigging, bearings, weather limits and contingency landing/recovery before any lift.

Both support B0 packets and beam receiving H1 records precede E0. Place and restrain beam 1 (E1); then place/securing beam 2 (E2). Survey and accept connection/strength/bearings/NCRs (E3), release advance loads and the next supports before moving the launcher (E4), then complete walkways/barriers/waterproofing/drainage/track interface and following-trade handover (E5). The next bay depends on the preceding advance release. Following trades may overlap only with protected, agreed load/access boundaries.

Run boundaries, stations and specials require a passage or dismantle/transport/reassembly/recommission package. No discontinuity becomes an assumed launchable gap. Junction-affected orders remain on a design hold until J2 freezes the profile, clearance, support/cap geometry and staged-load/utility/access design. A physical crossing does not create a rail switch.

Initial bay packages

Front	Sequence	Span	Product	Foundation A	Foundation B	Prior span
line-2-front-1	1	line-2-run-0001-span-0001	OSR-Pi25	FND:line-2-support-000002228800	FND:line-2-support-000002253800	mobilisation
line-2-front-1	2	line-2-run-0002-span-0001	OSR-Pi25	FND:line-2-support-000002446800	FND:line-2-support-000002471800	line-2-run-0001-span-0001
line-2-front-1	3	line-2-run-0004-span-0001	OSR-Pi20	FND:line-2-support-000004334700	FND:line-2-support-000004354700	line-2-run-0002-span-0001
line-2-front-1	4	line-2-run-0005-span-0001	OSR-Pi20	FND:line-2-support-000004572700	FND:line-2-support-000004592700	line-2-run-0004-span-0001
line-2-front-1	5	line-2-run-0008-span-0001	OSR-Pi20	FND:line-2-support-000009424200	FND:line-2-support-000009444200	line-2-run-0005-span-0001
line-2-front-1	6	line-2-run-0013-span-0001	OSR-Pi20	FND:line-2-support-000012035800	FND:line-2-support-000012055800	line-2-run-0008-span-0001
line-2-front-2	1	line-2-run-0101-span-0001	OSR-Pi20	FND:line-2-support-000051669800	FND:line-2-support-000051689800	mobilisation
line-2-front-2	2	line-2-run-0100-span-0002	OSR-Pi20	FND:line-2-support-000051509800	FND:line-2-support-000051529800	line-2-run-0101-span-0001
line-2-front-2	3	line-2-run-0100-span-0001	OSR-Pi20	FND:line-2-support-000051489800	FND:line-2-support-000051509800	line-2-run-0100-span-0002
line-2-front-2	4	line-2-run-0099-span-0001	OSR-Pi20	FND:line-2-support-000050274400	FND:line-2-support-000050294400	line-2-run-0100-span-0001
line-2-front-2	5	line-2-run-0097-span-0042	OSR-Pi20	FND:line-2-support-000049889800	FND:line-2-support-000049909800	line-2-run-0099-span-0001
line-2-front-2	6	line-2-run-0097-span-0041	OSR-Pi20	FND:line-2-support-000049869800	FND:line-2-support-000049889800	line-2-run-0097-span-0042

The compressed span register provides the full sequence, individual beam IDs, predecessors and junction holds. Stage library supplies the method dependencies. Front package lists every assigned span and actual disconnected working interval.

Junctions, stations and residential interfaces

Coordinated structural interface IDs on this line: crossing-1c3badfbb529, crossing-5564106417eb, crossing-6644faf065db, crossing-778f06209d30, crossing-868cd72fc737, crossing-9eea764fde61, crossing-c9cacdc51c5b, shared-corridor-ea625bfc9e95.

Evaluate grade-separated crossings, shared four-track civil footprints or separate deck levels, and bounded platform/concourse/access complexes with the station authority. Freeze actual vertical profiles and gradients before selecting supports and ordinary spans. Preserve rail/PSD/egress/waterproofing/earthing interfaces. Proposed residential infill sites on this line require station/approach structures and revised service, fleet, power and full installed prices; they are not existing paid journeys.

Candidate infill IDs: infill-line-2-022000, infill-line-2-032000, infill-line-2-046500.

Inspection, handover and programme

Use the foundation/production/lift/connection/concealed-work hold points from the master ITP. Capture installed component IDs, geometry, tests, personnel/equipment authority, NCR disposition and signed stage release. Update actual shift cycles, losses, stocks and resource reservations; do not infer an opening date from bay throughput. Station/special/track/energy/depot/fleet/testing/approval releases close the whole line. Unknown ground quantities, installed costs and dates stay unknown.

Integrated master package · Interactive support/junction inspection.

line-3 — foundation and assembly construction plan

Repository source: `engineering/network-planning/baghdad/line-3-construction-plan.md`

Line-specific design-development package; survey, ground, supplier and staged-load releases are still required.

Route length 49.496 km. 932 unique proposed support packets; 863 identified catalogue bay assemblies and 29 special packages. Each ordinary bay has two named beam components and both foundation parents. The 25 m average is a sizing basis; actual Pi20/Pi25/closure geometry controls installation.

Front	Launcher	Previous asset assignment	Direction	Initial assembly chainage m	Bay assemblies
line-3-front-1	launcher-05	initial mobilisation	1	3436.5	431
line-3-front-2	launcher-06	initial mobilisation	-1	48187.4	432

Foundation workface plan

Filter the foundation register by `line-3`. For every support, verify the proposed coordinate/chainage, rights, utilities and the referenced desktop sample; commission field ground, groundwater, durability and load investigations. Review shallow footing, bored shaft, driven bent and pile-group alternatives against actual ground, axial/lateral/settlement/scour and construction loads. Desktop topsoil does not select a pile depth, count or allowable bearing pressure.

Release trial and working-platform design (F1), install the selected foundation with actual geometry/material/installation records (F2), then verify integrity/load/settlement/dimensions and close the foundation packet (F3). Construct/erect column and cap with independently checked connections/temporary restraint (P0). Survey and inspect bearings, strength and the next launcher/beam/delivery stage before support release (B0). Keep 10–15 consecutive bays of accepted supports and beam stock ahead only where the run and storage permit.

Directed assembly and logistics sequence

Before mobilisation, release the predecessor asset assignment and its dismantle/transport/recommission package. The positive front installs toward increasing chainage; the negative front starts at its high-chainage end and installs backward. Factory/dispatch plans must follow this directed sequence, not the ascending raw span register. A foundation used by two bays is one packet; it is not built or priced twice.

For each front, identify the first complete run, permitted delivery/assembly site, temporary support/ground platform, configured plant and authorised shift/crew. Reserve both beam travellers, transport/receiving equipment, an operator, required riggers, lift supervision and inspection. Verify source strength, complete mass/CG, lifting points, rigging, bearings, weather limits and contingency landing/recovery before any lift.

Both support B0 packets and beam receiving H1 records precede E0. Place and restrain beam 1 (E1); then place/securing beam 2 (E2). Survey and accept connection/strength/bearings/NCRs (E3), release advance loads and the next supports before moving the launcher (E4), then complete walkways/barriers/waterproofing/drainage/track interface and following-trade handover (E5). The next bay depends on the preceding advance release. Following trades may overlap only with protected, agreed load/access boundaries.

Run boundaries, stations and specials require a passage or dismantle/transport/reassembly/recommission package. No discontinuity becomes an assumed launchable gap. Junction-affected orders remain on a design hold until J2 freezes the profile, clearance, support/cap geometry and staged-load/utility/access design. A physical crossing does not create a rail switch.

Initial bay packages

Front	Sequence	Span	Product	Foundation A	Foundation B	Prior span
line-3-front-1	1	line-3-run-0005-span-0001	OSR-Pi25	FND:line-3-support-000003436500	FND:line-3-support-000003461500	mobilisation
line-3-front-1	2	line-3-run-0007-span-0001	OSR-Pi25	FND:line-3-support-000004584200	FND:line-3-support-000004609200	line-3-run-0005-span-0001
line-3-front-1	3	line-3-run-0007-span-0002	OSR-Pi20	FND:line-3-support-000004609200	FND:line-3-support-000004629200	line-3-run-0007-span-0001
line-3-front-1	4	line-3-run-0007-span-0003	OSR-Pi20	FND:line-3-support-000004629200	FND:line-3-support-000004649200	line-3-run-0007-span-0002
line-3-front-1	5	line-3-run-0009-span-0001	OSR-Pi20	FND:line-3-support-000006777000	FND:line-3-support-000006797000	line-3-run-0007-span-0003
line-3-front-1	6	line-3-run-0009-span-0002	OSR-Pi20	FND:line-3-support-000006797000	FND:line-3-support-000006817000	line-3-run-0009-span-0001
line-3-front-2	1	line-3-run-0040-span-0003	OSR-Pi20	FND:line-3-support-000048167400	FND:line-3-support-000048187400	mobilisation
line-3-front-2	2	line-3-run-0040-span-0002	OSR-Pi20	FND:line-3-support-000048147400	FND:line-3-support-000048167400	line-3-run-0040-span-0003
line-3-front-2	3	line-3-run-0040-span-0001	OSR-Pi20	FND:line-3-support-000048127400	FND:line-3-support-000048147400	line-3-run-0040-span-0002
line-3-front-2	4	line-3-run-0039-span-0003	OSR-Pi20	FND:line-3-support-000048047400	FND:line-3-support-000048067400	line-3-run-0040-span-0001
line-3-front-2	5	line-3-run-0039-span-0002	OSR-Pi20	FND:line-3-support-000048027400	FND:line-3-support-000048047400	line-3-run-0039-span-0003
line-3-front-2	6	line-3-run-0039-span-0001	OSR-Pi20	FND:line-3-support-000048007400	FND:line-3-support-000048027400	line-3-run-0039-span-0002

The compressed span register provides the full sequence, individual beam IDs, predecessors and junction holds. Stage library supplies the method dependencies. Front package lists every assigned span and actual disconnected working interval.

Junctions, stations and residential interfaces

Coordinated structural interface IDs on this line: crossing-0fac546553f2, crossing-1fc40bbf9e02, crossing-42edbb5cfef1, crossing-658b4261f614, crossing-66c0e3e30d82, crossing-778f06209d30, crossing-86eb625bd399, crossing-bde9f0d16859.

Evaluate grade-separated crossings, shared four-track civil footprints or separate deck levels, and bounded platform/concourse/access complexes with the station authority. Freeze actual vertical profiles and gradients before selecting supports and ordinary spans. Preserve rail/PSD/egress/waterproofing/earthing interfaces. Proposed residential infill sites on this line require station/approach structures and revised service, fleet, power and full installed prices; they are not existing paid journeys.

Candidate infill IDs: infill-line-3-030000, infill-line-3-039500, infill-line-3-026500, infill-line-3-019500.

Inspection, handover and programme

Use the foundation/production/lift/connection/concealed-work hold points from the master ITP. Capture installed component IDs, geometry, tests, personnel/equipment authority, NCR disposition and signed stage release. Update actual shift cycles, losses, stocks and resource reservations; do not infer an opening date from bay throughput. Station/special/track/energy/depot/fleet/testing/approval releases close the whole line. Unknown ground quantities, installed costs and dates stay unknown.

Integrated master package · Interactive support/junction inspection.

line-4 — foundation and assembly construction plan

Repository source: `engineering/network-planning/baghdad/line-4-construction-plan.md`

Line-specific design-development package; survey, ground, supplier and staged-load releases are still required.

Route length 41.932 km. 1,040 unique proposed support packets; 943 identified catalogue bay assemblies and 44 special packages. Each ordinary bay has two named beam components and both foundation parents. The 25 m average is a sizing basis; actual Pi20/Pi25/closure geometry controls installation.

Front	Launcher	Previous asset assignment	Direction	Initial assembly chainage m	Bay assemblies
line-4-front-1	launcher-07	initial mobilisation	1	240.0	471
line-4-front-2	launcher-08	initial mobilisation	-1	41149.0	472

Foundation workface plan

Filter the foundation register by `line-4`. For every support, verify the proposed coordinate/chainage, rights, utilities and the referenced desktop sample; commission field ground, groundwater, durability and load investigations. Review shallow footing, bored shaft, driven bent and pile-group alternatives against actual ground, axial/lateral/settlement/scour and construction loads. Desktop topsoil does not select a pile depth, count or allowable bearing pressure.

Release trial and working-platform design (F1), install the selected foundation with actual geometry/material/installation records (F2), then verify integrity/load/settlement/dimensions and close the foundation packet (F3). Construct/erect column and cap with independently checked connections/temporary restraint (P0). Survey and inspect bearings, strength and the next launcher/beam/delivery stage before support release (B0). Keep 10–15 consecutive bays of accepted supports and beam stock ahead only where the run and storage permit.

Directed assembly and logistics sequence

Before mobilisation, release the predecessor asset assignment and its dismantle/transport/recommission package. The positive front installs toward increasing chainage; the negative front starts at its high-chainage end and installs backward. Factory/dispatch plans must follow this directed sequence, not the ascending raw span register. A foundation used by two bays is one packet; it is not built or priced twice.

For each front, identify the first complete run, permitted delivery/assembly site, temporary support/ground platform, configured plant and authorised shift/crew. Reserve both beam travellers, transport/receiving equipment, an operator, required riggers, lift supervision and inspection. Verify source strength, complete mass/CG, lifting points, rigging, bearings, weather limits and contingency landing/recovery before any lift.

Both support B0 packets and beam receiving H1 records precede E0. Place and restrain beam 1 (E1); then place/securing beam 2 (E2). Survey and accept connection/strength/bearings/NCRs (E3), release advance loads and the next supports before moving the launcher (E4), then complete walkways/barriers/waterproofing/drainage/track interface and following-trade handover (E5). The next bay depends on the preceding advance release. Following trades may overlap only with protected, agreed load/access boundaries.

Run boundaries, stations and specials require a passage or dismantle/transport/reassembly/recommission package. No discontinuity becomes an assumed launchable gap. Junction-affected orders remain on a design hold until J2 freezes the profile, clearance, support/cap geometry and staged-load/utility/access design. A physical crossing does not create a rail switch.

Initial bay packages

Front	Sequence	Span	Product	Foundation A	Foundation B	Prior span
line-4-front-1	1	line-4-run-0001-span-0001	OSR-Pi25	FND:line-4-support-000000240000	FND:line-4-support-000000265000	mobilisation
line-4-front-1	2	line-4-run-0002-span-0001	OSR-Pi25	FND:line-4-support-000000421400	FND:line-4-support-000000446400	line-4-run-0001-span-0001
line-4-front-1	3	line-4-run-0003-span-0001	OSR-Pi25	FND:line-4-support-000000546300	FND:line-4-support-000000571300	line-4-run-0002-span-0001
line-4-front-1	4	line-4-run-0003-span-0002	OSR-Pi25	FND:line-4-support-000000571300	FND:line-4-support-000000596300	line-4-run-0003-span-0001
line-4-front-1	5	line-4-run-0004-span-0001	OSR-Pi25	FND:line-4-support-000000642800	FND:line-4-support-000000667800	line-4-run-0003-span-0002
line-4-front-1	6	line-4-run-0004-span-0002	OSR-Pi20	FND:line-4-support-000000667800	FND:line-4-support-000000687800	line-4-run-0004-span-0001
line-4-front-2	1	line-4-run-0053-span-0002	OSR-Pi20	FND:line-4-support-000041129000	FND:line-4-support-000041149000	mobilisation
line-4-front-2	2	line-4-run-0053-span-0001	OSR-Pi20	FND:line-4-support-000041109000	FND:line-4-support-000041129000	line-4-run-0053-span-0002
line-4-front-2	3	line-4-run-0052-span-0001	OSR-Pi20	FND:line-4-support-000040884100	FND:line-4-support-000040904100	line-4-run-0053-span-0001
line-4-front-2	4	line-4-run-0051-span-0002	OSR-Pi20	FND:line-4-support-000040393100	FND:line-4-support-000040413100	line-4-run-0052-span-0001
line-4-front-2	5	line-4-run-0051-span-0001	OSR-Pi25	FND:line-4-support-000040368100	FND:line-4-support-000040393100	line-4-run-0051-span-0002
line-4-front-2	6	line-4-run-0049-span-0188	OSR-Pi25	FND:line-4-support-000039568400	FND:line-4-support-000039593400	line-4-run-0051-span-0001

The compressed span register provides the full sequence, individual beam IDs, predecessors and junction holds. Stage library supplies the method dependencies. Front package lists every assigned span and actual disconnected working interval.

Junctions, stations and residential interfaces

Coordinated structural interface IDs on this line: crossing-27b707b08a14, crossing-42edbb5cfef1, crossing-4761735b935e, crossing-69a6a5b5c4a4, crossing-6f9463d5811f, crossing-954a77cefa6e, crossing-a79096d48aef, crossing-dff9d62fd115.

Evaluate grade-separated crossings, shared four-track civil footprints or separate deck levels, and bounded platform/concourse/access complexes with the station authority. Freeze actual vertical profiles and gradients before selecting supports and ordinary spans. Preserve rail/PSD/egress/waterproofing/earthing interfaces. Proposed residential infill sites on this line require station/approach structures and revised service, fleet, power and full installed prices; they are not existing paid journeys.

Candidate infill IDs: infill-line-4-028000, infill-line-4-022000, infill-line-4-037000, infill-line-4-024000, infill-line-4-017000.

Inspection, handover and programme

Use the foundation/production/lift/connection/concealed-work hold points from the master ITP. Capture installed component IDs, geometry, tests, personnel/equipment authority, NCR disposition and signed stage release. Update actual shift cycles, losses, stocks and resource reservations; do not infer an opening date from bay throughput. Station/special/track/energy/depot/fleet/testing/approval releases close the whole line. Unknown ground quantities, installed costs and dates stay unknown.

Integrated master package · Interactive support/junction inspection.

line-5 — foundation and assembly construction plan

Repository source: `engineering/network-planning/baghdad/line-5-construction-plan.md`

Line-specific design-development package; survey, ground, supplier and staged-load releases are still required.

Route length 47.345 km. 952 unique proposed support packets; 874 identified catalogue bay assemblies and 35 special packages. Each ordinary bay has two named beam components and both foundation parents. The 25 m average is a sizing basis; actual Pi20/Pi25/closure geometry controls installation.

Front	Launcher	Previous asset assignment	Direction	Initial assembly chainage m	Bay assemblies
line-5-front-1	launcher-09	initial mobilisation	1	656.6	437
line-5-front-2	launcher-10	initial mobilisation	-1	46242.5	437

Foundation workface plan

Filter the foundation register by `line-5`. For every support, verify the proposed coordinate/chainage, rights, utilities and the referenced desktop sample; commission field ground, groundwater, durability and load investigations. Review shallow footing, bored shaft, driven bent and pile-group alternatives against actual ground, axial/lateral/settlement/scour and construction loads. Desktop topsoil does not select a pile depth, count or allowable bearing pressure.

Release trial and working-platform design (F1), install the selected foundation with actual geometry/material/installation records (F2), then verify integrity/load/settlement/dimensions and close the foundation packet (F3). Construct/erect column and cap with independently checked connections/temporary restraint (P0). Survey and inspect bearings, strength and the next launcher/beam/delivery stage before support release (B0). Keep 10–15 consecutive bays of accepted supports and beam stock ahead only where the run and storage permit.

Directed assembly and logistics sequence

Before mobilisation, release the predecessor asset assignment and its dismantle/transport/recommission package. The positive front installs toward increasing chainage; the negative front starts at its high-chainage end and installs backward. Factory/dispatch plans must follow this directed sequence, not the ascending raw span register. A foundation used by two bays is one packet; it is not built or priced twice.

For each front, identify the first complete run, permitted delivery/assembly site, temporary support/ground platform, configured plant and authorised shift/crew. Reserve both beam travellers, transport/receiving equipment, an operator, required riggers, lift supervision and inspection. Verify source strength, complete mass/CG, lifting points, rigging, bearings, weather limits and contingency landing/recovery before any lift.

Both support B0 packets and beam receiving H1 records precede E0. Place and restrain beam 1 (E1); then place/securing beam 2 (E2). Survey and accept connection/strength/bearings/NCRs (E3), release advance loads and the next supports before moving the launcher (E4), then complete walkways/barriers/waterproofing/drainage/track interface and following-trade handover (E5). The next bay depends on the preceding advance release. Following trades may overlap only with protected, agreed load/access boundaries.

Run boundaries, stations and specials require a passage or dismantle/transport/reassembly/recommission package. No discontinuity becomes an assumed launchable gap. Junction-affected orders remain on a design hold until J2 freezes the profile, clearance, support/cap geometry and staged-load/utility/access design. A physical crossing does not create a rail switch.

Initial bay packages

Front	Sequence	Span	Product	Foundation A	Foundation B	Prior span
line-5-front-1	1	line-5-run-0001-span-0001	OSR-Pi25	FND:line-5-support-000000656600	FND:line-5-support-000000681600	mobilisation
line-5-front-1	2	line-5-run-0001-span-0002	OSR-Pi25	FND:line-5-support-000000681600	FND:line-5-support-000000706600	line-5-run-0001-span-0001
line-5-front-1	3	line-5-run-0001-span-0003	OSR-Pi25	FND:line-5-support-000000706600	FND:line-5-support-000000731600	line-5-run-0001-span-0002
line-5-front-1	4	line-5-run-0001-span-0004	OSR-Pi25	FND:line-5-support-000000731600	FND:line-5-support-000000756600	line-5-run-0001-span-0003
line-5-front-1	5	line-5-run-0001-span-0005	OSR-Pi25	FND:line-5-support-000000756600	FND:line-5-support-000000781600	line-5-run-0001-span-0004
line-5-front-1	6	line-5-run-0001-span-0006	OSR-Pi25	FND:line-5-support-000000781600	FND:line-5-support-000000806600	line-5-run-0001-span-0005
line-5-front-2	1	line-5-run-0042-span-0001	OSR-Pi20	FND:line-5-support-000046222500	FND:line-5-support-000046242500	mobilisation
line-5-front-2	2	line-5-run-0041-span-0001	OSR-Pi25	FND:line-5-support-000045268000	FND:line-5-support-000045293000	line-5-run-0042-span-0001
line-5-front-2	3	line-5-run-0040-span-0001	OSR-Pi20	FND:line-5-support-000045003100	FND:line-5-support-000045023100	line-5-run-0041-span-0001
line-5-front-2	4	line-5-run-0038-span-0001	OSR-Pi25	FND:line-5-support-000044267100	FND:line-5-support-000044292100	line-5-run-0040-span-0001
line-5-front-2	5	line-5-run-0037-span-0002	OSR-Pi20	FND:line-5-support-000042635900	FND:line-5-support-000042655900	line-5-run-0038-span-0001
line-5-front-2	6	line-5-run-0037-span-0001	OSR-Pi25	FND:line-5-support-000042610900	FND:line-5-support-000042635900	line-5-run-0037-span-0002

The compressed span register provides the full sequence, individual beam IDs, predecessors and junction holds. Stage library supplies the method dependencies. Front package lists every assigned span and actual disconnected working interval.

Junctions, stations and residential interfaces

Coordinated structural interface IDs on this line: crossing-1c303c088c01, crossing-4e4707f7a13b, crossing-596aa3d726d4, crossing-66c0e3e30d82, crossing-6f9463d5811f, crossing-868cd72fc737, crossing-ec8cd65ba79b, shared-corridor-5c1c3ab579e9, shared-corridor-96ce2230245a.

Evaluate grade-separated crossings, shared four-track civil footprints or separate deck levels, and bounded platform/concourse/access complexes with the station authority. Freeze actual vertical profiles and gradients before selecting supports and ordinary spans. Preserve rail/PSD/egress/waterproofing/earthing interfaces. Proposed residential infill sites on this line require station/approach structures and revised service, fleet, power and full installed prices; they are not existing paid journeys.

Candidate infill IDs: infill-line-5-028000, infill-line-5-011000, infill-line-5-019500, infill-line-5-016000, infill-line-5-005500, infill-line-5-009500, infill-line-5-023000.

Inspection, handover and programme

Use the foundation/production/lift/connection/concealed-work hold points from the master ITP. Capture installed component IDs, geometry, tests, personnel/equipment authority, NCR disposition and signed stage release. Update actual shift cycles, losses, stocks and resource reservations; do not infer an opening date from bay throughput. Station/special/track/energy/depot/fleet/testing/approval releases close the whole line. Unknown ground quantities, installed costs and dates stay unknown.

Integrated master package · Interactive support/junction inspection.

line-6 — foundation and assembly construction plan

Repository source: `engineering/network-planning/baghdad/line-6-construction-plan.md`

Line-specific design-development package; survey, ground, supplier and staged-load releases are still required.

Route length 52.503 km. 974 unique proposed support packets; 846 identified catalogue bay assemblies and 60 special packages. Each ordinary bay has two named beam components and both foundation parents. The 25 m average is a sizing basis; actual Pi20/Pi25/closure geometry controls installation.

Front	Launcher	Previous asset assignment	Direction	Initial assembly chainage m	Bay assemblies
line-6-front-1	launcher-11	initial mobilisation	1	20.0	423
line-6-front-2	launcher-12	initial mobilisation	-1	46777.2	423

Foundation workface plan

Filter the foundation register by `line-6`. For every support, verify the proposed coordinate/chainage, rights, utilities and the referenced desktop sample; commission field ground, groundwater, durability and load investigations. Review shallow footing, bored shaft, driven bent and pile-group alternatives against actual ground, axial/lateral/settlement/scour and construction loads. Desktop topsoil does not select a pile depth, count or allowable bearing pressure.

Release trial and working-platform design (F1), install the selected foundation with actual geometry/material/installation records (F2), then verify integrity/load/settlement/dimensions and close the foundation packet (F3). Construct/erect column and cap with independently checked connections/temporary restraint (P0). Survey and inspect bearings, strength and the next launcher/beam/delivery stage before support release (B0). Keep 10–15 consecutive bays of accepted supports and beam stock ahead only where the run and storage permit.

Directed assembly and logistics sequence

Before mobilisation, release the predecessor asset assignment and its dismantle/transport/recommission package. The positive front installs toward increasing chainage; the negative front starts at its high-chainage end and installs backward. Factory/dispatch plans must follow this directed sequence, not the ascending raw span register. A foundation used by two bays is one packet; it is not built or priced twice.

For each front, identify the first complete run, permitted delivery/assembly site, temporary support/ground platform, configured plant and authorised shift/crew. Reserve both beam travellers, transport/receiving equipment, an operator, required riggers, lift supervision and inspection. Verify source strength, complete mass/CG, lifting points, rigging, bearings, weather limits and contingency landing/recovery before any lift.

Both support B0 packets and beam receiving H1 records precede E0. Place and restrain beam 1 (E1); then place/securing beam 2 (E2). Survey and accept connection/strength/bearings/NCRs (E3), release advance loads and the next supports before moving the launcher (E4), then complete walkways/barriers/waterproofing/drainage/track interface and following-trade handover (E5). The next bay depends on the preceding advance release. Following trades may overlap only with protected, agreed load/access boundaries.

Run boundaries, stations and specials require a passage or dismantle/transport/reassembly/recommission package. No discontinuity becomes an assumed launchable gap. Junction-affected orders remain on a design hold until J2 freezes the profile, clearance, support/cap geometry and staged-load/utility/access design. A physical crossing does not create a rail switch.

Initial bay packages

Front	Sequence	Span	Product	Foundation A	Foundation B	Prior span
line-6-front-1	1	line-6-run-0001-span-0001	OSR-Pi20	FND:line-6-support-000000020000	FND:line-6-support-000000040000	mobilisation
line-6-front-1	2	line-6-run-0001-span-0002	OSR-Pi20	FND:line-6-support-000000040000	FND:line-6-support-000000060000	line-6-run-0001-span-0001
line-6-front-1	3	line-6-run-0001-span-0003	OSR-Pi20	FND:line-6-support-000000060000	FND:line-6-support-000000080000	line-6-run-0001-span-0002
line-6-front-1	4	line-6-run-0001-span-0004	OSR-Pi20	FND:line-6-support-000000080000	FND:line-6-support-000000100000	line-6-run-0001-span-0003
line-6-front-1	5	line-6-run-0003-span-0001	OSR-Pi20	FND:line-6-support-000000346300	FND:line-6-support-000000366300	line-6-run-0001-span-0004
line-6-front-1	6	line-6-run-0004-span-0001	OSR-Pi25	FND:line-6-support-000000522800	FND:line-6-support-000000547800	line-6-run-0003-span-0001
line-6-front-2	1	line-6-run-0068-span-0001	OSR-Pi20	FND:line-6-support-000046757200	FND:line-6-support-000046777200	mobilisation
line-6-front-2	2	line-6-run-0067-span-0001	OSR-Pi20	FND:line-6-support-000046277200	FND:line-6-support-000046297200	line-6-run-0068-span-0001
line-6-front-2	3	line-6-run-0066-span-0036	OSR-Pi25	FND:line-6-support-000046087900	FND:line-6-support-000046112900	line-6-run-0067-span-0001
line-6-front-2	4	line-6-run-0066-span-0035	OSR-Pi25	FND:line-6-support-000046062900	FND:line-6-support-000046087900	line-6-run-0066-span-0036
line-6-front-2	5	line-6-run-0066-span-0034	OSR-Pi25	FND:line-6-support-000046037900	FND:line-6-support-000046062900	line-6-run-0066-span-0035
line-6-front-2	6	line-6-run-0066-span-0033	OSR-Pi25	FND:line-6-support-000046012900	FND:line-6-support-000046037900	line-6-run-0066-span-0034

The compressed span register provides the full sequence, individual beam IDs, predecessors and junction holds. Stage library supplies the method dependencies. Front package lists every assigned span and actual disconnected working interval.

Junctions, stations and residential interfaces

Coordinated structural interface IDs on this line: crossing-1c303c088c01, crossing-27b707b08a14, crossing-526dbc821706, crossing-658b4261f614, crossing-7bba02156870, crossing-8431d2567c58, crossing-9090e87568d6, crossing-9eea764fde61, crossing-c15126eb0d40.

Evaluate grade-separated crossings, shared four-track civil footprints or separate deck levels, and bounded platform/concourse/access complexes with the station authority. Freeze actual vertical profiles and gradients before selecting supports and ordinary spans. Preserve rail/PSD/egress/waterproofing/earthing interfaces. Proposed residential infill sites on this line require station/approach structures and revised service, fleet, power and full installed prices; they are not existing paid journeys.

Candidate infill IDs: infill-line-6-033000, infill-line-6-035000.

Inspection, handover and programme

Use the foundation/production/lift/connection/concealed-work hold points from the master ITP. Capture installed component IDs, geometry, tests, personnel/equipment authority, NCR disposition and signed stage release. Update actual shift cycles, losses, stocks and resource reservations; do not infer an opening date from bay throughput. Station/special/track/energy/depot/fleet/testing/approval releases close the whole line. Unknown ground quantities, installed costs and dates stay unknown.

Integrated master package · Interactive support/junction inspection.

line-7 — foundation and assembly construction plan

Repository source: `engineering/network-planning/baghdad/line-7-construction-plan.md`

Line-specific design-development package; survey, ground, supplier and staged-load releases are still required.

Route length 39.957 km. 867 unique proposed support packets; 813 identified catalogue bay assemblies and 21 special packages. Each ordinary bay has two named beam components and both foundation parents. The 25 m average is a sizing basis; actual Pi20/Pi25/closure geometry controls installation.

Front	Launcher	Previous asset assignment	Direction	Initial assembly chainage m	Bay assemblies
line-7-front-1	launcher-13	initial mobilisation	1	1036.6	406
line-7-front-2	launcher-14	initial mobilisation	-1	39723.9	407

Foundation workface plan

Filter the foundation register by `line-7`. For every support, verify the proposed coordinate/chainage, rights, utilities and the referenced desktop sample; commission field ground, groundwater, durability and load investigations. Review shallow footing, bored shaft, driven bent and pile-group alternatives against actual ground, axial/lateral/settlement/scour and construction loads. Desktop topsoil does not select a pile depth, count or allowable bearing pressure.

Release trial and working-platform design (F1), install the selected foundation with actual geometry/material/installation records (F2), then verify integrity/load/settlement/dimensions and close the foundation packet (F3). Construct/erect column and cap with independently checked connections/temporary restraint (P0). Survey and inspect bearings, strength and the next launcher/beam/delivery stage before support release (B0). Keep 10–15 consecutive bays of accepted supports and beam stock ahead only where the run and storage permit.

Directed assembly and logistics sequence

Before mobilisation, release the predecessor asset assignment and its dismantle/transport/recommission package. The positive front installs toward increasing chainage; the negative front starts at its high-chainage end and installs backward. Factory/dispatch plans must follow this directed sequence, not the ascending raw span register. A foundation used by two bays is one packet; it is not built or priced twice.

For each front, identify the first complete run, permitted delivery/assembly site, temporary support/ground platform, configured plant and authorised shift/crew. Reserve both beam travellers, transport/receiving equipment, an operator, required riggers, lift supervision and inspection. Verify source strength, complete mass/CG, lifting points, rigging, bearings, weather limits and contingency landing/recovery before any lift.

Both support B0 packets and beam receiving H1 records precede E0. Place and restrain beam 1 (E1); then place/securing beam 2 (E2). Survey and accept connection/strength/bearings/NCRs (E3), release advance loads and the next supports before moving the launcher (E4), then complete walkways/barriers/waterproofing/drainage/track interface and following-trade handover (E5). The next bay depends on the preceding advance release. Following trades may overlap only with protected, agreed load/access boundaries.

Run boundaries, stations and specials require a passage or dismantle/transport/reassembly/recommission package. No discontinuity becomes an assumed launchable gap. Junction-affected orders remain on a design hold until J2 freezes the profile, clearance, support/cap geometry and staged-load/utility/access design. A physical crossing does not create a rail switch.

Initial bay packages

Front	Sequence	Span	Product	Foundation A	Foundation B	Prior span
line-7-front-1	1	line-7-run-0001-span-0001	OSR-Pi20	FND:line-7-support-000001036600	FND:line-7-support-000001056600	mobilisation
line-7-front-1	2	line-7-run-0002-span-0001	OSR-Pi20	FND:line-7-support-000005141400	FND:line-7-support-000005161400	line-7-run-0001-span-0001
line-7-front-1	3	line-7-run-0003-span-0001	OSR-Pi20	FND:line-7-support-000007085100	FND:line-7-support-000007105100	line-7-run-0002-span-0001
line-7-front-1	4	line-7-run-0004-span-0001	OSR-Pi20	FND:line-7-support-000011133900	FND:line-7-support-000011153900	line-7-run-0003-span-0001
line-7-front-1	5	line-7-run-0005-span-0001	OSR-Pi25	FND:line-7-support-000011173900	FND:line-7-support-000011198900	line-7-run-0004-span-0001
line-7-front-1	6	line-7-run-0006-span-0001	OSR-Pi20	FND:line-7-support-000013002400	FND:line-7-support-000013022400	line-7-run-0005-span-0001
line-7-front-2	1	line-7-run-0033-span-0068	OSR-Pi25	FND:line-7-support-000039698900	FND:line-7-support-000039723900	mobilisation
line-7-front-2	2	line-7-run-0033-span-0067	OSR-Pi25	FND:line-7-support-000039673900	FND:line-7-support-000039698900	line-7-run-0033-span-0068
line-7-front-2	3	line-7-run-0033-span-0066	OSR-Pi25	FND:line-7-support-000039648900	FND:line-7-support-000039673900	line-7-run-0033-span-0067
line-7-front-2	4	line-7-run-0033-span-0065	OSR-Pi25	FND:line-7-support-000039623900	FND:line-7-support-000039648900	line-7-run-0033-span-0066
line-7-front-2	5	line-7-run-0033-span-0064	OSR-Pi25	FND:line-7-support-000039598900	FND:line-7-support-000039623900	line-7-run-0033-span-0065
line-7-front-2	6	line-7-run-0033-span-0063	OSR-Pi25	FND:line-7-support-000039573900	FND:line-7-support-000039598900	line-7-run-0033-span-0064

The compressed span register provides the full sequence, individual beam IDs, predecessors and junction holds. Stage library supplies the method dependencies. Front package lists every assigned span and actual disconnected working interval.

Junctions, stations and residential interfaces

Coordinated structural interface IDs on this line: crossing-66746bfdc84f, crossing-7bba02156870, crossing-80b4593a8386, crossing-bde9f0d16859, crossing-c7d529a495dc, crossing-c9cacdc51c5b, crossing-dff9d62fd115, crossing-e1b0b7941672.

Evaluate grade-separated crossings, shared four-track civil footprints or separate deck levels, and bounded platform/concourse/access complexes with the station authority. Freeze actual vertical profiles and gradients before selecting supports and ordinary spans. Preserve rail/PSD/egress/waterproofing/earthing interfaces. Proposed residential infill sites on this line require station/approach structures and revised service, fleet, power and full installed prices; they are not existing paid journeys.

Candidate infill IDs: infill-line-7-035500, infill-line-7-025000.

Inspection, handover and programme

Use the foundation/production/lift/connection/concealed-work hold points from the master ITP. Capture installed component IDs, geometry, tests, personnel/equipment authority, NCR disposition and signed stage release. Update actual shift cycles, losses, stocks and resource reservations; do not infer an opening date from bay throughput. Station/special/track/energy/depot/fleet/testing/approval releases close the whole line. Unknown ground quantities, installed costs and dates stay unknown.

Integrated master package · Interactive support/junction inspection.

line-8 — foundation and assembly construction plan

Repository source: `engineering/network-planning/baghdad/line-8-construction-plan.md`

Line-specific design-development package; survey, ground, supplier and staged-load releases are still required.

Route length 48.137 km. 915 unique proposed support packets; 811 identified catalogue bay assemblies and 39 special packages. Each ordinary bay has two named beam components and both foundation parents. The 25 m average is a sizing basis; actual Pi20/Pi25/closure geometry controls installation.

Front	Launcher	Previous asset assignment	Direction	Initial assembly chainage m	Bay assemblies
line-8-front-1	launcher-15	initial mobilisation	1	120.0	405
line-8-front-2	launcher-16	initial mobilisation	-1	47319.2	406

Foundation workface plan

Filter the foundation register by `line-8`. For every support, verify the proposed coordinate/chainage, rights, utilities and the referenced desktop sample; commission field ground, groundwater, durability and load investigations. Review shallow footing, bored shaft, driven bent and pile-group alternatives against actual ground, axial/lateral/settlement/scour and construction loads. Desktop topsoil does not select a pile depth, count or allowable bearing pressure.

Release trial and working-platform design (F1), install the selected foundation with actual geometry/material/installation records (F2), then verify integrity/load/settlement/dimensions and close the foundation packet (F3). Construct/erect column and cap with independently checked connections/temporary restraint (P0). Survey and inspect bearings, strength and the next launcher/beam/delivery stage before support release (B0). Keep 10–15 consecutive bays of accepted supports and beam stock ahead only where the run and storage permit.

Directed assembly and logistics sequence

Before mobilisation, release the predecessor asset assignment and its dismantle/transport/recommission package. The positive front installs toward increasing chainage; the negative front starts at its high-chainage end and installs backward. Factory/dispatch plans must follow this directed sequence, not the ascending raw span register. A foundation used by two bays is one packet; it is not built or priced twice.

For each front, identify the first complete run, permitted delivery/assembly site, temporary support/ground platform, configured plant and authorised shift/crew. Reserve both beam travellers, transport/receiving equipment, an operator, required riggers, lift supervision and inspection. Verify source strength, complete mass/CG, lifting points, rigging, bearings, weather limits and contingency landing/recovery before any lift.

Both support B0 packets and beam receiving H1 records precede E0. Place and restrain beam 1 (E1); then place/securing beam 2 (E2). Survey and accept connection/strength/bearings/NCRs (E3), release advance loads and the next supports before moving the launcher (E4), then complete walkways/barriers/waterproofing/drainage/track interface and following-trade handover (E5). The next bay depends on the preceding advance release. Following trades may overlap only with protected, agreed load/access boundaries.

Run boundaries, stations and specials require a passage or dismantle/transport/reassembly/recommission package. No discontinuity becomes an assumed launchable gap. Junction-affected orders remain on a design hold until J2 freezes the profile, clearance, support/cap geometry and staged-load/utility/access design. A physical crossing does not create a rail switch.

Initial bay packages

Front	Sequence	Span	Product	Foundation A	Foundation B	Prior span
line-8-front-1	1	line-8-run-0001-span-0001	OSR-Pi20	FND:line-8-support-000000120000	FND:line-8-support-000000140000	mobilisation
line-8-front-1	2	line-8-run-0002-span-0001	OSR-Pi25	FND:line-8-support-000001178000	FND:line-8-support-000001203000	line-8-run-0001-span-0001
line-8-front-1	3	line-8-run-0002-span-0002	OSR-Pi25	FND:line-8-support-000001203000	FND:line-8-support-000001228000	line-8-run-0002-span-0001
line-8-front-1	4	line-8-run-0002-span-0003	OSR-Pi25	FND:line-8-support-000001228000	FND:line-8-support-000001253000	line-8-run-0002-span-0002
line-8-front-1	5	line-8-run-0002-span-0004	OSR-Pi25	FND:line-8-support-000001253000	FND:line-8-support-000001278000	line-8-run-0002-span-0003
line-8-front-1	6	line-8-run-0002-span-0005	OSR-Pi20	FND:line-8-support-000001278000	FND:line-8-support-000001298000	line-8-run-0002-span-0004
line-8-front-2	1	line-8-run-0064-span-0002	OSR-Pi20	FND:line-8-support-000047299200	FND:line-8-support-000047319200	mobilisation
line-8-front-2	2	line-8-run-0064-span-0001	OSR-Pi20	FND:line-8-support-000047279200	FND:line-8-support-000047299200	line-8-run-0064-span-0002
line-8-front-2	3	line-8-run-0063-span-0002	OSR-Pi20	FND:line-8-support-000047022600	FND:line-8-support-000047042600	line-8-run-0064-span-0001
line-8-front-2	4	line-8-run-0063-span-0001	OSR-Pi20	FND:line-8-support-000047002600	FND:line-8-support-000047022600	line-8-run-0063-span-0002
line-8-front-2	5	line-8-run-0062-span-0003	OSR-Pi20	FND:line-8-support-000046894300	FND:line-8-support-000046914300	line-8-run-0063-span-0001
line-8-front-2	6	line-8-run-0062-span-0002	OSR-Pi20	FND:line-8-support-000046874300	FND:line-8-support-000046894300	line-8-run-0062-span-0003

The compressed span register provides the full sequence, individual beam IDs, predecessors and junction holds. Stage library supplies the method dependencies. Front package lists every assigned span and actual disconnected working interval.

Junctions, stations and residential interfaces

Coordinated structural interface IDs on this line: crossing-0fac546553f2, crossing-3ba74f14bd23, crossing-66746bfdc84f, crossing-8431d2567c58, crossing-ec8cd65ba79b, crossing-efa108fc0f05, shared-corridor-481ca8ac8acb.

Evaluate grade-separated crossings, shared four-track civil footprints or separate deck levels, and bounded platform/concourse/access complexes with the station authority. Freeze actual vertical profiles and gradients before selecting supports and ordinary spans. Preserve rail/PSD/egress/waterproofing/earthing interfaces. Proposed residential infill sites on this line require station/approach structures and revised service, fleet, power and full installed prices; they are not existing paid journeys.

Candidate infill IDs: infill-line-8-040000, infill-line-8-036000, infill-line-8-044000.

Inspection, handover and programme

Use the foundation/production/lift/connection/concealed-work hold points from the master ITP. Capture installed component IDs, geometry, tests, personnel/equipment authority, NCR disposition and signed stage release. Update actual shift cycles, losses, stocks and resource reservations; do not infer an opening date from bay throughput. Station/special/track/energy/depot/fleet/testing/approval releases close the whole line. Unknown ground quantities, installed costs and dates stay unknown.

Integrated master package · Interactive support/junction inspection.

line-9 — foundation and assembly construction plan

Repository source: `engineering/network-planning/baghdad/line-9-construction-plan.md`

Line-specific design-development package; survey, ground, supplier and staged-load releases are still required.

Route length 97.684 km. 2,423 unique proposed support packets; 2,280 identified catalogue bay assemblies and 61 special packages. Each ordinary bay has two named beam components and both foundation parents. The 25 m average is a sizing basis; actual Pi20/Pi25/closure geometry controls installation.

Front	Launcher	Previous asset assignment	Direction	Initial assembly chainage m	Bay assemblies
line-9-front-1	launcher-17	initial mobilisation	1	0.0	1140
line-9-front-2	launcher-18	initial mobilisation	-1	97660.1	1140

Foundation workface plan

Filter the foundation register by `line-9`. For every support, verify the proposed coordinate/chainage, rights, utilities and the referenced desktop sample; commission field ground, groundwater, durability and load investigations. Review shallow footing, bored shaft, driven bent and pile-group alternatives against actual ground, axial/lateral/settlement/scour and construction loads. Desktop topsoil does not select a pile depth, count or allowable bearing pressure.

Release trial and working-platform design (F1), install the selected foundation with actual geometry/material/installation records (F2), then verify integrity/load/settlement/dimensions and close the foundation packet (F3). Construct/erect column and cap with independently checked connections/temporary restraint (P0). Survey and inspect bearings, strength and the next launcher/beam/delivery stage before support release (B0). Keep 10–15 consecutive bays of accepted supports and beam stock ahead only where the run and storage permit.

Directed assembly and logistics sequence

Before mobilisation, release the predecessor asset assignment and its dismantle/transport/recommission package. The positive front installs toward increasing chainage; the negative front starts at its high-chainage end and installs backward. Factory/dispatch plans must follow this directed sequence, not the ascending raw span register. A foundation used by two bays is one packet; it is not built or priced twice.

For each front, identify the first complete run, permitted delivery/assembly site, temporary support/ground platform, configured plant and authorised shift/crew. Reserve both beam travellers, transport/receiving equipment, an operator, required riggers, lift supervision and inspection. Verify source strength, complete mass/CG, lifting points, rigging, bearings, weather limits and contingency landing/recovery before any lift.

Both support B0 packets and beam receiving H1 records precede E0. Place and restrain beam 1 (E1); then place/securing beam 2 (E2). Survey and accept connection/strength/bearings/NCRs (E3), release advance loads and the next supports before moving the launcher (E4), then complete walkways/barriers/waterproofing/drainage/track interface and following-trade handover (E5). The next bay depends on the preceding advance release. Following trades may overlap only with protected, agreed load/access boundaries.

Run boundaries, stations and specials require a passage or dismantle/transport/reassembly/recommission package. No discontinuity becomes an assumed launchable gap. Junction-affected orders remain on a design hold until J2 freezes the profile, clearance, support/cap geometry and staged-load/utility/access design. A physical crossing does not create a rail switch.

Initial bay packages

Front	Sequence	Span	Product	Foundation A	Foundation B	Prior span
line-9-front-1	1	line-9-run-0001-span-0001	OSR-Pi25	FND:line-9-support-000000000000	FND:line-9-support-000000025000	mobilisation
line-9-front-1	2	line-9-run-0001-span-0002	OSR-Pi25	FND:line-9-support-000000025000	FND:line-9-support-000000050000	line-9-run-0001-span-0001
line-9-front-1	3	line-9-run-0001-span-0003	OSR-Pi25	FND:line-9-support-000000050000	FND:line-9-support-000000075000	line-9-run-0001-span-0002
line-9-front-1	4	line-9-run-0001-span-0004	OSR-Pi25	FND:line-9-support-000000075000	FND:line-9-support-000000100000	line-9-run-0001-span-0003
line-9-front-1	5	line-9-run-0001-span-0005	OSR-Pi25	FND:line-9-support-000000100000	FND:line-9-support-000000125000	line-9-run-0001-span-0004
line-9-front-1	6	line-9-run-0001-span-0006	OSR-Pi25	FND:line-9-support-000000125000	FND:line-9-support-000000150000	line-9-run-0001-span-0005
line-9-front-2	1	line-9-run-0083-span-0006	OSR-Pi20	FND:line-9-support-000097640100	FND:line-9-support-000097660100	mobilisation
line-9-front-2	2	line-9-run-0083-span-0005	OSR-Pi20	FND:line-9-support-000097620100	FND:line-9-support-000097640100	line-9-run-0083-span-0006
line-9-front-2	3	line-9-run-0083-span-0004	OSR-Pi25	FND:line-9-support-000097595100	FND:line-9-support-000097620100	line-9-run-0083-span-0005
line-9-front-2	4	line-9-run-0083-span-0003	OSR-Pi25	FND:line-9-support-000097570100	FND:line-9-support-000097595100	line-9-run-0083-span-0004
line-9-front-2	5	line-9-run-0083-span-0002	OSR-Pi25	FND:line-9-support-000097545100	FND:line-9-support-000097570100	line-9-run-0083-span-0003
line-9-front-2	6	line-9-run-0083-span-0001	OSR-Pi25	FND:line-9-support-000097520100	FND:line-9-support-000097545100	line-9-run-0083-span-0002

The compressed span register provides the full sequence, individual beam IDs, predecessors and junction holds. Stage library supplies the method dependencies. Front package lists every assigned span and actual disconnected working interval.

Junctions, stations and residential interfaces

Coordinated structural interface IDs on this line: crossing-1c3badfbb529, crossing-1fc40bbf9e02, crossing-3ba74f14bd23, crossing-4e4707f7a13b, crossing-526dbc821706, crossing-532e0b6bb63c, crossing-6644faf065db, crossing-80b4593a8386, crossing-86eb625bd399, crossing-92b255c18da1, crossing-954a77cefa6e, crossing-a79096d48aef, crossing-c15126eb0d40, crossing-c4ea0b93414e, crossing-c6c859f1b2b3, crossing-c7d529a495dc, crossing-d02bcca1ecd2, shared-corridor-481ca8ac8acb, shared-corridor-5c1c3ab579e9.

Evaluate grade-separated crossings, shared four-track civil footprints or separate deck levels, and bounded platform/concourse/access complexes with the station authority. Freeze actual vertical profiles and gradients before selecting supports and ordinary spans. Preserve rail/PSD/egress/waterproofing/earthing interfaces. Proposed residential infill sites on this line require station/approach structures and revised service, fleet, power and full installed prices; they are not existing paid journeys.

Candidate infill IDs: infill-line-9-014500, infill-line-9-012000, infill-line-9-087500, infill-line-9-079000, infill-line-9-073500, infill-line-9-092000, infill-line-9-023000, infill-line-9-075000.

Inspection, handover and programme

Use the foundation/production/lift/connection/concealed-work hold points from the master ITP. Capture installed component IDs, geometry, tests, personnel/equipment authority, NCR disposition and signed stage release. Update actual shift cycles, losses, stocks and resource reservations; do not infer an opening date from bay throughput. Station/special/track/energy/depot/fleet/testing/approval releases close the whole line. Unknown ground quantities, installed costs and dates stay unknown.

Integrated master package · Interactive support/junction inspection.

Residential network additions across the catalogue

Repository source: `engineering/network-planning/catalogue/README.md`

Controlled planning inventories and actual emitted-station population accounting; field and operating release remain open.

All 266 city designs have a source-bound expansion/evaluation package. 234 inventories contain added lines. Native population evidence is available for 247 and unavailable for 19.

0 city inventories reach their 80% working station-circle target; every remaining gap is reported. The target is not declared met from proposed sites or a demand score.

The city register records original/current line counts, added route quantities and actual station-circle fractions. The country register counts city programmes and reports the median city fraction; it is not national resident coverage. Population is never summed across potentially overlapping city bboxes. Recorded jurisdiction/source conflicts retain their explicit review gates.

Use each city alignment directory for immutable original design bytes, priority cells, common corridor connections, controlled terminal changes, native site/geometry reviews and actual emitted coverage. Country wages, fleet, depot, production, energy and finance models remain local. No Baghdad financing mix is transferred to another country.

Construction quantities, finite machine queues and line-opening dependencies follow the changed inventory. Public money, land, utilities, ground design, profiles, accessible walking routes, suppliers, prices, qualified workers and independent acceptance remain separate releases. These retained 2020 circles are not current census, observed passengers, funded feeder service or safe walking catchments.

Regeneration method · Sources and complete city results

Operations Management: Planning, Field Work And Exchange

Repository source: `engineering/operations/README.md`

This implements the six functions identified in the commercial/open-source review. The code runs in the existing authenticated integration gateway and is available through Workbench Service & field work. Its outputs are management plans and execution evidence; railway command, work-start permission, inspection and handback remain with their existing authorities.

Review item	Implemented behaviour	Important scope
1. Network/resources/possessions	Explicit directed physical edges and paths; track, platform, junction, depot and charger capacities; occupation/headway sweeps; time-bounded possessions; departure and path choices	A passenger interchange does not create a physical crossover. Resources and route connectivity must be supplied and reviewed.
2. Fleet and coordinated recovery	Revenue and positioning/deadhead legs, charger reservations and energy reserves, fixed inspection/maintenance windows and deadlines; combined track/train/charger/staff recovery	Bounded service/train search uses a declared positioning and charging policy. It reports exhaustion and does not claim unrestricted global fleet optimality.
3. Staff roster optimisation	Weighted coverage, previous-assignment preference and workload balancing; explicit OCC, station, maintenance, depot, cleaning, emergency and optional driver roles; competence expiry/revocation, availability, travel, rest, duty-hour/shift-span and break limits	GoA4 services do not invent a driver duty. Results are proposed rosters; native employment, permissions and execution gates still apply.
4. Offline field execution	Assigned, expiring work packs; embedded instructions and typed/unit-qualified readings; hashed PNG/JPEG photos; lookup QR/asset filtering; persistent IndexedDB queue and offline shell; idempotent gateway/ERP delivery	Reconnection checks planning/engineering revision, installed serial, assignment and revocation. Native ERP re-resolves Task, Employee, method and allocation state; receipts await independent inspection.
5. Reliability and RUL	Controlled fault codes; kilometre/cycle exposure and approved resets; failure cohorts, MDBF/MTTR and period trends; linear degradation estimates, residual sensitivity and held-out validation	Missing exposure produces unknown MDBF. No failures produces observed failure-free exposure, not infinity. A forecast needs matching units/family and separately held-out assets; false-alert, missed-failure, error and lead-time gates precede maintenance-review recommendations.
6. Operational exchange	Source-bound operational JSON import/export with topology, possessions, timetables, fleet and roster; GTFS timetable import/export; binary GTFS-Realtime trip updates/cancellations; configured reversible country/operator identity adapters	Imported passenger timetables require physical path mapping. Changed stop patterns require a reviewed replacement static feed. The existing railML geometry subset remains limited; this is not full railML timetable/signalling conformance.

Reproduce The Pilot

The package is an explicitly synthetic simulation fixture, not surveyed Samawah infrastructure, real staff or commissioned assets. The combined disruption closes the direct track, removes T2 and W1, and disables the first destination charger. The revised plan retains all three services, uses the connected alternative route and backup charger, and assigns replacement staff. Inspection remains in the fleet diagram.

```
python3 -m pip install -r engineering/operations/test-requirements.txt
python3 tools/automation/operations-pilot.py --check
python3 tools/automation/operations-pilot.py
python3 -m pytest -q tools/automation/tests/test_operations_management.py tools/automation/tests/
test_field_operations_native.py
node deployment/workbench/tests/verify-operations.mjs
```

The CLI writes normal/recovery plans, operational JSON, GTFS ZIP and trip-update protobuf to `build/operations-pilot/`. The browser test starts its own gateway and Workbench on separate ports, downloads a revised feed, records readings/photo offline, reloads offline, then reconnects the durable queue. Its report explicitly records that native ERP delivery is not exercised by that isolated browser test. Python intake tests exercise native orchestration with controlled record doubles; a real operator/site must verify the provisioned native workflow after migration.

For an isolated interactive workspace:

```
python3 tools/automation/operations-pilot.py --serve
```

Open http://127.0.0.1:8197/docs/operating/operations/ and use the appropriate role credential from the private `build/operations-pilot/credentials.json` file. This workspace uses its own SQLite database and does not alter an installed city stack. Its work pack is test evidence; no ERP worker runs in this isolated server.

Controlled Inputs And API

The contracts validate `osr-operations/1`. All intervals are integer seconds relative to a timezone-qualified epoch, with a declared horizon of at most 14 days. The bounded contract permits up to 80 service requests, 100 trains and 200 workers; search has an explicit node budget of at most 100,000. These are bounded planning limits, not a demonstrated production scale or solve-time commitment.

Supply known nodes/coordinates and directed edges with durations, energy and physical resource IDs; connected path alternatives; service/departure choices; train location, charge/reserve, route compatibility and maintenance deadline; chargers

and fixed fleet jobs; assessed staff competencies, validity windows and local rest/travel/break policies. An inspection can explicitly extend the declared maintenance deadline. The input is an engineering snapshot, not a live permit.

The gateway endpoints are `/operations/<action>`; Workbench forwards the same allowlist at `/api/operations/<action>`. Every request requires an explicit bearer credential and authorised city/environment. Browser writes also require the Workbench origin. View-role selectors do not grant credentials or authority.

Method / action	Caller / result
GET `baseline`, `plan`, `export`	Scoped reads; exports reject stale plans
POST `baseline`	Engineer; validated package and compare-and-swap `expected` hash
POST `plan`	Engineer/operator/maintainer; expected baseline plus optional possessions/unavailable trains/workers
POST `work`, `revoke`	Engineer; issue or revoke a reviewed work pack
GET `work`, `field-events`	Technician sees their own records; engineer/operator sees scoped records
POST `complete`	Assigned maintainer; typed event queues native evidence exactly once
POST `analytics`, `calibrate`, `forecast`	Scoped management analysis; source hashes and validation outcomes retained in gateway audit
POST `import`, `import-gtfs`	Engineer; review-only import without changing the baseline
POST `operator-exchange`	Provisioned, accepted adapter; unknown or ambiguous mappings rejected

A work-pack issue supplies `id`, city/environment, `asset_id`, `assigned_to`, `method_revision`, `erp_task`, `erp_asset`, current `native_modified`, `expires_at`, `requires_photo`, and reading definitions. Each reading names a kind and required flag; numeric readings also name exact units and finite bounds. `instructions` can carry issued plain-text method instructions for offline use. Server-controlled baseline/revision/serial/hash fields are added at issue, not accepted from callers. A lookup QR is `osr://asset?city=<city>&environment=<environment>&id=<asset-id>`; it selects an existing assigned work pack and establishes no new physical binding.

Offline completion status is always `performed-awaiting-inspection`. A 409 conflict retains device evidence for review; it never overwrites a newer work pack. The ERP outbox retries transport/server failures; permission, scope and revision rejection is terminal and requires refreshed work. Lost replies return the original receipt.

Native ERP And Operator Configuration

Normal app install/migration creates the immutable OSR Field Execution DocType. The adapter intake uses the existing private `osr_integration_users` scope policy and native document permissions. An administrator must provision `osr_field_technicians`, mapping each gateway subject to its actual native `user` and `employee`. The user must have an open native ToDo assignment to the already Working Task. Task/asset/method and engineering revision, employee/company, serial and current allocation gates must match. Intake stores private photo attachments and an immutable receipt without completing/submitting the Task or granting railway handback. Generic Desk/REST writes cannot forge or alter an intake receipt.

Gateway configuration may contain `operations_adapters`, keyed by adapter ID. Each `osr-operator-adapter/1` profile names country, operator, revision, accepted review record, authorised cities/environments and a one-to-one `identities` map. There are no fabricated PKP PLK, ARTC or other national-system endpoints. A real transport/schema adapter and national/operator acceptance remain deployment work.

A predictive-model study supplies measured/synthetic provenance in its own controlled input records, rising-degradation units/family, threshold, warning horizon and error/false-alert/lead-time limits. Training and held-out asset IDs must be disjoint; observed non-failure cohorts must cover the warning horizon. Validation can fail or return insufficient data. Even a passing research model is marked uncommissioned and cannot automatically issue a maintenance order. The residual sensitivity band is not a calibrated statistical confidence interval.

Evidence And Remaining Deployment Work

The source-bound pilot review retains the actual combined recovery checks and hashes; regenerate it with `operations-pilot.py --review`. Code and negative tests cover all six management functions. The fixture is a software demonstration. Real topology/resources, calibrated running times and energy, approved labour policies, operator master data, issued methods/permits, measured condition histories, native-site acceptance, support and security configuration remain deployment inputs. The first adoptable product and existing operating acceptance retain their separate commissioning and railway-authority boundaries.

The passenger formats follow the primary GTFS schedule and GTFS-Realtime references; protobuf output is independently decoded with the official MobilityData bindings. Full railML timetable exchange remains a distinct extension rather than an implied capability of OSR's existing geometry export.

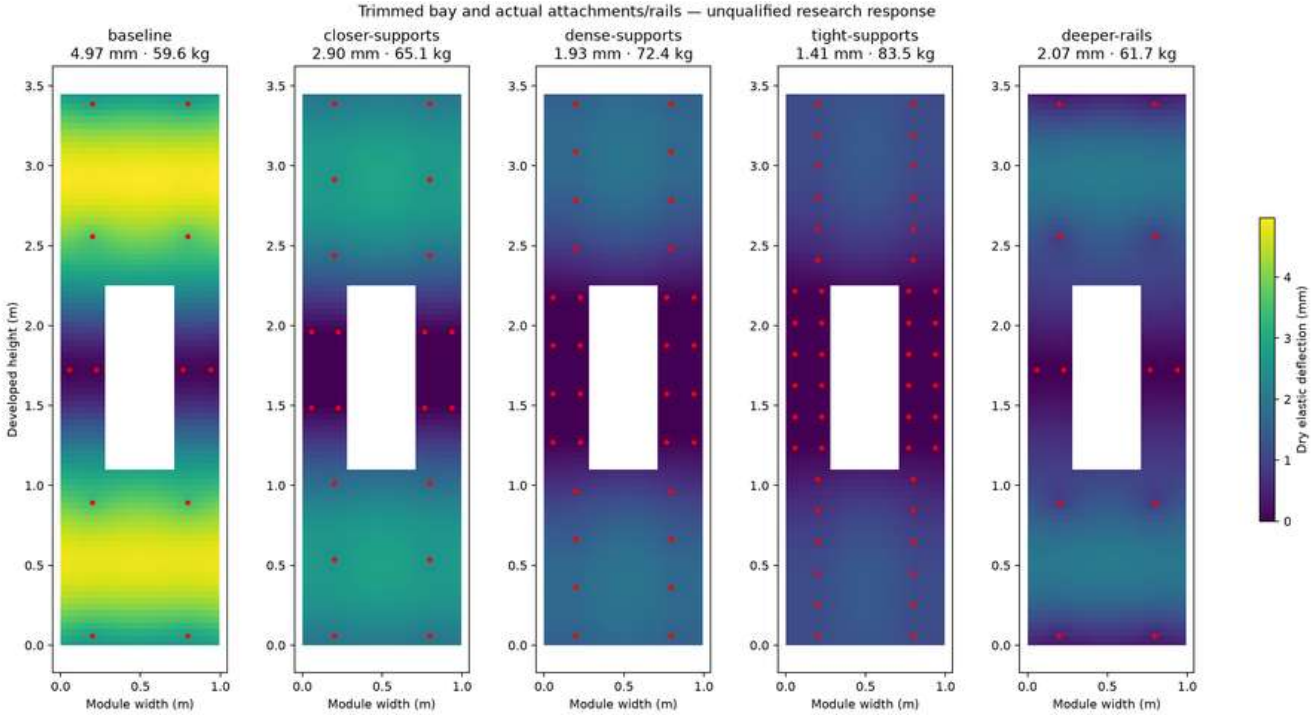
Clipped panel response refinement

Repository source: `engineering/panel_response/README.md`

This study continues the shared engineering demonstrators with a trimmed flat plate, individual attachments and actual hollow support rails. It addresses global pressure deflection and load sharing. Local contact, pull-out, debonding, fatigue, fire, curved-roof response and measured correlation remain open; no design is released.

The plate reads each manufactured ply's angle, thickness and coordinates, solid edge/aperture lands, core and insert regions from the same contract used by native FreeCAD. A rectangular MITC4 formulation uses mixed edge shear interpolation, with a homogeneous cylindrical plate checked against both an analytical solution and an independently executed CalculiX S8R model. That check verifies the element formulation, not the project laminate or joint properties. The method follows Bathe and Dvorkin's plate formulation; the native solver uses the CalculiX element/input definitions.

Each clip connects to a real rail beam through provisional bolt/potting compliance. Rail bending uses its manufactured hollow-section inertia. Bracket ends are idealised as simply supported, including assumed aperture-frame anchorage. The joint springs are bilateral elastic seeds; actual gaps, friction, peel, pull-out and weld resistance are not established. Bond/finish extensional stiffness is uncredited. Circular insert/bore regions use quadrature within the rectangular mesh; local stresses at those boundaries are not qualified. Membrane-free ABD condensation is a local plate assumption, not a complete membrane/contact model. Material values are dry nominal development inputs at room temperature.



Refined panel deflections

Results compare nominal spacing of 1.0, 0.5, 0.33 and 0.2 m, plus a 40 mm rail-depth alternative at the baseline spacing. The deeper alternative keeps the clip contact plane and moves brackets and the aperture frame behind the new rail depth. Both the geometry and material quantities change. Defined mass excludes the still-unresolved seal/secondary-retention and final connection details.

A 0.35 m proposal was rejected before ranking because two potting regions lacked matching foam subtractions; the retained rejection binds that failure to its proposed geometry. Native volume/overlap checks remain mandatory after this analytical preflight.

Every accepted arrangement is solved at two mesh sizes, at 2 kPa, with softer joint springs, at a separate 3.5 kPa pressure, with every individual clip removed, and with higher pressure combined with the worst clip loss. Exact rank-one updates of the linear system evaluate all loss locations; a direct reassembly independently checks the worst case. Selecting only the most-loaded clip would miss the governing end-attachment loss.

The baseline's dry global deflection is near the 5 mm study limit, while individual clip loss and the pressure holdout expose failures. Closer supports improve deflection but add rails, brackets, fasteners, inserts and inspection work. Deeper rails improve nominal stiffness with a smaller mass increase, but do not resolve the end-attachment loss. The 0.2 m arrangement meets these specified elastic deflection cases with substantially more mass and attachments. This is a conditional research comparison, not a verified railway solution or an optimum. Strength, fatigue, fire, installation access, repair time, supplier pricing and physical tests remain mandatory and unresolved.

Native geometry evidence checks occupied-material volumes, overlap, restored part identities and STEP reimport for all five arrangements. Numerical evidence binds complete meshes, responses, attachment forces, rail reactions, formulation verification and the comparison figure to source hashes. The older train structural failures and roof thermal failure remain unchanged.

Reproduce:

```
.venv/bin/python tools/automation/panel-response-study.py
flatpak run --filesystem=/home/hayder/Documents/OpenSourceRail --command=FreeCADCmd org.freecad.FreeCAD tools/
automation/verify-panel-response-cad.py
.venv/bin/python tools/automation/panel-response-study.py --check
.venv/bin/python -m pytest -q engineering/analysis/tests/test_panel_response.py
```

Use the existing research dependencies, plus Matplotlib for the standalone figure. FreeCAD and CalculiX are required for native evidence. The next refinement requires controlled constitutive/contact data and local joint/edge models before prototype correlation or any physical release.

Retained panel response study

Repository source: `engineering/panel_response/examples/README.md`

Summary compares five physical arrangements. Manifest binds their definitions, complete meshes, pressure responses, all single-clip-loss envelopes, pressure holdouts, combined loss/pressure cases and softer-joint sensitivities.

Comparison figure displays nominal dry deflection and the actual attachment locations. Native CAD verifies all five arrangements; native formulation verification checks the numerical plate element independently.

The filename prefix identifies the arrangement: `baseline`, `closer-supports`, `dense-supports`, `tight-supports` or `deeper-rails`. The suffix identifies the scenario. Core properties, manufacturing definitions and every response remain unqualified research inputs/results. Follow the parent documentation to reproduce or interpret them.

Manufacturing rejection retains the invalid 0.35 m arrangement's contract. Its potting/core overlap prevents ranking or physical release.

Native refined panel arrangements

Repository source: `engineering/panel_response/examples/cad/README.md`

Verification checks occupied material, nonoverlap, reopened native identities and STEP volumes against the shared contracts.

Arrangement	FreeCAD	STEP
Baseline 1.0 m supports	baseline.FCStd	baseline.step
0.5 m supports	closer-supports.FCStd	closer-supports.step
0.33 m supports	dense-supports.FCStd	dense-supports.step
0.2 m supports	tight-supports.FCStd	tight-supports.step
40 mm deeper rails	deeper-rails.FCStd	deeper-rails.step

Material/part identities and ply schedules are stored in each native document. All physical/manufacturing-release flags remain false. Proposed contact and frame/weld details require further engineering.

Independent plate formulation verification

Repository source: `engineering/panel_response/examples/native-benchmark/README.md`

Verification compares the rectangular MITC4 strip against an analytical cylindrical-bending solution and actual CalculiX S8R results. Input, output and solver log retain the native execution.

This homogeneous benchmark validates a numerical formulation within its recorded tolerance; it does not validate the composite's constitutive properties, clips or physical service performance.

Engineering Toolchain

Repository source: `engineering/toolchain/README.md`

This directory records reproducible inputs for the open-source design and simulation tools selected in `../../docs/engineering-design-simulation-plan.md`.

The installed environments are intentionally not stored in Git:

- FreeCAD, Blender/Bonsai, QGIS, CloudCompare, and SUMO are user Flatpaks;
- Python solvers and APIs use the repository `.venv`;
- EnergyPlus and FDS use versioned user-local directories beneath `${XDG_DATA_HOME:-$HOME/.local/share}/opensource-rail/native/`;
- generated checks and analysis scratch belong under `build/engineering/`.

`tool-manifest.toml` records the adopted releases, distribution channel, license identifier/source, Flatpak commit or binary checksum, and deliberately deferred tools.

RTKLIB is installed user-locally with the optional engineering applications, but remains unused until surveyed-pilot mobilisation because the public repository contains no raw observations. The deterministic `survey_control.py` gate validates controlled receipts and runs `rnx2rtkp` when the frozen deployment binary and real RINEX inputs are present; its automated quality result never records survey approval.

Use the root one-command setup and accept the optional engineering applications when prompted.

Run engineering workflows through the common launcher:

```
./osr engineering --check
./osr engineering --smoke
./osr engineering --baseline-assurance
./osr engineering --benchmarks
./osr engineering --station-ifc
./osr engineering --station-analysis
./osr engineering --flesh-out
./osr engineering --cities --city samawah,songea --jobs 2
./osr engineering --cities --all --generate-only --allow-input-gaps --jobs 8
./osr engineering --cities --all --allow-input-gaps --jobs 2 --resume
```

`python-requirements.txt` pins direct engineering packages and includes the smaller `ifc-python-requirements.txt` used by IFC CI and GUI jobs. The installer also writes a full transitive `pip-freeze.txt` manifest under `build/engineering/toolchain/` so evidence captures the exact environment that ran it.

The Flatpak applications remain independently versioned because they are GUI and desktop toolchains rather than Python library dependencies. The check command captures their installed commits and versions. The one-command installer selects these locations; users do not need to set paths or environment variables.

The smoke check performs calculations rather than import-only checks for IFC creation, an analytical axial bar in OpenSees, a two-bus pandapower load flow, pvlib plane-of-array irradiance, and a short PyBaMM single-particle battery model. It also runs a short SWMM rainfall/runoff fixture and rejects excessive continuity error. `--benchmarks` runs a deterministic JuPedSim normal/constrained station corridor, the all-variant station structure/flow/drainage screens, and a SUMO timetable directly from every Samawah line and station using the scenario's energy-derived dwell times.

`--baseline-assurance` is the fast, tracked regression layer. It executes seven independent analytical forms and verifies LandXML/OSR-ALN units and CRS, IFC source/index identity and finite envelopes, and a versioned duty-cycle projection into battery, grid and traffic rows. Reports are replaced atomically under `engineering/toolchain/`; planning or simulated duty remains explicitly ineligible as measured acceptance evidence. `--station-ifc` exports all seven positive-volume station product structures and checks that every BOM/traveler ID round-trips through IFC. The station analysis executes EnergyPlus and FDS when installed, preserving adverse results as open findings until project-specific climate/fire inputs and mitigations are independently reviewed.

`../analysis/analysis-register.toml` distinguishes planned, screening, calibrated, independently checked, and accepted analyses. Its validator requires review metadata at higher maturity and refuses image-only result artifacts.

Catalogue-wide city packages

`../../tools/automation/generate-city-engineering.py` discovers `cities/catalogue/*/*/*/design.toml`; it does not maintain a second city list. For every selected city it writes:

- a depot energy inventory and initial-dispatch capacity reconciliation under `engineering/depot-scope/`, with source hashes, PV module area and equipment cost sensitivity; physical placement, installed cost and stabling gates remain open until controlled evidence exists;
- a distributed-station overnight allocation candidate under `engineering/stabling/`, with a local runnable scenario and explicit physical-capacity gates;
- a QGIS/GDAL GeoPackage plus GeoJSON review layers for corridors, stations, civil segments, energy sites, depots and input issues;
- a SUMO node, edge and route deck containing every declared line and station, with edge shapes taken from the canonical corridor GeoJSON;
- pandapower grid-only and coordinated-daylight networks and a pvlib PV-yield envelope, with overload, voltage and per-site import/export limit findings;
- a machine-readable result with source hash, tool version, service arrivals, per-line journey times and explicit input-quality findings;
- a station occurrence map resolving every city station to its shared station variant, product-item count, assembly count, open-release count and IFC template; and
- per-city logs plus an aggregate `batch-summary.json` under `build/engineering/cities/batch-summary.json`.

The generator creates and checks the seven shared station-archetype IFC files once per batch. It also runs the shared station systems package once; identical archetype models and results are referenced by city station occurrences rather than copied thousands of times.

`--generate-only` creates reviewable SUMO decks without launching SUMO; GIS generation and energy screening still run. Use `--skip-gis` or `--skip-energy` to omit those package families. Use `--allow-input-gaps` only for catalogue auditing: findings remain in the summary and `passed` remains false even though generation is allowed to continue. Omitting that flag makes input gaps fail the command. A full solver batch is selected by omitting `--generate-only`. Use `--resume` after an interrupted or partially successful solver batch; it reuses only summaries whose design, corridor, scenario and station-manifest hashes still match and whose simulation, GIS generation, energy design screen and station-product map passed, and the depot-scope source hashes remain current. Transient launcher failures are retried automatically at reduced concurrency.

Samawah and Mosul carry the full pilot evidence scope, with open electrical connection findings; Songea exercises the same code path as a portability check. The generator discovers all 266 canonical designs, and every one currently has the required `<slug>.corridor.geojson` and `<slug>.toml` companions. GIS layers use EPSG:4326. SUMO uses a city-local metric projection for visual geometry while retaining canonical chainage as edge length. Electrical results are planning screens. Transformer sizing uses the greater of installed rectifier input and declared import with 25% headroom; each site's solved HV import/export is independently checked against its declared limit, including transformer and converter losses. `solver_passed` records convergence; `passed` additionally requires no design findings. The screen reports exceedances without changing connection limits, charging demand or export controls. Feeder and utility capacity remain unmodelled. pvlib clear-sky output is an envelope rather than measured weather.

These are screening packages. Surveyed geometry, city demand and dwell calibration, connected interchange/junction topology, road interactions, local climate/fire inputs and competent review cannot be generated from the current catalogue and therefore remain explicitly pending.

The September 2026 provenance review also identifies 264 retained native simulation reports requiring scenario/validator and resilience refresh. Their full-package manifests list the stale sources. Strict README generation continues to reject stale evidence; `osr_scenario.network_readme --allow-stale-evidence` writes an explicitly unverified audit view without changing solver results or package acceptance.

Each checked-in city deliverable uses only two generated subfolders: `engineering/` contains alignment, solver results, simulation, screenshots, and finance validation; `operations/` contains asset, manufacturing, maintenance, QA, and acceptance evidence. Retry logs and scratch results stay under `build/engineering/`.

The same command runs EnergyPlus's `1ZoneUncontrolled` design-day example and the tracked FDS empty-box fixture, then requires each solver's explicit successful-completion marker.

Regenerate depot reconciliation alone with `.venv/bin/python tools/automation/generate-depot-scope.py --all` (or `--design path/to/design.toml`). A successful command means the reports were written; their `passed: false` preserves the unresolved physical/cost/stabling gates. The equipment sensitivity is not applied to city CAPEX.

The distributed station-stabling workflow generates a station/depot plan plus a separate station-only native benchmark. Plans assign two revenue trains per selected station where fleet permits and all remaining stock to storage on its own line. `hybrid_allocation` records inventory, depot stabling positions and usable lengths separately from workshop bays; `allocation_passed` covers inventory and morning direction coverage. The separate `native_hybrid_candidate` records a runnable home-return candidate only where the allocation and same-line depot access permit it. The connected `lib/examples/hybrid-stabling.toml` fixture exercises depot returns and restart through two service cycles; yard geometry and city operating acceptance remain open. Samawah's plan is 40 station trains plus 37/16/15 stored on Lines 1/2/3. Retained canonical scenarios remain source evidence for their existing operations. `generate-stabling-plan.py --all` refreshes the plans; `screen-stabling-plan.py --design path/to/design.toml` builds the native simulator and compares the retained and candidate 01:30–06:00 operations. The combined result now requires both operating behavior and the two-train station-capacity check to pass. Candidates retain the design's spare and cold-reserve counts as parked roles. The replay checks each planned line/station/direction at morning opening and rejects routine reserve departures. Plans compare train lengths and clearances with reference platform berths, leaving actual track availability unverified.

`screen-stabling-cycles.py --design path/to/design.toml --days 2` runs continuous service days without resetting trains or site storage. It checks each following morning by elapsed time, rejects trains outside selected stabling locations and compares observed night allocations with reference platform space. Samawah's two-day holding/restart behavior passes after the departure gate was extended to preserve energy through to the next selected charger. Queues of 18–19 trains fail the two-train capacity gate, so the combined replay fails. The optional `service-cycle-screen.json` is included in package failure and source-drift checks. Neither replay proves daytime headway delivery or outage resilience.

`study-stabling-redistribution.py --design path/to/design.toml` consumes the source-checked continuous replay and its local native CSV. It produces a counterfactual station allocation, explicit transfer routes and nominal charging/time requirements while preserving reserve locations and every planned revenue direction. Samawah's seven-train target remains above a two-train station provision. The optional `redistribution-study.json` remains failed for physical/movement release and is included in package/source checks. It can analyse a replay with passing behavior but failed capacity; this does not waive the capacity gate for either the observed or target allocation.

`screen-hybrid-stabling.py --design path/to/design.toml --days 2` runs the separate native hybrid candidate. It checks explicit homes, station/depot capacity, passenger-service closure and station launch directions. Empty `ReturnToStabling` movements are counted separately from passenger departures. The optional `hybrid-cycle-screen.json` participates in package and source checks. `--analyze-existing` reuses a run only after matching candidate, binary and raw-output hashes. No interline depot connection is assumed.

Engineering Baseline Assurance

Repository source: `engineering/toolchain/baseline-assurance.md`

> Deterministic software/toolchain regression evidence only. This is not physical validation, design approval, certification, or authority acceptance.

- Overall result: PASS
- Analytical benchmarks: 7
- Interchange checks: 3

Analytical Benchmarks

ID	Case	Expected	Actual	Tolerance	Result
BENCH-STRUCT-001	cantilever tip displacement	0.0642857143 m	0.0642857143 m	1e-12	PASS
BENCH-THERM-001	thermal block midpoint	40 degC	40 degC	1e-12	PASS
BENCH-DRAIN-001	simple drainage mass balance	33.3333333 m3	33.3333333 m3	1e-12	PASS
BENCH-ENERGY-001	one-zone steady-state temperature	45 degC	45 degC	1e-09	PASS
BENCH-ELEC-001	four-bus terminal voltage	694.75 V	694.75 V	1e-09	PASS
BENCH-EGRESS-001	corridor evacuation clearance	85.5384615 s	85.5384615 s	1e-09	PASS
BENCH-TIME-001	one-line timetable traversal	1050 s	1050 s	1e-12	PASS

Interchange Drift

ID	Contract	Result
XCHG-ALN-001	LandXML to OSR-ALN units/CRS/golden round trip	PASS
XCHG-IFC-001	IFC asset identity and finite analysis envelope	PASS
XCHG-DUTY-001	Duty-cycle battery/grid/traffic semantic projection	PASS

The duty fixture is deliberately marked planning-only. Conversion cannot make it acceptance-eligible; measured data still requires controlled instrumentation and source evidence.

Retained TLC assurance executable

Repository source: `engineering/toolchain/tla/README.md`

`tla2tools-1.8.0.jar` is the exact executable used by the retained TACS formal evidence and the bounded-model CI job. The filename records the upstream release asset, rather than inferring its internal version: its manifest records TLC 2.0, build 2026-10-04 and source revision `1813307068bbc5759a441df820f3f548948d6b7a`.

Identity	Value
Bytes	4,500,445
SHA256	`c2fe4e56e43bde19f213b4a7e441d037297fda733e503579623e859b79348239`
Published release SHA1	`93d304f9c77595c8728c3ad939ed65b7a7149d1e`

The upstream release published the matching SHA1. On 5 October 2026 its download asset was replaced: the replacement has SHA256 `411ab54221cf0c9fa7ae18f07a3e0ebbdf9e5ba6254b79017e7007f1feb44e89` and manifest revision `610da1d073754ea8e0dd3c2af3eae8011d20191a`. That replacement is not adopted.

CI copies this retained input to its build directory, verifies the original SHA256, then runs the assurance consumer. The verification constant and model remain unchanged. Tool upgrades require verified upstream provenance and a new actual model execution.

License.txt is extracted unchanged from this JAR. Its embedded third-party notices also remain intact. This executable is a reproducible assurance input, not a deployed railway component or physical-release evidence.

Mechanical Design and Manufacturing

Design Sources

Repository source: `design/README.md`

This domain contains reusable design logic, not individual city revisions or site-approved engineering.

Area	Authority	Main outputs
`city-generation/`	GIS ingestion, demand/buildability rasters, route and station synthesis, scenario generation	Candidate city networks and simulator inputs
`component-catalogue/`	Parametric rolling-stock, track, station, depot and reusable civil components	CAD review models, product trees, BOMs and manufacturing templates

Editable projects are under `../cities/workspaces/`. Published city results are under `../cities/catalogue/`. Federated BIM and engineering validation are under `../engineering/`.

Parametric Component Catalogue

Repository source: `design/component-catalogue/README.md`

Reusable OpenSourceRail rolling-stock, track, station, and civil-kit components are expressed as parametric Python geometry. Deployment survey, railway alignment, ground conditions, structural release calculations, and the federated civil model have separate authorities.

Design philosophy

The catalogue is biased toward prefabrication and maintainable mechanical connections. Design choices use three priorities:

1. Reduce bespoke site design. Reusable families and stated planning envelopes reduce repeated work. A competent deployment engineer must still select, adapt, analyse, detail, and release every site-specific application.
2. Minimise site welding and wet work. Bolted steel and precast elements are preferred where the released structural, durability, transport, and construction design supports them. Foundations, closures, reinforcement, grouting, rail welding, and temporary works remain deployment-specific.
3. Design for transport and assembly. Catalogue modules target practical road transport and repeatable erection, but actual vehicle limits, lifting studies, crane selection, temporary works, crew, and programme are released for each deployment.

What's in scope

Module	Components
`osr_mech.track`	Rail profile (54E1 / 60E1), precast mono-block sleeper, Pandrol-style fastener, assembled track panel
`osr_mech.civil`	OSR-Pi20/Pi25 decked beams, special U-girders, OSR-ST6/elevated direct-fixation trackforms, soil/access foundation catalogue, resource production plan, substructures and platform edges (RFC 0011)
`osr_mech.station`	Seven station families: precast platform/guideway, platform and auxiliary solar canopies, passenger/MEP/access/charging systems, terminal turnout and main-heavy depot interfaces
`osr_mech.rolling_stock`	Car body, bogies, trainsets, doors, batteries, traction/electronics, T-OBS, COTS fit-out and the common service-rail/fastener/fixture system
`osr_mech.cad_templates`	Fabrication templates plus supplier-neutral COTS fixture envelopes

The CAD is an envelope and design-review package, not a homologated production drawing set. It now includes COTS-inspired fixture geometry, rolling-stock systems, and sheet-metal/chassis templates. Remaining production-detail work includes supplier-exact SKUs, weld maps, tolerance stacks, FEA-ready brackets, harness clamp locations, and controlled 2D manufacturing drawings.

Supplier-neutral fixture models intentionally remain generic until a procurement freeze. A selected SKU closes the CAD gap only when its datasheet envelope, mounting keep-outs, service-removal path, and revision are captured in the parametric source and matching drawing register.

The generated station/civil factory and deployment work packages cover all 45 stable product families with 18 controlled drawing/interface IDs. Their readiness register keeps every package open until reusable drawings, exact supplier configurations, site surveys/calculations, tools, tests and named approvals are recorded. The accompanying open-product reference defaults give all 29 unresolved families practical performance values and cost-conscious configurations without pretending that a supplier, site design or authority approval has been selected. Eighteen generated drawing-definition seeds then bind those product/default records to exact drawing ownership, views, source inputs, tools and verification tasks; every seed remains explicitly unissued.

The separate buildable civil catalogue reconciles all 19 deterministic federation types: nine civil-owned geometry types and ten controlled track, station, vehicle, or coordination interfaces. Its six release packages and nine drawing-definition briefs fail closed when an IFC type hash changes. They do not replace project survey, ground, reinforcement/prestress, structural, temporary-works, supplier, test, independent-check, or approval evidence. The civil fabrication and construction controls add repeatable workface sequences, crew/plant allowances, stop conditions and handback records beneath that unchanged release boundary.

Parametric inputs

Every component takes its parameters from the RFC-level choices the operator has already made in `design.toml`:

- Rolling-stock family platform length canopy bay count.
- Track-geometry preset rail profile + sleeper spacing.

- Station archetype bay count + facility mast count.
- Civil class (at-grade / elevated / bridge) ST6 or Pi20/Pi25 product, foundation zone and special-span interface
  - pier spacing or ballast depth.

`design.toml` supplies planning selections to both the Rust pipeline and the component catalogue. It is not the sole source for a deployment: accepted GIS and survey data, OSR-ALN alignment, released analysis and drawings, supplier records, and operational evidence retain their own controlled authorities.

Canonical Source

The Python CAD source files under `src/osr_mech/` are the design basis. Tracked FreeCAD `.FCStd` files under `models/cad/` are the compact assembly-review artifacts. Neutral CAD exports are local-only scratch files when a supplier specifically asks for them.

The source geometry uses the local `osr_mech.cad` facade. Under `FreeCADCmd` that facade emits native FreeCAD `Part` shapes directly; in ordinary Python it keeps lightweight volume and bounding-box metadata so the unit tests can run without a local FreeCAD install.

For a component-and-system integration review in FreeCAD, run `scripts/freecad_civil_systems_example.sh`. It creates `models/cad/civil-systems-integration-test.FCStd` with reference viaduct, elevated-station, ground-station, turnout, and two complete three-car trainset assemblies plus executable bearing, track-support, platform-height, and clearance checks. Its paired `models/cad/civil-systems-integration-test.json` provides stable asset IDs, transforms, relationships, design-reference operational state, validation results, and a hash of the native model.

Key rolling-stock source entry points:

Source	Scope
`src/osr_mech/rolling_stock/trainset.py`	Trainset family assembly and motorisation
`src/osr_mech/rolling_stock/car_body.py`	Concept-aligned layered car body: structure, exterior, interior, HVAC ducts, LV/data, HV/PV, thermal/fire routes
`src/osr_mech/rolling_stock/sensor_cowl.py`	Large glass end cowl and T-OBS nose envelope
`src/osr_mech/rolling_stock/systems.py`	Doors, batteries, couplers, electronics, charging, sensor packs
`src/osr_mech/rolling_stock/bogie/`	Powered and trailer bogie components
`src/osr_mech/cad_templates/rolling_stock.py`	Chassis/body sheet-metal templates
`src/osr_mech/design_definition.py`	Top-down design-space iteration, scoring, and candidate selection
`src/osr_mech/buildable_trainset.py`	Buildable product tree from parts through trainset assemblies

Buildable trainset handoff

The buildable trainset generator turns the optimized `light-metro-3car` baseline into a usable manufacturing handoff. It is still pre-release engineering data, but it is structured so a fabricator can start drawing, RFQ, routing, QA, and first-article planning without reverse-engineering the CAD tree.

Generated artifacts live in `catalog/buildable-trainset/`:

Artifact	Purpose
`buildable-trainset-manifest.md`	Product tree: fabricated parts, external components, subassemblies, assemblies, trainset
`current-design-buildability-review.md`	Green/yellow/red buildability findings and next closure actions
`small-component-standard.md`	Four fastener families, common rail, keyed connectors, modular illumination and serviceable door/window boundaries
`exterior-finish-system.md`	Mandatory base protection, rail-use livery film zones, and trial-only CaCO3 radiative roof-coating qualification gates
`manufacturing-and-assembly-controls.md`	Ten reference work instructions for datums, steel, joints, bonding/glazing, GFRP, roof, interiors, lifting, finish and NCR handback
`manufacturing-control-record-template.json`	Blank per-work-order execution, inspection, stop/NCR and configuration-handback record
`first-article-inspection-plan.md`	All 16 factory packages expanded into 416 uniquely identified hold, witness and record-review characteristics
`production-data-release-register.md`	Per-product route and 472 fail-closed production-artifact slots across all 62 locally made rows, linked to drawings, tooling, controls and inspection IDs
`factory-release-work-packages.md`	Sixteen controlled chassis, body, bogie, door/window, fascia, roof, interior, finish, electrical and recovery drawing/interface work packages
`factory-release-readiness.md`	Package-level drawing, product revision, tooling, verification and approval readiness; all 16 remain open
`factory-drawings/`	Twenty-nine product-, tooling-, source- and package-bound drafting/checking briefs covering all 62 locally made rows; none is issued for manufacture
`mass-budget.md`	Nine-category optimizer subtotal, engineering reserve and controlled planning tare

Artifact	Purpose
mass-closure-ledger.md	Product-level mass responsibility, evidence state, lightweight design-space comparison and recovery-reaction link
critical-path.md and factory-plan.md	First-article sequence, labour, work centres, space and machinery planning
cots-candidates.md and supplier-anchors.md	Exact catalogue/RFQ candidates and controlled local-equivalent boundaries for all bought-in rows
first-article-execution-pack.md and first-article-evidence-status.md	Ordered execution route and fail-closed status for the 13 supplier, analysis, physical-test and mass-properties gates
definitions/index.md	Drawing/RFQ seed: JSON + Markdown definition for every product-tree node, including structured material and process specs
travelers/index.md	Shop traveler seed: material/process controls, operation routers, labor estimates, tooling IDs, QA gates, approval/signoff blocks

The generated travelers are intentionally `unsigned-template` records. They are signable manufacturing forms, not already-approved build records; a real build cell must fill approvals, signatures, dates, inspection evidence, and NCR/deviation closure.

Usage

```
# One-time: install the package in editable mode.
pip install -e .[test]

# From this design/component-catalogue directory, check FreeCAD / Flatpak
# availability and the tracked output locations.
../../tools/automation/freecad-generate.sh --check

# From this design/component-catalogue directory, regenerate every tracked FreeCAD
# review document, FEM screening model, and screenshot set.
../../tools/automation/freecad-generate.sh --all

# Render the two-scene civil/rolling-stock twin and encode the front-page GIF.
../../tools/automation/freecad-generate.sh --digital-twin-animation

# Render the source-linked track/station/viaduct/train production twin.
../../tools/automation/freecad-generate.sh --fabrication-twin

# From this design/component-catalogue directory, iterate the rolling-stock design
# space and then generate the buildable product tree, definitions, and
# unsigned shop travelers.
../../tools/automation/design-iterate.sh
../../tools/automation/buildable-trainset.sh

# Build a FreeCAD review assembly.
# The launcher uses native FreeCADCmd or the FreeCAD Flatpak and handles
# FreeCAD's script argument quirks.
scripts/freecad_trainset.sh --family light-metro-3car

# Build the single-car review assembly; both bogies use the shared
# wheelbase/chassis placement datum.
scripts/freecad_trainset.sh --family urban-shuttle-1car --out models/cad/single-car-assembly.FCStd

# Build chassis/bogie and full-body FreeCAD documents with assembled
# and disassembled/exploded review states, plus geometry-check markdown.
scripts/freecad_assembly_review.sh

# Build the native FreeCAD replacement catalogue for civil, track,
# station, depot, and rolling-stock fabrication-template parts, plus
# at-grade/elevated platform and station-canopy assemblies.
scripts/freecad_catalog.sh

# Run first-pass FreeCAD/CalculiX screening FEA beam models and
# solver-result PNGs for chassis, bogie, and body load cases.
scripts/freecad_fea.sh

# Capture FreeCAD GUI screenshots from the review/FEA documents.
# Uses Xvfb automatically when available and refreshes the stable
# docs/screenshots/freecad/ latest image set.
scripts/freecad_screenshots.sh

# Build and capture FreeCAD station, ballastless track, and rolling-stock
# scene renders for at-grade, elevated, and interchange configurations.
scripts/freecad_station_scenes.sh
```

Testing

`pytest` validates volumes, masses, overall footprints, and solar-roof area against the published numbers in the RFCs. A geometry test that regresses mass outside ± 1 % fails — the RFC needs updating or the model needs fixing, but the two can't silently diverge.

The buildable-trainset tests also verify that every product-tree node has a definition and shop traveler, that all children/parents resolve, that material/process specs are present, and that traveler operations carry labor, tooling, approval, QA, and signoff fields.

The generated `mass-closure-ledger.json` maps all 120 LM3 product rows into the nine optimizer mass categories without treating simplified envelope geometry as a weighed production solid. It also keeps the lightest existing design-space candidate and its recovery reactions visible as an unpromoted study until production-CAD, supplier, calibrated weighing and weight/balance evidence close the design.

Handoff to structural engineering

The tracked handoff package is the Python source, FreeCAD `.FCStd` review assemblies, screenshots, and design notes. A deployment partner can generate local neutral CAD exports from the same source if their toolchain requires them, but those exports are not committed because they are bulky and reproducible.

The component docstrings carry common screening cases for early comparison:

- EN 1991-2 rail loading (LM71 × α = 0.83 for light metro).
- Eurocode 8 seismic screening at PGA 0.3 g.
- Wind: 50 m/s basic wind speed (typhoon-class).
- Thermal: ΔT = 70 K for steel, 35 K for concrete.

These values do not establish site suitability or structural capacity. Every deployment must derive its applicable actions, combinations, material data, ground response, fatigue and durability requirements, then release the chosen or adapted family through project calculations and drawings.

FreeCAD assembly bridge

FreeCAD MCP bridge

The supported installer provides FreeCAD through Flatpak. An optional MIT-licensed FreeCAD MCP addon/server may be used for interactive local review; restart FreeCAD, switch to the MCP Addon workbench, and start its RPC server before starting an MCP client. The optional repository-local MCP client example is in `../../.mcp.json`; it expects `freecad-mcp` on PATH and uses localhost only with text feedback by default. Machine-specific installation paths belong in the user's local configuration.

The MCP server is an interaction layer, not the design authority. New designs should be added as reproducible FreeCAD source scripts under `src/osr_mech/`, with `.FCStd` files treated as generated review artifacts.

The repository-level entry point is `../../tools/automation/freecad-generate.sh`. It orchestrates the package launchers below so CI, maintainers, and deployment partners can regenerate the same artifacts from one command:

- `--models models/cad/trainset-light-metro-3car.FCStd`
- `--assemblies` assembled/exploded chassis-bogie and body review documents plus `assembly-geometry-review.md`
- `--fem` CalculiX `.inp` / `.dat` / `.frd` study folders, result PNGs, `screening-summary.{md,json}`, and `models/cad/fea-screening-models.FCStd`
- `--screenshots` and `--station-scenes` stable documentation images under `docs/screenshots/freecad/` and `docs/screenshots/stations/`
- `--samawah-line-twin` the full-line FreeCAD/JSON asset twin plus the Blender perspective S5 operations scene, MP4, and README GIF

`osr_mech.freecad_trainset` builds a structured FreeCAD document directly from the parametric source geometry. The resulting `.FCStd` file is a review and handoff assembly: parts are grouped as car bodies, bogies, onboard systems, couplers, platform interfaces, and clearance references, with placements and display colours applied for inspection.

The bridge deliberately does not make FreeCAD the source of truth. Authoritative geometry stays in the Python source; FreeCAD is the compact tracked review format for assembly inspection, drawing generation, and partner mark-up.

Licence

Apache 2.0 for the Python code. Generated CAD review artifacts under `models/cad/` follow the same open-hardware intent as the rest of the `control-electronics/` tree.

Generated Mechanical Review Artifacts

Repository source: `design/component-catalogue/catalog/README.md`

This directory contains generated mechanical review artifacts from `design/component-catalogue`. The authoritative designs are the parametric source files under `../src/osr_mech/`. Tracked artifacts are FreeCAD `.FCStd` review documents, FEA screening output, and screenshots referenced from the docs. Neutral CAD interchange exports are local-only scratch files and are ignored by git.

Regenerate

FreeCAD review documents are regenerated with:

```
tools/automation/design-iterate.sh
tools/automation/buildable-trainset.sh
tools/automation/buildable-civil.sh
PYTHONPATH=design/component-catalogue/src design/component-catalogue/scripts/freecad_trainset.sh --family light-metro-
3car
PYTHONPATH=design/component-catalogue/src design/component-catalogue/scripts/freecad_assembly_review.sh
PYTHONPATH=design/component-catalogue/src design/component-catalogue/scripts/freecad_fea.sh
```

Supplier interchange exports, when needed, should be written outside the tracked tree or left ignored.

Sections

Folder	Scope
`design-system/`	Generated top-down / bottom-up design iteration scorecards and candidate shortlists
`buildable-trainset/`	Generated buildable product tree from parts through subassemblies, assemblies, and the trainset
`buildable-civil/`	Generated 19-type civil/IFC accountability register, six release packages, nine drawing briefs and empty authority record
`../models/cad/`	FreeCAD review assemblies, including trainset, assembled/exploded state documents, and FEA-screening model documents
`fea/`	FreeCAD/CalculiX first-pass beam-model screening inputs, solver outputs, and result summaries

Buildable Civil Release Catalogue

Repository source: `design/component-catalogue/catalog/buildable-civil/README.md`

This generated catalogue reconciles all 19 reusable IFC types (138 occurrences) into 6 bounded release packages and 9 drawing-definition briefs. It distinguishes the 9 civil-owned geometry types from 10 track, station, vehicle, and coordination interfaces.

Nothing here is issued for fabrication or construction. Site survey, geotechnics, per-span structural analysis, reinforcement/prestress, supplier certification, erection engineering, permits, independent check, and signed release remain open.

Outputs

File	Purpose
`reusable-type-release-register.md`	Exact one-to-one accountability for every deterministic IFC type
`factory-release-work-packages.md`	Outputs, tools/gauges, and open hold points for six release packages
`factory-drawings/index.md`	Nine controlled, non-issued drawing-definition briefs
`fabrication-and-construction-controls.md`	Ten practical moulding, survey, precast, transport, erection, track-interface and handback controls
`evidence/construction-control-record-template.json`	Blank workfront execution, hold-point, inspection and next-trade handback record
`inspection-and-test-plan.md`	Package-specific hold, witness and record-review characteristics covering all reusable IFC types
`evidence/civil-release-record-template.json`	Empty evidence record that project authorities must complete
`reusable-type-release-register.json`	Machine-readable register, packages, briefs, and validation flags

Regenerate

```
tools/automation/buildable-civil.sh
```

The generator deliberately fails if the reference IFC type hashes change without a corresponding ownership and release-path decision.

Civil Fabrication and Construction Reference Controls

Repository source: `design/component-catalogue/catalog/buildable-civil/fabrication-and-construction-controls.md`

> Status: reference defaults — not issued for fabrication or construction.

This workface handbook fills repeatable moulding, inspection, handling, erection and handover gaps around the reusable civil kit. It deliberately leaves site actions, ground, reinforcement/prestress, temporary works and statutory release with the competent project organisations.

Workface release rule

No work starts from coordination geometry alone: released drawings/method, accepted inputs, competent people/plant and open-hold-point authorization must all identify the same asset and revision.

Planning/RFQ definition only. Fabrication or construction release requires project survey, geotechnical model, checked calculations, reinforcement/prestress design, supplier data, temporary-works design and signed engineering acceptance.

CIV-MFG-010 — Survey, Datums And Set-Out

Default sequence:

1. establish protected project control outside the work zone.
2. transfer primary control independently to factory bed or work front.
3. set mould, foundation, bearing and rail-seat datums from primary control.
4. record as-built coordinates before the work is concealed or loaded.

Planning default: Use one project grid and vertical datum; close each set-out from two independent control points. Survey all first-article interfaces and every foundation, bearing seat, rail seat, platform edge and movement joint.

Typical first-shift crew/plant: two-person survey crew, total station, digital level and traceable check artefact. This is a resource-planning allowance, not authorisation or a minimum safe crew.

Hold the work when: control closure fails, benchmark disturbed, design/survey revision mismatch, or required accuracy is absent from the released survey plan.

Required handback: accepted survey control report, set-out sheets and signed as-built survey.

CIV-MFG-020 — Precast Mould, Reinforcement And Inserts

Default sequence:

1. clean and survey mould/bed datums.
2. apply release system and fit cages, ducts, drains, lifting anchors and inserts.
3. independently check cover, restraint and insert templates.
4. photograph and sign the pre-pour hold point.

Planning default: Use rigid reusable steel moulds with replaceable end boxes and positive insert templates. First mould and every repaired/reset mould receive a full dimensional survey; subsequent pours receive datum, closure and insert checks before each pour.

Typical first-shift crew/plant: mould/carpentry 2–3, reinforcement 4–6, one independent inspector, mould survey kit and cover/insert gauges. This is a resource-planning allowance, not authorisation or a minimum safe crew.

Hold the work when: unreleased reinforcement/prestress drawing, unidentified steel/insert, cover failure, dirty duct, loose insert, mould outside drawing tolerance, or unsigned pre-pour checklist.

Required handback: mould acceptance, material lots, cage/insert/cover report and signed pre-pour record.

CIV-MFG-030 — Concrete Placement, Curing And Prestress Transfer

Default sequence:

1. accept batch and fresh-concrete tests.

2 place in the released sequence without displacing cages or ducts.

3 finish and cure with temperature/maturity records.

4 confirm specified release/transfer strength before demoulding or prestress transfer.

Planning default: Reject site water addition unless the approved mix procedure permits and records it. Concrete class, exposure, w/c ratio, workability window, test frequency, curing cycle, release strength and prestress sequence remain calculation/specification values—not catalogue defaults.

Typical first-shift crew/plant: placement 6–8 plus pump/crane operator, finishing 2–3, laboratory technician and independent inspector at hold points. This is a resource-planning allowance, not authorisation or a minimum safe crew.

Hold the work when: batch ticket mismatch, workability/temperature outside specification, interrupted placement beyond permitted joint time, curing excursion, failed strength result, or prestress anomaly.

Required handback: batch/test records, curing history, transfer/release strength, prestress elongation/force record and approved deviation/NCR closure.

CIV-MFG-040 — Demould, Dimensional Acceptance And Storage

Default sequence:

1 confirm release strength and approved lifting arrangement.

2 demould without prying at unapproved points.

3 inspect surfaces, cracks, geometry, inserts, ducts and bearing/rail interfaces.

4 mark unique ID, orientation, mass and approved support points before storage.

Planning default: Inspect 100% of the first three units of each mould/type and every safety/interface feature thereafter. Routine non-interface dimensions may move to first/last plus 10% minimum three only through an approved control plan. Support units at the drawing-defined points on level, verified bearers.

Typical first-shift crew/plant: lift supervisor, certified crane operator, four riggers, two survey/inspection staff and rated lifting frame. This is a resource-planning allowance, not authorisation or a minimum safe crew.

Hold the work when: strength not accepted, crack/damage beyond acceptance limits, interface out of tolerance, blocked duct/drain, unidentified unit, or storage support not verified.

Required handback: dimensional/crack report, concrete and lift records, traceability mark and storage inspection.

CIV-MFG-050 — Transport And Delivery Acceptance

Default sequence:

1 match unit, mass and centre of gravity to transport schedule.

2 inspect route, trailer, temporary supports and restraints.

3 load only from approved points and record restraint installation.

4 inspect and survey interfaces again after delivery.

Planning default: Use the approved transport orientation and support spacing; do not support on bearing seats, walkway edges, drains or rail plinths. The route survey, permits, dynamic allowances and temporary-state calculation are deployment-specific.

Typical first-shift crew/plant: transport supervisor, certified operator, four riggers, escort/traffic staff as permitted, inspected trailer and rated lifting equipment. This is a resource-planning allowance, not authorisation or a minimum safe crew.

Hold the work when: route/permit absent, actual unit mass exceeds schedule, support/restraint mismatch, weather beyond method limits, transport damage, or delivery survey failure.

Required handback: route/permit pack, checked transport calculation, pre/post condition records and delivery acceptance.

CIV-MFG-060 — Foundation, Pier And Cap Construction

Default sequence:

1 verify utilities, ground horizon and formation.

2 construct/test foundation and survey starter interface.

3 erect or cast column with temporary stability maintained.

4 install cap, closure/grout and bearing-seat survey.

Planning default: Release one work front at a time from accepted survey, utility and geotechnical zone data. Never extrapolate a foundation selection across an uninvestigated ground boundary. Keep column/cap temporary stability independent of incomplete permanent connections.

Typical first-shift crew/plant: site engineer, supervisor, survey pair, foundation crew sized to selected method, 4–6 erection riggers and independent quality hold-point inspector. This is a resource-planning allowance, not authorisation or a minimum safe crew.

Hold the work when: unexpected ground/water/utility, foundation test failure, starter or bearing-seat survey failure, uncontrolled temporary state, grout/material mismatch, or open structural NCR.

Required handback: ground log, foundation tests, concrete/grout records, temporary-works inspection and pier/cap as-built survey.

CIV-MFG-070 — Girder Erection, Bearings And Closures

Default sequence:

1 accept bearing seats, bearings, unit and erection plant.

2 set bearing orientation/temperature and install temporary restraints.

3 lift or launch to the released sequence and survey before de-rigging.

4 complete diaphragms/closures and remove temporary restraints only at the authorised stage.

Planning default: Plan one lift controller, one command channel and a controlled exclusion zone. Set each bearing by ID and orientation; protect movement surfaces and retain access for inspection/replacement. No person enters beneath a suspended or incompletely restrained unit.

Typical first-shift crew/plant: lift supervisor, crane/launcher operator(s), four riggers, survey pair, two closure/grout workers and safety lead; plant capacity follows the checked lift study. This is a resource-planning allowance, not authorisation or a minimum safe crew.

Hold the work when: seat/bearing/unit not accepted, wind or visibility beyond method limits, plant configuration mismatch, loss of communication, unplanned movement, survey failure, or incomplete restraint.

Required handback: checked erection/lift plan, plant and personnel certificates, bearing schedule, lift record, closure strength and final survey.

CIV-MFG-080 — At-Grade Formation, Slab And Transition

Default sequence:

1 prove utilities, drainage outfall and formation level.

2 place and test ground treatment/subbase by lot.

3 set rail-seat/duct/drain templates and sign pre-pour hold.

4 cast/cure, survey and trial the first complete transition panel.

Planning default: Build a full-width trial panel including drainage, ducts and rail-seat fixings before repetitive work. Divide earthworks and pavement testing into traceable lots no larger than one shift or one material/source change; the project specification sets test values and lot size if smaller.

Typical first-shift crew/plant: site engineer, survey pair, earthworks 5–8 plus selected plant, concrete 6–8, laboratory technician and track-interface inspector. This is a resource-planning allowance, not authorisation or a minimum safe crew.

Hold the work when: soft spot or groundwater differs from model, drainage unavailable, compaction/plate test failure, template movement, concrete/cure failure, or transition survey outside released limits.

Required handback: formation lot tests, drainage proof, pre-pour record, concrete tests, trial-panel acceptance and as-built survey.

CIV-MFG-090 — Track, Platform And Systems Interface Handover

Default sequence:

1. clean and survey rail-seat/plinth, platform and equipment interfaces.
2. install accepted fastening/rail/turnout products to supplier procedures.
3. measure gauge, alignment, cant, stepping, clearances, drainage and electrical continuity.
4. hand over an as-built interface model with all NCRs and temporary conditions visible.

Planning default: Use a common chainage and asset ID across survey, IFC, inspection and maintenance records. Perform 100% measurement at turnouts, movement joints, stations, transitions and first article; routine interval and acceptance values come from the released track/vehicle/interface plan.

Typical first-shift crew/plant: permanent-way supervisor, 6–10 track workers, survey pair, signalling/power representatives, calibrated track gauge and clearance gauge. This is a resource-planning allowance, not authorisation or a minimum safe crew.

Hold the work when: supplier revision mismatch, missing vehicle envelope, failed gauge/cant/stepping/continuity, obstructed drainage/access, or interface NCR not accepted by both owners.

Required handback: supplier installation records, track/clearance survey, signed civil-track-station-vehicle ICD and configuration handover.

CIV-MFG-100 — Nonconformance, Repair And Work-Front Handback

Default sequence:

1. identify, contain and preserve the original evidence.
2. assess structural, durability, geometry and downstream interface effects.
3. obtain authorised use-as-is, rework, repair, return/scrap or design-change disposition.
4. perform repair and repeat every invalidated inspection before handback.

Planning default: No field cutting, drilling, heat straightening, reinforcement/strand exposure, concrete repair, bearing adjustment, grout substitution or tolerance concession without a written disposition by the accountable designer and checker where required.

Typical first-shift crew/plant: responsible construction engineer, quality lead, accountable designer and independent checker according to consequence; repair team only after disposition. This is a resource-planning allowance, not authorisation or a minimum safe crew.

Hold the work when: unapproved repair, repeat defect trend, hidden defect extent, missing retest, configuration mismatch, or any open safety/structural NCR.

Required handback: NCR, engineering assessment, approved repair method, repeat-test/survey result, as-built update and signed handback.

Civil Factory/Release Work Packages

Repository source: `design/component-catalogue/catalog/buildable-civil/factory-release-work-packages.md`

> Status: definition seed — not issued.

These packages define the smallest reusable handoffs around the current civil kit. Hold points remain open until real project evidence is recorded.

CIV-FRP-100 — precast superstructure and guideway edge

Lane: `reusable-product`

Drawing briefs: `CIV-SUP-100`
`CIV-SUP-110`
`CIV-EGR-120`

Tooling/gauges: `CIV-TOL-001`, `CIV-MLD-010`, `CIV-JIG-020`, `CIV-GGE-030`

Reference workface controls: `CIV-MFG-010`, `CIV-MFG-020`, `CIV-MFG-030`, `CIV-MFG-040`, `CIV-MFG-050`, `CIV-MFG-070`, `CIV-MFG-100`

Controlled outputs:

- checked product drawings
- mould and insert schedules
- inspection/test plan
- first-article dossier

Open hold points:

- [ ] project action set accepted
- [ ] independent check closed
- [ ] first mould and girder accepted

CIV-FRP-110 — pier column and precast cap

Lane: `hybrid-project-product`

Drawing briefs: `CIV-SUB-200`
`CIV-SUB-210`

Tooling/gauges: `CIV-TPL-100`, `CIV-MLD-110`, `CIV-GGE-120`

Reference workface controls: `CIV-MFG-010`, `CIV-MFG-020`, `CIV-MFG-030`, `CIV-MFG-040`, `CIV-MFG-050`, `CIV-MFG-060`, `CIV-MFG-100`

Controlled outputs:

- pier/foundation schedule
- column and cap drawings
- reinforcement and lifting schedules
- survey/test dossier

Open hold points:

- [ ] survey/geotechnics accepted
- [ ] foundation and pier calculations signed
- [ ] first cap accepted

CIV-FRP-120 — bearing, restraint and replacement interfaces

Lane: `supplier-configured`

Drawing briefs: `CIV-BRG-300`

Tooling/gauges: `CIV-TPL-200`, `CIV-JCK-210`, `CIV-GGE-220`

Reference workface controls: `CIV-MFG-010`, `CIV-MFG-060`, `CIV-MFG-070`, `CIV-MFG-100`

Controlled outputs:

- bearing and movement schedule
- supplier data/certificates
- seat and jacking details
- installed survey

Open hold points:

- [ ] movement/CWR model accepted
- [ ] supplier capacities approved
- [ ] trial jack/replacement method accepted

CIV-FRP-130 — at-grade slab and transition

Lane: `deployment-led-product`

Drawing briefs: `CIV-ATG-400`

Tooling/gauges: `CIV-RIG-300`, `CIV-TPL-310`, `CIV-GGE-320`

Reference workface controls: `CIV-MFG-010`, `CIV-MFG-030`, `CIV-MFG-080`, `CIV-MFG-090`, `CIV-MFG-100`

Controlled outputs:

- formation and slab drawings
- ground-treatment schedule
- drainage/utility details
- track survey and test record

Open hold points:

- [ ] survey/ground/drainage inputs accepted
- [ ] formation and slab checks signed
- [ ] trial panel accepted

CIV-INT-200 — track and station-deck interfaces

Lane: `coordination-interface`

Drawing briefs: `CIV-INT-500`

Tooling/gauges: `CIV-GGE-400`, `CIV-CHK-410`

Reference workface controls: `CIV-MFG-010`, `CIV-MFG-090`, `CIV-MFG-100`

Controlled outputs:

- supplier interface control document
- IFC/IDS issue
- clash and disposition report

Open hold points:

- [ ] rail/turnout data frozen
- [ ] station deck data frozen
- [ ] interdisciplinary review signed

CIV-INT-210 — station and vehicle envelope interfaces

Lane: `coordination-interface`

Drawing briefs: `CIV-INT-510`

Tooling/gauges: `CIV-GGE-500`, `CIV-CHK-510`

Reference workface controls: CIV-MFG-010, CIV-MFG-090, CIV-MFG-100

Controlled outputs:

- platform/canopy/vehicle ICD
- IFC/IDS issue
- kinematic and access report

Open hold points:

- [ ] vehicle envelope accepted
- [ ] station interface data frozen
- [ ] interdisciplinary review signed

Authority Boundary

Planning/RFQ definition only. Fabrication or construction release requires project survey, geotechnical model, checked calculations, reinforcement/prestress design, supplier data, temporary-works design and signed engineering acceptance.

Civil Inspection and Test Plan

Repository source: `design/component-catalogue/catalog/buildable-civil/inspection-and-test-plan.md`

> Status: unfilled protocol — not fabrication, construction, inspection or test evidence.

This plan assigns unique characteristics to every civil package prerequisite, drawing, reusable IFC type, tool/gauge, workface control and handover output. Project-specific values and results remain blank until competent teams execute an authorised workfront.

Packages: 6 · Characteristics: 114.

Point types: H mandatory hold, W witness, R record review.

Package	Lane	Characteristics	H	W	R	Status
`CIV-FRP-100` — precast superstructure and guideway edge	`reusable-product`	24	10	7	7	`open`
`CIV-FRP-110` — pier column and precast cap	`hybrid-project-product`	21	8	7	6	`open`
`CIV-FRP-120` — bearing, restraint and replacement interfaces	`supplier-configured`	16	7	4	5	`open`
`CIV-FRP-130` — at-grade slab and transition	`deployment-led-product`	18	8	5	5	`open`
`CIV-INT-200` — track and station-deck interfaces	`coordination-interface`	16	9	3	4	`open`
`CIV-INT-210` — station and vehicle envelope interfaces	`coordination-interface`	19	12	3	4	`open`

Execution Rules

- bind every result to project, asset/workfront, location, drawing/method revision, people, plant and date
- coordination IFC, an empty form or planned inspection is not performed evidence
- stop and contain affected work on failure; link the accepted NCR/RFI disposition and repeated inspection before release
- do not sample away survey control, foundations, lifting, bearings, movement joints, track geometry, platform stepping or other safety/interface characteristics
- handover requires accepted as-builts, tests, temporary-condition register, NCR status and receiving-party signature

Use `construction-control-record-template.json` for each asset/workfront and retain the ITP characteristic IDs when importing into a project QMS. All project release evidence remains governed by the deployment checklist.

Reusable Civil Type Release Register

Repository source: `design/component-catalogue/catalog/buildable-civil/reusable-type-release-register.md`

> Status: definition seed — not issued for fabrication or construction.

This register reconciles every reusable type in the reference civil IFC with one accountable release lane. Planning/RFQ definition only. Fabrication or construction release requires project survey, geotechnical model, checked calculations, reinforcement/prestress design, supplier data, temporary-works design and signed engineering acceptance.

Coverage

Measure	Count
ifc reusable types	19
ifc occurrences	138
civil owned types	9
controlled interface types	10
release packages	6
drawing definition briefs	9
tooling and gauge families	17

Type Register

IFC type	Asset class	Occurrences	Release lane	Package	Drawing brief	Disposition
OSR-TYPE-0B5DD2286469	civil.trackform	2	civil-structure-owned	CIV-FRP-130	CIV-ATG-400	long-panel transition/interface geometry
OSR-TYPE-15CA3C9B2913	station.platform-interface	1	station-interface	CIV-INT-210	CIV-INT-510	guideway edge and platform clearance interface
OSR-TYPE-239C28547990	civil.station-deck-interface	1	civil-structure-owned	CIV-INT-200	CIV-INT-500	coordination-only virtual deck interface; not a structural product
OSR-TYPE-24181B037817	station.platform-interface	1	station-interface	CIV-INT-210	CIV-INT-510	mirrored guideway edge and platform clearance interface
OSR-TYPE-33216A9A4A24	civil.trackform	2	civil-structure-owned	CIV-FRP-130	CIV-ATG-400	reusable at-grade slab envelope; project ground treatment
OSR-TYPE-38460DD866EF	track.turnout	1	track-supplier-interface	CIV-INT-200	CIV-INT-500	supplier turnout interface
OSR-TYPE-3FD9673EF391	civil.pier-cap	9	civil-structure-owned	CIV-FRP-110	CIV-SUB-210	reusable hollow/precast-shell cap definition
OSR-TYPE-4891C9E3C493	station.solar-canopy	2	station-interface	CIV-INT-210	CIV-INT-510	canopy clearance/load interface controlled by station package
OSR-TYPE-6336CC3F3205	station.platform-interface	1	station-interface	CIV-INT-210	CIV-INT-510	platform product interface controlled by station package
OSR-TYPE-661239622B15	rolling-stock.trainset	2	vehicle-envelope-interface	CIV-INT-210	CIV-INT-510	vehicle swept/envelope interface; not a civil product
OSR-TYPE-7406FA0925A0	station.platform-interface	1	station-interface	CIV-INT-210	CIV-INT-510	platform product interface controlled by station package
OSR-TYPE-7BF63285B921	station.solar-canopy	2	station-interface	CIV-INT-210	CIV-INT-510	canopy clearance/load interface controlled by station package
OSR-TYPE-99A11C1DC30D	civil.pier-column	9	civil-structure-owned	CIV-FRP-110	CIV-SUB-200	catalogue column envelope; project reinforcement and foundation
OSR-TYPE-C966BC7372EC	civil.bearing	36	civil-structure-owned	CIV-FRP-120	CIV-BRG-300	supplier-configured bearing and jacking interface
OSR-TYPE-D36A7871D9B7	civil.decked-pi-beam	12	civil-structure-owned	CIV-FRP-100	CIV-SUP-100	reusable precast product definition
OSR-TYPE-D4376465ACD5	civil.trackform	12	civil-structure-owned	CIV-FRP-100	CIV-SUP-110	reusable elevated deck/trackform definition
OSR-TYPE-EACE63284F07	track.rail	8	track-supplier-interface	CIV-INT-200	CIV-INT-500	supplier rail and fastening interface
OSR-TYPE-F2832317E733	track.rail	24	track-supplier-interface	CIV-INT-200	CIV-INT-500	supplier rail and fastening interface
OSR-TYPE-FDB074324FD0	civil.walkway-cassette	12	civil-structure-owned	CIV-FRP-100	CIV-EGR-120	reusable walkway cassette definition

Release Boundary

The hashes identify deterministic coordination geometry. They are not drawing revisions, certificates, approvals, or permission to build. A geometry change intentionally fails generation until its new type is classified.

Buildable station kit catalogue

Repository source: `design/component-catalogue/catalog/buildable-stations/README.md`

Generated station EBOM/MBOM and unsigned assembly travelers for the six base station shells and the controlled elevated-interchange variant.

Archetype	Platforms	Platform length m	Bays/platform	Product rows	Open product gaps	Auxiliary modules / installed m²	Definition	Traveler
halt	1	55.5	10	26	11	1 / 187.0	md	md
standard	2	59.5	10	29	13	7 / 1309.0	md	md
major	1	59.5	10	30	14	9 / 1683.0	md	md
interchange	2	59.5	10	30	14	11 / 2057.0	md	md
interchange-elevated	2	59.5	10	35	20	12 / 2244.0	md	md
terminal	2	59.5	10	37	21	11 / 2057.0	md	md
depot-terminal	2	59.5	10	44	28	6 / 1122.0	md	md

Auxiliary area is quantised upward into repeatable 8.5 m × 22 m solar-roof modules rather than left as an unbuildable square-metre allowance. Site structural, foundation, drainage, egress, and electrical approvals remain gates. See the generated `open release gap register` for the supplier, site, utility, and component-design closures behind these counts. The `factory/release work packages` classify all 45 unique products as reusable-definition, supplier-configuration or deployment-specific scope. The intentionally open `readiness register` and evidence template prevent catalogue maturity from being mistaken for fabrication or construction release. The 18 individual `drawing-definition seeds` provide controlled drafting/checking briefs; none is issued. The `reference defaults` give all 29 open product families practical concept/RFQ values plus mandatory override triggers. The generated `station product reconciliation` checks the BOM, traveler, drawing, FreeCAD and IFC identities in both directions. The shared station systems screening runs OpenSees, JuPedSim and SWMM across the family. EnergyPlus/FDS execute the baseline and proposed depot thermal/fire cases; the baseline failures, screened mitigation and open deployment work packages remain evidence for detailed design, not construction release.

Station open-product reference defaults

Repository source: `design/component-catalogue/catalog/buildable-stations/default-product-specifications.md`

Status: recommended concept/RFQ defaults; not a supplier selection or construction release.

These values close avoidable ambiguity in the open product rows with common, serviceable configurations. A default remains subordinate to every listed override trigger and the factory/release evidence gate.

Coverage: 34 defaults for 34 open product families; 13 source anchors.

`STN-CNP-P030` — factory-bonded solar roof sandwich panel with MC4 leads

Default: commodity mono-PERC/TOPCon modules on insulated standing-seam roof

Use class: `rfq-default-not-selected`

Values:

- 6,000 x 4,200 x 55 mm bay cassette
- 200 W/m2 gross design yield; about 5.0 kWp/bay
- 20 kg/m2 maximum complete cassette mass
- 1,000 VDC-rated string components
- IEC 61215/61730-qualified module; 450 W class reference
- 40 mm insulated core; white underside; factory MC4-compatible leads

Affordability: Uses high-volume framed modules and a repeated dry-fixed roof cassette rather than a bespoke BIPV laminate.

Must override when:

- site wind/snow/hail loads exceed qualified module values
- fire or roof warranty disallows the selected bond/rail system
- local module price or availability favours a different certified format

Sources: `OSR-STATION`, `IEC-61215`, `TRINA-450`.

`STN-CNP-P040` — platform-canopy PV string, combiner, isolation, bonding, and downlink kit

Default: one independently isolated PV string group per platform

Use class: `rfq-default-not-selected`

Values:

- 1,000 VDC equipment rating
- 6 mm2 UV-resistant copper PV cable baseline
- lockable DC isolator and Type 2 DC surge protection
- one monitored combiner/downlink per platform
- string voltage kept within selected inverter/MPPT window at local minimum temperature
- IEC 62446-1 inspection, polarity, insulation and handover record

Affordability: Standard string hardware and one downlink per platform minimize field joints and specialist site labour.

Must override when:

- cold-corrected open-circuit voltage exceeds equipment margin
- parallel-string current requires larger protection/cable
- lightning or utility study requires Type 1 protection

Sources: `OSR-STATION`, `IEC-62446`.

STN-CNP-P050 — 8.5 m × 22 m factory-bonded auxiliary solar-roof bay module

Default: repeatable auxiliary PV roof bay

Use class: `rfq-default-not-selected`

Values:

- 8,500 x 22,000 mm module planning grid
- 187 m2 gross roof area
- 31.8 kWp nominal after 15% packing allowance
- 20 kg/m2 roof/PV cassette mass allowance
- 3,850 mm minimum clear height
- dry-bolted panel rails and replaceable commodity PV modules

Affordability: One repeated 187 m2 grid shares frames between bays and uses the same commodity PV family as platform canopies.

Must override when:

- site circulation or fire access cannot fit the module grid
- wind/snow/seismic calculation changes frame spacing
- local solar yield makes partial PV coverage preferable

Sources: OSR-STATION, IEC-61215, TRINA-450.

STN-CNP-P060 — 22 m S355 transverse Warren-truss frame with two HSS 200 columns

Default: galvanised bolted/welded S355 auxiliary Warren truss

Use class: `preliminary-design-default-not-construction-release`

Values:

- 22,000 mm clear span planning envelope
- 2,000 mm truss depth
- S355 steel baseline
- HSS 200 mm column and HSS 150 mm chord planning envelopes
- 8,500 mm frame spacing
- hot-dip galvanised after fabrication; bolted site splices where transport requires

Affordability: Common structural hollow sections, repeated jigs and shared N+1 frames support local fabrication and low part variety.

Must override when:

- signed structural model changes any member
- transport envelope requires additional splices
- corrosivity category requires a duplex coating system

Sources: OSR-STATION.

STN-CNP-P070 — auxiliary-canopy pad footing, reinforcement, base plate, and anchor-bolt kit

Default: reinforced-concrete pad with reusable anchor cage

Use class: `preliminary-design-default-not-construction-release`

Values:

- 2.0 x 2.0 x 0.65 m pad allowance for estimating only
- C30/37 concrete and B500 reinforcement baseline
- 100 kPa preliminary allowable bearing pressure
- four M24 cast-in anchors on removable steel template
- 75 mm nominal soil-face cover
- top of pedestal kept 150 mm above finished drainage level

Affordability: A simple square pad, common bar grades and reusable anchor template are normally cheaper than piles on competent ground.

Must override when:

- geotechnical report gives less than 100 kPa or settlement risk
- uplift/sliding/seismic calculation fails
- buried utilities, flood level or expansive/aggressive soil affects the location

Sources: OSR-STATION.

STN-CNP-P080 — auxiliary-canopy PV string, combiner, isolation, bonding, and downlink kit

Default: modular auxiliary-canopy PV combiner groups

Use class: `rfq-default-not-selected`

Values:

- one independently isolated group per 10 or fewer roof modules
- 1,000 VDC equipment rating
- Type 2 DC surge protection baseline
- string current monitoring at combiner
- lockable local isolation plus remote inverter trip
- IEC 62446-1 commissioning dossier

Affordability: Repeating one protected group avoids a bespoke combiner for every station size and limits fault/maintenance isolation scope.

Must override when:

- selected module/inverter electrical window changes string length
- lightning study requires Type 1 SPD
- cable voltage-drop or fault study changes conductor size

Sources: OSR-STATION, IEC-62446.

STN-CNP-P090 — auxiliary-canopy gutter, downpipe, lightning, maintenance-access, and edge-protection kit

Default: gravity drainage, LPS and guarded maintenance kit

Use class: `preliminary-design-default-not-construction-release`

Values:

- 200 mm minimum eaves gutter planning size
- DN150 downpipe at each low point and no more than one bay per outlet baseline
- 600 mm clear anti-slip maintenance route

- 1,100 mm guardrail where fall exposure remains
- LPS Class III planning basis with bonded metalwork
- removable debris baskets and rodding access at every outlet

Affordability: Gravity drainage, standard pipe sizes and permanent simple access reduce pumps, bespoke parts and recurring cleaning labour.

Must override when:

- local design storm exceeds hydraulic capacity
- roof geometry creates additional low points
- lightning risk assessment changes LPS class
- fall-risk assessment permits or requires a different access system

Sources: `OSR-STATION`.

`STN-MEP-P020` — incoming switchboard, distribution board, metering, UPS, and earthing kit

Default: 400/230 V modular LV board with one-hour online UPS

Use class: `rfq-default-not-selected`

Values:

- 400/230 V 3-phase 4-wire 50 Hz default
- 25 kA minimum board short-circuit rating pending utility study
- 20% spare ways and 25% spare load capacity
- 10 kVA online double-conversion UPS starting point
- 60 minutes autonomy for life-safety/control loads only
- TN-S downstream earthing with monitored surge protection

Affordability: A standard modular IEC LV board and small centralized critical-load UPS avoid maintaining many proprietary batteries.

Must override when:

- utility voltage/frequency or fault level differs
- life-safety load schedule exceeds 8 kW
- local code requires a generator or different earthing arrangement

Sources: `OSR-STATION`.

`STN-PAX-P030` — CCTV, PA loudspeaker, help-point, radio, and station-LAN kit

Default: PoE fixed-camera, 100 V PA and SIP help-point platform kit

Use class: `rfq-default-not-selected`

Values:

- four 2 MP H.265 fixed cameras per 60 m platform starting layout
- outdoor devices IP66 and IK10; -30 to +55 degC baseline
- IEEE 802.3at PoE with 20% switch power reserve
- two weatherproof loudspeakers plus one help point per platform baseline
- managed gigabit fibre uplink with physically separate life-safety VLANs
- coverage and speech-intelligibility model governs final quantity

Affordability: Fixed PoE cameras and standard IP networking are cheaper to buy, cable and replace than PTZ/proprietary field networks.

Must override when:

- coverage model finds blind spots
- facial/incident evidential requirements need higher resolution
- PA is part of voice evacuation and must use the approved EN 54 system

Sources: `OSR-STATION`, `AXIS-M11`.

`STN-PAX-P040` — fare gate, accessible gate, and validator equipment kit

Default: compact bidirectional flap gate with one wide accessible lane

Use class: `rfq-default-not-selected`

Values:

- 600 mm clear standard lane
- 900 mm clear accessible lane minimum
- fail-safe open on loss of power or fire input
- contactless bank-card, account token and QR reader interfaces
- dry-contact/Ethernet controller interface
- indoor IP20 equipment only under weatherproof canopy; outdoor-rated gate otherwise

Affordability: Uses a conventional compact speed-gate envelope and open reader interfaces instead of custom mechanical gates.

Must override when:

- local accessibility rule requires a wider lane
- unpaid-area weather exposure requires outdoor equipment
- evacuation analysis prohibits barriers or demands more free-exit width

Sources: `OSR-STATION`, `PRM-TSI`, `GUNNEBO-FL`.

`STN-PAX-P050` — ticket-vending machine equipment kit

Default: cashless outdoor ticket/validation terminal

Use class: `rfq-default-not-selected`

Values:

- 15 inch sunlight-readable capacitive display baseline
- contactless EMV/account token and QR issuance/reading
- no cash module by default; staffed/offline exception process
- IP54 and IK10 minimum enclosure target
- 230 VAC input with internal 15 minute orderly-shutdown reserve
- wheelchair reach and knee/toe clearance governed by local access code

Affordability: Omitting cash acceptors, printers where e-ticketing permits, and proprietary fare media lowers purchase and field-maintenance cost.

Must override when:

- cash acceptance is a legal or inclusion requirement
- fare policy requires a printed ticket
- solar gain, dust, rain or vandalism needs a higher enclosure rating

Sources: `OSR-STATION`, `PRM-TSI`.

STN-CHG-P010 — station charging cabinet, protection, cable, and wayside connector kit

Default: shared 500 kW modular bidirectional DC charger

Use class: `rfq-default-not-selected`

Values:

- 500 kW continuous station cabinet
- 650-700 VDC normal train connection
- 825 A maximum equipment current
- 1,000 VDC insulation/equipment class
- two mechanically separate platform contacts interlocked so total cabinet power remains 500 kW
- IP55 outdoor cabinet; liquid-cooled power modules at high ambient

Affordability: One shared modular cabinet uses established DC power-stage building blocks and avoids duplicating 500 kW hardware per platform.

Must override when:

- timetable energy model needs simultaneous charging
- selected train voltage/current envelope changes
- utility/harmonic study or 50 degC duty requires derating/additional modules

Sources: OSR-STATION, ABB-TPSS.

STN-CHG-P020 — traction power substation transformer/rectifier and protection interface

Default: prefabricated dry-type transformer/rectifier substation

Use class: `rfq-default-not-selected`

Values:

- 11 kV 3-phase 50 Hz incoming planning default
- 750 VDC nominal traction/energy bus
- 12-pulse diode rectifier baseline
- catalogue ratings 800/1,000/1,500/2,000 kVA by archetype
- metal-enclosed MV, dry transformer, DC switchgear, protection and SCADA in one e-house
- 20% physical spare feeder capacity; no installed redundant transformer unless RAM study requires it

Affordability: A factory-integrated e-house and passive diode rectifier are widely used, locally installable and simpler than a bespoke building or active front end.

Must override when:

- utility supply voltage/fault level differs
- regeneration/export study requires active conversion
- RAM, fire or load-flow study requires N+1 capacity

Sources: OSR-STATION, ABB-TPSS.

STN-ACC-P020 — installed passenger lift

Default: machine-room-less through-car passenger lift core

Use class: `rfq-default-not-selected`

Values:

- 1,150 kg / 15-person nominal capacity
- 1.0 m/s nominal speed
- 1,100 mm minimum clear door target
- through-car doors preferred where they remove wheelchair turning
- two-way alarm, fire recall and automatic rescue/backup lowering
- external weatherproof shaft/enclosure where not inside a conditioned concourse

Affordability: A standard machine-room-less 1,150 kg catalogue lift avoids a plant room while supporting wheelchairs, attendants and luggage.

Must override when:

- local lift/access/fire code changes car or door dimensions
- travel or passenger demand needs higher speed/capacity
- flood, seismic or exposed climate conditions need a special package

Sources: OSR-STATION, KONE-MRL, PRM-TSI.

STN-ACC-P040 — installed escalator

Default: escalator study

Use class: `preliminary-design-default-not-construction-release`

Values:

- equipment identity, served levels and envelope from shared station layout
- separate installed equipment and associated works

Affordability: Quantify actual equipment instead of a blanket accessibility kit.

Must override when:

- passenger-flow, one-lift-out, evacuation or structural assessment changes geometry

Sources: OSR-STATION.

STN-ACC-P050 — installed staircase

Default: separated staircase study

Use class: `preliminary-design-default-not-construction-release`

Values:

- equipment identity, served levels and envelope from shared station layout
- separate installed equipment and associated works

Affordability: Quantify actual equipment instead of a blanket accessibility kit.

Must override when:

- passenger-flow, one-lift-out, evacuation or structural assessment changes geometry

Sources: OSR-STATION.

STN-ACC-P060 — installed shaft

Default: lift shaft study

Use class: `preliminary-design-default-not-construction-release`

Values:

- equipment identity, served levels and envelope from shared station layout
- separate installed equipment and associated works

Affordability: Quantify actual equipment instead of a blanket accessibility kit.

Must override when:

- passenger-flow, one-lift-out, evacuation or structural assessment changes geometry

Sources: OSR-STATION.

STN-CIV-P060 — station track-deck and running-rail interface

Default: station track-deck and rail interface study

Use class: `preliminary-design-default-not-construction-release`

Values:

- tracks and rail-head datums from shared layout
- station decks require separate support/foundation structural release

Affordability: Keep station track structures separate from running viaduct quantities.

Must override when:

- station or track-spreading geometry changes

Sources: OSR-STATION.

STN-CIV-P050 — island centre deck infill between edge L-unit strips

Default: island centre deck infill study

Use class: `preliminary-design-default-not-construction-release`

Values:

- 420 mm structural envelope; reinforcement and reaction design unreleased
- width equals island width minus two 3 m edge strips

Affordability: Price wider island deck explicitly alongside access equipment.

Must override when:

- station support, launcher clearance or structural design changes

Sources: OSR-STATION.

STN-ACC-P030 — pedestrian overbridge/concourse structural and enclosure kit

Default: simple weather-protected steel pedestrian bridge

Use class: `preliminary-design-default-not-construction-release`

Values:

- 4,000 mm clear walking width planning default
- 5,500 mm minimum rail clearance planning envelope pending infrastructure gauge
- 5.0 kPa crowd live-load starting case
- two simply supported transportable spans preferred
- anti-slip drained deck and 1,400 mm high non-climbable side enclosure
- lift/stair cores structurally independent where practical

Affordability: Straight repeated steel spans and independent vertical cores reduce bespoke geometry, erection possessions and future lift replacement risk.

Must override when:

- railway gauge/electrification requires more clearance
- crowd/egress analysis changes width
- span, wind, seismic, impact or foundation design changes the structural system

Sources: OSR-STATION, PRM-TSI.

STN-TRK-P010 — 1:9 UIC60 stock-rail, machined switch-blade, and closure-rail kit

Default: 60E1 1:9 fixed-frog turnout rail kit

Use class: `preliminary-design-default-not-construction-release`

Values:

- 1,435 mm gauge
- 60E1 rail family
- 1:9 tangent; 27,000 mm overall planning length
- 190 m nominal diverging radius
- 7,800 mm switch blade and 4,200 mm crossing envelopes
- 40 km/h maximum reverse-route planning speed

Affordability: One standard-gauge 1:9 family covers terminal and depot low-speed moves with common spares and training.

Must override when:

- wheel/rail study selects another profile
- required diverging speed exceeds 40 km/h
- supplier system geometry or slab-track interface differs

Sources: OSR-STATION, VOESTALPINE-TURNOUT.

STN-TRK-P020 — cast-manganese frog, check-rail, stretcher-bar, and mechanical-lock kit

Default: cast-manganese fixed crossing with external mechanical lock

Use class: `rfq-default-not-selected`

Values:

- fixed crossing for 40 km/h reverse-route duty
- cast manganese frog baseline
- adjustable check rails and replaceable wear components
- positive external blade lock independent of motor force
- one common stretcher/lock family across terminal and depot
- insulated joints omitted unless signalling design explicitly requires them

Affordability: A fixed frog and common mechanical lock are lower-cost and simpler to maintain than a movable-point crossing at this speed.

Must override when:

- RAM/wear analysis favours a different crossing
- axle load or wheel profile changes check-rail geometry
- signalling architecture requires insulated joints or another lock arrangement

Sources: OSR-STATION, VOESTALPINE-TURNOUT.

STN-TRK-P030 — prestressed turnout sleeper, slide-chair, and elastic-fastener set

Default: 42-position prestressed turnout bearer set

Use class: `rfq-default-not-selected`

Values:

- 42 numbered sleeper/bearer positions per turnout
- prestressed concrete baseline with supplier-drilled inserts
- elastic rail fastening with replaceable pads/insulators
- galvanised slide chairs with low-friction replaceable plates
- deliver as sequenced packs matching erection plan
- same fastener family as plain line where technically compatible

Affordability: Supplier-numbered concrete bearers and shared elastic fasteners reduce site drilling, inventory and installation error.

Must override when:

- trackform requires steel or composite bearers
- supplier layout changes bearer count
- stray-current or vibration study changes pads/insulation

Sources: `OSR-STATION`, `VOESTALPINE-TURNOUT`.

`STN-TRK-P040` — 6 kN nominal / 12 kN peak point-machine actuator, crank, and hand-wind kit

Default: compact electro-hydraulic point machine with external lock

Use class: `rfq-default-not-selected`

Values:

- 6 kN nominal and 12 kN peak operating-force requirement
- 18 kN ultimate proof requirement
- 3 second maximum normal throw
- integral hand-wind/manual release
- separate lock and dual end-position detection
- weatherproof low-profile assembly suitable for urban track

Affordability: A proven compact point-machine architecture fits multiple turnout types and reduces bespoke mechanisms and maintenance burden.

Must override when:

- measured switch force exceeds duty
- interlocking safety case requires another actuator/lock architecture
- flood, sand, temperature or tunnel envelope exceeds qualification

Sources: `OSR-STATION`, `VOESTALPINE-ECOSTAR`.

`STN-TRK-P050` — dual position detector, W-SBC interface, junction, and turnout harness kit

Default: two-channel fail-safe end-position detection kit

Use class: `rfq-default-not-selected`

Values:

- two electrically independent detector channels
- normal, reverse and neither/unknown states

- 24 VDC field circuits with isolated safety inputs baseline
- IP67 junction box and halogen-free armoured home-run cable
- no software-only inference of blade position
- bench and installed route-lock proof for every turnout

Affordability: Simple hardwired independent detectors keep the safety boundary inspectable and avoid an unnecessary proprietary fieldbus.

Must override when:

- interlocking interface voltage differs
- safety integrity allocation requires a certified fieldbus
- cable distance/EMC study changes transmission or protection

Sources: `OSR-STATION`, `VOESTALPINE-ECOSTAR`.

`STN-TRK-P060` — 3 kW points-heating strip, thermostat, IP67 cabinet, isolation, and cabling kit

Default: thermostatic resistance points-heating kit

Use class: `rfq-default-not-selected`

Values:

- 3 kW installed heat per turnout baseline
- two or more replaceable heater strips
- rail-temperature plus precipitation enable logic
- IP67 local isolation/control cabinet
- 30 mA personnel RCD only where permitted by railway earthing design; dedicated fault monitoring otherwise
- omit entirely in climates with no credible ice/snow case

Affordability: Basic resistance strips are widely serviceable; climate gating avoids buying and energising heaters where they provide no benefit.

Must override when:

- climate study removes the ice case
- thermal test needs more/less power
- railway earthing and protection rules prohibit the baseline RCD arrangement

Sources: `OSR-STATION`.

`STN-TRK-P070` — terminal stop-block, passive end marker, foundation, and fixing kit

Default: energy-absorbing terminal buffer stop and passive marker

Use class: `rfq-default-not-selected`

Values:

- friction/hydraulic energy-absorbing stop selected to site approach case
- 10 km/h maximum design-impact starting case for restricted terminal track
- bright passive end marker with retroreflective face
- independent reinforced-concrete foundation allowance
- minimum 2,000 mm protected overrun/service zone behind stop for planning
- no reliance on buffer stop as normal stopping control

Affordability: A proprietary tested buffer stop is safer and usually cheaper than locally designing impact hardware; only its foundation is localized.

Must override when:

- signalling hazard analysis sets another impact speed
- train mass/coupler interface changes
- available overrun or foundation conditions change

Sources: `OSR-STATION`, `VOESTALPINE-TURNOUT`.

`STN-DEP-P010` — main-heavy depot site formation, drainage, service-road, and secure-boundary kit

Default: compact main-heavy depot site shell

Use class: `preliminary-design-default-not-construction-release`

Values:

- 8,000 m2 reference footprint
- 1.0% minimum paved surface fall away from buildings
- separate clean stormwater and wash/contaminated drainage
- 6.0 m two-way service road planning width
- 2.4 m anti-climb perimeter fence with controlled vehicle/pedestrian gates
- 100-year local storm plus authority climate allowance for concept drainage

Affordability: A compact rectilinear site, gravity drainage and standard fencing minimize land, pumps and custom civil details.

Must override when:

- survey/ground/flood study changes levels or drainage
- fire access or swept-path study changes roads
- security or planning authority changes the boundary treatment

Sources: `OSR-STATION`.

`STN-DEP-P020` — stabling, inspection, wash, and workshop track-panel/stop-block package

Default: four-road 400 track-m inspection/stabling package

Use class: `preliminary-design-default-not-construction-release`

Values:

- 400 track-m total across four reference roads
- 60E1 rail and common elastic fastener baseline
- minimum 80 m useful clear length per road
- one drained wash road and one inspection/lift road
- 1,200 mm minimum side walkway planning width
- low-speed 10 km/h depot operating limit

Affordability: Four repeated roads match the base maintenance concurrency and reuse the line rail/fastener family.

Must override when:

- fleet/concurrency plan requires more roads
- site geometry changes road length or throat
- maintenance equipment requires pits, embedded rail or different trackform

Sources: OSR-STATION, VOESTALPINE-TURNOUT.

STN-DEP-P030 — 1:9 depot-throat turnout assembly replicated from the terminal turnout standard

Default: two common 1:9 depot-throat turnouts

Use class: `preliminary-design-default-not-construction-release`

Values:

- two turnout assemblies for four reference workshop roads
- same 60E1 1:9 geometry and spares as terminal
- shop-preassembled panels where transport permits
- 10 km/h depot operating limit
- independent lock/detection proof before route use
- manual hand-wind and clipped/scotched degraded mode

Affordability: Reusing the terminal turnout family avoids a depot-only geometry, spares stock and competence set.

Must override when:

- yard layout needs another turnout count/tangent
- transport prevents panel delivery
- signalling or degraded-operation rules differ

Sources: OSR-STATION, VOESTALPINE-TURNOUT.

STN-DEP-P040 — outdoor/open-sided per-stall plug-in charger, isolation, suspended cable, and data-dock kit

Default: 150 kW outdoor plug-in charger per maintenance stall

Use class: `rfq-default-not-selected`

Values:

- four 150 kW charger/data docks in the reference depot
- 650-700 VDC operating range
- 250 A connector/cable continuous rating minimum
- IP55 charger and IP67 stowed connector target
- overhead retractable/suspended cable with mechanical strain relief
- Ethernet maintenance-data dock kept electrically isolated from HV

Affordability: One modest modular charger per active bay supports 12-minute turnaround/overnight work without installing a 500 kW unit at every stall.

Must override when:

- maintenance timetable needs faster simultaneous charging
- selected connector or train voltage differs
- ambient/solar load causes supplier derating

Sources: OSR-STATION, ABB-TPSS.

STN-DEP-P050 — depot PV canopy, inverter, microgrid switchgear, and separated outdoor stationary-battery package

Default: 600 kWp PV plus four modular 500 kWh outdoor LFP blocks

Use class: `rfq-default-not-selected`

Values:

- 4,000 m2 PV canopy and 600 kWp nominal array
- 2,000 kWh gross stationary LFP storage
- four independently isolatable 500 kWh blocks
- 500 kW bidirectional PCS starting rating
- outdoor liquid-cooled containers/cabinets with off-gas detection and remote isolation
- 6 m preliminary separation zone from occupied/control buildings pending fire strategy

Affordability: Repeated outdoor LFP blocks can be procured competitively, phased and isolated individually; factory integration reduces site labour.

Must override when:

- energy simulation changes capacity/power
- cell-to-pack fire evidence or authority changes separation/containment
- utility export/protection rules change PCS architecture

Sources: `OSR-STATION`, `CATL-ENERONE`, `IEC-62446`.

STN-DEP-P060 — main workshop, synchronized LM3 lift/bogie-change bay, overhaul/inspection bays, 40 t crane, wash plant, stores, and wheel-lathe package

Default: four-bay modular workshop with synchronized vehicle lift

Use class: `rfq-default-not-selected`

Values:

- 1,800 m2 workshop reference envelope
- four concurrent inspection/maintenance bays
- 40 t overhead crane
- 40 t synchronized four-column vehicle lift minimum system SWL
- 10 t minimum rated capacity per column with mechanical locks
- one bogie-change/extraction path, wheel lathe and recycling wash plant

Affordability: A repeatable portal building and commercial synchronized columns avoid a bespoke buried lifting table while retaining full-car/bogie change capability.

Must override when:

- weighed car/recovery load case exceeds lift allowance
- supplier column reactions change foundations
- fleet maintenance plan removes/adds heavy equipment
- crane runway or building calculation changes geometry

Sources: `OSR-STATION`.

STN-DEP-P070 — depot cooled controls room, LV, compressed-air, fire, lighting, CCTV, LAN, access-control, and maintenance-data kit

Default: separate cooled controls room with N+1 packaged DX

Use class: `rfq-default-not-selected`

Values:

- two independent 30 kW nominal packaged DX units
- one unit carries the 30 kW design sensible duty
- automatic duty/standby rotation and high-temperature alarm
- 400/230 V modular LV, UPS, fire alarm, PoE CCTV/LAN and access control
- 8 bar oil-free compressed-air ring baseline
- no shared return air with workshop or battery/charger compound

Affordability: Two commodity packaged units provide maintainable N+1 cooling without a central chilled-water plant.

Must override when:

- room heat-load test exceeds 30 kW
- maximum ambient exceeds selected unit envelope
- fire strategy changes room separation/ventilation
- tool air demand changes compressor/ring size

Sources: `OSR-STATION`, `DAIKIN-ROOFTOP`, `AXIS-M11`.

Sources

Source `OSR-STATION`

OpenSourceRail station, energy, depot and turnout controlled design basis — `repository-controlled-design-basis`; checked 2026-09-04.

Source `IEC-61215`

IEC 61215-2 terrestrial photovoltaic module design qualification — `official-standard`; checked 2026-09-04.

Source `IEC-62446`

IEC 62446-1 PV system documentation, commissioning tests and inspection — `official-standard`; checked 2026-09-04.

Source `TRINA-450`

Trina Solar Vertex S+ 450 W module public data sheet — `manufacturer-architecture-anchor`; checked 2026-09-04.

Source `ABB-TPSS`

ABB DC railway traction power switchgear and conversion portfolio — `manufacturer-architecture-anchor`; checked 2026-09-04.

Source `KONE-MRL`

KONE MonoSpace 500 machine-room-less lift planning guide — `manufacturer-architecture-anchor`; checked 2026-09-04.

Source `PRM-TSI`

Commission Regulation (EU) 1300/2014 accessibility TSI — `official-regulation-reference`; checked 2026-09-04.

Source `VOESTALPINE-TURNOUT`

voestalpine urban-rail turnout system portfolio — `manufacturer-architecture-anchor`; checked 2026-09-04.

Source `VOESTALPINE-ECOSTAR`

voestalpine ECOSTAR compact electro-hydraulic point machine — `manufacturer-architecture-anchor`; checked 2026-09-04.

Source `AXIS-M11`

Axis M11 affordable outdoor network camera family — `manufacturer-architecture-anchor`; checked 2026-09-04.

Source GUNNEBO-FL

Gunnebo SpeedStile FL gate technical specification — manufacturer-architecture-anchor; checked 2026-09-04.

Source CATL-ENERONE

CATL EnerOne modular outdoor liquid-cooled LFP storage system — manufacturer-architecture-anchor; checked 2026-09-04.

Source DAIKIN-ROOFTOP

Daikin UATYA packaged rooftop unit catalogue — manufacturer-architecture-anchor; checked 2026-09-04.

Boundary: Defaults support concept design, RFQs and cost comparison. Exact supplier data, site survey, geotechnical/utility inputs, calculations, approvals and performed tests still govern fabrication, construction, energisation and operation.

Station and civil factory/release readiness

Repository source: `design/component-catalogue/catalog/buildable-stations/factory-release-readiness.md`

Generated with the station catalogue. Blank evidence fields and `open-unissued` states are deliberate; this is not construction release.

- Template status: `unfilled-not-release-evidence`
- Packages: 9; open: 9
- Drawing/interface IDs: 18
- Controlled unique products: 50
- Tool/gauge families: 22

Package	Delivery lane	Status	Drawings	Products	Inputs	Outputs	Tests
`STN-FRP-010` — precast platform, guideway edge, drainage and closure pack	`hybrid-prefabrication-and-site`	`open-unissued`	2	7	3	3	3
`STN-FRP-020` — platform canopy steel, footing and solar-roof pack	`hybrid-prefabrication-and-supplier`	`open-unissued`	2	4	3	3	3
`STN-FRP-030` — auxiliary solar-canopy module, truss and site-interface pack	`deployment-led-hybrid`	`open-unissued`	2	5	3	3	3
`STN-FRP-040` — station services, passenger equipment and plinth integration pack	`supplier-interface`	`open-unissued`	2	12	3	3	3
`STN-FRP-050` — pedestrian approach, lift/stair core and overbridge interface pack	`deployment-led-hybrid`	`open-unissued`	2	6	3	3	3
`STN-FRP-060` — wayside charging and traction substation interface pack	`supplier-and-utility-interface`	`open-unissued`	2	2	3	3	3
`STN-FRP-070` — 1:9 turnout, actuation, detection, heating and track-end pack	`hybrid-fabrication-and-supplier`	`open-unissued`	2	7	3	3	3
`STN-FRP-080` — depot site, drainage, track and throat-turnout pack	`deployment-specific`	`open-unissued`	2	3	3	3	3
`STN-FRP-090` — depot charging, energy, workshop and services integration pack	`deployment-and-supplier`	`open-unissued`	2	4	3	3	3

Populate `factory-release-record-template.json` under document control. Reusable factory information may be released independently only where its package boundary permits; site work still requires the deployment survey, calculations, supplier configuration, permits and named approvals. Use the individual `factory-drawings/` seeds as drafting/checking briefs, never as issued fabrication or construction drawings.

Boundary: These packages define reusable drafting and deployment handoff scope. They are not approved fabrication/construction drawings, supplier selections, signed calculations, permits, surveys or performed acceptance records.

Station and civil factory/release work packages

Repository source: `design/component-catalogue/catalog/buildable-stations/factory-release-work-packages.md`

These packages turn the 45 stable station product identities into bounded drawing, tooling, site-handoff and verification tasks. They deliberately separate reusable definitions from supplier configuration and deployment release.

- Packages: 9
- Unique product rows: 50
- Drawing/interface IDs: 18
- Tool/gauge families: 22
- Release paths: 18 reusable definition, 14 supplier configuration, 18 deployment-specific

`release-candidate` means the catalogue definition is mature enough to enter controlled detailing; it never means a part, structure or site is released.

`STN-FRP-010` — precast platform, guideway edge, drainage and closure pack

Delivery lane: `hybrid-prefabrication-and-site`.

Drawings/interfaces: `STN-CIV-100`, `STN-CIV-110`. Tools/gauges: `STN-TOOL-PRECAST-MOULD`, `STN-TOOL-EDGE-GAUGE`, `STN-TOOL-LIFTING-GAUGE`.

Controlled products

Product	Route	Catalogue maturity	Release path	Default	Variants
`STN-CIV-P010` — 6 m ground-level station slab and depressed double-track guideway-channel panel	`MAKE`	`release-candidate`	`reusable-definition`	`not-required`	7
`STN-CIV-P020` — platform sub-base, levelling pad, grout, and closure-pour kit	`MAKE`	`release-candidate`	`reusable-definition`	`not-required`	7
`STN-CIV-P030` — platform and track drainage channel, pipe, catch-pit, and outlet kit	`MAKE`	`release-candidate`	`reusable-definition`	`not-required`	7
`STN-CIV-P040` — 3 m at-grade guideway-channel edge beam, coping/tactile carrier, and drained service trough	`MAKE`	`release-candidate`	`reusable-definition`	`not-required`	6
`STN-CIV-P050` — island centre deck infill between edge L-unit strips	`MAKE`	`study-only`	`deployment-specific`	`specified`	1
`STN-CIV-P060` — station track-deck and running-rail interface	`MAKE`	`study-only`	`deployment-specific`	`specified`	7
`STN-PLT-P010` — platform coping, tactile strip, warning line, and edge-marker kit	`SOURCE`	`release-candidate`	`reusable-definition`	`not-required`	7

Frozen inputs

- accepted survey control, track alignment and platform stepping/gap envelope
- site geotechnical, drainage/outfall and foundation design
- released concrete, reinforcement, tactile and joint/seal systems

Controlled outputs

- repeatable precast mould, reinforcement, insert and lifting drawings
- site set-out, levelling, drainage and closure-pour schedule
- platform-edge datum, tolerance and interface-control plan

Verification

- mould and first-article dimensional survey
- concrete, reinforcement and lifting-insert records
- installed track/platform gap-step and drainage survey

Boundary: The reusable mould and panel definition does not release site excavation, foundations, drainage falls or track/platform geometry.

STN-FRP-020 — platform canopy steel, footing and solar-roof pack

Delivery lane: `hybrid-prefabrication-and-supplier`.

Drawings/interfaces: `STN-CNP-200`, `STN-CNP-210`. Tools/gauges: `STN-TOOL-PORTAL-FIXTURE`, `STN-TOOL-ANCHOR-TEMPLATE`, `STN-TOOL-ROOF-WATER-TEST`.

Controlled products

Product	Route	Catalogue maturity	Release path	Default	Variants
`STN-CNP-P010` — 6 m galvanised HEA portal-frame steel kit	`MAKE`	`release-candidate`	`reusable-definition`	`not-required`	7
`STN-CNP-P020` — canopy footing, reinforcement, base plate, and anchor-bolt kit	`MAKE`	`release-candidate`	`reusable-definition`	`not-required`	7
`STN-CNP-P030` — factory-bonded solar roof sandwich panel with MC4 leads	`BID`	`buildable-after-supplier-freeze`	`supplier-configuration`	`specified`	7
`STN-CNP-P040` — platform-canopy PV string, combiner, isolation, bonding, and downlink kit	`BID`	`buildable-after-supplier-freeze`	`supplier-configuration`	`specified`	7

Frozen inputs

- site wind, snow, seismic, thermal and maintenance load cases
- accepted steel, coating, roof-panel, PV and connector systems
- surveyed foundation, platform, electrical and drainage interfaces

Controlled outputs

- portal cut/weld, baseplate, anchor-template and erection drawings
- roof-panel layout, joints, gutters, penetrations and edge details
- PV string, isolation, bonding and cable-route schedule

Verification

- steel certificates, weld/NDT and frame survey
- anchor-template and erected-frame survey
- roof watertightness, PV insulation/polarity and bond-continuity tests

Boundary: Catalogue bay geometry is not a structural release; foundations and every site load case require the deployment design authority.

STN-FRP-030 — auxiliary solar-canopy module, truss and site-interface pack

Delivery lane: `deployment-led-hybrid`.

Drawings/interfaces: `STN-CNP-220`, `STN-CNP-230`. Tools/gauges: `STN-TOOL-TRUSS-FIXTURE`, `STN-TOOL-AUX-ANCHOR-TEMPLATE`, `STN-TOOL-AUX-ROOF-GAUGE`.

Controlled products

Product	Route	Catalogue maturity	Release path	Default	Variants
STN-CNP-P050 — 8.5 m × 22 m factory-bonded auxiliary solar-roof bay module	BID	buildable-after-supplier-and-structural-release	supplier-configuration	specified	7
STN-CNP-P060 — 22 m S355 transverse Warren-truss frame with two HSS 200 columns	MAKE	buildable-after-structural-calculation-and-drawing-release	reusable-definition	specified	7
STN-CNP-P070 — auxiliary-canopy pad footing, reinforcement, base plate, and anchor-bolt kit	MAKE	buildable-after-site-structural-release	deployment-specific	specified	7
STN-CNP-P080 — auxiliary-canopy PV string, combiner, isolation, bonding, and downlink kit	BID	buildable-after-electrical-and-supplier-freeze	supplier-configuration	specified	7
STN-CNP-P090 — auxiliary-canopy gutter, downpipe, lightning, maintenance-access, and edge-protection kit	SOURCE	buildable-after-site-and-supplier-freeze	deployment-specific	specified	7

Frozen inputs

- released site layout, egress, fire, drainage and maintenance-access plan
- site-specific structural calculation and foundation reactions
- selected roof/PV/lightning/edge-protection supplier configurations

Controlled outputs

- repeatable truss fabrication and roof-bay module drawings
- site footing, anchor, erection, gutter/downpipe and access drawings
- PV string, protection, bonding and commissioning schedule

Verification

- truss weld/NDT and dimensional survey
- foundation pre-pour, anchor and erected-geometry surveys
- water, electrical, lightning and edge-protection acceptance tests

Boundary: The 8.5 m by 22 m catalogue module is an area-planning unit only until site structure, foundations, egress and electrical integration are signed.

STN-FRP-040 — station services, passenger equipment and plinth integration pack

Delivery lane: supplier-interface.

Drawings/interfaces: STN-SYS-300, STN-SYS-310. Tools/gauges: STN-TOOL-CABINET-PLINTH-GAUGE, STN-TOOL-FARE-PLINTH-GAUGE, STN-TOOL-ACCESSIBILITY-GAUGE.

Controlled products

Product	Route	Catalogue maturity	Release path	Default	Variants
STN-MEP-P010 — weatherproof services cabinet, plinth, cooling, and maintenance-light kit	MAKE	release-candidate	reusable-definition	not-required	7
STN-MEP-P020 — incoming switchboard, distribution board, metering, UPS, and earthing kit	BID	buildable-after-supplier-freeze	supplier-configuration	specified	7
STN-MEP-P030 — platform and emergency LED luminaire, support, and cable kit	SOURCE	release-candidate	reusable-definition	not-required	7
STN-MEP-P040 — fire detection, alarm interface, extinguisher, and evacuation-sign kit	SOURCE	release-candidate	reusable-definition	not-required	7
STN-PAX-P010 — S-SBC station/depot host and rack enclosure	SOURCE	release-candidate	reusable-definition	not-required	7
STN-PAX-P020 — passenger-information display and route-strip kit	SOURCE	release-candidate	reusable-definition	not-required	7
STN-PAX-P030 — CCTV, PA loudspeaker, help-point, radio, and station-LAN kit	BID	buildable-after-supplier-freeze	supplier-configuration	specified	7
STN-PAX-P040 — fare gate, accessible gate, and validator equipment kit	BID	buildable-after-supplier-freeze	supplier-configuration	specified	7
STN-PAX-P050 — ticket-vending machine equipment kit	BID	buildable-after-supplier-freeze	supplier-configuration	specified	6
STN-PAX-P060 — seating, wheelchair-zone marking, wayfinding, and accessible-signage kit	SOURCE	release-candidate	reusable-definition	not-required	7
STN-PAX-P070 — anchored rolled-steel fare-lane / validator plinth with protected cable void	MAKE	release-candidate	reusable-definition	not-required	7
STN-PAX-P080 — anchored rolled-steel TVM plinth with protected power/data entry	MAKE	release-candidate	reusable-definition	not-required	6

Frozen inputs

- frozen operator equipment, communications, fare and cyber interfaces
- utility, UPS, cooling, earthing, fire and evacuation requirements
- released accessibility, sightline, coverage and maintainability zones

Controlled outputs

- cabinet/plinth fabrication and coordinated equipment-layout drawings
- power, data, containment, earth and fire-interface schedules
- equipment anchorage, accessible reach and replacement-clearance map

Verification

- plinth and anchorage dimensional/proof checks
- supplier FAT and station integrated functional tests
- accessibility, CCTV/PA coverage and power-loss survey

Boundary: Generic plinths may be prepared from controlled envelopes; holes, anchors and services may not be released against assumed supplier equipment.

STN-FRP-050 — pedestrian approach, lift/stair core and overbridge interface pack

Delivery lane: deployment-led-hybrid.

Drawings/interfaces: STN-ACC-400, STN-ACC-410. Tools/gauges: STN-TOOL-ACCESSIBILITY-GAUGE, STN-TOOL-STAIR-RISER-GAUGE.

Controlled products

Product	Route	Catalogue maturity	Release path	Default	Variants
STN-ACC-P010 — direct/protected pedestrian approach, kerb, ramp, and boundary kit	MAKE	release-candidate	reusable-definition	not-required	7
STN-ACC-P020 — installed passenger lift	BID	buildable-after-site-and-supplier-freeze	deployment-specific	specified	1
STN-ACC-P030 — pedestrian overbridge/concourse structural and enclosure kit	BID	buildable-after-site-and-supplier-freeze	deployment-specific	specified	1
STN-ACC-P040 — installed escalator	BID	study-only	deployment-specific	specified	1
STN-ACC-P050 — installed staircase	BID	study-only	deployment-specific	specified	1
STN-ACC-P060 — installed shaft	BID	study-only	deployment-specific	specified	1

Frozen inputs

- topographical survey, land boundary, pedestrian demand and road interfaces
- site structural, geotechnical, fire/egress and accessibility approvals
- selected lift, enclosure and emergency-power configuration

Controlled outputs

- approach, ramp, kerb, boundary and accessible-route drawings
- lift/stair/overbridge structure, enclosure and service-interface drawings
- evacuation, rescue, drainage and inspection-access plan

Verification

- route gradient, crossfall, width, surface and obstacle survey
- structural, clearance and weatherproofing acceptance
- lift certification, fire recall, backup power and egress test

Boundary: No catalogue access arrangement substitutes for a site accessibility, egress, highway or structural approval.

STN-FRP-060 — wayside charging and traction substation interface pack

Delivery lane: supplier-and-utility-interface.

Drawings/interfaces: STN-PWR-500, STN-PWR-510. Tools/gauges: STN-TOOL-CHARGER-ALIGNMENT, STN-TOOL-EARTH-BOND-TEST.

Controlled products

Product	Route	Catalogue maturity	Release path	Default	Variants
STN-CHG-P010 — station charging cabinet, protection, cable, and wayside connector kit	BID	buildable-after-supplier-freeze	supplier-configuration	specified	6
STN-CHG-P020 — traction power substation transformer/rectifier and protection interface	BID	buildable-after-utility-and-supplier-freeze	deployment-specific	specified	5

Frozen inputs

- utility fault level, capacity, metering and protection requirements
- selected charger and transformer/rectifier supplier data
- released vehicle docking envelope and operational charging duty

Controlled outputs

- equipment arrangement, foundation reaction and maintainability drawings
- single-line, protection, earthing, isolation and cable schedules
- vehicle/wayside datum, alignment, interlock and abort interface control

Verification

- supplier FAT and protection-coordination review
- earthing, insulation, isolation and utility witness tests
- vehicle alignment, charge, abort and emergency-isolation SAT

Boundary: Rated catalogue power is a requirement, not authority to connect; the utility and electrical design authority retain release.

STN-FRP-070 — 1:9 turnout, actuation, detection, heating and track-end pack

Delivery lane: `hybrid-fabrication-and-supplier`.

Drawings/interfaces: STN-TRK-600, STN-TRK-610. Tools/gauges: STN-TOOL-TURNOUT-BENCH, STN-TOOL-BLADE-PROFILE-GAUGE, STN-TOOL-TRACK-GEOMETRY.

Controlled products

Product	Route	Catalogue maturity	Release path	Default	Variants
STN-TRK-P010 — 1:9 UIC60 stock-rail, machined switch-blade, and closure-rail kit	MAKE	buildable-after-controlled-drawing-release	reusable-definition	specified	2
STN-TRK-P020 — cast-manganese frog, check-rail, stretcher-bar, and mechanical-lock kit	BID	buildable-after-supplier-freeze	supplier-configuration	specified	2
STN-TRK-P030 — prestressed turnout sleeper, slide-chair, and elastic-fastener set	SOURCE	buildable-after-supplier-freeze	supplier-configuration	specified	2
STN-TRK-P040 — 6 kN nominal / 12 kN peak point-machine actuator, crank, and hand-wind kit	BID	buildable-after-actuator-qualification	supplier-configuration	specified	2
STN-TRK-P050 — dual position detector, W-SBC interface, junction, and turnout harness kit	SOURCE	buildable-after-hardware-freeze	supplier-configuration	specified	2
STN-TRK-P060 — 3 kW points-heating strip, thermostat, IP67 cabinet, isolation, and cabling kit	SOURCE	buildable-after-climate-and-supplier-freeze	supplier-configuration	specified	2
STN-TRK-P070 — terminal stop-block, passive end marker, foundation, and fixing kit	SOURCE	buildable-after-site-geometry-freeze	deployment-specific	specified	2

Frozen inputs

- released wheel/rail interface, axle loads, route speed and climate envelope
- selected rail, frog, sleeper, actuator, detector and heater configurations
- site track alignment, signalling, drainage and track-end geometry

Controlled outputs

- rail machining, switch/closure, gauge and weld drawings

- complete turnout assembly, harness, detection, heating and bench-test schedule
- site set-out, installation, stop-block and commissioning drawings

Verification

- material, machining, weld/NDT and dimensional records
- bench throw, lock, detection, hand-wind and heating proof
- installed geometry, route/detection and stop-block acceptance

Boundary: The catalogue tangent and geometry do not release rail machining or site installation without controlled drawings and supplier qualifications.

STN-FRP-080 — depot site, drainage, track and throat-turnout pack

Delivery lane: `deployment-specific`.

Drawings/interfaces: STN-DEP-700, STN-DEP-710. Tools/gauges: STN-TOOL-DEPOT-SET-OUT, STN-TOOL-TRACK-GEOMETRY, STN-TOOL-DRAINAGE-TEST.

Controlled products

Product	Route	Catalogue maturity	Release path	Default	Variants
STN-DEP-P010 — main-heavy depot site formation, drainage, service-road, and secure-boundary kit	MAKE	buildable-after-site-design-release	deployment-specific	specified	1
STN-DEP-P020 — stabling, inspection, wash, and workshop track-panel/stop-block package	MAKE	buildable-after-controlled-layout-release	deployment-specific	specified	1
STN-DEP-P030 — 1:9 depot-throat turnout assembly replicated from the terminal turnout standard	MAKE	buildable-after-turnout-design-and-site-freeze	deployment-specific	specified	1

Frozen inputs

- boundary/topographical/utility/geotechnical surveys and environmental approvals
- released fleet plan, movements, swept paths and maintenance concept
- controlled depot layout, gradients, drainage/outfall and track standards

Controlled outputs

- earthworks, pavement, drainage, boundary and service-road drawings
- stabling, inspection, wash and workshop track-layout drawings
- turnout, stop-block, walkways, crossings and clearance-control schedule

Verification

- formation, compaction, drainage and pavement records
- track geometry, clearance and stop-block proof
- route, detection and vehicle swept-path demonstration

Boundary: These are deployment drawings: the reference depot quantities do not authorize site works or fix a universal depot layout.

STN-FRP-090 — depot charging, energy, workshop and services integration pack

Delivery lane: `deployment-and-supplier`.

Drawings/interfaces: STN-DEP-720, STN-DEP-730. Tools/gauges: STN-TOOL-VEHICLE-LIFT-GAUGE, STN-TOOL-CHARGER-ALIGNMENT, STN-TOOL-ENERGY-ISOLATION-TEST.

Controlled products

Product	Route	Catalogue maturity	Release path	Default	Variants
STN-DEP-P040 — outdoor/open-sided per-stall plug-in charger, isolation, suspended cable, and data-dock kit	BID	buildable-after-supplier-and-energy-freeze	deployment-specific	specified	1
STN-DEP-P050 — depot PV canopy, inverter, microgrid switchgear, and separated outdoor stationary-battery package	BID	buildable-after-energy-site-and-supplier-freeze	deployment-specific	specified	1
STN-DEP-P060 — main workshop, synchronized LM3 lift/bogie-change bay, overhaul/inspection bays, 40 t crane, wash plant, stores, and wheel-lathe package	BID	buildable-after-building-and-equipment-freeze	deployment-specific	specified	1
STN-DEP-P070 — depot cooled controls room, LV, compressed-air, fire, lighting, CCTV, LAN, access-control, and maintenance-data kit	BID	buildable-after-services-and-supplier-freeze	deployment-specific	specified	1

Frozen inputs

- selected equipment loads, heat rejection, utilities and maintenance envelopes
- site building, fire, structural, energy and environmental approvals
- released LM3 lift points, bogie extraction path and service requirements

Controlled outputs

- equipment layouts, foundations, clearances and replacement paths
- power/microgrid/battery isolation, fire, cooling and controls drawings
- workshop lift, crane, pit, wash, stores and maintenance-data schedules

Verification

- supplier FAT, certification and equipment foundation survey
- charging, energy, fire, cooling and emergency-isolation SAT
- synchronised lift, mechanical lock, bogie extraction and crane proof

Boundary: Reference duties and envelopes do not select suppliers or release a depot building, stationary battery compound, crane or lifting system.

Global boundary: These packages define reusable drafting and deployment handoff scope. They are not approved fabrication/construction drawings, supplier selections, signed calculations, permits, surveys or performed acceptance records.

Station open release gap register

Repository source: `design/component-catalogue/catalog/buildable-stations/open-release-gaps.md`

Generated from the same product rows as the station EBOMs and assembly travelers. A row disappears only when its product maturity is promoted to `release-candidate`; acceptance evidence remains mandatory at build time.

Engineering ID	Component	Route	Maturity / blocker	Variant quantities	Closure evidence
STN-ACC-P020	installed passenger lift	BID	buildable-after-site-and-supplier-freeze	interchange-elevated=4 core	lift certification; fire recall test; backup-power test; accessibility survey
STN-ACC-P030	pedestrian overbridge/concourse structural and enclosure kit	BID	buildable-after-site-and-supplier-freeze	interchange-elevated=2 ea	structural release; clearance survey; egress test; weatherproofing inspection
STN-ACC-P040	installed escalator	BID	study-only	interchange-elevated=4 ea	supplier and independent access/egress acceptance
STN-ACC-P050	installed staircase	BID	study-only	interchange-elevated=4 ea	supplier and independent access/egress acceptance
STN-ACC-P060	installed shaft	BID	study-only	interchange-elevated=4 ea	supplier and independent access/egress acceptance
STN-CHG-P010	station charging cabinet, protection, cable, and wayside connector kit	BID	buildable-after-supplier-freeze	standard=500 kW installed; major=500 kW installed; interchange=500 kW installed; interchange-elevated=500 kW installed; terminal=500 kW installed; depot-terminal=500 kW installed	FAT; insulation/protection test; vehicle alignment test; abort/interlock test
STN-CHG-P020	traction power substation transformer/rectifier and protection interface	BID	buildable-after-utility-and-supplier-freeze	major=800 kVA installed; interchange=1000 kVA installed; interchange-elevated=1000 kVA installed; terminal=1500 kVA installed; depot-terminal=2000 kVA installed	utility approval; FAT/SAT; protection coordination; earthing test
STN-CIV-P050	island centre deck infill between edge L-unit strips	MAKE	study-only	interchange-elevated=238 m2	station structural design release; deck lifting/installation release
STN-CIV-P060	station track-deck and running-rail interface	MAKE	study-only	halt=55.5 track m; standard=119 track m; major=119 track m; interchange=238 track m; interchange-elevated=238 track m; terminal=119 track m; depot-terminal=119 track m	station civil/support release; rail datum and gauge survey
STN-CNP-P030	factory-bonded solar roof sandwich panel with MC4 leads	BID	buildable-after-supplier-freeze	halt=10 ea; standard=20 ea; major=10 ea; interchange=20 ea; interchange-elevated=20 ea; terminal=20 ea; depot-terminal=20 ea	fire/material certificate; insulation test; connector inspection; watertightness test
STN-CNP-P040	platform-canopy PV string, combiner, isolation, bonding, and downlink kit	BID	buildable-after-supplier-freeze	halt=1 platform kit; standard=2 platform kit; major=1 platform kit; interchange=2 platform kit; interchange-elevated=2 platform kit; terminal=2 platform kit; depot-terminal=2 platform kit	polarity test; insulation resistance; bond continuity; string commissioning
STN-CNP-P050	8.5 m × 22 m factory-bonded auxiliary solar-roof bay module	BID	buildable-after-supplier-and-structural-release	halt=1 187 m2 module; standard=7 187 m2 module; major=9 187 m2 module; interchange=11 187 m2 module; interchange-elevated=12 187 m2 module; terminal=11 187 m2 module; depot-terminal=6 187 m2 module	panel fire/material certificate; module dimensional report; watertightness test; PV insulation/polarity test
STN-CNP-P060	22 m S355 transverse Warren-truss frame with two HSS 200 columns	MAKE	buildable-after-structural-calculation-and-drawing-release	halt=2 shared frame; standard=8 shared frame; major=10 shared frame; interchange=12 shared frame; interchange-elevated=13 shared frame; terminal=12 shared frame; depot-terminal=7 shared frame	controlled fabrication drawing; steel certificates; weld/NDT record; frame dimensional survey; galvanising report

Engineering ID	Component	Route	Maturity / blocker	Variant quantities	Closure evidence
STN-CNP-P070	auxiliary-canopy pad footing, reinforcement, base plate, and anchor-bolt kit	MAKE	buildable-after-site-structural-release	halt=4 column kit; standard=16 column kit; major=20 column kit; interchange=24 column kit; interchange-elevated=26 column kit; terminal=24 column kit; depot-terminal=14 column kit	site geotechnical release; foundation calculation; rebar/pre-pour inspection; anchor-template survey; concrete test
STN-CNP-P080	auxiliary-canopy PV string, combiner, isolation, bonding, and downlink kit	BID	buildable-after-electrical-and-supplier-freeze	halt=1 string group; standard=1 string group; major=1 string group; interchange=2 string group; interchange-elevated=2 string group; terminal=2 string group; depot-terminal=1 string group	string schedule; polarity/insulation test; bond continuity; protection settings; energy-site commissioning
STN-CNP-P090	auxiliary-canopy gutter, downpipe, lightning, maintenance-access, and edge-protection kit	SOURCE	buildable-after-site-and-supplier-freeze	halt=1 roof-bay kit; standard=7 roof-bay kit; major=9 roof-bay kit; interchange=11 roof-bay kit; interchange-elevated=12 roof-bay kit; terminal=11 roof-bay kit; depot-terminal=6 roof-bay kit	drainage flow test; lightning/bond test; edge-protection inspection; maintenance-access walkdown
STN-DEP-P010	main-heavy depot site formation, drainage, service-road, and secure-boundary kit	MAKE	buildable-after-site-design-release	depot-terminal=8000 m2 site	site/geotechnical release; drainage/outfall approval; road swept-path test; secure-boundary inspection
STN-DEP-P020	stabling, inspection, wash, and workshop track-panel/stop-block package	MAKE	buildable-after-controlled-layout-release	depot-terminal=400 track-m	track layout release; rail/fastener certificates; geometry survey; stop-block proof; wash-track drainage test
STN-DEP-P030	1:9 depot-throat turnout assembly replicated from the terminal turnout standard	MAKE	buildable-after-turnout-design-and-site-freeze	depot-terminal=2 turnout assembly	controlled STN-TRK-SA800 definition; shop bench proof; yard geometry survey; route/detection proof
STN-DEP-P040	outdoor/open-sided per-stall plug-in charger, isolation, suspended cable, and data-dock kit	BID	buildable-after-supplier-and-energy-freeze	depot-terminal=4 stall kit	supplier loss/duty and maximum-ambient declaration; supplier FAT; protection/isolation test; vehicle reach/alignment test; charge/abort/data-sync SAT; fire separation and emergency-isolation walkdown
STN-DEP-P050	depot PV canopy, inverter, microgrid switchgear, and separated outdoor stationary-battery package	BID	buildable-after-energy-site-and-supplier-freeze	depot-terminal=1 energy-site kit	structural/PV layout release; utility/protection approval; cell-to-pack propagation and heat-release evidence; outdoor compound separation/ventilation/containment review; remote isolation and gas/fire detection proof; energy-site SAT
STN-DEP-P060	main workshop, synchronized LM3 lift/bogie-change bay, overhaul/inspection bays, 40 t crane, wash plant, stores, and wheel-lathe package	BID	buildable-after-building-and-equipment-freeze	depot-terminal=1800 m2 workshop	building/egress release; foundation reaction calculation; four-point LM3 interface gauge; synchronized lift proof and mechanical-lock test; bogie extraction sweep; crane certification; pit/fall-protection inspection; lathe acceptance; wash/recycling SAT
STN-DEP-P070	depot cooled controls room, LV, compressed-air, fire, lighting, CCTV, LAN, access-control, and maintenance-data kit	BID	buildable-after-services-and-supplier-freeze	depot-terminal=1 depot services kit	utility capacity; 2 x 30 kW packaged-DX schedule and 30 kW single-unit thermal duty proof; cooling duty/standby changeover and high-temperature alarm; fire cause/effect and remote energy-isolation test; air-system pressure test; coverage/access test; maintenance-data sync
Engineering ID	**Component**	**Route**	**Maturity / blocker**	**Variant quantities**	**Closure evidence**
STN-MEP-P020	incoming switchboard, distribution board, metering, UPS, and earthing kit	BID	buildable-after-supplier-freeze	halt=1 station kit; standard=1 station kit; major=1 station kit; interchange=1 station kit; interchange-elevated=1 station kit;	factory acceptance test; protection test; earth test; one-hour UPS test

Engineering ID	Component	Route	Maturity / blocker	Variant quantities	Closure evidence
				terminal=1 station kit; depot-terminal=1 station kit	
STN-PAX-P030	CCTV, PA loudspeaker, help-point, radio, and station-LAN kit	BID	buildable-after-supplier-freeze	halt=1 platform kit; standard=2 platform kit; major=1 platform kit; interchange=2 platform kit; interchange-elevated=2 platform kit; terminal=2 platform kit; depot-terminal=2 platform kit	camera coverage; speech intelligibility; help-point call; OCC failover test
STN-PAX-P040	fare gate, accessible gate, and validator equipment kit	BID	buildable-after-supplier-freeze	halt=2 lane/validator; standard=4 lane/validator; major=6 lane/validator; interchange=8 lane/validator; interchange-elevated=8 lane/validator; terminal=6 lane/validator; depot-terminal=6 lane/validator	accessible-lane gauge; AFC transaction test; power-loss release; plinth anchor test
STN-PAX-P050	ticket-vending machine equipment kit	BID	buildable-after-supplier-freeze	standard=2 ea; major=4 ea; interchange=4 ea; interchange-elevated=4 ea; terminal=4 ea; depot-terminal=4 ea	transaction test; accessibility reach check; cabinet ingress test
STN-TRK-P010	1:9 UIC60 stock-rail, machined switch-blade, and closure-rail kit	MAKE	buildable-after-controlled-drawing-release	terminal=1 turnout kit; depot-terminal=1 turnout kit	rail certificates; blade profile/CMM report; rail geometry inspection; weld/NDT release
STN-TRK-P020	cast-manganese frog, check-rail, stretcher-bar, and mechanical-lock kit	BID	buildable-after-supplier-freeze	terminal=1 turnout kit; depot-terminal=1 turnout kit	material certificate; frog/check-rail gauge; lock proof; stretcher dimensional report
STN-TRK-P030	prestressed turnout sleeper, slide-chair, and elastic-fastener set	SOURCE	buildable-after-supplier-freeze	terminal=42 sleeper position; depot-terminal=42 sleeper position	EN 13145 certificate; fastener certificate; chair/fastener torque; sleeper-position survey
STN-TRK-P040	6 kN nominal / 12 kN peak point-machine actuator, crank, and hand-wind kit	BID	buildable-after-actuator-qualification	terminal=1 ea; depot-terminal=1 ea	18 kN ultimate proof; three-second throw test; manual operation; endurance/environment qualification
STN-TRK-P050	dual position detector, W-SBC interface, junction, and turnout harness kit	SOURCE	buildable-after-hardware-freeze	terminal=1 turnout kit; depot-terminal=1 turnout kit	channel independence inspection; normal/reverse detection proof; unknown-state test; W-SBC route-lock interface test
STN-TRK-P060	3 kW points-heating strip, thermostat, IP67 cabinet, isolation, and cabling kit	SOURCE	buildable-after-climate-and-supplier-freeze	terminal=1 turnout kit; depot-terminal=1 turnout kit	insulation/protection test; thermostat test; cabinet ingress inspection; hot/cold function test
STN-TRK-P070	terminal stop-block, passive end marker, foundation, and fixing kit	SOURCE	buildable-after-site-geometry-freeze	terminal=2 track-end kit; depot-terminal=2 track-end kit	track-end geometry release; foundation inspection; stop-block proof; marker visibility check

Package-level exclusions

These conditions are not product rows and therefore remain explicit package gates:

- platform screen doors are optional for light-metro-3car and are not included
- site survey, geotechnical design, utilities, permits, and stamped calculations remain deployment-specific
- auxiliary canopy requires deployment structural, foundation, drainage, egress, and electrical release

Station product reconciliation

Repository source: `design/component-catalogue/catalog/buildable-stations/station-product-reconciliation.md`

Status: PASS

This generated register proves stable identities in both directions across the station manifest, BOM, traveler, compact variant definition/drawing register, native FreeCAD installed/exploded states and IFC4.3 handoff. It proves configuration consistency, not construction readiness.

Variant	Products	Assemblies	Definition sheets	Connection controls	States	Result
`halt`	26	8	26	8	installed, exploded	PASS
`standard`	29	8	29	10	installed, exploded	PASS
`major`	30	8	30	10	installed, exploded	PASS
`interchange`	30	8	30	10	installed, exploded	PASS
`interchange-elevated`	35	8	35	10	installed, exploded	PASS
`terminal`	37	9	37	13	installed, exploded	PASS
`depot-terminal`	44	10	44	14	installed, exploded	PASS

A definition-sheet or connection-control identifier is a required deployment deliverable keyed to its product row. It is not evidence that a signed fabrication drawing, supplier data or site approval already exists. Those gates remain in the open-release register and each assembly traveler.

Generated buildable trainset artifacts

Repository source: `design/component-catalogue/catalog/buildable-trainset/README.md`

This folder is generated by:

```
tools/automation/buildable-trainset.sh
```

The manifest starts the buildable rolling-stock product structure: fabricated parts, external components, subassemblies, assemblies, and the final trainset. Generated definitions and travelers now carry structured material specs, process specs, evidence requirements, and inspection controls. The review calls out what is already solid in the current basic design and what must be closed before first steel cut.

How to use this folder

If you are…	Start with	Then use
Design authority	`current-design-buildability-review.md`	Close yellow/red findings before release
Configuration / release manager	`open-release-gaps.md`	Keep every supplier, drawing, analysis and approval gap visible against the controlled candidate
First-article lead	`first-article-baseline.json`	Hold `LM3-FA-001`, assign the 81 public work packages, and manage closure only through the controlled state file
Test / assurance lead	`first-article-evidence-status.md`	Separate current screening from the 13 still-open supplier, structural, physical, thermal, electrical, HIL, manufacturing and mass-properties release gates
Manufacturing engineer	`buildable-trainset-manifest.md`	Turn `MAKE` definitions into controlled drawings, fixtures, and NC/cut data
Drawing/interface lead	`factory-release-readiness.md`	Execute the 16 chassis, module, fascia, roof, interior, finish and recovery packages through issued drawings, released tooling and performed verification
Drafting/checking team	`factory-drawings/index.md`	Develop the 29 drawing-definition seeds covering all 62 locally made rows into checked production drawings without treating design envelopes as manufacturing dimensions
Train configuration engineer	`train-end-interface.md`	Select panoramic glass outer ends or the optional mid open train-to-train connection on the common end interface
Vehicle mechanical / interior engineer	`small-component-standard.md`	Freeze the common rail, four fastener families, adapters, keyed connectors, lighting modules, and door/window service boundaries
Exterior finish engineer	`exterior-finish-system.md`	Retain mandatory corrosion/fire/UV protection while using livery film and trial-only radiative roof coating only in permitted zones
Production planner	`critical-path.md`	Plan parallel part fabrication, subassembly release, final train assembly, space use, and rough labour
Factory planner	`factory-plan.md`	Size the pilot factory cells, enclosed area, yard, assembly-time rollups, machinery list, and rough equipment prices
Cost planner	`trainset-build-cost.md`	Review the recalculated 3-car trainset build cost with USD 10/h labour, 20% unexpected-cost premium, and included seats/floors/lighting/HVAC/windows/doors scope
Mass-properties engineer	`mass-closure-ledger.md`	Close all 117 active product responsibilities with production-CAD, supplier and calibrated weighed evidence; use `mass-budget.md` for the nine-category control total
Assembly engineer	`joint-control-schedule.md`	Close each joint with its named torque authority and interface drawing
Manufacturing / quality lead	`manufacturing-and-assembly-controls.md`	Adopt or replace the ten reference datum, steel, joint, bonding, moulding, roof, interior, lifting, finish and NCR controls
Build-record controller	`evidence/manufacturing-control-record-template.json`	Copy the blank ten-control execution record per work order; bind actual revisions, people, plant, lots, observations, NCRs and handback
Inspection lead	`first-article-inspection-plan.md`	Import the 416 package-specific hold, witness and record-review characteristics into the QMS without pre-claiming a result
Production-data controller	`production-data-release-register.md`	Issue and hash-bind the 472 drawing, material, cut/form/NC, mould, harness, finish and inspection artifacts required across all 62 locally made products
Buyer / supplier manager	`supplier-anchors.md`	Source all 56 bought-in rows from 27 real supplier/research families or qualify a controlled local equivalent, then issue the per-part RFQ/evidence definitions
Buyer preparing enquiries	`cots-candidates.md`	Use 41 manufacturer/research catalogue/RFQ candidates covering all 56 bought-in rows, while preserving the exact fit and evidence still required
Buyer needing a costable starting point	`default-product-specifications.md`	Start every 34 `BID` and 24 `SOURCE` product row from one controlled affordable reference, then override it when fit, duty, evidence or whole-life cost requires
First-article execution team	`first-article-execution-pack.md`	Follow one ordered procurement, rig, physical-test, mass-properties and independent-acceptance route across all 13 evidence gates
Shop lead	`travelers/index.md`	Assign work centers, fill approvals/signatures during a real build
QA / inspector	`travelers/index.md`	Use QA gates, hold points, signoff blocks, and NCR/deviation logs
Local factory/tooling team	`manufacturing-methods.md`	Review all 120 product links, 9 timed methods, 30 tooling families, joining parts, mould use, coating, seals, floor/fixture and motor/bogie steps
Factory supervisor	`../../../../deployment/supervision/README.md`	Generate the nine read-only method views on a real city plant, rehearse unavailable/quality-hold routing, and correlate only reviewed ERP Item/BOM mappings

Open the 120 native FreeCAD parts and 26 assembly files, the selectable tooling in `../../models/cad/lm3-manufacturing-tooling.FCStd`, and the matching IFC4.3 product/assembly and method

federation.

Generated manufacturing visuals

View	Use
trainset-factory-layout.png	Pilot factory cell arrangement: 55 m final bay, chassis fixtures, moulding/trim, bogies, interiors/HVAC kits, corrosion/finish, stores/QA, yard, and test access
trainset-assembly-method-flow.png	Parallel first-article critical-path work streams and labour/day envelopes
trainset-gfrp-moulding-method.png	Reusable one-metre glass-fibre mould route, CNC trim, insert/seal fit, master-frame dry fit, and clip-on carbody installation
trainset-bogie-marriage-method.png	Final-bay bogie-to-carbody marriage with lifting columns, surveyed chassis datums, accepted bogies, and static-release hold points
simplified-small-component-package.png	Common service rails, plug-in lighting, fixture adapters, adjustable door carriers, and open window pressure frames

The intended sequence is:

1. Hold the LM3-FA-001 configuration in first-article-baseline.json, then read the buildability review for release blockers.
2. Use the manifest to understand the 120-product-row / 26-assembly tree, then use default-product-specifications.md as the costable starting configuration, cots-candidates.md for exact catalogue samples and vehicle-specific RFQs, and supplier-anchors.md for the broader local-equivalent boundary.
3. Use train-end-interface.md to confirm whether each end position is dressed as a panoramic glass front/end or an open mid-train train-to-train connection.
4. Use definitions as the drawing/RFQ/material/process source of truth.
5. Use the critical path to reserve the long final assembly track only for accepted subassemblies and keep steel, GFRP, bogie, HV, and interior kit work off-line where possible.
6. Use factory-plan.md to reserve about 3,515 m2 of enclosed pilot factory, 2,200 m2 of yard/test apron, and the rough machinery package before lease/building work or RFQs start.
7. Use trainset-build-cost.md for the current rough-order build cost: direct material/supplier modules plus explicit labour at USD 10/h, with a 20% unexpected-cost premium. Its direct-module bucket already includes seats, floors, grab rails, interior lighting, roof HVAC, windows, side doors, sill/emergency kits, and panoramic end glass.
8. Qualify the small-component rail/fastener samples and close each fixture, lighting, door and glazing release gate in small-component-standard.md.
9. Close every active product row in mass-closure-ledger.md, reconcile the nine-category mass budget, then close joint-control rows as supplier, production-CAD and calibrated weighed evidence becomes available.
10. Use travelers as the shop-routing, material/process-control, and QA template, applying manufacturing-and-assembly-controls.md wherever a released procedure has not yet superseded its reference control.
11. Run the ordered procurement and rig sequence in first-article-execution-pack.md; begin supplier and test records from the controlled templates under evidence/.
12. Attach real inspection evidence and signatures only during an authorised build.
13. Run python3 tools/automation/validate-lm3-first-article-evidence.py before claiming a release gate. It verifies accountable submissions, calibrated-equipment metadata, artifact existence and SHA-256 values.

The work-package register is generated from the product manifest plus first-article-work-package-state.toml. Use python3 tools/automation/publish-lm3-work-packages.py for a read-only publication plan; --apply idempotently reconciles open rows with GitHub issues after maintainer review.

definitions/ is the generated definition pack. It contains JSON and Markdown files for every product-tree node:

- parts/ — fabricated parts plus BID/SOURCE external components.
- subassemblies/ — reusable build cells such as underframe, roof, HV, bogie, and interior subassemblies.
- assemblies/ — complete car, end, articulation, and control assemblies.
- trainsets/ — final trainset-level definition.

Use definitions/index.md as the drawing/RFQ worklist. These files are not stamped shop drawings; they are the traceable seed for the v2A drawing, supplier-freeze, material/process-control, and proof-evidence packages.

`travelers/` is the generated shop-traveler pack. It contains JSON and Markdown traveler templates for every product-tree node. Each traveler includes:

- document revision and pending approval blocks,
- material specifications, process specifications, and evidence requirements,
- estimated labor by operation,
- work centers and tooling IDs,
- QA gates linked to the component/assembly acceptance checks,
- operator, cell-lead, quality, and manufacturing-engineering signoff blocks,
- a nonconformance/deviation log.

These are signable manufacturing templates, not already-signed release records. A real build cell must fill the approval/signature fields and attach actual inspection evidence before first-article release.

Buildable trainset manifest

Repository source: `design/component-catalogue/catalog/buildable-trainset/buildable-trainset-manifest.md`

Generated by `tools/automation/buildable-trainset.sh`. This is the first structured product breakdown for a buildable OpenSourceRail trainset: parts subassemblies assemblies final trainset.

- Family: `light-metro-3car`
- Candidate: `light-metro-3car__16p5m__reference-body__reference-bogie__motor-350kw-hm47-class__battery-225kwh-lfp-800v__hvac-24kw-direct-hv-dc__pv12`
- Candidate score: `92.628`
- Candidate feasible: `true`
- Product item rows: `120`
- Assembly nodes: 26
- Open supplier/component rows: `81` (register)
- Bought-in product reference defaults: `58` (register)

Candidate metrics

Metric	Value
cars	3.0
trainset_length_m	49.5
platform_margin_m	10.0
mass_t	75.308
cost_usd	682431.18
axle_load_kg	6275.63
installed_continuous_kw	1500.0
traction_margin	2.845
battery_usable_kwh	540.0
offwire_range_km	68.915
hvac_margin	1.2
pv_peak_kw	15.12
manufacturing_complexity_index	1.0

Current CAD baseline vs optimizer target

Parameter	Current CAD baseline	Optimizer target
car_length_m	16.5	16.5
motor	motor-350kw-hm47-class	motor-350kw-hm47-class
battery	battery-225kwh-lfp-800v	battery-225kwh-lfp-800v
hvac	hvac-24kw-direct-hv-dc	hvac-24kw-direct-hv-dc
pv_modules_per_car	12.0	12.0

Product item route summary

Route	Quantity sum
MAKE	396
BID	116
SOURCE	493

Assembly tree

ID	Layer	Qty/trainset	Build cell	Children	Hold points
LM3-BDY-SA110	subassembly	3	weld and fixture cell	LM3-BDY-P010 LM3-BDY-P020 LM3-BDY-P021 LM3-BDY-P030 LM3-BDY-P120	material release fixture tack survey weld/NDT release post-weld datum survey
LM3-BDY-SA120	subassembly	3	weld and fixture cell	LM3-BDY-SA110 LM3-BDY-P060 LM3-BDY-P061 LM3-BDY-P070	door/window aperture survey roof rail survey carbody dimensional report
LM3-SHELL-A200	assembly	3	paint / clip-on body / glazing cells	LM3-BDY-SA120 LM3-BDY-P130 LM3-BDY-P131 LM3-BDY-P132 LM3-BDY-P133 LM3-BDY-P140 LM3-WIN-SA320 LM3-EXT-P080 LM3-EXT-P090 LM3-FIN-P010	corrosion report clip and anti-lift witness map eight-hour trainset body route water ingress pre-test
LM3-WIN-SA320	subassembly	18	composite / glazing cell	LM3-BDY-P110 LM3-WIN-P010 LM3-EXT-P020	aperture gauge bond/gasket procedure water ingress test
LM3-DOOR-SA310	subassembly	12	final assembly and commissioning cell	LM3-BDY-P100 LM3-DOOR-P010 LM3-EXT-P010	door gauge fit obstruction test closed-and-locked test
LM3-INT-SA330	subassembly	3	final assembly and commissioning cell	LM3-EXT-P060 LM3-EXT-P061 LM3-EXT-P062 LM3-EXT-P063 LM3-EXT-P064 LM3-EXT-P065 LM3-EXT-P066 LM3-INT-P010 LM3-INT-P020 LM3-INT-P021 LM3-INT-P022 LM3-INT-P030 LM3-INT-P031 LM3-INT-P032 LM3-INT-P040 LM3-INT-P041 LM3-INT-P050 LM3-INT-P051 LM3-INT-P052	egress check fire-material pack liner/trim fit survey lighting/PIS/CCTV static test
LM3-FIX-SA340	subassembly	3	interior pre-fit and final assembly cell	LM3-FIX-P010 LM3-FIX-P020 LM3-FIX-P030	rail datum survey fastener-family audit fixture load-evidence check service/removal demonstration
LM3-LGT-SA350	subassembly	3	interior pre-fit and commissioning cell	LM3-LGT-P010 LM3-LGT-P020	connector key audit lighting lux map emergency-feed isolation and duration test module replacement demonstration
LM3-ROOF-SA410	subassembly	3	final assembly and commissioning cell	LM3-BDY-P080 LM3-ROOF-P010 LM3-ROOF-P020 LM3-ROOF-P030 LM3-ROOF-P040 LM3-EXT-P040 LM3-EXT-P050 LM3-EXT-P070 LM3-FIN-P020 LM3-TRC-P050	roof leak test HVAC drain test PV isolation/bonding check
LM3-HV-SA510	subassembly	3	final assembly and commissioning cell	LM3-BDY-P050 LM3-HV-P010 LM3-HV-P020 LM3-HV-P030 LM3-TRC-P030 LM3-TRC-P040 LM3-TRC-P060 LM3-TRC-P070 LM3-SAF-P010	HVIL test insulation resistance coolant pressure test first energisation release
LM3-BOG-SA611	subassembly	3	bogie clean assembly and brake cell	LM3-BOG-P040 LM3-BOG-P042 LM3-BOG-P044 LM3-BOG-P048	wheelset identity bearing installation primary-height match static brake/WSP test free rotation
LM3-TRC-SA615	subassembly	3	traction drive clean assembly cell	LM3-TRC-P010 LM3-TRC-P020 LM3-BOG-P050	motor/gearbox serial match coupling alignment torque-link proof insulation/rotation test removal-envelope trial

ID	Layer	Qty/trainset	Build cell	Children	Hold points
LM3-BOG-SA610	subassembly	3	bogie weld and assembly cell	LM3-BOG-P010 LM3-BOG-P030 LM3-BOG-SA611 LM3-TRC-SA615 LM3-BOG-P046 LM3-BOG-P060	frame NDT wheelset/bearing certificate motor/gearbox alignment static brake test
LM3-BOG-SA621	subassembly	3	bogie clean assembly and brake cell	LM3-BOG-P041 LM3-BOG-P043 LM3-BOG-P045 LM3-BOG-P049	wheelset identity bearing installation primary-height match static brake/WSP test free rotation
LM3-BOG-SA620	subassembly	3	bogie weld and assembly cell	LM3-BOG-P020 LM3-BOG-P031 LM3-BOG-SA621 LM3-BOG-P047 LM3-BOG-P061	frame NDT wheelset/bearing certificate ride-height setup static brake test
LM3-CWL-SA710	subassembly	2	composite moulding and trim cell	LM3-CWL-P010 LM3-CWL-P011 LM3-CWL-P012 LM3-CWL-P013 LM3-CWL-P014 LM3-CWL-P015 LM3-CWL-P016 LM3-FAS-P010 LM3-FAS-P020 LM3-FAS-P030	laminate coupon release insert pull-out trim/drill survey A/B-end dry-build water test
LM3-EIF-SA650	subassembly	2	end-interface fixture / final assembly cell	LM3-END-P060 LM3-END-P061 LM3-END-P062	common bolt-grid survey selected end-option fit gauge seal and drain continuity panoramic-or-open-mid configuration record
LM3-END-SA700	assembly	2	composite / final assembly and commissioning cells	LM3-BDY-P040 LM3-BDY-P090 LM3-CWL-SA710 LM3-EXT-P030 LM3-END-P010 LM3-END-P020 LM3-END-P030 LM3-END-P040 LM3-END-P050	A/B end interchange coupler datum survey sensor calibration recovery interface check
LM3-TTART-SA850	assembly	0	final assembly and commissioning cell	LM3-EIF-SA650 LM3-ART-P040 LM3-ART-P041	open-end option configuration record train-to-train motion-envelope proof walk-through gangway continuity water ingress/drain test
LM3-ART-SA810	subassembly	2	articulation bench and proof-load cell	LM3-ART-P010 LM3-ART-P020 LM3-ART-P021	pin/bearing identity shimmed datum survey proof load lubrication/seal release motion sweep
LM3-ART-SA820	subassembly	2	gangway clean assembly cell	LM3-ART-P022 LM3-ART-P023	fire-material pack bridge load test gap/pinch gauge water test full-motion sweep
LM3-ART-SA830	subassembly	2	harness, hose and articulation bench	LM3-ART-P024 LM3-ART-P030	HV/LV segregation continuity/pressure test bend-radius sweep drain test replaceability trial
LM3-ART-SA800	assembly	2	final assembly and commissioning cell	LM3-ART-SA810 LM3-ART-SA820 LM3-ART-SA830	motion-envelope proof trainline continuity water ingress/drain test
LM3-SYS-SA900	assembly	1	final assembly and commissioning cell	LM3-CTRL-P010 LM3-CTRL-P020 LM3-CTRL-P030 LM3-CTRL-P040 LM3-CTRL-P050	network enumeration firmware record self-test event-recorder write/read test
LM3-CAR-A900	assembly	3	final assembly and commissioning cell	LM3-SHELL-A200 LM3-DOOR-SA310 LM3-INT-SA330 LM3-FIX-SA340 LM3-LGT-SA350 LM3-ROOF-SA410 LM3-HV-SA510 LM3-BOG-SA610 LM3-BOG-SA620 LM3-AUX-P010	car weigh door/HVAC/static systems test bogie marriage report low-speed yard movement
LM3-TRAINSET-A000	trainset	1	final assembly and commissioning cell	LM3-CAR-A900 LM3-EIF-SA650 LM3-END-SA700 LM3-ART-SA800 LM3-SYS-SA900	trainset weigh static brake/door/HVAC/HV tests FEM screening accepted dynamic-test release

Product items

ID	BOM lines	Layer	Route	Qty/trainset	Parent	Title	Acceptance
LM3-BDY-P010	B1	fabricated-part	MAKE	6 ea	LM3-BDY-SA110	laser-cut side sill beam, LH/RH	heat traceability dimensional check weld VT/MT where classed
LM3-BDY-P020	B1	fabricated-part	MAKE	3 kit	LM3-BDY-SA110	underframe centre spine and longitudinal load-path kit	tube/plate certs spine straightness survey bogie-centre datum report
LM3-BDY-P021	B1	fabricated-part	MAKE	3 car pack	LM3-BDY-SA110	underframe cross-bearer, door-bay outrigger, and equipment-bracket pack	cut-list identity cross-bearer profile gauge station map fixture tack survey
LM3-BDY-P030	B2	fabricated-part	MAKE	6 set	LM3-BDY-SA110	bolster box, air-spring pad, and centre-pivot insert set	line-bore report air-spring datum survey NDT report
LM3-BDY-P040	B2	fabricated-part	MAKE	2 set	LM3-END-SA700	coupler pocket, shear plate, and crash-can insert kit	coupler face datum bolt-hole survey crash-load drawing check
LM3-BDY-P050	B2 T17	fabricated-part	MAKE	3 kit	LM3-HV-SA510	battery tray rails, vent plenum, and service-lid gutter kit	battery gauge fit vent-path inspection gasket land check
LM3-BDY-P060	B1 B12	fabricated-part	MAKE	3 set	LM3-BDY-SA120	low-floor centre pan and removable service-floor support set	PRM floor height egress aisle gauge pan weld inspection
LM3-BDY-P061	B1	fabricated-part	MAKE	3 car set	LM3-BDY-SA120	raised bogie-end deck, transition ramp, and removable hatch-frame set	deck-height survey PRM transition gauge hatch fit/removal deck weld inspection
LM3-BDY-P070	B1	fabricated-part	MAKE	6 side	LM3-BDY-SA120	side-wall post, door portal, waist rail, and cant rail kit	door cassette gauge window cassette gauge side-frame survey
LM3-BDY-P080	B1	fabricated-part	MAKE	3 kit	LM3-ROOF-SA410	roof bow, HVAC rail, PV rail, and cable-tray bracket kit	roof rail pitch HVAC curb gauge PV clamp pull test
LM3-BDY-P090	B1	fabricated-part	MAKE	2 set	LM3-END-SA700	end ring frame and anti-climber beam set	A/B interchange check glass carrier land survey anti-climber datum
LM3-END-P060	B2	fabricated-part	MAKE	2 end position	LM3-EIF-SA650	common reversible end-interface carrier ring, option bolt grid, and sealing datum kit	option bolt-grid survey seal datum continuity A/B interchange check end-option fit gauge
LM3-END-P061	B2	fabricated-part	MAKE	2 option kit	LM3-EIF-SA650	panoramic-end option shim, cowl/glass carrier, and sensor datum closeout kit	panoramic option fit gauge glass/cowl datum transfer sensor datum check water-ingress pre-test
LM3-END-P062	B2	fabricated-part	MAKE	0 option kit	LM3-EIF-SA650	mid open-connection option portal trim, bellows clamp, threshold bridge, and drain kit	open-portal gauge bellows clamp fit threshold/turntable level check drain-path water test
LM3-BDY-P100	B2 B25	fabricated-part	MAKE	12 opening kit	LM3-DOOR-SA310	door portal reinforcement, threshold beam, and cassette shim kit	door aperture gauge threshold height survey cassette shim record water-drain path check
LM3-BDY-P110	B2	fabricated-part	MAKE	18 opening kit	LM3-WIN-SA320	window carrier ring, bonded-gasket land, and replacement jack-point inserts	aperture gauge bond-land surface check water-ingress witness replacement tool clearance

ID	BOM lines	Layer	Route	Qty/trainset	Parent	Title	Acceptance
LM3-BDY-P120	B26	fabricated-part	MAKE	3 car kit	LM3-BDY-SA110	jacking pad, lifting eye, towing lug, and recovery-label kit	material traceability weld/NDT record proof load four-point depot interface gauge datum and label inspection
LM3-BDY-P130	B6	fabricated-part	MAKE	48 module	LM3-SHELL-A200	one-metre clip-on solid-side fiberglass body module	material/fire certificate trim gauge insert pull-out master-frame dry fit
LM3-BDY-P131	B6	fabricated-part	MAKE	24 module	LM3-SHELL-A200	one-metre clip-on window-edge fiberglass side module	window trim nest sealed-edge inspection cassette removal sweep master-frame dry fit
LM3-BDY-P132	B6	fabricated-part	MAKE	24 module	LM3-SHELL-A200	one-metre clip-on door-edge fiberglass side module	door trim nest sealed-edge inspection door motion/service sweep master-frame dry fit
LM3-BDY-P133	B7	fabricated-part	MAKE	48 module	LM3-SHELL-A200	one-metre clip-on fiberglass roof skin and equipment-fairing module	roof mould/trim record anti-lift proof equipment keep-out gauge water/debris-ingress test
LM3-BDY-P140	B7	fabricated-part	MAKE	3 car kit	LM3-SHELL-A200	keyed clip rail, captive retainer, anti-lift, and dry-seal car kit	clip proof load anti-reversal gauge retainer witness-mark check water ingress test
LM3-ROOF-P010	T14	fabricated-part	MAKE	3 kit	LM3-ROOF-SA410	HVAC curb, drop-duct collar, condensate tray, and drain fitting kit	curb flatness drop-duct gauge condensate drain flow test roof leak test
LM3-ROOF-P020	T22	fabricated-part	MAKE	3 car kit	LM3-ROOF-SA410	PV bonded-pad lands, raised rail kit, bonding jumpers, and roof isolation labels	rail pitch survey bond pull coupon earth continuity module keep-out gauge
LM3-ROOF-P030	B7 T14	fabricated-part	MAKE	3 car set	LM3-ROOF-SA410	removable HVAC curb fairing, intake/exhaust skirt, and access-hatch moulding set	fairing trim gauge HVAC airflow keep-out hatch removal trial roof leak test
LM3-ROOF-P040	B7 T22	fabricated-part	MAKE	3 car set	LM3-ROOF-SA410	PV junction plinth, cable-gland cover, antenna closeout, and walkway edge set	gland/closeout trim gauge drain-path test bonding access check service-removal trial
LM3-CWL-P010	B8	fabricated-part	SOURCE	2 kit	LM3-CWL-SA710	end-cowl fiberglass laminate, insert, adhesive, and coupon material kit	material batch trace fire/smoke certificate coupon layup record adhesive shelf-life check
LM3-CWL-P011	B8	fabricated-part	MAKE	2 ea	LM3-CWL-SA710	CWL-FRP-01 upper brow and roof-cap fiberglass cast	mould release record laminate coupon trim-line gauge roof-flange fit
LM3-CWL-P012	B8	fabricated-part	MAKE	2 ea	LM3-CWL-SA710	CWL-FRP-02 left cheek fiberglass cast	mould release record laminate coupon insert pull-out split-gap gauge
LM3-CWL-P013	B8	fabricated-part	MAKE	2 ea	LM3-CWL-SA710	CWL-FRP-03 right cheek fiberglass cast	mould release record laminate coupon insert pull-out split-gap gauge
LM3-CWL-P014	B8	fabricated-part	MAKE	2 ea	LM3-CWL-SA710	CWL-FRP-04 lower apron and anti-climber cover fiberglass cast	mould release record laminate coupon lamp pocket gauge drain-path water test
ID	BOM lines	Layer	Route	Qty/trainset	Parent	Title	Acceptance
LM3-CWL-P015	B8	fabricated-part	MAKE	4 hatch	LM3-CWL-SA710	CWL-FRP-05 lamp, washer, and	mould release record

ID	BOM lines	Layer	Route	Qty/trainset	Parent	Title	Acceptance
						service-hatchfiberglass cast set	insert pull-out gasket compression check hatch removal trial
LM3-CWL-P016	B8	fabricated-part	MAKE	8 flange section	LM3-CWL-SA710	CWL-FRP-06 backing-ring flange fiberglass cast set	mould release record glass-carrier land survey bond-line witness A/B interchange check
LM3-FAS-P010	B8 B27	fabricated-part	MAKE	2 end set	LM3-CWL-SA710	panoramic front-glass carrier ring, setting-block pockets, and secondary-retention frame	carrier datum survey glass-edge clearance gauge secondary-retention proof pane removal trial
LM3-FAS-P020	B8 B17	fabricated-part	MAKE	2 end set	LM3-CWL-SA710	reversible front-lamp cassette tray, aiming adjusters, and retained service bracket	lamp datum gauge aiming range/retention test harness clearance cassette removal trial
LM3-FAS-P030	B8 B27	fabricated-part	SOURCE	2 end kit	LM3-CWL-SA710	front glazing/lamp EPDM seal, drain rail, washer sleeve, and edge-closeout kit	seal batch/certificate compression map drain flow test water-ingress test
LM3-ART-P010	B2 B29	fabricated-part	MAKE	2 kit	LM3-ART-SA810	articulation adapter frame, anti-lift keeper, and shim kit	motion envelope bearing proof shim pack record
LM3-BOG-P010	G1 G18	fabricated-part	MAKE	3 ea	LM3-BOG-SA610	powered bogie welded H-frame and motor-cradle weldment	bogie fixture survey weld/NDT record motor-cradle proof
LM3-BOG-P020	G2	fabricated-part	MAKE	3 ea	LM3-BOG-SA620	trailer bogie welded H-frame	bogie fixture survey weld/NDT record air-spring datum survey
LM3-BOG-P030	G13 G20	fabricated-part	MAKE	3 kit	LM3-BOG-SA610	powered-bogie guards, cable guides, WSP brackets, and inspection covers	service access check harness clearance fastener torque record
LM3-BOG-P031	G13 G20	fabricated-part	MAKE	3 kit	LM3-BOG-SA620	trailer-bogie guards, cable guides, WSP brackets, and inspection covers	service access check harness clearance fastener torque record
LM3-BOG-P050	G18	fabricated-part	MAKE	3 bogie kit	LM3-TRC-SA615	powered-bogie motor torque link, anti-rotation stop, and safety lanyard bracket kit	torque-link gauge bracket NDT motor removal clearance fastener locking record
LM3-BOG-P060	G14 G15 G17	fabricated-part	MAKE	3 bogie kit	LM3-BOG-SA610	powered-bogie brake/WSP/speed-sensor harness and junction-bracket kit	continuity test connector IP rating wheelset clearance dynamic cable sweep
LM3-BOG-P061	G14 G15 G17	fabricated-part	MAKE	3 bogie kit	LM3-BOG-SA620	trailer-bogie brake/WSP/speed-sensor harness and junction-bracket kit	continuity test connector IP rating wheelset clearance dynamic cable sweep
LM3-EXT-P010	B11 B25 E20	external-component	BID	12 ea	LM3-DOOR-SA310	electric plug/sliding door cassette	EN 14752 evidence obstruction test closed-and-locked loop test
LM3-EXT-P020	B10	external-component	SOURCE	18 ea	LM3-WIN-SA320	side laminated glazing cassette	glazing certificate water ingress test replacement method
LM3-EXT-P030	B27	external-component	BID	2 ea	LM3-END-SA700	single panoramic heated end-glass assembly	glass certificate heater test bond/gasket procedure
LM3-EXT-P040	T14	external-component	BID	3 ea	LM3-ROOF-SA410	hvac-24kw-direct-hv-dc roof HVAC	+50 C capacity evidence condensate drain test EMC/vibration evidence
LM3-EXT-P050	T21	external-component	SOURCE	36 module	LM3-ROOF-SA410	roof PV module and edge-clamp kit	module datasheet clamp pull test isolation/bonding check

ID	BOM lines	Layer	Route	Qty/trainset	Parent	Title	Acceptance
LM3-EXT-P060	B12	external-component	SOURCE	135 m2	LM3-INT-SA330	stepped floor-board and removable service-hatch system	fire certificate panel load and deflection evidence hatch removal trial level/step and egress survey
LM3-EXT-P061	B13	external-component	SOURCE	135 m2	LM3-INT-SA330	welded resilient floor covering, cove, nosing, and adhesive system	fire/smoke certificate adhesive compatibility and cure record welded-seam peel sample slip and cleanability evidence
LM3-EXT-P062	B14	external-component	SOURCE	60 seat	LM3-INT-SA330	longitudinal passenger and priority-seat modules	fire/smoke certificate seat/occupant load evidence fastener and anti-rotation record egress and cleaning-clearance gauge
LM3-EXT-P063	B15	external-component	SOURCE	3 car kit	LM3-INT-SA330	stainless grab-pole, handrail, joint, and insulated adapter kit	material/finish certificate joint locking record fixture-specific proof-load evidence reach, egress and snag survey
LM3-EXT-P064	B18 E14	external-component	SOURCE	3 car kit	LM3-INT-SA330	passenger-information display, speaker, amplifier, and mounting kit	fire/EMC evidence network enumeration audio/intelligibility test display visibility and service-removal trial
LM3-EXT-P065	B19 E15	external-component	SOURCE	3 car kit	LM3-INT-SA330	CCTV camera, passenger intercom, PoE/data, and mounting kit	fire/EMC/IP evidence network enumeration camera coverage/privacy review intercom call and service-removal trial
LM3-EXT-P066	A1 A2 A3 A4	external-component	SOURCE	3 car kit	LM3-INT-SA330	PRM, safety-signage, emergency-lighting, extinguisher, and first-aid kit	accessible reach/contrast review emergency-light duration test equipment certificate/expiry audit location and egress survey
LM3-FIX-P010	B2 B15 B21	fabricated-part	MAKE	3 car kit	LM3-FIX-SA340	OSR-RAIL-42 common ceiling, waist, and seat-zone service rail kit	rail datum survey end-deburr check isolation/finish inspection representative pull/slip test
LM3-FIX-P020	B2 B21	external-component	SOURCE	3 car kit	LM3-FIX-SA340	four-family captive fastener, floating nut, isolator, and access-fastener kit	supplier certificate batch/finish trace installed-grip gauge locking and captive-part audit
LM3-FIX-P030	B14 B15 E14	fabricated-part	MAKE	3 car kit	LM3-FIX-SA340	standard passenger-fixture saddle and equipment adapter kit	adapter gauge fixture-specific load calculation proof-load sample egress and snag check
LM3-WIN-P010	B10	fabricated-part	MAKE	18 opening kit	LM3-WIN-SA320	replaceable window pressure frame, dry seal, drain, and captive retention kit	pressure-frame gauge retention calculation seal compression record water-ingress and replacement trial
LM3-DOOR-P010	B11 B25	fabricated-part	MAKE	12 opening kit	LM3-DOOR-SA310	door four-point adjustable carrier, datum pin, dry seal, and keyed connector bracket kit	carrier datum gauge interface load calculation seal compression record connector keying and cassette replacement trial
LM3-LGT-P010	B16	external-component	SOURCE	66 module	LM3-LGT-SA350	1.2 m plug-in main LED lighting cassette and captive mounting kit	rail fire certificate shock/vibration evidence photometric/lux test plug polarity and retention test

ID	BOM lines	Layer	Route	Qty/trainset	Parent	Title	Acceptance
LM3-LGT-P020	B16 A4	external-component	BID	3 car kit	LM3-LGT-SA350	emergency and doorway lighting modules with independent keyed feeder kit	emergency duration /effectiveness evidence evacuation visibility test feed isolation test doorway illuminatio n test
LM3-EXT-P070	E21	external-component	SOURCE	3 car kit	LM3-ROOF-SA410	roof antennas, service walkway pads, lifting covers, and maintenance labels	antenna VSWR test walkway slip certificate lifting-cover fit roof bonding check
LM3-EXT-P080	B6	external-component	BID	6 side kit	LM3-SHELL-A200	fire-rated GFRP side-module laminate, core, gelcoat, and consumable kit	EN 45545 evidence laminate coupon resin/fibre batch trace mould release record
LM3-EXT-P090	B7	external-component	BID	3 car kit	LM3-SHELL-A200	fire-rated GFRP roof-module, dry-seal, and removable skirt material kit	EN 45545 evidence roof laminate coupon seal certificate service-removal trial water and debris-ing ress check
LM3-FIN-P010	B20 B28	external-component	BID	3 car kit	LM3-SHELL-A200	pre-cut exterior livery graphic film, edge-seal, datum-mark, and repair-patch kit	supplier rail-use statement substrate/cure acceptance adhesion coupon edge/overlap inspection cleaning/removal trial
LM3-FIN-P020	B20 B28	external-component	BID	3 car kit	LM3-ROOF-SA410	calcium-carbonate radiative roof-coatin g qualification and exposed-roof application kit	rail fire/chemical evidence GFRP adhesion and flexibility UV/abrasion/wash ageing initial and aged solar reflectance/e mittance one-car thermal/mai ntenance trial
LM3-INT-P010	T14	external-component	SOURCE	3 car kit	LM3-INT-SA330	HVAC diffusers, side return ducts, saloon grilles, and access panels	airflow balance rattle check access-panel removal fire-material certificate
LM3-INT-P020	B21	fabricated-part	MAKE	3 car kit	LM3-INT-SA330	FRP/phenolic ceiling liner panel set	fire-material certificate trim-line gauge fastener insert pull-out rattle check
LM3-INT-P021	B21	fabricated-part	MAKE	3 car set	LM3-INT-SA330	ceiling light-trough bezel, diffuser carrier, and emergency-luminair e closeout set	fire-material certificate light cassette gauge emergency-feed identification module removal trial
LM3-INT-P022	B21	fabricated-part	MAKE	3 car set	LM3-INT-SA330	HVAC plenum cover, diffuser transition, detector bezel, and ceiling service-hatch set	fire-material certificate airflow opening gauge detector access hatch retention/rattl e check
LM3-INT-P030	B21	fabricated-part	MAKE	6 side kit	LM3-INT-SA330	FRP/phenolic sidewall liner panel set	fire-material certificate visible-gap gauge liner removal edge-radius inspection
LM3-INT-P031	B21	fabricated-part	MAKE	6 side set	LM3-INT-SA330	window reveal, setting-block inspection cover, and blind/label land set	fire-material certificate glass-edge clearanc e cassette removal sweep sharp-edge inspecti on
ID	BOM lines	Layer	Route	Qty/trainset	Parent	Title	Acceptance
LM3-INT-P032	B21	fabricated-part	MAKE	6 side set	LM3-INT-SA330	waist cable-cover, low-voltage access lid, and retained	fire-material certificate segregation

ID	BOM lines	Layer	Route	Qty/trainset	Parent	Title	Acceptance
						quarter-turnfastener set	inspection captive-fastener audit cover removal/rattle check
LM3-INT-P040	B7 B21	fabricated-part	MAKE	3 car kit	LM3-INT-SA330	FRP battery strake cover panel set	fire-material certificate no-saloon-opening inspection HV warning label check sharp-edge inspection
LM3-INT-P041	B21	fabricated-part	MAKE	3 car set	LM3-INT-SA330	seat-base fairing, retained equipment hatch, and rail-saddle closeout set	fire-material certificate seat-load-path clearance hatch removal trial cleaning/sharp-edge inspection
LM3-INT-P050	B21	fabricated-part	MAKE	12 door-zone kit	LM3-INT-SA330	FRP vestibule kick-panel and threshold-closeout set	fire-material certificate threshold/drain gauge anti-slip witness kick-panel retention test
LM3-INT-P051	B21	fabricated-part	MAKE	6 end-zone set	LM3-INT-SA330	PRM transition-ramp, bogie-deck step-cover, contrast-nosing, and anti-slip set	fire-material certificate PRM transition gauge anti-slip/contrast evidence trip-edge inspection
LM3-INT-P052	B21	fabricated-part	MAKE	12 door-zone set	LM3-INT-SA330	door-pocket liner, sensitive-edge clearance trim, and removable jamb-cover set	fire-material certificate door sweep/pinch gauge jamb-cover removal obstruction-test witness
LM3-TRC-P010	T1	external-component	BID	6 ea	LM3-TRC-SA615	motor-350kw-hm47-class axle traction motor	motor datasheet thermal curve mounting-foot load proof EMC evidence
LM3-TRC-P020	G19 T2	external-component	BID	6 ea	LM3-TRC-SA615	single-stage reduction gearbox and flexible coupling	gear ratio certificate oil access check coupling alignment
LM3-TRC-P030	T3 T4 T7 T13 T20 T23	external-component	BID	3 car kit	LM3-HV-SA510	two independent motor controllers, isolated LV DC/DC, MPPT, station protection, and cooling-loop kit	HVIL test coolant pressure test EMC/bonding check
LM3-TRC-P040	T5 T6	external-component	BID	3 ea	LM3-HV-SA510	battery-225kwh-lfp-800v saloon-isolated side traction battery pack	cell/module certificate isolation test no-saloon-opening inspection outward vent/fire containment data
LM3-HV-P010	T8	fabricated-part	MAKE	3 car kit	LM3-HV-SA510	battery sliding trays, retention straps, service interlocks, and drain pans	battery module gauge retention pull test tray slide/removal test drain-path inspection
LM3-HV-P020	T18	fabricated-part	MAKE	3 car kit	LM3-HV-SA510	segregated HV cable tray, bonding studs, grommets, and orange cover set	bend-radius gauge bond continuity cover fastener torque orange-label inspection
LM3-HV-P030	T4 T7 T17 T20	fabricated-part	MAKE	3 car kit	LM3-HV-SA510	coolant manifold brackets, bleed/drain points, and insulated pipe clamp kit	pressure-test access bleed point height check pipe clamp pitch thermal isolation inspection
LM3-TRC-P050	T15	external-component	SOURCE	3 ea	LM3-ROOF-SA410	roof-mounted regen dump resistor and thermal shield kit	resistance certificate thermal clearance roof bonding hot-surface label

ID	BOM lines	Layer	Route	Qty/trainset	Parent	Title	Acceptance
LM3-TRC-P060	T12 T19	external-component	BID	3 car kit	LM3-HV-SA510	station side-pin charging connector, actuator, shutter, and alignment target	dock alignment test HVIL test shutter cycle test emergency release
LM3-TRC-P070	T11 T16	external-component	SOURCE	3 car kit	LM3-HV-SA510	HV contactor, fuse, pre-charge, service-disconnect, and current-sensor panel	isolation test pre-charge timing fuse rating evidence service-disconnect lockout
LM3-SAF-P010	T9 T10	external-component	SOURCE	3 car kit	LM3-HV-SA510	battery temperature /off-gas detection, electrical-enclosure smoke detection, and localized mist kit	detector certificate loop continuity mist proof-flow reservoir/pump/pressure diagnostic event-recorder input
LM3-BOG-P040	G3 G8	external-component	BID	6 wheelset	LM3-BOG-SA611	powered-bogie wheelset with axle-mounted brake discs	wheel/axle heat certificates press-force chart back-to-back and runout report ultrasonic inspection balance record
LM3-BOG-P041	G3 G8	external-component	BID	6 wheelset	LM3-BOG-SA621	trailer-bogie wheelset with axle-mounted brake discs	wheel/axle heat certificates press-force chart back-to-back and runout report ultrasonic inspection balance record
LM3-BOG-P042	G4 G14 G15	external-component	BID	6 wheelset set	LM3-BOG-SA611	powered-wheelset axlebox, sealed bearing unit, speed and temperature sensor set	bearing serial/clearance record grease and seal certificate axle journal fit speed/temperature sensor calibration rotation test
LM3-BOG-P043	G4 G14 G15	external-component	BID	6 wheelset set	LM3-BOG-SA621	trailer-wheelset axlebox, sealed bearing unit, speed and temperature sensor set	bearing serial/clearance record grease and seal certificate axle journal fit speed/temperature sensor calibration rotation test
LM3-BOG-P044	G5 G12	external-component	BID	3 bogie set	LM3-BOG-SA611	powered-bogie primary suspension spring, guide and bump-stop set	load-deflection curves matched-height report compound/batch certificates installed preload and clearance survey
LM3-BOG-P045	G5 G12	external-component	BID	3 bogie set	LM3-BOG-SA621	trailer-bogie primary suspension spring, guide and bump-stop set	load-deflection curves matched-height report compound/batch certificates installed preload and clearance survey
LM3-BOG-P046	G6 G7 G10 G11 G12	external-component	BID	3 bogie set	LM3-BOG-SA610	powered-bogie to carbody connection: air springs, emergency spring, centre pivot, yaw links and dampers	vertical/lateral load curves pivot proof and articulation limit damper curves hot/cold ride-height and anti-lift survey
LM3-BOG-P047	G6 G7 G10 G11 G12	external-component	BID	3 bogie set	LM3-BOG-SA620	trailer-bogie to carbody connection: air springs, emergency spring, centre pivot, yaw links and dampers	vertical/lateral load curves pivot proof and articulation limit damper curves hot/cold ride-height and anti-lift survey
LM3-BOG-P048	G8 G9 G16	external-component	BID	3 bogie set	LM3-BOG-SA611	powered-bogie brake calipers, parking actuators, pads and wheel-slide hardware	brake-force calculation friction-pair certificate thermal capacity parking holding test WSP functional test

ID	BOM lines	Layer	Route	Qty/trainset	Parent	Title	Acceptance
LM3-BOG-P049	G8 G9 G16	external-component	BID	3 bogie set	LM3-BOG-SA621	trailer-bogie brake calipers, parking actuators, pads and wheel-slide hardware	brake-force calculation friction-pair certificate thermal capacity parking holding test WSP functional test
LM3-AUX-P010	G21	external-component	SOURCE	3 car kit	LM3-CAR-A900	secondary-suspension compressor, dryer, reservoir, and isolation-manifold kit	pressure certificate leak test dryer function relief-valve test service-access check
LM3-END-P010	B22 B23	external-component	BID	2 ea	LM3-END-SA700	automatic end coupler and crash-energy absorber	EN 15227 absorber evidence recovery procedure bolt torque record
LM3-ART-P020	B9	external-component	BID	2 joint set	LM3-ART-SA810	articulation lower spherical pivot, bearing housing and pin set	bearing static/dynamic capacity pin material/NDT proof load lubrication/sealing plan motion-envelope proof
LM3-ART-P021	B9	external-component	BID	2 joint set	LM3-ART-SA810	articulation upper lateral/yaw links, spherical joints and retained pins	link buckling/fatigue proof joint angular capacity pin retention inspection full-motion sweep
LM3-ART-P022	B9	external-component	BID	2 gangway set	LM3-ART-SA820	inter-car double-wall corrugated bellows and clamp-frame set	fire/smoke evidence pressure/water ingress test fatigue-cycle evidence replaceable-clamp demonstration
LM3-ART-P023	B9	external-component	BID	2 gangway set	LM3-ART-SA820	inter-car passenger bridge, turntable and flexible interior-panel set	passenger load proof anti-slip evidence gap/step gauge pinch/shear hazard review full-motion sweep
LM3-ART-P024	B24	external-component	SOURCE	2 carrier set	LM3-ART-SA830	articulation trainline carrier, support arms, abrasion liners and drain path	rated bend radius dynamic sweep abrasion/fire evidence drain test service replacement trial
LM3-CTRL-P010	E1 E2 E6 E14 E15	external-component	BID	1 trainset kit	LM3-SYS-SA900	T-ECU/S and T-ECU/A compute and safety-control cabinet kit	hardware BOM check self-test network enumeration firmware record safety-output test
LM3-CTRL-P020	E3 E4 E5 E6 E7 E8 E21	external-component	SOURCE	1 trainset kit	LM3-SYS-SA900	navigation, balise, 5G, LoRa, GNSS, IMU, and roof-antenna kit	SKU/firmware record antenna VSWR GNSS/IMU test balise read radio link test
LM3-CTRL-P030	E10 E11 E12 E13 E16	external-component	BID	1 trainset kit	LM3-SYS-SA900	maintenance HMI, depot pendant, emergency controls, and safety-relay kit	key/guarded-control test emergency input test 2oo2 relay test stowage and access check
LM3-CTRL-P040	E17 E20 E22	fabricated-part	MAKE	3 car kit	LM3-SYS-SA900	pre-terminated LV trainline harness, DIN cabinet, and terminal-distribution kit	continuity/hipot pinout check label inspection segregation survey configuration record
LM3-CTRL-P050	E9 E23	external-component	SOURCE	1 trainset kit	LM3-SYS-SA900	operational and crashworthy event-recorder storage kit	write/read test retention configuration crashworthy certificate download/recovery test

ID	BOM lines	Layer	Route	Qty/trainset	Parent	Title	Acceptance
LM3-END-P020	E15 E18 E19	external-component	BID	2 ea	LM3-END-SA700	T-OBS nose sensor pack, heated window services, and washer kit	sensor calibration washer/heater test 2oo2 verdict interface test
LM3-END-P030	E19	fabricated-part	MAKE	2 end kit	LM3-END-SA700	cowl service hatch, sensor backing bracket, washer-tube, and heater-cable clip kit	hatch water test sensor datum check heater-cable separation washer tube leak test
LM3-END-P040	B22 E17	external-component	BID	2 end kit	LM3-END-SA700	e-coupler LV jumper, recovery trainline, and end harness breakaway kit	pinout test breakaway force check ingress protection rescue compatibility
LM3-END-P050	B17	external-component	SOURCE	2 end kit	LM3-END-SA700	sealed headlight, tail/marker light, threshold-warning, and end-lamp harness kit	photometric certificate function/polarity test ingress protection A/B-end interchange check
LM3-ART-P030	B24	external-component	BID	2 articulation kit	LM3-ART-SA830	inter-car HV/LV jumper, coolant hose loop, energy chain, and drain sleeve kit	bend-radius sweep trainline continuity coolant pressure test water-drain test
LM3-ART-P040	B9 B29	external-component	BID	0 joint kit	LM3-TTART-SA850	train-to-train open-end articulation, gangway, drawbar, turntable, and service-jumper cassette	train-to-train motion-envelope proof walk-through gangway fire evidence trainline continuity water ingress/drain test
LM3-ART-P041	B24	external-component	SOURCE	0 joint kit	LM3-TTART-SA850	train-to-train jumper blanking, transition harness, isolation label, and dust-cover kit	pinout test blanking cover ingress check isolation label inspection bend-radius sweep

LM3 off-the-shelf and RFQ candidate register

Repository source: `design/component-catalogue/catalog/buildable-trainset/cots-candidates.md`

Freeze real catalogue candidates and vehicle-specific RFQ baselines for every bought-in LM3 product row.

Release boundary: A catalogue candidate proves that a relevant product exists; it does not prove LM3 fit, safety acceptance, availability, price, or authority approval. Exact order codes and supplier-controlled drawings remain mandatory before purchase or release.

Manufacturer sources checked on 2026-08-31.

A complete system such as a door, bogie, HVAC unit or gangway is vehicle-specific even when its supplier uses a standard platform. Exact catalogue components are therefore separated from RFQ baselines and partial interface kits.

Coverage

Measure	Value
Bought-in LM3 rows with a sourcing candidate	56 / 56
Rows containing at least one exact catalogue component	10
Controlled candidates	41

Candidate components and systems

Candidate	Model and primary source	State	LM3 rows	Publicly established boundary
OSR-COTS-DOOR-IFE	Knorr-Bremse IFE — IFE modular sliding-plug entrance system	rfq-baseline	LM3-EXT-P010	rail entrance-system family; supplier engineering required for vehicle aperture and controls
OSR-COTS-GLASS-AGC	AGC Automotive — rail glazing with framing system	rfq-baseline	LM3-EXT-P020, LM3-EXT-P030	laminated rail glazing family; supplier framing-system option
OSR-COTS-FINISH-3M-180MC	3M — Controltac Graphic Film with Comply Adhesive 180mC family	sample-and-system-rfq	LM3-FIN-P010	listed for rail-car graphic applications; 50 micrometre cast opaque film; pressure-activated removable adhesive; up to seven-year MCS warranty subject to selected construction and conditions
OSR-COTS-COOLROOF-PURDUE-CACO3	Purdue University research team (reference only) — 2020 CaCO3-acrylic particle-matrix paint formulation	qualification-only	LM3-FIN-P020	95.5 percent initial solar reflectance reported; 0.94 normal emissivity in the atmospheric window reported; 60 percent particle concentration research formulation
OSR-COTS-GLAZING-SIKA268	Sika — Sikaflex-268 PowerCure, 600 ml	sample-ready	LM3-EXT-P020, LM3-EXT-P030	EN 45545-2 R1/R7 HL3; 4.5 MPa typical lap-shear strength; -50 to 90 deg C service range; 30 minute open time at stated reference conditions
OSR-COTS-HVAC-LONGERTEK	Longertek — direct-DC rail HVAC platform	rfq-baseline	LM3-EXT-P040, LM3-INT-P010	direct high-voltage DC architecture; rail HVAC product platform
OSR-COTS-PV-SUNMAN	Sunman Energy — eArc lightweight PV module	sample-ready	LM3-EXT-P050	lightweight flexible photovoltaic family
OSR-COTS-FLOOR-COMFLOOR	3A Composites Mobility — COMFLOOR rail floor panel	rfq-baseline	LM3-EXT-P060	lightweight sandwich floor-panel family; optional integrated heating; rail vehicle application
OSR-COTS-FLOOR-FORBO-6R3883	Forbo Flooring Systems — Marmoleum Fresco FR2 6R3883 moonstone, 2.5 mm	sample-ready	LM3-EXT-P061	2.5 mm thickness; 2.9 kg/m2; EN 45545-2 HL2 stated for the published system; roll width 2 m
OSR-COTS-SEAT-FAINSA	Fainsa — metro and urban-rail seating family	rfq-baseline	LM3-EXT-P062, LM3-EXT-P063	urban rail passenger seat family
OSR-COTS-PIS-LUMINATOR	Luminator Technology Group — RailKom passenger-information and communication platform	rfq-baseline	LM3-EXT-P064	rail passenger information, audio and communication family
OSR-COTS-CAMERA-MOXA-P06HC	Moxa — VPort P06HC-1MP-M12-CAM36	sample-ready	LM3-EXT-P065	EN 50155 application statement; IEC 61373 shock/vibration; IP66; PoE and M12 Ethernet; 1280 x 800 up to 30 fps
OSR-COTS-LIGHT-TEKNOWARE	Teknoware — rolling-stock interior and emergency lighting platform	rfq-baseline	LM3-EXT-P066, LM3-LGT-P010, LM3-LGT-P020, LM3-END-P050	rolling-stock LED emergency-lighting platform
OSR-COTS-FASTENER-RIVNUT	Boellhoff — RIVNUT metric M6/M8 steel or stainless blind rivet nuts	sample-ready	LM3-FIX-P020	published M6/M8 thread and upset-strength data; one-sided installation
OSR-COTS-ANTENNA-HS-85137717	HUBER+SUHNER — SENCITY Rail MIMO 5-port with GNSS/LNA, product 85137717	orderable-candidate	LM3-EXT-P070, LM3-CTRL-P020	2x2 cellular and Wi-Fi MIMO; GNSS with integrated LNA; EN 50155; EN 45545-2 and NFPA 130; -40 to 85 deg C
OSR-COTS-COMPOSITE-GURIT-130FR	Gurit — SE 130FR / ST 130FR prepreg with SA130FR and SF130FR films	sample-ready	LM3-EXT-P080, LM3-EXT-P090	EN 45545 HL2 stated for R1/R7/R17 monolithic laminates; 85 deg C low-temperature cure option; 45 minute cure option at 120 deg C; published reinforcement and film order codes
OSR-COTS-MOTOR-ABB-AMXE250	ABB — AMXE250 configurable traction motor	rfq-baseline	LM3-TRC-P010	up to 436 kW peak; up to 3353 Nm peak; up to 3500 rpm; configurable shaft and length
OSR-COTS-GEAR-VOITH-SE	Voith — SE single-stage axle-mounted metro gear unit	rfq-baseline	LM3-TRC-P020	single-stage axle-mounted rail gear family
OSR-COTS-CONVERTER-ABB-CC400	ABB — BORDLINE CC400 LRV compact converter	rfq-baseline	LM3-TRC-P030	600/750 VDC LRV platform; two independent propulsion converters; integrated auxiliary converter and battery charger; regenerative or resistor braking
OSR-COTS-BATTERY-HOPPECKE-LITRAC	HOPPECKE — LiTrac modular high-voltage LFP rail traction battery	rfq-baseline	LM3-TRC-P040	modular/scalable traction system; published LFP option; UN 38.3 base-module statement; supplier rail safety engineering
OSR-COTS-RESISTOR-DANOTHERM	Danotherm Electric — custom traction brake resistor, 600-3000 VDC family	rfq-baseline	LM3-TRC-P050	rail traction brake-resistor family; 600-3000 VDC published range; IEC 60322 and IEC 61373 product-family statements; roof/underframe installation options

Candidate	Model and primary source	State	LM3 rows	Publicly established boundary
OSR-COTS-CHARGER-SCHUNK	Schunk Carbon Technology — Smart Charging automated high-current collector platform	concept-alternative	LM3-TRC-P060	automated conductive charging platform
OSR-COTS-HV-SCHALTBAU-C360	Schaltbau — C360K/300 railway bidirectional DC contactor	sample-ready	LM3-TRC-P070	IEC 60077-2 category B; 1000 V at pollution degree 3 / 1500 V at pollution degree 2; 300 A at 70 deg C; 3000 A one-second withstand; 24/36-48/72-110 VDC coil options
OSR-COTS-HV-TE-HVP800	TE Connectivity — AMP+ HVP800 cable assembly 1-2286937-1	sample-ready	LM3-TRC-P070, LM3-ART-P030	1000 VDC; 1 m 50 mm2 cable assembly; integrated HVIL family; -40 to 140 deg C; published CAD and product drawings
OSR-COTS-FIRE-LIIONTAMER-3	Xtralis / Li-ion Tamer — Li-ion Tamer GEN 3	sample-ready	LM3-SAF-P010	electrolyte-vapour early detection; temperature and humidity at sensing nodes; relay and Modbus interfaces; calibration-free sensing claim
OSR-COTS-FIRE-FOGTEC	FOGTEC Rail Systems — rail high-pressure water-mist system	hazard-analysis-rfq	LM3-SAF-P010	rail water-mist system family; small pipe diameters and configurable vehicle installation
OSR-COTS-WHEELSET-GHH	GHH-BONATRANS — urban-rail wheelset with axle-mounted disc interfaces	rfq-baseline	LM3-BOG-P040, LM3-BOG-P041	urban-rail wheelset supplier family
OSR-COTS-BEARING-SKF-TBU	SKF — sensor-ready railway tapered roller bearing unit	calculation-then-rfq	LM3-BOG-P042, LM3-BOG-P043	sealed and pre-lubricated ready-to-mount TBU family
OSR-COTS-SUSPENSION-CONTI	Continental — rail half-bellow air spring with integrated auxiliary spring	dynamics-then-rfq	LM3-BOG-P044, LM3-BOG-P045, LM3-BOG-P046, LM3-BOG-P047	metro/local-transport air-spring family; integrated rubber-metal auxiliary spring option
OSR-COTS-BRAKE-KNORR-RZKT	Knorr-Bremse — WheelAct Compact RZKT with spring parking-brake option	brake-calculation-then-rfq	LM3-BOG-P048, LM3-BOG-P049	50 kN service pad force; 45 kN spring parking force; 68 kg; 305 mm width; plus/minus 25 mm lateral movement; 8-10 year stated overhaul interval
OSR-COTS-AIR-KNORR-VV120T	Knorr-Bremse — VV120-T 2.0 oil-free rail compressor / AirSupply Smart	air-demand-then-rfq	LM3-AUX-P010	oil-free rail compressor family; variable-speed smart-control option; published low-noise operating mode
OSR-COTS-COUPLER-DELLNER430	Dellner — automatic coupler Type 430 with pivot anchor	rfq-baseline	LM3-END-P010, LM3-END-P040	light-rail/tram/monorail application; 600 kN buff and 500 kN draft; gas-hydraulic damper; remote uncoupling and foldable options
OSR-COTS-JOINT-SCHAEFFLER	Schaeffler — ELGES maintenance-free spherical plain bearing and rod-end family	calculation-then-order-code	LM3-ART-P020, LM3-ART-P021	maintenance-free spherical plain-bearing family
OSR-COTS-GANGWAY-HUBNER	HUEBNER — modular metro double-wall corrugated gangway	kinematic-envelope-then-rfq	LM3-ART-P022, LM3-ART-P023, LM3-ART-P040	single/double-wall corrugated-bellows options; bridge and elastic interior-panel options; metro/urban-rail application
OSR-COTS-TRAINLINE-IGUS-HARTING	igus / HARTING — rail e-chain carrier with Han HPR connector system	sample-and-cycle-test	LM3-ART-P024, LM3-ART-P030, LM3-ART-P041	rail dynamic cable-carrier family; rail connector/fire variants available
OSR-COTS-CONTROL-DUAGON-MH50C	duagon — D-SC Controller with MH50C safe CPU and distributed safe I/O	safety-allocation-then-rfq	LM3-CTRL-P010	applications up to SIL4 stated by supplier; three-CPU safe controller; safe I/O; EN 50155 rolling-stock environment; certification packages available
OSR-COTS-COMPUTE-MOXA-V2406C	Moxa — V2406C-WL1-T railway computer	application-compute-candidate	LM3-CTRL-P010	EN 50155:2017 mandatory-test statement; EN 50121-4; IEC 61373; -40 to 70 deg C; 24 to 110 VDC isolated input family
OSR-COTS-ROUTER-WESTERMO-630-5G	Westermo — Ibex-RT-630-5G	orderable-candidate	LM3-CTRL-P020	3GPP Release 16 5G; 4x4 sub-6 GHz MIMO; 2x2 Wi-Fi; EN 50155 and EN 45545-2; M12 and IP66
OSR-COTS-SAFETYRELAY-PILZ-751184	Pilz — PNOZ s4, railway order 751184	bench-safety-relay-candidate	LM3-CTRL-P030	railway application note for order 751184; three safety contacts; redundant self-monitoring circuit
OSR-COTS-RECORDER-HASLER	HaslerRail — JRU/EVA event-recorder platform	requirements-then-rfq	LM3-CTRL-P050	rail event-recording product family
OSR-COTS-LIDAR-OUSTER-OS1	Ouster — OS1 industrial lidar sensor	development-sensor-candidate	LM3-END-P020	industrial digital lidar family

Integration and closure actions

OSR-COTS-DOOR-IFE — Knorr-Bremse IFE IFE modular sliding-plug entrance system

- Source class: vehicle-specific-rfq

- Local manufacture: Buy the first cassette and controller; manufacture portal, threshold, carrier, harness and service parts locally to the released interface.
- Before freeze:
  - freeze clear aperture, stroke, mass and four mounting datums
  - obtain EN 14752 evidence and emergency-release/cycle data
  - freeze keyed power, network and interlock interfaces

`OSR-COTS-GLASS-AGC` — AGC Automotive rail glazing with framing system

- Source class: `vehicle-specific-rfq`
- Local manufacture: Retain the pressure-frame and dry-seal replacement concept; qualify regional laminators against the released cassette drawing.
- Before freeze:
  - freeze laminate stack, optical zones and curvature
  - freeze heated end-glass voltage and power
  - obtain impact, fire, weather and bond-system evidence

`OSR-COTS-FINISH-3M-180MC` — 3M Controltac Graphic Film with Comply Adhesive 180mC family

- Source class: `catalogue-family-with-rail-application-guidance`
- Local manufacture: Plot, print, label and apply pre-cut one-metre-bay kits locally; accept another supplier only through complete-system equivalence rather than nominal film thickness.
- Before freeze:
  - select film colour/print, ink, overlaminate and edge system together
  - obtain project fire and hot-climate exposure evidence
  - run adhesion, wash, weather, repair and removal trials on the actual LM3 base finish

`OSR-COTS-COOLROOF-PURDUE-CACO3` — Purdue University research team (reference only) 2020 CaCO3-acrylic particle-matrix paint formulation

- Source class: `research-formulation-not-orderable`
- Local manufacture: Local coupon preparation is encouraged, but fleet application remains prohibited until an orderable controlled formulation and signed rail qualification pack exist.
- Before freeze:
  - identify a controlled commercial binder/pigment/additive formulation
  - obtain rail fire, chemical and application evidence
  - pass GFRP adhesion/flexibility plus UV, abrasion, wash and soiling ageing
  - measure new/aged optical performance and complete a one-car thermal/maintenance trial

`OSR-COTS-GLAZING-SIKA268` — Sika Sikaflex-268 PowerCure, 600 ml

- Source class: `exact-catalogue-product`
- Local manufacture: Use locally applied standard cartridges with supplier-approved pretreatment; retain a mechanical pressure frame so glass replacement is not adhesive-only.
- Before freeze:
  - Sika-approved substrate preparation for the actual glass/frame coatings
  - bondline calculation and physical coupon validation
  - lot and shelf-life control

OSR-COTS-HVAC-LONGERTEK — Longertek direct-DC rail HVAC platform

- Source class: `vehicle-specific-rfq`
- Local manufacture: Buy the first roof unit and controller; localise ducts, diffusers, drains, filters and later coils after calorimeter testing.
- Before freeze:
  - confirm 24 kW LM3 duty at worst ambient
  - freeze roof envelope, mass, coolant/refrigerant and drain interfaces
  - obtain EN 50155/50121 and fire evidence for quoted configuration

OSR-COTS-PV-SUNMAN — Sunman Energy eArc lightweight PV module

- Source class: `catalogue-family`
- Local manufacture: Use standard modules and make clamps, bonded carriers and replaceable roof harnesses locally.
- Before freeze:
  - freeze module size and electrical string
  - qualify roof retention, fire, isolation, impact and walk-zone separation
  - verify MPPT and safe isolation under all charging states

OSR-COTS-FLOOR-COMFLOOR — 3A Composites Mobility COMFLOOR rail floor panel

- Source class: `configurable-catalogue-system`
- Local manufacture: Import first panel stock if needed; machine panels, edge closeouts, hatch frames and inserts locally from released cut files.
- Before freeze:
  - freeze panel layup, inserts, hatch edges and fire level
  - prove concentrated, wheelchair and seat-rail loads
  - validate wet-zone sealing and repair method

OSR-COTS-FLOOR-FORBO-6R3883 — Forbo Flooring Systems Marmoleum Fresco FR2 6R3883 moonstone, 2.5 mm

- Source class: `exact-catalogue-product`
- Local manufacture: Use roll material initially; cut, cove, weld and repair locally with a controlled substrate/adhesive system.
- Before freeze:
  - obtain fire certificate for selected substrate/adhesive combination
  - freeze cove, weld-cord, nosing and hatch details
  - slip, cleaning and repair trials

OSR-COTS-SEAT-FAINSA — Fainsa metro and urban-rail seating family

- Source class: `vehicle-specific-rfq`
- Local manufacture: Make the common seat rail, grab poles and adapters locally; localise shells and frames only after fire and abuse-load qualification.
- Before freeze:
  - select exact shell, frame and upholstery
  - freeze common-rail attachment loads and PRM geometry
  - obtain fire, abuse-load, corrosion and maintainability evidence

`OSR-COTS-PIS-LUMINATOR` — Luminator Technology Group RailKom passenger-information and communication platform

- Source class: `configurable-catalogue-system`
- Local manufacture: Use local brackets, speaker grilles and wiring; permit display/amplifier substitution through open interfaces and regression tests.
- Before freeze:
  - freeze display, amplifier, speaker and software licences
  - require open Ethernet/IP data interfaces
  - obtain EMC, fire, accessibility and cybersecurity evidence

`OSR-COTS-CAMERA-MOXA-P06HC` — Moxa VPort P06HC-1MP-M12-CAM36

- Source class: `exact-catalogue-model`
- Local manufacture: Use the camera as a replaceable Ethernet appliance with locally made anti-vandal mount and harness.
- Before freeze:
  - confirm lifecycle availability and cybersecurity baseline
  - freeze viewing zones and privacy policy
  - qualify cable, storage and passenger-intercom integration

`OSR-COTS-LIGHT-TEKNOWARE` — Teknoware rolling-stock interior and emergency lighting platform

- Source class: `vehicle-specific-rfq`
- Local manufacture: Make the 1.2 m carrier, lens, brackets and harness locally around replaceable quoted LED modules.
- Before freeze:
  - freeze 24 V keyed cassette, optics and emergency duration
  - obtain rail fire, EMC, ingress and photometry evidence
  - separately select certified exterior head/tail lamps

`OSR-COTS-FASTENER-RIVNUT` — Boellhoff RIVNUT metric M6/M8 steel or stainless blind rivet nuts

- Source class: `catalogue-family-with-sizing`
- Local manufacture: Source from regional distributors or qualify metric equivalents by joint family; do not substitute solely by thread size.
- Before freeze:
  - select body, head, grip range and coating per joint drawing
  - run substrate-specific setting and proof-load trials
  - freeze setting-tool force and inspection gauge

`OSR-COTS-ANTENNA-HS-85137717` — HUBER+SUHNER SENCITY Rail MIMO 5-port with GNSS/LNA, product 85137717

- Source class: `exact-catalogue-product`
- Local manufacture: Buy antenna assemblies; manufacture roof doublers, cable conduits and replaceable RF jumpers locally.
- Before freeze:
  - confirm city radio bands and cable losses

- freeze roof ground/bond and lightning path
- perform installed antenna and EMC tests

`OSR-COTS-COMPOSITE-GURIT-130FR` — Gurit SE 130FR / ST 130FR prepreg with SA130FR and SF130FR films

- Source class: `exact-catalogue-family`
- Local manufacture: Use the catalogue prepreg as the first-article material; cut, vacuum-bag, cure, trim and repair locally with batch coupons.
- Before freeze:
  - select reinforcement/core/film stack per component
  - fire-test the actual sandwich laminate
  - run environmental, insert, impact and repair qualification

`OSR-COTS-MOTOR-ABB-AMXE250` — ABB AMXE250 configurable traction motor

- Source class: `configurable-catalogue-machine`
- Local manufacture: Buy first motors; local rewind and bearing service may follow an ABB-approved service procedure before attempting local manufacture.
- Before freeze:
  - quote a rail-duty 800 V winding and cooling configuration
  - freeze shaft, foot, mass and torque-speed map
  - obtain rail environmental, insulation, overspeed and endurance evidence

`OSR-COTS-GEAR-VOITH-SE` — Voith SE single-stage axle-mounted metro gear unit

- Source class: `vehicle-specific-rfq`
- Local manufacture: Buy first gear units; machine guards, torque link and mounting hardware locally.
- Before freeze:
  - freeze ratio, centre distance, torque and coupling
  - obtain bearing, lubrication, efficiency, noise, overspeed and endurance evidence
  - freeze reaction-link and wheelset interfaces

`OSR-COTS-CONVERTER-ABB-CC400` — ABB BORDLINE CC400 LRV compact converter

- Source class: `configurable-rail-product`
- Local manufacture: Buy first converters; localise cabinet, coolant manifold, harness and diagnostics integration behind frozen interfaces.
- Before freeze:
  - confirm operation from the selected battery/DC-link range
  - freeze motor, cooling, MVB/Ethernet and protection interfaces
  - quote exact mass, envelope, losses, software and rail evidence

`OSR-COTS-BATTERY-HOPPECKE-LITRAC` — HOPPECKE LiTrac modular high-voltage LFP rail traction battery

- Source class: `vehicle-specific-rfq`
- Local manufacture: Buy certified packs/BMS initially; make external trays, ducts, guards, coolant manifolds and service fixtures locally.
- Before freeze:

- confirm the frozen 225 kWh gross per-car target against the validated duty cycle or raise a controlled change request
- freeze voltage window, pack segmentation, mass and cooling
- obtain IEC 62928, propagation, fire, EMC and software evidence for the quote

`OSR-COTS-RESISTOR-DANOTHERM` — Danotherm Electric custom traction brake resistor, 600-3000 VDC family

- Source class: `vehicle-specific-rfq`
- Local manufacture: Buy the element pack and insulators; manufacture roof carrier, shield, duct and service guard locally after the hot-zone model is frozen.
- Before freeze:
  - derive resistance, continuous/pulse energy and duty cycle from simulation
  - freeze enclosure, airflow, thermal switch and touch guards
  - run worst-case stopped-train thermal test

`OSR-COTS-CHARGER-SCHUNK` — Schunk Carbon Technology Smart Charging automated high-current collector platform

- Source class: `interface-rfq-baseline`
- Local manufacture: Prefer a proven automated collector geometry if it reduces bespoke side-pin risk; make station frame, guards and alignment target locally.
- Before freeze:
  - compare roof/bottom collector against the LM3 side-pin concept
  - freeze current, voltage, touch safety, alignment and emergency release
  - qualify stationary and vehicle halves together

`OSR-COTS-HV-SCHALTBAU-C360` — Schaltbau C360K/300 railway bidirectional DC contactor

- Source class: `exact-catalogue-model`
- Local manufacture: Buy contactors and qualified fuses; fabricate busbars, enclosure, precharge assembly and service-disconnect panel locally.
- Before freeze:
  - confirm breaking duty and prospective fault current
  - select coil, auxiliary contacts and mounting orientation
  - coordinate fuse, busbar, precharge and enclosure clearances

`OSR-COTS-HV-TE-HVP800` — TE Connectivity AMP+ HVP800 cable assembly 1-2286937-1

- Source class: `exact-catalogue-part`
- Local manufacture: Buy terminals/connectors; cut and terminate qualified rail cable locally only with controlled tooling and 100 percent electrical test.
- Before freeze:
  - confirm current derating, fire and railway vibration for installed route
  - freeze coding and mating header
  - perform bend, ingress, HVIL and articulation-cycle tests

`OSR-COTS-FIRE-LIIONTAMER-3` — Xtralis / Li-ion Tamer Li-ion Tamer GEN 3

- Source class: `exact-catalogue-system`

- Local manufacture: Buy sensing nodes/controller; make brackets, protected Ethernet/relay harness and test-gas service access locally.
- Before freeze:
  - verify sensor response in selected battery airflow and contaminants
  - define safe-state action and false-alarm handling
  - do not treat detection as suppression or a standalone safety system

`OSR-COTS-FIRE-FOGTEC` — FOGTEC Rail Systems rail high-pressure water-mist system

- Source class: `vehicle-specific-rfq`
- Local manufacture: Buy validated nozzles/actuation/controller; localise approved pipework, brackets and refill/service tooling.
- Before freeze:
  - perform LM3 battery and passenger fire-hazard analysis
  - freeze nozzles, reservoir, gas/water cylinders, detection and drains
  - conduct representative full-scale fire and discharge tests

`OSR-COTS-WHEELSET-GHH` — GHH-BONATRANS urban-rail wheelset with axle-mounted disc interfaces

- Source class: `vehicle-specific-rfq`
- Local manufacture: Buy certified wheelsets initially; local wheel/axle manufacture requires a separately qualified metallurgical and NDT route.
- Before freeze:
  - freeze wheel profile/diameter, axle journals and disc seats
  - issue load spectrum and duty
  - obtain material, forging, heat-treatment, NDT, press-fit and fatigue evidence

`OSR-COTS-BEARING-SKF-TBU` — SKF sensor-ready railway tapered roller bearing unit

- Source class: `vehicle-specific-rfq`
- Local manufacture: Buy bearing units; machine housings, sensor guards and cable brackets locally to the supplier-controlled fits.
- Before freeze:
  - calculate bearing life from journal, load spectrum and speed
  - freeze exact designation, fit, seal, grease and grounding
  - freeze speed/temperature sensor and cable interface

`OSR-COTS-SUSPENSION-CONTI` — Continental rail half-bellow air spring with integrated auxiliary spring

- Source class: `vehicle-specific-rfq`
- Local manufacture: Buy elastomer/spring/damper elements; fabricate spring seats, guides, centre pivot, yaw links and guards locally after calculation.
- Before freeze:
  - freeze load-deflection/damping curves from the vehicle dynamics model
  - select primary springs, air springs, dampers and bump stops as one system
  - obtain ageing, fatigue, fire and ride evidence

`OSR-COTS-BRAKE-KNORR-RZKT` — Knorr-Bremse WheelAct Compact RZKT with spring parking-brake option

- Source class: `configurable-catalogue-model`
- Local manufacture: Buy first calipers/actuators; localise discs, hoses, brackets and overhaul parts only under brake-system approval.
- Before freeze:
  - confirm disc diameter/pad pair and required force from brake calculation
  - freeze actuator, parking, WSP and degraded-mode architecture
  - perform stopping, thermal, adhesion and endurance tests

`OSR-COTS-AIR-KNORR-VV120T` — Knorr-Bremse VV120-T 2.0 oil-free rail compressor / AirSupply Smart

- Source class: `configurable-rail-product`
- Local manufacture: Buy compressor/dryer; fabricate reservoir, manifold, isolation, drains, mounts and acoustic enclosure locally to approved pressure codes.
- Before freeze:
  - size from suspension, brake and auxiliary air demand
  - freeze dryer, reservoir, relief, noise and mounting package
  - run duty, thermal, condensate and degraded-mode tests

`OSR-COTS-COUPLER-DELLNER430` — Dellner automatic coupler Type 430 with pivot anchor

- Source class: `configurable-catalogue-model`
- Local manufacture: Buy coupler/head/absorber; make pocket, covers, brackets and harness locally to Dellner drawings.
- Before freeze:
  - freeze height, gathering range, fold, crash stroke and pocket loads
  - freeze electrical head and rescue jumper pinout
  - integrate into whole-vehicle crash-energy management

`OSR-COTS-JOINT-SCHAEFFLER` — Schaeffler ELGES maintenance-free spherical plain bearing and rod-end family

- Source class: `catalogue-family-with-calculation`
- Local manufacture: Buy bearing liners/rod ends; machine housings, pins, links and retained fasteners locally from released drawings.
- Before freeze:
  - derive static/dynamic load, pin diameter, misalignment and life
  - freeze liner, seal, corrosion and temperature variant
  - proof and fatigue-test the complete local housing/pin/link joint

`OSR-COTS-GANGWAY-HUBNER` — HUEBNER modular metro double-wall corrugated gangway

- Source class: `vehicle-specific-rfq`
- Local manufacture: Buy first bellows/bridge system; fabricate clamp frames, threshold, drains and service panels locally to released interfaces.
- Before freeze:

- issue full articulation sweep and end-frame geometry
- freeze bellows, bridge, turntable, clamp and drainage
- perform fire, fatigue, water, passenger-load and pinch-gap tests

`OSR-COTS-TRAINLINE-IGUS-HARTING` — igus / HARTING rail e-chain carrier with Han HPR connector system

- Source class: `catalogue-family-with-sizing`
- Local manufacture: Buy carriers/connectors; cut rail cable and hose, make support arms, drains and protective sleeves locally with controlled termination.
- Before freeze:
  - select carrier from bend radius, fill, travel and torsion
  - freeze Han HPR inserts, housings, coding and breakaway
  - cycle-test complete HV/LV/coolant bundle through the articulation envelope

`OSR-COTS-CONTROL-DUAGON-MH50C` — duagon D-SC Controller with MH50C safe CPU and distributed safe I/O

- Source class: `certified-configurable-platform`
- Local manufacture: Use duagon for the independent qualified channel; keep OSR application compute separate and portable. Build cabinet/harness locally.
- Before freeze:
  - allocate each LM3 safety function and required SIL
  - freeze CPU/I/O/BSP/application configuration
  - independently verify OSR integration and safe-state timing

`OSR-COTS-COMPUTE-MOXA-V2406C` — Moxa V2406C-WL1-T railway computer

- Source class: `exact-catalogue-model`
- Local manufacture: Buy fanless computers; build the cabinet, isolated supply, M12 train network and replaceable storage locally.
- Before freeze:
  - not a safety controller
  - freeze memory/storage and supported Linux image
  - cybersecurity hardening, lifecycle and application-load tests

`OSR-COTS-ROUTER-WESTERMO-630-5G` — Westermo Ibex-RT-630-5G

- Source class: `exact-catalogue-model`
- Local manufacture: Buy router; integrate local SIM service, cybersecurity configuration, roof cables and cabinet harness.
- Before freeze:
  - confirm local 5G bands/SIMs and cybersecurity lifecycle
  - freeze antenna/cable topology
  - prove handover, loss-of-comms and update behaviour

`OSR-COTS-SAFETYRELAY-PILZ-751184` — Pilz PNOZ s4, railway order 751184

- Source class: `exact-catalogue-order`
- Local manufacture: Buy relays; fabricate labelled DIN cabinet, E-stop/pendant harness and replaceable terminal blocks locally.
- Before freeze:

- use only for an allocated compatible function
- observe supplier accessibility restriction for railway use
- verify contact load, diagnostics and safe-state integration

`OSR-COTS-RECORDER-HASLER` — HaslerRail JRU/EVA event-recorder platform

- Source class: `vehicle-specific-rfq`
- Local manufacture: Buy protected recorder; use local commodity storage only for non-protected operational logs.
- Before freeze:
  - freeze jurisdictional data set, retention and clock
  - select protected memory and download interfaces
  - obtain crash, fire, environmental and chain-of-custody evidence

`OSR-COTS-LIDAR-OUSTER-OS1` — Ouster OS1 industrial lidar sensor

- Source class: `catalogue-sensor`
- Local manufacture: Buy sensors; manufacture datum carrier, sacrificial heated window, washer, guard and service alignment fixture locally.
- Before freeze:
  - not a certified railway obstacle-detection function
  - freeze field of view, range, latency, heater/washer and protective window
  - validate sensing, diagnostics and failure response over route conditions

Controlled inputs

- Candidate source: `lib/templates/trainset-cots-candidates.toml` (`0588d76029c2f426beadc09a5ba096848d2eb85a22fb4e1306c54d68931f3b94`)
- Product manifest: `design/component-catalogue/catalog/buildable-trainset/buildable-trainset-manifest.json` (`ca385a6d3c27ee343d7f4bc4c90b4f4e27058380ff82a0aa76ca573275149703`)

LM3 fabrication and final-assembly critical path

Repository source: `design/component-catalogue/catalog/buildable-trainset/critical-path.md`

Generated by `tools/automation/buildable-trainset.sh`. This is a rough-order first-train manufacturing plan for parts, subassemblies, final train assembly, internal furnishings, and commissioning. It is intended for layout and takt planning; it is not a released shop baseline.

- Family: `light-metro-3car`
- Candidate: `light-metro-3car__16p5m__reference-body__reference-bogie__motor-350kw-hm47-class__battery-225kwh-lfp-800v__hvac-24kw-direct-hv-dc__pv12`
- Planning basis: one first-article three-car LM3 trainset after design release/material availability; one long final assembly track; off-line steel, composite, bogie, and interior benches run in parallel
- Rough elapsed duration: 35.0 working days
- Rough direct labour: 5,524 h
- Critical path: `CP-010` -> `CP-020` -> `CP-030` -> `CP-040` -> `CP-070` -> `CP-080` -> `CP-090` -> `CP-110` -> `CP-130` -> `CP-140` -> `CP-150`

Critical-path table

ID	Task	Level	Work center	Pred	Duration d	Labour h	Crew eq	ES	EF	Float	Critical
CP-010	release traveler pack, work orders, QA gates, and material lots	program release	project controls / production planning	-	2.0	72	4.5	0.0	2.0	0.0	yes
CP-020	cut, form, drill, and kit structural steel chassis/body parts	parts fabrication	steel prep cell	CP-010	3.0	288	12.0	2.0	5.0	0.0	yes
CP-030	weld three underframe datum weldments	subassembly	weld and fixture cell	CP-020	4.0	384	12.0	5.0	9.0	0.0	yes
CP-040	add side/roof spaceframes, floors, portals, roof rails, and chassis interfaces	subassembly	weld and fixture cell	CP-030	4.0	420	13.1	9.0	13.0	0.0	yes
CP-050	assemble powered/trailer bogies, wheelsets, suspension, brake, and bogie harness kits	subassembly	bogie assembly cell	CP-020	8.0	520	8.1	5.0	13.0	7.0	
CP-060	mould, cure, demould, trim, drill, and kit 144 GFRP exterior modules	parts fabrication	composite moulding and trim cell	CP-010	5.0	420	10.5	2.0	7.0	8.0	
CP-065	prebuild internal furnishings and cabin trim kits	parts fabrication	interior bench / supplier receiving	CP-010	5.0	360	9.0	2.0	7.0	12.0	
CP-070	blast, prime, topcoat, cavity-wax, and release painted carbody frames	subassembly	paint and corrosion cell	CP-040	2.0	192	12.0	13.0	15.0	0.0	yes
CP-080	install clip-on GFRP side/roof modules and complete water/rattle pre-test	assembly	clip-on body cell	CP-060, CP-070	1.0	96	12.0	15.0	16.0	0.0	yes
CP-090	install doors, windows, roof HVAC/PV hardware, and exterior service details	assembly	final assembly track with glazing/roof access	CP-080	3.0	360	15.0	16.0	19.0	0.0	yes
CP-100	install HV battery, traction, LV/control, safety loops, coolant, and fire interfaces	assembly	final assembly electrical/HV cell	CP-080	4.0	480	15.0	16.0	20.0	5.0	
CP-110	install internal furnishings, flooring, liners, seats, grab rails, PIS, CCTV, lighting, and signage	assembly	final assembly interior fit-out cell	CP-065, CP-090	3.0	420	17.5	19.0	22.0	0.0	yes
CP-120	marry bogies to carbodies and close brake/ride-height static checks	assembly	final assembly bogie marriage station	CP-050, CP-080	2.0	240	15.0	16.0	18.0	4.0	

ID	Task	Level	Work center	Pred	Duration d	Labour h	Crew eq	ES	EF	Float	Critical
CP-130	install articulation, gangways, end cowls, couplers, sensor packs, and trainlines	final trainset assembly	final assembly track	CP-110, CP-120	3.0	360	15.0	22.0	25.0	0.0	yes
CP-140	static commissioning, insulation, door/HVAC/brake tests, snag closeout, and QA release	commissioning	final assembly / static test cell	CP-100, CP-130	4.0	480	15.0	25.0	29.0	0.0	yes
CP-150	dynamic commissioning and trial-running release	commissioning	test track / depot	CP-140	6.0	432	9.0	29.0	35.0	0.0	yes

Space and parallelism

Area	Minimum planning allowance	Why it is enough
Long final assembly track	1 x 55 m bay	Doors, roof systems, HV, interior, bogie marriage, articulation, and static test share one controlled train-length space.
Underframe fixtures	2 fixtures	Two fixtures let three underframes move through without a third full-length weld bay.
Side/roof frame fixtures	2 fixtures	Keeps side/roof spaceframe work ahead of paint without holding the final track.
Paint bay	1 car-length bay	Frames enter only after dimensional acceptance, then leave for clip-on body installation.
Bogie stands	3 stands	Bogies are assembled off-line while carbody welding is on the critical path.
Short GFRP moulds	4 moulds	Repeated 1 m modules are produced without a full-car mould or final-bay cure hold.
Interior trim benches	1 bench set	Furnishings are pre-kitted by car/door zone, then installed after leak-sensitive work closes.

Key minimisation rule: keep the 55 m final assembly track for accepted subassemblies only. Steel parts, GFRP modules, bogies, HVAC/PV kits, doors/windows, cabin liners, seats, grab rails, lighting, PIS, CCTV, flooring, signage, and labels are all prepared off-line and enter the train only as released kits.

Labour by work center

Work center	Labour h
bogie assembly cell	520
clip-on body cell	96
composite moulding and trim cell	420
final assembly / static test cell	480
final assembly bogie marriage station	240
final assembly electrical/HV cell	480
final assembly interior fit-out cell	420
final assembly track	360
final assembly track with glazing/roof access	360
interior bench / supplier receiving	360
paint and corrosion cell	192
project controls / production planning	72
steel prep cell	288
test track / depot	432
weld and fixture cell	804

Float use

Composite module fabrication, bogie assembly, HV install, and interior pre-kitting have float in this rough network. Use that float to absorb first-article rework without expanding the final assembly track. Do not spend the float by letting incomplete kits enter the long bay; that trades cheap bench work for expensive train-length blockage.

Current basic design buildability review

Repository source: `design/component-catalogue/catalog/buildable-trainset/current-design-buildability-review.md`

This review uses the generated design-system optimum and the current rolling-stock CAD/BOM/fabrication package to identify what is already buildable, what is only an envelope, and what must be closed before first steel cut.

Summary

Status	Meaning
green	current design is good enough to carry into v2A build package
yellow	concept is sound but supplier/CAD/BOM alignment must be closed
red	missing build-release evidence or shop-drawing detail

Findings

ID	Status	Scope	Finding	Action
BDR-001	green	architecture	The current design already uses repeated self-contained car modules with one powered and one trailer bogie per car.	Keep repeated-car architecture as the buildable baseline; it simplifies fixtures, spares, and training.
BDR-002	green	COTS/fabricated delineation	The procurement BOM now carries a controlled many-to-many crosswalk from all 100 commercial lines to LM3 product or assembly IDs; definitions and travelers carry the reverse references.	Keep the generated crosswalk mandatory as component kits are split and supplier routes are frozen.
BDR-003	green	candidate/CAD alignment	The promoted FreeCAD/product-tree baseline matches the optimizer target: 16.5 m cars, motor-350kw-hm47-class, battery-225kwh-lfp-800v, hvac-24kw-direct-hv-dc, and 12 PV modules per car.	Treat this as the v2A buildable seed: 250 kW continuous / 350 kW peak motors, 180 kWh usable / 225 kWh gross batteries per car, 24 kW/car HVAC, and supplier-freeze envelopes before drawings.
BDR-004	yellow	supplier freeze	All 58 BID/SOURCE rows now have a generated affordable reference default tied to 41 controlled manufacturer/reference sources and the current CAD envelopes. Buildability still depends on supplier-frozen doors, HVAC, batteries, traction, wheelsets, brakes, couplers, and gangways.	Use the reference-default register to issue comparable RFQs, accept alternates only through fit/power/mass/evidence checks, then lock v2A supplier interfaces.
BDR-005	yellow	definition package / shop travelers / shop drawings	Every product-tree node now has a generated definition and signable shop-traveler template with structured material specs, process specs, labor, tooling, QA gates, revision approvals, and signoff blocks; controlled cut lists, weld maps, tolerance stacks, flat patterns, and harness/plumbing drawings are still v2A drawing-package work.	Use the generated material/process definitions and travelers as the drawing/RFQ/traveler index, then promote each MAKE/BID/SOURCE node into controlled LM3-BDY/BOG/HV/ELC drawings before first steel cut.
BDR-006	yellow	proof evidence	FEA screening exists and the previously coarse local bracket, coupler-pocket, battery-tray, door-portal, roof-equipment, and bogie-frame items now have generated definitions and assembly integration steps; local proof cases are still not complete.	Attach proof load cases to the generated component definitions and make FEM/static-test acceptance a release gate for each affected subassembly.
BDR-007	yellow	mass properties	The generated mass budget reconciles the 75.308 t optimizer subtotal with the 78.75 t controlled planning tare through an explicit 3.442 t engineering reserve. A 120-row responsibility ledger now identifies the evidence route for every product and quantifies the lightest existing design-space option without claiming achieved mass; drawing-level and as-built closure remains open.	Close each active ledger row with supplier-frozen, production-CAD and/or calibrated weighed values, reconcile car/axle/centre-of-gravity results, and repeat affected substantiation before changing the controlled tare.
BDR-008	yellow	joint and fastener control	Every assembly integration step now carries machine-readable join classes and torque authority. Interior small parts have been rationalised into four fastener families and one common datum rail instead of treating every attachment as a bespoke structural joint; numerical values remain open until the applicable standard, supplier instruction, or load calculation is released.	Close the generated joint-control schedule by joint ID, qualify the common rail/fastener samples, and reference accepted values from interface drawings and shop travelers.
BDR-009	green	small components and serviceability	Door and glazing interfaces, passenger-fixture adapters, lighting cassettes, captive fasteners, seals, drain rails, keyed plugs, and emergency illumination are now explicit CAD/product-tree items rather than an opaque interior-kit allowance.	Keep the simplified interface families fixed while suppliers freeze detailed door, glass, luminaire, connector, and fastener parts; release safety-critical loads and tests through their stated gates.

Immediate build-package work

1. Treat the generated definition and shop-traveler packs as the product-tree index for parts, external components, subassemblies, assemblies, and trainsets.
2. Execute the generated factory drawing/interface work packages for chassis, modules, fascia, roof, interiors, bogies, trainlines, finishes and recovery; approve their frozen inputs, controlled outputs and verification ownership.
3. Convert every remaining `MAKE` definition into controlled drawings: cut list, flat pattern, weld class, datum scheme, tolerance, and inspection method.
4. Convert each `BID/SOURCE` definition into an RFQ envelope: mass, power, volume, mounting datum, service clearance, evidence pack, and alternate acceptance rule.
5. Close every active product in the generated mass-closure ledger, reconcile its nine mass-budget categories, and close joint-control rows as supplier, calculation, production-CAD, and calibrated weighing evidence becomes available.
6. Attach local proof cases to the structural subassemblies: underframe, bolsters, coupler pocket, door portals, battery tray, roof equipment, bogie frames, and articulation adapters.
7. Fill traveler approval/signoff blocks only during a real build; do not pre-sign generated templates.
8. Regenerate FreeCAD/FEM only after the promoted candidate parameters are reflected in the parametric source.

LM3 bought-in product reference defaults

Repository source: `design/component-catalogue/catalog/buildable-trainset/default-product-specifications.md`

Provide a concrete, affordable starting configuration for every bought-in LM3 product row while preserving supplier competition and override gates.

- Status: `concept-and-rfq-defaults-not-procurement-or-engineering-release`
- Defaults: 58 (BID 34; SOURCE 24)
- Manufacturer/reference sources: 41
- Candidate sources checked: 2026-08-31

A default is the affordable first configuration to draw, cost and request in an RFQ. It is not a nominated supplier, order code, quotation, safety approval or permission to manufacture. CAD envelopes are fit constraints, not claimed supplier dimensions.

Default index

Product	Reference default	Use class	CAD envelope X × Y × Z (mm)	Sources
LM3-CWL-P010 end-cowl fiberglass laminate, insert, adhesive, and coupon material kit	Gurit — SE 130FR / ST 130FR prepreg with SA130FR and SF130FR films	sample-ready-reference-default-not-selected	2900 × 2800 × 3050	OSR-COTS-COMPOSITE-GURIT-130FR
LM3-FAS-P030 front glazing/lamp EPDM seal, drain rail, washer sleeve, and edge-closeout kit	AGC Automotive — rail glazing with framing system	rfq-baseline-reference-default-not-selected	2500 × 420 × 1950	OSR-COTS-GLASS-AGC OSR-COTS-GLAZING-SIKA268
LM3-EXT-P010 electric plug/sliding door cassette	Knorr-Bremse IFE — IFE modular sliding-plug entrance system	rfq-baseline-reference-default-not-selected	1350 × 260 × 2050	OSR-COTS-DOOR-IFE
LM3-EXT-P020 side laminated glazing cassette	AGC Automotive — rail glazing with framing system	rfq-baseline-reference-default-not-selected	1520 × 90 × 1100	OSR-COTS-GLASS-AGC OSR-COTS-GLAZING-SIKA268
LM3-EXT-P030 single panoramic heated end-glass assembly	AGC Automotive — rail glazing with framing system	rfq-baseline-reference-default-not-selected	2300 × 110 × 1450	OSR-COTS-GLASS-AGC OSR-COTS-GLAZING-SIKA268
LM3-EXT-P040 hvac-24kw-direct-hv-dc roof HVAC	Longertek — direct-DC rail HVAC platform	rfq-baseline-reference-default-not-selected	2600 × 1900 × 620	OSR-COTS-HVAC-LONGERTEK
LM3-EXT-P050 roof PV module and edge-clamp kit	Sunman Energy — eArc lightweight PV module	sample-ready-reference-default-not-selected	1700 × 1050 × 45	OSR-COTS-PV-SUNMAN
LM3-EXT-P060 stepped floor-board and removable service-hatch system	3A Composites Mobility — COMFLOOR rail floor panel	rfq-baseline-reference-default-not-selected	1000 × 1300 × 35	OSR-COTS-FLOOR-COMFLOOR
LM3-EXT-P061 welded resilient floor covering, cove, nosing, and adhesive system	Forbo Flooring Systems — Marmoleum Fresco FR2 6R3883 moonstone, 2.5 mm	sample-ready-reference-default-not-selected	1000 × 1300 × 6	OSR-COTS-FLOOR-FORBO-6R3883
LM3-EXT-P062 longitudinal passenger and priority-seat modules	Fainsa — metro and urban-rail seating family	rfq-baseline-reference-default-not-selected	1550 × 620 × 980	OSR-COTS-SEAT-FAINSA
LM3-EXT-P063 stainless grab-pole, handrail, joint, and insulated adapter kit	Fainsa — metro and urban-rail seating family	rfq-baseline-reference-default-not-selected	2300 × 2300 × 2050	OSR-COTS-SEAT-FAINSA
LM3-EXT-P064 passenger-information display, speaker, amplifier, and mounting kit	Luminator Technology Group — RailKom passenger-information and communication platform	rfq-baseline-reference-default-not-selected	1100 × 220 × 420	OSR-COTS-PIS-LUMINATOR
LM3-EXT-P065 CCTV camera, passenger intercom, PoE/data, and mounting kit	Moxa — VPort P06HC-1MP-M12-CAM36	sample-ready-reference-default-not-selected	650 × 420 × 260	OSR-COTS-CAMERA-MOXA-P06HC
LM3-EXT-P066 PRM, safety-signage, emergency-lighting, extinguisher, and first-aid kit	Teknoware — rolling-stock interior and emergency lighting platform	rfq-baseline-reference-default-not-selected	1200 × 650 × 600	OSR-COTS-LIGHT-TEKNOWARE
LM3-FIX-P020 four-family captive fastener, floating nut, isolator, and access-fastener kit	Boellhoff — RIVNUT metric M6/M8 steel or stainless blind rivet nuts	sample-ready-reference-default-not-selected	520 × 360 × 160	OSR-COTS-FASTENER-RIVNUT
LM3-LGT-P010 1.2 m plug-in main LED lighting cassette and captive mounting kit	Teknoware — rolling-stock interior and emergency lighting platform	rfq-baseline-reference-default-not-selected	1200 × 95 × 45	OSR-COTS-LIGHT-TEKNOWARE
LM3-LGT-P020 emergency and doorway lighting modules with independent keyed feeder kit	Teknoware — rolling-stock interior and emergency lighting platform	rfq-baseline-reference-default-not-selected	900 × 360 × 120	OSR-COTS-LIGHT-TEKNOWARE
LM3-EXT-P070 roof antennas, service walkway pads, lifting covers, and maintenance labels	HUBER+SUHNER — SENCITY Rail MIMO 5-port with GNSS/LNA, product 85137717	orderable-candidate-reference-default-not-selected	2400 × 1500 × 260	OSR-COTS-ANTENNA-HS-85137717
LM3-EXT-P080 fire-rated GFRP side-module laminate, core, gelcoat, and consumable kit	Gurit — SE 130FR / ST 130FR prepreg with SA130FR and SF130FR films	sample-ready-reference-default-not-selected	1000 × 120 × 3050	OSR-COTS-COMPOSITE-GURIT-130FR
LM3-EXT-P090 fire-rated GFRP roof-module, dry-seal, and removable skirt material kit	Gurit — SE 130FR / ST 130FR prepreg with SA130FR and SF130FR films	sample-ready-reference-default-not-selected	1000 × 2700 × 180	OSR-COTS-COMPOSITE-GURIT-130FR
LM3-FIN-P010 pre-cut exterior livery graphic film, edge-seal, datum-mark, and repair-patch kit	3M — Controltac Graphic Film with Comply Adhesive 180mC family	sample-and-system-rfq-reference-default-not-selected	16000 × 1400 × 8	OSR-COTS-FINISH-3M-180MC
LM3-FIN-P020 calcium-carbonate radiative roof-coating qualification and exposed-roof application kit	qualified light-colour base roof finish; CaCO3 coating restricted to coupon and one-car trial	trial-only-reference-not-an-orderable-fleet-product	16000 × 2700 × 8	OSR-COTS-COOLROOF-PURDUE-CACO3
LM3-INT-P010 HVAC diffusers, side return ducts, saloon grilles, and access panels	Longertek — direct-DC rail HVAC platform	rfq-baseline-reference-default-not-selected	6000 × 2400 × 320	OSR-COTS-HVAC-LONGERTEK
LM3-TRC-P010 motor-350kw-hm47-class axle traction motor	ABB — AMXE250 configurable traction motor	rfq-baseline-reference-default-not-selected	1050 × 720 × 720	OSR-COTS-MOTOR-ABB-AMXE250
LM3-TRC-P020 single-stage reduction gearbox and flexible coupling	Voith — SE single-stage axle-mounted metro gear unit	rfq-baseline-reference-default-not-selected	760 × 620 × 620	OSR-COTS-GEAR-VOITH-SE
LM3-TRC-P030 two independent motor controllers, isolated LV DC/DC, MPPT, station protection, and cooling-loop kit	ABB — BORDLINE CC400 LRV compact converter	rfq-baseline-reference-default-not-selected	2200 × 1200 × 620	OSR-COTS-CONVERTER-ABB-CC400
LM3-TRC-P040 battery-225kwh-lfp-800v saloon-isolated side traction battery pack	HOPPECKE — LiTrac modular high-voltage LFP rail traction battery	rfq-baseline-reference-default-not-selected	5200 × 780 × 720	OSR-COTS-BATTERY-HOPPECKE-LITRAC

Product	Reference default	Use class	CAD envelope X × Y × Z (mm)	Sources
LM3-TRC-P050 roof-mounted regen dump resistor and thermal shield kit	Danotherm Electric — custom traction brake resistor, 600-3000 VDC family	rfq-baseline-reference-default-not-selected	1900 × 750 × 420	OSR-COTS-RESISTOR-DANOTHERM
LM3-TRC-P060 station side-pin charging connector, actuator, shutter, and alignment target	supplier-neutral 650-700 VDC side-contact RFQ; automated collector retained only as a concept alternative	interface-rfq-default-not-selected	1050 × 620 × 540	OSR-COTS-CHARGER-SCHUNK
LM3-TRC-P070 HV contactor, fuse, pre-charge, service-disconnect, and current-sensor panel	Schaltbau — C360K/300 railway bidirectional DC contactor	sample-ready-reference-default-not-selected	1500 × 420 × 1050	OSR-COTS-HV-SCHALTBAU-C360 OSR-COTS-HV-TE-HVP800
LM3-SAF-P010 battery temperature/off-gas detection, electrical-enclosure smoke detection, and localized mist kit	Xtralis / Li-ion Tamer — Li-ion Tamer GEN 3	sample-ready-reference-default-not-selected	1600 × 900 × 520	OSR-COTS-FIRE-FOGTEC OSR-COTS-FIRE-LIIONTAMER-3
LM3-BOG-P040 powered-bogie wheelset with axle-mounted brake discs	GHH-BONATRANS — urban-rail wheelset with axle-mounted disc interfaces	rfq-baseline-reference-default-not-selected	1850 × 2200 × 920	OSR-COTS-WHEELSET-GHH
LM3-BOG-P041 trailer-bogie wheelset with axle-mounted brake discs	GHH-BONATRANS — urban-rail wheelset with axle-mounted disc interfaces	rfq-baseline-reference-default-not-selected	1850 × 2200 × 920	OSR-COTS-WHEELSET-GHH
LM3-BOG-P042 powered-wheelset axlebox, sealed bearing unit, speed and temperature sensor set	SKF — sensor-ready railway tapered roller bearing unit	calculation-then-rfq-reference-default-not-selected	820 × 620 × 620	OSR-COTS-BEARING-SKF-TBU
LM3-BOG-P043 trailer-wheelset axlebox, sealed bearing unit, speed and temperature sensor set	SKF — sensor-ready railway tapered roller bearing unit	calculation-then-rfq-reference-default-not-selected	820 × 620 × 620	OSR-COTS-BEARING-SKF-TBU
LM3-BOG-P044 powered-bogie primary suspension spring, guide and bump-stop set	Continental — rail half-bellow air spring with integrated auxiliary spring	dynamics-then-rfq-reference-default-not-selected	1800 × 2100 × 520	OSR-COTS-SUSPENSION-CONTI
LM3-BOG-P045 trailer-bogie primary suspension spring, guide and bump-stop set	Continental — rail half-bellow air spring with integrated auxiliary spring	dynamics-then-rfq-reference-default-not-selected	1800 × 2100 × 520	OSR-COTS-SUSPENSION-CONTI
LM3-BOG-P046 powered-bogie to carbody connection: air springs, emergency spring, centre pivot, yaw links and dampers	Continental — rail half-bellow air spring with integrated auxiliary spring	dynamics-then-rfq-reference-default-not-selected	1900 × 2200 × 650	OSR-COTS-SUSPENSION-CONTI
LM3-BOG-P047 trailer-bogie to carbody connection: air springs, emergency spring, centre pivot, yaw links and dampers	Continental — rail half-bellow air spring with integrated auxiliary spring	dynamics-then-rfq-reference-default-not-selected	1900 × 2200 × 650	OSR-COTS-SUSPENSION-CONTI
LM3-BOG-P048 powered-bogie brake calipers, parking actuators, pads and wheel-slide hardware	Knorr-Bremse — WheelAct Compact RZKT with spring parking-brake option	brake-calculation-then-rfq-reference-default-not-selected	2100 × 2200 × 520	OSR-COTS-BRAKE-KNORR-RZKT
LM3-BOG-P049 trailer-bogie brake calipers, parking actuators, pads and wheel-slide hardware	Knorr-Bremse — WheelAct Compact RZKT with spring parking-brake option	brake-calculation-then-rfq-reference-default-not-selected	2100 × 2200 × 520	OSR-COTS-BRAKE-KNORR-RZKT
LM3-AUX-P010 secondary-suspension compressor, dryer, reservoir, and isolation-manifold kit	Knorr-Bremse — VV120-T 2.0 oil-free rail compressor / AirSupply Smart	air-demand-then-rfq-reference-default-not-selected	1300 × 850 × 720	OSR-COTS-AIR-KNORR-VV120T
LM3-END-P010 automatic end coupler and crash-energy absorber	Dellner — automatic coupler Type 430 with pivot anchor	rfq-baseline-reference-default-not-selected	1850 × 760 × 620	OSR-COTS-COUPLER-DELLNER430
LM3-ART-P020 articulation lower spherical pivot, bearing housing and pin set	Schaeffler — ELGES maintenance-free spherical plain bearing and rod-end family	calculation-then-order-code-reference-default-not-selected	780 × 780 × 520	OSR-COTS-JOINT-SCHAEFFLER
LM3-ART-P021 articulation upper lateral/yaw links, spherical joints and retained pins	Schaeffler — ELGES maintenance-free spherical plain bearing and rod-end family	calculation-then-order-code-reference-default-not-selected	1200 × 1500 × 420	OSR-COTS-JOINT-SCHAEFFLER
LM3-ART-P022 inter-car double-wall corrugated bellows and clamp-frame set	HUEBNER — modular metro double-wall corrugated gangway	kinematic-envelope-then-rfq-reference-default-not-selected	1100 × 2100 × 2300	OSR-COTS-GANGWAY-HUBNER
LM3-ART-P023 inter-car passenger bridge, turntable and flexible interior-panel set	HUEBNER — modular metro double-wall corrugated gangway	kinematic-envelope-then-rfq-reference-default-not-selected	1250 × 1850 × 260	OSR-COTS-GANGWAY-HUBNER
LM3-ART-P024 articulation trainline carrier, support arms, abrasion liners and drain path	igus / HARTING — rail e-chain carrier with Han HPR connector system	sample-and-cycle-test-reference-default-not-selected	1300 × 1600 × 320	OSR-COTS-TRAINLINE-IGUS-HARTING
LM3-CTRL-P010 T-ECU/S and T-ECU/A compute and safety-control cabinet kit	duagon — D-SC Controller with MH50C safe CPU and distributed safe I/O	safety-allocation-then-rfq-reference-default-not-selected	1200 × 550 × 1350	OSR-COTS-COMPUTE-MOXA-V2406C OSR-COTS-CONTROL-DUAGON-MH50C
LM3-CTRL-P020 navigation, balise, 5G, LoRa, GNSS, IMU, and roof-antenna kit	Westermo — Ibex-RT-630-5G	orderable-candidate-reference-default-not-selected	1500 × 1100 × 420	OSR-COTS-ANTENNA-HS-85137717 OSR-COTS-ROUTER-WESTERMO-630-5G
LM3-CTRL-P030 maintenance HMI, depot pendant, emergency controls, and safety-relay kit	Pilz — PNOZ s4, railway order 751184	bench-safety-relay-candidate-reference-default-not-selected	1100 × 550 × 650	OSR-COTS-SAFETYRELAY-PILZ-751184
LM3-CTRL-P050 operational and crashworthy event-recorder storage kit	HaslerRail — JRU/EVA event-recorder platform	requirements-then-rfq-reference-default-not-selected	520 × 420 × 280	OSR-COTS-RECORDER-HASLER

Product	Reference default	Use class	CAD envelope X × Y × Z (mm)	Sources
LM3-END-P020 T-OBS nose sensor pack, heated window services, and washer kit	Ouster — OS1 industrial lidar sensor	development-sensor-candidate-reference-default-not-selected	1050 × 1850 × 520	OSR-COTS-LIDAR-OUSTER-OS1
LM3-END-P040 e-coupler LV jumper, recovery trainline, and end harness breakaway kit	Dellner — automatic coupler Type 430 with pivot anchor	rfq-baseline-reference-default-not-selected	1700 × 900 × 260	OSR-COTS-COUPLER-DELLNER430
LM3-END-P050 sealed headlight, tail/marker light, threshold-warning, and end-lamp harness kit	Teknoware — rolling-stock interior and emergency lighting platform	rfq-baseline-reference-default-not-selected	1600 × 900 × 280	OSR-COTS-LIGHT-TEKNOWARE
LM3-ART-P030 inter-car HV/LV jumper, coolant hose loop, energy chain, and drain sleeve kit	igus / HARTING — rail e-chain carrier with Han HPR connector system	sample-and-cycle-test-reference-default-not-selected	1400 × 1600 × 360	OSR-COTS-HV-TE-HVP800 OSR-COTS-TRAINLINE-IGUS-HARTING
LM3-ART-P040 train-to-train open-end articulation, gangway, drawbar, turntable, and service-jumper cassette	HUEBNER — modular metro double-wall corrugated gangway	kinematic-envelope-then-rfq-reference-default-not-selected	1450 × 2250 × 2400	OSR-COTS-GANGWAY-HUBNER
LM3-ART-P041 train-to-train jumper blanking, transition harness, isolation label, and dust-cover kit	igus / HARTING — rail e-chain carrier with Han HPR connector system	sample-and-cycle-test-reference-default-not-selected	1200 × 850 × 260	OSR-COTS-TRAINLINE-IGUS-HARTING

Product requirements and override gates

LM3-CWL-P010 — end-cowl fiberglass laminate, insert, adhesive, and coupon material kit

Default: Gurit — SE 130FR / ST 130FR prepreg with SA130FR and SF130FR films.

Published reference facts:

- EN 45545 HL2 stated for R1/R7/R17 monolithic laminates
- 85 deg C low-temperature cure option
- 45 minute cure option at 120 deg C
- published reinforcement and film order codes

Affordability/localisation basis: Use the catalogue prepreg as the first-article material; cut, vacuum-bag, cure, trim and repair locally with batch coupons.

Close before supplier freeze:

- select reinforcement/core/film stack per component
- fire-test the actual sandwich laminate
- run environmental, insert, impact and repair qualification

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-FAS-P030 — front glazing/lamp EPDM seal, drain rail, washer sleeve, and edge-closeout kit

Default: AGC Automotive — rail glazing with framing system.

Published reference facts:

- laminated rail glazing family
- supplier framing-system option

Affordability/localisation basis: Retain the pressure-frame and dry-seal replacement concept; qualify regional laminators against the released cassette drawing.

Close before supplier freeze:

- freeze laminate stack, optical zones and curvature

- freeze heated end-glass voltage and power
- obtain impact, fire, weather and bond-system evidence

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-EXT-P010 — electric plug/sliding door cassette

Default: Knorr-Bremse IFE — IFE modular sliding-plug entrance system.

Published reference facts:

- rail entrance-system family
- supplier engineering required for vehicle aperture and controls

Affordability/localisation basis: Buy the first cassette and controller; manufacture portal, threshold, carrier, harness and service parts locally to the released interface.

Close before supplier freeze:

- freeze clear aperture, stroke, mass and four mounting datums
- obtain EN 14752 evidence and emergency-release/cycle data
- freeze keyed power, network and interlock interfaces

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-EXT-P020 — side laminated glazing cassette

Default: AGC Automotive — rail glazing with framing system.

Published reference facts:

- laminated rail glazing family
- supplier framing-system option

Affordability/localisation basis: Retain the pressure-frame and dry-seal replacement concept; qualify regional laminators against the released cassette drawing.

Close before supplier freeze:

- freeze laminate stack, optical zones and curvature
- freeze heated end-glass voltage and power
- obtain impact, fire, weather and bond-system evidence

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-EXT-P030 — single panoramic heated end-glass assembly

Default: AGC Automotive — rail glazing with framing system.

Published reference facts:

- laminated rail glazing family
- supplier framing-system option

Affordability/localisation basis: Retain the pressure-frame and dry-seal replacement concept; qualify regional laminators against the released cassette drawing.

Close before supplier freeze:

- freeze laminate stack, optical zones and curvature
- freeze heated end-glass voltage and power
- obtain impact, fire, weather and bond-system evidence

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

`LM3-EXT-P040` — hvac-24kw-direct-hv-dc roof HVAC

Default: Longertek — direct-DC rail HVAC platform.

Published reference facts:

- direct high-voltage DC architecture
- rail HVAC product platform

Affordability/localisation basis: Buy the first roof unit and controller; localise ducts, diffusers, drains, filters and later coils after calorimeter testing.

Close before supplier freeze:

- confirm 24 kW LM3 duty at worst ambient
- freeze roof envelope, mass, coolant/refrigerant and drain interfaces
- obtain EN 50155/50121 and fire evidence for quoted configuration

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

`LM3-EXT-P050` — roof PV module and edge-clamp kit

Default: Sunman Energy — eArc lightweight PV module.

Published reference facts:

- lightweight flexible photovoltaic family

Affordability/localisation basis: Use standard modules and make clamps, bonded carriers and replaceable roof harnesses locally.

Close before supplier freeze:

- freeze module size and electrical string
- qualify roof retention, fire, isolation, impact and walk-zone separation
- verify MPPT and safe isolation under all charging states

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-EXT-P060 — stepped floor-board and removable service-hatch system

Default: 3A Composites Mobility — COMFLOOR rail floor panel.

Published reference facts:

- lightweight sandwich floor-panel family
- optional integrated heating
- rail vehicle application

Affordability/localisation basis: Import first panel stock if needed; machine panels, edge closeouts, hatch frames and inserts locally from released cut files.

Close before supplier freeze:

- freeze panel layup, inserts, hatch edges and fire level
- prove concentrated, wheelchair and seat-rail loads
- validate wet-zone sealing and repair method

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-EXT-P061 — welded resilient floor covering, cove, nosing, and adhesive system

Default: Forbo Flooring Systems — Marmoleum Fresco FR2 6R3883 moonstone, 2.5 mm.

Published reference facts:

- 2.5 mm thickness
- 2.9 kg/m2
- EN 45545-2 HL2 stated for the published system
- roll width 2 m

Affordability/localisation basis: Use roll material initially; cut, cove, weld and repair locally with a controlled substrate/adhesive system.

Close before supplier freeze:

- obtain fire certificate for selected substrate/adhesive combination
- freeze cove, weld-cord, nosing and hatch details
- slip, cleaning and repair trials

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-EXT-P062 — longitudinal passenger and priority-seat modules

Default: Fainsa — metro and urban-rail seating family.

Published reference facts:

- urban rail passenger seat family

Affordability/localisation basis: Make the common seat rail, grab poles and adapters locally; localise shells and frames only after fire and abuse-load qualification.

Close before supplier freeze:

- select exact shell, frame and upholstery
- freeze common-rail attachment loads and PRM geometry
- obtain fire, abuse-load, corrosion and maintainability evidence

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

`LM3-EXT-P063` — stainless grab-pole, handrail, joint, and insulated adapter kit

Default: Fainsa — metro and urban-rail seating family.

Published reference facts:

- urban rail passenger seat family

Affordability/localisation basis: Make the common seat rail, grab poles and adapters locally; localise shells and frames only after fire and abuse-load qualification.

Close before supplier freeze:

- select exact shell, frame and upholstery
- freeze common-rail attachment loads and PRM geometry
- obtain fire, abuse-load, corrosion and maintainability evidence

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

`LM3-EXT-P064` — passenger-information display, speaker, amplifier, and mounting kit

Default: Luminator Technology Group — RailKom passenger-information and communication platform.

Published reference facts:

- rail passenger information, audio and communication family

Affordability/localisation basis: Use local brackets, speaker grilles and wiring; permit display/amplifier substitution through open interfaces and regression tests.

Close before supplier freeze:

- freeze display, amplifier, speaker and software licences
- require open Ethernet/IP data interfaces
- obtain EMC, fire, accessibility and cybersecurity evidence

Override the default when:

- the selected order code exceeds the controlled design-reference envelope

- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-EXT-P065 — CCTV camera, passenger intercom, PoE/data, and mounting kit

Default: Moxa — VPort P06HC-1MP-M12-CAM36.

Published reference facts:

- EN 50155 application statement
- IEC 61373 shock/vibration
- IP66
- PoE and M12 Ethernet
- 1280 x 800 up to 30 fps

Affordability/localisation basis: Use the camera as a replaceable Ethernet appliance with locally made anti-vandal mount and harness.

Close before supplier freeze:

- confirm lifecycle availability and cybersecurity baseline
- freeze viewing zones and privacy policy
- qualify cable, storage and passenger-intercom integration

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-EXT-P066 — PRM, safety-signage, emergency-lighting, extinguisher, and first-aid kit

Default: Teknoware — rolling-stock interior and emergency lighting platform.

Published reference facts:

- rolling-stock LED emergency-lighting platform

Affordability/localisation basis: Make the 1.2 m carrier, lens, brackets and harness locally around replaceable quoted LED modules.

Close before supplier freeze:

- freeze 24 V keyed cassette, optics and emergency duration
- obtain rail fire, EMC, ingress and photometry evidence
- separately select certified exterior head/tail lamps

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-FIX-P020 — four-family captive fastener, floating nut, isolator, and access-fastener kit

Default: Boellhoff — RIVNUT metric M6/M8 steel or stainless blind rivet nuts.

Published reference facts:

- published M6/M8 thread and upset-strength data
- one-sided installation

Affordability/localisation basis: Source from regional distributors or qualify metric equivalents by joint family; do not substitute solely by thread size.

Close before supplier freeze:

- select body, head, grip range and coating per joint drawing
- run substrate-specific setting and proof-load trials
- freeze setting-tool force and inspection gauge

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

`LM3-LGT-P010` — 1.2 m plug-in main LED lighting cassette and captive mounting kit

Default: Teknoware — rolling-stock interior and emergency lighting platform.

Published reference facts:

- rolling-stock LED emergency-lighting platform

Affordability/localisation basis: Make the 1.2 m carrier, lens, brackets and harness locally around replaceable quoted LED modules.

Close before supplier freeze:

- freeze 24 V keyed cassette, optics and emergency duration
- obtain rail fire, EMC, ingress and photometry evidence
- separately select certified exterior head/tail lamps

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

`LM3-LGT-P020` — emergency and doorway lighting modules with independent keyed feeder kit

Default: Teknoware — rolling-stock interior and emergency lighting platform.

Published reference facts:

- rolling-stock LED emergency-lighting platform

Affordability/localisation basis: Make the 1.2 m carrier, lens, brackets and harness locally around replaceable quoted LED modules.

Close before supplier freeze:

- freeze 24 V keyed cassette, optics and emergency duration
- obtain rail fire, EMC, ingress and photometry evidence
- separately select certified exterior head/tail lamps

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-EXT-P070 — roof antennas, service walkway pads, lifting covers, and maintenance labels

Default: HUBER+SUHNER — SENCITY Rail MIMO 5-port with GNSS/LNA, product 85137717.

Published reference facts:

- 2x2 cellular and Wi-Fi MIMO
- GNSS with integrated LNA
- EN 50155
- EN 45545-2 and NFPA 130
- -40 to 85 deg C

Affordability/localisation basis: Buy antenna assemblies; manufacture roof doublers, cable conduits and replaceable RF jumpers locally.

Close before supplier freeze:

- confirm city radio bands and cable losses
- freeze roof ground/bond and lightning path
- perform installed antenna and EMC tests

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-EXT-P080 — fire-rated GFRP side-module laminate, core, gelcoat, and consumable kit

Default: Gurit — SE 130FR / ST 130FR prepreg with SA130FR and SF130FR films.

Published reference facts:

- EN 45545 HL2 stated for R1/R7/R17 monolithic laminates
- 85 deg C low-temperature cure option
- 45 minute cure option at 120 deg C
- published reinforcement and film order codes

Affordability/localisation basis: Use the catalogue prepreg as the first-article material; cut, vacuum-bag, cure, trim and repair locally with batch coupons.

Close before supplier freeze:

- select reinforcement/core/film stack per component
- fire-test the actual sandwich laminate
- run environmental, insert, impact and repair qualification

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating

- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-EXT-P090 — fire-rated GFRP roof-module, dry-seal, and removable skirt material kit

Default: Gurit — SE 130FR / ST 130FR prepreg with SA130FR and SF130FR films.

Published reference facts:

- EN 45545 HL2 stated for R1/R7/R17 monolithic laminates
- 85 deg C low-temperature cure option
- 45 minute cure option at 120 deg C
- published reinforcement and film order codes

Affordability/localisation basis: Use the catalogue prepreg as the first-article material; cut, vacuum-bag, cure, trim and repair locally with batch coupons.

Close before supplier freeze:

- select reinforcement/core/film stack per component
- fire-test the actual sandwich laminate
- run environmental, insert, impact and repair qualification

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-FIN-P010 — pre-cut exterior livery graphic film, edge-seal, datum-mark, and repair-patch kit

Default: 3M — Controltac Graphic Film with Comply Adhesive 180mC family.

Published reference facts:

- listed for rail-car graphic applications
- 50 micrometre cast opaque film
- pressure-activated removable adhesive
- up to seven-year MCS warranty subject to selected construction and conditions

Affordability/localisation basis: Plot, print, label and apply pre-cut one-metre-bay kits locally; accept another supplier only through complete-system equivalence rather than nominal film thickness.

Close before supplier freeze:

- select film colour/print, ink, overlaminate and edge system together
- obtain project fire and hot-climate exposure evidence
- run adhesion, wash, weather, repair and removal trials on the actual LM3 base finish

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-FIN-P020 — calcium-carbonate radiative roof-coating qualification and exposed-roof application kit

Default: qualified light-colour base roof finish; CaCO3 coating restricted to coupon and one-car trial.

Published reference facts:

- 95.5 percent initial solar reflectance reported
- 0.94 normal emissivity in the atmospheric window reported
- 60 percent particle concentration research formulation

Affordability/localisation basis: Local coupon preparation is encouraged, but fleet application remains prohibited until an orderable controlled formulation and signed rail qualification pack exist.

Close before supplier freeze:

- identify a controlled commercial binder/pigment/additive formulation
- obtain rail fire, chemical and application evidence
- pass GFRP adhesion/flexibility plus UV, abrasion, wash and soiling ageing
- measure new/aged optical performance and complete a one-car thermal/maintenance trial

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-INT-P010 — HVAC diffusers, side return ducts, saloon grilles, and access panels

Default: Longertek — direct-DC rail HVAC platform.

Published reference facts:

- direct high-voltage DC architecture
- rail HVAC product platform

Affordability/localisation basis: Buy the first roof unit and controller; localise ducts, diffusers, drains, filters and later coils after calorimeter testing.

Close before supplier freeze:

- confirm 24 kW LM3 duty at worst ambient
- freeze roof envelope, mass, coolant/refrigerant and drain interfaces
- obtain EN 50155/50121 and fire evidence for quoted configuration

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-TRC-P010 — motor-350kw-hm47-class axle traction motor

Default: ABB — AMXE250 configurable traction motor.

Published reference facts:

- up to 436 kW peak
- up to 3353 Nm peak

- up to 3500 rpm
- configurable shaft and length

Affordability/localisation basis: Buy first motors; local rewind and bearing service may follow an ABB-approved service procedure before attempting local manufacture.

Close before supplier freeze:

- quote a rail-duty 800 V winding and cooling configuration
- freeze shaft, foot, mass and torque-speed map
- obtain rail environmental, insulation, overspeed and endurance evidence

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

`LM3-TRC-P020` — single-stage reduction gearbox and flexible coupling

Default: Voith — SE single-stage axle-mounted metro gear unit.

Published reference facts:

- single-stage axle-mounted rail gear family

Affordability/localisation basis: Buy first gear units; machine guards, torque link and mounting hardware locally.

Close before supplier freeze:

- freeze ratio, centre distance, torque and coupling
- obtain bearing, lubrication, efficiency, noise, overspeed and endurance evidence
- freeze reaction-link and wheelset interfaces

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

`LM3-TRC-P030` — two independent motor controllers, isolated LV DC/DC, MPPT, station protection, and cooling-loop kit

Default: ABB — BORDLINE CC400 LRV compact converter.

Published reference facts:

- 600/750 VDC LRV platform
- two independent propulsion converters
- integrated auxiliary converter and battery charger
- regenerative or resistor braking

Affordability/localisation basis: Buy first converters; localise cabinet, coolant manifold, harness and diagnostics integration behind frozen interfaces.

Close before supplier freeze:

- confirm operation from the selected battery/DC-link range
- freeze motor, cooling, MVB/Ethernet and protection interfaces
- quote exact mass, envelope, losses, software and rail evidence

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

`LM3-TRC-P040` — battery-225kwh-lfp-800v saloon-isolated side traction battery pack

Default: HOPPECKE — LiTrac modular high-voltage LFP rail traction battery.

Published reference facts:

- modular/scalable traction system
- published LFP option
- UN 38.3 base-module statement
- supplier rail safety engineering

Affordability/localisation basis: Buy certified packs/BMS initially; make external trays, ducts, guards, coolant manifolds and service fixtures locally.

Close before supplier freeze:

- confirm the frozen 225 kWh gross per-car target against the validated duty cycle or raise a controlled change request
- freeze voltage window, pack segmentation, mass and cooling
- obtain IEC 62928, propagation, fire, EMC and software evidence for the quote

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

`LM3-TRC-P050` — roof-mounted regen dump resistor and thermal shield kit

Default: Danotherm Electric — custom traction brake resistor, 600-3000 VDC family.

Published reference facts:

- rail traction brake-resistor family
- 600-3000 VDC published range
- IEC 60322 and IEC 61373 product-family statements
- roof/underframe installation options

Affordability/localisation basis: Buy the element pack and insulators; manufacture roof carrier, shield, duct and service guard locally after the hot-zone model is frozen.

Close before supplier freeze:

- derive resistance, continuous/pulse energy and duty cycle from simulation
- freeze enclosure, airflow, thermal switch and touch guards
- run worst-case stopped-train thermal test

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support

- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-TRC-P060 — station side-pin charging connector, actuator, shutter, and alignment target

Default: supplier-neutral 650-700 VDC side-contact RFQ; automated collector retained only as a concept alternative.

Published reference facts:

- automated conductive charging platform

Affordability/localisation basis: Prefer a proven automated collector geometry if it reduces bespoke side-pin risk; make station frame, guards and alignment target locally.

Close before supplier freeze:

- compare roof/bottom collector against the LM3 side-pin concept
- freeze current, voltage, touch safety, alignment and emergency release
- qualify stationary and vehicle halves together

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-TRC-P070 — HV contactor, fuse, pre-charge, service-disconnect, and current-sensor panel

Default: Schaltbau — C360K/300 railway bidirectional DC contactor.

Published reference facts:

- IEC 60077-2 category B
- 1000 V at pollution degree 3 / 1500 V at pollution degree 2
- 300 A at 70 deg C
- 3000 A one-second withstand
- 24/36-48/72-110 VDC coil options

Affordability/localisation basis: Buy contactors and qualified fuses; fabricate busbars, enclosure, precharge assembly and service-disconnect panel locally.

Close before supplier freeze:

- confirm breaking duty and prospective fault current
- select coil, auxiliary contacts and mounting orientation
- coordinate fuse, busbar, precharge and enclosure clearances

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-SAF-P010 — battery temperature/off-gas detection, electrical-enclosure smoke detection, and localized mist kit

Default: Xtralis / Li-ion Tamer — Li-ion Tamer GEN 3.

Published reference facts:

- electrolyte-vapour early detection
- temperature and humidity at sensing nodes
- relay and Modbus interfaces
- calibration-free sensing claim

Affordability/localisation basis: Buy sensing nodes/controller; make brackets, protected Ethernet/relay harness and test-gas service access locally.

Close before supplier freeze:

- verify sensor response in selected battery airflow and contaminants
- define safe-state action and false-alarm handling
- do not treat detection as suppression or a standalone safety system

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

`LM3-BOG-P040` — powered-bogie wheelset with axle-mounted brake discs

Default: GHH-BONATRANS — urban-rail wheelset with axle-mounted disc interfaces.

Published reference facts:

- urban-rail wheelset supplier family

Affordability/localisation basis: Buy certified wheelsets initially; local wheel/axle manufacture requires a separately qualified metallurgical and NDT route.

Close before supplier freeze:

- freeze wheel profile/diameter, axle journals and disc seats
- issue load spectrum and duty
- obtain material, forging, heat-treatment, NDT, press-fit and fatigue evidence

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

`LM3-BOG-P041` — trailer-bogie wheelset with axle-mounted brake discs

Default: GHH-BONATRANS — urban-rail wheelset with axle-mounted disc interfaces.

Published reference facts:

- urban-rail wheelset supplier family

Affordability/localisation basis: Buy certified wheelsets initially; local wheel/axle manufacture requires a separately qualified metallurgical and NDT route.

Close before supplier freeze:

- freeze wheel profile/diameter, axle journals and disc seats
- issue load spectrum and duty
- obtain material, forging, heat-treatment, NDT, press-fit and fatigue evidence

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-BOG-P042 — powered-wheelset axlebox, sealed bearing unit, speed and temperature sensor set

Default: SKF — sensor-ready railway tapered roller bearing unit.

Published reference facts:

- sealed and pre-lubricated ready-to-mount TBU family

Affordability/localisation basis: Buy bearing units; machine housings, sensor guards and cable brackets locally to the supplier-controlled fits.

Close before supplier freeze:

- calculate bearing life from journal, load spectrum and speed
- freeze exact designation, fit, seal, grease and grounding
- freeze speed/temperature sensor and cable interface

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-BOG-P043 — trailer-wheelset axlebox, sealed bearing unit, speed and temperature sensor set

Default: SKF — sensor-ready railway tapered roller bearing unit.

Published reference facts:

- sealed and pre-lubricated ready-to-mount TBU family

Affordability/localisation basis: Buy bearing units; machine housings, sensor guards and cable brackets locally to the supplier-controlled fits.

Close before supplier freeze:

- calculate bearing life from journal, load spectrum and speed
- freeze exact designation, fit, seal, grease and grounding
- freeze speed/temperature sensor and cable interface

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-BOG-P044 — powered-bogie primary suspension spring, guide and bump-stop set

Default: Continental — rail half-bellow air spring with integrated auxiliary spring.

Published reference facts:

- metro/local-transport air-spring family

- integrated rubber-metal auxiliary spring option

Affordability/localisation basis: Buy elastomer/spring/damper elements; fabricate spring seats, guides, centre pivot, yaw links and guards locally after calculation.

Close before supplier freeze:

- freeze load-deflection/damping curves from the vehicle dynamics model
- select primary springs, air springs, dampers and bump stops as one system
- obtain ageing, fatigue, fire and ride evidence

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-BOG-P045 — trailer-bogie primary suspension spring, guide and bump-stop set

Default: Continental — rail half-bellow air spring with integrated auxiliary spring.

Published reference facts:

- metro/local-transport air-spring family
- integrated rubber-metal auxiliary spring option

Affordability/localisation basis: Buy elastomer/spring/damper elements; fabricate spring seats, guides, centre pivot, yaw links and guards locally after calculation.

Close before supplier freeze:

- freeze load-deflection/damping curves from the vehicle dynamics model
- select primary springs, air springs, dampers and bump stops as one system
- obtain ageing, fatigue, fire and ride evidence

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-BOG-P046 — powered-bogie to carbody connection: air springs, emergency spring, centre pivot, yaw links and dampers

Default: Continental — rail half-bellow air spring with integrated auxiliary spring.

Published reference facts:

- metro/local-transport air-spring family
- integrated rubber-metal auxiliary spring option

Affordability/localisation basis: Buy elastomer/spring/damper elements; fabricate spring seats, guides, centre pivot, yaw links and guards locally after calculation.

Close before supplier freeze:

- freeze load-deflection/damping curves from the vehicle dynamics model
- select primary springs, air springs, dampers and bump stops as one system
- obtain ageing, fatigue, fire and ride evidence

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-BOG-P047 — trailer-bogie to carbody connection: air springs, emergency spring, centre pivot, yaw links and dampers

Default: Continental — rail half-bellow air spring with integrated auxiliary spring.

Published reference facts:

- metro/local-transport air-spring family
- integrated rubber-metal auxiliary spring option

Affordability/localisation basis: Buy elastomer/spring/damper elements; fabricate spring seats, guides, centre pivot, yaw links and guards locally after calculation.

Close before supplier freeze:

- freeze load-deflection/damping curves from the vehicle dynamics model
- select primary springs, air springs, dampers and bump stops as one system
- obtain ageing, fatigue, fire and ride evidence

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-BOG-P048 — powered-bogie brake calipers, parking actuators, pads and wheel-slide hardware

Default: Knorr-Bremse — WheelAct Compact RZKT with spring parking-brake option.

Published reference facts:

- 50 kN service pad force
- 45 kN spring parking force
- 68 kg
- 305 mm width
- plus/minus 25 mm lateral movement
- 8-10 year stated overhaul interval

Affordability/localisation basis: Buy first calipers/actuators; localise discs, hoses, brackets and overhaul parts only under brake-system approval.

Close before supplier freeze:

- confirm disc diameter/pad pair and required force from brake calculation
- freeze actuator, parking, WSP and degraded-mode architecture
- perform stopping, thermal, adhesion and endurance tests

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support

- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-BOG-P049 — trailer-bogie brake calipers, parking actuators, pads and wheel-slide hardware

Default: Knorr-Bremse — WheelAct Compact RZKT with spring parking-brake option.

Published reference facts:

- 50 kN service pad force
- 45 kN spring parking force
- 68 kg
- 305 mm width
- plus/minus 25 mm lateral movement
- 8-10 year stated overhaul interval

Affordability/localisation basis: Buy first calipers/actuators; localise discs, hoses, brackets and overhaul parts only under brake-system approval.

Close before supplier freeze:

- confirm disc diameter/pad pair and required force from brake calculation
- freeze actuator, parking, WSP and degraded-mode architecture
- perform stopping, thermal, adhesion and endurance tests

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-AUX-P010 — secondary-suspension compressor, dryer, reservoir, and isolation-manifold kit

Default: Knorr-Bremse — VV120-T 2.0 oil-free rail compressor / AirSupply Smart.

Published reference facts:

- oil-free rail compressor family
- variable-speed smart-control option
- published low-noise operating mode

Affordability/localisation basis: Buy compressor/dryer; fabricate reservoir, manifold, isolation, drains, mounts and acoustic enclosure locally to approved pressure codes.

Close before supplier freeze:

- size from suspension, brake and auxiliary air demand
- freeze dryer, reservoir, relief, noise and mounting package
- run duty, thermal, condensate and degraded-mode tests

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-END-P010 — automatic end coupler and crash-energy absorber

Default: Dellner — automatic coupler Type 430 with pivot anchor.

Published reference facts:

- light-rail/tram/monorail application
- 600 kN buff and 500 kN draft
- gas-hydraulic damper
- remote uncoupling and foldable options

Affordability/localisation basis: Buy coupler/head/absorber; make pocket, covers, brackets and harness locally to Dellner drawings.

Close before supplier freeze:

- freeze height, gathering range, fold, crash stroke and pocket loads
- freeze electrical head and rescue jumper pinout
- integrate into whole-vehicle crash-energy management

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-ART-P020 — articulation lower spherical pivot, bearing housing and pin set

Default: Schaeffler — ELGES maintenance-free spherical plain bearing and rod-end family.

Published reference facts:

- maintenance-free spherical plain-bearing family

Affordability/localisation basis: Buy bearing liners/rod ends; machine housings, pins, links and retained fasteners locally from released drawings.

Close before supplier freeze:

- derive static/dynamic load, pin diameter, misalignment and life
- freeze liner, seal, corrosion and temperature variant
- proof and fatigue-test the complete local housing/pin/link joint

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-ART-P021 — articulation upper lateral/yaw links, spherical joints and retained pins

Default: Schaeffler — ELGES maintenance-free spherical plain bearing and rod-end family.

Published reference facts:

- maintenance-free spherical plain-bearing family

Affordability/localisation basis: Buy bearing liners/rod ends; machine housings, pins, links and retained fasteners locally from released drawings.

Close before supplier freeze:

- derive static/dynamic load, pin diameter, misalignment and life
- freeze liner, seal, corrosion and temperature variant
- proof and fatigue-test the complete local housing/pin/link joint

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-ART-P022 — inter-car double-wall corrugated bellows and clamp-frame set

Default: HUEBNER — modular metro double-wall corrugated gangway.

Published reference facts:

- single/double-wall corrugated-bellows options
- bridge and elastic interior-panel options
- metro/urban-rail application

Affordability/localisation basis: Buy first bellows/bridge system; fabricate clamp frames, threshold, drains and service panels locally to released interfaces.

Close before supplier freeze:

- issue full articulation sweep and end-frame geometry
- freeze bellows, bridge, turntable, clamp and drainage
- perform fire, fatigue, water, passenger-load and pinch-gap tests

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-ART-P023 — inter-car passenger bridge, turntable and flexible interior-panel set

Default: HUEBNER — modular metro double-wall corrugated gangway.

Published reference facts:

- single/double-wall corrugated-bellows options
- bridge and elastic interior-panel options
- metro/urban-rail application

Affordability/localisation basis: Buy first bellows/bridge system; fabricate clamp frames, threshold, drains and service panels locally to released interfaces.

Close before supplier freeze:

- issue full articulation sweep and end-frame geometry
- freeze bellows, bridge, turntable, clamp and drainage
- perform fire, fatigue, water, passenger-load and pinch-gap tests

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support

- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-ART-P024 — articulation trainline carrier, support arms, abrasion liners and drain path

Default: igus / HARTING — rail e-chain carrier with Han HPR connector system.

Published reference facts:

- rail dynamic cable-carrier family
- rail connector/fire variants available

Affordability/localisation basis: Buy carriers/connectors; cut rail cable and hose, make support arms, drains and protective sleeves locally with controlled termination.

Close before supplier freeze:

- select carrier from bend radius, fill, travel and torsion
- freeze Han HPR inserts, housings, coding and breakaway
- cycle-test complete HV/LV/coolant bundle through the articulation envelope

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-CTRL-P010 — T-ECU/S and T-ECU/A compute and safety-control cabinet kit

Default: duagon — D-SC Controller with MH50C safe CPU and distributed safe I/O.

Published reference facts:

- applications up to SIL4 stated by supplier
- three-CPU safe controller
- safe I/O
- EN 50155 rolling-stock environment
- certification packages available

Affordability/localisation basis: Use duagon for the independent qualified channel; keep OSR application compute separate and portable. Build cabinet/harness locally.

Close before supplier freeze:

- allocate each LM3 safety function and required SIL
- freeze CPU/I/O/BSP/application configuration
- independently verify OSR integration and safe-state timing

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-CTRL-P020 — navigation, balise, 5G, LoRa, GNSS, IMU, and roof-antenna kit

Default: Westermo — Ibex-RT-630-5G.

Published reference facts:

- 3GPP Release 16 5G

- 4x4 sub-6 GHz MIMO
- 2x2 Wi-Fi
- EN 50155 and EN 45545-2
- M12 and IP66

Affordability/localisation basis: Buy router; integrate local SIM service, cybersecurity configuration, roof cables and cabinet harness.

Close before supplier freeze:

- confirm local 5G bands/SIMs and cybersecurity lifecycle
- freeze antenna/cable topology
- prove handover, loss-of-comms and update behaviour

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-CTRL-P030 — maintenance HMI, depot pendant, emergency controls, and safety-relay kit

Default: Pilz — PNOZ s4, railway order 751184.

Published reference facts:

- railway application note for order 751184
- three safety contacts
- redundant self-monitoring circuit

Affordability/localisation basis: Buy relays; fabricate labelled DIN cabinet, E-stop/pendant harness and replaceable terminal blocks locally.

Close before supplier freeze:

- use only for an allocated compatible function
- observe supplier accessibility restriction for railway use
- verify contact load, diagnostics and safe-state integration

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-CTRL-P050 — operational and crashworthy event-recorder storage kit

Default: HaslerRail — JRU/EVA event-recorder platform.

Published reference facts:

- rail event-recording product family

Affordability/localisation basis: Buy protected recorder; use local commodity storage only for non-protected operational logs.

Close before supplier freeze:

- freeze jurisdictional data set, retention and clock
- select protected memory and download interfaces

- obtain crash, fire, environmental and chain-of-custody evidence

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-END-P020 — T-OBS nose sensor pack, heated window services, and washer kit

Default: Ouster — OS1 industrial lidar sensor.

Published reference facts:

- industrial digital lidar family

Affordability/localisation basis: Buy sensors; manufacture datum carrier, sacrificial heated window, washer, guard and service alignment fixture locally.

Close before supplier freeze:

- not a certified railway obstacle-detection function
- freeze field of view, range, latency, heater/washer and protective window
- validate sensing, diagnostics and failure response over route conditions

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-END-P040 — e-coupler LV jumper, recovery trainline, and end harness breakaway kit

Default: Dellner — automatic coupler Type 430 with pivot anchor.

Published reference facts:

- light-rail/tram/monorail application
- 600 kN buff and 500 kN draft
- gas-hydraulic damper
- remote uncoupling and foldable options

Affordability/localisation basis: Buy coupler/head/absorber; make pocket, covers, brackets and harness locally to Dellner drawings.

Close before supplier freeze:

- freeze height, gathering range, fold, crash stroke and pocket loads
- freeze electrical head and rescue jumper pinout
- integrate into whole-vehicle crash-energy management

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-END-P050 — sealed headlight, tail/marker light, threshold-warning, and end-lamp harness kit

Default: Teknoware — rolling-stock interior and emergency lighting platform.

Published reference facts:

- rolling-stock LED emergency-lighting platform

Affordability/localisation basis: Make the 1.2 m carrier, lens, brackets and harness locally around replaceable quoted LED modules.

Close before supplier freeze:

- freeze 24 V keyed cassette, optics and emergency duration
- obtain rail fire, EMC, ingress and photometry evidence
- separately select certified exterior head/tail lamps

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-ART-P030 — inter-car HV/LV jumper, coolant hose loop, energy chain, and drain sleeve kit

Default: igus / HARTING — rail e-chain carrier with Han HPR connector system.

Published reference facts:

- rail dynamic cable-carrier family
- rail connector/fire variants available

Affordability/localisation basis: Buy carriers/connectors; cut rail cable and hose, make support arms, drains and protective sleeves locally with controlled termination.

Close before supplier freeze:

- select carrier from bend radius, fill, travel and torsion
- freeze Han HPR inserts, housings, coding and breakaway
- cycle-test complete HV/LV/coolant bundle through the articulation envelope

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

LM3-ART-P040 — train-to-train open-end articulation, gangway, drawbar, turntable, and service-jumper cassette

Default: HUEBNER — modular metro double-wall corrugated gangway.

Published reference facts:

- single/double-wall corrugated-bellows options
- bridge and elastic interior-panel options
- metro/urban-rail application

Affordability/localisation basis: Buy first bellows/bridge system; fabricate clamp frames, threshold, drains and service panels locally to released interfaces.

Close before supplier freeze:

- issue full articulation sweep and end-frame geometry
- freeze bellows, bridge, turntable, clamp and drainage
- perform fire, fatigue, water, passenger-load and pinch-gap tests

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

`LM3-ART-P041` — train-to-train jumper blanking, transition harness, isolation label, and dust-cover kit

Default: igus / HARTING — rail e-chain carrier with Han HPR connector system.

Published reference facts:

- rail dynamic cable-carrier family
- rail connector/fire variants available

Affordability/localisation basis: Buy carriers/connectors; cut rail cable and hose, make support arms, drains and protective sleeves locally with controlled termination.

Close before supplier freeze:

- select carrier from bend radius, fill, travel and torsion
- freeze Han HPR inserts, housings, coding and breakaway
- cycle-test complete HV/LV/coolant bundle through the articulation envelope

Override the default when:

- the selected order code exceeds the controlled design-reference envelope
- vehicle load, duty, thermal, EMC, fire, accessibility or RAM calculations require another rating
- the quoted configuration lacks the required rail evidence or lifecycle support
- a qualified local equivalent has lower whole-life cost after interface and regression testing

Controlled manufacturer/reference sources

`OSR-COTS-DOOR-IFE` — Knorr-Bremse IFE IFE modular sliding-plug entrance system

- Source class: `vehicle-specific-rfq`
- Selection state: `rfq-baseline`

`OSR-COTS-GLASS-AGC` — AGC Automotive rail glazing with framing system

- Source class: `vehicle-specific-rfq`
- Selection state: `rfq-baseline`

`OSR-COTS-FINISH-3M-180MC` — 3M Controltac Graphic Film with Comply Adhesive 180mC family

- Source class: `catalogue-family-with-rail-application-guidance`
- Selection state: `sample-and-system-rfq`

OSR-COTS-COOLROOF-PURDUE-CACO3 — Purdue University research team (reference only) 2020 CaCO3-acrylic particle-matrix paint formulation

- Source class: `research-formulation-not-orderable`
- Selection state: `qualification-only`

OSR-COTS-GLAZING-SIKA268 — Sika Sikaflex-268 PowerCure, 600 ml

- Source class: `exact-catalogue-product`
- Selection state: `sample-ready`

OSR-COTS-HVAC-LONGERTEK — Longertek direct-DC rail HVAC platform

- Source class: `vehicle-specific-rfq`
- Selection state: `rfq-baseline`

OSR-COTS-PV-SUNMAN — Sunman Energy eArc lightweight PV module

- Source class: `catalogue-family`
- Selection state: `sample-ready`

OSR-COTS-FLOOR-COMFLOOR — 3A Composites Mobility COMFLOOR rail floor panel

- Source class: `configurable-catalogue-system`
- Selection state: `rfq-baseline`

OSR-COTS-FLOOR-FORBO-6R3883 — Forbo Flooring Systems Marmoleum Fresco FR2 6R3883 moonstone, 2.5 mm

- Source class: `exact-catalogue-product`
- Selection state: `sample-ready`

OSR-COTS-SEAT-FAINSA — Fainsa metro and urban-rail seating family

- Source class: `vehicle-specific-rfq`
- Selection state: `rfq-baseline`

OSR-COTS-PIS-LUMINATOR — Luminator Technology Group RailKom passenger-information and communication platform

- Source class: `configurable-catalogue-system`
- Selection state: `rfq-baseline`

OSR-COTS-CAMERA-MOXA-P06HC — Moxa VPort P06HC-1MP-M12-CAM36

- Source class: `exact-catalogue-model`
- Selection state: `sample-ready`

OSR-COTS-LIGHT-TEKNOWARE — Teknoware rolling-stock interior and emergency lighting platform

- Source class: `vehicle-specific-rfq`
- Selection state: `rfq-baseline`

OSR-COTS-FASTENER-RIVNUT — Boellhoff RIVNUT metric M6/M8 steel or stainless blind rivet nuts

- Source class: `catalogue-family-with-sizing`
- Selection state: `sample-ready`

OSR-COTS-ANTENNA-HS-85137717 — HUBER+SUHNER SENCITY Rail MIMO 5-port with GNSS/LNA, product 85137717

- Source class: `exact-catalogue-product`
- Selection state: `orderable-candidate`

OSR-COTS-COMPOSITE-GURIT-130FR — Gurit SE 130FR / ST 130FR prepreg with SA130FR and SF130FR films

- Source class: `exact-catalogue-family`
- Selection state: `sample-ready`

OSR-COTS-MOTOR-ABB-AMXE250 — ABB AMXE250 configurable traction motor

- Source class: `configurable-catalogue-machine`
- Selection state: `rfq-baseline`

OSR-COTS-GEAR-VOITH-SE — Voith SE single-stage axle-mounted metro gear unit

- Source class: `vehicle-specific-rfq`
- Selection state: `rfq-baseline`

OSR-COTS-CONVERTER-ABB-CC400 — ABB BORDLINE CC400 LRV compact converter

- Source class: `configurable-rail-product`
- Selection state: `rfq-baseline`

OSR-COTS-BATTERY-HOPPECKE-LITRAC — HOPPECKE LiTrac modular high-voltage LFP rail traction battery

- Source class: `vehicle-specific-rfq`
- Selection state: `rfq-baseline`

OSR-COTS-RESISTOR-DANOTHERM — Danotherm Electric custom traction brake resistor, 600-3000 VDC family

- Source class: `vehicle-specific-rfq`
- Selection state: `rfq-baseline`

OSR-COTS-CHARGER-SCHUNK — Schunk Carbon Technology Smart Charging automated high-current collector platform

- Source class: `interface-rfq-baseline`
- Selection state: `concept-alternative`

OSR-COTS-HV-SCHALTBAU-C360 — Schaltbau C360K/300 railway bidirectional DC contactor

- Source class: `exact-catalogue-model`
- Selection state: `sample-ready`

OSR-COTS-HV-TE-HVP800 — TE Connectivity AMP+ HVP800 cable assembly 1-2286937-1

- Source class: `exact-catalogue-part`
- Selection state: `sample-ready`

OSR-COTS-FIRE-LIIONTAMER-3 — Xtralis / Li-ion Tamer Li-ion Tamer GEN 3

- Source class: `exact-catalogue-system`
- Selection state: `sample-ready`

OSR-COTS-FIRE-FOGTEC — FOGTEC Rail Systems rail high-pressure water-mist system

- Source class: `vehicle-specific-rfq`
- Selection state: `hazard-analysis-rfq`

OSR-COTS-WHEELSET-GHH — GHH-BONATRANS urban-rail wheelset with axle-mounted disc interfaces

- Source class: `vehicle-specific-rfq`
- Selection state: `rfq-baseline`

OSR-COTS-BEARING-SKF-TBU — SKF sensor-ready railway tapered roller bearing unit

- Source class: `vehicle-specific-rfq`
- Selection state: `calculation-then-rfq`

OSR-COTS-SUSPENSION-CONTI — Continental rail half-bellow air spring with integrated auxiliary spring

- Source class: `vehicle-specific-rfq`
- Selection state: `dynamics-then-rfq`

OSR-COTS-BRAKE-KNORR-RZKT — Knorr-Bremse WheelAct Compact RZKT with spring parking-brake option

- Source class: `configurable-catalogue-model`
- Selection state: `brake-calculation-then-rfq`

OSR-COTS-AIR-KNORR-VV120T — Knorr-Bremse VV120-T 2.0 oil-free rail compressor / AirSupply Smart

- Source class: `configurable-rail-product`
- Selection state: `air-demand-then-rfq`

OSR-COTS-COUPLER-DELLNER430 — Dellner automatic coupler Type 430 with pivot anchor

- Source class: `configurable-catalogue-model`
- Selection state: `rfq-baseline`

OSR-COTS-JOINT-SCHAEFFLER — Schaeffler ELGES maintenance-free spherical plain bearing and rod-end family

- Source class: `catalogue-family-with-calculation`
- Selection state: `calculation-then-order-code`

OSR-COTS-GANGWAY-HUBNER — HUEBNER modular metro double-wall corrugated gangway

- Source class: `vehicle-specific-rfq`

- Selection state: `kinematic-envelope-then-rfq`

`OSR-COTS-TRAINLINE-IGUS-HARTING` — igus / HARTING rail e-chain carrier with Han HPR connector system

- Source class: `catalogue-family-with-sizing`
- Selection state: `sample-and-cycle-test`

`OSR-COTS-CONTROL-DUAGON-MH50C` — duagon D-SC Controller with MH50C safe CPU and distributed safe I/O

- Source class: `certified-configurable-platform`
- Selection state: `safety-allocation-then-rfq`

`OSR-COTS-COMPUTE-MOXA-V2406C` — Moxa V2406C-WL1-T railway computer

- Source class: `exact-catalogue-model`
- Selection state: `application-compute-candidate`

`OSR-COTS-ROUTER-WESTERMO-630-5G` — Westermo Ibex-RT-630-5G

- Source class: `exact-catalogue-model`
- Selection state: `orderable-candidate`

`OSR-COTS-SAFETYRELAY-PILZ-751184` — Pilz PNOZ s4, railway order 751184

- Source class: `exact-catalogue-order`
- Selection state: `bench-safety-relay-candidate`

`OSR-COTS-RECORDER-HASLER` — HaslerRail JRU/EVA event-recorder platform

- Source class: `vehicle-specific-rfq`
- Selection state: `requirements-then-rfq`

`OSR-COTS-LIDAR-OUSTER-OS1` — Ouster OS1 industrial lidar sensor

- Source class: `catalogue-sensor`
- Selection state: `development-sensor-candidate`

Controlled source

- Candidate register: `lib/templates/trainset-cots-candidates.toml` (`0588d76029c2f426beadc09a5ba096848d2eb85a22fb4e1306c54d68931f3b94`)

LM3 simplified exterior finish system

Repository source: `design/component-catalogue/catalog/buildable-trainset/exterior-finish-system.md`

Generated by `tools/automation/buildable-trainset.sh`. This controls the separation between mandatory base protection, replaceable livery film and the trial-only calcium-carbonate radiative roof coating.

- Design ID: `LM3-FIN-260`
- Status: `design-reference; CaCO3 roof route is trial-only`
- Principle: film simplifies livery but never replaces corrosion, composite-surface or fire protection
- Film procurement boundary: rail-application product selected by current supplier bulletin and project fire/adhesion evidence
- CaCO3 research screen: initial solar reflectance `0.955`; sky-window emissivity `0.94`. research screening targets only; measure the selected LM3 system new and aged

Finish zones

ID	Substrate	Mandatory base	Simplified finish	Replaceable	Prohibited coverage	Release evidence
`LM3-FIN-Z01`	S355 primary and secondary steel	qualified blast/clean, corrosion primer, stripe coat, topcoat and cavity protection	single light neutral topcoat; optional livery film only after full cure	no	weld inspection lands until released earth studs machined datums drains labels	surface-preparation record coating batch dry-film thickness holiday/visual inspection adhesion/cure release
`LM3-FIN-Z02`	clip-on GFRP side modules and end-cowl casts	fire/UV-qualified in-mould white gelcoat or compatible primer/base topcoat with sealed cut edges	pre-cut rail-use graphic film for coloured livery bands, logos and replaceable local graphics	yes	EPDM seals drains retained fasteners service labels glass/lamp edges bond/seal lands	substrate cure/cleanliness film-system certificate adhesion coupon edge/overlap map removal/repair trial
`LM3-FIN-Z03`	exposed GFRP roof skins and removable HVAC/PV fairings	fire/UV-qualified in-mould white gelcoat or compatible rail coating	CaCO3-acrylic radiative coating candidate only after coupon and one-car qualification	yes	PV active area and clamps HVAC intake/exhaust and heat exchangers antennas and sensor apertures walkways and anti-slip pads lifting/jacking covers earth bonds glands, drains, seals, labels and service lands	fire/chemical compatibility GFRP adhesion/flexibility UV/abrasion/wash ageing new and aged solar reflectance/emittance glare/visibility review one-car thermal and maintenance trial

Controlled route

Seq.	Operation	Hold point
10	release corrosion, fire and composite base finish before decorative work	base finish records accepted
20	survey, repair, fully cure and clean each film/coating substrate	signed pre-installation inspection
30	cut film by module/bay and make film plus roof-coating witness coupons	coupon adhesion and appearance accepted
40	dry-apply livery film; mask roof keep-outs and apply qualified candidate coating	edges, overlaps, mask map and wet/dry film record accepted
50	inspect, water/wash test, retain coupons and complete removal/repair demonstration	new-build release or trial-only disposition signed
60	age coupons and one-car trial through UV, wash, abrasion and service exposure	aged optical/adhesion and maintenance evidence reviewed

The CaCO3 values come from the Purdue research paper linked in `exterior-finish-process.md`. They are screening targets, not LM3 performance claims.

LM3 pilot factory sizing and machinery plan

Repository source: `design/component-catalogue/catalog/buildable-trainset/factory-plan.md`

Generated by `tools/automation/buildable-trainset.sh`. This is a rough-order factory layout and machinery plan for the basic three-car LM3 trainset. It sizes a lean pilot plant, not a turnkey rolling-stock works.

- Family: `light-metro-3car`
- Candidate: `light-metro-3car__16p5m__reference-body__reference-bogie__motor-350kw-hm47-class__battery-225kwh-lfp-800v__hvac-24kw-direct-hv-dc__pv12`
- Planning basis: minimum pilot factory for one first-article LM3 three-car trainset at a time; one 55 m final assembly bay; chassis, bogies, moulded GFRP modules, and interiors run off-line in parallel
- Recommended enclosed factory: 3,515 m2 (37,835 ft2)
- Outside yard/test apron: 2,200 m2 (23,681 ft2)
- Dynamic test track: separate short depot/test track, nominal 150-300 m plus charging interface; not counted in enclosed factory area
- Rough-order machinery/setup: $1,021,200 including 20% setup contingency

Factory Area By Cell

Cell	Area	Supports	Included work
steel prep and chassis parts cell	180 m2 (18 x 10 m)	`CP-020`	saw/plasma cutting, press-brake forming, drilling, deburr, and three-car chassis/body kitting
chassis weld and body-frame fixture cell	640 m2 (32 x 20 m)	`CP-030`, `CP-040`	two underframe rotating fixtures, two side/roof frame fixtures, local bolster/coupler subfixtures, weld extraction, and fixture survey access
composite moulding, cure, trim, and dry-fit cell	264 m2 (22 x 12 m)	`CP-060`, `CP-080`	four short GFRP moulds, lay-up tables, cure racks, CNC trim/drill table, master-frame dry-fit stand, and module staging
paint and corrosion protection cell	336 m2 (28 x 12 m)	`CP-070`	one car-length paint booth, prep/mask lane, flash-off/inspection space, seam-seal and cavity-wax station
bogie assembly and test cell	240 m2 (20 x 12 m)	`CP-050`, `CP-120`	three bogie stands, wheelset lane, brake/sensor bench, ride-height setup, and bogie-to-car marriage support tools
interior, HVAC duct, harness, and supplier kit bench	160 m2 (16 x 10 m)	`CP-065`, `CP-110`	seat/grab-rail racks, floor and liner trim bench, lighting/PIS/CCTV label bench, quarantine shelves, and car-zone kitting
final assembly, bogie marriage, and static-test track	600 m2 (60 x 10 m)	`CP-090`, `CP-100`, `CP-110`, `CP-120`, `CP-130`, `CP-140`	one 55 m controlled train bay with side access, roof platforms, HV lockout, shore power, charge simulator, jacks/mobile columns, and end-cowl stands
stores, incoming inspection, QA, toolroom, and offices	392 m2 (28 x 14 m)	`CP-010`	controlled document desk, stores, receiving inspection, calibrated-tool cage, NCR quarantine, compressor/electrical room, and welfare/office space
Circulation, services, safety aisles	703 m2	all cells	25% allowance over net process cells for forklifts, carts, fire egress, utilities, compressors, and WIP buffering
Total enclosed factory	3,515 m2	all cells	Minimum recommended lease/building area before offices are expanded or production rate is increased

Assembly-Time Rollup

Scope	Tasks	Start d	Finish d	Elapsed d	Touch labour h	Peak crew eq
chassis and painted carbody frame fabrication	`CP-020`, `CP-030`, `CP-040`, `CP-070`	2.0	15.0	13.0	1,284	13.1
bogie build and bogie-to-carbody integration	`CP-050`, `CP-120`	5.0	18.0	13.0	760	15.0
GFRP moulding and clip-on body installation	`CP-060`, `CP-080`	2.0	16.0	14.0	516	12.0
interior furnishing pre-kit and installation	`CP-065`, `CP-110`	2.0	22.0	20.0	780	17.5
final assembly and static commissioning	`CP-090`, `CP-100`, `CP-110`, `CP-120`, `CP-130`, `CP-140`	16.0	29.0	13.0	2,340	45.0
dynamic commissioning and trial-running release	`CP-150`	29.0	35.0	6.0	432	9.0

Machinery And Setup Estimate

ID	Item	Qty	Unit	Extended	Cell	Basis
M-010	4x8 or 5x10 CNC plasma table with extraction	1	$20,000	$20,000	steel prep	CNC plasma listings show 4x8 tables roughly in the $7k-$17k range; allowance includes extraction and industrialization.
M-020	RHS/tube saw and general fabrication saws	2	$6,000	$12,000	steel prep	Shop-floor allowance for repeatable chassis tube cutting and backup saw capacity.
M-030	100-160 t CNC press brake	1	$45,000	$45,000	steel prep	Market check shows low FOB offers around $18k and local/dealer reality nearer $45k for 100 t class machines.
M-040	drill/mill/magnetic-drill and deburr package	1	$18,000	$18,000	steel prep	Datum holes, inserts, fixtures, bracket slots, and local machining support.
M-050	500 A MIG/MAG welding sets	8	$2,000	$16,000	weld cell	Industrial 500 A MIG/MAG machines are commonly listed from sub-$1k FOB to around $1.5k-$2.5k retail; allowance includes torches/leads.
M-060	weld fume extraction and screens	1	$25,000	$25,000	weld cell	Shared extraction, curtains, and local arms for two long fixtures plus bracket benches.
M-070	underframe rotating weld fixtures	2	$40,000	$80,000	weld cell	Custom datum tooling for 16.5 m chassis underframes.
M-080	side/roof frame fixtures	2	$30,000	$60,000	weld cell	Custom fixtures for door/window portals, roof bows, HVAC rails, and clip-grid datums.
M-090	bogie frame fixture, three stands, and brake/sensor bench	1	$45,000	$45,000	bogie cell	Local assembly tooling for powered/trailer bogie frames and supplier wheelset/brake installation.
M-100	10 t bridge/gantry crane coverage and hoists	2	$37,500	$75,000	lifting	Crane price guides show 10 t single-girder equipment in the high single-digit to low tens of thousands FOB; allowance includes runway/installation margin.
M-110	3 t electric forklift	1	$25,000	$25,000	material handling	2026 3 t electric forklift guide ranges around $18k-$30k for higher-spec lithium machines.
M-120	mobile columns/jacks or shallow-pit bogie marriage kit	1	$75,000	$75,000	final assembly	Heavy-vehicle mobile-column sets range from lower-cost import sets to about $60k+ for branded 72,000 lb sets; allowance covers trainset adaptation.
M-130	bus/truck-length paint booth and prep ventilation	1	$70,000	$70,000	paint cell	Truck/bus booth market checks show roughly $13k-$50k FOB and $80k-$90k for larger custom booths; allowance sits between.
M-140	blast/strip prep, seam-seal, and cavity-wax package	1	$35,000	$35,000	paint cell	Lean in-house prep package; full blast hall can be outsourced or added later.
M-150	four short GFRP moulds with trim/drill jigs	1	$40,000	$40,000	composite cell	Reusable one-metre side/roof mould set plus master trim gauges; no full-car mould.
ID	Item	Qty	Unit	Extended	Cell	Basis
M-160	vacuum infusion/wet-layup pumps, reusable	1	$20,000	$20,000	composite cell	Vacuum infusion equipment is low-cost

ID	Item	Qty	Unit	Extended	Cell	Basis
	membranes, tables,and cure racks					at tool level; allowancecovers shop-readypumps, membranes,gauges, and racks.
M-170	CNC router/trim table and composite dust extraction	1	$45,000	$45,000	composite cell	Trim/drill accuracy for side/window/door/roof variants and cabin liners.
M-180	metrology, laser level/track, torque tools, and calibrated gauges	1	$35,000	$35,000	QA	Factory datum surveys, torque control, clip gauges, door/window gauges, and calibration control.
M-190	HV insulation, continuity, shore-power, and charge-simulator tools	1	$70,000	$70,000	final assembly	Static commissioning of HV battery, traction, charging, door/HVAC, and train-control interfaces.
M-200	interior trim benches, carts, racks, and hand tools	1	$25,000	$25,000	interior cell	Pre-kitting seats, floors, lighting, liners, PIS/CCTV, and labels by car zone.
M-210	factory IT, traveler terminals, printers, labels, and document control	1	$15,000	$15,000	production control	Shop traveler, QR label, NCR, and QA evidence capture support.
Subtotal				$851,000		
Setup contingency	20% install/adaptation/ commissioning allowance			$170,200		
Rough-order machinery total	Excludes building/land/ taxes/freight/duty			$1,021,200		

Production Takt Scenarios

Scenario	Trainsets/year	Final bays	Basis
first article / learning build	4	1	35 working-day elapsed build with protected rework float and one final bay
low-rate repeat build	8	1	25 working-day repeat takt after fixtures, supplier kits, and QA gates are stable
steady modular local production	12	1	20 working-day repeat takt; add second final bay only if static commissioning or supplier rework blocks the first bay

Exclusions

- land, building shell, leasehold works, taxes, freight, duty, and utility connection upgrades
- full homologation test laboratory and destructive crash/fire testing rigs
- working capital, warranty spares, and production payroll

Market Anchor Sources

The equipment costs are rough 2026 planning anchors, not supplier quotes. Final procurement must replace them with landed, installed, warranted offers.

- press brake: Alibaba/ADHMT 2026 press-brake market notes
- CNC plasma table: StyleCNC 2026 4x8 CNC plasma table listings
- truck/bus paint booth: Made-in-China 2026 bus spray booth listings
- vacuum infusion equipment: Easy Composites vacuum bagging/infusion equipment listings
- cranes: Voitto Crane 2026 overhead crane price guide
- mobile column lifts: BendPak mobile column lift listings
- forklift: Hongli 2026 3-ton electric forklift guide

LM3 factory drawing and interface readiness

Repository source: `design/component-catalogue/catalog/buildable-trainset/factory-release-readiness.md`

Generated by `tools/automation/buildable-trainset.sh`. This register is the review surface for issuing the 16 factory packages; blank template fields and `open-unissued` states are deliberate and are not release evidence.

- Template status: `unfilled-not-release-evidence`
- Packages: 16; open: 16
- Unique drawing IDs: 29
- Controlled product rows: 80
- Referenced tooling families: 30

Package	Status	Drawings	Products	Prerequisites	Outputs	Verifications
`LM3-FRP-010` — primary chassis, transverse structure and stepped-floor drawing pack	`open-unissued`	3	8	3	4	4
`LM3-FRP-020` — one-metre exterior module variant and retention drawing pack	`open-unissued`	3	5	3	4	4
`LM3-FRP-030` — panoramic front-glass carrier, seal and drainage interface pack	`open-unissued`	2	4	3	4	4
`LM3-FRP-040` — reversible front-lamp cassette and fascia-service pack	`open-unissued`	2	4	3	4	4
`LM3-FRP-050` — roof curb, HVAC, PV, antenna, fairing and access-zone pack	`open-unissued`	3	10	3	4	4
`LM3-FRP-060` — interior moulding, floor, service-access and fitout pack	`open-unissued`	2	19	3	4	4
`LM3-FRP-070` — common service rail, fastener and fixture-adapter pack	`open-unissued`	2	3	3	4	4
`LM3-FRP-080` — pre-cut exterior film artwork, application and repair pack	`open-unissued`	1	9	3	4	4
`LM3-FRP-090` — radiative roof-coating coupon and one-car trial pack	`open-unissued`	1	4	3	4	4
`LM3-FRP-100` — vehicle jacking, lifting, towing and field-railing interface pack	`open-unissued`	2	1	3	4	4
`LM3-FRP-110` — primary end, battery, door and window structural-interface pack	`open-unissued`	4	6	3	4	4
`LM3-FRP-120` — configurable train-end and articulation structural-interface pack	`open-unissued`	3	6	3	4	4
`LM3-FRP-130` — powered-bogie local frame, bracket and harness drawing pack	`open-unissued`	1	4	3	4	4
`LM3-FRP-140` — trailer-bogie local frame, bracket and harness drawing pack	`open-unissued`	1	3	3	4	4
`LM3-FRP-150` — battery tray, high-voltage and coolant local-hardware pack	`open-unissued`	2	3	3	4	4
`LM3-FRP-160` — low-voltage trainline harness and terminal-distribution pack	`open-unissued`	1	1	3	4	4

Fill the machine-readable `factory-release-record-template.json` during controlled drawing production. A package remains open until every source input, issued drawing, exact product configuration, tool/gauge record, required output, performed verification and named approval is accepted. Use the per-drawing `factory-drawings/` seeds as drafting/checking briefs, never as issued manufacturing drawings.

Boundary: These work packages define drawing and evidence scope. They are not approved drawings, supplier selections, signed calculations or physical first-article records.

LM3 v2A factory drawing and interface-release work packages

Repository source: `design/component-catalogue/catalog/buildable-trainset/factory-release-work-packages.md`

Generated by `tools/automation/buildable-trainset.sh`. These packages turn the dedicated LM3 part families into an explicit drawing, datum, tooling and verification worklist without implying that supplier or physical evidence already exists. Package issue state is tracked in `factory-release-readiness.md`. Individual drafting briefs are under `factory-drawings/`.

- Design ID: `LM3-V2A-FACTORY-RELEASE`
- Status: `controlled-work-package-definitions; release evidence open`
- Work packages: `16`
- Controlled product rows represented: `80`
- Registered tooling referenced: `30`
- Reference work instructions: `manufacturing-and-assembly-controls.md`
- Boundary: These work packages define drawing and evidence scope. They are not approved drawings, supplier selections, signed calculations or physical first-article records.

Package index

Package	Scope	Drawings	Products	Tooling
LM3-FRP-010	primary chassis, transverse structure and stepped-floor drawing pack	LM3-BDY-100 LM3-BDY-110 LM3-BDY-120	8	2
LM3-FRP-020	one-metre exterior module variant and retention drawing pack	LM3-BDY-150 LM3-BDY-160 LM3-BDY-165	5	4
LM3-FRP-030	panoramic front-glass carrier, seal and drainage interface pack	LM3-BDY-155 LM3-FAS-180	4	3
LM3-FRP-040	reversible front-lamp cassette and fascia-service pack	LM3-BDY-155 LM3-FAS-185	4	2
LM3-FRP-050	roof curb, HVAC, PV, antenna, fairing and access-zone pack	LM3-HVAC-220 LM3-HV-325 LM3-ROOF-225	10	3
LM3-FRP-060	interior moulding, floor, service-access and fitout pack	LM3-INT-230 LM3-INT-231	19	6
LM3-FRP-070	common service rail, fastener and fixture-adapter pack	LM3-INT-230 LM3-FIX-235	3	2
LM3-FRP-080	pre-cut exterior film artwork, application and repair pack	LM3-FIN-240	9	2
LM3-FRP-090	radiative roof-coating coupon and one-car trial pack	LM3-FIN-245	4	2
LM3-FRP-100	vehicle jacking, lifting, towing and field-rerailing interface pack	LM3-BDY-100 LM3-REC-270	1	3
LM3-FRP-110	primary end, battery, door and window structural-interface pack	LM3-BDY-130 LM3-BDY-140 LM3-DOOR-200 LM3-WIN-210	6	5
LM3-FRP-120	configurable train-end and articulation structural-interface pack	LM3-BDY-130 LM3-END-650 LM3-SYS-170	6	4
LM3-FRP-130	powered-bogie local frame, bracket and harness drawing pack	LM3-BOG-400	4	4
LM3-FRP-140	trailer-bogie local frame, bracket and harness drawing pack	LM3-BOG-410	3	3
LM3-FRP-150	battery tray, high-voltage and coolant local-hardware pack	LM3-HV-310 LM3-HV-320	3	3
LM3-FRP-160	low-voltage trainline harness and terminal-distribution pack	LM3-ELC-300	1	2

LM3-FRP-010 — primary chassis, transverse structure and stepped-floor drawing pack

Release boundary: No steel cutting release until structural calculations, WPS/PQR, material certificates and drawing approval are present.

Controlled products and geometry

Product	Route / maturity	Qty	Parent	Design-reference envelope (mm)	Reference default	Representation
LM3-BDY-P010 — laser-cut side sill beam, LH/RH	MAKE / release-candidate	6 ea	LM3-BDY-SA110	16000 × 180 × 260	local manufacture	one side-sill member with end datum faces
LM3-BDY-P020 — underframe centre spine and longitudinal load-path kit	MAKE / release-candidate	3 kit	LM3-BDY-SA110	15800 × 2650 × 380	local manufacture	centre spine and repeated cross-bearers
LM3-BDY-P021 — underframe cross-bearer, door-bay outrigger, and equipment-bracket pack	MAKE / release-candidate	3 car pack	LM3-BDY-SA110	15200 × 2650 × 320	local manufacture	stationed cross-bearers, door outriggers and equipment brackets
LM3-BDY-P030 — bolster box, air-spring pad, and centre-pivot insert set	MAKE / release-candidate	6 set	LM3-BDY-SA110	1700 × 2650 × 420	local manufacture	bolster box, spring pads and pivot land
LM3-BDY-P060 — low-floor centre pan and removable service-floor support set	MAKE / release-candidate	3 set	LM3-BDY-SA120	15600 × 2650 × 460	local manufacture	low-floor pan and raised bogie-end decks
LM3-BDY-P061 — raised bogie-end deck, transition ramp, and removable hatch-frame set	MAKE / release-candidate	3 car set	LM3-BDY-SA120	5200 × 2650 × 460	local manufacture	raised end decks, transition ramps and removable hatch frames
LM3-BDY-P070 — side-wall post, door portal, waist rail, and cant rail kit	MAKE / release-candidate	6 side	LM3-BDY-SA120	15800 × 180 × 2850	local manufacture	posts, door portals, waist and cant rails
LM3-BDY-P080 — roof bow, HVAC rail, PV rail, and cable-tray bracket kit	MAKE / release-candidate	3 kit	LM3-ROOF-SA410	15600 × 2500 × 360	local manufacture	roof bows and equipment/cable rails

Frozen inputs

- released structural load cases and material grades
- door, window, bogie, battery, coupler and roof interface reactions
- weld-class, distortion and corrosion-process requirements

Controlled outputs

- member and plate drawings with datums, tolerances and material callouts
- cut, bend, machining and weld maps with heat/part trace fields
- fixture loading sequence and dimensional inspection characteristic list
- mass-properties contribution and controlled centre-of-gravity ledger rows

Verification and hold points

- independent calculation review
- fixture and tack survey
- post-weld datum/straightness survey
- weld/NDT and corrosion hold-point records

Tooling: LM3-TOOL-STEEL-FIXTURE, LM3-TOOL-DATUM-GAUGE.

Reference controls: LM3-MFG-010, LM3-MFG-020, LM3-MFG-100.

LM3-FRP-020 — one-metre exterior module variant and retention drawing pack

Release boundary: Module skins remain non-structural and may not close primary carbody or roof-equipment load paths.

Controlled products and geometry

Product	Route / maturity	Qty	Parent	Design-reference envelope (mm)	Reference default	Representation
LM3-BDY-P130 — one-metre clip-on solid-side fiberglass body module	MAKE / release-candidate	48 module	LM3-SHELL-A200	1000 × 120 × 3050	local manufacture	one-metre clip-on side/roof module with solid attachment lands
LM3-BDY-P131 — one-metre clip-on window-edge fiberglass side module	MAKE / release-candidate	24 module	LM3-SHELL-A200	1000 × 160 × 3050	local manufacture	window-edge side module with reveal and drain clearance
LM3-BDY-P132 — one-metre clip-on door-edge fiberglass side module	MAKE / release-candidate	24 module	LM3-SHELL-A200	1000 × 180 × 3050	local manufacture	door-edge side module with pocket and threshold closeout
LM3-BDY-P133 — one-metre clip-on fiberglass roof skin and equipment-fairing module	MAKE / release-candidate	48 module	LM3-SHELL-A200	1000 × 2850 × 420	local manufacture	roof skin and equipment-fairing trim variant
LM3-BDY-P140 — keyed clip rail, captive retainer, anti-lift, and dry-seal car kit	MAKE / release-candidate	3 car kit	LM3-SHELL-A200	15800 × 180 × 160	local manufacture	keyed clip rail, retainers and dry-seal route

Frozen inputs

- accepted laminate, core, gelcoat, insert, seal and fire-material system
- frozen car bay map and door/window/roof service-clearance model
- released clip, anti-lift, pressure and debris load cases

Controlled outputs

- A/B tool-face, split, draft, trim and insert drawings
- solid/window/door/roof variant CNC trim and drill definitions
- clip, seal, anti-lift and drain interface drawing
- serialized module/bay configuration and repair map

Verification and hold points

- mould survey and witness coupon
- trim/drill first article
- master-frame interchange and anti-lift proof
- water, vibration and timed module replacement trial

Tooling: LM3-TOOL-SIDE-MOULD, LM3-TOOL-SIDE-VARIANT-NEST, LM3-TOOL-ROOF-MOULD, LM3-TOOL-TRIM-DRILL.

Reference controls: LM3-MFG-010, LM3-MFG-040, LM3-MFG-050, LM3-MFG-100.

LM3-FRP-030 — panoramic front-glass carrier, seal and drainage interface pack

Release boundary: The selected glazing supplier retains responsibility for pane construction and its bonded cassette process.

Controlled products and geometry

Product	Route / maturity	Qty	Parent	Design-reference envelope (mm)	Reference default	Representation
LM3-CWL-P016 — CWL-FRP-06 backing-ring flange fiberglass cast set	MAKE / buildable-after-supplier-freeze	8 flange section	LM3-CWL-SA710	720 × 120 × 720	local manufacture	backing-ring flange cast
LM3-FAS-P010 — panoramic front-glass carrier ring, setting-block pockets, and secondary-retention frame	MAKE / buildable-after-supplier-freeze	2 end set	LM3-CWL-SA710	2300 × 260 × 1780	local manufacture	panoramic glass carrier, setting blocks and secondary retention
LM3-FAS-P030 — front glazing/lamp EPDM seal, drain rail, washer sleeve, and edge-closeout kit	SOURCE / buildable-after-supplier-freeze	2 end kit	LM3-CWL-SA710	2500 × 420 × 1950	specified	glazing/lamp seals, drains, washer sleeves and closeouts
LM3-EXT-P030 — single panoramic heated end-glass assembly	BID / buildable-after-supplier-freeze	2 ea	LM3-END-SA700	2300 × 110 × 1450	specified	heated panoramic end glazing

Frozen inputs

- supplier glass construction, edge, mass, heater and retention loads
- accepted glazing adhesive/gasket and EPDM compatibility data
- released steel backing-ring and cowl flange datums

Controlled outputs

- carrier segments, corner joints, setting blocks and secondary-retention drawing
- glass-edge clearance and seal-compression characteristic map
- drain rail, washer sleeve, earth and heater-service route
- protected removal path and lifting-tool access drawing

Verification and hold points

- carrier and backing-ring datum survey
- retention calculation and representative proof
- compression/drain map and controlled spray test
- heated-pane service isolation and timed removal/refit trial

Tooling: LM3-TOOL-COWL-MOULD, LM3-TOOL-GLASS-CARRIER-NEST, LM3-TOOL-WATER-TEST.

Reference controls: LM3-MFG-010, LM3-MFG-040, LM3-MFG-050, LM3-MFG-100.

LM3-FRP-040 — reversible front-lamp cassette and fascia-service pack

Release boundary: The jig establishes mechanical aim only; supplier and vehicle photometric evidence remains mandatory.

Controlled products and geometry

Product	Route / maturity	Qty	Parent	Design-reference envelope (mm)	Reference default	Representation
LM3-CWL-P014 — CWL-FRP-04 lower apron and anti-climber cover fiberglass cast	MAKE / buildable-after-supplier-freeze	2 ea	LM3-CWL-SA710	1000 × 2650 × 760	local manufacture	lower apron and anti-climber cover cast
LM3-CWL-P015 — CWL-FRP-05 lamp, washer, and service-hatch fiberglass cast set	MAKE / buildable-after-supplier-freeze	4 hatch	LM3-CWL-SA710	650 × 520 × 420	local manufacture	lamp/washer/service-hatch cast set
LM3-FAS-P020 — reversible front-lamp cassette tray, aiming adjusters, and retained service bracket	MAKE / buildable-after-supplier-freeze	2 end set	LM3-CWL-SA710	1850 × 720 × 420	local manufacture	reversible lamp cassette tray and aiming adjusters
LM3-END-P050 — sealed headlight, tail/marker light, threshold-warning, and end-lamp harness kit	SOURCE / buildable-after-supplier-freeze	2 end kit	LM3-END-SA700	1600 × 900 × 280	specified	head/tail/marker/threshold lamps and harness

Frozen inputs

- selected lamp photometric, thermal, EMC, IP and connector data
- A/B end configuration and cowl service-hatch envelope
- released lamp reaction, aiming and retention requirements

Controlled outputs

- common reversible tray, adjuster and retained service-bracket drawing
- lamp optical-axis datum and adjustment-limit schedule
- harness, earth, drip-loop and connector-access route
- A/B interchange and lamp-cassette configuration record

Verification and hold points

- tray and optical-axis datum gauge
- aim range, lock and vibration-retention test
- thermal/IP/functional evidence review
- service-hatch and cassette removal demonstration

Tooling: LM3-TOOL-COWL-MOULD, LM3-TOOL-LAMP-AIM.

Reference controls: LM3-MFG-010, LM3-MFG-030, LM3-MFG-050, LM3-MFG-100.

LM3-FRP-050 — roof curb, HVAC, PV, antenna, fairing and access-zone pack

Release boundary: No supplier unit or person load may be reacted through a clip-on skin or non-structural fairing.

Controlled products and geometry

Product	Route / maturity	Qty	Parent	Design-reference envelope (mm)	Reference default	Representation
LM3-BDY-P080 — roof bow, HVAC rail, PV rail, and cable-tray bracket kit	MAKE / release-candidate	3 kit	LM3-ROOF-SA410	15600 × 2500 × 360	local manufacture	roof bows and equipment/cable rails
LM3-BDY-P133 — one-metre clip-on fiberglass roof skin and equipment-fairing module	MAKE / release-candidate	48 module	LM3-SHELL-A200	1000 × 2850 × 420	local manufacture	roof skin and equipment-fairing trim variant
LM3-ROOF-P010 — HVAC curb, drop-duct collar, condensate tray, and drain fitting kit	MAKE / release-candidate	3 kit	LM3-ROOF-SA410	2800 × 1800 × 420	local manufacture	HVAC curb, duct collar, tray and drains
LM3-ROOF-P020 — PV bonded-pad lands, raised rail kit, bonding jumpers, and roof isolation labels	MAKE / release-candidate	3 car kit	LM3-ROOF-SA410	14000 × 2300 × 140	local manufacture	PV rails, bonded pads, jumpers and isolation datums
LM3-ROOF-P030 — removable HVAC curb fairing, intake/exhaust skirt, and access-hatch moulding set	MAKE / buildable-after-supplier-freeze	3 car set	LM3-ROOF-SA410	3200 × 2400 × 720	local manufacture	removable HVAC curb fairing, skirts and access hatches
LM3-ROOF-P040 — PV junction plinth, cable-gland cover, antenna closeout, and walkway edge set	MAKE / buildable-after-supplier-freeze	3 car set	LM3-ROOF-SA410	12000 × 2300 × 320	local manufacture	PV plinth, glands, antenna closeouts and walkway edges
LM3-EXT-P040 — hvac-24kw-direct-hv-dc roof HVAC	BID / buildable-after-supplier-freeze	3 ea	LM3-ROOF-SA410	2600 × 1900 × 620	specified	roof HVAC casing, fans and curb interface
LM3-EXT-P050 — roof PV module and edge-clamp kit	SOURCE / release-candidate	36 module	LM3-ROOF-SA410	1700 × 1050 × 45	specified	PV laminate and edge-clamp datums
LM3-EXT-P070 — roof antennas, service walkway pads, lifting covers, and maintenance labels	SOURCE / buildable-after-supplier-freeze	3 car kit	LM3-ROOF-SA410	2400 × 1500 × 260	specified	antennas, walkway pads, covers and labels
LM3-TRC-P050 — roof-mounted regen dump resistor and thermal shield kit	SOURCE / buildable-after-supplier-freeze	3 ea	LM3-ROOF-SA410	1900 × 750 × 420	specified	regen resistor bank and thermal shield

Frozen inputs

- supplier HVAC, PV, resistor and antenna mass/reaction/service envelopes
- released roof-rail capacity, vehicle gauge and lifting/fall-access constraints
- accepted sealing, bonding, fire and electrical-isolation systems

Controlled outputs

- roof equipment coordinate and penetration schedule
- curb, duct, drain, rail/pad, gland, bond and fairing drawings
- HVAC/PV/antenna removal paths and tool/worker access zones
- finish, anti-slip, heat and electrical keep-out mask map

Verification and hold points

- rail/curb datum and equipment fit survey
- attachment proof, bond continuity and isolation checks
- roof/condensate water tests and airflow clearance review
- service-removal demonstration with adjacent equipment retained

Tooling: LM3-TOOL-ROOF-MOULD, LM3-TOOL-ROOF-FAIRING-MOULD, LM3-TOOL-WATER-TEST.

Reference controls: LM3-MFG-010, LM3-MFG-030, LM3-MFG-040, LM3-MFG-060, LM3-MFG-100.

LM3-FRP-060 — interior moulding, floor, service-access and fitout pack

Release boundary: Interior finish panels carry no seat, handrail, equipment or passenger restraint load.

Controlled products and geometry

Product	Route / maturity	Qty	Parent	Design-reference envelope (mm)	Reference default	Representation
LM3-INT-P010 — HVAC diffusers, side return ducts, saloon grilles, and access panels	SOURCE / buildable-after-supplier-freeze	3 car kit	LM3-INT-SA330	6000 × 2400 × 320	specified	diffusers, returns, grilles and access panels
LM3-INT-P020 — FRP/phenolic ceiling liner panel set	MAKE / buildable-after-supplier-freeze	3 car kit	LM3-INT-SA330	1000 × 2650 × 260	local manufacture	ceiling liner, light trough and plenum cover
LM3-INT-P021 — ceiling light-trough bezel, diffuser carrier, and emergency-luminaire closeout set	MAKE / buildable-after-supplier-freeze	3 car set	LM3-INT-SA330	1200 × 240 × 110	local manufacture	light-trough bezels and diffuser carriers
LM3-INT-P022 — HVAC plenum cover, diffuser transition, detector bezel, and ceiling service-hatch set	MAKE / buildable-after-supplier-freeze	3 car set	LM3-INT-SA330	1100 × 900 × 220	local manufacture	HVAC transitions, detector bezels and ceiling service hatches
LM3-INT-P030 — FRP/phenolic sidewall liner panel set	MAKE / buildable-after-supplier-freeze	6 side kit	LM3-INT-SA330	1000 × 140 × 2200	local manufacture	sidewall liner, reveal and cable-cover panel
LM3-INT-P031 — window reveal, setting-block inspection cover, and blind/label land set	MAKE / buildable-after-supplier-freeze	6 side set	LM3-INT-SA330	1750 × 180 × 1300	local manufacture	four-piece removable window reveal and inspection covers
LM3-INT-P032 — waist cable-cover, low-voltage access lid, and retained quarter-turn fastener set	MAKE / buildable-after-supplier-freeze	6 side set	LM3-INT-SA330	1000 × 220 × 520	local manufacture	waist cable cover and retained access lid
LM3-INT-P040 — FRP battery strake cover panel set	MAKE / buildable-after-supplier-freeze	3 car kit	LM3-INT-SA330	4800 × 720 × 780	local manufacture	battery strake/seat-base/service-hatch shells
LM3-INT-P041 — seat-base fairing, retained equipment hatch, and rail-saddle closeout set	MAKE / buildable-after-supplier-freeze	3 car set	LM3-INT-SA330	1600 × 780 × 820	local manufacture	seat-base fairing and retained equipment hatch
LM3-INT-P050 — FRP vestibule kick-panel and threshold-closeout set	MAKE / buildable-after-supplier-freeze	12 door-zone kit	LM3-INT-SA330	1100 × 420 × 1050	local manufacture	vestibule, PRM and door-pocket trim set
LM3-INT-P051 — PRM transition-ramp, bogie-deck step-cover, contrast-nosing, and anti-slip set	MAKE / buildable-after-supplier-freeze	6 end-zone set	LM3-INT-SA330	2500 × 1200 × 460	local manufacture	PRM transition, step cover, nosing and anti-slip set
LM3-INT-P052 — door-pocket liner, sensitive-edge clearance trim, and removable jamb-cover set	MAKE / buildable-after-supplier-freeze	12 door-zone set	LM3-INT-SA330	1200 × 420 × 2100	local manufacture	door-pocket liner and removable jamb covers
LM3-EXT-P060 — stepped floor-board and removable service-hatch system	SOURCE / buildable-after-supplier-freeze	135 m2	LM3-INT-SA330	1000 × 1300 × 35	specified	numbered floor board/removable hatch module
LM3-EXT-P061 — welded resilient floor covering, cove, nosing, and adhesive system	SOURCE / buildable-after-supplier-freeze	135 m2	LM3-INT-SA330	1000 × 1300 × 6	specified	resilient covering and welded-seam allowance
LM3-EXT-P062 — longitudinal passenger and priority-seat modules	SOURCE / buildable-after-supplier-freeze	60 seat	LM3-INT-SA330	1550 × 620 × 980	specified	longitudinal passenger-seat module and rail saddles
LM3-EXT-P063 — stainless grab-pole, handrail, joint, and insulated adapter kit	SOURCE / buildable-after-supplier-freeze	3 car kit	LM3-INT-SA330	2300 × 2300 × 2050	specified	grab poles, rails and repeated adapters
LM3-EXT-P064 — passenger-information display, speaker, amplifier, and mounting kit	SOURCE / buildable-after-supplier-freeze	3 car kit	LM3-INT-SA330	1100 × 220 × 420	specified	display, loudspeaker and amplifier module
LM3-EXT-P065 — CCTV camera, passenger intercom, PoE/data, and mounting kit	SOURCE / buildable-after-supplier-freeze	3 car kit	LM3-INT-SA330	650 × 420 × 260	specified	CCTV/intercom modules and mounting datums

Product	Route / maturity	Qty	Parent	Design-reference envelope (mm)	Reference default	Representation
LM3-EXT-P066 — PRM, safety-signage, emergency-lighting, extinguisher, and first-aid kit	SOURCE / buildable-after-supplier-freeze	3 car kit	LM3-INT-SA330	1200 × 650 × 600	specified	PRM controls, signs, lamps and emergency equipment

Frozen inputs

- accepted fire/smoke material and supplier equipment evidence
- frozen door/window/HVAC/lighting/seat/PRM equipment envelopes
- released passenger load, egress, accessibility and cleaning requirements

Controlled outputs

- individual panel, mould, trim, edge-radius and insert drawings
- floor board/covering seam, hatch, ramp and threshold definitions
- service-access and sequential removal map for every concealed system
- serialized panel, fastener, equipment and label configuration schedule

Verification and hold points

- mould/trim first article and fire-material trace review
- dry fit, gap, rattle, sharp-edge and captive-fastener inspections
- door/window/equipment removal sweeps and accessibility survey
- floor load, slip, seam, cleaning and water-path evidence

Tooling: LM3-TOOL-INT-CEILING-MOULD, LM3-TOOL-INT-SIDE-MOULD, LM3-TOOL-INT-STRAKE-MOULD, LM3-TOOL-INT-DOOR-PRM-MOULD, LM3-TOOL-FLOOR-TEMPLATE, LM3-TOOL-SERVICE-RAIL.

Reference controls: LM3-MFG-010, LM3-MFG-030, LM3-MFG-050, LM3-MFG-070, LM3-MFG-100.

LM3-FRP-070 — common service rail, fastener and fixture-adapter pack

Release boundary: Each adapter needs its own released load case; one generic proof result cannot qualify every fixture.

Controlled products and geometry

Product	Route / maturity	Qty	Parent	Design-reference envelope (mm)	Reference default	Representation
LM3-FIX-P010 — OSR-RAIL-42 common ceiling, waist, and seat-zone service rail kit	MAKE / release-candidate	3 car kit	LM3-FIX-SA340	15000 × 42 × 18	local manufacture	OSR-RAIL-42 extrusion with repeated datum marks
LM3-FIX-P020 — four-family captive fastener, floating nut, isolator, and access-fastener kit	SOURCE / buildable-after-supplier-freeze	3 car kit	LM3-FIX-SA340	520 × 360 × 160	specified	four captive fastener families and isolators
LM3-FIX-P030 — standard passenger-fixture saddle and equipment adapter kit	MAKE / concept	3 car kit	LM3-FIX-SA340	520 × 420 × 180	local manufacture	seat, handrail and equipment adapter variants

Frozen inputs

- fixture-specific service/ultimate loads and attachment envelopes
- selected rail alloy/temper, fastener, isolator and finish data
- carbody attachment capacity and electrical/galvanic constraints

Controlled outputs

- rail extrusion, cut/drill, end-stop and foot drawing
- seat, handrail, PIS/CCTV and cable-support adapter variants

- fastener grip, locking, torque-authority and captive-part schedule
- installed coordinate, orientation and accessible-removal map

Verification and hold points

- rail/foot and fixture-specific calculation review
- adapter and installed-grip gauges
- representative pull/slip/rotation proof
- egress, snag, isolation and timed replacement inspection

Tooling: `LM3-TOOL-SERVICE-RAIL`, `LM3-TOOL-FIXTURE-PROOF`.

Reference controls: `LM3-MFG-010`, `LM3-MFG-030`, `LM3-MFG-070`, `LM3-MFG-100`.

LM3-FRP-080 — pre-cut exterior film artwork, application and repair pack

Release boundary: Film replaces decorative masking only and never substitutes for corrosion, fire, sealing or electrical protection.

Controlled products and geometry

Product	Route / maturity	Qty	Parent	Design-reference envelope (mm)	Reference default	Representation
`LM3-FIN-P010` — pre-cut exterior livery graphic film, edge-seal, datum-mark, and repair-patch kit	`BID` / `buildable-after-supplier-freeze`	3 car kit	`LM3-SHELL-A200`	16000 × 1400 × 8	`specified`	pre-cut side/cowl livery film, edge seals and repair patches
`LM3-BDY-P130` — one-metre clip-on solid-side fiberglass body module	`MAKE` / `release-candidate`	48 module	`LM3-SHELL-A200`	1000 × 120 × 3050	local manufacture	one-metre clip-on side/roof module with solid attachment lands
`LM3-BDY-P131` — one-metre clip-on window-edge fiberglass side module	`MAKE` / `release-candidate`	24 module	`LM3-SHELL-A200`	1000 × 160 × 3050	local manufacture	window-edge side module with reveal and drain clearance
`LM3-BDY-P132` — one-metre clip-on door-edge fiberglass side module	`MAKE` / `release-candidate`	24 module	`LM3-SHELL-A200`	1000 × 180 × 3050	local manufacture	door-edge side module with pocket and threshold closeout
`LM3-CWL-P011` — CWL-FRP-01 upper brow and roof-cap fiberglass cast	`MAKE` / `buildable-after-supplier-freeze`	2 ea	`LM3-CWL-SA710`	1250 × 2750 × 720	local manufacture	upper brow and roof-cap cast
`LM3-CWL-P012` — CWL-FRP-02 left cheek fiberglass cast	`MAKE` / `buildable-after-supplier-freeze`	2 ea	`LM3-CWL-SA710`	1150 × 700 × 2300	local manufacture	handed cheek cast
`LM3-CWL-P013` — CWL-FRP-03 right cheek fiberglass cast	`MAKE` / `buildable-after-supplier-freeze`	2 ea	`LM3-CWL-SA710`	1150 × 700 × 2300	local manufacture	handed cheek cast
`LM3-CWL-P014` — CWL-FRP-04 lower apron and anti-climber cover fiberglass cast	`MAKE` / `buildable-after-supplier-freeze`	2 ea	`LM3-CWL-SA710`	1000 × 2650 × 760	local manufacture	lower apron and anti-climber cover cast
`LM3-CWL-P015` — CWL-FRP-05 lamp, washer, and service-hatch fiberglass cast set	`MAKE` / `buildable-after-supplier-freeze`	4 hatch	`LM3-CWL-SA710`	650 × 520 × 420	local manufacture	lamp/washer/service-hatch cast set

Frozen inputs

- approved artwork, colours, logos and operator/legal markings
- selected rail-use film/ink/overlaminate/edge system
- accepted cured steel/GFRP base finishes and cleaning process

Controlled outputs

- bay-numbered cut files with controlled seams, overlaps and datum marks
- substrate acceptance, environment and installer record forms
- edge/keep-out inspection map and patch/one-metre module repair instruction

- film material, print batch and retained-coupon trace schedule

Verification and hold points

- actual-substrate adhesion/removal coupon
- first-car edge, seam and appearance inspection
- wash/water and cleaning-chemical compatibility test
- local patch and complete module removal demonstration

Tooling: `LM3-TOOL-FILM-TEMPLATE`, `LM3-TOOL-COATING-COUPON`.

Reference controls: `LM3-MFG-090`, `LM3-MFG-100`.

LM3-FRP-090 — radiative roof-coating coupon and one-car trial pack

Release boundary: Research values are screening targets only; the baseline qualified light roof finish remains available if the trial fails.

Controlled products and geometry

Product	Route / maturity	Qty	Parent	Design-reference envelope (mm)	Reference default	Representation
`LM3-FIN-P020` — calcium-carbonate radiative roof-coating qualification and exposed-roof application kit	`BID` / `concept`	3 car kit	`LM3-ROOF-SA410`	16000 × 2700 × 8	`specified`	masked exposed-roof radiative coating and witness coupons
`LM3-BDY-P133` — one-metre clip-on fiberglass roof skin and equipment-fairing module	`MAKE` / `release-candidate`	48 module	`LM3-SHELL-A200`	1000 × 2850 × 420	local manufacture	roof skin and equipment-fairing trim variant
`LM3-ROOF-P030` — removable HVAC curb fairing, intake/exhaust skirt, and access-hatch moulding set	`MAKE` / `buildable-after-supplier-freeze`	3 car set	`LM3-ROOF-SA410`	3200 × 2400 × 720	local manufacture	removable HVAC curb fairing, skirts and access hatches
`LM3-ROOF-P040` — PV junction plinth, cable-gland cover, antenna closeout, and walkway edge set	`MAKE` / `buildable-after-supplier-freeze`	3 car set	`LM3-ROOF-SA410`	12000 × 2300 × 320	local manufacture	PV plinth, glands, antenna closeouts and walkway edges

Frozen inputs

- controlled formulation and application process
- accepted rail fire/chemical and GFRP compatibility evidence
- frozen roof equipment, service, drainage and keep-out map

Controlled outputs

- coupon dimensions, substrate, coating thickness and exposure matrix
- new/aged/soiled/cleaned optical measurement procedure
- one-car paired roof-zone sensor and HVAC-energy trial plan
- inspection, cleaning, local repair and trial-removal instructions

Verification and hold points

- adhesion, flexibility, thermal-cycle, UV, abrasion and wash ageing
- new and aged solar reflectance/emittance measurement
- glare, runoff, soiling and service-access review
- representative hot-service one-car trial and independent disposition

Tooling: `LM3-TOOL-RADIATIVE-COUPON`, `LM3-TOOL-COATING-RACK`.

Reference controls: `LM3-MFG-060`, `LM3-MFG-090`, `LM3-MFG-100`.

LM3-FRP-100 — vehicle jacking, lifting, towing and field-rerailing interface pack

Release boundary: Automotive scissor jacks, unilateral lifts and work beneath an uncribbed hydraulic load remain prohibited.

Controlled products and geometry

Product	Route / maturity	Qty	Parent	Design-reference envelope (mm)	Reference default	Representation
`LM3-BDY-P120` — jacking pad, lifting eye, towing lug, and recovery-label kit	`MAKE` / `release-candidate`	3 car kit	`LM3-BDY-SA110`	1600 × 900 × 280	local manufacture	jacking, lifting and towing interface kit

Frozen inputs

- individual-car mass and centre-of-gravity envelopes
- released underframe, articulation, bogie-retention and coupler load cases
- selected depot and portable recovery equipment interfaces

Controlled outputs

- J1--J4 pad, lifting eye, tow/rerailing lug and keyed-adapter drawings
- permitted support combinations, reactions, stop conditions and labels
- vehicle isolation, brake release, bogie retention and recovery diagrams
- proof-load, inspection, NDT, maintenance and damage-rejection schedule

Verification and hold points

- structural and weld/NDT calculation review
- four-point datum/interchange gauge
- representative proof and asymmetric/loss-of-pressure trials
- timed depot lift and field rerailing demonstration by trained crews

Tooling: `LM3-TOOL-STEEL-FIXTURE`, `LM3-TOOL-DATUM-GAUGE`, `LM3-TOOL-LIFT-COLUMNS`.

Reference controls: `LM3-MFG-010`, `LM3-MFG-030`, `LM3-MFG-080`, `LM3-MFG-100`.

LM3-FRP-110 — primary end, battery, door and window structural-interface pack

Release boundary: Reference envelopes do not release load paths, glazing retention, door carriers or battery restraints for manufacture.

Controlled products and geometry

Product	Route / maturity	Qty	Parent	Design-reference envelope (mm)	Reference default	Representation
`LM3-BDY-P040` — coupler pocket, shear plate, and crash-can insert kit	`MAKE` / `release-candidate`	2 set	`LM3-END-SA700`	1250 × 1100 × 620	local manufacture	coupler pocket and crash-can shear structure
`LM3-BDY-P050` — battery tray rails, vent plenum, and service-lid gutter kit	`MAKE` / `release-candidate`	3 kit	`LM3-HV-SA510`	6200 × 900 × 520	local manufacture	battery support rails, vent plenum and service gutter
`LM3-BDY-P100` — door portal reinforcement, threshold beam, and cassette shim kit	`MAKE` / `release-candidate`	12 opening kit	`LM3-DOOR-SA310`	1500 × 260 × 2150	local manufacture	door carrier/portal with threshold and four-point datum
`LM3-BDY-P110` — window carrier ring, bonded-gasket land, and replacement jack-point inserts	`MAKE` / `release-candidate`	18 opening kit	`LM3-WIN-SA320`	1650 × 140 × 1250	local manufacture	replaceable glazing pressure frame, seal and drain
`LM3-DOOR-P010` — door four-point adjustable carrier, datum pin, dry seal, and keyed connector bracket kit	`MAKE` / `concept`	12 opening kit	`LM3-DOOR-SA310`	1500 × 260 × 2150	local manufacture	door carrier/portal with threshold and four-point datum
`LM3-WIN-P010` — replaceable window pressure frame, dry seal, drain, and captive retention kit	`MAKE` / `concept`	18 opening kit	`LM3-WIN-SA320`	1650 × 140 × 1250	local manufacture	replaceable glazing pressure frame, seal and drain

Frozen inputs

- released crash, battery-restraint, door and glazing interface reactions
- supplier-frozen coupler, battery, door and window installation envelopes
- released structural, seal/drain, corrosion and service-removal requirements

Controlled outputs

- coupler-pocket, battery-rail, door-portal and window-carrier detail drawings
- machined datum, shim/adjustment, seal compression and drain schedules
- supplier keep-out, installation and complete service-removal maps
- weld/fastener/bond authority, tolerance and inspection characteristic lists

Verification and hold points

- independent structural and retention calculation review
- post-weld datum plus door/window aperture survey
- carrier proof, dry fit and timed supplier-cassette replacement trials
- battery drain/vent and door/window water-ingress tests

Tooling: `LM3-TOOL-STEEL-FIXTURE`, `LM3-TOOL-DATUM-GAUGE`, `LM3-TOOL-DOOR-GAUGE`, `LM3-TOOL-WINDOW-GAUGE`, `LM3-TOOL-SEAL-GAUGE`.

Reference controls: `LM3-MFG-010`, `LM3-MFG-020`, `LM3-MFG-030`, `LM3-MFG-040`, `LM3-MFG-100`.

LM3-FRP-120 — configurable train-end and articulation structural-interface pack

Release boundary: Exactly one end option is fitted at each position; neither option is released by this definition seed alone.

Controlled products and geometry

Product	Route / maturity	Qty	Parent	Design-reference envelope (mm)	Reference default	Representation
LM3-BDY-P090 — end ring frame and anti-climber beam set	MAKE / release-candidate	2 set	LM3-END-SA700	2700 × 260 × 2900	local manufacture	end ring and anti-climber datum
LM3-END-P060 — common reversible end-interface carrier ring, option bolt grid, and sealing datum kit	MAKE / release-candidate	2 end position	LM3-EIF-SA650	2650 × 280 × 2650	local manufacture	reversible end carrier and option bolt grid
LM3-END-P061 — panoramic-end option shim, cowl/glass carrier, and sensor datum closeout kit	MAKE / release-candidate	2 option kit	LM3-EIF-SA650	1250 × 2650 × 2650	local manufacture	panoramic closeout, glass carrier and sensor datums
LM3-END-P062 — mid open-connection option portal trim, bellows clamp, threshold bridge, and drain kit	MAKE / release-candidate	0 option kit	LM3-EIF-SA650	1300 × 2100 × 2350	local manufacture	open portal, clamp, threshold and drain option
LM3-END-P030 — cowl service hatch, sensor backing bracket, washer-tube, and heater-cable clip kit	MAKE / release-candidate	2 end kit	LM3-END-SA700	900 × 700 × 420	local manufacture	service hatch, backing bracket and clipped services
LM3-ART-P010 — articulation adapter frame, anti-lift keeper, and shim kit	MAKE / release-candidate	2 kit	LM3-ART-SA810	2400 × 300 × 2400	local manufacture	articulation adapter and anti-lift interface

Frozen inputs

- released crash, articulation, anti-lift, gangway and end-equipment load cases
- frozen panoramic-end and open-mid supplier interface envelopes
- selected-end configuration, seal, drain, threshold and service-access requirements

Controlled outputs

- common end ring, option bolt grid and articulation-adapter production drawings
- panoramic and open-mid shim/closeout configuration drawings
- anti-lift, service-hatch, sensor-backing, drain and threshold details
- serialized end-position option and interchangeability record

Verification and hold points

- independent structural, fatigue and full-motion calculation review
- end-ring/bolt-grid datum survey and A/B interchange gauge
- articulation sweep plus keeper/pin-retention inspection
- configured-end water, drain, threshold and service-removal trials

Tooling: LM3-TOOL-STEEL-FIXTURE, LM3-TOOL-DATUM-GAUGE, LM3-TOOL-COWL-MOULD, LM3-TOOL-WATER-TEST.

Reference controls: LM3-MFG-010, LM3-MFG-030, LM3-MFG-040, LM3-MFG-080, LM3-MFG-100.

LM3-FRP-130 — powered-bogie local frame, bracket and harness drawing pack

Release boundary: The package prohibits reuse of freight bogie frames and does not release a bogie before supplier certificates and physical tests close.

Controlled products and geometry

Product	Route / maturity	Qty	Parent	Design-reference envelope (mm)	Reference default	Representation
LM3-BOG-P010 — powered bogie welded H-frame and motor-cradle weldment	MAKE / release-candidate	3 ea	LM3-BOG-SA610	3400 × 2500 × 620	local manufacture	welded H-frame with axlebox and bolster datums
LM3-BOG-P030 — powered-bogie guards, cable guides, WSP brackets, and inspection covers	MAKE / release-candidate	3 kit	LM3-BOG-SA610	3200 × 2350 × 460	local manufacture	guards, cable guides, brackets and covers
LM3-BOG-P050 — powered-bogie motor torque link, anti-rotation stop, and safety lanyard bracket kit	MAKE / release-candidate	3 bogie kit	LM3-TRC-SA615	1400 × 420 × 300	local manufacture	motor torque link, stop and lanyard bracket
LM3-BOG-P060 — powered-bogie brake/WSP/speed-sensor harness and junction-bracket kit	MAKE / release-candidate	3 bogie kit	LM3-BOG-SA610	3000 × 2100 × 220	local manufacture	bogie harness route and junction brackets

Frozen inputs

- released bogie-frame, motor, brake, suspension and fatigue load cases
- supplier-frozen wheelset, motor, gearbox, brake, spring and damper envelopes
- released weld class, machining datum, cable protection and motion keep-outs

Controlled outputs

- powered H-frame cut, bend, machining and weld drawings
- motor cradle, torque-link, guard, brake and sensor bracket drawings
- brake/WSP/speed-sensor harness board, clamp-coordinate and junction schedule
- weld/NDT, corrosion, serial trace and bogie configuration maps

Verification and hold points

- independent EN 13749 structural/fatigue calculation review
- fixture/tack, post-weld and machined-datum surveys
- classed weld NDT and torque-link/guard proof inspections
- supplier dry fit, harness motion sweep, continuity and insulation tests

Tooling: LM3-TOOL-BOGIE-STAND, LM3-TOOL-MOTOR-ALIGN, LM3-TOOL-HARNESS-BOARD, LM3-TOOL-DATUM-GAUGE.

Reference controls: LM3-MFG-010, LM3-MFG-020, LM3-MFG-030, LM3-MFG-070, LM3-MFG-100.

LM3-FRP-140 — trailer-bogie local frame, bracket and harness drawing pack

Release boundary: The package prohibits reuse of freight bogie frames and does not release a bogie before supplier certificates and physical tests close.

Controlled products and geometry

Product	Route / maturity	Qty	Parent	Design-reference envelope (mm)	Reference default	Representation
LM3-BOG-P020 — trailer bogie welded H-frame	MAKE / release-candidate	3 ea	LM3-BOG-SA620	3400 × 2500 × 620	local manufacture	welded H-frame with axlebox and bolster datums
LM3-BOG-P031 — trailer-bogie guards, cable guides, WSP brackets, and inspection covers	MAKE / release-candidate	3 kit	LM3-BOG-SA620	3200 × 2350 × 460	local manufacture	guards, cable guides, brackets and covers
LM3-BOG-P061 — trailer-bogie brake/WSP/speed-sensor harness and junction-bracket kit	MAKE / release-candidate	3 bogie kit	LM3-BOG-SA620	3000 × 2100 × 220	local manufacture	bogie harness route and junction brackets

Frozen inputs

- released bogie-frame, brake, suspension and fatigue load cases
- supplier-frozen wheelset, brake, spring and damper envelopes
- released weld class, machining datum, cable protection and motion keep-outs

Controlled outputs

- trailer H-frame cut, bend, machining and weld drawings
- guard, brake, WSP and speed-sensor bracket drawings
- brake/WSP/speed-sensor harness board, clamp-coordinate and junction schedule
- weld/NDT, corrosion, serial trace and bogie configuration maps

Verification and hold points

- independent EN 13749 structural/fatigue calculation review
- fixture/tack, post-weld and machined-datum surveys
- classed weld NDT and guard/bracket proof inspections
- supplier dry fit, harness motion sweep, continuity and insulation tests

Tooling: LM3-TOOL-BOGIE-STAND, LM3-TOOL-HARNESS-BOARD, LM3-TOOL-DATUM-GAUGE.

Reference controls: LM3-MFG-010, LM3-MFG-020, LM3-MFG-030, LM3-MFG-070, LM3-MFG-100.

LM3-FRP-150 — battery tray, high-voltage and coolant local-hardware pack

Release boundary: No battery or HV hardware may be energised from this drawing definition; the supplier safety case and controlled energisation procedure remain mandatory.

Controlled products and geometry

Product	Route / maturity	Qty	Parent	Design-reference envelope (mm)	Reference default	Representation
LM3-HV-P010 — battery sliding trays, retention straps, service interlocks, and drain pans	MAKE / release-candidate	3 car kit	LM3-HV-SA510	5400 × 900 × 320	local manufacture	sliding trays, retention and drain pan
LM3-HV-P020 — segregated HV cable tray, bonding studs, grommets, and orange cover set	MAKE / release-candidate	3 car kit	LM3-HV-SA510	14000 × 420 × 180	local manufacture	segregated HV tray, covers, grommets and studs
LM3-HV-P030 — coolant manifold brackets, bleed/drain points, and insulated pipe clamp kit	MAKE / release-candidate	3 car kit	LM3-HV-SA510	6500 × 950 × 260	local manufacture	coolant manifolds, clamps and bleed/drain points

Frozen inputs

- supplier-frozen pack mass, retention, vent, HV and coolant interfaces
- released crash/restraint, insulation, creepage, clearance and fire-barrier requirements
- released cable/pipe ratings, segregation, bend radius, drain and service rules

Controlled outputs

- sliding-tray, retention, interlock, drain-pan and lifting detail drawings
- segregated HV tray, orange cover, bonding-stud and penetration schedule
- coolant manifold, insulated clamp, bleed/drain and service-removal drawings
- clamp-coordinate, label, continuity, pressure-test and configuration records

Verification and hold points

- battery restraint and tray structural calculation review
- tray fit/removal, interlock and exterior-only vent-path inspection
- HV segregation, bond continuity, insulation and HVIL tests
- coolant pressure, bleed, drain and hot/cold movement checks

Tooling: `LM3-TOOL-HARNESS-BOARD`, `LM3-TOOL-FINAL-DATUM`, `LM3-TOOL-SEAL-GAUGE`.

Reference controls: `LM3-MFG-010`, `LM3-MFG-030`, `LM3-MFG-070`, `LM3-MFG-080`, `LM3-MFG-100`.

LM3-FRP-160 — low-voltage trainline harness and terminal-distribution pack

Release boundary: Harness routes remain unissued until supplier connector data, EMC/fire evidence and safety I/O ownership are frozen.

Controlled products and geometry

Product	Route / maturity	Qty	Parent	Design-reference envelope (mm)	Reference default	Representation
`LM3-CTRL-P040` — pre-terminated LV trainline harness, DIN cabinet, and terminal-distribution kit	`MAKE` / `release-candidate`	3 car kit	`LM3-SYS-SA900`	8000 × 2000 × 240	local manufacture	LV trainline harness and distribution cabinets

Frozen inputs

- frozen control, safety, PIS/CCTV, door, brake and inter-car I/O schedule
- selected rail cable, connector, cabinet and terminal component data
- released EMC, fire, segregation, service-loop and network requirements

Controlled outputs

- wire/connector/terminal schedule and harness-board drawing
- per-car route, branch, penetration, clamp-coordinate and service-loop drawings
- cabinet/terminal allocation, ferrule, label and keying schedule
- continuity, insulation, network-enumeration and as-built configuration forms

Verification and hold points

- independent schematic, pinout and segregation review
- first harness-board dimensional and connector-keying inspection
- 100 percent continuity/insulation test plus shield/bond audit
- installed bend-radius, chafe, service access and network-enumeration inspection

Tooling: `LM3-TOOL-HARNESS-BOARD`, `LM3-TOOL-FINAL-DATUM`.

Reference controls: LM3-MFG-010, LM3-MFG-030, LM3-MFG-070, LM3-MFG-100.

LM3 First-Article Evidence Status

Repository source: `design/component-catalogue/catalog/buildable-trainset/first-article-evidence-status.md`

This register distinguishes public supplier data, planned procedures and solver-backed screening from supplier-controlled, laboratory and independently accepted release evidence. Planned work is never represented as a performed test.

Release ready: no · Accepted: 0 · Open: 13

Evidence package	Repository status	Release gate	Planning / screening artifacts
`EVD-STR-001` — Carbody and chassis structural analysis	available-screening	open	present
`EVD-STR-002` — Bogie frame and equipment attachment structural analysis	available-screening	open	present
`EVD-TEST-001` — Carbody, bogie, articulation and local attachment proof tests	protocol-ready	open	present
`EVD-THM-001` — Battery, charger contact and cooling-system thermal test	protocol-ready	open	present
`EVD-ELEC-001` — Released HV/LV harness and protection definition	protocol-ready	open	present
`EVD-ELEC-002` — Harness continuity, insulation, dielectric and bonding tests	protocol-ready	open	present
`EVD-CTRL-001` — Safety-controller bench and hardware-in-the-loop evidence	protocol-ready	open	present
`EVD-SUP-001` — Bogie, wheelset, bearing, brake and suspension supplier data	protocol-ready	open	present
`EVD-SUP-002` — Door-system supplier data and type-test evidence	protocol-ready	open	present
`EVD-SUP-003` — Battery, converter, motor, gearbox and charger supplier data	protocol-ready	open	present
`EVD-SUP-004` — Glazing, HVAC, interior, lighting, passenger-system and fire-system supplier data	protocol-ready	open	present
`EVD-MFG-001` — Locally manufactured joints, inserts, glazing frames and interior modules	protocol-ready	open	present
`EVD-MASS-001` — Product, car, axle-load and complete-train mass properties	protocol-ready	open	present

A gate becomes accepted only through a submission under `evidence/submissions/` containing accountable people, configuration/procedure revisions, calibrated equipment, raw/review artifacts and verified SHA-256 values.

LM3 first-article execution pack

Repository source: `design/component-catalogue/catalog/buildable-trainset/first-article-execution-pack.md`

This is the single working page for converting `LM3-FA-001` from a controlled design candidate into a physically evidenced first article. It links the 41 sourced catalogue/RFQ/research candidates, 10 factory drawing/interface packages, 81 public work packages, shop travelers and the 13 evidence gates. The upstream supplier technical-support package controls legal/technical access, configuration-specific review, cross-supplier responsibility, separated quotation and local-capability evidence before those inputs enter this execution pack. Neither document claims that unperformed tests have passed.

Release sequence

1. Hold the `LM3-FA-001` dimensions, duty cycle and product tree. A proposed change needs a configuration record, affected analyses and a regenerated city twin.
2. Issue the candidate-specific RFQs from `cots-candidates.md`. Require an exact order code, controlled drawings, mass, price, lead time, spares, software, RAMS, maintenance and compliance evidence.
3. Complete interfaces before purchase: mounting datums/envelopes, loads, voltage/current, cooling, network/I/O, protection, access, lifting and replacement.
4. Review bids by evidence and lifecycle cost, not headline unit price. Record every deviation and local-equivalent proposal.
5. Release detailed drawings and analyses only after chosen supplier interfaces are frozen.
6. Build coupons and subsystem rigs before the vehicle. Failed gates create an NCR and block dependent assembly release.
7. Build the first article from serialized parts and signed travelers. Preserve raw data, photographs, calibration records and software hashes.
8. Weigh received components, completed subassemblies, all three individual cars and the configured trainset; reconcile the nine categories, axle loads, centres of gravity and engineering reserve without deriving mass from envelope geometry.
9. Independent reviewers accept evidence packages. Generation scripts never convert a planned or supplier-claimed result into an accepted result.

Evidence gates and executable work

Gate	Work to perform	Minimum controlled output	Hold point
`EVD-STR-001` carbody/chassis	Replace beam screening with released shell/solid geometry; apply the approved vehicle load specification; model welds, joints, doors/windows and equipment loads; run sensitivity and mesh-convergence studies; correlate to body proof test.	Native model, solver deck/version, load-case report, convergence results, weld/joint assessment, independent calculation review and correlation report.	No carbody production release before utilization, instability, fatigue and permanent-set dispositions are independently accepted.
`EVD-STR-002` bogie/attachments	Issue bogie load spectrum; analyse frame, motor cradle, brake brackets, spring seats, pivot/yaw links and exceptional loads; extract weld-class stresses; correlate to frame proof/fatigue rig.	Released model/decks, load spectrum, convergence, weld assessment, attachment reactions, reviewer record and rig correlation.	No powered-bogie vehicle test before frame and attachment proof evidence is accepted.
`EVD-TEST-001` physical proof	Approve fixtures and safe test method; identify specimen/as-built deviations; install calibrated force, displacement and strain channels; apply proof/fatigue/functi onal sequence; inspect before, during and after.	Procedure, risk assessment, fixture certificate, calibration register, raw time series, synchronized photographs, inspection/NDT, deviations, signed report and design disposition.	Test lead may stop at any unexpected deformation, noise, crack, leakage, temperature or instrumentation anomaly.
`EVD-THM-001` battery/charger/cooling	Use the frozen city ambient and worst repeatable duty cycle; test charging, traction, regeneration, parked soak and blocked/degraded cooling; measure cells/modules, contacts, busbars, coolant and enclosure; verify every protective trip.	Loss/limit maps, rig diagram, channel list, raw temperatures/flows/pressures/current /voltage, thermal images, alarm/trip log and signed margin report.	No unattended charge or passenger operation before worst-case and single-fault results are accepted.
`EVD-ELEC-001` harness/protection	Release the single-line, cable and connector schedules, pin maps, fuse/contactor/selectivity study, bonding plan, HVIL loop and routed harness drawings. Enforce HV, 110 V, 24 V, data and fluid segregation.	Drawing index plus supplier/order codes, conductor/temperature/fire ratings, termini/tooling, protection settings, load/voltage-drop calculations and configuration signature.	No harness cutting or energisation from an illustrative/preliminary drawing.
`EVD-ELEC-002` electrical test	Test every serialized harness pin-to-pin; insulation resistance and dielectric withstand only at approved values; verify shields, bonds, polarity, keying and HVIL; record rework and retest.	Tester identity/calibration, automated raw result file, signed summary, fault/rework log and released serial list.	Any cross-wire, missing shield/bond, insulation failure or unapproved repair blocks installation.
`EVD-CTRL-001` controller/HIL	Keep OSR application compute separate from the qualified safety channel. Freeze duagon/Pilz or accepted equivalents, firmware, BSP, I/O and timing. Run requirements-linked nominal, boundary, power, clock, bus, sensor, actuator, watchdog, reset and network fault tests.	Configuration hashes, requirements-to-test matrix, rig schematic, timestamped raw logs, safe-state measurements, coverage/deviation report and independent witness record.	OSR remains shadow/supervised until the allocated safety functions and complete vehicle interfaces are accepted.
`EVD-SUP-001` bogie supply	RFQ wheelsets, bearings, suspension, brakes and compressor using the released load/duty envelopes and common interface sheets.	Exact order codes, supplier drawings, certificates, RAMS/maintenance/spares, type tests, price/lead time and signed interface acceptance.	No substitution by approximate size or catalogue photo. Brake and running-gear system authority must accept the combined configuration.
`EVD-SUP-002` door supply	Issue aperture, kinematic, mounting, electrical/network and emergency-interface sheet to IFE and equivalents; request the exact cassette/controller and its type evidence.	Controlled GA/ICD, mass, order code, cycle/force/obstacle/emergency/water evidence, software/configuration and spares list.	Door portal/carrier detail and production opening remain conditional until the exact cassette is accepted.
`EVD-SUP-003` traction energy	RFQ the motor, gearbox, converter, LFP battery, resistor, collector and HV panel as one compatible system. Require loss/thermal/derating maps and short-circuit/protection coordination.	Exact bill of supply, all ICDs, energy and mass reconciliation, RAMS, maintenance, software, fire/propagation, EMC/vibration and quotation validity.	Reject a bid that closes one component while leaving voltage, shaft, cooling, protection or software responsibility undefined.
`EVD-SUP-004` fit-out/safety	Freeze glazing/bonding, floor, seats/rails, HVAC/ducts, PIS/CCTV, lighting and fire system using the published candidates and complete vehicle environment.	Exact products/configurations, fire/smoke matrix, abuse loads, EMC/IP where relevant, cleaning/maintenance, samples and first-fit records.	Public datasheets may start the RFQ; only controlled configuration evidence closes the gate.
`EVD-MFG-001` local manufacture	Release drawings/tolerances for body joints, GFRP, inserts, window pressure frame, door carrier, seat/light rails, harnesses and ducts. Qualify material batches, operators, fixtures and processes with coupons and first-article inspection.	Drawing/process revisions, batch trace, operator qualification, gauge calibration, coupon/joint results, dimensional report, NCR disposition and signed FAI.	Local manufacture is encouraged but cannot bypass material, fire, structural, electrical or dimensional qualification.
`EVD-MASS-001` mass properties	Fill the generated 120-row record using released production-CAD, exact supplier or calibrated weighed evidence; weigh each car and the complete configured trainset; reconcile AW0/AW2/AW3 axle loads, longitudinal/vertical CG, roof/battery distribution, category variance and reserve.	Completed checksummed mass record, scale/load-cell calibration, included/excluded scope, product/category reconciliation, three car reports, complete-train report, independent review and design-authority disposition.	No estimate or simplified-envelope calculation may close a row or reduce the 78.75 t controlled planning tare.

First purchase and rig priority

The lowest-risk sequence buys samples before committing to complete vehicle-specific systems:

1. Sika 268 PowerCure, Böllhoff RIVNUT samples, Forbo 6R3883, Gurit 130FR materials and TE HVP800 connector samples for local process coupons.

2 Schaltbau C360, Pilz 751184, Moxa V2406C/camera, Westermo router, Huber+Suhner antenna and Li-ion Tamer nodes for electrical/control benches.

3 One brake caliper, bearing unit, spring sample, articulation bearing and cable carrier for fit/load/cycle rigs after calculations select sizes.

4 Vehicle-specific supplier design/RFQ for wheelsets, door, glazing, HVAC, gangway, coupler, traction motor/gearbox/converter/battery/resistor/collector and fire suppression.

5 Only then place the serialized first-article order and update cost, lead time, purchase-order dates and cashflow in each generated city twin.

Required submission structure

Start supplier evidence from `evidence/rfq-response-template.json`, controlled drawing/tooling records from the generated `evidence/factory-release-record-template.json`, performed test evidence from `evidence/test-run-template.json`, and weight/balance evidence from the generated `evidence/mass-properties-record-template.json`. A reviewer moves a gate to accepted only through the existing checksummed submission format. Supplier pages remain cited design inputs, not accepted submissions.

LM3 First-Article Inspection and Test Plan

Repository source: `design/component-catalogue/catalog/buildable-trainset/first-article-inspection-plan.md`

> Status: unfilled protocol — not manufacturing, inspection or test evidence.

This plan converts every factory release package into uniquely identified hold, witness and record-review characteristics. Detailed result fields remain blank in the JSON until copied into an authorised first-article build record.

Packages: 16 · Characteristics: 416.

Point types: H mandatory hold, W witness, R record review.

Package Index

Package	Characteristics	H	W	R	Status
`LM3-FRP-010` — primary chassis, transverse structure and stepped-floor drawing pack	27	17	3	7	open
`LM3-FRP-020` — one-metre exterior module variant and retention drawing pack	27	16	4	7	open
`LM3-FRP-030` — panoramic front-glass carrier, seal and drainage interface pack	24	14	4	6	open
`LM3-FRP-040` — reversible front-lamp cassette and fascia-service pack	23	13	4	6	open
`LM3-FRP-050` — roof curb, HVAC, PV, antenna, fairing and access-zone pack	32	20	5	7	open
`LM3-FRP-060` — interior moulding, floor, service-access and fitout pack	43	32	5	6	open
`LM3-FRP-070` — common service rail, fastener and fixture-adapter pack	22	12	4	6	open
`LM3-FRP-080` — pre-cut exterior film artwork, application and repair pack	25	18	2	5	open
`LM3-FRP-090` — radiative roof-coating coupon and one-car trial pack	21	13	3	5	open
`LM3-FRP-100` — vehicle jacking, lifting, towing and field-rerailing interface pack	21	11	4	6	open
`LM3-FRP-110` — primary end, battery, door and window structural-interface pack	31	18	5	8	open
`LM3-FRP-120` — configurable train-end and articulation structural-interface pack	29	17	5	7	open
`LM3-FRP-130` — powered-bogie local frame, bracket and harness drawing pack	25	15	5	5	open
`LM3-FRP-140` — trailer-bogie local frame, bracket and harness drawing pack	23	13	5	5	open
`LM3-FRP-150` — battery tray, high-voltage and coolant local-hardware pack	24	13	5	6	open
`LM3-FRP-160` — low-voltage trainline harness and terminal-distribution pack	19	10	4	5	open

Execution Rules

- record actual results only while executing the identified work order and configuration

- a blank, planned or design-reference result does not satisfy a characteristic
- on failure stop, contain since the last accepted check, raise an NCR and extend inspection as the approved disposition requires
- do not waive a safety, structural, fire, braking, door, glazing, HV, lifting or recovery characteristic through routine sampling
- package acceptance requires every characteristic accepted, NCRs closed or expressly accepted, and named approvals recorded

Use `manufacturing-control-record-template.json` for the detailed operation record and retain characteristic IDs when importing this plan into a QMS. The factory release remains governed by `factory-release-readiness.md`.

LM3 First-Article Public Work Packages

Repository source: `design/component-catalogue/catalog/buildable-trainset/first-article-work-packages.md`

Generated from the controlled LM3 manifest and closure-state overrides. An issue may close only after its required evidence is reviewed; repository generation never converts a planned test into performed evidence.

Baseline: `LM3-FA-001` · Open: 81 · Accepted: 0

Work package	Status	Owner	Candidate / evidence route	Issue
`WP-LM3-ROOF-P030` — freeze removable HVAC curb fairing, intake/exhaust skirt, and access-hatch moulding set	open	local manufacturing engineer + design authority	`LM3-FRP-050`, `LM3-FRP-090`, `EVD-MASS-001`, `mass:roof HVAC`	ready to publish
`WP-LM3-ROOF-P040` — freeze PV junction plinth, cable-gland cover, antenna closeout, and walkway edge set	open	local manufacturing engineer + design authority	`LM3-FRP-050`, `LM3-FRP-090`, `EVD-MASS-001`, `mass:roof PV`	ready to publish
`WP-LM3-CWL-P010` — freeze end-cowl fiberglass laminate, insert, adhesive, and coupon material kit	open	materials/component engineer + procurement lead	`EVD-STR-001`, `EVD-MFG-001`, `EVD-MASS-001`, `mass:end cowls and interfaces`	ready to publish
`WP-LM3-CWL-P011` — freeze CWL-FRP-01 upper brow and roof-cap fiberglass cast	open	local manufacturing engineer + design authority	`LM3-FRP-080`, `EVD-STR-001`, `EVD-MFG-001`, `EVD-MASS-001`, `mass:end cowls and interfaces`	ready to publish
`WP-LM3-CWL-P012` — freeze CWL-FRP-02 left cheek fiberglass cast	open	local manufacturing engineer + design authority	`LM3-FRP-080`, `EVD-STR-001`, `EVD-MFG-001`, `EVD-MASS-001`, `mass:end cowls and interfaces`	ready to publish
`WP-LM3-CWL-P013` — freeze CWL-FRP-03 right cheek fiberglass cast	open	local manufacturing engineer + design authority	`LM3-FRP-080`, `EVD-STR-001`, `EVD-MFG-001`, `EVD-MASS-001`, `mass:end cowls and interfaces`	ready to publish
`WP-LM3-CWL-P014` — freeze CWL-FRP-04 lower apron and anti-climber cover fiberglass cast	open	local manufacturing engineer + design authority	`LM3-FRP-040`, `LM3-FRP-080`, `EVD-STR-001`, `EVD-MFG-001`, `EVD-MASS-001`, `mass:end cowls and interfaces`	ready to publish
`WP-LM3-CWL-P015` — freeze CWL-FRP-05 lamp, washer, and service-hatch fiberglass cast set	open	local manufacturing engineer + design authority	`LM3-FRP-040`, `LM3-FRP-080`, `EVD-STR-001`, `EVD-MFG-001`, `EVD-MASS-001`, `mass:end cowls and interfaces`	ready to publish
`WP-LM3-CWL-P016` — freeze CWL-FRP-06 backing-ring flange fiberglass cast set	open	local manufacturing engineer + design authority	`LM3-FRP-030`, `EVD-STR-001`, `EVD-MFG-001`, `EVD-MASS-001`, `mass:end cowls and interfaces`	ready to publish
`WP-LM3-FAS-P010` — freeze panoramic front-glass carrier ring, setting-block pockets, and secondary-retention frame	open	local manufacturing engineer + design authority	`LM3-FRP-030`, `EVD-MASS-001`, `mass:end cowls and interfaces`	ready to publish
`WP-LM3-FAS-P020` — freeze reversible front-lamp cassette tray, aiming adjusters, and retained service bracket	open	local manufacturing engineer + design authority	`LM3-FRP-040`, `EVD-MASS-001`, `mass:end cowls and interfaces`	ready to publish
`WP-LM3-FAS-P030` — freeze front glazing/lamp EPDM seal, drain rail, washer sleeve, and edge-closeout kit	open	materials/component engineer + procurement lead	`LM3-FRP-030`, `EVD-MASS-001`, `mass:end cowls and interfaces`	ready to publish
`WP-LM3-EXT-P010` — freeze electric plug/sliding door cassette	open	subsystem engineer + procurement lead	`OSR-COTS-DOOR-IFE`, `EVD-SUP-004`, `EVD-MASS-001`, `mass:doors, glazing, interior, and auxiliaries`	ready to publish
`WP-LM3-EXT-P020` — freeze side laminated glazing cassette	open	materials/component engineer + procurement lead	`OSR-COTS-GLASS-AGC`, `OSR-COTS-GLAZING-SIKA268`, `EVD-SUP-004`, `EVD-MASS-001`, `mass:doors, glazing, interior, and auxiliaries`	ready to publish
`WP-LM3-EXT-P030` — freeze single panoramic heated end-glass assembly	open	subsystem engineer + procurement lead	`OSR-COTS-GLASS-AGC`, `OSR-COTS-GLAZING-SIKA268`, `LM3-FRP-030`, `EVD-SUP-004`, `EVD-MASS-001`, `mass:doors, glazing, interior, and auxiliaries`	ready to publish
`WP-LM3-EXT-P040` — freeze hvac-24kw-direct-hv-dc roof HVAC	open	subsystem engineer + procurement lead	`OSR-COTS-HVAC-LONGERTEK`, `LM3-FRP-050`, `EVD-SUP-004`, `EVD-MASS-001`, `mass:roof HVAC`	ready to publish
`WP-LM3-EXT-P060` — freeze stepped floor-board and removable service-hatch system	open	materials/component engineer + procurement lead	`OSR-COTS-FLOOR-COMFLOOR`, `LM3-FRP-060`, `EVD-SUP-004`, `EVD-MASS-001`, `mass:doors, glazing, interior, and auxiliaries`	ready to publish

Work package	Status	Owner	Candidate / evidence route	Issue
WP-LM3-EXT-P061 — freeze welded resilient floor covering, cove, nosing, and adhesive system	open	materials/component engineer + procurement lead	OSR-COTS-FLOOR-FORBO-6R3883, LM3-FRP-060, EVD-SUP-004, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-EXT-P062 — freeze longitudinal passenger and priority-seat modules	open	materials/component engineer + procurement lead	OSR-COTS-SEAT-FAINSA, LM3-FRP-060, EVD-SUP-004, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-EXT-P063 — freeze stainless grab-pole, handrail, joint, and insulated adapter kit	open	materials/component engineer + procurement lead	OSR-COTS-SEAT-FAINSA, LM3-FRP-060, EVD-SUP-004, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-EXT-P064 — freeze passenger-information display, speaker, amplifier, and mounting kit	open	materials/component engineer + procurement lead	OSR-COTS-PIS-LUMINATOR, LM3-FRP-060, EVD-SUP-004, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-EXT-P065 — freeze CCTV camera, passenger intercom, PoE/data, and mounting kit	open	materials/component engineer + procurement lead	OSR-COTS-CAMERA-MOXA-P06HC, LM3-FRP-060, EVD-SUP-004, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-EXT-P066 — freeze PRM, safety-signage, emergency-lighting, extinguisher, and first-aid kit	open	materials/component engineer + procurement lead	OSR-COTS-LIGHT-TEKNOWARE, LM3-FRP-060, EVD-SUP-004, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-FIX-P020 — freeze four-family captive fastener, floating nut, isolator, and access-fastener kit	open	materials/component engineer + procurement lead	OSR-COTS-FASTENER-RIVNUT, LM3-FRP-070, EVD-MFG-001, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-FIX-P030 — freeze standard passenger-fixture saddle and equipment adapter kit	open	local manufacturing engineer + design authority	LM3-FRP-070, EVD-MFG-001, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-WIN-P010 — freeze replaceable window pressure frame, dry seal, drain, and captive retention kit	open	local manufacturing engineer + design authority	LM3-FRP-110, EVD-MFG-001, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-DOOR-P010 — freeze door four-point adjustable carrier, datum pin, dry seal, and keyed connector bracket kit	open	local manufacturing engineer + design authority	LM3-FRP-110, EVD-TEST-001, EVD-SUP-002, EVD-SUP-004, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-LGT-P010 — freeze 1.2 m plug-in main LED lighting cassette and captive mounting kit	open	materials/component engineer + procurement lead	OSR-COTS-LIGHT-TEKNOWARE, EVD-ELEC-001, EVD-ELEC-002, EVD-SUP-004, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-LGT-P020 — freeze emergency and doorway lighting modules with independent keyed feeder kit	open	subsystem engineer + procurement lead	OSR-COTS-LIGHT-TEKNOWARE, EVD-ELEC-001, EVD-ELEC-002, EVD-SUP-004, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-EXT-P070 — freeze roof antennas, service walkway pads, lifting covers, and maintenance labels	open	materials/component engineer + procurement lead	OSR-COTS-ANTENNA-HS-85137717, LM3-FRP-050, EVD-SUP-004, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-EXT-P080 — freeze fire-rated GFRP side-module laminate, core, gelcoat, and consumable kit	open	subsystem engineer + procurement lead	OSR-COTS-COMPOSITE-GURIT-130FR, EVD-SUP-004, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-EXT-P090 — freeze fire-rated GFRP roof-module, dry-seal, and removable skirt material kit	open	subsystem engineer + procurement lead	OSR-COTS-COMPOSITE-GURIT-130FR, EVD-SUP-004, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-FIN-P010 — freeze pre-cut exterior livery graphic film, edge-seal, datum-mark, and repair-patch kit	open	subsystem engineer + procurement lead	OSR-COTS-FINISH-3M-180MC, LM3-FRP-080, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-FIN-P020 — freeze calcium-carbonate radiative roof-coating qualification and exposed-roof application kit	open	subsystem engineer + procurement lead	OSR-COTS-COOLROOF-PURDUE-CACO3, LM3-FRP-090, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-INT-P010 — freeze HVAC diffusers, side return ducts, saloon grilles, and access panels	open	materials/component engineer + procurement lead	OSR-COTS-HVAC-LONGERTEK, LM3-FRP-060, EVD-SUP-004, EVD-MFG-001, EVD-MASS-001, mass:roof HVAC	ready to publish

Work package	Status	Owner	Candidate / evidence route	Issue
WP-LM3-INT-P020 — freeze FRP/phenolic ceiling liner panel set	open	local manufacturing engineer + design authority	LM3-FRP-060, EVD-SUP-004, EVD-MFG-001, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-INT-P021 — freeze ceiling light-trough bezel, diffuser carrier, and emergency-luminaire closeout set	open	local manufacturing engineer + design authority	LM3-FRP-060, EVD-SUP-004, EVD-MFG-001, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-INT-P022 — freeze HVAC plenum cover, diffuser transition, detector bezel, and ceiling service-hatch set	open	local manufacturing engineer + design authority	LM3-FRP-060, EVD-SUP-004, EVD-MFG-001, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-INT-P030 — freeze FRP/phenolic sidewall liner panel set	open	local manufacturing engineer + design authority	LM3-FRP-060, EVD-SUP-004, EVD-MFG-001, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-INT-P031 — freeze window reveal, setting-block inspection cover, and blind/label land set	open	local manufacturing engineer + design authority	LM3-FRP-060, EVD-SUP-004, EVD-MFG-001, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-INT-P032 — freeze waist cable-cover, low-voltage access lid, and retained quarter-turn fastener set	open	local manufacturing engineer + design authority	LM3-FRP-060, EVD-SUP-004, EVD-MFG-001, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-INT-P040 — freeze FRP battery strake cover panel set	open	local manufacturing engineer + design authority	LM3-FRP-060, EVD-SUP-004, EVD-MFG-001, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-INT-P041 — freeze seat-base fairing, retained equipment hatch, and rail-saddle closeout set	open	local manufacturing engineer + design authority	LM3-FRP-060, EVD-SUP-004, EVD-MFG-001, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-INT-P050 — freeze FRP vestibule kick-panel and threshold-closeout set	open	local manufacturing engineer + design authority	LM3-FRP-060, EVD-SUP-004, EVD-MFG-001, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-INT-P051 — freeze PRM transition-ramp, bogie-deck step-cover, contrast-nosing, and anti-slip set	open	local manufacturing engineer + design authority	LM3-FRP-060, EVD-SUP-004, EVD-MFG-001, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-INT-P052 — freeze door-pocket liner, sensitive-edge clearance trim, and removable jamb-cover set	open	local manufacturing engineer + design authority	LM3-FRP-060, EVD-SUP-004, EVD-MFG-001, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-TRC-P010 — freeze motor-350kw-hm47-class axle traction motor	open	subsystem engineer + procurement lead	OSR-COTS-MOTOR-ABB-AMXE250, EVD-STR-002, EVD-THM-001, EVD-SUP-003, EVD-MASS-001, mass:traction motors, gearboxes, and controllers	ready to publish
WP-LM3-TRC-P020 — freeze single-stage reduction gearbox and flexible coupling	open	subsystem engineer + procurement lead	OSR-COTS-GEAR-VOITH-SE, EVD-STR-002, EVD-THM-001, EVD-SUP-003, EVD-MASS-001, mass:traction motors, gearboxes, and controllers	ready to publish
WP-LM3-TRC-P030 — freeze two independent motor controllers, isolated LV DC/DC, MPPT, station protection, and cooling-loop kit	open	subsystem engineer + procurement lead	OSR-COTS-CONVERTER-ABB-CC400, EVD-STR-002, EVD-THM-001, EVD-SUP-003, EVD-MASS-001, mass:traction motors, gearboxes, and controllers	ready to publish
WP-LM3-TRC-P040 — freeze battery-225kwh-lfp-800v saloon-isolated side traction battery pack	open	subsystem engineer + procurement lead	OSR-COTS-BATTERY-HOPPECKE-LITRAC, EVD-STR-002, EVD-THM-001, EVD-SUP-003, EVD-MASS-001, mass:traction batteries	ready to publish
WP-LM3-TRC-P050 — freeze roof-mounted regen dump resistor and thermal shield kit	open	materials/component engineer + procurement lead	OSR-COTS-RESISTOR-DANOTHERM, LM3-FRP-050, EVD-STR-002, EVD-THM-001, EVD-SUP-003, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-TRC-P060 — freeze station side-pin charging connector, actuator, shutter, and alignment target	open	subsystem engineer + procurement lead	OSR-COTS-CHARGER-SCHUNK, EVD-STR-002, EVD-THM-001, EVD-SUP-003, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-TRC-P070 — freeze HV contactor, fuse, pre-charge, service-disconnect, and current-sensor panel	open	materials/component engineer + procurement lead	OSR-COTS-HV-SCHALTBAU-C360, OSR-COTS-HV-TE-HVP800, EVD-STR-002, EVD-THM-001, EVD-SUP-003, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish

Work package	Status	Owner	Candidate / evidence route	Issue
WP-LM3-SAF-P010 — freeze battery temperature/off-gas detection, electrical-enclosure smoke detection, and localized mist kit	open	materials/component engineer + procurement lead	OSR-COTS-FIRE-FOGTEC, OSR-COTS-FIRE-LIIONTAMER-3, EVD-SUP-004, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-BOG-P040 — freeze powered-bogie wheelset with axle-mounted brake discs	open	subsystem engineer + procurement lead	OSR-COTS-WHEELSET-GHH, EVD-STR-002, EVD-TEST-001, EVD-ELEC-002, EVD-SUP-001, EVD-MASS-001, mass:bogie frames, wheelsets, brakes, and suspension	ready to publish
WP-LM3-BOG-P041 — freeze trailer-bogie wheelset with axle-mounted brake discs	open	subsystem engineer + procurement lead	OSR-COTS-WHEELSET-GHH, EVD-STR-002, EVD-TEST-001, EVD-ELEC-002, EVD-SUP-001, EVD-MASS-001, mass:bogie frames, wheelsets, brakes, and suspension	ready to publish
WP-LM3-BOG-P042 — freeze powered-wheelset axlebox, sealed bearing unit, speed and temperature sensor set	open	subsystem engineer + procurement lead	OSR-COTS-BEARING-SKF-TBU, EVD-STR-002, EVD-TEST-001, EVD-ELEC-002, EVD-SUP-001, EVD-MASS-001, mass:bogie frames, wheelsets, brakes, and suspension	ready to publish
WP-LM3-BOG-P043 — freeze trailer-wheelset axlebox, sealed bearing unit, speed and temperature sensor set	open	subsystem engineer + procurement lead	OSR-COTS-BEARING-SKF-TBU, EVD-STR-002, EVD-TEST-001, EVD-ELEC-002, EVD-SUP-001, EVD-MASS-001, mass:bogie frames, wheelsets, brakes, and suspension	ready to publish
WP-LM3-BOG-P044 — freeze powered-bogie primary suspension spring, guide and bump-stop set	open	subsystem engineer + procurement lead	OSR-COTS-SUSPENSION-CONTI, EVD-STR-002, EVD-TEST-001, EVD-ELEC-002, EVD-SUP-001, EVD-MASS-001, mass:bogie frames, wheelsets, brakes, and suspension	ready to publish
WP-LM3-BOG-P045 — freeze trailer-bogie primary suspension spring, guide and bump-stop set	open	subsystem engineer + procurement lead	OSR-COTS-SUSPENSION-CONTI, EVD-STR-002, EVD-TEST-001, EVD-ELEC-002, EVD-SUP-001, EVD-MASS-001, mass:bogie frames, wheelsets, brakes, and suspension	ready to publish
WP-LM3-BOG-P046 — freeze powered-bogie to carbody connection: air springs, emergency spring, centre pivot, yaw links and dampers	open	subsystem engineer + procurement lead	OSR-COTS-SUSPENSION-CONTI, EVD-STR-002, EVD-TEST-001, EVD-ELEC-002, EVD-SUP-001, EVD-MASS-001, mass:bogie frames, wheelsets, brakes, and suspension	ready to publish
WP-LM3-BOG-P047 — freeze trailer-bogie to carbody connection: air springs, emergency spring, centre pivot, yaw links and dampers	open	subsystem engineer + procurement lead	OSR-COTS-SUSPENSION-CONTI, EVD-STR-002, EVD-TEST-001, EVD-ELEC-002, EVD-SUP-001, EVD-MASS-001, mass:bogie frames, wheelsets, brakes, and suspension	ready to publish
WP-LM3-BOG-P048 — freeze powered-bogie brake calipers, parking actuators, pads and wheel-slide hardware	open	subsystem engineer + procurement lead	OSR-COTS-BRAKE-KNORR-RZKT, EVD-STR-002, EVD-TEST-001, EVD-ELEC-002, EVD-SUP-001, EVD-MASS-001, mass:bogie frames, wheelsets, brakes, and suspension	ready to publish
WP-LM3-BOG-P049 — freeze trailer-bogie brake calipers, parking actuators, pads and wheel-slide hardware	open	subsystem engineer + procurement lead	OSR-COTS-BRAKE-KNORR-RZKT, EVD-STR-002, EVD-TEST-001, EVD-ELEC-002, EVD-SUP-001, EVD-MASS-001, mass:bogie frames, wheelsets, brakes, and suspension	ready to publish
WP-LM3-AUX-P010 — freeze secondary-suspension compressor, dryer, reservoir, and isolation-manifold kit	open	materials/component engineer + procurement lead	OSR-COTS-AIR-KNORR-VV120T, EVD-SUP-004, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-END-P010 — freeze automatic end coupler and crash-energy absorber	open	subsystem engineer + procurement lead	OSR-COTS-COUPLER-DELLNER430, EVD-STR-001, EVD-ELEC-001, EVD-ELEC-002, EVD-MASS-001, mass:end cowls and interfaces	ready to publish
WP-LM3-ART-P020 — freeze articulation lower spherical pivot, bearing housing and pin set	open	subsystem engineer + procurement lead	OSR-COTS-JOINT-SCHAEFFLER, EVD-TEST-001, EVD-ELEC-001, EVD-ELEC-002, EVD-MASS-001, mass:inter-car articulation	ready to publish
WP-LM3-ART-P021 — freeze articulation upper lateral/yaw links, spherical joints and retained pins	open	subsystem engineer + procurement lead	OSR-COTS-JOINT-SCHAEFFLER, EVD-TEST-001, EVD-ELEC-001, EVD-ELEC-002, EVD-MASS-001, mass:inter-car articulation	ready to publish

Work package	Status	Owner	Candidate / evidence route	Issue
WP-LM3-ART-P022 — freeze inter-car double-wall corrugated bellows and clamp-frame set	open	subsystem engineer + procurement lead	OSR-COTS-GANGWAY-HUBNER, EVD-TEST-001, EVD-ELEC-001, EVD-ELEC-002, EVD-MASS-001, mass:inter-car articulation	ready to publish
WP-LM3-ART-P023 — freeze inter-car passenger bridge, turntable and flexible interior-panel set	open	subsystem engineer + procurement lead	OSR-COTS-GANGWAY-HUBNER, EVD-TEST-001, EVD-ELEC-001, EVD-ELEC-002, EVD-MASS-001, mass:inter-car articulation	ready to publish
WP-LM3-ART-P024 — freeze articulation trainline carrier, support arms, abrasion liners and drain path	open	materials/component engineer + procurement lead	OSR-COTS-TRAINLINE-IGUS-HARTING, EVD-TEST-001, EVD-ELEC-001, EVD-ELEC-002, EVD-MASS-001, mass:inter-car articulation	ready to publish
WP-LM3-CTRL-P010 — freeze T-ECU/S and T-ECU/A compute and safety-control cabinet kit	open	subsystem engineer + procurement lead	OSR-COTS-COMPUTE-MOXA-V2406C, OSR-COTS-CONTROL-DUAGON-MH50C, EVD-ELEC-001, EVD-ELEC-002, EVD-CTRL-001, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-CTRL-P020 — freeze navigation, balise, 5G, LoRa, GNSS, IMU, and roof-antenna kit	open	materials/component engineer + procurement lead	OSR-COTS-ANTENNA-HS-85137717, OSR-COTS-ROUTER-WESTERMO-630-5G, EVD-ELEC-001, EVD-ELEC-002, EVD-CTRL-001, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-CTRL-P030 — freeze maintenance HMI, depot pendant, emergency controls, and safety-relay kit	open	subsystem engineer + procurement lead	OSR-COTS-SAFETYRELAY-PILZ-751184, EVD-ELEC-001, EVD-ELEC-002, EVD-CTRL-001, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-CTRL-P050 — freeze operational and crashworthy event-recorder storage kit	open	materials/component engineer + procurement lead	OSR-COTS-RECORDER-HASLER, EVD-ELEC-001, EVD-ELEC-002, EVD-CTRL-001, EVD-MASS-001, mass:doors, glazing, interior, and auxiliaries	ready to publish
WP-LM3-END-P020 — freeze T-OBS nose sensor pack, heated window services, and washer kit	open	subsystem engineer + procurement lead	OSR-COTS-LIDAR-OUSTER-OS1, EVD-STR-001, EVD-ELEC-001, EVD-ELEC-002, EVD-MASS-001, mass:end cowls and interfaces	ready to publish
WP-LM3-END-P040 — freeze e-coupler LV jumper, recovery trainline, and end harness breakaway kit	open	subsystem engineer + procurement lead	OSR-COTS-COUPLER-DELLNER430, EVD-STR-001, EVD-ELEC-001, EVD-ELEC-002, EVD-MASS-001, mass:end cowls and interfaces	ready to publish
WP-LM3-END-P050 — freeze sealed headlight, tail/marker light, threshold-warning, and end-lamp harness kit	open	materials/component engineer + procurement lead	OSR-COTS-LIGHT-TEKNOWARE, LM3-FRP-040, EVD-STR-001, EVD-ELEC-001, EVD-ELEC-002, EVD-MASS-001, mass:end cowls and interfaces	ready to publish
WP-LM3-ART-P030 — freeze inter-car HV/LV jumper, coolant hose loop, energy chain, and drain sleeve kit	open	subsystem engineer + procurement lead	OSR-COTS-HV-TE-HVP800, OSR-COTS-TRAINLINE-IGUS-HARTING, EVD-TEST-001, EVD-ELEC-001, EVD-ELEC-002, EVD-MASS-001, mass:inter-car articulation	ready to publish
WP-LM3-ART-P040 — freeze train-to-train open-end articulation, gangway, drawbar, turntable, and service-jumper cassette	open	subsystem engineer + procurement lead	OSR-COTS-GANGWAY-HUBNER, EVD-TEST-001, EVD-ELEC-001, EVD-ELEC-002, EVD-MASS-001, mass:inter-car articulation	ready to publish
WP-LM3-ART-P041 — freeze train-to-train jumper blanking, transition harness, isolation label, and dust-cover kit	open	materials/component engineer + procurement lead	OSR-COTS-TRAINLINE-IGUS-HARTING, EVD-TEST-001, EVD-ELEC-001, EVD-ELEC-002, EVD-MASS-001, mass:inter-car articulation	ready to publish

Machine-readable authority: `first-article-work-packages.json`.

LM3 joint and fastener control schedule

Repository source: `design/component-catalogue/catalog/buildable-trainset/joint-control-schedule.md`

Generated from every parent/child integration step in the trainset product tree. It makes joining method and torque authority machine-readable. Numeric torques are intentionally prohibited until the named supplier instruction or joint calculation is accepted; nominal bolt diameter alone is not a safe preload specification.

Controlled integration joints: 145.

Joint ID	Parent child	Join classes	Torque authority	Release status
J-LM3-BDY-SA110-01	LM3-BDY-SA110 LM3-BDY-P010	structural-weld bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-BDY-SA110-02	LM3-BDY-SA110 LM3-BDY-P020	structural-weld bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-BDY-SA110-03	LM3-BDY-SA110 LM3-BDY-P021	structural-weld bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-BDY-SA110-04	LM3-BDY-SA110 LM3-BDY-P030	structural-weld bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-BDY-SA110-05	LM3-BDY-SA110 LM3-BDY-P120	bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-BDY-SA120-01	LM3-BDY-SA120 LM3-BDY-SA110	structural-weld bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-BDY-SA120-02	LM3-BDY-SA120 LM3-BDY-P060	bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-BDY-SA120-03	LM3-BDY-SA120 LM3-BDY-P061	gasketed-removable-panel bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-BDY-SA120-04	LM3-BDY-SA120 LM3-BDY-P070	bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-SHELL-A200-01	LM3-SHELL-A200 LM3-BDY-SA120	structural-weld bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-SHELL-A200-02	LM3-SHELL-A200 LM3-BDY-P130	gasketed-removable-panel bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-SHELL-A200-03	LM3-SHELL-A200 LM3-BDY-P131	adhesive-bonded-panel bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-SHELL-A200-04	LM3-SHELL-A200 LM3-BDY-P132	adhesive-bonded-panel bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-SHELL-A200-05	LM3-SHELL-A200 LM3-BDY-P133	adhesive-bonded-panel bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-SHELL-A200-06	LM3-SHELL-A200 LM3-BDY-P140	gasketed-removable-panel bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-SHELL-A200-07	LM3-SHELL-A200 LM3-WIN-SA320	adhesive-bonded-panel bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-SHELL-A200-08	LM3-SHELL-A200 LM3-EXT-P080	bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-SHELL-A200-09	LM3-SHELL-A200 LM3-EXT-P090	gasketed-removable-panel bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-SHELL-A200-10	LM3-SHELL-A200 LM3-FIN-P010	bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-WIN-SA320-01	LM3-WIN-SA320 LM3-BDY-P110	adhesive-bonded-panel bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-WIN-SA320-02	LM3-WIN-SA320 LM3-WIN-P010	cassette-floating-fastener fluid-thermal	released cassette interface drawing and calculation plus supplier installation manual	cassette-interface-release-required
J-LM3-WIN-SA320-03	LM3-WIN-SA320 LM3-EXT-P020	adhesive-bonded-panel bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-DOOR-SA310-01	LM3-DOOR-SA310 LM3-BDY-P100	structural-weld bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required

Joint ID	Parent child	Join classes	Torque authority	Release status
J-LM3-DOOR-SA310-02	LM3-DOOR-SA310 LM3-DOOR-P010	cassette-floating-fastener electrical-data	released cassette interface drawing and calculation plus supplier installation manual	cassette-interface-release-required
J-LM3-DOOR-SA310-03	LM3-DOOR-SA310 LM3-EXT-P010	gasketed-removable-panel bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-INT-SA330-01	LM3-INT-SA330 LM3-EXT-P060	gasketed-removable-panel bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-INT-SA330-02	LM3-INT-SA330 LM3-EXT-P061	bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-INT-SA330-03	LM3-INT-SA330 LM3-EXT-P062	bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-INT-SA330-04	LM3-INT-SA330 LM3-EXT-P063	bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-INT-SA330-05	LM3-INT-SA330 LM3-EXT-P064	bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-INT-SA330-06	LM3-INT-SA330 LM3-EXT-P065	bolted-structural-datum electrical-data	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-INT-SA330-07	LM3-INT-SA330 LM3-EXT-P066	bolted-structural-datum electrical-data	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-INT-SA330-08	LM3-INT-SA330 LM3-INT-P010	gasketed-removable-panel bolted-structural-datum electrical-data fluid-thermal	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-INT-SA330-09	LM3-INT-SA330 LM3-INT-P020	bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-INT-SA330-10	LM3-INT-SA330 LM3-INT-P021	bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-INT-SA330-11	LM3-INT-SA330 LM3-INT-P022	gasketed-removable-panel bolted-structural-datum electrical-data fluid-thermal	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-INT-SA330-12	LM3-INT-SA330 LM3-INT-P030	bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-INT-SA330-13	LM3-INT-SA330 LM3-INT-P031	adhesive-bonded-panel bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-INT-SA330-14	LM3-INT-SA330 LM3-INT-P032	bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-INT-SA330-15	LM3-INT-SA330 LM3-INT-P040	bolted-structural-datum electrical-data	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-INT-SA330-16	LM3-INT-SA330 LM3-INT-P041	gasketed-removable-panel bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-INT-SA330-17	LM3-INT-SA330 LM3-INT-P050	bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-INT-SA330-18	LM3-INT-SA330 LM3-INT-P051	bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-INT-SA330-19	LM3-INT-SA330 LM3-INT-P052	bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-FIX-SA340-01	LM3-FIX-SA340 LM3-FIX-P010	service-rail-captive-fastener	released OSR small-component standard plus accepted hardware batch and calibrated-tool procedure	standard-hardware-release-required
J-LM3-FIX-SA340-02	LM3-FIX-SA340 LM3-FIX-P020	service-rail-captive-fastener	released OSR small-component standard plus accepted hardware batch and calibrated-tool procedure	standard-hardware-release-required
J-LM3-FIX-SA340-03	LM3-FIX-SA340 LM3-FIX-P030	service-rail-captive-fastener	released OSR small-component standard plus accepted hardware batch and calibrated-tool procedure	standard-hardware-release-required
J-LM3-LGT-SA350-01	LM3-LGT-SA350 LM3-LGT-P010	service-rail-captive-fastener electrical-data	released OSR small-component standard plus accepted hardware batch and calibrated-tool procedure	standard-hardware-release-required

Joint ID	Parent child	Join classes	Torque authority	Release status
J-LM3-LGT-SA350-02	LM3-LGT-SA350 LM3-LGT-P020	service-rail-captive-fastener electrical-data	released OSR small-component standard plus accepted hardware batch and calibrated-tool procedure	standard-hardware-release-required
J-LM3-ROOF-SA410-01	LM3-ROOF-SA410 LM3-BDY-P080	structural-weld gasketed-removable-panel bolted-structural-datum electrical-data fluid-thermal	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-ROOF-SA410-02	LM3-ROOF-SA410 LM3-ROOF-P010	gasketed-removable-panel bolted-structural-datum electrical-data fluid-thermal	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-ROOF-SA410-03	LM3-ROOF-SA410 LM3-ROOF-P020	adhesive-bonded-panel bolted-structural-datum electrical-data	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-ROOF-SA410-04	LM3-ROOF-SA410 LM3-ROOF-P030	gasketed-removable-panel bolted-structural-datum electrical-data fluid-thermal	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-ROOF-SA410-05	LM3-ROOF-SA410 LM3-ROOF-P040	bolted-structural-datum electrical-data	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-ROOF-SA410-06	LM3-ROOF-SA410 LM3-EXT-P040	gasketed-removable-panel bolted-structural-datum electrical-data fluid-thermal	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-ROOF-SA410-07	LM3-ROOF-SA410 LM3-EXT-P050	bolted-structural-datum electrical-data	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-ROOF-SA410-08	LM3-ROOF-SA410 LM3-EXT-P070	bolted-structural-datum electrical-data	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-ROOF-SA410-09	LM3-ROOF-SA410 LM3-FIN-P020	bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-ROOF-SA410-10	LM3-ROOF-SA410 LM3-TRC-P050	bolted-structural-datum electrical-data fluid-thermal	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-HV-SA510-01	LM3-HV-SA510 LM3-BDY-P050	bolted-structural-datum electrical-data	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-HV-SA510-02	LM3-HV-SA510 LM3-HV-P010	bolted-structural-datum electrical-data fluid-thermal	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-HV-SA510-03	LM3-HV-SA510 LM3-HV-P020	bolted-structural-datum electrical-data	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-HV-SA510-04	LM3-HV-SA510 LM3-HV-P030	bolted-structural-datum electrical-data fluid-thermal	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-HV-SA510-05	LM3-HV-SA510 LM3-TRC-P030	bolted-structural-datum electrical-data	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-HV-SA510-06	LM3-HV-SA510 LM3-TRC-P040	bolted-structural-datum electrical-data	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-HV-SA510-07	LM3-HV-SA510 LM3-TRC-P060	bolted-structural-datum electrical-data	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-HV-SA510-08	LM3-HV-SA510 LM3-TRC-P070	bolted-structural-datum electrical-data	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-HV-SA510-09	LM3-HV-SA510 LM3-SAF-P010	bolted-structural-datum electrical-data fluid-thermal	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-BOG-SA611-01	LM3-BOG-SA611 LM3-BOG-P040	bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-BOG-SA611-02	LM3-BOG-SA611 LM3-BOG-P042	bolted-structural-datum electrical-data	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-BOG-SA611-03	LM3-BOG-SA611 LM3-BOG-P044	bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-BOG-SA611-04	LM3-BOG-SA611 LM3-BOG-P048	bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-TRC-SA615-01	LM3-TRC-SA615 LM3-TRC-P010	bolted-structural-datum electrical-data	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required

Joint ID	Parent child	Join classes	Torque authority	Release status
J-LM3-TRC-SA615-02	LM3-TRC-SA615 LM3-TRC-P020	bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-TRC-SA615-03	LM3-TRC-SA615 LM3-BOG-P050	bolted-structural-datum electrical-data	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-BOG-SA610-01	LM3-BOG-SA610 LM3-BOG-P010	structural-weld bolted-structural-datum electrical-data	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-BOG-SA610-02	LM3-BOG-SA610 LM3-BOG-P030	bolted-structural-datum electrical-data	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-BOG-SA610-03	LM3-BOG-SA610 LM3-BOG-SA611	bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-BOG-SA610-04	LM3-BOG-SA610 LM3-TRC-SA615	bolted-structural-datum electrical-data	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-BOG-SA610-05	LM3-BOG-SA610 LM3-BOG-P046	bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-BOG-SA610-06	LM3-BOG-SA610 LM3-BOG-P060	bolted-structural-datum electrical-data	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-BOG-SA621-01	LM3-BOG-SA621 LM3-BOG-P041	bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-BOG-SA621-02	LM3-BOG-SA621 LM3-BOG-P043	bolted-structural-datum electrical-data	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-BOG-SA621-03	LM3-BOG-SA621 LM3-BOG-P045	bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-BOG-SA621-04	LM3-BOG-SA621 LM3-BOG-P049	bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-BOG-SA620-01	LM3-BOG-SA620 LM3-BOG-P020	structural-weld bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-BOG-SA620-02	LM3-BOG-SA620 LM3-BOG-P031	bolted-structural-datum electrical-data	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-BOG-SA620-03	LM3-BOG-SA620 LM3-BOG-SA621	bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-BOG-SA620-04	LM3-BOG-SA620 LM3-BOG-P047	bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-BOG-SA620-05	LM3-BOG-SA620 LM3-BOG-P061	bolted-structural-datum electrical-data	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-CWL-SA710-01	LM3-CWL-SA710 LM3-CWL-P010	adhesive-bonded-panel gasketed-removable-panel bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-CWL-SA710-02	LM3-CWL-SA710 LM3-CWL-P011	adhesive-bonded-panel bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-CWL-SA710-03	LM3-CWL-SA710 LM3-CWL-P012	adhesive-bonded-panel bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-CWL-SA710-04	LM3-CWL-SA710 LM3-CWL-P013	adhesive-bonded-panel bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-CWL-SA710-05	LM3-CWL-SA710 LM3-CWL-P014	adhesive-bonded-panel bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-CWL-SA710-06	LM3-CWL-SA710 LM3-CWL-P015	adhesive-bonded-panel gasketed-removable-panel bolted-structural-datum fluid-thermal	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-CWL-SA710-07	LM3-CWL-SA710 LM3-CWL-P016	adhesive-bonded-panel bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-CWL-SA710-08	LM3-CWL-SA710 LM3-FAS-P010	bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-CWL-SA710-09	LM3-CWL-SA710 LM3-FAS-P020	bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-CWL-SA710-10	LM3-CWL-SA710 LM3-FAS-P030	adhesive-bonded-panel bolted-structural-datum fluid-thermal	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required

Joint ID	Parent child	Join classes	Torque authority	Release status
J-LM3-EIF-SA650-01	LM3-EIF-SA650 LM3-END-P060	bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-EIF-SA650-02	LM3-EIF-SA650 LM3-END-P061	gasketed-removable-panel bolted-structural-datum electrical-data	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-EIF-SA650-03	LM3-EIF-SA650 LM3-END-P062	gasketed-removable-panel bolted-structural-datum fluid-thermal	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-END-SA700-01	LM3-END-SA700 LM3-BDY-P040	bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-END-SA700-02	LM3-END-SA700 LM3-BDY-P090	structural-weld bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-END-SA700-03	LM3-END-SA700 LM3-CWL-SA710	adhesive-bonded-panel gasketed-removable-panel bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-END-SA700-04	LM3-END-SA700 LM3-EXT-P030	bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-END-SA700-05	LM3-END-SA700 LM3-END-P010	bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-END-SA700-06	LM3-END-SA700 LM3-END-P020	adhesive-bonded-panel bolted-structural-datum electrical-data fluid-thermal	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-END-SA700-07	LM3-END-SA700 LM3-END-P030	gasketed-removable-panel bolted-structural-datum electrical-data fluid-thermal	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-END-SA700-08	LM3-END-SA700 LM3-END-P040	bolted-structural-datum electrical-data	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-END-SA700-09	LM3-END-SA700 LM3-END-P050	bolted-structural-datum electrical-data	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-TTART-SA850-01	LM3-TTART-SA850 LM3-EIF-SA650	bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-TTART-SA850-02	LM3-TTART-SA850 LM3-ART-P040	bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-TTART-SA850-03	LM3-TTART-SA850 LM3-ART-P041	bolted-structural-datum electrical-data	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-ART-SA810-01	LM3-ART-SA810 LM3-ART-P010	bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-ART-SA810-02	LM3-ART-SA810 LM3-ART-P020	bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-ART-SA810-03	LM3-ART-SA810 LM3-ART-P021	bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-ART-SA820-01	LM3-ART-SA820 LM3-ART-P022	gasketed-removable-panel bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-ART-SA820-02	LM3-ART-SA820 LM3-ART-P023	bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-ART-SA830-01	LM3-ART-SA830 LM3-ART-P024	bolted-structural-datum electrical-data fluid-thermal	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-ART-SA830-02	LM3-ART-SA830 LM3-ART-P030	bolted-structural-datum electrical-data fluid-thermal	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-ART-SA800-01	LM3-ART-SA800 LM3-ART-SA810	bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-ART-SA800-02	LM3-ART-SA800 LM3-ART-SA820	gasketed-removable-panel bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-ART-SA800-03	LM3-ART-SA800 LM3-ART-SA830	bolted-structural-datum electrical-data	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-SYS-SA900-01	LM3-SYS-SA900 LM3-CTRL-P010	bolted-structural-datum electrical-data	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-SYS-SA900-02	LM3-SYS-SA900 LM3-CTRL-P020	bolted-structural-datum electrical-data	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required

Joint ID	Parent child	Join classes	Torque authority	Release status
J-LM3-SYS-SA900-03	LM3-SYS-SA900 LM3-CTRL-P030	bolted-structural-datum electrical-data	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-SYS-SA900-04	LM3-SYS-SA900 LM3-CTRL-P040	bolted-structural-datum electrical-data	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-SYS-SA900-05	LM3-SYS-SA900 LM3-CTRL-P050	bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-CAR-A900-01	LM3-CAR-A900 LM3-SHELL-A200	adhesive-bonded-panel bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-CAR-A900-02	LM3-CAR-A900 LM3-DOOR-SA310	gasketed-removable-panel bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-CAR-A900-03	LM3-CAR-A900 LM3-INT-SA330	bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-CAR-A900-04	LM3-CAR-A900 LM3-FIX-SA340	service-rail-captive-fastener	released OSR small-component standard plus accepted hardware batch and calibrated-tool procedure	standard-hardware-release-required
J-LM3-CAR-A900-05	LM3-CAR-A900 LM3-LGT-SA350	service-rail-captive-fastener electrical-data	released OSR small-component standard plus accepted hardware batch and calibrated-tool procedure	standard-hardware-release-required
J-LM3-CAR-A900-06	LM3-CAR-A900 LM3-ROOF-SA410	gasketed-removable-panel bolted-structural-datum electrical-data fluid-thermal	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-CAR-A900-07	LM3-CAR-A900 LM3-HV-SA510	bolted-structural-datum electrical-data fluid-thermal	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-CAR-A900-08	LM3-CAR-A900 LM3-BOG-SA610	bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-CAR-A900-09	LM3-CAR-A900 LM3-BOG-SA620	bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-CAR-A900-10	LM3-CAR-A900 LM3-AUX-P010	bolted-structural-datum	accepted supplier installation manual plus released OSR interface-control drawing	supplier-freeze-required
J-LM3-TRAINSET-A000-01	LM3-TRAINSET-A000 LM3-CAR-A900	bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-TRAINSET-A000-02	LM3-TRAINSET-A000 LM3-EIF-SA650	bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-TRAINSET-A000-03	LM3-TRAINSET-A000 LM3-END-SA700	gasketed-removable-panel bolted-structural-datum electrical-data	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-TRAINSET-A000-04	LM3-TRAINSET-A000 LM3-ART-SA800	bolted-structural-datum	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required
J-LM3-TRAINSET-A000-05	LM3-TRAINSET-A000 LM3-SYS-SA900	bolted-structural-datum electrical-data	released joint calculation plus interface-control drawing and calibrated-tool procedure	joint-calculation-required

Release use

A traveler may reference a joint ID, but no fastener is released for final torque until its status is closed by the authority named above. The released record must also state fastener grade, coating/lubrication condition, locking method, calibrated tool range, witness marking, and re-torque rule. Supplier wheelset, bearing, brake, coupler, door, HVAC, battery, traction, and gangway values remain supplier-controlled and must not be replaced with generic tables.

LM3 Manufacturing and Assembly Reference Controls

Repository source: `design/component-catalogue/catalog/buildable-trainset/manufacturing-and-assembly-controls.md`

> Status: reference defaults — not a production or vehicle release.

These are affordable pilot-factory defaults for work that should not wait for a bespoke invention. They make assembly planning, quotations, travelers and first-article inspection concrete while preserving every supplier, calculation, drawing and approval gate.

How authority works

Reference defaults for first-article planning only. A released drawing, joint calculation, qualified process specification, supplier instruction, or authorised deviation takes precedence. These values do not release a vehicle, structural joint, glazing system, pressure boundary, HV system, lifting operation, or safety-critical installation.

The word must below controls use of this reference package; it does not certify a component. Where a numerical product value is safety- or supplier-dependent, the default states the required authority instead of guessing it.

Common inspection escalation

- First article: 100% of defined characteristics.
- Safety and interfaces: 100% unless an approved control plan says more.
- Routine non-safety: first-off, last-off and 10%, minimum three per batch.
- Failure: on any failure, stop the batch, contain since last accepted check and inspect 100% until the approved control plan restores sampling.

LM3-MFG-010 — Drawing, Datum And Measurement Control

Applies to: all MAKE parts, all assembly fixtures, all marriage interfaces.

Reference work instructions:

1. Use vehicle coordinates X forward, Y left and Z up; identify the car centre plane, bogie-centre planes and underframe top plane on every assembly drawing.
2. Locate work by a 3-2-1 datum scheme. Do not dimension serial features from one another or use an unverified body panel as a master datum.
3. Treat CAD nominal geometry as basic only where the released drawing provides a geometric tolerance; otherwise the dimension is not released for manufacture.
4. For non-safety brackets and trim only, use a provisional general shop tolerance of ±0.5 mm to 300 mm, ±1.0 mm from 300–1,000 mm and ±2.0 mm above 1,000 mm; hole position ±0.5 mm. A drawing-specific tolerance is mandatory at bogie, coupler, door, glazing, bearing, articulation and crash interfaces.
5. Verify each fixture at installation, after any move or repair, and before each first-article batch against traceable artefacts; record as-found and as-left results.

Inspection: 100% first article and all safety/interface datums; routine non-safety dimensions use first-off, last-off and at least one in ten, with a minimum of three per batch.

Stop and raise an NCR when: missing drawing revision; damaged or out-of-calibration gauge; datum disagreement above released tolerance; unrecorded CAD or NC revision.

Release evidence that supersedes/closes this default: checked detail/assembly drawing, tolerance analysis, inspection plan and calibrated measurement record.

LM3-MFG-020 — Steel Preparation, Fit-Up And Welding

Applies to: underframe, body frame, bolsters, end rings, roof frame, bogie frame if locally manufactured.

Reference work instructions:

1. Receive material by heat/lot and preserve traceability through cutting and kitting; quarantine unidentified offcuts from structural use.

2 Deburr and remove cutting dross; keep weld lands free of oil, moisture, scale, coating and zinc unless the qualified procedure explicitly permits it.

3 Dry-build in the surveyed fixture, restrain only at released hard points, and record diagonals, twist, bogie centres, coupler centreline and door-portal datums before welding.

4 Use a balanced weld sequence from the centre outward, with inter-pass temperature, consumables, tack removal and distortion correction controlled by the qualified WPS.

5 No flame straightening, weld repair, weld-size substitution or hole enlargement is permitted without an approved repair/deviation instruction.

Inspection: 100% visual inspection and dimensional survey of first article; NDT method, extent and acceptance follow the released weld map and qualified procedure, never this default.

Stop and raise an NCR when: material certificate mismatch; unqualified welder or WPS; fit-up outside drawing tolerance; arc strike or crack; fixture movement.

Release evidence that supersedes/closes this default: released weld map/WPS, welder qualifications, material certificates, NDT plan and signed dimensional report.

LM3-MFG-030 — Bolted And Captive-Fastener Joints

Applies to: service rails, interior modules, removable panels, roof equipment, structural bolted interfaces.

Reference work instructions:

1 Kit fasteners by joint ID, grade, coating, locking device and lot; do not mix visually similar hardware or substitute stainless/plated pairs without galvanic review.

2 Bring multi-fastener joints to contact by hand, seat in a cross-pattern, then apply the released final torque in at least two stages.

3 Use sockets within the calibrated tool range, record actual tool ID and setting, and apply a visible witness mark after final acceptance.

4 Do not publish or infer numerical torque from nominal diameter. Lubrication, coating, insert strength, prevailing-torque devices and supplier bearing limits control preload.

5 Replace single-use prevailing-torque fasteners and any fastener rejected by the released reuse rule; never re-torque a witness-marked joint unless the procedure requires it.

Inspection: 100% identity and witness-mark check on safety, roof, door, glazing, bogie, coupler, articulation and HV enclosure joints; normal service-rail joints first/last and 10% minimum three.

Stop and raise an NCR when: no numerical torque authority; tool outside calibration; cross-threading or spinning insert; witness mark broken; joint gap after seating.

Release evidence that supersedes/closes this default: joint calculation or supplier instruction, released joint schedule, batch record and signed torque trace.

LM3-MFG-040 — Adhesive Bonding, Sealing And Glazing

Applies to: panoramic end glass, side glazing, bonded GFRP, roof PV laminates, sealed penetrations.

Reference work instructions:

1 Work in an enclosed clean zone at 18–30 °C, 30–75% RH and at least 3 °C above dew point unless the qualified product procedure is narrower.

2 Record substrate, cleaner, primer, adhesive/sealant and applicator batch/expiry; perform the supplier surface-preparation route and a witnessed adhesion coupon at the start of each shift/material combination.

3 Dry-fit with hard locators before priming. The released drawing must state glass edge clearance, adhesive bite/thickness, spacer hardness, drain path and installation gap.

4 Record start/end application times, environmental readings, bead continuity and clamp/removal time. Supplier open-time and cure limits are mandatory and are not replaced by this default.

5 Protect bond lines from movement, water, dust and coating until released cure; water-test glazing and roof penetrations before interior closure.

Inspection: 100% first article, every safety glazing perimeter and every roof penetration; retain witness coupons by material lot and shift through the defined evidence-retention period.

Stop and raise an NCR when: substrate below dew-point margin; expired or unmatched batch; missed open time; primer contamination; bead discontinuity; glass edge contact.

Release evidence that supersedes/closes this default: supplier-qualified bonding procedure, released glazing/bond drawing, coupon results, environment log and leak-test record.

LM3-MFG-050 — Gfrp Body, Fascia And Roof Moulded Parts

Applies to: side and roof panels, front fascia pieces, lamp bezels, interior liners, fairings.

Reference work instructions:

1. Survey each mould from tooling datums before first use and after repair; clean, release and inspect it before lay-up.
2. Issue a ply/cut kit and resin/gelcoat batch traveler. Fibre type, orientation, overlap, core, resin ratio, vacuum level and cure cycle come from the qualified laminate schedule.
3. Use moulded or drill-jigged locator features; CNC-trim only from released datums and trial-fit the first part in the master body/fascia frame before drilling production parts.
4. Bond or mould inserts only with positive anti-rotation/pull-out features and witness samples; prevent exposed fibre and seal all trimmed edges and penetrations.
5. The fascia master gauge must simultaneously accept the glass dummy, lamp gauges, washer/sensor keep-outs and removable service panels without forced fit.

Inspection: 100% first article visual, thickness, mass, trim, insert and master-gauge inspection; production sampling follows the qualified control plan, while every safety insert remains individually traceable.

Stop and raise an NCR when: mould datum out; unreleased laminate schedule; cure excursion; dry fibre/void/delamination; insert movement; forced master-gauge fit.

Release evidence that supersedes/closes this default: qualified laminate/process specification, mould acceptance, coupon/mechanical evidence, trim program revision and signed first-article report.

LM3-MFG-060 — Roof Hvac, Solar And Weatherproofing Assembly

Applies to: HVAC modules, PV rails and laminates, fairings, cable glands, condensate and roof drains.

Reference work instructions:

1. Complete a bench dry-build of rails, HVAC support frame, fairings, glands, drain paths and removable envelopes before the car roof is closed.
2. Lift equipment only from supplier points with a released lift plan; do not use PV rails, fairings or duct flanges as lifting points.
3. Maintain physical segregation between HV/PV, LV/data, condensate and fresh-air paths; route drains continuously downhill and provide cleanable traps only where shown.
4. Install replaceable equipment without permanent adhesive across its removal boundary; preserve roof access, fall-protection and emergency-isolation zones.
5. Perform staged water tests after bare roof sealing, after equipment installation and after fairing closure; electrically test bonding, insulation and PV isolation before energisation.

Inspection: 100% roof fasteners, penetrations, drains, isolation labels and leak-test zones on every car.

Stop and raise an NCR when: standing water; unsealed penetration; blocked removal path; missing lift point; failed bond/insulation test; supplier mass/CG absent.

Release evidence that supersedes/closes this default: released roof arrangement and load calculation, supplier manuals, lift plan, water-test map, electrical test and mass/CG record.

LM3-MFG-070 — Interior Fit-Out And Service Routing

Applies to: floors, seats, panels, lighting, grab rails, PIS/CCTV, harnesses, hoses.

Reference work instructions:

1. Complete concealed harness, duct, drain and fire-barrier inspection before floors, ceilings or service panels close the zone.
2. Use the common service rail and keyed connectors for non-structural modular fit-out; passenger load-bearing fixtures require a released load path and proof evidence.
3. Route cables and hoses in dedicated trays, protect every edge crossing, support branches close to connectors, and preserve supplier minimum bend radius and service loop.
4. Apply unique labels at both ends and every branch; run continuity/insulation/network checks before energisation and again after panel closure.
5. Dry-fit floors and panels from the centre datum outward, preserve drainage/inspection hatches, eliminate trip lips and sharp edges, then gauge PRM aisle, door circulation and emergency access.

Inspection: 100% concealed-route photo/map, connector identity, safety fixture and accessibility check; cosmetic modules use first/last plus 10% minimum three.

Stop and raise an NCR when: unreleased routing drawing; chafing or unsupported span; mixed HV/LV zone; blocked access panel; failed fire barrier; PRM gauge obstruction.

Release evidence that supersedes/closes this default: released routing and fit-out drawings, fixture load calculation/test, cable test report, fire-material evidence and dimensional gauge record.

LM3-MFG-080 — Module Marriage, Lifting And Recovery Interfaces

Applies to: body-to-bogie marriage, battery/HVAC modules, train ends, articulation, jacking and rerailing points.

Reference work instructions:

1. Weigh each accepted module and record centre of gravity before planning the lift; compare cumulative body mass and reactions with the controlled mass ledger.
2. Use only marked, rated and proof-accepted lifting/jacking points with positive cribbing and a released synchronous lift plan; establish an exclusion zone and appoint one lift controller.
3. Mobile lifting columns are the factory default for carbody/bogie marriage. Portable screw or scissor jacks are a recovery option only when the selected units, foundations/packing, side-load restraint and vehicle jacking pads are specifically engineered and proof tested.
4. Raise in defined increments while monitoring level and pad load; stop on unequal load, unexpected deformation, loss of communication or support movement.
5. After marriage, record ride heights, axle loads, clearances, hoses/cables, retention devices and fastener witness marks before static energisation.

Inspection: 100% lift-point identity, equipment certificates, pre-use check, lift log and post-marriage geometry for every vehicle movement.

Stop and raise an NCR when: unknown mass/CG; expired lifting certificate; unverified ground or cribbing; side-loaded jack; pad load imbalance; person beneath unsupported vehicle.

Release evidence that supersedes/closes this default: checked lift/recovery plan, equipment schedule/certificates, jacking-pad calculation and proof test, lift log and post-marriage survey.

LM3-MFG-090 — Finish, Corrosion Protection And Livery

Applies to: steel structure, GFRP exterior, livery film, roof radiative coating trial.

Reference work instructions:

1. Inspect and record hidden steel preparation and base corrosion system before panels, floors or equipment conceal it.
2. Use pre-coloured GFRP and replaceable rail-rated livery film on approved smooth zones to reduce spray finishing; seal cut edges and keep film clear of joints, vents, drains, labels, sensors and inspection areas.
3. Apply film to a clean dry surface within the film supplier temperature window, use knifeless tape at visible boundaries and record batch, installer and adhesion trial.
4. Calcium-carbonate nanoparticle roof coating remains a controlled trial over the mandatory fire/UV/weathering-compatible base system; it must not cover PV, bonded joints, walkways, vents, drains or earth bonds.

5 Compare trial and control coupons for adhesion, fire/smoke, UV/weathering, cleanability, solar reflectance and thermal benefit before fleet use.

Inspection: 100% hidden base protection, prohibited zones and first livery/roof trial; routine film appearance and edge adhesion by first/last plus 10% minimum three zones.

Stop and raise an NCR when: unprotected steel; surface contamination; film over inspection/safety feature; roof coating without compatible base; failed coupon or peel edge.

Release evidence that supersedes/closes this default: released finish-zone map, coating/film supplier system, environmental log, coupon qualification and signed appearance/corrosion inspection.

LM3-MFG-100 — Ncr, Rework And Configuration Handback

Applies to: all parts, all suppliers, all build cells, all test stages.

Reference work instructions:

1 Tag and segregate nonconforming material or digitally inhibit its next operation; never erase the original result or repair before disposition.

2 Disposition is use-as-is, rework to the original requirement, repair to an approved instruction, return/scrap, or design change. Only the named design authority may approve use-as-is, repair or design change.

3 Repeat all inspections invalidated by rework and extend the check to adjacent features that could have been affected.

4 Before cell handback, reconcile installed serial/lot numbers, software/configuration IDs, concessions, open work, tool records and measured mass with the as-built record.

5 A red-line is temporary build evidence, not a revised design; incorporate accepted changes into controlled source and reissue before repetition.

Inspection: 100% NCR and concession traceability; quality independently verifies closure and configuration reconciliation before the next release gate.

Stop and raise an NCR when: unidentified part; unapproved repair; repeat defect trend; missing retest; as-built/configuration mismatch; open safety NCR.

Release evidence that supersedes/closes this default: approved NCR disposition, repair instruction, repeat-inspection evidence, updated design revision and signed configuration handback.

LM3 manufacturing methods and tooling

Repository source: `design/component-catalogue/catalog/buildable-trainset/manufacturing-methods.md`

Pilot local-manufacture route for one LM3 three-car trainset; elapsed allowances support factory planning but never override a released material datasheet, supplier manual, WPS, adhesive open/cure time, or competent engineering decision.

Release boundary: Not a construction release. Supplier freeze, controlled drawings, structural calculations, fire/rail evidence, qualified special processes, first-article tests, independent safety assessment, and national authorisation remain mandatory.

Controlled coverage	Value
Product rows linked to at least one method	120 / 120
Manufacturing methods	9
Named tooling/fixture families	30

Method index

Method	Work centre	Crew	Planning cycle	Products	Tooling
LM3-MFG-010 — Locally fabricated steel body, bogie and interface structures	local steel preparation, weld and metrology cells	6	72 h	19	2
LM3-MFG-020 — Reusable GFRP body, roof, end-cowl and interior panel mould families	local composite moulding, cure, trim and dry-fit cell	3	24 h	27	10
LM3-MFG-030 — In-mould base finish, applied livery film, and qualified radiative roof-coating route	local surface preparation, film application and controlled coating cell	2	16 h	12	4
LM3-MFG-040 — Replaceable side-window cassette, pressure frame, seal and drain installation	local glazing preparation and vehicle fit-out cell	2	3 h	6	3
LM3-MFG-050 — Supplier door cassette on locally made four-point adjustable carrier	local door preparation, installation and static-test cell	3	5 h	3	2
LM3-MFG-060 — Removable floor panels, transport flooring and common passenger-fixture rail	local interior panel, floor and fixture preassembly cell	4	16 h	21	2
LM3-MFG-070 — Supplier traction motor/gearbox integration into locally fabricated powered bogie	local bogie build, alignment and static-test cell	4	12 h	18	2
LM3-MFG-080 — Lighting, HVAC, HV/LV services and keyed replaceable modules	local harness, interior systems and roof-module preassembly cells	4	16 h	23	3
LM3-MFG-090 — Three-car final assembly, static test and manufacturing release	55 m local final assembly and static-test road	8	48 h	19	2

LM3-MFG-010 — Locally fabricated steel body, bogie and interface structures

- Work centre: local steel preparation, weld and metrology cells
- Crew / planning cycle: 6 people / 72 hours elapsed
- Product IDs: `LM3-BDY-P010`, `LM3-BDY-P020`, `LM3-BDY-P021`, `LM3-BDY-P030`, `LM3-BDY-P040`, `LM3-BDY-P050`, `LM3-BDY-P060`, `LM3-BDY-P061`, `LM3-BDY-P070`, `LM3-BDY-P080`, `LM3-BDY-P090`, `LM3-BDY-P100`, `LM3-BDY-P110`, `LM3-BDY-P120`, `LM3-FAS-P010`, `LM3-FAS-P020`, `LM3-ART-P010`, `LM3-BOG-P010`, `LM3-BOG-P020`
- Tooling: `LM3-TOOL-STEEL-FIXTURE`, `LM3-TOOL-DATUM-GAUGE`
- Joining parts: released structural welds; machined datum inserts; OSR-FST-M8-FLOAT only for calculated secondary interfaces; galvanic-isolation shims
- Release gate: Controlled cut/flat-pattern pack, WPS/WPQR, qualified welders, weld map, NDT plan, dimensional survey and released structural proof.

Seq.	Planning time	Instruction	Hold point
10 — Release material and fixture	120 min	Verify drawing revision, heat certificates, section/plate identity, fixture calibration	yes

Seq.	Planning time	Instruction	Hold point
		and datum targets before cutting.Quarantine substitutions.	
20 — Cut, form, machine and deburr	960 min	Produce the controlled cut list and flat patterns; protect datum and bond lands; identify every primary part to heat and drawing position.	no
30 — Fixture, tack and survey	480 min	Load parts against the 3-2-1 datum scheme, install calibrated shims/stops, tack to the released sequence and survey before production welding.	yes
40 — Qualified weld and controlled cool	1680 min	Execute only the released weld map/WPS with heat-input and consumable records; use a balanced sequence and controlled cooling or stress relief where specified.	no
50 — NDT, machine datums and coat	1080 min	Complete VT and classed MT/UT, repair through an NCR, machine released interfaces, survey the frame, restore edge/cavity protection and record coating batches.	yes

LM3-MFG-020 — Reusable GFRP body, roof, end-cowl and interior panel mould families

- Work centre: local composite moulding, cure, trim and dry-fit cell
- Crew / planning cycle: 3 people / 24 hours elapsed
- Product IDs: `LM3-BDY-P130`, `LM3-BDY-P131`, `LM3-BDY-P132`, `LM3-BDY-P133`, `LM3-BDY-P140`, `LM3-CWL-P010`, `LM3-CWL-P011`, `LM3-CWL-P012`, `LM3-CWL-P013`, `LM3-CWL-P014`, `LM3-CWL-P015`, `LM3-CWL-P016`, `LM3-ROOF-P030`, `LM3-ROOF-P040`, `LM3-EXT-P080`, `LM3-EXT-P090`, `LM3-INT-P020`, `LM3-INT-P021`, `LM3-INT-P022`, `LM3-INT-P030`, `LM3-INT-P031`, `LM3-INT-P032`, `LM3-INT-P040`, `LM3-INT-P041`, `LM3-INT-P050`, `LM3-INT-P051`, `LM3-INT-P052`
- Tooling: `LM3-TOOL-SIDE-MOULD`, `LM3-TOOL-SIDE-VARIANT-NEST`, `LM3-TOOL-ROOF-MOULD`, `LM3-TOOL-ROOF-FAIRING-MOULD`, `LM3-TOOL-COWL-MOULD`, `LM3-TOOL-TRIM-DRILL`, `LM3-TOOL-INT-CEILING-MOULD`, `LM3-TOOL-INT-SIDE-MOULD`, `LM3-TOOL-INT-STRAKE-MOULD`, `LM3-TOOL-INT-DOOR-PRM-MOULD`
- Joining parts: potted threaded inserts; solid laminate clip lands; OSR-FST-M6-CAPTIVE; keyed over-centre clips; replaceable EPDM dry seals
- Release gate: Released laminate/core/insert map, fire evidence, material shelf life, mould survey, vacuum leak check, cure record, witness coupon, trim report, insert pull-out sample and master-frame dry fit.

Seq.	Planning time	Instruction	Hold point
10 — Inspect, clean and release mould	60 min	Inspect mould A-surface, split line, vacuum ports and datums; clean with the qualified system and apply the released semi-permanent release process. Record mould and release lots.	yes
20 — Apply exterior finish primer or gelcoat	75 min	Apply the released UV/fire-compatible finish layer at controlled wet-film thickness. The supplier minimum/maximum overcoat window overrides this planning allowance.	no
30 — Load reinforcement, core, solid lands and inserts	180 min	Place identified dry plies, local core and solid laminate lands to the released ply book; pot or fixture inserts and build a co-cured witness coupon.	yes
40 — Bag, leak-test and infuse	120 min	Install peel ply/flow media/bag and resin lines, record a stable vacuum leak test, then infuse within supplier temperature, viscosity and working-time limits.	yes
50 — Cure in mould	720 min	Hold the supplier-qualified time/temperature cure and record thermocouples. Do not demould on elapsed time alone; cure acceptance or hardness/Tg evidence controls release.	yes
60 — Demould, trim, seal and dry-fit	285 min	Demould without prying on show surfaces; inspect/tap-check, CNC trim and drill in the datum fixture, seal every cut edge, install inserts/seals, gauge the part and dry-fit it to the master frame.	yes

LM3-MFG-030 — In-mould base finish, applied livery film, and qualified radiative roof-coating route

- Work centre: local surface preparation, film application and controlled coating cell
- Crew / planning cycle: 2 people / 16 hours elapsed
- Product IDs: `LM3-BDY-P130`, `LM3-BDY-P131`, `LM3-BDY-P132`, `LM3-BDY-P133`, `LM3-CWL-P011`, `LM3-CWL-P012`, `LM3-CWL-P013`, `LM3-CWL-P014`, `LM3-CWL-P015`, `LM3-CWL-P016`, `LM3-FIN-P010`, `LM3-FIN-P020`
- Tooling: `LM3-TOOL-COATING-RACK`, `LM3-TOOL-COATING-COUPON`, `LM3-TOOL-FILM-TEMPLATE`, `LM3-TOOL-RADIATIVE-COUPON`
- Joining parts: in-mould white gelcoat or qualified base topcoat; rail-application livery film and edge treatment; sealed GFRP edges; masked CaCO3-acrylic candidate roof coating; traceable new/aged witness coupons
- Release gate: Base corrosion/fire protection accepted first; then film substrate/adhesion/edge/removal evidence and radiative-coating fire, adhesion, weathering, cleanability, glare and measured new/aged solar reflectance/emittance. No cooling or energy-saving claim before one-car thermal and maintenance trials.

Seq.	Planning time	Instruction	Hold point
10 — Condition and inspect substrate	60 min	Verify cured laminate, sealed edges, moisture/temperature limits and defect repair; mask bonds, seals, datums, windows, sensors and electrical earths.	yes
20 — Prepare base finish and cut film	120 min	Accept the fully cured corrosion/gelcoat base, make adhesion coupons, and plot/cut the livery film from controlled bay templates. Do not apply film over seals, drains, earths, labels, service lands or unresolved defects.	no
30 — Apply livery film and candidate roof coating	120 min	Dry-apply and squeegee the selected rail-use film with controlled seams/edges. On qualified exposed-roof zones only, mask all keep-outs and apply the CaCO3 candidate beside retained witness coupons to the released wet/dry-film range.	no
40 — Controlled cure	600 min	Maintain the film/coating suppliers' temperature, humidity and ventilation limits; post-heat or edge-treat film only where the selected system requires it, and let the roof coating reach its released cure before handling.	yes
50 — Inspect and retain coupons	60 min	Inspect bubbles, wrinkles, seams, edges and prohibited-zone masking; record coating thickness/adhesion/cure, colour/gloss and measured solar reflectance/emittance, and retain labelled coupons for UV, abrasion and wash ageing.	yes

LM3-MFG-040 — Replaceable side-window cassette, pressure frame, seal and drain installation

- Work centre: local glazing preparation and vehicle fit-out cell
- Crew / planning cycle: 2 people / 3 hours elapsed
- Product IDs: `LM3-BDY-P110`, `LM3-WIN-P010`, `LM3-EXT-P020`, `LM3-EXT-P030`, `LM3-FAS-P010`, `LM3-FAS-P030`
- Tooling: `LM3-TOOL-WINDOW-GAUGE`, `LM3-TOOL-GLASS-CARRIER-NEST`, `LM3-TOOL-WATER-TEST`
- Joining parts: supplier-bonded aluminium glazing cassette; replaceable closed-cell/EPDM outer seal; aluminium pressure frame; OSR-FST-M6-CAPTIVE; drain rail and jack points
- Release gate: Supplier glazing/edge certificate, retained frame calculation, approved surface preparation, seal compression map, drain test, water-ingress test, heater/isolation test where fitted and timed replacement demonstration.

Seq.	Planning time	Instruction	Hold point
10 — Receive and inspect cassette	25 min	Confirm pane/cassette identity and orientation; inspect laminate, edges, frame bond and heater terminals. Quarantine any edge impact, delamination or expired adhesive evidence.	yes
Seq.	**Planning time**	**Instruction**	**Hold point**
20 — Gauge aperture and dry-fit	35 min	Gauge carrier-ring datums and drains, protect glass edges, place the cassette on	yes

Seq.	Planning time	Instruction	Hold point
		nonmetallic setting blocks and confirmuniform design clearance without forcingthe glass.	
30 — Install dry seal and pressure frame	55 min	Fit the one-piece keyed seal without stretch, lubricants or unapproved sealant; install the pressure frame and tighten captive fasteners in the released cross-pattern to the joint schedule.	no
40 — Verify compression, drains and services	25 min	Measure seal compression at the released witness points; confirm drain continuity, secondary retention, electrical isolation and heater continuity where fitted.	yes
50 — Water and replacement test	40 min	Run the controlled spray test, inspect the dry side and drains, then remove/refit the first article with the documented lifting/jacking method and record elapsed replacement time.	yes

LM3-MFG-050 — Supplier door cassette on locally made four-point adjustable carrier

- Work centre: local door preparation, installation and static-test cell
- Crew / planning cycle: 3 people / 5 hours elapsed
- Product IDs: `LM3-BDY-P100`, `LM3-DOOR-P010`, `LM3-EXT-P010`
- Tooling: `LM3-TOOL-DOOR-GAUGE`, `LM3-TOOL-SEAL-GAUGE`
- Joining parts: four adjustable carrier shoes; two hardened datum pins; replaceable dry perimeter seal; OSR-FST-M10-SEAL; keyed body-side connector bracket
- Release gate: Supplier and interface freeze, carrier load proof, EN 14752/national evidence as applicable, datum and seal maps, closed-and-locked loop, obstacle detection, emergency release, water test and timed cassette replacement.

Seq.	Planning time	Instruction	Hold point
10 — Accept supplier cassette and portal	45 min	Verify serial/revision/evidence, inspect seals/leaf/drive and gauge the locally made portal and four carrier datums before lifting the cassette.	yes
20 — Lift, pin and adjust carrier	75 min	Use the approved lift points; engage datum pins, install four carrier shoes and adjust only within their released range to achieve leaf/aperture geometry without distorting the cassette.	yes
30 — Join, seal and connect	60 min	Install sealed fasteners to the controlled joint schedule, fit the keyed replaceable seal and connector, retain service loops and record final adjuster positions and witness marks.	no
40 — Static functional and safety tests	75 min	Test open/close cycles, obstruction detection, closed-and-locked circuit, traction interlock, emergency release, local isolation and degraded/manual operation under the supplier procedure.	yes
50 — Ingress and replacement demonstration	45 min	Run the controlled water test and first-article removal/refit; repeat safety tests after replacement and record time, tooling and adjustments.	yes

LM3-MFG-060 — Removable floor panels, transport flooring and common passenger-fixture rail

- Work centre: local interior panel, floor and fixture preassembly cell
- Crew / planning cycle: 4 people / 16 hours elapsed
- Product IDs: `LM3-EXT-P060`, `LM3-EXT-P061`, `LM3-EXT-P062`, `LM3-EXT-P063`, `LM3-EXT-P064`, `LM3-EXT-P065`, `LM3-EXT-P066`, `LM3-FIX-P010`, `LM3-FIX-P020`, `LM3-FIX-P030`, `LM3-INT-P020`, `LM3-INT-P021`, `LM3-INT-P022`, `LM3-INT-P030`, `LM3-INT-P031`, `LM3-INT-P032`, `LM3-INT-P040`, `LM3-INT-P041`, `LM3-INT-P050`, `LM3-INT-P051`, `LM3-INT-P052`
- Tooling: `LM3-TOOL-FLOOR-TEMPLATE`, `LM3-TOOL-FIXTURE-PROOF`
- Joining parts: removable sealed floor boards/hatches; rail-qualified resilient floor covering; OSR-RAIL-42; OSR-FST-M8-FLOAT; calculated seat/handrail saddles; OSR-FST-QT-CAPTIVE access panels

- Release gate: Fire/smoke certificates, substrate moisture/flatness, adhesive batch and open time, seam/edge detail, slip resistance, hatch removal, fixture load calculations and proof samples, PRM/egress gauge and snag/rattle inspection.

Seq.	Planning time	Instruction	Hold point
10 — Dry-build removable floor and service hatches	180 min	Gauge floor supports, dry-fit numbered boards/hatches, verify flush joints, edge radii, lift points and uninterrupted access to battery/HV inspection zones.	yes
20 — Prepare floor and pre-cut covering	150 min	Verify substrate flatness, dryness and cleanliness; cut labelled floor pieces from controlled CAD templates and preserve access-hatch breaks, coved edges and drainage paths.	no
30 — Bond covering	210 min	Condition material, apply only the selected supplier adhesive at specified spread rate/open time, lay without trapped air, roll in both directions and complete released seam/edge treatment.	yes
40 — Install service rails and inserts	180 min	Set blind threaded inserts perpendicular with the supplier force-controlled tool and sampled pull checks; gauge OSR-RAIL-42 pitch/alignment and preserve galvanic isolation.	yes
50 — Mount and proof passenger fixtures	180 min	Install seats, handrails and equipment through released saddles/joints; record torque/locking, then proof-load first articles to the approved fixture calculation without using trim as a load path.	yes
60 — Finish, cure and inspect	60 min	Respect adhesive return-to-service time, inspect adhesion/seams/slip and PRM geometry, remove/refit every service hatch and complete snag, sharp-edge and rattle checks.	yes

LM3-MFG-070 — Supplier traction motor/gearbox integration into locally fabricated powered bogie

- Work centre: local bogie build, alignment and static-test cell
- Crew / planning cycle: 4 people / 12 hours elapsed
- Product IDs: `LM3-BOG-P010`, `LM3-BOG-P030`, `LM3-BOG-P031`, `LM3-BOG-P040`, `LM3-BOG-P041`, `LM3-BOG-P042`, `LM3-BOG-P043`, `LM3-BOG-P044`, `LM3-BOG-P045`, `LM3-BOG-P046`, `LM3-BOG-P047`, `LM3-BOG-P048`, `LM3-BOG-P049`, `LM3-BOG-P050`, `LM3-BOG-P060`, `LM3-BOG-P061`, `LM3-TRC-P010`, `LM3-TRC-P020`
- Tooling: `LM3-TOOL-BOGIE-STAND`, `LM3-TOOL-MOTOR-ALIGN`
- Joining parts: supplier motor and gearbox mounting kit; calculated high-integrity bolted joints; torque link and safety lanyard; flexible coupling; bonding braid; sealed harness brackets
- Release gate: Supplier motor/gearbox/wheelset/brake freeze, bogie-frame proof/NDT, mount-load proof, certified fastener/joint schedule, alignment and backlash, lubrication, insulation/bonding, rotation, brake and thermal instrumentation tests.

Seq.	Planning time	Instruction	Hold point
10 — Accept frame and supplier kits	90 min	Verify bogie frame NDT/datum survey and all serialized motor, gearbox, coupling, wheelset, bearing, brake and suspension evidence before unpacking.	yes
20 — Set wheelsets and running gear	150 min	Load the controlled bogie stand, install axlebox/suspension/brake kits under supplier instructions and record orientation, clearances, lubrication and safety locking.	yes
30 — Install and align motor/gearbox	180 min	Use the dedicated alignment fixture; lift from supplier points, set mount shims, coupling/gear mesh and torque link, then close released high-integrity joints with calibrated tools and independent witness.	yes
40 — Route services and guards	90 min	Install bonding, temperature/speed/WSP harnesses, junctions, guards and safety lanyards with released bend radii, movement envelope and service access.	no
50 — Static rotation, brake and electrical tests	150 min	Check free rotation, alignment/backlash, oil level/leaks, brake application/release, insulation resistance, bonding and sensor polarity before powered testing.	yes
60 — Release and preserve	60 min	Record mass/ride-height setup, close inspection covers, apply witness marks and preservation, and quarantine the bogie until vehicle marriage.	yes

LM3-MFG-080 — Lighting, HVAC, HV/LV services and keyed replaceable modules

- Work centre: local harness, interior systems and roof-module preassembly cells
- Crew / planning cycle: 4 people / 16 hours elapsed
- Product IDs: `LM3-LGT-P010`, `LM3-LGT-P020`, `LM3-INT-P010`, `LM3-EXT-P040`, `LM3-EXT-P050`, `LM3-EXT-P070`, `LM3-ROOF-P010`, `LM3-ROOF-P020`, `LM3-ROOF-P030`, `LM3-ROOF-P040`, `LM3-HV-P010`, `LM3-HV-P020`, `LM3-HV-P030`, `LM3-TRC-P030`, `LM3-TRC-P040`, `LM3-TRC-P050`, `LM3-TRC-P060`, `LM3-TRC-P070`, `LM3-CTRL-P040`, `LM3-END-P020`, `LM3-END-P030`, `LM3-END-P050`, `LM3-FAS-P020`
- Tooling: `LM3-TOOL-SERVICE-RAIL`, `LM3-TOOL-HARNESS-BOARD`, `LM3-TOOL-LAMP-AIM`
- Joining parts: OSR-RAIL-42; OSR-FST-M6-CAPTIVE; OSR-CON-LV4; OSR-CON-LVDATA12; sealed HV connectors; segregated service trays; removable HVAC curb seal
- Release gate: Supplier evidence, fire/EMC/vibration/IP checks, harness 100% continuity and insulation, keyed/polarity audit, HVIL, fluid pressure/drain tests, lighting lux/emergency-duration test, HVAC airflow and condensate test.

Seq.	Planning time	Instruction	Hold point
10 — Preassemble and test modules off vehicle	180 min	Build lighting, harness, duct, tray and cabinet modules on dedicated boards/rails; record serials, connectors, firmware/configuration and incoming evidence.	yes
20 — Install rails, trays, ducts and curb seals	180 min	Gauge common datums, install isolated rails/trays and supplier-approved seals, protect fire barriers and preserve removable access zones.	no
30 — Install keyed LV/data and segregated HV services	240 min	Route pre-terminated looms to released clips and bend radii; keep HV, LV/data and fluids segregated, fit service loops and blanking caps, and prevent field-cut lighting wires.	yes
40 — Install HVAC, batteries and power modules	180 min	Use approved lifts and supplier mounting kits; close coolant, drain, bond, service-disconnect and HVIL interfaces under LOTO.	yes
50 — Cold commissioning	120 min	Complete continuity, insulation, bonding, connector keying, sensor enumeration, pressure/leak and drain-flow tests before energisation.	yes
60 — Functional release	60 min	Run lighting lux/emergency tests, HVAC airflow/condensate tests and controlled HVIL/pre-charge checks; attach actual records to the train configuration.	yes

LM3-MFG-090 — Three-car final assembly, static test and manufacturing release

- Work centre: 55 m local final assembly and static-test road
- Crew / planning cycle: 8 people / 48 hours elapsed
- Product IDs: `LM3-END-P010`, `LM3-END-P040`, `LM3-END-P060`, `LM3-END-P061`, `LM3-END-P062`, `LM3-ART-P020`, `LM3-ART-P021`, `LM3-ART-P022`, `LM3-ART-P023`, `LM3-ART-P024`, `LM3-ART-P030`, `LM3-ART-P040`, `LM3-ART-P041`, `LM3-AUX-P010`, `LM3-SAF-P010`, `LM3-CTRL-P010`, `LM3-CTRL-P020`, `LM3-CTRL-P030`, `LM3-CTRL-P050`
- Tooling: `LM3-TOOL-LIFT-COLUMNS`, `LM3-TOOL-FINAL-DATUM`
- Joining parts: released bogie-to-body joints; articulation/coupler supplier kits; segregated trainline jumpers; bonding network; configuration-controlled safety/control equipment
- Release gate: All child travelers accepted, train mass/geometry/clearance, brake/door/HV/communications/fire-detection tests, software/configuration hashes, static and low-speed test evidence, open NCR review and independent manufacturing release.

Seq.	Planning time	Instruction	Hold point
10 — Accept child assemblies and final road	240 min	Verify signed child travelers, serial/configuration records, lifting-column certification, track level and vehicle datum survey; unresolved safety NCRs block marriage.	yes

Seq.	Planning time	Instruction	Hold point
20 — Bogie marriage and articulation	600 min	Lift only at released points, roll accepted bogies into surveyed positions, lower onto interfaces, close body/bogie and articulation joints, and independently witness safety-critical locking.	yes
30 — Close trainlines, couplers and remaining fit-out	480 min	Connect labelled HV/LV/data/coolant/air paths, install end equipment and remaining service panels, then perform continuity, leak, movement-envelope and access checks.	yes
40 — Static systems commissioning	720 min	Run brake, door/interlock, lighting, HVAC, battery/HVIL, detection, communications, passenger systems, emergency and degraded-mode procedures against released test scripts.	yes
50 — Geometry, mass and water tests	360 min	Survey ride height/body/articulation/clearance, weigh the train, run roof/door/window ingress tests and reconcile the as-built mass and configuration records.	yes
60 — Independent review and controlled release	480 min	Review all records and NCRs, freeze software/configuration hashes and issue only a manufacturing release to the authorised low-speed/static test programme—not passenger service approval.	yes

External design references

These links explain the engineering basis; they are not copied supplier instructions or a substitute for the selected product's current technical datasheet.

- ERA LOC&PAS TSI application guide — Passenger-door, glazing, structure and rolling-stock evidence gates; deployment authority must select the legally applicable national/current standards.
- Composites One vacuum infusion process — Generic closed-mould sequence: dry reinforcement, vacuum/process materials, resin infusion, controlled cure and demould. Product-specific laminate and resin instructions remain controlling.
- Sika rail bonding, sealing and glazing application overview — Surface preparation, controlled bead geometry, glazing, floor/interior bonding and exterior sealing process categories; the selected adhesive technical datasheet controls primers, open time, cure and safe handling.
- Forbo rail flooring installation and maintenance download centre — Rail-floor material, installation, cleaning and maintenance evidence route; selected floor and adhesive documents control actual installation.
- US Department of Energy guidelines for high-reflectance, high-emittance surfaces — Screening rationale for measuring solar reflectance and thermal emittance. Building-roof thresholds are not transferred as rail certification or an LM3 performance claim.
- 3M Application: Rail instruction bulletin — Rail-vehicle graphic-film substrate inspection, cleaning, adhesion trial, dry application, edge treatment, inspection, maintenance and removal route; the selected film and overlaminate product bulletins remain controlling.
- Purdue full daytime sub-ambient CaCO3-acrylic radiative-cooling paint paper — Research basis for the CaCO3-acrylic candidate and initial optical targets. It is not rail qualification, a commercial product release, or evidence of durability on moving-vehicle GFRP.
- Böllhoff RIVNUT installation and setting guidance — One-sided threaded insert installation, perpendicularity, supplier setting force and tool-control principles. Released joint loads and supplier data control selection.
- Knorr-Bremse IFE modular rail entrance systems — Complete supplier-owned door cassette, drive, controller, obstacle detection, commissioning and lifecycle boundary.
- ABB AMXM railway traction motor family — Supplier-frozen railway traction-motor envelope and evidence route; no vendor model or rating is selected by this reference.

Controlled inputs

- Method source: `lib/templates/trainset-manufacturing-methods.toml` (`814542c64f387f4ba32a2126552ec5636f0839dfdaf547fdea6ae56b0ab1b06d`)

- Product manifest: `design/component-catalogue/catalog/buildable-trainset/buildable-trainset-manifest.json` (`ca385a6d3c27ee343d7f4bc4c90b4f4e27058380ff82a0aa76ca573275149703`)

LM3 controlled mass budget

Repository source: `design/component-catalogue/catalog/buildable-trainset/mass-budget.md`

Generated from the promoted design candidate and baseline constants. This record resolves the earlier ambiguity between the optimizer subtotal and the planning tare; it is not a substitute for drawing-level or as-built weighing.

- Candidate: `light-metro-3car__16p5m__reference-body__reference-bogie__motor-350kw-hm47-class__battery-225kwh-lfp-800v__hvac-24kw-direct-hv-dc__pv12`
- Document revision: `A-DRAFT`
- Release status: `planning-control`

Category	Modeled mass (kg)
carbody primary structure	25,623.53
bogie frames, wheelsets, brakes, and suspension	20,190.00
traction motors, gearboxes, and controllers	3,360.00
traction batteries	4,350.00
roof HVAC	1,530.00
roof PV	864.00
doors, glazing, interior, and auxiliaries	16,350.00
end cowls and interfaces	1,240.00
inter-car articulation	1,800.00
Modeled subtotal	75,308
Engineering reserve (4.57 %)	3,442
Controlled planning tare	78,750

Closure rule

Replace category estimates with weighed, supplier-frozen, or CAD-derived masses; transfer consumed reserve to the affected category; do not reduce controlled tare until the signed trainset weigh establishes the as-built value.

The reserve currently covers unclosed wiring, fluids, fasteners, coatings, production tolerances, and supplier mass growth. Every released drawing or supplier selection must update its category without hiding growth in another line. Final closure requires individual car weights and a complete trainset weight recorded by serial number.

LM3 product-level mass closure and lightweighting ledger

Repository source: `design/component-catalogue/catalog/buildable-trainset/mass-closure-ledger.md`

This ledger assigns every product row to one controlled planning category. It does not divide the category allowance into invented part masses: the current product geometry is an envelope/representation set, not production-solid mass evidence.

- Controlled planning tare: 78,750 kg
- Modeled exact subtotal: 75,307.53 kg
- Engineering reserve: 3,442 kg
- Coverage: 120/120 rows mapped; 0/117 active rows mass-closed

Category responsibility

Planning category	Budget (kg)	Rows	Active	Closed active	Status
carbody primary structure	25,623.53	19	19	0	`open-product-mass-evidence`
bogie frames, wheelsets, brakes, and suspension	20,190.00	17	17	0	`open-product-mass-evidence`
traction motors, gearboxes, and controllers	3,360.00	3	3	0	`open-product-mass-evidence`
traction batteries	4,350.00	4	4	0	`open-product-mass-evidence`
roof HVAC	1,530.00	4	4	0	`open-product-mass-evidence`
roof PV	864.00	3	3	0	`open-product-mass-evidence`
doors, glazing, interior, and auxiliaries	16,350.00	43	43	0	`open-product-mass-evidence`
end cowls and interfaces	1,240.00	18	17	0	`open-product-mass-evidence`
inter-car articulation	1,800.00	9	7	0	`open-product-mass-evidence`

Lightweighting decision

The lightest feasible candidate already present in the declared design space is `light-metro-3car__16p5m__light-body__light-bogie__motor-350kw-hm47-class__battery-225kwh-lfp-800v__hvac-24kw-direct-hv-dc__pv12` at 73,375.62 kg before reserve. It saves 1,931.91 kg (2.57%) against the selected modeled candidate. Keeping the reserve unchanged gives 76,817.62 kg, not a closed new tare.

Decision: retain the 78,750 kg controlled planning tare; the lighter candidate is a study option, not achieved mass.

Promotion gates:

- close every active product row with accepted CAD, supplier and/or calibrated weigh evidence
- reconcile individual-car and complete-train weights, axle loads and longitudinal/vertical centre of gravity
- repeat structural, crash, fatigue, dynamics, braking, thermal and manufacturability substantiation for changed gauges
- approve reserve drawdown and configuration change through the design authority

Recovery reaction link

All cases retain the 200 kN/point rail-rated portable-cylinder envelope.

Mass scenario	Status	Train (kg)	Full car, 4-point (kN/point)	One end, 2-point (kN/point)
`controlled-planning-tare`	controlled design basis; product mass evidence open	78,750.00	130.32	156.39
`lightest-existing-feasible-design-space-with-same-reserve`	unpromoted candidate; structural, dynamic, manufacturing and mass closure required	76,817.62	127.12	152.55
`tare-minus-10-percent`	sensitivity only; no corresponding closed design candidate	70,875.00	117.29	140.75
`tare-minus-20-percent`	sensitivity only; no corresponding closed design candidate	63,000.00	104.26	125.11

Lower mass reduces calculated reactions, but it does not authorise scissor jacks, unilateral lifting or a reduced-capacity recovery kit.

Product closure rows

Product	Route	Qty	Mass responsibility	Evidence state	Minimum evidence
LM3-BDY-P010 — laser-cut side sill beam, LH/RH	MAKE	6 ea	carbody primary structure	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-BDY-P020 — underframe centre spine and longitudinal load-path kit	MAKE	3 kit	carbody primary structure	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-BDY-P021 — underframe cross-bearer, door-bay outrigger, and equipment-bracket pack	MAKE	3 car pack	carbody primary structure	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-BDY-P030 — bolster box, air-spring pad, and centre-pivot insert set	MAKE	6 set	carbody primary structure	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-BDY-P040 — coupler pocket, shear plate, and crash-can insert kit	MAKE	2 set	carbody primary structure	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-BDY-P050 — battery tray rails, vent plenum, and service-lid gutter kit	MAKE	3 kit	carbody primary structure	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-BDY-P060 — low-floor centre pan and removable service-floor support set	MAKE	3 set	carbody primary structure	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-BDY-P061 — raised bogie-end deck, transition ramp, and removable hatch-frame set	MAKE	3 car set	carbody primary structure	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-BDY-P070 — side-wall post, door portal, waist rail, and cant rail kit	MAKE	6 side	carbody primary structure	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-BDY-P080 — roof bow, HVAC rail, PV rail, and cable-tray bracket kit	MAKE	3 kit	carbody primary structure	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-BDY-P090 — end ring frame and anti-climber beam set	MAKE	2 set	carbody primary structure	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-END-P060 — common reversible end-interface carrier ring, option bolt grid, and sealing datum kit	MAKE	2 end position	end cowls and interfaces	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record

Product	Route	Qty	Mass responsibility	Evidence state	Minimum evidence
`LM3-END-P061` — panoramic-end option shim, cowl/glass carrier, and sensor datum closeout kit	MAKE	2 option kit	end cowls and interfaces	`unclosed-evidence-required`	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
`LM3-END-P062` — mid open-connection option portal trim, bellows clamp, threshold bridge, and drain kit	MAKE	0 option kit	end cowls and interfaces	`inactive-option-not-weighed`	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
`LM3-BDY-P100` — door portal reinforcement, threshold beam, and cassette shim kit	MAKE	12 opening kit	carbody primary structure	`unclosed-evidence-required`	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
`LM3-BDY-P110` — window carrier ring, bonded-gasket land, and replacement jack-point inserts	MAKE	18 opening kit	carbody primary structure	`unclosed-evidence-required`	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
`LM3-BDY-P120` — jacking pad, lifting eye, towing lug, and recovery-label kit	MAKE	3 car kit	carbody primary structure	`unclosed-evidence-required`	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
`LM3-BDY-P130` — one-metre clip-on solid-side fiberglass body module	MAKE	48 module	carbody primary structure	`unclosed-evidence-required`	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
`LM3-BDY-P131` — one-metre clip-on window-edge fiberglass side module	MAKE	24 module	carbody primary structure	`unclosed-evidence-required`	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
`LM3-BDY-P132` — one-metre clip-on door-edge fiberglass side module	MAKE	24 module	carbody primary structure	`unclosed-evidence-required`	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
`LM3-BDY-P133` — one-metre clip-on fiberglass roof skin and equipment-fairing module	MAKE	48 module	carbody primary structure	`unclosed-evidence-required`	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
`LM3-BDY-P140` — keyed clip rail, captive retainer, anti-lift, and dry-seal car kit	MAKE	3 car kit	carbody primary structure	`unclosed-evidence-required`	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
`LM3-ROOF-P010` — HVAC curb, drop-duct collar, condensate tray, and drain fitting kit	MAKE	3 kit	roof HVAC	`unclosed-evidence-required`	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
`LM3-ROOF-P020` — PV bonded-pad lands, raised rail kit, bonding jumpers, and roof isolation labels	MAKE	3 car kit	roof PV	`unclosed-evidence-required`	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
Product	**Route**	**Qty**	**Mass responsibility**	**Evidence state**	**Minimum evidence**
`LM3-ROOF-P030` — removable HVAC curb fairing, intake/exhaust skirt, and access-hatch	MAKE	3 car set	roof HVAC	`unclosed-evidence-required`	released production-solid CAD mass-property report with material, thickness and finish density;

Product	Route	Qty	Mass responsibility	Evidence state	Minimum evidence
moulding set					first-article calibratedcomponent or completed-subassembly weigh record
`LM3-ROOF-P040` — PV junction plinth, cable-gland cover, antenna closeout, and walkway edge set	`MAKE`	3 car set	roof PV	`unclosed-evidence-required`	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
`LM3-CWL-P010` — end-cowl fiberglass laminate, insert, adhesive, and coupon material kit	`SOURCE`	2 kit	end cowls and interfaces	`unclosed-evidence-required`	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
`LM3-CWL-P011` — CWL-FRP-01 upper brow and roof-cap fiberglass cast	`MAKE`	2 ea	end cowls and interfaces	`unclosed-evidence-required`	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
`LM3-CWL-P012` — CWL-FRP-02 left cheek fiberglass cast	`MAKE`	2 ea	end cowls and interfaces	`unclosed-evidence-required`	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
`LM3-CWL-P013` — CWL-FRP-03 right cheek fiberglass cast	`MAKE`	2 ea	end cowls and interfaces	`unclosed-evidence-required`	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
`LM3-CWL-P014` — CWL-FRP-04 lower apron and anti-climber cover fiberglass cast	`MAKE`	2 ea	end cowls and interfaces	`unclosed-evidence-required`	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
`LM3-CWL-P015` — CWL-FRP-05 lamp, washer, and service-hatch fiberglass cast set	`MAKE`	4 hatch	end cowls and interfaces	`unclosed-evidence-required`	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
`LM3-CWL-P016` — CWL-FRP-06 backing-ring flange fiberglass cast set	`MAKE`	8 flange section	end cowls and interfaces	`unclosed-evidence-required`	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
`LM3-FAS-P010` — panoramic front-glass carrier ring, setting-block pockets, and secondary-retention frame	`MAKE`	2 end set	end cowls and interfaces	`unclosed-evidence-required`	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
`LM3-FAS-P020` — reversible front-lamp cassette tray, aiming adjusters, and retained service bracket	`MAKE`	2 end set	end cowls and interfaces	`unclosed-evidence-required`	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
`LM3-FAS-P030` — front glazing/lamp EPDM seal, drain rail, washer sleeve, and edge-closeout kit	`SOURCE`	2 end kit	end cowls and interfaces	`unclosed-evidence-required`	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
`LM3-ART-P010` — articulation adapter frame, anti-lift keeper, and shim kit	`MAKE`	2 kit	inter-car articulation	`unclosed-evidence-required`	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record

Product	Route	Qty	Mass responsibility	Evidence state	Minimum evidence
LM3-BOG-P010 — powered bogie welded H-frame and motor-cradle weldment	MAKE	3 ea	bogie frames, wheelsets, brakes, and suspension	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-BOG-P020 — trailer bogie welded H-frame	MAKE	3 ea	bogie frames, wheelsets, brakes, and suspension	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-BOG-P030 — powered-bogie guards, cable guides, WSP brackets, and inspection covers	MAKE	3 kit	bogie frames, wheelsets, brakes, and suspension	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-BOG-P031 — trailer-bogie guards, cable guides, WSP brackets, and inspection covers	MAKE	3 kit	bogie frames, wheelsets, brakes, and suspension	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-BOG-P050 — powered-bogie motor torque link, anti-rotation stop, and safety lanyard bracket kit	MAKE	3 bogie kit	bogie frames, wheelsets, brakes, and suspension	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-BOG-P060 — powered-bogie brake/WSP/speed-sensor harness and junction-bracket kit	MAKE	3 bogie kit	bogie frames, wheelsets, brakes, and suspension	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-BOG-P061 — trailer-bogie brake/WSP/speed-sensor harness and junction-bracket kit	MAKE	3 bogie kit	bogie frames, wheelsets, brakes, and suspension	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-EXT-P010 — electric plug/sliding door cassette	BID	12 ea	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-EXT-P020 — side laminated glazing cassette	SOURCE	18 ea	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-EXT-P030 — single panoramic heated end-glass assembly	BID	2 ea	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-EXT-P040 — hvac-24kw-direct-hv-dc roof HVAC	BID	3 ea	roof HVAC	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-EXT-P050 — roof PV module and edge-clamp kit	SOURCE	36 module	roof PV	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation

Product	Route	Qty	Mass responsibility	Evidence state	Minimum evidence
LM3-EXT-P060 — stepped floor-board and removable service-hatch system	SOURCE	135 m2	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-EXT-P061 — welded resilient floor covering, cove, nosing, and adhesive system	SOURCE	135 m2	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-EXT-P062 — longitudinal passenger and priority-seat modules	SOURCE	60 seat	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-EXT-P063 — stainless grab-pole, handrail, joint, and insulated adapter kit	SOURCE	3 car kit	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-EXT-P064 — passenger-information display, speaker, amplifier, and mounting kit	SOURCE	3 car kit	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-EXT-P065 — CCTV camera, passenger intercom, PoE/data, and mounting kit	SOURCE	3 car kit	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-EXT-P066 — PRM, safety-signage, emergency-lighting, extinguisher, and first-aid kit	SOURCE	3 car kit	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-FIX-P010 — OSR-RAIL-42 common ceiling, waist, and seat-zone service rail kit	MAKE	3 car kit	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-FIX-P020 — four-family captive fastener, floating nut, isolator, and access-fastener kit	SOURCE	3 car kit	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-FIX-P030 — standard passenger-fixture saddle and equipment adapter kit	MAKE	3 car kit	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-WIN-P010 — replaceable window pressure frame, dry seal, drain, and captive retention kit	MAKE	18 opening kit	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
Product	**Route**	**Qty**	**Mass responsibility**	**Evidence state**	**Minimum evidence**
LM3-DOOR-P010 — door four-point adjustable carrier, datum pin, dry seal,	MAKE	12 opening kit	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness

Product	Route	Qty	Mass responsibility	Evidence state	Minimum evidence
and keyed connector bracket kit					and finish density;first-article calibratedcomponent or completed-subassembly weigh record
`LM3-LGT-P010` — 1.2 m plug-in main LED lighting cassette and captive mounting kit	`SOURCE`	66 module	doors, glazing, interior, and auxiliaries	`unclosed-evidence-required`	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
`LM3-LGT-P020` — emergency and doorway lighting modules with independent keyed feeder kit	`BID`	3 car kit	doors, glazing, interior, and auxiliaries	`unclosed-evidence-required`	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
`LM3-EXT-P070` — roof antennas, service walkway pads, lifting covers, and maintenance labels	`SOURCE`	3 car kit	doors, glazing, interior, and auxiliaries	`unclosed-evidence-required`	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
`LM3-EXT-P080` — fire-rated GFRP side-module laminate, core, gelcoat, and consumable kit	`BID`	6 side kit	doors, glazing, interior, and auxiliaries	`unclosed-evidence-required`	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
`LM3-EXT-P090` — fire-rated GFRP roof-module, dry-seal, and removable skirt material kit	`BID`	3 car kit	doors, glazing, interior, and auxiliaries	`unclosed-evidence-required`	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
`LM3-FIN-P010` — pre-cut exterior livery graphic film, edge-seal, datum-mark, and repair-patch kit	`BID`	3 car kit	doors, glazing, interior, and auxiliaries	`unclosed-evidence-required`	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
`LM3-FIN-P020` — calcium-carbonate radiative roof-coating qualification and exposed-roof application kit	`BID`	3 car kit	doors, glazing, interior, and auxiliaries	`unclosed-evidence-required`	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
`LM3-INT-P010` — HVAC diffusers, side return ducts, saloon grilles, and access panels	`SOURCE`	3 car kit	roof HVAC	`unclosed-evidence-required`	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
`LM3-INT-P020` — FRP/phenolic ceiling liner panel set	`MAKE`	3 car kit	doors, glazing, interior, and auxiliaries	`unclosed-evidence-required`	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
`LM3-INT-P021` — ceiling light-trough bezel, diffuser carrier, and emergency-luminaire closeout set	`MAKE`	3 car set	doors, glazing, interior, and auxiliaries	`unclosed-evidence-required`	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
`LM3-INT-P022` — HVAC plenum cover, diffuser transition, detector bezel, and ceiling service-hatch set	`MAKE`	3 car set	doors, glazing, interior, and auxiliaries	`unclosed-evidence-required`	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record

Product	Route	Qty	Mass responsibility	Evidence state	Minimum evidence
LM3-INT-P030 — FRP/phenolic sidewall liner panel set	MAKE	6 side kit	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-INT-P031 — window reveal, setting-block inspection cover, and blind/label land set	MAKE	6 side set	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-INT-P032 — waist cable-cover, low-voltage access lid, and retained quarter-turn fastener set	MAKE	6 side set	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-INT-P040 — FRP battery strake cover panel set	MAKE	3 car kit	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-INT-P041 — seat-base fairing, retained equipment hatch, and rail-saddle closeout set	MAKE	3 car set	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-INT-P050 — FRP vestibule kick-panel and threshold-closeout set	MAKE	12 door-zone kit	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-INT-P051 — PRM transition-ramp, bogie-deck step-cover, contrast-nosing, and anti-slip set	MAKE	6 end-zone set	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-INT-P052 — door-pocket liner, sensitive-edge clearance trim, and removable jamb-cover set	MAKE	12 door-zone set	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-TRC-P010 — motor-350kw-hm47-class axle traction motor	BID	6 ea	traction motors, gearboxes, and controllers	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-TRC-P020 — single-stage reduction gearbox and flexible coupling	BID	6 ea	traction motors, gearboxes, and controllers	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-TRC-P030 — two independent motor controllers, isolated LV DC/DC, MPPT, station protection, and cooling-loop kit	BID	3 car kit	traction motors, gearboxes, and controllers	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-TRC-P040 — battery-225kwh-lfp-800v saloon-isolated side traction battery pack	BID	3 ea	traction batteries	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation

Product	Route	Qty	Mass responsibility	Evidence state	Minimum evidence
LM3-HV-P010 — battery sliding trays, retention straps, service interlocks, and drain pans	MAKE	3 car kit	traction batteries	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-HV-P020 — segregated HV cable tray, bonding studs, grommets, and orange cover set	MAKE	3 car kit	traction batteries	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-HV-P030 — coolant manifold brackets, bleed/drain points, and insulated pipe clamp kit	MAKE	3 car kit	traction batteries	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-TRC-P050 — roof-mounted regen dump resistor and thermal shield kit	SOURCE	3 ea	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-TRC-P060 — station side-pin charging connector, actuator, shutter, and alignment target	BID	3 car kit	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-TRC-P070 — HV contactor, fuse, pre-charge, service-disconnect, and current-sensor panel	SOURCE	3 car kit	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-SAF-P010 — battery temperature/off-gas detection, electrical-enclosure smoke detection, and localized mist kit	SOURCE	3 car kit	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-BOG-P040 — powered-bogie wheelset with axle-mounted brake discs	BID	6 wheelset	bogie frames, wheelsets, brakes, and suspension	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-BOG-P041 — trailer-bogie wheelset with axle-mounted brake discs	BID	6 wheelset	bogie frames, wheelsets, brakes, and suspension	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-BOG-P042 — powered-wheelset axlebox, sealed bearing unit, speed and temperature sensor set	BID	6 wheelset set	bogie frames, wheelsets, brakes, and suspension	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-BOG-P043 — trailer-wheelset axlebox, sealed bearing unit, speed and temperature sensor set	BID	6 wheelset set	bogie frames, wheelsets, brakes, and suspension	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
Product	**Route**	**Qty**	**Mass responsibility**	**Evidence state**	**Minimum evidence**
LM3-BOG-P044 — powered-bogie primary suspension spring, guide	BID	3 bogie set	bogie frames, wheelsets, brakes, and suspension	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate

Product	Route	Qty	Mass responsibility	Evidence state	Minimum evidence
and bump-stop set					with included/excludedscope; calibrated receiptweigh or acceptedinstalled-systemreconciliation
`LM3-BOG-P045` — trailer-bogie primary suspension spring, guide and bump-stop set	`BID`	3 bogie set	bogie frames, wheelsets, brakes, and suspension	`unclosed-evidence-required`	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
`LM3-BOG-P046` — powered-bogie to carbody connection: air springs, emergency spring, centre pivot, yaw links and dampers	`BID`	3 bogie set	bogie frames, wheelsets, brakes, and suspension	`unclosed-evidence-required`	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
`LM3-BOG-P047` — trailer-bogie to carbody connection: air springs, emergency spring, centre pivot, yaw links and dampers	`BID`	3 bogie set	bogie frames, wheelsets, brakes, and suspension	`unclosed-evidence-required`	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
`LM3-BOG-P048` — powered-bogie brake calipers, parking actuators, pads and wheel-slide hardware	`BID`	3 bogie set	bogie frames, wheelsets, brakes, and suspension	`unclosed-evidence-required`	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
`LM3-BOG-P049` — trailer-bogie brake calipers, parking actuators, pads and wheel-slide hardware	`BID`	3 bogie set	bogie frames, wheelsets, brakes, and suspension	`unclosed-evidence-required`	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
`LM3-AUX-P010` — secondary-suspension compressor, dryer, reservoir, and isolation-manifold kit	SOURCE	3 car kit	doors, glazing, interior, and auxiliaries	`unclosed-evidence-required`	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
`LM3-END-P010` — automatic end coupler and crash-energy absorber	`BID`	2 ea	end cowls and interfaces	`unclosed-evidence-required`	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
`LM3-ART-P020` — articulation lower spherical pivot, bearing housing and pin set	`BID`	2 joint set	inter-car articulation	`unclosed-evidence-required`	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
`LM3-ART-P021` — articulation upper lateral/yaw links, spherical joints and retained pins	`BID`	2 joint set	inter-car articulation	`unclosed-evidence-required`	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
`LM3-ART-P022` — inter-car double-wall corrugated bellows and clamp-frame set	`BID`	2 gangway set	inter-car articulation	`unclosed-evidence-required`	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
Product	**Route**	**Qty**	**Mass responsibility**	**Evidence state**	**Minimum evidence**
`LM3-ART-P023` — inter-car passenger bridge, turntable and flexible interior-panel set	`BID`	2 gangway set	inter-car articulation	`unclosed-evidence-required`	selected exact supplier configuration mass declaration or certificate with included/excluded

Product	Route	Qty	Mass responsibility	Evidence state	Minimum evidence
					scope; calibrated receiptweigh or acceptedinstalled-systemreconciliation
LM3-ART-P024 — articulation trainline carrier, support arms, abrasion liners and drain path	SOURCE	2 carrier set	inter-car articulation	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-CTRL-P010 — T-ECU/S and T-ECU/A compute and safety-control cabinet kit	BID	1 trainset kit	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-CTRL-P020 — navigation, balise, 5G, LoRa, GNSS, IMU, and roof-antenna kit	SOURCE	1 trainset kit	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-CTRL-P030 — maintenance HMI, depot pendant, emergency controls, and safety-relay kit	BID	1 trainset kit	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-CTRL-P040 — pre-terminated LV trainline harness, DIN cabinet, and terminal-distribution kit	MAKE	3 car kit	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-CTRL-P050 — operational and crashworthy event-recorder storage kit	SOURCE	1 trainset kit	doors, glazing, interior, and auxiliaries	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-END-P020 — T-OBS nose sensor pack, heated window services, and washer kit	BID	2 ea	end cowls and interfaces	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-END-P030 — cowl service hatch, sensor backing bracket, washer-tube, and heater-cable clip kit	MAKE	2 end kit	end cowls and interfaces	unclosed-evidence-required	released production-solid CAD mass-property report with material, thickness and finish density; first-article calibrated component or completed-subassembly weigh record
LM3-END-P040 — e-coupler LV jumper, recovery trainline, and end harness breakaway kit	BID	2 end kit	end cowls and interfaces	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-END-P050 — sealed headlight, tail/marker light, threshold-warning, and end-lamp harness kit	SOURCE	2 end kit	end cowls and interfaces	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
LM3-ART-P030 — inter-car HV/LV jumper, coolant hose loop, energy chain, and drain sleeve kit	BID	2 articulation kit	inter-car articulation	unclosed-evidence-required	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation

Product	Route	Qty	Mass responsibility	Evidence state	Minimum evidence
`LM3-ART-P040` — train-to-train open-end articulation, gangway, drawbar, turntable, and service-jumper cassette	`BID`	0 joint kit	inter-car articulation	`inactive-option-not-weighed`	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation
`LM3-ART-P041` — train-to-train jumper blanking, transition harness, isolation label, and dust-cover kit	`SOURCE`	0 joint kit	inter-car articulation	`inactive-option-not-weighed`	selected exact supplier configuration mass declaration or certificate with included/excluded scope; calibrated receipt weigh or accepted installed-system reconciliation

Closure rule

Do not reduce controlled tare until all active row evidence is accepted and a signed trainset weight-and-balance report closes category totals, individual-car weights, axle loads and centre of gravity.

Trainset open release gap register

Repository source: `design/component-catalogue/catalog/buildable-trainset/open-release-gaps.md`

Generated from the same controlled product rows used by the definition and shop-traveler packs. Product ownership and procurement-BOM links are already resolved; the rows below still require supplier, component, or interface evidence before their maturity can become `release-candidate`.

Open product rows: 81 of 120.

Engineering ID	BOM lines	Route	Maturity / blocker	Component	Parent assembly	Closure evidence
LM3-ROOF-P030	B7, T14	MAKE	buildable-after-supplier-freeze	removable HVAC curb fairing, intake/exhaust skirt, and access-hatch moulding set	LM3-ROOF-SA410	fairing trim gauge; HVAC airflow keep-out; hatch removal trial; roof leak test
LM3-ROOF-P040	B7, T22	MAKE	buildable-after-supplier-freeze	PV junction plinth, cable-gland cover, antenna closeout, and walkway edge set	LM3-ROOF-SA410	gland/closeout trim gauge; drain-path test; bonding access check; service-removal trial
LM3-CWL-P010	B8	SOURCE	buildable-after-supplier-freeze	end-cowl fiberglass laminate, insert, adhesive, and coupon material kit	LM3-CWL-SA710	material batch trace; fire/smoke certificate; coupon layup record; adhesive shelf-life check
LM3-CWL-P011	B8	MAKE	buildable-after-supplier-freeze	CWL-FRP-01 upper brow and roof-cap fiberglass cast	LM3-CWL-SA710	mould release record; laminate coupon; trim-line gauge; roof-flange fit
LM3-CWL-P012	B8	MAKE	buildable-after-supplier-freeze	CWL-FRP-02 left cheek fiberglass cast	LM3-CWL-SA710	mould release record; laminate coupon; insert pull-out; split-gap gauge
LM3-CWL-P013	B8	MAKE	buildable-after-supplier-freeze	CWL-FRP-03 right cheek fiberglass cast	LM3-CWL-SA710	mould release record; laminate coupon; insert pull-out; split-gap gauge
LM3-CWL-P014	B8	MAKE	buildable-after-supplier-freeze	CWL-FRP-04 lower apron and anti-climber cover fiberglass cast	LM3-CWL-SA710	mould release record; laminate coupon; lamp pocket gauge; drain-path water test
LM3-CWL-P015	B8	MAKE	buildable-after-supplier-freeze	CWL-FRP-05 lamp, washer, and service-hatch fiberglass cast set	LM3-CWL-SA710	mould release record; insert pull-out; gasket compression check; hatch removal trial
LM3-CWL-P016	B8	MAKE	buildable-after-supplier-freeze	CWL-FRP-06 backing-ring flange fiberglass cast set	LM3-CWL-SA710	mould release record; glass-carrier land survey; bond-line witness; A/B interchange check
LM3-FAS-P010	B8, B27	MAKE	buildable-after-supplier-freeze	panoramic front-glass carrier ring, setting-block pockets, and secondary-retention frame	LM3-CWL-SA710	carrier datum survey; glass-edge clearance gauge; secondary-retention proof; pane removal trial
LM3-FAS-P020	B8, B17	MAKE	buildable-after-supplier-freeze	reversible front-lamp cassette tray, aiming adjusters, and retained service bracket	LM3-CWL-SA710	lamp datum gauge; aiming range/retention test; harness clearance; cassette removal trial
LM3-FAS-P030	B8, B27	SOURCE	buildable-after-supplier-freeze	front glazing/lamp EPDM seal, drain rail, washer sleeve, and edge-closeout kit	LM3-CWL-SA710	seal batch/certificate; compression map; drain flow test; water-ingress test
LM3-EXT-P010	B11, B25, E20	BID	buildable-after-supplier-freeze	electric plug/sliding door cassette	LM3-DOOR-SA310	EN 14752 evidence; obstruction test; closed-and-locked loop test
LM3-EXT-P020	B10	SOURCE	buildable-after-supplier-freeze	side laminated glazing cassette	LM3-WIN-SA320	glazing certificate; water ingress test; replacement method
LM3-EXT-P030	B27	BID	buildable-after-supplier-freeze	single panoramic heated end-glass assembly	LM3-END-SA700	glass certificate; heater test; bond/gasket procedure
LM3-EXT-P040	T14	BID	buildable-after-supplier-freeze	hvac-24kw-direct-hv-dc roof HVAC	LM3-ROOF-SA410	+50 C capacity evidence; condensate drain test; EMC/vibration evidence
LM3-EXT-P060	B12	SOURCE	buildable-after-supplier-freeze	stepped floor-board and removable service-hatch system	LM3-INT-SA330	fire certificate; panel load and deflection evidence; hatch removal trial; level/step and egress survey

Engineering ID	BOM lines	Route	Maturity / blocker	Component	Parent assembly	Closure evidence
LM3-EXT-P061	B13	SOURCE	buildable-after-supplier-freeze	welded resilient floor covering, cove, nosing, and adhesive system	LM3-INT-SA330	fire/smoke certificate; adhesive compatibility and cure record; welded-seam peel sample; slip and cleanability evidence
LM3-EXT-P062	B14	SOURCE	buildable-after-supplier-freeze	longitudinal passenger and priority-seat modules	LM3-INT-SA330	fire/smoke certificate; seat/occupant load evidence; fastener and anti-rotation record; egress and cleaning-clearance gauge
LM3-EXT-P063	B15	SOURCE	buildable-after-supplier-freeze	stainless grab-pole, handrail, joint, and insulated adapter kit	LM3-INT-SA330	material/finish certificate; joint locking record; fixture-specific proof-load evidence; reach, egress and snag survey
LM3-EXT-P064	B18, E14	SOURCE	buildable-after-supplier-freeze	passenger-information display, speaker, amplifier, and mounting kit	LM3-INT-SA330	fire/EMC evidence; network enumeration; audio/intelligibility test; display visibility and service-removal trial
LM3-EXT-P065	B19, E15	SOURCE	buildable-after-supplier-freeze	CCTV camera, passenger intercom, PoE/data, and mounting kit	LM3-INT-SA330	fire/EMC/IP evidence; network enumeration; camera coverage/privacy review; intercom call and service-removal trial
LM3-EXT-P066	A1, A2, A3, A4	SOURCE	buildable-after-supplier-freeze	PRM, safety-signage, emergency-lighting, extinguisher, and first-aid kit	LM3-INT-SA330	accessible reach/contrast review; emergency-light duration test; equipment certificate/expiry audit; location and egress survey
LM3-FIX-P020	B2, B21	SOURCE	buildable-after-supplier-freeze	four-family captive fastener, floating nut, isolator, and access-fastener kit	LM3-FIX-SA340	supplier certificate; batch/finish trace; installed-grip gauge; locking and captive-part audit
LM3-FIX-P030	B14, B15, E14	MAKE	concept	standard passenger-fixture saddle and equipment adapter kit	LM3-FIX-SA340	adapter gauge; fixture-specific load calculation; proof-load sample; egress and snag check
LM3-WIN-P010	B10	MAKE	concept	replaceable window pressure frame, dry seal, drain, and captive retention kit	LM3-WIN-SA320	pressure-frame gauge; retention calculation; seal compression record; water-ingress and replacement trial
LM3-DOOR-P010	B11, B25	MAKE	concept	door four-point adjustable carrier, datum pin, dry seal, and keyed connector bracket kit	LM3-DOOR-SA310	carrier datum gauge; interface load calculation; seal compression record; connector keying and cassette replacement trial
LM3-LGT-P010	B16	SOURCE	buildable-after-supplier-freeze	1.2 m plug-in main LED lighting cassette and captive mounting kit	LM3-LGT-SA350	rail fire certificate; shock/vibration evidence; photometric/lux test; plug polarity and retention test
LM3-LGT-P020	B16, A4	BID	buildable-after-supplier-freeze	emergency and doorway lighting modules with independent keyed feeder kit	LM3-LGT-SA350	emergency duration/effectiveness evidence; evacuation visibility test; feed isolation test; doorway illumination test
LM3-EXT-P070	E21	SOURCE	buildable-after-supplier-freeze	roof antennas, service walkway pads, lifting covers, and maintenance labels	LM3-ROOF-SA410	antenna VSWR test; walkway slip certificate; lifting-cover fit; roof bonding check
LM3-EXT-P080	B6	BID	buildable-after-supplier-freeze	fire-rated GFRP side-module laminate, core, gelcoat, and consumable kit	LM3-SHELL-A200	EN 45545 evidence; laminate coupon; resin/fibre batch trace; mould release record
LM3-EXT-P090	B7	BID	buildable-after-supplier-freeze	fire-rated GFRP roof-module, dry-seal, and removable skirt material kit	LM3-SHELL-A200	EN 45545 evidence; roof laminate coupon; seal certificate; service-removal trial; water and debris-ingress check
Engineering ID	BOM lines	Route	Maturity / blocker	Component	Parent assembly	Closure evidence
LM3-FIN-P010	B20, B28	BID	buildable-after-supplier-freeze	pre-cut exterior livery graphic film, edge-seal, datum-mark, and repair-patch kit	LM3-SHELL-A200	supplier rail-use statement; substrate/cure acceptance; adhesion coupon;

Engineering ID	BOM lines	Route	Maturity / blocker	Component	Parent assembly	Closure evidence
						edge/overlapinspection; cleaning/removal trial
LM3-FIN-P020	B20, B28	BID	concept	calcium-carbonate radiative roof-coating qualification and exposed-roof application kit	LM3-ROOF-SA410	rail fire/chemical evidence; GFRP adhesion and flexibility; UV/abrasion/wash ageing; initial and aged solar reflectance/emittance; one-car thermal/maintenance trial
LM3-INT-P010	T14	SOURCE	buildable-after-supplier-freeze	HVAC diffusers, side return ducts, saloon grilles, and access panels	LM3-INT-SA330	airflow balance; rattle check; access-panel removal; fire-material certificate
LM3-INT-P020	B21	MAKE	buildable-after-supplier-freeze	FRP/phenolic ceiling liner panel set	LM3-INT-SA330	fire-material certificate; trim-line gauge; fastener insert pull-out; rattle check
LM3-INT-P021	B21	MAKE	buildable-after-supplier-freeze	ceiling light-trough bezel, diffuser carrier, and emergency-luminaire closeout set	LM3-INT-SA330	fire-material certificate; light cassette gauge; emergency-feed identification; module removal trial
LM3-INT-P022	B21	MAKE	buildable-after-supplier-freeze	HVAC plenum cover, diffuser transition, detector bezel, and ceiling service-hatch set	LM3-INT-SA330	fire-material certificate; airflow opening gauge; detector access; hatch retention/rattle check
LM3-INT-P030	B21	MAKE	buildable-after-supplier-freeze	FRP/phenolic sidewall liner panel set	LM3-INT-SA330	fire-material certificate; visible-gap gauge; liner removal; edge-radius inspection
LM3-INT-P031	B21	MAKE	buildable-after-supplier-freeze	window reveal, setting-block inspection cover, and blind/label land set	LM3-INT-SA330	fire-material certificate; glass-edge clearance; cassette removal sweep; sharp-edge inspection
LM3-INT-P032	B21	MAKE	buildable-after-supplier-freeze	waist cable-cover, low-voltage access lid, and retained quarter-turn fastener set	LM3-INT-SA330	fire-material certificate; segregation inspection; captive-fastener audit; cover removal/rattle check
LM3-INT-P040	B7, B21	MAKE	buildable-after-supplier-freeze	FRP battery strake cover panel set	LM3-INT-SA330	fire-material certificate; no-saloon-opening inspection; HV warning label check; sharp-edge inspection
LM3-INT-P041	B21	MAKE	buildable-after-supplier-freeze	seat-base fairing, retained equipment hatch, and rail-saddle closeout set	LM3-INT-SA330	fire-material certificate; seat-load-path clearance; hatch removal trial; cleaning/sharp-edge inspection
LM3-INT-P050	B21	MAKE	buildable-after-supplier-freeze	FRP vestibule kick-panel and threshold-closeout set	LM3-INT-SA330	fire-material certificate; threshold/drain gauge; anti-slip witness; kick-panel retention test
LM3-INT-P051	B21	MAKE	buildable-after-supplier-freeze	PRM transition-ramp, bogie-deck step-cover, contrast-nosing, and anti-slip set	LM3-INT-SA330	fire-material certificate; PRM transition gauge; anti-slip/contrast evidence; trip-edge inspection
LM3-INT-P052	B21	MAKE	buildable-after-supplier-freeze	door-pocket liner, sensitive-edge clearance trim, and removable jamb-cover set	LM3-INT-SA330	fire-material certificate; door sweep/pinch gauge; jamb-cover removal; obstruction-test witness
LM3-TRC-P010	T1	BID	buildable-after-supplier-freeze	motor-350kw-hm47-class axle traction motor	LM3-TRC-SA615	motor datasheet; thermal curve; mounting-foot load proof; EMC evidence
LM3-TRC-P020	G19, T2	BID	buildable-after-supplier-freeze	single-stage reduction gearbox and flexible coupling	LM3-TRC-SA615	gear ratio certificate; oil access check; coupling alignment
LM3-TRC-P030	T3, T4, T7, T13, T20, T23	BID	buildable-after-supplier-freeze	two independent motor controllers, isolated LV DC/DC, MPPT, station protection, and cooling-loop kit	LM3-HV-SA510	HVIL test; coolant pressure test; EMC/bonding check
Engineering ID	**BOM lines**	**Route**	**Maturity / blocker**	**Component**	**Parent assembly**	**Closure evidence**
LM3-TRC-P040	T5, T6	BID	buildable-after-supplier-freeze	battery-225kwh-lfp-800v saloon-isolated side traction battery pack	LM3-HV-SA510	cell/module certificate; isolation test; no-saloon-opening

Engineering ID	BOM lines	Route	Maturity / blocker	Component	Parent assembly	Closure evidence
						inspection; outwardvent/fire containmentdata
LM3-TRC-P050	T15	SOURCE	buildable-after-supplier-freeze	roof-mounted regen dump resistor and thermal shield kit	LM3-ROOF-SA410	resistance certificate; thermal clearance; roof bonding; hot-surface label
LM3-TRC-P060	T12, T19	BID	buildable-after-supplier-freeze	station side-pin charging connector, actuator, shutter, and alignment target	LM3-HV-SA510	dock alignment test; HVIL test; shutter cycle test; emergency release
LM3-TRC-P070	T11, T16	SOURCE	buildable-after-supplier-freeze	HV contactor, fuse, pre-charge, service-disconnect, and current-sensor panel	LM3-HV-SA510	isolation test; pre-charge timing; fuse rating evidence; service-disconnect lockout
LM3-SAF-P010	T9, T10	SOURCE	buildable-after-supplier-freeze	battery temperature/off-gas detection, electrical-enclosure smoke detection, and localized mist kit	LM3-HV-SA510	detector certificate; loop continuity; mist proof-flow; reservoir/pump/pressure diagnostic; event-recorder input
LM3-BOG-P040	G3, G8	BID	buildable-after-supplier-freeze	powered-bogie wheelset with axle-mounted brake discs	LM3-BOG-SA611	wheel/axle heat certificates; press-force chart; back-to-back and runout report; ultrasonic inspection; balance record
LM3-BOG-P041	G3, G8	BID	buildable-after-supplier-freeze	trailer-bogie wheelset with axle-mounted brake discs	LM3-BOG-SA621	wheel/axle heat certificates; press-force chart; back-to-back and runout report; ultrasonic inspection; balance record
LM3-BOG-P042	G4, G14, G15	BID	buildable-after-supplier-freeze	powered-wheelset axlebox, sealed bearing unit, speed and temperature sensor set	LM3-BOG-SA611	bearing serial/clearance record; grease and seal certificate; axle journal fit; speed/temperature sensor calibration; rotation test
LM3-BOG-P043	G4, G14, G15	BID	buildable-after-supplier-freeze	trailer-wheelset axlebox, sealed bearing unit, speed and temperature sensor set	LM3-BOG-SA621	bearing serial/clearance record; grease and seal certificate; axle journal fit; speed/temperature sensor calibration; rotation test
LM3-BOG-P044	G5, G12	BID	buildable-after-supplier-freeze	powered-bogie primary suspension spring, guide and bump-stop set	LM3-BOG-SA611	load-deflection curves; matched-height report; compound/batch certificates; installed preload and clearance survey
LM3-BOG-P045	G5, G12	BID	buildable-after-supplier-freeze	trailer-bogie primary suspension spring, guide and bump-stop set	LM3-BOG-SA621	load-deflection curves; matched-height report; compound/batch certificates; installed preload and clearance survey
LM3-BOG-P046	G6, G7, G10, G11, G12	BID	buildable-after-supplier-freeze	powered-bogie to carbody connection: air springs, emergency spring, centre pivot, yaw links and dampers	LM3-BOG-SA610	vertical/lateral load curves; pivot proof and articulation limit; damper curves hot/cold; ride-height and anti-lift survey
LM3-BOG-P047	G6, G7, G10, G11, G12	BID	buildable-after-supplier-freeze	trailer-bogie to carbody connection: air springs, emergency spring, centre pivot, yaw links and dampers	LM3-BOG-SA620	vertical/lateral load curves; pivot proof and articulation limit; damper curves hot/cold; ride-height and anti-lift survey
LM3-BOG-P048	G8, G9, G16	BID	buildable-after-supplier-freeze	powered-bogie brake calipers, parking actuators, pads and wheel-slide hardware	LM3-BOG-SA611	brake-force calculation; friction-pair certificate; thermal capacity; parking holding test; WSP functional test
LM3-BOG-P049	G8, G9, G16	BID	buildable-after-supplier-freeze	trailer-bogie brake calipers, parking actuators, pads and wheel-slide hardware	LM3-BOG-SA621	brake-force calculation; friction-pair certificate; thermal capacity; parking holding test; WSP functional test
LM3-AUX-P010	G21	SOURCE	buildable-after-supplier-freeze	secondary-suspension compressor, dryer, reservoir, and isolation-manifold kit	LM3-CAR-A900	pressure certificate; leak test; dryer function; relief-valve test; service-access check
LM3-END-P010	B22, B23	BID	buildable-after-supplier-freeze	automatic end coupler and crash-energy absorber	LM3-END-SA700	EN 15227 absorber evidence; recovery procedure; bolt torque record

Engineering ID	BOM lines	Route	Maturity / blocker	Component	Parent assembly	Closure evidence
LM3-ART-P020	B9	BID	buildable-after-supplier-freeze	articulation lower spherical pivot, bearing housing and pin set	LM3-ART-SA810	bearing static/dynamic capacity; pin material/NDT; proof load; lubrication/sealing plan; motion-envelope proof
LM3-ART-P021	B9	BID	buildable-after-supplier-freeze	articulation upper lateral/yaw links, spherical joints and retained pins	LM3-ART-SA810	link buckling/fatigue proof; joint angular capacity; pin retention inspection; full-motion sweep
LM3-ART-P022	B9	BID	buildable-after-supplier-freeze	inter-car double-wall corrugated bellows and clamp-frame set	LM3-ART-SA820	fire/smoke evidence; pressure/water ingress test; fatigue-cycle evidence; replaceable-clamp demonstration
LM3-ART-P023	B9	BID	buildable-after-supplier-freeze	inter-car passenger bridge, turntable and flexible interior-panel set	LM3-ART-SA820	passenger load proof; anti-slip evidence; gap/step gauge; pinch/shear hazard review; full-motion sweep
LM3-ART-P024	B24	SOURCE	buildable-after-supplier-freeze	articulation trainline carrier, support arms, abrasion liners and drain path	LM3-ART-SA830	rated bend radius; dynamic sweep; abrasion/fire evidence; drain test; service replacement trial
LM3-CTRL-P010	E1, E2, E6, E14, E15	BID	buildable-after-supplier-freeze	T-ECU/S and T-ECU/A compute and safety-control cabinet kit	LM3-SYS-SA900	hardware BOM check; self-test; network enumeration; firmware record; safety-output test
LM3-CTRL-P020	E3, E4, E5, E6, E7, E8, E21	SOURCE	buildable-after-supplier-freeze	navigation, balise, 5G, LoRa, GNSS, IMU, and roof-antenna kit	LM3-SYS-SA900	SKU/firmware record; antenna VSWR; GNSS/IMU test; balise read; radio link test
LM3-CTRL-P030	E10, E11, E12, E13, E16	BID	buildable-after-supplier-freeze	maintenance HMI, depot pendant, emergency controls, and safety-relay kit	LM3-SYS-SA900	key/guarded-control test; emergency input test; 2oo2 relay test; stowage and access check
LM3-CTRL-P050	E9, E23	SOURCE	buildable-after-supplier-freeze	operational and crashworthy event-recorder storage kit	LM3-SYS-SA900	write/read test; retention configuration; crashworthy certificate; download/recovery test
LM3-END-P020	E15, E18, E19	BID	buildable-after-supplier-freeze	T-OBS nose sensor pack, heated window services, and washer kit	LM3-END-SA700	sensor calibration; washer/heater test; 2oo2 verdict interface test
LM3-END-P040	B22, E17	BID	buildable-after-supplier-freeze	e-coupler LV jumper, recovery trainline, and end harness breakaway kit	LM3-END-SA700	pinout test; breakaway force check; ingress protection; rescue compatibility
LM3-END-P050	B17	SOURCE	buildable-after-supplier-freeze	sealed headlight, tail/marker light, threshold-warning, and end-lamp harness kit	LM3-END-SA700	photometric certificate; function/polarity test; ingress protection; A/B-end interchange check
LM3-ART-P030	B24	BID	buildable-after-supplier-freeze	inter-car HV/LV jumper, coolant hose loop, energy chain, and drain sleeve kit	LM3-ART-SA830	bend-radius sweep; trainline continuity; coolant pressure test; water-drain test
LM3-ART-P040	B9, B29	BID	buildable-after-supplier-freeze	train-to-train open-end articulation, gangway, drawbar, turntable, and service-jumper cassette	LM3-TTART-SA850	train-to-train motion-envelope proof; walk-through gangway fire evidence; trainline continuity; water ingress/drain test
LM3-ART-P041	B24	SOURCE	buildable-after-supplier-freeze	train-to-train jumper blanking, transition harness, isolation label, and dust-cover kit	LM3-TTART-SA850	pinout test; blanking cover ingress check; isolation label inspection; bend-radius sweep

Non-product release gates

The product rows above do not close controlled manufacturing drawings, supplier-specific mass/power/envelope freezes, structural proof, physical first-article inspection, or signed shop-traveler evidence.

LM3 Production-Data Release Register

Repository source: `design/component-catalogue/catalog/buildable-trainset/production-data-release-register.md`

> Status: all locally made production data open and unissued.

This register connects every locally made product to its drawing package, planning route, tooling, work instruction, inspection characteristics and required production artifacts. Fill the JSON records only in a controlled project copy.

Products: 62 · Open: 62 · Required artifact slots: 472.

Planning Route Summary

Route	Products
`composite-mould-trim-and-fit`	16
`cut-form-machine-fixture-and-join`	19
`harness-cut-terminate-formboard-and-test`	3
`mixed-kit-cut-form-machine-and-fit`	13
`section-cut-machine-drill-and-fit`	9
`tube-cut-bend-bracket-and-pressure-test`	2

Per-Product Ownership

Product	Planning route	Packages	Drawings	Artifacts	Inspection links	Status
LM3-ART-P010 — articulation adapter frame, anti-lift keeper, and shim kit	cut-form-machine-fixture-and-join	LM3-FRP-120	1	8	7	open-unissued
LM3-BDY-P010 — laser-cut side sill beam, LH/RH	section-cut-machine-drill-and-fit	LM3-FRP-010	3	7	7	open-unissued
LM3-BDY-P020 — underframe centre spine and longitudinal load-path kit	cut-form-machine-fixture-and-join	LM3-FRP-010	3	8	7	open-unissued
LM3-BDY-P021 — underframe cross-bearer, door-bay outrigger, and equipment-bracket pack	cut-form-machine-fixture-and-join	LM3-FRP-010	3	8	7	open-unissued
LM3-BDY-P030 — bolster box, air-spring pad, and centre-pivot insert set	cut-form-machine-fixture-and-join	LM3-FRP-010	3	8	7	open-unissued
LM3-BDY-P040 — coupler pocket, shear plate, and crash-can insert kit	cut-form-machine-fixture-and-join	LM3-FRP-110	1	8	7	open-unissued
LM3-BDY-P050 — battery tray rails, vent plenum, and service-lid gutter kit	mixed-kit-cut-form-machine-and-fit	LM3-FRP-110	1	7	7	open-unissued
LM3-BDY-P060 — low-floor centre pan and removable service-floor support set	cut-form-machine-fixture-and-join	LM3-FRP-010	3	8	7	open-unissued
LM3-BDY-P061 — raised bogie-end deck, transition ramp, and removable hatch-frame set	cut-form-machine-fixture-and-join	LM3-FRP-010	3	8	7	open-unissued
LM3-BDY-P070 — side-wall post, door portal, waist rail, and cant rail kit	cut-form-machine-fixture-and-join	LM3-FRP-010	3	8	7	open-unissued
LM3-BDY-P080 — roof bow, HVAC rail, PV rail, and cable-tray bracket kit	cut-form-machine-fixture-and-join	LM3-FRP-010, LM3-FRP-050	6	8	16	open-unissued
LM3-BDY-P090 — end ring frame and anti-climber beam set	cut-form-machine-fixture-and-join	LM3-FRP-120	1	8	7	open-unissued
LM3-BDY-P100 — door portal reinforcement, threshold beam, and cassette shim kit	section-cut-machine-drill-and-fit	LM3-FRP-110	1	7	7	open-unissued
LM3-BDY-P110 — window carrier ring, bonded-gasket land, and replacement jack-point inserts	section-cut-machine-drill-and-fit	LM3-FRP-110	1	7	7	open-unissued
LM3-BDY-P120 — jacking pad, lifting eye, towing lug, and recovery-label kit	cut-form-machine-fixture-and-join	LM3-FRP-100	2	8	7	open-unissued
LM3-BDY-P130 — one-metre clip-on solid-side fiberglass body module	composite-mould-trim-and-fit	LM3-FRP-020, LM3-FRP-080	4	8	12	open-unissued
LM3-BDY-P131 — one-metre clip-on window-edge fiberglass side module	composite-mould-trim-and-fit	LM3-FRP-020, LM3-FRP-080	4	8	12	open-unissued
LM3-BDY-P132 — one-metre clip-on door-edge fiberglass side module	composite-mould-trim-and-fit	LM3-FRP-020, LM3-FRP-080	4	8	12	open-unissued
LM3-BDY-P133 — one-metre clip-on fiberglass roof skin and equipment-fairing module	composite-mould-trim-and-fit	LM3-FRP-020, LM3-FRP-050, LM3-FRP-090	7	8	22	open-unissued
LM3-BDY-P140 — keyed clip rail, captive retainer, anti-lift, and dry-seal car kit	section-cut-machine-drill-and-fit	LM3-FRP-020	3	7	8	open-unissued

Product	Planning route	Packages	Drawings	Artifacts	Inspection links	Status
LM3-BOG-P010 — powered bogie welded H-frame and motor-cradle weldment	cut-form-machine-fixture-and-join	LM3-FRP-130	1	8	7	open-unissued
LM3-BOG-P020 — trailer bogie welded H-frame	cut-form-machine-fixture-and-join	LM3-FRP-140	1	8	7	open-unissued
LM3-BOG-P030 — powered-bogie guards, cable guides, WSP brackets, and inspection covers	cut-form-machine-fixture-and-join	LM3-FRP-130	1	8	7	open-unissued
LM3-BOG-P031 — trailer-bogie guards, cable guides, WSP brackets, and inspection covers	cut-form-machine-fixture-and-join	LM3-FRP-140	1	8	7	open-unissued
LM3-BOG-P050 — powered-bogie motor torque link, anti-rotation stop, and safety lanyard bracket kit	cut-form-machine-fixture-and-join	LM3-FRP-130	1	8	7	open-unissued
LM3-BOG-P060 — powered-bogie brake/WSP/speed-sensor harness and junction-bracket kit	harness-cut-terminate-formboard-and-test	LM3-FRP-130	1	8	7	open-unissued
LM3-BOG-P061 — trailer-bogie brake/WSP/speed-sensor harness and junction-bracket kit	harness-cut-terminate-formboard-and-test	LM3-FRP-140	1	8	7	open-unissued
LM3-CTRL-P040 — pre-terminated LV trainline harness, DIN cabinet, and terminal-distribution kit	harness-cut-terminate-formboard-and-test	LM3-FRP-160	1	8	6	open-unissued
LM3-CWL-P011 — CWL-FRP-01 upper brow and roof-cap fiberglass cast	composite-mould-trim-and-fit	LM3-FRP-080	1	8	4	open-unissued
LM3-CWL-P012 — CWL-FRP-02 left cheek fiberglass cast	composite-mould-trim-and-fit	LM3-FRP-080	1	8	4	open-unissued
LM3-CWL-P013 — CWL-FRP-03 right cheek fiberglass cast	composite-mould-trim-and-fit	LM3-FRP-080	1	8	4	open-unissued
LM3-CWL-P014 — CWL-FRP-04 lower apron and anti-climber cover fiberglass cast	composite-mould-trim-and-fit	LM3-FRP-040, LM3-FRP-080	3	8	11	open-unissued
LM3-CWL-P015 — CWL-FRP-05 lamp, washer, and service-hatch fiberglass cast set	composite-mould-trim-and-fit	LM3-FRP-040, LM3-FRP-080	3	8	11	open-unissued
LM3-CWL-P016 — CWL-FRP-06 backing-ring flange fiberglass cast set	composite-mould-trim-and-fit	LM3-FRP-030	2	8	7	open-unissued
LM3-DOOR-P010 — door four-point adjustable carrier, datum pin, dry seal, and keyed connector bracket kit	section-cut-machine-drill-and-fit	LM3-FRP-110	1	7	7	open-unissued
LM3-END-P030 — cowl service hatch, sensor backing bracket, washer-tube, and heater-cable clip kit	tube-cut-bend-bracket-and-pressure-test	LM3-FRP-120	1	7	7	open-unissued
LM3-END-P060 — common reversible end-interface carrier ring, option bolt grid, and sealing datum kit	cut-form-machine-fixture-and-join	LM3-FRP-120	1	8	7	open-unissued
LM3-END-P061 — panoramic-end option shim, cowl/glass carrier, and sensor datum closeout kit	composite-mould-trim-and-fit	LM3-FRP-120	1	8	7	open-unissued
LM3-END-P062 — mid open-connection option portal trim, bellows clamp, threshold bridge, and drain kit	mixed-kit-cut-form-machine-and-fit	LM3-FRP-120	1	7	7	open-unissued

Product	Planning route	Packages	Drawings	Artifacts	Inspection links	Status
LM3-FAS-P010 — panoramic front-glass carrier ring, setting-block pockets, and secondary-retention frame	cut-form-machine-fixture-and-join	LM3-FRP-030	2	8	7	open-unissued
LM3-FAS-P020 — reversible front-lamp cassette tray, aiming adjusters, and retained service bracket	cut-form-machine-fixture-and-join	LM3-FRP-040	2	8	7	open-unissued
LM3-FIX-P010 — OSR-RAIL-42 common ceiling, waist, and seat-zone service rail kit	section-cut-machine-drill-and-fit	LM3-FRP-070	2	7	7	open-unissued
LM3-FIX-P030 — standard passenger-fixture saddle and equipment adapter kit	section-cut-machine-drill-and-fit	LM3-FRP-070	2	7	7	open-unissued
LM3-HV-P010 — battery sliding trays, retention straps, service interlocks, and drain pans	mixed-kit-cut-form-machine-and-fit	LM3-FRP-150	1	7	7	open-unissued
LM3-HV-P020 — segregated HV cable tray, bonding studs, grommets, and orange cover set	mixed-kit-cut-form-machine-and-fit	LM3-FRP-150	1	7	7	open-unissued
LM3-HV-P030 — coolant manifold brackets, bleed/drain points, and insulated pipe clamp kit	tube-cut-bend-bracket-and-pressure-test	LM3-FRP-150	1	7	7	open-unissued
LM3-INT-P020 — FRP/phenolic ceiling liner panel set	composite-mould-trim-and-fit	LM3-FRP-060	2	8	8	open-unissued
LM3-INT-P021 — ceiling light-trough bezel, diffuser carrier, and emergency-luminaire closeout set	mixed-kit-cut-form-machine-and-fit	LM3-FRP-060	2	7	8	open-unissued
LM3-INT-P022 — HVAC plenum cover, diffuser transition, detector bezel, and ceiling service-hatch set	mixed-kit-cut-form-machine-and-fit	LM3-FRP-060	2	7	8	open-unissued
LM3-INT-P030 — FRP/phenolic sidewall liner panel set	composite-mould-trim-and-fit	LM3-FRP-060	2	8	8	open-unissued
LM3-INT-P031 — window reveal, setting-block inspection cover, and blind/label land set	mixed-kit-cut-form-machine-and-fit	LM3-FRP-060	2	7	8	open-unissued
LM3-INT-P032 — waist cable-cover, low-voltage access lid, and retained quarter-turn fastener set	mixed-kit-cut-form-machine-and-fit	LM3-FRP-060	2	7	8	open-unissued
LM3-INT-P040 — FRP battery strake cover panel set	composite-mould-trim-and-fit	LM3-FRP-060	2	8	8	open-unissued
LM3-INT-P041 — seat-base fairing, retained equipment hatch, and rail-saddle closeout set	mixed-kit-cut-form-machine-and-fit	LM3-FRP-060	2	7	8	open-unissued
LM3-INT-P050 — FRP vestibule kick-panel and threshold-closeout set	composite-mould-trim-and-fit	LM3-FRP-060	2	8	8	open-unissued
LM3-INT-P051 — PRM transition-ramp, bogie-deck step-cover, contrast-nosing, and anti-slip set	mixed-kit-cut-form-machine-and-fit	LM3-FRP-060	2	7	8	open-unissued
LM3-INT-P052 — door-pocket liner, sensitive-edge clearance trim, and removable jamb-cover set	mixed-kit-cut-form-machine-and-fit	LM3-FRP-060	2	7	8	open-unissued

Product	Planning route	Packages	Drawings	Artifacts	Inspection links	Status
`LM3-ROOF-P010` — HVAC curb, drop-duct collar, condensate tray, and drain fitting kit	`mixed-kit-cut-form-machine-and-fit`	`LM3-FRP-050`	3	7	9	`open-unissued`
`LM3-ROOF-P020` — PV bonded-pad lands, raised rail kit, bonding jumpers, and roof isolation labels	`section-cut-machine-drill-and-fit`	`LM3-FRP-050`	3	7	9	`open-unissued`
`LM3-ROOF-P030` — removable HVAC curb fairing, intake/exhaust skirt, and access-hatch moulding set	`composite-mould-trim-and-fit`	`LM3-FRP-050`, `LM3-FRP-090`	4	8	14	`open-unissued`
`LM3-ROOF-P040` — PV junction plinth, cable-gland cover, antenna closeout, and walkway edge set	`mixed-kit-cut-form-machine-and-fit`	`LM3-FRP-050`, `LM3-FRP-090`	4	7	14	`open-unissued`
`LM3-WIN-P010` — replaceable window pressure frame, dry seal, drain, and captive retention kit	`section-cut-machine-drill-and-fit`	`LM3-FRP-110`	1	7	7	`open-unissued`

Release Rule

No locally made product may enter production until every required artifact is issued, revision-compatible, hash-bound, checked and approved and its package hold points permit work.

Use the drawing seeds to develop the issued drawings, the manufacturing controls for work execution, and the first-article ITP for actual acceptance results.

Boundary: This register assigns production-data ownership and evidence slots. Geometry envelopes, planning routes and blank artifact records are not shop dimensions, approved process data, inspection results or authority to manufacture. A competent design authority must issue the exact material, datums, tolerances, NC/flat-pattern or mould data and special-process controls.

LM3 small-component and fixture standard

Repository source: `design/component-catalogue/catalog/buildable-trainset/small-component-standard.md`

Generated by `tools/automation/buildable-trainset.sh`. This is the controlled supplier-neutral design reference for repeatable interior and cassette hardware. It does not release structural loads or numeric torque values.

- Revision: `A-DRAFT`
- Status: `design-reference-not-released`
- Principle: Use one datum rail and the smallest qualified fastener/connector family that matches load and service frequency; preserve supplier ownership of doors, glazing and safety-critical internal mechanisms.

Common datum rail

- ID: `OSR-RAIL-42`
- Section: `42.0 × 18.0 mm`
- Datum pitch: `50.0 mm`
- Material/interface: extruded aluminium rail with electrically isolated body attachments where required
- Functions: repeatable mounting datum; floating-nut capture; cable and connector bracket attachment; trim edge alignment
- Release evidence: fixture-specific load path calculation; rail pull-out and slip test; shock/vibration evidence; galvanic-isolation and fire-material review

Four fastener families

ID	Nominal arrangement	Intended uses	Prohibited uses	Installation control	Release authority
`OSR-FST-M6-CAPTIVE`	M6 captive flanged screw into floating service-rail nut	light cassettes; trim and service panels; small PIS and cable supports	seat or handhold primary load paths; door primary retention; glazing retention	positive seating, released torque table, witness mark and visual captive-part check	small-component standard plus accepted fastener batch and calibrated-tool procedure
`OSR-FST-M8-FLOAT`	M8 flanged bolt into floating nutplate or qualified blind-rivet nut	calculated seat and handrail adapters; door/window secondary carriers; PIS equipment trays	uncalculated passenger-restraint joints; primary crash structure; bogie or coupler joints	joint-specific load/preload calculation, released torque, locking feature and witness mark	released interface calculation and drawing plus accepted fastener batch
`OSR-FST-QT-CAPTIVE`	tool-operated captive quarter-turn stud and receptacle	frequently opened trim; lighting diffuser; inspection and access covers	structural retention; passenger handholds; door or glazing retention	supplier grip-range gauge, closed-position indication and pull-check sampling	accepted supplier part and receptacle/grip schedule
`OSR-FST-M10-SEAL`	M10 isolating flanged bolt into sealed floating nutplate	external cassette carriers; removable exterior equipment adapters	direct glass clamping; unsealed wet cavities; primary crash joints without calculation	interface calculation, seal compression gauge, released torque and water-ingress test	released cassette interface drawing and calculation plus supplier installation manual

Keyed connector families

ID	Service	Interface	Keying	Release evidence
`OSR-CON-LV4`	24 V/48 V main and emergency lighting	touch-safe keyed four-pole plug with retained strain relief	main and emergency feeds use mechanically incompatible keys	rail fire/smoke evidence; shock/vibration evidence; pinout and polarity test; connector retention test
`OSR-CON-LVDATA12`	door, PIS, CCTV and local low-voltage/data services	keyed twelve-position plug family with blanking cap and fixed body-side bracket	different voltage/data duties use unique shell keys and labels	rail fire/smoke evidence; shock/vibration evidence; EMC-compatible termination; 100% continuity test

Modular illumination

- Main modules per car: `22 × 1200.0 mm`.
- Independent emergency modules per car: `4`.
- Door-threshold modules per car: `4`.
- Replacement rule: one module, two captive screws and one keyed plug; no field-cut strip or loose wire termination

Door cassette boundary

- Supplier boundary: complete certified supplier cassette including leaf, operator, lock, seals and controller
- OSR interface: four adjustable carrier shoes, two datum pins, dry perimeter seal and one keyed service bracket
- Release gate: supplier freeze, structural interface calculation and all applicable door safety tests

Window cassette boundary

- Supplier boundary: glass remains bonded to its supplier aluminium frame
- OSR interface: replaceable pressure frame, dry outer seal, drain path and captive secondary retention
- Replacement rule: remove internal pressure frame and withdraw cassette; do not cut adhesive at the carbody
- Release gate: glazing certificate, retention calculation, seal compression and water-ingress test

Design sources

- EU LOC&PAS TSI, Commission Regulation (EU) No 1302/2014: interior fixture retention, glazing, passenger doors and fire-safety release gates.
- ERA guide for the PRM TSI: emergency-lighting and evacuation-visibility intent.
- Knorr-Bremse IFE entrance systems: complete supplier-owned modular door system boundary.
- AGC glass with framing system: supplier-bonded framed glazing and rapid cassette replacement.
- Bollhoff RIVNUT blind-rivet nuts: controlled one-side thread installation in thin sections.
- Southco DZUS quarter-turn fasteners: captive, vibration-resistant repeated-access panel retention.
- HARTING compact railway connectors: compact rail-qualified plug interfaces for lights, doors and displays.

LM3 supplier anchors and local equivalents

Repository source: `design/component-catalogue/catalog/buildable-trainset/supplier-anchors.md`

Real supplier product families provide a purchasable/RFQ baseline and a reproducible interface target. They are anchors, not mandatory brands: a local equivalent is accepted only after the same controlled equivalence dossier passes.

Release boundary: A web page is not a purchase specification. Design authority must freeze the exact supplier part number, configuration, drawings, software, evidence, commercial terms and spares through RFQ before release. Safety-critical substitutes require calculation, test, independent assessment and authority acceptance.

Checked against manufacturer sources on 2026-08-30.

How to use this register

1. Use the named family to obtain an RFQ, sample, interface drawing and evidence pack; do not infer an order code from the family page.
2. Freeze the chosen order code/configuration and its controlled supplier documents in the city revision.
3. A local supplier may propose an equivalent. Compare it against every mandatory rule and product acceptance item, then repeat affected subassembly and vehicle tests.
4. Preserve the original anchor, accepted-equivalent record and hashes so later regeneration changes cost and geometry without silently changing safety assumptions.

Mandatory equivalence dossier:

- same or better released fit, mounting datums, connector keying and service envelope
- same or better mass, load, duty-cycle, thermal, electrical and environmental ratings
- same or better functional safety, fire, EMC, cybersecurity and applicable rail evidence
- documented failure modes, maintenance intervals, spares and obsolescence route
- first-article inspection plus component, subassembly and vehicle regression tests
- signed design-authority substitution record preserving the original anchor and evidence hashes

Controlled coverage

Measure	Value
Bought-in product rows with one primary anchor	56 / 56
Primary anchor families	27

Anchor families

Anchor	Manufacturer product family	State	Product rows	Local-equivalent route
OSR-ANC-DOOR-IFE	Knorr-Bremse IFE — IFE modular entrance systems	rfq-required	LM3-EXT-P010	Buy the first fleet cassettes and controls; localise portal, threshold, harness and service parts first; replace the cassette only after EN 14752-equivalent validation.
OSR-ANC-GLASS-AGC	AGC Automotive — rail glazing with framing system	rfq-required	LM3-EXT-P020, LM3-EXT-P030	Local glass processors may replace AGC after laminate, heater, optical, edge, fire, impact and bond-system qualification against the frozen cassette drawing.
OSR-ANC-FINISH-3M	3M Commercial Branding and Transportation — rail-application graphic films including Controltac 180mC	sample-and-system-rfq	LM3-FIN-P010	Print, plot, kit, apply, inspect and repair locally using the selected film/ink/overlaminate system; qualify a regional equivalent only as a complete system on the released LM3 base finish.
OSR-ANC-COOLROOF-PURDUE	Purdue University research reference — no nominated commercial supplier — 2020 CaCO3-acrylic passive radiative-cooling paint formulation	coupon-development-and-one-car-trial-only	LM3-FIN-P020	Develop/mix trial coupons locally under competent coatings and occupational-hygiene control; do not purchase or apply fleet-wide until a controlled commercial formulation and rail evidence are released.
OSR-ANC-HVAC-LONGERTEK	Longertek — direct-DC rail HVAC platform	rfq-required	LM3-EXT-P040, LM3-INT-P010	Keep the released 800 V-class input, thermal duty, refrigerant, drains, duct and CAN/Ethernet interface; source local coils, ducts and service parts after performance testing.
OSR-ANC-PV-SUNMAN	Sunman Energy — eArc lightweight photovoltaic modules	sample-and-rfq	LM3-EXT-P050, LM3-EXT-P070	Equivalent local modules are acceptable after roof fire, isolation, vibration, hail/impact, walk-zone and bonded/clamped retention tests.
OSR-ANC-FLOOR-FORBO	Forbo Flooring Systems — Transport Flooring rail systems	sample-and-rfq	LM3-EXT-P060, LM3-EXT-P061	Use the supplier system for first articles; qualify local board/covering as a complete substrate, adhesive, weld, cove, hatch and fire-tested system rather than swapping one layer.
OSR-ANC-SEAT-FAINSA	Fainsa — Metro and urban rail passenger seating	rfq-required	LM3-EXT-P062, LM3-EXT-P063	The OSR common rail and adapters are local; seat shells, frames and handrails may be locally made after fire, abuse-load, sharp-edge, corrosion and accessibility validation.
OSR-ANC-PASSENGER-LUMINATOR	Luminator Technology Group — rail passenger information, video and communication systems	rfq-required	LM3-EXT-P064, LM3-EXT-P065	Use open Ethernet/IP interfaces and OSR mounting adapters so displays, cameras, audio and intercoms can be substituted individually after EMC, cybersecurity, accessibility and system regression tests.
OSR-ANC-LIGHT-TEKNOWARE	Teknoware — rail interior, emergency and exterior lighting	rfq-required	LM3-EXT-P066, LM3-LGT-P010, LM3-LGT-P020, LM3-END-P050	Local luminaires are acceptable at the standard plug and mounting cassette after photometry, glare, fire, EMC, IP, emergency-duration and temperature tests.
OSR-ANC-FASTENER-BOLLHOFF	Böllhoff — RIVNUT blind rivet nuts and setting systems	catalogue-sourceable-after-sizing	LM3-FIX-P020	Equivalent inserts and captive hardware are allowed only by joint family after substrate-specific setting trials, proof loads, corrosion and removal-cycle tests.
OSR-ANC-COMPOSITE-GURIT	Gurit — fire-retardant composite laminating and core materials	sample-and-rfq	LM3-EXT-P080, LM3-EXT-P090	Resin, reinforcement, core and coating may be sourced locally only as a qualified laminate system with coupon, fire, weathering, insert and full-module evidence.
OSR-ANC-MOTOR-ABB-AMXM	ABB — AMXM railway traction motor	rfq-required	LM3-TRC-P010	Buy the initial motor; local rewind, bearings and eventually a locally manufactured equivalent require the frozen torque-speed, cooling, insulation, shaft, foot, mass and EMC interfaces plus type tests.
Anchor	**Manufacturer product family**	**State**	**Product rows**	**Local-equivalent route**
OSR-ANC-GEAR-VOITH-SE	Voith — SE-type single-stage axle-mounted metro gear unit and gear coupling	rfq-required	LM3-TRC-P020	Local housings/gears may replace the anchor only after tooth, bearing, lubrication, fatigue, efficiency, noise,

Anchor	Manufacturer product family	State	Product rows	Local-equivalent route
				overspeed and endurancequalification with the frozenmotor/wheel set interfaces.
OSR-ANC-TRACTION-ABB	ABB — BORDLINE CC400/ESC/PB and Traction Battery Pro reference platform	architecture-rfq-required	LM3-TRC-P030, LM3-TRC-P040, LM3-TRC-P050, LM3-TRC-P060, LM3-TRC-P070	Treat each converter, battery, charger/contact system and protection panel as separable behind frozen DC-link, cooling, HVIL, network and mechanical interfaces; qualify lower-cost local modules one at a time.
OSR-ANC-FIRE-FOGTEC	FOGTEC — rail fire detection and high-pressure water-mist systems	rfq-and-hazard-test-required	LM3-SAF-P010	Local detectors, pipework, reservoir and pumps may be introduced only within a vehicle fire-hazard analysis and propagation-tested suppression/detection system.
OSR-ANC-WHEELSET-GHH	GHH-BONATRANS — urban-rail wheelsets	rfq-required	LM3-BOG-P040, LM3-BOG-P041	Local wheel/axle manufacture is encouraged only after metallurgy, forging, heat treatment, NDT, press-fit, profile, fatigue and route-specific standards are qualified and independently accepted.
OSR-ANC-AXLEBOX-SKF	SKF — ready-to-mount railway axlebox bearing units with sensors	rfq-required	LM3-BOG-P042, LM3-BOG-P043	Local housings and sensor brackets may be made to the supplier-controlled bearing unit; an alternative bearing unit needs L10/load, fit, seal, grease, thermal, shock and field-monitoring evidence.
OSR-ANC-SUSPENSION-CONTI	Continental — rail primary suspension and secondary air-spring systems	rfq-and-dynamics-freeze-required	LM3-BOG-P044, LM3-BOG-P045, LM3-BOG-P046, LM3-BOG-P047	Springs, bushes and dampers may be sourced regionally after the multibody model, load-deflection/damping curves, compound ageing, fatigue and ride tests are repeated; pivot/anti-lift hardware stays separately calculated.
OSR-ANC-BRAKE-KNORR	Knorr-Bremse — WheelAct/AxleAct rail brake actuation and PistonSupply/DrySupply air supply	system-rfq-required	LM3-BOG-P048, LM3-BOG-P049, LM3-AUX-P010	Pads, discs, hoses and overhaul parts can localise first under brake-supplier approval; caliper/actuator or compressor substitutes require stopping, parking, thermal, WSP, degraded-mode and endurance revalidation.
OSR-ANC-COUPLER-DELLNER430	Dellner — automatic coupler Type 430 with pivot anchor	rfq-required	LM3-END-P010, LM3-END-P040	Local pocket, covers and harness may be manufactured to the Dellner interface; another coupler requires buff/draft, crash, rescue, gathering range, electrical head and fatigue compatibility evidence.
OSR-ANC-JOINT-SCHAEFFLER	Schaeffler — ELGES maintenance-free spherical plain bearings and rod ends	calculation-and-rfq-required	LM3-ART-P020, LM3-ART-P021	Local housings, pins and links are preferred; bearing or rod-end substitution requires static/dynamic capacity, misalignment, liner, corrosion, temperature, sealing and fatigue equivalence.
OSR-ANC-GANGWAY-HUBNER	HÜBNER — modular metro/urban-rail gangway systems	rfq-required	LM3-ART-P022, LM3-ART-P023, LM3-ART-P040	Bellows textile, panels and bridge pieces may localise after fire, fatigue, water, passenger load, pinch-gap and full-motion testing to the frozen clamp frames.
OSR-ANC-TRAINLINE-IGUS-HARTING	igus / HARTING — e-chain dynamic cable carriers with Han rail connectors	sample-and-rfq	LM3-ART-P024, LM3-ART-P030, LM3-ART-P041	Hoses and cables may be made locally to controlled drawings; carrier, connector or jacket substitutions require bend-cycle, separation, IP, fire, current, pressure, EMC and breakaway tests.
OSR-ANC-CONTROL-MOXA	Moxa — EN 50155 railway computers, Ethernet switches and wireless gateways	architecture-rfq-required	LM3-CTRL-P010, LM3-CTRL-P020, LM3-CTRL-P030	OSR software remains portable behind published I/O, TSN/Ethernet and update interfaces; alternate compute/network hardware must pass timing, safety allocation, EMC, environmental, cybersecurity and whole-train hardware-in-loop tests.
OSR-ANC-RECORDER-HASLER	HaslerRail — JRU/EVA event recording systems	rfq-required	LM3-CTRL-P050	Operational storage can be local commodity hardware; the protected recorder, data set, clock, crash/fire resistance and download chain require independently accepted railway evidence.

Anchor	Manufacturer product family	State	Product rows	Local-equivalent route
OSR-ANC-OBS-OUSTER	Ouster — OS industrial digital lidar sensor family	sample-and-safety-validation-required	LM3-END-P020	The nose carrier accepts alternative lidar/camera/radar modules after field of view, latency, heating/washer, ingress, EMC, failure-detection and complete obstacle-detection safety validation.

Parts subassemblies final trainset

These generated chains make the structural and mechanical ownership explicit. Repeated car/end/joint quantities remain in the product manifest.

Starting part	Controlled assembly chain
LM3-BDY-P010 laser-cut side sill beam, LH/RH	LM3-BDY-P010 laser-cut side sill beam, LH/RH LM3-BDY-SA110 underframe datum weldment LM3-BDY-SA120 carbody spaceframe and floor assembly LM3-SHELL-A200 painted carbody frame with one-metre clip-on fiberglass exterior LM3-CAR-A900 complete repeated car module LM3-TRAINSET-A000 complete light-metro trainset
LM3-BDY-P020 underframe centre spine and longitudinal load-path kit	LM3-BDY-P020 underframe centre spine and longitudinal load-path kit LM3-BDY-SA110 underframe datum weldment LM3-BDY-SA120 carbody spaceframe and floor assembly LM3-SHELL-A200 painted carbody frame with one-metre clip-on fiberglass exterior LM3-CAR-A900 complete repeated car module LM3-TRAINSET-A000 complete light-metro trainset
LM3-BDY-P030 bolster box, air-spring pad, and centre-pivot insert set	LM3-BDY-P030 bolster box, air-spring pad, and centre-pivot insert set LM3-BDY-SA110 underframe datum weldment LM3-BDY-SA120 carbody spaceframe and floor assembly LM3-SHELL-A200 painted carbody frame with one-metre clip-on fiberglass exterior LM3-CAR-A900 complete repeated car module LM3-TRAINSET-A000 complete light-metro trainset
LM3-CWL-P010 end-cowl fiberglass laminate, insert, adhesive, and coupon material kit	LM3-CWL-P010 end-cowl fiberglass laminate, insert, adhesive, and coupon material kit LM3-CWL-SA710 front/back fiberglass cowl cast kit LM3-END-SA700 train-end cowl, coupler, crash, and sensor assembly LM3-TRAINSET-A000 complete light-metro trainset
LM3-END-P010 automatic end coupler and crash-energy absorber	LM3-END-P010 automatic end coupler and crash-energy absorber LM3-END-SA700 train-end cowl, coupler, crash, and sensor assembly LM3-TRAINSET-A000 complete light-metro trainset
LM3-ART-P020 articulation lower spherical pivot, bearing housing and pin set	LM3-ART-P020 articulation lower spherical pivot, bearing housing and pin set LM3-ART-SA810 structural articulation joint and anti-lift load path LM3-ART-SA800 complete inter-car structural articulation, passenger gangway and service transfer LM3-TRAINSET-A000 complete light-metro trainset
LM3-ART-P022 inter-car double-wall corrugated bellows and clamp-frame set	LM3-ART-P022 inter-car double-wall corrugated bellows and clamp-frame set LM3-ART-SA820 passenger gangway bellows, bridge and turntable subassembly LM3-ART-SA800 complete inter-car structural articulation, passenger gangway and service transfer LM3-TRAINSET-A000 complete light-metro trainset
LM3-ART-P024 articulation trainline carrier, support arms, abrasion liners and drain path	LM3-ART-P024 articulation trainline carrier, support arms, abrasion liners and drain path LM3-ART-SA830 articulation service-transfer and segregated trainline subassembly LM3-ART-SA800 complete inter-car structural articulation, passenger gangway and service transfer LM3-TRAINSET-A000 complete light-metro trainset
LM3-BOG-P040 powered-bogie wheelset with axle-mounted brake discs	LM3-BOG-P040 powered-bogie wheelset with axle-mounted brake discs LM3-BOG-SA611 powered-bogie running unit: wheelsets, axleboxes, primary suspension and brakes LM3-BOG-SA610 complete powered bogie with running unit, bogie-mounted drive and body connection LM3-CAR-A900 complete repeated car module LM3-TRAINSET-A000 complete light-metro trainset
LM3-BOG-P042 powered-wheelset axlebox, sealed bearing unit, speed and temperature sensor set	LM3-BOG-P042 powered-wheelset axlebox, sealed bearing unit, speed and temperature sensor set LM3-BOG-SA611 powered-bogie running unit: wheelsets, axleboxes, primary suspension and brakes LM3-BOG-SA610 complete powered bogie with running unit, bogie-mounted drive and body connection LM3-CAR-A900 complete repeated car module LM3-TRAINSET-A000 complete light-metro trainset
LM3-BOG-P046 powered-bogie to carbody connection: air springs, emergency spring, centre pivot, yaw links and dampers	LM3-BOG-P046 powered-bogie to carbody connection: air springs, emergency spring, centre pivot, yaw links and dampers LM3-BOG-SA610 complete powered bogie with running unit, bogie-mounted drive and body connection LM3-CAR-A900 complete repeated car module LM3-TRAINSET-A000 complete light-metro trainset
LM3-BOG-P048 powered-bogie brake calipers, parking actuators, pads and wheel-slide hardware	LM3-BOG-P048 powered-bogie brake calipers, parking actuators, pads and wheel-slide hardware LM3-BOG-SA611 powered-bogie running unit: wheelsets, axleboxes, primary suspension and brakes LM3-BOG-SA610 complete powered bogie with running unit, bogie-mounted drive and body connection LM3-CAR-A900 complete repeated car module LM3-TRAINSET-A000 complete light-metro trainset
LM3-TRC-P010 motor-350kw-hm47-class axle traction motor	LM3-TRC-P010 motor-350kw-hm47-class axle traction motor LM3-TRC-SA615 bogie-mounted motor, gearbox, flexible coupling and torque-reaction drive unit LM3-BOG-SA610 complete powered bogie with running unit, bogie-mounted drive and body connection LM3-CAR-A900 complete repeated car module LM3-TRAINSET-A000 complete light-metro trainset
LM3-TRC-P020 single-stage reduction gearbox and flexible coupling	LM3-TRC-P020 single-stage reduction gearbox and flexible coupling LM3-TRC-SA615 bogie-mounted motor, gearbox, flexible coupling and torque-reaction drive unit LM3-BOG-SA610 complete powered bogie with running unit, bogie-mounted drive and body connection LM3-CAR-A900 complete repeated car module LM3-TRAINSET-A000 complete light-metro trainset

Known fit gaps

- OSR-ANC-DOOR-IFE: Exact aperture, stroke, voltage, software and certification configuration remains to be quoted.

- OSR-ANC-GLASS-AGC: Curvature, heater power, optical zones, laminate stack and frame/bond geometry remain project-specific.
- OSR-ANC-FINISH-3M: Exact film, colour/ink, overlaminate, edge treatment, fire evidence, hot-climate durability and warranty territory remain to be selected and tested.
- OSR-ANC-COOLROOF-PURDUE: No commercial rail-qualified product, vehicle durability pack, fire evidence or released LM3 application procedure is established by the research paper.
- OSR-ANC-HVAC-LONGERTEK: Public material does not freeze the LM3 24 kW duty, rail evidence or roof envelope.
- OSR-ANC-PV-SUNMAN: Select an exact module and rail-compatible clamp/bond stack after the roof curvature and electrical string are frozen.
- OSR-ANC-FLOOR-FORBO: Structural board selection and the final covering/adhesive combination remain to be frozen.
- OSR-ANC-SEAT-FAINSA: Exact seat model, upholstery, spacing, handrail geometry and calculated attachment loads remain open.
- OSR-ANC-PASSENGER-LUMINATOR: Exact display, camera, amplifier, intercom, recorder and software licences remain to be selected.
- OSR-ANC-LIGHT-TEKNOWARE: Exact lamp models, optics, voltage variants and emergency battery arrangement remain open.
- OSR-ANC-FASTENER-BOLLHOFF: Exact grip range, head, thread, material and setting force must be selected from each released joint drawing.
- OSR-ANC-COMPOSITE-GURIT: No public product family alone establishes the complete LM3 rail fire/structural laminate; formulation and process qualification remain open.
- OSR-ANC-MOTOR-ABB-AMXM: The current HM47-class planning envelope is not an ABB order code; ABB must confirm a 250 kW continuous / 350 kW short-peak, 800 V-compatible configuration and mass envelope.
- OSR-ANC-GEAR-VOITH-SE: Ratio, centre distance, torque, coupling, reaction-link and lubrication configuration remain to be quoted.
- OSR-ANC-TRACTION-ABB: The ABB platform is a rail-qualified architecture anchor, not proof that one published configuration meets the LM3 800 V LFP, 225 kWh/car, side-pin charging, packaging or price targets.
- OSR-ANC-FIRE-FOGTEC: Battery chemistry, enclosure, detection thresholds, nozzle layout, reservoir and control integration remain project-specific.
- OSR-ANC-WHEELSET-GHH: Wheel profile, diameter, axle journal, disc seats, material grades and load spectrum remain to be frozen.
- OSR-ANC-AXLEBOX-SKF: Bearing size, journal fit, housing, grounding, speed/temperature sensor and cable interfaces remain to be frozen.
- OSR-ANC-SUSPENSION-CONTI: Continental anchors spring systems; centre pivot, yaw-link and damper suppliers still require project RFQ and a common body-to-bogie interface control drawing.
- OSR-ANC-BRAKE-KNORR: The LM3 brake concept needs supplier confirmation of actuation energy, caliper variant, disc/pad pair, parking brake and whether the suspension-only compressor can support the selected brake architecture.
- OSR-ANC-COUPLER-DELLNER430: The published 600/500 kN family is an anchor; height, articulation, fold, electrical head, crash stroke and OSR pocket loads remain to be frozen.
- OSR-ANC-JOINT-SCHAEFFLER: No catalogue bearing is selected until the articulation load spectrum, pin diameter, angular motion and safety factors are released.
- OSR-ANC-GANGWAY-HUBNER: The vehicle-specific bellows, bridge, turntable, clamp and open-end drawbar geometry remains to be engineered with the supplier.
- OSR-ANC-TRAINLINE-IGUS-HARTING: The final carrier series, connector inserts, cable compounds, hose lengths and articulation sweep remain to be frozen.
- OSR-ANC-CONTROL-MOXA: Moxa anchors computing/network hardware only; the independent safety controller, navigation sensors, HMI and safety relays need exact part selection and safety allocation.
- OSR-ANC-RECORDER-HASLER: Exact recorder family, protected memory, interfaces and national event-data requirements remain open.

- `OSR-ANC-OBS-OUSTER`: The commercial sensor is an envelope/data anchor, not railway obstacle-detection certification or a complete T-OBS function.

Controlled inputs

- Anchor source: `lib/templates/trainset-supplier-anchors.toml` (`4621199776db6c8c0074555b71bfb03657967cd7d563e204accd009d647ea820`)
- Product manifest: `design/component-catalogue/catalog/buildable-trainset/buildable-trainset-manifest.json` (`ca385a6d3c27ee343d7f4bc4c90b4f4e27058380ff82a0aa76ca573275149703`)

LM3 configurable train-end interface

Repository source: `design/component-catalogue/catalog/buildable-trainset/train-end-interface.md`

Generated by `tools/automation/buildable-trainset.sh`. This design note records the single train end structure that can be dressed as either the normal panoramic glass front/end or a mid-train open connection for joining two train modules into a longer walk-through consist.

- Candidate: `light-metro-3car__16p5m__reference-body__reference-bogie__motor-350kw-hm47-class__battery-225kwh-lfp-800v__hvac-24kw-direct-hv-dc__pv12`
- Document revision: `A-DRAFT`
- Release status: `interface-design-basis`
- Principle: One train/car end structure is carried as a common configurable interface. Each end position is dressed with exactly one option: the closed panoramic glass end or the open mid-train connection.

Common Interface

- Assembly: `LM3-EIF-SA650`
- Owned items: `LM3-END-P060`, `LM3-END-P061`, `LM3-END-P062`

Interfaces:

- common S355 end carrier ring and option bolt grid
- replaceable EPDM sealing datum and drain lands
- shared underframe/coupler/articulation load-path datums
- protected HV/LV/trainline/coolant/HVAC service routes
- configuration record that locks each end as panoramic or open-mid

Manufacturing route:

- jig-weld and machine the common end carrier ring
- survey the common bolt grid and seal lands
- fit either panoramic closeout hardware or open-portal clamp/threshold hardware
- record selected end option before final trainset integration

Selectable End Options

Option	Assembly	Ref qty	Use case	Uses	Omits	Acceptance
`panoramic-glass-front-end`	`LM3-END-SA700`	2	outer A and C ends of the reference three-car trainset	LM3-END-P061 panoramic option shim/closeout kit LM3-CWL-SA710 multi-part fiberglass cowl LM3-EXT-P030 heated panoramic glass LM3-END-P010 automatic coupler/crash absorber LM3-END-P020 T-OBS sensor pack	open passenger portal train-to-train bellows and turntable through-passenger threshold bridge	panoramic option fit gauge glass/cowl datum transfer sensor calibration coupler datum survey water-ingress pre-test
`mid-open-train-to-train-connection`	`LM3-TTART-SA850`	0	optional end treatment when two complete train modules are semi-permanently married into one walk-through consist	LM3-END-P062 open-portal clamp/threshold/drain kit LM3-ART-P040 train-to-train articulation and gangway cassette LM3-ART-P041 service-jumper blanking and transition kit	panoramic glass fiberglass nose cowl nose-mounted T-OBS sensor pack at the joined end	open-portal gauge bellows clamp fit threshold/turntable level check train-to-train motion-envelope proof trainline continuity water ingress/drain test

Selection And Assembly Rules

- Each end position must select one option only. A panoramic end may not carry the open-portal threshold hardware, and an open-mid end may not carry the panoramic cowl/glass/sensor stack.

- Keep the common end-interface survey before option fit-out. Install panoramic hardware during outer-end assembly, or install the open gangway cassette only after both train modules are on the final assembly track and aligned at the train-to-train joint.

LM3 trainset build cost estimate

Repository source: `design/component-catalogue/catalog/buildable-trainset/trainset-build-cost.md`

Generated by `tools/automation/buildable-trainset.sh`. This is a rough-order build cost for one three-car LM3 trainset using the updated modular glass-fibre body, explicit assembly labour plan, and requested unexpected-cost premium.

- Candidate: `light-metro-3car__16p5m__reference-body__reference-bogie__motor-350kw-hm47-class__battery-225kwh-lfp-800v__hvac-24kw-direct-hv-dc__pv12`
- Family: `light-metro-3car`
- Release status: rough-order planning; not a supplier quote or signed production budget

Cost bucket	Basis	Cost
Direct material and supplier modules	Promoted design candidate cost metric	$682,431
Direct labour	5,524 h at $10/h	$55,240
Subtotal before premium	Material/supplier modules plus direct labour	$737,671
Unexpected-cost premium	20% for rework, logistics, consumables, local fabrication variation, and shop learning	$147,534
Total per 3-car trainset	Recalculated build estimate	$885,205
Rounded local-owner planning unit	Retained city-CAPEX reporting bucket	$900,000

Basis

Direct material and supplier-module cost comes from the promoted design candidate; labour hours come from the generated critical-path plan; the 20% premium covers unexpected local fabrication, rework, logistics, consumables, and shop-learning costs.

Included Fit-Out, Doors, And Glazing Scope

The direct material and supplier-module bucket already includes the interior fit-out, HVAC, windows, and doors below; do not add a second $20k interior allowance unless a separate contingency is intentionally being carried.

Included scope	BOM lines	Included base cost
seats, floors, grab rails, and interior lighting	B12, B13, B14, B15, B16	$17,500
roof HVAC	T14	$17,250
side windows, side doors, door sill/emergency kits, and panoramic end glass	B10, B11, B25, B27	$112,000
Requested scope total	Seats/floors/lighting + HVAC + doors/windows/end glass	$146,750

The seats/floors/grab-rail/lighting subset is $17,500, close to the $20k check value. HVAC adds $17,250, and doors/glazing add $112,000 before labour and premium.

Exclusions:

- route/platform compatibility changes
- supplier certification and homologation campaigns
- warranty and initial spares
- production-plant fixtures and tooling carried in the separate plant allowance

Generated design-system artifacts

Repository source: `design/component-catalogue/catalog/design-system/README.md`

This folder is generated by:

```
tools/automation/design-iterate.sh
```

The JSON and Markdown outputs record the best feasible rolling-stock candidate inside the declared design space, plus ranked shortlists for review. They are planning artefacts that guide which candidate should be promoted into FreeCAD/FEM, not homologated design releases.

Rolling-stock design iteration summary

Repository source: `design/component-catalogue/catalog/design-system/design-iteration-summary.md`

This file is generated by `tools/automation/design-iterate.sh`. It records the best candidate inside the declared discrete design space; it is not a supplier freeze or homologated drawing release.

- Iterations evaluated: 81
- Best candidate: `light-metro-3car__16p5m__reference-body__reference-bogie__motor-350kw-hm47-class__battery-225kwh-lfp-800v__hvac-24kw-direct-hv-dc__pv12`
- Feasible: `true`
- Score: `92.628`

Best-candidate metrics

Metric	Value
`cars`	3.0
`trainset_length_m`	49.5
`platform_margin_m`	10.0
`mass_t`	75.308
`cost_usd`	682431.18
`axle_load_kg`	6275.63
`installed_continuous_kw`	1500.0
`traction_margin`	2.845
`battery_usable_kwh`	540.0
`offwire_range_km`	68.915
`hvac_margin`	1.2
`pv_peak_kw`	15.12
`manufacturing_complexity_index`	1.0

Best-candidate modeled mass breakdown

This is the optimizer subtotal before the promoted engineering reserve.

Category	Mass (kg)
carbody primary structure	25623.53
bogie frames, wheelsets, brakes, and suspension	20190.00
traction motors, gearboxes, and controllers	3360.00
traction batteries	4350.00
roof HVAC	1530.00
roof PV	864.00
doors, glazing, interior, and auxiliaries	16350.00
end cowls and interfaces	1240.00
inter-car articulation	1800.00
Modeled subtotal	75307.53

Best-candidate parameters

Parameter	Value
family	`light-metro-3car`
car length	`16.5 m`
body structure gauge	`reference`
bogie frame gauge	`reference`
motor	`motor-350kw-hm47-class`
battery	`battery-225kwh-lfp-800v`
HVAC	`hvac-24kw-direct-hv-dc`
PV modules per car	`12`

Design hierarchy

External components

ID	Role	Selection class
`motor-350kw-hm47-class`	traction motor	350 kW peak heavy-vehicle PMSM, HM47-class qualified reference envelope
`battery-225kwh-lfp-800v`	traction battery pack per car	225 kWh gross / 180 kWh usable liquid-cooled LFP, 650–700 V nominal
`hvac-24kw-direct-hv-dc`	roof HVAC per car	24 kW direct 650–700 V DC packaged roof HVAC for high ambient

Fabricated parts

ID	Role	Process
`fab-carbody-primary-shell`	primary car body shell, side frames, floor pan, roof bows	CNC cut, press-brake, MIG/MAG weld, adhesive bond
`fab-bogie-frame-powered`	powered bogie frame and motor cradle weldment	cut plate, fixture weld, machine pivot/air-spring datums
`fab-bogie-frame-trailer`	trailer bogie frame weldment	cut plate, fixture weld, machine pivot/air-spring datums
`fab-cowl-and-interface-kit`	fiberglass end cowl casts, crash ring, backing ring, lamp/sensor hatches	low-volume moulding, trim/drill jig, adhesive bond
`fab-articulation-adapter-kit`	inter-car adapter frames, anti-lift keepers, service-loop brackets	cut, weld, machine shims and datum faces

Subassemblies

ID	Role	Acceptance checks
`sub-car-module`	one repeated 16.5 m car module	body datum survey door cassette gauge battery vent and service-lid gauge roof HVAC/PV clearance
`sub-powered-bogie`	powered bogie with motor, gearbox, brakes, wheelsets, and suspension	weld records wheelbase/gauge survey motor cradle load proof
`sub-final-trainset`	complete cableless consist assembled from repeated cars	trainset envelope mass budget traction/brake thermal budget FEM screening

Top ten feasible candidates

Rank	Candidate	Score	Mass t	Cost USD	Range km	Traction margin
1	light-metro-3car__16p5m__reference-body__reference-bogie__motor-350kw-hm47-class__battery-225kwh-lfp-800v__hvac-24kw-direct-hv-dc__pv12	92.628	75.308	682431	68.915	2.845
2	light-metro-3car__16p5m__reference-body__reference-bogie__motor-350kw-hm47-class__battery-225kwh-lfp-800v__hvac-24kw-direct-hv-dc__pv16	92.287	75.596	685311	68.826	2.835
3	light-metro-3car__16p5m__reference-body__light-bogie__motor-350kw-hm47-class__battery-225kwh-lfp-800v__hvac-24kw-direct-hv-dc__pv12	92.011	74.913	680601	69.036	2.86
4	light-metro-3car__16p5m__light-body__reference-bogie__motor-350kw-hm47-class__battery-225kwh-lfp-800v__hvac-24kw-direct-hv-dc__pv12	91.908	73.77	681226	69.391	2.905
5	light-metro-3car__16p5m__reference-body__light-bogie__motor-350kw-hm47-class__battery-225kwh-lfp-800v__hvac-24kw-direct-hv-dc__pv16	91.705	75.201	683481	68.948	2.85
6	light-metro-3car__16p5m__light-body__reference-bogie__motor-350kw-hm47-class__battery-225kwh-lfp-800v__hvac-24kw-direct-hv-dc__pv16	91.682	74.058	684106	69.302	2.893
7	light-metro-3car__16p5m__light-body__light-bogie__motor-350kw-hm47-class__battery-225kwh-lfp-800v__hvac-24kw-direct-hv-dc__pv12	91.169	73.376	679396	69.515	2.92
8	light-metro-3car__16p5m__reference-body__reference-bogie__motor-350kw-hm47-class__battery-225kwh-lfp-800v__hvac-24kw-direct-hv-dc__pv20	91.032	75.884	688191	68.738	2.824
9	light-metro-3car__16p5m__light-body__light-bogie__motor-350kw-hm47-class__battery-225kwh-lfp-800v__hvac-24kw-direct-hv-dc__pv16	90.942	73.664	682276	69.425	2.909
10	light-metro-3car__16p5m__light-body__reference-bogie__motor-350kw-hm47-class__battery-225kwh-lfp-800v__hvac-24kw-direct-hv-dc__pv20	90.541	74.346	686986	69.212	2.882

FEA Screening Outputs

Repository source: `design/component-catalogue/catalog/fea/README.md`

This folder contains generated FreeCAD/CalculiX first-pass screening artifacts. The Python source under `../../src/osr_mech/freecad_fea.py` defines the beam models, loads, supports, result parsing, and FreeCAD visual document generation. Each regeneration clears the old solver output folders and summaries first, then rewrites the stable latest paths documented below. Each study folder also includes a solver-derived `*-result.png` plot: deformed beam shape, support/load markers, and CalculiX von Mises stress colour scale parsed from the `.dat` output.

Outputs

Path	Purpose
`screening-summary.md`	Human-readable result summary
`screening-summary.json`	Machine-readable result summary
`dependency-check.json`	FreeCAD/FEM/CalculiX/gmsh availability captured at generation time
`chassis-bogie-screen/`	Low-floor chassis supported at bogie connectors: `.inp`, `.dat`, `.frd`, `.sta`, `.cvg`, `.12d`, and solver log
`chassis-aw3-proof-screen/`	1.5 x vertical AW3 proof-load screen for the low-floor chassis
`chassis-track-twist-screen/`	Asymmetric track-twist / uneven-load chassis screen
`bogie-frame-screen/`	Motor/trailer bogie H-frame screening model raw CalculiX inputs and outputs
`bogie-brake-traction-screen/`	Bogie vertical plus longitudinal brake/traction reaction screen
`full-body-frame-screen/`	Full car body side/roof frame screening model raw CalculiX inputs and outputs
`full-body-lateral-sway-screen/`	Full body side-frame lateral sway/racking screen
`train-to-train-joint-vertical-screen/`	Local open train-to-train joint vertical passenger/gangway load screen
`train-to-train-joint-lateral-sway-screen/`	Local open train-to-train joint lateral/racking load screen

Raw File Index

Every study folder contains the same generated file set: `{slug}.inp`, `{slug}.dat`, `{slug}.frd`, `{slug}.sta`, `{slug}.cvg`, `{slug}.12d`, `{slug}.ccx.log`, and `{slug}-result.png`.

Study	Result PNG
Chassis service gravity	`chassis-bogie-screen-result.png`
Chassis AW3 proof	`chassis-aw3-proof-screen-result.png`
Chassis track twist	`chassis-track-twist-screen-result.png`
Bogie vertical	`bogie-frame-screen-result.png`
Bogie brake/traction	`bogie-brake-traction-screen-result.png`
Full body vertical	`full-body-frame-screen-result.png`
Full body lateral sway	`full-body-lateral-sway-screen-result.png`
Train-to-train joint vertical	`train-to-train-joint-vertical-screen-result.png`
Train-to-train joint lateral sway	`train-to-train-joint-lateral-sway-screen-result.png`

Regenerate from the repository root with:

```
tools/automation/freecad-generate.sh --fem --screenshots
```

Or regenerate from `design/component-catalogue/` with the lower-level launchers:

```
scripts/freecad_fea.sh
scripts/freecad_screenshots.sh
```

The FreeCAD visual document is `../../models/cad/fea-screening-models.FCStd`, and the rendered review screenshots are embedded in `docs/rolling-stock/light-metro-3car/README.md`.

Rolling-Stock FEA Screening Summary

Repository source: `design/component-catalogue/catalog/fea/screening-summary.md`

This is a first-pass FreeCAD/CalculiX beam-model screen for gross load paths. It is not a homologation, fatigue, crashworthiness, weld, shell-buckling, or supplier-final mesh.

FEA Stack

- FreeCAD importable: True
- FreeCAD version: ['1', '1', '4', '45039 (Git)', 'https://github.com/FreeCAD/FreeCAD.git', '2026/09/28 13:46:34', '(HEAD detached at 1.1.4)', '4fd3bf320d9566a27e60069fc8387448aaa3a094']
- Modules: {'Fem': True, 'femtools': True, 'femmesh': True, 'Assembly': True}
- Executables: {'ccx': {'path': '/app/bin/ccx', 'version': ['This is Version 2.23']}, 'gmsh': {'path': '/app/bin/gmsh', 'version': ['4.15.0-git']}}

Results

Study	Load case	Nodes	Elements	Applied load kN	Vertical load kN	Max displacement mm	Target mm	Max von Mises MPa	SF to S355 yield	Result PNG	Status
Low-floor chassis supported at bogie connectors	360 kN vertical service load distributed over low-floor and high-floor bearer grid	155	159	360.0	360.0	20.885	25.0	101.1	3.51	PNG	OK
Low-floor chassis AW3 vertical proof-load screen	540 kN vertical proof load: 1.5 x the baseline 360 kN service gravity case	155	159	540.0	540.0	31.327	35.0	151.6	2.34	PNG	OK
Low-floor chassis asymmetric track-twist screen	Asymmetric 65/110/125 % side load bias plus 25 kN diagonal equipment offset	155	159	385.0	385.0	24.205	30.0	121.2	2.93	PNG	OK
Motor/trailer bogie H-frame screening model	160 kN bogie vertical load applied through two secondary-air-spring seats	29	24	160.0	160.0	2.110	5.0	57.6	6.16	PNG	OK
Bogie frame brake/traction longitudinal load screen	160 kN vertical bogie load plus 60 kN longitudinal brake/traction reaction	29	24	220.0	160.0	22.210	6.0	160.3	2.21	PNG	Review: screening deflection exceeds 6.0 mm target
Full car body side/roof frame screening model	420 kN body/payload plus 60 kN installed systems and 30 kN roof equipment allowance	72	168	510.0	510.0	3.035	15.0	41.2	8.61	PNG	OK
Full car body lateral sway screen	165 kN lateral body/interior/installed-systems equivalent load through side posts, waist rails, and cant rails	72	168	165.0	0.0	726.825	20.0	587.6	0.60	PNG	Review: screening deflection exceeds 20.0 mm target
Train-to-train open joint vertical load screen	90 kN vertical load through open portal, threshold bridge, lower drawbar, and upper links	48	51	90.0	90.0	18.332	12.0	255.1	1.39	PNG	Review: screening deflection exceeds 12.0 mm target
Train-to-train open joint lateral/racking screen	55 kN lateral load through open portal clamp frames, upper links, and threshold bridge	48	51	55.0	0.0	9.542	16.0	120.3	2.95	PNG	OK

Study Notes

Low-floor chassis supported at bogie connectors

- Bogie support points represent four secondary-air-spring/chassis interface pads.
- Reworked chassis uses deep side torsion boxes, twin keel beams, upper battery-zone chords, and stiffer cross-bearers.

Low-floor chassis AW3 vertical proof-load screen

- Uses the same bogie support pads as the service screen with a 1.5 x vertical load multiplier.
- This is a static proof-load screen only; fatigue and local weld toe stresses remain v2 work.

Low-floor chassis asymmetric track-twist screen

- Represents uneven passenger/load distribution during a track-twist or low-speed ramp transition.
- The diagonal equipment allowance biases the roof-side service mass onto one chassis side.

Motor/trailer bogie H-frame screening model

- Axlebox/primary-suspension support nodes constrain vertical displacement.
- Motor reaction brackets are not included in this load case; use the motor connector CAD for interface review.

Bogie frame brake/traction longitudinal load screen

- Adds a longitudinal force path through the secondary-seat structure on top of the vertical bogie case.
- Gearbox, axle, and detailed motor-bracket local stresses are still outside this beam-model screen.

Full car body side/roof frame screening model

- Side posts, side sills, waist rails, cant rails, and roof bows are idealised as S355 beams.
- Revised body includes diagonal side-frame racking members added after the initial lateral-sway exceedance.
- Composite panels and glazing are treated as non-structural for this screening pass.

Full car body lateral sway screen

- Applies an approximate 0.15 g lateral inertial load through the occupied side-frame height.
- The current candidate includes diagonal side-frame racking members and enlarged side/roof bearers.
- This complements the vertical body frame case; it is not a modal, ride, or fatigue analysis.

Train-to-train open joint vertical load screen

- Screens the local common end-interface carrier rings and open gangway cassette.
- The fixed-side ring is supported along its full moulded end-frame interface, matching the chassis/body pick-up concept.
- Passenger threshold bridge and gangway loads are included as vertical distributed loads.

Train-to-train open joint lateral/racking screen

- Complements the open-end vertical screen with a local racking case.
- The fixed-side ring is supported laterally along its full moulded end-frame interface.
- Supplier bellows fabric, rubber fatigue, clamps, and fastener details still require supplier proof evidence.

Geometry/Model Caveats

- Native study walls and B32R BOX sections share explicit dimensions and proposed 8 mm walls. These research sections are not supplier/CAD-catalogue acceptance; old solid-envelope results are historical diagnostics.
- Composite body panels, glazing, adhesive lands, local brackets, bolt holes, weld toes, and notches are not meshed.
- Loads are static screening loads only; no crash, fatigue spectrum, modal, thermal, or derailment cases are included.
- Any stress above the 0.6 x S355 yield service screen should trigger a detailed shell/solid mesh and local joint design.

One-metre clip-on glass-fibre body design

Repository source: `design/component-catalogue/catalog/modular-fiberglass-body/README.md`

Generated from `osr_mech.rolling_stock.modular_fiberglass_body`. The moulded GFRP panels are a non-structural weather skin over the welded steel frame.

Parameter	Value
Longitudinal module pitch	1,000 mm
Bays per car	16
Modules per car	48
Modules per 3-car trainset	144
Exterior-body assembly elapsed time	8.0 h

Moulded module fabrication route

Seq	Activity	Work center	Release check
10	Release mould, trim fixture, laminate schedule, core map, insert map, and module serial range	production control	traveler revision, material certificates, and mould release are accepted
20	Clean mould, inspect A-surface, apply release system, and apply UV-stable gelcoat or paint-primer layer	composite moulding cell	mould surface record and release-system lot recorded
30	Cut glass reinforcement and local core, lay up solid clip lands, pot insert bosses, and bag or close mould	composite moulding cell	ply/core/insert checklist matches the released module variant
40	Infuse or wet-lay laminate, control cure, demould, post-cure where specified, and keep witness coupons with the batch	controlled cure area	cure time/temperature record, coupon ID, and demould inspection complete
50	CNC trim door/window/roof variant, drill clip grid from datum, seal all cut edges, and mark serial/revision	trim and drill cell	trim gauge, hole-position gauge, and edge-seal record accepted
60	Fit captive inserts, clips, anti-lift features, drain details, and EPDM seals; dry-fit to master frame	module fit-up cell	insert pull-out lot, gasket compression witness, and master-frame fit accepted

Side, window-edge, door-edge, and roof variants share the same 1,000 mm mould pitch, clip datum, trim datum, edge-seal rule, and master-frame dry-fit check. The moulded module batch is complete before the eight-hour trainset installation shift starts.

Required fabrication evidence includes the mould release record, laminate schedule, resin and fibre batch trace, cure time/temperature record, witness coupon, CNC trim report, insert pull-out lot, sealed-edge record, and master-frame dry-fit record.

One-shift route

Seq	Activity	Elapsed	Parallel crews	Release check
10	Receive three released frames and verify clip-rail datums	0.5 h	3	frame and kit revision match
20	Fit dry EPDM seals and inspect keyed clip receptacles	1.0 h	6	continuous seal and open drains
30	Hang and latch left/right 1 m side modules	3.0 h	6	every captive witness mark visible
40	Hang and latch 1 m roof modules from mobile access stands	1.5 h	6	anti-lift retainers closed
50	Fit corner closures, skirts, labels, and earth bonds	1.0 h	6	bond continuity and service access
60	Water test, rattle check, snag closeout, and QA release	1.0 h	3	dry interior and signed module map

The eight-hour claim applies to installation and release of the exterior body modules on three completed, painted, dimensionally accepted frames. Doors, glazing, traction equipment, bogies, commissioning, and homologation remain separate controlled work.

Mechanical FreeCAD Library

Repository source: `design/component-catalogue/models/cad/README.md`

This folder contains `.FCStd` review assemblies generated directly from the parametric source under `../../src/osr_mech/`, which remains the authoritative geometry.

Documents

File	Purpose
`lm3-parts/`	120 separate native FreeCAD part documents: one controlled product-tree item per file, with selectable solids, envelope, route, maturity and source-manifest hash
`lm3-assemblies/`	26 subassembly/car/final-train documents containing every descendant part on a deterministic inspection fixture; start with `LM3-TRAINSET-A000.FCStd`
`lm3-product-library.index.json`	Exact file inventory, hashes, primitive counts and final-assembly reachability result
`trainset-light-metro-3car.FCStd`	Full light-metro trainset review assembly generated from source geometry
`trainset-light-metro-3car.installed-coordinate.json`	Source-to-controlled-ID occurrence map, installed positions and nominal/curve/twist review states for that assembly
`stations/`	Seven native station product assemblies covering all 45 controlled station products, with installed/exploded geometry, hierarchy, controlled datums/zones, semantic class and fidelity metadata
`../manufacturing-reference/`	STEP, DXF inspection projections and browser-viewable three-view SVG references for all 46 LM3 `MAKE` rows
`single-car-assembly.FCStd`	Corrected urban-shuttle single-car assembly with both bogies on the shared ±6,150 mm chassis datums
`chassis-bogie-assembly-states.FCStd`	Chassis and bogie connector review with assembled and exploded state groups
`full-body-assembly-states.FCStd`	Body frame, roof, windows, doors, floor, battery, bench, HVAC, lighting, and sensor attachment review states
`lm3-manufacturing-tooling.FCStd`	Twenty separately selectable local-manufacture mould, fixture, gauge, harness-board, bogie-alignment and final-assembly tooling families with product IDs, method steps, timings and release gates
`fea-screening-models.FCStd`	Visual FreeCAD document for the expanded CalculiX screening models and support/load markers
`station-scenes.FCStd`	Station, ballastless track, and driverless train review scenes for at-grade, elevated, and elevated interchange configurations
`civil-systems-integration-test.FCStd`	Source-driven civil/rolling-stock twin containing a ground station, 1:9 junction, six double-track viaduct spans, an elevated station, and two complete trains
`civil-systems-integration-test.json`	Portable twin asset, transform, relationship, state, validation, and FCStd-hash snapshot
`../../../../docs/assets/digital-twin-animation.gif`	160-frame FreeCAD/Coin3D operating animation with correctly framed LM3 ground and elevated movements
`native-catalogue-parts.FCStd`	Native FreeCAD replacements for civil, track, station, depot, and fabrication-template source families
`battery-pack-set.FCStd`	Per-car battery-pack source-geometry review document
`car-body-17m.FCStd`	Canonical car-body source-geometry review document
`motor-bogie.FCStd`	Powered-bogie source-geometry review document
`trailer-bogie.FCStd`	Trailer-bogie source-geometry review document
`raked-sensor-cowl.FCStd`	Raked obstacle-sensor end-cowl review document
`platform-at-grade-assembly.FCStd`	At-grade slab, platform edge, and complete track-panel assembly
`platform-elevated-assembly.FCStd`	U-girder, elevated deck, and complete track-panel assembly
`station-platform-canopy-assembly.FCStd`	Platform edge, track, canopy portals, roof, and PV assembly
`assembly-geometry-review.md`	Geometry-review notes for assembled and exploded FreeCAD documents

The matching captured review images are documented in `docs/rolling-stock/light-metro-3car/README.md` and `docs/stations/README.md`. The scripts below replace their stable output filenames on each run, so the README screenshots and `.FCStd` links always point at the latest generated review set.

The split library is generated from one shared 120-item geometry registry and contains 619 native OCC primitives. Assembly documents preserve all 26 EBOM/ MBOM hierarchy nodes and include every active product in the final trainset. Their spaced layout remains an inspection fixture. Installed source occurrences and four kinematic review states are recorded separately in the trainset installed-coordinate sidecar; neither artifact is a released tolerance stack. The matching dependency-timed `fabrication-assembly` twin tests the build order and provides the animated review.

Add-on and Render Toolchain

The supported launcher checks the `org.freecad.FreeCAD` Flatpak and records the actual runtime in generated evidence. Optional workbenches are review aids; they do not change the Python geometry or IFC/Bonsai authority boundaries.

Capability	Installed tool	Current use
Assembly review	FreeCAD 1.1 built-in Assembly, Assembly4, A2plus	Existing generated review states remain source-driven; Assembly4/A2plus are available for GUI datum, constraint, and kinematic inspection experiments

Capability	Installed tool	Current use
Structural testing	FreeCAD FEM plus CalculiX	Already used for chassis, bogie, body, and train-to-train joint screening cases in `../../catalog/fea/`
Mould/manufacturing checks	DFM workbench with `OCP`, `vtk`, and `gmsh` in FreeCAD's Flatpak Python	Available for draft, undercut, wall/thickness, bridge-span, and process checks on moulded GFRP/end-cowl candidates; not yet a release gate
High-quality stills	Render workbench plus the optional Blender/Cycles export path	`--high-quality-renders` uses Blender when installed; the GIF path does not require it

For repeatable repository animation, the selected path is FreeCAD's built-in Coin3D view renderer plus ImageMagick GIF encoding. The runner checks both dependencies and operates directly on the generated FCStd review file. Blender's Python animation API is the stronger future route for cinematic lighting and materials, while Godot's glTF/3D scene importer is suitable for a user-driven interactive twin. Neither Blender nor Godot is needed for the tracked GIF. FreeCAD's optional Animation workbench is also not required because the repository script drives placements and frames directly.

The MBDyn-style multibody dynamics add-ons were reviewed as candidates for rail dynamic testing, but are not installed as a default repo dependency because the available public workbench path is not yet a clean, maintained, headless pipeline for these models. Dynamic release therefore remains represented here by CalculiX quasi-static structural screens plus operational/dynamic commissioning plans, not by validated multibody ride simulation.

Regenerate

From the repository root, the canonical orchestrator is:

```
tools/automation/freecad-generate.sh --models --assemblies --station-library --neutral-exports --civil-systems --fem --
screenshots --station-scenes --digital-twin-animation --high-quality-renders
```

When iterating on one document, run the lower-level package launchers from `design/component-catalogue/`:

```
scripts/freecad_trainset.sh --family light-metro-3car
scripts/freecad_assembly_review.sh
scripts/freecad_product_library.sh
scripts/freecad_station_library.sh
scripts/freecad_neutral_exports.sh
scripts/freecad_fea.sh
scripts/freecad_screenshots.sh
scripts/freecad_station_scenes.sh
scripts/freecad_catalog.sh
scripts/freecad_civil_systems_example.sh
scripts/freecad_digital_twin_animation.sh
scripts/freecad_mesh_exports.sh
scripts/blender_freecad_renders.sh
```

Civil systems integration test

The integration document is organized into five independently selectable FreeCAD groups: an at-grade ground station, an at-grade 1:9 junction, viaduct approaches/substructure, an elevated station, and rolling stock. It assembles 82 placed source components and 13,861 native solids, including 12 U-girders on nine shared piers, direct-fixation track, platform edge kits, four solar canopies, translucent controlled kinematic envelopes, and two complete three-car light-metro trainsets.

Every placed feature carries a stable asset ID, asset class, parent zone, geometry role, and JSON operational state in its FreeCAD properties. The paired JSON snapshot records the same 82 assets, five zones, 86 relationships, two train states, nine interface checks, nine native clearance-clash checks, and a SHA-256 link back to the exact FCStd artifact. It is suitable for deterministic design review and as an initial-state handoff to OCC/CBM tooling; it does not claim to be a live telemetry feed.

Generation stops before saving if any native shape is null or invalid, or if an interface check fails. The checked datums are pier bearing-to-girder soffit, viaduct-to-station track support, 350 mm platform height above top of rail, and 75 mm platform clearance from the tangent dynamic envelope; both complete rolling-stock assemblies must also be present. Open the document's `IntegrationReviewNotes` object to see the recorded PASS results.

The four zones form a review test site, not an operational alignment. In particular, no vertical transition is implied between its at-grade and elevated zones. The wide elevated-station deck is explicitly a project-specific coordination envelope; structural, seismic, bearing, erection, and geotechnical design remain release work.

The animated review reopens the unchanged FCStd model for each scene, selects the relevant integration zones, moves each complete LM3 train along its checked track Y/Z datum, and computes the axonometric camera from visible source geometry. FreeCAD/Coin3D renders the PNG frames and ImageMagick encodes the looping GIF. No FreeCAD animation workbench is required. Regenerate and enforce the repository's strict sub-20-MB media limit with:

```
tools/automation/freecad-generate.sh --digital-twin-animation
```

The repository contains no Build123d implementation or dependency. The portable `osr_mech.cad` layer remains as the tested parametric source so geometry can be checked without FreeCAD; the documents above are the native FreeCAD replacement and review handoff.

The high-quality render path writes transient STL files to `design/component-catalogue/catalog/render-meshes/`. That directory is intentionally ignored because it is large and fully reproducible from the tracked FreeCAD documents and render scripts.

FreeCAD Assembly Geometry Review

Repository source: `design/component-catalogue/models/cad/assembly-geometry-review.md`

Generated directly from parametric source geometry. The checks below use FreeCAD/OCC `Shape.isValid()`, `Shape.check(True)`, solid counts, volume, and bounding-box sanity checks on each assembled-state input.

Chassis + Bogie Assembly

Item	Source	Valid	OCC check	Solids	Volume mm^3	Bounding box mm	Issue
Low-floor centre chassis	`low-floor-chassis`	True	True	41	13641594000	15940 x 2790 x 755	
Bogie-to-chassis connector package	`bogie-to-chassis-connector`	True	True	38	1472149073	13760 x 2440 x 370	
A-end motor bogie	`motor-bogie`	True	True	385	2229700152	3604 x 2690 x 1100	
B-end trailer bogie	`trailer-bogie`	True	True	339	1591914328	3604 x 2644 x 1074	
A-end bogie-to-motor connector	`bogie-to-motor-connector`	True	True	22	43773013	3354 x 817 x 327	

Full Body Assembly

Item	Source	Valid	OCC check	Solids	Volume mm^3	Bounding box mm	Issue
Body primary structure	car-body-structure	True	True	61	150759318286	16500 x 3110 x 3630	
Body exterior layer	car-body-exterior	True	True	528	8412698884	16320 x 2970 x 4396	
Body interior layer	car-body-interior	True	True	44	5842345400	15880 x 2770 x 2725	
Body service layers	car-body-services	True	True	38	3585089000	14630 x 2800 x 4005	
Car systems package	car-systems	True	True	365	6919568742	15595 x 3212 x 4325	
Mechanical interface package	mechanical-interface-package	True	True	1072	37298470307	18490 x 3076 x 3984	
Window cassettes and glazing installation	window-installations	True	True	72	1562806176	15560 x 2954 x 1197	
Door leaf design package	door-design	True	True	13	208207248	1382 x 168 x 2203	
Door portal and mount package	door-mounts	True	True	44	229984266	7700 x 2875 x 2365	
Door installation package	door-installations	True	True	28	2649559564	8020 x 3076 x 2520	
Door-to-body seal and interlock package	door-to-body-installations	True	True	56	153756033	7660 x 2780 x 2584	
Low-floor centre cabin flooring	cabin-flooring	True	True	30	1715407200	15080 x 2830 x 523	
Passenger bench and battery-strake mounts	bench-on-battery-installations	True	True	48	2074585760	15180 x 2508 x 1435	
HVAC roof ducting and supply plenums	hvac-roof-ducting-installation	True	True	38	3873196842	13780 x 2150 x 1144	
Interior lighting and emergency luminaires	internal-lighting-installation	True	True	96	120455279	13182 x 2124 x 543	
Universal ceiling, waist, and fixture rails	universal-service-rail-installation	True	True	112	84279200	13500 x 2590 x 2378	
Standard seat, handrail, and equipment adapters	standard-fixture-adapters	True	True	52	73716075	11780 x 2568 x 1955	
Simplified door and window cassette hardware	door-window-cassette-hardware	True	True	114	152148312	15558 x 2808 x 2110	
Passenger information screens and speakers	screen-speaker-mountings	True	True	39	220555641	13410 x 2542 x 506	
External lighting, lidar, radar, and cameras	external-lighting-lidar-system	True	True	32	399423624	17512 x 1310 x 1518	
Battery installation and contactor interfaces	battery-installations	True	True	58	617259840	13830 x 2856 x 476	
Side body frame and fixture attachments	side-body-frame-attachments	True	True	52	1603635840	15720 x 2836 x 2620	
Composite body and roof fixture attachments	composite-body-roof-attachments	True	True	33	3381460370	15350 x 3018 x 2441	

LM3 Manufacturing Tooling

Item	Source	Valid	OCC check	Solids	Volume mm^3	Bounding box mm	Issue
LM3-TOOL-STEEL-FIXTURE — Locally fabricated steel body, bogie and interface structures	manufacturing-tool:LM3-TOOL-STEEL-FIXTURE	True	True	22	13914104529	17000 x 3600 x 1350	
LM3-TOOL-DATUM-GAUGE — Locally fabricated steel body, bogie and interface structures	manufacturing-tool:LM3-TOOL-DATUM-GAUGE	True	True	4	75812028	3400 x 2744 x 620	
LM3-TOOL-SIDE-MOULD — Reusable GFRP body, roof, end-cowl and interior panel mould families	manufacturing-tool:LM3-TOOL-SIDE-MOULD	True	True	6	684647041	1300 x 3600 x 245	
LM3-TOOL-SIDE-VARIANT-NEST — Reusable GFRP body, roof, end-cowl and interior panel mould families	manufacturing-tool:LM3-TOOL-SIDE-VARIANT-NEST	True	True	15	584009957	1350 x 3550 x 225	
LM3-TOOL-ROOF-MOULD — Reusable GFRP body, roof, end-cowl and interior panel mould families	manufacturing-tool:LM3-TOOL-ROOF-MOULD	True	True	9	767202000	1300 x 3200 x 504	
LM3-TOOL-ROOF-FAIRING-MOULD — Reusable GFRP body, roof, end-cowl and interior panel mould families	manufacturing-tool:LM3-TOOL-ROOF-FAIRING-MOULD	True	True	7	2612115000	3600 x 2900 x 780	
LM3-TOOL-COWL-MOULD — Reusable GFRP body, roof, end-cowl and interior panel mould families	manufacturing-tool:LM3-TOOL-COWL-MOULD	True	True	8	4136400000	3300 x 3400 x 2410	
LM3-TOOL-TRIM-DRILL — Reusable GFRP body, roof, end-cowl and interior panel mould families	manufacturing-tool:LM3-TOOL-TRIM-DRILL	True	True	15	852438541	1400 x 3800 x 270	
LM3-TOOL-INT-CEILING-MOULD — Reusable GFRP body, roof, end-cowl and interior panel mould families	manufacturing-tool:LM3-TOOL-INT-CEILING-MOULD	True	True	7	641700000	1350 x 2950 x 550	
LM3-TOOL-INT-SIDE-MOULD — Reusable GFRP body, roof, end-cowl and interior panel mould families	manufacturing-tool:LM3-TOOL-INT-SIDE-MOULD	True	True	3	594627500	1350 x 2450 x 275	
LM3-TOOL-INT-STRAKE-MOULD — Reusable GFRP body, roof, end-cowl and interior panel mould families	manufacturing-tool:LM3-TOOL-INT-STRAKE-MOULD	True	True	3	2909400000	5200 x 1050 x 850	
LM3-TOOL-INT-DOOR-PRM-MOULD — Reusable GFRP body, roof, end-cowl and interior panel mould families	manufacturing-tool:LM3-TOOL-INT-DOOR-PRM-MOULD	True	True	4	2014730000	2800 x 1600 x 1920	
LM3-TOOL-COATING-RACK — In-mould base finish, applied livery film, and qualified radiative roof-coating route	manufacturing-tool:LM3-TOOL-COATING-RACK	True	True	7	899840000	4000 x 3160 x 2690	
Item	Source	Valid	OCC check	Solids	Volume mm^3	Bounding box mm	Issue
LM3-TOOL-COATING-COUPON — In-mould base	manufacturing-tool:LM3-TOOL-COATING-COUPON	True	True	3	540000	960 x 200 x 3	

Item	Source	Valid	OCC check	Solids	Volume mm^3	Bounding box mm	Issue
finish, applied liveryfilm, and qualifiedradiative roof-coating route							
LM3-TOOL-FILM-TEMPLATE — In-mould base finish, applied livery film, and qualified radiative roof-coating route	`manufacturing-tool:LM3-TOOL-FILM-TEMPLATE`	True	True	18	828225189	8500 x 1800 x 150	
LM3-TOOL-RADIATIVE-COUPON — In-mould base finish, applied livery film, and qualified radiative roof-coating route	`manufacturing-tool:LM3-TOOL-RADIATIVE-COUPON`	True	True	4	35040000	1100 x 520 x 67	
LM3-TOOL-WINDOW-GAUGE — Replaceable side-window cassette, pressure frame, seal and drain installation	`manufacturing-tool:LM3-TOOL-WINDOW-GAUGE`	True	True	8	265830881	2150 x 240 x 1590	
LM3-TOOL-GLASS-CARRIER-NEST — Replaceable side-window cassette, pressure frame, seal and drain installation	`manufacturing-tool:LM3-TOOL-GLASS-CARRIER-NEST`	True	True	14	379900828	2800 x 240 x 2120	
LM3-TOOL-WATER-TEST — Replaceable side-window cassette, pressure frame, seal and drain installation	`manufacturing-tool:LM3-TOOL-WATER-TEST`	True	True	12	314015660	2400 x 640 x 1940	
LM3-TOOL-DOOR-GAUGE — Supplier door cassette on locally made four-point adjustable carrier	`manufacturing-tool:LM3-TOOL-DOOR-GAUGE`	True	True	8	315030881	2000 x 240 x 2390	
LM3-TOOL-SEAL-GAUGE — Supplier door cassette on locally made four-point adjustable carrier	`manufacturing-tool:LM3-TOOL-SEAL-GAUGE`	True	True	6	1152000	1480 x 80 x 1530	
LM3-TOOL-FLOOR-TEMPLATE — Removable floor panels, transport flooring and common passenger-fixture rail	`manufacturing-tool:LM3-TOOL-FLOOR-TEMPLATE`	True	True	31	1465835734	15500 x 2700 x 112	
LM3-TOOL-FIXTURE-PROOF — Removable floor panels, transport flooring and common passenger-fixture rail	`manufacturing-tool:LM3-TOOL-FIXTURE-PROOF`	True	True	3	178685574	1200 x 900 x 2360	
LM3-TOOL-BOGIE-STAND — Supplier traction motor/gear box integration into locally fabricated powered bogie	`manufacturing-tool:LM3-TOOL-BOGIE-STAND`	True	True	6	847392000	4200 x 2620 x 740	
LM3-TOOL-MOTOR-ALIGN — Supplier traction motor/gear box integration into locally fabricated powered bogie	`manufacturing-tool:LM3-TOOL-MOTOR-ALIGN`	True	True	4	794542273	2200 x 1300 x 1220	
LM3-TOOL-SERVICE-RAIL — Lighting, HVAC, HV/LV services and keyed replaceable modules	`manufacturing-tool:LM3-TOOL-SERVICE-RAIL`	True	True	16	192471239	4000 x 600 x 180	
LM3-TOOL-HARNESS-BOARD — Lighting, HVAC, HV/LV services and keyed replaceable modules	`manufacturing-tool:LM3-TOOL-HARNESS-BOARD`	True	True	46	1155619115	8000 x 2400 x 160	

Item	Source	Valid	OCC check	Solids	Volume mm^3	Bounding box mm	Issue
LM3-TOOL-LAMP-AIM — Lighting, HVAC, HV/LV services and keyed replaceable modules	`manufacturing-tool:LM3-TOOL-LAMP-AIM`	True	True	5	497259710	2200 x 1100 x 1680	
LM3-TOOL-LIFT-COLUMNS — Three-car final assembly, static test and manufacturing release	`manufacturing-tool:LM3-TOOL-LIFT-COLUMNS`	True	True	16	11160960000	14900 x 4900 x 4800	
LM3-TOOL-FINAL-DATUM — Three-car final assembly, static test and manufacturing release	`manufacturing-tool:LM3-TOOL-FINAL-DATUM`	True	True	18	485191168	50000 x 120 x 380	

Geometry Issues

- No invalid source shapes or zero-size bounding boxes detected.

Developing-Country Planning Briefs

DR Congo National OpenSourceRail Strategy

Repository source: `cities/catalogue/central-africa/DR Congo/NATIONAL-BRIEF.md`

This page contains only DR Congo-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $40.72 B (87.8%) of external capital and $52.61 B of external interest. Capital plus saved interest totals $93.33 B.

National Programme

Local measure	Planning value
Catalogue cities	7
Represented population	27,007,000
Trainsets / vehicle modules	2,678 / 13,835
City infrastructure and fleet CAPEX	$25.00 B
Shared national factory	$703.2 M
Factory sizing basis	10,650 modules for Kinshasa, then reused nationally
Total national programme	$25.75 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$5.63 B (21.9%)
Domestic / local capital	$20.12 B (78.1%)
Annual external capital draw	$563.3 M / yr
Annual local capital draw	$2.01 B / yr
Annual public construction commitment	$2.75 B / yr for 10 years
Annual post-grace debt service	$2.49 B / yr
Default foreign-turnkey external capital	$46.36 B
External capital saved	$40.72 B
Capital + lifetime external interest saved	$93.33 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$10.69 B	$1.60 B	$9.08 B
Stations	$5.89 B	$1.18 B	$4.71 B
Depots	$1.13 B	$281.3 M	$843.8 M
Rolling stock	$3.90 B	$1.36 B	$2.53 B
Dedicated solar plants	$1.52 B	$682.4 M	$834.1 M
Residual train control	$55.9 M	$27.9 M	$27.9 M
Charging microgrids	$291.7 M	$116.7 M	$175.0 M
EPC / project services	$1.59 B	$237.8 M	$1.35 B
Shared national trainset factory	$703.2 M	$140.6 M	$562.6 M
Total	$25.75 B	$5.63 B	$20.12 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Kinshasa	17,178,000	1,775	$16.87 B	$3.83 B	$13.04 B	14	5	51.6%	54.8%
Lubumbashi	2,829,000	190	$2.21 B	$447.4 M	$1.76 B	8	3	21.9%	23.3%
Mbuji Mayi	2,500,000	165	$1.93 B	$390.7 M	$1.54 B	6	2	47.0%	37.5%
Kisangani	1,300,000	66	$751.8 M	$152.4 M	$599.4 M	3	1	12.8%	13.7%
Kananga	1,200,000	55	$746.3 M	$146.6 M	$599.8 M	3	1	14.3%	16.9%
Bukavu	1,000,000	241	$1.46 B	$301.4 M	$1.15 B	5	2	33.3%	47.1%
Goma	1,000,000	186	$1.04 B	$218.0 M	$818.7 M	5	2	30.2%	36.5%

Local Basis And Regeneration

Country finance parameters use `CD` in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Angola National OpenSourceRail Strategy

Repository source: `cities/catalogue/east-africa/Angola/NATIONAL-BRIEF.md`

This page contains only Angola-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $22.03 B (88.0%) of external capital and $27.09 B of external interest. Capital plus saved interest totals $49.12 B.

National Programme

Local measure	Planning value
Catalogue cities	9
Represented population	13,135,000
Trainsets / vehicle modules	1,678 / 7,045
City infrastructure and fleet CAPEX	$13.29 B
Shared national factory	$573.3 M
Factory sizing basis	4,356 modules for Luanda, then reused nationally
Total national programme	$13.90 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$2.99 B (21.5%)
Domestic / local capital	$10.91 B (78.5%)
Annual external capital draw	$599.0 M / yr
Annual local capital draw	$2.18 B / yr
Annual public construction commitment	$1.57 B / yr for 5 years
Annual post-grace debt service	$1.20 B / yr
Default foreign-turnkey external capital	$25.03 B
External capital saved	$22.03 B
Capital + lifetime external interest saved	$49.12 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$6.83 B	$1.02 B	$5.81 B
Stations	$1.78 B	$355.4 M	$1.42 B
Depots	$831.6 M	$207.9 M	$623.7 M
Rolling stock	$2.02 B	$706.9 M	$1.31 B
Dedicated solar plants	$905.0 M	$407.2 M	$497.7 M
Residual train control	$37.4 M	$18.7 M	$18.7 M
Charging microgrids	$80.0 M	$32.0 M	$48.0 M
EPC / project services	$850.4 M	$127.6 M	$722.8 M
Shared national trainset factory	$573.3 M	$114.7 M	$458.6 M
Total	$13.90 B	$2.99 B	$10.91 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Luanda	9,085,000	726	$7.70 B	$1.75 B	$5.95 B	14	5	40.8%	35.8%
Huambo	800,000	194	$1.09 B	$222.1 M	$865.3 M	5	2	20.5%	21.1%
Lubango	700,000	188	$1.09 B	$223.5 M	$863.3 M	4	1	15.3%	26.0%
Benguela	600,000	185	$982.9 M	$203.2 M	$779.7 M	5	2	30.4%	30.0%
Lobito	500,000	133	$761.3 M	$155.5 M	$605.7 M	4	1	18.2%	26.6%
Malanje	500,000	46	$280.9 M	$57.9 M	$223.0 M	2	0	12.9%	11.5%
Uige	400,000	39	$211.8 M	$44.3 M	$167.5 M	1	0	6.2%	14.0%
Namibe	300,000	103	$730.2 M	$135.8 M	$594.4 M	5	2	43.6%	35.6%
Soyo	250,000	64	$447.4 M	$84.5 M	$362.9 M	4	1	32.8%	24.2%

Local Basis And Regeneration

Country finance parameters use A0 in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Kenya National OpenSourceRail Strategy

Repository source: `cities/catalogue/east-africa/Kenya/NATIONAL-BRIEF.md`

This page contains only Kenya-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $44.42 B (88.3%) of external capital and $55.69 B of external interest. Capital plus saved interest totals $100.11 B.

National Programme

Local measure	Planning value
Catalogue cities	15
Represented population	11,750,000
Trainsets / vehicle modules	3,181 / 12,630
City infrastructure and fleet CAPEX	$27.14 B
Shared national factory	$750.0 M
Factory sizing basis	6,768 modules for Nairobi, then reused nationally
Total national programme	$27.95 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$5.88 B (21.1%)
Domestic / local capital	$22.06 B (78.9%)
Annual external capital draw	$840.5 M / yr
Annual local capital draw	$3.15 B / yr
Annual public construction commitment	$2.92 B / yr for 7 years
Annual post-grace debt service	$2.43 B / yr
Default foreign-turnkey external capital	$50.31 B
External capital saved	$44.42 B
Capital + lifetime external interest saved	$100.11 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$13.81 B	$2.07 B	$11.74 B
Stations	$4.71 B	$942.1 M	$3.77 B
Depots	$1.51 B	$377.2 M	$1.13 B
Rolling stock	$3.59 B	$1.26 B	$2.33 B
Dedicated solar plants	$1.58 B	$711.3 M	$869.4 M
Residual train control	$73.9 M	$37.0 M	$37.0 M
Charging microgrids	$198.1 M	$79.2 M	$118.9 M
EPC / project services	$1.72 B	$258.7 M	$1.47 B
Shared national trainset factory	$750.0 M	$150.0 M	$600.0 M
Total	$27.95 B	$5.88 B	$22.06 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Nairobi	5,700,000	1,128	$11.72 B	$2.65 B	$9.07 B	14	5	39.3%	38.4%
Mombasa	1,350,000	423	$5.18 B	$1.05 B	$4.13 B	9	3	49.5%	48.0%
Nakuru	700,000	202	$1.09 B	$225.1 M	$865.8 M	5	2	28.0%	48.0%
Kisumu	600,000	191	$1.08 B	$224.0 M	$851.2 M	5	2	26.2%	32.4%
Eldoret	500,000	261	$1.45 B	$302.6 M	$1.15 B	5	2	33.1%	44.0%
Thika	350,000	256	$1.34 B	$284.9 M	$1.05 B	5	2	28.8%	29.0%
Garissa	300,000	57	$394.6 M	$74.9 M	$319.7 M	4	1	55.0%	61.8%
Kakamega	300,000	98	$754.4 M	$141.2 M	$613.1 M	4	1	31.8%	35.7%
Kisii	300,000	69	$512.2 M	$95.9 M	$416.3 M	4	1	21.4%	26.4%
Kitale	300,000	99	$727.1 M	$137.5 M	$589.6 M	4	1	15.9%	37.1%
Machakos	300,000	68	$480.0 M	$89.2 M	$390.8 M	4	1	28.4%	34.7%
Malindi	300,000	68	$497.2 M	$92.9 M	$404.4 M	4	1	41.1%	48.8%
Meru Ke	250,000	65	$473.8 M	$87.8 M	$386.0 M	4	1	25.7%	30.3%
Naivasha	250,000	110	$796.6 M	$150.5 M	$646.1 M	4	1	49.3%	45.5%
Nyeri	250,000	86	$656.4 M	$122.4 M	$534.0 M	4	1	29.0%	30.5%

Local Basis And Regeneration

Country finance parameters use KE in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Madagascar National OpenSourceRail Strategy

Repository source: `cities/catalogue/east-africa/Madagascar/NATIONAL-BRIEF.md`

This page contains only Madagascar-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $12.86 B (87.7%) of external capital and $16.61 B of external interest. Capital plus saved interest totals $29.47 B.

National Programme

Local measure	Planning value
Catalogue cities	1
Represented population	3,058,000
Trainsets / vehicle modules	752 / 4,512
City infrastructure and fleet CAPEX	$7.77 B
Shared national factory	$354.4 M
Factory sizing basis	4,512 modules for Antananarivo, then reused nationally
Total national programme	$8.15 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$1.81 B (22.2%)
Domestic / local capital	$6.34 B (77.8%)
Annual external capital draw	$181.2 M / yr
Annual local capital draw	$634.0 M / yr
Annual public construction commitment	$766.2 M / yr for 10 years
Annual post-grace debt service	$694.6 M / yr
Default foreign-turnkey external capital	$14.67 B
External capital saved	$12.86 B
Capital + lifetime external interest saved	$29.47 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$3.57 B	$536.0 M	$3.04 B
Stations	$1.39 B	$277.8 M	$1.11 B
Depots	$358.1 M	$89.5 M	$268.6 M
Rolling stock	$1.26 B	$442.2 M	$821.2 M
Dedicated solar plants	$627.3 M	$282.3 M	$345.0 M
Residual train control	$17.8 M	$8.9 M	$8.9 M
Charging microgrids	$76.0 M	$30.4 M	$45.6 M
EPC / project services	$492.2 M	$73.8 M	$418.4 M
Shared national trainset factory	$354.4 M	$70.9 M	$283.5 M
Total	$8.15 B	$1.81 B	$6.34 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Antananarivo	3,058,000	752	$7.77 B	$1.74 B	$6.04 B	14	5	35.7%	36.2%

Local Basis And Regeneration

Country finance parameters use `MG` in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Mozambique National OpenSourceRail Strategy

Repository source: `cities/catalogue/east-africa/Mozambique/NATIONAL-BRIEF.md`

This page contains only Mozambique-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $16.03 B (88.7%) of external capital and $20.71 B of external interest. Capital plus saved interest totals $36.75 B.

National Programme

Local measure	Planning value
Catalogue cities	10
Represented population	5,015,000
Trainsets / vehicle modules	1,229 / 3,726
City infrastructure and fleet CAPEX	$9.06 B
Shared national factory	$917.1 M
Factory sizing basis	1,208 modules for Maputo, then reused nationally
Total national programme	$10.04 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$2.04 B (20.3%)
Domestic / local capital	$8.00 B (79.7%)
Annual external capital draw	$203.7 M / yr
Annual local capital draw	$800.3 M / yr
Annual public construction commitment	$1.12 B / yr for 10 years
Annual post-grace debt service	$1.01 B / yr
Default foreign-turnkey external capital	$18.07 B
External capital saved	$16.03 B
Capital + lifetime external interest saved	$36.75 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$4.80 B	$719.7 M	$4.08 B
Stations	$1.52 B	$303.8 M	$1.22 B
Depots	$663.5 M	$165.9 M	$497.6 M
Rolling stock	$1.08 B	$379.1 M	$704.0 M
Dedicated solar plants	$353.0 M	$158.8 M	$194.1 M
Residual train control	$26.5 M	$13.3 M	$13.3 M
Charging microgrids	$45.4 M	$18.1 M	$27.2 M
EPC / project services	$633.7 M	$95.1 M	$538.7 M
Shared national trainset factory	$917.1 M	$183.4 M	$733.7 M
Total	$10.04 B	$2.04 B	$8.00 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Maputo	1,530,000	302	$3.27 B	$681.7 M	$2.59 B	9	3	40.0%	45.5%
Nampula	800,000	182	$1.02 B	$213.0 M	$806.8 M	5	2	23.3%	31.6%
Beira	535,000	164	$897.8 M	$189.8 M	$708.0 M	4	1	35.8%	35.0%
Chimoio	400,000	116	$666.7 M	$138.5 M	$528.2 M	3	1	25.0%	26.3%
Quelimane	350,000	32	$174.4 M	$36.5 M	$137.8 M	1	0	18.5%	26.6%
Tete	350,000	170	$1.17 B	$230.7 M	$942.5 M	5	2	23.0%	24.6%
Nacala	300,000	113	$771.0 M	$148.3 M	$622.7 M	4	1	38.0%	47.5%
Lichinga	250,000	21	$142.1 M	$27.6 M	$114.5 M	2	0	17.9%	17.9%
Pemba Mz	250,000	75	$533.1 M	$101.1 M	$431.9 M	4	1	37.7%	50.1%
Xai Xai	250,000	54	$407.3 M	$77.0 M	$330.4 M	4	1	41.7%	38.5%

Local Basis And Regeneration

Country finance parameters use `MZ` in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Rwanda National OpenSourceRail Strategy

Repository source: `cities/catalogue/east-africa/Rwanda/NATIONAL-BRIEF.md`

This page contains only Rwanda-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $8.76 B (89.1%) of external capital and $10.98 B of external interest. Capital plus saved interest totals $19.73 B.

National Programme

Local measure	Planning value
Catalogue cities	3
Represented population	1,708,000
Trainsets / vehicle modules	460 / 1,434
City infrastructure and fleet CAPEX	$4.61 B
Shared national factory	$794.6 M
Factory sizing basis	1,028 modules for Kigali, then reused nationally
Total national programme	$5.46 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$1.08 B (19.7%)
Domestic / local capital	$4.39 B (80.3%)
Annual external capital draw	$153.6 M / yr
Annual local capital draw	$626.7 M / yr
Annual public construction commitment	$472.1 M / yr for 7 years
Annual post-grace debt service	$383.2 M / yr
Default foreign-turnkey external capital	$9.83 B
External capital saved	$8.76 B
Capital + lifetime external interest saved	$19.73 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$2.79 B	$418.4 M	$2.37 B
Stations	$587.9 M	$117.6 M	$470.3 M
Depots	$284.8 M	$71.2 M	$213.6 M
Rolling stock	$401.5 M	$140.5 M	$261.0 M
Dedicated solar plants	$224.4 M	$101.0 M	$123.4 M
Residual train control	$14.7 M	$7.3 M	$7.3 M
Charging microgrids	$22.9 M	$9.1 M	$13.7 M
EPC / project services	$342.7 M	$51.4 M	$291.3 M
Shared national trainset factory	$794.6 M	$158.9 M	$635.7 M
Total	$5.46 B	$1.08 B	$4.39 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Kigali	1,208,000	257	$3.15 B	$631.9 M	$2.52 B	9	3	37.0%	41.1%
Huye	250,000	108	$804.4 M	$150.9 M	$653.6 M	5	2	26.0%	27.5%
Rubavu	250,000	95	$660.1 M	$125.4 M	$534.7 M	4	1	34.9%	32.6%

Local Basis And Regeneration

Country finance parameters use `RW` in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Somalia National OpenSourceRail Strategy

Repository source: `cities/catalogue/east-africa/Somalia/NATIONAL-BRIEF.md`

This page contains only Somalia-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $5.05 B (88.6%) of external capital and $6.52 B of external interest. Capital plus saved interest totals $11.57 B.

National Programme

Local measure	Planning value
Catalogue cities	1
Represented population	2,610,000
Trainsets / vehicle modules	284 / 1,136
City infrastructure and fleet CAPEX	$2.75 B
Shared national factory	$386.4 M
Factory sizing basis	1,136 modules for Mogadishu, then reused nationally
Total national programme	$3.16 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$647.0 M (20.5%)
Domestic / local capital	$2.52 B (79.5%)
Annual external capital draw	$64.7 M / yr
Annual local capital draw	$251.6 M / yr
Annual public construction commitment	$381.4 M / yr for 10 years
Annual post-grace debt service	$346.3 M / yr
Default foreign-turnkey external capital	$5.69 B
External capital saved	$5.05 B
Capital + lifetime external interest saved	$11.57 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$1.16 B	$174.3 M	$987.4 M
Stations	$850.3 M	$170.1 M	$680.2 M
Depots	$124.7 M	$31.2 M	$93.6 M
Rolling stock	$318.1 M	$111.3 M	$206.8 M
Dedicated solar plants	$77.9 M	$35.1 M	$42.9 M
Residual train control	$6.7 M	$3.3 M	$3.3 M
Charging microgrids	$35.5 M	$14.2 M	$21.3 M
EPC / project services	$201.8 M	$30.3 M	$171.6 M
Shared national trainset factory	$386.4 M	$77.3 M	$309.1 M
Total	$3.16 B	$647.0 M	$2.52 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Mogadishu	2,610,000	284	$2.75 B	$565.7 M	$2.18 B	6	2	37.7%	35.5%

Local Basis And Regeneration

Country finance parameters use SO in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

South Sudan National OpenSourceRail Strategy

Repository source: `cities/catalogue/east-africa/South Sudan/NATIONAL-BRIEF.md`

Uncalibrated country scenario. Uncalibrated generic XX scenario; obtain South Sudan income, lending, FX and market-access evidence before appraisal. Numerical XX defaults are illustrative, not South Sudan country estimates or available financing.

This page contains only South Sudan-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $826.7 M (89.2%) of external capital and $1.02 B of external interest. Capital plus saved interest totals $1.84 B.

National Programme

Local measure	Planning value
Catalogue cities	1
Represented population	300,000
Trainsets / vehicle modules	45 / 90
City infrastructure and fleet CAPEX	$310.3 M
Shared national factory	$191.0 M
Factory sizing basis	90 modules for Wau, then reused nationally
Total national programme	$514.6 M

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$99.6 M (19.4%)
Domestic / local capital	$415.0 M (80.6%)
Annual external capital draw	$19.9 M / yr
Annual local capital draw	$83.0 M / yr
Annual public construction commitment	$44.3 M / yr for 5 years
Annual post-grace debt service	$31.3 M / yr
Default foreign-turnkey external capital	$926.4 M
External capital saved	$826.7 M
Capital + lifetime external interest saved	$1.84 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$158.4 M	$23.8 M	$134.6 M
Stations	$64.5 M	$12.9 M	$51.6 M
Depots	$39.7 M	$9.9 M	$29.8 M
Rolling stock	$25.2 M	$8.8 M	$16.4 M
Residual train control	$841 k	$420 k	$420 k
Charging microgrids	$1.4 M	$540 k	$810 k
EPC / project services	$33.7 M	$5.1 M	$28.6 M
Shared national trainset factory	$191.0 M	$38.2 M	$152.8 M
Total	$514.6 M	$99.6 M	$415.0 M

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Wau	300,000	45	$310.3 M	$59.4 M	$250.9 M	3	0	9.9%	9.9%

Local Basis And Regeneration

Country finance parameters use SS in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Tanzania National OpenSourceRail Strategy

Repository source: `cities/catalogue/east-africa/Tanzania/NATIONAL-BRIEF.md`

This page contains only Tanzania-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $41.29 B (88.3%) of external capital and $51.76 B of external interest. Capital plus saved interest totals $93.05 B.

National Programme

Local measure	Planning value
Catalogue cities	15
Represented population	13,854,689
Trainsets / vehicle modules	2,918 / 11,768
City infrastructure and fleet CAPEX	$25.00 B
Shared national factory	$910.8 M
Factory sizing basis	6,318 modules for Dar Es Salaam, then reused nationally
Total national programme	$25.98 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$5.47 B (21.1%)
Domestic / local capital	$20.51 B (78.9%)
Annual external capital draw	$781.6 M / yr
Annual local capital draw	$2.93 B / yr
Annual public construction commitment	$2.39 B / yr for 7 years
Annual post-grace debt service	$1.96 B / yr
Default foreign-turnkey external capital	$46.76 B
External capital saved	$41.29 B
Capital + lifetime external interest saved	$93.05 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$12.78 B	$1.92 B	$10.87 B
Stations	$4.18 B	$836.2 M	$3.34 B
Depots	$1.44 B	$361.1 M	$1.08 B
Rolling stock	$3.36 B	$1.18 B	$2.19 B
Dedicated solar plants	$1.46 B	$655.6 M	$801.3 M
Residual train control	$67.4 M	$33.7 M	$33.7 M
Charging microgrids	$168.9 M	$67.6 M	$101.4 M
EPC / project services	$1.60 B	$240.6 M	$1.36 B
Shared national trainset factory	$910.8 M	$182.2 M	$728.6 M
Total	$25.98 B	$5.47 B	$20.51 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Dar Es Salaam	7,404,689	1,053	$12.15 B	$2.69 B	$9.46 B	14	5	44.3%	45.5%
Mwanza	1,100,000	298	$3.44 B	$700.5 M	$2.74 B	9	3	50.2%	43.9%
Dodoma	800,000	182	$1.00 B	$205.4 M	$795.9 M	5	2	32.4%	36.8%
Arusha	700,000	218	$1.18 B	$242.0 M	$936.0 M	4	1	33.7%	37.0%
Mbeya	550,000	182	$983.0 M	$202.8 M	$780.1 M	5	2	37.4%	37.3%
Morogoro	500,000	178	$1.05 B	$216.7 M	$831.2 M	5	2	30.7%	36.1%
Zanzibar City	500,000	186	$1.04 B	$218.1 M	$819.4 M	5	2	40.2%	37.7%
Tanga	400,000	178	$1.02 B	$213.8 M	$808.6 M	5	2	29.3%	48.7%
Kigoma	300,000	75	$522.2 M	$99.4 M	$422.8 M	4	1	44.4%	45.4%
Moshi	300,000	79	$623.8 M	$116.0 M	$507.8 M	4	1	28.8%	38.9%
Tabora	300,000	65	$458.3 M	$87.5 M	$370.8 M	4	1	38.4%	48.5%
Iringa	250,000	86	$622.3 M	$118.3 M	$503.9 M	4	1	51.2%	49.9%
Shinyanga	250,000	77	$512.3 M	$95.5 M	$416.8 M	4	1	38.9%	43.5%
Songea	250,000	29	$184.7 M	$35.3 M	$149.4 M	2	0	24.3%	24.3%
Sumbawanga	250,000	32	$219.0 M	$42.9 M	$176.1 M	3	0	42.4%	42.4%

Local Basis And Regeneration

Country finance parameters use `TZ` in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Uganda National OpenSourceRail Strategy

Repository source: `cities/catalogue/east-africa/Uganda/NATIONAL-BRIEF.md`

This page contains only Uganda-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $21.14 B (89.0%) of external capital and $26.50 B of external interest. Capital plus saved interest totals $47.64 B.

National Programme

Local measure	Planning value
Catalogue cities	12
Represented population	4,925,000
Trainsets / vehicle modules	1,609 / 4,403
City infrastructure and fleet CAPEX	$12.40 B
Shared national factory	$750.0 M
Factory sizing basis	1,444 modules for Kampala, then reused nationally
Total national programme	$13.20 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$2.62 B (19.9%)
Domestic / local capital	$10.58 B (80.1%)
Annual external capital draw	$374.8 M / yr
Annual local capital draw	$1.51 B / yr
Annual public construction commitment	$1.61 B / yr for 7 years
Annual post-grace debt service	$1.35 B / yr
Default foreign-turnkey external capital	$23.76 B
External capital saved	$21.14 B
Capital + lifetime external interest saved	$47.64 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$6.74 B	$1.01 B	$5.73 B
Stations	$2.24 B	$448.3 M	$1.79 B
Depots	$848.9 M	$212.2 M	$636.7 M
Rolling stock	$1.26 B	$441.2 M	$819.4 M
Dedicated solar plants	$428.9 M	$193.0 M	$235.9 M
Residual train control	$36.3 M	$18.1 M	$18.1 M
Charging microgrids	$60.9 M	$24.4 M	$36.6 M
EPC / project services	$835.7 M	$125.3 M	$710.3 M
Shared national trainset factory	$750.0 M	$150.0 M	$600.0 M
Total	$13.20 B	$2.62 B	$10.58 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Kampala	1,875,000	361	$4.03 B	$824.4 M	$3.20 B	9	3	29.5%	30.9%
Mbarara	500,000	202	$1.15 B	$239.7 M	$906.1 M	4	1	32.1%	39.4%
Gulu	350,000	261	$1.47 B	$311.4 M	$1.16 B	5	2	31.4%	36.6%
Jinja	300,000	121	$930.8 M	$176.4 M	$754.3 M	4	1	30.9%	36.7%
Mbale	300,000	95	$705.3 M	$132.4 M	$572.9 M	5	2	28.1%	33.1%
Arua	250,000	98	$721.9 M	$136.7 M	$585.2 M	4	1	33.3%	31.7%
Entebbe	250,000	110	$797.6 M	$152.6 M	$645.0 M	4	1	46.6%	54.6%
Lira	250,000	92	$627.8 M	$119.3 M	$508.5 M	4	1	41.8%	42.6%
Masaka	250,000	83	$626.0 M	$117.9 M	$508.1 M	4	1	37.2%	36.3%
Fort Portal	200,000	85	$621.0 M	$116.4 M	$504.6 M	4	1	30.0%	35.2%
Hoima	200,000	71	$523.9 M	$98.4 M	$425.5 M	4	1	40.0%	44.4%
Soroti	200,000	30	$201.3 M	$40.0 M	$161.4 M	2	0	18.3%	21.2%

Local Basis And Regeneration

Country finance parameters use `UG` in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Zambia National OpenSourceRail Strategy

Repository source: `cities/catalogue/east-africa/Zambia/NATIONAL-BRIEF.md`

This page contains only Zambia-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $9.15 B (87.7%) of external capital and $11.48 B of external interest. Capital plus saved interest totals $20.63 B.

National Programme

Local measure	Planning value
Catalogue cities	1
Represented population	3,037,000
Trainsets / vehicle modules	472 / 2,832
City infrastructure and fleet CAPEX	$5.25 B
Shared national factory	$516.9 M
Factory sizing basis	2,832 modules for Lusaka, then reused nationally
Total national programme	$5.80 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$1.29 B (22.2%)
Domestic / local capital	$4.51 B (77.8%)
Annual external capital draw	$183.7 M / yr
Annual local capital draw	$644.9 M / yr
Annual public construction commitment	$782.7 M / yr for 7 years
Annual post-grace debt service	$675.3 M / yr
Default foreign-turnkey external capital	$10.44 B
External capital saved	$9.15 B
Capital + lifetime external interest saved	$20.63 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$2.59 B	$388.4 M	$2.20 B
Stations	$724.3 M	$144.9 M	$579.4 M
Depots	$262.8 M	$65.7 M	$197.1 M
Rolling stock	$793.0 M	$277.5 M	$515.4 M
Dedicated solar plants	$510.1 M	$229.5 M	$280.6 M
Residual train control	$14.7 M	$7.3 M	$7.3 M
Charging microgrids	$42.6 M	$17.0 M	$25.6 M
EPC / project services	$346.1 M	$51.9 M	$294.2 M
Shared national trainset factory	$516.9 M	$103.4 M	$413.5 M
Total	$5.80 B	$1.29 B	$4.51 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Lusaka	3,037,000	472	$5.25 B	$1.18 B	$4.07 B	12	4	39.6%	35.3%

Local Basis And Regeneration

Country finance parameters use `ZM` in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Bolivia National OpenSourceRail Strategy

Repository source: `cities/catalogue/latin-america/Bolivia/NATIONAL-BRIEF.md`

This page contains only Bolivia-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $7.41 B (88.8%) of external capital and $9.12 B of external interest. Capital plus saved interest totals $16.53 B.

National Programme

Local measure	Planning value
Catalogue cities	1
Represented population	1,815,000
Trainsets / vehicle modules	317 / 1,268
City infrastructure and fleet CAPEX	$3.75 B
Shared national factory	$827.2 M
Factory sizing basis	1,268 modules for La Paz, then reused nationally
Total national programme	$4.64 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$931.1 M (20.1%)
Domestic / local capital	$3.71 B (79.9%)
Annual external capital draw	$186.2 M / yr
Annual local capital draw	$741.0 M / yr
Annual public construction commitment	$501.3 M / yr for 5 years
Annual post-grace debt service	$374.3 M / yr
Default foreign-turnkey external capital	$8.35 B
External capital saved	$7.41 B
Capital + lifetime external interest saved	$16.53 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$2.28 B	$341.4 M	$1.93 B
Stations	$443.6 M	$88.7 M	$354.9 M
Depots	$164.1 M	$41.0 M	$123.1 M
Rolling stock	$355.0 M	$124.3 M	$230.8 M
Dedicated solar plants	$251.6 M	$113.2 M	$138.4 M
Residual train control	$11.8 M	$5.9 M	$5.9 M
Charging microgrids	$20.3 M	$8.1 M	$12.2 M
EPC / project services	$286.9 M	$43.0 M	$243.8 M
Shared national trainset factory	$827.2 M	$165.4 M	$661.8 M
Total	$4.64 B	$931.1 M	$3.71 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
La Paz	1,815,000	317	$3.75 B	$757.0 M	$2.99 B	9	3	39.0%	37.3%

Local Basis And Regeneration

Country finance parameters use `BO` in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Ecuador National OpenSourceRail Strategy

Repository source: `cities/catalogue/latin-america/Ecuador/NATIONAL-BRIEF.md`

This page contains only Ecuador-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $3.31 B (88.6%) of external capital and $4.07 B of external interest. Capital plus saved interest totals $7.37 B.

National Programme

Local measure	Planning value
Catalogue cities	1
Represented population	817,100
Trainsets / vehicle modules	236 / 708
City infrastructure and fleet CAPEX	$1.34 B
Shared national factory	$688.0 M
Factory sizing basis	708 modules for Cuenca, then reused nationally
Total national programme	$2.07 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$424.1 M (20.5%)
Domestic / local capital	$1.65 B (79.5%)
Annual external capital draw	$84.8 M / yr
Annual local capital draw	$329.7 M / yr
Annual public construction commitment	$210.3 M / yr for 5 years
Annual post-grace debt service	$155.0 M / yr
Default foreign-turnkey external capital	$3.73 B
External capital saved	$3.31 B
Capital + lifetime external interest saved	$7.37 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$722.8 M	$108.4 M	$614.3 M
Stations	$161.9 M	$32.4 M	$129.5 M
Depots	$92.6 M	$23.1 M	$69.4 M
Rolling stock	$212.4 M	$74.3 M	$138.1 M
Dedicated solar plants	$56.6 M	$25.5 M	$31.1 M
Residual train control	$3.7 M	$1.9 M	$1.9 M
Charging microgrids	$2.9 M	$1.1 M	$1.7 M
EPC / project services	$131.9 M	$19.8 M	$112.1 M
Shared national trainset factory	$688.0 M	$137.6 M	$550.4 M
Total	$2.07 B	$424.1 M	$1.65 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Cuenca	817,100	236	$1.34 B	$279.3 M	$1.06 B	5	2	34.0%	33.7%

Local Basis And Regeneration

Country finance parameters use EC in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

El Salvador National OpenSourceRail Strategy

Repository source: `cities/catalogue/latin-america/El Salvador/NATIONAL-BRIEF.md`

This page contains only El Salvador-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $7.62 B (88.7%) of external capital and $9.36 B of external interest. Capital plus saved interest totals $16.98 B.

National Programme

Local measure	Planning value
Catalogue cities	1
Represented population	1,800,000
Trainsets / vehicle modules	359 / 1,436
City infrastructure and fleet CAPEX	$4.25 B
Shared national factory	$484.4 M
Factory sizing basis	1,436 modules for San Salvador, then reused nationally
Total national programme	$4.77 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$966.0 M (20.3%)
Domestic / local capital	$3.80 B (79.7%)
Annual external capital draw	$193.2 M / yr
Annual local capital draw	$760.3 M / yr
Annual public construction commitment	$545.2 M / yr for 5 years
Annual post-grace debt service	$413.0 M / yr
Default foreign-turnkey external capital	$8.58 B
External capital saved	$7.62 B
Capital + lifetime external interest saved	$16.98 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$2.52 B	$377.4 M	$2.14 B
Stations	$560.9 M	$112.2 M	$448.7 M
Depots	$173.3 M	$43.3 M	$130.0 M
Rolling stock	$402.1 M	$140.7 M	$261.4 M
Dedicated solar plants	$299.0 M	$134.6 M	$164.5 M
Residual train control	$13.6 M	$6.8 M	$6.8 M
Charging microgrids	$25.6 M	$10.2 M	$15.3 M
EPC / project services	$292.3 M	$43.8 M	$248.5 M
Shared national trainset factory	$484.4 M	$96.9 M	$387.5 M
Total	$4.77 B	$966.0 M	$3.80 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
San Salvador	1,800,000	359	$4.25 B	$864.0 M	$3.39 B	9	3	38.0%	36.5%

Local Basis And Regeneration

Country finance parameters use SV in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Morocco National OpenSourceRail Strategy

Repository source: `cities/catalogue/north-africa/Morocco/NATIONAL-BRIEF.md`

This page contains only Morocco-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $29.80 B (88.8%) of external capital and $36.63 B of external interest. Capital plus saved interest totals $66.43 B.

National Programme

Local measure	Planning value
Catalogue cities	12
Represented population	8,050,000
Trainsets / vehicle modules	2,251 / 7,443
City infrastructure and fleet CAPEX	$17.67 B
Shared national factory	$912.3 M
Factory sizing basis	1,492 modules for Marrakech, then reused nationally
Total national programme	$18.65 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$3.77 B (20.2%)
Domestic / local capital	$14.88 B (79.8%)
Annual external capital draw	$755.0 M / yr
Annual local capital draw	$2.98 B / yr
Annual public construction commitment	$1.30 B / yr for 5 years
Annual post-grace debt service	$897.7 M / yr
Default foreign-turnkey external capital	$33.57 B
External capital saved	$29.80 B
Capital + lifetime external interest saved	$66.43 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$9.45 B	$1.42 B	$8.03 B
Stations	$3.08 B	$616.5 M	$2.47 B
Depots	$1.06 B	$265.6 M	$796.9 M
Rolling stock	$2.15 B	$753.8 M	$1.40 B
Dedicated solar plants	$653.3 M	$294.0 M	$359.3 M
Residual train control	$51.0 M	$25.5 M	$25.5 M
Charging microgrids	$107.0 M	$42.8 M	$64.2 M
EPC / project services	$1.18 B	$176.6 M	$1.00 B
Shared national trainset factory	$912.3 M	$182.5 M	$729.9 M
Total	$18.65 B	$3.77 B	$14.88 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Fez	1,300,000	218	$2.63 B	$523.8 M	$2.11 B	6	2	46.0%	53.7%
Marrakech	1,200,000	373	$3.89 B	$787.5 M	$3.10 B	9	3	41.2%	57.5%
Tangier	1,200,000	299	$3.33 B	$671.5 M	$2.66 B	8	3	57.5%	49.8%
Agadir	900,000	298	$1.53 B	$321.0 M	$1.21 B	5	2	46.8%	51.7%
Meknes	700,000	158	$925.7 M	$191.1 M	$734.5 M	5	2	38.9%	37.4%
Oujda	600,000	123	$711.9 M	$145.5 M	$566.4 M	4	1	51.7%	47.1%
Kenitra	500,000	229	$1.23 B	$259.2 M	$970.1 M	4	1	50.8%	51.2%
Tetouan	500,000	205	$1.22 B	$251.2 M	$972.3 M	4	1	40.4%	56.1%
Safi	350,000	148	$789.6 M	$164.8 M	$624.8 M	4	1	67.8%	69.7%
Beni Mellal	300,000	69	$468.7 M	$88.3 M	$380.4 M	4	1	59.0%	59.4%
Khouribga	250,000	39	$286.0 M	$54.1 M	$231.9 M	3	0	49.3%	49.6%
Nador	250,000	92	$663.5 M	$124.7 M	$538.8 M	4	1	47.2%	53.6%

Local Basis And Regeneration

Country finance parameters use `MA` in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Sudan National OpenSourceRail Strategy

Repository source: `cities/catalogue/north-africa/Sudan/NATIONAL-BRIEF.md`

This page contains only Sudan-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $28.06 B (88.4%) of external capital and $36.25 B of external interest. Capital plus saved interest totals $64.31 B.

National Programme

Local measure	Planning value
Catalogue cities	6
Represented population	10,729,000
Trainsets / vehicle modules	1,820 / 8,287
City infrastructure and fleet CAPEX	$17.10 B
Shared national factory	$507.2 M
Factory sizing basis	4,584 modules for Khartoum, then reused nationally
Total national programme	$17.64 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$3.70 B (21.0%)
Domestic / local capital	$13.95 B (79.0%)
Annual external capital draw	$369.6 M / yr
Annual local capital draw	$1.39 B / yr
Annual public construction commitment	$2.12 B / yr for 10 years
Annual post-grace debt service	$1.93 B / yr
Default foreign-turnkey external capital	$31.76 B
External capital saved	$28.06 B
Capital + lifetime external interest saved	$64.31 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$8.98 B	$1.35 B	$7.64 B
Stations	$2.73 B	$546.8 M	$2.19 B
Depots	$849.2 M	$212.3 M	$636.9 M
Rolling stock	$2.35 B	$823.1 M	$1.53 B
Dedicated solar plants	$947.5 M	$426.4 M	$521.1 M
Residual train control	$47.3 M	$23.6 M	$23.6 M
Charging microgrids	$128.0 M	$51.2 M	$76.8 M
EPC / project services	$1.09 B	$163.8 M	$928.3 M
Shared national trainset factory	$507.2 M	$101.4 M	$405.8 M
Total	$17.64 B	$3.70 B	$13.95 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Khartoum	5,829,000	764	$8.48 B	$1.84 B	$6.64 B	14	5	22.4%	22.4%
Omdurman	2,800,000	535	$5.51 B	$1.12 B	$4.39 B	9	3	24.5%	23.3%
Nyala	600,000	164	$1.01 B	$203.7 M	$811.1 M	5	2	18.4%	19.3%
El Obeid	500,000	166	$908.4 M	$188.1 M	$720.3 M	5	2	18.7%	24.6%
Kassala	500,000	86	$584.4 M	$114.5 M	$469.8 M	4	1	21.7%	21.6%
Port Sudan	500,000	105	$598.1 M	$121.7 M	$476.4 M	4	1	18.2%	19.4%

Local Basis And Regeneration

Country finance parameters use SD in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Tunisia National OpenSourceRail Strategy

Repository source: `cities/catalogue/north-africa/Tunisia/NATIONAL-BRIEF.md`

This page contains only Tunisia-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $7.17 B (88.9%) of external capital and $8.82 B of external interest. Capital plus saved interest totals $15.99 B.

National Programme

Local measure	Planning value
Catalogue cities	1
Represented population	2,900,000
Trainsets / vehicle modules	344 / 1,376
City infrastructure and fleet CAPEX	$3.99 B
Shared national factory	$457.2 M
Factory sizing basis	1,376 modules for Tunis, then reused nationally
Total national programme	$4.48 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$896.6 M (20.0%)
Domestic / local capital	$3.59 B (80.0%)
Annual external capital draw	$179.3 M / yr
Annual local capital draw	$717.1 M / yr
Annual public construction commitment	$441.9 M / yr for 5 years
Annual post-grace debt service	$322.8 M / yr
Default foreign-turnkey external capital	$8.07 B
External capital saved	$7.17 B
Capital + lifetime external interest saved	$15.99 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$2.46 B	$368.8 M	$2.09 B
Stations	$457.4 M	$91.5 M	$365.9 M
Depots	$159.4 M	$39.8 M	$119.5 M
Rolling stock	$385.3 M	$134.8 M	$250.4 M
Dedicated solar plants	$250.2 M	$112.6 M	$137.6 M
Residual train control	$13.4 M	$6.7 M	$6.7 M
Charging microgrids	$23.3 M	$9.3 M	$14.0 M
EPC / project services	$276.8 M	$41.5 M	$235.3 M
Shared national trainset factory	$457.2 M	$91.4 M	$365.8 M
Total	$4.48 B	$896.6 M	$3.59 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Tunis	2,900,000	344	$3.99 B	$800.4 M	$3.19 B	8	3	35.0%	32.8%

Local Basis And Regeneration

Country finance parameters use `TN` in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

South Africa National OpenSourceRail Strategy

Repository source: `cities/catalogue/south-africa/South Africa/NATIONAL-BRIEF.md`

This page contains only South Africa-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $23.47 B (88.0%) of external capital and $28.85 B of external interest. Capital plus saved interest totals $52.32 B.

National Programme

Local measure	Planning value
Catalogue cities	5
Represented population	6,200,000
Trainsets / vehicle modules	1,674 / 7,370
City infrastructure and fleet CAPEX	$13.90 B
Shared national factory	$863.1 M
Factory sizing basis	4,872 modules for Durban, then reused nationally
Total national programme	$14.82 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$3.21 B (21.7%)
Domestic / local capital	$11.61 B (78.3%)
Annual external capital draw	$642.5 M / yr
Annual local capital draw	$2.32 B / yr
Annual public construction commitment	$1.58 B / yr for 5 years
Annual post-grace debt service	$1.19 B / yr
Default foreign-turnkey external capital	$26.68 B
External capital saved	$23.47 B
Capital + lifetime external interest saved	$52.32 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$6.85 B	$1.03 B	$5.82 B
Stations	$2.24 B	$447.0 M	$1.79 B
Depots	$718.3 M	$179.6 M	$538.8 M
Rolling stock	$2.11 B	$738.5 M	$1.37 B
Dedicated solar plants	$1.02 B	$457.5 M	$559.2 M
Residual train control	$36.2 M	$18.1 M	$18.1 M
Charging microgrids	$91.5 M	$36.6 M	$54.9 M
EPC / project services	$903.1 M	$135.5 M	$767.6 M
Shared national trainset factory	$863.1 M	$172.6 M	$690.5 M
Total	$14.82 B	$3.21 B	$11.61 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Durban	3,900,000	812	$9.05 B	$2.01 B	$7.03 B	14	5	28.1%	31.0%
East London Za	800,000	221	$1.25 B	$265.1 M	$989.0 M	5	2	34.3%	35.6%
Bloemfontein	600,000	354	$1.92 B	$415.4 M	$1.50 B	5	2	39.1%	51.5%
Polokwane	600,000	199	$1.03 B	$214.5 M	$812.1 M	5	2	23.0%	36.5%
Nelspruit	300,000	88	$653.6 M	$122.4 M	$531.1 M	4	1	40.1%	37.8%

Local Basis And Regeneration

Country finance parameters use ZA in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Afghanistan National OpenSourceRail Strategy

Repository source: `cities/catalogue/south-asia/Afghanistan/NATIONAL-BRIEF.md`

This page contains only Afghanistan-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $17.39 B (88.1%) of external capital and $22.46 B of external interest. Capital plus saved interest totals $39.85 B.

National Programme

Local measure	Planning value
Catalogue cities	5
Represented population	7,051,000
Trainsets / vehicle modules	1,388 / 6,036
City infrastructure and fleet CAPEX	$10.22 B
Shared national factory	$696.0 M
Factory sizing basis	3,744 modules for Kabul, then reused nationally
Total national programme	$10.97 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$2.35 B (21.4%)
Domestic / local capital	$8.62 B (78.6%)
Annual external capital draw	$235.1 M / yr
Annual local capital draw	$861.5 M / yr
Annual public construction commitment	$1.52 B / yr for 10 years
Annual post-grace debt service	$1.39 B / yr
Default foreign-turnkey external capital	$19.74 B
External capital saved	$17.39 B
Capital + lifetime external interest saved	$39.85 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$4.60 B	$690.3 M	$3.91 B
Stations	$2.09 B	$417.1 M	$1.67 B
Depots	$620.3 M	$155.1 M	$465.2 M
Rolling stock	$1.74 B	$607.6 M	$1.13 B
Dedicated solar plants	$418.5 M	$188.3 M	$230.2 M
Residual train control	$25.1 M	$12.5 M	$12.5 M
Charging microgrids	$92.2 M	$36.9 M	$55.4 M
EPC / project services	$690.0 M	$103.5 M	$586.5 M
Shared national trainset factory	$696.0 M	$139.2 M	$556.8 M
Total	$10.97 B	$2.35 B	$8.62 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Kabul	4,601,000	624	$6.02 B	$1.33 B	$4.68 B	11	4	40.8%	40.7%
Herat	800,000	153	$822.5 M	$170.0 M	$652.5 M	4	1	18.8%	19.9%
Kandahar	700,000	204	$1.10 B	$229.1 M	$874.1 M	5	2	27.0%	42.5%
Mazar E Sharif	600,000	237	$1.29 B	$269.7 M	$1.02 B	5	2	44.3%	57.8%
Jalalabad Af	350,000	170	$984.2 M	$200.8 M	$783.5 M	5	2	31.3%	36.8%

Local Basis And Regeneration

Country finance parameters use AF in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Bangladesh National OpenSourceRail Strategy

Repository source: `cities/catalogue/south-asia/Bangladesh/NATIONAL-BRIEF.md`

This page contains only Bangladesh-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $40.80 B (88.3%) of external capital and $51.14 B of external interest. Capital plus saved interest totals $91.94 B.

National Programme

Local measure	Planning value
Catalogue cities	10
Represented population	13,550,000
Trainsets / vehicle modules	2,924 / 11,654
City infrastructure and fleet CAPEX	$24.68 B
Shared national factory	$927.4 M
Factory sizing basis	4,350 modules for Chittagong, then reused nationally
Total national programme	$25.67 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$5.42 B (21.1%)
Domestic / local capital	$20.26 B (78.9%)
Annual external capital draw	$773.9 M / yr
Annual local capital draw	$2.89 B / yr
Annual public construction commitment	$2.20 B / yr for 7 years
Annual post-grace debt service	$1.80 B / yr
Default foreign-turnkey external capital	$46.21 B
External capital saved	$40.80 B
Capital + lifetime external interest saved	$91.94 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$12.93 B	$1.94 B	$10.99 B
Stations	$3.85 B	$769.5 M	$3.08 B
Depots	$1.26 B	$315.0 M	$945.1 M
Rolling stock	$3.35 B	$1.17 B	$2.18 B
Dedicated solar plants	$1.57 B	$705.2 M	$861.9 M
Residual train control	$66.5 M	$33.3 M	$33.3 M
Charging microgrids	$148.9 M	$59.6 M	$89.4 M
EPC / project services	$1.58 B	$236.6 M	$1.34 B
Shared national trainset factory	$927.4 M	$185.5 M	$741.9 M
Total	$25.67 B	$5.42 B	$20.26 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Chittagong	5,200,000	725	$7.79 B	$1.74 B	$6.05 B	12	4	46.6%	48.1%
Khulna	1,500,000	313	$3.74 B	$753.3 M	$2.99 B	9	3	51.8%	41.9%
Gazipur	1,400,000	394	$4.40 B	$908.6 M	$3.49 B	9	3	29.1%	30.3%
Narayanganj	950,000	251	$1.41 B	$293.7 M	$1.12 B	5	2	34.2%	31.1%
Rajshahi	950,000	203	$1.13 B	$240.8 M	$893.0 M	4	1	36.1%	37.8%
Sylhet	900,000	192	$1.12 B	$232.1 M	$889.6 M	4	1	25.1%	33.6%
Rangpur	800,000	164	$916.6 M	$191.4 M	$725.1 M	5	2	22.4%	25.2%
Mymensingh	700,000	164	$1.13 B	$227.1 M	$907.0 M	4	1	18.8%	21.6%
Comilla	600,000	277	$1.63 B	$340.9 M	$1.29 B	5	2	28.7%	33.1%
Barisal	550,000	241	$1.41 B	$293.3 M	$1.11 B	5	2	39.6%	52.4%

Local Basis And Regeneration

Country finance parameters use BD in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

India National OpenSourceRail Strategy

Repository source: `cities/catalogue/south-asia/India/NATIONAL-BRIEF.md`

This page contains only India-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $107.89 B (88.5%) of external capital and $132.64 B of external interest. Capital plus saved interest totals $240.53 B.

National Programme

Local measure	Planning value
Catalogue cities	17
Represented population	36,304,000
Trainsets / vehicle modules	5,657 / 27,138
City infrastructure and fleet CAPEX	$66.79 B
Shared national factory	$865.5 M
Factory sizing basis	3,636 modules for Coimbatore, then reused nationally
Total national programme	$67.71 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$13.99 B (20.7%)
Domestic / local capital	$53.72 B (79.3%)
Annual external capital draw	$2.80 B / yr
Annual local capital draw	$10.74 B / yr
Annual public construction commitment	$5.87 B / yr for 5 years
Annual post-grace debt service	$4.19 B / yr
Default foreign-turnkey external capital	$121.88 B
External capital saved	$107.89 B
Capital + lifetime external interest saved	$240.53 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$37.80 B	$5.67 B	$32.13 B
Stations	$9.65 B	$1.93 B	$7.72 B
Depots	$2.48 B	$619.0 M	$1.86 B
Rolling stock	$7.60 B	$2.66 B	$4.94 B
Dedicated solar plants	$4.52 B	$2.03 B	$2.49 B
Residual train control	$196.7 M	$98.4 M	$98.4 M
Charging microgrids	$468.9 M	$187.6 M	$281.4 M
EPC / project services	$4.13 B	$620.1 M	$3.51 B
Shared national trainset factory	$865.5 M	$173.1 M	$692.4 M
Total	$67.71 B	$13.99 B	$53.72 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Lucknow	3,500,000	532	$6.05 B	$1.31 B	$4.74 B	8	0	Unavailable	Unavailable
Indore	3,200,000	581	$6.17 B	$1.36 B	$4.81 B	8	0	Unavailable	Unavailable
Kanpur	3,200,000	536	$6.35 B	$1.36 B	$4.99 B	8	0	Unavailable	Unavailable
Coimbatore	3,084,000	606	$6.58 B	$1.47 B	$5.11 B	9	0	Unavailable	Unavailable
Patna	2,520,000	473	$5.94 B	$1.19 B	$4.75 B	6	0	Unavailable	Unavailable
Bhopal	2,400,000	281	$3.29 B	$656.5 M	$2.64 B	6	0	Unavailable	Unavailable
Visakhapatnam	2,300,000	326	$3.91 B	$793.1 M	$3.11 B	6	0	Unavailable	Unavailable
Vadodara	2,200,000	233	$2.63 B	$526.9 M	$2.11 B	6	0	Unavailable	Unavailable
Rajkot	1,800,000	153	$1.97 B	$382.3 M	$1.59 B	5	0	Unavailable	Unavailable
Agra	1,700,000	191	$2.52 B	$488.7 M	$2.03 B	5	0	Unavailable	Unavailable
Madurai	1,600,000	295	$3.49 B	$708.8 M	$2.78 B	6	0	Unavailable	Unavailable
Meerut	1,600,000	184	$2.28 B	$447.1 M	$1.83 B	4	0	Unavailable	Unavailable
Raipur	1,500,000	195	$2.20 B	$450.6 M	$1.75 B	6	0	Unavailable	Unavailable
Varanasi	1,500,000	270	$3.59 B	$697.1 M	$2.90 B	6	0	Unavailable	Unavailable
Vijayawada	1,500,000	254	$3.18 B	$640.5 M	$2.53 B	6	0	Unavailable	Unavailable
Ranchi	1,400,000	336	$3.87 B	$787.9 M	$3.08 B	6	0	Unavailable	Unavailable
Jodhpur	1,300,000	211	$2.76 B	$534.2 M	$2.23 B	5	0	Unavailable	Unavailable

Local Basis And Regeneration

Country finance parameters use `IN` in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Nepal National OpenSourceRail Strategy

Repository source: `cities/catalogue/south-asia/Nepal/NATIONAL-BRIEF.md`

This page contains only Nepal-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $12.71 B (88.7%) of external capital and $15.93 B of external interest. Capital plus saved interest totals $28.63 B.

National Programme

Local measure	Planning value
Catalogue cities	3
Represented population	2,342,000
Trainsets / vehicle modules	811 / 2,809
City infrastructure and fleet CAPEX	$7.06 B
Shared national factory	$836.7 M
Factory sizing basis	1,788 modules for Kathmandu, then reused nationally
Total national programme	$7.96 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$1.62 B (20.4%)
Domestic / local capital	$6.34 B (79.6%)
Annual external capital draw	$231.6 M / yr
Annual local capital draw	$905.4 M / yr
Annual public construction commitment	$634.3 M / yr for 7 years
Annual post-grace debt service	$514.0 M / yr
Default foreign-turnkey external capital	$14.33 B
External capital saved	$12.71 B
Capital + lifetime external interest saved	$28.63 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$3.71 B	$557.1 M	$3.16 B
Stations	$1.35 B	$269.2 M	$1.08 B
Depots	$347.0 M	$86.7 M	$260.2 M
Rolling stock	$804.1 M	$281.4 M	$522.7 M
Dedicated solar plants	$347.6 M	$156.4 M	$191.2 M
Residual train control	$18.2 M	$9.1 M	$9.1 M
Charging microgrids	$47.6 M	$19.1 M	$28.6 M
EPC / project services	$498.0 M	$74.7 M	$423.3 M
Shared national trainset factory	$836.7 M	$167.3 M	$669.3 M
Total	$7.96 B	$1.62 B	$6.34 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Kathmandu	1,442,000	447	$4.74 B	$984.4 M	$3.75 B	9	3	53.0%	46.5%
Pokhara	600,000	293	$1.62 B	$335.5 M	$1.29 B	5	2	43.2%	45.5%
Biratnagar	300,000	71	$704.1 M	$125.1 M	$579.1 M	4	1	26.2%	32.0%

Local Basis And Regeneration

Country finance parameters use `NP` in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Pakistan National OpenSourceRail Strategy

Repository source: `cities/catalogue/south-asia/Pakistan/NATIONAL-BRIEF.md`

This page contains only Pakistan-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $55.74 B (88.6%) of external capital and $69.88 B of external interest. Capital plus saved interest totals $125.62 B.

National Programme

Local measure	Planning value
Catalogue cities	13
Represented population	37,603,000
Trainsets / vehicle modules	3,469 / 15,345
City infrastructure and fleet CAPEX	$33.88 B
Shared national factory	$1.02 B
Factory sizing basis	5,526 modules for Karachi, then reused nationally
Total national programme	$34.97 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$7.20 B (20.6%)
Domestic / local capital	$27.77 B (79.4%)
Annual external capital draw	$1.03 B / yr
Annual local capital draw	$3.97 B / yr
Annual public construction commitment	$4.78 B / yr for 7 years
Annual post-grace debt service	$4.11 B / yr
Default foreign-turnkey external capital	$62.94 B
External capital saved	$55.74 B
Capital + lifetime external interest saved	$125.62 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$18.60 B	$2.79 B	$15.81 B
Stations	$5.06 B	$1.01 B	$4.05 B
Depots	$1.71 B	$426.9 M	$1.28 B
Rolling stock	$4.36 B	$1.52 B	$2.83 B
Dedicated solar plants	$1.72 B	$772.2 M	$943.8 M
Residual train control	$96.6 M	$48.3 M	$48.3 M
Charging microgrids	$238.9 M	$95.6 M	$143.4 M
EPC / project services	$2.18 B	$326.3 M	$1.85 B
Shared national trainset factory	$1.02 B	$203.6 M	$814.5 M
Total	$34.97 B	$7.20 B	$27.77 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Karachi	20,300,000	921	$10.17 B	$2.20 B	$7.97 B	14	5	47.7%	44.8%
Faisalabad	3,556,000	317	$3.42 B	$742.1 M	$2.68 B	9	3	40.9%	45.8%
Gujranwala	2,300,000	261	$2.84 B	$572.0 M	$2.27 B	8	3	29.1%	32.4%
Peshawar	2,300,000	257	$3.09 B	$608.6 M	$2.49 B	8	3	24.3%	21.5%
Multan	2,197,000	199	$2.29 B	$453.3 M	$1.83 B	8	3	34.3%	38.8%
Hyderabad Pk	1,900,000	362	$3.90 B	$782.1 M	$3.12 B	9	3	49.4%	56.1%
Quetta	1,200,000	145	$1.92 B	$372.1 M	$1.55 B	6	2	32.0%	34.5%
Bahawalpur	900,000	171	$973.4 M	$200.6 M	$772.8 M	5	2	35.6%	39.3%
Sialkot	750,000	197	$1.38 B	$267.4 M	$1.11 B	5	2	28.1%	31.5%
Sheikhupura	600,000	118	$603.3 M	$130.3 M	$473.1 M	2	0	14.9%	14.9%
Sukkur	600,000	165	$1.16 B	$225.1 M	$933.4 M	4	1	28.9%	40.5%
Larkana	500,000	127	$918.9 M	$176.6 M	$742.3 M	3	1	22.2%	35.2%
Rahim Yar Khan	500,000	229	$1.21 B	$252.5 M	$954.7 M	5	2	20.7%	41.3%

Local Basis And Regeneration

Country finance parameters use PK in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Sri Lanka National OpenSourceRail Strategy

Repository source: `cities/catalogue/south-asia/Sri Lanka/NATIONAL-BRIEF.md`

This page contains only Sri Lanka-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $18.65 B (87.9%) of external capital and $23.38 B of external interest. Capital plus saved interest totals $42.03 B.

National Programme

Local measure	Planning value
Catalogue cities	4
Represented population	7,398,000
Trainsets / vehicle modules	1,371 / 6,069
City infrastructure and fleet CAPEX	$10.98 B
Shared national factory	$759.7 M
Factory sizing basis	3,912 modules for Colombo, then reused nationally
Total national programme	$11.79 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$2.57 B (21.8%)
Domestic / local capital	$9.22 B (78.2%)
Annual external capital draw	$367.5 M / yr
Annual local capital draw	$1.32 B / yr
Annual public construction commitment	$1.37 B / yr for 7 years
Annual post-grace debt service	$1.16 B / yr
Default foreign-turnkey external capital	$21.22 B
External capital saved	$18.65 B
Capital + lifetime external interest saved	$42.03 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$5.39 B	$807.8 M	$4.58 B
Stations	$1.66 B	$331.2 M	$1.32 B
Depots	$609.4 M	$152.4 M	$457.1 M
Rolling stock	$1.74 B	$609.9 M	$1.13 B
Dedicated solar plants	$814.1 M	$366.3 M	$447.7 M
Residual train control	$29.3 M	$14.6 M	$14.6 M
Charging microgrids	$75.8 M	$30.3 M	$45.5 M
EPC / project services	$718.1 M	$107.7 M	$610.4 M
Shared national trainset factory	$759.7 M	$151.9 M	$607.8 M
Total	$11.79 B	$2.57 B	$9.22 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Colombo	5,648,000	652	$7.03 B	$1.58 B	$5.45 B	14	5	40.2%	37.9%
Kandy	650,000	265	$1.50 B	$312.9 M	$1.18 B	5	2	22.2%	24.9%
Jaffna	600,000	212	$1.15 B	$243.4 M	$906.1 M	5	2	25.3%	40.0%
Galle	500,000	242	$1.31 B	$275.2 M	$1.03 B	5	2	35.1%	33.8%

Local Basis And Regeneration

Country finance parameters use `LK` in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Cambodia National OpenSourceRail Strategy

Repository source: `cities/catalogue/southeast-asia/Cambodia/NATIONAL-BRIEF.md`

This page contains only Cambodia-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $9.65 B (88.7%) of external capital and $12.09 B of external interest. Capital plus saved interest totals $21.74 B.

National Programme

Local measure	Planning value
Catalogue cities	1
Represented population	2,281,000
Trainsets / vehicle modules	515 / 2,060
City infrastructure and fleet CAPEX	$5.67 B
Shared national factory	$351.7 M
Factory sizing basis	2,060 modules for Phnom Penh, then reused nationally
Total national programme	$6.04 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$1.23 B (20.3%)
Domestic / local capital	$4.81 B (79.7%)
Annual external capital draw	$175.4 M / yr
Annual local capital draw	$687.7 M / yr
Annual public construction commitment	$500.9 M / yr for 7 years
Annual post-grace debt service	$406.6 M / yr
Default foreign-turnkey external capital	$10.87 B
External capital saved	$9.65 B
Capital + lifetime external interest saved	$21.74 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$3.08 B	$462.4 M	$2.62 B
Stations	$1.07 B	$213.1 M	$852.4 M
Depots	$203.2 M	$50.8 M	$152.4 M
Rolling stock	$576.8 M	$201.9 M	$374.9 M
Dedicated solar plants	$329.1 M	$148.1 M	$181.0 M
Residual train control	$14.9 M	$7.5 M	$7.5 M
Charging microgrids	$43.7 M	$17.5 M	$26.2 M
EPC / project services	$373.7 M	$56.1 M	$317.6 M
Shared national trainset factory	$351.7 M	$70.3 M	$281.4 M
Total	$6.04 B	$1.23 B	$4.81 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Phnom Penh	2,281,000	515	$5.67 B	$1.15 B	$4.51 B	9	3	34.2%	44.4%

Local Basis And Regeneration

Country finance parameters use KH in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Indonesia National OpenSourceRail Strategy

Repository source: `cities/catalogue/southeast-asia/Indonesia/NATIONAL-BRIEF.md`

This page contains only Indonesia-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $19.46 B (88.0%) of external capital and $23.92 B of external interest. Capital plus saved interest totals $43.38 B.

National Programme

Local measure	Planning value
Catalogue cities	2
Represented population	5,624,000
Trainsets / vehicle modules	1,139 / 5,820
City infrastructure and fleet CAPEX	$11.91 B
Shared national factory	$347.0 M
Factory sizing basis	3,792 modules for Surabaya, then reused nationally
Total national programme	$12.28 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$2.65 B (21.6%)
Domestic / local capital	$9.63 B (78.4%)
Annual external capital draw	$530.5 M / yr
Annual local capital draw	$1.93 B / yr
Annual public construction commitment	$1.02 B / yr for 5 years
Annual post-grace debt service	$727.7 M / yr
Default foreign-turnkey external capital	$22.11 B
External capital saved	$19.46 B
Capital + lifetime external interest saved	$43.38 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$5.47 B	$820.8 M	$4.65 B
Stations	$2.67 B	$533.6 M	$2.13 B
Depots	$437.7 M	$109.4 M	$328.2 M
Rolling stock	$1.63 B	$570.4 M	$1.06 B
Dedicated solar plants	$826.4 M	$371.9 M	$454.5 M
Residual train control	$28.3 M	$14.2 M	$14.2 M
Charging microgrids	$126.0 M	$50.4 M	$75.6 M
EPC / project services	$749.6 M	$112.4 M	$637.1 M
Shared national trainset factory	$347.0 M	$69.4 M	$277.6 M
Total	$12.28 B	$2.65 B	$9.63 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Surabaya	3,009,000	632	$6.73 B	$1.51 B	$5.23 B	9	0	Unavailable	Unavailable
Bandung	2,615,000	507	$5.18 B	$1.07 B	$4.11 B	6	0	Unavailable	Unavailable

Local Basis And Regeneration

Country finance parameters use `ID` in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Laos National OpenSourceRail Strategy

Repository source: `cities/catalogue/southeast-asia/Laos/NATIONAL-BRIEF.md`

This page contains only Laos-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $3.66 B (88.5%) of external capital and $4.59 B of external interest. Capital plus saved interest totals $8.26 B.

National Programme

Local measure	Planning value
Catalogue cities	1
Represented population	948,000
Trainsets / vehicle modules	270 / 810
City infrastructure and fleet CAPEX	$1.47 B
Shared national factory	$770.6 M
Factory sizing basis	810 modules for Vientiane, then reused nationally
Total national programme	$2.30 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$474.1 M (20.6%)
Domestic / local capital	$1.82 B (79.4%)
Annual external capital draw	$67.7 M / yr
Annual local capital draw	$260.7 M / yr
Annual public construction commitment	$226.8 M / yr for 7 years
Annual post-grace debt service	$187.1 M / yr
Default foreign-turnkey external capital	$4.14 B
External capital saved	$3.66 B
Capital + lifetime external interest saved	$8.26 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$785.1 M	$117.8 M	$667.4 M
Stations	$178.7 M	$35.7 M	$143.0 M
Depots	$98.0 M	$24.5 M	$73.5 M
Rolling stock	$243.0 M	$85.0 M	$157.9 M
Dedicated solar plants	$70.2 M	$31.6 M	$38.6 M
Residual train control	$4.4 M	$2.2 M	$2.2 M
Charging microgrids	$3.4 M	$1.3 M	$2.0 M
EPC / project services	$145.8 M	$21.9 M	$123.9 M
Shared national trainset factory	$770.6 M	$154.1 M	$616.5 M
Total	$2.30 B	$474.1 M	$1.82 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Vientiane	948,000	270	$1.47 B	$311.9 M	$1.16 B	5	2	26.4%	35.3%

Local Basis And Regeneration

Country finance parameters use LA in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Myanmar National OpenSourceRail Strategy

Repository source: `cities/catalogue/southeast-asia/Myanmar/NATIONAL-BRIEF.md`

This page contains only Myanmar-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $23.75 B (88.0%) of external capital and $30.68 B of external interest. Capital plus saved interest totals $54.43 B.

National Programme

Local measure	Planning value
Catalogue cities	2
Represented population	6,926,000
Trainsets / vehicle modules	1,275 / 7,012
City infrastructure and fleet CAPEX	$14.37 B
Shared national factory	$579.8 M
Factory sizing basis	5,736 modules for Yangon, then reused nationally
Total national programme	$14.99 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$3.24 B (21.6%)
Domestic / local capital	$11.75 B (78.4%)
Annual external capital draw	$324.1 M / yr
Annual local capital draw	$1.18 B / yr
Annual public construction commitment	$1.60 B / yr for 10 years
Annual post-grace debt service	$1.45 B / yr
Default foreign-turnkey external capital	$26.99 B
External capital saved	$23.75 B
Capital + lifetime external interest saved	$54.43 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$7.39 B	$1.11 B	$6.28 B
Stations	$2.24 B	$447.3 M	$1.79 B
Depots	$577.7 M	$144.4 M	$433.3 M
Rolling stock	$1.96 B	$687.2 M	$1.28 B
Dedicated solar plants	$1.19 B	$536.4 M	$655.6 M
Residual train control	$39.8 M	$19.9 M	$19.9 M
Charging microgrids	$117.7 M	$47.1 M	$70.6 M
EPC / project services	$903.0 M	$135.4 M	$767.5 M
Shared national trainset factory	$579.8 M	$116.0 M	$463.8 M
Total	$14.99 B	$3.24 B	$11.75 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Yangon	5,200,000	956	$10.53 B	$2.37 B	$8.16 B	14	5	46.6%	44.8%
Mandalay	1,726,000	319	$3.85 B	$753.1 M	$3.09 B	9	3	40.5%	42.1%

Local Basis And Regeneration

Country finance parameters use `MM` in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Philippines National OpenSourceRail Strategy

Repository source: `cities/catalogue/southeast-asia/Philippines/NATIONAL-BRIEF.md`

This page contains only Philippines-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $7.87 B (88.7%) of external capital and $9.67 B of external interest. Capital plus saved interest totals $17.54 B.

National Programme

Local measure	Planning value
Catalogue cities	1
Represented population	1,827,000
Trainsets / vehicle modules	379 / 1,516
City infrastructure and fleet CAPEX	$4.53 B
Shared national factory	$369.9 M
Factory sizing basis	1,516 modules for Davao, then reused nationally
Total national programme	$4.93 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$1.00 B (20.3%)
Domestic / local capital	$3.93 B (79.7%)
Annual external capital draw	$200.3 M / yr
Annual local capital draw	$785.0 M / yr
Annual public construction commitment	$396.8 M / yr for 5 years
Annual post-grace debt service	$279.0 M / yr
Default foreign-turnkey external capital	$8.87 B
External capital saved	$7.87 B
Capital + lifetime external interest saved	$17.54 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$2.68 B	$401.8 M	$2.28 B
Stations	$602.9 M	$120.6 M	$482.3 M
Depots	$177.5 M	$44.4 M	$133.1 M
Rolling stock	$424.5 M	$148.6 M	$275.9 M
Dedicated solar plants	$328.8 M	$147.9 M	$180.8 M
Residual train control	$14.9 M	$7.5 M	$7.5 M
Charging microgrids	$28.7 M	$11.5 M	$17.2 M
EPC / project services	$300.8 M	$45.1 M	$255.7 M
Shared national trainset factory	$369.9 M	$74.0 M	$295.9 M
Total	$4.93 B	$1.00 B	$3.93 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Davao	1,827,000	379	$4.53 B	$923.4 M	$3.61 B	9	3	42.8%	40.5%

Local Basis And Regeneration

Country finance parameters use `PH` in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Burkina Faso National OpenSourceRail Strategy

Repository source: `cities/catalogue/west-africa/Burkina Faso/NATIONAL-BRIEF.md`

This page contains only Burkina Faso-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $7.30 B (88.9%) of external capital and $9.43 B of external interest. Capital plus saved interest totals $16.73 B.

National Programme

Local measure	Planning value
Catalogue cities	1
Represented population	2,531,000
Trainsets / vehicle modules	351 / 1,404
City infrastructure and fleet CAPEX	$3.78 B
Shared national factory	$735.7 M
Factory sizing basis	1,404 modules for Ouagadougou, then reused nationally
Total national programme	$4.56 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$912.0 M (20.0%)
Domestic / local capital	$3.65 B (80.0%)
Annual external capital draw	$91.2 M / yr
Annual local capital draw	$365.1 M / yr
Annual public construction commitment	$391.5 M / yr for 10 years
Annual post-grace debt service	$353.0 M / yr
Default foreign-turnkey external capital	$8.21 B
External capital saved	$7.30 B
Capital + lifetime external interest saved	$16.73 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$2.06 B	$309.6 M	$1.75 B
Stations	$707.9 M	$141.6 M	$566.3 M
Depots	$172.6 M	$43.2 M	$129.5 M
Rolling stock	$393.1 M	$137.6 M	$255.5 M
Dedicated solar plants	$159.0 M	$71.6 M	$87.5 M
Residual train control	$11.6 M	$5.8 M	$5.8 M
Charging microgrids	$30.8 M	$12.3 M	$18.5 M
EPC / project services	$288.1 M	$43.2 M	$244.9 M
Shared national trainset factory	$735.7 M	$147.1 M	$588.6 M
Total	$4.56 B	$912.0 M	$3.65 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Ouagadougou	2,531,000	351	$3.78 B	$757.1 M	$3.02 B	9	3	29.1%	24.3%

Local Basis And Regeneration

Country finance parameters use BF in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Cameroon National OpenSourceRail Strategy

Repository source: `cities/catalogue/west-africa/Cameroon/NATIONAL-BRIEF.md`

This page contains only Cameroon-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $25.63 B (88.2%) of external capital and $32.13 B of external interest. Capital plus saved interest totals $57.76 B.

National Programme

Local measure	Planning value
Catalogue cities	10
Represented population	11,650,000
Trainsets / vehicle modules	1,815 / 7,756
City infrastructure and fleet CAPEX	$15.06 B
Shared national factory	$1.02 B
Factory sizing basis	2,604 modules for Yaounde, then reused nationally
Total national programme	$16.15 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$3.44 B (21.3%)
Domestic / local capital	$12.71 B (78.7%)
Annual external capital draw	$491.9 M / yr
Annual local capital draw	$1.82 B / yr
Annual public construction commitment	$1.38 B / yr for 7 years
Annual post-grace debt service	$1.13 B / yr
Default foreign-turnkey external capital	$29.07 B
External capital saved	$25.63 B
Capital + lifetime external interest saved	$57.76 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$7.86 B	$1.18 B	$6.68 B
Stations	$2.04 B	$408.7 M	$1.63 B
Depots	$895.7 M	$223.9 M	$671.8 M
Rolling stock	$2.23 B	$781.5 M	$1.45 B
Dedicated solar plants	$979.2 M	$440.6 M	$538.5 M
Residual train control	$39.7 M	$19.9 M	$19.9 M
Charging microgrids	$93.3 M	$37.3 M	$56.0 M
EPC / project services	$992.6 M	$148.9 M	$843.7 M
Shared national trainset factory	$1.02 B	$203.6 M	$814.5 M
Total	$16.15 B	$3.44 B	$12.71 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Yaounde	4,100,000	434	$4.95 B	$1.09 B	$3.86 B	8	3	27.3%	30.3%
Douala	3,900,000	342	$4.12 B	$903.5 M	$3.22 B	8	3	30.3%	28.5%
Bafoussam	600,000	231	$1.29 B	$270.2 M	$1.02 B	5	2	21.2%	23.6%
Bamenda	600,000	168	$965.7 M	$201.6 M	$764.1 M	5	2	24.3%	23.5%
Garoua	600,000	109	$649.4 M	$129.9 M	$519.5 M	4	1	18.6%	21.2%
Maroua	500,000	193	$1.11 B	$224.6 M	$881.0 M	5	2	30.1%	31.2%
Kumba	400,000	150	$851.1 M	$177.5 M	$673.6 M	4	1	19.2%	25.5%
Bertoua	350,000	80	$461.1 M	$95.7 M	$365.4 M	3	0	18.4%	16.3%
Ngaoundere	350,000	91	$545.8 M	$110.1 M	$435.7 M	4	1	23.9%	25.0%
Edea	250,000	17	$125.8 M	$23.3 M	$102.5 M	1	0	15.2%	15.2%

Local Basis And Regeneration

Country finance parameters use `CM` in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Guinea National OpenSourceRail Strategy

Repository source: `cities/catalogue/west-africa/Guinea/NATIONAL-BRIEF.md`

This page contains only Guinea-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $3.86 B (88.8%) of external capital and $4.99 B of external interest. Capital plus saved interest totals $8.85 B.

National Programme

Local measure	Planning value
Catalogue cities	1
Represented population	2,010,000
Trainsets / vehicle modules	165 / 660
City infrastructure and fleet CAPEX	$1.84 B
Shared national factory	$542.3 M
Factory sizing basis	660 modules for Conakry, then reused nationally
Total national programme	$2.42 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$488.6 M (20.2%)
Domestic / local capital	$1.93 B (79.8%)
Annual external capital draw	$48.9 M / yr
Annual local capital draw	$192.9 M / yr
Annual public construction commitment	$214.9 M / yr for 10 years
Annual post-grace debt service	$193.7 M / yr
Default foreign-turnkey external capital	$4.35 B
External capital saved	$3.86 B
Capital + lifetime external interest saved	$8.85 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$970.6 M	$145.6 M	$825.0 M
Stations	$359.6 M	$71.9 M	$287.7 M
Depots	$91.0 M	$22.7 M	$68.2 M
Rolling stock	$184.8 M	$64.7 M	$120.1 M
Dedicated solar plants	$95.9 M	$43.1 M	$52.7 M
Residual train control	$5.2 M	$2.6 M	$2.6 M
Charging microgrids	$16.6 M	$6.6 M	$9.9 M
EPC / project services	$151.9 M	$22.8 M	$129.1 M
Shared national trainset factory	$542.3 M	$108.5 M	$433.9 M
Total	$2.42 B	$488.6 M	$1.93 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Conakry	2,010,000	165	$1.84 B	$374.4 M	$1.46 B	5	2	26.0%	36.2%

Local Basis And Regeneration

Country finance parameters use `GN` in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Mali National OpenSourceRail Strategy

Repository source: `cities/catalogue/west-africa/Mali/NATIONAL-BRIEF.md`

This page contains only Mali-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $10.40 B (89.3%) of external capital and $13.43 B of external interest. Capital plus saved interest totals $23.83 B.

National Programme

Local measure	Planning value
Catalogue cities	1
Represented population	2,929,000
Trainsets / vehicle modules	474 / 1,896
City infrastructure and fleet CAPEX	$6.17 B
Shared national factory	$280.8 M
Factory sizing basis	1,896 modules for Bamako, then reused nationally
Total national programme	$6.47 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$1.25 B (19.3%)
Domestic / local capital	$5.22 B (80.7%)
Annual external capital draw	$124.9 M / yr
Annual local capital draw	$522.1 M / yr
Annual public construction commitment	$557.5 M / yr for 10 years
Annual post-grace debt service	$501.4 M / yr
Default foreign-turnkey external capital	$11.65 B
External capital saved	$10.40 B
Capital + lifetime external interest saved	$23.83 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$3.60 B	$540.7 M	$3.06 B
Stations	$1.18 B	$235.2 M	$940.6 M
Depots	$195.8 M	$49.0 M	$146.9 M
Rolling stock	$530.9 M	$185.8 M	$345.1 M
Dedicated solar plants	$207.6 M	$93.4 M	$114.2 M
Residual train control	$15.6 M	$7.8 M	$7.8 M
Charging microgrids	$50.0 M	$20.0 M	$30.0 M
EPC / project services	$409.7 M	$61.5 M	$348.3 M
Shared national trainset factory	$280.8 M	$56.2 M	$224.7 M
Total	$6.47 B	$1.25 B	$5.22 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Bamako	2,929,000	474	$6.17 B	$1.19 B	$4.98 B	9	3	40.3%	48.9%

Local Basis And Regeneration

Country finance parameters use `ML` in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Niger National OpenSourceRail Strategy

Repository source: `cities/catalogue/west-africa/Niger/NATIONAL-BRIEF.md`

This page contains only Niger-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $6.63 B (89.0%) of external capital and $8.56 B of external interest. Capital plus saved interest totals $15.19 B.

National Programme

Local measure	Planning value
Catalogue cities	1
Represented population	1,407,635
Trainsets / vehicle modules	284 / 1,136
City infrastructure and fleet CAPEX	$3.20 B
Shared national factory	$875.6 M
Factory sizing basis	1,136 modules for Niamey, then reused nationally
Total national programme	$4.14 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$818.6 M (19.8%)
Domestic / local capital	$3.32 B (80.2%)
Annual external capital draw	$81.9 M / yr
Annual local capital draw	$331.7 M / yr
Annual public construction commitment	$342.0 M / yr for 10 years
Annual post-grace debt service	$308.6 M / yr
Default foreign-turnkey external capital	$7.44 B
External capital saved	$6.63 B
Capital + lifetime external interest saved	$15.19 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$1.78 B	$267.3 M	$1.51 B
Stations	$582.1 M	$116.4 M	$465.7 M
Depots	$160.0 M	$40.0 M	$120.0 M
Rolling stock	$318.1 M	$111.3 M	$206.8 M
Dedicated solar plants	$120.4 M	$54.2 M	$66.2 M
Residual train control	$9.4 M	$4.7 M	$4.7 M
Charging microgrids	$25.4 M	$10.2 M	$15.3 M
EPC / project services	$262.7 M	$39.4 M	$223.3 M
Shared national trainset factory	$875.6 M	$175.1 M	$700.5 M
Total	$4.14 B	$818.6 M	$3.32 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Niamey	1,407,635	284	$3.20 B	$634.3 M	$2.56 B	9	3	35.0%	31.3%

Local Basis And Regeneration

Country finance parameters use `NE` in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Nigeria National OpenSourceRail Strategy

Repository source: `cities/catalogue/west-africa/Nigeria/NATIONAL-BRIEF.md`

This page contains only Nigeria-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $46.35 B (88.4%) of external capital and $58.11 B of external interest. Capital plus saved interest totals $104.46 B.

National Programme

Local measure	Planning value
Catalogue cities	10
Represented population	19,200,000
Trainsets / vehicle modules	2,811 / 12,880
City infrastructure and fleet CAPEX	$28.19 B
Shared national factory	$877.6 M
Factory sizing basis	4,662 modules for Kano, then reused nationally
Total national programme	$29.13 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$6.08 B (20.9%)
Domestic / local capital	$23.05 B (79.1%)
Annual external capital draw	$868.3 M / yr
Annual local capital draw	$3.29 B / yr
Annual public construction commitment	$3.42 B / yr for 7 years
Annual post-grace debt service	$2.89 B / yr
Default foreign-turnkey external capital	$52.43 B
External capital saved	$46.35 B
Capital + lifetime external interest saved	$104.46 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$15.17 B	$2.28 B	$12.89 B
Stations	$4.26 B	$851.7 M	$3.41 B
Depots	$1.43 B	$357.0 M	$1.07 B
Rolling stock	$3.64 B	$1.27 B	$2.37 B
Dedicated solar plants	$1.67 B	$752.3 M	$919.5 M
Residual train control	$78.7 M	$39.4 M	$39.4 M
Charging microgrids	$207.2 M	$82.9 M	$124.3 M
EPC / project services	$1.80 B	$269.4 M	$1.53 B
Shared national trainset factory	$877.6 M	$175.5 M	$702.1 M
Total	$29.13 B	$6.08 B	$23.05 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Kano	4,200,000	777	$8.41 B	$1.82 B	$6.59 B	14	5	38.8%	43.0%
Ibadan	3,900,000	343	$3.52 B	$787.1 M	$2.74 B	8	3	23.1%	29.4%
Port Harcourt	3,000,000	270	$3.38 B	$680.0 M	$2.70 B	8	3	23.8%	25.8%
Benin City	1,800,000	286	$2.77 B	$576.5 M	$2.19 B	8	3	34.4%	34.9%
Onitsha	1,500,000	278	$3.46 B	$694.3 M	$2.77 B	8	3	25.1%	29.5%
Maiduguri	1,200,000	253	$3.23 B	$623.9 M	$2.61 B	8	3	30.7%	36.8%
Ilorin	1,000,000	180	$1.01 B	$211.6 M	$802.2 M	5	2	20.9%	24.7%
Aba Ng	900,000	115	$673.8 M	$139.8 M	$534.0 M	4	1	27.2%	30.0%
Jos	900,000	185	$989.5 M	$203.4 M	$786.2 M	5	2	21.6%	28.1%
Uyo	800,000	124	$735.8 M	$152.9 M	$582.8 M	4	1	14.7%	17.9%

Local Basis And Regeneration

Country finance parameters use `NG` in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Senegal National OpenSourceRail Strategy

Repository source: `cities/catalogue/west-africa/Senegal/NATIONAL-BRIEF.md`

This page contains only Senegal-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $9.12 B (87.9%) of external capital and $11.43 B of external interest. Capital plus saved interest totals $20.55 B.

National Programme

Local measure	Planning value
Catalogue cities	1
Represented population	4,030,000
Trainsets / vehicle modules	526 / 3,156
City infrastructure and fleet CAPEX	$5.40 B
Shared national factory	$336.1 M
Factory sizing basis	3,156 modules for Dakar, then reused nationally
Total national programme	$5.76 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$1.25 B (21.8%)
Domestic / local capital	$4.51 B (78.2%)
Annual external capital draw	$179.3 M / yr
Annual local capital draw	$644.1 M / yr
Annual public construction commitment	$491.9 M / yr for 7 years
Annual post-grace debt service	$402.6 M / yr
Default foreign-turnkey external capital	$10.37 B
External capital saved	$9.12 B
Capital + lifetime external interest saved	$20.55 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$2.39 B	$358.5 M	$2.03 B
Stations	$1.17 B	$233.4 M	$933.7 M
Depots	$240.3 M	$60.1 M	$180.2 M
Rolling stock	$883.7 M	$309.3 M	$574.4 M
Dedicated solar plants	$312.0 M	$140.4 M	$171.6 M
Residual train control	$13.4 M	$6.7 M	$6.7 M
Charging microgrids	$64.2 M	$25.7 M	$38.5 M
EPC / project services	$356.6 M	$53.5 M	$303.1 M
Shared national trainset factory	$336.1 M	$67.2 M	$268.9 M
Total	$5.76 B	$1.25 B	$4.51 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Dakar	4,030,000	526	$5.40 B	$1.18 B	$4.22 B	9	3	56.0%	56.9%

Local Basis And Regeneration

Country finance parameters use `SN` in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Egypt National OpenSourceRail Strategy

Repository source: `cities/catalogue/west-asia/Egypt/NATIONAL-BRIEF.md`

This page contains only Egypt-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $28.70 B (88.6%) of external capital and $35.28 B of external interest. Capital plus saved interest totals $63.98 B.

National Programme

Local measure	Planning value
Catalogue cities	19
Represented population	10,600,000
Trainsets / vehicle modules	3,108 / 9,129
City infrastructure and fleet CAPEX	$17.17 B
Shared national factory	$769.1 M
Factory sizing basis	807 modules for Tanta, then reused nationally
Total national programme	$17.99 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$3.69 B (20.5%)
Domestic / local capital	$14.30 B (79.5%)
Annual external capital draw	$738.3 M / yr
Annual local capital draw	$2.86 B / yr
Annual public construction commitment	$1.94 B / yr for 5 years
Annual post-grace debt service	$1.45 B / yr
Default foreign-turnkey external capital	$32.39 B
External capital saved	$28.70 B
Capital + lifetime external interest saved	$63.98 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$9.36 B	$1.40 B	$7.96 B
Stations	$2.04 B	$407.6 M	$1.63 B
Depots	$1.39 B	$347.2 M	$1.04 B
Rolling stock	$2.73 B	$955.8 M	$1.78 B
Dedicated solar plants	$454.4 M	$204.5 M	$249.9 M
Residual train control	$50.2 M	$25.1 M	$25.1 M
Charging microgrids	$52.6 M	$21.1 M	$31.6 M
EPC / project services	$1.15 B	$172.1 M	$975.3 M
Shared national trainset factory	$769.1 M	$153.8 M	$615.2 M
Total	$17.99 B	$3.69 B	$14.30 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Mansoura Eg	1,000,000	177	$964.0 M	$199.2 M	$764.7 M	4	1	25.7%	27.5%
Port Said	800,000	106	$652.7 M	$132.3 M	$520.4 M	4	1	53.9%	38.9%
Suez	800,000	222	$1.17 B	$243.3 M	$923.1 M	5	2	47.5%	46.4%
Tanta	750,000	269	$1.33 B	$283.9 M	$1.05 B	5	2	36.6%	32.9%
Ismailia	700,000	191	$1.10 B	$226.4 M	$877.1 M	4	1	48.0%	57.0%
Zagazig	700,000	147	$788.6 M	$163.5 M	$625.2 M	4	1	36.2%	33.9%
Asyut	600,000	196	$961.1 M	$204.5 M	$756.6 M	4	1	29.9%	44.7%
Mahalla	600,000	131	$682.7 M	$142.3 M	$540.4 M	4	1	26.9%	28.6%
Minya	600,000	191	$1.18 B	$239.2 M	$945.2 M	4	1	14.1%	27.0%
Sohag	550,000	127	$689.4 M	$141.6 M	$547.8 M	4	1	23.7%	29.2%
Damanhur	500,000	190	$1.00 B	$208.9 M	$793.8 M	5	2	39.5%	46.7%
Fayoum	500,000	233	$1.19 B	$249.8 M	$936.8 M	5	2	31.7%	36.2%
Luxor	500,000	177	$965.0 M	$198.8 M	$766.2 M	5	2	32.5%	48.2%
Damietta	400,000	264	$1.38 B	$289.5 M	$1.09 B	5	2	37.3%	45.6%
Beni Suef	350,000	123	$749.8 M	$150.6 M	$599.2 M	4	1	28.3%	27.7%
Qena	350,000	169	$1.04 B	$208.2 M	$833.1 M	5	2	32.5%	46.7%
Arish	300,000	39	$241.5 M	$45.9 M	$195.6 M	2	0	28.5%	33.5%
Hurghada	300,000	84	$585.5 M	$109.4 M	$476.1 M	4	1	65.1%	63.4%
Kafr El Sheikh	300,000	72	$493.7 M	$92.2 M	$401.6 M	4	1	45.7%	43.3%

Local Basis And Regeneration

Country finance parameters use `EG` in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Iraq National OpenSourceRail Strategy

Repository source: `cities/catalogue/west-asia/Iraq/NATIONAL-BRIEF.md`

Current Iraq planning basis — 2026-10-06

Current country basis · Baghdad summary · Connected design, production and energy study.

The retained `local_positive` Baghdad financial comparator is USD 8.667bn, with IQD 0.000tn unsourced support and IQD 0.000tn terminal debt. All debt clears at model month 295, conditional on the recorded funding assumptions. Its prices and facilities are unquoted and uncommitted. It does not price the complete accelerated construction and battery scenario.

The accelerated scenario's complete CAPEX, financing, net island savings and accepted opening dates remain unknown. Its USD 9m / 18-launcher allowance covers purchase cash only; freight, assembly, commissioning, transporters, lifting frames, temporary supports, spares, crews and relocations require separate scope-matched prices. The earlier delivered-and-commissioned interpretation is not used. No launcher saving is added to an installed civil rate before removing a verified matching embedded allowance.

Iraq has existing precast facilities and expertise; qualify contracted civil supply and price tooling/upgrades. Baghdad trainset assembly and component-process allowances remain in the financial comparator. Additional national plant capacity and transfer costs require separate scope reconciliation.

The coordinated industrialisation programme joins vehicle interfaces, staged bogie localisation and the standard viaduct system. Its possible vendor register records official-source component and equipment leads for comparison. Iraq already has many precast facilities and relevant expertise; plant capability, compatible tooling and available production slots require qualification. No listed vendor is a selected supplier or a manufacturing-rights agreement.

The coupled service–OD–cost–finance study now derives capacity-limited passenger itineraries from completed train sections and carries the same grid purchases and train distance into conditional monthly finance. Its demand and retained prices/opening months are explicit sensitivities. Complete line openings additionally require stations, specials, track, energy, depots, fleet, testing and approvals; missing dates or prices remain unknown. No coupled case is an accepted forecast or an adopted national budget.

No current consolidated national budget or funded rollout is established. Other Iraqi city catalogue estimates and the earlier generic foreign-turnkey calculation remain separately identified comparators. Their differences are scenario arithmetic, not realised capital or interest savings. Baghdad's later scope cannot be mixed into an older national aggregate and presented as one complete budget.

Construction sensitivities

Case	Days for modelled running bays	Scope
full-doubling	1022	identified catalogue planning spans only; unresolved closures, station structures and special crossings remain separate gates
higher-productivity	1098	identified catalogue planning spans only; unresolved closures, station structures and special crossings remain separate gates
initial-accelerated	1218	identified catalogue planning spans only; unresolved closures, station structures and special crossings remain separate gates
initial-accelerated-reassigned	966	identified catalogue planning spans only; unresolved closures, station structures and special crossings remain separate gates
initial-six-day-calendar	1348	identified catalogue planning spans only; unresolved closures, station structures and special crossings remain separate gates

These are conditional resource schedules. Station structures, special/closure spans, approved pier locations, delivery access and release evidence remain opening gates. Empty supplier/contract registers indicate missing qualifying evidence; they do not imply that Iraqi industry lacks capability.

Chronological energy study

Case, lower-solar duty	Minimum SOC	Unserved traction kWh
lfp-throughout	0.0%	676,286
sodium-both	0.0%	4,024,245
sodium-stationary	0.0%	676,383

The fixed-assignment diagnostic above uses both ring directions, declared dispatch stations and native rest-to-rest section timing. The current screen spans 4,440 continuous model minutes (3.08 days), including overnight state carry-over and a

study strategy permitting grid replenishment within remaining site import capacity. Charge-rate, thermal and life inputs are unqualified profiles, so these results do not establish a chemistry advantage or suitability.

The separate coupled energy/service controller keeps reserve-limited trains at their actual stations, continues charging, and reflects their availability in subsequent dispatch opportunities.

Controlled case	Completed journeys	Missed dispatch opportunities	Distinct energy-held journeys
lfp-throughout	17,491	909	706
sodium-both	8,536	9,605	2,761
sodium-stationary	17,492	908	719

Geometry/operating-distance findings remain recorded. Conflict-capable timetable, berth/depot access, transient traction and operational multi-day acceptance remain open. Current station passenger demand is unknown where no surveyed station/OD assignment has been entered; the common 3,000 pax/h value is retained only as a separate stress case.

City estimates and their scope

| City | Population | Fleet | City-case capital | Assumed imported purchases | Basis | | --- | ---: | ---: | ---: | ---: | --- | | Baghdad | 9,780,429 | 809 | $8.67 B | $2.02 B | Retained local_positive comparator; accelerated scope unpriced | 14 | 5 | 40.3% | 27.7% | | Basra | 3,955,000 | 557 | $6.58 B | $1.40 B | Catalogue base estimate; scope and field releases open | 11 | 4 | 42.9% | 38.3% | | Sulaymaniyah | 2,150,000 | 144 | $2.00 B | $398.7 M | Catalogue base estimate; scope and field releases open | 6 | 2 | 35.5% | 31.1% | | Erbil | 1,952,000 | 261 | $2.35 B | $489.5 M | Catalogue base estimate; scope and field releases open | 8 | 3 | 31.8% | 29.8% | | Mosul | 1,940,000 | 308 | $3.93 B | $763.6 M | Catalogue base estimate; scope and field releases open | 9 | 3 | 40.6% | 41.0% | | Kirkuk | 1,780,000 | 235 | $2.81 B | $556.2 M | Catalogue base estimate; scope and field releases open | 8 | 3 | 41.2% | 41.1% | | Najaf | 1,540,000 | 309 | $3.33 B | $674.7 M | Catalogue base estimate; scope and field releases open | 9 | 3 | 33.3% | 46.2% | | Karbala | 1,390,000 | 253 | $2.84 B | $567.5 M | Catalogue base estimate; scope and field releases open | 9 | 3 | 46.1% | 42.0% | | Nasiriyah | 705,000 | 184 | $1.09 B | $221.0 M | Catalogue base estimate; scope and field releases open | 5 | 2 | 41.8% | 34.8% | | Hillah | 700,000 | 236 | $1.19 B | $250.8 M | Catalogue base estimate; scope and field releases open | 5 | 2 | 29.0% | 37.7% | | Amarah | 660,000 | 177 | $1.13 B | $224.6 M | Catalogue base estimate; scope and field releases open | 5 | 2 | 37.4% | 35.1% | | Ramadi | 525,000 | 181 | $978.5 M | $204.3 M | Catalogue base estimate; scope and field releases open | 5 | 2 | 30.0% | 36.9% | | Baqubah | 470,000 | 243 | $1.32 B | $275.3 M | Catalogue base estimate; scope and field releases open | 5 | 2 | 37.2% | 33.8% | | Diwaniyah | 440,000 | 171 | $935.4 M | $192.3 M | Catalogue base estimate; scope and field releases open | 5 | 2 | 30.6% | 26.8% | | Kut | 410,000 | 166 | $1.05 B | $209.6 M | Catalogue base estimate; scope and field releases open | 5 | 2 | 40.8% | 42.7% | | Samawah | 373,770 | 182 | $1.03 B | $211.9 M | Catalogue base estimate; scope and field releases open | 5 | 2 | 40.9% | 47.8% | | Duhok | 360,000 | 213 | $1.08 B | $235.9 M | Catalogue base estimate; scope and field releases open | 4 | 1 | 46.8% | 53.3% | | Fallujah | 360,000 | 195 | $1.06 B | $218.6 M | Catalogue base estimate; scope and field releases open | 5 | 2 | 35.4% | 30.3% |

These case envelopes have different appraisal scopes. They are not summed into a current national project budget. Baghdad's included assembly/component facilities must be reconciled with any national shared-factory allowance before aggregation.

Shared assumptions and interpretation are in the deployment planning reference.

Regeneration

Run `python3 tools/automation/generate-national-briefs.py --country IQ` after regenerating the Baghdad study, then publish the current city/country documents and proposal archive.

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Jordan National OpenSourceRail Strategy

Repository source: `cities/catalogue/west-asia/Jordan/NATIONAL-BRIEF.md`

This page contains only Jordan-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $17.39 B (88.2%) of external capital and $21.38 B of external interest. Capital plus saved interest totals $38.77 B.

National Programme

Local measure	Planning value
Catalogue cities	4
Represented population	5,557,000
Trainsets / vehicle modules	1,167 / 5,394
City infrastructure and fleet CAPEX	$10.18 B
Shared national factory	$723.2 M
Factory sizing basis	3,954 modules for Amman, then reused nationally
Total national programme	$10.95 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$2.33 B (21.3%)
Domestic / local capital	$8.63 B (78.7%)
Annual external capital draw	$465.5 M / yr
Annual local capital draw	$1.73 B / yr
Annual public construction commitment	$967.4 M / yr for 5 years
Annual post-grace debt service	$695.6 M / yr
Default foreign-turnkey external capital	$19.72 B
External capital saved	$17.39 B
Capital + lifetime external interest saved	$38.77 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$5.50 B	$825.5 M	$4.68 B
Stations	$1.20 B	$239.2 M	$956.8 M
Depots	$569.8 M	$142.4 M	$427.3 M
Rolling stock	$1.54 B	$537.5 M	$998.2 M
Dedicated solar plants	$667.9 M	$300.6 M	$367.4 M
Residual train control	$29.3 M	$14.7 M	$14.7 M
Charging microgrids	$55.6 M	$22.3 M	$33.4 M
EPC / project services	$672.9 M	$100.9 M	$572.0 M
Shared national trainset factory	$723.2 M	$144.6 M	$578.6 M
Total	$10.95 B	$2.33 B	$8.63 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Amman	4,007,000	659	$7.16 B	$1.57 B	$5.60 B	14	5	46.2%	37.7%
Zarqa	700,000	250	$1.39 B	$286.2 M	$1.10 B	5	2	33.6%	33.1%
Irbid	600,000	174	$985.1 M	$203.4 M	$781.7 M	5	2	27.7%	32.5%
Aqaba	250,000	84	$641.3 M	$118.5 M	$522.7 M	4	1	58.2%	50.4%

Local Basis And Regeneration

Country finance parameters use `JO` in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Lebanon National OpenSourceRail Strategy

Repository source: `cities/catalogue/west-asia/Lebanon/NATIONAL-BRIEF.md`

This page contains only Lebanon-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $8.50 B (88.9%) of external capital and $10.76 B of external interest. Capital plus saved interest totals $19.27 B.

National Programme

Local measure	Planning value
Catalogue cities	3
Represented population	3,230,000
Trainsets / vehicle modules	520 / 1,739
City infrastructure and fleet CAPEX	$4.45 B
Shared national factory	$811.1 M
Factory sizing basis	1,052 modules for Beirut, then reused nationally
Total national programme	$5.31 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$1.06 B (20.0%)
Domestic / local capital	$4.25 B (80.0%)
Annual external capital draw	$132.9 M / yr
Annual local capital draw	$531.4 M / yr
Annual public construction commitment	$1.00 B / yr for 8 years
Annual post-grace debt service	$914.3 M / yr
Default foreign-turnkey external capital	$9.57 B
External capital saved	$8.50 B
Capital + lifetime external interest saved	$19.27 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$2.49 B	$373.6 M	$2.12 B
Stations	$683.0 M	$136.6 M	$546.4 M
Depots	$282.6 M	$70.6 M	$211.9 M
Rolling stock	$497.3 M	$174.1 M	$323.2 M
Dedicated solar plants	$174.8 M	$78.7 M	$96.1 M
Residual train control	$12.9 M	$6.4 M	$6.4 M
Charging microgrids	$25.9 M	$10.3 M	$15.5 M
EPC / project services	$336.2 M	$50.4 M	$285.8 M
Shared national trainset factory	$811.1 M	$162.2 M	$648.8 M
Total	$5.31 B	$1.06 B	$4.25 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Beirut	2,200,000	263	$2.89 B	$583.5 M	$2.31 B	9	3	61.8%	64.5%
Tripoli Lb	730,000	173	$952.9 M	$197.5 M	$755.3 M	4	1	34.6%	62.8%
Sidon	300,000	84	$600.4 M	$111.3 M	$489.1 M	4	1	48.5%	68.8%

Local Basis And Regeneration

Country finance parameters use `LB` in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Palestine National OpenSourceRail Strategy

Repository source: `cities/catalogue/west-asia/Palestine/NATIONAL-BRIEF.md`

This page contains only Palestine-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $6.67 B (88.6%) of external capital and $8.36 B of external interest. Capital plus saved interest totals $15.02 B.

National Programme

Local measure	Planning value
Catalogue cities	3
Represented population	1,850,000
Trainsets / vehicle modules	602 / 1,806
City infrastructure and fleet CAPEX	$3.37 B
Shared national factory	$759.7 M
Factory sizing basis	798 modules for Nablus, then reused nationally
Total national programme	$4.18 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$858.9 M (20.5%)
Domestic / local capital	$3.32 B (79.5%)
Annual external capital draw	$122.7 M / yr
Annual local capital draw	$474.6 M / yr
Annual public construction commitment	$359.5 M / yr for 7 years
Annual post-grace debt service	$292.8 M / yr
Default foreign-turnkey external capital	$7.53 B
External capital saved	$6.67 B
Capital + lifetime external interest saved	$15.02 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$1.81 B	$272.2 M	$1.54 B
Stations	$426.6 M	$85.3 M	$341.3 M
Depots	$238.2 M	$59.5 M	$178.6 M
Rolling stock	$541.8 M	$189.6 M	$352.2 M
Dedicated solar plants	$117.0 M	$52.7 M	$64.4 M
Residual train control	$9.6 M	$4.8 M	$4.8 M
Charging microgrids	$7.5 M	$3.0 M	$4.5 M
EPC / project services	$265.9 M	$39.9 M	$226.0 M
Shared national trainset factory	$759.7 M	$151.9 M	$607.8 M
Total	$4.18 B	$858.9 M	$3.32 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Hebron	800,000	227	$1.33 B	$275.1 M	$1.06 B	4	1	29.4%	51.5%
Gaza City	600,000	109	$579.6 M	$121.3 M	$458.3 M	4	1	57.2%	46.2%
Nablus	450,000	266	$1.46 B	$302.6 M	$1.15 B	5	2	37.9%	51.4%

Local Basis And Regeneration

Country finance parameters use PS in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Saudi Arabia National OpenSourceRail Strategy

Repository source: `cities/catalogue/west-asia/Saudi Arabia/NATIONAL-BRIEF.md`

This page contains only Saudi Arabia-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $51.06 B (88.4%) of external capital and $62.78 B of external interest. Capital plus saved interest totals $113.83 B.

National Programme

Local measure	Planning value
Catalogue cities	14
Represented population	15,600,000
Trainsets / vehicle modules	4,060 / 15,237
City infrastructure and fleet CAPEX	$31.15 B
Shared national factory	$863.1 M
Factory sizing basis	4,212 modules for Jeddah, then reused nationally
Total national programme	$32.07 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$6.67 B (20.8%)
Domestic / local capital	$25.40 B (79.2%)
Annual external capital draw	$1.33 B / yr
Annual local capital draw	$5.08 B / yr
Annual public construction commitment	$2.23 B / yr for 5 years
Annual post-grace debt service	$1.55 B / yr
Default foreign-turnkey external capital	$57.73 B
External capital saved	$51.06 B
Capital + lifetime external interest saved	$113.83 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$17.23 B	$2.59 B	$14.65 B
Stations	$3.93 B	$785.6 M	$3.14 B
Depots	$1.76 B	$440.1 M	$1.32 B
Rolling stock	$4.41 B	$1.54 B	$2.87 B
Dedicated solar plants	$1.63 B	$731.3 M	$893.8 M
Residual train control	$97.0 M	$48.5 M	$48.5 M
Charging microgrids	$161.8 M	$64.7 M	$97.0 M
EPC / project services	$1.99 B	$298.8 M	$1.69 B
Shared national trainset factory	$863.1 M	$172.6 M	$690.5 M
Total	$32.07 B	$6.67 B	$25.40 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Jeddah	4,700,000	702	$7.29 B	$1.61 B	$5.68 B	14	5	31.9%	34.0%
Mecca	2,200,000	315	$3.68 B	$731.3 M	$2.95 B	9	3	35.7%	31.8%
Dammam	1,500,000	381	$4.41 B	$886.4 M	$3.53 B	9	3	39.3%	31.7%
Medina	1,500,000	255	$3.05 B	$604.9 M	$2.44 B	9	3	25.1%	25.8%
Hofuf	800,000	267	$1.32 B	$280.1 M	$1.04 B	5	2	28.1%	29.8%
Buraidah	700,000	249	$1.30 B	$271.2 M	$1.03 B	5	2	24.6%	24.2%
Taif	700,000	220	$1.14 B	$239.3 M	$902.6 M	5	2	22.8%	22.4%
Tabuk	650,000	237	$1.24 B	$258.1 M	$977.3 M	5	2	28.2%	26.8%
Khamis Mushait	600,000	320	$1.72 B	$359.9 M	$1.36 B	5	2	32.3%	40.8%
Hail	500,000	246	$1.34 B	$280.1 M	$1.06 B	5	2	26.6%	31.3%
Najran	500,000	213	$1.28 B	$258.6 M	$1.02 B	5	2	25.3%	30.0%
Abha	450,000	229	$1.15 B	$241.4 M	$908.6 M	4	1	16.2%	24.2%
Al Kharj	400,000	232	$1.20 B	$251.8 M	$949.9 M	5	2	34.9%	34.8%
Jizan	400,000	194	$1.02 B	$214.9 M	$805.6 M	5	2	32.6%	32.5%

Local Basis And Regeneration

Country finance parameters use SA in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Syria National OpenSourceRail Strategy

Repository source: `cities/catalogue/west-asia/Syria/NATIONAL-BRIEF.md`

This page contains only Syria-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $23.73 B (88.7%) of external capital and $30.65 B of external interest. Capital plus saved interest totals $54.39 B.

National Programme

Local measure	Planning value
Catalogue cities	9
Represented population	7,617,000
Trainsets / vehicle modules	1,853 / 6,206
City infrastructure and fleet CAPEX	$14.15 B
Shared national factory	$667.5 M
Factory sizing basis	1,864 modules for Aleppo, then reused nationally
Total national programme	$14.86 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$3.02 B (20.3%)
Domestic / local capital	$11.84 B (79.7%)
Annual external capital draw	$302.4 M / yr
Annual local capital draw	$1.18 B / yr
Annual public construction commitment	$2.27 B / yr for 10 years
Annual post-grace debt service	$2.09 B / yr
Default foreign-turnkey external capital	$26.76 B
External capital saved	$23.73 B
Capital + lifetime external interest saved	$54.39 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$7.15 B	$1.07 B	$6.08 B
Stations	$2.85 B	$569.8 M	$2.28 B
Depots	$874.6 M	$218.7 M	$656.0 M
Rolling stock	$1.79 B	$626.5 M	$1.16 B
Dedicated solar plants	$451.3 M	$203.1 M	$248.2 M
Residual train control	$38.6 M	$19.3 M	$19.3 M
Charging microgrids	$97.7 M	$39.1 M	$58.6 M
EPC / project services	$942.9 M	$141.4 M	$801.5 M
Shared national trainset factory	$667.5 M	$133.5 M	$534.0 M
Total	$14.86 B	$3.02 B	$11.84 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Damascus	2,503,000	348	$3.88 B	$777.1 M	$3.10 B	9	3	38.4%	41.2%
Aleppo	1,639,000	466	$4.31 B	$892.1 M	$3.42 B	9	3	44.1%	40.4%
Homs	775,000	142	$846.6 M	$173.4 M	$673.2 M	4	1	29.8%	29.9%
Latakia	700,000	135	$742.9 M	$153.4 M	$589.5 M	5	2	37.5%	41.0%
Hama	600,000	229	$1.26 B	$262.2 M	$1.00 B	5	2	33.5%	43.6%
Deir Ez Zor	500,000	210	$1.12 B	$232.2 M	$883.2 M	5	2	43.1%	45.2%
Raqqa	350,000	156	$834.9 M	$173.8 M	$661.2 M	4	1	42.1%	45.2%
Idlib	300,000	87	$617.1 M	$116.2 M	$500.8 M	4	1	55.7%	60.4%
Tartus	250,000	80	$542.4 M	$103.3 M	$439.1 M	4	1	43.7%	49.0%

Local Basis And Regeneration

Country finance parameters use SY in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
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```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Yemen National OpenSourceRail Strategy

Repository source: `cities/catalogue/west-asia/Yemen/NATIONAL-BRIEF.md`

This page contains only Yemen-specific aggregation. Shared network, service, energy, civil, cost, finance, QA and validation methods are defined once in the deployment planning reference.

> Important
> Foreign-capital advantage: against the default equivalent foreign-turnkey sensitivity, this national programme avoids $20.42 B (88.3%) of external capital and $26.38 B of external interest. Capital plus saved interest totals $46.80 B.

National Programme

Local measure	Planning value
Catalogue cities	9
Represented population	8,337,500
Trainsets / vehicle modules	1,643 / 6,413
City infrastructure and fleet CAPEX	$12.06 B
Shared national factory	$733.3 M
Factory sizing basis	3,384 modules for Sanaa, then reused nationally
Total national programme	$12.84 B

Capital And Funding

Local funding measure	Planning value
Imported / external capital	$2.70 B (21.0%)
Domestic / local capital	$10.15 B (79.0%)
Annual external capital draw	$269.7 M / yr
Annual local capital draw	$1.01 B / yr
Annual public construction commitment	$1.79 B / yr for 10 years
Annual post-grace debt service	$1.64 B / yr
Default foreign-turnkey external capital	$23.12 B
External capital saved	$20.42 B
Capital + lifetime external interest saved	$46.80 B

Procurement-Origin Composition

CAPEX bucket	Total	Imported	Local value
Civil works	$5.92 B	$887.6 M	$5.03 B
Stations	$2.10 B	$420.8 M	$1.68 B
Depots	$843.6 M	$210.9 M	$632.7 M
Rolling stock	$1.85 B	$646.8 M	$1.20 B
Dedicated solar plants	$479.0 M	$215.6 M	$263.5 M
Residual train control	$32.3 M	$16.1 M	$16.1 M
Charging microgrids	$77.0 M	$30.8 M	$46.2 M
EPC / project services	$808.9 M	$121.3 M	$687.6 M
Shared national trainset factory	$733.3 M	$146.7 M	$586.6 M
Total	$12.84 B	$2.70 B	$10.15 B

City Programme

City	Population	Fleet	City CAPEX	External capital	Local capital	Lines	Added lines	Original circles	Current circles
Sanaa	3,937,500	564	$5.69 B	$1.25 B	$4.44 B	11	4	61.4%	53.6%
Aden	985,000	182	$966.9 M	$202.6 M	$764.3 M	5	2	45.6%	50.7%
Hodeidah	750,000	106	$608.7 M	$124.3 M	$484.4 M	4	1	39.3%	45.1%
Ibb	750,000	157	$887.6 M	$181.4 M	$706.2 M	5	2	21.0%	14.6%
Taiz	615,000	172	$1.02 B	$207.9 M	$809.0 M	5	2	36.3%	36.6%
Mukalla	550,000	254	$1.41 B	$293.9 M	$1.12 B	5	2	49.7%	59.7%
Dhamar	300,000	77	$524.3 M	$98.9 M	$425.4 M	4	1	34.5%	51.0%
Lahij	250,000	68	$512.4 M	$95.6 M	$416.8 M	4	1	54.9%	62.1%
Sayun	200,000	63	$441.8 M	$83.3 M	$358.5 M	4	1	57.4%	48.7%

Local Basis And Regeneration

Country finance parameters use `YE` in `lib/templates/country-finance.toml`. The factory is counted once nationally and excluded from city CAPEX. City values come from each local `design.toml` and expanded scenario; common limitations and interpretation are not repeated here.

```
python3 tools/automation/generate-national-briefs.py
```

Population-led route regeneration (2026-10-09)

The city inventories include additional lines selected from retained unserved residential areas, continuous corridor junctions and bounded station spacing. Fleet, depots, construction, energy, staffing and country finance use those revised inventories. The working target is 80% within 1 km circles of emitted stations; remaining gaps and missing data stay explicit.

Coverage uses retained 2020 city-bbox population counts. Overlapping city populations are not added to claim national coverage. These circles are not current census, surveyed walking catchments, observed fare demand or construction approval. The catalogue comparison records targets and evidence limits.

Baghdad Structured Funding Appraisal

Baghdad — Urban Rail Network

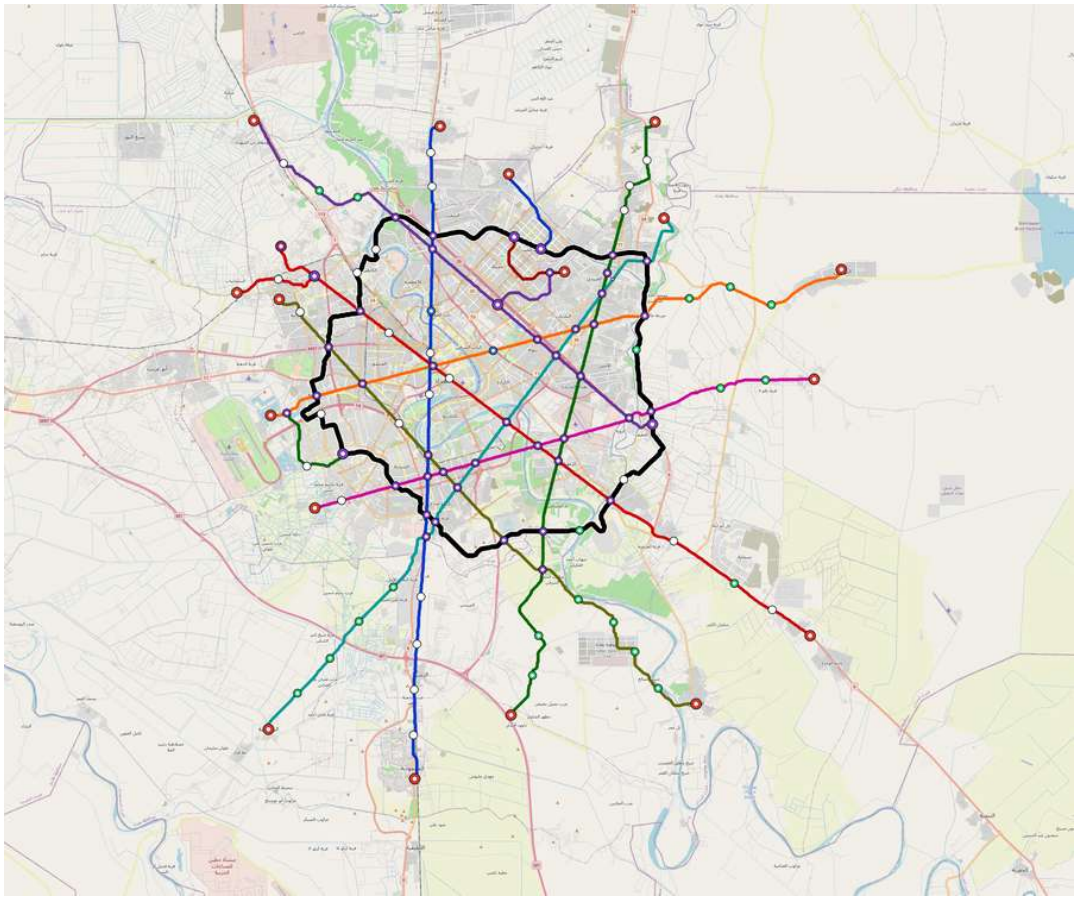
Repository source: `cities/catalogue/west-asia/Iraq/Baghdad/README.md`

Country: IQ · Population: 9,780,429 · Current Iraq planning basis

Current planning basis: 2026-10-04 programme recalculation, `local_positive` conditional local-production case. The main route is the reworked city-centre elevated planning alignment; service remains a capacity-led assumption. Revised scope is unquoted and uncommitted; this is not a construction design or an operating release.

Connected construction and battery study, 2026-10-06 now reconciles island topology, 18 launchers/two shifts, supplier/logistics constraints, equipment cash and sodium alternatives. Its full-network energy duties report service shortfalls; supplier contracts, installed-rate credits and accessible-entrance coverage remain unqualified. The financial figures below are retained comparators and do not include an accepted accelerated-build saving. Base programme planning allowance is USD 8.667bn, including line-local depots, final assembly and selected upstream component plants. Assumed facilities cover the modelled cash deficits; their placement is uncommitted, and the case retains IQD 0.000tn terminal debt. Local special/segmental structures, installed grid/charging upgrades, actual foundations, land and utilities remain unpriced. Removing search penalties establishes no realised saving. Older catalogue financial passes establish arithmetic for their own assumptions, not viability of this revised scope.

Network



Baghdad rail network on OpenStreetMap

Lines / unique stations / interchanges: 14 / 159 / 45; Route length: 513.3 km double track; Direct transfers / reachable line pairs: 39.6% / 100.0%; Residents within 800 m radial station catchments: 1,129,864 (2020 raster; 18.9% of bbox).

Service span / peak headway: 05:30–02:00 / 3 min; Fleet: 809 × 6-car `metro-6car` trainsets (728 peak revenue); Peak network throughput: 403,200 passengers/hour; Practical service capacity: 3,615,840 passenger-trips/day.

Annual paid-trip planning range: 659.9–1055.8 M.

Lines

Line	Length	Stations	Trainsets	Termini
line-1	49.1 km	17	86	S Outer N Outer
line-2	52.9 km	14	95	SE Outer NW Mid
line-3	49.5 km	16	90	S Outer NE Outer
line-4	41.9 km	12	76	NW Outer E Mid
line-5	47.3 km	16	86	W Mid E Outer
line-6	52.5 km	16	95	SW Outer NE Mid
line-7	40.0 km	13	72	E Outer SW Mid
line-8	48.1 km	15	89	SE Outer NW Mid
line-9	97.7 km	26	45	NW Mid NW Mid
line-10	6.8 km	3	15	N Mid NE Mid
line-11	6.9 km	2	14	N Mid N Mid
line-12	6.6 km	3	15	NW Mid NW Outer
line-13	8.0 km	4	18	W Mid W Mid
line-14	6.0 km	2	13	N Inner NE Mid
Total	513.3 km	159 unique	809	

Population access uses retained native count pixels where available; radial catchments require pedestrian/feeder validation. The former demand-score resident proxy is retired. Access and transfers · Building, support and terrain clearance.

Energy

Scheduled service: 6,278 one-way journeys / 215,989 train-km/day; Annual traction demand: 2,043.4 GWh; Station/depot PV / storage: 46.4 MW / 316.0 MWh; Aggregate charging power: 278.0 MW.

Dedicated solar plant: 1,019.7 MW; Residual grid/PPA import: 0.0 GWh/yr; Worst powered-stop gap: line-6: 18.9 km / 305 kWh; Lowest traversal charging margin: line-11: 74 kWh. These energy quantities describe the current regenerated scenario. Zero annual residual grid import is an accounting balance, not accepted hourly autonomy. Depot charging/grid upgrades, duty and launch conflicts remain open.

Current capital and operating people

Revised capital scope	USD equivalent million
Civil and bearing allowance	4,058.152
Stations and core elevated access	1,269.500
Rolling stock	1,219.590
Final assembly and component plants	416.253
Line-local depots	289.645
Solar and charging	874.140
Signalling and programme overhead	539.271
Total programme	8,666.552

There are 14 depots, one per line, with 809 storage slots for all 111 m six-car trains plus clearance. Depot reference capital is USD 289.645m, replacing the old USD 8m once. Workshop bays are sized separately by workload. The current case gives no capacity credit to station stabling. Actual land, foundations, connected access, installed charging and morning launch acceptance remain open. Depot quantities · Items.

The operating establishment is 3,523 FTE, including 1,572 station-cover FTE. Two staff per station and two normal eight-hour shifts give 636 daily shift assignments; retained 20.5-hour service also funds late cover and weekly/leave/training/sickness relief. Loaded annual payroll is IQD 83.415bn, with subsequent OPEX inflation. General pay starts at IQD 1,175,455/month, 50% above the historical employee median indexed to 2026. Technical/supervisor/senior/director grades are separate; this is not a newly measured median. Roles and wages. Final assembly has 1215 production and 183 support FTE; selected upstream plants add 290 production and 45 support FTE. Their 59 paid production months are separate from permanent railway jobs. Construction crew screen is incomplete; contractor labour is already inside contract rates.

Iraqi manufacture and USD capital exposure

bogie: 9708 (Local process option); motor-inverter-set: 9708 (Local process option); battery-225kwh-pack: 4854 (Local process option); door-cassette: 19416 (Local process option); window-cassette: 29124 (Bought component).

Imported process machinery supports the selected Iraqi fabrication and assembly options shown above. Cells/BMS, wheels/axles/bearings, inverters and other critical inputs retain imports. Products with negative Baghdad-only whole-order margins remain bought in this case. Plant readiness/qualification within 18 months is assumed, not demonstrated. Make/buy

appraisal.

Compared with the matched bought-component case, capital changes from USD 8.795bn to USD 8.667bn; imported invoice exposure changes from USD 2.358bn (26.82%) to USD 2.016bn (23.27%). The historical 148 km / USD 18bn proposal has a different scope and assumed all-USD funding; it is not a matched tender saving.

Current funding and cashflows

Government capital is exactly 25%. Imports use 50% government USD cash / 50% proposed Chinese USD credit. Only Chinese debt is USD; the rest of government cash, bonds, bank/gap credit and mezzanine is IQD. Government invoice downpayments are scheduled within the total 25% contribution. FX is the historical planning anchor of IQD 1,300/USD.

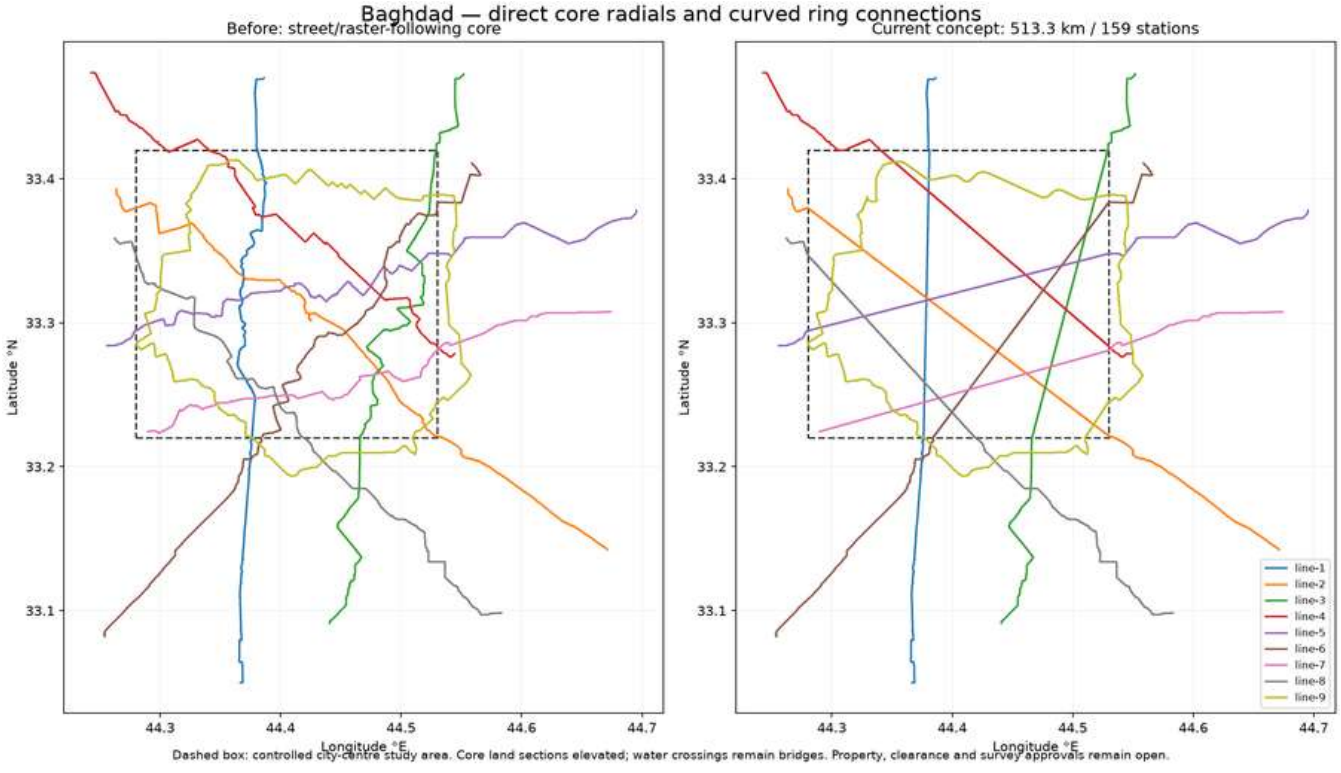
Capital-only source	Currency	Native amount
Government import cash	USD	1,008,216,375
Government local cash	IQD	1,505,948,192,086
Chinese capital credit	USD	1,008,216,375
Ordinary capital bonds	IQD	4,182,186,517,697
Green capital bonds	IQD	1,147,843,845,665
Senior bank capital credit	IQD	1,776,676,787,787
Conditional climate grant	IQD	32,500,000,000

Interest/fees, reserve cash, OPEX and gap facilities are additional cashflows, not capital added twice. Fares, kiosks, advertising, additional receipts and fare/OPEX indexation are included. Green/grant/rights terms and concessional gap credit remain uncommitted. Conditional first/full line revenue is month 40 / 81; physical and financing gates are open.

The tested IQD mezzanine leaves 81 defaulted draw vintages and increases terminal debt to IQD 0.000tn. It does not establish sustainable repayment. Monthly cashflow · Six-month bond/loan placements · All 13 cases · Cost and demand review.

City-centre elevated alignment

The main design uses straight core radial tangents and broad curved ring connections where the retained water evidence permits them, inside the explicit 33.22–33.42°N / 44.28–44.53°E study area. Shoreline detours retain their separate curve and structure review gates. Land sections there are elevated; water crossings remain bridges. The final water-constrained core routes change from 301.670 km to 286.225 km. The main network is 513.336 route km, with 56.73% elevated across the whole system. Maps, station placement, fleet, civil quantities, staff and finance use that reworked design. Analytical controls and limitations.



Earlier corridors and current central alignment

Property/air rights, obstacles, protected sites, surveyed heights, utilities, piers, foundations, transition curves and vertical alignment remain open. Existing outer approaches retain street/raster bends; exceptional geometry still requires realignment

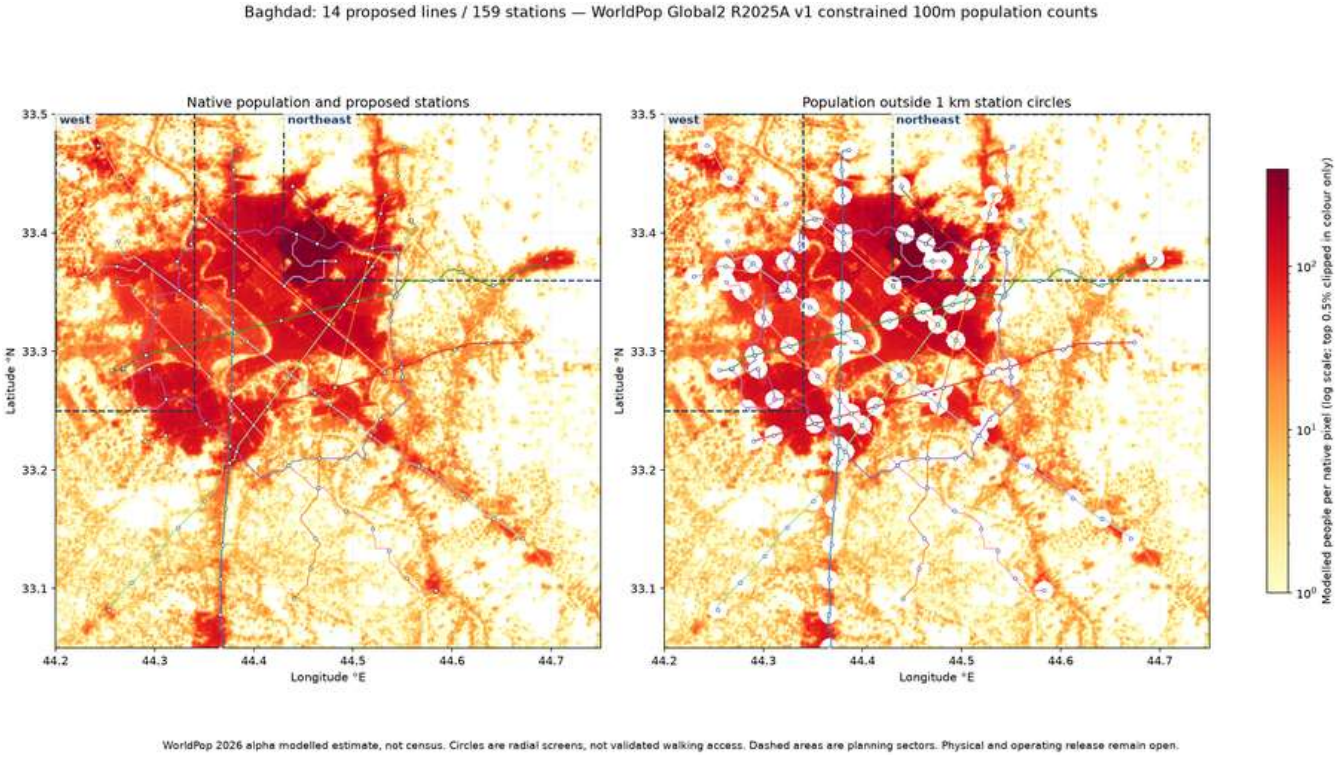
or special products. Additional outer grade-separation sensitivities add 7.924 km and retain current dates; they are not the adopted core geometry or an achieved routing-penalty saving.

Neighbourhood population and mapping sources

Native WorldPop 2026 counts guide the compact corridor additions, with separate GHSL 2025 and retained WorldPop 2020 comparisons. The 2026 epoch is an alpha modelled projection using older population inputs, not the 2024 census. Planning sectors are explicit rectangles, not administrative boundaries.

Sector	Modelled native population	Within 1 km of emitted stations	Outside station circles
northeast	2,197,150	38.4%	1,353,322
west	1,459,449	29.6%	1,027,430

Measured Baghdad traffic counts, speeds and OD demand remain unavailable. Pedestrian topology is screened on a dated OSM graph; station entrances, snap connections, sidewalks and accessible paths remain unverified. Station circles are potential radial access, not validated walking coverage or paid trips. Source comparisons, licences and neighbourhood gaps.



Population and underserved planning sectors

Evidence and regeneration

Complete proposal PDF · Editable proposal · Supporting data · Planning-only ERP drafts · Original catalogue finance · Original depot/stabling screen. Simulation, energy and asset evidence uses the current reworked geometry; the new staffing/depot/factory plans do not create accepted sites, appointed employees or operational assets.

Auto-planned by the OpenSourceRail design pipeline. Shared assumptions are in the deployment planning reference. Inputs: design, scenario, map, package evidence. City-local simulation, energy, GIS, operations and delivery retain their recorded planning/release gates. For presentation-only updates, run `.venv/bin/python tools/automation/publish-city-summary.py`; add `--check` to detect drift. The city regeneration pipeline uses this publisher. A stale scope study stops publication rather than silently restoring the original cost headline. Publication provenance is in publication-manifest.json.

Baghdad programme recalculation

Repository source: `cities/catalogue/west-asia/Iraq/Baghdad/engineering/programme-recalculation/README.md`

Controlled planning basis, 2026-10-04; unquoted and uncommitted. This is the latest staffing/depot/local-production review. Existing cost and financing cases remain historical comparators; no survey, manufacturing qualification or lender commitment is created.

Operating people and wages

The 159 stations require two concurrent staff during two normal eight-hour shifts: 636 daily shift assignments, not a headcount of people working every day. Retaining 20.5 service hours also requires 4.5 hours of separately funded late cover per station. Weekly cover, leave, training, sickness and handovers give 1572 station-cover FTE, within 3523 permanent operating FTE. Full-operation loaded payroll is IQD 83.415bn/year (USD 64.165m equivalent), indexed thereafter with OPEX.

The employee median is IQD 614,000 from the 2021 Labour Force Survey, cited in the IMF 2023 report. Applying an assumed 5% annual index to 2026 gives IQD 783,637; the general-worker floor is IQD 1,175,455/month, 50% higher. Technical, supervisor, senior and director roles use 2.25/3/4/5 times that indexed proxy. This is a historical observation plus an editable index, not a measured current national median. Employer costs and overtime are priced separately. Roles and recruitment/cohorts retain appointment and competence gates.

ERP planning drafts map the revised Staffing Plan, depot Project/Asset budgets and factory Workstations to native business records; they create no Employee, appointment, transaction or approval.

Construction workers, factory production/support workers and permanent operators are separate populations. Construction applies explicit trade-crew allowances to actual work-package resource lanes/dates, counting overlapping use of the same lane once; its peak is 429 FTE. It is not a measured complete contractor crew schedule; revised depot and upstream factory construction crews still require their own work-package schedules. Construction labour is already inside contract rates; its unknown payroll-content overlap remains unpriced, preventing a duplicate charge. A separate wage-content stress increases assumed labour content (20% of affected contracts) by 50%, with incremental EPC once; it is an overlap sensitivity, not proof of contractor wage contents. Existing assembly wages are replaced using an explicitly assumed embedded payroll credit; component process labour is inside component unit prices. Paid production capacity above those productive hours is added separately, never charged twice.

The final-assembly establishment is 1215 production FTE plus 183 support FTE. The scheduled order requires 59 paid months from month 18 through month 76; payroll follows this calendar, rather than dividing the order by a theoretical steady-state rate and losing ramp-up/idle months. Each upstream plant also funds its whole production establishment for that period, with the productive labour already in unit prices deducted once. These jobs are additional to permanent railway operations; no national order or permanent post-order subsidy is assumed.

One depot per line, full-size trains and fleet

There are 14 line-local depots, providing 809 storage slots for all six-car revenue/spare/reserve trains. Slots use the actual 111 m train plus 10 m clearance. Workshop bays follow annual bay-hour demand and usable bay capacity; parking slots are not maintenance bays. Station storage is an option without any capacity credit in this case. Full overnight-to-morning launch and throat conflicts remain unvalidated.

Line	Trainsets/storage slots	Tracks	Workshop bays	Land envelope ha	Reference USD m
line-1	86	29	12	11.50	27.045
line-2	95	32	14	12.72	36.828
line-3	90	30	13	11.94	28.589
line-4	76	26	11	10.40	24.991
line-5	86	29	12	11.50	27.045
line-6	95	32	14	12.72	30.328
line-7	72	24	10	9.62	23.282
line-8	89	30	13	11.94	28.559
line-9	45	15	7	6.32	17.210
line-10	15	5	3	2.57	9.624
line-11	14	5	2	2.45	8.335
line-12	15	5	3	2.57	9.624
line-13	18	6	3	2.89	9.879
line-14	13	5	2	2.45	8.305

Total depot reference: USD 289.645m, replacing the original USD 8m exactly once. Storage, workshop and access tracks, points, drainage/access, workshop shell/equipment/services, yard lighting/fire services, wash plant, wheel lathe,

offices/stores and rescue/quarantine are visible. Access-track length is an editable allowance, not a connected site alignment. Existing depot PV/BESS is retained once. Land/title, utility relocation, actual soil/foundations, installed charger/grid upgrades and tax/duties remain unpriced. Item register.

Core elevated stations

The reworked core has 84 station platforms with raised-structure/access scope. The register replaces their catalogue allowance with a separately stated elevated reference, adding USD 292.500m direct, with incremental EPC and operating maintenance once. Elevated interchange allowances are already present where catalogued; they receive no duplicate uplift. Other core stations use a 30% structure allowance plus USD 1m for vertical access. These are unquoted allowances, not released multi-level junction or station designs. Core station register.

Iraqi component manufacture

Product	Network quantity	Required units/year	Cells	Production/support FTE	Factory capital USD m	Unit make/buy USD	Whole-order margin USD m
bogie	9708	2582	30	140/21	42.568	15156/25000	41.595
motor-inverter-set	9708	2582	20	70/11	29.253	11078/15000	2.088
battery-225kwh-pack	4854	1291	8	38/6	25.577	29828/45000	42.977
door-cassette	19416	5164	12	42/7	15.351	2023/3333	6.269
window-cassette	29124	7746	9	32/5	22.392	687/1111	-14.256

Make/buy inputs price imported process machinery, Iraqi buildings/site work, qualification, residual imported inputs, local materials, graded labour, process overhead, fixed support and plant maintenance. Cells are sized to the existing train factory's required production rate, not a small demonstration line. The order is 1.092 GWh of gross onboard packs. Battery pack assembly is evaluated; imported cells/BMS remain. Bogie wheels/axles/bearings, motor inverters/magnets, door safety electronics and glazing feedstock also retain imports. Local manufacture does not mean zero USD input.

The buy case itemises these five imported completed products; the old blanket vehicle import percentage is retained only for other parent scope. This prevents replacing already-local scope with a second import credit. The all-product and positive-margin selections are separate unquoted cases. Whole-order margins are before finance/tax/risk; vendor prices, license, QA, process yield, supply commitments and first articles must validate them. No future national order pays Baghdad debt or makes an uneconomic line look profitable. Serial component qualification is assumed within the 18-month readiness target, not proven. A six-month supplier delay shifts rolling-stock invoices and opening/service cash, extends the full operating horizon and prices idle production payroll. A 25% raw-input price stress preserves the same selected facilities. Rejected/reworked product still needs a measured production replay. Existing final assembly tooling remains priced; upstream machinery is added without an unproven overlap credit.

Paid journeys and transfers

Revenue remains a capacity-led sensitivity rather than a surveyed OD forecast. The reference uses one boarding per paid journey, a zero-transfer upper-bound assumption. Integrated-fare sensitivities use 1.25, 1.5 and 2 boardings per journey: only fare receipts are divided; kiosk/rental/advertising receipts, service, staffing and energy are retained. These are uncalibrated factors, not estimates of Baghdad travel. Physical access and surveyed OD/section loads must qualify any adopted demand forecast. Demand handoff.

More elevation and fewer bends

The current main design adopts the straight central elevated alignment. The screening policy allows up to 65% elevated and at least 25% at grade, subject to site/design acceptance. Baghdad currently has 56.73% elevated. Investigation windows around all 49 exceptional segments add 7.924 km of candidate at-grade conversion, reaching 58.27%; overlapping intervals are merged and existing viaduct/bridge lengths excluded. Approach length is at least 300.0 m from assumed height/gradient.

Elevation alone removes no horizontal bend. Wider-radius geometry, station moves, ROW, vertical alignment, ramps, egress, ground/utility evidence, crossings and whole-life costs must be designed together. The additional-elevation case includes only its base allowance; routing scores are excluded from money and the old 25%/50% penalty-removal cases are retired. Special designs and consequential installed costs remain unknown. The added standard civil allowance uses the conservative simple-span bearing index. Candidate intervals.

Funding and mezzanine comparison

Government capital remains exactly 25% of the scenario total. Imported invoices receive 50% government USD cash / 50% Chinese USD credit. Remaining government cash, bonds, senior bank/gap credit and mezzanine are IQD. Government USD payment dates follow machinery/input invoices; the remaining appropriation is allocated proportionally to local invoices, so a machinery-heavy month need not be falsely limited to 25% government cash. Chinese supplier origin and export-credit eligibility are assumptions requiring vendor/lender confirmation. China Exim describes buyer credit for Chinese products, technologies and services; machinery eligibility is not a loan commitment (official product description).

The positive-margin local-production senior case's capital-only sources are shown below. They sum to its capital uses; gap credit, interest/fees, reserve funding and operating receipts are additional cashflows in its monthly ledger and six-month placement schedule. Ordinary and green bonds are separate placements, never added again to a combined bond figure. Bond face units are IQD 1m; rounded placement envelopes are not additional cash raised.

Capital source	Currency	Native amount	USD equivalent m
Government import cash	USD	1,008,216,375	1008.216
Government local cash	IQD	1,505,948,192,086	1158.422
Chinese capital credit	USD	1,008,216,375	1008.216
Ordinary capital bonds	IQD	4,182,186,517,697	3217.067
Green capital bonds	IQD	1,147,843,845,665	882.957
Senior bank capital credit	IQD	1,776,676,787,787	1366.674
Conditional climate capital grant	IQD	32,500,000,000	25.000

Mezzanine replaces 10% of domestic residual capital borrowing; it is not extra capital on top of the uses. The IQD case has 6% cash coupon, 4% PIK, 2% arrangement fee and a 15-year balloon from each draw. Deferred cash coupon and PIK increase outstanding debt until maturity. Thereafter the overdue principal is retained with separately disclosed simple 10% contractual-interest sensitivity; arrears are not compounded and the balloon is not silently extended. Cash junior payments require senior DSCR of 1.20, funded reserves and actual residual cash; new gap draws/unfunded support cannot pay junior debt. No conversion, fresh equity or automatic refinancing is invented. Unpaid maturity balances/defaults remain visible. Mezzanine is structurally junior to senior debt and senior to equity, as described by UNCITRAL; the rates here are project sensitivities.

Matched case / six-month placement schedule	CAPEX USD bn	USD capital intensity	Peak IQD gap tn	Terminal all debt IQD tn	Unfunded IQD tn	Junior defaulted vintages
revised_scope_buy	8.795	26.82%	5.109	0.000	0.000	0
local_all	8.677	23.12%	5.059	0.000	0.000	0
local_positive	8.667	23.27%	5.033	0.000	0.000	0
local_positive_mezzanine	8.667	23.27%	4.267	0.000	0.000	81
local_positive_commercial_gap	8.667	23.27%	12.041	0.000	0.000	0
local_positive_mezzanine_stress	8.667	23.27%	9.117	21.346	0.000	83
additional_elevation_base_allowance	8.731	23.20%	5.114	0.000	0.000	0
local_positive_raw_price_stress	8.770	23.74%	5.128	0.000	0.000	0
local_positive_supplier_delay	8.667	23.27%	5.366	0.000	0.000	0
construction_wage_content_stress	9.267	22.77%	5.927	0.000	0.000	0
integrated_fare_1_25_boardings	8.667	23.27%	9.220	0.000	0.000	0
integrated_fare_1_5_boardings	8.667	23.27%	13.000	10.554	0.406	0
integrated_fare_2_boardings	8.667	23.27%	13.000	13.000	12.070	0

The 4 positive-margin process options reduce capital from USD 8.795bn to USD 8.667bn, and imported invoice exposure from USD 2.358bn to USD 2.016bn. Half of that exposure is government USD cash and half Chinese USD credit; all other capital funding is IQD. The historical 148 km / USD 18bn third-party benchmark has a different scope and assumed full foreign-currency financing; this is a planning comparison, not a like-for-like tender saving. The reworked 513.3 km network's 40.5% legacy routing-demand cell fraction is not population access. Its former resident multiplication is retired; see the native population and transfer audit.

Current financial conclusion: the positive-margin senior case records IQD 0.000tn of residual unsourced support after assumed facilities and retains IQD 0.000tn of debt at the horizon. Its company NPV before finance is USD -3.246bn at the assumed nominal discount rate. Adding mezzanine does not change the underlying operating return: it leaves IQD 0.000tn of

total debt and 81 unpaid junior vintages. It is not recommended as a cure for the funding deficit. Fare and OPEX indexation, kiosks/advertising and inherited additional receipts are already included; new verified capital, affordable revenue or accepted scope savings are still needed.

All cases use the same revised staff, depot and indexed fare/OPEX assumptions. Six months of scheduled senior service are reserved from the first draw; three months of OPEX plus explicit industrial working capital are restricted. This revised reserve policy differs from older reference cases, so compare matched cases in this table when judging mezzanine. Concessional 2% gap funding, grants/rights/additional income and enhanced green coupons are uncommitted; the 8% gap and high mezzanine-rate case expose that dependence. There are zero dividends. Cash/principal/PIK identities, maturity risk, monthly draws and six-month bond denominations are retained for every case.

Regenerate with `.venv/bin/python tools/automation/baghdad_programme_recalculation.py`; verify with `--check`. Complete installed scope, construction wage-content adjustment, local-pay survey, validated demand/timetable and supplier/lender agreements remain open.

Iraq funding programme and Baghdad planning status

Repository source: `cities/catalogue/west-asia/Iraq/IRAQ-FUNDING-PROGRAMME.md`

Current Iraq planning basis — 2026-10-06

Current country basis · Baghdad summary · Connected design, production and energy study.

The retained `local_positive` Baghdad financial comparator is USD 8.667bn, with IQD 0.000tn unsourced support and IQD 0.000tn terminal debt. All debt clears at model month 295, conditional on the recorded funding assumptions. Its prices and facilities are unquoted and uncommitted. It does not price the complete accelerated construction and battery scenario.

The accelerated scenario's complete CAPEX, financing, net island savings and accepted opening dates remain unknown. Its USD 9m / 18-launcher allowance covers purchase cash only; freight, assembly, commissioning, transporters, lifting frames, temporary supports, spares, crews and relocations require separate scope-matched prices. The earlier delivered-and-commissioned interpretation is not used. No launcher saving is added to an installed civil rate before removing a verified matching embedded allowance.

Iraq has existing precast facilities and expertise; qualify contracted civil supply and price tooling/upgrades. Baghdad trainset assembly and component-process allowances remain in the financial comparator. Additional national plant capacity and transfer costs require separate scope reconciliation.

The coordinated industrialisation programme joins vehicle interfaces, staged bogie localisation and the standard viaduct system. Its possible vendor register records official-source component and equipment leads for comparison. Iraq already has many precast facilities and relevant expertise; plant capability, compatible tooling and available production slots require qualification. No listed vendor is a selected supplier or a manufacturing-rights agreement.

The coupled service–OD–cost–finance study now derives capacity-limited passenger itineraries from completed train sections and carries the same grid purchases and train distance into conditional monthly finance. Its demand and retained prices/opening months are explicit sensitivities. Complete line openings additionally require stations, specials, track, energy, depots, fleet, testing and approvals; missing dates or prices remain unknown. No coupled case is an accepted forecast or an adopted national budget.

No current consolidated national budget or funded rollout is established. Other Iraqi city catalogue estimates and the earlier generic foreign-turnkey calculation remain separately identified comparators. Their differences are scenario arithmetic, not realised capital or interest savings. Baghdad's later scope cannot be mixed into an older national aggregate and presented as one complete budget.

Construction sensitivities

Case	Days for modelled running bays	Scope
full-doubling	1022	identified catalogue planning spans only; unresolved closures, station structures and special crossings remain separate gates
higher-productivity	1098	identified catalogue planning spans only; unresolved closures, station structures and special crossings remain separate gates
initial-accelerated	1218	identified catalogue planning spans only; unresolved closures, station structures and special crossings remain separate gates
initial-accelerated-reassigned	966	identified catalogue planning spans only; unresolved closures, station structures and special crossings remain separate gates
initial-six-day-calendar	1348	identified catalogue planning spans only; unresolved closures, station structures and special crossings remain separate gates

These are conditional resource schedules. Station structures, special/closure spans, approved pier locations, delivery access and release evidence remain opening gates. Empty supplier/contract registers indicate missing qualifying evidence; they do not imply that Iraqi industry lacks capability.

Chronological energy study

Case, lower-solar duty	Minimum SOC	Unserved traction kWh
lfp-throughout	0.0%	676,286
sodium-both	0.0%	4,024,245
sodium-stationary	0.0%	676,383

The fixed-assignment diagnostic above uses both ring directions, declared dispatch stations and native rest-to-rest section timing. The current screen spans 4,440 continuous model minutes (3.08 days), including overnight state carry-over and a

study strategy permitting grid replenishment within remaining site import capacity. Charge-rate, thermal and life inputs are unqualified profiles, so these results do not establish a chemistry advantage or suitability.

The separate coupled energy/service controller keeps reserve-limited trains at their actual stations, continues charging, and reflects their availability in subsequent dispatch opportunities.

Controlled case	Completed journeys	Missed dispatch opportunities	Distinct energy-held journeys
lfp-throughout	17,491	909	706
sodium-both	8,536	9,605	2,761
sodium-stationary	17,492	908	719

Geometry/operating-distance findings remain recorded. Conflict-capable timetable, berth/depot access, transient traction and operational multi-day acceptance remain open. Current station passenger demand is unknown where no surveyed station/OD assignment has been entered; the common 3,000 pax/h value is retained only as a separate stress case.

Historical funding calculation — retained comparator

The calculation below retains the earlier USD 7.581bn scope and its own assumptions. Its debt-clearance months and financial differences do not describe the current `local_positive` comparator or the unpriced accelerated scenario.

Scope: Baghdad and one manufacturing plant sized for Baghdad. Samawah, Mosul and every other city are excluded. All funding is proposed and uncommitted.

Independent reconciliation, priced gap finance and six-month tranches

Capital sources still total USD 7,951.287m. The independent pooled-cash reconstruction requires USD 3,193.940m of gross extra cash and retains USD 7,867.053m later, leaving a USD -4,673.113m nominal net deficit before pricing additional gap finance. These three amounts answer different questions. The original capital principal is repaid in the lifetime cash ledger, not added to CAPEX a second time. City EPC is now spread over direct works; the initial baseline-freeze task no longer receives the full programme overhead allowance.

Detailed arithmetic and Iraqi financing routes · six-month reference bond/loan requirements · six-month blended candidate · independent calculation.

Green debt replaces qualified conventional borrowing; grants replace domestic capital debt; guarantees supply credit enhancement rather than cash. The new sensitivity charges supplemental IQD funding for interest and fees, sweeps later surplus to repayment, and exposes cash beyond its illustrative IQD 13tn cap and any unpaid terminal loan. The earlier gross cash-support figures below assume external contributions; they are not a priced bridge-loan requirement.

Priced gap-finance sensitivity	Gross gap draws, IQD tn	Uncovered cash, IQD tn	Unpaid terminal loan, IQD tn
unfunded reference	0.000	4.152	0.000
green label only	0.000	4.152	0.000
zero cost liquidity bound	4.152	0.000	0.000
commercial gap credit	13.000	7.067	13.000
concessional gap credit	5.318	0.000	0.000
blended candidate	3.574	0.000	0.000
green concessional only	4.433	0.000	0.000
fare 5pct flat costs fixed demand	2.003	0.000	0.000
fare 5pct flat costs elastic	2.232	0.000	0.000
fixed fare 5pct opex	13.185	0.486	13.000
fare 5pct opex 5pct	2.293	0.000	0.000
fare 5pct opex 7pct	2.471	0.000	0.000
fare 5pct opex 5pct income 2pct	2.491	0.000	0.000
variable fare 5pct opex 5pct	2.216	0.000	0.000
fare 5pct opex 5pct rents indexed	2.254	0.000	0.000

The candidate mix is uncommitted: eligible IQD green bonds at an assumed 4% plus enhancement fees; USD 25m equivalent climate capital grant; USD 300m equivalent net development-rights proceeds; and USD 25m equivalent annual new net local receipts at full opening. The grant, valuation, legal powers and IQD concessional facility need evidence. Existing rents/fare receipts cannot be counted again. With constant nominal fares and OPEX, a terminal unpaid loan means that sensitivity has not achieved self-financing.

Additional pricing and OPEX inflation sensitivities

The requested paired sensitivity increases fares and OPEX 5% annually from financial close, while testing income growth separately. At 5% income growth and assumed -0.30 real-price elasticity, the illustrative blended case peaks at IQD 2.102tn supplemental debt and ends with IQD 0.000tn unpaid. The case repays the facility within the model horizon under these conditional assumptions. Placed early financing, candidate grant/rights receipts, fixed nominal debt terms and income growth remain assumptions. It is not a committed funding outcome.

Average nominal tickets move from IQD 1,525 at first opening to IQD 1,765 at full opening; 44 trips remain 11.7% of the indexed income proxy when incomes grow 5%. Separate cases test 2% income growth, 7% OPEX inflation, peak/off-peak tiers, fixed demand, and rental indexation. Capital escalation and future FX changes remain outside these sensitivities.

Paired 5% six-month financing · monthly tickets and affordability · variable-ticket six-month financing. The detailed report contains the NPV, all assumptions and downside cases.

Surplus cash and early repayment

These cases use identical paired 5% fare/OPEX/income assumptions and the same conditional blended capital sources. Each pays OPEX, scheduled principal, interest and fees, then funds the debt-service reserve and a three-month current-OPEX buffer before voluntary repayment. Borrowed gap proceeds and uncovered external cash cannot fund early payments. The extra buffer remains cash held at the terminal horizon and earns no interest.

Compare strategies against buffered gap-only, rather than attributing the buffer change to repayment savings. Loans-first pays bank credit, Chinese credit and gap credit before bonds. Cost priority pays bank (9%), ordinary bonds (8%), Chinese credit (5%), green bonds (4% plus 0.5% annual guarantee), then gap credit (2%). It is an interest-rate heuristic, not a proof of the globally best strategy.

Surplus strategy	All debt cleared, month from close	Finance cost saving vs buffered gap-only, USD equivalent m	Early-payment premiums, USD equivalent m	Peak gap debt, IQD tn	Terminal unrestricted cash, IQD tn
gap only buffered	369	0.000	0.000	2.208	44.706
gap then core	209	506.359	36.995	2.208	45.364
loans then bonds	207	653.848	42.809	2.158	45.556
cost priority	203	988.754	56.843	2.158	45.991
cost priority zero premium	203	1,060.010	0.000	2.156	46.084
cost priority noncallable bonds	369	260.705	12.395	2.158	45.045

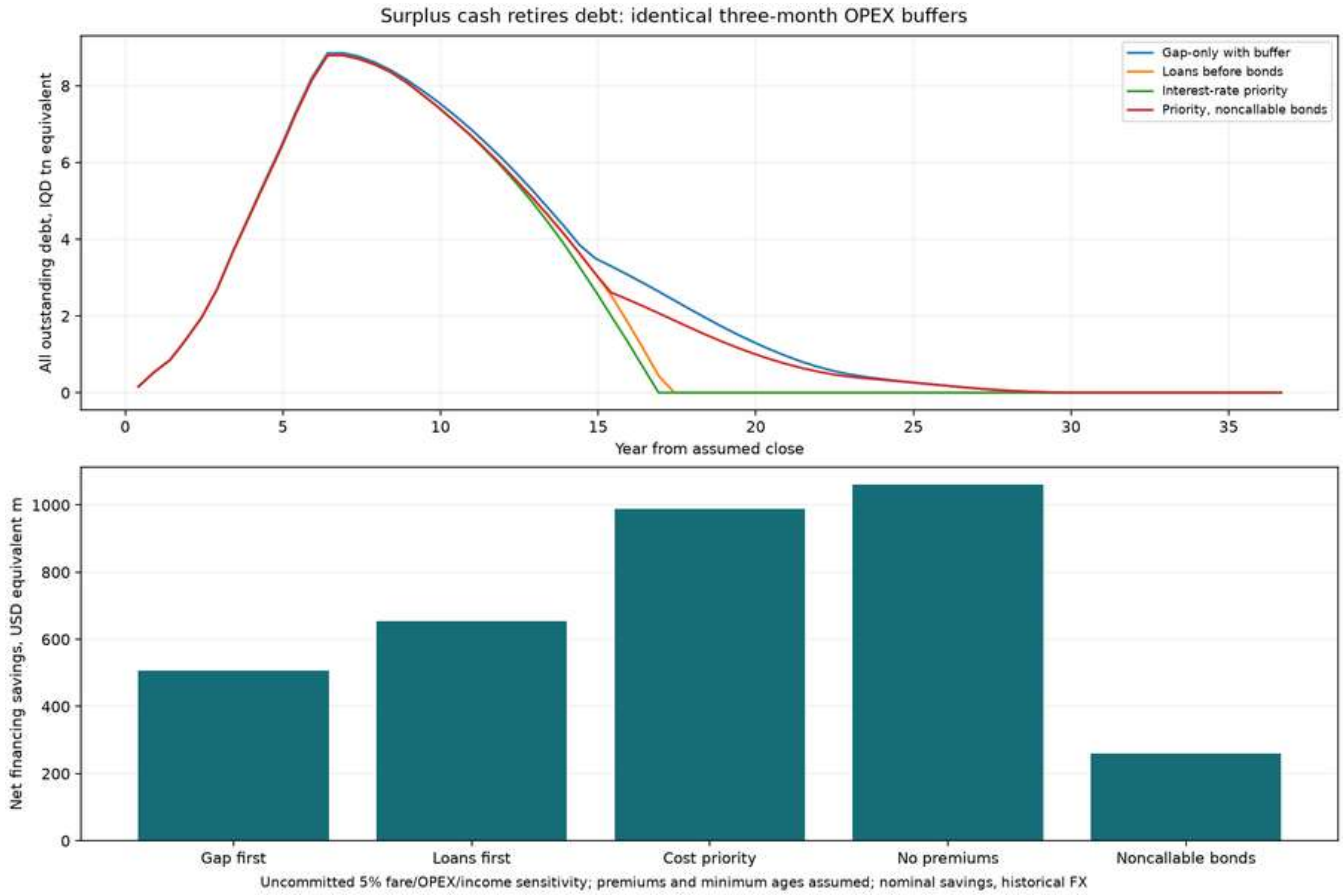
Cost-priority debt is cleared in month 203; buffered gap-only debt is cleared in month 369. Net nominal financing savings are USD 988.754m equivalent, after USD 56.843m assumed early-payment premiums. Savings include core interest, annual green guarantee charges and supplemental interest/draw fees; they exclude principal, which is returned once. All cases retain IQD 0.386tn operating buffer separately from unrestricted cash. There is no additional government contribution above 25% of CAPEX in these cases.

Assumed premiums are 1% of bank/Chinese/green principal and 2% of ordinary bond principal. A minimum draw age of 6 months (bank), 12 (Chinese) and 24 (both bonds) prevents immediate issue-and-redemption. Eligible vintages are repaid oldest first; contractual instalments are kept and maturity shortens. Notice, issuer call rights, investor consent, buyback price, remaining-maturity compensation, tax and FX require actual agreements. Noncallable-bond sensitivity makes no voluntary bond payments; zero-premium sensitivity removes only the assumed premium, retaining minimum ages.

Facility	Buffered gap-only final principal payment month	Cost-priority final principal payment month	Early principal, native currency	Premium, native currency
bank credit	165	122	IQD 636.771bn	IQD 6.368bn
domestic bonds	285	167	IQD 2,547.814bn	IQD 50.956bn
chinese export credit	309	175	USD 592.389m	USD 5.924m
green bonds	369	184	IQD 887.075bn	IQD 8.871bn

The gap facility's final principal payment is in month 203. Later surplus remains unrestricted cash after debt retirement. Each month's native debt balance equals prior balance plus draws minus scheduled and early principal. Each six-month closing balance is its final month's balance; payments, premiums, interest and reserve movements are period sums. Contractual repayment windows in tranche files describe the original draw terms; the actual final-payment months above incorporate early repayment.

Monthly cost-priority repayments · six-month repayment tranches · loans-first tranches · all six complete calculations



Surplus repayment comparison

Loan prepayment can carry redeployment/unwind charges: the World Bank Treasury FAQ illustrates the concept, without establishing Chinese or Iraqi loan terms. The US Treasury buyback FAQ and published purchase results illustrate issuer authority and priced buybacks rather than automatic par redemption; they do not supply an Iraqi legal mandate.

This repayment allocation does not change the unlevered project NPV or validate demand. Nominal savings over several decades are not present-value gains. CAPEX escalation, FX, lifecycle replacement, floating-rate risk and the uncommitted early funding still require appraisal.

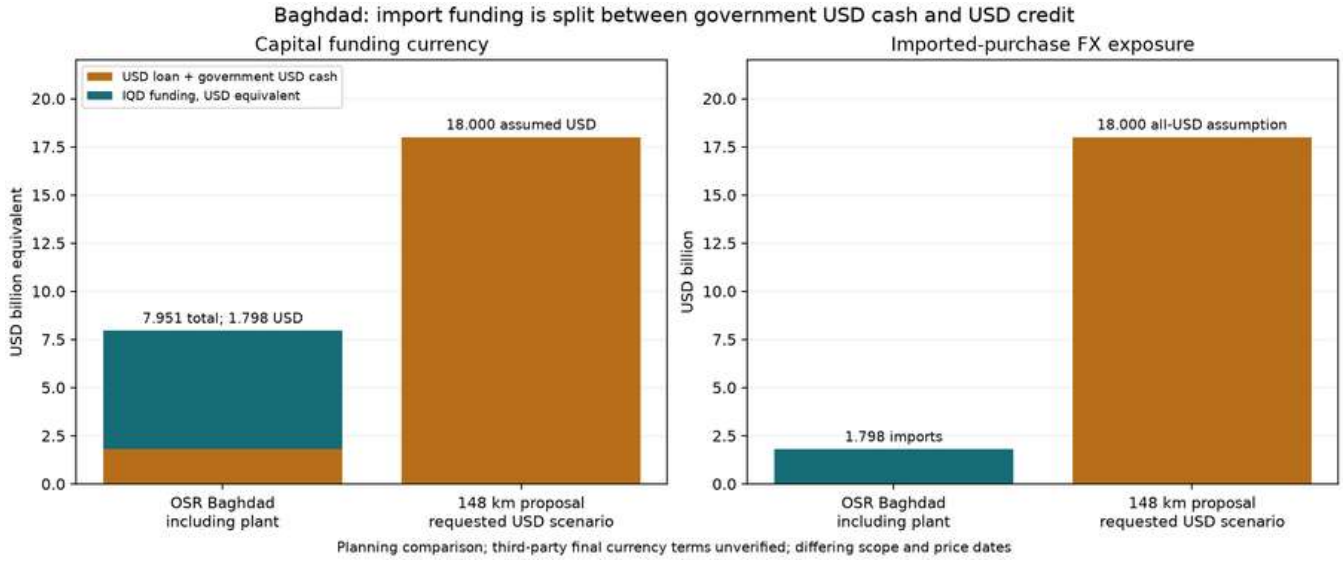
Consolidated sources and uses

Government contributes 25% of total capital uses, including the plant and EPC. Imported purchases are split 50% government USD cash and 50% proposed Chinese USD credit, assuming the full imported basket can qualify. That USD government cash is inside the 25% total contribution; the rest of the government contribution is IQD. The remaining balance after those two sources is split 75% domestic IQD bonds and 25% IQD bank term credit.

Proposed capital source	Currency	Native amount, billions	USD equivalent, million	Share of total capital
chinese export credit	USD	0.899	898.82	11.30%
government import cash	USD	0.899	898.82	11.30%
government local cash	IQD	1,415.697	1,089.00	13.70%
domestic bonds	IQD	4,938.025	3,798.48	47.77%
bank credit	IQD	1,646.008	1,266.16	15.92%
Total capital uses	Mixed	—	7,951.29	100%

Baghdad city CAPEX is USD 7,626.54 million. The plant is counted once at USD 303.50 million plus USD 21.25 million EPC. Its sizing basis is 4,854 vehicle/car modules from Baghdad's controlled fleet and car count. Imported tooling receives proposed Chinese credit within that plant budget. The physical cell/floor/tooling envelope replaces the smaller module allowance where necessary; both remain unquoted engineering assumptions.

Currency and USD capital intensity



Funding currency and imported-purchase FX comparison

Only Chinese credit is USD-denominated debt: USD 898.82 million. Government additionally provides USD 898.82 million cash for imports. Combined USD capital funding is 22.61% of uses; the remaining 77.39% is IQD government cash, IQD bonds and IQD bank credit. Revenue and local OPEX are also budgeted in IQD. At the historical planning conversion of 1,300 IQD/USD, the local government cash allowance is IQD 1.416 trillion, alongside its separate USD import cash. Together they equal 25% of total capital. USD columns are comparison equivalents, not a requirement to borrow or appropriate those domestic amounts in dollars.

Funding currency and procurement currency differ. Estimated imported purchases total USD 1,797.65 million (22.61% of capital), including plant imports. The government supplies half directly in USD cash; proposed Chinese credit supplies half as USD debt. The full import pool, including categories beyond the initially selected solar, bogies, batteries, windows, doors and tooling, is assumed lender-eligible pending origin qualification. If that expanded basket cannot qualify, its loan funding is uncovered; it is not automatically replaced by IQD bank credit or another grant. IQD borrowing reduces the revenue/debt currency mismatch; it does not remove Chinese debt-service FX risk, imported maintenance costs, domestic interest, inflation or placement constraints.

Eligibility sensitivity: the original named-component categories cover USD 849.64 million of invoices. At a 50% advance, they alone support USD 424.82 million of proposed credit. If the additional imported categories cannot qualify, USD 474.00 million of planned capital loan proceeds remains uncovered, in addition to the base support requirement below. Even the original categories remain unqualified; this screen is not a lender approval. No extra government contribution is assumed to replace that missing loan.

What the 25% government limit leaves unfunded

The 25% capital contribution is USD 1,987.82 million. If that is also the limit on all public cash, the model leaves USD 4,013.14 million of additional funding requirements across the full construction, operating and debt horizon. These are interest, fees, reserve and operating/debt cash needs after modelled revenue; they are not additional approved government contributions.

Peak annual capital contribution is USD 394.38 million. Peak annual additional funding requirement is USD 986.21 million. If all additional support were provided publicly, conditional lifetime public cash would be USD 6,000.96 million, with a combined annual peak of USD 1,135.99 million. This exceeds the requested contribution and is shown only to expose the funding gap.

Across the 36.8-year nominal model horizon, additional requirements divide into USD 1,888.68 million before Baghdad's full-network opening, USD 1,699.66 million during Baghdad operations/debt tail, and USD 424.80 million for plant capital financing. Principal repayment before opening is part of the pre-opening requirement. This base withholds all fares until full-network completion; the separate phased sensitivity below tests earlier revenue. Gross additional support is a liquidity requirement over time, not net lifetime loss: later retained cash cannot repay earlier obligations without an approved bridge.

The programme CSVs distinguish the capped capital contribution, additional funding requirement and conditional payment requirements. Underlying city and factory ledgers calculate the support needed to pay scheduled obligations; their positive reserve balances and completed repayments are conditional on that support being raised. They are not a cash-solvent forecast under the public cash cap. No bridge, equity investor, rollover or extra appropriation is invented to close the gap.

Cash and contractual structure

1. An Iraqi public sponsor seeks MoF authority for proposed sovereign IQD bonds through the established MoF/CBI issuance route, subject to legal review and placement. Municipal borrowing powers are not assumed. The 25% limit is the direct capital contribution, not a limit on sovereign bond liabilities, guarantees or lifetime public exposure.
2. Chinese export buyer credit is allocated to qualified Chinese invoices for solar equipment, bogies, batteries, windows, doors and Baghdad plant tooling. The requested 50% loan advance is matched by 50% government USD cash. All other imported categories also require Chinese supplier-origin and lender qualification before the full-basket financing assumption is valid. Noneligible imports still need foreign currency. Eligibility and origin require supplier evidence.
3. Government's 25% capital contribution covers the planned eligible downpayment; bonds and bank credit cover the remaining capital balance. Fees, construction interest and reserve funding are separate cash requirements, not silently capitalised or counted as capital-source proceeds.
4. Each draw starts its own grace and repayment clock. Planned receipts match procurement milestones; no actual payment or lender commitment is asserted. Chinese credit uses USD, bonds and bank credit IQD. Terms remain unconfirmed appraisal assumptions: 5%/48-month grace/180-month repayment, 8%/24/180, and 9%/0/84 respectively.
5. Revenue pays OPEX first and debt service next; subsidy is excluded from DSCR. Additional support shown by the city and plant models is an unfunded requirement under the cap. The factory model covers capital financing only; manufacturing income and factory OPEX need a separate business case.
6. Declined Chinese finance, delayed contributions, FX changes and bullet redemption are separately tested in the Baghdad appraisal. None creates an automatic facility or refinancing source.

Rollout and financial close

Baghdad's current capital milestones span 82 calendar months. Full-network fare revenue begins after all budget milestones. Lowering the capital grant increases borrowing and cash-service needs; it does not shorten this schedule. The city-sized Baghdad plant becomes available after 390 working days (18 months from NTP), followed by the separately scheduled first-article series-release gate. The separate phased sensitivity uses this integrated planning schedule; actual plant and line acceptance and a commissioned operating baseline remain pending.

Required before financial close: approved sponsor and borrowing powers; appropriation limited to the agreed government contribution; a funded solution for the additional cash gap; supplier quotations and origin evidence; signed term sheets, guarantees and insurance; IQD placement/redemption plan; lender draw windows and FX access; surveyed demand/fare policy; tax, duty, land and utility pricing; accepted resource calendar and reserve covenants.

Model and evidence

Programme monthly cashflow · annual cashflow · machine-readable programme · Baghdad city appraisal · editable assumptions. Other Iraqi city appraisals remain standalone examples outside this programme.

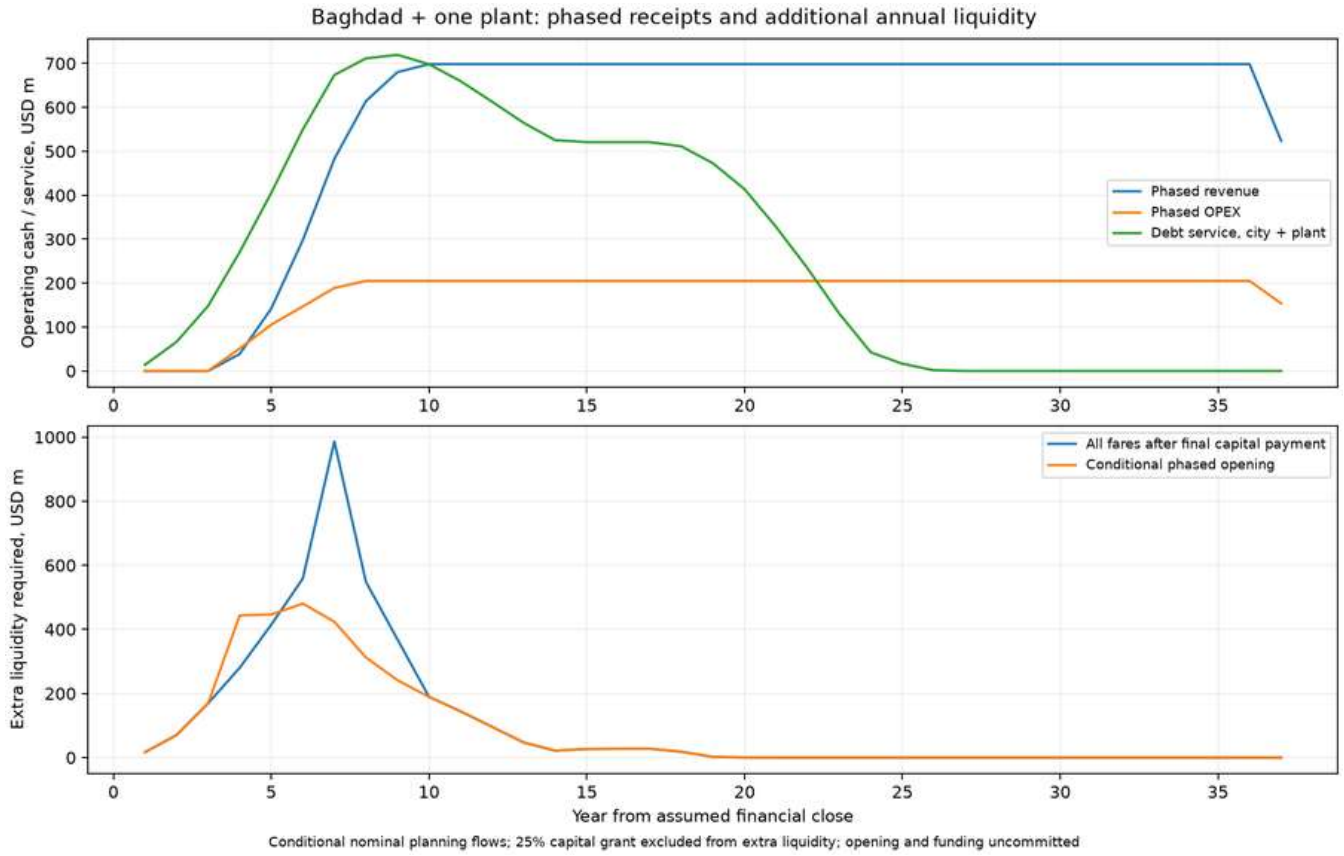
IMF 2025 Article IV supplies the historical 2024 FX anchor, not a current dealing rate. CBI fiscal-agent description and Injaz issue example support sovereign-bond context. China Exim export buyer credit supports the instrument concept; numeric terms and eligibility are unconfirmed assumptions.

Recalculation: conditional phased opening

The earlier USD 9.890 billion result combined whole-fleet stage batching, no fares before the final capital payment and reserve deposits always funded with additional public cash. Ready-task dispatch now pipelines work within the same resource limits, gates train production on the new plant, and funds reserves from available project cash before seeking support. The full-network-only result above remains a conservative comparator.

With line openings tied to completed line and shared/depot work plus a 3-month commissioning allowance, first revenue begins in month 40, and all nine lines operate from month 81 after financial close. These dates are conditional planning milestones, not authorisation to run trains.

Each line's controlled trainset share allocates revenue and variable OPEX; its own 50% / 75% / 100% ramp applies. From first opening, 25% of full-network OPEX is fixed and 75% scales with opened fleet. This proxy can misstate early demand, central staffing and transfer benefits; a surveyed phase-specific operating plan is required. The same nominal fares, capital total, debt terms and 25% grant apply in every case.



Phased operating cash and additional liquidity

Opening / demand scenario	Additional funding beyond 25% capital, USD bn	Conditional lifetime public cash, USD bn
All fares after final capital payment	4.013	6.001
Phased: commissioning delay two years	4.192	6.180
Phased: demand minus 40 percent	7.076	9.063
Phased: low demand	3.205	5.193

Line	Planned opening month	Fleet / variable-cost share
line-1	40	10.63%
line-10	41	1.85%
line-11	43	1.73%
line-12	42	1.85%
line-13	44	2.22%
line-14	44	1.61%
line-2	50	11.74%
line-3	55	11.12%
line-4	59	9.39%
line-5	64	10.63%
line-6	69	11.74%
line-7	73	8.90%
line-8	78	11.00%
line-9	81	5.56%

All these are gross nominal liquidity contributions, conditional on funding. Later operating surplus is retained and is not netted against earlier required injections. The phased city ledger ends with USD 7.878 billion of unrestricted cash; no distribution or return to the sponsor is assumed. The plant ledger has no manufacturing income/OPEX: it remains a capital-financing allowance. A 25% direct grant also does not cap sovereign IQD bond liabilities or guarantees.

Phased programme monthly cashflow · phased programme annual cashflow

Comparison with the historical 148 km third-party proposal

The July 2024 report, citing the prime minister's adviser, gives USD 18 billion, 148 km, seven lines and 64 stations. The NIC procurement notice describes DBOMFT and requires bidders to submit a funding plan. Neither establishes a final all-USD financing contract. The 100% USD foreign-loan/government-cash comparator below is the user's requested scenario, not a verified split of debt and grants. The older USD estimate has not been restated to a common price date.

Measure	Current OpenSourceRail Baghdad plan	Historical proposal / requested USD scenario
Route km (not track km)	513.3	148
Lines / stations	14 / 159	7 / 64
Capital uses, USD equivalent bn	7.951, including one plant	18.000, reported estimate
Capital intensity, USD equivalent m / route km	15.49	121.62
USD-denominated capital funding, USD bn	1.798, half loan / half government cash	18.000, assumed USD cash + loans
IQD-denominated capital funding, IQD tn	8.000	Zero in requested scenario
Imported-purchase FX exposure, USD bn	1.798	18.000 only under all-USD purchase assumption
USD purchase intensity, USD m / route km	3.50	121.62 under that assumption
Contracted fares / debt terms	Planning fare and uncommitted facilities	Not established by these sources

The OSR plan has 3.47 times the route length and 2.48 times the stations. Its planning capital estimate is 55.8% below USD 18 billion. Under the requested all-USD funding scenario, USD-denominated capital funding is 90.0% lower; under an all-USD purchase scenario, imported-purchase exposure is 90.0% lower. These measure different exposures and are not interchangeable debt-service savings.

The planning comparison is not a like-for-like qualified bid: route geometry, tunnelling/structures, land, utilities, taxes/duties, contingency, escalation, supplier qualification and acceptance maturity differ or remain unresolved. The older report's four-year completion expectation also differs substantially from OSR's current 82-month capital schedule. A lower capital estimate does not establish an earlier or more sustainable delivered service.

Fares, population access and financial sustainability

The modelled average paid-trip fare is IQD 1,317 (USD 1.01 comparison equivalent), denominated in IQD rather than automatically indexed to the USD loan. It derives from the retained USD 380/month income proxy, equivalent to IQD 494,000/month at the planning anchor. Thirty paid trips consume 8.0% of that proxy income; 44 commuter trips cost IQD 57,963, or 11.7%. This is an average-trip yield assumption, not an adopted tariff or unlimited monthly pass; concessions, transfers and family affordability need an explicit tariff and household survey.

Low/high capacity-use cases assume 1,807,920 / 2,892,672 paid trips/day, not unique people or surveyed demand. Low-case steady annual revenue is IQD 908.0 billion; OPEX is IQD 266.5 billion. At that trip volume, the OPEX-only neutral fare is IQD 345, with nonfare receipts held constant. It excludes capital, debt and reserve funding.

In the first complete operating year, the 50% revenue ramp produces 329.9 million paid trips. Holding those trips and nonfare receipts fixed, covering OPEX, scheduled debt service and fees requires approximately IQD 3,417 per paid trip, excluding reserve deposits and factory debt. This is a cash threshold, not a recommended tariff; higher fares can reduce demand. Minimum annual operating DSCR before support is 0.04. Higher utilisation improves later cashflow but cannot fund the long pre-opening debt-service period. Neither this fare nor IQD denomination closes the additional funding gap shown above.

The design retains a planning population of 9,780,429. The legacy 40.5% routing-demand cell fraction is not a population percentage; its former multiplication into a resident estimate is retired. See the native population-count and multi-hop transfer audit for radius sensitivities, date, bbox denominator and access limitations. The older report's 80% city-coverage ambition uses no published comparable denominator or access method; it cannot establish that either plan reaches more residents. The 159 station platforms and 14 routes offer a network to test, not proof of superior walking coverage.

Direct Iraqi labour and industrial benefits

Approximately USD 6.154 billion equivalent (77.4%) of programme procurement is assigned to local suppliers under the editable origin assumptions. This is potential local expenditure, not wages, GDP added or guaranteed Iraqi content. City civil works alone allocate USD 3.346 billion locally; rolling stock allocates USD 883.43 million locally, with plant local expenditure separately counted.

Local train assembly and fabrication create work in body modules, fit-out, wiring, coatings, systems integration, inspection and maintenance while bogies, batteries, windows and doors remain imported inputs requiring supplier and process qualification. Infrastructure work supports Iraqi concrete/precast production, civil erection, stations, utilities, solar installation and supervision. Tooling, training, process qualification and supplier access can leave reusable industrial capacity, shorter repair chains and retained skills after construction. Local employment and supplier income circulate in IQD and can generate Iraqi tax receipts; no multiplier or tax recovery is booked without evidence.

The operating allowance supports 3,422 indicative FTE with annual labour cost of IQD 50.29 billion. These are operating positions, not construction or manufacturing job counts. Construction employment requires validated work hours, productivity, wage rates, shift cover and local-content contracts; no fabricated job total is assigned. The third-party proposal may also use Iraqi civil labour, so its local share cannot be assumed zero.

A sustainable appraisal must demonstrate phased service before full-network completion, realistic paid demand and affordable tariffs, funded debt/interest/reserves within the public limit, placed IQD facilities, supplier and labour qualification,

lifecycle replacement funding and an accepted environmental/physical design. Current software and ledger checks establish planning consistency, not those outcomes.

Baghdad financing reconciliation and six-month placement programme

Repository source: `cities/catalogue/west-asia/Iraq/Baghdad/engineering/finance/FUNDING-RECONCILIATION.md`

> Original catalogue or earlier scope reference. The figures and policies below retain their original assumptions; they are not the latest Baghdad staffing, depot, procurement or funding basis. See the current Baghdad recalculation and current city summary. National/portfolio totals have not been repriced with that conditional Baghdad option.

Scope: Baghdad city and one Baghdad manufacturing plant. Samawah, Mosul and every other city are excluded. Month zero is an assumed financial close, not an approved date. All financing and additional revenues remain uncommitted.

The numbers that must add up

The original capital sources and uses reconcile. The confusion was combining upfront capital, repayment of that same capital debt, gross liquidity injections and later retained cash. The revised schedule also spreads city EPC overhead across direct works instead of paying the entire programme allowance against the initial five-day baseline-freeze task. Capital cost is unchanged. An independent implementation rebuilds every debt vintage, interest charge, principal payment and financing fee from native draws; it does not copy the reported support total.

Capital source	Native currency	Native amount, billion	USD equivalent, million
chinese export credit	USD	0.898824	898.824
government import cash	USD	0.898824	898.824
government local cash	IQD	1,415.697277	1,088.998
domestic bonds	IQD	4,938.025474	3,798.481
bank credit	IQD	1,646.008491	1,266.160
Capital uses / sources	Mixed	—	7,951.287

Government cash remains 25% of capital, with government USD import cash inside that contribution. Imports are financed 50% government USD cash / 50% Chinese USD loan. Every other loan and bond is IQD. An international green lender must provide confirmed IQD on-lending or a priced currency hedge to preserve that strategy. A green label does not supply additional money.

Lifetime operating / finance reconciliation	USD equivalent, million
Fare and existing nonfare receipts	21,638.485
OPEX	6,591.251
Repayment of originally drawn capital principal	5,963.465
Interest on original capital financing	4,367.991
Original financing fees	42.664
Net nominal lifetime liquidity deficit	-4,673.113
Gross extra cash required as bills fall due	3,193.940
Later unrestricted cash retained if those requirements are met	7,867.053

USD 3,193.940m gross injections minus USD 7,867.053m retained cash equals USD -4,673.113m net deficit. Capital purchases are already financed by their capital sources; principal repayments belong to the lifetime cash ledger and must not be added to the construction budget again. Net reserve deposits are zero at the end. This is a nominal identity, not discounted viability, and assumes city/plant cash can legally be pooled

A zero-interest, zero-fee revolving bridge is only a mathematical lower bound. Real gap funding must charge interest and fees on the borrowing needed to carry early deficits, repay itself from later surplus and show any terminal debt. Later surplus cannot be counted both as retained cash and as loan repayment.

Tickets, kiosk rents and advertising already included

Full-network steady annual receipts	IQD billion	USD equivalent, million
Passenger tickets	869.296	668.689
Station shops and kiosk leases	15.538	11.952
Advertising space	23.180	17.831
Total existing operating receipts	908.015	698.473

Tickets use 659.89m annual paid trips at IQD 1,317 average yield. This is assumed capacity use, not calibrated demand. Shops/kiosks use 37,232 m² at IQD 39,520/m²/month and 88% occupancy; advertising uses 6,572 boards at IQD 345,800/month and 85% occupancy. Rates derive from the retained income proxy, not Baghdad lease quotes. These are gross

rental/advertising receipts; dedicated concession costs, tenant demand and collection losses require appraisal.

All three receipts enter the existing operating cashflows and therefore already reduce the financing gap. The phased sensitivity scales the combined receipts by opened-fleet share and the separate 50% / 75% / 100% revenue ramp for each line. Retail and advertising demand may follow different ramps in practice. Proposed naming rights, telecom leases, parking, development proceeds or new commercial receipts must be genuinely additional and net of costs; existing rents cannot be booked again or sold upfront while also retained in future revenue.

What gap finance actually does

The illustrative maximum outstanding facility is IQD 13.000 trillion (USD 10.000bn equivalent). This is a size assumption, not proof that the Iraqi banking market can place it. All available surplus after core obligations and reserves is swept to its repayment. No refinancing or debt forgiveness is assumed at the final maturity.

Scenario	Gross new liquidity draws, IQD tn	Peak facility balance, IQD tn	Uncovered cash beyond facility, IQD tn	Terminal unpaid facility, IQD tn
unfunded reference	0.000	0.000	4.152	0.000
green label only	0.000	0.000	4.152	0.000
zero cost liquidity bound	4.152	4.152	0.000	0.000
commercial gap credit	13.000	13.000	7.067	13.000
concessional gap credit	5.318	5.317	0.000	0.000
blended candidate	3.574	3.384	0.000	0.000
green concessional only	4.433	4.428	0.000	0.000

Reference and label-only cases have no gap facility: their uncovered column is the required external cash contribution, not an approved government top-up. With a facility, uncovered cash excludes the separate terminal loan balance; both need a solution. Gross draws can exceed the maximum outstanding commitment when a revolving facility is repaid and redrawn. Conditional ledgers do not establish that an unfunded payment occurred.

Commercial gap credit assumes 8% interest and 1.0% draw fees. Concessional gap credit assumes 2% interest and 0.5% draw fees. Neither is a term sheet or an available GCF line. Funding old debt-service deficits is not automatically eligible green use of proceeds.

Explicit blended sensitivity

Candidate climate expenditure is USD 2,233.26m of electric rolling stock, solar and charging provision, including USD 1,367.13m assigned locally. Climate benefit, supplier qualification and green eligibility require independent review. Candidate green bonds replace only domestic bonds attributable to those expenditures; civil works and unrelated overhead are not automatically relabelled.

The blended sensitivity replaces that eligible bond portion with proposed IQD green bonds/sukuk at 4%, 48 months' grace and 240 months' amortisation, with 0.75% arrangement fees and 0.5%/year enhancement fees on outstanding principal. The lower rate depends on a concessional or guarantee structure; a green label alone retains the original 8% terms and changes no financial outcome.

Additional targets are a USD 25m climate capital grant translated into IQD receipts, replacing domestic borrowing against eligible local capital after month 24; USD 300m net development-rights proceeds in 6 six-month instalments from month 24; and USD 25m/year genuinely new net local receipts at full opening, scaled with the opened fleet. These are editable targets, not appraisals, GCF approvals, developer contracts or statutory levies. Existing rents, advertising and fares cannot be counted again; transaction/development costs and displaced future rents must be deducted. Public land contribution also needs economic valuation.

Solving the model rather than declaring the targets sufficient requires approximately USD 0.00m/year additional NET receipts (IQD 0.00bn/year at full opening) to eliminate both uncovered cash and the terminal facility under the other blended assumptions. This is a break-even funding requirement, not a forecast or a recommended passenger fare increase. Any covenant, inflation, FX or demand change can raise it.

Variable tickets, annual price changes and OPEX inflation

The following cases retain the same conditional blended capital/grant/rights/2% IQD facility assumptions. They change fares, OPEX and income separately. A 5% tariff increase with flat costs is an optimistic sensitivity; the paired 5% OPEX case is the requested inflation comparison. None is an adopted fare, a lender commitment or an Iraqi inflation forecast.

Indices compound in annual steps from financial close, including the years before first opening. No tickets are sold before service starts; the nominal launch price reflects elapsed inflation. Existing retail/advertising are held flat unless the case explicitly indexes those rents. New net income and rights targets remain fixed nominal amounts. CAPEX escalation and future

FX changes still require a separate full nominal appraisal.

Demand responds to fare relative to the indexed income proxy with an illustrative elasticity of -0.30, except the explicitly fixed-demand cases. Income growth is a separate assumption, not evidence that wages will keep pace. Discount-led demand growth is capped at 2.0 times the baseline, within the original practical capacity. There is no calibrated Baghdad peak-switching or household demand model.

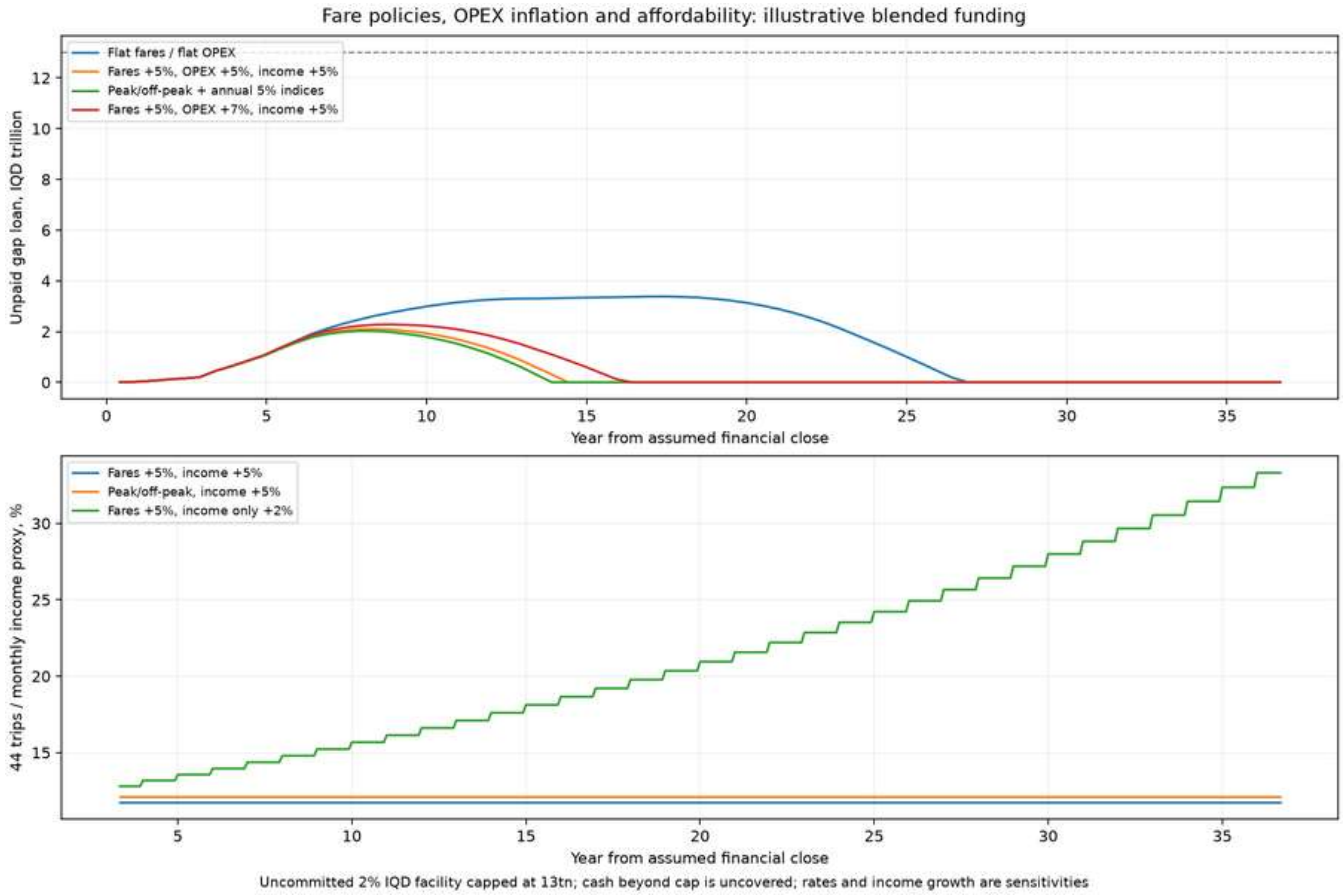
Variable tickets assume 40% of baseline trips in a peak tier at 1.25 times the standard fare, and remaining trips at 0.90 times it. Each tier gets its own demand response before revenues are combined. Concessions, commuter caps, integrated transfers and off-peak eligibility require an actual tariff design and funding contract; their costs are not silently waived.

Pricing sensitivity	Fare / OPEX / income growth	Full-opening average fare, IQD	44 trips / income at full opening	Peak facility, IQD tn	Uncovered cash, IQD tn	Terminal debt, IQD tn
fare 5pct flat costs fixed demand	5% / 0% / 0%	1,765	15.7%	1.807	0.000	0.000
fare 5pct flat costs elastic	5% / 0% / 0%	1,765	15.7%	2.037	0.000	0.000
fixed fare 5pct opex	0% / 5% / 5%	1,317	8.8%	13.000	0.486	13.000
fare 5pct opex 5pct	5% / 5% / 5%	1,765	11.7%	2.102	0.000	0.000
fare 5pct opex 7pct	5% / 7% / 5%	1,765	11.7%	2.283	0.000	0.000
fare 5pct opex 5pct income 2pct	5% / 5% / 2%	1,765	14.0%	2.302	0.000	0.000
variable fare 5pct opex 5pct	5% / 5% / 5%	1,822	12.1%	2.025	0.000	0.000
fare 5pct opex 5pct rents indexed	5% / 5% / 5%	1,765	11.7%	2.063	0.000	0.000

In the paired 5% case, the first line's average nominal fare is IQD 1,525 in month 40; at full opening in month 81 it is IQD 1,765. Its maximum supplemental facility is IQD 2.102tn, with IQD 0.000tn uncovered cash and IQD 0.000tn unpaid at the end. The case repays the facility within the model horizon under these conditional assumptions. Early borrowing still needs placed facilities.

At an 5% general-price assumption, the 8% real discount assumption becomes 13.4% nominal. The paired-case unlevered NPV is USD -1.670bn equivalent before grants/new rights/net-income targets and with capital still un-escalated. Large distant nominal balances are not present-value wealth or proof of project viability.

If income rises with fares, the modelled commuting share stays broadly constant. With income rising only 2%, the same ticket policy becomes progressively less affordable and reduces paid trips. If OPEX rises 7% against 5% fares, the tested financing again leaves uncovered requirements and terminal debt. Maintaining an affordable real tariff, collecting revenue and placing the required early IQD facility matter as much as the nominal annual increase.



Fare and OPEX inflation sensitivities

Paired 5% six-month funding requirements · paired monthly prices and affordability · variable-ticket funding requirements · variable monthly tickets

The initial real-fare multiplier required to remove both terminal debt and uncovered cash under paired 5% indices and the other blended assumptions is approximately 1.000 times the existing baseline. This is a conditional break-even sensitivity, not a tariff recommendation; it does not guarantee the facility can be placed or that households' incomes will grow.

Surplus cash and early repayment

These cases use identical paired 5% fare/OPEX/income assumptions and the same conditional blended capital sources. Each pays OPEX, scheduled principal, interest and fees, then funds the debt-service reserve and a three-month current-OPEX buffer before voluntary repayment. Borrowed gap proceeds and uncovered external cash cannot fund early payments. The extra buffer remains cash held at the terminal horizon and earns no interest.

Compare strategies against buffered gap-only, rather than attributing the buffer change to repayment savings. Loans-first pays bank credit, Chinese credit and gap credit before bonds. Cost priority pays bank (9%), ordinary bonds (8%), Chinese credit (5%), green bonds (4% plus 0.5% annual guarantee), then gap credit (2%). It is an interest-rate heuristic, not a proof of the globally best strategy.

Surplus strategy	All debt cleared, month from close	Finance cost saving vs buffered gap-only, USD equivalent m	Early-payment premiums, USD equivalent m	Peak gap debt, IQD tn	Terminal unrestricted cash, IQD tn
gap only buffered	369	0.000	0.000	2.208	44.706
gap then core	209	506.359	36.995	2.208	45.364
loans then bonds	207	653.848	42.809	2.158	45.556
cost priority	203	988.754	56.843	2.158	45.991
cost priority zero premium	203	1,060.010	0.000	2.156	46.084
cost priority noncallable bonds	369	260.705	12.395	2.158	45.045

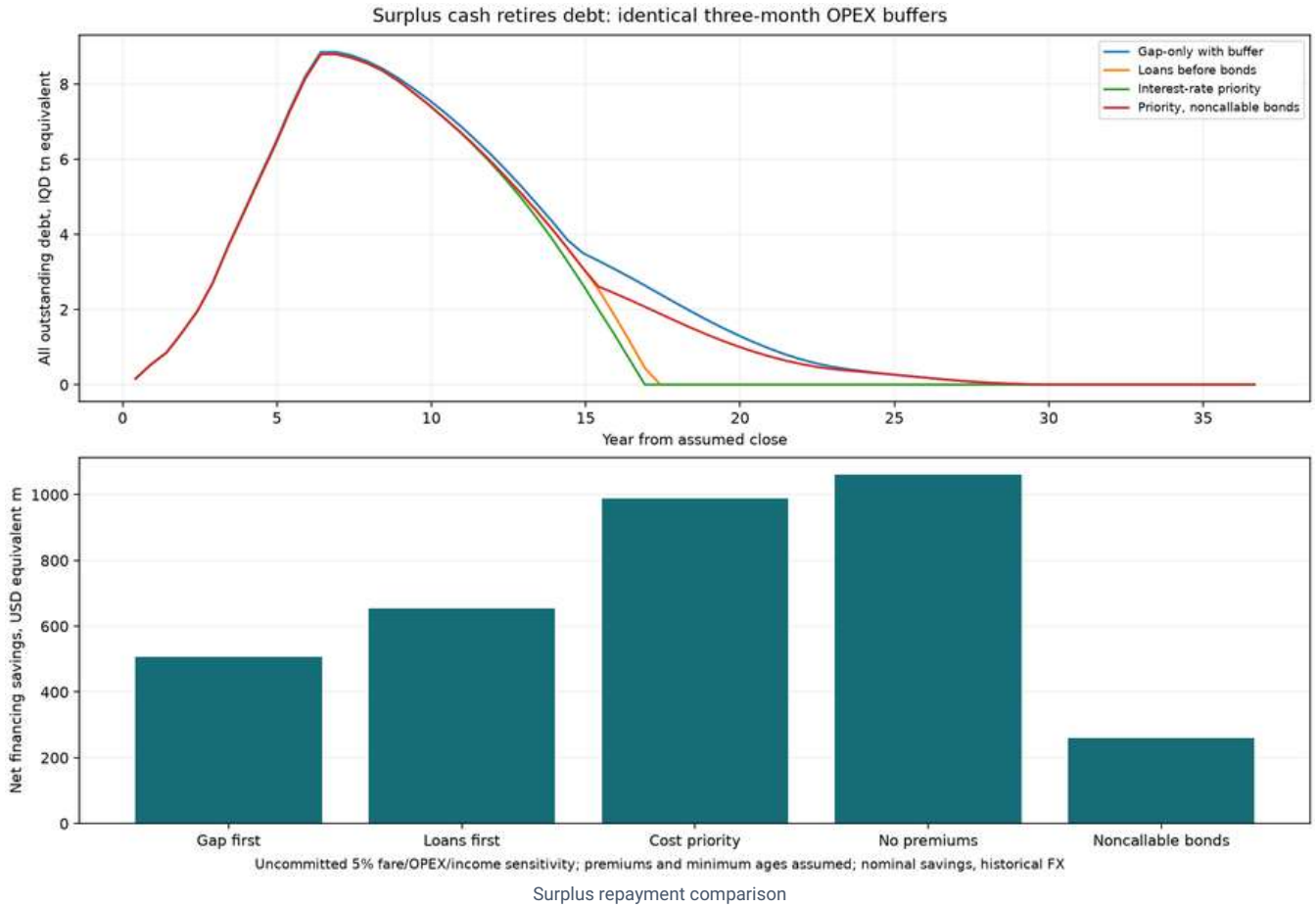
Cost-priority debt is cleared in month 203; buffered gap-only debt is cleared in month 369. Net nominal financing savings are USD 988.754m equivalent, after USD 56.843m assumed early-payment premiums. Savings include core interest, annual green guarantee charges and supplemental interest/draw fees; they exclude principal, which is returned once. All cases retain IQD 0.386tn operating buffer separately from unrestricted cash. There is no additional government contribution above 25% of CAPEX in these cases.

Assumed premiums are 1% of bank/Chinese/green principal and 2% of ordinary bond principal. A minimum draw age of 6 months (bank), 12 (Chinese) and 24 (both bonds) prevents immediate issue-and-redemption. Eligible vintages are repaid oldest first; contractual instalments are kept and maturity shortens. Notice, issuer call rights, investor consent, buyback price, remaining-maturity compensation, tax and FX require actual agreements. Noncallable-bond sensitivity makes no voluntary bond payments; zero-premium sensitivity removes only the assumed premium, retaining minimum ages.

Facility	Buffered gap-only final principal payment month	Cost-priority final principal payment month	Early principal, native currency	Premium, native currency
bank credit	165	122	IQD 636.771bn	IQD 6.368bn
domestic bonds	285	167	IQD 2,547.814bn	IQD 50.956bn
chinese export credit	309	175	USD 592.389m	USD 5.924m
green bonds	369	184	IQD 887.075bn	IQD 8.871bn

The gap facility's final principal payment is in month 203. Later surplus remains unrestricted cash after debt retirement. Each month's native debt balance equals prior balance plus draws minus scheduled and early principal. Each six-month closing balance is its final month's balance; payments, premiums, interest and reserve movements are period sums. Contractual repayment windows in tranche files describe the original draw terms; the actual final-payment months above incorporate early repayment.

Monthly cost-priority repayments · six-month repayment tranches · loans-first tranches · all six complete calculations

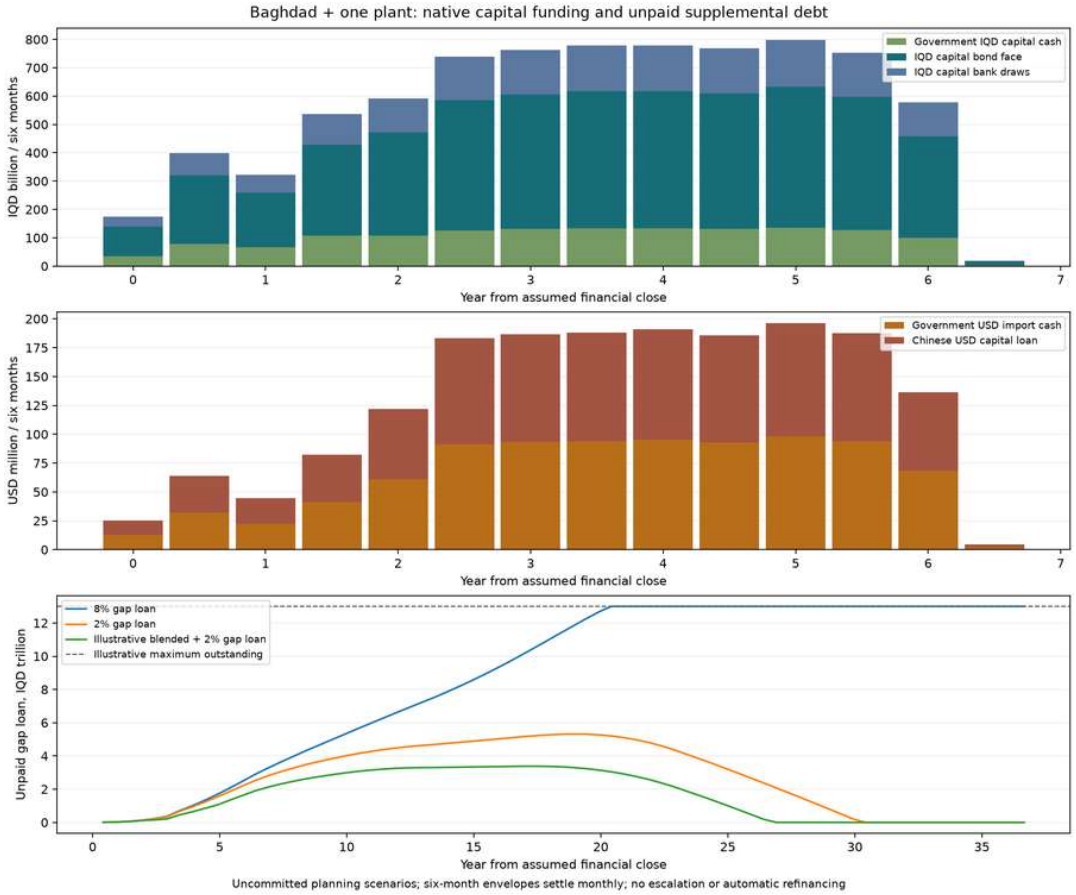


Surplus repayment comparison

Loan prepayment can carry redeployment/unwind charges: the World Bank Treasury FAQ illustrates the concept, without establishing Chinese or Iraqi loan terms. The US Treasury buyback FAQ and published purchase results illustrate issuer authority and priced buybacks rather than automatic par redemption; they do not supply an Iraqi legal mandate.

This repayment allocation does not change the unlevered project NPV or validate demand. Nominal savings over several decades are not present-value gains. CAPEX escalation, FX, lifecycle replacement, floating-rate risk and the uncommitted early funding still require appraisal.

Six-month bond sales and capital loan requirements



Native six-month capital funding and supplemental loan balances

Each half-year is a placement/draw envelope, with monthly settlement against capital milestones. Bonds assume par proceeds and an illustrative IQD 1,000,000 denomination. Face-value requirements below use unrounded values; unit counts in CSV round upwards and separately disclose the resulting indicative face amount. Sale discounts, issuance fees and investor capacity need actual bookbuilding. Selling every half-year amount upfront changes interest/carry and grace clocks and is not simulated here.

The government USD cash and Chinese USD loan are separate; government local cash, bonds, bank draws and uncovered liquidity are native IQD. Chinese facility commitment fees retain the existing annual procurement-cohort assumption; six-month envelopes do not assert a new signing schedule. Grace starts on each actual monthly settlement. First/last repayments for a half-year series therefore differ by draw date.

Tranche	Months from close	Government USD m	Government IQD bn	Chinese loan USD m	Bond face IQD bn	Bank loan IQD bn	Additional cash need IQD bn
BGD-H001	0–5	12.71	35.27	12.71	104.13	34.71	4.94
BGD-H002	6–11	32.04	79.08	32.04	240.40	80.13	16.96
BGD-H003	12–17	22.28	66.12	22.28	192.21	64.07	34.43
BGD-H004	18–23	41.20	107.37	41.20	321.91	107.30	56.92
BGD-H005	24–29	61.05	108.22	61.05	362.54	120.85	98.95
BGD-H006	30–35	91.69	125.17	91.69	460.44	153.48	121.11
BGD-H007	36–41	93.19	130.29	93.19	474.88	158.29	320.33
BGD-H008	42–47	94.06	133.32	94.06	483.38	161.13	256.29
BGD-H009	48–53	95.55	132.44	95.55	484.31	161.44	283.82
BGD-H010	54–59	92.92	131.79	92.92	477.73	159.24	296.42
BGD-H011	60–65	98.05	135.98	98.05	497.15	165.72	313.81
BGD-H012	66–71	93.72	127.41	93.72	469.42	156.47	310.71
BGD-H013	72–77	68.21	100.06	68.21	358.14	119.38	306.16
BGD-H014	78–83	2.14	3.20	2.14	11.38	3.79	244.86
Total capital envelopes	—	898.82	1415.70	898.82	4938.03	1646.01	—

Displayed two-decimal values can differ from full-precision totals by rounding. The six-month CSVs continue through the entire operating/repayment horizon, including periods with no new capital but interest, principal, liquidity draws, reserve movements and cash sweeps. Every tranche independently reconciles its capital sources and uses.

Reference six-month requirements · blended six-month requirements · commercial gap-credit six-month requirements · independent machine-readable calculation

Financing routes to investigate in Iraq

Route	What it can finance	Cashflow treatment and next requirement
IQD green bonds / green sukuk	Qualified electric transport, solar and charging capital	Replace conventional debt. Define eligible use, segregate proceeds, obtain independent review and report allocation/impact. Sukuk requires an approved asset/usufruct, legal and Sharia structure; the model does not assume a qualified sukuk issue.
GCF / GEF preparation and climate capital support	Climate studies, proven incremental mitigation/adaptation expenditure and eligible capital	Grants reduce debt only when approved; concessional debt still repays. Seek an accredited entity and Iraq NDA support with an emissions/resilience baseline. No grant is booked in the reference.
CBI renewable-energy initiative	Qualifying solar installations under the latest participating-bank rules	Obtain bank confirmation of borrower/project eligibility, current caps and terms. This existing Iraqi route is not assumed available for the whole metro or booked as a new capital source.
MDB / IFC credit enhancement and IQD syndication	Mobilising Iraqi banks and institutional bond investors	Guarantees improve placement or terms, supply no standalone cash, cost fees and may create sovereign contingent liabilities. Confirm IQD availability, hedge pricing and lender limits.
Station-area joint development and development rights	A contracted contribution towards stations/access or unrestricted net proceeds	Independently value parcels and rights, confirm title and revenue powers, tender transparently, deduct costs and future rent foregone. Early rights proceeds can reduce peak liquidity more than late rents.
Parking/congestion proceeds, employer transport subscriptions and business improvement charges	Incremental local recurring cash after authorised net collection costs	Separate each revenue stream from existing fare/nonfare receipts; validate legal powers, demand, affordability, exemptions and enforcement. No unverified tax power is assumed.
Solar/BESS project company, energy-service or availability contracts	Private financing of an accepted energy package	Remove transferred CAPEX and related city debt once; add the PPA/lease/service payments and guarantees. Compare their NPV with existing electricity/OPEX to avoid a free-CAPEX saving. No assumed saving is booked here.
Manufacturer / tooling leases and supplier finance	Qualified equipment and assembly capacity	Deferred invoices or leases are debt-like obligations with priced future payments. Do not duplicate Chinese invoice credit. Factory customer receipts and actual margins need a consolidated business case.
Diaspora and institutional IQD issues	Local capital subscription within the bond programme	These are purchaser channels for existing bond face, not another capital source. Disclose investor FX risk, maturity and redemption; no automatic rollover or compulsory pension purchase.
Carbon / results-based climate receipts	Verified additional emission reductions after delivery	Treat as contingent upside until methodology, monitoring, ownership and an offtake are contracted; no carbon revenue in the reference or blended case.

Priority: first align civil spending and production with accepted openings, then seek eligible cheaper/longer capital financing and contractual early rights proceeds. Obtain a funded recurring-revenue plan before borrowing a multi-trillion-dinar gap facility. Green finance and creative structures do not cure an unfunded timing mismatch or unsupported demand on their own.

Primary evidence checked 3 October 2026

The GCF Iraq country profile lists two projects with USD 49.5m total GCF finance and USD 5.2m approved readiness support when checked; these are existing country programmes, not Baghdad rail allocations. The Iraq country programme provides national priorities. An additional USD 25m application target is therefore a sensitivity requiring a new proposal, not money already available.

GCF investment framework and financial instruments support the grant/concessional/guarantee concepts and require viable revenue-generating activities for loans. ICMA Green Bond Principles require a transparent use-of-proceeds process, allocation controls and impact reporting; they do not promise a reduced coupon.

The CBI renewable-energy amendment of April 2025 establishes a local financing route to investigate for eligible energy packages. Its limits and bank eligibility cannot be extrapolated into a multi-trillion-dinar metro facility. IsDB green/sustainability sukuk guidance supplies a structuring reference, not a Baghdad issuance mandate.

World Bank guarantee programme supports foreign/local-currency obligations subject to project and sovereign approvals. IFC local-currency syndications explains domestic parallel loans and hedge structures; it does not establish an IQD loan offer for OSR. IFC Iraq investment activity establishes an existing country presence, not project eligibility or earmarked money.

World Bank land-value-capture guidance and innovative revenues provide structuring examples, not Iraq borrowing/tax authority or a Baghdad parcel valuation.

World Bank integrated-fare guidance supports examining peak/off-peak, transfer and integrated ticket structures. Its Bogotá affordability study illustrates the need for household-specific demand evidence; neither calibrates the -0.30 sensitivity for Baghdad or endorses a 5% annual increase.

Remaining modelling and term-sheet limits

- Six-month bond placement envelopes settle monthly at par; they are not advance-funded bond sales. Advance issuance changes carry, reserve and interest costs and needs a separate cash forecast.
- Full unrounded placement values reconcile; illustrative IQD 1 million unit counts round upwards and are not executable orders. No auction price, investor demand or mandate is established.
- A green label alone uses unchanged domestic pricing. The lower coupon/longer tenor and 0.5% annual guarantee fee are explicit sensitivities, not donor or lender terms.
- Climate capital grants replace domestic capital borrowing only, against candidate local expenditure. They do not pay old debt interest, imported invoices already funded 50:50, or increase government cash above 25%.
- Incremental land/rights/local receipts are zero in the reference and must be genuinely additional to existing fare, retail, rent and advertising forecasts. Net proceeds after costs, title, statutory powers and independent valuation require qualification.
- The supplemental IQD facility pays its own interest and origination fees. Surplus is swept to repayment; terminal balances are exposed rather than assumed refinanced or forgiven.
- The illustrative IQD facility has a 13 trillion maximum outstanding commitment; residual cash requirements beyond that cap remain explicitly uncovered. No committed lender or future market capacity is assumed.
- Supplemental facilities have no extra DSRA or undrawn commitment fee and no fixed amortisation before the terminal maturity; term sheets can materially increase costs. Long draw availability and the terminal balloon are uncommitted assumptions.
- City/plant cash is pooled subject to legally permitted transfers and creditor consent. Factory income/OPEX remain outside the capital-only plant appraisal; no free factory profit is credited.
- All non-Chinese debt is modelled in IQD. Foreign-funded concessional programmes need confirmed IQD on-lending or a priced hedge; an unhedged USD development loan would change the currency strategy.
- The reference has constant nominal fares, OPEX and FX. Dedicated pricing cases explicitly index fares, OPEX and income independently from financial close. Capital escalation, FX inflation pass-through and separate lifecycle replacement inflation remain outside this sensitivity; full nominal appraisal must include them.
- Pricing cases retain fixed nominal core and gap debt terms; supplier terms, floating rates and imported OPEX exposure need qualification. New net revenue targets are held nominal rather than assumed inflation-protected.
- Price elasticity is an uncalibrated response to fare relative to indexed income; no transfer, peak switching or low-income household demand survey is modelled. Off-peak trip increases are bounded by the same practical capacity, not new physical capacity.
- Early repayment is a contractual sensitivity, not an exercised call right. Bank/Chinese loan eligibility, bond call or voluntary buyback rights, notice, price and premiums require actual terms. The noncallable case forbids all early bond principal. Illustrative minimum draw ages are 6 months for bank credit, 12 for Chinese credit and 24 for bonds; these are not actual covenants.
- Cost priority is a coupon/guarantee-fee heuristic, not a globally optimal treasury strategy. Premiums, tax, remaining maturities, future draws and FX can change the best order; no refinancing or capital-issue cancellation is modelled.
- A three-month OPEX buffer is funded before discretionary repayment in every early-policy comparison. It is retained at the terminal horizon and earns no income. Original baseline cases have no such additional buffer; savings are compared with a buffered gap-only case as well as the original.

Editable options · existing capital assumptions · original city appraisal · Baghdad-only programme

Baghdad — Iraq funding appraisal

Repository source: `cities/catalogue/west-asia/Iraq/Baghdad/engineering/finance/FUNDING-MODEL.md`

> Original catalogue or earlier scope reference. The figures and policies below retain their original assumptions; they are not the latest Baghdad staffing, depot, procurement or funding basis. See the current Baghdad recalculation and current city summary. National/portfolio totals have not been repriced with that conditional Baghdad option.

Generated from current city CAPEX and procurement milestones. All facilities and appropriations remain uncommitted.

Government capital is 25% of total city CAPEX, before Chinese credit is deducted. The residual after that contribution and proposed Chinese proceeds is split 75% IQD bonds / 25% IQD bank credit. Imported purchases use 50% government USD cash and 50% proposed Chinese credit, assuming full-basket eligibility is later qualified; the remaining government contribution is IQD. Financing fees, construction interest, reserves and operating/debt support are additional funding requirements, not included in the 25% capital contribution. See the Baghdad-only programme for the uncovered cash requirements if no additional government cash is available.

Construction cash runs through month 82; full-network operations start in month 82 after retention. This conservative rollout follows the current resource-constrained CPM, not a five-year promise.

Capital sources and uses

USD is the comparison unit below. Chinese credit is USD debt; domestic bonds and bank credit are IQD debt. Baghdad imported purchases are funded 50% government USD cash and 50% proposed Chinese loan; the remaining government capital and local cash are IQD. The full Baghdad import basket is assumed eligible pending supplier-origin and lender qualification. Other city appraisals retain their own assumptions. Fares and local operating payments are budgeted in IQD.

Capital source	Contract / cash currency	Native amount at model FX	USD equivalent
chinese export credit	USD	866,880,047.75	866,880,047.75
domestic bonds	IQD	4,731,697,412,249.30	3,639,767,240.19
bank credit	IQD	1,577,232,470,749.77	1,213,255,746.73
government import cash	USD	866,880,047.75	866,880,047.75
government local cash	IQD	1,351,680,586,280.91	1,039,754,297.14
Total city capital uses	Mixed	—	7,626,537,379.57

Chinese buyer credit is proposed for eligible Chinese component invoices only. Government contributes its configured capital share, including the eligible-invoice downpayment; IQD bonds and bank credit finance the residual. The conditional ledger also calculates cash needed for fees, construction interest, reserves and operating/debt shortfalls. That additional support is uncommitted and is an unfunded requirement if Baghdad public cash is capped at its 25% capital contribution.

Proposed instruments

Instrument	Currency	Rate assumed	Grace from draw	Repayment
chinese export credit	USD	5.0%	48 months	180 months
domestic bonds	IQD	8.0%	24 months	180 months
bank credit	IQD	9.0%	0 months	84 months

Chinese component allocation

These allocations divide existing imported budgets; they are not additional costs, supplier quotes or confirmed origin.

Bucket	Component	Assumed eligible invoice USD
civil	other_imports_pending_lender_and_origin_qualification	590,556,608.25
stations	other_imports_pending_lender_and_origin_qualification	195,400,000.00
depots	other_imports_pending_lender_and_origin_qualification	2,000,000.00
rolling_stock	bogies	190,276,800.00
rolling_stock	traction_batteries	142,707,600.00
rolling_stock	windows	23,784,600.00
rolling_stock	doors	47,569,200.00
rolling_stock	other_imports_pending_lender_and_origin_qualification	71,353,800.00
solar_plant	solar_panels_and_inverters	367,083,177.55
signalling	other_imports_pending_lender_and_origin_qualification	12,833,386.00
charging_microgrid	solar_panels_and_inverters	8,176,000.00
charging_microgrid	station_batteries	9,344,000.00

Bucket	Component	Assumed eligible invoice USD
charging_microgrid	other_imports_pending_lender_and_origin_qualification	5,840,000.00
epc_overhead	other_imports_pending_lender_and_origin_qualification	66,834,923.70

Annual cash requirements

Year 1 begins at assumed financial close; NTP follows 30 working days later. Amounts below are USD millions. DSCR excludes subsidy; public cash includes capital, fees, interest, reserves and support.

Year	CAPEX	Revenue	OPEX	Debt service	Public cash	DSCR before support
1	319.74	0.00	0.00	10.17	91.75	0.00
2	674.06	0.00	0.00	47.54	219.55	0.00
3	1329.10	0.00	0.00	122.77	462.24	0.00
4	1560.09	0.00	0.00	242.13	640.60	0.00
5	1566.91	0.00	0.00	375.12	775.33	0.00
6	1577.50	0.00	0.00	518.61	921.53	0.00
7	599.13	58.21	34.17	642.28	1104.51	0.04
8	0.00	378.34	205.01	681.51	520.88	0.25
9	0.00	552.96	205.01	695.76	347.23	0.50
10	0.00	698.47	205.01	676.85	168.32	0.73
11	0.00	698.47	205.01	638.55	123.21	0.77
12	0.00	698.47	205.01	591.60	74.58	0.83
13	0.00	698.47	205.01	543.50	25.75	0.91
14	0.00	698.47	205.01	504.07	0.32	0.98
15	0.00	698.47	205.01	499.66	5.61	0.99
16	0.00	698.47	205.01	499.66	6.20	0.99
17	0.00	698.47	205.01	499.66	6.20	0.99
18	0.00	698.47	205.01	492.58	1.16	1.00
19	0.00	698.47	205.01	465.02	0.00	1.06
20	0.00	698.47	205.01	410.94	0.00	1.20
21	0.00	698.47	205.01	326.82	0.00	1.51
22	0.00	698.47	205.01	233.15	0.00	2.12
23	0.00	698.47	205.01	130.53	0.00	3.78
24	0.00	698.47	205.01	42.37	0.00	11.65
25	0.00	698.47	205.01	16.40	0.00	30.09
26	0.00	698.47	205.01	1.62	0.00	304.26
27	0.00	698.47	205.01	0.00	0.00	—
28	0.00	698.47	205.01	0.00	0.00	—
29	0.00	698.47	205.01	0.00	0.00	—
30	0.00	698.47	205.01	0.00	0.00	—
31	0.00	698.47	205.01	0.00	0.00	—
32	0.00	698.47	205.01	0.00	0.00	—
33	0.00	698.47	205.01	0.00	0.00	—
34	0.00	698.47	205.01	0.00	0.00	—
35	0.00	698.47	205.01	0.00	0.00	—
36	0.00	698.47	205.01	0.00	0.00	—
37	0.00	582.06	170.84	0.00	0.00	—

Sensitivities

Scenario	Peak annual public cash USD m	Minimum operating DSCR	Peak uncovered monthly capital USD m
base low demand	1104.51	0.04	0.00
high capacity use	1071.07	0.09	0.00
capex plus 25 percent	1386.64	0.03	0.00
demand minus 40 percent	1127.79	0.00	0.00
iqd depreciation 35 percent	932.77	0.03	0.00
commissioning delay two years	1017.20	0.03	0.00
china credit unavailable	1028.52	0.04	18.14
four year bullet bonds	1862.45	0.02	0.00
government payment delay six months	1291.63	0.04	15.65
interest plus three points	1307.66	0.03	0.00
combined downside	1276.42	0.00	0.00

Assumptions and evidence

- Working-day milestones use 260 working days/year and 30 working days before NTP; holidays and an approved local calendar are pending.
- No signed Chinese loan, supplier-origin qualification, guarantee, bond mandate, bank facility, appropriation or tax/duty assessment.
- Chinese origin and lender eligibility remain unqualified proxies. Default advance is 85%; Baghdad assumes all imports can qualify for 50% USD loan / 50% government USD cash. Additional imported categories require lender and supplier approval; neither assumption is a verified lender rule.
- Domestic bonds assume a proposed MoF sovereign IQD programme; municipal borrowing powers and market demand are not assumed.
- Long amortising domestic bonds are an appraisal target; the four-year bullet sensitivity exposes redemption without automatic refinancing.
- Construction cohorts need separately approved facilities. A long rollout does not imply a lender offers decades of draw availability.
- Interest is paid on drawn native balances; mid-month draw convention, monthly repayment, fees paid by government; no interest capitalisation.
- All cash is nominal; no inflation/escalation in base. IQD revenue and OPEX are translated at scenario FX; imported energy/parts OPEX needs a separate currency survey.
- Unquoted CAPEX budgets remain nominal USD planning values. IQD tranches and appropriations convert at draw-date FX; no automatic local supplier price benefit from depreciation is assumed.
- Income used for labour and fares is the retained country-finance planning proxy, not a verified current Iraqi household median or an agreed wage/fare contract.
- Government payment delays and unavailable China credit produce explicit capital cash gaps; no committed bridge credit is assumed.
- Restricted DSRA targets six times current monthly service and uses available project cash before requesting support. Support cash is a required contribution, not a commitment; repayments and reserve balances are conditional on it. Baghdad programme caps government capital at 25% and separately discloses the additional unfunded requirement. Future-service covenant testing is pending.
- Battery renewal reserve remains inside existing rolling-stock maintenance OPEX; no second battery CAPEX is added.
- Demand is capacity-led, not a surveyed forecast. Passing reconciliation does not demonstrate affordability or bankability.

Long construction schedules can leave much of the debt already repaid by government before full-network opening. An operating DSCR above one therefore does not establish self-financing or remove the construction-period public cash burden. Uncovered capital gaps are conditional funding requirements, not actual cash receipts.

See editable assumptions, monthly cashflow, annual cashflow, and machine-readable model.

The IMF Article IV provides the historical FX anchor. CBI describes its role as fiscal agent for MoF bonds. China Exim describes export buyer credit; numeric project terms remain assumptions.

Conditional phased-opening sensitivity

The base above withholds all fares until every capital milestone. This separate sensitivity opens completed lines after the editable commissioning lag and gates every line on shared system/depot work. The city-sized Baghdad production plant is unavailable for its first 390 working days (18 months from NTP), followed by the scheduled first-article qualification gate. Opening also requires actual civil, train, depot, power and safety acceptance, which is not established by this calculation.

Revenue and variable OPEX use each line's share of controlled trainsets. Each line has its own 50% / 75% / 100% revenue ramp. Fixed OPEX is 25% of the full-network annual budget from first opening; remaining OPEX scales with opened fleet. These are explicit uncalibrated sensitivity assumptions, not a ridership survey or verified staffing plan.

First / last planned opening: month 40 / 81 from financial close (month zero). City additional funding requirement: USD 2,780.13 million. The plant remains separate. Future surplus is retained; gross additional support is not net lifetime loss or discounted cost.

Line	Opening month	Revenue / variable OPEX share
line-1	40	10.63%

Line	Opening month	Revenue / variable OPEX share
line-10	41	1.85%
line-11	43	1.73%
line-12	42	1.85%
line-13	44	2.22%
line-14	44	1.61%
line-2	50	11.74%
line-3	55	11.12%
line-4	59	9.39%
line-5	64	10.63%
line-6	69	11.74%
line-7	73	8.90%
line-8	78	11.00%
line-9	81	5.56%

Phased monthly cashflow · phased annual cashflow

Standalone Iraqi City Appraisals — Outside the Baghdad Programme

Samawah — Iraq funding appraisal

Repository source: `cities/catalogue/west-asia/Iraq/Samawah/engineering/finance/FUNDING-MODEL.md`

Generated from current city CAPEX and procurement milestones. All facilities and appropriations remain uncommitted.

This standalone city appraisal is outside the Baghdad-only funding programme.

Construction cash runs through month 41; full-network operations start in month 41 after retention. This conservative rollout follows the current resource-constrained CPM, not a five-year promise.

Capital sources and uses

USD is the comparison unit below. Chinese credit is USD debt; domestic bonds and bank credit are IQD debt. Baghdad imported purchases are funded 50% government USD cash and 50% proposed Chinese loan; the remaining government capital and local cash are IQD. The full Baghdad import basket is assumed eligible pending supplier-origin and lender qualification. Other city appraisals retain their own assumptions. Fares and local operating payments are budgeted in IQD.

Capital source	Contract / cash currency	Native amount at model FX	USD equivalent
chinese export credit	USD	51,727,804.18	51,727,804.18
domestic bonds	IQD	380,609,521,570.51	292,776,555.05
bank credit	IQD	126,869,840,523.50	97,592,185.02
government	IQD	761,219,043,141.01	585,553,110.11
Total city capital uses	Mixed	—	1,027,649,654.36

Chinese buyer credit is proposed for eligible Chinese component invoices only. Government contributes its configured capital share, including the eligible-invoice downpayment; IQD bonds and bank credit finance the residual. The conditional ledger also calculates cash needed for fees, construction interest, reserves and operating/debt shortfalls. That additional support is uncommitted and is an unfunded requirement if Baghdad public cash is capped at its 25% capital contribution.

Proposed instruments

Instrument	Currency	Rate assumed	Grace from draw	Repayment
chinese export credit	USD	5.0%	48 months	180 months
domestic bonds	IQD	8.0%	24 months	180 months
bank credit	IQD	9.0%	0 months	84 months

Chinese component allocation

These allocations divide existing imported budgets; they are not additional costs, supplier quotes or confirmed origin.

Bucket	Component	Assumed eligible invoice USD
rolling_stock	bogies	22,932,000.00
rolling_stock	traction_batteries	17,199,000.00
rolling_stock	windows	2,866,500.00
rolling_stock	doors	5,733,000.00
solar_plant	solar_panels_and_inverters	11,360,740.21
charging_microgrid	solar_panels_and_inverters	357,000.00
charging_microgrid	station_batteries	408,000.00

Annual cash requirements

Year 1 begins at assumed financial close; NTP follows 30 working days later. Amounts below are USD millions. DSCR excludes subsidy; public cash includes capital, fees, interest, reserves and support.

Year	CAPEX	Revenue	OPEX	Debt service	Public cash	DSCR before support
1	0.00	0.00	0.00	0.00	0.00	—
2	196.00	0.00	0.00	1.04	118.68	0.00
3	826.54	0.00	0.00	30.31	498.92	0.00

Year	CAPEX	Revenue	OPEX	Debt service	Public cash	DSCR before support
4	5.11	37.55	17.68	45.00	50.56	0.44
5	0.00	83.15	30.31	51.61	8.26	1.02
6	0.00	115.34	30.31	55.00	0.00	1.55
7	0.00	128.75	30.31	55.91	0.00	1.76
8	0.00	128.75	30.31	57.32	0.00	1.72
9	0.00	128.75	30.31	56.99	0.00	1.73
10	0.00	128.75	30.31	44.78	0.00	2.20
11	0.00	128.75	30.31	38.50	0.00	2.56
12	0.00	128.75	30.31	38.48	0.00	2.56
13	0.00	128.75	30.31	38.48	0.00	2.56
14	0.00	128.75	30.31	38.48	0.00	2.56
15	0.00	128.75	30.31	38.48	0.00	2.56
16	0.00	128.75	30.31	38.48	0.00	2.56
17	0.00	128.75	30.31	38.48	0.00	2.56
18	0.00	128.75	30.31	38.48	0.00	2.56
19	0.00	128.75	30.31	37.88	0.00	2.60
20	0.00	128.75	30.31	16.13	0.00	6.10
21	0.00	128.75	30.31	4.93	0.00	19.98
22	0.00	128.75	30.31	3.00	0.00	32.81
23	0.00	128.75	30.31	0.02	0.00	5385.20
24	0.00	128.75	30.31	0.00	0.00	—
25	0.00	128.75	30.31	0.00	0.00	—
26	0.00	128.75	30.31	0.00	0.00	—
27	0.00	128.75	30.31	0.00	0.00	—
28	0.00	128.75	30.31	0.00	0.00	—
29	0.00	128.75	30.31	0.00	0.00	—
30	0.00	128.75	30.31	0.00	0.00	—
31	0.00	128.75	30.31	0.00	0.00	—
32	0.00	128.75	30.31	0.00	0.00	—
33	0.00	128.75	30.31	0.00	0.00	—
34	0.00	53.65	12.63	0.00	0.00	—

Sensitivities

Scenario	Peak annual public cash USD m	Minimum operating DSCR	Peak uncovered monthly capital USD m
base low demand	498.92	0.44	0.00
high capacity use	498.92	0.92	0.00
capex plus 25 percent	623.65	0.35	0.00
demand minus 40 percent	498.92	0.11	0.00
iqd depreciation 35 percent	498.92	0.44	0.00
commissioning delay two years	498.92	0.36	0.00
china credit unavailable	497.18	0.47	8.84
four year bullet bonds	498.92	0.37	0.00
government payment delay six months	484.49	0.44	49.68
interest plus three points	507.04	0.35	0.00
combined downside	633.80	0.06	0.00

Assumptions and evidence

- Working-day milestones use 260 working days/year and 30 working days before NTP; holidays and an approved local calendar are pending.
- No signed Chinese loan, supplier-origin qualification, guarantee, bond mandate, bank facility, appropriation or tax/duty assessment.
- Chinese origin and lender eligibility remain unqualified proxies. Default advance is 85%; Baghdad assumes all imports can qualify for 50% USD loan / 50% government USD cash. Additional imported categories require lender and supplier approval; neither assumption is a verified lender rule.
- Domestic bonds assume a proposed MoF sovereign IQD programme; municipal borrowing powers and market demand are not assumed.
- Long amortising domestic bonds are an appraisal target; the four-year bullet sensitivity exposes redemption without automatic refinancing.

- Construction cohorts need separately approved facilities. A long rollout does not imply a lender offers decades of draw availability.
- Interest is paid on drawn native balances; mid-month draw convention, monthly repayment, fees paid by government; no interest capitalisation.
- All cash is nominal; no inflation/escalation in base. IQD revenue and OPEX are translated at scenario FX; imported energy/parts OPEX needs a separate currency survey.
- Unquoted CAPEX budgets remain nominal USD planning values. IQD tranches and appropriations convert at draw-date FX; no automatic local supplier price benefit from depreciation is assumed.
- Income used for labour and fares is the retained country-finance planning proxy, not a verified current Iraqi household median or an agreed wage/fare contract.
- Government payment delays and unavailable China credit produce explicit capital cash gaps; no committed bridge credit is assumed.
- Restricted DSRA targets six times current monthly service and uses available project cash before requesting support. Support cash is a required contribution, not a commitment; repayments and reserve balances are conditional on it. Baghdad programme caps government capital at 25% and separately discloses the additional unfunded requirement. Future-service covenant testing is pending.
- Battery renewal reserve remains inside existing rolling-stock maintenance OPEX; no second battery CAPEX is added.
- Demand is capacity-led, not a surveyed forecast. Passing reconciliation does not demonstrate affordability or bankability.

Long construction schedules can leave much of the debt already repaid by government before full-network opening. An operating DSCR above one therefore does not establish self-financing or remove the construction-period public cash burden. Uncovered capital gaps are conditional funding requirements, not actual cash receipts.

See editable assumptions, monthly cashflow, annual cashflow, and machine-readable model.

The IMF Article IV provides the historical FX anchor. CBI describes its role as fiscal agent for MoF bonds. China Exim describes export buyer credit; numeric project terms remain assumptions.

Mosul — Iraq funding appraisal

Repository source: `cities/catalogue/west-asia/Iraq/Mosul/engineering/finance/FUNDING-MODEL.md`

Generated from current city CAPEX and procurement milestones. All facilities and appropriations remain uncommitted.

This standalone city appraisal is outside the Baghdad-only funding programme.

Construction cash runs through month 41; full-network operations start in month 41 after retention. This conservative rollout follows the current resource-constrained CPM, not a five-year promise.

Capital sources and uses

USD is the comparison unit below. Chinese credit is USD debt; domestic bonds and bank credit are IQD debt. Baghdad imported purchases are funded 50% government USD cash and 50% proposed Chinese loan; the remaining government capital and local cash are IQD. The full Baghdad import basket is assumed eligible pending supplier-origin and lender qualification. Other city appraisals retain their own assumptions. Fares and local operating payments are budgeted in IQD.

Capital source	Contract / cash currency	Native amount at model FX	USD equivalent
chinese export credit	USD	162,437,544.94	162,437,544.94
domestic bonds	IQD	1,468,487,700,917.59	1,129,605,923.78
bank credit	IQD	489,495,900,305.86	376,535,307.93
government	IQD	2,936,975,401,835.19	2,259,211,847.57
Total city capital uses	Mixed	—	3,927,790,624.22

Chinese buyer credit is proposed for eligible Chinese component invoices only. Government contributes its configured capital share, including the eligible-invoice downpayment; IQD bonds and bank credit finance the residual. The conditional ledger also calculates cash needed for fees, construction interest, reserves and operating/debt shortfalls. That additional support is uncommitted and is an unfunded requirement if Baghdad public cash is capped at its 25% capital contribution.

Proposed instruments

Instrument	Currency	Rate assumed	Grace from draw	Repayment
chinese export credit	USD	5.0%	48 months	180 months
domestic bonds	IQD	8.0%	24 months	180 months
bank credit	IQD	9.0%	0 months	84 months

Chinese component allocation

These allocations divide existing imported budgets; they are not additional costs, supplier quotes or confirmed origin.

Bucket	Component	Assumed eligible invoice USD
rolling_stock	bogies	48,294,400.00
rolling_stock	traction_batteries	36,220,800.00
rolling_stock	windows	6,036,800.00
rolling_stock	doors	12,073,600.00
solar_plant	solar_panels_and_inverters	81,412,394.05
charging_microgrid	solar_panels_and_inverters	3,297,000.00
charging_microgrid	station_batteries	3,768,000.00

Annual cash requirements

Year 1 begins at assumed financial close; NTP follows 30 working days later. Amounts below are USD millions. DSCR excludes subsidy; public cash includes capital, fees, interest, reserves and support.

Year	CAPEX	Revenue	OPEX	Debt service	Public cash	DSCR before support
1	675.44	0.00	0.00	9.78	412.49	0.00
2	1475.01	0.00	0.00	59.83	928.67	0.00
3	1728.34	0.00	0.00	132.56	1104.67	0.00
4	49.00	86.59	57.72	184.50	278.59	0.16
5	0.00	191.74	98.96	200.95	116.93	0.46
6	0.00	265.96	98.96	211.29	45.63	0.79
7	0.00	296.89	98.96	214.13	18.76	0.92
8	0.00	296.89	98.96	213.69	10.83	0.93
9	0.00	296.89	98.96	192.40	0.00	1.03

Year	CAPEX	Revenue	OPEX	Debt service	Public cash	DSCR before support
10	0.00	296.89	98.96	162.47	0.00	1.22
11	0.00	296.89	98.96	145.10	0.00	1.36
12	0.00	296.89	98.96	144.96	0.00	1.37
13	0.00	296.89	98.96	144.96	0.00	1.37
14	0.00	296.89	98.96	144.96	0.00	1.37
15	0.00	296.89	98.96	144.96	0.00	1.37
16	0.00	296.89	98.96	144.96	0.00	1.37
17	0.00	296.89	98.96	144.96	0.00	1.37
18	0.00	296.89	98.96	137.95	0.00	1.43
19	0.00	296.89	98.96	99.95	0.00	1.98
20	0.00	296.89	98.96	46.57	0.00	4.25
21	0.00	296.89	98.96	13.54	0.00	14.61
22	0.00	296.89	98.96	7.45	0.00	26.58
23	0.00	296.89	98.96	0.06	0.00	3121.16
24	0.00	296.89	98.96	0.00	0.00	—
25	0.00	296.89	98.96	0.00	0.00	—
26	0.00	296.89	98.96	0.00	0.00	—
27	0.00	296.89	98.96	0.00	0.00	—
28	0.00	296.89	98.96	0.00	0.00	—
29	0.00	296.89	98.96	0.00	0.00	—
30	0.00	296.89	98.96	0.00	0.00	—
31	0.00	296.89	98.96	0.00	0.00	—
32	0.00	296.89	98.96	0.00	0.00	—
33	0.00	296.89	98.96	0.00	0.00	—
34	0.00	123.70	41.23	0.00	0.00	—

Sensitivities

Scenario	Peak annual public cash USD m	Minimum operating DSCR	Peak uncovered monthly capital USD m
base low demand	1104.67	0.16	0.00
high capacity use	1104.67	0.42	0.00
capex plus 25 percent	1380.84	0.13	0.00
demand minus 40 percent	1104.67	-0.03	0.00
iqd depreciation 35 percent	1104.67	0.16	0.00
commissioning delay two years	1104.67	0.14	0.00
china credit unavailable	1098.73	0.16	18.67
four year bullet bonds	1102.55	0.17	0.00
government payment delay six months	1070.37	0.16	43.71
interest plus three points	1138.72	0.13	0.00
combined downside	1423.40	-0.02	0.00

Assumptions and evidence

- Working-day milestones use 260 working days/year and 30 working days before NTP; holidays and an approved local calendar are pending.
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The IMF Article IV provides the historical FX anchor. CBI describes its role as fiscal agent for MoF bonds. China Exim describes export buyer credit; numeric project terms remain assumptions.

Generated City Models

City Model Directory

City models are grouped as cities/catalogue////. These generated city models are the source of truth for city-specific network, fleet, cost, and energy figures. RFCs and worked examples provide context or methods, not competing city plans. The book uses concise city briefs to avoid repeating the generated README boilerplate.

Region	Countries	Cities
central-africa	DR Congo (7)	7
east-africa	Angola (9), Kenya (15), Madagascar (1), Mozambique (10), Rwanda (3), Somalia (1), South Sudan (1), Tanzania (15), Uganda (12), Zambia (1)	68
latin-america	Bolivia (1), Ecuador (1), El Salvador (1)	3
north-africa	Morocco (12), Sudan (6), Tunisia (1)	19
south-africa	South Africa (5)	5
south-asia	Afghanistan (5), Bangladesh (10), India (17), Nepal (3), Pakistan (13), Sri Lanka (4)	52
southeast-asia	Cambodia (1), Indonesia (2), Laos (1), Myanmar (2), Philippines (1)	7
west-africa	Burkina Faso (1), Cameroon (10), Guinea (1), Mali (1), Niger (1), Nigeria (10), Senegal (1)	25
west-asia	Egypt (19), Iraq (18), Jordan (4), Lebanon (3), Palestine (3), Saudi Arabia (14), Syria (9), Yemen (9)	79

central-africa

DR Congo

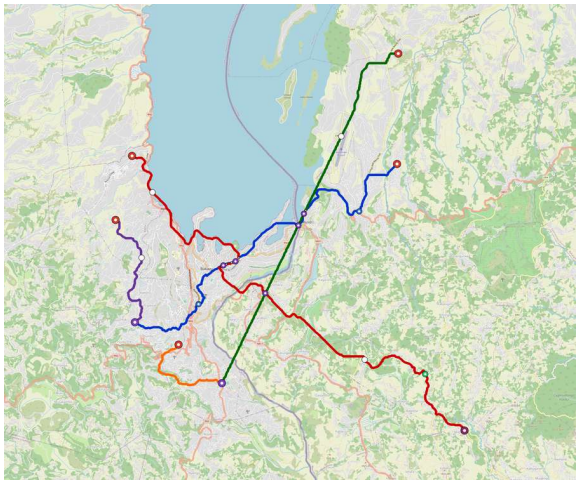
Generated City Model - Bukavu

central-africa / DR Congo / Bukavu

Metric	Value
Path	cities/catalogue/central-africa/DR Congo/Bukavu
Recorded ISO / planning population	CD / 1,000,000
Family	light-metro-3car
Lines / stations	5 / 41
Route length	72.4 km
Fleet	241 trainsets
Routing-demand cells (legacy)	60%; not population access
City CAPEX incl. dedicated solar	USD 1.46bn (USD 20M / km); shared national factory excluded
Residents within 800 m station circles	346,383 (2020; 32.4% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Bukavu network map

Line	Length	Stations	Trainsets	Shape
line-1	17.6 km	15	63	radial
line-2	26.6 km	12	84	radial
line-3	17.4 km	9	58	radial
line-4	6.0 km	3	20	radial
line-5	4.7 km	2	16	radial

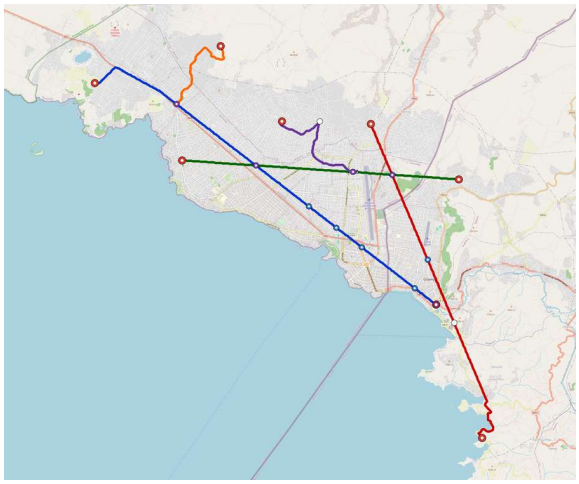
Generated City Model - Goma

central-africa / DR Congo / Goma

Metric	Value
Path	cities/catalogue/central-africa/DR Congo/Goma
Recorded ISO / planning population	CD / 1,000,000
Family	light-metro-3car
Lines / stations	5 / 27
Route length	57.2 km
Fleet	186 trainsets
Routing-demand cells (legacy)	42%; not population access
City CAPEX incl. dedicated solar	USD 1.04bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	170,000 (2020; 25.4% of raster bbox)
Direct / reachable line pairs	40.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Goma network map

Line	Length	Stations	Trainsets	Shape
line-1	19.4 km	8	60	radial
line-2	16.1 km	5	49	radial
line-3	11.4 km	7	38	radial
line-4	5.8 km	5	23	radial
line-5	4.5 km	2	16	radial

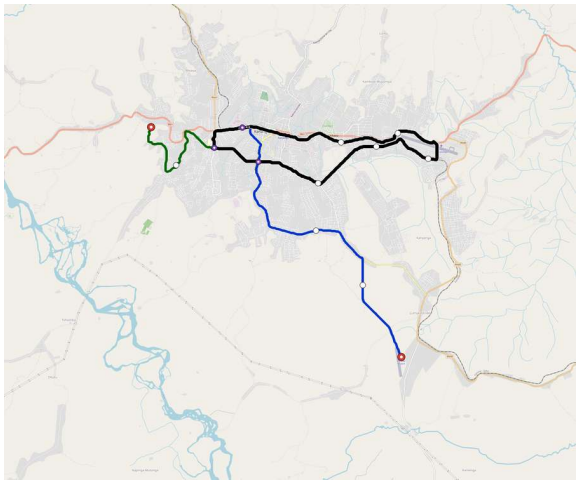
Generated City Model - Kananga

central-africa / DR Congo / Kananga

Metric	Value
Path	cities/catalogue/central-africa/DR Congo/Kananga
Recorded ISO / planning population	CD / 1,200,000
Family	metro-4car
Lines / stations	3 / 22
Route length	44.5 km
Fleet	55 trainsets
Routing-demand cells (legacy)	80%; not population access
City CAPEX incl. dedicated solar	USD 746M (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	210,531 (2020; 12.2% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Kananga network map

Line	Length	Stations	Trainsets	Shape
line-1	14.2 km	8	30	radial
line-2	24.1 km	11	12	ring
line-3	6.2 km	3	13	radial

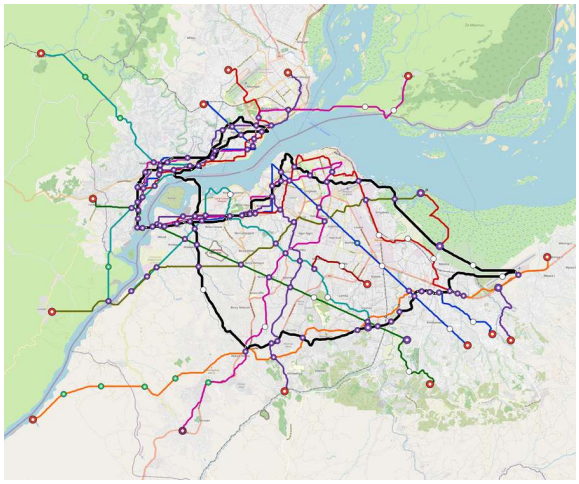
Generated City Model - Kinshasa

central-africa / DR Congo / Kinshasa

Metric	Value
Path	cities/catalogue/central-africa/DR Congo/Kinshasa
Recorded ISO / planning population	CD / 17,178,000
Family	metro-6car
Lines / stations	14 / 617
Route length	639.8 km
Fleet	1775 trainsets
Routing-demand cells (legacy)	62%; not population access
City CAPEX incl. dedicated solar	USD 16.87bn (USD 26M / km); shared national factory excluded
Residents within 800 m station circles	3,036,815 (2020; 40.4% of raster bbox)
Direct / reachable line pairs	46.2% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Kinshasa network map

Line	Length	Stations	Trainsets	Shape
line-1	62.6 km	85	265	radial
line-2	71.5 km	86	276	radial
line-3	31.8 km	29	101	radial
line-4	61.7 km	72	233	radial
line-5	55.1 km	60	193	radial
line-6	70.0 km	40	158	radial
line-7	91.8 km	87	294	radial
line-8	36.7 km	16	73	radial
line-9	125.5 km	128	107	ring
line-10	6.4 km	3	15	radial
line-11	6.5 km	3	15	radial
line-12	6.5 km	2	14	radial
line-13	7.4 km	4	18	radial
line-14	6.3 km	2	13	radial

Generated City Model - Kisangani

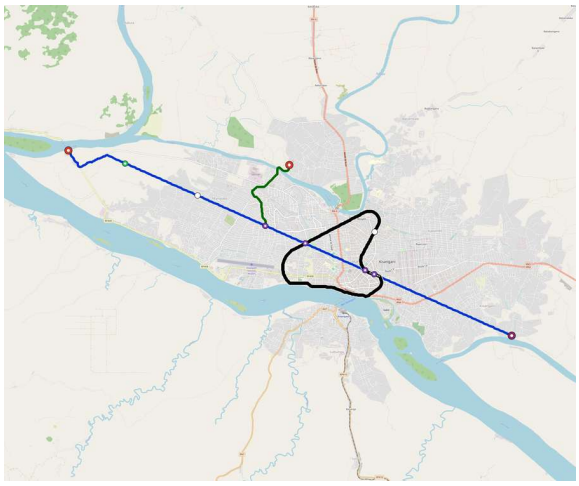
central-africa / DR Congo / Kisangani

Metric	Value
Path	cities/catalogue/central-africa/DR Congo/Kisangani

Metric	Value
Recorded ISO / planning population	CD / 1,300,000
Family	metro-4car
Lines / stations	3 / 24
Route length	40.0 km
Fleet	66 trainsets
Routing-demand cells (legacy)	48%; not population access
City CAPEX incl. dedicated solar	USD 752M (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	79,354 (2020; 9.5% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Kisangani network map

Line	Length	Stations	Trainsets	Shape
line-1	22.2 km	13	47	radial
line-2	13.8 km	9	9	ring
line-3	4.0 km	2	10	radial

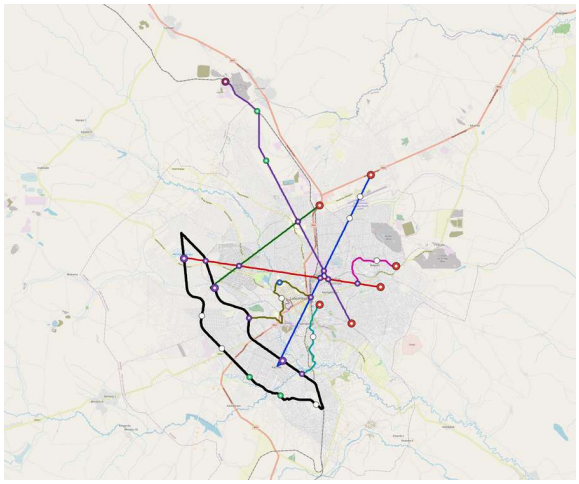
Generated City Model - Lubumbashi

central-africa / DR Congo / Lubumbashi

Metric	Value
Path	cities/catalogue/central-africa/DR Congo/Lubumbashi
Recorded ISO / planning population	CD / 2,829,000
Family	metro-4car
Lines / stations	8 / 50
Route length	140.3 km
Fleet	190 trainsets
Routing-demand cells (legacy)	58%; not population access
City CAPEX incl. dedicated solar	USD 2.21bn (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	654,559 (2020; 15.6% of raster bbox)
Direct / reachable line pairs	42.9% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Lubumbashi network map

Line	Length	Stations	Trainsets	Shape
line-1	19.5 km	8	35	radial
line-2	16.4 km	7	30	radial
line-3	12.5 km	5	23	radial
line-4	24.5 km	7	39	radial
line-5	43.8 km	13	18	ring
line-6	7.4 km	3	14	radial
line-7	5.9 km	3	13	radial
line-8	10.4 km	4	18	radial

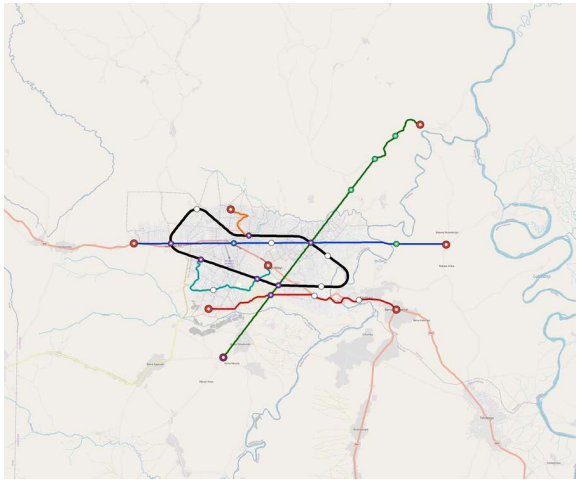
Generated City Model - Mbuji Mayi

central-africa / DR Congo / Mbuji-Mayi

Metric	Value
Path	cities/catalogue/central-africa/DR Congo/Mbuji-Mayi
Recorded ISO / planning population	CD / 2,500,000
Family	metro-4car
Lines / stations	6 / 40
Route length	122.8 km
Fleet	165 trainsets
Routing-demand cells (legacy)	62%; not population access
City CAPEX incl. dedicated solar	USD 1.93bn (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	1,162,457 (2020; 26.0% of raster bbox)
Direct / reachable line pairs	40.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Mbuji Mayi network map

Line	Length	Stations	Trainsets	Shape
line-1	25.1 km	7	41	radial
line-2	16.8 km	6	28	radial
line-3	29.3 km	10	50	radial
line-4	35.3 km	10	15	ring
line-5	4.4 km	2	10	radial
line-6	11.9 km	5	21	radial

east-africa

Angola

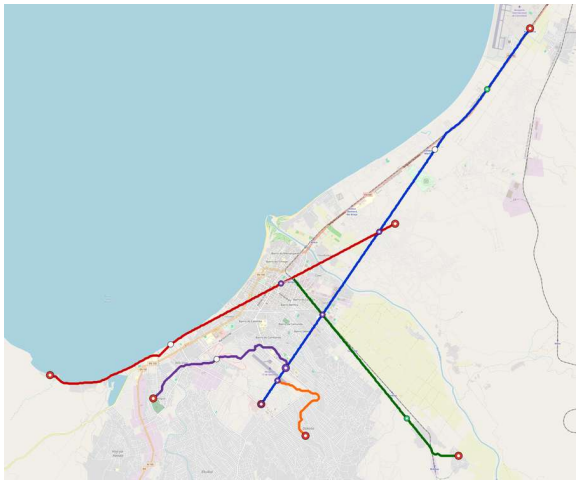
Generated City Model - Benguela

east-africa / Angola / Benguela

Metric	Value
Path	cities/catalogue/east-africa/Angola/Benguela
Recorded ISO / planning population	AO / 600,000
Family	light-metro-3car
Lines / stations	5 / 22
Route length	59.0 km
Fleet	185 trainsets
Routing-demand cells (legacy)	43%; not population access
City CAPEX incl. dedicated solar	USD 983M (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	92,975 (2020; 20.2% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Benguela network map

Line	Length	Stations	Trainsets	Shape
line-1	20.5 km	8	63	radial
line-2	16.6 km	5	49	radial
line-3	10.8 km	4	35	radial
line-4	7.0 km	3	24	radial
line-5	4.0 km	2	14	radial

Generated City Model - Huambo

east-africa / Angola / Huambo

Metric	Value
Path	cities/catalogue/east-africa/Angola/Huambo
Recorded ISO / planning population	AO / 800,000
Family	light-metro-3car
Lines / stations	5 / 22
Route length	62.3 km
Fleet	194 trainsets
Routing-demand cells (legacy)	71%; not population access
City CAPEX incl. dedicated solar	USD 1.09bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	52,809 (2020; 14.6% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

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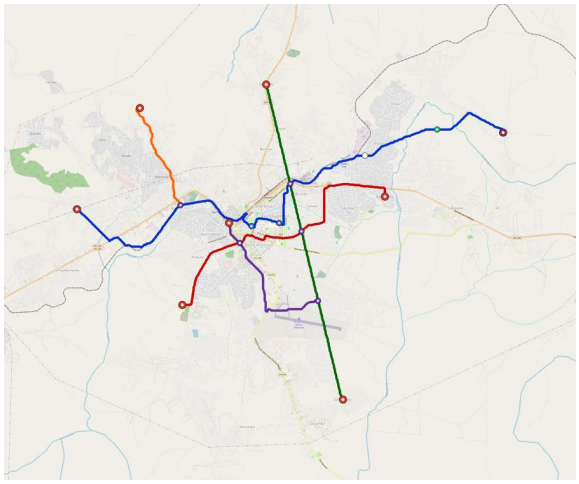


Huambo network map

Line	Length	Stations	Trainsets	Shape
line-1	25.4 km	8	76	radial
line-2	12.2 km	4	39	radial
line-3	13.6 km	5	43	radial
line-4	6.4 km	3	20	radial
line-5	4.7 km	2	16	radial

Generated City Model - Lobito

east-africa / Angola / Lobito

Metric	Value
Path	cities/catalogue/east-africa/Angola/Lobito
Recorded ISO / planning population	AO / 500,000
Family	light-metro-3car
Lines / stations	4 / 19
Route length	41.8 km
Fleet	133 trainsets
Routing-demand cells (legacy)	72%; not population access
City CAPEX incl. dedicated solar	USD 761M (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	47,834 (2020; 18.7% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.

Lobito network map

Line	Length	Stations	Trainsets	Shape
line-1	17.5 km	6	52	radial
line-2	10.0 km	6	34	radial
line-3	7.7 km	4	26	radial
line-4	6.7 km	3	21	radial

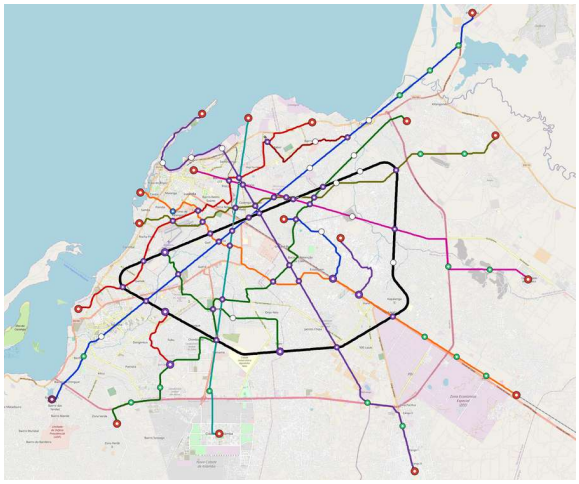
Generated City Model - Luanda

east-africa / Angola / Luanda

Metric	Value
Path	cities/catalogue/east-africa/Angola/Luanda
Recorded ISO / planning population	AO / 9,085,000
Family	metro-6car
Lines / stations	14 / 160
Route length	424.9 km
Fleet	726 trainsets
Routing-demand cells (legacy)	61%; not population access
City CAPEX incl. dedicated solar	USD 7.70bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	3,951,233 (2020; 25.4% of raster bbox)
Direct / reachable line pairs	47.3% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Luanda network map

Line	Length	Stations	Trainsets	Shape
line-1	52.4 km	18	103	radial
line-2	31.2 km	10	59	radial
line-3	46.5 km	18	84	radial
line-4	47.0 km	16	87	radial
line-5	45.2 km	16	84	radial
line-6	27.1 km	16	63	radial
line-7	31.7 km	13	59	radial
line-8	33.2 km	14	63	radial
line-9	65.8 km	21	31	ring
line-10	9.0 km	4	19	radial
line-11	8.0 km	4	18	radial
line-12	6.1 km	2	13	radial
line-13	15.2 km	6	29	radial
line-14	6.6 km	2	14	radial

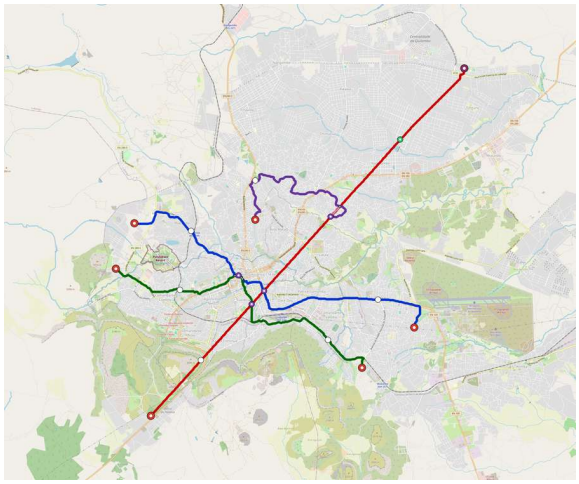
Generated City Model - Lubango

east-africa / Angola / Lubango

Metric	Value
Path	cities/catalogue/east-africa/Angola/Lubango
Recorded ISO / planning population	AO / 700,000
Family	light-metro-3car
Lines / stations	4 / 32
Route length	58.3 km
Fleet	188 trainsets
Routing-demand cells (legacy)	54%; not population access
City CAPEX incl. dedicated solar	USD 1.09bn (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	98,833 (2020; 17.6% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Lubango network map

Line	Length	Stations	Trainsets	Shape
line-1	15.8 km	10	53	radial
line-2	20.1 km	8	62	radial
line-3	14.0 km	10	46	radial
line-4	8.4 km	4	27	radial

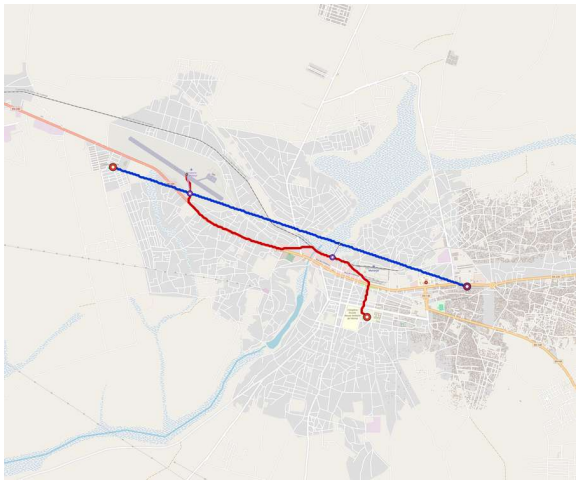
Generated City Model - Malanje

east-africa / Angola / Malanje

Metric	Value
Path	cities/catalogue/east-africa/Angola/Malanje
Recorded ISO / planning population	AO / 500,000
Family	light-metro-3car
Lines / stations	2 / 8
Route length	13.8 km
Fleet	46 trainsets
Routing-demand cells (legacy)	58%; not population access
City CAPEX incl. dedicated solar	USD 281M (USD 20M / km); shared national factory excluded
Residents within 800 m station circles	24,799 (2020; 8.0% of raster bbox)
Direct / reachable line pairs	100.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Malanje network map

Line	Length	Stations	Trainsets	Shape
line-1	8.1 km	4	27	radial
line-2	5.7 km	4	19	radial

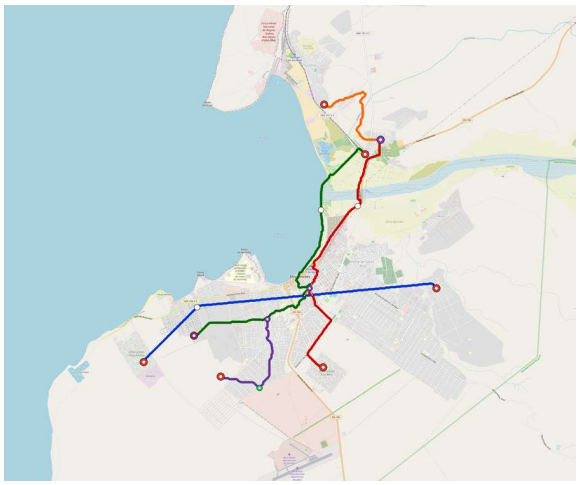
Generated City Model - Namibe

east-africa / Angola / Namibe

Metric	Value
Path	cities/catalogue/east-africa/Angola/Namibe
Recorded ISO / planning population	AO / 300,000
Family	tram-2car
Lines / stations	5 / 23
Route length	47.8 km
Fleet	103 trainsets
Routing-demand cells (legacy)	53%; not population access
City CAPEX incl. dedicated solar	USD 730M (USD 15M / km); shared national factory excluded
Residents within 800 m station circles	52,363 (2020; 24.9% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Namibe network map

Line	Length	Stations	Trainsets	Shape
line-1	12.8 km	5	26	radial
line-2	12.0 km	6	25	radial
line-3	13.5 km	7	29	radial
line-4	4.7 km	3	12	radial
line-5	4.8 km	2	11	radial

Generated City Model - Soyo

east-africa / Angola / Soyo

Metric	Value
Path	cities/catalogue/east-africa/Angola/Soyo
Recorded ISO / planning population	AO / 250,000
Family	tram-2car
Lines / stations	4 / 12
Route length	28.0 km
Fleet	64 trainsets
Routing-demand cells (legacy)	62%; not population access
City CAPEX incl. dedicated solar	USD 447M (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	20,836 (2020; 16.8% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.

Soyo network map

Line	Length	Stations	Trainsets	Shape
line-1	9.4 km	4	20	radial
line-2	3.9 km	3	12	radial
line-3	10.2 km	3	21	radial
line-4	4.5 km	2	11	radial

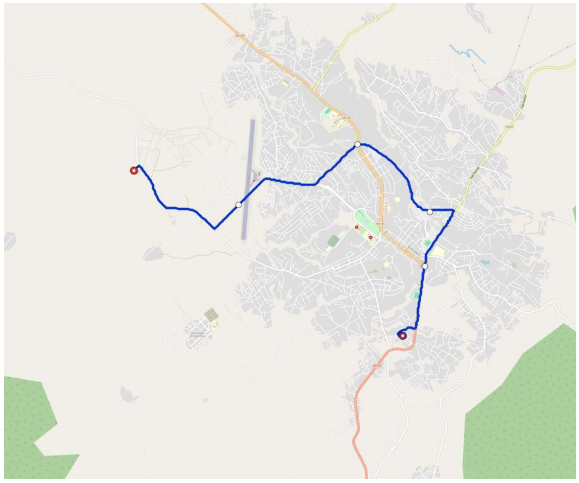
Generated City Model - Uige

east-africa / Angola / Uige

Metric	Value
Path	cities/catalogue/east-africa/Angola/Uige
Recorded ISO / planning population	AO / 400,000
Family	light-metro-3car
Lines / stations	1 / 6
Route length	12.0 km
Fleet	39 trainsets
Routing-demand cells (legacy)	47%; not population access
City CAPEX incl. dedicated solar	USD 212M (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	23,573 (2020; 9.7% of raster bbox)
Direct / reachable line pairs	100.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Uige network map

Line	Length	Stations	Trainsets	Shape
line-1	12.0 km	6	39	radial

Kenya

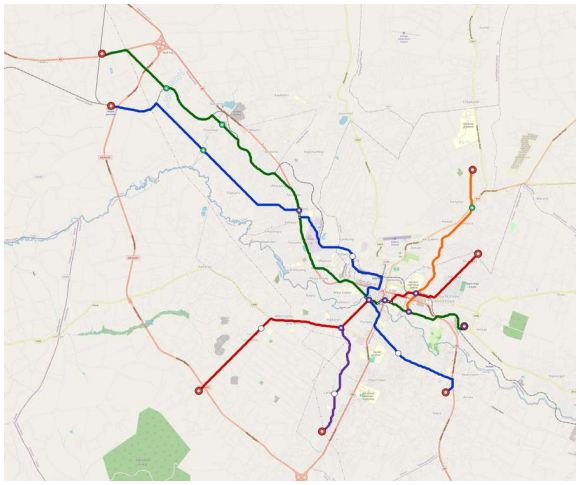
Generated City Model - Eldoret

east-africa / Kenya / Eldoret

Metric	Value
Path	cities/catalogue/east-africa/Kenya/Eldoret
Recorded ISO / planning population	KE / 500,000
Family	light-metro-3car
Lines / stations	5 / 51
Route length	74.6 km
Fleet	261 trainsets
Routing-demand cells (legacy)	79%; not population access
City CAPEX incl. dedicated solar	USD 1.45bn (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	170,799 (2020; 32.4% of raster bbox)
Direct / reachable line pairs	60.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Eldoret network map

Line	Length	Stations	Trainsets	Shape
line-1	24.1 km	9	76	radial
line-2	14.3 km	15	60	radial
line-3	24.2 km	20	83	radial
line-4	4.9 km	3	17	radial

Line	Length	Stations	Trainsets	Shape
line-5	7.1 km	4	25	radial

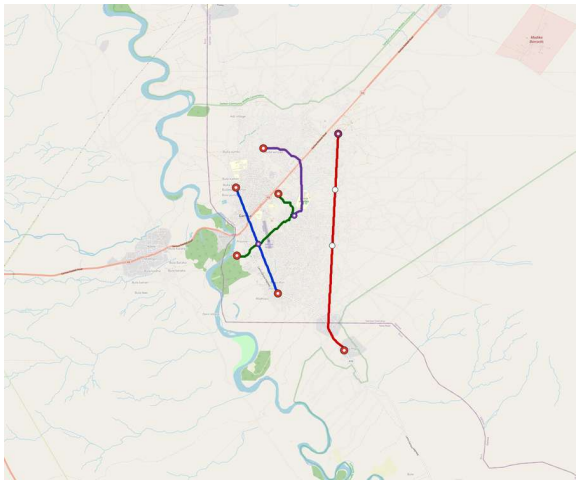
Generated City Model - Garissa

east-africa / Kenya / Garissa

Metric	Value
Path	cities/catalogue/east-africa/Kenya/Garissa
Recorded ISO / planning population	KE / 300,000
Family	tram-2car
Lines / stations	4 / 13
Route length	23.0 km
Fleet	57 trainsets
Routing-demand cells (legacy)	69%; not population access
City CAPEX incl. dedicated solar	USD 395M (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	131,566 (2020; 42.4% of raster bbox)
Direct / reachable line pairs	33.3% / 50.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Garissa network map

Line	Length	Stations	Trainsets	Shape
line-1	5.0 km	3	13	radial
line-2	9.2 km	4	20	radial
line-3	4.7 km	4	14	radial
line-4	4.1 km	2	10	radial

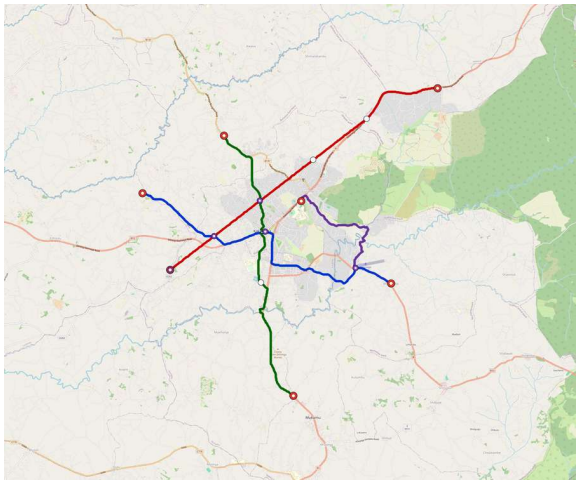
Generated City Model - Kakamega

east-africa / Kenya / Kakamega

Metric	Value
Path	cities/catalogue/east-africa/Kenya/Kakamega
Recorded ISO / planning population	KE / 300,000
Family	tram-2car
Lines / stations	4 / 20
Route length	47.5 km
Fleet	98 trainsets
Routing-demand cells (legacy)	72%; not population access
City CAPEX incl. dedicated solar	USD 754M (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	73,061 (2020; 25.2% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Kakamega network map

Line	Length	Stations	Trainsets	Shape
line-1	13.6 km	6	29	radial
line-2	14.7 km	6	30	radial
line-3	13.2 km	6	26	radial
line-4	5.9 km	2	13	radial

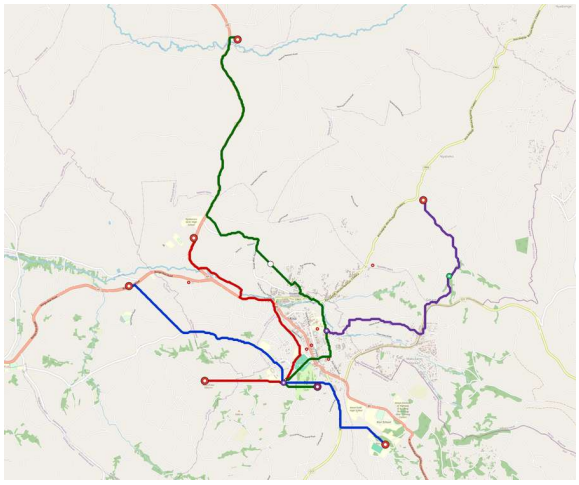
Generated City Model - Kisii

east-africa / Kenya / Kisii

Metric	Value
Path	cities/catalogue/east-africa/Kenya/Kisii
Recorded ISO / planning population	KE / 300,000
Family	tram-2car
Lines / stations	4 / 16
Route length	30.2 km
Fleet	69 trainsets
Routing-demand cells (legacy)	96%; not population access
City CAPEX incl. dedicated solar	USD 512M (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	67,243 (2020; 19.0% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Kisii network map

Line	Length	Stations	Trainsets	Shape
line-1	7.1 km	3	16	radial
line-2	6.3 km	4	15	radial

Line	Length	Stations	Trainsets	Shape
line-3	10.9 km	6	24	radial
line-4	5.9 km	3	14	radial

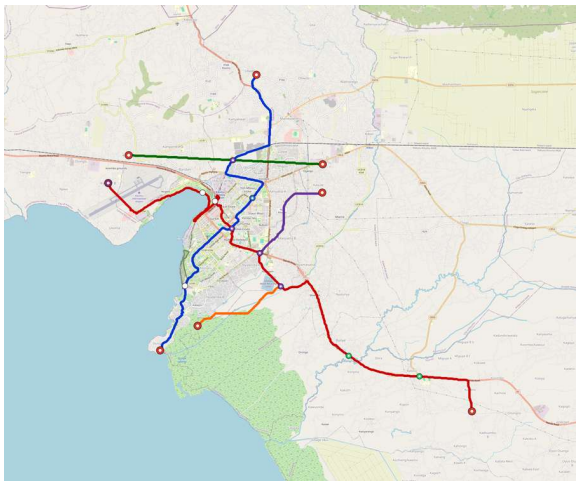
Generated City Model - Kisumu

east-africa / Kenya / Kisumu

Metric	Value
Path	cities/catalogue/east-africa/Kenya/Kisumu
Recorded ISO / planning population	KE / 600,000
Family	light-metro-3car
Lines / stations	5 / 22
Route length	60.8 km
Fleet	191 trainsets
Routing-demand cells (legacy)	72%; not population access
City CAPEX incl. dedicated solar	USD 1.08bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	120,752 (2020; 21.9% of raster bbox)
Direct / reachable line pairs	40.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Kisumu network map

Line	Length	Stations	Trainsets	Shape
line-1	16.9 km	6	52	radial
line-2	27.7 km	9	85	radial
line-3	7.9 km	3	26	radial
line-4	4.1 km	2	14	radial
line-5	4.1 km	2	14	radial

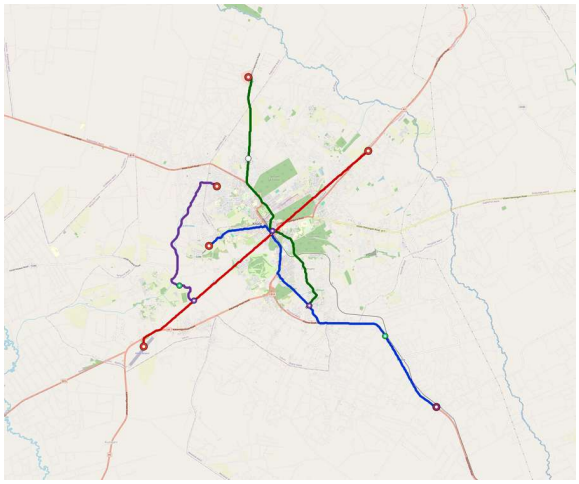
Generated City Model - Kitale

east-africa / Kenya / Kitale

Metric	Value
Path	cities/catalogue/east-africa/Kenya/Kitale
Recorded ISO / planning population	KE / 300,000
Family	tram-2car
Lines / stations	4 / 20
Route length	46.5 km
Fleet	99 trainsets
Routing-demand cells (legacy)	79%; not population access
City CAPEX incl. dedicated solar	USD 727M (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	82,400 (2020; 27.8% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Kitale network map

Line	Length	Stations	Trainsets	Shape
line-1	14.6 km	6	31	radial
line-2	13.4 km	5	27	radial
line-3	11.4 km	6	25	radial
line-4	7.1 km	3	16	radial

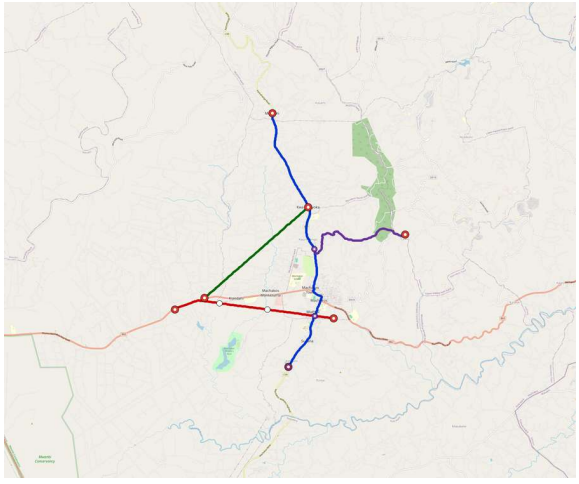
Generated City Model - Machakos

east-africa / Kenya / Machakos

Metric	Value
Path	cities/catalogue/east-africa/Kenya/Machakos
Recorded ISO / planning population	KE / 300,000
Family	tram-2car
Lines / stations	4 / 13
Route length	30.3 km
Fleet	68 trainsets
Routing-demand cells (legacy)	67%; not population access
City CAPEX incl. dedicated solar	USD 480M (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	44,563 (2020; 25.9% of raster bbox)
Direct / reachable line pairs	33.3% / 50.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Machakos network map

Line	Length	Stations	Trainsets	Shape
line-1	12.0 km	4	25	radial
line-2	6.8 km	5	18	radial
line-3	6.2 km	2	13	radial
line-4	5.3 km	2	12	radial

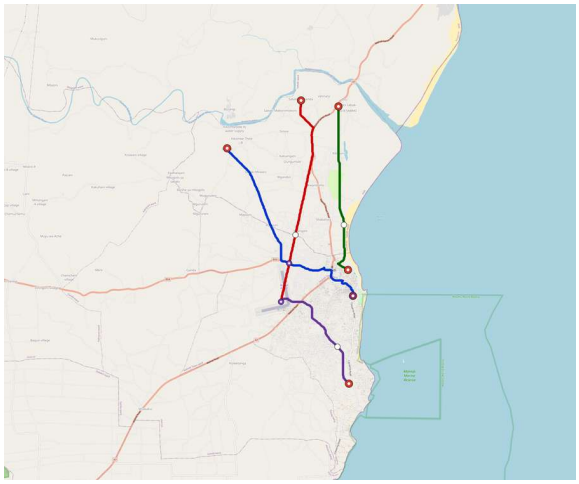
Generated City Model - Malindi

east-africa / Kenya / Malindi

Metric	Value
Path	cities/catalogue/east-africa/Kenya/Malindi
Recorded ISO / planning population	KE / 300,000
Family	tram-2car
Lines / stations	4 / 14
Route length	31.3 km
Fleet	68 trainsets
Routing-demand cells (legacy)	66%; not population access
City CAPEX incl. dedicated solar	USD 497M (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	69,731 (2020; 36.0% of raster bbox)
Direct / reachable line pairs	33.3% / 50.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Malindi network map

Line	Length	Stations	Trainsets	Shape
line-1	10.0 km	3	19	radial
line-2	9.0 km	5	20	radial
line-3	7.1 km	3	16	radial
line-4	5.2 km	3	13	radial

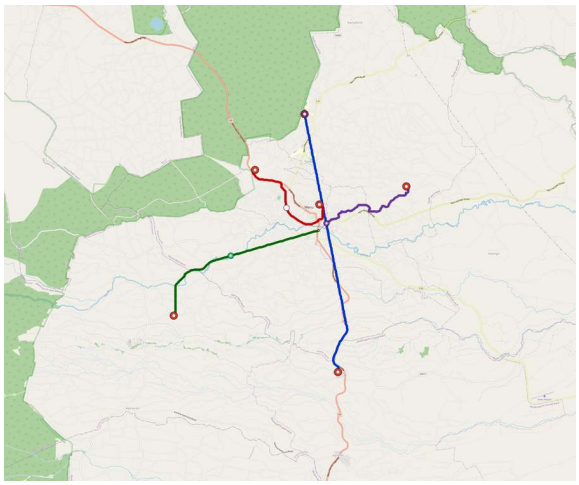
Generated City Model - Meru Ke

east-africa / Kenya / Meru-Ke

Metric	Value
Path	cities/catalogue/east-africa/Kenya/Meru-Ke
Recorded ISO / planning population	KE / 250,000
Family	tram-2car
Lines / stations	4 / 11
Route length	29.7 km
Fleet	65 trainsets
Routing-demand cells (legacy)	83%; not population access
City CAPEX incl. dedicated solar	USD 474M (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	45,948 (2020; 22.6% of raster bbox)
Direct / reachable line pairs	50.0% / 50.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Meru Ke network map

Line	Length	Stations	Trainsets	Shape
line-1	11.5 km	3	23	radial
line-2	5.2 km	3	13	radial
line-3	8.1 km	3	18	radial
line-4	4.9 km	2	11	radial

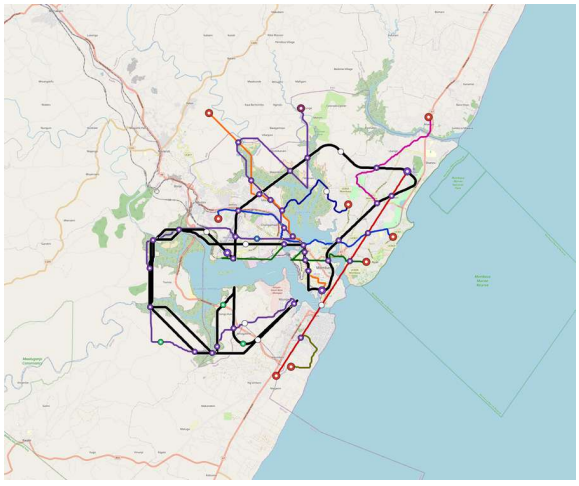
Generated City Model - Mombasa

east-africa / Kenya / Mombasa

Metric	Value
Path	cities/catalogue/east-africa/Kenya/Mombasa
Recorded ISO / planning population	KE / 1,350,000
Family	metro-4car
Lines / stations	9 / 159
Route length	284.5 km
Fleet	423 trainsets
Routing-demand cells (legacy)	74%; not population access
City CAPEX incl. dedicated solar	USD 5.18bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	521,831 (2020; 33.7% of raster bbox)
Direct / reachable line pairs	44.4% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Mombasa network map

Line	Length	Stations	Trainsets	Shape
line-1	20.9 km	11	42	radial
line-2	21.6 km	7	34	radial
line-3	17.0 km	14	46	radial
line-4	63.7 km	37	133	radial
line-5	21.1 km	21	64	radial
line-6	112.9 km	61	57	ring
line-7	11.9 km	3	19	radial
line-8	5.3 km	2	11	radial
line-9	10.0 km	3	17	radial

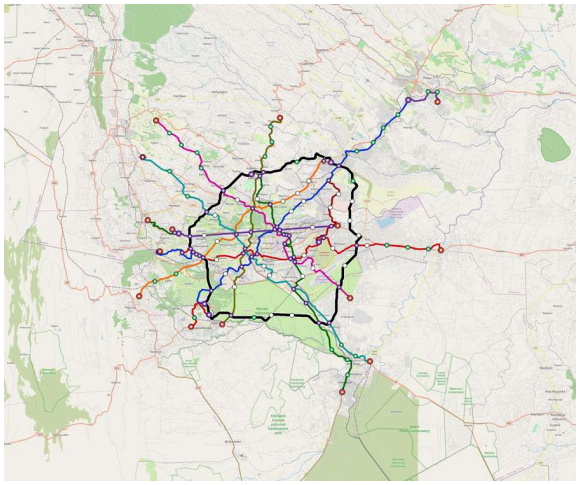
Generated City Model - Nairobi

east-africa / Kenya / Nairobi

Metric	Value
Path	cities/catalogue/east-africa/Kenya/Nairobi
Recorded ISO / planning population	KE / 5,700,000
Family	metro-6car
Lines / stations	14 / 304
Route length	589.5 km
Fleet	1128 trainsets
Routing-demand cells (legacy)	55%; not population access
City CAPEX incl. dedicated solar	USD 11.72bn (USD 20M / km); shared national factory excluded
Residents within 800 m station circles	1,776,673 (2020; 27.2% of raster bbox)
Direct / reachable line pairs	45.1% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Nairobi network map

Line	Length	Stations	Trainsets	Shape
line-1	63.2 km	20	119	radial
line-2	57.6 km	54	181	radial
line-3	55.7 km	37	138	radial
line-4	28.1 km	13	58	radial
line-5	51.3 km	23	101	radial
line-6	65.6 km	23	122	radial
line-7	58.1 km	39	145	radial
line-8	47.2 km	15	90	radial
line-9	108.5 km	59	60	ring
line-10	17.9 km	6	36	radial
line-11	10.5 km	5	23	radial
line-12	8.7 km	3	18	radial
line-13	7.6 km	3	16	radial
line-14	9.7 km	4	21	radial

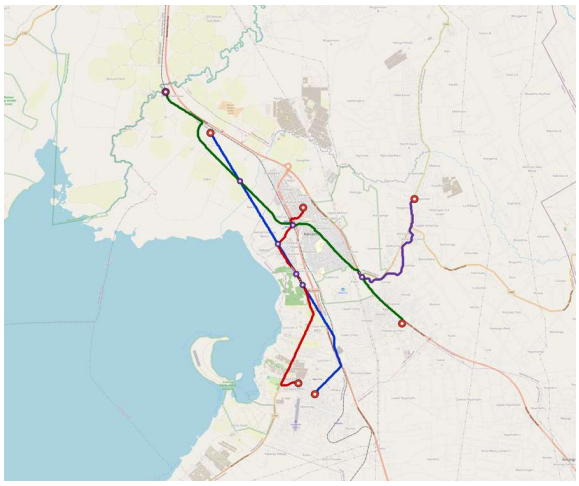
Generated City Model - Naivasha

east-africa / Kenya / Naivasha

Metric	Value
Path	cities/catalogue/east-africa/Kenya/Naivasha
Recorded ISO / planning population	KE / 250,000
Family	tram-2car
Lines / stations	4 / 31
Route length	43.9 km
Fleet	110 trainsets
Routing-demand cells (legacy)	70%; not population access
City CAPEX incl. dedicated solar	USD 797M (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	58,377 (2020; 35.5% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Naivasha network map

Line	Length	Stations	Trainsets	Shape
line-1	13.7 km	11	37	radial
line-2	9.7 km	10	30	radial
line-3	14.7 km	7	30	radial
line-4	5.8 km	3	13	radial

Generated City Model - Nakuru

east-africa / Kenya / Nakuru

Metric	Value
Path	cities/catalogue/east-africa/Kenya/Nakuru
Recorded ISO / planning population	KE / 700,000
Family	light-metro-3car
Lines / stations	5 / 23
Route length	64.3 km
Fleet	202 trainsets
Routing-demand cells (legacy)	55%; not population access
City CAPEX incl. dedicated solar	USD 1.09bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	223,283 (2020; 35.1% of raster bbox)
Direct / reachable line pairs	40.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.

Nakuru network map

Line	Length	Stations	Trainsets	Shape
line-1	11.5 km	4	36	radial
line-2	22.5 km	8	70	radial
line-3	18.5 km	6	58	radial
line-4	6.0 km	3	20	radial
line-5	5.8 km	2	18	radial

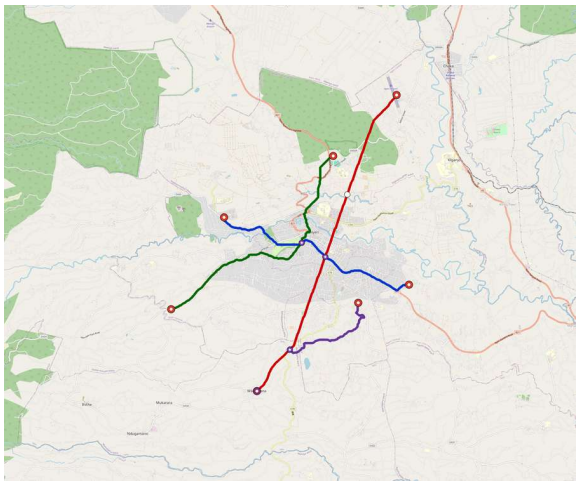
Generated City Model - Nyeri

east-africa / Kenya / Nyeri

Metric	Value
Path	cities/catalogue/east-africa/Kenya/Nyeri
Recorded ISO / planning population	KE / 250,000
Family	tram-2car
Lines / stations	4 / 16
Route length	40.0 km
Fleet	86 trainsets
Routing-demand cells (legacy)	71%; not population access
City CAPEX incl. dedicated solar	USD 656M (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	36,363 (2020; 21.2% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Nyeri network map

Line	Length	Stations	Trainsets	Shape
line-1	9.4 km	4	20	radial
line-2	14.4 km	6	30	radial

Line	Length	Stations	Trainsets	Shape
line-3	11.2 km	3	23	radial
line-4	5.1 km	3	13	radial

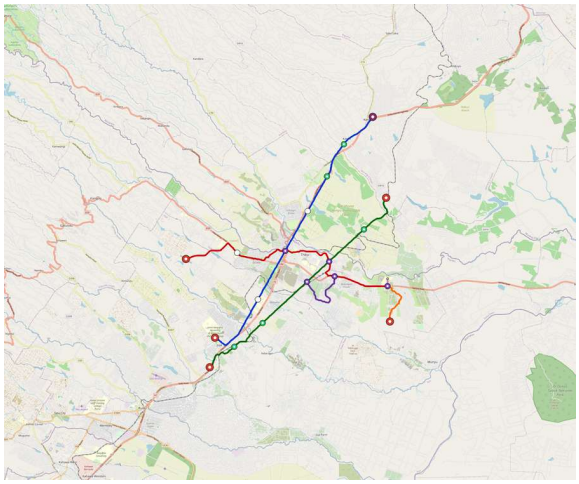
Generated City Model - Thika

east-africa / Kenya / Thika

Metric	Value
Path	cities/catalogue/east-africa/Kenya/Thika
Recorded ISO / planning population	KE / 350,000
Family	light-metro-3car
Lines / stations	5 / 25
Route length	82.7 km
Fleet	256 trainsets
Routing-demand cells (legacy)	60%; not population access
City CAPEX incl. dedicated solar	USD 1.34bn (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	77,553 (2020; 20.6% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Thika network map

Line	Length	Stations	Trainsets	Shape
line-1	26.1 km	7	81	radial
line-2	22.9 km	7	68	radial
line-3	23.8 km	7	75	radial
line-4	5.7 km	2	18	radial
line-5	4.2 km	2	14	radial

Madagascar

Generated City Model - Antananarivo

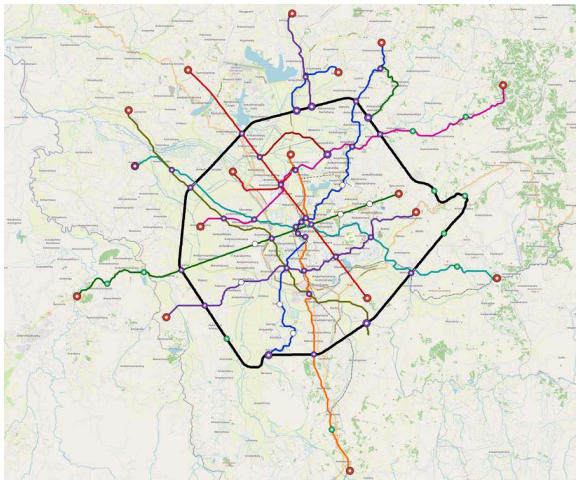
east-africa / Madagascar / Antananarivo

Metric	Value
Path	cities/catalogue/east-africa/Madagascar/Antananarivo
Recorded ISO / planning population	MG / 3,058,000
Family	metro-6car
Lines / stations	14 / 198
Route length	356.1 km
Fleet	752 trainsets
Routing-demand cells (legacy)	76%; not population access
City CAPEX incl. dedicated solar	USD 7.77bn (USD 22M / km); shared national factory excluded
Residents within 800 m station circles	854,506 (2020; 24.6% of raster bbox)

Metric	Value
Direct / reachable line pairs	46.2% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Antananarivo network map

Line	Length	Stations	Trainsets	Shape
line-1	36.0 km	45	144	radial
line-2	24.4 km	10	48	radial
line-3	29.6 km	12	54	radial
line-4	27.4 km	8	50	radial
line-5	30.1 km	41	127	radial
line-6	37.3 km	22	87	radial
line-7	33.3 km	13	67	radial
line-8	31.0 km	10	59	radial
line-9	71.7 km	20	35	ring
line-10	6.8 km	2	14	radial
line-11	6.8 km	3	15	radial
line-12	6.6 km	5	19	radial
line-13	6.7 km	4	17	radial
line-14	8.3 km	3	16	radial

Mozambique

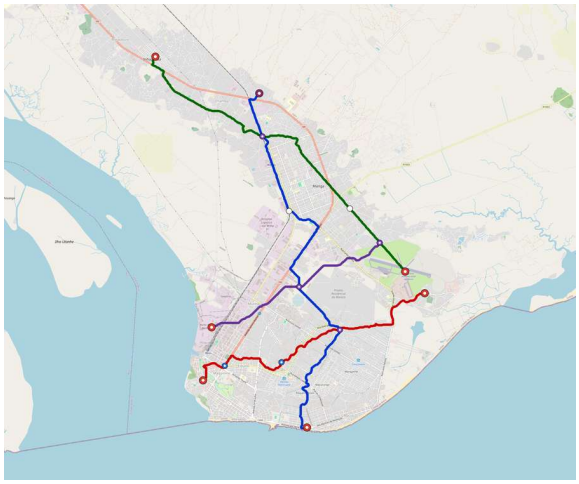
Generated City Model - Beira

east-africa / Mozambique / Beira

Metric	Value
Path	cities/catalogue/east-africa/Mozambique/Beira
Recorded ISO / planning population	MZ / 535,000
Family	light-metro-3car
Lines / stations	4 / 19
Route length	52.5 km
Fleet	164 trainsets
Routing-demand cells (legacy)	40%; not population access
City CAPEX incl. dedicated solar	USD 898M (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	100,193 (2020; 24.2% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Beira network map

Line	Length	Stations	Trainsets	Shape
line-1	17.5 km	6	52	radial
line-2	12.2 km	5	38	radial
line-3	15.0 km	5	48	radial
line-4	7.8 km	3	26	radial

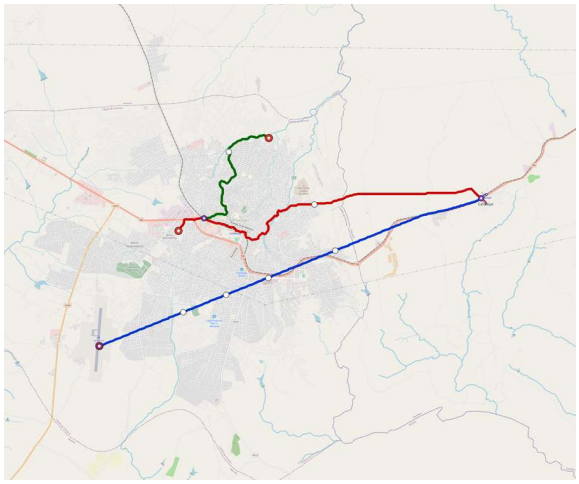
Generated City Model - Chimoio

east-africa / Mozambique / Chimoio

Metric	Value
Path	cities/catalogue/east-africa/Mozambique/Chimoio
Recorded ISO / planning population	MZ / 400,000
Family	light-metro-3car
Lines / stations	3 / 15
Route length	38.1 km
Fleet	116 trainsets
Routing-demand cells (legacy)	52%; not population access
City CAPEX incl. dedicated solar	USD 667M (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	75,032 (2020; 17.9% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Chimoio network map

Line	Length	Stations	Trainsets	Shape
line-1	17.6 km	7	53	radial
line-2	14.2 km	5	43	radial
line-3	6.3 km	3	20	radial

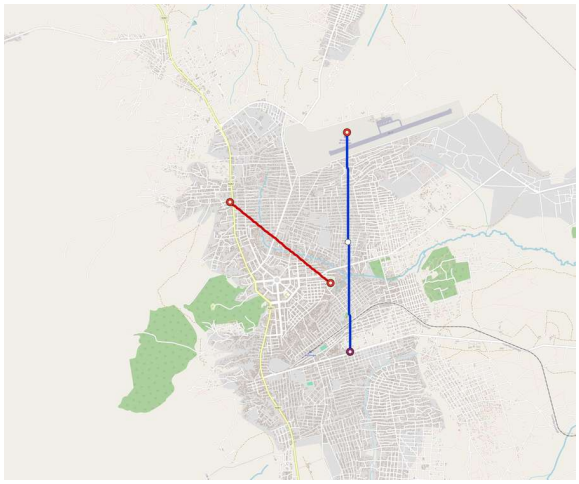
Generated City Model - Lichinga

east-africa / Mozambique / Lichinga

Metric	Value
Path	cities/catalogue/east-africa/Mozambique/Lichinga
Recorded ISO / planning population	MZ / 250,000
Family	tram-2car
Lines / stations	2 / 5
Route length	7.1 km
Fleet	21 trainsets
Routing-demand cells (legacy)	47%; not population access
City CAPEX incl. dedicated solar	USD 142M (USD 20M / km); shared national factory excluded
Residents within 800 m station circles	30,103 (2020; 12.4% of raster bbox)
Direct / reachable line pairs	0.0% / 0.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Lichinga network map

Line	Length	Stations	Trainsets	Shape
line-1	4.3 km	3	12	radial
line-2	2.8 km	2	9	radial

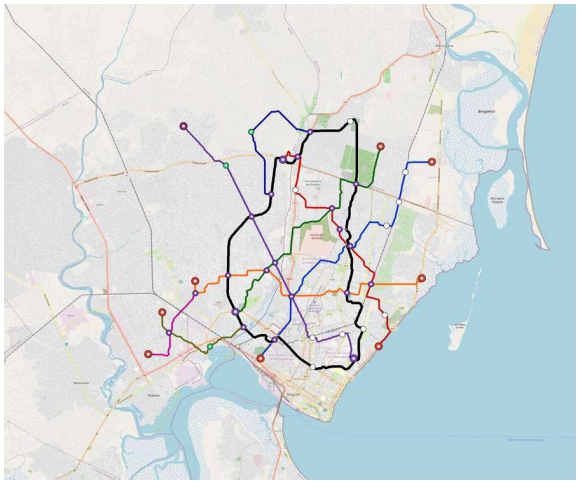
Generated City Model - Maputo

east-africa / Mozambique / Maputo

Metric	Value
Path	cities/catalogue/east-africa/Mozambique/Maputo
Recorded ISO / planning population	MZ / 1,530,000
Family	metro-4car
Lines / stations	9 / 92
Route length	204.7 km
Fleet	302 trainsets
Routing-demand cells (legacy)	67%; not population access
City CAPEX incl. dedicated solar	USD 3.27bn (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	647,938 (2020; 31.3% of raster bbox)
Direct / reachable line pairs	47.2% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Maputo network map

Line	Length	Stations	Trainsets	Shape
line-1	26.0 km	19	63	radial
line-2	23.0 km	12	46	radial
line-3	22.7 km	6	36	radial
line-4	26.4 km	9	41	radial
line-5	22.9 km	9	39	radial
line-6	56.8 km	26	27	ring
line-7	6.9 km	3	14	radial
line-8	9.0 km	5	18	radial
line-9	10.9 km	3	18	radial

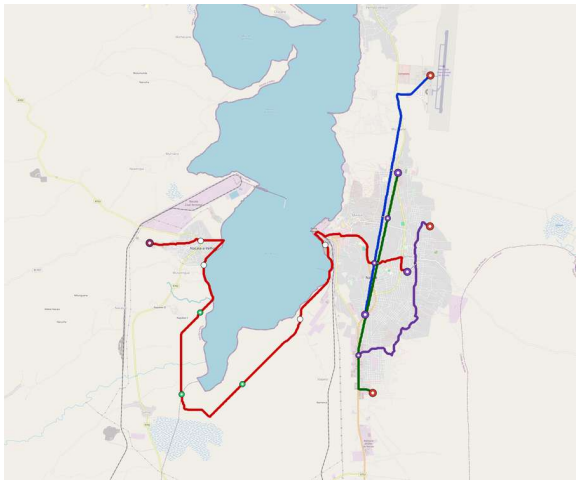
Generated City Model - Nacala

east-africa / Mozambique / Nacala

Metric	Value
Path	cities/catalogue/east-africa/Mozambique/Nacala
Recorded ISO / planning population	MZ / 300,000
Family	tram-2car
Lines / stations	4 / 25
Route length	55.2 km
Fleet	113 trainsets
Routing-demand cells (legacy)	71%; not population access
City CAPEX incl. dedicated solar	USD 771M (USD 14M / km); shared national factory excluded
Residents within 800 m station circles	78,340 (2020; 34.3% of raster bbox)
Direct / reachable line pairs	83.3% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Nacala network map

Line	Length	Stations	Trainsets	Shape
line-1	10.8 km	5	21	radial
line-2	26.6 km	11	53	radial
line-3	9.7 km	6	23	radial
line-4	8.1 km	3	16	radial

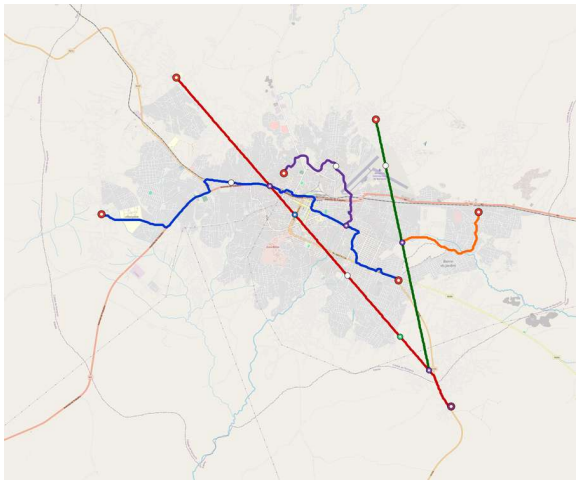
Generated City Model - Nampula

east-africa / Mozambique / Nampula

Metric	Value
Path	cities/catalogue/east-africa/Mozambique/Nampula
Recorded ISO / planning population	MZ / 800,000
Family	light-metro-3car
Lines / stations	5 / 22
Route length	57.8 km
Fleet	182 trainsets
Routing-demand cells (legacy)	69%; not population access
City CAPEX incl. dedicated solar	USD 1.02bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	190,740 (2020; 21.8% of raster bbox)
Direct / reachable line pairs	40.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Nampula network map

Line	Length	Stations	Trainsets	Shape
line-1	16.9 km	5	52	radial
line-2	18.9 km	7	60	radial
line-3	10.6 km	5	34	radial
line-4	6.5 km	3	20	radial
line-5	4.8 km	2	16	radial

Generated City Model - Pemba Mz

east-africa / Mozambique / Pemba-Mz

Metric	Value
Path	cities/catalogue/east-africa/Mozambique/Pemba-Mz
Recorded ISO / planning population	MZ / 250,000
Family	tram-2car
Lines / stations	4 / 19
Route length	31.3 km
Fleet	75 trainsets
Routing-demand cells (legacy)	52%; not population access
City CAPEX incl. dedicated solar	USD 533M (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	77,927 (2020; 35.2% of raster bbox)

Metric	Value
Direct / reachable line pairs	33.3% / 33.3%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.

Pemba Mz network map

Line	Length	Stations	Trainsets	Shape
line-1	11.1 km	7	26	radial
line-2	6.9 km	3	15	radial
line-3	7.6 km	3	16	radial
line-4	5.6 km	6	18	radial

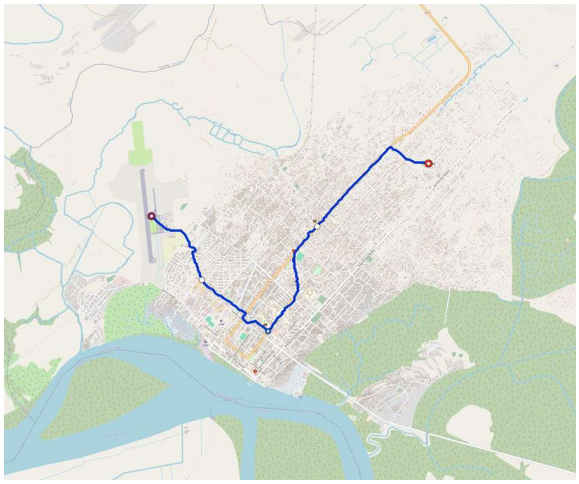
Generated City Model - Quelimane

east-africa / Mozambique / Quelimane

Metric	Value
Path	cities/catalogue/east-africa/Mozambique/Quelimane
Recorded ISO / planning population	MZ / 350,000
Family	light-metro-3car
Lines / stations	1 / 5
Route length	9.4 km
Fleet	32 trainsets
Routing-demand cells (legacy)	44%; not population access
City CAPEX incl. dedicated solar	USD 174M (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	62,344 (2020; 18.5% of raster bbox)
Direct / reachable line pairs	100.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Quelimane network map

Line	Length	Stations	Trainsets	Shape
line-1	9.4 km	5	32	radial

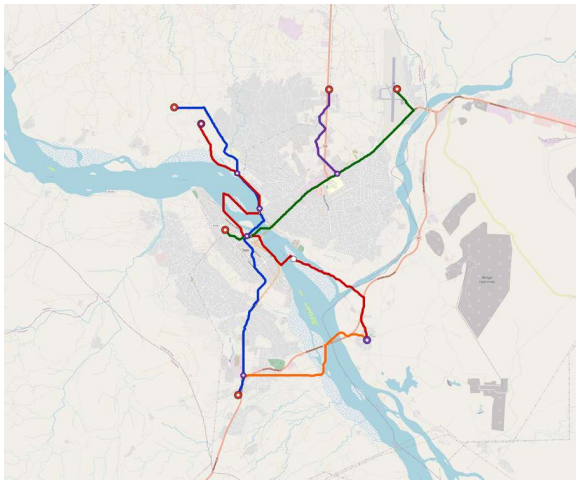
Generated City Model - Tete

east-africa / Mozambique / Tete

Metric	Value
Path	cities/catalogue/east-africa/Mozambique/Tete
Recorded ISO / planning population	MZ / 350,000
Family	light-metro-3car
Lines / stations	5 / 29
Route length	52.2 km
Fleet	170 trainsets
Routing-demand cells (legacy)	78%; not population access
City CAPEX incl. dedicated solar	USD 1.17bn (USD 22M / km); shared national factory excluded
Residents within 800 m station circles	53,349 (2020; 17.0% of raster bbox)
Direct / reachable line pairs	60.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Tete network map

Line	Length	Stations	Trainsets	Shape
line-1	14.8 km	8	46	radial
line-2	16.3 km	9	51	radial
line-3	10.8 km	7	38	radial
line-4	4.0 km	3	15	radial
line-5	6.2 km	2	20	radial

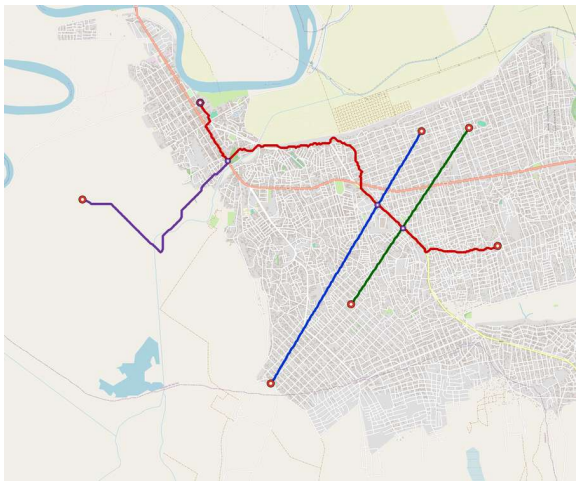
Generated City Model - Xai Xai

east-africa / Mozambique / Xai-Xai

Metric	Value
Path	cities/catalogue/east-africa/Mozambique/Xai-Xai
Recorded ISO / planning population	MZ / 250,000
Family	tram-2car
Lines / stations	4 / 13
Route length	22.5 km
Fleet	54 trainsets
Routing-demand cells (legacy)	50%; not population access
City CAPEX incl. dedicated solar	USD 407M (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	40,342 (2020; 27.8% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Xai Xai network map

Line	Length	Stations	Trainsets	Shape
line-1	6.0 km	3	13	radial
line-2	8.2 km	5	19	radial
line-3	4.3 km	3	12	radial
line-4	4.0 km	2	10	radial

Rwanda

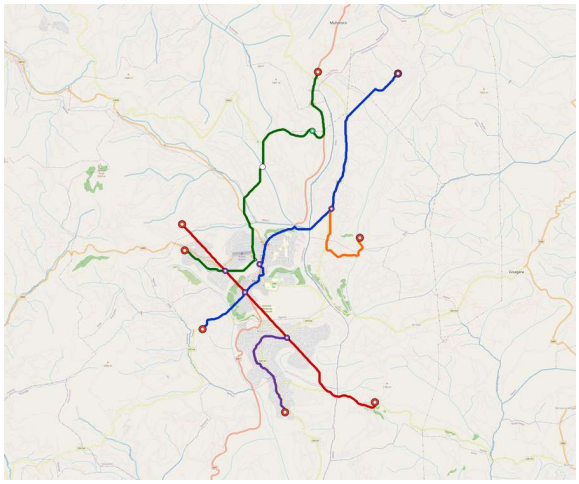
Generated City Model - Huye

east-africa / Rwanda / Huye

Metric	Value
Path	cities/catalogue/east-africa/Rwanda/Huye
Recorded ISO / planning population	RW / 250,000
Family	tram-2car
Lines / stations	5 / 20
Route length	51.7 km
Fleet	108 trainsets
Routing-demand cells (legacy)	64%; not population access
City CAPEX incl. dedicated solar	USD 804M (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	40,528 (2020; 18.9% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Huye network map

Line	Length	Stations	Trainsets	Shape
line-1	15.3 km	5	30	radial
line-2	12.5 km	5	26	radial
line-3	14.8 km	6	31	radial
line-4	5.0 km	2	11	radial
line-5	4.1 km	2	10	radial

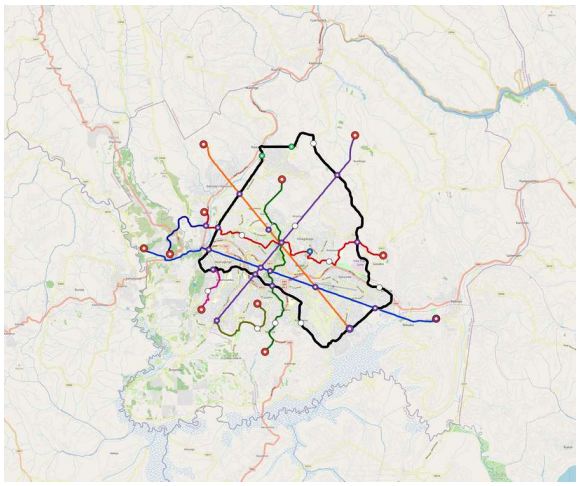
Generated City Model - Kigali

east-africa / Rwanda / Kigali

Metric	Value
Path	cities/catalogue/east-africa/Rwanda/Kigali
Recorded ISO / planning population	RW / 1,208,000
Family	metro-4car
Lines / stations	9 / 64
Route length	194.9 km
Fleet	257 trainsets
Routing-demand cells (legacy)	67%; not population access
City CAPEX incl. dedicated solar	USD 3.15bn (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	483,284 (2020; 29.2% of raster bbox)
Direct / reachable line pairs	47.2% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Kigali network map

Line	Length	Stations	Trainsets	Shape
line-1	27.6 km	7	42	radial
line-2	25.2 km	11	46	radial
line-3	18.5 km	7	31	radial
line-4	21.3 km	10	40	radial
line-5	21.4 km	6	35	radial
line-6	61.2 km	15	24	ring
line-7	5.1 km	3	13	radial
line-8	8.1 km	3	15	radial
line-9	6.4 km	2	11	radial

Generated City Model - Rubavu

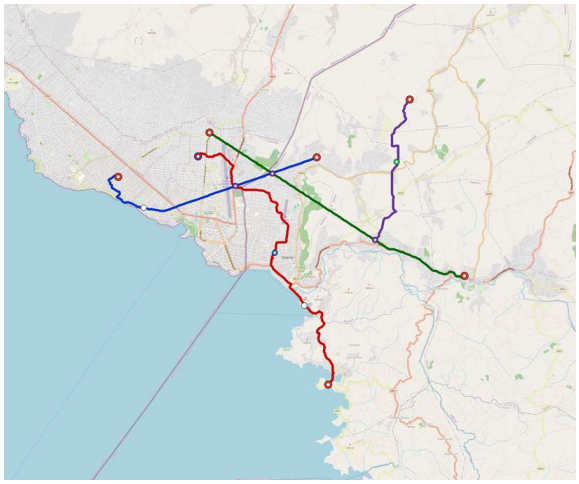
east-africa / Rwanda / Rubavu

Metric	Value
Path	cities/catalogue/east-africa/Rwanda/Rubavu
Recorded ISO / planning population	RW / 250,000
Family	tram-2car

Metric	Value
Lines / stations	4 / 17
Route length	46.9 km
Fleet	95 trainsets
Routing-demand cells (legacy)	44%; not population access
City CAPEX incl. dedicated solar	USD 660M (USD 14M / km); shared national factory excluded
Residents within 800 m station circles	64,034 (2020; 21.2% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Rubavu network map

Line	Length	Stations	Trainsets	Shape
line-1	11.0 km	5	23	radial
line-2	15.4 km	5	30	radial
line-3	13.3 km	4	26	radial
line-4	7.1 km	3	16	radial

Somalia

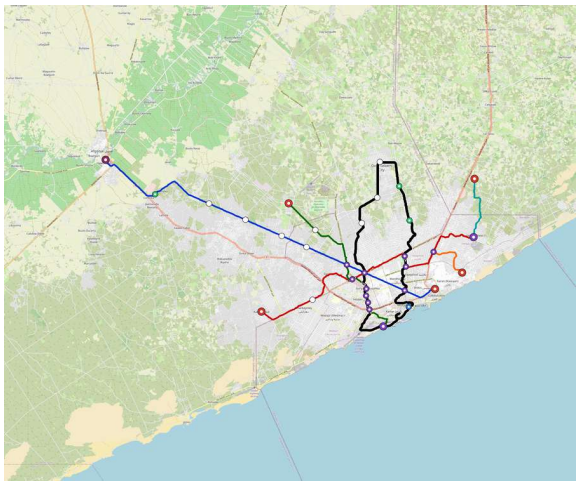
Generated City Model - Mogadishu

east-africa / Somalia / Mogadishu

Metric	Value
Path	cities/catalogue/east-africa/Somalia/Mogadishu
Recorded ISO / planning population	SO / 2,610,000
Family	metro-4car
Lines / stations	6 / 120
Route length	133.1 km
Fleet	284 trainsets
Routing-demand cells (legacy)	61%; not population access
City CAPEX incl. dedicated solar	USD 2.75bn (USD 21M / km); shared national factory excluded
Residents within 800 m station circles	215,178 (2020; 24.7% of raster bbox)
Direct / reachable line pairs	53.3% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Mogadishu network map

Line	Length	Stations	Trainsets	Shape
line-1	33.4 km	11	56	radial
line-2	28.2 km	20	68	radial
line-3	16.8 km	41	105	radial
line-4	45.1 km	44	34	ring
line-5	4.1 km	2	10	radial
line-6	5.4 km	2	11	radial

South Sudan

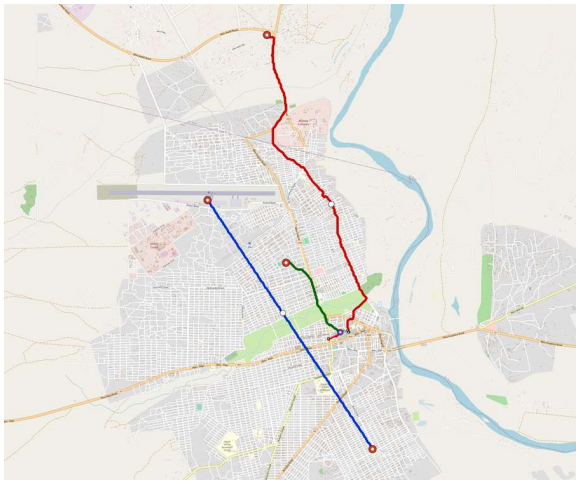
Generated City Model - Wau

east-africa / South Sudan / Wau

Metric	Value
Path	cities/catalogue/east-africa/South Sudan/Wau
Recorded ISO / planning population	SS / 300,000
Family	tram-2car
Lines / stations	3 / 12
Route length	16.8 km
Fleet	45 trainsets
Routing-demand cells (legacy)	80%; not population access
City CAPEX incl. dedicated solar	USD 310M (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	734 (2020; 6.8% of raster bbox)
Direct / reachable line pairs	33.3% / 33.3%
Input discrepancy	Wau is in South Sudan. The canonical jurisdiction and national aggregation now use SS; the historical slug waw remains an alias. Finance uses an explicitly uncalibrated generic scenario, with SSP currency; Sudan SD/SDG assumptions are excluded.

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Wau network map

Line	Length	Stations	Trainsets	Shape
line-1	6.6 km	3	14	radial
line-2	7.9 km	5	19	radial
line-3	2.3 km	4	12	radial

Tanzania

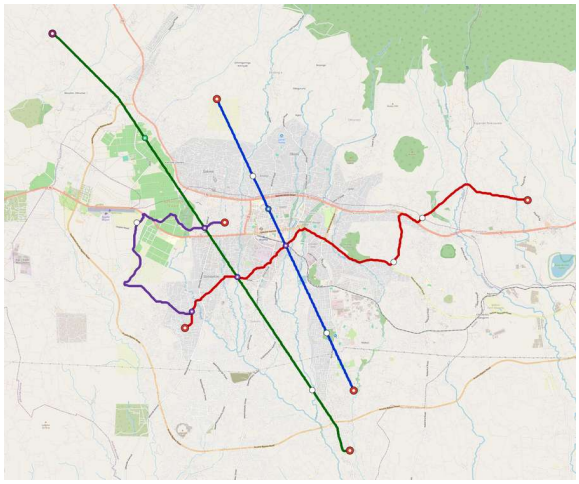
Generated City Model - Arusha

east-africa / Tanzania / Arusha

Metric	Value
Path	cities/catalogue/east-africa/Tanzania/Arusha
Recorded ISO / planning population	TZ / 700,000
Family	light-metro-3car
Lines / stations	4 / 23
Route length	69.6 km
Fleet	218 trainsets
Routing-demand cells (legacy)	46%; not population access
City CAPEX incl. dedicated solar	USD 1.18bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	207,263 (2020; 25.5% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Arusha network map

Line	Length	Stations	Trainsets	Shape
line-1	14.3 km	6	46	radial
line-2	21.6 km	7	67	radial

Line	Length	Stations	Trainsets	Shape
line-3	22.9 km	6	70	radial
line-4	10.9 km	4	35	radial

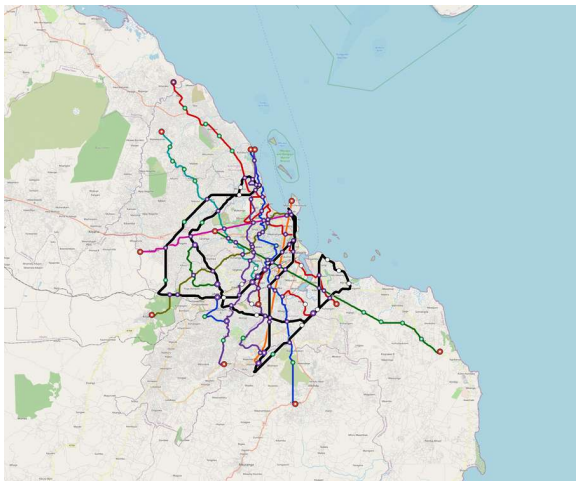
Generated City Model - Dar Es Salaam

east-africa / Tanzania / Dar-Es-Salaam

Metric	Value
Path	cities/catalogue/east-africa/Tanzania/Dar-Es-Salaam
Recorded ISO / planning population	TZ / 7,404,689
Family	metro-6car
Lines / stations	14 / 309
Route length	611.0 km
Fleet	1053 trainsets
Routing-demand cells (legacy)	65%; not population access
City CAPEX incl. dedicated solar	USD 12.15bn (USD 20M / km); shared national factory excluded
Residents within 800 m station circles	2,304,454 (2020; 33.4% of raster bbox)
Direct / reachable line pairs	45.1% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Dar Es Salaam network map

Line	Length	Stations	Trainsets	Shape
line-1	57.5 km	32	129	radial
line-2	72.0 km	34	145	radial
line-3	46.6 km	18	90	radial
line-4	52.0 km	44	155	radial
line-5	29.1 km	15	64	radial
line-6	43.1 km	18	87	radial
line-7	27.6 km	12	56	radial
line-8	37.7 km	36	123	radial
line-9	187.2 km	81	91	ring
line-10	6.2 km	3	15	radial
line-11	8.2 km	3	17	radial
line-12	8.8 km	3	17	radial
line-13	13.1 km	3	24	radial
line-14	21.9 km	7	40	radial

Generated City Model - Dodoma

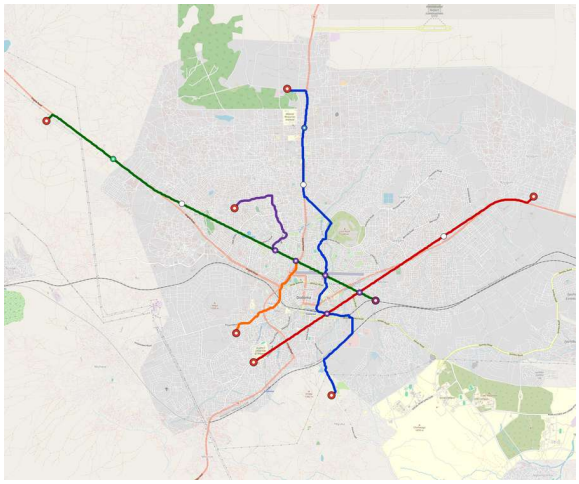
east-africa / Tanzania / Dodoma

Metric	Value
Path	cities/catalogue/east-africa/Tanzania/Dodoma
Recorded ISO / planning population	TZ / 800,000

Metric	Value
Family	light-metro-3car
Lines / stations	5 / 23
Route length	56.6 km
Fleet	182 trainsets
Routing-demand cells (legacy)	69%; not population access
City CAPEX incl. dedicated solar	USD 1.00bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	78,650 (2020; 25.8% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Dodoma network map

Line	Length	Stations	Trainsets	Shape
line-1	16.5 km	6	51	radial
line-2	14.6 km	5	45	radial
line-3	16.5 km	8	54	radial
line-4	4.5 km	2	16	radial
line-5	4.4 km	2	16	radial

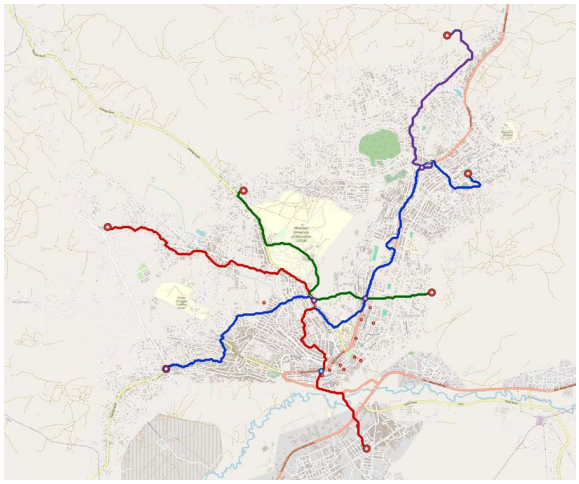
Generated City Model - Iringa

east-africa / Tanzania / Iringa

Metric	Value
Path	cities/catalogue/east-africa/Tanzania/Iringa
Recorded ISO / planning population	TZ / 250,000
Family	tram-2car
Lines / stations	4 / 25
Route length	31.4 km
Fleet	86 trainsets
Routing-demand cells (legacy)	73%; not population access
City CAPEX incl. dedicated solar	USD 622M (USD 20M / km); shared national factory excluded
Residents within 800 m station circles	75,683 (2020; 38.2% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Iringa network map

Line	Length	Stations	Trainsets	Shape
line-1	11.4 km	9	30	radial
line-2	9.7 km	7	25	radial
line-3	6.0 km	6	19	radial
line-4	4.3 km	3	12	radial

Generated City Model - Kigoma

east-africa / Tanzania / Kigoma

Metric	Value
Path	cities/catalogue/east-africa/Tanzania/Kigoma
Recorded ISO / planning population	TZ / 300,000
Family	tram-2car
Lines / stations	4 / 17
Route length	33.2 km
Fleet	75 trainsets
Routing-demand cells (legacy)	52%; not population access
City CAPEX incl. dedicated solar	USD 522M (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	90,212 (2020; 32.1% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.

Kigoma network map

Line	Length	Stations	Trainsets	Shape
line-1	11.6 km	6	25	radial
line-2	11.5 km	5	25	radial
line-3	6.1 km	4	15	radial

Line	Length	Stations	Trainsets	Shape
line-4	4.0 km	2	10	radial

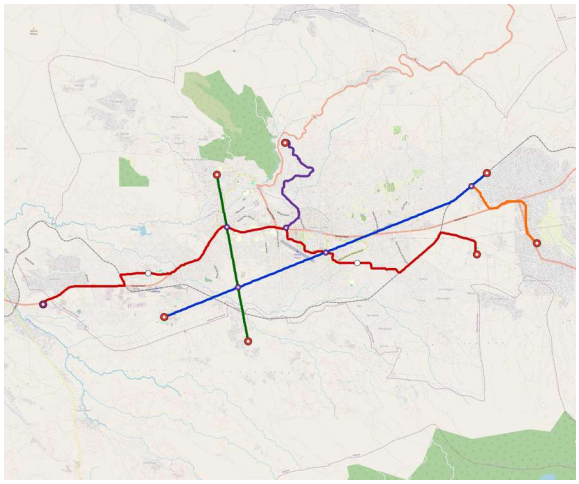
Generated City Model - Mbeya

east-africa / Tanzania / Mbeya

Metric	Value
Path	cities/catalogue/east-africa/Tanzania/Mbeya
Recorded ISO / planning population	TZ / 550,000
Family	light-metro-3car
Lines / stations	5 / 20
Route length	57.8 km
Fleet	182 trainsets
Routing-demand cells (legacy)	48%; not population access
City CAPEX incl. dedicated solar	USD 983M (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	139,298 (2020; 25.5% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Mbeya network map

Line	Length	Stations	Trainsets	Shape
line-1	15.5 km	5	48	radial
line-2	24.7 km	7	74	radial
line-3	7.1 km	4	24	radial
line-4	6.0 km	2	20	radial
line-5	4.5 km	2	16	radial

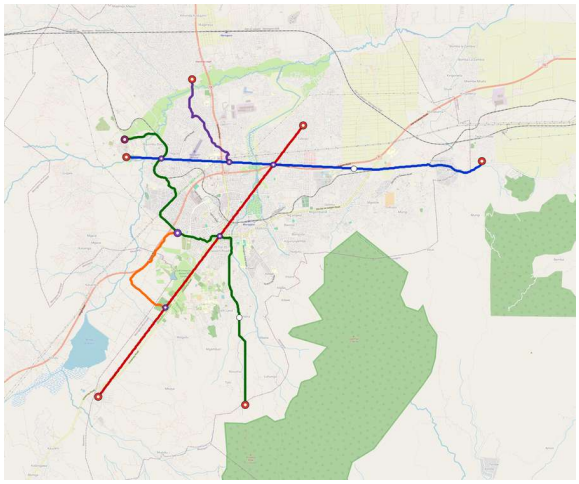
Generated City Model - Morogoro

east-africa / Tanzania / Morogoro

Metric	Value
Path	cities/catalogue/east-africa/Tanzania/Morogoro
Recorded ISO / planning population	TZ / 500,000
Family	light-metro-3car
Lines / stations	5 / 21
Route length	56.4 km
Fleet	178 trainsets
Routing-demand cells (legacy)	49%; not population access
City CAPEX incl. dedicated solar	USD 1.05bn (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	105,178 (2020; 24.6% of raster bbox)
Direct / reachable line pairs	60.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Morogoro network map

Line	Length	Stations	Trainsets	Shape
line-1	15.3 km	6	47	radial
line-2	15.1 km	5	48	radial
line-3	16.5 km	6	51	radial
line-4	4.5 km	2	16	radial
line-5	4.9 km	2	16	radial

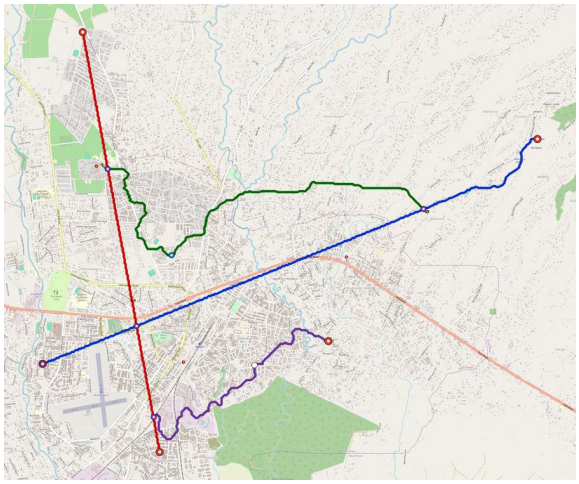
Generated City Model - Moshi

east-africa / Tanzania / Moshi

Metric	Value
Path	cities/catalogue/east-africa/Tanzania/Moshi
Recorded ISO / planning population	TZ / 300,000
Family	tram-2car
Lines / stations	4 / 17
Route length	37.0 km
Fleet	79 trainsets
Routing-demand cells (legacy)	70%; not population access
City CAPEX incl. dedicated solar	USD 624M (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	99,138 (2020; 28.4% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Moshi network map

Line	Length	Stations	Trainsets	Shape
line-1	12.2 km	4	25	radial

Line	Length	Stations	Trainsets	Shape
line-2	9.1 km	5	20	radial
line-3	9.8 km	5	21	radial
line-4	5.9 km	3	13	radial

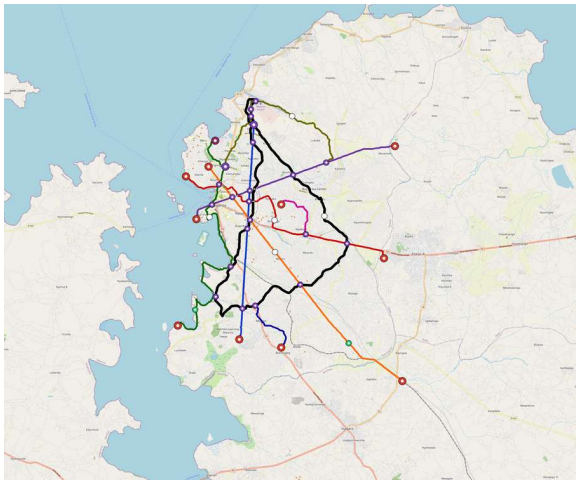
Generated City Model - Mwanza

east-africa / Tanzania / Mwanza

Metric	Value
Path	cities/catalogue/east-africa/Tanzania/Mwanza
Recorded ISO / planning population	TZ / 1,100,000
Family	metro-4car
Lines / stations	9 / 88
Route length	199.5 km
Fleet	298 trainsets
Routing-demand cells (legacy)	76%; not population access
City CAPEX incl. dedicated solar	USD 3.44bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	338,238 (2020; 32.1% of raster bbox)
Direct / reachable line pairs	55.6% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Mwanza network map

Line	Length	Stations	Trainsets	Shape
line-1	17.8 km	8	34	radial
line-2	24.2 km	13	49	radial
line-3	27.4 km	13	51	radial
line-4	19.0 km	10	38	radial
line-5	26.0 km	10	43	radial
line-6	56.1 km	20	24	ring
line-7	4.9 km	2	10	radial
line-8	18.7 km	10	38	radial
line-9	5.5 km	2	11	radial

Generated City Model - Shinyanga

east-africa / Tanzania / Shinyanga

Metric	Value
Path	cities/catalogue/east-africa/Tanzania/Shinyanga
Recorded ISO / planning population	TZ / 250,000
Family	tram-2car
Lines / stations	4 / 15
Route length	35.8 km
Fleet	77 trainsets

Metric	Value
Routing-demand cells (legacy)	80%; not population access
City CAPEX incl. dedicated solar	USD 512M (USD 14M / km); shared national factory excluded
Residents within 800 m station circles	46,640 (2020; 32.0% of raster bbox)
Direct / reachable line pairs	33.3% / 50.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.

Shinyanga network map

Line	Length	Stations	Trainsets	Shape
line-1	15.9 km	6	32	radial
line-2	7.8 km	4	17	radial
line-3	6.7 km	2	15	radial
line-4	5.3 km	3	13	radial

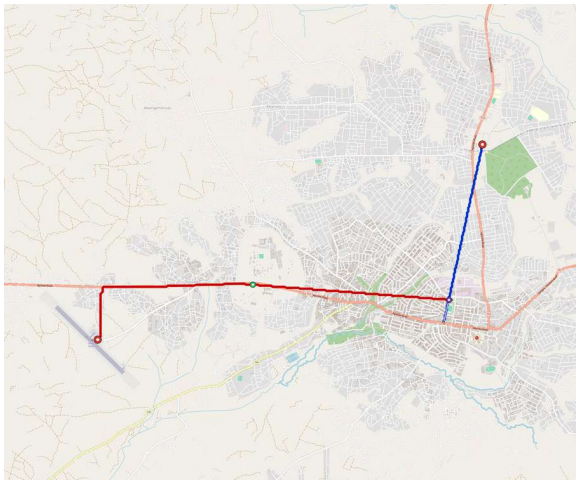
Generated City Model - Songea

east-africa / Tanzania / Songea

Metric	Value
Path	cities/catalogue/east-africa/Tanzania/Songea
Recorded ISO / planning population	TZ / 250,000
Family	tram-2car
Lines / stations	2 / 6
Route length	11.9 km
Fleet	29 trainsets
Routing-demand cells (legacy)	38%; not population access
City CAPEX incl. dedicated solar	USD 185M (USD 15M / km); shared national factory excluded
Residents within 800 m station circles	40,457 (2020; 16.5% of raster bbox)
Direct / reachable line pairs	100.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Songea network map

Line	Length	Stations	Trainsets	Shape
line-1	3.8 km	3	11	radial
line-2	8.1 km	3	18	radial

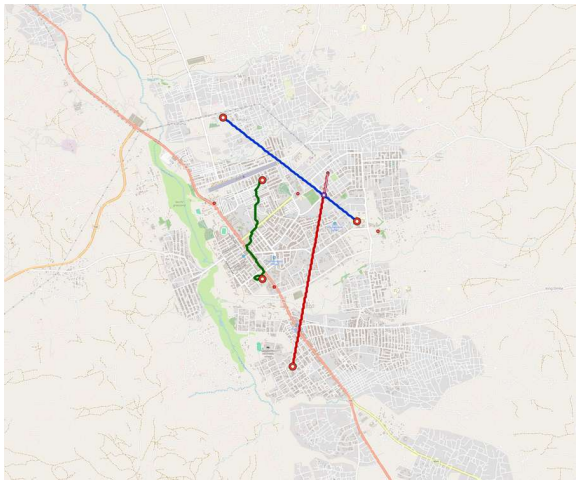
Generated City Model - Sumbawanga

east-africa / Tanzania / Sumbawanga

Metric	Value
Path	cities/catalogue/east-africa/Tanzania/Sumbawanga
Recorded ISO / planning population	TZ / 250,000
Family	tram-2car
Lines / stations	3 / 8
Route length	10.6 km
Fleet	32 trainsets
Routing-demand cells (legacy)	62%; not population access
City CAPEX incl. dedicated solar	USD 219M (USD 21M / km); shared national factory excluded
Residents within 800 m station circles	54,643 (2020; 30.8% of raster bbox)
Direct / reachable line pairs	33.3% / 33.3%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Sumbawanga network map

Line	Length	Stations	Trainsets	Shape
line-1	3.8 km	3	11	radial
line-2	4.1 km	3	12	radial
line-3	2.7 km	2	9	radial

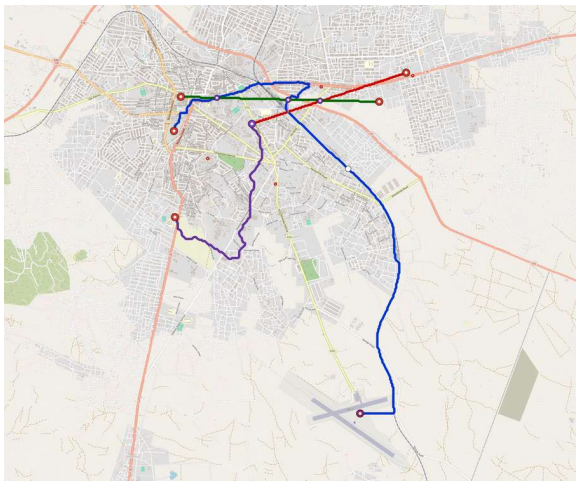
Generated City Model - Tabora

east-africa / Tanzania / Tabora

Metric	Value
Path	cities/catalogue/east-africa/Tanzania/Tabora
Recorded ISO / planning population	TZ / 300,000
Family	tram-2car
Lines / stations	4 / 17
Route length	24.8 km
Fleet	65 trainsets
Routing-demand cells (legacy)	64%; not population access
City CAPEX incl. dedicated solar	USD 458M (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	74,586 (2020; 38.3% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Tabora network map

Line	Length	Stations	Trainsets	Shape
line-1	12.3 km	6	26	radial
line-2	3.5 km	4	13	radial
line-3	4.0 km	5	15	radial
line-4	5.0 km	2	11	radial

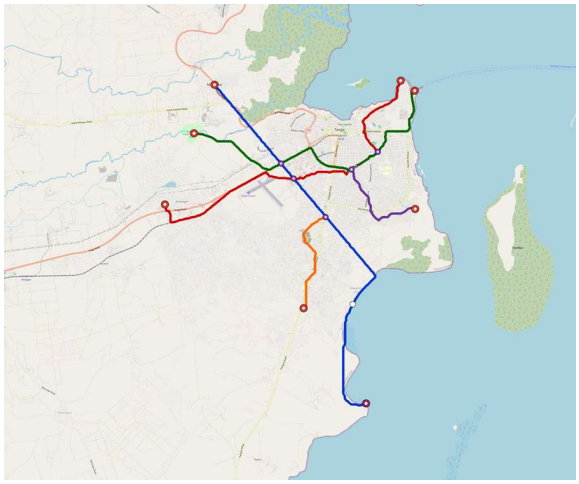
Generated City Model - Tanga

east-africa / Tanzania / Tanga

Metric	Value
Path	cities/catalogue/east-africa/Tanzania/Tanga
Recorded ISO / planning population	TZ / 400,000
Family	light-metro-3car
Lines / stations	5 / 25
Route length	54.6 km
Fleet	178 trainsets
Routing-demand cells (legacy)	53%; not population access
City CAPEX incl. dedicated solar	USD 1.02bn (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	90,341 (2020; 34.4% of raster bbox)
Direct / reachable line pairs	60.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Tanga network map

Line	Length	Stations	Trainsets	Shape
line-1	17.7 km	6	53	radial
line-2	15.0 km	8	50	radial
line-3	13.1 km	7	45	radial
line-4	4.2 km	2	14	radial
line-5	4.6 km	2	16	radial

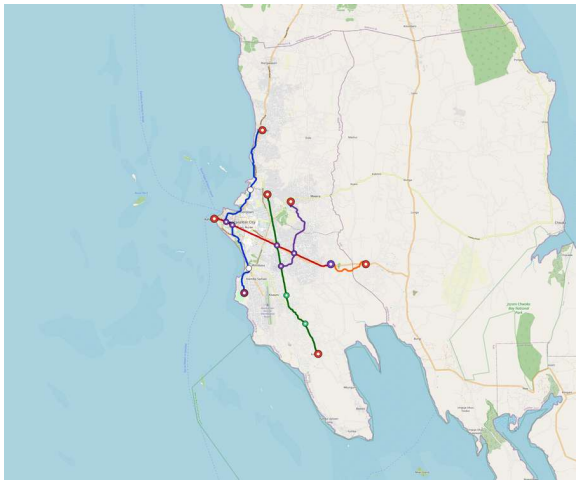
Generated City Model - Zanzibar City

east-africa / Tanzania / Zanzibar-City

Metric	Value
Path	cities/catalogue/east-africa/Tanzania/Zanzibar-City
Recorded ISO / planning population	TZ / 500,000
Family	light-metro-3car
Lines / stations	5 / 25
Route length	58.4 km
Fleet	186 trainsets
Routing-demand cells (legacy)	59%; not population access
City CAPEX incl. dedicated solar	USD 1.04bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	226,838 (2020; 27.6% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Zanzibar City network map

Line	Length	Stations	Trainsets	Shape
line-1	19.8 km	7	60	radial
line-2	11.3 km	7	38	radial

Line	Length	Stations	Trainsets	Shape
line-3	15.7 km	6	50	radial
line-4	7.5 km	3	24	radial
line-5	4.1 km	2	14	radial

Uganda

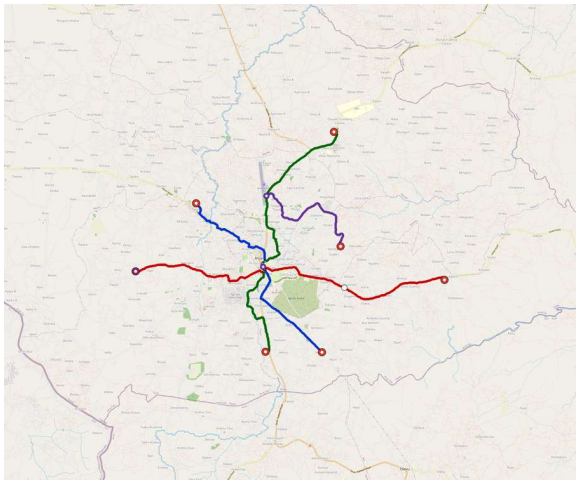
Generated City Model - Arua

east-africa / Uganda / Arua

Metric	Value
Path	cities/catalogue/east-africa/Uganda/Arua
Recorded ISO / planning population	UG / 250,000
Family	tram-2car
Lines / stations	4 / 22
Route length	43.5 km
Fleet	98 trainsets
Routing-demand cells (legacy)	82%; not population access
City CAPEX incl. dedicated solar	USD 722M (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	59,749 (2020; 21.6% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Arua network map

Line	Length	Stations	Trainsets	Shape
line-1	10.0 km	7	25	radial
line-2	14.8 km	5	30	radial
line-3	12.9 km	8	30	radial
line-4	5.9 km	2	13	radial

Generated City Model - Entebbe

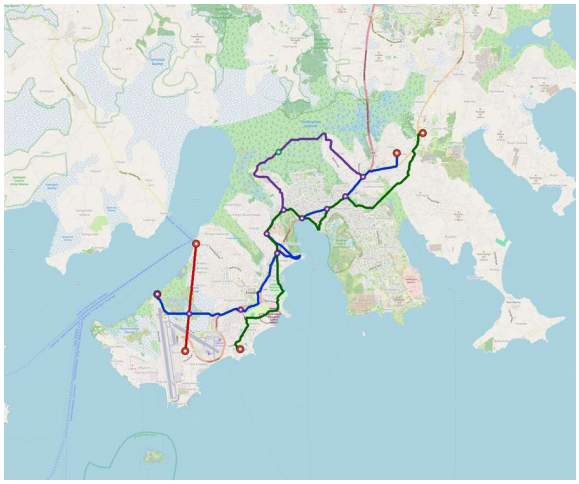
east-africa / Uganda / Entebbe

Metric	Value
Path	cities/catalogue/east-africa/Uganda/Entebbe
Recorded ISO / planning population	UG / 250,000
Family	tram-2car
Lines / stations	4 / 27
Route length	48.5 km
Fleet	110 trainsets
Routing-demand cells (legacy)	57%; not population access
City CAPEX incl. dedicated solar	USD 798M (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	87,078 (2020; 42.3% of raster bbox)

Metric	Value
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Entebbe network map

Line	Length	Stations	Trainsets	Shape
line-1	18.1 km	12	43	radial
line-2	4.5 km	3	12	radial
line-3	17.4 km	9	37	radial
line-4	8.6 km	3	18	radial

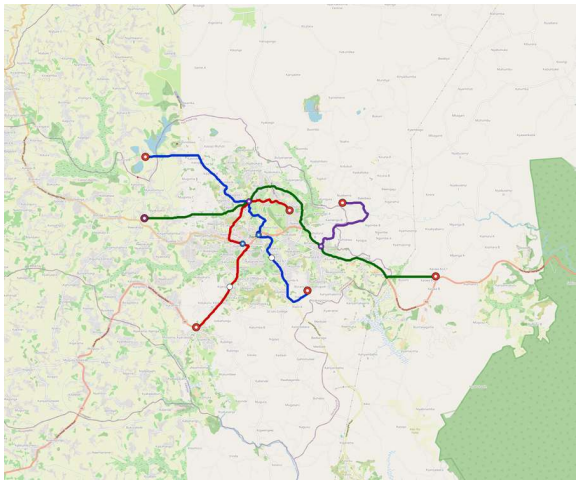
Generated City Model - Fort Portal

east-africa / Uganda / Fort-Portal

Metric	Value
Path	cities/catalogue/east-africa/Uganda/Fort-Portal
Recorded ISO / planning population	UG / 200,000
Family	tram-2car
Lines / stations	4 / 16
Route length	40.5 km
Fleet	85 trainsets
Routing-demand cells (legacy)	84%; not population access
City CAPEX incl. dedicated solar	USD 621M (USD 15M / km); shared national factory excluded
Residents within 800 m station circles	39,900 (2020; 26.5% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Fort Portal network map

Line	Length	Stations	Trainsets	Shape
line-1	12.2 km	5	26	radial
line-2	9.2 km	5	20	radial
line-3	14.9 km	4	29	radial
line-4	4.3 km	2	10	radial

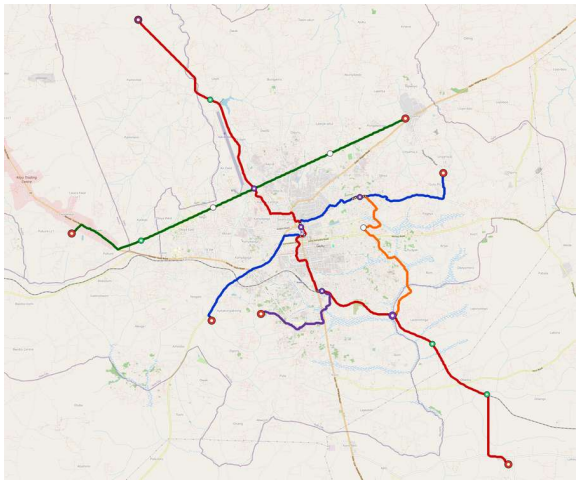
Generated City Model - Gulu

east-africa / Uganda / Gulu

Metric	Value
Path	cities/catalogue/east-africa/Uganda/Gulu
Recorded ISO / planning population	UG / 350,000
Family	light-metro-3car
Lines / stations	5 / 48
Route length	73.9 km
Fleet	261 trainsets
Routing-demand cells (legacy)	68%; not population access
City CAPEX incl. dedicated solar	USD 1.47bn (USD 20M / km); shared national factory excluded
Residents within 800 m station circles	70,641 (2020; 25.0% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Gulu network map

Line	Length	Stations	Trainsets	Shape
line-1	14.8 km	14	58	radial
line-2	29.3 km	16	100	radial
line-3	16.4 km	6	51	radial
line-4	5.2 km	4	18	radial
line-5	8.1 km	8	34	radial

Generated City Model - Hoima

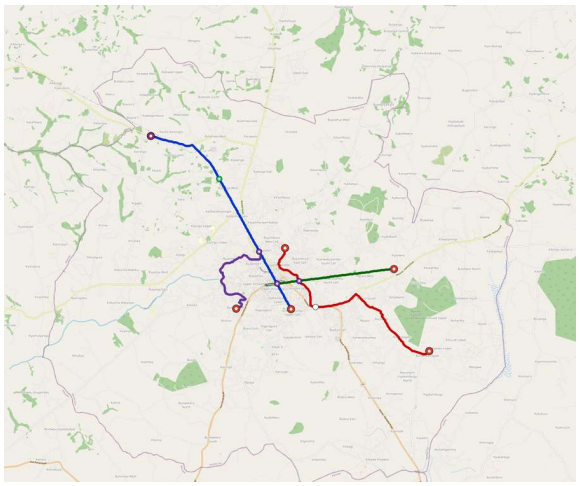
east-africa / Uganda / Hoima

Metric	Value
Path	cities/catalogue/east-africa/Uganda/Hoima
Recorded ISO / planning population	UG / 200,000
Family	tram-2car
Lines / stations	4 / 17
Route length	30.8 km
Fleet	71 trainsets
Routing-demand cells (legacy)	82%; not population access
City CAPEX incl. dedicated solar	USD 524M (USD 17M / km); shared national factory excluded

Metric	Value
Residents within 800 m station circles	48,536 (2020; 37.7% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Hoima network map

Line	Length	Stations	Trainsets	Shape
line-1	10.3 km	6	23	radial
line-2	9.7 km	4	20	radial
line-3	5.4 km	4	15	radial
line-4	5.4 km	3	13	radial

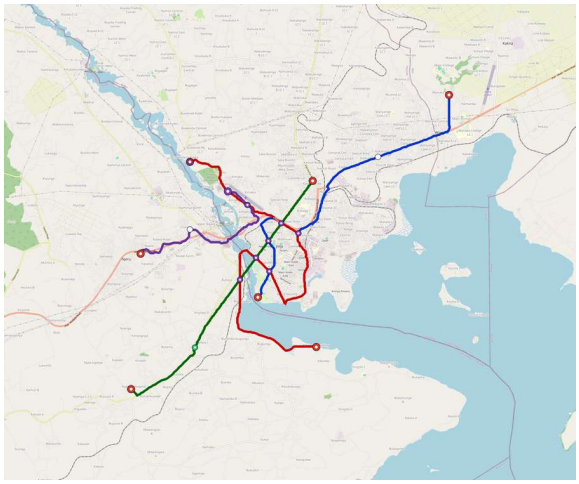
Generated City Model - Jinja

east-africa / Uganda / Jinja

Metric	Value
Path	cities/catalogue/east-africa/Uganda/Jinja
Recorded ISO / planning population	UG / 300,000
Family	tram-2car
Lines / stations	4 / 31
Route length	56.1 km
Fleet	121 trainsets
Routing-demand cells (legacy)	83%; not population access
City CAPEX incl. dedicated solar	USD 931M (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	92,904 (2020; 27.3% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Jinja network map

Line	Length	Stations	Trainsets	Shape
line-1	15.2 km	8	32	radial
line-2	19.9 km	11	43	radial
line-3	12.8 km	7	27	radial
line-4	8.2 km	5	19	radial

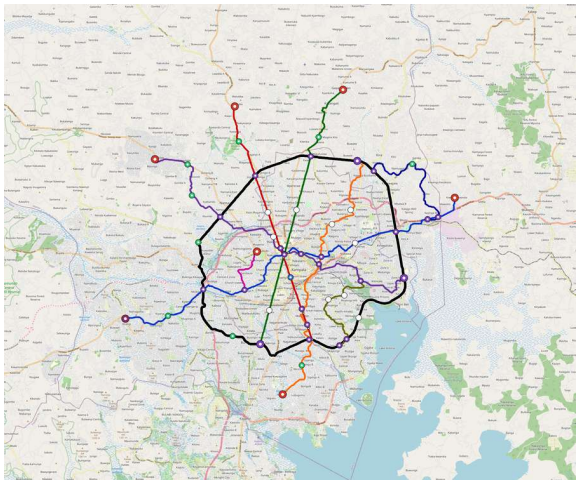
Generated City Model - Kampala

east-africa / Uganda / Kampala

Metric	Value
Path	cities/catalogue/east-africa/Uganda/Kampala
Recorded ISO / planning population	UG / 1,875,000
Family	metro-4car
Lines / stations	9 / 108
Route length	237.0 km
Fleet	361 trainsets
Routing-demand cells (legacy)	61%; not population access
City CAPEX incl. dedicated solar	USD 4.03bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	783,172 (2020; 20.9% of raster bbox)
Direct / reachable line pairs	52.8% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Kampala network map

Line	Length	Stations	Trainsets	Shape
line-1	37.2 km	18	70	radial
line-2	22.3 km	13	47	radial
line-3	25.0 km	9	40	radial

Line	Length	Stations	Trainsets	Shape
line-4	32.6 km	17	63	radial
line-5	30.8 km	12	51	radial
line-6	60.4 km	21	24	ring
line-7	4.6 km	2	10	radial
line-8	11.7 km	11	35	radial
line-9	12.5 km	5	21	radial

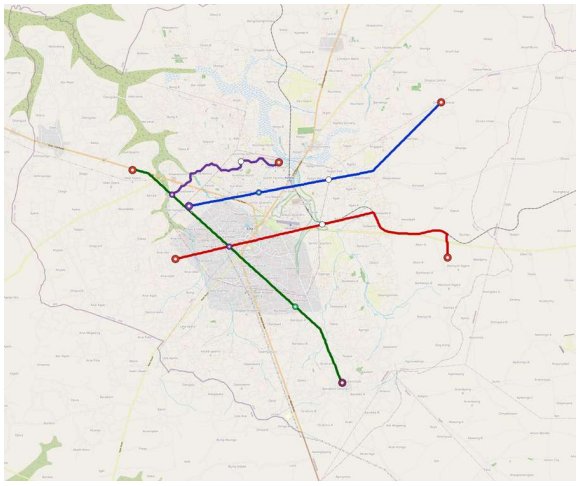
Generated City Model - Lira

east-africa / Uganda / Lira

Metric	Value
Path	cities/catalogue/east-africa/Uganda/Lira
Recorded ISO / planning population	UG / 250,000
Family	tram-2car
Lines / stations	4 / 17
Route length	44.6 km
Fleet	92 trainsets
Routing-demand cells (legacy)	58%; not population access
City CAPEX incl. dedicated solar	USD 628M (USD 14M / km); shared national factory excluded
Residents within 800 m station circles	67,839 (2020; 31.1% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Lira network map

Line	Length	Stations	Trainsets	Shape
line-1	11.8 km	4	25	radial
line-2	13.3 km	4	26	radial
line-3	13.7 km	6	28	radial
line-4	5.8 km	3	13	radial

Generated City Model - Masaka

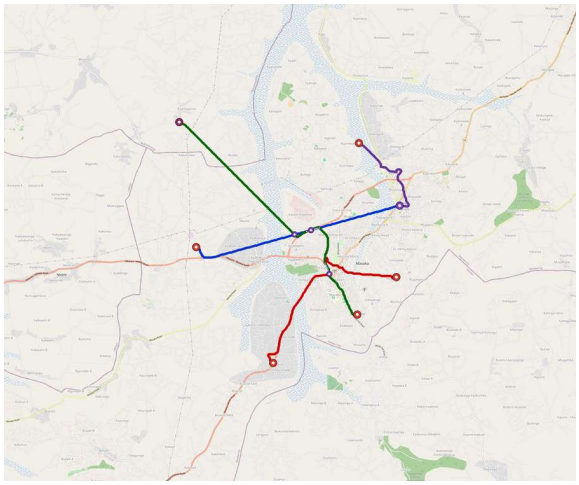
east-africa / Uganda / Masaka

Metric	Value
Path	cities/catalogue/east-africa/Uganda/Masaka
Recorded ISO / planning population	UG / 250,000
Family	tram-2car
Lines / stations	4 / 20
Route length	35.3 km
Fleet	83 trainsets
Routing-demand cells (legacy)	61%; not population access
City CAPEX incl. dedicated solar	USD 626M (USD 18M / km); shared national factory excluded

Metric	Value
Residents within 800 m station circles	51,068 (2020; 25.8% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Masaka network map

Line	Length	Stations	Trainsets	Shape
line-1	9.3 km	6	23	radial
line-2	9.3 km	4	20	radial
line-3	12.1 km	8	29	radial
line-4	4.5 km	2	11	radial

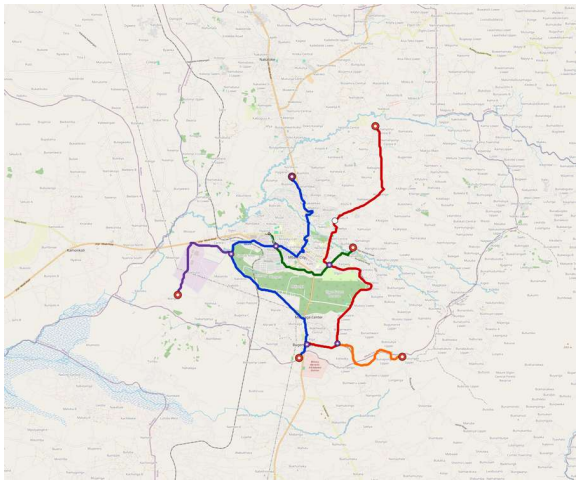
Generated City Model - Mbale

east-africa / Uganda / Mbale

Metric	Value
Path	cities/catalogue/east-africa/Uganda/Mbale
Recorded ISO / planning population	UG / 300,000
Family	tram-2car
Lines / stations	5 / 20
Route length	42.4 km
Fleet	95 trainsets
Routing-demand cells (legacy)	85%; not population access
City CAPEX incl. dedicated solar	USD 705M (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	82,297 (2020; 23.1% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Mbale network map

Line	Length	Stations	Trainsets	Shape
line-1	14.5 km	6	30	radial
line-2	14.4 km	6	30	radial
line-3	5.2 km	4	15	radial
line-4	4.2 km	2	10	radial
line-5	4.1 km	2	10	radial

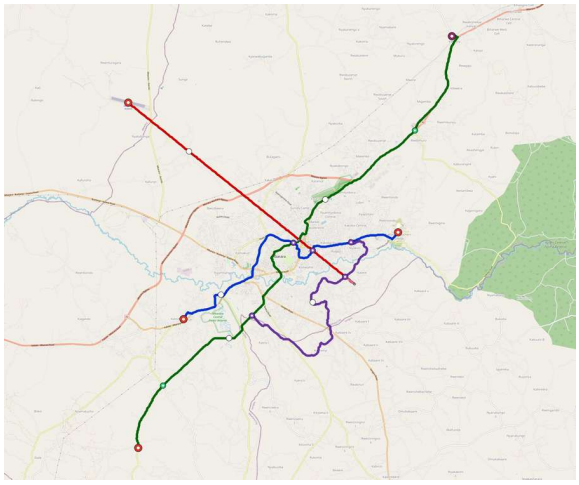
Generated City Model - Mbarara

east-africa / Uganda / Mbarara

Metric	Value
Path	cities/catalogue/east-africa/Uganda/Mbarara
Recorded ISO / planning population	UG / 500,000
Family	light-metro-3car
Lines / stations	4 / 26
Route length	64.6 km
Fleet	202 trainsets
Routing-demand cells (legacy)	77%; not population access
City CAPEX incl. dedicated solar	USD 1.15bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	71,924 (2020; 29.2% of raster bbox)
Direct / reachable line pairs	100.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Mbarara network map

Line	Length	Stations	Trainsets	Shape
line-1	12.9 km	6	40	radial
line-2	13.0 km	6	40	radial

Line	Length	Stations	Trainsets	Shape
line-3	26.7 km	9	84	radial
line-4	12.0 km	5	38	radial

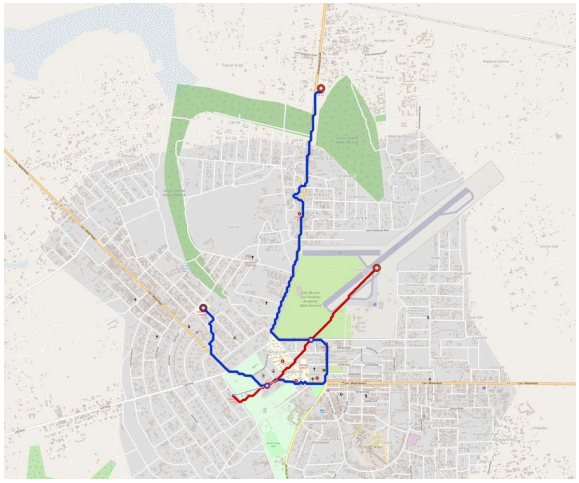
Generated City Model - Soroti

east-africa / Uganda / Soroti

Metric	Value
Path	cities/catalogue/east-africa/Uganda/Soroti
Recorded ISO / planning population	UG / 200,000
Family	tram-2car
Lines / stations	2 / 10
Route length	7.9 km
Fleet	30 trainsets
Routing-demand cells (legacy)	69%; not population access
City CAPEX incl. dedicated solar	USD 201M (USD 26M / km); shared national factory excluded
Residents within 800 m station circles	22,370 (2020; 15.6% of raster bbox)
Direct / reachable line pairs	100.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Soroti network map

Line	Length	Stations	Trainsets	Shape
line-1	5.8 km	5	17	radial
line-2	2.1 km	5	13	radial

Zambia

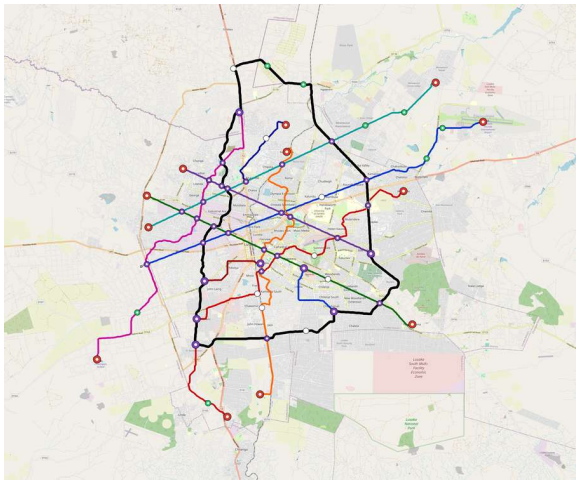
Generated City Model - Lusaka

east-africa / Zambia / Lusaka

Metric	Value
Path	cities/catalogue/east-africa/Zambia/Lusaka
Recorded ISO / planning population	ZM / 3,037,000
Family	metro-6car
Lines / stations	12 / 113
Route length	293.3 km
Fleet	472 trainsets
Routing-demand cells (legacy)	62%; not population access
City CAPEX incl. dedicated solar	USD 5.25bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	777,158 (2020; 24.5% of raster bbox)
Direct / reachable line pairs	42.4% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Lusaka network map

Line	Length	Stations	Trainsets	Shape
line-1	33.6 km	11	65	radial
line-2	29.2 km	11	54	radial
line-3	25.5 km	11	51	radial
line-4	18.5 km	13	50	radial
line-5	29.1 km	10	52	radial
line-6	27.7 km	11	56	radial
line-7	28.9 km	8	53	radial
line-8	74.9 km	28	35	ring
line-9	6.9 km	3	15	radial
line-10	5.7 km	2	12	radial
line-11	5.7 km	2	12	radial
line-12	7.6 km	3	17	radial

latin-america

Bolivia

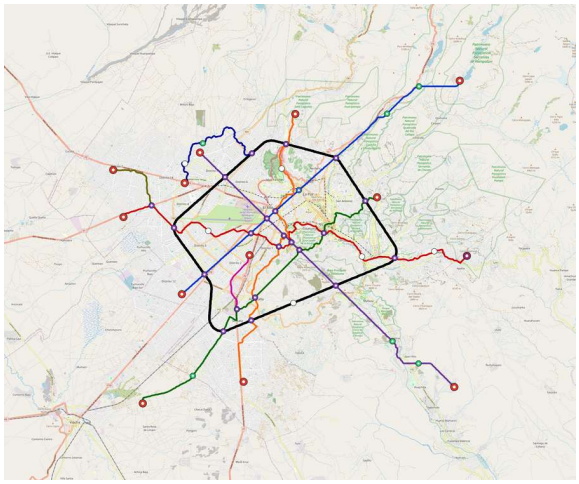
Generated City Model - La Paz

latin-america / Bolivia / La-Paz

Metric	Value
Path	cities/catalogue/latin-america/Bolivia/La-Paz
Recorded ISO / planning population	BO / 1,815,000
Family	metro-4car
Lines / stations	9 / 72
Route length	235.4 km
Fleet	317 trainsets
Routing-demand cells (legacy)	53%; not population access
City CAPEX incl. dedicated solar	USD 3.75bn (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	484,555 (2020; 25.8% of raster bbox)
Direct / reachable line pairs	47.2% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



La Paz network map

Line	Length	Stations	Trainsets	Shape
line-1	30.5 km	10	50	radial
line-2	37.3 km	12	58	radial
line-3	31.3 km	9	49	radial
line-4	30.3 km	10	49	radial
line-5	30.1 km	9	49	radial
line-6	53.1 km	13	20	ring
line-7	5.0 km	2	10	radial
line-8	5.3 km	2	11	radial
line-9	12.3 km	5	21	radial

Ecuador

Generated City Model - Cuenca

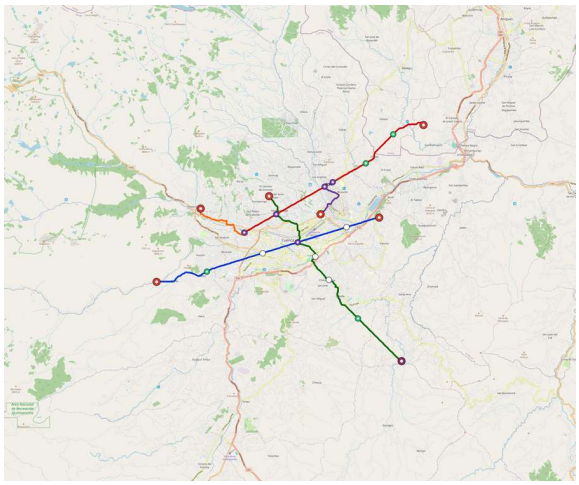
latin-america / Ecuador / Cuenca

Metric	Value
Path	cities/catalogue/latin-america/Ecuador/Cuenca
Recorded ISO / planning population	EC / 817,100
Family	light-metro-3car
Lines / stations	5 / 31

Metric	Value
Route length	74.2 km
Fleet	236 trainsets
Routing-demand cells (legacy)	48%; not population access
City CAPEX incl. dedicated solar	USD 1.34bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	131,723 (2020; 23.3% of raster bbox)
Direct / reachable line pairs	40.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Cuenca network map

Line	Length	Stations	Trainsets	Shape
line-1	21.1 km	6	65	radial
line-2	19.6 km	10	63	radial
line-3	21.9 km	7	67	radial
line-4	5.2 km	4	18	radial
line-5	6.4 km	4	23	radial

El Salvador

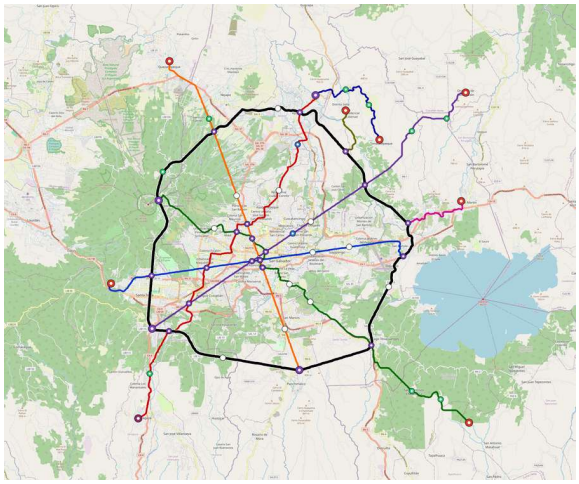
Generated City Model - San Salvador

latin-america / El Salvador / San-Salvador

Metric	Value
Path	cities/catalogue/latin-america/El Salvador/San-Salvador
Recorded ISO / planning population	SV / 1,800,000
Family	metro-4car
Lines / stations	9 / 91
Route length	271.7 km
Fleet	359 trainsets
Routing-demand cells (legacy)	52%; not population access
City CAPEX incl. dedicated solar	USD 4.25bn (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	561,682 (2020; 26.1% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



San Salvador network map

Line	Length	Stations	Trainsets	Shape
line-1	27.2 km	12	49	radial
line-2	40.9 km	11	62	radial
line-3	40.6 km	16	67	radial
line-4	35.9 km	11	57	radial
line-5	29.5 km	14	54	radial
line-6	74.8 km	19	28	ring
line-7	6.6 km	2	12	radial
line-8	5.1 km	2	10	radial
line-9	11.0 km	4	20	radial

north-africa

Morocco

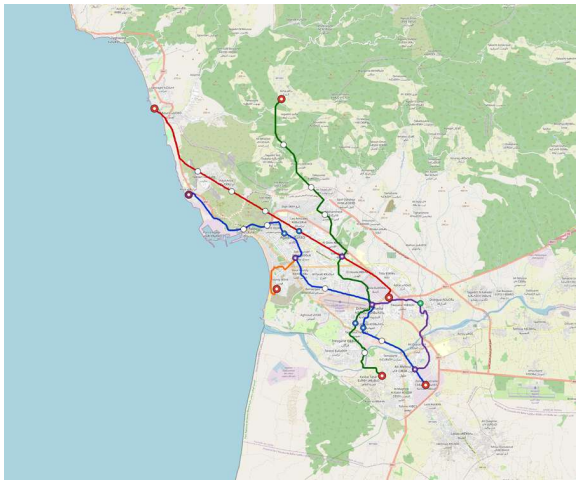
Generated City Model - Agadir

north-africa / Morocco / Agadir

Metric	Value
Path	cities/catalogue/north-africa/Morocco/Agadir
Recorded ISO / planning population	MA / 900,000
Family	light-metro-3car
Lines / stations	5 / 33
Route length	97.5 km
Fleet	298 trainsets
Routing-demand cells (legacy)	56%; not population access
City CAPEX incl. dedicated solar	USD 1.53bn (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	354,995 (2020; 37.2% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Agadir network map

Line	Length	Stations	Trainsets	Shape
line-1	29.3 km	12	90	radial
line-2	24.5 km	7	74	radial
line-3	28.7 km	9	86	radial
line-4	10.2 km	3	32	radial
line-5	4.9 km	2	16	radial

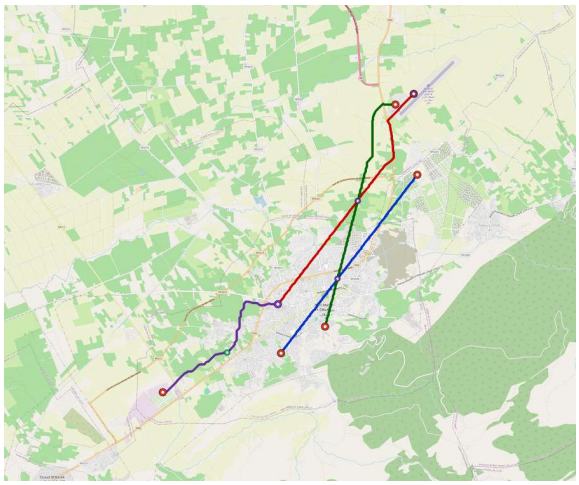
Generated City Model - Beni Mellal

north-africa / Morocco / Beni-Mellal

Metric	Value
Path	cities/catalogue/north-africa/Morocco/Beni-Mellal
Recorded ISO / planning population	MA / 300,000
Family	tram-2car
Lines / stations	4 / 13
Route length	33.0 km
Fleet	69 trainsets
Routing-demand cells (legacy)	80%; not population access
City CAPEX incl. dedicated solar	USD 469M (USD 14M / km); shared national factory excluded
Residents within 800 m station circles	104,097 (2020; 42.7% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Beni Mellal network map

Line	Length	Stations	Trainsets	Shape
line-1	8.5 km	3	18	radial
line-2	9.7 km	3	19	radial
line-3	8.8 km	4	18	radial
line-4	6.0 km	3	14	radial

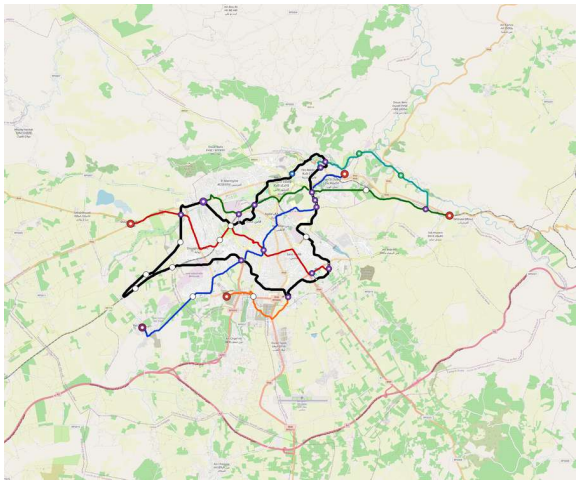
Generated City Model - Fez

north-africa / Morocco / Fez

Metric	Value
Path	cities/catalogue/north-africa/Morocco/Fez
Recorded ISO / planning population	MA / 1,300,000
Family	metro-4car
Lines / stations	6 / 85
Route length	142.0 km
Fleet	218 trainsets
Routing-demand cells (legacy)	73%; not population access
City CAPEX incl. dedicated solar	USD 2.63bn (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	526,707 (2020; 37.7% of raster bbox)
Direct / reachable line pairs	53.3% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Fez network map

Line	Length	Stations	Trainsets	Shape
line-1	24.5 km	23	71	radial
line-2	22.6 km	12	46	radial
line-3	21.5 km	7	34	radial
line-4	55.0 km	34	30	ring
line-5	6.6 km	3	14	radial
line-6	11.8 km	6	23	radial

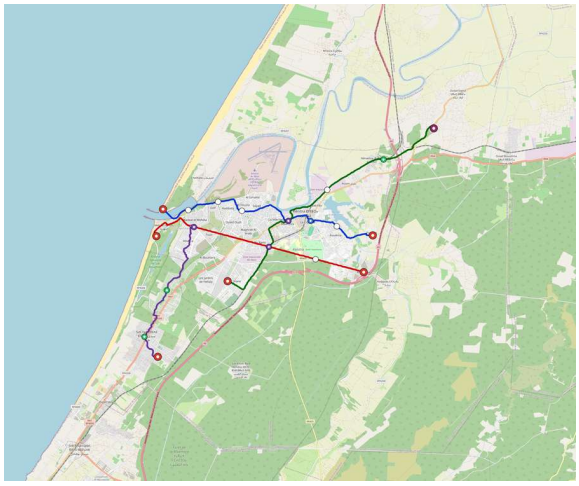
Generated City Model - Kenitra

north-africa / Morocco / Kenitra

Metric	Value
Path	cities/catalogue/north-africa/Morocco/Kenitra
Recorded ISO / planning population	MA / 500,000
Family	light-metro-3car
Lines / stations	4 / 36
Route length	70.7 km
Fleet	229 trainsets
Routing-demand cells (legacy)	59%; not population access
City CAPEX incl. dedicated solar	USD 1.23bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	221,104 (2020; 36.8% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Kenitra network map

Line	Length	Stations	Trainsets	Shape
line-1	17.8 km	13	58	radial
line-2	16.7 km	7	52	radial
line-3	24.0 km	10	78	radial
line-4	12.2 km	6	41	radial

Generated City Model - Khouribga

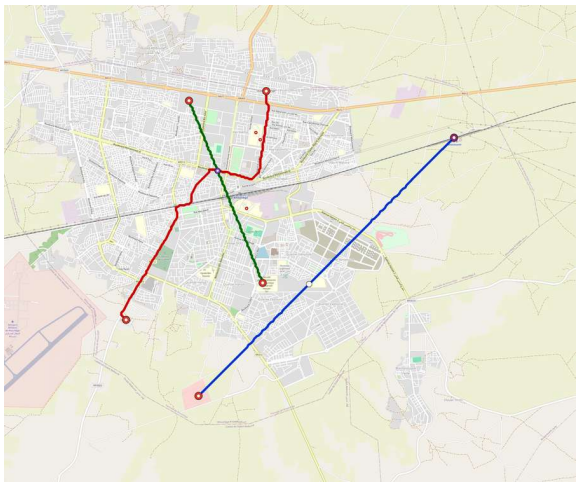
north-africa / Morocco / Khouribga

Metric	Value
Path	cities/catalogue/north-africa/Morocco/Khouribga
Recorded ISO / planning population	MA / 250,000
Family	tram-2car
Lines / stations	3 / 9
Route length	16.2 km
Fleet	39 trainsets
Routing-demand cells (legacy)	64%; not population access
City CAPEX incl. dedicated solar	USD 286M (USD 18M / km); shared national factory excluded

Metric	Value
Residents within 800 m station circles	82,676 (2020; 35.6% of raster bbox)
Direct / reachable line pairs	33.3% / 33.3%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Khouribga network map

Line	Length	Stations	Trainsets	Shape
line-1	6.8 km	3	15	radial
line-2	5.7 km	3	13	radial
line-3	3.7 km	3	11	radial

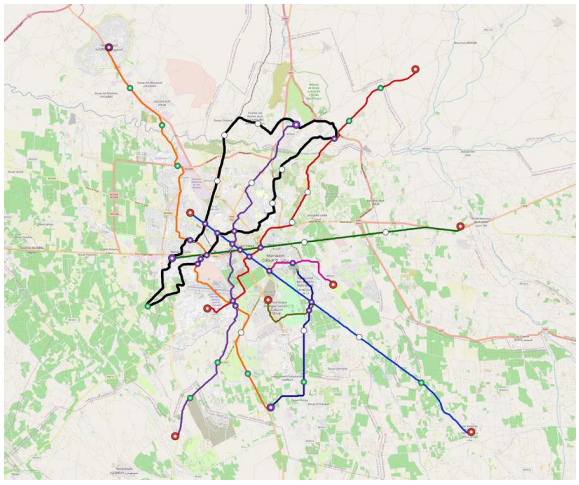
Generated City Model - Marrakech

north-africa / Morocco / Marrakech

Metric	Value
Path	cities/catalogue/north-africa/Morocco/Marrakech
Recorded ISO / planning population	MA / 1,200,000
Family	metro-4car
Lines / stations	9 / 128
Route length	219.7 km
Fleet	373 trainsets
Routing-demand cells (legacy)	72%; not population access
City CAPEX incl. dedicated solar	USD 3.89bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	569,176 (2020; 41.4% of raster bbox)
Direct / reachable line pairs	55.6% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Marrakech network map

Line	Length	Stations	Trainsets	Shape
line-1	27.8 km	12	49	radial
line-2	30.1 km	13	52	radial
line-3	21.1 km	9	38	radial
line-4	29.2 km	16	58	radial
line-5	35.8 km	33	98	radial
line-6	53.0 km	33	29	ring
line-7	5.9 km	3	13	radial
line-8	4.9 km	3	12	radial
line-9	11.9 km	6	24	radial

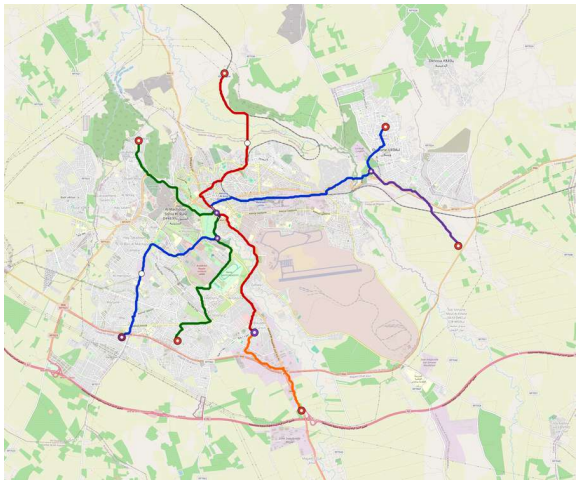
Generated City Model - Meknes

north-africa / Morocco / Meknes

Metric	Value
Path	cities/catalogue/north-africa/Morocco/Meknes
Recorded ISO / planning population	MA / 700,000
Family	light-metro-3car
Lines / stations	5 / 27
Route length	47.7 km
Fleet	158 trainsets
Routing-demand cells (legacy)	58%; not population access
City CAPEX incl. dedicated solar	USD 926M (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	194,413 (2020; 26.2% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Meknes network map

Line	Length	Stations	Trainsets	Shape
line-1	15.8 km	10	53	radial
line-2	12.2 km	7	39	radial
line-3	11.4 km	6	38	radial
line-4	4.2 km	2	14	radial
line-5	4.0 km	2	14	radial

Generated City Model - Nador

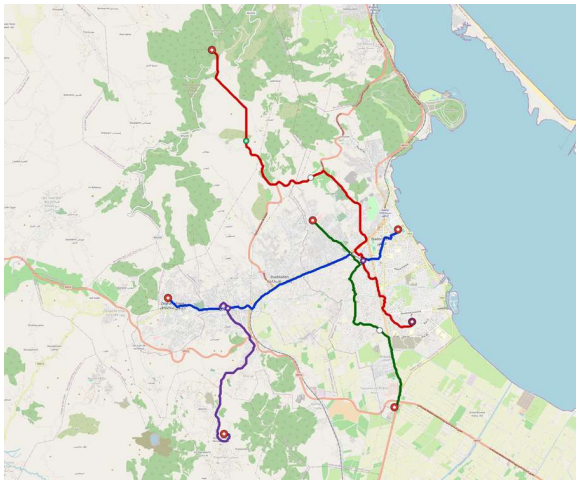
north-africa / Morocco / Nador

Metric	Value
Path	cities/catalogue/north-africa/Morocco/Nador
Recorded ISO / planning population	MA / 250,000
Family	tram-2car

Metric	Value
Lines / stations	4 / 22
Route length	40.8 km
Fleet	92 trainsets
Routing-demand cells (legacy)	70%; not population access
City CAPEX incl. dedicated solar	USD 664M (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	102,614 (2020; 41.4% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Nador network map

Line	Length	Stations	Trainsets	Shape
line-1	10.1 km	8	27	radial
line-2	15.4 km	6	31	radial
line-3	8.5 km	5	19	radial
line-4	6.9 km	3	15	radial

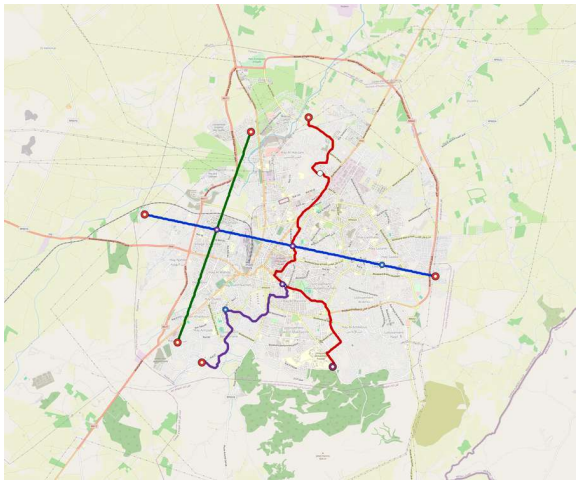
Generated City Model - Oujda

north-africa / Morocco / Oujda

Metric	Value
Path	cities/catalogue/north-africa/Morocco/Oujda
Recorded ISO / planning population	MA / 600,000
Family	light-metro-3car
Lines / stations	4 / 18
Route length	37.8 km
Fleet	123 trainsets
Routing-demand cells (legacy)	44%; not population access
City CAPEX incl. dedicated solar	USD 712M (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	177,254 (2020; 31.9% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Oujda network map

Line	Length	Stations	Trainsets	Shape
line-1	10.4 km	5	34	radial
line-2	12.8 km	6	40	radial
line-3	8.0 km	3	26	radial
line-4	6.6 km	4	23	radial

Generated City Model - Safi

north-africa / Morocco / Safi

Metric	Value
Path	cities/catalogue/north-africa/Morocco/Safi
Recorded ISO / planning population	MA / 350,000
Family	light-metro-3car
Lines / stations	4 / 20
Route length	45.3 km
Fleet	148 trainsets
Routing-demand cells (legacy)	65%; not population access
City CAPEX incl. dedicated solar	USD 790M (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	202,094 (2020; 55.1% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.

Safi network map

Line	Length	Stations	Trainsets	Shape
line-1	18.7 km	8	61	radial
line-2	10.4 km	5	34	radial
line-3	7.6 km	4	26	radial

Line	Length	Stations	Trainsets	Shape
line-4	8.6 km	3	27	radial

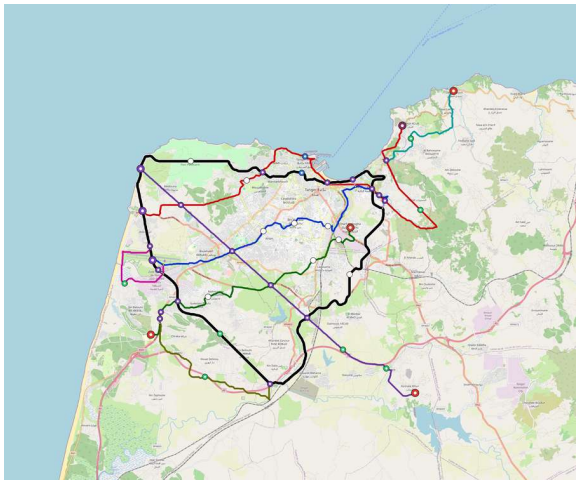
Generated City Model - Tangier

north-africa / Morocco / Tangier

Metric	Value
Path	cities/catalogue/north-africa/Morocco/Tangier
Recorded ISO / planning population	MA / 1,200,000
Family	metro-4car
Lines / stations	8 / 88
Route length	203.6 km
Fleet	299 trainsets
Routing-demand cells (legacy)	61%; not population access
City CAPEX incl. dedicated solar	USD 3.33bn (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	416,640 (2020; 34.4% of raster bbox)
Direct / reachable line pairs	46.4% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Tangier network map

Line	Length	Stations	Trainsets	Shape
line-1	25.0 km	16	57	radial
line-2	36.9 km	17	68	radial
line-3	22.7 km	11	43	radial
line-4	26.8 km	8	45	radial
line-5	61.1 km	21	25	ring
line-6	9.5 km	3	17	radial
line-7	8.5 km	6	20	radial
line-8	13.1 km	6	24	radial

Generated City Model - Tetouan

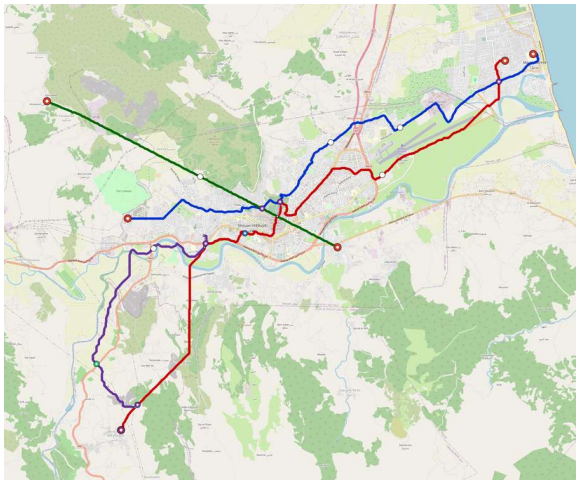
north-africa / Morocco / Tetouan

Metric	Value
Path	cities/catalogue/north-africa/Morocco/Tetouan
Recorded ISO / planning population	MA / 500,000
Family	light-metro-3car
Lines / stations	4 / 30
Route length	65.0 km
Fleet	205 trainsets
Routing-demand cells (legacy)	69%; not population access
City CAPEX incl. dedicated solar	USD 1.22bn (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	236,791 (2020; 41.8% of raster bbox)

Metric	Value
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Tetouan network map

Line	Length	Stations	Trainsets	Shape
line-1	18.6 km	7	59	radial
line-2	24.4 km	11	73	radial
line-3	11.7 km	5	38	radial
line-4	10.3 km	7	35	radial

Sudan

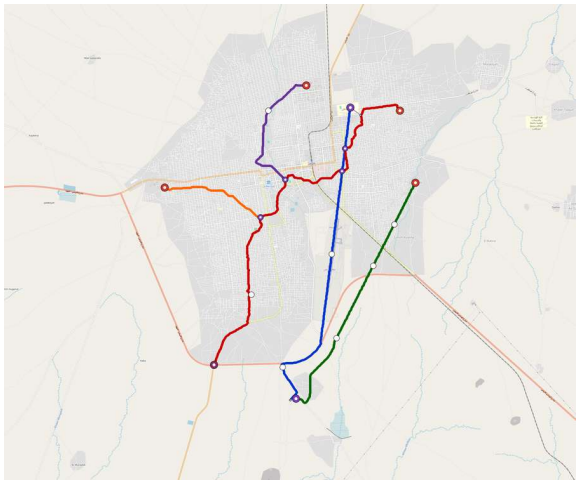
Generated City Model - El Obeid

north-africa / Sudan / El-Obeid

Metric	Value
Path	cities/catalogue/north-africa/Sudan/El-Obeid
Recorded ISO / planning population	SD / 500,000
Family	light-metro-3car
Lines / stations	5 / 25
Route length	51.6 km
Fleet	166 trainsets
Routing-demand cells (legacy)	57%; not population access
City CAPEX incl. dedicated solar	USD 908M (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	43,376 (2020; 16.7% of raster bbox)
Direct / reachable line pairs	40.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



El Obeid network map

Line	Length	Stations	Trainsets	Shape
line-1	13.4 km	6	45	radial
line-2	17.3 km	8	53	radial
line-3	11.0 km	5	34	radial
line-4	5.6 km	4	19	radial
line-5	4.3 km	2	15	radial

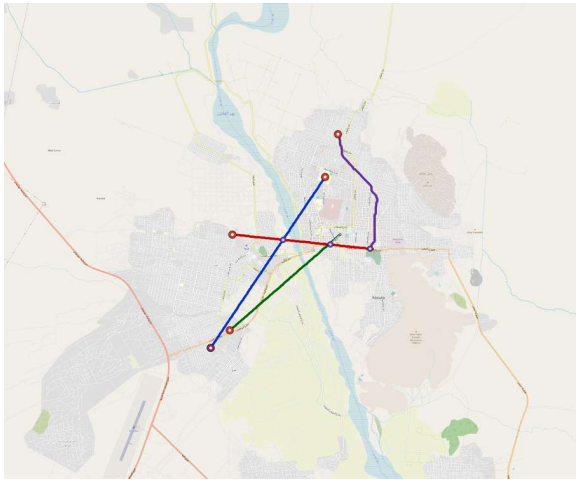
Generated City Model - Kassala

north-africa / Sudan / Kassala

Metric	Value
Path	cities/catalogue/north-africa/Sudan/Kassala
Recorded ISO / planning population	SD / 500,000
Family	light-metro-3car
Lines / stations	4 / 13
Route length	26.0 km
Fleet	86 trainsets
Routing-demand cells (legacy)	69%; not population access
City CAPEX incl. dedicated solar	USD 584M (USD 22M / km); shared national factory excluded
Residents within 800 m station circles	73,728 (2020; 15.0% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Kassala network map

Line	Length	Stations	Trainsets	Shape
line-1	8.9 km	3	27	radial
line-2	5.7 km	5	21	radial

Line	Length	Stations	Trainsets	Shape
line-3	6.3 km	3	20	radial
line-4	5.2 km	2	18	radial

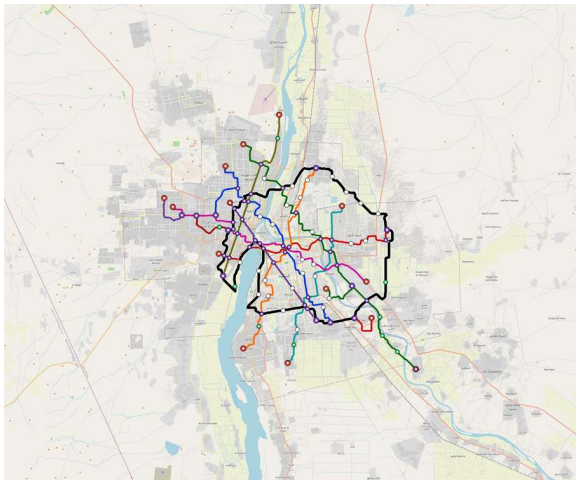
Generated City Model - Khartoum

north-africa / Sudan / Khartoum

Metric	Value
Path	cities/catalogue/north-africa/Sudan/Khartoum
Recorded ISO / planning population	SD / 5,829,000
Family	metro-6car
Lines / stations	14 / 181
Route length	468.1 km
Fleet	764 trainsets
Routing-demand cells (legacy)	54%; not population access
City CAPEX incl. dedicated solar	USD 8.48bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	624,587 (2020; 15.1% of raster bbox)
Direct / reachable line pairs	39.6% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Khartoum network map

Line	Length	Stations	Trainsets	Shape
line-1	43.4 km	19	86	radial
line-2	36.1 km	19	80	radial
line-3	58.7 km	16	111	radial
line-4	25.4 km	15	61	radial
line-5	44.0 km	19	86	radial
line-6	34.2 km	11	63	radial
line-7	43.2 km	20	89	radial
line-8	30.1 km	9	58	radial
line-9	118.7 km	41	57	ring
line-10	7.7 km	3	17	radial
line-11	6.7 km	2	14	radial
line-12	6.1 km	2	13	radial
line-13	7.2 km	3	15	radial
line-14	6.6 km	2	14	radial

Generated City Model - Nyala

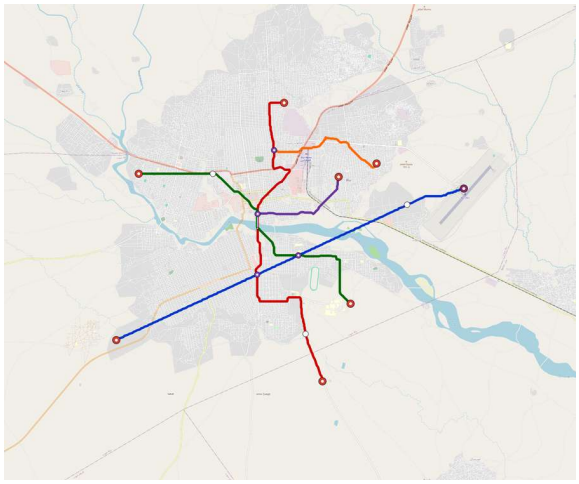
north-africa / Sudan / Nyala

Metric	Value
Path	cities/catalogue/north-africa/Sudan/Nyala
Recorded ISO / planning population	SD / 600,000

Metric	Value
Family	light-metro-3car
Lines / stations	5 / 21
Route length	52.9 km
Fleet	164 trainsets
Routing-demand cells (legacy)	58%; not population access
City CAPEX incl. dedicated solar	USD 1.01bn (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	35,207 (2020; 12.8% of raster bbox)
Direct / reachable line pairs	60.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Nyala network map

Line	Length	Stations	Trainsets	Shape
line-1	16.4 km	5	49	radial
line-2	15.3 km	7	47	radial
line-3	11.9 km	5	38	radial
line-4	4.2 km	2	14	radial
line-5	5.0 km	2	16	radial

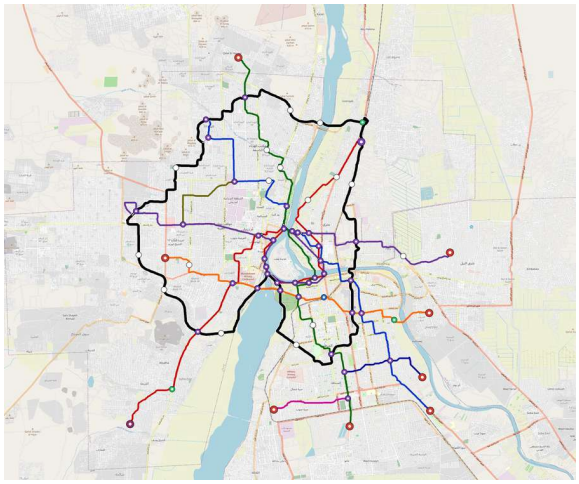
Generated City Model - Omdurman

north-africa / Sudan / Omdurman

Metric	Value
Path	cities/catalogue/north-africa/Sudan/Omdurman
Recorded ISO / planning population	SD / 2,800,000
Family	metro-4car
Lines / stations	9 / 168
Route length	313.4 km
Fleet	535 trainsets
Routing-demand cells (legacy)	59%; not population access
City CAPEX incl. dedicated solar	USD 5.51bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	442,625 (2020; 16.0% of raster bbox)
Direct / reachable line pairs	52.8% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Omdurman network map

Line	Length	Stations	Trainsets	Shape
line-1	41.7 km	29	98	radial
line-2	49.0 km	36	118	radial
line-3	42.1 km	14	65	radial
line-4	44.0 km	43	130	radial
line-5	26.8 km	13	50	radial
line-6	89.2 km	26	36	ring
line-7	6.4 km	2	11	radial
line-8	6.7 km	2	13	radial
line-9	7.4 km	3	14	radial

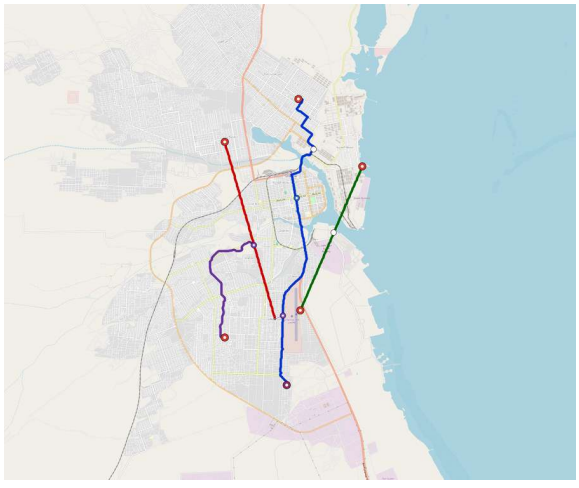
Generated City Model - Port Sudan

north-africa / Sudan / Port-Sudan

Metric	Value
Path	cities/catalogue/north-africa/Sudan/Port-Sudan
Recorded ISO / planning population	SD / 500,000
Family	light-metro-3car
Lines / stations	4 / 13
Route length	33.2 km
Fleet	105 trainsets
Routing-demand cells (legacy)	42%; not population access
City CAPEX incl. dedicated solar	USD 598M (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	61,247 (2020; 12.7% of raster bbox)
Direct / reachable line pairs	33.3% / 50.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Port Sudan network map

Line	Length	Stations	Trainsets	Shape
line-1	13.4 km	5	42	radial
line-2	7.5 km	3	24	radial
line-3	6.5 km	3	21	radial
line-4	5.6 km	2	18	radial

Tunisia

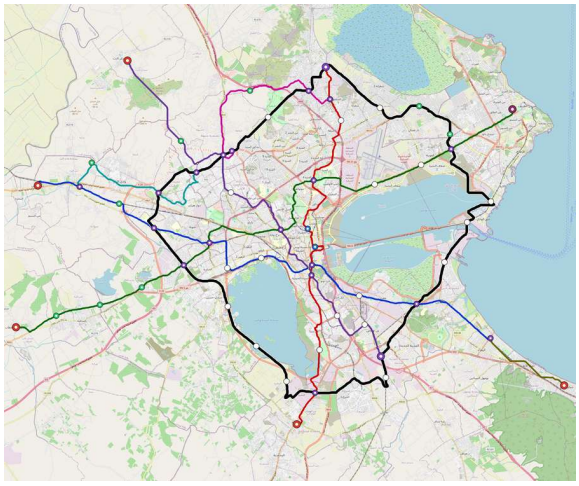
Generated City Model - Tunis

north-africa / Tunisia / Tunis

Metric	Value
Path	cities/catalogue/north-africa/Tunisia/Tunis
Recorded ISO / planning population	TN / 2,900,000
Family	metro-4car
Lines / stations	8 / 85
Route length	268.4 km
Fleet	344 trainsets
Routing-demand cells (legacy)	60%; not population access
City CAPEX incl. dedicated solar	USD 3.99bn (USD 15M / km); shared national factory excluded
Residents within 800 m station circles	609,677 (2020; 22.2% of raster bbox)
Direct / reachable line pairs	60.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Tunis network map

Line	Length	Stations	Trainsets	Shape
line-1	41.0 km	13	68	radial
line-2	32.0 km	13	56	radial
line-3	48.3 km	12	76	radial
line-4	33.3 km	13	56	radial
line-5	81.7 km	23	32	ring
line-6	12.4 km	4	20	radial
line-7	13.6 km	5	25	radial
line-8	6.1 km	2	11	radial

south-africa

South Africa

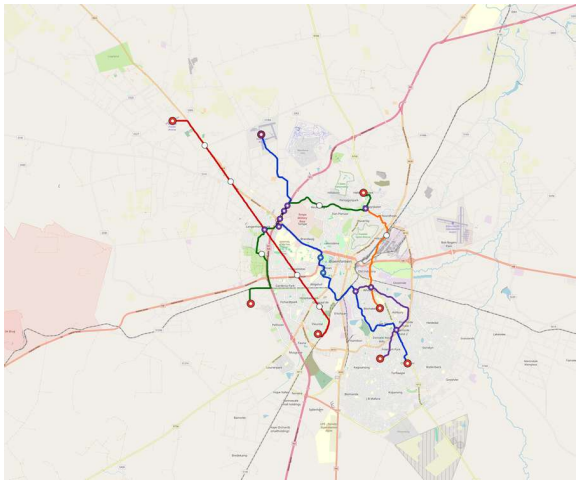
Generated City Model - Bloemfontein

south-africa / South Africa / Bloemfontein

Metric	Value
Path	cities/catalogue/south-africa/South Africa/Bloemfontein
Recorded ISO / planning population	ZA / 600,000
Family	light-metro-3car
Lines / stations	5 / 74
Route length	91.3 km
Fleet	354 trainsets
Routing-demand cells (legacy)	46%; not population access
City CAPEX incl. dedicated solar	USD 1.92bn (USD 21M / km); shared national factory excluded
Residents within 800 m station circles	125,625 (2020; 36.9% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Bloemfontein network map

Line	Length	Stations	Trainsets	Shape
line-1	30.3 km	29	117	radial
line-2	22.3 km	7	68	radial
line-3	20.6 km	30	109	radial
line-4	9.1 km	4	30	radial
line-5	9.0 km	4	30	radial

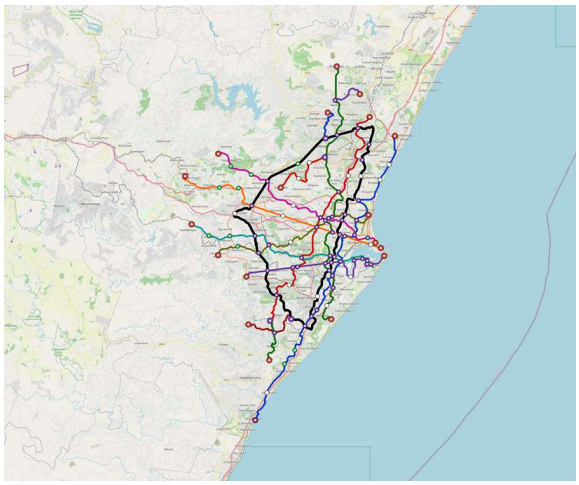
Generated City Model - Durban

south-africa / South Africa / Durban

Metric	Value
Path	cities/catalogue/south-africa/South Africa/Durban
Recorded ISO / planning population	ZA / 3,900,000
Family	metro-6car
Lines / stations	14 / 201
Route length	458.3 km
Fleet	812 trainsets
Routing-demand cells (legacy)	76%; not population access
City CAPEX incl. dedicated solar	USD 9.05bn (USD 20M / km); shared national factory excluded
Residents within 800 m station circles	709,731 (2020; 21.4% of raster bbox)
Direct / reachable line pairs	38.5% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Durban network map

Line	Length	Stations	Trainsets	Shape
line-1	62.1 km	22	119	radial
line-2	47.6 km	20	93	radial
line-3	52.1 km	28	116	radial
line-4	27.4 km	15	62	radial
line-5	35.9 km	12	65	radial
line-6	32.9 km	16	68	radial
line-7	34.1 km	18	73	radial
line-8	30.2 km	24	86	radial
line-9	92.8 km	29	42	ring
line-10	10.3 km	4	20	radial
line-11	6.3 km	2	13	radial
line-12	11.7 km	4	21	radial
line-13	8.4 km	4	19	radial
line-14	6.4 km	3	15	radial

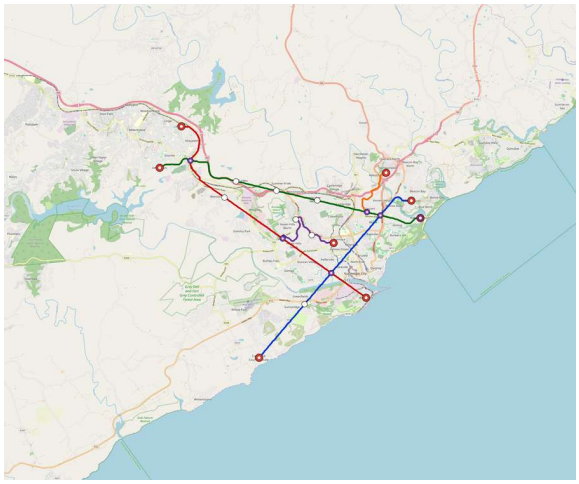
Generated City Model - East London Za

south-africa / South Africa / East-London-Za

Metric	Value
Path	cities/catalogue/south-africa/South Africa/East-London-Za
Recorded ISO / planning population	ZA / 800,000
Family	light-metro-3car
Lines / stations	5 / 26
Route length	70.8 km
Fleet	221 trainsets
Routing-demand cells (legacy)	40%; not population access
City CAPEX incl. dedicated solar	USD 1.25bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	91,582 (2020; 23.9% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



East London Za network map

Line	Length	Stations	Trainsets	Shape
line-1	17.5 km	6	52	radial
line-2	20.7 km	6	63	radial
line-3	21.0 km	8	67	radial
line-4	7.2 km	4	24	radial
line-5	4.4 km	2	15	radial

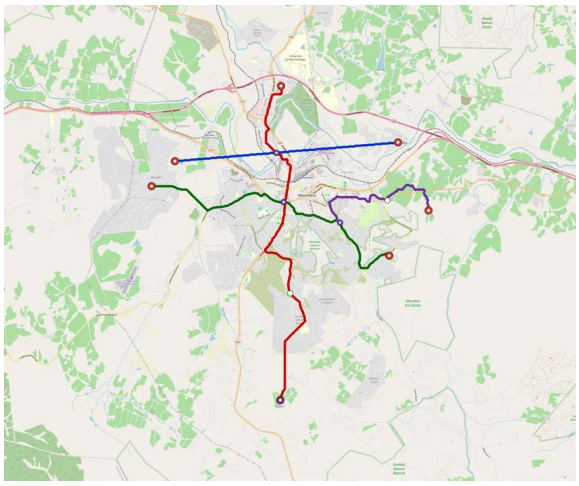
Generated City Model - Nelspruit

south-africa / South Africa / Nelspruit

Metric	Value
Path	cities/catalogue/south-africa/South Africa/Nelspruit
Recorded ISO / planning population	ZA / 300,000
Family	tram-2car
Lines / stations	4 / 17
Route length	40.6 km
Fleet	88 trainsets
Routing-demand cells (legacy)	55%; not population access
City CAPEX incl. dedicated solar	USD 654M (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	21,838 (2020; 27.5% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Nelspruit network map

Line	Length	Stations	Trainsets	Shape
line-1	8.4 km	3	18	radial
line-2	14.4 km	5	29	radial

Line	Length	Stations	Trainsets	Shape
line-3	11.4 km	5	25	radial
line-4	6.4 km	4	16	radial

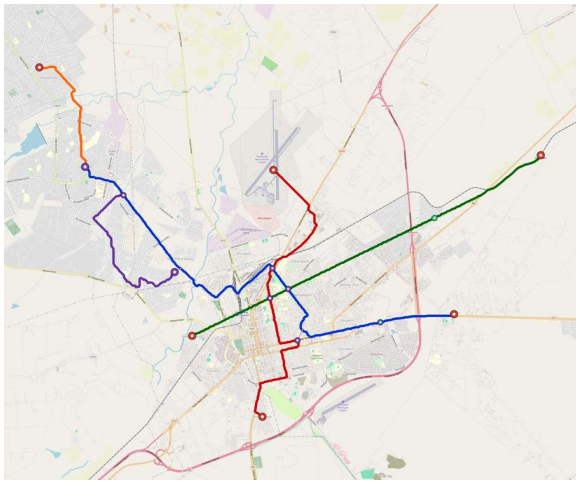
Generated City Model - Polokwane

south-africa / South Africa / Polokwane

Metric	Value
Path	cities/catalogue/south-africa/South Africa/Polokwane
Recorded ISO / planning population	ZA / 600,000
Family	light-metro-3car
Lines / stations	5 / 22
Route length	62.6 km
Fleet	199 trainsets
Routing-demand cells (legacy)	57%; not population access
City CAPEX incl. dedicated solar	USD 1.03bn (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	73,278 (2020; 25.9% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Polokwane network map

Line	Length	Stations	Trainsets	Shape
line-1	21.2 km	8	68	radial
line-2	13.6 km	5	43	radial
line-3	15.9 km	5	49	radial
line-4	7.0 km	2	23	radial
line-5	5.0 km	2	16	radial

south-asia

Afghanistan

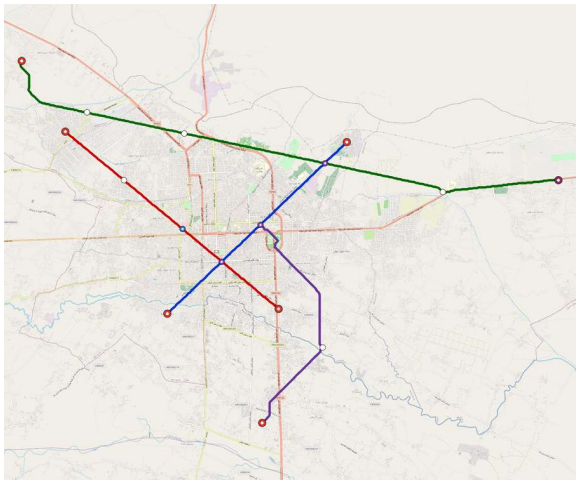
Generated City Model - Herat

south-asia / Afghanistan / Herat

Metric	Value
Path	cities/catalogue/south-asia/Afghanistan/Herat
Recorded ISO / planning population	AF / 800,000
Family	light-metro-3car
Lines / stations	4 / 19
Route length	48.6 km
Fleet	153 trainsets
Routing-demand cells (legacy)	51%; not population access
City CAPEX incl. dedicated solar	USD 823M (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	94,679 (2020; 13.5% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Herat network map

Line	Length	Stations	Trainsets	Shape
line-1	9.2 km	5	29	radial
line-2	10.3 km	5	34	radial
line-3	20.7 km	6	63	radial
line-4	8.4 km	3	27	radial

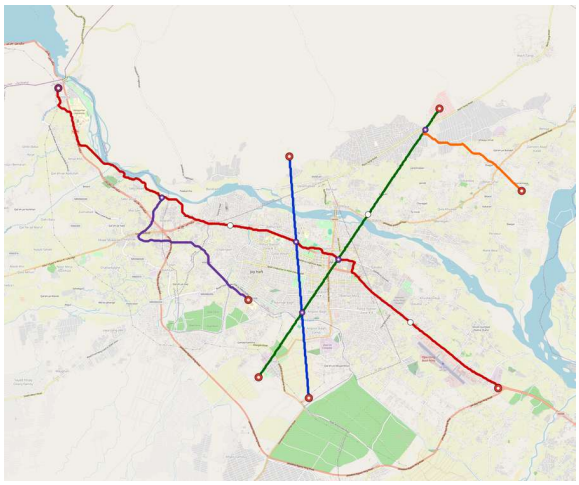
Generated City Model - Jalalabad Af

south-asia / Afghanistan / Jalalabad-Af

Metric	Value
Path	cities/catalogue/south-asia/Afghanistan/Jalalabad-Af
Recorded ISO / planning population	AF / 350,000
Family	light-metro-3car
Lines / stations	5 / 23
Route length	53.8 km
Fleet	170 trainsets
Routing-demand cells (legacy)	60%; not population access
City CAPEX incl. dedicated solar	USD 984M (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	148,581 (2020; 25.6% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Jalalabad Af network map

Line	Length	Stations	Trainsets	Shape
line-1	8.2 km	4	27	radial
line-2	23.0 km	7	69	radial
line-3	11.9 km	7	39	radial
line-4	6.4 km	2	20	radial
line-5	4.2 km	3	15	radial

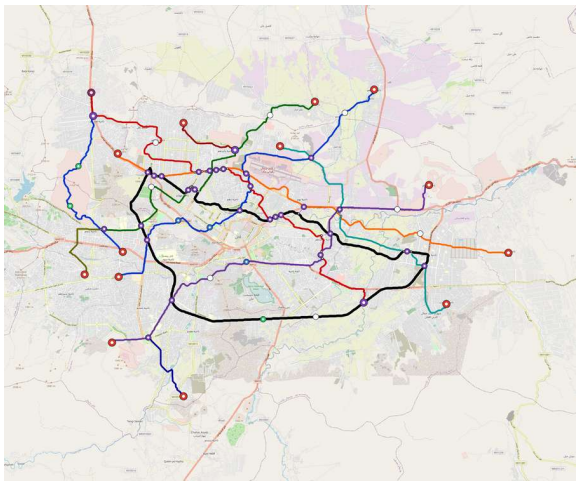
Generated City Model - Kabul

south-asia / Afghanistan / Kabul

Metric	Value
Path	cities/catalogue/south-asia/Afghanistan/Kabul
Recorded ISO / planning population	AF / 4,601,000
Family	metro-6car
Lines / stations	11 / 205
Route length	260.1 km
Fleet	624 trainsets
Routing-demand cells (legacy)	57%; not population access
City CAPEX incl. dedicated solar	USD 6.02bn (USD 23M / km); shared national factory excluded
Residents within 800 m station circles	1,183,877 (2020; 28.6% of raster bbox)
Direct / reachable line pairs	40.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Kabul network map

Line	Length	Stations	Trainsets	Shape
line-1	30.0 km	16	68	radial
line-2	34.4 km	40	130	radial
line-3	24.9 km	23	80	radial
line-4	30.2 km	19	74	radial
line-5	31.7 km	34	113	radial
line-6	22.7 km	11	49	radial
line-7	56.9 km	50	45	ring
line-8	5.6 km	3	14	radial
line-9	5.4 km	2	12	radial
line-10	4.9 km	2	11	radial
line-11	13.4 km	5	28	radial

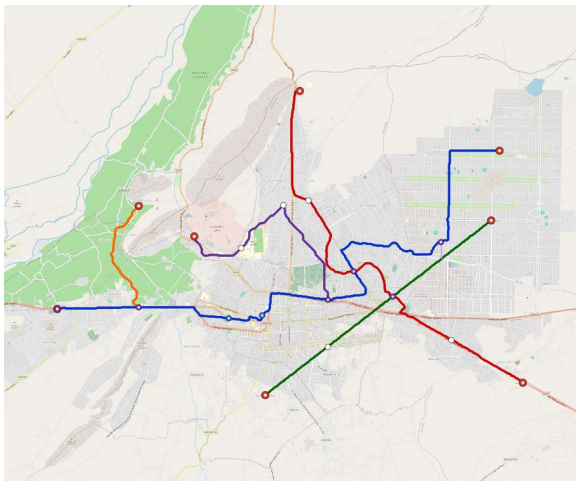
Generated City Model - Kandahar

south-asia / Afghanistan / Kandahar

Metric	Value
Path	cities/catalogue/south-asia/Afghanistan/Kandahar
Recorded ISO / planning population	AF / 700,000
Family	light-metro-3car
Lines / stations	5 / 26
Route length	64.8 km
Fleet	204 trainsets
Routing-demand cells (legacy)	64%; not population access
City CAPEX incl. dedicated solar	USD 1.10bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	169,786 (2020; 29.8% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Kandahar network map

Line	Length	Stations	Trainsets	Shape
line-1	24.5 km	9	76	radial
line-2	16.5 km	6	51	radial
line-3	10.9 km	5	34	radial
line-4	8.3 km	4	27	radial
line-5	4.5 km	2	16	radial

Generated City Model - Mazar E Sharif

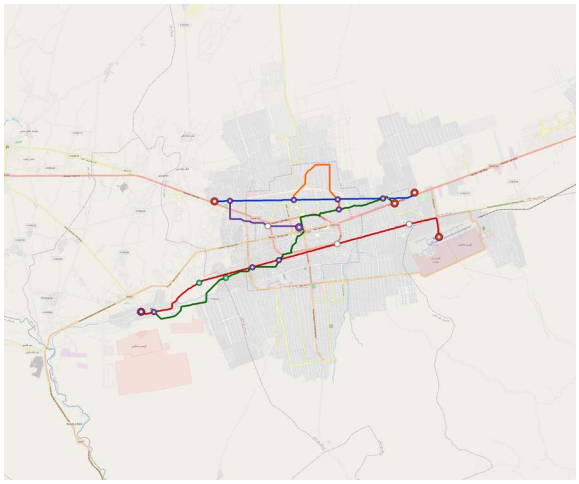
south-asia / Afghanistan / Mazar-E-Sharif

Metric	Value
Path	cities/catalogue/south-asia/Afghanistan/Mazar-E-Sharif
Recorded ISO / planning population	AF / 600,000

Metric	Value
Family	light-metro-3car
Lines / stations	5 / 35
Route length	73.9 km
Fleet	237 trainsets
Routing-demand cells (legacy)	58%; not population access
City CAPEX incl. dedicated solar	USD 1.29bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	182,917 (2020; 39.8% of raster bbox)
Direct / reachable line pairs	60.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Mazar E Sharif network map

Line	Length	Stations	Trainsets	Shape
line-1	13.2 km	8	43	radial
line-2	24.2 km	9	76	radial
line-3	23.8 km	12	74	radial
line-4	5.8 km	3	20	radial
line-5	6.9 km	3	24	radial

Bangladesh

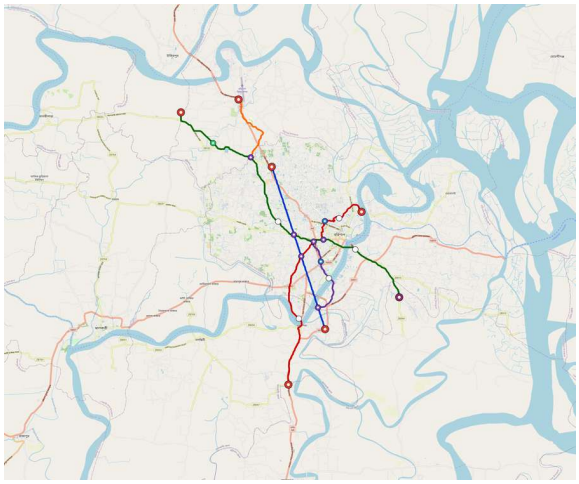
Generated City Model - Barisal

south-asia / Bangladesh / Barisal

Metric	Value
Path	cities/catalogue/south-asia/Bangladesh/Barisal
Recorded ISO / planning population	BD / 550,000
Family	light-metro-3car
Lines / stations	5 / 33
Route length	75.5 km
Fleet	241 trainsets
Routing-demand cells (legacy)	57%; not population access
City CAPEX incl. dedicated solar	USD 1.41bn (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	311,319 (2020; 41.1% of raster bbox)
Direct / reachable line pairs	70.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Barisal network map

Line	Length	Stations	Trainsets	Shape
line-1	13.7 km	5	43	radial
line-2	21.4 km	11	70	radial
line-3	28.1 km	11	87	radial
line-4	6.4 km	4	23	radial
line-5	5.9 km	2	18	radial

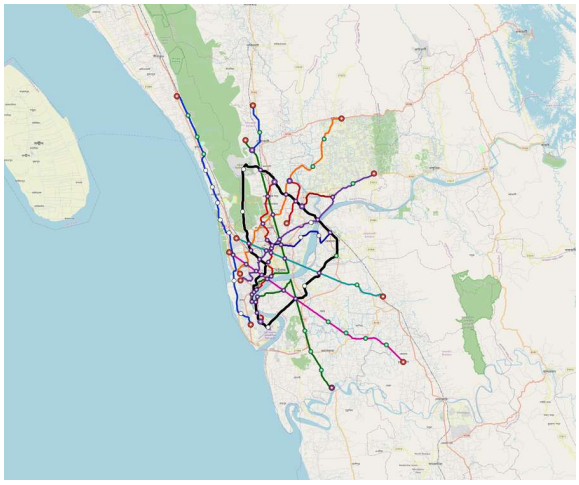
Generated City Model - Chittagong

south-asia / Bangladesh / Chittagong

Metric	Value
Path	cities/catalogue/south-asia/Bangladesh/Chittagong
Recorded ISO / planning population	BD / 5,200,000
Family	metro-6car
Lines / stations	12 / 191
Route length	380.6 km
Fleet	725 trainsets
Routing-demand cells (legacy)	75%; not population access
City CAPEX incl. dedicated solar	USD 7.79bn (USD 20M / km); shared national factory excluded
Residents within 800 m station circles	1,967,609 (2020; 35.7% of raster bbox)
Direct / reachable line pairs	43.9% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Chittagong network map

Line	Length	Stations	Trainsets	Shape
line-1	47.1 km	13	87	radial
line-2	31.9 km	35	115	radial

Line	Length	Stations	Trainsets	Shape
line-3	61.2 km	40	150	radial
line-4	34.1 km	15	68	radial
line-5	36.6 km	14	72	radial
line-6	26.5 km	9	50	radial
line-7	34.4 km	14	68	radial
line-8	72.8 km	36	39	ring
line-9	13.5 km	7	30	radial
line-10	6.4 km	2	13	radial
line-11	7.8 km	3	17	radial
line-12	8.4 km	3	16	radial

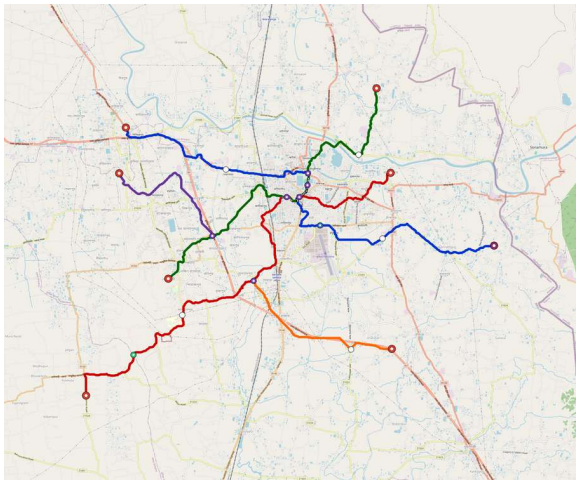
Generated City Model - Comilla

south-asia / Bangladesh / Comilla

Metric	Value
Path	cities/catalogue/south-asia/Bangladesh/Comilla
Recorded ISO / planning population	BD / 600,000
Family	light-metro-3car
Lines / stations	5 / 62
Route length	71.8 km
Fleet	277 trainsets
Routing-demand cells (legacy)	70%; not population access
City CAPEX incl. dedicated solar	USD 1.63bn (USD 23M / km); shared national factory excluded
Residents within 800 m station circles	319,162 (2020; 23.8% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Comilla network map

Line	Length	Stations	Trainsets	Shape
line-1	22.1 km	20	83	radial
line-2	22.0 km	15	75	radial
line-3	15.1 km	22	80	radial
line-4	5.9 km	2	18	radial
line-5	6.7 km	3	21	radial

Generated City Model - Gazipur

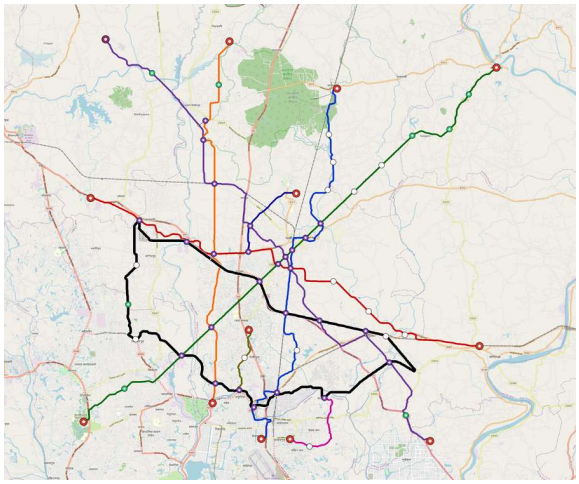
south-asia / Bangladesh / Gazipur

Metric	Value
Path	cities/catalogue/south-asia/Bangladesh/Gazipur
Recorded ISO / planning population	BD / 1,400,000
Family	metro-4car

Metric	Value
Lines / stations	9 / 97
Route length	289.7 km
Fleet	394 trainsets
Routing-demand cells (legacy)	47%; not population access
City CAPEX incl. dedicated solar	USD 4.40bn (USD 15M / km); shared national factory excluded
Residents within 800 m station circles	1,123,643 (2020; 20.7% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Gazipur network map

Line	Length	Stations	Trainsets	Shape
line-1	39.7 km	17	70	radial
line-2	36.2 km	11	54	radial
line-3	46.9 km	16	75	radial
line-4	50.5 km	15	79	radial
line-5	30.4 km	9	48	radial
line-6	66.5 km	20	27	ring
line-7	7.5 km	3	14	radial
line-8	5.5 km	3	13	radial
line-9	6.6 km	3	14	radial

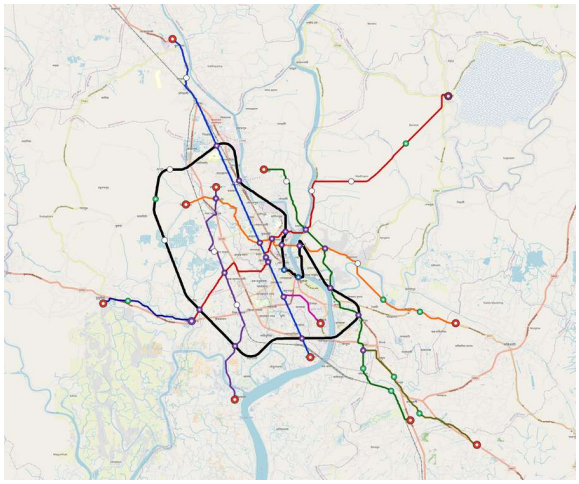
Generated City Model - Khulna

south-asia / Bangladesh / Khulna

Metric	Value
Path	cities/catalogue/south-asia/Bangladesh/Khulna
Recorded ISO / planning population	BD / 1,500,000
Family	metro-4car
Lines / stations	9 / 82
Route length	223.7 km
Fleet	313 trainsets
Routing-demand cells (legacy)	60%; not population access
City CAPEX incl. dedicated solar	USD 3.74bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	513,642 (2020; 30.6% of raster bbox)
Direct / reachable line pairs	41.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Khulna network map

Line	Length	Stations	Trainsets	Shape
line-1	28.7 km	12	51	radial
line-2	33.1 km	15	60	radial
line-3	28.5 km	10	48	radial
line-4	21.2 km	7	34	radial
line-5	29.2 km	13	51	radial
line-6	59.1 km	16	24	ring
line-7	4.0 km	2	10	radial
line-8	12.3 km	4	21	radial
line-9	7.5 km	3	14	radial

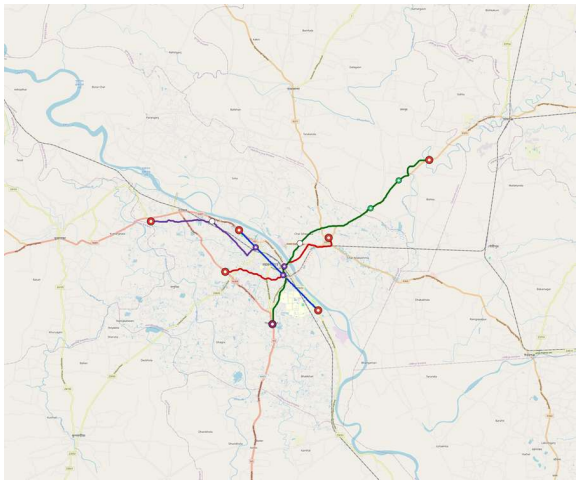
Generated City Model - Mymensingh

south-asia / Bangladesh / Mymensingh

Metric	Value
Path	cities/catalogue/south-asia/Bangladesh/Mymensingh
Recorded ISO / planning population	BD / 700,000
Family	light-metro-3car
Lines / stations	4 / 28
Route length	51.0 km
Fleet	164 trainsets
Routing-demand cells (legacy)	68%; not population access
City CAPEX incl. dedicated solar	USD 1.13bn (USD 22M / km); shared national factory excluded
Residents within 800 m station circles	162,177 (2020; 14.8% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Mymensingh network map

Line	Length	Stations	Trainsets	Shape
line-1	9.2 km	5	29	radial
line-2	11.0 km	8	37	radial
line-3	21.4 km	12	69	radial
line-4	9.4 km	3	29	radial

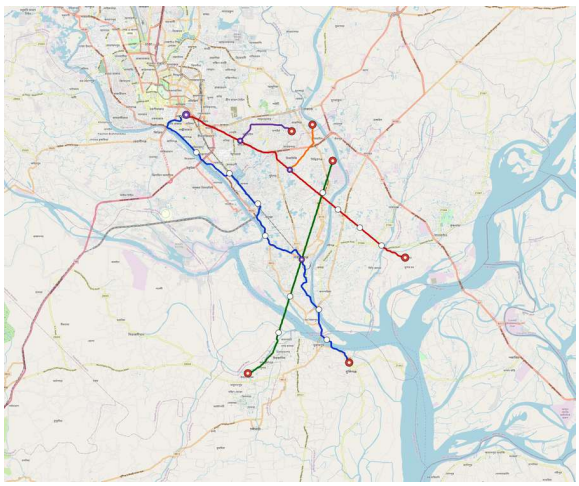
Generated City Model - Narayanganj

south-asia / Bangladesh / Narayanganj

Metric	Value
Path	cities/catalogue/south-asia/Bangladesh/Narayanganj
Recorded ISO / planning population	BD / 950,000
Family	light-metro-3car
Lines / stations	5 / 26
Route length	81.1 km
Fleet	251 trainsets
Routing-demand cells (legacy)	52%; not population access
City CAPEX incl. dedicated solar	USD 1.41bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	1,277,967 (2020; 21.1% of raster bbox)
Direct / reachable line pairs	40.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Narayanganj network map

Line	Length	Stations	Trainsets	Shape
line-1	30.1 km	9	93	radial
line-2	23.0 km	7	69	radial
line-3	19.0 km	6	58	radial
line-4	4.8 km	2	16	radial
line-5	4.3 km	2	15	radial

Generated City Model - Rajshahi

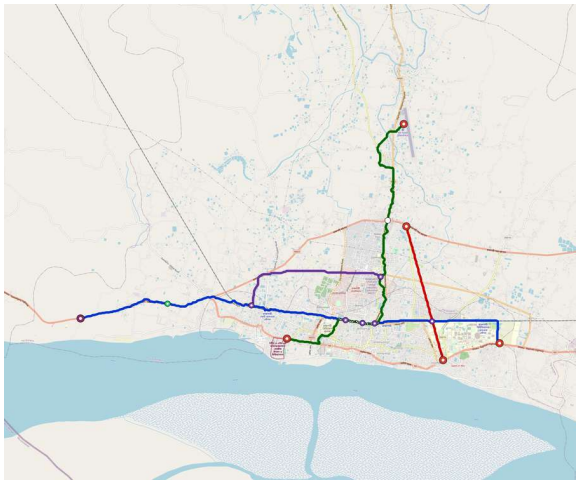
south-asia / Bangladesh / Rajshahi

Metric	Value
Path	cities/catalogue/south-asia/Bangladesh/Rajshahi
Recorded ISO / planning population	BD / 950,000
Family	light-metro-3car
Lines / stations	4 / 51
Route length	44.7 km
Fleet	203 trainsets
Routing-demand cells (legacy)	61%; not population access
City CAPEX incl. dedicated solar	USD 1.13bn (USD 25M / km); shared national factory excluded
Residents within 800 m station circles	246,705 (2020; 27.5% of raster bbox)

Metric	Value
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Rajshahi network map

Line	Length	Stations	Trainsets	Shape
line-1	19.4 km	23	86	radial
line-2	5.4 km	3	18	radial
line-3	14.2 km	22	79	radial
line-4	5.8 km	3	20	radial

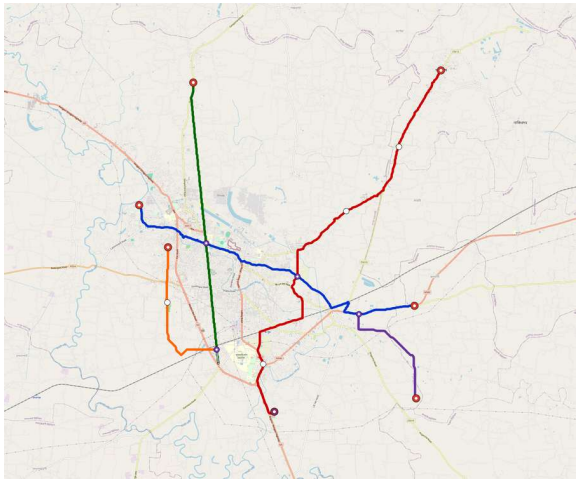
Generated City Model - Rangpur

south-asia / Bangladesh / Rangpur

Metric	Value
Path	cities/catalogue/south-asia/Bangladesh/Rangpur
Recorded ISO / planning population	BD / 800,000
Family	light-metro-3car
Lines / stations	5 / 20
Route length	51.3 km
Fleet	164 trainsets
Routing-demand cells (legacy)	71%; not population access
City CAPEX incl. dedicated solar	USD 917M (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	202,506 (2020; 17.3% of raster bbox)
Direct / reachable line pairs	40.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Rangpur network map

Line	Length	Stations	Trainsets	Shape
line-1	13.0 km	5	42	radial
line-2	17.7 km	6	53	radial
line-3	10.6 km	4	35	radial
line-4	4.5 km	2	16	radial
line-5	5.5 km	3	18	radial

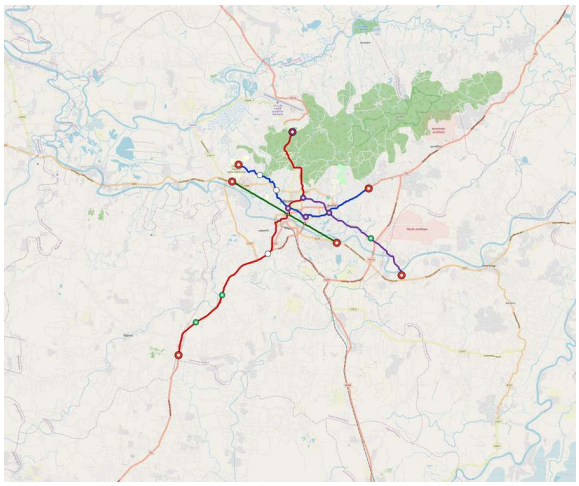
Generated City Model - Sylhet

south-asia / Bangladesh / Sylhet

Metric	Value
Path	cities/catalogue/south-asia/Bangladesh/Sylhet
Recorded ISO / planning population	BD / 900,000
Family	light-metro-3car
Lines / stations	4 / 23
Route length	60.9 km
Fleet	192 trainsets
Routing-demand cells (legacy)	83%; not population access
City CAPEX incl. dedicated solar	USD 1.12bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	358,949 (2020; 25.6% of raster bbox)
Direct / reachable line pairs	83.3% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Sylhet network map

Line	Length	Stations	Trainsets	Shape
line-1	14.2 km	7	46	radial
line-2	26.6 km	8	81	radial
line-3	9.7 km	4	31	radial
line-4	10.4 km	4	34	radial

India

Generated City Model - Agra

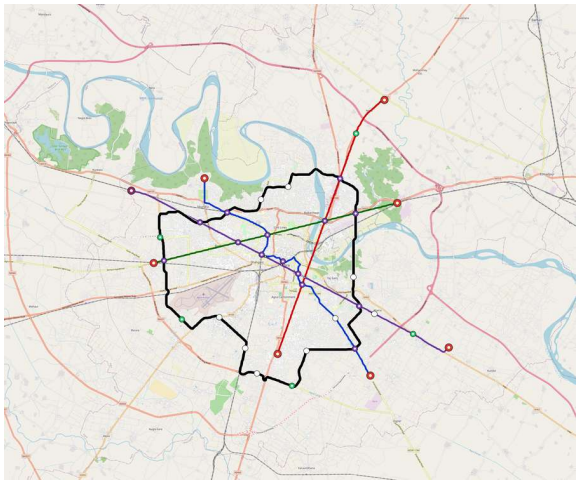
south-asia / India / Agra

Metric	Value
Path	cities/catalogue/south-asia/India/Agra
Recorded ISO / planning population	IN / 1,700,000
Family	metro-4car
Lines / stations	5 / 54
Route length	158.6 km

Metric	Value
Fleet	191 trainsets
Routing-demand cells (legacy)	47%; not population access
City CAPEX incl. dedicated solar	USD 2.52bn (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	Unavailable native population evidence
Direct / reachable line pairs	100.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Agra network map

Line	Length	Stations	Trainsets	Shape
line-1	26.1 km	11	47	radial
line-2	21.9 km	7	36	radial
line-3	19.2 km	7	32	radial
line-4	28.5 km	12	50	radial
line-5	62.8 km	17	26	ring

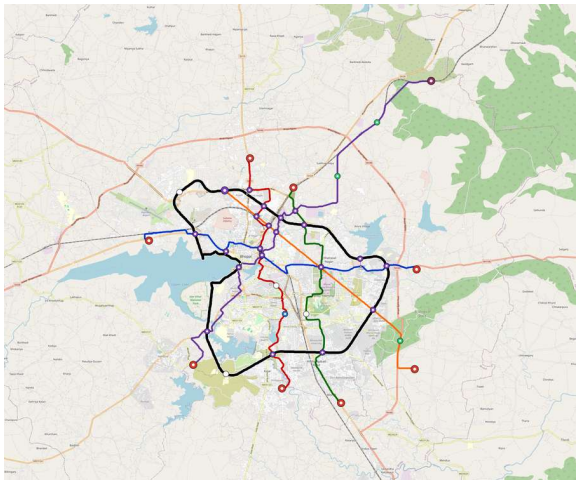
Generated City Model - Bhopal

south-asia / India / Bhopal

Metric	Value
Path	cities/catalogue/south-asia/India/Bhopal
Recorded ISO / planning population	IN / 2,400,000
Family	metro-4car
Lines / stations	6 / 84
Route length	196.9 km
Fleet	281 trainsets
Routing-demand cells (legacy)	59%; not population access
City CAPEX incl. dedicated solar	USD 3.29bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	Unavailable native population evidence
Direct / reachable line pairs	93.3% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Bhopal network map

Line	Length	Stations	Trainsets	Shape
line-1	27.8 km	13	52	radial
line-2	27.9 km	16	59	radial
line-3	23.7 km	8	38	radial
line-4	36.9 km	18	69	radial
line-5	22.8 km	10	40	radial
line-6	57.7 km	19	23	ring

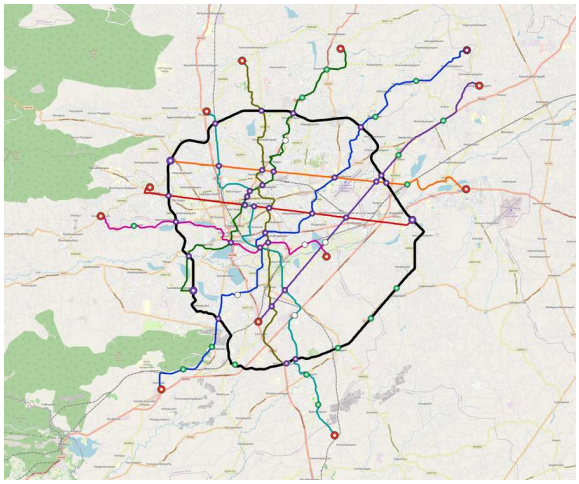
Generated City Model - Coimbatore

south-asia / India / Coimbatore

Metric	Value
Path	cities/catalogue/south-asia/India/Coimbatore
Recorded ISO / planning population	IN / 3,084,000
Family	metro-6car
Lines / stations	9 / 158
Route length	335.0 km
Fleet	606 trainsets
Routing-demand cells (legacy)	70%; not population access
City CAPEX incl. dedicated solar	USD 6.58bn (USD 20M / km); shared national factory excluded
Residents within 800 m station circles	Unavailable native population evidence
Direct / reachable line pairs	80.6% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Coimbatore network map

Line	Length	Stations	Trainsets	Shape
line-1	49.2 km	18	92	radial

Line	Length	Stations	Trainsets	Shape
line-2	23.2 km	10	47	radial
line-3	33.9 km	24	89	radial
line-4	29.5 km	10	54	radial
line-5	26.3 km	9	50	radial
line-6	40.7 km	33	116	radial
line-7	25.4 km	9	49	radial
line-8	32.3 km	17	72	radial
line-9	74.4 km	28	37	ring

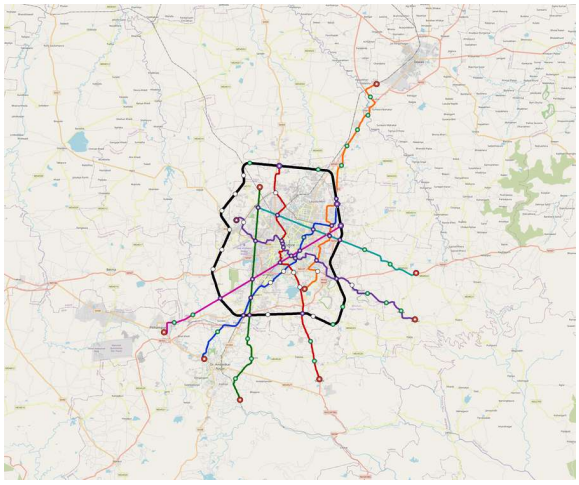
Generated City Model - Indore

south-asia / India / Indore

Metric	Value
Path	cities/catalogue/south-asia/India/Indore
Recorded ISO / planning population	IN / 3,200,000
Family	metro-6car
Lines / stations	8 / 133
Route length	360.9 km
Fleet	581 trainsets
Routing-demand cells (legacy)	48%; not population access
City CAPEX incl. dedicated solar	USD 6.17bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	Unavailable native population evidence
Direct / reachable line pairs	82.1% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Indore network map

Line	Length	Stations	Trainsets	Shape
line-1	42.2 km	13	76	radial
line-2	43.3 km	23	94	radial
line-3	35.5 km	10	65	radial
line-4	46.5 km	26	103	radial
line-5	45.7 km	14	85	radial
line-6	29.2 km	10	54	radial
line-7	35.8 km	12	65	radial
line-8	82.8 km	25	39	ring

Generated City Model - Jodhpur

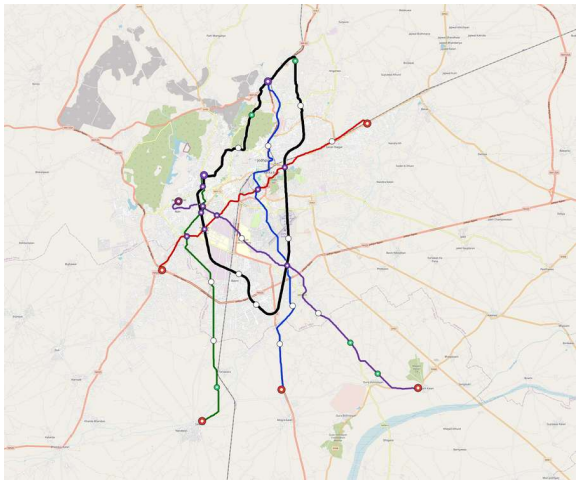
south-asia / India / Jodhpur

Metric	Value
Path	cities/catalogue/south-asia/India/Jodhpur
Recorded ISO / planning population	IN / 1,300,000

Metric	Value
Family	metro-4car
Lines / stations	5 / 66
Route length	151.2 km
Fleet	211 trainsets
Routing-demand cells (legacy)	54%; not population access
City CAPEX incl. dedicated solar	USD 2.76bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	Unavailable native population evidence
Direct / reachable line pairs	90.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Jodhpur network map

Line	Length	Stations	Trainsets	Shape
line-1	28.1 km	11	48	radial
line-2	23.7 km	11	45	radial
line-3	22.2 km	13	47	radial
line-4	28.2 km	14	52	radial
line-5	49.0 km	17	19	ring

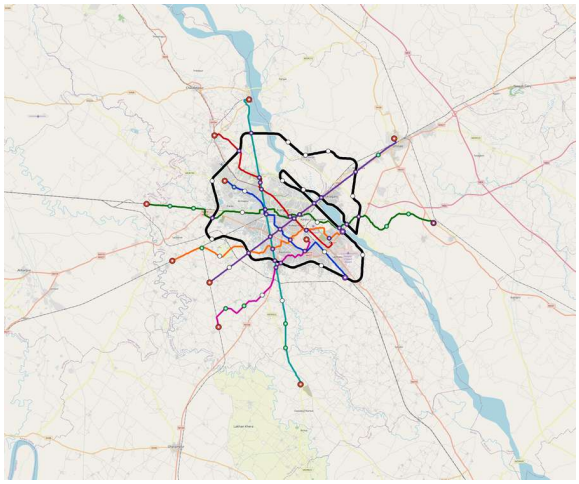
Generated City Model - Kanpur

south-asia / India / Kanpur

Metric	Value
Path	cities/catalogue/south-asia/India/Kanpur
Recorded ISO / planning population	IN / 3,200,000
Family	metro-6car
Lines / stations	8 / 115
Route length	372.2 km
Fleet	536 trainsets
Routing-demand cells (legacy)	51%; not population access
City CAPEX incl. dedicated solar	USD 6.35bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	Unavailable native population evidence
Direct / reachable line pairs	78.6% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Kanpur network map

Line	Length	Stations	Trainsets	Shape
line-1	30.0 km	12	58	radial
line-2	31.6 km	11	57	radial
line-3	54.3 km	17	101	radial
line-4	38.5 km	13	73	radial
line-5	34.5 km	14	65	radial
line-6	45.9 km	15	84	radial
line-7	24.4 km	8	47	radial
line-8	113.0 km	25	51	ring

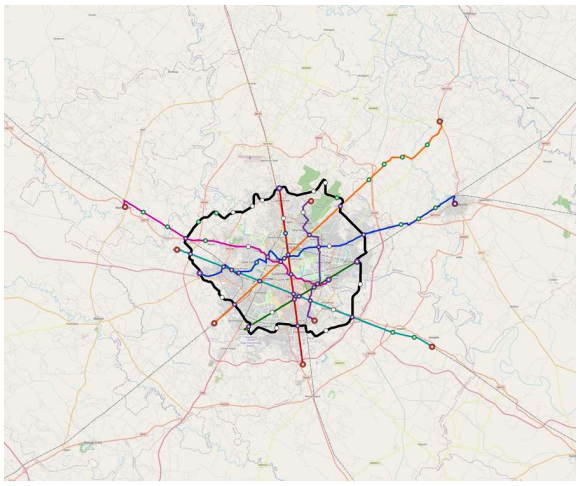
Generated City Model - Lucknow

south-asia / India / Lucknow

Metric	Value
Path	cities/catalogue/south-asia/India/Lucknow
Recorded ISO / planning population	IN / 3,500,000
Family	metro-6car
Lines / stations	8 / 111
Route length	360.3 km
Fleet	532 trainsets
Routing-demand cells (legacy)	45%; not population access
City CAPEX incl. dedicated solar	USD 6.05bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	Unavailable native population evidence
Direct / reachable line pairs	82.1% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Lucknow network map

Line	Length	Stations	Trainsets	Shape
line-1	52.6 km	18	97	radial
line-2	26.6 km	10	50	radial
line-3	21.8 km	9	42	radial
line-4	23.2 km	8	41	radial
line-5	51.2 km	15	97	radial
line-6	43.4 km	14	79	radial
line-7	41.4 km	12	79	radial
line-8	100.1 km	25	47	ring

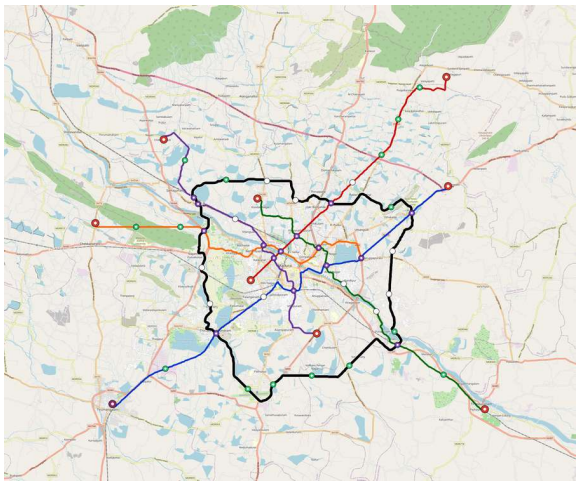
Generated City Model - Madurai

south-asia / India / Madurai

Metric	Value
Path	cities/catalogue/south-asia/India/Madurai
Recorded ISO / planning population	IN / 1,600,000
Family	metro-4car
Lines / stations	6 / 84
Route length	226.8 km
Fleet	295 trainsets
Routing-demand cells (legacy)	71%; not population access
City CAPEX incl. dedicated solar	USD 3.49bn (USD 15M / km); shared national factory excluded
Residents within 800 m station circles	Unavailable native population evidence
Direct / reachable line pairs	86.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Madurai network map

Line	Length	Stations	Trainsets	Shape
line-1	38.8 km	9	59	radial
line-2	26.5 km	10	45	radial
line-3	30.7 km	9	49	radial
line-4	29.4 km	18	63	radial
line-5	27.4 km	10	47	radial
line-6	74.0 km	28	32	ring

Generated City Model - Meerut

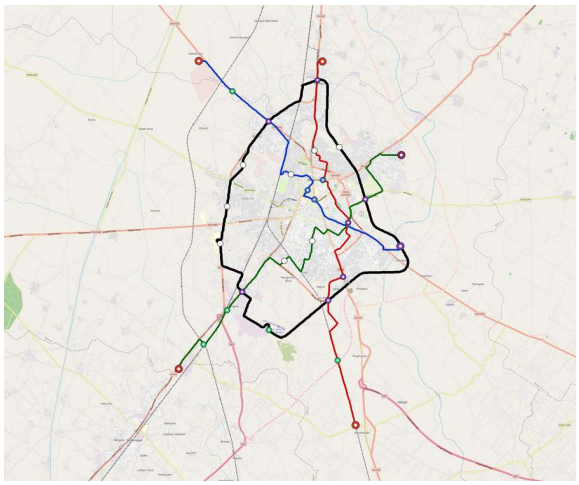
south-asia / India / Meerut

Metric	Value
Path	cities/catalogue/south-asia/India/Meerut
Recorded ISO / planning population	IN / 1,600,000
Family	metro-4car
Lines / stations	4 / 50

Metric	Value
Route length	149.6 km
Fleet	184 trainsets
Routing-demand cells (legacy)	68%; not population access
City CAPEX incl. dedicated solar	USD 2.28bn (USD 15M / km); shared national factory excluded
Residents within 800 m station circles	Unavailable native population evidence
Direct / reachable line pairs	100.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Meerut network map

Line	Length	Stations	Trainsets	Shape
line-1	27.4 km	8	43	radial
line-2	32.1 km	15	59	radial
line-3	33.0 km	15	59	radial
line-4	57.1 km	12	23	ring

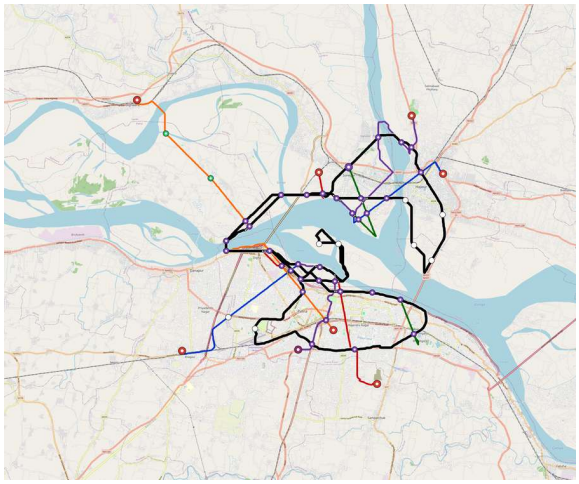
Generated City Model - Patna

south-asia / India / Patna

Metric	Value
Path	cities/catalogue/south-asia/India/Patna
Recorded ISO / planning population	IN / 2,520,000
Family	metro-4car
Lines / stations	6 / 173
Route length	299.0 km
Fleet	473 trainsets
Routing-demand cells (legacy)	54%; not population access
City CAPEX incl. dedicated solar	USD 5.94bn (USD 20M / km); shared national factory excluded
Residents within 800 m station circles	Unavailable native population evidence
Direct / reachable line pairs	100.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Patna network map

Line	Length	Stations	Trainsets	Shape
line-1	40.7 km	26	91	radial
line-2	30.3 km	23	76	radial
line-3	39.1 km	27	92	radial
line-4	44.7 km	32	107	radial
line-5	29.2 km	15	56	radial
line-6	115.0 km	50	51	ring

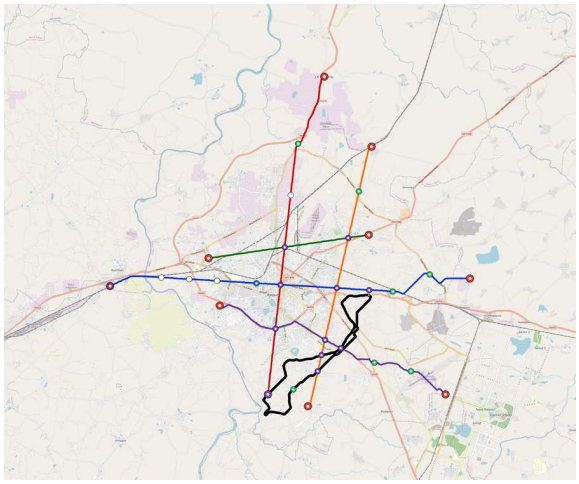
Generated City Model - Raipur

south-asia / India / Raipur

Metric	Value
Path	cities/catalogue/south-asia/India/Raipur
Recorded ISO / planning population	IN / 1,500,000
Family	metro-4car
Lines / stations	6 / 45
Route length	143.1 km
Fleet	195 trainsets
Routing-demand cells (legacy)	41%; not population access
City CAPEX incl. dedicated solar	USD 2.20bn (USD 15M / km); shared national factory excluded
Residents within 800 m station circles	Unavailable native population evidence
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Raipur network map

Line	Length	Stations	Trainsets	Shape
line-1	29.0 km	11	47	radial

Line	Length	Stations	Trainsets	Shape
line-2	25.8 km	7	42	radial
line-3	12.7 km	4	20	radial
line-4	22.9 km	8	38	radial
line-5	21.5 km	8	35	radial
line-6	31.1 km	7	13	ring

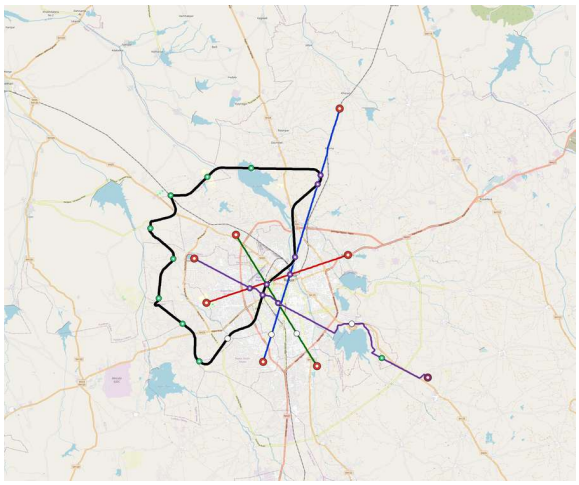
Generated City Model - Rajkot

south-asia / India / Rajkot

Metric	Value
Path	cities/catalogue/south-asia/India/Rajkot
Recorded ISO / planning population	IN / 1,800,000
Family	metro-4car
Lines / stations	5 / 47
Route length	121.1 km
Fleet	153 trainsets
Routing-demand cells (legacy)	64%; not population access
City CAPEX incl. dedicated solar	USD 1.97bn (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	Unavailable native population evidence
Direct / reachable line pairs	100.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Rajkot network map

Line	Length	Stations	Trainsets	Shape
line-1	21.8 km	9	39	radial
line-2	12.1 km	6	25	radial
line-3	12.7 km	8	29	radial
line-4	23.5 km	9	39	radial
line-5	50.9 km	15	21	ring

Generated City Model - Ranchi

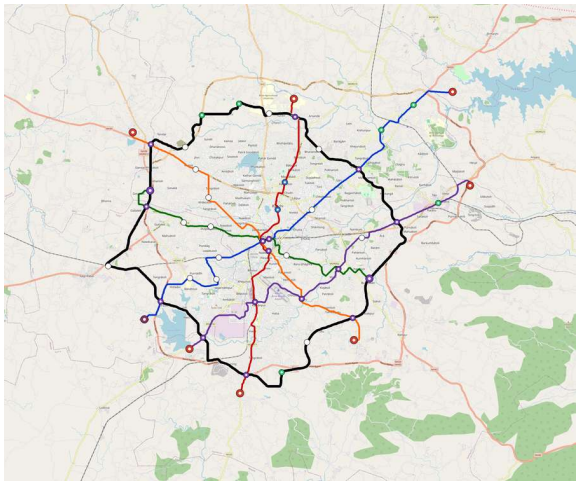
south-asia / India / Ranchi

Metric	Value
Path	cities/catalogue/south-asia/India/Ranchi
Recorded ISO / planning population	IN / 1,400,000
Family	metro-4car
Lines / stations	6 / 106
Route length	233.0 km
Fleet	336 trainsets
Routing-demand cells (legacy)	49%; not population access
City CAPEX incl. dedicated solar	USD 3.87bn (USD 17M / km); shared national factory excluded

Metric	Value
Residents within 800 m station circles	Unavailable native population evidence
Direct / reachable line pairs	93.3% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Ranchi network map

Line	Length	Stations	Trainsets	Shape
line-1	36.9 km	19	71	radial
line-2	29.2 km	19	67	radial
line-3	26.0 km	18	62	radial
line-4	31.4 km	10	52	radial
line-5	29.6 km	13	53	radial
line-6	79.8 km	27	31	ring

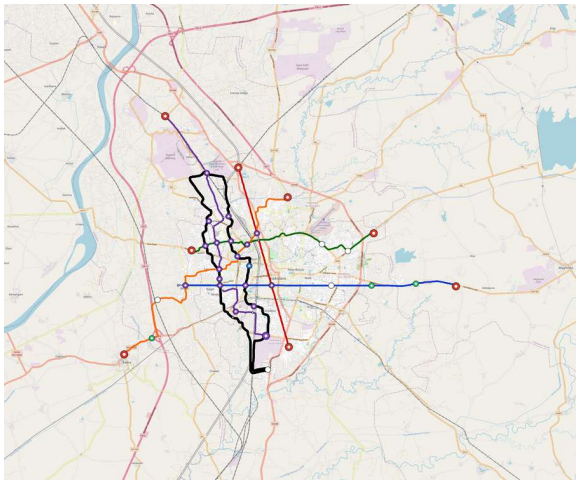
Generated City Model - Vadodara

south-asia / India / Vadodara

Metric	Value
Path	cities/catalogue/south-asia/India/Vadodara
Recorded ISO / planning population	IN / 2,200,000
Family	metro-4car
Lines / stations	6 / 79
Route length	149.5 km
Fleet	233 trainsets
Routing-demand cells (legacy)	61%; not population access
City CAPEX incl. dedicated solar	USD 2.63bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	Unavailable native population evidence
Direct / reachable line pairs	80.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Vadodara network map

Line	Length	Stations	Trainsets	Shape
line-1	21.4 km	9	37	radial
line-2	14.9 km	5	25	radial
line-3	16.7 km	13	43	radial
line-4	25.2 km	15	54	radial
line-5	25.0 km	14	51	radial
line-6	46.2 km	23	23	ring

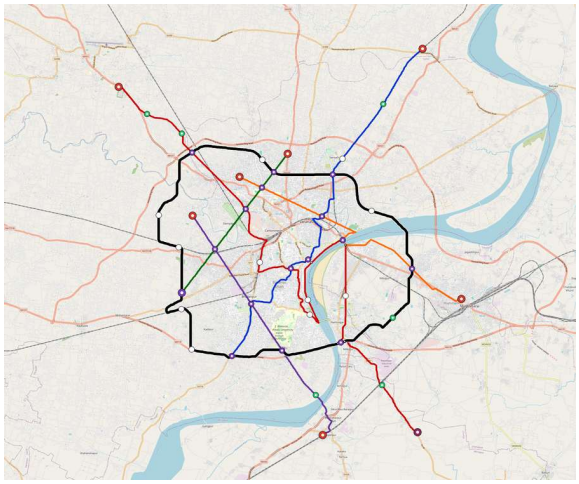
Generated City Model - Varanasi

south-asia / India / Varanasi

Metric	Value
Path	cities/catalogue/south-asia/India/Varanasi
Recorded ISO / planning population	IN / 1,500,000
Family	metro-4car
Lines / stations	6 / 75
Route length	213.7 km
Fleet	270 trainsets
Routing-demand cells (legacy)	62%; not population access
City CAPEX incl. dedicated solar	USD 3.59bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	Unavailable native population evidence
Direct / reachable line pairs	80.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Varanasi network map

Line	Length	Stations	Trainsets	Shape
line-1	33.3 km	10	53	radial

Line	Length	Stations	Trainsets	Shape
line-2	57.1 km	23	94	radial
line-3	14.1 km	6	26	radial
line-4	22.0 km	6	36	radial
line-5	22.9 km	7	35	radial
line-6	64.2 km	23	26	ring

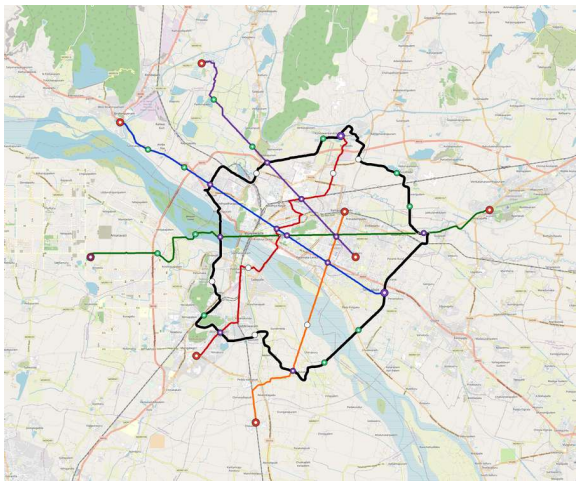
Generated City Model - Vijayawada

south-asia / India / Vijayawada

Metric	Value
Path	cities/catalogue/south-asia/India/Vijayawada
Recorded ISO / planning population	IN / 1,500,000
Family	metro-4car
Lines / stations	6 / 63
Route length	212.8 km
Fleet	254 trainsets
Routing-demand cells (legacy)	78%; not population access
City CAPEX incl. dedicated solar	USD 3.18bn (USD 15M / km); shared national factory excluded
Residents within 800 m station circles	Unavailable native population evidence
Direct / reachable line pairs	86.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Vijayawada network map

Line	Length	Stations	Trainsets	Shape
line-1	27.4 km	8	45	radial
line-2	29.5 km	8	47	radial
line-3	35.5 km	12	59	radial
line-4	23.6 km	8	39	radial
line-5	21.1 km	7	34	radial
line-6	75.8 km	20	30	ring

Generated City Model - Visakhapatnam

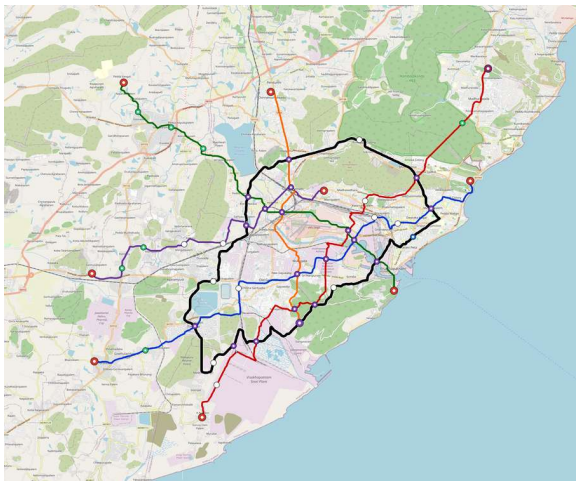
south-asia / India / Visakhapatnam

Metric	Value
Path	cities/catalogue/south-asia/India/Visakhapatnam
Recorded ISO / planning population	IN / 2,300,000
Family	metro-4car
Lines / stations	6 / 82
Route length	251.0 km
Fleet	326 trainsets
Routing-demand cells (legacy)	50%; not population access

Metric	Value
City CAPEX incl. dedicated solar	USD 3.91bn (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	Unavailable native population evidence
Direct / reachable line pairs	86.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Visakhapatnam network map

Line	Length	Stations	Trainsets	Shape
line-1	45.5 km	13	71	radial
line-2	49.7 km	13	74	radial
line-3	33.3 km	12	56	radial
line-4	28.0 km	14	52	radial
line-5	22.9 km	12	46	radial
line-6	71.5 km	18	27	ring

Nepal

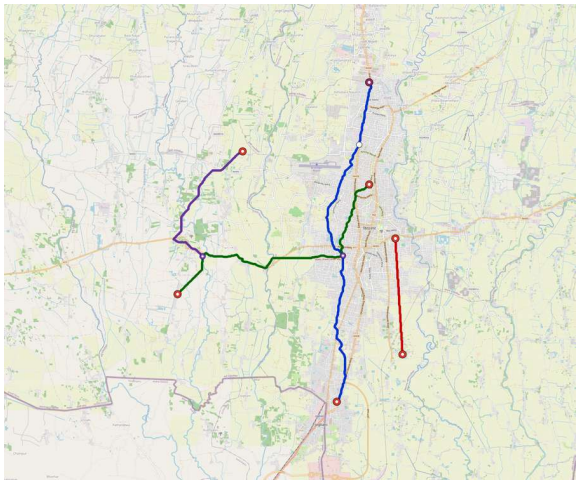
Generated City Model - Biratnagar

south-asia / Nepal / Biratnagar

Metric	Value
Path	cities/catalogue/south-asia/Nepal/Biratnagar
Recorded ISO / planning population	NP / 300,000
Family	tram-2car
Lines / stations	4 / 12
Route length	34.3 km
Fleet	71 trainsets
Routing-demand cells (legacy)	49%; not population access
City CAPEX incl. dedicated solar	USD 704M (USD 21M / km); shared national factory excluded
Residents within 800 m station circles	101,842 (2020; 22.0% of raster bbox)
Direct / reachable line pairs	33.3% / 50.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Biratnagar network map

Line	Length	Stations	Trainsets	Shape
line-1	13.4 km	4	27	radial
line-2	4.3 km	2	10	radial
line-3	10.9 km	4	21	radial
line-4	5.7 km	2	13	radial

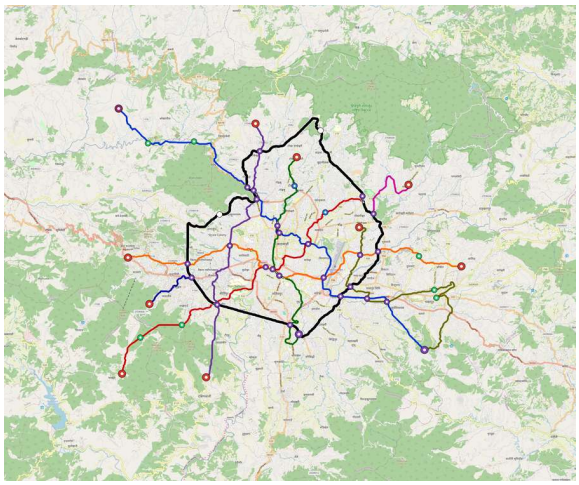
Generated City Model - Kathmandu

south-asia / Nepal / Kathmandu

Metric	Value
Path	cities/catalogue/south-asia/Nepal/Kathmandu
Recorded ISO / planning population	NP / 1,442,000
Family	metro-4car
Lines / stations	9 / 147
Route length	237.9 km
Fleet	447 trainsets
Routing-demand cells (legacy)	55%; not population access
City CAPEX incl. dedicated solar	USD 4.74bn (USD 20M / km); shared national factory excluded
Residents within 800 m station circles	2,423,591 (2020; 32.1% of raster bbox)
Direct / reachable line pairs	52.8% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Kathmandu network map

Line	Length	Stations	Trainsets	Shape
line-1	38.1 km	35	106	radial
line-2	29.4 km	17	60	radial
line-3	22.0 km	23	70	radial

Line	Length	Stations	Trainsets	Shape
line-4	25.1 km	16	57	radial
line-5	31.5 km	19	68	radial
line-6	56.4 km	25	26	ring
line-7	6.3 km	2	11	radial
line-8	24.1 km	8	39	radial
line-9	4.9 km	2	10	radial

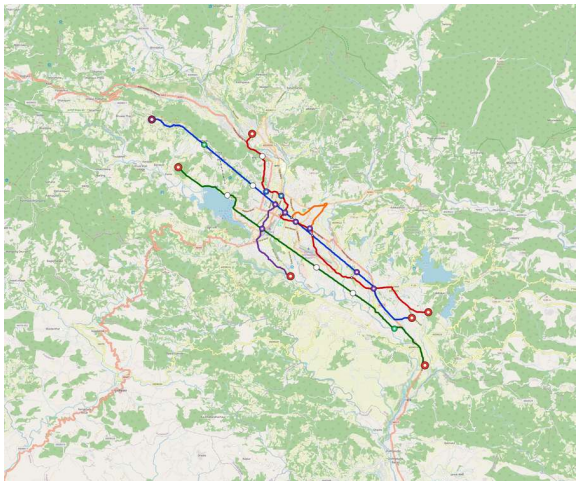
Generated City Model - Pokhara

south-asia / Nepal / Pokhara

Metric	Value
Path	cities/catalogue/south-asia/Nepal/Pokhara
Recorded ISO / planning population	NP / 600,000
Family	light-metro-3car
Lines / stations	5 / 38
Route length	91.5 km
Fleet	293 trainsets
Routing-demand cells (legacy)	59%; not population access
City CAPEX incl. dedicated solar	USD 1.62bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	132,866 (2020; 33.0% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Pokhara network map

Line	Length	Stations	Trainsets	Shape
line-1	26.7 km	13	89	radial
line-2	25.7 km	13	83	radial
line-3	26.3 km	7	79	radial
line-4	7.2 km	3	24	radial
line-5	5.5 km	2	18	radial

Pakistan

Generated City Model - Bahawalpur

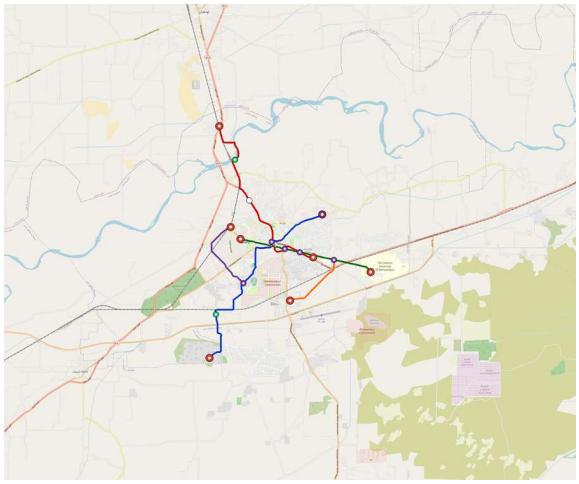
south-asia / Pakistan / Bahawalpur

Metric	Value
Path	cities/catalogue/south-asia/Pakistan/Bahawalpur
Recorded ISO / planning population	PK / 900,000
Family	light-metro-3car
Lines / stations	5 / 27
Route length	52.0 km

Metric	Value
Fleet	171 trainsets
Routing-demand cells (legacy)	54%; not population access
City CAPEX incl. dedicated solar	USD 973M (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	344,075 (2020; 29.4% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Bahawalpur network map

Line	Length	Stations	Trainsets	Shape
line-1	16.7 km	7	53	radial
line-2	14.9 km	9	52	radial
line-3	10.1 km	7	32	radial
line-4	5.7 km	2	18	radial
line-5	4.6 km	2	16	radial

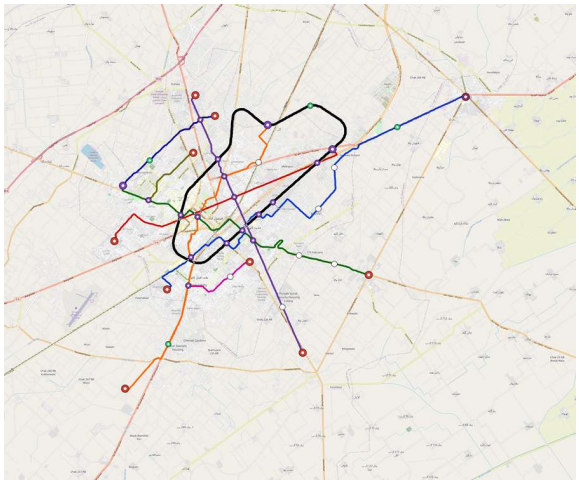
Generated City Model - Faisalabad

south-asia / Pakistan / Faisalabad

Metric	Value
Path	cities/catalogue/south-asia/Pakistan/Faisalabad
Recorded ISO / planning population	PK / 3,556,000
Family	metro-6car
Lines / stations	9 / 79
Route length	184.7 km
Fleet	317 trainsets
Routing-demand cells (legacy)	76%; not population access
City CAPEX incl. dedicated solar	USD 3.42bn (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	1,646,062 (2020; 33.2% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Faisalabad network map

Line	Length	Stations	Trainsets	Shape
line-1	34.0 km	14	64	radial
line-2	20.0 km	8	39	radial
line-3	24.3 km	13	54	radial
line-4	20.9 km	10	45	radial
line-5	27.7 km	10	54	radial
line-6	39.2 km	15	18	ring
line-7	5.2 km	3	14	radial
line-8	4.8 km	2	11	radial
line-9	8.4 km	4	18	radial

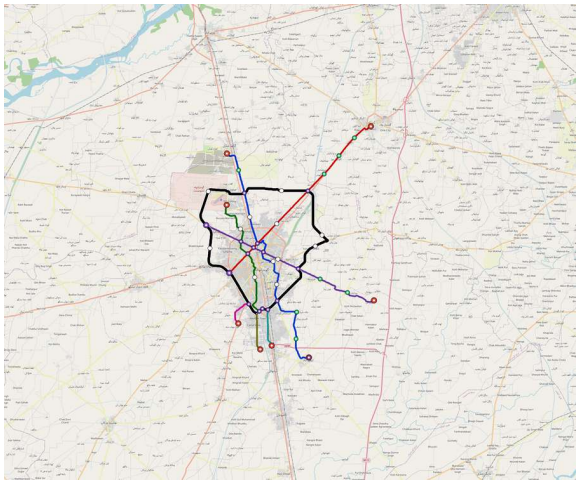
Generated City Model - Gujranwala

south-asia / Pakistan / Gujranwala

Metric	Value
Path	cities/catalogue/south-asia/Pakistan/Gujranwala
Recorded ISO / planning population	PK / 2,300,000
Family	metro-4car
Lines / stations	8 / 63
Route length	202.4 km
Fleet	261 trainsets
Routing-demand cells (legacy)	69%; not population access
City CAPEX incl. dedicated solar	USD 2.84bn (USD 14M / km); shared national factory excluded
Residents within 800 m station circles	795,530 (2020; 22.6% of raster bbox)
Direct / reachable line pairs	46.4% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Gujranwala network map

Line	Length	Stations	Trainsets	Shape
line-1	44.0 km	12	69	radial
line-2	31.4 km	10	52	radial
line-3	21.5 km	9	38	radial
line-4	28.7 km	9	46	radial
line-5	62.4 km	17	25	ring
line-6	5.2 km	2	11	radial
line-7	4.1 km	2	10	radial
line-8	5.1 km	2	10	radial

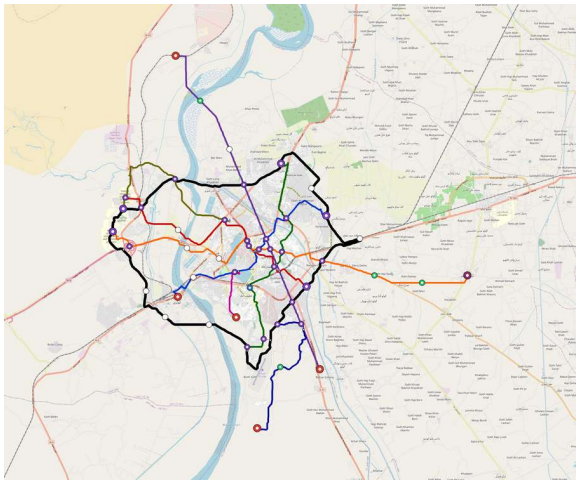
Generated City Model - Hyderabad Pk

south-asia / Pakistan / Hyderabad-Pk

Metric	Value
Path	cities/catalogue/south-asia/Pakistan/Hyderabad-Pk
Recorded ISO / planning population	PK / 1,900,000
Family	metro-4car
Lines / stations	9 / 115
Route length	216.0 km
Fleet	362 trainsets
Routing-demand cells (legacy)	68%; not population access
City CAPEX incl. dedicated solar	USD 3.90bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	1,148,808 (2020; 41.8% of raster bbox)
Direct / reachable line pairs	55.6% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Hyderabad Pk network map

Line	Length	Stations	Trainsets	Shape
line-1	17.9 km	13	45	radial
line-2	22.3 km	21	65	radial
line-3	21.0 km	9	38	radial
line-4	27.9 km	12	49	radial
line-5	32.2 km	24	78	radial
line-6	64.1 km	20	25	ring
line-7	4.4 km	4	14	radial
line-8	13.3 km	7	27	radial
line-9	12.9 km	5	21	radial

Generated City Model - Karachi

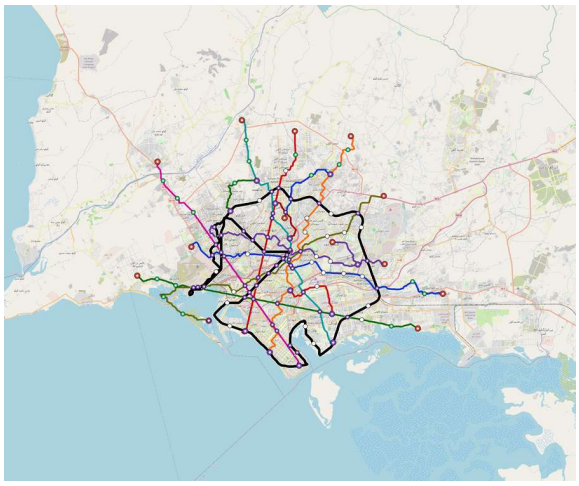
south-asia / Pakistan / Karachi

Metric	Value
Path	cities/catalogue/south-asia/Pakistan/Karachi

Metric	Value
Recorded ISO / planning population	PK / 20,300,000
Family	metro-6car
Lines / stations	14 / 242
Route length	565.5 km
Fleet	921 trainsets
Routing-demand cells (legacy)	70%; not population access
City CAPEX incl. dedicated solar	USD 10.17bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	4,999,890 (2020; 31.5% of raster bbox)
Direct / reachable line pairs	45.1% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Karachi network map

Line	Length	Stations	Trainsets	Shape
line-1	51.6 km	29	116	radial
line-2	34.9 km	17	73	radial
line-3	46.5 km	16	85	radial
line-4	41.4 km	18	82	radial
line-5	50.7 km	15	95	radial
line-6	45.8 km	29	111	radial
line-7	41.0 km	17	76	radial
line-8	60.0 km	27	118	radial
line-9	146.0 km	56	68	ring
line-10	7.5 km	3	16	radial
line-11	12.3 km	5	26	radial
line-12	11.5 km	4	21	radial
line-13	9.7 km	3	19	radial
line-14	6.5 km	3	15	radial

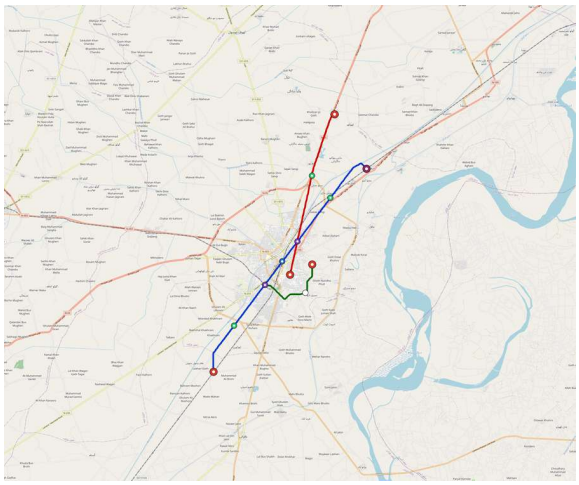
Generated City Model - Larkana

south-asia / Pakistan / Larkana

Metric	Value
Path	cities/catalogue/south-asia/Pakistan/Larkana
Recorded ISO / planning population	PK / 500,000
Family	light-metro-3car
Lines / stations	3 / 14
Route length	39.8 km
Fleet	127 trainsets
Routing-demand cells (legacy)	45%; not population access
City CAPEX incl. dedicated solar	USD 919M (USD 23M / km); shared national factory excluded
Residents within 800 m station circles	193,638 (2020; 24.4% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Larkana network map

Line	Length	Stations	Trainsets	Shape
line-1	21.3 km	7	67	radial
line-2	12.8 km	4	40	radial
line-3	5.7 km	3	20	radial

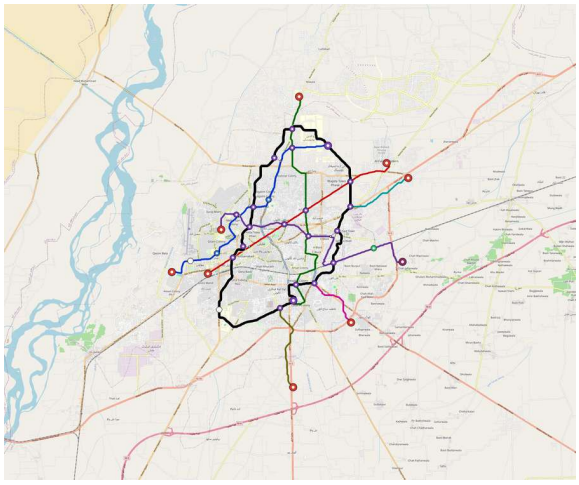
Generated City Model - Multan

south-asia / Pakistan / Multan

Metric	Value
Path	cities/catalogue/south-asia/Pakistan/Multan
Recorded ISO / planning population	PK / 2,197,000
Family	metro-4car
Lines / stations	8 / 62
Route length	139.4 km
Fleet	199 trainsets
Routing-demand cells (legacy)	61%; not population access
City CAPEX incl. dedicated solar	USD 2.29bn (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	952,500 (2020; 28.1% of raster bbox)
Direct / reachable line pairs	42.9% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Multan network map

Line	Length	Stations	Trainsets	Shape
line-1	19.8 km	10	39	radial

Line	Length	Stations	Trainsets	Shape
line-2	16.9 km	8	32	radial
line-3	20.3 km	7	32	radial
line-4	21.9 km	12	45	radial
line-5	44.9 km	19	20	ring
line-6	4.9 km	2	10	radial
line-7	4.3 km	2	10	radial
line-8	6.4 km	2	11	radial

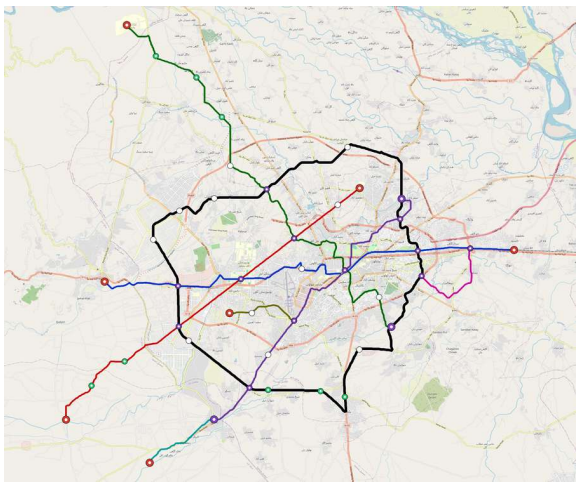
Generated City Model - Peshawar

south-asia / Pakistan / Peshawar

Metric	Value
Path	cities/catalogue/south-asia/Pakistan/Peshawar
Recorded ISO / planning population	PK / 2,300,000
Family	metro-4car
Lines / stations	8 / 61
Route length	206.2 km
Fleet	257 trainsets
Routing-demand cells (legacy)	61%; not population access
City CAPEX incl. dedicated solar	USD 3.09bn (USD 15M / km); shared national factory excluded
Residents within 800 m station circles	546,767 (2020; 14.5% of raster bbox)
Direct / reachable line pairs	46.4% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Peshawar network map

Line	Length	Stations	Trainsets	Shape
line-1	31.6 km	9	50	radial
line-2	29.0 km	8	46	radial
line-3	36.0 km	11	59	radial
line-4	25.5 km	8	39	radial
line-5	66.7 km	18	27	ring
line-6	5.6 km	2	11	radial
line-7	6.8 km	2	13	radial
line-8	5.1 km	3	12	radial

Generated City Model - Quetta

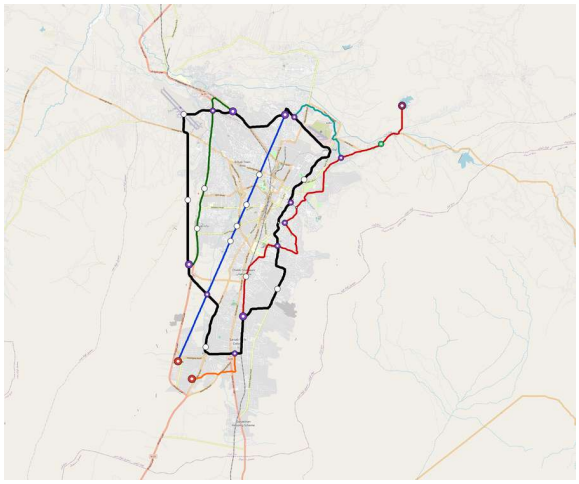
south-asia / Pakistan / Quetta

Metric	Value
Path	cities/catalogue/south-asia/Pakistan/Quetta
Recorded ISO / planning population	PK / 1,200,000
Family	metro-4car

Metric	Value
Lines / stations	6 / 41
Route length	128.8 km
Fleet	145 trainsets
Routing-demand cells (legacy)	42%; not population access
City CAPEX incl. dedicated solar	USD 1.92bn (USD 15M / km); shared national factory excluded
Residents within 800 m station circles	210,031 (2020; 23.3% of raster bbox)
Direct / reachable line pairs	40.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Quetta network map

Line	Length	Stations	Trainsets	Shape
line-1	20.6 km	7	34	radial
line-2	27.2 km	8	43	radial
line-3	14.4 km	5	24	radial
line-4	54.9 km	17	21	ring
line-5	4.8 km	2	10	radial
line-6	7.0 km	2	13	radial

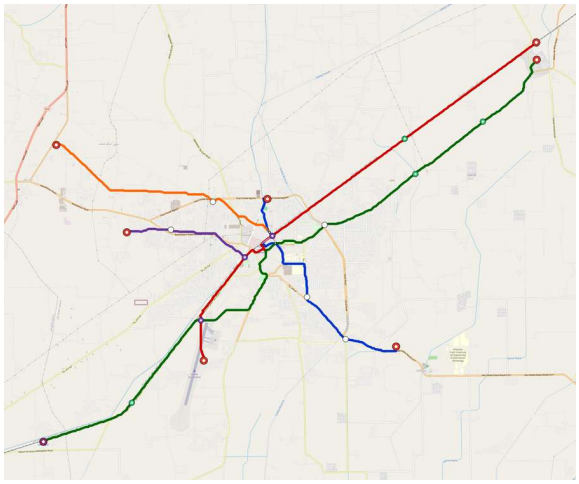
Generated City Model - Rahim Yar Khan

south-asia / Pakistan / Rahim-Yar-Khan

Metric	Value
Path	cities/catalogue/south-asia/Pakistan/Rahim-Yar-Khan
Recorded ISO / planning population	PK / 500,000
Family	light-metro-3car
Lines / stations	5 / 31
Route length	69.5 km
Fleet	229 trainsets
Routing-demand cells (legacy)	88%; not population access
City CAPEX incl. dedicated solar	USD 1.21bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	342,399 (2020; 29.7% of raster bbox)
Direct / reachable line pairs	70.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Rahim Yar Khan network map

Line	Length	Stations	Trainsets	Shape
line-1	9.8 km	9	38	radial
line-2	18.4 km	7	59	radial
line-3	27.8 km	8	85	radial
line-4	4.7 km	3	17	radial
line-5	8.9 km	4	30	radial

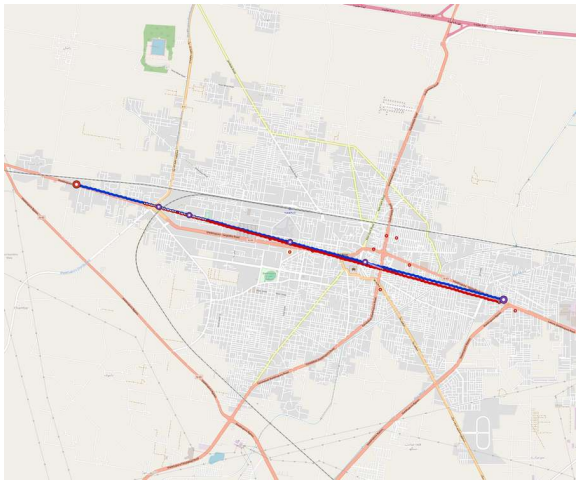
Generated City Model - Sheikhupura

south-asia / Pakistan / Sheikhupura

Metric	Value
Path	cities/catalogue/south-asia/Pakistan/Sheikhupura
Recorded ISO / planning population	PK / 600,000
Family	light-metro-3car
Lines / stations	2 / 35
Route length	15.0 km
Fleet	118 trainsets
Routing-demand cells (legacy)	54%; not population access
City CAPEX incl. dedicated solar	USD 603M (USD 40M / km); shared national factory excluded
Residents within 800 m station circles	131,640 (2020; 11.2% of raster bbox)
Direct / reachable line pairs	100.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Sheikhupura network map

Line	Length	Stations	Trainsets	Shape
line-1	8.1 km	18	61	radial
line-2	6.9 km	17	57	radial

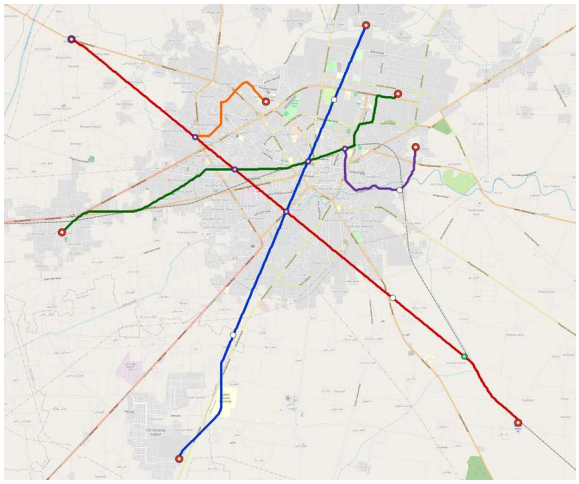
Generated City Model - Sialkot

south-asia / Pakistan / Sialkot

Metric	Value
Path	cities/catalogue/south-asia/Pakistan/Sialkot
Recorded ISO / planning population	PK / 750,000
Family	light-metro-3car
Lines / stations	5 / 23
Route length	63.4 km
Fleet	197 trainsets
Routing-demand cells (legacy)	50%; not population access
City CAPEX incl. dedicated solar	USD 1.38bn (USD 22M / km); shared national factory excluded
Residents within 800 m station circles	405,453 (2020; 21.2% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Sialkot network map

Line	Length	Stations	Trainsets	Shape
line-1	17.5 km	6	52	radial
line-2	22.7 km	7	71	radial
line-3	14.2 km	5	43	radial
line-4	5.0 km	3	17	radial
line-5	4.0 km	2	14	radial

Generated City Model - Sukkur

south-asia / Pakistan / Sukkur

Metric	Value
Path	cities/catalogue/south-asia/Pakistan/Sukkur
Recorded ISO / planning population	PK / 600,000
Family	light-metro-3car
Lines / stations	4 / 23
Route length	50.3 km
Fleet	165 trainsets
Routing-demand cells (legacy)	57%; not population access
City CAPEX incl. dedicated solar	USD 1.16bn (USD 23M / km); shared national factory excluded
Residents within 800 m station circles	267,724 (2020; 27.9% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.

Sukkur network map

Line	Length	Stations	Trainsets	Shape
line-1	20.8 km	8	68	radial
line-2	12.0 km	5	38	radial
line-3	9.0 km	6	32	radial
line-4	8.4 km	4	27	radial

Sri Lanka

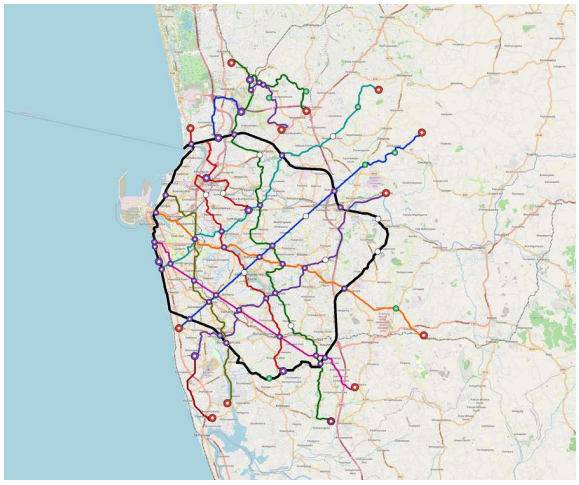
Generated City Model - Colombo

south-asia / Sri Lanka / Colombo

Metric	Value
Path	cities/catalogue/south-asia/Sri Lanka/Colombo
Recorded ISO / planning population	LK / 5,648,000
Family	metro-6car
Lines / stations	14 / 172
Route length	355.7 km
Fleet	652 trainsets
Routing-demand cells (legacy)	67%; not population access
City CAPEX incl. dedicated solar	USD 7.03bn (USD 20M / km); shared national factory excluded
Residents within 800 m station circles	773,297 (2020; 26.3% of raster bbox)
Direct / reachable line pairs	39.6% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Colombo network map

Line	Length	Stations	Trainsets	Shape
line-1	29.4 km	11	57	radial

Line	Length	Stations	Trainsets	Shape
line-2	29.9 km	10	58	radial
line-3	47.8 km	22	97	radial
line-4	29.9 km	10	58	radial
line-5	28.9 km	16	65	radial
line-6	30.5 km	14	62	radial
line-7	25.1 km	14	58	radial
line-8	25.4 km	13	56	radial
line-9	71.0 km	38	40	ring
line-10	6.5 km	2	14	radial
line-11	6.6 km	2	14	radial
line-12	6.2 km	3	15	radial
line-13	9.5 km	8	28	radial
line-14	9.0 km	9	30	radial

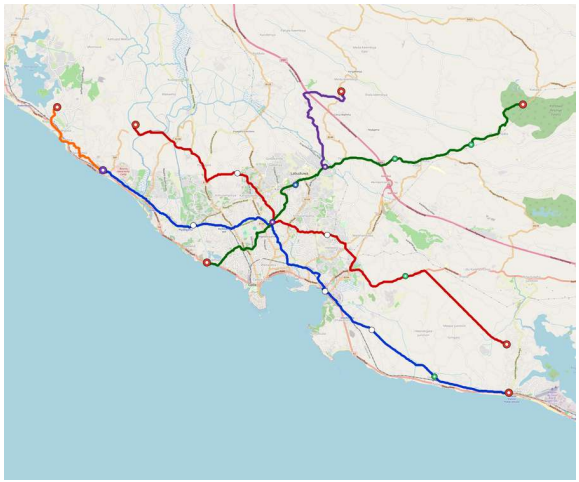
Generated City Model - Galle

south-asia / Sri Lanka / Galle

Metric	Value
Path	cities/catalogue/south-asia/Sri Lanka/Galle
Recorded ISO / planning population	LK / 500,000
Family	light-metro-3car
Lines / stations	5 / 27
Route length	77.5 km
Fleet	242 trainsets
Routing-demand cells (legacy)	54%; not population access
City CAPEX incl. dedicated solar	USD 1.31bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	87,467 (2020; 21.8% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Galle network map

Line	Length	Stations	Trainsets	Shape
line-1	24.4 km	7	73	radial
line-2	23.9 km	7	72	radial
line-3	18.6 km	9	62	radial
line-4	6.2 km	2	20	radial
line-5	4.4 km	2	15	radial

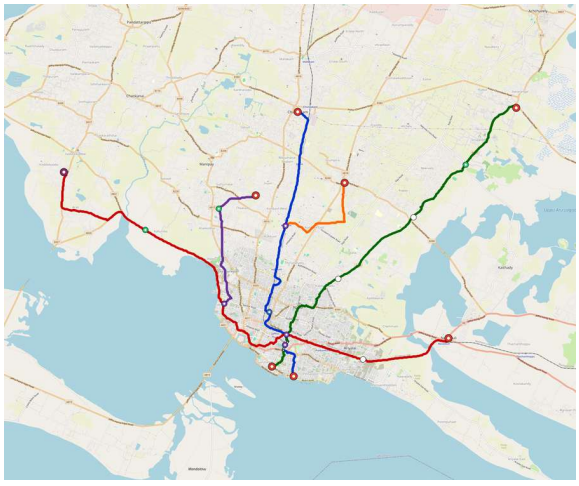
Generated City Model - Jaffna

south-asia / Sri Lanka / Jaffna

Metric	Value
Path	cities/catalogue/south-asia/Sri Lanka/Jaffna
Recorded ISO / planning population	LK / 600,000
Family	light-metro-3car
Lines / stations	5 / 29
Route length	66.2 km
Fleet	212 trainsets
Routing-demand cells (legacy)	73%; not population access
City CAPEX incl. dedicated solar	USD 1.15bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	98,042 (2020; 30.5% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Jaffna network map

Line	Length	Stations	Trainsets	Shape
line-1	13.7 km	8	45	radial
line-2	25.2 km	7	78	radial
line-3	17.1 km	9	54	radial
line-4	5.9 km	3	20	radial
line-5	4.4 km	2	15	radial

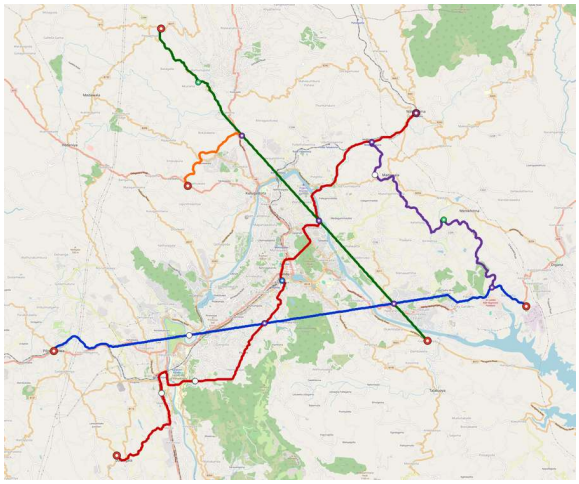
Generated City Model - Kandy

south-asia / Sri Lanka / Kandy

Metric	Value
Path	cities/catalogue/south-asia/Sri Lanka/Kandy
Recorded ISO / planning population	LK / 650,000
Family	light-metro-3car
Lines / stations	5 / 28
Route length	86.6 km
Fleet	265 trainsets
Routing-demand cells (legacy)	37%; not population access
City CAPEX incl. dedicated solar	USD 1.50bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	114,868 (2020; 16.5% of raster bbox)
Direct / reachable line pairs	60.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Kandy network map

Line	Length	Stations	Trainsets	Shape
line-1	22.2 km	6	65	radial
line-2	28.2 km	9	85	radial
line-3	19.4 km	7	60	radial
line-4	12.6 km	4	40	radial
line-5	4.3 km	2	15	radial

southeast-asia

Cambodia

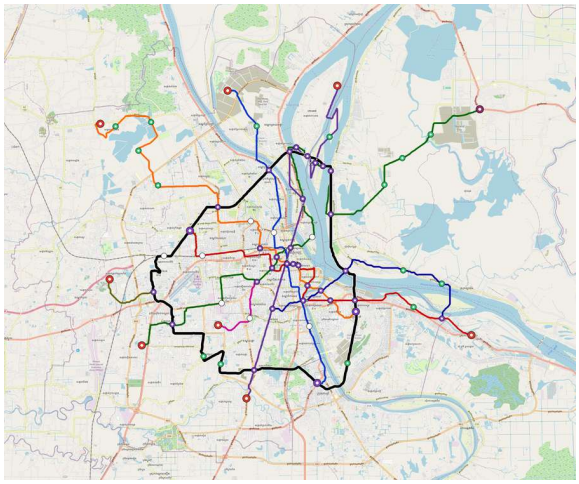
Generated City Model - Phnom Penh

southeast-asia / Cambodia / Phnom-Penh

Metric	Value
Path	cities/catalogue/southeast-asia/Cambodia/Phnom-Penh
Recorded ISO / planning population	KH / 2,281,000
Family	metro-4car
Lines / stations	9 / 159
Route length	298.5 km
Fleet	515 trainsets
Routing-demand cells (legacy)	72%; not population access
City CAPEX incl. dedicated solar	USD 5.67bn (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	935,744 (2020; 32.3% of raster bbox)
Direct / reachable line pairs	61.1% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Phnom Penh network map

Line	Length	Stations	Trainsets	Shape
line-1	30.9 km	15	58	radial
line-2	33.0 km	29	91	radial
line-3	59.4 km	27	105	radial
line-4	36.8 km	19	71	radial
line-5	39.4 km	32	100	radial
line-6	67.5 km	21	27	ring
line-7	6.5 km	3	13	radial
line-8	5.4 km	2	11	radial
line-9	19.8 km	11	39	radial

Indonesia

Generated City Model - Bandung

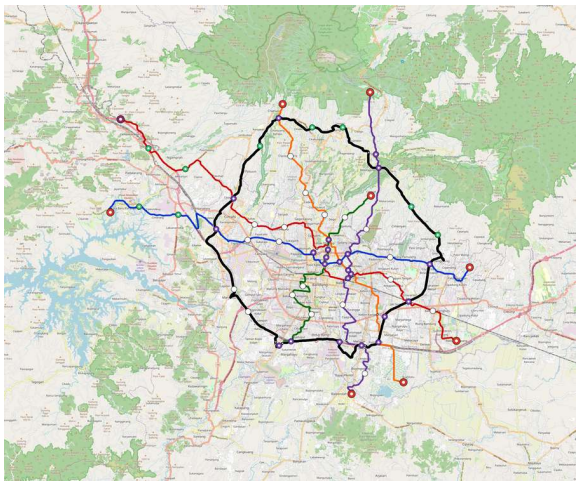
southeast-asia / Indonesia / Bandung

Metric	Value
Path	cities/catalogue/southeast-asia/Indonesia/Bandung
Recorded ISO / planning population	ID / 2,615,000
Family	metro-4car
Lines / stations	6 / 189

Metric	Value
Route length	254.7 km
Fleet	507 trainsets
Routing-demand cells (legacy)	60%; not population access
City CAPEX incl. dedicated solar	USD 5.18bn (USD 20M / km); shared national factory excluded
Residents within 800 m station circles	Unavailable native population evidence
Direct / reachable line pairs	93.3% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Bandung network map

Line	Length	Stations	Trainsets	Shape
line-1	41.6 km	20	76	radial
line-2	48.6 km	36	117	radial
line-3	24.6 km	35	97	radial
line-4	34.8 km	33	102	radial
line-5	32.3 km	23	78	radial
line-6	72.8 km	42	37	ring

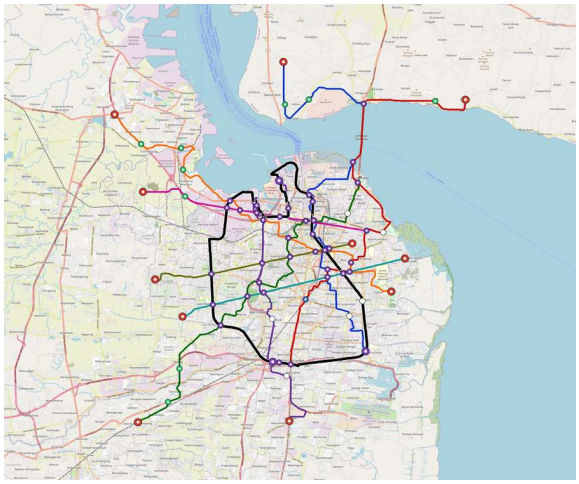
Generated City Model - Surabaya

southeast-asia / Indonesia / Surabaya

Metric	Value
Path	cities/catalogue/southeast-asia/Indonesia/Surabaya
Recorded ISO / planning population	ID / 3,009,000
Family	metro-6car
Lines / stations	9 / 184
Route length	311.8 km
Fleet	632 trainsets
Routing-demand cells (legacy)	63%; not population access
City CAPEX incl. dedicated solar	USD 6.73bn (USD 22M / km); shared national factory excluded
Residents within 800 m station circles	Unavailable native population evidence
Direct / reachable line pairs	86.1% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Surabaya network map

Line	Length	Stations	Trainsets	Shape
line-1	43.8 km	28	106	radial
line-2	41.3 km	20	86	radial
line-3	40.8 km	17	78	radial
line-4	33.8 km	24	90	radial
line-5	36.3 km	30	104	radial
line-6	19.8 km	9	41	radial
line-7	19.7 km	15	54	radial
line-8	18.1 km	9	39	radial
line-9	58.3 km	32	34	ring

Laos

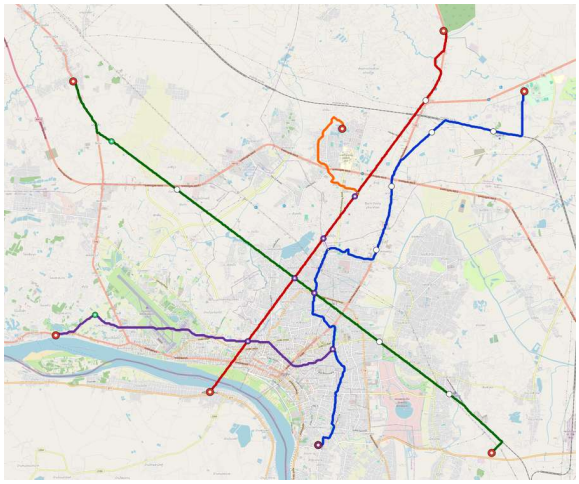
Generated City Model - Vientiane

southeast-asia / Laos / Vientiane

Metric	Value
Path	cities/catalogue/southeast-asia/Laos/Vientiane
Recorded ISO / planning population	LA / 948,000
Family	light-metro-3car
Lines / stations	5 / 34
Route length	87.1 km
Fleet	270 trainsets
Routing-demand cells (legacy)	51%; not population access
City CAPEX incl. dedicated solar	USD 1.47bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	138,646 (2020; 23.4% of raster bbox)
Direct / reachable line pairs	60.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Vientiane network map

Line	Length	Stations	Trainsets	Shape
line-1	25.6 km	9	78	radial
line-2	19.1 km	9	59	radial
line-3	25.1 km	8	76	radial
line-4	12.0 km	4	38	radial
line-5	5.3 km	4	19	radial

Myanmar

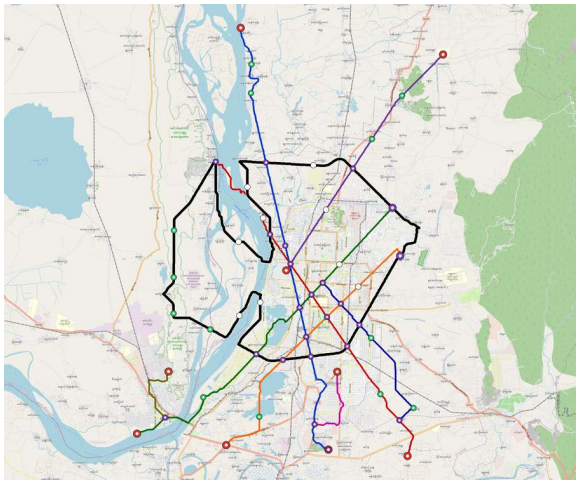
Generated City Model - Mandalay

southeast-asia / Myanmar / Mandalay

Metric	Value
Path	cities/catalogue/southeast-asia/Myanmar/Mandalay
Recorded ISO / planning population	MM / 1,726,000
Family	metro-4car
Lines / stations	9 / 81
Route length	254.7 km
Fleet	319 trainsets
Routing-demand cells (legacy)	55%; not population access
City CAPEX incl. dedicated solar	USD 3.85bn (USD 15M / km); shared national factory excluded
Residents within 800 m station circles	511,164 (2020; 29.3% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Mandalay network map

Line	Length	Stations	Trainsets	Shape
line-1	38.3 km	12	61	radial
line-2	30.6 km	13	53	radial
line-3	30.2 km	9	48	radial
line-4	21.8 km	8	37	radial
line-5	23.4 km	8	40	radial
line-6	84.9 km	21	32	ring
line-7	6.2 km	2	11	radial
line-8	5.7 km	2	11	radial
line-9	13.6 km	6	26	radial

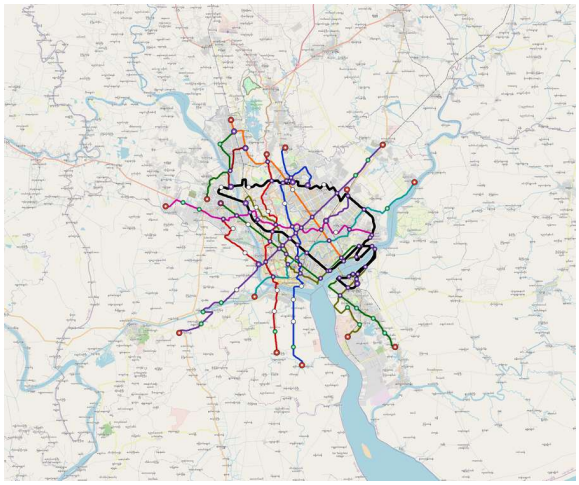
Generated City Model - Yangon

southeast-asia / Myanmar / Yangon

Metric	Value
Path	cities/catalogue/southeast-asia/Myanmar/Yangon
Recorded ISO / planning population	MM / 5,200,000
Family	metro-6car
Lines / stations	14 / 252
Route length	542.2 km
Fleet	956 trainsets
Routing-demand cells (legacy)	58%; not population access
City CAPEX incl. dedicated solar	USD 10.53bn (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	1,988,986 (2020; 32.5% of raster bbox)
Direct / reachable line pairs	48.4% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Yangon network map

Line	Length	Stations	Trainsets	Shape
line-1	44.9 km	19	86	radial
line-2	40.4 km	16	75	radial
line-3	59.4 km	31	127	radial
line-4	52.1 km	17	96	radial
line-5	48.5 km	25	106	radial
line-6	38.6 km	12	70	radial
line-7	48.4 km	24	101	radial
line-8	56.2 km	32	127	radial
line-9	102.6 km	53	57	ring
line-10	8.3 km	4	18	radial
line-11	7.6 km	5	20	radial
line-12	12.3 km	6	27	radial
line-13	15.1 km	5	30	radial
line-14	7.8 km	3	16	radial

Philippines

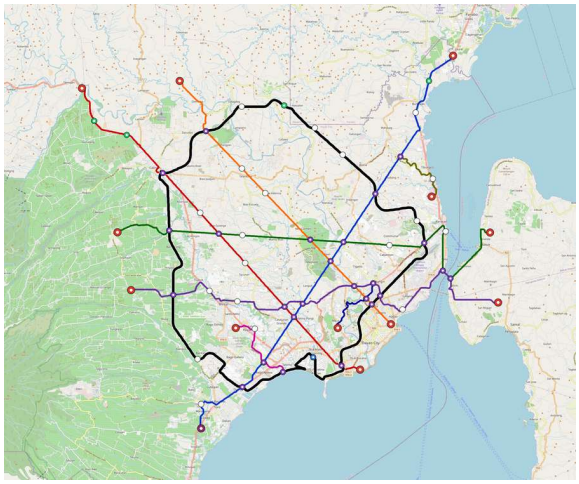
Generated City Model - Davao

southeast-asia / Philippines / Davao

Metric	Value
Path	cities/catalogue/southeast-asia/Philippines/Davao
Recorded ISO / planning population	PH / 1,827,000
Family	metro-4car
Lines / stations	9 / 97
Route length	298.3 km
Fleet	379 trainsets
Routing-demand cells (legacy)	48%; not population access
City CAPEX incl. dedicated solar	USD 4.53bn (USD 15M / km); shared national factory excluded
Residents within 800 m station circles	467,235 (2020; 29.3% of raster bbox)
Direct / reachable line pairs	52.8% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Davao network map

Line	Length	Stations	Trainsets	Shape
line-1	44.1 km	13	70	radial
line-2	37.7 km	15	62	radial
line-3	39.9 km	12	60	radial
line-4	41.2 km	13	63	radial
line-5	29.2 km	10	47	radial
line-6	85.7 km	24	34	ring
line-7	7.5 km	4	16	radial
line-8	6.5 km	3	14	radial
line-9	6.4 km	3	13	radial

west-africa

Burkina Faso

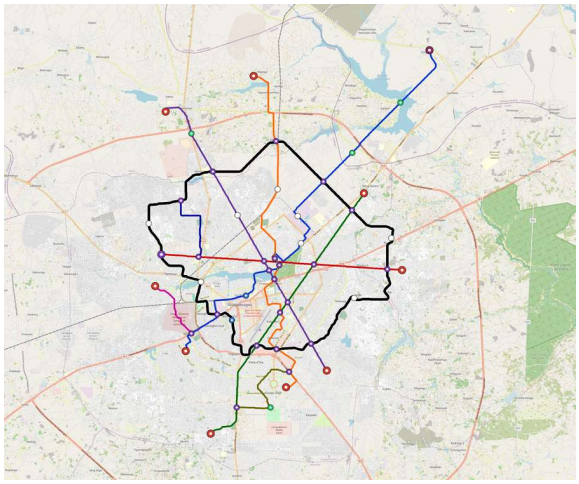
Generated City Model - Ouagadougou

west-africa / Burkina Faso / Ouagadougou

Metric	Value
Path	cities/catalogue/west-africa/Burkina Faso/Ouagadougou
Recorded ISO / planning population	BF / 2,531,000
Family	metro-4car
Lines / stations	9 / 105
Route length	232.2 km
Fleet	351 trainsets
Routing-demand cells (legacy)	57%; not population access
City CAPEX incl. dedicated solar	USD 3.78bn (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	635,857 (2020; 16.3% of raster bbox)
Direct / reachable line pairs	52.8% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Ouagadougou network map

Line	Length	Stations	Trainsets	Shape
line-1	39.4 km	26	87	radial
line-2	19.3 km	7	32	radial
line-3	25.5 km	8	39	radial
line-4	27.7 km	10	45	radial
line-5	33.3 km	26	85	radial
line-6	67.3 km	21	26	ring
line-7	5.5 km	2	11	radial
line-8	8.1 km	3	15	radial
line-9	6.1 km	2	11	radial

Cameroon

Generated City Model - Bafoussam

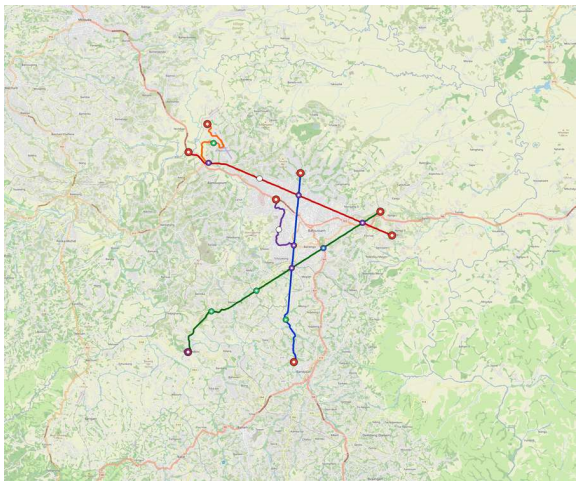
west-africa / Cameroon / Bafoussam

Metric	Value
Path	cities/catalogue/west-africa/Cameroon/Bafoussam
Recorded ISO / planning population	CM / 600,000
Family	light-metro-3car
Lines / stations	5 / 27

Metric	Value
Route length	73.5 km
Fleet	231 trainsets
Routing-demand cells (legacy)	30%; not population access
City CAPEX incl. dedicated solar	USD 1.29bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	60,797 (2020; 16.0% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Bafoussam network map

Line	Length	Stations	Trainsets	Shape
line-1	17.0 km	6	54	radial
line-2	19.6 km	7	60	radial
line-3	22.7 km	7	71	radial
line-4	6.5 km	3	20	radial
line-5	7.7 km	4	26	radial

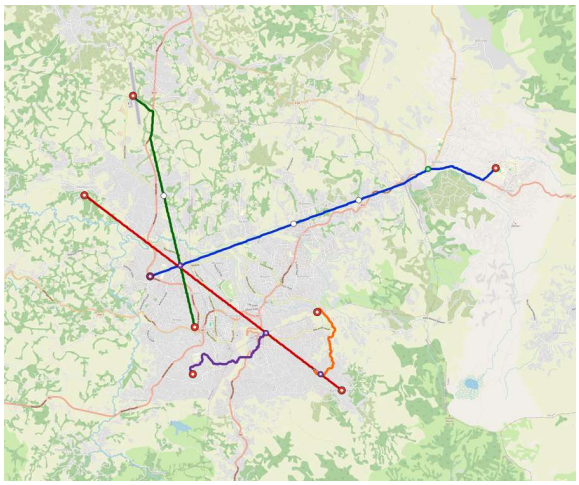
Generated City Model - Bamenda

west-africa / Cameroon / Bamenda

Metric	Value
Path	cities/catalogue/west-africa/Cameroon/Bamenda
Recorded ISO / planning population	CM / 600,000
Family	light-metro-3car
Lines / stations	5 / 26
Route length	51.2 km
Fleet	168 trainsets
Routing-demand cells (legacy)	41%; not population access
City CAPEX incl. dedicated solar	USD 966M (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	61,411 (2020; 15.9% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Bamenda network map

Line	Length	Stations	Trainsets	Shape
line-1	16.7 km	7	53	radial
line-2	14.4 km	8	45	radial
line-3	10.5 km	5	34	radial
line-4	5.3 km	2	18	radial
line-5	4.3 km	4	18	radial

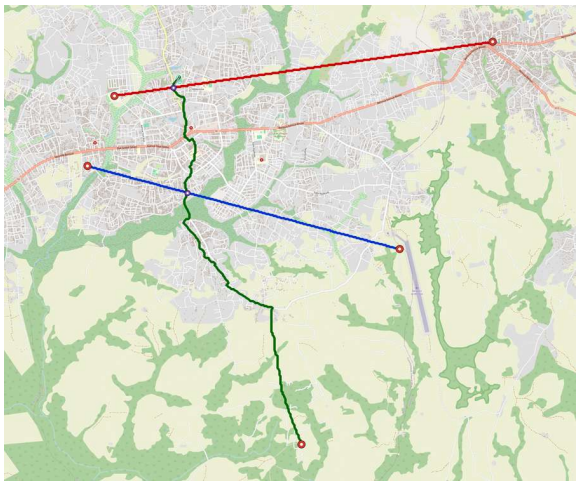
Generated City Model - Bertoua

west-africa / Cameroon / Bertoua

Metric	Value
Path	cities/catalogue/west-africa/Cameroon/Bertoua
Recorded ISO / planning population	CM / 350,000
Family	light-metro-3car
Lines / stations	3 / 10
Route length	24.5 km
Fleet	80 trainsets
Routing-demand cells (legacy)	55%; not population access
City CAPEX incl. dedicated solar	USD 461M (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	4,705 (2020; 11.3% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Bertoua network map

Line	Length	Stations	Trainsets	Shape
line-1	7.0 km	3	24	radial
line-2	8.0 km	3	26	radial

Line	Length	Stations	Trainsets	Shape
line-3	9.5 km	4	30	radial

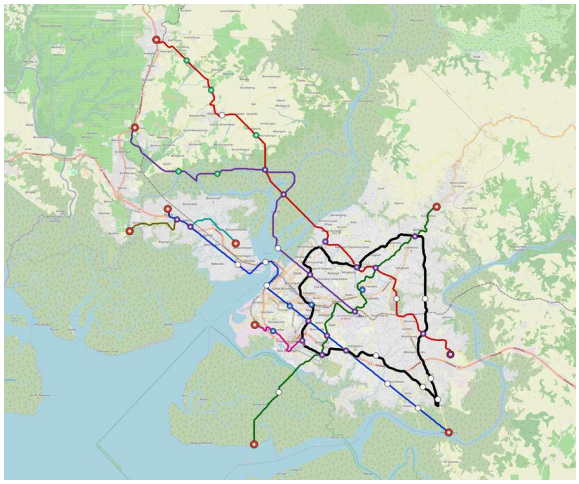
Generated City Model - Douala

west-africa / Cameroon / Douala

Metric	Value
Path	cities/catalogue/west-africa/Cameroon/Douala
Recorded ISO / planning population	CM / 3,900,000
Family	metro-6car
Lines / stations	8 / 78
Route length	222.1 km
Fleet	342 trainsets
Routing-demand cells (legacy)	57%; not population access
City CAPEX incl. dedicated solar	USD 4.12bn (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	720,062 (2020; 19.4% of raster bbox)
Direct / reachable line pairs	39.3% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Douala network map

Line	Length	Stations	Trainsets	Shape
line-1	35.7 km	13	65	radial
line-2	45.0 km	16	84	radial
line-3	34.8 km	11	65	radial
line-4	33.9 km	9	62	radial
line-5	55.6 km	21	26	ring
line-6	5.0 km	2	11	radial
line-7	6.9 km	4	17	radial
line-8	5.2 km	2	12	radial

Generated City Model - Edea

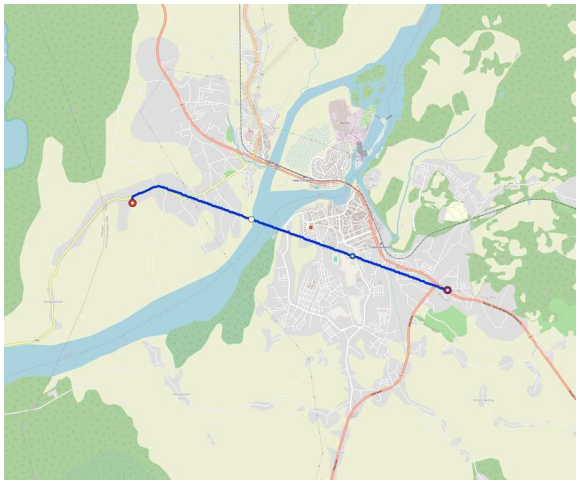
west-africa / Cameroon / Edea

Metric	Value
Path	cities/catalogue/west-africa/Cameroon/Edea
Recorded ISO / planning population	CM / 250,000
Family	tram-2car
Lines / stations	1 / 4
Route length	7.4 km
Fleet	17 trainsets
Routing-demand cells (legacy)	50%; not population access
City CAPEX incl. dedicated solar	USD 126M (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	2,154 (2020; 9.7% of raster bbox)

Metric	Value
Direct / reachable line pairs	100.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Edea network map

Line	Length	Stations	Trainsets	Shape
line-1	7.4 km	4	17	radial

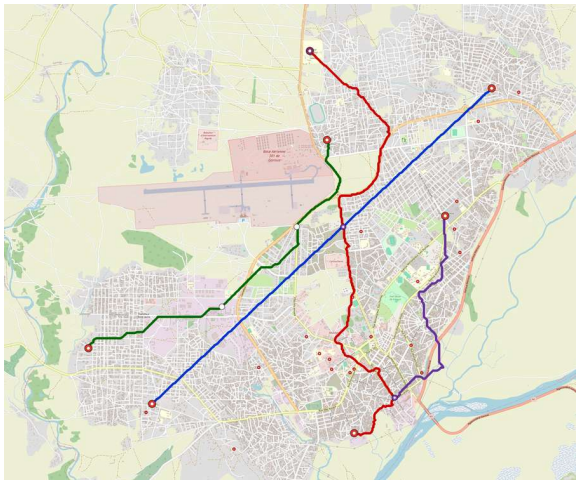
Generated City Model - Garoua

west-africa / Cameroon / Garoua

Metric	Value
Path	cities/catalogue/west-africa/Cameroon/Garoua
Recorded ISO / planning population	CM / 600,000
Family	light-metro-3car
Lines / stations	4 / 13
Route length	33.8 km
Fleet	109 trainsets
Routing-demand cells (legacy)	57%; not population access
City CAPEX incl. dedicated solar	USD 649M (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	44,757 (2020; 14.6% of raster bbox)
Direct / reachable line pairs	33.3% / 50.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Garoua network map

Line	Length	Stations	Trainsets	Shape
line-1	10.2 km	3	32	radial
line-2	10.9 km	4	35	radial
line-3	7.8 km	4	26	radial
line-4	4.9 km	2	16	radial

Generated City Model - Kumba

west-africa / Cameroon / Kumba

Metric	Value
Path	cities/catalogue/west-africa/Cameroon/Kumba
Recorded ISO / planning population	CM / 400,000
Family	light-metro-3car
Lines / stations	4 / 18
Route length	47.3 km
Fleet	150 trainsets
Routing-demand cells (legacy)	57%; not population access
City CAPEX incl. dedicated solar	USD 851M (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	28,022 (2020; 17.9% of raster bbox)
Direct / reachable line pairs	83.3% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.

Kumba network map

Line	Length	Stations	Trainsets	Shape
line-1	20.6 km	6	64	radial
line-2	12.7 km	5	39	radial
line-3	8.1 km	4	27	radial
line-4	5.8 km	3	20	radial

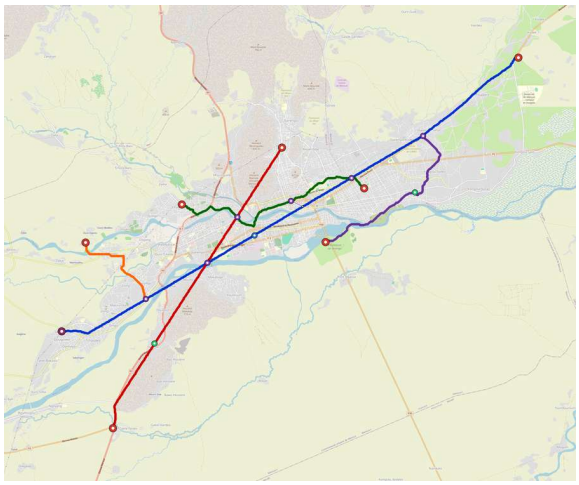
Generated City Model - Maroua

west-africa / Cameroon / Maroua

Metric	Value
Path	cities/catalogue/west-africa/Cameroon/Maroua
Recorded ISO / planning population	CM / 500,000
Family	light-metro-3car
Lines / stations	5 / 23
Route length	59.7 km
Fleet	193 trainsets
Routing-demand cells (legacy)	69%; not population access
City CAPEX incl. dedicated solar	USD 1.11bn (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	60,668 (2020; 21.1% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Maroua network map

Line	Length	Stations	Trainsets	Shape
line-1	23.8 km	8	75	radial
line-2	14.5 km	5	46	radial
line-3	9.5 km	5	32	radial
line-4	7.7 km	3	26	radial
line-5	4.1 km	2	14	radial

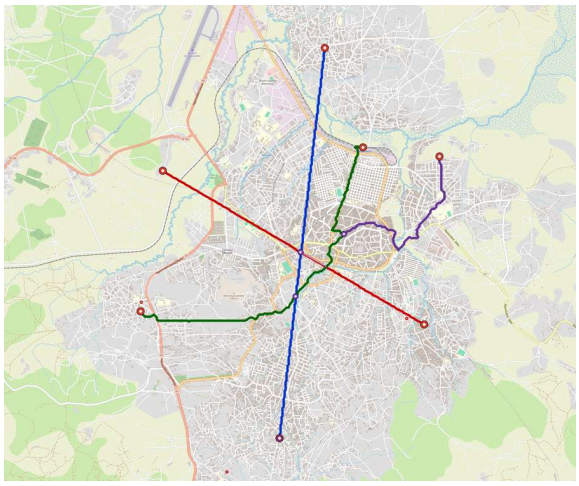
Generated City Model - Ngaoundere

west-africa / Cameroon / Ngaoundere

Metric	Value
Path	cities/catalogue/west-africa/Cameroon/Ngaoundere
Recorded ISO / planning population	CM / 350,000
Family	light-metro-3car
Lines / stations	4 / 15
Route length	26.9 km
Fleet	91 trainsets
Routing-demand cells (legacy)	84%; not population access
City CAPEX incl. dedicated solar	USD 546M (USD 20M / km); shared national factory excluded
Residents within 800 m station circles	16,050 (2020; 17.7% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Ngaoundere network map

Line	Length	Stations	Trainsets	Shape
line-1	8.2 km	4	27	radial
line-2	6.7 km	4	23	radial
line-3	7.9 km	5	27	radial
line-4	4.1 km	2	14	radial

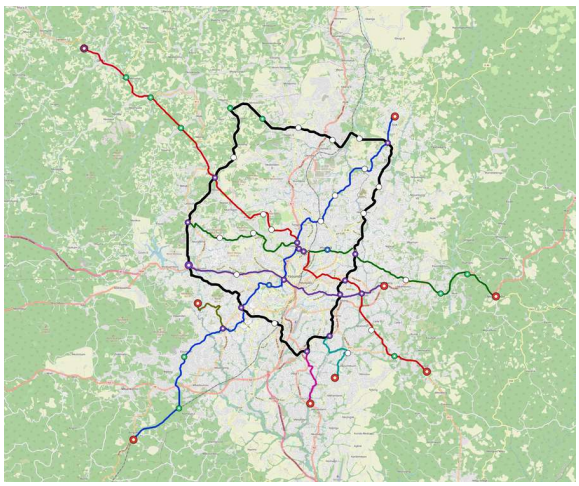
Generated City Model - Yaounde

west-africa / Cameroon / Yaounde

Metric	Value
Path	cities/catalogue/west-africa/Cameroon/Yaounde
Recorded ISO / planning population	CM / 4,100,000
Family	metro-6car
Lines / stations	8 / 132
Route length	248.0 km
Fleet	434 trainsets
Routing-demand cells (legacy)	56%; not population access
City CAPEX incl. dedicated solar	USD 4.95bn (USD 20M / km); shared national factory excluded
Residents within 800 m station circles	892,013 (2020; 20.1% of raster bbox)
Direct / reachable line pairs	42.9% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Yaounde network map

Line	Length	Stations	Trainsets	Shape
line-1	48.2 km	23	98	radial
line-2	50.9 km	27	108	radial
line-3	35.5 km	30	104	radial
line-4	21.0 km	9	42	radial
line-5	73.9 km	34	38	ring
line-6	7.4 km	4	18	radial
line-7	5.6 km	3	14	radial
line-8	5.6 km	2	12	radial

Guinea

Generated City Model - Conakry

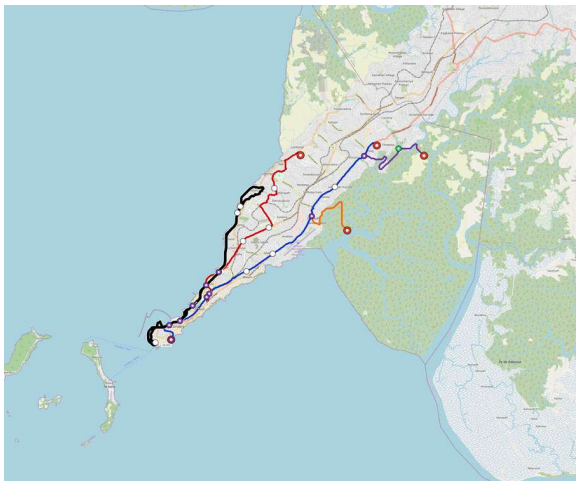
west-africa / Guinea / Conakry

Metric	Value
Path	cities/catalogue/west-africa/Guinea/Conakry
Recorded ISO / planning population	GN / 2,010,000
Family	metro-4car
Lines / stations	5 / 54

Metric	Value
Route length	104.5 km
Fleet	165 trainsets
Routing-demand cells (legacy)	86%; not population access
City CAPEX incl. dedicated solar	USD 1.84bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	262,332 (2020; 24.8% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Conakry network map

Line	Length	Stations	Trainsets	Shape
line-1	30.2 km	20	70	radial
line-2	22.0 km	15	52	radial
line-3	37.4 km	13	16	ring
line-4	8.4 km	4	16	radial
line-5	6.5 km	2	11	radial

Mali

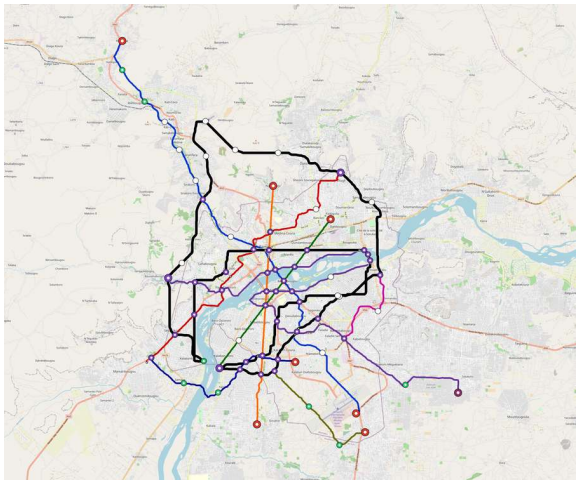
Generated City Model - Bamako

west-africa / Mali / Bamako

Metric	Value
Path	cities/catalogue/west-africa/Mali/Bamako
Recorded ISO / planning population	ML / 2,929,000
Family	metro-4car
Lines / stations	9 / 173
Route length	312.8 km
Fleet	474 trainsets
Routing-demand cells (legacy)	76%; not population access
City CAPEX incl. dedicated solar	USD 6.17bn (USD 20M / km); shared national factory excluded
Residents within 800 m station circles	1,744,343 (2020; 35.4% of raster bbox)
Direct / reachable line pairs	55.6% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Bamako network map

Line	Length	Stations	Trainsets	Shape
line-1	47.8 km	31	105	radial
line-2	29.9 km	16	60	radial
line-3	16.3 km	9	35	radial
line-4	53.7 km	30	109	radial
line-5	19.8 km	11	41	radial
line-6	111.9 km	59	56	ring
line-7	8.0 km	3	15	radial
line-8	10.5 km	4	19	radial
line-9	14.8 km	10	34	radial

Niger

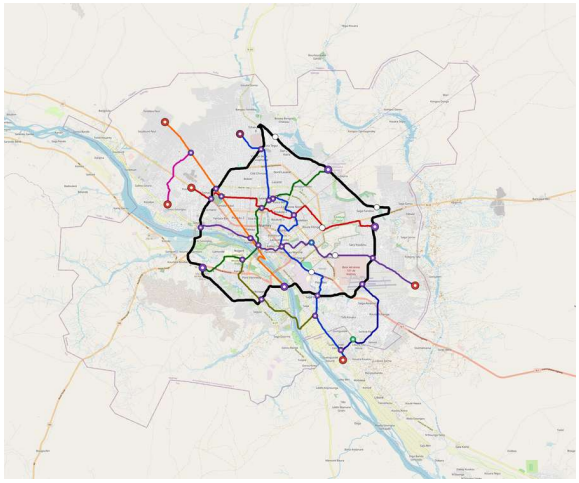
Generated City Model - Niamey

west-africa / Niger / Niamey

Metric	Value
Path	cities/catalogue/west-africa/Niger/Niamey
Recorded ISO / planning population	NE / 1,407,635
Family	metro-4car
Lines / stations	9 / 84
Route length	188.4 km
Fleet	284 trainsets
Routing-demand cells (legacy)	67%; not population access
City CAPEX incl. dedicated solar	USD 3.20bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	296,892 (2020; 21.2% of raster bbox)
Direct / reachable line pairs	52.8% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Niamey network map

Line	Length	Stations	Trainsets	Shape
line-1	29.7 km	17	62	radial
line-2	20.8 km	10	40	radial
line-3	16.8 km	13	43	radial
line-4	22.8 km	10	41	radial
line-5	18.6 km	7	31	radial
line-6	56.3 km	19	23	ring
line-7	5.1 km	2	11	radial
line-8	10.3 km	3	18	radial
line-9	8.0 km	3	15	radial

Nigeria

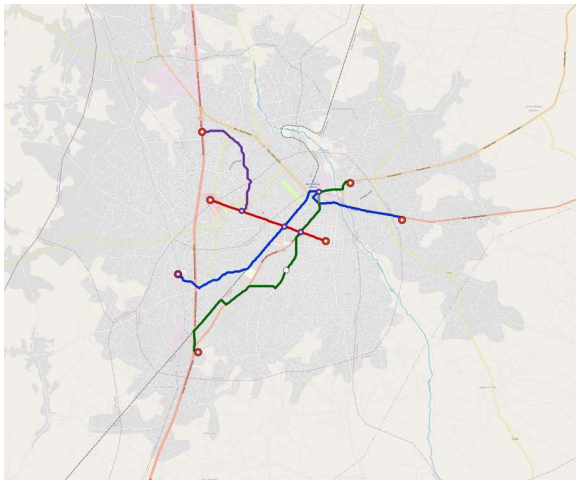
Generated City Model - Aba Ng

west-africa / Nigeria / Aba-Ng

Metric	Value
Path	cities/catalogue/west-africa/Nigeria/Aba-Ng
Recorded ISO / planning population	NG / 900,000
Family	light-metro-3car
Lines / stations	4 / 18
Route length	34.1 km
Fleet	115 trainsets
Routing-demand cells (legacy)	68%; not population access
City CAPEX incl. dedicated solar	USD 674M (USD 20M / km); shared national factory excluded
Residents within 800 m station circles	293,975 (2020; 22.1% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Aba Ng network map

Line	Length	Stations	Trainsets	Shape
line-1	12.4 km	5	39	radial
line-2	5.4 km	5	21	radial
line-3	11.5 km	6	39	radial
line-4	4.8 km	2	16	radial

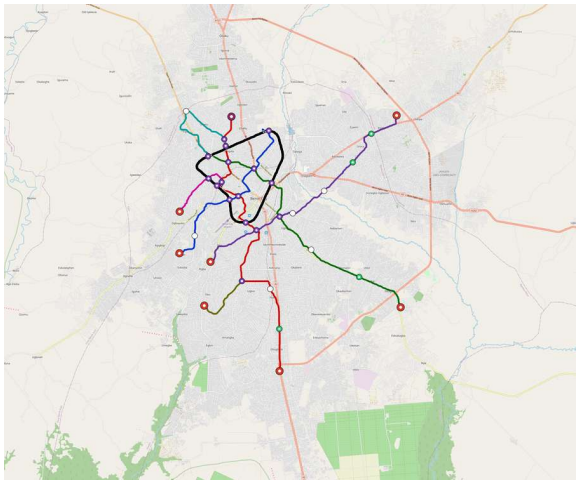
Generated City Model - Benin City

west-africa / Nigeria / Benin-City

Metric	Value
Path	cities/catalogue/west-africa/Nigeria/Benin-City
Recorded ISO / planning population	NG / 1,800,000
Family	metro-4car
Lines / stations	8 / 89
Route length	143.0 km
Fleet	286 trainsets
Routing-demand cells (legacy)	75%; not population access
City CAPEX incl. dedicated solar	USD 2.77bn (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	382,864 (2020; 25.0% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Benin City network map

Line	Length	Stations	Trainsets	Shape
line-1	17.9 km	10	38	radial
line-2	28.8 km	26	81	radial
line-3	27.1 km	11	46	radial

Line	Length	Stations	Trainsets	Shape
line-4	23.8 km	8	39	radial
line-5	24.0 km	12	13	ring
line-6	10.4 km	13	38	radial
line-7	6.2 km	7	21	radial
line-8	4.9 km	2	10	radial

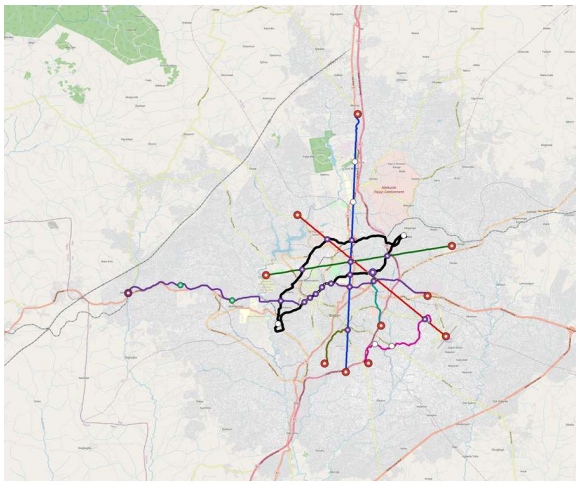
Generated City Model - Ibadan

west-africa / Nigeria / Ibadan

Metric	Value
Path	cities/catalogue/west-africa/Nigeria/Ibadan
Recorded ISO / planning population	NG / 3,900,000
Family	metro-6car
Lines / stations	8 / 127
Route length	139.5 km
Fleet	343 trainsets
Routing-demand cells (legacy)	41%; not population access
City CAPEX incl. dedicated solar	USD 3.52bn (USD 25M / km); shared national factory excluded
Residents within 800 m station circles	841,913 (2020; 21.3% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Ibadan network map

Line	Length	Stations	Trainsets	Shape
line-1	21.5 km	10	45	radial
line-2	17.2 km	10	41	radial
line-3	15.8 km	6	30	radial
line-4	30.3 km	44	134	radial
line-5	34.8 km	43	35	ring
line-6	5.2 km	7	24	radial
line-7	10.3 km	5	23	radial
line-8	4.4 km	2	11	radial

Generated City Model - Ilorin

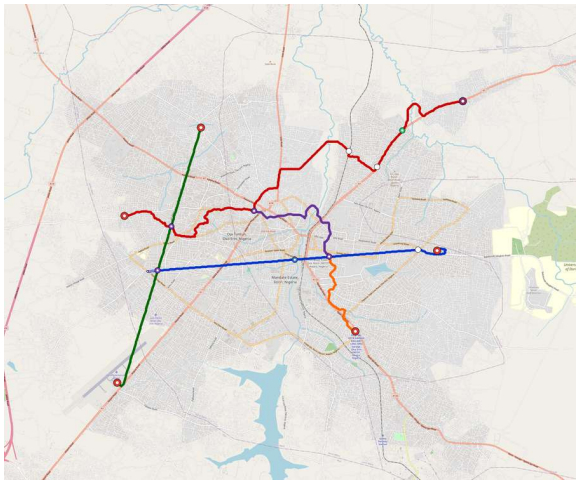
west-africa / Nigeria / Ilorin

Metric	Value
Path	cities/catalogue/west-africa/Nigeria/Ilorin
Recorded ISO / planning population	NG / 1,000,000
Family	light-metro-3car
Lines / stations	5 / 21
Route length	57.8 km

Metric	Value
Fleet	180 trainsets
Routing-demand cells (legacy)	55%; not population access
City CAPEX incl. dedicated solar	USD 1.01bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	171,254 (2020; 17.2% of raster bbox)
Direct / reachable line pairs	60.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Ilorin network map

Line	Length	Stations	Trainsets	Shape
line-1	13.1 km	6	40	radial
line-2	22.8 km	7	71	radial
line-3	11.8 km	4	36	radial
line-4	5.7 km	2	18	radial
line-5	4.4 km	2	15	radial

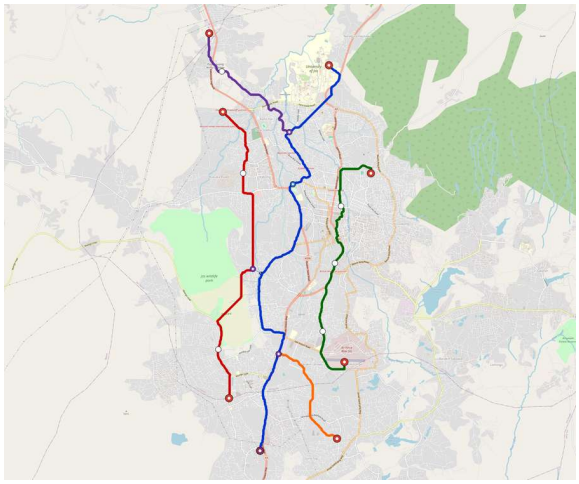
Generated City Model - Jos

west-africa / Nigeria / Jos

Metric	Value
Path	cities/catalogue/west-africa/Nigeria/Jos
Recorded ISO / planning population	NG / 900,000
Family	light-metro-3car
Lines / stations	5 / 21
Route length	58.4 km
Fleet	185 trainsets
Routing-demand cells (legacy)	43%; not population access
City CAPEX incl. dedicated solar	USD 990M (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	138,443 (2020; 18.7% of raster bbox)
Direct / reachable line pairs	30.0% / 60.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Jos network map

Line	Length	Stations	Trainsets	Shape
line-1	22.4 km	6	69	radial
line-2	13.3 km	5	42	radial
line-3	11.6 km	5	38	radial
line-4	6.2 km	3	20	radial
line-5	4.8 km	2	16	radial

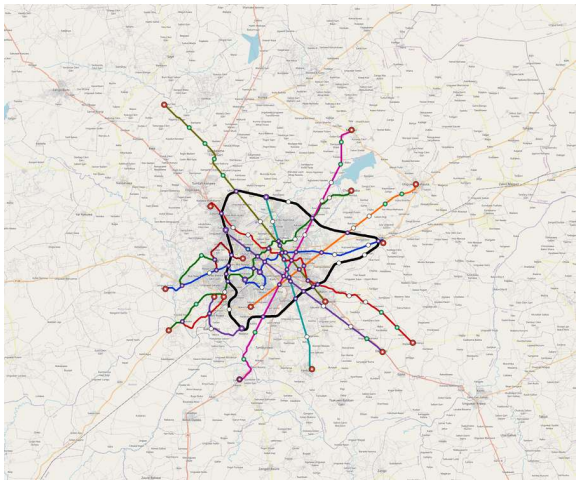
Generated City Model - Kano

west-africa / Nigeria / Kano

Metric	Value
Path	cities/catalogue/west-africa/Nigeria/Kano
Recorded ISO / planning population	NG / 4,200,000
Family	metro-6car
Lines / stations	14 / 168
Route length	473.7 km
Fleet	777 trainsets
Routing-demand cells (legacy)	56%; not population access
City CAPEX incl. dedicated solar	USD 8.41bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	1,882,923 (2020; 30.8% of raster bbox)
Direct / reachable line pairs	46.2% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Kano network map

Line	Length	Stations	Trainsets	Shape
line-1	47.5 km	17	85	radial
line-2	49.5 km	18	96	radial

Line	Length	Stations	Trainsets	Shape
line-3	48.5 km	19	90	radial
line-4	35.5 km	13	63	radial
line-5	36.7 km	13	70	radial
line-6	30.3 km	12	58	radial
line-7	50.1 km	17	97	radial
line-8	44.7 km	17	85	radial
line-9	83.2 km	23	37	ring
line-10	10.2 km	4	20	radial
line-11	6.8 km	3	15	radial
line-12	9.9 km	4	20	radial
line-13	9.6 km	4	20	radial
line-14	11.3 km	4	21	radial

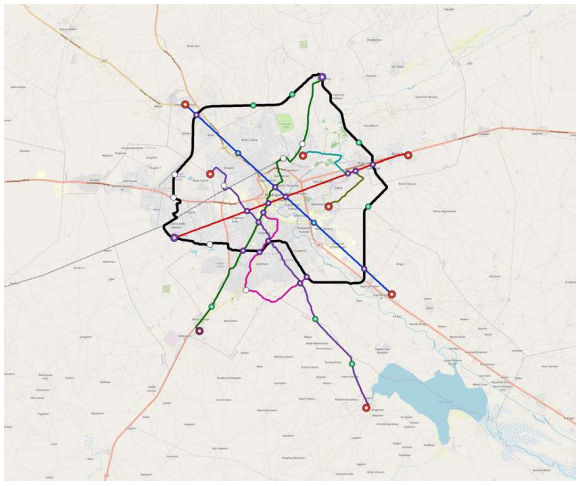
Generated City Model - Maiduguri

west-africa / Nigeria / Maiduguri

Metric	Value
Path	cities/catalogue/west-africa/Nigeria/Maiduguri
Recorded ISO / planning population	NG / 1,200,000
Family	metro-4car
Lines / stations	8 / 68
Route length	194.5 km
Fleet	253 trainsets
Routing-demand cells (legacy)	47%; not population access
City CAPEX incl. dedicated solar	USD 3.23bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	323,645 (2020; 26.6% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Maiduguri network map

Line	Length	Stations	Trainsets	Shape
line-1	24.6 km	8	39	radial
line-2	21.6 km	10	40	radial
line-3	30.4 km	13	53	radial
line-4	28.7 km	10	48	radial
line-5	64.6 km	17	26	ring
line-6	5.6 km	2	11	radial
line-7	14.3 km	6	26	radial
line-8	4.8 km	2	10	radial

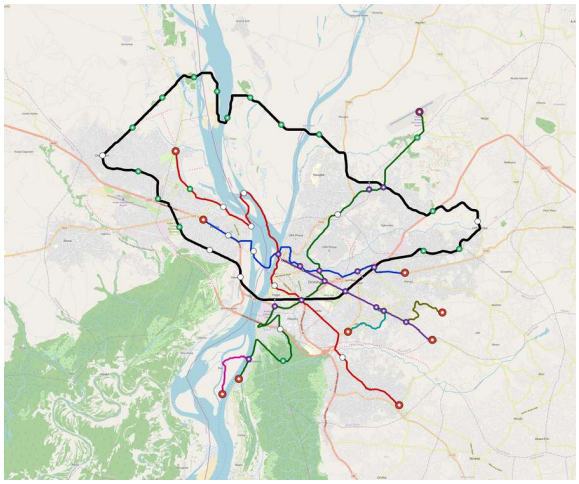
Generated City Model - Onitsha

west-africa / Nigeria / Onitsha

Metric	Value
Path	cities/catalogue/west-africa/Nigeria/Onitsha
Recorded ISO / planning population	NG / 1,500,000
Family	metro-4car
Lines / stations	8 / 79
Route length	230.8 km
Fleet	278 trainsets
Routing-demand cells (legacy)	80%; not population access
City CAPEX incl. dedicated solar	USD 3.46bn (USD 15M / km); shared national factory excluded
Residents within 800 m station circles	396,481 (2020; 19.6% of raster bbox)
Direct / reachable line pairs	46.4% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Onitsha network map

Line	Length	Stations	Trainsets	Shape
line-1	25.2 km	9	41	radial
line-2	40.0 km	11	61	radial
line-3	42.6 km	15	67	radial
line-4	15.0 km	9	34	radial
line-5	91.5 km	27	38	ring
line-6	5.6 km	4	15	radial
line-7	5.3 km	2	11	radial
line-8	5.5 km	2	11	radial

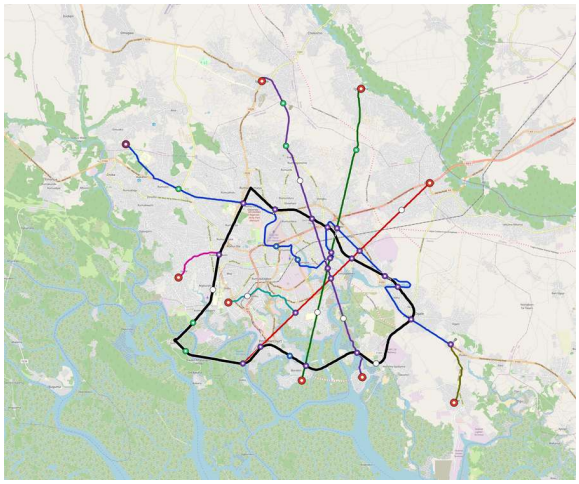
Generated City Model - Port Harcourt

west-africa / Nigeria / Port-Harcourt

Metric	Value
Path	cities/catalogue/west-africa/Nigeria/Port-Harcourt
Recorded ISO / planning population	NG / 3,000,000
Family	metro-4car
Lines / stations	8 / 77
Route length	206.4 km
Fleet	270 trainsets
Routing-demand cells (legacy)	37%; not population access
City CAPEX incl. dedicated solar	USD 3.38bn (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	415,470 (2020; 18.0% of raster bbox)
Direct / reachable line pairs	42.9% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Port Harcourt network map

Line	Length	Stations	Trainsets	Shape
line-1	51.1 km	18	81	radial
line-2	23.5 km	9	40	radial
line-3	25.6 km	10	42	radial
line-4	29.2 km	11	47	radial
line-5	60.0 km	22	25	ring
line-6	7.0 km	3	14	radial
line-7	4.5 km	2	10	radial
line-8	5.3 km	2	11	radial

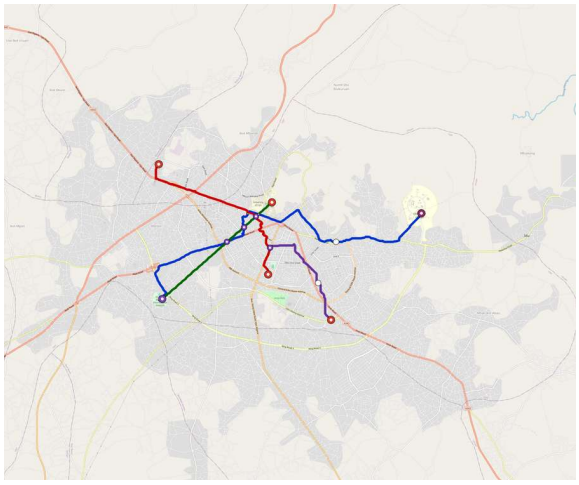
Generated City Model - Uyo

west-africa / Nigeria / Uyo

Metric	Value
Path	cities/catalogue/west-africa/Nigeria/Uyo
Recorded ISO / planning population	NG / 800,000
Family	light-metro-3car
Lines / stations	4 / 23
Route length	35.7 km
Fleet	124 trainsets
Routing-demand cells (legacy)	66%; not population access
City CAPEX incl. dedicated solar	USD 736M (USD 21M / km); shared national factory excluded
Residents within 800 m station circles	105,953 (2020; 13.0% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Uyo network map

Line	Length	Stations	Trainsets	Shape
line-1	15.8 km	8	51	radial
line-2	8.6 km	5	28	radial
line-3	6.6 km	7	28	radial
line-4	4.7 km	3	17	radial

Senegal

Generated City Model - Dakar

west-africa / Senegal / Dakar

Metric	Value
Path	cities/catalogue/west-africa/Senegal/Dakar
Recorded ISO / planning population	SN / 4,030,000
Family	metro-6car
Lines / stations	9 / 165
Route length	267.2 km
Fleet	526 trainsets
Routing-demand cells (legacy)	67%; not population access
City CAPEX incl. dedicated solar	USD 5.40bn (USD 20M / km); shared national factory excluded
Residents within 800 m station circles	1,536,179 (2020; 42.0% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.

Dakar network map

Line	Length	Stations	Trainsets	Shape
line-1	45.2 km	11	81	radial
line-2	28.5 km	21	79	radial
line-3	32.5 km	10	60	radial
line-4	29.3 km	36	116	radial
line-5	35.9 km	23	87	radial
line-6	71.6 km	53	49	ring
line-7	7.5 km	4	18	radial
line-8	11.4 km	5	24	radial
line-9	5.3 km	2	12	radial

west-asia

Egypt

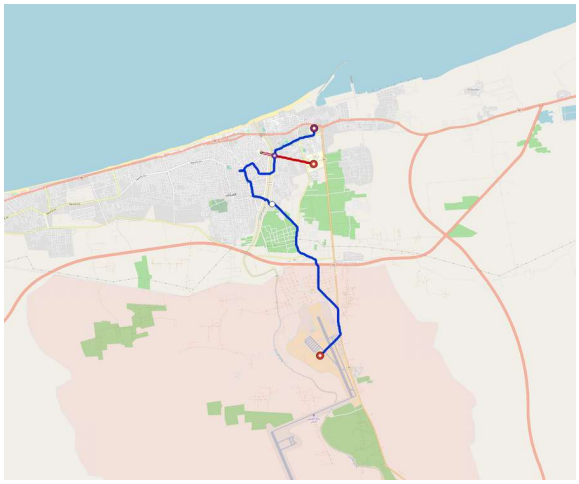
Generated City Model - Arish

west-asia / Egypt / Arish

Metric	Value
Path	cities/catalogue/west-asia/Egypt/Arish
Recorded ISO / planning population	EG / 300,000
Family	tram-2car
Lines / stations	2 / 7
Route length	14.8 km
Fleet	39 trainsets
Routing-demand cells (legacy)	56%; not population access
City CAPEX incl. dedicated solar	USD 241M (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	29,632 (2020; 25.1% of raster bbox)
Direct / reachable line pairs	100.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Arish network map

Line	Length	Stations	Trainsets	Shape
line-1	12.9 km	4	29	radial
line-2	2.0 km	3	10	radial

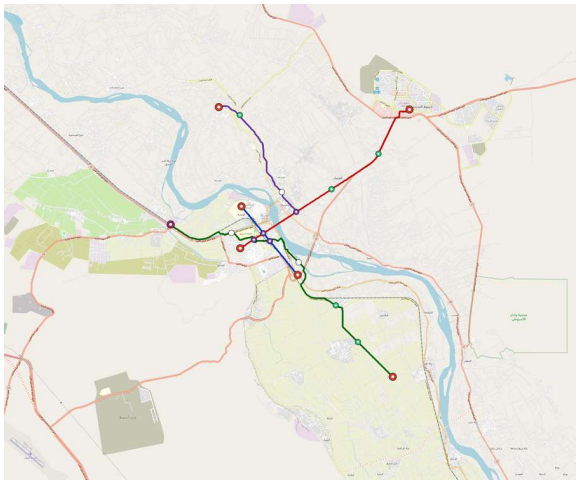
Generated City Model - Asyut

west-asia / Egypt / Asyut

Metric	Value
Path	cities/catalogue/west-asia/Egypt/Asyut
Recorded ISO / planning population	EG / 600,000
Family	light-metro-3car
Lines / stations	4 / 23
Route length	61.7 km
Fleet	196 trainsets
Routing-demand cells (legacy)	61%; not population access
City CAPEX incl. dedicated solar	USD 961M (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	498,502 (2020; 33.9% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Asyut network map

Line	Length	Stations	Trainsets	Shape
line-1	7.0 km	4	24	radial
line-2	17.9 km	7	57	radial
line-3	26.4 km	8	80	radial
line-4	10.4 km	4	35	radial

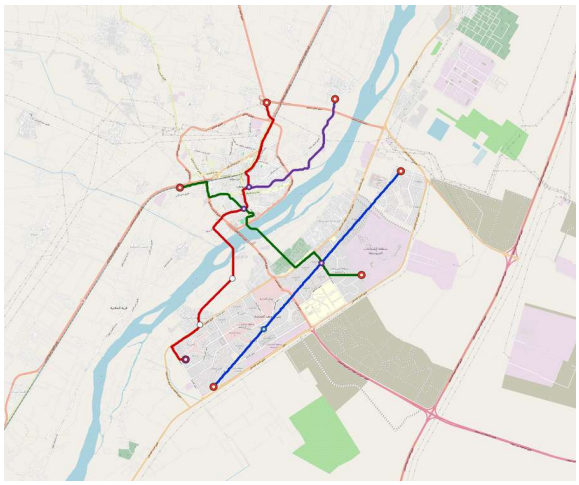
Generated City Model - Beni Suef

west-asia / Egypt / Beni-Suef

Metric	Value
Path	cities/catalogue/west-asia/Egypt/Beni-Suef
Recorded ISO / planning population	EG / 350,000
Family	light-metro-3car
Lines / stations	4 / 16
Route length	38.4 km
Fleet	123 trainsets
Routing-demand cells (legacy)	59%; not population access
City CAPEX incl. dedicated solar	USD 750M (USD 20M / km); shared national factory excluded
Residents within 800 m station circles	183,316 (2020; 19.8% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Beni Suef network map

Line	Length	Stations	Trainsets	Shape
line-1	11.2 km	4	35	radial
line-2	12.5 km	6	40	radial
line-3	9.5 km	4	30	radial

Line	Length	Stations	Trainsets	Shape
line-4	5.2 km	2	18	radial

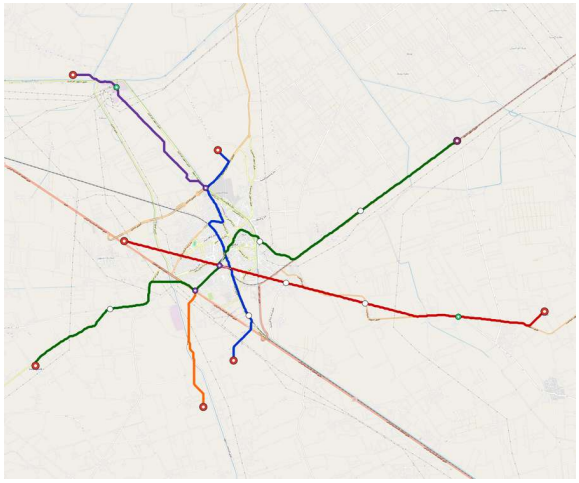
Generated City Model - Damanhur

west-asia / Egypt / Damanhur

Metric	Value
Path	cities/catalogue/west-asia/Egypt/Damanhur
Recorded ISO / planning population	EG / 500,000
Family	light-metro-3car
Lines / stations	5 / 26
Route length	58.2 km
Fleet	190 trainsets
Routing-demand cells (legacy)	82%; not population access
City CAPEX incl. dedicated solar	USD 1.00bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	355,599 (2020; 36.7% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Damanhur network map

Line	Length	Stations	Trainsets	Shape
line-1	9.4 km	6	32	radial
line-2	16.2 km	7	52	radial
line-3	21.7 km	8	68	radial
line-4	6.7 km	3	23	radial
line-5	4.3 km	2	15	radial

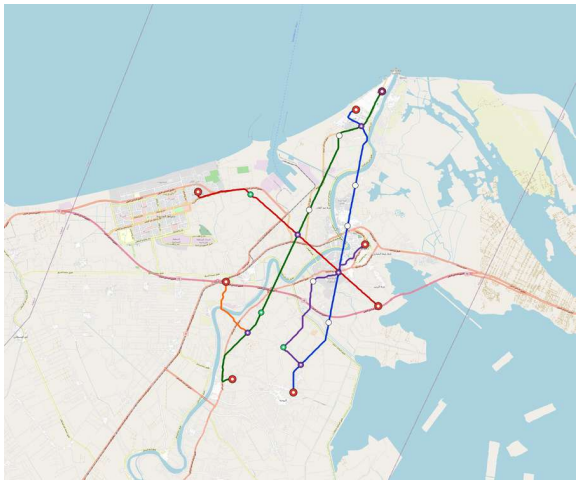
Generated City Model - Damietta

west-asia / Egypt / Damietta

Metric	Value
Path	cities/catalogue/west-asia/Egypt/Damietta
Recorded ISO / planning population	EG / 400,000
Family	light-metro-3car
Lines / stations	5 / 31
Route length	85.1 km
Fleet	264 trainsets
Routing-demand cells (legacy)	62%; not population access
City CAPEX incl. dedicated solar	USD 1.38bn (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	280,497 (2020; 32.1% of raster bbox)
Direct / reachable line pairs	60.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Damietta network map

Line	Length	Stations	Trainsets	Shape
line-1	23.1 km	9	69	radial
line-2	17.8 km	6	56	radial
line-3	27.3 km	8	83	radial
line-4	12.3 km	6	40	radial
line-5	4.7 km	2	16	radial

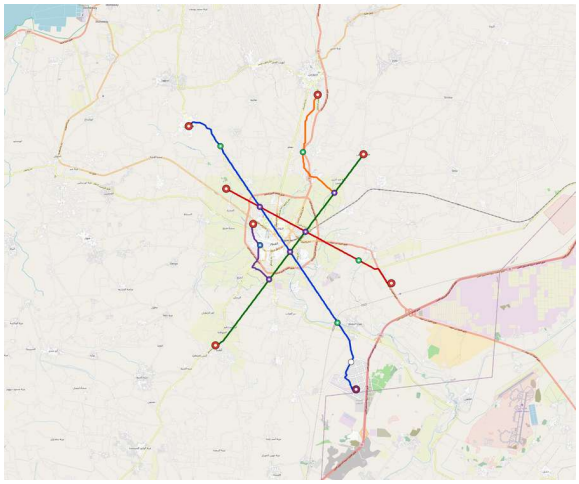
Generated City Model - Fayoum

west-asia / Egypt / Fayoum

Metric	Value
Path	cities/catalogue/west-asia/Egypt/Fayoum
Recorded ISO / planning population	EG / 500,000
Family	light-metro-3car
Lines / stations	5 / 24
Route length	74.1 km
Fleet	233 trainsets
Routing-demand cells (legacy)	70%; not population access
City CAPEX incl. dedicated solar	USD 1.19bn (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	387,518 (2020; 25.9% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Fayoum network map

Line	Length	Stations	Trainsets	Shape
line-1	26.2 km	7	82	radial

Line	Length	Stations	Trainsets	Shape
line-2	15.2 km	5	48	radial
line-3	18.9 km	6	58	radial
line-4	5.1 km	3	17	radial
line-5	8.6 km	3	28	radial

Generated City Model - Hurghada

west-asia / Egypt / Hurghada

Metric	Value
Path	cities/catalogue/west-asia/Egypt/Hurghada
Recorded ISO / planning population	EG / 300,000
Family	tram-2car
Lines / stations	4 / 17
Route length	38.1 km
Fleet	84 trainsets
Routing-demand cells (legacy)	55%; not population access
City CAPEX incl. dedicated solar	USD 586M (USD 15M / km); shared national factory excluded
Residents within 800 m station circles	55,902 (2020; 49.7% of raster bbox)
Direct / reachable line pairs	83.3% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.

Hurghada network map

Line	Length	Stations	Trainsets	Shape
line-1	13.7 km	6	29	radial
line-2	14.2 km	5	29	radial
line-3	5.3 km	4	15	radial
line-4	4.8 km	2	11	radial

Generated City Model - Ismailia

west-asia / Egypt / Ismailia

Metric	Value
Path	cities/catalogue/west-asia/Egypt/Ismailia
Recorded ISO / planning population	EG / 700,000
Family	light-metro-3car
Lines / stations	4 / 28
Route length	60.1 km
Fleet	191 trainsets
Routing-demand cells (legacy)	68%; not population access
City CAPEX incl. dedicated solar	USD 1.10bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	294,095 (2020; 43.7% of raster bbox)
Direct / reachable line pairs	83.3% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.

Ismailia network map

Line	Length	Stations	Trainsets	Shape
line-1	14.8 km	6	46	radial
line-2	25.5 km	12	80	radial
line-3	9.7 km	4	31	radial
line-4	10.1 km	6	34	radial

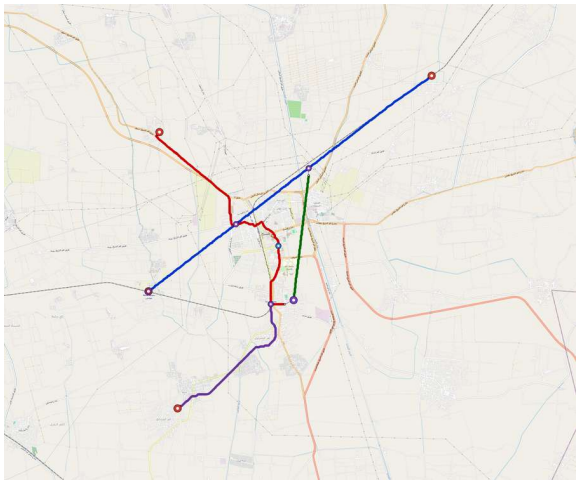
Generated City Model - Kafr El Sheikh

west-asia / Egypt / Kafr-El-Sheikh

Metric	Value
Path	cities/catalogue/west-asia/Egypt/Kafr-El-Sheikh
Recorded ISO / planning population	EG / 300,000
Family	tram-2car
Lines / stations	4 / 13
Route length	33.2 km
Fleet	72 trainsets
Routing-demand cells (legacy)	78%; not population access
City CAPEX incl. dedicated solar	USD 494M (USD 15M / km); shared national factory excluded
Residents within 800 m station circles	164,506 (2020; 32.3% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Kafr El Sheikh network map

Line	Length	Stations	Trainsets	Shape
line-1	13.6 km	4	28	radial
line-2	9.5 km	5	20	radial
line-3	4.5 km	2	11	radial
line-4	5.6 km	2	13	radial

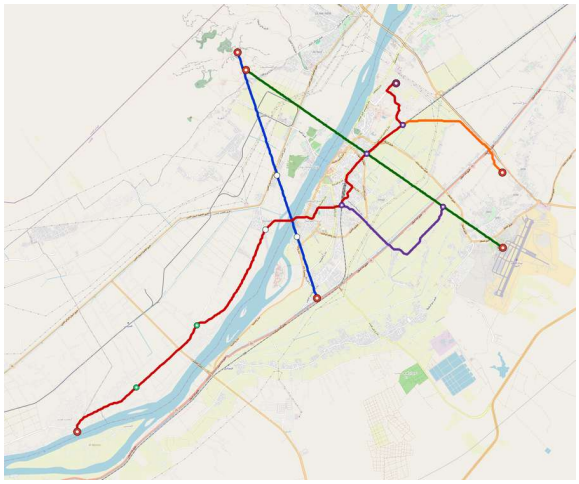
Generated City Model - Luxor

west-asia / Egypt / Luxor

Metric	Value
Path	cities/catalogue/west-asia/Egypt/Luxor
Recorded ISO / planning population	EG / 500,000
Family	light-metro-3car
Lines / stations	5 / 20
Route length	55.9 km
Fleet	177 trainsets
Routing-demand cells (legacy)	55%; not population access
City CAPEX incl. dedicated solar	USD 965M (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	1,029,508 (2020; 35.1% of raster bbox)
Direct / reachable line pairs	40.0% / 60.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Luxor network map

Line	Length	Stations	Trainsets	Shape
line-1	10.1 km	4	31	radial
line-2	23.1 km	8	73	radial
line-3	12.6 km	4	39	radial
line-4	5.4 km	2	18	radial
line-5	4.7 km	2	16	radial

Generated City Model - Mahalla

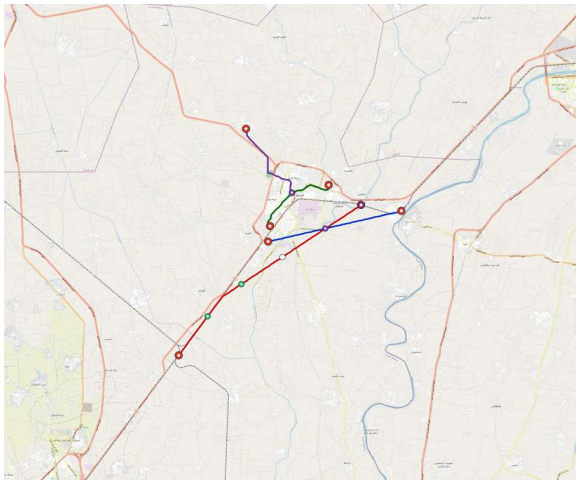
west-asia / Egypt / Mahalla

Metric	Value
Path	cities/catalogue/west-asia/Egypt/Mahalla
Recorded ISO / planning population	EG / 600,000
Family	light-metro-3car
Lines / stations	4 / 14
Route length	41.1 km
Fleet	131 trainsets
Routing-demand cells (legacy)	47%; not population access
City CAPEX incl. dedicated solar	USD 683M (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	313,764 (2020; 19.6% of raster bbox)

Metric	Value
Direct / reachable line pairs	33.3% / 33.3%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Mahalla network map

Line	Length	Stations	Trainsets	Shape
line-1	10.0 km	3	32	radial
line-2	18.1 km	6	58	radial
line-3	6.6 km	3	21	radial
line-4	6.3 km	2	20	radial

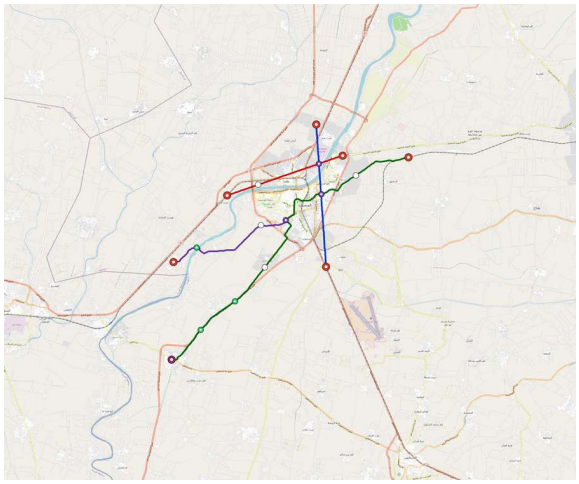
Generated City Model - Mansoura Eg

west-asia / Egypt / Mansoura-Eg

Metric	Value
Path	cities/catalogue/west-asia/Egypt/Mansoura-Eg
Recorded ISO / planning population	EG / 1,000,000
Family	light-metro-3car
Lines / stations	4 / 20
Route length	56.9 km
Fleet	177 trainsets
Routing-demand cells (legacy)	45%; not population access
City CAPEX incl. dedicated solar	USD 964M (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	328,522 (2020; 18.8% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Mansoura Eg network map

Line	Length	Stations	Trainsets	Shape
line-1	10.1 km	4	31	radial
line-2	9.2 km	4	30	radial
line-3	27.9 km	8	85	radial
line-4	9.7 km	4	31	radial

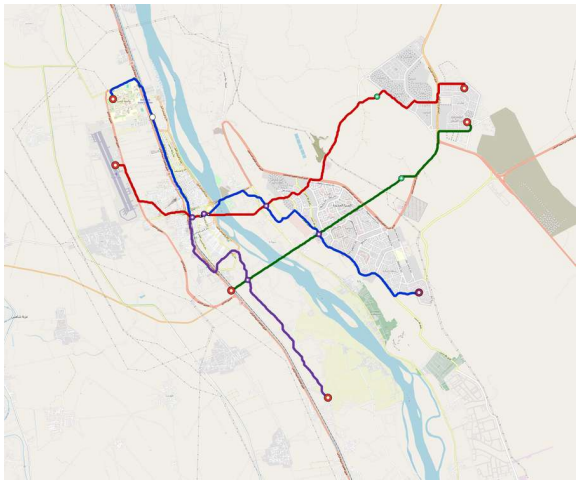
Generated City Model - Minya

west-asia / Egypt / Minya

Metric	Value
Path	cities/catalogue/west-asia/Egypt/Minya
Recorded ISO / planning population	EG / 600,000
Family	light-metro-3car
Lines / stations	4 / 29
Route length	58.4 km
Fleet	191 trainsets
Routing-demand cells (legacy)	64%; not population access
City CAPEX incl. dedicated solar	USD 1.18bn (USD 20M / km); shared national factory excluded
Residents within 800 m station circles	215,945 (2020; 20.5% of raster bbox)
Direct / reachable line pairs	83.3% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Minya network map

Line	Length	Stations	Trainsets	Shape
line-1	18.4 km	11	60	radial
line-2	17.7 km	10	59	radial
line-3	12.0 km	5	40	radial
line-4	10.3 km	3	32	radial

Generated City Model - Port Said

west-asia / Egypt / Port-Said

Metric	Value
Path	cities/catalogue/west-asia/Egypt/Port-Said
Recorded ISO / planning population	EG / 800,000
Family	light-metro-3car
Lines / stations	4 / 19
Route length	31.1 km
Fleet	106 trainsets
Routing-demand cells (legacy)	64%; not population access
City CAPEX incl. dedicated solar	USD 653M (USD 21M / km); shared national factory excluded
Residents within 800 m station circles	180,882 (2020; 26.8% of raster bbox)

Metric	Value
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.

Port Said network map

Line	Length	Stations	Trainsets	Shape
line-1	11.3 km	6	38	radial
line-2	6.0 km	5	23	radial
line-3	8.8 km	5	28	radial
line-4	5.0 km	3	17	radial

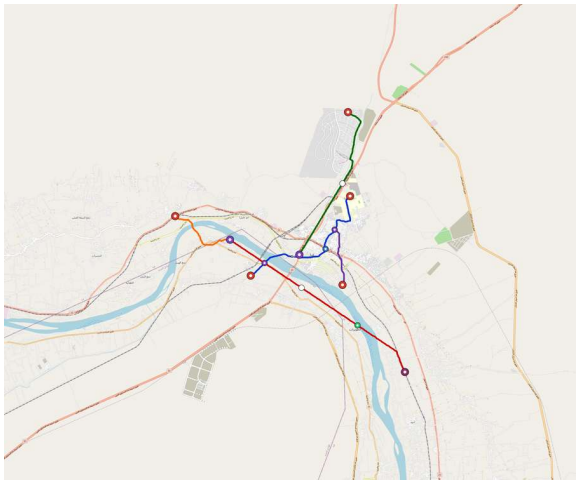
Generated City Model - Qena

west-asia / Egypt / Qena

Metric	Value
Path	cities/catalogue/west-asia/Egypt/Qena
Recorded ISO / planning population	EG / 350,000
Family	light-metro-3car
Lines / stations	5 / 18
Route length	54.0 km
Fleet	169 trainsets
Routing-demand cells (legacy)	73%; not population access
City CAPEX incl. dedicated solar	USD 1.04bn (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	105,661 (2020; 34.0% of raster bbox)
Direct / reachable line pairs	40.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Qena network map

Line	Length	Stations	Trainsets	Shape
line-1	12.4 km	6	39	radial
line-2	18.4 km	5	57	radial
line-3	13.2 km	3	39	radial
line-4	4.6 km	2	16	radial
line-5	5.4 km	2	18	radial

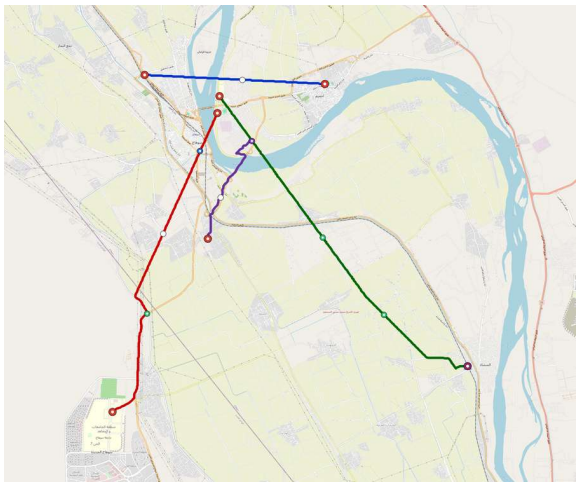
Generated City Model - Sohag

west-asia / Egypt / Sohag

Metric	Value
Path	cities/catalogue/west-asia/Egypt/Sohag
Recorded ISO / planning population	EG / 550,000
Family	light-metro-3car
Lines / stations	4 / 16
Route length	39.0 km
Fleet	127 trainsets
Routing-demand cells (legacy)	47%; not population access
City CAPEX incl. dedicated solar	USD 689M (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	327,400 (2020; 21.6% of raster bbox)
Direct / reachable line pairs	16.7% / 16.7%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Sohag network map

Line	Length	Stations	Trainsets	Shape
line-1	6.6 km	3	21	radial
line-2	12.9 km	5	41	radial
line-3	14.8 km	5	48	radial
line-4	4.8 km	3	17	radial

Generated City Model - Suez

west-asia / Egypt / Suez

Metric	Value
Path	cities/catalogue/west-asia/Egypt/Suez
Recorded ISO / planning population	EG / 800,000
Family	light-metro-3car
Lines / stations	5 / 24
Route length	71.3 km
Fleet	222 trainsets
Routing-demand cells (legacy)	67%; not population access
City CAPEX incl. dedicated solar	USD 1.17bn (USD 16M / km); shared national factory excluded

Metric	Value
Residents within 800 m station circles	190,828 (2020; 34.2% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.

Suez network map

Line	Length	Stations	Trainsets	Shape
line-1	25.0 km	9	78	radial
line-2	16.2 km	5	49	radial
line-3	17.0 km	5	52	radial
line-4	4.7 km	2	16	radial
line-5	8.4 km	3	27	radial

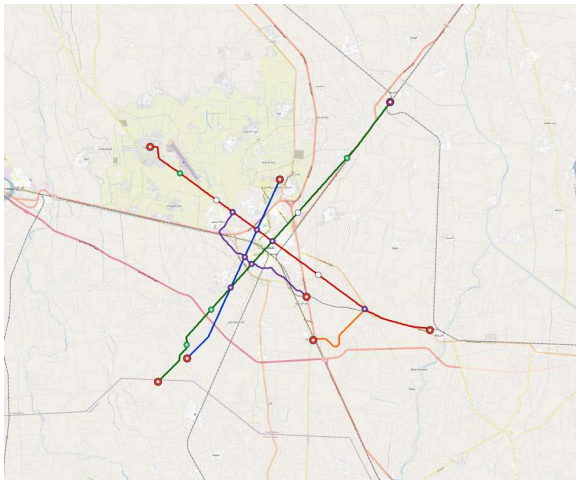
Generated City Model - Tanta

west-asia / Egypt / Tanta

Metric	Value
Path	cities/catalogue/west-asia/Egypt/Tanta
Recorded ISO / planning population	EG / 750,000
Family	light-metro-3car
Lines / stations	5 / 29
Route length	85.9 km
Fleet	269 trainsets
Routing-demand cells (legacy)	68%; not population access
City CAPEX incl. dedicated solar	USD 1.33bn (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	356,753 (2020; 23.5% of raster bbox)
Direct / reachable line pairs	70.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Tanta network map

Line	Length	Stations	Trainsets	Shape
line-1	15.3 km	5	48	radial
line-2	27.0 km	9	85	radial
line-3	28.4 km	9	87	radial
line-4	10.0 km	4	31	radial
line-5	5.3 km	2	18	radial

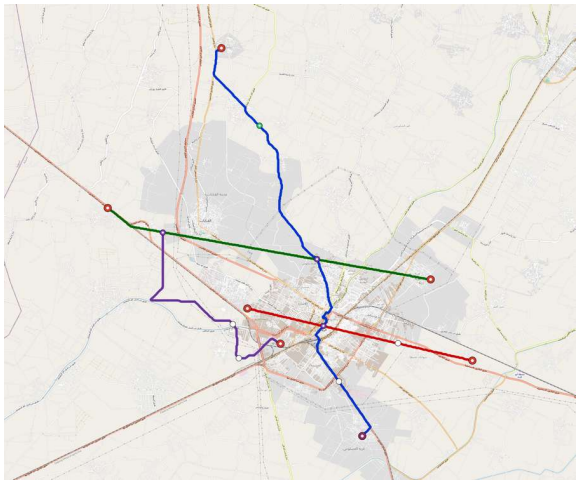
Generated City Model - Zagazig

west-asia / Egypt / Zagazig

Metric	Value
Path	cities/catalogue/west-asia/Egypt/Zagazig
Recorded ISO / planning population	EG / 700,000
Family	light-metro-3car
Lines / stations	4 / 18
Route length	46.5 km
Fleet	147 trainsets
Routing-demand cells (legacy)	59%; not population access
City CAPEX incl. dedicated solar	USD 789M (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	380,624 (2020; 23.7% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Zagazig network map

Line	Length	Stations	Trainsets	Shape
line-1	17.0 km	6	54	radial
line-2	8.6 km	4	27	radial

Line	Length	Stations	Trainsets	Shape
line-3	12.2 km	4	38	radial
line-4	8.8 km	4	28	radial

Iraq

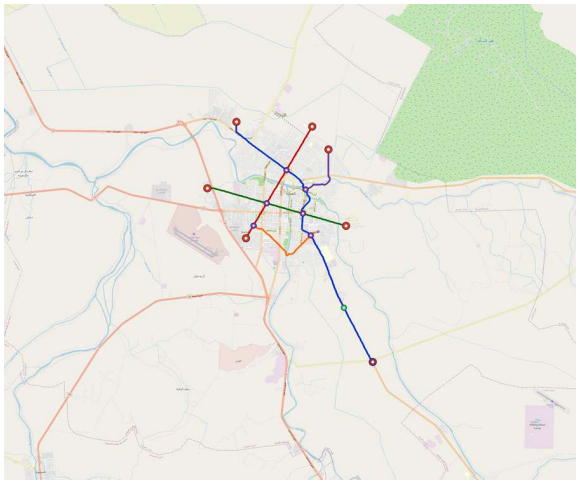
Generated City Model - Amarah

west-asia / Iraq / Amarah

Metric	Value
Path	cities/catalogue/west-asia/Iraq/Amarah
Recorded ISO / planning population	IQ / 660,000
Family	light-metro-3car
Lines / stations	5 / 21
Route length	55.3 km
Fleet	177 trainsets
Routing-demand cells (legacy)	56%; not population access
City CAPEX incl. dedicated solar	USD 1.13bn (USD 20M / km); shared national factory excluded
Residents within 800 m station circles	155,109 (2020; 24.9% of raster bbox)
Direct / reachable line pairs	60.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Amarah network map

Line	Length	Stations	Trainsets	Shape
line-1	24.6 km	7	76	radial
line-2	9.8 km	5	32	radial
line-3	10.5 km	4	35	radial
line-4	4.2 km	2	14	radial
line-5	6.1 km	3	20	radial

Generated City Model - Baghdad

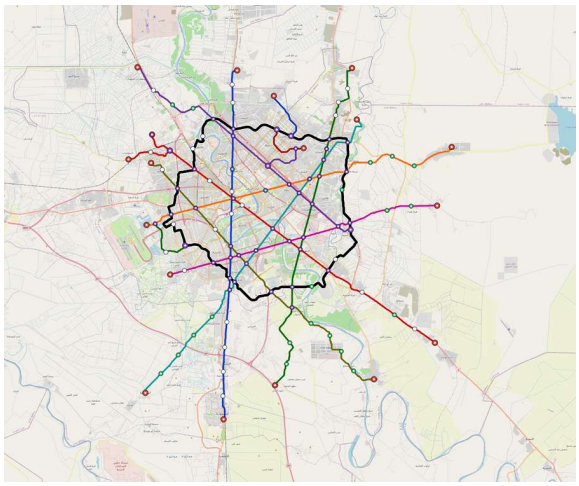
west-asia / Iraq / Baghdad

Metric	Value
Path	cities/catalogue/west-asia/Iraq/Baghdad
Recorded ISO / planning population	IQ / 9,780,429
Family	metro-6car
Lines / stations	14 / 159
Route length	513.3 km
Fleet	809 trainsets
Routing-demand cells (legacy)	40%; not population access
Original catalogue city CAPEX	USD 7.63bn (USD 15M / km); shared national factory excluded
Residents within 800 m station circles	1,129,864 (2020; 18.9% of raster bbox)

Metric	Value
Direct / reachable line pairs	39.6% / 100.0%
Current conditional programme: local_positive	USD 8.67bn including final assembly/component plants; unpriced scope and funding gaps remain

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Baghdad network map

Line	Length	Stations	Trainsets	Shape
line-1	49.1 km	17	86	radial
line-2	52.9 km	14	95	radial
line-3	49.5 km	16	90	radial
line-4	41.9 km	12	76	radial
line-5	47.3 km	16	86	radial
line-6	52.5 km	16	95	radial
line-7	40.0 km	13	72	radial
line-8	48.1 km	15	89	radial
line-9	97.7 km	26	45	ring
line-10	6.8 km	3	15	radial
line-11	6.9 km	2	14	radial
line-12	6.6 km	3	15	radial
line-13	8.0 km	4	18	radial
line-14	6.0 km	2	13	radial

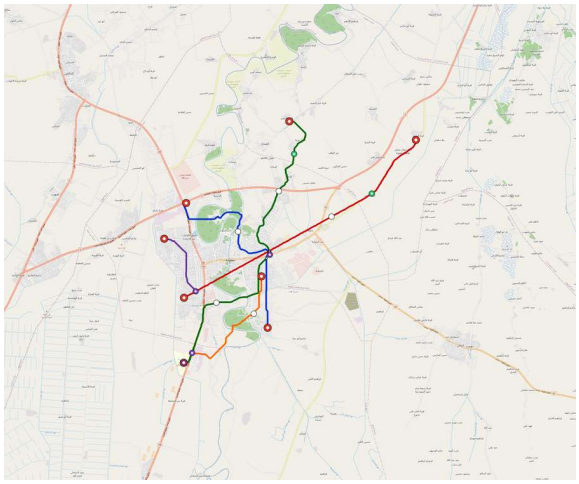
Generated City Model - Baqubah

west-asia / Iraq / Baqubah

Metric	Value
Path	cities/catalogue/west-asia/Iraq/Baqubah
Recorded ISO / planning population	IQ / 470,000
Family	light-metro-3car
Lines / stations	5 / 32
Route length	76.7 km
Fleet	243 trainsets
Routing-demand cells (legacy)	56%; not population access
City CAPEX incl. dedicated solar	USD 1.32bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	40,361 (2020; 25.2% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Baqubah network map

Line	Length	Stations	Trainsets	Shape
line-1	15.6 km	9	50	radial
line-2	22.0 km	6	67	radial
line-3	25.8 km	12	83	radial
line-4	4.7 km	2	16	radial
line-5	8.7 km	3	27	radial

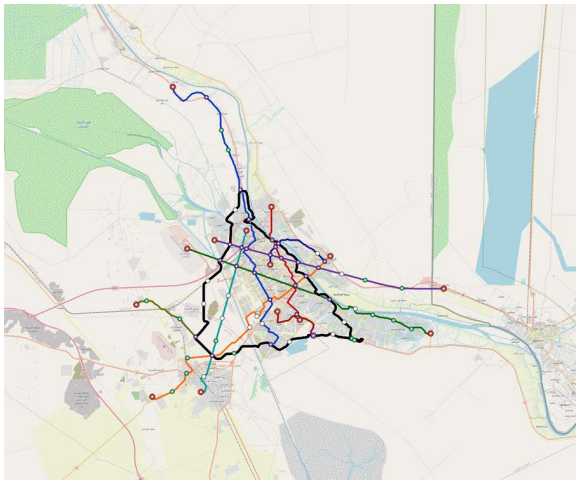
Generated City Model - Basra

west-asia / Iraq / Basra

Metric	Value
Path	cities/catalogue/west-asia/Iraq/Basra
Recorded ISO / planning population	IQ / 3,955,000
Family	metro-6car
Lines / stations	11 / 137
Route length	348.0 km
Fleet	557 trainsets
Routing-demand cells (legacy)	72%; not population access
City CAPEX incl. dedicated solar	USD 6.58bn (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	626,018 (2020; 27.0% of raster bbox)
Direct / reachable line pairs	45.5% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Basra network map

Line	Length	Stations	Trainsets	Shape
line-1	48.3 km	18	95	radial
line-2	21.9 km	17	62	radial

Line	Length	Stations	Trainsets	Shape
line-3	41.5 km	11	79	radial
line-4	35.0 km	12	65	radial
line-5	41.2 km	16	81	radial
line-6	26.4 km	10	48	radial
line-7	95.2 km	38	45	ring
line-8	10.7 km	4	23	radial
line-9	11.4 km	6	26	radial
line-10	9.9 km	3	19	radial
line-11	6.6 km	2	14	radial

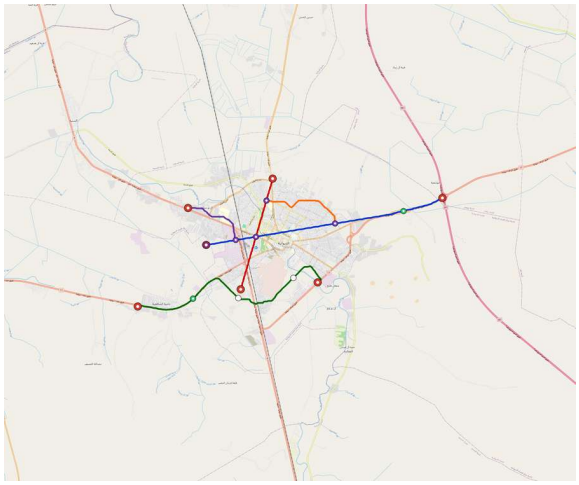
Generated City Model - Diwaniyah

west-asia / Iraq / Diwaniyah

Metric	Value
Path	cities/catalogue/west-asia/Iraq/Diwaniyah
Recorded ISO / planning population	IQ / 440,000
Family	light-metro-3car
Lines / stations	5 / 19
Route length	53.5 km
Fleet	171 trainsets
Routing-demand cells (legacy)	46%; not population access
City CAPEX incl. dedicated solar	USD 935M (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	127,356 (2020; 18.5% of raster bbox)
Direct / reachable line pairs	40.0% / 60.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Diwaniyah network map

Line	Length	Stations	Trainsets	Shape
line-1	18.0 km	6	56	radial
line-2	8.5 km	4	27	radial
line-3	16.4 km	5	52	radial
line-4	4.7 km	2	16	radial
line-5	6.0 km	2	20	radial

Generated City Model - Duhok

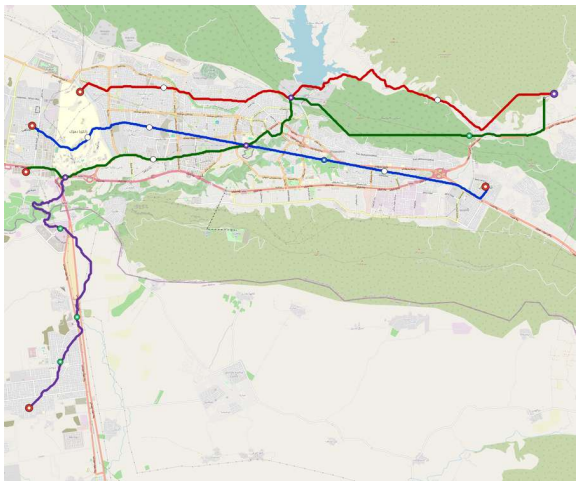
west-asia / Iraq / Duhok

Metric	Value
Path	cities/catalogue/west-asia/Iraq/Duhok
Recorded ISO / planning population	IQ / 360,000
Family	light-metro-3car
Lines / stations	4 / 24

Metric	Value
Route length	68.4 km
Fleet	213 trainsets
Routing-demand cells (legacy)	58%; not population access
City CAPEX incl. dedicated solar	USD 1.08bn (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	266,787 (2020; 38.4% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Duhok network map

Line	Length	Stations	Trainsets	Shape
line-1	17.4 km	7	53	radial
line-2	18.3 km	5	54	radial
line-3	21.6 km	7	67	radial
line-4	11.1 km	5	39	radial

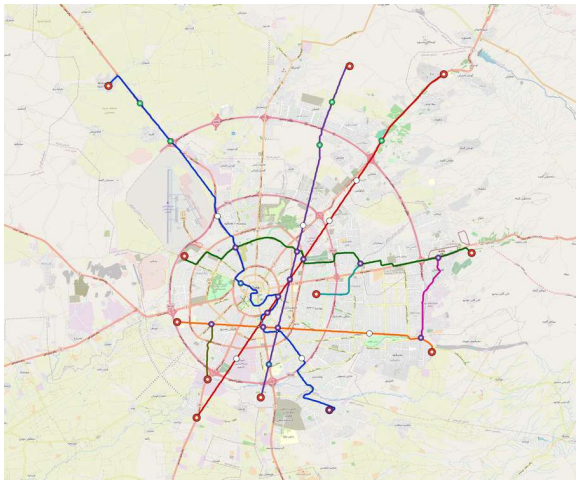
Generated City Model - Erbil

west-asia / Iraq / Erbil

Metric	Value
Path	cities/catalogue/west-asia/Iraq/Erbil
Recorded ISO / planning population	IQ / 1,952,000
Family	metro-4car
Lines / stations	8 / 56
Route length	152.3 km
Fleet	261 trainsets
Routing-demand cells (legacy)	57%; not population access
City CAPEX incl. dedicated solar	USD 2.35bn (USD 15M / km); shared national factory excluded
Residents within 800 m station circles	250,544 (2020; 20.3% of raster bbox)
Direct / reachable line pairs	46.4% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Erbil network map

Line	Length	Stations	Trainsets	Shape
line-1	38.5 km	13	64	radial
line-2	30.8 km	12	51	radial
line-3	25.8 km	7	40	radial
line-4	25.0 km	10	41	radial
line-5	18.0 km	8	34	radial
line-6	4.4 km	2	10	radial
line-7	5.8 km	2	11	radial
line-8	4.0 km	2	10	radial

Generated City Model - Fallujah

west-asia / Iraq / Fallujah

Metric	Value
Path	cities/catalogue/west-asia/Iraq/Fallujah
Recorded ISO / planning population	IQ / 360,000
Family	light-metro-3car
Lines / stations	5 / 23
Route length	60.2 km
Fleet	195 trainsets
Routing-demand cells (legacy)	60%; not population access
City CAPEX incl. dedicated solar	USD 1.06bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	39,150 (2020; 20.5% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.

Fallujah network map

Line	Length	Stations	Trainsets	Shape
line-1	18.0 km	6	56	radial
line-2	15.8 km	6	50	radial
line-3	14.9 km	6	49	radial
line-4	6.9 km	3	24	radial
line-5	4.6 km	2	16	radial

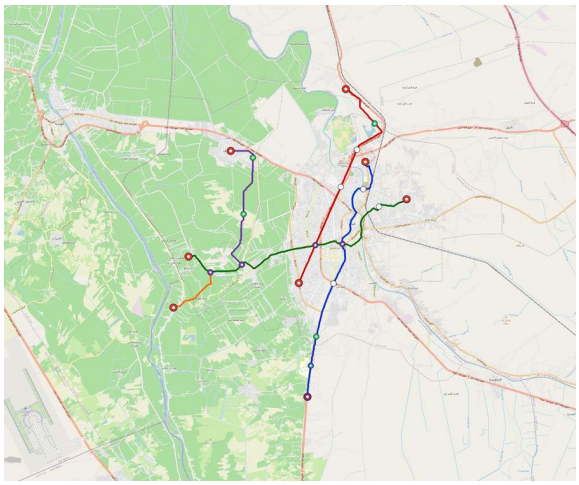
Generated City Model - Hillah

west-asia / Iraq / Hillah

Metric	Value
Path	cities/catalogue/west-asia/Iraq/Hillah
Recorded ISO / planning population	IQ / 700,000
Family	light-metro-3car
Lines / stations	5 / 26
Route length	74.7 km
Fleet	236 trainsets
Routing-demand cells (legacy)	67%; not population access
City CAPEX incl. dedicated solar	USD 1.19bn (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	245,550 (2020; 27.0% of raster bbox)
Direct / reachable line pairs	40.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Hillah network map

Line	Length	Stations	Trainsets	Shape
line-1	21.4 km	7	65	radial
line-2	17.8 km	6	56	radial
line-3	21.2 km	7	67	radial
line-4	10.1 km	4	34	radial
line-5	4.1 km	2	14	radial

Generated City Model - Karbala

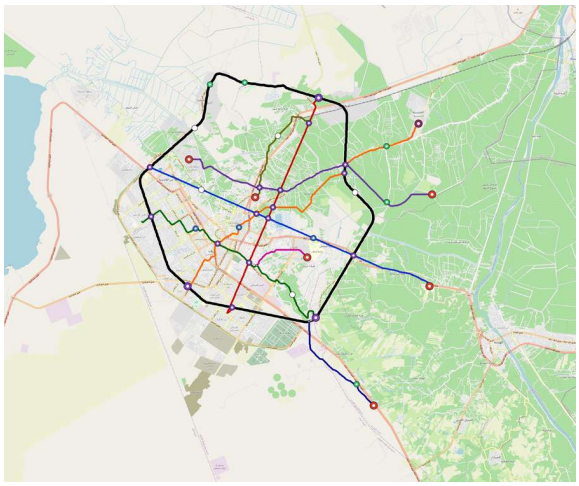
west-asia / Iraq / Karbala

Metric	Value
Path	cities/catalogue/west-asia/Iraq/Karbala
Recorded ISO / planning population	IQ / 1,390,000
Family	metro-4car
Lines / stations	9 / 69
Route length	184.4 km
Fleet	253 trainsets
Routing-demand cells (legacy)	54%; not population access
City CAPEX incl. dedicated solar	USD 2.84bn (USD 15M / km); shared national factory excluded

Metric	Value
Residents within 800 m station circles	370,504 (2020; 30.1% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Karbala network map

Line	Length	Stations	Trainsets	Shape
line-1	23.6 km	8	38	radial
line-2	18.0 km	8	34	radial
line-3	19.2 km	8	35	radial
line-4	21.9 km	7	36	radial
line-5	25.2 km	11	45	radial
line-6	56.6 km	18	24	ring
line-7	4.1 km	2	10	radial
line-8	7.8 km	4	16	radial
line-9	8.0 km	3	15	radial

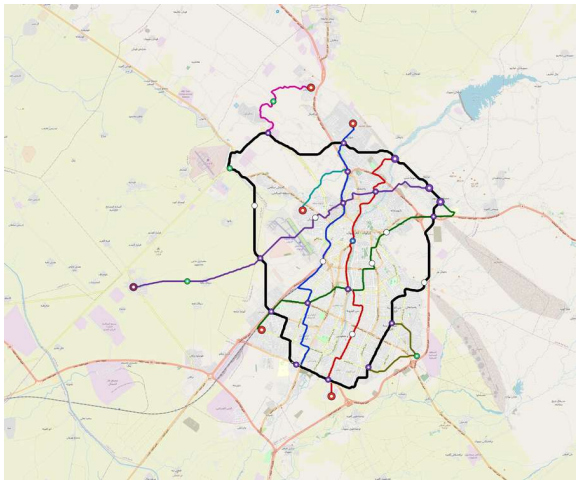
Generated City Model - Kirkuk

west-asia / Iraq / Kirkuk

Metric	Value
Path	cities/catalogue/west-asia/Iraq/Kirkuk
Recorded ISO / planning population	IQ / 1,780,000
Family	metro-4car
Lines / stations	8 / 71
Route length	174.0 km
Fleet	235 trainsets
Routing-demand cells (legacy)	61%; not population access
City CAPEX incl. dedicated solar	USD 2.81bn (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	316,565 (2020; 27.9% of raster bbox)
Direct / reachable line pairs	39.3% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Kirkuk network map

Line	Length	Stations	Trainsets	Shape
line-1	24.2 km	12	47	radial
line-2	24.0 km	9	40	radial
line-3	22.0 km	9	39	radial
line-4	25.3 km	7	41	radial
line-5	58.6 km	24	26	ring
line-6	4.7 km	2	10	radial
line-7	8.0 km	3	15	radial
line-8	7.2 km	5	17	radial

Generated City Model - Kut

west-asia / Iraq / Kut

Metric	Value
Path	cities/catalogue/west-asia/Iraq/Kut
Recorded ISO / planning population	IQ / 410,000
Family	light-metro-3car
Lines / stations	5 / 19
Route length	53.1 km
Fleet	166 trainsets
Routing-demand cells (legacy)	45%; not population access
City CAPEX incl. dedicated solar	USD 1.05bn (USD 20M / km); shared national factory excluded
Residents within 800 m station circles	136,094 (2020; 28.9% of raster bbox)
Direct / reachable line pairs	40.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.

Kut network map

Line	Length	Stations	Trainsets	Shape
line-1	22.2 km	6	67	radial
line-2	7.0 km	4	23	radial
line-3	13.7 km	4	42	radial
line-4	6.2 km	3	20	radial
line-5	4.0 km	2	14	radial

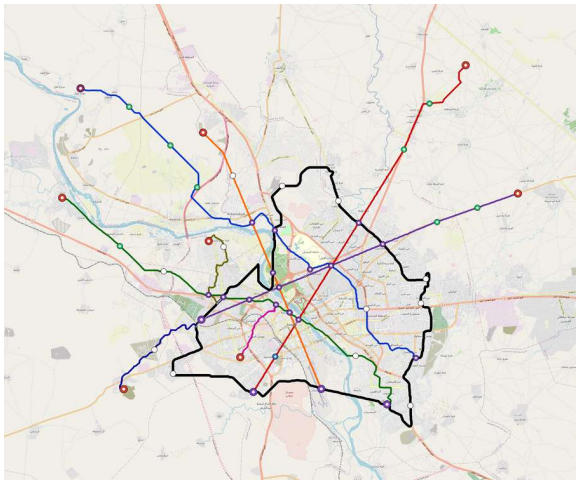
Generated City Model - Mosul

west-asia / Iraq / Mosul

Metric	Value
Path	cities/catalogue/west-asia/Iraq/Mosul
Recorded ISO / planning population	IQ / 1,940,000
Family	metro-4car
Lines / stations	9 / 85
Route length	230.4 km
Fleet	308 trainsets
Routing-demand cells (legacy)	51%; not population access
City CAPEX incl. dedicated solar	USD 3.93bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	310,851 (2020; 29.9% of raster bbox)
Direct / reachable line pairs	52.8% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Mosul network map

Line	Length	Stations	Trainsets	Shape
line-1	35.6 km	13	60	radial
line-2	28.7 km	10	48	radial
line-3	30.4 km	13	53	radial
line-4	25.6 km	10	42	radial
line-5	20.0 km	8	35	radial
line-6	69.4 km	21	27	ring
line-7	4.8 km	2	10	radial
line-8	7.5 km	5	18	radial
line-9	8.3 km	3	15	radial

Generated City Model - Najaf

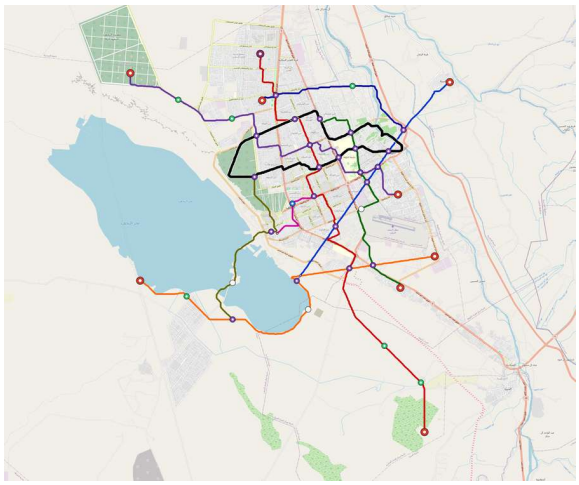
west-asia / Iraq / Najaf

Metric	Value
Path	cities/catalogue/west-asia/Iraq/Najaf
Recorded ISO / planning population	IQ / 1,540,000
Family	metro-4car
Lines / stations	9 / 84

Metric	Value
Route length	193.3 km
Fleet	309 trainsets
Routing-demand cells (legacy)	55%; not population access
City CAPEX incl. dedicated solar	USD 3.33bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	437,722 (2020; 32.1% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Najaf network map

Line	Length	Stations	Trainsets	Shape
line-1	19.3 km	9	37	radial
line-2	37.2 km	18	69	radial
line-3	17.6 km	10	37	radial
line-4	28.4 km	9	46	radial
line-5	27.3 km	9	46	radial
line-6	32.2 km	18	17	ring
line-7	5.1 km	3	13	radial
line-8	14.1 km	4	24	radial
line-9	12.2 km	4	20	radial

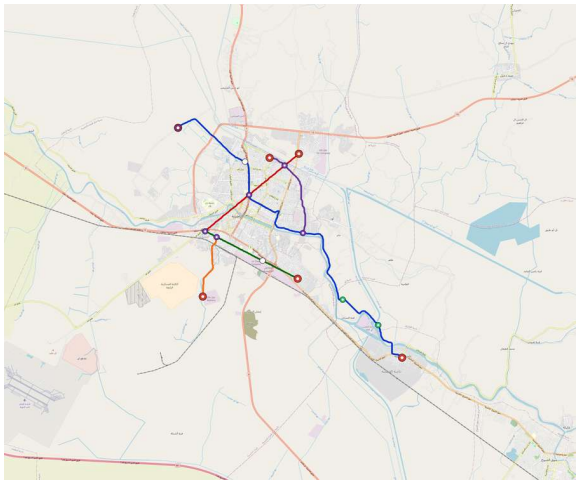
Generated City Model - Nasiriyah

west-asia / Iraq / Nasiriyah

Metric	Value
Path	cities/catalogue/west-asia/Iraq/Nasiriyah
Recorded ISO / planning population	IQ / 705,000
Family	light-metro-3car
Lines / stations	5 / 22
Route length	57.6 km
Fleet	184 trainsets
Routing-demand cells (legacy)	46%; not population access
City CAPEX incl. dedicated solar	USD 1.09bn (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	211,518 (2020; 24.9% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Nasiriyah network map

Line	Length	Stations	Trainsets	Shape
line-1	28.8 km	7	87	radial
line-2	9.5 km	5	32	radial
line-3	8.1 km	5	28	radial
line-4	6.6 km	3	21	radial
line-5	4.6 km	2	16	radial

Generated City Model - Ramadi

west-asia / Iraq / Ramadi

Metric	Value
Path	cities/catalogue/west-asia/Iraq/Ramadi
Recorded ISO / planning population	IQ / 525,000
Family	light-metro-3car
Lines / stations	5 / 29
Route length	54.9 km
Fleet	181 trainsets
Routing-demand cells (legacy)	60%; not population access
City CAPEX incl. dedicated solar	USD 978M (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	97,424 (2020; 26.4% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.

Ramadi network map

Line	Length	Stations	Trainsets	Shape
line-1	12.6 km	8	43	radial
line-2	14.3 km	6	46	radial

Line	Length	Stations	Trainsets	Shape
line-3	17.3 km	10	56	radial
line-4	5.6 km	2	18	radial
line-5	5.1 km	3	18	radial

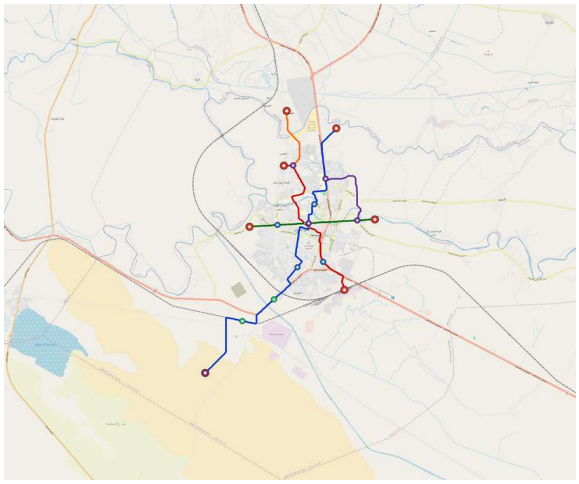
Generated City Model - Samawah

west-asia / Iraq / Samawah

Metric	Value
Path	cities/catalogue/west-asia/Iraq/Samawah
Recorded ISO / planning population	IQ / 373,770
Family	light-metro-3car
Lines / stations	5 / 26
Route length	57.9 km
Fleet	182 trainsets
Routing-demand cells (legacy)	54%; not population access
City CAPEX incl. dedicated solar	USD 1.03bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	158,459 (2020; 33.1% of raster bbox)
Direct / reachable line pairs	60.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Samawah network map

Line	Length	Stations	Trainsets	Shape
line-1	26.6 km	10	81	radial
line-2	12.8 km	6	40	radial
line-3	8.8 km	6	28	radial
line-4	5.4 km	2	18	radial
line-5	4.3 km	2	15	radial

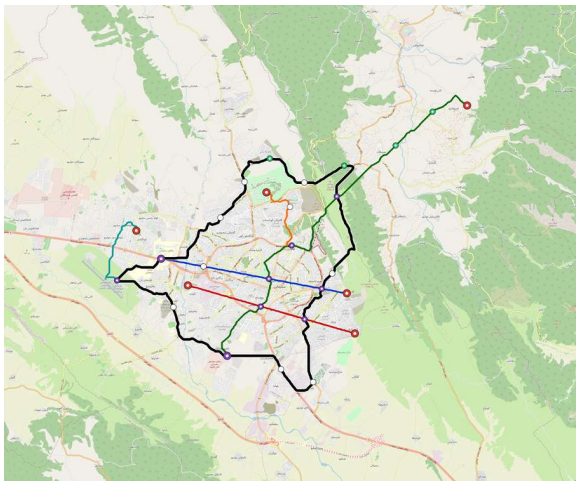
Generated City Model - Sulaymaniyah

west-asia / Iraq / Sulaymaniyah

Metric	Value
Path	cities/catalogue/west-asia/Iraq/Sulaymaniyah
Recorded ISO / planning population	IQ / 2,150,000
Family	metro-4car
Lines / stations	6 / 39
Route length	131.5 km
Fleet	144 trainsets
Routing-demand cells (legacy)	50%; not population access
City CAPEX incl. dedicated solar	USD 2.00bn (USD 15M / km); shared national factory excluded
Residents within 800 m station circles	187,519 (2020; 20.8% of raster bbox)
Direct / reachable line pairs	46.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Sulaymaniyah network map

Line	Length	Stations	Trainsets	Shape
line-1	13.1 km	5	23	radial
line-2	12.3 km	4	20	radial
line-3	30.9 km	9	50	radial
line-4	62.8 km	15	25	ring
line-5	6.1 km	4	15	radial
line-6	6.3 km	2	11	radial

Jordan

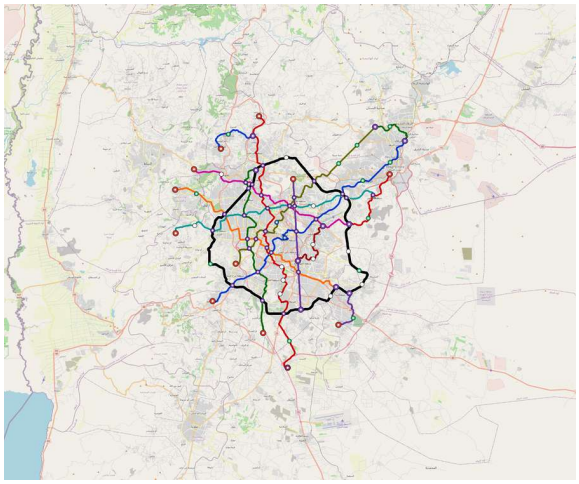
Generated City Model - Amman

west-asia / Jordan / Amman

Metric	Value
Path	cities/catalogue/west-asia/Jordan/Amman
Recorded ISO / planning population	JO / 4,007,000
Family	metro-6car
Lines / stations	14 / 132
Route length	412.9 km
Fleet	659 trainsets
Routing-demand cells (legacy)	67%; not population access
City CAPEX incl. dedicated solar	USD 7.16bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	919,907 (2020; 25.7% of raster bbox)
Direct / reachable line pairs	44.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Amman network map

Line	Length	Stations	Trainsets	Shape
line-1	48.2 km	13	85	radial
line-2	52.6 km	14	95	radial
line-3	30.3 km	9	56	radial
line-4	18.2 km	8	37	radial
line-5	33.0 km	10	59	radial
line-6	35.8 km	14	68	radial
line-7	28.8 km	12	57	radial
line-8	34.9 km	12	65	radial
line-9	82.4 km	23	38	ring
line-10	11.1 km	4	21	radial
line-11	8.4 km	3	17	radial
line-12	13.3 km	4	27	radial
line-13	8.3 km	3	17	radial
line-14	7.6 km	3	17	radial

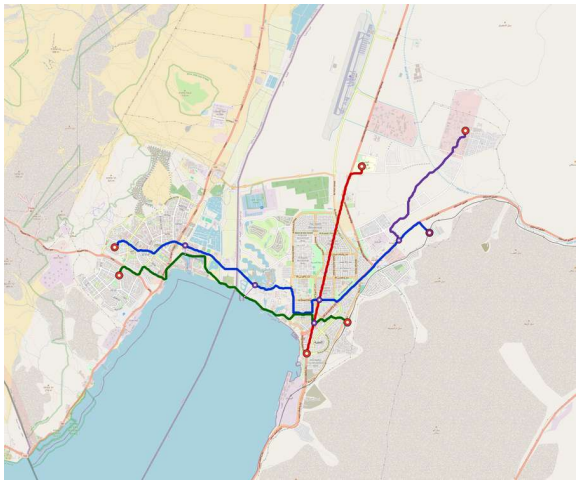
Generated City Model - Aqaba

west-asia / Jordan / Aqaba

Metric	Value
Path	cities/catalogue/west-asia/Jordan/Aqaba
Recorded ISO / planning population	JO / 250,000
Family	tram-2car
Lines / stations	4 / 17
Route length	39.2 km
Fleet	84 trainsets
Routing-demand cells (legacy)	62%; not population access
City CAPEX incl. dedicated solar	USD 641M (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	32,044 (2020; 39.1% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Aqaba network map

Line	Length	Stations	Trainsets	Shape
line-1	15.0 km	6	30	radial
line-2	7.5 km	4	17	radial
line-3	11.2 km	5	25	radial
line-4	5.5 km	2	12	radial

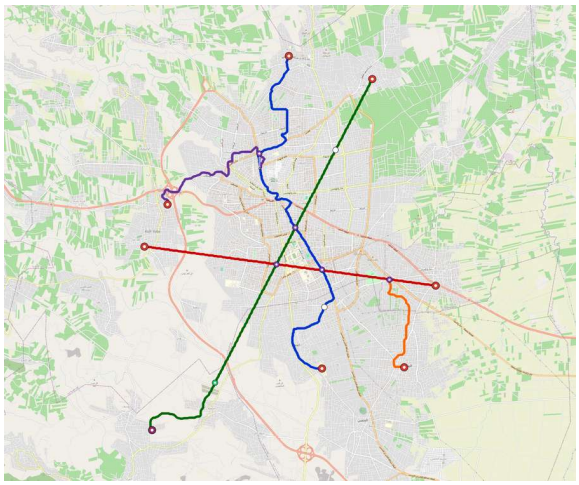
Generated City Model - Irbid

west-asia / Jordan / Irbid

Metric	Value
Path	cities/catalogue/west-asia/Jordan/Irbid
Recorded ISO / planning population	JO / 600,000
Family	light-metro-3car
Lines / stations	5 / 23
Route length	53.4 km
Fleet	174 trainsets
Routing-demand cells (legacy)	57%; not population access
City CAPEX incl. dedicated solar	USD 985M (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	174,972 (2020; 23.2% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Irbid network map

Line	Length	Stations	Trainsets	Shape
line-1	15.8 km	7	51	radial
line-2	10.4 km	5	34	radial
line-3	16.7 km	6	54	radial

Line	Length	Stations	Trainsets	Shape
line-4	6.2 km	3	20	radial
line-5	4.3 km	2	15	radial

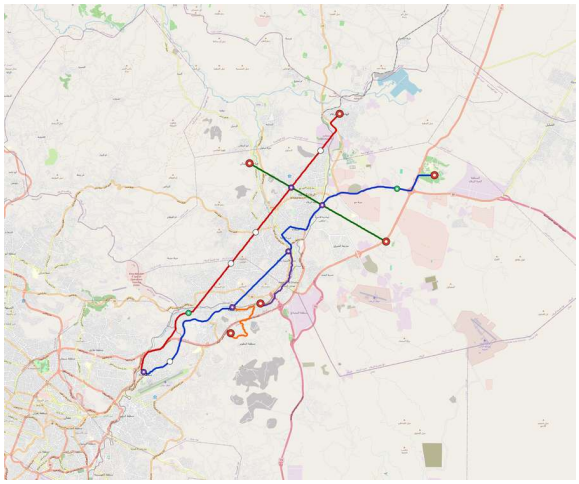
Generated City Model - Zarqa

west-asia / Jordan / Zarqa

Metric	Value
Path	cities/catalogue/west-asia/Jordan/Zarqa
Recorded ISO / planning population	JO / 700,000
Family	light-metro-3car
Lines / stations	5 / 27
Route length	81.0 km
Fleet	250 trainsets
Routing-demand cells (legacy)	51%; not population access
City CAPEX incl. dedicated solar	USD 1.39bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	268,149 (2020; 22.3% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Zarqa network map

Line	Length	Stations	Trainsets	Shape
line-1	30.4 km	9	94	radial
line-2	26.6 km	9	80	radial
line-3	11.9 km	4	38	radial
line-4	5.7 km	2	18	radial
line-5	6.4 km	3	20	radial

Lebanon

Generated City Model - Beirut

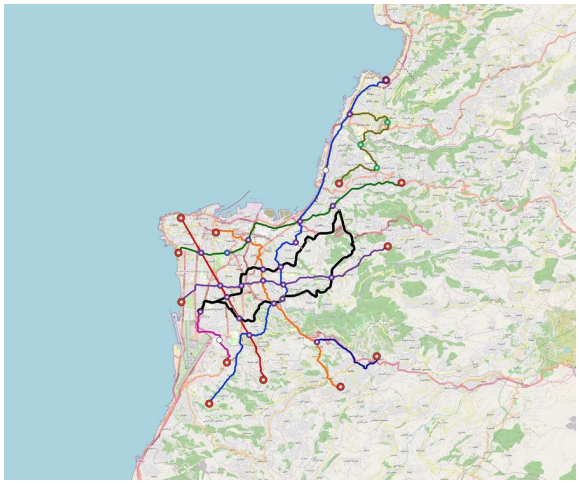
west-asia / Lebanon / Beirut

Metric	Value
Path	cities/catalogue/west-asia/Lebanon/Beirut
Recorded ISO / planning population	LB / 2,200,000
Family	metro-4car
Lines / stations	9 / 75
Route length	164.8 km
Fleet	263 trainsets
Routing-demand cells (legacy)	57%; not population access
City CAPEX incl. dedicated solar	USD 2.89bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	2,086,508 (2020; 46.7% of raster bbox)

Metric	Value
Direct / reachable line pairs	44.4% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Beirut network map

Line	Length	Stations	Trainsets	Shape
line-1	32.9 km	16	63	radial
line-2	13.7 km	7	28	radial
line-3	21.1 km	8	36	radial
line-4	17.4 km	9	35	radial
line-5	18.5 km	9	36	radial
line-6	36.7 km	16	17	ring
line-7	5.6 km	3	13	radial
line-8	12.8 km	5	24	radial
line-9	6.0 km	2	11	radial

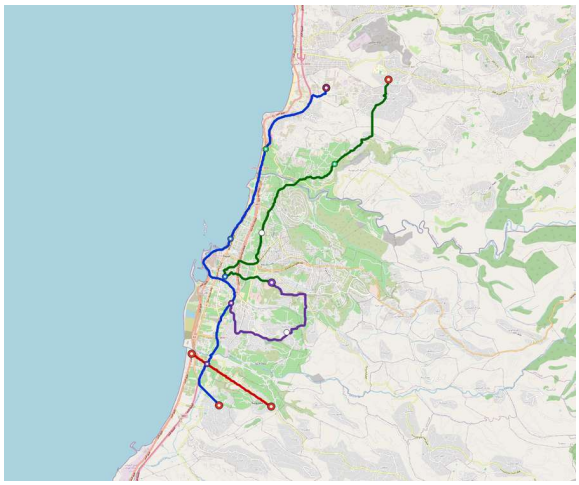
Generated City Model - Sidon

west-asia / Lebanon / Sidon

Metric	Value
Path	cities/catalogue/west-asia/Lebanon/Sidon
Recorded ISO / planning population	LB / 300,000
Family	tram-2car
Lines / stations	4 / 17
Route length	38.1 km
Fleet	84 trainsets
Routing-demand cells (legacy)	75%; not population access
City CAPEX incl. dedicated solar	USD 600M (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	262,566 (2020; 52.9% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Sidon network map

Line	Length	Stations	Trainsets	Shape
line-1	14.7 km	6	31	radial
line-2	3.6 km	3	11	radial
line-3	12.9 km	5	27	radial
line-4	6.9 km	3	15	radial

Generated City Model - Tripoli Lb

west-asia / Lebanon / Tripoli-Lb

Metric	Value
Path	cities/catalogue/west-asia/Lebanon/Tripoli-Lb
Recorded ISO / planning population	LB / 730,000
Family	light-metro-3car
Lines / stations	4 / 20
Route length	55.0 km
Fleet	173 trainsets
Routing-demand cells (legacy)	66%; not population access
City CAPEX incl. dedicated solar	USD 953M (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	378,368 (2020; 49.4% of raster bbox)
Direct / reachable line pairs	100.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.

Tripoli Lb network map

Line	Length	Stations	Trainsets	Shape
line-1	12.4 km	4	39	radial
line-2	18.8 km	6	58	radial
line-3	14.2 km	6	46	radial

Line	Length	Stations	Trainsets	Shape
line-4	9.6 km	4	30	radial

Palestine

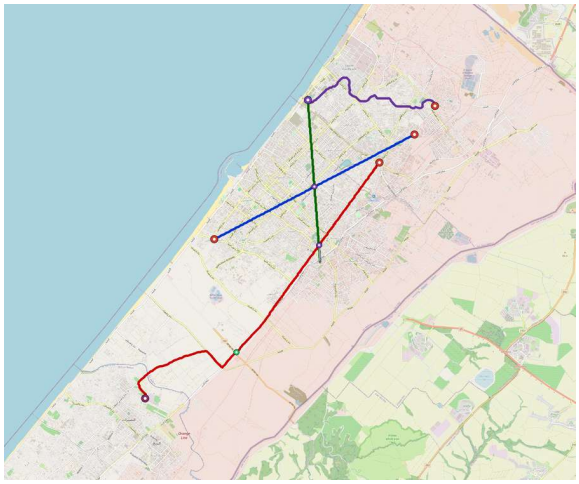
Generated City Model - Gaza City

west-asia / Palestine / Gaza-City

Metric	Value
Path	cities/catalogue/west-asia/Palestine/Gaza-City
Recorded ISO / planning population	PS / 600,000
Family	light-metro-3car
Lines / stations	4 / 13
Route length	34.4 km
Fleet	109 trainsets
Routing-demand cells (legacy)	44%; not population access
City CAPEX incl. dedicated solar	USD 580M (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	398,374 (2020; 32.9% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Gaza City network map

Line	Length	Stations	Trainsets	Shape
line-1	8.6 km	3	27	radial
line-2	13.9 km	4	43	radial
line-3	5.7 km	4	19	radial
line-4	6.2 km	2	20	radial

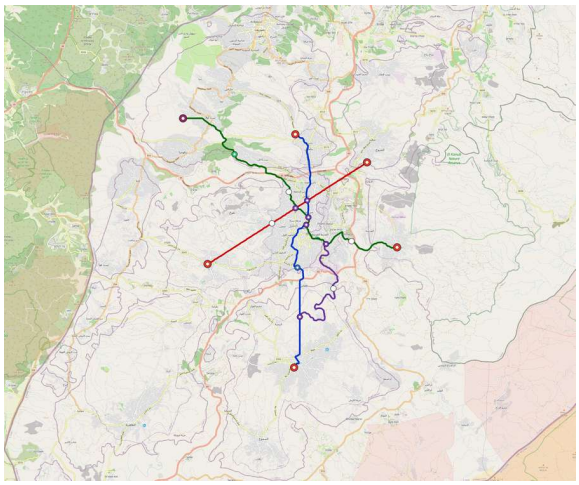
Generated City Model - Hebron

west-asia / Palestine / Hebron

Metric	Value
Path	cities/catalogue/west-asia/Palestine/Hebron
Recorded ISO / planning population	PS / 800,000
Family	light-metro-3car
Lines / stations	4 / 33
Route length	72.8 km
Fleet	227 trainsets
Routing-demand cells (legacy)	67%; not population access
City CAPEX incl. dedicated solar	USD 1.33bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	205,942 (2020; 39.9% of raster bbox)
Direct / reachable line pairs	83.3% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Hebron network map

Line	Length	Stations	Trainsets	Shape
line-1	20.9 km	12	67	radial
line-2	14.4 km	5	43	radial
line-3	25.8 km	11	79	radial
line-4	11.7 km	5	38	radial

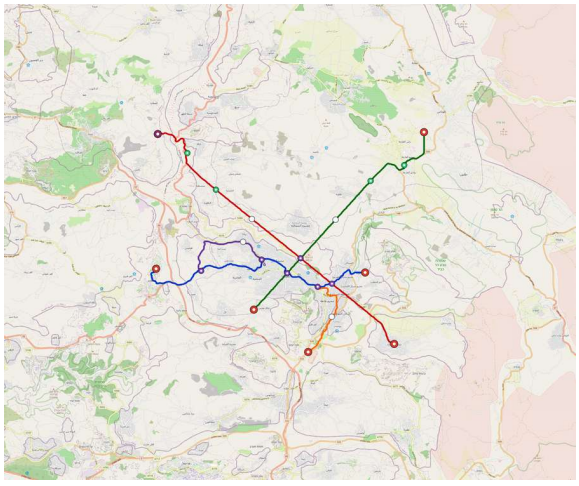
Generated City Model - Nablus

west-asia / Palestine / Nablus

Metric	Value
Path	cities/catalogue/west-asia/Palestine/Nablus
Recorded ISO / planning population	PS / 450,000
Family	light-metro-3car
Lines / stations	5 / 30
Route length	83.8 km
Fleet	266 trainsets
Routing-demand cells (legacy)	61%; not population access
City CAPEX incl. dedicated solar	USD 1.46bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	122,586 (2020; 40.5% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Nablus network map

Line	Length	Stations	Trainsets	Shape
line-1	22.4 km	8	69	radial
line-2	25.3 km	7	78	radial
line-3	20.2 km	8	67	radial
line-4	7.8 km	4	26	radial
line-5	8.1 km	3	26	radial

Saudi Arabia

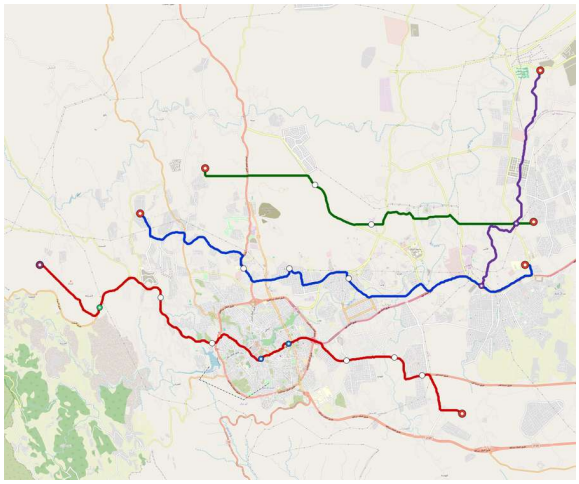
Generated City Model - Abha

west-asia / Saudi Arabia / Abha

Metric	Value
Path	cities/catalogue/west-asia/Saudi Arabia/Abha
Recorded ISO / planning population	SA / 450,000
Family	light-metro-3car
Lines / stations	4 / 24
Route length	74.3 km
Fleet	229 trainsets
Routing-demand cells (legacy)	29%; not population access
City CAPEX incl. dedicated solar	USD 1.15bn (USD 15M / km); shared national factory excluded
Residents within 800 m station circles	23,117 (2020; 16.1% of raster bbox)
Direct / reachable line pairs	33.3% / 50.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Abha network map

Line	Length	Stations	Trainsets	Shape
line-1	23.0 km	6	70	radial
line-2	25.8 km	10	79	radial
line-3	14.8 km	5	48	radial
line-4	10.7 km	3	32	radial

Generated City Model - Al Kharj

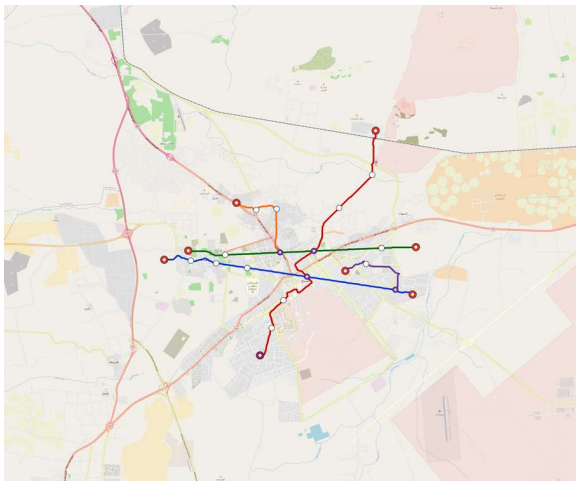
west-asia / Saudi Arabia / Al-Kharj

Metric	Value
Path	cities/catalogue/west-asia/Saudi Arabia/Al-Kharj
Recorded ISO / planning population	SA / 400,000
Family	light-metro-3car
Lines / stations	5 / 28
Route length	75.4 km
Fleet	232 trainsets
Routing-demand cells (legacy)	55%; not population access

Metric	Value
City CAPEX incl. dedicated solar	USD 1.20bn (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	71,612 (2020; 23.8% of raster bbox)
Direct / reachable line pairs	40.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Al Kharj network map

Line	Length	Stations	Trainsets	Shape
line-1	20.4 km	7	61	radial
line-2	24.9 km	8	76	radial
line-3	17.2 km	6	52	radial
line-4	6.0 km	3	20	radial
line-5	6.9 km	4	23	radial

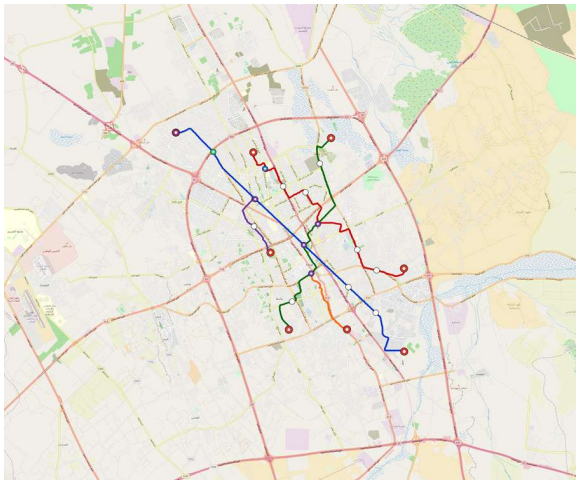
Generated City Model - Buraidah

west-asia / Saudi Arabia / Buraidah

Metric	Value
Path	cities/catalogue/west-asia/Saudi Arabia/Buraidah
Recorded ISO / planning population	SA / 700,000
Family	light-metro-3car
Lines / stations	5 / 27
Route length	80.3 km
Fleet	249 trainsets
Routing-demand cells (legacy)	46%; not population access
City CAPEX incl. dedicated solar	USD 1.30bn (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	48,113 (2020; 15.8% of raster bbox)
Direct / reachable line pairs	40.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Buraidah network map

Line	Length	Stations	Trainsets	Shape
line-1	27.0 km	7	83	radial
line-2	23.2 km	8	70	radial
line-3	19.0 km	7	60	radial
line-4	5.5 km	3	18	radial
line-5	5.6 km	2	18	radial

Generated City Model - Dammam

west-asia / Saudi Arabia / Dammam

Metric	Value
Path	cities/catalogue/west-asia/Saudi Arabia/Dammam
Recorded ISO / planning population	SA / 1,500,000
Family	metro-4car
Lines / stations	9 / 96
Route length	306.1 km
Fleet	381 trainsets
Routing-demand cells (legacy)	54%; not population access
City CAPEX incl. dedicated solar	USD 4.41bn (USD 14M / km); shared national factory excluded
Residents within 800 m station circles	485,710 (2020; 21.4% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.

Dammam network map

Line	Length	Stations	Trainsets	Shape
line-1	45.2 km	13	72	radial
line-2	39.3 km	13	64	radial

Line	Length	Stations	Trainsets	Shape
line-3	31.4 km	11	50	radial
line-4	32.2 km	10	49	radial
line-5	43.7 km	15	68	radial
line-6	91.8 km	26	37	ring
line-7	7.0 km	3	14	radial
line-8	9.4 km	3	16	radial
line-9	6.1 km	2	11	radial

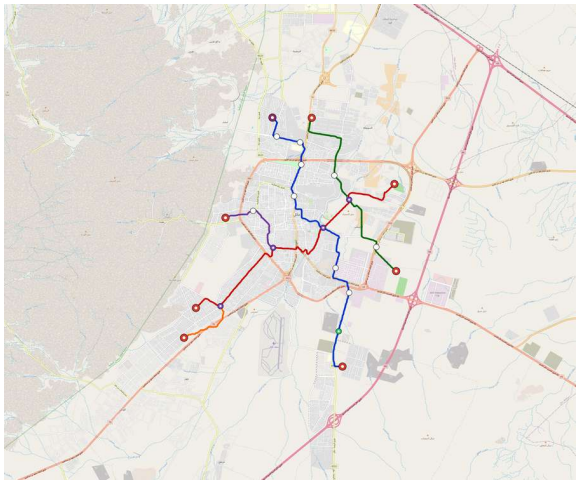
Generated City Model - Hail

west-asia / Saudi Arabia / Hail

Metric	Value
Path	cities/catalogue/west-asia/Saudi Arabia/Hail
Recorded ISO / planning population	SA / 500,000
Family	light-metro-3car
Lines / stations	5 / 36
Route length	78.3 km
Fleet	246 trainsets
Routing-demand cells (legacy)	47%; not population access
City CAPEX incl. dedicated solar	USD 1.34bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	47,914 (2020; 20.3% of raster bbox)
Direct / reachable line pairs	40.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Hail network map

Line	Length	Stations	Trainsets	Shape
line-1	28.3 km	10	87	radial
line-2	23.6 km	11	72	radial
line-3	15.9 km	9	50	radial
line-4	5.9 km	3	20	radial
line-5	4.6 km	3	17	radial

Generated City Model - Hofuf

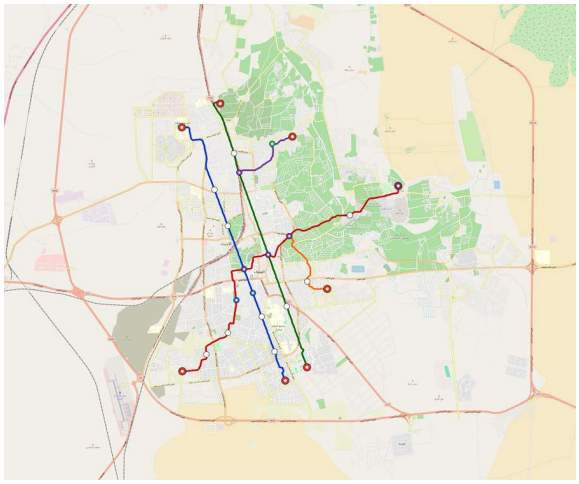
west-asia / Saudi Arabia / Hofuf

Metric	Value
Path	cities/catalogue/west-asia/Saudi Arabia/Hofuf
Recorded ISO / planning population	SA / 800,000
Family	light-metro-3car
Lines / stations	5 / 29
Route length	88.5 km
Fleet	267 trainsets

Metric	Value
Routing-demand cells (legacy)	48%; not population access
City CAPEX incl. dedicated solar	USD 1.32bn (USD 15M / km); shared national factory excluded
Residents within 800 m station circles	189,174 (2020; 20.1% of raster bbox)
Direct / reachable line pairs	40.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Hofuf network map

Line	Length	Stations	Trainsets	Shape
line-1	23.5 km	8	70	radial
line-2	29.2 km	9	86	radial
line-3	24.0 km	6	71	radial
line-4	5.7 km	3	20	radial
line-5	6.1 km	3	20	radial

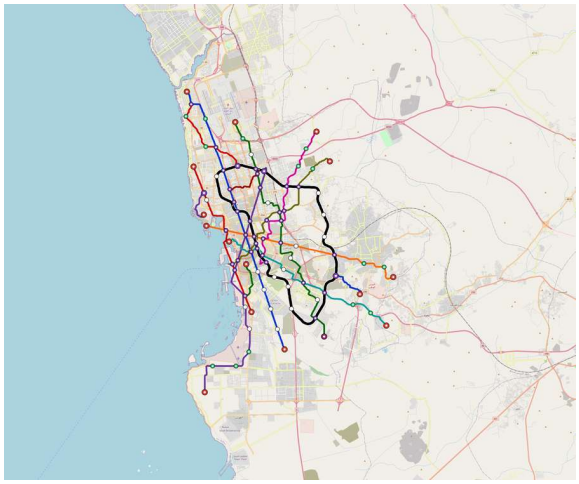
Generated City Model - Jeddah

west-asia / Saudi Arabia / Jeddah

Metric	Value
Path	cities/catalogue/west-asia/Saudi Arabia/Jeddah
Recorded ISO / planning population	SA / 4,700,000
Family	metro-6car
Lines / stations	14 / 159
Route length	408.4 km
Fleet	702 trainsets
Routing-demand cells (legacy)	76%; not population access
City CAPEX incl. dedicated solar	USD 7.29bn (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	896,030 (2020; 23.4% of raster bbox)
Direct / reachable line pairs	37.4% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Jeddah network map

Line	Length	Stations	Trainsets	Shape
line-1	47.3 km	17	84	radial
line-2	26.9 km	9	52	radial
line-3	49.2 km	22	98	radial
line-4	44.5 km	14	84	radial
line-5	33.2 km	13	67	radial
line-6	33.9 km	11	65	radial
line-7	29.7 km	11	58	radial
line-8	29.2 km	13	58	radial
line-9	74.2 km	27	34	ring
line-10	7.4 km	10	32	radial
line-11	6.2 km	2	13	radial
line-12	14.2 km	5	29	radial
line-13	6.3 km	3	15	radial
line-14	6.4 km	2	13	radial

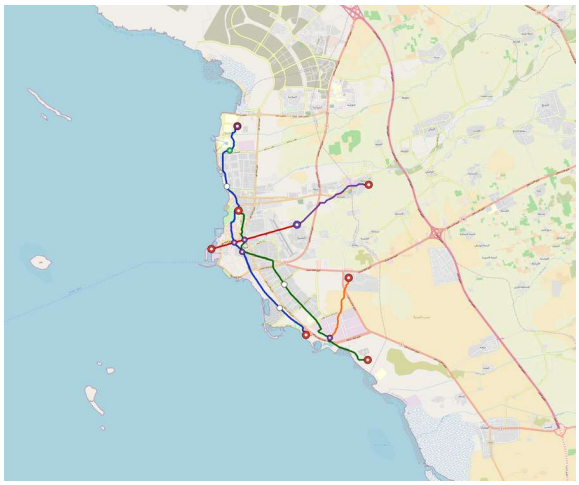
Generated City Model - Jizan

west-asia / Saudi Arabia / Jizan

Metric	Value
Path	cities/catalogue/west-asia/Saudi Arabia/Jizan
Recorded ISO / planning population	SA / 400,000
Family	light-metro-3car
Lines / stations	5 / 27
Route length	61.3 km
Fleet	194 trainsets
Routing-demand cells (legacy)	62%; not population access
City CAPEX incl. dedicated solar	USD 1.02bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	20,786 (2020; 22.9% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Jizan network map

Line	Length	Stations	Trainsets	Shape
line-1	22.7 km	10	70	radial
line-2	7.4 km	4	24	radial
line-3	19.1 km	9	59	radial
line-4	6.9 km	2	23	radial
line-5	5.2 km	2	18	radial

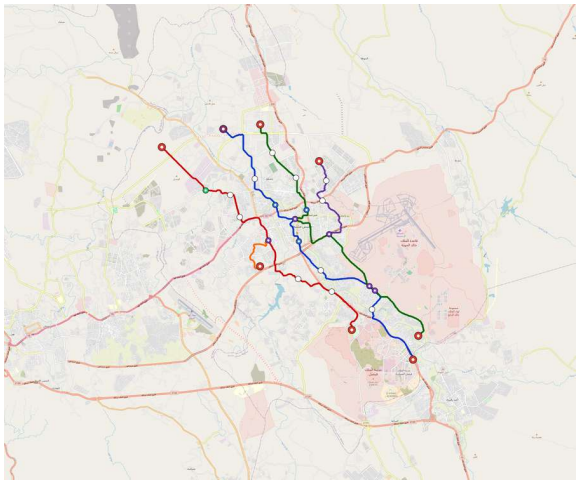
Generated City Model - Khamis Mushait

west-asia / Saudi Arabia / Khamis-Mushait

Metric	Value
Path	cities/catalogue/west-asia/Saudi Arabia/Khamis-Mushait
Recorded ISO / planning population	SA / 600,000
Family	light-metro-3car
Lines / stations	5 / 45
Route length	99.3 km
Fleet	320 trainsets
Routing-demand cells (legacy)	56%; not population access
City CAPEX incl. dedicated solar	USD 1.72bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	148,409 (2020; 28.3% of raster bbox)
Direct / reachable line pairs	30.0% / 40.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Khamis Mushait network map

Line	Length	Stations	Trainsets	Shape
line-1	30.0 km	16	100	radial
line-2	27.9 km	8	84	radial

Line	Length	Stations	Trainsets	Shape
line-3	28.7 km	15	94	radial
line-4	8.4 km	4	27	radial
line-5	4.4 km	2	15	radial

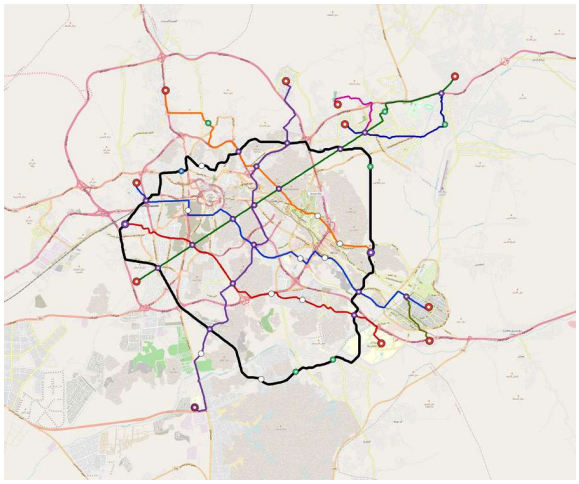
Generated City Model - Mecca

west-asia / Saudi Arabia / Mecca

Metric	Value
Path	cities/catalogue/west-asia/Saudi Arabia/Mecca
Recorded ISO / planning population	SA / 2,200,000
Family	metro-4car
Lines / stations	9 / 73
Route length	250.1 km
Fleet	315 trainsets
Routing-demand cells (legacy)	41%; not population access
City CAPEX incl. dedicated solar	USD 3.68bn (USD 15M / km); shared national factory excluded
Residents within 800 m station circles	393,589 (2020; 21.6% of raster bbox)
Direct / reachable line pairs	44.4% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Mecca network map

Line	Length	Stations	Trainsets	Shape
line-1	32.9 km	12	53	radial
line-2	27.8 km	7	42	radial
line-3	32.6 km	11	50	radial
line-4	35.0 km	10	56	radial
line-5	26.1 km	8	42	radial
line-6	72.8 km	16	28	ring
line-7	6.5 km	2	12	radial
line-8	5.0 km	3	12	radial
line-9	11.5 km	4	20	radial

Generated City Model - Medina

west-asia / Saudi Arabia / Medina

Metric	Value
Path	cities/catalogue/west-asia/Saudi Arabia/Medina
Recorded ISO / planning population	SA / 1,500,000
Family	metro-4car
Lines / stations	9 / 63
Route length	202.8 km
Fleet	255 trainsets

Metric	Value
Routing-demand cells (legacy)	46%; not population access
City CAPEX incl. dedicated solar	USD 3.05bn (USD 15M / km); shared national factory excluded
Residents within 800 m station circles	213,798 (2020; 17.1% of raster bbox)
Direct / reachable line pairs	38.9% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.

Medina network map

Line	Length	Stations	Trainsets	Shape
line-1	32.7 km	9	50	radial
line-2	19.1 km	7	32	radial
line-3	22.7 km	8	37	radial
line-4	25.5 km	8	39	radial
line-5	19.4 km	7	32	radial
line-6	62.2 km	16	24	ring
line-7	5.7 km	3	13	radial
line-8	9.7 km	3	17	radial
line-9	5.6 km	2	11	radial

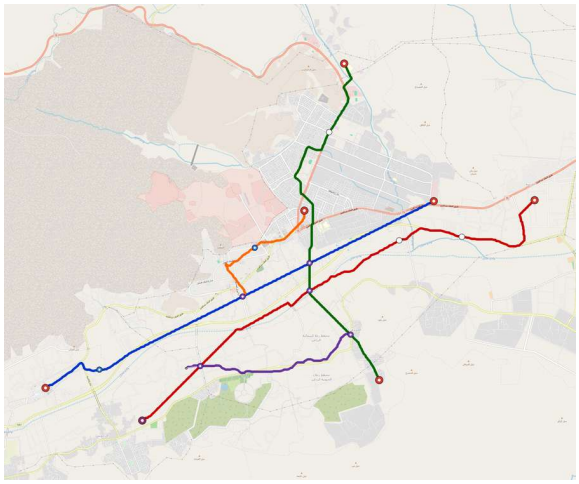
Generated City Model - Najran

west-asia / Saudi Arabia / Najran

Metric	Value
Path	cities/catalogue/west-asia/Saudi Arabia/Najran
Recorded ISO / planning population	SA / 500,000
Family	light-metro-3car
Lines / stations	5 / 23
Route length	69.2 km
Fleet	213 trainsets
Routing-demand cells (legacy)	54%; not population access
City CAPEX incl. dedicated solar	USD 1.28bn (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	54,499 (2020; 20.0% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Najran network map

Line	Length	Stations	Trainsets	Shape
line-1	19.0 km	5	58	radial
line-2	21.8 km	6	64	radial
line-3	15.3 km	6	47	radial
line-4	7.4 km	3	24	radial
line-5	5.7 km	3	20	radial

Generated City Model - Tabuk

west-asia / Saudi Arabia / Tabuk

Metric	Value
Path	cities/catalogue/west-asia/Saudi Arabia/Tabuk
Recorded ISO / planning population	SA / 650,000
Family	light-metro-3car
Lines / stations	5 / 27
Route length	74.2 km
Fleet	237 trainsets
Routing-demand cells (legacy)	40%; not population access
City CAPEX incl. dedicated solar	USD 1.24bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	100,788 (2020; 18.1% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.

Tabuk network map

Line	Length	Stations	Trainsets	Shape
line-1	19.0 km	8	60	radial
line-2	18.2 km	7	59	radial

Line	Length	Stations	Trainsets	Shape
line-3	25.2 km	7	78	radial
line-4	4.6 km	2	16	radial
line-5	7.2 km	3	24	radial

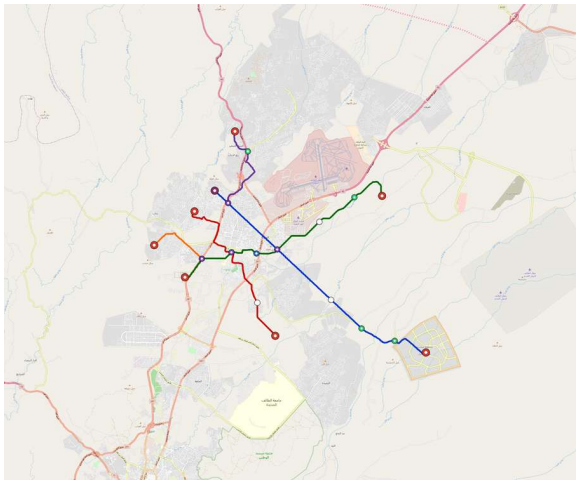
Generated City Model - Taif

west-asia / Saudi Arabia / Taif

Metric	Value
Path	cities/catalogue/west-asia/Saudi Arabia/Taif
Recorded ISO / planning population	SA / 700,000
Family	light-metro-3car
Lines / stations	5 / 26
Route length	70.9 km
Fleet	220 trainsets
Routing-demand cells (legacy)	56%; not population access
City CAPEX incl. dedicated solar	USD 1.14bn (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	68,207 (2020; 15.4% of raster bbox)
Direct / reachable line pairs	40.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Taif network map

Line	Length	Stations	Trainsets	Shape
line-1	23.9 km	7	73	radial
line-2	14.5 km	5	43	radial
line-3	20.0 km	9	62	radial
line-4	7.6 km	3	26	radial
line-5	4.9 km	2	16	radial

Syria

Generated City Model - Aleppo

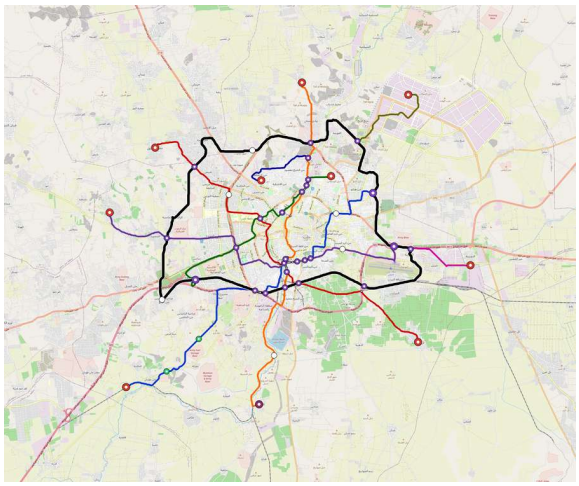
west-asia / Syria / Aleppo

Metric	Value
Path	cities/catalogue/west-asia/Syria/Aleppo
Recorded ISO / planning population	SY / 1,639,000
Family	metro-4car
Lines / stations	9 / 163
Route length	205.2 km
Fleet	466 trainsets
Routing-demand cells (legacy)	60%; not population access
City CAPEX incl. dedicated solar	USD 4.31bn (USD 21M / km); shared national factory excluded

Metric	Value
Residents within 800 m station circles	970,902 (2020; 29.0% of raster bbox)
Direct / reachable line pairs	47.2% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Aleppo network map

Line	Length	Stations	Trainsets	Shape
line-1	28.7 km	40	111	radial
line-2	27.9 km	17	61	radial
line-3	15.7 km	18	53	radial
line-4	27.4 km	31	91	radial
line-5	29.2 km	30	91	radial
line-6	59.5 km	21	25	ring
line-7	4.4 km	2	10	radial
line-8	7.0 km	2	13	radial
line-9	5.5 km	2	11	radial

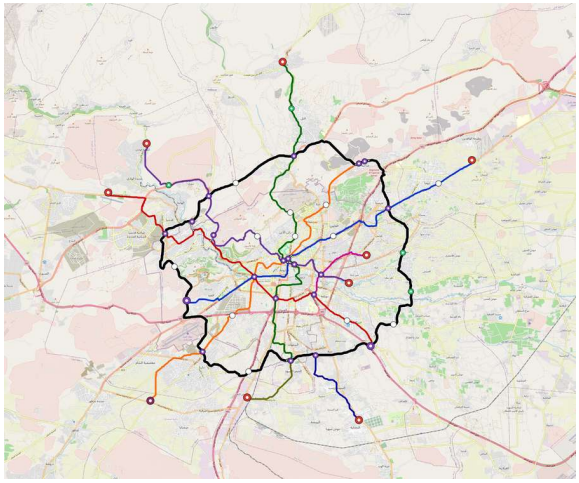
Generated City Model - Damascus

west-asia / Syria / Damascus

Metric	Value
Path	cities/catalogue/west-asia/Syria/Damascus
Recorded ISO / planning population	SY / 2,503,000
Family	metro-4car
Lines / stations	9 / 111
Route length	228.2 km
Fleet	348 trainsets
Routing-demand cells (legacy)	78%; not population access
City CAPEX incl. dedicated solar	USD 3.88bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	1,220,380 (2020; 28.6% of raster bbox)
Direct / reachable line pairs	55.6% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Damascus network map

Line	Length	Stations	Trainsets	Shape
line-1	28.5 km	17	61	radial
line-2	28.9 km	10	47	radial
line-3	29.4 km	10	46	radial
line-4	26.7 km	13	50	radial
line-5	30.1 km	24	78	radial
line-6	67.7 km	29	30	ring
line-7	5.5 km	3	13	radial
line-8	4.9 km	3	12	radial
line-9	6.3 km	2	11	radial

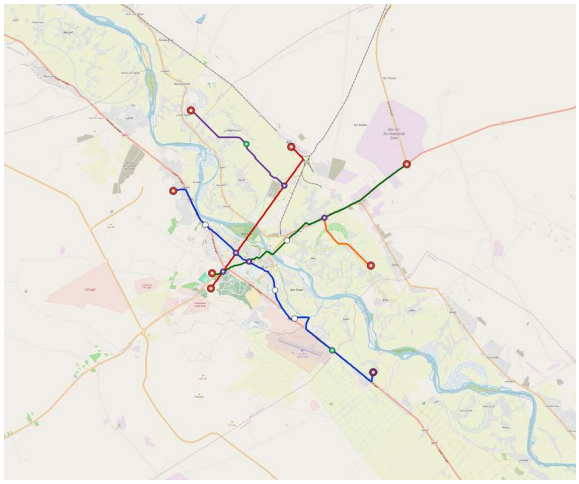
Generated City Model - Deir Ez Zor

west-asia / Syria / Deir-Ez-Zor

Metric	Value
Path	cities/catalogue/west-asia/Syria/Deir-Ez-Zor
Recorded ISO / planning population	SY / 500,000
Family	light-metro-3car
Lines / stations	5 / 25
Route length	66.1 km
Fleet	210 trainsets
Routing-demand cells (legacy)	60%; not population access
City CAPEX incl. dedicated solar	USD 1.12bn (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	137,047 (2020; 32.8% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Deir Ez Zor network map

Line	Length	Stations	Trainsets	Shape
line-1	23.8 km	8	75	radial
line-2	12.8 km	5	39	radial
line-3	16.3 km	7	52	radial
line-4	8.4 km	3	28	radial
line-5	4.8 km	2	16	radial

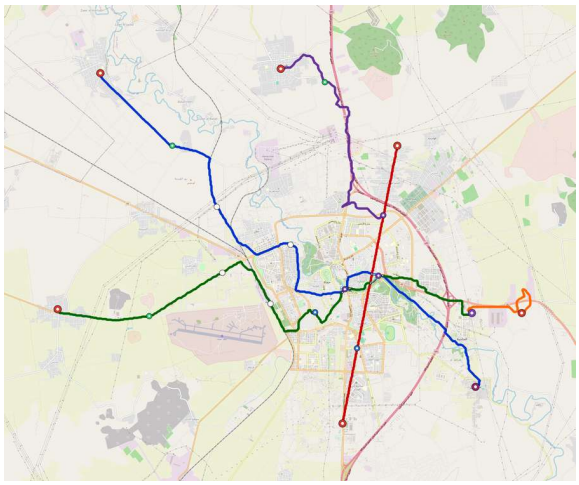
Generated City Model - Hama

west-asia / Syria / Hama

Metric	Value
Path	cities/catalogue/west-asia/Syria/Hama
Recorded ISO / planning population	SY / 600,000
Family	light-metro-3car
Lines / stations	5 / 39
Route length	67.0 km
Fleet	229 trainsets
Routing-demand cells (legacy)	60%; not population access
City CAPEX incl. dedicated solar	USD 1.26bn (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	181,001 (2020; 32.1% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Hama network map

Line	Length	Stations	Trainsets	Shape
line-1	22.8 km	13	78	radial
line-2	9.8 km	6	34	radial
line-3	21.3 km	14	72	radial
line-4	9.1 km	3	30	radial
line-5	4.0 km	3	15	radial

Generated City Model - Homs

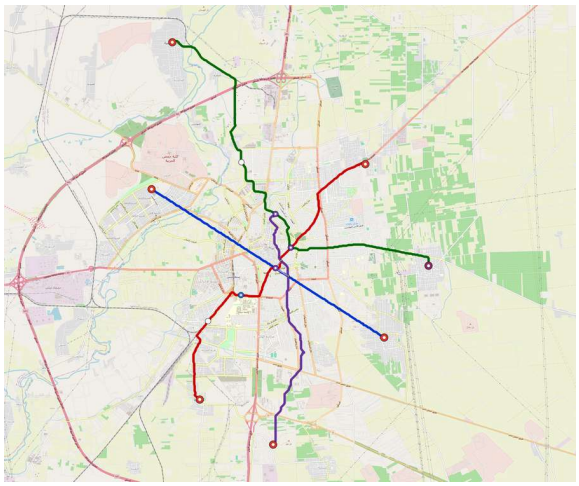
west-asia / Syria / Homs

Metric	Value
Path	cities/catalogue/west-asia/Syria/Homs
Recorded ISO / planning population	SY / 775,000
Family	light-metro-3car
Lines / stations	4 / 20
Route length	45.4 km
Fleet	142 trainsets
Routing-demand cells (legacy)	61%; not population access
City CAPEX incl. dedicated solar	USD 847M (USD 19M / km); shared national factory excluded

Metric	Value
Residents within 800 m station circles	249,246 (2020; 21.9% of raster bbox)
Direct / reachable line pairs	83.3% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Homs network map

Line	Length	Stations	Trainsets	Shape
line-1	10.1 km	4	31	radial
line-2	11.8 km	7	39	radial
line-3	14.8 km	5	45	radial
line-4	8.8 km	4	27	radial

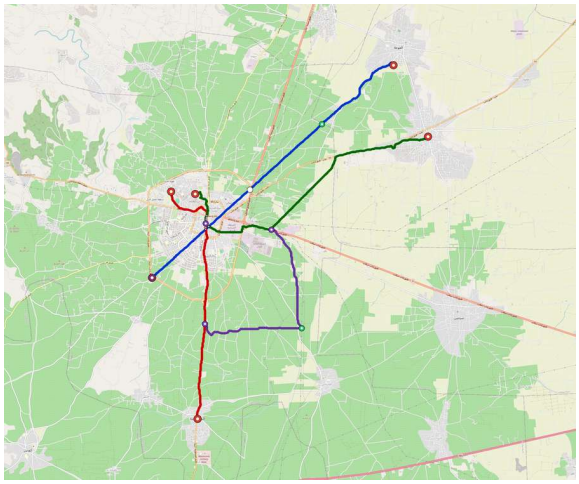
Generated City Model - Idlib

west-asia / Syria / Idlib

Metric	Value
Path	cities/catalogue/west-asia/Syria/Idlib
Recorded ISO / planning population	SY / 300,000
Family	tram-2car
Lines / stations	4 / 21
Route length	38.6 km
Fleet	87 trainsets
Routing-demand cells (legacy)	72%; not population access
City CAPEX incl. dedicated solar	USD 617M (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	96,001 (2020; 46.6% of raster bbox)
Direct / reachable line pairs	83.3% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Idlib network map

Line	Length	Stations	Trainsets	Shape
line-1	11.7 km	6	26	radial
line-2	8.9 km	6	21	radial
line-3	10.8 km	6	24	radial
line-4	7.2 km	3	16	radial

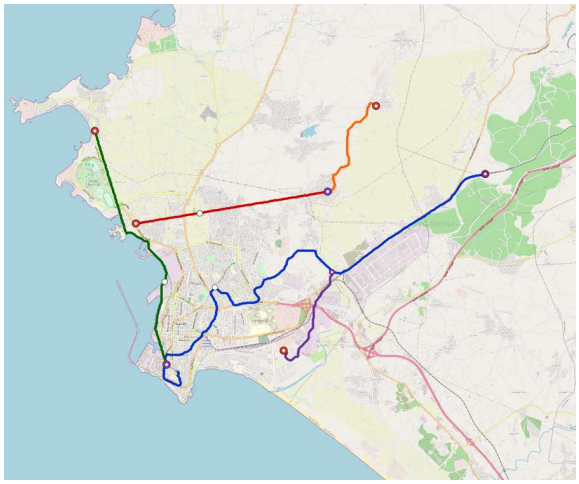
Generated City Model - Latakia

west-asia / Syria / Latakia

Metric	Value
Path	cities/catalogue/west-asia/Syria/Latakia
Recorded ISO / planning population	SY / 700,000
Family	light-metro-3car
Lines / stations	5 / 16
Route length	42.0 km
Fleet	135 trainsets
Routing-demand cells (legacy)	51%; not population access
City CAPEX incl. dedicated solar	USD 743M (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	180,114 (2020; 28.7% of raster bbox)
Direct / reachable line pairs	30.0% / 40.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Latakia network map

Line	Length	Stations	Trainsets	Shape
line-1	18.0 km	5	54	radial
line-2	6.7 km	3	23	radial
line-3	9.3 km	4	30	radial

Line	Length	Stations	Trainsets	Shape
line-4	4.0 km	2	14	radial
line-5	4.1 km	2	14	radial

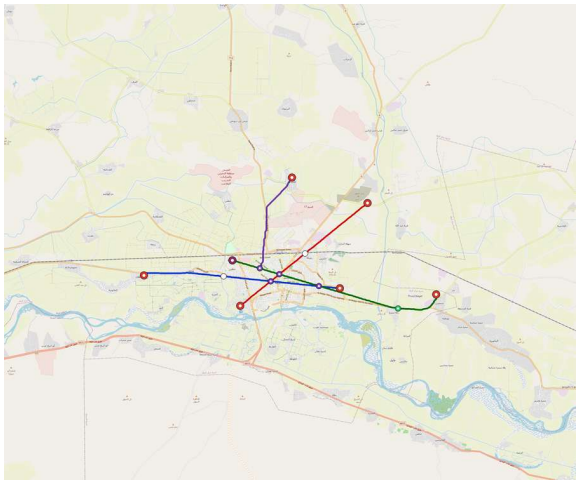
Generated City Model - Raqqa

west-asia / Syria / Raqqa

Metric	Value
Path	cities/catalogue/west-asia/Syria/Raqqa
Recorded ISO / planning population	SY / 350,000
Family	light-metro-3car
Lines / stations	4 / 20
Route length	47.7 km
Fleet	156 trainsets
Routing-demand cells (legacy)	65%; not population access
City CAPEX incl. dedicated solar	USD 835M (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	112,193 (2020; 34.0% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Raqqa network map

Line	Length	Stations	Trainsets	Shape
line-1	13.3 km	6	45	radial
line-2	11.9 km	5	38	radial
line-3	15.7 km	7	52	radial
line-4	6.8 km	2	21	radial

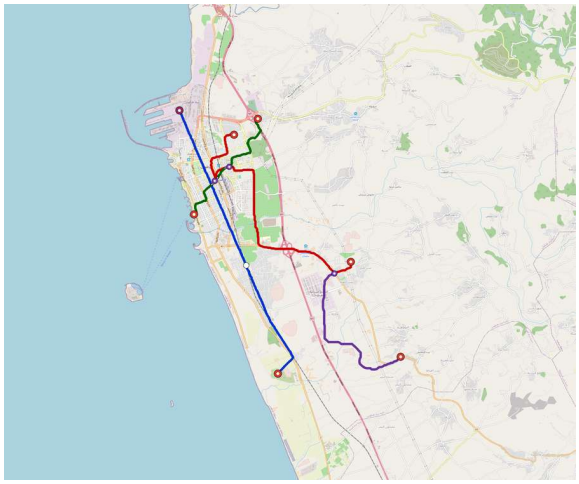
Generated City Model - Tartus

west-asia / Syria / Tartus

Metric	Value
Path	cities/catalogue/west-asia/Syria/Tartus
Recorded ISO / planning population	SY / 250,000
Family	tram-2car
Lines / stations	4 / 20
Route length	31.7 km
Fleet	80 trainsets
Routing-demand cells (legacy)	78%; not population access
City CAPEX incl. dedicated solar	USD 542M (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	75,433 (2020; 36.5% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Tartus network map

Line	Length	Stations	Trainsets	Shape
line-1	10.6 km	4	21	radial
line-2	10.4 km	7	26	radial
line-3	5.0 km	7	20	radial
line-4	5.8 km	2	13	radial

Yemen

Generated City Model - Aden

west-asia / Yemen / Aden

Metric	Value
Path	cities/catalogue/west-asia/Yemen/Aden
Recorded ISO / planning population	YE / 985,000
Family	light-metro-3car
Lines / stations	5 / 26
Route length	56.4 km
Fleet	182 trainsets
Routing-demand cells (legacy)	60%; not population access
City CAPEX incl. dedicated solar	USD 967M (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	277,949 (2020; 37.3% of raster bbox)
Direct / reachable line pairs	40.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.

Aden network map

Line	Length	Stations	Trainsets	Shape
line-1	17.1 km	8	53	radial
line-2	13.7 km	6	45	radial
line-3	15.1 km	6	47	radial
line-4	4.6 km	3	17	radial
line-5	5.9 km	3	20	radial

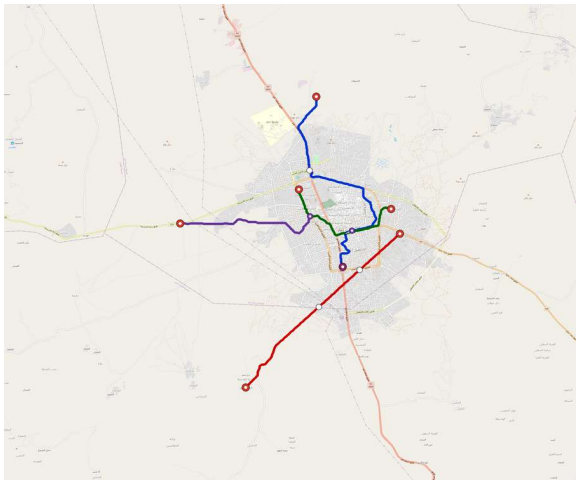
Generated City Model - Dhamar

west-asia / Yemen / Dhamar

Metric	Value
Path	cities/catalogue/west-asia/Yemen/Dhamar
Recorded ISO / planning population	YE / 300,000
Family	tram-2car
Lines / stations	4 / 18
Route length	32.4 km
Fleet	77 trainsets
Routing-demand cells (legacy)	68%; not population access
City CAPEX incl. dedicated solar	USD 524M (USD 16M / km); shared national factory excluded
Residents within 800 m station circles	83,270 (2020; 40.9% of raster bbox)
Direct / reachable line pairs	33.3% / 50.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Dhamar network map

Line	Length	Stations	Trainsets	Shape
line-1	11.1 km	6	25	radial
line-2	9.4 km	4	20	radial
line-3	6.0 km	6	19	radial
line-4	5.8 km	2	13	radial

Generated City Model - Hodeidah

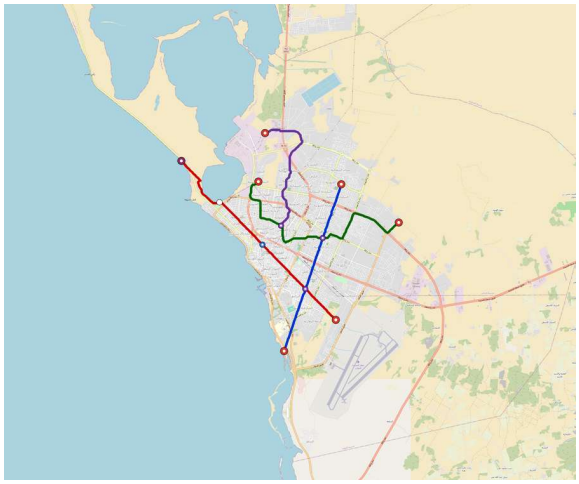
west-asia / Yemen / Hodeidah

Metric	Value
Path	cities/catalogue/west-asia/Yemen/Hodeidah
Recorded ISO / planning population	YE / 750,000
Family	light-metro-3car
Lines / stations	4 / 15
Route length	32.4 km
Fleet	106 trainsets
Routing-demand cells (legacy)	65%; not population access
City CAPEX incl. dedicated solar	USD 609M (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	201,501 (2020; 32.0% of raster bbox)

Metric	Value
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Hodeidah network map

Line	Length	Stations	Trainsets	Shape
line-1	7.5 km	4	26	radial
line-2	9.8 km	5	32	radial
line-3	9.5 km	4	30	radial
line-4	5.7 km	2	18	radial

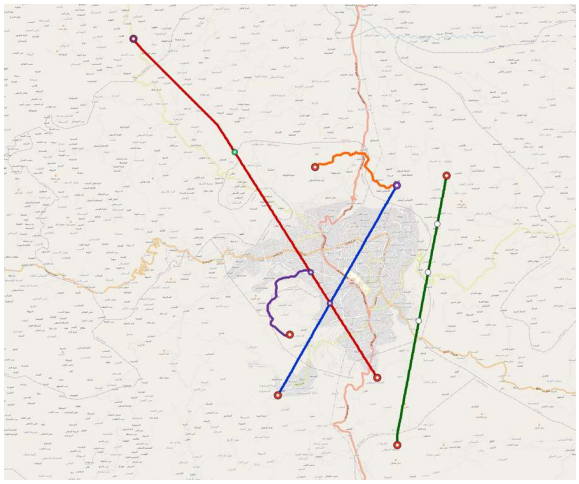
Generated City Model - Ibb

west-asia / Yemen / Ibb

Metric	Value
Path	cities/catalogue/west-asia/Yemen/Ibb
Recorded ISO / planning population	YE / 750,000
Family	light-metro-3car
Lines / stations	5 / 21
Route length	49.0 km
Fleet	157 trainsets
Routing-demand cells (legacy)	23%; not population access
City CAPEX incl. dedicated solar	USD 888M (USD 18M / km); shared national factory excluded
Residents within 800 m station circles	81,642 (2020; 9.4% of raster bbox)
Direct / reachable line pairs	30.0% / 60.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Ibb network map

Line	Length	Stations	Trainsets	Shape
line-1	10.4 km	3	32	radial
line-2	17.8 km	7	54	radial
line-3	11.3 km	5	37	radial
line-4	4.6 km	4	18	radial
line-5	4.9 km	2	16	radial

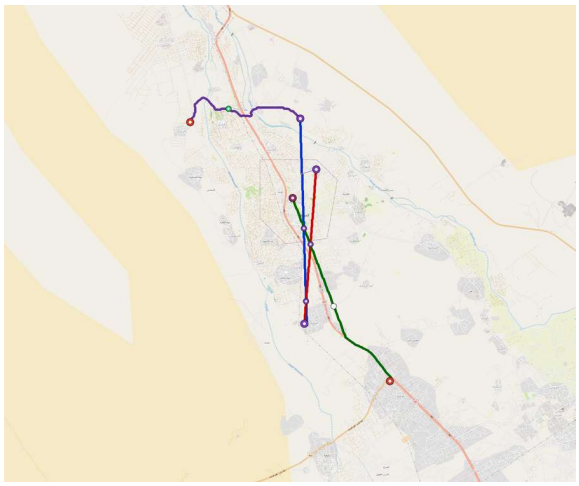
Generated City Model - Lahij

west-asia / Yemen / Lahij

Metric	Value
Path	cities/catalogue/west-asia/Yemen/Lahij
Recorded ISO / planning population	YE / 250,000
Family	tram-2car
Lines / stations	4 / 17
Route length	29.5 km
Fleet	68 trainsets
Routing-demand cells (legacy)	64%; not population access
City CAPEX incl. dedicated solar	USD 512M (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	51,713 (2020; 55.1% of raster bbox)
Direct / reachable line pairs	66.7% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Lahij network map

Line	Length	Stations	Trainsets	Shape
line-1	8.2 km	5	19	radial
line-2	6.2 km	4	15	radial
line-3	9.2 km	5	20	radial
line-4	5.9 km	3	14	radial

Generated City Model - Mukalla

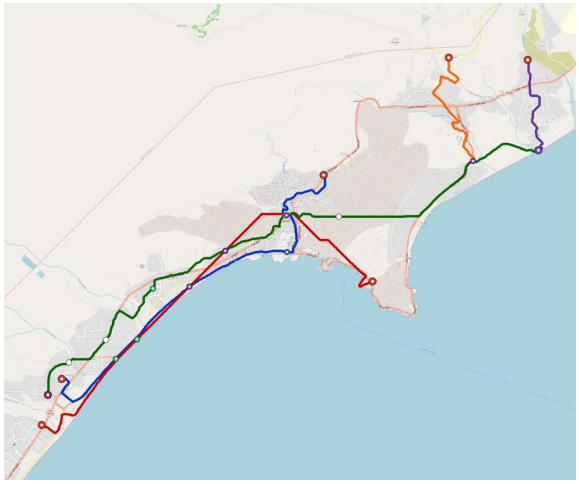
west-asia / Yemen / Mukalla

Metric	Value
Path	cities/catalogue/west-asia/Yemen/Mukalla
Recorded ISO / planning population	YE / 550,000
Family	light-metro-3car
Lines / stations	5 / 41
Route length	76.1 km
Fleet	254 trainsets
Routing-demand cells (legacy)	67%; not population access
City CAPEX incl. dedicated solar	USD 1.41bn (USD 19M / km); shared national factory excluded

Metric	Value
Residents within 800 m station circles	117,022 (2020; 45.0% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Mukalla network map

Line	Length	Stations	Trainsets	Shape
line-1	18.0 km	13	64	radial
line-2	19.0 km	8	61	radial
line-3	27.8 km	15	91	radial
line-4	4.4 km	2	15	radial
line-5	6.8 km	3	23	radial

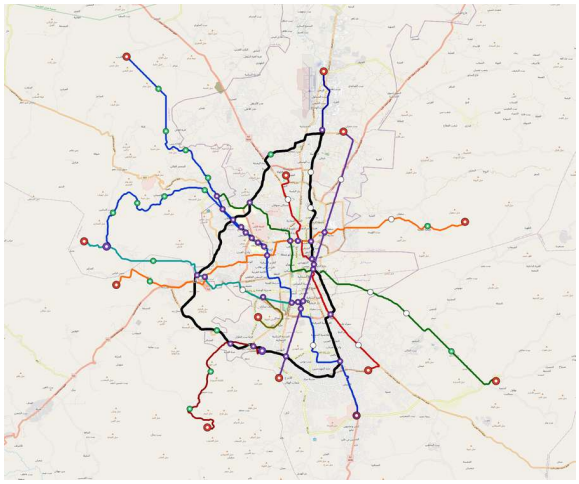
Generated City Model - Sanaa

west-asia / Yemen / Sanaa

Metric	Value
Path	cities/catalogue/west-asia/Yemen/Sanaa
Recorded ISO / planning population	YE / 3,937,500
Family	metro-6car
Lines / stations	11 / 160
Route length	289.3 km
Fleet	564 trainsets
Routing-demand cells (legacy)	71%; not population access
City CAPEX incl. dedicated solar	USD 5.69bn (USD 20M / km); shared national factory excluded
Residents within 800 m station circles	1,106,176 (2020; 39.5% of raster bbox)
Direct / reachable line pairs	43.6% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Sanaa network map

Line	Length	Stations	Trainsets	Shape
line-1	44.6 km	33	117	radial
line-2	22.9 km	19	69	radial
line-3	32.6 km	8	60	radial
line-4	22.1 km	8	40	radial
line-5	34.6 km	13	68	radial
line-6	23.5 km	14	56	radial
line-7	62.9 km	34	36	ring
line-8	5.1 km	2	11	radial
line-9	5.0 km	3	14	radial
line-10	15.0 km	9	35	radial
line-11	20.9 km	17	58	radial

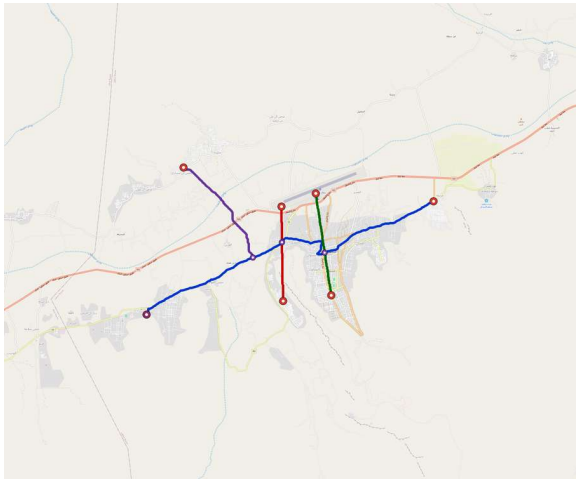
Generated City Model - Sayun

west-asia / Yemen / Sayun

Metric	Value
Path	cities/catalogue/west-asia/Yemen/Sayun
Recorded ISO / planning population	YE / 200,000
Family	tram-2car
Lines / stations	4 / 13
Route length	26.7 km
Fleet	63 trainsets
Routing-demand cells (legacy)	64%; not population access
City CAPEX incl. dedicated solar	USD 442M (USD 17M / km); shared national factory excluded
Residents within 800 m station circles	46,368 (2020; 35.3% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Sayun network map

Line	Length	Stations	Trainsets	Shape
line-1	14.1 km	5	29	radial
line-2	3.7 km	3	11	radial
line-3	4.2 km	3	12	radial
line-4	4.7 km	2	11	radial

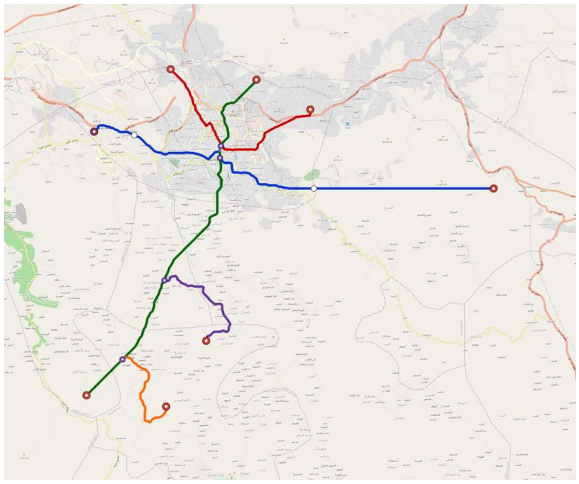
Generated City Model - Taiz

west-asia / Yemen / Taiz

Metric	Value
Path	cities/catalogue/west-asia/Yemen/Taiz
Recorded ISO / planning population	YE / 615,000
Family	light-metro-3car
Lines / stations	5 / 28
Route length	53.3 km
Fleet	172 trainsets
Routing-demand cells (legacy)	65%; not population access
City CAPEX incl. dedicated solar	USD 1.02bn (USD 19M / km); shared national factory excluded
Residents within 800 m station circles	306,919 (2020; 25.2% of raster bbox)
Direct / reachable line pairs	50.0% / 100.0%

Native station-circle populations are historical radial access, not verified walksheds or paid-trip estimates. Reachability includes intermediate lines; physical interchange access and timetable acceptance remain open.

City evidence: population and transfers · building, support and terrain clearance.



Taiz network map

Line	Length	Stations	Trainsets	Shape
line-1	18.4 km	7	59	radial
line-2	8.7 km	4	27	radial
line-3	15.9 km	11	50	radial

Line	Length	Stations	Trainsets	Shape
line-4	5.4 km	2	18	radial
line-5	5.0 km	4	18	radial